ClearPass Policy Manager 6.11 Online Help

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ClearPass Policy Manager 6.11 Online Help

Abstract

The ClearPass Policy Manager Online Help is available to assist Aruba customers and partners in securely configuring ClearPass within their unique deployment environment. The intended audience is customers, partners, and field System Engineers. The online help is not a training guide, and it is assumed the reader has a minimum foundational training in Policy Manager essentials and, if possible, Certified Policy Manager Professional (ACCP) certification.

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ClearPass Policy Manager Deployment Guide

The Policy Manager Deployment Guide is intended to assist field System Engineers, network administrators, customers and partners in deploying Policy Manager. This guide presents the recommended sequence in which Policy Manager deployment should take place, and makes the major deployment tasks as easy as possible to implement.

For information on installation and basic setup, tasks, including steps to Install and configure Policy Manager hardware and virtual appliances, refer to the ClearPass Installation Guide.

The Policy Manager Deployment Guide includes the following information:

- Prepare the Mobility Controller for integration with Policy Manager
- Integrate Policy Manager with Microsoft Active Directory
- Set up 802.1X wireless authentication with Active Directory
- Design and deploy Policy Manager clusters
- Integrating the ArubaOS switch with Policy Manager
- Integrating the Cisco Switch with Policy Manager
- Configure a Mobility Access Switch for 802.1X wired authentication
- Prepare Policy Manager for LDAP and SQL authentication
- 802.1X EAP-PEAP Reference

Subtopics

Intended Audience
Firewall Ports Recommended and Required to Be Open
Preparing the Controller or Gateway for Policy Manager Integration
Deploying Policy Manager Clusters
Preparing ClearPass for Active Directory Authentication
Preparing for 802.1X Wireless Authentication with Active Directory
Integrating the AOS Switch with ClearPass
Integrating with a Cisco Switch
New Central Integration
Mobility Access Switch Configuration for 802.1X Authentication
Preparing for LDAP and SQL Authentication Sources
802.1X EAP-PEAP Reference
Continued Guidance Configuration

Intended Audience

The intended audience for the Policy Manager Deployment Guide includes customers, partners, and field System Engineers.

This document is not a training guide, and it is assumed the reader has at minimum foundational training in Policy Manager essentials and, if possible, Aruba Certified Policy Manager Professional (ACCP) certification.

The user of this guide should have a working knowledge of the following:

- AAA technologies (RADIUS, TACACS, 802.1X, MAC address authentication, and Web authentication)
- Layer-2 and Layer-3 networking
- Microsoft Active Directory
- Switch technologies: ArubaOS, Cisco switches, Aruba Mobility Access Switch

NOTE

Providing information about network device configurations and capabilities is outside the scope of this guide. For information on these topics, refer to the documentation provided by the vendor of your network equipment.

Firewall Ports Recommended and Required to Be Open

Most ClearPass Policy Manager communication for updates is through HTTPS (TCP 443) to **clearpass.arubanetworks.com**. To allow devices to communicate over a network firewall, ensure communication between the (publisher/servers) is allowed to that URL over port 443.

Table 1 lists the ports recommended by Policy Manager to correctly operate within your environment, as well as the ports recommended to be open.

Table 1. Required Open Ports

Port	Protocol	Service/Application Use
Ports Required to Be Open		
22	TCP	Secure Shell (SSH)

Port	Protocol	Service/Application Use
123	UDP	Network Time Protocol (NTP)
443	TCP	HTTPS
1645	UDP	RADIUS (Authenticatio n)
1646	UDP	RADIUS (Accounting)
1812	UDP	RADIUS (Authenticatio n)
1813	UDP	RADIUS (Accounting)
5432	TCP	Database System, in cl uster deployments, Pos tgreSQL for DB replicat ion - encrypted using SSL, and time synchro nization
5433	TCP	Ensure port 5433 is op en in order to access t he Insight database. Po rt 5433 is also used fo r connecting to the tip sLogDb.

Table 2. Additional ports used by ClearPass

Port	Protocol
443	TCP
22	TCP
443	TCP
443	TCP

Port	Protocol
443	TCP
6658	TCP **NOTE** Recommended to be open
7432	TCP **NOTE** The **cluster diagnostics** command uses the ort number 7432 to collect diagnostics data
3799	TCP/UDP
49	TCP/UDP **NOTE** Recommended to be open
	ICMP echo (8) and echo - reply (0) between CPPM hc ng domain join
389	TCP/UDP
636	TCP/UDP
445	TCP/UDP
49152 - 65535	TCP
1025 - 5000	TCP

Port	Protocol
88	UDP
464	TCP
139	TCP
161	UDP **NOTE** Recommended to be open
162	UDP **NOTE** Recommended to be open
135	TCP
25	TCP
465	TCP
53	TCP
67	UDP **NOTE** Recommended to be open
2055	UDP
6343	UDP
514	UDP
2083	TCP

Preparing the Controller or Gateway for Policy Manager Integration

Use the following procedures to prepare the mobility controller or gateway to integrate with Policy Manager.

- Add a controller or gateway to Policy Manager: The mobility controller or gateway is responsible for managing access to the Wireless LAN.
- Add a Policy Manager/RADIUS Server to the controller or gateway and then assign the server to a server group: The Policy Manager server is a RADIUS server. You must add a Policy Manager/RADIUS server to the mobility controller or gateway because doing so allows Policy Manager to be integrated with the mobility controller or gateway and the wireless LAN authentication process.
- Configure an AAA profile on the Mobility controller or gateway for 802.1X authentication: The AAA profile configures the authentication for a Wireless LAN. The AAA profile defines the type of authentication (in this example, 802.1x), the authentication server group, and the default user role for authenticated users.
- Configure the Policy Manager RADIUS server as an RFC 3576 (CoA) Server: You can configure the Policy Manager RADIUS server to send user disconnect, change of authorization (CoA), and session-timeout messages,
- Add a SSID to the mobiity controller or gateway for 802.1X authentication: An SSID profile defines the name of the network, authentication type for the network, basic rates, transmit rates, SSID cloaking, and certain wireless multimedia settings for the network.
- Configure a Virtual AP (VAP) Profile: An access point uses a unique BSSID for each Wireless LAN. Thus, a physical AP can support multiple WLANs. The WLAN configuration applied to a BSSID on an AP is called a *virtual AP*. You can configure and apply multiple virtual APs to an AP group or to an individual AP by defining one or more *virtual AP profiles*. You can configure virtual AP profiles to provide different network access or services to users on the same physical network.

Subtopics

Adding a Mobility Controller or Gateway
Adding a RADIUS Server to the Mobility Controller
Configuring an AAA Profile for 802.1X Authentication
Configuring Policy Manager as an RFC 3576 (CoA) Server
Adding an SSID to the Mobility Controller for 802.1X Authentication

Adding a Mobility Controller or Gateway

About this task

The mobility controller or gateway is responsible for managing access to the Wireless LAN.

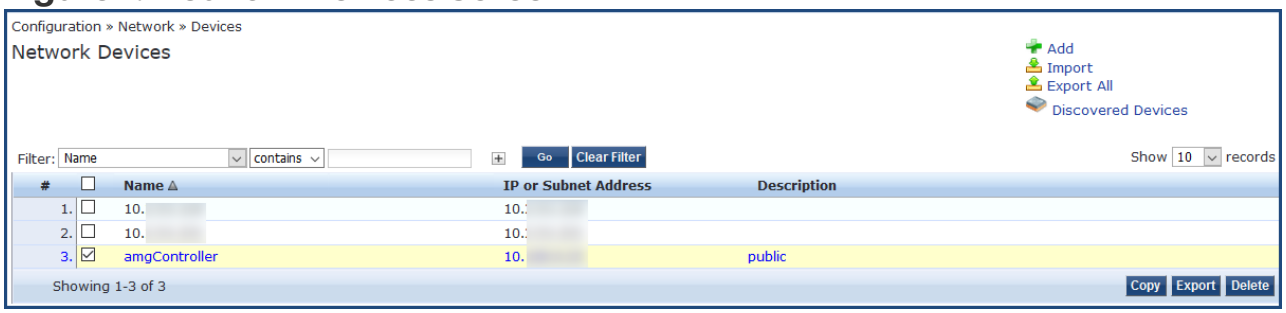
> **NOTE**
> You can use this procedure to add any network device from any vendor that supports RADIUS or TACACS+ to Policy Manager.

To define a new mobility controller or gateway in Policy Manager:

Procedure

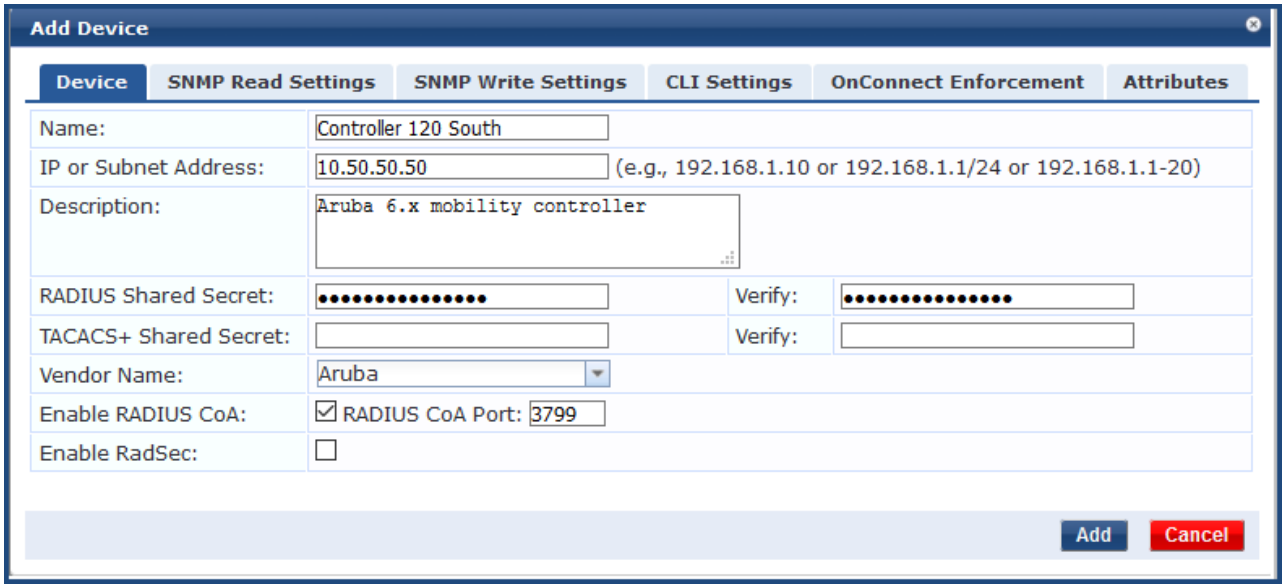
1. In Policy Manager, navigate to **Configuration > Network > Devices**. The **Network Devices** screen opens:

Figure 1. Network Devices Screen



2. Click the **Add** link. The **Add Device** page opens: You can also import a list of devices from a file. For details, see Importing a List of Network Devices .

Figure 2. Add Device Page > Device Tab



For a complete description of adding a device to ClearPass Policy Manager, refer to "Adding and Modifying Network Devices" in the ClearPass Policy Manager 6.11 User Guide:

3. Populate the **Network Device** parameters as described in Table 1 .

Table 1. Defining a Mobility Controller

Parameter	Action/Description
Name	Enter the name of the controller or gateway.
IP or Subnet Address	Enter the IP address or subnet address of the controller or gateway.
Description	Enter a description of the device (recommended).
RADIUS Shared Secret	Specify the RADIUS Shared Secret for the current Policy Manager server.

NOTE

Make sure that the value of the **Key** parameter for the RADIUS server configured on the controller or gateway is identical to the RADIUS Shared Secret you specify here for the current Policy Manager server (see).

Parameter	Action/Description
TACACS+ Shared Secret	If adding a device because you want Policy Manager to manage access to that device with TACACS+, specify the TACACS+ Shared Secret.
Vendor Name	From the drop - down, select the manufacturer of the controller or gateway.
Enable RADIUS CoA	To enable RADIUS - initiated Change of Authorization (CoA) on the controller or gateway, select the check box for this parameter. This parameter is enabled by default.
RADIUS CoA Port	If RADIUS CoA is enabled, this specifies the default port **3799**. Change this value only if you defined a custom port on the mobility controller or gateway. For related information, see Configuring Policy Manager as an RFC 3576 (CoA) Server.
Enable RadSec	To enable RadSec, click the **Enable Radsec** check box. When RadSec is enabled, the RADIUS shared secret is populated with a default shared secret named "*radsec*." **NOTE** It is important that the controller is configured with the same shared secret.

4. Click **Add**.

   You return to the **Network Devices** page. The new mobility controller or gateway is now present in the list of network devices.

Importing a List of Network Devices

About this task

To import a list of network devices from a file:

> **NOTE**
> The import file must be in XML format. See the next section for an example of the import file XML format.

Procedure

1. In Policy Manager, navigate to **Configuration > Network > Devices**. The **Network Devices** page opens.

   2. From the **Network Devices** page, click **Import**, then click **Import from file**. The **Import from File** dialog opens.

2. To browse to the file, click **Browse**.
3. Enter the shared secret if required, then click **Import**. The list of network devices is imported into Policy Manager.

Generating an Example of Import File XML Format

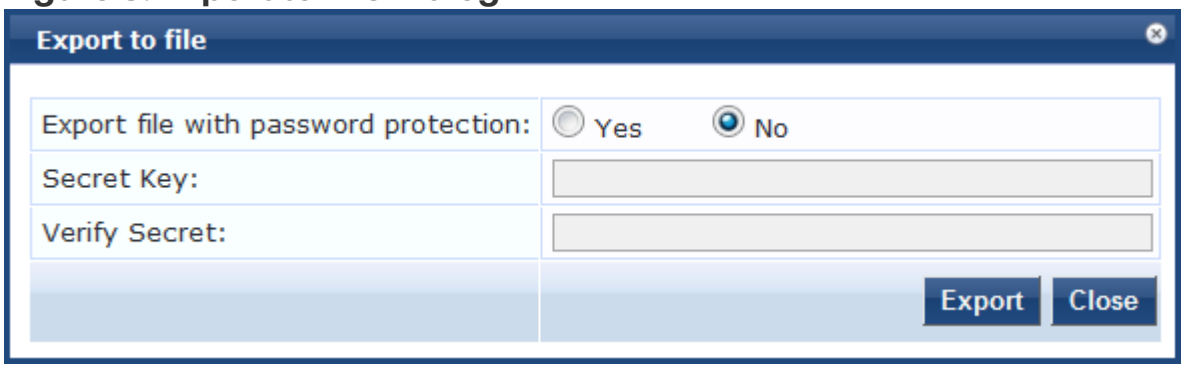
About this task

To generate an example of the import file XML format:

Procedure

1. From the **Network Devices** dialog, click the **Add** link. The **Add Device** dialog opens.
2. In the **Device** tab, define your network device, then click **Add**. You return to the **Network Devices** dialog, where the new device is listed.
3. Click **Export All**. The **Export to File** dialog opens.

Figure 3. Export to File Dialog



4. **Export file with password protection**: Select **No**.

5. Click **Export**.
6. Download the XML file to your system.
7. Open the XML file in a text editor to view the format.

Adding a RADIUS Server to the Mobility Controller

The Policy Manager server is a RADIUS server. You must add a Policy Manager/RADIUS server to the mobility controller or gateway because doing so allows Policy Manager to be integrated with the mobility controller or gateway and the wireless LAN authentication process. By adding the Policy Manager/RADIUS server and then associating the server to a server group the mobility controller or gateway, you are configuring the mobility controller or gateway to send authentication requests to the Policy Manager/RADIUS server.

Before you can reference the Policy Manager/RADIUS server in the configuration, you must add the Policy Manager/RADIUS server to a server group.

- You can add multiple RADIUS servers in a server group. You can configure the same server in more than one server group. Note that you must configure a server before you can include it in a server group. Server names must be unique.
- You can create groups of RADIUS servers for specific types of authentication—for example, you can specify one or more RADIUS servers to be used for 802.1x authentication.
- You can also configure servers of different types in one server group. For example, you can include the internal database as a backup to a RADIUS server.

NOTE

Even if there is only one RADIUS server, you must add it to a RADIUS server group.

Adding a RADIUS Server

The following sections describe the procedure to define the Policy Manager/RADIUS server in the mobility controller or gateway so that it can be used for any RADIUS authentication task. The procedure to add the Policy Manager/RADIUS server varies, depending upon the version of ArubaOS running on the mobility controller or gateway.

ArubaOS 8.x

About this task

For mobility controllers running ArubaOS 8.x:

Procedure

1. In the **Managed Network** node hierarchy, navigate to the C**onfiguration > Authentication > Auth Servers** tab.
2. In the **All Servers** table, click **+** to add a new server.
3. Enter a Name and IP address/ hostname for the new Policy Manager server and set the Type to RADIUS. Click Submit.
4. In the All Servers table, select the server created to configure server parameters.
5. Configure the server parameters, making sure to select the **Mode** check box to activate the authentication server. For more information, refer to the ArubaOS 8.x User Guides.

ArubaOS 6.x

About this task

For mobility controllers running ArubaOS 6.x:

Procedure

1. Navigate to the **Configuration > Security > Authentication > Servers** page.
2. Select **RADIUS Server** to display the RADIUS Server List.
3. To configure a Policy Manager RADIUS server, enter the name for the server and click **Add**.
4. Select the server name and configure server parameters, while making sure to select the **Mode** check box to activate the authentication server.

Adding the Policy Manager/RADIUS Server to a Server Group

The procedure to add the Policy Manager/RADIUS server to a server group varies, depending upon the version of ArubaOS running on the mobility controller or gateway.

ArubaOS 8.x

About this task

For mobility controllers or gateways running ArubaOS 8.x:

Procedure

1. In the **Mobility Master** node hierarchy, navigate to the **Configuration > Authentication > Auth Servers** tab.

2. The **Server Groups** table displays the server group list.
3. Click **+** in the **Server Groups**. Enter the name of the new server group and click Submit.
4. Select the new server group created.
5. In **Server Group <server group name>**, click the **Servers** tab and click **+** to add a server to the group.
6. To add a new server, select **Add new server**. Specify the **RADIUS** type from the **Type** drop-down list, and enter a Name and IP address/hostname for the Policy Manager server. For more information, refer to the ArubaOS 8.x User Guides.

ArubaOS 6.x

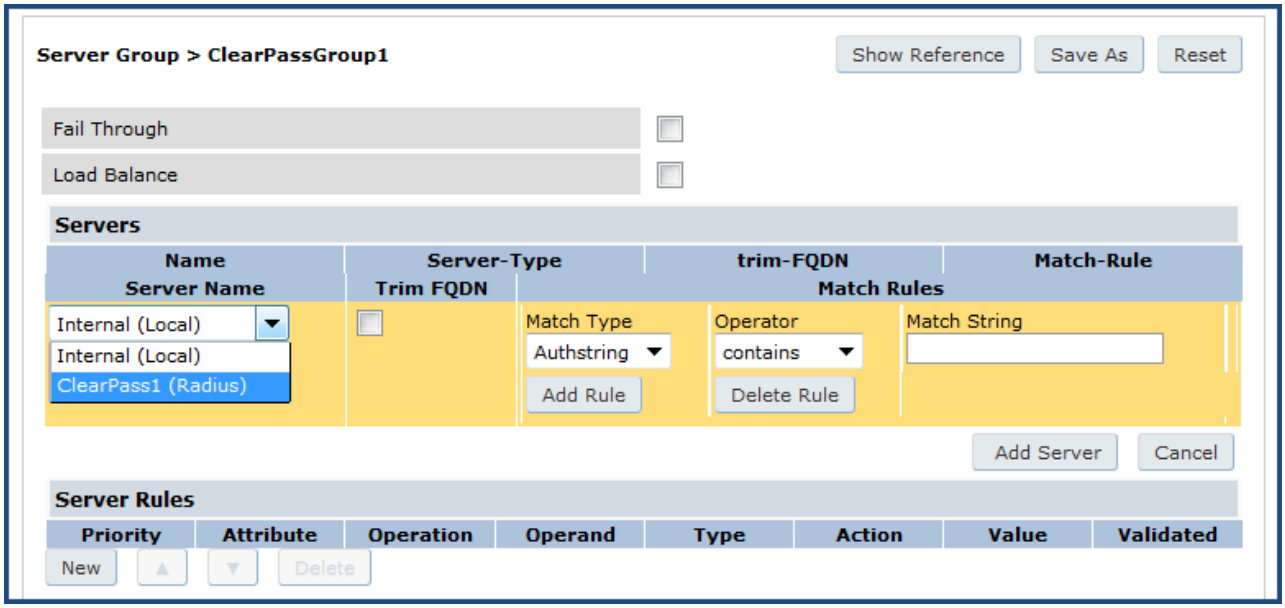
About this task

For mobility controllers running ArubaOS 6.x:

Procedure

1. On the mobility controller or gateway, select the **Configuration** tab.
2. In the navigation pane, select **SECURITY > Authentication**. The **Authentication > Servers** screen opens.
3. From the list of server types on the left side of the screen, select **Server Group**. The **Server Group** page opens.
4. To add a server group, enter the name of the server group in the **Add** field, then click **Add**. The new server group you defined is now included in the **Server Group** list.
5. To configure the server group, click the **name of the new server group**. The configuration screen for the selected server group opens.
6. To add a Policy Manager server to the server group, in the **Servers** section, click **New**. The **Servers** configuration screen opens.
7. To choose the Policy Manager server for inclusion in the RADIUS server group, select the Policy Manager (RADIUS) server name from the drop-down list (see Figure 3). The new RADIUS server name is now displayed in the **Server Name** list.

Figure 1. Selecting thePolicy ManagerServer for Inclusion in the RADIUS Server Group



8. If necessary, modify the **Servers** settings as needed, then click **Add Server**. You return to the **Server Group** configuration screen. The Policy Manager server is now included in the RADIUS server group.

9. Click **Apply**, then from the top of the screen, click **Save Configuration**. You have now defined the Policy Manager server as a RADIUS server, and the RADIUS server is a member of a RADIUS server group. These tasks are required before you can use the Policy Manager server as a RADIUS server in the network. For more information, refer to the ArubaOS 6.x User Guides.

Configuring an AAA Profile for 802.1X Authentication

AAA stands for *authentication*, *authorization*, and *accounting*. AAA is a framework that allows network administrators to access to computer resources, enforce policies, and audit usage. These processes all work together for effective network management and security.

With the RADIUS server and RADIUS server group configured, you can now configure an AAA profile that will refer to that server group, which, in turn, refers to a server in that group. The procedure to create a AAA profile on the mobility controller or gateway varies, depending upon the version of ArubaOS running on that device.

NOTE

Be sure to assign a unique name to each virtual AP, SSID, and AAA profile that you modify.

ArubaOS 8.x

About this task

For mobility controllers or gateways running ArubaOS 8.x:

Procedure

1. In the **Managed Network** node hierarchy, navigate to the **Configuration > Authentication > AAA Profiles** tab.
2. From the **AAA Profiles** list, select **AAA**.
3. To edit an existing AAA profile, select the AAA profile you want to edit. To create a new AAA profile, click **+** and enter a name for the new AAA profile in the **Profile name** field.
4. Configure the AAA profile parameters as necessary for the deployment. For more information, refer to the ArubaOS 8.x User Guides.

ArubaOS 6.x

About this task

For mobility controllers running ArubaOS 6.x:

Procedure

1. Navigate to **Configuration > Security > Authentication > Profiles**, then select the **AAA Profiles** tab.
2. Scroll down to the bottom of the **AAA Profiles Summary** pane, then click **Add**. An entry blank appears.
3. Enter the AAA profile name, then click **Add**.
4. In the profiles list, and select the AAA profile you just created.
5. Configure the AAA profile parameters as necessary for the deployment. For more information, refer to the ArubaOS 6.x User Guides.

Configuring Policy Manager as an RFC 3576 (CoA) Server

Introduction

You can configure the Policy Manager RADIUS server to send user disconnect, change of authorization (CoA), and session-timeout messages as described in RFC 3576, “Dynamic Authorization Extensions to Remote Dial In User Service (RADIUS).” The disconnect, session timeout, and change of authorization messages sent from the server to the mobility controller or gateway contain information to identify the user for whom the message is sent.

A mobility controller or gateway supports the following attributes for identifying the users who authenticate with an RFC 3576 server:

- **user-name**: name of the user to be authenticated
- **framed-ip-address**: user IPv4 address
- **framed-ipv6-address**: user IPv6 address
- **calling-station-id**: phone number of a station that originated a call
- **accounting-session-id**: unique accounting ID for the user session.

  If the authentication server sends both supported and unsupported attributes to the mobility controller or gateway, the unknown or unsupported attributes are ignored. If no matching user is found, the mobility controller or gateway sends a *503: Session Not Found* error message back to the RFC 3576 server.

Configuring the Server as a CoA Server

The procedure to configure the Policy Manager server as a CoA server varies, depending upon the version of ArubaOS running on your mobility controller or gateway:

NOTE

Before you configure any server as a CoA server, RADIUS CoA must be enabled on the device (for details, see Adding a Mobility Controller or Gateway to Policy Manager).

To enable communication between the mobility controller or gateway and the Policy Manager server, the values for RADIUS key configured on the Mobility master or controller and the RADIUS shared secret configured on the Policy Manager server must be identical.

ArubaOS 8.x

About this task

For ArubaOS 8.x deployments:

Procedure

1. In the **Mobility Master** node hierarchy, navigate to the **Configuration > Authentication > Auth Servers** tab.

2. To define a new RFC 3576 RADIUS server, click + under **All Servers**.
3. Select the Type as **Dynamic Authorization** from the drop-down list.
4. In **IP address version**, select either **IPv4** or **IPv6** radio button based on your preference.
5. In **IP address**, enter the IPv4 or IPv6 address of the Policy Manager server.
6. Click **Submit**.
7. From the **All Servers** list, select the server that you created to configure the server parameters.
8. Under **Server Options**, enter the server authentication key into the **Key** and **Retype key** fields.
9. Click **Submit**. For more information, refer to the ArubaOS 8.x User Guides.

ArubaOS 6.x

For ArubaOS 6.x deployments, access the mobility controller command-line interface and issue the **aaa rfc-3576-server** command.

```
aaa rfc-3576-server <ipaddr>
clone <source>
key <psk>
where
```

Parameter	Description
<ipaddr>	IP address of the Policy Manager server.
clone <source>	Name of an existing RFC 3576 server confi guration from which parameter values are copied
key <psk>	Shared secret to authenticate communicat ion between the RADIUS client and server.

For more information, refer to the ArubaOS 6.x User Guides.

Adding an SSID to the Mobility Controller for 802.1X Authentication

An SSID (Service Set Identifier) is the name of the network or Wireless LAN that clients see. An SSID profile defines the name of the network, authentication type for the network, basic rates, transmit rates, SSID cloaking, and certain wireless multimedia settings for the network. ArubaOS supports different types of the Advanced Encryption Standard (AES), Temporal Key Integrity Protocol (TKIP), and wired equivalent privacy (WEP) encryption.

AES is the most secure and the recommended encryption method. Modern devices are AES-capable, and therefore AES should be the default encryption method. Use TKIP only when the network includes devices that do not support AESn and n these situations, use a separate SSID for devices that are only capable of TKIP.

The procedure to create a WLAN using the ArubaOS WLAN wizard on the mobility controller or gateway varies, depending upon the version of ArubaOS running on that device.

ArubaOS 8.x

For Mobiilty Master, managed devices, mobility controllers or gateways running ArubaOS 8.x, navigate to the **Configuration > WLANs** section of the ArubaOS 8.x WebUI and create a new SSID using the **New WLAN** wizard. This wizard will walk you through the basic steps for adding a WLAN SSID.

ArubaOS 6.x

For mobility controllers running ArubaOS 6.x, navigate to the **Configuration > WIZARDS > Campus WLAN** section of the ArubaOS 6.x WebUI to create a new SSID using the Campus WLAN wizard. Help is available for this wizard using the **Help** tab in the WebUI.

Configuring a Virtual AP Profile

Access points (APs) advertise Wireless LANs to wireless clients by sending out beacons and probing responses that contain the Wireless LAN's SSID and the supported authentication and data rates. When a wireless client associates to an AP, it sends traffic to the AP's Basic Service Set Identifier (BSSID), which is usually the AP's MAC address.

An access point uses a unique BSSID for each Wireless LAN. Thus, a physical AP can support multiple WLANs. The WLAN configuration applied to a BSSID on an AP is called a *virtual AP*. You can configure and apply multiple virtual APs to an AP group or to an individual AP by defining one or more *virtual AP profiles*. You can configure virtual AP profiles to provide different network access or services to users on the same physical network.

- For example, you can configure a Wireless LAN to provide access to guest users and another WLAN to provide access to employee users through the same APs.
- You can also configure a Wireless LAN that offers open authentication and Captive Portal access with data rates of 1 MBps (Megabytes per second) and 2 MBps, and another Wireless LAN that requires Wi-Fi Protected Access (WPA) authentication with data rates of up to 11 MBps.

  As an example, suppose there are users in both Edmonton and Toronto that access the same "Corpnet" Wireless LAN.

  If the Wireless LAN required authentication to an external server, users who associate with the APs in Toronto would want to authenticate with their local servers. In this case, you can configure two virtual APs that each reference a slightly different AAA profile—one AAA profile that references authentication servers in Edmonton and the other AAA profile that references servers in Toronto (see Table 1).

  When you create a Wireless LAN using the WLAN wizard, the mobility controller or gateway automatically creates a Virtual AP profile (VAP) based on the Wireless LAN's configuration. You can apply multiple virtual AP profiles to individual APs. You can also apply the same virtual AP profile to one or more AP groups.

NOTE

The name the mobility controller assigns to the VAP is the name of the WLAN with "*-vap_prof*" appended to the name. For example, the VAP for a Wireless LAN named "802.1X-CP" would be named "802.1X-CP-vap_prof."

Table 1. Applying WLAN Profiles to AP Groups

Profile	Default AP Group	Toronto AP Group
Virtual AP	Corpnet - Ed	Corpnet - Tr
SSID	Corpnet	Corpnet
AAA	Ed - Servers	Tr - Servers

The procedure to create a WLAN using the ArubaOS WLAN wizard on the mobility controller or gateway varies, depending upon the version of ArubaOS running on that device.

ArubaOS 8.x

About this task

For mobility controllers or gateways running ArubaOS 8.x:

Procedure

1. In the **Managed Network** node hierarchy, navigate to the **Configuration > System > Profiles** tab.
2. From the **All Profiles** list, select **Wireless LAN > Virtual AP**.
3. To edit an existing Virtual AP profile, select the Virtual AP profile you want to edit. To create a new Virtual AP profile, click **+** and enter a name for the new Virtual AP profile in the **Profile name** field.
4. Configure the profile parameters. For more information, refer to the ArubaOS 8.x User Guides.

ArubaOS 6.x

About this task

For mobility controllers running ArubaOS 6.x:

Procedure

1. Navigate to the **Configuration > ADVANCED SERVICES > All Profiles** section of the ArubaOS 6.x WebUI
2. Expand the *Wireless LAN* profile and select **Virtual AP**. The list of existing Virtual AP profiles appears in the **Profiles** pane.
3. To edit an existing Virtual AP profile, select the name of the profile in the **Profile Details** pane. To create a new Virtual AP profile, enter a name for the profile in the entry field at the bottom of the **Profile Details** pane, click **Add**, then select the name of the new profile in the **Profile Details** pane
4. Configure the profile parameters. For more information, refer to the ArubaOS 6.x User Guides.

Deploying Policy Manager Clusters

A *cluster* is a logical connection of any combination of Policy Manager hardware or virtual appliances. This section of the deployment guide provides guidance on how to

design and deploy Policy Manager clusters, how to complete major tasks such as adding a Subscriber server and deploying a standby Publisher, as well as how to rejoin a down server to the cluster. Finally, the set of cluster-specific CLI commands is included.

Subtopics

Policy Manager Cluster Overview
Policy Manager Databases
Publisher/Subscriber Model
Network Ports That Must Be Enabled
Cluster Scaling Limitations
Cluster Design Considerations
About Large Scale Deployments
Deploying the Standby Publisher
Adding a Subscriber to the Publisher
Rejoining a Down or Disabled Server to the Cluster
Deploying Policy Manager Insight in a Cluster
Configuring Cluster File-Backup Servers
Cluster CLI Commands

Policy Manager Cluster Overview

Policy Manager can be deployed either as a dedicated hardware appliance or a virtual machine running on top of VMware vSphere Hypervisor or Microsoft Hyper-V.

When demand exceeds the capacity of a single instance, or you have a requirement for a High Availability deployment, you have the option of logically joining multiple instances to process the workload from the network. You can logically join physical and virtual instances and also join Policy Manager instances that are dissimilar in size. However, careful planning must be taken, especially if you plan to utilize the failover capabilities within the clustering feature. The cluster feature allows for shared configuration and databases. However, it does not provide a virtual IP address for the cluster, so failover/redundancy for captive portal for Guest relies on Domain Name System (DNS) lookup or load balancing. RADIUS clients must define a primary and backup RADIUS server.

Subtopics

Authentication Requests in a Cluster

Authentication Requests in a Cluster

The typical use case for Policy Manager is to process authentication requests using the policy framework. The policy framework is a selection of services that work to process

authentication requests, but the policy framework also determines authentication, authorization, posture, enforcement, role, etc. of the endpoint/end-user.

In the context of cluster operations, authentication typically involves a read-only operation from the configuration database. A cluster server receives an authentication request, determines the appropriate policies to apply, and responds appropriately. This does not require a configuration change, and can therefore be scaled across the entire cluster.

NOTE

Authentication is performed from the server itself to the configured identity store, whether locally (as synchronized by the Publisher; for example, a Guest account) or externally, such as with Microsoft Active Directory.

Logs relevant to each authentication request are recorded separately on each server, using that server's log database. Centralized reporting is handled by generating a Netevent from the server, which is sent to all Insight servers and recorded in the Insight database (for related information, see Deploying Policy Manager Insight in a Cluster).

Policy Manager Databases

Each Policy Manager server makes use of the following databases:

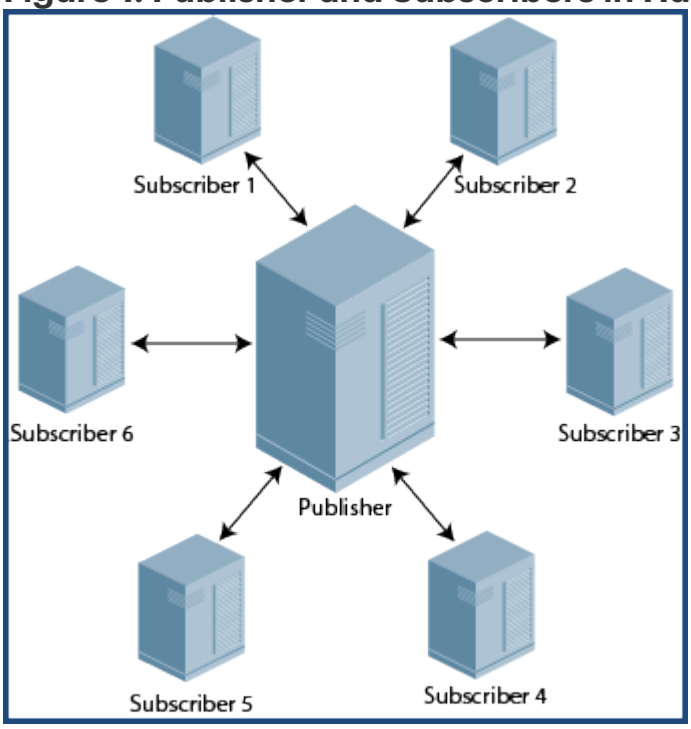
- **Configuration database**. Contains most of the editable entries that can be seen in the Policy Manager user interface. This includes, but is not limited to:
  - Administrative user accounts
  - Local user accounts
  - Service definitions
  - Role definitions
  - Enforcement policies and profiles
  - Network access devices
  - Guest accounts
  - Onboard certificates
  - Most of the configuration shown within Guest and Onboard
- **Log database**. Contains activity logs generated by typical usage of the system. This includes information shown in Access Tracker and the Event Viewer.

- **Insight database**. Records historical information generated by the Netevents framework. This database is used to generate reports.

Publisher/Subscriber Model

Policy Manager uses a Publisher/Subscriber model to provide multiple-box clustering. Another term for this model is *hub and spoke*, where the hub corresponds to the Publisher, and the spokes correspond to the Subscribers.

Figure 1. Publisher and Subscribers in Hub and Spoke Configuration



- The **Publisher server** functions as the master controller in a cluster. The Publisher is your central point of configuration, monitoring, and reporting. It is also the central point of database replication. All the databases are managed through the Publisher.
    - There is at most one active Publisher in this model, and a potentially unlimited number of Subscribers.
    - The Publisher server has full read/write access to the configuration database. All configuration changes must be made on the Publisher. The Publisher server sends configuration changes to each Subscriber server.
- The **Subscriber servers** are worker servers. All the AAA load, all RADIUS requests, and the server where policy decisions are being made are on the Subscriber servers.

- Subscriber servers maintain a local copy of the configuration database, and each Subscriber has read-only access to a local copy of the configuration database.

  Network Address Translation (NAT) is not supported between the Publisher and Subscriber servers.

Subtopics

What Information Is Replicated?

What Information Is Replicated?

A background replication process handles the task of updating the configuration database based on the configuration changes received from the Publisher.

Multiple entities exist within a Policy Manager server cluster that must be shared to ensure successful operation of the cluster. Only the configuration database is replicated.

NOTE

The Log and Insight databases are not replicated across the cluster.

However, certain elements are server-specific and these must be configured separately for each server, which you can achieve directly on the Publisher or individually on the Subscriber server.

Subtopics

Elements Replicated
Elements Not Replicated

Elements Replicated

Cluster replication is delta-based; that is, only changed information is replicated.

The cluster elements that are replicated across all the servers in the cluster are as follows:

- All policy configuration elements
- All audit data
- All identity store data

    - Guest accounts, endpoints, and profile data
- Runtime information
    - Authorization status, posture status, and roles
    - Connectivity information, NAS details
- Database replication on port 5432 over SSL
- Runtime replication on port 443 over SSL

Elements Not Replicated

The following elements are not replicated:

- Access Tracker logs and Session logs
- Authentication records
- Accounting records
- System events (Event Viewer data)
- System monitoring data

Network Ports That Must Be Enabled

Table 1 lists the network ports that must be opened between the Publisher and the Subscriber servers.

Table 1. Network Ports to Be Enabled

Port	Protocol	Desc
80	HTTP	Inter
123	UDP	TNTF
443	TCP	HTTF - se
5432	TCP	Post

Because any Subscriber server can be promoted to be the Publisher server, all port/protocol combinations listed in Table 1 should be:

- Bidirectional
- Open between any two servers in the cluster

Cluster Scaling Limitations

Due to the design requirements of the cluster Publisher/Subscriber model, various Policy Manager components scale differently (see Table 2).

Table 1. Policy Manager Cluster Scaling Limitations

Component	Scaling Limitation
Authentication capacity	Scales linearly according to the number of Subscriber servers. Add more servers as necessary to provide additional capacity to service authenticatio n requests.
Configuration changes (Guest/ Onboard)	These configuration changes do not scale with additional servers as they are centrali zed. Requires the Publisher be scaled to suppo rt write traffic from the maximum number of Subscribers that would be active concur rently.
Configuration changes (Policy Manager)	As the total size of the configuration set is bounded, these configuration changes are assumed to be infrequent and therefore no t a significant limit to scaling.
Insight reports	Because this function is centralized, reporti ng does not scale with additional servers. Use a separate Insight server sufficient to h andle the incoming Netevents traffic from all servers in the cluster. In a very large - s cale deployment, the Publisher server sho uld not be used as the Insight reporting se rver.

Component	Scaling Limitation
Logging capacity	Scales linearly according to the number of Subscriber servers, as each server handles its own logging operations.
Replication load on Publisher	Scales linearly according to the number of Subscriber servers. The replication is effici ent as only changed information is sent.

Cluster Design Considerations

This section contains the following information:

- Cluster Deployment Sizing Guidance
- Publisher Node Guidelines
- Subscriber Node Guidelines
- Providing Sufficient Bandwidth Between Publisher and Subscribers
- Round-Trip Time Considerations for Geographically Distributed Clusters
- Implementing Zones for Geographical Regions
- Maximum DB Connection Considerations
- Synchronize Cluster Password
- IPSec Cluster Considerations

  This section contains recommendations on how to optimize the Publisher and Subscriber constraints when deploying a Policy Manager cluster.

Cluster Deployment Sizing Guidance

About this task

Cluster deployment sizing should not be based on raw performance numbers.

> **NOTE**
> In ClearPass 6.11.0 and later, the maximum single cluster size is 32 servers. This includes the publisher, standby publisher, subscribers, dedicated insight server, and standby insight servers.

> **NOTE**
> When designing large clusters, we recommend contacting your Aruba account team to discuss options. For details on selecting ClearPass servers, refer to the **Policy Manager Ordering and Sizing Guide**, available for download from the **HPE Networking Support Portal**.

To determine the optimum sizing for a Policy Manager cluster:

Procedure

1. Determine how many endpoints need to be authenticated.
   a. The number of authenticating endpoints can be determined by taking the number of users times the number of devices per user.
   b. To this total, add the other endpoints that just perform MAC authentication, such as printers and other non-authenticating endpoints.
2. Take into account the following factors:
   a. Number and type of authentications and authorizations:
      - MAC authentication/authorizations vs. PAP vs. EAP-MSCHAPv2 vs. PEAP-MSCHAPv2 vs. PEAP-GTC vs. EAP-TLS
      - Active Directory vs. local database vs. external SQL datastore
      - No posture assessment vs. in-band posture assessment in the PEAP tunnel vs. HTTPS-based posture assessment done by OnGuard.
   b. RADIUS accounting load.
   c. Operational tasks taking place during authentications, such as configuration activities, administrative tasks, replication load, periodic report generation, and so on.
   d. Disk space consumed.

Note that Policy Manager writes copious amounts of data for each transaction (this data is displayed in the Access Tracker).

3. Then pick the number of Policy Manager hardware appliances you would need, with redundancy ranging from (N+1) to full redundancy, depending on the needs of the customer.

Publisher Guidelines

Setting Up a Standby Publisher

Policy Manager allows you to designate one of the Subscriber servers in a cluster to be the *Standby Publisher*, thereby providing for that Subscriber to be automatically promoted to active Publisher status in the event that the Publisher goes out of service. This ensures that any service degradation is limited to an absolute minimum. For details, see Deploying the Standby Publisher .

Publisher Sizing

The Publisher server must be sized appropriately because it handles database write operations from all Subscribers simultaneously. The Publisher must also be capable of handling the total-number of endpoints within the cluster and be capable of processing remote work directed to it when guest-account creation and onboarding are occurring.

Publisher Deployment Guidance

- In a world-wide large-scale deployment, not all Subscriber servers are equally busy. To determine the maximum request rate that must be handled by the Publisher server, examine the cluster's traffic pattern for busy hours and estimate the traffic load for each Subscriber server, adjusting for time zone differences.
- In a large-scale deployment, isolate the Publisher server, to allow it to handle the maximum amount of traffic possible.
- To help reduce the maximum amount of traffic possible in a large-scale deployment (ignoring API requests from Subscribers as well as the outbound replication traffic to Subscribers), the Publisher should not receive any authentication requests or Guest/Onboard requests directly .
- If the worker traffic sent from the Subscriber servers is expected to fully saturate the capacity of the Publisher server, Insight should not be enabled on the Publisher server. If the Publisher server has spare capacity, it can be used to support the Policy Manager Insight database. However, take care to carefully monitor the Publisher server's capacity and performance.

Subscriber Guidelines

Guidelines for a Subscriber deployment are as follows:

- Use The Nearest Subscriber. Guests and Onboard clients should be directed to the nearest Subscriber server. From the client's point of view, the internal API call to the Publisher is handled transparently. The best response times for static resources is obtained if the server is nearby.
- Use Subscribers as Workers. Subscriber should be used as workers that process the following:
  - Authentication requests (for example, RADIUS, TACACS+, Web-Auth)
  - Online Certificate Status Protocol (OCSP) requests
  - Static content delivery (for example, images, CSS, JavaScript)
- Avoid Sending Worker Traffic to the Publisher. Best practices is to avoid sending "worker traffic" to the Publisher, as the Publisher services API requests from Subscribers, handles the resulting database writes, and generates replication changes to send back to the Subscribers.
- Verify OSCP checks for EAP-TLS If Onboard is Being Used. If Onboard is used, ensure that the EAP-TLS authentication method in Policy Manager is configured to perform */localhost* OCSP (Online Certificate Status Protocol) checks.
- When a configuration backup file from an earlier ClearPass version that includes a virtual IP (VIP) address is restored on a 6.11 installation, the restoration is successful on the standalone ClearPass server and includes the VIP address. However, when a subscriber is rejoined to the cluster the VIP address is not reflected in the subscriber's configuration because it is not enabled by default. After the subscriber has rejoined the cluster, go to **Administration > Server Manager > Server Configuration > Virtual IP Settings**. Select the subscriber in the **Secondary Node** drop-down list, select the **Enabled** check box, and then **Save**.

Providing Sufficient Bandwidth Between Publisher and Subscribers

In a large-scale deployment, reduced bandwidth or high latency on the link (greater than 200 ms) delivers a lower-quality user experience for all users of that Subscriber, even though static content is delivered locally almost instantaneously.

For reliable operation of each Subscriber, ensure that there is sufficient bandwidth available for communications with the Publisher. For basic authentication operations, there is no specific requirement for high bandwidth. However, the number of round-trips to complete an EAP authentication could cause delay for the end user.

Traffic Flows Between Publisher and Subscriber

The traffic flows between the Publisher and Subscriber servers include:

- **Basic monitoring of the cluster**. Monitoring operations generate a small amount of traffic.
- **Time synchronization for clustering**. Generates standard Network Time Protocol (NTP) traffic.
- **Policy Manager configuration changes**. This is not a significant bandwidth consumer.
- **Zone Cache**. The amount of traffic depends on the authentication load and other details of the deployment. Cached information is metadata and is not large. This data is replicated only within the Policy Manager zone.
- Guest/Onboard dynamic content proxy requests. This is essentially a web page and averages approximately 100 KB.
- Guest/Onboard configuration changes. Only the changes to the database configuration are sent, and this information is typically small in size (approximately 10 KB).

Round-Trip Time Considerations for Geographically Distributed Clusters

It's important to take the delay between a Policy Manager server and a NAD/NAS (a controller or switch) into consideration when building geographically distributed clusters.

In a large geographically dispersed cluster, the worst case round-trip time (RTT) between a NAS /NAD and all potential servers in the cluster that might handle authentication is a design consideration.

- Aruba recommends that the round-trip time between the NAD/NAS and a Policy Manager server should not exceed 600 ms.
- The acceptable delay between cluster servers is less than 100 ms (RTT less than 200 ms).
- The link bandwidth should be greater than 10 Mbps.

  It's possible to configure a NAD/NAS to point at multiple RADIUS servers, either for load balancing or failover.

  For example, a NAD/NAS in Paris could point to a Policy Manager server in London as a backup RADIUS server. That's not a problem as long as the round-trip time guidelines are adhered to.

Implementing Policy Manager Zones for Geographical Regions

Policy Manager zones exist to control the replication of information between servers in a cluster. Included in this control is the replication of the *Zone Cache*, which contains the endpoints' run-time state information.

The Zone Cache is replicated across all servers in a zone—not all servers in the cluster. If zoning has not been configured, traffic flows between the Publisher and Subscriber servers, as well as between all the Subscriber servers in the cluster.

Run-Time Information

The run-time state information includes:

- Roles and postures of the connected entities
- Connection status of all endpoints running OnGuard
- Machine authentication state
- Session information used for Change of Authorization (CoA)
- Information about which endpoints are on which NAS/NAD

  Policy Manager uses run-time state information to make policy decisions across multiple transactions.

When a Cluster Spans WAN Boundaries and Geographic Zones

In a deployment where a cluster spans WAN boundaries and multiple geographic zones, it's not necessary to share run-time state information across all the servers in the cluster.

For example, endpoints present in one geographical area are not likely to authenticate or be present in another area. It's therefore more efficient from a network usage and processing perspective to restrict the sharing of such run-time state information to a specific geographical area.

Certain cached information is replicated only on the servers within a Policy Manager zone. In a large-scale deployment with multiple geographical areas, multiple zones should be used to reduce the amount of data that needs to be replicated over a wide-area network.

Zones and the Persistent Agent

A persistent agent attempts to establish communications with a Policy Manager server in the same zone; if that is not possible, it contacts a server in another zone.

Zone configurations allow for fairly deterministic control of where the persistent agent will send its health information. At minimum, the agent health information should go to a server in the same zone as the authentication request.

From a design perspective, for large geographically dispersed deployments, the design goal should be for agent health information and authentication requests to be sent to the same cluster server. Targeting authentication requests to a specific server is easily accomplished with NAS configuration.

Creating Geographical Zones in Policy Manager

About this task

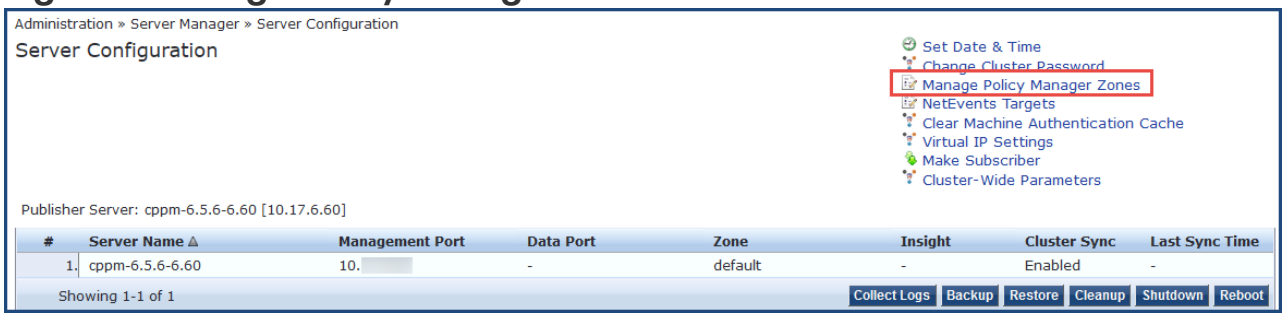
You can configure zones in Policy Manager to match with the geographical areas in your deployment. You can define multiple zones per cluster. Each zone has a number of Policy Manager servers that share their runtime state.

To create geographical zones in Policy Manager:

Procedure

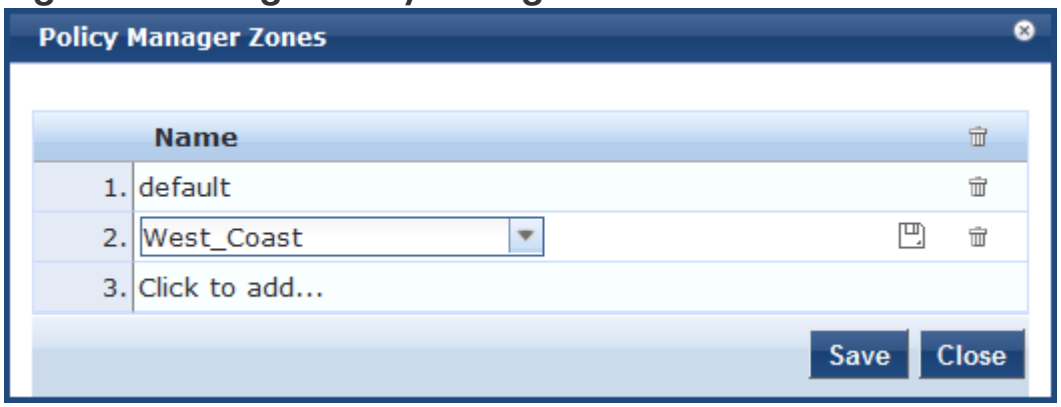
1. Navigate to the **Administration > Server Manager > Server Configuration** page.

Figure 1. Manage Policy Manager Zones Link



2. Click the **Manage Policy Manager Zones** link. The Policy Manager Zones dialog opens.
3. Select **Click to add...**. A blank field appears in the dialog.

Figure 2. Adding a Policy Manager Zone



4. Enter the name of the new Policy Manager zone.
5. To create additional Policy Manager zones, repeat Steps 3 and 4.
6. When finished, click **Save**. You see the message, "*Policy Manager Zones modified successfully.*"

Zone Deployment Guidance

About this task

Guidance for deploying Policy Manager zones is as follows:

Procedure

1. In a large-scale deployment, create one Policy Manager zone for each major geographical area of the deployment.
2. To handle RADIUS authentication traffic in each region, configure the region's networking devices with the Policy Manager servers in the same zone.
3. If additional authentication servers are required for backup, you can specify one or more Policy Manager servers located in a different zone, but Arubarecommends that you deploy remote servers that have the best connection, that is, the lowest latency, highest bandwidth, and highest reliability.
4. There may be cases in which the RADIUS server on the network infrastructure is configured to use remote Policy Manager server servers that are outside of their primary geographic area.

   In this scenario, the replication of the runtime states might be relevant. Consider this behavior during the design and deployment of a distributed cluster of Policy Manager server servers.

Maximum DB Connection Considerations

HPEAruba Networking recommends loading appliances (C30XX=2000 , C20XX,C10XX=400) according to their specific capacity.

Setting the max_connection = 2000 for a C1000 model does not impact the performance of the appliance, as long as the C1000's load is within its recommended capacity limits.

Synchronize Cluster Password

About this task

In deployments where a change cluster password operation failed for a Subscriber, HPE Aruba Networking recommends using the existing ClearPass Policy Manager **Synchronize Cluster Password** functionality to update the Subscriber with the cluster password configured in the Publisher.

NOTE

The following procedure should be on the Subscriber in which the cluster password update failed for it not to enter an OUT_OF_SYNC cluster state.

To synchronize the Subscriber's cluster password:

Procedure

1. Login to the UI of the Subscriber. This is the Subscriber where the cluster password change failed.
2. Navigate to **Administration > Server Manager > Server Configuration**.
3. In the table containing the list of servers in cluster, click on the row corresponding to the Subscriber with the failed password update.
4. Select the **Synchronize Cluster Password** link from the top, right-hand, side of the screen. A pop-up window displays that navigates to the destination required to update the Subscriber cluster password.

IPSec Cluster Considerations

When configuring IPSec in a cluster deployment, be sure to configure the Subscriber first, then the Publisher. Additionally, remove the IPSec tunnel before updating or upgrading the current Policy Manager version. When removing the configuration, remove the Publisher first, then the Subscriber. Once the update or upgrade operation is completed, reconfigure as required for the deployment.

About Large Scale Deployments

This section contains the following information:

- What Is a Large Scale Deployment?
- Design Guidelines
- Examples of Medium-Scale and Large-Scale Cluster Deployments

Subtopics

What Is a Large Scale Deployment?
Design Guidelines
Examples of Medium-Scale and Large-Scale Cluster Deployments

What Is a Large Scale Deployment?

Large-scale deployments are defined as those clusters that require the Publisher server to be dedicated to servicing the Subscriber servers.

This occurs when the volume of configuration changes generated by all the Subscriber servers in the cluster limits the Publisher server's capacity to handle other important tasks, such as authentication.

Note that not every clustering scenario is a large-scale deployment. Policy Manager clustering can also be performed for other reasons, for example, to distribute several Policy Manager servers geographically for policy reasons, or to have an off-site disaster recovery system.

Design Guidelines

About this task

Policy Manager cluster design guidelines are as follows:

Procedure

1. The dedicated Publisher should be a Policy Manager C3000 or c30100 hardware appliance or a Policy Manager virtual appliance that matches the minimum specification for the C3000/C3010 virtual appliance.
2. Configuration changes that should be considered in the context of a large-scale deployment include:
    - Creating, modifying, or deleting a guest account.
    - Issuing or revoking an Onboard certificate.
    - Modifying Policy Manager configuration; for example, adding a network access device, defining a new service, and updating an enforcement profile.
    - Adding new endpoints (including automatically created endpoints) in Policy Manager.
    - Making modifications to guest accounts or endpoint records with a Policy Manager post-authentication profile.

Examples of Medium-Scale and Large-Scale Cluster Deployments

This section provides two examples of typical customer cluster deployments.

Subtopics

Authenticating Corporate Users with Guest Access (Medium-Scale Cluster Deployment)
Authenticating Conference Center Users (Large-Scale Cluster Deployment)
Recommendations for a Dedicated Publisher

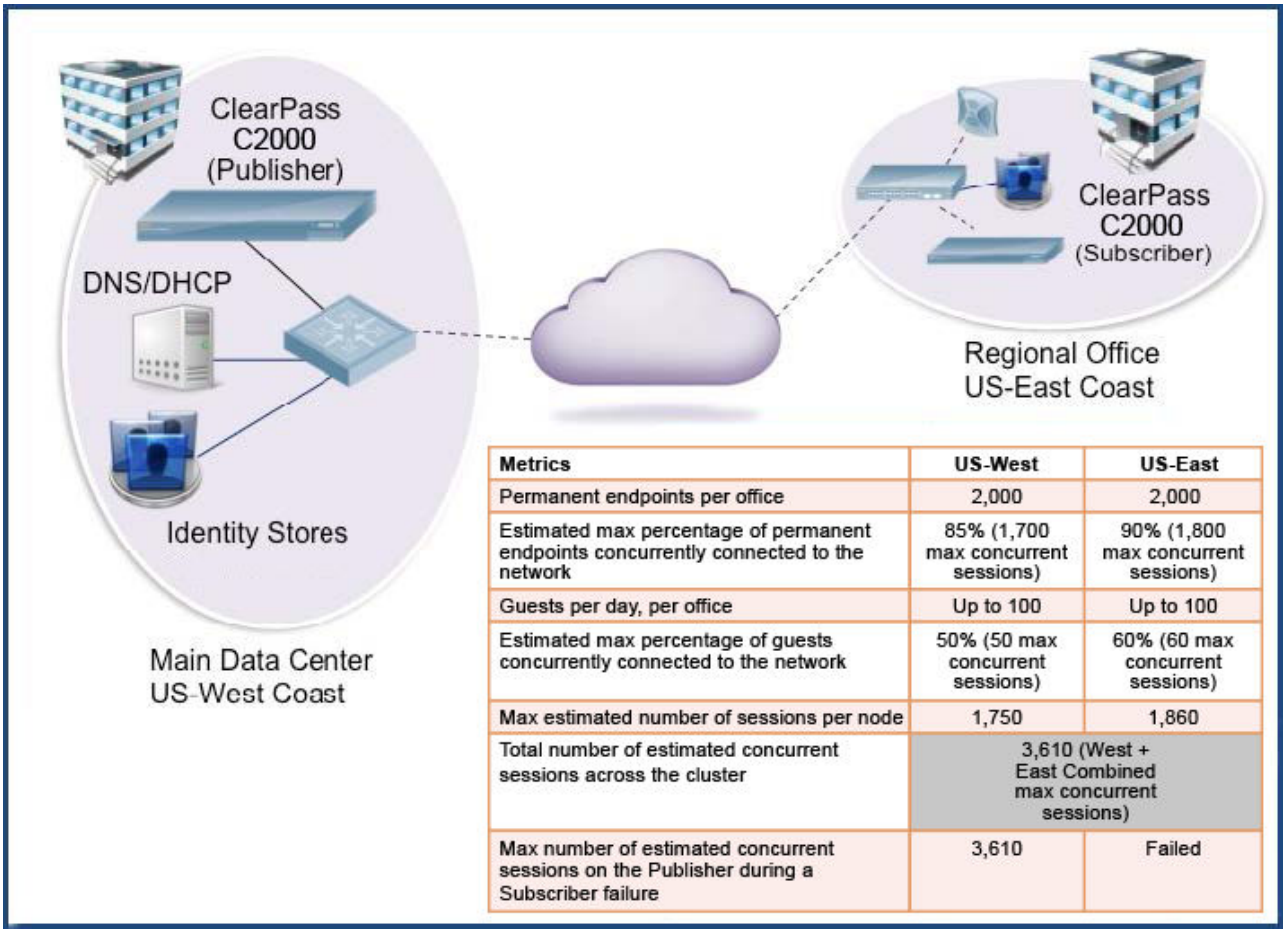
Authenticating Corporate Users with Guest Access (Medium-Scale Cluster Deployment)

In this example, a cluster of Policy Manager C2000 hardware appliances has two servers —U.S. East Coast and U.S. West Coast (see Figure 1).

- *US-West* is the Publisher server.
- *US-East* is the Subscriber server.
- Each server handles the authentication traffic for 2,000 corporate endpoints. Each server also registers 100 guests per day.
- There are few configuration updates in the network.

  In this example, each server could be used as the backup for the other server. In the event of a server failure, the other server could handle the authentication requirements of all 4,000 endpoints in addition to 200 guest registrations per day.

Figure 1. Example of a Medium-Scale Cluster Deployment



Metrics	US-West	US-East
Permanent endpoints per office	2,000	2,000
Estimated max percentage of permanent endpoints concurrently connected to the network	85% (1,700 max concurrent sessions)	90% (1,800 max concurrent sessions)
Guests per day, per office	Up to 100	Up to 100
Estimated max percentage of guests concurrently connected to the network	50% (50 max concurrent sessions)	60% (60 max concurrent sessions)
Max estimated number of sessions per node	1,750	1,860
Total number of estimated concurrent sessions across the cluster	3,610 (West + East Combined max concurrent sessions)	
Max number of estimated concurrent sessions on the Publisher during a Subscriber failure	3,610	Failed

This fictitious customer example would not be considered a large-scale cluster deployment, for the following reasons:

- The additional load on the Publisher due to clustering can be estimated at 100 guest accounts created per day.
- The authentication traffic on the Subscriber server does not impose any additional load on the Publisher and the new endpoints registered (in the order of 100 per day, assuming new guests each day) does also not add any significant load.
- The workload on the Publisher is small and represents a fraction of its capacity.

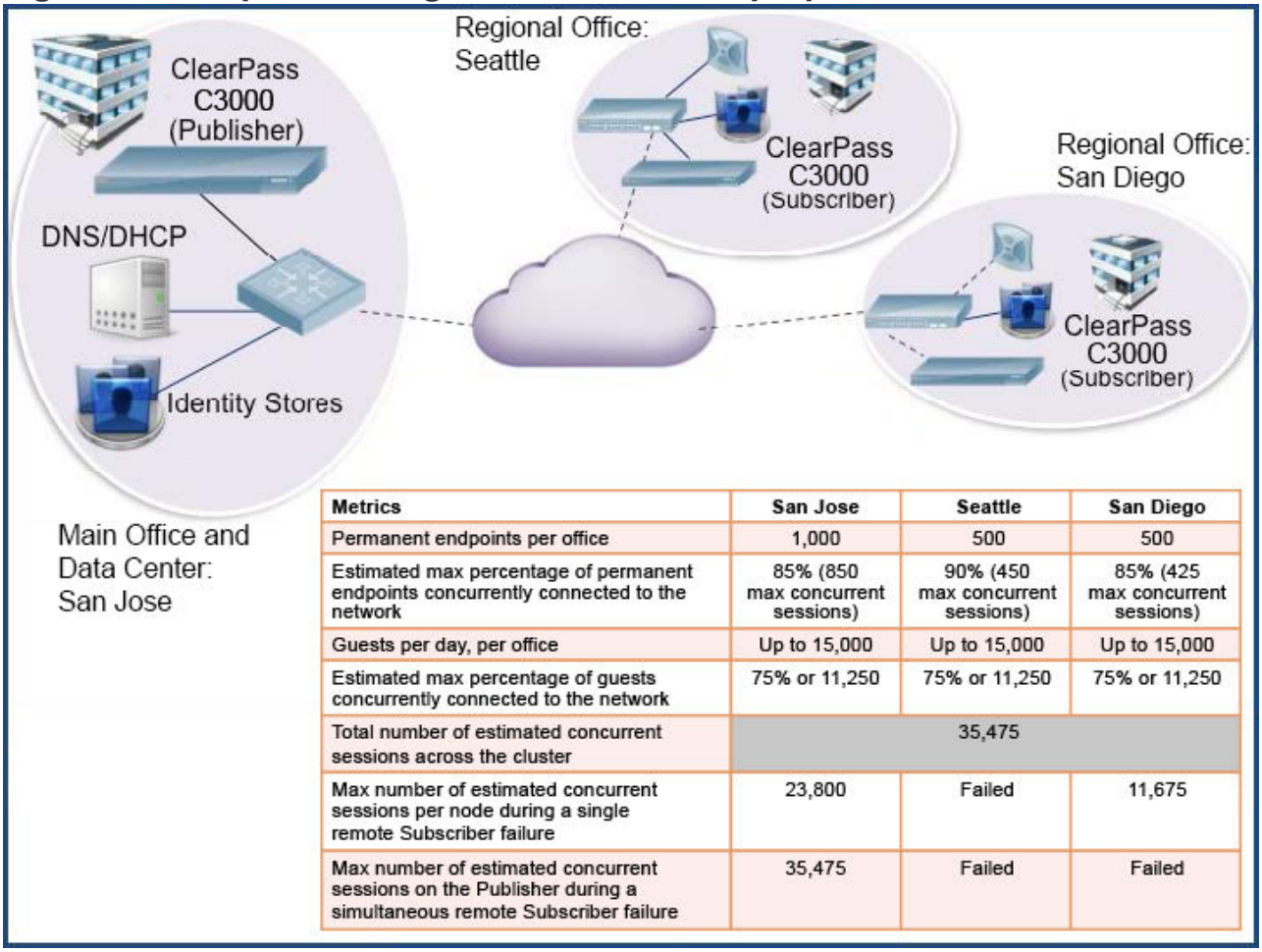
Authenticating Conference Center Users (Large-Scale Cluster Deployment)

In this example, the cluster has three Policy Manager C3000 hardware appliance servers in the same time-zone (see Figure 2).

- These servers are located in San Jose (Publisher), San Diego (Subscriber), and Seattle (Subscriber).

- Each server can register up to 15,000 guests per day, often in short bursts.
- There is constant authentication traffic through the day from the onsite employees and guest.
- On some days, a server may be idle, but there are days where all servers are busy.

Figure 1. Example of a Large-Scale Cluster Deployment



Metrics	San Jose	Seattle	San Diego
Permanent endpoints per office	1,000	500	500
Estimated max percentage of permanent endpoints concurrently connected to the network	85% (850 max concurrent sessions)	90% (450 max concurrent sessions)	85% (425 max concurrent sessions)
Guests per day, per office	Up to 15,000	Up to 15,000	Up to 15,000
Estimated max percentage of guests concurrently connected to the network	75% or 11,250	75% or 11,250	75% or 11,250
Total number of estimated concurrent sessions across the cluster	35,475		
Max number of estimated concurrent sessions per node during a single remote Subscriber failure	23,800	Failed	11,675
Max number of estimated concurrent sessions on the Publisher during a simultaneous remote Subscriber failure	35,475	Failed	Failed

The cluster illustrated in Figure 2 would be considered a large-scale deployment, for the following reasons:

- Each of the three sites manages more simultaneous connections than are possible on a single Policy Manager C1000 or Policy Manager C2000 appliance.
- The vast majority of the users are temporary and mobile, which means there is a much heavier load on the Publisher since a) RADIUS accounting updates are sent from the network, and b) endpoints are modified to reflect the captive portal's state.
- This is a significant load on the Publisher.

Recommendations for a Dedicated Publisher

Assuming that the guests are using a self-registration workflow with MAC caching, there would be a significant amount of database writes on the Publisher. With the additional Publisher load of the replication of these accounts to each of the Subscriber servers, this is an example of a deployment warranting a dedicated Publisher. A Policy Manager C3000 hardware appliance is the recommended appliance for this role.

Additionally, the Policy Manager C3000 hardware appliance can support up to 50,000 concurrent sessions. With a maximum of 23,800 sessions during a single server failure being approximately 48% of the maximum supported, there is still headroom left over on the Publisher to process the Guest account creation workload.

In the event of both remote Subscriber servers failing and authentications failing over to the Publisher, this would result in 35,475 concurrent sessions, or 71% of the maximum supported. Without the large volume of Guest account creations per second, 71% of the maximum number of concurrent sessions would be sustainable, but with it, performance may deteriorate.

Deploying the Standby Publisher

This section contains the following information:

- Setting Up the Standby Publisher
- About the Fail-Over Process
- Mitigation Strategies
- Virtual IP Address Considerations
- Functions Lost When the Publisher Is Down

Subtopics

Setting Up the Standby Publisher
About the Fail-Over Process
Managing Certificates
Mitigation Strategies
Virtual IP Address Considerations
Functions Lost When the Publisher Is Down

Setting Up the Standby Publisher

About this task

Policy Manager allows you to designate one of the Subscriber servers in a cluster to be the *Standby Publisher*, thereby providing for that Subscriber server to be automatically promoted to active Publisher status in the event that the Publisher goes out of service. This ensures that any service degradation is limited to an absolute minimum.

During the period when a cluster does not have an active Publisher, some functions across the cluster are not available, such as being able to create guest accounts (for details, see Functions Lost When the Publisher Is Down).

> **NOTE**
> Before you can designate a Policy Manager server as a Standby Publisher, the designated server must be in a cluster.

The Standby Publisher can function as a fully operational Subscriber server. However, in a large cluster deployment, the Publisher and Standby Publisher might need to be dedicated servers, in which case the Standby Publisher will not be available to handle authentication requests.

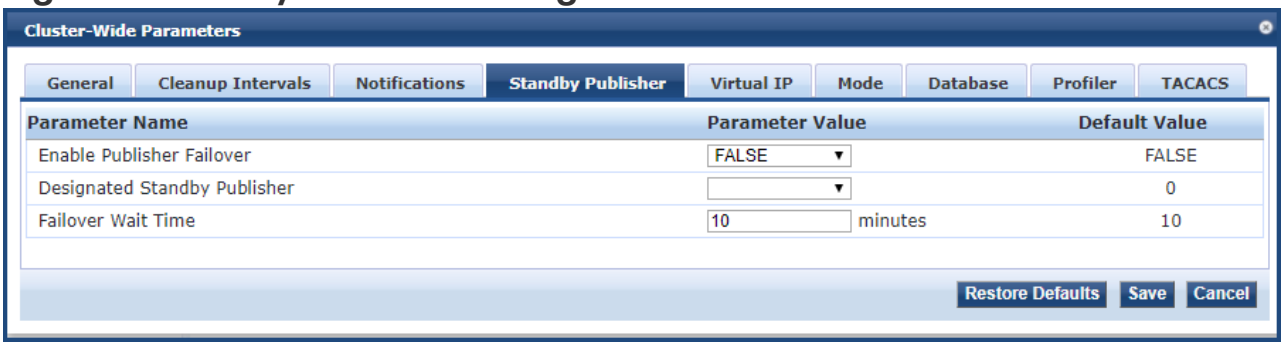
If the Standby Publisher is on a different subnet than the Publisher, ensure that a reliable connection between the two subnets is established. This avoids network segmentation and potential data loss from a false failover.

To designate and configure the Standby Publisher:

Procedure

1. From the server to be designated the Standby Publisher, navigate to **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters > Standby Publisher**.

Figure 1. Standby Publisher Dialog



2. Configure the **Standby Publisher** parameters as described in Table 1, then click **Save**.

Table 1. Configuring Standby Publisher Parameters

Parameter	Action/Description
Enable Publisher Failover	To authorize a server in a cluster to act a s a Publisher if the primary Publisher fails , select **TRUE.** The default value is **FALSE**. **NOTE** The nodes which were not reachable to standby publis her at the time of failover (w hen the publisher goes dow n due to a network issue or power down) are marked as disabled after the failover to the standby publisher. Those disabled nodes should be fo rce dropped from the cluste r and added back using ma ke subscriber. **NOTE** To avoid false failover trigge rs, the **Enable Publisher Fail over** value should be set to **false** before starting a cluste r update.
Designated Standby Publisher	Select the server in the cluster to act as t he Standby Publisher. The default value is **0**. **NOTE** If the Standby Publisher is o n a different subnet than the Publisher, then ensure that a reliable connection between the two subnets is available t o avoid unwanted network s

Parameter	Action/Description
	egmentation and potential d ata loss from a false failover.
Failover Wait Time	Specify the time (in minutes) that the Sta ndby Publisher server must wait before it assumes the role of Publisher after the pr imary Publisher server becomes unreach able. The default failover wait time is **10 minutes**. This parameter prevents the Standby Pub lisher server from taking over when the P ublisher is temporarily unavailable during a restart. . **NOTE** Failover wait times vary base d on the size of the data ne eding to synchronize, as wel l as the number of Subscrib ers that need to be checked . The failover wait time will n ever be less than 10 minutes and, depending on circums tances, can take longer.

About the Fail-Over Process

The Standby Publisher health-checks the primary Publisher server every 60 seconds by making an SQL call to the active Publisher. If this SQL call fails, after ten additional attempts (one per minute), the Standby Publisher begins the process of promoting itself to be the active Publisher server.

The process used to verify the reachability of the remote Policy Manager servers uses an outbound HTTPS call. As noted in Network Ports That Must Be Enabled , **port 443/TCP** must be open between all the servers in the cluster. Utilizing this HTTPS health check provides for a more robust and predictable failover process.

When a Publisher failure is detected, the designated Subscriber server is promoted to active Publisher status. The other Subscriber servers automatically update and replicate their configuration with the new Publisher, which resolves the issue.

Managing Certificates

If a cluster is configured with a standby publisher, add the HTTPS server certificate of the standby publisher to the Trust list and ensure sure all the servers in the cluster have this certificate in the Trust list. Similarly, before you promote a subscriber to publisher, add the HTTPS server certificate of the subscriber to the Trust list and ensure sure all the servers in the cluster have this certificate in the Trust list. This step is not required if the HTTPS server certificates for all the nodes in the cluster are signed by a certificate authority (CA)

Mitigation Strategies

The recommended mitigation strategies for deploying a Standby Publisher are as follows:

- Use a virtual IP address for the Publisher.

  Doing so reduces the potential for a prolonged service outage while the active Publisher is out of service or promoting the Standby Publisher (for related information, see Virtual IP Address Considerations.

  NOTE

  When you configure a Standby Publisher and deploy a virtual IP address, the Standby Publisher should be paired with the active Publisher in the VIP group.

- Ensure that the cluster servers are being monitored.

  Determine if a Publisher server is no longer reachable or not providing service (for example, by SNMP host checking).

- Set up the network access devices (NADs) to point to a primary server, backup server, and a tertiary server.

  Doing so provides for continuity of the RADIUS authentication and accounting traffic until the Standby Publisher transitions to the active state.

Virtual IP Address Considerations

Using a virtual IP address allows for the deployment of a highly available pair of servers. This reduces the amount of down-time in the event of a server failure. If one of the servers in a high-availability pair fails, the other server can take over the virtual IP address and continue providing service to clients. This is particularly useful if the network access server (NAS) devices are processing basic RADIUS authentications to a Policy Manager server.

The Standby Publisher server cannot take over immediately as the failure may be transient and the minimum time for a Standby Publisher to become active is about eight minutes. This duration is due to five attempts (one per minute) to connect to the active Publisher's database, then about four minutes for the server to promote itself to an active state.

Thus, there will always be a delay before the virtual IP address on the transitioning active Publisher the NAS clients are communicating with is back in service and able to process RADIUS authentication requests.

During this eight-minute window, requests from Subscribers to write to the Publisher's database will fail as there will be no Publisher available that can write to the database (for related information, refer to the description of the **Failover Wait Time** parameter in Table 1).

Functions Lost When the Publisher Is Down

When the active Publisher server goes out of service, the following Policy Manager functions are temporarily lost:

- Device Registration (including Aruba AirGroup and MPSK)
- Certificate creation and revocation
- Certificate revocation list updates
- Policy Manager Exchange outbound enforcement
- General Policy Manager and Policy Manager Guest configuration changes
- Policy Manager Guest account creation
- Mobile device management endpoint polling and ingestion
- Onboarding functionality
- The **ActiveSessionDisconnect** API function
- The POST /session/ {id} /reauthorize function of the **ActiveSessionReauthorize** API

Adding a Subscriber to the Publisher

This section contains the following information:

- Introduction
- Using the WebUI to Add a Subscriber Node
- Using the CLI to Create a Subscriber Node

Subtopics

Introduction
Using the WebUI to Add a Subscriber
Using the CLI to Create a Subscriber

Introduction

In the Policy Manager cluster environment, the Publisher server acts as the cluster master. A Policy Manager cluster can contain only one Publisher. Administration, configuration, and database write operations can occur only on the Publisher.

The Policy Manager hardware or virtual appliance defaults to a Publisher unless it is made a Subscriber. You can demote the Publisher to Subscriber status.

NOTE
When the current server is a Subscriber, the **Make Subscriber** link isn't displayed.

Using the WebUI to Add a Subscriber

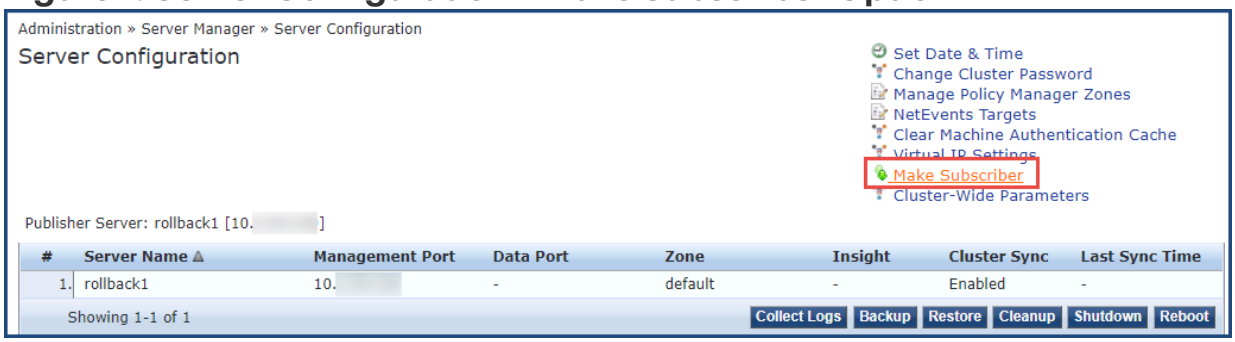
When you make a Policy Manager server a Subscriber in a cluster, keep in mind the following:

- As part of this operation, configuration changes will be blocked on the Publisher during initial cluster sync.
- all application licenses on this server will be removed. To add and activate these licenses, contact Support (**Administration > Support > Contact Support**).

To add a Subscriber to a Publisher via the WebUI:

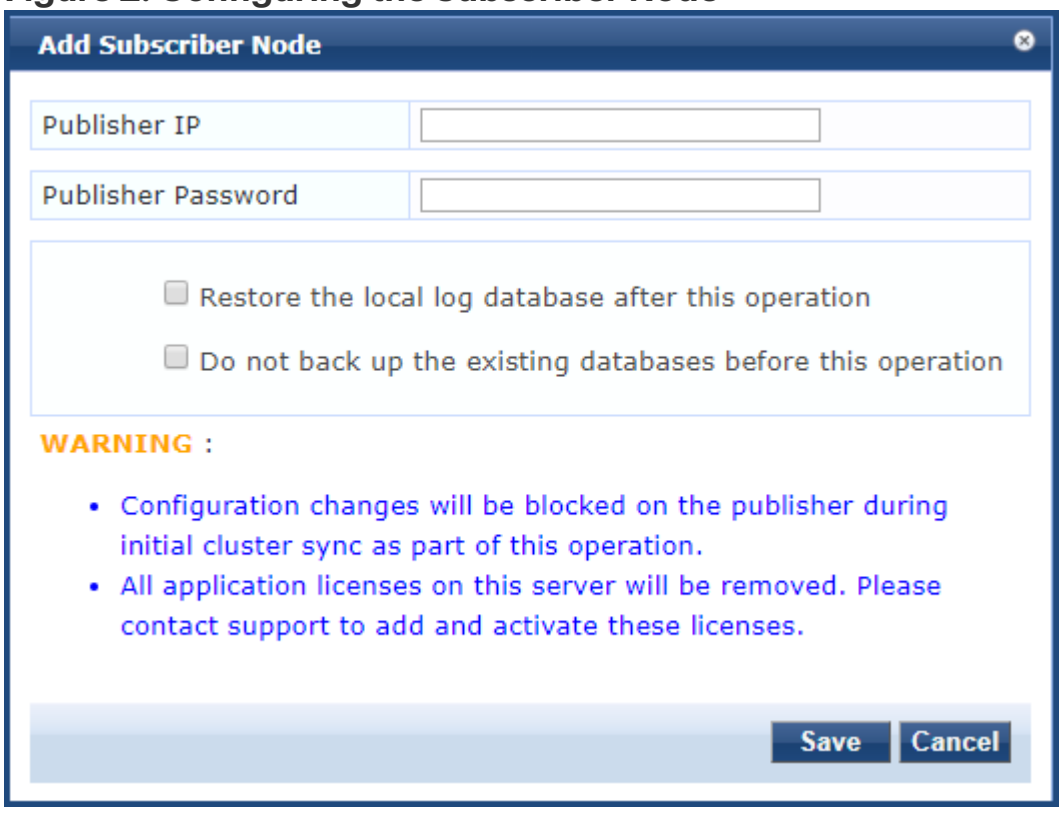
1. Log onto the Policy Manager server that you want to make a Subscriber.
2. Navigate to **Administration > Server Manager > Server Configuration**. The **Server Configuration** page opens.

Figure 1. Server Configuration > Make Subscriber Option



3. Click **Make Subscriber**. The **Add Subscriber Node** dialog opens.

Figure 2. Configuring the Subscriber Node



4. Specify the **Add Subscriber Node** parameters as described in Table 1.

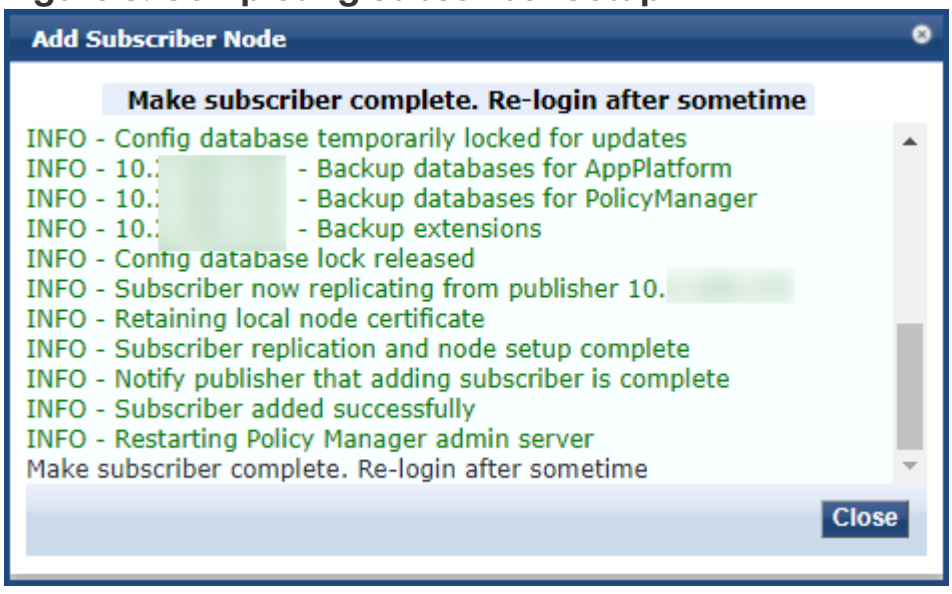
Table 1. Configuring Add Subscriber Node Parameters

Parameter	Action/Description
Publisher IP	Enter the Publisher server's IP addr
Publisher Password	Enter the **appadmin** (CLI) passwor
Restore the local log database after this ope ration	To restore the log database followi Subscriber, select this check box.
Do not backup the existing databases before this operation	Select this check box only if you d kup to the existing database.

5. Be sure to note the warnings on this dialog and respond as needed.
6. When finished, click **Save**. You will see the message: *Adding node as subscriber to <Publisher_IP_address>'s cluster*. When the process completes, the following messages are displayed:

> **NOTE**
>
> The process of adding a server as a Subscriber takes several minutes.

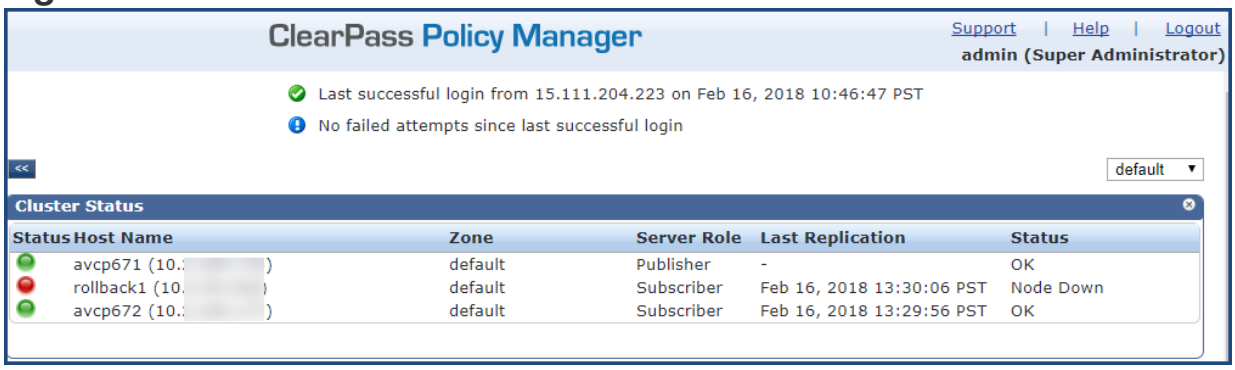
Figure 3. Completing Subscriber Setup



7. Click **Close**.

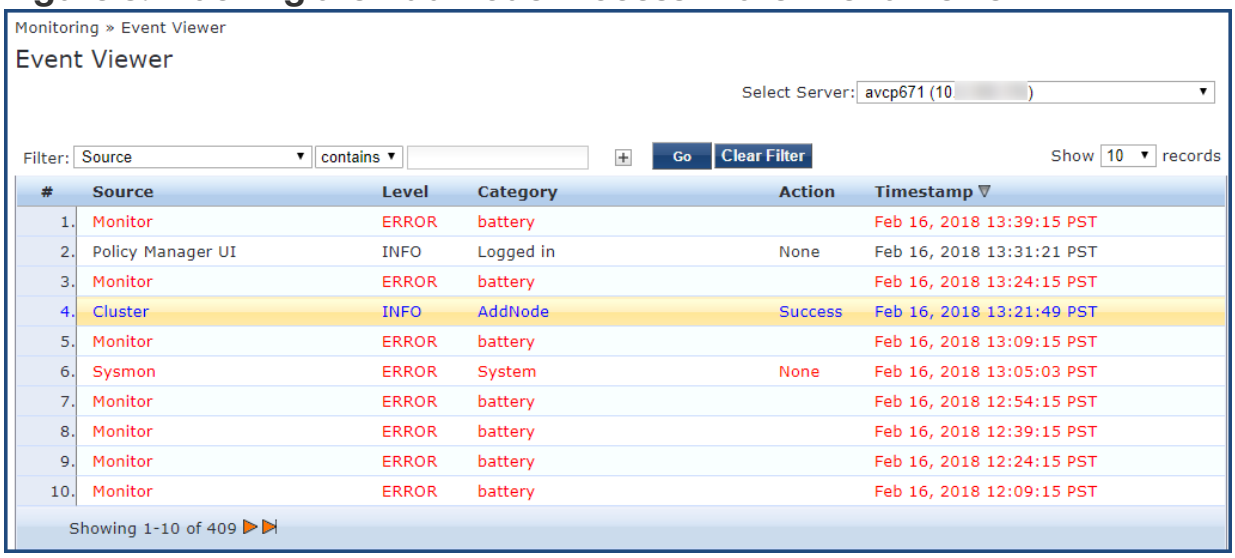
8. To complete the Subscriber setup, log back into the new Subscriber. When you log into the Publisher or the Subscriber , the Policy Manager Dashboard presents the updated cluster status:

Figure 4. Cluster Status: Subscriber Added to Publisher



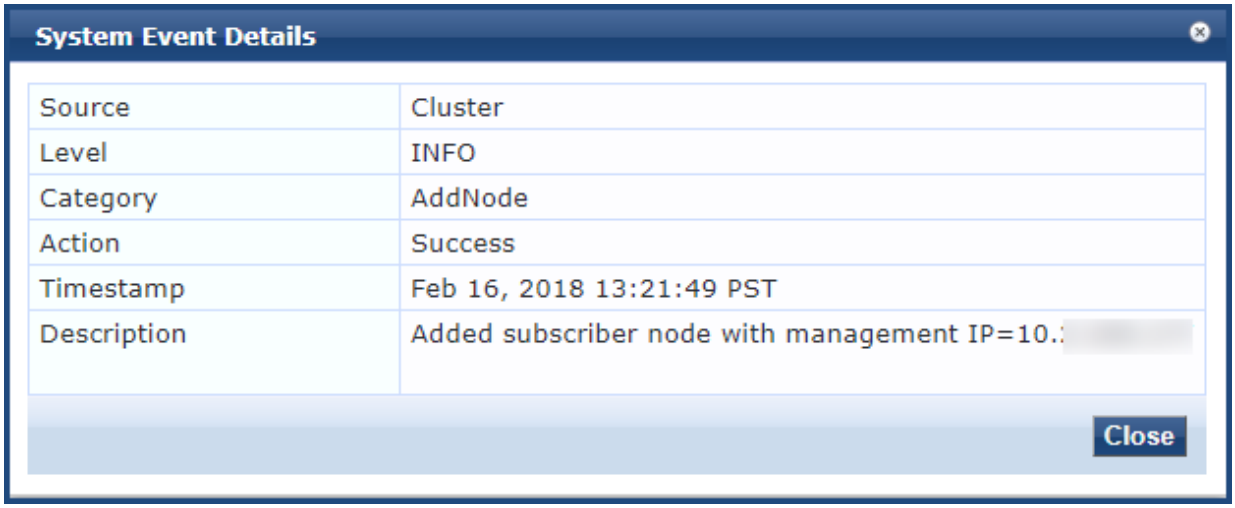
9. You can also track this process in the Event Viewer (**Monitoring > Event Viewer**) following a successful Subscriber addition, as shown in Figure 5.

Figure 5. Tracking the Add Node Process in the Event Viewer



10. To view the system event details for the Add Subscriber event, click the appropriate Event Viewer item in the list. The **System Event Details** page for the Add Node event opens.

Figure 6. Viewing the Add Subscriber Node Event System Event Details



Using the CLI to Create a Subscriber

About this task

You can perform multiple cluster-related administrative functions from the CLI. The CLI provides additional functionality that cannot be accomplished from the user interface. In addition to the WebUI, you can use the command-line interface (CLI) to make a Subscriber. To use the CLI to make a cluster server a Subscriber:

Procedure

1. Log in as the **appadmin** user to the Policy Manager server using SSH client software (such as PuTTY).
2. Issue the following command:

```
cluster make-subscriber –i [publisher_IP_address]
```

Table 1. Cluster make - subscriber Command Parameters

Parameter	Action/Description
- i <*publisher_IP_address*>	The **- i** option specifies the Publi ddress.
- l	Use the **- l** option to restore the l e after the Make Subscriber oper e.

Parameter	Action/Description
- b	Use the **- b** option to skip genera efore the Make Subscriber operat .

After you enter the IP address of the Publisher, you will see the following warning message:

> **WARNING**
>
> *Executing this command will make the current machine subscriber to the publisher host specified. Current configuration and application licenses installed (if any) on this node will be lost when the operation is complete.*

Do not close the shell or interrupt this command execution.

3. To confirm that you want to continue, enter **y**.
4. Enter the cluster (**appadmin**) password for the Publisher. The process to downgrade the server to a Subscriber begins.

Rejoining a Down or Disabled Server to the Cluster

This section contains the following information:

- Introduction
- Rejoining a Down or Disabled Server Back Into the Cluster
- Removing a Server from the Cluster

Subtopics

Introduction
Rejoining a Disabled Server Back Into the Cluster
Removing a Server from the Cluster

Introduction

When a server loses communication with the cluster for a period greater than 24 hours, the Publisher designates that server as *down*.

To rejoin a down or disabled server to the cluster, you can simply rejoin the server back to the publisher. Alternatively, you can remove the server from the cluster, reset the configuration on the out-of-sync server, then rejoin the server to the cluster.

Rejoining a Disabled Server Back Into the Cluster

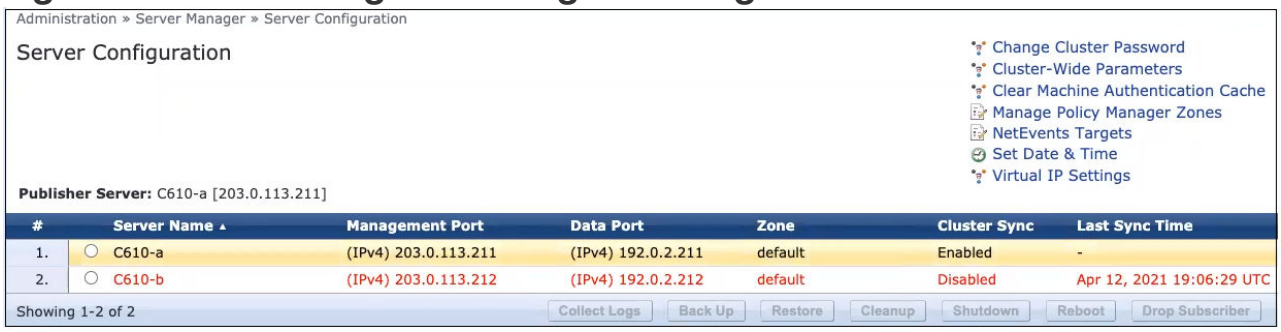
About this task

To rejoin a disabled server back into the cluster:

Procedure

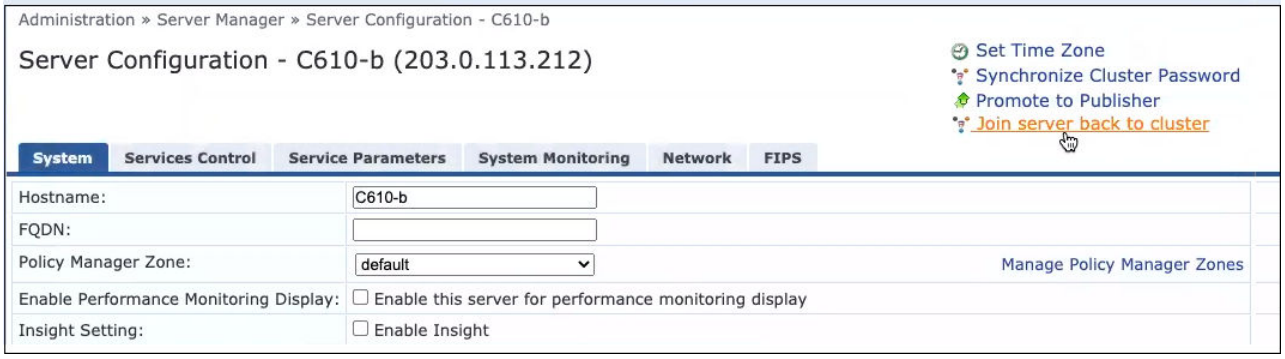
1. Access the Publisher server and navigate to the **Administration > Server Manager > Server Configuration** page . Figure 4 shows that one of the Subscribers in the cluster is disabled.

Figure 1. Server Configuration Page Showing Disabled Cluster Server



2. Select the disabled Subscriber server that you want to rejoin to the cluster. The **Server Configuration > System** dialog opens for the selected server. As shown in Figure 5, the dialog includes the **Join server back to cluster** option.

Figure 2. Join Server Back to Cluster Option Displayed



3. Click **Join server back to cluster**. A confirmation window appears, providing the option to promote the current server to Publisher status.

Figure 3. Option to Promote Disabled Server to Publisher

4. To proceed (without promoting the disabled server to Publisher), click **Yes**. The **Join server back to cluster** window displays the log entries for each completed task during this process.

> **NOTE**
>
> The process to rejoin a disabled server back to the cluster may take several minutes.

Figure 4. Example log entries showing successful rejoin operation

Removing a Server from the Cluster

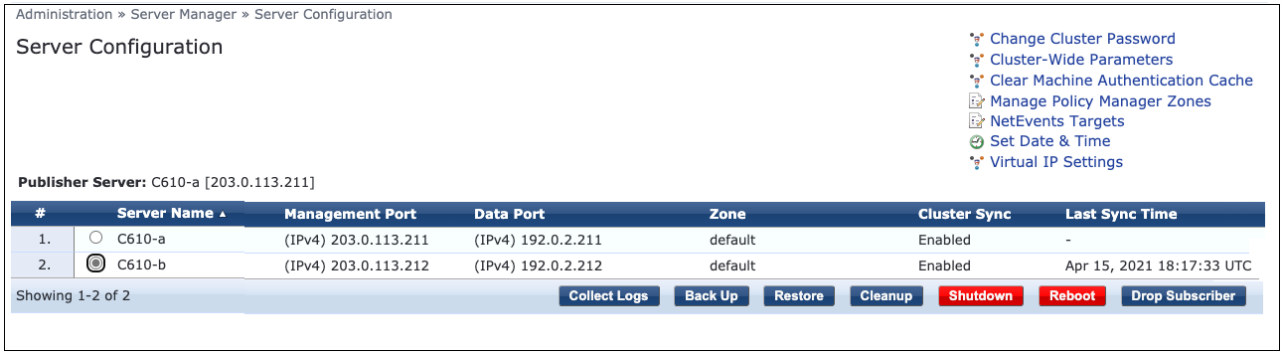
About this task

To remove a Subscriber server from the cluster:

Procedure

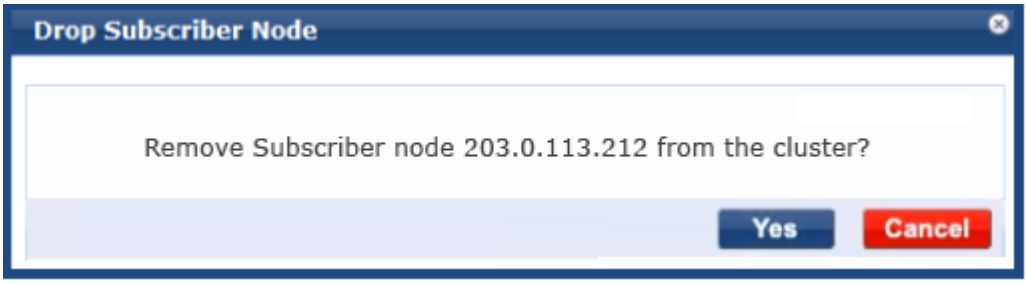
1. From the Publisher server, navigate to **Administration > Server Manager > Server Configuration**.
2. From the **Server Configuration** screen, select the Subscriber you want to remove.

Figure 1. Selecting the Subscriber to Remove



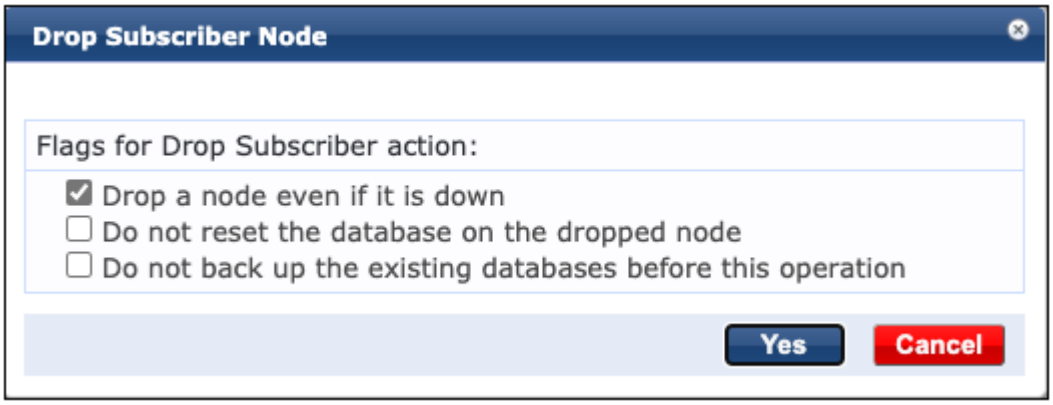
3. Click **Drop Subscriber**. You are prompted to confirm the drop action.

Figure 2. Confirming the Drop Subscriber Operation



4. To remove the selected Subscriber server, click **Yes** (or press **Cancel** to cancel the operation). When you proceed, you are presented with a set of options to further refine the Drop Subscriber operation:

Figure 3. Drop Subscriber Confirmation Options



This window includes the following **Drop Subscriber Node** options:

- Drop a server even if it is down.
- Do not reset the database on the dropped server.
- Do not back up the existing databases before this operation.

5. Click the check box for each confirmation option you wish to enable, then click **Yes**. The **Drop Subscriber Node** status screen is displayed.

> **NOTE**
>
> If you plan on resetting the configuration on the down node and then rejoining it back to the cluster, select just the first option, **drop a node even if it is down**.

6. When the status is complete, click **Close**. The server is removed from the cluster.
7. To rejoin the server to the cluster, refer to Rejoining a Disabled Server Back Into the Cluster

Deploying Policy Manager Insight in a Cluster

This section contains the following information:

- Introduction
- Insight Placement Considerations
- When a Insight-Enabled server Is Down
- Enabling Insight

Introduction

Multiple functions are dependent on Policy Manager Insight. For example, to use MAC caching, Policy Manager Insight must be enabled on at least one server within a cluster.

NOTE

Enabling Policy Manager Insight on at least two servers in the cluster is recommended.

As you enable Policy Manager Insight on additional servers in the cluster, Policy Manager automatically adds these servers to the Policy Manager Insight database authentication source definition. Policy Manager Insight does not replicate data to any other servers within the cluster—it is an entirely stand-alone database.

Insight Deployment Considerations

Having Policy Manager Insight enabled on multiple servers within the cluster provides for a level of resilience, but you need to carefully consider where you enable Policy Manager Insight. For every server where Policy Manager Insight is enabled, all the other servers within the cluster subscribe through *NetEvents* to send data to the Policy Manager Insight database.

The amount of data sent to the Policy Manager Insight database can be extremely high, and if you use Insight for processing authentication requests within your cluster, where you enable Policy Manager Insight is an important design consideration:

- If you are running a large Policy Managernetwork in which the subscriber traffic is *not* consuming all the publisher's resources, enable Policy Manager Insight on the dedicated publisher and the standby publisher.
- If you are running a very large CPPM network in which the subscriber traffic will consume the publisher's resources, you could enable Policy Manager Insight on the dedicated publisher and the standby publisher, but only if both of these servers are

dedicated to cluster operations—the publisher and standby publisher should not be processing authentication requests.

- In a very large-scale deployment, Policy Manager Insight should be placed on its own dedicated server. This removes a lot of processing and I/O from the publisher, allowing it to handle the maximum amount of worker traffic.
- Policy Manager Insight data is valuable and could be used as part of policy evaluation. If this is the case, Aruba recommends that you enable redundant Policy Manager Insight servers for fault tolerance.
- If the worker traffic sent from the subscriber servers is expected to fully saturate the capacity of the publisher server, Policy Manager Insight should not be enabled on the publisher server. However, if the publisher server has spare capacity, it can be used to support the Policy Manager Insight database. However, take care to carefully monitor the publisher server's capacity and performance.

When a Policy Manager Insight-Enabled Server Is Down

When a Policy Manager Insight-enabled server in a cluster is down or out-of-sync for more than 30 minutes, the Policy Manager Insight server is moved to be the last Policy Manager Insight server in the fall-back list. This allows for fail-though to other Policy Manager Insight servers.

When a Policy Manager Insight-enabled server is dropped from the cluster, the corresponding server entry in the Policy Manager Insight repository is removed.

Enabling Policy Manager Insight

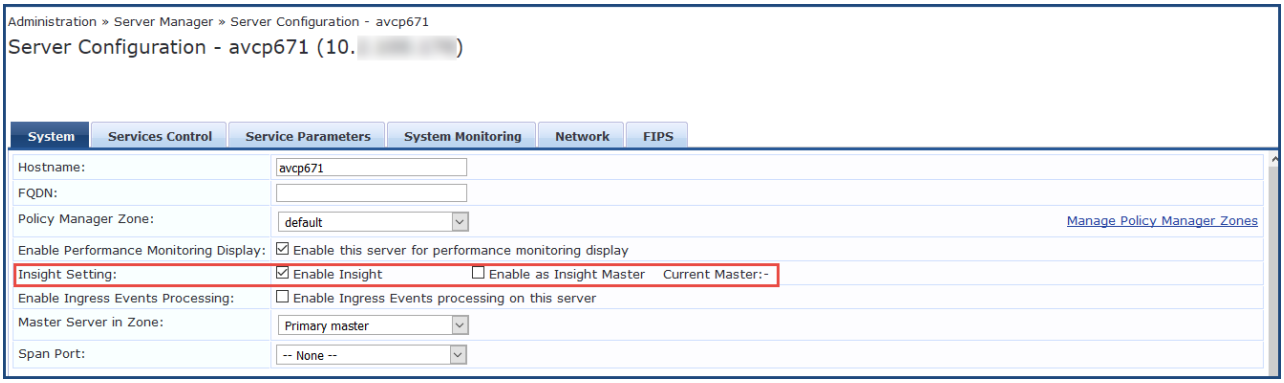
About this task

Policy Manager Insight is not enabled by default, so you must manually enable it.

To enable Policy Manager Insight:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. From the **Server Configuration** page, select the Policy Manager server you want to configure. The **Server Configuration** dialog opens.



3. To enable the Policy Manager Insight reporting tool on this server, select the **Enable Insight** check box.

   - When you enable this check box on a cluster server, the Policy Manager Insight Repository configuration is automatically updated to point to the server's management IP address.
   - When you enable this check box for other servers in the cluster, those servers are added as backups for the same authentication source.
   - The order of the primary and backup servers in the Policy Manager Insight Repository is the same order in which Policy Manager Insight was enabled on those servers.

4. To specify the current cluster server as an Insight Master, click the **Enable as Insight Master** check box. Enabling a cluster server as an Insight Master allows other servers where Insight has been enabled to subscribe to this server 's Insight Report configuration. In the event that this server fails, the reports will still be produced because all the servers in the cluster send a copy of their NetEvents data to all the servers that have Policy Manager Insight enabled.

5. When finished with enabling Policy Manager Insight and configuring any other elements in the **Server Configuration** dialog, click **Save**.

Configuring Cluster File-Backup Servers

This section contains the following information:

- Adding Cluster File-Backup Servers
- Backing Up Configuration and Access Tracker Log Information

Adding Cluster File-Backup Servers

About this task

To add cluster file-backup servers:

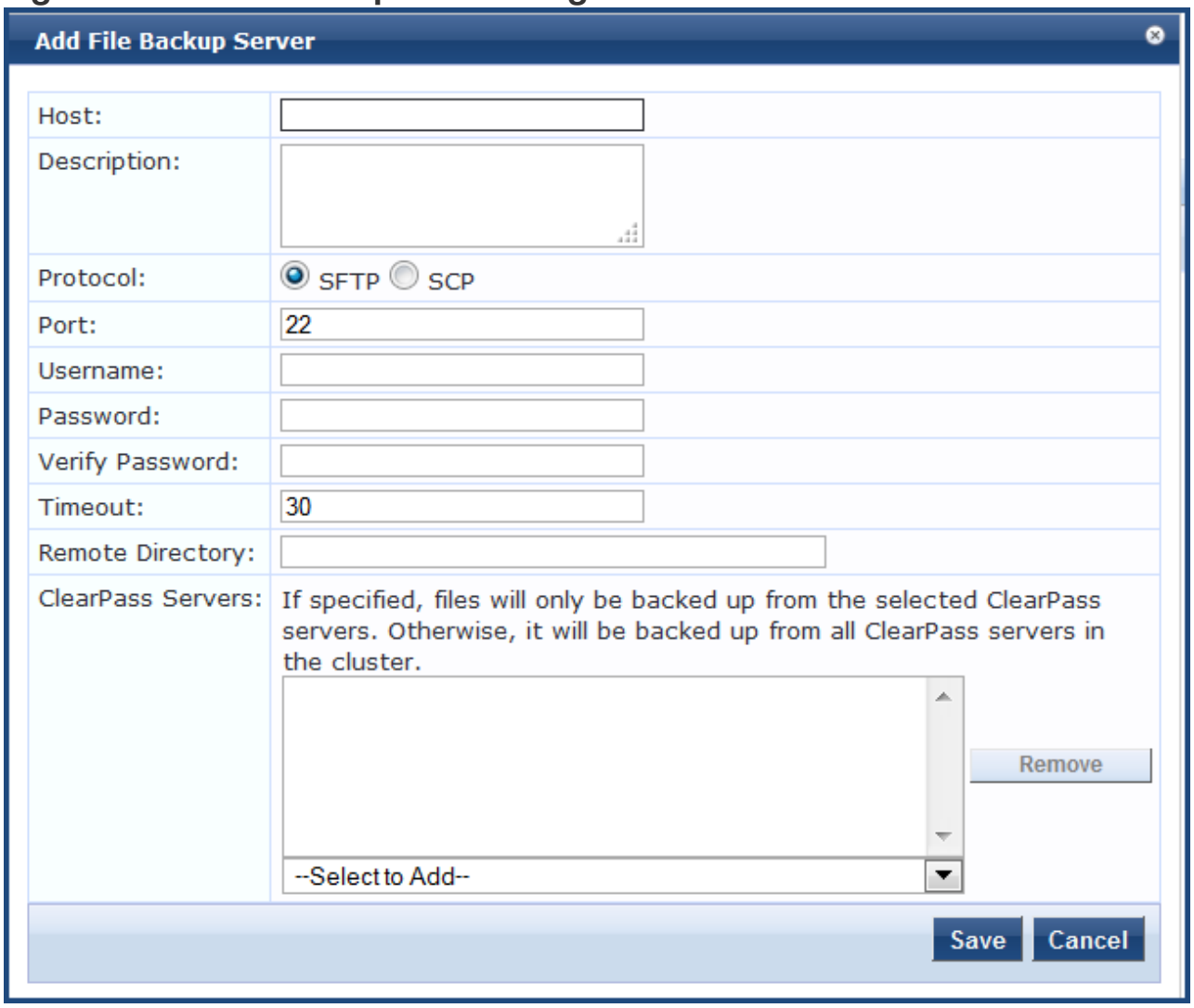
Policy Manager provides the ability to push scheduled data securely to an external server. You can push the data using the SFTP (SSH File Transfer Protocol)and SCP (Session Control Protocol) protocols.

To configure cluster file-backup servers in Policy Manager:

Procedure

1. Navigate to the **Administration > External Servers > File Backup Servers** page. The **File Backup Server** page opens.
2. Click the **Add** link (at the top-right). The **Add File Backup Server** page opens.

Figure 1. Add File Backup Servers Page

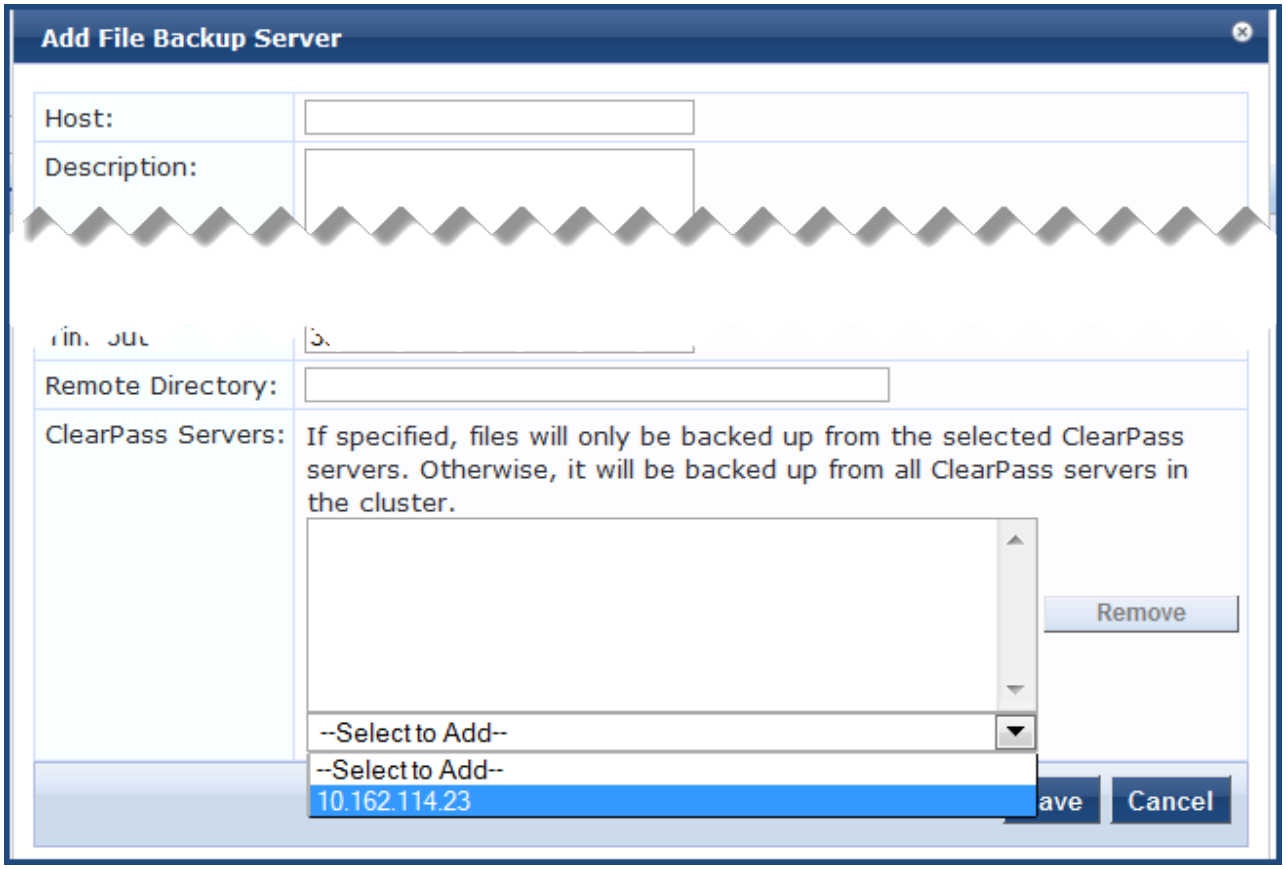


3. Specify the **Add File Backup Server** page parameters as described in the following table:

Table 1. Add File Backup Page Server Page Parameters

Parameter	Action/Description
Host	Enter the name or IP address of the
Description	Enter the description that provides formation about the File Backup ser
Protocol	Specify the protocol to be used to nerated reports to an external serve the following protocols: • SFTP (SSH File Transfer Protocol • SCP (Session Control Protocol)
Port	Specify the port number. The defau
Username	Enter the user name and password rver, then verify the password.
Password	
Timeout	Specify the timeout value in secon value is **30** seconds.
Remote Directory	Specify the location where the files ied. A folder will be automatically create ath that you specify based on the s Manager servers in the **Policy Mana** eld.
ClearPass Servers	When you select specific Policy Ma files are backed up from the selecte ager servers only. Otherwise, the fil Policy Manager servers in the cluste up.

Figure 2. Specifying the File Backup Server



4. From the **Select to Add** drop-down, select the cluster-file backup server(s) to be backed up.
5. When finished, click **Save**.

Backing Up Configuration and Access Tracker Log Information

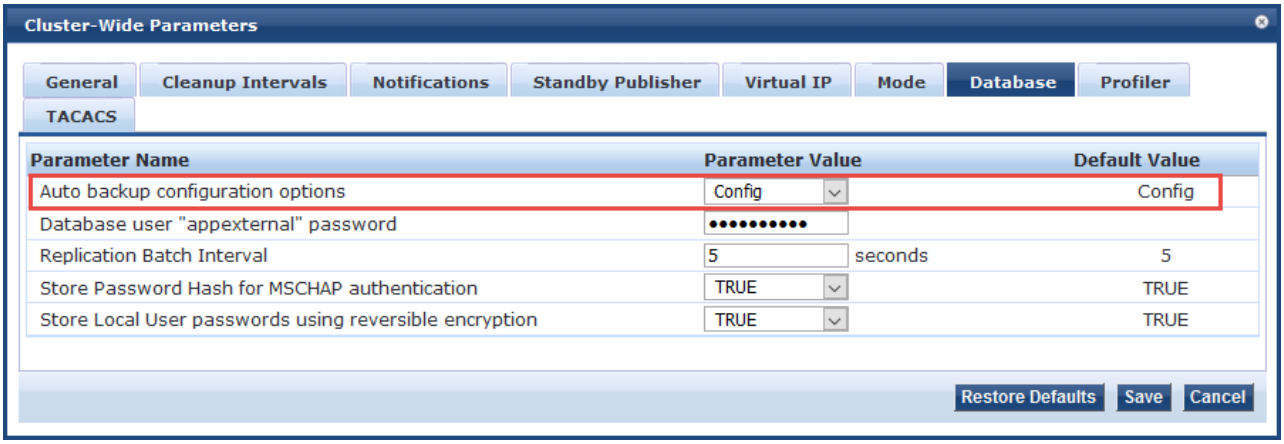
About this task

By default, only cluster configuration information is sent for backup. However, if you need cluster log information to be backed up as well, enter the following change. To back up both configuration and Access Tracker log information:

Procedure

1. On the Publisher, navigate to **Administration > Server Manager > Server Configuration**.
2. From the **Server Configuration** page, choose **Cluster-Wide Parameters**.
3. Select the **Database** tab. The **Database** page opens.

Figure 3. Setting the Auto Backup Configuration Options



4. From the **Auto backup configuration options** drop-down, choose **Config| SessionInfo**.
5. When finished with changes to the cluster-wide parameters, click **Save**.

Cluster CLI Commands

To view the latest and most complete list of cluster configuration commands in the Policy Manager command-line interface, refer to Cluster Commands.

Preparing ClearPass for Active Directory Authentication

Subtopics

Preparing for Active Directory Authentication
Joining a Policy Manager Server to an Active Directory Domain
Adding Active Directory as an Authentication Source to Policy Manager
Obtaining and Installing a Signed Certificate From Active Directory
Manually Testing Login Credentials Against Active Directory

Preparing for Active Directory Authentication

This section of the Deployment Guide describes the required steps to integrate Policy Manager and Microsoft Active Directory. For some use cases, Policy Manager is required to join the Active Directory—802.1X authentication with EAP-PEAP-MSCHAPv2 is one

such use case. 802.1X authentication with Active Directory as the primary authentication source is the focus of this chapter. In other use cases, such as with Captive Portal authentication, joining Policy Manager to Active Directory is optional.

The first task in preparing Policy Manager for Active Directory® (AD) authentication via EAP-PEAP-CHAP-v2 is to join the Policy Manager server to an Active Directory domain. Joining Policy Manager to an Active Directory domain allows you to authenticate users and computers that are members of an Active Directory domain. Joining Policy Manager to an Active Directory domain creates a computer account for the Policy Manager node in the Active Directory database. Users can then authenticate to the network using 802.1X and EAP methods, such as PEAP-MSCHAPv2, with their own Active Directory credentials.

Joining a Policy Manager Server to an Active Directory Domain

When joining an Active Directory domain and doing PEAPv0+MSCHAPv2 authentication, Policy Manager negotiates and uses the highest Server Message Block (SMB) protocol version that is supported by the Policy Manager server. Policy Manager supports SMBv1, v2, and v3.

A one-time procedure to join Policy Manager to the domain must be performed from an account that has the ability to join a computer to the domain; if you are unsure whether the administrator account has the ability to do so, check with your Windows administrator.

Why does Policy Manager need to join Active Directory to perform EAP-PEAP-MS-CHAPv2 authentication for 802.1x? Policy Manager needs to be joined to Active Directory because when performing authentication for a client using EAP-PEAP-MS-CHAPv2, only the password hashes supplied by the user are used to authenticate against Active Directory. This is done using NT LAN Manager (NTLM) authentication, which requires Active Directory domain membership.

If you need to authenticate users that belong to multiple Active Directory forests or domains in your network, and there is no trust relationship between these entities, then you must join Policy Manager to each of these untrusting forests or domains.

NOTE

You do not need to join Policy Manager to multiple domains belonging to the same Active Directory forest, because a one-way trust relationship exists between these domains. In this case, you should join CPPM to the root domain.

About the Domain Controller

A *domain* is defined as a logical group of network objects (computers, users, and devices) that share the same Active Directory database. The domain controller is the Microsoft Active Directory server responsible for responding to requests for authentication from users and computer accounts (for example, logging in and checking permissions) within the Windows Server domain. The Active Directory server contains the domain controller.

It's common for an Active Directory domain controller to function as a DNS server. Active Directory domain controllers can also be LDAP servers, as well as perform any number of additional functions that are loaded on the same server. By default, a domain controller stores one domain directory partition consisting of information about the domain in which it is located, plus the schema and configuration directory partitions for the entire forest.

Synchronizing the Cluster Date and Time with the NTP Server

About this task

Assuming that this Policy Manager server has never been joined to the Active Directory domain before, first make sure that the date and time are correct and in sync on both the Policy Manager server and the Active Directory domain controller that you will use for the join domain operation.

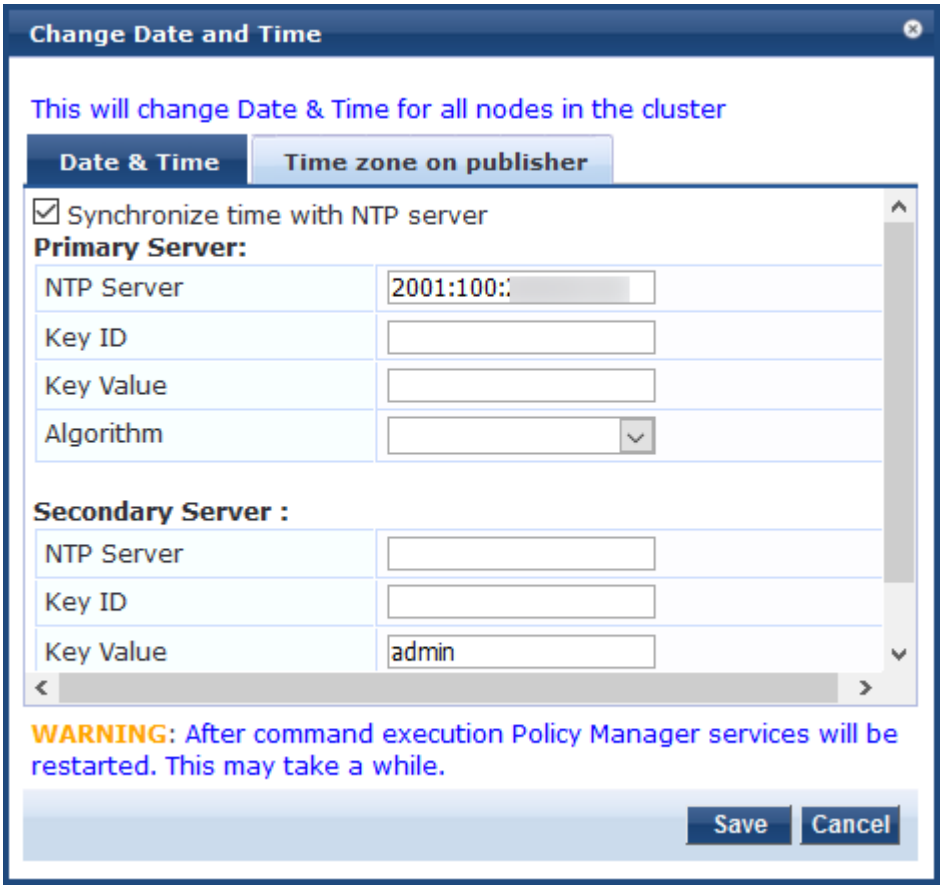
> **NOTE**
> The option to change the date and time for the cluster is available only on the Publisher node. Subscriber nodes in a cluster will synchronize the date and time from the Publisher node.

To synchronize the date and time on the nodes in a cluster with an NTP (Network Time Protocol) server:

Procedure

1. Log in to the Publisher node.
2. Navigate to the **Administration > Server Manager > Server Configuration** page.
3. Select the **Set Date and Time** link. The **Change Date and Time** dialog opens to the **Date & Time** tab.

Figure 1. Confirming NTP Server Synchronization



4. Specify the **Date & Time** parameters as described in the following table:

Table 1. Change Date and Time > Date & Time Parameters

Parameter	Description
Synchronize time with NTP server	To synchronize with a Network Time Protocol (NTP) server, enable this check box (enabled by default). **NOTE** You can also specify the date and time for the cluster manually by disabling the **Synchronize time with NTP server** check box and entering the current date and time in the dialog provided:
Primary Server and Secondary Server	

Parameter	Description
NTP Server	Specify the IP address or host name for the Primary NTP server and the Secondary NTP server. The IP address can be an IPv4 or IPv6 address. Users should be aware that ClearPass can time sync with any of the configured NTP servers in any order. It does not need to prefer the primary NTP server first and then the secondary as per configuration.
Key ID	The **Key ID** is a number that specifies the index for key values. The **Key ID** value can be from 1 to 65534 inclusive. For authentication to succeed, typically an NTP client and server have to trust the same key ID and key value pair.
Key Value	The **Key Value** is a form of shared secret, which both the client and the Policy Manager server use for authenticating NTP messages. The **Key Value** can be: • Up to 20 - character printable ASCII string • Up to 40 - character hex value When entering an ASCII string for the **Key Value**, note that it *cannot* contain the following characters: • **&** (ampersand) • **;** (semicolon) • ` (grave accent) • **|** (pipe) • **<** (left angle bracket) • **>** (right angle bracket)

Parameter	Description
	• **(** (left parenthesis) • **)** (right parenthesis) The **Key Value** ASCII string must start and end with one of the following characters: • - (hyphen) • **'** (apostrophe) • **"** (quotation mark)
Algorithm	Select one of the following encryption ty pes: • SHA • SHA1 **NOTE** In FIPS mode, only the SHA1 hashing algorithm is support ed.

5. Click **Save.**
6. Return to the **Server Configuration** page by clicking **Cancel**.
7. Compare the clock time displayed at the bottom of the Policy Manager **Server Configuration** page against the clock time on the Active Directory server.

   NOTE
   The maximum allowed clock skew between the Policy Manager server and the Active Directory server is **five minutes**.

8. If the time on the two systems doesn't exceed the clock skew limit, then proceed.

Restarting Services

About this task

Once you have saved the **Date & Time** configuration, you must restart Policy Manager services.

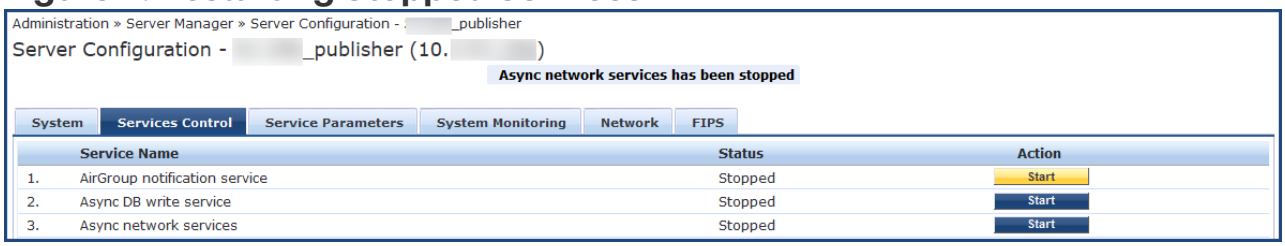
> **NOTE**
> The Audit Viewer (**Monitoring > Audit Viewer**) tracks NTP configuration changes.

To restart Policy Manager services:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager Publisher node.
3. From the **Server Configuration** page, select the **Services Control** tab.

Figure 2. Restarting Stopped Services



4. From the **Action** column, click **Start** for each service that needs to be restarted.

   For each restarted service, the **Start** button is changed to **Stop**.

Specifying the Time Zone on the Publisher Node

About this task

To specify the time zone on the Publisher node:

Procedure

1. Click the **Time Zone on Publisher** tab.

Figure 3. Setting the Time Zone on the Publisher

The time zones are listed in alphabetical order.

2. Select the time zone where the Publisher node resides, then click **Save**.

> **NOTE**
>
> This option is available only on the Publisher. To set the time zone on a Subscriber node, select the specific server and set the time zone from the server-specific page.

Joining an Active Directory Domain

You can join Policy Manager to an Active Directory (AD) domain to authenticate users and computers that are members of an Active Directory domain. If you join Policy Manager to an Active Directory domain, it creates an account for the Policy Manager node in the Active Directory database.

Users can then authenticate into the network using 802.1X and EAP methods, such as PEAP-MSCHAPv2, with their own their own Active Directory credentials.

If you need to authenticate users belonging to multiple Active Directory forests or domains in your network, and there is no trust relationship between these entities, then you must join Policy Manager to each of these untrusted forests or domains.

> **NOTE**
> Policy Manager is not required to join multiple domains belonging to the same Active Directory forest because a one-way trust relationship exists between those domains. In this case, Policy Manager can join the root domain.

Policy Manager can join or leave an Active Directory domain by using the following two buttons in the **Server Configuration** page > **System** tab:

- **Join Domain**: Click **Join Domain** to join this Policy Manager appliance to an Active Directory domain. Password servers can be configured after Policy Manager is successfully joined. For more information on adding a password server, see Manually Configuring Active Directory Password Servers.
- **Leave Domain**: If the server is already part of multiple Active Directory domains, click **Leave Domain** to disassociate this Policy Manager appliance from an Active Directory domain (for details, see Disassociating a Policy Manager Server From an Active Directory Domain).

To Join an Active Directory Domain

About this task

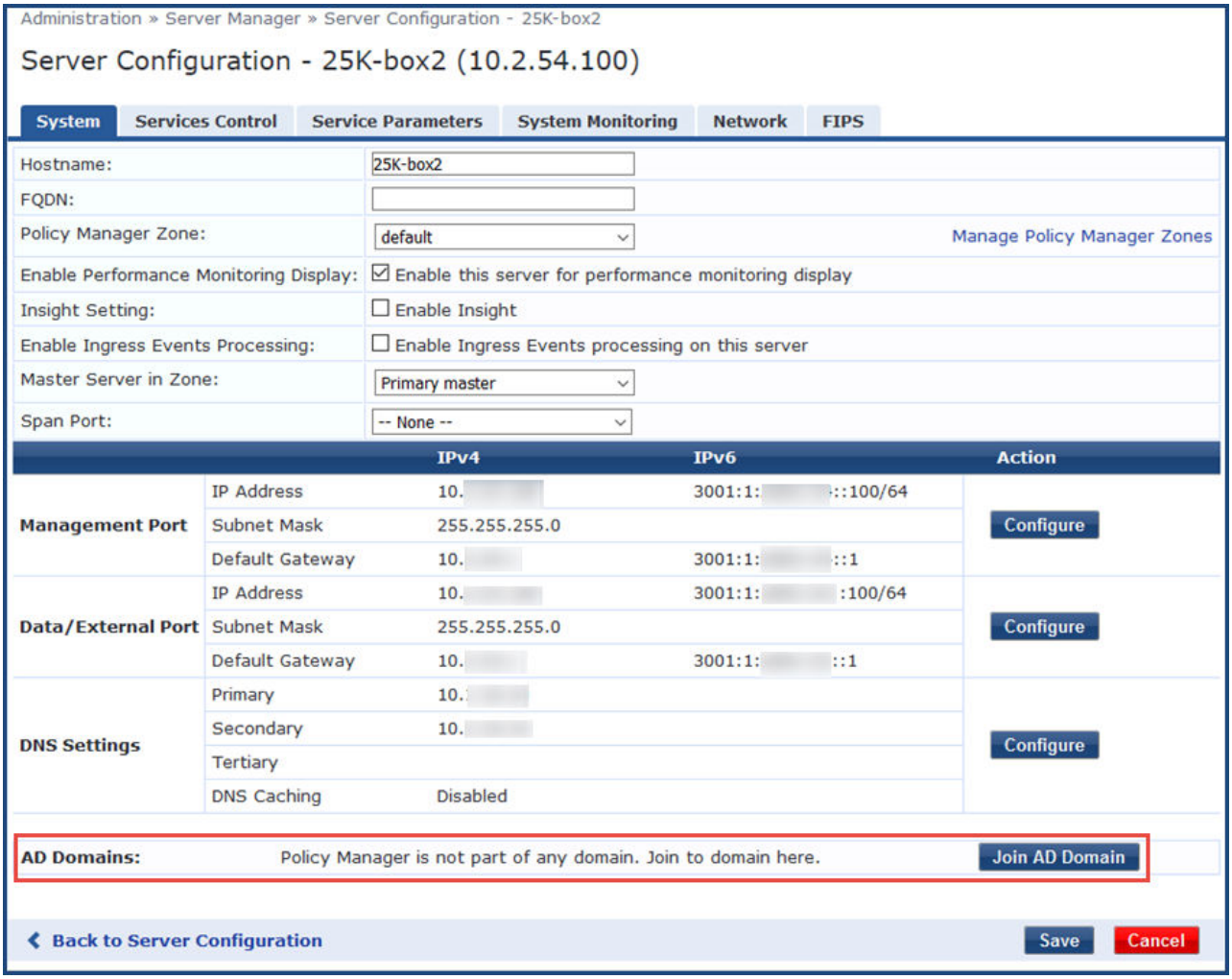
To join a Policy Manager server to an Active Directory domain:

Procedure

1. In the **Server Configuration** screen, click the **name of the server** that you want to join to the domain.

   The **Server Configuration** screen for the selected server opens.

Figure 4. Join AD Domain Option for Selected Policy ManagerServer



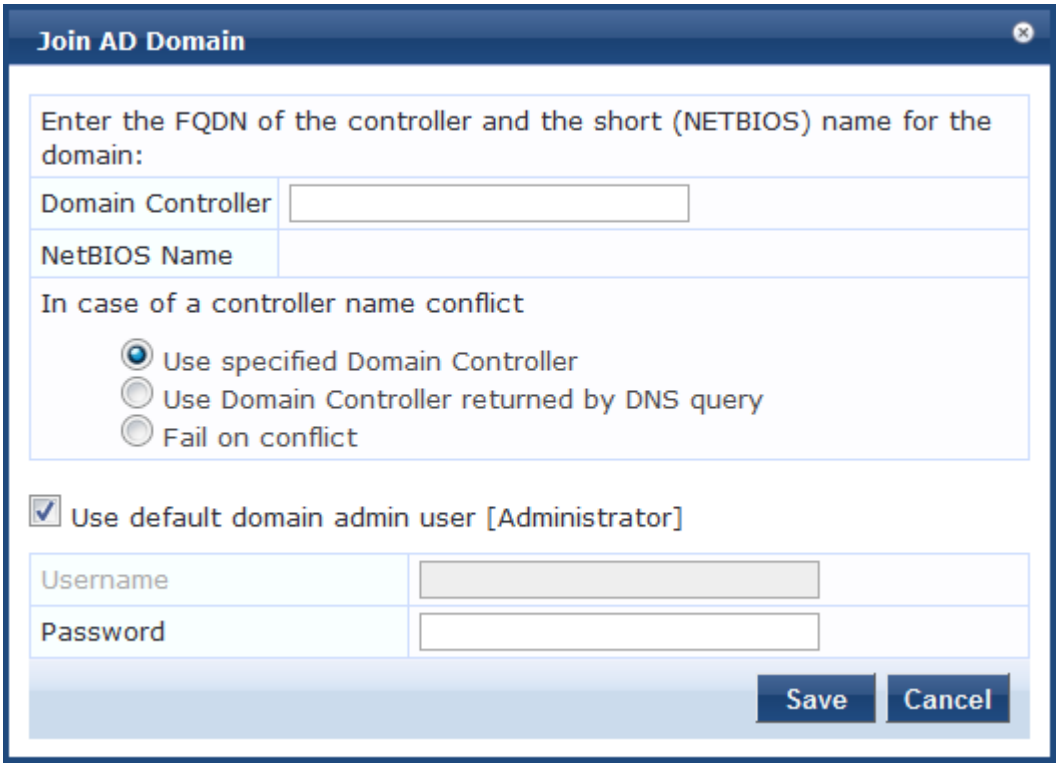
You can now join the Active Directory domain.

> **NOTE**
>
> The primary DNS server IP address is also the IP address of the Active Directory domain controller.

2. Click **Join AD Domain**.

   The **Join AD Domain** dialog opens.

Figure 5. Join AD Domain Dialog



3. Specify the **Join AD Domain** parameters as described in the following table.

Table 2. Join AD Domain Parameters

Parameter	Action/Description
Domain Controller	Enter the Fully Qualified Domain Name (FQDN) of the domain controller, then p ress **Tab**. The following message is displayed: *Tryin g to determine the NetBIOS name...* Policy Manager searches for the NetBIOS name for the domain. Policy Manager displays a warning messa ge if the domain controller is not the clos est one. **NOTE** You should always join to th e domain controller that is p hysically the closest, otherw ise authentication might fail

Parameter	Action/Description
	or the controller's performan ce might be impacted.
NetBIOS name	The **NetBIOS Name** is automatically pop ulated after you enter the FQDN of the **D omain Controller** and it is not editable. **NOTE** If you enter an incorrect valu e for the NetBIOS name, you see a warning message in th e user interface. If you see t his warning message, leave t he domain by clicking on th e **Leave Domain** button (whi ch replaces the **Join Domain** button once you join the do main). After leaving the dom ain, join again with the corre ct NetBIOS name.
Domain Controller name conflict	Specify the action to take in the event of a domain controller name conflict. In some deployments (especially if there are multiple domain controllers, or if the domain name has been wrongly entered i n the last step), the domain controller FQ DN returned by the DNS query can be dif ferent from what was entered. In this cas e, you can: • **Use specified Domain Controller**: Co ntinue to use the domain controller n ame that you entered. • **Use Domain Controller returned by D NS query**: Use the domain controller name returned by the DNS query.

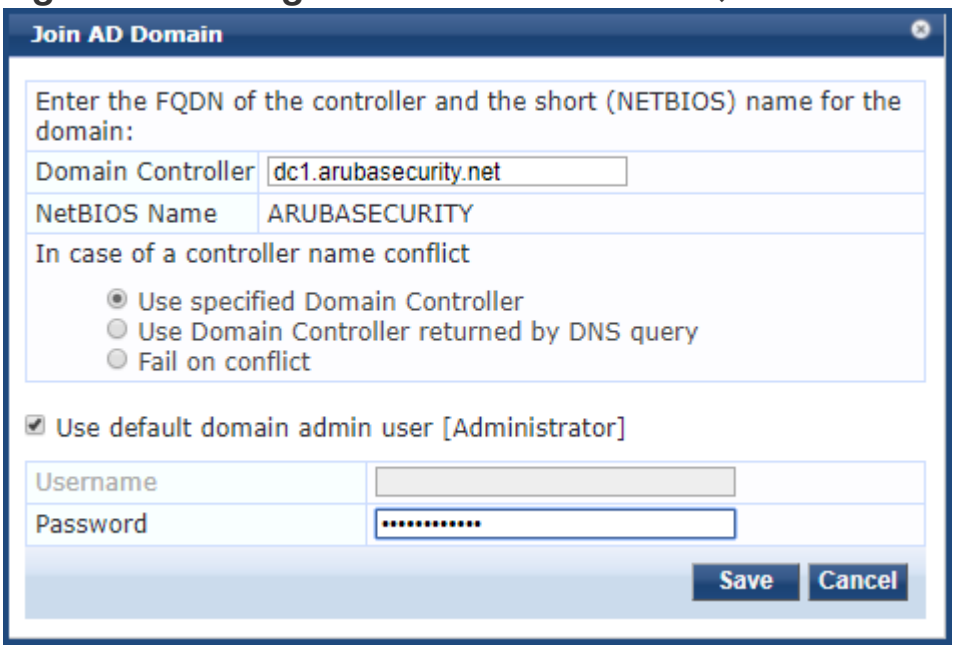
Parameter	Action/Description
	• **Fail on conflict**: Abort the Join Domai n operation.
Use default domain admin user	Check this box to use the Administrator u ser name to join the domain. **NOTE** In a production environment , it is likely that an Administ rative username that has pe rmissions to join machines t o the domain would be used for the default domain admi n user. In that case, 1) disab le (that is, uncheck) the **Use default domain admin user [Administrator]** check box a nd 2) enter the Admin usern ame and password in the fiel ds provided.
Username	Enter the user ID of the domain administ rator account. **NOTE** This field is disabled if the **Use default domain admin u ser** check box is selected.
Password	Enter the password for the user account that will join Policy Manager with the do main (for related information, see Table 3 , which displays the characters that are al lowed and not allowed for the Active Dire ctory username and password).

Table 3. Active Directory UserName and Password Characters Allowed

Field	Characters Allowed
Username	~ ! @ # $ % ^ * _ - + = { } , . \ ' " ? /
Password	! @ # $ % ^ & * ( ) _ - + = { } < , > . ? /

The figure below shows the NetBIOS domain name and the **NetBIOS Name** field with the correct name.

Figure 6. Entering the DomainControllerFQDN



4. Click **Save**.

   The **Join AD Domain** screen opens.

   During the joining process, the screen displays the messages "Checking whether <name> domain controller is co-located with this ClearPass Policy Manager server." and then "*Adding host to AD domain*," The screen displays status during the joining process.

   When the joining process completes successfully, you see the message "*Added host to the domain*."

   - If Policy Manager cannot identify the domain controller because Active Directory has not been configured with site information, this form displays the warning "Unable to identify which Active Directory site this Policy Manager server is

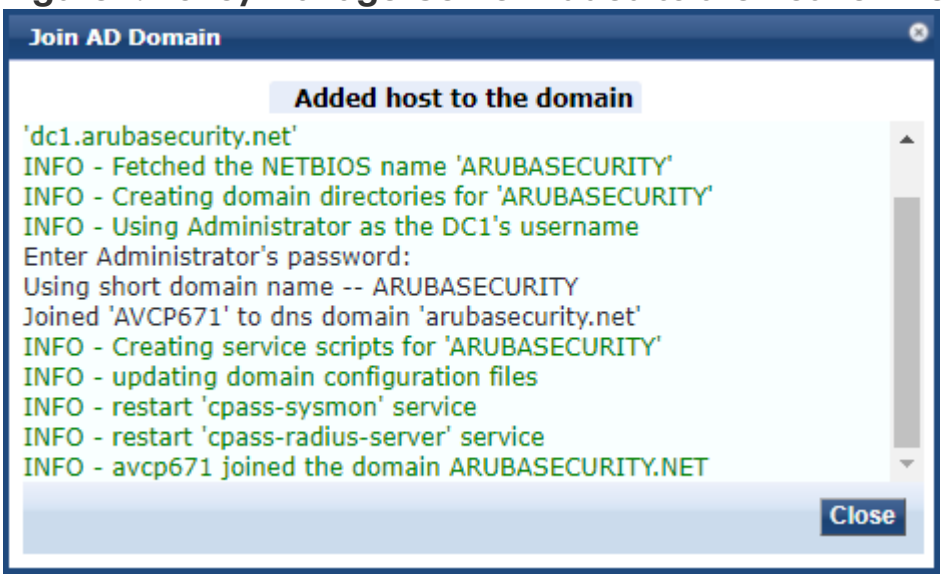
located in. Authentications might fail or be impacted if the selected domain controller is not co-located."

- If Policy Manager determines that you are not joining the closest domain controller, the form displays both a warning message and a **Find Domain Controller** button that you can use to find the closest domain controllers.

5. To display a list of available controllers, click the **Find Domain Controller** button .

   You can then select a controller from the available list, or proceed with the domain Controller you originally selected.

Figure 7. Policy ManagerServer Added to the Active Directory Domain



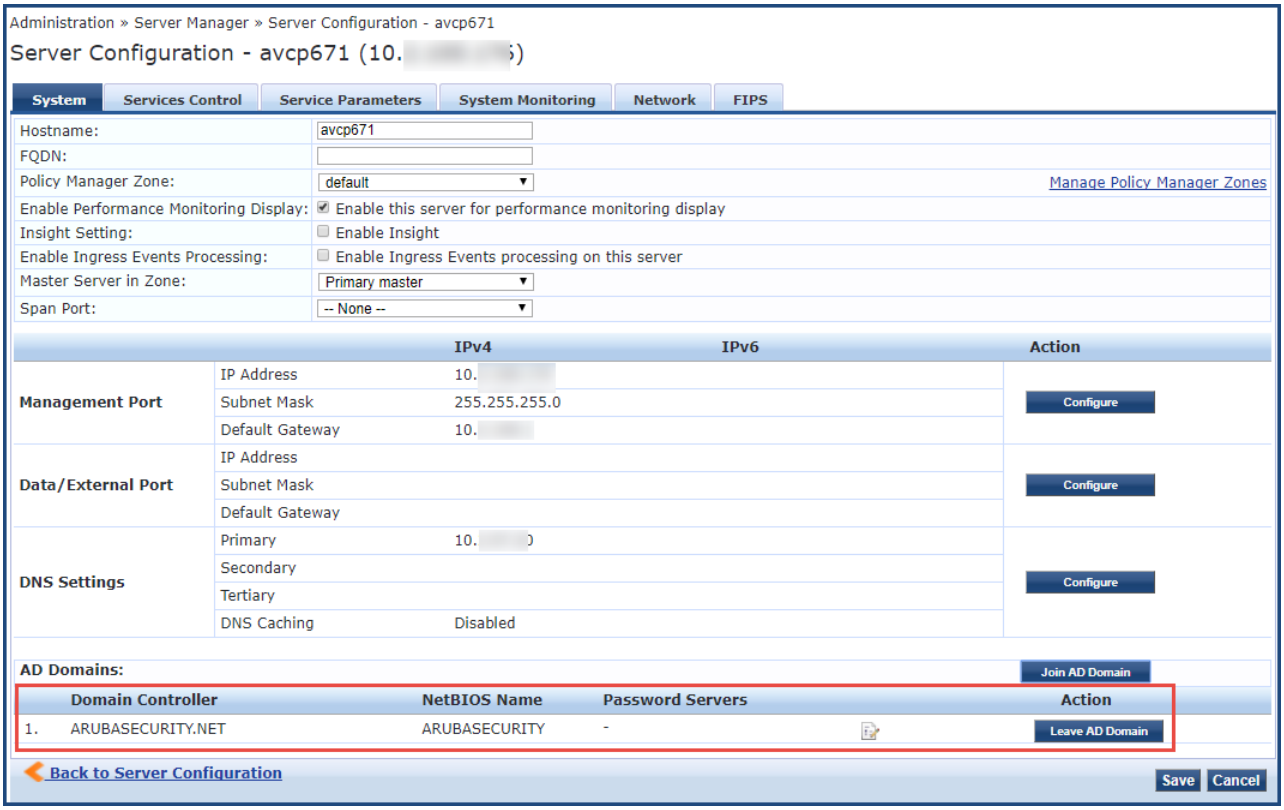
The **Join AD Domain** status screen indicates that the services have restarted. As shown in Figure 7, the final INFO line states that the selected Policy Manager server joined the domain.

6. Click **Close**.

   You return to the **Server Configuration** page, and it now shows that the Policy Manager server is joined to the domain.

   Now that the Policy Manager server has joined the domain, the server can authenticate users with Active Directory.

Figure 8. Policy ManagerServer Joined to Domain



Now that the server has joined the domain, the server can authenticate users with Active Directory.

About the Authentication Source and the Authorization Process

During the NT LAN Manager authentication process, Policy Manager queries Active Directory for a suitable domain controller to use to handle the authentication.

Please note that when used with 802.1x EAP-PEAP-MSCHAPv2 services, the authentication process is separate from the Active Directory authentication source in Policy Manager, which in this context only handles authorization.

Optionally, you can configure a list of domain controllers to be used for MSCHAPv2 authentication, as described in the next section, Manually Configuring Active Directory Password Servers.

If you do not specify this list of domain controllers, all available domain controllers obtained from DNS will be used for authentication.

Manually Configuring Active Directory Password Servers

About this task

To manually specify Active Directory domain controllers for authentication:

Procedure

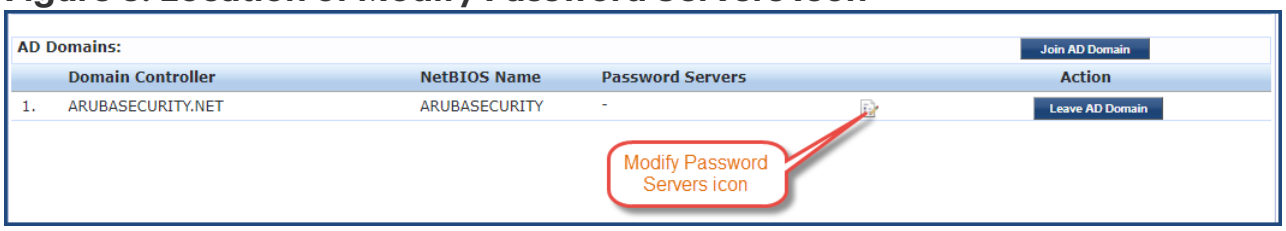
1. Navigate to **Administration > Server Manager > Server Configuration**.

2. Select the Policy Manager server name.

   The **Server Configuration** page for the selected server opens to the **System** tab.

3. Click the **Modify Password Servers** icon (located at the bottom of the **System** page).

Figure 9. Location of Modify Password Servers Icon



The **Configure AD Passwords Servers** screen opens.

Figure 10. Configuring Active Directory Password Servers

4. In the **Password Servers** text box, enter the names of the domain controllers that will be used for authentication (one entry per line).

5. When finished, click **Save**.

Disassociating a Policy Manager Server From an Active Directory Domain

About this task

When leaving an Active Directory domain, Policy Manager uses the supplied credentials to remove the “Computer Account” from Active Directory before removing its own relationship with that AD domain that exists on Policy Manager itself.

If a Policy Manager server is already part of multiple Active Directory domains, follow this procedure to disassociate this Policy Manager appliance from an Active Directory domain.

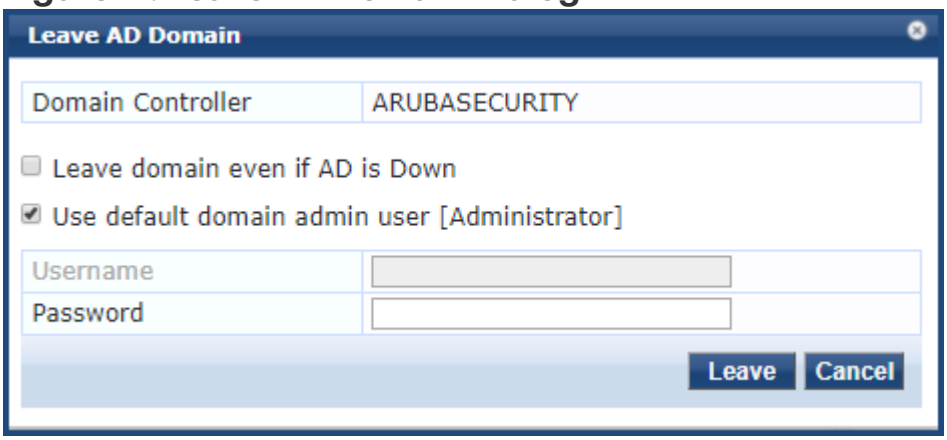
To disassociate a Policy Manager server from an Active Directory domain:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. Select the name of the Policy Manager server that you want to disassociate from the domain.
3. Click **Leave AD Domain**.

   The **Leave AD Domain** dialog opens.

Figure 11. Leave AD Domain Dialog



4. Specify the **Leave AD Domain** parameters as described in the following table:

Table 4. Leave AD Domain Parameters

Parameter	Action/Description
Leave domain even if AD is Down	When you enable **Leave domain even if AD is Down**, Policy Manager tries to delete the Computer Account from Active Directory, but instead of stopping the leave process if Active Directory is unreachable, Policy Manager removes its own links to Active Directory. In that event, the Active Directory administrator should manually remove the Computer Account.

Parameter	Action/Description
Use default domain admin user [Administrator]	This option allows you to use either the *administrator* user account or another account in Active Directory that has the same privileges. This option is enabled by default. **NOTE** The Adminstrator account doesn't have to be the same account that is used to join the server to the domain—it only has to be an account that has permissions to do this operation.
Password	Enter the Administrator account password.

5. Click **Leave**.

   The **Leave AD Domain** status screen appears, with the heading message: "*Removing host from the AD domain.*"

   When the process is complete, the status screen displays the message: "*Removed host from the domain.*"

6. Click **Close**.

   When you return to the **Server Configuration > System** page, the Policy Manager server is no longer listed in the AD Domains section.

7. Click **Save**.

Adding Active Directory as an Authentication Source to Policy Manager

After you have joined Policy Manager to the domain, add an authentication source to Policy Manager in order to process authentication and authorization against this

Active Directory. This section describes how to add the Active Directory server as an authentication source in Policy Manager.

If using EAP-PEAP-MS-CHAPv2, you must join Policy Manager to the Active Directory domain. Joining the Active Directory domain is necessary for Policy Manager to access the user credential information stored in the Active Directory.

> **NOTE**
> If using EAP-TLS for checking client certificates, you don't need to join the Policy Manager server to the domain.

About Authorization

Authorization determines access rights related to computer security in general, and access control in particular. In functional terms, "to authorize" is to define an access policy. In the context of 802.1X authentication, authorization is accomplished using LDAP (Lightweight Directory Access Protocol). LDAP is a protocol for accessing directories. It provides a means to search, retrieve, and manipulate directory content, and provide access to a rich set of security functions. LDAP provides the ability to locate organizations, individuals, and other resources, such as files and devices in a network, whether on the Internet or on a corporate intranet.

> **NOTE**
> When authenticating users via EAP-PEAP-MSCHAPv2 to Active Directory, the authentication source created in Policy Manager only serves for authorization and not authentication. When authenticating users via Captive Portal, the authentication source created in Policy Manager serves both authorization and authentication functions.

User Objects

The directory is simply a list of objects. One of those types of objects is a "user" object, and that user object has a number of different attributes, such as last name, first name, group membership, phone number, and so on. There is a default set of attributes, however, the list of user attributes is customizable.

An authentication source of type Active Directory is essentially an LDAP query that Policy Manager runs. When a user is authenticating, they give Policy Manager their username. After authentication is successfully completed, Policy Manager takes the username and, using Active Directory via LDAP, looks up the user and finds all the LDAP attributes pertaining to that user.

About the Bind Operation

The Bind operation allows authentication information to be exchanged between the client and server to establish a new authorization state. In the Active Directory context, *bind* is a term that indicates authenticating to an LDAP server, which Active Directory

must do before it can run any queries against the LDAP server. Active Directory must provide credentials to prove to the LDAP server that it is authorized to make queries against it. Only entities and devices that have an account can make queries against Active Directory.

Adding Active Directory as an Authentication Source

This procedure creates an enforcement policy based on information Active Directory has about users in the domain.

Group Membership

About this task

The most commonly applied user attribute is *group membership*. In Active Directory, you can define groups and put users into the groups you define. For example, a college might have groups for students, faculty, and contractors. For example, the enforcement policy can dictate that students are given a limited level of access to the network, whereas members of the faculty are typically given a higher level of access to the network, though faculty access would be less network access than that granted to network administrators and operators.

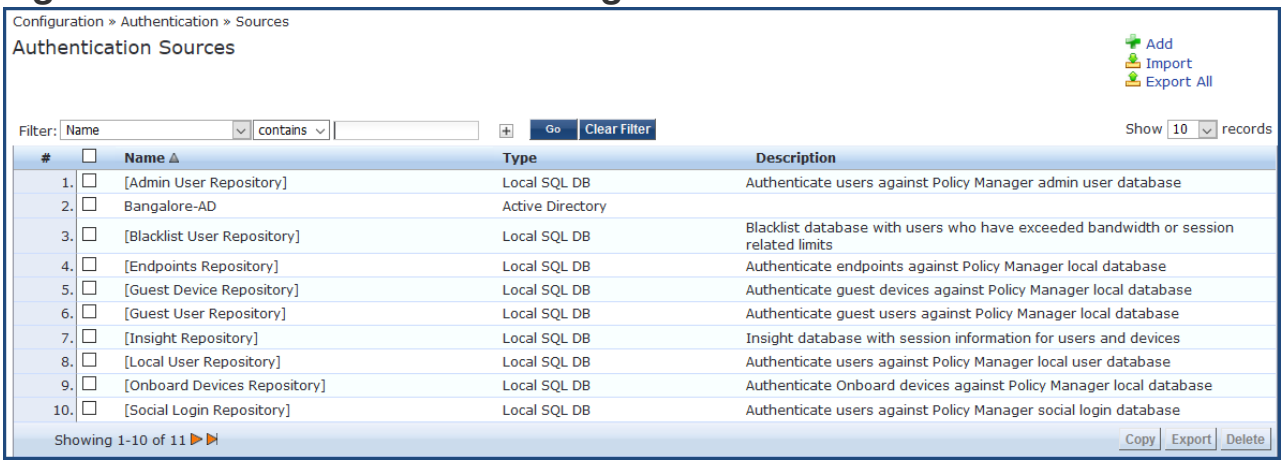
Active Directory needs to know which group each user who is trying to authenticate is a member of. This allows Policy Manager to do *enforcement*, which is the process of specifying what each user will be allowed to do on the network. After authentication takes place, there are usually additional enforcement details provided to the controller, such as VLAN assignment and user membership.

To add Active Directory as an authentication source:

Procedure

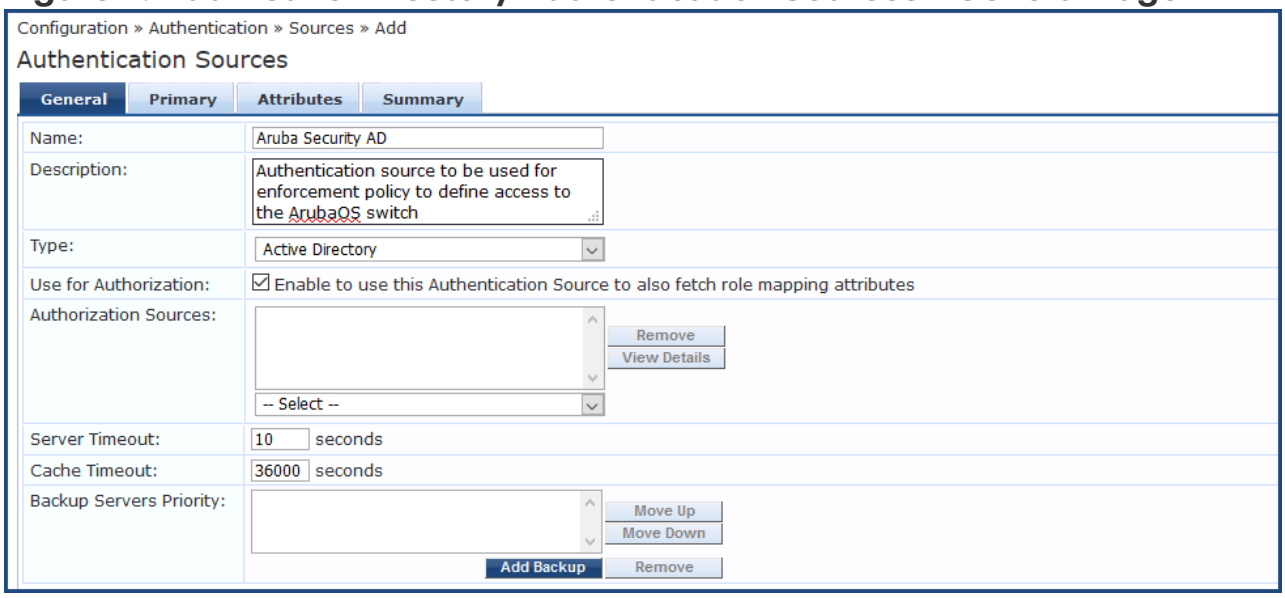
1. In Policy Manager, navigate to **Configuration > Authentication > Sources**. The following page opens:

Figure 1. Authentication Sources Page



2. Click the **Add** link. The **Add Authentication Sources > General** page opens.

Figure 2. Add Active Directory Authentication Sources > General Page



3. Enter the **Add Authentication Sources**, as described in Table 1.

Table 1. General Parameters for Adding an AD Authentication Source

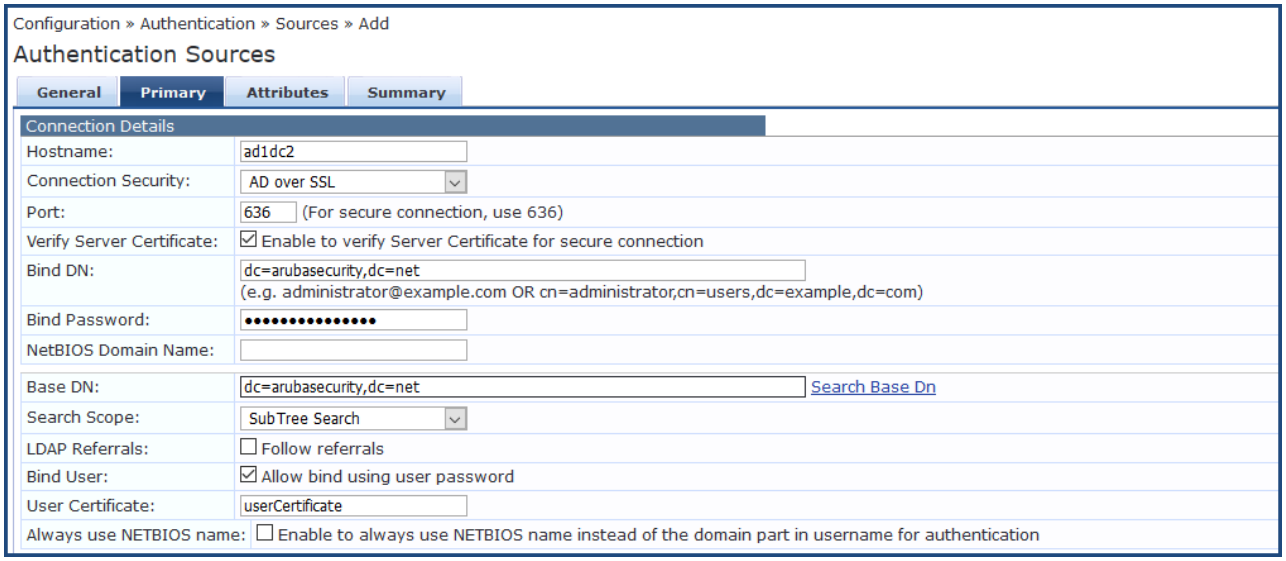
Parameter	Action/Description
Name	Enter the name of the Active Directory authentication source. The name of this authentication source is needed when you create the enforcement policy (see Switch Management TACACS+) and the role - mapping policy. **NOTE** In this example, we assign the name of the Active Directory authentication source as "*Aruba Security AD*." Assign a name to the authentication source that is appropriate for your network or installation.
Description	Provide the additional information to help identify the Active Directory authentication source.

Parameter	Action/Description
Type	If not already selected, select **Active Directory**.
Use for Authorization	When *Use for Authorization* is enabled, Policy Manager can use this authentication source to fetch role - mapping attributes. This option is enabled by default.
Authorization Sources	Specifies additional sources from which role - mapping attributes may be fetched . Select a previously configured authentication source from the drop - down list. • To add authentication source to the list of authorization sources, click **Add**. • To remove the authentication source from the list, click **Remove**. If Policy Manager authenticates the user or device from this authentication source, it also fetches role mapping attributes from these additional authorization sources.
Server Timeout	Specifies the duration in seconds Policy Manager waits before considering this server unreachable. If multiple backup servers are available, then this value indicates the duration in seconds Policy Manager waits before attempting to fail over from the primary to the backup servers in the order they are configured.
Cache Timeout	Policy Manager caches attributes fetched for an authenticating entity. This parameter controls the duration in number of seconds for which the attributes are cached.

Parameter	Action/Description
Backup Servers Priority	a. To add a backup server, click **Add Bac kup**. • The **Backup 1** tab appears. The **Pri mary** page parameters are prepop ulated in the **Backup 1** page. b. To complete the configuration for the backup server, specify the hostname for the backup server. • To remove a backup server, select t he server name and click **Remove**. • To change the server priority of the backup servers, select **Move Up** or **Move Down**. ◦ The server priority is the order Policy Manager attempts to con nect to the backup servers whe n the primary server is unreach able. **NOTE** Aruba recommends settin g up one or more backup servers.

4. When satisfied with these settings, click **Next**. The **Add Authentication Sources > Primary** page opens.

Figure 3. Add Active Directory Authentication Sources > Primary Page



5. Enter the information for each of the required parameters as described in Table 2.

Table 2. Primary Parameters for an Active Directory Authentication Source

Parameter	Action/Description
Hostname	Enter the name or IP addres The hostname entered mus Policy Manager uses LDAP
Connection Security	Set **Connection Security** to This enables the secure soc rectory. Selecting **AD over** **NOTE** In a production en ds to an LDAP ser over the network h the port set to **6** **NOTE** To ensure success Directory/LDAP se porting the Root C
Port	Specify the TCP port at whi

Parameter	Action/Description
	For a single domain Active • Default port for LDAP: **3** • Default port for LDAP ov When you set the *Connecti* **6**. For a multi - domain Active obal catalog are: • Default port without SSI • Default port with SSL: **3**
Verify Server Certificate	Enable this option to verify
Bind DN	Enter the Distinguished Nai for records. This is a require The Bind DN text box speci of an Active Directory user rator account). For example `CN=Administrator,CN=U` **NOTE** You may need to This user account must hav The Bind DN user, such as A ount. • For a single domain Act he same branch and be • For a multi - domain Ac Base DN text box empty lti - domain forest. Policy Manager fills in the c Policy Manager also popula nformation, see About the

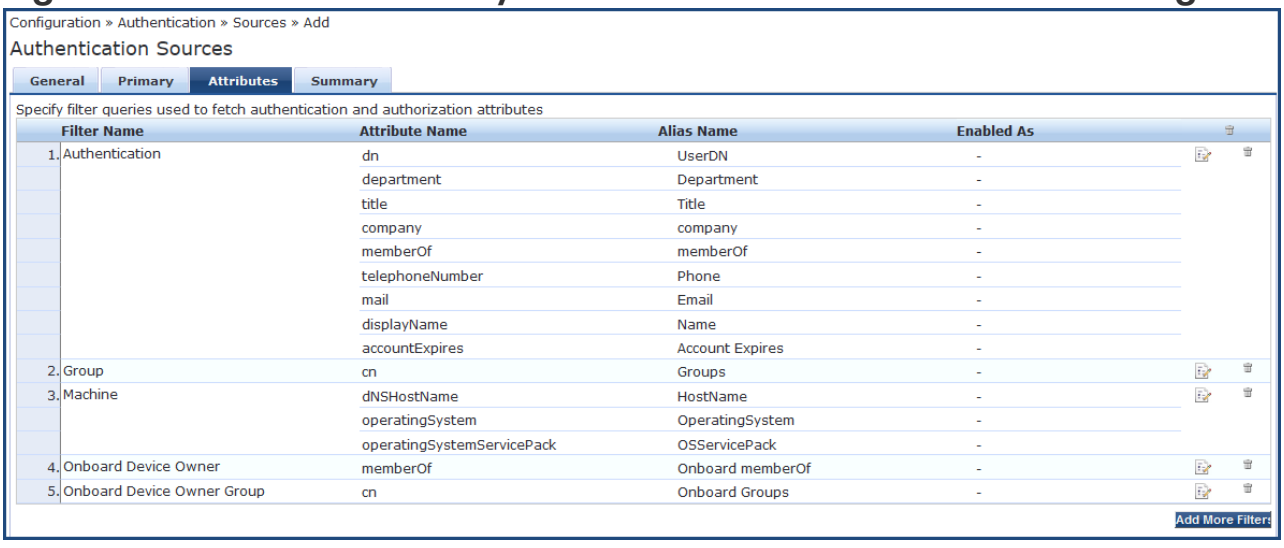
Parameter	Action/Description
Bind Password	This is the text box for the A This is a required paramete Enter the **Bind Password**. **NOTE** The Bind passwor account.
NetBIOS Domain Name	This field is automatically p
Base DN	For a single domain Active ame (DN) of the starting po ample: **DC=mycompany,DC** m which you can later filter **NOTE** Best practices rec atically populated ase DN as far as p r example, if all yo the Base DN to se To browse the LDAP directo a. Click **Search Base DN**. T b. Navigate to the DN you v c. Click the appropriate no Active Directory Domain N text box blank **NOTE** This is also a w lues entered fo browse the dire
Search Scope	Search scope is related to t rch for your objects.

Parameter	Action/Description
	Specify the search scope y • Subtree Search: Search • One - Level Search: Loc • Base Object: Searches a
LDAP Referrals	Aruba recommends *not* en P server to find a specific u LDAP server, which can cau
Bind User	This option allows the bind heck box is enabled by def
User Certificate	Leave the value that is auto rectory administrator has a
Always use NetBIOS name	Enable this option only if yo to authenticate the user ins tribute.

6. When satisfied with the **Add Authentication Sources Primary** page settings, click **Next**. The **Add Active Directory Authentication Sources > Attributes** page opens.

   The **Attributes** page defines the Active Directory or LDAP Directory query filters and the attributes fetched using those filters.

Figure 4. Add Active Directory Authentication Sources > Attributes Page



7. Click **Save**. You return to the **Authentication Sources** page where the new authentication source is now listed. The following message is displayed: *Authentication source* <authentication source name> *added*

Obtaining and Installing a Signed Certificate From Active Directory

This section describes how to obtain and install a signed server certificate from Active Directory for 802.1X authentication. This section contains the following information:

- About Certificates in Policy Manager Deployments
- Tasks to Obtain a Signed Certificate from Active Directory
- Creating a Certificate Signing Request
- Importing the Root CA Files to the Certificate Trust List
- Obtaining a Signed Certificate from Active Directory
- Importing a Server Certificate into ClearPass

About Certificates in Policy Manager Deployments

A certificate is a file that makes it possible for network devices to communicate with each other securely. For example, in Policy Manager deployments, certificates are provided for all devices involved in authentication, such as client laptops, smart phones, Mobility controllers, Mobility Access Switches, ArubaOSes, Policy Manager servers, and so on.

Certificates help you communicate securely in two ways:

- Certificates help devices verify the identity of other devices.
- Certificates enable devices to use encryption to securely communicate with each other.

  When a certificate is created, two keys are generated; the private key and the public key. The private key is always stored securely and never sent out. If the private key is compromised, the entire security framework established by the certificate is compromised. The public key contains important information about the certificate owner. The public key is inside the file sent to all devices that wish to communicate with the certificate owner. This file contains additional information about the identity of the certificate owner's device.

Public and private key pairs are generated so that any data encrypted by one of these keys can only be decrypted by the other corresponding key. Any data encrypted by the private key can only be decrypted by the corresponding public key. Conversely, any data encrypted by the public key can only be decrypted by the corresponding private key.

When Certificate Usage Is Necessary

There are three common Policy Manager deployments in which certificates are necessary:

- When using HTTPS to manage network devices such as mobility controllers, mobility access switches, ArubaOSes or Policy Manager servers.
- During captive portal authentication.
- When using 802.1X authentication.

How 802.1X Authentication Uses Server Certificates

About this task

When an employee attempts to log into his laptop, the EAP-PEAP authentication process begins:

Procedure

1. The Policy Manager server sends the server certificate to the employee's device.
2. The employee sends his encrypted username and password to the server.
3. The server verifies the employee's credentials, and the employee is connected to the network.

Using Both Client and Server Certificates

There is a potential problem in this authentication sequence—the employee verified the server's identity, but the server didn't verify the employee's identity. It is possible that the user stole the username and password from another employee and is using these stolen credentials on his own device.

This problem can be solved by using both a client certificate and a server certificate. Because EAP-TLS authentication employs both server and client certificates, when the employee begins authentication, the Policy Manager server sends the server certificate to the employee's laptop. The employee's laptop then sends the client certificate to the server.

Both the client and the server can then verify the identity of the other party and are ready to proceed: The employee sends the encrypted username and password to the server, the server verifies the employee's credentials, and the employee is connected to the network. This access process is secure.

Tasks to Obtain a Signed Certificate from Active Directory

About this task

The tasks to obtain a signed certificate from Active Directory are as follows:

Procedure

1. Create a Certificate Signing Request.
2. Import the root Certificate Authority file to the Certificate Trust List.
3. Obtain a signed certificate from Active Directory.
4. Import the server certificate into the Policy Manager server.

   These tasks are described in the following sections:

Creating a Certificate Signing Request

About this task

This task creates a Certificate Signing Request to be signed by a Certificate Authority (CA).

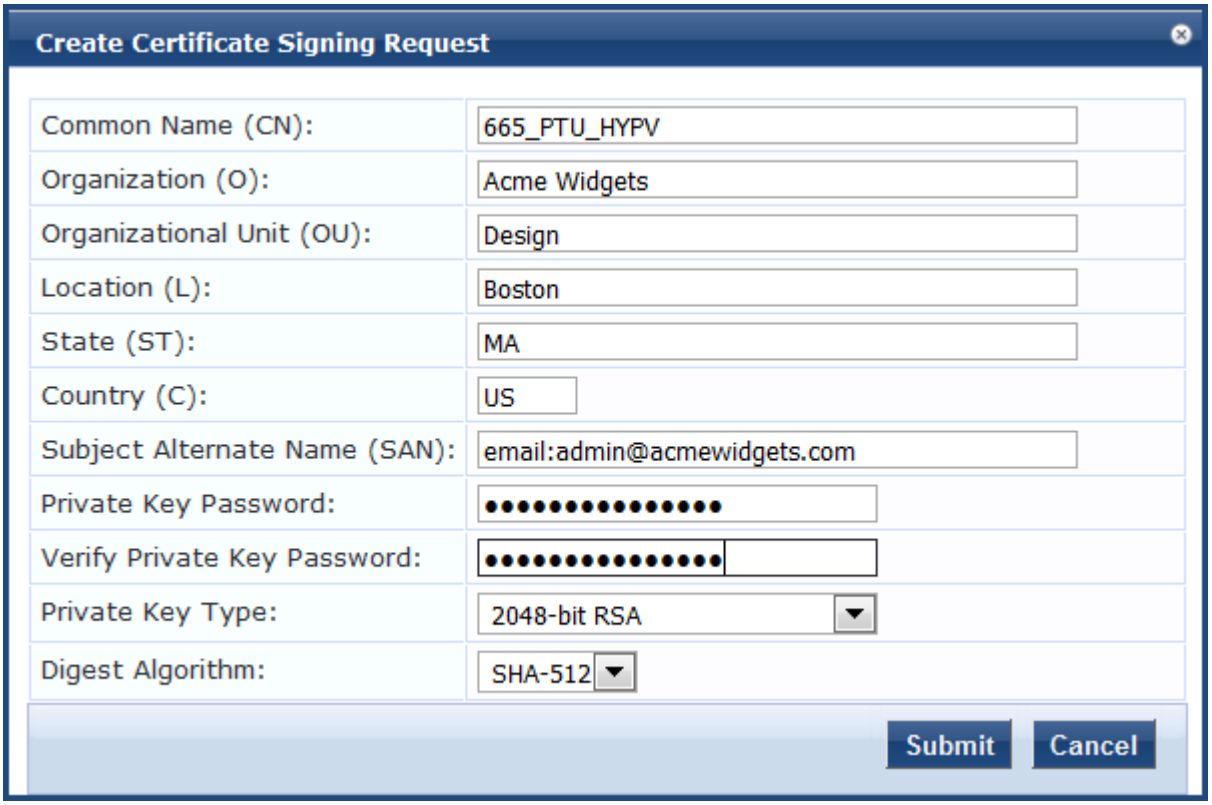
Figure 1 shows an example of the **Create Certificate Signing Request** page, followed by descriptions of each parameter (see Table 1).

To create a Certificate Signing Request:

Procedure

1. In Policy Manager, navigate to **Administration** > **Certificates** > **Server Certificates**.
2. Select the **Create Certificate Signing Request** link.

Figure 1. Create Certificate Signing Request Dialog



3. Enter the information for each of the required parameters as described in Table 1.

Table 1. Parameters for Creating a Certificate Signing Request

Parameter	Action/Description
Common Name	Displays the name associated with this e ntity. This can be a host name, IP addres s, or other name. The default is the fully - qualified domain name (FQDN). This fie ld is mandatory.
Organization (O)	Specify the name of the organization. Thi s field is optional.
Organizational Unit (OU)	Specify the name of the department, divi sion, or section. This field is optional.
Location (L) State (ST) Country (C)	Specify the name of the state, country, a nd/or another location. These fields are o ptional

Parameter	Action/Description
Subject Alternate Name (SAN)	Specify the alternative names for the specified Common Name. **NOTE** Specify the SAN in the following formats: • email: *email_address* • URI: *url* • IP: *ip_address* • dns: *dns_name* • rid: *id* This field is optional.
Private Key Password	a. Enter the private key password, then reenter it to verify the password.
Private Key Type	a. Select the length for the generated private key types from the following options: • 1024 - bit RSA • 2048 - bit RSA • 4096 - bit RSA • X9.62/SECG curve over a 256 bit prime field • NIST/SECG curve over a 384 bit prime field The default private key type is **2048 - bit RSA**.
Digest Algorithm	a. Select one of the following message digest algorithms:

Parameter	Action/Description
	• MD5 • SHA - 1 • SHA - 224 • SHA - 256 • SHA - 384 • SHA - 512 **NOTE** The MD5 algorithm is not available in **FIPS** mode.

4. When satisfied with the certificate signing request settings, click **Submit**. The **Create Certificate Signing Request** generates and displays (see Figure 2).

Figure 2. Displayed View of the Create Certificate Signing Request

5. Copy the contents of the certificate request into a text file so you can paste it into the Directory Certificate Services web form as described in Obtaining a Signed Certificate from Active Directory.

6. To save the Certificate Signing Request file, click **Download CSR**.

Importing the Root CA Files to the Certificate Trust List

About this task

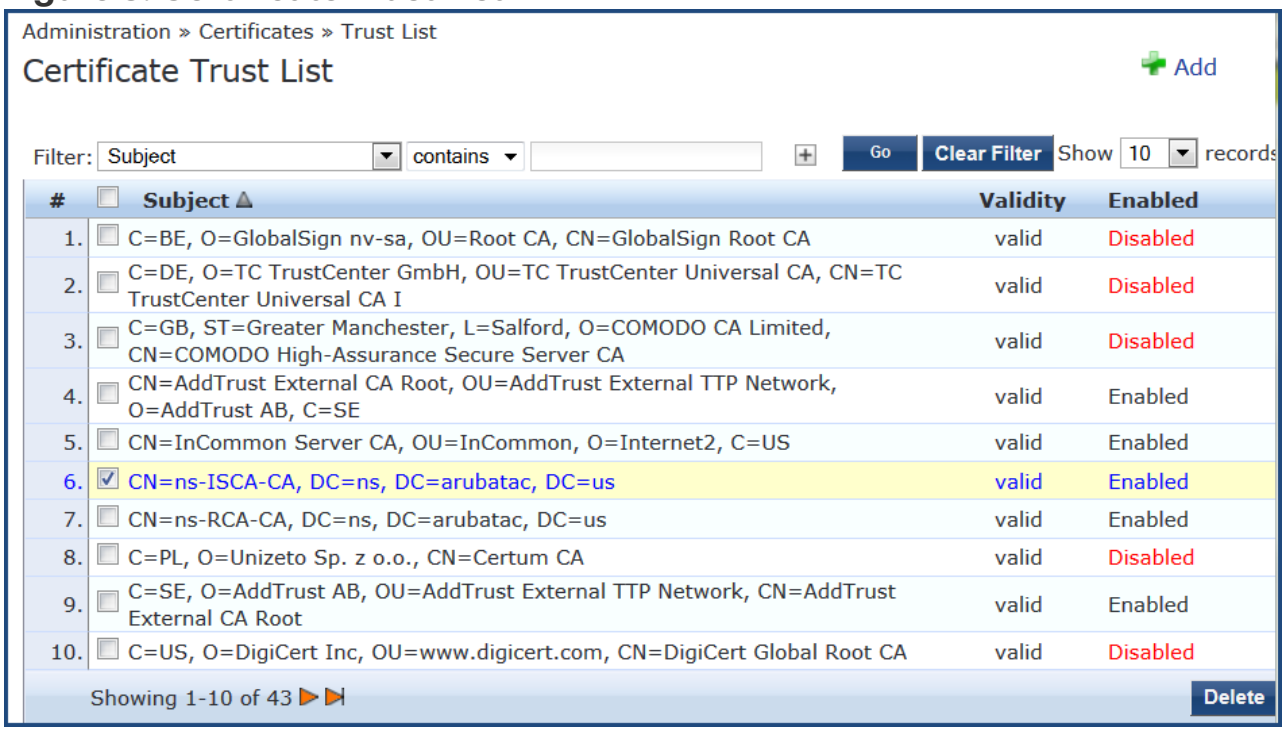
Make sure the root Certificate Authority (CA) certificate and any intermediate CA certificates are downloaded as separate base-64-encoded files and imported into the Certificate Trust List in Policy Manager *before* starting this operation.

To import the root CA files into the Policy Manager server Certificate Trusted List:

Procedure

1. Get the root CA certificate and any intermediate CA certificates from your Active Directory administrator. This typically consists of a root CA certificate and one or more intermediate CA certificates.
2. In Policy Manager, navigate to **Administration > Certificates > Trust List**.

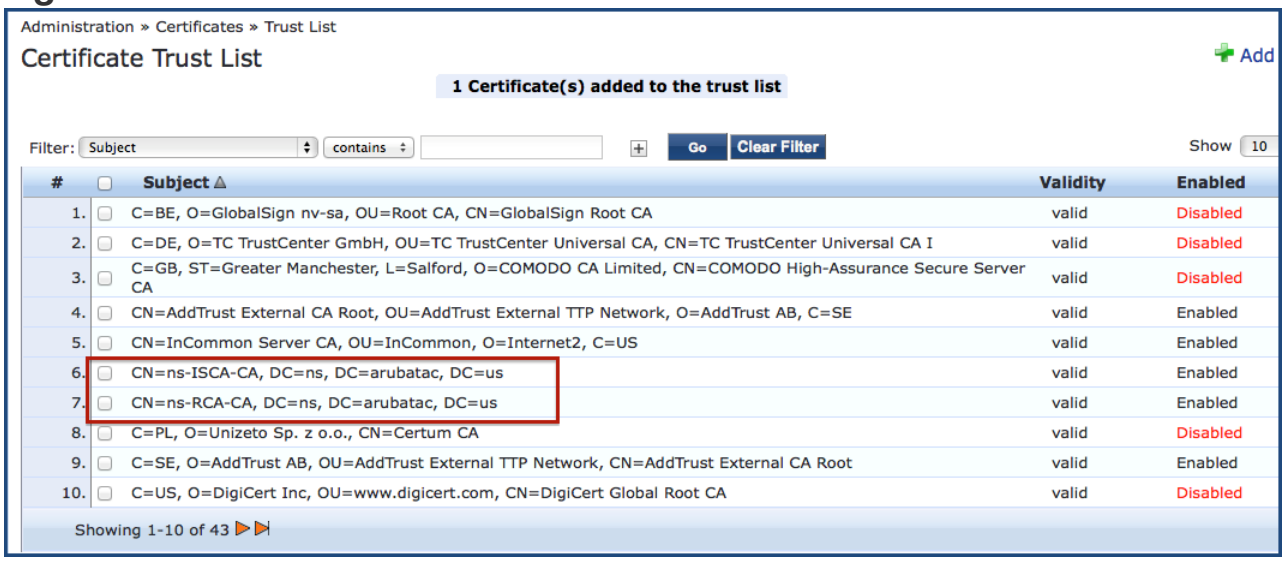
Figure 3. Certificate Trust List



3. To add the certificate file(s) to the Certificate Trust List, click **Add**, then browse to the root CA certificate file on your computer. The root CA certificate file is now listed in the Certificate Trust List.

> **NOTE**
>
> Be sure to add the root CA file first, then add the intermediate CA files after you've added the root CA file.

Figure 4. New Root CA Files Added to the Certificate Trust List



4. Make sure the **Enabled** column for the newly added certificate says *Enabled*, which is the status displayed when you successfully import a certificate manually.

5. Repeat steps 2, 3, and 4 for each certificate you received from your Active Directory administrator.

Obtaining a Signed Certificate from Active Directory

About this task

This section describes how to obtain a signed server certificate from Active Directory.

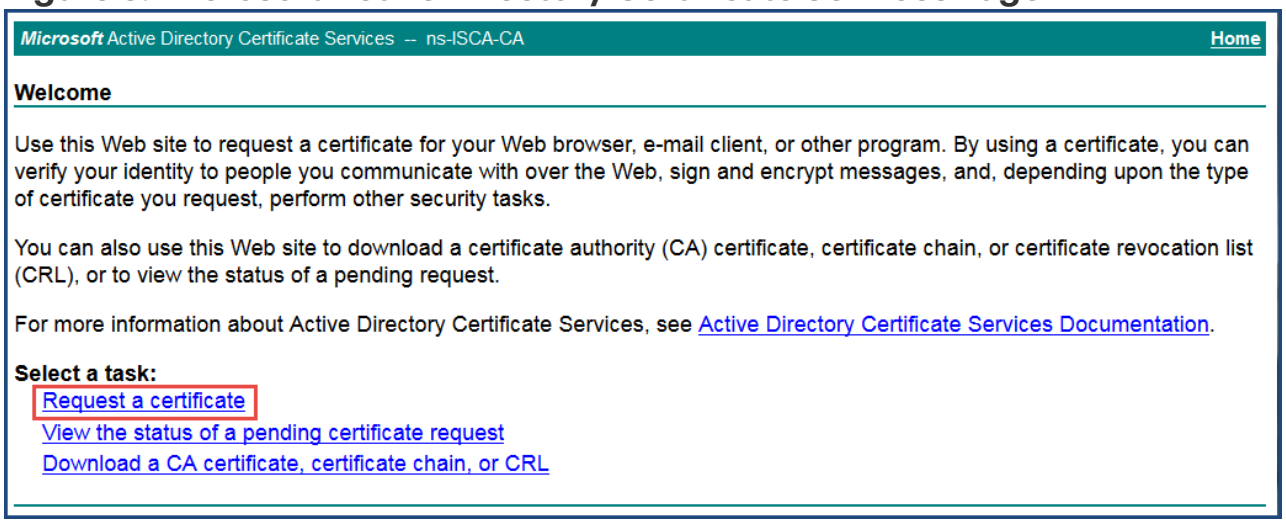
> **NOTE**
>
> Before you begin this operation, have the copy of the Certificate Signing Request at hand, as described in Step 4 of Creating a Certificate Signing Request .
>
> Also note the location where you saved the Certificate Signing Request and the private key password files, as you will need to retrieve these items to complete this operation.

To obtain a signed certificate from Active Directory:

Procedure

1. Navigate to the **Microsoft Active Directory Certificate Services** page:

Figure 5. Microsoft Active Directory Certificate Services Page

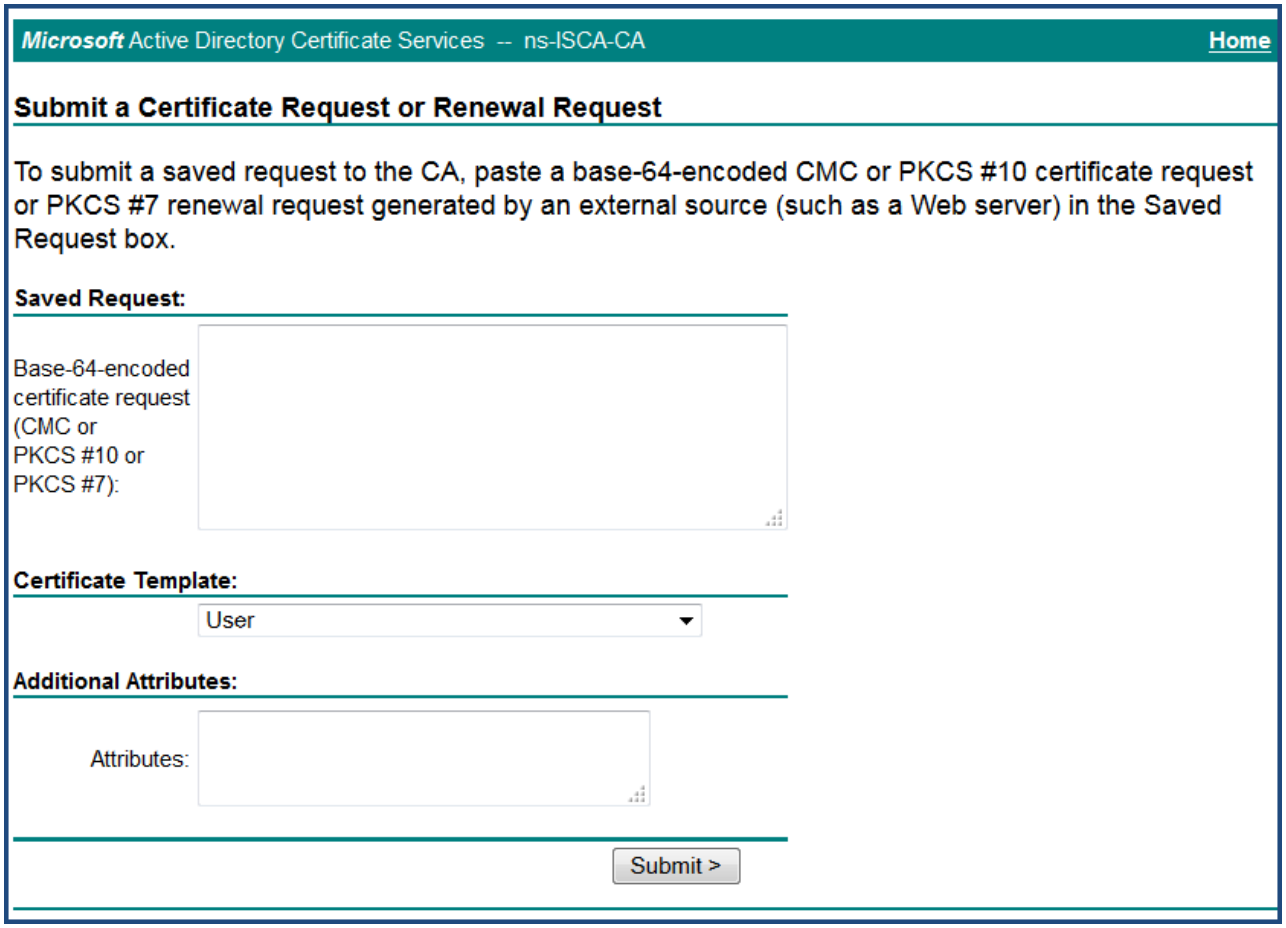


2. Click **Request a certificate**.

Figure 6. Request a Certificate

3. Choose **advanced certificate request**. The **Submit a Certificate Request or Renewal Request** dialog opens. This request operation submits a saved certificate request to the Certificate Authority.

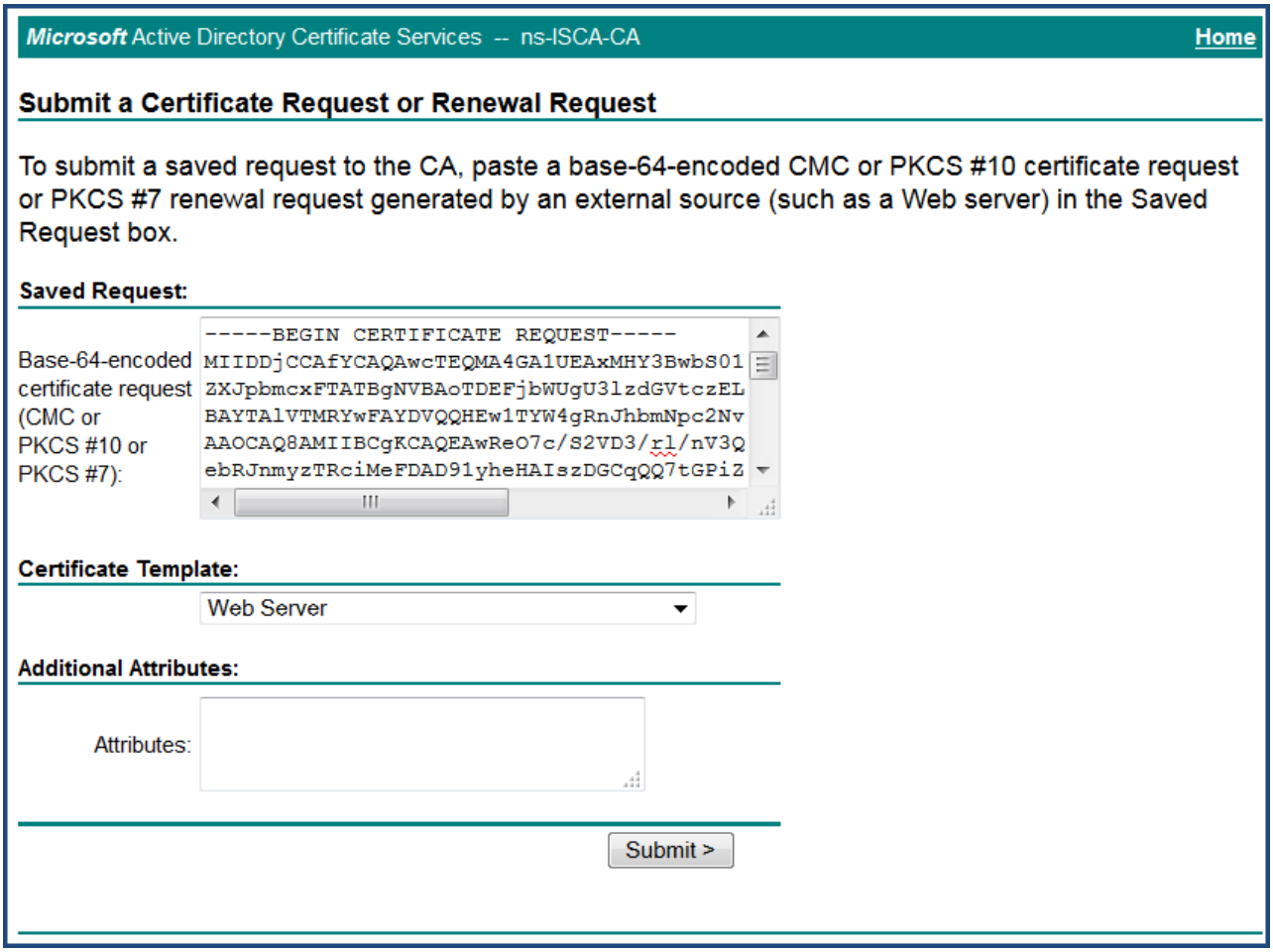
Figure 7. Submit a Certificate Request



4. Copy the contents of the Certificate Signing Request into the **Saved Request** text box.

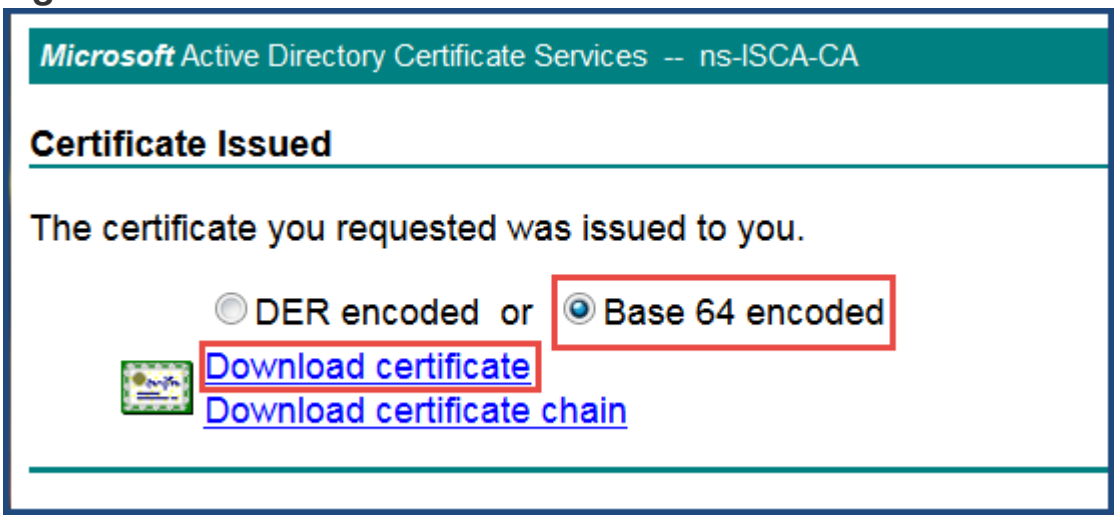
5. In the **Certificate Template** drop-down menu, select **Web Server**. Figure 8 shows an example of the completed Certificate Request web form.

Figure 8. Completed Submit a Certificate Request Dialog



6. Click **Submit**. The **Certificate Issued** dialog opens.

Figure 9. Certificate Issued



7. Do the following:

   a. Select **Base 64 encoded**. Base-64 encoding is used for 802.1X authentication.

   b. Click **Download certificate**. The server certificate is downloaded to your system.

c. Be sure to note the name of the downloaded certificate so that you can identify it when you import the server certificate into the Policy Manager server.

Importing a Server Certificate into ClearPass

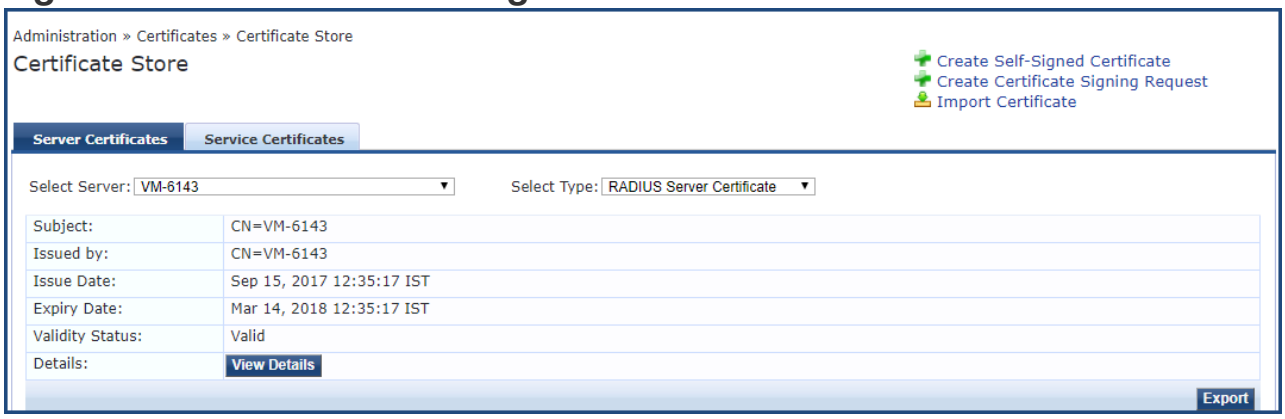
About this task

To import a server certificate into Policy Manager:

Procedure

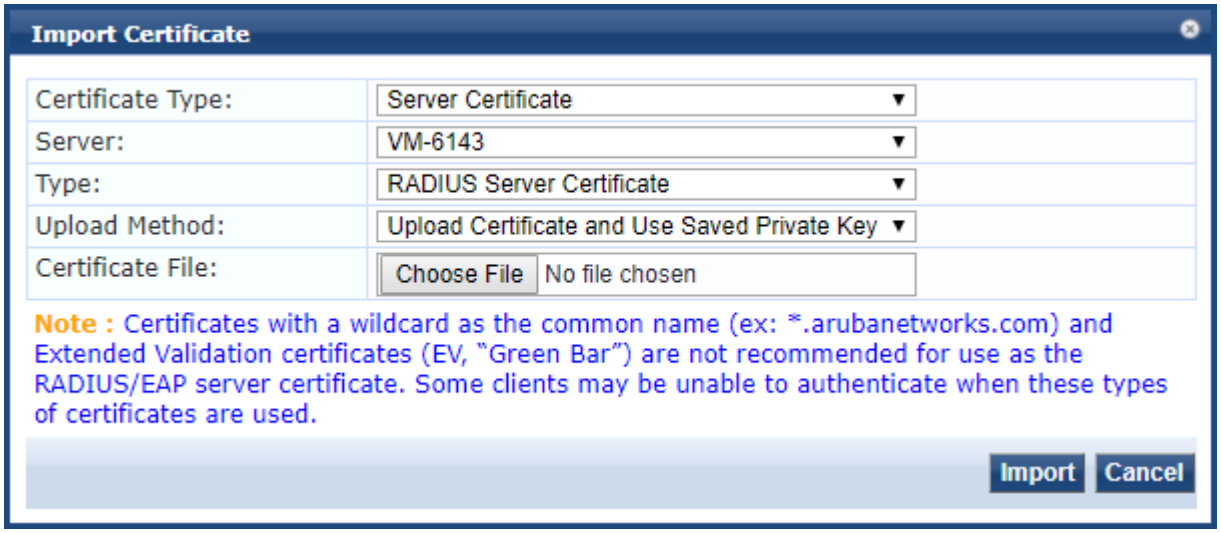
1. Navigate to **Administration > Certificates > Certificate Store**. The **Certificate Store** page opens.

Figure 10. Certificate Store Page



2. From the **Server Certificates** tab, click the **Import Certificate** link. The **Import Certificate** dialog opens:

Figure 11. Import Certificate Dialog



NOTE

For security reasons, certificates signed using SHA1RSA are not recommended. Importing certificates signed with stronger keys, such as RSA with a length of more than 1024 bits, is recommended.

NOTE

Policy Manager does not support importing the HTTPS Server Certificate chain or RADIUS Server Certificate chain in P7b Base64 format. A P7B file contains only certificates and chain certificates (intermediate certificate authorities), not the private key.

3. Specify the **Import Certificate** parameters as described in the following table:

Table 2. Import Server Certificate Parameters

Parameter	Action/Description
Certificate Type	**Server Certificate** is select
Server	Displays the name of the se
Type	Displays the type of server • RADIUS Server Certifica • HTTPS Service Certifica

Parameter	Action/Description
Upload Method	Select one of the following • The **Upload Certificate** o upload only the certif aved on the Policy Mana • With the **Upload PKCS#** oads the PKCS#12 file ar • If you select **Upload Ce** o upload the private key
Certificate File	Browse to the certificate fil **NOTE** Both certificates v tion certificates a tificate. Some clie rtificates are used

4. Click **Import**. The selected server certificate is imported into Policy Manager. The Server Certificate screen displays the message: *Server Certificate updated successfully. Please log in again to continue.*
5. Log out of the Policy Manager server, then log in again to resume operations on this server.

Manually Testing Login Credentials Against Active Directory

About this task

To test a username and password against the Active Directory, run the **ad auth** command in the Policy Manager CLI.

This command manually checks against Active Directory to indicate whether or not a username and password are valid.

Procedure

1. Enter the following CLI command:

   (`server`) # `ad auth –u <username> –n <NetBIOS_domain_name>`

   - **–u** indicates the username.
   - **–n** indicates the NetBIOS domain name.

   For example: (server) # ad auth –u administrator –n COLLEGE

2. Enter the password. If the username and password you provide in this command are correct, the following message is displayed:

   INFO – NT_STATUS_OK: Success (0x0)

   This message indicates that NT LAN Manager (NTLM) authentication (NTLM being the mechanism that Policy Manager uses to authenticate users) has succeeded.

Preparing for 802.1X Wireless Authentication with Active Directory

This section describes how to configure 802.1X wireless authentication with Active Directory® in a ClearPass managed network.

802.1X is an IEEE standard and method for authenticating user identity before providing a user network access. 802.1X provides an authentication mechanism to devices that need to attach to a wireless LAN or a wired LAN.

RADIUS (Remote Authentication Dial In User Service) is a protocol that provides centralized authentication, authorization, and accounting management.

For authentication, the wireless client can associate with a network access server (NAS) or a RADIUS client. Policy Manager is a RADIUS server. The wireless client can pass data traffic only after successful 802.1X authentication.

- 802.1X offers the capability to permit or deny network connectivity based on the identity of the end user or device.
- 802.1X enables port-based access control using authentication. An 802.1X-enabled port can be dynamically enabled or disabled based on the identity of the user or device that connects to it.

  Before authentication, the identity of the endpoint is unknown and all traffic is blocked. After authentication, the identity of the endpoint is known, and all traffic from that endpoint is allowed.

Subtopics

802.1X Authentication Components
Configuring 802.1X Wireless Authentication with Active Directory
Walking Through a 802.1X Authentication Scenario
Troubleshooting 802.1X Configuration Issues

802.1X Authentication Components

802.1x authentication consists of three components—a *supplicant*, an *authenticator*, and an *authentication server* (see Figure 1).

- The *supplicant*, or client, is the device attempting to gain access to the network. You can configure the user-centric network to support 802.1x authentication for wired users as well as wireless users.
- The *authenticator* is the gatekeeper to the network and permits or denies access to the supplicants.

  The mobility controller acts as the authenticator, relaying information between the authentication/Policy Manager server and the supplicant. The EAP type must be consistent between the authentication server and supplicant and is transparent to the mobility controller.
- The *authentication server* is typically a host running software supporting the RADIUS and EAP protocols. It provides a database of information required for authentication and informs the authenticator to deny or permit access to the supplicant. In this guide, the authentication server is the Policy Manager server.

Figure 1. 802.1X Authentication Network Components

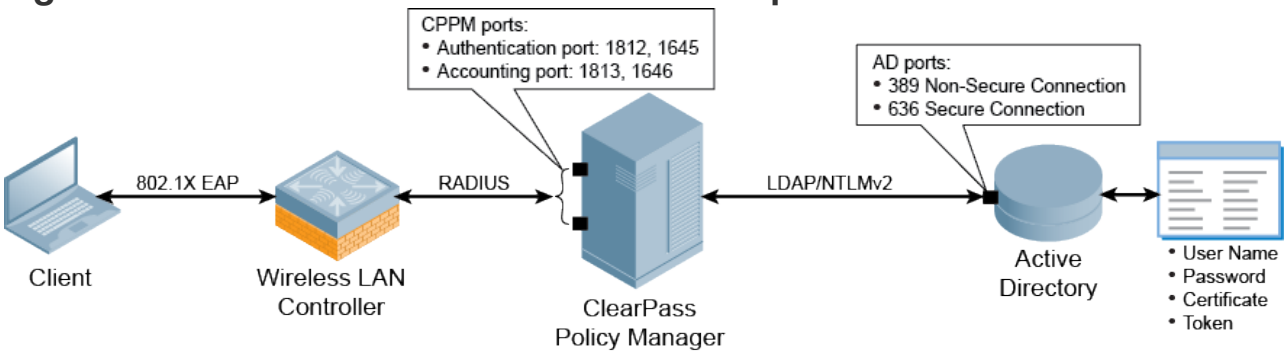


Table 1 describes each of the Policy Manager firewall ports that are used by Active Directory.

Table 1. Active Directory Policy Manager Firewall Ports

Firewall Port	Description
UDP Port 88	Used for Kerberos authentic
TCP and UDP Port 135	Used for domain controller - d client - to - domain contro
UDP Port 389	Used for LDAP to handle nor mputers to the domain cont
TCP and UDP Port 445	Used for Kerberos password
TCP Ports 3268 and 3269	Used for Global Catalog dist the domain controller. The Global Catalog makes th in a forest transparent to use . In a multidomain Active Dir rest, the Global Catalog prov f domain information for the plicas of all domain director replicas are distributed by m l Global Catalog servers in a
TCP and UDP Port 53	Used for DNS from the client and from the domain contro troller.
ICMP types echo (8) and echo - reply (0)	The Internet Control Messag ny messages identified by **Ty** nd **echo - reply (0)** are used ger host and the domain cor join operation (see Joining a ory Domain).

Configuring 802.1X Wireless Authentication with Active Directory

This section contains the following information:

- Authenticating Against Active Directory
- About the 802.1X Wireless Service
- Creating the 802.1X Wireless Service
- Deleting a Policy Manager Service

  This section describes how to configure 802.1X authentication with Active Directory in a ClearPass managed network.

Authenticating Against Active Directory

802.1x authentication can be used to authenticate users or computers against a user database or domain such as Microsoft Active Directory. For more information, go to: Preparing for Active Directory Authentication.

The supplicant (wireless client) authenticates against the RADIUS server (which is the authentication server/Policy Manager server) using EAP configured on both the supplicant and RADIUS server. They will, in turn, negotiate which EAP method to use based on the list of EAP methods each supports.

The mobility controller's (authenticator) role is to send authentication messages between the supplicant and authentication server. This means the RADIUS server is responsible for authenticating users.

Mobility controllers perform EAP exchanges between the supplicant and convert these to RADIUS access-request messages sent to the RADIUS server's IP address and the specified UDP port. For more information, go to: 802.1X EAP-PEAP Reference.

About the 802.1X Wireless Service

The basic Policy Manager use case configures a Policy Manager Service to identify and evaluate a RADIUS request from a user logging into a Mobility Controller.

Figure 1 illustrates the authentication process flow for an 802.1X Wireless Service.

Figure 1. 802.1X Wireless Service Authentication Process Flow

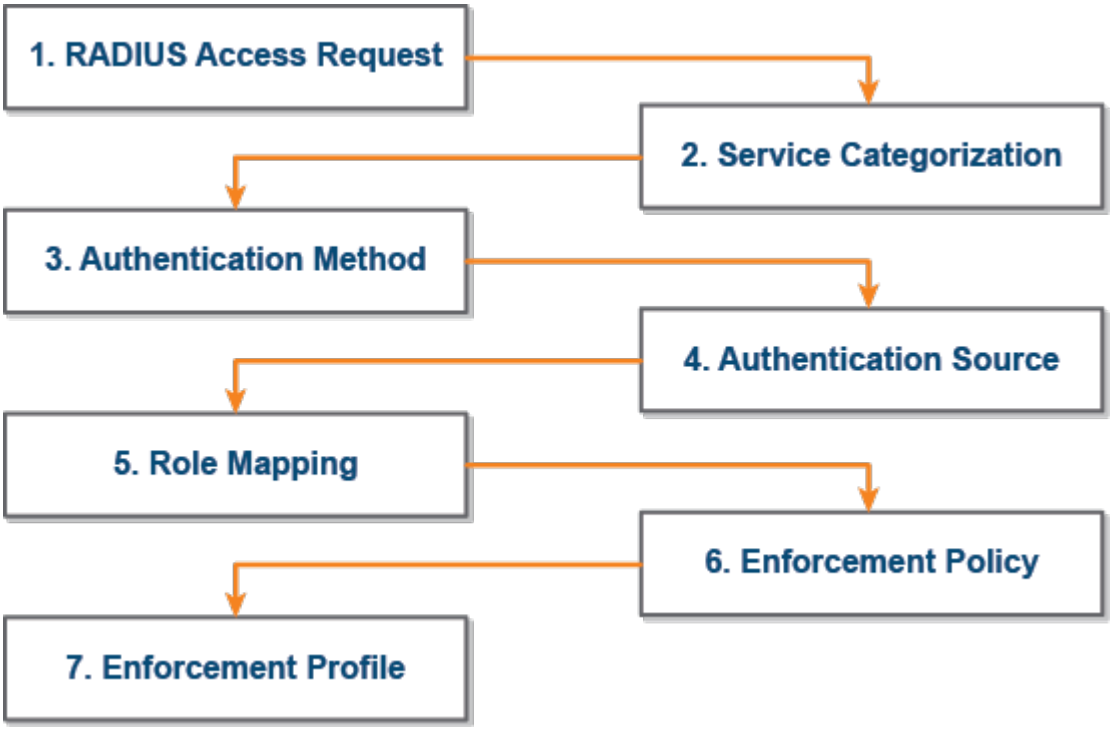


Table 1 provides descriptions of each of the 802.1X authentication processes illustrated in Figure 1.

Table 1. Description of the 802.1X Authentication Processes

	Authentication Process	Description
1	RADIUS Access - Request	The Network Access Server (NAS) sends a RADIUS acc ess request to Policy Manag er, which then evaluates th e request and identifies RAD IUS connection control attri butes.
2	Service Categorization	Based on the RADIUS conne ction control attributes iden tified by Policy Manager, the request will be categorized i nto a Policy Manager servic e.
3	Authentication Method	Policy Manager attempts to authenticate the user (in ord er of priority) using the auth entication method defined i n the Policy Manager servic e.

	Authentication Process	Description
4	Authentication Source	After negotiating an authentication method with the user, Policy Manager authenticates the user (in order of priority) against the authentication sources defined in the Policy Manager service.
5	Role Mapping	Any roles defined in role - mapping policies or automatically assigned by Policy Manager based on several sources of information, including RADIUS connection control attributes, authentication sources, or authorization attributes.
6	Enforcement Policy	An enforcement policy is a way to organize enforcement profiles and apply them to users or Policy Manager roles. Based on the enforcement policy assigned to the role, enforcement profiles are applied to the service request.
7	Enforcement Profile	Enforcement profiles are the building blocks that control network access and define types of access. Multiple enforcement profiles can be used in an enforcement policy.

For a detailed description of the EAP-PEAP-MSCHAPV2 process, go to: A Tour of the EAP-PEAP-MSCHAPv2 Ladder.

Creating the 802.1X Wireless Service

The 802.1X Wireless Service provides a method for wireless end-hosts connecting through an 802.1X wireless access device or mobility controller, with authentication using IEEE 802.1X and with service rules customized for mobiilty controllers.

This ClearPass Policy Manager 802.1X template guides you through the following tasks:

- Selecting an Active Directory Authentication Source.

This guide assumes the Active Directory Authentication Source has already been configured. For more information, go to: Preparing for Active Directory Authentication.

- Selecting a Mobility Controller.

  This guide assumes the mobility controller used for 802.1X authentication has already been configured. For more information, go to: Preparing the Controller or Gateway for Integration.

- Creating an Enforcement Policy for Active Directory-based attributes.

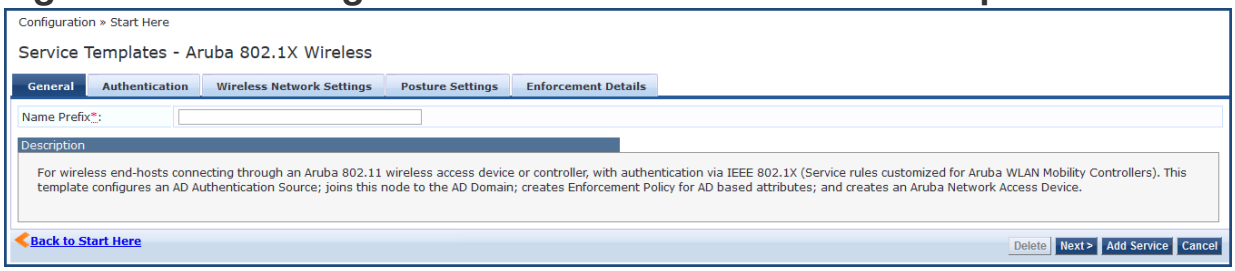
  The procedure for creating an Enforcement Policy is described in this section.

  To create an 802.1X wireless service:

  1. From Policy Manager, navigate to **Configuration > Start Here > Aruba 802.1x Wireless**.

     The **General** page for the Policy Manager 802.1X Wireless Service template opens.

     Figure 2. General Page in the 802.1X Wireless Service Template

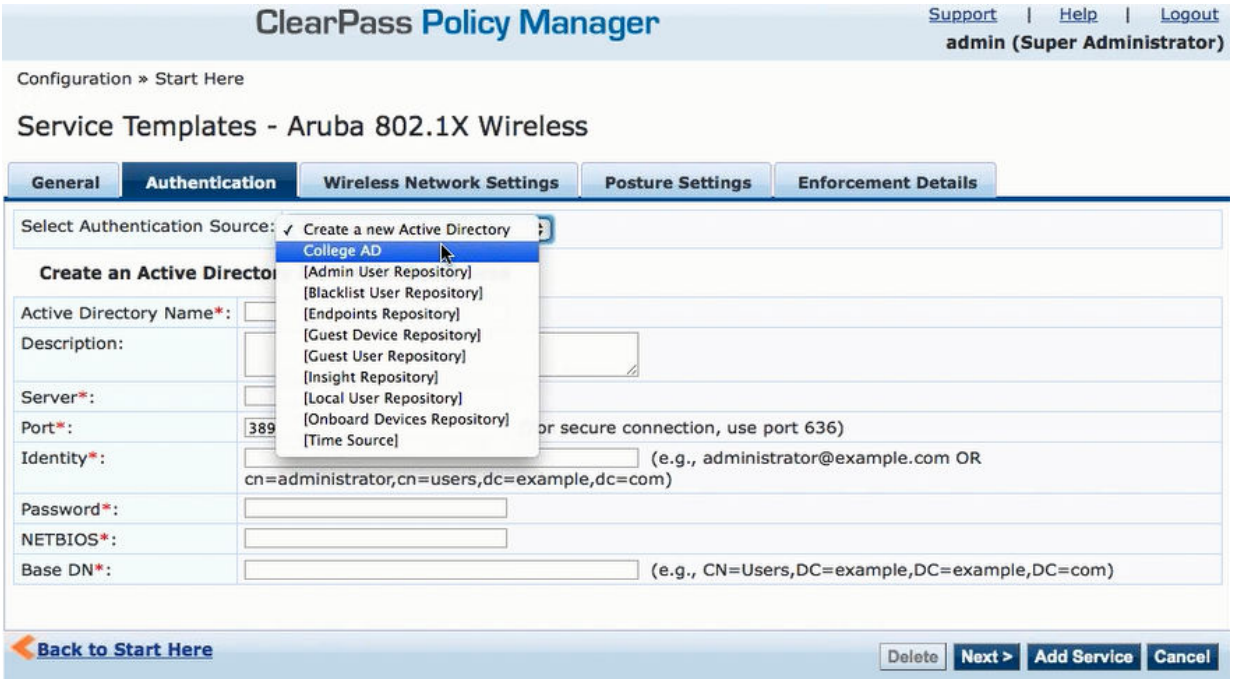
     

  2. In the *Name Prefix* field, enter a prefix that is appended to services using this template, then click **Next**.

     The **Authentication** page opens.

  3. From the **Select Authentication Source** drop-down list, select the name of the Active Directory, as shown in Figure 3, then click **Next**.

     Figure 3. Selecting the Active Directory

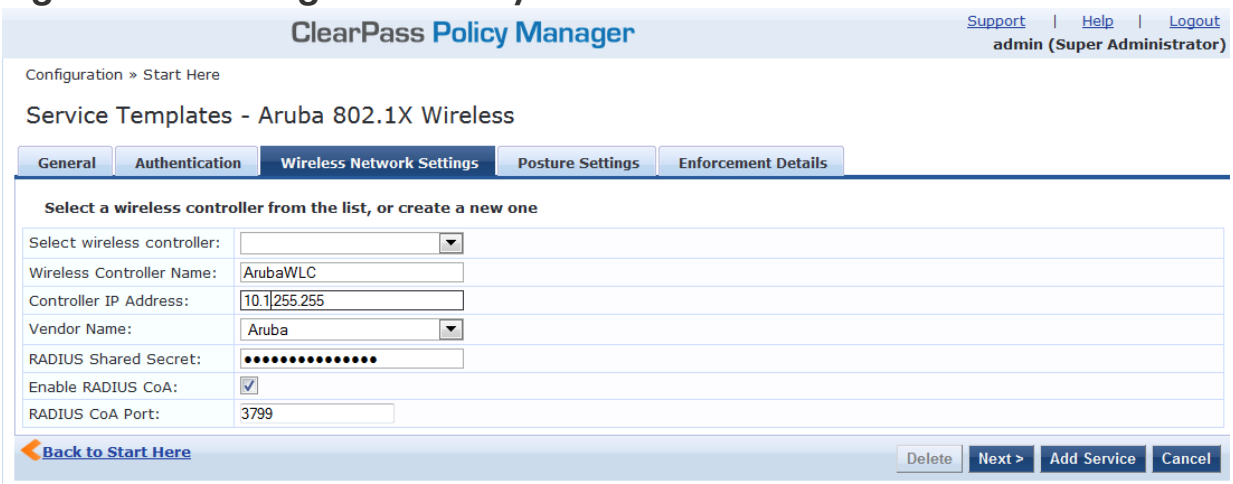


When you choose an existing Authentication Source, the information in the **Authentication** and **Enforcement Details** pages is populated automatically.

The **Wireless Network Settings** page opens.

4. Select the mobility controller you defined earlier. For more information, go to: Preparing the Controller or Gateway for Integration.

Figure 4. Selecting the MobilityController

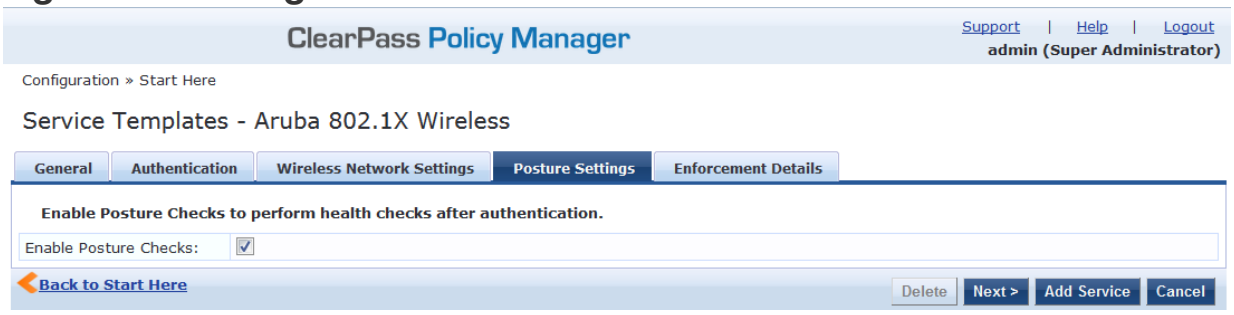


The fields in the **Wireless Network Settings** page are automatically populated with the selected mobility controller's configuration information.

5. Click **Next**.

The **Posture Settings** page opens.

Figure 5. Enabling Posture Checks



Policy Manager performs automated endpoint health checks and posture assessments to ensure that devices are compliant before they connect to mobile networks.

6. To enable posture checks to be performed after the authentication process completes, click the **Enable Posture Checks** check box, then click **Next**.

   The **Enforcement Details** page opens.

   Figure 6 shows an example of a new Enforcement Policy, with three attributes defined:

   - If **memberOf** equals **Faculty**, then assign Role **Faculty**.
   - If **memberOf** equals **Students**, then assign Role **Students**.
   - If **memberOf** equals **Contractors**, then assign Role **Contractors**.

Figure 6. Creating a New Enforcement Policy

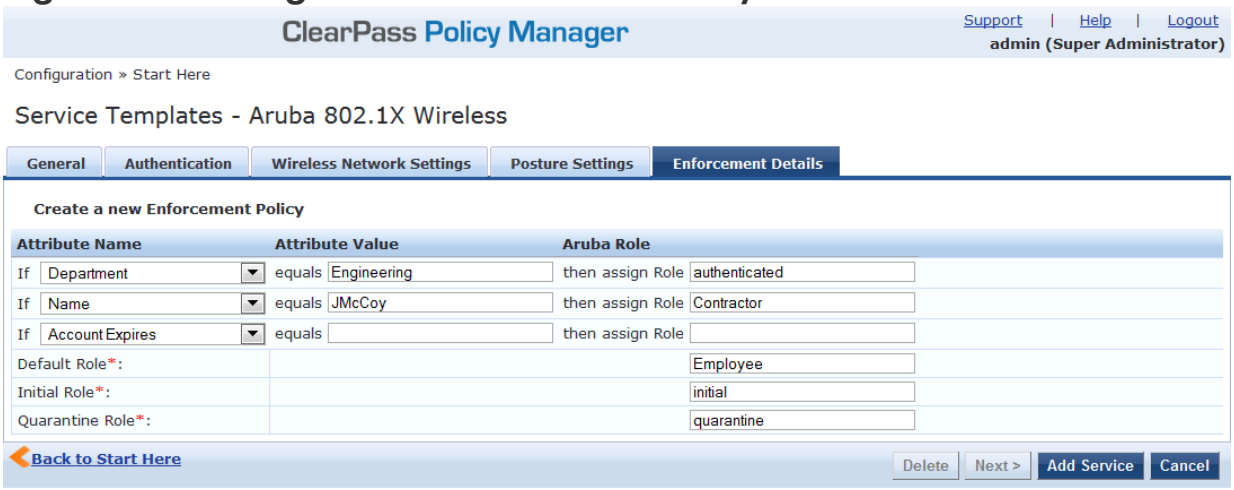


Table 2. Enforcement Policy Configuration Settings

Parameter	Action/Description
Attribute Name	The attributes defined in the Authentication Source are listed here. a. Configure an optional enforcement policy based on the following attributes: ◦ Department ◦ Email ◦ Name ◦ Phone ◦ Title ◦ UserDN ◦ company ◦ member of
Attribute Value	a. Enter the Active Directory attribute value for the selected name in the *Attribute Name* field.
Aruba Role	a. Assign a user role to the Enforcement Policy. ◦ The configured user roles are defined in the mobility controller specified for this service. To see the list of configured user roles defined in the mobility controller: i. Log in to the Mobility Controller. ii. Navigate to **Configuration > SECURITY > Access Control**. The User Roles page is displayed.

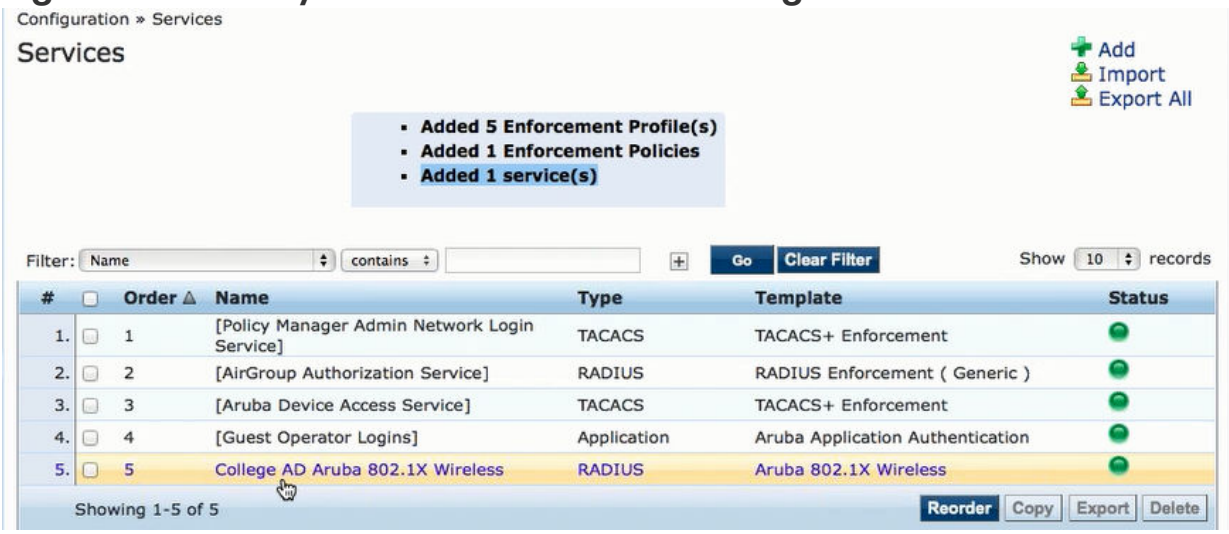
This completes the base configuration for a new 802.1X Wireless Service.

7. Click **Add Service**.

   An entry for the new set of configurations is created under the Services, Roles, Role Mapping, Enforcement Policies, and Profiles menus.

   A summary for the 802.X service you configured is displayed.

Figure 7. Summary of the 802.1X Service Configuration



Deleting a Policy Manager Service

About this task

You can only delete Policy Manager services that have been created by an administrator. Default services cannot be deleted.

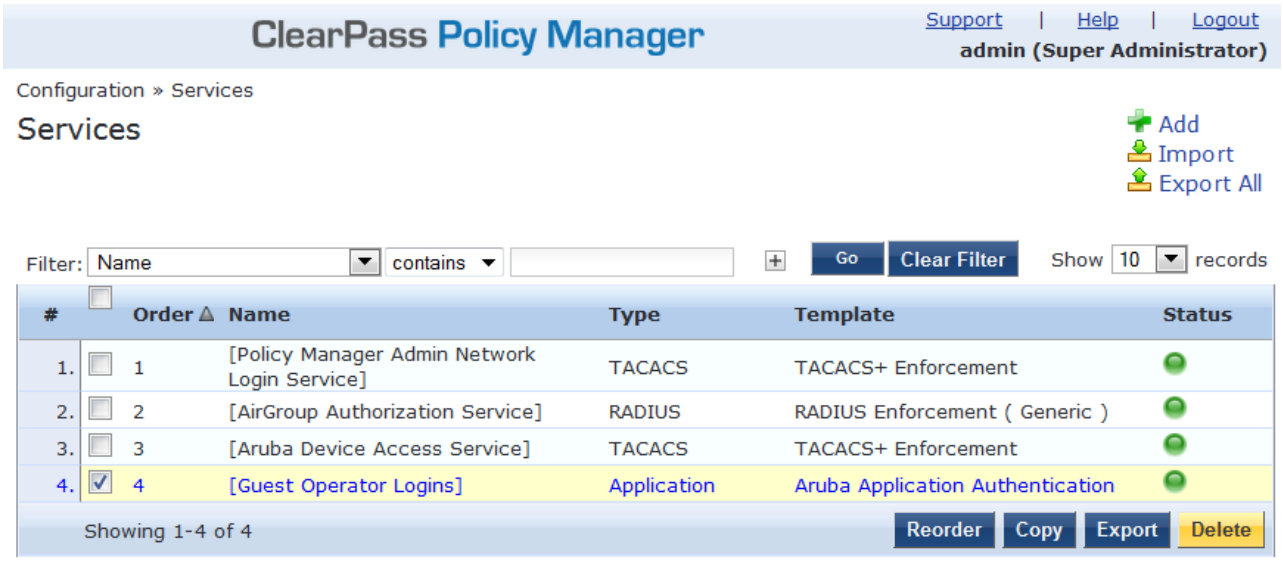
To delete a Policy Manager service:

Procedure

1. Navigate to **Configuration > Services**.

   The **Configuration > Services** page opens.

Figure 8. Deleting aPolicy ManagerService



2. Select the appropriate service's check box, then click **Delete** to remove the service.

   All the configured entries under the Services, Authentication Source, Roles, Role Mapping, Enforcement Policies, and Profiles menus are deleted (if these entities were created from the Service Template).

> **NOTE**
>
> Do not delete entities used in service configurations that were not created using the Service Template.

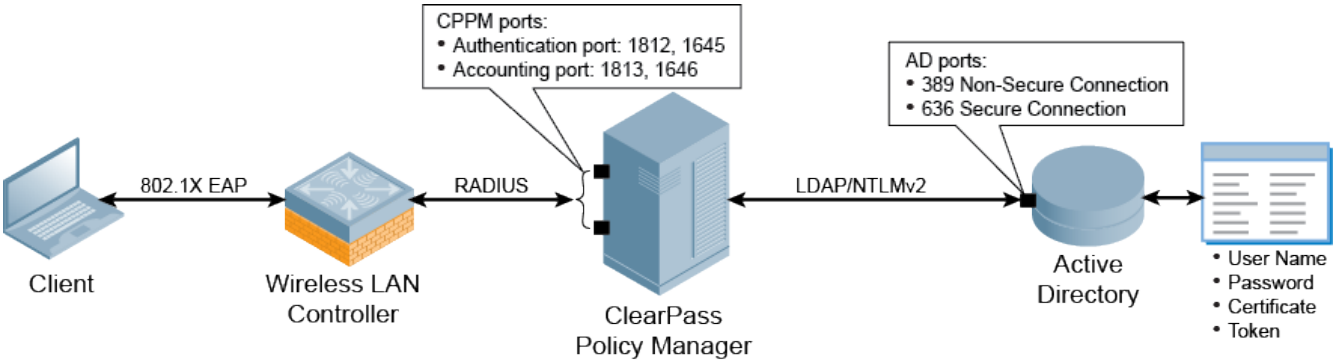
Walking Through a 802.1X Authentication Scenario

This section shows the 802.1X authentication traffic flow for wireless and wired authentication scenarios and provides a typical example of the 802.1X authentication process.

802.1X Wireless Authentication Traffic Flow

Figure 1 shows the traffic flow for 802.1X authentication using Active Directory.

Figure 1. Traffic Flow for 802.1X Wireless Authentication with Active Directory



Walking Through the 802.1X Authentication Process

About this task

The following is an example of the authentication process illustrated in Figure 1.

Procedure

1. A Sales Department employee connects to the Aruba wireless network from their laptop, and the 802.1X EAP-PEAP authentication process begins automatically.

   EAP-PEAP (Protected Extensible Authentication Protocol) is the protocol used to communicate between the client and the network device, in this case, a mobility controller.

2. The client's authentication request is sent to the mobility controller.

3. When the mobility controller receives the authentication request, it sends a RADIUS access-request packet to the Policy Manager server with the encrypted user name and password.

   RADIUS is the protocol network access device (NAD) authenticators use to communicate with the Policy Manager server to look up information in the RADIUS database, which in this example is Active Directory.

4. The Policy Manager server checks the Active Directory database for a matching user name and password.

   The communication between the Policy Manager server and Active Directory is via NT LAN Manager (NTLM) for authentication in conjunction with Lightweight Directory Access Protocol (LDAP) for search and directory lookup.

   - If there is not a match, the Policy Manager server sends an *access-reject* message to the mobility controller, and the Sales Department employee is denied access to the network.
   - If there is a match, the Policy Manager server sends an *access-accept* message to the mobility controller, and the Sales Department employee is granted access to the network.

User Role Attribute Information

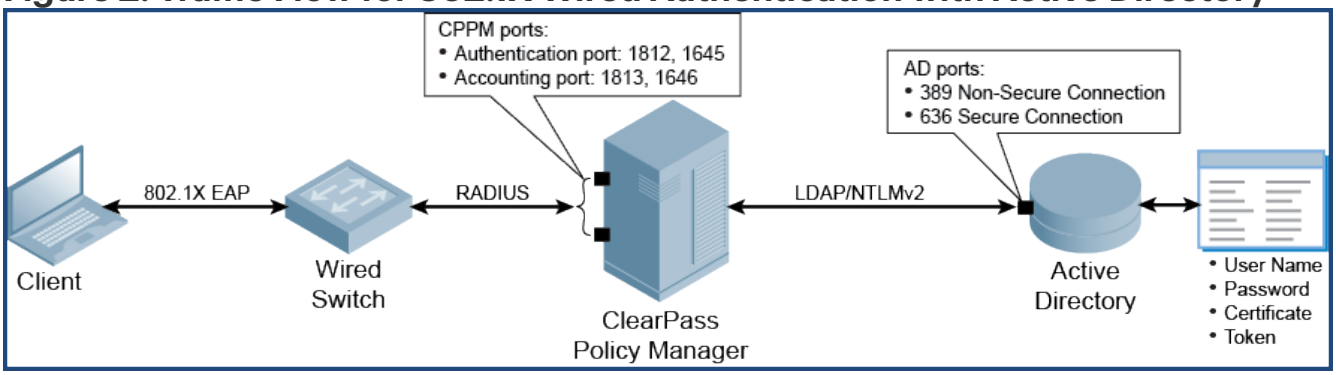
The Policy Manager server can also send attribute information about the user (for example, User Role) to the mobility controller. In this example, the server uses the User Role attribute, which indicates the user is in the Sales Department.

The mobility controller applies a Sales Department firewall role to this user's traffic. Typically for such a role, the firewall rule applied would be *IP any permit*, which permits all IP traffic.

802.1X Wired Authentication Traffic Flow

This same process applies to wired clients that connect to an ArubaOS or third-party switch and perform 802.1X authentication to the Policy Manager server (see Figure 2).

Figure 2. Traffic Flow for 802.1X Wired Authentication with Active Directory



For more information about Aruba AOS Switch and 802.1X authentication, go to: 802.1X Authentication Configuration.

Troubleshooting 802.1X Configuration Issues

This section provides troubleshooting information when configuring Active Directory and the Mobility Controller.

Active Directory Authentication Source Configuration Issues

Procedure

1. If you have configured a host name instead of an IP address for the Active Directory server in the **Primary** tab > **Hostname** field, ensure the Active Directory hostname is resolved to an IP address by the Domain Name System (DNS).
2. Ensure the Bind DN credentials have read access to the Active Directory locations where users and computers are present.
3. Verify the user name used for Bind DN is not locked in the Active Directory.

4. While joining Policy Manager to the Active Directory domain, use the *Fully Qualified Domain Name* (FQDN) of the Active Directory host, and not just the Domain Name.
5. Verify the Policy Manager server's time is synchronized with Active Directory, as a clock skew can cause the join domain operation to fail (formore information, go to: Joining a Server to an Active Directory Domain.

> **NOTE**
>
> The maximum allowed clock skew between the Policy Manager server and the Active Directory server is five minutes.

Mobility Controller Configuration Issues

Procedure

1. Ensure the Role information was sent to the mobility controller via enforcement matches the role defined in the mobility controller.
2. If authentication requests are not visible in the Access Tracker (**Monitoring > Live Monitoring > Access Tracker)**, verify the following:
   a. Verify the shared secret in the mobility controller and Policy Manager's Network Access Device configuration. Shared secret errors are shown in the Policy Manager Event Viewer (**Monitoring > Event Viewer**).
   b. Ensure the mobility controller's IP address is configured correctly in Policy Manager.

      Any mismatch displays ERROR/WARN events in the Event Viewer stating an authentication request is received from an unknown IP address.

Integrating the AOS Switch with ClearPass

This chapter describes the process of integrating AOS Switches with ClearPass Policy Manager. This chapter includes the following information:

- Initial ArubaOS Switch Configuration
- Setting Up RADIUS Authentication, Authorization, and Accounting
- Authenticating Users to the ArubaOS Switch
- Switch Management Using TACACS+

- Switch Management Using RADIUS
- OnGuard Authentication Configuration
- Monitoring and Troubleshooting

Subtopics

Initial ArubaOS Switch Configuration
Setting Up RADIUS Authentication, Authorization, and Accounting
Authenticating Users to the ArubaOS Switch
Switch Management Using TACACS+
Switch Management Using RADIUS
OnGuard Authentication Configuration
Monitoring and Troubleshooting

Initial ArubaOS Switch Configuration

This section describes the tasks required for initial AOS Switch configuration. This section contains the following information:

- Configuring Administrator Credentials
- Configuring the IP Address of the Out-of-Band Management Port
- Configuring SNMPv3
- Configuring a Policy Manager/RADIUS Server on the Switch
- Defining the ArubaOS Switch in ClearPass

Configuring Administrator Credentials

About this task

The first task in the initial setup for the AOS Switch is to configure the administrator credentials.

To configure the AOS Switch administrator credentials:

Procedure

1. Enter configuration mode.

```
AOS-Switch# configure
```

2. Create a local administrative account to be used when RADIUS or TACACS+ authentication is unavailable.

```
AOS-Switch(config)# password manager user-name ladmin
```

Configuring the IP Address of the Out-of-Band Management Port

About this task

> **NOTE**
> This procedure assumes that switch management will be done using out-of-band management.

To configure the AOS Switch IP address of the out-of-band management port:

Procedure

1. Enable out-of-band management (oobm).

   ```
   AOS-Switch(config)# oobm enable
   ```

2. Assign an IP address to the oobm port.

   ```
   AOS-Switch(config)# oobm ip address 10.2.100.68 255.255.255.0
   ```

3. Add a default gateway to the oobm port.

   ```
   AOS-Switch(config)# oobm ip default-gateway 10.2.100.1
   ```

4. Save the configuration

   ```
   AOS-Switch(config)# save
   ```

Configuring SNMPv3

Aruba recommends using SNMPv3. SNMPv3 access requires an IP address and subnet mask configured on the switch. If you are using DHCP/Bootp to configure the switch, ensure that the DHCP/Bootp process provides the IP address.

- You can (optionally) restrict access to SNMPv3 agents only by using the **snmpv3 only** command.
- To restrict write-access to only SNMPv3 agents, use the **snmpv3 restricted-access** command.

Adding a New User When SNMPv3 Is Already Configured

If SNMPv3 has already been configured on this switch, then the only task required is to add an SNMP v3 user.

To add a new user to an AOS Switch that already has SNMPv3 configured, issue the following command:

```
AOS-Switch (config)# snmpv3 user <user_name> auth sha <authentication_password priv aes <privacy_password>
```

Configuring SNMPv3 for the First Time

About this task

When you enable SNMPv3 operation on the switch, an initial user entry is generated with the authentication protocol set to **MD5** and the privacy protocol set to **DES** (Data Encryption Standard). This section describes how to change the privacy protocol from default setting of **DES** to the more secure protocol **AES** (Advanced Encryption Standard). This includes the creation of the initial user record.

When SNMPv3 is enabled, the AOS Switch begins the initialization process by creating an account named “initial."

To configure SNMPv3 for the first time on the AOS Switch:

Procedure

1. `ArubOS-switch (config)# snmpv3 enable`

   SNMPv3 Initialization process.

   Creating user 'initial'

   Authentication Protocol: MD5

2. Enter authentication password: **************

   Privacy protocol is DES.

3. Enter privacy password: **************

   User 'initial' has been created.

4. Would you like to create a user that uses SHA? [y/n] **y**

5. Enter user name: <*user_name*>

   Authentication Protocol: SHA

6. Enter authentication password: **************

   Privacy protocol is DES.

7. Enter privacy password: **************

   User creation is done. SNMPv3 is now functional.

8. Would you like to restrict SNMPv1 and SNMPv2c messages to have read-only access (you can set this later by the command '**snmpv3 restricted-access**')? [y/n] **n**

> **NOTE**
>
> Restricting access to only SNMPv3 messages makes the community named "public" inaccessible to network management applications (such as autodiscovery, traffic monitoring, SNMP trap generation, and threshold setting) from operating in the AOS Switch. For that reason, Aruba recommends answering **no** to Step 8.

Changing the Privacy Protocol to Secure Protocol AES

After enabling SNMPv3 for the first time, the user that is created during this process (in our example, **cp-snmp**) has the privacy protocol set to **DES**.

Because **AES** (Advanced Encryption Standard) is more secure, Aruba recommends changing the privacy protocol to **AES**. To change the privacy protocol to **AES**, issue the following command:

```
AOS-Switch (config)# snmpv3 user <user_name> auth sha <authentication_password priv aes <privacy_password>
```

Configuring a Policy Manager/RADIUS Server on the Switch

Before you begin this procedure, have at hand the following information:

- IP address of the target Policy Manager server(s)
- RADIUS Shared Secret configured for the Policy Manager server(s)

To configure a Policy Manager/RADIUS server on the AOS Switch (also referred to as the Network Access Device (NAD)):

1. Log in to the AOS Switch via the console port or via SSH.
2. To add a Policy Manager server node, enter the following commands:

```
AOS-Switch # config t

AOS-Switch(config)# radius-server host <ClearPass_IP_address> key <RADIUS_shared_secret>

AOS-Switch(config)# radius-server host <ClearPass_IP_address> dyn-authorization
```

```
AOS-Switch(config) # aaa server-group radius <GroupName> host <
ClearPass_IP_address >
```

3. Repeat these commands to add additional Policy Manager/RADIUS servers.
4. To view the current status of the Policy Manager/RADIUS server configuration:

```
AOS-Switch(config) # show radius
```

Figure 1. show radius Command Output

As shown in Figure 1, running a **show radius** command displays the following information:

- The switch is listening for inbound CoA commands on the default UDP port of **3799**.
- Three Policy Manager nodes have been defined to this switch.
- Authentication and Accounting are running on their default UDP ports **1812** and **1813**.

Defining the AOS Switch in ClearPass

About this task

To define the Network Access Device (NAD)—that is, the AOS Switch—in Policy Manager:

Procedure

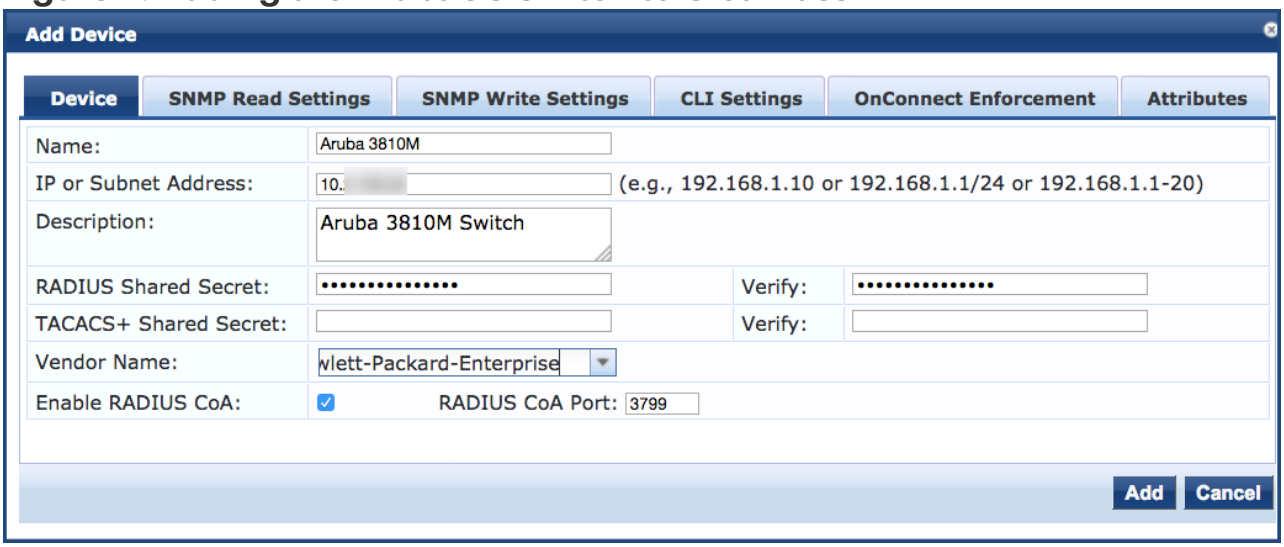
1. In Policy Manager, navigate to **Configuration > Network > Devices**.

   The **Network Devices** page opens.

2. Click the **Add** link.

The **Add Device** dialog opens.

Figure 2. Adding the ArubaOS Switch to ClearPass



3. Specify the **Add Device** parameters as described in the following table:

Table 1. Add Device Parameters for an AOS Switch

Parameter	Action/Description
Name	Enter a descriptive name to identify the n etwork device being added.
IP or Subnet Address	• If this is a single device, enter its **IP ad dress**. • If there are multiple devices in the sa me subnet that are being defined, ent er the **subnet address**.
Description	Provide a detailed set of information des cribing the network device (recommende d).
RADIUS Shared Secret	Enter the password that will be used on b oth the Policy Manager server and on the network device(s) to authenticate each o ther.
TACACS+ Shared Secret	If TACACS+ is used on this device, enter t he password used on both the Policy Ma

Parameter	Action/Description
	nager server and on the network device(s) to authenticate each other.
Vendor Name	For AOS Switches, choose **Hewlett - Pac kard - Enterprise** as the vendor name.
Enable RADIUS CoA	Make sure this check box is enabled. This parameter is enabled by default.
RADIUS CoA Port	If RADIUS CoA is enabled, this specifies t he default port as **3799**. Change this val ue only if you defined a custom port. Fo r related information, see Configuring Pol icy Manager as an RFC 3576 (CoA) Server .

4. Click **Add**.

   For more information on configuring Network Access Devices, refer to Adding and Modifying Network Devices

Setting Up RADIUS Authentication, Authorization, and Accounting

This section contains the following information:

- About AAA Services
- About the RADIUS Protocol
- About the TACACS+ Protocol
- About RADIUS Authentication and Authorization
- Setting Up RADIUS Accounting

About AAA Services

AAA network security services provide the primary framework through which a network administrator can set up access control on network points of entry or network access servers.

- *Authentication* identifies a user.
- *Authorization* determines what that user can do on the network.
- *Accounting* monitors the network usage time for billing purposes.

  AAA information is typically stored in an external database or remote server such as a RADIUS or TACACS+ server. The information can also be stored locally on the access server or router.

  Remote security servers, such as RADIUS and TACACS+ servers, assign users specific privileges by associating attribute-value pairs, which define the access rights with the appropriate user. All authorization methods must be defined through AAA.

About RADIUS Authentication and Authorization

Authentication is the process by which a system or network verifies the identity of a user who wishes to access it. Authentication ensures that the individual is who he or she claims to be, but says nothing about the access rights of the individual—that is the role of authorization.

Authorization is the process of giving individuals specific access rights to system or network resources based on their identity. Authorization employs *access control rules* to determine whether access requests from authenticated users are approved (granted) or disapproved (rejected).

The RADIUS protocol combines user authentication and authorization steps into one phase. The user must be successfully authenticated before the RADIUS server sends authorization information (from the user's profile) to the Network Access Server (NAS).

Commands authorization assigns a list of CLI commands that can be executed by a specified user. The permitted CLI commands are defined on the remote RADIUS server in a user's profile.When authentication is successful, the RADIUS server returns the permitted list of CLI commands that the authenticated user is authorized to execute. By default, all users can execute a minimal set of commands regardless of their authorization status, for example, "exit" and "logout." This minimal set of commands can prevent deadlock on the switch due to an error in the user's authorization profile on the RADIUS server. The user's profile is encoded into Vendor-Specific Attributes (VSAs).

The list of permitted commands is used to filter all the commands executed by the user until the end of the session. This allows greater authorization control, where different rights can be given to different manager or operator users.

About the RADIUS Protocol

The RADIUS (Remote Authentication Dial-In User Service) protocol carries authentication, authorization, and configuration information between a network access server (NAS) and a RADIUS authentication server.

Authentication with RADIUS allows for a unique password for each user, instead of the need to maintain and distribute switch-specific passwords to all users. RADIUS verifies identity for the following types of primary password access to the switch:

- Serial port (console)
- Telnet
- SSH
- SFTP/SCP
- WebAgent
- Port-Access (802.1X)

  AOS Switch es support RADIUS accounting for web-based authentication and MAC authentication sessions, collecting resource consumption data and forwarding it to the RADIUS server. This data can be used for trend analysis, capacity planning, billing, auditing, and cost analysis.

  Requests and responses carried by the RADIUS protocol are called RADIUS *attributes*. These attributes provide the information needed by a RADIUS server to authenticate users and to establish authorized network service for them. The RADIUS protocol also carries accounting information between a network access server and a RADIUS accounting server.

  RADIUS is a client/server protocol. The RADIUS client is typically a network access server. The client passes user information to designated RADIUS servers and acts on the response that is returned. RADIUS servers receive user connection requests, authenticate the user, and then return the configuration information necessary for the client to deliver service to the user.

About the TACACS+ Protocol

TACACS+ AAA systems are used as a single point of management to configuring and store user accounts. They are often coupled with directories and management repositories, simplifying the set up and maintenance of the end-user accounts.

In the authorization function of the AAA system, network devices with Authentication Services can provide fine-grained control over user capabilities for the duration of the user's session; for example, setting access control or session duration. Enforcement of restrictions to a user account can limit available commands and levels of access.

TACACS+ authentication provides a central server in which you can allow or deny access to switches and other TACACS+-aware devices in your network. TACACS+ employs a central database that creates multiple unique user name and password sets with their associated privilege levels. This central database can be accessed via the ArubaOS switch from either a console port or via Telnet.

TACACS+ uses an authentication hierarchy consisting of:

- Remote passwords assigned in a TACACS+ server
- Local passwords configured on the switch

> **NOTE**
> In the event of a connection failure, a TACACS+ server defaults to locally assigned passwords for authentication control.

A TACACS+ server is able to:

- Configure login authentication for read/write or read-only privileges.
- Manage the authentication of login attempts by either the console port or via Telnet.

The list of supported TACACS+ authentication types includes:

- TAC_PLUS_AUTHEN_TYPE_ASCII (0x01)
- TAC_PLUS_AUTHEN_TYPE_PAP (0x02)
- TAC_PLUS_AUTHEN_TYPE_CHAP (0x03)

The list of supported TACACS+ authentication services includes:

- TAC_PLUS_AUTHEN_SVC_NONE (0x00)
- TAC_PLUS_AUTHEN_SVC_LOGIN (0x01)
- TAC_PLUS_AUTHEN_SVC_ENABLE (0x02)
- TAC_PLUS_AUTHEN_SVC_PPP (0x03)

The list of supported TACACS+ authentication actions includes:

- TAC_PLUS_AUTHEN_LOGIN (0x01)

Setting Up RADIUS Accounting

This section provides the following information:

- Accounting Services
- RADIUS Accounting Server
- Enabling RADIUS Accounting
- Operating Rules for RADIUS Accounting
- Operating Rules for RADIUS

Accounting Services

RADIUS accounting on the switch collects resource consumption data and forwards it to the RADIUS server. This data can be used for trend analysis, capacity planning, billing, auditing, and cost analysis. Accounting support is provided for WebAgent sessions on the switch.

RADIUS accounting collects data about user activity and system events and sends it to a RADIUS server when specified events occur on the switch, such as a logoff or a reboot.

Accounting Service Types

The switch supports four types of accounting services:

- Network accounting: Provides records containing information on clients directly connected to the switch and operating under Port-Based Access Control (802.1X).
- Executive accounting: Provides records holding the information about login sessions (console, Telnet, and SSH) on the switch.
- System accounting: Provides information regarding system events that occur on the switch, including system reset, system boot, and enabling or disabling system accounting.
- Commands accounting: Provides records containing information on CLI-command execution during user sessions.

RADIUS Accounting Server

A Network Access Server (NAS) operates as a client of the RADIUS accounting server. The client is responsible for passing user accounting information to a designated RADIUS accounting server. The RADIUS accounting server is responsible for receiving the accounting request and returning a response to the client indicating that it has successfully received the request. The RADIUS accounting server can act as a proxy client to other kinds of accounting servers. Transactions between the client and RADIUS accounting server are authenticated through the use of a shared secret, which is never sent over the network.

Enabling RADIUS Accounting

You can enable RADIUS Accounting for multiple features within the switch accounting configuration. Additionally you can configure **accounting start-stop** for other components. You also need to configure the accounting interval update timer—**aaa accounting update periodic** parameter (set to **2** minutes in the example below) . To set up RADIUS accounting, run the following commands:

```
   AOS-Switch(config)# aaa accounting network start-stop radius server-
group CP-cluster
   AOS-Switch(config)# aaa accounting update periodic 2
   AOS-Switch(config)# show accounting
```

Figure 1. show accounting Command Output

```
Status and Counters - Accounting Information

 Interval(min) : 2
 Suppress Empty User : No
 Sessions Identification : Unique

 Type      | Method Mode           Server Group
 --------- + ------ -------------- ------------
 Network   | Radius Start-Stop     radius
 Exec      | None
 System    | None
 Commands  | None
```

Operating Rules for RADIUS Accounting

The operating rules for RADIUS accounting are as follows:

- You can configure up to four types of accounting to run simultaneously: executive, system, network, and command.
- RADIUS servers used for accounting are also used for authentication.
- The switch must be configured to access at least one RADIUS server.
- RADIUS servers are accessed in the order in which their IP addresses were configured in the switch. To view the order of the RADIUS servers, use the **show radius** command .As long as the first server is accessible and responding to authentication requests from the switch, a second or third server cannot be accessed.
- If access to a RADIUS server fails during a session, but after the client has been authenticated, the switch continues to assume the server is available to receive accounting data. Thus, if server access fails during a session, it doesn't receive accounting data transmitted from the switch.

Operating Rules for RADIUS

The AOS Switch operating rules for RADIUS are as follows:

- You must have at least one RADIUS server accessible to the switch.
- The switch supports authentication and accounting using up to fifteen RADIUS servers. The switch accesses the servers in the order in which they are listed by show radius. If the first server does not respond, the switch tries the next one, and so on.

- You can select RADIUS as the primary authentication method for each type of access.

> **NOTE**
>
> Only one primary and one secondary access method is allowed for each access type.

- In the switch, EAP RADIUS uses MD5 and TLS to encrypt a response to a challenge from a RADIUS server.
- When primary/secondary authentication is set to **Radius/Local** (for either **Login** or **Enable**) and the RADIUS server fails to respond to a client attempt to authenticate, the failure is noted in the Event Log with the message:

  radius: Can't reach RADIUS server <server-ip-address>.

  When this type of failure occurs, the switch prompts the client again to enter a user name and password. In this case, use the local user name (if any) and password configured on the switch itself.
- Zero-length user names or passwords are not allowed for RADIUS authentication, even though this is allowed by some RADIUS servers.

Additional Configuration Considerations

Beyond the 802.1X configuration basics described above, there are many additional parameters you may choose to configure across the switch ports, such as the following recommendations.

Limiting Access for Unauthorized Clients

On the AOS Switch, a switch port with a static VLAN ID and an unauthenticated client VLAN ID is automatically part of the Unauthenticated-client VLAN as soon as a device connects. If the device passes authentication, the port becomes an untagged member of the static VAN. This behavior helps guest and other devices with 802.1X supplicants to connect more quickly.

To set an unauthenticated-client VLAN for one or more interfaces, issue the following command:

```
AOS-Switch (config) # aaa port-access authenticator <port ID list> u
nauth-vid <VLAN ID>
```

The **unauth-vid** parameter configures the VLAN to keep the specified ports while there is an unauthenticated client connected to the network.

Preventing Connectivity Delays for 802.1X Devices

For users who use 802.1X to log in, setting an unauthenticated-client VLAN might lose connectivity. If the user's device allows non-EAP traffic before authentication, it might

receive a DHCP address that is in the unauthenticated-client VLAN, which would cause the user's device to lose connectivity after the port moves to the VLAN for authenticated users.

To prevent connectivity delays based on this scenario, issue the following command:

```
AOS-Switch (config) # aaa port-access authenticator <port ID list> u
nauth-period <seconds>
```

Authenticating Users to the ArubaOS Switch

This section contains the following information

-
- What Are the Differences Between RADIUS vs TACACS+?

Why Use ClearPass to Perform Authentication and Authorization for the ArubaOS Switch?

Using Policy Manager for handling authentication to the ArubaOS switch allows for a centralized source of user account information that can be used for controlling access to the ArubaOS switch. This helps to minimize the number of changes that need to be made to the network when switch administrators come and go or change access levels.

In addition to authenticating users, Policy Manager can also perform command authorization. Some users logged in as administrators to the AOS Switch might only be allowed to send certain commands to the command line interface (CLI) to perform their job functions. With command authorization, each and every command that is entered into the switch CLI is validated against the enforcement policy that Policy Manager has assigned to that user.

Policy Manager provides a robust platform for defining access policies based on contextual information about the authentication request, as well as orchestration of possible follow-up actions based on the policy outcome.

What Are the Differences Between RADIUS vs TACACS+?

The ArubaOS switch supports authentication and authorization checks to an AAA server using either the RADIUS or TACACS+ protocols. There are some inherent differences between the two protocols that might determine which protocol to choose in a given environment.

The differences between RADIUS vs TACACS+ for authentication and authorization services can be summarized by the following table:

Table 1. Comparing RADIUS vs TACACS+ Authentication Services

Protocol Services	RADIUS	TACA
Authentication	Yes	Yes
Command Authorization	Yes	Yes
Command Accounting	Yes (during Authentication)	Yes (
Encryption	Yes	Yes

Both protocols work for authentication and for command authorization, but TACACS+ sends a request to the AAA server for each and every command being performed. This can be performance-intensive if there are more than 1,000 simultaneous operators.

On the other hand, RADIUS sends the list of allowed commands back to the ArubaOS switch during authentication. This removes the requirement that the AAA server remain online for the duration of the session, while also removing much of the performance overhead that comes from processing a TACACS+ request each time a command is run.

Command accounting is handled by the AAA server when TACACS+ is being used, but not when RADIUS is being used. In order to see a list of commands being issued by the administrator, the switch must be configured to log each command being run to a syslog server.

Both protocols offer some level of encryption. For RADIUS, only passwords are encrypted. For TACACS+, the entire body of the packet is encrypted. Both protocols use shared secrets for encryption.

Switch Management Using TACACS+

This section contains the following information:

- Overview
- Setting Up Switch Management Using TACACS+
- Creating Enforcement Profiles to Provide Manager Access and Command Control to the ArubaOS Switch
- Creating an Enforcement Policy to Define Access to the Switch
- Creating a Service to Support TACACS+ Authentication Requests from the Switch
- Setting Up the Switch for Command Authorization Using TACACS+

- Setting Up Enforcement Profiles in ClearPass to Support TACACS+ Command Authorization Requests from the Switch

Overview

To use TACACS+ authentication, you need the following:

- A TACACS+ server application installed and configured on one or more servers or management stations in your network.
- An AOS Switch configured for TACACS+ authentication, with access to one or more TACACS+ servers.

  The effectiveness of TACACS+ security depends on correctly using your TACACS+ server application. For this reason, Hewlett Packard Enterprise recommends that you thoroughly test all TACACS+ configurations used in your network.

  TACACS+-aware switches include the capability of configuring multiple backup TACACS+ servers. Hewlett Packard Enterprise recommends that you use a TACACS+ server application that supports a redundant backup installation. This allows you to configure the switch to use a backup TACACS+ server if it loses access to the first-choice TACACS+ server. TACACS+ does not affect WebAgent access.

Setting Up Switch Management Using TACACS+

This section describes how to set up AOS Switch management using TACACS+. The following methods are described:

- SSH (Secure Shell)
- Telnet
- Console

Initial TACACS+ Management Configuration

About this task

> **NOTE**
> Out-of-band management (oobm) is only required if the ArubaOS switch will be using the out-of-band management interface to communicate with the TACACS+ server.

To provide initial TACACS+ management configuration:

Procedure

1. Define the TACACS+ server in the AOS Switch.

```
ArubaOS-switch(config)#tacacs-server host 10.2.97.10 oobm key
supersecretkey123
```

2. Optionally, adjust the TACACS+ server timeout period as needed. The default is 3 seconds.

```
ArubaOS-switch(config)#tacacs-server timeout 5
```

3. Configure TACACS+ single login capability.

```
ArubaOS-switch(config)#aaa authentication login privilege-mode
```

Configuring SSH Login for TACACS+ Authentication

About this task

To configure the SSH (Secure Shell) login for TACACS+ authentication:

Procedure

1. Configure TACACS+ authentication for SSH login with read-only (operator) access:

```
ArubaOS-switch(config)#aaa authentication ssh login tacacs local
```

2. Configure TACACS+ authentication for SSH login with access to privileged (manager) access

```
ArubaOS-switch(config)#aaa authentication ssh enable tacacs local
```

Configuring Telnet Login for TACACS+ Authentication

About this task

To configure the Telnet login for TACACS+ authentication:

Procedure

1. Configure TACACS+ authentication for Telnet login with read-only (operator) access:

```
ArubaOS-switch(config)#aaa authentication telnet login tacacs local
```

2. Configure TACACS+ authentication for Telnet login with access to privileged (manager) access

```
ArubaOS-switch(config)#aaa authentication telnet enable tacacs local
```

Configuring the Console Login for TACACS+ Authentication

About this task

To configure the Console login for TACACS+ authentication:

Procedure

1. Configure TACACS+ authentication for Console login with read-only (operator) access:

```
ArubaOS-switch(config)# aaa authentication console login tacacs local
```

2. Configure TACACS+ authentication for Console login with access to privileged (manager) access

```
ArubaOS-switch(config)#aaa authentication console enable tacacs local
```

Creating Enforcement Profiles to Provide Manager Access and Command Control to the ArubaOS Switch

- Adding Active Directory as an Authentication Source
- Creating an Enforcement Profile to Provide Manager-Level Access
- Creating an Enforcement Profile to Provide Operator-Level Access

The service to authenticate TACACS+ users against Active Directory incorporates enforcement profiles that define manager-level access and operator-level access to the ArubaOS switch. For this reason, we recommend that the necessary enforcement profiles be created *before* the service is created.

Adding Active Directory as an Authentication Source

Before you create the enforcement profiles as described below, we recommend that you configure Active Directory as an authentication source first, as you will need to specify the authentication source when you create the RADIUS enforcement policy.

For details, see Adding Active Directory as an Authentication Source to Policy Manager .

Creating an Enforcement Profile to Provide Manager-Level Access

About this task

In Policy Manager, an enforcement policy provides the rules that tells Policy Manager when to use specific enforcement profiles. Enforcement profiles consist of actions that are taken by Policy Manager, for example, assigning a certain role to a user.

The actions in an enforcement policy are the enforcement profiles to be activated when specific conditions are met or rules are enforced. Then an enforcement policy is associated with a service—a service ties all the elements together: authentication sources, authorization sources, enforcement policies, and role-mapping.

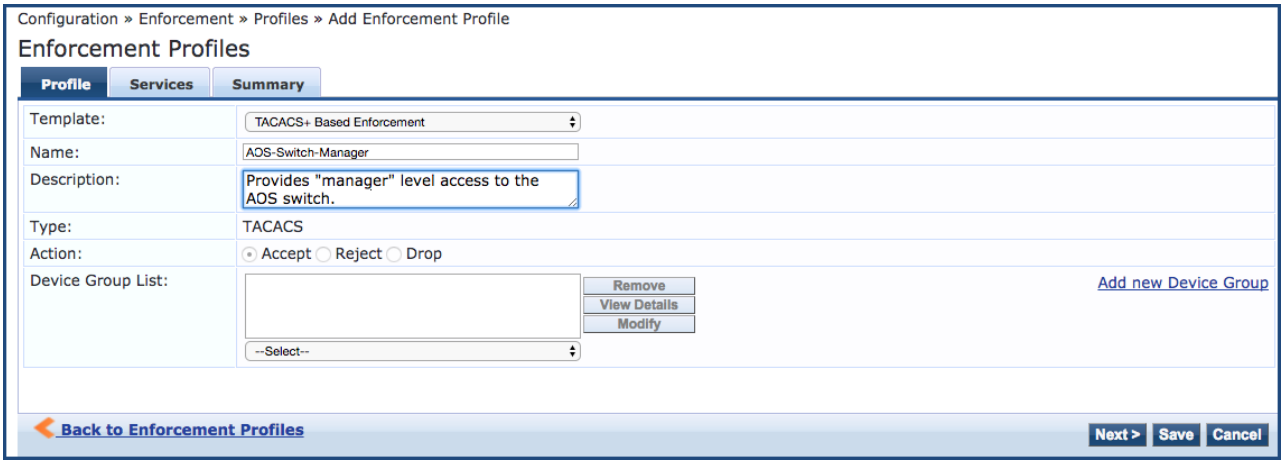
To create an enforcement profile to provide manager-level access to the AOS Switch:

Procedure

1. On the Policy Manager server, navigate to **Configuration > Enforcement > Profiles**.
2. From the **Enforcement Profiles** page, click **Add**.

   The **Add Enforcement Profiles** dialog opens.

Figure 1. Adding a TACACS+ Based Enforcement Profile for Manager-Level Access



3. Specify the **Add Enforcement Profile > Profile** tab parameters as described in the following table, then click **Next**:

Table 1. Add TACACS+ Based Manager - Level Enforcement Profile > Profile T

Parameter	Action/Description
Template	From the Template drop - down, **nforcement**.
Name	Enter the name of this enforcem **Switch - Manager**.
Description	Add a description of this enforce anager - level access to the AOS

Parameter	Action/Description
Type	When you select **TACACS+ Base** rcement profile Type is set to **TA**
Action	Set to **Accept** (the default).

Table 2. Add TACACS+ - Based Manager - Level Enforcement Profile > Profile

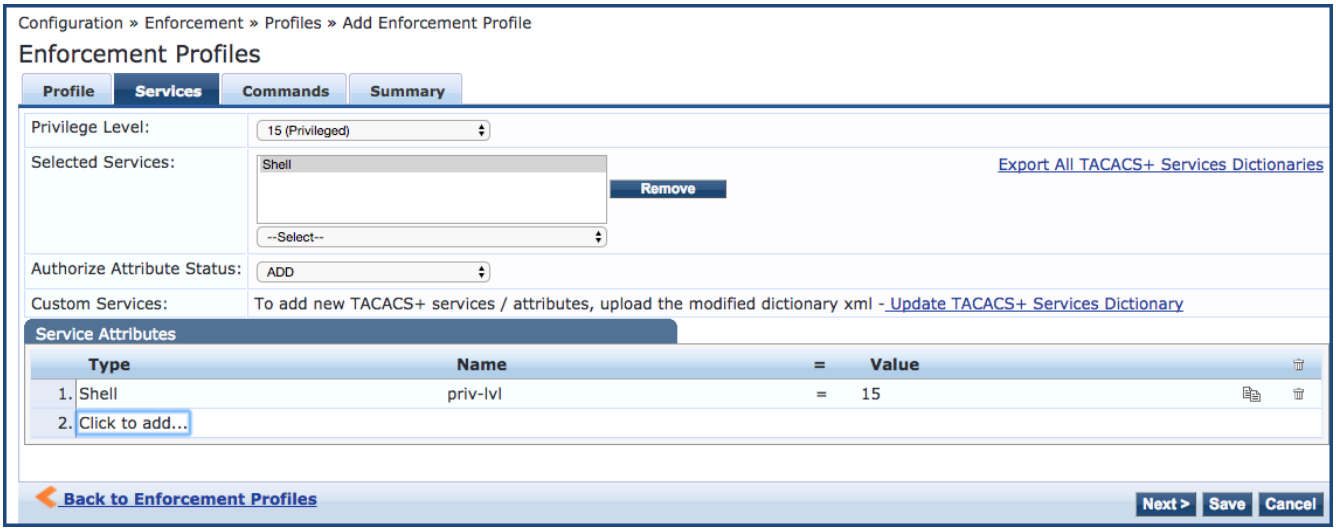
Parameter	Action/Description
Template	From the Template drop - down **nforcement**.
Name	Enter the name of this enforcem **itch - Manager**.
Description	Add a description of this enforc anager - level access to the AO
Type	When you select **TACACS+ Base** rcement profile Type is set to **TA**
Action	Set to **Accept** (the default).

Services Tab

About this task

The next step is to configure the **Services** parameters.

Figure 2. TACACS+ Enforcement Profile for Manager-Level Access > Services Tab



Procedure

Specify the **Add Enforcement Profile > Services** tab parameters as described in the following table, then click **Next**:

Table 3. TACACS+ Manager - Level Enforcement Profile > Services Parameters

Parameter	Action/Description
Privilege Level	Select **15 (Privileged)**.
Selected Services	Select **Shell**. Within a TACACS+ enforcer access services that are av device, such as the AOS Sv a service that is available o rcement profile will use.
Authorize Attribute Status	Specify **Add** (the default).
Service Attributes	
Type	Select **Shell**.
Name	Select **priv - lvl**.
Value	Enter a value of **15**.

Table 4. TACACS+ Manager - Level Enforcement Profile > Services Parameters

Parameter	Action/Description
Privilege Level	Select **15 (Privileged)**.

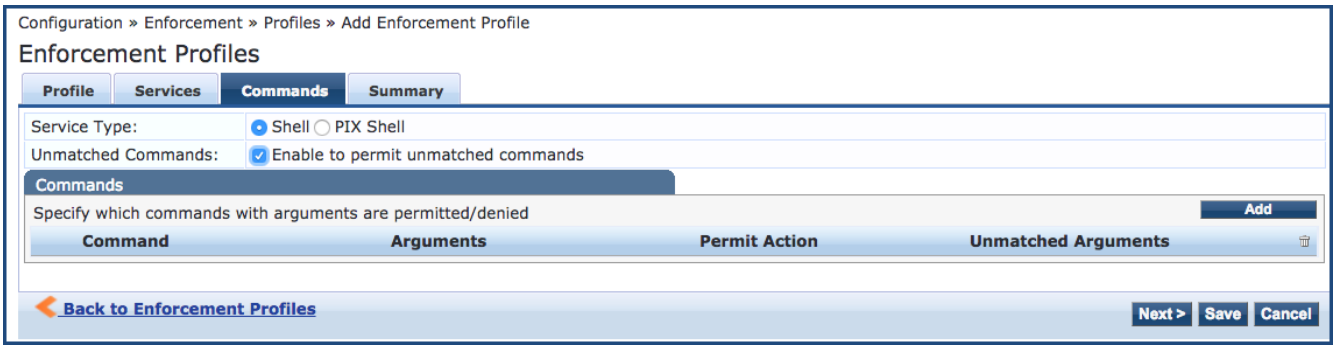
Parameter	Action/Description
Selected Services	Select **Shell**. Within a TACACS+ enforcer access services that are av device, such as the AOS Sv a service that is available o rcement profile will use.
Authorize Attribute Status	Specify **Add** (the default).
Service Attributes	
Type	Select **Shell**.
Name	Select **priv - lvl**.
Value	Enter a value of **15**.

Commands Tab

About this task

The next step is to configure the **Commands** parameters.

Figure 3. TACACS+ Enforcement Profile for Manager-Level Access > Commands Tab



Procedure

Specify the **Add Enforcement Profile > Commands** tab parameters as described in the following table, then click **Next**:

Table 5. Add Enforcement Profile > Commands Parameters

Parameter	Action/Description
Service Type	Select **Shell**.

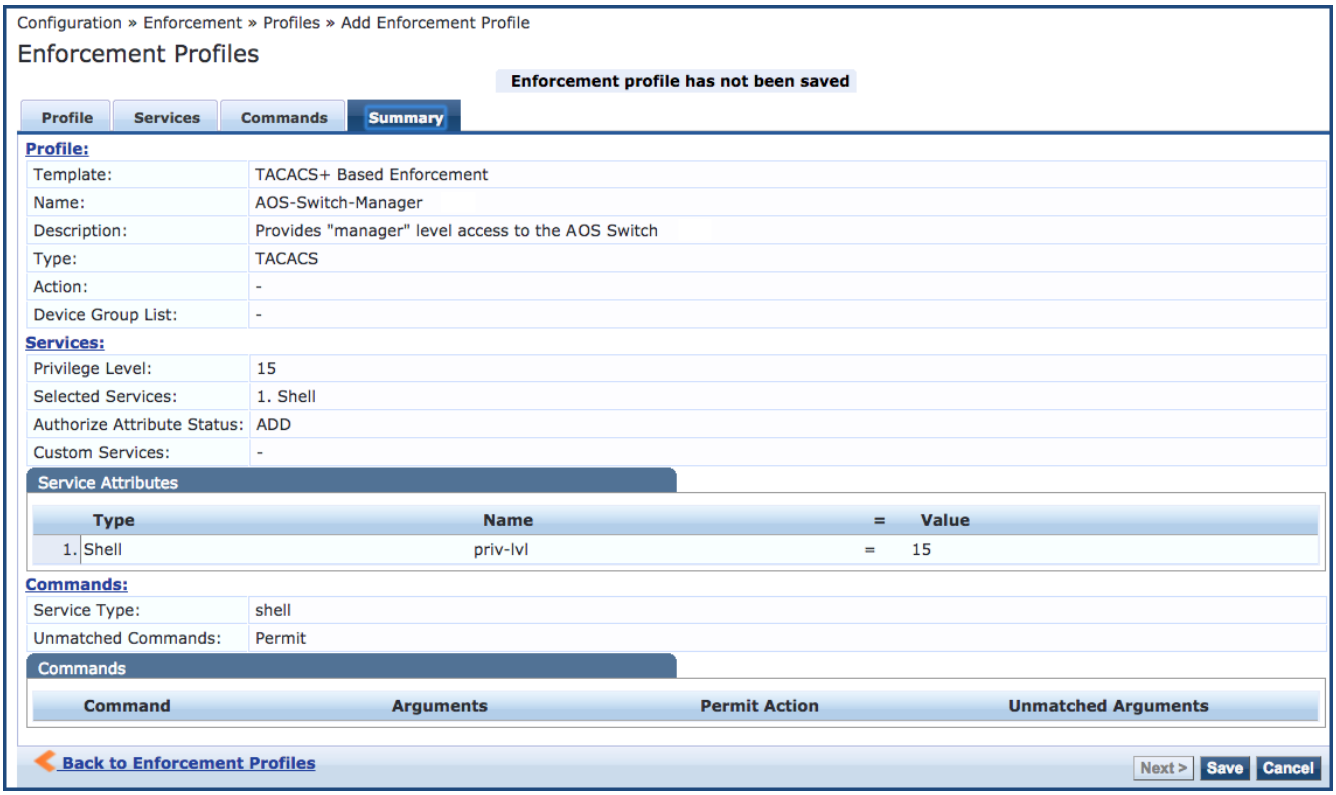
Parameter	Action/Description
Unmatched Commands	Click the check box for **Enable to permit u nmatched commands**.

Summary Tab

About this task

The following figure displays the **Summary** tab for the enforcement profile for manager-level access.

Figure 4. Summary of the TACACS+ Enforcement Profile for Manager-Level Access



Procedure

Click **Save**.

You return to the **Enforcement Profiles** page where you receive the message:

Enforcement profile "ArubaOS Switch-Manager" added

Creating an Enforcement Profile to Provide Operator-Level Access

About this task

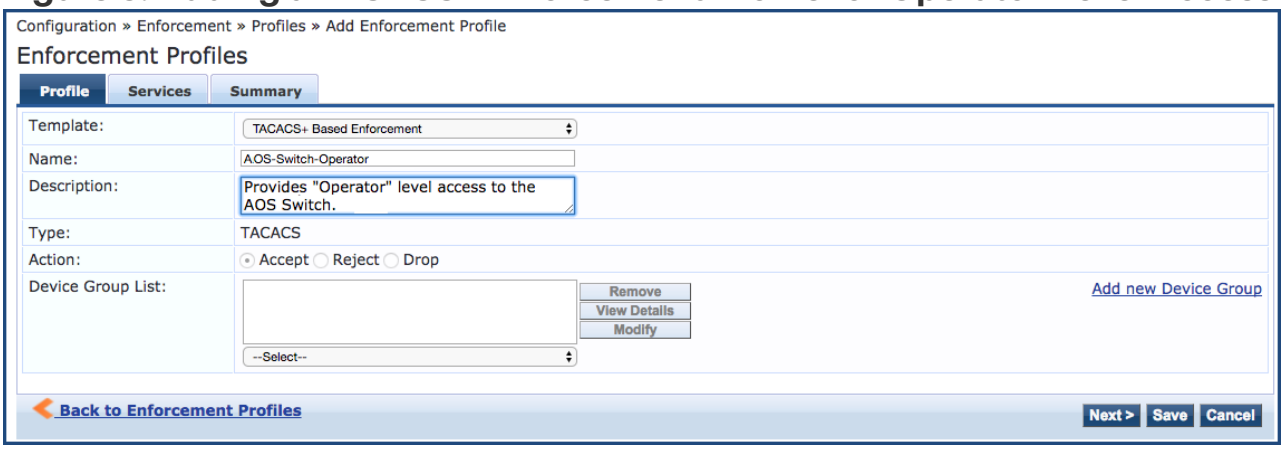
To create an enforcement profile to provide operator-level access:

Procedure

1. On the Policy Manager server, navigate to **Configuration > Enforcement > Profiles**.
2. From the **Enforcement Profiles** page, click **Add**.

   The **Add Enforcement Profiles** dialog opens.

Figure 5. Adding a TACACS+ Enforcement Profile for Operator-Level Access



3. Specify the **Add Enforcement Profile > Profile** tab parameters as described in the following table:

Table 6. Add Operator - Level Enforcement Profile > Profile Tab Parameters

Parameter	Action/Description
Template	From the Template drop - do **nforcement**.
Name	Enter the name of this enforc **Switch - Operator**.
Description	Add a description of this Ope ofile: Provides Operator - lev
Type	When you select **TACACS+ B** rcement profile type is set to
Action	Set to **Accept** the default.

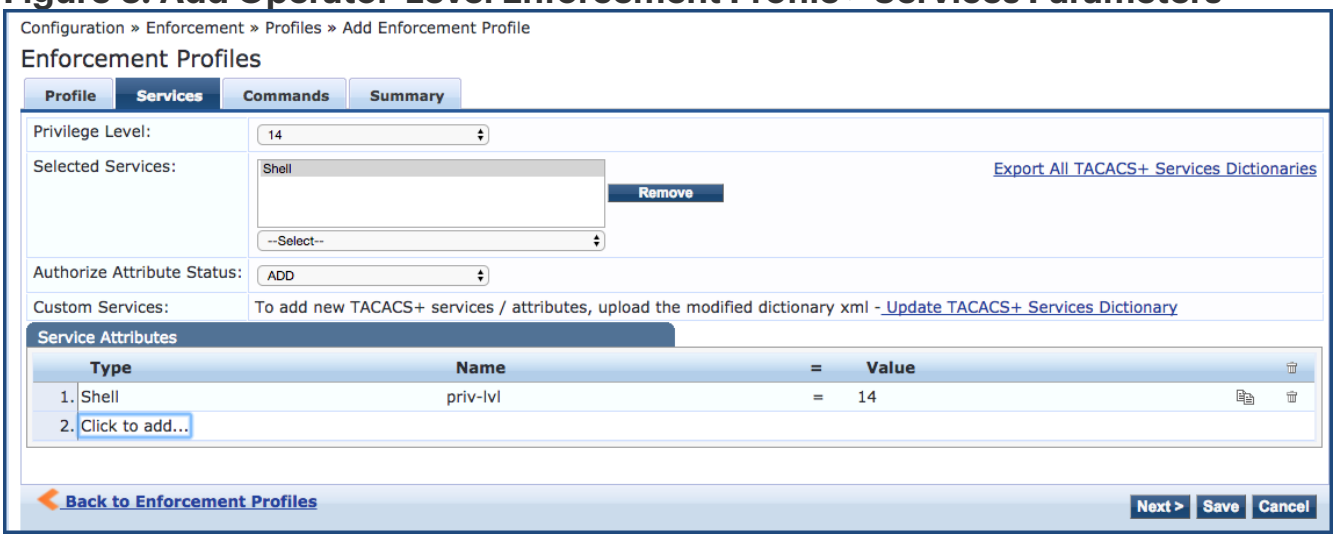
4. Click **Next**.

Services Tab

About this task

The **Services** tab opens.

Figure 6. Add Operator-Level Enforcement Profile > Services Parameters



Procedure

Specify the **Add Enforcement Profile > Services** tab parameters as described in the following table, then click **Next**:

Table 7. Add Operator - Level Enforcement Profile > Services Parameters

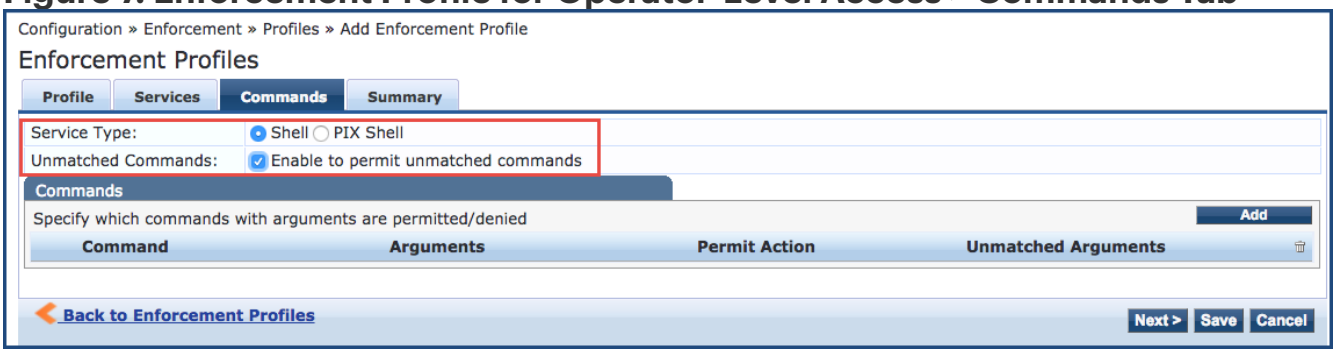
Parameter	Action/Description
Privilege Level	Select **14**.
Selected Service	Select **Shell**.
Authorize Attribute Status	Specify **ADD** (the default).
Service Attributes	
Type	Select **Shell**.
Name	Select **priv - lvl**.
Value	Enter a value of **14**.

Commands Tab

About this task

The next step is to configure the enforcement profile **Commands** parameters.

Figure 7. Enforcement Profile for Operator-Level Access > Commands Tab



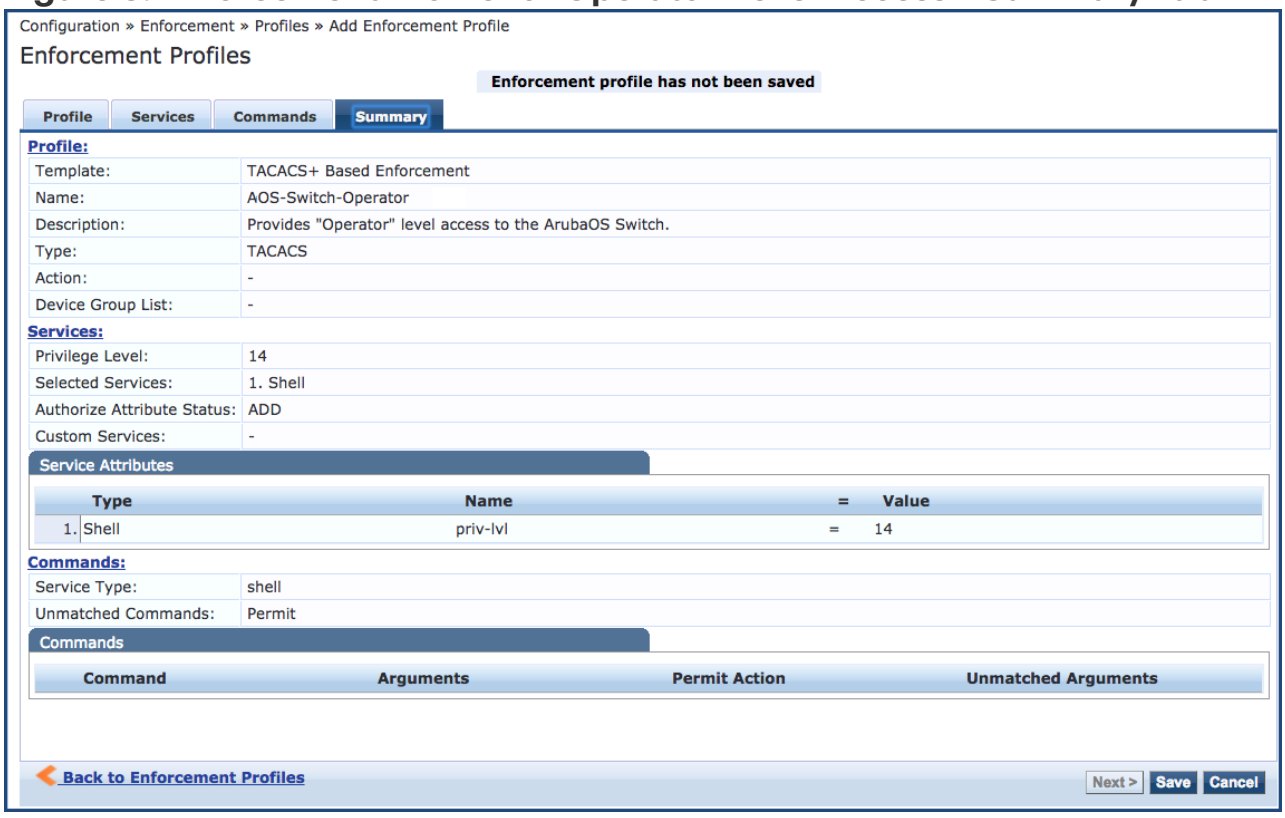
Procedure

1. Specify the **Commands** parameters as described in the following table, then click **Next**:

Table 8. Add Operator - Level Enforcement Profile > Commands Parameters

Parameter	Action/Description
Service Type	Specify **Shell**.
Unmatched Commands	Click the check box to **Enable t mmands**.

Figure 8. Enforcement Profile for Operator-Level Access > Summary Tab



2. Click **Save**.

   You return to the **Enforcement Profiles** page where you receive the message:

   Enforcement profile "ArubaOS Switch-Operator" added

Creating an Enforcement Policy to Define Access to the Switch

About this task

The service to authenticate TACACS+ users against Active Directory incorporates an enforcement policy that defines access to the ArubaOS switch. For this reason, we recommend that you create the enforcement policy before the service is created.

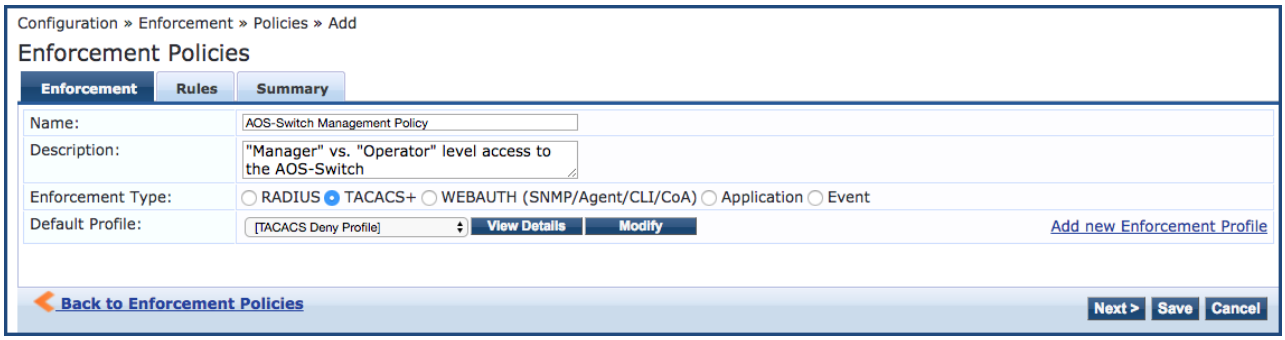
To create an enforcement policy to define manager-level and operator-level access to the AOS Switch:

Procedure

1. Navigate to **Configuration > Enforcement > Policies**.
2. From the **Enforcement Policies** page, click **Add**.

   The **Add Enforcement Policy** page opens.

Figure 9. Defining a Policy for Manager-Level and Operator-Level Access to the Switch



3. Specify the **Add Enforcement Policy** parameters as described in the following table, then click **Next**.

Table 9. Add Enforcement Policy Parameters

Parameter	Action/Description
Name	Enter a name for this enforcement policy; for example, "ArubaOS - Switch Management Policy."

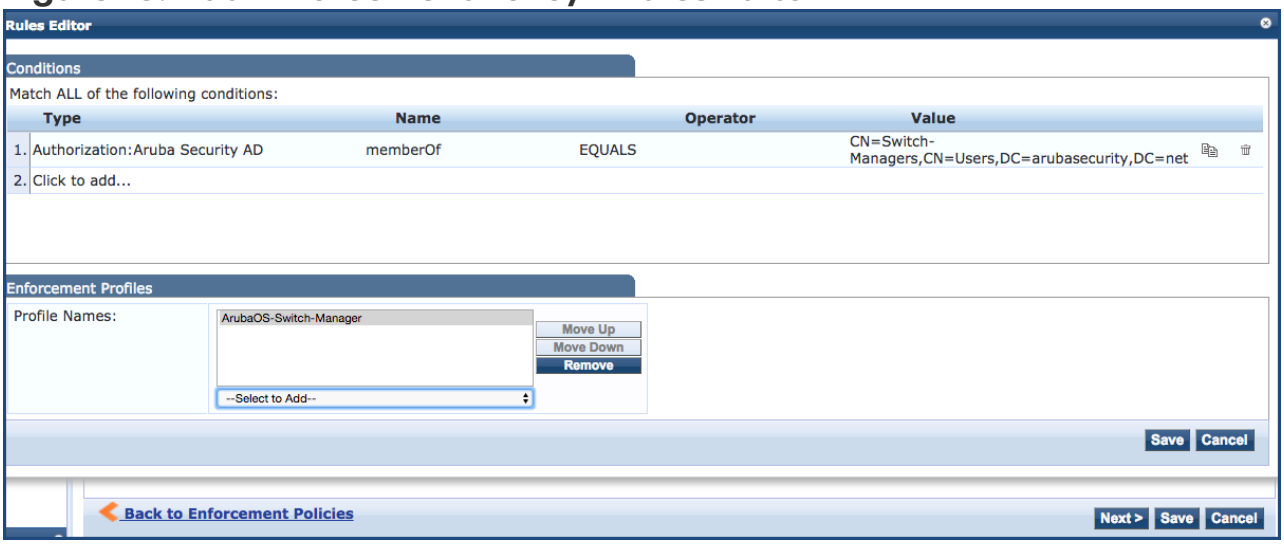
Parameter	Action/Description
Description	Optionally (but recommended), add a d escription of this enforcement policy; for example, " Policy to assign manager vso Operator - level access to the AOS Switc h."
Enforcement Type	Select **TACACS+**. Based on this selection, the **Default Prof ile** drop - down lists the associated enfo rcement profiles. In this example, it is the **TACACS+ Deny** profile. **NOTE** Web - based Authentication or WebAuth (HTTPS) is the mechanism used by authen tications performed via a b rowser, and authentications performed via ClearPass On Guard. Both SNMP - and C LI - (SSH/Telnet) based enfo rcement profiles can be sent to the network device based on the type of device and th e use case.
Default Profile	Select the **TACACS+ Deny** profile. An enforcement policy applies condition s (roles, health, and time attributes) again st specific values associated with those a ttributes to determine the enforcement p rofile. If none of the rules matches, Clear Pass applies the default profile.

When you click **Next**, the **Rules** dialog opens.

4. Click **Add Rule**.

5. The **Rules Editor** opens:

Figure 10. Add Enforcement Policy > Rules Editor



6. Specify the **Add Enforcement Policy > Rules** tab parameters as described in the following table:

Table 10. Configuring Rules for ArubaOS Switch—Manager and Operator

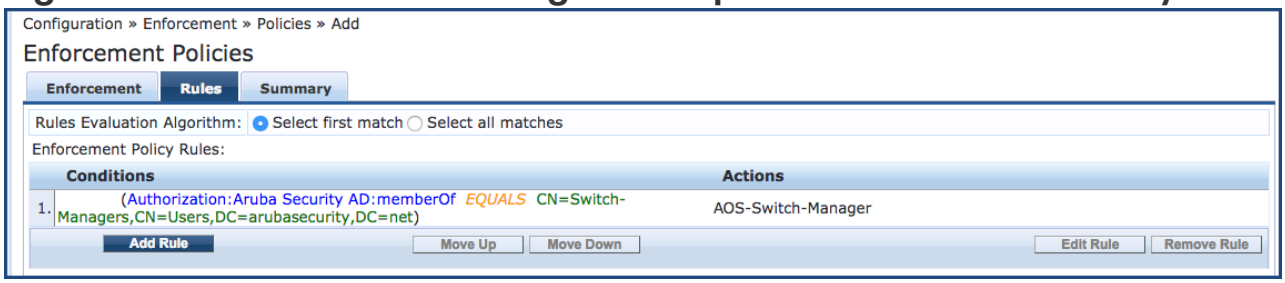
Parameter	Action/Description
Conditions	
Type	Select your Active Directory authorization source. For example: a. Select **Authorization:Aruba Security AD**. **NOTE** The Active Directory authorization source must be added manually. For details, see Adding Active Directory as an Authentication Source to Policy Manager .
Name	Select **memberOf**.

Parameter	Action/Description
Operator	Select **EQUALS**.
Value	Type in the Distinguished Name of your Active Directory group that the AD user is a member of. For example: • For a *Manager* role, enter **CN=Switch - Managers,CN=Users,DC=arubasecurity,DC=net**. • For an *Operator* role, enter **CN=Switch - Operators,CN=Users,DC=arubasecurity,DC=net**.
Enforcement Profiles	
Profile Names	• For a *Manager* role, select **ArubaOS Switch - Manager**. • For an *Operator*, role, select **ArubaOS Switch - Operator**.

7. Click **Save**.

   You return to the **Add Enforcement Policies > Rules** page, where the new enforcement rules are displayed:

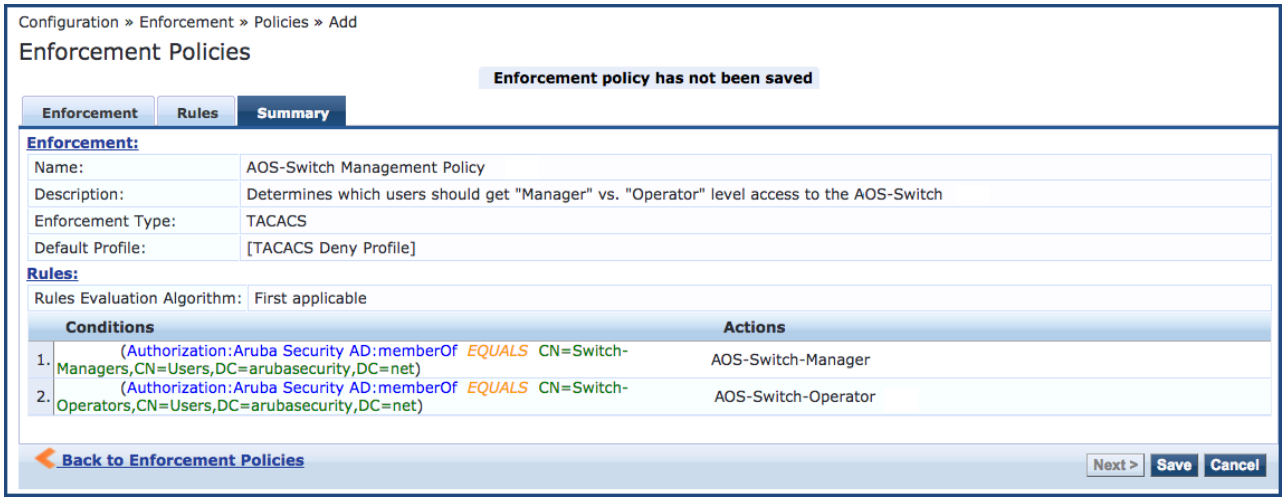
Figure 11. ArubaOS Switch Manager and Operator Enforcement Policy Rules



8. Click **Next**.

   The **Add Enforcement Policies > Summary** page opens:

Figure 12. ArubaOS TACACS+ Enforcement Policy Summary



9. Click **Save**.

   You return to the **Enforcement Policies** page where the following message is displayed:

   Enforcement policy "ArubaOS Switch Mgmt Enforcement Policy" has been added.

Creating a Service to Support TACACS+ Authentication Requests from the Switch

The service to support TACACS+ authentication requests from the ArubaOS switch incorporates the previously-created elements:

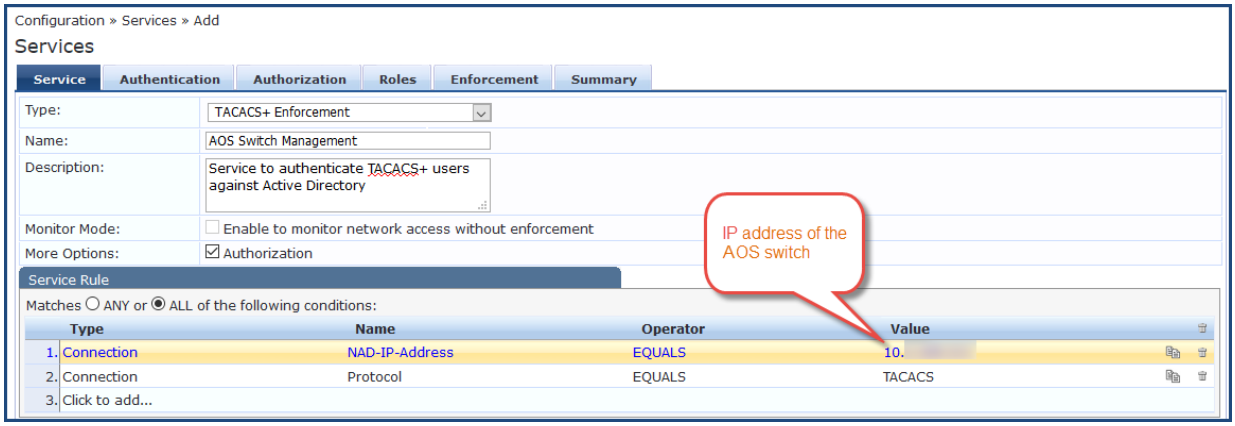
- Enforcement profiles that define manager-level access and operator-level access to the ArubaOS switch (see Creating Enforcement Profiles to Provide Manager Access and Command Control to the ArubaOS Switch).
- Enforcement policy that defines access to the ArubaOS switch (see Creating an Enforcement Policy to Define Access to the Switch).

To create a service to authenticate TACACS+ users against Active Directory:

1. Navigate to **Configuration > Services**.
2. From the **Services** page, click the **Add** link.

   The **Add Configuration Services** page opens.

Figure 13. Adding a TACACS+ Enforcement Service



- The service rule states that if the NAD-IP-Address value in the TACACS+ request = the IP address value specified in the **Value** parameter.
- And if the protocol of the request = TACACS+, the TACACS+ request will match the TACACS+ Enforcement service.

3. Specify the **Add TACACS+ Enforcement Service** parameters as described in the following table:

Table 11. Add TACACS+ Enforcement Service Parameters

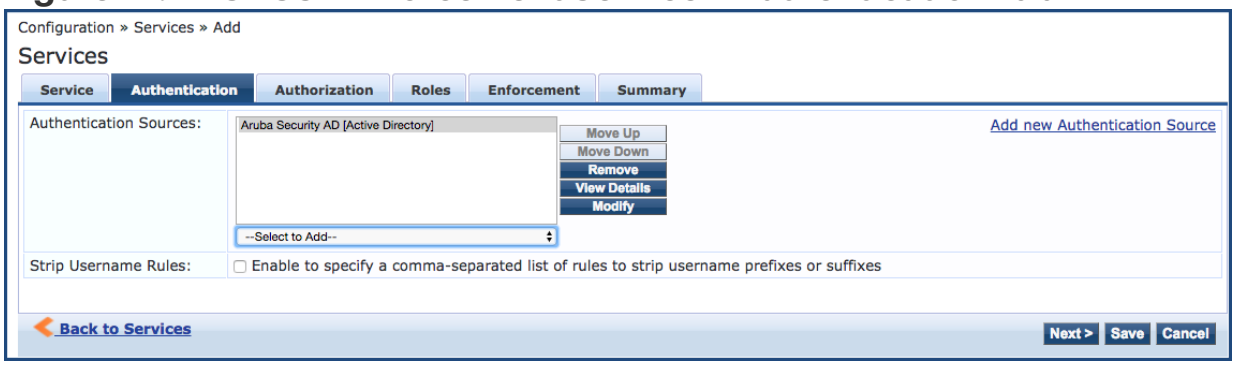
Parameter	Action/Description
Type	Select **TACACS+ Enforcement**.
Name	Enter a name for this service.
Description	Optionally (but recommended), add a d escription of this enforcement service.
Monitor Mode	This option is disabled by default. Do n ot enable Monitor Mode.
More Options	Click the check box to enable **Authori zation**.
Service Rule	
Matches	Select **ALL of the following conditions**.
Rule condition 1: Connecting to the Acti ve Directory server	Select **Click to add**, then specify the fol lowing attributes: ◦ **Type**: **Connection**

Parameter	Action/Description
	◦ Name: **NAD - IP - Address** ◦ Operator: **EQUALS** ◦ Value: <*IP address of the ArubaOS switch*>
Rule condition 2: Specifying the TACACS+ protocol	Select **Click to add**, then specify the following attributes: ◦ **Type**: **Connection** ◦ Name: **Protocol** ◦ Operator: **EQUALS** ◦ Value: **TACACS+**

4. Click **Next**.

   The **Authentication** tab opens.

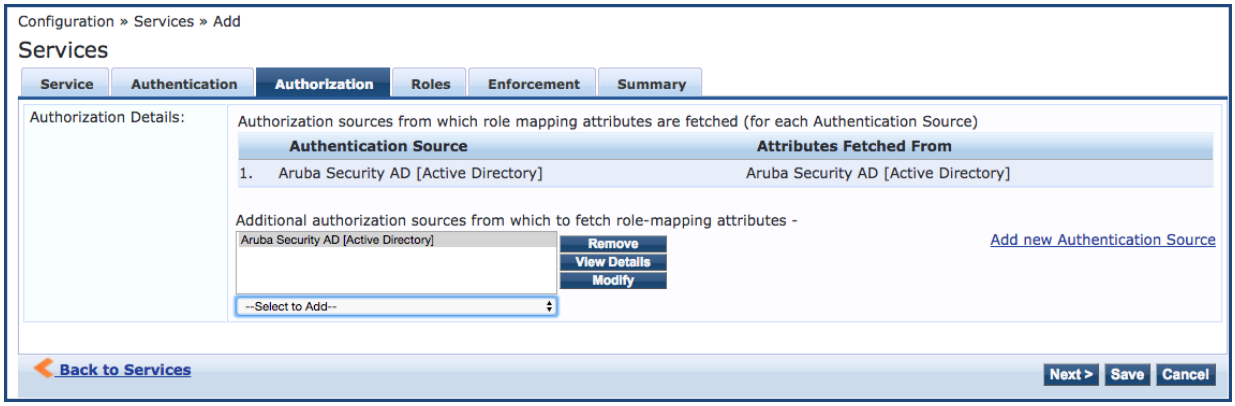
Figure 14. TACACS+ Enforcement Service > Authentication Tab



5. From the **Authentication Sources** drop-down, select **Aruba Security AD** (or whatever name was assigned to this authentication source), then click **Next**.

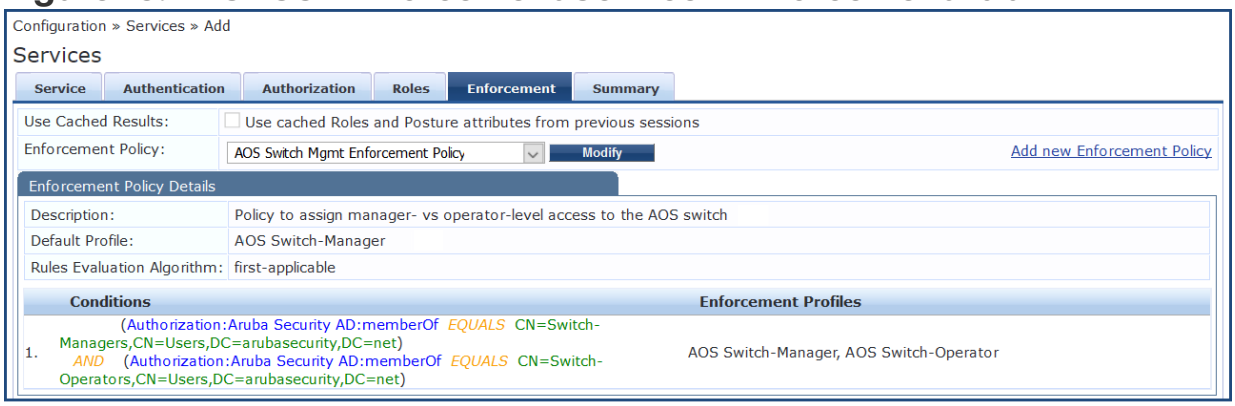
   The **Authorization** tab opens.

Figure 15. TACACS+ Enforcement Service > Authorization Tab



6. From the *Additional authorization sources from which to fetch role-mapping attributes* drop-down, select **Aruba Security AD (Active Directory)**.
7. Select the **Enforcement** tab.

Figure 16. TACACS+ Enforcement Service > Enforcement Tab



8. From the **Enforcement Policy** drop-down, select **ArubaOS Switch Mgmt Enforcement Policy**.
9. Click **Save**.

   This completes the service for ArubaOS switch management.

Setting Up the Switch for Command Authorization Using TACACS+

From the AOS Switch, enable command authorization using the same protocol that was used for authentication:

```
ArubaOS-Switch (config)# aaa authorization commands auto
```

Setting Up Enforcement Profiles in ClearPass to Support TACACS+ Command Authorization Requests from the Switch

About this task

On the Policy Manager server, implementing command authorization using TACACS+ is achieved by creating an enforcement profile that defines commands that are either allowed or denied.

The sample enforcement profile created here will permit all commands to be run on the switch, except for those under the AAA configuration hierarchy. This is useful for situations in which the user's role does not include making changes to anything related to authentication, authorization, or accounting.

To set up enforcement profiles in Policy Manager to support TACACS+ command authorization requests from the AOS Switch:

Procedure

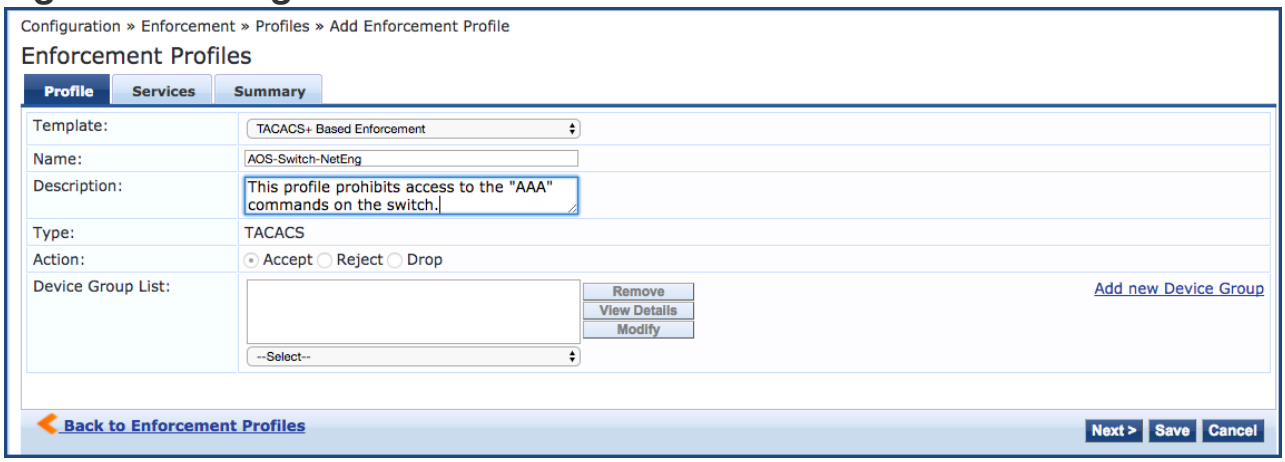
1. Navigate to **Configuration > Enforcement > Profiles**.

   The **Enforcement Profiles** page opens.

2. Click the **Add** link.

   The **Add Enforcement Profile** page opens.

Figure 17. Adding a TACACS+ Command Authorization Enforcement Profile



3. Specify the **TACACS+-Based Enforcement Profile** parameters as described in the following table:

Table 12. Add TACACS+ Enforcement Profile Parameters

Parameter	Action/Description
Template	Select **TACACS+ - Based Enforcement**.
Name	Enter a name for this enforcement profile .

Parameter	Action/Description
Description	Optionally (but recommended), add a d escription of this enforcement profile.
Type	When you select the *TACACS+ - Based En forcement* template, the enforcement pr ofile **Type** is set automatically to **TACACS +**.
Action	Keep the default action: **Accept**.
Device Group List	The **Device Group List** is no longer pertin ent and this option is grayed out.

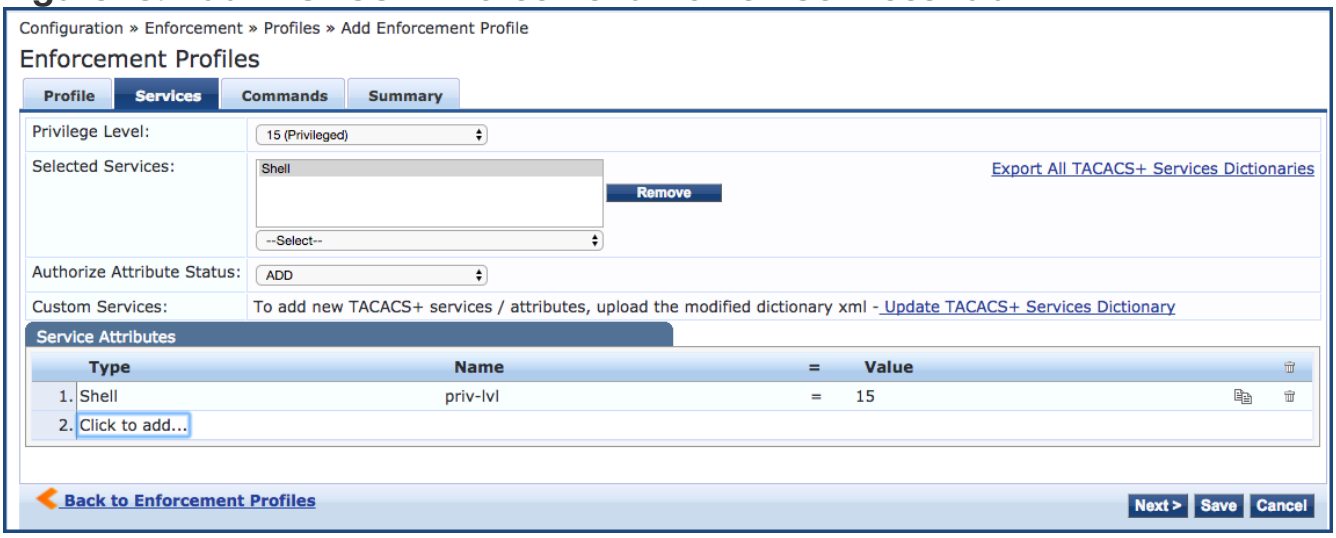
4. Click **Next**.

Add Enforcement Profile > Services Tab

About this task

When you click **Next**, the **Services** tab opens.

Figure 18. Add TACACS+ Enforcement Profile > Services Tab



Procedure

1. Specify the **TACACS+-Based Enforcement Profile > Services** parameters as described in the following table:

Table 13. Add TACACS+ Enforcement Profile > Services Parameters

Parameter	Action/Description
Privilege Level	Select **15 (Privileged)**.
Selected Services	From the **Select** drop - down, choose **Shell**.
Authorize Attribute Status	Specify **ADD** (which is the default).
Custom Services	When you select the *TACACS+ - Based Enforcement* template, the enforcement profile **Type** is set automatically to **TACACS+**.
Service Attributes	
Type	Click **Click to add**, then select **Shell**.
Name	Select **priv - lvl**.
Value	Enter a value of **15**.

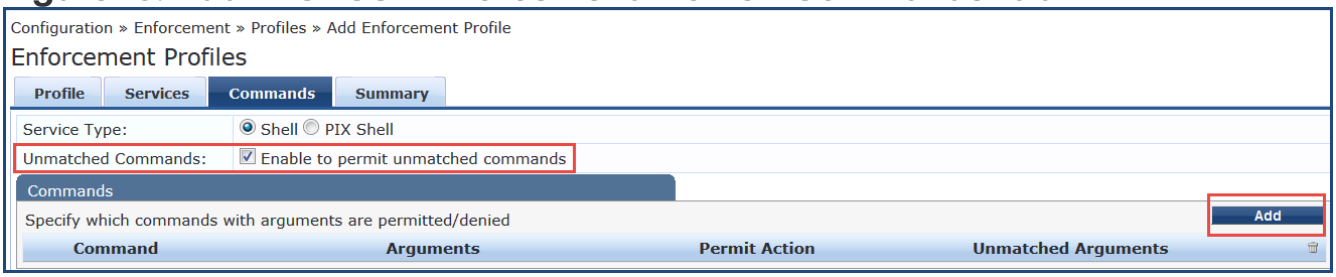
2. Click **Next**.

Add Enforcement Profile > Commands Tab

About this task

When you click **Next**, the **Commands** tab opens.

Figure 19. Add TACACS+ Enforcement Profile > Commands Tab



Procedure

1. Specify the **TACACS+-Based Enforcement Profile > Commands** parameters as described in the following table:

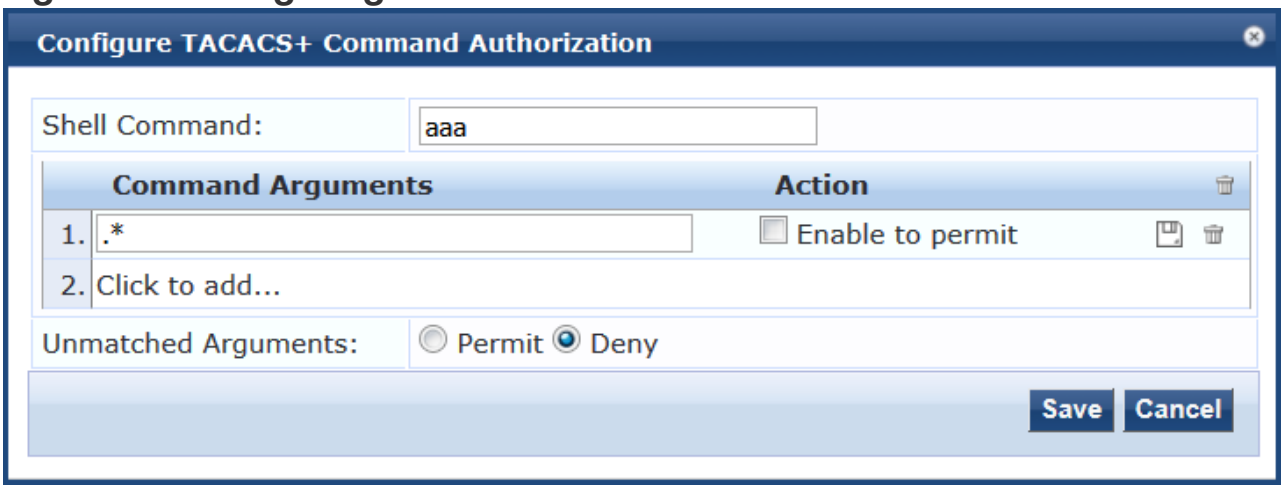
Table 14. Add TACACS+ Enforcement Profile > Commands Parameters

Parameter	Action/Description
Service Type	The **Service Type** is automatically set to **Shell**.
Unmatched Commands	Select the check box to permit unmatched records.

2. From the **Commands** panel, click **Add**.

   The **Configure TACACS+ Command Authorization** configuration dialog opens.

Figure 20. Configuring TACACS+ Command Authorization



3. Specify the **TACACS+-Based Enforcement Profile > Commands** parameters as described in the following table:

Table 15. Add TACACS+ Enforcement Profile > Commands Parameters

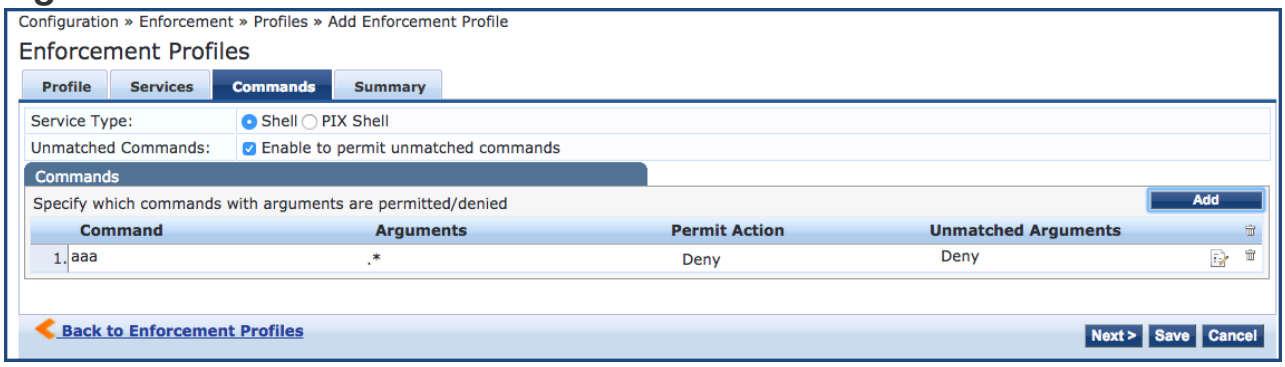
Parameter	Action/Description
Shell Command	The **Service Type** is automatically set to **Shell**.
Command Arguments	Select **Click to add**, then enter **.*** By entering a period and an asterisk (.*), any string that follows the shell command listed in the *Shell Command* field will b

Parameter	Action/Description
	e matched against the enforcement poli cy.
Action	Do not enable to permit—leave the check box unchecked.
Unmatched Arguments	Select **Deny**.

4. Click **Save**.

   You return to the **Commands** page where it shows that the **aaa** command requests to the Policy Manager server from the AOS Switch are denied.

Figure 21. ArubaOS Command Denied



5. Click **Save**.

   This enforcement profile is added to the **Enforcement Profiles** page.

Switch Management Using RADIUS

This section contains the following information

- Setting Up Switch Management Using RADIUS
- Using RADIUS-Based Authentication and Command Authorization
- Creating Enforcement Profiles to Provide Manager Access and Command Authorization to the ArubaOS Switch
- Creating an Enforcement Policy to Define Access to the Switch

Setting Up Switch Management Using RADIUS

This section describes how to set up AOS Switch management using RADIUS . The following login methods are described:

- Initial configuration
- SSH
- Telnet
- Console
- Web Agent

Initial RADIUS Management Configuration

About this task

> **NOTE**
> Out-of-band management (oobm) is only required if the ArubaOS switch will be using the out-of-band management interface to communicate with the RADIUS server.

To provide the initial RADIUS management configuration:

Procedure

1. Define the RADIUS server in the AOS Switch:

```
   AOS-Switch(config)#radius-server host 10.x.x.x oobm key
supersecretkey123
```

2. Optionally, adjust the RADIUS server timeout period as needed. The default is 3 seconds.

```
   AOS-Switch(config)#radius-server timeout 30
```

3. Configure RADIUS single login capability:

```
   AOS-Switch(config)#aaa authentication login privilege-mode
```

Configuring SSH Login for RADIUS Authentication

About this task

To configure the SSH (Secure Shell) login for RADIUS authentication:

Procedure

1. Configure RADIUS authentication for SSH login with read-only (operator) access:

   ```
   AOS-Switch(config)#aaa authentication ssh login radius local
   ```

2. Configure RADIUS authentication for SSH login with access to privileged (manager) access:

   ```
   AOS-Switch(config)#aaa authentication ssh enable radius local
   ```

Configuring Telnet Login for RADIUS Authentication

About this task

To configure the Telnet login for RADIUS authentication:

Procedure

1. Configure RADIUS authentication for Telnet login with read-only (operator) access:

   ```
   AOS-Switch(config)#aaa authentication telnet login radius local
   ```

2. Configure RADIUS authentication for Telnet login with access to privileged (manager) access:

   ```
   AOS-Switch(config)#aaa authentication telnet enable radius local
   ```

Configuring the Console Login for RADIUS Authentication

About this task

To configure the Console login for RADIUS authentication:

Procedure

1. Configure RADIUS authentication for the Console login with read-only (operator) access:

   ```
   AOS-Switch(config)# aaa authentication console login radius local
   ```

2. Configure RADIUS authentication for the Console login with access to privileged (manager) access:

   ```
   AOS-Switch(config)#aaa authentication console enable radius local
   ```

Configuring Web Agent Login for RADIUS Authentication

About this task

To configure the Web Agent login for RADIUS authentication:

Procedure

1. Configure Web Agent authentication for the Web Agent login with read-only (operator) access:

```
AOS-Switch(config)#aaa authentication web login radius local
```

2. Configure RADIUS authentication for the Web Agent login with access to privileged (manager) access:

```
AOS-Switch(config)#aaa authentication web enable radius local
```

Using RADIUS-Based Authentication and Command Authorization

When using RADIUS-based command authorization on an AOS Switch, the list of commands that the user is authorized to run are supplied at authentication time. This is in contrast to TACACS+, where each command being run by the user is sent to the AAA server to be authorized.

The RADIUS-based method of command authorization requires less overhead on the AAA server and affords higher performance in environments that have many concurrent management sessions on the switches.

The examples in the following section describe the following:

- Both Manager and "ViewOnly" levels of access are actually the same as far as the ArubaOS switch is concerned, with the only difference being which commands can be run.
- The Manager-level of access still has command authorization as a RADIUS Vendor Specific Attribute (VSA), but there is no list of commands accompanying that role. When there is no list of commands, all commands can be run.
- The ViewOnly-level of access has command authorization as a *RADIUS Vendor Specific Attribute* (VSA) and a command list with a regular expression indicating that only commands that begin with the word "*show*" may be run.

Creating Enforcement Profiles to Provide Manager Access and Command Authorization to the ArubaOS Switch

The service to authenticate RADIUS users against Active Directory incorporates enforcement profiles that define manager-level access and command authorization to

the ArubaOS switch. For this reason, we recommend that the necessary enforcement profiles be created *before* the service is created.

Adding Active Directory as an Authentication Source

Before you create the enforcement profiles as described below, we recommend that you configure Active Directory as an authentication source first, as you will need to specify the authentication source when you create the RADIUS enforcement policy.

For details, see Adding Active Directory as an Authentication Source to Policy Manager .

Creating an Enforcement Profile to Provide Manager-Level Access

About this task

In Policy Manager Policy Manager, an enforcement policy provides the rules that tells Policy Manager when to use specific enforcement profiles. Enforcement profiles consist of actions that are taken by Policy Manager, for example, assigning a certain role to a user.

The actions in an enforcement policy are the enforcement profiles to be activated when specific conditions are met or rules are enforced. Then an enforcement policy is associated with a service—a service ties all the elements together: authentication sources, authorization sources, enforcement policies, and role-mapping.

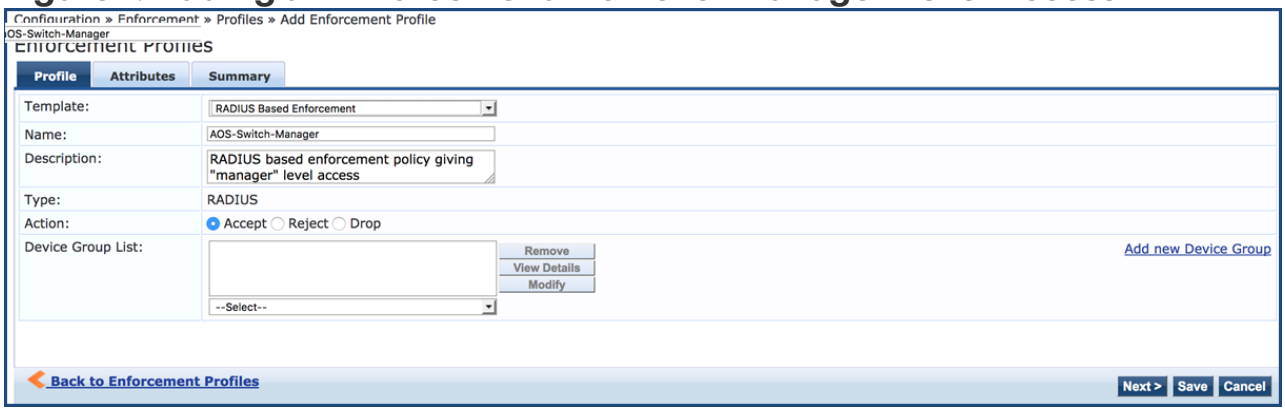
To create an enforcement profile to provide manager-level access to the AOS Switch:

Procedure

1. On the Policy Manager server, navigate to **Configuration > Enforcement > Profiles**.
2. From the **Enforcement Profiles** page, click **Add**.

   The **Add Enforcement Profiles** dialog opens.

Figure 1. Adding an Enforcement Profile for Manager-Level Access



3. Specify the **Add Enforcement Profile > Profile** tab parameters as described in the following table, then click **Next**:

Table 1. Add Manager - Level Enforcement Profile > Profile Tab Parameters

Parameter	Action/Descript
Template	From the Templa **rcement**.
Name	Enter the name **- Manager**.
Description	Add a descriptic ager - level acce
Type	When you select ment profile Typ
Action	Set to **Accept** (t

Table 2. Add Manager - Level Enforcement Profile > Profile Tab Parameters

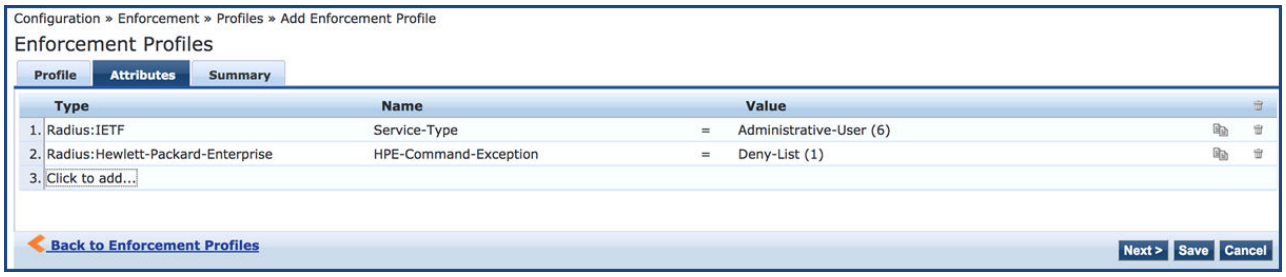
Parameter	Action/Descript
Template	From the Templa **rcement**.
Name	Enter the name **- Manager**.
Description	Add a descriptic ager - level acce
Type	When you select ment profile Typ
Action	Set to **Accept** (t

Specifying Enforcement Profile Attributes

Procedure

1. From the Enforcement Profiles **Attributes** tab, select **Click to add**.

Figure 2. Enforcement Profile for Manager-Level Access >Specifying Attributes



2. Specify the **Add Enforcement Profile > Attributes** as described in the following table:

Table 3. Manager - Level Enforcement Profile > Attributes

Attribute	Action/Des
Service - Type Attribute	
Type	Select **Rad**
Name	Select **Ser**
Value	Select **Adn** The value c h instructs - level acc
Service - Type Attribute	
Type	Select **Rad**
Name	Select **HPE**
Value	Select **Den** **HPE - Com** r the comn attribute ar ns permit a) means de adding a va - String VS

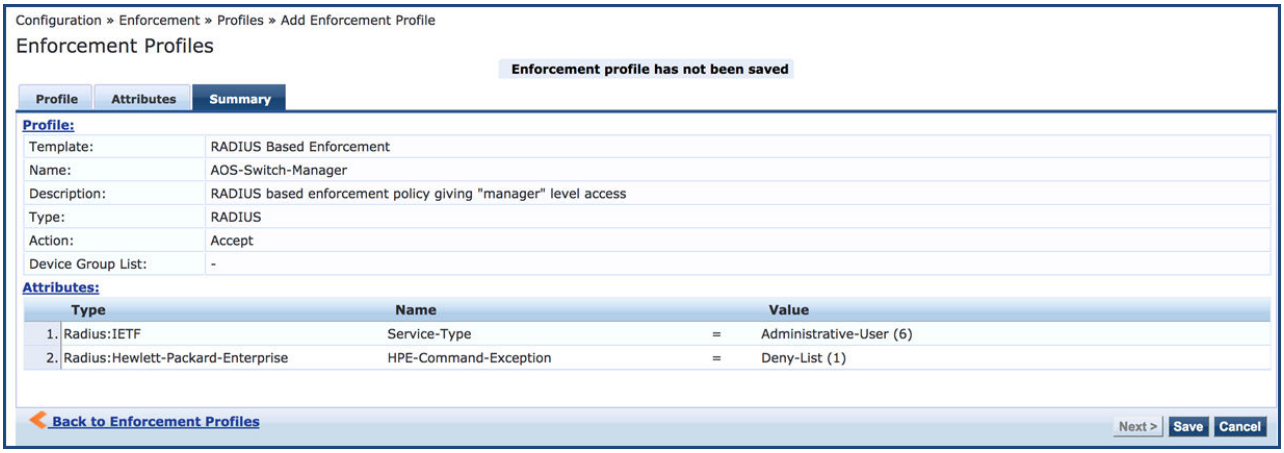
Table 4. Manager - Level Enforcement Profile > Attributes

Attribute	Action/Des
Service - Type Attribute	
Type	Select **Rad**
Name	Select **Ser**
Value	Select **Adn** The value c h instructs - level acc
Service - Type Attribute	
Type	Select **Rad**
Name	Select **HPE**
Value	Select **Den** **HPE - Com** r the comn attribute ar ns permit a) means de adding a va - String VS

3. Click **Next**:

   The **Summary** for the RADIUS enforcement profile for manager-level access is displayed.

Figure 3. Summary of the RADIUS Enforcement Profile for Manager-Level Access



4. Click **Save**.

   You return to the **Enforcement Profiles** page where you receive the message:

   Enforcement profile "ArubaOS Switch-Manager" added

Creating an Enforcement Profile to Provide Command Authorization

About this task

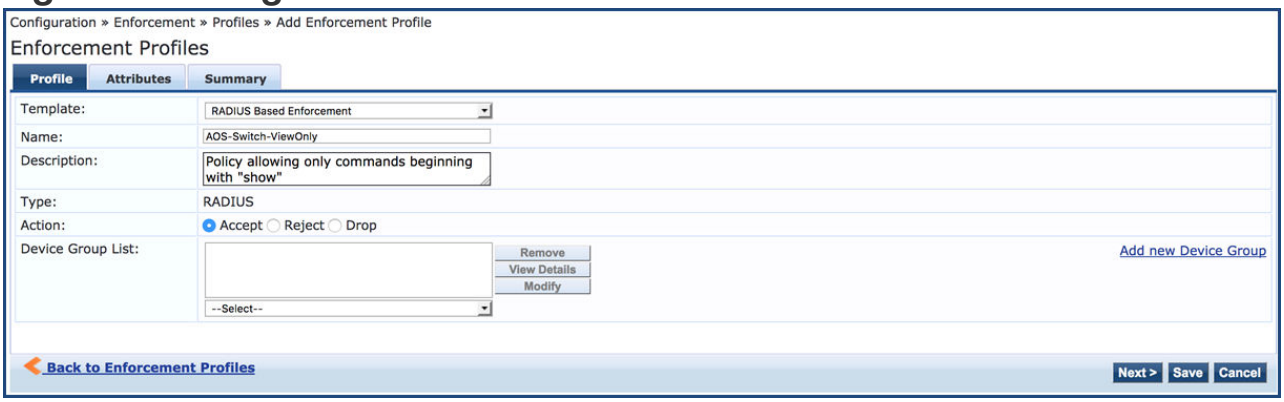
To create a RADIUS-based enforcement profile to provide command authorization:

Procedure

1. On the Policy Manager server, navigate to **Configuration > Enforcement > Profiles**.
2. From the **Enforcement Profiles** page, click **Add**.

   The **Add Enforcement Profiles** dialog opens.

Figure 4. Adding a RADIUS Enforcement Profile for Command Authorization



3. Specify the **Add Enforcement Profile > Profile** tab parameters as described in the following table:

Table 5. RADIUS Enforcement Profile > Profile Tab Parameters

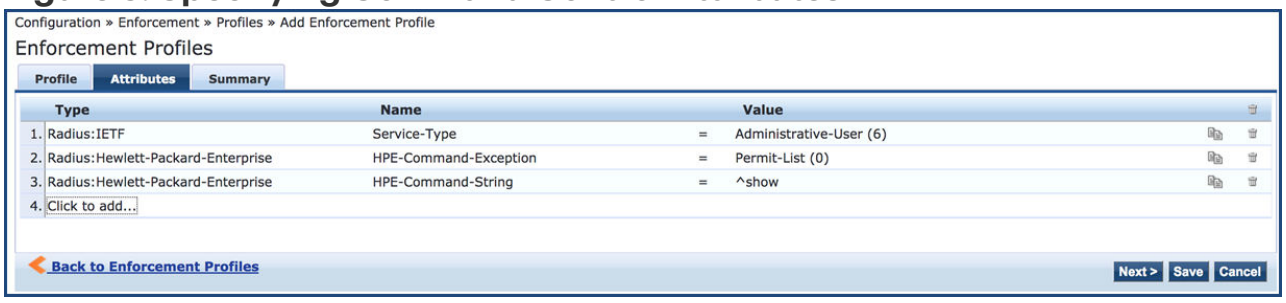
Parameter	Action/Description
Template	From the Template drop - down, select **RADIUS Based Enforcement**.
Name	Enter the name of this enforcement profile: **AOS - Switch - ViewOnly**.
Description	Add a description of this command control enforcement profile: Policy allowing only commands beginning with "show."
Type	When you select **RADIUS Based Enforcement**, the enforcement profile type is set to **RADIUS** .
Action	Set to **Accept** (the default).

Specifying Enforcement Profile Attributes

Procedure

1. From the Enforcement Profiles **Attributes** tab, select **Click to add**.

Figure 5. Specifying Command Control Attributes



2. Specify the **Add Enforcement Profile > Attributes** as described in the following table:

Table 6. Manager - Level Enforcement Profile > Attributes

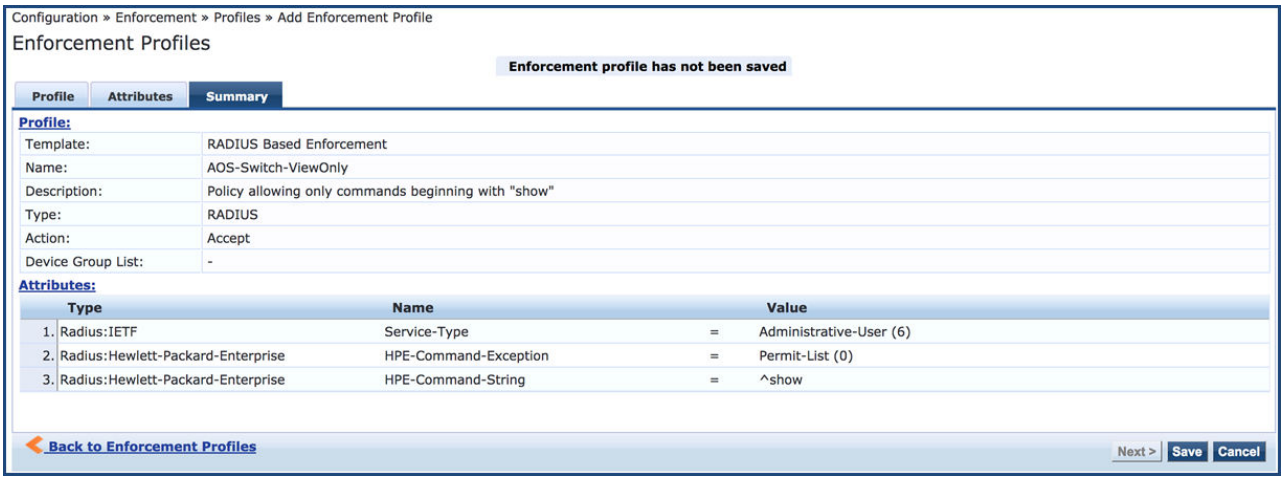
Attribute	Action/Description
Service - Type	
Type	Select **Radius:IETF**.

Attribute	Action/Description
Name	Select **Service - Type**.
Value	Select **Administrative - User (6)**. The value of the **Administrative - user** p arameter is **6**, which instructs the Aruba OS switch to grant the user manager - l evel access.
HPE - Command Exception	
Type	Select **Radius: Hewlett - Packard_Enterp rise**.
Name	Select **HPE - Command - Exception**.
Value	Select **Permit - List**.
HPE - Command String	
Type	Select **Radius: Hewlett - Packard_Enterp rise**.
Name	Select **HPE - Command String**.
Value	Enter **^show** **HPE - Command - String** is a list of com mands (regular expressions) that are per mitted or denied execution by the user. T he commands are delimited by semicolo ns and must be between 1 and 249 chara cters in length. Multiple instances of this attribute can be present in Access - Acc ept packets. (A single instance can be p resent in Accounting - Request packets.) In this example, the regular expression ^s how means that commands that begin w ith the word “show” are allowed. All other commands are denied.

3. Click **Next**:

   The **Summary** for the RADIUS enforcement profile for command authorization is displayed.

Figure 6. Summary of the RADIUS Enforcement Profile for Command Authorization



4. Click **Save**.

   You return to the **Enforcement Profiles**page where you receive the message:

   Enforcement profile "ArubaOS Switch-ViewOnly" added

Creating an Enforcement Policy to Define Access to the Switch

About this task

The service to authenticate RADIUS users against Active Directory incorporates an enforcement policy that defines access to the ArubaOS switch. For this reason, we recommend that you create the enforcement policy before the service is created.

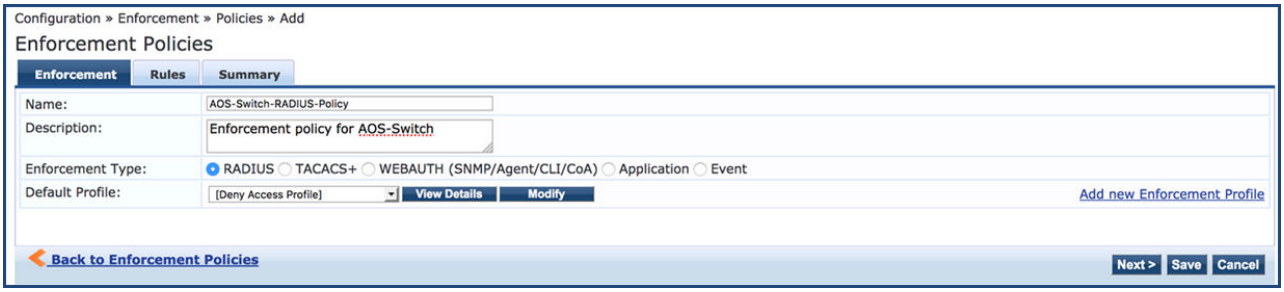
To create an enforcement policy to define manager-level access and command authorization to the AOS Switch:

Procedure

1. Navigate to **Configuration > Enforcement > Policies**.
2. From the **Enforcement Policies** page, click **Add**.

   The **Add Enforcement Policy** page opens.

Figure 7. Defining an Enforcement Policy for the ArubaOS Switch



3. Specify the **Add Enforcement Policy** parameters as described in the following table, then click **Next**.

Table 7. Add RADIUS Enforcement Policy Parameters

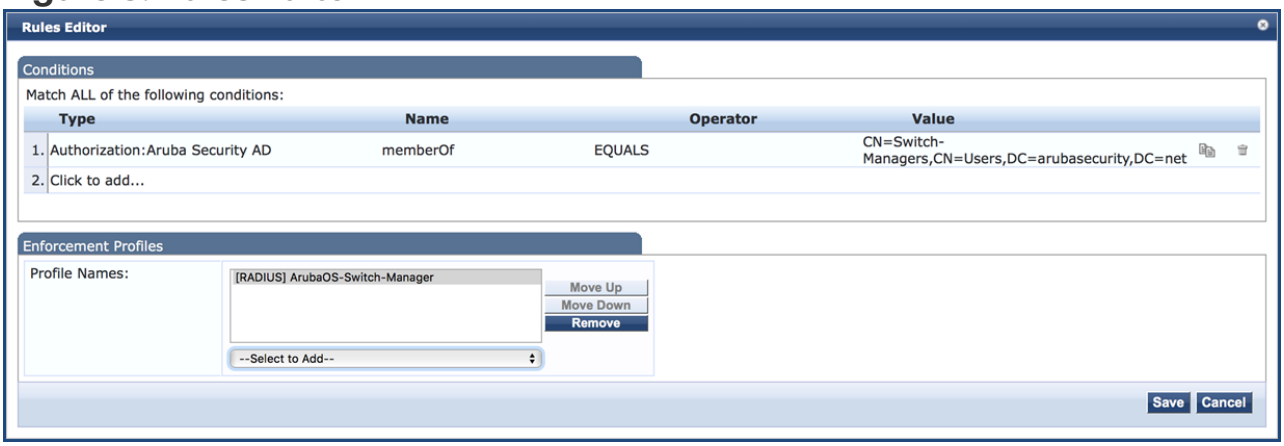
Parameter	Action/Description
Name	Enter a name for this enforcement policy; for example, "AOS - Switch - RADIUS - Policy."
Description	Optionally (but recommended), add a description of this enforcement policy; for example, " Enforcement policy for AOS Switch."
Enforcement Type	Select **RADIUS**. Based on this selection, the **Default Profile** drop - down lists the associated enforcement profiles.
Default Profile	Select the **Deny Access Profile**. An enforcement policy applies conditions (roles, health, and time attributes) against specific values associated with those attributes to determine the enforcement profile. If none of the rules matches, Clear Pass applies the default profile. To add a new profile, click **Add New Enforcement Profile**.

When you click **Next**, the **Rules** dialog opens.

4. Click **Add Rule**.

The **Rules Editor** opens.

Figure 8. Rules Editor



5. Specify the **Rules Editor** parameters as described in the following table:

Table 8. Configuring Rules for ArubaOS Switch—Manager and ViewOnly

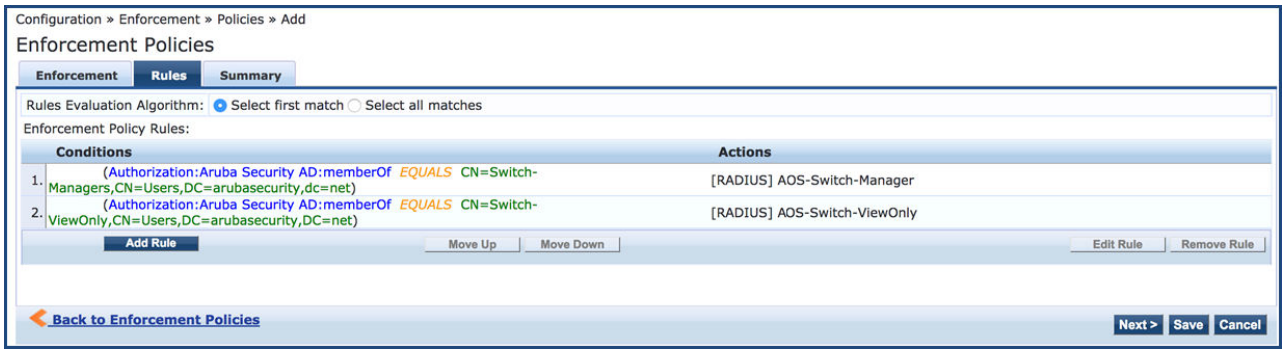
Parameter	Action/Description
Conditions	
Type	• *Manager*: Specify the authorization source that was created for your Active Directory domain. In this example, the name of the authorization source is **Authorization:Aruba Security AD**. • *ViewOnly*: Specify the authorization source that was created for your Active Directory domain. **NOTE** The authorization source must be added manually. For details, see Adding Active Directory as an Authentication Source to Policy Manager .
Name	• *Manager*: Select **memberOf**.

Parameter	Action/Description
Operator	• *Manager*: Select **EQUALS**.
Value	Fill out the Active Directory Distinguished Name (DN) of the group that contains the user who will be subjected to the enforce ment profile below. For example: • **CN=Switch - Managers,CN=Users,DC =arubasecurity,DC=net**
Enforcement Profiles	
Profile Names	Choose the enforcement profile that will be applied when members of the Active Directory group listed above are matche d. For example: • **ArubaOS Switch - Manager**.

6. Click **Save**.

   You return to the **Add Enforcement Policies > Rules** page, where the new enforcement rules are displayed:

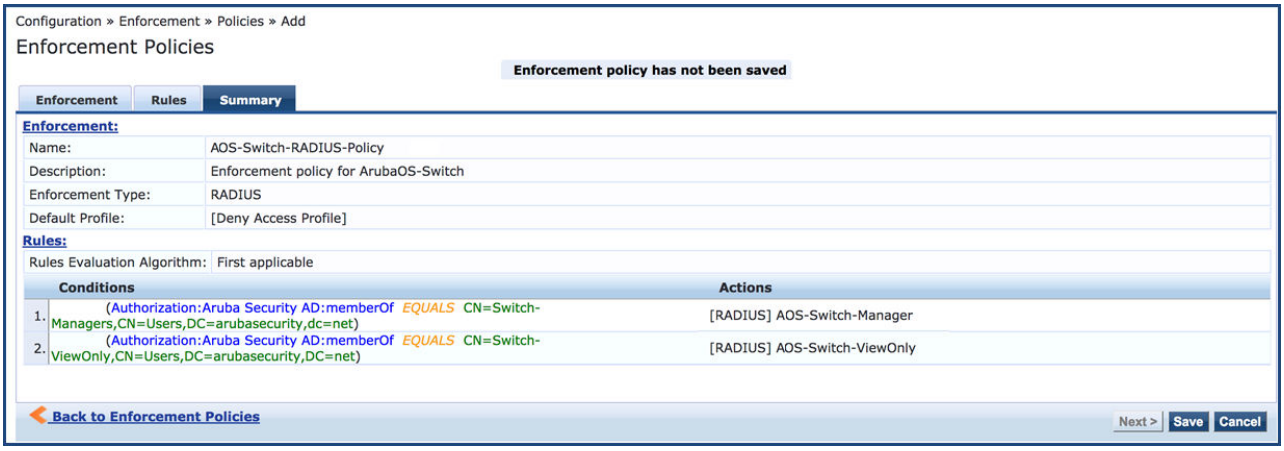
Figure 9. ArubaOS Switch- Manager and ViewOnly Enforcement Policy Rules



7. Click **Next**.

   The **Add Enforcement Policies > Summary** page opens:

Figure 10. ArubaOS RADIUS Enforcement Policy Summary



8. Repeat Step 4 through Step 7 for each role you need to add to the enforcement policy (such as *ViewOnly*), with the only differences being the **memberOf** value and **Enforcement Profile** value.

9. Click **Save**.

   You return to the **Enforcement Policies** page where the following message is displayed:

   Enforcement policy "AOS-Switch-RADIUS-Policy" has been added.

Setting Up a Service in ClearPass to Support RADIUS Authentication Requests From the Switch

The service to support RADIUS authentication requests from the ArubaOS switch incorporates the previously-created elements:

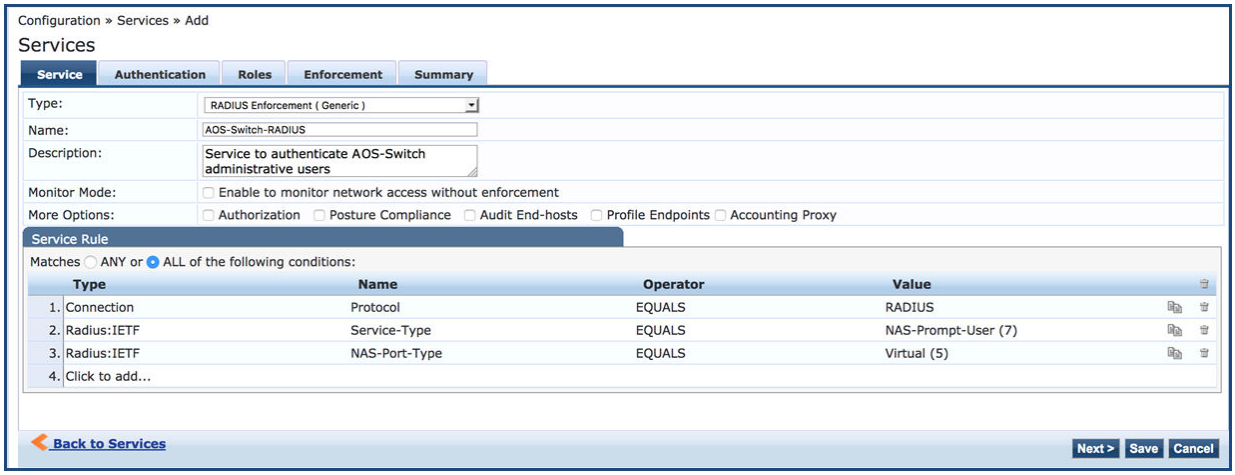
- Enforcement profiles that define manager-level access and operator-level access to the ArubaOS switch (see Creating Enforcement Profiles to Provide Manager Access and Command Authorization to the ArubaOS Switch).
- Enforcement policy that defines access to the ArubaOS switch (see Creating an Enforcement Policy to Define Access to the Switch).

To create a service to authenticate RADIUS users against Active Directory:

1. Navigate to **Configuration > Services**.
2. From the **Services** page, click the **Add** link.

   The **Add Configuration Services** page opens.

   Figure 11. Adding a RADIUS Enforcement Service



3. Specify the **Add RADIUS Enforcement Service** parameters as described in the following table:

Table 9. Add RADIUS Enforcement Service Parameters

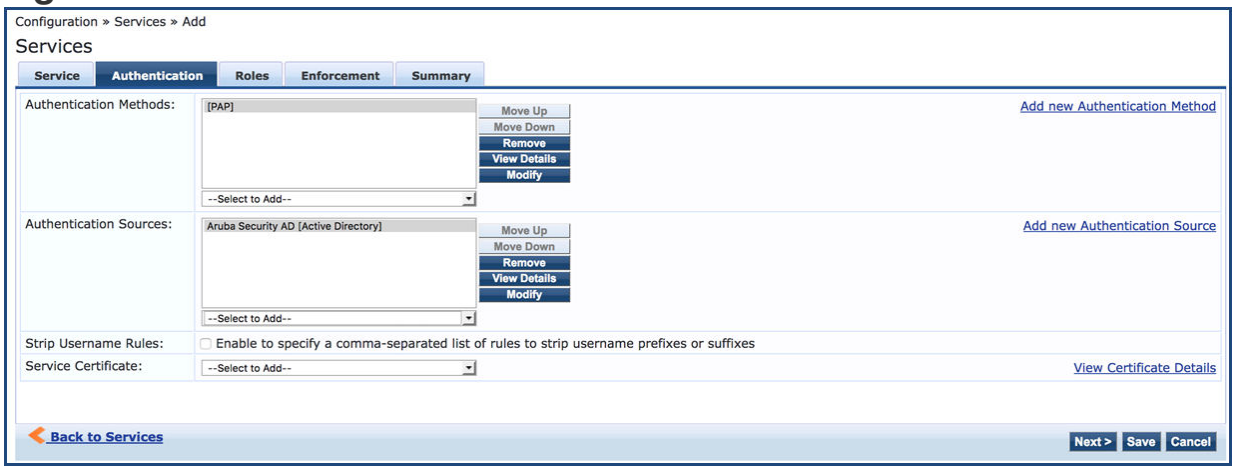
Parameter	Action/Description
Type	Select **RADIUS Enforcement (Generic)**.
Name	Enter a name for this service.
Description	Optionally (but recommended), add a description of this enforcement service.
Monitor Mode	This option is disabled by default. Do not enable Monitor Mode.
More Options	No additional options are required for this service.
Service Rule	
Matches	Select **ALL of the following conditions**.
Rule condition 1: Specifying the RADIUS protocol	Select **Click to add**, then specify the following attributes: ◦ **Type**: **Connection** ◦ Name: **Protocol** ◦ Operator: **EQUALS**

Parameter	Action/Description
	◦ Value: **RADIUS**
Rule condition 2: Specifying the Servic e - Type	Select **Click to add**, then specify the fol lowing attributes: ◦ **Type**: **Radius:IETF** ◦ Name: **Service - Type** ◦ Operator: **EQUALS** ◦ Value: **NAS - Prompt - User**
Rule condition 3: Specifying the NAS - Port - Type	Select **Click to add**, then specify the fol lowing attributes: ◦ **Type**: **Radius:IETF** ◦ Name: **NAS - Port - Type** ◦ Operator: **EQUALS** ◦ Value: **Virtual**

4. Click **Next**.

   The **Authentication** tab opens.

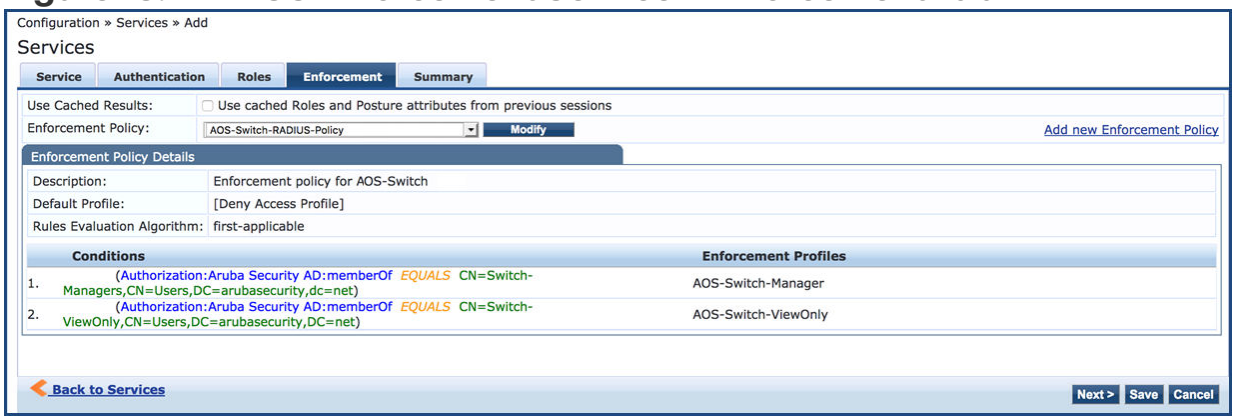
Figure 12. RADIUS Enforcement Service > Authentication Tab



5. From the **Authentication Sources** drop-down, select **Aruba Security AD** (or whatever name was assigned to this authentication source).

6. Select the **Enforcement** tab.

Figure 13. RADIUS Enforcement Service > Enforcement Tab



7. From the **Enforcement Policy** drop-down, select **AOS-Switch-RADIUS-Policy**.
8. Click **Save**.

   This completes the service for ArubaOS switch management.

OnGuard Authentication Configuration

This section contains the following information:

- Overview
- OnGuard Configuration Workflow
- ClearPass Configuration

Overview

This section describes a basic OnGuard deployment using the persistent agent to provide posture checks on centrally managed Windows devices that are connected via Ethernet to an ArubaOS switch.

> **NOTE**
>
> For detailed information about the OnGuard health checks for Windows, macOS, and Linux, refer to "Configuring Posture Policy Agents and Hosts" in Chapter 6, "Posture Policies and Audit Servers" in the Policy Manager 6.11 *Policy Manager User Guide*.

Policy Manager provides several methods for assessing the health posture of clients requesting access: OnGuard Agents, NAP Agents, and NESSUS or NMap scans. All of these methods return posture tokens (for example, *Healthy*, *Quarantine*) that Policy Manager uses to provide input into enforcement policies. One or more of these posture methods can be associated with a service. Policy Manager OnGuard performs advanced

endpoint posture assessments by running checks on the endpoints that are attempting to gain access to the network.

You can use information provided by the OnGuard agent about endpoint integrity (such as status of antivirus, firewall, and peer-to-peer applications) to enhance authorization policies. Automatic remediation services are also available for noncompliant devices.

OnGuard Agent

The OnGuard Agent enables more extensive health checks then those available in the NAP Agent. Both persistent and dissolvable agents are available for Windows, macOS, and Linux operating systems.

- The *persistent agent* is installed on the end system and runs in the background. It requires network connectivity. The persistent agent regularly reports health information to a Policy Manager Webauth posture check service.
- The *dissolvable agent* does not permanently install anything on the end system. The user is redirected to the Policy Manager Web Login page created by the administrator. The OnGuard agent is run on-demand in the web browser.

Posture Policy Checks

Both the persistent and dissolvable agents cache the health results in the Endpoint Database. The latest health posture token can be used by Policy Manager services. The persistent and dissolvable agents perform the same health checks, but auto-remediation is only available with the persistent agent.

Depending on the operating system where OnGuard is installed, there are multiple health checks that can be performed by the OnGuard agent. Some health checks are not applicable on all operating systems. For example, when the macOS operating system is selected, the available health checks do not include "Windows Hotfixes" or "Registry Keys" as possible checks for OnGuard to perform.

OnGuard Configuration Workflow

About this task

The following Policy Manager components must be configured to configure OnGuard:

Procedure

1. Define the posture policy (see Defining the Posture Policy below).
2. Policy Manager Configuration
    - Agent settings
    - Policy Manager Zone mapping
    - Policy Manager service
    - Policy Manager Web Authorization page

ArubaOS switch configuration

ClearPass Configuration

This section describes the Policy Manager configuration required for configuring OnGuard authentication.

Defining the Posture Policy

Before you configure the posture policy, determine the following elements:

- Which end systems the OnGuard agents will be installed on
- What health checks need to be performed on the end system(s)
- What results will be required to return a Healthy Token

  These decisions will be the basis for the Policy Manager posture policy that you will configure later in this chapter (see Configuring Posture Policies).

Defining Policy Manager Zones

To define the Policy Manager Zones:

Policy Manager shares a distributed cache of run-time states across all nodes in a cluster. These run-time states include:

- Roles and postures of connected entities
- Connection status of all endpoints running OnGuard
- Endpoint details gathered by OnGuard Agent

  Policy Manager uses this run-time state information to make policy decisions across multiple transactions.

  In a deployment where a cluster spans WAN boundaries and multiple geographic zones, it is not necessary to share all of this run-time state across all nodes in the cluster.

  For example, when endpoints present in one geographical area are not likely to authenticate or be present in another area, it is more efficient from a network bandwidth usage and processing perspective to restrict the sharing of such run-time state information to a given geographical area.

  You can configure zones in Policy Manager to match the geographical areas in your deployment. There can be multiple zones per cluster, and each zone has a number of Policy Manager nodes that share their run-time state.

Adding Policy Manager Zones

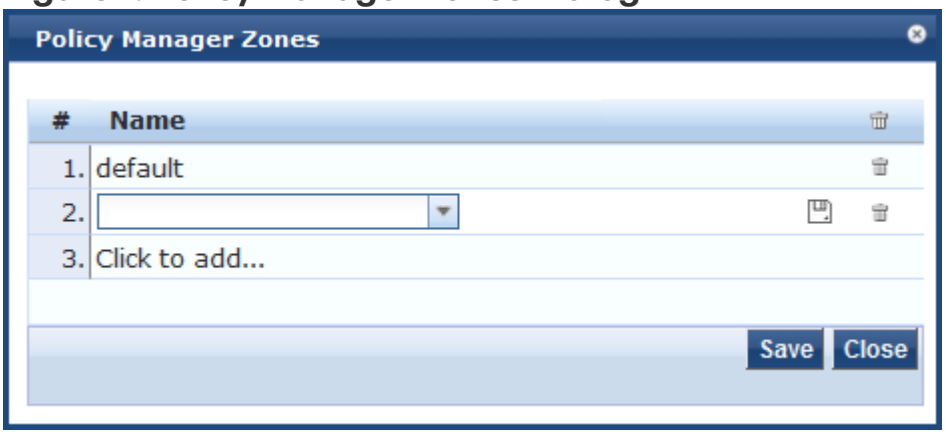
About this task

To add a Policy Manager zone:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Click the **Manage Policy Manager Zones** link. The **Policy Manager Zones** dialog opens:

Figure 1. Policy Manager Zones Dialog



3. To add a new Policy Manager Zone, click **Click to add...** and enter the name of the Policy Manager Zone to be added.
4. Be sure to click the **Save** icon, then click **Save .**
5. To delete a zone, click the trash can icon (🗑).

Mapping Policy Manager Zones to OnGuard Clients

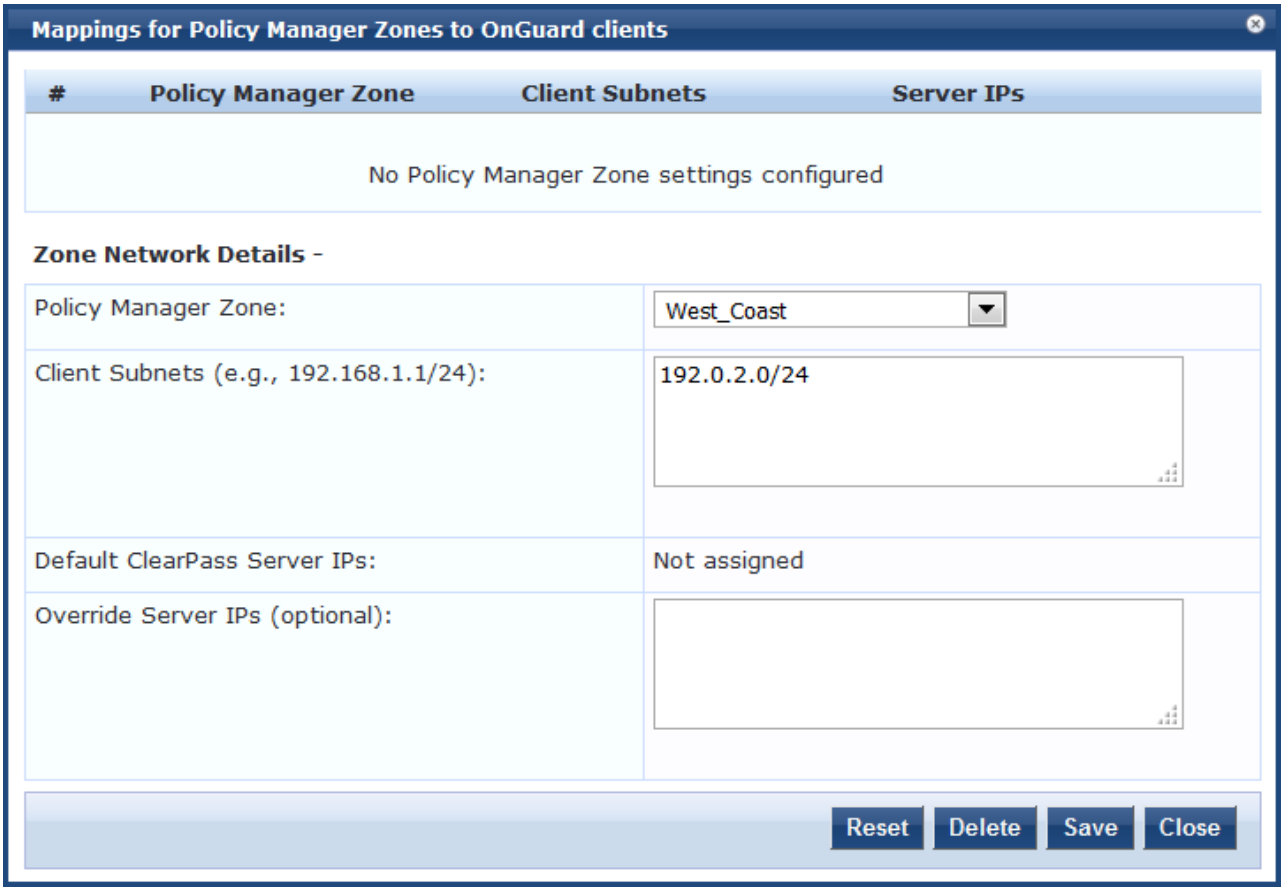
About this task

To configure the Policy Manager Zone you created:

Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**. The **OnGuard Settings** page opens.
2. Click the **Policy Manager Zones** link. The **Mappings for Policy Manager Zones to OnGuard Clients** page opens.

Figure 2. Mappings forPolicy ManagerZones to OnGuard Clients Page



3. Specify the **Mappings for Policy Manager Zones to OnGuard Clients** parameters as described in the following table:

Table 1. OnGuard Settings > Zones Parameters

Parameter	Action/Description
ClearPass Policy Manager Zone	Select the Policy Manager Zone from the drop - down list. If no Policy Manager zone is configured, the default Policy Manager zone is displ ayed in this field.
Client Subnets	Displays the client subnet addresses spe cified for the selected Policy Manager zo ne.
Server IPs	Displays the server IP addresses specific to the selected Policy Manager zone.
Zone Network Details	

Parameter	Action/Description
Policy Manager Zone	Select the Policy Manager zone from th e drop - down list that is created from th e **Administration > Server Manager > Se rver Configuration > Manage Policy Man ager Zones** page.
Client Subnets	Specify the client subnets that are config ured for the selected zone.
Default Policy Manager Server IPs	Specify the IP address of the default Poli cy Manager server.
Override Server IPs	Optionally, specify the IP addresses or th e Fully Qualified Domain Name (FQDN) to which you want the OnGuard agent to se nd the request in the sequence. You can specify the data port or load bal ancer IP address in this field. The IP addresses configured here will ove rride the IP address configured in the *De fault ClearPass Server IPs* field. For example, if you have configured the I P addresses 10.17.XXX.1, 10.17.XXX.2, and 1 0.17.XXX.3, OnGuard agent will send the request in the same sequence.

Configuring OnGuard Settings

About this task

Use the **OnGuard Settings** page to configure the agent deployment packages. When you save the OnGuard configuration, Policy Manager creates agent deployment packages for the Windows, macOS, and Ubuntu operating systems and provides the packages at a fixed URL on the Policy Manager hardware or virtual appliance.

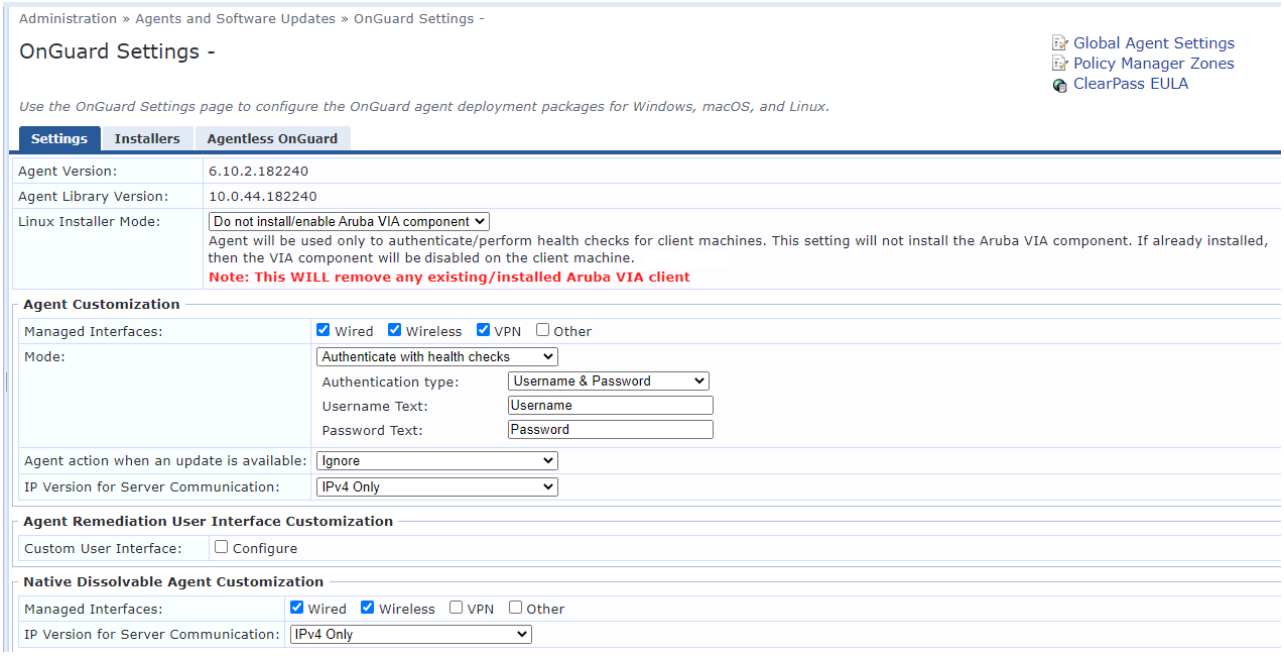
You can then publish this URL to the user community or download the agent deployment packages to another location.

To configure the OnGuard settings:

Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**. The **OnGuard Settings** page opens to the **Settings** tab:

Figure 3. OnGuard Settings Page > Settings Tab



2. Configure the **OnGuard Settings** parameters as described in the following table, then click **Save**.

Table 2. OnGuard Settings Parameters

Parameter	Action/Description
Global Agent Settings	Click the **Global Agent Settings** link to c onfigure the global agent settings for On Guard agents.
Policy Manager Zones	Click the **Policy Manager Zones** link to c onfigure the client subnets that comprise the Policy Manager Zone.
Agent Version	Indicates the current version of the OnG uard agent.
Agent Library Version	Indicates the version of the OnGuard Age nt Library. OnGuard Agent uses this libra

Parameter	Action/Description
	ry for health collection and auto - remedi ation.
Installer Mode	Specify the action to be taken from the following options when the Policy Manag er VIA component is used to provide VPN - based access: • **Do not install/enable Aruba VIA com ponent** **NOTE** Selecting this option will aut omatically remove any existi ng and installed Policy Mana ger VIA client software. • **Install and enable Aruba VIA compon ent** **NOTE** Selecting this option will aut omatically upgrade any exist ing and installed Policy Man ager VIA client software.
Agent Customization	
Managed Interfaces	Select the type(s) of interfaces that OnG uard will manage on the endpoint. Select from the following options: • **Wired** • **Wireless** • **VPN** • **Other**
Mode	Select one of the following options:

Parameter	Action/Description
	• **Authenticate with health checks**: On Guard collects the username and pas sword and also performs health chec ks on the endpoint. This is the default setting. • **Authenticate - no health checks**: On Guard collects username and passwo rd but does not perform health chec ks on the endpoint. • **Check health - no authentication**: O nGuard does not collect the usernam e and password. • **Username/Password Text**: The label for the *Username* and *Password* fields on the OnGuard agent user interface. **NOTE** This setting is not valid for th e **Check health - no authen tication** mode.
Agent action when an update is available	Determines what the Policy Manager On Guard Agent does when an update of On Guard Agent Library is available. Select o ne of the following options: • **Ignore**: Policy Manager OnGuard Age nt ignores the available OnGuard Age nt Library update. • **Notify User**: Policy Manager OnGuard Agent notifies the user that a new ve rsion of the OnGuard Agent Library is available on the Policy Manager serve r. • **Download and Install**: When this opti on is selected and a new version of O nGuard Agent Library is available on t

Parameter	Action/Description
	he Policy Manager server, Policy Man ager OnGuard Agent automatically d ownloads and installs the new version of OnGuard Agent Library from the Po licy Manager server.
Agent Remediation User Interface Customization	
Custom User Interface	When you select the **Configure** check b ox, the **Agent Remediation User Interfac e Customization** dialog opens. • **Web Pages**: To create the OnGuard cu stom web pages and define the prope rties for the web pages, click the **Crea te** link for the corresponding web pag e.
Native Dissolvable Agent Customization	
Managed Interfaces	The Native Dissolvable Agent performs h ealth checks for one of the selected inte rfaces. This feature ensures that, if both wired a nd wireless interfaces are connected, the OnGuard Agent will send health requests through the correct interface. Select the type(s) of managed interfaces that are supported for the Native Dissolva ble Agent. Select from the following options: • **Wired** • **Wireless** • **VPN** • **Other**

Configuring Posture Policies

About this task

Posture policies can be associated with Policy Manager services to verify the security posture of end systems prior to granting network access. The policy defines the end-system operating system and the type of agent to deploy. The posture policy also tells the OnGuard agent which health checks to perform and defines the rules that determine what is required to return a Healthy Token to the Policy Manager service.

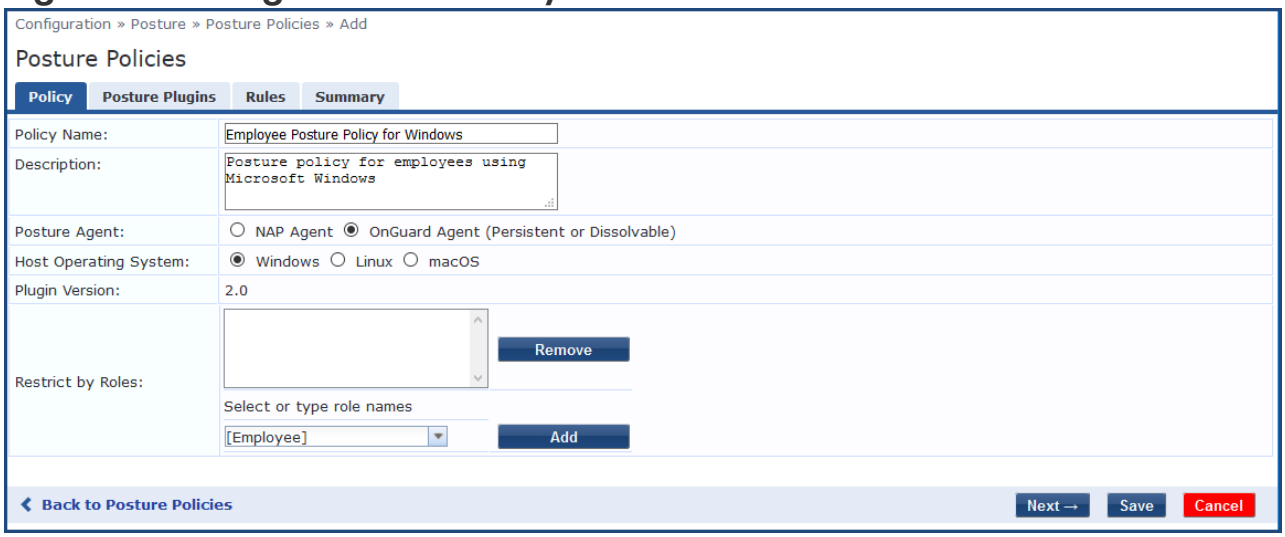
For Windows end systems, the Microsoft NAP (Network Access Protection) agent and the OnGuard agent are both available. For Linux and macOS, only the OnGuard agent is available.

To create a new posture policy:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies.** The **Posture Policies** page opens.
2. Click the **Add** link. The **Add Posture Policies** configuration dialog opens.

Figure 4. Adding a Posture Policy for Windows



3. For detailed information about configuring posture policies, agents, and hosts, refer to Chapter 6, "Posture Policies and Audit Servers" in the Policy Manager 6.11 Policy Manager User Guide.

Restricting Policies

About this task

The **Restrict by Roles** section of the **Policy** tab allows the administrator to apply the posture policy only to end systems that authenticate with selected roles. Typically users with access to sensitive information would authenticate with roles associated with more restrictive posture policies.

For example, users with access to research data might have a posture policy that does not permit the mounting of USB storage devices; or users with access to employee information, customer personal information, or health data could configure a posture policy that requires full disk encryption.

To restrict the posture policy by roles:

Procedure

1. From the **Policy** tab, click the **Select or type role names** drop-down. The list of available roles opens.
2. Select the roles you wish to restrict access by, then click **Add**. The selected roles are added to the **Restrict by Roles** field.

Monitoring and Troubleshooting

This section contains the following information:

- Monitoring Active 802.1X Sessions
- Monitoring RADIUS Messages

Monitoring Active 802.1X Sessions

You can examine any active 802.1X sessions, and capture the users' Domain/User ID and the MAC address of the endpoint.

To monitor active 802.1X sessions:

```
AOS-Switch# show port-access authenticator clients
```

This command displays the following information:

- Port
- Client name
- MAC address
- IP address
- Client status

Monitoring RADIUS Messages

To monitor the RADIUS messages between the switch and the Policy Manager server, use the following command:

```
AOS-Switch# show radius authentication
```

Figure 1. Monitoring Overview of Activity Between the NAS and thePolicy ManagerServer

```
Status and Counters - RADIUS Authentication Information

 NAS Identifier           : CPPM-2920
 Invalid Server Addresses : 0
                  UDP
 Server IP Addr  Port Timeouts   Requests   Challenges Accepts    Rejects
 --------------- ---- ---------- ---------- ---------- ---------- ----------
 10.             1812 4          2000       1457       542        1
 10.             1812 0          0          0          0          0
```

Using the show radius host Command

For a more detailed view of the interaction between the NAS and Policy Manager, the **show radius host** command provides insight into the activity at the RADIUS level.

```
AOS-Switch# show radius host 10.x.x.x
```

Figure 2. Monitoring Details of Activity Between the Switch and thePolicy ManagerServer

```
Status and Counters - RADIUS Server Information
 Server IP Addr : 10.

 Authentication UDP Port : 1812          Accounting UDP Port  : 1813
 Round Trip Time         : 4             Round Trip Time      : 0
 Pending Requests        : 0             Pending Requests     : 0
 Retransmissions         : 4             Retransmissions      : 1
 Timeouts                : 4             Timeouts             : 1
 Malformed Responses     : 0             Malformed Responses  : 0
 Bad Authenticators      : 0             Bad Authenticators   : 0
 Unknown Types           : 0             Unknown Types        : 0
 Packets Dropped         : 0             Packets Dropped      : 0
 Access Requests         : 2024          Accounting Requests  : 3058
 Access Challenges       : 1475          Accounting Responses : 3058
 Access Accepts          : 548
 Access Rejects          : 1
```

As shown in Figure 2, the break-out of port message information by **Authentication UDP Port** and **Accounting UDP Port** is of particular interest.

Integrating with a Cisco Switch

This section of the Deployment Guide provides the set-up instructions for integrating a Cisco switch with Policy Manager. This includes 802.1x, MAC address, and downloadable Access Control List (dACL) authentications.

Cisco switches support multiple authentication methods and many RADIUS options that are passed to the switch. This chapter discusses only the subset of Cisco switch configuration features that are required for integration with Policy Manager. Basic familiarity with most Cisco switches is assumed. For in-depth information about the features and functions of Policy Manager, refer to the *ClearPass Policy Manager User Guide*.

Requirements

- Cisco LAN switch that supports 802.1X and MAC Authentication Bypass
- DHCP server for the registration VLAN and the mandatory VLANs (see VLAN Numbers)
- Current Policy Manager release
- Verify that a basic configuration of Policy Manager has been completed, which consists at minimum of initial set up and configuring a generic RADIUS service.

New Central Integration

Prerequisites

ClearPass Policy Manager supports integration with HPE Engineering New Central. New Central support is enabled in Policy Manager using the command line interface (CLI). Currently, Policy Manager is integrated with Classic Central with policy mapping using Device Insight tags. However, Policy Manager must be updated to use different tags to avoid authentication failures when switching from Classic Central to New Central.

To integrate Policy Manager with the correct Device Insight tags:

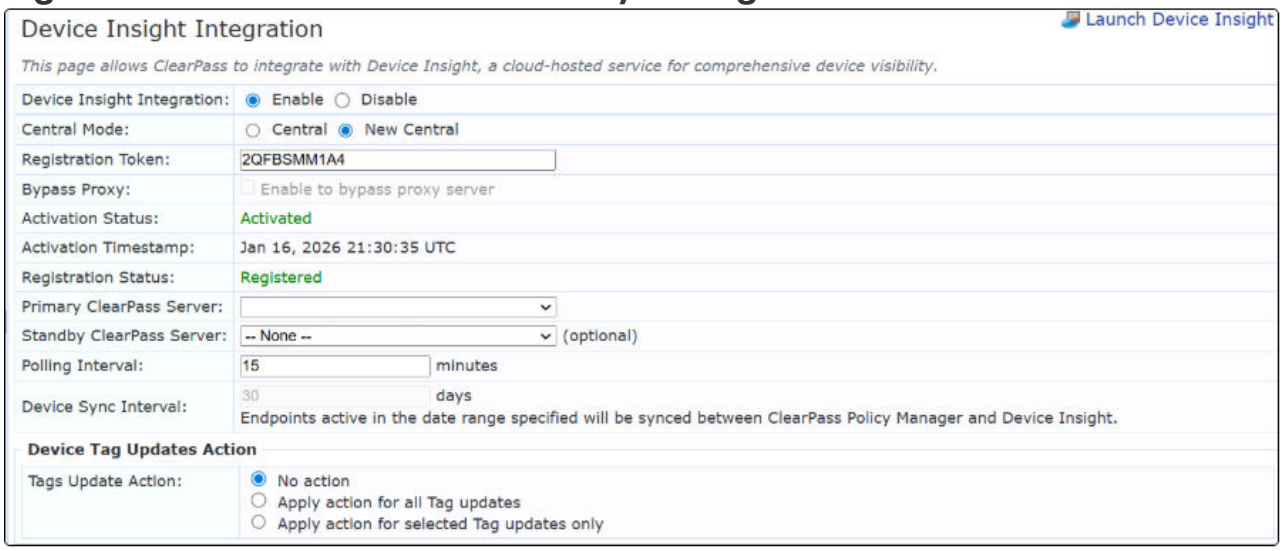
Procedure

1. Navigate to the Policy Manager CLI's Configure Commands.
2. Enable the central-communication command by setting the mode to 1. When the central-communication command is enabled, the Policy Manager UI displays an option to allow Policy Manager to select either the Central or New Central interface. For more information on using the central-communication command, see.

```
[appadmin]# configure central-communication 1
 Central Communication Mode is Enabled.
```

3. Verify New Central is enabled by navigating to **Administration > Server Manager > Device Insight** and ensure **New Central** is selected as the **Central Mode**.

Figure 1. Enable New Central in Policy Manager



4. Ensure Classic Central Device Insight tags are updated to use the correct New Central tag mappings.

> **NOTE**
>
> For the purposes of the following example, **Computer** will be used as the client's Classic Central Category with **Computers & Servers** as the tag category.

Figure 2. Endpoint details in Classic Central

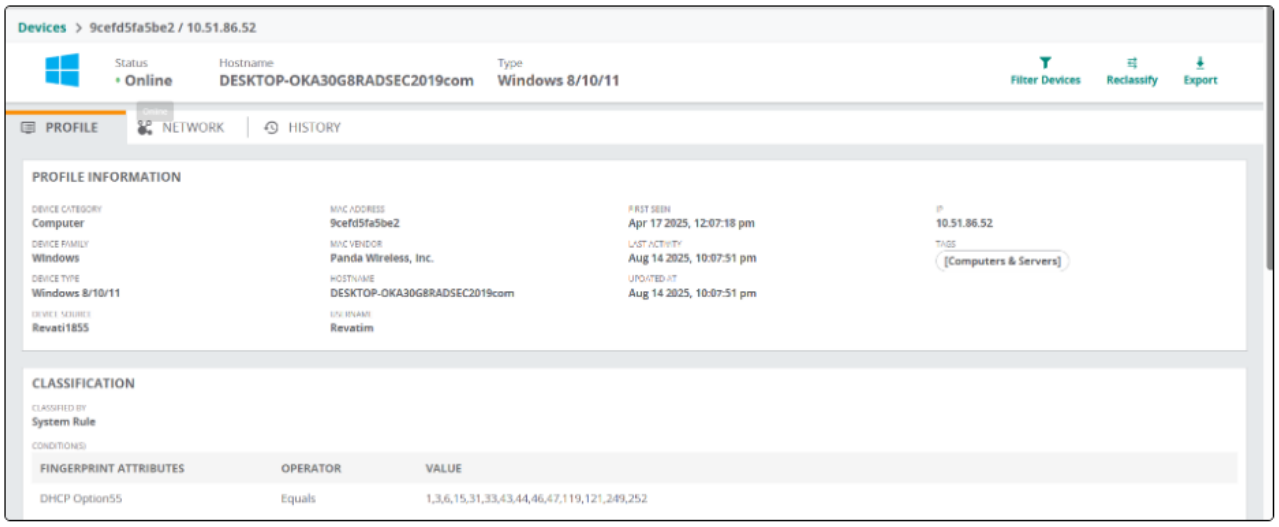
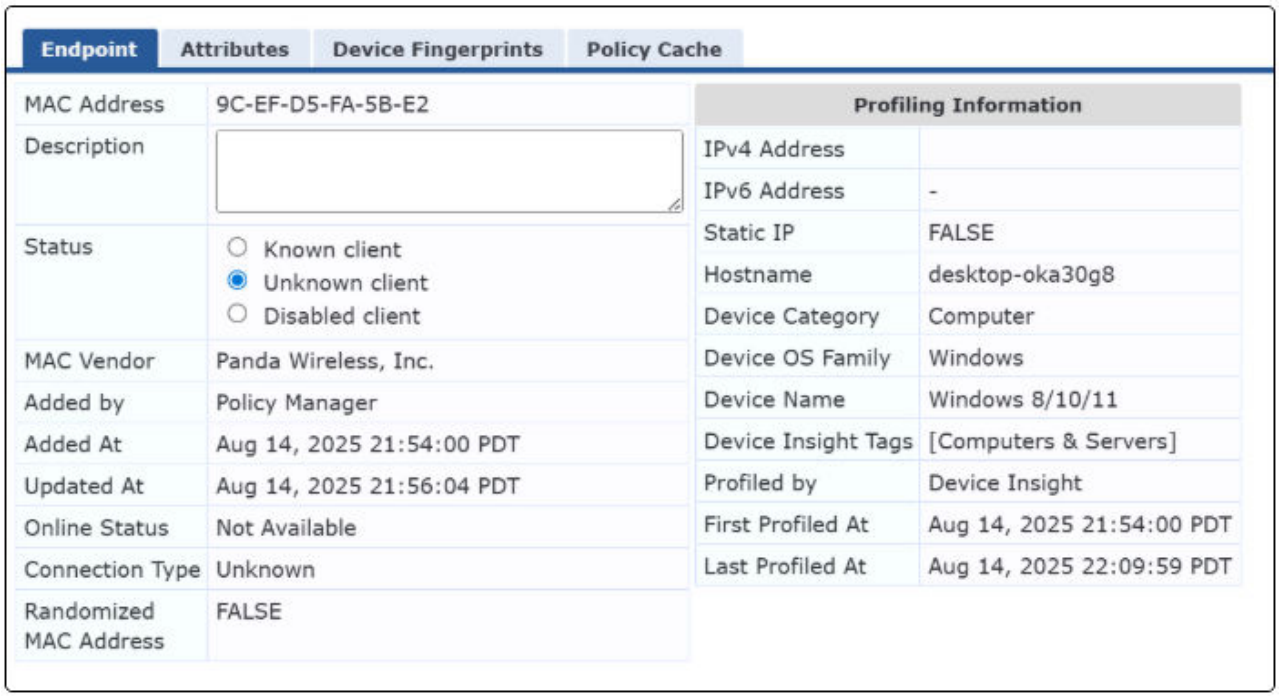


Figure 3. Endpoint details in ClearPass Policy Manager



5. Refer to the following screenshot to review how Classic Central uses **Computer Systems** as the Category.

Figure 4. Endpoint details

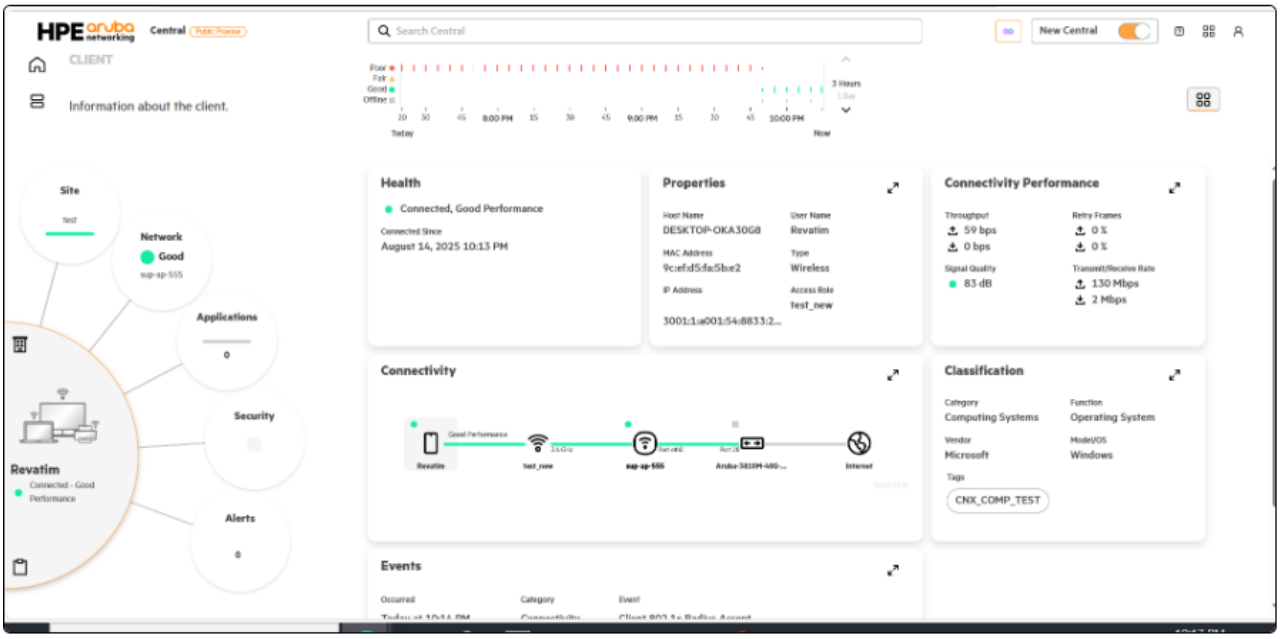


Figure 5. Endpoint details in ClearPass Policy Manager

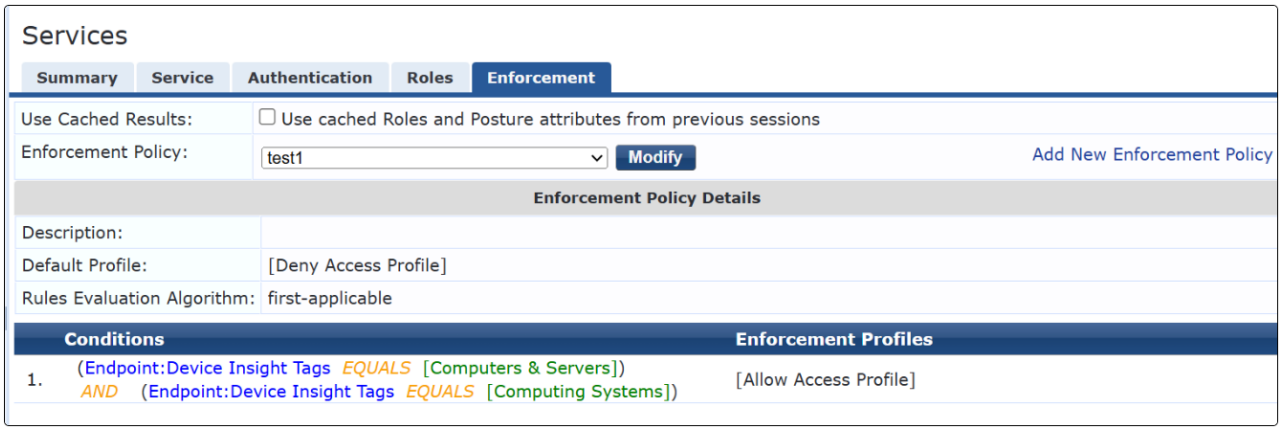
Endpoint | Attributes | Device Fingerprints | Policy Cache

MAC Address	9C-EF-D5-FA-5B-E2
Description	
Status	○ Known client ◉ Unknown client ○ Disabled client
MAC Vendor	Panda Wireless, Inc.
Added by	Policy Manager
Added At	Aug 14, 2025 22:00:20 PDT
Updated At	Aug 14, 2025 22:03:09 PDT
Online Status	Not Available
Connection Type	Unknown
Randomized MAC Address	FALSE

Profiling Information	
IPv4 Address	
IPv6 Address	-
Static IP	FALSE
Hostname	desktop-oka30g8
Device Category	Computer
Device OS Family	Windows
Device Name	Windows 10/11
Device Insight Tags	CNX_COMP_TEST [Computing Systems]
Profiled by	Policy Manager
First Profiled At	Aug 14, 2025 22:00:19 PDT
Last Profiled At	Aug 14, 2025 22:00:19 PDT

6. When configuring an enforcement policy in Policy Manager, ensure the Device Insight tags are correctly mapped for both Classic Central and New Central. In this example, the Classic Central mapping was based on the **Computers and Servers** category and the New Central mapping was based on the **Computing Systems** category.

Figure 6. Policy Manager Enforcement Policy Configuration



7. To review the differences that must be accounted for when transitioning ClearPass Policy Manager support from Classic Central to New Central, see Classic Central and New Central Tag Mappings.

Subtopics

Classic Central and New Central Tag Mappings

Classic Central and New Central Tag Mappings

Use to the following table to review the differences that must be accounted for when transitioning ClearPass Policy Manager support from Classic Central to New Central:

Table 1. Classic Central and New Central Tag Mappings

System Tag (Central)	Device Category (Central)	New Central Category	Comments
Facilities & Building Automation	Network Camera	Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure	Maps to Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure
Facilities & Building Automation	UPS	Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure	Maps to Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure
Facilities & Building Automation	Embedded	Facilities & Building Automation, Computin	Maps to Facilities & Building Automation, Compu

System Tag (Central)	Device Category (Central)	New Central Category	Comments
		g Systems, Network Firewall and Networking Infrastructure	ting Systems, Network Firewall and Networking Infrastructure
Facilities & Building Automation	Security	Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure	Maps to Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure
Facilities & Building Automation	Power	Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure	Maps to Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure
Facilities & Building Automation	Building Automation	Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure	Maps to Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure
Facilities & Building Automation	Automobile	Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure	Maps to Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure
Facilities & Building Automation	Biometric Devices	Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure	Maps to Facilities & Building Automation, Computing Systems, Network Firewall and Networking Infrastructure
Point Of Sale Devices	Point of Sale Devices	Point Of Sale Devices, Smart Buildings	The device categories in Point of Sale devices map to Point of Sale devices and Smart Buildings
Point Of Sale Devices	Cash Drawer	Point Of Sale Devices, Smart Buildings	The device categories in Point of Sale devices map to Point of Sale devices and Smart Buildings
Point Of Sale Devices	Barcode Scanner	Point Of Sale Devices, Smart Buildings	The device categories in Point of Sale devices ma

System Tag (Central)	Device Category (Central)	New Central Category	Comments
			p to Point of Sale devices and Smart Buildings
Medical Devices	Medical Devices	Medical Devices	Medical Devices map to Medical Systems
Computers & Servers	Computer	Computing Systems	Computers & Servers map to Computing Systems
Computers & Servers	Server	Computing Systems	Computers & Servers map to Computing Systems
Computers & Servers	Trading Systems	Computing Systems	Computers & Servers map to Computing Systems
Computers & Servers	Clock	Computing Systems	Computers & Servers map to Computing Systems
Computers & Servers	Thin Client	Computing Systems	Computers & Servers map to Computing Systems
Computers & Servers	KVM	Computing Systems	Computers & Servers map to Computing Systems
Mobile and Gadgets	Smart Device	Mobile & Gadgets,Computing Systems	Mobile & Gadgets map to both Mobile & Gadgets as well as Computing Systems as Mobile Phones are part of Computing Systems
Networking And Switching	Router	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services
Networking And Switching	Access Points	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services
Networking And Switching	Network Infrastructure	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services

System Tag (Central)	Device Category (Central)	New Central Category	Comments
Networking And Switching	ATM	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services
Networking And Switching	Network Boot Agents	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services
Networking And Switching	Monitoring Agents	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services
Networking And Switching	Storage Device	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services
Networking And Switching	Switch	Network Infrastructure, Business Systems & Services	Networking & Switching is mapped to Network Infrastructure and Business Systems and Services
Audio And Video	Game Console	Audio & Video, Gaming & Casino	Audio & Video is mapped to Gaming & Casino and Audio & Video
Audio And Video	SetTop Box	Audio & Video, Gaming & Casino	Audio & Video is mapped to Gaming & Casino and Audio & Video
Audio And Video	Home Audio/Video Equipment	Audio & Video, Gaming & Casino	Audio & Video is mapped to Gaming & Casino and Audio & Video
Audio And Video	Audio/Video Devices	Audio & Video, Gaming & Casino	Audio & Video is mapped to Gaming & Casino and Audio & Video
Printers	Printer	Computing Systems	Printer is mapped to Computing Systems which is a broader category
Collaboration And Communication	Projectors	Collaboration & Communication	Collaboration & Communication is same

System Tag (Central)	Device Category (Central)	New Central Category	Comments
Collaboration And Communication	VoIP Phones	Collaboration & Communication	Collaboration & Communication is same
Collaboration And Communication	VOIP Adapters	Collaboration & Communication	Collaboration & Communication is same
Collaboration And Communication	Video Conferencing	Collaboration & Communication	Collaboration & Communication is same
Collaboration And Communication	Presentation	Collaboration & Communication	Collaboration & Communication is same
IOT	IoT	IOT tag is deprecated. Now only the ml-IOT tag is applied.	There is no IoT function in New Central, it is deprecated.
IOT	Vending	IOT tag is deprecated. Now only the ml-IOT tag is applied.	Vending is mapped to IoT behavior tag
IOT	Traffic And Parking Mgmt	IOT tag is deprecated. Now only the ml-IOT tag is applied.	Mapped to IoT behavior tag
IOT	Programmable Logic Controller	IOT tag is deprecated. Now only the ml-IOT tag is applied.	Mapped to IoT behavior tag

Mobility Access Switch Configuration for 802.1X Authentication

This section of the Deployment Guide describes how to configure an Mobility Access Switch for 802.1X authentication, and includes the following information:

- Mobility Access Switch Configuration for 802.1X Wired Authentication
- Configuring 802.1X Authentication with Machine Authentication
- CLI-Based Configuration for Mobility Access Switch 802.1X Authentication

Subtopics

Mobility Access Switch Configuration for 802.1X Wired Authentication

CLI-Based Configuration for Mobility Access Switch 802.1X Authentication
Configuring 802.1X Authentication with Machine Authentication

Mobility Access Switch Configuration for 802.1X Wired Authentication

Subtopics

About Defining Wired 802.1X Authentication
Configuring Authentication with a RADIUS Server
Authentication Terminated on the Mobility Access Switch
Configuring Access Control Lists

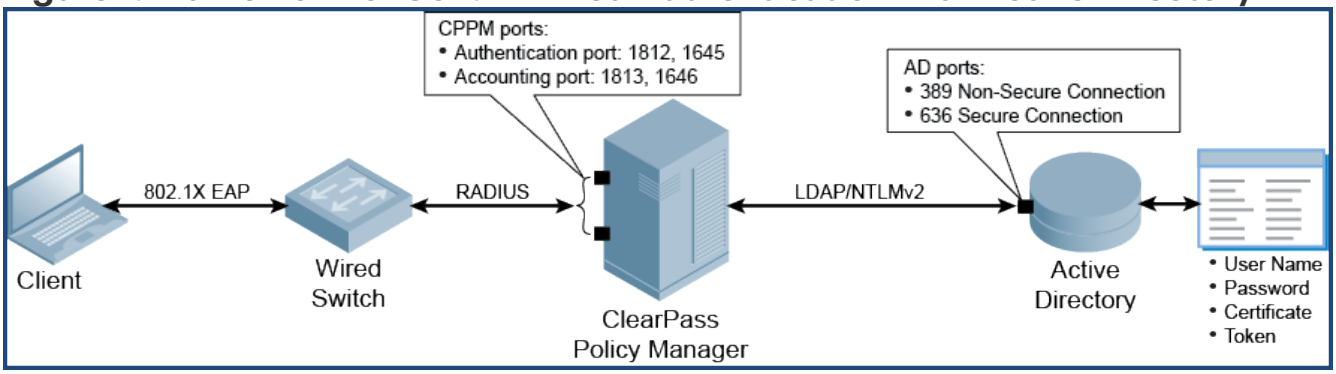
About Defining Wired 802.1X Authentication

About this task

Port-based 802.1X authentication on the Mobility Access Switch is configured similarly to how it's done on the mobility controller, the main difference being the AAA profile is applied on a wired interface or interface-group, as opposed to a Virtual Access Point (VAP) on the mobility controller.

Figure 1 shows the network traffic flow for wired clients that connect to an Aruba Mobility Access Switch or a third-party switch and perform 802.1X authentication to the Policy Manager server.

Figure 1. Traffic flow for 802.1X Wired Authentication with Active Directory



The configuration process is as follows:

Procedure

1. Define an external RADIUS server or create an internal database.

2. Define a server group and apply one of the servers above to this server group.
3. Create 802.1X authentication profiles.
4. Apply the server group to each of the 802.1X authentication profiles.
5. Apply the 802.1X authentication profiles to an AAA profile.
6. Apply the AAA profile to the physical interface or interface group.

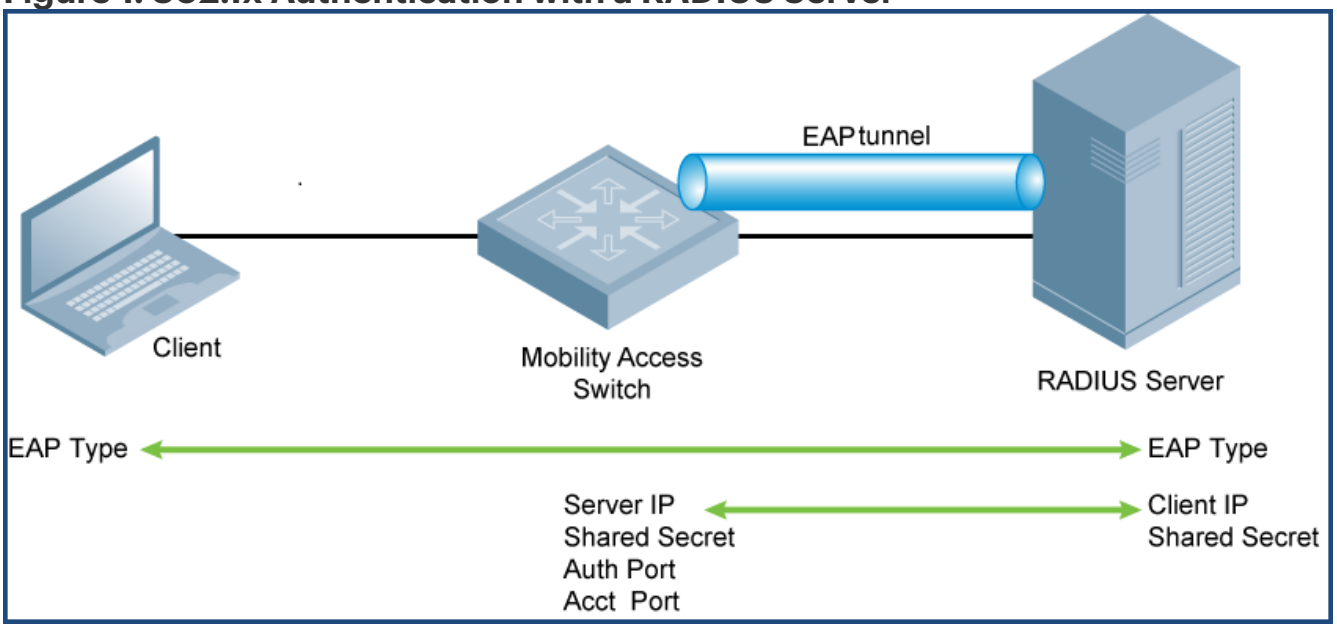
   You can now configure an interface for 802.1X authentication.

Configuring Authentication with a RADIUS Server

About this task

In order to authenticate to the network, the client communicates with the Mobility Access Switch through an EAP tunnel (see Figure 2). Therefore, the network authentication and encryption configured must be the same on both the client and the Mobility Access Switch.

Figure 1. 802.1x Authentication with a RADIUS Server



To configure 802.1X authentication with a RADIUS server:

Procedure

1. For the Mobility Access Switch to communicate with the authentication server, you must configure the following parameters on the Mobility Access Switch:

Parameter	Action/Description
IP address	Enter the IP address of the authentication server.
Authentication port	Enter the Authentication port number on the authentication server. Default: **1812**.
Accounting port	Enter the Accounting port number on the authentication server. Default: **1813**.

2. You must configure the supplicant (the client device) and authentication server (the Mobility Access Switch) to use the same EAP type.

   The Mobility Access Switch doesn't need to know the EAP type used between the supplicant and authentication server.

3. You must configure the authentication server with the IP address of the RADIUS client, which in this case is the Mobility Access Switch.

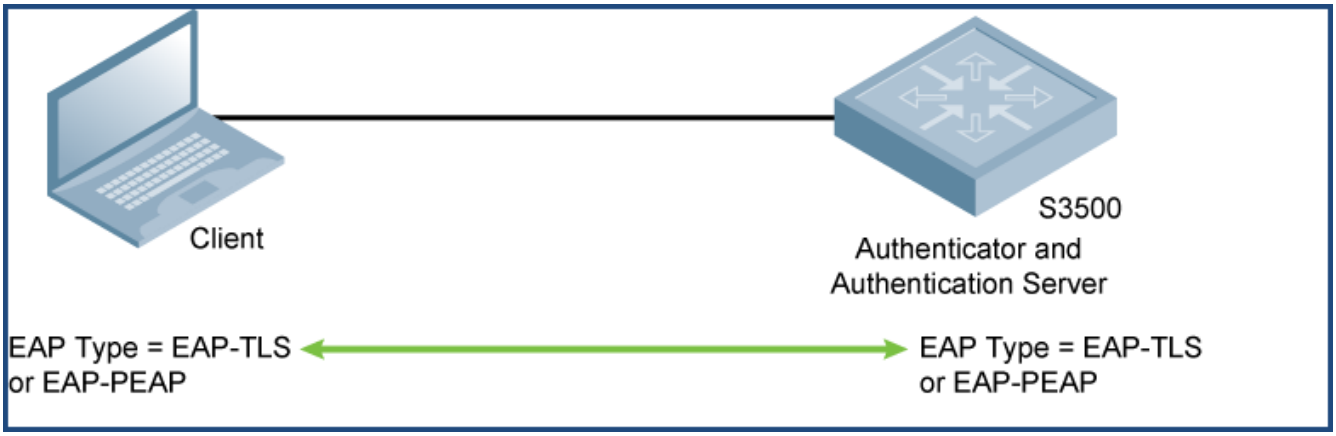
4. Be sure to configure both the Mobility Access Switch and the authentication server to use the same shared secret.

> **NOTE**
>
> Additional information on EAP types supported in a Windows environment for Microsoft supplicants and the authentication server is available at **http://technet.microsoft.com/en-us/library/cc782851(WS.10).aspx**.

Authentication Terminated on the Mobility Access Switch

User authentication is performed either via the Mobility Access Switch's internal database or a non-802.1x server.

Figure 1. 802.1x Authentication with Termination on the Mobility Access Switch



In this scenario, the supplicant is configured for EAP-Protected EAP (PEAP) or EAP-Transport Layer Security (TLS).

Subtopics

EAP-PEAP
EAP-TLS
Internal Database Configuration Task
LDAP Server Configuration Task
RADIUS Server Configuration Task

EAP-PEAP

EAP-PEAP uses TLS to create an encrypted tunnel. Within the tunnel, one of the following "inner EAP" methods is used:

- EAP-Generic Token Card (GTC)

  Described in RFC 2284, this EAP method permits the transfer of unencrypted usernames and passwords from client to server. The main uses for EAP-GTC are one-time token cards such as SecureID and the use of an LDAP or RADIUS server as the user authentication server.

  You can also enable caching of user credentials on the Mobility Access Switch as a backup to an external authentication server.

- EAP-Microsoft Challenge Handshake Authentication Protocol version 2 (MS-CHAPv2)

  Described in RFC 2759, this EAP method is widely supported by Microsoft clients. A RADIUS server must be used as the backend authentication server.

EAP-TLS

EAP-TLS is used with smart-card user authentication. A smart card holds a digital certificate which, with the user-entered personal identification number (PIN), allows the user to be authenticated on the network. EAP-TLS relies on digital certificates to verify the identities of both the client and server.

EAP-TLS requires that you import server and certification authority (CA) certificates onto the Mobility Access Switch. The client certificate is verified on the Mobility Access Switch (the client certificate must be signed by a known CA) before the user name is checked on the authentication server.

Internal Database Configuration Task

If you are using the Mobility Access Switch's internal database for user authentication, you need to add the names and passwords of the users to be authenticated.

LDAP Server Configuration Task

If you are using an LDAP server for user authentication, you need to configure the LDAP server on the Mobility Access Switch, and configure user IDs and passwords.

RADIUS Server Configuration Task

If you are using a RADIUS server for user authentication, you need to configure the RADIUS server on the Mobility Access Switch:

- For more information, see Configuring Authentication with a RADIUS Server .

Configuring Access Control Lists

To provide flexibility for controlling traffic, ArubaOS in Mobility Access Switches supports multiple types of Access Control Lists (ACLs).

- Ethertype ACL

  Ethertype ACLs filter based on the *Ethertype* field in the frame header. Ethertype ACLs can be either named or numbered, with valid numbers in the range from 200

to 299. These ACLs can be used to permit IP, while blocking other non-IP protocols, such as IPX or AppleTalk.

- MAC ACL

  MAC ACLs filter traffic on a specific source MAC address or range of MAC addresses. MAC ACLs can be either named or numbered, with valid numbers in the range from 700 to 799 and 1,200 to 1,299.

- Standard IP ACL

  Standard ACLs permit or deny traffic based on the source IP address of the packet. Standard ACLS can be either named or numbered, with valid numbers in the range from 1 to 99 and 1,300 to 1,399. Standard ACLs use a bit-wise mask to specify the portion of the source IP address to be matched.

- Extended IP ACL

  Extended ACLs permit or deny traffic based on the source or destination IP address, or the IP protocol. Extended ACLs can be named or numbered, with valid numbers in the range from 100 to 199 and 2,000 to 2,699.

- Stateless ACL

  Stateless ACLs define stateless packet filtering and quality of service (QoS). A stateless ACL statically evaluates packet contents. The traffic in the reverse direction is allowed unconditionally.

  Note that you can use names only when configuring stateless ACLs.

Subtopics

Configuring a Stateless ACL
Verifying Stateless ACL Configuration

Configuring a Stateless ACL

To configure a stateless ACL:

```
(ArubaSwitch)(config) #'''ip access-list stateless STATELESS'''
```

```
(ArubaSwitch)(config-stateless-STATELESS)#'''any host 192.16.0.100 t
cp 0 65535 permit'''
```

Subtopics

Applying a Stateless ACL on a Physical Interface

Applying a Stateless ACL to a User Role

Applying a Stateless ACL on a Physical Interface

To apply a stateless ACL on a physical interface:

```
(ArubaSwitch) (config) #'''interface gigabitethernet 0/0/8'''
```

```
(ArubaSwitch) (gigabitethernet "0/0/8") #'''ip access-group in STATE
LESS'''
```

Applying a Stateless ACL to a User Role

To apply a stateless ACL to a user role:

```
(ArubaSwitch)(config) #'''user-role EMPLOYEE_1'''
```

```
(ArubaSwitch)(config-role) #'''access-list stateless STATELESS'''
```

NOTE

You can also apply MAC and Ethertype ACLs to a user role. However, these ACLs apply only to a user's non-IP traffic.

Verifying Stateless ACL Configuration

Subtopics

To verify a stateless ACL configuration:
Verifying Stateless ACL Traffic Hits
Verifying Stateless ACL Operation

To verify a stateless ACL configuration:

```
(ArubaSwitch) #'''show ip access-list STATELESS'''
```

Verifying Stateless ACL Traffic Hits

To verify stateless traffic hits:

```
(ArubaSwitch) #'''show acl hits'''
```

Verifying Stateless ACL Operation

To verify stateless ACL operation:

```
(ArubaSwitch) # '''show acl acl-table'''
```

CLI-Based Configuration for Mobility Access Switch 802.1X Authentication

Subtopics

Termination Options
Configuring a Server Rule Using the CLI
Setting Variables for LDAP Servers
Configuring Certificates with Authentication Termination

Termination Options

The Mobility Access Switch supports 802.1x authentication, including *termination*. For example, the list of termination options for the profile name *FacultyAuth* is shown below.

```
(host) (802.1X Authentication Profile "FacultyAuth") # termination ?
eap-type                Configure the EAP method.Default method is EAP-PEAP
enable                  Enable Dot1x Termination. Default is disabled
enable-token-caching    Enable Token Caching.Default is disabled
inner-eap-type          Configure the inner EAP method.Default method is
                        EAP-MSCHAPV2
token-caching-period    Configure the Token Caching Period
```

Subtopics

802.1x Authentication Profile Configuration Examples
Verifying Configurations

802.1x Authentication Profile Configuration Examples

The following example configures various options for the 802.1x Authentication profile *FacultyAuth.*

```
(host) (802.1X Authentication Profile "FacultyAuth") #termination enable
(host) (802.1X Authentication Profile "FacultyAuth") #termination eap-type
eap-peap
(host) (802.1X Authentication Profile "FacultyAuth") #max-authentication-
failures 2
(host) (802.1X Authentication Profile "FacultyAuth") #timer reauth-period
3600
(host) (802.1X Authentication Profile "FacultyAuth") #framed-mtu 1500
(host) (802.1X Authentication Profile "FacultyAuth") #reauth-max 2
(host) (802.1X Authentication Profile "FacultyAuth") #reauthentication
```

Verifying Configurations

To verify the above configurations, execute the following **show** command:

```
(host) (config) #show aaa authentication dot1x FacultyAuth
802.1X Authentication Profile "FacultyAuth"
-------------------------------------------
Parameter                                                    Value
---------                                                    -----
Max authentication failures                                  2
<--
Enforce Machine Authentication                               Disabled
Machine Authentication: Default Machine Role                 guest
Machine Authentication Cache Timeout                         24 hr(s)
Blacklist on Machine Authentication Failure                  Disabled
Machine Authentication: Default User Role                    guest
Interval between Identity Requests                           30 sec
Quiet Period after Failed Authentication                     30 sec
Reauthentication Interval                                    3600 sec
<--
Use Server provided Reauthentication Interval                Disabled
Authentication Server Retry Interval                         30 sec
Authentication Server Retry Count                            2
```

```
Framed MTU                                                    1500 bytes
<--
Number of times ID-Requests are retried                       3
Maximum Number of Reauthentication Attempts                   2
<--
Maximum number of times Held State can be bypassed            0
Reauthentication                                              Enabled
<--
Termination                                                   Enabled
<--
Termination EAP-Type                                          eap-peap
<--
Termination Inner EAP-Type                                    N/A
Enforce Suite-B 128 bit or more security level Authentication Disabled
Enforce Suite-B 192 bit security level Authentication         Disabled
Token Caching                                                 Disabled
Token Caching Period                                          24 hr(s)
CA-Certificate                                                N/A
Server-Certificate                                            N/A
TLS Guest Access                                              Disabled
TLS Guest Role                                                guest
Ignore EAPOL-START after authentication                       Disabled
Handle EAPOL-Logoff                                           Disabled
Ignore EAP ID during negotiation.                             Disabled
Check certificate common name against AAA server              Enabled
```

NOTE

Use the privileged mode in the CLI to configure users in the Mobility Access Switch's internal database.

Adding Users to the Local Database

To add users to the local database, use the following command:

```
local-userdb add username <user> password <password> role <user_role>
```

Configuring a Server Rule Using the CLI

To configure a server rule using the CLI:

```
aaa server-group dot1x_internal
set role condition Role value-of
```

Setting Variables for LDAP Servers

If you are using a LDAP server for authentication, the following variables should be set:

- Termination enabled
- EAP type of PEAP (with inner-EAP-type set to **GTC**) or TLS

Subtopics

LDAP Server Example Configuration
Verifying the Configuration

LDAP Server Example Configuration

Below is an example configuration for the profile *FacultyAuth* for an LDAP server:

```
(host) (802.1X Authentication Profile "FacultyAuth") #termination enable
(host) (802.1X Authentication Profile "FacultyAuth") #termination eap-type
eap-peap
(host) (802.1X Authentication Profile "FacultyAuth") # termination inner-
eap-type eap-gtc
```

Verifying the Configuration

To verify the configuration, execute the **show aaa authentication dot1x** <*profile_name*> command.

Configuring Certificates with Authentication Termination

The Mobility Access Switch supports 802.1x authentication using digital certificates for authentication termination.

- Server Certificate

A server certificate installed in the Mobility Access Switch verifies the authenticity of the Mobility Access Switch for 802.1x authentication. Mobility Access Switches ship with a demonstration digital certificate.

Until you install a customer-specific server certificate in the Mobility Access Switch, this demonstration certificate is used by default for all secure HTTP connections and auth termination. This certificate is included primarily for feature demonstration and convenience and is not intended for long-term use in production networks.

Users in a production environment are urged to obtain and install a certificate issued for their site or domain by a well-known certificate authority (CA). You can generate a Certificate Signing Request (CSR) on the Mobility Access Switch to submit to a CA.

- Client Certificates

  Client certificates are verified on the Mobility Access Switch (the client certificate must be signed by a known CA) before the user name is checked on the authentication server. To use client certificate authentication for auth termination you need to import the following certificates into the Mobility Access Switch:

  - Mobility Access Switch's server certificate
  - CA certificate for the CA that signed the client certificates

    To use the CLI to configure certificates with authentication termination:

```
aaa authentication dot1x <profile>
   termination enable
   server-cert <certificate>
   ca-cert <certificate>
```

Configuring 802.1X Authentication with Machine Authentication

This section contains the following information:

- About Machine Authentication
- Enabling the Enforce Machine Authentication Option
- Role Assignment with Machine Authentication Enabled
- VLAN Assignments
- Authentication with an 802.1x RADIUS Server

- Examples of Common 802.1X Configuration Tasks Via the CLI

Subtopics

About Machine Authentication
Enabling the Enforce Machine Authentication Option
Role Assignment with Machine Authentication Enabled
VLAN Assignments
Authentication with an 802.1x RADIUS Server
Examples of Common 802.1X Configuration Tasks Via the CLI

About Machine Authentication

When a Windows device boots, it logs onto the network domain using a machine account. Within the domain, the device is authenticated before computer group policies and software settings can be executed; this process is known as *machine authentication*. Machine authentication ensures that only authorized devices are allowed on the network.

Enabling the Enforce Machine Authentication Option

About this task

You can configure 802.1X authentication for both user and machine authentication (for Windows environments only). This strengthens the authentication process further since both the device and user need to be authenticated.

Select the **Enforce Machine Authentication** option in the ArubaOS WebUI to enforce machine authentication before user authentication. When selected, either **the Machine Authentication Default Role** or the **User Authentication Default Role** is assigned to the user, depending on which authentication is successful. This option is disabled by default.

NOTE

This option may require a Policy Enforcement Firewall Next Generation (PEFNG) or Policy Enforcement Firewall Module (PEFV) license.

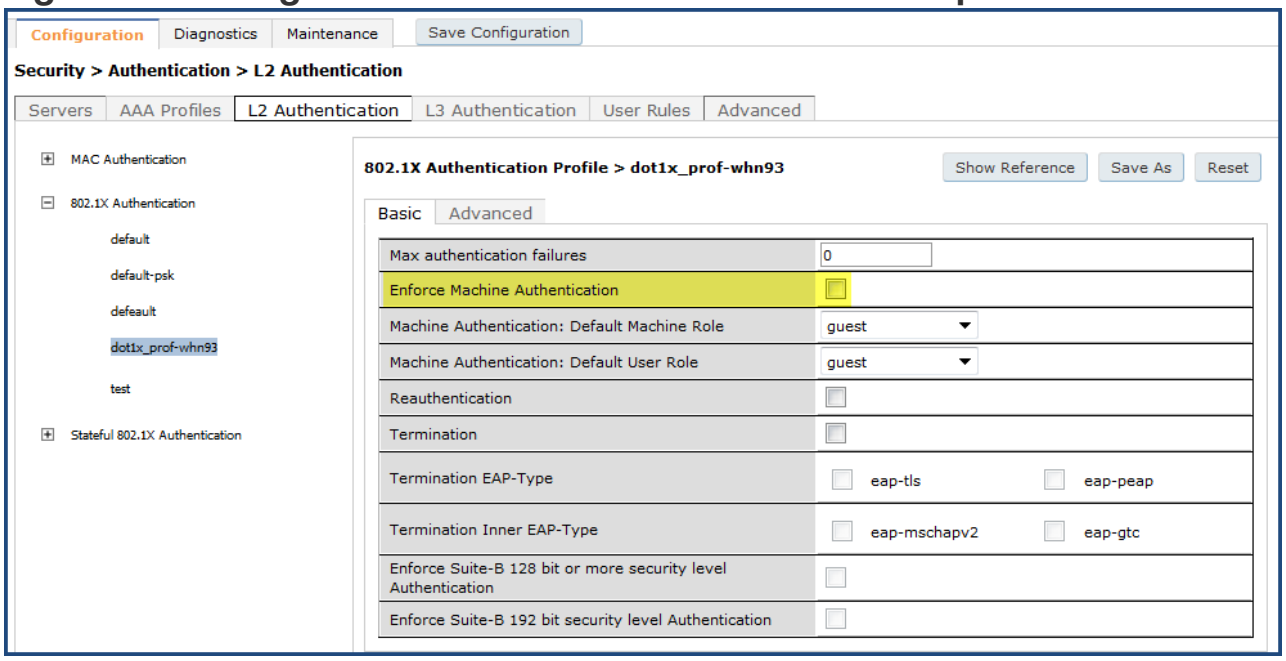
To enable **Enforce Machine Authentication**:

Procedure

1. On the Mobility Access Switch, navigate to the **Configuration > SECURITY > Authentication > L2 Authentication** page.
2. In the Profiles list, expand the **802.1x Authentication** list and select the 802.1X authentication profile of interest.

   The selected 802.1X authentication profile is displayed.

Figure 1. Enabling the Enforce Machine Authentication Option



3. To enable the option, select the **Enforce Machine Authentication** check box.

Role Assignment with Machine Authentication Enabled

When you enable machine authentication, there are two additional roles you can define in the 802.1x authentication profile:

- Machine authentication: default machine role
- Machine authentication: default user role

  While you can select the same role for both options, you should define the roles according to the policies that need to be enforced. Also, these machine authentication roles can be different from the 802.1x authentication default role configured in the AAA profile.

With machine authentication enabled, the assigned role depends upon the success or failure of the machine and user authentications. In certain cases, the role that is ultimately assigned to a client can also depend upon attributes returned by the authentication server or server derivation rules configured on the Mobility Access Switch.

Table 1 describes role assignment based on the results of the machine and user authentications.

Table 1. Role Assignments for User and Machine Authentication

Machine Auth Status	User Auth Status	Description
Failed	Failed	Both machine authentic n and user authenticatio led. Layer 2 authenticati iled.
Failed	Passed	Machine authentication (for example, the machi formation is not present he server) and user auth cation succeeds. Server rived roles do not apply.
Passed	Failed	Machine authentication ceeds and user authenti on has not been initiate rver - derived roles do n pply.
Passed	Passed	Both machine and user a uccessfully authenticate there are server - derive es, the role assigned via derivation take precede This is the *only* case whe erver - derived roles are ied.

Subtopics

Role Assignments Example

For example, if the following roles are configured:

- 802.1x authentication default role (in AAA profile): **dot1x_user**
- Machine authentication default machine role (in 802.1x authentication profile): **dot1x_mc**
- Machine authentication default user role (in 802.1x authentication profile): **guest**

The Role assignments would be as follows:

- If both machine and user authentication succeed, the role is **dot1x_user**.

  If there is a server-derived role, the server-derived role takes precedence.
- If only machine authentication succeeds, the role is **dot1x_mc**.
- If only user authentication succeeds, the role is **guest**.
- On failure of both machine and user authentication, the initial role defined in the AAA profile is assigned.

VLAN Assignments

With machine authentication enabled, the VLAN to which a client is assigned (and from which the client obtains its IP address) depends upon the success or failure of the machine and user authentications.

The VLAN that is ultimately assigned to a client can also depend upon attributes returned by the authentication server or server derivation rules configured on the Mobility Access Switch.

If machine authentication is successful, the client is associated to the VLAN configured on the interface. However, the client can be assigned a derived VLAN upon successful user authentication.

NOTE

You can optionally assign a VLAN as part of a user role configuration. It is recommended not to use VLAN derivation if user roles are configured with VLAN assignments.

Table 2 describes VLAN assignment based on the results of the machine and user authentications when VLAN derivation is used.

Table 1. VLAN Assignments for User and Machine Authentication

Machine Auth Status	User Auth Status	Description	VL
Failed	Failed	Both machine authenticati on and user authentication failed. Layer 2 authenticati on failed.	• •
Failed	Passed	Machine authentication fail s (for example, the machin e information is not present on the server) and user aut hentication succeeds.	• •
Passed	Failed	Machine authentication su cceeds and user authentic ation has not been initiated	• •
Passed	Passed	Both machine and user are successfully authenticated.	• •

Authentication with an 802.1x RADIUS Server

When authenticating with an 802.1X RADIUS server:

- An EAP-compliant RADIUS server provides the 802.1x authentication.

  The RADIUS server administrator must configure the server to support this authentication. The administrator must also configure the server to handle all communications with the Mobility Access Switch.

- 802.1x authentication based on PEAP with MS-CHAPv2 provides both computer and user authentication.

  If a user attempts to log in without the computer being authenticated first, the user is placed into a limited guest user role.

  Windows domain credentials are used for computer authentication, and the user's Windows login and password are used for user authentication. A single user sign-on facilitates both authentication to the network and access to the Windows server resources.

  You can create the following policies and user roles for:

- Student
- Faculty
- Guest
- Sysadmin
- Computer

Examples of Common 802.1X Configuration Tasks Via the CLI

This section provides several examples of common configuration tasks via the command line interface (CLI):

Subtopics

Creating an Alias for the Internal Network
Creating the Student Role and Policy
Creating the Faculty Role and Policy
Creating the Guest Role and Policy
Configuring the RADIUS Authentication Server
Configuring 802.1x Authentication Profile
Configuring the AAA Profile

Creating an Alias for the Internal Network

To create an alias for the internal network:

```
netdestination "Internal Network"
   network 10.0.0.0 255.0.0.0
   network 172.16.0.0 255.255.0.0
```

Creating the Student Role and Policy

The *student* policy prevents students from using Telnet, POP3, FTP, SMTP, SNMP, or using SSH to access the wired portion of the network. The *student* policy is mapped to the *student* user role.

To create the Student role and policy:

```
ip access-list stateless student
   any alias "Internal Network" svc-telnet deny
   any alias "Internal Network" svc-pop3 deny
   any alias "Internal Network" svc-ftp deny
   any alias "Internal Network" svc-smtp deny
   any alias "Internal Network" svc-snmp deny
   any alias "Internal Network" svc-ssh deny
user-role student
access-list stateless student
access-list stateless allowall
```

Creating the Faculty Role and Policy

The *faculty* policy is similar to the **student** policy. However, the faculty members are allowed to use POP3 and SMTP. The *faculty* policy is mapped to the *faculty* user role.

To create the Faculty role and policy:

```
ip access-list stateless faculty
   any alias "Internal Network" svc-telnet deny
   any alias "Internal Network" svc-ftp deny
   any alias "Internal Network" svc-snmp deny
   any alias "Internal Network" svc-ssh deny
user-role faculty
access-list stateless faculty
access-list stateless allowall
```

Creating the Guest Role and Policy

The *guest* policy permits only access to the Internet (via HTTP or HTTPS) and only during daytime working hours. The *guest* policy is mapped to the *guest* user role.

To create the guest role and policy:

```
time-range working-hours periodic
   weekday 07:30 to 17:00
ip access-list stateless guest
   any host 10.1.1.25 svc-dhcp permit time-range working-hours
   any host 10.1.1.25 svc-dns permit time-range working-hours
   any alias "Internal Network" any deny
   any any svc-http permit time-range working-hours
   any any svc-https permit time-range working-hours
   any any any deny
user-role guest
access-list stateless guest
```

Configuring the RADIUS Authentication Server

You can set the role condition to identify the user's group. The Mobility Access Switch uses the literal value of this attribute to determine the role name.

The following example uses the RADIUS server name *radiusFaculty* to configure the RADIUS server.

To configure the RADIUS authentication server to identify the user's group:

```
(host) (config) #aaa authentication-server radius radiusTechPubs
(host) (RADIUS Server "radiusFaculty") #host 10.41.255.30
(host) (RADIUS Server "radiusFaculty") #key hometown
(host) (RADIUS Server "radiusFaculty") #exit
(host) (config) #aaa server-group radiusTechpubs
(host) (Server Group "radiusFaculty") #auth-server radiusTechpubs
(host) (Server Group "radiusFaculty") #set role condition Class Value-of
```

Configuring 802.1x Authentication Profile

In the 802.1x authentication profile, configure enforcement of machine authentication before user authentication (see Enabling the Enforce Machine Authentication Option).

If a user attempts to log in without machine authentication taking place first, the user is placed in the guest role.

To configure the 802.1X authentication profile:

```
aaa authentication dot1x dot1x
   machine-authentication enable
   machine-authentication machine-default-role student
   machine-authentication user-default-role guest
```

Configuring the AAA Profile

An AAA profile specifies the 802.1x authentication profile and 802.1x server group to be used for authenticating clients. The AAA profile also specifies the default user roles for 802.1x authentication.

To configure the AAA profile:

```
aaa profile aaa_dot1x
   dot1x-default-role guest
   authentication-dot1x dot1x
   dot1x-server-group radiusGuest
```

Preparing for LDAP and SQL Authentication Sources

This section describes how to prepare Policy Manager for LDAP Authentication Source Configuration and SQL Authentication Source Configuration.

LDAP Authentication Source Configuration

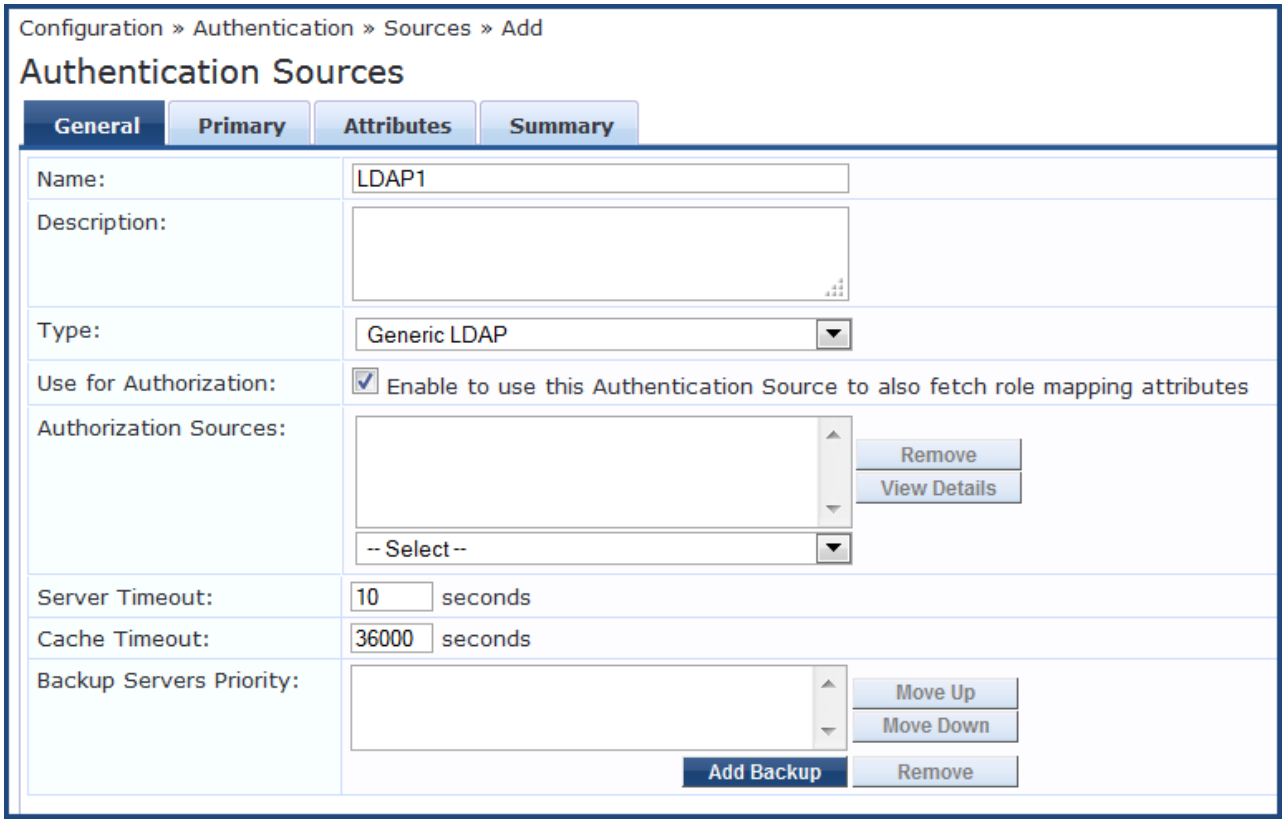
About this task

Policy Manager can perform NTLM/MSCHAPv2, PAP/GTC, and certificate-based authentications against any LDAP-compliant directory (for example, Novell eDirectory, OpenLDAP, and Sun Directory Server). LDAP and Active Directory-based server configurations are similar. You can retrieve role-mapping attributes by using filters. To configure Generic LDAP authentication sources in Policy Manager:

Procedure

1. Navigate to the **Configuration > Authentication > Sources** page. The **Authentication Sources > General** page opens. The **General** page labels the authentication source and defines session details.
2. Click **Add**. The **Add Authentication Source** page opens.

Figure 1. Adding a Generic LDAP Authentication Database



3. Enter the values for these parameters as described in Table 1.

Table 1. General Page Parameters for Generic LDAP Database

Parameter	Action/Description
Name	Enter the name of the LDAP authenticatio n source.
Description	Provide the additional information that he lps to identify the LDAP authentication so urce.
Type	Select **Generic LDAP**.
Use for Authorization	When **Use for Authorization** is enabled, Policy Manager can use this authenticati on source to fetch role - mapping attribu tes. This option is enabled by default.

Parameter	Action/Description
Backup Servers Priority	To add a backup server in the event the main server goes down, click **Add Backup**. **NOTE** Aruba recommends setting up one or more backup servers.
Authorization Sources	Specifies additional sources from which role - mapping attributes may be fetched . a. Select a previously configured authentication source from the drop - down list. b. To add authentication source to the list of authorization sources, click **Add**. • To remove the authentication source from the list, click **Remove**. If Policy Manager authenticates the user or device from this authentication source, it also fetches role mapping attributes from these additional authorization sources.
Cache Timeout	Policy Manager caches attributes fetched for an authenticating entity. This parameter controls the duration in number of seconds for which the attributes are cached. The default is **36000** seconds (one hour) .
Backup Servers Priority	To add a backup server, click **Add Backup** .

Parameter	Action/Description
	• If the **Backup 1** tab appears, you can s pecify connection details for a backu p server. • To remove a backup server, select the server name and click **Remove**. • To change the server priority of the ba ckup servers, select **Move Up** or **Move Down**. ◦ This is the order in which Policy M anager attempts to connect to the backup servers when the primary server is unreachable.

4. When satisfied with these settings, click **Next**. The **Authentication Sources Primary** page opens.

Figure 2. Primary Page: Generic LDAP Authentication Database

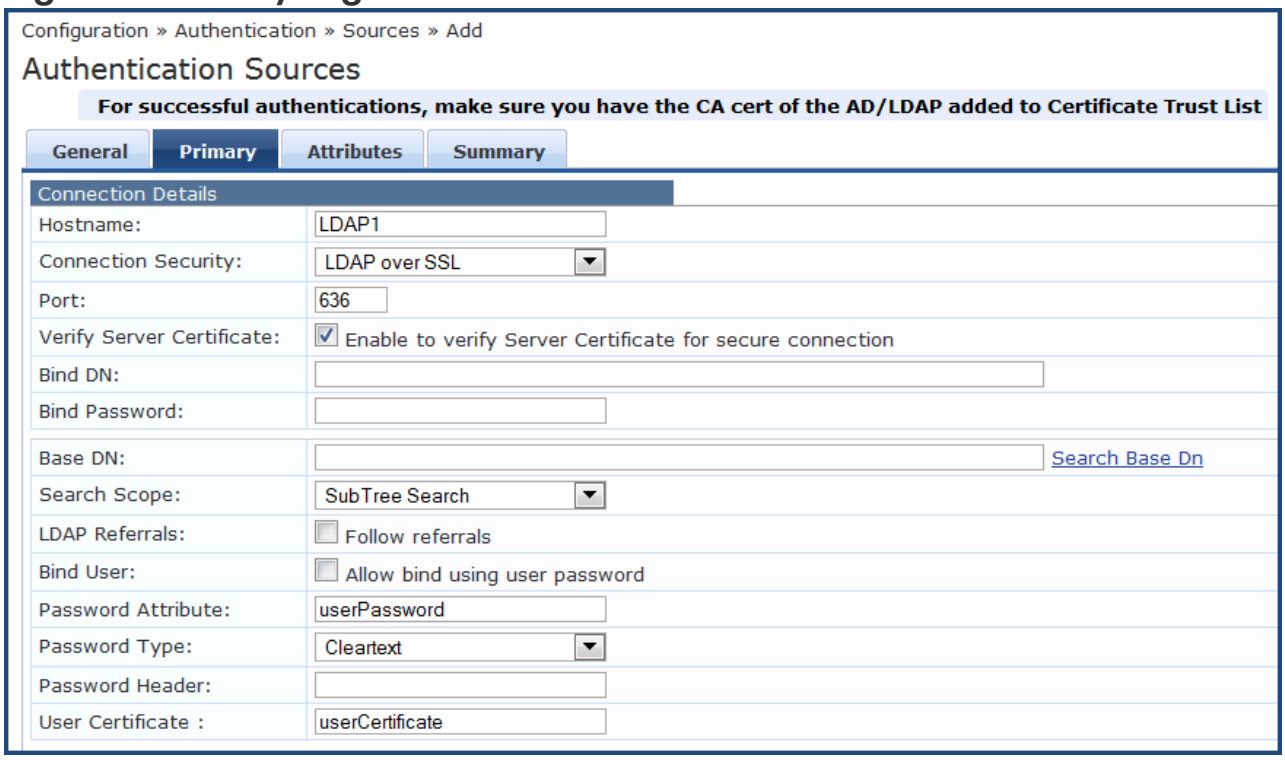


Table 2. Primary Parameters for an LDAP Authentication Source

Parameter	Action/Description
Hostname	Enter the name or IP address of the LDAP server you're going to use for authenticati on. Note that most domain controllers are als o LDAP servers. Policy Manager uses LDA P to talk to the domain controller.
Connection Security	Set **Connection Security** to: **LDAP over SSL**. This enables the secure sockets layer (SS L) cryptographic protocol to connect to your Active Directory. Selecting **LDAP ove r SSL** automatically populates the *Port* fie ld to **636**. **NOTE** In a production environment , security is a concern beca use when Policy Manager bi nds to an LDAP server, it su bmits the username and pas sword for that account over the network under clear text unless you protect it using C onnection Security and set t he port to **636**. **NOTE** To ensure successful authen tication, be sure to add the CA certificate of the LDAP se rver to the Certificate Trust L ist.

Parameter	Action/Description
Port	Specify the TCP port at which the LDAP s erver is listening for connections. For a single domain LDAP Domain Servic e: • Default port for LDAP: **389** • Default port for LDAP over SSL: **636** When you set the *Connection Security* fi eld to **AD over SSL**, this port is automatic ally set to **636**. For a multi - domain LDAP Domain Servic e forest, the default ports for the global c atalog are: • Default port without SSL: **3268** • Default port with SSL: **3269**
Verify Server Certificate	Enable this option to verify the Server Ce rtificate for a secure connection.
Bind DN	Enter the Distinguished Name of the nod e in your directory tree from which to sta rt searching for records. The Bind DN text box specifies the full d istinguished name (DN), including comm on name (CN), of an LDAP user account that has privileges to search for users (u sually the Administrator account). For ex ample: **CN=Administrator,CN=Users,DC= mycompany,DC=com** This user account must have at least dom ain user privileges. The Bind DN user, such as Administrator, i s the username associated with the Bind DN user account. • For a single domain LDAP Domain Se rvice, the Bind DN entry must be loca

Parameter	Action/Description
	ted in the same branch and below the Base DN. • For a multi - domain LDAP Domain Service forest, because you leave the Base DN text box empty, the restrictions that apply for a single domain do not apply for a multi - domain forest. Policy Manager fills in the domain portion of the Bind DN. Policy Manager also populates the *Base DN*, and the *NetBIOS Domain Name* fields. **NOTE** You may need to get the Bind DN from the LDAP administrator.
Bind Password	This is the text box for the Active Directory password for the account that can search for users. Enter the Bind password. **NOTE** The Bind password is the same password used in association with the Bind DN user account.
NetBIOS Domain Name	Specify the DN (Distinguised Name) of the administrator account. Policy Manager uses this account to access all other records in the directory. **NOTE** For Active Directory, the bind DN can also be in the administrator@domain format (fo

Parameter	Action/Description
	r example, administrator@acme.com).
Base DN	For a single domain Active Directory Domain Service, this is the text box for the Distinguished Name (DN) of the starting point for directory server searches. For example:**DC=mycompany,DC=com**. The LDAP server starts from this DN to create master lists from which you can later filter out individual users and groups. **NOTE** The Base DN value that is automatically populated in this instance is *not* the best practice Base DN value. Aruba recommends that you narrow down the Base DN as far as possible to reduce the load on the Active Directory LDAP server. For example, if all your users are in the AD Users and Computer Users folder, then set the Base DN to search in the Users folder. a. To browse the LDAP directory hierarchy, click **Search Base DN**.The LDAP Browser opens. b. Navigate to the DN you want to use as the Base DN. c. To select a node as a Base DN, click the appropriate node in the tree structure. For a multi - domain Active Directory Domain Service (AD DS) forest, the

Parameter	Action/Description
	appropriate action is to leave the Base DN text box blank. **NOTE** **Search Base DN**.
Search Scope	Search scope is related to the Base DN. T he search scope defines how LDAP will s earch for your objects. Select one of the foloiwng **Search Scope** types. • Subtree Search: Searches every objec t and sub - object in the LDAP directo ry. • One - Level Search: Looks directly un der the Base DN. • Base Object: Searches any object und er the Base DN.
LDAP Referrals	Aruba does *not* recommend enabling the "Follow Referrals" check box. This function directs the LDAP server to fi nd a specific user in its tree, but it's possi ble for the user to be included on anothe r LDAP server, which can cause a search l oop.
Bind User	Enable this option to allow a bind operati on using the user password. For clients to be authenticated by using t he LDAP bind method, Policy Manager m ust receive the password in clear text.
Password Attribute	Enter the name of the attribute in the use r record from which the user password c an be retrieved.

Parameter	Action/Description
Password Type	
Password Header	Oracle's LDAP implementation prepends a header to a hashed password string. If you are using Oracle LDAP, enter the h eader in this field so the hashed passwo rd can be correctly identified and read.
User Certificate	Leave the value that is automatically pop ulated in this field as the default unless your LDAP administrator has a different at tribute for storing the user certificate.
Always use NetBIOS name	Check this option to always use the NetBI OS name instead of the domain part in th e username for authentication. **NOTE** This field is available only if you select **Active Directory** as an authentication source.

5. When satisfied with these settings, click **Next**. The **Summary** page is displayed, which shows all the settings you have entered for the LDAP authentication source.

SQL Authentication Source Configuration

Policy Manager can perform MSCHAPv2 and PAP/GTC authentication against any Open Database Connectivity (ODBC) compliant SQL database such as Microsoft SQL Server, Oracle, or PostgrSQL.

- You can specify a stored procedure to query the relevant tables and retrieve role-mapping attributes by using filters.
- You can configure the primary and backup servers, session details, filter query, and role mapping attributes to fetch the generic SQL authentication sources.

Configuring a Generic SQL Authentication Source

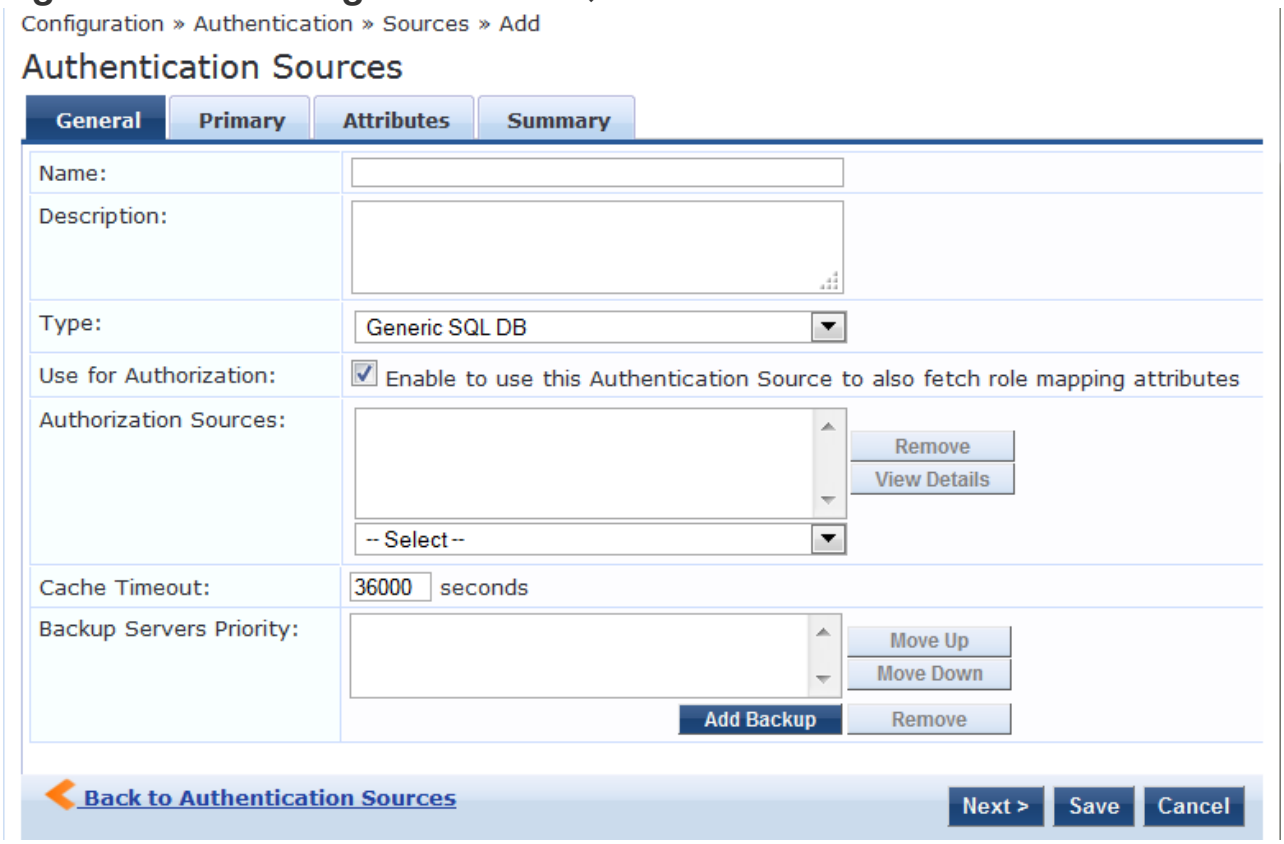
About this task

To configure a generic SQL authentication source:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click **Add**. The **Authentication Sources > General** page opens. The **General** page labels the authentication source and defines session details.

Figure 3. General Page: Generic SQL Authentication Database



3. Enter the information for each of the required parameters as described in Table 1.

Table 3. General Page Parameters for Generic SQL Database

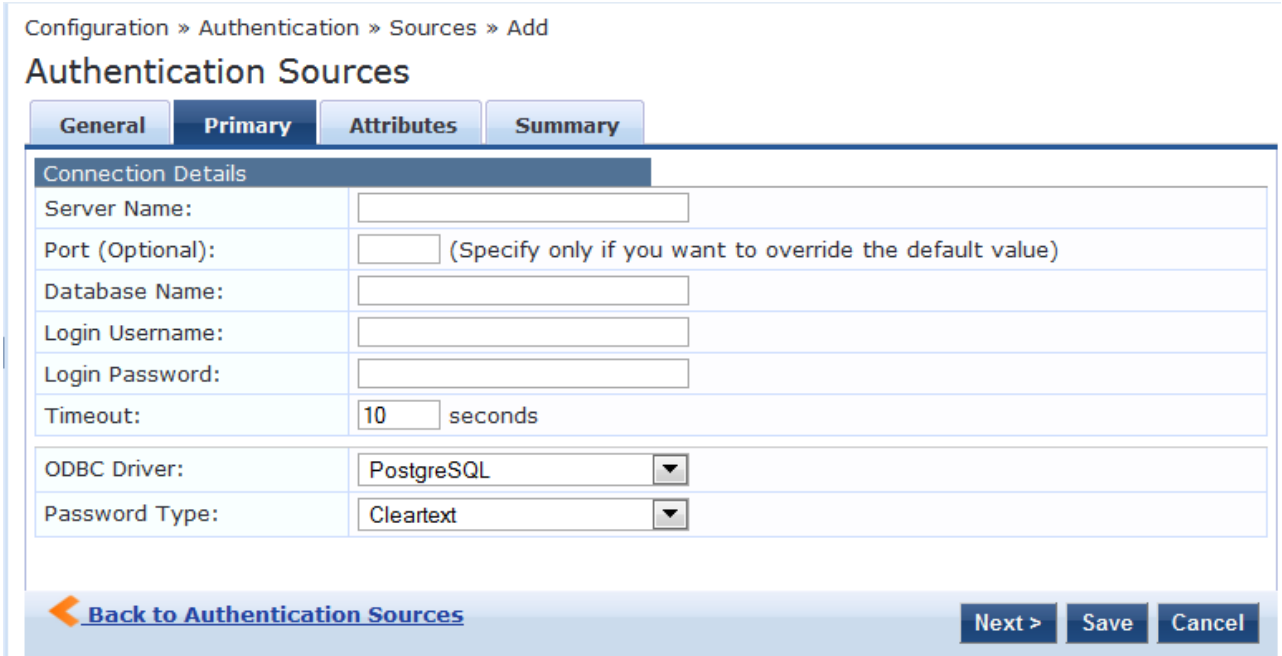
Parameter	Action/Description
Name	Enter the name of the SQL authenticatio n source.
Description	Provide the additional information that he lps to identify the authentication source.

Parameter	Action/Description
Type	Select **Generic SQL DB**.
Use for Authorization	Leave the **Use for Authorization** setting e nabled. When **Use for Authorization** is enabled, Policy Manager can use this authenticati on source to fetch role - mapping attribu tes. This option is enabled by default.
Backup Servers Priority	To add a backup server in the event the main server goes down, click **Add Backu p**. **NOTE** Aruba recommends setting up one or more backup se rvers.
Authorization Sources	Specify additional sources from which ro le - mapping attributes can be fetched. • Select a previously configured authen tication source from the drop - down l ist. • To add authentication source to the lis t of authorization sources, click **Add**. • To remove the authentication source f rom the list, click **Remove**. If Policy Manager authenticates the user or device from this authentication sourc e, it also fetches role mapping attributes from these additional authorization sourc es.
Cache Timeout	Specify the number of seconds for the **Cache Timeout**. Policy Manager caches attributes fetched for an authenticating entity. This paramet

Parameter	Action/Description
	er controls the duration in number of sec onds for which the attributes are cached.
Backup Servers Priority	To add a backup server, click **Add Backup** . • If the **Backup 1** tab appears, you can s pecify connection details for a backu p server. • To remove a backup server, select the server name and click **Remove**. • To change the server priority of the ba ckup servers, select **Move Up** or **Move Down**. ◦ This is the order in which Policy M anager attempts to connect to the backup servers when the primary server is unreachable.

4. When satisfied with these settings, click **Next**. The Authentication Sources **Primary** page opens.

Figure 4. Primary Page: Generic SQL Authentication Source



5. Enter the information for each of the required parameters as described in Table 2.

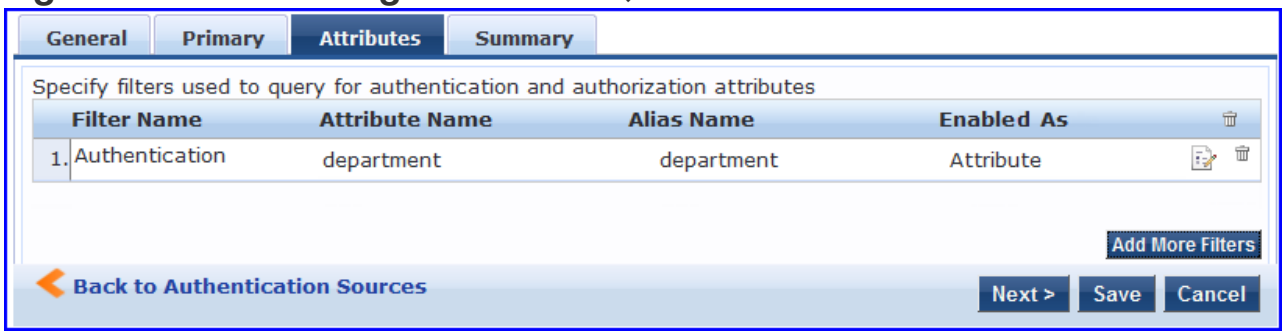
Table 4. Primary Page Parameters for Generic SQL Database

Parameter	Action/Description
Server Name	Enter the name or IP adc ation.
Port	Optionally, you can spec
Database Name	Enter the name of the da
Login Username	Enter the name of the us ccess to all the attribute
Password	Enter the password for tl
Timeout	Enter the duration in sec m the primary to the bac
ODBC Driver	Select the ODBC driver t • PostgreSQL • Oracle 11g • MariaDB • MSSQL **NOTE** MySQL is no lo replaced by Ma If you connect to a Micro name in the authenticati rincipal Name), the follo • Backslash (\ ) • At - sign (@) • Hyphen • Underscore

Parameter	Action/Description
Password Type	Specify how the user pa • Cleartext : Password • NT Hash: Password is • LM Hash : Password i • SHA: Password is sto • SHA256: Password is • MD5

6. When satisfied with the **Primary** page settings, click **Next**. The Attributes page appears. The **Attributes** page defines the SQL database query filters and the attributes to be fetched when using those filters.

Figure 5. Attributes Page: Generic SQL Authentication Source



7. Enter the information for each of the required parameters as described in Table 3.

Table 5. Attributes Page Parameters for Generic SQL Database

Parameter	Action/Description
Filter Name	Displays the name of the fil
Attribute Name	Specifies the name of the S ed for this filter.
Alias Name	Specifies an alias name for for the filter.
Enabled As	Optionally, indicates wheth e or an attribute type. This

Parameter	Action/Description
Add More Filters	Click this button to open th etails, see the next section, Use this page to define a fi butes to be fetched from th

8. When satisfied with the **Attribute** page settings, click **Next**. The **Summary** page appears.

Defining a Filter Query

About this task

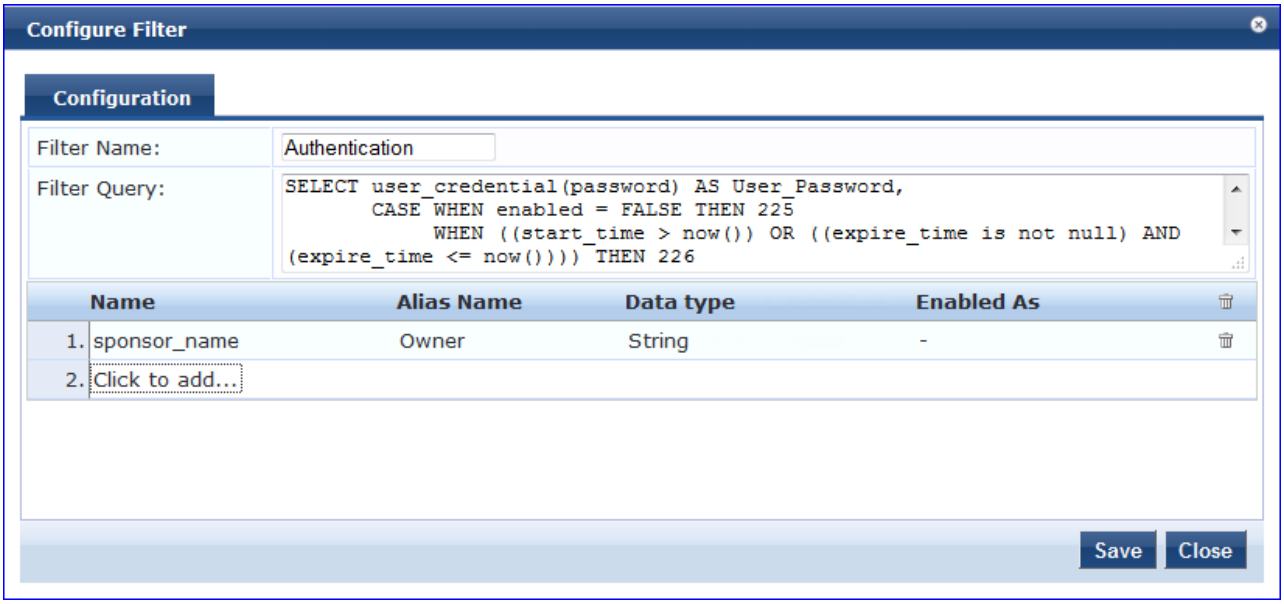
The **Configure Filter** page allows you to define a filter query and the related attributes to be fetched from the SQL DB store.

To define a filter query:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Source**s page opens.
   a. If you're defining a new filter for an existing authentication source, click the name of the authentication source, then select the **Attributes** tab.
   b. If you're defining a new filter query for a newly configured authentication source, follow the steps described in the previous section.
2. From the **Attributes** page, click **Add More Filters**. The **Configure Filter** page opens.

Figure 6. Configure Filter Page: Generic SQL Authentication Source



3. Enter the information for each of the required parameters as described in Table 4.

Table 6. Configure Filter Page Parameters for Generic SQL Database

Parameter	Action/Description
Filter Name	Enter the name of the new filter.
Filter Query	Specify an SQL query to fetch the attributes from the user or device record in the database.
Name	Select Click to add to specify the name of the attribute.
Alias Name	Specify the alias name for the attribute. By default, this is the same value as the attribute name.
Data Type	Specify the data type for this attribute, such as String, Integer, or Boolean.
Enabled As	Specify whether this value is to be used directly as a role or an attribute in an Enforcement Policy. This option bypasses having to assign a role in Policy Manager through a Role Mapping Policy.

4. **onfigure Filter Page Parameters for Generic SQL DatabaseWhen satisfied with the Configure Filter page settings, click Save.** When satisfied with the **Configure Filter** page settings, click **Save**.

802.1X EAP-PEAP Reference

About EAP-PEAP MSCHAPv2

The authenticated wireless access design based on the Protected Extensible Authentication Protocol Microsoft Challenge Handshake Authentication Protocol version 2 (PEAP-MS-CHAPv2) utilizes user account credentials (user name and password) stored in Active Directory Domain Services to authenticate wireless clients, instead of using smart cards or user and computer certificates for client authentication.

EAP-PEAP MSCHAPv2 Handshake Exchange Summary

Table 1 describes how a typical 802.1X authentication session flows when using Policy Manager as the authentication server with Microsoft Active Directory as the back-end user identity repository.

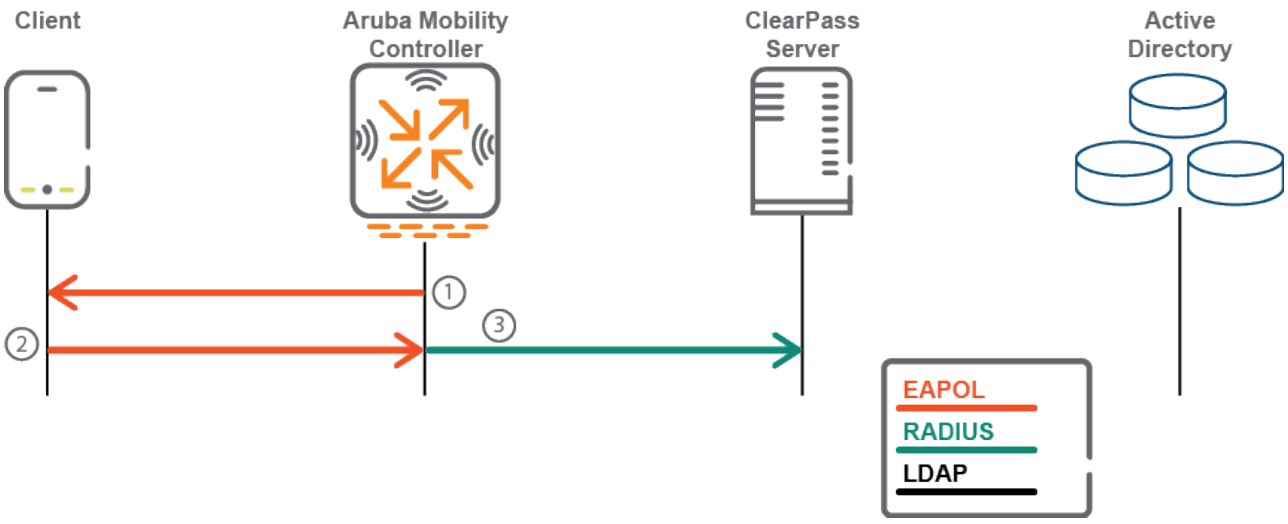
- The term **supplicant** refers to a client device, such as a laptop, tablet, or mobile phone requesting access to a network.
- The term **authenticator** refers to a network device, such as an Aruba Mobility Controller or an Access Point (AP), which controls access to a network resource.
- The term **authentication server** refers to the Policy Manager server, which processes the authentication requests and provides either an accept or reject response.

  Each section of Table 1 is followed by a diagram that illustrates the communication steps between the devices described in the table. The numbers of each step in the table correspond to the numbers assigned to the handshake sequences in the accompanying illustrations.

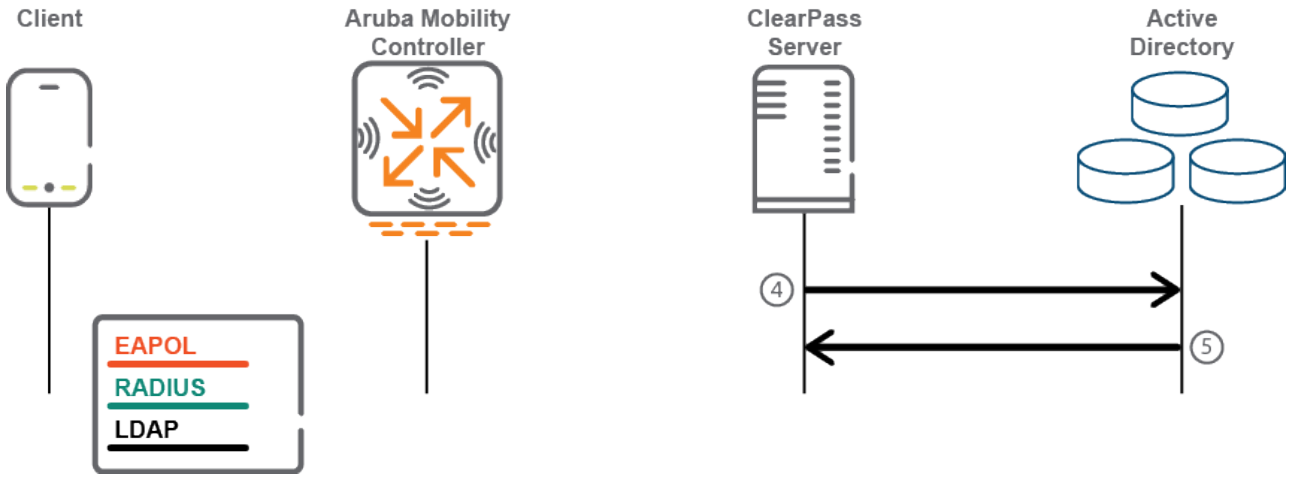
Table 1. Detailed Sequence of the EAP - PEAP - Active Directory Handshake Exchange

Extensible Authentication Protocol over LAN (EAPOL) Start	
1	The authenticator sends an EAP - Request for the identi ty of the connecting supplicant (client).

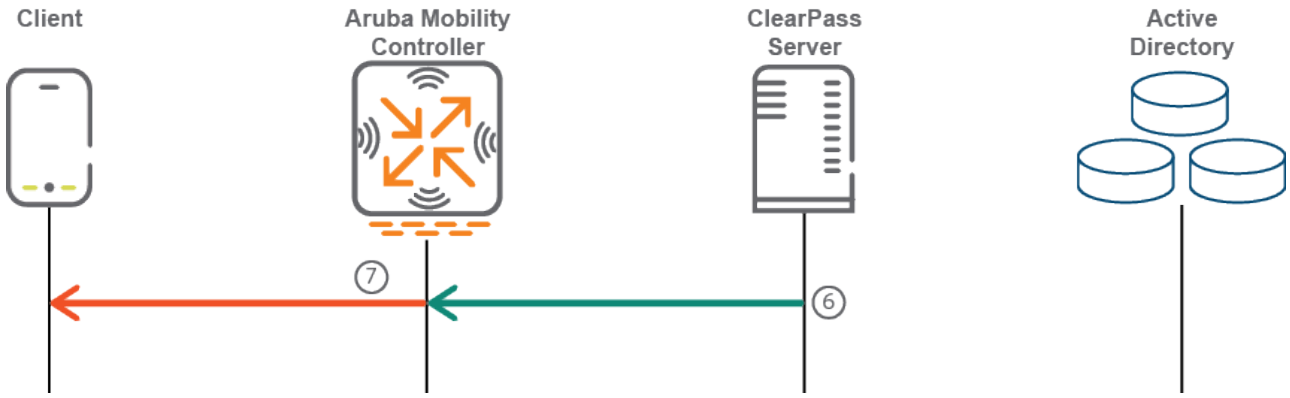
Extensible Authentication Protocol over LAN (EAPOL) Start	
2	The supplicant responds to the authenticator with an E AP Identity Response that contains the identity (userna me) used for authentication. This is referred to as the "O uter Identity."
3	The authenticator forwards the EAP Identity Response with the identity of the user to the authentication server (Policy Manager).



Active Directory	
4	The authentication server performs an LDAP lookup aga inst its configured Active Directory authentication sourc es to try to find the user's name in the directory, along with some basic LDAP attributes, such as *sAMAccountN ame*.
5	The LDAP server responds to the authentication server's LDAP search request with the appropriate repsonses to the LDAP lookup.



EAPOL	
6	The authentication server responds to the supplicant th rough the authenticator with an EAP - Request messag e indicating that it would like to initiate EAP - PEAP.
7	The authenticator passes the EAP - Request message t o the supplicant.

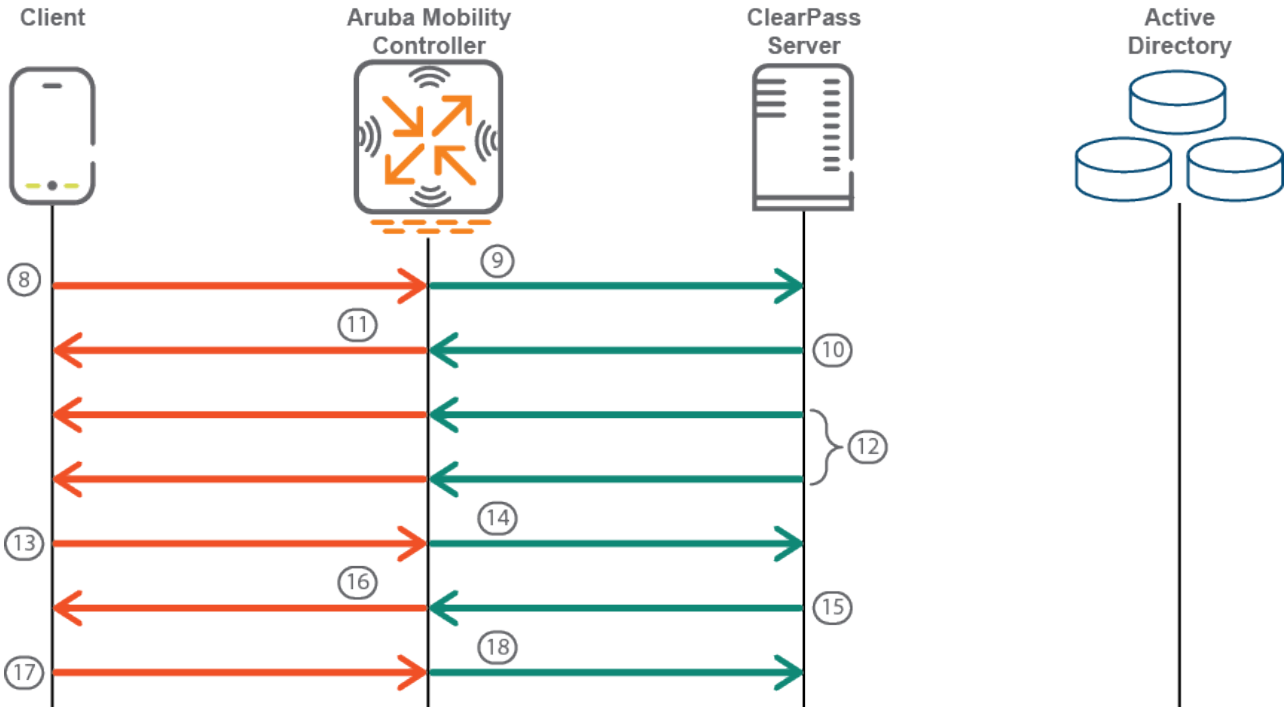


Transport Layer Security (TLS) Tunnel Setup	
8	The supplicant sends a Transport Layer S ecurity (TLS) *"Client Hello"* message with in an EAP - response message through t he authenticator to the authentication se rver.
9	The authenticator passes the EAP - Resp onse message containing the TLS Clien

	t Hello message to the authentication se rver.
10	The authentication server responds with a TLS Handshake message of types "Se rver Hello," "Certificate," "Server Key Exch ange," and "Server Hello Done" to the aut henticator.
11	The authenticator forwards the TLS hand shake messages between the authenticat ion server and the supplicant inside of EA P Request (server) and EAP Response (su pplicant) messages.
12	Steps 10 and 11 repeat until the authentic ation server has transmitted all of its han dshake messages. This may take several s teps due to having to dismantle the certif icates into fragments that fit within the si ze limits of an EAP message.
13	The supplicant sends another TLS Hands hake message inside an EAP - Response message of types "Client Key Exchange," "Change Cipher Spec," "Handshake," and "Client Finished" to the authenticator.
14	The authenticator sends this EAP - Respo nse to the authentication server.
14	The authentication server responds to th e authenticator with an EAP - Request for the supplicant that contains the message types "Change Cipher Spec" and "Server Finished."
16	The authenticator passes the EAP messa ge to the supplicant.

Transport Layer Security (TLS) Tunnel Setup

17	The supplicant sends an EAP - Response for the authentication server to the authenticator.
18	The authenticator sends the EAP - Response to the authentication server.

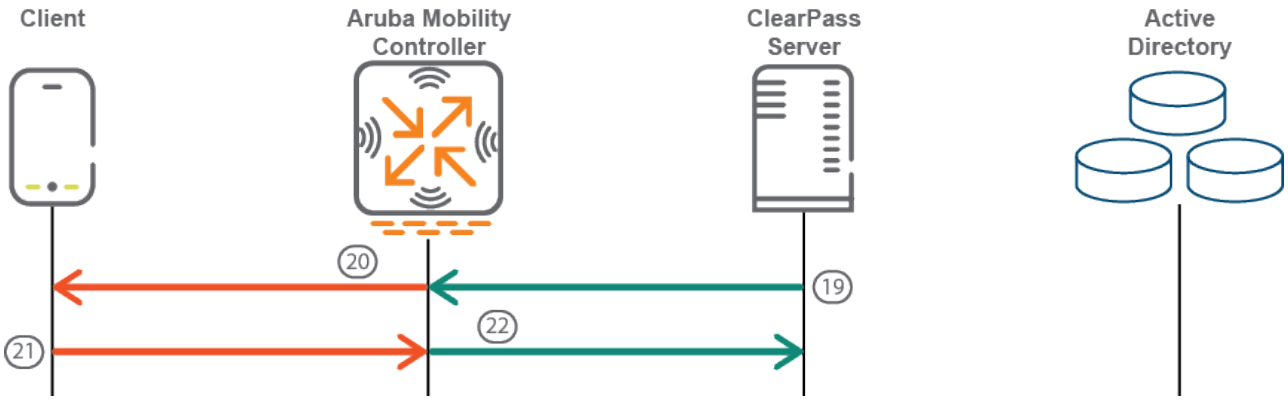


Inner EAP MSCHAPv2

19	Inside the TLS tunnel, the EAP process starts again with the authentication server sending an EAP Identity Request to the supplicant requesting the client's identity.
20	The authenticator sends the EAP Identity Request message to the supplicant requesting the client's identity.
21	The supplicant responds with an EAP Identity Response containing its identity to the authenticator.

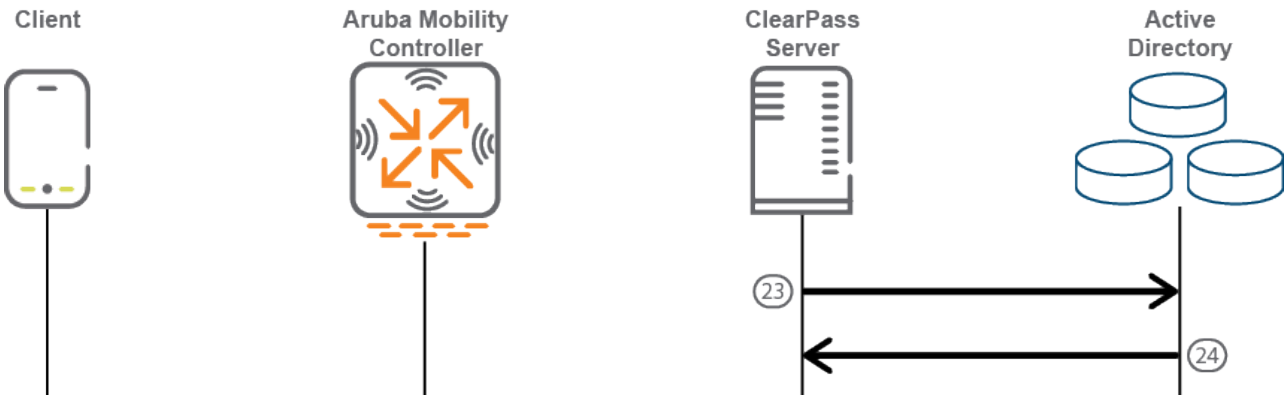
Inner EAP MSCHAPv2

22	The authenticator forwards this EAP Identity Response to the authentication server.

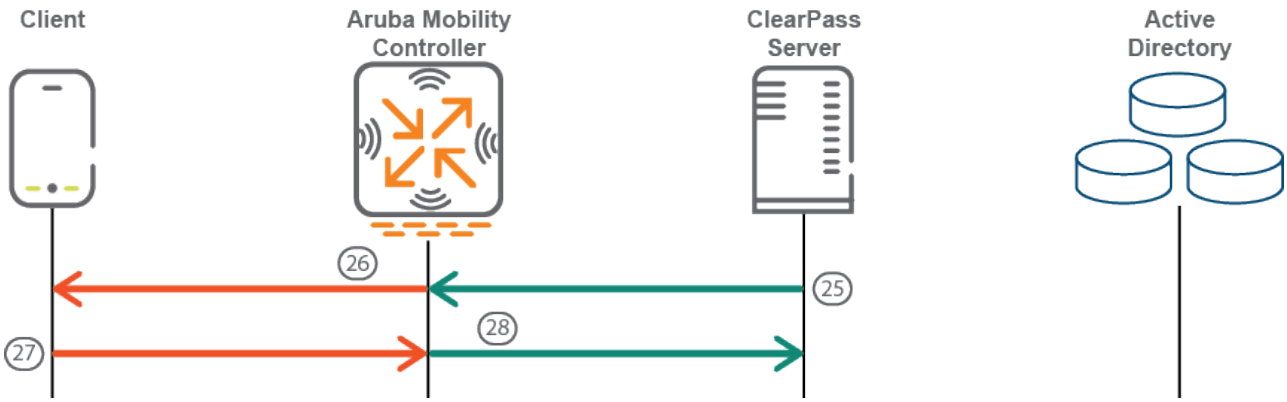


Active Directory

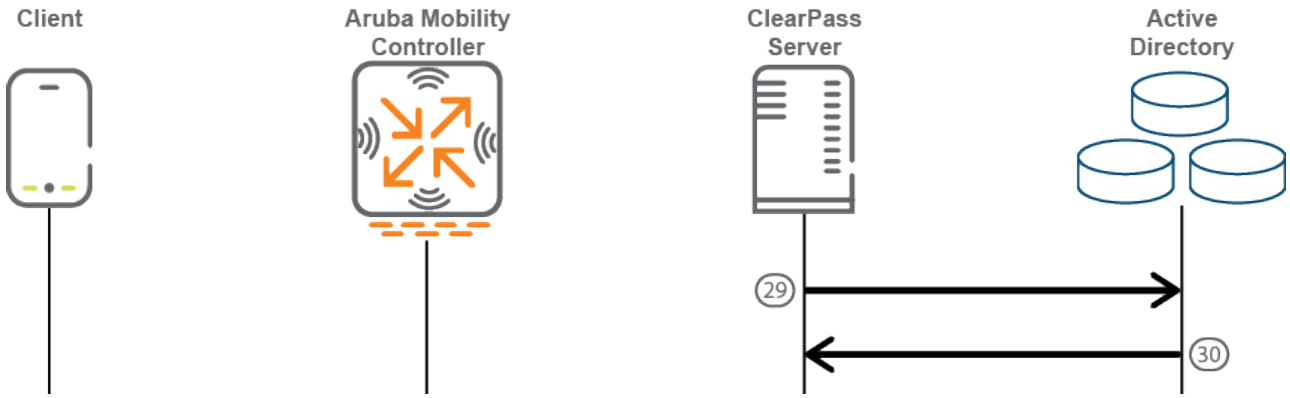
23	The authentication server performs an LDAP lookup against its configured Active Directory authentication sources to try to find the user's name in the directory, along with some basic LDAP attributes, such as *sAMAccountName*.
24	The LDAP server responds to the LDAP search request with the appropriate answers to the query.



Inner EAP MSCHAPv2	
25	The authentication server sends an EAP request to the supplicant containing an MS - CHAPv2 challenge.
26	The authenticator forwards the EAP requ est to the supplicant.
27	The supplicant responds with an EAP Ide ntity Response containing its identity to t he authenticator.
28	The authenticator forwards this EAP Ident ity Response to the authentication server.

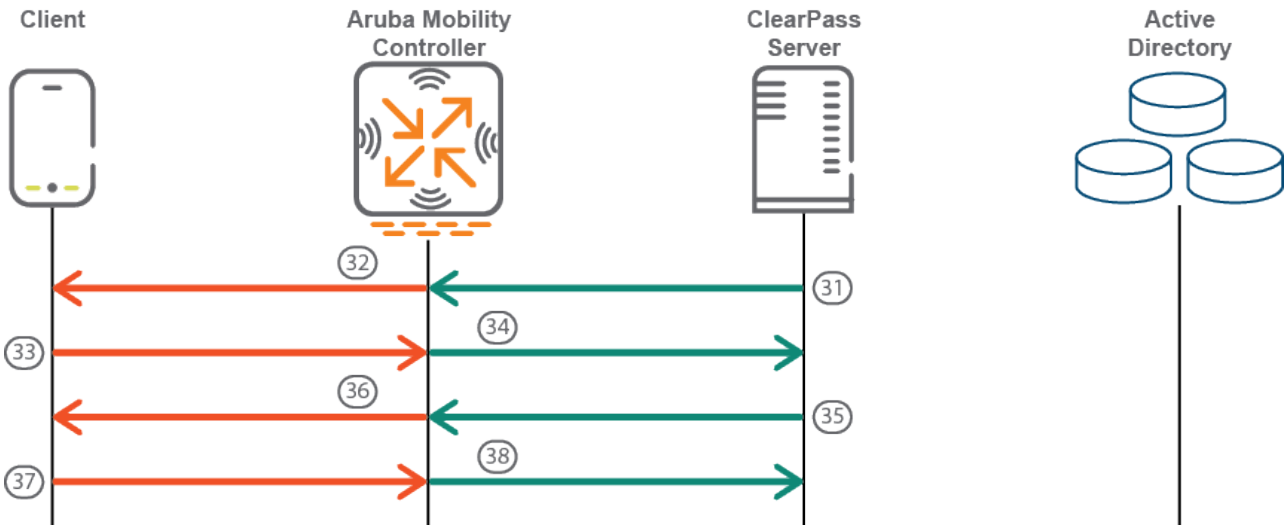


Active Directory	
29	The authentication server takes the username and the MSCHAPv2 response from the supplicant and combin es it with the MSCHAPv2 challenge and the NetBIOS na me of the Active Directory domain and submits this set of information to the Active Directory domain controller for authentication. This is done via NT LAN Manager (NT LM).
30	The Active Directory domain controller lets the authenti cation server know authentication was successful.

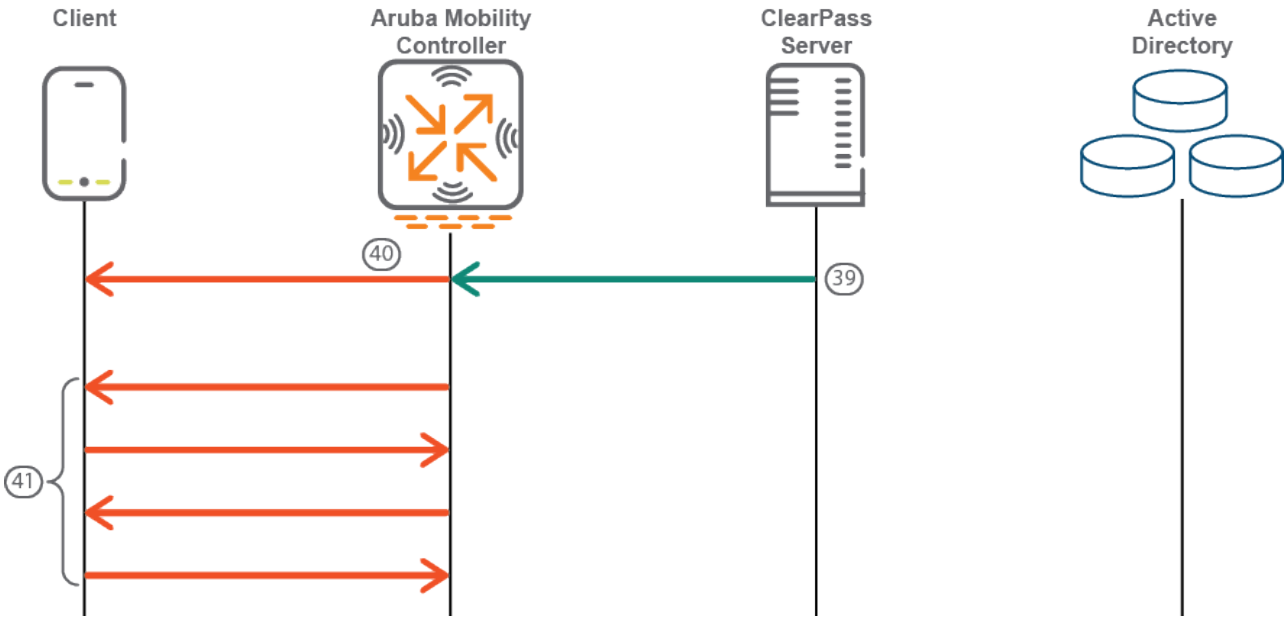


Inner EAP MSCHAPv2	
31	The authentication server sends an EAP - Request message for the supplicant with an MSCHAPv2 success message and an authenticator response string from the Active Directory Domain Controller to the authenticator.
32	The authenticator passes the EAP - Request with an MSCHAPv2 success message and the authenticator response to the supplicant.
33	The supplicant sends an EAP - Response message for the authentication server with an MSCHAPv2 success message to the authenticator.
34	The authenticator sends the EAP - Response message from the supplicant with the MSCHAPv2 success message to the authentication server.
35	The authentication server sends an EAP - Request message to the authenticator indicating the Inner EAP method was successful.
36	The authenticator forwards this EAP - Request to the supplicant.

Inner EAP MSCHAPv2	
37	The supplicant sends an EAP - Response to the authentication server, acknowledging the Inner EAP method was successful.
38	The authenticator forwards the EAP - Response from the supplicant to the authentication server.



EAPOL	
39	The authentication server sends a RADIUS access - accept message to the authenticator with an EAPOL success message along with the key material.
40	The authenticator sends an EAPOL success message to the supplicant.
41	The authenticator and supplicant complete a four - way handshake to start the flow of encrypted wireless traffic.



Continued Guidance Configuration

This section describes guidance criteria for ClearPass Policy Manager Common Criteria acceptance.

ClearPass Policy Manager User Guide

Subtopics

About ClearPass Policy Manager
Overview of ClearPass Policy Manager features, licensing, dashboard use, configuration, system information, and data import or export.
Services
Live Monitoring
Live Monitoring provides real-time visibility into system activity and status for faster awareness and response.
Authentication Methods and Sources
Posture Policies, Audit Servers, Agentless OnGuard
Configuring Identity Settings
Configuring Enforcement Profiles and Policies
Adding and Modifying Network Devices
Configuring Network and Subnet Scans
Configuring Policy Simulation
Policy Manager Profiler
Administration

ClearPass Licensing Management
Cluster Update and Upgrade
Configuring Processing for Ingress Events
Agentless OnGuard
Insight Reports
Command Line Interface
SNMP, System Events & Error Codes
Cipher Suites
Common Criteria (CC)
Use Cases
Rules Editing and Namespaces

About ClearPass Policy Manager

Overview of ClearPass Policy Manager features, licensing, dashboard use, configuration, system information, and data import or export.

This section provides an overview of the ClearPass Policy Manager Access Management System and includes the following information:

- ClearPass Policy Manager Overview
- Initial Login and Activation of the Policy Manager Platform License
- Using the Policy Manager Dashboard
- Accessing Configuration Information
- Accessing System Information
- Importing and Exporting Information

Subtopics

Using the Policy Manager Dashboard
Initial Login and Activation of the Policy Manager Platform License
Initial login and platform license activation enable access to Policy Manager features and prepare the system for use.
Accessing Configuration Information
Importing and Exporting Information
Accessing System Information
Support Services
Copyright Information

Using the Policy Manager Dashboard

The Policy Manager Dashboard organizes and presents key information about the status and performance of the current Policy Manager server or cluster, as well as a set of Quick Links to the most commonly used functions, such as configuring policies, and viewing the Access Tracker. The Dashboard information is illustrated in interactive bar chart, graph, and table formats.

To customize the Dashboard layout, drag and drop from the list of Widget elements on the left pane to one of the available Dashboard slots in the right pane.

Table 1. Dashboard Widget Summary

Alerts	To view the table with latest system level events, drag and drop the **Alerts** widget t o the **Dashboard**. Clicking on a row drills d own to the **Event Viewer**.
All Requests	To view the graph that displays all request s (such as RADIUS, TACACS+, and WebAuth requests) processed by Policy Manager ove r the past week, drag and drop the **All Req uests** widget. Clicking on each bar in the graph drills d own to the **Access Tracker** page and show s the requests for the selected day.
Applications	To view the links to the Aruba Insight, Gues t, and Onboard applications that are integ rated with Policy Manager, drag and drop t he **Applications** widget to the **Dashboard**.
Authentication Status	To view a graph of the failed and successfu l requests over the past week, drag and dro p the **Authentication Status** to the **Dashbo ard**. This graph includes RADIUS, WebAuth , and TACACS+ requests. The default data fi lters **Failed Requests** and **Successful Requ ests** are used to plot this graph. Clicking on each circle on the line graph dr ills down to the **Access Tracker** page that s hows the failed and successful requests for the day specified.
Cluster Status	To view the status of all nodes in a cluster, drag and drop the **Cluster Status** widget t o the **Dashboard**. The following fields are s hown for each node:

	• **Status**: Shows the overall health status of the cluster. Green indicates healthy s tatus. Red indicates connectivity proble ms or high CPU or high memory utilizat ion. The status also shows red when a n ode is out - of - sync with the rest of th e cluster. • **Host Name**: Specifies the name of the host and IP address of the node. • **Zone**: The configured cluster zone. • **Server Role**: Indicates whether the clus ter node is a publisher or subscriber. If a ClearPass appliance is a subscriber in a cluster, all configuration options throu ghout the user interface specific to the publisher are grayed - out. • **Last Replication**: Date of the last replic ation. • **Status**: Indicates the status of the clust er node.
Device Category	To view the chart that shows the graph of a ll profiled devices categorized into the foll owing categories: • Access Points • Building Automation • Computer • Conflict ◦ Indicates a conflict occurred in the categorization of the device. • Datacenter Appliance • Game Console • Monitoring Devices • Network Boot Agents • Physical Security

	• Printer • Router • Server • Smart Device • Switch • Unknown ◦ **Indicates** devices that are not inclu ded in the Profiler database. • VOIP phone
Device Family	To view the device family of a particular d evice category, drag and drop the **Device Category** widget to the **Dashboard, then s elect the device category from the drop d own menu.** For example, selecting **Compu ter** would show that the device family is *Wi ndows*.
Device Insight Tags	To view the Device Insight tags, drag and d rop the **Device Insight Tags** widget to the Dashboard.
Endpoint Profiler Summary	To view a display that shows the number o f smart devices, computers, and unmanag ed devices, as well as the total number of devices defined by the Endpoint Profiler fo r this Policy Manager server, drag and drop the **Endpoint Profiler Summary** widget to t he **Dashboard**.
Failed Authentications	To view the table with the latest failed auth entications, drag and drop the **Failed Auth entications** widget to the **Dashboard**. Clicking on a row drills down to the **Acces s Tracker** page and shows failed requests s orted by timestamp, with the latest request displayed on the top.
Health Status	To view the graph of the healthy and unhea lthy requests over the past week, drag and drop the **Health Status** widget. • *Healthy requests* are the requests to wh ich the health state was deemed to be

	healthy based on the posture data sent from the client. • *Unhealthy requests* are the requests to which the health state was deemed to be *quarantined* (posture data received but health status is not compliant) or *unknown* (no posture data received). ◦ This includes RADIUS and WebAuth requests. The default data filters **Health Requests** and **Unhealthy Requests** are used to plot this graph. • Clicking on each circle on the line graph drills down to the **Access Tracker** page that shows the healthy and unhealthy requests for the last week.
Latest Authentications	To view the table with the latest authentications, drag and drop the **Latest Authentications** widget to the **Dashboard**. Clicking on a row in the table drills down to the **Access Tracker** page that shows requests sorted by timestamp with the latest request displayed on the top.
License Usage	To view the table that displays the **Available Count** and **Used Count** of the application licenses installed on the current Policy Manager server, drag and drop the **License Usage** widget to the **Dashboard**. • From the **Select Application** drop - down, select the Application License of interest. • Verify each license indicates the current support end date, as only supported customers can upgrade their system(s) to the 6.11 baseline.
MDM Discovery Summary	To view the charts that show the endpoints discovered, drag and drop the **MDM Discovery Summary** widget to the **Dashboard**.

	• The endpoints are displayed in separate charts based on the endpoint's operatin g system. • Clicking a chart drills down to the **Confi guration > Identity > Endpoints** page. T he results depends on the operating sy stem selected. ◦ For example, if you click the Android devices chart, you can view the list of only Android devices in the **Endp oints** page.
OnGuard Clients Summary	To view a display that shows the number o f Linux, macOS, and Windows OnGuard cli ents, as well as the total number of OnGu ard clients for this Policy Manager server, d rag and drop the **OnGuard Clients Summa ry** to the **Dashboard**.
Quick Links	To view the links to the following configurat ion tasks, drag and drop the **Quick Links** wi dget to the **Dashboard**: • **Start Configuring Policies** • **Manage Services** • **Access Tracker** • **Analysis and Trending** • **Network Devices** • **Server Manager** • **Policy Manager Guest** • **Policy Manager Onboard** • **Policy Manager Insight** • **Policy Manager Extensions**
Request Processing Time	To view the trend of total request processin g time, drag and drop the **Request Process ing Time** widget to the **Dashboard**.
Service Categorization	To view the bar chart with each bar repres enting a categorized Policy Manager servic

	e request, drag and drop the **Service Categorization** widget to the **Dashboard**. • Clicking on a bar drills down to the **Access Tracker** that shows the requests that were categorized into a specific service.
Successful Authentications	To view a table with the latest successful authentications, drag and drop the **Successful Authentications** widget to the **Dashboard**. • Clicking on a row in the table drills down to the **Access Tracker** page that shows successful requests sorted by timestamp, with the latest request displayed on the top.
System CPU Utilization	To view the CPU usage for the last 30 minutes, drag and drop the **System CPU Utilization** widget to the **Dashboard**. • The widget displays the CPU utilization time in minutes and percentage for System, User, and IO Wait time, indicated by color. • CPU utilization is presented in one - minute increments.
System Summary	To view the **Percentage Used** statistics for the following components, drag and drop the **System Summary** widget to **Dashboard**: • Main Memory • Swap Memory • Disk • Swap Disk
Authentication Source Status	The **Authentication Source Status** table indicates if an Active Directory or LDAP authentiation source is reachable by the primary server and all backup servers. Data in this table is updated every 15 minutes, thoug

h clicking the **Refresh** link at the top of thi s table automatically refreshes the data. Cl icking any row in this table opens the **Auth entication Sources** page where you can vi ew information on all available authenticati on sources.

Initial Login and Activation of the Policy Manager Platform License

Initial login and platform license activation enable access to Policy Manager features and prepare the system for use.

Initial Login and Activation

About this task

Upon initial login to a Policy Manager server and later, you are prompted to enter the Policy Manager Platform License Key. The Policy Manager Platform License provides a platform activation code that is installed on all the nodes in a Policy Manager cluster. The Policy Manager Platform License is the base-level license. Each Policy Manager server has one Policy Manager Platform License for the physical hardware. Virtual devices have a Policy Manager Platform License as well on a per-expected device level.

Additionally, ClearPass continues to display the Mandatory DoD Notice and Consent Banner until the usage conditions are acknowledged. Only when acknowledged can you log into ClearPass for further access to the ClearPass feature set.

NOTE

License keys are encoded for a specific application type (for example, *Platform*, *Onboard*, or *Access*). For this reason, you are not prompted to select the application type.

If you are upgrading to Policy Manager 6.11.x from Policy Manager 6.6.x or later, your existing Policy Manager License Key will be automatically converted to a Platform Activation Key (PAK). You will not need to do anything to make the conversion happen, and the Platform Activation Key is preactivated.

The Policy Manager licenses on each cluster node are converted to Policy Manager Platform Licenses. The Policy Manager Platform License provides a platform activation code that is installed on all the nodes in a Policy Manager cluster. For complete upgrade information, refer to theClearPass Release Notes. Additionally, a warning displays in

6.11 supported clusters if all nodes are not compliant with PAK license checks and are running version 6.11.

> **NOTE**
> You cannot have more than one Policy Manager Platform license installed on a Policy Manager node.

> **NOTE**
> You cannot upgrade to the 6.11 Policy Manager Platform without a valid PAK and the appropriate license(s) associated. Systems without support can continue to operate without service interruption, but are not able to upgrade to 6.11.

To specify the Policy Manager Platform License Key upon initial login:

Procedure

1. Navigate to the Policy Manager publisher: **https://x.x.x.x/tips/**

   where **x.x.x.x** is the IP address of the management interface defined for the server.

   > **NOTE**
   > View or reconfigure the management IP address using the appliance command-line interface (CLI) via serial console or SSH.

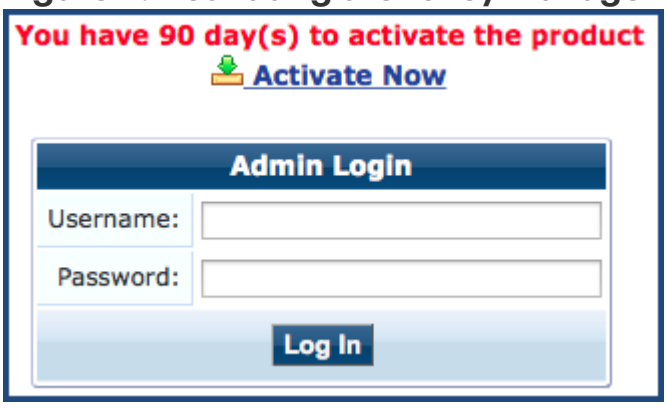
2. Log in to the Policy Manager 6.11 or later server. The Policy Manager End-User Software License Agreement dialog appears. The **Select Application** field is preset to **Policy Manager Platform**.

   Figure 1. Entering the Platform License Key

3. Enter the **Policy Manager Platform License Key**.
4. Click the check box for **I agree to the above terms and conditions**. The **Add License** button is now enabled.
5. Click **Add License**.

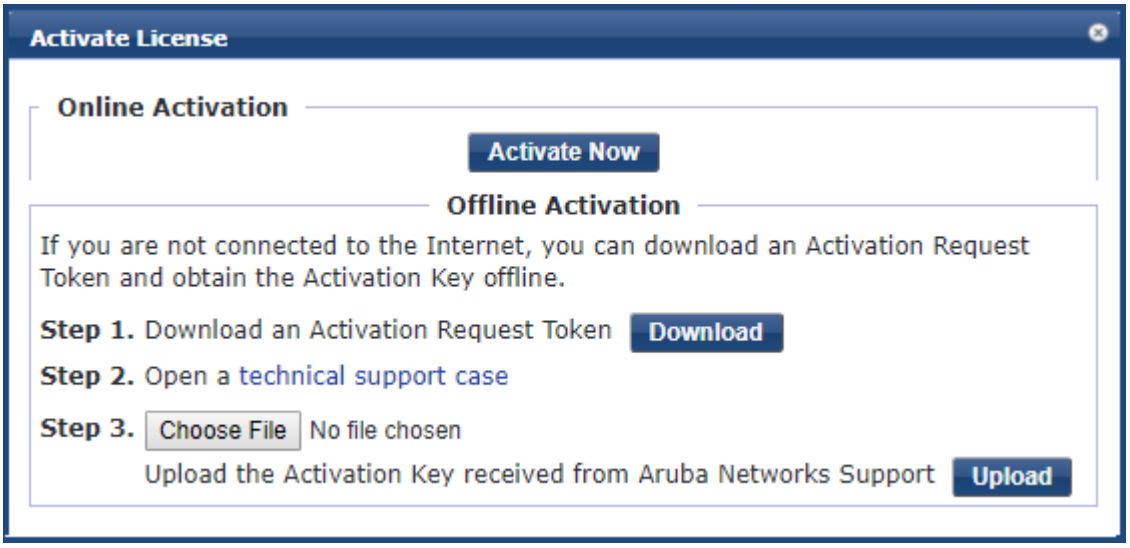
   The **Admin Login** screen appears with a message indicating that you have 90 days to activate the product and a link to activate the product.

Figure 2. Activating thePolicy ManagerNode



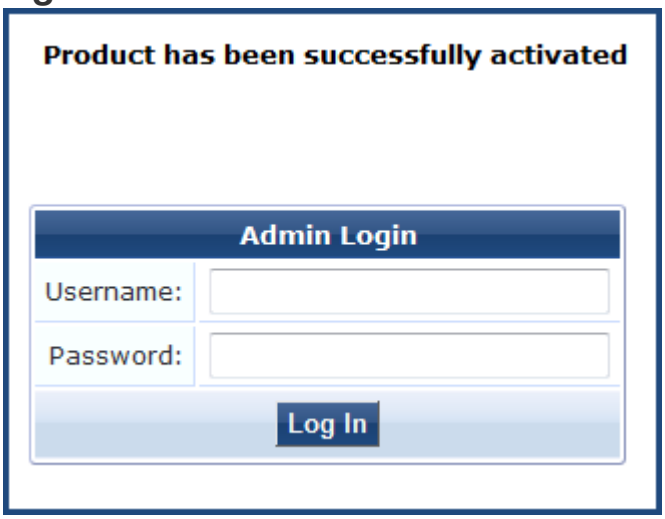
6. To activate the Policy Manager Platform License on this appliance, click the **Activate Now** link. The **Activate License** dialog opens:

Figure 3. Activating thePolicy ManagerPlatform License



7. If the Policy Manager server is connected to the Internet, click the **Activate Now** button. The message, "*Product has been successfully activated*" appears, and the **Admin Login** dialog is displayed.

Figure 4. Activation Successful: Admin Login Available



Logging in to the Policy Manager Server

Access to the Policy Manager server requires admin credentials and opens the landing page after initial sign-in prompts.

Procedure

1. Log in to the Policy Manager server with the following credentials:
   - **Username**: admin
   - **Password**: Enter the Policy Manager cluster password.

2. Click **Log In**. The *What's New in ClearPass Policy Manager* modal is displayed. This modal provides a list of new features for the release, a link to the release notes, and any changes of behavior that would be important to a system administrator.
3. If you don't wish to see the *What's New* modal when you log in to this server again, click the **Do not show again** check box.
4. Click the **Close** button.

   The **Policy Manager Landing Page** is displayed.

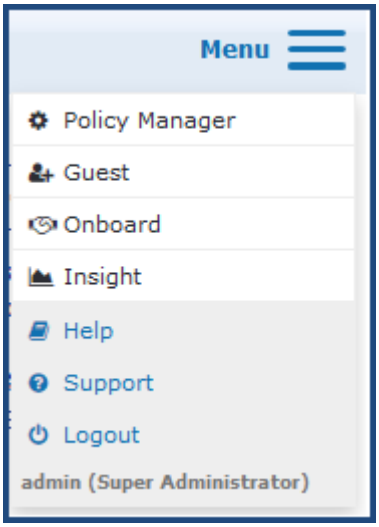
Customizing the Landing Page Layout

To customize the Landing Page layout to display the information you most want to see, drag and drop from the list of the Widget elements on Dashboard to one of the available Dashboard slots in the right pane.

ClearPass Menu

The Policy Manager Menu is available on the upper-right corner of every page in the Policy Manager user interface.

Figure 5. Policy ManagerMenu



From the Policy Manager Menu, you can:

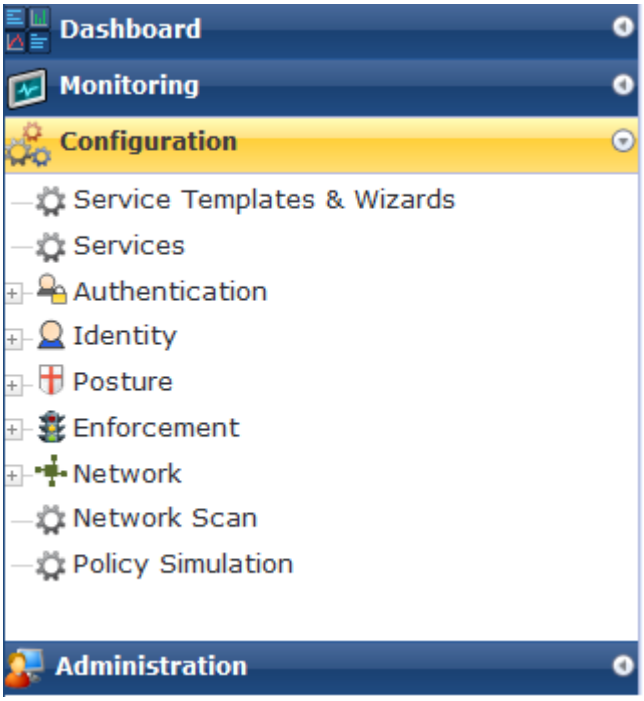
- Navigate to any of the Policy Manager modules: *Policy Manager*, *Guest*, *Onboard*, and *Insight* (Report tool).
- Bring up online Help for the current page of the Policy Manager user interface.
- Access the **Support** information: *Contact Support*, *Remote Assistance*, and *Documentation*.
- Log out of the current Policy Manager server.

Accessing Configuration Information

This section provides pointers to information on how to configure the primary configuration tasks in ClearPass Policy Manager. You can access all these configuration tasks via the **Policy Manager Configuration** menu.

To access the **Policy Manager Configuration** menu, select **Configuration**. The **Policy Manager Configuration** menu opens (see Figure 1):

Figure 1. Policy ManagerPolicy Manager Configuration Menu



Service Templates & Wizards

The ClearPass Policy Manager **Service Templates & Wizards** page provides the ability to create templates for services that will allows you to define baseline policies and require specific data when you create services.

For more information, see Configuring Common Services Using Service Templates .

Services

The **Services** page provides options to add, modify, and remove a service. For more information, refer to the following sections:

- Modifying and Managing Services

  This page also shows the current list and order of services that ClearPass Policy Manager keeps track of during authentication and authorization.

Authentication and Authorization

The **Authentication** page provides options to configure the following components:

- Adding and Modifying Authentication Methods
- Adding and Configuring Authentication Sources
- Modifying Authentication Methods or Sources for an Existing Service

Identity

The **Identity** page provides options on the settings required to configure ClearPass Policy Manager Identity settings. For more information, refer to the following sections:

- Configuring Single Sign-On
- Managing Local Users
- Adding and Modifying Endpoints
- Managing Static Host Lists
- Adding and Modifying Roles
- Adding and Modifying Role-Mapping Policies

Posture

The **Posture** page provides options to configure posture policies and audit servers. For more information, refer to the following sections:

- Posture Architecture and Flow
- Creating a New Posture Policy
- Configuring Audit Servers

Enforcement

The **Enforcement** page provides options to configure the Enforcement Profiles globally and to reference in an enforcement policy that is associated with a service.

For more information, refer to the following sections:

- Configuring Enforcement Profiles
- Configuring Enforcement Policies

Network

The **Network** page provides options to configure the Network Access Device (NAD) that sends network access requests to Policy Manager using the supported RADIUS, TACACS+, or SNMP protocol. The NAD in this context is usually a mobility controller or a switch.

For more information, refer to the following sections :

- Adding a Network Device
- Adding and Modifying Device Groups
- Adding and Modifying Proxy Targets
- Configuring the Ingress Event Source

Network Scan

The **Network Scan** page provides options to configure the following elements:

- SNMP Configuration: See SNMP Credentials Configuration.
- SSH Configuration: See SSH Credentials Configuration.
- WMI Configuration: See WMI Credentials Configuration.

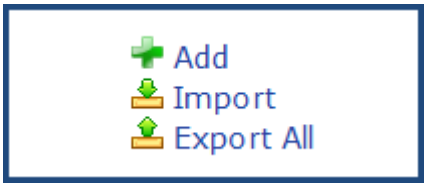
Policy Simulation

The **Policy Simulation** page provides options to configure the **Policy Simulation** utility that applies a set of request parameters as input against a given policy component.

- For more information, refer to Configuring Policy Simulation.

Importing and Exporting Information

The option to import information into or export information from a Policy Manager server is available from every top-level page in the Policy Manager user interface. The **Add**, **Import**, and **Export All** options are displayed in the top-right corner of the configuration and administration pages:



NOTE

There are some workflows supported in Policy Manager where endpoints without a real MAC address are added to Policy Manager. (These workflows can include IP subnet-based scans and integration with ClearPass Device Insight, which may send endpoints discovered by span traffic.)

When information is exported from Policy Manager, Policy Manager ignores any endpoints that have a generated / synthetic MAC address because their real MAC address is unavailable. This is because a MAC address helps uniquely identify an endpoint and so, without having this unique value, its not useful to share information Policy Manager has for an IP address alone.

Importing Information Into Policy Manager

About this task

Policy Manager allows you to import configuration and administration-related information. This information is stored as an XML file, which can be password protected.

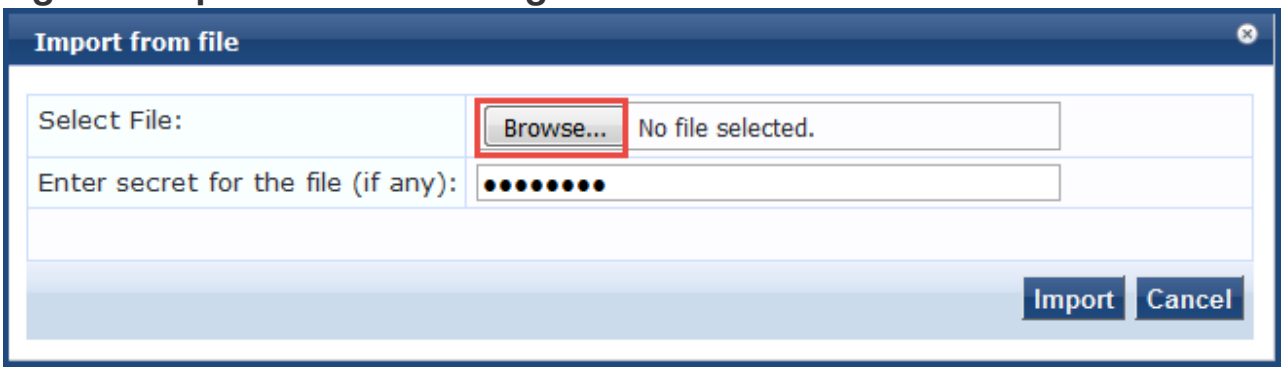
> **NOTE**
> To view ClearPass API information, access the ClearPass Guest WebUI and navigate to **Administration > API Services > API Explorer**.

To import information into the current Policy Manager server:

Procedure

1. Click the Import link. The **Import from file** dialog opens.

Figure 1. Import From File Dialog



2. Click Browse.
3. Browse to the file you want to import.
   - Be sure to select an XML file that is in the correct format.
4. **Enter secret for the file (if any)**: If you entered a secret key to encrypt the exported file, enter the same secret key to import the device back into this Policy Manager server.
5. Click Import.

Exporting Information From Policy Manager

About this task

Policy Manager allows you to export configuration and administration-related information from the current Policy Manager server to an XML file. You can set this file to be password protected (see Table 1 for details).

To export multiple items, select the check boxes in the rows of the specific items that you want to export.

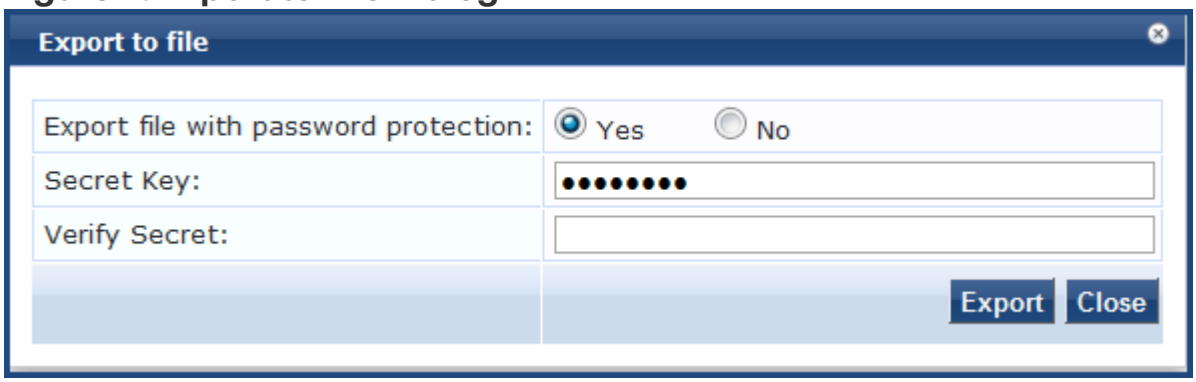
To export information from the current Policy Manager server:

Procedure

1. Click the **Export All** link.

   The **Export to File** dialog opens.

Figure 2. Export to File Dialog



2. Specify the export parameters based on Table 1.

   Specify the **Export to file** parameters as described in Table 1:

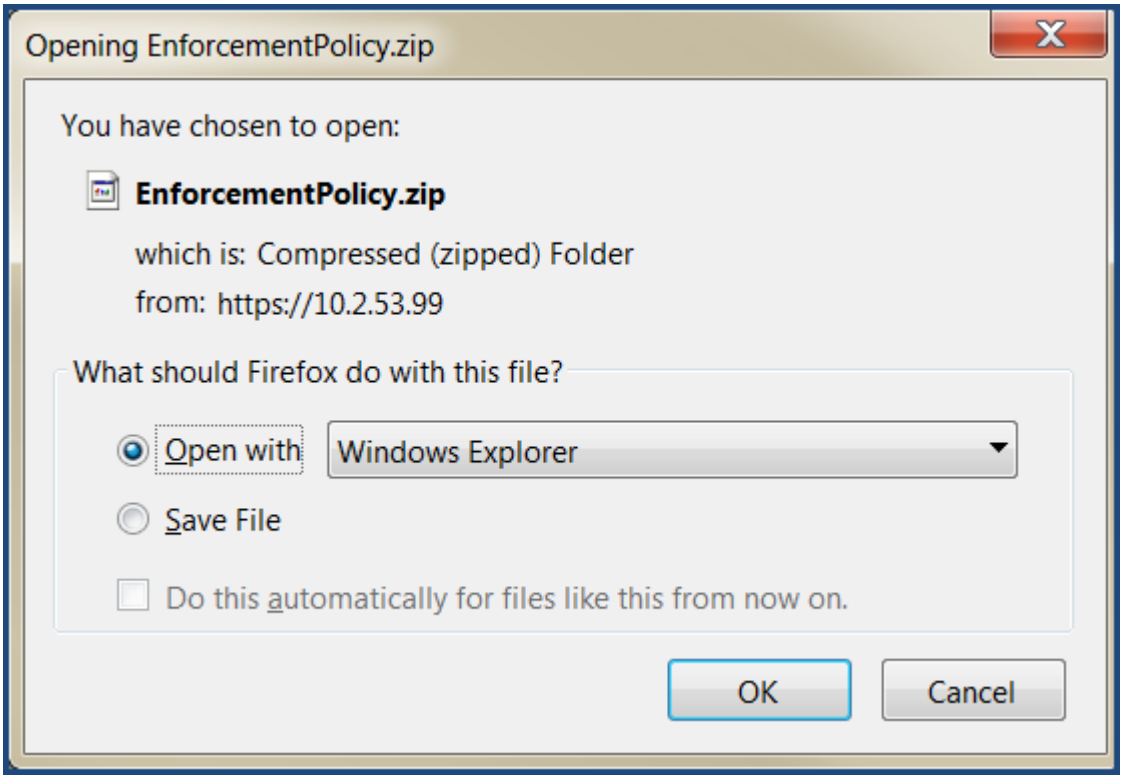
Table 1. Export to File Dialog Parameters

Parameter	Action/Description
Export file with password protection	To export the file with password protectio n, choose **Yes**.
Secret Key/ Verify Secret	You can choose to encrypt the exported data with a key. This protects data such a s shared secret from being visible in the exported file. To import it back, you speci fy the same key with which you exported Enter the secret key. Then reenter the sec ret key.

3. Click **Export**.

   An export dialog named for the selected Policy Manager function opens:

Figure 3. Export XML File to Zip File Dialog



4. Specify to open the zip file or save the XML file to your system.
5. Click **OK** to proceed.

Export Considerations

The XML file generated from an export operation has a specific layout that is unique for each function in the Policy Manager user interface. If you import an XML file with an incorrect layout (usually because it's from the wrong function), it will be rejected. Aruba recommends that you note the message displayed when the XML operation is incorrect and what the issue is—typically either because it's from the wrong export file or because it was modified incorrectly before being imported.

Accessing System Information

About this task

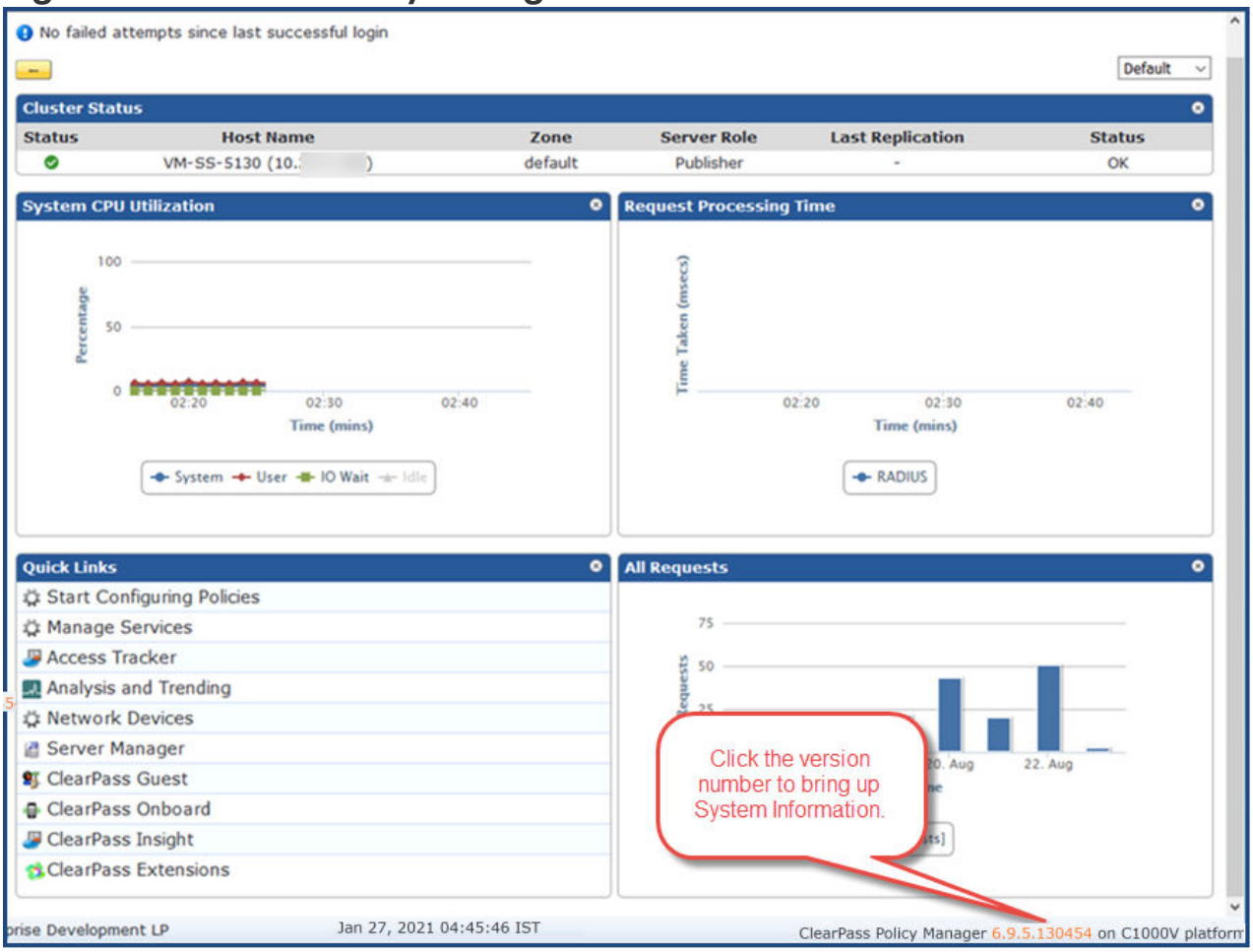
Policy Manager provides a summary of the current Policy Manager server's system status, hardware information, and the software patches that have been installed on the server.

To access the Policy Manager system information:

Procedure

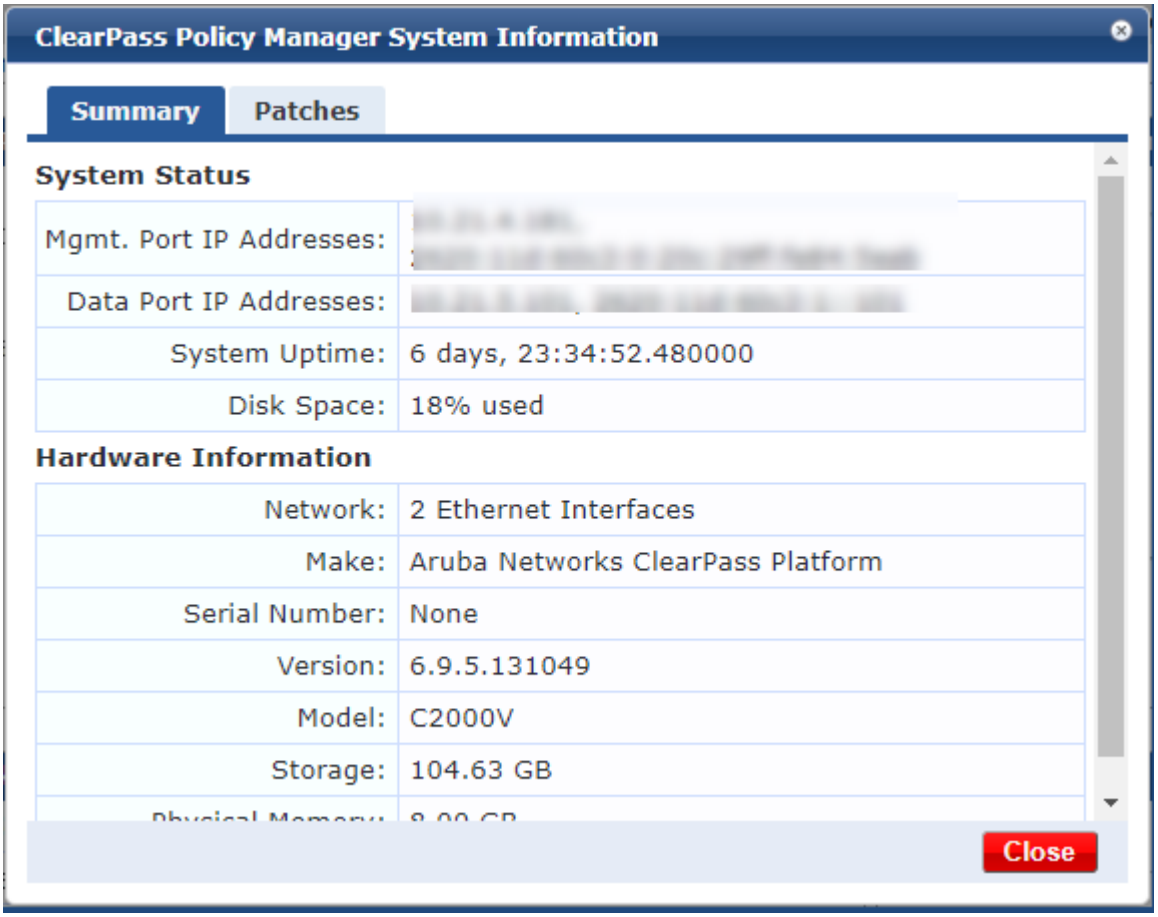
1. Navigate to the **Dashboard**. The Policy Manager version number and hardware platform are displayed in the lower-right corner of the **Dashboard** window.

Figure 1. ClearPass Policy Manager Dashboard



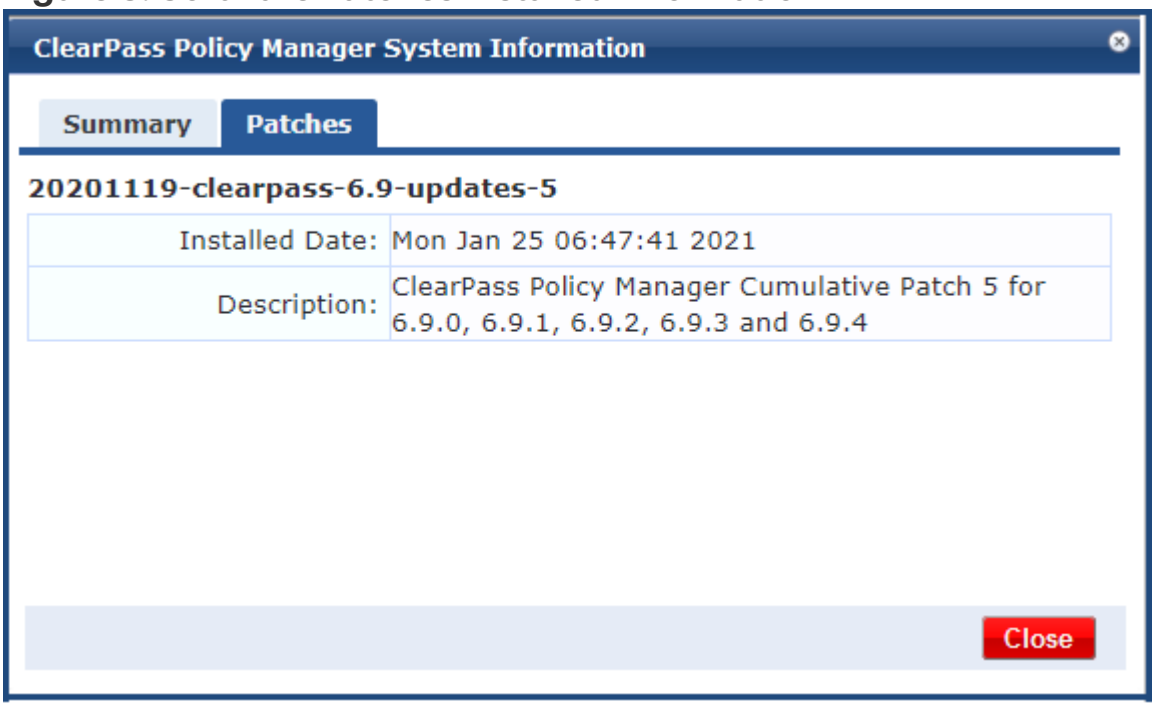
2. Click the version number link. The Policy Manager **System Information** window opens, displaying the system status and hardware information for this server.

Figure 2. ClearPass Policy Manager System Information Window



3. To view the software patch information, click the **Patches** tab. The **Patches** tab lists when each patch was installed on this Policy Manager server.

Figure 3. Software Patches Installed Information



4. Click **Close** to return to the **Dashboard**.

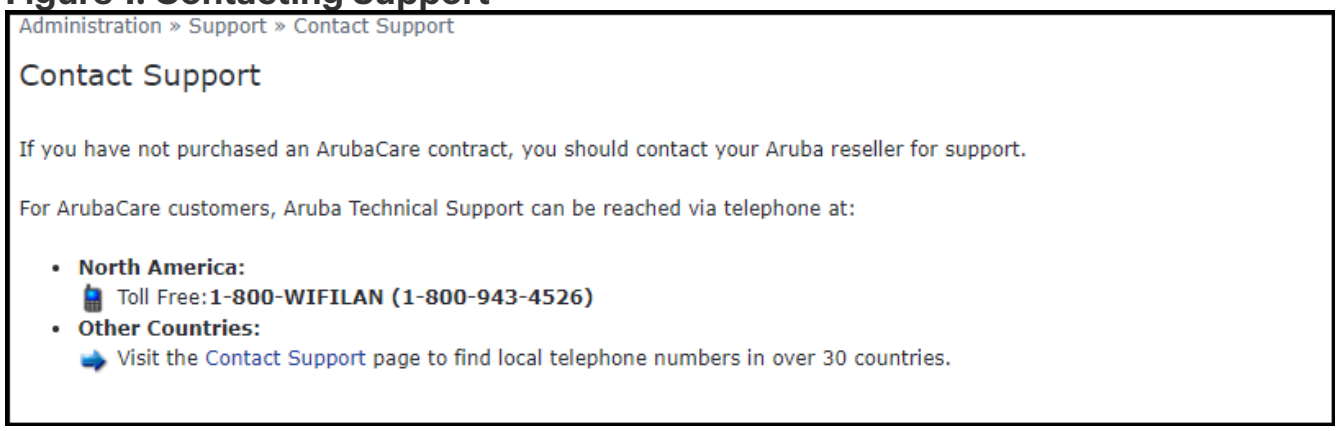
Support Services

The **Administration > Contact Support** page provide information for contacting Support, setting up a Remote Assistance session, and viewing Policy Manager documentation:

Contacting Support

The **Administration > Support > Contact Support** describes how to contact ArubaCare.

Figure 1. Contacting Support



Contacting North America Support

To contact ArubaCare Support in North America (United States and Canada), call the following number:

1-800-943-5426

Contacting ArubaCare Support in Other Countries

About this task

You can either call ArubaCare in the country of interest, contact Support, or create a Support case by clicking the **HPE Networking Support Portal** website link.

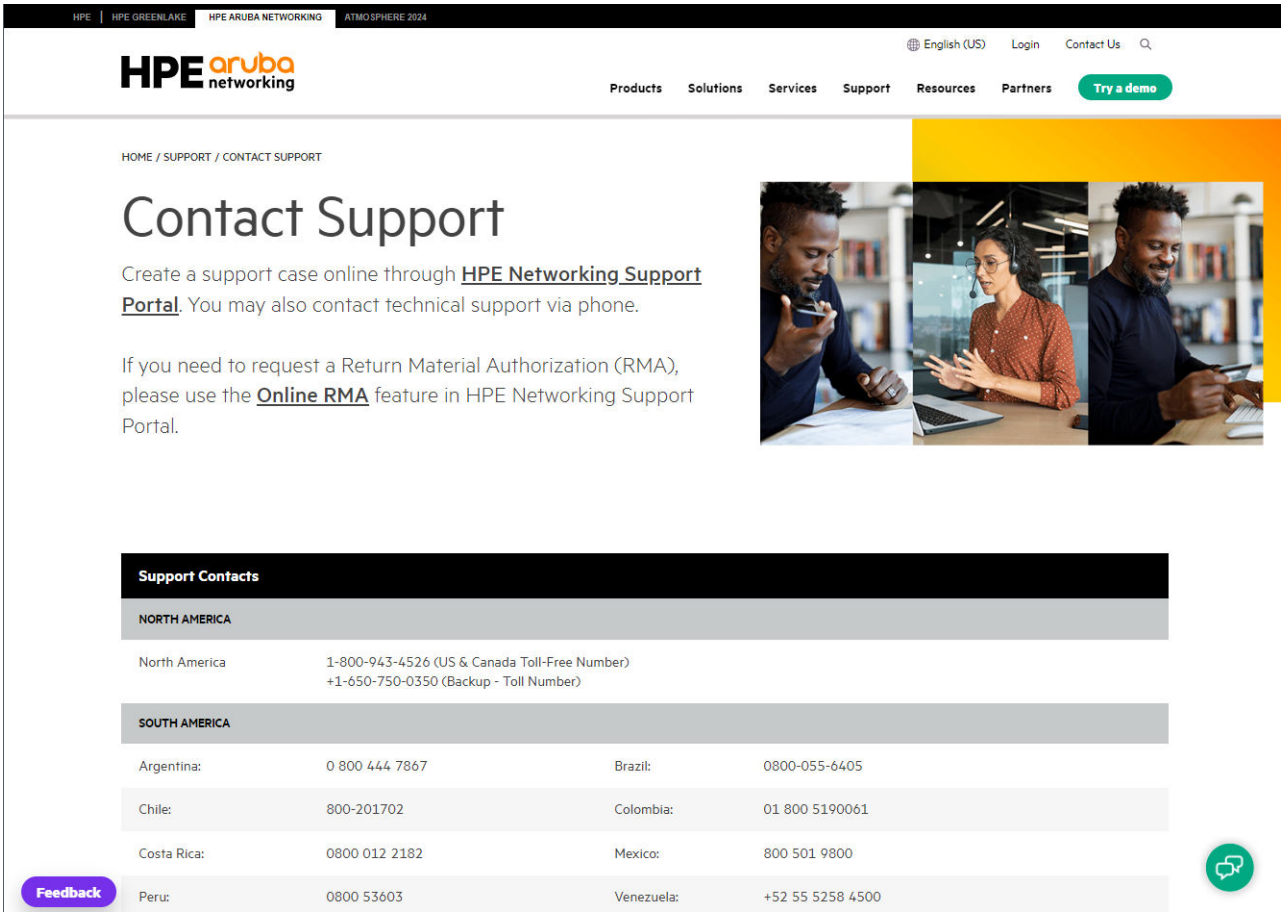
To contact ArubaCare Support in countries other than the United States and Canada:

Procedure

1. Click the **Contact Support** link.

   The **Contact Support** page for other countries opens.

   Figure 2. Contacting Support in Other Countries



2. Scroll down to find the country of interest.

Remote Assistance

The **Remote Assistance** feature allows a Support engineer to log in remotely using Secured Shell (SSH) to a customer's Policy Manager server and perform the following tasks:

- View and operate the Policy Manager user interface to debug any issues the customer is experiencing.
- Perform proactive monitoring of the Policy Manager server.

Remote Assistance Process

Remote Assistance lets administrators grant temporary support access and provides an alternate CLI when the user interface is unavailable.

About this task

The Remote Assistance process is as follows:

Procedure

1. The Administrator sets up a Remote Assistance session.

2. The Support contact receives an email with instructions and credentials to log in to the remote system.
3. The Remote Assistance session is terminated at the end of the specified duration.

> **NOTE**
>
> You can terminate a session before its stipulated duration.

4. if the ClearPass Policy Manager user interface at the customer site is inaccessible, you can configure a Remote Assistance session through the followilng Policy Manager CLI commands:
   - `system start-rasession`
   - `system status-rasession`
   - `system terminate-rasession`

Setting Up a Remote Assistance Session

About this task

Optionally set up a Remote Assistance session on a Policy Manager server, in either a single node or cluster setup.

Sessions can be saved and deleted only from the Publisher node. Sessions can be terminated from both Publisher and Subscriber nodes.

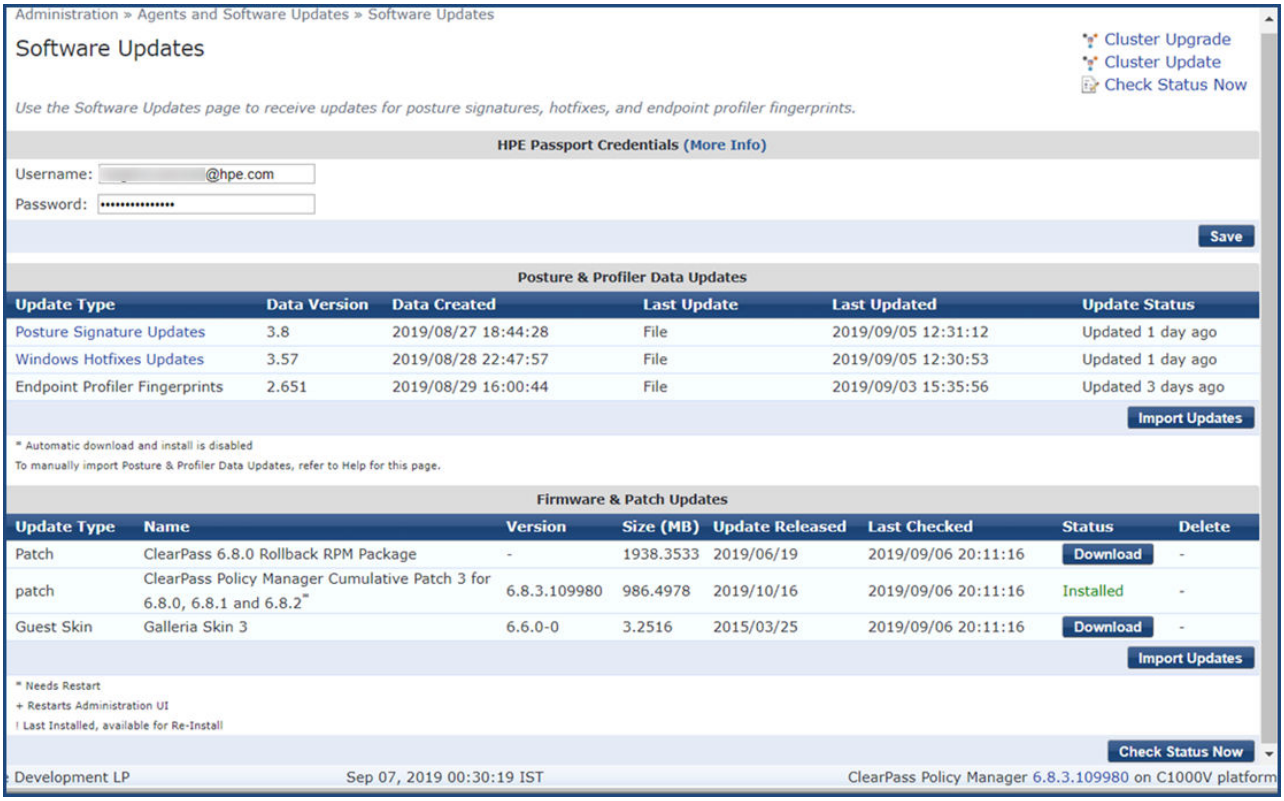
To set up a Remote Assistance session:

Procedure

1. Prior to setting up a Remote Assistance session, you must enter valid *HPE Passport Credentials*. To do so, navigate to **Administration > Agents and Software Updates > Software Updates**.

   The **Software Updates** page opens.

   Figure 3. Software Updates Page

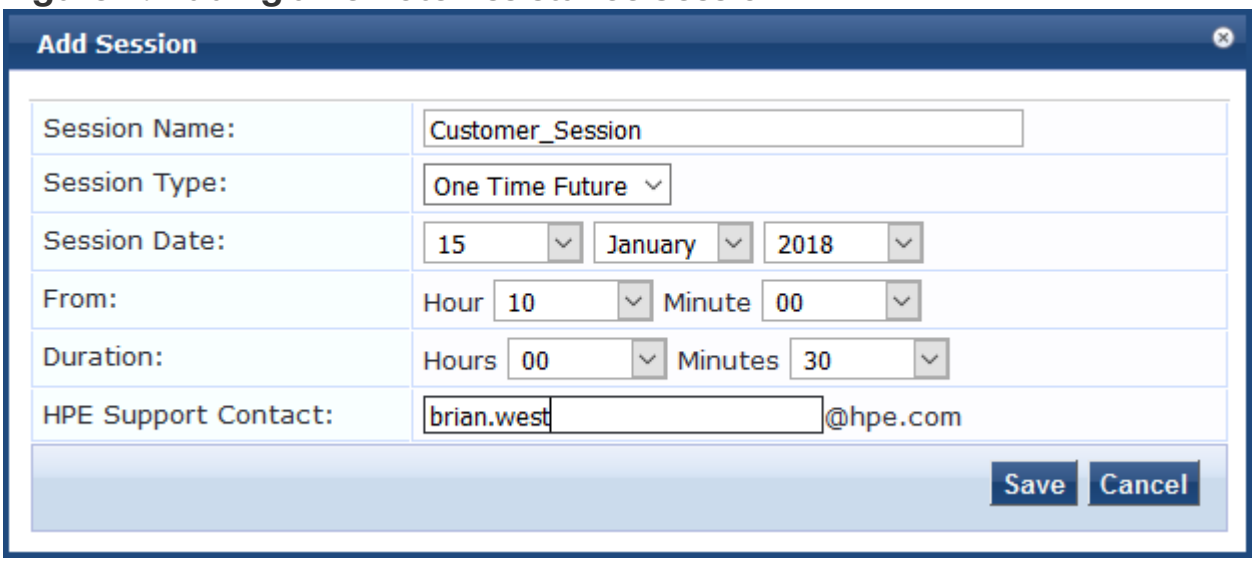


2. Enter the **HPE Passport Credentials**, then click **Save**.
3. Navigate to **Administration > Support > Remote Assistance**.

   The **Remote Assistance** page opens.
4. Click the **Add Session** link.

   The **Add Session** dialog opens:

Figure 4. Adding a Remote Assistance Session



5. Specify the **Add Session** parameters as described in the following table:

   Add Session Parameters

Parameter	Action/Description
Session Name	Enter the name of the Remote Assistance session.
Session Type	Specify the **Session Type**: • **One Time Now** • **One Time Future**: Initiates a session a t a specified date and time. • **Weekly**: Initiates a session on a specif ied weekday at the selected time. • **Monthly**: Initiates a session on a speci fied day of every month at the selecte d time.
Duration	Specify the duration of the Remote Assis tance session in hours and minutes. The "session begin" time is the time relati ve to the Policy Manager server's time, an d is specified in a 24 - hour format.
Status	Indicates the session state. Available stat es are: • Saving • Scheduled • Initiated • Running • Terminated • Failed
HPE Support Contact	Enter the email ID of the Hewlett Packard Enterprise (HPE) Support contact.

6. Click **Save**.

   You return to the **Remote Assistance** page where the new session has been added.

Accessing ClearPass Documentation

About this task

The Policy Manager **Documentation** page includes links to frequently-used sections of the Policy Manager Online Help system.

This page also provides links to the PDF version of the Policy Manager User Guide, the OnGuard Agent Support Charts for Plugin Version 2.0 (V4 SDK), and the cipher suites that are supported in Policy Manager.

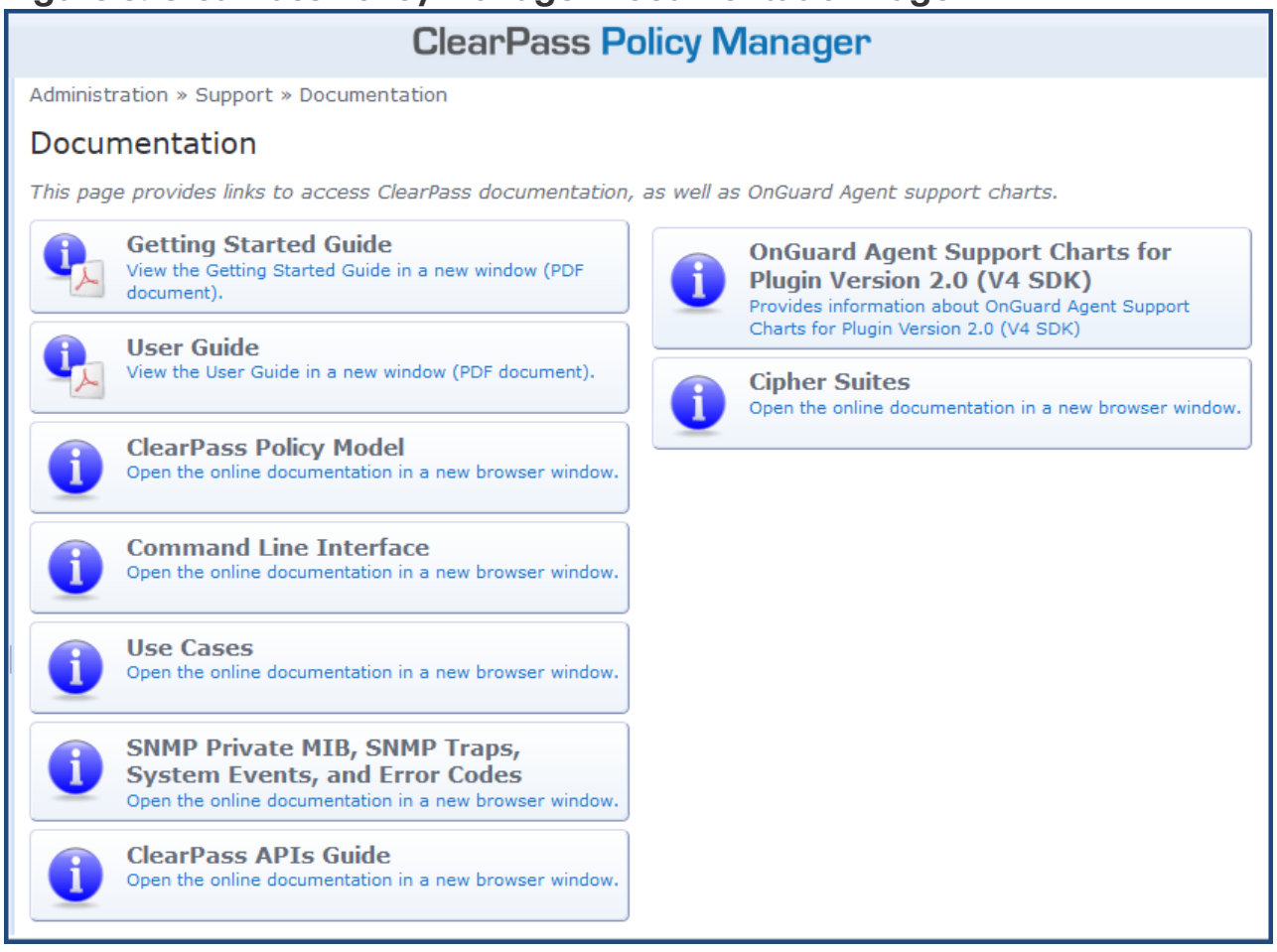
To access the **Documentation** page:

Procedure

1. Navigate to **Administration > Support > Documentation**.

   The **Documentation** page opens.

Figure 5. ClearPass Policy Manager Documentation Page



2. Select the specific ClearPass documentation you wish to view.

   On some browsers (for example, *Google Chrome*), the selected documentation displays on your screen. On other browsers (for example, *Mozilla Firefox*), the selected documentation is downloaded in PDF format to the local *Downloads* folder.

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USA

Services

The **Services** page shows the current list and order of services that ClearPass Policy Manager follows during authentication and authorization. You can use the configured default service types or you can add additional services. Policy Manager provides a set of service templates to help you create services for common use cases. Other service types that are used less frequently can be configured either manually or with a service wizard that guides you through the manual steps. The three most common service types, Aruba 802.1X Wireless, 802.1X Wireless and 802.1X Wired, can be configured using all three methods.

This chapter describes the following topics:

- Policy Manager Service Components
- Viewing the List of Services

- Configuring Common Services Using Service Templates
- Configuring Services Manually or with Wizards
- Modifying and Managing Services

Subtopics

Policy Manager Service Components
Viewing the List of Services
Adding Services
Configuring Common Services Using Service Templates
Configuring Services Manually or with Wizards
Modifying and Managing Services

Policy Manager Service Components

The network devices or other entities that need authentication and authorization services view Policy Manager as a RADIUS, TACACS+, or HTTP/S based authentication server. However, Policy Manager's rich and extensible policy model allows it to broker security functions across a range of existing network infrastructure, identity stores, health/posture services, and client technologies within an enterprise. Services are the highest level element in the Policy Manager's policy model. Services have two purposes:

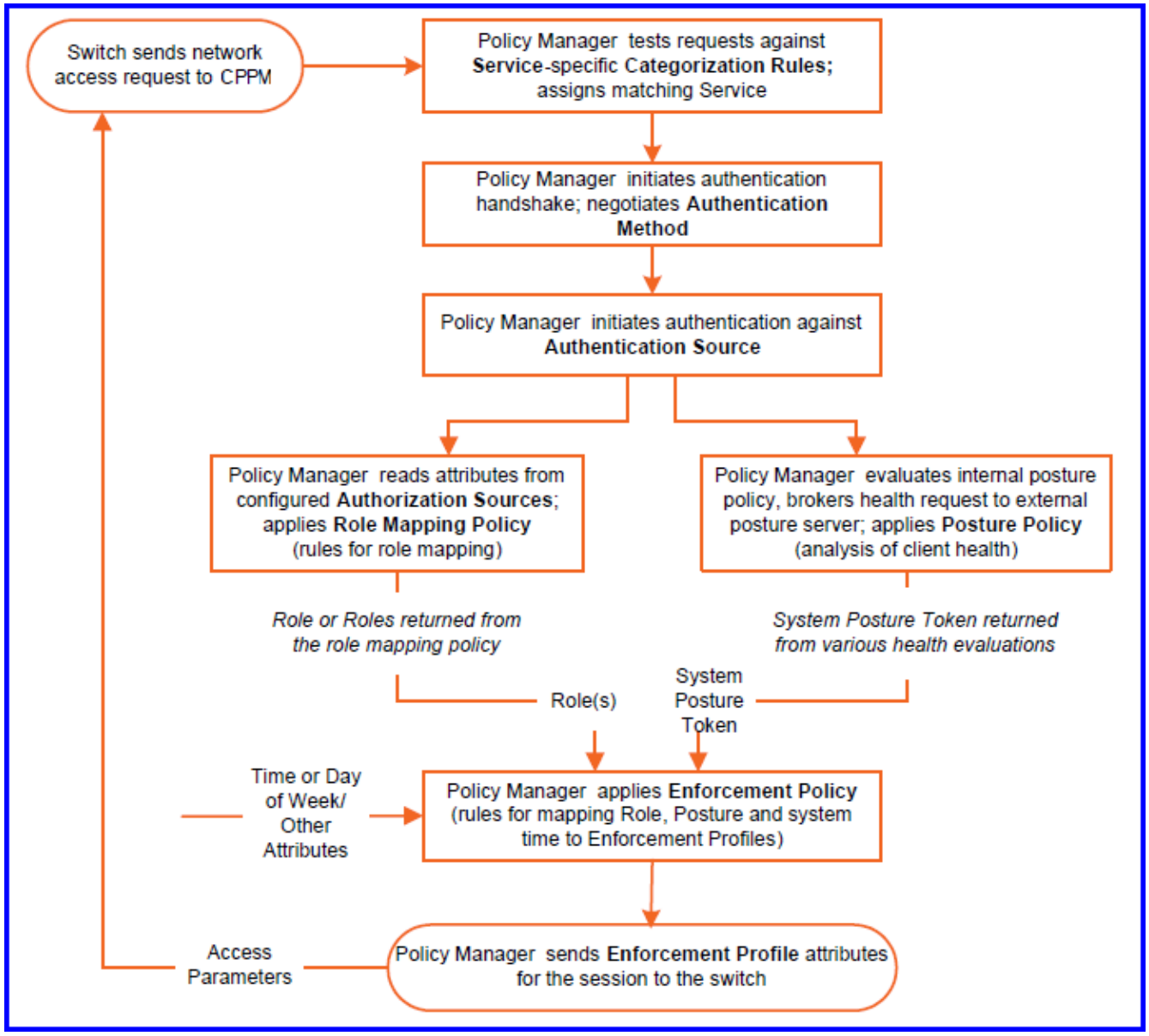
- Unique categorization rules (per service) enable Policy Manager to test access requests (requests) against available services to provide robust differentiation of requests by access method, location, or other network vendor-specific attributes.

> **NOTE**
>
> Policy Manager is shipped with a number of basic service types configured. You can extend these service types to copy them and use as templates, import other service types from another implementation (from which you have previously exported them), or develop new services from scratch.

- By wrapping a specific set of policy components, a service can coordinate the flow of a request from authentication to role and health evaluation to determine the enforcement parameters for network access.

The following figure and table illustrate the basic Policy Manager flow of control and its underlying architecture:

Figure 1. Generic Policy Manager Service Flow of Control



The following table describes the Policy Manager service components:

Table 1. Policy Manager Service Components

Component	Service: Component Ratio	Description
A - Authentication Method	Zero or more per service	Specifies the E s the following • **EAP, tunne** • **EAP, non -** • **Non - EAP,** • **MAC_AUTH** the MAC_AU

Component	Service: Component Ratio	Description
		◦ perform t (and n ◦ ensures e Some services cases, Policy M **NOTE** The E Policy
B - Authentication Source	Zero or more per service	An authenticat fies an identity. • Microsoft A • RSA or othe • SQL databa • Static host
C - Authorization Source	Zero or more per service, and one or more authentication source	An authorizatic ibutes you wan r supports the • Microsoft A • RSA or othe • SQL databa
C - Role Mapping Policy	Zero or one per service	Policy Manage s to role(s). All no rules match Some services rently: • For MAC - b an authenti udit rules a
D - Internal Posture Policies	Zero or more per service	An internal pos Posture rule co es.

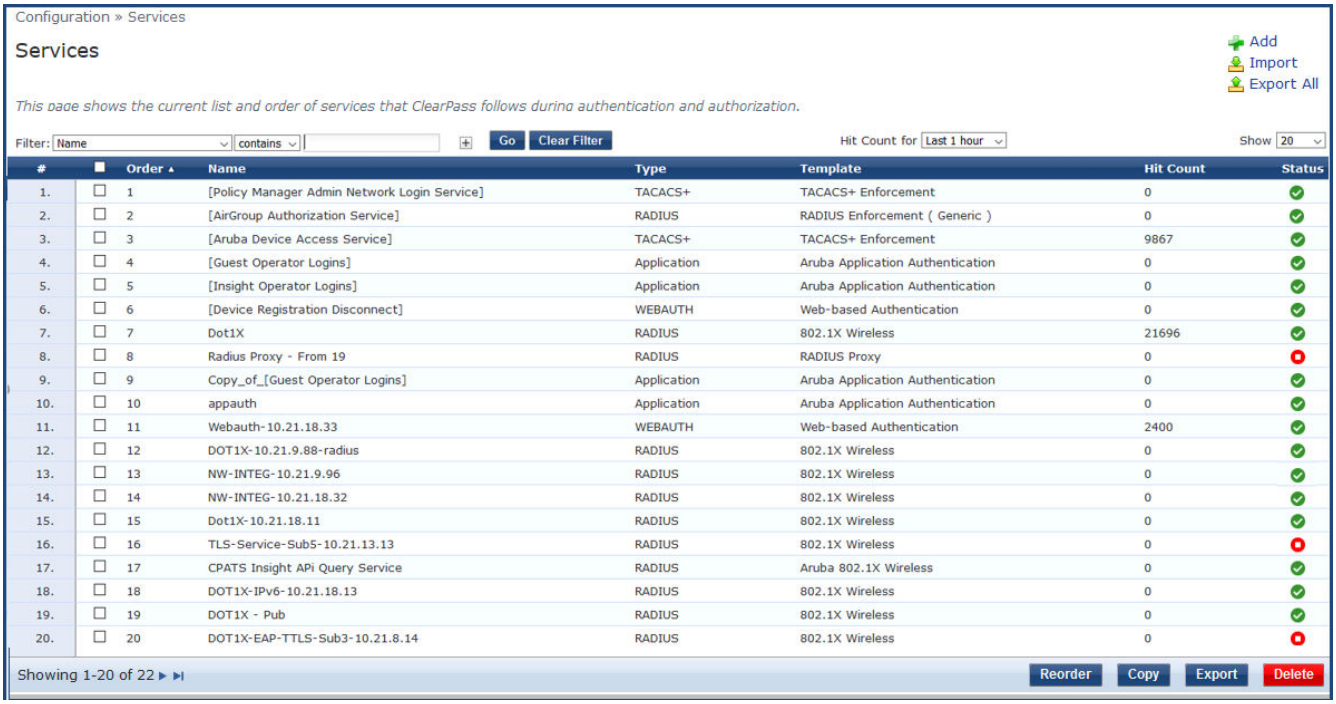
Component	Service: Component Ratio	Description
E - Posture Servers	Zero or more per service	Posture servers ntials. These p s, not by intern Currently, Polic • HCAP • RADIUS • GAMEv2
F - Audit Servers	Zero or more per service	Audit servers e nnot respond t uthentication r rver. In addition to r map results fro
G - Enforcement Policy	One per service (mandatory)	Policy Manage icy rules to retu or more match
H - Enforcement Profile	One or more per service	Enforcement p ion. Policy Man

Viewing the List of Services

The **Services** page shows the current list and order of services that ClearPass Policy Manager follows during authentication and authorization. You can use the configured default service types or you can add additional services.

To view the list of services on the current Policy Manager server, navigate to **Configuration > Services**. The **Services** page opens, which lists the current list and order of services that Policy Manager follows during authentication and authorization:

Figure 1. Services Page



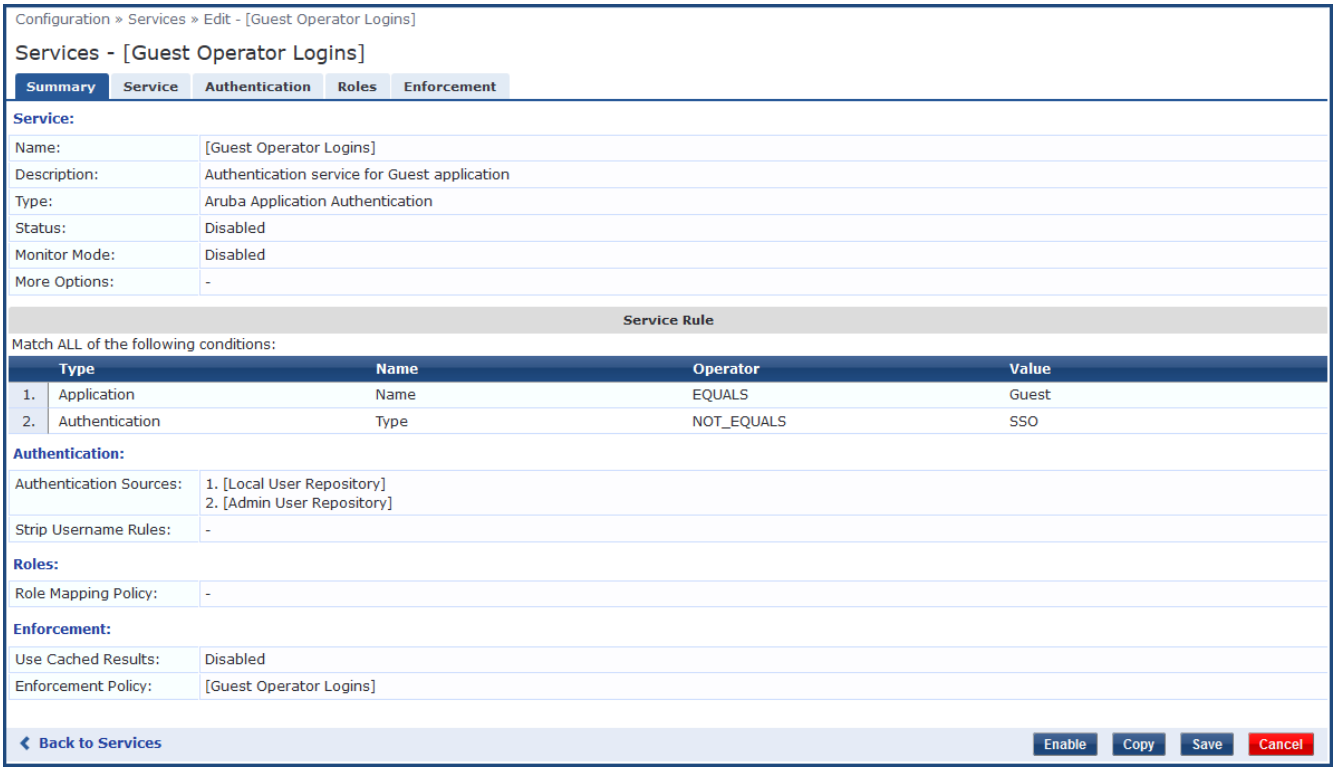
NOTE

Services included in square brackets "[ ]" indicate default services.

If your deployment has Insight enabled, this page also includes the **Hit Count** column, which displays the frequency at which policies are matching specific services during authentication. The **Hit Count For** menu at the top of the page allows you do view hit count numbers for the previous hour, day, three days, or week.

To view a service's details, select the **service of interest**. As shown in Figure 2, the **Edit Services** page opens to the **Summary** page, which provides the detailed information about the selected service's configuration.

Figure 2. Viewing Details for an Individual Service



For information about additional common service tasks, refer to Modifying and Managing Services.

Adding Services

To add a new service, you can define an entirely new service, or create a new service based off of a copy of an existing service.

Adding a New Service

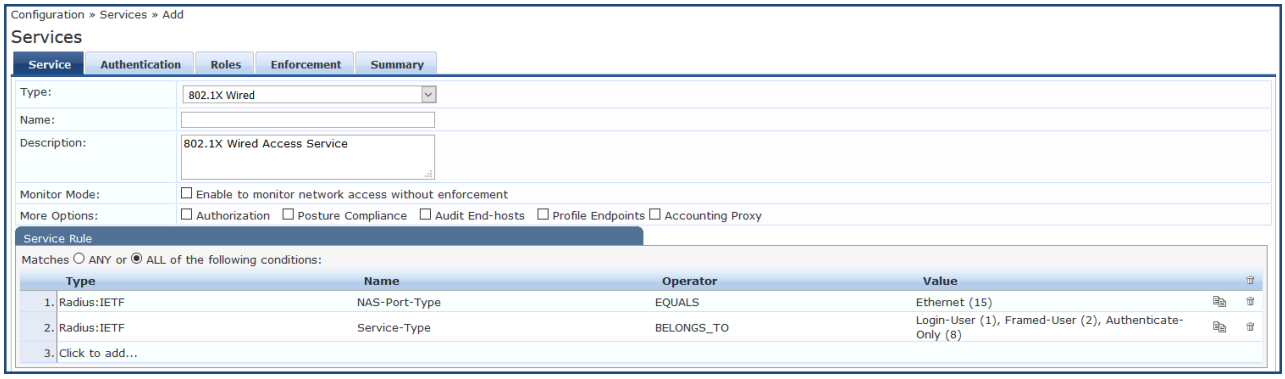
About this task

To add a new service to the Policy Manager server:

Procedure

1. Navigate to the **Configuration > Services** page. Alternatively, you can go to the **Configuration** > **Service Templates & Wizards** page.
2. From the **Services** page, click **Add**. The **Add Services** page opens.

Figure 1. Add Service Page



3. Configure the **Service Page > General** parameters as described in the following table:

Table 1. Service Page > General Tab Parameters

Label	Action/Description
Type	Select the desired service ty st. To create new services, you ervices for use as is or as ter a new service.
Name	Enter the name or label for t reate.
Description	Enter the description that pr on to identify the service. Th
Monitor Mode	Optionally, check the **Enable** **ess without enforcement** ch ation and health validation e tween the endpoint and Polic nforcement. In **Monitor Mod** (and associated attributes) a ice. Policy Manager also allows P **ng > Policy Simulation**), whe est the results of a particular mponents.
More Options	Select any of the available ch onfiguration tabs for those o

Label	Action/Description
	k boxes varies based on the t cted and may include one or • **Authorization**: Select an the drop - down list to ad e **Add new Authenticatio** ew source. • **Posture Compliance:** Sel he drop - down list to adc ew policy by clicking the sture token. Specify whet diation of non - complian led, then enter the Remec fy the **Posture Server** fro dd a new server by clickir **erver** link. • **Audit End - hosts:** Select uilt - in or customized. Re ervers for audit server co type of service, you can p ◦ Always ◦ When posture is not a ◦ For MAC address auth **MAC authentication r** you can perform an au *s only* or *For unknown* *end hosts*. Known end e clients that are foun urce(s) associated wit ◦ Performing an audit o ous task, which means ed only after the MAC s been completed anc n IP address through D lts are available, Policy es on the network dev g ways:

Label	Action/Description
	◦ **No Action:** The audit c the network device aft ◦ **Do SNMP bounce:** Thi ch port or forces an 8(both done using SNMI gers a new 802.1X or M t by the client. If the au posture token and attr s client in its cache, it attributes to Policy Ma ◦ **Trigger RADIUS CoA a** a RADIUS CoA comma by Policy Manager. • Optionally configure **Prof** r more endpoint classifica down list, then select the an also create a new actic **ew RADIUS CoA Action** li
Service Rule	When working with service ru he following namespace dict • **Application**: The type of a e. • **Authentication**: The Auth sed for this service. • **Connection:** Originator a Src - Port), Destination ac , Dest - Port), and Protocc • **Device:** Filter the service ce type, vendor, operatin roller ID. • **Date:** Time - of - Day, Day - Year. • **Endpoint**: Filter based on h as enabled/disabled, de ore.

Label	Action/Description
	• **Host**: Filter based on host erAgent, CheckType, Unic stalledSHAs. • **RADIUS:** Policy Manager ndor - specific namespac uishes vendor - specific F the notation RADIUS:venc ditional suffix for a particu onary for a vendor - spec avigate to **Administratior Import**. ◦ The notation **RADIUS:** attributes defined in R FCs. As the name sug is only available if the • **Any other supported nan** g and Namespaces for an aces and their descriptior

Creating a Service Template by Copying an Existing Service

About this task

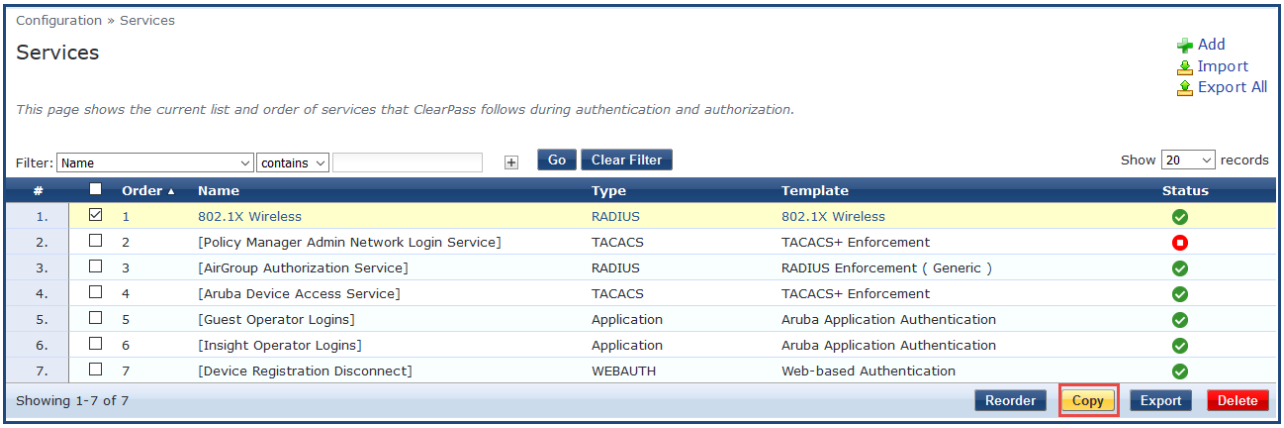
> **NOTE**
> You can perform a service-copy operation only on a publisher.

To create a service template by making a copying an existing service:

Procedure

1. From the **Services** page, select the check box by the service you wish to copy.

Figure 2. Copying a Service



2. Click **Copy**. Policy Manager displays the message: *Service Copy_of_* <service_name> *has been added*.

Configuring Common Services Using Service Templates

ClearPass Policy Manager Service templates provide a way to step through the service-creation process for the most common use cases, allowing you to easily create services and configure their specific components, such as role-mapping policies, enforcement policies, and associated network devices. Other service types that are used less frequently can be configured manually, or using the service wizards.

Using Templates to Create Policy Manager Services

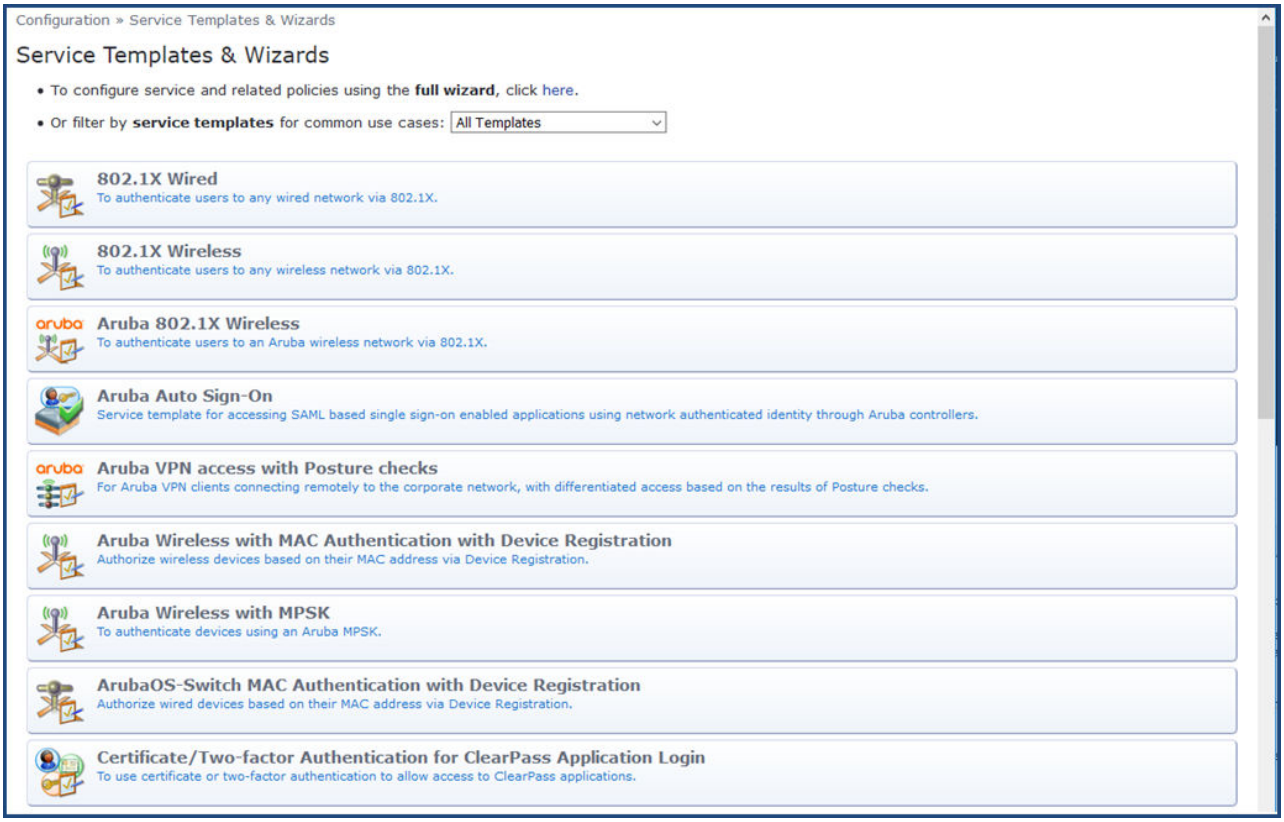
About this task

You can use templates to create services for common use cases for by defining baseline policies and required data:

Procedure

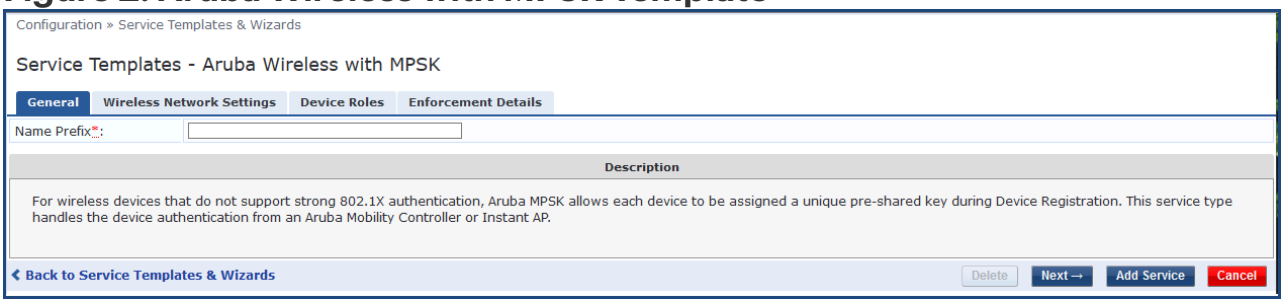
1. Navigate to the **Configuration > Service Templates & Wizards** page. The following page opens:

Figure 1. Service Templates & Wizards Page (Partial View)



2. Select the desired service template. The configuration dialog for the selected service template opens, as shown in the following example:

Figure 2. Aruba Wireless with MPSK Template



3. Fill in the various fields that are presented in the templates—Policy Manager then creates the configuration elements that are needed for that particular service.

List of Provided Service Templates

Refer to the following descriptions of the Policy Manager service templates for configuration details:

- 802.1X Wired, 802.1X Wireless, and 802.1X Wireless Service Template
- Aruba Auto Sign-On Service Template
- Aruba VPN Access with Posture Checks Service Template

- Aruba Wireless with MAC Authentication with Device Registration Service Template
- Aruba Wireless with MPSK Service Template
- ArubaOS-Switch MAC Authentication with Device Registration Service Template
- Certificate/Two-Factor Authentication for Application Login Service Template
- Cisco Wireless with iPSK
- Policy Manager Admin Access Service Template
- Policy Manager Admin SSO Login (SAML SP Service) Service Template
- Policy Manager Identity Provider (SAML IdP Service) Service Template
- Cloud Identity/Social Media Authentication Service Template
- Device MAC Authentication Service Template
- Eduroam Service Template
- Encrypted Wireless Access via 802.1X Public PEAP Method Service Template
- Guest Access Service Template
- Guest Access Web Login Service Template
- Guest Authentication with MAC Caching Service Template
- OAuth2 API User Access Service Template
- Onboard Service Template
- Onboard Services Only Service Template

Using Wizards to Create ClearPass Services

Although Policy Manager includes templates to define services for the most common use cases, you can manually configure services for any of the other supported service types using service Wizards, or by manually creating your services on the **Configuration > Services > Add** page.

For more information on wizards and the procedures to manually define these other service types, see Configuring Other Policy Manager Services.

Aruba Auto Sign-On Service Template

About this task

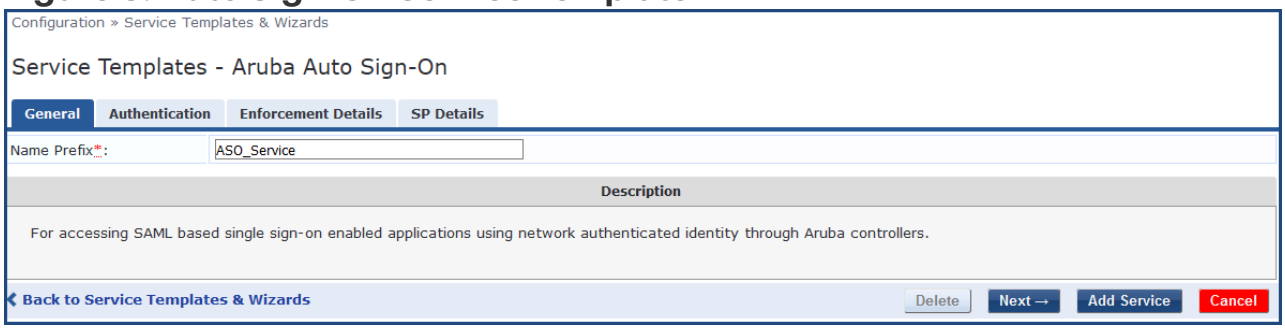
The Aruba **Auto Sign-On** service template allows you to access the SAML-based single sign-on-enabled applications (such as Policy Manager, Guest, Onboard, and Policy Manager Insight) using a network authenticated (802.1X) identity through controllers.

To access the **Auto Sign-On Service** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Aruba Auto Sign-On Service**. The **Service Templates - Auto Sign-On Service** page opens to the **General** tab.

Figure 3. Auto Sign-On Service Template



3. Specify the **Auto Sign-On** service template parameters as described in the following table:

Table 1. Aruba Auto Sign - On Service Template Parameters

Parameter	Action/Description
General	
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to identify services that use this service template.
Authentication	
Select Authentication Source	Select an authentication source from the list. The information provided in the **Authent ication**, **Enforcement Details**, and **SP de tails** tabs are auto - populated.
Active Directory Name	Enter the hostname or the IP address of t he Active Directory server. This field is mandatory.

Parameter	Action/Description
Description	Enter a description that helps you to iden tify the characteristics of this template. This field is mandatory.
Server	Enter the hostname or the IP address of t he Active Directory server. This field is ma ndatory.
Port	Enter the TCP port where the server is list ening for a connection. This value defaults to **389**. This field is m andatory.
Identity	Enter the Distinguished Name (DN) of the administrator account. This field is mand atory.
Password	Enter the account password. This field is mandatory.
NETBios	Enter the server Active Directory domain name. This field is mandatory.
Base DN	Enter the Base DN of the administrator ac count. This field is mandatory.
Enforcement Details	
Create new Enforcement Policy	Configure an optional enforcement poli cy based on the following attributes: • Department • Email • Name • Phone • UserDN • company

Parameter	Action/Description
	• memberOf • Title
SP Details	
SP URL	Enter the Service Provider (SP) URL.
Attribute Name Attribute Value	Enter attribute names and assign values t o those names. These name/value pairs are included in SAML responses.

Aruba Wireless with MAC Authentication with Device Registration Service Template

About this task

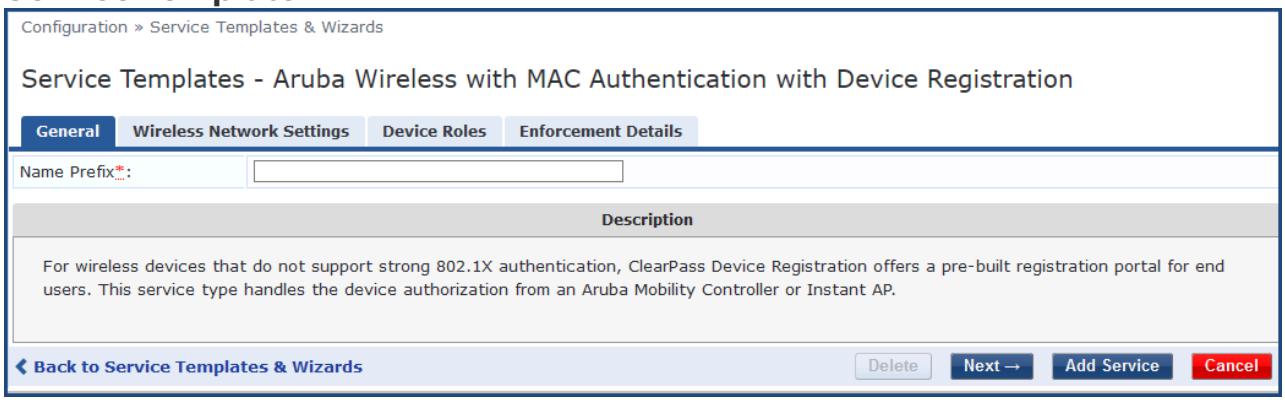
For wireless devices that do not support strong 802.1X authentication, Policy Manager Device Registration offers a prebuilt registration portal for end users. This service type handles the device authorization from an Aruba Mobility or Instant AP.

To access the Aruba Wireless with MAC Authentication with Device Registration service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Aruba Wireless with MAC Authentication with Device Registration**. The following page opens:

Figure 4. Aruba Wireless with MAC Authentication with Device Registration Service Template



General Tab

Procedure

1. Specify the **General** tab service template parameter as described in the following table:

Table 2. General Tab Parameters

Parameter	Action/Description
General	
Name Prefix	Enter a unique prefix that is appended to services using this template. Use this to identify the services that use t his template.

2. Click **Next** or select the **Wireless Network Settings** tab.

Wireless Network Settings

About this task

When you select the **Wireless Network Settings** tab, the following configuration dialog opens:

Figure 5. Wireless Network Settings Configuration Dialog

Procedure

1. Specify the **Wireless Network Settings** tab service template parameters as described in the following table:

Table 3. Wireless Network Settings Parameters

Parameter	Action/Description
Wireless Network Settings	
Wireless Controller Name	The Wireless Controller Name is automatically populated when you select a wireless controller. **NOTE** The controller must exist in the Policy Manager server's list of network devices (see Adding a Network Device).
Controller IP Address	The Controller IP Address is automatically populated when you select a wireless controller.
Vendor Name	The vendor name is set to: **Aruba**.
RADIUS Shared Secret	The RADIUS Shared Secret is automatically populated when you select a wireless controller.
Enable RADIUS Dynamic Authorization	If **RADIUS Dynamic Authorization** has not been automatically enabled, click the check box to enable this option. RADIUS Dynamic Authorization allows dynamic changes to a user session, as implemented by network access server products. This includes support for disconnecting users and changing authorizations applicable to a user session.

Parameter	Action/Description
Dynamic Authorization Port	The access point's UDP Port for Dynamic Authorization must be reachable from yo ur RADIUS server. The Dynamic Authorization Port is set by default to **3799**. This value may not be ch anged.
Enable RadSec	To enable RadSec, click the **Enable RadS ec** check box. When RadSec is enabled, the RADIUS sha red secret is populated with a default sha red secret named "*radsec*." **NOTE** It's important that the wirel ess controller is configured with the same shared secret.

2. Click **Next** or select the **Device Roles** tab.

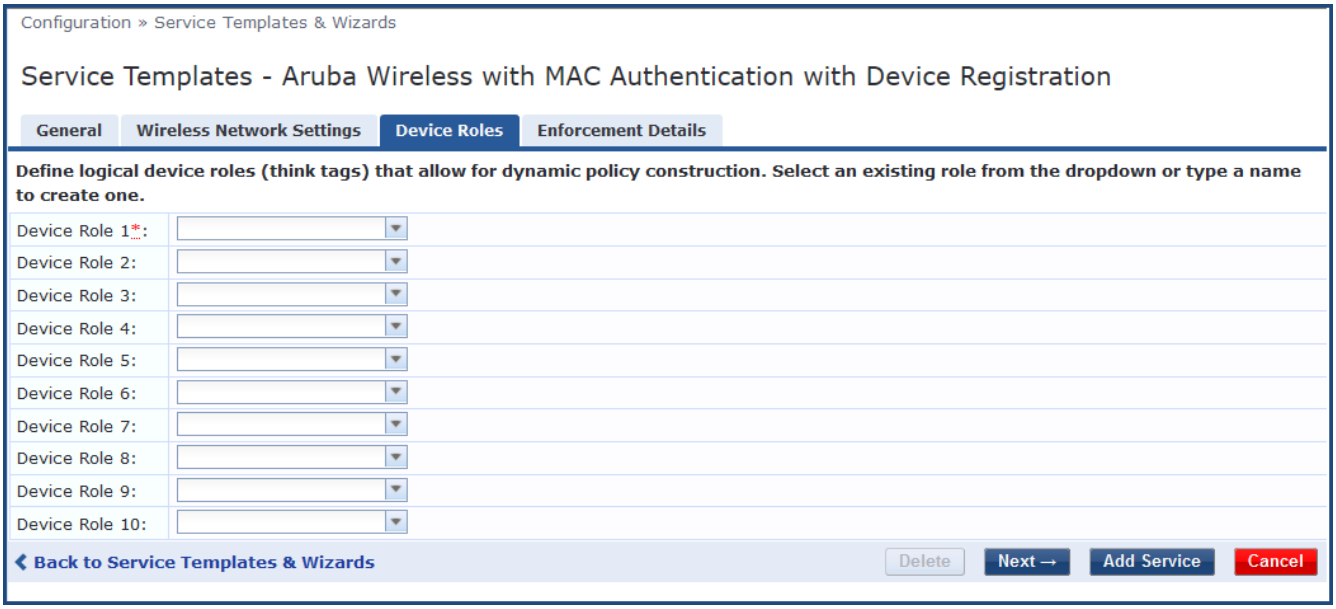
Device Roles

About this task

Define logical device roles (think tags) that allow for dynamic policy construction; for example, *Media Player*, *Printer*, *Game Console*, *Building Controls*, etc.

When you select the **Device Roles** tab, the following configuration dialog opens:

Figure 6. Device Roles Configuration Dialog



Procedure

1. Select one or more existing roles from the drop-down or type in a role name to create a new one.
2. Click **Next** or select the **Enforcement Details** tab.

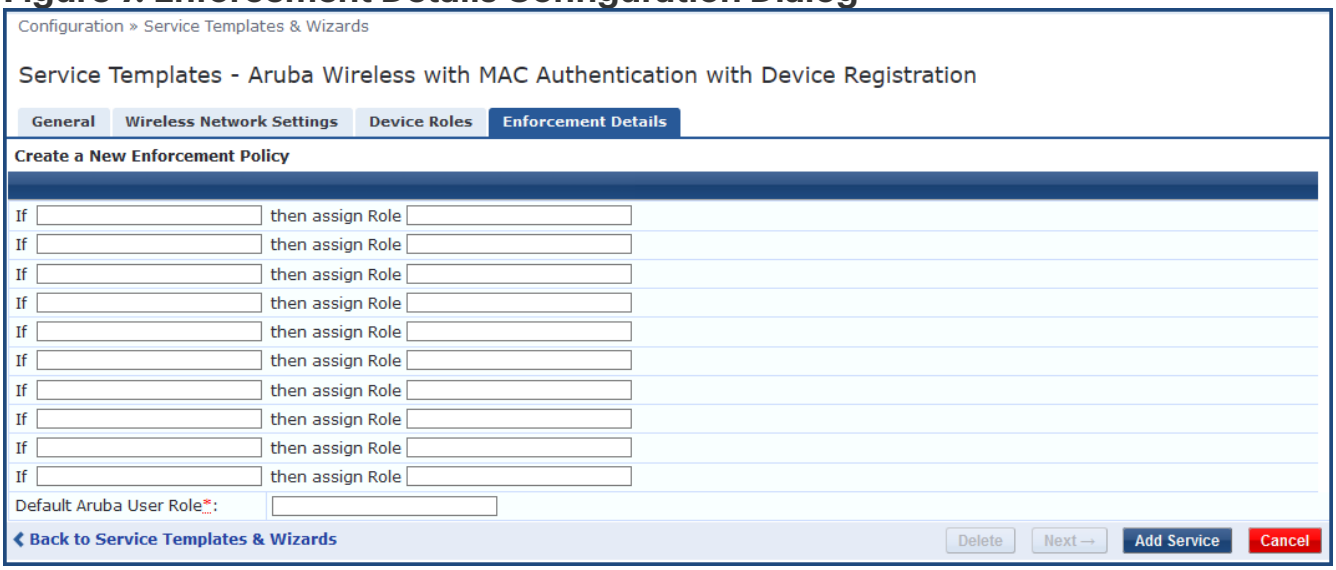
Enforcement Details

About this task

The device roles selected in the **Device Roles** dialog are populated into the new Enforcement policy defined in the **Enforcement Details** configuration dialog.

Aruba Roles are configured on the controller.

Figure 7. Enforcement Details Configuration Dialog



Procedure

1. *Aruba Role*: For each **Device Role**, specify the corresponding **Aruba Role** that is configured on your Aruba wireless controller(s) or access points.
2. *Default Aruba User Role*: Enter the default Aruba User role that is configured on your Aruba wireless controller(s) or access points.
3. Click **Add Service**.

   The **Aruba Wireless with MAC Authentication with Device Registration** service is created. You return to the **Services** page where the new service is now listed.

Aruba Wireless with MPSK Service Template

The Aruba Wireless with MPSK service template allows you to authenticate devices using an Aruba MPSK. For wireless devices that do not support strong 802.1X authentication, Aruba MPSK allows each device to be assigned a unique preshared key during Device Registration. The service type handles the device authentication from an Aruba Mobility controller or Instant Access Point (AP).

Prerequisites

About this task

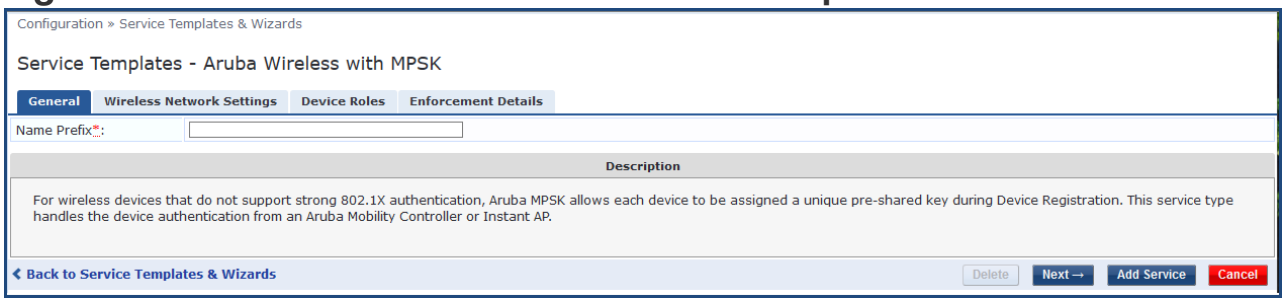
ClearPass Policy Manager Insight must be enabled for the Aruba MPSK feature to work. For information on enabling Insight, see Table 1, *Server Configuration > System Page Parameters*.

To access the service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Aruba Wireless with MPSK**. The following page opens:

Figure 8. Aruba Wireless with MPSK Service Template



General Tab

Procedure

1. Specify the **General** tab service template parameters as described in the following table:

Table 4. General Tab Parameters

Parameter	Action/Description
General	
Name Prefix	Enter a unique prefix that is appended to the services using this template. Use this to identify the services that use t his template.

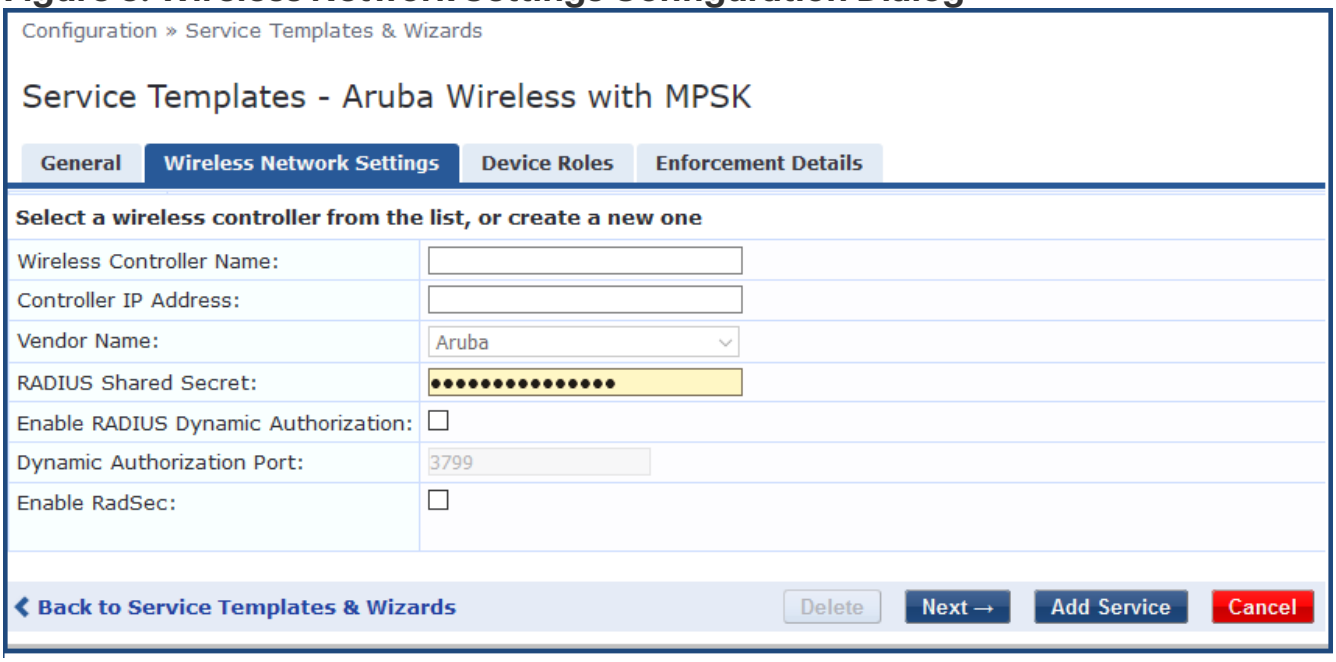
2. Click **Next** or select the **Wireless Network Settings** tab.

Wireless Network Settings

About this task

When you select the **Wireless Network Settings** tab, the following configuration dialog opens:

Figure 9. Wireless Network Settings Configuration Dialog



Procedure

1. Specify the **Wireless Network Settings** tab service template parameters as described in the following table:

Table 5. Wireless Network Settings Parameters

Parameter	Action/Description
Wireless Network Settings	
Wireless Controller Name	Specify the name of a wireless controller. **NOTE** The controller must exist in the Policy Manager server's list of network devices (see Adding a Network Device).
Controller IP Address	The Controller IP Address is automaticall y populated when you select a wireless c ontroller.
Vendor Name	The vendor name is set to: **Aruba**.
RADIUS Shared Secret	The RADIUS Shared Secret is automatical ly populated when you select a wireless c ontroller.
Enable RADIUS Dynamic Authorization	If **RADIUS Dynamic Authorization** has no t been automatically enabled, click the c heck box to enable this option. RADIUS Dynamic Authorization allows dy namic changes to a user session, as impl emented by network access server produ cts. This includes support for disconnect ing users and changing authorizations ap plicable to a user session.
Dynamic Authorization Port	The access point's UDP Port for Dynamic Authorization must be reachable from yo ur RADIUS server.

Parameter	Action/Description
	The Dynamic Authorization Port is set by default to **3799**. This value may not be ch anged.
Enable RadSec	Aruba MPSK works with RadSec for the A ruba Controller. Enable as needed for you r deployment requirements.

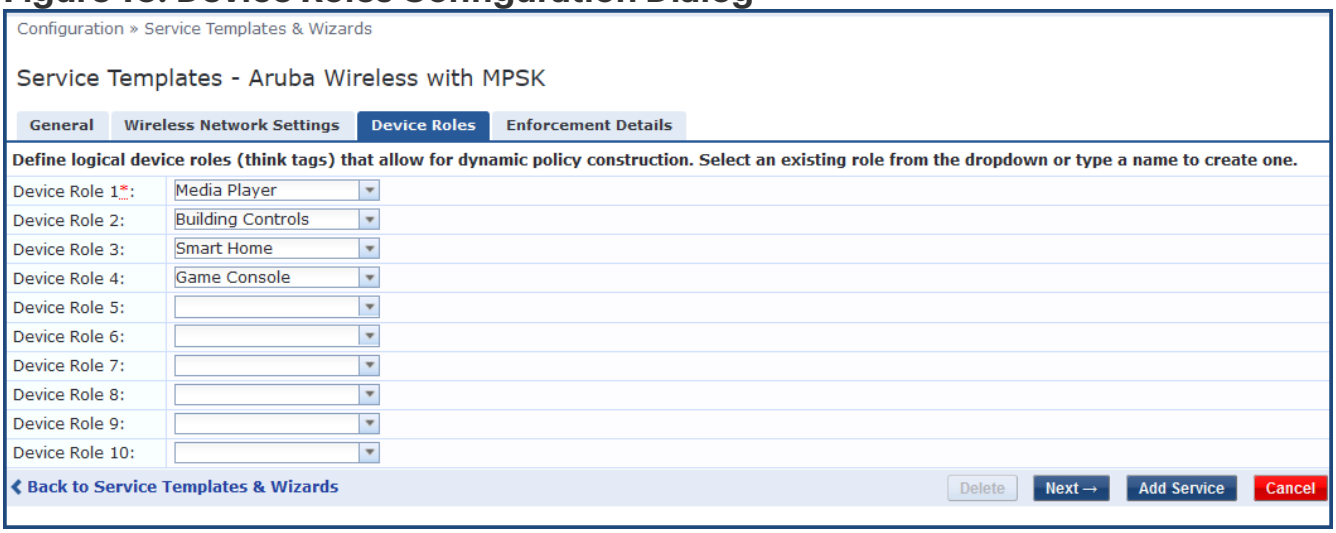
2. Click **Next** or select the **Device Roles** tab.

Device Roles

About this task

Define logical device roles (think tags) that allow for dynamic policy construction; for example, *Media Player*, *Printer*, *Game Console*, *Building Controls*, etc. Enter up to ten device roles. When you select the **Device Roles** tab, the following configuration dialog opens:

Figure 10. Device Roles Configuration Dialog



Procedure

1. Select one or more existing roles from the drop-down or type in a role name to create a new one.
2. Click **Next** or select the **Enforcement Details** tab.

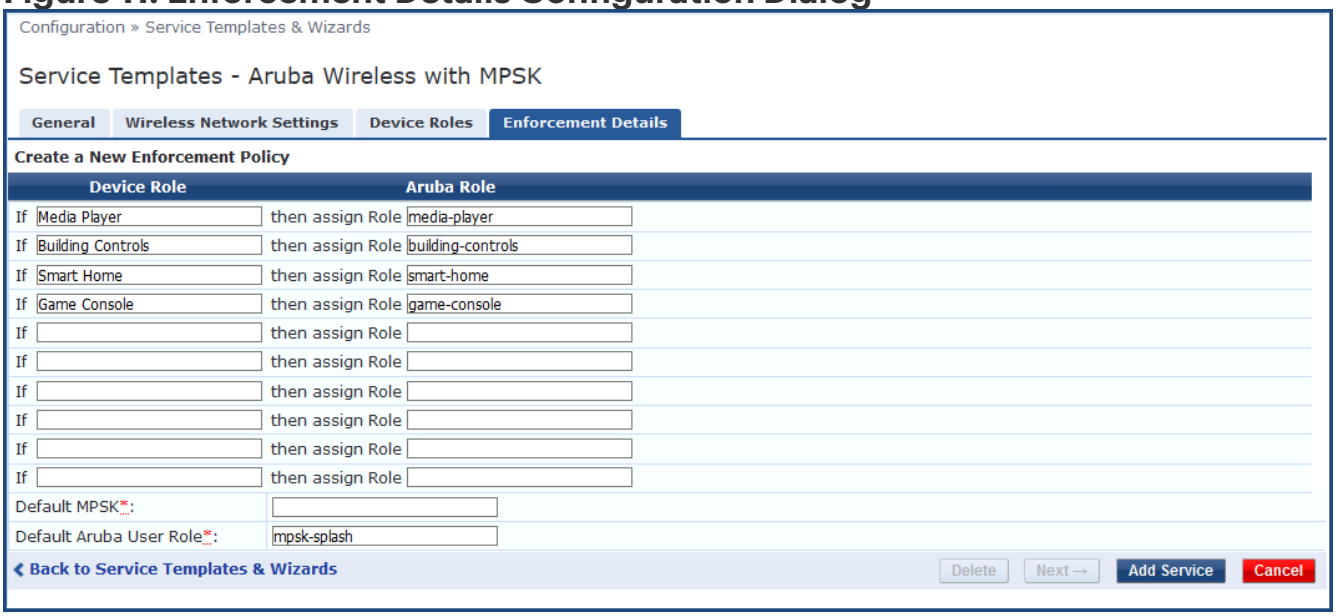
Enforcement Details

About this task

The device roles selected in the **Device Roles** dialog are populated into the new Enforcement policy defined in the **Enforcement Details** configuration dialog.

Aruba Roles are configured on the controller.

Figure 11. Enforcement Details Configuration Dialog



Procedure

1. *Aruba Role*: For each **Device Role**, specify the corresponding **Aruba Role** configured on your Aruba controller(s) or access points.
2. *Default MPSK*: Enter the default MPSK for new or unregistered devices. The Default MPSK is a static (constant) password (passphrase) that is provided while using this service template.

   The Default MPSK is returned when a device does not have a unique MPSK and is used to lock users into a locked-down role.
3. *Default Aruba User Role*: Enter the Default Aruba User Role. This is the user role that is sent back with the Default MPSK. In our example, the Default Aruba User Role is set to **mpsk-splash**.
4. Click **Add Service**.

   The **Aruba Wireless with MPSK** service is created. You return to the **Services** page where the new service is now listed.

Policy Manager Admin Access Service Template

About this task

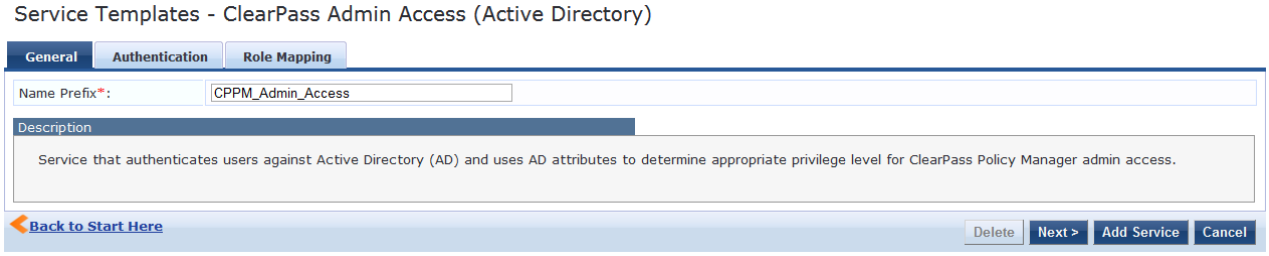
This template is designed for services that authenticate users against Active Directory. Use Active Directory attributes to determine appropriate privilege levels for ClearPass Policy Manager admin access.

To access the **Policy Manager Admin Access** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Policy Manager Admin Access**. The **Service Templates - Policy Manager Admin Access** page opens to the **General** tab.

Figure 12. Policy ManagerAdmin Access Service Template



Specify the **Policy Manager Admin Access** service template parameters as described in the following table:

Table 6. Policy Manager Admin Access Service Template Parameters

Parameter	Action/Description
General	
Select Prefix	Select a prefix from the existing list of prefixes. This populates the preconfigured information in the **Authentication** and **Role Mapping** sections. The **Name Prefix** field is not editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to identify services that use templates.
Authentication: Create an Active Directory Authentication Source	

Parameter	Action/Description
Select Authentication Source	Select an authentication source from the list. The information updated in the **Authent ication** and **Role Mapping** tabs are auto - populated.
Active Directory Name	Enter the hostname or the IP address of t he Active Directory server. This field is ma ndatory.
Description	Enter a description that helps to identify the characteristics of this template. This f ield is mandatory.
Server	Enter the hostname or the IP address of t he Active Directory server. This field is ma ndatory.
Port	Enter the TCP port where the server is list ening for a connection. This field is mand atory.
Identity	Enter the Distinguished Name (DN) of the administrator account. This field is mand atory.
Password	Enter the account password. This field is mandatory.
NetBIOS	Enter the server Active Directory domain name. This field is mandatory.
Base DN	Enter the DN of the administrator accoun t. This field is mandatory.
Role Mapping	
Attribute Name	Select the Active Directory attribute.
Super Admin Condition	Defines the various privilege levels.

Parameter	Action/Description
Read Only Admin Condition	
Help Desk Condition	

Policy Manager Admin SSO Login (SAML SP Service) Service Template

About this task

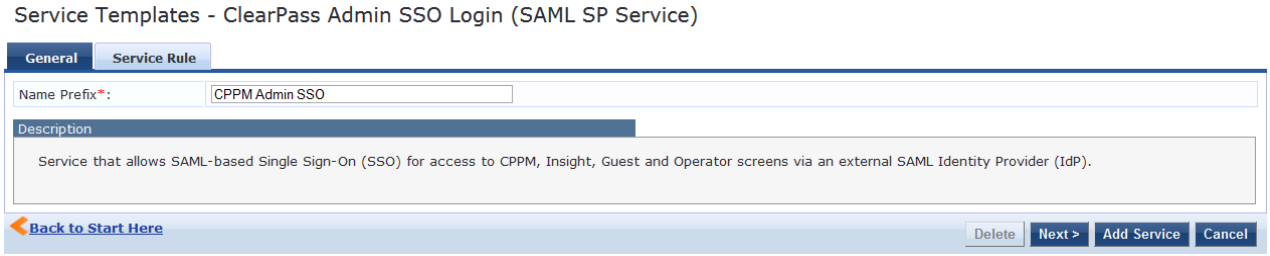
This application service template allows Security Asserting Markup Language (SAML)-based Single Sign-On (SSO) authenticated users to access Policy Manager, Guest, Insight, and Operator pages.

To access the **Policy Manager Admin SSO Login** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Policy Manager Admin SSO Login**.The **Service Templates - Policy Manager Admin SSO Login** page opens to the **General** tab.

Figure 13. Policy ManagerAdmin SSO Login (SAML SP Service) Service Template



Specify the **Policy Manager Admin SSO Login** service template parameters as described in the following table:

Table 7. ClearPass Admin SSO Login Service Template Parameters\

Parameter	Action/Description
General	
Select Prefix	Select a prefix from the existing list of pr efixes.

Parameter	Action/Description
	This populates the preconfigured inform ation in the **Service Rule** tab. The **Name Prefix** field is not editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to identify services that use tem plates.
Service Rule	
Application	Select the application that single - sign - on - authenticated administrative users can access.

Policy Manager Identity Provider (SAML IdP Service) Service Template

About this task

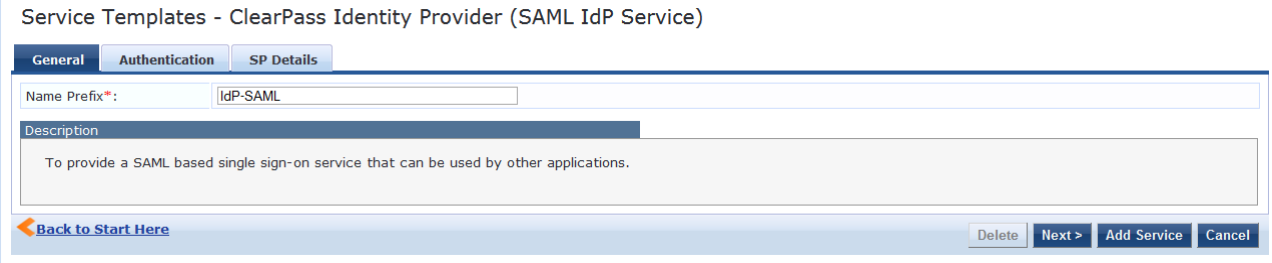
This template is designed for services that act as an Identity Provider (IdP). This Identity Provider feature allows the Layer-2 device, RADIUS server, and SAML Identity Provider to work together and deliver application-based single sign-on using network authentication information.

To access the **Policy Manager Identity Provider (SAML IdP Service)** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Policy Manager Identity Provider (SAML IdP Service)**. The **Service Templates - Policy Manager Identity Provider (SAML IdP Service)** page opens to the **General** tab.

Figure 14. Policy ManagerIdentity Provider (SAML IdP Service)



Specify the **Policy Manager Identity Provider (SAML IdP Service)** service template parameters:

Table 8. Policy Manager Identity Provider (SAML IdP Service) Service Template Parameters

Parameter	Action/Description
General	
Select Prefix	Select a prefix from the existing list of pr efixes. This populates the pre - configured infor mation in the **Authentication** and **SP De tails** sections. The **Name Prefix** field is no t editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to i dentify services that use templates.
Authentication	
Select Authentication Source	Select an authentication source from the list, the information updated in the **Authe ntication** and **SP Details** tabs are auto - p opulated.
Active Directory Name	Enter the hostname or the IP address of t he Active Directory server. This field is ma ndatory.
Description	Enter a description that helps you to iden tify the characteristics of this template. T his field is mandatory.
Server	Enter the hostname or the IP address of t he Active Directory server. This field is ma ndatory.
Identity	Enter the Distinguished Name (DN) of the administrator account. This field is mand atory.

Parameter	Action/Description
NetBIOS	Enter the Active Directory server domain name. This field is mandatory.
Base DN	Enter the Distinguished Name (DN) of the administrator account. This field is mand atory.
Password	Enter the account password. This field is mandatory.
Port	Enter the TCP port where the server is list ening for a connection. This field is mand atory.
SP Details	
SP URL	Enter the Service Provider (SP) URL.
Attribute Name	Enter the name of the attributes and ass ign values to those names. These name /value pairs are included in SAML respon ses.
Attribute Value	

Cloud Identity/Social Media Authentication Service Template

About this task

You can use this service template to authenticate guest users who log in to the network via captive portal with their social media accounts. Guests must reauthenticate after their session ends. You can limit the network access for guest devices that do not have a user directly associated with them for a specific duration in days or by a bandwidth limit. Services created using the Cloud Identity / Social Media Authentication template now have single sign-on (SSO) as the authentication method Instead of CHAP, MSCHAP, or PAP.

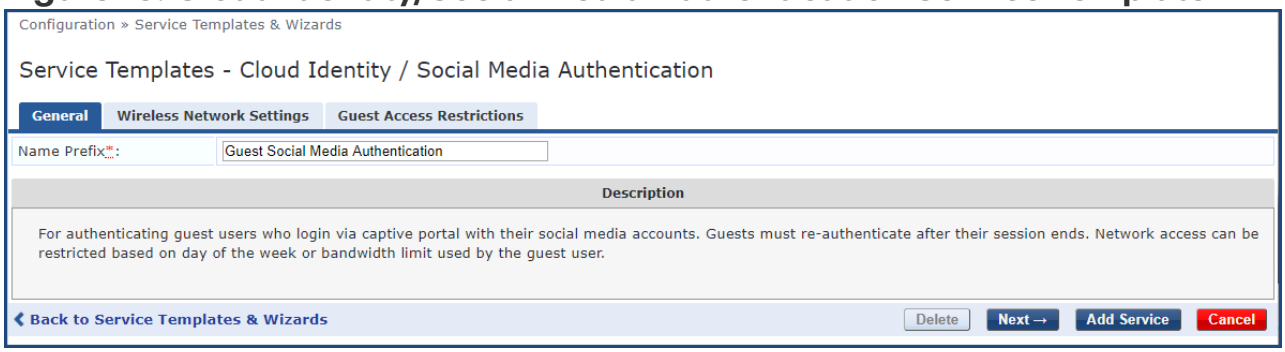
To access the **Cloud Identity/Social Media Authentication** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Cloud Identity/Social Media Authentication**.

The **Service Templates - Cloud Identity/Social Media Authentication** page opens to the **General** tab.

Figure 15. Cloud Identity/Social Media Authentication Service Template



3. Specify the parameters in the **Service Templates - Cloud Identity/Social Media Authentication** service template as described in the following table:

Table 9. Cloud Identity/Social Media Authentication Service Template Parameters

Parameter	Description
General	
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to i dentify services that use this service tem plates.
Wireless Network Settings	
Select Wireless Controller	From the drop - down, select the IP add ress of the wireless controller.
Wireless Controller Name	When you select the Wireless controller, Policy Manager automatically populates t his field..
Controller IP Address	When you select the Wireless controller, Policy Manager automatically populates t his field...

Parameter	Description
Vendor Name	When you select the Wireless controller, Policy Manager automatically populates t his field.
RADIUS Shared Secret	When you select the Wireless controller, Policy Manager automatically populates t he shared secret that is configured on th e controller and in Policy Manager to sen d and receive RADIUS requests.
Enable RADIUS CoA	When you select the wireless controller , Policy Manager automatically enables R ADIUS - initiated CoA (Change of Authori zation) on the network device.
RADIUS CoA Port	Specifies the default port **3799** when RA DIUS CoA is enabled. **NOTE** Change this value only if you defined a custom port on th e network device.
Guest Access Restrictions	
Social Login Provider	Select the social media network provider (s): *Google*, *Facebook*, *LinkedIn*, and X.
Days allowed for access	Deselect the days of the week that the gu est users are not allowed network access. By default, all seven days of the week are enabled for guest access.
Maximum bandwidth allowed per user	Specify the maximum amount of data in Megabytes a user is allowed per day. A va lue of **0** (zero) means no limit is set (the d efault).

4. Click **Add Service.**

You return to the Service page where the new service created by the service template has been added and enabled. A message like the following is also displayed:

- *Added 6 Enforcement Profile(s)*
- *Added 1 Enforcement Policies*
- *Added 1 service(s)*

Device MAC Authentication Service Template

About this task

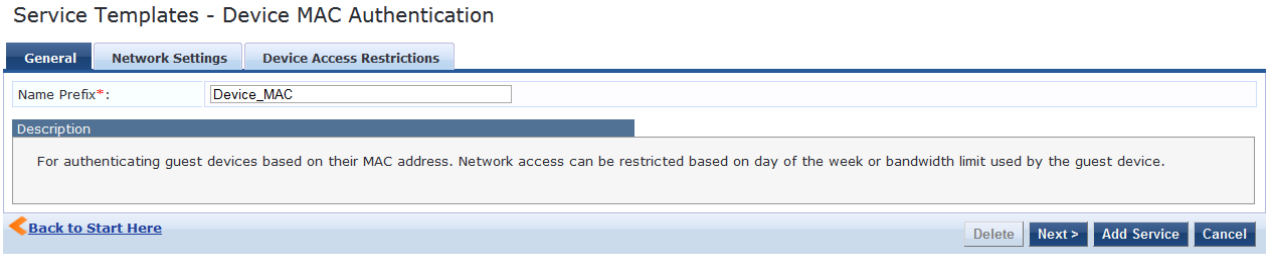
This template is designed for authenticating guest devices based on their MAC address. You can limit the network access for guest devices that do not have user directly associated with them for a specific duration in days or by the bandwidth limit.

To access the **Device MAC Authentication** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Device MAC Authentication**.The **Service Templates - Device MAC Authentication** page opens to the **General** tab.

Figure 16. Device MAC Authentication Service Template



Specify the parameters in the **Device MAC Authentication** service template as described in the following table:

Table 10. Device MAC Authentication Template Parameters

Parameter	Action/Description
General	
Select Prefix	Select a prefix from the existing list of pr efixes. This populates the preconfigured i nformation in the **Authentication** and **SP**

Parameter	Action/Description
	Details sections. The **Name Prefix** field is not editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to i dentify services that use templates.
Network Settings	
Select Device	Select a preconfigured device from the d rop - down list. To create a new device, l eave this field blank and enter the remain ing fields.
Device Name	The name of the device is populated aut omatically based on the device selected from the **Select Device** field. If you creat e a new device, enter the name of the d evice.
IP Address	The IP address of the device is populated automatically based on the device select ed from the **Select Device** field. If you c reate a new device, enter the name of th e device.
Vendor Name	The name of the manufacturer of the dev ice is populated automatically based on t he device selected from the **Select Devic e** field. If you create a new device, enter t he name of the manufacturer of the devi ce.
RADIUS Shared Secret	Enter the shared secret that is configured on the controllerand in Policy Manager to send and receive RADIUS requests.
Enable RADIUS CoA	Select to enable RADIUS initiated Chang e of Authorization (CoA) on the network device.

Parameter	Action/Description
RADIUS CoA Port	Specifies the default port **3799** if RADIUS CoA is enabled. Change this value only if you defined a custom port on the networ k device.
Device Access Restrictions	
Days allowed for access	Select the days on which network access is allowed.
Maximum bandwidth allowed per device	Enter a number to set an upper limit for t he amount of data in megabytes to whic h a device is allowed per day. A value of **0** (zero), the default, means no limit is set .

Eduroam Service Template

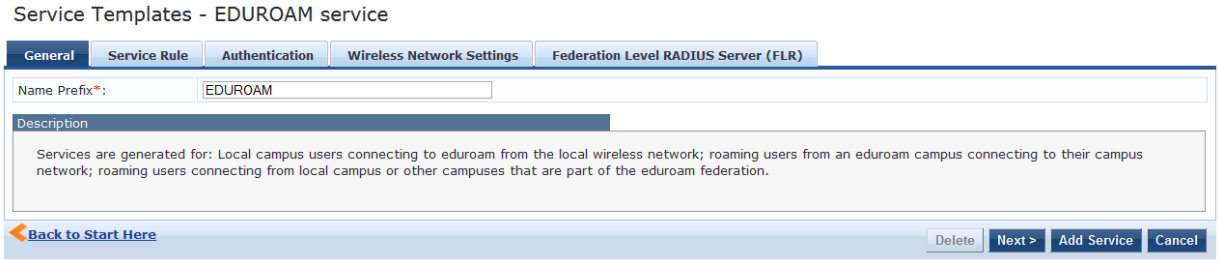
This template is designed for the following scenarios:

- Local campus users connecting to eduroam from the local wireless network.
- Roaming users from an eduroam campus connecting to their campus network.
- Roaming users connecting from a local campus or other campuses that are part of the eduroam federation.

To access the **Eduroam** service template:

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select Eduroam. The **Service Templates - Eduroam** page opens to the **General** tab.

Figure 17. Eduroam Service Template



Specify the parameters used in the **Eduroam** service template as described in the following table:

Table 11. Eduroam Service Template Parameters

Parameter	Action/Description
General	
Select Prefix	Select a prefix from the existing list of p refixes. This populates the pre - configured info rmation in the **Authentication, Service Rule, Wireless,** and**Federation Level Ra dius Server (FLR) tabs**. The **Name Prefi x** field is not editable.
Name Prefix	Enter a prefix that you want to append t o services using this template. Use this to identify services that use te mplates.
Service Rule	
Enter domain details	Enter the domain name of the network. For example, @edunet.ucla.com. This fi eld is mandatory.
Select Vendor	Select the vendor of the network devic e. This field is mandatory.
Authentication	
Select Active Directory	Select an authentication source from th e list, the information updated in the **Au thentication**, **Wireless**, and **Federation Level Radius Server (FLR)** tabs are auto - populated.
Active Directory Name	Enter the hostname or the IP address o f the Active Directory server. This field i s mandatory.

Parameter	Action/Description
Description	Enter a description that helps you iden tify the characteristics of this template. This field is mandatory.
Server	Enter the host name or the IP address o f the Active Directory server. This field i s mandatory.
Identity	Enter the Distinguished Name (DN) of t he administrator account. This field is mandatory.
NetBIOS	Enter the server Active Directory domai n name. This field is mandatory.
Base DN	Enter the Distinguished Name (DN) of t he administrator account. This field is mandatory.
Password	Enter the account password. This field i s mandatory.
Port	Enter the TCP port where the server is li stening for a connection. This field is m andatory.
Wireless Network Settings	
Select wireless controller	Select a wireless controller from the d rop - down list.
Wireless controller name	Enter the name given to the wireless co ntroller.
Controller IP Address	Enter the IP address of the wireless con troller.
Vendor Name	Select the manufacturer of the wireless controller.
RADIUS Shared Secret	Enter the shared secret that is configur ed on the controllerand inside Policy M

Parameter	Action/Description
	anager to send and receive RADIUS req uests.
Enable RADIUS CoA	Select to enable RADIUS initiated CoA on the network device.
RADIUS CoA Port	Specifies the default port 3799 if RADIU S CoA is enabled. Change this value on ly if you defined a custom port on the n etwork device.
Federation Level RADIUS Server (FLR)	
Host Name	Enter the host name of the federation R ADIUS server.
IP Address	Enter the IP address of the federation R ADIUS server.
Vendor Name	Select the manufacturer of the wireless controller.
RADIUS Shared Secret	Enter the shared secret that is configur ed on the controllerand inside Policy M anager to send and receive RADIUS req uests.
Enable RADIUS CoA	Select to enable RADIUS initiated CoA on the network device.
RADIUS CoA Port	Specifies the default port **3799** if RADIU S CoA is enabled. **NOTE** Change this value only if you defined a custom port on the network device.
RADIUS Authentication Port	Enter the port number for the RADIUS Authentication Port.

Parameter	Action/Description
RADIUS Accounting Port	Enter the port number for the RADIUS Accounting Port.

Encrypted Wireless Access via 802.1X Public PEAP Method Service Template

About this task

This template is designed for providing encrypted wireless access to users using fixed 802.1X PEAP credentials. It configures an **EAP PEAP Public** type authentication method and creates an enforcement policy for network access.

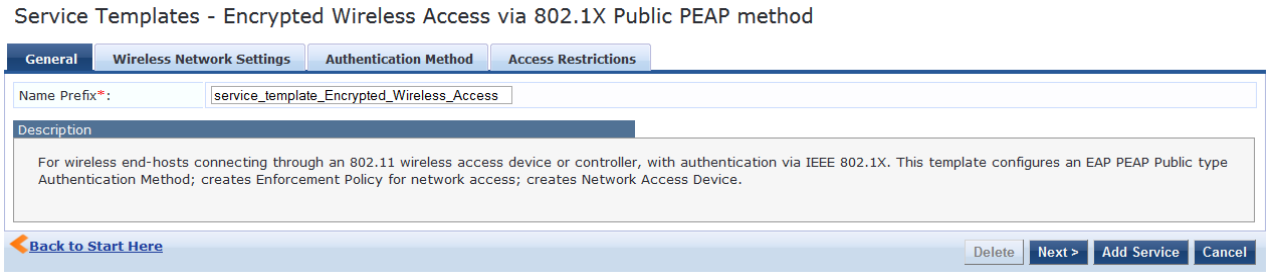
To access the **Encrypted Wireless Access via 802.1X Public PEAP Method** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Encrypted Wireless Access via 802.1X Public PEAP Method.**

   The **Service Templates - Encrypted Wireless Access via 802.1X Public PEAP Method** page opens to the **General** tab.

Figure 18. Encrypted Wireless Access via 802.1X Public PEAP Method Service Template



Specify the parameters used in the **Encrypted Wireless Access via 802.1X Public PEAP method** service template s described in the following table:

Table 12. Encrypted Wireless Access via 802.1X Public PEAP Method Service Template Parameters

Parameter	Action/Description
General	

Parameter	Action/Description
Name Prefix	Enter a prefix that you want to append to services using this template. You can use this to identify services that use templat es.
Wireless Network Settings	
Select wireless controller	Select a wireless controller from the drop - down list.
Wireless controller name	Enter the name given to the wireless cont roller.
Controller IP Address	Enter the IP address of the wireless cont roller.
Vendor Name	Select the manufacturer of the wireless c ontroller.
RADIUS Shared Secret	Enter the shared secret that is configured on the controllerand inside Policy Manag er to send and receive RADIUS requests.
Enable RADIUS CoA	Select to enable RADIUS initiated CoA on the network device.
RADIUS CoA Port	Specifies the default port **3799** if RADIUS CoA is enabled. **NOTE** Change this value only if you defined a custom port on th e network device.
Authentication Method	
Public Username	Enter the public username for the EAP PE AP Public type authentication method.

Parameter	Action/Description
Public Password	Enter the password for the EAP PEAP Public type authentication method.
Access Restrictions	
Days allowed for access	Select the days on which network access is allowed.

Guest Access Service Template

About this task

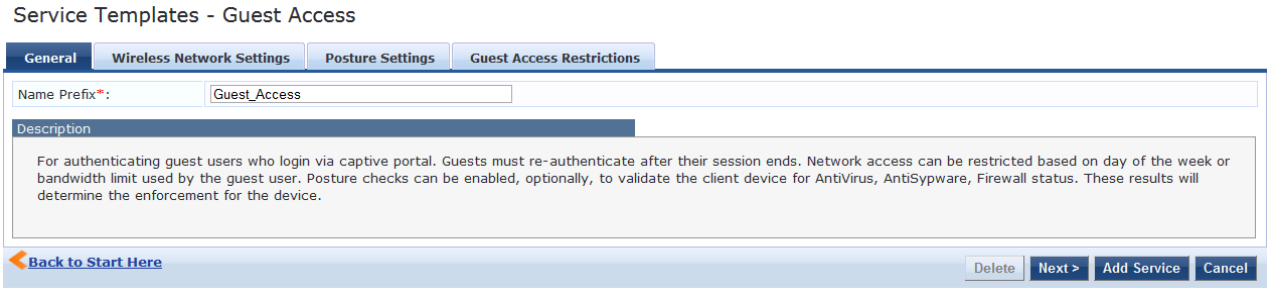
This template is designed for authenticating guest users who log in using captive portal. Guests must reauthenticate after session expiry. Guest access can be restricted based on day of the week, bandwidth limit, and number of unique devices used by the guest user.

To access the **Guest Access** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Guest Access** . The **Service Templates - Guest Access** page opens to the **General** tab.

Figure 19. Guest Access Service Template



Specify the parameters used in the **Guest Access** service template as described in the following table:

Table 13. Guest Access Service Template Parameters

Parameter	Action/Description
General	

Parameter	Action/Description
Select Prefix	Select any one prefix from the existing lis t of prefixes. This populates the pre - configured infor mation in the **Wireless Network Settings** and **Guest Access Restrictions** sections. The **Name Prefix** field is not editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to i dentify services that use templates.
Wireless Network Settings	
Wireless SSID for Guest access	Enter the SSID value here.
Select wireless controller	Select the wireless controller from the d rop - down list if you already configured.
Wireless controller name	Enter the name of the wireless controller.
Controller IP Address	Enter the wireless controller's IP address.
Vendor Name	Select the manufacturer of the wireless c ontroller.
RADIUS Shared Secret	Enter the shared secret that is configured on the controller and inside Policy Manag er to send and receive RADIUS requests.
Enable RADIUS CoA	Select to enable RADIUS initiated CoA on the network device.
RADIUS CoA Port	Specifies the default port **3799** if RADIUS CoA is enabled. **NOTE** Change this value only if you defined a custom port on th e network device.

Parameter	Action/Description
Posture Settings	
Enable Posture Checks	Select the check box to perform health c hecks post authentication. This enables the **Host Operating System** and **Quarantine Message** fields.
Host Operating System	Select the operating system: Windows, Li nux, or macOS.
Quarantine Message	Specify the quarantine message that will appear on the client.
Initial Role/ VLAN	Enter the initial role of the client before p osture checks are performed.
Quarantine Role/VLAN	Enter the role of clients that fail posture c hecks.
Guest Access Restrictions	
Days allowed for access	Select the days of the week that guest us ers are allowed network access.
Maximum bandwidth allowed per user	Enter a number to set an upper limit for t he amount of data in Megabytes (MB) to which a user is allowed per day. A value of 0 (zero), the default, means no limit is set.

Guest Access Web Login Service Template

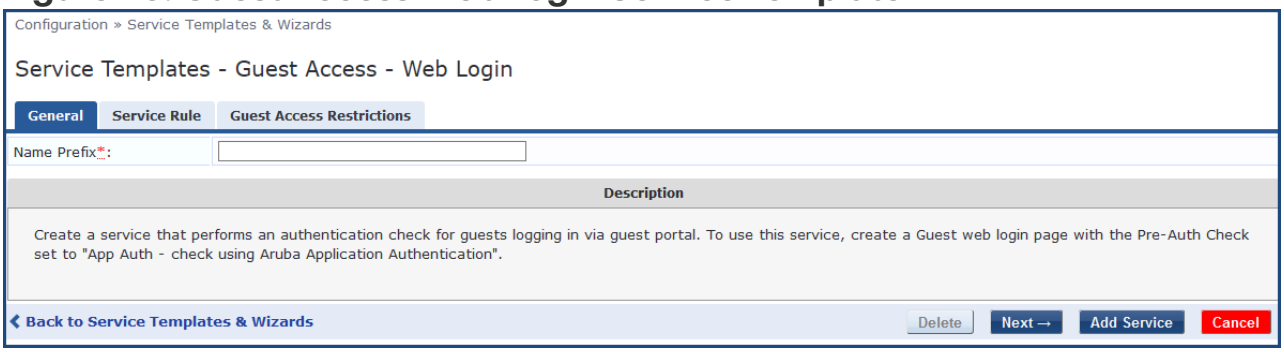
About this task

This service performs an authentication check for guests logging in using the Guest Portal. To use this service:

Procedure

1. First create a **Guest Web Login** page that sets the **Pre-Auth Check** option to **AppAuth - Check** using Aruba **Application Authentication** (for details, see Authentication in Guest Web Login Using OnGuard Native Dissolvable Agent).
2. Navigate to **Configuration > Start Here**. The list of Policy Manager Service Templates opens.
3. Select **Guest Access - Web Login**. The **Guest Access Web Login** service template opens:

Figure 20. Guest Access-Web Login Service Template



4. Specify the **Guest Access - Web Login** service template parameters as described in the following table:

Table 14. Guest Web Login Service Template Parameters

Parameter	Action/Description
General	
Select Prefix	From the drop - down, select **Guest Web Login**. This prefix is appended to services that u se this template.
Name Prefix	The prefix you selected is automatically p opulated into this field.
Service Rule	
Page name	From the **Page** name drop - down, select the name of the appropriate **Guest Web Login** page.

Parameter	Action/Description
Add New Guest Web Login page	To launch a new Web session to add a n ew Guest Web Login page, click this link.
Guest Access Restrictions	
Enable days allowed for guest access	Select the days of the week that guest us ers are allowed network access. **NOTE** All seven days of the week a re enabled by default.

5. Click **Add Service**.

Guest Authentication with MAC Caching Service Template

About this task

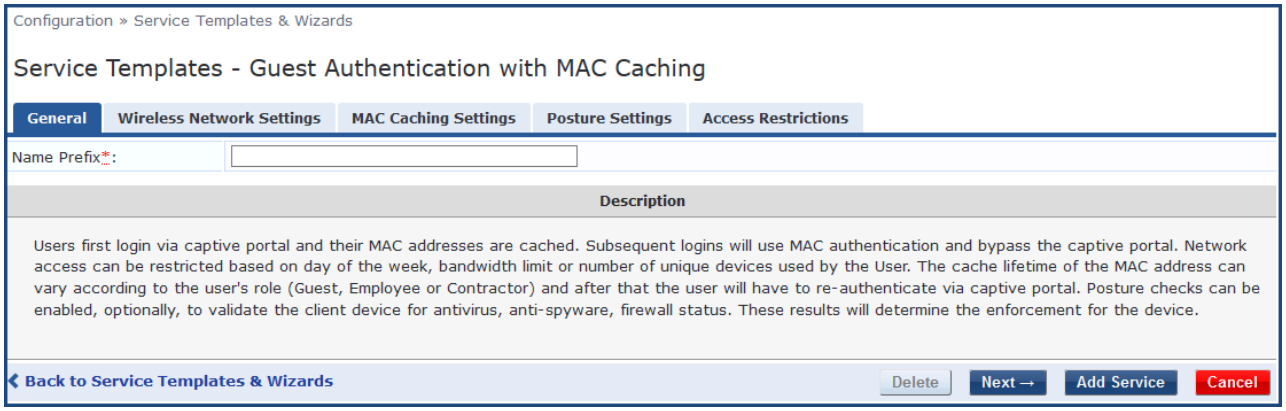
This template is designed for authenticating guest accounts based on the cached MAC addresses used during authentication. When users first log in via the Captive Portal, their MAC addresses are cached. Subsequent logins use MAC authentication and bypass the Captive Portal. A guest can belong to a specific role such as *Contractor*, *Guest*, or *Employee*, and each role can have a different lifetime for the cached MAC address. The cache lifetime of the MAC address can vary according to the user's role (*Guest*, *Employee*, or *Contractor*) and after that the user will have to reauthenticate via the Captive Portal. Network access can be restricted based on day of the week, bandwidth limit, or number of unique devices used by the user. Optionally, posture checks can be enabled to validate the client device for antivirus or firewall status. These results determine the enforcement for the device.

To configure the **Guest Authentication with MAC Caching** service template:

Procedure

1. Navigate to the **Configuration > Service Templates & Wizards** page. The **Service Templates & Wizards** page opens.
2. Scroll down to and select the **Guest Authentication with MAC Caching** service template:

Figure 21. Guest Authentication with MAC Caching Service Template



3. Specify the **Guest Authentication with MAC Caching** service template parameters as described in the following table:

Table 15. Guest Authentication with MAC Caching Service Template Parameters

Parameter	Action/Description
General	
Select Prefix	Select a prefix from the existing list of prefixes. This populates the preconfigured information in the **Wireless Network Settings, MAC Caching Settings,** and **Access restrictions** tabs. The **Name Prefix** field is not editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to identify services that use templates.
Wireless Network Settings	
Wireless SSID	Enter the SSID name of your network.
Select Wireless Controller	Select the wireless controller from the drop - down list if you already configured the controller for Policy Manager.
Wireless Controller Name	Enter the name of the wireless controller.
Controller IP Address	Enter the wireless controller's IP address.

Parameter	Action/Description
Vendor Name	Select the manufacturer of the wireless c ontroller.
RADIUS Shared Secret	Enter the shared secret that is configured on the controller and in Policy Manager t o send and receive RADIUS requests.
Enable RADIUS CoA	Select this check box to enable RADIUS i nitiated CoA (Change of Authorization) o n the network device.
RADIUS CoA Port	Specifies the default port **3799** if RADIUS CoA is enabled. **NOTE** Change this value only if you defined a custom port on th e network device.
MAC Caching Settings	
Cache Duration for Employee	From the **Account Expiry Time** drop - d own, select the MAC caching duration for employees: • One Day • One Week • One Month • Six Months **NOTE** When this duration expires, users must reauthenticate vi a the captive portal. **NOTE**

Parameter	Action/Description
	You must specify the cache duration for at least one role.
Cache Duration for Guest	From the **Account Expiry Time** drop - down, select the MAC caching duration for guests: • One Day • One Week • One Month • Six Months
Cache Duration for Contractor	From the **Account Expiry Time** drop - down, select the MAC caching duration for contractors: • One Day • One Week • One Month • Six Months
Posture Settings	
Enable Posture Checks	Select the **Enable Posture Checks** check box to perform health checks post authentication. This enables the **Host Operating System** and **Quarantine Message** fields.
Host Operating System	Select the operating system(s): Windows, Linux, or macOS.
Quarantine Message	Specify the quarantine message that will appear on the client.
Initial Role/VLAN	Enter the initial role of the client before posture checks are performed.

Parameter	Action/Description
Quarantine Role/VLAN	Enter the role of clients that fail posture c hecks.
Access Restrictions	
Enforcement Type	Select one of the following enforcement t ypes: • Aruba Role Enforcement • VLAN Enforcement • Filter ID - Based Enforcement **NOTE** Enforcement Type applies to the **Captive Portal Access**, **E mployee Access**, **Guest Acc ess**, and **Contractor Access** fields. **NOTE** At least one of *Employee*, *G uest*, or *Contractor Access* must be specified.
Captive Portal Access	Used for unauthenticated users and after the MAC caching has expired. • If you selected **Aruba Role Enforceme nt** as the **Enforcement Type**, enter th e **Aruba Role Name**. • If you selected **VLAN Enforcement** as the **Enforcement Type**, enter the **VLA N ID**. • If you selected **Filter ID - Based Enfo rcement** as the **Enforcement Type**, en ter the **Filter ID**.

Parameter	Action/Description
Days allowed for access	Select the days of the week that guest users are allowed network access. By default, all seven days of the week are enabled.
Maximum number of devices allowed per user	Enter the maximum number of devices that users can connect to the network.
Maximum bandwidth allowed per user	Enter a number to set an upper limit for the amount of data in megabytes to which a user is allowed per day. A value of **0** (zero), the default, means no limit is set.
Employee Access	• If you selected **Aruba Role Enforcement** as the **Enforcement Type**, enter the **Aruba Role Name**. • If you selected **VLAN Enforcement** as the **Enforcement Type**, enter the **VLAN ID**. • If you selected **Filter ID - Based Enforcement** as the **Enforcement Type**, enter the **Filter ID**.
Guest Access	• If you selected **Aruba Role Enforcement** as the **Enforcement Type**, enter the **Aruba Role Name**. • If you selected **VLAN Enforcement** as the **Enforcement Type**, enter the **VLAN ID**. • If you selected **Filter ID - Based Enforcement** as the **Enforcement Type**, enter the **Filter ID**.
Contractor Access	• If you selected **Aruba Role Enforcement** as the **Enforcement Type**, enter the **Aruba Role Name**.

Parameter	Action/Description
	• If you selected **VLAN Enforcement** as the **Enforcement Type**, enter the **VLAN ID**. • If you selected **Filter ID - Based Enforcement** as the **Enforcement Type**, enter the **Filter ID**.

Guest Social Media Authentication Service Template

About this task

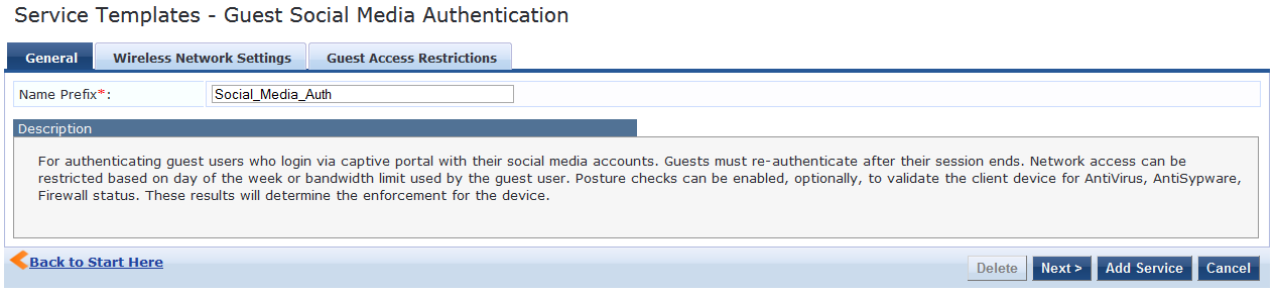
This template is designed for authenticating guest users logging in through the captive portal with their social media accounts, such as Google, Facebook, LinkedIn, and Twitter. Guests must reauthenticate after the session ends.

To access the **Guest Social Media Authentication** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Guest Social Media Authentication**. The **Service Templates - Guest Social Media Authentication** page opens to the **General** tab.

Figure 22. Guest Social Media Authentication Service Template



Specify the **Guest Social Media Authentication** service template parameters as described in the following table:

Table 16. Guest Social Media Service Template Parameters

Parameter	Description
General	

Parameter	Description
Select Prefix	Select a prefix from the existing list of prefixes. This populates the preconfigured information in the **Wireless Network Settings, MAC Caching Settings,** and **Guest Access restrictions** tabs. The **Name Prefix** field is not editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to identify services that use templates.
Wireless Network Settings	
Wireless Controller Name	Enter the name of the wireless controller.
Controller IP Address	Enter the wireless controller's IP address.
Vendor Name	Select the manufacturer of the wireless controller.
RADIUS Shared Secret	Enter the shared secret that is configured on the controller and inside Policy Manager to send and receive RADIUS requests.
Enable RADIUS CoA	Select to enable RADIUS initiated CoA (Change of Authorization) on the network device.
RADIUS CoA Port	Specifies the default port **3799** if RADIUS CoA is enabled. **NOTE** Change this value only if you defined a custom port on the network device.
Guest Access Restrictions	

Parameter	Description
Social Login Provider	Select the social media network options: Google, Facebook, LinkedIn, and Twitter.
Days allowed for access	Select the days of the week that the gues t users are allowed network access. By default, all seven days of the week are enabled.
Maximum bandwidth allowed per user	Specify the maximum amount of data in Megabytes a user is allowed per day. A va lue of **0** (zero) means no limit is set (the d efault).

OAuth2 API User Access Service Template

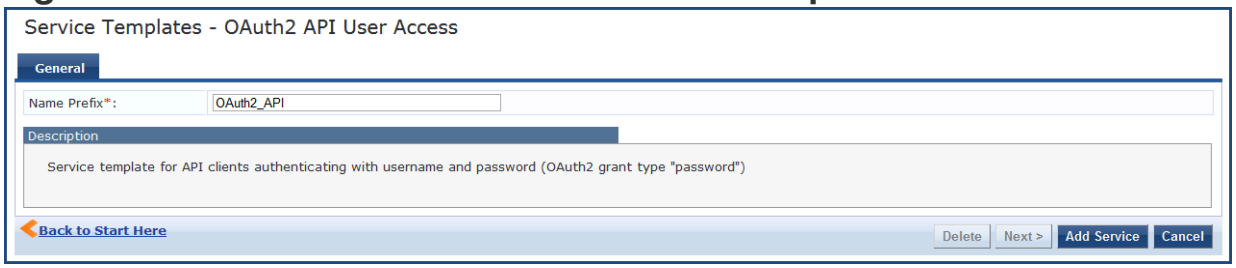
This template is designed for configurations that supports ClearPass Policy Manager authenticating API clients with the username and OAuth2 grant-type password.

- The **OAuth2 API User Access** service template uses the **Guest Operator Logins** as the default enforcement policy.
- The **Local User Repository** and **Admin User Repository** repositories are used as the default authentication sources.

To access the **OAuth2 API User Access** service template:

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **OAuth2 API User Access**. The **Service Templates - OAuth2 API User Access** page opens.

Figure 23. OAuth2 API User Access Service Template



3. Specify the **OAuth2 API User Access** service template parameter as described in the following table:

Table 17. OAuth2 API User Access Service Template Parameter

Parameter	Description
Select Prefix	Select a prefix from the existing list of p refixes.
Name Prefix	Enter a prefix that is appended to servi ces using this template. You can use this prefix to identify the se rvices that use templates.

4. Click **Add Service**.

Onboard Services Only Service Template

About this task

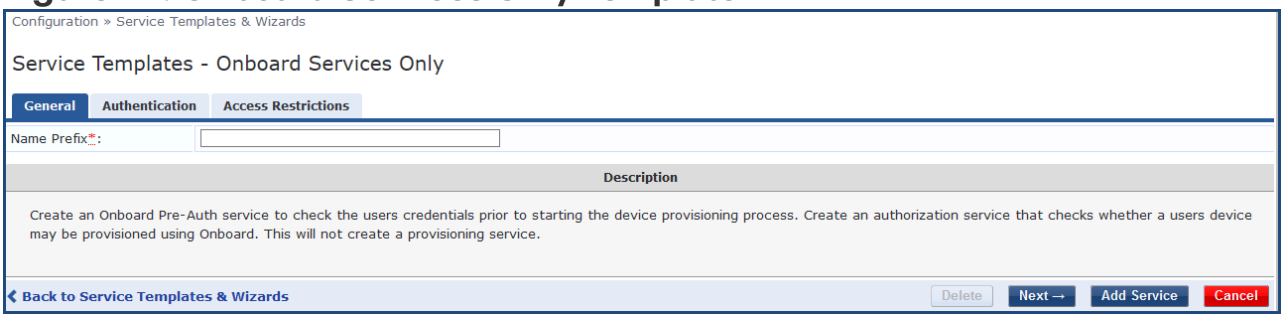
Use this service template to create the application services required to support Policy Manager Onboard assisted provisioning. This service template can be used when an existing 802.1X service already exists and just the Onboard application services are required.

To configure the **Onboard Services Only** template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Onboard Services Only**.

Figure 24. Onboard Services Only Template



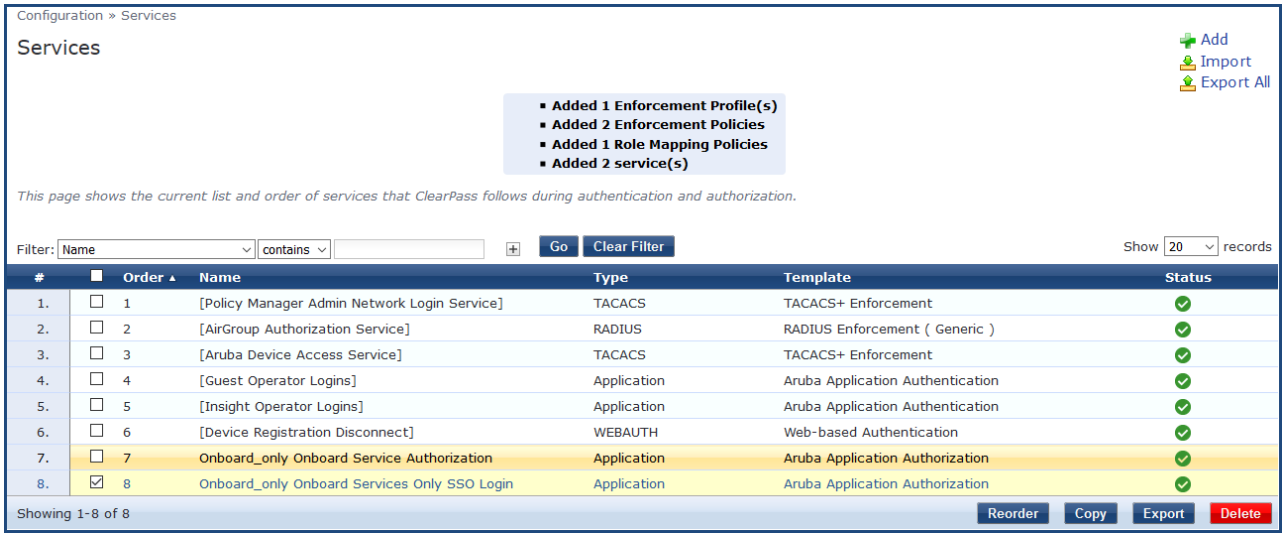
3. Specify the **Onboard Services Only** service template parameters as described in the following table:

Table 18. Onboard Services Only Template Parameters

Parameter	Description
General	
Select Prefix	Select a prefix from the existing list of pr efixes or enter the name of a new prefix.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to i dentify services that use templates.
Authentication	
Enable SSO	Select the check box to enable Single Si gn - On. When this option is selected, guests are redirected to the Identity Provider (IdP) c onfigured in Policy Manager, where they authenticate themselves.
Access Restrictions	
Days allowed for access	Select the days of the week that onboard ed devices are allowed network access.

4. Click **Add Service**. You return to the **Services** page, where two Onboard Only services have been added:

Figure 25. Onboard Only Services Added to ClearPass



Onboard Service Template

About this task

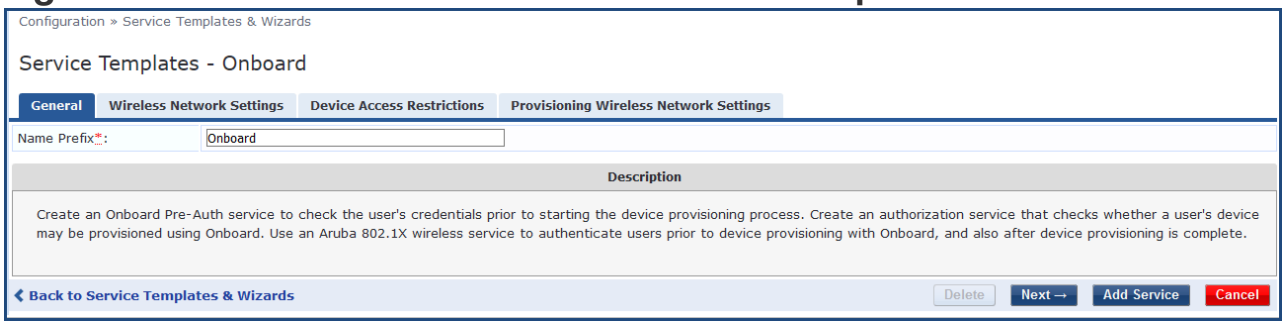
This service creates an Onboard Pre-Authentication service to check the user's credentials before starting the device provisioning process. This service template also creates an authorization service that checks whether a user's device can be provisioned using Onboard. To authenticate users prior to device provisioning with Onboard, as well as after device provisioning is completed, use an **Aruba 802.1X Wireless** service .

To access the **Onboard** service template:

Procedure

1. Navigate to **Configuration > Service Templates & Wizards**.
2. From the **Service Templates & Wizards** page, select **Onboard**. The **Service Templates - Onboard** page opens to the **General** tab.

Figure 26. Onboard Pre-Authorization Service Template



3. Specify the **Onboard Authorization** service template parameters as described in the following table:

Table 19. Onboard Authorization Service Template Parameters

Parameter	Description
General	
Select Prefix	Select a prefix from the existing list of pr efixes or enter the name of a new prefix. This populates the preconfigured inform ation in the **Wireless Network Settings**, **D evice Access Restrictions**, and **Provision ing Wireless Network Settings** sections. The **Name Prefix** field is not editable.
Name Prefix	Enter a prefix that you want to append to services using this template. Use this to i dentify services that use templates.
Wireless Network Settings	
Wireless Controller Name	Enter the name of the wireless controller.
Controller IP Address	Enter the wireless controller's IP address.
Vendor Name	Select the manufacturer of the wireless c ontroller.
RADIUS Shared Secret	Enter the shared secret that is configured on the controller and inside Policy Manag er to send and receive RADIUS requests.
Enable RADIUS CoA.	Select to enable RADIUS initiated CoA (C hange of Authorization) on the network d evice.
RADIUS CoA Port	Specifies the default port **3799** if RADIUS CoA is enabled. **NOTE**

Parameter	Description
	Change this value only if you defined a custom port on th e network device.
Device Access Restrictions	
Days allowed for access	Select the days of the week that onboard ed devices are allowed network access.
Provisioning Wireless Network Settings	
Wireless SSID for Onboard Provisioning	Enter the SSID of your network.
Add New Onboard Network Settings	Click the **Add New Onboard Network Set tings** link to launch the Web UI to modify the Onboard network settings.

4. Click **Add Service**.

   You return to the **Services** page, where the Onboard service has been added.

Configuring Services Manually or with Wizards

Although Policy Manager provides service templates for the most common use cases, you can configure services for all of the other supported service types either manually, or by using a Policy Manager wizard. You will need to provide the same information for both the wizard and manual configuration processes, though the wizard does provide an enhanced User Interface to guide you through the configuration steps.

- To manually configure a service, navigate to **Configuration > Services** and click the **Add** link
- To use a wizard to configure a service, navigate to **Configuration > Service Templates & Wizards,** and click the option **To configure service and related policies using the full wizard, click here.**

  Click any of the links below for details on defining a particular service type:

- Aruba 802.1X Wireless Service

- 802.1X Wireless Service
- 802.1X Wired Service
- 802.1X Wired—Identity Only Service
- 802.1X Wireless—Identity Only Service
- Cisco Web Authentication Proxy Service
- MAC Authentication Service
- RADIUS Authorization Service
- RADIUS Enforcement (Generic) Service
- RADIUS Proxy Service
- Aruba Application Authentication Service
- Aruba Application Authorization Service
- Policy Manager OnConnect Enforcement Service
- Event-Based Enforcement Service
- TACACS+ Enforcement Service
- Web-Based Authentication Service
- Web-based Health Check Only Service
- Web-Based Open Network Access Service

Subtopics

Aruba 802.1X Wireless Service
802.1X Wireless Service
802.1X Wired Service
802.1X Wired—Identity Only Service
802.1X Wireless—Identity Only Service
Aruba 802.1X Wireless Service
Cisco Web Authentication Proxy Service
MAC Authentication Service
RADIUS Authorization Service
RADIUS Enforcement (Generic) Service
RADIUS Proxy Service
Aruba Application Authentication Service
Aruba Application Authorization Service
OnConnect Enforcement Service
Event-Based Enforcement Service

TACACS+ Enforcement Service
Web-Based Authentication Service
Web-based Health Check Only Service
Web-Based Open Network Access Service

Aruba 802.1X Wireless Service

Configure this service for wireless hosts that are connecting through an Aruba 802.1X wireless access device or controller using IEEE 802.1X authentication. Service rules are customized for a typical Aruba Controller deployment. The Aruba 802.1X service includes a rule that specifies that an Aruba ESSID exists.

NOTE

If you want to administer the same set of policies for wired and wireless access, you can combine the service rule to define one single service.

The other option is to keep two services for wired and wireless access, but reuse the policy components (authentication methods, authentication source, authorization source, role mapping policies, posture policies, and enforcement policies) in both services.

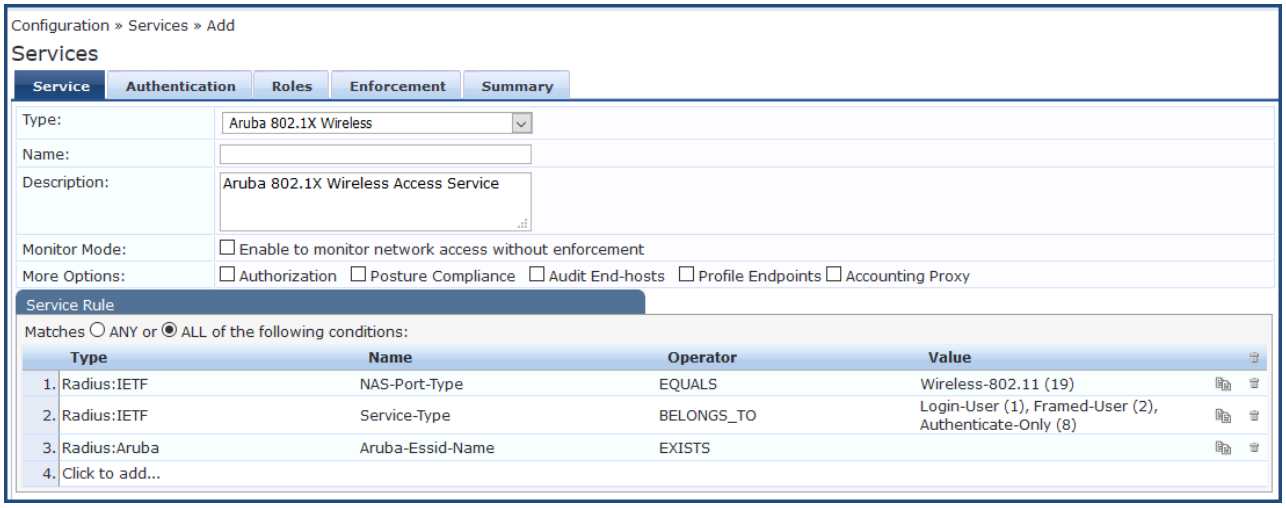
Service Configuration

The **Service** tab provides basic configuration parameters for the service. The **Service Rules** section defines a set of criteria that supplicants must match to trigger the service. Some service templates have one or more rules predefined. You can click on a service rule to modify any of its options.

To configure a 802.1X Wireless service:

1. Navigate to **Configuration > Services**, then click the **Add** link. The **Add Services** dialog opens.
2. From the **Type** drop-down, select **Aruba 802.1X Wireless**.The **Aruba 802.1X Wireless** service configuration dialog opens:

Figure 1. Aruba802.1X Wireless Service > Service Dialog



3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add Aruba 802.1X Wireless Service > Service Tab Parameters

Parameter	Action/Description
Type	Select **Aruba 802.1X Wire**
Name	Enter the name of this ser
Description	Policy Manager autofills t "Aruba 802.1X Wireless Ac ange the description if yc
Monitor Mode	Select this check box to r ctivity without enforceme
More Options	Check these boxes to acc uration tabs: • Authorization: See Au • Posture Compliance: S • Audit End - hosts: See • Profile Endpoints: See uration • Accounting Proxy: See guration
Service Rule	

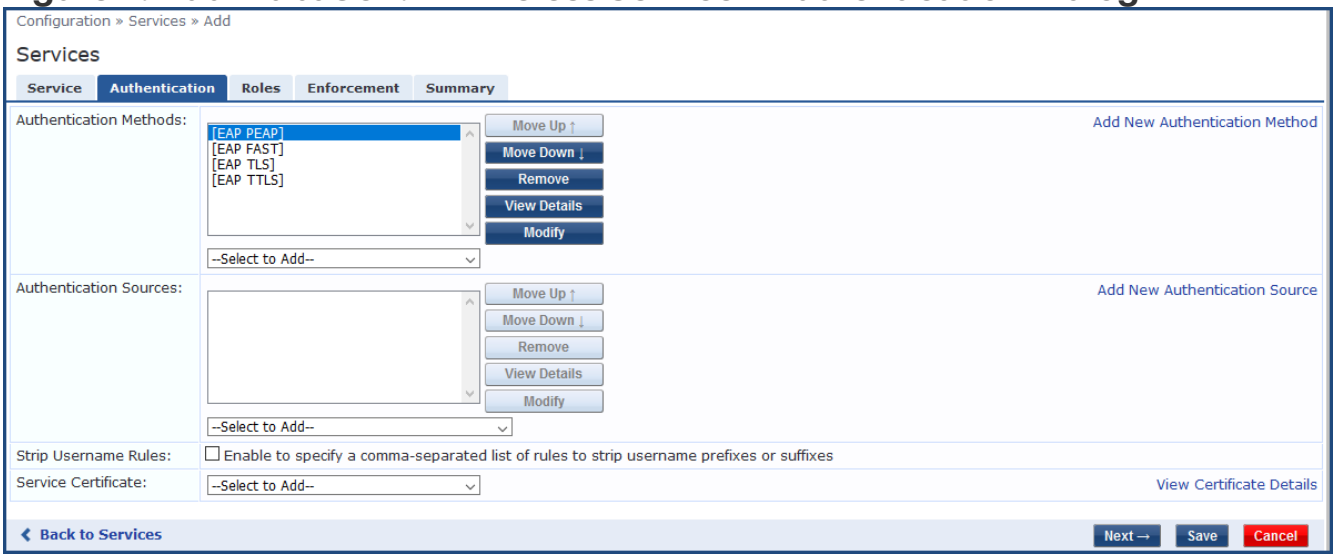
Parameter	Action/Description
Matches	Select the match conditic • Matches ANY • Matches ALL of the fo
Type	Select **Click to add** to sel
Name	Select the name of the se - down list.
Operator	Select an appropriate ope rators for the data type of
Value	Enter the value or select t - down list. The value list d **Operator** selected.

4. Click a service rule to modify its options.
5. Click **Next** to view the **Authentication** tab.

Authentication Configuration

The **Authentication** tab contains options for configuring authentication methods and authentication sources. The following figure displays the **Authentication** dialog:

Figure 2. AddAruba802.1X Wireless Service > Authentication Dialog



1. Specify the **Authentication** tab parameters as described in the following table:

Table 2. Add Aruba 802.1X Wireless Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	The authentication metho and the type of authentic ates this field with the foll • EAP PEAP • EAP FAST • EAP TLS with CN Chec • EAP TTLS Click the **Select to Add** fi uthentication method, sel The **EAP - MD5** authentic ode. The order of authenticatic tication. Policy Manager p he client can accept the a ue authentication or send uthentication method. If t hen the authentication pr If most of the clients in th ation method should be c IUS packets exchanged. For more information, see
Authentication Sources	Specify the authenticatio This can be one or more i d by default: • Admin User Repositor • Denylist Repository • Endpoints Repository • Guest Device Repositc • Guest User Repository

Parameter	Action/Description
	• Insight Repository • Local User Repository • Zone Cache Repositor • Onboard Devices Repo • Social Login Repositor • Time Source **NOTE** When you attem the following err *not exceed 23*. For more information, see
Strip Username Rules	Select the check box to p before authenticating and
Service Certificate	Optionally, select the Serv es.

Authorization Configuration

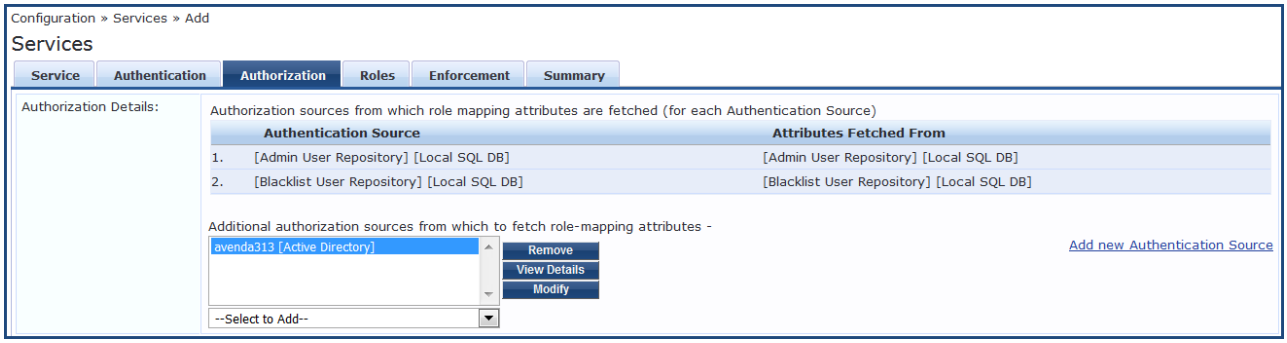
Use the **Authorization** tab to select the authorization sources for this service. The **Authorization** tab is not displayed by default. To access this tab, select the **More Options** > **Authorization**check box. Policy Manager fetches role-mapping attributes from the authorization sources associated with the service, regardless of which authentication source was used to authenticate the user.

For a given service, role-mapping attributes are fetched from the following authorization sources:

- Authorization sources associated with the authentication source
- Authorization sources associated with the service

The following figure displays the **Authorization** dialog:

Figure 3. AddAruba802.1X Wireless Service > Authorization Dialog



Specify the **Authorization** parameters as described in the following table:

Table 3. Add Aruba 802.1X Wireless Service > Authorization Parameters

Parameter	Action/Description
Authentication Source	Displays the authorization sources from which role mapping attributes are fetche d for each authentication source.
Attributes Fetched From	Displays the source of attributes.
Additional authorization sources from wh ich to fetch role - mapping attributes	Specify the authorization sources using t he **Select to Add** field. There can be one or more instances of th e following list of authorization sources: ◦ Admin User Repository ◦ Denylist Repository ◦ Endpoints Repository ◦ Guest Device Repository ◦ Guest User Repository ◦ Insight Repository ◦ Local User Repository ◦ Zone Cache Repository ◦ Onboard Devices Repository ◦ Social Login Repository

Parameter	Action/Description
	◦ Time Source **NOTE** *No. of Authorization Sources cannot exceed 23*

Roles Configuration

Use the **Roles** tab to associate a role-mapping policy with this service. The following figure displays the **Aruba 802.1X Wireless Service > Roles** dialog:

Figure 4. AddAruba802.1X Wireless Service > Roles Dialog

Specify the **Roles** parameters as described in the following table:

Table 4. Add Aruba 802.1X Wireless Service > Roles Tab Parameters

Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the drop - down list. Policy Manager ships a number of preconfi gured roles. **NOTE**
Role Mapping Policy Details	
Description	Provide additional information about the se lected role - mapping policy.

Parameter	Action/Description
Default Role	Specify the role to which Policy Manager d efaults when the role - mapping policy do es not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

For information on configuring role-mapping policies, see Configuring a Role and Role-Mapping Policy .

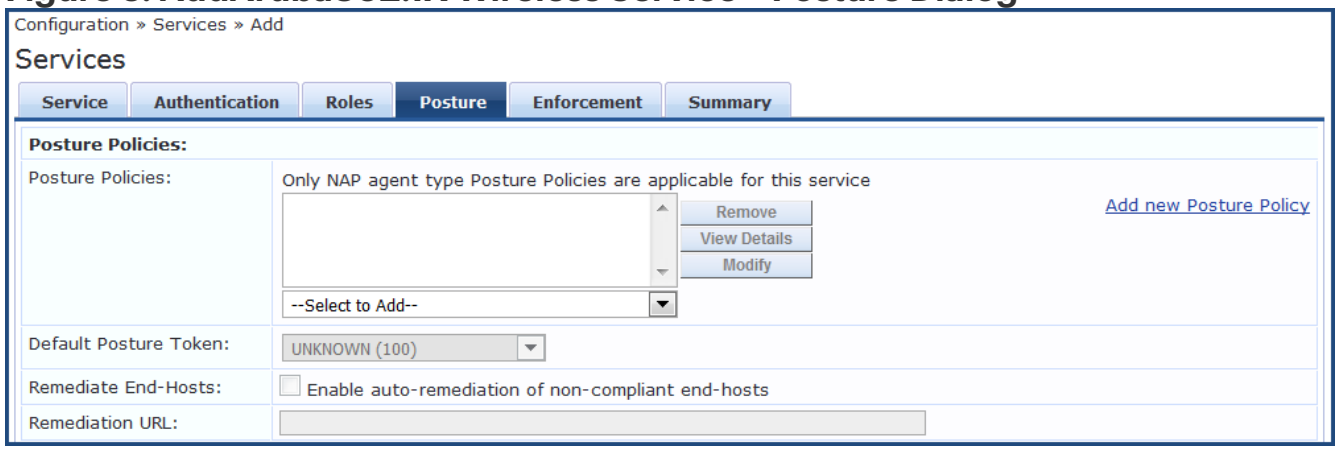
Posture Configuration

The **Posture** tab is not enabled by default. To enable posture configuration options, return to the **Service** tab and select **More Options** > **Posture Compliance**. You can enable the posture checking for this kind of service if you deploy any of the following:

- ClearPass Policy Manager in a Microsoft Network Access Protection (NAP) environment
- Cisco Network Admission Control (NAC) Framework environment
- Aruba hosted captive portal that performs posture checks through a dissolvable agent

The following figure displays the **Posture** dialog:

Figure 5. AddAruba802.1X Wireless Service > Posture Dialog



Specify the Wireless Service **Posture** parameters as described in Table 5:

Table 5. Add Aruba 802.1X Wireless Service > Posture Parameters

Parameter	Action/Description
Posture Policies	

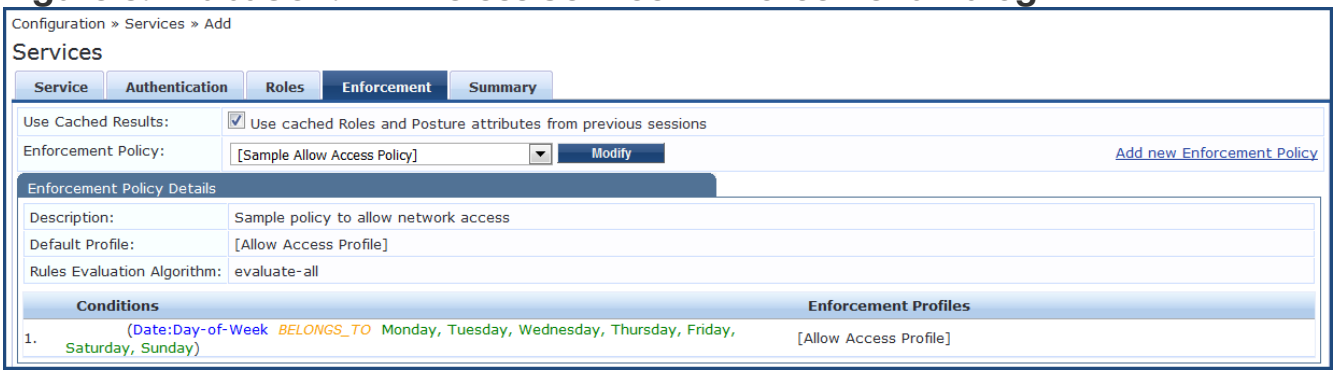
Parameter	Action/Description
Posture Policies	Select the posture policy from the **Select to Add** drop - down list. If you do not have any preconfigured posture policies, click **Add New Posture Policy** to create a new posture policy. **NOTE**
Default Posture Token	Select the default posture token from the drop - down list.
Remediate End - Hosts	To perform remediation action, when a client is quarantined, select the **Enable auto - remediation of non - compliant end - hosts** check box.
Remediation URL	To perform the remediation, enter the web link of a server resource.

For more information on configuring posture polices, see Configuring Posture Policy Agents and Hosts .

Enforcement Configuration

Use this tab to select an enforcement policy for a service. The following figure displays the **Enforcement** dialog:

Figure 6. Aruba802.1X Wireless Service > Enforcement Dialog



Specify the **Enforcement** parameters as described in the following table:

Table 6. Aruba 802.1X Wireless Service > Enforcement Parameters

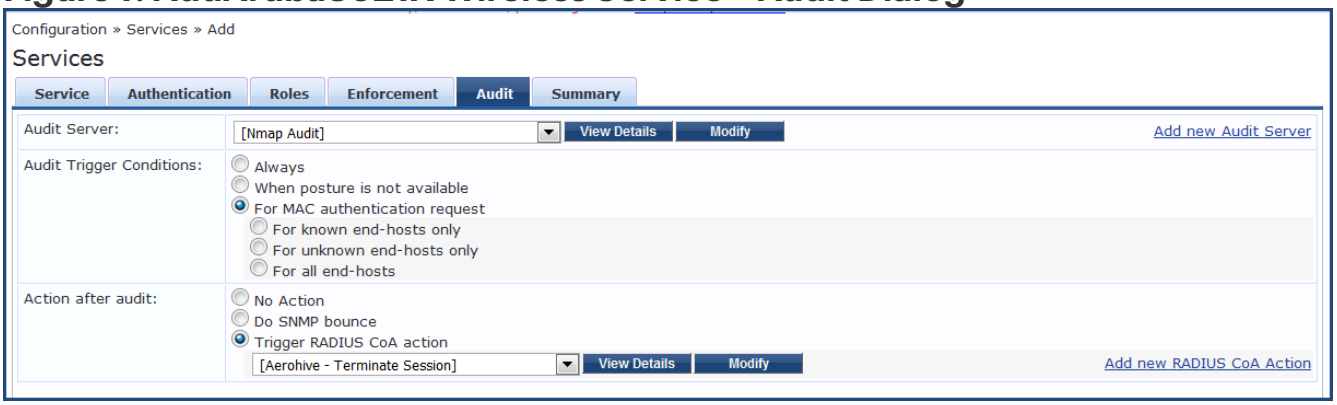
Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles a nd posture attributes from previous sessio ns.
Enforcement Policy	Select the preconfigured enforcement poli cy from the drop - down list. This is manda tory. If you do not have any preconfigured enfo rcement policies, click **Add New Enforcem ent Policy** to create a new enforcement pol icy.
Enforcement Policy Details	
Description	Displays additional information about the s elected enforcement policy.
Default Profile	Displays a default profile applied by Policy Manager Policy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

For related information, see Configuring Enforcement Policies .

Audit Configuration

The **Audit** tab is not enabled by default. To enable this tab, return to the **Service** tab and select **More Options** > **Audit End-hosts**. The **Audit** tab opens, as shown in the figure below.

Figure 7. AddAruba8021X Wireless Service > Audit Dialog



To enable audit checking for this service, specify the **Audit End-Hosts** parameters as described in the following table:

Table 7. Add Aruba 802.1X Wireless Service > Audit End - Hosts Parameters

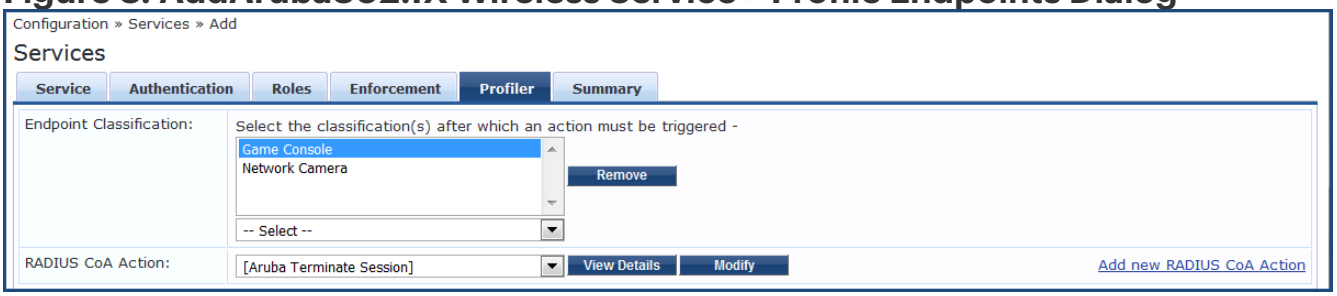
Parameter	Action/Description
Audit Server	Select the audit server from the following options: • **Nessus Server**: Interfaces with Policy Manager primarily to perform vulnerability scanning. • **Nmap Audit**: Performs specific Nmap audit functions. ◦ To view the **Policy Manager Entity Details** dialog with the summary of audit server details, click the **View Details** button. ◦ To view the **Summary** tab with audit server details, click the **Modify** button.
Audit Trigger Conditions	Select an audit trigger condition: • **Always**: Always perform an audit. • **When posture is not available**: Perform audit only when posture credentials are not available in the request. • **For MAC Authentication Request**: If you select this option, Policy Manager presents the following three additional settings: ◦ **For known end - hosts only:** Select this option when you want to reject unknown end - hosts and to audit known clients. *Known end - hosts* are defined as clients that are found in the authentication source(s) associated with this service. ◦ **For unknown end - hosts only:** Select this option when the known end - hosts are assumed to be healthy,

Parameter	Action/Description
	but you want to establish the identi ty of unknown end - hosts and assig n roles. *Unknown end - hosts* are en d - hosts that are not found in any o f the authentication sources associa ted with this service. ◦ **For all end - hosts:** For both known and unknown end - hosts.
Action After Audit	Specify the audit that can be performed on ly after the MAC authentication request is c ompleted and the client has acquired an IP address through DHCP. Once the audit results are available, Policy Manager reapplies policies on the network device in one of the following ways: • **No Action**: The audit does not apply pol icies on the network device after compl eting this audit. • **Do SNMP bounce**: This option bounces the switch port or forces an 802.1X reau thentication (both done using SNMP). ◦ Bouncing the port triggers a new 80 2.1X or MAC authentication request by the client. If the audit server alrea dy has the posture token and attribu tes associated with this client in its cache, it returns the token and the a ttributes to Policy Manager. • **Trigger RADIUS CoA action**: This optio n sends a RADIUS CoA command from Policy Manager to the network device.

Profile Endpoints Configuration

The **Profiler** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Profile Endpoints**. The **Add Profile Endpoints** dialog opens, as shown in the figure below.

Figure 8. AddAruba802.1X Wireless Service > Profile Endpoints Dialog



Specify the **Profile Endpoints** parameters as described in the following table:

Table 8. Add Aruba 802.1X Wireless Service > Profile Endpoints Parameters

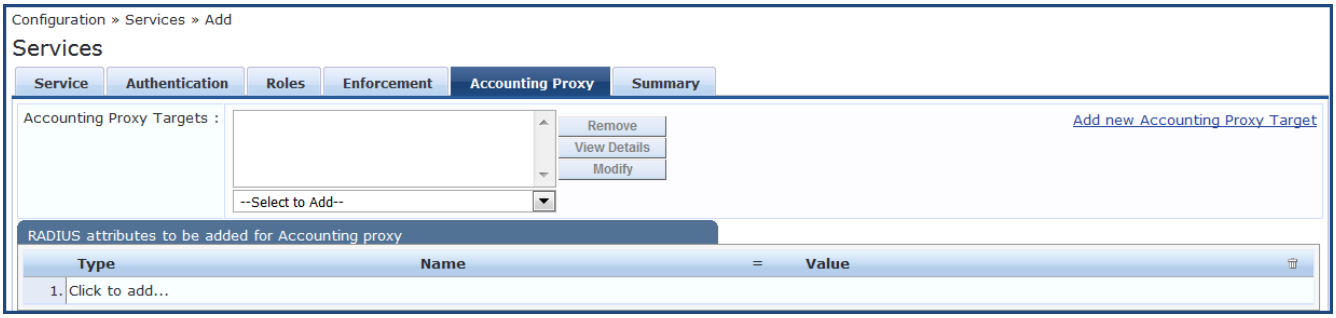
Parameter	Action/Description
Endpoint Classification	Select one or more endpoint classification items from the drop - down list.
RADIUS CoA Action	Select the RADIUS CoA action from the drop - down list. • To view the **Policy Manager Entity Details** page with the summary of enforcement profile details, click the **View Details** button. • To view the **Summary** tab with profile details, click the **Modify** button. • To create a new RADIUS CoA action, click the **Add New RADIUS CoA Action** link.

Accounting Proxy Configuration

The **Accounting Proxy** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Acconting Proxy**. Use the **Accounting Proxy** tab to broadcast the RADIUS accounting packets to all the proxy targets. You can configure the proxy targets to which RADIUS server should be forwarded and the attributes to be added in the accounting. This enables the external security solutions to use the RADIUS account event to detect when a user connects and disconnects to the server.

The following figure displays the **Accounting Proxy** dialog:

Figure 9. Add 802.1X Wireless > Accounting Proxy Dialog



Specify the **Accounting Proxy** parameters as described in the following table:

Table 9. Add Aruba 802.1X Wireless Service > Accounting Proxy Tab Parameters

Parameter	Action/Description
Accounting Proxy Targets	Specify the proxy targets to which the RADIUS server should be forwarded and the attributes to be added in the accounting. Select the accounting proxy target from the **Select to Add** drop - down list.
Add New Accounting Proxy Target	Click this link to add a new accounting proxy target.
RADIUS attributes to be added for Accounting Proxy	
Type	Select the RADIUS attribute type from the drop - down list: • Radius:IETF • Radius:Cisco • Radius:Hewlett - Packard - Enterprise • Radius:Alcatel - Lucent - Enterprise • Radius:Microsoft • Radius:Avenda • Radius:Aruba
Name	Select the name of the RADIUS attribute from the drop - down list.
Value	Select the value from the **Value** drop - down list.

Parameter	Action/Description
	The values displayed here depend on the RADIUS attribute **Type** selected.

Summary Information

The **Summary** page presents the summary of parameters defined when you created a new service.

802.1X Wireless Service

Configure this service for wireless end-hosts connecting through any 802.11 wireless access device or controller, with authentication via IEEE 802.1X. To create a wireless service with rules are customized for a typical Aruba Controller deployment, see also Aruba 802.1X Wireless Service.

> **NOTE**
> If you want to administer the same set of policies for wired and wireless access, you can combine the service rule to define one single service. The other option is to keep two services for wired and wireless access, but reuse the policy components (authentication methods, authentication source, authorization source, role mapping policies, posture policies, and enforcement policies) in both services.

Service Configuration

The **Service** tab provides basic configuration parameters for the service. The **Service Rules** section defines a set of criteria that supplicants must match to trigger the service. Some service templates have one or more rules predefined. You can click on a service rule to modify any of its options.

To configure a 802.1X Wireless service:

1. Navigate to **Configuration > Services**, then click the **Add** link. The **Add Services** dialog opens.
2. From the **Type** drop-down, select **802.1X Wireless**.
3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add 802.1X Wireless Service > Service Tab Parameters

Parameter	Action/Description
Type	Select **802.1X Wireless**.
Name	Enter the name of this se
Description	Policy Manager autofills t "802.1X Wireless Access he description if you wis
Monitor Mode	Select this check box to ctivity without enforceme
More Options	Check these boxes to ac uration tabs: • Authorization: See Au • Posture Compliance: • Audit End - hosts: See • Profile Endpoints: See uration • Accounting Proxy: Se guration
Service Rule	
Matches	Select the match conditi • Matches ANY • Matches ALL of the fo
Type	Select **Click to add** to sel
Name	Select the name of the se - down list.
Operator	Select an appropriate op rators for the data type o

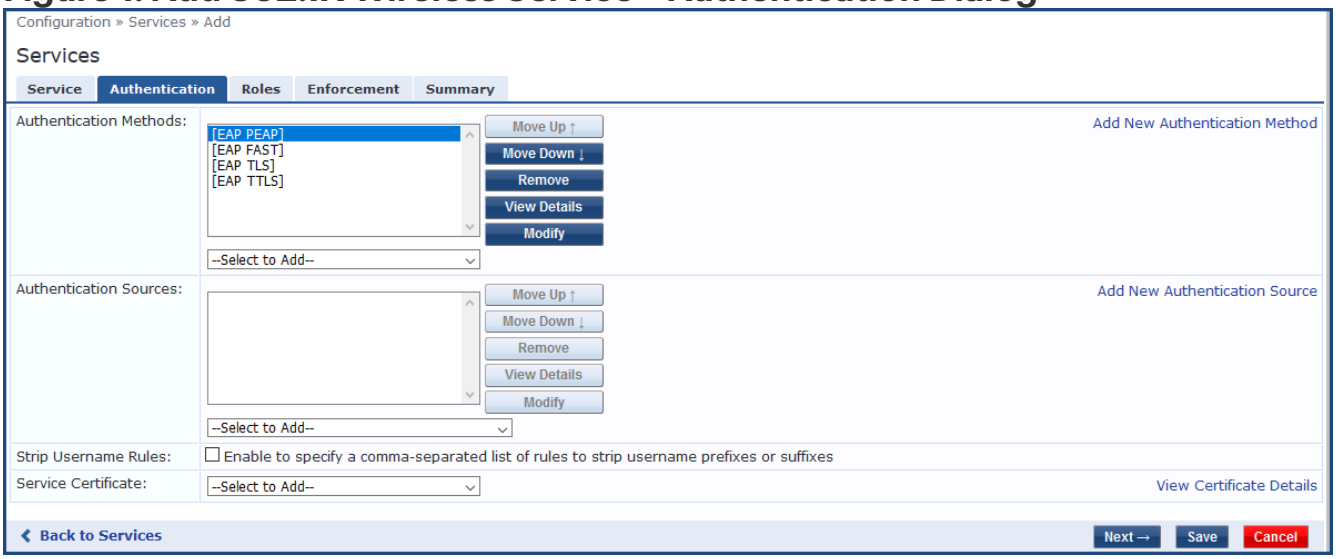
Parameter	Action/Description
Value	Enter the value or select t - down list. The value list d **Operator** selected.

4. Click a service rule to modify its options.
5. Click **Next** to view the **Authentication** tab.

Authentication Configuration

The **Authentication** tab contains options for configuring authentication methods and authentication sources. The following figure displays the **Authentication** dialog:

Figure 1. Add 802.1X Wireless Service > Authentication Dialog



1. Specify the **Authentication** tab parameters as described in the following table:

Table 2. Add 802.1X Wireless Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	The authentication methods available for this service depend on the 802.1X supplic ants and the type of authentication meth ods you choose to deploy. Policy Manage r propopulates this field with the followin g common authentication methods for th is service type.

Parameter	Action/Description
	• EAP PEAP • EAP FAST • EAP TLS with CN Check • EAP TTLS • EAP MSCHAPv2 Click the **Select to Add** field to add an a dditional authentication method. To rem ove an authentication method, select it t hen click **Remove**. The **EAP - MD5** authentication type is not supported if you use ClearPass Policy Ma nager in **FIPS** mode. The order of authentication is significant , when a client tries to perform an 802.1X authentication. Policy Manager proposes the first authentication method configure d. However, the client can accept the au thentication method proposed by Policy Manager and continue authentication or send a Negative - Acknowledgment (NAK) and propose a different authentication method. If the newly proposed authentic ation method is also configured, then th e authentication proceeds, otherwise aut hentication fails. If most of the clients in the network use a specific authentication method, that au thentication method should be configure d first in the list. This would reduce the n umber of RADIUS packets exchanged. For more information, see Adding and Mo difying Authentication Methods .
Authentication Sources	Specify the authentication sources using the **Select to Add** field.

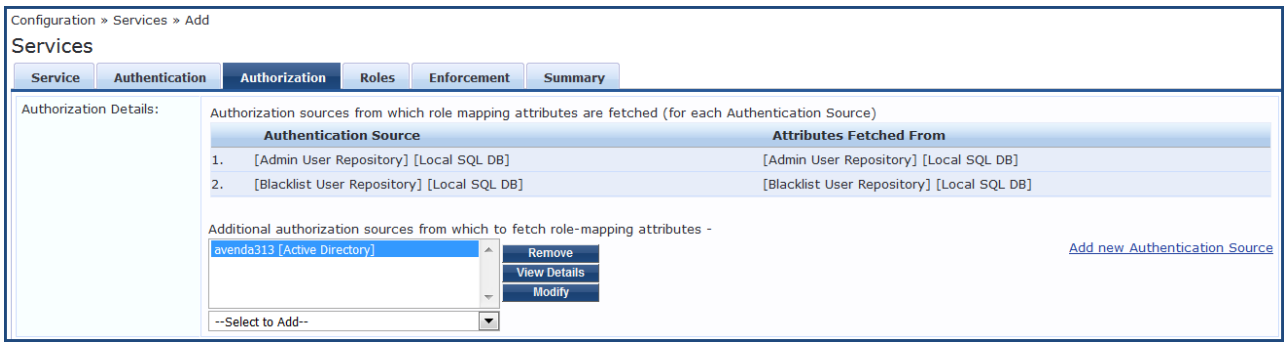
Parameter	Action/Description
	This can be one or more instances of the following list of authentication sources p rovided by default: • Admin User Repository • Denylist Repository • Endpoints Repository • Guest Device Repository • Guest User Repository • Insight Repository • Local User Repository • Zone Cache Repository • Onboard Devices Repository • Social Login Repository • Time Source **NOTE** When you attempt to speci fy more than 23 Services au thentication sources, the foll owing error message is disp layed: *No. of Authentication Sources cannot exceed 23*. For more information, see Adding and Co nfiguring Authentication Sources.
Strip Username Rules	Select the check box to preprocess the u sername (that is, remove prefixes and suf fixes) before authenticating and authoriz ing against the authentication source.
Service Certificate	Optionally, select the Service Certificate. For more information, refer to Service an d Client Certificates.

Authorization Configuration

Use the **Authorization** tab to select the authorization sources for this service. The **Authorization** tab is not displayed by default. To access this tab, select the **More Options** > **Authorization** check box. Policy Manager fetches role-mapping attributes from the authorization sources associated with the service, regardless of which authentication source was used to authenticate the user.

For a given service, role-mapping attributes are fetched from the following authorization sources:

- Authorization sources associated with the authentication source
- Authorization sources associated with the service



Specify the **Authorization** parameters as described in the following table:

Table 3. Add Aruba 802.1X Wireless Service > Authorization Parameters

Parameter	Action/Description
Authentication Source	Displays the authorization sources from which role mapping attributes are fetche d for each authentication source.
Attributes Fetched From	Displays the source of attributes.
Additional authorization sources from wh ich to fetch role - mapping attributes	Specify the authorization sources using t he **Select to Add** field. There can be one or more instances of th e following list of authorization sources: ◦ Admin User Repository ◦ Denylist Repository ◦ Endpoints Repository ◦ Guest Device Repository

Parameter	Action/Description
	◦ Guest User Repository ◦ Insight Repository ◦ Local User Repository ◦ Zone Cache Repository ◦ Onboard Devices Repository ◦ Social Login Repository ◦ Time Source **NOTE** *No. of Authorization Sources cannot exceed 23*

Roles Configuration

Use the **Roles** tab to associate a role-mapping policy with this service. Specify the **Roles** parameters as described in the following table:

Table 4. Add Aruba 802.1X Wireless Service > Roles Tab Parameters

Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the drop - down list. Policy Manager ships a number of preconfi gured roles. **NOTE**
Role Mapping Policy Details	
Description	Provide additional information about the se lected role - mapping policy.
Default Role	Specify the role to which Policy Manager d efaults when the role - mapping policy do es not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

For information on configuring role-mapping policies, see Configuring a Role and Role-Mapping Policy.

Posture Configuration

The **Posture** tab is not enabled by default. To enable posture configuration options, return to the **Service** tab and select **More Options** > **Posture Compliance**. You can enable the posture checking for this kind of service if you deploy any of the following:

- ClearPass Policy Manager in a Microsoft Network Access Protection (NAP) environment
- Cisco Network Admission Control (NAC) Framework environment
- Aruba hosted captive portal that performs posture checks through a dissolvable agent

Specify the Wireless Service **Posture** parameters as described in Table 5:

Table 5. Add 802.1X Wireless Service > Posture Parameters

Parameter	Action/Description
Posture Policies	
Posture Policies	Select the posture policy from the **Select to Add** drop - down list. If you do not have any preconfigured posture policies, click **Add New Posture Policy** to create a new posture policy. **NOTE**
Default Posture Token	Select the default posture token from the drop - down list.
Remediate End - Hosts	To perform remediation action, when a client is quarantined, select the **Enable auto - remediation of non - compliant end - hosts** check box.
Remediation URL	To perform the remediation, enter the web link of a server resource.

For more information on configuring posture polices, see Configuring Posture Policy Agents and Hosts.

Enforcement Configuration

Use this tab to select an enforcement policy for a service. Specify the **Enforcement** parameters as described in the following table:

Table 6. 802.1X Wireless Service > Enforcement Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles and posture attributes from previous sessions.
Enforcement Policy	Select the preconfigured enforcement policy from the drop - down list. This is mandatory. If you do not have any preconfigured enforcement policies, click **Add New Enforcement Policy** to create a new enforcement policy.
Enforcement Policy Details	
Description	Displays additional information about the selected enforcement policy.
Default Profile	Displays a default profile applied by Policy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

For related information, see Configuring Enforcement Policies.

Audit Configuration

The **Audit** tab is not enabled by default. To enable this tab, return to the **Service** tab and select **More Options** > **Audit End-hosts**. To enable audit checking for this service, specify the **Audit End-Hosts** parameters as described in the following table:

Table 7. Add Aruba 802.1X Wireless Service > Audit End - Hosts Parameters

Parameter	Action/Description
Audit Server	Select the audit server from the following options: • **Nessus Server**: Interfaces with Policy Manager primarily to perform vulnerability scanning.

Parameter	Action/Description
	• **Nmap Audit**: Performs specific Nmap a udit functions. • To view the **Policy Manager Entity Deta ils** dialog with the summary of audit se rver details, click the **View Details** butt on. • To view the **Summary** tab with audit se rver details, click the **Modify** button.
Audit Trigger Conditions	Select an audit trigger condition: • **Always**: Always perform an audit. • **When posture is not available**: Perform audit only when posture credentials are not available in the request. • **For MAC Authentication Request**: If yo u select this option, Policy Manager p resents the following three additional s ettings: ◦ **For known end - hosts only:** Select this option when you want to reject unknown end - hosts and to audit k nown clients. *Known end - hosts* are defined as clients that are found in t he authentication source(s) associat ed with this service. ◦ **For unknown end - hosts only:** Sele ct this option when the known end - hosts are assumed to be healthy, but you want to establish the identi ty of unknown end - hosts and assig n roles. *Unknown end - hosts* are en d - hosts that are not found in any o f the authentication sources associa ted with this service. ◦ **For all end - hosts:** For both known and unknown end - hosts.

Parameter	Action/Description
Action After Audit	Specify the audit that can be performed on ly after the MAC authentication request is c ompleted and the client has acquired an IP address through DHCP. Once the audit results are available, Policy Manager reapplies policies on the network device in one of the following ways: • **No Action**: The audit does not apply pol icies on the network device after compl eting this audit. • **Do SNMP bounce**: This option bounces the switch port or forces an 802.1X reau thentication (both done using SNMP). ◦ Bouncing the port triggers a new 80 2.1X or MAC authentication request by the client. If the audit server alrea dy has the posture token and attribu tes associated with this client in its cache, it returns the token and the a ttributes to Policy Manager. • **Trigger RADIUS CoA action**: This optio n sends a RADIUS CoA command from Policy Manager to the network device.

Profiler Configuration

The **Profiler** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Profile Endpoints**. Specify the **Profiler** parameters as described in the following table:

Table 8. Add 802.1X Wireless Service > Profile Endpoints Parameters

Parameter	Action/Description
Endpoint Classification	Select one or more endpoint classification items from the drop - down list.
RADIUS CoA Action	Select the RADIUS CoA action from the dro p - down list. • To view the **Policy Manager Entity Deta ils** page with the summary of enforcem

Parameter	Action/Description
	ent profile details, click the **View Detail s** button. • To view the **Summary** tab with profile d etails, click the **Modify** button. • To create a new RADIUS CoA action, cli ck the **Add New RADIUS CoA Action** lin k.

Accounting Proxy Configuration

The **Accounting Proxy** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Acconting Proxy**. Use the **Accounting Proxy** tab to broadcast the RADIUS accounting packets to all the proxy targets. You can configure the proxy targets to which RADIUS server should be forwarded and the attributes to be added in the accounting. This enables the external security solutions to use the RADIUS account event to detect when a user connects and disconnects to the server.

Specify the **Accounting Proxy** parameters as described in the following table:

Table 9. Add 802.1X Wireless Service > Accounting Proxy Tab Parameters

Parameter	Action/Description
Accounting Proxy Targets	Specify the proxy targets to which the RAD IUS server should be forwarded and the att ributes to be added in the accounting. Select the accounting proxy target from th e **Select to Add** drop - down list.
Add New Accounting Proxy Target	Click this link to add a new accounting p roxy target.
RADIUS attributes to be added for Accounting Proxy	
Type	Select the RADIUS attribute type from the drop - down list: • Radius:IETF • Radius:Cisco • Radius:Hewlett - Packard - Enterprise • Radius:Alcatel - Lucent - Enterprise • Radius:Microsoft

Parameter	Action/Description
	• Radius:Avenda • Radius:Aruba
Name	Select the name of the RADIUS attribute f rom the drop - down list.
Value	Select the value from the **Value** drop - dow n list. The values displayed here depend on the R ADIUS attribute **Type** selected.

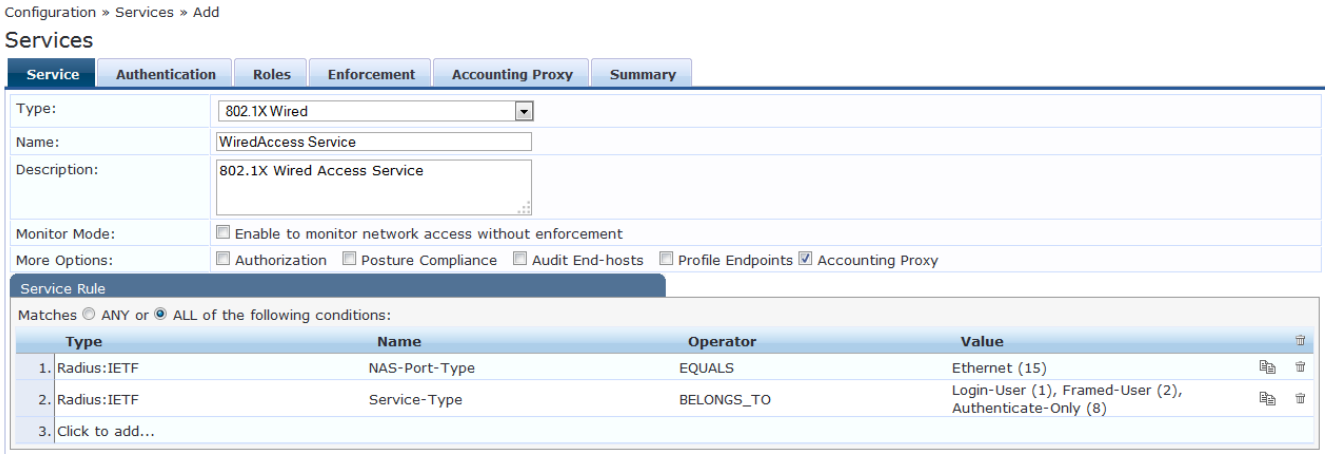
Summary Information

The **Summary** page presents the summary of parameters defined when you created a new service.

802.1X Wired Service

Navigate to **Configuration > Services** then select the **Add** link to onfigure this service for clients connecting through an Ethernet LAN with authentication using IEEE 802.1X. The following figure displays the **Add 802.1X Wired Service** page:

Figure 1. Add 802.1X Wired Service Dialog



Service Configuration

The **Service** tab provides basic configuration parameters for the service. The **Service Rules** section defines a set of criteria that supplicants must match to trigger the service. Some service templates have one or more rules predefined. You can click on a service rule to modify any of its options.

To configure a 802.1X Wired service:

1. Navigate to **Configuration > Services**, then click the **Add** link. The **Add Services** dialog opens.
2. From the **Type** drop-down, select **Aruba 802.1X Wired**.The **Aruba 802.1X Wired** service configuration dialog opens:
3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add Aruba 802.1X Wired Service > Service Tab Parameters

Parameter	Action/Description
Type	Select **Aruba 802.1X Wired**.
Name	Enter the name of this service.
Description	Policy Manager autofills the *Description* field with "Aruba 802.1X Wired Access Service." You can change the description if you wish.
Monitor Mode	Select this check box to monitor network access activity without enforcement.
More Options	Check these boxes to access the additional configuration tabs: • Authorization: See Authorization Configuration • Posture Compliance: See Posture Configuration • Audit End - hosts: See Audit Configuration • Profile Endpoints: See Profile Endpoints Configuration • Accounting Proxy: See Accounting Proxy Configuration
Service Rule	

Parameter	Action/Description
Matches	Select the match condition for this service: • Matches ANY • Matches ALL of the following conditions
Type	Select **Click to add** to select the service rule type.
Name	Select the name of the service rule from the drop - down list.
Operator	Select an appropriate operator from the list of operators for the data type of the attribute.
Value	Enter the value or select the value from the drop - down list. The value list depends on the **Type** and **Operator** selected.

4. Click a service rule to modify its options.
5. Click **Next** to view the **Authentication** tab.

Authentication Configuration

The **Authentication** tab contains options for configuring authentication methods and authentication sources. Specify the **Authentication** tab parameters as described in the following table:

Table 2. Add Aruba 802.1X Wired Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	The authentication methods available for this service depend on the 802.1X supplicants and the type of authentication methods you choose to deploy. Policy Manager propopulates this field with the following common authentication methods for this service type. • EAP PEAP

Parameter	Action/Description
	• EAP FAST • EAP TLS with CN check • EAP TTLS • EAP MSCHAPV2 Click the **Select to Add** field to add an additional authentication method. To remove an authentication method, select it then click **Remove**. The **EAP - MD5** authentication type is not supported if you use Policy Manager in **FIPS** mode. The order of authentication is significant, when a client tries to perform an 802.1X authentication. Policy Manager proposes the first authentication method configured. However, the client can accept the authentication method proposed by Policy Manager and continue authentication or send a Negative - Acknowledgment (NAK) and propose a different authentication method. If the newly proposed authentication method is also configured, then the authentication proceeds, otherwise authentication fails. If most of the clients in the network use a specific authentication method, that authentication method should be configured first in the list. This would reduce the number of RADIUS packets exchanged. For more information, see Adding and Configuring Authentication Methods .
Authentication Sources	Specify the authentication sources using the **Select to Add** field. This can be one or more instances of the following list of authentication sources provided by default: • Admin User Repository • Denylist Repository • Endpoints Repository • Guest Device Repository

Parameter	Action/Description
	• Guest User Repository • Insight Repository • Local User Repository • Zone Cache Repository • Onboard Devices Repository • Social Login Repository • Time Source **NOTE** When you attempt to specify more than 23 Services authentication sources, the following error message is displayed: *No. of Authentication Sources cannot exceed 23*. For more information, see Adding and Configuring Authentication Sources .
Strip Username Rules	Select the check box to preprocess the username (that is, remove prefixes and suffixes) before authenticating and authorizing against the authentication source.
Service Certificate	Optionally, select the Service Certificate. For more information, refer to Service Certificates.

Authorization Configuration

The **Authorization** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Authorization**. Use the **Authorization** tab to select the authorization sources for this service. Policy Manager fetches role-mapping attributes from the authorization sources associated with the service, regardless of which authentication source was used to authenticate the user.

For a given service, role-mapping attributes are fetched from the following authorization sources:

- Authorization sources associated with the authentication source
- Authorization sources associated with the service

Specify the **Authorization** parameters as described in the following table:

Table 3. Add Aruba 802.1X Wired Service > Authorization Parameters

Parameter	Action/Description
Authentication Source	Displays the authorization sources from which role mapping attributes are fetche d for each authentication source.
Attributes Fetched From	Displays the source of attributes.
Additional authorization sources from wh ich to fetch role - mapping attributes	Specify the authorization sources using t he **Select to Add** field. There can be one or more instances of th e following list of authorization sources: ◦ Admin User Repository ◦ Denylist Repository ◦ Endpoints Repository ◦ Guest Device Repository ◦ Guest User Repository ◦ Insight Repository ◦ Local User Repository ◦ Zone Cache Repository ◦ Onboard Devices Repository ◦ Social Login Repository ◦ Time Source **NOTE** *No. of Authorization Sources cannot exceed 23*

Roles Configuration

Use the **Roles** tab to associate a role-mapping policy with this service. Specify the **Roles** parameters as described in the following table:

Table 4. Add Aruba 802.1X Wired Service > Roles Tab Parameters

Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the drop - down list. Policy Manager ships a numbe r of preconfigured roles. **NOTE**
Role Mapping Policy Details	
Description	Provide additional information about the se lected role - mapping policy.
Default Role	Specify the role to which Policy Manager d efaults when the role - mapping policy do es not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

For information on configuring role-mapping policies, see Configuring a Role and Role-Mapping Policy .

Posture Configuration

The **Posture** tab is not enabled by default. To access this tab, return to the **Service** tab and select **More Options** > **Posture Compliance**. Enable the posture checking for this kind of service if you deploy any of the following:

- ClearPass Policy Manager in a Microsoft Network Access Protection (NAP) environment
- Cisco Network Admission Control (NAC) Framework environment
- Aruba hosted captive portal that performs posture checks through a dissolvable agent

To enable posture checking for this service, specify the posture parameters as described in Table 5:

Table 5. Add Aruba 802.1X Wired Service > Posture Parameters

Parameter	Action/Description
Posture Policies	

Parameter	Action/Description
Posture Policies	Select the posture policy from the **Select to Add** drop - down list. If you do not have any preconfigured posture policies, click **Add New Posture Policy** to create a new posture policy. **NOTE**
Default Posture Token	Select the default posture token from the drop - down list.
Remediate End - Hosts	To perform remediation action, when a client is quarantined, select the **Enable auto - remediation of non - compliant end - hosts** check box.
Remediation URL	To perform the remediation, enter the web link of a server resource.

For more information on configuring posture polices, see Configuring Posture Policy Agents and Hosts .

Enforcement Configuration

Use this tab to select an enforcement policy for a service. Specify the **Enforcement** parameters as described in the following table:

Table 6. Aruba 802.1X Wired Service > Enforcement Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles and posture attributes from previous sessions.
Enforcement Policy	Select the preconfigured enforcement policy from the drop - down list. This is mandatory. If you do not have any preconfigured enforcement policies, click **Add New Enforcement Policy** to create a new enforcement policy.
Enforcement Policy Details	

Parameter	Action/Description
Description	Displays additional information about the s elected enforcement policy.
Default Profile	Displays a default profile applied by Policy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

For related information, see Configuring Enforcement Policies .

Audit Configuration

The **Audit** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Audit End-hosts**.

Use the **Audit** tab to enable the Audit checking for this service by specifying the **AuditEnd-Hosts** parameters as described in the following table:

Table 7. Add Aruba 802.1X Wired Service > Audit End - Hosts Parameters

Parameter	Action/Description
Audit Server	Select the audit server from the following o ptions: • **Nessus Server**: Interfaces with Policy M anager primarily to perform vulnerabilit y scanning. • **Nmap Audit**: Performs specific Nmap a udit functions. ◦ To view the **Policy Manager Entity D etails** dialog with the summary of a udit server details, click the **View De tails** button. ◦ To view the **Summary** tab with audit server details, click the **Modify** butt on.
Audit Trigger Conditions	Select an audit trigger condition: • **Always**: Always perform an audit.

Parameter	Action/Description
	• **When posture is not available**: Perform audit only when posture credentials are not available in the request. • **For MAC Authentication Request**: If yo u select this option, Policy Manager p resents the following three additional s ettings: ◦ **For known end - hosts only:** Select this option when you want to reject unknown end - hosts and to audit k nown clients. *Known end - hosts* are defined as clients that are found in t he authentication source(s) associat ed with this service. ◦ **For unknown end - hosts only:** Sele ct this option when the known end - hosts are assumed to be healthy, but you want to establish the identi ty of unknown end - hosts and assig n roles. *Unknown end - hosts* are en d - hosts that are not found in any o f the authentication sources associa ted with this service. ◦ **For all end - hosts:** For both known and unknown end - hosts.
Action After Audit	Specify the audit that can be performed on ly after the MAC authentication request is c ompleted and the client has acquired an IP address through DHCP. Once the audit results are available, Policy Manager reapplies policies on the network device in one of the following ways: • **No Action**: The audit does not apply pol icies on the network device after compl eting this audit.

Parameter	Action/Description
	• **Do SNMP bounce**: This option bounces the switch port or forces an 802.1X reau thentication (both done using SNMP). ◦ Bouncing the port triggers a new 80 2.1X or MAC authentication request by the client. If the audit server alrea dy has the posture token and attribu tes associated with this client in its cache, it returns the token and the a ttributes to Policy Manager. • **Trigger RADIUS CoA action**: This optio n sends a RADIUS CoA command from Policy Manager to the network device.

Profile Endpoints Configuration

The **Profiler** tab is not displayed by default. To access this tab, return to the **Services** tab and select **More Options > Profile Endpoints**.

Specify the **Profile Endpoints** parameters as described in the following table:

Table 8. Add Aruba 802.1X Wired Service > Profile Endpoints Parameters

Parameter	Action/Description
Endpoint Classification	Select one or more endpoint classification items from the drop - down list.
RADIUS CoA Action	Select the RADIUS CoA action from the dro p - down list. • To view the **Policy Manager Entity Deta ils** page with the summary of enforcem ent profile details, click the **View Detail s** button. • To view the **Summary** tab with profile d etails, click the **Modify** button. • To create a new RADIUS CoA action, cli ck the **Add New RADIUS CoA Action** lin k.

Accounting Proxy Configuration

The **Accounting Proxy** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Acounting Proxy**. Use the **Accounting Proxy** tab to broadcast the RADIUS accounting packets to all the proxy targets. You can configure the proxy targets to which RADIUS server should be forwarded and the attributes to be added in the accounting. This enables the external security solutions to use the RADIUS account event to detect when a user connects and disconnects to the server.

Specify the **Accounting Proxy** parameters as described in the following table:

Table 9. Add Aruba 802.1X Wired Service > Accounting Proxy Tab Parameters

Parameter	Action/Description
Accounting Proxy Targets	Specify the proxy targets to which the RAD IUS server should be forwarded and the att ributes to be added in the accounting. Select the accounting proxy target from th e **Select to Add** drop - down list.
Add New Accounting Proxy Target	Click this link to add a new accounting p roxy target.
RADIUS attributes to be added for Accounting Proxy	
Type	Select the RADIUS attribute type from the drop - down list: • Radius:IETF • Radius:Cisco • Radius:Hewlett - Packard - Enterprise • Radius:Alcatel - Lucent - Enterprise • Radius:Microsoft • Radius:Avenda • Radius:Aruba
Name	Select the name of the RADIUS attribute f rom the drop - down list.
Value	Select the value from the **Value** drop - dow n list. The values displayed here depend on the R ADIUS attribute **Type** selected.

Summary Information

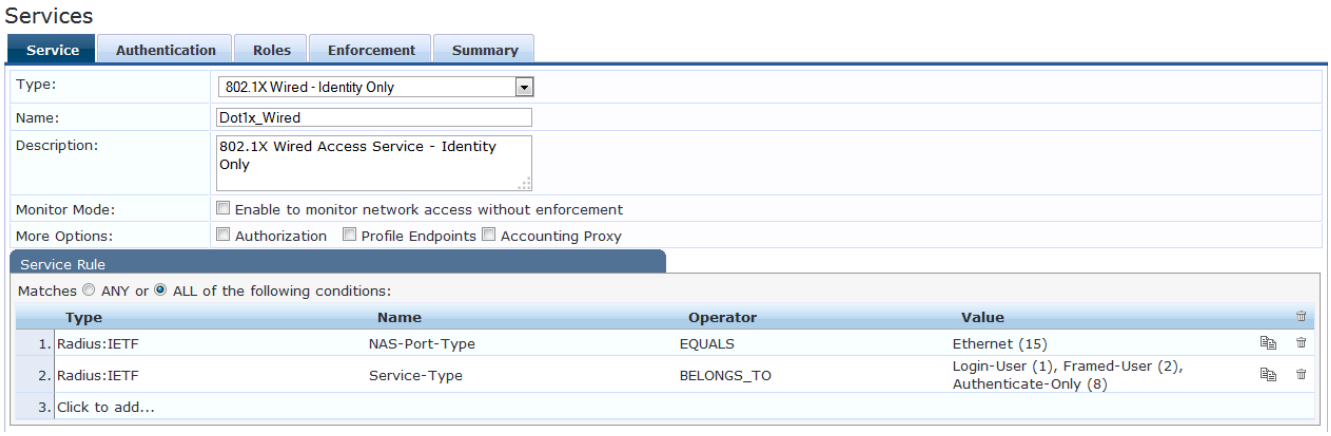
The **Summary** page presents the summary of parameters defined when you created a new service.

802.1X Wired—Identity Only Service

Configure this service for clients connecting through an Ethernet LAN with authentication using IEEE 802.1X. Configuration for this type of service is the same as the **Aruba 802.1X Wired** service, except that **posture** and **audit** policies are not configurable when you use this template.

The following figure displays the **802.1X Wired—Identity Only Service** dialog:

Figure 1. 802.1X Wired—Identity Only Service



Service Configuration

The **Service** tab provides basic configuration parameters for the service. The **Service Rules** section defines a set of criteria that supplicants must match to trigger the service. Some service templates have one or more rules predefined. You can click on a service rule to modify any of its options.

To configure a 802.1X Wired -Identity Only service:

1. Navigate to **Configuration > Services**, then click the **Add** link. The **Add Services** dialog opens.
2. From the **Type** drop-down, select **Aruba 802.1X Wired - Identity Only**.
3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add Aruba 802.1X Wired - Identity Only Service > Service Tab Parameters

Parameter	Action/Description
Type	Select **Aruba 802.1X Wired - Identity Only** .
Name	Enter the name of this service.
Description	Policy Manager autofills the *Description* field with "802.1X Wired Access Service - Identity Only." You can change the description if you wish.
Monitor Mode	Select this check box to monitor network access activity without enforcement.
More Options	Check these boxes to access the additional configuration tabs: • Authorization: See Authorization Configuration • Profile Endpoints: See Profile Endpoints Configuration • Accounting Proxy: See Accounting Proxy Configuration
Service Rule	
Matches	Select the match condition for this service: • Matches ANY • Matches ALL of the following conditions
Type	Select **Click to add** to select the service rule type.
Name	Select the name of the service rule from the drop - down list.

Parameter	Action/Description
Operator	Select an appropriate operator from the l ist of operators for the data type of the at tribute.
Value	Enter the value or select the value from t he drop - down list. The value list depend s on the **Type** and **Operator** selected.

4. Click a service rule to modify its options.
5. Click **Next** to view the **Authentication** tab.

Authentication Configuration

The **Authentication** tab contains options for configuring authentication methods and authentication sources. Specify the **Authentication** tab parameters as described in the following table:

Table 2. Add Aruba 802.1X Wired - Identity Only Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	The authentication methods available for t his service depend on the 802.1X supplican ts and the type of authentication methods you choose to deploy. Policy Manager prop opulates this field with the following comm on authentication methods for this service type. • EAP PEAP • EAP FAST • EAP TLS with CN Check • EAP TTLS • EAP MSCHAPv2 Click the **Select to Add** field to add an add itional authentication method. To remove a n authentication method, select it then clic k **Remove**.

Parameter	Action/Description
	The **EAP - MD5** authentication type is not supported if you use Policy Manager in **FIPS** mode. The order of authentication is significant, when a client tries to perform an 802.1X authentication. Policy Manager proposes the first authentication method configured. However, the client can accept the authentication method proposed by Policy Manager and continue authentication or send a Negative - Acknowledgment (NAK) and propose a different authentication method. If the newly proposed authentication method is also configured, then the authentication proceeds, otherwise authentication fails. If most of the clients in the network use a specific authentication method, that authentication method should be configured first in the list. This would reduce the number of RADIUS packets exchanged. For more information, see Adding and Configuring Authentication Methods .
Authentication Sources	Specify the authentication sources using the **Select to Add** field. This can be one or more instances of the following list of authentication sources provided by default: • Admin User Repository • Denylist Repository • Endpoints Repository • Guest Device Repository • Guest User Repository • Insight Repository • Local User Repository • Zone Cache Repository • Onboard Devices Repository • Social Login Repository

Parameter	Action/Description
	• Time Source **NOTE** When you attempt to specify more than 23 Services authentication sources, the following error message is displayed: *No. of Authentication Sources cannot exceed 23*. For more information, see Adding and Configuring Authentication Sources .
Strip Username Rules	Select the check box to preprocess the username (that is, remove prefixes and suffixes) before authenticating and authorizing against the authentication source.
Service Certificate	Optionally, select the Service Certificate. For more information, refer to Service Certificates.

Authorization Configuration

Use the **Authorization** tab to select the authorization sources for this service. The **Authorization** tab is not displayed by default. To access this tab, select the **More Options** > **Authorization**check box. Policy Manager fetches role-mapping attributes from the authorization sources associated with the service, regardless of which authentication source was used to authenticate the user.

For a given service, role-mapping attributes are fetched from the following authorization sources:

- Authorization sources associated with the authentication source
- Authorization sources associated with the service

  Specify the **Authorization** parameters as described in the following table:

Table 3. Add Aruba 802.1X Wired - Identity Only Service > Authorization Parameters

Parameter	Action/Description
Authentication Source	Displays the authorization sources from which role mapping attributes are fetched for each authentication source.
Attributes Fetched From	Displays the source of attributes.
Additional authorization sources from which to fetch role - mapping attributes	Specify the authorization sources using the **Select to Add** field. There can be one or more instances of the following list of authorization sources: ◦ Admin User Repository ◦ Denylist Repository ◦ Endpoints Repository ◦ Guest Device Repository ◦ Guest User Repository ◦ Insight Repository ◦ Local User Repository ◦ Zone Cache Repository ◦ Onboard Devices Repository ◦ Social Login Repository ◦ Time Source **NOTE** *No. of Authorization Sources cannot exceed 23*

Roles Configuration

Use the **Roles** tab to associate a role-mapping policy with this service.

Specify the **Roles** parameters as described in the following table:

Table 4. Add Aruba 802.1X Wired - Identity Only Service > Roles Tab Parameters

Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the drop - down list. Policy Manager ships a number of preconfi gured roles. **NOTE** A service can be configured w ithout a role - mapping policy, but only one role - mapping p olicy can be configured for ea ch service.
Role Mapping Policy Details	
Description	Provide additional information about the se lected role - mapping policy.
Default Role	Specify the role to which Policy Manager d efaults when the role - mapping policy do es not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

For information on configuring role-mapping policies, see Configuring a Role and Role-Mapping Policy .

Enforcement Configuration

Use this tab to select an enforcement policy for a service by specifying the **Enforcement** parameters as described in the following table:

Table 5. Aruba 802.1X Wired - Identity Only Service > Enforcement Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles a nd posture attributes from previous sessio ns.
Enforcement Policy	Select the preconfigured enforcement poli cy from the drop - down list. This is manda tory.

Parameter	Action/Description
	If you do not have any preconfigured enfo rcement policies, click **Add New Enforcem ent Policy** to create a new enforcement pol icy.
Enforcement Policy Details	
Description	Displays additional information about the s elected enforcement policy.
Default Profile	Displays a default profile applied by Policy Manager Policy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

For related information, see Configuring Enforcement Policies .

Profile Endpoints Configuration

The **Profiler** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Profile Endpoints**.

Specify the **Profile Endpoints** parameters as described in the following table:

Table 6. Add 802.1X Wired - Identity Only Service > Profile Endpoints Parameters

Parameter	Action/Description
Endpoint Classification	Select one or more endpoint classification items from the drop - down list.
RADIUS CoA Action	Select the RADIUS CoA action from the dro p - down list. • To view the **Policy Manager Entity Deta ils** page with the summary of enforcem ent profile details, click the **View Detail s** button. • To view the **Summary** tab with profile d etails, click the **Modify** button. • To create a new RADIUS CoA action, cli ck the **Add New RADIUS CoA Action** lin k.

Accounting Proxy Configuration

The **Accounting Proxy** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Acconting Proxy**. Use the **Accounting Proxy** tab to broadcast the RADIUS accounting packets to all the proxy targets. You can configure the proxy targets to which RADIUS server should be forwarded and the attributes to be added in the accounting. This enables the external security solutions to use the RADIUS account event to detect when a user connects and disconnects to the server.

Specify the **Accounting Proxy** parameters as described in the following table:

Table 7. Add Aruba 802.1X Wired - Identity Only Service > Accounting Proxy Tab Parameters

Parameter	Action/Description
Accounting Proxy Targets	Specify the proxy targets to which the RADIUS server should be forwarded and the attributes to be added in the accounting. Select the accounting proxy target from the **Select to Add** drop - down list.
Add New Accounting Proxy Target	Click this link to add a new accounting proxy target.
RADIUS attributes to be added for Accounting Proxy	
Type	Select the RADIUS attribute type from the drop - down list: • Radius:IETF • Radius:Cisco • Radius:Hewlett - Packard - Enterprise • Radius:Alcatel - Lucent - Enterprise • Radius:Microsoft • Radius:Avenda • Radius:Aruba
Name	Select the name of the RADIUS attribute from the drop - down list.
Value	Select the value from the **Value** drop - down list. The values displayed here depend on the RADIUS attribute **Type** selected.

Summary Information

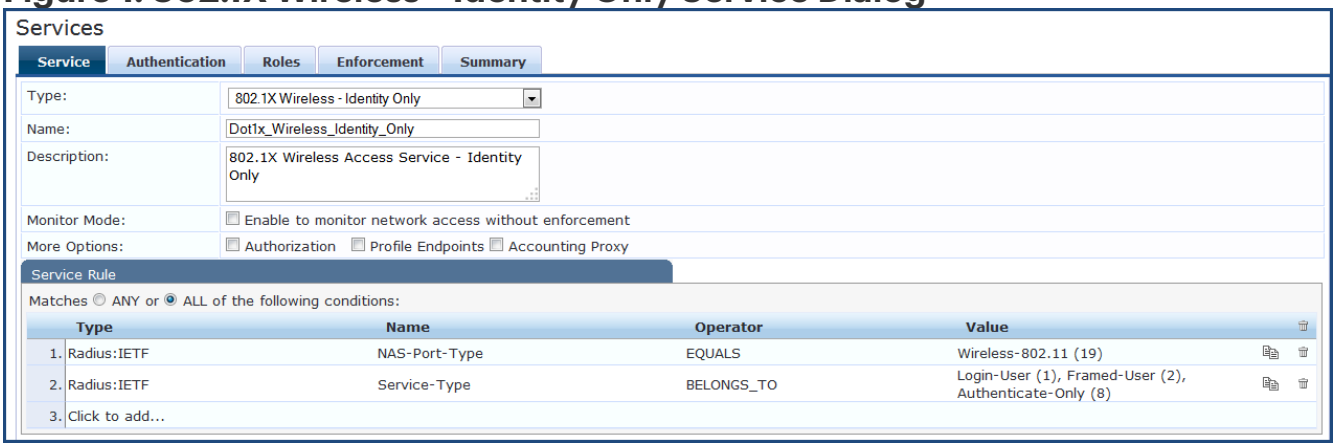
The **Summary** page presents the summary of parameters defined when you created a new service.

802.1X Wireless—Identity Only Service

Configuration for this type of service is the same as the **802.1X Wireless Service**, except that **Posture** and **Audit** policies are not configurable when you use this template. For more information, see Aruba 802.1X Wireless Service.

The following figure displays the **Configuration > Services > Add 802.1X Wireless—Identity Only Service** dialog:

Figure 1. 802.1X Wireless—Identity Only Service Dialog



Service Configuration

The **Service** tab provides basic configuration parameters for the service. The **Service Rules** section defines a set of criteria that supplicants must match to trigger the service. Some service templates have one or more rules predefined. You can click on a service rule to modify any of its options.

To configure a 802.1X Wireless -Identity Only service:

1. Navigate to **Configuration > Services**, then click the **Add** link. The **Add Services** dialog opens.
2. From the **Type** drop-down, select **Aruba 802.1X Wireless - Identity Only**.
3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add Aruba 802.1X Wireless - Identity Only Service > Service Tab Parameters

Parameter	Action/Description
Type	Select **Aruba 802.1X Wireless - Identity Only** .
Name	Enter the name of this service.
Description	Policy Manager autofills the *Description* field with "802.1X Wireless Access Service - Identity Only." You can change the description if you wish.
Monitor Mode	Select this check box to monitor network access activity without enforcement.
More Options	Check these boxes to access the additional configuration tabs: • Authorization: See Authorization Configuration • Profile Endpoints: See Profile Endpoints Configuration • Accounting Proxy: See Accounting Proxy Configuration
Service Rule	
Matches	Select the match condition for this service: • Matches ANY • Matches ALL of the following conditions
Type	Select **Click to add** to select the service rule type.
Name	Select the name of the service rule from the drop - down list.

Parameter	Action/Description
Operator	Select an appropriate operator from the l ist of operators for the data type of the at tribute.
Value	Enter the value or select the value from t he drop - down list. The value list depend s on the **Type** and **Operator** selected.

4. Click a service rule to modify its options.
5. Click **Next** to view the **Authentication** tab.

Authentication Configuration

The **Authentication** tab contains options for configuring authentication methods and authentication sources. Specify the **Authentication** tab parameters as described in the following table:

Table 2. Add Aruba 802.1X Wireless - Identity Only Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	The authentication methods available for t his service depend on the 802.1X supplican ts and the type of authentication methods you choose to deploy. Policy Manager prop opulates this field with the following comm on authentication methods for this service type. • EAP PEAP • EAP FAST • EAP TLS with CN Check • EAP TTLS Click the **Select to Add** field to add an add itional authentication method. To remove a n authentication method, select it then clic k **Remove**. The **EAP - MD5** authentication type is not s upported if you use Policy Manager in **FIPS** mode.

Parameter	Action/Description
	The order of authentication is significant, when a client tries to perform an 802.1X a uthentication. Policy Manager proposes th e first authentication method configured. However, the client can accept the authen tication method proposed by Policy Manag er and continue authentication or send a N egative - Acknowledgment (NAK) and prop ose a different authentication method. If th e newly proposed authentication method i s also configured, then the authentication proceeds, otherwise authentication fails. If most of the clients in the network use a s pecific authentication method, that authen tication method should be configured first in the list. This would reduce the number o f RADIUS packets exchanged. For more information, see Adding and Mod ifying Authentication Methods .
Authentication Sources	Specify the authentication sources using t he **Select to Add** field. This can be one or more instances of the fo llowing list of authentication sources provi ded by default: • Admin User Repository • Denylist Repository • Endpoints Repository • Guest Device Repository • Guest User Repository • Insight Repository • Local User Repository • Zone Cache Repository • Onboard Devices Repository • Social Login Repository • Time Source **NOTE**

Parameter	Action/Description
	When you attempt to specify more than 23 Services authentication sources, the following error message is displayed: *No. of Authentication Sources cannot exceed 23*. For more information, see Adding and Configuring Authentication Sources.
Strip Username Rules	Select the check box to preprocess the username (that is, remove prefixes and suffixes) before authenticating and authorizing against the authentication source.
Service Certificate	Optionally, select the Service Certificate. For more information, refer to Service and Client Certificates.

Authorization Configuration

Use the **Authorization** tab to select the authorization sources for this service. The **Authorization** tab is not displayed by default. To access this tab, select the **More Options** > **Authorization**check box. Policy Manager fetches role-mapping attributes from the authorization sources associated with the service, regardless of which authentication source was used to authenticate the user.

For a given service, role-mapping attributes are fetched from the following authorization sources:

- Authorization sources associated with the authentication source
- Authorization sources associated with the service

Specify the **Authorization** parameters as described in the following table:

Table 3. Add Aruba 802.1X Wireless - Identity Only Service > Authorization Parameters

Parameter	Action/Description
Authentication Source	Displays the authorization sources from which role mapping attributes are fetched for each authentication source.
Attributes Fetched From	Displays the source of attributes.

Parameter	Action/Description
Additional authorization sources from wh ich to fetch role - mapping attributes	Specify the authorization sources using t he **Select to Add** field. There can be one or more instances of th e following list of authorization sources: ◦ Admin User Repository ◦ Denylist Repository ◦ Endpoints Repository ◦ Guest Device Repository ◦ Guest User Repository ◦ Insight Repository ◦ Local User Repository ◦ Zone Cache Repository ◦ Onboard Devices Repository ◦ Social Login Repository ◦ Time Source **NOTE** *No. of Authorization Sources cannot exceed 23*

Roles Configuration

Use the **Roles** tab to associate a role-mapping policy with this service.

Specify the **Roles** parameters as described in the following table:

Table 4. Add Aruba 802.1X Wireless - Identity Only Service > Roles Tab Parameters

Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the drop - down list.

Parameter	Action/Description
	Policy Manager ships a number of preconfi gured roles. **NOTE** A service can be configured w ithout a role - mapping policy, but only one role - mapping p olicy can be configured for ea ch service.
Role Mapping Policy Details	
Description	Provide additional information about the se lected role - mapping policy.
Default Role	Specify the role to which Policy Manager d efaults when the role - mapping policy do es not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

For information on configuring role-mapping policies, see Configuring a Role and Role-Mapping Policy.

Enforcement Configuration

Use this tab to select an enforcement policy for a service by specifying the **Enforcement** parameters as described in the following table:

Table 5. Aruba 802.1X Wireless - Identity Only Service > Enforcement Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles a nd posture attributes from previous sessio ns.
Enforcement Policy	Select the preconfigured enforcement poli cy from the drop - down list. This is manda tory. If you do not have any preconfigured enfo rcement policies, click **Add New Enforcem ent Policy** to create a new enforcement pol icy.

Parameter	Action/Description
Enforcement Policy Details	
Description	Displays additional information about the s elected enforcement policy.
Default Profile	Displays a default profile applied by Policy Manager Policy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

For related information, see Configuring Enforcement Policies.

Profile Endpoints Configuration

The **Profiler** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Profile Endpoints**.

Specify the **Profile Endpoints** parameters as described in the following table:

Table 6. Add 802.1X Wireless - Identity Only Service > Profile Endpoints Parameters

Parameter	Action/Description
Endpoint Classification	Select one or more endpoint classification items from the drop - down list.
RADIUS CoA Action	Select the RADIUS CoA action from the dro p - down list. • To view the **Policy Manager Entity Deta ils** page with the summary of enforcem ent profile details, click the **View Detail s** button. • To view the **Summary** tab with profile d etails, click the **Modify** button. • To create a new RADIUS CoA action, cli ck the **Add New RADIUS CoA Action** lin k.

Accounting Proxy Configuration

The **Accounting Proxy** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Acconting Proxy**. Use the **Accounting Proxy** tab to broadcast the RADIUS accounting packets to all the proxy targets. You can configure

the proxy targets to which RADIUS server should be forwarded and the attributes to be added in the accounting. This enables the external security solutions to use the RADIUS account event to detect when a user connects and disconnects to the server.

Specify the **Accounting Proxy** parameters as described in the following table:

Table 7. Add Aruba 802.1X Wireless - Identity Only Service > Accounting Proxy Tab Parameters

Parameter	Action/Description
Accounting Proxy Targets	Specify the proxy targets to which the RADIUS server should be forwarded and the attributes to be added in the accounting. Select the accounting proxy target from the **Select to Add** drop - down list.
Add New Accounting Proxy Target	Click this link to add a new accounting proxy target.
RADIUS attributes to be added for Accounting Proxy	
Type	Select the RADIUS attribute type from the drop - down list: • Radius:IETF • Radius:Cisco • Radius:Hewlett - Packard - Enterprise • Radius:Alcatel - Lucent - Enterprise • Radius:Microsoft • Radius:Avenda • Radius:Aruba
Name	Select the name of the RADIUS attribute from the drop - down list.
Value	Select the value from the **Value** drop - down list. The values displayed here depend on the RADIUS attribute **Type** selected.

Summary Information

The **Summary** page presents the summary of parameters defined when you created a new service.

Aruba 802.1X Wireless Service

Configure this service for wireless hosts that are connecting through an Aruba 802.1X wireless access device or controller using IEEE 802.1X authentication. Service rules are customized for a typical Aruba Controller deployment. The Aruba 802.1X service includes a rule that specifies that an Aruba ESSID exists.

> **NOTE**
>
> If you want to administer the same set of policies for wired and wireless access, you can combine the service rule to define one single service.
>
> The other option is to keep two services for wired and wireless access, but reuse the policy components (authentication methods, authentication source, authorization source, role mapping policies, posture policies, and enforcement policies) in both services.

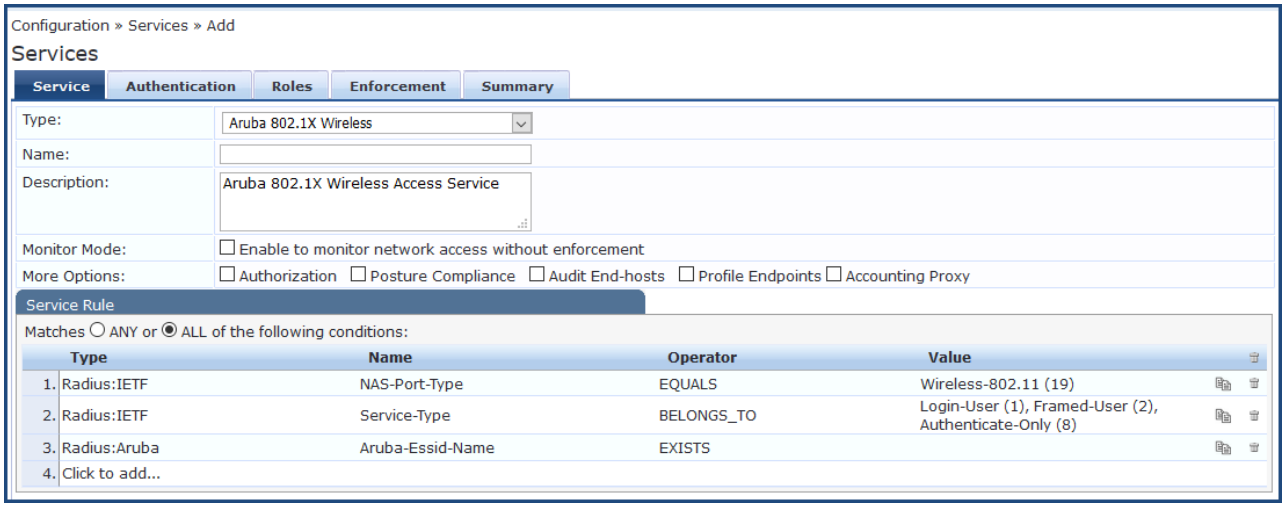
Service Configuration

The **Service** tab provides basic configuration parameters for the service. The **Service Rules** section defines a set of criteria that supplicants must match to trigger the service. Some service templates have one or more rules predefined. You can click on a service rule to modify any of its options.

To configure a 802.1X Wireless service:

1. Navigate to **Configuration > Services**, then click the **Add** link. The **Add Services** dialog opens.
2. From the **Type** drop-down, select **Aruba 802.1X Wireless**.The **Aruba 802.1X Wireless** service configuration dialog opens:

Figure 1. Aruba802.1X Wireless Service > Service Dialog



3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add Aruba 802.1X Wireless Service > Service Tab Parameters

Parameter	Action/Description
Type	Select **Aruba 802.1X Wire**
Name	Enter the name of this ser
Description	Policy Manager autofills t "Aruba 802.1X Wireless Ac ange the description if yc
Monitor Mode	Select this check box to r ctivity without enforceme
More Options	Check these boxes to acc uration tabs: • Authorization: See Au • Posture Compliance: S • Audit End - hosts: See • Profile Endpoints: See uration • Accounting Proxy: See guration
Service Rule	

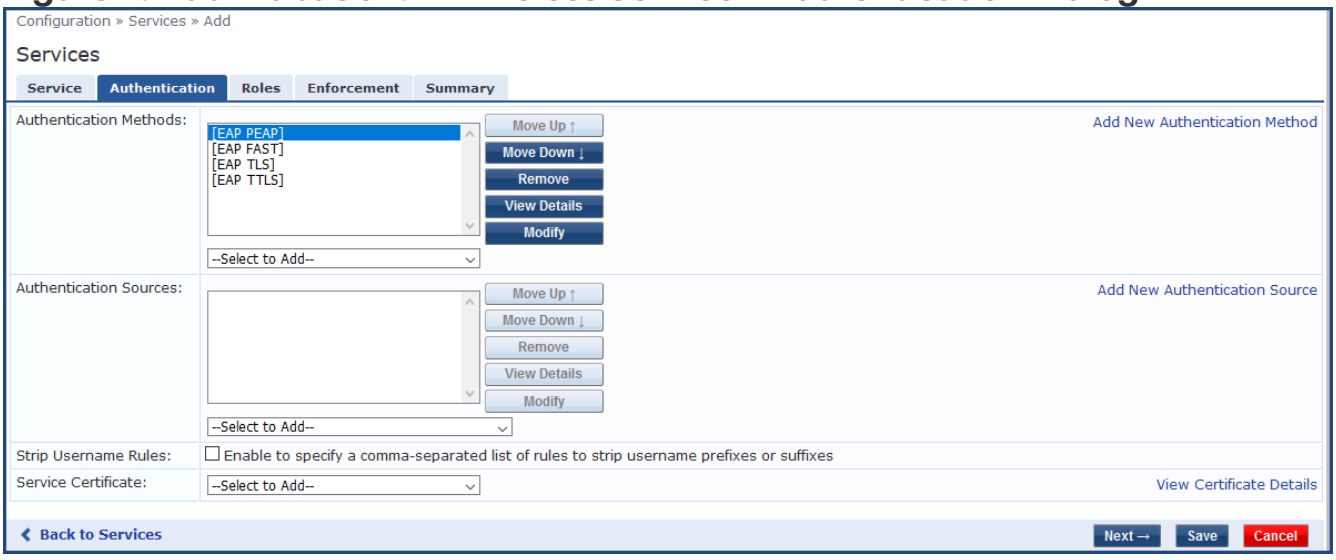
Parameter	Action/Description
Matches	Select the match conditic • Matches ANY • Matches ALL of the fo
Type	Select **Click to add** to sel
Name	Select the name of the se - down list.
Operator	Select an appropriate ope rators for the data type of
Value	Enter the value or select t - down list. The value list d **Operator** selected.

4. Click a service rule to modify its options.
5. Click **Next** to view the **Authentication** tab.

Authentication Configuration

The **Authentication** tab contains options for configuring authentication methods and authentication sources. The following figure displays the **Authentication** dialog:

Figure 2. AddAruba802.1X Wireless Service > Authentication Dialog



1. Specify the **Authentication** tab parameters as described in the following table:

Table 2. Add Aruba 802.1X Wireless Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	The authentication metho and the type of authentic ates this field with the foll • EAP PEAP • EAP FAST • EAP TLS with CN Chec • EAP TTLS Click the **Select to Add** fi uthentication method, sel The **EAP - MD5** authentic ode. The order of authenticatic tication. Policy Manager p he client can accept the a ue authentication or send uthentication method. If t hen the authentication pr If most of the clients in th ation method should be c IUS packets exchanged. For more information, see
Authentication Sources	Specify the authenticatio This can be one or more i d by default: • Admin User Repositor • Denylist Repository • Endpoints Repository • Guest Device Repositc • Guest User Repository

Parameter	Action/Description
	• Insight Repository • Local User Repository • Zone Cache Repositor • Onboard Devices Rep • Social Login Repositor • Time Source **NOTE** When you attem the following err *not exceed 23*. For more information, see
Strip Username Rules	Select the check box to p before authenticating and
Service Certificate	Optionally, select the Serv es.

Authorization Configuration

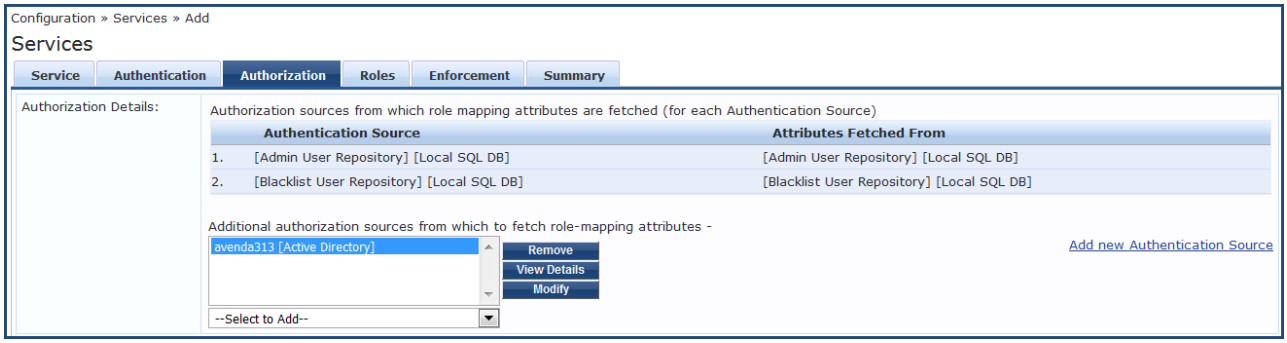
Use the **Authorization** tab to select the authorization sources for this service. The **Authorization** tab is not displayed by default. To access this tab, select the **More Options** > **Authorization**check box. Policy Manager fetches role-mapping attributes from the authorization sources associated with the service, regardless of which authentication source was used to authenticate the user.

For a given service, role-mapping attributes are fetched from the following authorization sources:

- Authorization sources associated with the authentication source
- Authorization sources associated with the service

The following figure displays the **Authorization** dialog:

Figure 3. AddAruba802.1X Wireless Service > Authorization Dialog



Specify the **Authorization** parameters as described in the following table:

Table 3. Add Aruba 802.1X Wireless Service > Authorization Parameters

Parameter	Action/Description
Authentication Source	Displays the authorization sources from which role mapping attributes are fetche d for each authentication source.
Attributes Fetched From	Displays the source of attributes.
Additional authorization sources from wh ich to fetch role - mapping attributes	Specify the authorization sources using t he **Select to Add** field. There can be one or more instances of th e following list of authorization sources: ◦ Admin User Repository ◦ Denylist Repository ◦ Endpoints Repository ◦ Guest Device Repository ◦ Guest User Repository ◦ Insight Repository ◦ Local User Repository ◦ Zone Cache Repository ◦ Onboard Devices Repository ◦ Social Login Repository

Parameter	Action/Description
	◦ Time Source **NOTE** *No. of Authorization Sources cannot exceed 23*

Roles Configuration

Use the **Roles** tab to associate a role-mapping policy with this service. The following figure displays the **Aruba 802.1X Wireless Service > Roles** dialog:

Figure 4. AddAruba802.1X Wireless Service > Roles Dialog

Specify the **Roles** parameters as described in the following table:

Table 4. Add Aruba 802.1X Wireless Service > Roles Tab Parameters

Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the drop - down list. Policy Manager ships a number of preconfi gured roles. **NOTE**
Role Mapping Policy Details	
Description	Provide additional information about the se lected role - mapping policy.

Parameter	Action/Description
Default Role	Specify the role to which Policy Manager d efaults when the role - mapping policy do es not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

For information on configuring role-mapping policies, see Configuring a Role and Role-Mapping Policy .

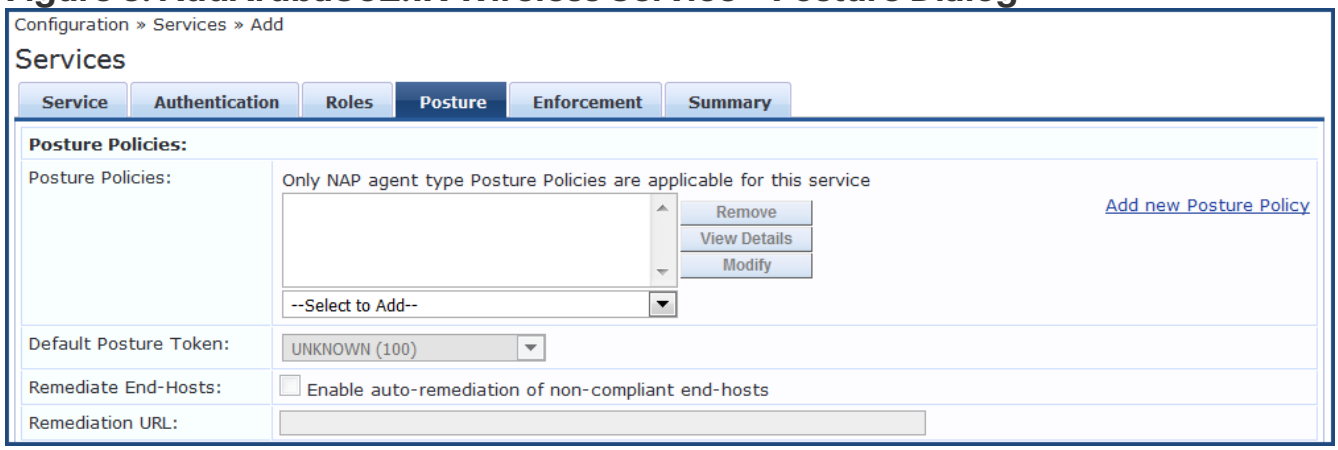
Posture Configuration

The **Posture** tab is not enabled by default. To enable posture configuration options, return to the **Service** tab and select **More Options** > **Posture Compliance**. You can enable the posture checking for this kind of service if you deploy any of the following:

- ClearPass Policy Manager in a Microsoft Network Access Protection (NAP) environment
- Cisco Network Admission Control (NAC) Framework environment
- Aruba hosted captive portal that performs posture checks through a dissolvable agent

The following figure displays the **Posture** dialog:

Figure 5. AddAruba802.1X Wireless Service > Posture Dialog



Specify the Wireless Service **Posture** parameters as described in Table 5:

Table 5. Add Aruba 802.1X Wireless Service > Posture Parameters

Parameter	Action/Description
Posture Policies	

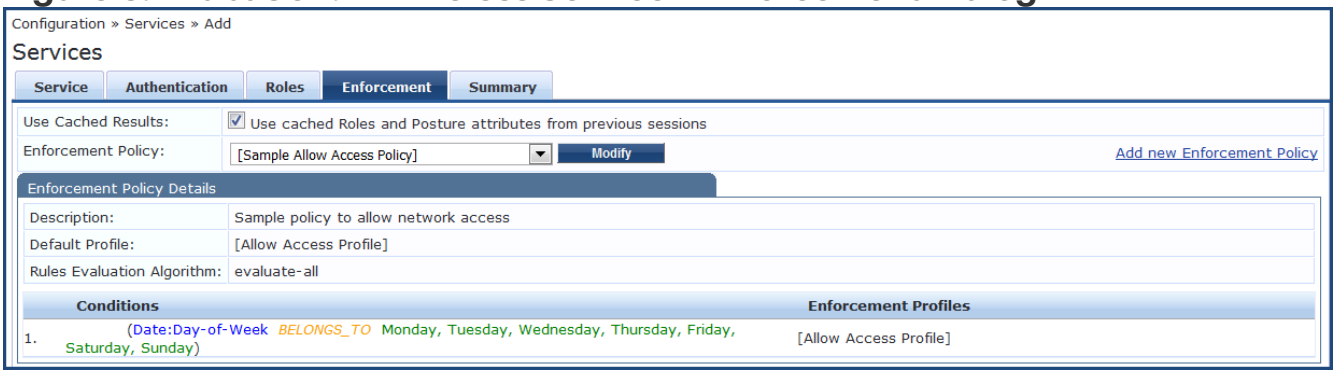
Parameter	Action/Description
Posture Policies	Select the posture policy from the **Select to Add** drop - down list. If you do not have any preconfigured posture policies, click **Add New Posture Policy** to create a new posture policy. **NOTE**
Default Posture Token	Select the default posture token from the drop - down list.
Remediate End - Hosts	To perform remediation action, when a client is quarantined, select the **Enable auto - remediation of non - compliant end - hosts** check box.
Remediation URL	To perform the remediation, enter the web link of a server resource.

For more information on configuring posture polices, see Configuring Posture Policy Agents and Hosts .

Enforcement Configuration

Use this tab to select an enforcement policy for a service. The following figure displays the **Enforcement** dialog:

Figure 6. Aruba802.1X Wireless Service > Enforcement Dialog



Specify the **Enforcement** parameters as described in the following table:

Table 6. Aruba 802.1X Wireless Service > Enforcement Parameters

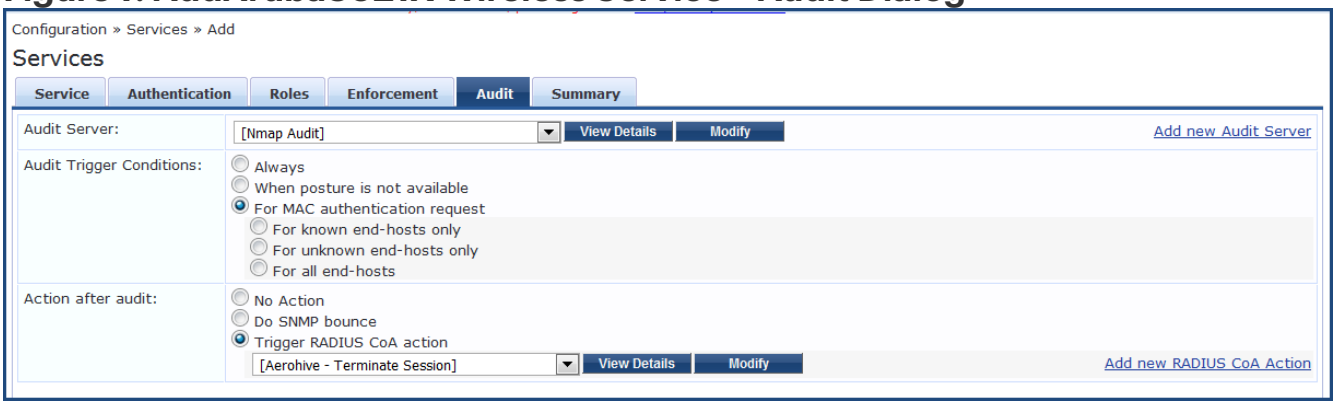
Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles a nd posture attributes from previous sessio ns.
Enforcement Policy	Select the preconfigured enforcement poli cy from the drop - down list. This is manda tory. If you do not have any preconfigured enfo rcement policies, click **Add New Enforcem ent Policy** to create a new enforcement pol icy.
Enforcement Policy Details	
Description	Displays additional information about the s elected enforcement policy.
Default Profile	Displays a default profile applied by Policy Manager Policy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

For related information, see Configuring Enforcement Policies .

Audit Configuration

The **Audit** tab is not enabled by default. To enable this tab, return to the **Service** tab and select **More Options** > **Audit End-hosts**. The **Audit** tab opens, as shown in the figure below.

Figure 7. AddAruba8021X Wireless Service > Audit Dialog



To enable audit checking for this service, specify the **Audit End-Hosts** parameters as described in the following table:

Table 7. Add Aruba 802.1X Wireless Service > Audit End - Hosts Parameters

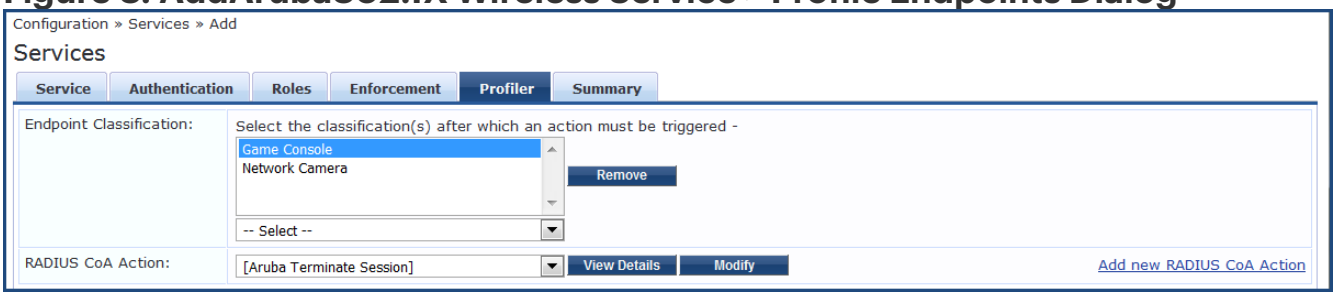
Parameter	Action/Description
Audit Server	Select the audit server from the following options: • **Nessus Server**: Interfaces with Policy Manager primarily to perform vulnerability scanning. • **Nmap Audit**: Performs specific Nmap audit functions. ◦ To view the **Policy Manager Entity Details** dialog with the summary of audit server details, click the **View Details** button. ◦ To view the **Summary** tab with audit server details, click the **Modify** button.
Audit Trigger Conditions	Select an audit trigger condition: • **Always**: Always perform an audit. • **When posture is not available**: Perform audit only when posture credentials are not available in the request. • **For MAC Authentication Request**: If you select this option, Policy Manager presents the following three additional settings: ◦ **For known end - hosts only:** Select this option when you want to reject unknown end - hosts and to audit known clients. *Known end - hosts* are defined as clients that are found in the authentication source(s) associated with this service. ◦ **For unknown end - hosts only:** Select this option when the known end - hosts are assumed to be healthy,

Parameter	Action/Description
	but you want to establish the identi ty of unknown end - hosts and assig n roles. *Unknown end - hosts* are en d - hosts that are not found in any o f the authentication sources associa ted with this service. ◦ **For all end - hosts:** For both known and unknown end - hosts.
Action After Audit	Specify the audit that can be performed on ly after the MAC authentication request is c ompleted and the client has acquired an IP address through DHCP. Once the audit results are available, Policy Manager reapplies policies on the network device in one of the following ways: • **No Action**: The audit does not apply pol icies on the network device after compl eting this audit. • **Do SNMP bounce**: This option bounces the switch port or forces an 802.1X reau thentication (both done using SNMP). ◦ Bouncing the port triggers a new 80 2.1X or MAC authentication request by the client. If the audit server alrea dy has the posture token and attribu tes associated with this client in its cache, it returns the token and the a ttributes to Policy Manager. • **Trigger RADIUS CoA action**: This optio n sends a RADIUS CoA command from Policy Manager to the network device.

Profile Endpoints Configuration

The **Profiler** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Profile Endpoints**. The **Add Profile Endpoints** dialog opens, as shown in the figure below.

Figure 8. AddAruba802.1X Wireless Service > Profile Endpoints Dialog



Specify the **Profile Endpoints** parameters as described in the following table:

Table 8. Add Aruba 802.1X Wireless Service > Profile Endpoints Parameters

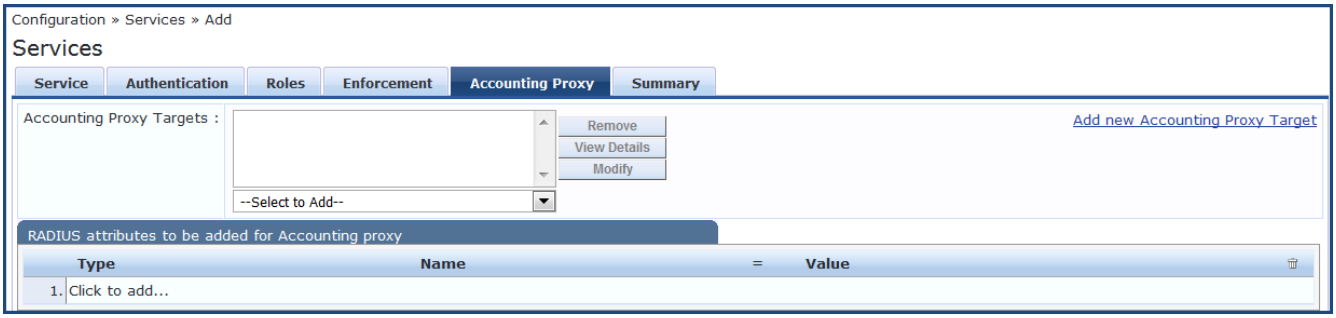
Parameter	Action/Description
Endpoint Classification	Select one or more endpoint classification items from the drop - down list.
RADIUS CoA Action	Select the RADIUS CoA action from the dro p - down list. • To view the **Policy Manager Entity Deta ils** page with the summary of enforcem ent profile details, click the **View Detail s** button. • To view the **Summary** tab with profile d etails, click the **Modify** button. • To create a new RADIUS CoA action, cli ck the **Add New RADIUS CoA Action** lin k.

Accounting Proxy Configuration

The **Accounting Proxy** tab is not displayed by default. To access this tab, return to the **Service** tab and select **More Options > Acconting Proxy**. Use the **Accounting Proxy** tab to broadcast the RADIUS accounting packets to all the proxy targets. You can configure the proxy targets to which RADIUS server should be forwarded and the attributes to be added in the accounting. This enables the external security solutions to use the RADIUS account event to detect when a user connects and disconnects to the server.

The following figure displays the **Accounting Proxy** dialog:

Figure 9. Add 802.1X Wireless > Accounting Proxy Dialog



Specify the **Accounting Proxy** parameters as described in the following table:

Table 9. Add Aruba 802.1X Wireless Service > Accounting Proxy Tab Parameters

Parameter	Action/Description
Accounting Proxy Targets	Specify the proxy targets to which the RADIUS server should be forwarded and the attributes to be added in the accounting. Select the accounting proxy target from the **Select to Add** drop - down list.
Add New Accounting Proxy Target	Click this link to add a new accounting proxy target.
RADIUS attributes to be added for Accounting Proxy	
Type	Select the RADIUS attribute type from the drop - down list: • Radius:IETF • Radius:Cisco • Radius:Hewlett - Packard - Enterprise • Radius:Alcatel - Lucent - Enterprise • Radius:Microsoft • Radius:Avenda • Radius:Aruba
Name	Select the name of the RADIUS attribute from the drop - down list.
Value	Select the value from the **Value** drop - down list.

Parameter	Action/Description
	The values displayed here depend on the RADIUS attribute **Type** selected.

Summary Information

The **Summary** page presents the summary of parameters defined when you created a new service.

Cisco Web Authentication Proxy Service

About this task

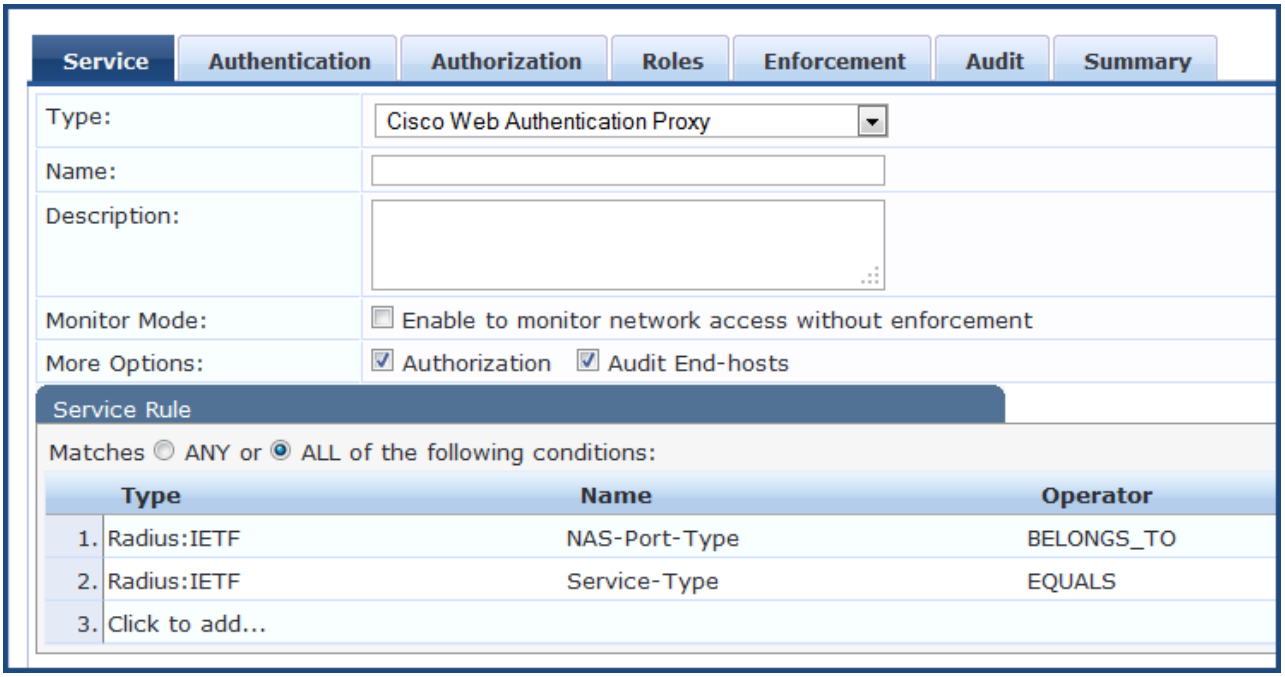
This service is a web-based authentication service for guests or agent-less hosts. The Cisco switch hosts a captive portal and the portal web page that collects username and password information. Subsequently, the switch sends a RADIUS request in the form of a Password Authentication Protocol (PAP) authentication request to Policy Manager. By default, this service uses the **PAP** authentication method. You can click on the **Authorization** and **Audit End-hosts** options to enable additional tabs.

To create a **Cisco Web Authentication Proxy** service:

Procedure

1. Navigate to **Configuration > Services**, then select the **Add** link. The **Add Services** page opens.
2. From the Type drop down, select **Cisco Web Authentication Proxy**.The **Cisco Web Authentication Proxy** service configuration dialog opens:

Figure 1. Cisco Web Authentication Proxy Service



Configuring the **Cisco Web Authentication Proxy** service is similar to configuring the **Aruba 802.1X Wireless** service except that the **Posture Compliance** and **Profile Endpoints** options are not available. For more information on configuration, see Aruba 802.1X Wireless Service.

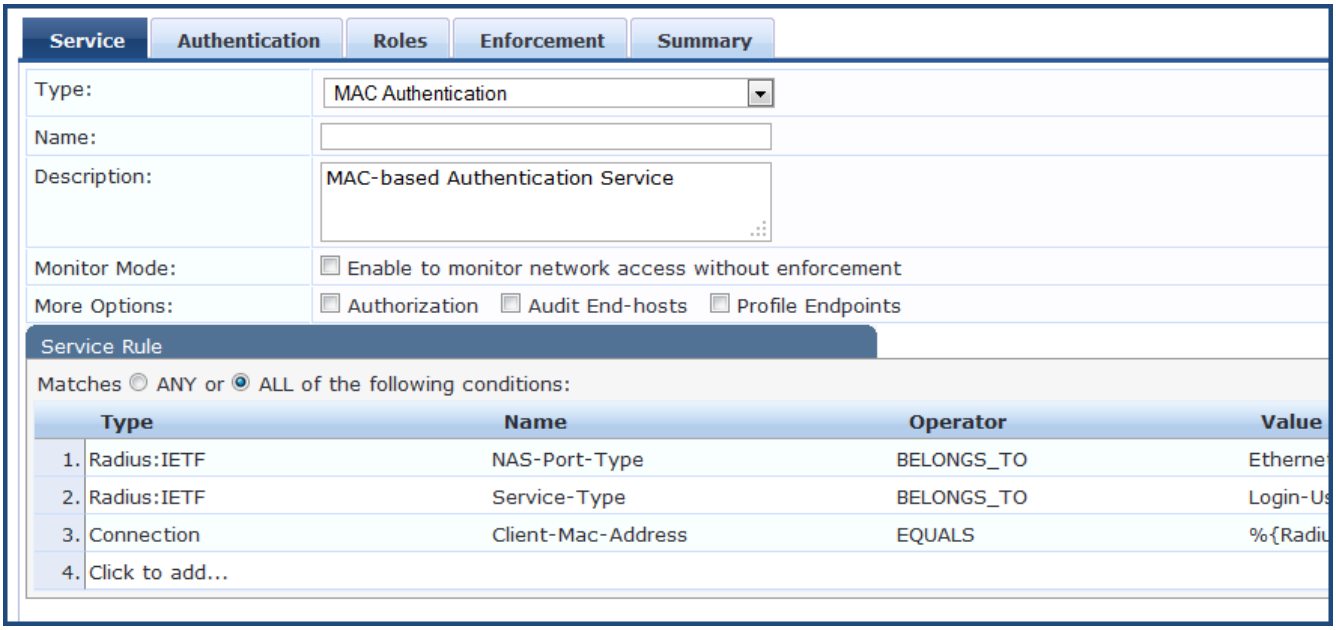
MAC Authentication Service

The MAC-based authentication service is used for clients without an 802.1X supplicant or a posture agent (such as printers, other embedded devices, and computers owned by guests or contractors). The network access device sends a MAC authentication request to Policy Manager. Policy Manager can look up the client in an allowed list or blocked list, authenticate, and authorize the client against an external authentication or authorization source, and optionally perform an audit on the client.

NOTE
You cannot configure posture for this type of service.

The following figure displays the **MAC Authentication** service configuration dialog.

Figure 1. MAC Authentication Service Configuration Dialog

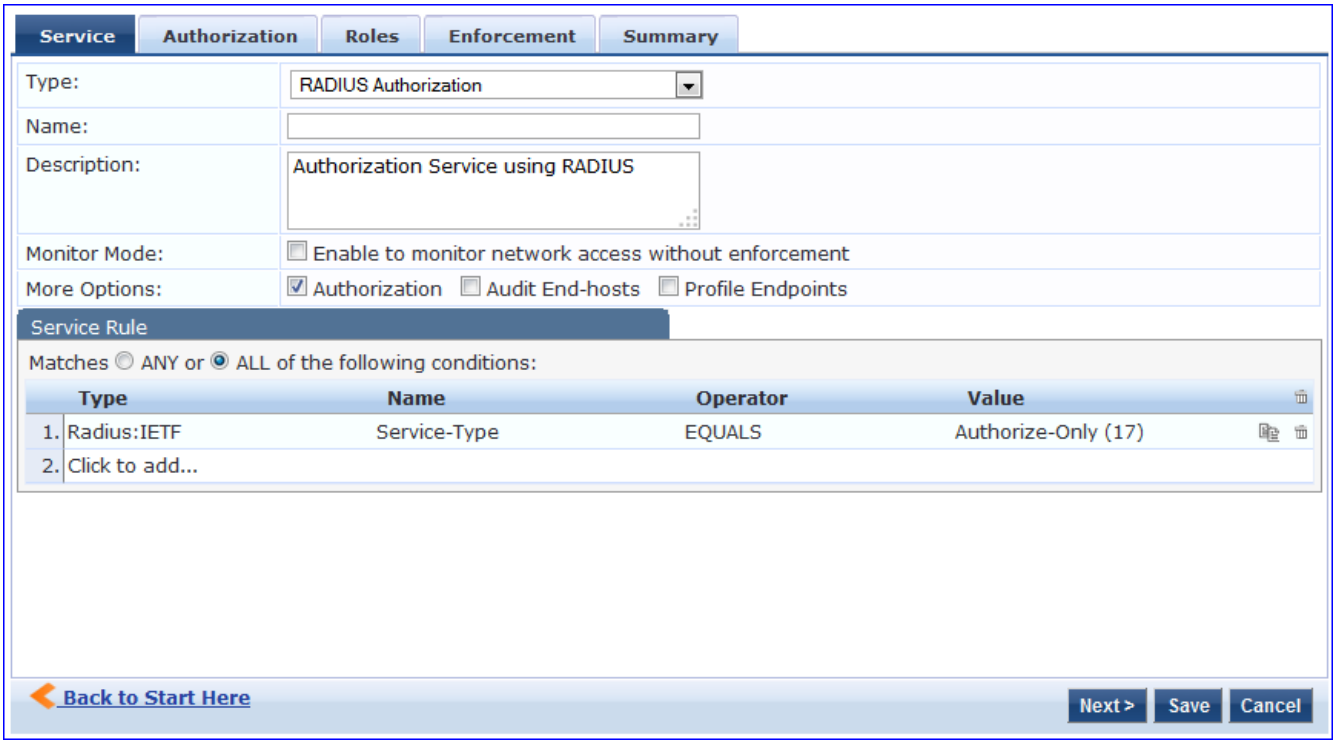


Configuration for the rest of the tabs is similar to the **Aruba 802.1X Wireless** service configuration. For details on this service's configuration, see 802.1X Wireless .

RADIUS Authorization Service

Configure the **RADIUS Authorization** service type for services that perform authorization using RADIUS. When you select this service, the **Authorization** tab is enabled. The following figure displays the **RADIUS Authorization** service configuration dialog:

Figure 1. RADIUS Authorization Service Configuration Dialog



Configuration for this service is the same as the **RADIUS Enforcement (Generic)** service, except that you do not configure authentication or posture with this service type. Refer to RADIUS Enforcement (Generic) for more information.

RADIUS Enforcement (Generic) Service

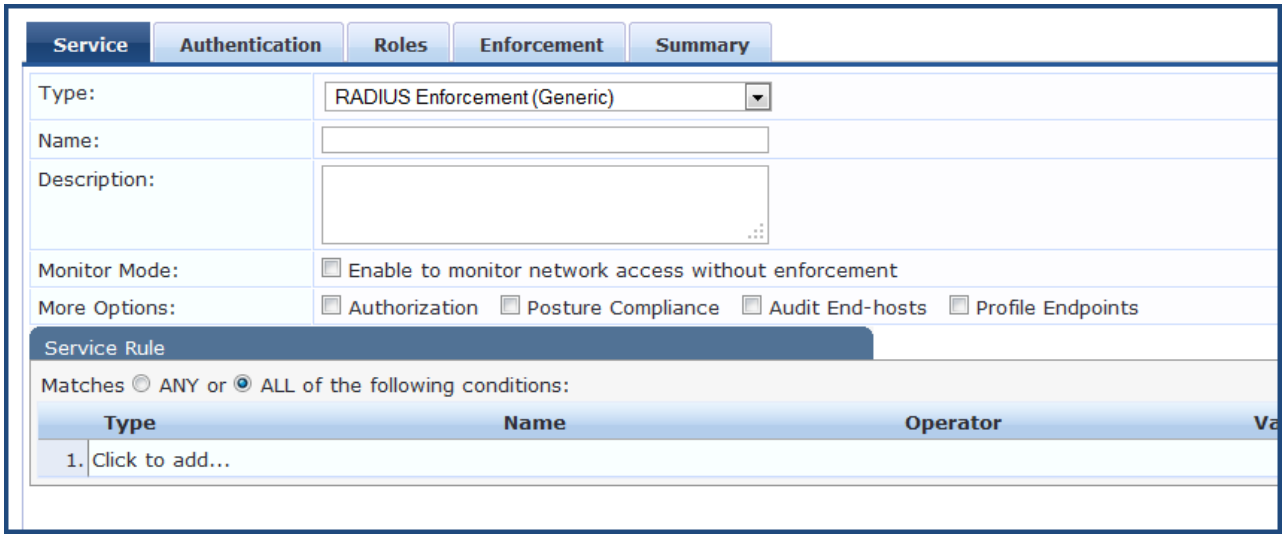
About this task

Configure the **RADIUS Enforcement (Generic)** service for any kind of RADIUS request. There are no default rules associated with this service type. You can add rules to handle any type of standard or vendor-specific RADIUS attributes (that is, any attribute that is loaded through the prepackaged vendor-specific or standard RADIUS dictionaries, or through other dictionaries imported into Policy Manager). For related information, see Configuring Enforcement Policies .

Service Configuration To create a RADIUS Enforcement (Generic) service:

1. Navigate to **Configuration > Services**, then click the **Add** link.The **Add Services** dialog opens.
2. From the **Type** drop-down, select **RADIUS Enforcement (Generic)**.The **RADIUS Enforcement (Generic)** service configuration dialog opens:

Figure 1. Add RADIUS Enforcement (Generic) Service Configuration Dialog



3. Specify the **Service** tab parameters as described in the following table.

Table 1. Aruba RADIUS Enforcement (Generic) Service > Service Tab Parameters

Parameter	Action/Description
Type	Select a service from the drop - down lis t that defines what type of service can be configured.
Name	Enter the name of the service.
Description	Provide additional information that helps to identify the service.
Monitor Mode	Select this check box to monitor network access activity without enforcement.
More Options	Check these boxes to access the additio nal configuration tabs: • Authorization • Posture Compliance • Audit End - hosts • Profile Endpoints • Accounting Proxy

Parameter	Action/Description
Service Rule	
Matches	Select the match condition for this servic e: • Matches ANY • Matches ALL of the following conditio ns
Type	Select **Click to add** to select the service r ule type.
Name	Select the name of the service rule from t he drop - down list.
Operator	Select an appropriate operator from the l ist of operators for the data type of the at tribute.
Value	Enter the value or select the value from t he drop - down list. The value list depend s on the **Type** and **Operator** selected.
Name	Select the name of the service rule from t he drop - down list.
Operator	Select an appropriate operator from the l ist of operators for the data type of the at tribute.
Value	Enter the value or select the value from t he drop - down list. The value list depend s on the **Type** and **Operator** selected.

4. Click **Next**.

Authentication Configuration

The **Authentication** tab contains options for configuring authentication methods and authentication sources. The following figure displays the **Authentication** dialog:

Figure 2. AddArubaRADIUS Enforcement (Generic) Service > Authentication Dialog

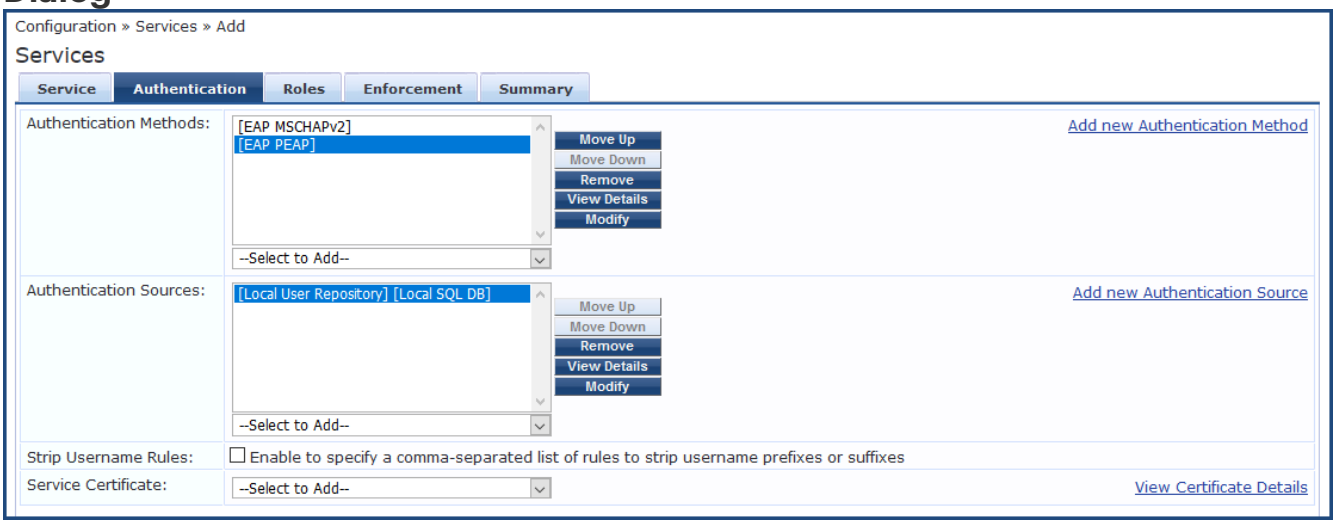


Table 2. Add Aruba RADIUS Enforcement (Generic) Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	Select authentication methods using the **Select to Add** field. The authentication methods available for this service depend on the 802.1X supplicants and the type of authentication methods you choose to deploy. Policy Manager automatically selects the appropriate method for authentication, when a user attempts to connect. The common types, which are automatically selected include the following examples: • EAP PEAP • EAP FAST • EAP TLS • EAP TTLS • EAP MSCHAPV2 The order of authentication is significant, when a client tries to perform an 802.1X authentication. Policy Manager proposes the first authentication method configured. However, the client can accept the authentication method proposed by Policy Manager and continue authentication or send a Negative - Acknowledgment (NAK) and prop

Parameter	Action/Description
	ose a different authentication method. If th e newly proposed authentication method i s also configured, then the authentication proceeds, otherwise authentication fails. If most of the clients in the network use a s pecific authentication method, that authen tication method should be configured first in the list. This would reduce the number o f RADIUS packets exchanged. For more information, see the following: • Adding and Configuring Authentication Methods • Adding and Configuring Authentication Sources .
Authentication Sources	Specify the authentication sources using t he **Select to Add** field. This can be one or more instances of the fo llowing list of authentication sources: • AD_avenda313 [Active Directory] • Admin User Repository • Denylist Repository • Endpoints Repository • Guest Device Repository • Guest User Repository • http_3702 • HTTP_Auth_Source • Insight Repository • Local User Repository • Onboard Devices Repository • Social Login Repository • Time Source **NOTE**

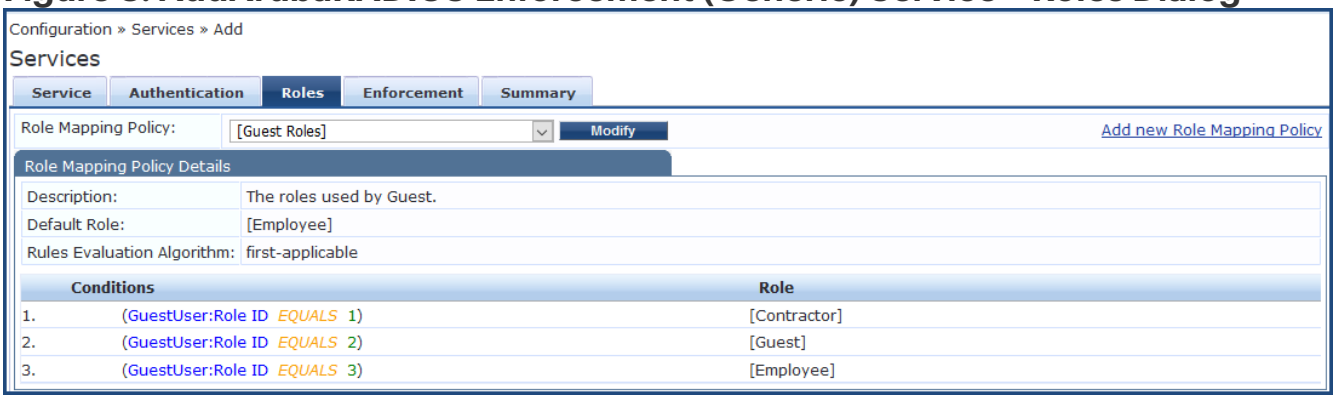
Parameter	Action/Description
	When you attempt to specify more than 23 Services authentication sources, the following error message is displayed: *No. of Authentication Sources cannot exceed 23*.
Strip Username Rules	Select the check box to preprocess the username (to remove prefixes and suffixes) before authenticating and authorizing against the authentication source.
Service Certificate	Select the Service Certificate. Once you select the Service Certificate, click the **View Certificate Details** link to view information about the certificate.

Procedure

Click **Next**.

Roles Configuration Use the **Roles** tab to associate a role-mapping policy with this service. The following figure displays the **Aruba RADIUS Enforcement (Generic) > Roles** dialog:

Figure 3. AddArubaRADIUS Enforcement (Generic) Service > Roles Dialog



1. Specify the **Roles** parameters as described in the following table:

Table 3. Add Aruba 802.1X Wireless Service > Roles Tab Parameters

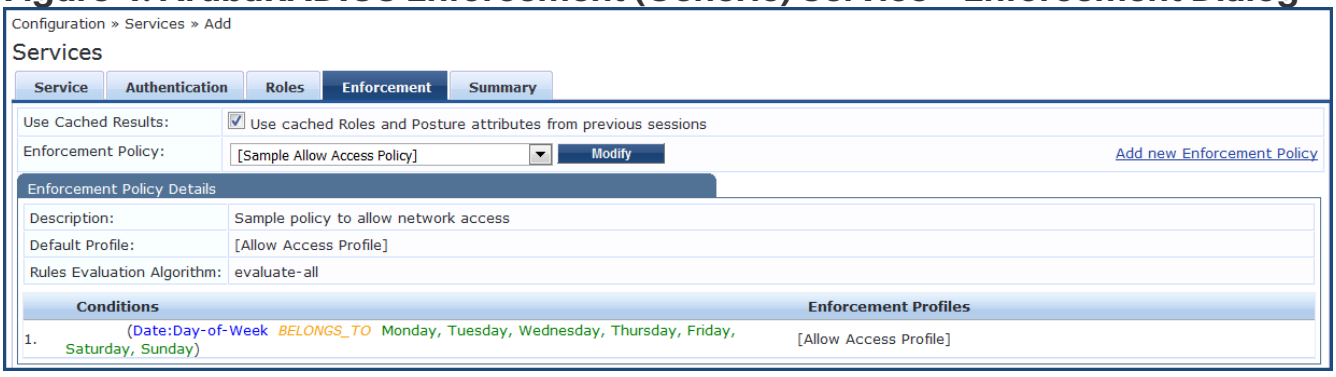
Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the drop - down list.

Parameter	Action/Description
	Policy Manager ships a number of precon figured roles. **NOTE** A service can be configured without a role - mapping pol icy, but only one role - mapp ing policy can be configured for each service. For information on configuring role - ma pping policies, see Configuring a Role an d Role - Mapping Policy .
Role Mapping Policy Details	
Description	Provide additional information about the selected role - mapping policy.
Default Role	Specify the role to which Policy Manage r defaults when the role - mapping policy does not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

2. Click **Next**.

Enforcement Configuration Use this tab to select an enforcement policy for a service. The following figure displays the **Enforcement** dialog:

Figure 4. ArubaRADIUS Enforcement (Generic) Service > Enforcement Dialog



1. Specify the **Enforcement** parameters as described in the following table:

Table 4. Aruba RADIUS Enforcement (Generic) > Enforcement Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles and posture attributes from previous sess ions.
Enforcement Policy	Select the preconfigured enforcement p olicy from the drop - down list. This is ma ndatory. If you do not have any preconfigured en forcement policies, click **Add New Enforc ement Policy** to create a new enforceme nt policy.
Enforcement Policy Details	
Description	Displays additional information about the selected enforcement policy.
Default Profile	Displays a default profile applied by Poli cy Manager Policy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

2. Click **Save**. You return to the **Services** page. The following message is displayed: *Service "RADIUS Enforcement service" has been added.*

RADIUS Proxy Service

About this task

Configure the **RADIUS Proxy** service for any kind of RADIUS request that needs to be proxied to another RADIUS server (that is, a proxy target). There are no default rules associated with this service type. You can add rules to handle any type of standard or vendor-specific RADIUS attributes. Typically, proxying is based on the realm or the domain of the user who is trying to access the network.

To configure a RADIUS Proxy service:

Procedure

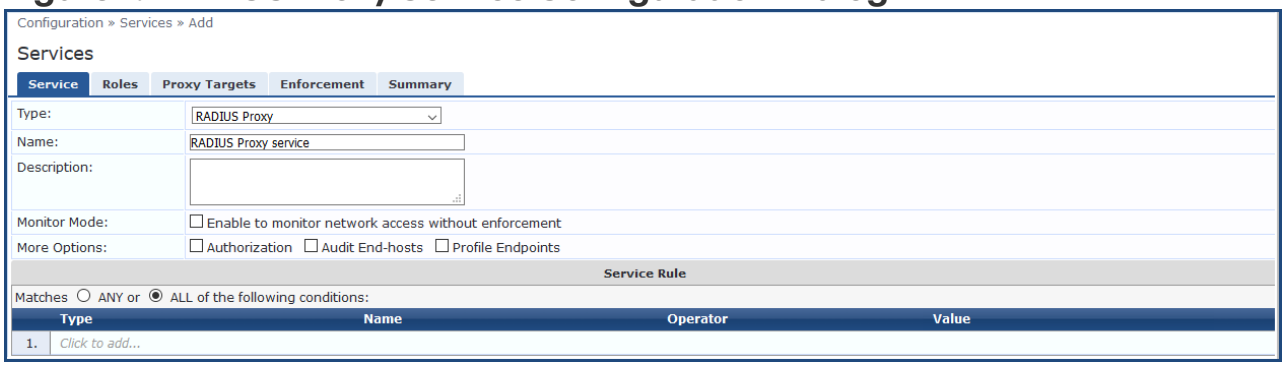
1. Navigate to **Configuration > Services**, then click the **Add** link.

   The **Add Services** page opens.

2. From the **Service** tab, select **RADIUS Proxy** from the **Type** drop down.

   The **RADIUS Proxy** service configuration dialog opens:

Figure 1. RADIUS Proxy Service Configuration Dialog



3. Specify the **Service** tab parameters as described in the following table:

Table 1. RADIUS Proxy Service Tab Parameters

Parameter	Action/Description
Type	Select RADIUS Proxy.
Name	Enter the name of the service.
Description	Optionally, provide additional information that helps to identify the service.
Monitor Mode	Select this check box to monitor network access activity without enforcement.
More Options	Check these boxes to access the additio nal configuration tabs: • Authorization • Audit End - hosts • Profile Endpoints

Parameter	Action/Description
Service Rule	
Matches	Select the match condition for this servic e: • Matches ANY • Matches ALL of the following conditio ns
Type	Select **Click to add** to select the service r ule type.
Name	Select the name of the service rule from t he drop - down list.
Operator	Select an appropriate operator from the l ist of operators for the data type of the at tribute.
Value	Enter the value or select the value from t he drop - down list. The value list depend s on the **Type** and **Operator** selected.

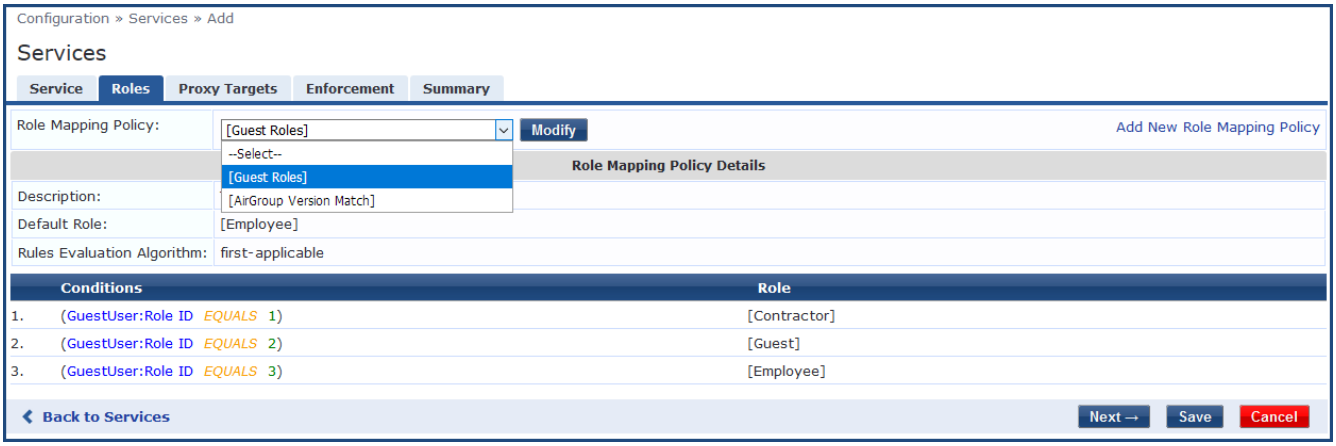
4. Click **Next**.

Roles Tab

About this task

Use the **Roles** tab to associate a role-mapping policy with this service.

Figure 2. RADIUS Proxy Roles Configuration Dialog



Procedure

1. Specify the **Roles** tab parameters as described in the following table:

Table 2. RADIUS Proxy Service > Roles Tab Parameters

Parameter	Action/Description
Role Mapping Policy	Select a role mapping policy from the d rop - down list. Policy Manager ships a number of precon figured roles. **NOTE** A service can be configured without a role - mapping pol icy, but only one role - mapp ing policy can be configured for each service. For information on configuring role - ma pping policies, see Configuring a Role an d Role - Mapping Policy .
Role Mapping Policy Details	
Description	When you select a Role Mapping Policy, Policy Manager populates the *Descriptio n* field.

Parameter	Action/Description
Default Role	When you select a Role Mapping Policy, Policy Manager populates the *Default Role* field. The Default Roleis the role to which Policy Manager defaults when the role - mapping policy does not produce a match.
Rules Evaluation Algorithm	Shows the first matched rule.

2. Click **Next**.

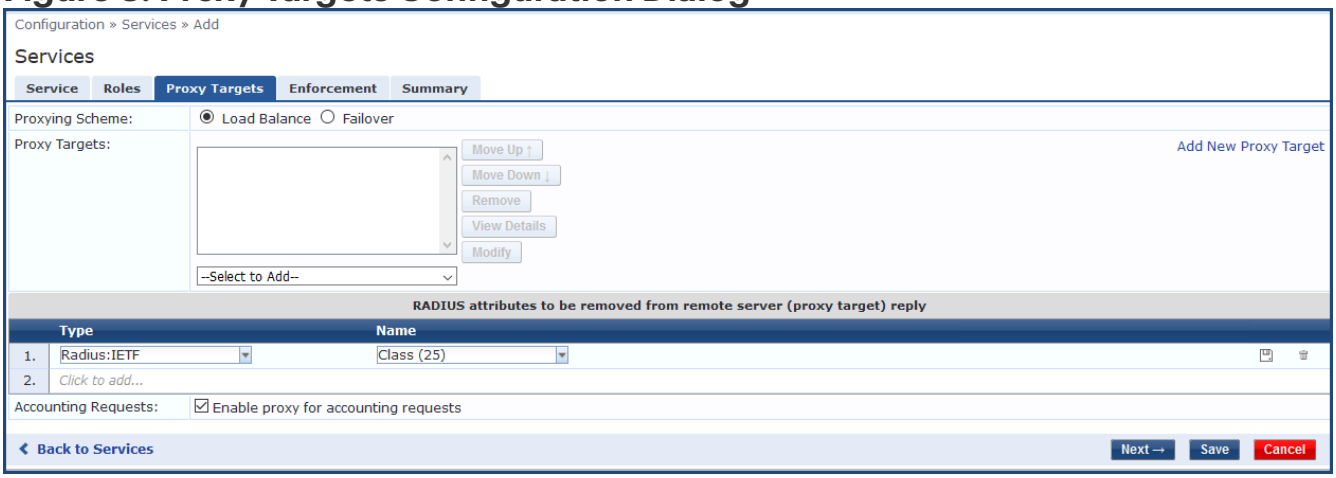
Proxy Targets Tab

About this task

In Policy Manager, a proxy target represents a RADIUS server (Policy Manager or a third party) that is the target of a proxied RADIUS request.

For example, when a branch office employee visits a main office and logs into the network, Policy Manager assigns the request to the first service in priority order that contains a service rule for RADIUS proxy services and appends the domain to the username.

Figure 3. Proxy Targets Configuration Dialog



Procedure

1. Specify the **Proxy Targets** parameters as described in the following table:

Table 3. RADIUS Proxy Service > Proxy Targets Parameters

Parameter	Action/Description
Proxying Scheme	Select one of the following proxying schemes: • *Load Balance*: When you select **Load Balance**, requests can be dispatched to the proxy targets randomly and load balanced. • *Failover*: In **Failover** mode, requests can be dispatched to the first proxy target in the ordered list of targets and subsequently to the other proxy targets if the prior requests failed.
Proxy Targets	From the **Select to Add** drop down, select one or more proxy targets.
RADIUS attributes to be removed from remote server (proxy target) reply	
Type	Select **Radius: IETF**.
Name	Select **Class (25)**.
Accounting Requests	Note the configuration below when working with Policy Manager or a proxy target that sends back a *Class* attribute in *Access - Accept*. This is required for the Accounting Proxy to work properly. The Policy Manager Proxy/Proxy server should be configured to remove those *Class* attributes. To do so, select the **Enable proxy for accounting requests** check box and select the **IETF:Class** attribute from the drop - down.

2. Click **Next**.

   The **Enforcement** tab opens.

Enforcement Tab

About this task

Use this tab to select an enforcement policy for a service.

Figure 4. Enforcement Configuration Dialog

Procedure

1. Specify the **Enforcement** parameters as described in the following table:

Table 4. RADIUS Proxy Service > Enforcement Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles and posture attributes from previous sessions.
Enforcement Policy	Select the preconfigured enforcement policy from the drop - down list. This is mandatory. If you do not have any preconfigured enforcement policies, click **Add New Enforcement Policy** to create a new enforcement policy.
Enforcement Policy Details	
Description	Displays additional information about the selected enforcement policy.
Default Profile	Displays a default profile applied by Policy Manager Policy Manager.

Parameter	Action/Description
Rules Evaluation Algorithm	Shows the first matched rule.

2. Click **Save**.

Aruba Application Authentication Service

This type of service provides authentication and authorization to users of Policy Manager Guest and Policy Manager Insight. You can send Generic Application Enforcement to these or other generic applications for authenticating and authorizing the users. The following figure displays the **Aruba Application Authentication** service configuration dialog:

Configuring the **Aruba Application Authentication** service is similar to configuring the **Aruba 802.1X Wireless** service except that the *Posture Compliance*, *Audit End-hosts*, and *Profile Endpoints* options are not available. For configuration details, see Aruba 802.1X Wireless Service.

Aruba Application Authorization Service

This type of service provides authorization for users of Aruba applications: Policy Manager Guest and Policy Manager Insight. You can send Generic Application Enforcement to these or other generic applications for authorizing the users. The following figure displays the **Aruba Application Authorization** service configuration dialog:

Configuring the Aruba Application Authorization service is similar to configuring the Aruba 802.1X Wireless service except that the *Posture Compliance*, *Audit End-hosts*, and *Profile Endpoints* options are not available. For configuration details, see 802.1X Wireless .

OnConnect Enforcement Service

This section provides the following information:

- Configuring the Service
- Configure an Enforcement Policy

Policy Manager OnConnect Enforcement is an enforcement model that allows you to use non-802.1X methods for device scans, VLAN placement, and so on. **Policy Manager OnConnect Enforcement** allows enforcement in non-802.1X environments without the need for an agent, such as OnGuard, on the endpoint. For related information, see:

- Enabling OnConnect Enforcement on a Network Device
- System Page

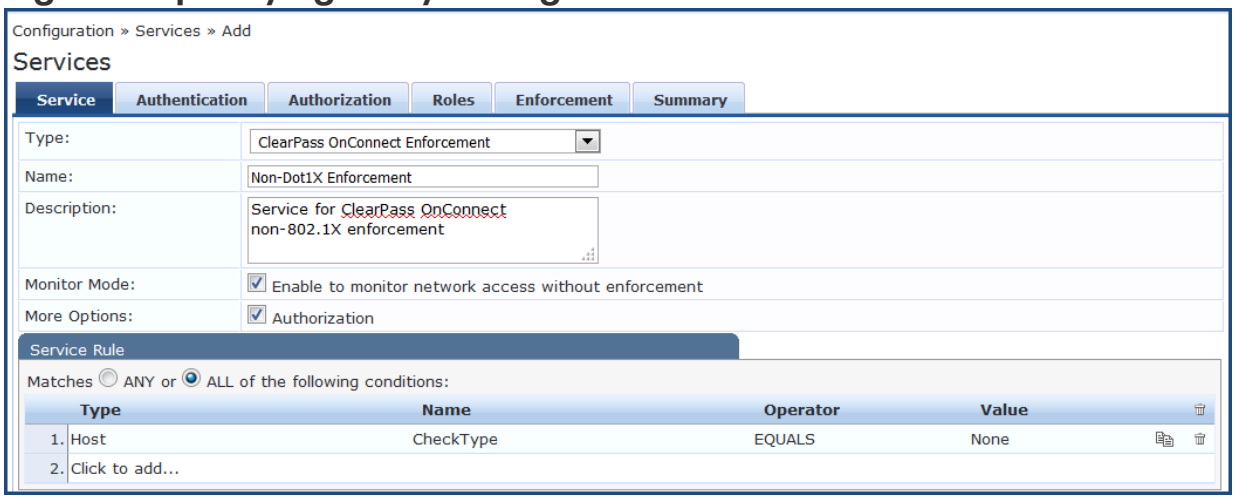
When **Policy Manager OnConnect Enforcement** is enabled, Policy Manager performs the following actions:

- Detects when a new endpoint connects to the network.
- Scans the endpoint to identify the logged-in user and other device-specific information.
- Triggers a Web-based authentication (WebAuth) for the device.
- Performs SNMP-based enforcement to change the network access profile for the device.

Configuring the Service To add an OnConnect Enforcement service:

1. Navigate to **Configuration > Services**.The **Services** page opens.
2. To add the service, click **Add**.The **Add Services** dialog opens.
3. From the **Type** drop-down list, select **Policy Manager OnConnect Enforcement** (see Figure 1).

Figure 1. SpecifyingPolicy ManagerOnConnect Enforcement



4. Enter the name or label of the OnConnect Enforcement service.

5. Enter the values for any other parameters, including service rules, required for this service.

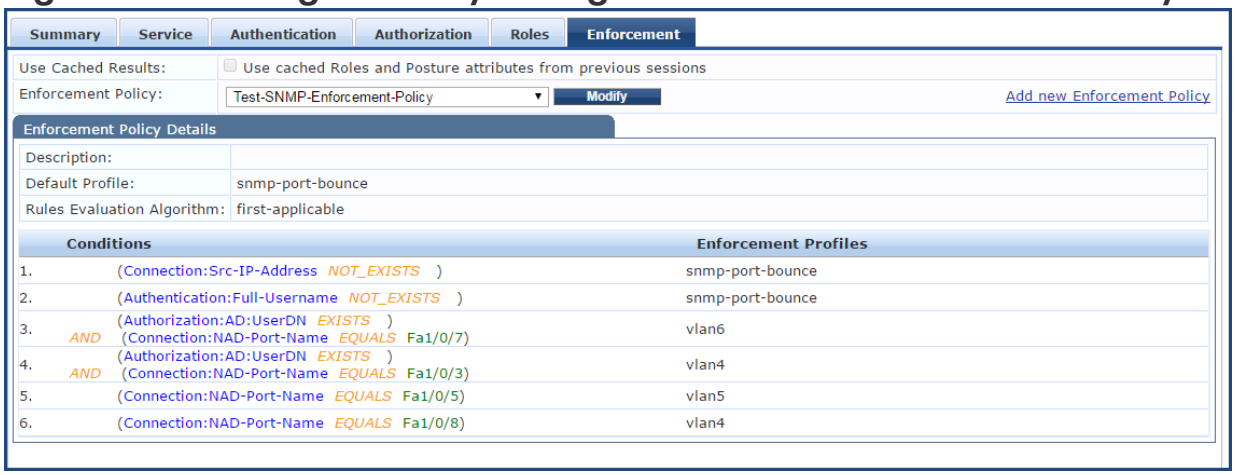
   For a description of all the parameters in the **Service** page, see Adding Services.

Configure an Enforcement Policy After you create the Policy Manager OnConnect Enforcement service, you must associate the service with an enforcement policy. WMI (Windows Management Instrumentation) configuration is used to retrieve the Loggedin User information. Whenever a domain-joined Windows client connects to an OnConnect-enabled port with the domain user logged in, Authorization attributes for this user are fetched from authorization sources to determine the role of the user; this information is then used in configuring policy enforcement. For details on configuring WMI credentials, see WMI Credentials Configuration. To associate a Policy Manager OnConnect Enforcement service with an enforcement policy:

1. When finished with the parameter settings on the **Add Services > Service** page, click **Next**.The **Services > Enforcement** page opens.

2. From the **Services > Add > Enforcement** page, you can either select an existing enforcement policy or create a new one.

Figure 2. Selecting thePolicy ManagerOnConnect Enforcement Policy



3. From the **Enforcement Policy** drop-down list, select the appropriate OnConnect Enforcement policy.If you have not configured an OnConnect-type Enforcement policy, click **Add New Enforcement Policy** to create a new enforcement policy to associate with this OnConnect service.

4. Specify the values for the remaining parameters as described in Table 1, then click **Save**.

Table 1. Service Enforcement Page Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached rol es and posture attributes from previous sessions.
Enforcement Policy	From the drop - down list, select the pr econfigured enforcement policy. This is a mandatory step.
Enforcement Policy Details	
Description	Displays additional information about th e selected enforcement policy.
Default Profile	Displays a default profile applied by .
Rules Evaluation Algorithm	Shows first matched rule and return the role or select all matched rules and retu rn a set of roles.

Event-Based Enforcement Service

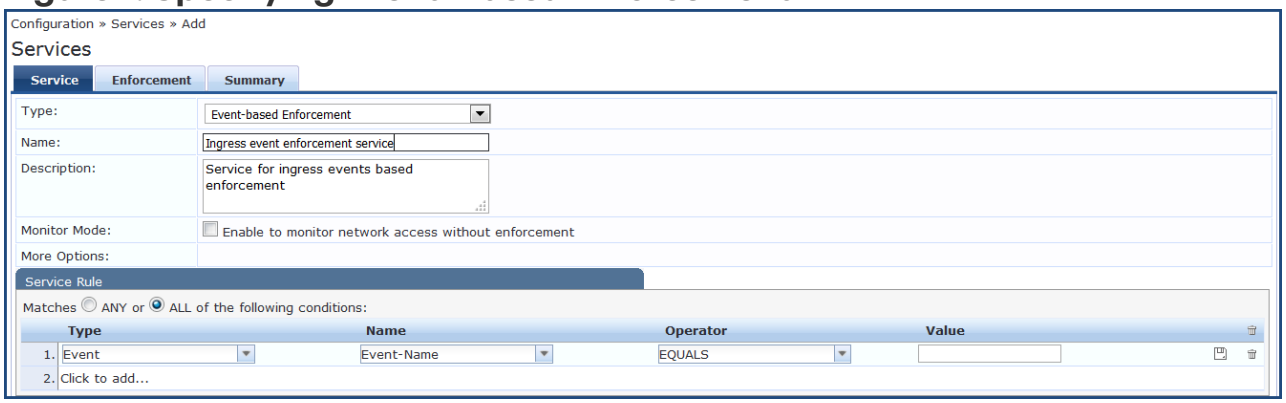
About this task

The **Event-Based Enforcement** service manages enforcement actions in response to threat-event processing. When there is a suspicious user, this user could represent a common DOS attack or some other threat. When a threat is detected, Policy Manager performs enforcement operations as configured; for example, executing a change of authorization (COA) to disconnect a suspicious user from the network. To add an event-based enforcement service:

Procedure

1. Navigate to **Configuration > Services**. The **Services** page appears. The **Services** page provides options to add, modify, and remove a service.
2. To add the service, click **Add**. The **Add Services** dialog appears.
3. From the **Type** drop-down list, select **Event-based Enforcement** (see Figure 1).

Figure 1. Specifying Event -Based Enforcement



4. Enter the name or label of the event-based enforcement service.
5. Enter the values for any other parameters, including service rules, required for this service.

Associating the Service with an Enforcement Policy

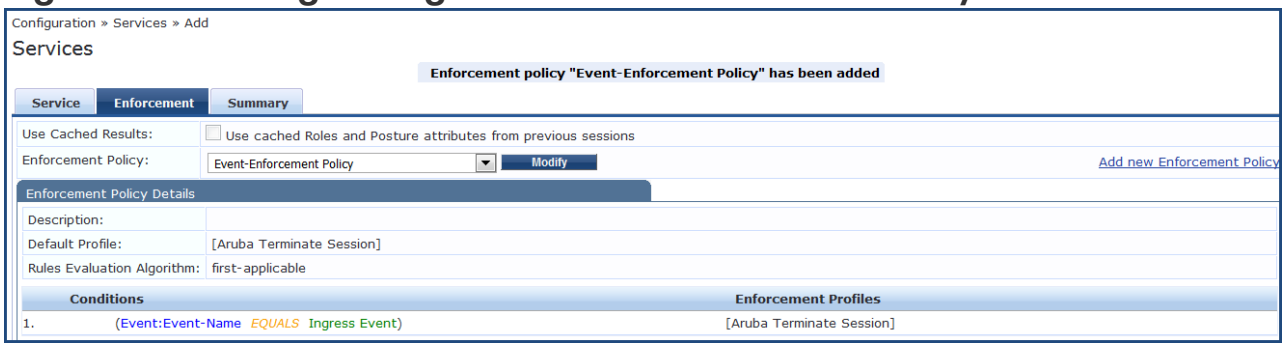
About this task

After you create the event-based enforcement service, you must associate the service with an enforcement policy. You can do this from the **Services > Add > Enforcement** page.

Procedure

1. When finished with the parameter settings on the **Add Services > Service** page, click **Next**.The **Services > Enforcement** page appears.
2. From the **Services > Add > Enforcement** page, you can either select an existing enforcement policy or create a new one.

Figure 2. Selecting the Ingress Events Enforcement Policy



3. From the **Enforcement Policy** drop-down list, select the appropriate Event Enforcement policy.

4. If you have not configured Event-type Enforcement policies, click **Add New Enforcement Policy** to create a new enforcement policy.
5. Specify the values for the remaining parameters as described in Table 1, then click **Save**.

Table 1. Service Enforcement Page Parameters

Parameter	Description
Use Cached Results	Select this check box to use cached roles and posture attributes from previous sess ions.
Enforcement Policy	From the drop - down list, select the prec onfigured enforcement policy. This is ma ndatory.
Enforcement Policy Details	
Description	Displays additional information about the selected enforcement policy.
Default Profile	Displays a default profile applied by .
Rules Evaluation Algorithm	Shows first matched rule and return the role or select all matched rules and retur n a set of roles.

TACACS+ Enforcement Service

About TACACS+ Enforcement

Configure the **TACACS+ Enforcement** service for any kind of TACACS+ request. TACACS+ users can be authenticated against any of the supported authentication source types:

- Local DB
- SQL DB
- Active Directory

- LDAP Directory
- Token Servers with a RADIUS interface

Similarly, you can specify service-level authorization sources from the **Authorization** tab. You can associate a role-mapping policy with this service via the **Roles** tab.

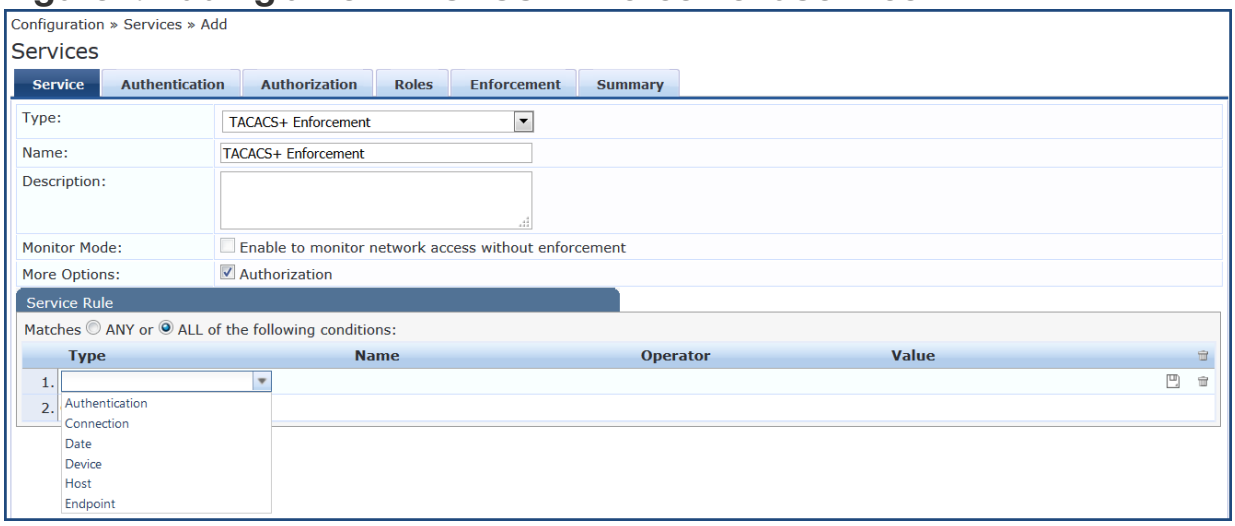
> **NOTE**
>
> If the Admin User repository is used against the TACACS+ service to support Admin UI login functionality, an administrator must ensure this service is at the top of the service order list, and the service rules for this particular service do not contain a username filter. This is required for the Admin user lockout functionality to take effect.

The result of evaluating a TACACS+ enforcement policy is one or more TACACS+ enforcement profiles. For more information on TACACS+ enforcement profiles, see TACACS+ Based Enforcement Profile .

Configuring a TACACS+ Emforcement Service To create a **TACACS+ Enforcement** service:

1. Navigate to **Configuration > Services**, then select the **Add** link. The **Add Services** page opens.
2. From the Type drop down, select **TACACS+ Enforcement**. The **TACACS+ Enforcement** service configuration dialog opens:

Figure 1. Adding a New TACACS+ Enforcement Service



3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add TACACS+ Enforcement > Service Tab Parameters

Parameter	Action/Description
Type	From the drop - down list, select **TACACS+ Enforcement**.
Name	Enter the name of the service.
Description	Provide additional information that helps to identify the service.
Monitor Mode	The **Monitor Mode** option is disabled for an enforcement policy.
More Options	The **Authorization** tab is not enabled by default. To bring up the **Authorization** configuration tab, check the **Authorization** check box.
Service Rule	
Type	Select one of the following service rule types: ◦ Authentication ◦ Connection ◦ Date ◦ Device ◦ Host ◦ Endpoint
Name	Select the name of the service rule from the drop - down list.
Operator	Select an appropriate operator from the list of operators for the data type of the attribute.

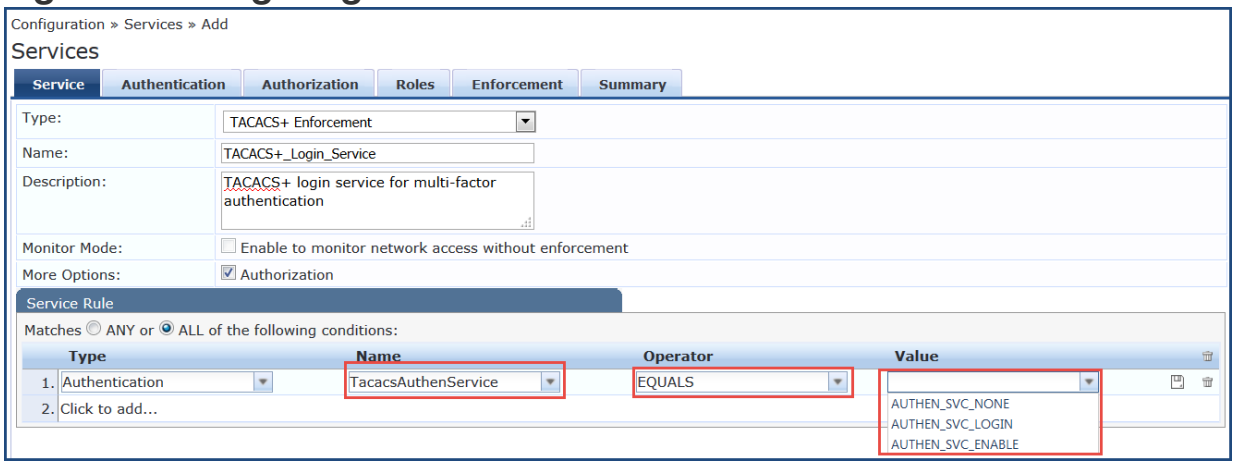
Parameter	Action/Description
	For example, you can select from BELONGS_TO, NOT_BELONGS_TO, CONTAINS, or EQUALS.
Value	Select the value from the drop - down list. The value list depends on the operator selected.

Configuring a TACACS+ Multi-Factor Authentication Service Policy Manager provides a way to differentiate between a TACACS+ login authentication and a TACACS+ enable authentication. To do so, in Service and Policy rule configuration, you can use the **Authentication** type **TacacsAuthenService** parameter. This parameter accepts three values: **None**, **Login**, or **Enable**. Note that an Enforcement policy can also use the **Authentication: TacacsAuthenService** attribute in the **Rules Editor > Conditions** section of the policy configuration. **Creating a TACACS+ Multi-Factor Authentication Service** To apply the **TacacsAuthenService** parameter in a TACACS+ Enforcement service:

1. Navigate to **Configuration > Services**, then select the **Add** link.

   The **TACACS+ Enforcement** service configuration dialog opens:

Figure 2. Configuring a TACACS+ Multi-Factor Authentication Service



2. Specify the **TACACS+ Enforcement Service** parameters as described in the following table:

Table 2. TACACS+ Enforcement > Service Tab Parameters

Parameter	Action/Description
Type	From the drop - down list, select **TACACS+ Enforcement**.
Name	Enter the name of the service.
Description	Provide additional information that help s to identify the service.
Monitor Mode	The **Monitor Mode** option is disabled fo r an enforcement policy.
More Options	The **Authorization** tab is not enabled by default. To bring up the **Authorization** configura tion tab, check the **Authorization** check box.
Service Rule	
Type	Select **Click to add**, then select the **Aut hentication** type.
Name	Select **TacacsAuthenService**.
Operator	Select **EQUALS**.
Value	Select one of the following values: ◦ **AUTHEN_SVC_NONE**: Choose this o ption when the user does not requi re authentication. ◦ **AUTHEN_SVC_LOGIN**: Choose this option when the user is authentica ting in Login mode. ◦ **AUTHEN_SVC_ENABLE**: Choose this option when the user is authenticati ng in Privileged mode.

Web-Based Authentication Service

Configure a web-based authentication service for guests or agentless hosts that connect through the Policy Manager Portal. The user is redirected to the Policy Manager captive portal by the network device or by a DNS server that is set up to redirect traffic on a subnet to a specific URL.

The web page collects the user name and password, and also optionally collects health information on supported Windows operation systems. (See the *Attribute Name* **OSType** in Table 1, *Service Rule > Web-Based Authentication Host Attributes.*)

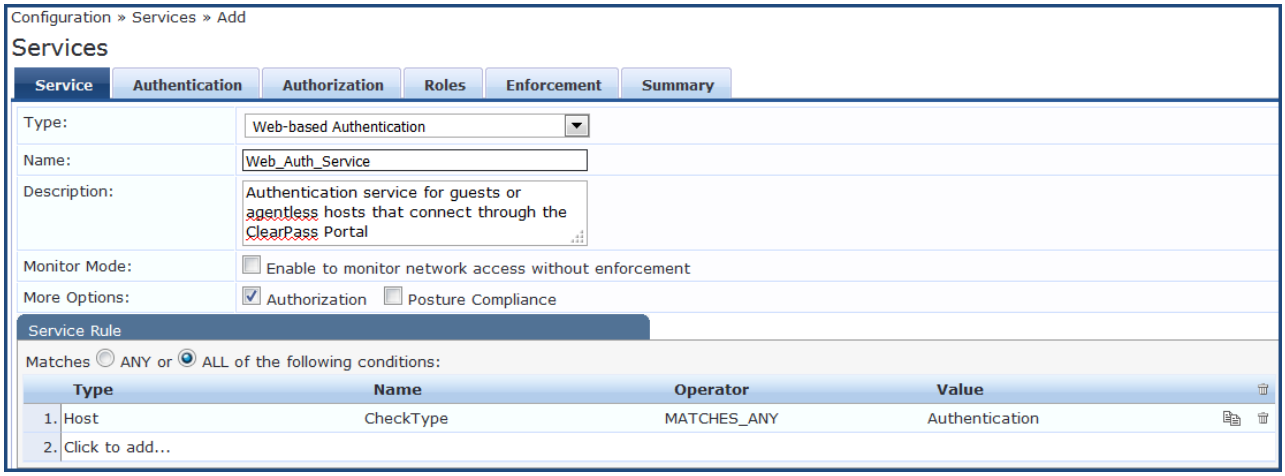
An internal service rule—*Connection:Protocol EQUALS WebAuth*—categorizes requests into this type of service. You can add additional rules if needed. In addition, you can select a Web-based Authentication service based on the operating system name. You can specify or exclude specific OS versions (for details, see the next section, Selecting a Web-Based Authentication Service by the OS Name).

Configuring a Web-Based Authentication Service

To configure a web-based authentication service:

1. Navigate to **Configuration > Services**. The **Services** page opens.
2. Select the **Add** link. The **Add Services** page opens.
3. From the Type drop-down list, select **Web-Based Authentication**. The following service configuration dialog opens:

Figure 1. Web-Based Authentication Service Configuration Dialog



NOTE

The **Audit End-hosts** and **Profile Endpoints** options are not available for a **Web-based Authentication** service.

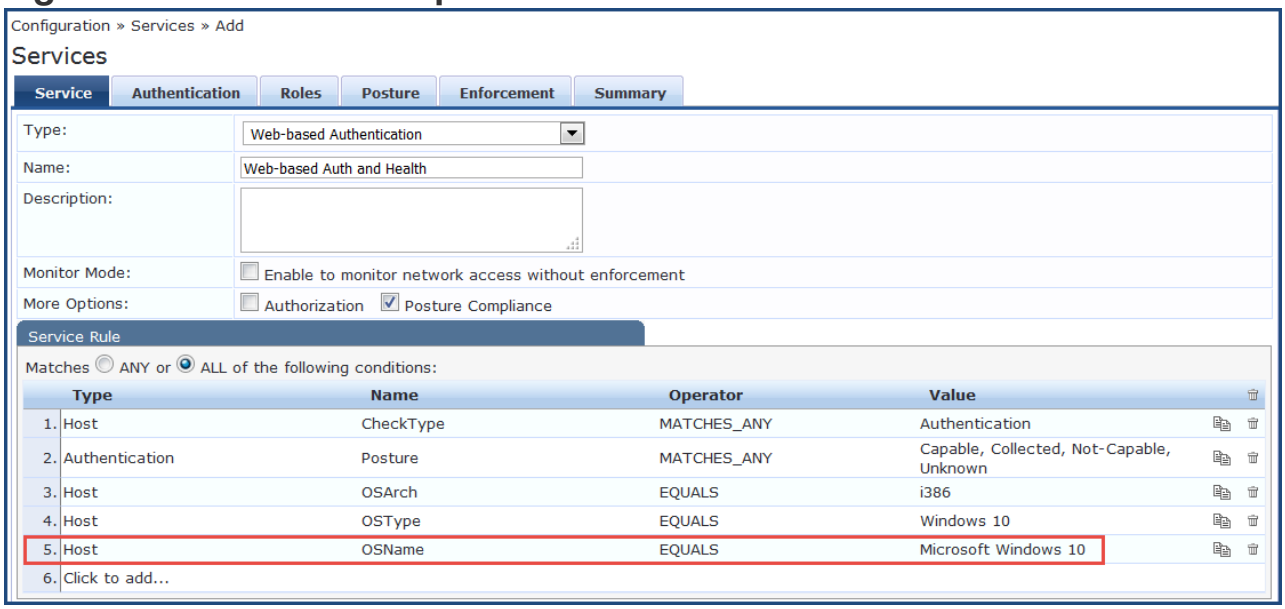
Selecting a Web-Based Authentication Service by the OS Name

The **Service Rule > Host:OSName** attribute allows you to select a Web-based Authentication service based on the OS name. You can specify or exclude specific OS versions.

To select a web-based authentication service by the OS name and version:

1. From the **Service** tab > **Service Rule** area, select **Click to add**.

Figure 2. Host OS Name Specified in the Web-Based Authentication Service



2. Specify the **Host OS Architecture** attribute as follows:

- Type=**Host**
- **Name=OSArch**
- Operator=**EQUALS**
- Value=**i386** or **x86_64**
- Specify the Host OS Type attribute as follows:
- Type=**Host**
- **Name=OSType**
- Operator=**EQUALS**
- Value=**Windows 10**
- Specify the Host OS Name attribute as follows:
- Type=**Host**
- **Name=OSName**
- Operator=**EQUALS**
- Value=**Microsoft Windows 10**

Example Showing How to Differentiate Between Windows 8 and Windows 8.1

- Type=**Host**
- Name=**OSName**
- Operator=**CONTAINS**
- Value=**Microsoft Windows 8.1**

Service Rule > Web-Based Authentication Host Attributes

The following table describes the list of other attributes that can be used to create services based on the client's information.

Table 1. Service Rule > Web - Based Authentication Host Attributes

	Attribute Name	
Host	AgentType	Specifies the type of OnGuard Agent. This attribute provides a way to define a separate service for each OnGuard Agent Type. The supported values are:

Attribute Name		
		• OnGuardAgent: OnGuard Agent • OnGuardAgentService: OnGuard Agent running as a service • **NativeWebAgent**: Native Dissolvable Agent • **JavaWebAgent**: Java Dissolvable Agent
	Agent Version	OnGuard Agent version. This attribute can be used to create a service based on the OnGuard Agent version.
	CheckType	Specifies the type of check OnGuard Agent is performing based on the **Mode** setting in the OnGuard Settings page (for details, see OnGuard Settings and OnGuard Custom Web Pages). For **Authenticate with health checks**, the value of this attribute is **Authentication, Health**. The supported values are: • **Authentication**: OnGuard Agent is performing authentication; that is, the request contains credentials. • **Health**: OnGuard Agent is performing health checks; that is, the request contains Posture information. • **None**

Attribute Name	
FQDN	Indicates the Fully Qualified Domain Name of the client.
HealthCheckLevel	Indicates the level of health checks performed by OnGuard Agent; that is, whether the user is logged in at the time of health check or not. This attribute can be used to see the health check level when OnGuard Agent is running as **Service** or **BothServiceAndUser**. • **System**: The user is *not* logged in when health checks are being run. • **User**: The user is logged in when health checks are being run.
Installed SHAs	Specifies the SHAs installed on the client.
InterfaceType	Specifies the type of **Network Interface**. This attribute can be used to define different services based on Network Interface type. The supported values are: • **Wired** • **Wireless** • **VPN**
LastHealthStatus	Most recent Health status. Value can be from one of the following • Healthy • Quarrantine • Unknown

Attribute Name	
LastHealthStatusTimeStamp	Timestamp of last health status in epoch format, For example,12 PM on 2 December, 2021 would be 1638446400.
LastHealthStatusDateTime	Timestamp of last health status in date/time format.
Machine Type	Identify the device as one of the following types: • Desktop • Laptop • Virtual Machine • Server • Other • Unknown
Name	This is the host name of the client (without the domain name).
OSArch	Specifies whether the client is running a 32 - bit or 64 - bit OS. The supported values are: • **i386**: 32 - bit OS • **x86_64**: 64 - bit OS
OSName	Indicates the full Operating System name. This attribute can be used to create services for a specific OS. For example, you can use this attribute to differentiate between Windows 8 and Windows 8.1
OSNameVersion	Provides the Windows OS name and the build version. This attribute can be used to create different Posture policies for different Windows 1

Attribute Name	
	0 versions such as **2015 LTSB** or **2016 LTSB**.
OSType	Specifies the Operating System type. The supported values are: • **Linux** • **macOS** • **Windows 8** • **Windows 10** • **Windows Server 2016** • **Windows Server 2019**
SDKType	Specifies the SDK type. For example, you can specify that the SDK type equals **V4**.
SDKVersion	Specifies the SDK version.
ServerCertificateCheck	This attribute's value shows the status of the Policy Manager Server Certificate Check performed by OnGuard agent while sending a WebAuth request to the Policy Manager server.This attribute can also be used in a Service Classification. The value of this attribute can be one of the following: • **Passed**: OnGuard Agent successfully verified the Policy Manager Server Certificate. • **Failed**: OnGuard Agent failed to verify the Policy Manager Server Certificate.

	Attribute Name	
	UserAgent	The value of this attribute c ontains both **Agent Type** an d **Agent Version**. For exampl e, OnGuard 6.6.5.89660.

Table 2. Service Rule > Web - Based Authentication Host Attributes

	Attribute Name	
Host	AgentType	Specifies the type of OnGua rd Agent. This attribute prov ides a way to define a sepa rate service for each OnGua rd Agent Type. The supporte d values are: • **OnGuardAgent**: OnGua rd Agent • **OnGuardAgentService**: OnGuard Agent running as a service • **NativeWebAgent**: Native Dissolvable Agent • **JavaWebAgent**: Java Dis solvable Agent
	Agent Version	OnGuard Agent version. Thi s attribute can be used to c reate a service based on the OnGuard Agent version.
	CheckType	Specifies the type of check OnGuard Agent is performin g based on the **Mode** settin g in the OnGuard Settings p age (for details, see OnGua rd Settings and OnGuard Cu stom Web Pages). For **Authenticate with healt h checks**, the value of this attribute is **Authentication,**

Attribute Name	
	Health. The supported values are: • **Authentication**: OnGuard Agent is performing authentication; that is, the request contains credentials. • **Health**: OnGuard Agent is performing health checks; that is, the request contains Posture information. • **None**
FQDN	Indicates the Fully Qualified Domain Name of the client.
HealthCheckLevel	Indicates the level of health checks performed by OnGuard Agent; that is, whether the user is logged in at the time of health check or not. This attribute can be used to see the health check level when OnGuard Agent is running as **Service** or **BothServiceAndUser**. • **System**: The user is *not* logged in when health checks are being run. • **User**: The user is logged in when health checks are being run.
Installed SHAs	Specifies the SHAs installed on the client.
InterfaceType	Specifies the type of **Network Interface**. This attribute can be used to define different services based on Netwo

Attribute Name	
	rk Interface type. The suppo rted values are: • **Wired** • **Wireless** • **VPN**
Name	This is the host name of the client (without the domain n ame).
OSArch	Specifies whether the clien t is running a 32 - bit or 64 - bit OS. The supported val ues are: • **i386**: 32 - bit OS • **x86_64**: 64 - bit OS
OSName	Indicates the full Operating System name. This attribute can be used to create servic es for a specific OS. For example, you can use th is attribute to differentiate b etween Windows 8 and Win dows 8.1
OSNameVersion	Provides the Windows OS n ame and the build version. T his attribute can be used to create different Posture poli cies for different Windows 1 0 versions such as **2015 LTS B** or **2016 LTSB**.
OSType	Specifies the Operating Sys tem type. The supported val ues are: • **Linux** • **macOS** • **Windows 8** • **Windows 10**

Attribute Name	
	• **Windows Server 2016** • **Windows Server 2019**
SDKType	Specifies the SDK type. For example, you can specify that the SDK type equals **V4**. For related information, see Upgrading From OnGuard Plugin Version 1.0 to 2.0 .
SDKVersion	Specifies the SDK version.
ServerCertificateCheck	This attribute's value shows the status of the Policy Manager Server Certificate Check performed by OnGuard agent while sending a WebAuth request to the Policy Manager server. This attribute can also be used in a Service Classification. The value of this attribute can be one of the following: • **Passed**: OnGuard Agent successfully verified the Policy Manager Server Certificate. • **Failed**: OnGuard Agent failed to verify the Policy Manager Server Certificate.
UserAgent	The value of this attribute contains both **Agent Type** and **Agent Version**. For example, OnGuard 6.7.0.89660.

Web-based Health Check Only Service

This type of service is the same as the **Web-based Authentication** service except that there is no authentication performed; only health check are performed.

- The internal service rule **Connection:Protocol EQUALS WebAuth** categorizes requests into this type of service.
- The external service rule **Host:CheckType EQUALS Health** is automatically added when you select this type of service.

  For more information, see Web-Based Authentication Service .

Creating the Service

About this task

To create a **Web-based Health Check Only** service:

Procedure

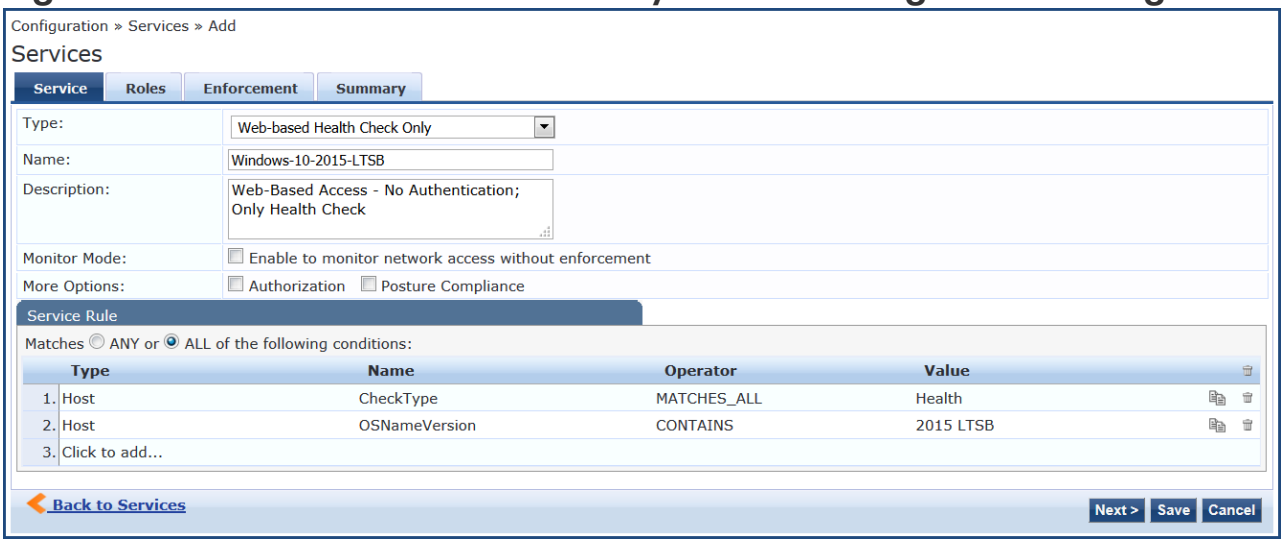
1. Navigate to **Configuration > Services**, then select the **Add** link.

   The **Add Services** page opens.

2. From the **Type** drop-down, select the **Web-Based Health Check Only** service.

   The **Web-Based Health Check Only** service configuration dialog opens:

Figure 1. Web-Based Health Check Only Service Configuration Dialog



3. Specify the **Service** tab parameters as described in the following table:

Table 1. Add Web - based Health Check Only Service > Service Tab Parameters

Parameter	Action/Description
Type	From the drop - down list, select **Web - based Health Check Only**.
Name	Enter the name of the service.
Description	Provide additional information that helps to identify the service.
Monitor Mode	Select the check box to monitor network access activity without enforcement.
More Options	Check these boxes to access the additional configuration tabs: • Authorization • Posture Compliance
Service Rule	
Type	Select one of the following service rule types: • Authentication • Connection • Date • Device • Host • Endpoint
Name	Select the name of the service rule from the drop - down list.
Operator	Select an appropriate operator from the list of operators for the data type of the attribute.

Parameter	Action/Description
Value	Select the value from the drop - down lis t. The value list depends on the operator selected.

Service Rule > Web-Based Authentication Host Attributes

The following table describes the list of other attributes that can be used to create host services based on the client's information.

Table 2. Service Rule > Web - Based Health Check Only Host Attributes

Attribute Name	Description
AgentType	Specifies the type of OnGuard Agent. This attribute provides a way to define a separat e service for each OnGuard Agent Type. Th e supported values are: • **OnGuardAgent**: OnGuard Agent • **OnGuardAgentService**: OnGuard Agen t running as a service • **NativeWebAgent**: Native Dissolvable Agent • **JavaWebAgent**: Java Dissolvable Agent
Agent Version	OnGuard Agent version. This attribute can be used to create a service based on the O nGuard Agent version.
CheckType	Specifies the type of check OnGuard Agen t is performing based on the **Mode** setting in the OnGuard Settings page (for details, see OnGuard Settings and OnGuard Custo m Web Pages). For **Authenticate with health checks**, the value of this attribute is **Authentication, Health**. The supported values are: • **Authentication**: OnGuard Agent is pe rforming authentication; that is, the re quest contains credentials.

Attribute Name	Description
	• **Health**: OnGuard Agent is performing h ealth checks; that is, the request contai ns Posture information. • **None**
FQDN	Indicates the Fully Qualified Domain Name of the client.
HealthCheckLevel	Indicates the level of health checks perfor med by OnGuard Agent; that is, whether th e user is logged in at the time of health ch eck or not. This attribute can be used to see the healt h check level when OnGuard Agent is runn ing as **Service** or **BothServiceAndUser**. • **System**: The user is *not* logged in when health checks are being run. • **User**: The user is logged in when health checks are being run.
Installed SHAs	Specifies the SHAs installed on the client.
InterfaceType	Specifies the type of **Network Interface**. T his attribute can be used to define differen t services based on Network Interface typ e. The supported values are: • **Wired** • **Wireless** • **VPN**
LastHealthStatus	Most recent Health status. Value can be fro m one of the following • Healthy • Quarrantine • Unknown
LastHealthStatusTimeStamp	Timestamp of last health status in epoch format, For example,12 PM on 2 December, 2021 would be 1638446400.

Attribute Name	Description
LastHealthStatusDateTime	Timestamp of last health status in date/tim e format.
Machine Type	Identify the device as one of the following t ypes: • Desktop • Laptop • Virtual Machine • Server • Other • Unknown
Name	This is the host name of the client (without the domain name).
OSArch	Specifies whether the client is running a 3 2 - bit or 64 - bit OS. The supported values are: • **i386**: 32 - bit OS • **x86_64**: 64 - bit OS
OSName	Indicates the full Operating System name. This attribute can be used to create servic es for a specific OS. For example, you can use this attribute to differentiate between Windows 8 and Wind ows 8.1
OSNameVersion	Provides the Windows OS name and the bu ild version. This attribute can be used to c reate different Posture policies for different Windows 10 versions such as **2015 LTSB** or **2016 LTSB**.
OSType	Specifies the Operating System type. The s upported values are: • **Linux** • **macOS** • **Windows 8**

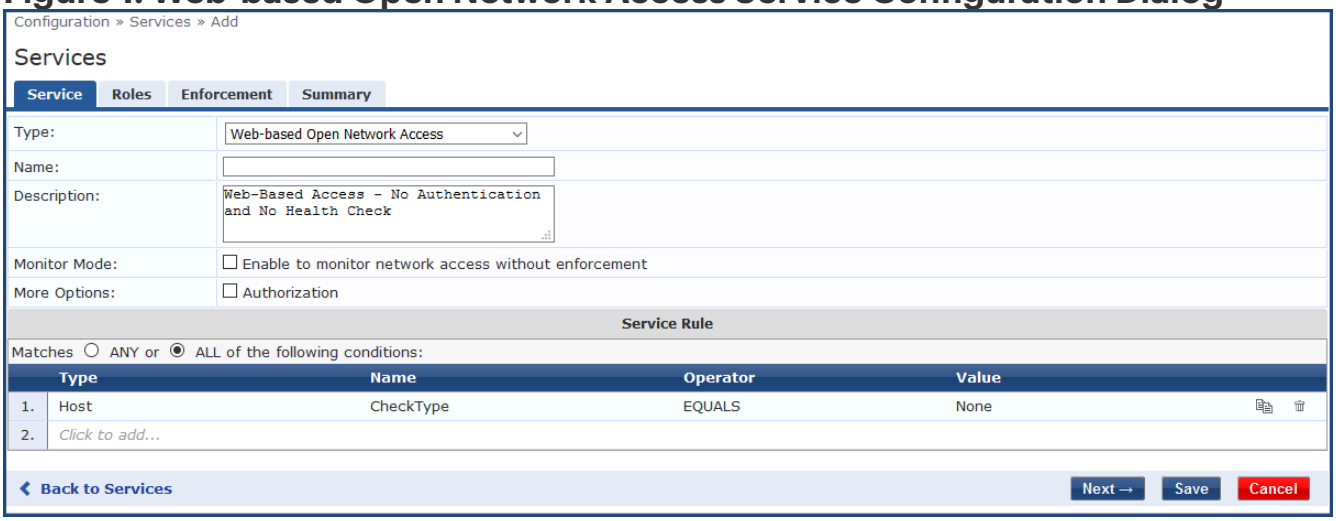
Attribute Name	Description
	• **Window 10** • **Windows Server 2016** • **Windows Server 2019**
SDKType	Specifies the SDK type. For example, you can specify that the SDK type equals **V4**. For related information, see Upgrading From OnGuard Plugin Version 1.0 to 2.0 .
SDKVersion	Specifies the SDK version.
ServerCertificateCheck	This attribute's value shows the status of the Policy Manager Server Certificate Check performed by OnGuard agent while sending a WebAuth request to the Policy Manager server.This attribute can also be used in a Service Classification. The value of this attribute can be one of the following: • **Passed**: OnGuard Agent successfully verified the Policy Manager Server Certificate. • **Failed**: OnGuard Agent failed to verify the Policy Manager Server Certificate.
UserAgent	The value of this attribute contains both the **Agent Type** and **Agent Version**. For example, OnGuard 6.7.0.89660.

Web-Based Open Network Access Service

Configuration for this service is the same as **Web-based Authentication** service, except that a health check is not performed on the endpoints. A **Terms of Service** page (as configured on the Policy Manager Guest Portal page) is presented to the user. Network access is granted when you click **Submit Action**. For more information, see Web-Based Authentication Service .

The following figure displays the **Web-based Open Network Access** service page configuration dialog:

Figure 1. Web-based Open Network Access Service Configuration Dialog



Modifying and Managing Services

Following the procedures described in this section to modify and manage your services.

Modifying an Existing Service

About this task

You can use service types as configured, or you can edit their settings. To modify an existing service:

Procedure

1. Navigate to the **Configuration** > **Services** page.
2. Select the service you wish to modify. The **Services** > *[service_name* **]** dialog opens.
3. Edit the service parameters, then click **Save**.

   For complete information on the settings supported by each service type, see Configuring Common Services Using Service Templates.

Creating a Service by Copying an Existing Service

About this task

You can perform a service-copy operation only on a publisher. To create a service template by copying an existing service:

Procedure

1. From the **Services** page, select the check box by the desired service.
2. Click **Copy**. You receive the message *Service Copy_of_<service_name> has been added*.

Removing a Non-Default Service

About this task

You cannot remove the default Policy Manager services. However, you can remove any service that has been created in Policy Manager. To remove a non-default service:

Procedure

1. From the **Services** page, select the check box for the non-default service you want to remove.
2. Click **Delete**. You receive the prompt *Are you sure you want to delete this service?*
3. To cancel the delete operation, click **No**. To delete the selected service, click **Yes**.

Starting or Stopping Policy Manager Services

About this task

From the **Services Control** page, you can view the status of a service (that is, see whether a service is running or not), and stop or start Policy Manager services, including any Active Directory domains to which the current server is now joined.

To access the **Services Control** page:

Procedure

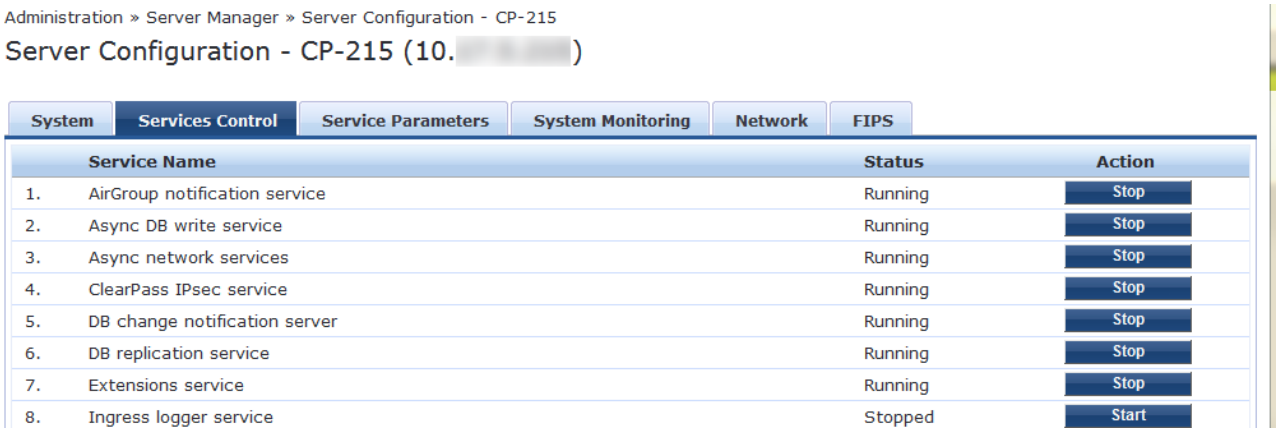
1. In Policy Manager, navigate to **Administration > Server Manager > Server Configuration**.

   The **Server Configuration** page opens.
2. Click the row that lists the Policy Manager server of interest.

   The Server Configuration screen for the selected Policy Manager server appears.
3. Select the **Services Control** tab.

   The **Services Control** page opens.

   Figure 1. Server Configuration > Services Control Page



NOTE

You will notice that the **Virtual IP** service is the only service that is not running. It's normal for the **Virtual IP** service to be stopped when this service is not being used.

- If a service is stopped, you can use its **Start** button to restart it.
- You can also start an individual service from the command line:

  `service start` <*service_name*>

- You can start all the services from the command line:

  `service start all`

Restarting Policy Manager Services

About this task

NOTE

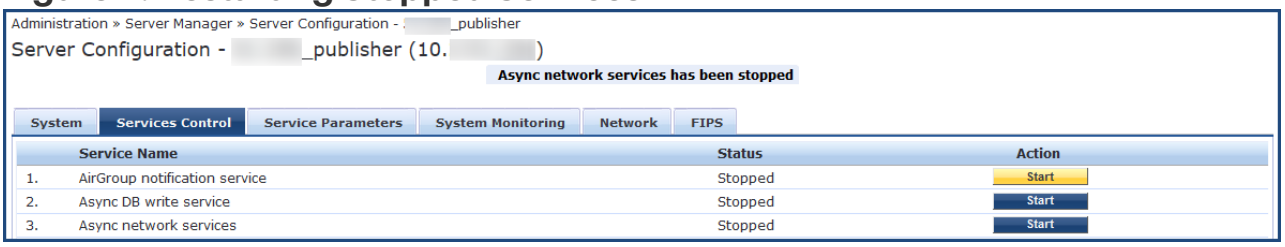
The Audit Viewer (**Monitoring > Audit Viewer**) tracks NTP configuration changes.

To restart Policy Manager services:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager publisher.
3. From the **Server Configuration** page, select the **Services Control** tab.
4. From the **Action** column, click **Start** for each service that needs to be restarted.

Figure 2. Restarting Stopped Services



Once the service is restarted, the **Start** button is changed to **Stop**.

Reordering Services

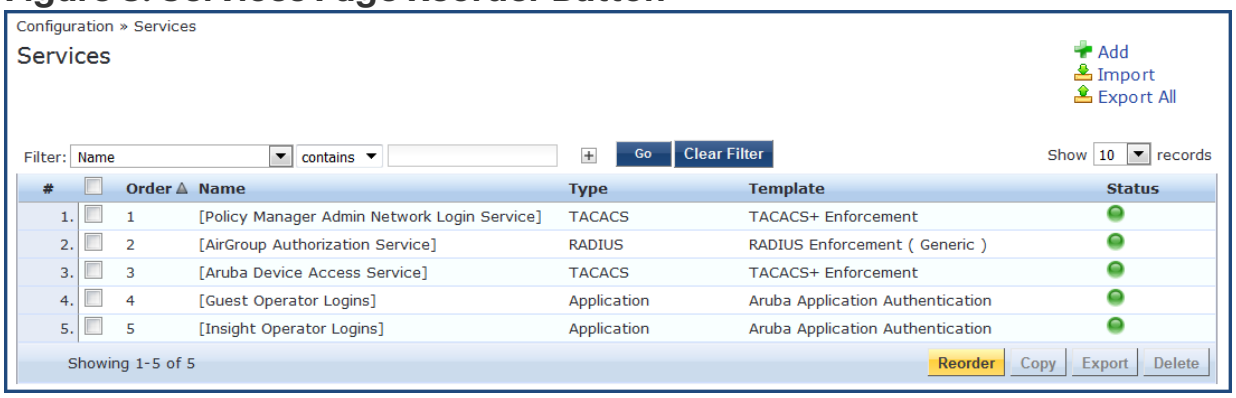
Policy Manager evaluates requests against the service rules of each service that is configured, in the order in which these services are defined.

- The service associated with the first matching service rule is associated with the request.
- To change the order in which service rules are processed, you can change the order of services.

  To change the order in which services are executed:

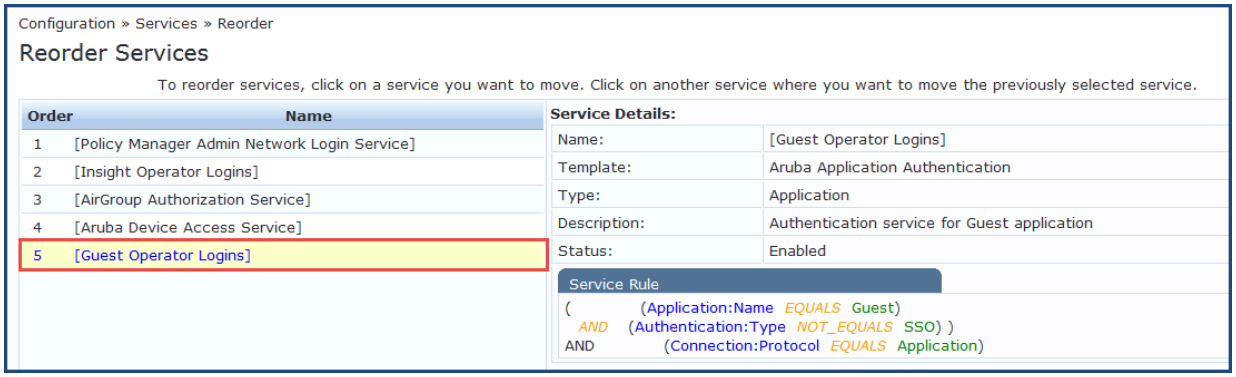
  1. Navigate to the **Configuration > Services** page. The **Services** page opens.

     ### Figure 3. Services Page Reorder Button

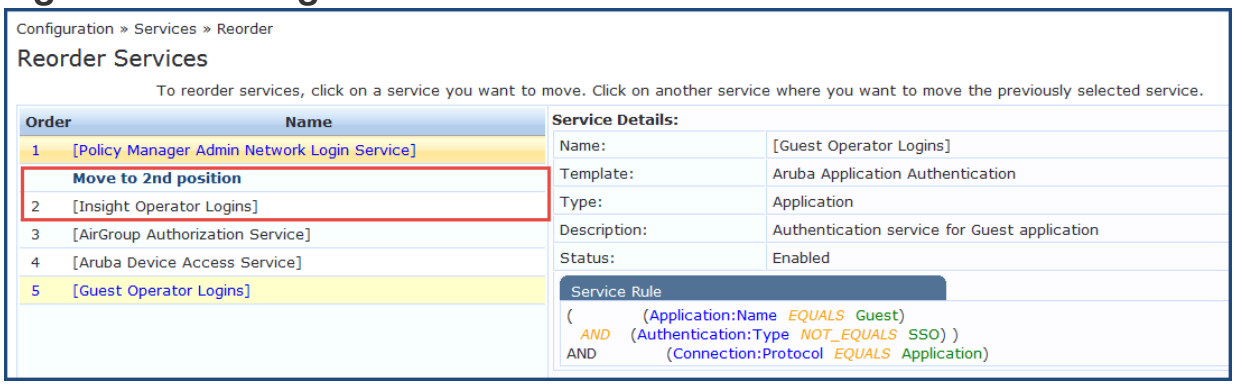
  2. Click the **Reorder** button. The **Reorder Services** page opens.
  3. Click the service you want to move to another position in the order (see Figure 3). In this example, we will move **Guest Operator Logins** from the 5th position to the 2nd position.

     ### Figure 4. Selecting the Service to Be Reordered



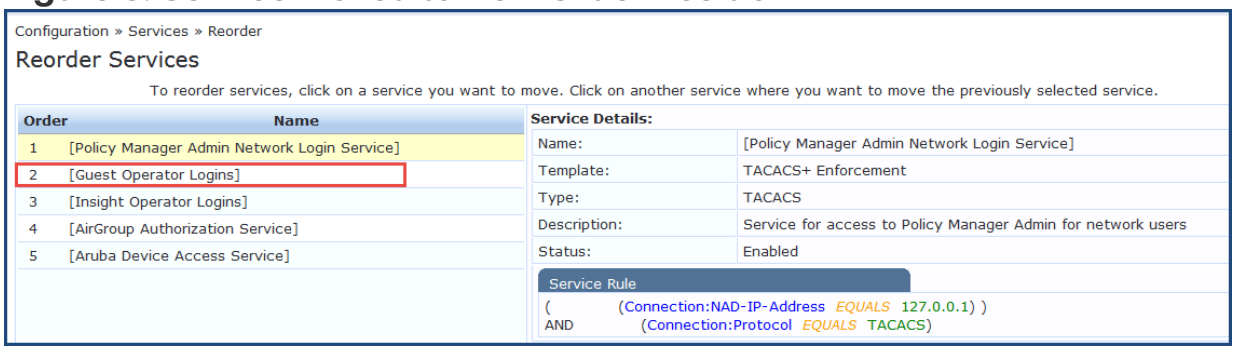
4. Select the position where you want to move the service (see Figure 4).

Figure 5. Selecting the Destination Order Position



5. Click the selected destination position (see Figure 5). The service is now moved to its new position in the services order.

Figure 6. Service Moved to New Order Position



6. Click **Save**.You return to the **Services** page, which shows the service in its new order and displays the message:

   Services have been reordered successfully.

Live Monitoring

Live Monitoring provides real-time visibility into system activity and status for faster awareness and response.

Subtopics

Live Monitoring
Live Monitoring provides access to component status and related monitoring functions.
Additional Monitoring Functions
Live Monitoring: Access Tracker
Live Monitoring: Accounting
Live Monitoring: OnGuard Activity
Live Monitoring Analysis and Trending
Live Monitoring: System Monitor
Monitoring Endpoint Profiler and Network Scan
Audit Viewer
Event Viewer
Data Filters
Restoring Users on a Blocked List to the Network

Live Monitoring

Live Monitoring provides access to component status and related monitoring functions.

The Live Monitoring features in Policy Manager provide access to live monitoring of components and other functions. ClearPass Policy Manager includes the following Live Monitoring features:

- Live Monitoring: Access Tracker
- Live Monitoring: Accounting
- Live Monitoring: Analysis and Trending
- Live Monitoring: OnGuard Activity
- Live Monitoring: System Monitor

Additional Monitoring Functions

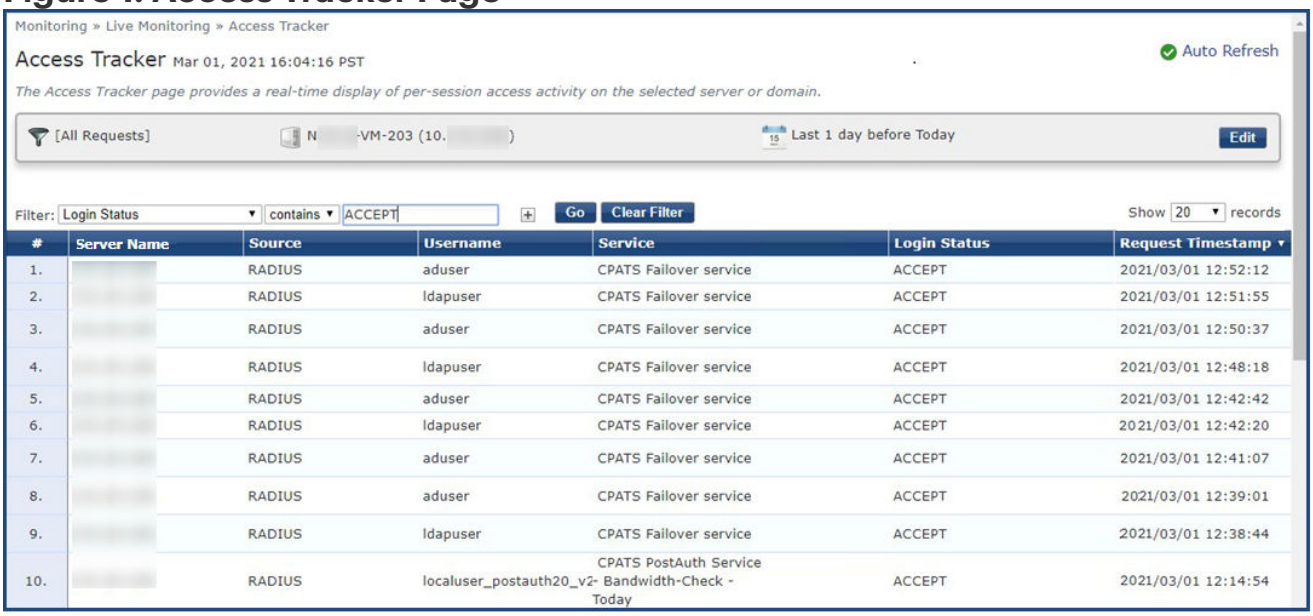
The following additional monitoring features are provided:

- Audit Viewer
- Event Viewer
- Data Filters
- Restoring Users on a Blocked List to the Network

Live Monitoring: Access Tracker

The **Access Tracker** provides a real-time display of per-session access activity on the selected server or domain. To access this page, navigate to **Monitoring** > **Live Monitoring** > **Access Tracker**.

Figure 1. Access Tracker Page



NOTE

The Access Tracker's columns automatically resize as new columns are added, and can be manually resized as needed so displayed data aligns correctly in the columns. Use the mouse to grab the column separator to the left of the column title, then move the column left or right to change the column width as needed to optimize the Access Tracker display.

Table 1 describes the information provided by default on the **Access Tracker** page. Data columns not displayed by default can be manually added by editing the Access Tracker page.

Table 1. Access Tracker Page Default Columns

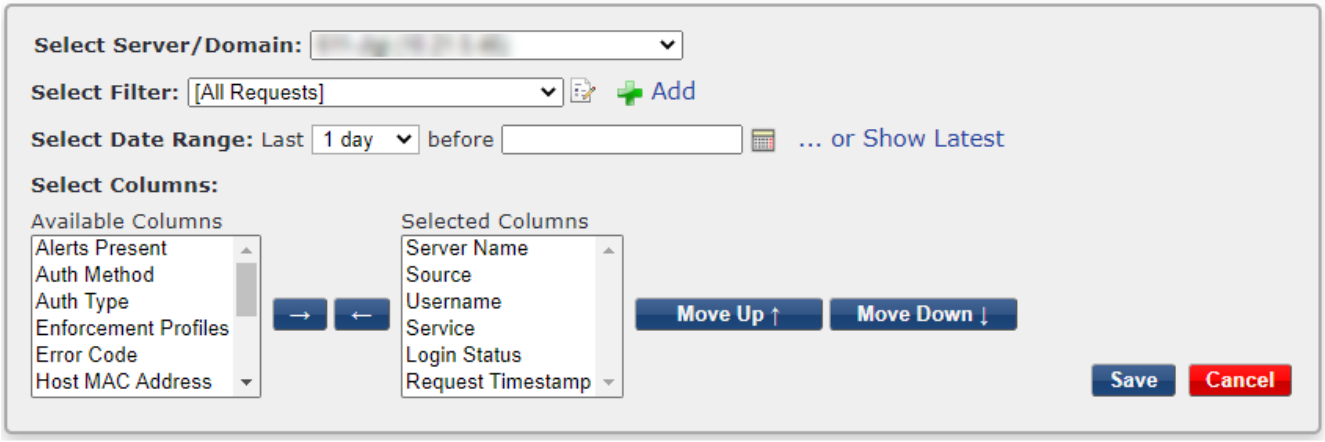
Column	Description
Server Name	Displays the name of the Policy Manager server. The server most recently selected remains highlighted, even if the mouse is moved away and the details window is closed. The highlighting is removed when the page is refreshed, or the filter is reset.
Source	Displays the authentication source for the session. For example, *RADIUS* or web authentication (*WEBAUTH*).
Username	Displays the username or MAC address of the host.
Service	Displays the name of the service.
Login Status	Displays the status of the request, such as **Accept**, **Reject**, or **Timeout**.
Request Timestamp	Displays the date and time when the request process completed.

Editing the Access Tracker Page

You can modify the columns displayed in this table by clicking the **Edit** button at the top of the page to display the **Access Tracker Edit** Window. This page allows you to specify the server, filter, and date range displayed in the Access Tracker.

To add an additional data columns to the access tracker, select any entry in the **Available Columns** field then click the arrow to add that column to the table. To remove a column, select that entry from the **Selected Columns** field then click the arrow to move that entry back to the **Available Columns** field.

Figure 2. Editing the Access Tracker



The following list of available columns are not displayed by default, but can be added to the Access Tracker through the **Access Tracker Edit** window.

- Auth Method
- Auth Type
- Enforcement Profiles
- Error Code
- Host MAC Address
- Monitor Mode
- NAS IP address
- NAS Name
- Request ID
- Tips Role
- Server IP

> **NOTE**
>
> The **Alerts Present** column has been removed from the **Access Tracker**, **Data Filter** > **Rules configuration**, and **Syslog Export Filter** column selectors. However, if a backup is restored that includes Alerts Present in the selected columns for the Syslog Export Filter, the Alerts Present value will still display in syslog messages.

Using the Filter

The **Filter** field above the table provides a variety of predefined filter options you can use to refine and customize the information on the Access Tracker page. ClearPass supports Access Tracker filtering by RADIUS input and output attributes, authorization attributes, computed input and output attributes, and posture request and response attributes. The following lists describes the complete set of Access Tracker filtering options.

- Request ID
- Source
- Username
- NAS IP Address (This field supports IP address filtering in both IPv4 and IPv6 formats)
- NAS Name
- Service
- Login Status
- Error Code
- Host MAC Address
- Alerts
- Monitor Mode
- Auth Type
- Auth Method
- Roles
- Enforcement Profiles
- System Posture Token
- Audit Posture Token
- Authorization Attributes
- Computed Input Attributes
- Computed Output Attributes
- Posture Request Attributes
- Posture Response Attributes
- RADIUS Input Attributes
    - RADIUS:IETF

- RADIUS:Cisco
- RADIUS:Hewlett-Packard-Enterprise
- RADIUS:Alcatel-Lucent-Enterprise allows searching for a MAC address in any of the known formats:112233AABBCC11:22:33:aa:bb:cc11-22-33-AA-BB-CC1111-2222-33331111.2222.3333
- RADIUS:Microsoft
- RADIUS:Aruba

- RADIUS Output Attributes
  - RADIUS:IETF
  - RADIUS:Cisco
  - RADIUS:Hewlett-Packard-Enterprise
  - RADIUS:Alcatel-Lucent-Enterprise
  - RADIUS:Microsoft
  - RADIUS:Aruba

Filtering Access Tracker Information

About this task

To filter Access Tracker information:

Procedure

1. Click the **Filter** drop down menu and select a filter data type.
2. Click the **contains** drop down menu and specify **contains** or **equals.** To exlude Access Tracker filter results, consider selecting **not contains** or **not equals**. This type of negative filtering is only available on the client browser side.
3. Enter the filter text string in the entry field.
4. To filter the contents of the Access Tracker page, click **Go**.
   - To add an additional filter, click the **+** icon and define another filter.
   - To remove a filter, click the **-** icon by that filter.
   - To remove all filters and display all available information in the **Access Tracker** table, click **Clear Filter**.

Customizing the Access Tracker

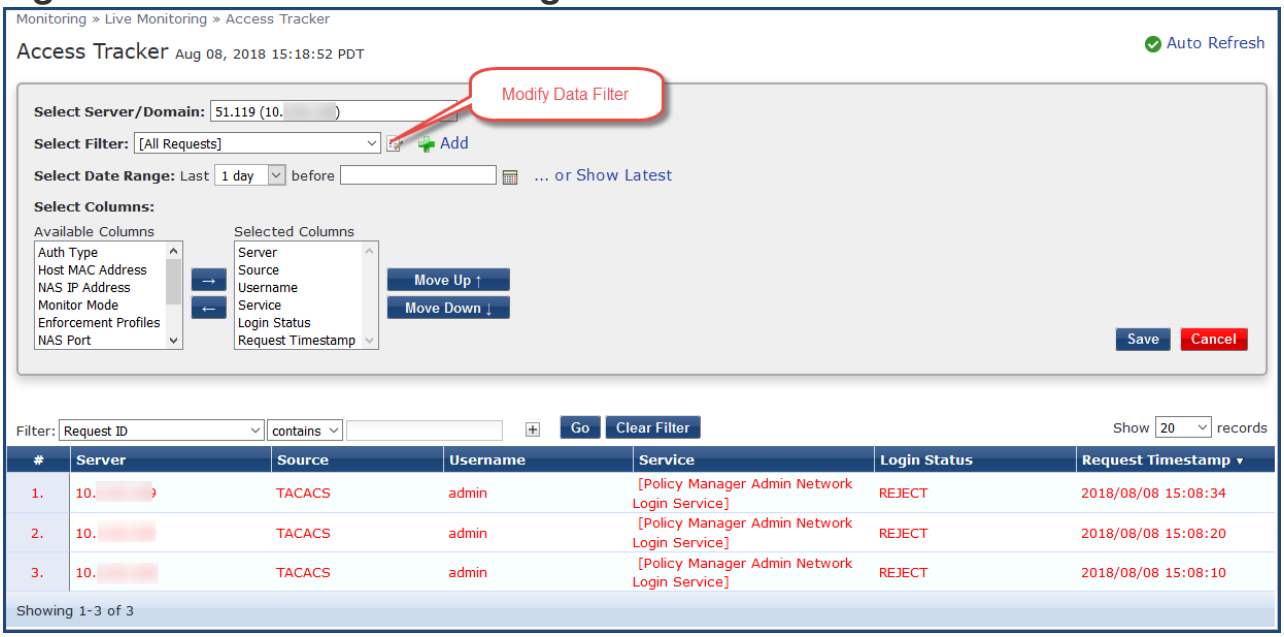
About this task

To customize the **Access Tracker** parameters:

Procedure

1. From the **Access Tracker** page, click the **Edit** button (in the upper-right corner of the page). The **Edit Access Tracker** page opens.

Figure 3. Edit Access Tracker Page



2. Modify the **Edit Access Tracker** page parameters as described in the following table, then click **Save**:

Table 2. Edit Access Tracker Page Parameters

Parameter	Action/Description
Select Server/ Domain	From the drop - down, select the server o r domain on the **Access Tracker** page. To display transactions from all nodes in the Policy Manager cluster, select the **de fault** option.
Select Filter	Select a filter category to filter the displ ayed data.

Parameter	Action/Description
	For a description of available filters, see Data Filters .
Modify Filter	To modify the current data filter, click the icon. For more information, see Data Filters .
Add Filter	To add a data filter, click the icon. For more information, see Data Filters .
Select Date Range	To select the start of the range of dates for which the **Access Tracker** table displays data, click the **Last** drop - down list. Available options are one to six days, or one week.
Select Date	To select a date, click the icon.
Show Latest	To set the date in the **before** field to the current date, click **Show Latest**.
Select Columns	This section displays the following two fields: • **Available Columns**: Displays the data column available to be displayed in an **Access Tracker** table. • **Selected Columns:** Displays the data columns currently selected for display. To move a column name from one field to another, select the column name and click the left or right arrows. To change the order in which the columns are displayed, click a column name in the **Selected Col**

Parameter	Action/Description
	umns field, then click the **Up** or **Down** bu ttons.

NOTE

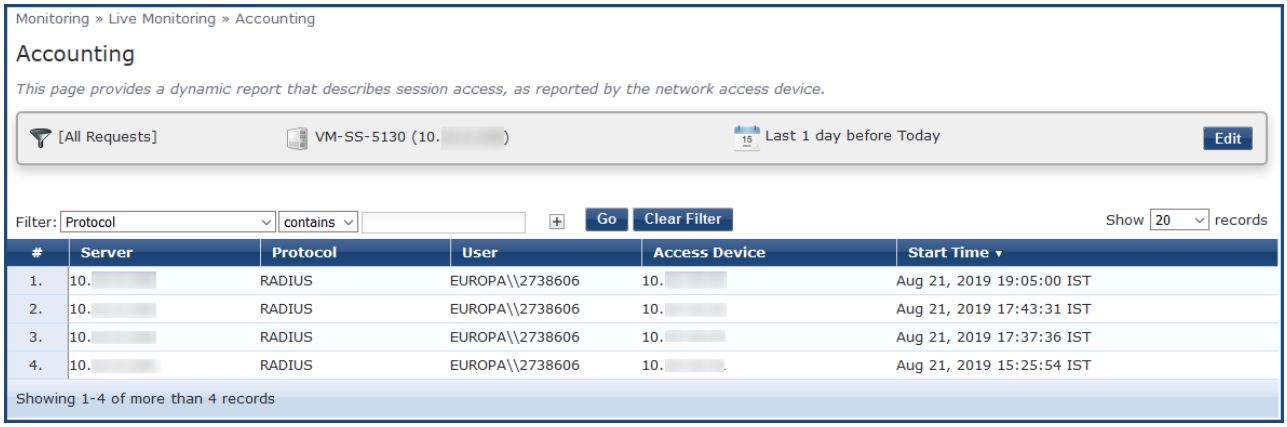
The **Alerts Present** column has been removed from the **Access Tracker**, **Data Filter > Rules configuration**, and **Syslog Export Filter** column selectors in the 6.11 release. However, if a backup is restored that includes Alerts Present in the selected columns for the Syslog Export Filter, the Alerts Present value will still display in syslog messages.

Live Monitoring: Accounting

- Modifying the Accounting Page Parameters
- RADIUS Accounting **Details > Summary**Tab
- RADIUS Accounting Record **Details > Auth**Sessions Tab
- RADIUS Accounting Record **Details > Utilization**Tab
- RADIUS Accounting Record **Details > Details**Tab
- TACACS+ Accounting Record **Details > Request**Tab
- TACACS+ Accounting Record **Details > Auth**Sessions Tab
- TACACS+ Accounting Record **Details > Details**Tab

The **Accounting** page provides a dynamic report that describes session access, as reported by the network access device by means of RADIUS or TACACS+ accounting records. To bring up the **Accounting** page, navigate to **Monitoring > Live Monitoring > Accounting**.

Figure 1. Live Monitoring > Accounting Page



NOTE

You can click any row in the Accounting page to drill down and display the corresponding **Accounting Record Details** page for the selected session. For details, see RADIUS Accounting **Details > Summary**Tab on page 1 and TACACS+ Accounting Details .

Modifying the Accounting Page Parameters

About this task

You can filter or modify the information displayed in this table by creating a filter, or selecting a different server, domain, or time range. To filter the data currently displayed in the **Accounting** page:

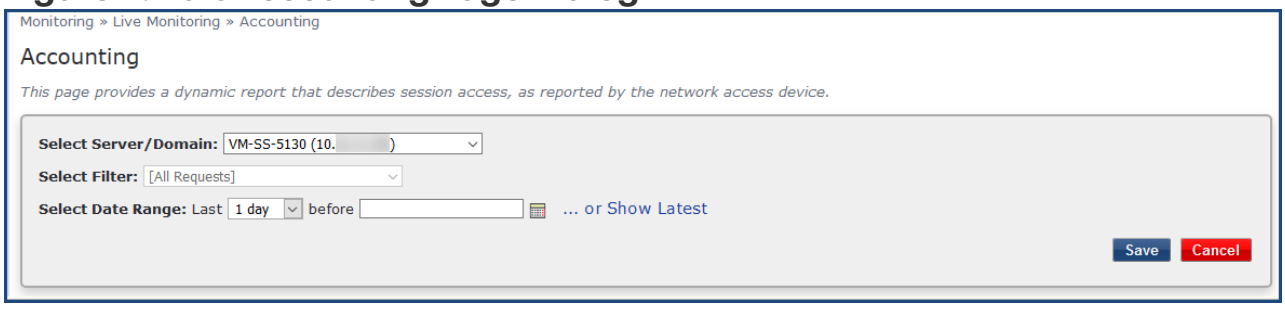
Procedure

1. Navigate to the **Monitoring > Live Monitoring > Accounting** page.
2. Click **Edit**.

   The **Edit Accounting Page** dialog opens.

   This page provides a dynamic report that describes session access as reported by the network access device.

Figure 2. Edit Accounting Page Dialog



3. Specify the **Edit Accounting Page** parameters as described in Table 2:

Table 1. Edit Accounting Page Parameters

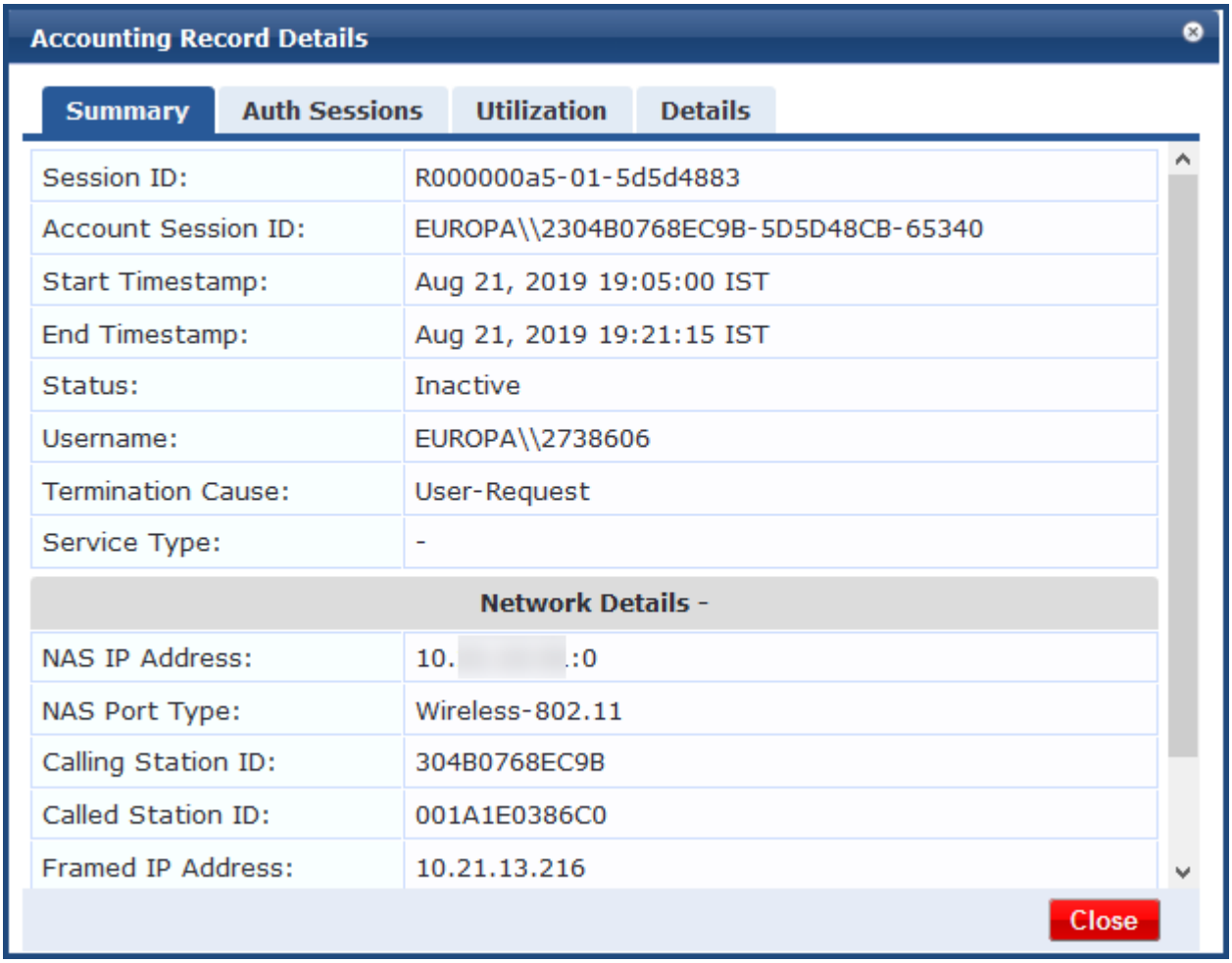
Parameter	Action/Description
Select Server/ Domain	Select the Policy Manager server for the dashboard data to be displayed.
Select Filter	To constrain the data display, select a filte r from the drop - down list. You can select one of the following filters : • Protocol • User • Access Device
Select Date Range	Select the number of days prior to the co nfigured date for which the accounting d ata is to be displayed. You can specify th e duration from one day to a week.
or Show Latest	To view the latest information, click the **.. .or Show Latest** link.

RADIUS Accounting Details > Summary Tab

To drill down and display the corresponding **Accounting Record Details** page for the session, click any row in the **Accounting** page. The **Accounting Record Details > Summary** tab shows a summary view of the transaction, including session IDs, timestamp, and network details such as the framed IPv6 addresses for the RADIUS protocol.

The following figure displays the **RADIUS Accounting Record Details > Summary** page:

Figure 3. RADIUS Accounting Record Details Summary Page



The following table describes the configuration parameters on the **RADIUS Accounting Record Details > Summary** page:

Table 2. RADIUS Accounting Record Details Summary Tab Parameters

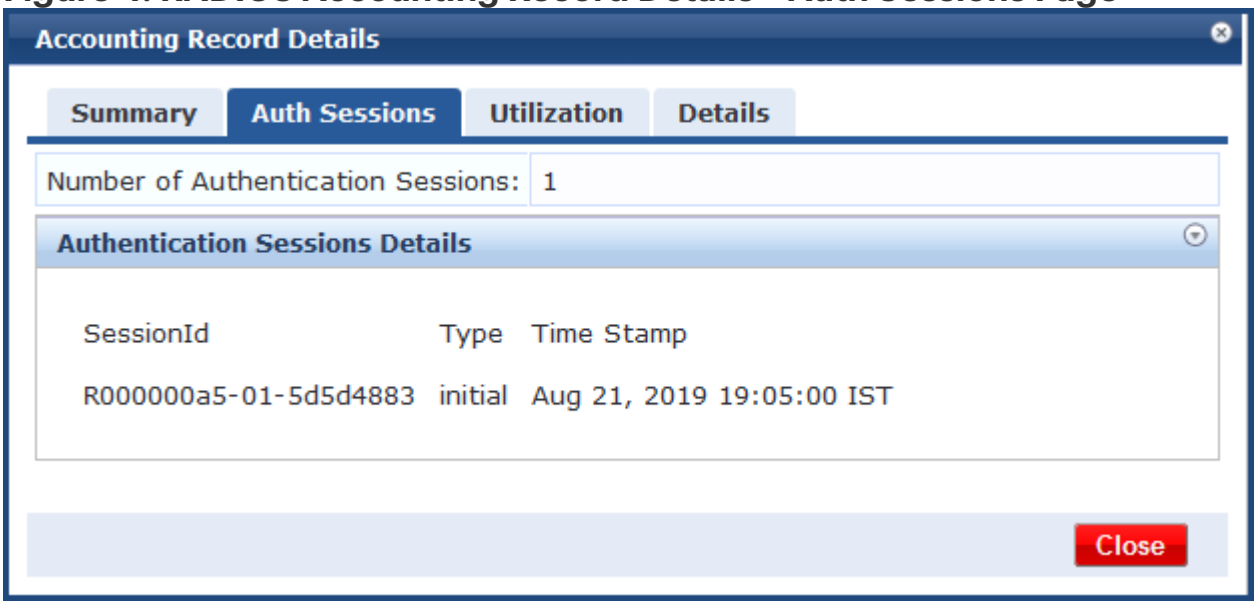
Parameter	Description
Session ID	Specifies the Policy Manager session identifier. You can correlate this record with a record in **Access Tracker**.
Account Session ID	Specifies a unique ID for this accounting record.
Start Timestamp End Timestamp	Shows the start time and end time of the session.
Status	Shows the current connection status of the session.

Parameter	Description
Username	Username associated with this record.
Termination Cause	Specifies the reason for termination of this session.
Service Type	Shows the value of the standard RADIUS at tribute service type.
Network Details	
NAS IP Address	Shows the IP address of the network devic e.
NAS Port Type	Shows the access methods. For example, E thernet, or 802.11 Wireless.
Calling Station ID	Specifies the MAC address of the client tha t is supported by Policy Manager. **NOTE** Policy Manager supports Calli ng Station ID MAC addresses i n the following formats: • XXXX - XXXX - XXXX • aa:bb:cc:dd:ee:ff • aa - bb - cc - dd - ee - ff • aaaa.bbbb.cccc • aabbccddeeff • aaaa - bbbb - cccc
Called Station ID	Shows the MAC address of the network d evice.
Framed IP Address	Shows the IP address of the client (if availa ble).
Framed IPv6 Address	Displays a list of the IPv6 addresses from t he Framed - IPv6 - Address vendor - specif ic attributes.
Account Auth	Specifies the type of authentication; for ex ample, RADIUS authentication.

RADIUS Accounting Record Details > Auth Sessions Tab

This section describes the parameters of the **Accounting Record Details > Auth Sessions** tab for the RADIUS protocol. The following figure displays the **Accounting Record Details > Auth Sessions** page:

Figure 4. RADIUS Accounting Record Details > Auth Sessions Page



The following table describes the **RADIUS Accounting Record Details > Auth Sessions** parameters:

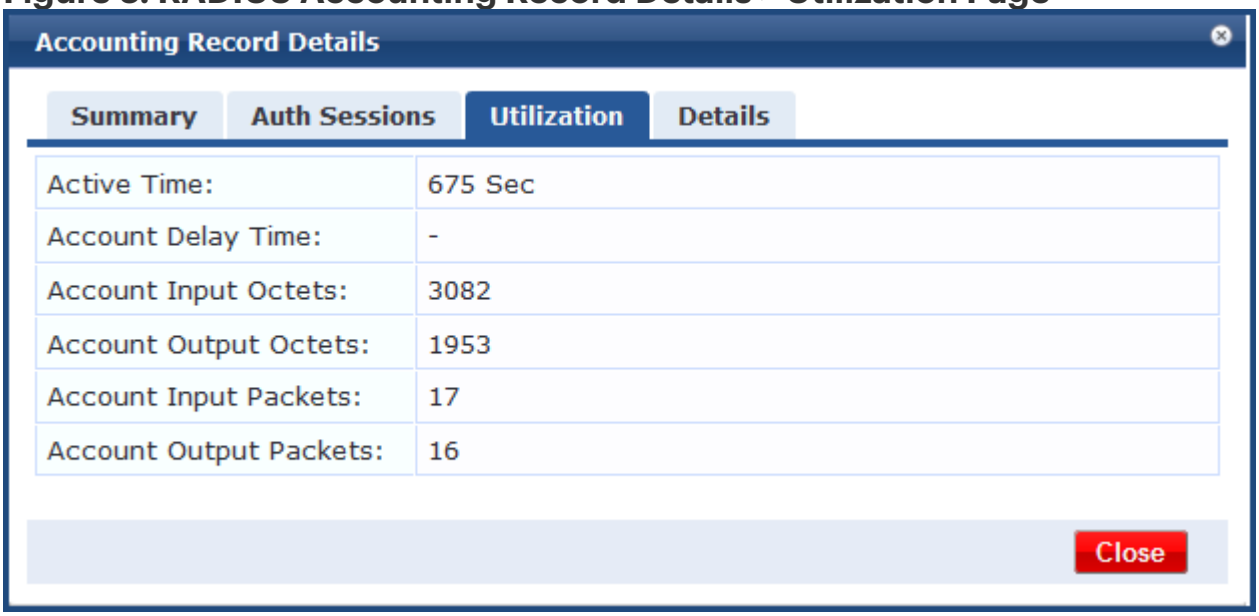
Table 3. RADIUS Accounting Record Details Auth Sessions Tab Parameters

Parameter	Description
Number of Authentication Sessions	Specifies the total number of authentications (always 1) and authorizations in this session.
Authentication Sessions Details	
Session ID	Displays the Policy Manager session ID.
Type	Specifies the type of authentication: Initial authentication or reauthentication.
Time Stamp	Specifies the time when the event occurred.

RADIUS Accounting Record Details > Utilization Tab

This section describes the parameters of the **Accounting Record Details > Utilization** tab for the RADIUS protocol. The following figure displays the **RADIUS Accounting Record Details > Utilization** page:

Figure 5. RADIUS Accounting Record Details > Utilization Page



The following table describes the configuration parameters on the **RADIUS Accounting Record Details - Utilization** tab:

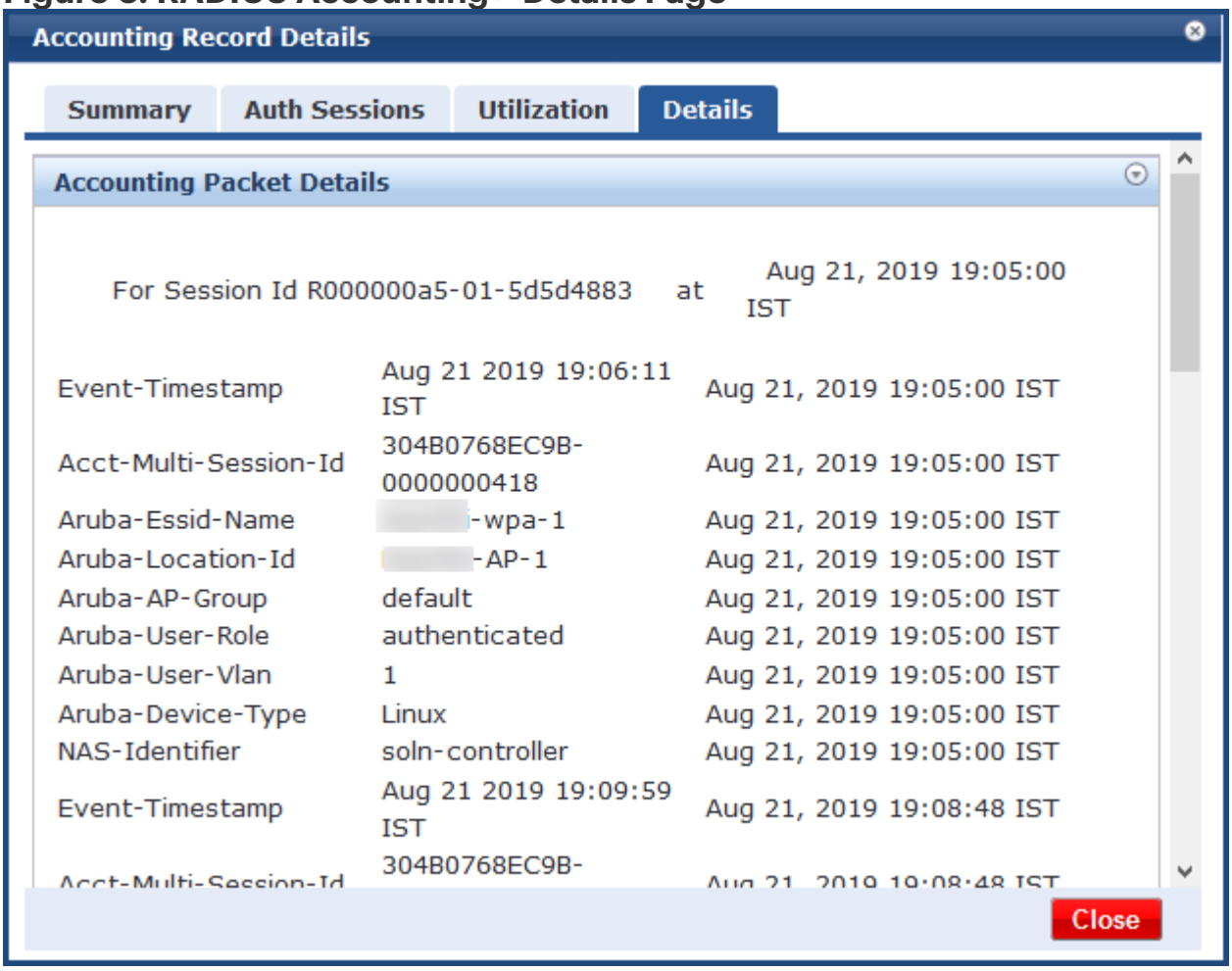
Table 4. RADIUS Accounting Record Details > Utilization Tab Parameters

Parameter	Description
Active Time	Displays the duration of the session that was active.
Account Delay Time	Displays how many seconds the network d evice has been trying to send this record fo r (subtract from record time stamp to dete rmine the time this record was actually gen erated by the device).
Account Input Octets	Specifies the quantity of octets sent to and received from the device port during the s ession.
Account Output Octets	
Account Input Packets	Specifies the packets sent and received fro m the device port during the session.
Account Output Packets	

RADIUS Accounting Record Details > Details Tab

This section describes the parameters of the **Accounting Record Details > Details** tab for the RADIUS protocol. The following figure displays the example of the **RADIUS Accounting Record Details > Details** page:

Figure 6. RADIUS Accounting > Details Page



The following table summarizes the configuration information provided on the **RADIUS Accounting Record Details > Details** page:

Table 5. RADIUS Accounting Record > Details Page Summary

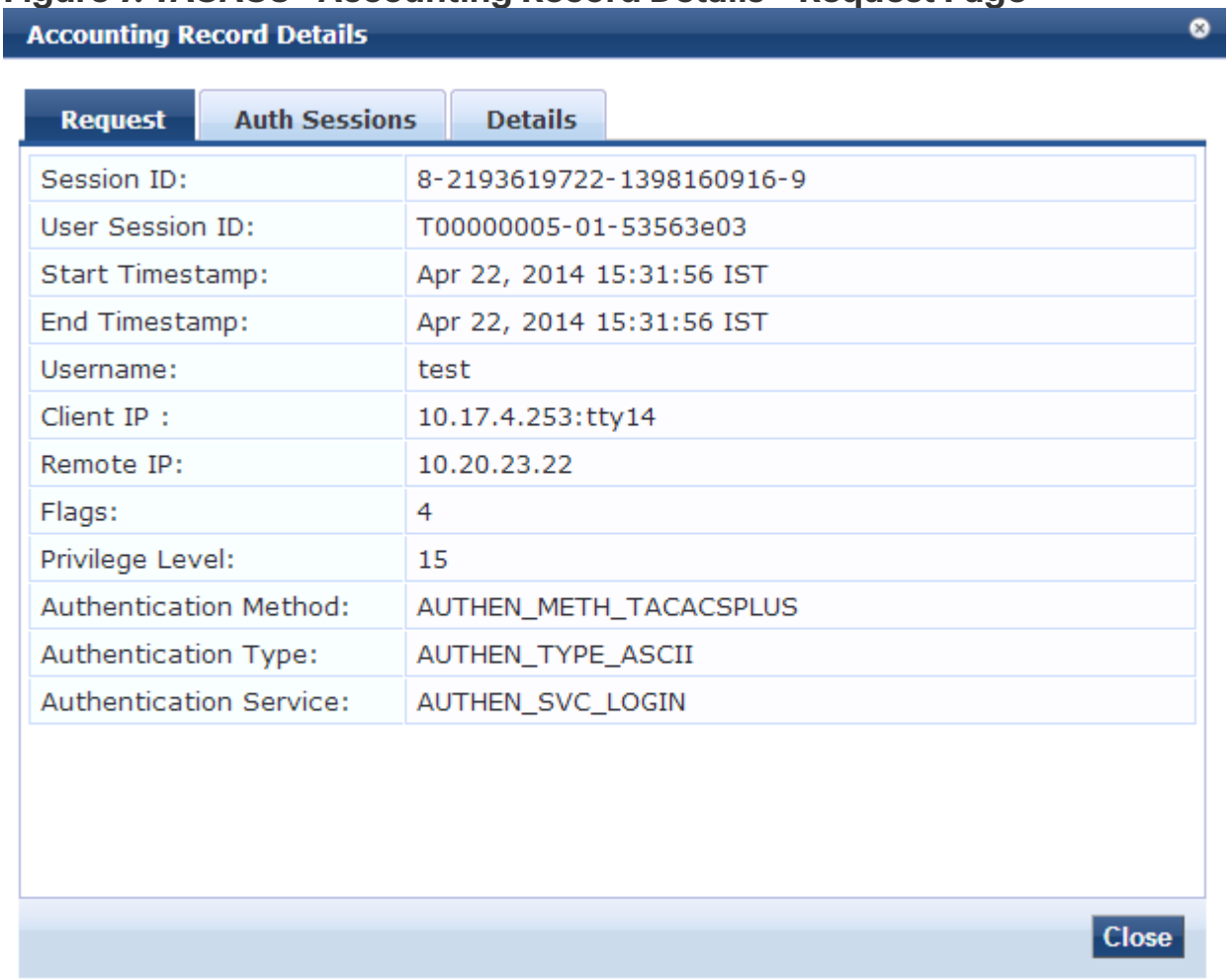
Parameter	Description
Accounting Packet Details	Shows the details of RADIUS attributes sen t and received from the network device du ring an initial authentication and subseque nt reauthentications. Each section in the **D etails** page corresponds to a session in Poli cy Manager.

TACACS+ Accounting Record Details > Request Tab

When you navigate to the **Monitoring > Live Monitoring > Accounting** page and select a TACACS+ Accounting record, the **Accounting Record Details** page opens to the **Request** page.

The following figure displays the **TACACS+ Accounting Record Details > Request** page:

Figure 7. TACACS+ Accounting Record Details > Request Page



The following table describes the configuration parameters on the **TACACS+ Accounting Record > Request** page:

Table 6. TACACS+ Accounting Record Request Page Parameters

Parameter	Description
Session ID	Specifies the **Session ID**, which is a unique ID associated with a request.
User Session ID	Specifies a session ID that correlates authe ntication, authorization, and accounting re cords.

Parameter	Description
Start and End Timestamp	Shows the start and end times of the sessi on.
Username	Shows the username associated with this record.
Client IP	Shows the IP address and tty (text terminal) of the device interface.
Remote IP	Shows the IP address from which the Admi nistrator is logged in.
Flags	Shows the identifier corresponding to start ing, stopping, or updating the accounting record.
Privilege Level	Specifies the privilege level of the Administ rator. The range is from **1** (lowest) to **15** (highest).
Authentication Method	Identifies the authentication method used for network access.
Authentication Type	Identifies the authentication type used for network access.
Authentication Service	Identifies the authentication service used for network access.

TACACS+ Accounting Record Details > Auth Sessions Tab

About this task

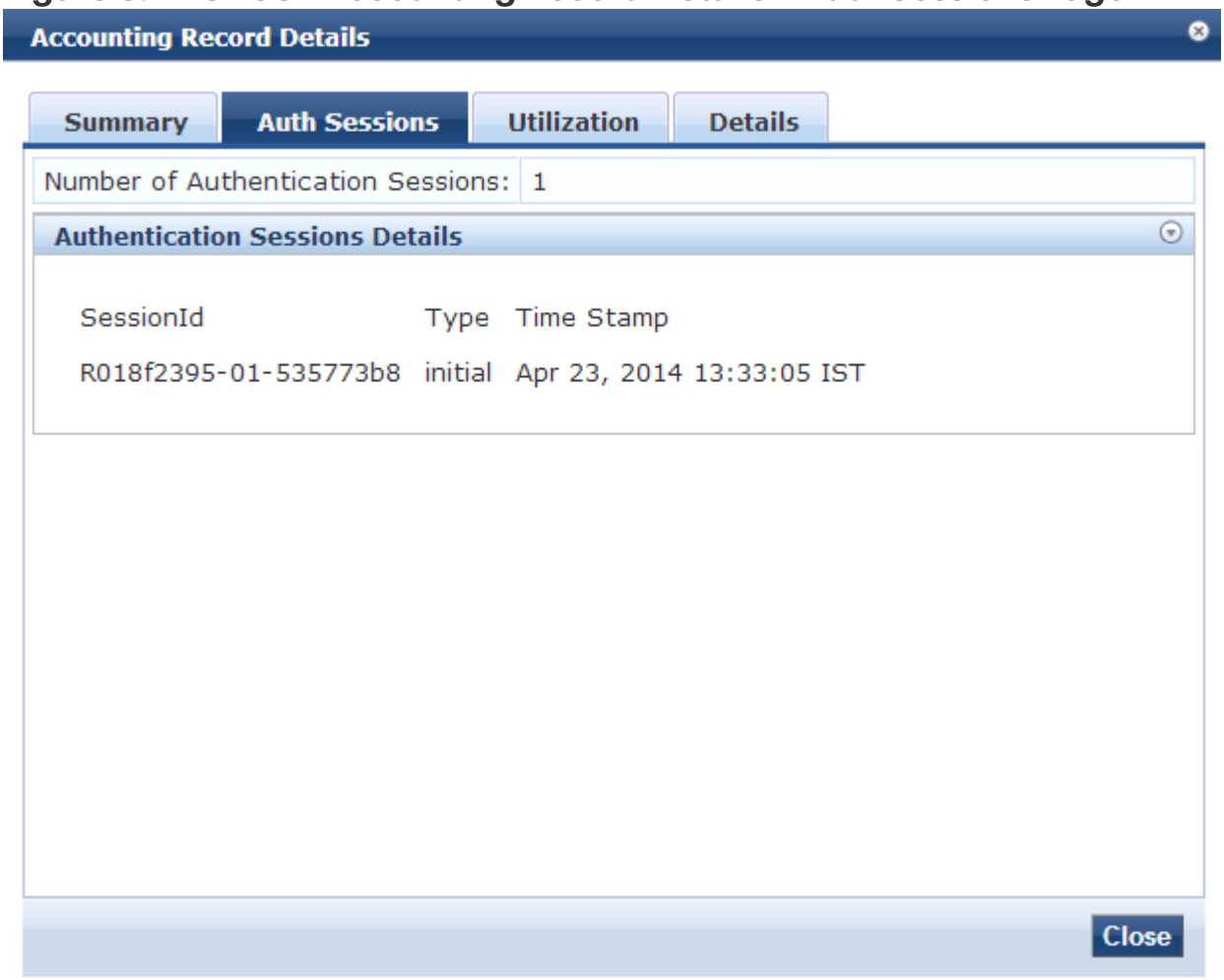
This section describes the parameters of the **Accounting Record Details > Auth Sessions** tab for the TACACS+ protocol.

You can click any row in the **Accounting** page to display the corresponding **Accounting Record Details** page for the session.

Procedure

1. Navigate to the **Monitoring > Live Monitoring > Accounting** page.
2. Select a TACACS+ Accounting record. The **Accounting Record Details** page opens.
3. To view the Authentication Sessions details, click the **Auth Sessions** tab.The following figure displays the **TACACS+ Accounting Record Details > Auth Sessions** page:

Figure 8. TACACS+ Accounting Record Details > Auth Sessions Page



The following table summarizes the information available on the **TACACS+ Accounting Record Details > Auth Sessions** page:

Table 7. TACACS+ Accounting Record Details > Authentication Sessions Page Parameters

Parameter	Description
Number of Authentication Sessions	Specifies the total number of authentications (always 1) and authorizations in this session.
Authentication Sessions Details	Denotes whether the request is an authentication or authorization request, and the time at which the request was sent for each request ID.

TACACS+ Accounting Record Details > Details Tab

About this task

This section describes the parameters of the **Accounting Record Details > Details** page for the TACACS+ protocol.

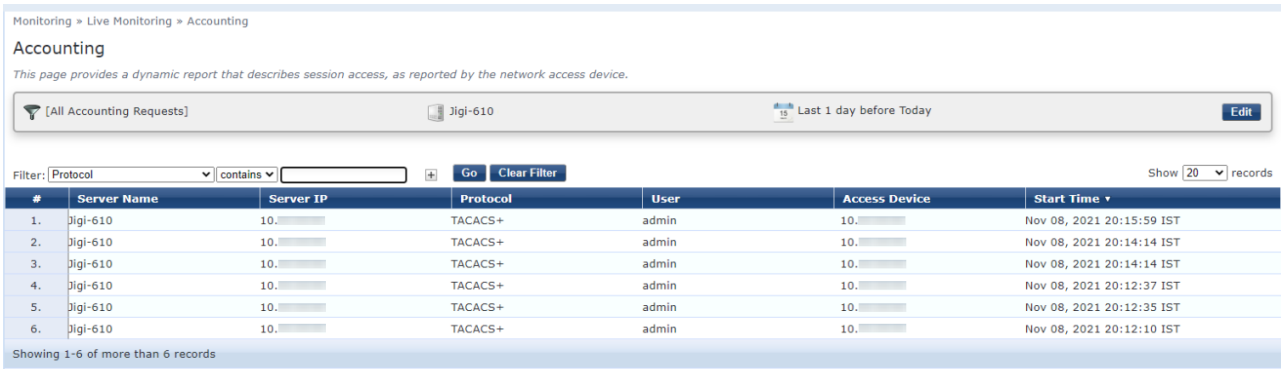
> **NOTE**
> With TACACS+ accounting, when ClearPass receives standalone accounting packets (without authentication and authorization) each command in the same login is logged as a new session and every command has an entry.

You can click any row in the **Accounting** page to display the corresponding **Accounting Record Details** page for the session.

Procedure

1. Navigate to **Monitoring** > **Live Monitoring** > **Accounting**.

   The **Live Monitoring > Accounting** page opens.

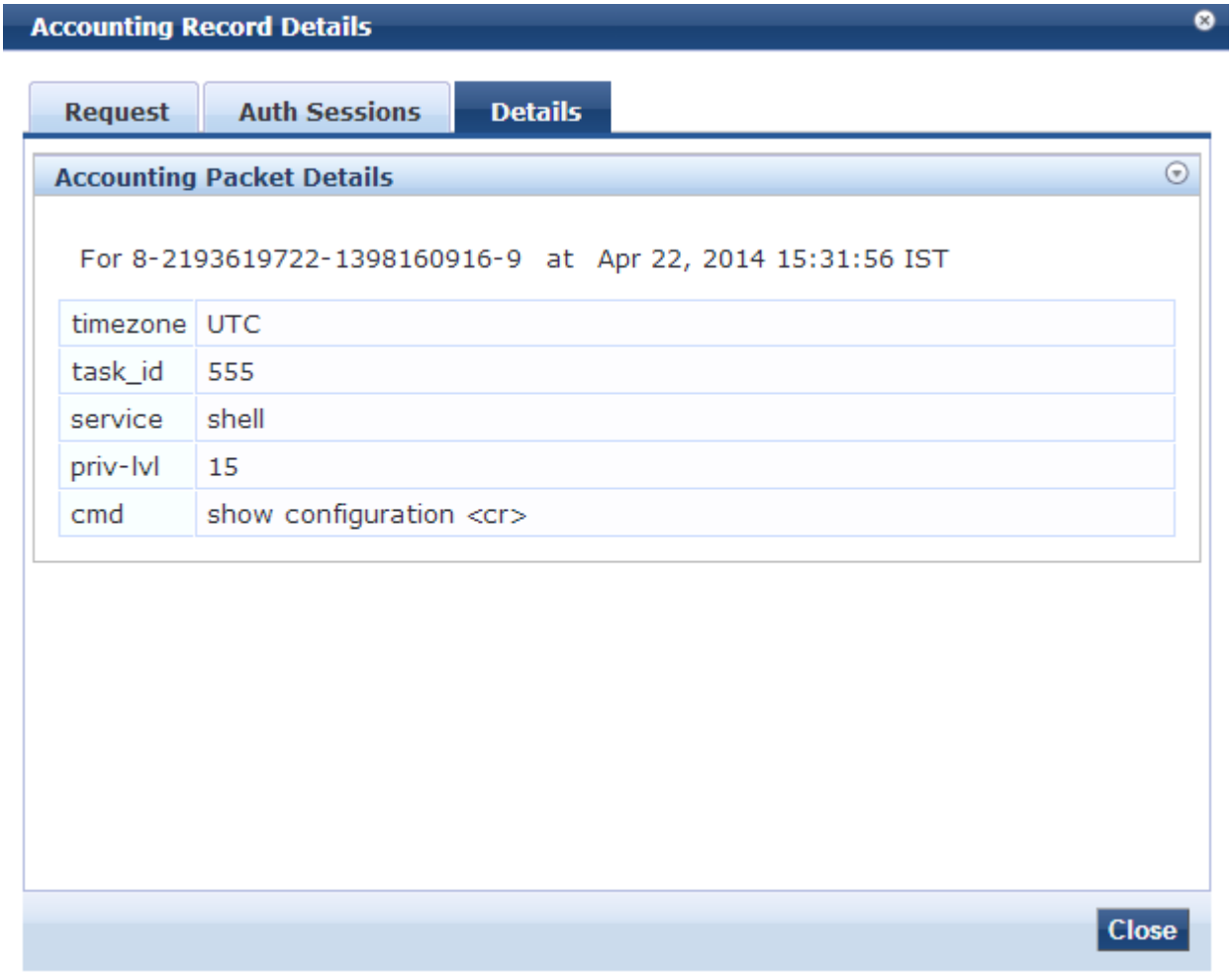


2. Select a TACACS+ Accounting record.

   The **Accounting Record Details** page opens.

3. To view the sessions details, click the **Details** tab.

Figure 9. TACACS+ Accounting Record Details > Details Page



The following table summarizes the configuration parameters provided on the **TACACS+ Accounting Record > Details** tab:

Table 8. TACACS+ Accounting Record > Details Page Parameters

Parameter	Description
Accounting Packet Details	Shows command typed (**cmd**), privilege l evel of the administrator executing the c ommand (**priv - lvl**) and service (**shell**) fo r each authorization request, as well as th e start time, task ID, and the time zone.

Live Monitoring: OnGuard Activity

About this task

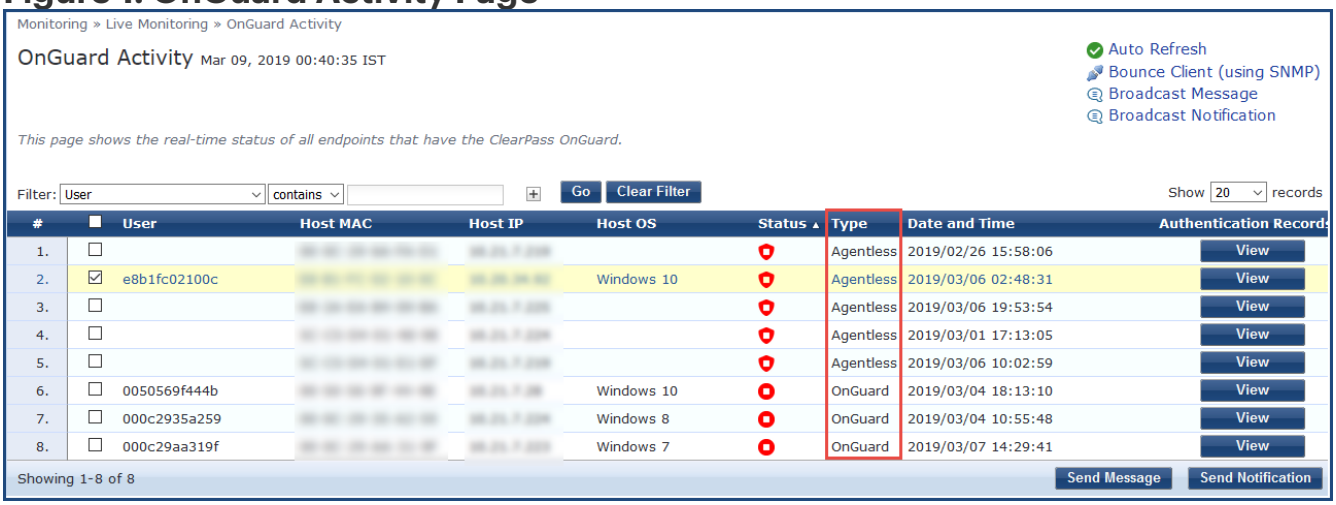
The **OnGuard Activity** page shows the real-time status of all endpoints that have Policy Manager OnGuard. This page also presents configuration tools to bounce an endpoint, restart a session, and send unicast or broadcast messages to all endpoints running the OnGuard agent.

To access the **OnGuard Activity** page:

Procedure

Navigate to **Monitoring > Live Monitoring > OnGuard Activity**. The **OnGuard Activity** page opens:

Figure 1. OnGuard Activity Page



Subtopics

Filtering by MAC Address
About OnGuard Agent Type
OnGuard Activity Action Buttons
OnGuard Activity Status Icons
Agentless OnGuard Log
Viewing Authentication Records
Broadcasting Notifications to OnGuard Agents
Sending a Message to Selected Endpoints
Bouncing a Client Using SNMP
Bouncing an Agent Not Using SNMP
Defining Categories of Updates for a Windows Update Agent in a Patch Management Health Class
Bouncing the Network Interface After Waking Up From Sleep Mode

Filtering by MAC Address

Policy Manager allows you to search for a MAC address in any of the following formats:

- 112233AABBCC
- 11:22:33:aa:bb:cc
- 11-22-33-AA-BB-CC
- 1111-2222-3333
- 1111.2222.3333

About OnGuard Agent Type

The **Type** column in the **OnGuard Activity** page indicates *Agent Type*: Persistent Agent, Agentless OnGuard, etc. This column shows the Agent Type of the last request. For details on the **Host:AgentType** attribute, see Table 2: Service Rule > Web-Based Health Check Only Host Attributes.

OnGuard Activity Action Buttons

The following table describes the action buttons on the **OnGuard Activity** page:

Table 1. OnGuard Activity Action Buttons

Action Button	Description
Send Message	Click **Send Message** to send a message to the selected endpoints. For details, see Sending a Message to Selected Endpoints.
Send Notification	Click **Send Notification** to apply the *Bounce* or *Restart Session* notification options to one or more endpoints that you selected in the OnGuard Activity list. For details, see Broadcasting Notifications to OnGuard Agents.

OnGuard Activity Status Icons

The OnGuard Activity **Status** column shows status of OnGuard Agents: *Persistent*, *Agentless*, and *Native Dissolvable Agent*.

Agentless OnGuard Launch Status

When Agentless OnGuard is launched, it shows the status as one of the following:

- **Success**: Agentless OnGuard was launched successfully.
- **Failure**: Policy Manager server failed to launch Agentless OnGuard.

Agent Online/Offline Status

- **Online**: Persistent or Agentless OnGuard is connected and is sending Keep-alive messages.
- **Offline**: Persistent or Agentless OnGuard is not connected to the network.

> **NOTE**
>
> The status for Native Dissolvable Agent is always *Offline* because the Native Dissolvable Agent shuts down after performing health checks and does not send Keep-alive messages to the Policy Manager server.

The following table describes the status icons that are displayed in the **Status** column of the **OnGuard Activity** page:

Table 1. OnGuard Activity Status Icons

OnGuard Icon	Status	Description
	Failure	Failed to launch Agentless OnGuard on this client.
	Success	Successfully launched Age ntless OnGuard on this clie nt.
	Online	OnGuard Agent (Persistent or Agentless) is currently o

OnGuard Icon	Status	Description
		nline or connected to a Pol icy Manager server.
	Offline	OnGuard Agent (Persistent or Agentless) is currently of fline or not connected to a Policy Manager server.

Agentless OnGuard Log

About this task

The Agentless OnGuard Log provides a log of all Agentless OnGuard activity for an endpoint.

To access the Agentless OnGuard Log:

Procedure

1. From the **OnGuard Activity** page, click the row for the desired endpoint.

   The **Agent and Endpoint Details** page opens.

2. Select the **Agentless OnGuard Log** tab.

Figure 1. Agentless OnGuard Log

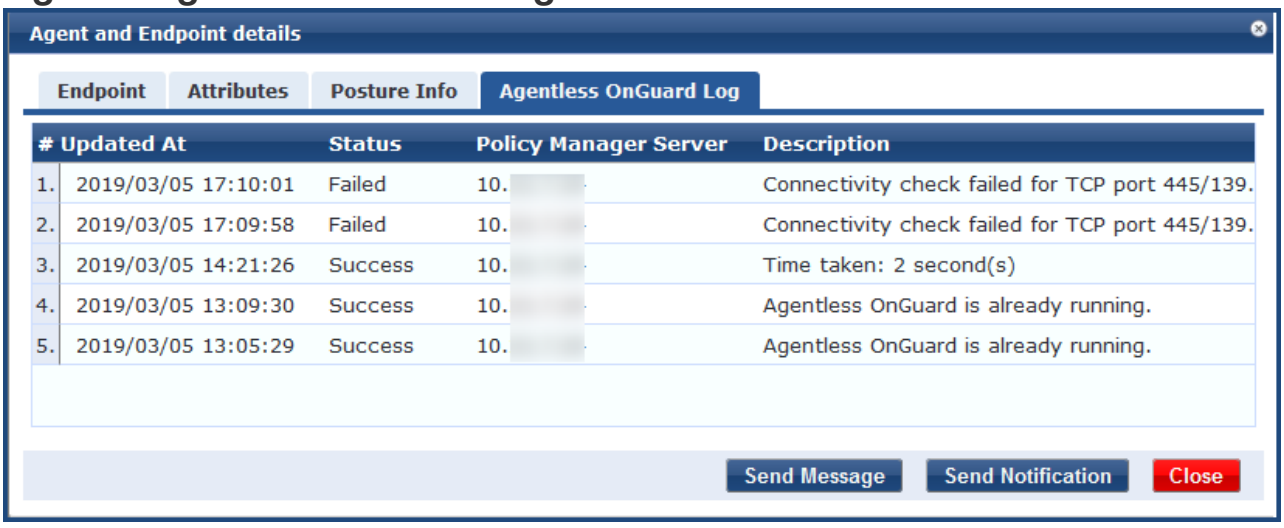


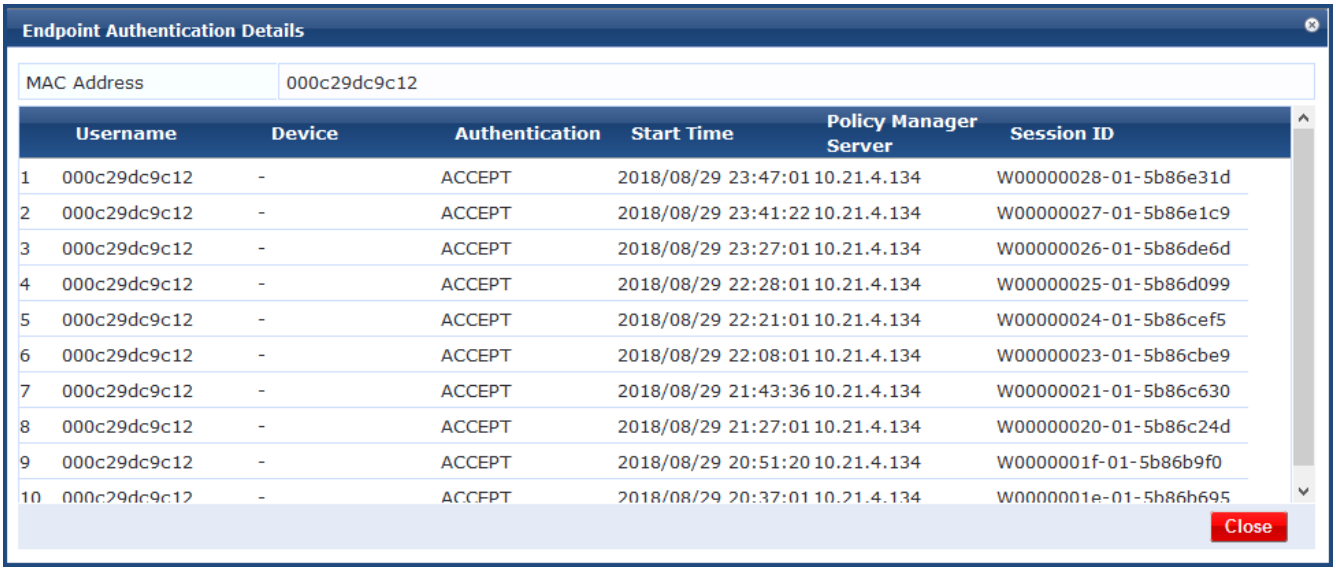
Table 1. Agentless OnGuard Log Description

Parameter	Action/Description
Updated At	Indicates the time when a request to launch Agentless OnGuard was processed by the Policy Manager server.
Status	Indicates whether launching Agentless OnGuard was a success or it failed.
Policy Manager Server	IP address of Policy Manager server that processed the request to launch Agentless OnGuard.
Description	Shows detailed output of launching Agentless OnGuard. For a **Failure** status, it describes why the Policy Manager server was not able to launch Agentless OnGuard.

Viewing Authentication Records

From the **OnGuard Activity** page, you can view the authentication records for the online devices that are listed.To view the authentication records for a device that is online, click the **View** button on the pertinent row. The **Endpoint Authentication Details** page opens.

Figure 1. Viewing an Endpoint's Authentication Records



Broadcasting Notifications to OnGuard Agents

This option allows administrators to send bounce or restart notifications to every connected OnGuard Agent. A custom message can be included with the notifications.

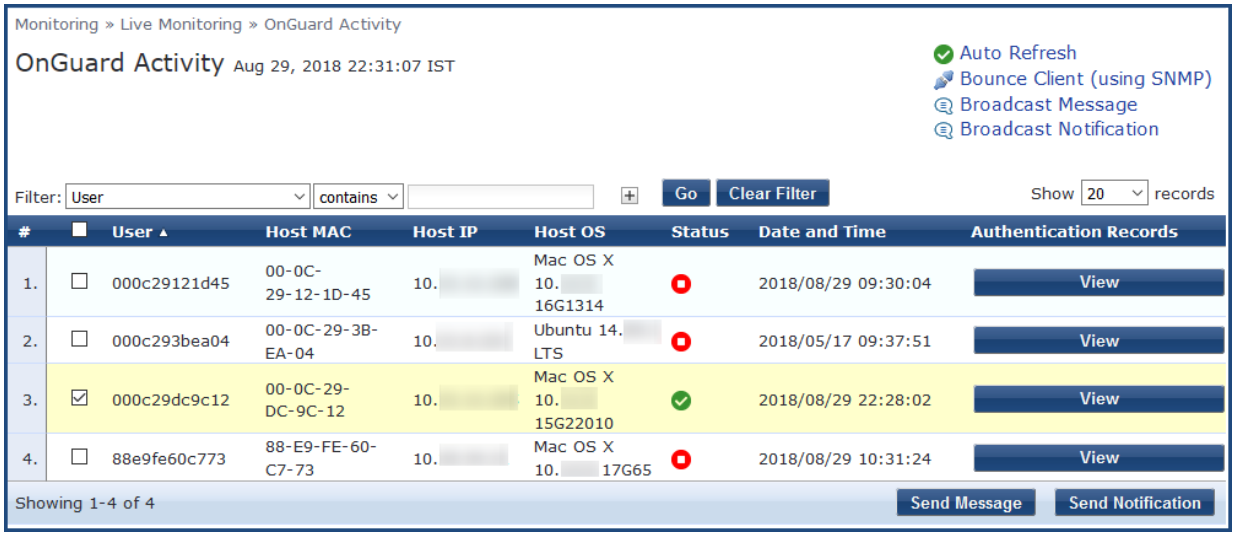
Sending **broadcast messages** involves a different behavior than sending **broadcast notifications**.

- Use broadcast messages to send a message to selected endpoints that are currently visible in the OnGuard Activity table.
- Use broadcast notifications to send a notification to all connected/online Agents, regardless of the list of endpoints currently visible or selected in the OnGuard Activity table. **Broadcasting notifications to a large number of Agents (> 500) may place a heavy load on the system and may cause a network outage.**For more information about broadcast notifications, see Broadcasting Notifications to OnGuard Agents.

  To send bounce or restart notifications to selected OnGuard Agents:

  1. Navigate to **Monitoring > Live Monitoring > OnGuard Activity**.The **OnGuard Activity** page opens.

     Figure 1. OnGuard Activity Page

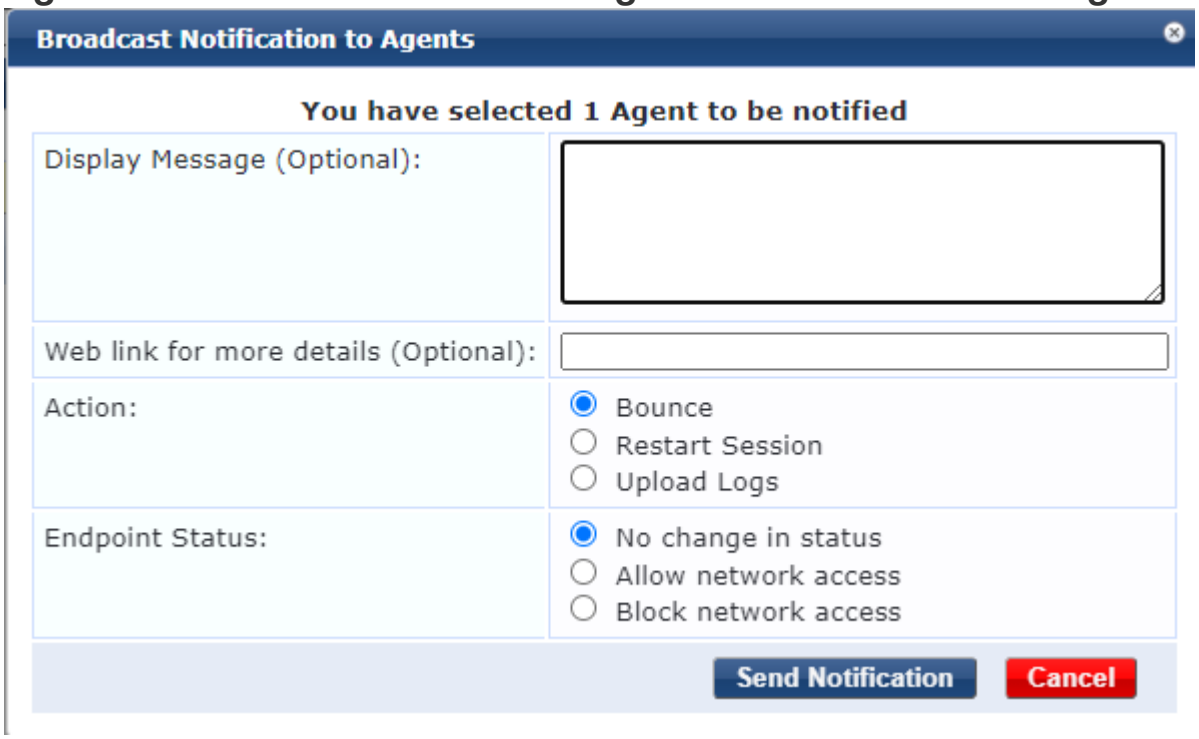


2. Select the endpoint(s) to which you want to broadcast notifications.

> **NOTE**
>
> In order for a message or notification to be sent, the endpoint status must be *Online* (green check icon).

3. Click the **Broadcast Notification** link. The **Broadcast Notification to Agents** dialog opens.

Figure 2. Broadcast Notification to Agents > Bounce Action Dialog



4. Specify the *Broadcast Notification to Agents* parameters as described in the following table:

Table 1. Send Notification to Agents Parameters

Parameter	Action/Description
Display Message (Optional)	Enter the text of the message you want to send to the selected OnGuard Agent s.
Web link for more details (optional)	Optionally, enter a URL to be included with the message specified for **Display Message**.
Action	Select one of the following actions: ◦ *Bounce*: Select this option to have t he OnGuard Agent bounce the net work interface. ◦ *Restart Session*: Select this option t o have the OnGuard Agent restart th e session in order to perform authen tication and health checks. ◦ *Upload Logs*: Select this option to u pload logs from an agent to a prope rly configured log collection server.
Endpoint Status	When you select **Bounce**, specify one o f the following options: ◦ *No change in status*: No change is made to the status of the endpoint. The existing status of **Known**, **Unkn own**, or **Disabled** continues to be ap plied. Access control is granted or d enied based on the existing status o f an endpoint. ◦ *Allow network access*: Allow netwo rk access by white - listing this end point. Clicking **Allow network acces**

Parameter	Action/Description
	s sets the status of the endpoint as **Known**. **NOTE** **Known**. ◦ *Block network access*: Block networ k access by adding this endpoint to a blocked list. Clicking **Block netwo rk access** sets the status of the end point to **Disabled**. **NOTE** **Disabled**.

5. Click **Send Notification**. The notification, along with specified endpoint status, is sent to selected endpoints.

Sending a Message to Selected Endpoints

Sending broadcast messagesis a separate activity from sending broadcast notifications.

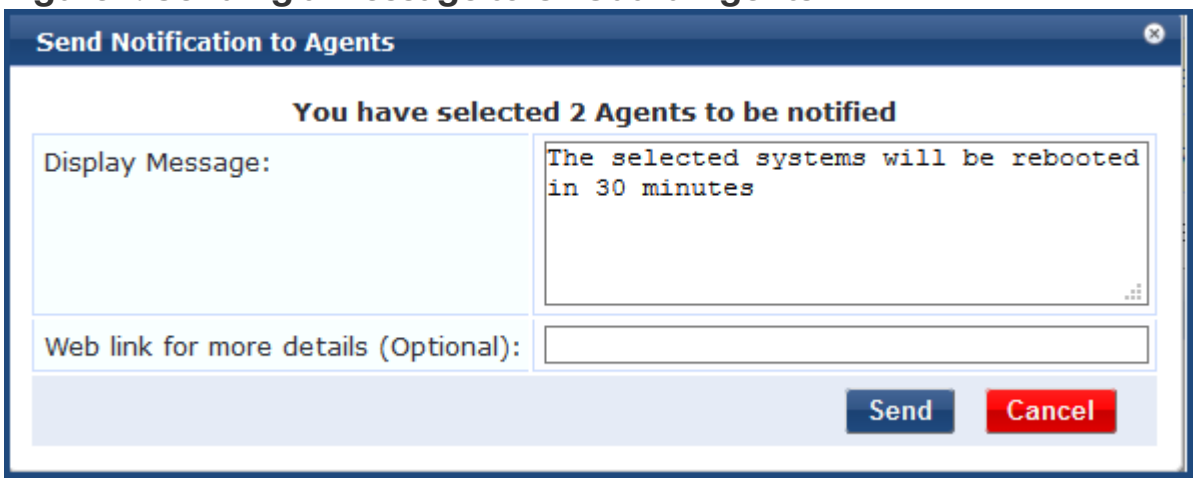
- Use broadcast messages to send a message to selected endpoints that are currently visible in the OnGuard Activity table.
- Use broadcast notifications to send a notification to all connected/online Agents, regardless of the list of endpoints that are currently visible or selected in the OnGuard Activity table. **Broadcasting notifications to a large number of Agents (> 500) may place a heavy load on the system and may cause a network outage.**For more information on broadcast notifications, see Broadcasting Notifications to OnGuard Agents

  To send a message to selected endpoints:

  1. Navigate to Navigate to **Monitoring > Live Monitoring > OnGuard Activity**. The **OnGuard Activity** page opens.
  2. Select one or more devices listed on the **OnGuard Activity** page.

3. Click the **Send Message** button. The **Send Notification to Agents** dialog opens.

Figure 1. Sending a Message to OnGuard Agents



4. Specify the *Send Notification to Agents* parameters as described in the following table:

Table 1. Send Notification to Agents Parameters

Parameter	Action/Description
Display Message	Enter the text of the message you want to send to the selected OnGuard Agent s.
Web link for more details (optional)	Optionally, enter a URL to be included with the text specified for **Display Mess age**.

5. Click **Send**. The message is sent to the selected devices.

Bouncing a Client Using SNMP

The **Bounce Client (using SNMP)** dialog is used to initiate a bounce operation using SNMP with wired Ethernet switches.

> **NOTE**
> Endpoint bounce only works with endpoints that run the OnGuard persistent agent.

Requirements

To successfully bounce a client using SNMP, the following conditions are required:

- The network device must be added to Policy Manager and SNMP read and write parameters must be configured.
  - For information about adding a network device to ClearPass Policy Manager, see Adding a Network Device .
  - For details about configuring SNMP settings, see SNMP Read Settings Parameters and SNMP Write Settings Parameters .
- SNMP traps (link up and/or MAC notification) must be enabled on the switch port (see SNMP Traps and OIDs and SNMP Trap Details).
- The DHCP snooper service on a Policy Manager server must receive DHCP packets from the endpoint to specify the IP address of the endpoint to bounce.
  - For information about configuring the **DhcpSnooper** service, see Network Services Options .
  - For information about configuring the IP helper address, see DHCP Collector .

Bouncing a Client

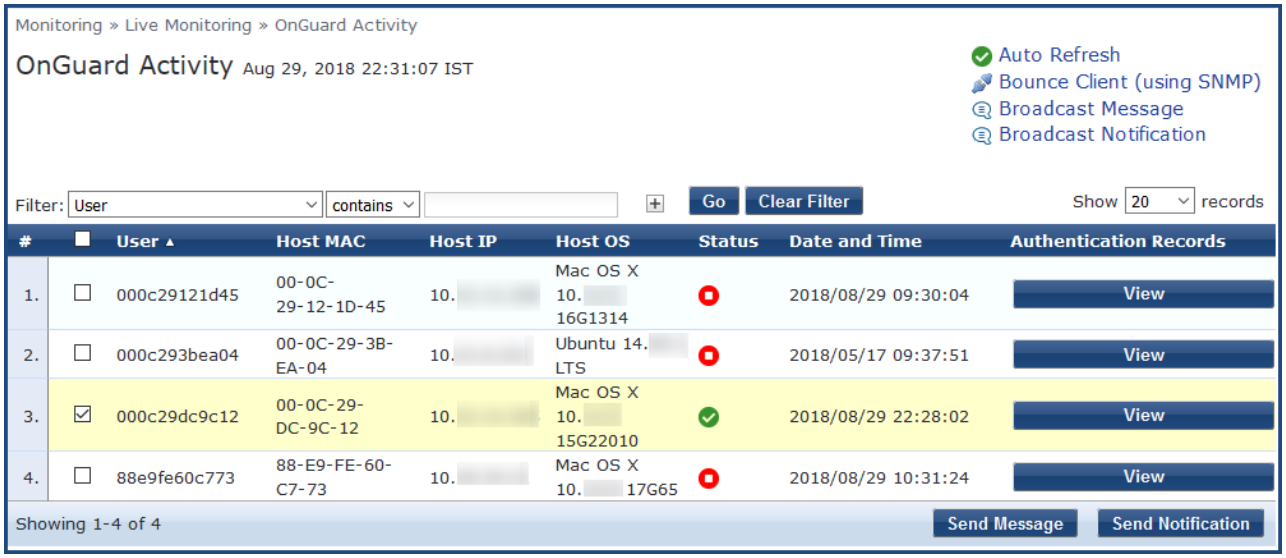
About this task

To bounce a client using SNMP:

Procedure

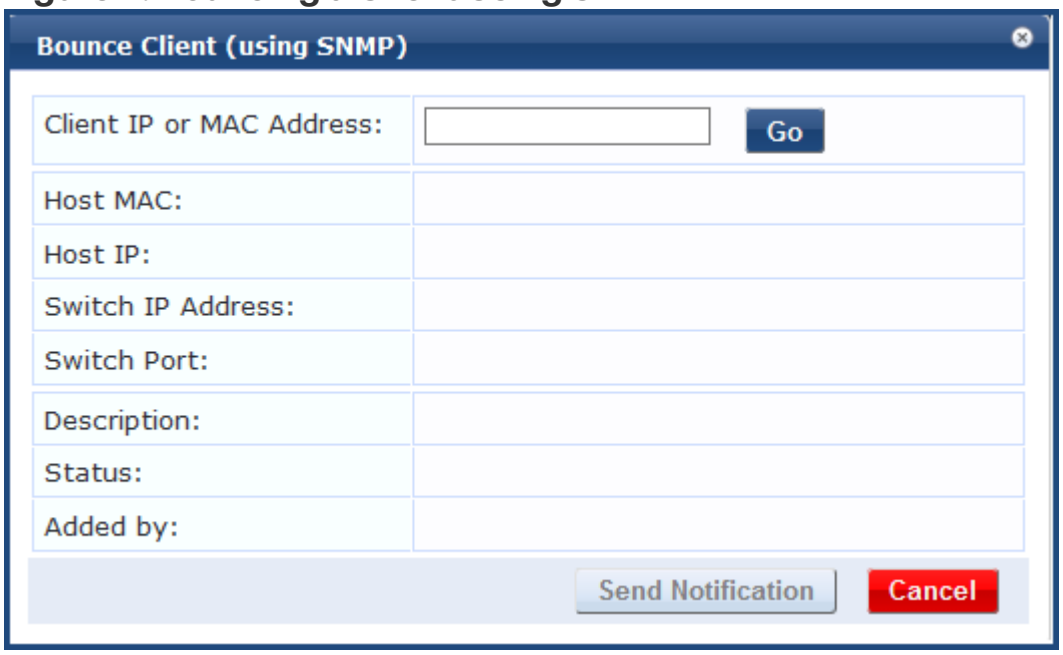
1. Navigate to **Monitoring > Live Monitoring > OnGuard Activity**. The **OnGuard Activity** page opens.

Figure 1. OnGuard Activity Page



2. Click the **Bounce Client (using SNMP)** link. The **Bounce Client (using SNMP)** dialog opens.

Figure 2. Bouncing a Client Using SNMP



3. Enter the client IP address or MAC address of the client to be bounced, then click **Go**. The rest of the fields in the *Bounce Client (Using SNMP)* dialog are automatically populated (*Host MAC*, *Host IP*, etc.) and the **Send Notification** button is enabled.

4. Click **Go**, then click **Send Notification**. The OnGuard Agent bounces the network interface of the specified client device.

Bouncing an Agent Not Using SNMP

This section describes how to initiate a bounce on an endpoint's managed interface.

> **NOTE**
> Endpoint bounce works only with endpoints that run the OnGuard persistent agent.

Initiating a bounce on the managed interface on the endpoint results in creating tags for the specified endpoint in the **Endpoints** page (navigate to **Configuration** > **Identity** > **Endpoints**).

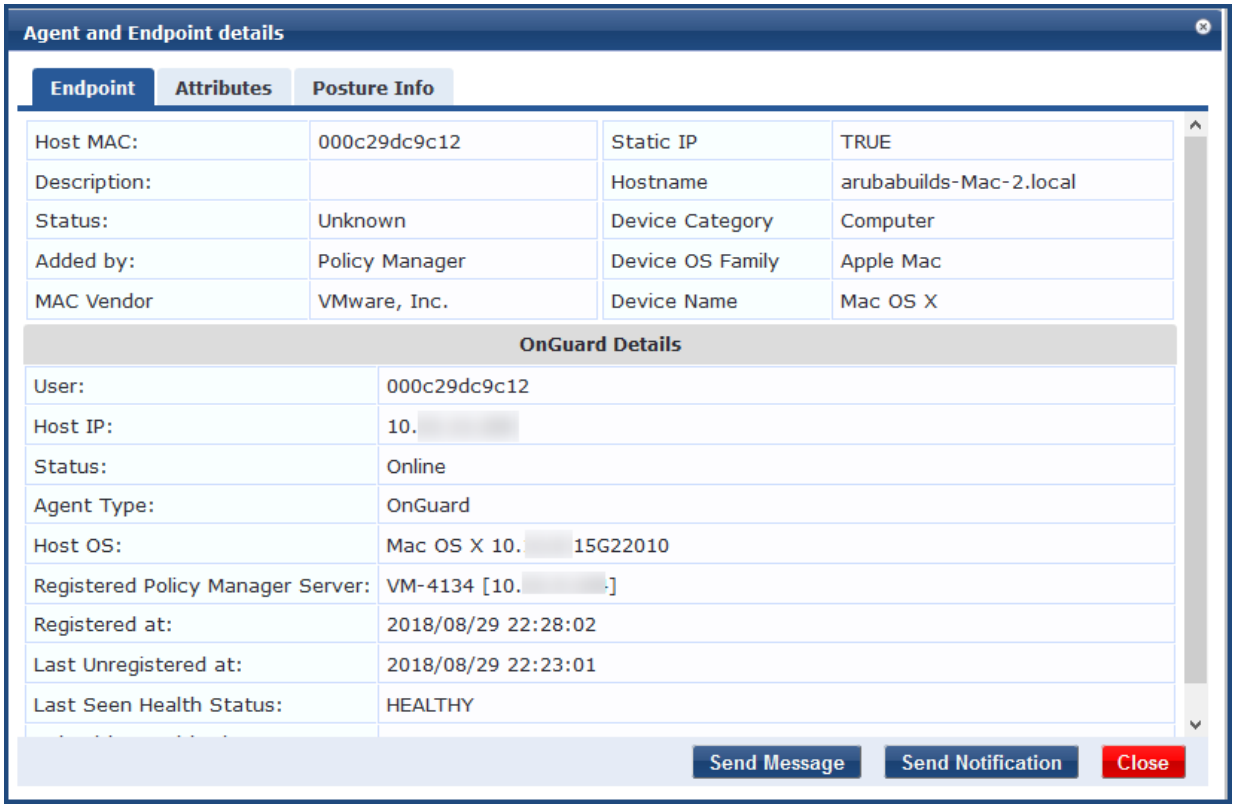
One or more of the following tags are created:

- Disabled by
- Disabled Reason
- Enabled by
- Enabled Reason
- Info URL

  To bounce an agent:

  1. Navigate to **Monitoring > Live Monitoring > OnGuard Activity**. The **OnGuard Activity** page opens.
  2. Click a device listed on the **OnGuard Activity** page. The **Agent and Endpoint Details** page opens.

     Figure 1. Agent and Endpoint Details



The following table describes the configuration parameters on the **Agent and Endpoint Details** page:

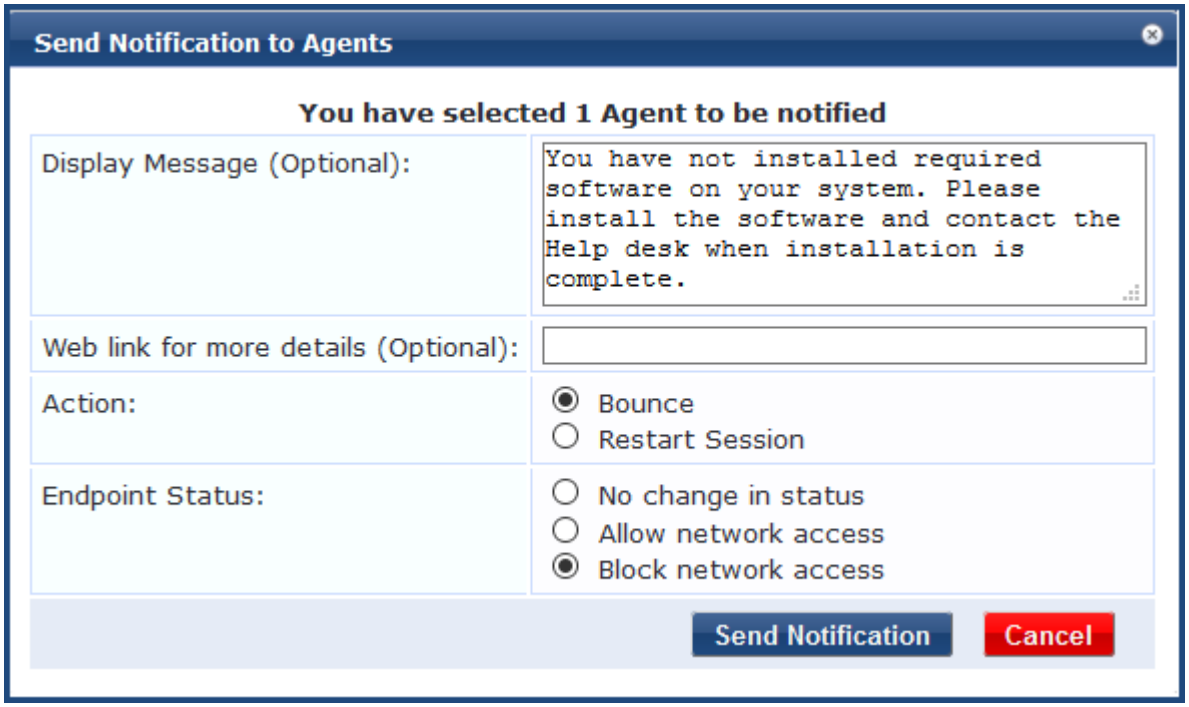
Table 1. Agent and Endpoint Details Parameters

Parameter	Description
Host MAC	Displays the MAC address of the user.
Description	Optional description of the endpoint.
Status	Displays the status of the endpoint.
Added by	Displays the server name.
MAC Vendor	Vendor name and OS of the endpoint d evice.
OnGuard Details	
User	Displays the MAC address of the host.
Host IP	Displays the IP address of the host.

Parameter	Description
Status	Shows the status of the agent as either *Online* or *Offline*.
Agent Type	Specifies the agent as an OnGuard age nt.
Host OS	Displays the operating system that runs on the endpoint.
Registered Policy Manager Server	Displays the name and IP address of the Policy Manager server.
Registered at	Displays the date and time at which th e Policy Manager installation was regist ered.
Last Unregistered at	Displays the date and time at which th e Policy Manager installation was last u nregistered.
Last Seen Health Status	Displays the health status of the endpo int. For example, QUARANTINED or HEA LTHY.
Unhealthy Health Classes	Displays the health classes that are unh ealthy.

3. Click **Send Notification**.The **Send Notification to Agents** dialog opens.

Figure 2. Sending Notification to OnGuard Agents



The following table describes the configuration parameters on the **Send Notification to Agents** dialog:

Table 2. Send Notification to Agents Parameters

Parameter	Action/Description
Display Message (Optional)	An optional message to display on the endpoint using the OnGuard interface.
Web link for more details (Optional)	An optional clickable URL that is displ ayed along with the Display Message.
Action	Select one of the following actions: ◦ *Bounce*: Select this option to have t he OnGuard Agent bounce the net work interface. ◦ *Restart Session*: Select this option t o have the OnGuard Agent restart th e session in order to perform authen tication and health checks. ◦ *Upload Logs*: Select this option to u pload logs from an agent to a prope rly configured log collection server.

Parameter	Action/Description
Endpoint Status	When you select **Bounce**, you must spe cify one of the following options: ◦ **No change in status**: No change is made to the status of the endpoint. The existing status of **Known**, **Unkn own**, or **Disabled** continues to be ap plied. Access control is granted or d enied based on the existing status o f an endpoint. ◦ **Allow network access**: Allow netwo rk access by white - listing this end point. Clicking **Allow network acces s** sets the status of the endpoint to **Known**. **NOTE** You must configure Enforc ement Policy Rules to all ow access to the endpoin ts with the status **Known**. For details, see Configurin g Enforcement Policies. ◦ **Block network access**: Block netwo rk access by adding this endpoint to a blocked list. Clicking **Block netwo rk access** sets the status of the end point to **Disabled**. **NOTE** You must configure **Enforc ement Policy Rules** to all ow access to the endpoin ts with the status **Disabled** .

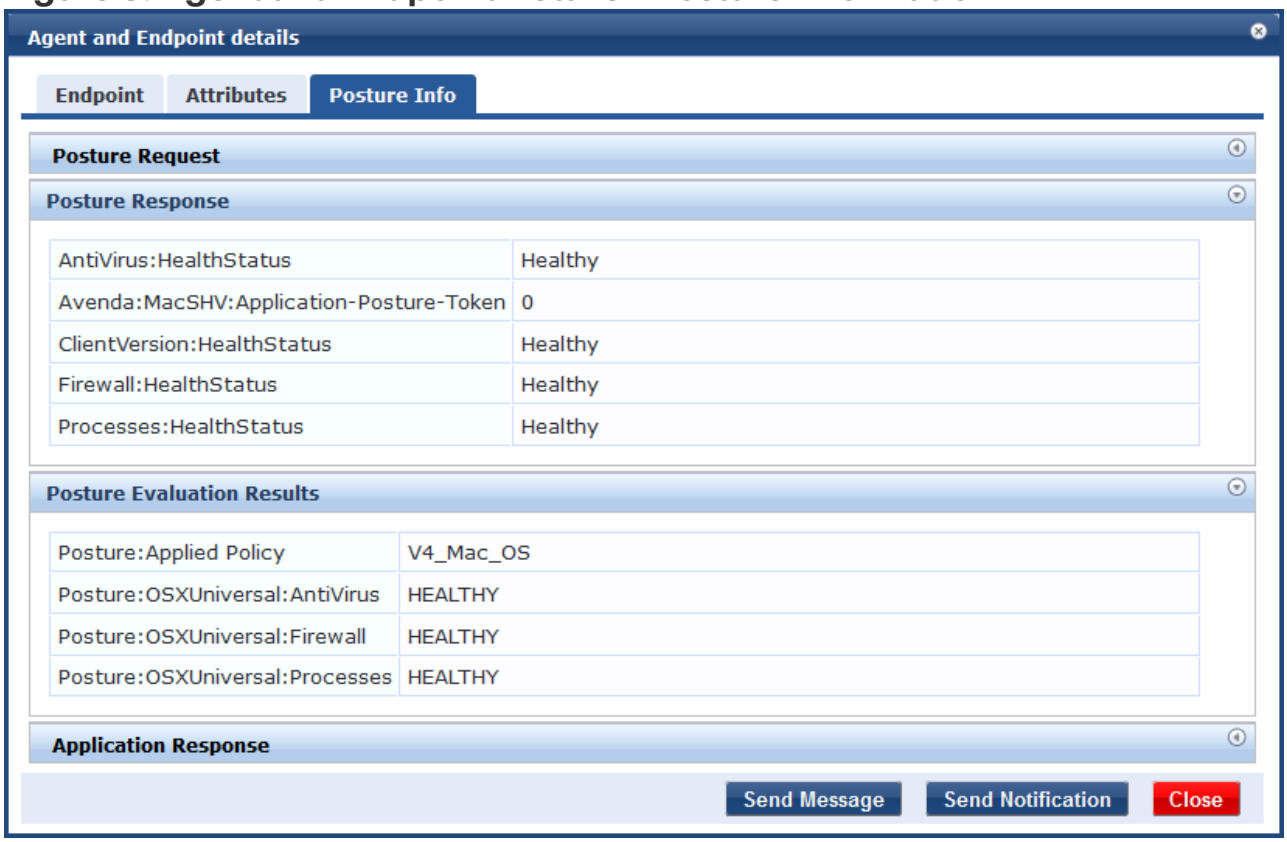
4. To proceed, click **Send Notification**.The notification is sent to the selected device.

Posture Info

The **Agent and Endpoint Details** page includes the **Posture Info** tab. This page provides posture information for the selected device in the following categories:

- Posture Request
- Posture Response
- Posture Evaluation Results
- Application Response

Figure 3. Agent and Endpoint Details > Posture Information



Defining Categories of Updates for a Windows Update Agent in a Patch Management Health Class

ClearPass OnGuard Agent can use information in the Windows registry key **MissingPatchesCategoryIncludeList** to determine which types of updates it should include in a OnGuard agent check for missing updates when Windows Update Agent is configured with value **All** for **Install Level Check Type**. This registry key is located at HKLM\SOFTWARE\Aruba Networks\ClearPassOnGuard, and is not automatically added by OnGuard agent during installation—administrators must add it manually. If the

registry key is not present, OnGuard agent will include all software update category types when checking for missing updates.

Network administrators can define a value of the MissingPatchesCategoryIncludeList registry key as a comma-separated list of any of the following Microsoft software update category types.

- security_update: A widely released fix for a product-specific, security-related vulnerability.
- update_rollup: A tested, cumulative set of hotfixes, security updates, critical updates, and updates that are packaged together for easy deployment.
- critical_update: A widely released fix for a specific problem that addresses a critical, non-security-related bug.
- update: A widely released fix for a specific problem. An update addresses a non-critical, non-security-related bug.
- driver: A driver defined in the Windows Server Update Services (WSUS) driver category. This list includes drivers from a wide variety of vendors. service_pack: A tested, cumulative set of all hotfixes, security updates, critical updates, and updates that are applied to a product.
- unknown: An update not matching any of the categories listed above.

  For example, if the value of the **MissingPatchesCategoryIncludeList** registry key is set to "security_update, critical_update" then OnGuard agent will include only security and critical updates in missing patches list and will not report all other updates like 'driver' as missing updates.

Bouncing the Network Interface After Waking Up From Sleep Mode

OnGuard Agent for Windows

Policy Manager OnGuard Agent for Windows can bounce the network interface after waking up from sleep mode, using the value of the new **DisableBounceAfterWakeup** registry key at the following location:

HKLM|Software|Aruba Networks|ClearPassOnGuard

This registry key is not automatically added to OnGuard during installation—administrators must add it manually:

- If the value of **DisableBounceAfterWakeup = 0**, or if the key is not present, OnGuard will bounce the network interface after waking up from sleep mode.
- If the value of **DisableBounceAfterWakeup = 1**, OnGuard will not bounce the network interface after waking up from sleep mode.

  The type for **DisableBounceAfterWakeup** should be **REG_DWORD**.

OnGuard Agent for macOS

Policy Manager OnGuard Agent for macOS can bounce the network interface after waking up from sleep mode by checking the presence of the following file:

/Library/Application Support/ClearPassOnGuard/.disableBounceAfterWakeup

This file is not automatically created during installation—administrators must add it manually:

- If the file *.disableBounceAfterWakeup* is not present, OnGuard will bounce the network interface after waking up from sleep mode.
- If the file *.disableBounceAfterWakeup* is present, OnGuard will not bounce the network interface after waking up from sleep mode.

Live Monitoring Analysis and Trending

The **Analysis and Trending** page displays requests for the subset of components included in the selected filters over a selected time period: one month, two weeks, one week, one day, 12 hours, 6 hours, 3 hours, or one hour. The data can be aggregated by **minute**, **hour**, **day**, or **week**. The list at the end of this section shows the per-filter count for the aggregated data. Each bar corresponding to each filter in the bar graph is clickable. Clicking a bar drills down into the Access Tracker (see Live Monitoring: Access Tracker) that shows session data for the specific time-slice and for the specific requests. To access this page, navigate to **Monitoring > Live Monitoring > Analysis and Trending**.

Figure 1. Analysis and Trending

Use the following components in the user interface to customize and filter the **Analysis and Trending** page:

Table 1. Analysis and Trending page

Component	Action/Description
Select Server	Select a Policy Manager node from the cluster.
Update Now!	Click to update the display with the latest available data.
Customize This!	Click to customize the display by adding any of the following filters. You can add a maximum of four filters. • All requests • RADIUS requests • TACACS+ requests • Webauth requests • Event requests • Failed requests • Successful requests

Component	Action/Description
	• Healthy requests • ClearPass application requests • RADIUS accounting requests • Unhealthy requests
Toggle Chart Type	Click to toggle the chart display between li ne and bar type.
Add New Data Filter	Click to add a data filter in the global filter list. For more information on adding filters, refer to Data Filters .

Live Monitoring: System Monitor

The **System Monitor** page has four tabs. Each tab provides one or more charts or graphs that give real-time information about various components.

> **NOTE**
> Auto refresh ensures that the **System Monitor** page is updated for every two minutes. You can see the last updated time in the **Last updated at** field in the **System Monitor** page.

- System Monitor Page
- Process Monitor Page
- Network Monitor Page
- ClearPass Monitor Page

Subtopics

System Monitor Page
Process Monitor Page
Network Monitor Page
ClearPass Monitor Page

System Monitor Page

About this task

The **System Monitor** page displays charts and graphs that display information about CPU load and usage, memory usage, and disk usage for the selected Policy Manager server.

To access the **System Monitor** page for the selected Policy Manager server:

Procedure

1. Navigate to **Monitoring > Live Monitoring > System Monitor**.
2. Select the desired server from the **Select Server** drop-down menu. This is the specific server resource trended.

Figure 1. System Monitor Page

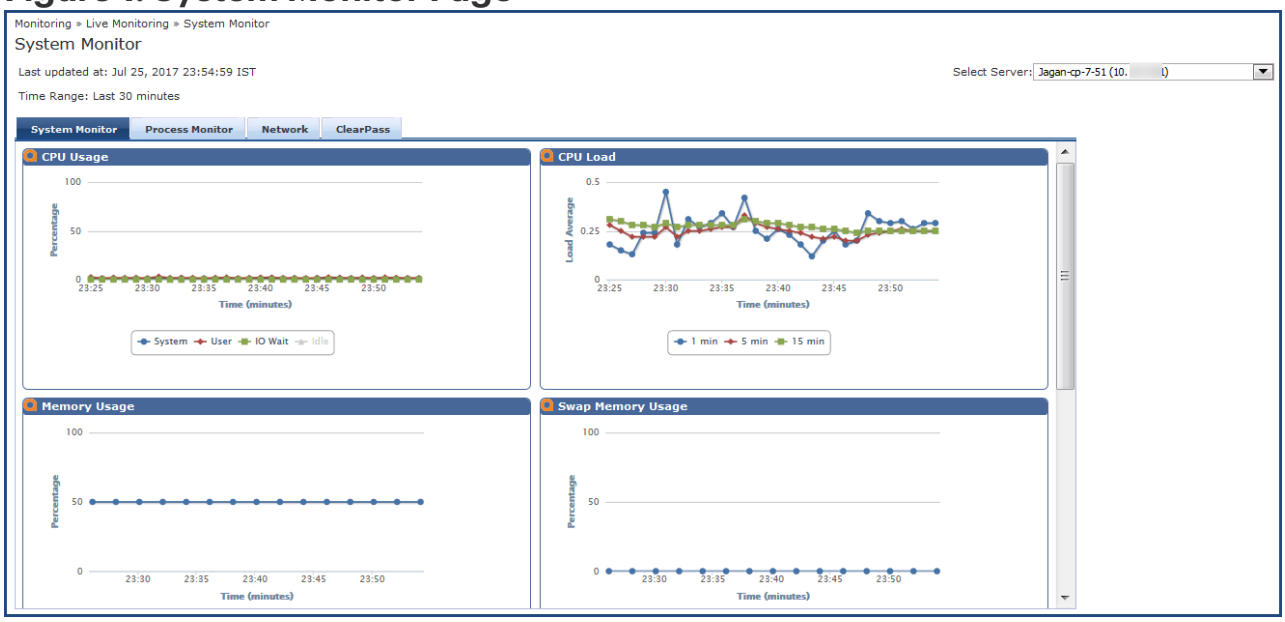


Table 1. System Monitor Page Report Widgets

Widget	Description
CPU Usage	Percentage of CPU usage for the **System** , **User**, **I/O Wait**, and **Idle** time. Each disp lay option can be displayed separately or collectively.
CPU Load	Percentage of CPU load averages in incre ments of 1, 5, and 15 minutes on the selec ted ClearPass server. Each display option

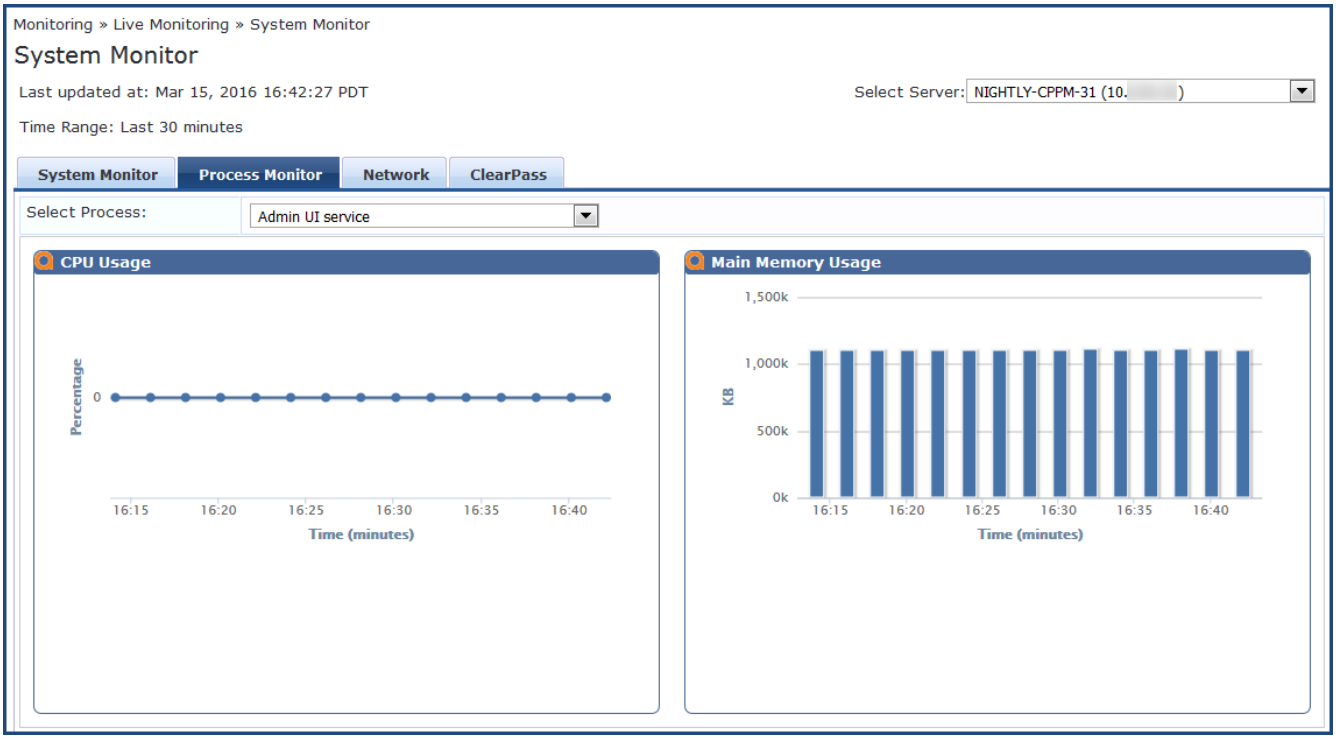
Widget	Description
	can be displayed separately or collectively.
Memory Usage	Percentage of free and total memory in Gigabytes.
Swap Memory Usage (Trend)	Percentage of free and total swap memory in Gigabytes.
Disk - / Usage	Percentage of used and free disk space on the selected ClearPass server.
Disk - / var Usage	Percentage of used and free disk space available in the /var partition of the selected ClearPass server.
Disk - /var/log Usage	Percentage of used and free disk space available in the /var/log partition of the selected ClearPass server.
Swap Memory Usage (%)	Percentage of free and total memory available to swap as needed amongst all partitions in Gigabytes.

Process Monitor Page

The **Process Monitor** page displays CPU Usage, Main Memory Usage for a selected process or service resource.

To access the Process Monitor page, navigate to **Monitoring** > **Live Monitoring** > **System Monitor** > **Process Monitor**.

Figure 1. System Monitoring: Process Monitor Page



Click the **Select Process** drop down list and select a process to view **CPU Usage** and **Main Memory Usage** charts for the selected process or service.

System Monitor Process List

The following processes can be monitored on the **Monitoring > Live Monitoring > System Monitor** page

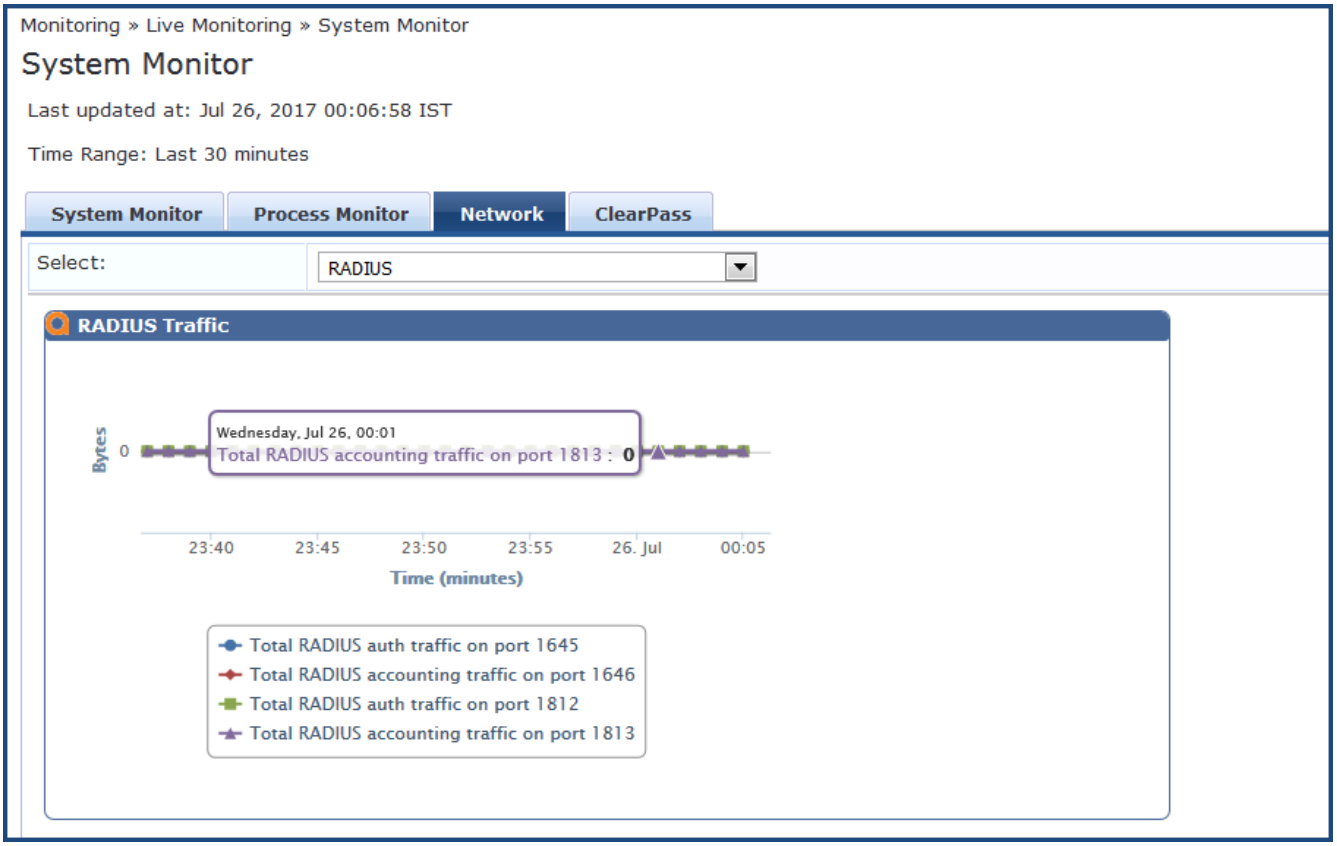
- Admin UI service
- AirGroup notification service
- Apache web server
- Async DB write service
- Async network services
- Policy Manager IPsec service
- DB change notification server
- DB replication service
- Domain service
- Extensions service
- Ingress logger service
- Ingress logrepo service

- Micros Fidelio FIAS
- Zone cache
- Policy server
- Radius server
- Stats aggregation service
- Stats collection service
- System auxiliary services
- System monitor service
- TACACS+ server
- Virtual IP service

Network Monitor Page

The **Network Monitor** page displays information about the selected network traffic type. To view network activity, navigate to **Monitoring > Live Monitoring > System Monitor > Network** tab, click the **Select** menu, and select the desired traffic type.

Figure 1. Network Monitor Page



Network Monitor Traffic List

The **Network Monitor** page displays network activity (in bytes) for the following supported traffic types:

- OnGuard
- Database Traffic
- Database Connections
- Web Traffic
- RADIUS
- TACACS +
- SSH
- NTP

ClearPass Monitor Page

About this task

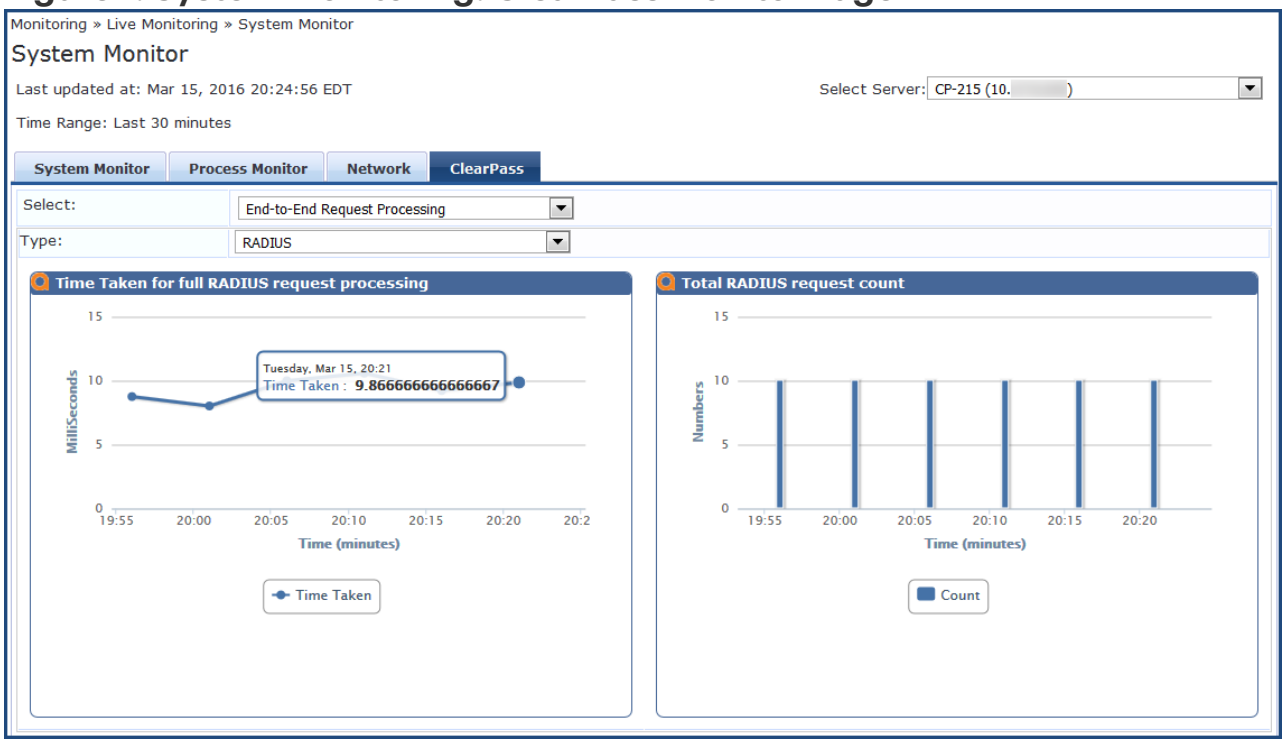
The **ClearPass** Monitoring page displays performance monitoring counters and timers for the last 30 minutes of activity on several key Policy Manager components.

To access the **ClearPass Monitor** page:

Procedure

1. Navigate to **Monitoring > Live Monitoring > System Monitor > ClearPass** tab.
2. Click the **Select** drop-down.
3. Select the desired Policy Manager performance monitoring counter.

Figure 1. System Monitoring: ClearPass Monitor Page



Subtopics

ClearPass Component List

ClearPass Component List

You can view statistics for the following Policy Manager components on this page.

- Service Categorization
- Authentication (RADIUS, TACACS+, or WebAuth)

- Authorization
- Role Mapping
- Posture Evaluation
- Audit Scan
- Enforcement
- End-to-End Request Processing (RADIUS, TACACS+, or WebAuth)
- Advanced

Monitoring Endpoint Profiler and Network Scan

This section provides the following information:

- Monitoring Endpoint Profiler
- Monitoring Network Scans and Subnet Scans
- Monitoring Discovered Devices

NOTE

When **Device Insight Integration** is enabled, the monitoring options for Endpoint Profiler, network and subnet scans, and discovered devices are removed from the Policy Manager user interface because Device Insight functionality replaces those Policy Manager functions (for more information, see Device Insight Integration Page).

Subtopics

Monitoring Endpoint Profiler
Monitoring Network Scans and Subnet Scans
Monitoring Discovered Devices

Monitoring Endpoint Profiler

Viewing Discovered Endpoints

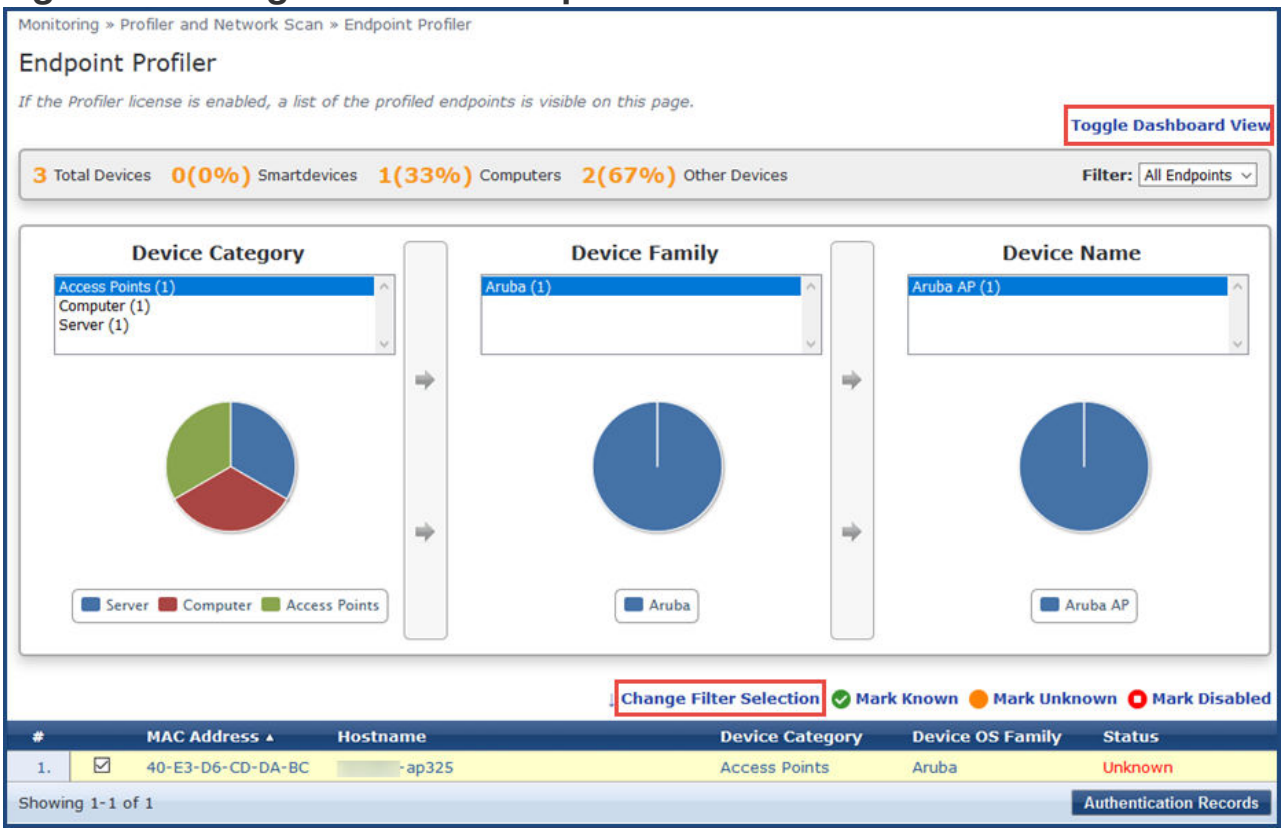
About this task

To view the discovered endpoints that are connected to the network:

Procedure

1. Navigate to **Monitoring > Profiler and Network Scan > Endpoint Profiler**. The **Endpoint Profiler** page opens and a list of the profiled endpoints is displayed.

Figure 1. Viewing Discovered Endpoints Information



2. Specify the **Device Category**, **Device Family**, and **Device Name** settings that you wish to view.

   The list of endpoints that are presented is based on the **Device Category**, **Device Family**, and **Device Name** items that you select.

3. To modify the selection criteria used to list the devices, click **Change Filter Selection**.

4. To see graphs that show information about distribution and update frequency for devices and computers, click **Toggle Dashboard View**.

Filtering by MAC Address

Policy Manager allows you to search for a MAC address in any of the following formats:

- 112233AABBCC
- 11:22:33:aa:bb:cc
- 11-22-33-AA-BB-CC
- 1111-2222-3333
- 1111.2222.3333

Viewing Details About a Specific Endpoint

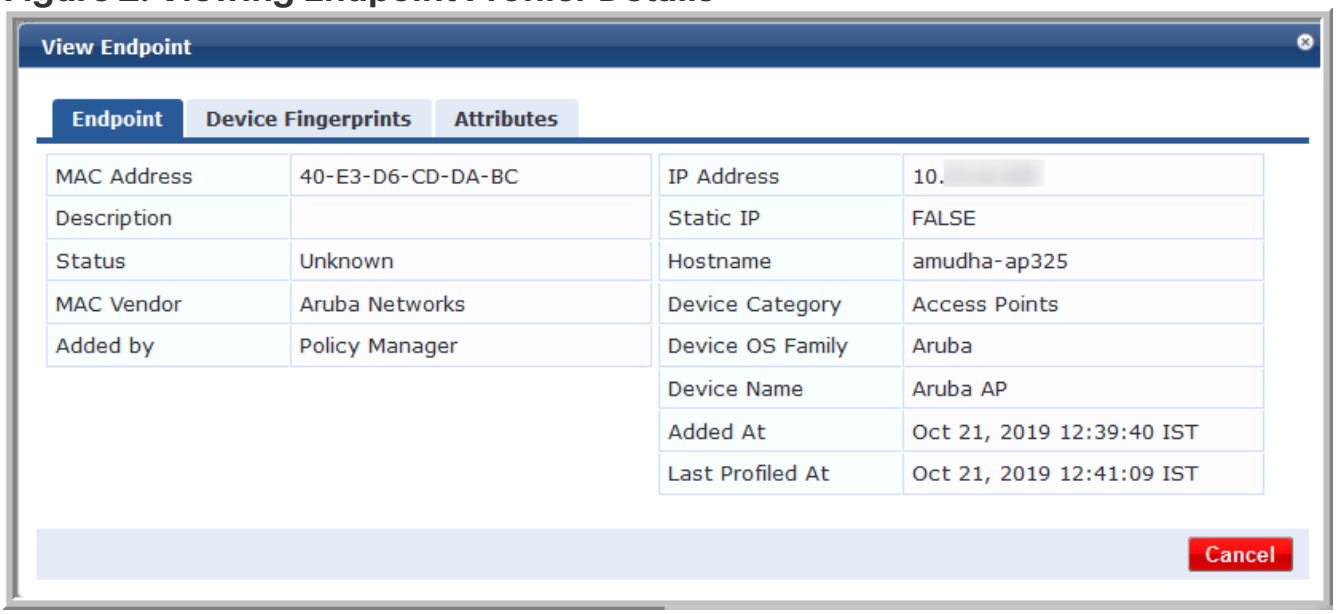
About this task

To view details about a specific endpoint:

Procedure

Click a device in the device table at the bottom of the page. The **View Endpoint** page opens:

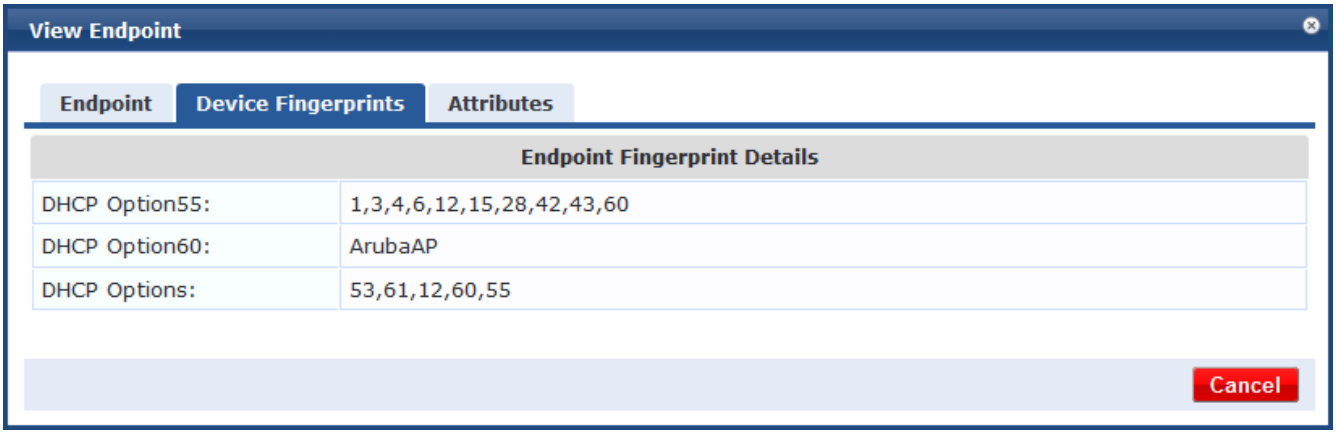
Figure 2. Viewing Endpoint Profiler Details



Viewing Endpoint Device Fingerprint Details

To view the endpoint device fingerprint details, select the **Device Fingerprints** tab. To return to the **Endpoint Profiler** page, click the **Cancel** button.

Figure 3. Viewing Endpoint Device Fingerprint Details



Monitoring Network Scans and Subnet Scans

- Viewing Network Scan and Subnet Scan Status
- Updating the Data on Network Scans and Subnet Scans

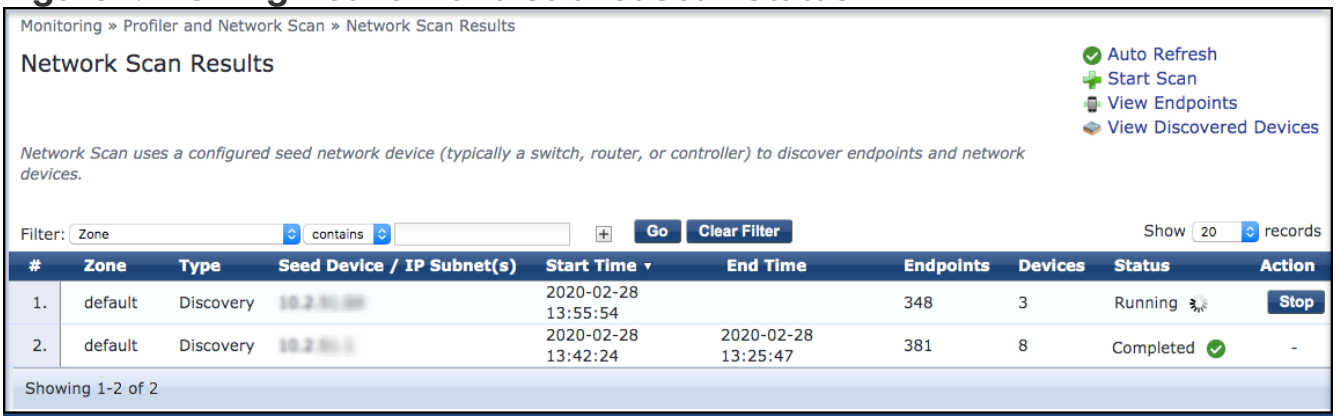
Viewing Network Scan and Subnet Scan Status

After you have , you can view the status of all scans run by the current Policy Manager server.

> **NOTE**
> Policy Manager retains the scan status information for the last ten network scans and subnet scans that were run.

To view network scan and subnet scan status, navigate to **Monitoring > Profile and Network Scan > Network Scan**.

Figure 1. Viewing Network and Subnet Scan Status



You can filter the **Network Scan Results** table by clicking the **Filter** drop down list and selecting one of the available filter options (**Zone**, **Type** , **Seed Device/IP Subnet(s)**, **Devices** or **Status**), selecting whether the filter **contains** or **equals** the search string, then entering the search term. You can also stop in-progress scans by clicking the **Stop** button in the **Action** column.

Updating the Data on Network Scans and Subnet Scans

About this task

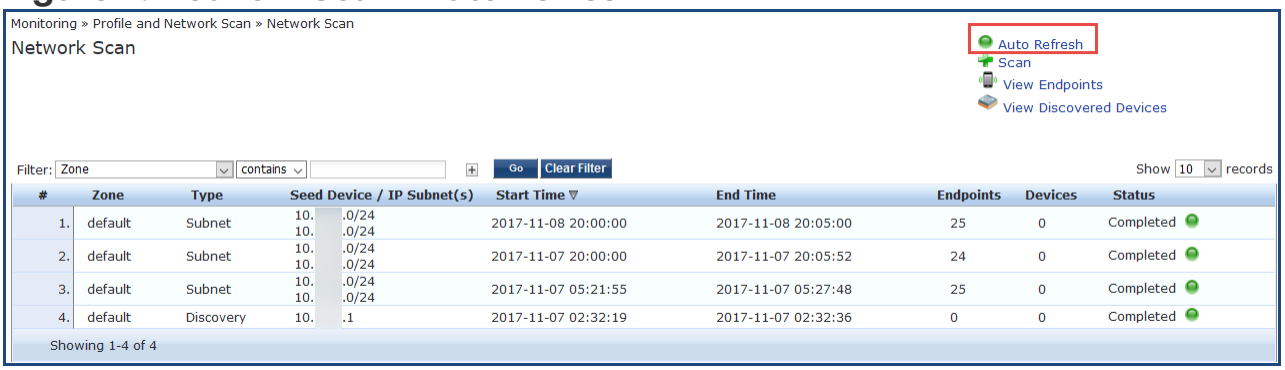
When **Auto-Refresh** is enabled (it is enabled by default), Policy Manager fetches fresh data every few seconds to ensure that the network scan and subnet scan status is always current. Every **Auto-Refresh** operation accesses the database and reads the data. When there is no network scan occurring, you can disable **Auto-Refresh** as there is no need to access the database so frequently in that case.

To enable or disable **Auto Refresh**:

Procedure

1. Navigate to **Monitoring > Profile and Network Scan > Network Scan**. When **Auto-Refresh** is enabled, the button in front of the **Auto Refresh** link is green.

Figure 2. Network Scan> Auto Refresh



2. To disable **Auto-Refresh**, click the green button, which then turns red to indicate that the **Auto Refresh** is disabled.

Figure 3. Auto Refresh Disabled

3. To reenable **Auto Refresh**, click the red button in front of the **Auto Refresh** link. The button turns green and **Auto Refresh** is now reenabled.

Monitoring Discovered Devices

This section provides the following information:

- Importing Network Devices
- Viewing Details on a Discovered Device

To access the **Discovered Devices** page:

1. Navigate to **Monitoring > Profiler and Network Scan > Discovered Devices**. The **Discovered Devices** page opens.

Figure 1. Discovered Devices Page

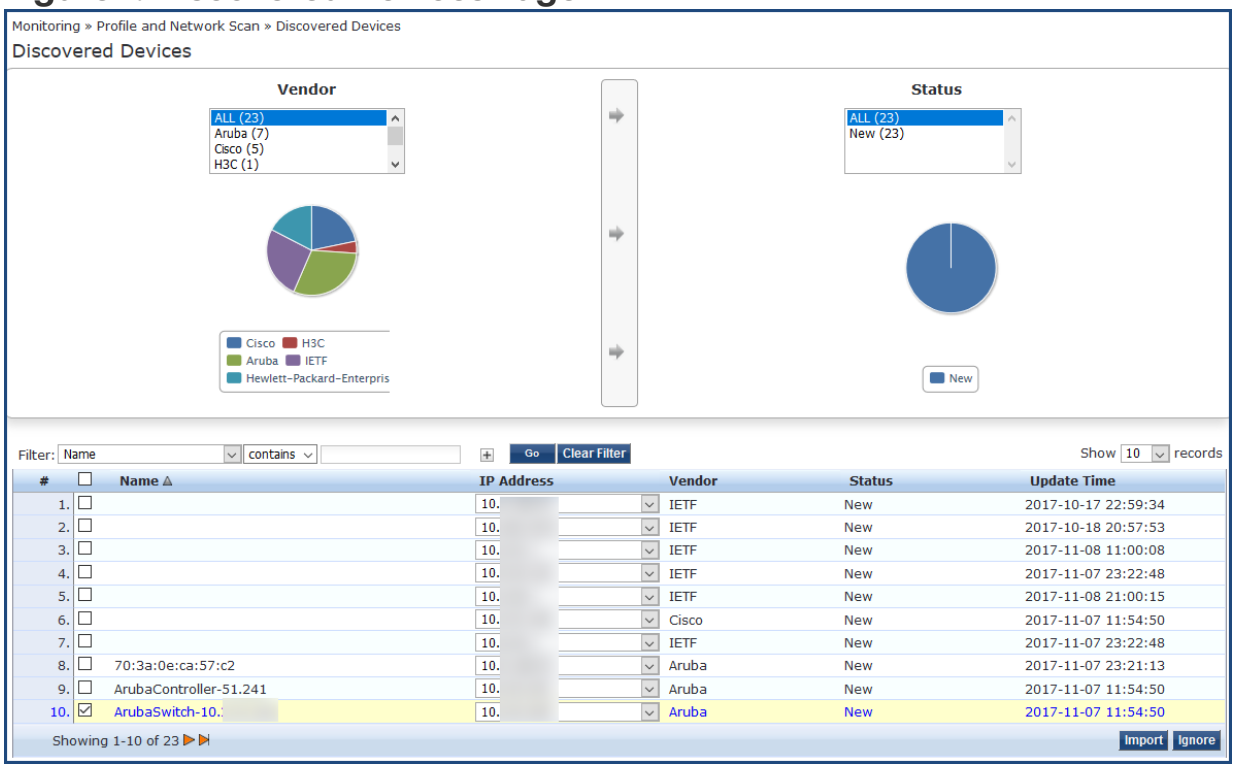


Table 1. Discovered Devices Action Buttons

Action Button	Action/Description
Import	Select the discovered device you want t o import, then click the **Import** button. For details, see the next section, Importing Network Devices.
Ignore	Select the network device you want to i gnore, then click the **Ignore** button.

Action Button	Action/Description
	Ignored devices will not be displayed o n the Discovered Devices list or the Net work Devices list.
Delete	To delete all previously discovered net work access device scan records befo re triggering a new scan, click the **Dele te** button.

Importing Network Devices

About this task

The devices that you import are added to the set of network devices known to Policy Manager.

> **NOTE**
> You can import devices from the publisher only.

To import and add discovered devices to the set of Policy Manager Network Devices:

Procedure

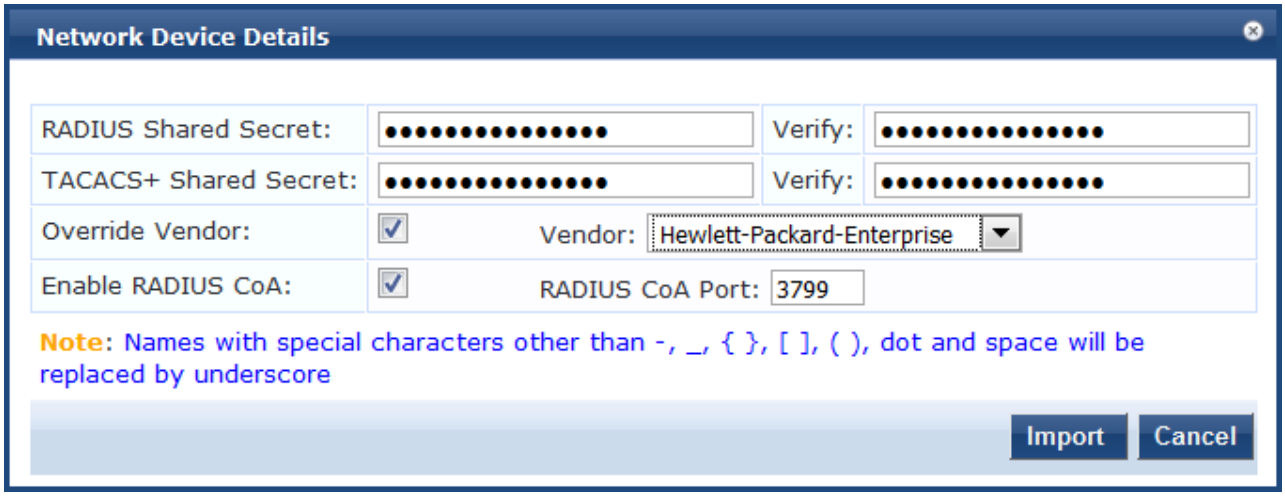
1. From the list of discovered devices, select a device you wish to import.

   Optionally select all of the discovered devices at once by clicking the **Name** check box.

2. Click the **Import** button.

   The **Network Device Details** dialog opens.

Figure 2. Importing a Network Device



3. Enter the appropriate information in the **Network Device Details** dialog.

Table 2. Specifying Network Device Details for Importing Devices

Field	Action/Description
RADIUS Shared Secret	If using RADIUS, enter the RADIUS Share d Secret for the selected discovered devi ce.
TACACS+ Shared Secret	If using TACACS+, enter the TACACS+ Sha red Secret for the selected discovered d evice.
Override Vendor	Optionally, to override the discovered ve ndor type, select this check box.
Vendor	This field is displayed when you select **O verride Vendor**. From the **Vendor** drop - down, select the name of the vendor type to override the discovered vendor type.
Enable RADIUS Dynamic Authorization	Select this check box to enable RADIUS Dynamic Authorization.
Port	The default port is **3799**. To change the port number, enter the n ew port number.

4. Click **Import**.

The selected network device has been added to Policy Manager. To see the network device listed, navigate to **Configuration > Network > Devices**.

Figure 3. Network Device Added toPolicy Manager

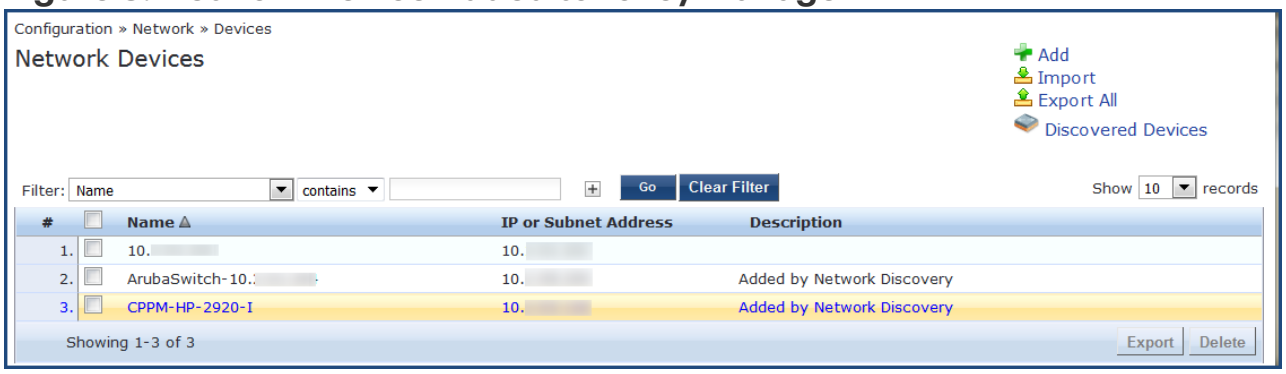


Table 3. Network Devices Action Buttons

Action Button	Action/Description
Copy	Select a network device to copy, then cli ck the **Copy** button. The device name ca n now be updated during the copy opera tion. With legacy releases, a device copy was created with name Copy_of_<device Name> appended, and users were only p ermitted to update the device name duri ng an edit operation. To modify the copied device, select the d evice; the **Edit Device Details** configurati on dialog opens.
Export	Select the network device you want to ex port, then click the **Export**button. The **Export to File** dialog opens. Enter th e fields as required, then click **Export**.
Delete	Select the network device you want to de lete, then click the **Delete**button. You are prompted to confirm that you want to del ete the device. To proceed, click **Yes**.

Viewing Details on a Discovered Device

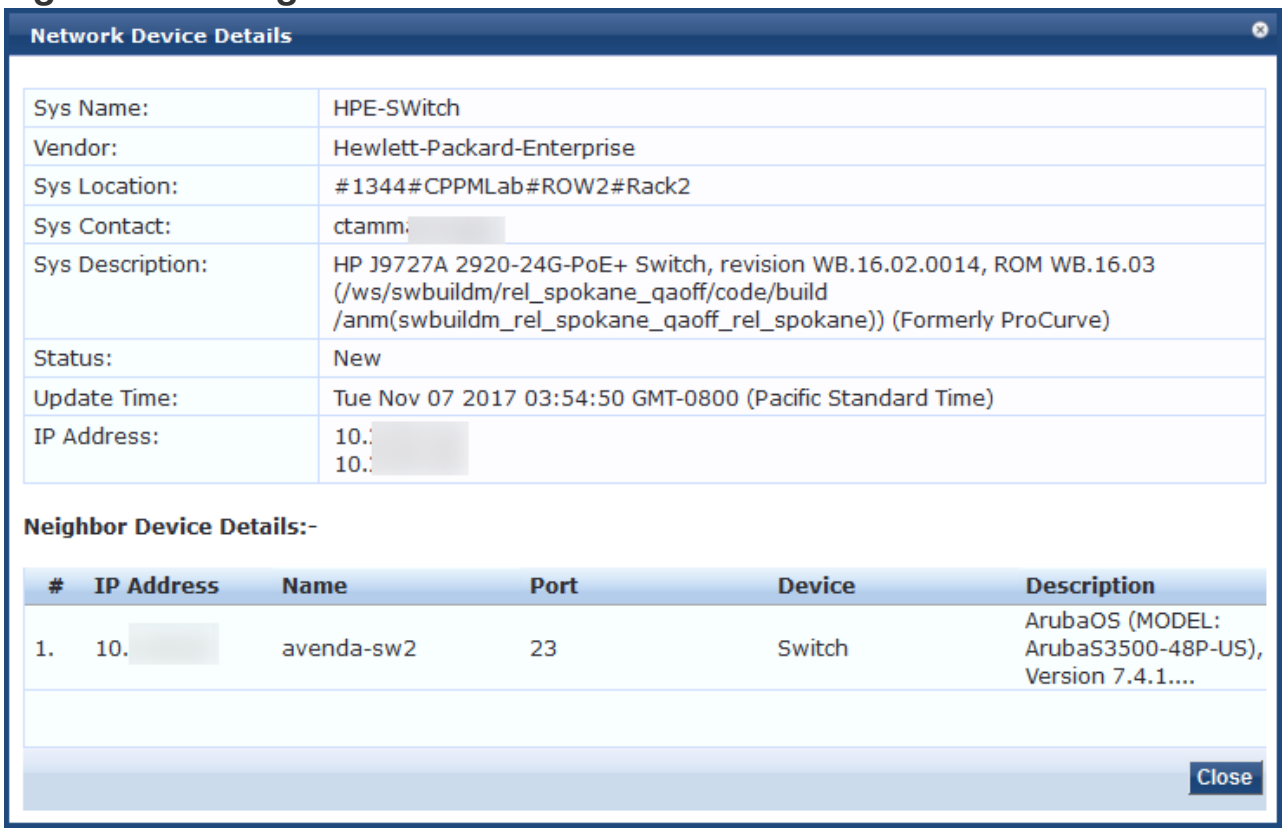
About this task

To view detailed information about a discovered network device, including a list of its neighbors in the network:

Procedure

1. From the **Network Devices** page, click the **Discovered Devices** link. The **Discovered Devices** page opens.
2. Click the name of the device of interest. The **Network Device Details** page opens.

Figure 4. Viewing Details fora Discovered Device



3. When finished, click **Close**.

Audit Viewer

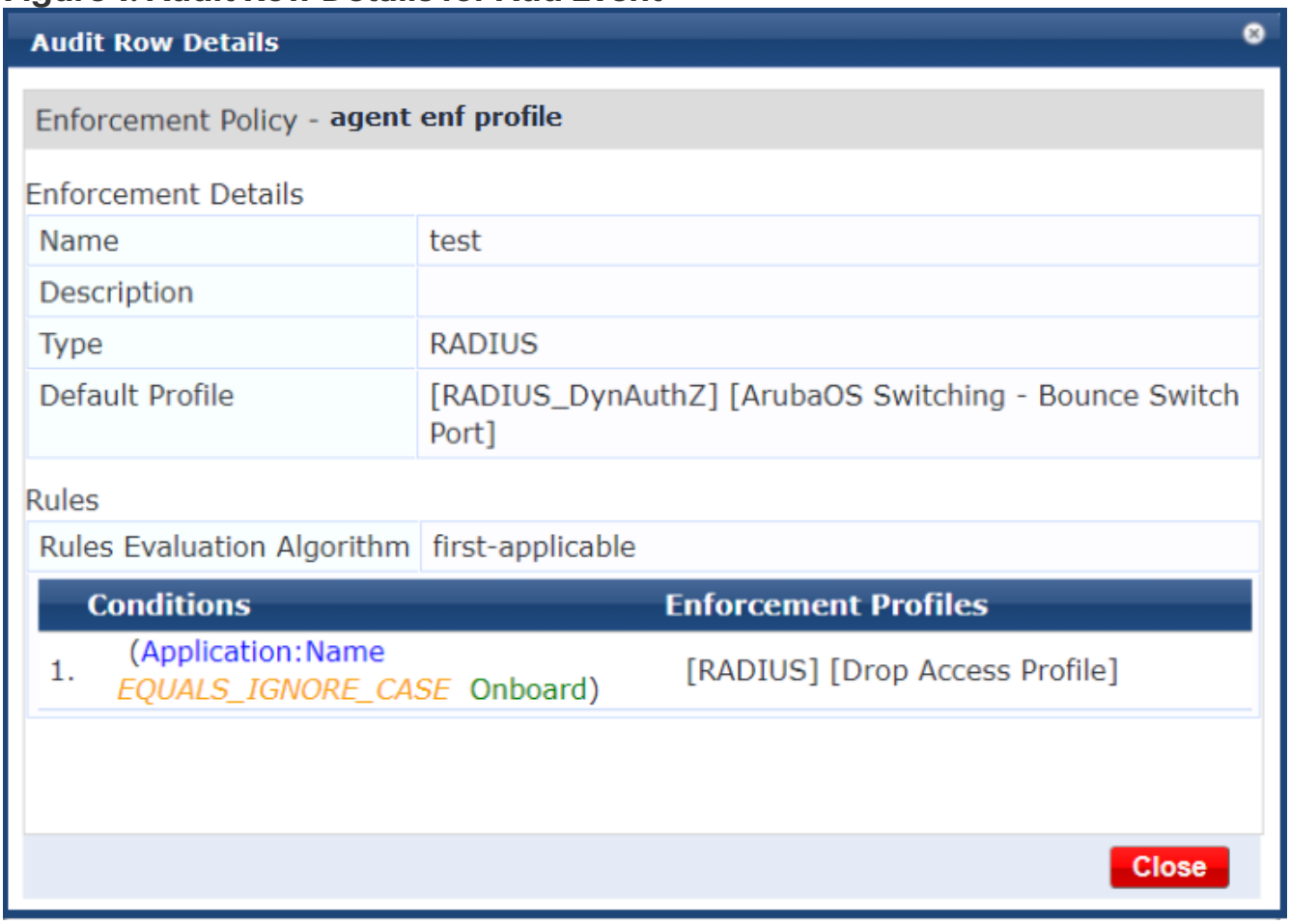
- Introduction
- Add Events

- Modify Events
- Remove Events

Add Events

To display additional details that are specific to the new policy component, click a row with the **Add** action type to display the **Audit Row Details** page.

Figure 1. Audit Row Details for Add Event



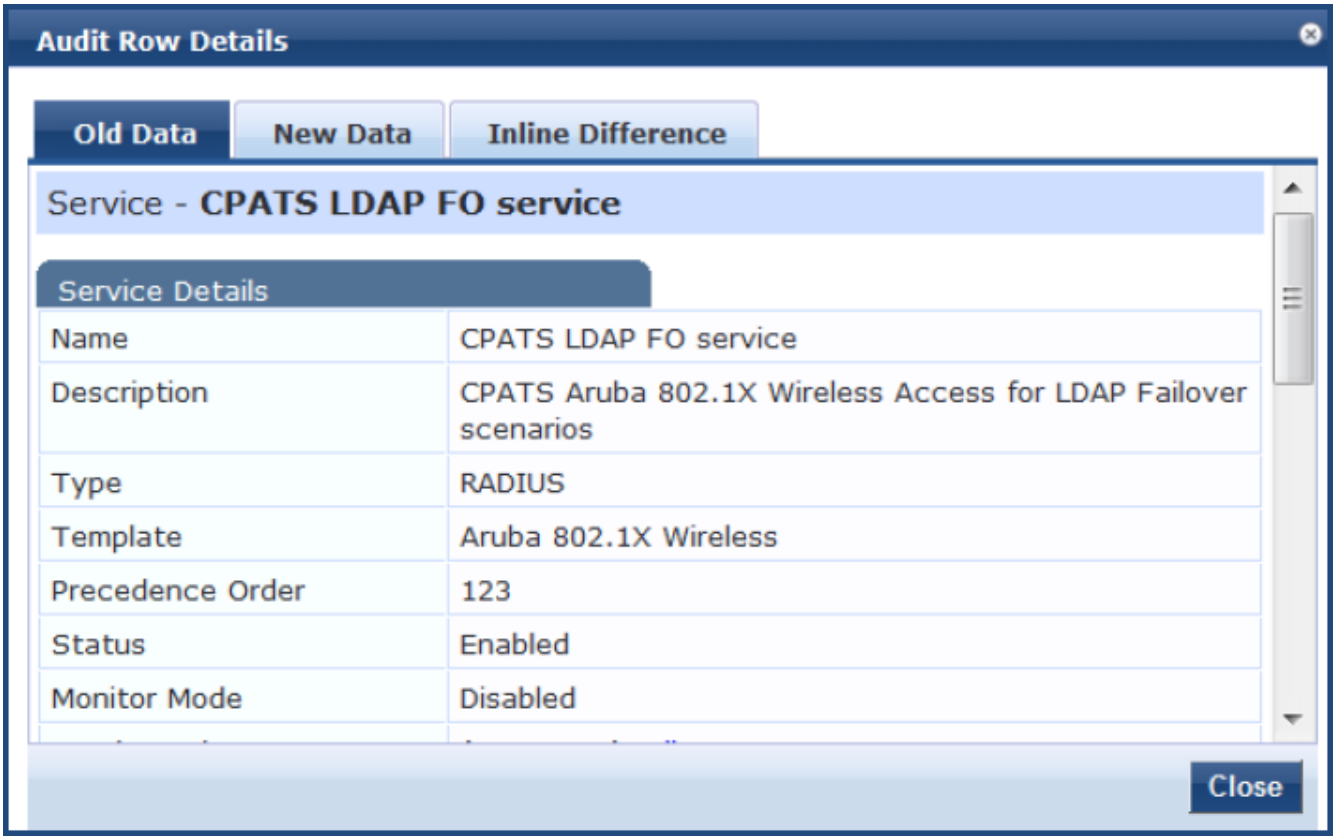
For example, if a RADIUS enforcement profile is added, the **Audit Row Details** page displays detailed information about that profile. If a policy is created, the **Audit Row Details** page displays information about the policy.

Modify Events

To display additional details information about the change, including the previous values, the latest, updated values, and the differences between the two, click a row with the **Modify** action type.

Figure 3 shows the **Audit Row Details** page for a **Modify Event**.

Figure 2. Audit Row Details for Modify Event



When you view a modify event, the **Audit Row Details** window contains the following three tabs:

Table 1. Audit Row Details > Modify Event Page

Parameter	Description
Old Data	Displays a summary of details about the ori ginal data values. • The **Profile** section shows a summary o f the profile values. • The **Attributes** section shows data abo ut the original attributes and values.
New Data	Displays a summary of details about the n ew data values. • The **Profile** section shows a summary o f the profile values. • The **Attributes** section shows data abo ut the original attributes and values.

Parameter	Description
Inline Difference	Displays information about what changed. The information is color - coded to indicat e the following types of changes: • Modified • Added • Deleted • Moved up • Moved down

Remove Events

To display details about attributes that were removed, click a row with the **Remove** action type.

Event Viewer

Unsupported Admin Access Attempts Logged to the Event Viewer

About this task

If an attempt is made to access the ClearPass Policy Manager administration command-line interface (CLI) with unsupported SSH protocol versions, or unsupported encryption or cryptographic hash algorithms, Policy Manager logs those alerts in the Event Viewer. This feature requires the **Ingress Event Engine** option and services to be enabled (for details, see Enabling Ingress Events Processing).

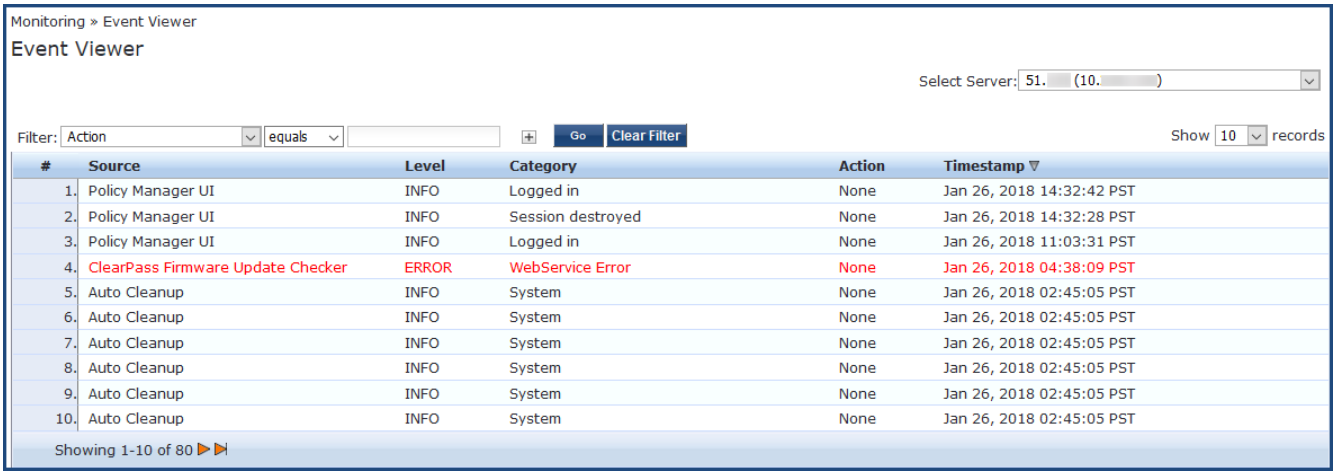
If an attempt is made to access ClearPass Policy Manager administration WebUI with unsupported SSL protocol versions, or unsupported ciphers, Policy Manager logs those alerts in the Event Viewer. This feature requires the **Ingress Event Engine** option and services to be enabled for the Policy Manager server (for details, see Enabling Ingress Events Processing).

To access the Event Viewer:

Procedure

Navigate to **Monitoring > Event Viewer**. The **Event Viewer** page opens.

Figure 1. Event Viewer Page



The following table describes the **Event Viewer** parameters:

Table 1. Event Viewer Page Parameters

Parameter	Description
Source	Displays the source of the event. For example, **OnGuard, Policy Manager UI** or Policy Manager **Firmware Update Checker**.
Level	Displays the level of the event from the following options: • INFO • WARN • ERROR
Category	Displays the category of the event. For example, **Message**, **Logged in**, **System**, or **AV/AS Updates**.
Action	Displays the status of the event action. For example, **Success**, **Failed**, **Unknown**, or **None**.
Timestamp	Displays the date and time when the event occurred.

Creating an Event Viewer Report Using Default Values

Procedure

1. In the **Filter** field, select **Source** as the filter parameter.

2. Click **Go**. Policy Manager returns all event records.

Creating an Event Viewer Report Using Custom Values

About this task

To create an Event Viewer report using custom values:

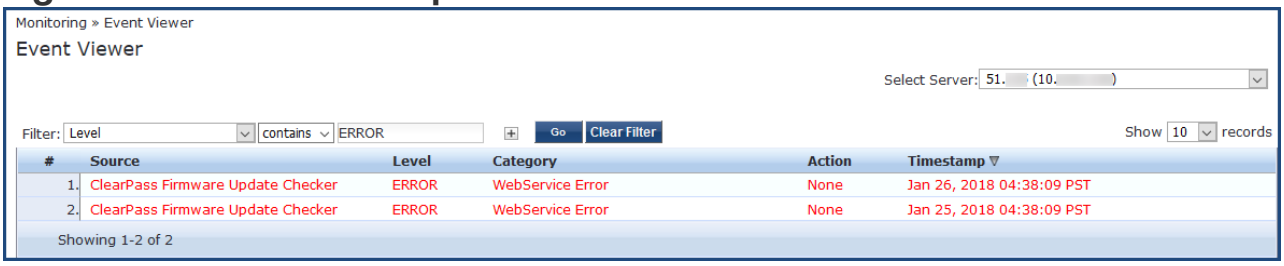
Procedure

1. Click the ⊞ icon. A new **filter** is added. You can add up to four **filters**.

Figure 2. Adding an Event Viewer Filter

2. In the **Filter** field, select the element on which you wish to filter:
   - Source
   - Level
   - Category
   - Action
   - Description
3. Specify **equals** or **contains**.
4. In the text field, enter the type of event for which you are searching.
5. (Optional) Create up to three additional filters if needed.
6. (Optional) Change the **Show records** value as desired.
7. Click **Go**. The following figure displays the **Event Viewer** report with custom values:

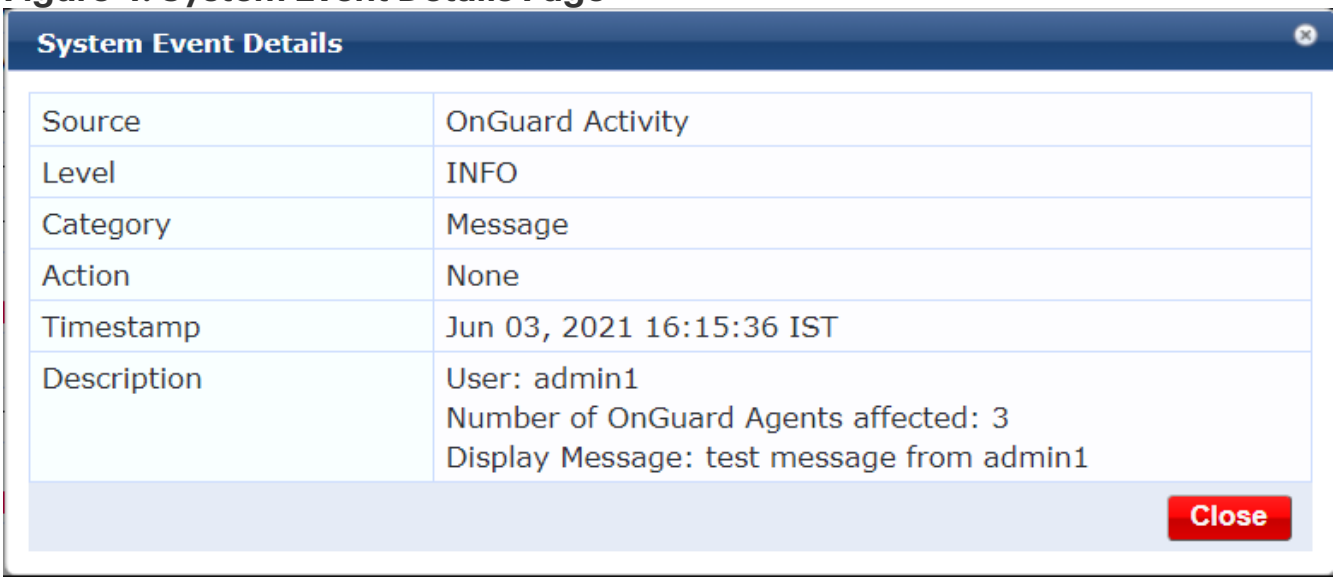
Figure 3. Event Viewer Report with Customized Filter



Viewing Report Details

To display the **System Event Details** page, click a row in the **Event Viewer** page.

Figure 4. System Event Details Page



The following table describes the **System Event Details** parameters:

Table 2. System Event Details Page Parameters

Parameter	Description
Source	Displays the source of the event. For example, Admin UI, RADIUS, and SnmpService.
Level	Displays the level of the event from the following options: • INFO • WARN • ERROR
Category	Displays the category of the event. For example, **Import**, **Export**, **Request**, **Authentication**, and **System**.
Action	Displays the action of the events. For example, **Success**, **Failed**, **Unknown**, and **None**.
Timestamp	Displays the date and time when the event occurred.

Parameter	Description
Description	Displays additional information about the event, including User, Role, Entity, Client IP Address, summary of the event, and when pertinent, the session inactive expiry time.

Data Filters

- About Data Filters
- Adding a Data Filter

The **Data Filters** page provides a way to limit the number of rows of data shown by defining custom criteria or rules in the following components in Policy Manager:

- Live Monitoring: Access Tracker
- Syslog Export Filters
- Live Monitoring: Analysis and Trending
- Live Monitoring: Accounting

Preconfigured Data Filters

Policy Manager is preconfigured with the following data filters:

Table 1. Available Data filters

Data Filter	Description
[Active sessions]	Show all Active sessions
[Show all Requests]	Show all session log requests
[Application Authentication]	Show all Application Authentications
[ClearPass Application Requests]	Show all Application session log requests
[ClearPass Configuration Audit]	Show all ClearPass Configuration Audit
[ClearPass Guest]	Show all ClearPass Guest
[ClearPass System Events]	Show all ClearPass System Events
[Endpoints]	Show all Endpoints
[Event Requests]	Show all Event requests

Data Filter	Description
[Failed Application Authentication]	Show all Failed Application Authentication
[Failed Requests]	Show all Failed session log requests
[Guest Access Requests]	Show all Guest Login requests
[Healthy Requests]	Show all Healthy session log requests
[Onboard Certificate]	Show all Onboard Certificate
[Onboard Enrollment]	Show all Onboard Enrollment
[Onboard OCSP]	Show all Onboard OCSP
[Posture Antispyware Summary]	Show Posture Antispyware Summary
[Posture Antivirus Summary]	Show Posture Antivirus Summary
[Posture DiskEncryption Summary]	Show Posture DiskEncryption Summary
[Posture Firewall Summary]	Show Posture Firewall Summary
[Posture Summary]	Show Posture Summary
[Posture Windows HotFixes Summary]	Show Posture Windows HotFixes Summary
[RADIUS Accounting]	Show all RADIUS Accounting
[RADIUS Accounting Requests]	Show all RADIUS accounting requests
[RADIUS Authentications]	Show all RADIUS Authentications
[RADIUS Failed Authentications]	Show all RADIUS Failed Authentications
[RADIUS Requests]	Show all RADIUS requests
[Successful Requests]	Show all Successful session log requests
[TACACS+ Authentication]	Show all TACACS+ Authentication
[TACACS+ Failed Authentication]	Show all TACACS+ Failed Authentication
[TACACS+ Requests]	Show all TACACS+ requests
[Unhealthy Requests]	Show all Unhealthy session log requests
[WEBAUTH Authentication]	Show all WEBAUTH Authentication
[WEBAUTH Failed Authentications]	Show all WEBAUTH Failed Authentications
[Webauth Requests]	Show all Webauth Requests

Table 2. Access Tracker Edit Page Parameters

Data Filter	Description
[Show all Requests]	Show all session log requests
[Application Authentication]	Show all Application Authentications
[ClearPass Application Requests]	Show all Application session log requests

Data Filter	Description
[ClearPass Configuration Audit]	Show all ClearPass Configuration Audit
[ClearPass Guest]	Show all ClearPass Guest
[ClearPass System Events]	Show all ClearPass System Events
[Endpoints]	Show all Endpoints
[Event Requests]	Show all Event requests
[Failed Application Authentication]	Show all Failed Application Authentication
[Failed Requests]	Show all Failed session log requests
[Guest Access Requests]	Show all Guest Login requests
[Healthy Requests]	Show all Healthy session log requests
[Onboard Certificate]	Show all Onboard Certificate
[Onboard Enrollment]	Show all Onboard Enrollment
[Onboard OCSP]	Show all Onboard OCSP
[Posture Antispyware Summary]	Posture Antispyware Summary
[Posture Antivirus Summary]	Posture Antivirus Summary
[Posture DiskEncryption Summary]	Posture DiskEncryption Summary
[Posture Firewall Summary]	Posture Firewall Summary
[Posture Summary]	Posture Summary
[Posture Windows HotFixes Summary]	Posture Windows HotFixes Summary
[RADIUS Accounting]	Show all RADIUS Accounting
[RADIUS Accounting Requests]	Show all RADIUS accounting requests
[RADIUS Authentications]	Show all RADIUS Authentications
[RADIUS Failed Authentications]	Show all RADIUS Failed Authentications
[RADIUS Requests]	Show all RADIUS requests
[Successful Requests]	Show all Successful session log requests
[TACACS+ Authentication]	Show all TACACS+ Authentication
[TACACS+ Failed Authentication]	Show all TACACS+ Failed Authentication
[TACACS+ Requests]	Show all TACACS+ requests
[Unhealthy Requests]	Show all Unhealthy session log requests
[WEBAUTH Authentication]	Show all WEBAUTH Authentication
[WEBAUTH Failed Authentications]	All WEBAUTH Failed Authentications
[Webauth Requests]	All Webauth Requests

Accessing the Data Filters Page

About this task

To access the **Data Filters** page:

Procedure

Navigate to **Monitoring > Data Filters**. The **Data Filters** page opens.

Figure 1. Data Filters Page

For each data filter, the **Data Filter**s table lists the name, description and query type, which can be either **Session**, **Accounting** or **Insight**.

Adding a Data Filter

About this task

Users should be aware that the following words and characters are not allowed in custom SQL queries:

• - -	• delete	• commit
• ;	• create	• overlay
• ?	• postgres	• update
• *	• appadmin	• truncate
• *	• appuser	• reindex
• <>	• password	• btrim
• alter	• schema	• lpad
• script	• replace	• rpad
• drop	• trim	• ltrim
• insert	• declare	• rtrim

To add a data filter:

Procedure

1. Click the **Add** link.

   The **Add Data Filters** page opens to the **Filter** tab.

   Figure 2 shows the **Filter** dialog when you choose **Select Attributes** (the default setting).

Figure 2. Add Data Filter > Filter Tab > Select Attributes

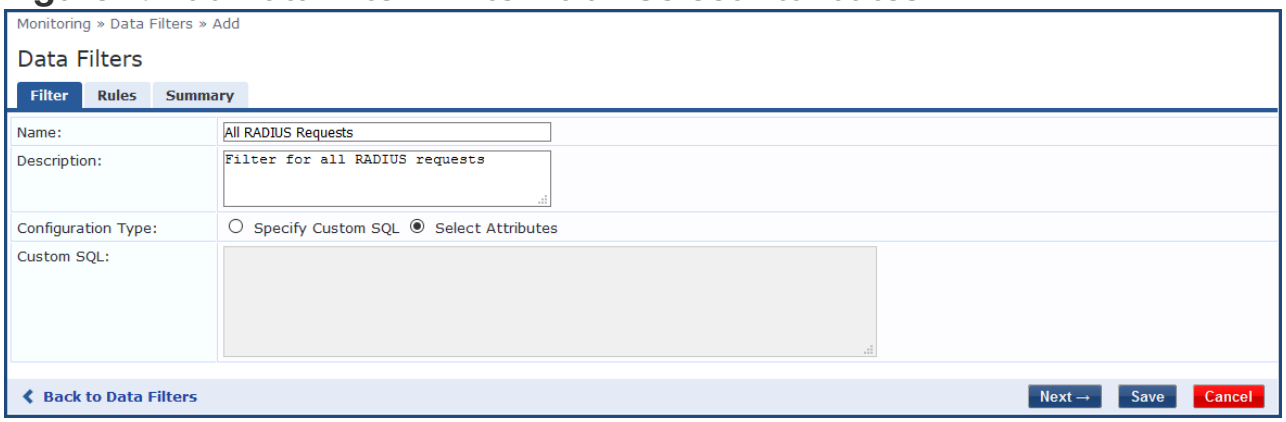


Figure 3 shows the **Filter** dialog when you choose **Specify Custom SQL**.

2. Specify the **Add Data Filters** parameters as described in the following table.

Table 3. Add Data Filters Page > Filter Tab Parameters

Parameter	Action/Description
Name	Enter a name for the data filter. **Note**: Use the default **Unique Count For Device** filter to fetch the unique device count for the MAC Caching workflow in the Endpoints Repository when MAC Caching is enabled.
Description	Optionally, enter a description of this data filter (recommended).
Configuration Type	Choose one of the following configuration types: • **Select Attributes** • **Specify Custom SQL**
Select Attributes	This option is selected by default. When you specify **Select Attributes**, the **Rules** tab appears. Use the **Rules** tab to configure rules for this filter.

Parameter	Action/Description
Specify Custom SQL	When you choose **Specify Custom SQL**, a default SQL template is displayed. In the text entry field, enter the attributes for the type, attribute name, and attribute value. **NOTE** Aruba does not recommend that you enable this option without first consulting Support (navigate to **Administration > Support > Contact Support**).

Rules Tab

The **Rules** tab displays when you choose the **Select Attributes** configuration type on the **Filter** dialog.

Figure 3. Add Data Filter > Rules Dialog

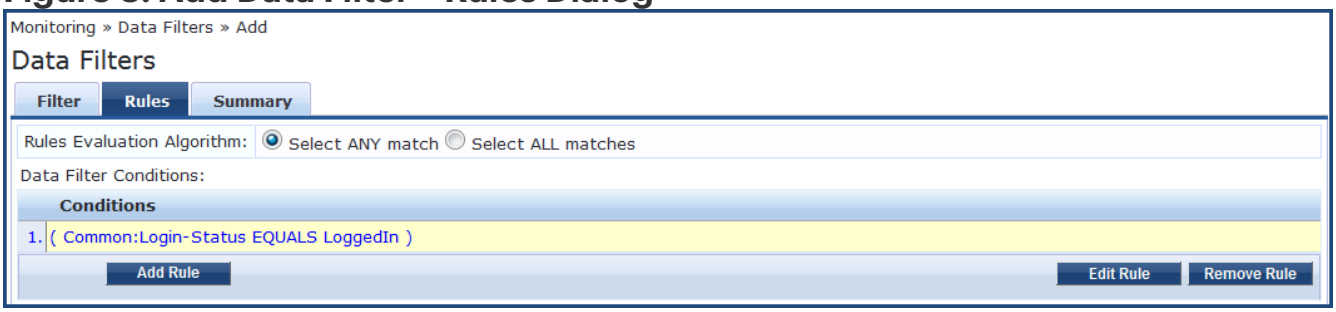


Table 2 describes the **Add Filter > Rules** tab parameters:

Table 4. Add Filter > Rules Tab

Parameter	Action/Description
Rule Evaluation Algorithm	• **Select ANY match** is a logical OR operation of all the rules. • **Select ALL matches** is a logical AND operation of all the rules.

Parameter	Action/Description
Add Rule	Add a rule to the filter.
Edit Rule	Edit an existing rule.
Remove Rule	When you select an existing rule and click **Remove Rule**, the selected Rule is deleted immediately (no confirmation prompt app ears).

When you click **Add Rule** or **Edit Rule**, the **Dashboard Filter Rules Editor** dialog opens.

Figure 4. Dashboard Filters > Rules Editor

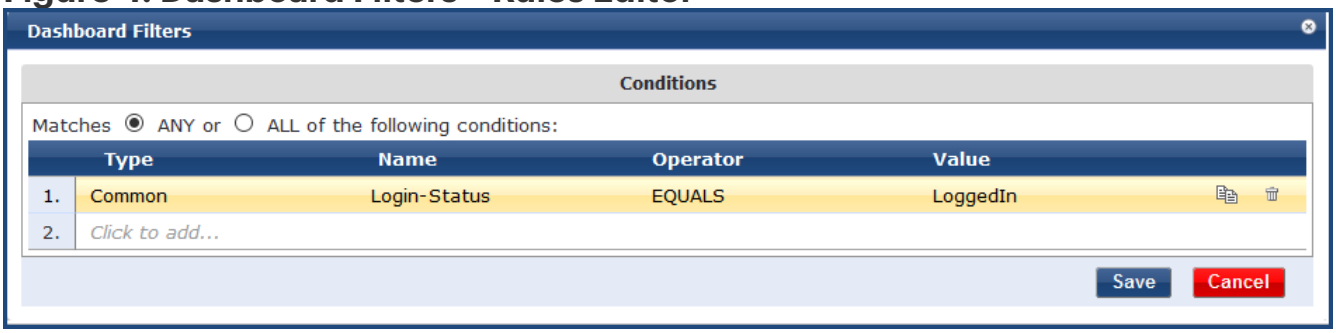


Table 3 describes the **Dashboard Filters > Rules Editor** parameters:

Table 5. Dashboard Filters > Rules Editor Configuration Parameters

Parameter	Action/Description
Matches	Specify the match conditions: • **ANY** matches one of the configured co nditions. • **ALL** specifies to match all of the config ured conditions.
Type	Select the type of data filter. • **Common**: Attributes common to RADIU S, TACACS+, and WebAuth requests and responses. • **RADIUS**: Attributes associated with R ADIUS authentication, accounting requ ests, and responses.

Parameter	Action/Description
	• **TACACS+**: Attributes associated with TACACS+ authentication, accounting, p olicy requests, and responses. • **Web Authentication Policy**: Policy Man ager policy objects assigned after the evaluation of policies associated with Web Authentication requests. For exam ple, Auth Method, Auth Source, and En forcement Profiles.
Name	Select the name of the attribute from the **Name** drop - down list. The **Name** list varies according to which **Ty pe** you selected.
Operator	Select any subset of string data type opera tors from the following list: • EQUALS • NOT_EQUALS • LESS_THAN • LESS_THAN_OR_EQUALS • GREATER_THAN • GREATER_THAN_OR_EQUALS • CONTAINS • NOT_CONTAINS • EXISTS • NOT_EXISTS
Value	The value of the attribute.

Restoring Users on a Blocked List to the Network

About this task

The **Denylisted Users** page lists the MAC address and user name of all users on a blocked list, the authentication source for that user, and indicates whether the bandwidth limit or session duration limits were exceeded by each user on that blocked list. After a user entry is removed from the blacklisted users list, the user is eligible to access the network.

To access the **Denylisted Users** page:

Procedure

1. Navigate to **Monitoring > Denylisted Users**.
2. To delete a user from this list, select the user row and click **Delete**.

   The deleted Denylisted user is then eligible to access the network.

Filtering by MAC Address

Policy Manager allows you to search for a MAC address in any of the following formats:

- 112233AABBCC
- 11:22:33:aa:bb:cc
- 11-22-33-AA-BB-CC
- 1111-2222-3333
- 1111.2222.3333

Authentication Methods and Sources

As a first step in the service-based processing, Policy Manager uses an authentication method to authenticate the user or device against an authentication source. Policy Manager includes several predefined Policy Manager authentication sources.

- Default Authentication Methods
- Tunneled and Non-Tunneled Authentication Methods
- Default Authentication Sources
- Adding and Modifying Authentication Methods
- Adding and Configuring Authentication Sources
- Modifying Authentication Methods or Sources for an Existing Service

Subtopics

Default Authentication Methods
Default Authentication Sources
Authentication and Authorization Architecture and Flow
Adding and Modifying Authentication Methods
Adding and Configuring Authentication Sources
Modifying Authentication Methods or Sources for an Existing Service

Default Authentication Methods

After the user or device is authenticated, Policy Manager fetches attributes for role-mapping policies from the authorization sources associated with this authentication source. For a general overview of Policy Manager authentication and authorization, see Authentication and Authorization Architecture and Flow.

Policy Manager provides the following default authentication methods:

Table 1. Default Authentication Methods

Authentication Methods	Description
[Allow All MAC AUTH]	Provides default settings for *Allow All MAC - AUTH* authentication method.
[Aruba EAP GTC]	Provides the EAP - GTC (Generic Token Ca rd) authentication method to use with the Aruba EAP - GTC plugin for Windows.
[Authorize]	Provides default settings for *Authorize* auth entication method.
[CHAP]	Provides default settings for *CHAP* authenti cation method.
[EAP FAST]	Provides default settings for *EAP - FAST* aut hentication method.
[EAP GTC]	Provides default settings for *EAP - GTC* aut hentication method.
[EAP MD5]	Provides default settings for *EAP - MD5* aut hentication method.
[EAP MSCHAPv2]	Provides default settings for *EAP - MSCHA Pv2* authentication method.

Authentication Methods	Description
[EAP PEAP]	Provides default settings for *EAP - PEAP* authentication method.
[EAP PEAP Public]	Provides default settings for *EAP - PEAP - Public* authentication method.
[EAP PEAP Without Fast Reconnect]	Provides *EAP - PEAP with Fast Reconnect* disabled; recommended for Onboard.
[EAP PWD]	Provides default settings for *EAP - PWD* (EAP Password) authentication method.
[EAP TLS]	Provides default settings for *EAP - TLS* authentication method.
[EAP TLS with OCSP Enabled]	Provides *EAP - TLS with OCSP* (Online Certificate Status Protocol) enabled; recommended for Onboard.
[EAP TTLS]	Provides default settings for *EAP - TTLS* authentication method.
[MAC AUTH]	Provides default settings for *MAC - AUTH* authentication method.
[MSCHAP]	Provides default settings for *MSCHAP* authentication method.
[PAP]	Provides default settings for *PAP* (Password Authentication Protocol) authentication method.
[SSO]	Provides default settings for *SSO* (Single Sign - On) authentication method.

Subtopics

Tunneled and Non-Tunneled Authentication Methods
Tunneled EAP Authentication Methods
Non-Tunneled Authentication Methods

Tunneled and Non-Tunneled Authentication Methods

Refer to the following list of tunneled and non-tunneled authentication methods Policy Manager supports:

Tunneled EAP Authentication Methods

The tunneled EAP authentication methods are:

- EAP Protected EAP (EAP-PEAP)
- EAP Flexible Authentication Secure Tunnel (EAP-FAST)
- EAP Transport Layer Security (EAP-TLS)
- EAP Tunneled TLS (EAP-TTLS)

Non-Tunneled Authentication Methods

The non-tunneled authentication methods are:

- EAP Message Digest 5 (EAP-MD5)
- EAP Microsoft Challenge Handshake Authentication Protocol version 2 (EAP-MSCHAPv2)
- EAP Generic Token Card (EAP-GTC)
- Challenge Handshake Authentication Protocol (CHAP)
- Password Authentication Protocol (PAP)
- Microsoft CHAP version 1 and 2
- MAC authentication method (MAC-AUTH)
- Authorize authentication

Default Authentication Sources

The following table describes the predefined Policy Manager authentication sources that are part of the Policy Manager local SQL database. These default authentication sources appear in brackets; they cannot be copied or deleted.

Table 1. Local Authentication Sources

Authentication Source Name	Description
[Admin User Repository]	Users are authenticated against this Policy Manager admin user database.
[Denylist User Repository]	This database includes blocked users who have exceeded bandwidth or session - rela ted limits.
[Endpoints Repository]	Authenticates endpoints against the local Policy Manager database. This database su pports the following filter categories: • Authentication • Status • Profile • Fingerprint • MAC caching
[Guest Device Repository]	Authenticates Guest devices against the lo cal Policy Manager database. This databas e supports the following filter queries: • Authentication • Device Role ID • Device MPSK • Authorization
[Guest User Repository]	Authenticates Guest users against the loca l Policy Manager database. This database s upports the following filter queries: • Authentication • Authorization
[Insight Repository]	Insight database with session information for users and devices. This database suppo rts the following filter queries: • WebAuth check • MAC caching

Authentication Source Name	Description
	• Successful login count since midnight • Active sessions • Online status • Daily duration • Weekly duration • Monthly duration
[Local User Repository]	Authenticates users against the Policy Man ager local user database.
[Onboard Devices Repository]	Authenticates Onboard devices against th e local Policy Manager database.
[Social Login Repository]	Authenticates users againstPolicy Manager Social Login database. This database supp orts the following filter queries: • Authentication • Social
[Time Source]	Authorization source for implementing vari ous time functions.

Authentication and Authorization Architecture and Flow

- Authentication Method
- Authentication Source
- Authorization Source
- Authentication and Authorization Flow of Control

Policy Manager divides the architecture of authentication and authorization into the following three components:

- Authentication method
- Authentication source

- Authorization source

  Authentication Method Policy Manager initiates the authentication handshake by sending available methods in a priority order until the client accepts a method or until the client rejects the last method with the following possible outcomes:

  - Successful negotiation returns a method, which is used to authenticate the client against the authentication source.
  - Where no method is specified (for example, for unmanageable devices), Policy Manager passes the request to the next configured policy component for this service.
  - Policy Manager rejects the connection.

  > **NOTE**
  >
  > An authentication method is configurable only for some service types. For more information, see Policy Manager Service Types . All 802.1X wired and wireless services have an associated authentication method.

  Authentication Source In Policy Manager, an authentication source is the identity store (Active Directory, LDAP directory, SQL DB, token server, etc.) against which users and devices are authenticated. Policy Manager first tests whether the connecting entity (the device or user) is present in the ordered list of configured authentication sources. Policy Manager looks for the device or user by executing the first filter associated with the authentication source. After the device or user is found, Policy Managerthen authenticates this entity against this authentication source. The flow is as follows:

  - On successful authentication, Policy Manager moves on to the next stage of policy evaluation, which collects role mapping attributes from the authorization sources.
  - Where no authentication source is specified (for example, for unmanageable devices), Policy Manager passes the request to the next configured policy component for this service.

- If Policy Manager does not find the connecting entity in any of the configured authentication sources, it rejects the request.

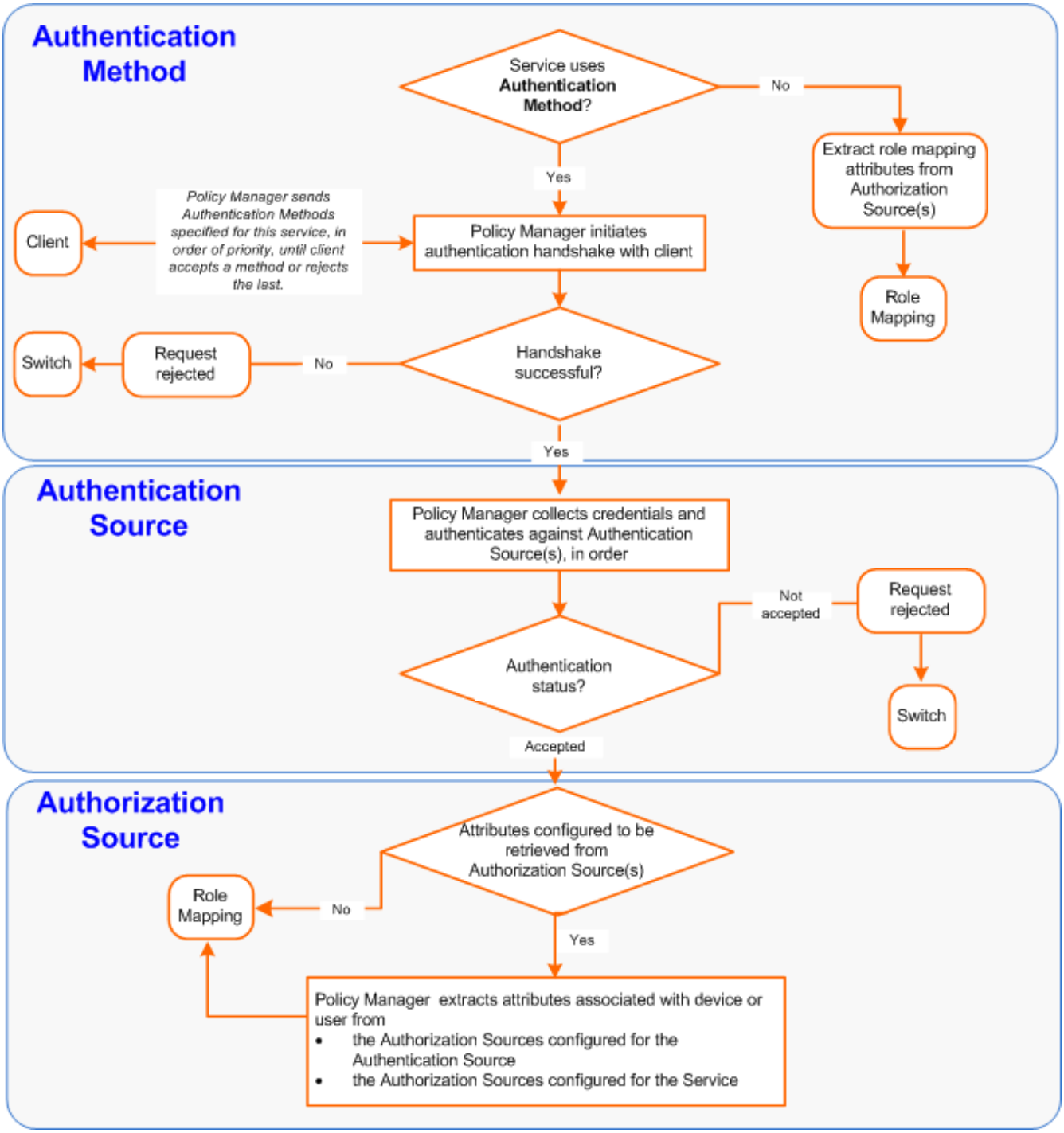
> **NOTE**
>
> Starting with the ClearPass 6.11.0 release, the database system also uses port **5433** in addition to port **5432**. Users should be aware that ports 5433 and 5432 must both be open when attempting to remotely connect to the Policy Manager database.

Authorization Source After Policy Manager successfully authenticates the user or device against an authentication source, it retrieves role-mapping attributes from each of the authorization sources configured for that authentication source. It also, optionally, can retrieve attributes from authorization sources configured for the service.

Authentication and Authorization Flow of Control

The flow of control for authentication takes the following components in sequence:

Figure 1. Authentication and Authorization Flow of Control



Adding and Modifying Authentication Methods

As a first step in the service-based processing, Policy Manager uses an authentication method to authenticate the user or device against an authentication source. You can create one or more instances of a default authentication method by assigning a unique name to each one. This customized authentication method can also be associated with a service .

Subtopics

Adding a Customized Authentication Method
Modifying an Existing Authentication Method
Authorize Authentication Method
CHAP Authentication Method
EAP-FAST Authentication Method
EAP-GTC Authentication Method
EAP-MD5 Authentication Method
EAP-MSCHAPv2
EAP-PEAP Authentication Method
EAP-PEAP-Public
EAP-PWD
EAP-TLS
EAP-TTLS
MAC-AUTH Authentication Method
MSCHAP
PAP Authentication Method
TEAP Authentication Method

Adding a Customized Authentication Method

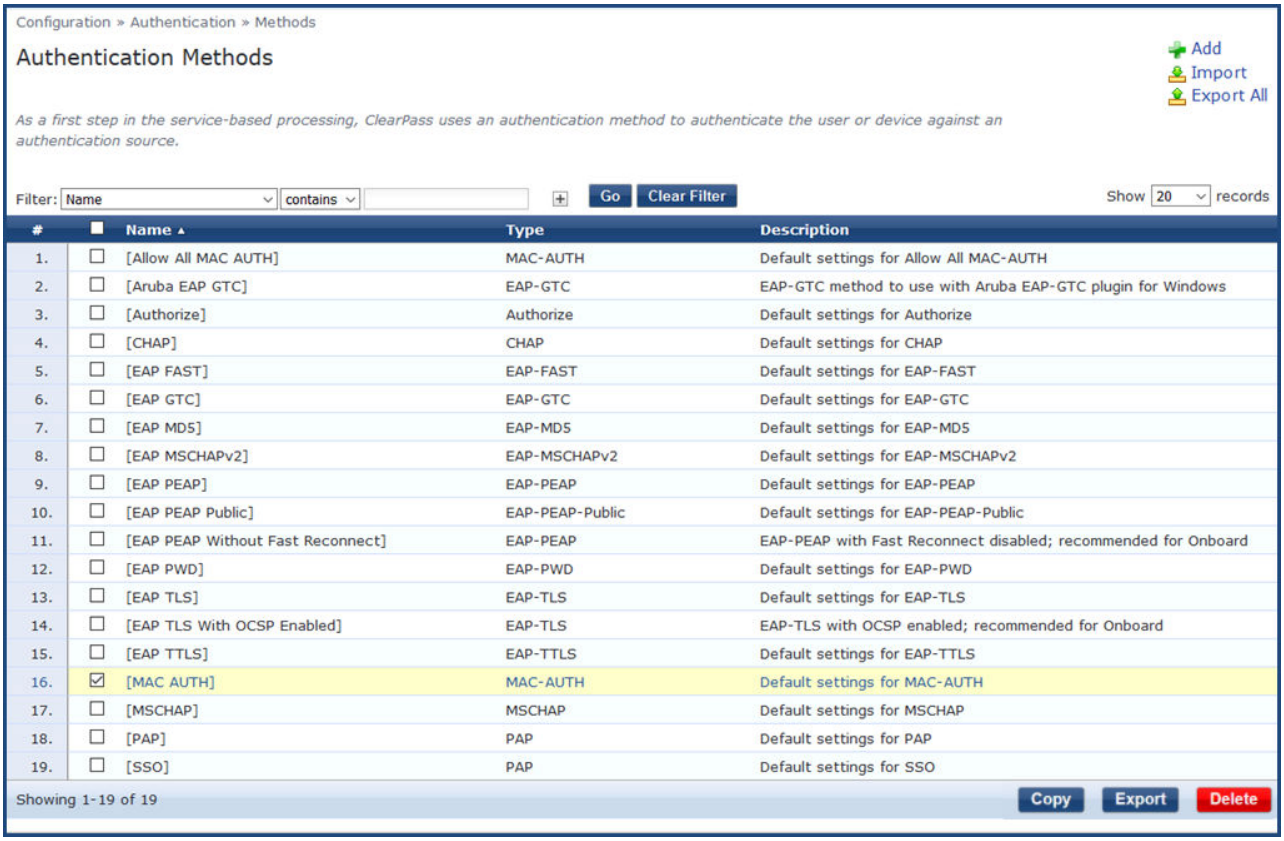
About this task

To add a customized authentication method:

Procedure

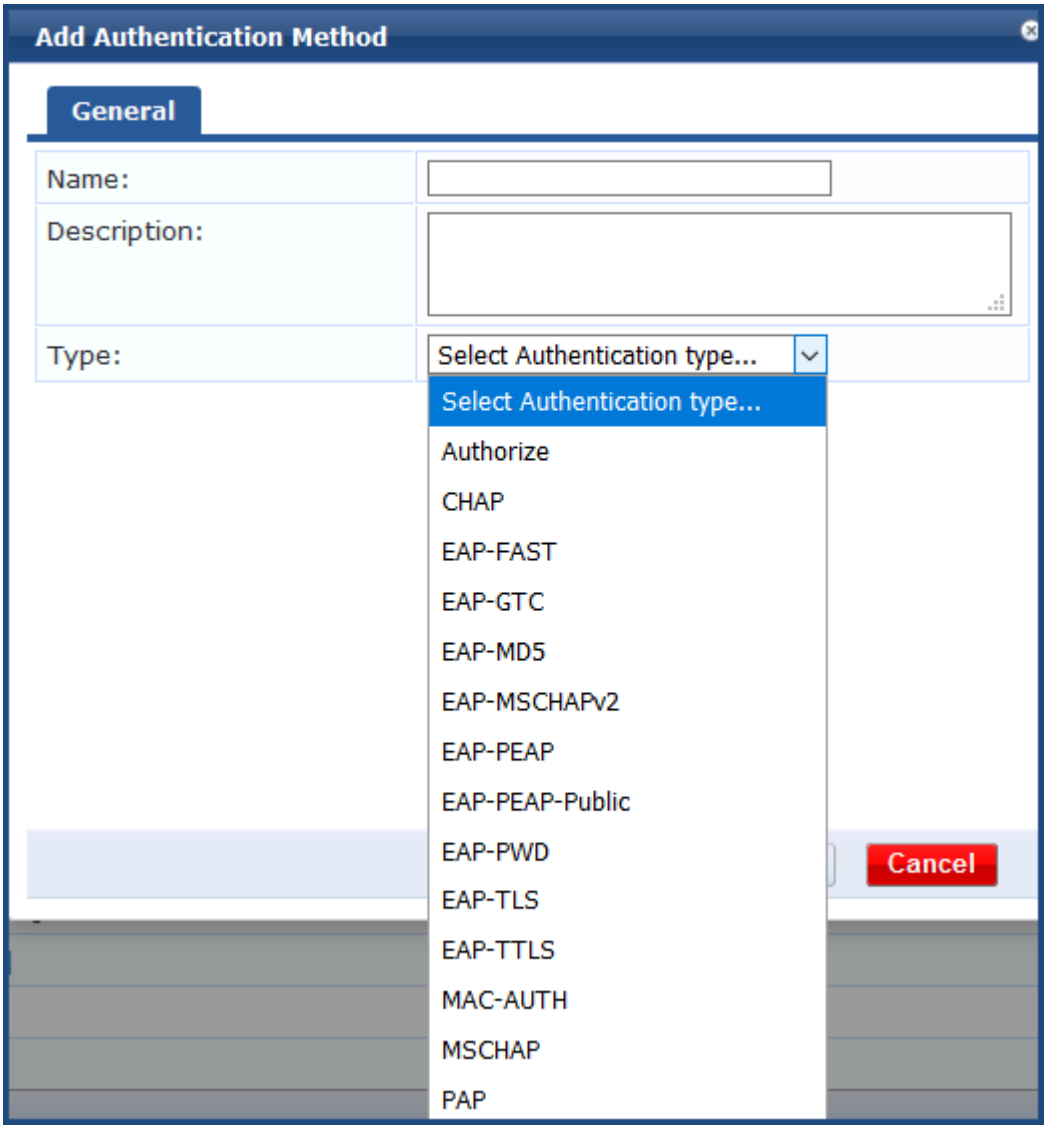
1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens. This page displays the list of authentication methods that Policy Manager provides by default.

Figure 1. Authentication Methods Page



2. Click **Add**. The **Add Authentication Method** page opens.

   Add Authentication Method Page



3. Enter the name and description of the new customized authentication method.
4. From the **Type** drop-down, select the authentication type.
5. Configure the customized authentication method of interest as described in the following sections:
   - Authorize Authentication Method
   - CHAP Authentication Method
   - EAP-FAST Authentication Method
   - EAP-GTC Authentication Method
   - EAP-MD5 Authentication Method
   - EAP-MSCHAPv2
   - EAP-PEAP Authentication Method

- EAP-PEAP-Public
- EAP-PWD
- EAP-TLS
- EAP-TTLS
- MAC-AUTH Authentication Method
- MSCHAP
- PAP Authentication Method
- TEAP Authentication Method

You can also create a new authentication method based off of an existing method by selecting any method, clicking **Copy**, and then modifying the copy as described in Modifying an Existing Authentication Method.

Modifying an Existing Authentication Method

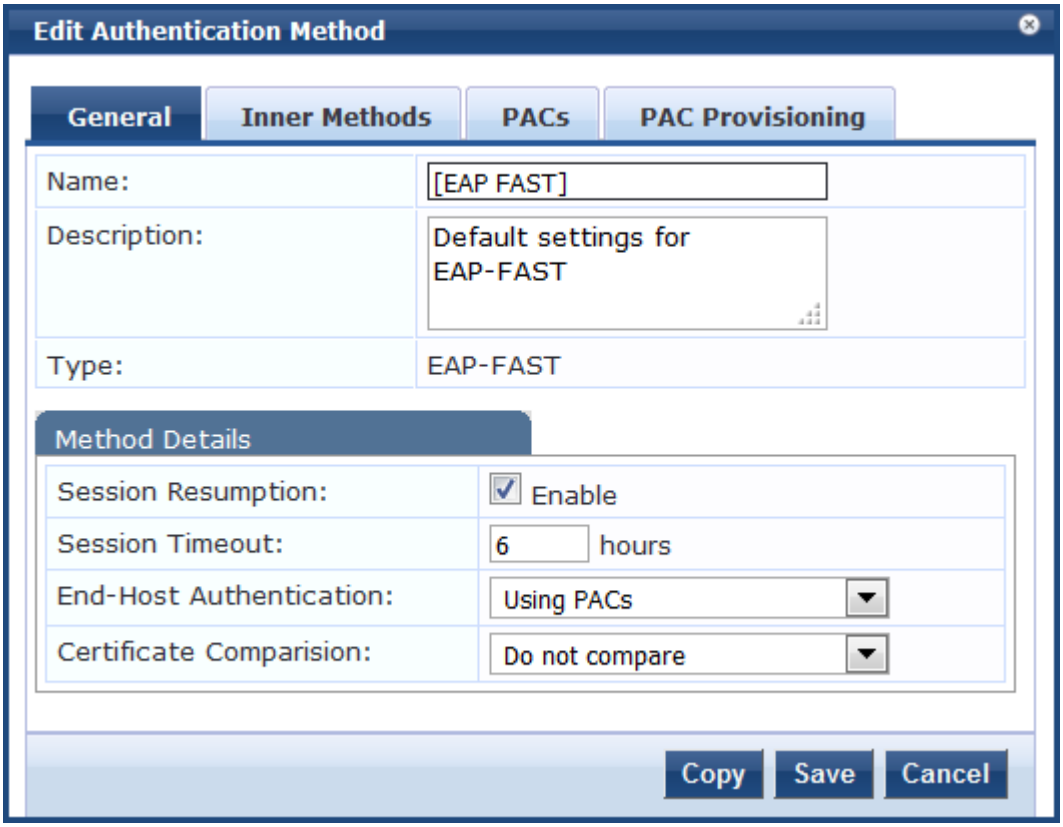
About this task

To modify an existing authentication method:

Procedure

1. Navigate to **Configuration** > **Authentication** > **Methods**. The **Authentication Methods** page opens.
2. Click the authentication method of interest. The **Edit Authentication Method** page opens.
3. Modify the selected authentication method fields as necessary.

Figure 1. Edit Authentication Method Page for EAP-FAST



4. Click **Save**.

Authorize Authentication Method

About this task

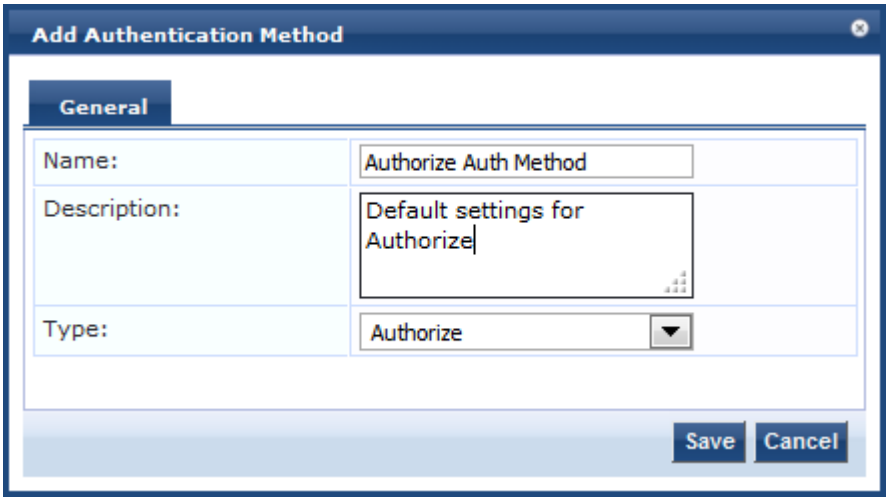
This is an authorization-only method that you can add with a custom name.

To add the Authorize authentication method:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Click **Add**. The **Add Authentication Method** configuration dialog opens:

Figure 1. Add Authorize Authentication Method Configuration Dialog



3. Specify the **Authorize Authentication Method** parameters as described in the following table:

Table 1. Authorize Authentication Method Parameters

Parameter	Action/Description
Name	Specify the label of the authentication m ethod.
Description	Provide additional information that helps to identify the authentication method.
Type	Select authentication method type **Auth orize.**

4. Click **Save**.

CHAP Authentication Method

About this task

Policy Manager includes the *Challenge-Handshake Authentication Protocol (CHAP)* authentication method. You can create one or more instances of the CHAP authentication method by assigning a customized name to each one. This method can also be associated to a service as an authentication method. TACACS+ authentication supports CHAP, ASCII, and PAP.

To add a customized instance of the CHAP authentication method to Policy Manager:

Procedure

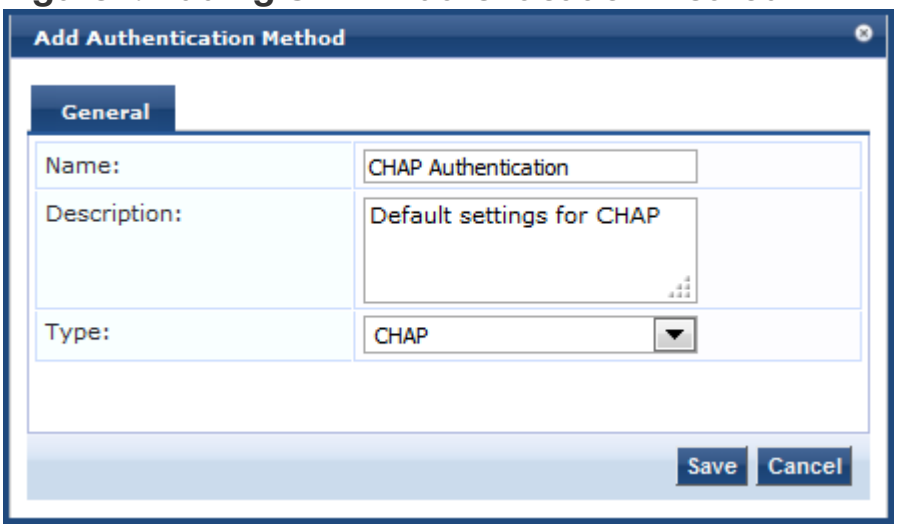
1. Navigate to **Configuration > Authentication > Methods**.

   The **Authentication Methods** page opens.

2. Click **Add**.

   The **Add Authentication Method** configuration dialog opens:

Figure 1. Adding CHAP Authentication Method



3. Specify the **CHAP** authentication method parameters as described in the following table:

Parameter	Description
Name	Specify a unique name for t authentication method.
Description	Provide the additional infor entify the authentication m
Type	Select **CHAP**.

4. Click **Save**.

EAP-FAST Authentication Method

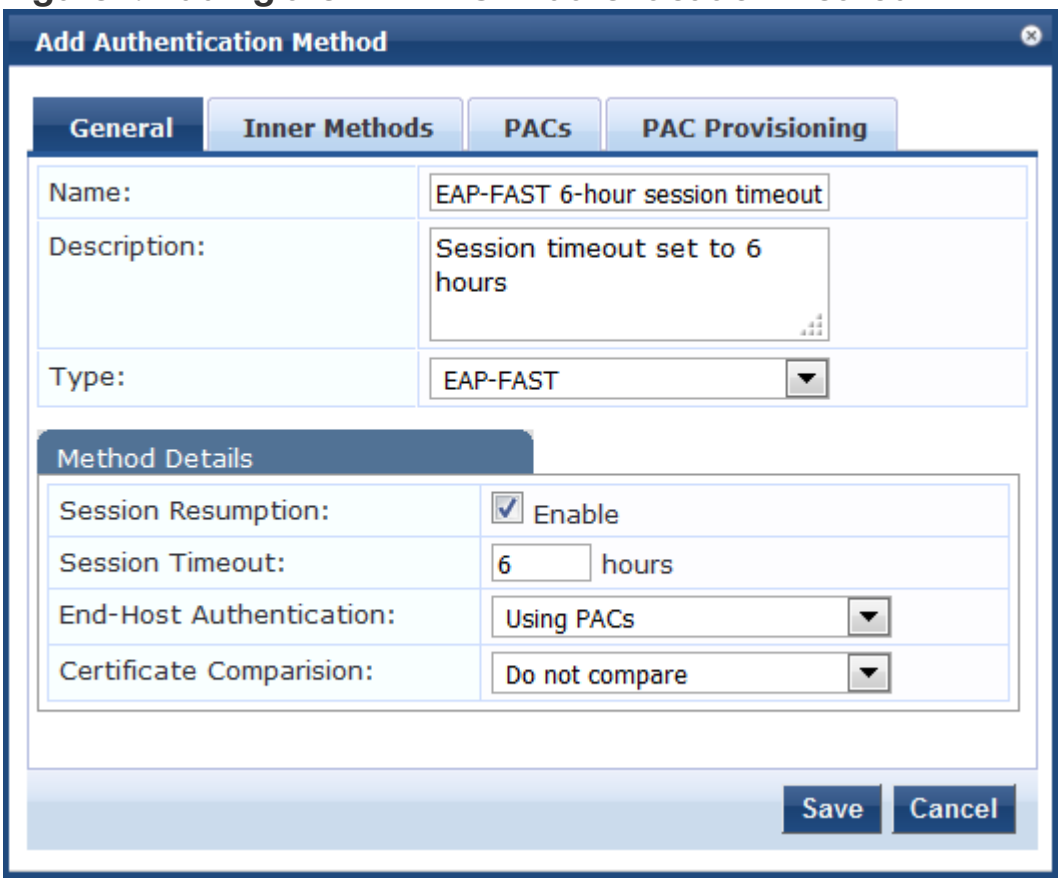
EAP-FAST (Flexible Authentication through Secure Tunneling) is an authentication method that encrypts EAP transactions within a TLS (Transport Layer Security) tunnel. Use the **General**, **Inner Methods**, **PACs**, and **PAC Provisioning** tabs to configure EAP-Fast authentication.

General Tab

To add the EAP-FAST authentication method to Policy Manager:

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens to the **General** tab.

Figure 1. Adding the EAP-FAST Authentication Method



3. Configure the EAP-FAST authentication service as described in Table 1 then click **Save**.

Table 1. Specifying the EAP - FAST > General Parameters

Parameter	Action/Description
Name	Specify the name of the authentication method.
Description	Provide the additional information that he lps to identify the authentication method .
Type	Select **EAP - FAST**.
Session Resumption	Caches EAP - FAST sessions on Policy Ma nager for reuse if the user/end - host reco nnects to the Policy Manager server withi n the session - timeout interval. By defaul t, this option is enabled.
Session Timeout	Caches EAP - FAST sessions on Policy Ma nager for reuse if the user/end - host reco nnects to Policy Manager within the sessi on - timeout interval. Specify the **Session Timeout** in the num ber of hours. • The default is **6** hours. • If the **Session Timeout** value is set to **0**, the cached sessions are not purge d.
End - Host Authentication	Specify one of the following end - host a uthentication methods: • **Using PACS** (Protected Access Crede ntials) • **Using Client Certificate** **NOTE** **Using PACs**.

Parameter	Action/Description
Certificate Comparison	Specify one of the following Certificate C omparison actions: • **Do not compare** • **Compare Distinguished Name (DN)** • **Compare Common Name (CN)** • **Compare Subject Alternative Name (SAN)** • **Compare CN or SAN** • **Compare Binary**

Table 2. Specifying the EAP - FAST > General Parameters

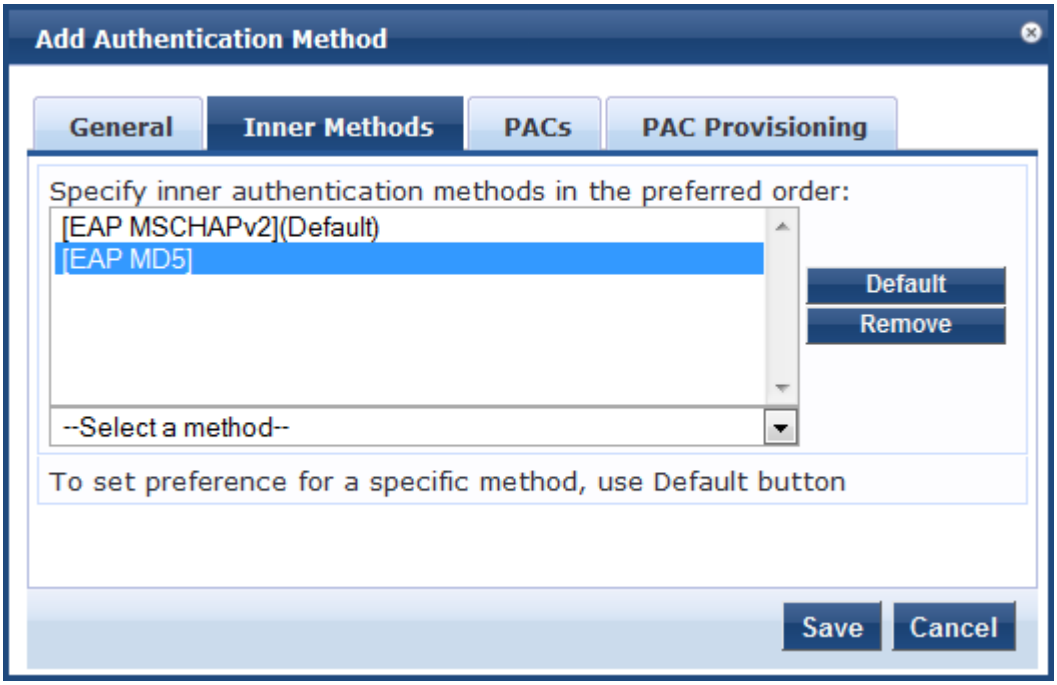
Parameter	Action/Description
Name	Specify the name of the authentication method.
Description	Provide the additional information that he lps to identify the authentication method .
Type	Select **EAP - FAST**.
Session Resumption	Caches EAP - FAST sessions on Policy Ma nager for reuse if the user/end - host reco nnects to the Policy Manager server withi n the session - timeout interval. By defaul t, this option is enabled.
Session Timeout	Caches EAP - FAST sessions on Policy Ma nager for reuse if the user/end - host reco nnects to Policy Manager within the sessi on - timeout interval. Specify the **Session Timeout** in the num ber of hours. • The default is **6** hours.

Parameter	Action/Description
	• If the **Session Timeout** value is set to **0**, the cached sessions are not purge d.
End - Host Authentication	Specify one of the following end - host a uthentication methods: • **Using PACS** (Protected Access Crede ntials) • **Using Client Certificate** **NOTE** **Using PACs**.
Certificate Comparison	Specify one of the following Certificate C omparison actions: • **Do not compare** • **Compare Distinguished Name (DN)** • **Compare Common Name (CN)** • **Compare Subject Alternname Subject Name (SAN)** • **Compare CN or SAN** • **Compare Binary**

Inner Methods Tab

The following figure displays the **EAP-FAST > Inner Methods** dialog:

Figure 2. EAP-FAST Authentication Method > Inner Methods Dialog



NOTE

In FIPS mode, the EAP-MD5 authentication method is not supported.

Table 3. EAP - FAST > Inner Methods Tab Parameters

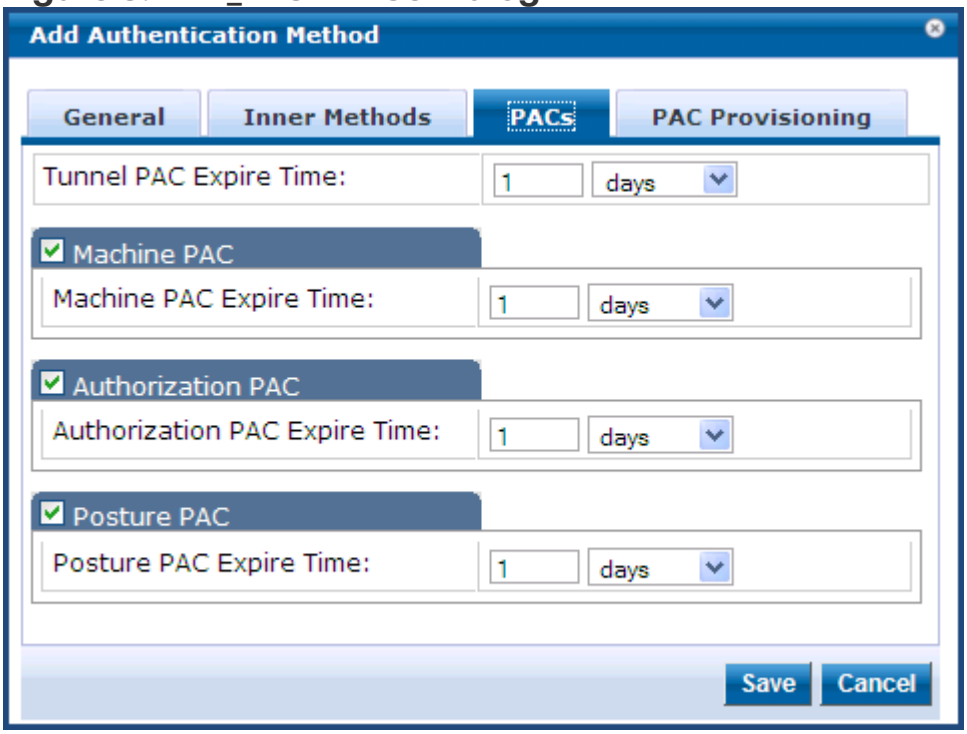
Parameter	Action/Description
Specify inner authentication methods in th e preferred order	Select a method from the drop - down list: • Aruba EAP GTC • EAP GTC • EAP MD5 • EAP MSCHAPV2 (Default) • EAP PWD • EAP TLS with OSCP Enabled • EAP TLS Functions available in this tab include: • To append an inner method to the displ ayed list, select from the **Select a meth od** drop - down list. The list can contain multiple inner methods, which Policy M

Parameter	Action/Description
	anager sends in priority order until neg otiation succeeds. • To remove an inner method from the dis played list, select the method and click **Remove**. • To set an inner method as the default in ner method (the method tried first), sel ect a method and click **Default**.

PACs (Protected Access Credential) Tab

The **PACs** dialog enables or disables Protected Access Credential (PAC) types. The following figure displays the **EAP-FAST > PACs** dialog:

Figure 3. EAP_FAST PACs Dialog



1. Specify the **Expiration Time** (number of **hours**, **days**, **weeks**, **months**, or **years**) for each of the Protected Access Credentials:
   a. Tunnel PAC Expire Time
   b. Machine PAC Expire Time
   c. Authorization PAC Expire Time

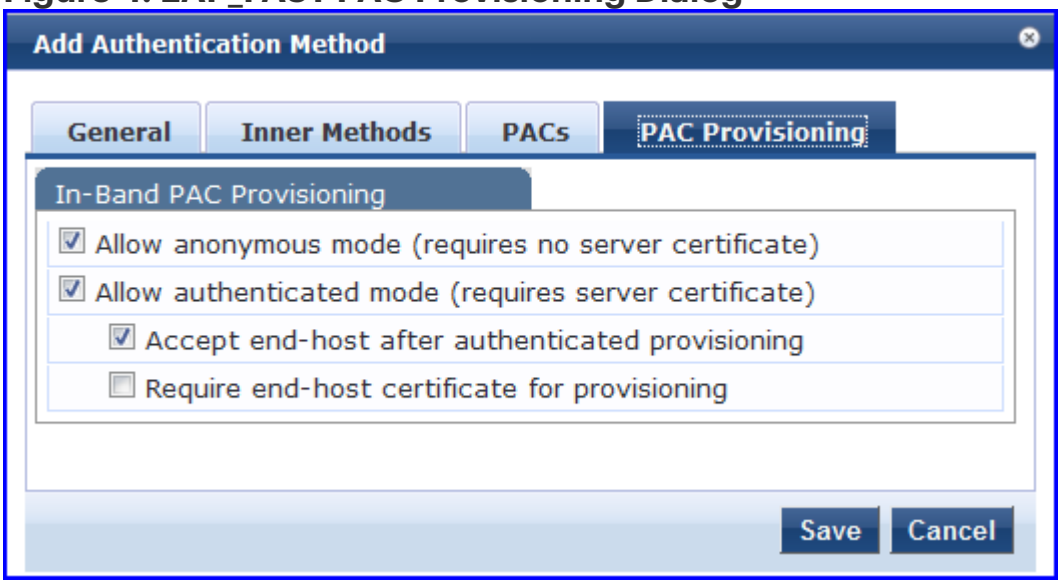
d. Posture PAC Expire Time

2. Select the **PAC Provisioning** tab.

PAC Provisioning Tab

The **PAC Provisioning** dialog controls anonymous and authenticated modes. The following figure displays the **EAP-FAST PAC > Provisioning** dialog:

Figure 4. EAP_FAST PAC Provisioning Dialog



1. Configure the PAC Provisioning parameters as described in Table 3.

2. When finished, click **Save**.

Table 4. EAP_FAST PAC Provisioning Parameters

Parameter	Action/Description	Considerations
In - Band PAC Provisioning		
Allow anonymous mode	When in anonymous mode, phase 0 of EAP_FAST provi sioning establishes an oute r tunnel without end - host /Policy Manager authentica tion. **NOTE** This mode is n ot as secure a	Authenticated mode is mo re secure than anonymous provisioning mode. After t he server is authenticated, the phase 0 tunnel is estab lished. The end - host and Policy Manager perform m utual authentication and p rovision on the end - host with an appropriate PAC (tu nnel or machine):

Parameter	Action/Description	Considerations
	s the authenti cated mode. After an outer tunnel is es tablished, the end - host a nd Policy Manager perform mutual authentication usin g MSCHAPv2, then Policy Manager provisions the en d - host with an appropriat e PAC (tunnel or machine).	• If both anonymous and authenticated provision ing modes are enabled and the end - host send s a cipher suite that su pports server authentic ation, Policy Manager pi cks the authenticated p rovisioning mode. • If the appropriate ciphe r suite is supported by the end - host, Policy M anager performs anony mous provisioning.
Allow authenticated mode	Enable to allow authenticat ed mode provisioning. When **Allow authenticated mode** is in phase 0, Policy Manager establishes the o uter tunnel inside a server - authenticated tunnel. Th e end - host authenticates the server by validating the Policy Manager certificate.	
Accept end - host after aut henticated provisioning	After the authenticated p rovisioning mode is compl ete and the end - host is p rovisioned with a PAC, Poli cy Manager rejects the end - host authentication. The end - host subsequent ly re - authenticates using t he newly provisioned PAC. When this field is enabled , Policy Manager accepts t he end - host authenticatio n in the provisioning mode itself. The end - host does	None.

Parameter	Action/Description	Considerations
	not have to re - authenticat e.	
Required end - host certifi cate for provisioning	In authenticated provisioni ng mode, the end - host a uthenticates the server by validating the server certifi cate, which results in a pro tected outer tunnel. The end - host is authentic ated by the server inside th is tunnel. When this field is enabled, the server can req uire the end - host to send a certificate inside the tun nel for the purpose of auth enticating the end - host.	None.

EAP-GTC Authentication Method

About this task

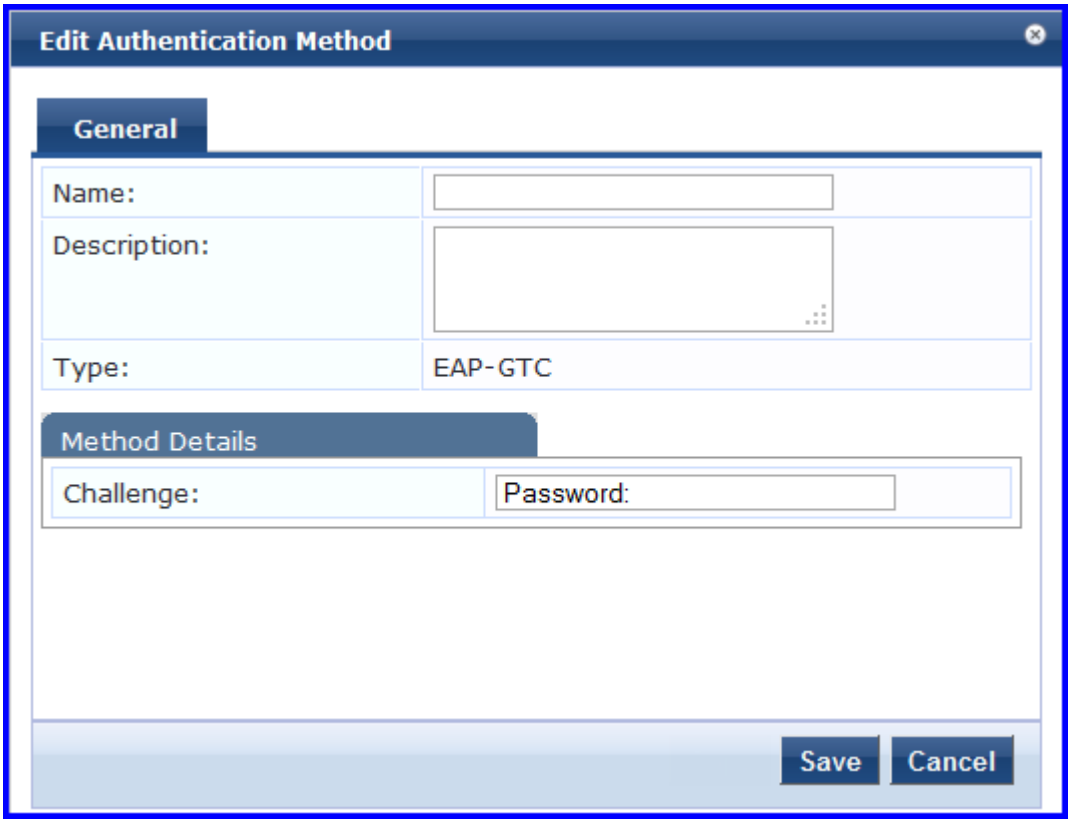
EAP-GTC (Generic Token Card) enables the exchange of clear-text authentication credentials across the network. EAP-GTC carries a text challenge from the authentication server and a reply generated by a security token.

To modify the EAP-GTC authentication method:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. From the list of Authentication Methods, select **EAP GTC**. The **Edit Authentication Method** dialog for **EAP-GTC** opens.

Figure 1. Edit EAP-GTC Authentication Method



3. Specify the **EAP-GTC > General** parameters as described in the following table:

Table 1. EAP - GTC Authentication Method Parameters

Parameter	Action/Description
Name	Specify the name of the au
Description	Optionally, provide the add t helps to identify this auth
Type	**EAP - GTC** is preselected.
Method Details	
Challenge	Optionally, specify or chan

4. Click **Save**.

EAP-MD5 Authentication Method

About this task

Policy Manager is packaged with the **EAP_MD5** authentication method. You can create one or more instances of the EAP-MD5 authentication method by assigning a customized name to each one. This method can also be associated to a service as an authentication method.

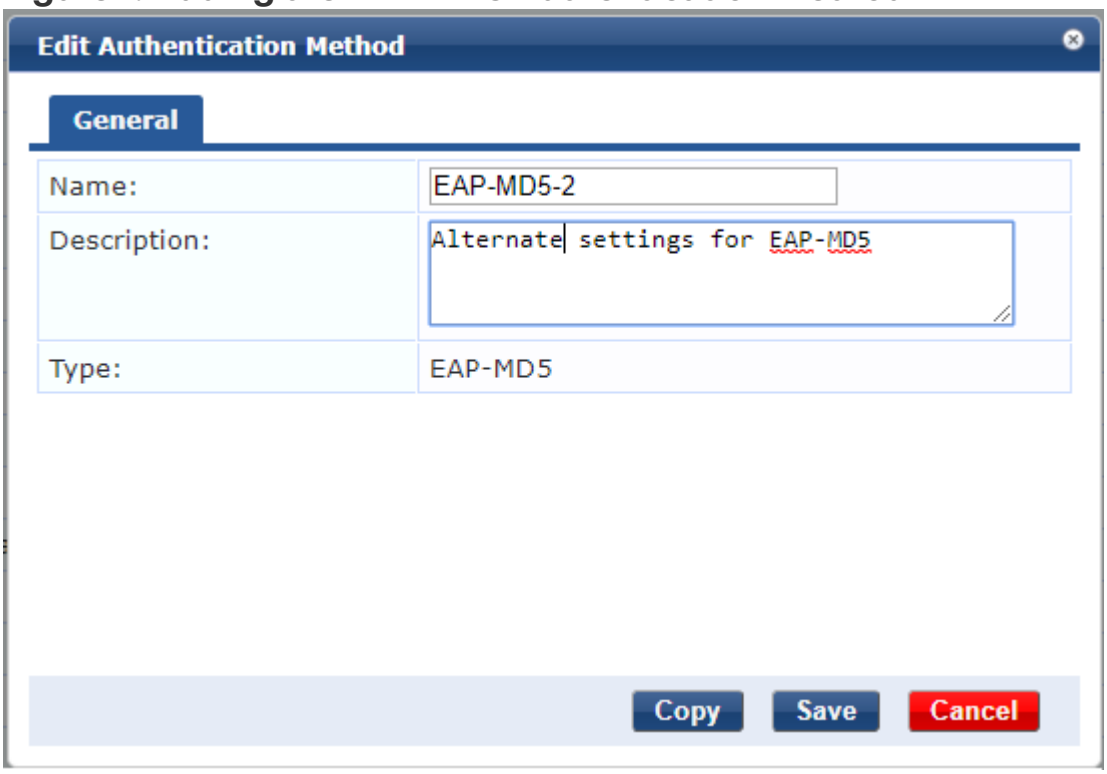
> **NOTE**
> In FIPS mode, the EAP-MD5 authentication method is not supported.

To add the EAP-MD5 authentication method:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Click **Add**. The **Add Authentication Method** configuration dialog opens:

Figure 1. Adding the EAP-MD5 Authentication Method



3. Specify the **EAP-MD5** parameters as described in the following table:

Table 1. CHAP Parameters

Parameter	Description
Name	Specify the name of the EAP n method.
Description	Provide the additional inform entify the authentication me
Type	Select **EAP - MD5**.

4. Click **Save**.

EAP-MSCHAPv2

About this task

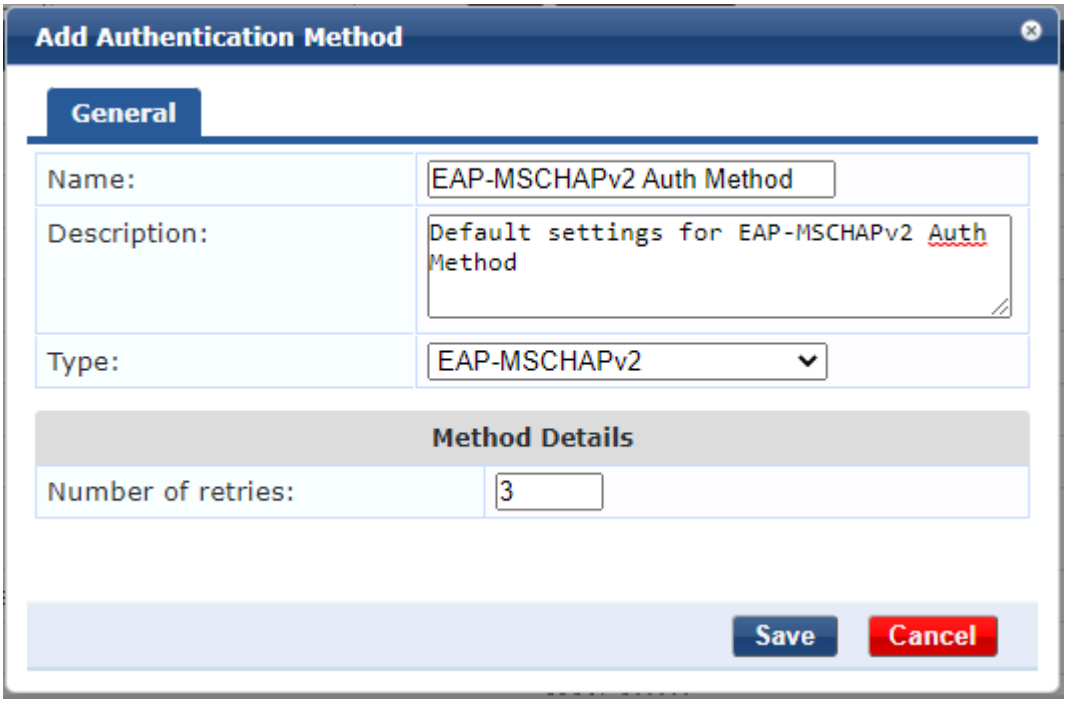
MS-CHAPv2 is an authentication protocol that Microsoft introduced with NT4.0 SP4 and Windows 98. The inner authentication protocol is Microsoft's CHAP (Challenge Handshake Authentication Protocol), meaning it allows authentication to databases that support the MS-CHAPv2 format, including Microsoft NT and Microsoft Active Directory.

To add the EAP-MSCHAPv2 authentication method to Policy Manager:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens:

Figure 1. Adding the EAP-MSCHAPv2 Authentication Method



3. Specify the **EAP-MSCHAPv2** parameters as described in the table below, then click **Save**.

Table 1. EAP - MSCHAPv2 Parameters

Parameter	Description
Name	Specify the name of the authentication method.
Description	Optionally, provide the additional information that helps to identify the authentication method.
Type	Select **EAP - MSCHAPv2**.
Number of retries	The number of authentication retries in response to a login failure or if the request contains an incorrect password. The default value is **3** retries. This field supports a range of **0 - 5** retries, with a setting of **0** retries disabling the authentication retry feature.

EAP-PEAP Authentication Method

EAP-Protected Extensible Authentication Protocol (EAP-PEAP) is a protocol that creates an encrypted (and more secure) channel before the password-based authentication occurs. PEAP is an 802.1X authentication method that uses server-side public key certificate to establish a secure tunnel in which the client authenticates with server. The PEAP authentication creates an encrypted SSL/TLS tunnel between client and authentication server. The exchange of information is encrypted and stored in the tunnel ensuring that the user credentials are kept secure.

General Tab

To add the EAP-PEAP authentication method to Policy Manager:

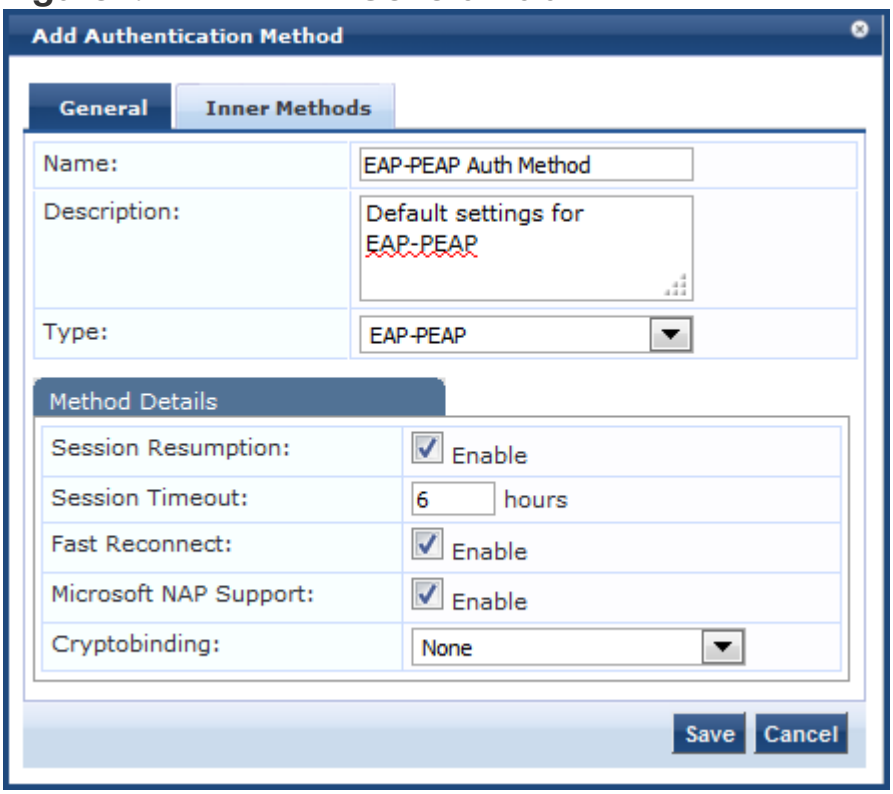
1. Navigate to **Configuration > Authentication > Methods**.

   The **Authentication Methods** page opens.

2. Select the **Add** link.

   The **Add Authentication Method** dialog opens:

Figure 1. EAP-PEAP > General Tab



3. Specify the **EAP-PEAP > General** parameters as described in the following table, then click **Save**.

Table 1. EAP - PEAP > General Parameters

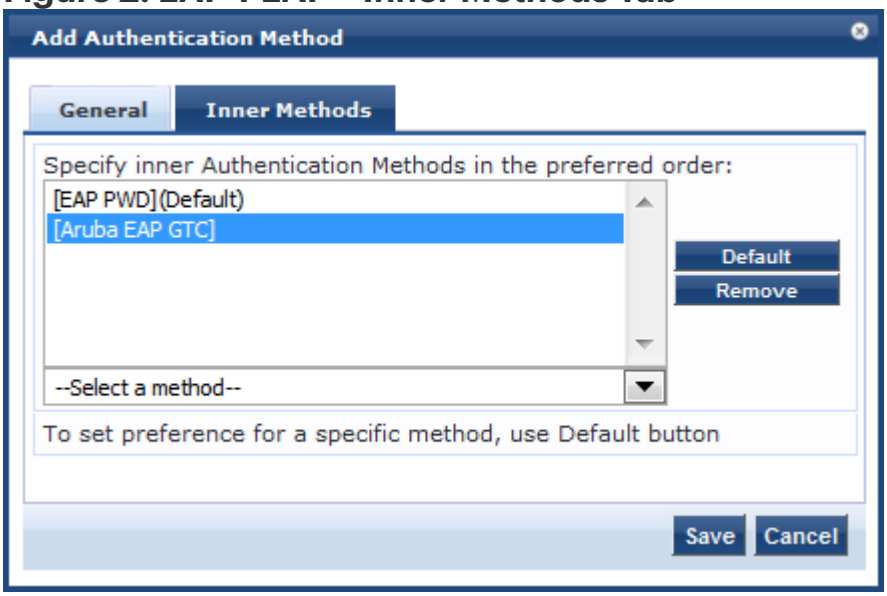
Parameter	Action/Description
Name	Specify the name of the authentication method.
Description	Optionally, provide the additional informa tion that helps to identify the authenticati on method.
Type	Select **EAP - PEAP**.
Method Details	
Session Resumption	Check the **Session Resumption** check b ox if you intend to enable **Fast Reconnec t**.
Session Timeout	Caches EAP - PEAP sessions on the Poli cy Manager server for reuse if the user/cl ient reconnects to the Policy Manager se rver within the session timeout interval. If session timeout value is set to 0, the cac hed sessions are not purged.
Fast Reconnect	Enable this check box to allow fast recon nect. When **Fast Reconnect** is enabled, the inn er method that takes place inside the se rver authenticated outer tunnel is also by passed. This makes the process of reaut hentication faster. For **Fast Reconnect** to work, session resumption must be enable d.
Microsoft NAP Support	Check this check box to enable Network Access Protection (NAP) on this Policy M anager server.
Cryptobinding	Cryptobinding protects tunnel methods against man - in - the - middle attacks. C

Parameter	Action/Description
	ryptographic binding focuses on protecting the server rather than the peer. Specify one of the following cryptobinding options: • None (the default) • Optional • Required

Inner Methods Tab

The tunneled method is frequently referred to as the "inner method." The **Inner Methods** tab controls the inner methods for the **EAP-PEAP** authentication method.

Figure 2. EAP-PEAP > Inner Methods Tab



NOTE

In FIPS mode, the EAP-MD5 authentication method is not supported.

Specify the **EAP-PEAP Inner Methods** parameters as described in the following table:

Table 2. EAP - PEAP Inner Methods Tab Parameters

Parameter	Description
Specify inner authentication methods in the preferred order	To append an inner method to the displayed list, select it from the **Select a method** drop - down list. The list can contain multiple inner methods, which Policy Manager sends in priority order until negotiation succeeds. Select any method available in the current context from the drop - down list. Inner methods available include: • Aruba EAP - GTC • EAP - GTC • EAP - MD5 • EAP - MSCHAPv2 • EAP - PWD • EAP - TLS with OSCP enabled • EAP - TLS
Default	To set an inner method as the default (the method tried first), select it and click **Default**.
Remove	To remove an inner method from the displayed list, select the method and click **Remove**.

EAP-PEAP-Public

The **EAP-PEAP-Public** method is used for authenticating and providing a secured wireless guest access to the endpoints. To provide a secured wireless guest access, the Wi-Fi Protected Access (WPA) is provided for a publicly known username and password. This ensures that every device gets a unique wireless session key that is used to encrypt the traffic and provide secured wireless access without intruding into the privacy of others, though the same username and password are shared to all devices.

General

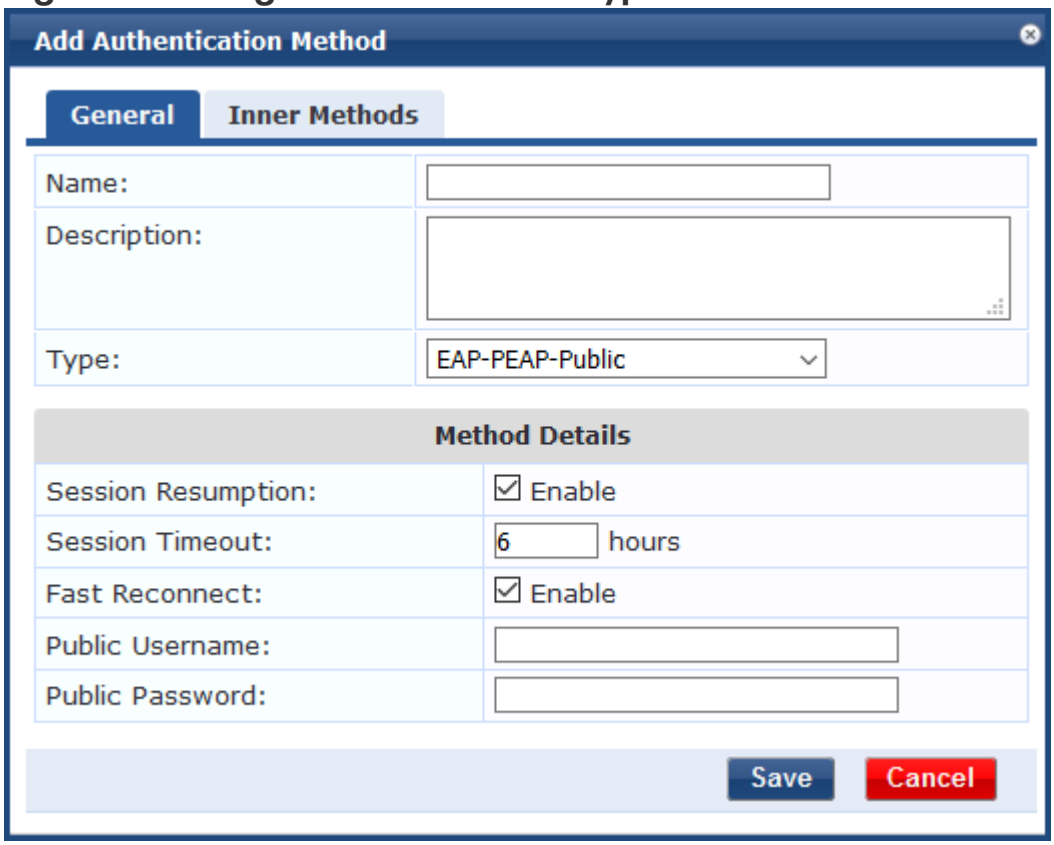
About this task

The **General** tab labels the authentication method and defines session details. To add a customized instance of the EAP-PEAP-Public authentication method to Policy Manager:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens to the **General** tab.
3. In the **Type** field, select **EAP-PEAP**. The **Add EAP-PEAP-Public - General** tab opens:

Figure 1. Adding EAP-PEAP-Public Type - General Tab



4. Specify the customized **EAP-PEAP-Public - General** parameters as needed:

Table 1. EAP - PEAP - Public - General Tab Parameters

Parameter	Action/Description
Name	Specify the name of the authentication method.

Parameter	Action/Description
Description	Provide the additional information that helps to identify the authentication method .
Type	Select the **EAP - PEAP - Public** authentication method type.
Session Resumption	Caches EAP - PEAP - Public sessions on Policy Manager for reuse if the user/client reconnects to Policy Manager within the session timeout interval. By default, this option is enabled.
Session Timeout	Caches EAP - PEAP - Public sessions on Policy Manager for reuse if the user/client reconnects to Policy Manager within the session timeout interval in hours. If session timeout value is set to 0, the cached sessions are not purged. The default session timeout is 6 hours.
Fast Reconnect	Enable this check box to allow fast reconnect. When fast reconnect is enabled, the inner method in the server - authenticated outer tunnel is also bypassed. This makes the process of reauthentication faster. For the fast reconnect to work, **Session Resumption** must be enabled.
Public Username	Enter the Guest username. In this context, enter '*public*'.
Public Password	Enter the Guest password. In this context, enter '*public*'.

Inner Methods

The **Inner Methods** tab controls the inner methods for the **EAP-PEAP-Public** authentication method. The following figure is an example of the **EAP-PEAP-Public - Inner Methods** tab:

Figure 2. Adding EAP-PEAP-Public Type - Inner Methods Tab

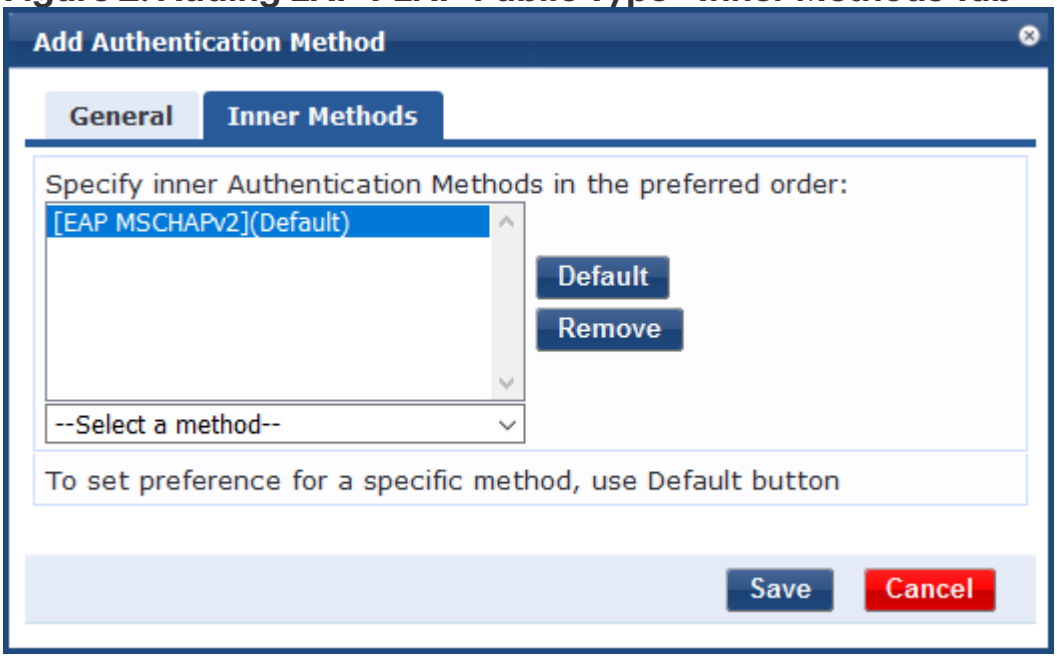


Table 2. EAP - PEAP - Public Inner Methods Tab Parameters

Parameter	Description
Specify inner Authentication Methods in th e preferred order	Select the inner authentication method ava ilable from the drop - down list. In this con text, only the **EAP - MSCHAPv2** method is available. The following functions are availa ble in this tab: • To append an inner method to the disp layed list, select it from the drop - dow n list. The list can contain multiple inner methods, which Policy Manager sends i n priority order until negotiation succee ds. • To remove an inner method from the dis played list, select the method and click **Remove**. • To set an inner method as the default (t he method tried first), select it and click **Default**.

EAP-PWD

About this task

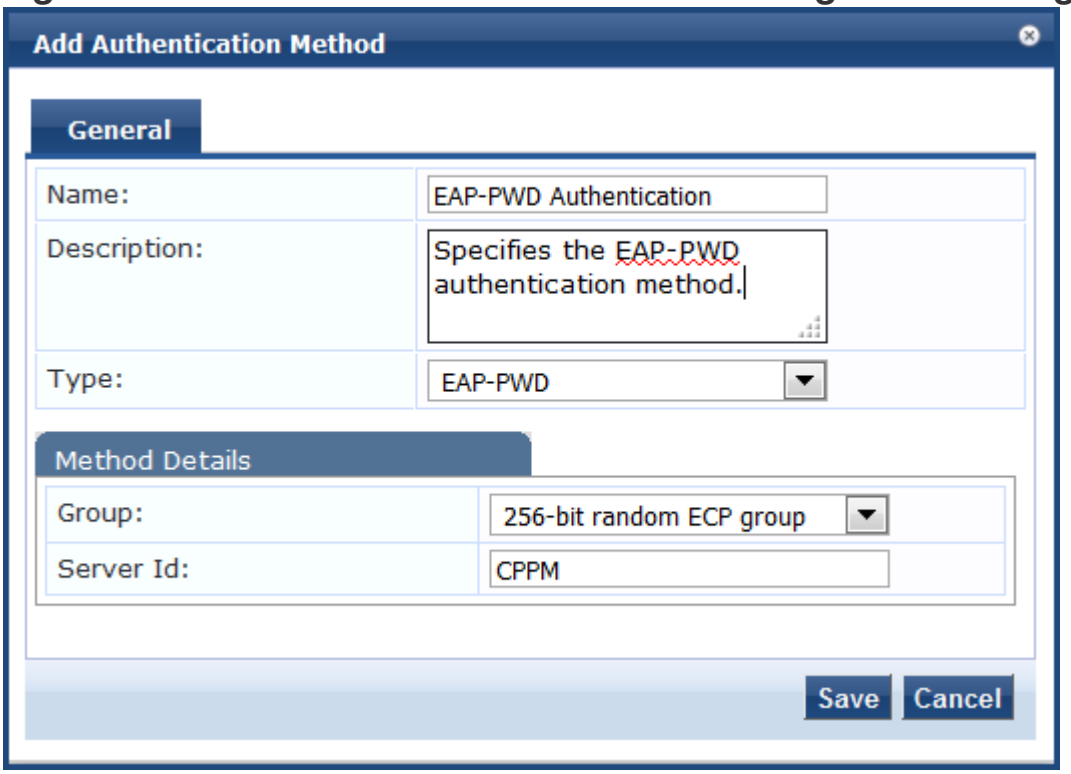
EAP-PWD (Password) is an EAP authentication method that uses a shared password for authentication. EAP-PWD addresses the problem of password-based authenticated key exchange using a possibly weak password for authentication to derive an authenticated and cryptographically strong shared secret.

To add the EAP-PWD authentication method:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens:

Figure 1. EAP-PWD Authentication Method Configuration Dialog



3. Specify the **EAP-PWD** parameters as described in the following table, then select **Save**.

Table 1. EAP - PWD Parameters

Parameter	Description
Name	Enter the name of the authentication r

Parameter	Description
Description	Provide the additional information tha entify the authentication method.
Type	Select **EAP - PWD**.
Method Details	
Group	Select the group from the drop - dow Each party to the exchange derives ep keys with respect to a particular set of rameters that is a *group*. A group can be based on Finite Field C hy (FFC) or Elliptic Curve Cryptograph
Server ID	Specify the string that identifies the P ger server to the peer.

EAP-TLS

About this task

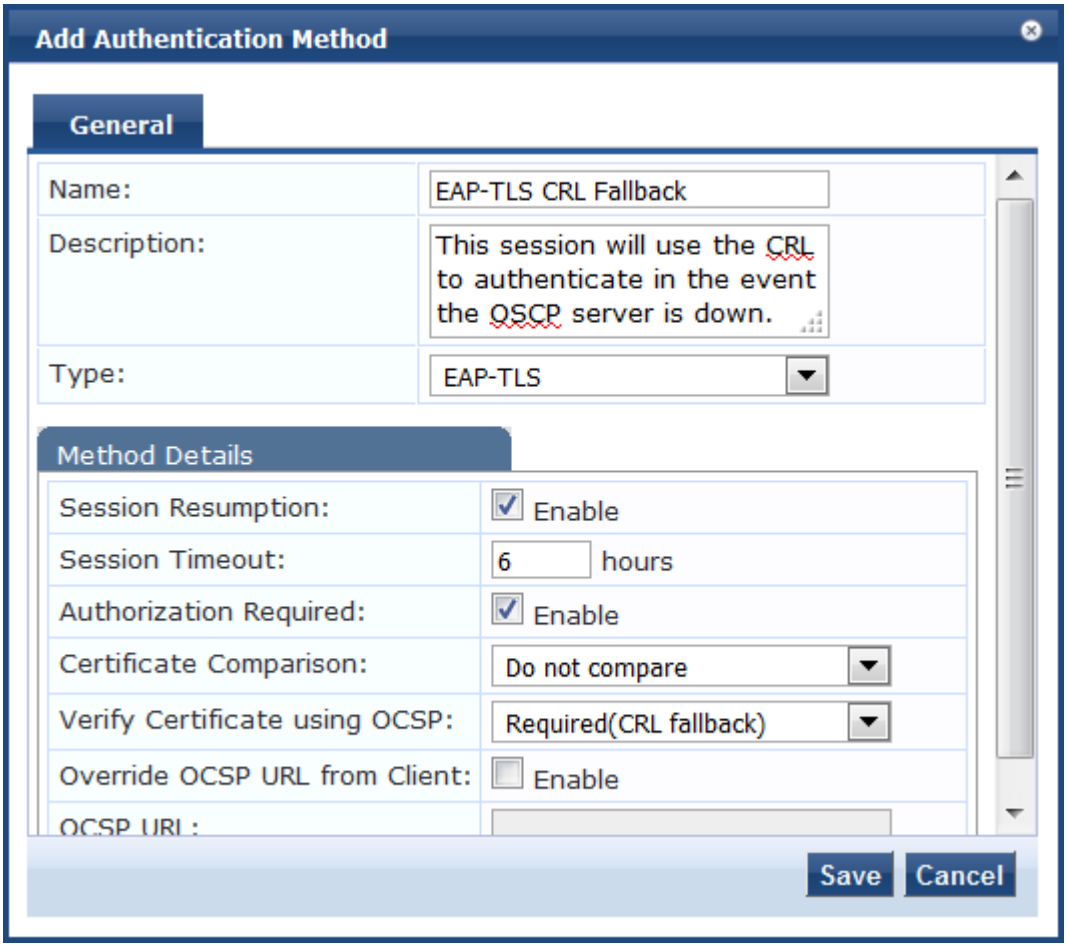
EAP-Transport Layer Security (EAP-TLS) requires an exchange of proof of identities through public key cryptography (such as digital certificates). EAP-TLS secures this exchange with an encrypted TLS tunnel, which helps to resist dictionary or other attacks.

To add the EAP-TLS authentication method:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Click **Add**. The **Add Authentication Method** dialog opens.

Figure 1. EAP-TLS Authentication Method Dialog



3. Specify the **Add Authentication Method** parameters as described in the following table, then click **Save**.

Table 1. EAP_TLS Authentication Method Parameters

Parameter	Action/Description
Name	Specify the name of the authentication method.
Description	Provide the additional information that he lps to identify the authentication method (recommended).
Method Details	
Type	Select **EAP_TLS**.
Session Resumption	Enable this option to cache EAP - TLS ses sions on the Policy Manager server for re use if the user or client reconnects to th

Parameter	Action/Description
	e Policy Manager server within the sessio n timeout interval. This option is enabled by default.
Session Timeout	Specify the duration in hours for the cac hed EAP - TLS sessions to be retained. Th e default is 6 hours.
Authorization Required	This parameter is enabled by default. Spe cify whether to perform an authorization check.
Certificate Comparison	Specify the type of certificate compariso n (identity matching) upon presenting Po licy Manager with a client certificate: • To skip the certificate comparison, ch oose **Do not compare.** • To compare specific attributes, choos e **Compare Common Name (CN)**, **Co mpare Subject Alternate Name (SAN)** , or **Compare CN or SAN**. • To perform a binary comparison of th e stored (in the client record in Active Directory or another LDAP - complian t directory) and presented certificates , choose **Compare Binary**.
Verify Certificate using OCSP	• If the certificate is to be verified by t he Online Certificate Status Protocol (OCSP), select **Optional** or **Required**. • To not verify the certificate, select **No ne**. • If an OCSP server is not accessible to perform certificate validation, you can skip the OCSP check and proceed wit h the CRL validation result by selectin g **Required (CRL fallback)**.

Parameter	Action/Description
Override OCSP URL from the Client	Select this option to use a different URL for OCSP. After this option is enabled, you can ente r a new URL in the **OCSP URL** field.
OCSP URL	If the **Override OCSP URL from the Clien t** field is enabled, enter the replacement URL.

EAP-TTLS

EAP-Tunneled Transport Layer Security (EAP-TTLS) is designed to provide authentication that is similar to EAP-TLS, but each user does not require a certificate be issued. The certificates are issued only to authentication servers.

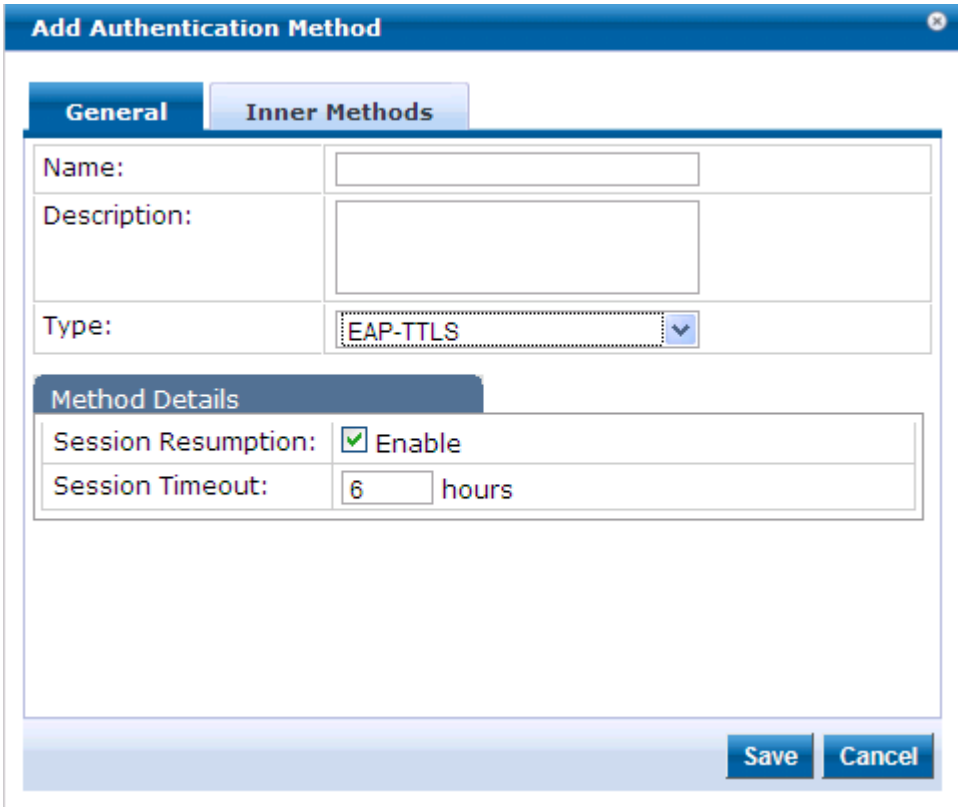
Use the **General** and **Inner Methods** tabs to configure the **EAP-TTLS** authentication method.

General Tab

The **General** tab labels the method and defines session details.

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens:

Figure 1. EAP-TTLS - General Tab



3. Specify the **EAP-TTLS** parameters as described in the table below, then click **Save**.

Table 1. EAP - TTLS - General Tab Parameters

Parameter	Description
Name	Specify the name of the authentication method.
Description	Provide the additional information that helps to identify the authentication method.
Type	Select the type of authentication. In this context, select **EAP - TTLS**.

NOTE

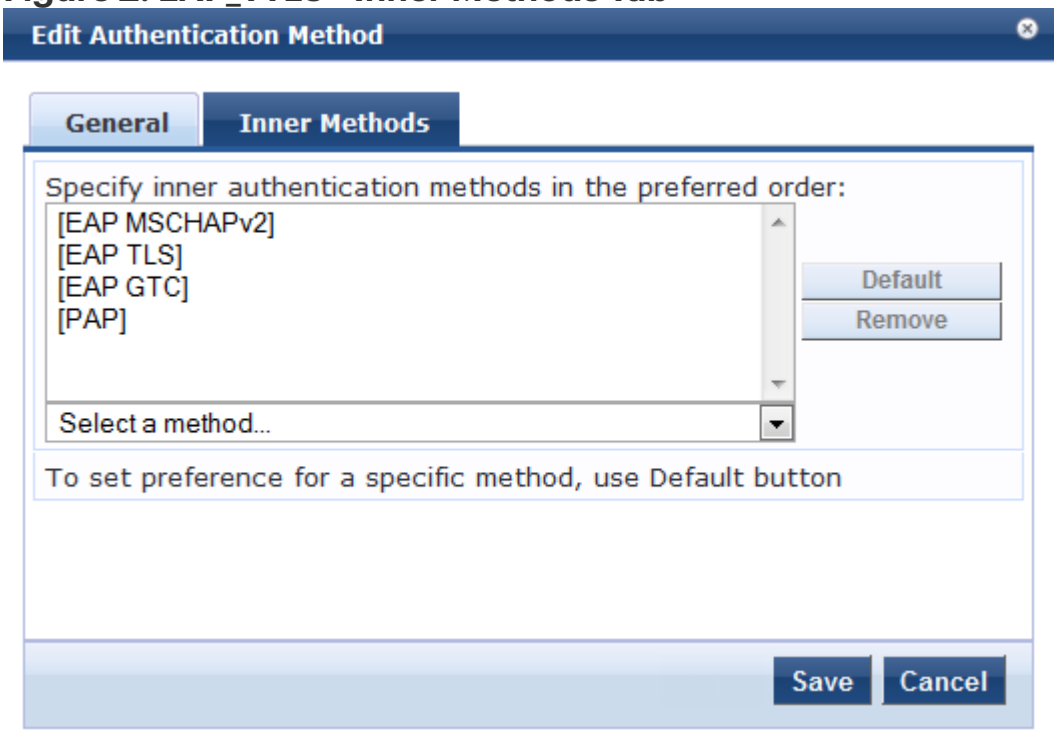
ClearPass Policy Manager in the FIPS (**Administration > Server Manager > Server Configuration > FIPS** tab) mode.

Parameter	Description
Method Details	
Session Resumption	Caches EAP - TTLS sessions on Policy Manager for reuse if the user/client reconnects to Policy Manager within the session timeout interval.
Session Timeout	Specify the duration in hours for the EAP - TTLS sessions to be cached.

Inner Methods Tab

The **Inner Methods** tab controls the inner methods for the **EAP-TTLS** method. The following figure is an example of the **EAP-TTLS - Inner Methods** tab:

Figure 2. EAP_TTLS - Inner Methods Tab



The following table describes the **EAP-TTLS - Inner Methods** parameters:

Table 2. EAP - TTLS - Inner Methods Tab Parameters

Parameter	Description
Specify inner authentication methods in the preferred order	Select any method available in the current context from the drop - down list.

Parameter	Description
	• To append an inner method to the displayed list, select it from the drop - down list. The list can contain multiple inner methods, which Policy Manager sends in priority order until negotiation succeeds. • To remove an inner method from the displayed list, select the method and click **Remove**. • To set an inner method as the default (the method that tried first), select it and click **Default**. **NOTE** ClearPass Policy Manager in the FIPS (**Administration > Server Manager > Server Configuration > FIPS** tab) mode.

MAC-AUTH Authentication Method

About this task

The **MAC_AUTH** authentication type must be used exclusively in a MAC-based authentication service. When the MAC_AUTH method is selected, Policy Manager makes internal checks to verify that the request is a **MAC_Authentication** request and not a spoofed request. In tunneled EAP methods, authentication and posture credential exchanges occur inside a protected outer tunnel. The MAC-AUTH method contains the **General** dialog that labels the authentication method and defines session details.

To add the MAC-AUTH authentication method to Policy Manager:

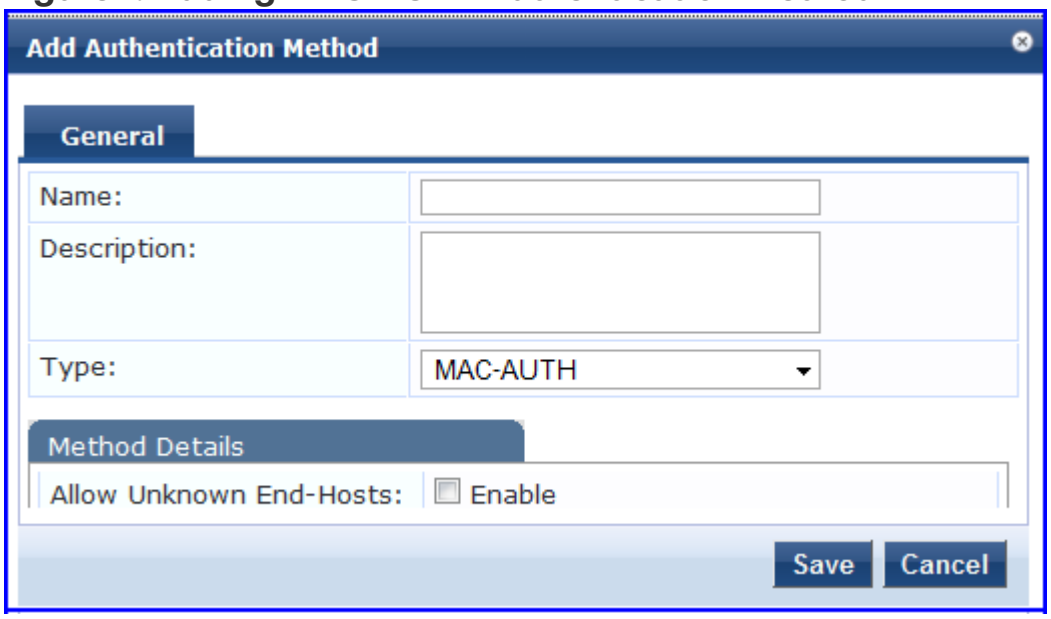
Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens to the **General** tab.

3. In the **Type** field, select **MAC-AUTH**.

   The following figure is an example of the **MAC-AUTH > General** dialog:

Figure 1. Adding MAC-AUTH Authentication Method



The following table describes the **MAC-Auth** parameters:

Table 1. MAC - Auth Parameters

Parameter	Action/Description
General	
Name	Specify the name of the authentication method.
Description	Provide the additional information that he lps to identify the authentication method (recommended).
Type	Select the **MAC - AUTH** type of authentic ation.
Method Details	
Allow Unknown End - Hosts	Select this check box to enable further policy processing of MAC authentication requests of unknown clients.

Parameter	Action/Description
	If this is not enabled, Policy Manager aut omatically rejects a request whose MAC a ddress is not in a configured authenticati on source. This setting is enabled, for exa mple, when you want Policy Manager to t rigger an audit for an unknown client. By selecting this check box and enabling au dit (see Configuring Audit Servers), you can trigger an audit of an unknown client .

MSCHAP

About this task

The MS-CHAP authentication method authenticates remote Windows-based workstations, integrating the functionality to which LAN-based users are accustomed with the hashing algorithms used on Windows networks. MS-CHAP uses a challenge-response mechanism to authenticate connections without sending any passwords. The MSCHAP method contains the **General** tab that labels the authentication method and defines session details.

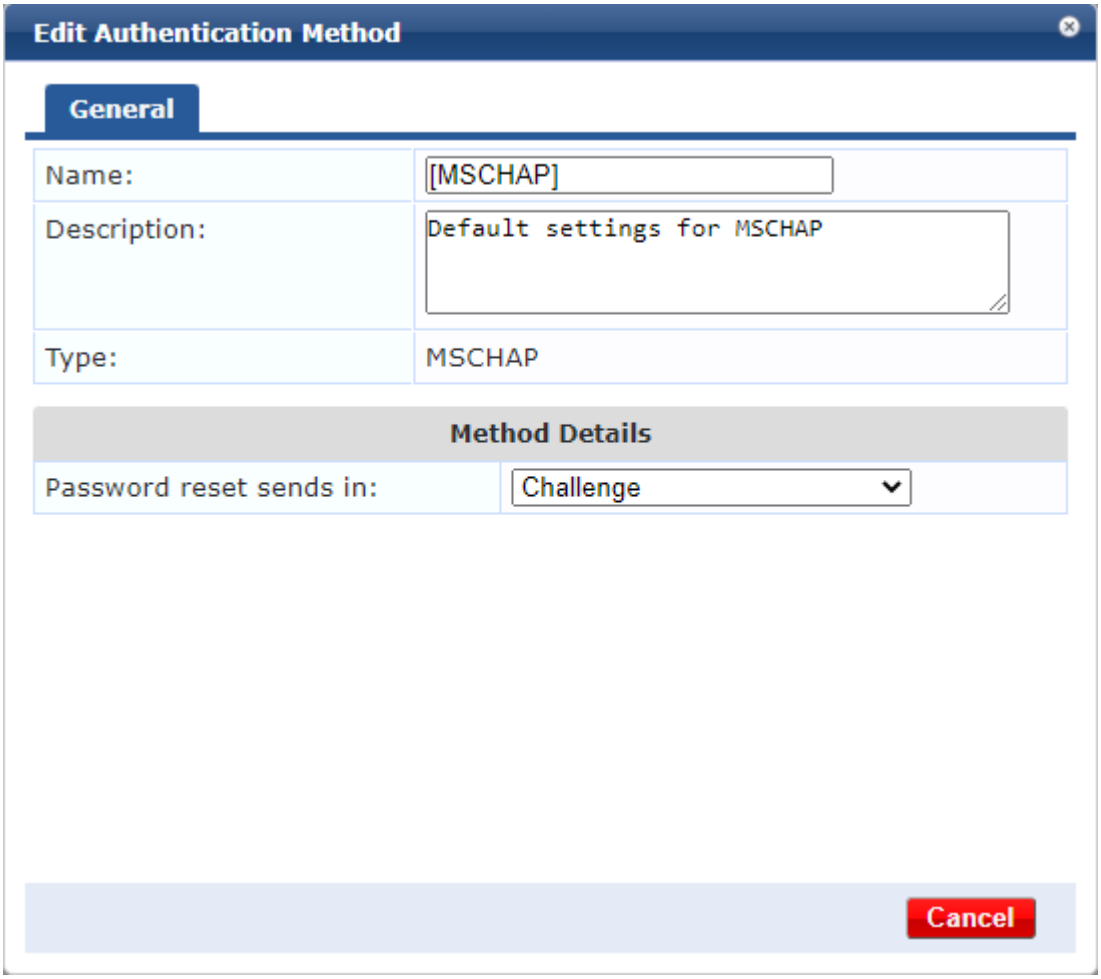
To add the MSCHAP authentication method to Policy Manager:

Procedure

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens to the **General** tab.
3. In the **Type** field, select **MSCHAP**.

   The following figure is an example of the **MSCHAP - General** tab:

   Figure 1. MSCHAP - General Tab



The following table describes the **MSCHAP - General** parameters:

Table 1. MSCHAP - General Tab Parameters

Parameter	Description
Name	Specify the name of the authentication method.
Description	Provide the additional information that helps to identify the authentication method.
Type	Select the type of authentication. In this context, select **MSCHAP**.
Password reset sends in	Select a password reset response for MSCHAP. The default value for the **Password reset sends in** field is **Challenge**, which triggers typical password reset capabil

Parameter	Description
	ities. Select **Reject** for services that requ ire a non - standard response to trigger t his behavior, such as those using Cisco A nyConnect clients with Cisco Adaptive S ecurity Appliance (ASA) workflows with C learPass.

PAP Authentication Method

About this task

The Password Authentication Protocol (PAP) is an authentication protocol in which the user name and password are sent to the remote access server in unencrypted form.

To add the PAP authentication method to Policy Manager:

Procedure

1. Navigate to **Configuration > Authentication > Methods**.

   The **Authentication Methods** page opens.

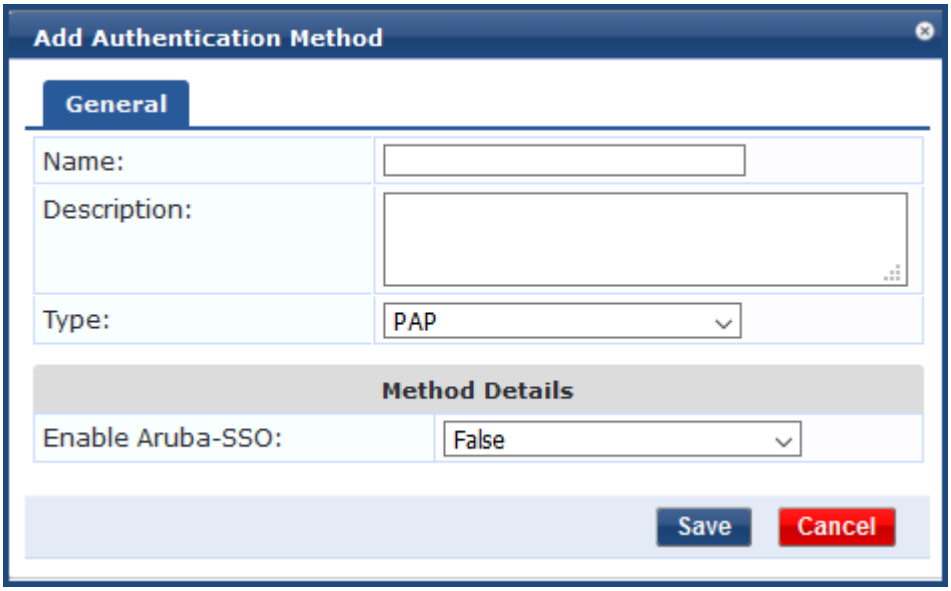
2. Select the **Add** link.

   The **Add Authentication Method** dialog opens to the **General** tab.

3. In the *Type* field, select **PAP**.

   The **Add PAP Authentication Method** dialog opens.

Figure 1. Adding the PAP Authentication Method



4. Specify the **PAP** parameters parameters as described in the following table, then click **Save**:

Table 1. PAP Authentication Method Parameters

Parameter	Action/Description
Name	Specify the name of the authentication method.
Description	Provide the additional information that he lps to identify the authentication method .
Type	Select **PAP** as the **Type** of authentication.
Method Details	
Enable Aruba - SSO	Enable or disable **Aruba - SSO** (Single Si gn - On) by specifying **True** or **False**. The default is **False**.

TEAP Authentication Method

Tunnel Extensible Authentication Protocol (TEAP) is a tunnel-based EAP method that enables secure communication between a peer and a server by using the Transport

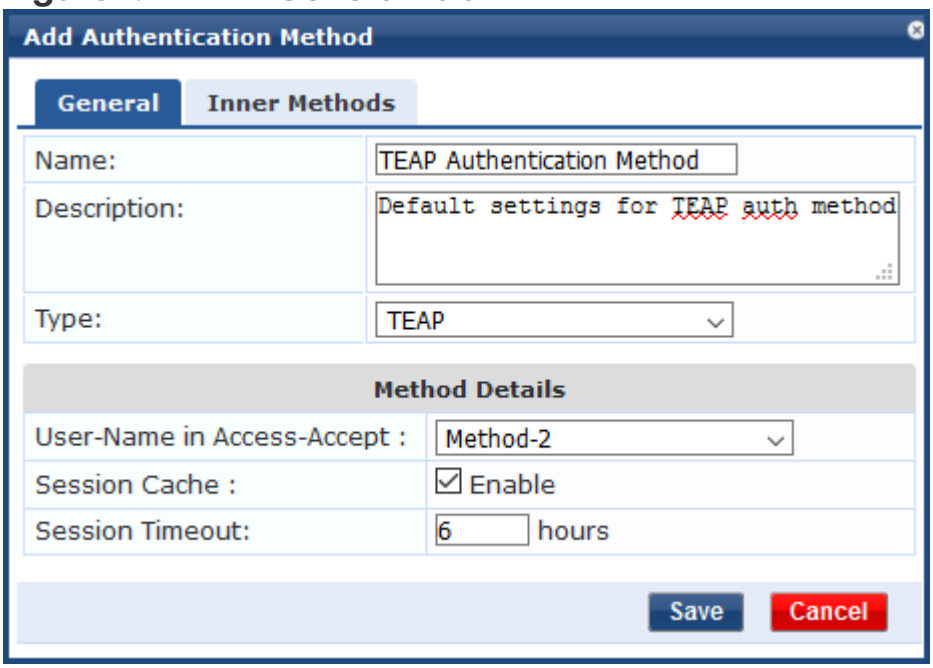
Layer Security (TLS) protocol to establish a mutually authenticated tunnel. Within the tunnel, TLV objects are used to convey authentication-related data between the EAP peer and the EAP server.

General Tab

To add the TEAP authentication method to Policy Manager:

1. Navigate to **Configuration > Authentication > Methods**. The **Authentication Methods** page opens.
2. Select the **Add** link. The **Add Authentication Method** dialog opens.
3. In the *Type* field, select **TEAP**.

Figure 1. TEAP > General Tab



4. Specify the **TEAP > General** parameters as described in the following table, then click **Save**.

Table 1. TEAP General Parameters

Parameter	Action/Description
Name	Specify the name of this TEAP authentication method.
Description	Optionally, provide the additional information that helps to identify the authentication method.

Parameter	Action/Description
Type	Select **TEAP**.
Method Details	
User - Name in Access - Accept	From the drop - down, select one of thes e options: • Method - 1 • Method - 2 This parameter deterines which usernam e should be shown in the **User - Name** c olumn in the access tracker. If you selec t **Method - 1**, the access tracker shows th e username for machine authentication. I f you select **Method - 2**, the acccess trac ker shows the username for user authenti cation.
Session Cache	The **Session Cache** option is enabled by default. Caches TEAP sessions on the Policy Man ager server for reuse if the user/client rec onnects to the Policy Manager server wit hin the session timeout interval.
Session Timeout	Specify the session timeout in hours. The default is **6** hours. If the session timeout value is set to **0**, th e cached sessions are not purged.

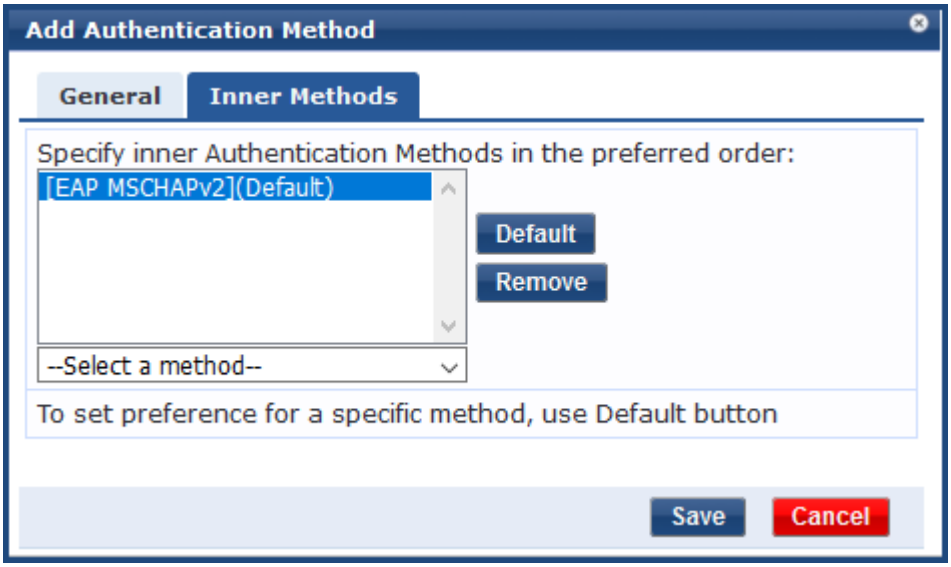
Inner Methods Tab

The tunneled method is frequently referred to as the "inner method."

1. Select the **Inner Methods** tab.

   The **Inner Methods** tab controls the tunneled methods for the **TEAP** authentication method.

Figure 2. TEAP > Inner Methods Tab



2. Specify the **TEAP Inner Methods** parameters as described in the following table:

Table 2. TEAP Inner Methods Tab Parameters

Parameter	Description
Specify inner authentication methods in the preferred order	To append an inner method to the displayed list, select it from the **Select a method** drop - down list. The list can contain multiple inner methods, which Policy Manager sends in priority order until negotiation succeeds. Select any method available in the current context from the drop - down list. Inner methods available include: • EAP MSCHAPV2 (Default) • EAP - TLS with OSCP enabled • EAP - TLS
Default	To set an inner method as the default (the method tried first), select it and click **Default**.
Remove	To remove an inner method from the displayed list, select the method and click **Remove**.

3. Click **Save**.

Adding and Configuring Authentication Sources

About this task

An *authentication source* is the identity store (Active Directory, LDAP directory, etc.) against which users and devices are authenticated.

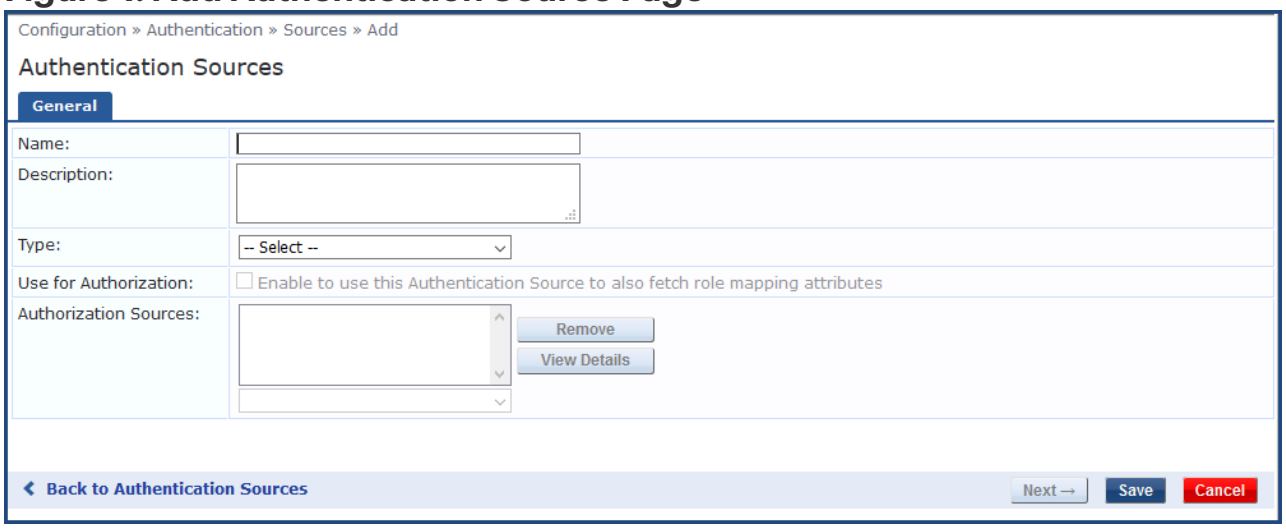
> **NOTE**
> Starting with the ClearPass 6.11.0 release, the database system also uses port **5433** in addition to port **5432**. Users should be aware that ports 5433 and 5432 must both be open when attempting to remotely connect to the Policy Manager database.

To configure an authentication source:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click the **Add** link. The **Add Authentication Sources** page opens. Different tabs and fields appear, depending on the authentication source selected.

Figure 1. Add Authentication Source Page



Refer to the following to configure supported authentication sources:

- Generic LDAP and Active Directory
- Azure

- Generic SQL DB
- HTTP
- Kerberos
- RADIUS/Radsec Server
- Adding a Static Host List as an Authentication Source
- Token Server

Subtopics

Generic LDAP and Active Directory
Azure
Generic SQL DB
HTTP
Kerberos
RADIUS/Radsec Server
Adding a Static Host List as an Authentication Source
Token Server

Generic LDAP and Active Directory

ClearPass Policy Manager can perform NTLM/MSCHAPv2, PAP/GTC, and certificate-based authentications against Microsoft Active Directory and against any LDAP-compliant directory (for example, Novell eDirectory, OpenLDAP, or Sun Directory Server). The LDAP and Active Directory-based server configurations are similar. You can retrieve role-mapping attributes by using filters. For configuration details, see Adding and Modifying Role Mapping Policies .

NOTE

ClearPass only supports integration with Active Directory using windows Server releases actively supported by Microsoft. For end-of-support information for the various of Windows 10, refer to the **Microsoft Windows lifecycle fact sheet**. ClearPass follows the Microsoft's Enterprise and Extended Support dates. As of April 2023, Active Directory releases supported by ClearPass 6.11.x are Active Directory 2012, Active Directory 2016, and Active Directory 2022. Active Directory releases supported by ClearPass 6.9.x include Active Directory 2012, Active Directory 2016, Active Directory 2019, and Active Directory 2022. Active Directory support for LDAP Server consists of OpenLDAP only.

Subtopics

Configure Generic LDAP and Active Directory authentication sources
Adding and Managing Generic LDAP and Active Directory Filters

Configure Generic LDAP and Active Directory authentication sources

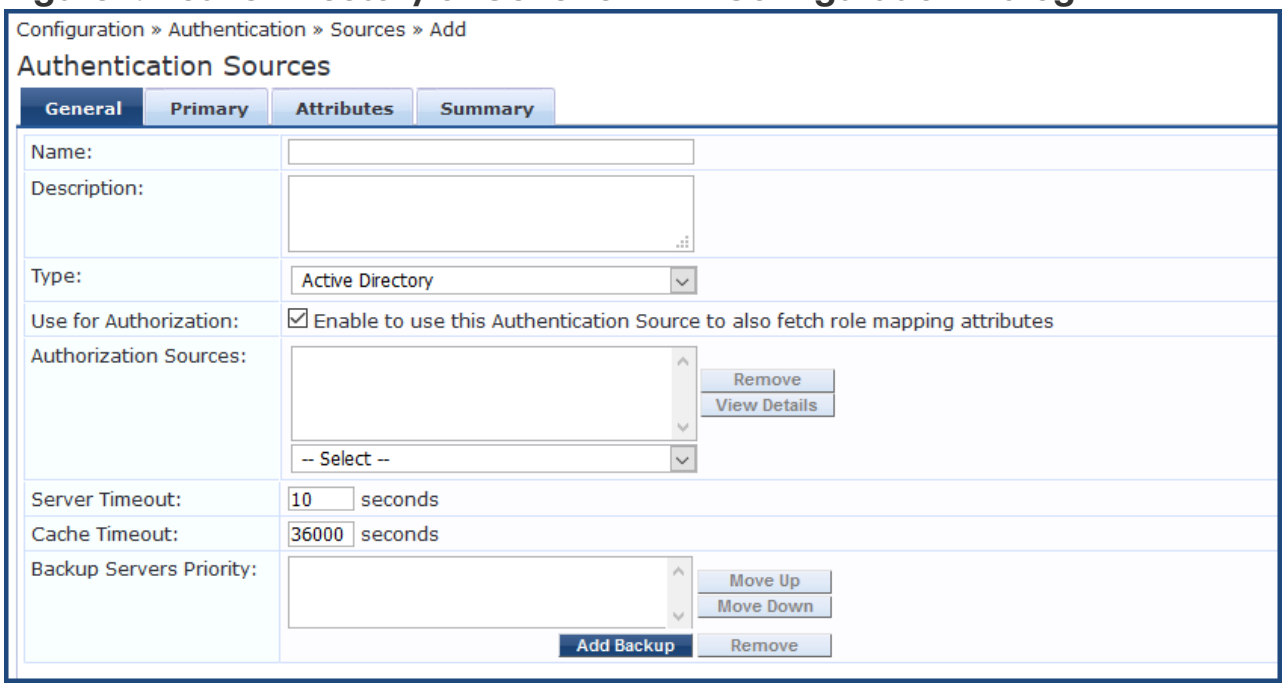
About this task

Use the following tabs to configure new Generic LDAP and Active Directory authentication sources:

General Configuration To add a new Generic LDAP or Active Directory authentication source:

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click **Add**. The **Add Authentication Sources** dialog opens. In this example, **Type** is set to **Active Directory**.

Figure 1. Active Directory or Generic LDAP Configuration Dialog



3. Specify the **Generic Active Directory** or **LDAP > General** parameters as described in the following table.

Table 1. Active Directory or Generic LDAP Authentication Source> General Parameters

Parameter	Action/Description
Name	Enter the name of the authentication sou rce.
Description	Provide the additional information that he lps to identify the authentication source (recommended).
Type	Select **Active Directory** or **Generic LDA P**.
Use for Authorization	Enable this check box to instruct Policy M anager to fetch role - mapping attributes (or authorization attributes) from this aut hentication source. If a user or device su ccessfully authenticates against this auth entication source, then Policy Manager al so fetches role - mapping attributes from the same source if the **Use for Authorizat ion** field is enabled. This check box is che cked (enabled) by default.
Authorization Sources	Specify additional sources from which ro le - mapping attributes are to be fetched. a. Select a previously configured authent ication source from the drop - down li st. b. To add the authentication source to th e list of authorization sources, click **Ad d**. • To remove the authentication sourc e from the list, click **Remove**. If Policy Manager authenticates the us er or device from this authentication s ource, then also fetches role - mappin

Parameter	Action/Description
	g attributes from these additional auth orization sources. **NOTE** Policy Manager fetches ro le - mapping attributes re gardless of which authent ication source the user or device was authenticated against.
Server Timeout	The Policy Manager server will automatic ally time out if the Active Directory or L DAP authentication source fails to send a response to an authorization query after 10 seconds. To specify a different duratio n for the timeout, enter the preferred tim eout value in the **Server Timeout** field. a. Specify the duration in number of sec onds that Policy Manager waits before considering this server unreachable. If multiple backup servers are availabl e, this value indicates the duration in n umber of seconds that Policy Manager waits before attempting to fail over fro m the primary to backup servers in the order in which they were configured.
Cache Timeout	Specify the duration in number of secon ds for which the attributes are cached. Policy Manager caches attributes fetched for an authenticating entity.
Backup Servers Priority	• To add a backup server, click **Add Bac kup**.

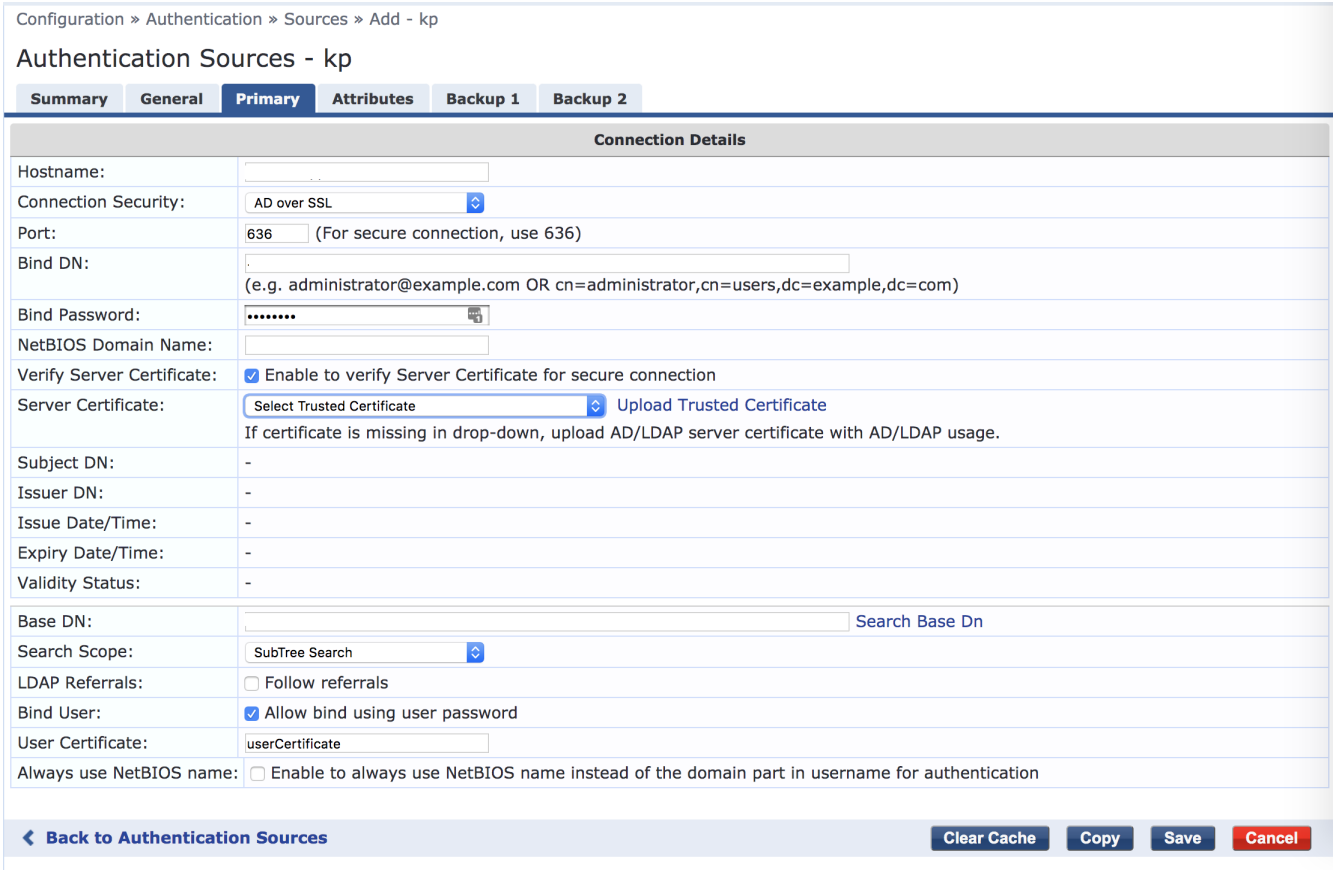
Parameter	Action/Description
	◦ When the **Backup 1** tab appears, you can specify connection details for a backup server. • To remove a backup server, select the server name and click **Remove**. • To change the server priority of the b ackup servers, select a server, then se lect **Move Up** or **Move Down**. ◦ This is the order in which Policy M anager attempts to connect to th e backup servers if the primary se rver is unreachable. **NOTE** Policy Manager logs an alert message in the Event Viewer if the primary Active Directo ry/LDAP server of an authent ication source for RADIUS an d TACACS+ authentication fa ils over to a designated bac kup server.

4. Click **Next** to continue to the **Primary** tab.

Primary Server Configuration

The **Primary** tab defines the settings for the primary server. The following figure is an example of the **Generic Active Directory > Primary** tab:

Figure 2. Generic LDAP or Active Directory > Primary Tab



Specify the **Active Directory** or **Generic LDAP > Primary** parameters as described in the following table:

Table 2. Active Directory or Generic LDAP > Primary Parameters

Parameter	Action/Description
Hostname	Specify the hostname or the IPv4/IPv6 address of the LDAP or Active Directory server.
Connection Security	• For a default nonsecure connection (usually port 389), select **None**. • For a secure connection that is negotiated over the standard LDAP port, select **StartTLS**. This is the preferred way to connect to an LDAP directory securely. • To choose the legacy way of securely connecting to an LDAP directory, select **LDAP over SSL** or **AD over SSL**. You mus

Parameter	Action/Description
	t use port **636** for this type of connecti on. **NOTE** In a production environment, security is a concern because when Policy Manager binds to an LDAP server, it submits the username and password for th at account over the network u nder clear text unless you prot ect it using Connection Securi ty and set the port to **636**. **NOTE** To ensure successful authenti cation, be sure to add the CA certificate of the LDAP server t o the Certificate Trust List.
Port	Specify the TCP port at which the LDAP or Active Directory server is listening for conn ections. **NOTE** The default TCP port for LDAP connections is **389** and the d efault port for LDAP over SSL is **636**. After restoring a confi guration backup on 6.11.0 that was taken from an older versio n (6.8.X, 6.9.X. or 6.10.X), any c ustom (non default) configura tion that uses the insightdb o r tipsLogDb database as an au thentication source must be u pdated to use port number **54 33** instead of **5432**.
Verify Server Certificate	Select this check box to verify the server c ertificate as part of authentication. If this o ption is enabled, you must select a server c

Parameter	Action/Description
	ertificate in the **Server Certificate** field bel ow. **NOTE** This field does not appear if you select the **None** option in the **Connection Security** field described at the top of this ta ble.
Server Certificate	If you selected **LDAP over SSL** or **AD over SSL** in the **Connection Security** field desc ribed at the top of this table, and selected the **Verify Server Certificate** option, use t he **Server Certificate** dropdown menu to s elect a trusted certificate. If the certificate you want to use is not currently in the trust list, you can click the **Upload Trusted Cert ificate** link and upload a new trusted certifi cate. **NOTE** During the server certificate u pload process, the server cert ificate is added to the trust lis t soon after the upload operat ion, and does not have to wait until the **Save** button is clicke d to save the authentication s ource settings. As a part of th e upload server certificate pro cess, Policy Manager will also enable the certificate and add the AD/LDAP Servers Usage to it.
Bind DN	Specify the DN (Distinguished Name) of th e administrator account. Policy Manager us es this account to access all other records in the directory. For Active Directory, the bind DN can also b e in the administrator@domain format (for example, administrator@acme.com).

Parameter	Action/Description
Bind Password	Specify the password for the administrator DN entered in the **Bind DN** field.
NetBIOS Domain Name	Specify the Active Directory domain name for this server. Policy Manager prepends thi s name to the user ID to authenticate users found in this Active Directory. **NOTE** This setting is available only fo r Active Directory.
Base DN	Enter the Base DN (Distinguished Name) of the node in your directory tree from which to start searching for records. 1. After entering the values for the fields d escribed above, click **Search Base DN** t o browse the directory hierarchy. The L DAP browser opens. 2. Navigate to the DN that you want to use as the base DN. 3. To select a node as a base DN, click the preferred node in the displayed tree str ucture. • The base DN displays at the top of t he LDAP browser. **NOTE** **Search Base DN**. The LDA P browser is able to search the subdomain, but canno t be further expanded to vi ew additional details.
Search Scope	This is setting is used to query and find the user or machine record given a unique attr ibute for that record in LDAP/Active Directo ry. Select the scope of the search you want to perform, starting at the base DN.

Parameter	Action/Description
	• **Base Object Search** allows you to search at the level specified by the base DN. • **One Level Search** allows you to search up to one level lesser to the immediate children of the base DN. This is a common option used by network admins that have selected the correct **Base DN** setting, allowing lookups to be performed with greater efficiency. • **Subtree Search** allows you to search the entire subtree under the base DN (including at the base DN level). This default setting can make it easier for network admins to configure role - mappings or authentication source filters, but this setting is resource - intensive if performed for each authentication lookup. **NOTE** Even though the Admin UI can show the User / Machine record when the search is performed, the real auth / authorization checks may fail.
LDAP Referrals	Enable this check box to automatically follow referrals returned by your directory server in search results. Refer to your directory documentation for more information on referrals.
Bind User	Enable this check box to authenticate users by performing a bind operation on the directory using the credentials (user name and password) obtained during authentication. For clients to be authenticated by using the LDAP bind method, Policy Manager must receive the password in clear text.

Parameter	Action/Description
User Certificate	Enter the name of the attribute in the user record from which the user certificate can be retrieved.
Always use NetBIOS name	Check this option to always use the NetBIOS name instead of the domain part in the username for authentication. **NOTE** This field is available only if you select **Active Directory** as an authentication source.
Special Character Handling for LDAP Query	Select this option to allow LDAP queries to handle special characters in a user Distinguished Name (DN), such as a blackslash, comma, bracket, or parenthesis. This setting is enabled by default.

Procedure

Click **Next** to continue to the **Attributes** tab.

Attributes Configuration The **Attributes** tab defines the Active Directory or LDAP Directory query filters and the attributes to be fetched by using those filters. The following figures are the examples of the **Active Directory > Attributes** tab and the **Generic LDAP Directory > Attributes** tabs:

Figure 3. Active Directory Attributes Dialog

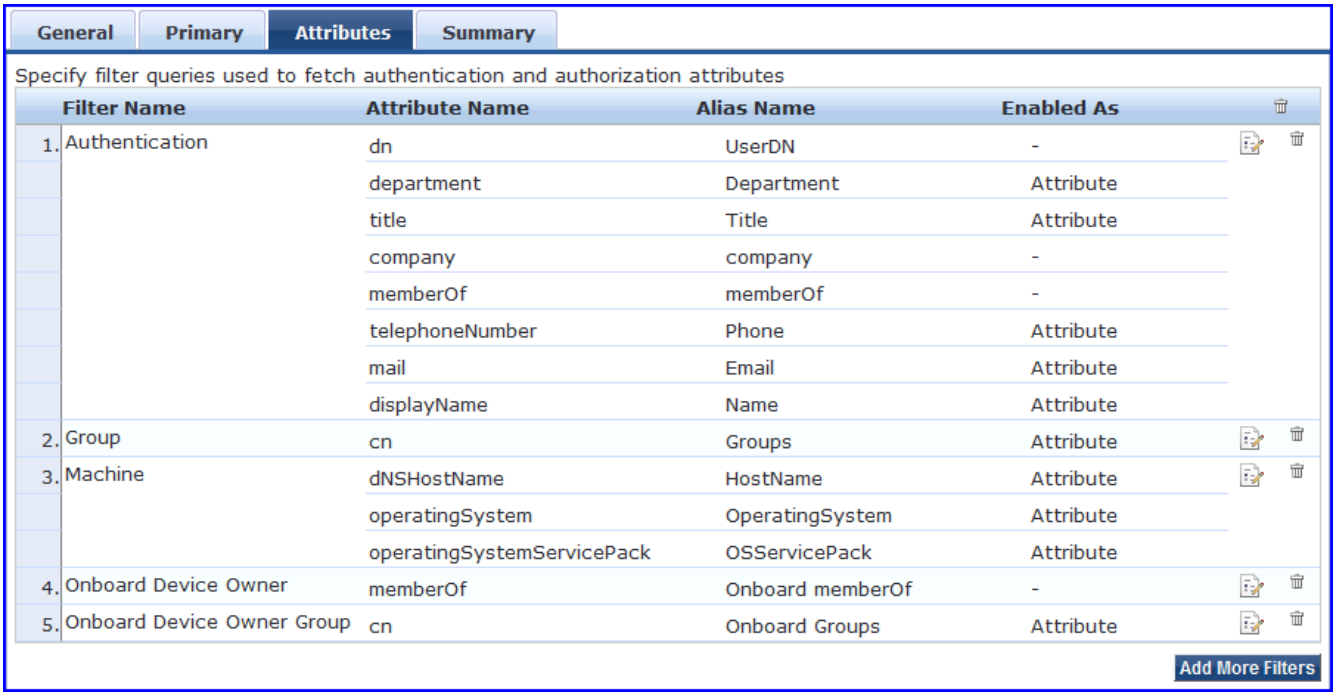


Figure 4. Generic LDAP Directory > Attributes Dialog

The **Attributes** tab displays the following information for the filters applied to the Generic LDAP Directory or Active Directory authentication source

Table 3. Active Directory or Generic LDAP Attributes tab Parameters

Parameter	Action/Description
Filter Name	Name of the filter.
Attribute Name	Name of the LDAP or Active Directory attributes defined for this filter.
Alias Name	Alias name for each attribute name selected for the filter.
Enable As	Indicates whether this value to be used directly as a role or attribute in an enforcem

Parameter	Action/Description
	ent policy. This bypasses the step to assign a role in Policy Manager through a role - m apping policy.

The following table describes the default filters for the Active Directory/Generic LDAP directories.

NOTE

You can select any filters in the **Attributes** tab to modify the parameters of that filter. For details, see Modifying Default Filters

Table 4. Active Directory/Generic LDAP Default Filters

Directory	Default Filters
Active Directory	• **Authentication**: The query searches in t he objectClass of the type **user**. This qu ery finds both user and machine accou nts in Active Directory: `(&(objectClass=user) (sAMAccountName=% {Authentication:Username}))` • After a request arrives, Policy Manage r populates **%{Authentication:Usernam e}** with the authenticating user or mach ine. This filter is also configured to fetc h the following attributes based on this filter query: • **dn** (alias of UserDN): This is an internal attribute that is populated with the user or machine record's DN. • **department** • **title** • **company** • **memberOf**: In Active Directory, this attr ibute is populated with the groups that

the user or machine belongs to. This attribute can have multiple values.

- **telephoneNumber**
- **mail**
- **displayName**
- **accountExpires**
- **Group**: This is a filter used for retrieving the name of the groups a user or machine belongs to.

```
(distinguishedName=%{memberOf})
```

- This query fetches all group records, where the distinguished name is the value returned by the **memberOf** attribute. The values for the **memberOf** attribute are fetched by the first filter (authentication) described above. The attribute fetched with this filter query is **cn**, which is the name of the group.
- **Machine**: This query fetches the machine record in Active Directory.

```
(&(objectClass=computer)
(sAMAccountName=%{Host:Name}$))
```

- %{Host:Name} is populated by Policy Manager with the name of the connecting host if available. dNSHostName,operatingSystem, and operatingSystemServicePack attributes are fetched with this filter query.
- **Onboard Device Owner:** This is the filter for retrieving the name of the owner the onboard device belongs to. This query finds the user in the Active Directory

```
(&(sAMAccountName=%
{Onboard:Owner})(objectClass=user))
```

Directory	Default Filters
	• %{Onboard:Owner} is populated by Policy Manager with the name of the onboarded user. • **Onboard Device Owner Group:** This filter is used for retrieving the name of the group the onboarded device owner belongs to. `(distinguishedName=%{Onboard memberOf})` • This query fetches all group records where the DN is the value returned by the Onboard **memberOf** attribute. The attribute fetched with this filter query is **cn**, which is the name of the Onboard group.

Generic LDAP Directory	• **Authentication**: This is the filter used for authentication:
	`(&(objectClass=*)(uid=%{Authentication:Username}))`
	• When a request arrives, Policy Manager populates %{Authentication:Username} with the authenticating user or machine. This filter is also set up to fetch the following attributes based on this filter query:
	• **dn** (aliased to UserDN): This is an internal attribute that is populated with the user record's DN.
	• **Group**: This is the filter used for retrieving the name of the groups to which a user belongs.
	`(&(objectClass=groupOfNames)(member=%{UserDn}))`
	• This query fetches all group records (of objectClass groupOfNames), where th

Directory	Default Filters
	e member field contains the DN of the user record (UserDN, which is populate d after the authentication filter query is executed. The attribute fetched with thi s filter query is **cn**, which is the name of the group (this is aliased to a more read able name: **groupName**)).
Add More Filters	Click this button to open the **Authenticatio n Sources > Add** page to open the **Configu re Filter** page. From this page, you can defi ne a filter query and the related attributes t o be fetched. For more information, see Ad ding and Managing Generic LDAP and Acti ve Directory Filters

Configuration Summary

Use the **Summary** tab to see a summarized view of the parameters configured in the **General**, **Primary**, and **Attributes** tab.

Adding and Managing Generic LDAP and Active Directory Filters

Policy Manager is preconfigured with filters and selected attributes for Active Directory and generic LDAP directory. Create new filters only if you need Policy Manager to fetch role-mapping attributes from a new type of record.The **Filter** tab provides an LDAP browser interface to define the filter search query. You can fetch different types of records by specifying multiple filters that use different dynamic session attributes. For example, Policy Manager can fetch the user record associated with %{Authentication:Username} and a machine record associated with %{RADIUS:IETF:Calling-Station-ID} for a given request.

To view or configure filters, navigate to the **Configuration** > **Authentication** > **Sources** > **Add** page, and define an LDAP or Active Directory authentication source according to the procedures described in the previous steps. The **Attributes** tab on the **Configuration** > **Authentication** > **Sources** > **Add** page includes the **Add More Filters** button, which opens the Configure Filters dialog. The four tabs on the **Configure Filter** page allow you to browse the nodes in the Active Directory or LDAP directory, and create and manage filters

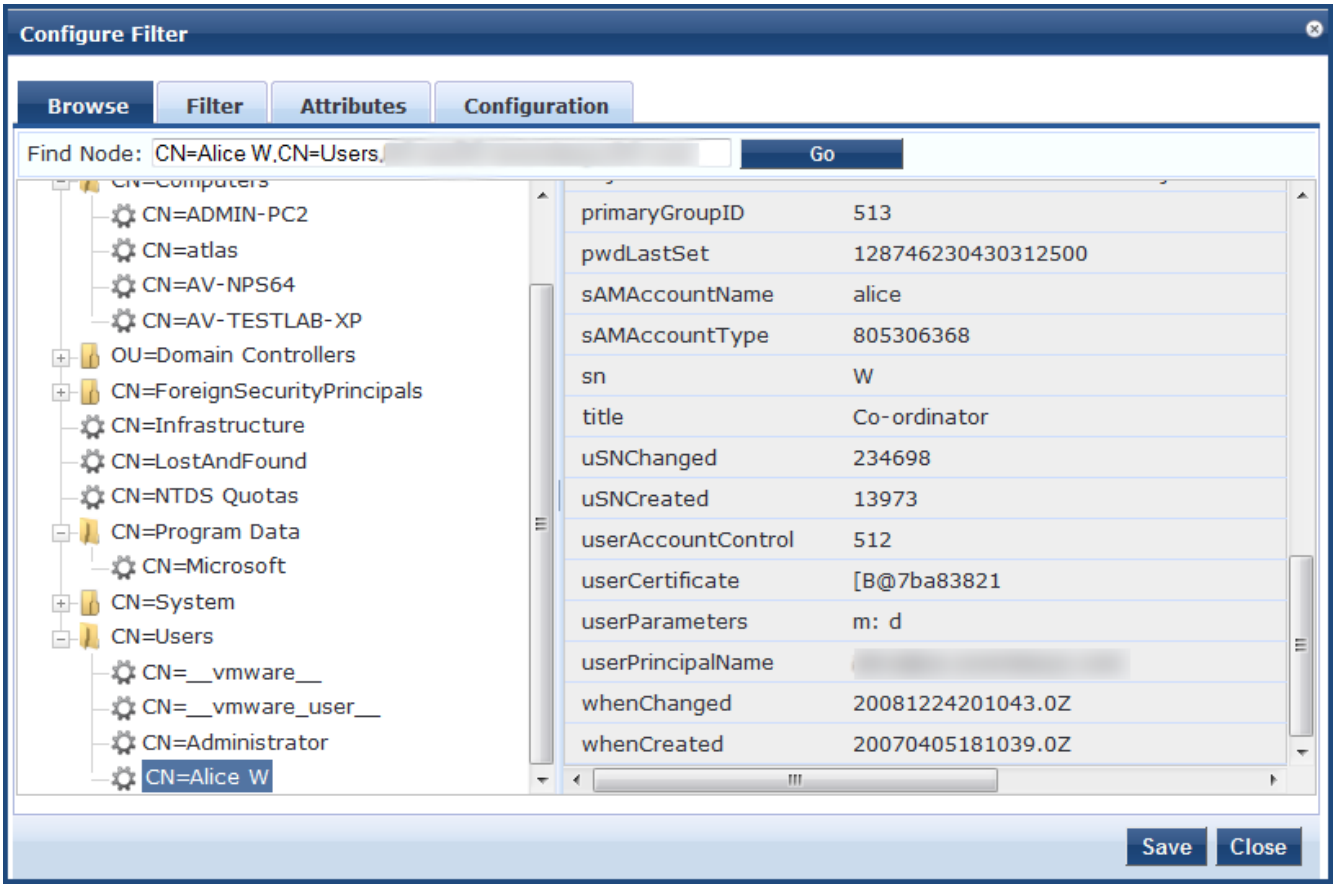
Subtopics

Browse Tab Settings
Filters Tab Settings
Attributes Tab Settings
Configuration Tab Settings
Modifying Default Filters

Browse Tab Settings

By default, The **Configure Filter** page opens on the **Browse** tab, which allows you to browse the nodes in the Active Directory or LDAP directory, starting at the base DN. This is presented in the read-only mode. Selecting a leaf node (a node that has no children) displays the attributes associated with that node.

Figure 1. Configure Filter > Browseconfiguration dialog for Active Directory or Generic LDAP authentication sources:



Specify the Active Directory or Generic LDAP Configure **Configure Filter > Browse** tab parameter as described in the following table:

Table 1. Configure Filter > Browse Actions for Active Directory or Generic LDAP

Navigation	Action/Description
Find Node	To find the node, enter th .

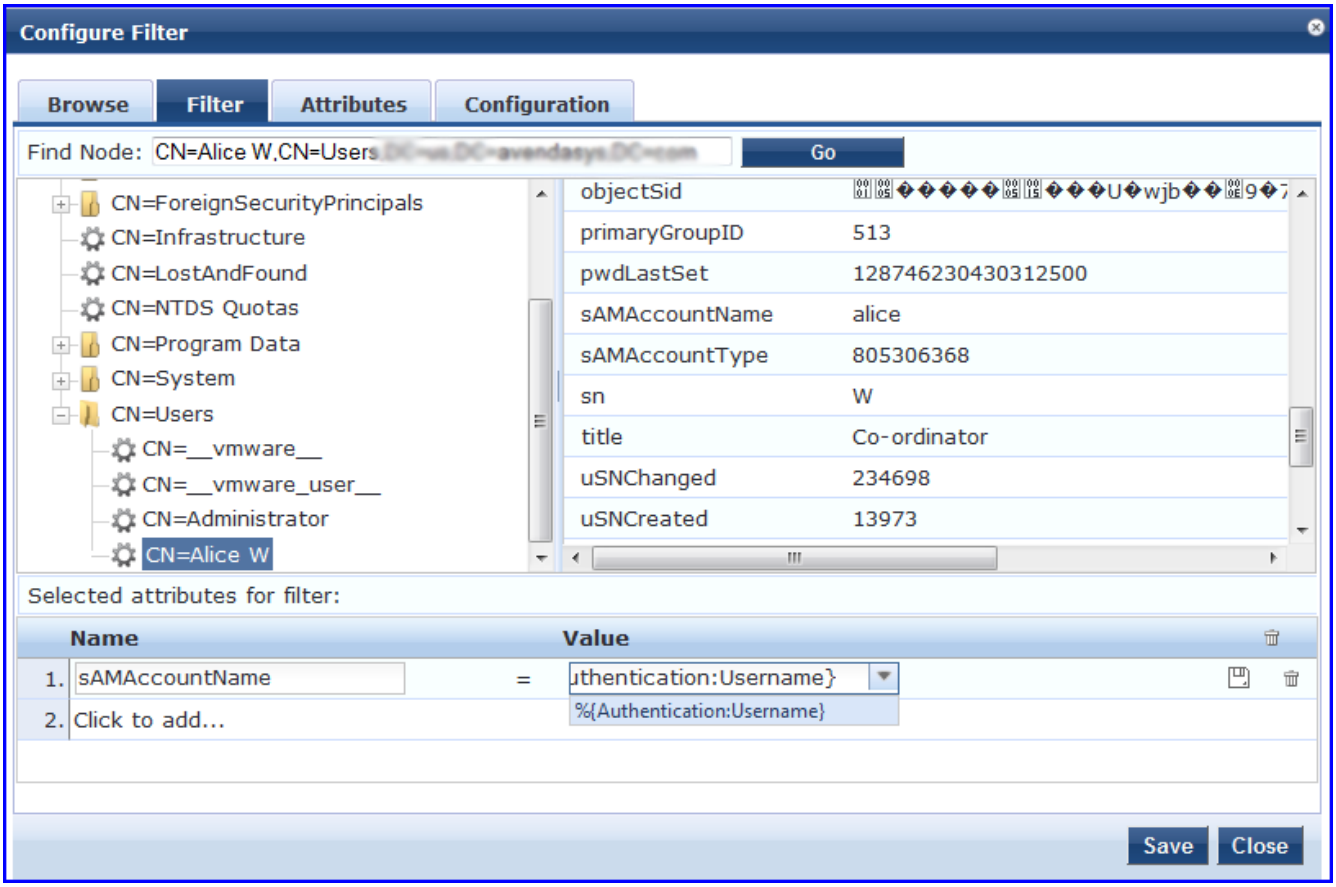
Filters Tab Settings

Use this tab to view existing filters, or create new ones

Viewing Existing Filters

The following figure is an example of the **Configure Filter > Filter** configuration dialog for Active Directory or Generic LDAP authentication sources:

Figure 1. Configure Filter > Filterconfiguration dialog for Active Directory or Generic LDAP Authentication Sources



The following table describes the **Configure Filter > Filter** tab parameters:

Table 1. Configure Filter > Filter parameters for Active Directory or Generic LDAP Authentication Sources

Parameter	Action/Description
Find Node	To find a node, enter the DN, then click the **Go** button.
Select the attributes for filter	This table has Name and Value columns. You can enter the attribute name in the following ways: • By selecting a node, inspecting the attributes, and then manually entering the attribute name by clicking on **Click to add...** in the table row. • By selecting an attribute on the right - hand side of the LDAP browser. The attribute name and value are automatically populated in the table. The attribute value can be a value that is automatically populated by selecting an attribute from the browser, or it can be manually populated. To aid in populating the value with dynamic session attribute values, a drop - down with the commonly used namespace and attribute names is presented.

Creating Filters

About this task

The goal of filter creation is to help Policy Manager find a user or device connecting to the network in LDAP or Active Directory.

To create a filter:

Procedure

1. From the **Filter** tab, click on a node that you want to extract user or device information from.

   For example, browse the **Users** container in Active Directory and select the node for a user (Alice, for example). On the right-hand side, you can view the attributes associated with that user.

2. Select the attributes that help Policy Manager identify the user or device. For example, in Active Directory, an attribute called **sAMAccountName** stores the user ID.

   The attributes that you select are automatically populated in the **Filter** table displayed below the browser section with their values.

   In this example, if you select **sAMAccountName**, the row in the **Filter** table shows this attribute with a value of Alice (assuming you picked Alice's record as a sample user node).

   After Step 2, you can have values for a specific record (in this example, Alice's record).

3. Change the value to a dynamic session attribute that helps Policy Manager associate a session with a specific record in LDAP/Active Directory. For example, if you selected the **sAMAccountName** attribute in Active Directory, click the **Value** field and select **%{Authentication:Username}**.

   When Policy Manager processes an authentication request, **%{Authentication:Username}** is populated with the user ID of the user connecting to the network.

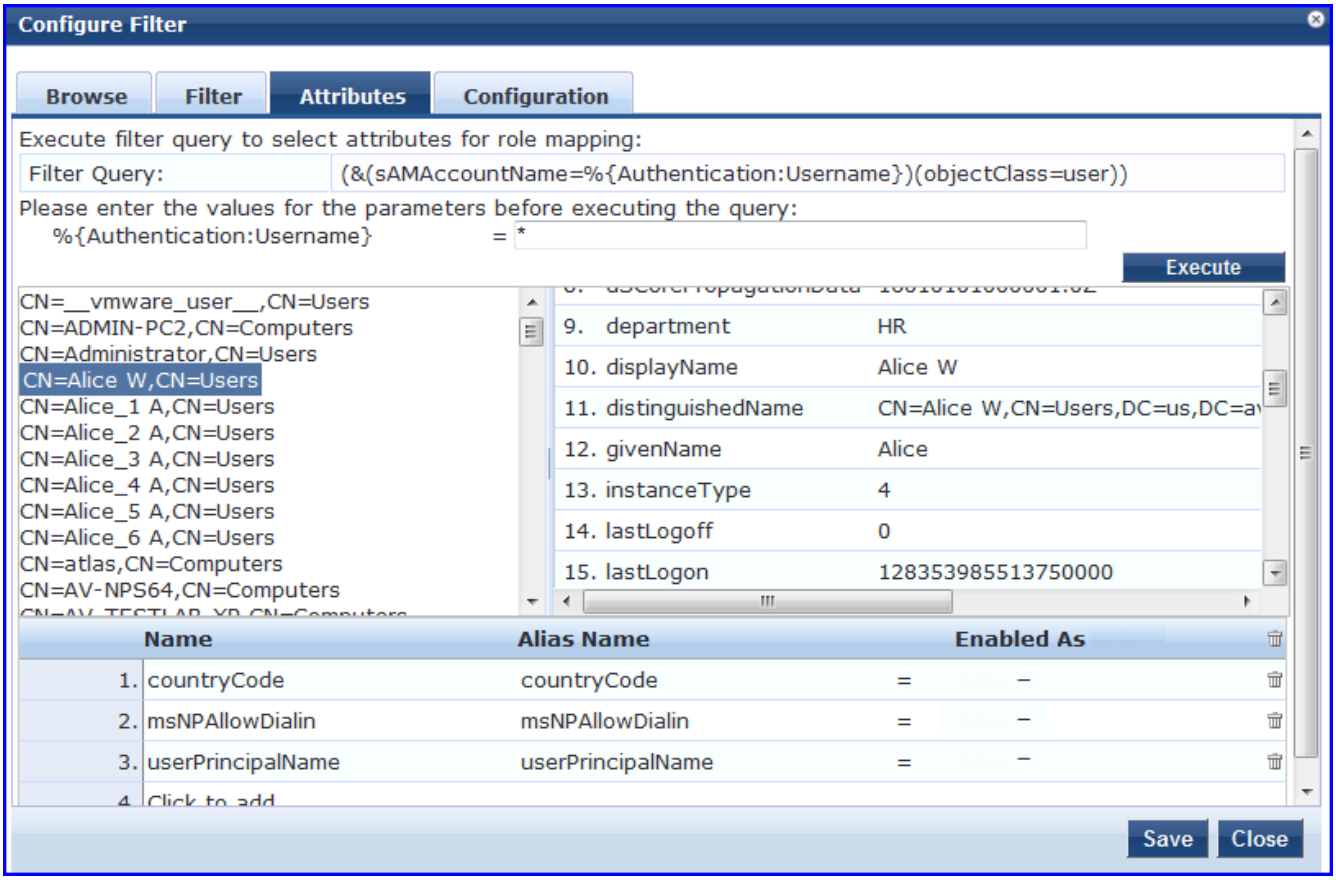
4. Add more attributes from the selected node and continue with Step 2.

Attributes Tab Settings

The **Attributes** tab defines the attributes to be fetched from the Active Directory or LDAP directory. You can also enable each attribute as a role, which means the value fetched for this attribute can be used directly in enforcement policies. For more information, see Configuring Enforcement Policies .

The following figure displays the Active Directory or Generic LDAP Configure Filter > Attributes tab:

Figure 1. Configure Filter > AttributesConfiguration Dialog for Active Directory or Generic LDAP Authentication Sources



Specify the **Active Directory/LDAP Configure Filter Page > Attributes** tab parameters as described in the following table:

Table 1. Configure Filter > Attributes Settings for Active Directory or Generic LDAP Authentication Sources

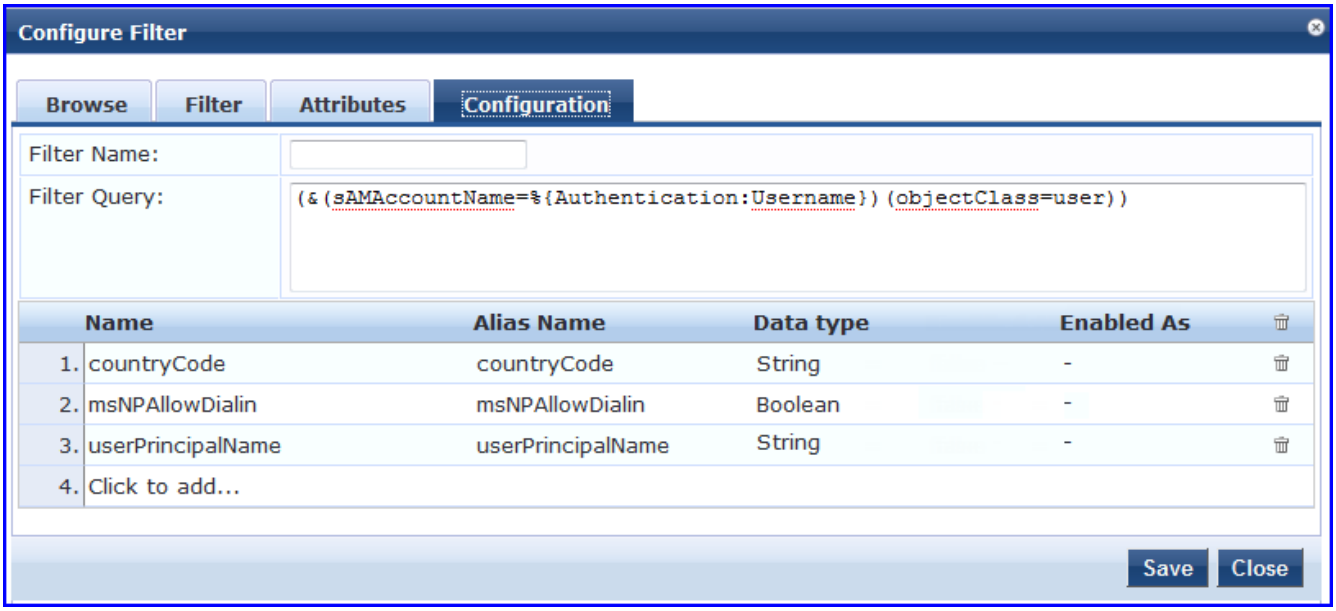
Parameter	Action/Description
Enter values for parameters	Policy Manager parses the filter query (c reated in the **Filter** tab and shown at the t op of the **Attributes** tab) and prompts to e nter the values for all dynamic session para meters in the query. For example, if you have **%{Authentication :Username}** in the filter query, you are pro mpted to enter the value for it. You can en ter wildcard character (*) here to match all entries. **NOTE** If there are thousands of entri es in the directory, entering th e wildcard character (*) can ta

Parameter	Action/Description
	ke a while to fetch all matchin g entries.
Execute	1. After entering the values for all dynami c parameters, click **Execute** to execute the filter query. • You can see all entries that match th e filter query. 2. Click one of the entries (nodes) to view the list of attributes for that node. 3. Click the attribute names that you want to use as role - mapping attributes.
Name	Specify the name of the attribute.
Alias Name	Specify the alternative name for the attribu te. By default, this is the same as the attrib ute name.
Enable As	Click this check box to enable this attribut e value to be used directly as a role in an enforcement policy. This bypasses the step of assigning a role in Policy Manager throu gh a role - mapping policy.

Configuration Tab Settings

The **Configuration** tab shows the filter and attributes configured in the **Filter** and **Attributes** tabs respectively. From this tab, you can also manually edit the filter query and the attributes to be fetched.

Figure 1. Configure Filter > ConfigurationDialog for Active Directory or Generic LDAP Authentication Sources



Modifying Default Filters

About this task

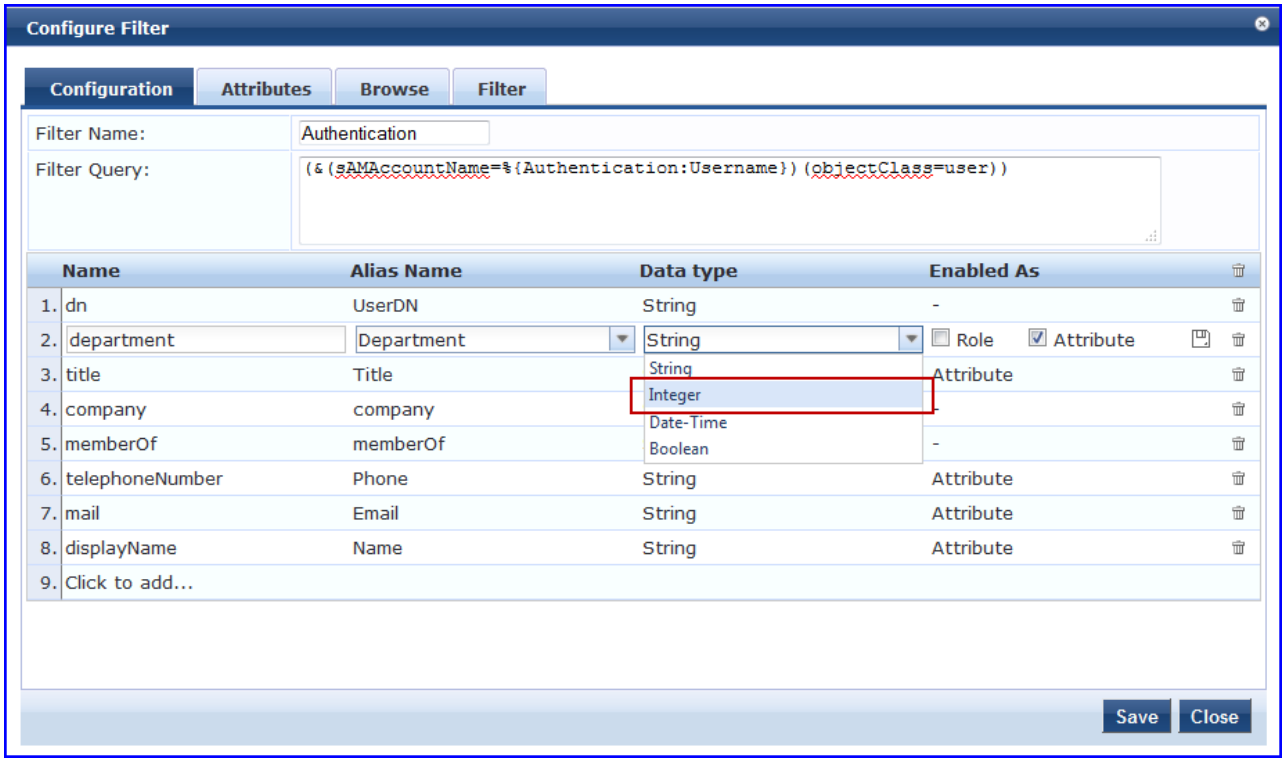
When you add a new authentication source of type Active Directory or LDAP, a few default filters and attributes are populated. To modify these predefined filters:

Procedure

1. Select a filter on the **Authentication > Sources > Attributes** dialog. The **Configure Filter** page for the specified filter opens.

> **NOTE**
>
> A minimum of one filter must be specified for the LDAP and Active Directory authentication source. This filter is used by Policy Manager to search for the user or device record. If not specified, authentication requests are rejected.

Figure 1. Modifying Default filters for Active Directory or Generic LDAP authentication sources:



The attributes that are defined for the authentication source display as attributes in role-mapping policy Rules Editor under the authorization source namespace.

2. From the **Configure Filter > Configuration** dialog, select the attribute you wish to modify.

3. Change the attribute operator values as needed, then click **Save**.

   The operator values that display are based on the **Data Type** specified here.

   For example, if you modify the Active Directory **department** to be an integer rather than a string, then the list of operator values populate with values that are specific to integers.

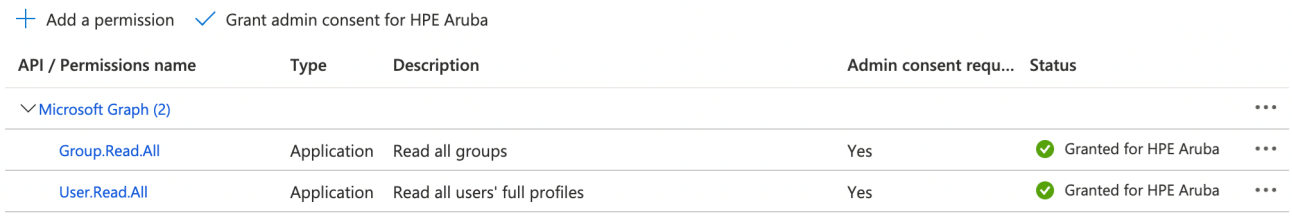
> **NOTE**
>
> This functionality allows you to modify the data type available for Generic SQL DB, Generic LDAP, Active Directory, and HTTP authentication source types.

Azure

About this task

ClearPass can interact with Azure to retrieve user group details and perform policy enforcement. This source is only capable of authorization, not authentication.

The following permissions must be granted to the Azure App for ClearPass to be able to fetch user and group information:



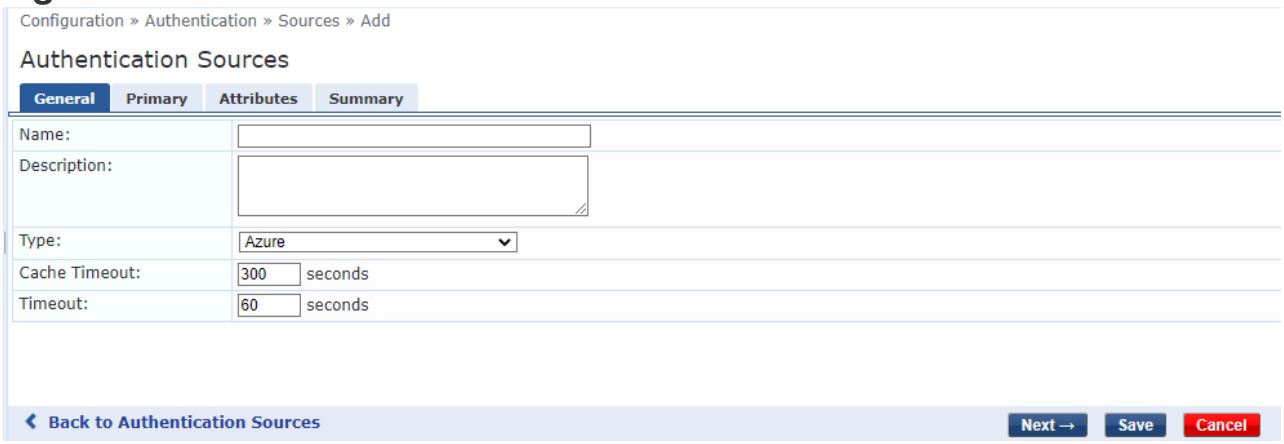
To configure the Azure service:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click the **Add** link. The **Add Authentication Sources** page opens with the **General** tab displayed. Each configuration parameter is empty, and the authentication source type is undefined.

   General Tab When **Azure** is selected as the source type, the **General** tab displays session timeout information unique to Azure.

Figure 1. Azure > General Tab



   a. Specify the following **Azure > General** tab parameters:

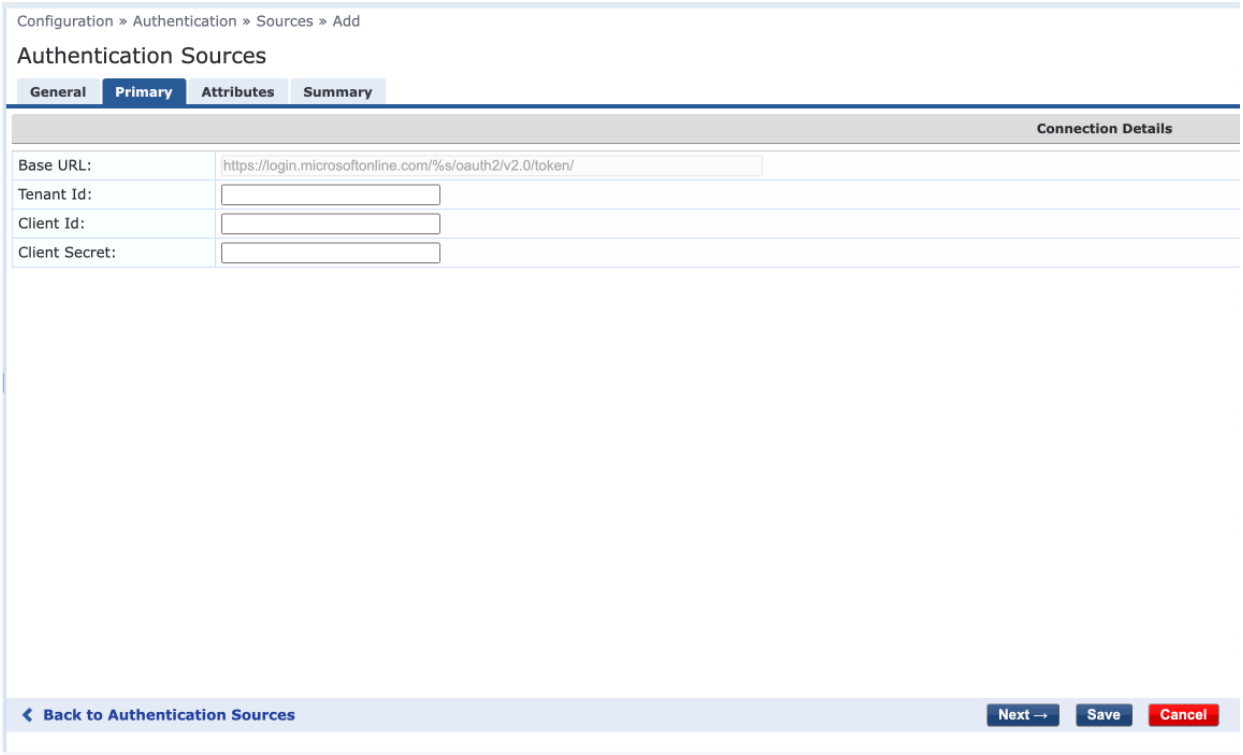
Table 1. Azure > General Tab Parameters

Parameter	Action/Description
Name	Specify a unique name of the Azure se rvice.
Description	Provide additional information to identi fy and differentiate the Azure source fro m others with similar attributes.
Type	Select the type of source. In this contex t, select **Azure**.
Cache Timeout	Sets the time (in seconds) Azure sessio n data remains in policy server cache b efore it is removed.
Timeout	Set the time an Azure request should wait for a response from the server befo re it can terminate.

Primary Tab

a. Use the **Primary** tab to define settings for the Azure primary server resource.

Figure 2. Azure > Primary Tab



b. Specify the following **Azure > Primary** tab parameters:

> **NOTE**
>
> For ClearPass to access user details from Azure, a ClearPass administrator needs to create an application and register it. Once registered, obtain **Tenant ID** and **Client ID** details from the application's Overview page. The application also requires certain permissions in order for ClearPass to integrate smoothly.

Table 2. Azure > Primary Tab Parameters

Parameter	Action/Description
Base URL	The Base URL is the default OA uthentication access tokens fo ase URL is not an editable fiel
Tenant ID	Enter the Azure tenant ID iden tance. The ID represents a res nstance an organization recei ns up for Azure. Each Azure Al parate from other Azure AD te

Parameter	Action/Description
Client ID	Enter an Azure Client ID as a u plication created in Active Dir
Client Secret	A secret string that the applic entity when requesting a toke lso be referred to as the applic

Attributes Tab Select the **Attributes** tab to set Azure query filters and the attributes fetched by using the filters.

> **NOTE**
>
> The current Azure release only retrieves user group details from Azure. It is the only attribute currently used for authorization. Any configurations and additional filters are not currently supported.

Figure 3. Azure > Attributes Tab

a. Select **Add More Filters** as needed, and set a filter query and related attributes fetched from the SQL DB store.

Figure 4. Azure Configure Filter Screen

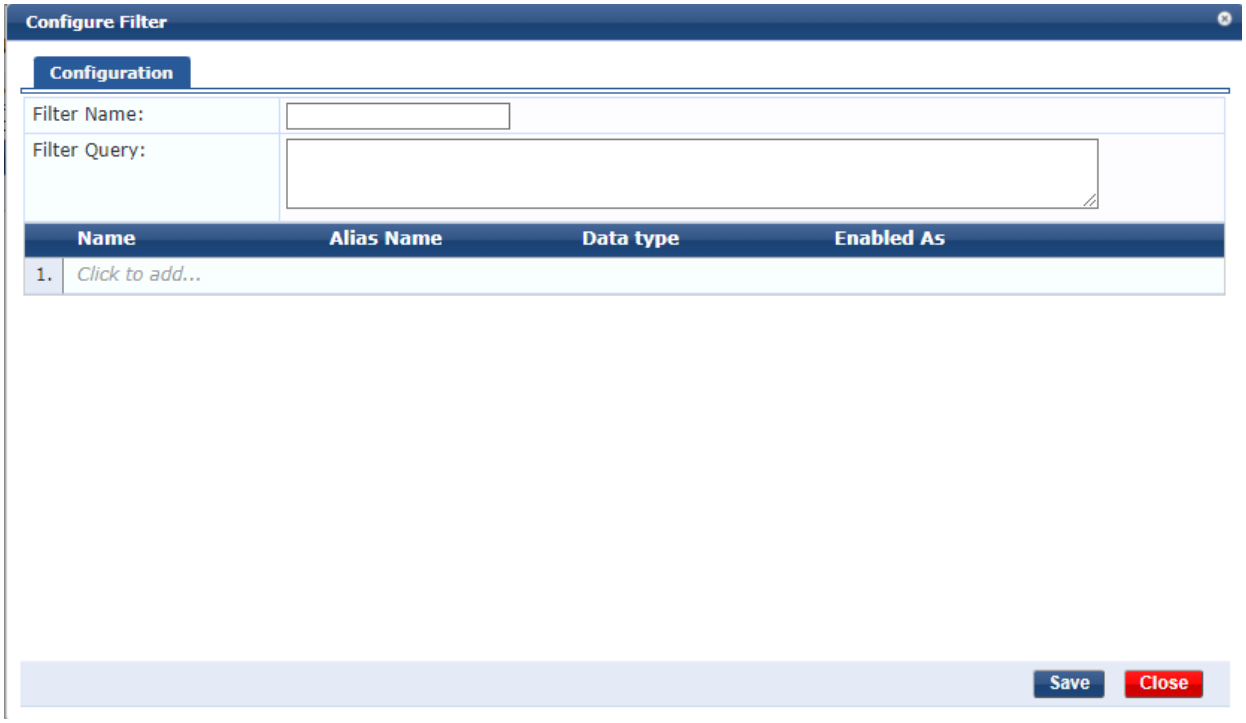


Table 3. Azure Configure Filter Page Parameters

Parameter	Action/Description
Filter Name	Enter the name of the selecte
Filter Query	List the specific user details t efault, the query includes an the user field is described be users/?$select=mail,id,depar er Filters> The above query fetches the countEnabled attributes. Eve supports fetching these attri hin the enforcement profile.
Name	Specify the name of the attri
Alias Name	Specifies the alias name for t is the same as the attribute n
Data Type	Specifies the data type for th nteger, and Boolean.

Parameter	Action/Description
Enabled As	Specify whether the value to attribute in an enforcement p p of assigning a role in Policy apping policy.

Summary Tab Select the **Summary** tab to review the parameters configured in the previous three tabs.

Generic SQL DB

Configure the primary and backup servers, session details, filter query, and role-mapping attributes to fetch the Generic SQL authentication sources. Policy Manager can perform MSCHAPv2 and PAP/GTC authentication against any Open Database Connectivity (ODBC)-compliant SQL database such as Microsoft SQL Server, Oracle, or PostgrSQL. You can specify a stored procedure to query the relevant tables and retrieve role-mapping attributes by using filters.

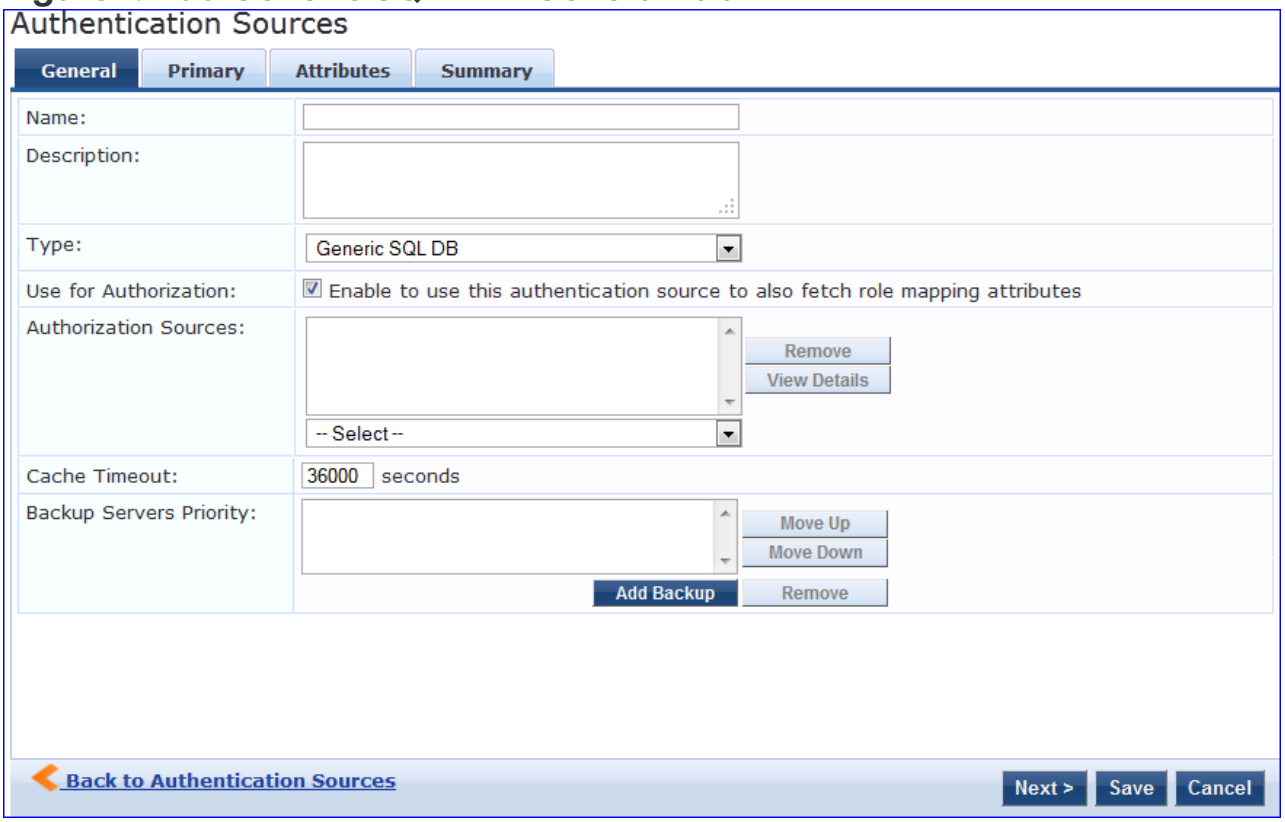
NOTE

The MySQL database is supported in Policy Manager 6.11 and later. The plugin/connector used to connect to MySQL database is MariaDB.

General Tab

1. Navigate to the **Configuration > Authentication > Sources**, then click **Add**. The **General** tab labels the authentication source and defines session details, authorization sources, and backup server details.

Figure 1. Add Generic SQL DB > General Tab



2. Specify the **Add General SQL DB > General** parameters as described in the following table.

Table 1. Add Generic SQL DB > General Parameters

Parameter	Action/Description
Name	Specify the name of the authentication s ource.
Description	Provide the additional information that he lps to identify the authentication source.
Type	Select **Generic SQL DB**.
Use for Authorization	Enable this option to request Policy Mana ger to fetch role - mapping attributes (or

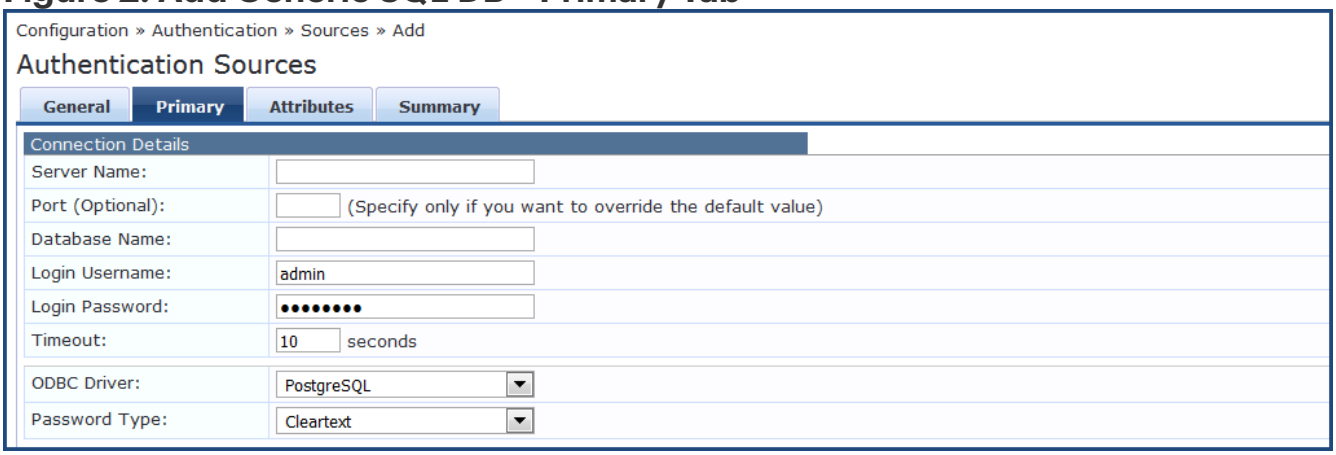
Parameter	Action/Description
	authorization attributes) from this authentication source. If a user or device successfully authenticates against this authentication source, then Policy Manager also fetches role - mapping attributes from the same source if the **Use for Authorization** field is enabled. This check box is enabled by default.
Authorization Sources	Specify additional sources from which role - mapping attributes are to be fetched. a. Select a previously configured authentication source from the drop - down list. b. To add the authentication source to the list of authorization sources, click **Add**. • To remove the authentication source from the list, click **Remove**. If Policy Manager authenticates the user or device from this authentication source, then also fetches role - mapping attributes from these additional authorization sources. **NOTE** Policy Manager fetches role - mapping attributes regardless of which authentication source the user or device was authenticated against.
Backup Servers	To add a backup server, click **Add Backup**.

Parameter	Action/Description
	From the **Backup 1** tab, you can specify c onnection details for a backup server (sa me fields as for primary server that are sp ecified below). To remove a backup server, select the se rver name and click **Remove**. Select **Move Up** or **Move Down** to chang e the server priority of the backup serve rs. This is the order in which Policy Mana ger attempts to connect to the backup s ervers.
Cache Timeout	Policy Manager caches attributes fetched for an authenticating entity. This paramet er controls the time period for which the attributes are cached.

Primary Tab

The **Primary** tab defines the settings for the primary server.

Figure 2. Add Generic SQL DB > Primary Tab



Specify the **Generic SQL DB > Primary** parameters as described in the following table:

Table 2. Generic SQL DB > Primary Tab Parameters

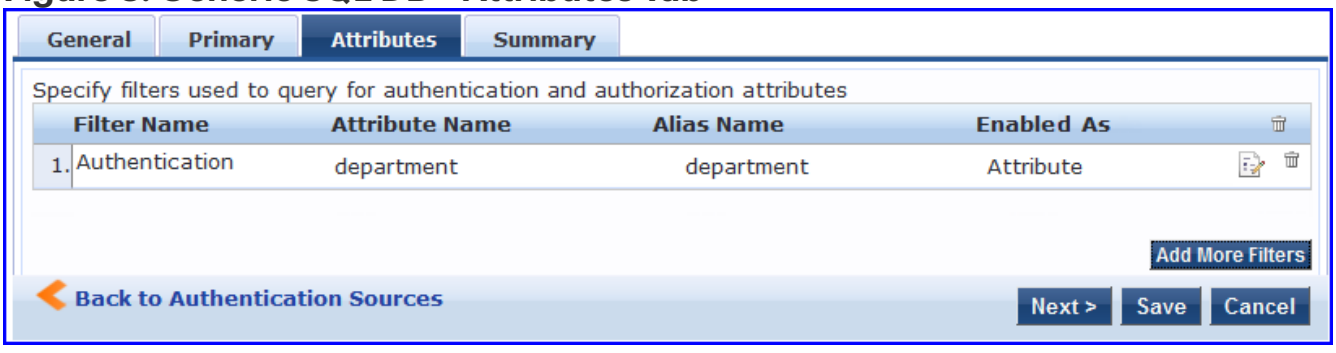
Parameter	Action/Description
Server Name	Enter the Fully Qualified Domain Name (FQDN) or the IPv4/IPv6 address of the database server.
Port (Optional)	Specify a port value to override the default port.
Database Name	Enter the name of the database from which records can be retrieved.
Login Username	Enter the name of the user used to log into the database. This account must have read access to all the attributes that need to be retrieved by the specified filters.
Password	Enter the password for the user account entered in the **Login Username** field.
Timeout	Enter the duration in seconds that Policy Manager waits before attempting to fail over from primary to backup servers (in the order in which they are configured).
ODBC Driver	Select the ODBC (Open Database Connectivity) driver to connect to the database: • PostgreSQL • Oracle 11g • MariaDB • MSSQL If you connect to a Microsoft SQL server using Integrated Authentication, the login username in the authentication source, formatted as either domain/username or UPN (User Principal Name), the following characters are supported: • Backslash (\ ) • At - sign (@) • Hyphen • Underscore

Parameter	Action/Description
Password Type	Specify how the user password is stored in the database: • Cleartext : Stored as clear, unencrypted text. • NT Hash: Stored with an NT hash using MD4. • LM Hash : Stored with a LAN Manager H ash using DES. • SHA: Stored with a Secure Hash Algorig hm (SHA) hash. • SHA256: Stored with an SHA - 256 hash function. • MD5 **NOTE** SHA1 - Base64, SHA256 - Bas e64 and SHA512 - Base64 are a Base64 encoding of their co rresponding hex encoded valu e or hex encoded value conve rted to bytes.

Attributes Tab

The **Attributes** tab defines the SQL DB query filters and the attributes to be fetched by using those filters.

Figure 3. Generic SQL DB > Attributes Tab



Viewing Existing Filters The following table describes the **Generic SQL DB > Attributes (Filter List)** parameters:

Table 3. Generic SQL DB > Attributes Tab (Filter List) Parameters

Parameter	Action/Description
Filter Name	Specifies the name of the filter.
Attribute Name	Specifies the name of the SQL DB attributes defined for this filter.
Alias Name	Specifies an alias name for each attribute name selected for the filter.
Enabled As	Indicates whether the filter is enabled as a role or attribute type. This can also be blank.
Add More Filters	Click this button to open the **Configure Filter** page. Use this page to define a filter query and the related attributes to be fetched from the SQL DB store. Figure 4 displays the **Generic SQL DB > Configure Filter** page.

Adding More Filters

To add more filter queries and their related attributes:

1. Click **Add More Filters**. The **Configure Filter** page opens:

Figure 4. Generic SQL DB > Configure Filter Page

2. Select **Click to add**. A new row opens.
3. Specify the **Name** of the filter, **Alias Name**, **Data Type**, and optionally, **Enabled As** attributes.
4. Repeat to add additional filters, then click **Save**.
5. Specify the **Add Generic SQL DB > Configure Filter** parameters as described in the following table:

Table 4. Add Generic SQL DB > Configure Filter Page Parameters

Parameter	Action/Description
Filter Name	Enter the name of the filter.
Filter Query	Specify an SQL query to fet he user or device record in D
Name	Specify the name of the attr
Alias Name	Specify the name for the att s the same as the attribute n
Data Type	Specify the data type for thi ng, Integer, or Boolean.

Parameter	Action/Description
Enabled As	Specify whether this value t role or attribute in an enforc asses the step of having to a anager through a role - map

Summary Tab

Use the **Summary** tab to view the Generic SQL DB authentication source configuration.

HTTP

About this task

The HTTP authentication source relies on the GET method to retrieve information. The client submits a request, and then the server returns a response. All request parameters are included in the URL. For example:

URL: https//hostname/webservice/.../%{Auth:Username}?param1=%{...}&param2=value2

HTTP relies on the assumption that the connection between the client and server is secure and can be trusted.

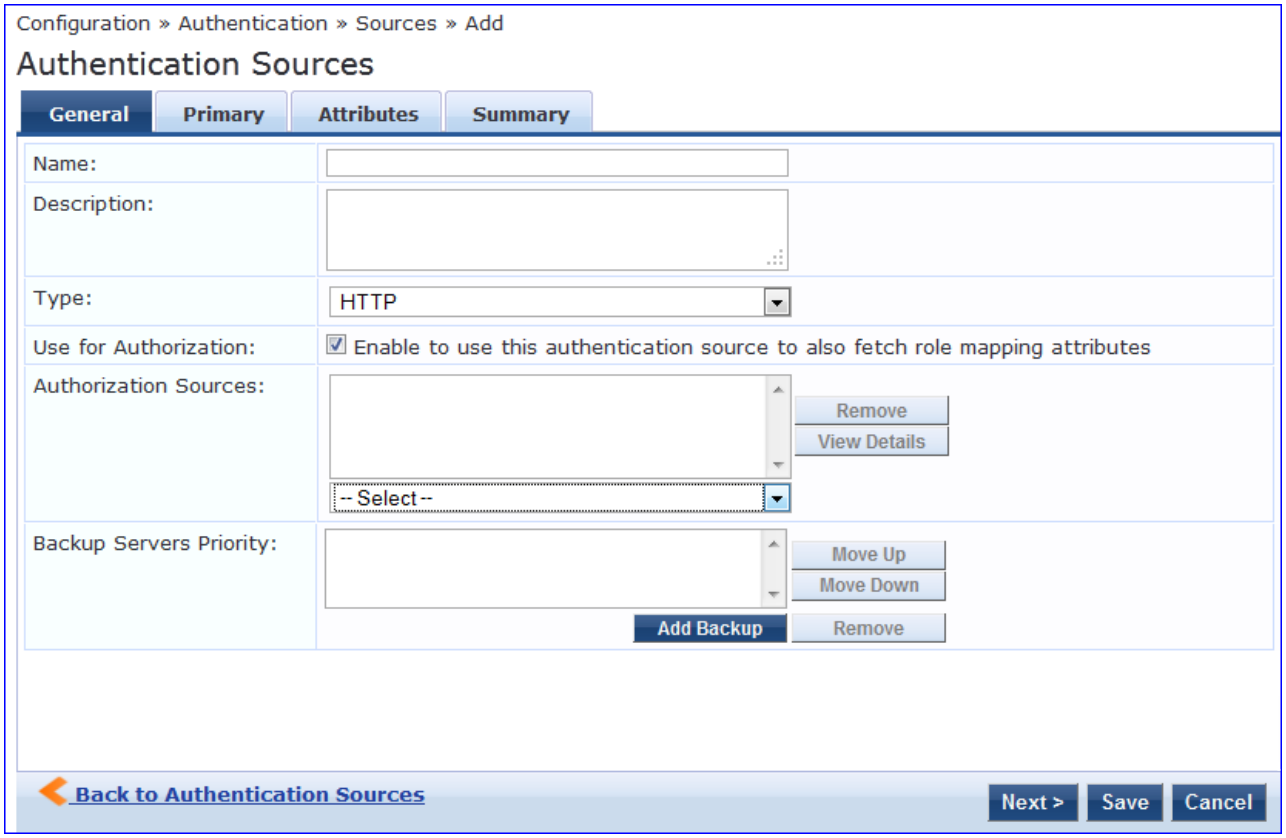
To configure an authentication source for an HTTP service:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click the **Add** link. The **Add Authentication Sources** page opens.
3. Configure primary and backup servers, session details, filter query, and role mapping attributes to fetch HTTP authentication sources.

   General Tab The **General** tab labels the authentication source and defines session details, authorization sources, and backup server details.

   Figure 1. HTTP > General Tab



a. Specify the **HTTP > General** tab parameters as described in the following table:

Table 1. HTTP > General Tab Parameters

Parameter	Action/Description
Name	Specify the name of the authentication source.
Description	Provide the additional information that helps to identify the authentication sou rce.
Type	Select the type of source. In this contex t, select **HTTP**.
Use for Authorization	Enable this option to request Policy Ma nager to fetch role - mapping attributes (or authorization attributes) from this a uthentication source. If a user or devic e successfully authenticates against thi s authentication source, then Policy Ma nager also fetches role - mapping attrib

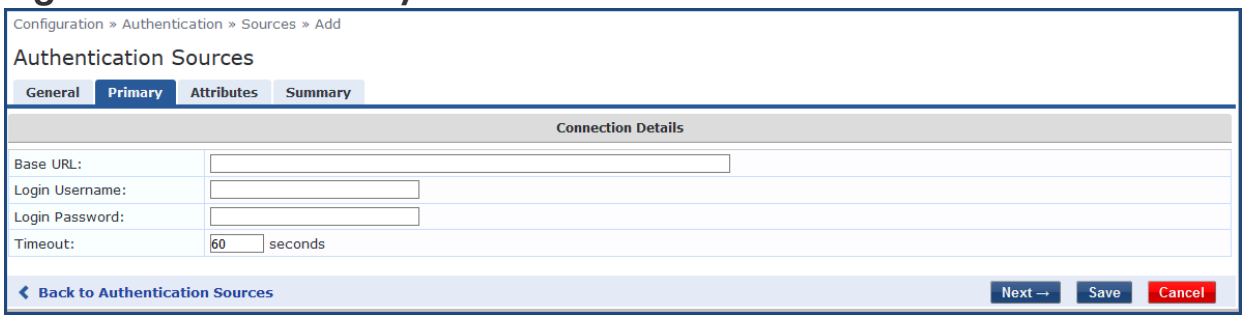
Parameter	Action/Description
	utes from the same source if the **Use for Authorization** field is enabled. This check box is enabled by default.
Authorization Sources	Specify additional sources from which to fetch role - mapping attributes. Select a previously configured authentication source from the drop - down list and click **Add** to add it to the list of authorization sources. Click **Remove** to remove the selected additional resource from the list. If Policy Manager authenticates the user or device from this authentication source, then also fetches role - mapping attributes from these additional authorization sources. The default authorization sources are as follows: • [Denylist User Repository] [Local] • [Endpoints Repository] [Local] • [Guest Device Repository] [Local] • [Guest User Repository] [Local] • [Insight Repository] [Local] • [Local User Repository] [Local] • [zone cache Repository] [HTTP] ◦ This authorization source of type **HTTP** fetches the **store numbers list** attribute stored in the cache against the local host. • [Onboard Devices Repository] [Local] • [Social Login Repository] [Local]

Parameter	Action/Description
	• [Time Source] [Local] **NOTE** You can specify additional authorization sources at th e service level. Policy Mana ger fetches role - mapping attributes irrespective of w hich authentication source the user or device was auth enticated against.
Backup Servers	To add a backup server, click **Add Backu p**. From the **Backup 1** tab, you can specify connection details for a backup server. To remove a backup server, select the se rver name and click **Remove**. Select **Move Up** or **Move Down** to chan ge the server priority of the backup se rvers. This is the order in which Policy M anager attempts to connect to the bac kup servers.

Primary Tab

a. The **Primary** tab defines the settings for the primary server.

Figure 2. HTTP > Primary Tab



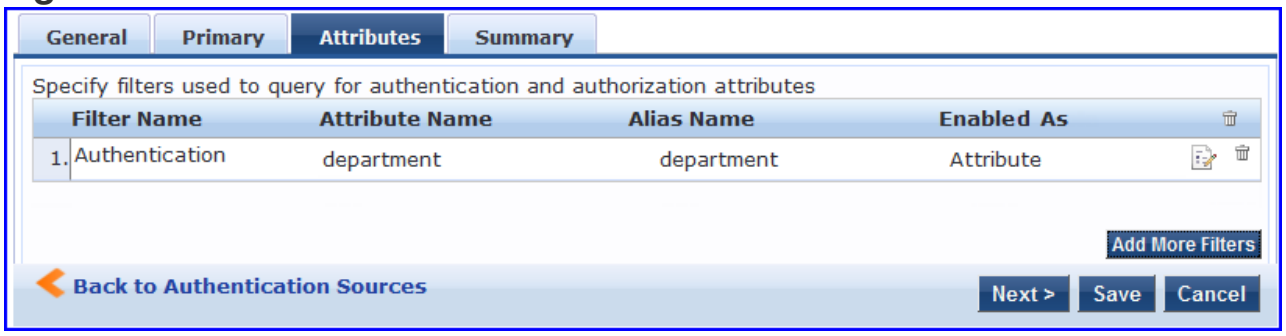
b. Specify the **HTTP > Primary** tab parameters as described in the following table:

Table 2. HTTP > Primary Tab Parameters

Parameter	Action/Description
Base URL	Enter the base URL (host name) or IP ad dress of the HTTP server. For example, http://<hostname> or <full y - qualified domain name>:xxxx, where xxxx is the port to access the HTTP se rver.
Login Username	Enter the name of the user used to log i nto the database. This account must have read access t o all the attributes that need to be retri eved by the specified filters.
Login Password	Enter the password for the user account entered in the **Login Username** field.
Timeout	You can configure a timeout interval. Po licy Manager will close the connection when the specified timeout interval is reached. The default is **60** seconds.

Attributes Tab The **Attributes** tab defines the HTTP query filters and the attributes to be fetched by using those filters.

Figure 3. HTTP > Attributes Tab



a. Specify the **HTTP > Attributes** tab parameters as described in the following table:

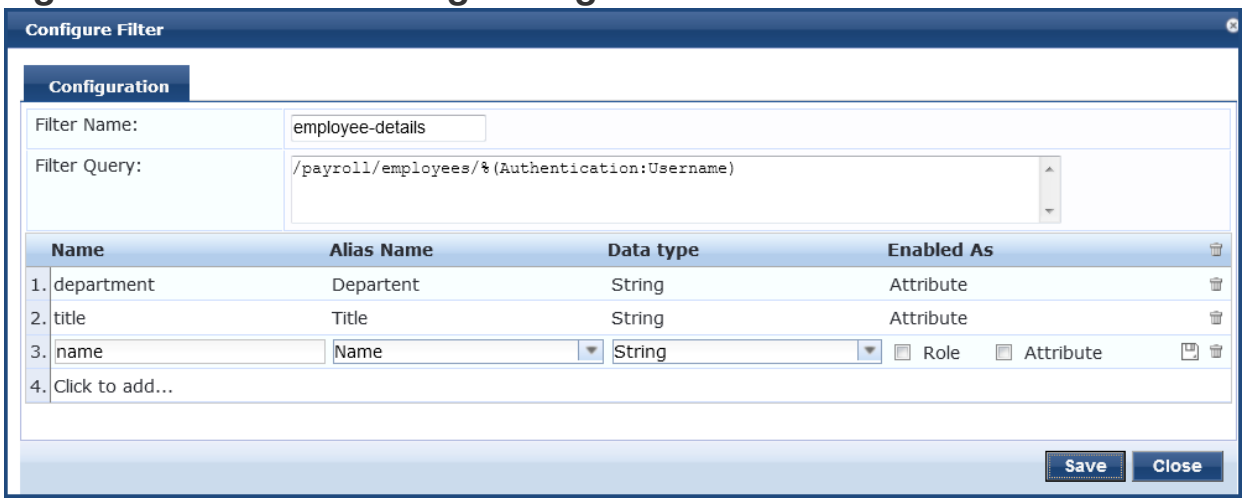
Table 3. HTTP > Attributes tab (Filter List) Parameters

Parameter	Action/Description
Filter Name	Displays the name of the filter.
Attribute Name	Specifies the name of the SQL DB attributes defined for this filter.
Alias Name	Specifies the name of an alias name for each attribute name selected for the filter.
Enabled As	Indicates whether an attribute is enabled as a role.
Add More Filters	Opens the **Configure Filter** page. For more information, see Add More Filters .

Add More Filters

The **Configure Filter** page defines a filter query and the related attributes to be fetched from the SQL DB store.

Figure 4. HTTP Filter Configure Page



b. Specify the **HTTP Configure > Filter** parameters as described in the following table:

Table 4. HTTP Configure Filter Page Parameters

Parameter	Action/Description
Filter Name	Enter the name of the selected filter.
Filter Query	Specify the HTTP path (without the se rver name) to fetch the attributes from t he HTTP server. For example, if the full path name to the filter is http server *URL = http://<hostna me* or *fqdn>:xxxx/abc/def/xyz*, where yo u enter **/abc/def/xyz**.
Name	Specify the name of the attribute.
Alias Name	Specifies the alias name for the attribut e. By default, this is the same as the attr ibute name.
Data Type	Specifies the data type for this attribute such as String, Integer, and Boolean.
Enabled As	Specify whether the value to be used di rectly as a role or attribute in an enforce ment policy. This bypasses the step of a ssigning a role in Policy Manager throug h a role - mapping policy.

Summary Tab You can use the **Summary** tab to view configured parameters.

Kerberos

About this task

Policy Manager can perform standard PAP/GTC or tunneled PAP/GTC (for example, EAP-PEAP[EAP-GTC]) authentication against any Kerberos 5-compliant server, such as the Microsoft Active Directory server. It is mandatory to pair this source type with an authorization source (identity store) containing user records.

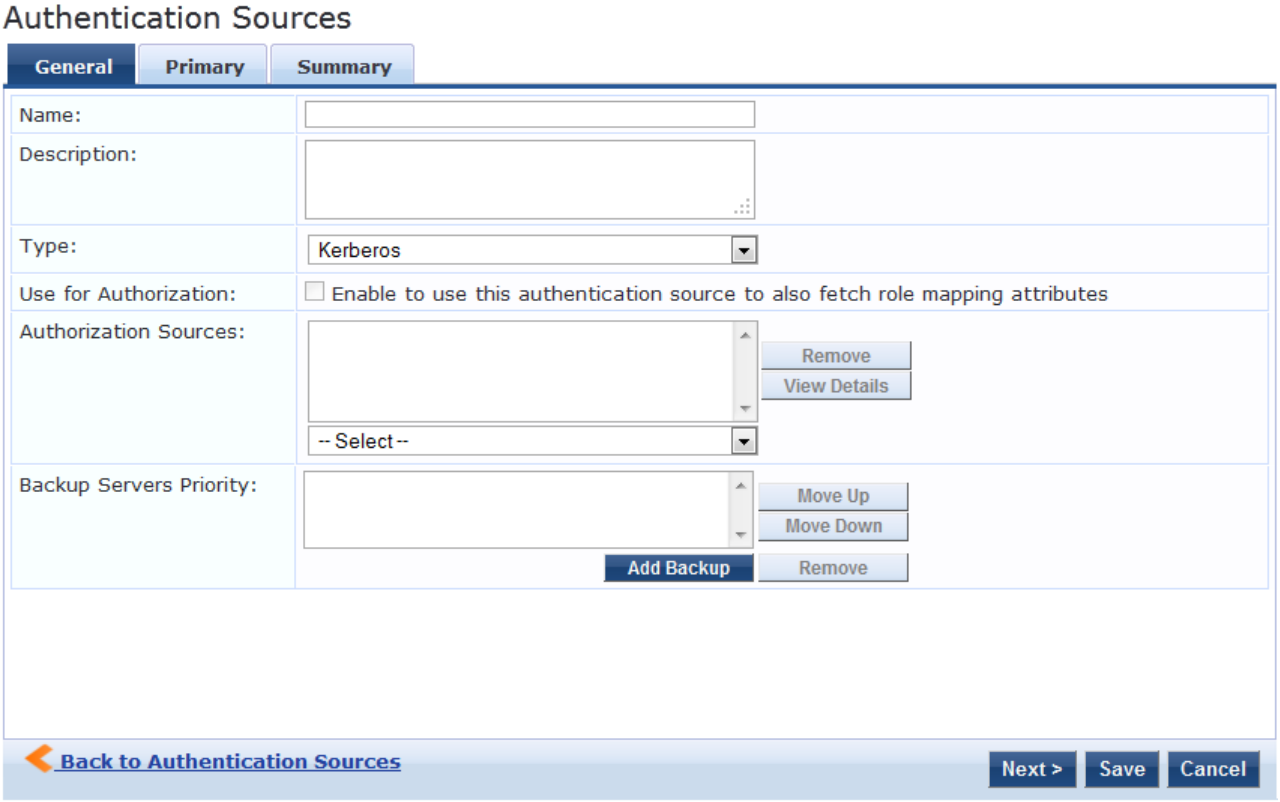
To configure an authentication source for a Kerboros service:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click the **Add** link. The **Add Authentication Sources** page opens.

   General Tab The **General** tab labels the authentication source and defines session details, authorization sources, and backup server details. The following figure displays the **Kerberos > General** tab:

Figure 1. Kerberos > General Tab



The following table describes the **Kerberos > General** parameters:

Table 1. Kerberos > General Tab Parameters

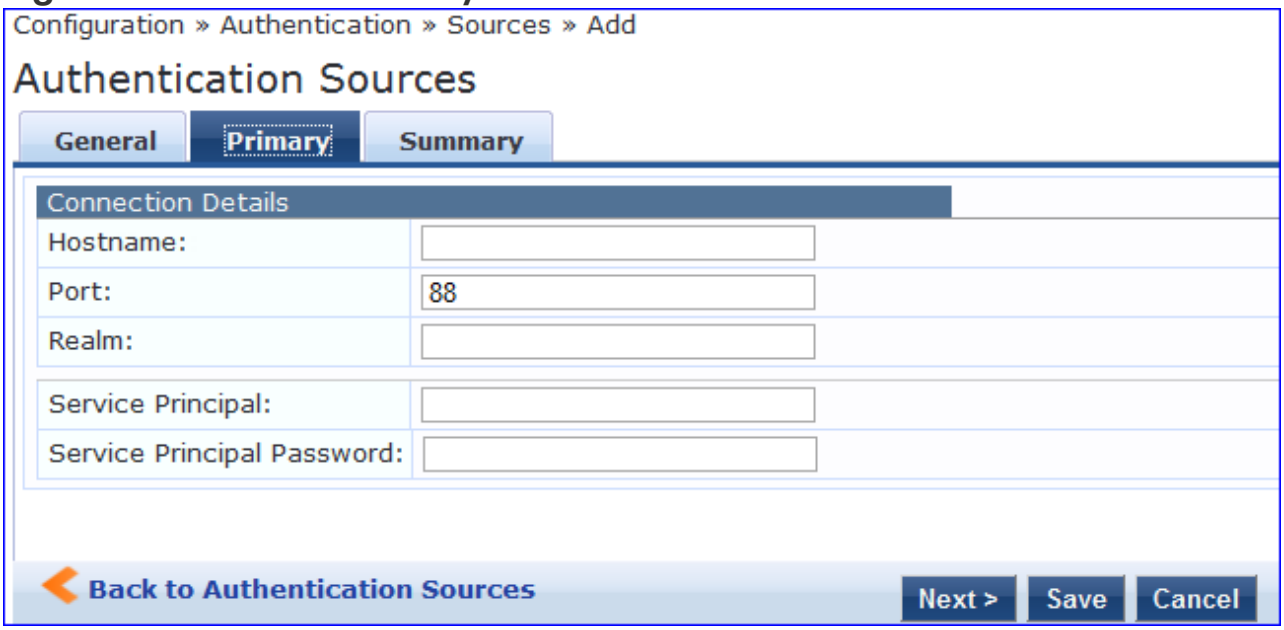
Parameter	Description
Name	Specify the name of the authentication s ource.
Description	Provide the additional information that he lps to identify the authentication source.

Parameter	Description
Type	From the **Type** drop - down, select **Kerberos**.
Use for Authorization	Leave disabled in this context.
Authorization Sources	Specify one or more authorization sources from which role - mapping attributes are to be fetched. a. Select an authentication source from the drop - down list . b. Click **Add** to add it to the list of authorization sources. • Click **Remove** to remove the selected authentication source from the list. **NOTE** You can specify additional authorization sources at the service level. Policy Manager fetches role - mapping attributes irrespective of which authentication source the user or device was authenticated against.
Backup Servers	a. To add a backup Kerberos server, click **Add Backup**. b. From the **Backup 1** tab, you can specify connection details for a backup server (same fields applicable for primary server specified below). • To remove a backup server, select the server name and click **Remove**.

Parameter	Description
	• Select **Move Up** or **Move Down** to change the server priority of the backup servers. This is the order in which Policy Manager attempts to connect to the backup servers.

Primary Tab The **Primary** tab defines the settings for the primary server. The following figure displays the **Kerberos > Primary** tab:

Figure 2. Kerberos > Primary Tab



The following table describes the **Kerberos > Primary** parameters:

Table 2. Kerberos > Primary Tab Parameters

Parameter	Description
Hostname	Specify the name of the host or the IP address of the Kerberos server.
Port	Specify the port at which the token server listens for kerberos connections. The default port is **88**.
Realm	Specify the **Kerberos** domain.

Parameter	Description
Service Principal Name	Enter the identity of the service principal as configured in the Kerberos server.
Service Principal Password	Enter the password for the service principal.

Summary Tab Use the **Summary** tab to view the authentication source configuration.

RADIUS/Radsec Server

About this task

Use the **RADIUS/RadSec Server** as an authentication source to allow Policy Manager to query a third-party **RADIUS/RadSec Server** for authentication.

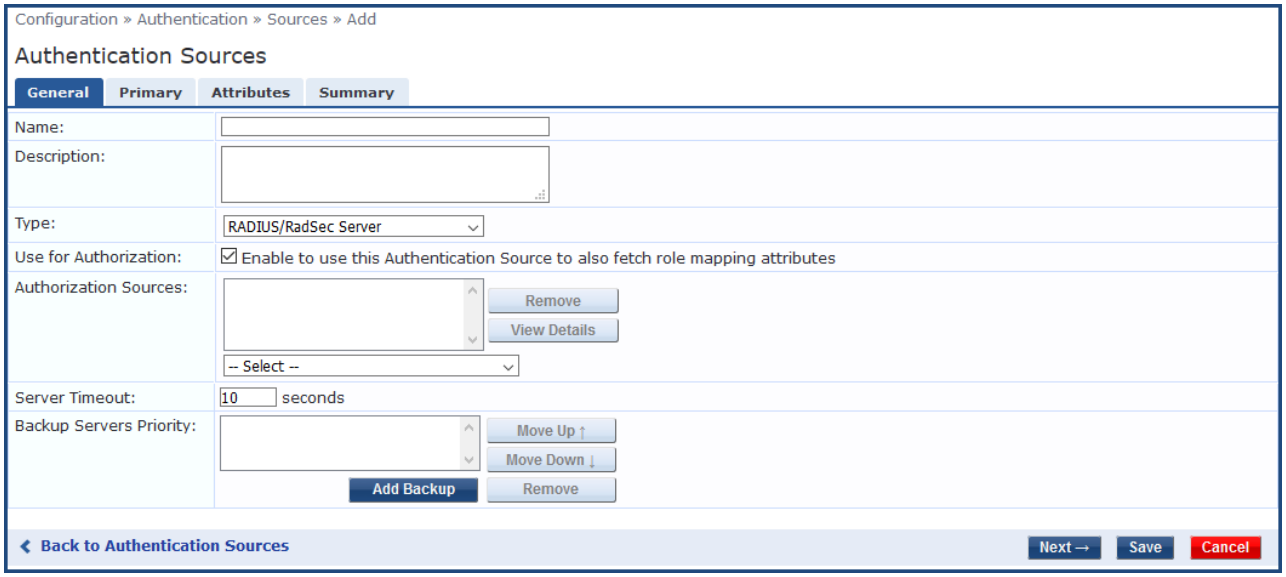
To configure an authentication source for a RADIUS service:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click the **Add** link. The **Add Authentication Sources** page opens.
3. From the **General** tab, click the **Type** drop-down list and select the **RADIUS/RadSec server** option.

   General Tab The **General** tab labels the authentication source and defines session details, authorization sources, and backup server details.

   Figure 1. RADIUS/RadSec Server > General Tab



To define a RADIUS/RadSec server as an authentication source specify the **RADIUS/ RadSec Server > General** parameters as described in the following table:

Table 1. RADIUS/RadSec Server > General Tab Parameters

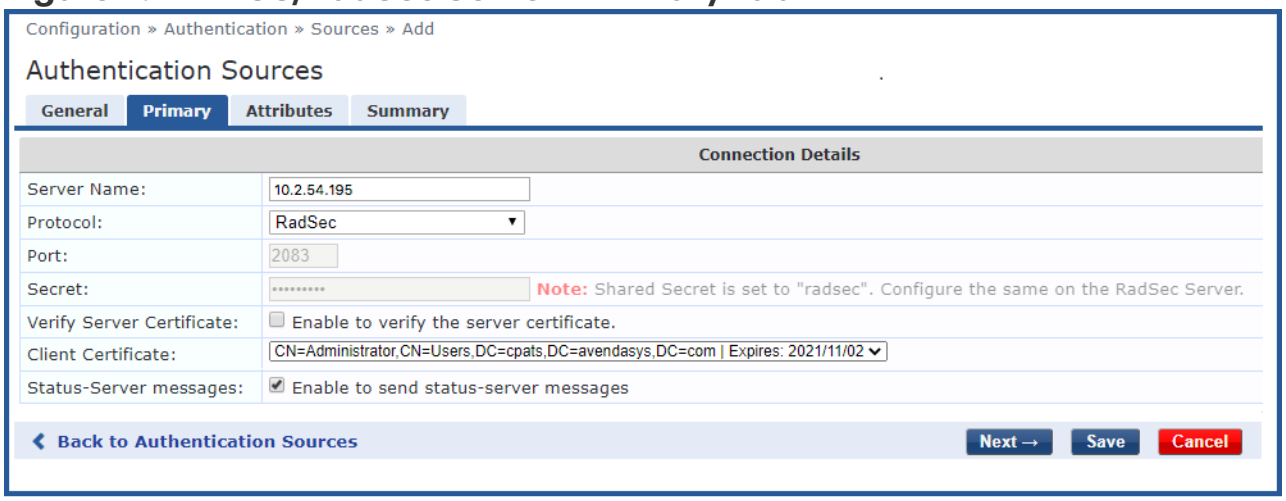
Parameter	Action/Description
Name	Specify the name of the authentication s ource.
Description	Provide the additional information that he lps to identify the authentication source.
Type	From the Type drop - down, select **RADI US/RadSec Server**.
Use for Authorization	Enable this check box to request Policy M anager to fetch role - mapping attributes (or authorization attributes) from this aut hentication source. If a user or device successfully authenti cates against this authentication source, Policy Manager also fetches role - mappi ng attributes from the same source if the **Use for Authorization** field is enabled. Th is check box is enabled by default.

Parameter	Action/Description
Authorization Sources	Specify additional sources from which ro le - mapping attributes are to be fetched. a. Select a previously configured authent ication source from the drop - down li st. b. To add the authentication source to th e list of authorization sources, click **Ad d**. To remove the authentication sourc e from the list, click **Remove**. If Policy Manager authenticates the us er or device from this authentication s ource, then also fetches role - mappin g attributes from these additional auth orization sources. **NOTE** Policy Manager fetches ro le - mapping attributes re gardless of which authent ication source the user or device was authenticated against.
Server Timeout	Specify the duration in number of secon ds that Policy Manager waits before consi dering this server unreachable. If multiple backup servers are available, t his value indicates the duration in numbe r of seconds that Policy Manager waits b efore attempting to fail over from the pri mary to the backup servers in the order i n which they are configured.
Backup Servers Priority	To add a backup server, click **Add Backup** . From the **Backup 1** tab, you can specify connection details for a backup server (s

Parameter	Action/Description
	ame fields as for primary server that are s pecified below). • To remove a backup server, select the server name and click **Remove**. • Select **Move Up** or **Move Down** to cha nge the server priority of the backup s ervers. This is the order Policy Manag er attempts to connect to the backup servers.

Primary Tab The **Primary** tab defines the settings of the primary server.

Figure 2. RADIUS/RadSec Server > Primary Tab



Specify the **Radius/RadSec Server > Primary** parameters as described in the following table. Note that the available parameters in this dialog depend upon whether you are creating a **RADIUS** or **RadSec** authentication source.

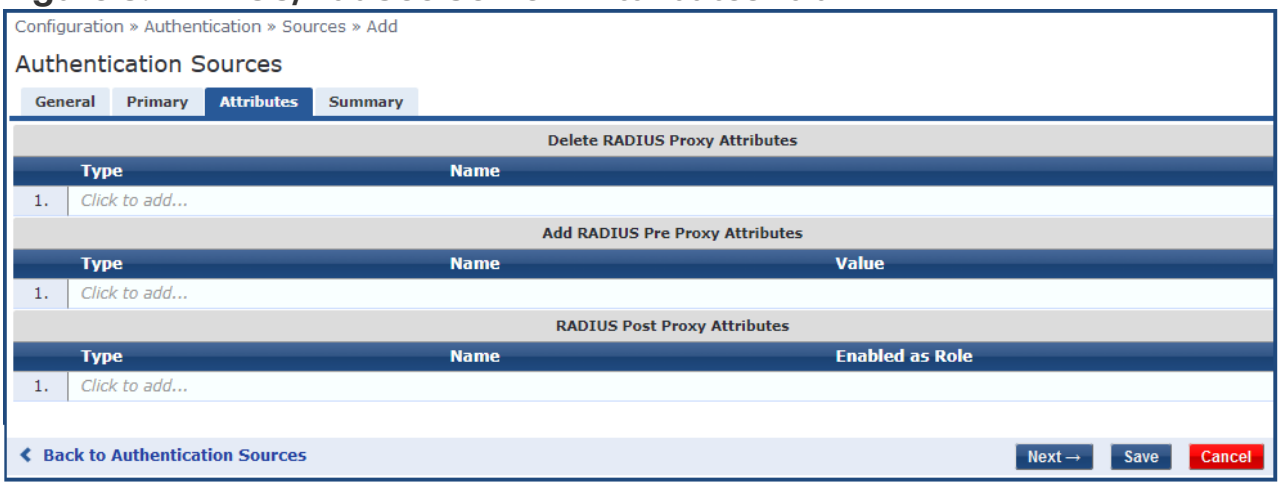
Table 2. RADIUS/RadSec Server > Primary Tab Parameters

Parameter	Action/Description
Connection Details	
Server Names	Enter the name of the RADIUS server.
Protocol	Select either **RADIUS** or **RadSec**

Parameter	Action/Description
Port	The default port number for the RADIUS protocol is **1812**, and the default port for RadSec is 2083. You may enter a different port number if required.
Secret	Enter the secret key for authentication.
Verify Server Certificate	This setting is available when you select t he **RadSec** option. When enabled, the Ra dSec proxy performs checks to validate t he AAA server certificate, including chec ks for valid start and end dates, and chec ks to verify the certificate chain of validit y back to the issuing CA.
Client Certificate	This setting is available when you select the **RadSec** protocol option. Click this d rop - down list and select a client certific ate for RADIUS authentication from the l ist of client certificates uploaded via the **Administration > Certificates > Certifica te** Store page, then defined as a RadSec certificate via the **Administration > Certi ficates > Trust** List page. This drop - dow n list displays both the subject name and certificate expiry date, allowing you to di stingush between certificates with the sa me name but with different expiration da tes.
Status - Server Messages	This setting is available when you select the **RadSec** protocol option. When enabl ed, the proxy will send regular status - se rver messages to the server during idle p eriods to verify that the proxy is still alive and active. This option is enabled by defa ult.

Attributes Tab The **Attributes** tab defines the RADIUS/RadSec query filters and the attributes to be fetched by using those filters.

Figure 3. RADIUS/RadSec Server > Attributes Tab



Specify the **RADIUS/RadSec Server > Attributes** parameters as described in the following table:

Table 3. RADIUS/RadSec Server > Attributes Tab Parameters

Parameter	Description
Delete RADIUS Proxy Attributes	Specify an attribute to be removed from Access - Request messages before those messages are sent to the proxy authentication server. • **Type**: Select a type from the drop - down. • **Name**: Select a name from the drop - down. • **Value**: Enter a value in the text box. This feature can be used when ClearPass Policy Manager is used as a RADIUS proxy and is sending undesirable attributes. For example, if ClearPass Policy Manager is acting as a RADIUS proxy for guest network authentication and sends requests to an openRADIUS server configured to accept RADIUS requests only from the ClearPass Policy Manager public IP address, when ClearPass passes the NAS IP address in the request, the RADIUS request may be blocked. This feature could be u

Parameter	Description
	sed in this scenario to remove this unexp ected NAS IP address and then modify th e attribute to be an allowed IP address.
Add RADIUS Pre - Proxy attributes	The following attributes that can be set p rior to the proxy authentication: • **Type**: Select a type from the drop - d own. • **Name**: Select a name from the drop - down. • **Value**: Enter a value in the text box.
RADIUS Post - Proxy attributes	The attributes for the post - proxy authen tication are identical except that these ca n be set after the proxy authentication. • **Type**: Select a type from the drop - d own. • **Name**: Select a name from the drop - down. • **Value**: Enter a value in the text box. Save the changes by clicking the **Save** ic on that appears at the end of the row.

Summary Tab Use the **Summary** tab to view the RADIUS Server authentication source configuration.

Adding a Static Host List as an Authentication Source

About Static Host Lists

You can configure primary and backup servers, session details, and the list of static hosts for **Static Host List** authentication sources. A static host list often functions, in the context of the service, as an allowed list or blocked list. Therefore, static host lists are

configured independently at the global level. A static host list comprises a named list of MAC addresses or IP addresses, which can be invoked in the following ways:

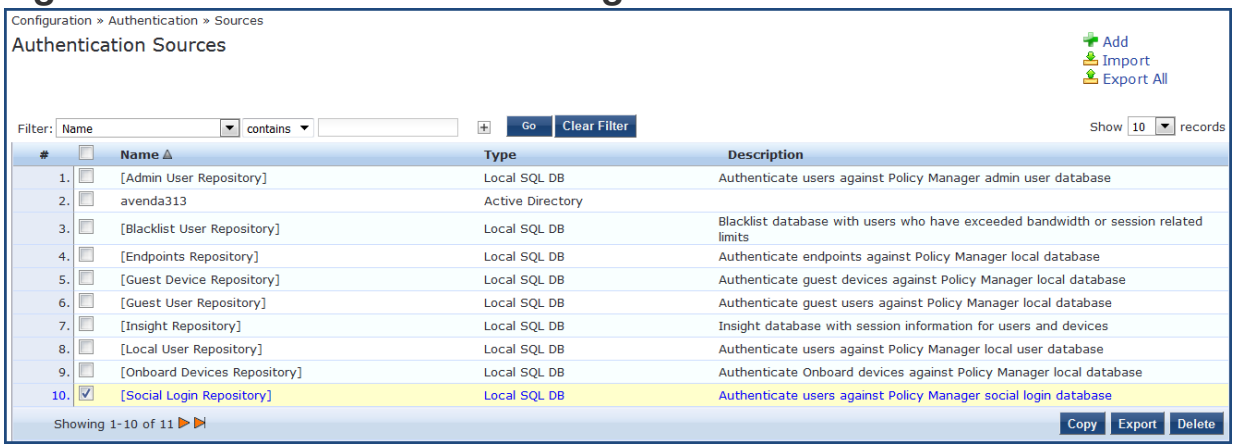
- In service and role-mapping rules as a component.
- For non-responsive services on the network (for example, printers or scanners), as an authentication source.

> **NOTE**
>
> Only static host lists of type **MAC Address** are available as authentication sources. For more information about static host lists, see Managing Static Host Lists .

To add a static host list as an authentication source:

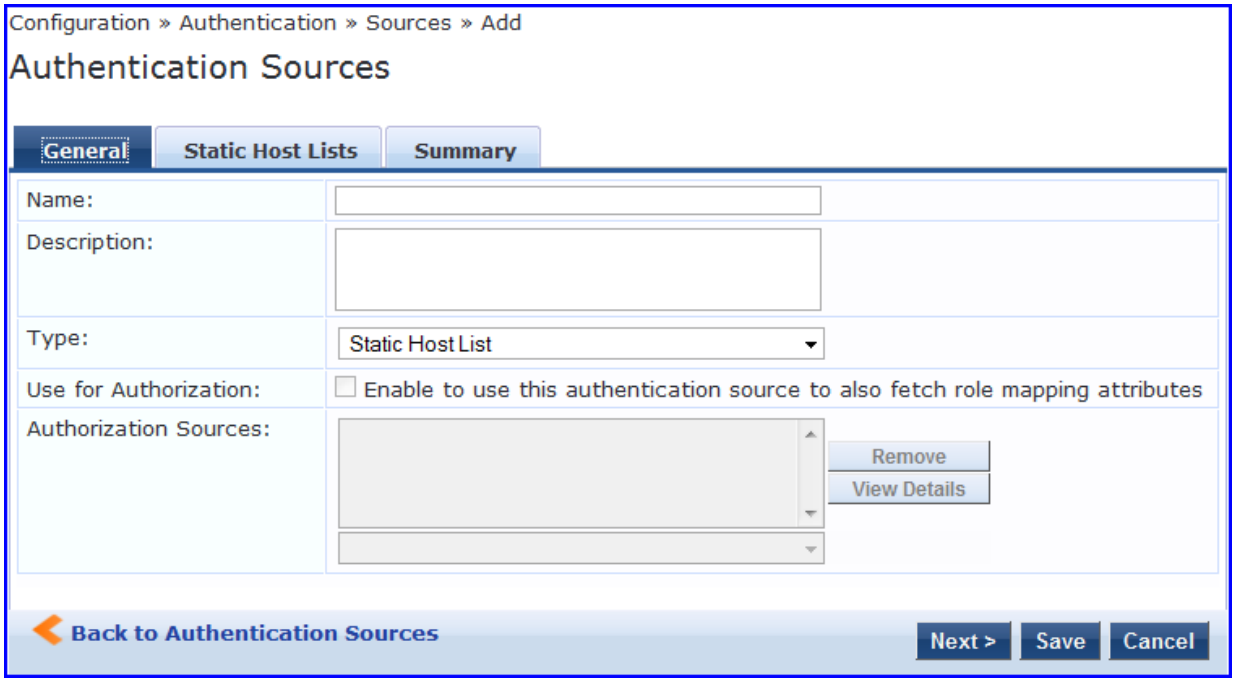
1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.

Figure 1. Authentication Sources Page



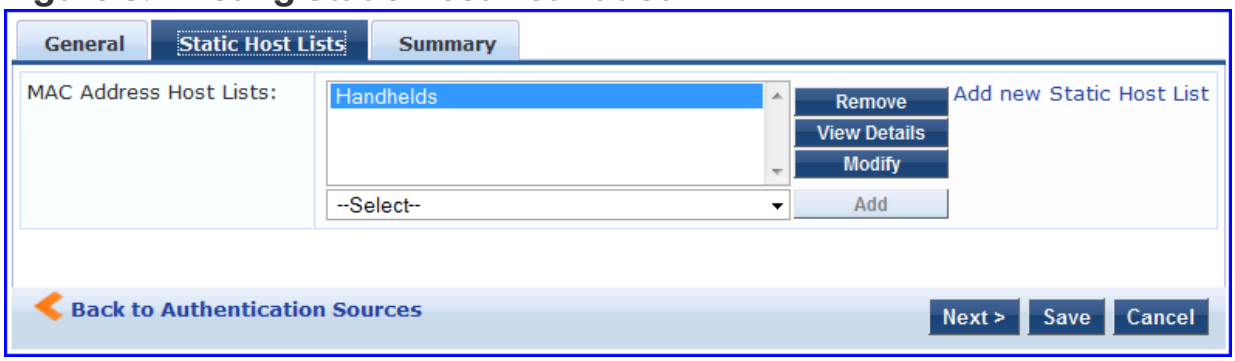
2. Click the **Add** link. The **Add Authentication Sources** dialog opens.

Figure 2. Specifying a Static Host List as Authentication Source



3. Enter the name and description of the static host list.
4. In the **Type** field, select **Static Host List**. In this context, the **Use for Authorization** and **Authorization Sources** fields are not configurable.
5. Click **Next**. The **Static Hosts Lists** dialog opens.
6. From the **Static Host Lists** tab, select a static host list from the drop-down list. The selected static host list is added to the MAC Address Host Lists (see Figure 3).

Figure 3. Existing Static Host List Added



NOTE

Only static host lists of type **MAC Address Host Lists** or **MAC Address Regular Expression** can be configured as authentication sources.

a. To remove the selected static host list, click **Remove**.

b. To view the contents of the selected static host list, click **View Details**.

c. To modify the selected static host list, click **Modify**.

7. Click **Save**.

Token Server

About this task

Policy Manager can perform GTC authentication against any token server than can authenticate users by acting as a RADIUS server (for example, RSA SecurID Token Server) and can authenticate users against a token server and fetch role mapping attributes from any other configured authorization source. Pair this source type with an authorization source (identity store) containing user records. When using a token server as an authentication source, use the administrative interface to optionally configure a separate authorization server. Policy Manager can also use the RADIUS attributes returned from a token server to create role mapping policies. For more information, see Namespaces . You configure primary and backup servers, session details, and the filter query and role mapping attributes to fetch for token server authentication sources.

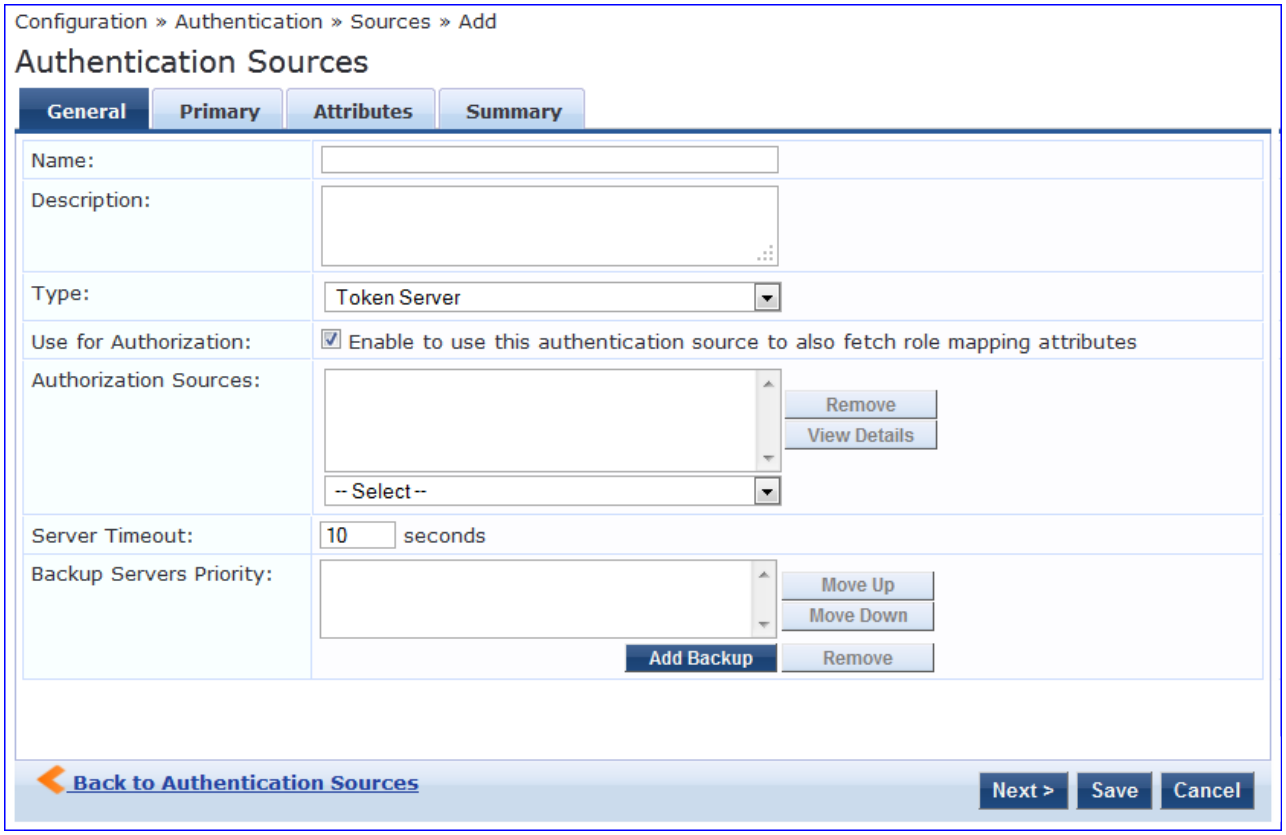
To configure an authentication source for a Token server:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click the **Add** link. The **Add Authentication Sources** page opens.

   General Tab The **General** tab labels the authentication source and defines session details, authorization sources, and backup server details. The following figure displays the **Token Server > General** tab:

   Figure 1. Token Server > General Tab



Specify the **Token Server > General** parameters as described in the following table:

Table 1. Token Server > General Tab Parameters

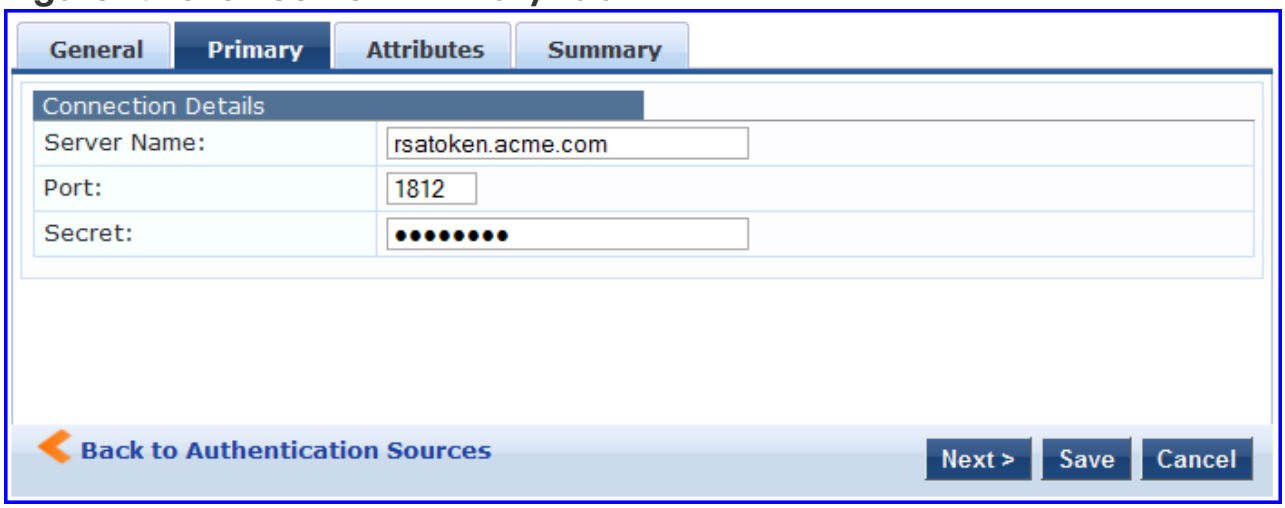
Parameter	Action/Description
Name	Specify the label of the authentication so urce.
Description	Provide the additional information that he lps to identify the authentication source.
Type	Select the type of authentication. In this context, select **Token Server**.
Use for Authorization	Enable this check box to instruct Policy Manager to fetch role mapping attributes (or authorization attributes) from this aut hentication source. If a user or device su ccessfully authenticates against this auth entication source, then Policy Manager al so fetches role mapping attributes from t he same source if the **Use for Authorizati**

Parameter	Action/Description
	on field is enabled. This check box is ena bled by default
Authorization Sources	Specify additional sources from which t o fetch role mapping attributes. Select a previously configured authentication sou rce from the drop - down list, and click **Add** to add it to the list of authorization s ources. Click **Remove** to remove it from t he list. If Policy Manager authenticates the user or device from this authentication sourc e, then it also fetches role mapping attri butes from these additional authorization sources. **NOTE** You can specify additional a uthorization sources at the s ervice level. Policy Manager fetches role mapping attribu tes irrespective of which au thentication source the user or device was authenticated against.
Server Timeout	Specify the duration in seconds that Poli cy Manager waits before attempting to fa il over from primary to backup servers (in the order in which they are configured).
Backup Servers Priority	To add a backup server, click **Add Backup** . From the **Backup 1** tab, you can specify connection details for a backup server (s ame fields as for primary server that are s pecified below). To remove a backup server, select the server name and click **Remove**. Select **Move Up** or **Move Down** to change the

Parameter	Action/Description
	server priority of the backup servers. This is the order in which Policy Manager atte mpts to connect to the backup servers.

Primary Tab The **Primary** tab defines the settings for the primary server.

Figure 2. Token Server > Primary Tab



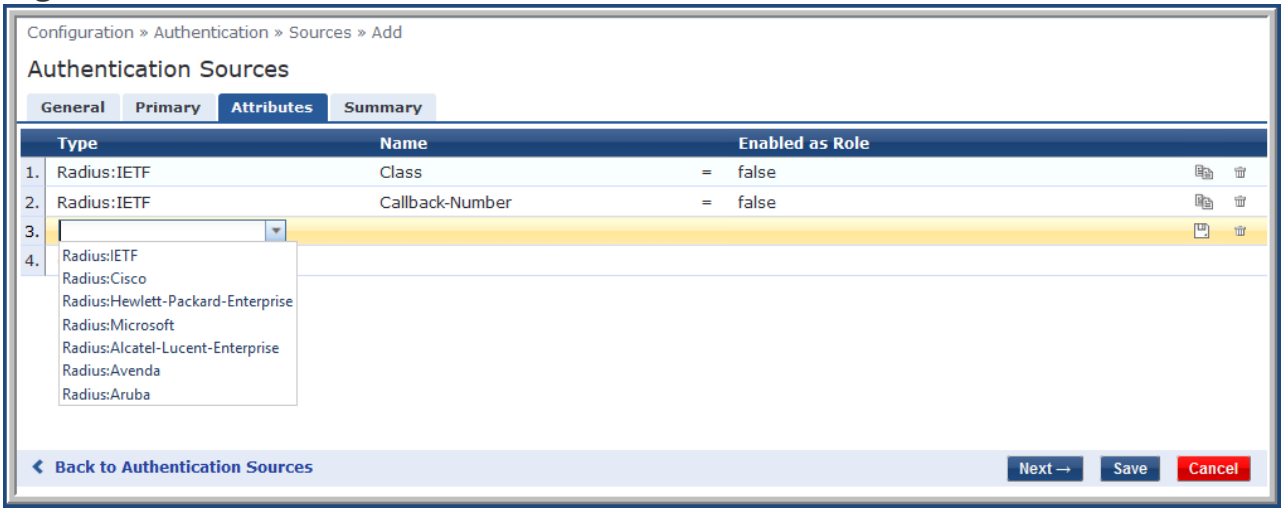
Specify the **Token Server > Primary** parameters as described in the following table:

Table 2. Token Server > Primary Tab Parameters

Parameter	Action/Description
Server Name	Specify the host name or the IP address of the token server,
Port	Specify the UDP port at which the token server listens for RADIUS connections. Th e default port is **1812**.
Secret	Specify the RADIUS shared secret to con nect to the token server.

Attributes Tab The **Attributes** tab defines the RADIUS attributes to be fetched from the token server. These attributes can be used in role-mapping policies. Policy Manager loads all RADIUS vendor dictionaries in the **Type** drop-down list with attributes.

Figure 3. Token Server > Attributes Tab



For more information, see Configuring a Role and Role Mapping Policy . Specify the **Token Server > Attribute** parameters as described in the following table:

Table 3. Token Server > Attribute Tab Parameters

Parameter	Action/Description
Type	Select the type of authentication source f rom the drop - down list.
Name	Specifies the name of the token server at tributes.
Enabled as Role	Specifies whether value is to be used di rectly as a *role* or as an *attribute* in an en forcement policy. This bypasses the step of assigning a role in Policy Manager thro ugh a role - mapping policy.

Summary Tab Use the **Summary** tab to see a summarized view of the parameters configured in the **General**, **Primary**, and **Attributes** tab.

Modifying Authentication Methods or Sources for an Existing Service

About this task

To add or modify the authentication methods or sources for an existing service:

Procedure

1. Navigate to the **Configuration** > **Services** page.

   The **Services** page opens.

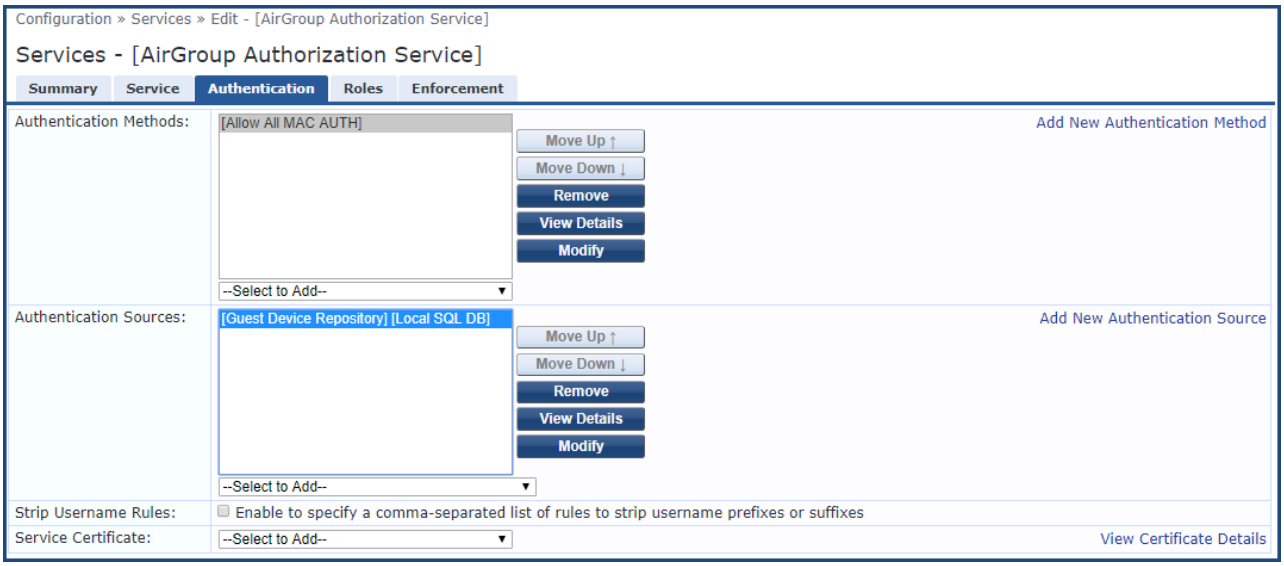
2. Select the service you want to modify.

   The **Edit Service** dialog for the selected service opens.

3. Select the **Authentication** tab.

   The **Edit Services > Authentication** dialog opens:

Figure 1. Modifying Authentication Methods or Sources for a Selected Service



4. Specify the Authentication method or source(s) for the selected service as described in the following table.

Table 1. Authentication Methods or Sources Configuration for a Service

Component	Configuration Steps
Authentication Methods	• Select a method, then select **Move Up** , **Move Down**, or **Remove**. • Select **View Details** to view the detail s of the selected method.

Component	Configuration Steps
	• Select **Modify** to modify the selected authentication method. This displays a dialog with the edit page for the sel ect authentication method. To add a p reviously configured authentication m ethod, select from the **Select to Add** drop - down list. To configure a new method, click the **Add new Authentic ation Method** link. For more informati on about authentication methods, see Adding and Modifying Authentication Methods . **NOTE** An authentication method is only configurable for some s ervice types. For more infor mation, refer to Configuring Services Manually or with Wi zards .
Authentication Sources	• Select an authentication source, then **Move Up**, **Move Down**, or **Remove** as needed. • Select **View Details** to view the detail s of the selected authentication sourc e. • Select **Modify** to modify the selected authentication source. This displays t he **Authentication Source Configurat ion** wizard for the selected authentica tion source. • To add a previously configured authe ntication source, select from the **Sele ct to Add** drop - down list. • To configure a new authentication sou rce, click the **Add new Authenticatio**

Component	Configuration Steps
	n Source link. For more information a bout authentication sources, see Addi ng and Configuring Authentication So urces .
Strip Username Rules	Select the **Enable to specify a comma - s eparated list of rules to strip username p refixes or suffixes** check box to preproc ess the username and to remove prefixes and suffixes before authenticating it to th e authentication source.
Service Certificate	If there is a Service Certificate to associat e with this service, select it from the drop - down list.

Posture Policies, Audit Servers, Agentless OnGuard

This chapter provides the following information:

- Posture Architecture and Flow
- Creating a New Posture Policy
- Configuring Posture Policy Agents and Hosts
- Configuring Posture Policy Plug-ins
- Configuring Posture Policy Rules
- Configuring Audit Servers
- Configuring Agentless OnGuard

Subtopics

Posture Architecture and Flow
Creating a New Posture Policy
Configuring Posture Policy Agents and Hosts
Configuring Posture Policy Plug-ins
Configuring Posture Policy Rules

Configuring Audit Servers
Configuring Agentless OnGuard
Policy Manager supports Agentless OnGuard for posture analysis of Windows, macOS, and Linux clients, simplifying endpoint software management.

Posture Architecture and Flow

Policy Manager supports two types of posture checking: *posture policies* and *audit servers*.

Subtopics

Posture Policy
Audit Servers
Assessing Client Consistency
Application Token
System Token

Posture Policy

Policy Manager supports four preconfigured posture plug-ins for Windows, one plug-in for Linux®, and one plug-in for macOS®. Administrators can configure rules against these policies that test for specific attributes of client health and correlate the results to return application posture tokens for processing by enforcement policies.

NOTE
A service can be configured without any posture policy.

Audit Servers

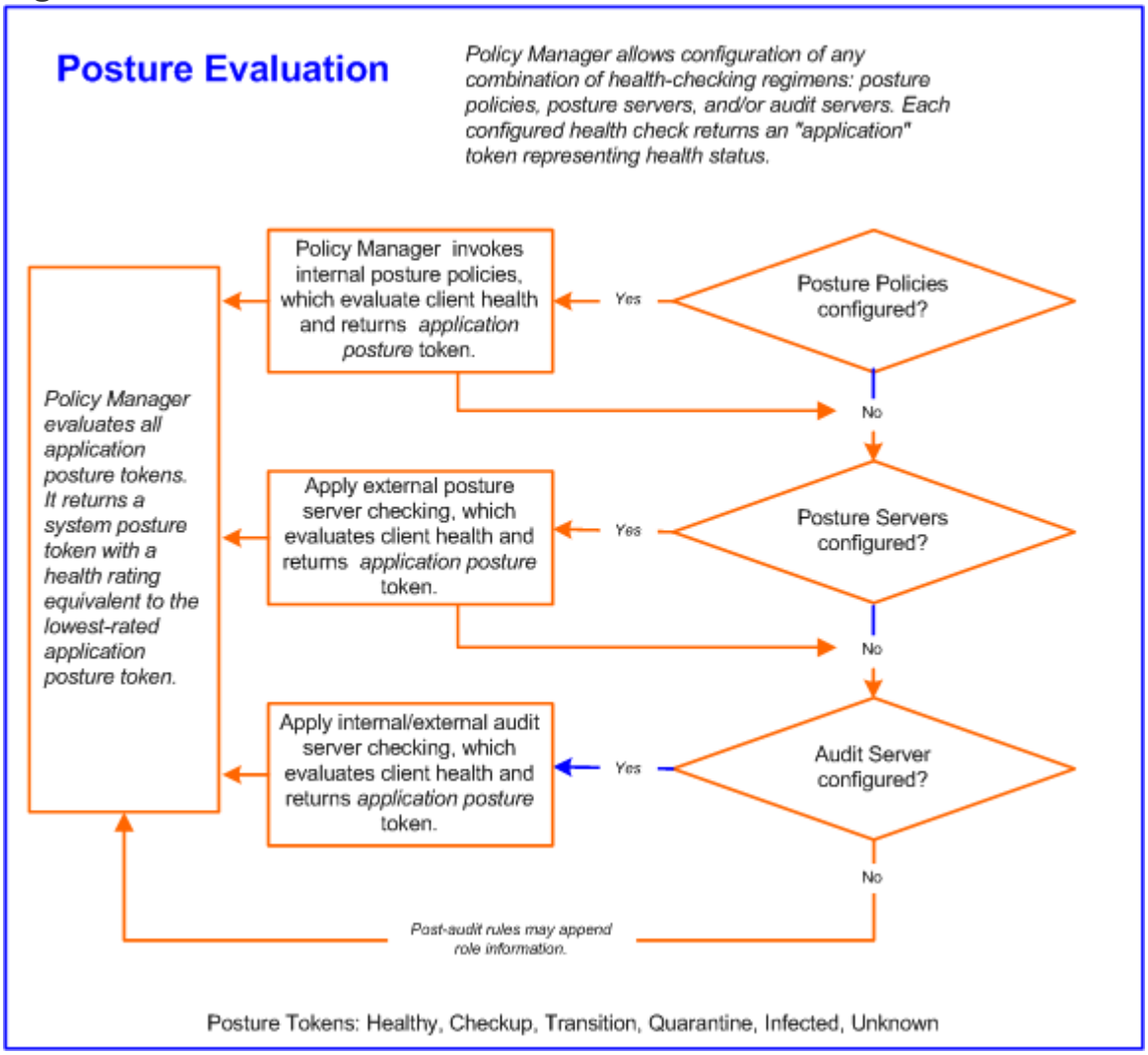
Audit servers provide posture checking for unmanageable devices, such as devices lacking adequate posture agents or supplicants. In the case of such clients, the audit server's post-audit rules map clients to roles.

Policy Manager supports two types of audit servers:

- **NMAP audit server**: Primarily used to derive roles from post-audit rules.

- **NESSUS audit server**: Primarily used for vulnerability scans (and, optionally, post-audit rules).

Figure 1. Posture Evaluation Process



Assessing Client Consistency

ClearPass Policy Manager uses posture evaluation to assess client consistency with enterprise endpoint health policies, specifically with respect to:

- Operating system version/type
- Registry keys/services present (or absent)
- Antivirus or firewall configuration
- Patch level of software components

- Peer-to-Peer (P2P) application checks
- Services to be running or not running
- Processes to be running or not running

Application Token

Each configured health check returns an application token representing health:

- **Healthy.** Client is compliant: there are no restrictions on network access.
- **Checkup.** Client is compliant; however, there is an update available. This can be used proactively to remediate to a healthy state.
- **Transient.** Client evaluation is in progress; typically associated with auditing a client. The network access granted is interim.
- **Quarantine.** Client is out of compliance; restrict network access so the client only has access to the remediation servers.
- **Infected.** Client is infected and is a threat to other systems in the network; network access should be denied or severely restricted.
- **Unknown.** The posture token of the client is unknown.

System Token

Upon completion of the configured posture checks, Policy Manager evaluates all application tokens and calculates a **system token** , equivalent to the most restrictive rating for all returned application tokens. The system token provides the health posture component for input to the enforcement policy.

Creating a New Posture Policy

Subtopics

About Posture Policies

About Posture Policies

About this task

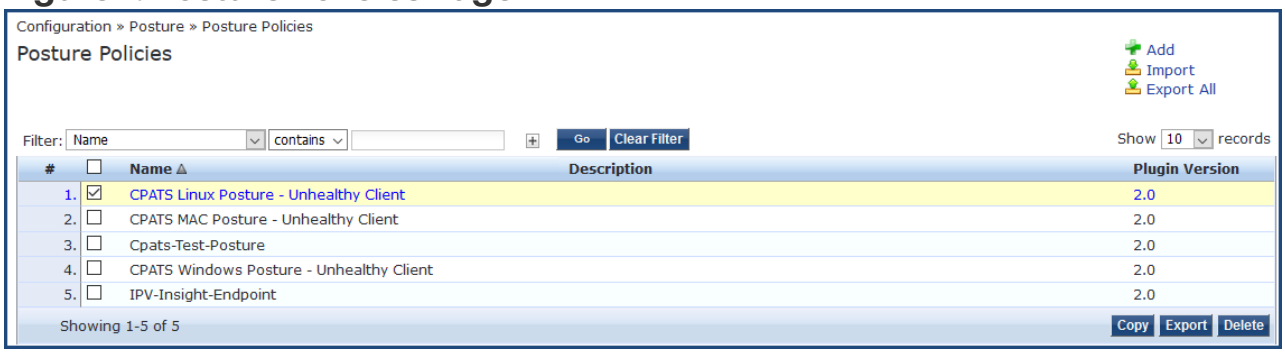
ClearPass Policy Manager evaluates the health of the clients that request access using posture policies and an audit server. All these methods return Posture Tokens (for example, *Healthy* and *Quarantine*) for use by Policy Manager as input into an enforcement policy. You can associate one or more posture methods with a single service. Policy Manager forwards all or part of the posture data received from the client to a posture server. Nmap (Network Mapper) or Nessus audit servers provide posture checking for unmanaged devices, such as devices lacking adequate posture agents or supplicants. For more information on audit servers, see Configuring Audit Servers . From the **Posture Policies** page, you can create a new policy or edit an existing policy.

To create a new posture policy:

Procedure

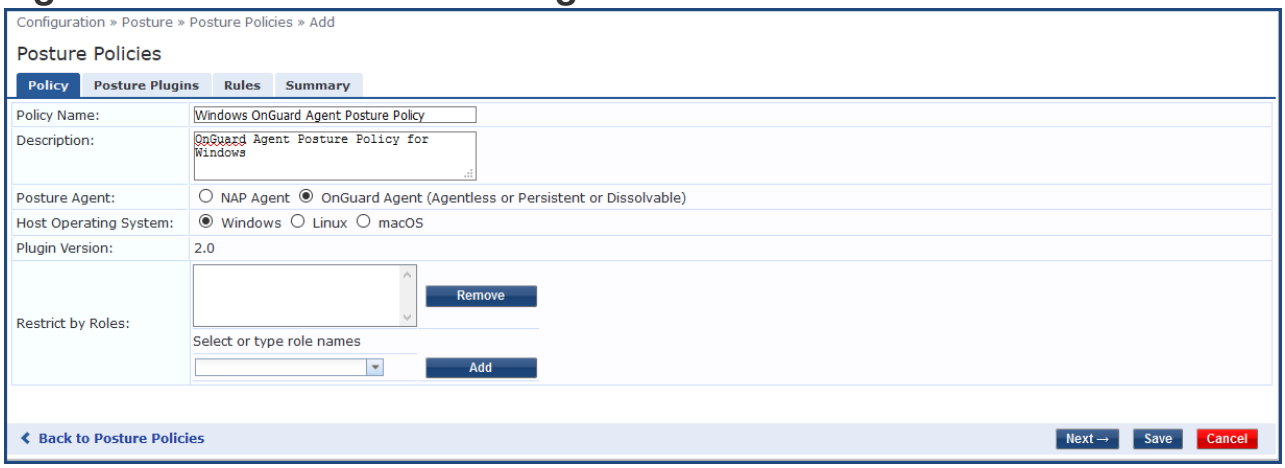
1. Navigate to **Configuration > Posture > Posture Policies.** The **Posture Policies** page displays a list of all existing posture policies configured for the current Policy Manager server.

Figure 1. Posture Policies Page



2. Click the **Add** link. The **Posture Policies Add** page opens.

Figure 2. Posture Policies Add Page



3. Configure the information in the **Policy Polices** page as described in Configuring Posture Policy Agents and Hosts .
    - For information on configuring the posture policy plug-ins, see Configuring Posture Policy Plug-ins .
    - For information on configuring posture policy rules, see Configuring Posture Policy Rules .

Configuring Posture Policy Agents and Hosts

Subtopics

Adding a New Posture Policy
Configure NAP Agent Posture Plug-ins
Configure OnGuard Agent Posture Plug-ins

Adding a New Posture Policy

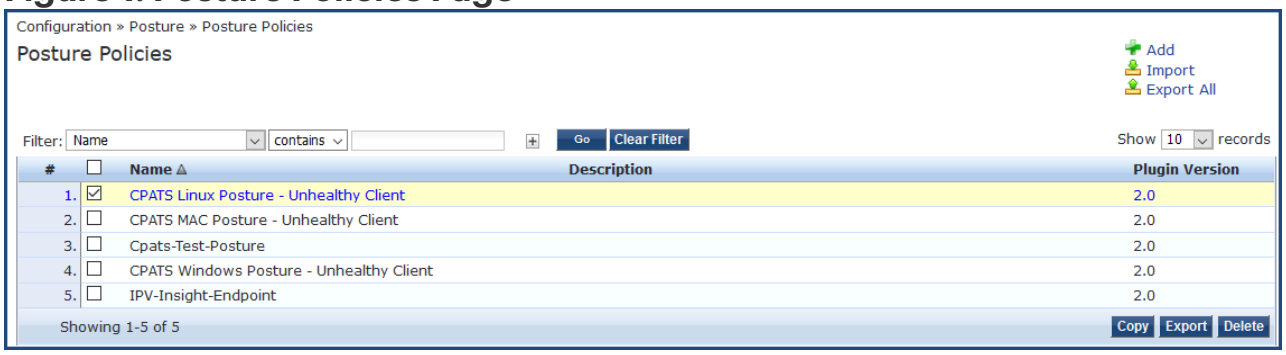
About this task

To configure posture policy agents and hosts:

Procedure

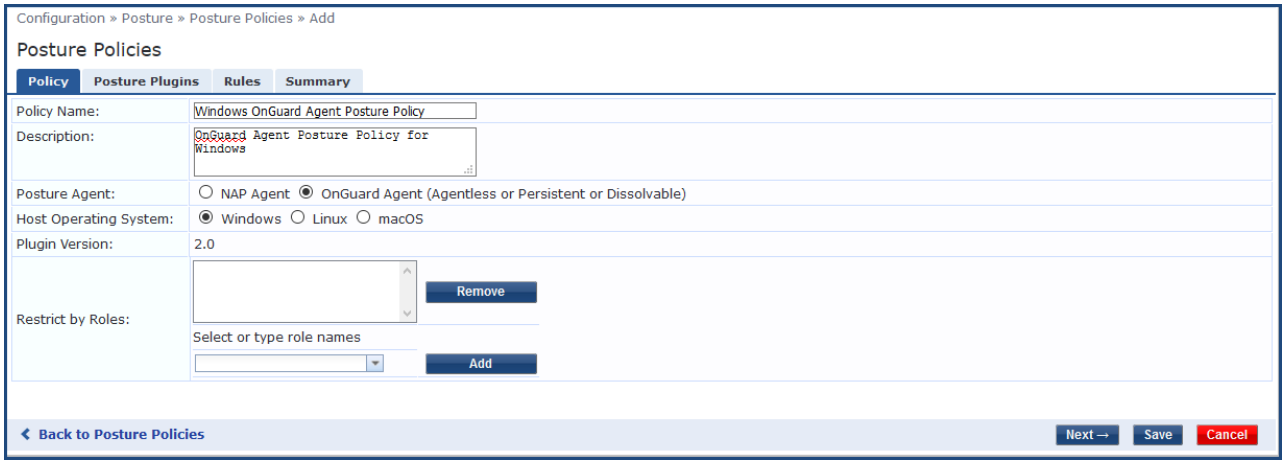
1. Navigate to **Configuration > Posture > Posture Policies.** The **Posture Policies** page displays a list of all existing posture policies.

Figure 1. Posture Policies Page



2. Click the **Add** link. The **Add Posture Policies** page opens.

Figure 2. Add Posture Policies Page.



3. Use the **Posture Policies > Policy** dialog to configure the policy name and description, select a posture agent and host operating system, and specify role restrictions.
4. Specify the **Add Posture Policy** parameters as described in the following table:

Table 1. Add Posture Policy Parameters

Parameter	Action/Description
Policy Name	Enter the name assigned to the policy by the ClearPass Policy Manager administrator.
Description	Specify the description that provides additional information about the posture policy.
Posture Agent	Select the posture agent type. • NAP Agent • OnGuard Agent (Agentless or Persistent or Dissolvable) For detailed information on these agents, see NAP Agent Posture Plug - ins and On Guard Agent Posture Plug - ins .
Host Operating System	Specify whether the host is using a Windows, Linux, or macOS operating system.

Parameter	Action/Description
Restrict by Roles	Apply the posture policy to the selected roles. To restrict the posture policy by roles: a. From the **Policy** tab, click the **Select o r type role names** drop - down. The lis t of available roles opens. b. Select the roles you wish to restrict ac cess by, then click **Add**. The selected roles are added to the *Restrict by Roles* field. For example, us ers with access to research data might have a posture policy that does not pe rmit the mounting of USB storage devi ces. Or users with access to employee or customer personal or health data m ight have a policy that requires full dis k encryption.

Configure NAP Agent Posture Plug-ins

When you select **NAP Agent** as the Posture agent, you can configure the posture plug-ins as described in the following table:

Table 1. NAP Agent: Windows OS Posture Plug - ins Support

Plug - in	Description/Health Checks**
Windows System Health Validator	The Windows System Health Validator per mits or denies client computers to connec t to your network. The Windows System H ealth Validator also restricts client access t o computers that have a service pack less t han service pack x.
Windows Security Health Validator	The Windows Security Health Validator per mits or denies client computers access to

Plug - in	Description/Health Checks**
	your network, subject to checks of the clie nt's system: • Firewall • Virus Protection • Automatic Updates • Security Updates

**As of January 14, 2020, Microsoft has reached End of Support for Windows 7, Windows Server 2008, and Windows Server 2008 R2. These operating systems are deprecated starting with ClearPass 6.9.0.

Configure OnGuard Agent Posture Plug-ins

Select **OnGuard Agent (Agentless or Persistent or Dissolvable)** as the **Posture Agent** for use in the following scenarios:

- An environment that does not support 802.1X-based authentication. For example, some legacy Microsoft Windows operating systems or legacy network devices.
- An environment configured with an operating system that provides support for 802.1X natively, but does not have a built-in health agent. macOS is an example of this type of environment.

  When you select **Posture Agent: OnGuard Agent (Agentless or Persistent or Dissolvable)**, you can configure the posture plug-ins for the following operating systems:

- Windows (see Table 3)

  NOTE

  Agentless OnGuard is supported in the Core versions of Windows Server OS.

- macOS (see Table 4)
- Linux (see Table 5)

Table 1. OnGuard Agent Validator Posture Plug - in Windows OS Support

Plug - in	Description/Health Checks**
Policy Manager Windows Universal System Health Validator	The configurable health checks for this validator are: ◦ Services ◦ Processes ◦ Registry Keys ◦ AntiVirus ◦ Patch Management ◦ Windows HotFixes ◦ USB Devices ◦ Virtual Machines ◦ Network Connections ◦ Disk Encryption ◦ Installed Applications ◦ File Check
Windows System Health Validator	The Windows System Health Validator allows you to configure client computers that can connect to your network, and clients that are restricted from your network. Access is determined by a check of the service pack level. You can determine the service pack level.
Windows Security Health Validator	Use the configurable health checks for this validator to permit or deny a client device network access, subject to checks of the client's system: ◦ Firewall ◦ Virus Protection ◦ Automatic Updates

Plug - in	Description/Health Checks**
	◦ Security Updates

**As of January 14, 2020, Microsoft has reached End of Support for Windows 7, Windows Server 2008, and Windows Server 2008 R2. These operating systems are deprecated starting with ClearPass 6.9.0.

Table 2. OnGuard Agent (Persistent or Dissolvable) Posture Plug - ins for macOS

Plug - in Name	Health Checks
Policy Manager macOS Universal System Health Validator	The configurable health checks for this h ealth validator are: ◦ Services ◦ Processes ◦ AntiVirus ◦ Firewall ◦ Patch Management ◦ Peer - to - Peer ◦ USB Devices ◦ Virtual Machines ◦ Network Connections ◦ Disk Encryption ◦ Installed Applications ◦ File Check

Table 3. OnGuard Agent (Persistent or Dissolvable) Posture Plug - in for Linux

Plug - in Name	Health Check
Policy Manager Linux Universal System H ealth Validator	The configurable health check for this h ealth validator is: *Services*.

Configuring Posture Policy Plug-ins

About this task

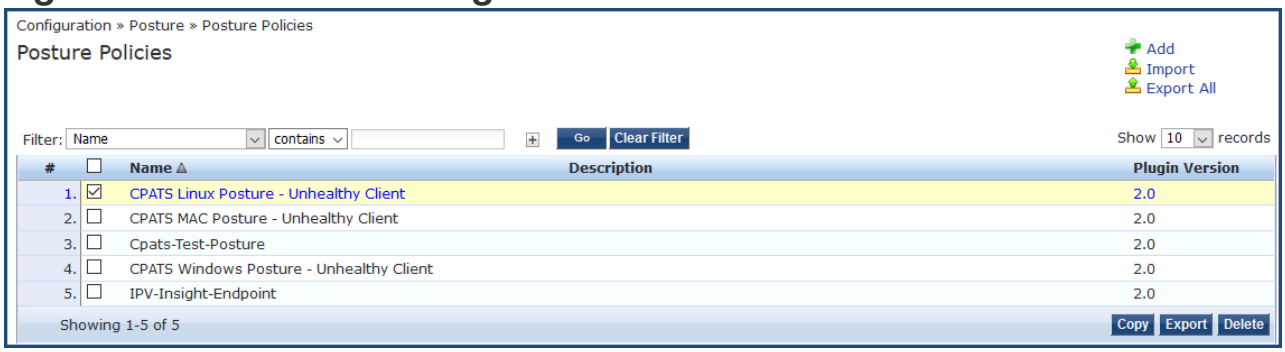
The **Posture Plugins** dialog of the **Posture Policies** page allows you to configure plug-ins for the posture policy. The plug-ins available on this tab vary, depending upon whether the policy is using a Network Access Protection (NAP) agent or the OnGuard Agent (Persistent or Dissolvable) plug-in.

To configure posture policy plug-ins:

Procedure

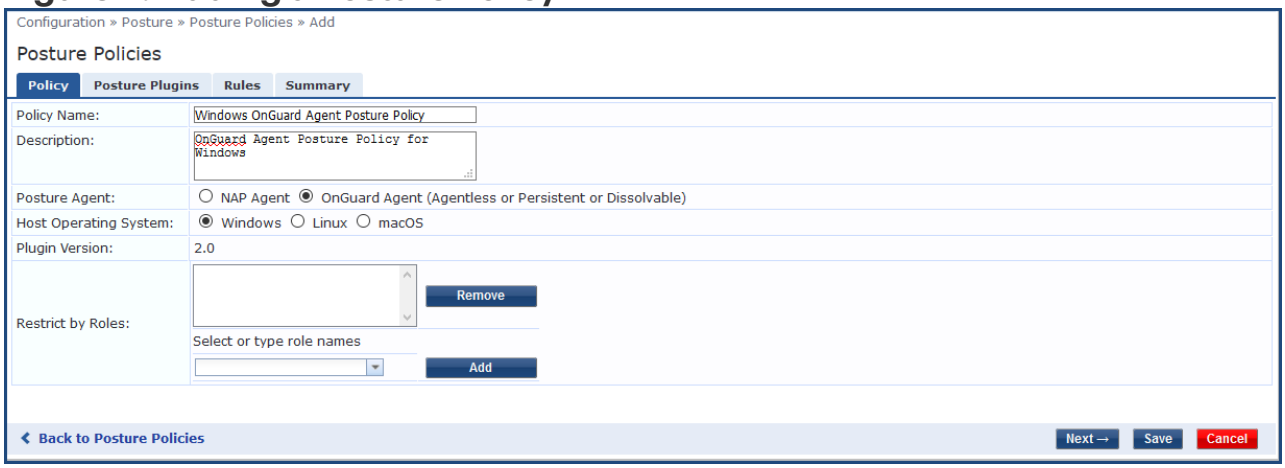
1. Navigate to **Configuration > Posture > Posture Policies**. The **Posture Policies** page opens.

Figure 1. Posture Policies Page



2. Click **Add**. The **Add Posture Policies** page opens.

Figure 2. Adding a Posture Policy



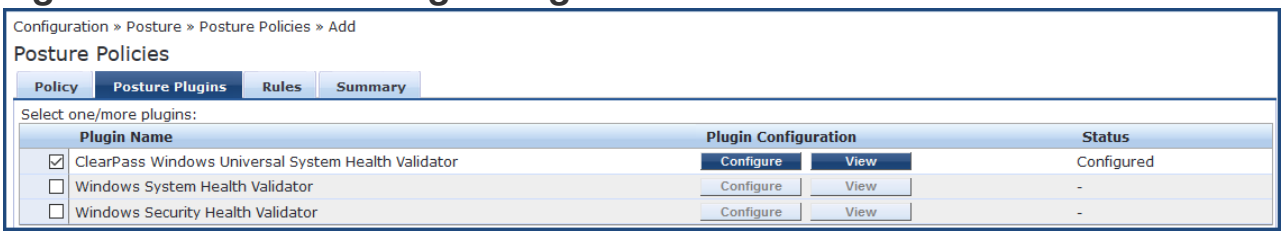
3. In the **Policy** tab, specify the following:

   - Policy Name

- Description
- Posture Agent
- Host Operating System

4. Select the **Posture Plugins** tab. The **Add Posture Plugins** page opens.

Figure 3. Add Posture Plugins Page



You can configure the following posture plug-ins in the **Posture Policies** page:

- ClearPass Windows Universal System Health Validator - OnGuard Agent
- Windows System Health Validator: NAP Agent
- Windows System Health Validator: OnGuard Agent
- Windows Security Health Validator: NAP Agent

5. Select the check box of the plug-in you wish to configure.
6. To view the configuration page for the selected plug-in, click **Configure**.

Subtopics

Configuring OnGuard Agent Plugins
Configuring NAP Agent Plugins

Configuring OnGuard Agent Plugins

If you select the **OnGuard Agent** option in the **Policy** tab of the **Posture Policies** page, the **Posture Plugins** tab allows you to configure different plugin types for hosts running Windows, Linux, or macOS operating systems. Refer to the following topics for details on each plugin type:

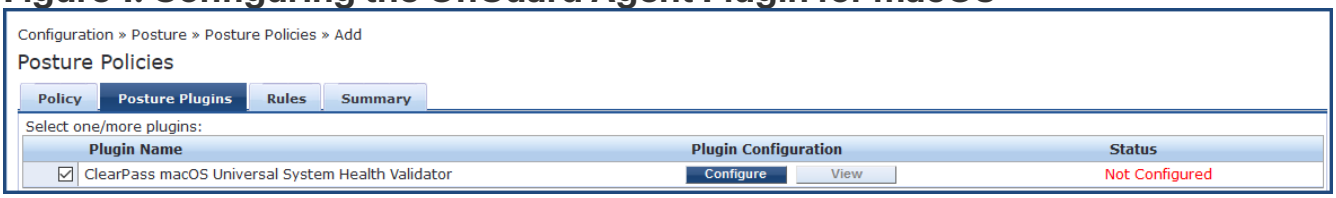
- Windows:
  - ClearPass Windows Universal System Health Validator
  - Windows System Health Validator OnGuard Agent

- Windows Security Health Validator OnGuard Agent

- Linux: ClearPass Linux Universal System Health Validator OnGuard Agent
- macOS: Policy Manager macOS Universal System Health Validator: OnGuard Agent

The following figure displays the **Posture Policies > Posture Plugins** tab:

Figure 1. Configuring the OnGuard Agent Plugin for macOS



Subtopics

ClearPass Windows Universal System Health Validator
Windows System Health Validator OnGuard Agent
Windows Security Health Validator OnGuard Agent
ClearPass Linux Universal System Health Validator OnGuard Agent
Policy Manager macOS Universal System Health Validator: OnGuard Agent

ClearPass Windows Universal System Health Validator

About this task

To configure the Policy Manager Windows Universal Health System Health Validator (OnGuard Agent):

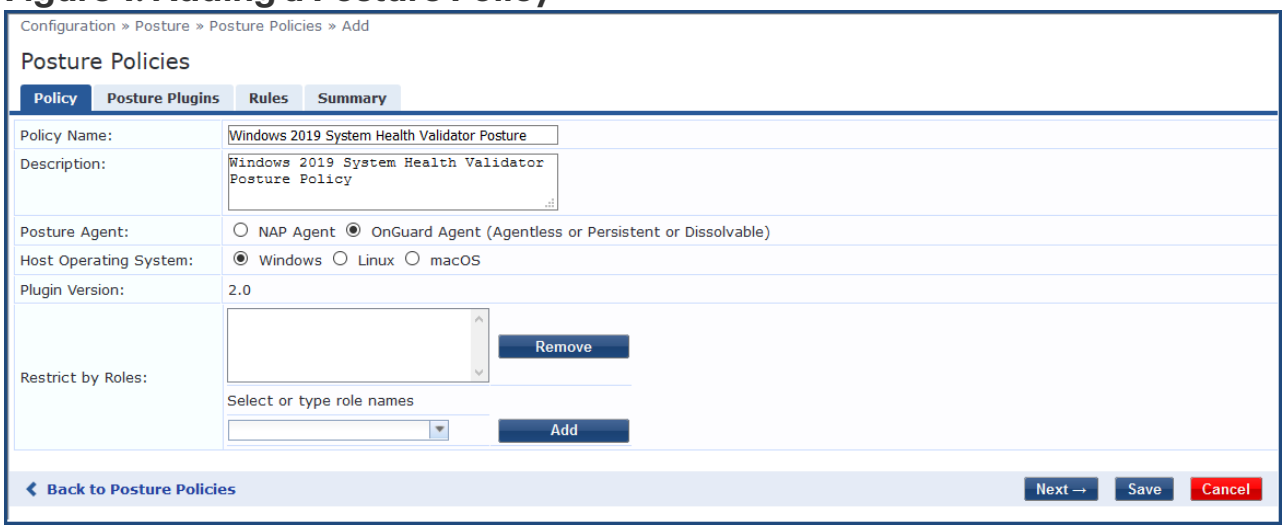
ClearPass 6.12 is the last major version that will support following Windows OS versions:

- Windows XP
- Windows Vista
- Windows 7
- Windows 8
- Windows 8.1
- Windows Server 2003
- Windows Server 2008
- Windows Server 2012
- Windows Server 2012 R2

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.

Figure 1. Adding a Posture Policy



   a. Enter the name and a description of the posture policy.

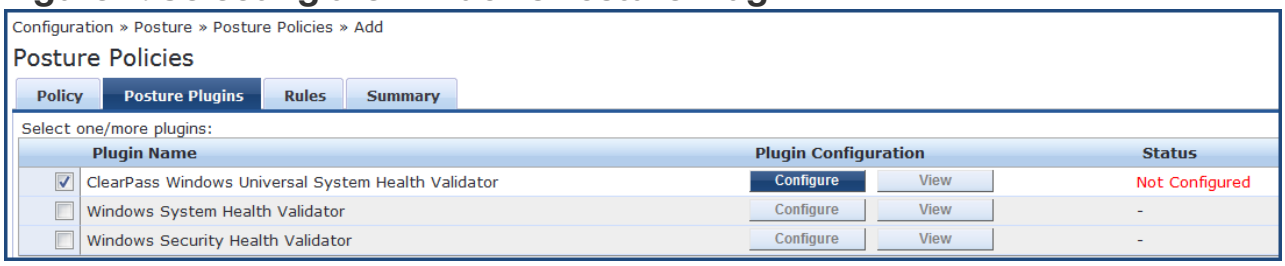
   b. **Posture Agent**: Choose **OnGuard Agent (Persistent or Dissolvable)**.

   c. **Host Operating System: Windows** is selected by default.

   d. **Plugin Version**: OnGuard plugin version **2.0** (V4 SDK) is set by default.

2. Click **Next**. The **Posture Plugins** dialog opens.

Figure 2. Selecting the Windows Posture Plugin



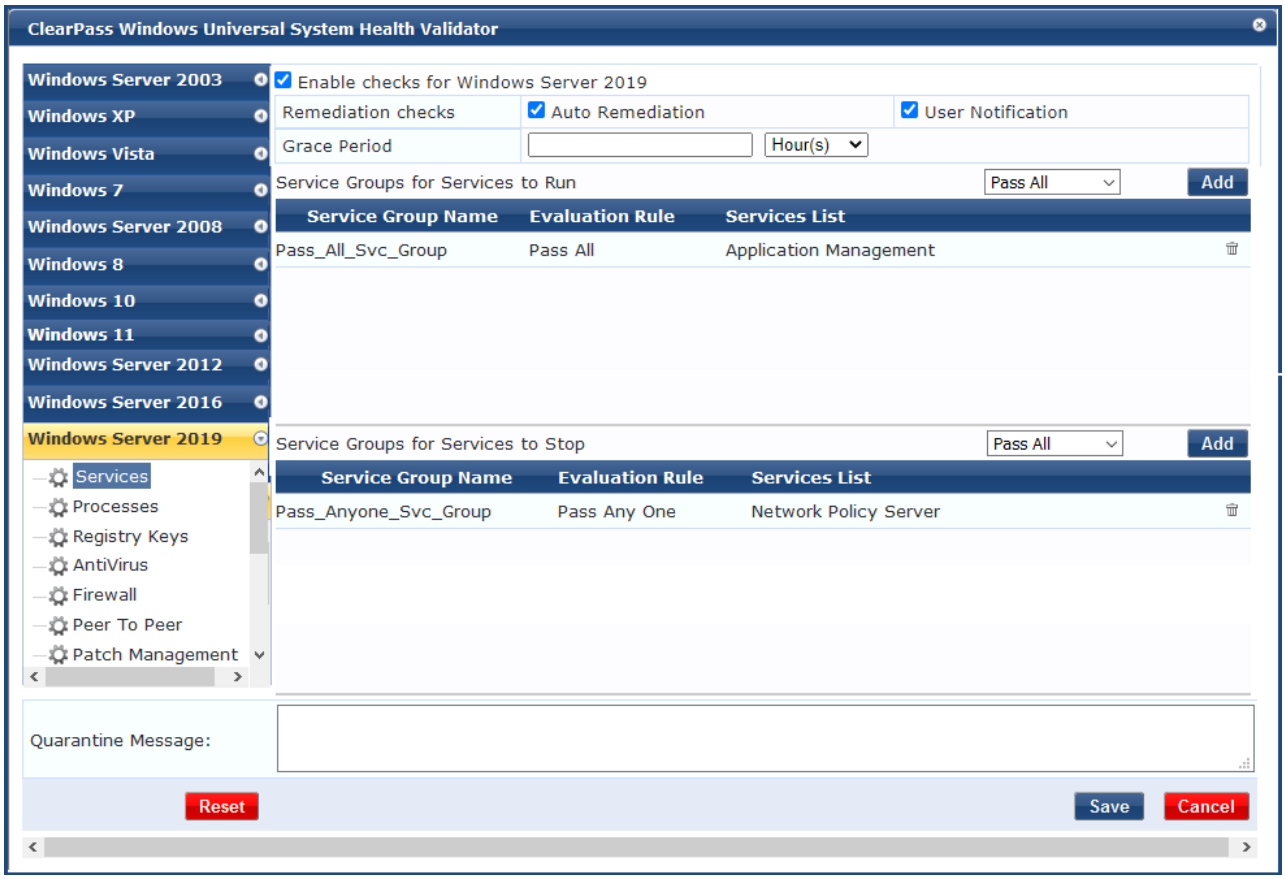
3. In the **Posture Plugins** page, click the check box for **ClearPass Windows Universal System Health Validator**.

4. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.

5. Select the desired version of Windows.

6. To enable checks for the selected version, click the **Enable checks for Windows <version>** check box.

Figure 3. ClearPass Windows Universal System Health Validator for Services



7. Configure health checks for the selected version of Windows. The list of available health checks (including **Services**, **Processes**, **Registry Keys** and **AntiVirus**) appears below that Windows version. Click any of the links below to view configuration details for that health check type.

   - Services
   - Processes
   - Registry Keys
   - Antivirus Health Check
   - Firewall
   - Peer To Peer
   - Patch Management
   - Windows Hotfixes
   - USB Devices
   - Virtual Machines

- Network Connections
- Disk Encryption
- Installed Applications
- File Check
- Custom Scripts

Subtopics

Posture Policies > Rules Tab
Services
Processes
Registry Keys
Antivirus Health Check
Firewall
Peer To Peer
Patch Management
Windows Hotfixes
USB Devices
Virtual Machines
Network Connections
Disk Encryption
Installed Applications
File Check
Custom Scripts

Posture Policies > Rules Tab

About this task

To configure the rules for this posture policy:

Procedure

1. From the **Posture Policies > Add** page, select the **Rules** tab.

Figure 1. Configuring Posture Policy Rules

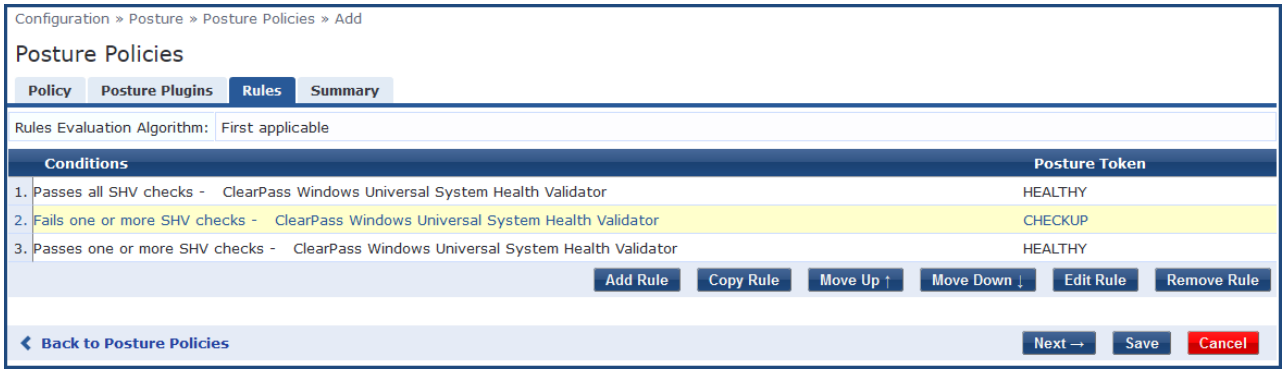


Table 1. Rules Configuration Action Buttons

Button	Action/Description
Add Rule	Click the **Add Rule** action button to bring up the Rules Editor and add a new rule.
Copy Rule	Select the rule you want to copy, then cli ck the **Copy Rule** action button. The cop ied rule is added to the existing list of rul es.
Move Up/ Move Down	Select an enforcement policy rule, then c lick **Move Up** or **Move Down** to change th e order that the rules are executed in the policy,
Edit Rule	Select the rule you want to edit, then clic k the **Edit Rule** action button.
Remove Rule	Select the rule you want to delete, then c lick the **Remove Rule** action button.

2. When finished, click **Save**. You return to the **Posture Policies > Posture Plugins** page, where the Status column for the **ClearPass Windows Universal System Health Validator** now shows *Configured*.

Services

About this task

The **Services** feature allows network admins to determine how the overall health status of the Services health class is determined—whether by using an AND condition (for example, *Group1 AND Group2*) or an OR condition (for example, *Group1 OR Group2*). Regarding services, for example, admins can run checks such as *Service1 AND Service2 OR Service3 AND Service4*. You can also use the **Services** page to verify the group of services to be present or absent and specify the service groups and services to run on a client.

To define Windows Service Groups, specify the evaluation rules, and add or remove specific Windows services on the endpoint:

Procedure

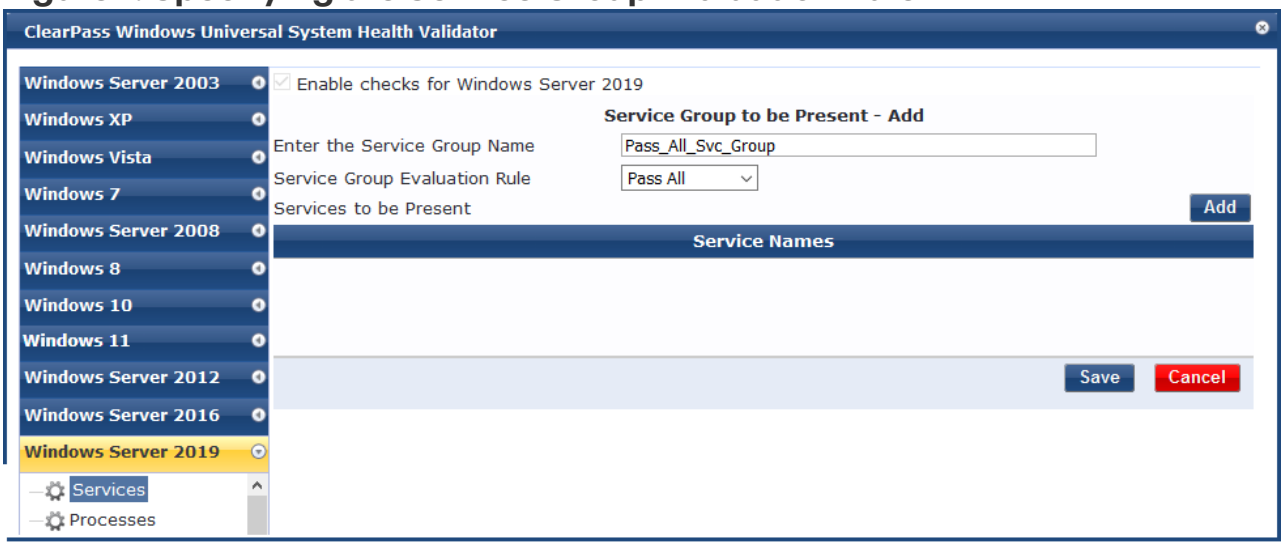
1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select **Services**. The **Services** health class configuration page opens.
8. Select the **Enable checks for <Windows OS>** and **A firewall application is on** options.
9. Specify the **Policy Manager Widows Universal System Health Validator > Services** configuration parameters as described in the following table:

   Defining the Service Group to Be Present

   You can configure the name of the service group and specify the evaluation rule for the service group.

0. To configure the **Service Groups for Services to Run**, click **Add.** The **Add Service Group to Be Present** dialog opens.

Figure 1. Specifying the Service Group Evaluation Rule



1. Specify the **Add Service Group to Be Present** parameters as described in the following table:

Table 1. Add Service Group to Be Present Parameters

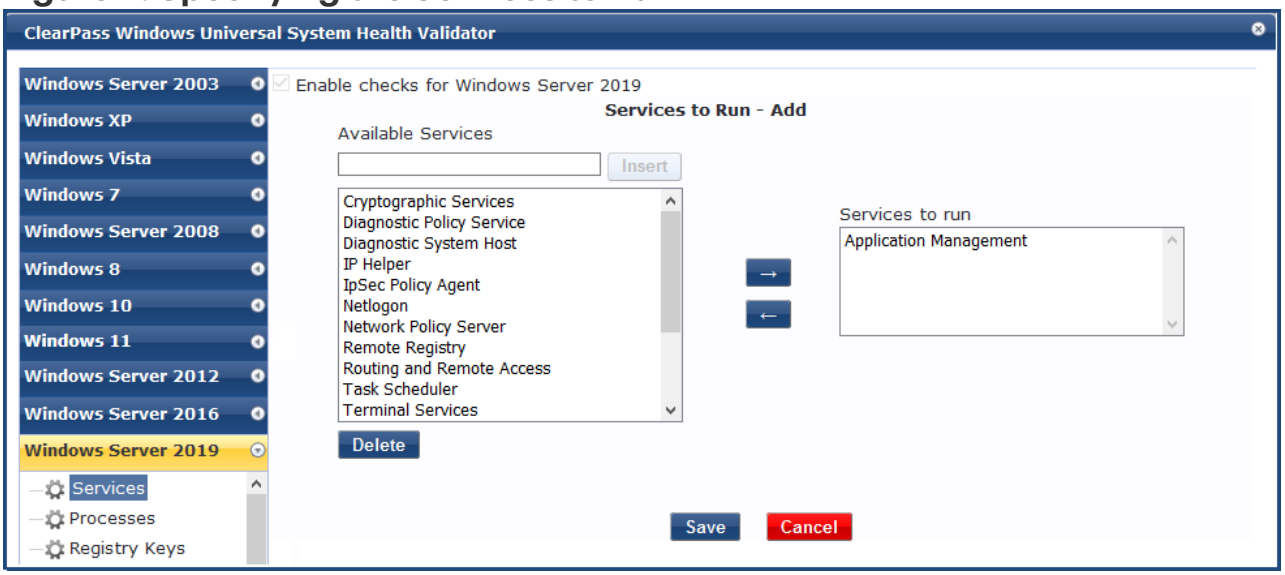
Parameter	Action/Description
Enter the Service Group Name	Enter the name of the Service Group.
Service Group Evaluation Rule	Select the appropriate Service Group Evaluation Rule: • **Pass All**: Select this evaluation rule if you want the **Services** health class to be deemed as *healthy* only if all the configured service groups are present. **Pass All** is the equivalent of an **AND** condition. • **Pass Any One**: Select this evaluation rule if you want the **Services Check** health class to be deemed as *healthy* even if any one of the configured service groups are present. **Pass Any One** is the equivalent of an **OR** condition.

Specifying the Services to Be Present

To specify the services to be present:

2. Click the **Services to Be Present > Add** button. The **Add Services to Run** dialog opens.

Figure 2. Specifying the Services to Run



3. Select one or more of the desired services from the **Available Services** list.
4. To move the desired services to the **Services to Run** box, click **> >** , then click **Save**.
5. You can also add a service to the list of available services. To do so, enter the service name in the **Insert** text box, then click **Insert**.

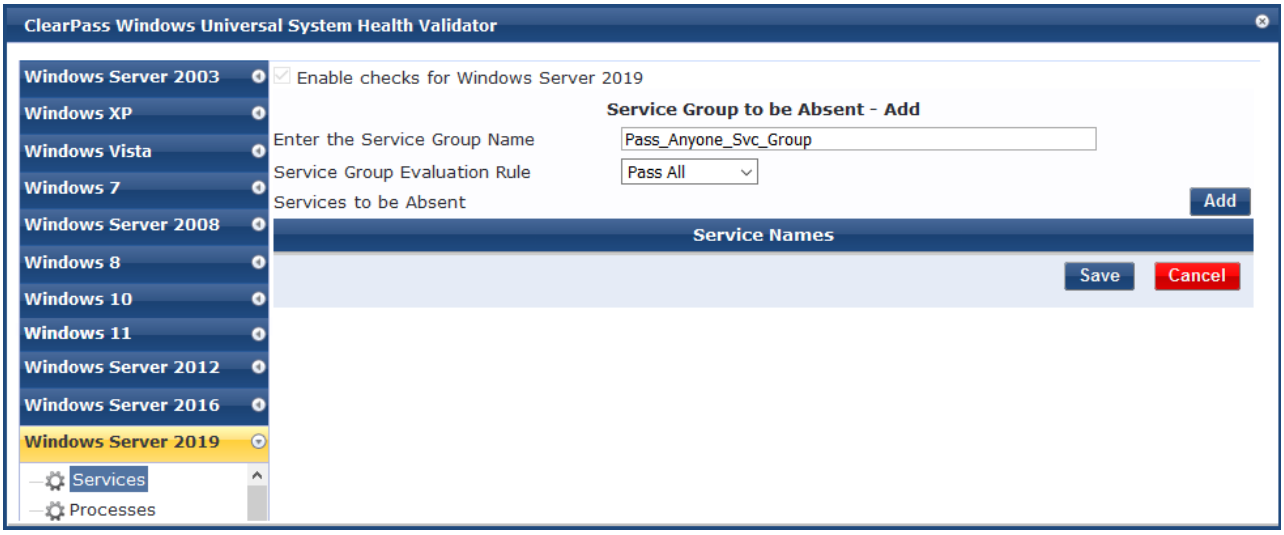
Defining the Service Group to Be Absent

You can configure the name of the service group and specify the evaluation rule for the service group.

6. To configure the **Service Groups for Services to Be Absent**, click **Add.**

The **Add Service Group to Be Absent** dialog opens.

Figure 3. Add Services to Be Absent Dialog



7. Specify the **Service Groups to Be Absent** parameters as described in the following table:

Table 2. Add Service Group to Be Absent Parameters

Parameter	Action/Description
Enter the Service Group Name	Enter the name of the Service Group.
Service Group Evaluation Rule	Select the appropriate Service Group Evaluation Rule: • **Pass All**: Select this evaluation rule if you want all service groups to be stopped. **Pass All** is the equivalent of an **AND** condition. • **Pass Any One**: Select this evaluation rule if you want any one of the service groups to be stopped. **Pass Any One** is the equivalent of an **OR** condition.

8. Click **Add**. The **Add Services to Stop** dialog opens.

Figure 4. Specifying the Services to Stop

9. Select one or more of the desired services from the **Available Services** list.
0. To move the desired services to the **Services to Stop** box, select the desired services, then click **> >**. To remove services from the **Services to Stop** box, select the services to be removed, then click **<<**.
1. When finished configuring the services to stop, click **Save**.

Processes

About this task

The **Processes** page provides a set of parameters to specify which processes to be explicitly present or absent on the system.

To configure Processes:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).

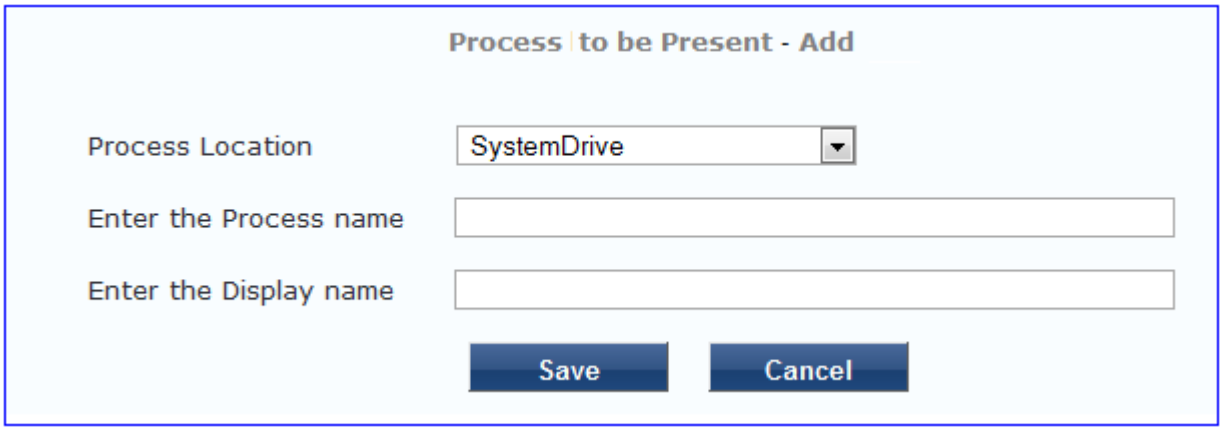
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select **Processes**. The **Processes** health class configuration page opens.
8. Select the **Enable checks for <Windows OS>** option.
9. Specify the **Processes** configuration parameters as described in the following table:

   Processes to be Present Parameters.

0. In the **Processes to be Present** section, click **Add.**

   The **Add Processes to be Present** page opens.

Figure 1. Add Processes to be Present Page



1. Specify the Processes to be Present parameters as described in the following table, then click **Save**:

Table 1. Processes to be Present Page Parameters

Parameter	Action/Description
Process Location	Choose from the following locations: • System Drive • System root • Program Files

Parameter	Action/Description
	• HOMEDRIVE • HOMEPATH • None
Enter the Process name	Specify the path name containing the p rocess executable name.
Enter the Display name	Enter a user - friendly name for the proc ess. This is displayed in end - user facing messages.

After you save the Processes parameters, the information appears in the **Processes to be Present** section.

Processes to be Absent Parameters

2. In the **Processes to be Absent** section, click **Add.** The **Add Processes to be Absent** page opens. Figure 3 shows the configuration parameters when you select **Process Name** and **MD5 Sum**.

Figure 2. Process to be Absent Pages

3. Specify the **Processes to be Absent** parameters as described in the following table, then click **Save**:

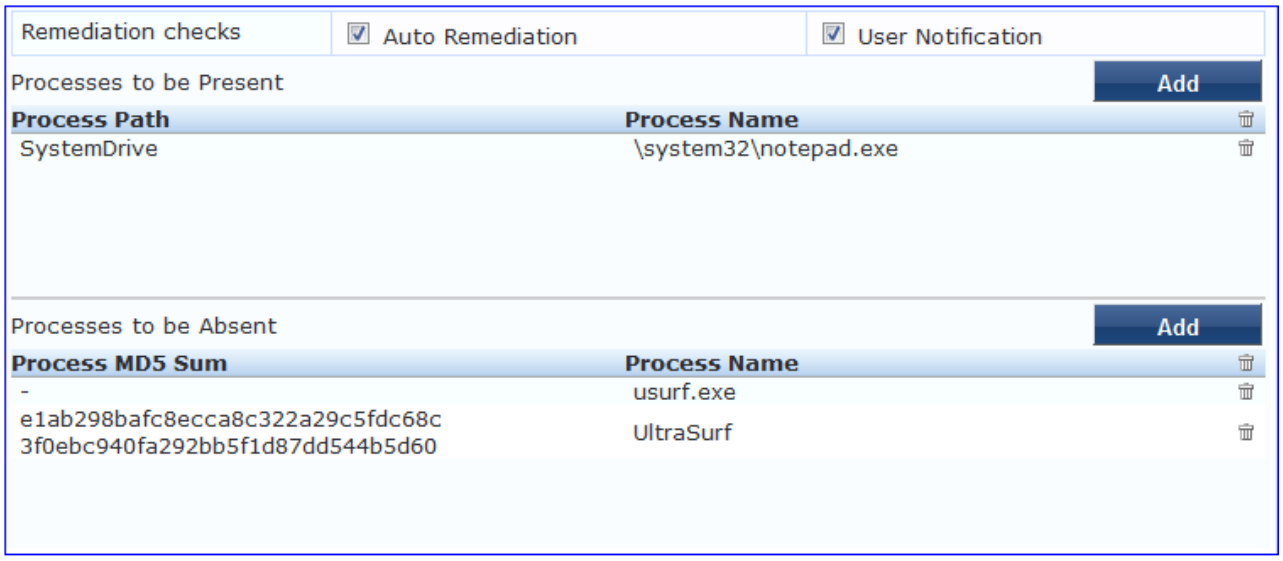
Table 2. Processes to be Absent Page Parameters

Parameter	Action/Description
Check Type	Select one of the following process optio ns: • **Process Name**: The agent looks for all processes that matches with the give n name. For example, if notepad.exe is specifie d, the agent kills all processes whose name matches, regardless of the loca tion from which these processes were started.

Parameter	Action/Description
	• **MD5 Sum**: This specifies one or more (comma - separated) MD5 checksums of the process executable file. f there are multiple versions of the process executable, you can specify the MD5 sums of all versions here. The agent c ounts all running processes on the sy stem, computes the MD5 sum of the process executable file, and matches this with the specified list. One or mo re of the matching processes are then terminated. • **SHA256 Sum:** This specifies one or m ore (comma - separated) SHA256 che cksums of the process executable file . If there are multiple versions of the p rocess executable, you can specify th e SHA256 sums of all versions here. T he agent counts all running processes on the system, computes the SHA256 sum of the process executable file, an d matches this with the specified list. One or more of the matching process es are then terminated. **NOTE** **MD5 Sum** option is not avail able if Policy Manager is ope rating in FIPS mode.
Enter the Display name	Enter a user - friendly name for the proc ess. This display name is displayed in end - user facing messages.

You return to the **Processes Configuration** page, which now shows the values for the processes that were configured.

Figure 3. Processes Configured



Registry Keys

About this task

The **Registry Keys** page allows you to specify which registry keys are to be explicitly present or absent. To define the registry keys:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.

7. Select **Registry Keys**. The **Registry Keys** health class configuration page opens.
8. Select the **Enable checks for <Windows OS>** option.
9. Specify the **Registry Keys** page parameters as described in the following table:

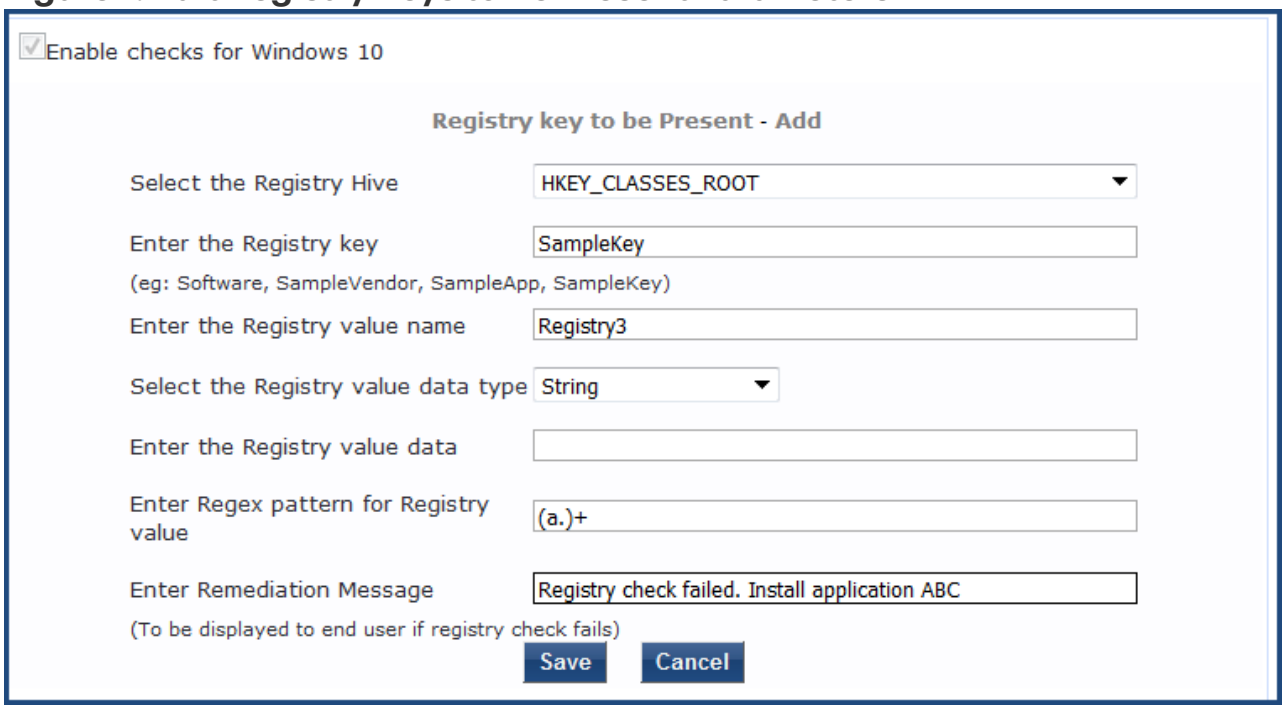
Table 1. Registry Keys Page Parameters

Parameter	Action/Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for proces ses. • **User Notification**: Enable to allow use r notifications in the event of process policy violations.
Monitor Mode	Enable this to set the health status of the **Registry Keys** health class healthy. This allows administrators to collect info rmation related to missing registry keys without marking the clients as unhealthy even if some registry keys are missing.
Registry keys to be present	To specify a registry key to be added to t he **Registry keys to be present** list, click **Add**. If the specified registry key is not present , the remediation message that is added i n the **Registry Keys Page (Detail)** window is displayed on **OnGuard Agent**.
Registry keys to be absent	To add a registry key to the **Registry keys to be absent** list, click **Add**. If the specified registry key is not absent, the remediation message that is added in

Parameter	Action/Description
	the **Registry Keys Page (Detail)** window i s displayed on **OnGuard Agent**.

0. To configure the **Registry key to Be Present**, click **Add**. The **Edit Registry Key to Be Present** dialog opens.

Figure 1. Edit Registry Keys to Be Present Parameters



1. Specify the **Registry Key to be Present** parameters as described in Table 2, then click **Save**.

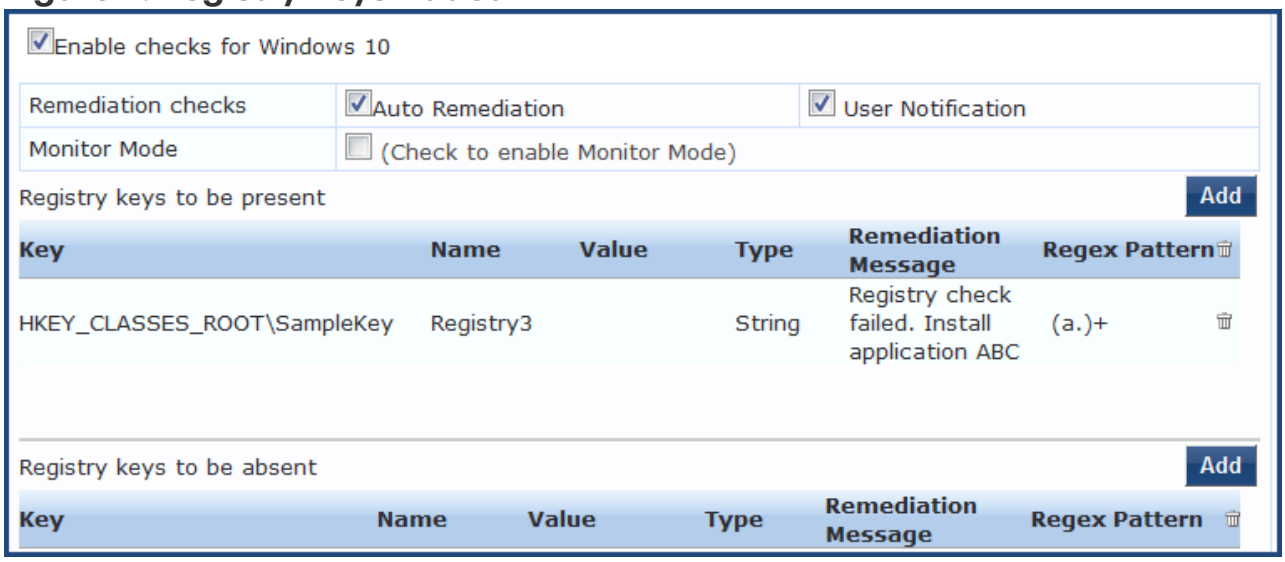
Table 2. Registry Keys Page (Detail)

Parameter	Action/Description
Select the Registry Hive	Specify the registry hive from the followi ng options: • HKEY_CLASSES_ROOT • HKEY_CURRENT_USER • HKEY_LOCAL_MACHINE • HKEY_USERS

Parameter	Action/Description
	• HKEY_CURRENT_CONFIG
Enter the Registry key	Specify the registry key using the exampl es given in the GUI.
Enter the Registry value name	Specify the name of the registry value.
Select the Registry value data type	Specify the registry value data types. The data type can be any of the following: • Multi String • String • DWORD • QWORD • Expandable String
Enter the Registry value data	Specify the registry value.
Enter Regex pattern for Registry value	Enter the Regular Expression (Regex) patt ern for the Registry value. A regular expression is a pattern that the regular expression engine attempts to m atch in input text. A pattern consists of o ne or more character literals, operators, o r constructs. **NOTE** Perl regular expressions are supported.
Enter Remediation Message	Specify the custom remediation messag e to be displayed to end users if the regis try check fails.

As shown in Figure 3, after you save the registry configuration settings, the remediation message and *Regular Expression* pattern appears in the **Registry** page.

Figure 2. Registry Keys Added



Antivirus Health Check

About this task

On the **Antivirus** page, you can turn on an antivirus application.

In ClearPass 6.10.1 and earlier releases, products added to an antivirus health class were sorted in alphabetical order, and if the configured products were missing during OnGuard health checks, the error message to install an antivirus was only shown for the first product alphabetically in the list.

To configure the Windows antivirus health class:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.

5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. From the list of health checks, select **AntiVirus**. The **antivirus** health class configuration page opens.
8. Select the **Enable checks for <Windows OS>** and the **An antivirus application is on** options.
9. Specify the **Policy Manager Widows Universal System Health Validator > AntiVirus** configuration parameters as described in the following table:

Table 1. AntiVirus Configuration Parameters

Parameter	Action/Description
Auto Remediation	Enable to allow auto - remediation for service checks. Enabling this option automatically stops or starts services based on the entries in **Service to Run** and **Services to Stop** configuration. Auto - remediation for the **Services** health class is enabled by default.
User Notification	When enabled, a remediation message that includes the groups of services to be present or absent is displayed to the end user.
Remediation checks	Disable or enable one or both of the following remediation checks. By default, both are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for antivirus policy violations. • **User Notification**: Enable to allow user notifications in the event of antivirus policy violations.
Grace Period	Use the **Grace Period** drop - down menu to configure the time period for which th

Parameter	Action/Description
	e Windows Universal System Health Valid ator plugin should ignore devices that fa il to pass antivirus health checks. You ca n specify the grace period in hours, days, weeks, or months.
Product Evaluation Rule	Select one of the following options: • **Pass All**: Select this product evaluatio n rule if you want the devices running the selected version of Windows to be deemed as *healthy* only if all the conf igured antivirus products are present. **Pass All** is the equivalent of an **AND** c ondition. • **Pass Any One**: Select this product eva luation rule if you want the **AntiVirus** health class to be deemed as *healthy* i f any one of the configured antivirus p roducts are present. **Pass Any One** is t he equivalent of an **OR** condition. **Pas s Any One** is the default.

To specify product and version-check information:

0. Click **Add**.
1. Specify the antivirus health class parameters as described in the following table:

Table 2. AntiVirus Health Class Parameters

Parameter	Action/Description
An antivirus - application is on	Click **An antivirus - application is on** to e nable testing of health data for configure d antivirus application(s).
Auto Remediation	Check the **Auto Remediation** check box t o enable auto remediation of antivirus sta tus. This option is enabled by default.

Parameter	Action/Description
User Notification	Check the **User Notification** check box t o enable user notification of policy violati on of antivirus status. This option is enabl ed by default.
Display Update URL	Check the **Display Update URL** check b ox to show the origination URL of the up date. This option is disabled by default. B eginning with the 6.11 release, the Display Update URL option is removed from the Windows Antivirus health class.
Product - specific checks	To configure for product - specific chec ks, leave the check box checked (the de fault setting). To allow any antivirus product, uncheck t he check box. All of these checks may not be available for some products. Where checks are not available, they are shown in disabled stat e.
Select the antivirus product	Select an antivirus product from the drop - down list.
Product version check	Select one of the following options: • **No Check** • **Is Latest** • **At Least**: When you select **At Least**, th e **Version is at Least** field appears. Ent er the At Least version you require for the antivirus product version. • **In Last N Updates**: Requires registrati on with Policy Manager Portal.
Data file version check	Select one of the following options: • **No Check**

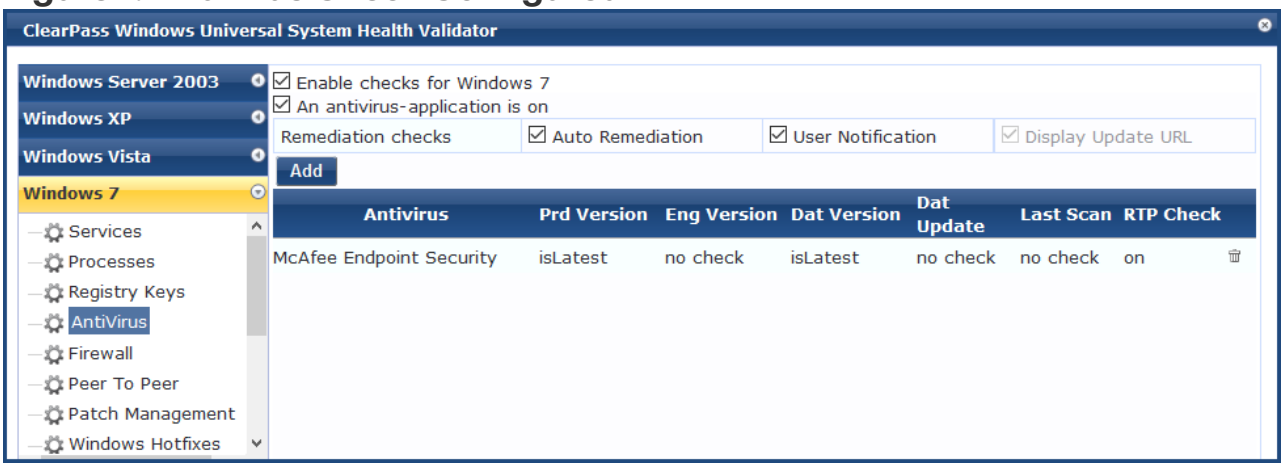
Parameter	Action/Description
	• **Is Latest**: Requires registration with Policy Manager portal. • **At Least**: You can specify the least version that you require for the data file. Enter the least version in the *Version is At Least* field. When you select **At Least**, the **Data file version format** option appears. • **In Last N Updates**: Requires registration with Policy Manager Portal.
Data file version format	This field allows admins to specify the data file version format when the **At Least** check is selected for **Data File Version check**. The configured format is used to compare the configured data file version value with value received from the client. This field should be configured when the **Data file version** value contains a *date*. The following format specifiers are supported: • **mm**: Month • **dd**: Day of the month • **yy** or **yyyy**: Year • **a**: Alphabetic • **n**: Numeric **NOTE** Characters other than alphabetic and numeric characters are considered as separators and are not compared. For example: • Data file version: *10/14/2018 rev. 10*

Parameter	Action/Description
	◦ Specify this data file version format: *mm/dd/yyyy aaa. n* • Data file version: *2018.10.08* ◦ Specify this data file version format: *yyyy.mm.dd* • Data file version: *20181014194001* ◦ Specify this data file version format: yyyymmddn • Data file version: *2018 - 10 - 12_04* ◦ Specify this data file version format: *yyyy - mm - dd_n* **NOTE** For antivirus products that do not include a date in the **Data file version check** value, this field can be left blank. In such cases, Policy Manager compares the value of the configured data file version with the value received from the client, character - by - character.
Data file has been updated in	Enter the number, then specify the interval in hours, days, weeks, or months.
Last scan has been done before	Enter the number, then specify the interval in hours, days, weeks, or months.
Real - time Protection Status Check	• **No Check**: Policy Manager does not use the Real - time Protection status value for health evaluation. This means that the client is treated as *healthy* irrespective of the value of its Real - time Protection status.

Parameter	Action/Description
	• **On**: Marked as *healthy* only if the value of Real - time Protection status is **On**.
Adaptive Threat Protection	When **McAfee Endpoint Security** is selected as the antivirus product, the **Adaptive Threat Protection** option is made available. Status checks for Adaptive Threat Protection are not enabled by default. • To enable checking the status of McAfee Endpoint Security Adaptive Threat Protection, select the **On** button. ◦ Auto - remediation action to enable Adaptive Threat Protection is not supported. • If **McAfee Endpoint Security Adaptive Threat Protection** is disabled, OnGuard will mark the client as *Unhealthy*.

2. Click **Save**. After you save your antivirus configuration, it is listed on the **Antivirus** page.

Figure 1. Antivirus Check Configured



Firewall

About this task

In the **Firewall** page, you can specify that a firewall application must be on and specify information about the firewall application.

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select the **Enable checks for <Windows OS>** and **A firewall application is on** options.
8. Specify the firewall configuration parameters as described in the following table:

Table 1. Firewall Configuration Parameters

Parameter	Action/Description
Auto Remediation	Enable to allow auto - remediation for se rvice checks. Enabling this option autom atically stops or starts services based on the entries in **Service to Run** and **Servic es to Stop** configuration. Auto - remediation for the **Services** healt h class is enabled by default.

Parameter	Action/Description
User Notification	When enabled, a remediation message th at includes the groups of services to be p resent or absent is displayed to the end u ser.
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for firewall policy violations. • **User Notification**: Enable to allow use r notifications in the event of firewall policy violations.
Grace Period	Use the **Grace Period** drop - down menu to configure the time period for which th e Windows Universal System Health Valid ator plugin should ignore devices that fa il to pass firewall health checks. You can specify the grace period in hours, days, weeks, or months.
Product - specific checks	To specify Firewall product - specific che cks, leave this parameter as is. To specify whether any firewall applicatio n (any vendor) is running on the end host , uncheck the **Uncheck to allow any prod uct** option.

Adding Firewall Products

When you select the **Product-specific checks** option, you can create list of firewall products which should be checked by the Windows Universal System Health Validator.

To create a list of firewall products for product-specific checks:

9. Click **Add**. The **Firewall Details** page opens.

0. Click the Select the firewall product menu and select a firewall product
1. In the **Product Version is at least** field, specify the minimum version of the product that must be present.
2. Click **Save**.

Peer To Peer

About this task

The **Peer To Peer** page provides a set of widgets for stopping specific peer to peer applications or networks. When you select a peer to peer network, all applications that use that network are stopped. The following figure shows the **Peer To Peer** health class configuration page:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select the **Enable checks for <Windows OS>** and the **A peer to peer application is on** options.
8. Specify the Peer to Peer configuration parameters as described in the following table:

Table 1. Peer to Peer Page

Parameter	Description
By Application / By Network	Select the appropriate radio button to sel ect individual peer - to - peer application s or a group of applications that use spec ific peer - to - peer networks.
Available Applications	This scrolling list contains a list of applica tions or networks that you can select and move to the **Applications to stop** panel. • Click the **> >** to add the applications o r networks to the **Applications to stop** pane. • Click the **<<** to remove the application s or networks from the **Applications t o stop** pane.

Patch Management

About this task

Use **Patch Management** page to specify indicate a patch management application must be on and active. You can also specify information about the patch management application, configure the product evaluation rule, and configure patch management application checks.

To configure Policy Manager Windows patch-management application:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).

d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).

3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the desired version of Windows.
7. From the selected version of the Windows health check list, select **Patch Management**.
8. Select the **Enable checks for <Windows OS>** and the **A patch management application is on** option.
9. Specify the Windows Patch Management parameters as described in the following table.

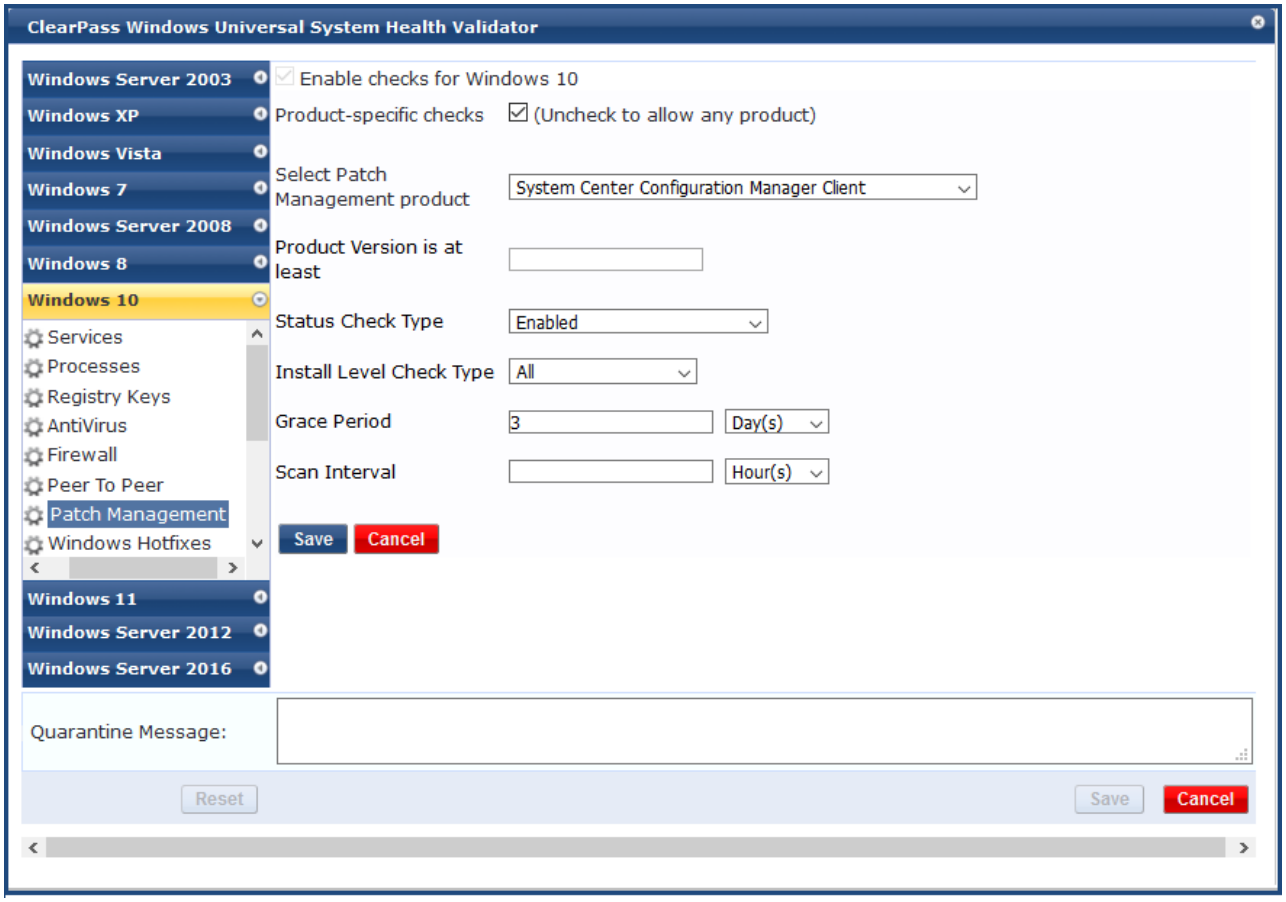
Table 1. Wiindows Patch Management Parameters

Parameter	Action/Description
A patch management application is on	Select the **A patch management applica tion is on** option to enable testing of hea lth data for configured Antivirus applicati ons.
Remediation Checks	
Auto Remediation	Enable auto - remediation of patch mana gement status.
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for patch management violations. • **User Notification**: Enable to allow use r notifications in the event of patch m anagement violations.

Parameter	Action/Description
Grace Period	Use the **Grace Period** drop - down menu to configure the time period for which th e Windows Universal System Health Valid ator plugin should ignore devices that fai l to pass patch management health chec ks. You can specify the grace period in h ours, days, weeks, or months.
Product Evaluation Rule	Select the appropriate **Product Evaluatio n Rule**: • **Pass All**: Select this product evaluatio n rule if you want the **Patch Managem ent** health class to be deemed as *heal thy* only if all the configured patch ma nagement products are present. **Pass All** is the equivalent of an **AND** conditi on. • **Pass Any One**: Select this product eva luation rule if you want the **Patch Man agement** health class to be deemed a s *healthy* if any one of the configured patch management products are pres ent. **Pass Any One** is the equivalent of an **OR** condition. **Pass Any One** is the default.

0. To configure patch management application checks, click **Add**.The **Patch Management Health Checks** configuration page opens:

Figure 1. Configuration Page for the Windows Patch Management Application



1. Specify these parameters as described in the following table:

> **NOTE**
>
> All checks might not be available for some products. Where checks are not available, they are shown in a disabled state.

Table 2. Windows Patch Management

Parameter	Action/Description
Product - specific checks	To check whether any patch managemen t application (any vendor) is running on t he end host, clear the **Uncheck to allow any product** check box.
Select Patch Management Product	Select a patch management product ven dor. This option is enabled *only* if the **Product - specific checks** check box is checked.

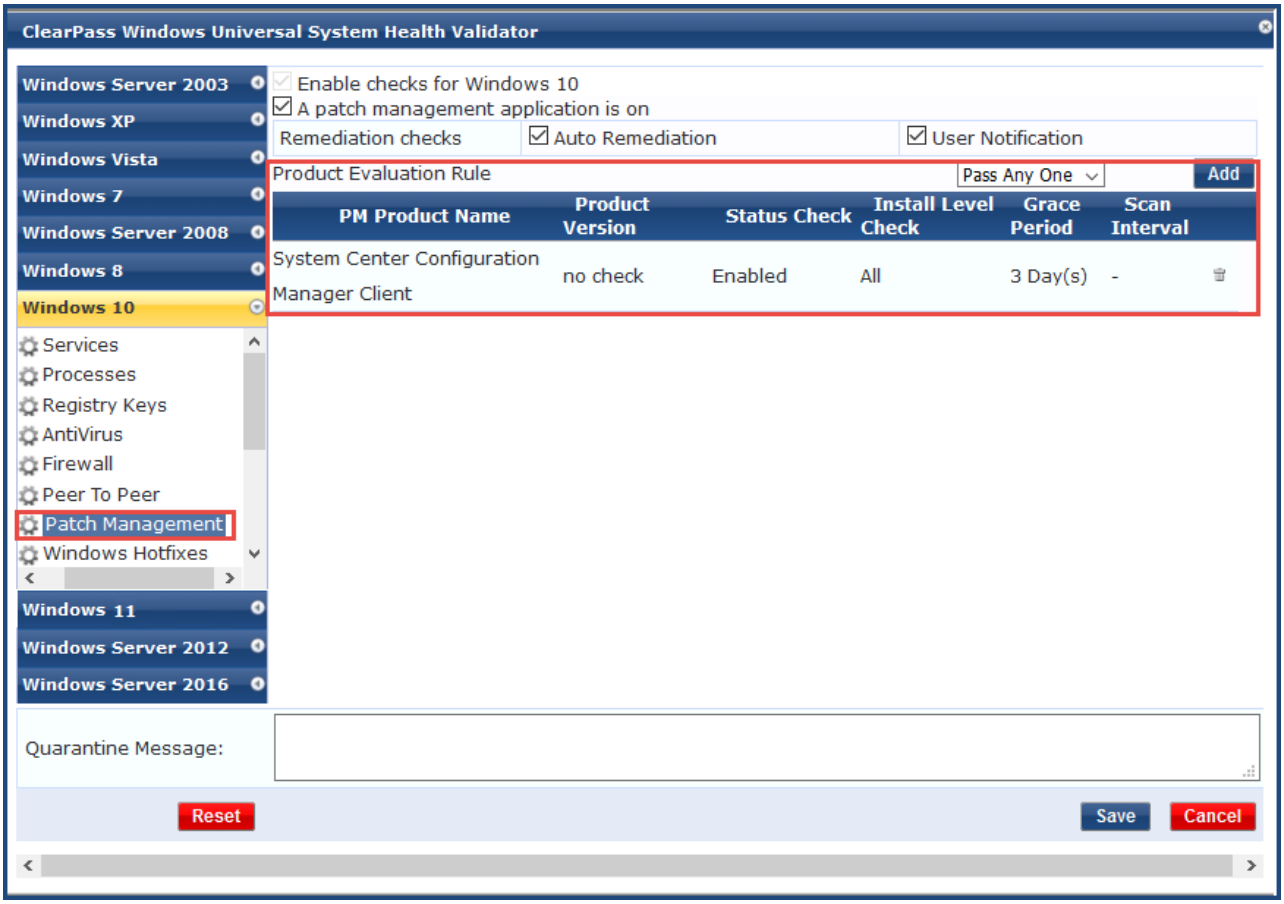
Parameter	Action/Description
Product version is at least	Enter the minimally recommended product version number. This option is enabled only if the **Product - specific checks** check box is checked.
Status Check Type	Specify the **Status Check Type** to check whether the Patch Agent is enabled. The ClearPass Policy Manager server compares the Patch Agent Status sent by OnGuard Agent with the configured value. If the Patch Agent Status value is different from configured value, the client is treated as unhealthy. If **Auto - remediation** is enabled, OnGuard Agent changes the Patch Agent Status on the client to the configured value. Select any of the following options: • **No Check**: ClearPass Policy Manager server ignores the Patch Agent Status value. This means it will not check the status of the Patch Agent application on the client. • **Enabled**: Patch Agent is turned on and it automatically updates the client. • **Disabled**: Patch Agent is disabled and it *will not* check for missing patches and update the client. **NOTE** The values specific to the selected patch management product are displayed in the **Status Check Type** field.
Install Level Check Type	This option is only enabled if the **Product - specific checks** check box is checked.

Parameter	Action/Description
	For Microsoft SCCM, selecting **All** will ret urn the full list of all missing patches. • **No Check**: ClearPass Policy Manager server ignores the Patch Agent Status value. This means it will not check the status of the Patch Agent application on the client. • **All**: Checks for all missing patches an d searches for all available patches. **NOTE** If you select the **Microsoft Windows Update Agent** fro m the **Select Patch Manage ment product** list and you s elect an option from the **Ins tall Level Check Type** list, th e results are as follows: • **All**: Returns the full list of missing pat ches. • **No Check**: Disables the **Grace Period** and **Scan Interval** fields.
Grace Period	Configure the time period for which OnG uard Agent should ignore missing patch es. You can specify the grace period in h ours, days, weeks, or months. After installing OnGuard Agent on an old and outdated client, the first time OnGua rd Agent detects missing patches, the ag ent sets the Grace Period start time. The grace period will last from the designate d start time to the end of the Grace Perio d value set for the **Grace Period** paramet er. For example:

Parameter	Action/Description
	a. **Grace Period** parameter is set to 3 days. b. **10/09 09:00 a.m.**: OnGuard Agent did not find any missing patches for System Center Configuration Manager (SCCM). c. **10/09 10:00 a.m.**: OnGuard Agent detects missing patches for SCCM. The grace period start time is set to 10:00 a.m. on 10/09. d. **10/12 10:00 a.m.**: The grace period expires after three days (In this example, 10:00 a.m. on 10/12). OnGuard Agent checks to see if patches are missing for SCCM. If patches are still missing and **Auto - remediation** is enabled, OnGuard Agent starts downloading and installing missing patches. **NOTE** **No Check** from the **Install Level Check Type** field, **Grace Period** is disabled.
Scan Interval	Specify the **Scan Interval** by specifying the number of hours, days, weeks, or months. If you selected **No Check** from the **Install Level Check Type** field, **Scan Interval** is disabled.

2. Click **Save**. When you save your patch management configuration, the configuration information is displayed on the Windows **Patch Management** page:

Figure 2. Windows Patch Management Configuration Summary



3. Click **Save**. You return to the **Posture Plugins** tab. The status of the Policy Manager Windows Universal Health Validator plugin configuration is now displayed as "*Configured*."

4. Click **Next** to configure the Rules for this Windows posture policy.

Windows Hotfixes

There are two Hotfixes evaluation rules that can be applied:

- The **Windows Hotfixes Groups to be Present Evaluation Rule** specifies how to evaluate the health among multiple Hotfixes groups (see Table 1 for details).

  For example, if the status of *Group1* is [Hotfix1] AND [Hotfix 2], and the status of *Group2* is [Hotfix3] OR [Hotfix4], the overall health status of the two groups is calculated as follows:

  - **Pass All**: ([Hotfix1] AND [Hotfix2]) AND ([Hotfix3] OR [Hotfix4])
  - **Pass Any**: ([Hotfix1] AND [Hotfix2]) OR ([Hotfix3] OR [Hotfix4])

- The **Windows Hotfixes Group Evaluation Rule** specifies how to evaluate the health for a specific hotfixes group (see Table 2 for details).

  To define Windows Hotfixes Groups, specify the evaluation rules for multiple groups or a single group, and add or remove specific Windows hotfixes on the endpoint:

  1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
  2. Select the **Posture Plugins** tab and specify the following:
     a. **Policy Name/Description**: Enter the name and a description of the posture policy.
     b. **Posture Agent**: Select **OnGuard Agent.**
     c. **Host Operating System**: Select **Windows** (set by default).
     d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
  3. Click **Next**. The **Posture Plugins** dialog opens.
  4. Click the check box for **ClearPass Windows Universal System Health Validator**.
  5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
  6. Select the operating system for which you want to configure health checks for Windows Hotfixes.
  7. Select **Windows Hotfixes**. The **Windows Hotfixes** health class configuration page opens.
  8. Select the **Enable checks for <Windows OS>** option.
  9. Specify the **Windows Hotfixes** parameters as described in the following table:

Table 1. Windows Hotfixes Page Parameters

Parameter	Action/Description
Auto Remediation	Allow auto - remediation for hotfix chec ks. Enabling this automatically triggers updates of the specified hotfixes. Auto - remediation for the Windows Hot fixes health class is enabled by default.

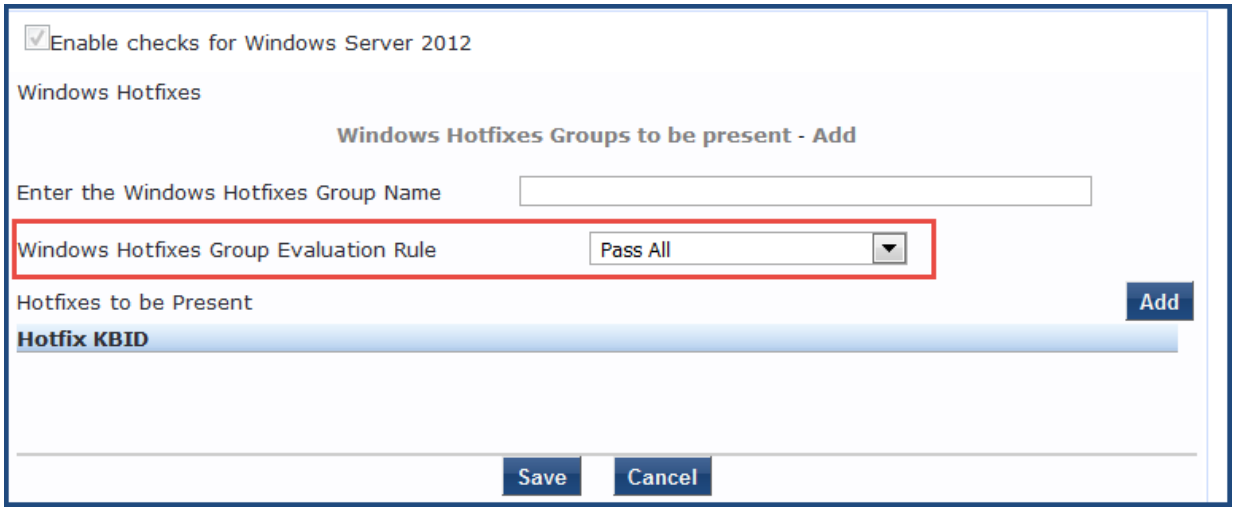
Parameter	Action/Description
User Notification	Allow user notifications to check for hot fix policy violations.
Monitor Mode	Enable **Monitor Mode**.
Grace Period	Use the **Grace Period** drop - down men u to configure the time period for whic h the Windows Universal System Health Validator plugin should ignore devices t hat fail to pass Windows Hotfix health c hecks. You can specify the grace period in hours, days, weeks, or months.
Windows Hotfixes Groups to be Present Evaluation Rule	This evaluation rule specifies whether a ll groups should be healthy or any one group should be healthy. For example, if there are two Hotfixes G roups: ◦ **Pass All**: Select this evaluation rule t o calculate the two group's health s tatus as *Group1* **AND** *Group2*. ◦ **Pass Any One**: Select this evaluation rule to calculate the two group's hea lth status as *Group1* **OR** *Group2*.

Configuring Windows Hotfixes Groups to Be Present

To configure the Windows Hotfixes Groups to be present:

10. From the **Windows Hotfixes** health class configuration page, click **Add**. The following Hotfixes dialog opens:

Figure 1. Defining the Hotfix Group

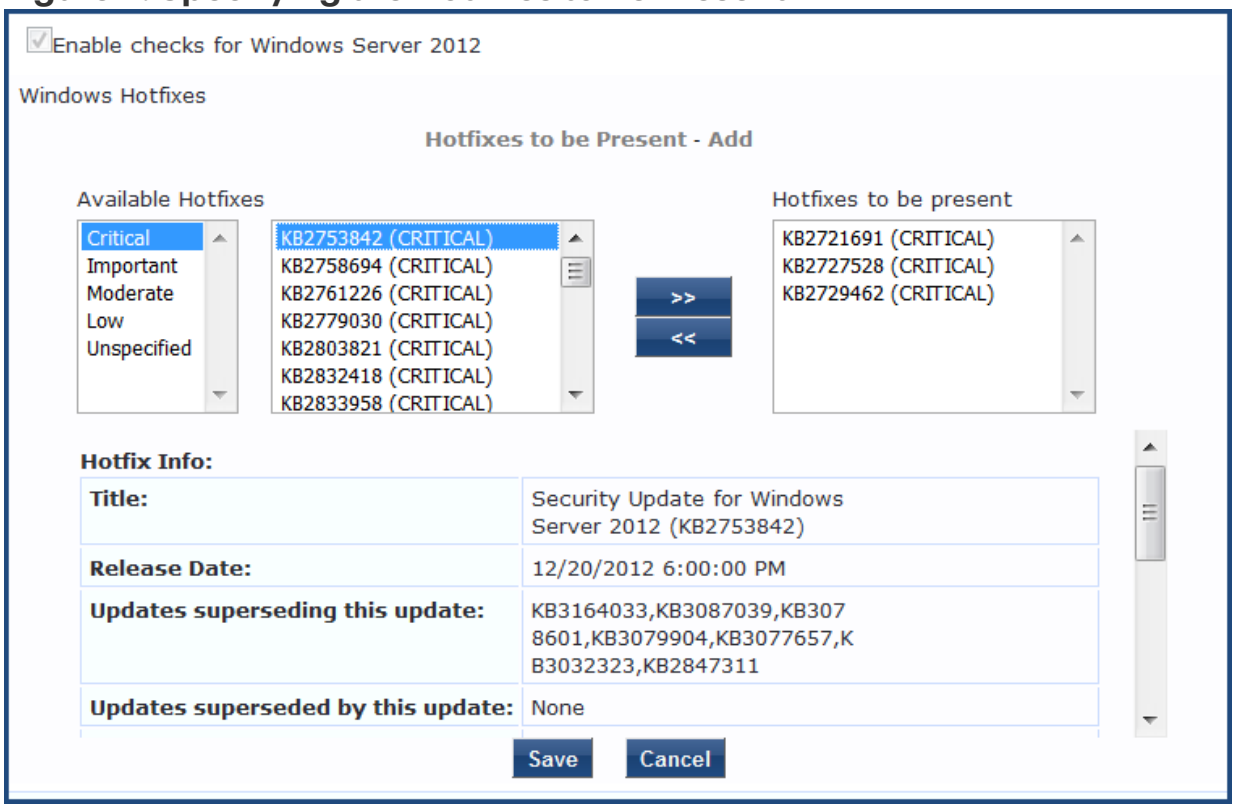


11. Specify the Hotfixes Group parameters as described in the following table:

Table 2. Specifying Hotfixes Group to Be Present Parameters

Parameter	Action/Description
Enter the Windows Hotfixes Group Name	Enter the name of the Hotfixes Group.
Windows Hotfixes Group Evaluation Rule	This evaluation rule specifies how to evaluate the health of a specific Hotfixes Group. Select the appropriate Hotfix Group Evaluation Rule: ◦ **Pass All**: Select this evaluation rule if you want all hotfixes groups to be present. **Pass All** is the equivalent of an **AND** condition. ◦ **Pass Any One**: Select this evaluation rule if you want any one of the hotfixes groups to be present. **Pass Any One** is the equivalent of an **OR** condition.
Hotfixes to be Present	To add hotfixes to the Hotfixes Group, click **Add**. The dialog shown in Figure 3 opens.

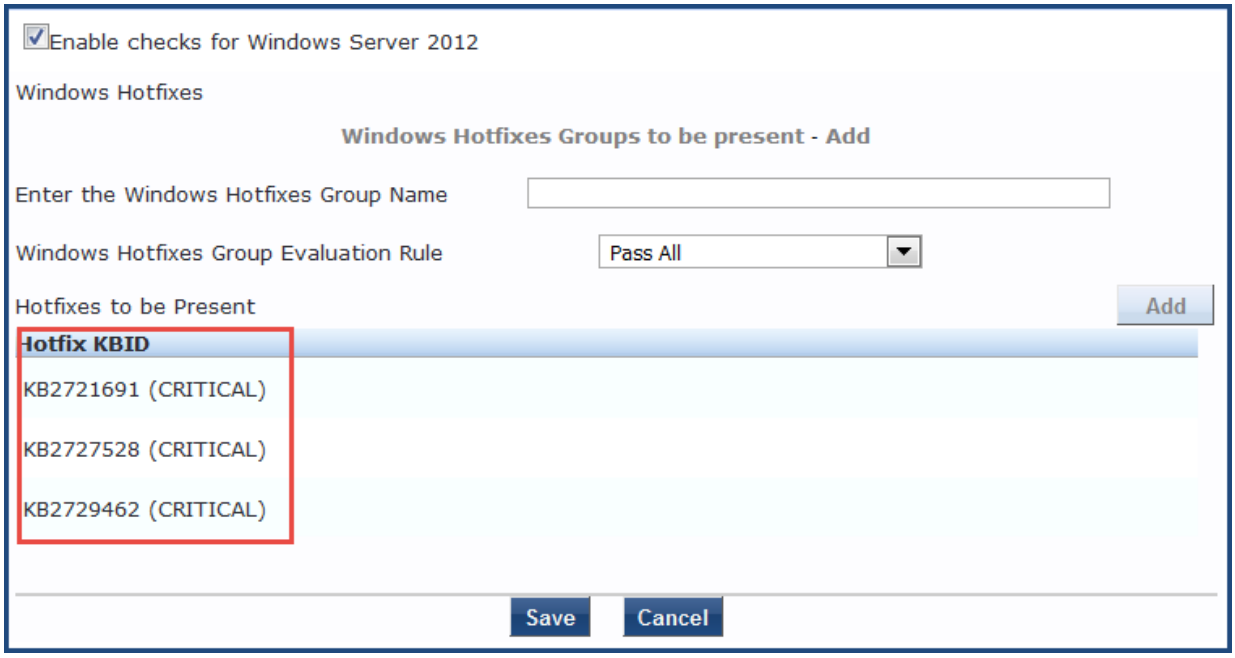
Figure 2. Specifying the Hotfixes to Be Present



12. From the first list, specify the criticality of the hotfixes:
    - Critical
    - Important
    - Moderate
    - Low
    - Unspecified

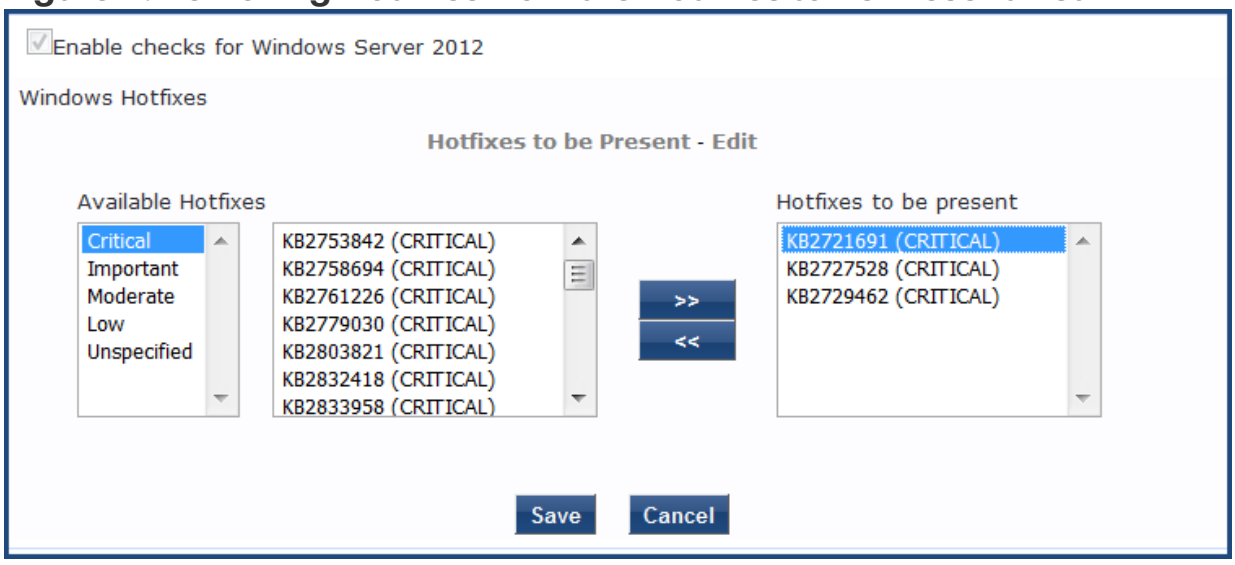
    As shown in Figure 3, the list of hotfixes for the selected criticality is displayed.
13. Select one or more of the desired hotfixes from the **Available Hotfixes** list. As shown in Figure 3, when you select a Hotfix from the list of available hotfixes, Policy Manager displays information about that hotfix.
14. To move hotfixes to the **Hotfixes to be present** box, select the desired hotfixes, then click **> >** .

Figure 3. Windows Hotfixes Added



15. To remove a hotfix from the **Hotfixes to be present** list, select the hotfix to be removed. The **Edit Hotfixes to be Present** dialog opens.

Figure 4. Removing Hotfixes from the Hotfixes to Be Present List



16. From the Hotfixes to be present list, select the hotfix you wish to remove and click **<<**. The selected hotfix is removed from the Hotfixes to be present list.
17. When finished, click **Save**. You return to the **Windows Hotfixes** dialog, where the **Hotfixes to be Present > Hotfix KBID** (Knowledge Base Article ID) list displays the updated list.
18. Click **Save**.

USB Devices

About this task

The **USB Devices** page provides configuration to control USB mass storage devices attached to an endpoint.To configure health checks for USB devices:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select **USB Devices**.
8. Select the **Enable checks for <Windows OS>**. The following table describes the configuration options on this page.

Table 1. USB Devices

Parameter	Description
Auto Remediation	Enable to allow auto remediation for USB mass storage devices attached to the en dpoint (Automatically stop or eject the dr ive).
User Notification	Enable to allow user notifications for USB devices policy violations.

Parameter	Description
Grace Period	Use the **Grace Period** drop - down menu to configure the time period for which th e Windows Universal System Health Valid ator plugin should ignore USB devices th at fail to pass health checks. You can spe cify the grace period in hours, days, wee ks, or months.
Remediation Action for USB Mass Storag e Devices	• No Action - Take no action; do not ej ect or disable the attached devices. • Remove USB Mass Storage devices - Eject the attached devices. • Disable USB Mass Storage devices - Disable the USB port to which the US B device is attached. Enable the USB port manually to use the disabled USB port again.

Virtual Machines

About this task

The **Virtual Machines** page provides configuration to Virtual Machines utilized by your network. To configure Virtual Machines.

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).

3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select **Virtual Machines**.
8. Select the **Enable checks for <Windows OS>** and **Virtual Machine Detection is on** options. The following table describes the **Virtual Machines** parameters:

Table 1. Virtual Machines

Parameter	Description
Auto Remediation	Allow auto remediation for virtual machin es connected to the endpoint.
User Notification	Allow user notifications for virtual machin e policy violations.
Allow access to clients running on Virtual Machine	Allow clients that running a VM to be acc essed and validated.
Allow access to clients hosting Virtual Ma chine	Allow clients that hosting a VM to be acc essed and validated.
Remediation Action for clients hosting Vi rtual Machines	• No Action - Take no action; do not st op or pause virtual machines. • Stop all Virtual Machines running on Host - Stop the VM clients that are r unning on Host. • Pause all Virtual Machines running on Host - Pause the VM clients that are r unning on Host.

Network Connections

About this task

The **Network Connections** page provides configuration to control network connections based on connection type. To configure Network Connections:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select **Network Connections**.
8. Select the **Enable checks for <Windows OS>** and **Network Connection Check is on** options. The following table describes the configuration options on this page.

Table 1. Network Connections Configuration Parameters

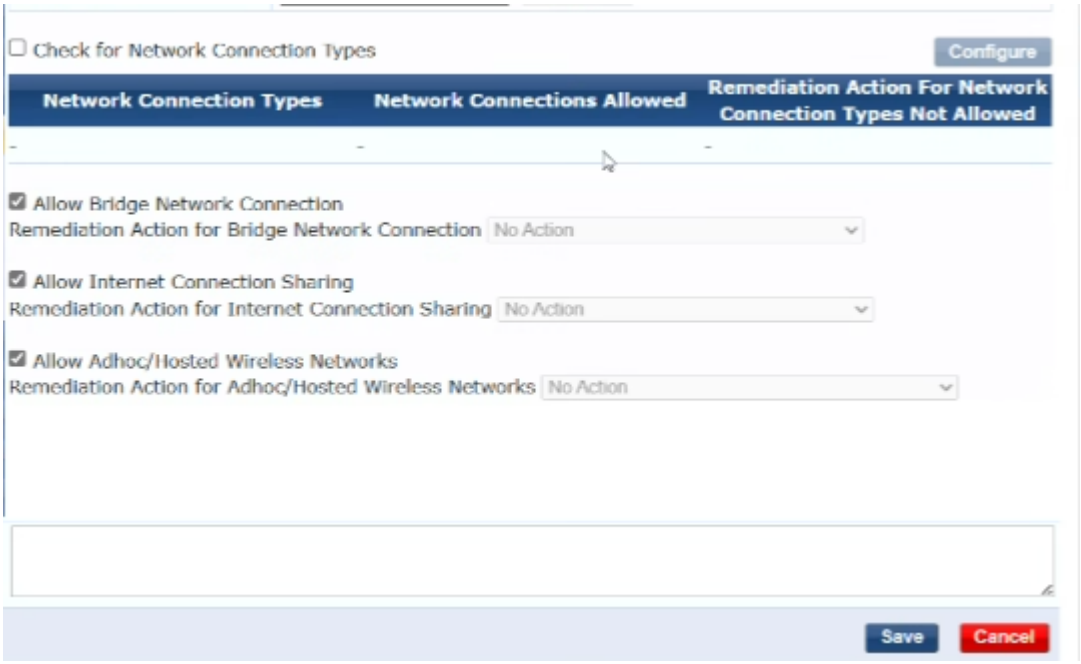
Parameter	Action/Description
Auto Remediation	Allow auto - remediation for network con nections.
User Notification	Allow user notifications in the event of ne twork connection policy violations.
Grace Period	Use the **Grace Period** drop - down menu to configure the time period for which th e Windows Universal System Health Valid

Parameter	Action/Description
	ator plugin should ignore network conne ctions that fail to pass health checks. You can specify the grace period in hours, d ays, weeks, or months.
Check for Network Connection Types	Select this option to define allowed net work connection types.
Remediation Action for Bridge Network C onnection	If **Allow Bridge Network Connection** is di sabled, then specify whether to take no a ction when a bridge network connection exists or to disable all bridge network co nnections.
Remediation Action for Internet Connecti on Sharing	If **Allow Internet Connection Sharing** is disabled, then specify whether to take no action when Internet connection sharing exists or to disable Internet connection s haring.
Remediation Action for Adhoc/Hosted Wi reless Networks	If **Allow Adhoc/Hosted Wireless Networ ks** is disabled, then specify whether to ta ke no action when an adhoc wireless net work exists or to disable all adhoc/hoste d wireless networks.

To check for specific network types:

9. Select **Check for Network Connection Types**, then click the **Configure** button.

Figure 1. Network Connections Configuration Page



0. Specify the **Network Connection Types** configuration parameters as described in the following table:

Figure 2. Network Connection Types Configuration Page

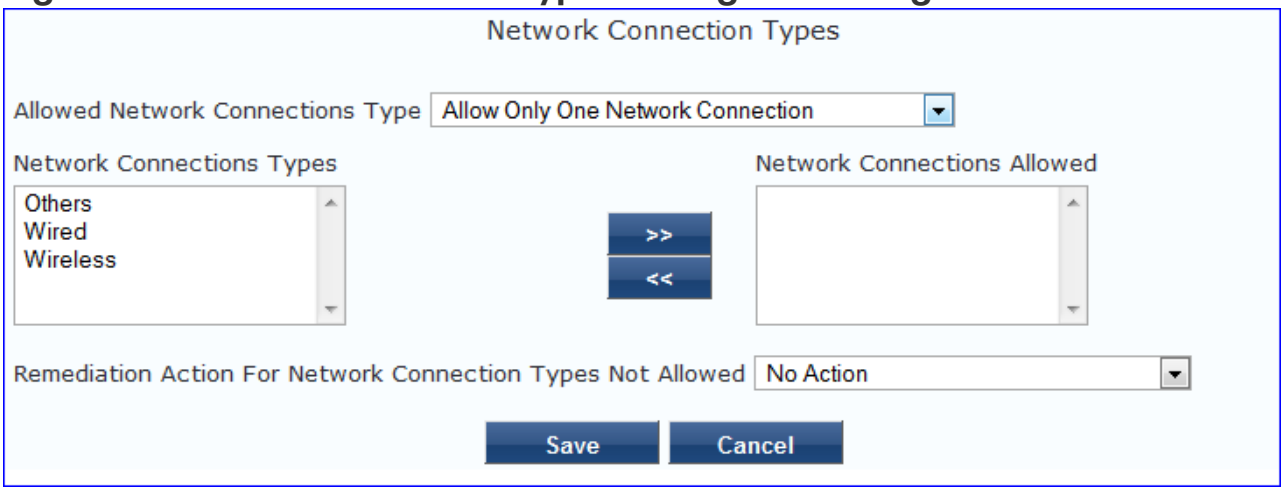


Table 2. Network Connection Type Configuration Parameters

Parameter	Action/Description
Allowed Network Connections Type	Select one of the following options: • Allow Only One Network Connection • Allow One Network Connection with VPN • Allow Multiple Network Connections

Parameter	Action/Description
Network Connection Types	To add or remove **Others**, **Wired**, and **Wireless** network connection types, click **>** **>** or **<<**.
Remediation Action for Network Connection Types Not Allowed	Specify one of the following • No Action: Take no action. Do not eject or disable the attached devices. • Disable Network Connections • Disconnect Network Connections

1. Click **Save**. This returns you to the **Network Connections** configuration page.

Disk Encryption

About this task

Disk encryption is a technology that protects information by converting it into unreadable code that cannot be deciphered easily by unauthorized people. Disk encryption uses encryption software or hardware to encrypt every bit of data that goes on a disk or disk volume. Disk encryption prevents unauthorized access to data storage.

To configure disk encryption for Windows:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**. The **Posture Policies** page opens.
2. Click the **Add** link. The **Posture Policies > Add** page opens.
3. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Leave the setting as **Windows**(the default).
   d. **Plugin Version**: OnGuard Plugin version **2.0** (V4 SDK) is set by default.
4. Click **Next**. The **Posture Plugins** dialog opens.

5. Select the check box for the Policy Manager Universal System Health Validator, then click the **Configure** button.
6. Select the operating system for which you want to configure health checks.
7. Select **Disk Encryption**.
8. Select the **Enable checks for <Windows OS>** and **A disk encryption application is on** options to specify the remediation checks, auto remediation, and user notification.
9. (Optional) Select **User Notification** to allow user notifications for disk encryption policy violations.
0. (Optional) Use the **Grace Period** drop-down menu to configure the time period for which the Windows Universal System Health Validator plugin should ignore disck encryption products that fail to pass health checks. You can specify the grace period in hours, days, weeks, or months.
1. Click **Add** to select a disk encryption product, then configure health check settings described in the following table.

Table 1. Disk Encryption Parameters

Parameter	Description
Product - specific checks	If you wish to select a specific disk encry ption product, do not change the default setting. To allow any encryption product to check disk encryption, select the **Uncheck to al low any product** check box. The **Select Disk Encryption product** and **Product Version is at least** fields are disa bled after you uncheck the**Uncheck to all ow any product** check box.
Select Disk Encryption product	Select a specific disk encryption product .
Product Version is at least	Search for the production version of the selected product.
Locations to Check	Select location to check. The options are:

Parameter	Description
	• None • System Root Drive • All Drives • Specific Locations

2. Click **Save**.

Installed Applications

About this task

The **Installed Applications** category groups classes that represent software-related objects. Access to these objects is supported by the Windows Installer. Examples of objects in this category are installed products, file specifications, and registration actions. In the **Installed Applications Configuration** page, you can turn on the installed applications check and specify information about which installed applications you want to monitor.

To configure the installed applications for Windows:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.

6. Select the operating system for which you want to configure health checks.
7. Select **Installed Applications**, then enable the **Installed Application Check is on** checkbox.
8. Specify the **Installed Applications Configuration** page parameters as described in the following table:

Table 1. Installed Applications for Windows Configuration Page Parameters

Parameter	Action/Description
Remediation checks	**Auto - Remediation** for the Installed Appl ications health class is not supported.
User Notification	Enable sending a remediation message w ith a list of applications to install or unins tall to the user.
Grace Period	Use the **Grace Period** drop - down menu to configure the time period for which th e Windows Universal System Health Valid ator plugin should ignore installed applic ations that fail to pass health checks. You can specify the grace period in hours, d ays, weeks, or months.
Monitor Mode	Enable **Monitor Mode** to treat all the ins talled applications as always healthy.
Applications Allowed (Mandatory)	Specify installed applications to be monit ored on a mandatory basis. **NOTE**
Applications Allowed (Optional)	Specify installed applications to be monit ored on an optional basis. **NOTE**

Parameter	Action/Description
Allow only Mandatory and Optional Applications	Specify that only the mandatory and optional applications are monitored.

NOTE

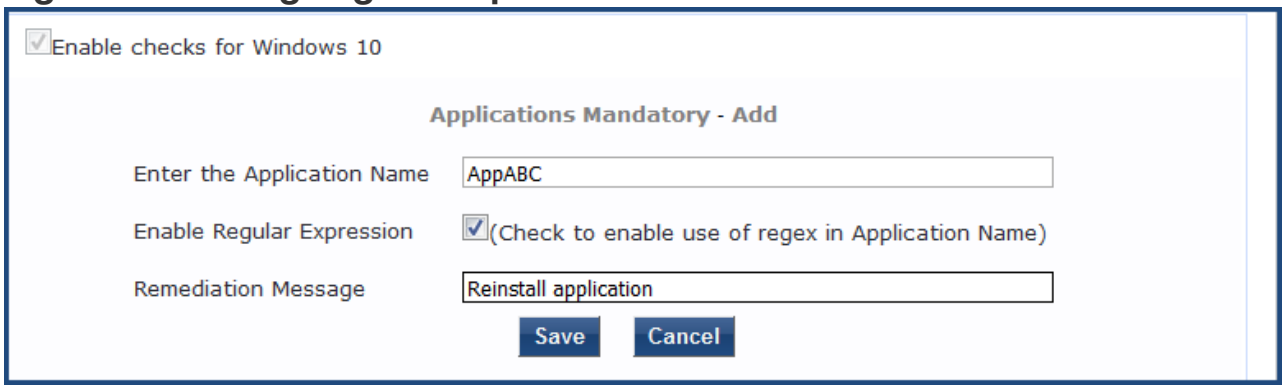
Enabling Regular Expressions for a Windows Application

To enable regular expressions for an application:

9. From the **ClearPass Windows Universal System Health Validator > Configure** page (see Windows System Health Validator OnGuard Agent), then select the desired version of Windows.
0. Select **Installed Applications**.
1. Click (enable) the **Installed Applications Check is on** check box. The **Installed Applications** dialog opens (see).
2. From the desired **Applications Allowed** category, click **Add**.

   The **Add Mandatory Applications** dialog for the selected **Applications Allowed** category opens.

Figure 1. Enabling Regular Expression



3. Configure the **Add Mandatory Applications** parameters as described in Table 2.

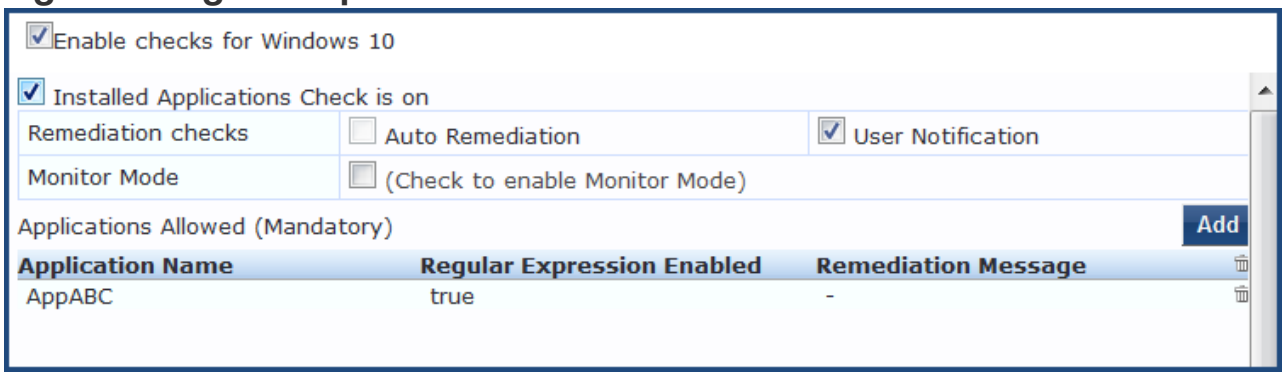
Table 2. Mandatory Applications Parameters

Parameter	Action/Description
Enter the Application Name	Enter the name of the application.

Parameter	Action/Description
Enable Regular Expression	Check (enable) this check box to enable t he use of regular expressions in the **Appli cation Name**. When this field is enabled, Policy Manage r treats the **Application Name** as a regula r expression when comparing application names.
Remediation Message	Optionally, enter the remediation messag e that will be displayed to the user in the event of an error. **NOTE**

4. Click **Save**. You return to the **Installed Applications** dialog, where **Regular Expression Enabled** is set to **true** for the specified application (see Figure 4).

Figure 2. Regular Expression Enabled



File Check

About this task

Use the **File Check** page to verify the group of files to present or absent. In the **File Check** page, you can turn on the file check and specify information about which the files you want to check.

To configure the **File Check** Health Class:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select the **File Check** option for that operating system. The **FIle Check** configuration page opens.
8. Select the **Enable checks for <Windows OS>** and **File Check is On** options.
9. Specify the **File Check Configuration** parameters as described in the following table:

Table 1. File Check Configuration Parameters

Parameter	Action/Description
Remediation checks	Auto - remediation for the **File Check** hea lth class is not supported. Select User N otification to send users a remediation m essage that includes the groups of files t o be present or absent
Monitor Mode	To treat all the file check health classes a s always healthy, enable **Monitor Mode**.
Grace Period	Use the **Grace Period** drop - down menu to configure the time period for which th e Windows Universal System Health Valid ator plugin should ignore file groups that fail to pass health checks. You can speci

Parameter	Action/Description
	fy the grace period in hours, days, weeks, or months.
File Groups to be Present	Click **Add** to create a list of files included in the **File Check** health class.
File Groups to be Absent	Click **Add** to create a list of files should b e absent from the **File Check** health class .

Defining the File Group to Be Present

0. To open the **File Group to be Present > Add** page, click **Add**. You can configure the name of the file group and specify the evaluation rule for the file group.
1. Specify the **File Group to be Present > Add** parameters as described in the following table:

Table 2. File Group to be Present - Add Parameters

Parameter	Action/Description
Enter the File Group Name	Enter the name of the file group.
File Group Evaluation Rule	Select the appropriate File Group Evaluati on Rule: • **Pass All**: Select this rule if you want th e **File Check** health class to be deeme d as 'healthy' only if all the configured file groups are present. • **Pass Any One**: Select this rule if you want the **File Check** health class to be deemed as 'healthy' even if any one of the configured file groups are present .

2. To configure the name of the file group and the evaluation rule for the file group, from **File Groups to be Present**, click **Add**. The **File to Be Present > Add** page opens:
3. Specify the **File to be Present > Add** parameters as described in the following table:

Table 3. File to be Present > Add Parameters

Parameter	Action/Description
File Location	Select any location of the file from the d rop - down list: • SystemDrive • Systemroot • ProgramFiles • ProgramFiles (x86) • HOMEDRIVE • HOMEPATH • None
Enter the File Path	Enter the file path as described in the exa mples from the user interface.
Enter the File Name	Enter the name of the file.
Enter the MD5 Sum	Optionally, specify one or more (comma separated) MD5 checksums of the proces s executable file.
Remediation Message	Specify the custom remediation messag e to be displayed to end users if **File che ck** fails.

4. When finished, click **Save**. The parameters configured in the **File to be Present > Add** dialog are reflected in the **File Groups to be Present** page.

Custom Scripts

About this task

Use the **Custom Scripts** health class to execute a custom script or auto remediation script on endpoints running the selected version of Windows. In the **Custom Scripts** page, you can enable custom scripts and select the scripts you want to run. Before you

can add scripts to the Universal System Health Validator plugin, you must have already defined your health collection or auto remediation scripts. For details, refer to OnGuard Custom Scripts.

To configure the **Custom Scripts** Health Class:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **Windows** (set by default).
   d. **Plugin Version**: OnGuard Plugin version 2.0 (set by default).
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Click the check box for **ClearPass Windows Universal System Health Validator**.
5. Click **Configure**. The **ClearPass Windows Universal System Health Validator** page opens.
6. Select the operating system for which you want to configure health checks.
7. Select the **Custom Scripts** option.
8. Select the **Enable checks for <Windows OS>** and **Custom Scripts is On** options.
9. Specify the Custom Script Health Class parameters as described in the following table:

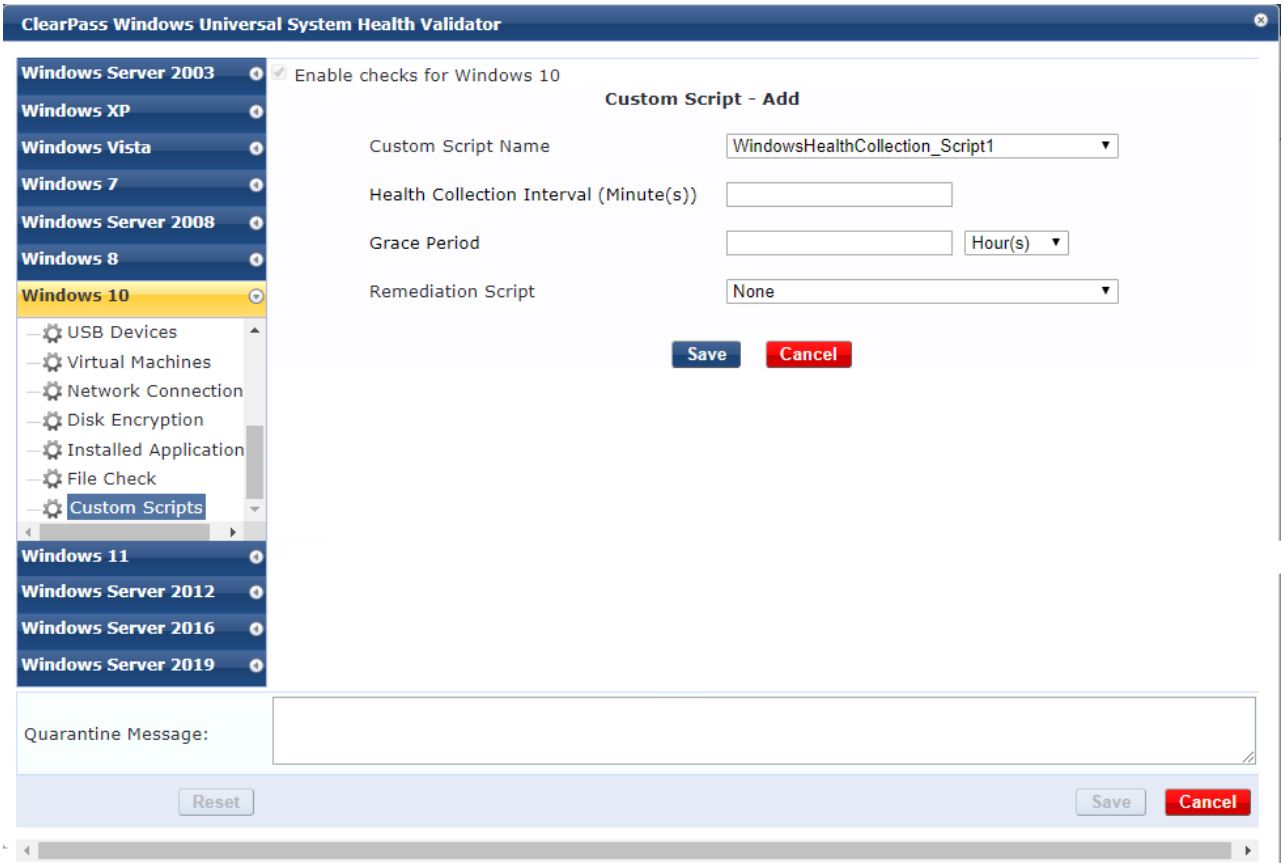
Table 1. Custom Scripts Configuration Parameters

Parameter	Action/Description
Auto Remediation	Allow auto - remediation for custom scr ipts. Enabling this option allows you to execute an auto remediation script for an endpoint that does not pass the health c heck

Parameter	Action/Description
User Notification	When enabled, a remediation message th at includes the results of the health chec k script is displayed to the end user.
Monitor Mode	If you want the custom script health class to treat all endpoints as always healthy, e nable **Monitor Mode**.
Evaluation Rule	The Windows Universal System Health Va lidator plugin can consider an endpoint h ealthy if it passes the health collection cu stom script checks from all groups of scr ipts, or if it passes any one of them. The default **Evaluation Rule** setting is **Pass A ny One.** Click the dropdown list and sele ct **Pass All** to modify this setting.

0. Click **Add** to select the custom scripts you want to use for health collection for the selected version of Windows. The **Custom Scripts Group - Add** dialog opens.
1. In the **Enter the Custom Scripts Group Name** field, enter a name for the collection of one or more custom scripts.
2. Click the **Custom Script Group Evaluation Rule** dropdown menu and select **Pass all** to consider an endpoint healthy if it passes health checks from all health collection scripts in the group, or select **Pass Any One** to consider the endpoint healthy if it passes the health checks by only one of the scripts in the group. This setting is different than the **Evaluation Rule** setting described in Table 1, in that this setting applies to individual scripts within a group, while the **Evaluation Rule** setting in Table 1 determines whether one or all groups of scripts need to pass to consider the device healthy.
3. Click **Add** to open the **Custom Script - Add** page.

Figure 1. Add a Custom Script



4. Click the **Custom Script Name** dropdown menu and select a custom health collection script defined using OnGuard Custom Scripts.

5. In the **Health Collection Interval (Minute(s)** field, choose how often the Windows Universal System Health Validator plugin should execute this script to collect health data. The minimum allowed value is **5** minutes. Please use caution while setting value of this field because if the custom script is CPU intensive then executing it frequently will cause extra CPU load on clients.

6. (Optional) Use the Grace Period drop-down menu to configure the time period for which the Windows Universal System Health Validator plugin should ignore devices that fail to pass health checks by custom health collection scripts. You can specify the grace period in hours, days, weeks, or months.

7. (Optional) Click the **Remediation Script** dropdown menu and select an auto remediation custom script that should run if the device does not pass the health check by the health collection script. For more information on creating a custom auto remediation script, see OnGuard Custom Scripts.

8. Click **Save** to return to the **Custom Scripts Group - Add** page. You can return to step 11 and click **Add** again to add another custom script to the scripts group, if desired.

9. When you have added all the scripts to the group, click **Save** again. The **Custom Scripts** dialog will show a list of all custom script groups, and the scripts contained in each group.

Windows System Health Validator OnGuard Agent

About this task

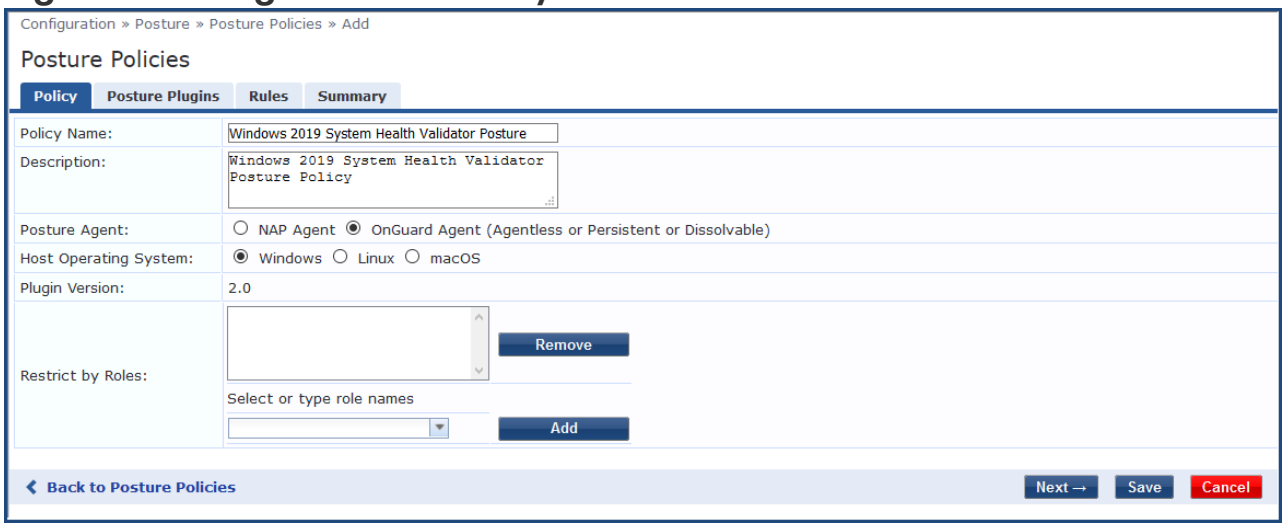
This Windows System Health Validator checks for current Windows Service Packs. Use the check boxes to enable support of specific Windows operating systems and to restrict access based on the Service Pack level.

To configure the **Windows System Health Validator: OnGuard Agent**:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**.
2. Click the **Add** link. The **Add Posture Policies** dialog opens.

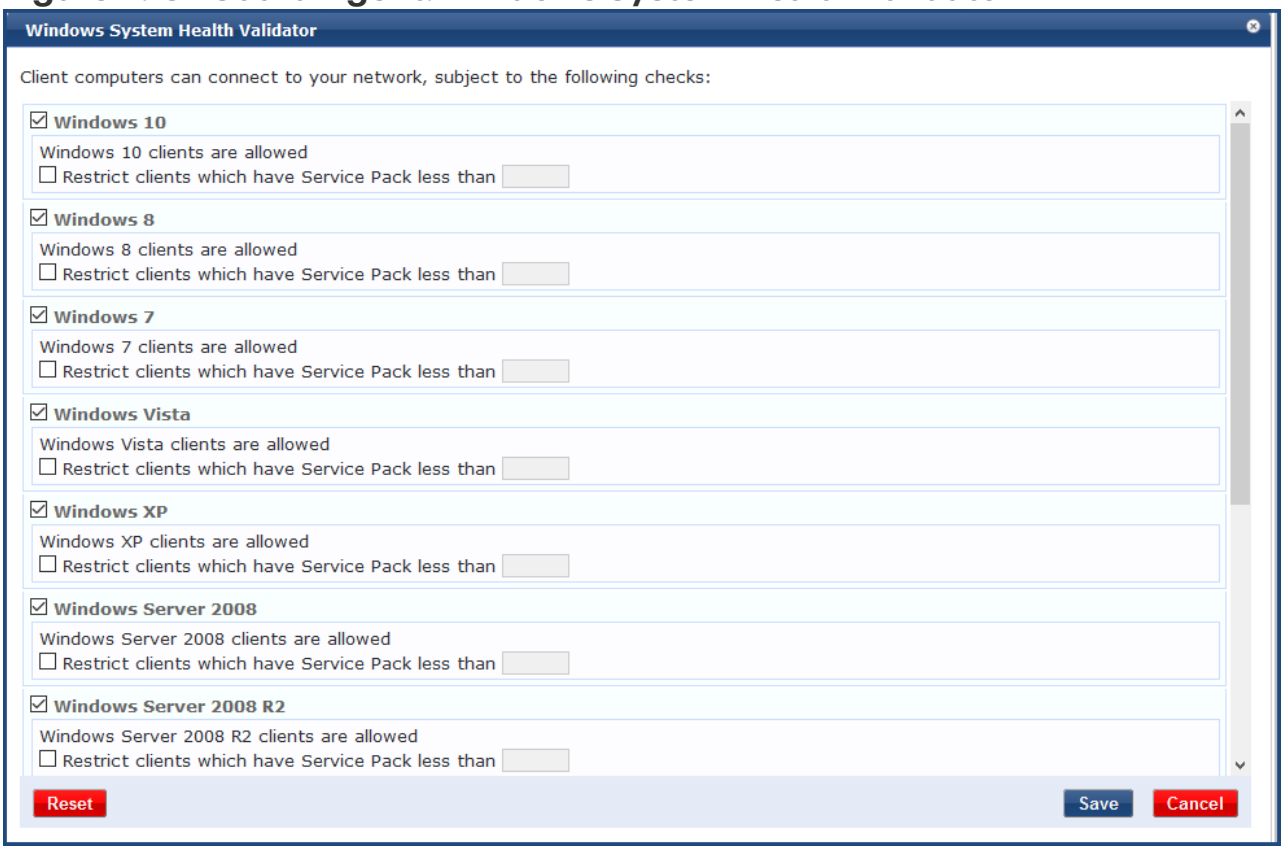
Figure 1. Adding a Posture Policy



3. Specify the following:
   a. **Policy Name**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Choose **OnGuard Agent (Persistent or Dissolvable)**.
   c. **Host Operating System: Windows** is selected by default.
   d. **Plugin Version**: OnGuard plugin version **2.0** (V4 SDK) is set by default.
4. Click **Next**. The **Posture Plugins** tab opens.

5. From the **Posture Plugins** tab, select **Windows System Health Validator**, then click the **Configure** button. The **Windows System Health Validator** page opens:

Figure 2. OnGuard Agent: Windows System Health Validator



6. To enable support of specific operating systems, click the corresponding check box.
7. Click the appropriate *Restrict clients which have Service Pack less than* check box, then enter the minimum *Service Pack* level required on the client computer to connect to your network.
8. Click **Save**. You return to the **Posture Policies > Posture Plugins** page, where the Status column for the **Windows System Health Validator** now shows *Configured*.

Windows Security Health Validator OnGuard Agent

About this task

The **Windows Security Health Validator** checks for the presence of specific types of security applications. You can use the options to restrict access based on the absence of the selected security application types.

To configure the **Windows Security Health Validator**:

Procedure

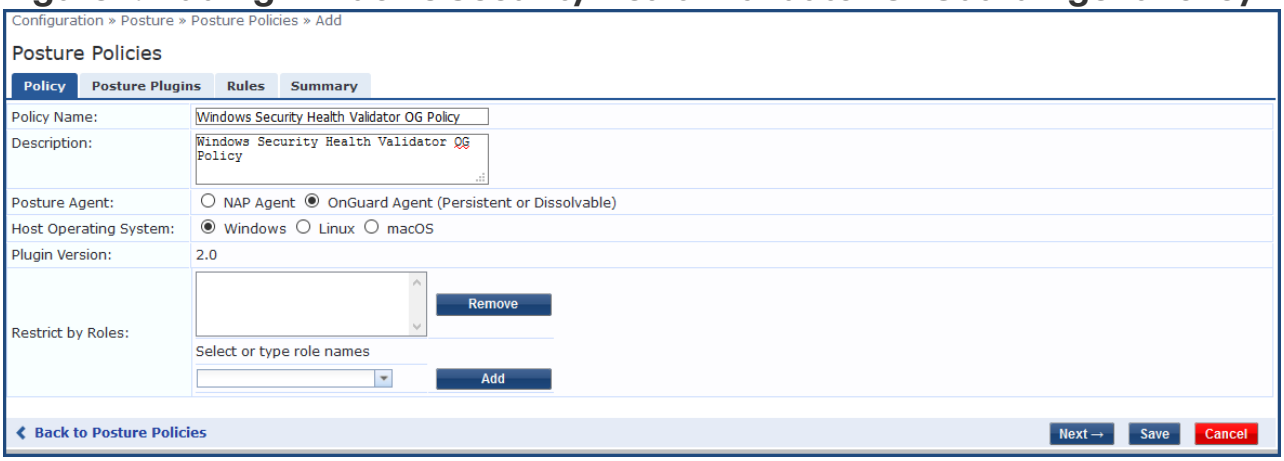
1. Navigate to **Configuration > Posture > Posture Policies**.

   The **Posture Policies** page opens.

2. Click the **Add** link.

   The **Add Posture Policies** dialog opens.

Figure 1. Adding Windows Security Health Validator OnGuard Agent Policy



3. Specify the following:

   a. **Policy Name**: Enter the name of the posture policy.

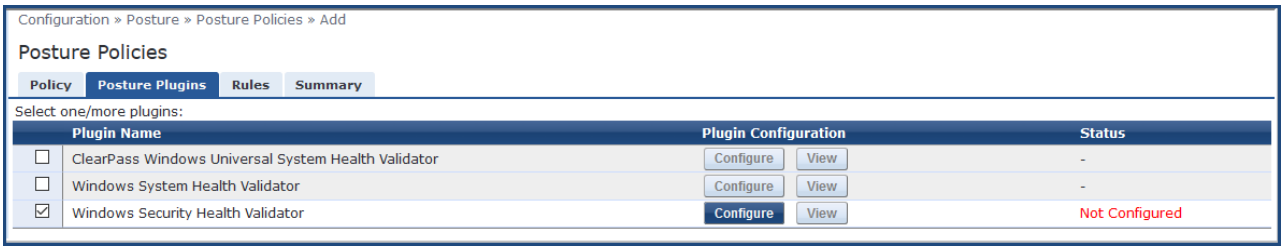
   b. **Posture Agent**: **Onguard Agent**

   c. **Host Operating System**: **Windows**

   d. **Plugin Version**: Set to **2.0** by default.

4. Click **Next**.

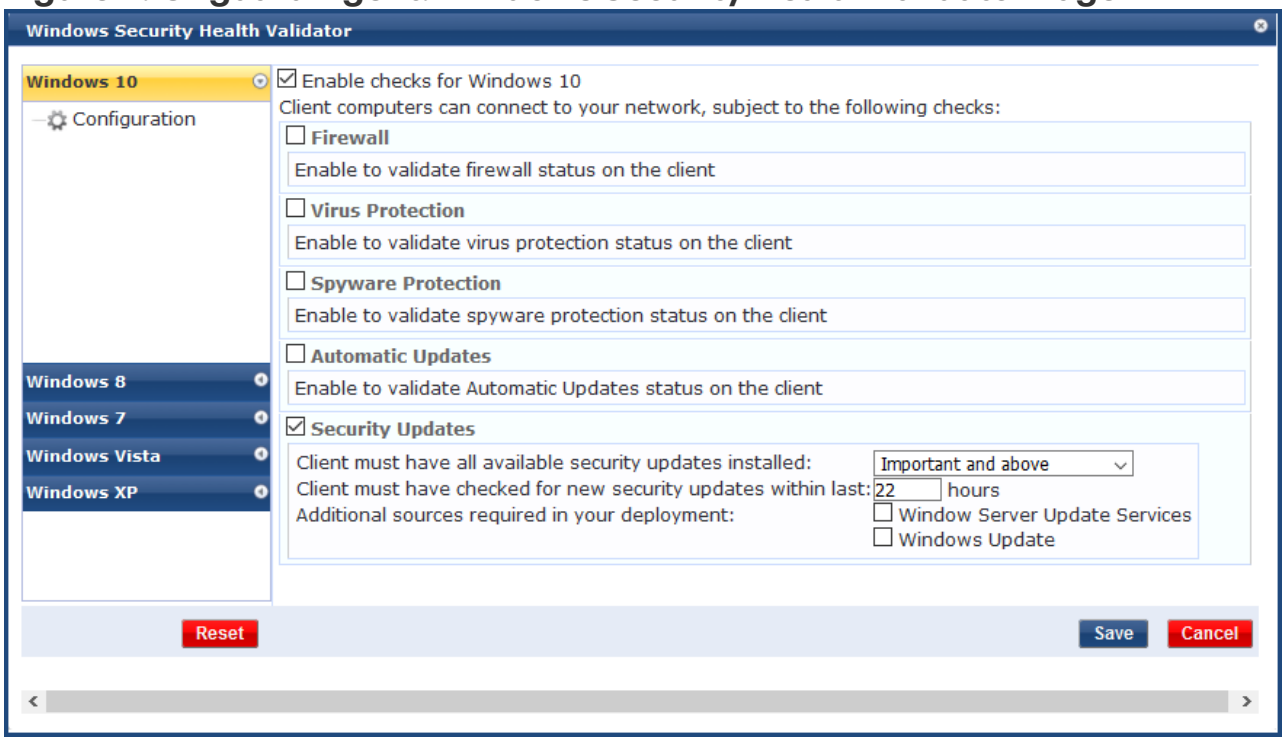
   The **Posture Plugins** dialog opens.



5. Select the **Windows Security Health Validator** check box, then click the **Configure** button.

   The following screen opens:

Figure 2. Onguard Agent: Windows Security Health Validator Page



6. Select the Windows operating system of interest.
7. Click the check box for *Enable checks for Windows n*.
8. To enable support of specific health checks for the selected version of Windows, click the corresponding check boxes.
9. Click **Save**.

ClearPass Linux Universal System Health Validator OnGuard Agent

About this task

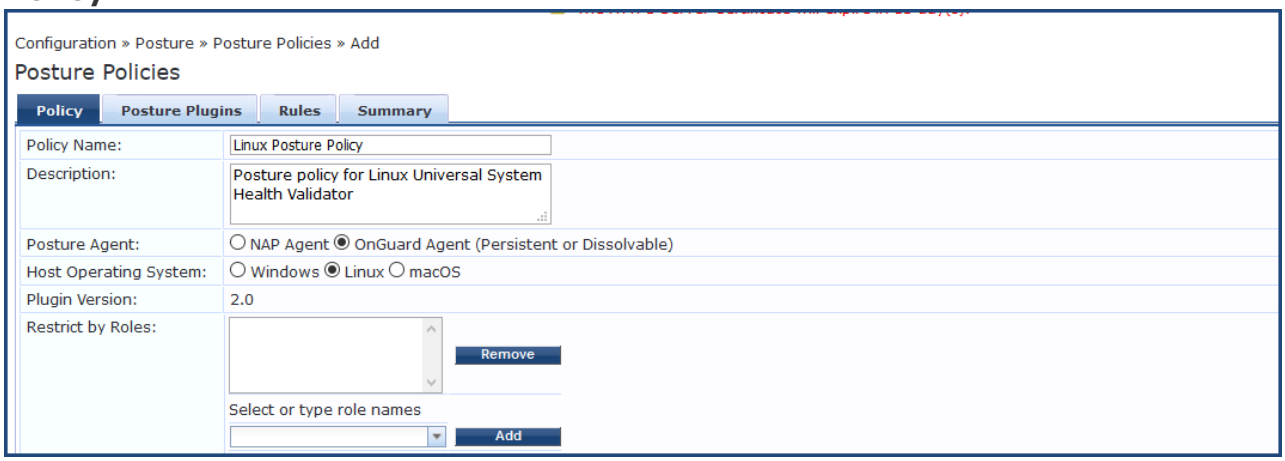
The OnGuard Dissolvable Agent version of the **ClearPass Linux Universal System Health Validator** plug-in supports the Antivirus and health classes:

To configure the **Policy Manager Linux Universal System Health Validator** plugin:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**.
2. Click the **Add** link. The **Add Posture Policies** dialog opens.

Figure 1. Adding a Universal System Health Validator for a Linux Posture Policy



3. Specify the following:

   a. **Policy Name/Description**: Enter the name and a description of the posture policy.

   b. **Posture Agent**: Select **OnGuard Agent.**

   c. **Host Operating System**: Select **Linux**.

   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.

4. Click **Next**. The **Posture Plugins** dialog opens.

Figure 2. Selecting thePolicy ManagerLinux Universal System Health Validator

5. Select the **Policy Manager Linux Universal System Health Validator**, then click **Configure** to configure antivirus settings and service types. The Linux posture plugin configuration dialog opens.

6. From the left pane, select the appropriate Linux OS:

   - CentOS
   - Fedora
   - Red Hat Enterprise Linux
   - SUSE Linux Enterprise
   - Ubuntu

7. Select a health check option to enable heath checks for Services, AntiVirus Applications or Custom Scripts.

Services Health Check

About this task

The **Services** page provides a set of widgets for specifying services to run or stop.

Procedure

1. From the left pane of the ClearPass Linux Universal System Health Validator, select **Services**. The **Services** configuration page opens.
2. Select the **Enable Checks for <Linux Operating System>** option.
3. Specify the **Services** page parameters as described in the following table:

Table 1. Services Health Class Configuration Parameters

Parameter	Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Allow auto - reme diation for processes. • **User Notification**: Allow user notificat ions in the event of process policy vio lations.
Available Services	Select a service from this list and move it to the **Services to run** or **Services to sto p** panels (using their associated widgets). This list varies, depending on OS type. • To add a service to the **Services to ru n** pane, select the service in the **Avail able Services** pane, then click the **> >** arrows. • To add a service to the **Services to s top** pane, select the services in the **Available Services** pane, then click th e **> >** arrows.

Parameter	Description
	• To move services from the **Services t o run** or **Services to stop** panes, selec t the appropriate services, then click t he corresponding **<<** arrows.
Insert	To add a service to the list of available se rvices, enter its name in the text box next to this button, then click **Insert**.
Delete	To remove a service from the list of availa ble services, select it and click **Delete**.

4. Click **Save**. After you save your Services health check configuration, it is listed on the **Services** page.

Antivirus Health Check

About this task

When configuring an antivirus health class, configure the primary antivirus product first. Starting with ClearPass 6.10.2, antivirusAntivirus products retain the order in which they were added in the antivirus health class. If none of the configured antivirus products is installed on the client, the installation message in the OnGuard Agent appears for the first product in the list

With 6.10.1 and earlier releases, products added to an antivirus health class were sorted in alphabetical order. If the configured products were missing during OnGuard health checks, the error message to install an antivirus was only shown for the first product alphabetically in the list.

Use the **Antivirus** page to activate an antivirus application.

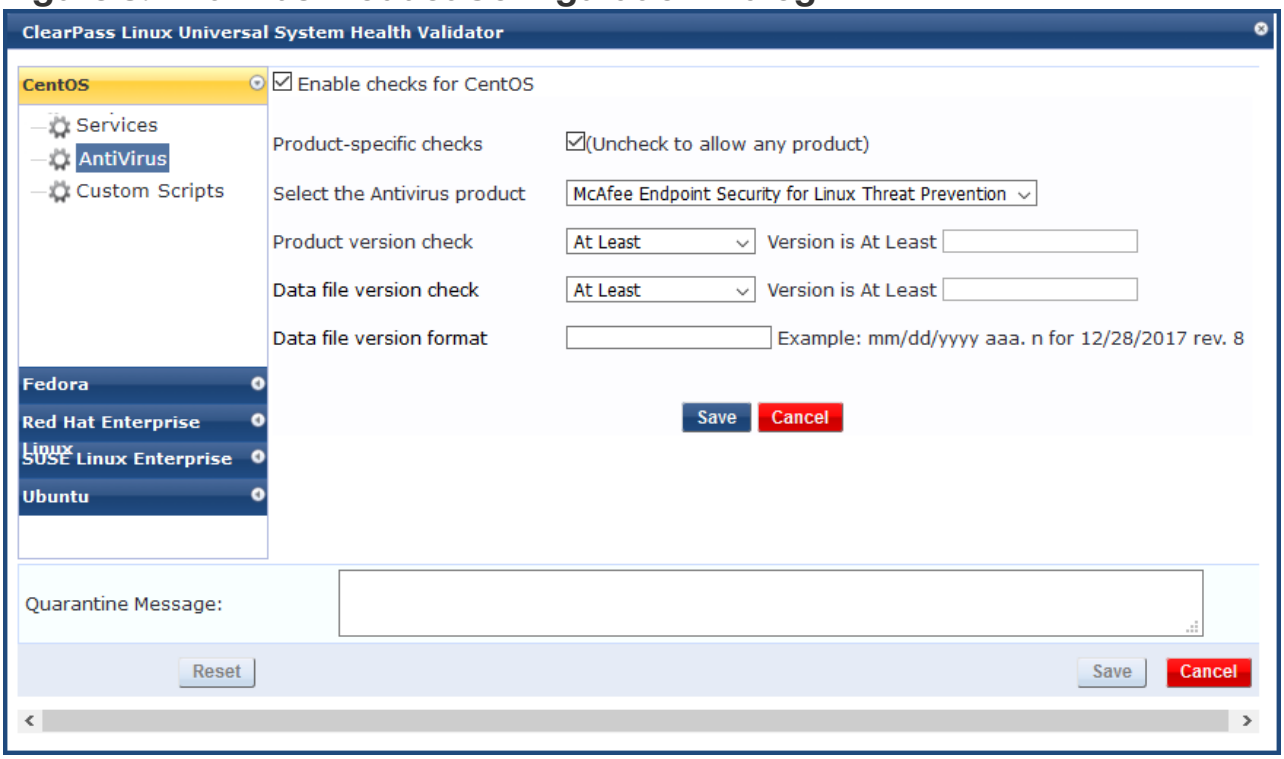
Procedure

1. From the left pane of the ClearPass Linux Universal System Health Validator, select **AntiVirus**.
2. Select the **Enable Checks for <Linux Operating System>** and **An antivirus-application is on** options.
3. Specify the **antivirus** health class parameters as described in the following table:

Table 2. Antivirus Health Class Configuration Parameters

4. To configure the antivirus product-specific checks, click **Add**. The antivirus product configuration dialog opens.

Figure 3. Antivirus Product Configuration Dialog



5. Specify the **antivirus product configuration** parameters described in the following table. Note that the available configuration settings may vary, depending upon the type of antivirus product being configured.

Table 3. Linux Antivirus Product Configuration Parameters

Parameter	Description
Product - specific checks	Select this check box if you want to con figure a specific antivirus product. If you want to allow any antivirus product, do n ot enable this field.
Select the Antivirus product	Select the antivirus from the drop - down list.
Product version check	Select one of the following options to ch eck the product version: • No Check

Parameter	Description
	• Is Latest • At Least. When you select **At Least**, th e **Version is at Least** field appears. Ent er the At Least version you require for the antivirus product version. • In Last N Updates.
Data file version check	Select one of the following options to ch eck the data file version: • No Check • Is Latest • At Least: You can specify the least ve rsion that you require for the data file . Enter the least version in the *Versio n is At Least* field. When you select **At Least**, the **Data file version format** op tion appears. • In Last N Updates.
Data file version format	This field allows admins to specify the da ta file version format when the **At Least** c heck is selected for **Data File Version che ck**. The configured format is used to compa re the configured data file version value with value received from the client. This f ield should be configured when the **Data file version** value contains a *date*. The following format specifiers are suppo rted: • **mm**: Month • **dd**: Day of the month • **yy** or **yyyy**: Year • **a**: Alphabetic

Parameter	Description
	• **n**: Numeric **NOTE** For example: • Data file version: *10/14/2021 rev. 10* ◦ Specify this data file version format: *mm/dd/yyyy aaa. n* • Data file version: *2021.10.08* ◦ Specify this data file version format: *yyyy.mm.dd* • Data file version: *20211014194001* ◦ Specify this data file version format: yyyymmddn • Data file version: *2021 - 10 - 12_04* ◦ Specify this data file version format: *yyyy - mm - dd_n* **NOTE** For antivirus products that do not include a date in the **Data file version check** value, this field can be left blank. In such cases, Policy Manager compares the value of the configured data file version with the value received from the client, character by character.

6. Click **Save**. After you save your antivirus health check configuration, it is listed on the **Antivirus** page.

Custom Scripts Health Check

About this task

Use the **Custom Scripts** page to execute a custom health check or auto remediation script on endpoints running the selected version of Linux. In the **Custom Scripts** page, you can enable custom scripts and select the scripts you want to run.

Before you can add scripts to the Linux Universal System Health Validator plugin, you must have already defined your health collection or auto remediation scripts. for details, refer to OnGuard Custom Scripts.

To configure the **Custom Scripts** Health Class:

Procedure

1. From the left pane of the ClearPass Linux Universal System Health Validator, select the **Custom Scripts** option. The **Custom Script** Health Class configuration page opens.
2. If it is not already enabled, select **Enable Checks for Linux <version>**, then select **Custom Scripts is on**.
3. Specify the custom script parameters as described in the following table:

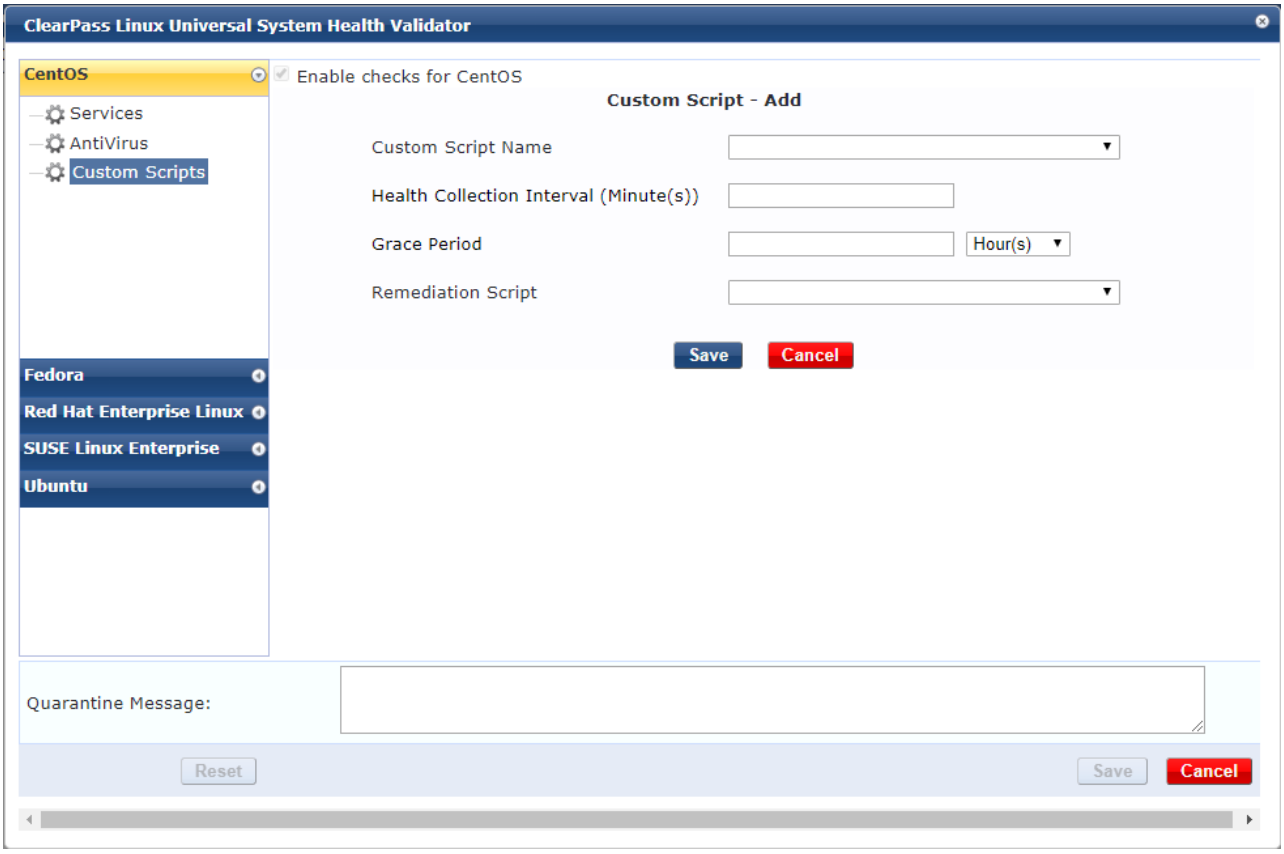
Table 4. Custom Scripts Configuration Parameters

Parameter	Action/Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Enable to allow au to - remediation for custom scripts. E nabling this option allows you to selec t an auto remediation script for an en dpoint that does not pass the health c heck • **User Notification**:When enabled, a re mediation message that includes the results of the health check script is di splayed to the end user.
Monitor Mode	If you want the custom script to tread all endpoints as always healthy, enable **Moni tor Mode**.

Parameter	Action/Description
Evaluation Rule	The Linux Universal System Health Validator plugin can consider an endpoint healthy if it passes the health collection script checks from all groups of scripts, or if it passes any one of them. The default **Evaluation Rule** setting is **Pass Any One.** Click the dropdown list and select **Pass All** to modify this setting.
Quarantine Message	(Optional) In the **Quarantine Message** field at the bottom of the dialog window, enter a quarantine message that will appear on a client that fails to pass any health check from the Linux Universal System Health Validator Plugin,

4. Click **Add** to select the custom scripts you want to use for health checks for the selected version of Linux. The **Custom Scripts Group - Add** dialog opens.
5. In the **Enter the Custom Scripts Group Name** field, enter a name for the collection of one or more custom scripts.
6. Click the **Custom Script Group Evaluation Rule** dropdown menu and select **Pass all** to consider an endpoint healthy if it passes health checks from all health collection scripts in the group, or select **Pass Any One** to consider the endpoint healthy if it passes the healthy checks by only one of the scripts in the group. This setting is different than the **Evaluation Rule** setting described in Table 1, in that this setting applies to individual scripts within a group, while the **Evaluation Rule** setting in Table 1 determines whether one or all groups of scripts need to pass to consider the device healthy.
7. Click **Add** to open the **Custom Script - Add** page.

Figure 4. Add a Custom Script



8. Click the **Custom Script Name** dropdown menu and select a custom health collection script defined using OnGuard Custom Scripts.

9. In the **Health Collection Interval (Minute(s)** field, choose how often the Linux Universal System Health Validator plugin should issue this script to collect health data. The recommended setting is every **5** minutes.

0. (Optional) Use the Grace Period drop-down menu to coonfigure the time period for which the Linux Universal System Health Validator plugin should ignore devices that fail to pass health checks by custom health collection scripts. You can specify the grace period in hours, days, weeks, or months.

1. (Optional) Click the **Remediation Script** dropdown menu and select an auto remediation custom script that should run if the device does not pass the health check by the health collection script. For more information on creating a custom auto remediation script, see OnGuard Custom Scripts.

2. Click **Save** to return to the **Custom Scripts Group - Add** page. You can return to step 11 and click **Add** again to add another custom script to the scripts group, if desired.

3. When you have added all the scripts to the group, click **Save** again. The **Custom Scripts** dialog will show a list of all custom script groups, and the scripts contained in each group.

Policy Manager macOS Universal System Health Validator: OnGuard Agent

About this task

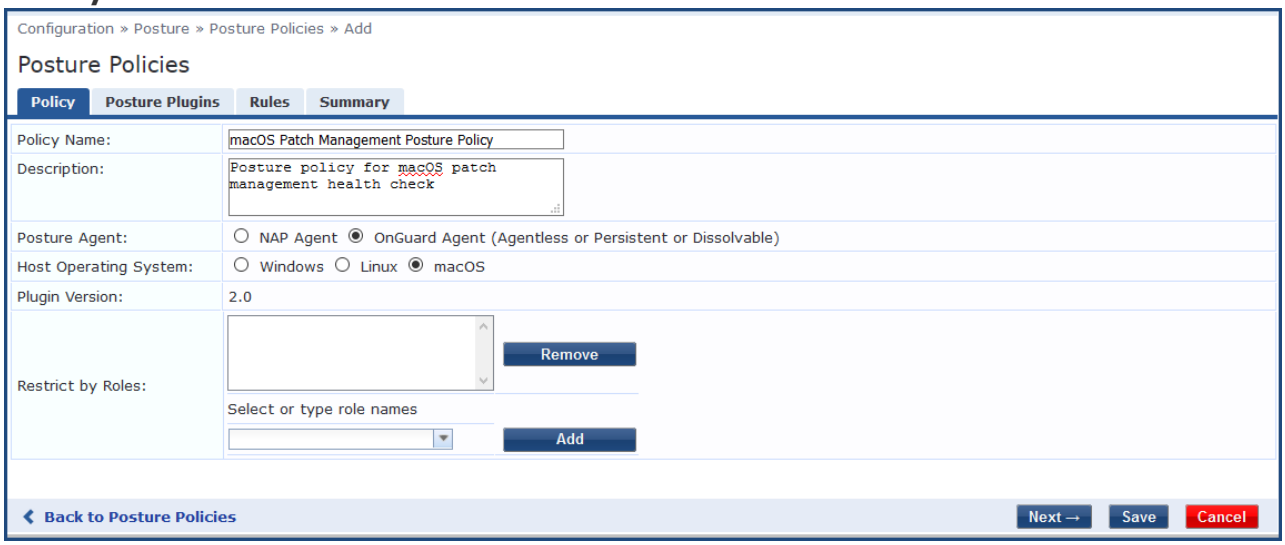
To configure the Policy Manager Universal System Health Validator for macOS:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**.

   The **Add Posture Policies** dialog opens.

   Figure 1. Adding a Universal System Health Validator for a macOS Posture Policy



   Specify the following:

   a. **Policy Name/Description**: Enter the name and a description of the posture policy.

   b. **Posture Agent**: Select **OnGuard Agent.**

   c. **Host Operating System**: Select **macOS.**

   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.

2. Click **Next**.

   The **Posture Plugins** dialog opens.

   Figure 2. Selecting the macOS Universal System Health Validator Posture Plug-in

3. In the **Posture Plugins** page, click the check box for **ClearPass macOS Universal System Health Validator**.
4. Click **Configure**.

   The **ClearPass macOS Universal System Health Validator** configuration page is displayed.
5. To enable checks for mac OS, select the **Enable checks for macOS** check box.

   The following configuration page opens:

Subtopics

Configuring macOS Health Checks
Services
Processes
Antivirus
Firewall
Patch Management
Peer to Peer
USB Devices
Virtual Machine
Network Connections
Disk Encryption
Installed Applications
File Check
Custom Scripts

Configuring macOS Health Checks

For details on configuring health checks for macOS, refer to the set of health check configuration pages described in the following sections:

- Services
- Processes
- Antivirus

- Firewall
- Patch Management
- USB Devices
- Virtual Machine
- Network Connections
- Disk Encryption
- Installed Applications
- File Check

Services

About this task

From the **Services** page, you can configure which services to run and which services to stop.

To add or remove specific services on the endpoint:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**.

   The **Posture Plugins** dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, enable the **Enable checks for macOS** checkbox.

6. Select **Services**. The **Services** health class configuration page opens:
7. Specify the **Policy Manager macOS Universal System Health Validator > Services** configuration parameters as described in the following table:

Table 1. Services Configuration Parameters

Parameter	Action/Description
Auto Remediation	Allow auto - remediation for services.
User Notification	Allow user notifications in the event of se rvice policy violations.
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for servic es. • **User Notification**: Enable to allow use r notifications in the event of service policy violations.
Grace Period	The **Grace Period** setting defines the tim e period for which the macOS Universal System Health Validator plugin ignores d evices that fail service health checks. You can specify the grace period in hours, d ays, weeks, or months.

Specifying the Services to Run

To specify the services to run:

8. Select one or more of the desired services from the **Available Services** list.
9. To move the desired services to the **Services to Run** box, click **> >** , then click **Save**.
0. You can also add a service to the list of available services. To do so, enter the service name in the **Insert** text box, then click **Insert**.

Specifying the Services to Stop

To specify the services to stop:

1. Select one or more of the desired services from the **Available Services** list.
2. To move the desired services to the **Services to Stop** box, click **> >** , then click **Save**.
3. You can also add a service to the list of available services. To do so, enter the service name in the **Insert** text box, then click **Insert**.

Processes

About this task

The **Processes** page provides a set of parameters to specify which processes to be explicitly present or absent on the system.

To view and configure processes:

Procedure

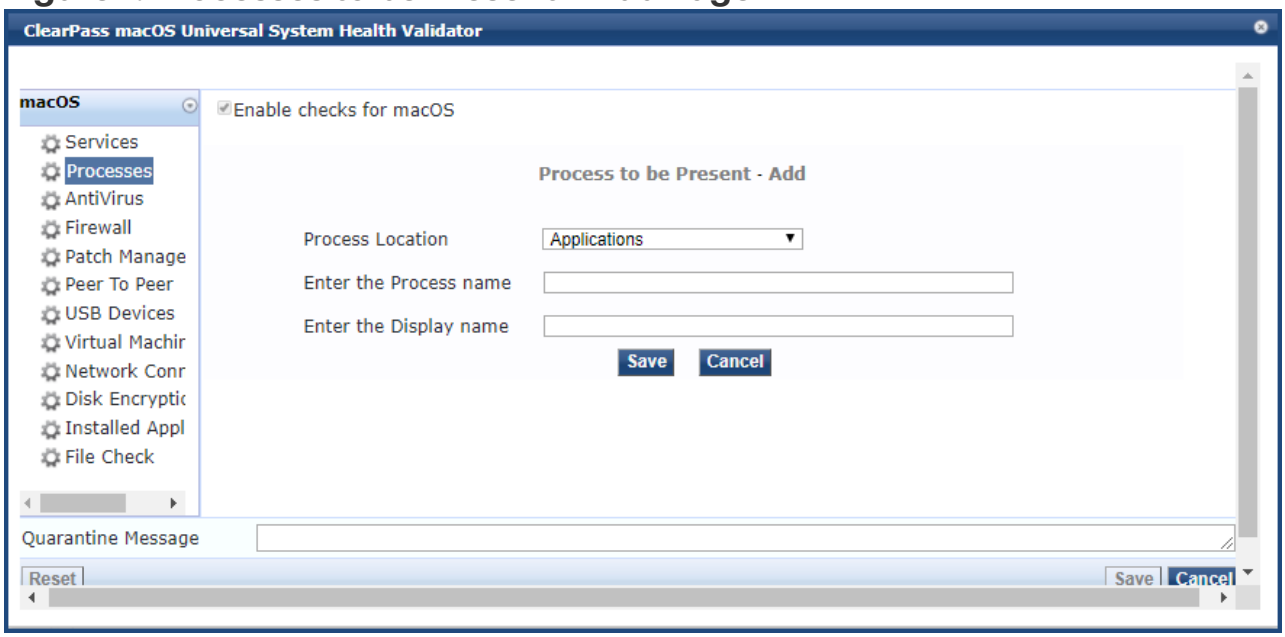
1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**.

   The **Posture Plugins** dialog opens.
4. Select the **ClearPass macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, enable the **Enable checks for macOS** checkbox.
6. Select **Processes**. The **Processes** health class configuration page opens:
7. Specify the **Processes** configuration parameters as described in the following table:

   Processes to be Present Parameters

8. In the **Processes to be Present** section, click **Add.** The **Add Processes to be Present** page opens.
9. To open the page with options to configure the name, location, and display name of the processes, click **Add**. The **Process to be Present > Add** page opens.

Figure 1. Processes to be Present > Add Page



0. Specify the **Processes to be Present** parameters as described in the following table, then click **Save**:

Table 1. Processes to be Present Page Parameters

Parameter	Action/Description
Process Location	Choose from the following locations: • Applications • UserBin • UserLocalBin • UserSBin • None
Enter the Process name	Specify the path name containing the p rocess executable name.

Parameter	Action/Description
Enter the Display name	Enter a user - friendly name for the proc ess. This is displayed in end - user facing messages.

After you save the Processes parameters, the information appears in the **Processes to be Present** section.

Processes to Be Absent Parameters

1. In the **Processes to be Absent** section, click **Add.** The **Add Processes to be Absent** page opens. Figure 4 shows the configuration parameters when you select **Process Name** and select **MD5 Sum**.

Figure 2. Process to Be Absent Pages: Process Name and MD5 Sum

2. Specify the **Processes to be Absent** parameters as described in the following table, then click **Save**:

Table 2. Processes to Be Absent Page Parameters

Parameter	Action/Description
Check Type	Select one of the following process optio ns: • **Process Name**: The agent looks for all processes that matches with the give n name. For example, if notepad.exe is specifie d, the agent kills all processes whose name matches, regardless of the loca tion from which these processes were started. • **MD5 Sum**: This specifies one or more (comma - separated) MD5 checksums of the process executable file. f there are multiple versions of the process executable, you can specify the MD5 sums of all versions here. The agent e numerates all running processes on t

Parameter	Action/Description
	he system, computes the MD5 sum of the process executable file, and matc hes this with the specified list. One or more of the matching processes are t hen terminated. • **SHA256 Sum:** This specifies one or m ore (comma - separated) SHA256 che cksums of the process executable file . f there are multiple versions of the p rocess executable, you can specify th e SHA256 sums of all versions here. T he agent enumerates all running proc esses on the system, computes the S HA256 sum of the process executable file, and matches this with the specifi ed list. One or more of the matching processes are then terminated.
Enter the Display name	Enter a user - friendly name for the proc ess. This display name is displayed in end - user facing messages.

You return to the **Processes Configuration** page, which now shows the values for the processes that were configured.

Antivirus

About this task

On the **Antivirus** page, you can activate an antivirus application.

To configure the macOS antivirus health class:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:

a. **Policy Name/Description**: Enter the name and a description of the posture policy.

b. **Posture Agent**: Select **OnGuard Agent.**

c. **Host Operating System**: Select **macOS.**

d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.

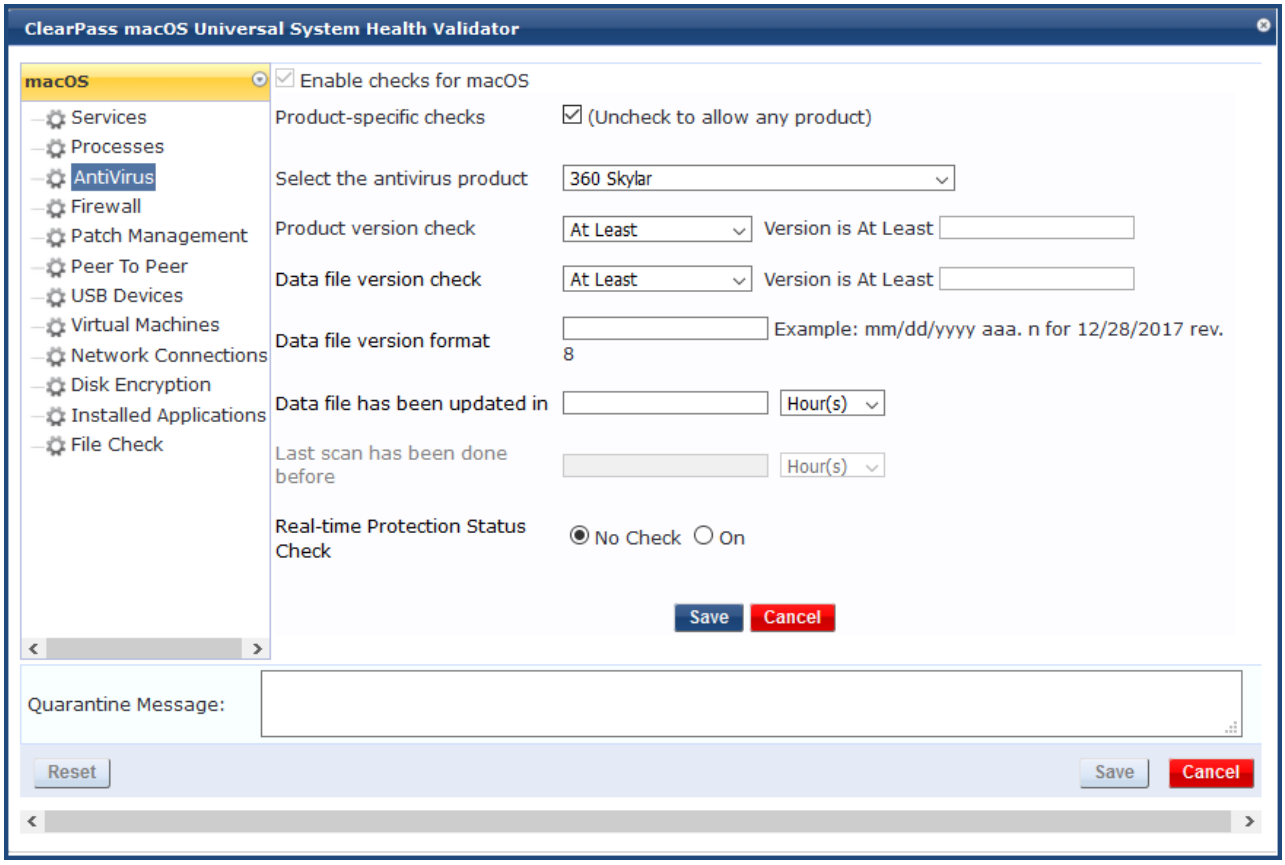
3. Click **Next**. The **Posture Plugins** configuration dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, select **Enable checks for macOS**.
6. Select **Antivirus**. The **antivirus** health class configuration page opens:
7. Configure the initial antivirus parameters as described in the following table:

Table 1. Initial Antivirus Health Class Parameters

Parameter	Action/Description
An antivirus - application is on	To enable testing of health data for confi gured antivirus applications, click **An anti virus - application is on**.

8. To specify product and version check information in the antivirus configuration page, click **Add**.

Figure 1. Configuring a macOS Antivirus Health Check



9. Specify the antivirus health class parameters as described in the following table:

Table 2. Add Antivirus Health Class Parameters

Parameter	Action/Description
Product - specific checks	To configure for product - specific chec ks, leave the check box checked (the de fault setting). To allow any antivirus product, uncheck t he check box. All of these checks may not be available for some products. Where checks are not available, they are shown in the disabled state.
Select the antivirus product	Select an antivirus product from the drop - down list.
Product version check	Select one of the following options:

Parameter	Action/Description
	• **No Check** • **Is Latest**: Requires registration with Policy Manager portal. • **At Least**: When you select **At Least**, th e **Version is at Least** field appears. Ent er the At Least version you require for the antivirus product version. • **In Last N Updates**: Requires registrati on with Policy Manager Portal.
Data file version check	Select one of the following options: • **No Check** • **Is Latest**: Requires registration with Policy Manager portal. • **At Least**: You can specify the least ve rsion that you require for the data file . Enter the least version in the *Versio n is At Least* field. When you select **At Least**, the **Data file version format** op tion appears. • **In Last N Updates**: Requires registrati on with Policy Manager Portal.
Data file version format	This field allows admins to specify the da ta file version format when the **At Least** c heck is selected for **Data File version che ck**. The configured format is used to compa re the configured data file version value with value received from the client. This f ield should be configured when the **Data file version** value contains a *date*. The foll owing format specifiers are supported: • **mm**: Month • **dd**: Day of the month

Parameter	Action/Description
	• **yy** or **yyyy**: Year • **a**: Alphabetic • **n**: Numeric **NOTE** For example: • Data file version: *10/14/2018 rev. 10* ◦ Specify this data file version format: *mm/dd/yyyy aaa. n* • Data file version: *2018.10.08* ◦ Specify this data file version format: *yyyy.mm.dd* • Data file version: *20181014194001* ◦ Specify this data file version format: yyyymmddn • Data file version: *2018 - 10 - 12_04* ◦ Specify this data file version format: *yyyy - mm - dd_n* **NOTE** You can leave this field blank for antivirus products that do not include a date in the **Data file version check** value. In these cases, Policy Manager compares the configured data file version with the value received from the client, character by character.
Data file has been updated in	Enter the number, then specify the interval in hours, days, weeks, or months.

Parameter	Action/Description
Last scan has been done before	Enter the number, then specify the inte rval in hours, days, weeks, or months.
Real - time Protection Status Check	• **No Check**: Policy Manager does not u se the real - time protection status val ue for health evaluation. The client is t reated as *healthy* regardless of the val ue of its real - time protection status. • **On**: Marked as *healthy* only if the real - time protection status is **On**.

0. Click **Save**. When you save your antivirus configuration, it appears in the **Antivirus** page list.

Firewall

About this task

In the **Firewall** page, you can specify that a firewall application must be on and specify information about the firewall application.

To view and configure processes:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**. The **Posture Plugins** dialog opens.

4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, enable the **Enable checks for macOS** checkbox.
6. Select **Firewall**. The **Firewall** health class configuration page opens:
7. To configure the firewall application information, click **A Firewall Application is On**.

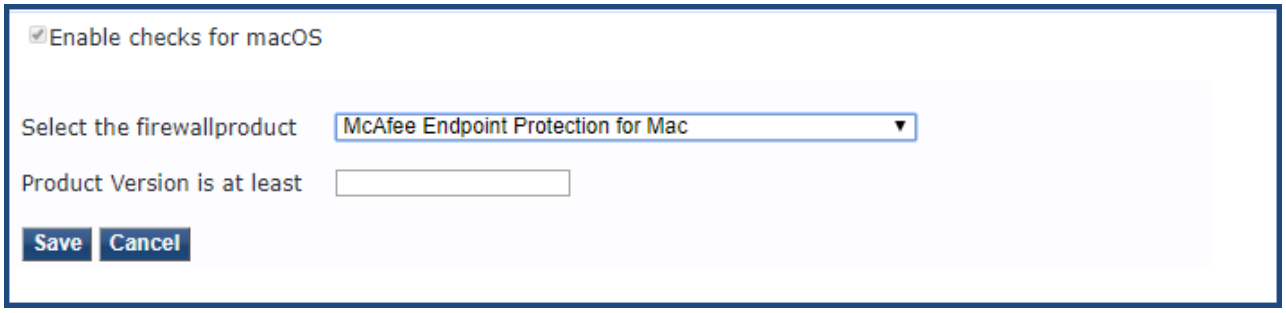
   The **Firewall** page opens:
8. Specify the initial Firewall parameters as described in the following table:

Table 1. Firewall Page Parameters

Parameter	Description
A Firewall Application is On	Enable or disable testing of health data for configured firewall applications.
Auto Remediation	Enable or disable auto remediation of fire wall status.
User Notification	Enable or disable user notification of poli cy violation of firewall status.
Product - specific checks	To specify Firewall product - specific che cks, leave this parameter as is. To specify whether any firewall applicatio n (any vendor) is running on the end host , uncheck the **Uncheck to allow any prod uct** check box.

9. To configure different configuration elements specific to the firewall product that you select, from the **Firewall** page, click **Add.** The **Add macOS Firewall** page opens.

Figure 1. macOS Firewall > Add Page



0. Specify the parameters as described in the following table:

Table 2. Firewall Page Parameters

Parameter	Description
Select the firewall product	From the drop - down list, select the fire wall product.
Product Version is at least	Optionally, specify the **Product Version i s at least**. Check for the version of the lat est product release.

1. , Click **Save**. When you save the firewall configuration, it appears in the **Firewall** page list.

Patch Management

About this task

The **Patch Management** page provides a way to specify that a patch management application must be on. You can also specify information about the patch management application, configure the product evaluation rule, and configure patch management application checks.

To configure the macOS patch-management application:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Select the **Posture Plugins** tab and specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.

b. **Posture Agent**: Select **OnGuard Agent.**

c. **Host Operating System**: Select **macOS.**

d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.

3. Click **Next**. The **Posture Plugins** dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator** check box, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, enable the **Enable checks for macOS** checkbox.
6. From the list of macOS health checks, select **Patch Management**.
7. Select the check box for **A patch management application is on**. The **Patch Management** health class configuration page opens:
8. Specify the initial macOS Patch Management parameters as described in the following table.

Table 1. macOS Initial Patch Management Parameter

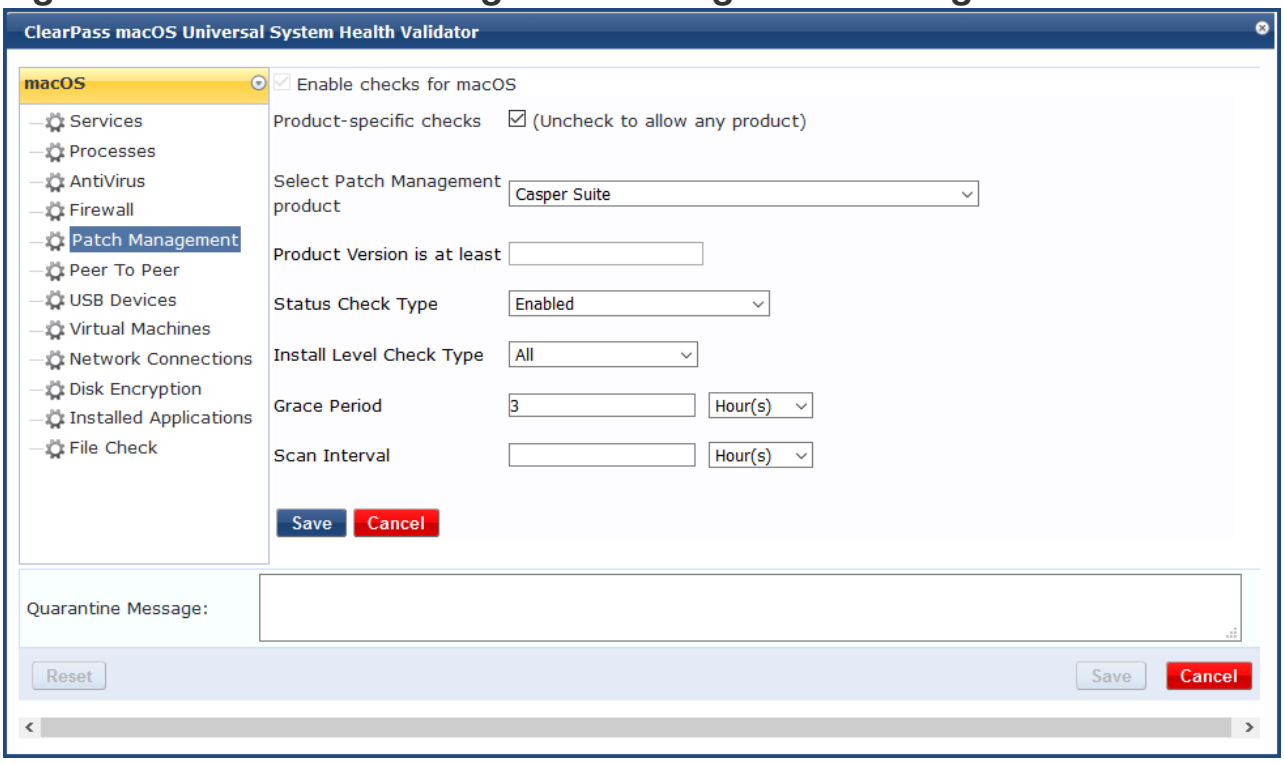
Parameter	Action/Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for patch management policy violations. • **User Notification**: Enable to allow use r notifications in the event of patch m anagement policy violations.
Product Evaluation Rule	Select the appropriate **Product Evaluatio n Rule**: • **Pass All**: Select this product evaluatio n rule if you want the **Patch Managem ent** health class to be deemed as *heal thy* only if all the configured patch ma nagement products are present. **Pass**

Parameter	Action/Description
	All is the equivalent of an **AND** conditi on. • **Pass Any One**: Select this product eva luation rule if you want the **Patch Man agement** health class to be deemed a s *healthy* if any one of the configured patch management products are pres ent. **Pass Any One** is the equivalent of an **OR** condition. **Pass Any One** is the default.

9. To configure the options for the specific patch management product, click **Add**.

   The **Patch Management > Add** page opens:

Figure 1. macOS Patch Management Configuration Dialog



0. Specify the macOS Patch Management parameters as described in the following table:

Table 2. MacOS Patch Management Plugin Parameters

Parameter	Action/Description
Product - specific checks	To configure product - specific checks, l eave this option as is. To check whether any patch managemen t application (any vendor) is running on t he end host, clear the **Uncheck to allow any product** check box.
Select Patch Management Product	Select a patch management product ven dor. This option is enabled *only* if the **Product - specific checks** check box is checked.
Product version is at least	Enter the minimally recommended produ ct version number. This option is enabled only if the **Product - specific checks** check box is checked.
Status Check Type	Specify the **Status Check Type** to check whether the Patch Agent is enabled. The ClearPass Policy Manager server co mpares the Patch Agent Status sent by O nGuard Agent with the configured value. If the Patch Agent Status value is differen t from configured value, the client is treat ed as unhealthy. If **Auto - remediation** is enabled, OnGua rd Agent changes the Patch Agent Status on the client to the configured value. Sel ect any of the following options: • **No Check**: ClearPass Policy Manager server ignores the Patch Agent Status value. This means it will not check the status of the Patch Agent application on the client.

Parameter	Action/Description
	• **Enabled**: Patch Agent is turned on and it automatically updates the client. • **Disabled**: Patch Agent is disabled and it *will not* check for missing patches and update the client.
Install Level Check Type	This option is only enabled if the **Product - specific checks** checkbox is selected. For Microsoft System Center Configuration Manager (SCCM), selecting **All** will return the full list of all missing patches. • **No Check**: The ClearPass Policy Manager server ignores the Patch Agent Status value. This means it will not check the status of the Patch Agent application on the client. • **All**: Checks for all missing patches and searches for all available patches. **NOTE** If you select the **Microsoft Windows Update Agent** from the **Select Patch Management product** list and you select an option from the **Install Level Check Type** list, the results are as follows: • **All**: Returns the full list of missing patches. • **No Check**: Disables the **Grace Period** and **Scan Interval** fields.
Grace Period	Configure the time period for which OnGuard Agent should ignore missing patches. You can specify the grace period in hours, days, weeks, or months.

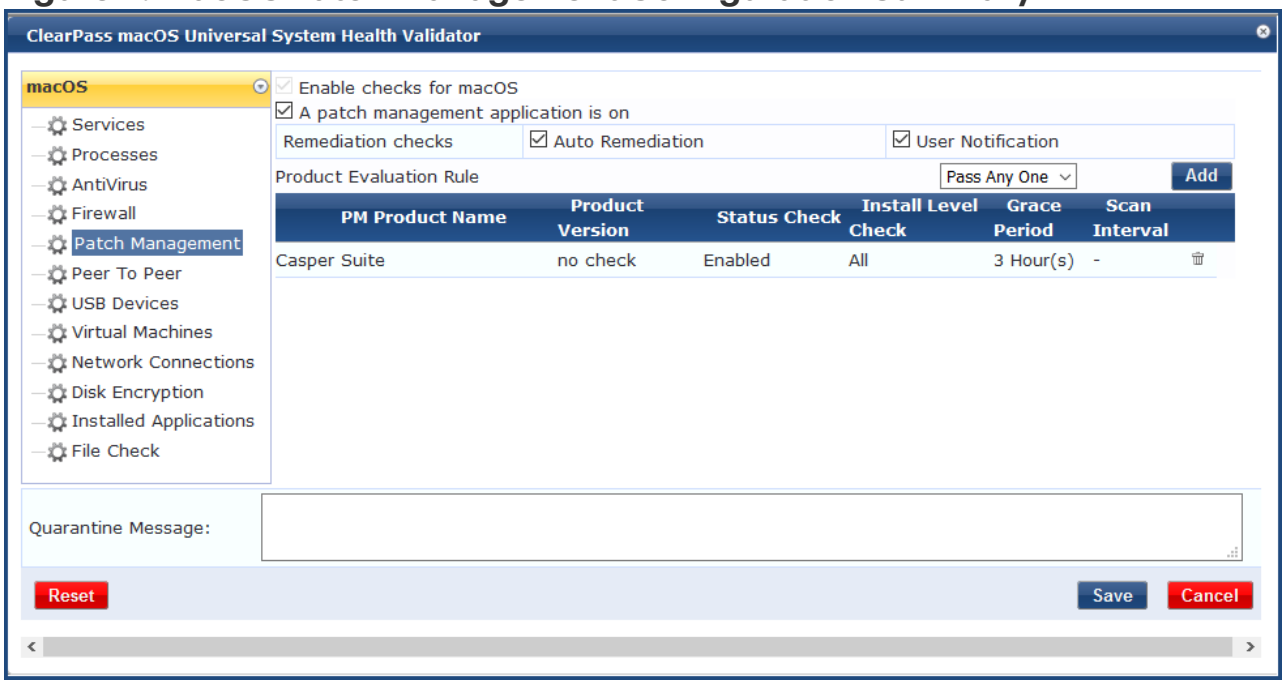
Parameter	Action/Description
	After installing OnGuard Agent on an out dated client, the first time OnGuard Agen t detects missing patches, the agent sets the Grace Period start time. The grace p eriod will last from the designated start ti me to the end of the Grace Period value s et for the **Grace Period** parameter. For ex ample: a. **Grace Period** parameter is set to 3 day s. b. **10/09 09:00 a.m.**: OnGuard Agent did not find any missing patches for Syste m Center Configuration Manager (SCC M). c. **10/09 10:00 a.m.**: OnGuard Agent det ects missing patches for SCCM. The g race period start time is set to 10:00 a. m. on 10/09. d. **10/12 10:00 a.m.**: The grace period ex pires after three days (In this example, 10:00 a.m. on 10/12). OnGuard Agent c hecks to see if patches are missing for SCCM. If patches are still missing and **Auto - remediation** is enabled, OnGua rd Agent starts downloading and instal ling missing patches. **NOTE** **No Check** from the **Install Level Check Type** field, **G race Period** is disabled.
Scan Interval	Specify the **Scan Interval** by specifying t he number of hours, days, weeks, or mon ths.

Parameter	Action/Description
	If you selected **No Check** from the **Install Level Check Type** field, **Scan Interval** is d isabled.

1. Click **Save**.

   When you save the patch management configuration, the configuration information is displayed on the **Patch Management** page.

Figure 2. macOS Patch Management Configuration Summary



2. Click **Save**.

   You return to the **Posture Plugins** tab. The status of the Policy Manager macOS Universal Health Validator plugin configuration is now displayed as "*Configured*."

3. Click **Next** to configure the Rules for this macS posture policy.

Peer to Peer

About this task

From the **Peer To Peer** page, you can view and add peer-to-peer applications.

To view and configure processes:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**.

   The **Posture Plugins** dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, enable the **Enable checks for macOS** checkbox.
6. Select **Peer to Peer**.
7. To provide configuration options to specify peer-to-peer applications or networks that need to be explicitly stopped, click **A Peer to Peer application is on**.

   The **Peer to Peer** health class configuration page opens:
8. Specify the **Peer to Peer** parameters as described in the following table:

Table 1. Peer - to Peer - Page Parameters

Parameter	Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for policy violations in peer - to - peer applicati ons.

Parameter	Description
	• **User Notification**: Enable to allow user notifications in the event of policy violations in peer - to - peer applications.
Grace Period	Configure the time period for which the macOS Universal System Health Validator plugin should ignore devices that fail peer - to - peer application health checks. You can specify the grace period in hours, days, weeks, or months.
By Application By Network	Select the appropriate radio button to select individual peer - to - peer applications or a group of applications that use specific peer - to - peer networks. When you select a peer - to - peer network, all applications that make use of that network are stopped.
Available Applications	This scrolling list contains a list of applications or networks that you can select and move to the **Applications to stop** pane. • Select the applications or networks in the **Available Applications** pane, then click the **> >** to add the applications or networks to the **Applications to stop** pane.
Applications to Stop	• Select the applications or networks in the **Applications to stop** pane, then click the **<<** to move the applications or networks to the **Available Applications** pane.

9. Click **Save**.

USB Devices

About this task

The **USB Devices** page provides settings to control USB mass storage devices attached to an endpoint. Use this page to configure the **Auto Remediation** and **User Notification** parameters. You can also configure the options to take remediation action for USB mass storage devices, or to remove USB mass storage devices from the **Remediation Action for USB Mass Storage Devices** drop-down menu.

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**. The **Posture Plugins** configuration dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, select **Enable checks for macOS**.
6. Select **USB Devices**.

   The following table describes the **USB Devices** parameters:

Table 1. macOS USB Devices Parameters

Parameter	Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for policy violations in USB devices

Parameter	Description
	• **User Notification**: Enable to allow user notifications in the event of policy violations in USB Devices.
Remediation Action for USB Mass Storage Devices	• No Action - Take no action; do not eject or disable the attached devices. • Remove USB Mass Storage Devices - Eject the attached devices.

Virtual Machine

About this task

The **Virtual Machines** page provides configuration options to virtual machines utilized by the network.

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**. The **Posture Plugins** configuration dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, select **Enable checks for macOS**.
6. Select **Virtual Machines**.

7. Select the **Virtual Machine Detection is on** option. The macOS **Virtual Machine** page opens:
8. Configure the virtual machine health check parameters as described in the following table:

Table 1. Virtual Machines

Parameter	Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled. • **Auto Remediation**: Select this option to allow auto - remediation for virtual machine policy violations. • **User Notification**: Enable to allow use r notifications in the event of virtual m achine policy violations.
Allow access to clients running on Virtual Machine	Allow clients that running a VM to be acc essed and validated.
Allow access to clients hosting Virtual Ma chine	Allow clients that hosting a VM to be acc essed and validated.
Remediation Action for clients hosting Vi rtual Machines	• No Action: Take no action; do not stop or pause virtual machines. • Stop all virtual machines running on Host: Stop the VM clients that are run ning on the host. • Pause all virtual machines running on Host: Pause the VM clients that are ru nning on the host.

9. Click **Save**.

Network Connections

About this task

The **Network Connections** page provides configuration options to control network connections based on connection type. To configure Network Connections:

Procedure

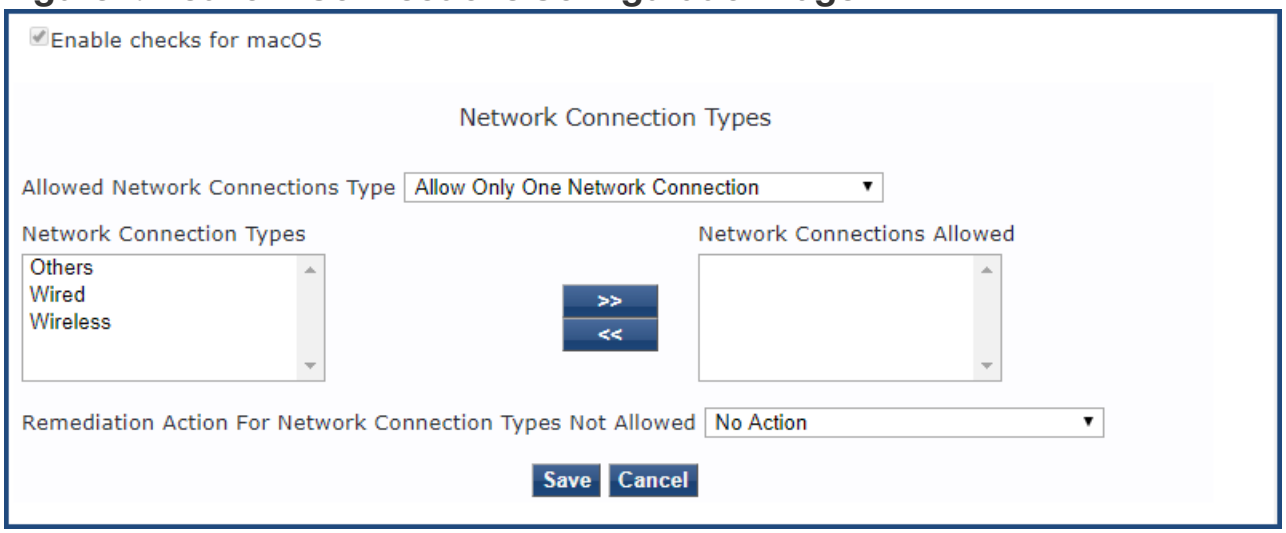
1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**. The **Posture Plugins** configuration dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, select **Enable checks for macOS**.
6. Select **Network Connections**.
7. Enable the **Network Connection Check is on** check box to provide the options to specify the remediation checks or user notification.
8. Select the **Check for Network Connection Types** check box.
9. Configure the network connection health check parameters as described in the following table:

Parameter	Description
Remediation checks	Disable or enable one or both of the foll owing remediation checks. By default, bo th are enabled.

Parameter	Description
	• **Auto Remediation**: Select this option to allow auto - remediation for networ k connection policy violations. • **User Notification**: Enable to allow use r notifications in the event of network connection policy violations.
Grace Period	Configure the time period for which the macOS Universal System Health Validator plugin should ignore network connection s that fail health checks. You can specify the grace period in hours, days, weeks, o r months.

0. To specify type of network connection, select **Check for Network Connection Types** then click **Configure**. The **Network Connection Types** configuration page opens.

Figure 1. Network Connections Configuration Page



1. Specify the **Network Connection Types** configuration parameters as described in the following table:

Parameter	Action/Description
Allowed Network Connections Type	Select one of the following options: • *Allow Only One Network Connection*

Parameter	Action/Description
	• *Allow One Network Connection with VPN* • *Allow Multiple Network Connections*
Network Connection Types	To add or remove **Others**, **Wired**, and **Wireless** network connection types, click **>** **>** or **<<**.
Remediation Action for Network Connection Types Not Allowed	Specify one of the following • *No Action*: Take no action. Do not eject or disable the attached devices. • *Disable Network Connections* • *Disconnect Network Connections*

2. Click **Save**. This returns you to the **Network Connections** configuration page.

Disk Encryption

About this task

Disk encryption is a technology that protects information by converting it into unreadable code that cannot be deciphered easily by unauthorized people. Disk encryption uses disk encryption software or hardware to encrypt every bit of data that goes on a disk or disk volume. Disk encryption prevents unauthorized access to data storage.

To configure disk encryption for macOS:

Procedure

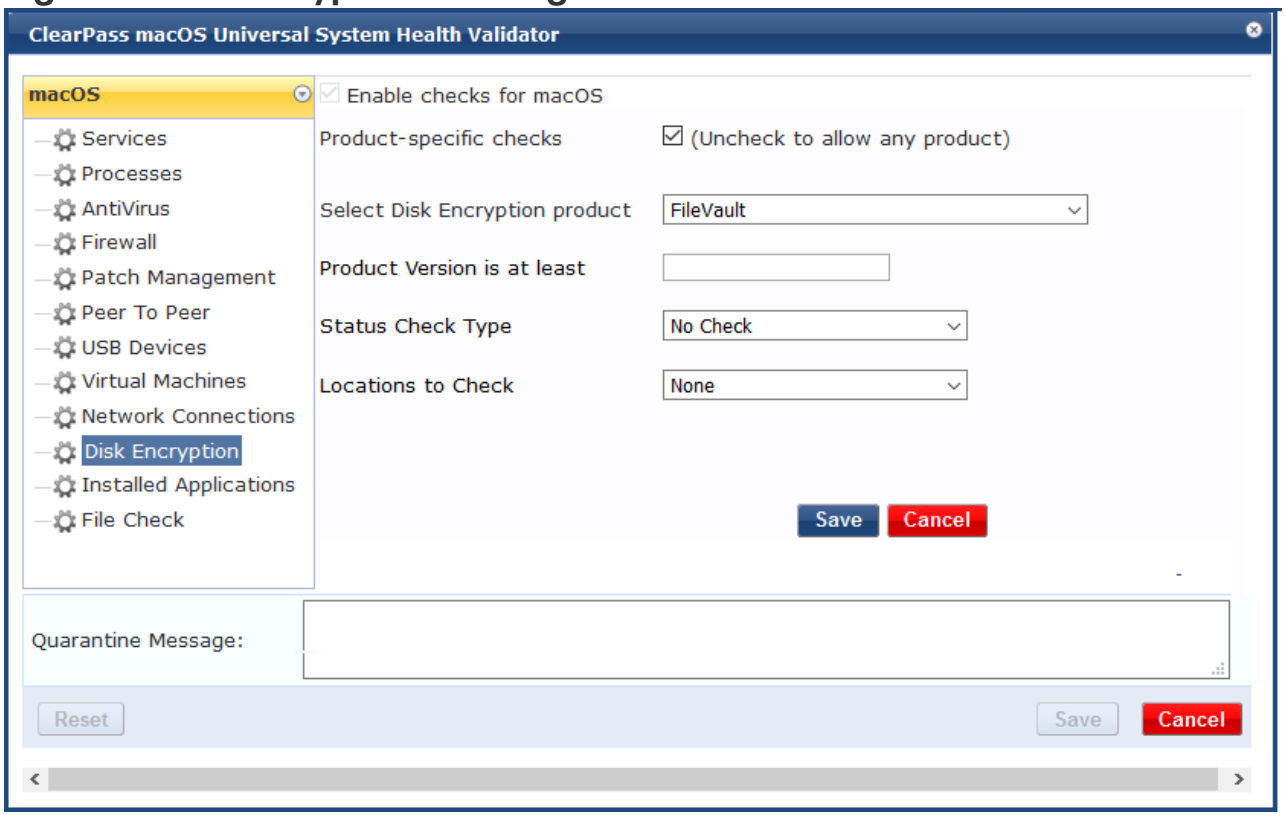
1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**

c. **Host Operating System**: Select **macOS.**

d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.

3. Click **Next**. The **Posture Plugins** configuration dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, select **Enable checks for macOS**.
6. Select **Disk Encryption**.
7. Enable the **A disk encryption application is on** option.The macOS **Disk Encryption** page opens.
8. (Optional) In the **Grace Period** field, configure the time period for which the macOS Universal System Health Validator plugin should ignore disk encryption applications that fail health checks. You can specify the grace period in hours, days, weeks, or months.
9. To configure the product-specific encryption checks, click **Add**. The **Disk Encryption Add** page opens.

Figure 1. Disk Encryption Add Page



0. Specify the **macOS Disk Encryption** health check parameters as described in the following table:

Table 1. macOS Disk Encryption Health Check Parameters

Parameter	Action/Description
Product - specific checks	If you wish to select a specific disk encry ption product, do not change the default setting. To allow any encryption product to check disk encryption, select the **Uncheck to al low any product** check box. The *Select Disk Encryption product* and *P roduct Version is at least* fields are disab led after you uncheck the**Uncheck to all ow any product** check box.
Select Disk Encryption product	Select a specific disk encryption product .
Product Version is at least	Search for the production version of the selected product.
Status Check Type	**NOTE** This option is enabled only when you specify **FileVault** a s the Disk Encryption produc t. Select one of the following options: • *No Check*: Policy Manager does not p erform any checks for the status of Fi leVault on the client. The status of Fil eVault will not be used in determining client health. • *Enabled*: The FileVault application sho uld be enabled on the client. Policy M anager marks the client as healthy if t he client reports the status of the File

Parameter	Action/Description
	Vault application as *Enabled*; otherwis e the client will be marked as unhealt hy. • *Disabled*: The FileVault application sh ould be disabled on the client. Policy Manager marks the clients as healthy i f the client reports the FileVault applic ation status as *Disabled*; otherwise th e client will be marked as unhealthy. **NOTE** Auto - remediation is not su pported for the **Status Type Check** option. If the client is marked unhealthy because t he reported status does not match the status configured in this option, OnGuard Age nt will not change the status of the FileVault application.
Locations to Check	Select the locations to check disk encryp tion. The options are: • None • System Root Drive • All Drives • Specific Locations
Locations to Exclude	Specify locations that should be exclude d from disk encryption checks. This field accepts a list of locations separated by a new line. During posture evaluation, Policy Manage r does not check the encryption state of the drives specified in **Locations to Exclu de**. This field is optional and is shown onl

Parameter	Action/Description
	y when **Locations to Check** is set to *All D rives* or Specific Locations. **NOTE** On macOS, the Policy Mana ger OnGuard Agent automat ically excludes the *Recovery* and *Virtual Memory (VM)* pa rtitions from disk encryption checks.

1. Click **Save**.

Installed Applications

About this task

The **Installed Applications** category groups classes that represent software-related objects. From the **Installed Applications** page, you can specify information about which installed applications you want to monitor.

To configure the installed applications for macOS:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**.

   The **Add Posture Policies** dialog opens.

   Specify the following:

   a. **Policy Name/Description**: Enter the name and a description of the posture policy.

   b. **Posture Agent**: Select **OnGuard Agent.**

   c. **Host Operating System**: Select **macOS.**

   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.

2. Click **Next**.

   The **Posture Plugins** dialog opens.

3. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
4. To provide the options to specify the remediation checks, auto remediation, or user notification, enable the **Enable checks for macOS** checkbox.
5. Select the **Installed Applications** and **Installed Application Check is on** options.
6. Specify the **Installed Applications for macOS Configuration** page parameters as described in the following table:

Table 1. Installed Applications for macOS Configuration Page Parameters

Parameter	Action/Description
Remediation checks	**Auto - Remediation** for the Installed Appl ications health class is not supported.
User Notification	Enable sending a remediation message w ith a list of applications to install or unins tall to the user.
Monitor Mode	Enable **Monitor Mode** to treat all the ins talled applications as always healthy.
Applications Allowed (Mandatory)	Specify installed applications to be monit ored on a mandatory basis. **NOTE**
Applications Allowed (Optional)	Specify installed applications to be monit ored on an optional basis. **NOTE**
Allow only Mandatory and Optional Appli cations	Specify that only the mandatory and opti onal applications are monitored. **NOTE**

7. Click **Save**.

Enabling Regular Expressions for a macOS Application

To enable regular expressions for an application:

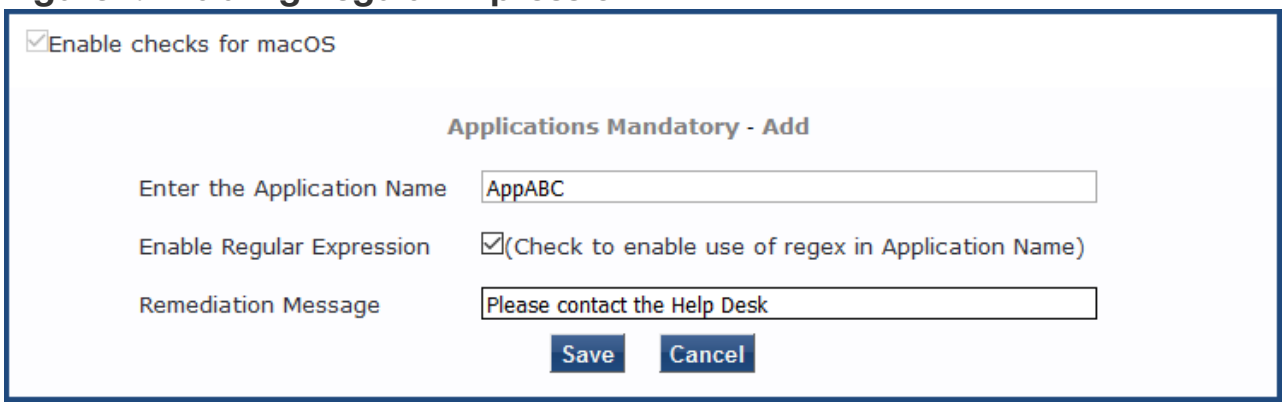
8. From the **ClearPass macOS Universal System Health Validator > Configure** page (see Policy Manager macOS Universal System Health Validator: OnGuard Agent), select **Installed Applications**.
9. Click (enable) the **Installed Applications Check is on** check box.

   The Installed Applications dialog displays.

0. From the **Applications Allowed (Mandatory)** category, click **Add**.

   The **Add Mandatory Applications** dialog displays.

Figure 1. Enabling Regular Expression



1. Configure the **Add Mandatory Applications** parameters as described in Table 2.

Table 2. Add Mandatory Applications Parameters

Parameter	Action/Description
Enter the Application Name	Enter the name of the application.
Enable Regular Expression	Check (enable) this check box to enable t he use of regular expressions in the **Appli cation Name**. When this field is enabled, Policy Manag er treats the **Application Name** as regula r expression when comparing application names.

Parameter	Action/Description
Remediation Message	Optionally, enter the remediation messag e that will be displayed to the user in the event of an error. **NOTE** Remediation messages can i nclude reasons for remediati on, links to helpful URLs and helpdesk contact informatio n.

2. Click **Save**.You return to the **Installed Applications** dialog, where **Regular Expression Enabled** is set to **true** for the specified application.

File Check

About this task

Use the **File Check** page to turn on the file check feature and specify information about the files you want to check. You can also use the **File Check** page to verify the group of files to be present or absent.

To configure macOS File Check:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**. The **Posture Plugins** dialog opens.

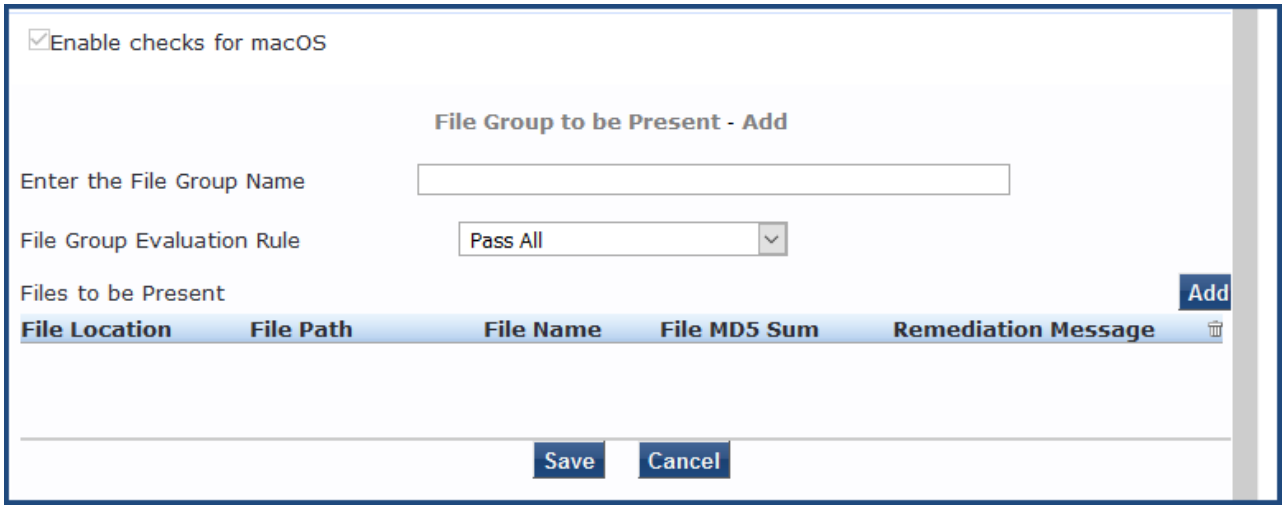
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, select the **Enable checks for macOS** checkbox.
6. Select **File Check**. The **File Check** health class configuration page opens.
7. Select the **File Check is On** option and configure parameters as described in the following table:

Table 1. File Check Configuration Parameters

Parameter	Action/Description
Remediation checks	Auto - remediation for the **File Check** hea lth class is not supported.
User Notification	When enabled, a remediation message th at includes the groups of files to be prese nt or absent is displayed to the end user.
Monitor Mode	To treat all the file check health classes a s always healthy, enable **Monitor Mode**.
File Groups to be Present	To add files that will be present in the **File Check** health class, click **Add**.
File Groups to be Absent	To add files that will be absent in the **File Check** health class, click **Add**.

8. To open the **File Group to be Present > Add** page, click **Add.** The **File Group to Be Present > Add** dialog opens:

Figure 1. macOS File Group to Be Present > Add Dialog



Specify the **File Group to Be Present > Add** parameters as described in the following table:

Table 2. File Group to Be Present > Add Parameters

Parameter	Action/Description
Enter the File Group Name	Enter the name of the file group.
File Group Evaluation Rule	Select the appropriate File Group Evaluation Rule: • **Pass All**: Select this option to mark the **File Check** health class as 'healthy' only if all the configured file groups are present. • **Pass Any One**: Select this option to mark the **File Check** health class as 'healthy' even if any one of the configured file groups are present.

9. To configure the name of the file group and the evaluation rule for the file group, from **File Groups to be Present**, click **Add**.

   The **File to Be Present > Add** page opens:

0. Specify the **File to Be Present > Add** parameters as described in the following table:

Table 3. File to Be Present > Add Parameters

Parameter	Action/Description
File Location	Select any location of the file from the d rop - down list: • Applications • UserBin • UserLocalBin • UserSBin • None
Enter the File Path	Enter the file path as described in the exa mples provided.
Enter the File Name	Enter the name of the file.
Enter the MD5 Sum	Optionally, you can specify one or more (comma - separated) MD5 checksums of the process executable file.
Remediation Message	Specify the custom remediation messag e to be displayed to end users if **File che ck** fails.

1. When finished, click **Save**.

   The parameters configured in the **File to Be Present > Add** dialog are reflected in the **File Groups to be Present** dialog.

Custom Scripts

About this task

Use the **Custom Scripts** dialog to execute a custom health check or auto remediation script on endpoints running the selected version of macOS. In the **Custom Scripts** page, you can enable custom scripts and select the scripts you want to run. Before you can add scripts to the Universal System Health Validator plugin, you must have already

defined your health collection or auto remediation scripts. For details, refer to OnGuard Custom Scripts.

To configure the **Custom Scripts** Health Class:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click **Add**. The **Add Posture Policies** dialog opens.
2. Specify the following:
   a. **Policy Name/Description**: Enter the name and a description of the posture policy.
   b. **Posture Agent**: Select **OnGuard Agent.**
   c. **Host Operating System**: Select **macOS.**
   d. **Plugin Version**: OnGuard Plugin version 2.0 (V4 SDK) is set by default.
3. Click **Next**. The **Posture Plugins** dialog opens.
4. Select the **Policy Manager macOS Universal System Health Validator**, then click **Configure**.
5. To provide the options to specify the remediation checks, auto remediation, or user notification, select the **Enable checks for macOS** checkbox.
6. Select the **Custom Scripts** option in the configuration settings menu. The **Custom Script** Health Class configuration page opens:
7. If it is not already enabled, select **Custom Scripts is On**.
8. Specify the custom script parameters as described in the following table:

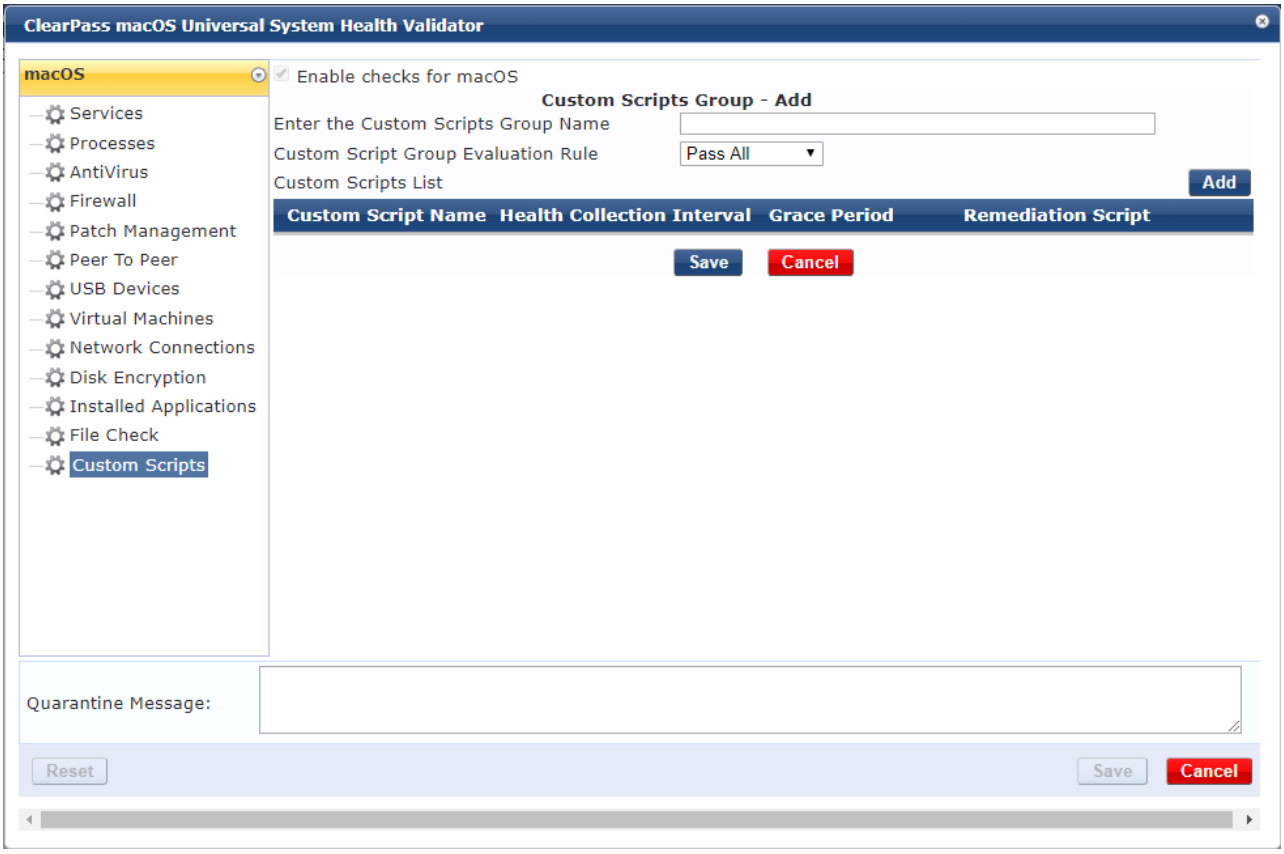
Table 1. Custom Scripts Configuration Parameters

Parameter	Action/Description
Auto Remediation	Allow auto - remediation for custom scri pts. Enabling this option allows you to sel ect an auto remediation script for an end point that does not pass the health check
User Notification	When enabled, a remediation message th at includes the results of the health chec k script is displayed to the end user.

Parameter	Action/Description
Monitor Mode	If you want the custom script to tread all endpoints as always healthy, enable **Moni tor Mode**.
Custom Script Group Evaluation Rule Rul e	The macOS Universal System Health Vali dator plugin considers an endpoint healt hy if it passes the health collection script checks from all groups of scripts, or any one of them. The default **Evaluation Rule** setting is **Pass Any One.** Click the dropd own list and select **Pass All** to modify this setting.
Quarantine Message	(Optional) Enter a quarantine message th at will appear on a client that fails any h ealth check from the macOS Universal Sy stem Health Validator Plugin,

9. Click **Add** to select the custom scripts you want to use for health checks for the selected version of macOS. The **Custom Scripts Group - Add** dialog opens.

Figure 1. Custom Scripts Group - Add dialog



0. In the **Enter the Custom Scripts Group Name** field, enter a name for the collection of one or more custom scripts.

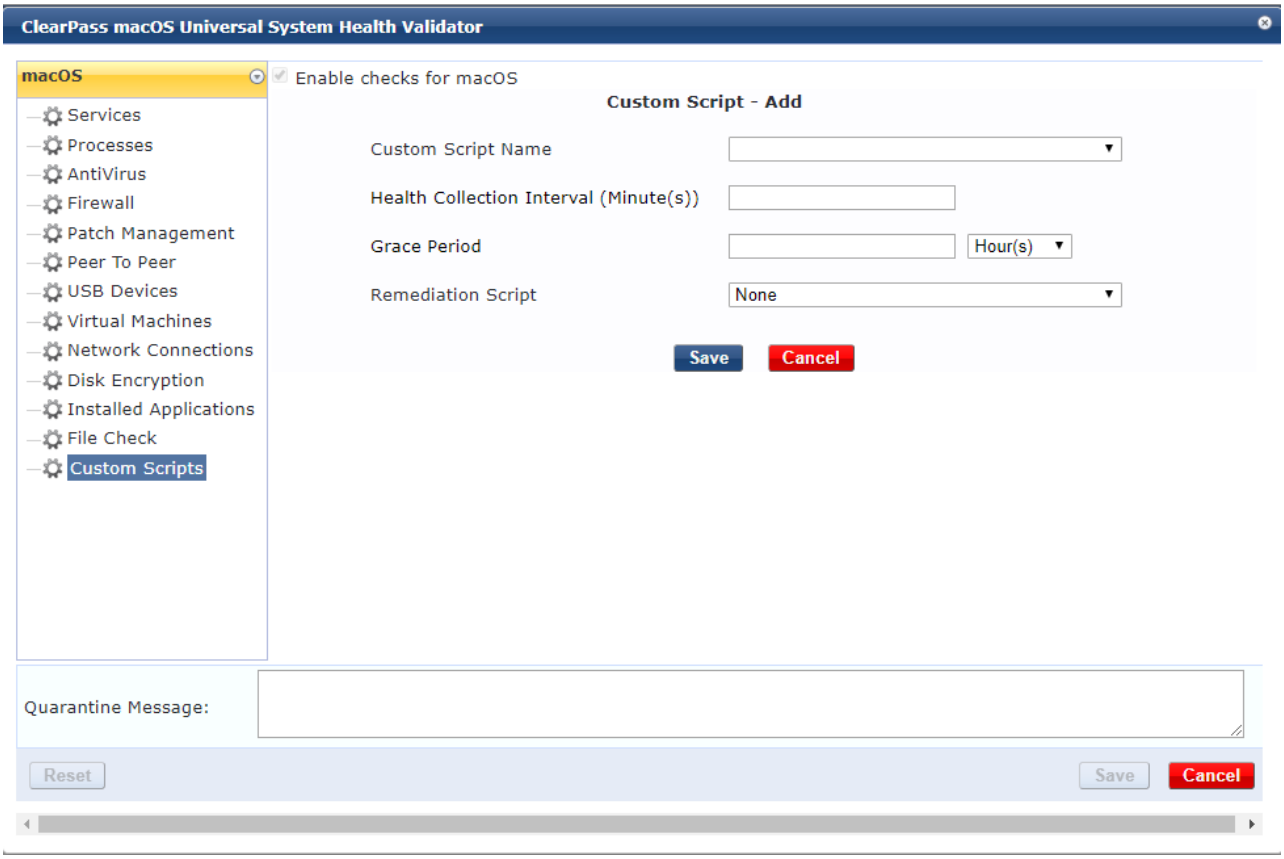
1. Click the **Custom Script Group Evaluation Rule** menu and select one of the following options:

   - **Pass all**: Consider an endpoint healthy if it passes health checks from all health collection scripts in the group.
   - **Pass Any One:** Consider the endpoint healthy if it passes the health checks by one of the scripts in the group.

   This setting is different than the **Evaluation Rule** setting described in Table 1. This setting applies to individual scripts within a group. The **Evaluation Rule** setting in Table 1 determines whether one or all groups of scripts need to pass to consider the device healthy.

2. Click **Add** to open the **Custom Script - Add** page.

Figure 2. Add a Custom Script



3. Click **Custom Script Name** and select a custom health collection script defined using OnGuard Custom Scripts.
4. In the **Health Collection Interval (Minute** field, choose how often the macOS Universal System Health Validator plugin should issue this script to collect health data. The recommended setting is every **5** minutes.
5. (Optional) Use the **Grace Period** field to configure the time period for which the macOS Universal System Health Validator plugin ignores devices that fail health checks by custom health collection scripts. You can specify the grace period in hours, days, weeks, or months.
6. (Optional) Click the **Remediation Script** dropdown menu and select an auto remediation custom script that should run if the device does not pass the health check by the health collection script. For more information on creating a custom auto remediation script, see OnGuard Custom Scripts.
7. Click **Save** to return to the **Custom Scripts Group - Add** page. You can return to step 11 and click **Add** again to add another custom script to the scripts group, if desired.
8. When you have added all the scripts to the group, click **Save** again. The **Custom Scripts** dialog will show a list of all custom script groups, and the scripts contained in each group.

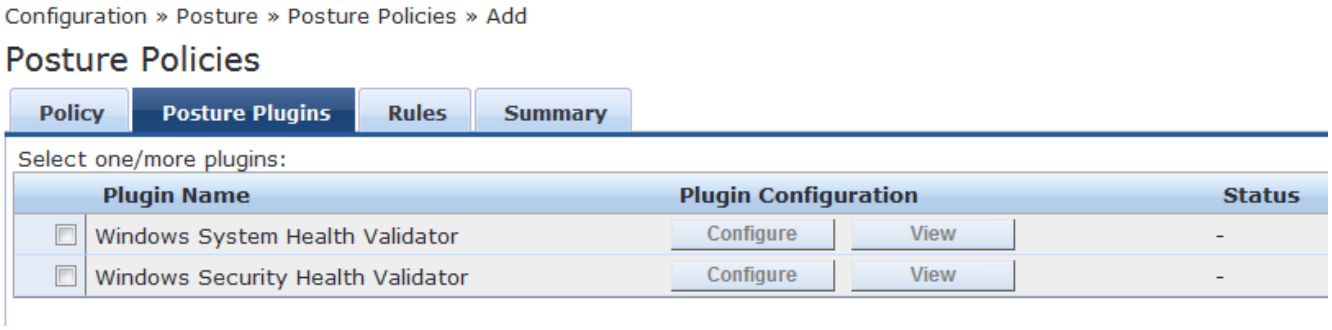
Configuring NAP Agent Plugins

If your posture policy is using a NAP agent, the **Posture Plugins** tab allows you to configure the following plug-in types:

- Windows System Health Validator - NAP Agent
- Windows Security Health Validator - NAP Agent

The following figure displays the **NAP Agent - Posture Plugins** tab:

Figure 1. NAP Agent - Posture Plugins Options



Subtopics

Windows System Health Validator: NAP Agent
Windows Security Health Validator: NAP Agent

Windows System Health Validator: NAP Agent

About this task

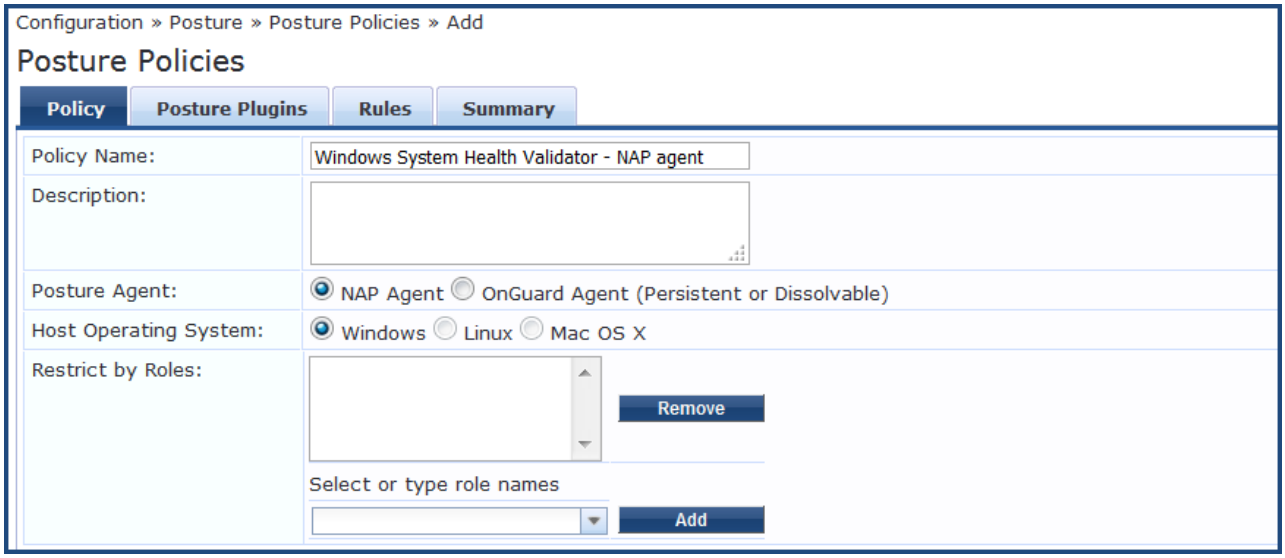
The Windows System Health Validator NAP (Network Access Protection) Agent checks for the level of Windows Service Packs.

To configure the minimum service pack level required, perform the following steps:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**. The **Posture Policies** page appears.
2. Click **Add**. The **Add Posture Policies > Policy** dialog opens.

Figure 1. Adding a Windows NAP Agent Posture Policy



3. Specify the following:

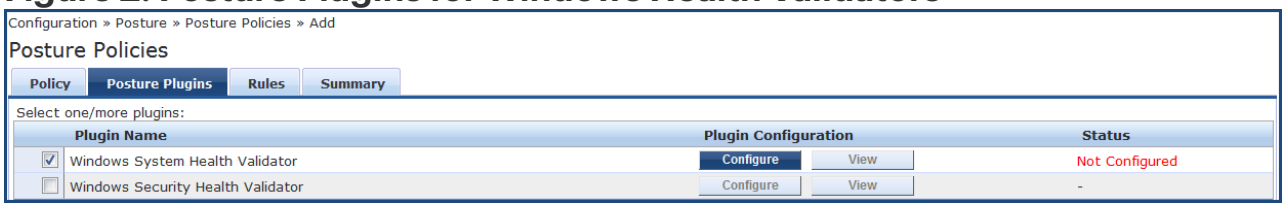
   a. **Policy Name/Description**: Enter the name and description of the posture policy.

   b. **Posture Agent**: Select **NAP Agent.**

   c. **Host Operating System**: Select **Windows.**

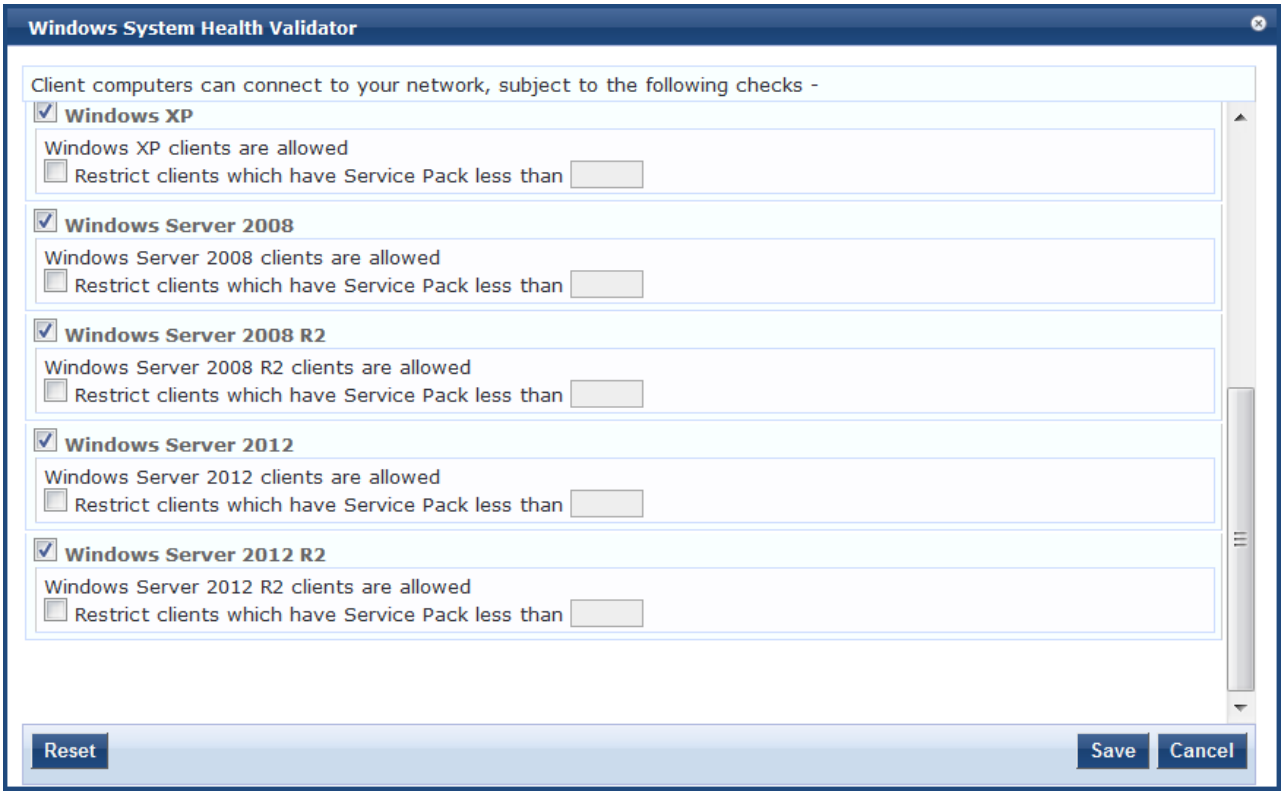
4. Click **Next**. The **Posture Policies > Posture Plugins** page opens.

Figure 2. Posture Plugins for Windows Health Validators



5. From the **Posture Plugins** tab, select **Windows System Health Validator**, then click the **Configure** button. The **Windows System Health Validator** page appears:

Figure 3. OnGuard NAP Agent: Windows System Health Validator



6. To enable support of specific Windows operating systems, click the corresponding check boxes.
7. Enable the **Restrict clients...** check box and specify the minimum Service Pack level required on the client computer to connect to your network.
8. Click **Save**. You return to the **Posture Plugins** page where the status of the plug-in is now set to **Configured**.

Windows Security Health Validator: NAP Agent

About this task

The Windows Security Health Validator: NAP (Network Access Protection) Agent checks for the presence of specific types of security applications. You can use the check boxes to restrict access based on the absence of the selected security application types.

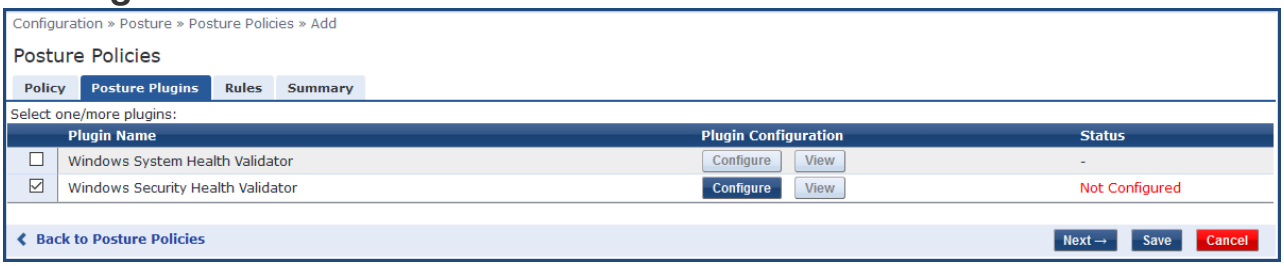
To configure the minimum service-pack level required, perform the following steps:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**. The **Posture Policies** page opens.

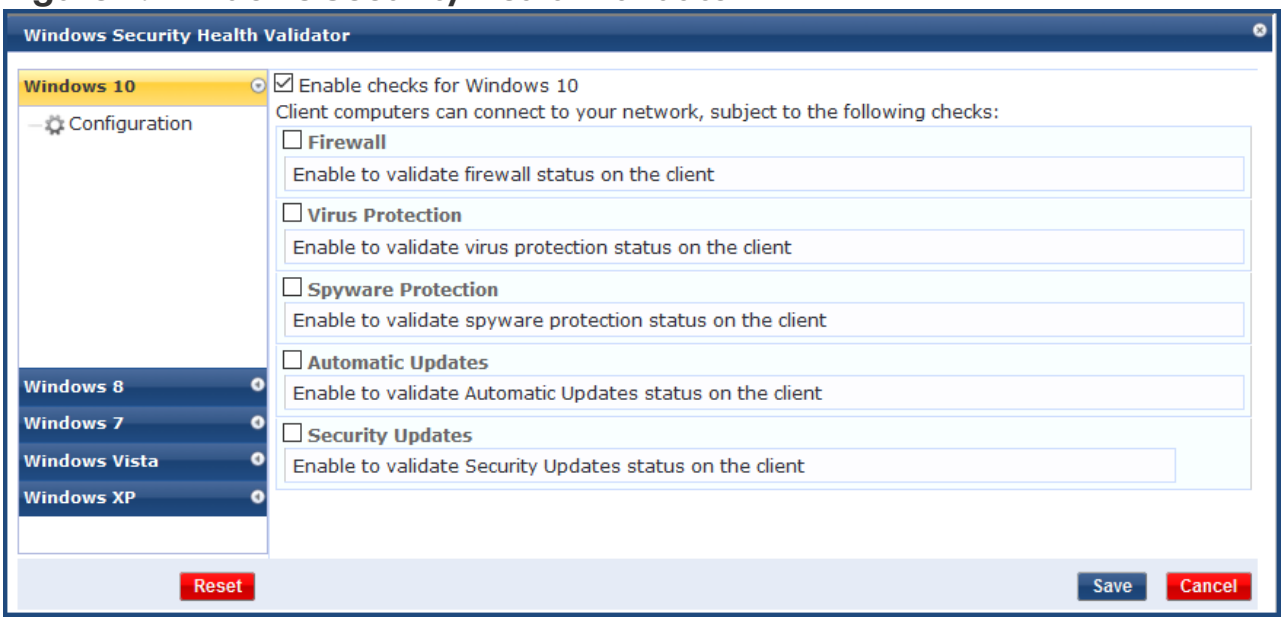
2. Click the **Add** link. The **Add Posture Policies** dialog opens.
3. Specify the following:
   a. **Policy Name**: Enter the name of the posture policy.
   b. **Posture Agent**: Select **NAP Agent**.
   c. **Host Operating System**: **Windows** is selected automatically. *Linux* and *macOS* are grayed out.
4. Click **Next**. The **Posture Policies > Posture Plugins** page opens.

Figure 1. Selecting Posture Plugins for Windows Security Health Validator: NAP Agent



5. From the **Posture Plugins** page, select **Windows Security Health Validator**, then click **Configure**. The **Windows System Health Validator** page opens. **Windows 10** is selected by default.
6. Click the **Enable checks for Windows 10** check box. The **Windows Security Health Validator** configuration page displays.

Figure 2. Windows Security Health Validator



7. To enable support of specific health checks (for example, *Firewall* and *Virus Protection*), click the corresponding check boxes.
8. Click **Save**. You return to the **Posture Plugins** page where the status of the Windows Security Health Validator plug-in is now *Configured*.

Configuring Posture Policy Rules

About this task

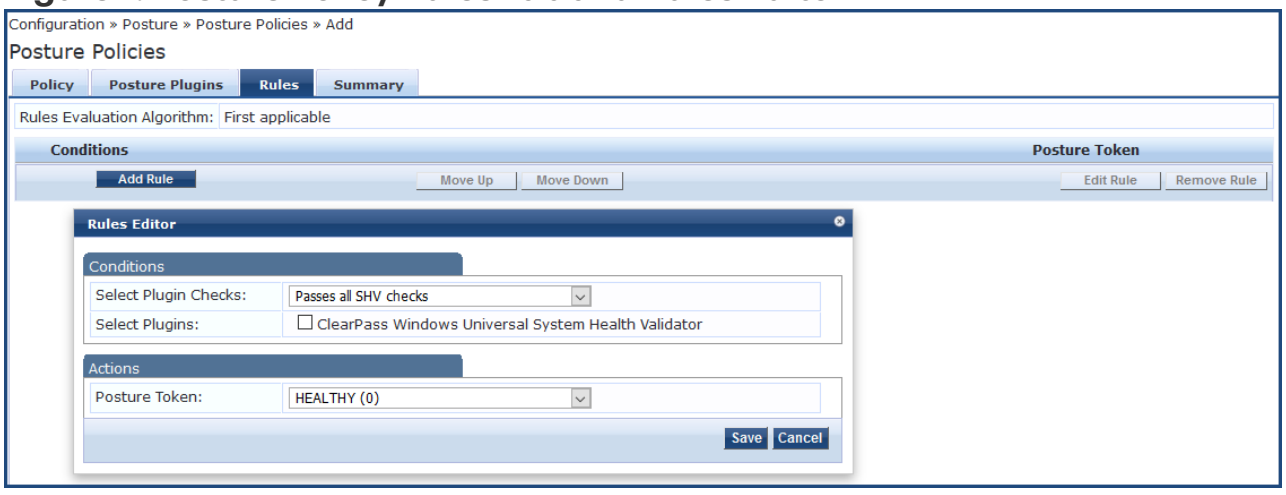
Once you have defined the posture hosts, agents, and plug-ins, you must configure the rules for the posture policy.

To configure posture policy rules:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**, then click the **Add** link. The **Add Posture Policies** dialog opens.
2. Click the **Rules** tab. The **Posture Policy Rules Editor** opens.
3. Click **Add Rule**. The **Rules Editor** opens.

Figure 1. Posture Policy Rules Tab and Rules Editor



4. Specify the **Rules Editor** configuration parameters as described in the following table:

Table 1. Posture Policy Rules Editor Parameters

Parameter	Action/Description
Select Plugin Checks	Click select one of the following plugin c heck types for System Health Validators: • Passes all System Health Validators ch ecks • Passes one or more System Health Val idators checks • Fails all System Health Validators chec ks • Fails one or more System Health Valid ators checks
Select Plugins	Select the plug - in to which the plug - in checks should apply.
Posture Token	Select one of the following posture token s: • HEALTHY (0). This is the default. • CHECKUP (10) • TRANSITION (15) • QUARANTINE (20) • INFECTED (30) • UNKNOWN (100)

Configuring Audit Servers

The Policy Manager server contains default Nessus (v2.X through v6.x) and Nmap (Network Mapping) servers. For enterprises with existing audit server infrastructure, or with external audit servers, Policy Manager supports these servers externally.

For more information, see:

- Audit Service Flow Control
- Default Audit Servers
- Custom Audit Servers

Subtopics

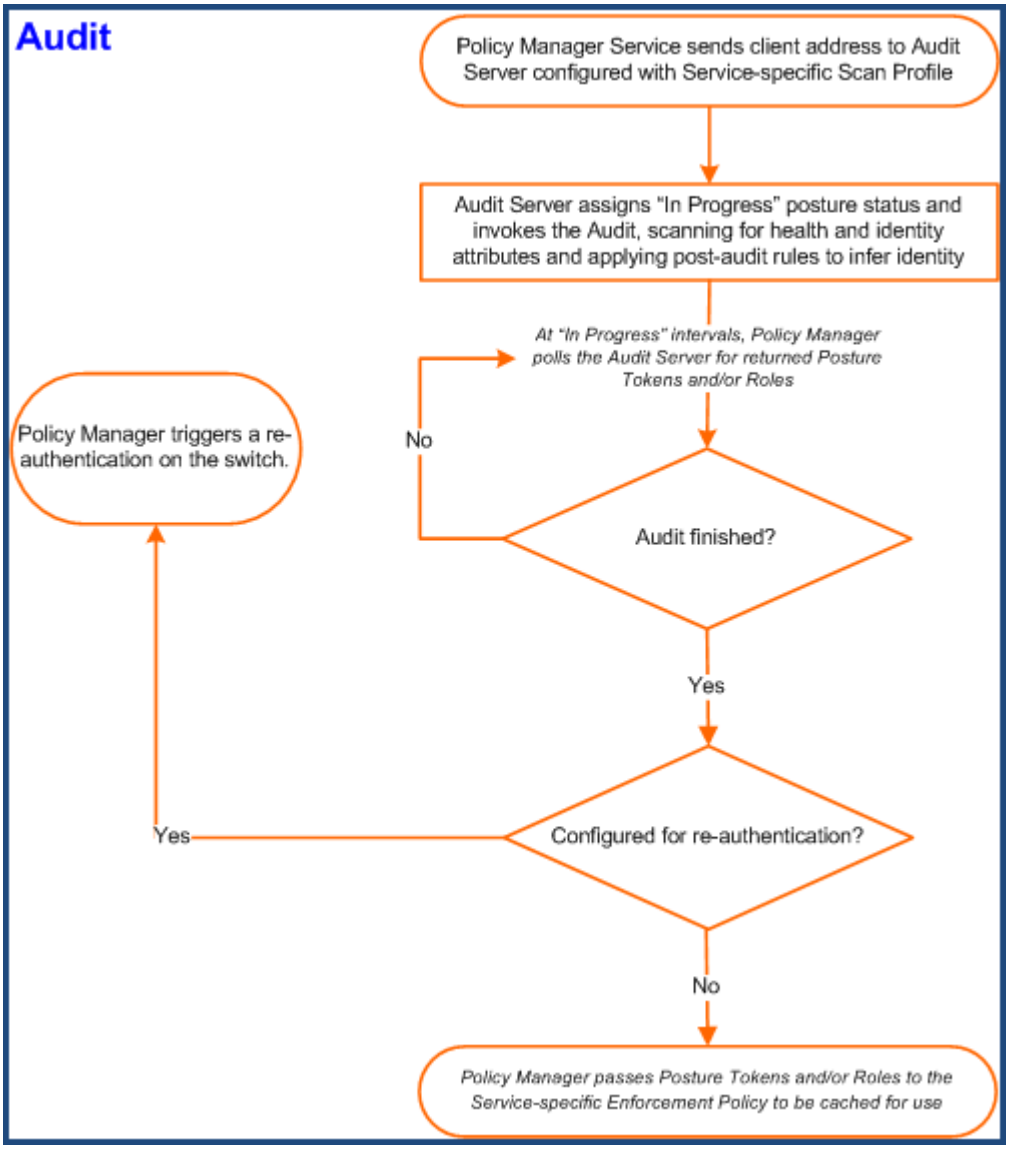
Audit Service Flow Control
Default Audit Servers
Custom Audit Servers

Audit Service Flow Control

Audit servers evaluate posture, role, or both for unmanaged or unmanageable clients. One example is clients that lack an adequate posture agent or an 802.1X supplicant. For example, printers, PDAs, or guest users might not be able to send posture credentials or identify themselves.

A Policy Manager service can trigger an audit by sending a client ID to a preconfigured audit server. The server then returns attributes for role mapping and posture evaluation. Audit servers are configured at a global level, and only one audit server can be associated with a service. The flow-of-control of the audit process is shown in the following figure.

Figure 1. Flow of Policy Manager Auditing Control



Default Audit Servers

When you configure an audit as part of a Policy Manager service, you can select the default Nessus server or Nmap audit configuration.

NOTE

Starting with Policy Manager 6.10.2, the Policy Manager server no longer needs to have Device Insight integration disabled in order to audit end hosts. In Policy Manager 6.10.0 and 6.10.1, when **Device Insight Integration** is enabled, the **Audit End-Hosts** option is not available. (for more information, see Device Insight Integration Page).

Adding Auditing to a Service

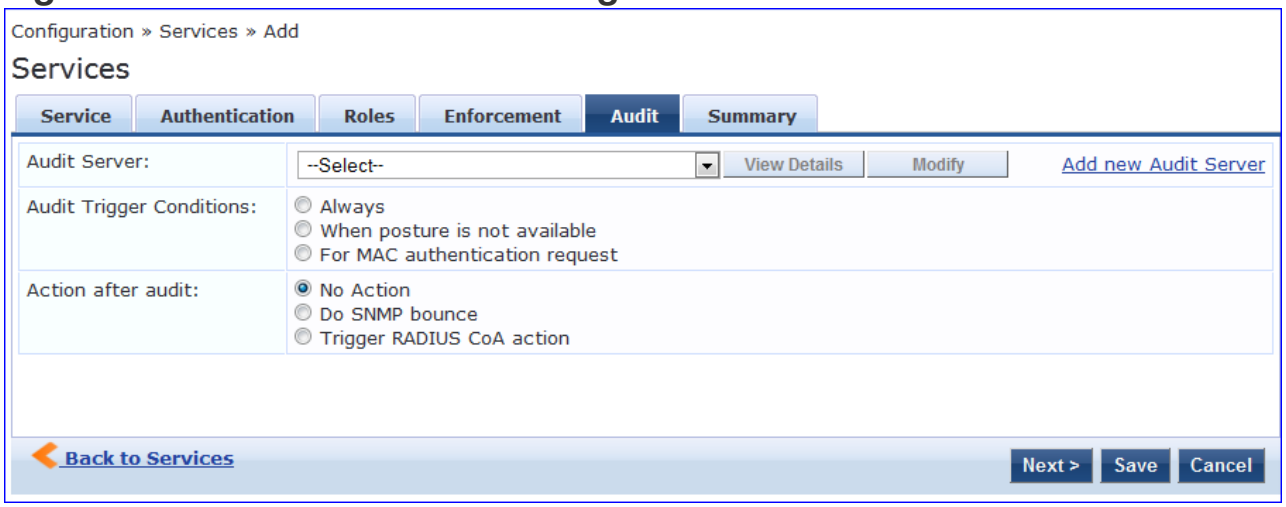
About this task

To configure an audit server for a new service:

Procedure

1. Navigate to **Configuration > Services**. The **Services** page opens.
2. Select the **Add** link. The **Add Services** dialog opens.
3. To display the **Audit** tab, select the **More Options > Audit End-Hosts** check box. The **Add Services > Audit** dialog opens.

Figure 1. Add Services > Audit Dialog



4. Specify the parameters in the **Add Services** > **Audit** tab as described in Table 1, then click **Save**.

Table 1. Add Services > Audit Dialog Parameters

Parameter	Action/Description
Audit Server	Select a server profile from the list: • **Audit Servers**: Performs vulnerability scanning and returns a Healthy/Quarantine result. • **Nmap Audit**: Performs network port scans. The health evaluation always returns a **Healthy** result. The port scan gathers attributes that allow determina

Parameter	Action/Description
	tion of role(s) through post - audit rul es. You can click the **View Details** button to view the **Policy Manager Entity Details** di alog with the summary of audit server de tails. To view the **Summary** tab with audit se rver details, click the **Modify** button. For Policy Manager to trigger an audit on an end - host, it needs to get the IP add ress of the end - host. The IP address of t he end - host is not available at the time of initial authentication for 802.1X and M AC authentication requests. Policy Manag er's DHCP snooping service examines the DHCP request and response packets to d erive the IP address of the end - host. For this to work, you need to use this serv ice, Policy Manager must be configured a s a DHCP “IP Helper” on your router/swit ch in addition to your main DHCP server. Refer to your switch documentation for “I P Helper” configuration. To audit devices that have a **static IP add ress** assigned, it is recommended that yo u create a static binding between the M AC address and IP address of the endpoi nt in your DHCP server. Refer to your DHC P server documentation for configuring s tatic bindings. **NOTE** Policy Manager does not iss ue the IP address; it only exa mines the DHCP traffic to de

Parameter	Action/Description
	rive the IP address of the en d - host.
Audit Trigger Conditions	Select from the following audit trigger co nditions: • **Always**: Always perform an audit. • **When posture is not available**: Perfor m audit only when posture credential s are not available in the request. • **For MAC Authentication Request**: If you select this option, then Policy Ma nager presents the following three ad ditional settings: ◦ **For known end - hosts only:** Selec t this option when you want to rej ect unknown end - hosts and to a udit known clients. Known end - h osts are defined as clients that are found in the authentication source (s) associated with this service. ◦ **For unknown end - hosts only:** Se lect this option when known end - hosts are assumed to be healthy, but you want to establish the ident ity of unknown end - hosts and ass ign roles. Unknown end - hosts are end - hosts that are not found in a ny of the authentication sources a ssociated with this service. ◦ **For all end - hosts:** For both know n and unknown end - hosts.
Action after audit	Select an Action after audit. Performing an audit on a client is an asyn chronous task, which means the audit ca n be performed only after the MAC authe

Parameter	Action/Description
	ntication request is completed and the c lient has acquired an IP address through DHCP. Once the audit results are availabl e, there should be a way for Policy Manag er to reapply policies on the network dev ice. This can be accomplished in one of t he following ways: • **No Action:** The audit will not apply po licies on the network device after this audit. • **Do SNMP bounce:** This option will bo unce the switch port or force an 802.1 X reauthentication (both done using S NMP). Bouncing the port triggers a n ew 802.1X/MAC authentication reques t by the client. ◦ If the audit server already has the posture token and attributes assoc iated with this client in its cache, i t returns the token and the attribut es to Policy Manager. • **Trigger RADIUS CoA action:** This opti on sends a RADIUS CoA command to the network device.

Modifying Default Audit Servers

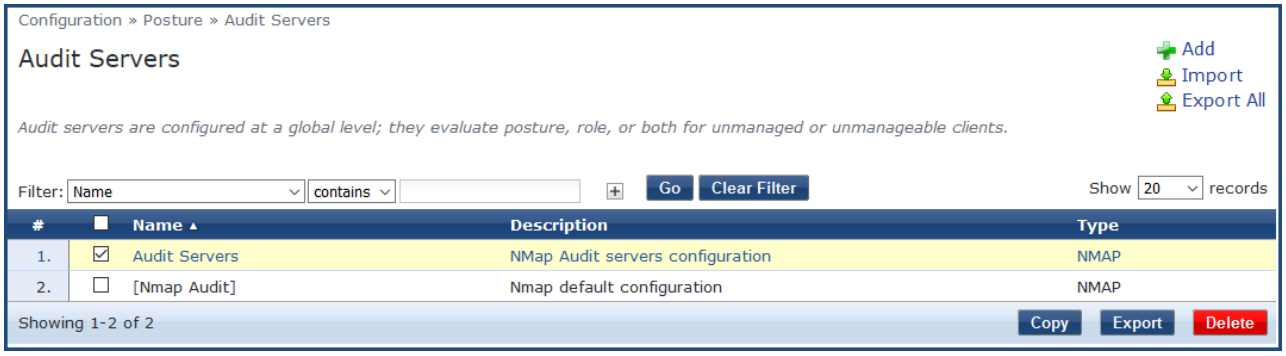
About this task

To reconfigure default Policy Manager audit servers:

Procedure

1. Navigate to **Configuration > Posture > Audit Servers**.

Figure 2. Audit Servers Page



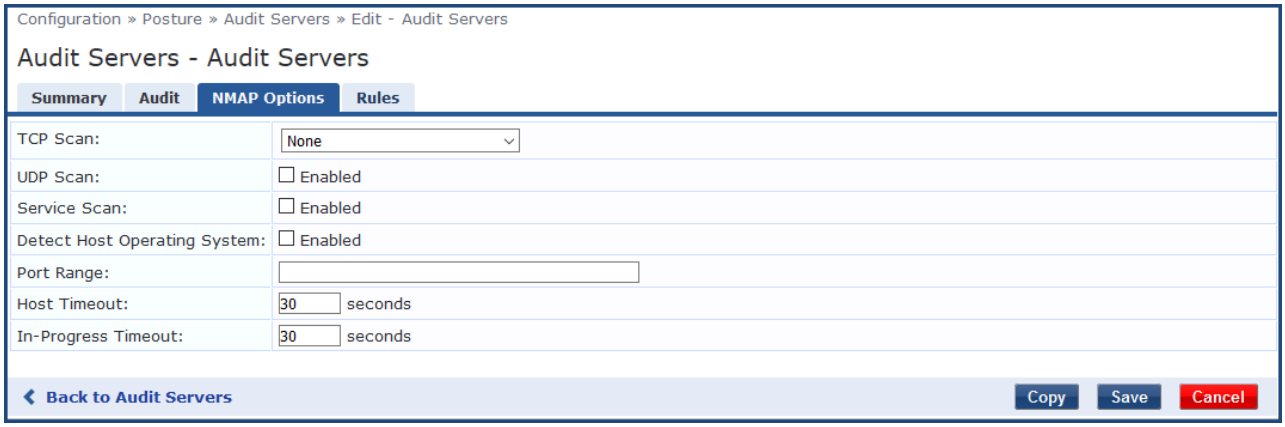
2. Select an audit server from the list of available servers. The **Edit Audit Servers** page opens.

Figure 3. Edit Audit Servers Page

3. Modify the profile, plugins, and/or preferences.

   - In the **Audit** tab, you can modify the *In-Progress Posture Status* and *Default Posture Status*.
   - In the **NMAP Options** tab, you can modify the following parameters:

Figure 4. Editing Audit Server Nmap Options



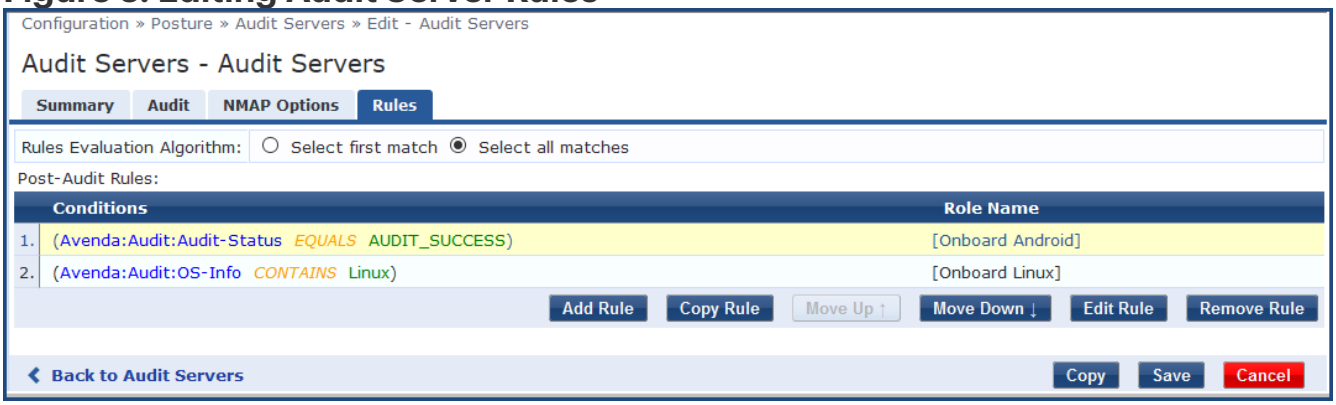
- If you selected a Nessus Server, the **Primary Server** and **Backup Server** tabs allow you to specify a scan profile. In addition, when you add a new scan profile, you can select plugins and preferences for the profile. Refer to Nessus Scan Profiles for more information.

The default Policy Manager Nessus audit server ships with approximately 1,000 of the most commonly used Nessus plugins.

Rules Tab

In the **Rules** tab, you can add new rules or edit the existing post-audit rules.

Figure 5. Editing Audit Server Rules



Custom Audit Servers

This section provides the following information:

- Adding a Nessus Audit Server
- Required Configuration Updates for External Nessus Servers
- Adding an Nmap Audit Server

For enterprises with existing audit server infrastructure or preferring custom audit servers, Policy Manager supports Nessus (v2.x through v6.x) and Nmap scans using the NMAP plug-in on external Nessus servers.

Subtopics

Adding a Nessus Audit Server
Adding an Nmap Audit Server

Adding a Nessus Audit Server

Policy Manager uses the Nessus audit server interface primarily to perform vulnerability scanning. It returns a result of *Healthy* or *Quarantine*. Nessus can be used to log into Unix and Windows servers, Cisco devices, SCADA systems, IBM iSeries servers, and databases to determine if they have been configured in accordance to the local site security policy. Nessus can also search the entire hard drive of Windows and Unix systems, for unauthorized content. Nessus can perform vulnerability scans of network services, as well as log into servers to discover any missing patches. The advantage of using Nessus to perform vulnerability scans and compliance audits is that all of this data can be obtained at one time. Knowing how a server is configured, how it is patched and what vulnerabilities are present can help determine measures to mitigate risk.

To add a Nessus audit server:

Audit Tab

Procedure

1. Navigate to **Configuration > Posture > Audit Servers**, then click the **Add** link. The **Add Audit Servers** dialog opens to the **Audit** tab.

Figure 1. Add Nessus Audit Server > Audit Tab

2. Specify the **Nessus Audit Server > Audit** tab parameters as described in the following table:

Table 1. Add Nessus Audit Server > Audit Tab Parameters

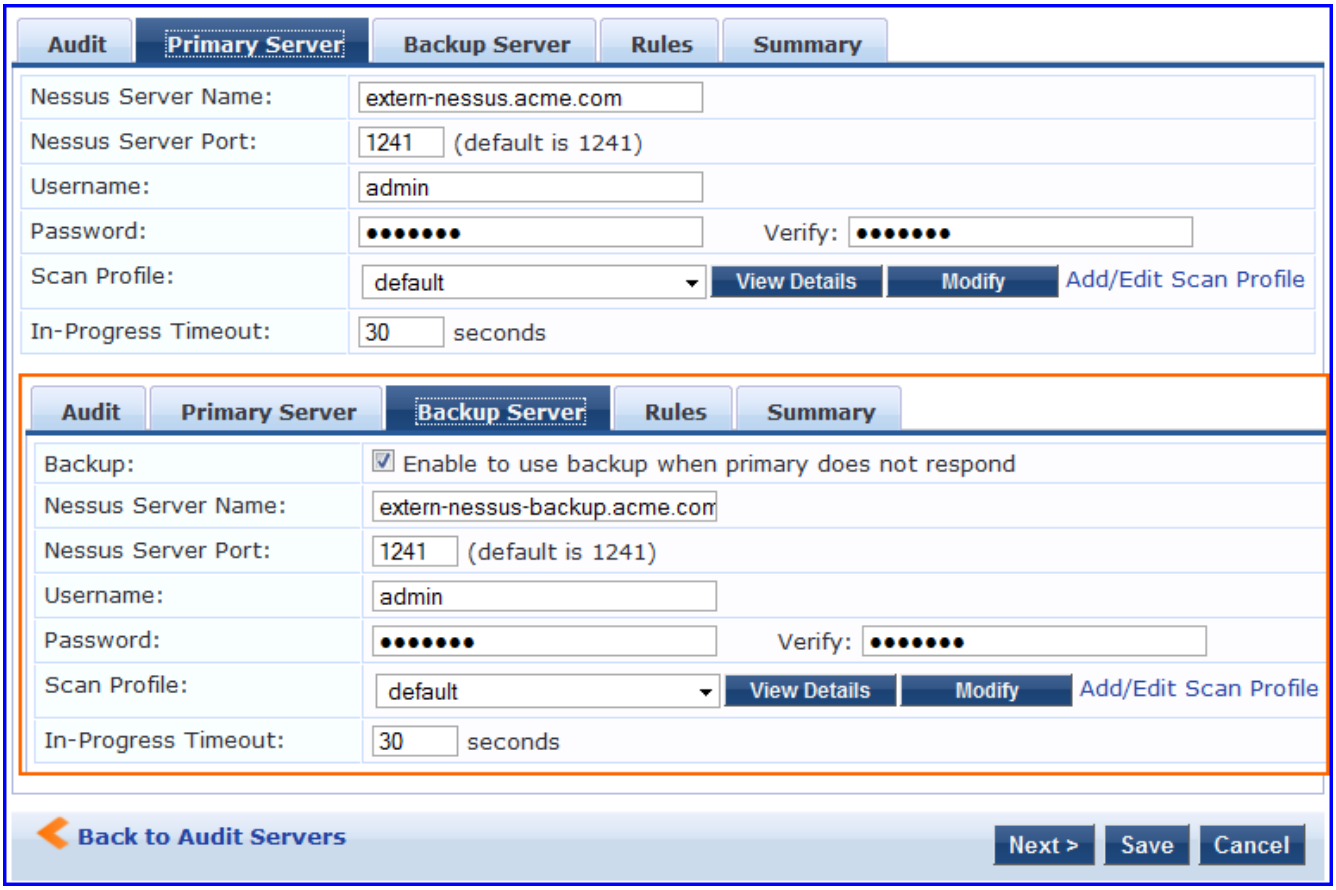
Parameter	Action/Description
Name	Specify the name of the audit server.
Description	Optionally (and recommended), enter th e description that provides additional in formation about the audit server.
Type	Specify the type of audit server: **Nessus**.
In - Progress Posture Status	Specify the posture status during audit: • Healthy (0) • Checkup (10) • Transition (15). This is the default setti ng. • Quarantine (20) • Infected (30) • Unknown (100)
Default Posture Status	Specify the posture status if evaluation d oes not return a condition or action matc h. The default setting is *UNKNOWN*.

Primary Server and Backup Server Tabs

About this task

The **Primary Server** and **Backup Server** tabs specify connection information for the Nessus audit server.

Figure 2. Add Nessus Audit Server > Primary and Backup Server Tabs



Procedure

1. Specify the **Nessus Audit Server > Primary Server** tab and **Backup Server** tab parameters as described in Table 2.

Table 2. Add Nessus Audit Server > Primary and Backup Server Tabs Parameters

Parameter	Action/Description
Backup	On the Backup Server dialog: For the backup server to be invoked on p rimary server failover, check the **Enable t o use backup when primary does not res pond** check box.
Nessus Server Name	Enter the name of the Nessus server
Nessus Server Port	Specify the Nessus Server port. The defa ult is **1241**.

Parameter	Action/Description
Username	Enter the username for the primary and b ackup Nessus server
Password	Enter the password for the primary and b ackup Nessus servers.
Scan Profile	You can accept the default scan profile o r select **Add > Edit Scan Profile** to create other profiles and add them to the scan p rofile list. For details, refer to Nessus Sca n Profiles.
In - Progress Timeout	Specify the duration (in seconds) before polling for Nmap results. The default is **3 0** seconds.

2. Configure the audit server Rules. The **Rules** tab specifies rules for post-audit evaluation of the request to assign a role.

Modifying a Nessus Audit Server

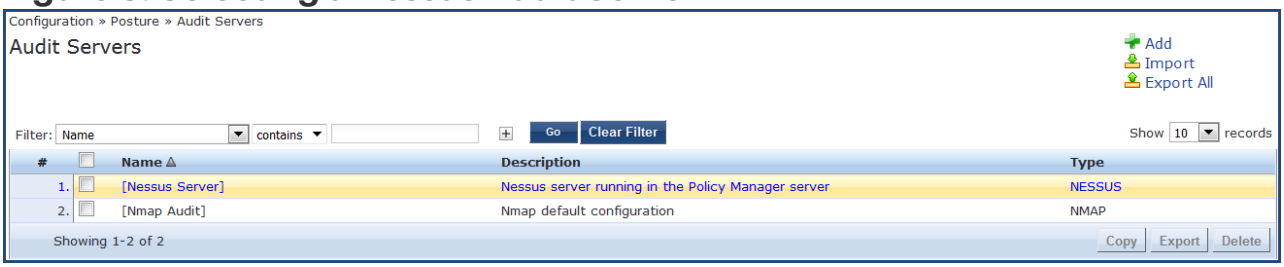
About this task

To modify an existing Nessus audit server:

Procedure

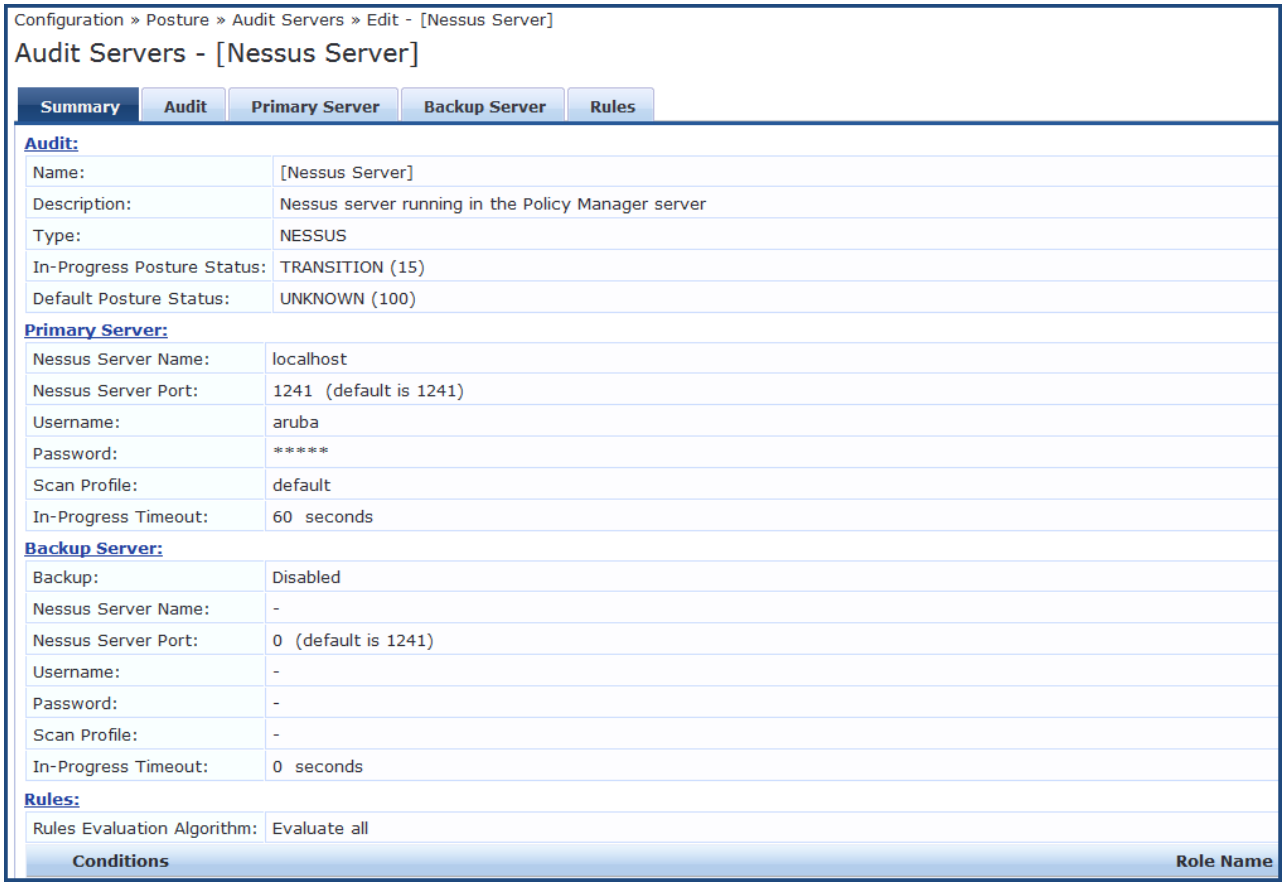
1. Navigate to **Configuration > Posture > Audit Server**. The **Audit Servers** dialog opens.

Figure 3. Selecting a Nessus Audit Server



2. Select the Nessus audit server you wish to modify. The **Edit Nessus Server** dialog opens to the **Summary** tab, which displays the configuration settings for the selected Nessus server.

Figure 4. Edit Nessus Server > Summary Page



3. Make any necessary configuration changes, then click **Save**.

Required Configuration Updates for External Nessus Servers

About this task

To properly support Nessus server configuration on Policy Manager servers, you must make the following configuration settings on the external Nessus server:

Procedure

1. On the external Nessus server, set the value for the **disable_ntp** parameter to **no**.

   For example, on a CENTOS/RHEL server running Nessus, you would enter the following command:

   ```
   centos# /opt/nessus/sbin/nessuscli fix --set disable_ntp=no
   ```

2. Restart the Nessus service.

   For example:

   ```
   centos# service nessusd restart
   ```

3. If the external Nessus server has Transport Layer Security (TLS) enabled, add the Nessus CA Certificate to the Policy Manager Certificate Trust List (see Certificate Trust List).

   You can download the Nessus CA certificate from:

   https://<nessus_server_name>:8834/getcert

Nessus Scan Profiles

About this task

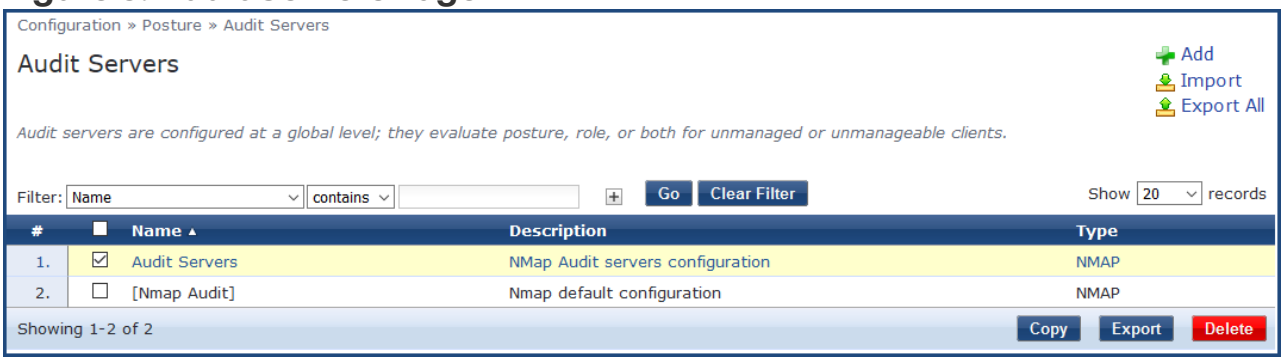
A scan profile contains a set of scripts (plugins) that perform specific audit functions.

To add or edit scan profiles:

Procedure

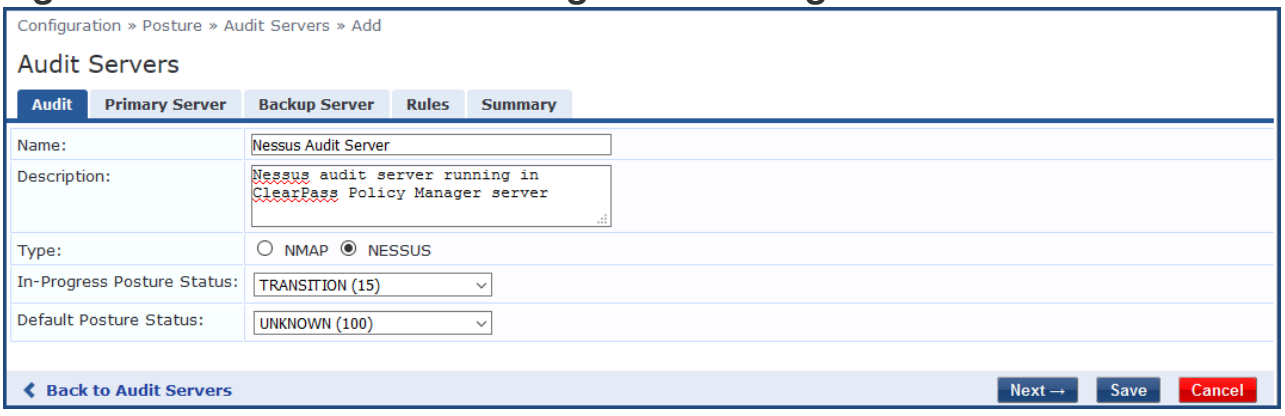
1. Navigate to **Configuration > Posture > Audit Servers**. The **Audit Servers** page opens.

Figure 5. Audit Servers Page



2. Click **Add**. The **Add Audit Servers** configuration dialog opens.

Figure 6. Add Audit Servers Configuration Dialog



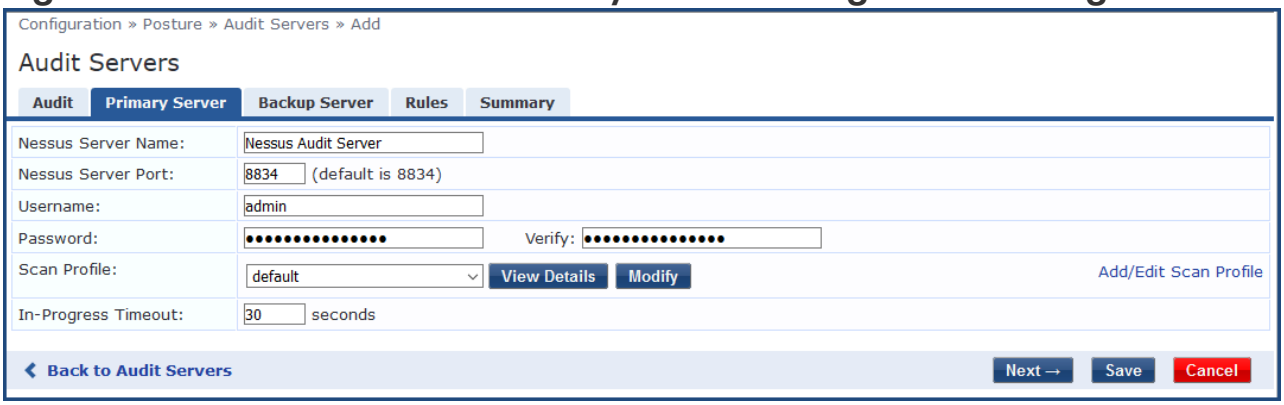
3. Specify the **Audit Servers > Audit** tab parameters as described in the following table:

Table 3. Audit Server > Audit Tab Configuration Parameters

Parameter	Action/Description
Name	Specify the host name of the Audit server.
Description	Enter a brief description of this server.
Type	
In - Progress Posture Status	Specify the posture status during audit: • Healthy (0) • Checkup (10) • Transition (15). This is the default setting. • Quarantine (20)
Default Posture Status	Specify the posture status if evaluation does not return a condition or action match. The default setting is *UNKNOWN*.

4. Select the **Primary Server** tab.

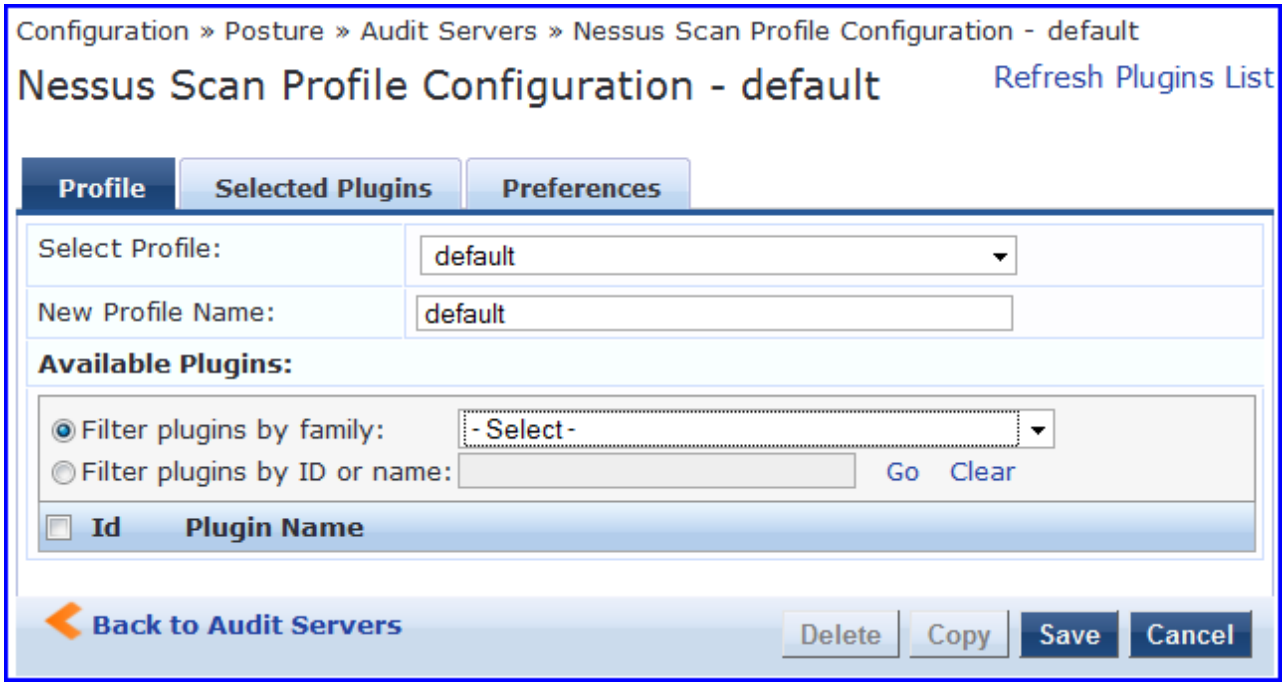
Figure 7. Add Audit Servers > Primary Server Configuration Dialog



5. From the **Primary Server** tab, click the **Add/Edit Scan Profile** link.

   The **Nessus Scan Profile Configuration** page opens.

Figure 8. Nessus Scan Profile Configuration Page



6. You can refresh the plugins list (after uploading plugins into Policy Manager, or after refreshing the plugins on your external Nessus server) by clicking **Refresh Plugins List**.

   The **Nessus Scan Profile Configuration** page provides three views for scan profile configuration:

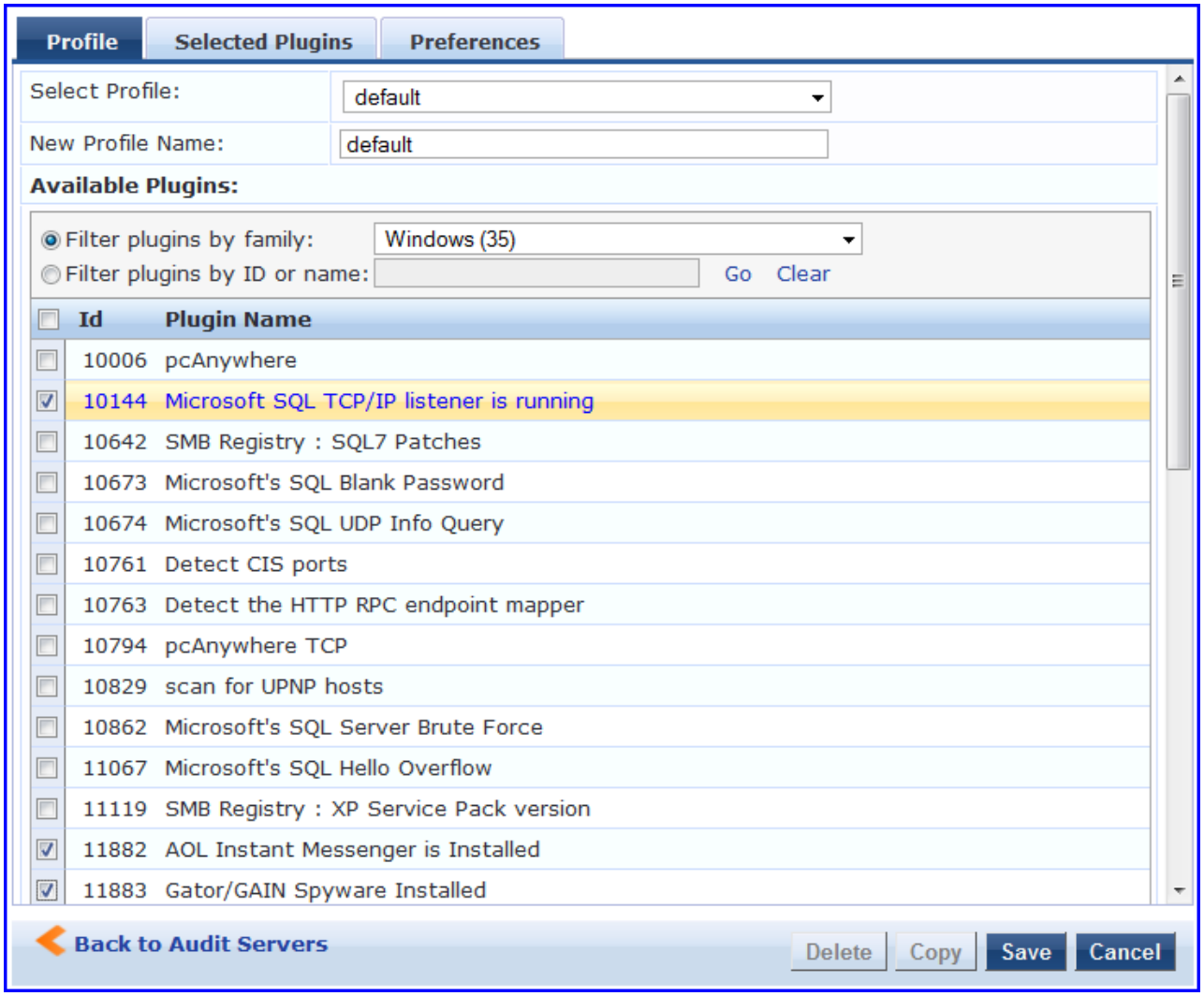
   The **Profile** tab identifies the profile and provides a mechanism for selection of plugins:

7. From the **Filter plugins by family** drop-down list, select a family to display all available member plugins in the list below. You may also enter the name of a plugin in **Filter plugins by ID** or name text box.

8. Select one or more plugins by enabling their corresponding check boxes (at left). Policy Manager will remember selections as you select other plugins from other plugin families.

9. When finished, click the **Selected Plugins** tab.

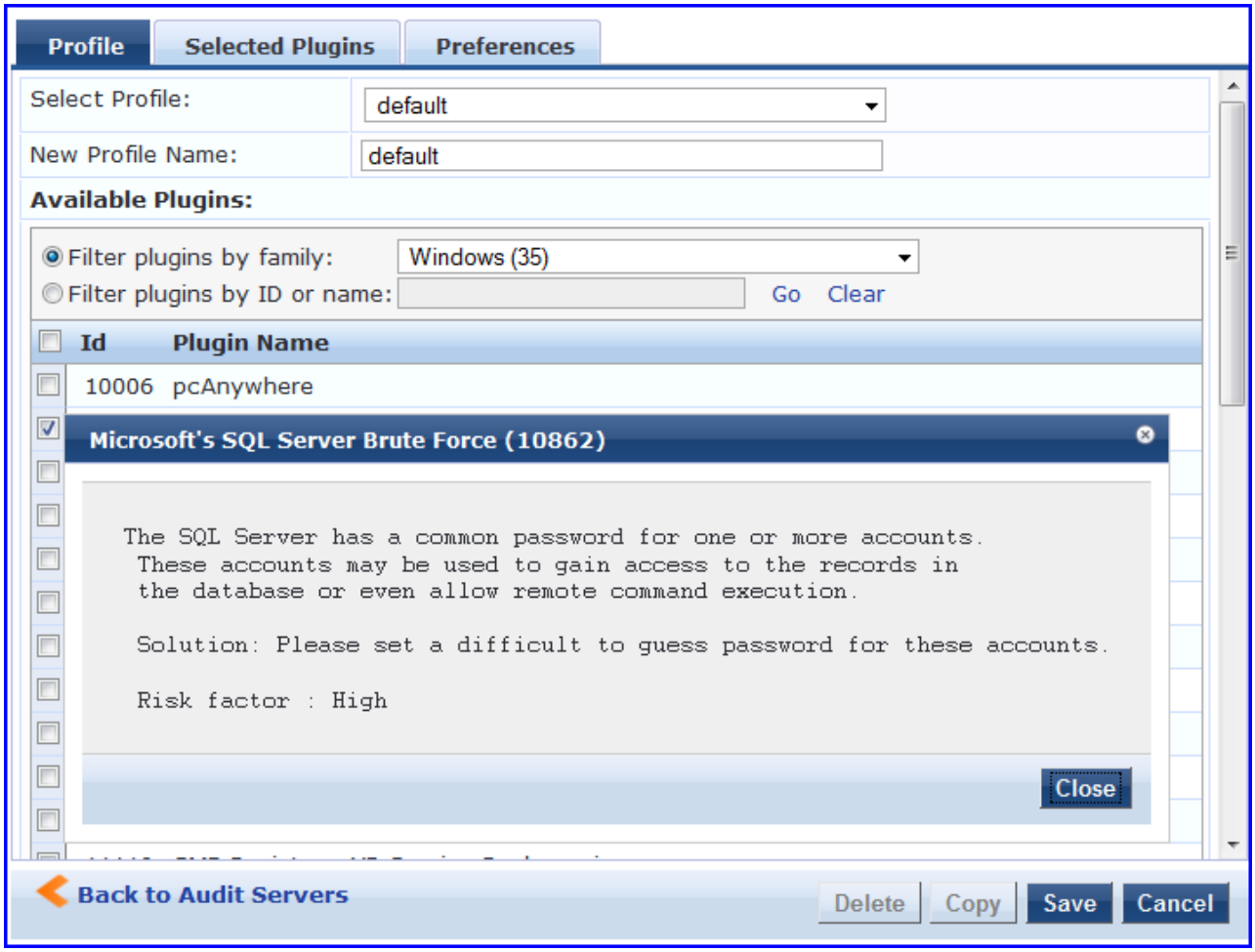
Figure 9. Nessus Scan Profile Configuration > Profile Tab



The **Selected Plugins** tab displays all selected plugins, plus any dependencies.

0. To display a synopsis of any listed plugin, click on its row.

Figure 10. Nessus Scan Profile Configuration Profile Tab > Plugin Synopsis



NOTE

Of special interest is the section of the synopsis entitled **Risk factor**. To delete any listed plugin, click on its corresponding trashcan icon. To change the vulnerability level of any listed plugin, click on the link to change the level to one of HOLE, WARN, or INFO. This action tells Policy Manager the vulnerability level that is considered to be assigned QUARANTINE status.

Figure 11. Nessus Scan Profile Configuration > Selected Plugins Tab

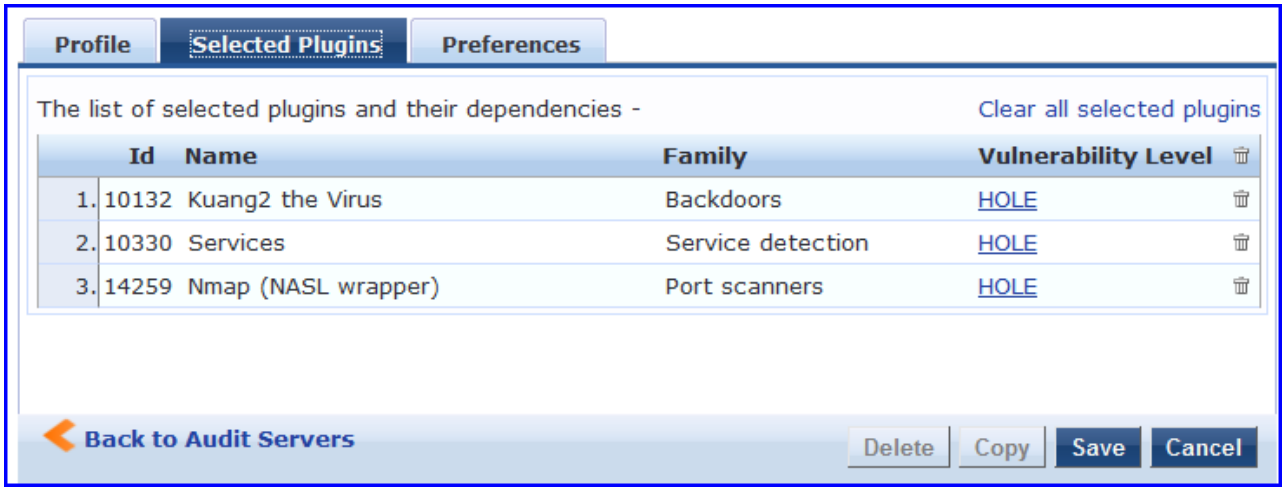


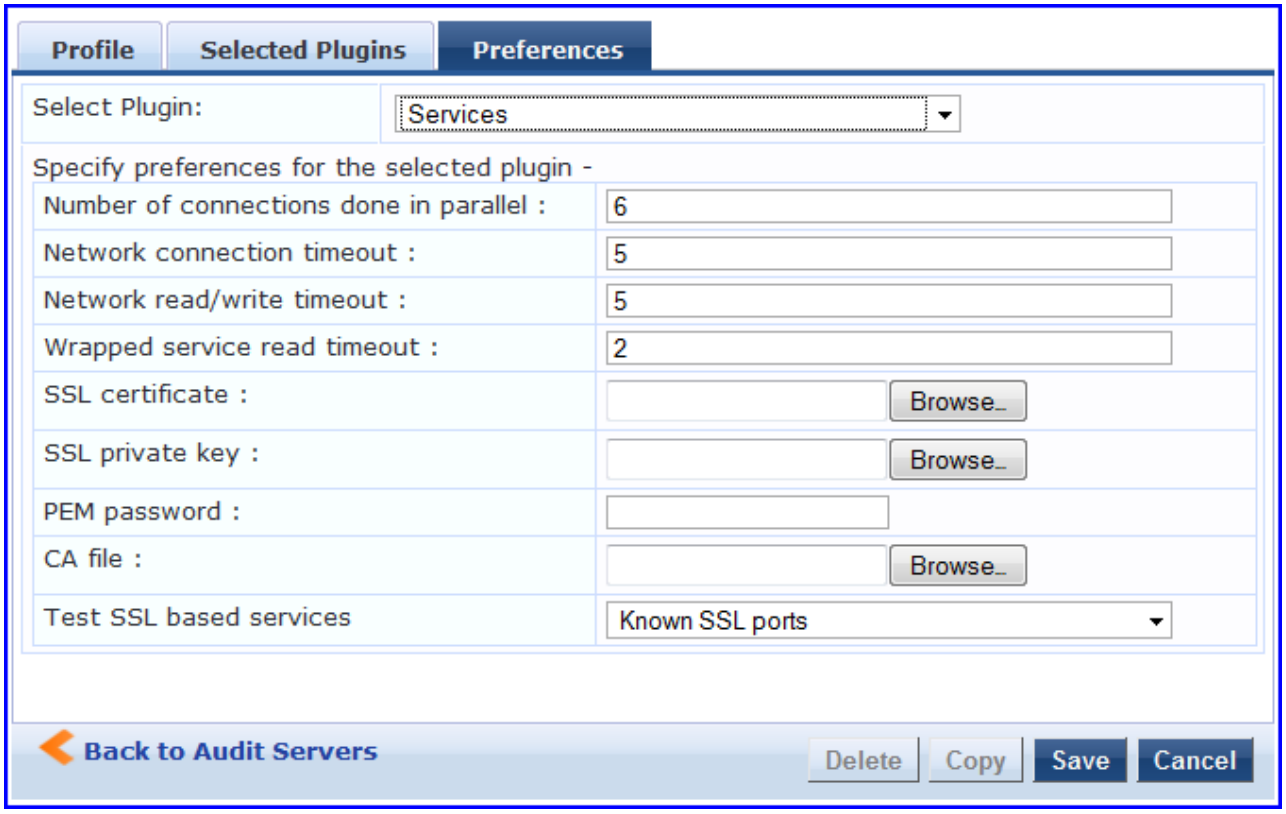
Figure 12. Nessus Scan Profile Configuration Selected Plugins Tab > Vulnerability Level

For each selected plugin, the **Preferences** tab contains a list of fields that require entries.

In many cases, these fields will be pre-populated. In other cases, you must provide information required for the operation of the plugin.

By way of example of how plugins use this information, consider a plugin that must access a particular service, in order to determine some aspect of the client's status; in such cases, login information might be among the preference fields.

Figure 13. Nessus Scan Profile Configuration > Preferences Tab



After saving the profile, plugin, and preference information for your new (or modified) plugin, you can go to the **Primary/Backup Servers** tabs and select it from the **Scan Profile** drop-down list.

Adding an Nmap Audit Server

Policy Manager uses the Nmap audit server interface exclusively for network port scans. Nmap (Network Mapper) uses raw IP packets to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters and firewalls are in use, as well as other characteristics.

- The Health evaluation always returns a status of **Healthy**.
- The port scan gathers attributes that allow determination of role(s) through post-audit rules.

To create an Nmap (Network Mapping) audit server:

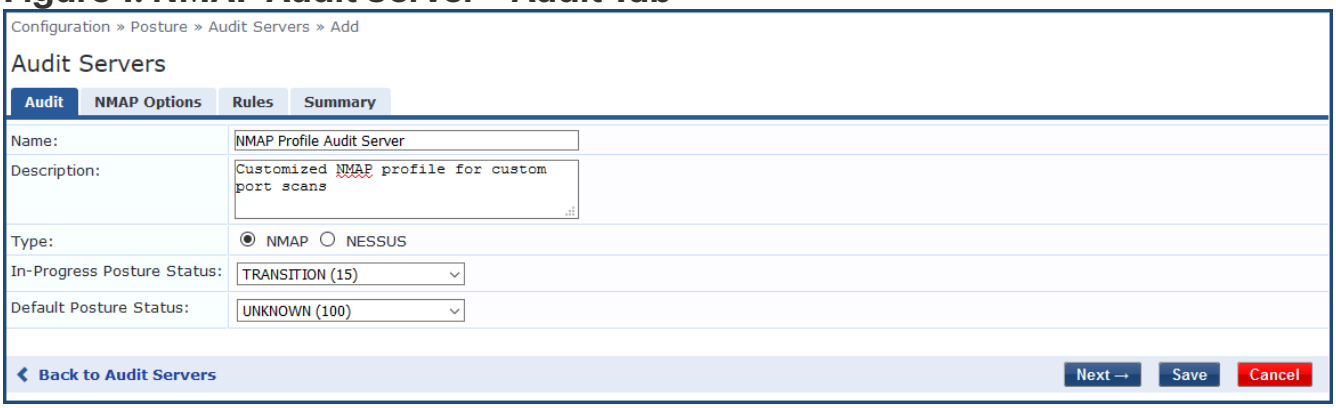
1. Navigate to the **Configuration > Posture > Audit Servers** page, then click **Add**.
2. From the **Audit** tab, select the **NMAP** radio button in the **Type** field.

Audit Tab

About this task

You can use the **Audit** tab to identify the server and define configuration details.

Figure 1. NMAP Audit Server > Audit Tab



Procedure

Specify the parameters configured in the **Audit** tab as described in the following table:

Table 1. Audit Tab Parameters

Parameter	Action/Description
Name	Enter the name of the Nmap audit server.
Description	Optionally (but recommended), enter the description of the Nmap audit server.
Type	Select **NMAP** (the default setting).
In - Progress Posture Status	Specify the posture status during audit: • Healthy (0) • Checkup (10) • Transition (15). This is the default setting. • Quarantine (20) • Infected (30) • Unknown (100)
Default Posture Status	Select the posture status if evaluation does not return a condition/action match. The default setting is *UNKNOWN*.

NMAP Options Tab

Use the **NMAP Options** tab to specify the type of scan configuration.

Figure 2. Nmap Server > NMAP Options Tab

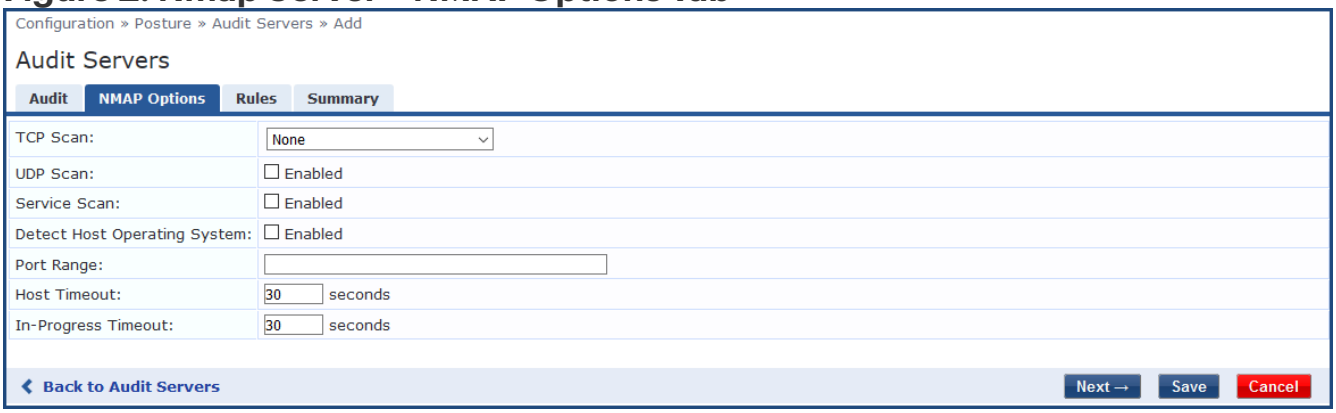


Table 2. NMAP Options Tab

Parameter	Action/Description
TCP Scan	Specify the type of TCP (Transmission Cont rol Protocol) scan: • TCP SYN scan • TCP Connect scan • TCP Null Scan • TCP FIN scan • TCP Xmas scan • TCP ACK scan • TCP Window scan • TCP Maimon scan Refer to Nmap documentation for more in formation on the TCP scan options. Nmap option: **scanflags**.
UDP Scan	To enable UDP (User Datagram Protocol) sc anning, check the **UDP Scan** check box. Nmap option: **sU**.
Service Scan	Enable this scan to identify all the services and their versions running on your target a pplication's server.

Parameter	Action/Description
	To enable Service scanning, check the **Service Scan** check box Nmap option: **sV**.
Detect Host Operating System	To enable host OS detection, check the **Detect Host Operating System** check box. NMAP option: **A**.
Port Range	Specify the range of ports to scan. Specify the range with a hyphen. For example, to scan ports 25 to 50, you would enter **25 - 50**. NMAP option: **p**.
Host Timeout	Specify the time in seconds for the target host to timeout. Nmap option: **host - timeout**
In - Progress Timeout	Specify the duration (in seconds) before polling for Nmap results.

Rules Tab

About this task

The **Rules** tab specifies rules for post-audit evaluation of the request to assign a role.

To add a post-audit rule:

Procedure

1. Select the **Rules** tab.

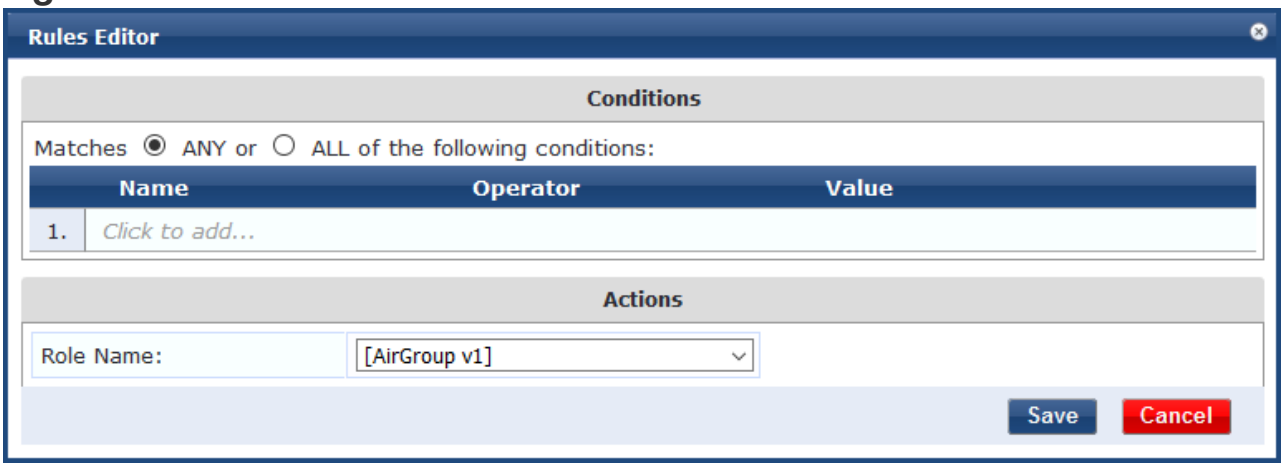
Figure 3. Configuring Audit Server Rules

Table 3. Rules Configuration Action Buttons

Button	Action/Description
Add Rule	Click the **Add Rule** action button to bring up the Rules Editor and add a new rule.
Copy Rule	Select the rule you want to copy, then click the **Copy Rule** action button. The copied rule is added to the existing list of rules.
Move Up/ Move Down	To change the order that the rules are executed in the enforcement policy, select an enforcement policy rule you want to move, then click **Move Up** or **Move Down** as desired.
Edit Rule	Select the rule you want to edit, then click the **Edit Rule** action button.
Remove Rule	Select the rule you want to delete, then click the **Remove Rule** action button.

2. Click **Add Rule**.
3. The **Rules Editor** opens.

Figure 4. Audit Server Rules Editor



4. Specify the **Rules Editor** page parameters as described in the following table.

Table 4. Rules Configuration Action Buttons

Parameter	Action/Description
Matches	Select the match conditions for audit se rver: • Matches ANY • Matches all of the following condition s:
Name	The Name list includes the following attri bute: • OS - Info • Open - Ports • Network - Apps • Mac - Vendor • Device - Type • Output - Msgs • Audit Status
Operator	Select the appropriate operator from the list of operators.
Value	Enter the value or select the value from t he drop - down list. The **Value** depends o n the **Name** that you selected.
Actions: Role Name	From the **Role Name** drop - down, select the appropriate role. **NOTE** The **Actions** list includes the names of the roles configure d in Policy Manager.

5. Click **Save**.

You return to the **Audit Servers** page where the new rule(s) have been added.

Figure 5. Audit Server Rules Added

Configuring Agentless OnGuard

Policy Manager supports Agentless OnGuard for posture analysis of Windows, macOS, and Linux clients, simplifying endpoint software management.

Configuring Identity Settings

The Policy Manager database supports storage of user records when a particular class of users is not present in a central user repository (for example, when there is neither an Active Directory nor any other database).

NOTE

To authenticate local users from a particular service, include **Local User Repository** among the authentication sources.

This section provides information on the following identity topics:

- Configuring Single Sign-On
- Managing Local Users
- Adding and Modifying Endpoints
- Managing Static Host Lists
- Configuring a Role and Role-Mapping Policy

Subtopics

Configuring Single Sign-On
Managing Local Users
Adding and Modifying Endpoints
Managing Static Host Lists
Configuring a Role and Role-Mapping Policy

Configuring Single Sign-On

The Single Sign-On (SSO) settings on the Single Sign-On page allows Policy Manager users that have signed in to ClearPass Policy Manager to access the Onboard, Guest, and Insight applications and Policy Manager administration settings without reauthenticating. Policy Manager provides single sign-on support using the Security Assertion Markup Language (SAML). This feature also provides differentiated single sign-on access for Guest web logins and Guest Operator logins .

NOTE

Before selecting SAML signing and encryption certificates for use in Single Sign-On, be sure to add a SAML signing certificate to the Certificate Trust List and a SAML encryption certificate to Service & Client Certificates list.

SAML Service Provider (SP) Configuration

About this task

To configure single sign-on service provider settings:

Procedure

1. Navigate to **Configuration > Identity > Single Sign-On**. The **Single Sign-On > SAML SP Configuration** dialog opens.
2. **NOTE**

   If configuring TLSv1.3 support within a cluster, enter the IDP URL of the second FQDN used for mTLS, as two FQDNs are required to support TLS1.3 certificate authentication. For information on configuring TLSv1.3 support for a cluster, see .

   In the **Identity Provider (IdP) URL** field, enter the URL used to access the IdP storing and managing your digital identity.

3. in the **Enable SSO for** field, select one or more applications you want users to access with single sign-on. Note the following behaviors when selecting the **GuestOperators** and **Guest** options.
    - If you select only the **GuestOperators** option, SSO will be enabled for Operator logins only, and Web logins will use standard non-SSO authentication.
    - If you select only the **Guest** option, SSO will be enabled for Web logins only, and Operator logins will use standard non-SSO authentication.
    - If you select both the **GuestOperators** and the **Guest** options, Operator logins and Web logins will both use SSO authentication.
4. (Optional) In the **ClearPass Service Provider (SP) Signing Certificate** section, click **Select Certificate** and choose a certificate to sign SAML requests from the service provider. When you select a certificate, the page displays the following information about the certificate:
    - Subject DN
    - Issuer DN
    - Issue Date/Time
    - Expiry Date/Time
    - Validity Status
    - Serial Number
5. (Optional) In the **ClearPass Service Provider (SP) Metadata** field, click **Download** to download and view an XML file containing metadata for the Service Provider URI (Uniform Resource Identifier). The **Metadata URI** field displays the URI for the Service Provider metadata resource.
6. Use the **Identity Provider Signing Certificate** field select a certificate to verify the signed SAML response from the IdP. This information is required. Click the **Select Certificate** drop-down list and select the Identity Provider (IdP) certificate to use for single sign-on. When you select a certificate, the page displays the following information about the certificate:
7. (Optional) Use the **Identity Provider Encryption Certificate** field, select a certificate to decrypt the encrypted SAML assertion from the IdP. Click the **Select Certificate** drop-down list and select the Identity Provider (IdP) certificate to use for single sign-on.

8. (Optional) If the Force Authentication option is enabled based on the IdP configuration, then it will always re-authenticate the user, even if the user already has an active session. This setting is disabled by default.

SAML Identity Provider (IdP) Configuration

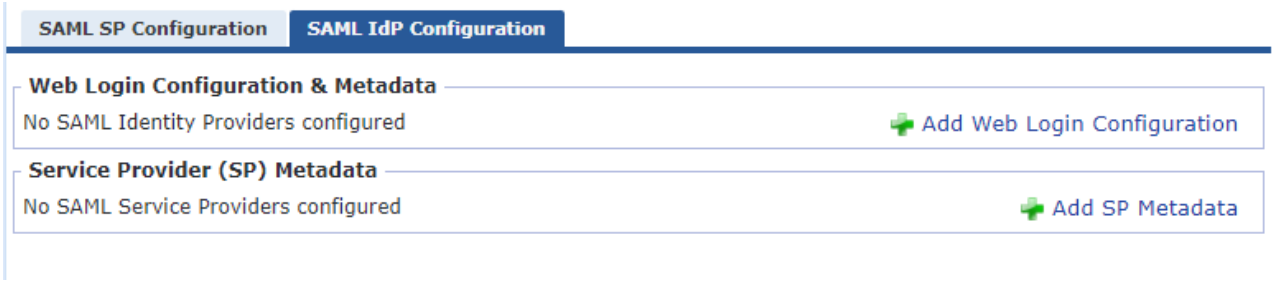
About this task

To configure single sign-on identity provider settings for Web Login pages or add SAML service provider metadata:

Procedure

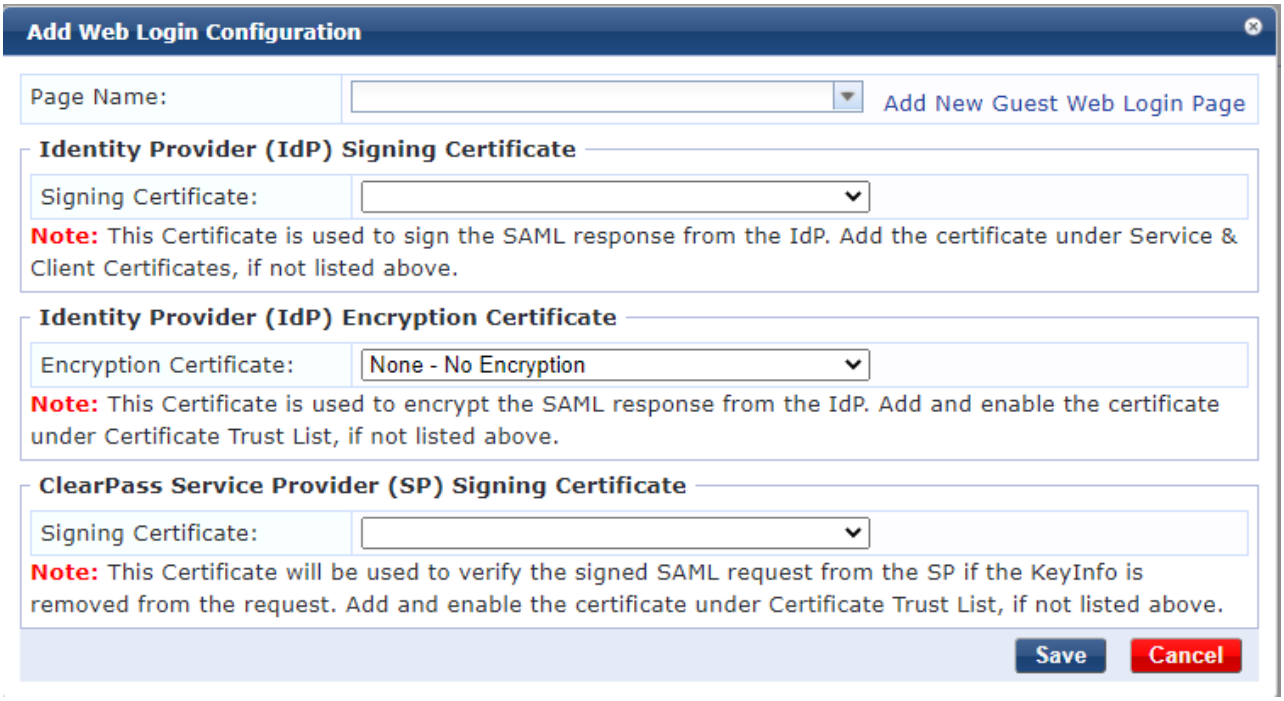
1. Navigate to **Configuration > Identity > Single Sign-On**.
2. Select the **SAML IdP Configuration** tab.

Figure 1. Configuring Single Sign-On > Identity Provider Parameters



3. To configure a Identity Provider (IdP) Signing Certificate and/or Identity Provider (IdP) certificate for Web Logins, Click the **Add Web Login Configuration** link at the right side of the table. The **Add Web Login Configuration** window opens.

Figure 2. Web Web Login Configuration & Metadata > Add Web Login Certificate



4. Click the **Page Name** menu to select the Web Login page to which you want to add signing or encryption certificate data.

   The value in the **Page Name** field must match the value in the **Page Name** field in ClearPass Guest at **Guest > Configuration > Pages > Web Logins**. For example, if the value "page_name_123" is provided for **Page Name** in Guest's **Web Login Editor**, then the value "page_name_123" must also be entered in Policy Manager's **Web Login Configuration**.

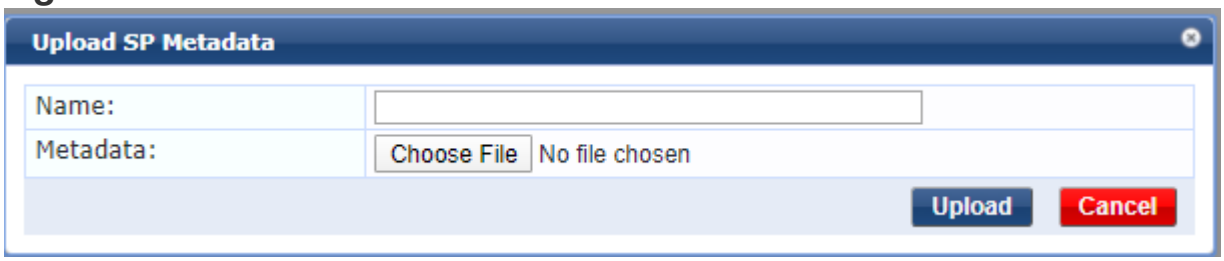
   If a value for **Page Name** is not specified (field is empty) in ClearPass Guest's **Web Login Editor**, then the value for **Page Name** fiel**n** must match the page ID of the web login page's URL. For example, if the **Page Name** field in Guest's **Web Login Editor** is empty, then the value in the **Page Name** field in Policy Manager's **Web Login Configuration** must match **just** the ID part of the default URL that is automatically generated for the web login page. In the default URL https://10.21.11.100/guest/weblogin.php/2?_browser=1, the ID is the number "2" immediately following "weblogin.php/", so you would enter 2 in the **Page Name** field on the **Add Web Login Configuration** form.

   NOTE

   Use the **Add New Guest Web Login Page** link by the **Page Name** field to open the **Guest > Configuration > Pages > Web Logins** form in ClearPass Guest and create a new **Web Login** page or verify an existing **Page Name** setting.

5. Click the **Signing Certificate** drop-down menu to select a certificate to sign the SAML response from the IdP. This field is required. When you select a certificate, the page displays the following information about the certificate, if available:
   - Subject DN
   - Issuer DN
   - Issue Date/Time
   - Expiry Date/Time
   - Validity Status
   - Serial Number
6. Optionally, click the **Encryption Certificate** drop-down menu to select a certificate to encrypt the SAML response from the IdP.
7. If you upload metadata for SAML service providers, Policy Manager can upload the service provider (SP) metadata for validation during the single-sign on process.
   a. Click **Add SP Metadata**. The **Upload SP Metadata** dialog opens.
   b. In the **Name** field, enter the name of the service provider.
   c. In the **Metadata** field, click **Choose File** and browse to the service provider metadata file. Once you have selected the file, click **Upload**.

Figure 3. Service Provider SP Metadata



Managing Local Users

- Adding a Local User
- Modifying a Local User Account
- Importing and Exporting Local Users

- Setting Password Policy for Local Users
- Disabling Local User Accounts

  Policy Manager lists all local users in the **Local Users** page. You can also add, import, export, set password policies, and configure the conditions for disabling accounts for the local users using the links provided at the top-right corner of the **Local Users** page.

Subtopics

Adding a Local User
Modifying a Local User Account
Importing and Exporting Local Users
Setting a Password Policy for Local Users
Disabling Local User Accounts

Adding a Local User

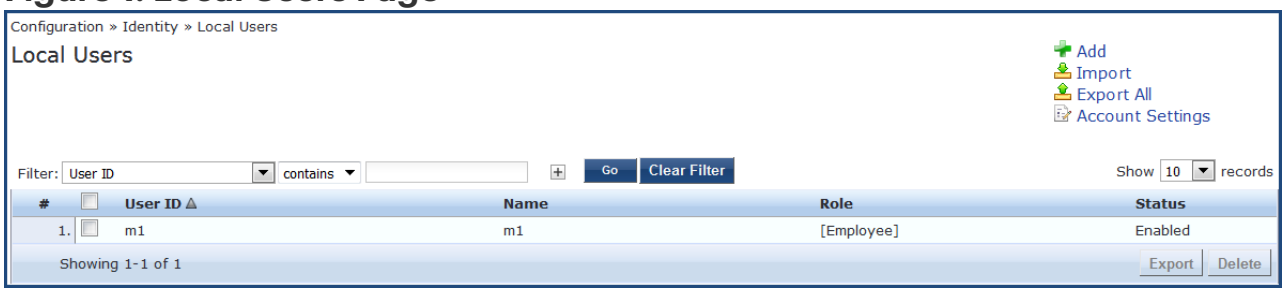
About this task

To add a local user in the **Local Users** table:

Procedure

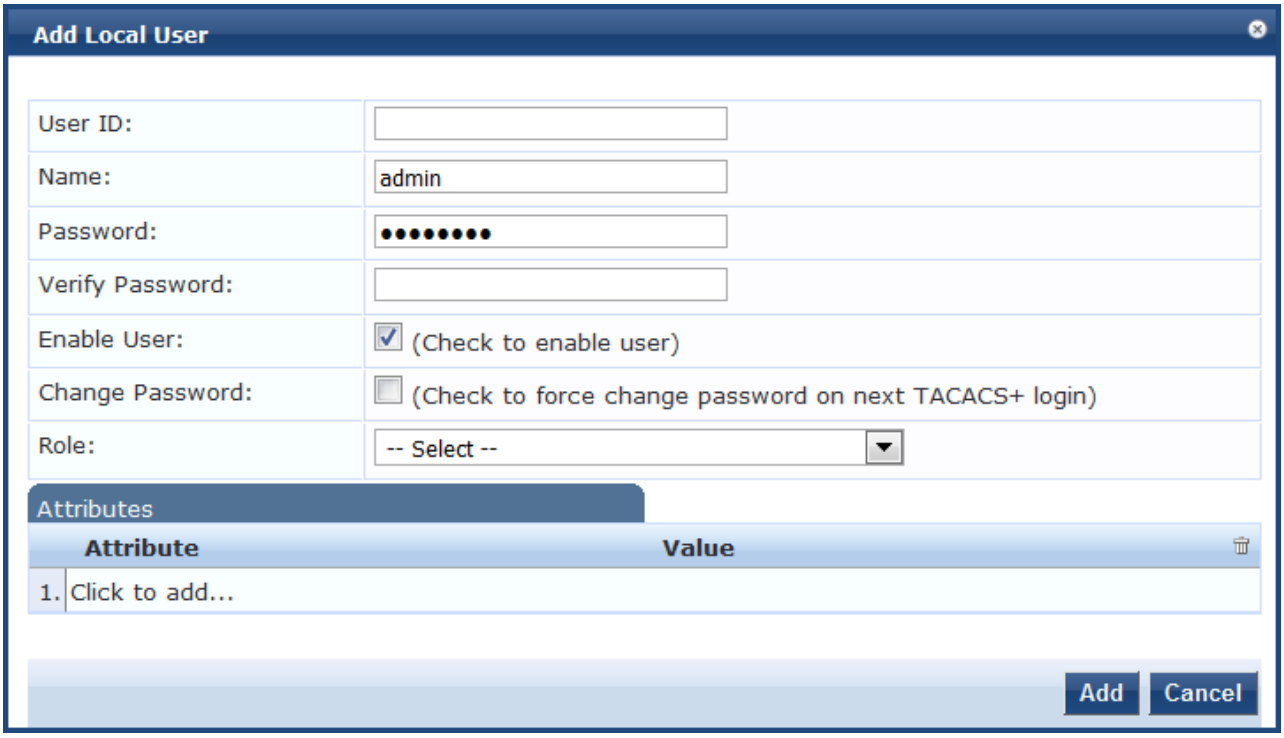
1. Navigate to **Configuration > Identity > Local Users**.The **Local Users** page opens.

Figure 1. Local Users Page



2. Click the **Add** link at the top-right corner the page. The **Add Local User** page opens (see Figure 2).

Figure 2. Adding a Local User



3. Specify the **Add Local User** parameters as described in the following table, then click **Add**:

Table 1. Adding a Local User Parameters

Parameter	Action/Description
User ID	Specify the local user's user ID.
Name	Enter the local user's name.
Password/ Verify Password	Specify a password for the local user, the n verify the password.
Enable User	You must enable this check box to enable the local user account. Otherwise, the lo cal user account is disabled.
Change Password	Enable this check box to allow the user to change the password at the next TACACS + login (after authenticating with the old password).

Parameter	Action/Description
	Once the password is changed successf ully, this option is automatically disabled. **NOTE** The option to change the p assword on the next login i s applicable for network devi ce administration logins usin g TACACS+ only.
Role	Select a static role to be assigned to the user from the **Role** drop - down list.
Attributes	
	To add attributes for the local users, click **Click to add...** A new row is created with a drop - down l ist in the **Attribute** column. This field is o ptional. The list of local user attributes a re: • Department • Designation • Email • Phone • Sponsor • Title a. To add a custom attribute in the **Attrib ute** column, select an attribute from th e drop - down list or enter any string. **NOTE** If you add a new custom attribute, it is available for selection in the **Attribute**

Parameter	Action/Description
	drop - down list for all loc al users. b. In the **Value** column, enter a value for t he attribute specified in the correspon ding row. **NOTE** All attributes entered for a local user are available in the role - mapping rules e ditor under the **LocalUser** namespace. When you cli ck **Add**, you return to the **Local User** page where th e following message is di splayed: *User <username> added successfully*

Modifying a Local User Account

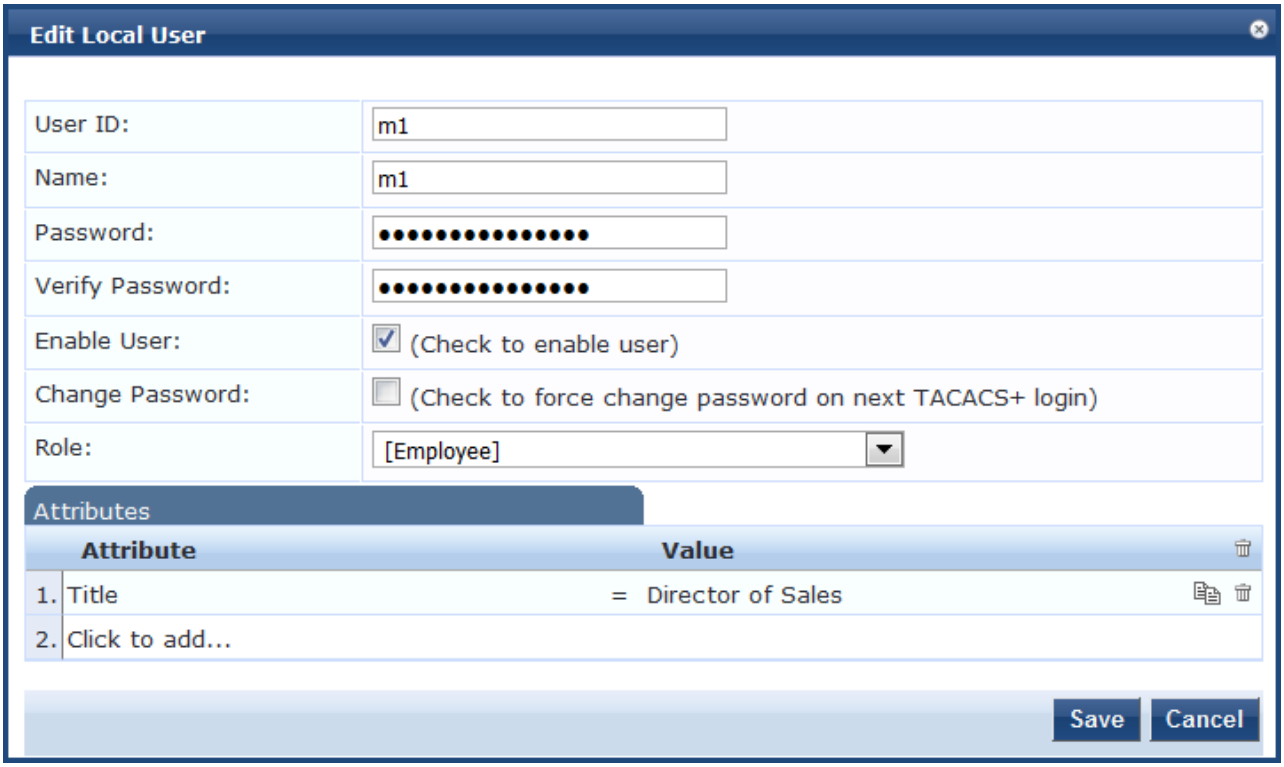
About this task

To modify a local user account:

Procedure

1. Navigate to the **Configuration > Identity > Local Users** page.
2. Click the **User ID** row that you want to edit. The **Edit Local User** window opens.

Figure 1. Modifying a Local User



3. Modify any values as necessary in the **Edit Local User** dialog.
4. Click **Save**.

Importing and Exporting Local Users

You can import or export the admin user accounts by using the **Import** and **Export All** links at the top-right corner of the **Local Users** page. (For more information, see Importing and Exporting Information .) After selecting one or more user accounts from the list, you can also export specific user accounts by clicking the **Export** button .

> **NOTE**
> The passwords of the local user accounts are not stored in clear text when exported to an XML file.

Setting a Password Policy for Local Users

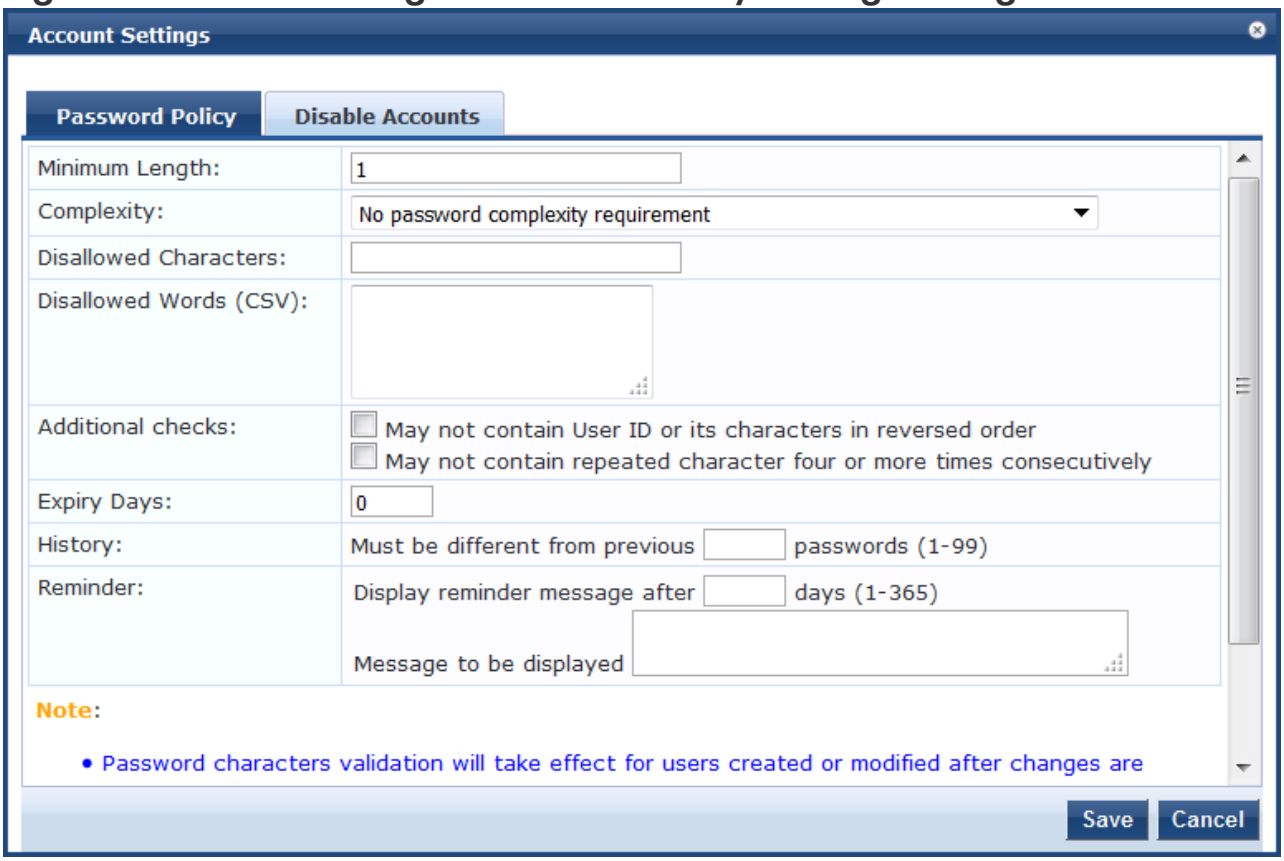
About this task

To set password policies for the local users:

Procedure

1. Navigate to the **Configuration > Identity > Local Users** page.
2. Click the **Account Settings** link. The **Account Settings** page opens.

Figure 1. Account Settings > Password Policy Settings Dialog



3. Specify the **Password Policy** parameters as described in Table 2, then click **Save**.

Table 1. Password Policy Parameters

Parameter	Action/Description
Minimum Length	Specify the minimum length required for the password.
Complexity	Select the complexity setting from the **Complexity** drop - down list. The complexity settings can be one of the following: • No password complexity requirement • At least one uppercase and one lowercase letter

Parameter	Action/Description
	• At least one digit • At lease one letter and one digit • At least one of each: uppercase letter, lowercase letter, digit • At least one symbol • At least one of each: uppercase letter, lowercase letter, digit, and symbol
Disallowed Characters	Specify the characters not to be allowed in the password. **NOTE** Password characters validati on takes effect for users crea ted or modified after chang es are saved.
Disallowed Words (CSV)	Specify the words not to be allowed in t he password. Separate the disallowed wo rds with commas.
Additional Checks	Select any additional checks, if required. The options are: • May not contain User ID or its charact ers in reversed order. • May not contain a repeated character four or more times consecutively.
Expiry Days	Set the password expiration time for local users. The allowed range is **0** to **500** day s. The default value is **0**. **NOTE** If the value is set to **0**, the p assword never expires. For a ny other value, local users w ill have to reset the expired

Parameter	Action/Description
	password through the Policy Manager user interface. Poli cy Manager alerts users five days before the password ex pires.
History	Specify the number of previous passwo rds for this user to be compared against. This option prevents users from setting a password that was used recently. Valid op tions are from 1 to 99.
Reminder	Configure the reminder message. Setting this option displays a reminder after n d ays to change the password. The valid op tions are from **1** to **365**. When set, this op tion only displays a reminder; it does not prompt for a new password. The messag e to be displayed can be set accordingly. **NOTE** The **Reminder** parameter is applicable for TACACS+ auth entication only. The other se ttings are applied to all users .

Table 2. Password Policy Parameters

Parameter	Action/Description
Minimum Length	Specify the minimum length required for the password.
Complexity	Select the complexity setting from the **C omplexity** drop - down list. The complexi ty settings can be one of the following:

Parameter	Action/Description
	• No password complexity requirement • At least one uppercase and one lowe rcase letter • At least one digit • At lease one letter and one digit • At least one of each: uppercase letter, lowercase letter, digit • At least one symbol • At least one of each: uppercase letter, lowercase letter, digit, and symbol
Disallowed Characters	Specify the characters not to be allowed in the password. **NOTE** Password characters validati on takes effect for users crea ted or modified after chang es are saved.
Disallowed Words (CSV)	Specify the words not to be allowed in t he password. Separate the disallowed wo rds with commas.
Additional Checks	Select any additional checks, if required. The options are: • May not contain User ID or its charact ers in reversed order. • May not contain a repeated character four or more times consecutively.
Expiry Days	Set the password expiration time for local users.

Parameter	Action/Description
	The allowed range is **0** to **500** days. The default value is **0**. **NOTE** If the value is set to **0**, the password never expires. For any other value, local users must reset the expired password when they log in and local users will have to reset the expired password through the Policy Manager user interface. Policy Manager alerts users five days before the password expires.
History	Specify the number of previous passwords for this user to be compared against. This option prevents users from setting a password that was used recently. Valid options are from 1 to 99.
Reminder	Configure the reminder message. Setting this option displays a reminder after n days to change the password. The valid options are from **1** to **365**. When set, this option only displays a reminder; it does not prompt for a new password. The message to be displayed can be set accordingly. **NOTE** The **Reminder** parameter is applicable for TACACS+ authentication only. The other settings are applied to all users.

Disabling Local User Accounts

Disabling a local user account can happen in two ways:

- When a local user tries to log in with an invalid password for a configured number of times defined by the **Failed attempts count** parameter, the local user account is locked.

  > **NOTE**
  >
  > If the mechanism for logging in to Policy Manager Policy Manager is Certificate + Password, the local user is allowed to enter the password even if the certificate is invalid.

- When the local user tries to log in with an invalid user certificate for a configured number of times defined by the **Failed attempts count** parameter, the local user account is disabled.

  > **NOTE**
  >
  > A local user's failed login attempts are counted only when the Password_Mismatch, Password_Not_Available, and User_Authentication_Failed error messages occur.

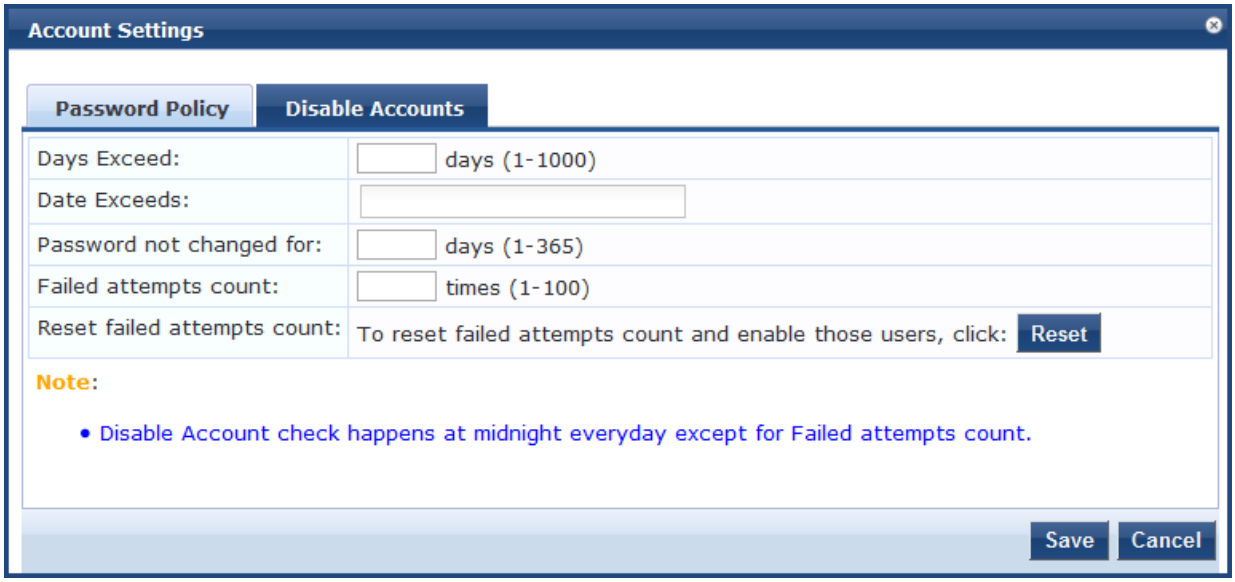
  To reset the **Failed attempts count** and enable a disabled local user account, click the **Reset** button (see Table 3). For Local users whose accounts are locked due to account settings validations, and whose accounts are enabled again after being locked out, entries are logged in both the Audit Viewer (see Audit Viewer) and the Event Viewer (see Event Viewer).

  The **Disable Account** check occurs every day at midnight, except for the **Failed attempts count**. Other local user configuration settings are applied to all local users.

  To specify the conditions for disabling local user accounts:

  1. Navigate to **Configuration > Identity > Local Users**.
  2. Click the **Account Settings** link. The **Account Settings** page opens.
  3. Select the **Disable Accounts** tab. The **Disable Accounts** dialog opens.

  Figure 1. Disable Accounts Dialog



4. Specify the **Disable Accounts** parameters as described in Table 3, then click **Save**.

Table 1. Disable Accounts Parameters

Parameter	Action/Description
Days Exceed	Specify the number of days before the account is disabled. The range is from **1** to **100** days.
Date Exceeds	Specify the date when local users are disabled when the current date exceeds the configured date. The configured date can either be the current system date or a future date. Entering a date prior to the current date is not supported.
Password not changed for	Specify the number of days allowed before the password must be changed The range is from **1** to **365** days.
Failed attempts count	Specify the number of failed log - in attempts are allowed before the account is disabled. The range is from **1** to **100** attempts.

Parameter	Action/Description
Reset failed attempts count	To reset the failed attempts count to ze ro and reenable those local users who were disabled after exceeding the faile d attempts count, click **Reset**.

Adding and Modifying Endpoints

- Viewing the List of Authentication Endpoints
- Manually Adding an Endpoint
- Modifying an Endpoint
- Viewing the Endpoint Device Fingerprint Details
- Viewing Endpoint Authentication Details
- Performing Bulk Updates of Endpoint Attributes
- Performing Bulk Deletions of Endpoint Attributes
- Triggering Actions to Be Performed on Endpoints
- Updating Device Fingerprints From a Hosted Portal
- Configuring Endpoint Context Server Actions
- Adding Vendor-Specific Endpoint Context Servers

Subtopics

Viewing the List of Authentication Endpoints
Manually Adding an Endpoint
Modifying an Endpoint
Viewing the Endpoint Device Fingerprint Details
Viewing Endpoint Authentication Details
Performing Bulk Updates of Endpoint Attributes
Performing Bulk Deletions of Endpoint Attributes
Triggering Actions to Be Performed on Endpoints
Updating Device Fingerprints From a Hosted Portal

Viewing the List of Authentication Endpoints

Policy Manager lists all authenticated endpoints. An endpoint device is an Internet-capable hardware device on a TCP/IP network (for example, laptops, smart phones, and tablets).

To access this page, navigate to **Configuration** > **Identity** > **Endpoints**.

Figure 1. Endpoints Page

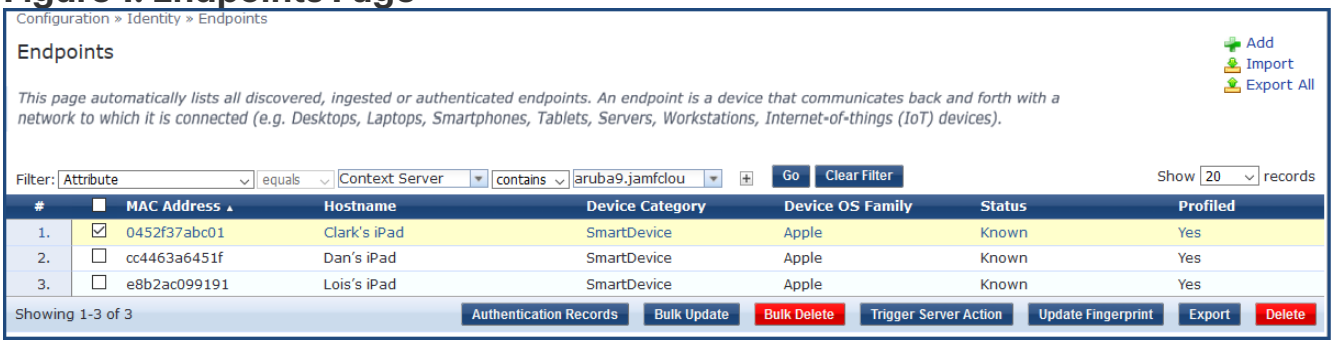


Table 1. Endpoint Page Parameters

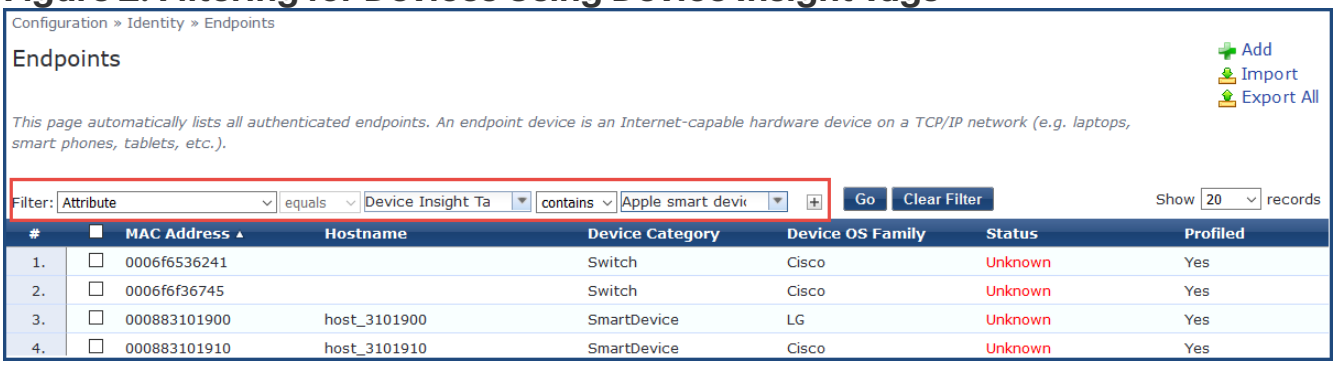
Parameter	Action/Description
Filter	You can choose to select **ALL matches** or **ANY matches**. Then you can specify from one to three d evice filters to refine the endpoint informat ion you wish to view. Available filter types include: MAC Addres s, Hostname, IPv4 Address, IPv6 Address , Device Category, Device OS Family, Device Name, Status, Profiled, Static IP, Profiler Co nflict, Added by, Description, Attribute and Randomized MAC Address. Policy Manager allows searching for a MAC address in any of the known formats: • 112233AABBCC • 11:22:33:aa:bb:cc • 11 - 22 - 33 - AA - BB - CC • 1111 - 2222 - 3333 • 1111.2222.3333
MAC Address	Displays the MAC address of the endpoint.

Parameter	Action/Description
Hostname	Specifies the host name of the endpoint.
Device Category	Indicates the category of the profiled device. For example, Access Points, Computer, Smart Device, VoIP phone, and so on.
Device OS Family	Specifies the operating system the device runs on. For example, when the category is Computer, Policy Manager shows a Device OS Family of *Windows*, *Linux*, or *macOS*.
Status	Displays the status of the endpoint: • Unknown • Known client • Unknown client • Disabled client
Profiled	Indicates whether the device has been added to the Policy Manager Profile.

Filtering for Specific Devices Using Device Insight Tags

Policy Manager allows you to use Device Insight tags to filter for specific device types, as illustrated in the following example:

Figure 2. Filtering for Devices Using Device Insight Tags



The current list of devices you can use Device Insight tags to filter are as follows:

- Apple Mac
- Apple smart device, Smart Device
- Gaming, Nintendo Gaming

- Gaming, Sony Gaming
- Kindle, Smart Device
- Smart Device
- Blackberry, Blackberry smart device, Smart Device

Manually Adding an Endpoint

About this task

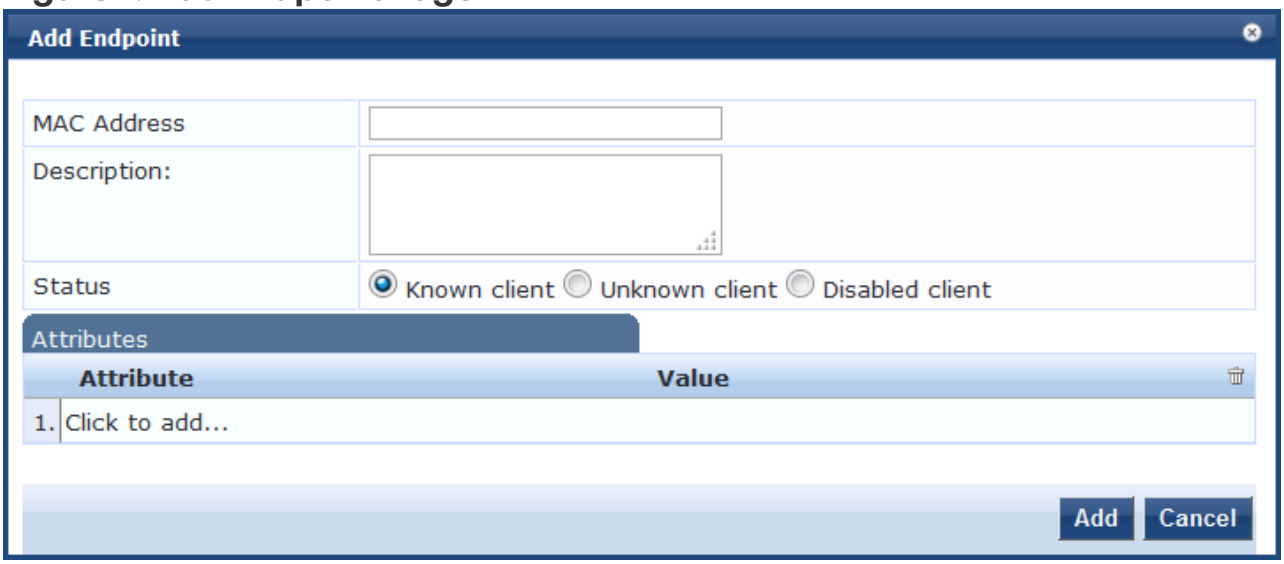
To manually add an endpoint:

Procedure

1. From the **Configuration > Identity > Endpoints** page, click the **Add** link.

   The **Add Endpoint** page opens.

Figure 1. Add Endpoint Page



2. Specify the **Add Endpoint** page parameters as described in the following table, then click **Save**:

Table 1. Add Endpoint Page Parameters

Parameter	Action/Description
MAC Address	Specify the MAC address of the endpoint .

Parameter	Action/Description
Description	Enter a description that provides additional information about the endpoint (recommended).
Status	Specify the client status as: • Known client • Unknown client • Disabled client ◦ You can use the **Known client** and **Unknown client** status in role - mapping rules by specifying the **Authentication:MacAuth** attribute. ◦ You can use the **Disabled client** status to block access to a specific endpoint. This status is automatically set when an endpoint is blocked from the**Endpoint Profiler**.
Attributes	Add custom attributes for this endpoint. Select the **Click to add...** row to add custom attributes. You can enter any name in the attribute field. All attributes are of String datatype. The *Value* field can also be populated with any string. Each time you enter a new custom attribute, it is available for selection in the **Attribute** drop - down list for all endpoints. All attributes entered for an endpoint are available in the role - mapping Rules Editor.

Support for Empty Values for Optional Endpoint Attributes

Policy Manager supports empty values for optional endpoint attributes when they are added through the user interface, an XML import, or the API:

- When empty optional attributes are added to an endpoint through the user interface at **Configuration > Identity > Endpoints**, all attribute data types are supported except *Boolean*, *Day*, *List*, and *TimeOfDay*.
- When empty optional attributes are added through an XML import or through the API, all data types are supported.
- Empty attributes are not valid if the **Is Mandatory** attribute is set to **true**.
- The *EXISTS* operator will not work in rule conditions for an attribute of the integer data type if the attribute has empty values. It does, however, work for all other data types.
- If the value of an optional endpoint attribute is set to **-For-Delete-** in the import file, the attribute is marked for delete.

Modifying an Endpoint

About this task

- Modifying an Endpoint
- Configuring the Attributes for the Selected Endpoint
- Endpoint Device Fingerprint Details Page

To modify an endpoint:

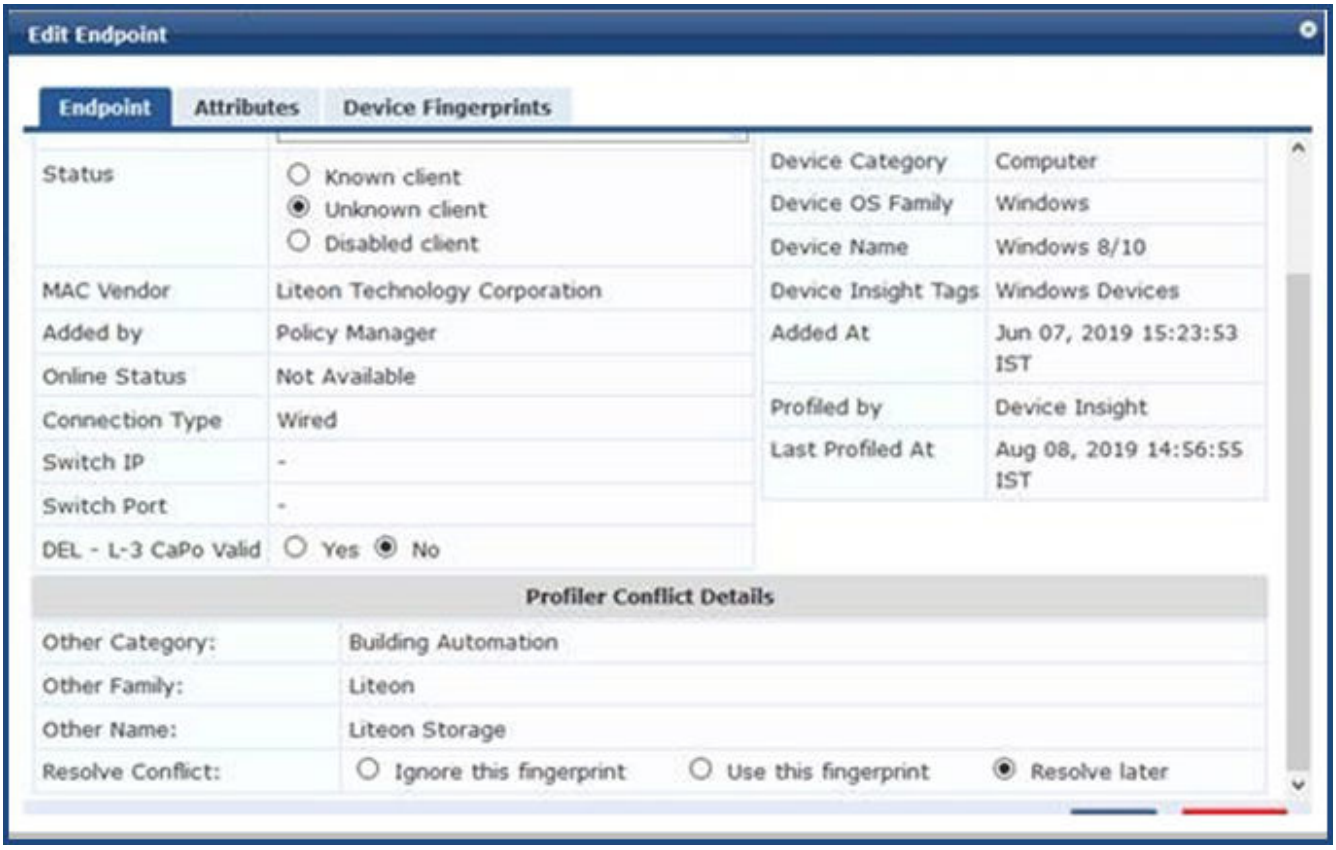
Procedure

From the **Configuration > Identity > Endpoints** page, click the endpoint of interest from the list of endpoints.

The **Edit Endpoint** page opens.

Modifying an Endpoint

Figure 1. Edit Endpoint Page



The **Edit Endpoint** page parameters are described in the following table:

Table 1. Edit Endpoint Page Parameters

Parameter	Action/Description
MAC Address	Displays the MAC address of the endpoint.
Description	Enter a description that provides additional information about the endpoint (recommended).
Status	Indicates the status of the selected endpoint as one of the following: • **Known client** ◦ You can use the **Known** client and **Unknown** client status in role - mapping rules by applying the **Authentication:MacAuth** attribute. • **Unknown client** • **Disabled client**

Parameter	Action/Description
	◦ You can use the **Disabled client** status to block access to a specific endpoint. This status is automatically set when an endpoint is blocked from the Endpoint Profiler.
MAC Vendor	Displays the MAC OUI (Organizationally Unique Identifier) information for all endpoints even when no other profiling information is available for an endpoint.
Added by	Displays the name of the admin user ID that added the endpoint.
Online Status	Displays the online status of the endpoint: • **Online** • **Offline** • **Not Available**
Connection Type	Indicates the connection type; for example, **Wireless**. If the connection type is not known, the connection type is displayed as *Unknown*.
Switch IP	Indicates the switch IP address.
Switch Port	Indicates the port on the switch.
IP Address	Displays the IP address that is associated with the endpoint.
Static IP	Indicates whether the IP address of the endpoint is a static IP address (**True** or **False**).
Hostname	Displays the host name or the IP address of the endpoint.
Device Category	From the drop - down list, select the device category that the endpoint belongs to.
Device OS Family	Specify the operating system that the endpoint runs on.

Parameter	Action/Description
Device Name	Select the name of the device from the dro p - down list.
Device Insight Tags	Displays the Device Insight tags created on this server. Device Insight tags are used to group simi lar devices together (for more information, seeAbout Device Insight).
Added At	Displays the date and time at which the en dpoint was added.
Last Profiled At	Displays the date and time at which the en dpoint was added to the Policy Manager Pr ofile.
Profiler Conflict Details	
NOTE The Profiler Conflict Details section is shown when Policy Manager detects a different Device OS Family and/or Device Category than what was previo usly specified.	
Other Category	Indicates the new *Device Category* that wa s detected but does not match.
Other Family	Indicates the new *Device OS Family* that was detected but does not match.
Other Name	Indicates the new *MAC Vendor* that was de tected but does not match.
Resolve Conflict	Select one of the following conflict resoluti on options: • *Ignore this fingerprint*: Ignore the new Device Category and/or the Device OS Family and retain the current Category and/or Family. • *Use this fingerprint*: Use the new Categ ory and/or Family and reject the previou sly profiled Category and/or Family spe cified for this device.

Parameter	Action/Description
	• *Resolve later*: Postpone the decision and let the device be in conflict.

Configuring the Attributes for the Selected Endpoint

About this task

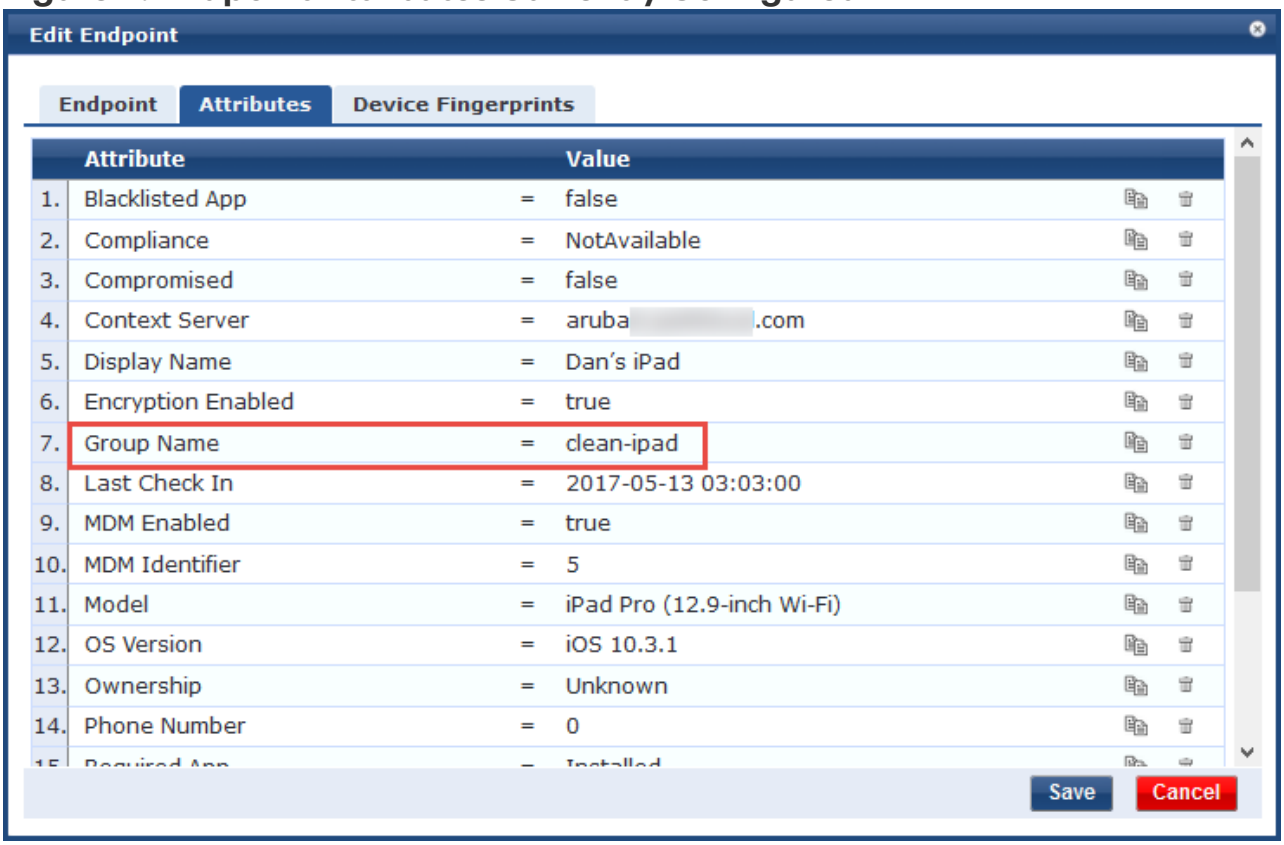
To configure the endpoint attributes for the selected endpoint:

Procedure

1. From the **Edit Endpoint** page, select the **Attributes** tab.

   The **Edit Endpoint Attributes** tab displays the set of endpoint attributes currently configured on the Policy Manager server.

Figure 2. Endpoint Attributes Currently Configured



2. To add attributes for the selected endpoint, scroll to the bottom of the list, then select **Click to add...**

   A new row is created with a drop-down list in the **Attribute** column.

3. To add an **attribute** to the endpoint, select one or more attributes from the drop-down list, then click **Save**.

JAMF Endpoint Context Server "Group Name" Attribute

For JAMF endpoint context servers, the **Group Name** endpoint attribute allows a device's contextual data to be used during session authorization to determine the device's access.

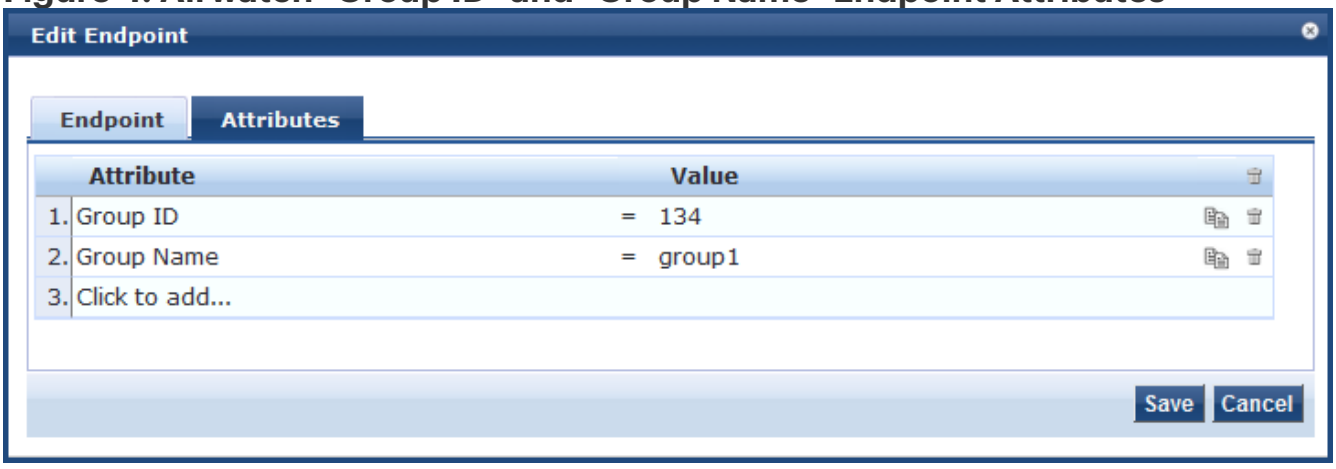
The **Group Name** endpoint attribute is dynamically retrieved from the JAMF Endpoint Context Server and made available in the Endpoint attributes list. The information is mapped to comma-separated values of the **Group Name** attribute.

Figure 3. JAMF Endpoint Context Server "Group Name" Attribute

Airwatch "Group ID" and "Group Name" Endpoint Attributes

The **LocationGroupName** and **LocationGroupID** attributes are fetched from an Airwatch server. These attributes are mapped to the corresponding **Group Name** and **Group ID** endpoint attributes.

Figure 4. Airwatch "Group ID" and "Group Name" Endpoint Attributes



Airwatch "Compliance" Attribute

The **Compliance** attribute is fetched from an Airwatch server and made available in the Endpoint attributes list. The values for the **Compliance** attribute are:

- **NotAvailable**

- **NonCompliant**
- **Compliant**

Viewing the Endpoint Device Fingerprint Details

About this task

The information displayed in the **Edit Endpoint > Device Fingerprints** page varies depending on what type of device is selected and whether an Nmap-based network discovery scan has been run (see Configuring SNMP, SSH, and WMI Credentials).

To obtain the list of processes running on the Windows endpoint:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.

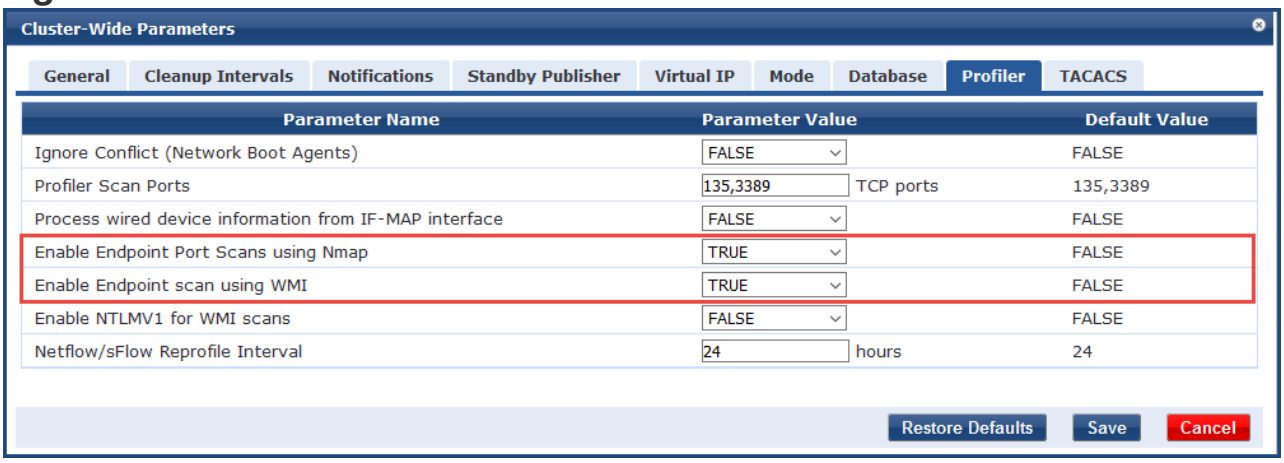
   The **Server Configuration** page opens.

2. Click the **Cluster-Wide Parameters** link.

   The **Cluster-Wide Parameters** page opens.

3. As shown in Figure 8, ensure that the Cluster-Wide parameters *Enable Endpoint Port Scans Using Nmap* and *Enable Endpoint Scan Using WMI* are set to **True**, then click **Save** (for details on these parameters, see Profiler Parameters).

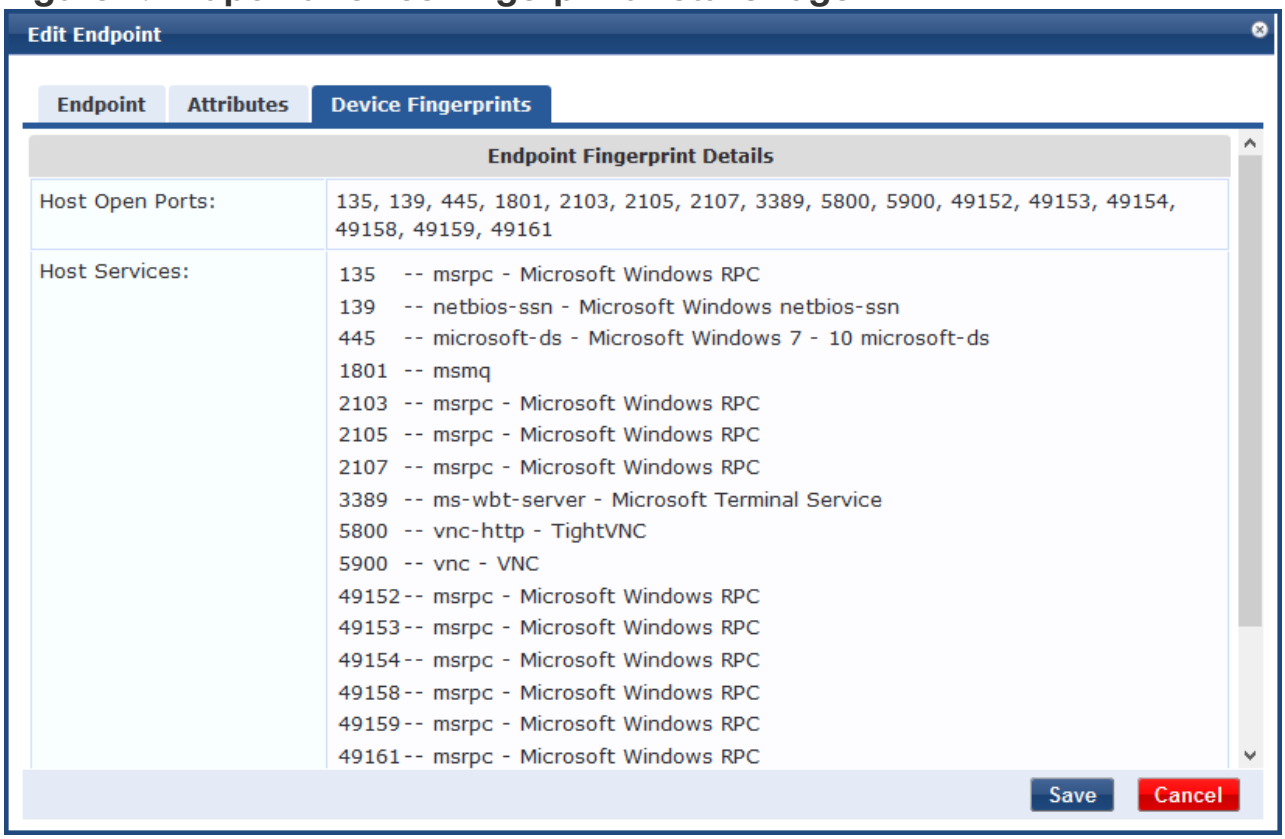
   Figure 1. Cluster-Wide Parameters > Profiler Tab



4. Navigate to the **Configuration > Identity > Endpoints** page.
5. Click the endpoint of interest from the list of endpoints.

As shown in Figure 9, the **Endpoint Fingerprints Details** page shows the open ports and services running for the selected Windows endpoint:

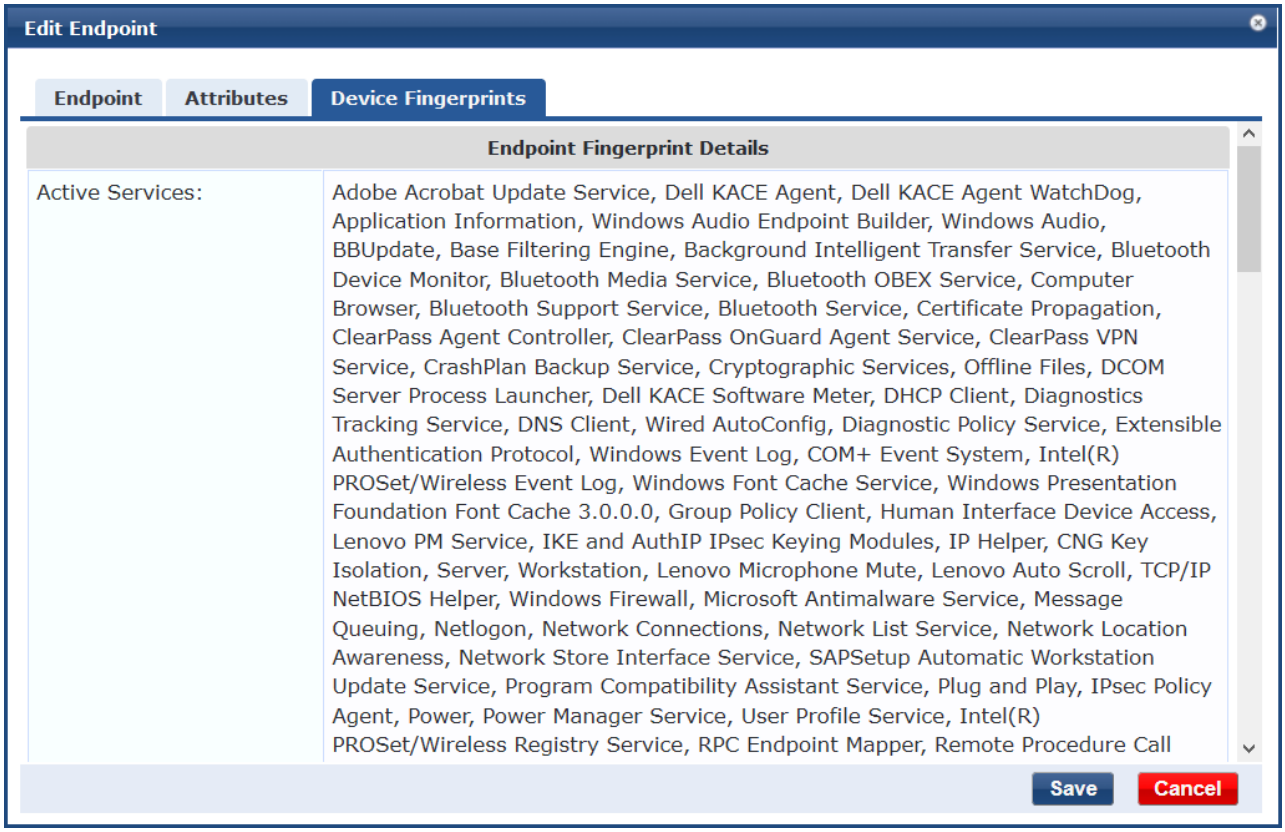
Figure 2. Endpoint Device Fingerprint Details Page



6. Additionally, to obtain the list of active and inactive Windows services on the Windows endpoint, Policy Manager Insight must be enabled on the Policy Manager server (for more information, refer to Table 1 : *Server Configuration > System Page Parameters* in System Page).

   Figure 10 shows the **Endpoint Fingerprint Details** *Active Services* information:

Figure 3. Endpoint Fingerprint Details with Active and Inactive Services Listed



In addition to the **Active Services** shown in Figure 10, the **Device Fingerprints > Endpoint Fingerprint Details** screen includes the following information:

- Host Open Ports
- Host Services
- Inactive Services
- MAC Vendor
- Open Ports
- Processes
- WMI OS Name

Viewing Endpoint Authentication Details

About this task

To view the authentication details of an endpoint:

Procedure

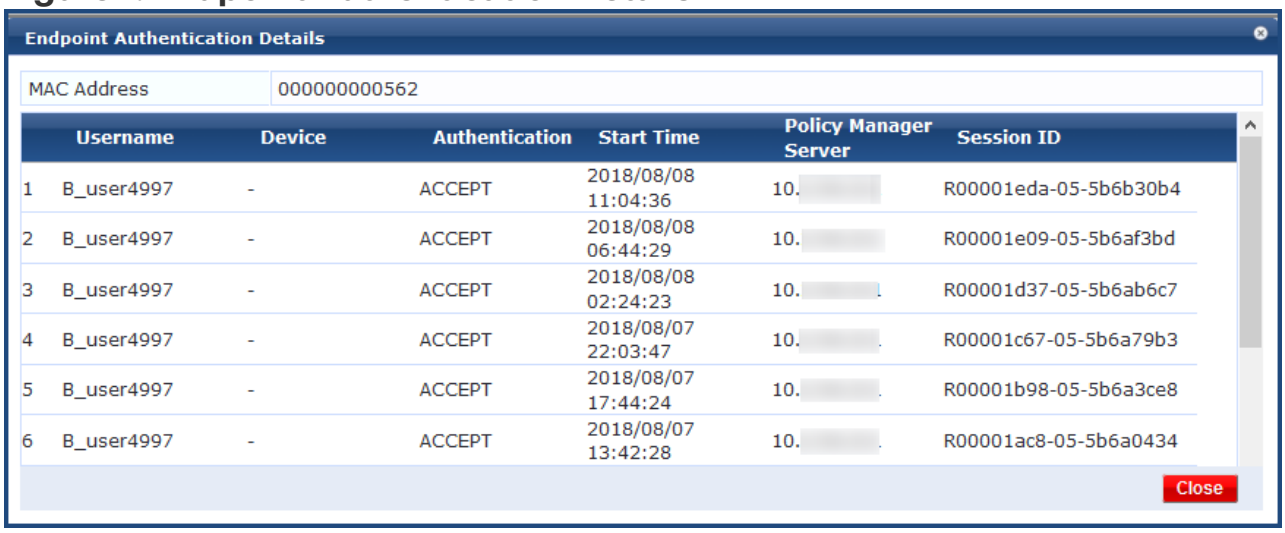
1. From the **Configuration > Identity > Endpoints** page, select an endpoint by the clicking the corresponding check box.

   The **Authentication Records** button is now enabled.

2. Click the **Authentication Records** button.

   The **Endpoint Authentication Details** page opens.

Figure 1. Endpoint Authentication Details



Performing Bulk Updates of Endpoint Attributes

About this task

You can perform bulk updates of endpoint attributes, either for a single endpoint or for multiple endpoints simultaneously.

To perform bulk updates of endpoint attributes:

Procedure

1. From the **Configuration > Identity > Endpoints** page, select one or more endpoints by the clicking the corresponding check boxes.

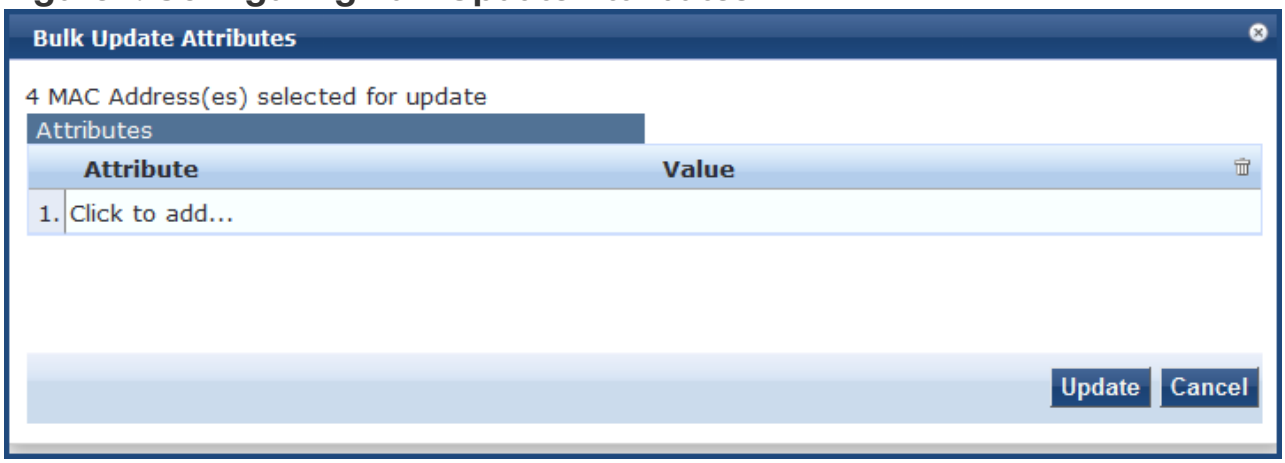
The **Bulk Update** button is now enabled.

> **NOTE**
>
> In network discovery, when endpoints do not have a MAC address, Policy Manager creates MAC addresses for them that include the prefix **xa**.

2. Click the **Bulk Update** button.

   The **Bulk Update Attributes** dialog opens.

Figure 1. Configuring Bulk Update Attributes



3. To select an attribute you want to update, select **Click to add**, select the attribute from the **Attribute** list, and then specify its **Value**.
4. Repeat the selection process for all the attributes you want to update, then click **Update**.

   The multiple attributes that were configured will be updated on all the selected endpoints at once.

Performing Bulk Deletions of Endpoint Attributes

About this task

Policy Manager supports bulk deletion of endpoint attributes.

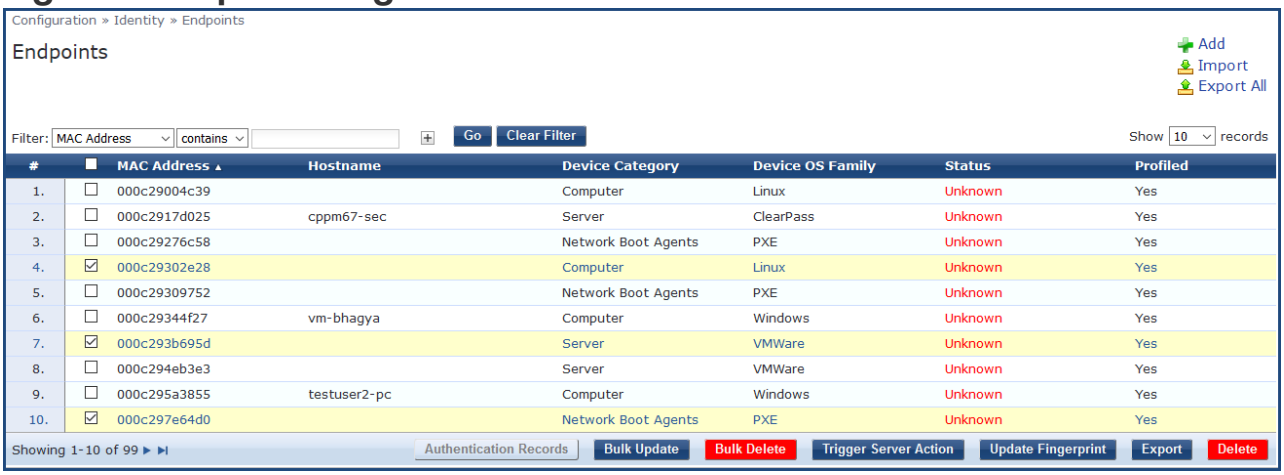
To use this feature:

Procedure

1. Navigate to **Configuration > Identity > Endpoints**.

   The **Endpoints** page opens.

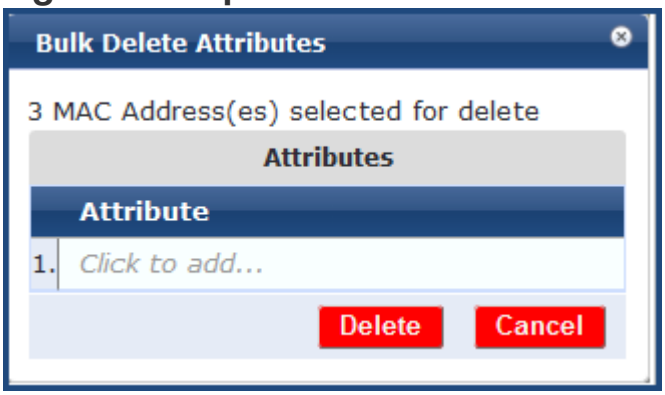
Figure 1. Endpoints Page and Bulk Delete Button



2. Select the check box(es) of one or more endpoints in the list from which you wish to delete multiple attributes.
3. Click the **Bulk Delete** button.

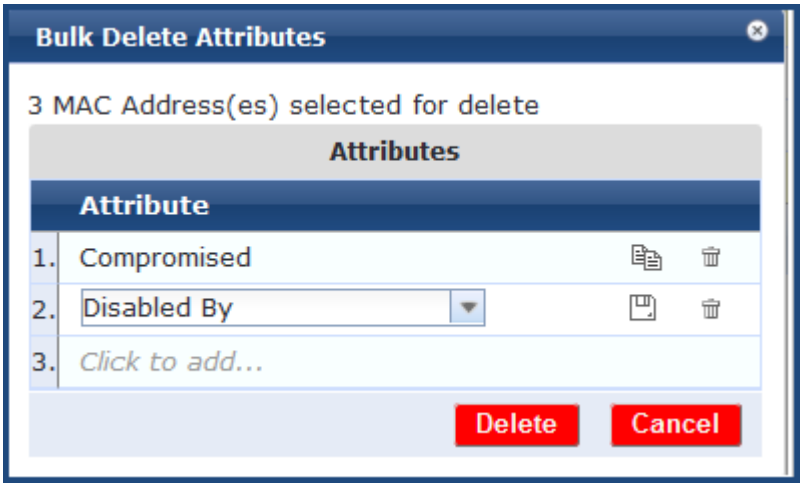
   The **Bulk Delete Attributes** dialog is displayed.

Figure 2. Endpoints: Bulk Delete Attributes Dialog



4. To select the endpoint attributes to be deleted, select the **Click to add** link.

Figure 3. Selecting Endpoint Attributes for Bulk Deletion



5. From the **Attribute** drop-down, select the attributes (one attribute per line) that you wish to delete from the specified endpoints, then click **Delete**.

Triggering Actions to Be Performed on Endpoints

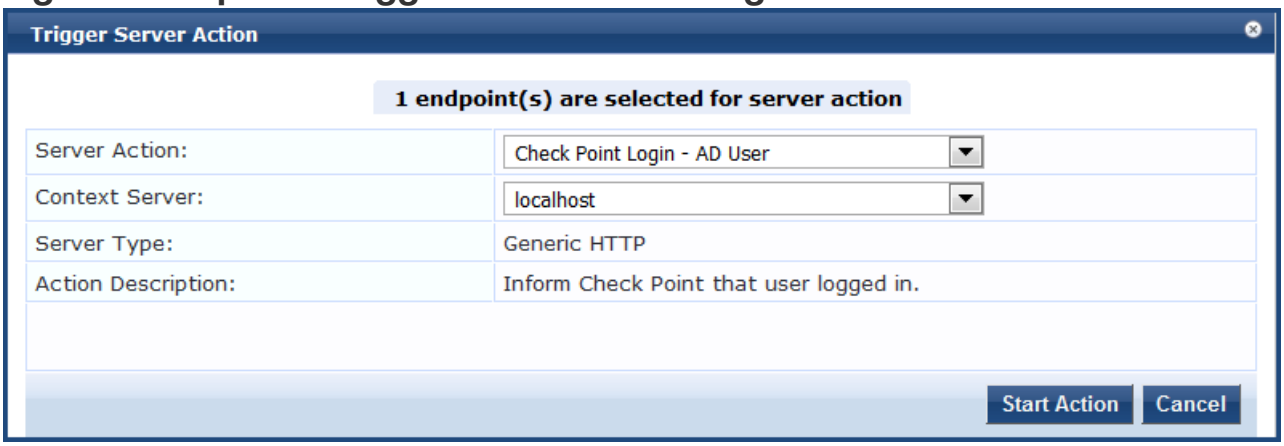
About this task

You can trigger endpoint actions for a single endpoint or for multiple endpoints simultaneously. To trigger actions that are to be performed on selected endpoints:

Procedure

1. From the **Configuration > Identity > Endpoints** page, select one or more endpoints from the **Endpoints** page by clicking the corresponding check boxes. The **Trigger Server Action** button is now enabled.
2. Click the **Trigger Server Action** button. The **Trigger Server Action** page opens.

Figure 1. Endpoints Trigger Server Action Page



3. Specify the **Trigger Server Action** page parameters as described in the following table, then click **Start Action**.

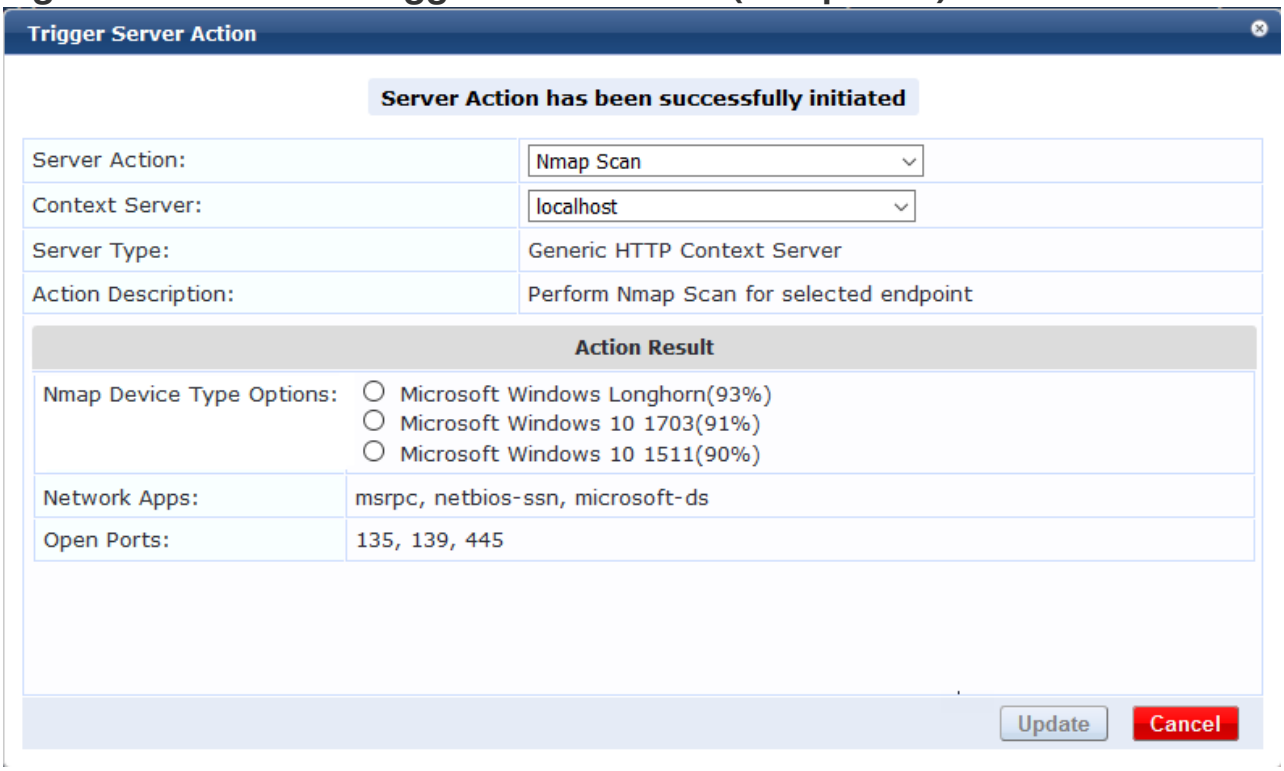
Table 1. Trigger Server Action Page Parameters

Parameter	Action/Description
Server Action	Select the server action from the drop - down list. The available server actions are as follows: • Check Point Login - AD User • Check Point Login - Guest User • Check Point Logout • Fortinet Login • Fortinet Logout • Handle AirGroup Time Sharing • Infoblox Login • Nmap Scan • OnDemand Endpoint Scan • SNMP Scan • Store Attributes in zone cache **NOTE** Starting with Policy Manager 6.10.2, the Policy Manager se rver no longer needs to have Device Insight integration di sabled in order to perform a n OnDemand Endpont Scan. In Policy Manager 6.10.0 and 6.10.1, when **Device Insight I ntegration** is enabled, the O nDemand EndPoint Scan op tion is not available. (for mo

Parameter	Action/Description
	re information, see Device Insight Integration Page).
Context Server	Enter a valid context server name. You can enter an IP address or a domain name.
Server Type	Indicates the server type specified when the server was configured (for example, *Generic HTTP Context Server*).
Action Description	Describes the action that will take place on the endpoint (for example, "Inform Check Point that user logged in.").

Once the action is complete, a popup window displays the results of the action. The following image shows the results of an Nmap scan.

Figure 2. Results of a Trigger Server Action (Nmap Scan)



Updating Device Fingerprints From a Hosted Portal

About this task

You can update device fingerprints for a single endpoint or for multiple endpoints simultaneously. If you configure custom fingerprints, the custom fingerprint always takes precedence over the dynamically pushed system fingerprint.

To update device fingerprints from a hosted portal:

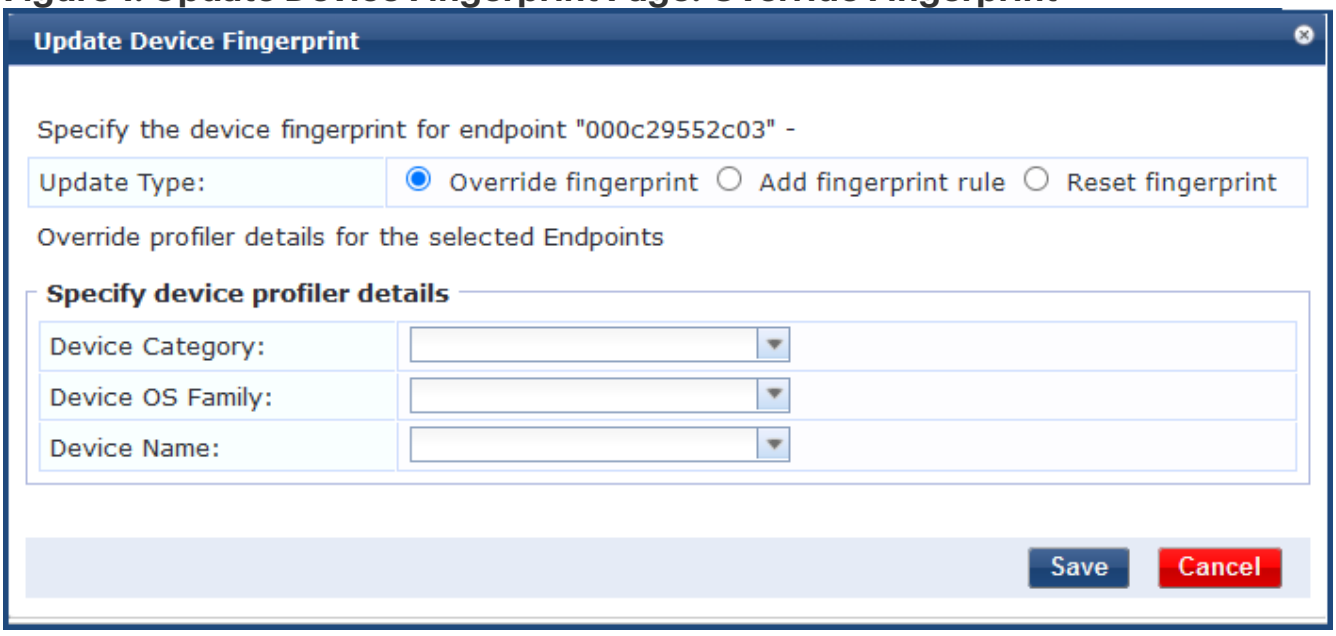
Procedure

1. From the **Configuration > Identity > Endpoints** page, select one or more endpoints by clicking the corresponding check boxes. Selecting an endpoint enables the **Update Fingerprint** button.
2. Click the **Update Fingerprint** button.
3. The **Update Device Fingerprint** page opens, displaying two update types, **Override fingerprint** and **Add fingerprint rule**.

Override Fingerprint Option

By default, the **Update Type** is set to **Override fingerprint** (see Figure 17).

Figure 1. Update Device Fingerprint Page: Override Fingerprint



NOTE
When a new fingerprint override rule is defined, an entry is added to the Audit Viewer.

Specify the **Update Device Fingerprint** page parameters as described in the following table, then click **Save**:

Table 1. Update Device Fingerprint Parameters

Parameter	Action/Description
Update Type	Select one of the following update types: • **Override fingerprint**: Update the devic e profile details (device category, devic e OS family, and device name) for the s elected endpoint. • **Add fingerprint rule**: Update the devic e profile with a new fingerprint rule. Thi s information is displayed at the bottom of the **Update Device Fingerprint** page, as shown in Figure 18. • **Reset Fingerprint**: Reset fingerprint da ta for the selected endpoints.
Specify Device Profile Details	
Device Category	Select the category the profiled device bel ongs to.
Device OS Family	Select the operating system configured on the device.
Device Name	Enter the name of the device or select the name of the device from the drop - down li st.

Add Fingerprint Rule Option

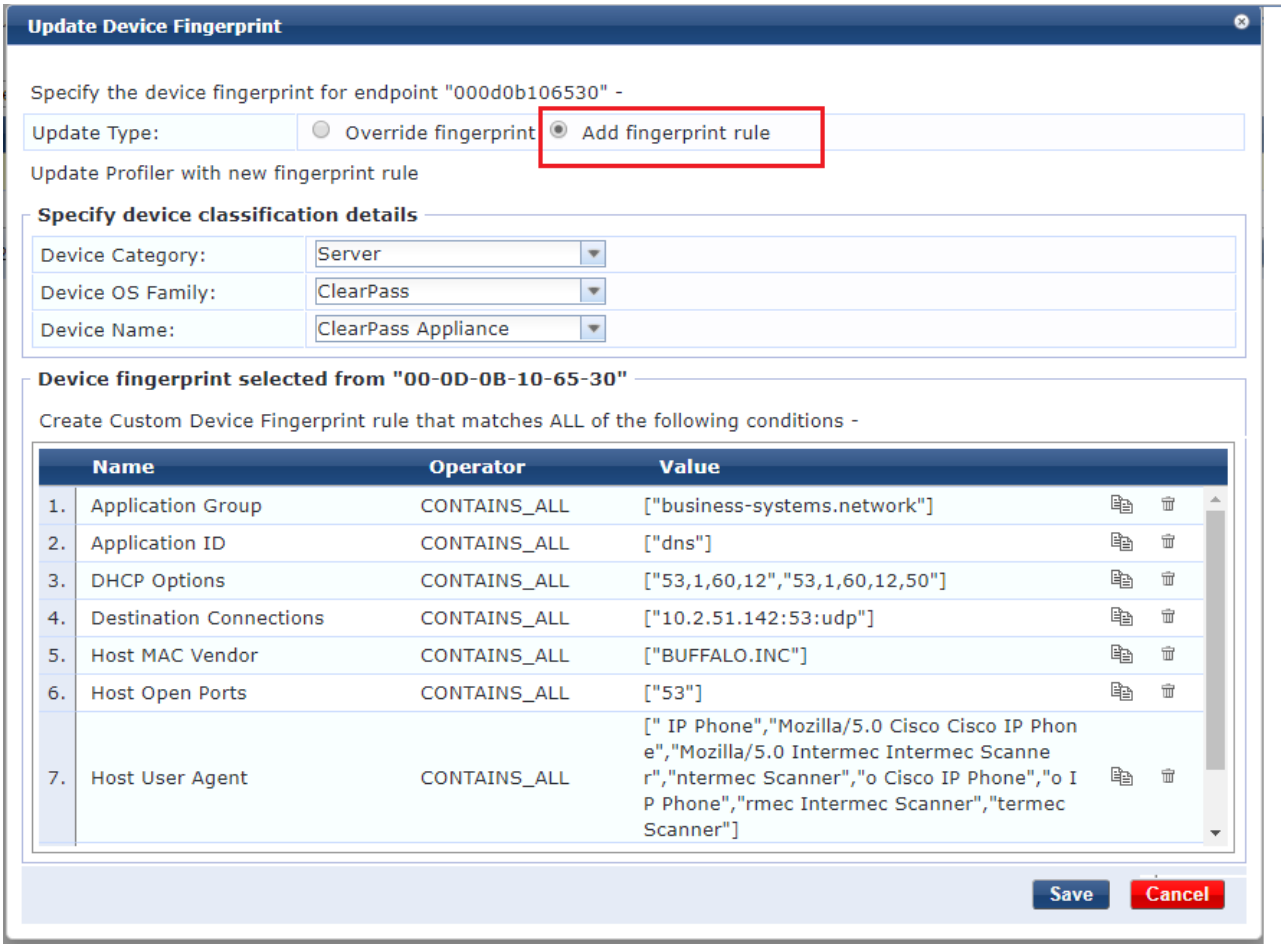
About this task

To add a fingerprint rule:

Procedure

1. In the **Update Device Fingerprint** screen, set the *Update Type* field to **Add Fingerprint Rule**. The **Device fingerprint selected from** section appears, as shown in Figure 18.

Figure 2. Update Device Fingerprint Page: Add Fingerprint Rule



2. In the **Specify device classification details** section, select device details as described in Table 5.
3. In the **Device fingerprint selected from** section, Select the fingerprint rule you want to edit. You can edit the following components of each fingerprint rule:
   - **Name**: Click the drop-down menu and select the name of a device fingerprint type.
   - **Operator**: Select from the options **Contains all**, **Contains** and **Not Contains** to allow the specified values to be included in or excluded from the rule.
   - **Value**: Add values to or delete values from the selected rule.
4. Click **Save**.

> **NOTE**
>
> When a new fingerprint rule is defined, an entry is added to the Audit Viewer.

Managing Static Host Lists

You can configure primary and backup servers, session details, and the list of static hosts for **Static Host List** authentication sources. A static host list often functions, in the context of the service, as an allowed list or blocked list. Therefore, static host lists are configured independently at the global level. A static host list comprises a named list of MAC addresses or IP addresses, which can be invoked in the following ways:

- In service and role-mapping rules as a component.
- For non-responsive services on the network (for example, printers or scanners), as an authentication source. For more information, see Adding a Static Host List as an Authentication Source.

> **NOTE**
>
> Only static host lists of type **MAC address** are available as authentication sources.

Subtopics

Internal Relational Database
Role Statically Assigned
Preconfigured Filter
Adding a Static Host List
Editing a Static Host List
Importing and Exporting Static Host Lists

Internal Relational Database

An internal relational database stores the ClearPass Policy Manager configuration data as well as locally configured user and device accounts.

The following predefined authentication sources represent the databases used to store local users, guest users, and registered devices respectively:

- [Local User Repository]
- [Guest User Repository]
- [Guest Device Repository]

While regular users reside in an authentication source such as Active Directory (or in other LDAP-compliant stores), you can configure the temporary users, including guest users, in the Policy Manager local repositories.

Role Statically Assigned

For a user account created in a local database, the role is statically assigned to that account. This means you do not need to specify a role-mapping policy for user accounts in the local database. However, if new custom attributes are assigned to a user account (local or guest) in the local database, these can be used in role-mapping policies.

Preconfigured Filter

The local user database is preconfigured with a filter to retrieve the password and the expiry time for the account. Policy Manager can perform MSCHAPv2 and PAP/GTC authentication against the local database.

Adding a Static Host List

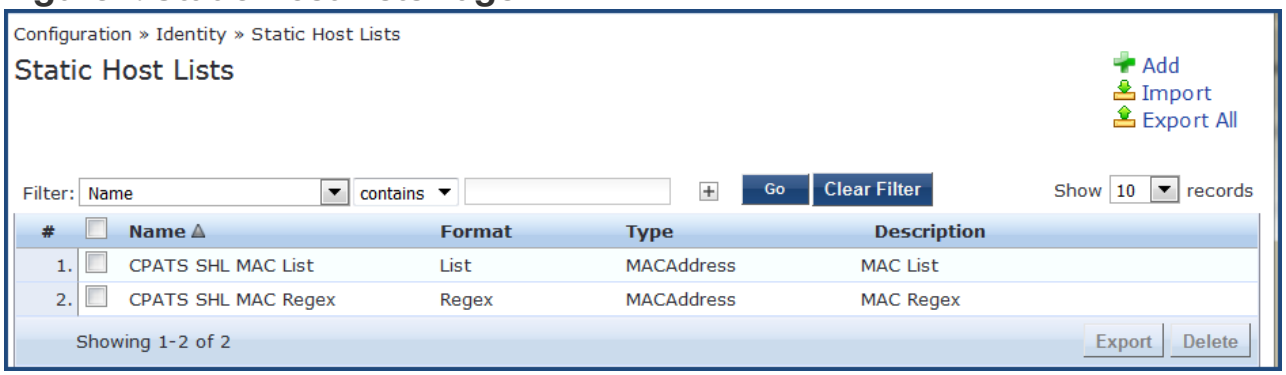
About this task

To add a static host list to Policy Manager:

Procedure

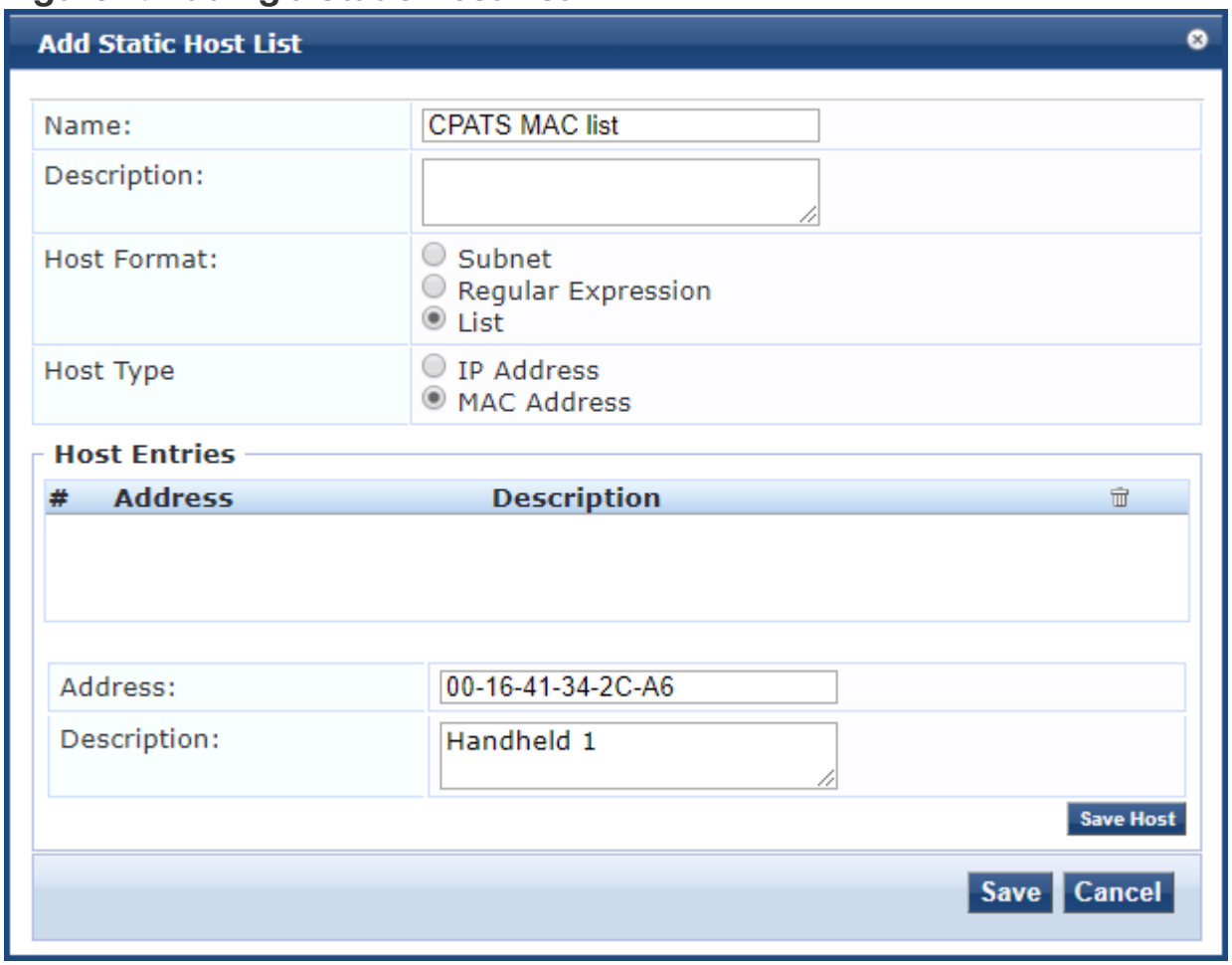
1. Navigate to the **Configuration > Identity > Static Host Lists** page. The **Static Hosts Lists** page opens.

Figure 1. Static Host Lists Page



2. Click **Add**. The **Add Static Host List** dialog opens.

Figure 2. Adding a Static Host List



3. Specify a host format:

   a. For the **Subnet** host format, specify the IP address and subnet.

   b. For the **Regular Expression** host format, select either **IP address** or **MAC address** as the host type, then enter a regular expression of the IP address or MAC address into the **Regular Expression** field.

   c. For the **List** host format, select either **IP address** or **MAC address** as the Host type, then enter the IP Address or MAC address into the **Address** field, and add a description of the host in the **Description** field. Click **Save Host** to save the host information.

4. Click **Save**.

   The new static host list is now available to be added as an authentication source. For details, see Adding a Static Host List as an Authentication Source.

Editing a Static Host List

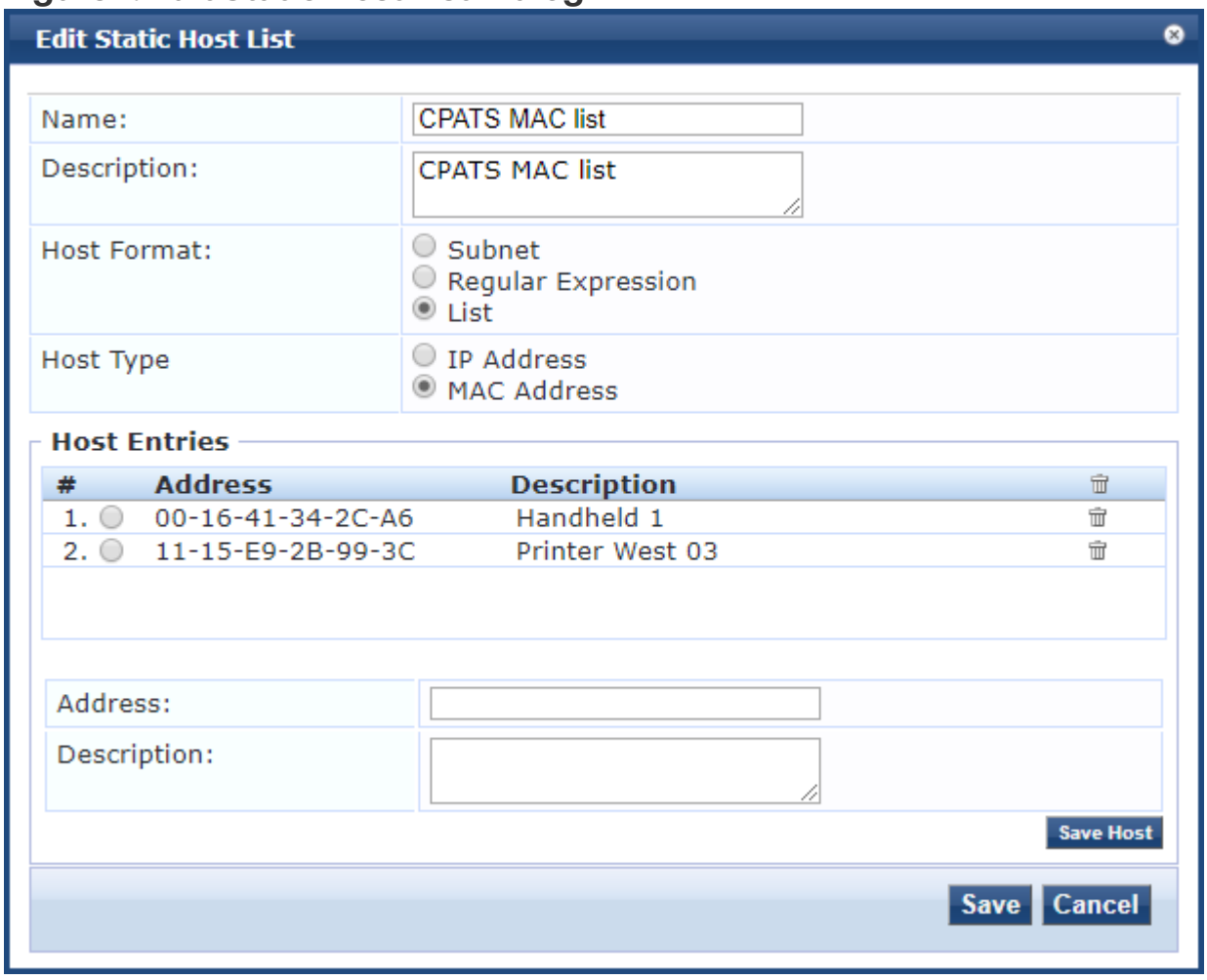
About this task

To edit a static host list:

Procedure

1. Navigate to the **Configuration > Identity > Static Host Lists** page . The **Static Hosts Lists** page opens.
2. Click on the name of the static hosts list you want to edit. The **Edit Static Host List** dialog opens.

Figure 1. Edit Static Host List Dialog



3. Make any required IP address or MAC address changes, then click **Save**.

> **NOTE**
>
> Starting with Policy Manager 6.10.2, you cannot edit the **host format** or **host type** on an existing static host list.

Importing and Exporting Static Host Lists

About this task

You can import static host lists into Policy Manager or export them to a file.

Procedure

1. Navigate to the **Configuration > Identity > Static Host Lists** page . The **Static Hosts Lists** page opens.
2. Click on the name of the static hosts list you want to import or export.
3. To import a static host list into Policy Manager, click the **Import** link.
4. To export a static host list to a file, click the **Export All** link. For further details, see Importing and Exporting Information .

Configuring a Role and Role-Mapping Policy

This section includes the following information:

- Preconfigured Roles
- Adding and Modifying Roles
- Adding and Modifying Role-Mapping Policies

After authenticating a request, a Policy Manager service invokes its role-mapping policy, resulting in assignment of a role(s) to the client. This role becomes the identity component of enforcement policy decisions.

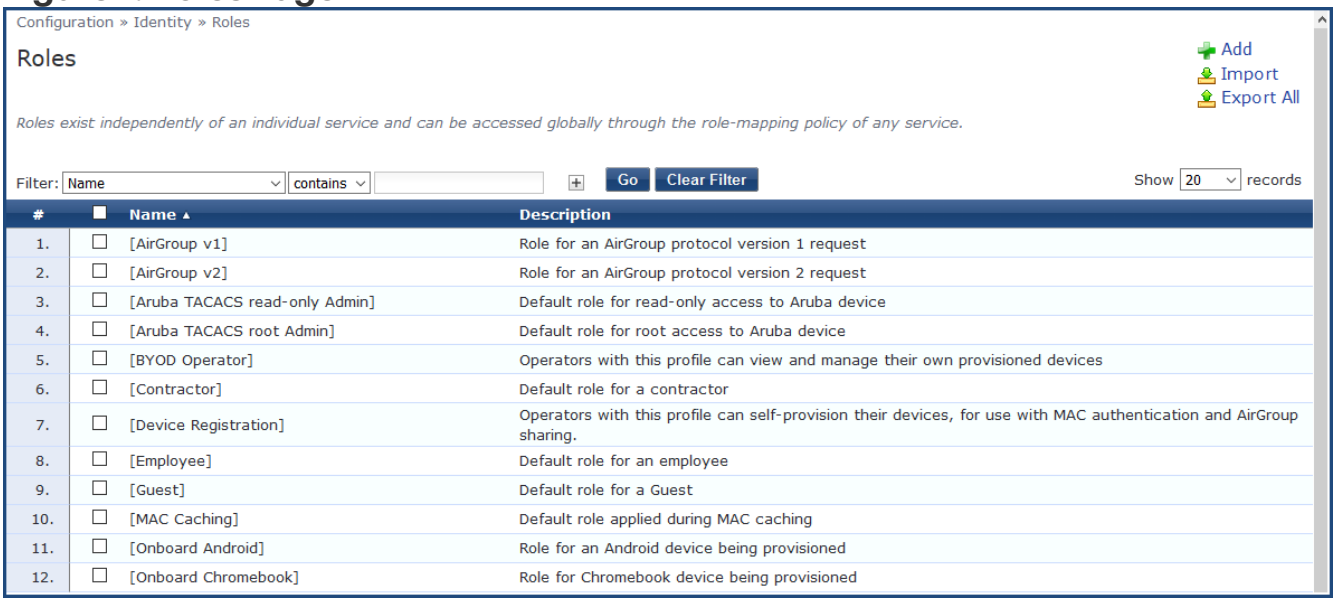
> **NOTE**
>
> A service can be configured without a role-mapping policy, but only one role-mapping policy can be configured for each service.

Preconfigured Roles

Roles exist independently of an individual service. Roles can be accessed globally through the role-mapping policy of any service.

To view the set of preconfigured roles in Policy Manager, navigate to **Configuration > Identity > Roles**. The **Roles** page opens.

Figure 1. Roles Page



Policy Manager provides the following preconfigured roles:

Table 1. Preconfigured Roles in ClearPass Policy Manager

Parameter	Action/Description
AirGroup v1	Role for an AirGroup protocol version 1 req uest
AirGroup v2	Role for an AirGroup protocol version 2 req uest

Parameter	Action/Description
Aruba TACACS+ read - only Admin	Default role for read - only access to an Ar uba device.
Aruba TACACS+ root Admin	Default role for root access to an Aruba d evice
BYOD Operator	Operators with this profile can view and m anage their own provisioned devices.
Contractor	Default role for a contractor
Device Registration	Operators with this profile can self - provisi on their devices for MAC authentication an d AirGroup sharing.
Employee	Default role for an employee.
Guest	Default role for guest access.
MAC Caching	Default role applied during MAC caching.
Onboard Android	Role for an Android devicethat is being p rovisioned.
Onboard Chromebook	Role for a ChromeOS device that is being p rovisioned.
Onboard iOS	Role for an iOS device that is being provisi oned.
Onboard Linux	Role for a Linux device that is being provisi oned.
Onboard macOS	Role for a macOS device that is being provi sioned.
Onboard Windows	Role for a Windows device that is being p rovisioned.
Other	Default role for another user or device
TACACS+ API Admin	API administrator role for Policy Manager Admin
TACACS+ Help Desk	Admin role, limited to views of the **Monitor ing** screens

Parameter	Action/Description
TACACS+ Network Admin	Policy Manager Admin role, limited to **Configuration** and **Monitoring** screens
TACACS+ Read - only Admin	Read - only administrator role for Policy Manager Admin
TACACS+ Receptionist	Policy Manager Guest provisioning role
TACACS+ Super Admin	Policy Manager Admin role with unlimited access to all user interface screens

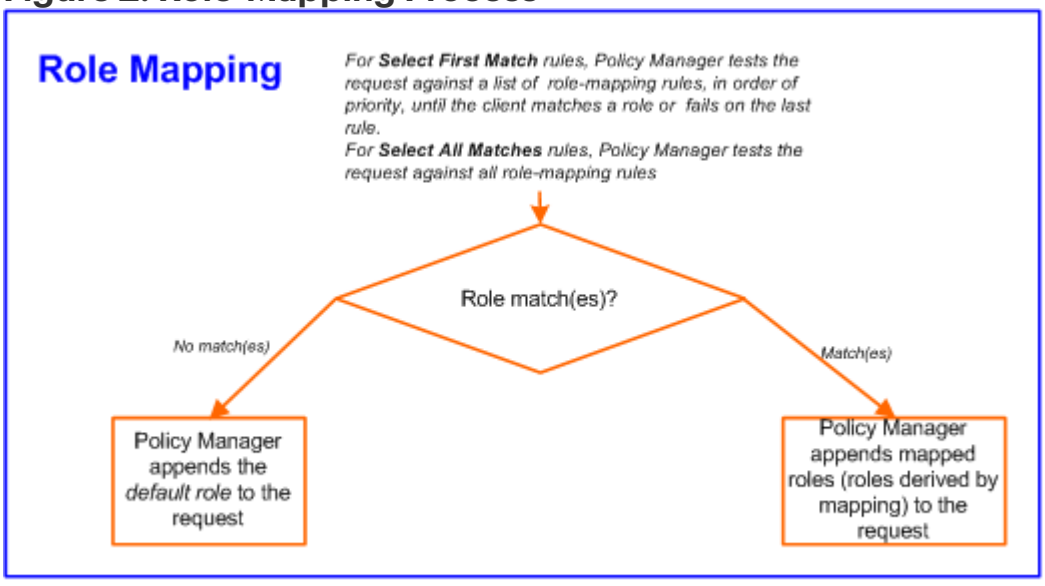
Identity Roles Architecture and Workflow

Roles can range in complexity from a simple user group (for example, Finance, Engineering, or Human Resources) to a combination of a user group with some dynamic constraints (for example, "Night Shift Worker," an employee in the Engineering department who logs in through the network device between 8:00 p.m. and 5:00 a.m on weekdays). It can also apply to a list of users. A Role-Mapping Policy reduces client (user or device) identity or attributes associated with the request to Role(s) for Enforcement Policy evaluation. The roles ultimately determine differentiated access.

For more information, see Configuring a Role and Role-Mapping Policy .

Figure 1 illustrates the role-mapping process and workflow.

Figure 2. Role-Mapping Process



A role can be:

- Authenticated through predefined single sign-on rules.
- Associated directly with a user in the Policy Manager local user database.

- Authenticated based on predefined allowed endpoints.
- Associated directly with a static host list, again through role mapping.
- Discovered by Policy Manager through role mapping.

  Roles are typically discovered by Policy Manager by retrieving attributes from the authentication source.

  Filter rules associated with the authentication source tell Policy Manager where to retrieve these attributes.
- Assigned automatically when retrieving attributes from the authentication source. Any attribute in the authentication source can be mapped directly to a role.

Configuring Enforcement Profiles and Policies

Policy Manager controls network access by sending a set of access-control attributes to the Network Access Device (NAD) from which the network-access request originates. Policy Manager sends these attributes by evaluating an enforcement policy associated with the service. Each enforcement policy comprises a rule or set of rules that matches *conditions* (role, posture, and so on) to *actions*, which are specified in enforcement profiles.

Subtopics

Configuring Enforcement Profiles
Configuring Enforcement Policies
Configuring Deep-Nested Active Directory Queries

Configuring Enforcement Profiles

You can configure Policy Manager enforcement profiles globally, but they must be referenced to an enforcement policy that is associated with a service. Policy Manager includes the following enforcement profiles by default.

Table 1. Default Enforcement Profiles

Enforcement Profile	Type	Description
[Aerohive - Terminate Session]	RADIUS_CoA	System - defined profile to disconnect user (Aerohive).

Enforcement Profile	Type	Description
[AirGroup Personal Device]	RADIUS	System - defined profile for an AirGroup personal devic e request.
[AirGroup Response]	RADIUS	System - defined profile for any AirGroup request.
[AirGroup Shared Device]	RADIUS	System - defined profile for an AirGroup shared device request.
[Allow Access Profile]	RADIUS	System - defined profile to allow network access.
[AOS - CX - Bounce Switch Port]	RADIUS_CoA	System - defined profile to bounce the switch port on AOS - CX switches.
[AOS - CX - Disconnect]	RADIUS_CoA	System - defined profile to disconnect a device on AOS - CX switches.
[Allow Application Access P rofile]	Application	System - defined profile to allow access to an applicati on.
ArubaOS Switching - Boun ce Switch Port	RADIUS_CoA	System - defined profile to bounce the switch port on A rubaOS Switching products.
ArubaOS Switching - Termi nate Session	RADIUS_CoA	System - defined profile to disconnect the user on Aru baOS Switching, HP ProCu rve, and HP UWW (Unified Wired - WLAN) products.
ArubaOS Wireless - Bounc e Switch Port	RADIUS_CoA	System - defined profile to bounce the switch port on A rubaOS Mobility controllers, Multi - Port APs, and Mobilit y Access Switches.
[ArubaOS Wireless - TACAC S+ Read - Only Access]	TACACS+	System - defined profile for TACACS+ read - only access on ArubaOS Mobility contro llers, Aruba Instant APs, and Mobility Access Switches.

Enforcement Profile	Type	Description
[ArubaOS Wireless - TACACS+ root Access]	TACACS+	System - defined profile for TACACS+ root access on ArubaOS Mobility controllers, Aruba Instant APs, and Mobility Access Switches.
[Aruba Wireless - Terminate Session]	RADIUS_CoA	System - defined profile to disconnect the user on ArubaOS Mobility controllers, Aruba Instant APs, and Mobility Access Switches.
[Cisco - Bounce - Host - Port]	RADIUS_CoA	System - defined profile to bounce the host port (Cisco).
[Cisco - Disable Host - Port]	RADIUS_CoA	System - defined profile to disable the host port (Cisco).
[Cisco - Reauthenticate - Session]	RADIUS_CoA	System - defined profile to re - authenticate session (Cisco).
[Cisco - Terminate - Session]	RADIUS_CoA	System - defined profile to disconnect a user (Cisco).
[Deny Access Profile]	RADIUS	System - defined profile to deny network access.
[Deny Application Access Profile]	Application	System - defined profile to deny access to an application.
[Drop Access Profile]	RADIUS	System - defined profile to drop the request.
[H3C - Bounce Switch Port]	RADIUS_CoA	System - defined profile to bounce the switch port on H3C products (including HPE FlexNetwork/Comware)
[H3C - Disable Switch Port]	RADIUS_CoA	System - defined profile to disable the switch port on H3C products (including HPE FlexNetwork/Comware).

Enforcement Profile	Type	Description
[H3C - Terminate Session]	RADIUS_CoA	System - defined profile to disconnect the user on H3 C products (including HPE F lexNetwork/Comware).
[Handle AirGroup Time Shar ing]	HTTP	System - defined profile to s end time - based sharing po licy to the AirGroup notificat ion service.
[Juniper Terminate Session]	RADIUS_CoA	System - defined profile to disconnect a user (Juniper).
[Motorola - Terminate Sess ion]	RADIUS_CoA	System - defined profile to disconnect a user (Motorola).
[Operator Login - Admin Users]	Application	Enforcement profile for Gu est admin logins.
[Operator Login - Local Us ers]	Application	Enforcement profile for Gu est operator logins.
Registered Device MPSK	RADIUS	Enforcement profile for Mult iple Pre - Shared Key (MPSK) Device Registration. Returns a device's assigned MPSK th at was generated automatic ally during Device Registrati on.
[Return Device Sponsor Na me - RADIUS User - Name]	RADIUS	Returns the [Guest Devic e Repository]:SponsorName value as the RADIUS:IETF:Us er - Name value when Policy Manager is configured to us e Multiple Pre - Shared Key (MPSK) authentication.
[TACACS+ API Admin]	TACACS+	API admin access for Policy Manager Policy Manager Ad min.
[TACACS+ Deny Profile]	TACACS+	System - defined profile to deny network access.

Enforcement Profile	Type	Description
[TACACS+ Help Desk]	TACACS+	Help desk access for Policy Manager Policy Manager Admin.
[TACACS+ Network Admin]	TACACS+	Network admin access for Policy Manager Policy Manager Admin.
[TACACS+ Read - only Admin]	TACACS+	Read - only admin access for Policy ManagerPolicy Manager Admin.
[TACACS+ Receptionist]	TACACS+	Receptionist access for Policy ManagerPolicy Manager Admin
[TACACS+ Super Admin]	TACACS+	Super admin access for Policy Manager Policy Manager Admin.
[Trapeze - Terminate Session]	RADIUS_CoA	System - defined profile to disconnect a user (Trapeze).
[Update Endpoint Known]	Post - Authentication	System - defined profile to change an Endpoint's status to *Known*.

NOTE

The **Framed-IPv6-Prefix** attribute can now be set or sent from a ClearPass Enforcement-Profile. The **Framed-IPv6-Prefix** attribute indicates the IPv6 prefix (and corresponding route) configured for a user, and is often used in an ISP environment, where a AAA server enforces a mobile gateway's allocation of IPv6 addresses within a prefix range. The **Framed-IPv6-Prefix** attribute displays the IPv6/Prefix-Length in readable format and is used in the Service Rule/Enforcement profile.

Subtopics

Adding a New Enforcement Profile
Modifying an Existing Enforcement Profile
Deleting an Existing Enforcement Profile
AOS-CX Management Access via RADIUS
Agent Enforcement Profile
Agent Script Enforcement Profile

Aruba Downloadable Role Enforcement Profiles
Aruba RADIUS Enforcement Profile
Cisco Downloadable ACL Enforcement Profile
Cisco Web Authentication Enforcement Profile
Policy Manager Entity Update Enforcement Profile
CLI-Based Enforcement Profile
Filter ID Based Enforcement Profile
Generic Application Enforcement Profile
HTTP Based Enforcement Profile
RADIUS Based Enforcement Profile
RADIUS Dynamic Authorization Enforcement Profile
Session Notification Enforcement Profile
Session Restrictions Enforcement Profile
SNMP-Based Enforcement Profile
TACACS+ Based Enforcement Profile
VLAN Enforcement Profile

Adding a New Enforcement Profile

About this task

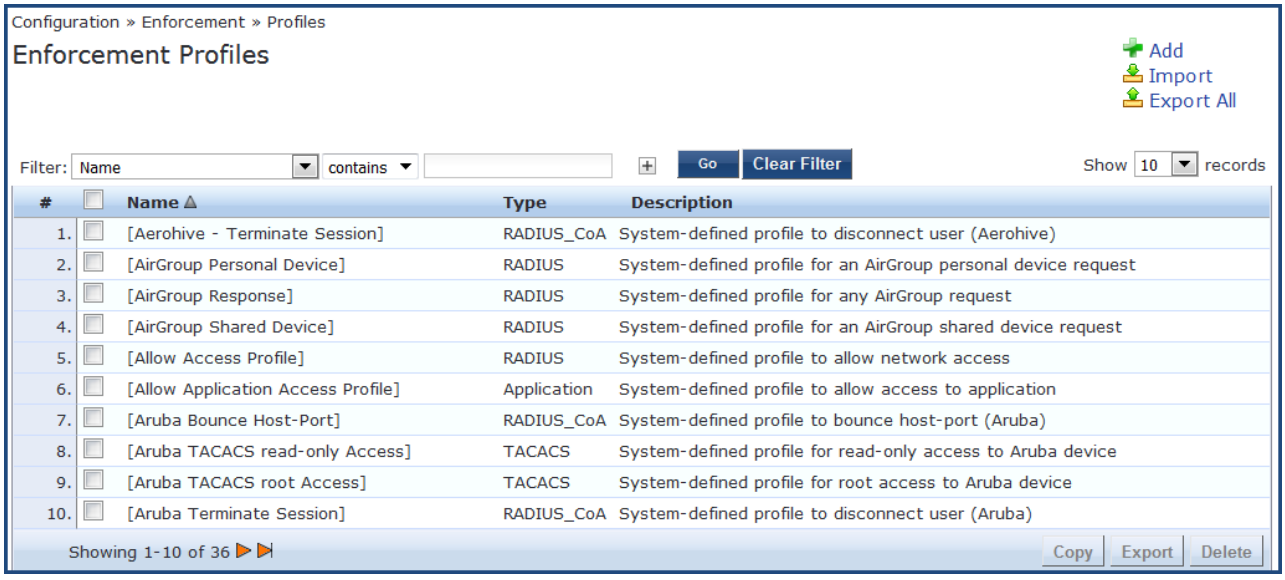
Each enforcement policy contains enforcement profiles that match conditions (role, posture, and time) to actions (enforcement profiles).

To create an enforcement profile:

Procedure

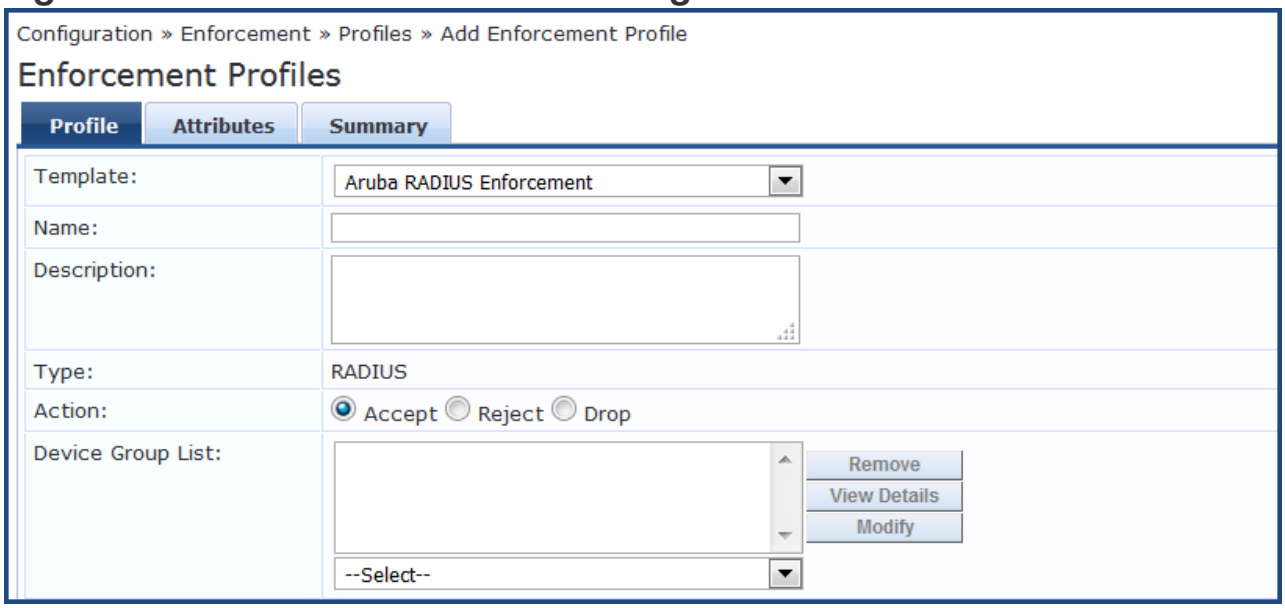
1. Navigate to **Configuration > Enforcement > Profiles**. The **Enforcement Profiles** page opens:

Figure 1. Enforcement Profiles Page



2. Click the **Add** link. The **Add Enforcement Profile** dialog opens.

Figure 2. Add Enforcement Profile Dialog



Select any of the following enforcement profile templates to create a profile based off of that template type.

Table 1. Enforcment Profile templates

Template	Description
Aruba Downloadable Role Enforcement Profiles	When Policy Manager successfully authenticates a user, the user is assigned a role by Policy Manager. However, if the role is not defined on the Aruba controller or sw

Template	Description
	itch, you can use downloadable role enfo rcement profiles to allow the role attribut es to be downloaded automatically.
Aruba RADIUS Enforcement Profile	Define an enforcement profile based on Aruba vendor - specific RADIUS attribut es.
Cisco Downloadable ACL Enforcement Pr ofile	Integrate a Cisco switch with Policy Mana ger by defining a Cisco Downloadable Ac cess Control List (dACL) profile based o n Cisco vendor - specific RADIUS attribut es.
Cisco Web Authentication Enforcement P rofile	Integrate a Cisco switch with Policy Man ager by defining a Web authentication en forcement profile based on Cisco vendor - specific RADIUS attributes.
Filter ID Based Enforcement Profile	Define an enforcement profile based on R ADIUS Internet Engineering Task Force (IE TF) attributes
RADIUS Based Enforcement Profile	Define an enforcement profile based on values from any of the following vendor - specific RADIUS attribute types. • Radius:Aruba • Radius:IETF • Radius:Cisco • Radius: Hewlett - Packared - Enterpris e • Radius: Lucent - Alcatel - Enterprise • Radius:Microsoft • Radius:Avenda
RADIUS Dynamic Authorization Enforcem ent Profile	RADIUS dynamic authorization enforcem ent profile configuration pages contain a large variety of templates for different act

Template	Description
	ions that are automatically populated wit h default RADIUS settings and values app ropriate for that template type.
VLAN Enforcement Profile	Define an enforcement profile that assign s VLAN settings.
Agent Enforcement Profile	Agent Enforcement profiles allows Policy Manager to define actions to be execute d by OnGuard agents.
Agent Script Enforcement Profile	Agent Script Enforcement profiles allows Policy Manager to execute custom script s on endpoint devices as part of agent en forcement.
CLI - Based Enforcement Profile	Agent Script Enforcement profiles allows Policy Manager to execute CLI command s on endpoint devices as part of agent en forcement.
Policy Manager Entity Update Enforceme nt Profile	Entity Update enforcement profiles can p ush endpoint information or status updat es to devices after they have been authe nticated.
Generic Application Enforcement Profile	Define an enforcement profile for an appl ication.
HTTP Based Enforcement Profile	Define an HTTP - based enforcement poli cy.
SNMP - Based Enforcement Profile	Define SNMP - based enforcement profil es with attributes for a VLAN ID or sessio n timeout period, or to reset the connecti on.
Session Notification Enforcement Profile	Use a Session Notification Enforcement p rofile to send notification of a change in IP address to any external context server (such as a firewall) by configuring that se

Template	Description
	rver as a generic HTTP server and adding the appropriate generic HTTP context se rver actions.
Session Restrictions Enforcement Profile	If OnGuard Agent is disconnected, Policy Manager uses Keep - Alive messages to i ssue Disconnect Messages for a Session Restrictions Enforcement Profile for agen t - based enforcement.
TACACS+ Based Enforcement Profile	Define TACACS+ - based enforcement pr ofiles

Modifying an Existing Enforcement Profile

About this task

To modify an existing enforcement profile:

Procedure

1. Navigate to the **Configuration > Enforcement > Profiles** page.
2. Select a profile name from the profiles list, then click **Edit** or **Delete**
3. Make the necessary changes in the **Profile** and **Attributes** dialogs, then click **Save**.

Deleting an Existing Enforcement Profile

About this task

To delete an existing enforcement profile:

Procedure

1. Navigate to the **Configuration > Enforcement > Profiles** page.
2. Click the checkbox by a profile name from the profiles list, then click **Edit** or **Delete**

3. Make the necessary changes in the **Profile** and **Attributes** dialogs, then click **Save**.

AOS-CX Management Access via RADIUS

Going forward, Policy Manager includes AOS-CX management access using RADIUS on AOS-CX versions 10.11 and above. User access is granted by the Vendor-specific attributes values sent over Access-Accept from Policy Manager. Vendor-specific attributes are not sent for denied users.

Vendor-specific attributes that must be configured to grant user access include:

- Aruba-User-Management-Interfaces - Supported AOS-CX interfaces include SSH, Console, Telnet, and HTTPS-Server
- Aruba-Priv-Admin-User - The privilege level granted to the user

Agent Enforcement Profile

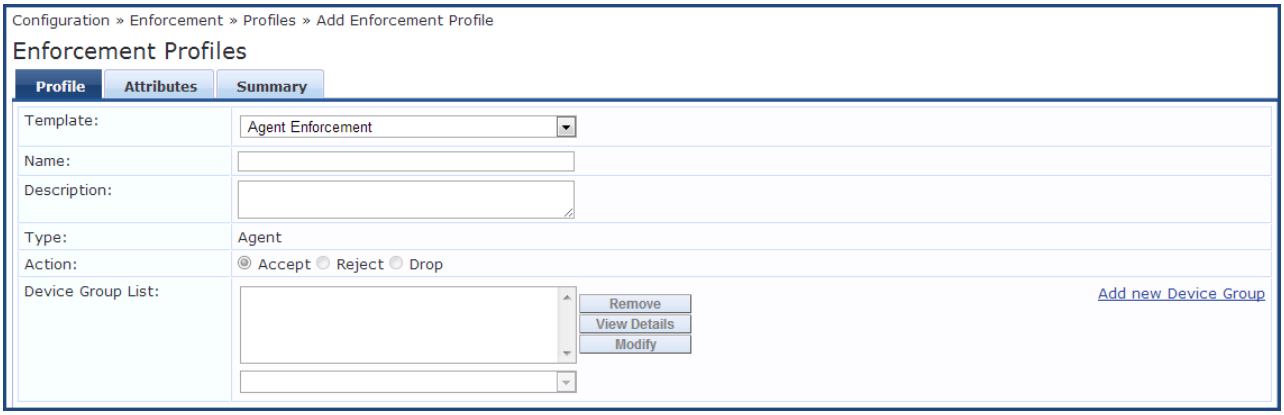
About this task

To configure profile and attribute parameters for an Agent Enforcement profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**. The **Enforcement Profiles** page opens.
2. Click **Add**. The **Add Enforcement Profiles > Profile** tab opens.
3. From the **Template** drop-down, select **Agent Enforcement**. The **Agent Enforcement > Profile** dialog opens.

Figure 1. Agent Enforcement > Profile Tab



4. Specify the **Add Agent Enforcement > Profile** parameters as described in the following table:

Table 1. Add Agent Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select **Agent Enforcement** from the drop - down list.
Name	Enter the name of the enforcement profil e.
Description	Optionally, enter a description of the en forcement profile (recommended).
Type	This field is populated automatically with type **Agent**.
Action	By default, this field is disabled. It is enab led only when **RADIUS** type is specified.
Device Group List	Select a device group from the drop - d own list. The list displays all configured d evice groups. All configured device groups are listed in the **Configuration > Network > Device G roups** page. After you add one or more d evice groups, you can select a group and take one of the following actions:

Parameter	Action/Description
	• To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**. • To change the parameters of the selected device group, click **Modify**.
Add New Device Group	To add a new device group, click the **Add New Device Group** link. For more information, see Adding and Modifying Device Groups .

5. Click **Next**.

   The **Attributes** tab opens.

Configuring Agent Enforcement Attributes

About this task

Use the **Attributes** tab to configure the attribute name and attribute value for each attribute you add.

Figure 2. Agent Enforcement > Attributes Dialog

Procedure

1. Specify the **Agent Enforcement > Attributes** parameters as described in the following table:

Table 2. Agent Enforcement > Attributes Tab Parameters

Attribute Name	Action/Description
Bounce Client	To bounce the network interface and ter minate the network connection, set the value to **True**.
Message	Enter the message that needs to be displ ayed on the endpoint.
Session Timeout (in seconds)	Configure the agent session timeout inte rval to periodically evaluate the endpoint 's health. OnGuard Agent performs health checks after the specified session timeout inte rval and updates the health status of the endpoint in Policy Cache. The default val ue is **0**. **NOTE** Setting the lower value for t he session timeout interval results in numerous authenti cation requests in the **Acces s Tracker** page.
Health Check interval (in hours)	Specify the health - check interval value in hours for different Agent Enforcement Profiles for different users. The allowed ra nge is **0** to **1000** hours. **NOTE** The value of the **Policy resul t cache timeout** parameter (found in *Server Configuratio n > Cluster - wide Paramete rs*) must be greater than the highest value of all the **Heal**

Attribute Name	Action/Description
	th Check Interval (in hours) values. Note the following information: • You can set the **Health Check Interval** if OnGuard mode is set to **Health only** . • This parameter is valid only for *wired* and *wireless* interface types. • This parameter is not applicable for th e *OnGuard Dissolvable Agent*, *VPN*, a nd *Other* interface types.
Enable to hide Retry button	To hide the **Retry** button in the OnGuard Agent, click the check box to set the valu e to **True**.
Enable to hide Logout button	To hide the **Logout** button in the OnGua rd Agent, click the check box to set the value to **True**.
Enable to hide Quit option	To hide all Quit options in OnGuard Agen t, click the check box to set the value to **True**.
Bounce Delay (in seconds)	When **Bounce Delay** is configured, the n etwork interface is bounced after the spe cified delay.
Show Custom UI for Custom Scripts	Click the check box to set this attribute t o **True** and enable the OnGuard Agent Re mediation User Interface for Custom Scri pts (for related information, refer to the **C ustom User Interface** parameter in Creat ing OnGuard Custom Web Pages).
SDK Type	Allows you to specify the V4 OnGuard D etection SDK (for related information, see

Attribute Name	Action/Description
	Upgrading From OnGuard Plugin Version 1.0 to 2.0).
Enable to auto update OnGuard Agent Library	When this option is enabled and a new version of OnGuard Agent Library is available on the Policy Manager server, Policy Manager OnGuard Agent automatically downloads and installs the new version of OnGuard Agent Library from the Policy Manager server. OnGuard modules, including detection libraries for client programs (V4 SDK), can be upgraded without having to upgrade your Policy Manager or OnGuard installations (for more information, see OnGuard Agent Library Updates).
Show Posture results in Guest page	To have OnGuard show the user posture results in a web page when the client is Unhealthy, click the check box to set this attribute to **True**.
Enable Evil Twin Detection	To enable OnGuard Agent to check for the presence of an "evil twin" with a wired interface, click the check box to set this attribute to **True**. When OnGuard Agent detects an evil twin on a wired interface, it sends a WebAuth request with the *Host:EvilTwin* attribute having the value "*MayExist*," indicating that an evil twin may exist. The *Host:EvilTwin* attribute is available in Service rules, Role Mapping, and Enforcement profiles. For more information, see Evil Twin Detection with OnGuard.
Enable to bounce client if Evil Twin is detected	This attribute allows administrators to bounce the client interface if an evil twin (rogue AP) is detected, enabling the client t

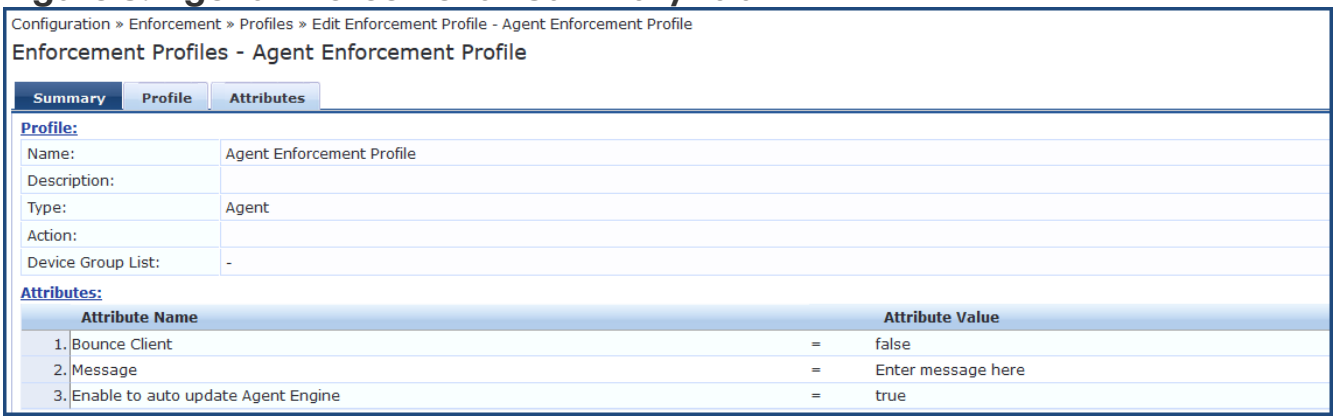
Attribute Name	Action/Description
	o re - authenticate and establish a conne ction to the server to report the evil twin. To use this feature, add both the **Enable Evil Twin Detection** and the **Enable to bo unce client if Evil Twin is detected** attrib utes, and set the value to **true** for each o f them. OnGuard will only bounce the int erface if both attributes are enabled with their values set to **true**.

2. Click **Save**.

Summary Information

The **Summary** tab summarizes the parameters configured in the **Profile** and **Attribute** tabs.

Figure 3. Agent Enforcement > Summary Tab



Agent Script Enforcement Profile

Agent Script Enforcement profiles allows Policy Manager to execute custom scripts on endpoint devices as part of agent enforcement. All the details of custom script configuration, such as the path of the custom script, the command to be executed, execution level, and so on, are configured in the Agent Script Enforcement profile. You can select multiple Agent Script Enforcement Profiles in a rule in an enforcement policy. OnGuard Agent executes them one after another.

OnGuard Agent applies the Agent Script Enforcement profile (that is, it executes a custom script) after first applying Agent Enforcement profiles (that is, after **Agent**

Bounce is executed, if configured). While applying an Agent Script Enforcement profile, OnGuard Agent does not check to see if a script is already running. It is possible for OnGuard Agent to launch the script multiple times if a previously launched script is still running. This can occur if OnGuard Agent performs multiple health checks (either manually triggered or caused by a change in health status). The script exits after performing its task.

> **NOTE**
> The Agent Script Enforcement profile is currently supported only with the OnGuard Agent for Windows.

Mandatory Agent Script Attributes

The following attributes are mandatory when configuring Agent Script Enforcement:

- Path of the Script
- Command to Execute
- Execution Level

Optional Agent Script Attributes

The following attributes are optional when configuring Agent Script Enforcement:

- SHA256 Checksum
- Wait Time (Seconds) Before Executing Script
- Pass Health Evaluation Results to Script
- Success Message
- Failure Message
- Progress Message
- Description
- Download URL
- Script Execution Timeout (Seconds)

Configuring the Agent Script Enforcement Profile

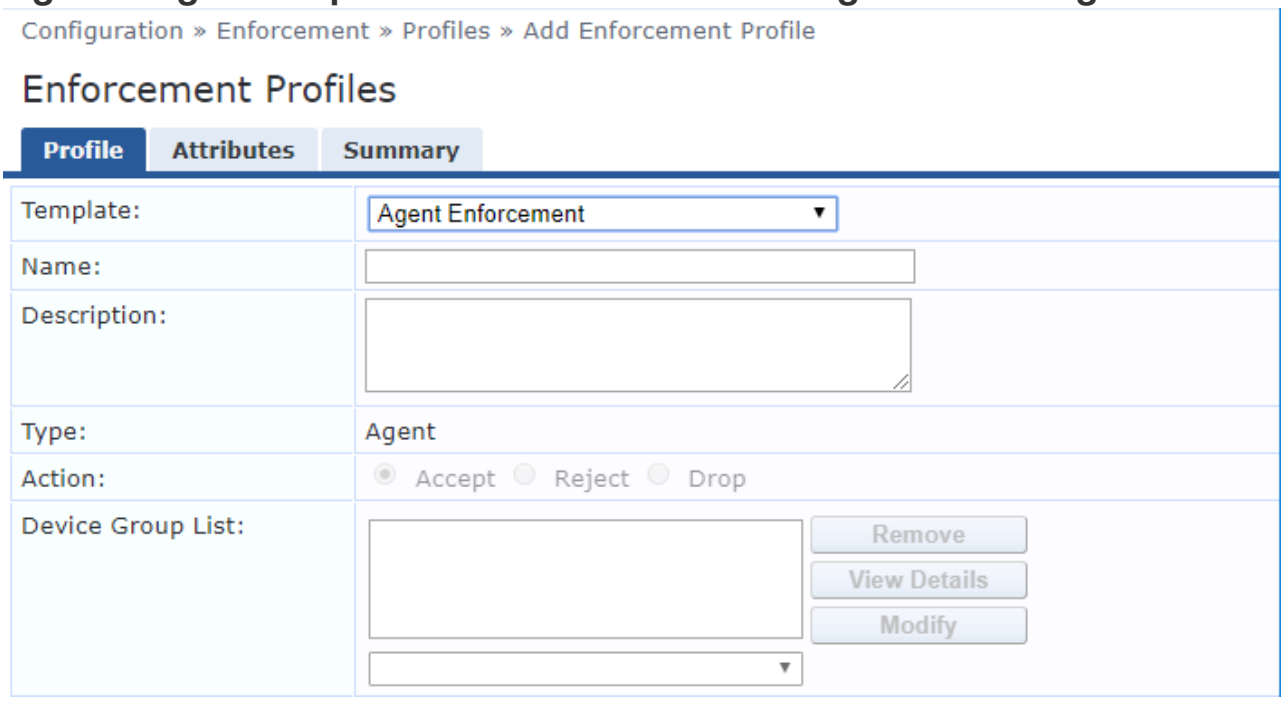
About this task

To configure an Agent Script Enforcement profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**. The **Enforcement Profiles** page opens.
2. Click the **Add** link. The **Add Enforcement Profiles** dialog opens.
3. From the **Template** drop-down, select **Agent Script Enforcement**.

Figure 1. Agent Script Enforcement > Profile Configuration Dialog



4. Specify the **Add Agent Script Enforcement > Profile** parameters as described in the following table:

Table 1. Add Agent Script Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select the **Agent Script Enforcement** te mplate.
Name	Enter the name of the enforcement profil e.
Description	Optionally, enter a description of the en forcement profile (recommended).

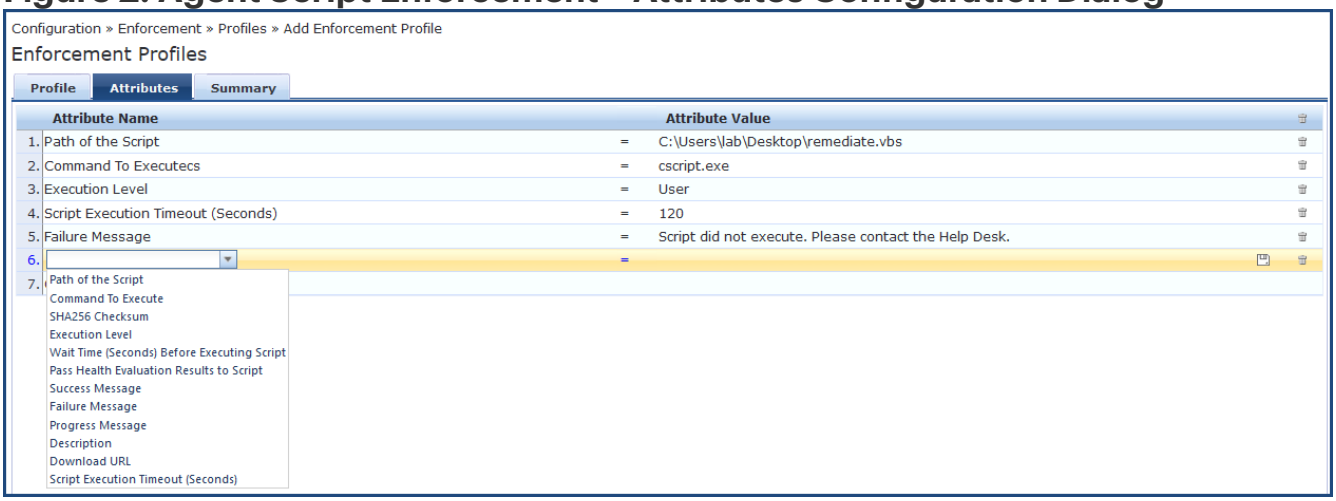
Parameter	Action/Description
Type	This field is populated automatically with type **Agent**.
Action / Device Group List	These parameters are disabled because t hey are not applicable to the Agent Scrip t Enforcement Profile.

Configuring Agent Script Enforcement Attributes

About this task

Use the **Attributes** tab to configure the attribute name and attribute value for each attribute you add. The following figure displays the **Agent Script Enforcement > Attributes** dialog:

Figure 2. Agent Script Enforcement > Attributes Configuration Dialog



Procedure

1. Specify the **Agent Script Enforcement > Attributes** parameters as described in the following table:

Table 2. Agent Script Enforcement > Attributes Parameters

Attribute	Action/Description
Path of the Script	Complete the path of the script/program, including the filename. This attribute che cks for the existence of a file on an endp oint device and also verifies the SHA256 Checksum.

Attribute	Action/Description
Command to Execute	Specify the complete command that On Guard Agent should execute. You can use the command to launch scripts or pass c ommand line arguments. For example, to launch VBScript (InstallH otfixes.vbs) and pass **All** as an argument, you would enter the following: `cscript /nologo C:\Test\Install Hotfixes.vbs All` If it is not required to pass arguments, set the value of this attribute to the same val ue specified for **Path of the Script**.
SHA256 Checksum	Specify the SHA256 checksum of the sc ript or program. This attribute accepts c omma - separated multiple SHA256 chec ksums to allow execution of different vers ions of the same script or program.
Execution Level	The attribute values are: **User** and **Syste m**. • To launch the script/program as the c urrent logged - on user, select **User**. • To launch the script/program as the sy stem user with admin rights, select **Sy stem**.
Wait Time (Seconds) Before Executing Sc ript	Specify the time (in seconds) after which OnGuard Agent should launch the script /program. When **Wait Time Before Executing Script** is configured, the OnGuard Agent does n ot process events such as Interface Up/In terface Down and health changes during the wait time.
Pass Health Evaluation Results to Script	Check the check box (which sets the valu e to **true**) to enable OnGuard Agent to p

Attribute	Action/Description
	ass health evaluation results to the scrip t/program as an argument. The default is **false**. • When the **Pass Health Evaluation Res ults to Script** attribute is set to **true**, OnGuard Agent passes health evaluat ion results to the script in a URL Enco ded JSON format. • URL Encode replaces double quotes, spaces, and Unicode characters with their ASCII value in %XX format. For example, spaces are replaced by %20 and double quotes are replaced by % 22.
Success Message	Enter the message to be shown to the en d user when the script or program is laun ched successfully.
Failure Message	Enter the message to be shown to the en d user when execution of the script or p rogram fails.
Progress Message	This message will be shown on the OnGu ard Progress Page (see OnGuard Settings and Agent Library Updates) while OnGua rd Agent is executing the custom script a nd states what action is being performed .
Description	Provides a description of the custom use r interface window or script.
Download URL	If the script or program configured in the **Path of the Script** attribute is not presen t on the client machine, enter the URL o f the remote server from which OnGuard Agent can download the script or progra m.

Attribute	Action/Description
	• OnGuard Agent supports downloadin g scripts only from HTTP and HTTPS U RLs. • For HTTPS URLs, OnGuard skips serve r certificate verification. **NOTE** OnGuard Agent does not s upport downloading files fro m URLs that require credenti als.
Script Execution Timeout (Seconds)	Specify the **Script Execution Timeout** att ribute value in the number of seconds. T he value must be an integer between **60** and **3600**. When this attribute is configured, OnGua rd Agent waits for the specified time for t he custom script to finish execution. If th e custom script does not finish execution within the specified time, OnGuard Agen t returns failure for that script. When this attribute is not configured, On Guard Agent uses varying timeout values , depending upon whether or not custom remediation is enabled. • When Custom Remediation is enabled : the default value is 20 minutes. • When Custom Remediation is not ena bled: 1 minute.

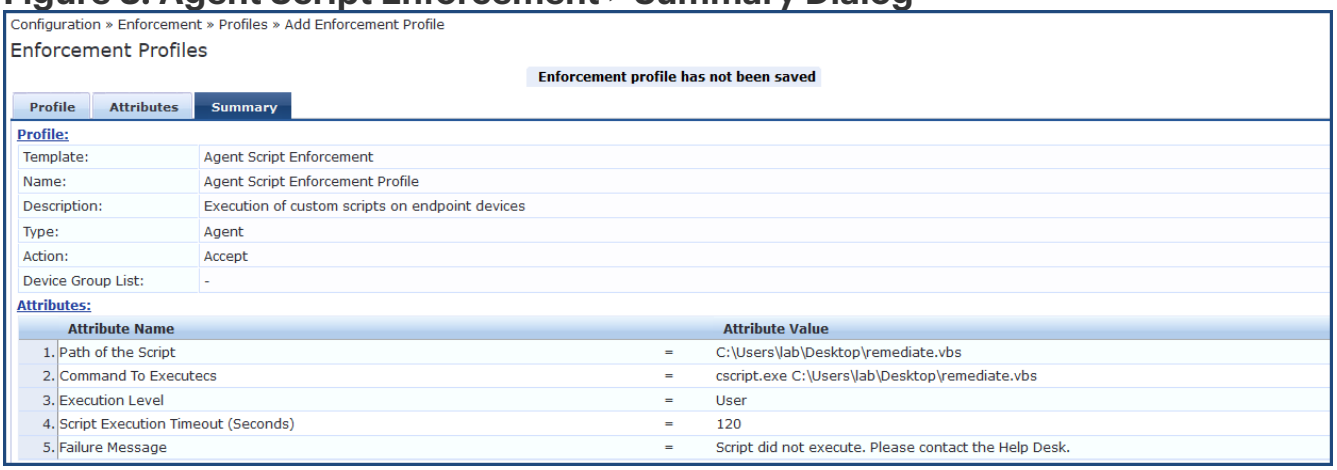
2. Click **Save**.

Viewing the Configuration Summary

The **Summary** page summarizes the parameters configured in the **Profile** and **Attribute** tabs.

The following figure displays the **Agent Script Enforcement > Summary** page:

Figure 3. Agent Script Enforcement > Summary Dialog



Aruba Downloadable Role Enforcement Profiles

Policy Manager includes support for centralized policy definition and distribution. When ClearPass Policy Manager successfully authenticates a user, the user is assigned a role by ClearPass Policy Manager. However, if the role is not defined on the Aruba controller or switch, the role attributes can also be downloaded automatically. The following validations are performed for a Downloadable User Role (DUR):

- Standard Mode elements are validated based on schema contents (syntax validation) prior to saving the DUR.
- After a schema update, existing DURs remain unchanged until the admin user edits the DUR profile.
- When an admin user edits an existing DUR profile, the syntax/configuration is re-validated, and an alert is shown to the admin to verify the changes.

> **NOTE**
>
> A DUR is not supported in AOS 10.

Basic Profile Settings

About this task

Use the **Profiles** tab on the **Downloadable User Role Enforcement Profile** dialog to configure the template, type of the profile, and the device group list, and specify the **Role Configuration Mode** as either **Standard** or **Advanced**.

- **Standard mode**: User-provided options to configure individual components of a role (for example, Policer Profile, Stateless ACL configuration, etc.). The user role is generated based on components added to the configuration.
- **Advanced mode**: You can enter the entire role configuration as a text under a single attribute.

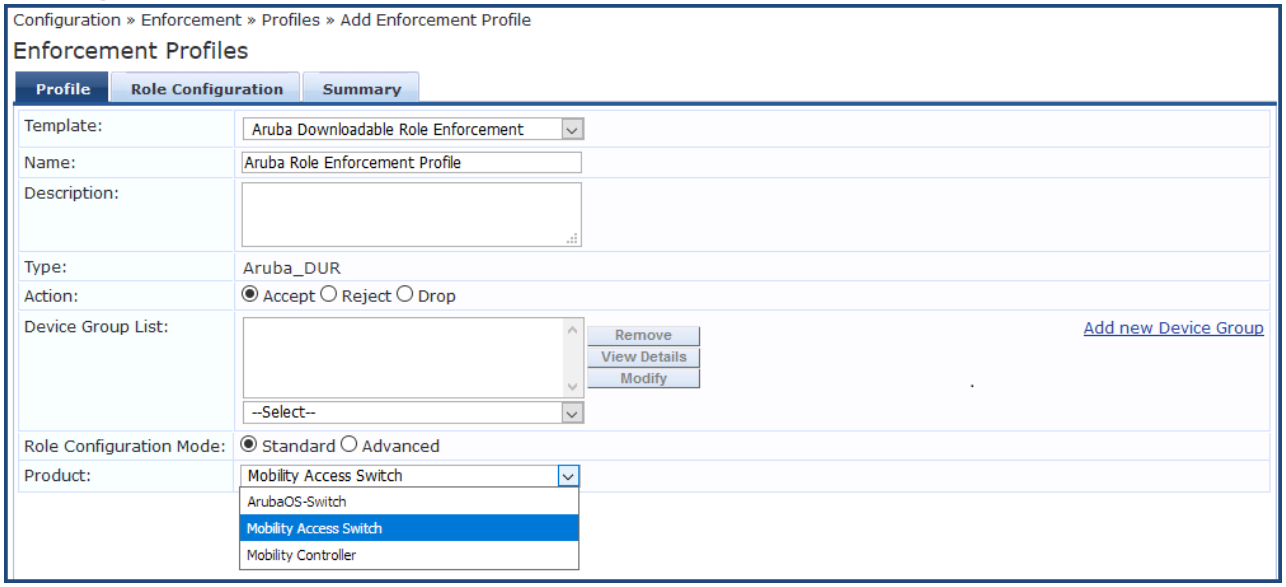
Events are logged in the **Audit Viewer** for create, update, and delete operations in the *Captive Portal*, *Policy*, and *Class* configurations. Events are also logged for generated user roles and import/export operations in enforcement profiles.

To configure the Aruba Downloadable Role Enforcement Profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**. The **Enforcement Profiles** page opens.
2. Click the **Add** link. The **Add Enforcement Profile** page opens.
3. From the Template drop-down, select **Aruba Downloadable Role Enforcement**.

Figure 1. ArubaDownloadable Role Enforcement > Profile Page (Standard Mode)



4. Specify the **Aruba Downloadable Role Enforcement > Profile** parameters as described in the following table:

Table 1. Aruba Downloadable Role Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select the **Aruba Downloadable Role Enforcement** template.
Name	Enter the name of the enforcement profile.
Description	Enter a description of the enforcement profile.
Type	This field is automatically populated with **Aruba_DUR**.
Action	Click **Accept**, **Reject**, or **Drop** to define the action taken on the request. The default action is **Accept**.
Device Group List	Select a device group from the drop - down list. The list displays all configured device groups. All configured device groups are listed in the **Configuration > Network > Device Groups** page. After adding one or more device group(s), you can select a group and perform one of the following actions: • To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**. • To change the parameters of the selected device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Group** link and see Adding

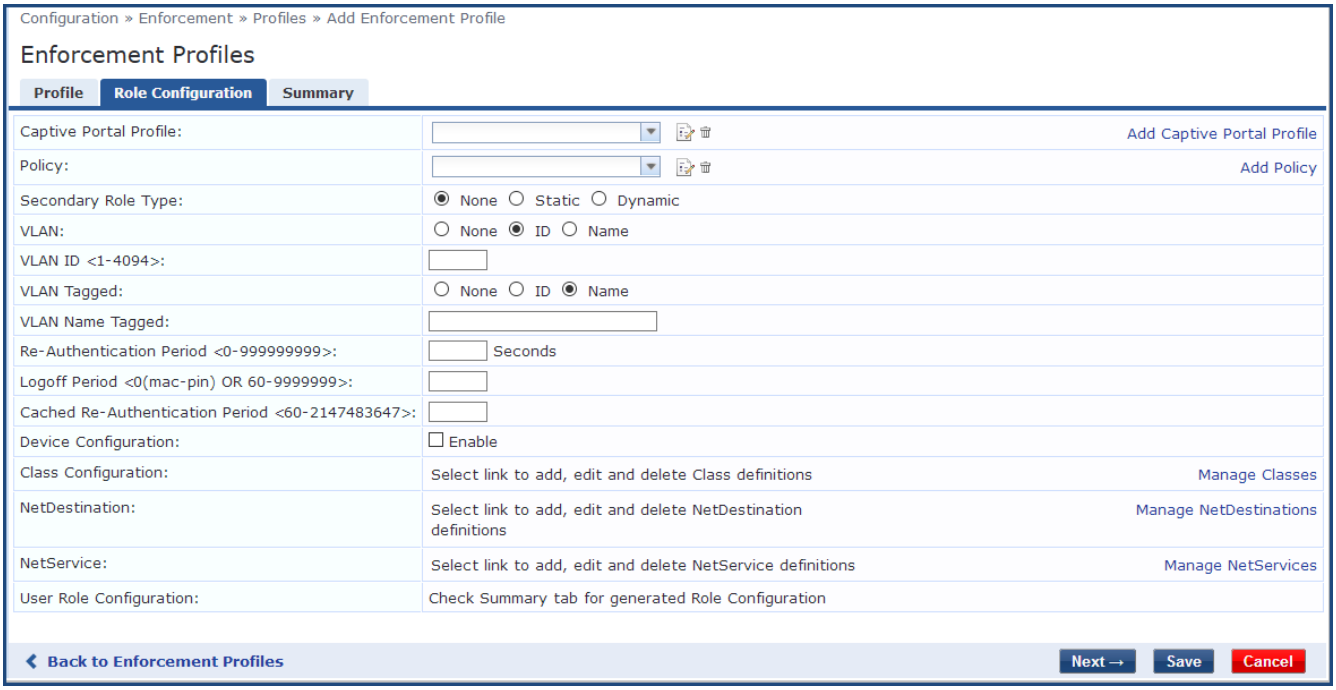
Parameter	Action/Description
	and Modifying Device Groups.
Role Configuration Mode	Select one of the following modes: • Standard (the default) • Advanced
Product	Specify one of the following products: • ArubaOS - Switch • Mobility Access Switch (MAS) • Mobility Controller • AOS - CX

Role Configuration: Standard Mode

When the **Role Configuration Mode** setting on the **Profile** tab is set to **Standard** (the default), the **Role Configuration** tab appears. In **Standard** mode, the **Role Configuration** tab includes only the options that are appropriate for the selected product.

The fields on the **Role Configuration** tab vary according to which product you specify. The example below displays the **Role Configuration** page when you specify the **ArubaOS Switch** product. This page provides support for class configuration for the ArubaOS-Switch, and the **Role Configuration** tab includes a **Manage Classes** link that does not appear when you configure enforcement profile roles for Mobility Access switch or Mobility Controller Aruba products.

Figure 2. Role Configuration Page: ArubaOS-Switch



The following tables describe the **Role Configuration** parameters for the Aruba Downloadable Role Enforcement Profile for Mobility Access Switch, Mobility Controller, and AOS-CX:

Table 2. Mobility Access Switch Role Configuration Parameters

Parameter	Action/Description
Captive Portal Profile	Select the captive portal profile from the d rop - down list if already configured. To add a new captive portal profile, click th e **Add Captive Portal Profile** link. For more information, see Captive Portal Profile.
Policer Profile	This parameter defines a Policer profile to manage the transmission rate of a class of traffic based on user - defined criteria. Sel ect the Policer profile from the drop - dow n list if already configured. Click **Add Polic er Profile** link to add a new Policer profile. For more information, see Policer Profile.
QoS Profile	This parameter defines a QoS profile to as sign Traffic - Class/Drop - Precedence, Diff erentiated Services Code Point (DSCP), an d 802.1p values to an interface or Policer p rofile of a Mobility Access Switch. Select th e QoS profile from the drop - down list if a lready configured. Click **Add QoS Profile** li

Parameter	Action/Description
	nk to add a new QoS profile. For more infor mation, see QoS Profile.
VoIP Profile	This parameter defines a VoIP profile that c an be applied to any interface, interface g roup, or a port - channel of a Mobility Acc ess Switch. Select the VoIP profile from the drop - down list if already configured. Clic k **Add VoIP Profile** link to add a new VoIP p rofile. For more information, see VoIP Profil e .
Reauthentication Interval Time (0 - 4096)	Enter the number of minutes between reau thentication intervals. You can select the ra nge between 0 to 4096 minutes.
VLAN To Be Assigned (1 - 4904)	Enter a number between 1 and 4094 that d efines when the VLAN is to be assigned.
NetService Configuration	Select the **Manage NetServices** link to add , edit, and delete the NetService definition s. For more information, see NetService Co nfiguration.
NetDestination Configuration	Select the **Manage NetDestinations** link to add, edit, and delete the NetDestinations d efinitions. For more information, see NetD estination Configuration.
Time Range Configuration	Select the **Manage Time Ranges** link to ad d, edit, and delete time range definitions. For more information, see Time Range Con figuration.
NAT Pool Configuration	Select the **Manage NAT Pool** link to add, e dit and delete NAT Pool definitions. For mo re information, see NAT Pool Configuration.
ACL Type	Select from the following ACL types: • **Ethertype**: Ethertype ACLs filter based on the Ethertype field in the frame head er. • **MAC**: MAC ACLs filter traffic on a specif ic source MAC address or range of MAC addresses.

Parameter	Action/Description
	• **Session:** Session ACLs define traffic an d firewall policies**.** • **Stateless**: Stateless ACLs define statel ess packet filtering and quality of servic e (QoS). A stateless ACL statically evalu ates packet contents
ACL Name	Click the name of the ACL type. • To move the **ACL Name** to the **ACL** field , click **Add**. • To modify the order of names in the AC L list, click **Move Up**, **Move Down**. • To delete an ACL from the list, click **Re move**.
User Role Configuration	Check the **Summary** tab for the generated role configuration.

Table 3. Mobility Controller Role Configuration Parameters

Parameter	Action/Description
Captive Portal Profile	Select the captive portal profile from the d rop - down menu if already configured. Click the **Add Captive Portal Profile** link to add a new captive portal profile. For more i nformation, see Captive Portal Profile.
Reauthentication Interval Time (0 - 4096)	Enter the number of minutes between reau thentication intervals. Select the range bet ween 0 to 4096 minutes.
VLAN	Select either the **ID**, **Name**, **Parameterized ID**, or **Parameterized Name** radio button t o set how the assigned VLAN is provided. Define the VLAN as follows based on the se lected option: • If using an **ID**, the **VLAN ID <1 - 4094>** parameter displays by default to set the VLAN ID from 1 to 4094.

Parameter	Action/Description
	• If assigning a **Name**, enter the characte rs comprising the **VLAN Name**. • If using a **Parameterized ID**, select a **Pa rameterized VLAN ID** from the drop - d own menu. • If using a **Parameterized Name**, select i t from the **Parameterized VLAN Name** drop - down menu.
NetService Configuration	Select the **Manage NetServices** link to add , edit, and delete the NetService definition s. For more information, see NetService Co nfiguration.
NetDestination Configuration	Select the **Manage NetDestinations** link to add, edit, and delete the NetDestinations d efinitions. For more information, see NetD estination Configuration.
Time Range Configuration	Select the **Manage Time Ranges** link to ad d, edit, and delete time range definitions. For more information, see Time Range Con figuration.
NAT Pool Configuration	Select the **Manage NAT Pool** link to add, e dit and delete NAT Pool definitions. For mo re information, see NAT Pool Configuration.
ACL Type	Select from the following ACL types: • **Ethertype**: Ethertype ACLs filter based on the Ethertype field in the frame head er. • **MAC**: MAC ACLs filter traffic on a specif ic source MAC address or range of MAC addresses. • **Session:** Session ACLs define traffic an d firewall policies**.** • **Stateless**: Stateless ACLs define statel ess packet filtering and quality of servic

Parameter	Action/Description
	e (QoS). A stateless ACL statically evalu ates packet contents
ACL Name	Click the name of the ACL type. • To move the **ACL Name** to the **ACL** field , click **Add**. • To modify the order of the names in the ACL list, click **Move Up**, **Move Down**. • To delete an ACL from the list, click **Re move**.
User Role Configuration	Check the **Summary** tab for the generated role configuration.

Table 4. AOS - CX Role Configuration Parameters

Parameter	Action/Description
Role Name	Enter a name for this AOS - CX role.
Description	Enter a description of this AOS - CX role.
Captive Portal	Select the Captive Portal profile from the d rop - down list if it is already configured. To add a new Captive Portal profile, click th e **Add Captive Portal Profile** link. For more information, see Captive Portal Profile.
Authentication Mode	Select an authentication mode: • **client - mode**: Supports access contro l on a per - user basis. • **device - mode**: Supports access only on ports where a 802.1X - capable devi ce has entered authorized user credenti als. • **multi - domain - mode**: In this mode, o nly one voice device is allowed to be a uthenticated in addition to the any data devices authenticated on a port. By de

Parameter	Action/Description
	fault, only one voice device and one da ta device can be authenticated in multi - domain mode.
Device Traffic Class	Specify the traffic class for the device: • **None**: No traffic class is configured for t he device. • **Voice**: Specifies that the client to asso ciate with the role is a voice device. T his attribute is applicable only to the **c ritical - voice - role** role in AOS - CX. I t is not applicable to other special AOS - CX roles such as **preauth - role**, **rej ect - role**, and **fallback - role**. In multi - domain mode, a client with a role tha t does not have the **device - traffic - cl ass** set to voice is considered a data d evice.
Session Timeout Type	The session timeout type can either of the following: • **Static**: Timeout value is a static number of seconds defined in the **Session Time out** field. • **Parameterized**: Policy Manager passes a configuration variable as a timeout val ue.
Parameterized Session Timeout	Select a parameter from the list, and the va lue of the parameter will be passed as the session - timeout value. This field is only av ilable when the **Session Timeout Type** fiel d is set to **Parameterized.**
Session Timeout <1 - 4294967295>	Number of seconds the enforcement profil e should assign the role before the device session ends. This field is only avilable whe n the **Session Timeout Type** field is set to **Static**.
Parameterized Session Timeout	Click the drop - down menu and select a c onfiguration variable to be used as a sessio

Parameter	Action/Description
	n timeout value. This field is only avilable w hen the **Session Timeout Type** field is set t o **Parameterized**.
Cached Re - Authentication period <30 - 4 294967295>	Cached reauthentication allows 802.1X, We b - based, or MAC reauthentication to succ eed when the RADIUS server is unavailable. The supported range for this parameter is 30 - 4294967295 seconds. **NOTE** This feature is not supported in AOS - CX switches running AOS - CX 10.5 or earlier releas es.
Re - Authentication Period Type	Select one of the following options: • **Manual:** All devices using this enforce ment profile will use the reauthenticatio n period specified in the **Re - authentic ation period <1 - 4294967295> field**. • **Random**: Allow ClearPass to create a downloadable role with a randomized - reauthentication timer. When enable d, Policy Manager sends a Radius:IETF s ession - timeout message with the valu e of %{Authorization:[Time Source]:Ran dom Time. This feature prevents Policy Manager from becoming overloaded by a very large number of clients using the same 802.1x reauthentication timer. Use the **Minimum Re - Authentication Period <1 - 4294967295>** field to defin e the shortest allowed random reauthe ntication period. T • **Parameterized**: Policy Manager passes a selected configuration variable as a reauthentication period value.

Parameter	Action/Description
Re - Authentication Period <1 - 4294967295>	Number of seconds after which the device is required to reauthenticate. This parameter supports a range of 1 - 4294967295 seconds. This field is only available if the **Re - Authentication Period Type** option is set to **Manual**.
Minimum Re - Authentication period <1 - 4294967295>	Minimum amount of time before a device using a random reauthentication period is required to authenticate. The maximum random reauthentication period is the <Minimum Re - Authentication Period> time plus 24 hours. This field is only available if the **Re - Authentication Period Type** option is set to **Random**.
Parameterized Re - Authentication	Click the drop - down menu and select a variable to be used as a reauthentication period value. This field is only available if the **Re - Authentication Period Type** option is set to **Parameterized**.
Client Inactivity Timeout Type	Select one of the following options: • **Value:** Enable a client inactivity timeout period with the value defined in the **Client Inactivity Timeout <300 - 4294967295>** field. • **None**: Disable the client activity timeout. • **Parameterized**: Policy Manager passes a selected configuration variable as a client inactivity timeout value.
Client Inactivity Timeout <300 - 4294967295>	Configures the time the switch waits for client activity before removing an inactive client from the port. Supported times are 300 - 4294967295 seconds. This field is only available if you select the **value** option in the **Client Inactivity Timeout Type** field.
Parameterized Client Inactivity Timeout	Click the drop - down menu and select a variable to be used as a client inactivity timeout value. This field is only available if the

Parameter	Action/Description
	Client Inactivity Timeout Type option is s et to **Parameterized**.
Allowed VLANs on Trunk <1 - 4094>	Specify a VLAN ID defines to indicate whic h VLAN has traffic allowed across the trunk interface.
Allowed VLAN Names on Trunk (one per lin e)	Assigns a VLAN to a trunk interface based on the VLAN name. This fields supports mu ltiple VLAN names.
Parameterized Allowed VLAN Type	To define an allowed VLAN type through a configuration variable, select one of the fol lowing options: • ID: The variable parameter is passed as a VLAN ID. • Name: The variable parameter is passe d as a VLAN name.
Parameterized Allowed VLAN	Click the drop - down menu and select a variable to be used as an allowed VLAN. Th is value can be passed as either a VLAN ID or VLAN name, depending upon the selecti on option in the **Parameterized Allowed V LAN Type** field.
Native Trunk VLAN Type	Assigns a native VLAN to a trunk interface based on a VLAN ID, VLAN name, or the val ue of a configuration variable. Select one of the following options: • **ID:** Assigns a native VLAN to a trunk int erface based on a VLAN ID. • **Name**: Assigns a native VLAN to a trunk interface based on a VLAN name. • **Parameterized**: Policy Manager passes a selected configuration variable as a n ative trunk VLAN.
Native Trunk VLAN ID <1 - 4094>	Enter a VLAN ID to assign a native VLAN to a trunk interface using the specified VLAN ID.

Parameter	Action/Description
Native Trunk VLAN Name	Enter an VLAN name to assign a native VLAN to a trunk interface with the specified VLAN name.
Parameterized Native Trunk VLAN	Click the drop - down menu to assign a native VLAN to a trunk interface based on the value of the selected configuration variable.
Access VLAN Type	Select one of the following options: • **ID:** Defines an access VLAN based on a VLAN ID. • **Name**: Defines an access VLAN based on a VLAN name. • **Parameterized**: Policy Manager passes a selected configuration variable as an access VLAN.
Access VLAN ID <1 - 4094>	Defines an access VLAN with the specified VLAN ID.
Access VLAN Name	Defines an access VLAN with the specified VLAN name.
Parameterized Native Trunk VLAN	Click the drop - down menu to define an access VLAN based on the value of the selected configuration variable.
Gateway Zone	If you selected the **Static** or **Dynamic** secondary role type, specify the per - role gateway zone needed for user - based tunneling (UBT).
Secondary Role Type	Policy Manager supports setting a controller Downloadable User Role (DUR) as a Secondary Role. Options are **none** if you do not want a secondary role, **static** for a static role, or **dynamic** for a dynamic role. The secondary role option allows network admins to configure roles that are not allowed on a Layer 3 switch by creating a User Based Tunnel (UBT) to the controller associated with the AOS - CX switch, and allowing the controller to download the secondary role from ClearPass.

Parameter	Action/Description
	• If you select the **static** option, roles and policies must be manually configured o n the controller. Clearpass just passes t he role name to the AOS - CX switch, w hich in turn sends it to the controller. • If you select the **dynamic** option, roles can be configured in Policy Manager, an d no additional role configuration is nec essary the controller side. Support for d ynamic secondary roles in AOS - CX wa s introduced in AOS - CX 10.08.
Gateway Static Role	If you selected the **Static** secondary role ty pe, specify the gateway static role.
Gateway Downloadable Role	If you selected the **Dynamic** secondary rol e type, specify the gateway dynamic role. **NOTE** Support for dynamic seconda ry roles in AOS - CX was introd uced in Policy Manager 6.10.2 for switches running AOS - CX 10.08 and later releases.
MTU <68 - 9198>	Sets the MTU (Maximum Transmission Unit) for an interface. This defines the maximu m size of a layer 2 (Ethernet) frame. Packet s larger than the MTU are dropped and cau se an ICMP fragmentation - needed messa ge to be sent back to the originator.
Trust Mode	Configures QoS trust mode for the role: • **None**: Do not trust any priority fields. • **COS**: Trust DSCP and preserve the 802.1 p priority. • **DHCP**: Trust 802.1p priority and retain DSCP or IP - ToS.
PoE Priority	Select the priority level for PoE; **High**, **Low** or **Critical**.

Parameter	Action/Description
Private VLAN Port Type	Private VLANs partition an existing VLAN in to multiple sets of ports for traffic isolation. The partitioned VLAN is referred to as the P rimary VLAN. To define a VLAN port type fo r the role, select one of the following port t ypes: • **Promiscuous**: A promiscuous port is a switch port that is connected to an upli nk router, firewall, or other common gat eway device, and can communicate wit h all ports within a private VLAN, includi ng the ports in the isolated and commu nity VLANs. By default, every primary V LAN port acts as a promiscuous port. I f the specified VLAN is a primary VLAN that has been associated with seconda ry VLANs, the command also assigns th e port to the associated secondary VLA Ns. • **Secondary**: Secondary VLAN trunk po rts carry secondary VLAN traffic. A seco ndary VLAN trunk port can carry traffic for multiple secondary VLANs only if ea ch secondary VLAN is a member of a dif ferent primary VLAN.
Class Configuration	Select the **Manage Classes** link to add, edi t, or delete class definitions. For more infor mation, see Manage Classes.
Policy	Select the Enforcement Policy from the dro p - down list if already configured. To add a new Enforcement Policy, click the **Add Policy** link. For more information, see Add Policy.
User Role Configuration	Select the **Summary** tab to view the gene rated user role configuration.

The **Role Configuration** tab also allows you to associate other profile types with your selected user role, or to create these profiles if they are not yet defined. Refer to the following sections for information on creating those profiles.

Captive Portal Profiles

About this task

To define the Captive Portal Profile for a downloadable Role Enforcement profile for any supported device type:

Procedure

1. From the **Role Configuration** tab, click the **Add Captive Portal Profile** link. The **Add Captive Portal Profile** dialog opens:
2. Enter a name of the profile and configure the required attributes. Note that the available parameters for configuring a Captive Portal profile vary by device type. The following tables describe the Captive Portal Attributes available for Mobility Controller and Mobility Access Switch, ArubaOS-Switch, and AOS-CX.

Table 5. Mobility Controller and Mobility Access Switch Captive Portal Attributes

Parameter	Action/Description
Server Group	Enter the name of a server group to be us ed for captive portal authentication
Default role	Specify the role assigned to the Captive Portal user upon login. When both user a nd guest logon are enabled, the default role applies to the user logon; users loggi ng in using the guest interface are assign ed the guest role.
Default Guest Role	Specify the role assigned to a guest.
Redirect Pause	Time, in seconds, that the system remain s in the initial welcome page before redi recting the user to the final web URL. If s et to 0, the welcome page displays until t he user clicks on the indicated link.
User Login	Select one of the following options: • **Yes**: Enables Captive Portal with authe ntication of user credentials.

Parameter	Action/Description
	• **No**: Disables Captive Portal with auth entication of user credentials.
Guest Login	Select one of the following options: • **Yes**: Enables Captive Portal logon with out authentication. • **No**: Disables Captive Portal logon wit hout authentication.
Login Popup Window	Enables a pop - up window with the Logo ut link for the user to log out after loggin g on . If this is disabled, the user remains logged in until the user timeout period h as elapsed or the station reloads.
Use HTTP for Authentication	Use HTTP protocol on redirection to the Captive Portal page. If you use this optio n, modify the captive portal policy to all ow HTTP traffic.
Logon Wait Minimum Delay	Minimum time, in seconds, the user will h ave to wait for the logon page to pop up if the CPU load is high. This works in con junction with the Logon wait CPU utilizati on threshold parameter.
Logon Wait Maximum Delay	Configure parameters for the logon wait i nterval
Logon Wait CPU Utilization Threshold (1 - 100%):	CPU utilization percentage above which t he Logon wait interval is applied when p resenting the user with the logon page.
Max Authentication Failures (0 - 10):	Maximum number of authentication failu res before the user is added to a blocked list
Show FQDN:	Allows the user to see and select the ful ly - qualified domain name (FQDN) on th e login page. The FQDNs shown are spec

Parameter	Action/Description
	ified when configuring individual servers for the server group used with captive po rtal authentication.
Use CHAP (Non - standard):	Select this option to use CHAP authentic ation. This option is disabled by default a nd should be enabled only under the sup ervision of an Aruba support representati ve.
Login Page:	URL of the page that appears before logo n. This can be set to any URL.
Welcome Page:	URL of the page that appears after logon and before redirection to the web URL. T his can be set to any URL.
Show Welcome Page:	Displays the configured welcome page b efore the user is redirected to their origin al URL. If this option is disabled, users are redirected to the web URL immediately a fter they log in.
Adding Switch IP Address in Redirection URL:	Sends the managed device's IP address i n the redirection URL when external capti ve portal servers are used. An external ca ptive portal server can determine the ma naged devices from which a request origi nated by parsing the 'switchip' variable in the URL.
Allow Only One Active User Session:	Allows only one active user session at a ti me.
Show the Acceptable Use Policy Page:	Show the acceptable use policy page be fore the logon page.
Add an IP Interface in the Redirection UR L:	Sends the interface IP address of the ma naged device in the redirection URL whe n external captive portal servers are used . An external captive portal server can de

Parameter	Action/Description
	termine the managed devices from whic h a request originated by parsing the 'swi tchip' variable in the URL.
Add User VLAN in Redirection URL:	Sends the user's VLAN ID in the redirectio n URL when external captive portal serve rs are used.
White List Net Destinations:	To add a netdestination to the list of all owed captive portal netdestinations, ent er the destination host or subnet, then cli ck Add. The netdestination will be added to the allowed list. To remove a netdestin ation from the allowed list, select it in this field, then click **Delete**.

Table 6. ArubaOS - Switch Captive Portal Profile Attributes

Parameter	Action/Description
Name	Name of the Captive portal Configuration
URL	URL of the page that appears for the user logon. This can be set to any URL.
URL - hash - key (Plain Text) / URL Hash - Key (Cipher Text)	You can optionally configure a URL hash key to provide some security for the Capt ive Portal exchange with CPPM. The key i s a shared secret between Policy Manage r and ArubaOS - Switch . When configure d, the switch generates a HMAC - SHA1 h ash of the entire redirect URL, and appen ds the hash to the URL to be sent to Poli cy Manager as part of the HTTP redirect. If Policy Manager is configured to check the hash, it will generate the hash of the URL using its version of the URL hash key and compare against the value provided by the switch. The action taken by CPPM

Parameter	Action/Description
	upon a match or mismatch is determined by what is configured on Policy Manager.

Table 7. AOS - CX Captive Portal Profile Attributes

Parameter	Action/Description
Name	Name of the Captive portal Configuration
URL	URL of the page that appears for the user logon. This can be set to any URL.
URL - hash - key (Plain Text) / URL Hash - Key (Cipher Text)	**NOTE** URL hash keys **are not allowed in Aruba Downloadable User Roles (DUR) for the AOS - CX switch**. If they are configured for the switch, it causes command executions to fail. To successfully enforce the captive portal configurations, leave the URL Hash-Key (PlainText) and URL Hash-Key (CipherText) fields empty when you configure the AOS - CX captive portal. These fields appear in Policy Manager 6.9.3 and earlier releases.

Policer Profiles

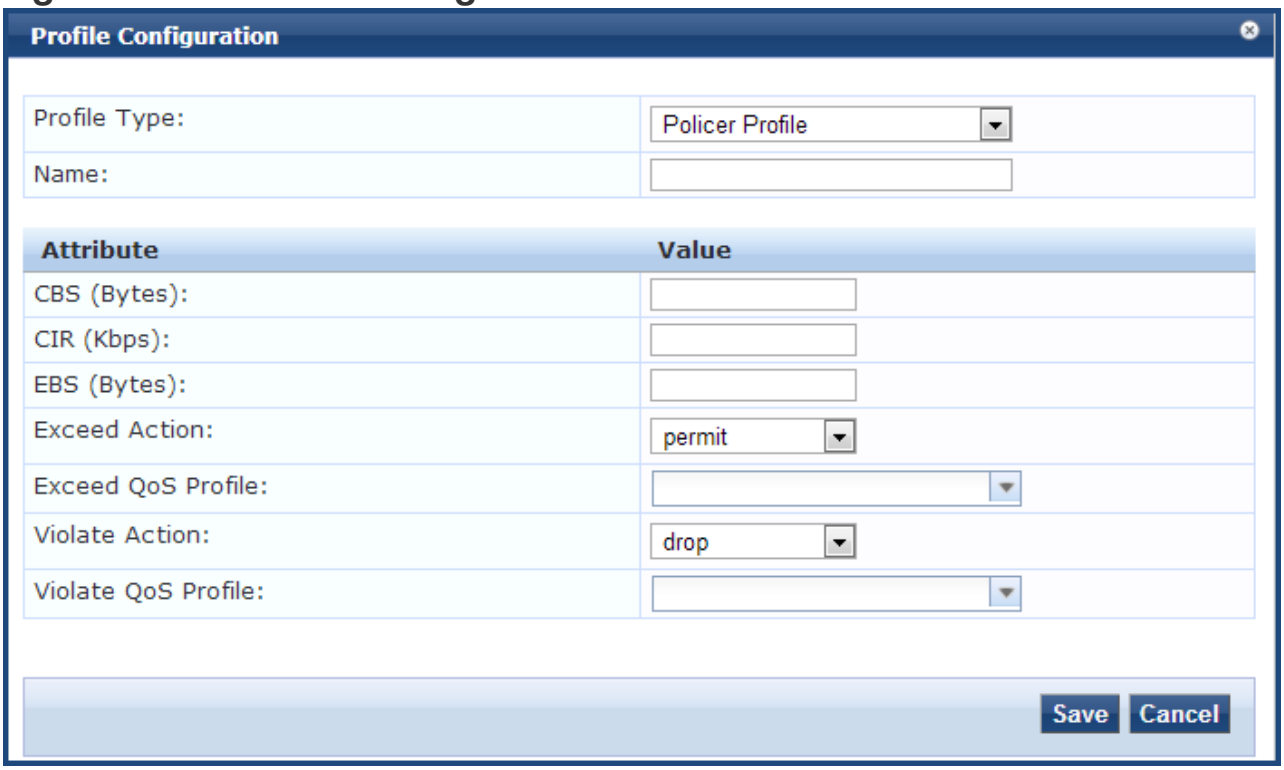
About this task

This profile limits inbound transmission rate of a class of traffic on the basis of user-defined criteria. To define a Policer Profile for an Aruba Mobility Access Switch:

Procedure

1. From the **Role Configuration** tab, click the **Add Policer Profile** link. The **Add Policer Profile** dialog opens:

Figure 3. Add Policer Configuration Profile



2. Enter a name of the profile and configure the required attributes.

Table 8. Policer Profile Attributes

Parameter	Action/Description
CBS (bytes)	Specify the Committed Burst Size (CBS) for this enforcement policy. Traffic rates b elow CIR or bursts below CBS limits are c onsidered “conforming” and are allowed t o pass through.
CIR (Kbps)	Specify the Committed Information Rate (CIR) for this enforcement policy. Traffic rates below CIR or bursts below CBS limit s are considered “conforming” and are all owed to pass through.

Parameter	Action/Description
EBS (bytes)	Specify the burst size, in bytes, conside red to be an Excessive Burst Size (EBS)
Exceed Action	Select an action for traffic rate exceeding CIR and CBS, yet bursting below the EBS limit. (**permit**, **drop**, **remark**). By default, t his traffic is permitted.
Exceed QoS Profile	Select a QoS profile to be assigned to tra ffic that exceeds CIR and CBS, yet bursts below the EBS limit.
Violate Action	Select an action for traffic rate exceeding CIR and CBS, and also bursting above the EBS limit. (**permit**, **drop**, **remark**). By defa ult, this traffic is dropped.
Violate QoS Profile	Select a QoS profile to be assigned to tra ffic that exceeds CIR and CBS and also b ursts above the EBS limit.

QoS Profiles

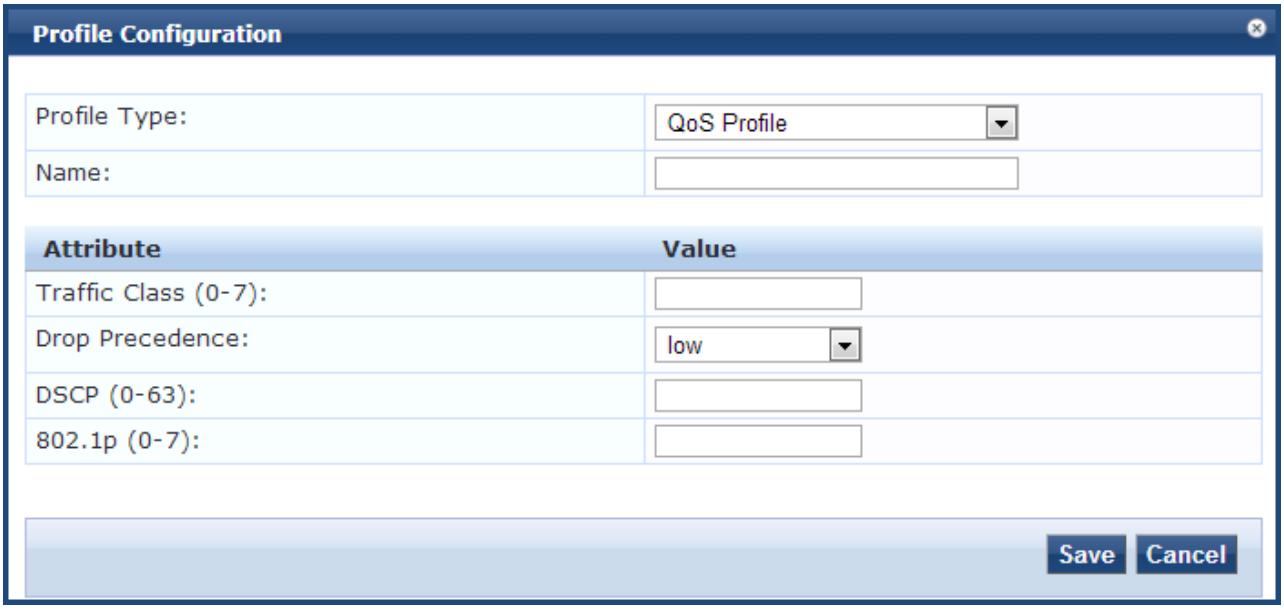
About this task

To define a QoS Profile for an Aruba Mobility Access Switch:

Procedure

1. Click the **Add QoS Profile** link. The **Add QoS Profile** opens:

Figure 4. Add QoS Profile Configuration Profile



2. Enter a name of the profile and configure the required attributes.

Table 9. QoS Profile Attributes

Parameter	Action/Description
Traffic Class (0 - 7)	Enter a value from 0 to 7 to specify whic h of the eight traffic classes (TCs), map to the corresponding queue.
Drop Precedence	Select Low or High to select the drop pre cedence for controlling tail - drop. Traffic that is at a lower precedence will encoun ter a higher drop rate, and is more likely t o be throttled.
DSCP (0 - 63)	DSCP value mapped to traffic matching t his enforcement profile.
802.1p (0 - 7)	Value of the 802.1p priority for trafic this matching this enforcement profile.

VoIP Profiles

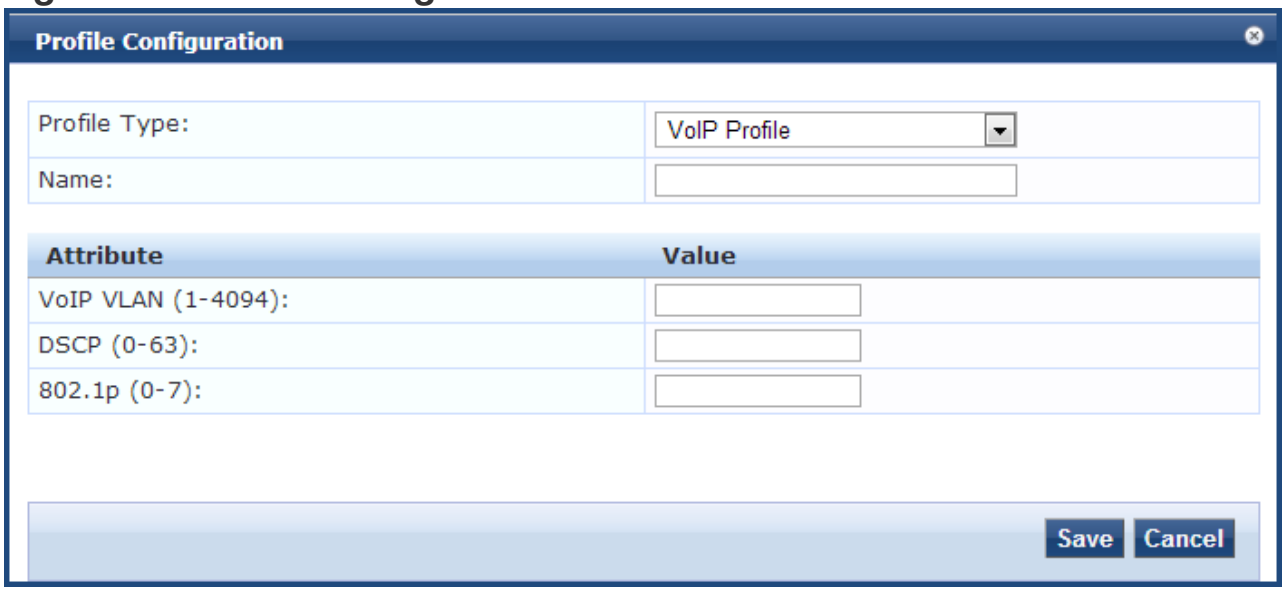
About this task

To define a VoIP Profile for an Aruba Mobility Access Switch:

Procedure

1. From the **Role Configuration** tab, click the **Add VoIP Profile** link. The **Add VoIP Profile** dialog opens:

Figure 5. Add VoIP Configuration Profile



2. Enter a name for the profile and configure the required attributes.

Table 10. VoIP Profile Attributes

Parameter	Action/Description
VoIP VLAN	VLAN to be assigned to VoIP traffic matc hing this enforcement profile.
DSCP (0 - 63)	DSCP value mapped to traffic matching t his enforcement profile.
802.1p (0 - 7)	Value of the 802.1p priority for this match ing this enforcement profile.

NetService Profiles

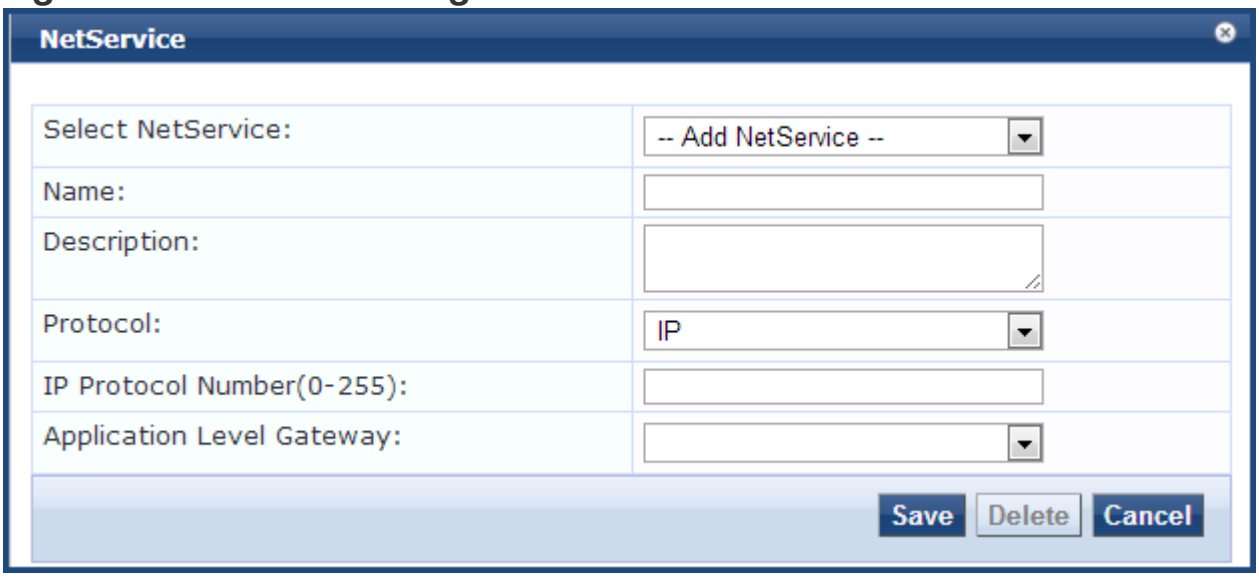
About this task

The NetService settings in a Downloadable Enforcement Profile vary, dependng upon whether you are definingNetService settings for an ArubaOS-Switch, or for a Mobiilty Controller or Mobiity Access Switch. To define a NetService Configuration profile for an Aruba Mobility Access Switch or ArubaOS-Switch:

Procedure

1. From the **Role Configuration** tab, click the **Manage NetServices** link. The **NetService** dialog opens:

Figure 6. NetService Configuration Profile



2. Enter a name for the profile and configure the required attributes.

Table 11. NetServices Attributes

Protocol	Action/Description
Protocol	Select a protocol: • IP • TCP • UDP
IP Protocol Number	If you selected **IP** in the **Protocol** field, en ter an IP protocol number (0 - 255)
Port Selection	If you selected **UDP** or **TCP** in the **Protoc ol** field, click this drop - down menu and select **List** or **Range** to identify how you will select the ports for that protocol.

Protocol	Action/Description
Port List	If you selected **List** in the **Port Selection** field, enter a comma - separated list of ports. **NOTE** If you are defining NetService settings for an ArubaOS - Switch, the **Port List** field is in sync with the maximum number of ports supported by the switch, and only accepts six entries. An error message is displayed if more than six ports are entered.
Port (1 - 65535) Max Port (1 - 65535)	If you selected **Range** in the **Port Selection** field, enter the starting port in the range in the **Port** field, and the ending port in the range in the **Max Port** field.
Application Level Gateway	Click this drop - down menu and select one of the following options: • DHCP • DNS • FTP • NOE • RTSP • SCCP • SIP • SIPS • SVP • TFTP • Vocera

NetDestination Profiles

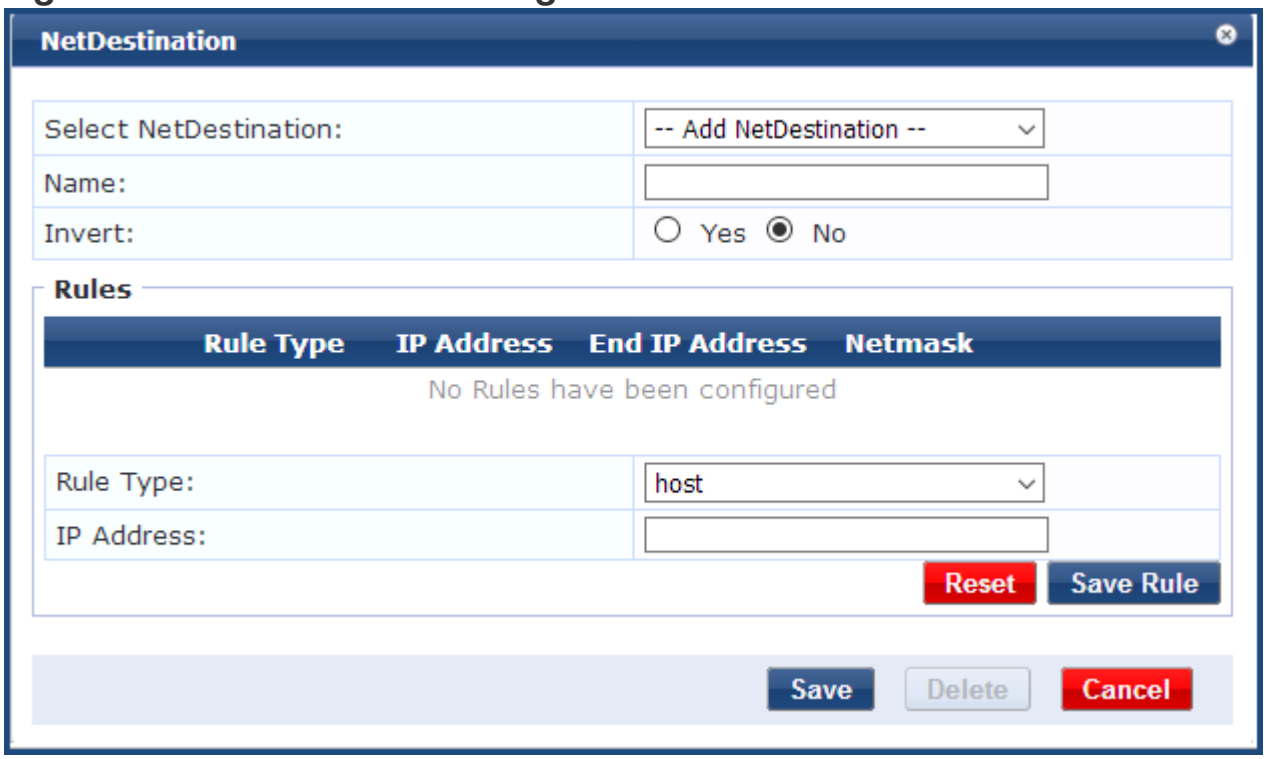
About this task

To define a NetDestination Configuration profile for an Aruba Mobility Access Switch or ArubaOS-Switch:

Procedure

1. Click the **Manage NetDestinations** link. The **NetDestinations** dialog opens:

Figure 7. Net Destinations Configuration Profile



2. Enter a name for the profile and configure the required attributes then click **Save Rule**.

Table 12. NetDestination Attributes

Parameter	Action/Description
Invert	Specifies that the inverse of the network addresses configured are used. For example, if a network of 172.16.0.0 is configured, this parameter specifies that the alias matches everything except this IP address.
Rule type	Select whether this netdestination rule type is defined by a **name** (identified by ho

Parameter	Action/Description
	stname or domain), a **host** (identified by an IPv4 or Ipv6 address) a **network** (ident ified by an IPv4 or IPv6 address and a net mask) or a **range** (identified by starting a nd ending IPv4/IPv6 addresses.)
Host Name or Domain	If you selected the **name** option in the **Ru le Type** field, enter a host or domain nam e. This field supports a single wildcard ch aracter (*) at the beginning of a host or d omain name for an Aruba Mobility Access Switch or controller.
IP address	If you selected the **host**, **network** or **rang e** option in the **Rule Type** field, enter the I P address of the host or network, or the s tarting IP address of the IP range.
Netmask	If you selected the **network** option in the **Rule Type** field, enter the subnetwork net mask.
End IP address	If you selected the **range** option in the **Rule Type** field, enter the end of an IP ran ge.

Time Range Profiles

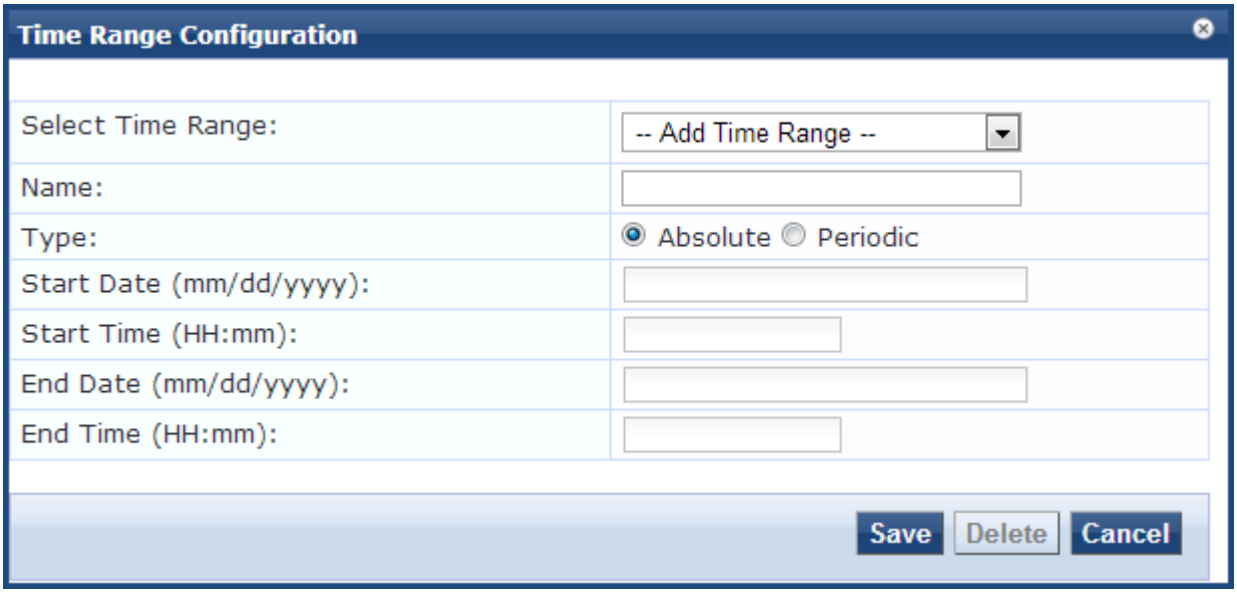
About this task

To define a Time Range Configuration profile for an Aruba Mobility Controller or Mobility Access Switch:

Procedure

1. From the **Role Configuration** tab, click the **Manage Time Ranges** link. The **Time Range Configuration** dialog opens:

Figure 8. Time Range Configuration Profile



2. Enter a name for the profile and configure the required attributes.

Table 13. Time Range Attributes

Parameter	Action/Description
Type	Select the type of time range profile. • Periodic— When configured, the enforcement policy role changes based on the time range configured in the profile. • Absolute—When configured, the enforcement policy role changes during a specific date / day and time.
Absolute Time Range Parameters	
Start Date (mm/dd/yyyy)	Enter the start date in **mm/dd/yyyy** format, or use the calendar function to browse to and select a date.
Start Time (HH:mm):	Enter the start time in **HH:mm** format, or use the time dropdown menu to select a time.

Parameter	Action/Description
End Date (mm/dd/yyyy)	Enter the end date in **mm/dd/yyyy** forma t, or use the calendar function to browse to and select a date.
End Time (HH:mm):	Enter the end time in **HH:mm** format, or use the time dropdown menu to select a time.
Periodic Time Range Parameters	
Start Day	Click the dropdown menu and select any of the **Start Day** options: • daily • weekend • weekday • Sunday • Monday • Tuesday • Wednesday • Thursday • Friday • Saturday
Start Time (HH:mm)	Enter the start time date in **HH:mm** forma t, or use the time dropdown menu to sele ct a time.
End Time (HH:mm)	Enter the end time in **HH:mm** format, or use the time dropdown menu to select a time.

NAT Pools

About this task

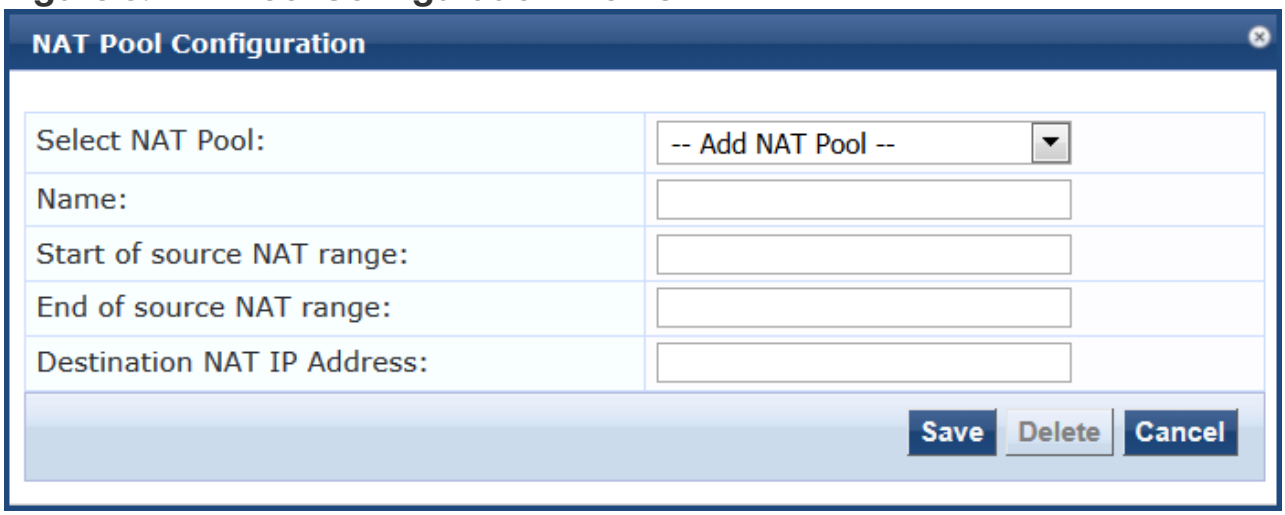
You can create a NAT pool associated with source NAT option or dual NAT option. When a pool is created with dual NAT option, both source IP and destination IP of the packet

are changed. To define a NAT (Network Address Translation) Pool Configuration profile for an Aruba Mobility Controller or Mobility Access Switch:

Procedure

1. From the **Role Configuration** tab, click the **Manage NAT Pool Configuration** link. The **NAT Pool Configuration** dialog opens:

Figure 9. NAT Pool Configuration Profile



2. Enter a name for the profile and configure the required attributes.

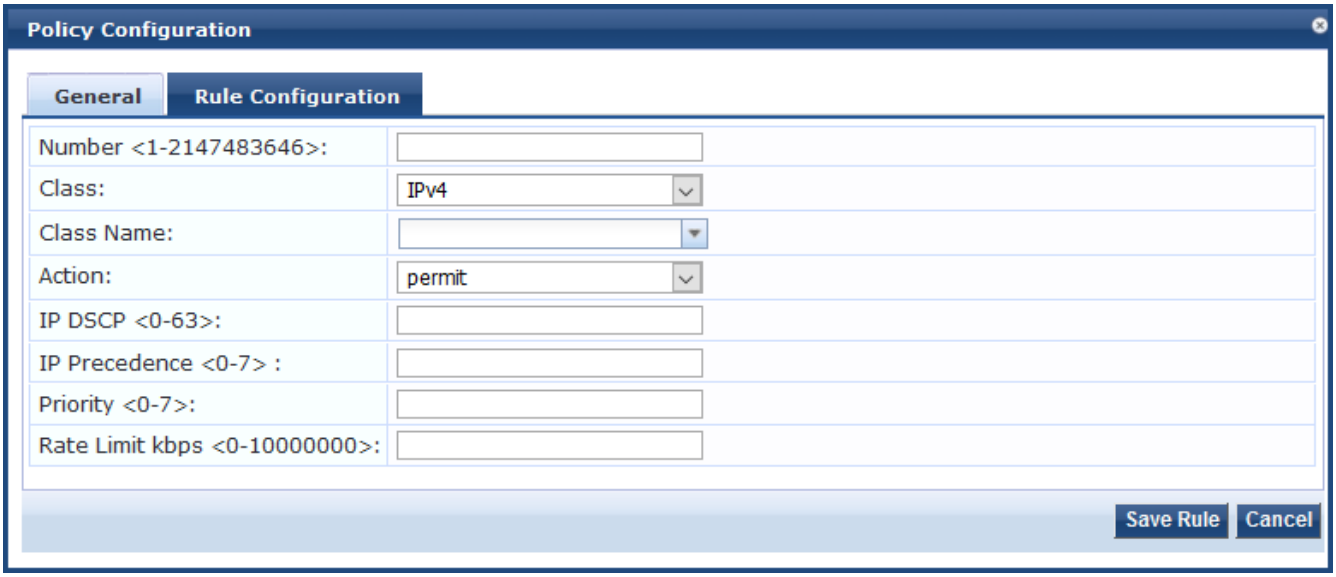
Table 14. Nat Pool Attributes

Parameter	Action/Description
Start of source NAT range	IP address at the start of the NAT range
End of source NAT range	IP address at the end of the NAT range
Destination IP address	If this parameter is configured, the destination IP changes to the IP from the NAT pool.

Policies

This setting appears on the **Role Configuration** tab when you configure an enforcement profile for an ArubaOS-Switch or AOS-CX device. When you select the **Add Policy** link on the **Role Configuration** tab, the following configuration dialog opens:

Figure 10. Policy Configuration Page



Enter the appropriate values for each of the fields in the **Policy Configuration > Rule Configuration** dialog, then click **Save Rule**. Table 15 describes the parameters for an ArubaOS Switch, and Table 16 describes the parameters for AOS-CX.

Table 15. Policy Configuration Page: ArubaOS - Switch Parameters

Parameter	Action/Description
Number	Enter a number.
Class	Select one of the following: • IPv4 • IPv6
Class Name	Specify the appropriate class name from t he drop - down menu.
Action	Select the desired action from the drop - d own list.
IP DSCP <0 - 63>	Enter a number from 0 to 63 to indicate th e IP DSCP (Differentiated Services Code Po int).
IP Precedence <0 - 7>	Specify the IP precedence by entering a nu mber from 0 to 7.
Priority	Enter a number from 0 to 7 to indicate QoS priority.

Parameter	Action/Description
Rate Limit Kpbs	Specify the rate limit in kilobits per second .

Table 16. Policy Configuration Page: AOS - CX Parameters

Parameter	Action/Description
Number	Enter a number.
Class	Select one of the following: • IPv4 • IPv6
Class Name	Specify the appropriate class name from t he drop - down menu.
Action	Select the desired action from the drop - d own list.
IP Precedence <0 - 7>	Specify the IP precedence by entering a nu mber from 0 to 7.
DSCP <0 - 63>	Enter a number from 0 to 63 to indicate th e IP DSCP (Differentiated Services Code Po int).
Committed Information Rate <1 - 42949672 95>	Specify the Committed Information Rate (CIR) for this policy. Traffic rates below CIR or bursts below CBS limits are considered “conforming” and are allowed to pass throu gh.
Committed Burst Size <1 - 4294967295>	Specify the Committed Burst Size (CBS) in bytes for this enforcement policy. Traffic ra tes below CIR or bursts below CBS limits a re considered “conforming” and are allowe d to pass through
Local Priority	Enter a number from 0 to 7 to indicate QoS priority.
Priority Code Point	Priority Code Point (PCP) value for traffic m atching this policy.

Classes

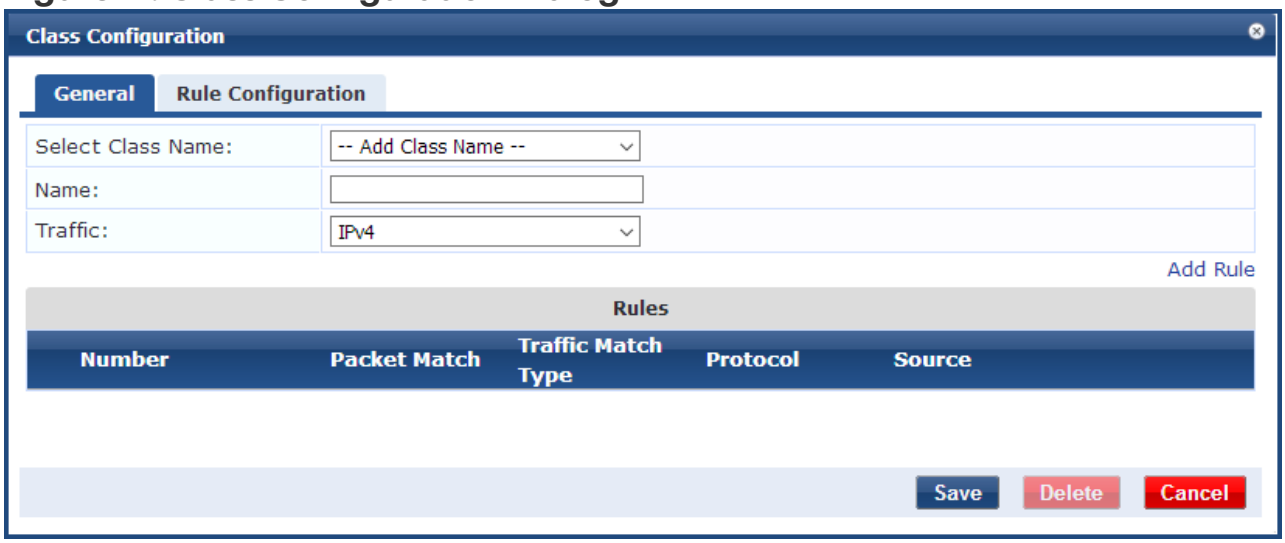
About this task

The **Manage Classes** link appears on the **Role Configuration** tab when you configure an enforcement profile for an ArubaOS-Switch or AOS-CX device. You can create and configure traffic classes and you can map the enforcement policy to the traffic classes.

Procedure

1. Select the **Manage Classes** link on the **Role Configuration** tab. The following configuration dialog opens:

Figure 11. Class Configuration Dialog



2. Specify the **Class Configuration** parameters as described in the following table.

Table 17. Class Configuration Page

Parameter	Action/Description
Select Class Name	Select the class name from the drop - d own list if it is already configured.
Name	Enter the Class name.
Traffic	Specify one of the following traffic types: • **IPv4** • **IPv6**

3. Click the **Rule Configuration** tab. The rule configuration dialog opens. The parameters displayed this dialog vary, depending upon whether you are creating a rule for ArubaOS-Switch or an AOS-CX device.
4. Specify the **Rule Configuration** parameters to create a rule for traffic that matches the selected conditions. The following tables describe the class rule parameters available for ArubaOS-Switch and AOS-CX.

Table 18. ArubaOS - Switch Class Rule Configuration Parameters

Parameter	Action/Description
Number	(Optional) Specify a sequence number fo r the rule in the access control list.
Packet Match	Select whether the rule should **match** or **ignore** the specified packet characteristi cs in this rule.
Protocol	Specify the appropriate traffic protocol f rom the drop - down.
Source	Select **Any** as the source option supporte d for ArubaOS.
Source Port	Select one of the following options: • **Equal to Port** • **Greater than Port** • **Less than Port** • **Except Port** • **Port Range <1 - 65535>**
Source Port Value	Specify the source port value.
Destination	Select one of the following destinations: • **Any** • **Host**

Parameter	Action/Description
	• **IP Subnet**. When you select **IP Subnet** , you are prompted for the **Source Val ue** (the IP subnet address).
Destination Port	Select one of the following options: • **Equal to Port** • **Greater than Port** • **Less than Port** • **Except Port** • **Port Range <1 - 65535>**
Destination Port Value	Specify the destination port value.
IP DSCP	Specify the Differentiated Services Code Point (DSCP) value.
VLAN ID	Specify the VLAN ID
IP Precedence	Specify the IP precedence: • Routine • Priority • Immediate • Flash • Flash Override • Critical • Internet • Network
IP Type of Service	Specify one of the following for the type of service: • Normal • Max Reliability • Max Throughput

Parameter	Action/Description
	• Minimum Delay
TCP Packets Bit Sets	Select one of the following: • **Established** • **Fin**: The FIN flag indicates the end of data transmission to finish a TCP conn ection. • **Rst**: Sets the Reset flag. • **Syn**: The SYN flag synchronizes seque nce numbers to initiate a TCP connect ion.

Table 19. AOS - CX Rule Configuration Parameters

Parameter	Action/Description
Number	(Optional) Specify a sequence number fo r the rule in the access control list.
Packet Match	Select whether the rule should **match** or **ignore** the specified packet characteristi cs in this rule.
Protocol	Specify the appropriate traffic protocol f rom the drop - down.
Protocol Number	Enter a protocol number from 0 - 255 to apply the rule to the traffic class using th at protocol.
Source IP	Specify if the traffic to which the rule is applied will come from a specific IP add ress, or from any IP address. If you select the IP option, specify the IPv4 or IPv6 so urce IP address in the **Source IP Address** field.

Parameter	Action/Description
Source Port	Specify an option to determine the sourc e port of the traffic to which the rule is ap plied. • Equal to Port • Less than Port • Greater than Port • Port Range
Source port value	Specify the source port value for the **Sou rce Port** setting above. To specify a port range, enter the beginning and ending p ort numbers separated by a space.
Destination	Specify if the traffic to which the rule is a pplied will be sent to a specific IP addres s or host, or from any IP address or host. I f you select the **IP Subnet** or **Host** option, specify the IPv4 or IPv6 destination IP ad dress in the **Destination IP Address** field.
Destination Port	Specify an option to determine the desti nation port of the traffic to which the rule is applied. • Equal to Port • Less than Port • Greater than Port • Port Range
Destination port value	Specify the destination port value for th e **Destination Port** setting above. To spec ify a port range, enter the beginning and ending port numbers separated by a spa ce.
DSCP	Specify the differentiated services code point (DSCP) value for the traffic.
VLAN ID	Specify the VLAN ID.

Parameter	Action/Description
ECN	Specify an Explicit Congestion Notificati on (ECN) value.
IP Precedence	Specify the IP precedence: • Routine • Priority • Immediate • Flash • Flash Override • Critical • Internet • Network
IP Type of Service	Specify the value of the ToS header. (0 - 31)
Fragment	This option in the header means that pac ket can fragmented into smaller packets and later reassembled later if part of that packet's route cannot manage the size of the packet.
Count Packet	If enabled, the AOS - CX switch keeps th e hit counts of the number of packets ma tching this ACE.
TCP Flags	Specify matching TCP flags. • **crw**: Specifies matching on the TCP Fl ag the Congestion Window Reduced (RFC3168). • **ece**: Specifies matching on the TCP Fl ag ECN - Echo (RFC3168). • **urg**: Specifies matching on the TCP Fl ag Urgent. • **ack**: Specifies matching on the TCP Fl ag Acknowledgment.

Parameter	Action/Description
	• **psh**: Specifies matching on the TCP Flag: Push buffered data to receiving application. • **rst**: Specifies matching on the TCP Flag Reset the connection. • **syn**: Specifies matching on the TCP Flag: Synchronize sequence numbers. • **fin**: Specifies matching on the TCP Flag Finish connection. • **established**: Specifies matching on the TCP Flag Established connection.
ICMP Type	Specify if the rule should match ICMP **echo** or **echo - reply** traffic, or a specific value in the ICMP code field (**<0 - 255>**).
ICMP Code	Specify the ICMP code.

Stateless Access Control Lists

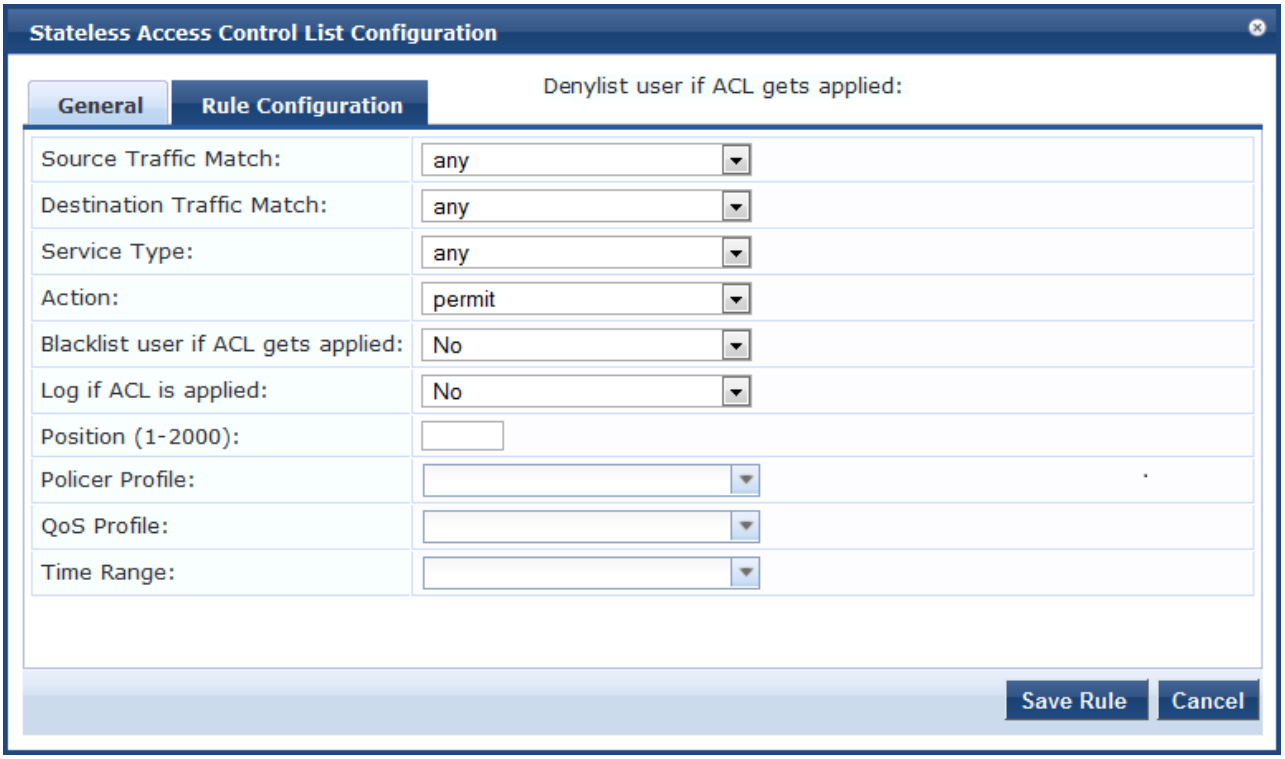
About this task

Stateless Access Control Lists (ACLs) are used to define stateless packet filtering and quality of service (QoS). A stateless ACL statically evaluates packet contents. The traffic in the reverse direction will be allowed unconditionally. To add a Stateless Access Control List to a Mobility Access Controller:

Procedure

1. From the **Role Configuration** tab, click the **Add Stateless Access Control List** link. The **Stateless Access Control List Configuration** dialog opens:

Figure 12. Stateless Access Control List Configuration Profile



2. On the **General** tab, enter a name for the Stateless ACL.
3. Click the **Add Rule** link. The **Rule Configuration** dialog opens.
4. Specify the rule attributes as described in the table below, then click **Save Rule**.

Table 20. Stateless Access Control List Configuration

Parameter	Action/Description
Source Traffic Match	Select one of the following options: • **alias**: Match an IP network resource • **any**: Match any source traffic • **host**: Match a single host address • **network**: Match an IP subnet
Source IP address	IP address of the source host or network.
Source Netmask	If you selected the **network** option in the **Source Traffic Match** field, enter the net mask of the source network.
Destination Traffic Match	Select one of the following options: • **alias**: Match an IP network resource

Parameter	Action/Description
	• **any**: Match any source traffic • **host**: Match a single host address • **network**: Match an IP subnet
Destination IP address	IP address of the destination host or net work.
Destination Netmask	If you selected the **network** option in the **Destination Traffic Match** field, enter the netmask of the destination network.
Service Type	Specify the name of the service to match the rule. • **any**: Match any traffic • **arp**: Match ARP traffic • **service**: Match a service defined by a Policy Manager NetService Configurat ion • **tcp** Match TCP traffic • **udp** Match UDP traffic
Action	Choose one of the following actions to ta ke on matching traffic: • **deny**: Reject packets. • **permit**: Forward packets. • **redirect**: Specify the location to whic h packets are redirected. If you select this option, you are prompted to spec ify one of the following redirect types: ◦ **ipsec**: Specify the name of an IPSe c map to redirect matching traffic over a VPN tunnel. ◦ **tunnel**: Specify a tunnel ID to redi rect packets over an L3 GRE tunnel .

Parameter	Action/Description
Denylist user if ACL gets applied	If the ACE entry is matched, the traffic fro m that particular user is denied and the u ser is added to a blocked list for 3600 se conds.
Log if ACL is applied	If enabled, log information is displayed w hen the ACL is applied.
Position (1 - 2000)	Manages the position of the rule within t he Acces Control List (ACL).
Policer Profile	This parameter defines a policer profile t o manage the transmission rate of a class of traffic based on user - defined criteria. For more information, see Policer Profile.
QoS Profile	Click the QoS profile dropdown menu to select a set of Traffic - Class/Drop - Prece dence, DSCP, and 802.1p values for a Mob ility Access Switch. For more information, see QoS Profile.
Time Range	Select the **Time Ranges** dropdown menu to create a rule matching time range defi nitions. For more information, see Time R ange Configuration.

Session Access Control Lists

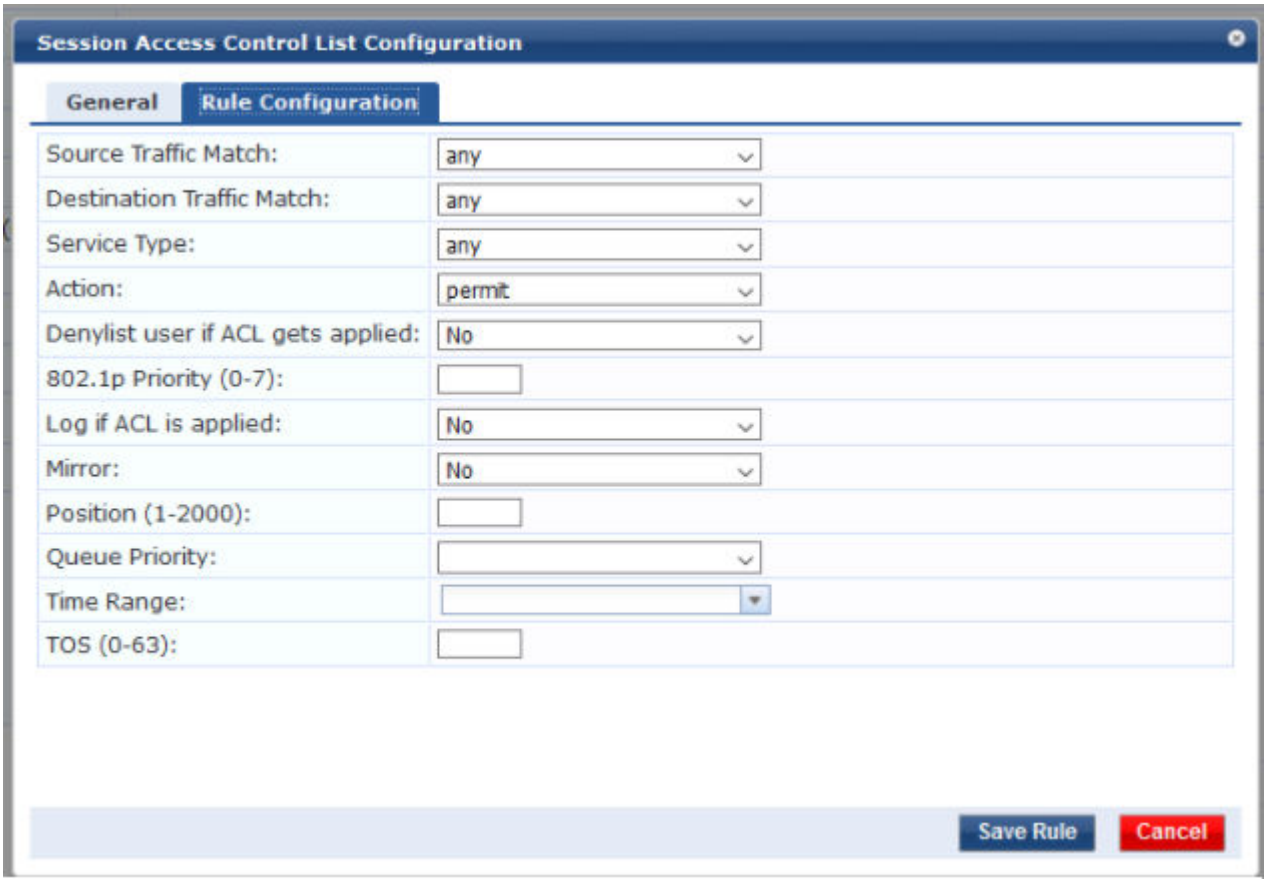
About this task

To add a Session Access Control List to a downloadable role enforcement profile for a mobility controller or mobility access switch:

Procedure

1. From the **Role Configuration** tab, click the **Add Session Access Control List** link. The **Session Access Control List Configuration** dialog opens.
2. Enter a name for the Session ACL.
3. On the **General** tab, click the **Add Rule** link. The **Rule Configuration** dialog opens.

Figure 13. Session Access Control List Rule Configuration Profile



4. Enter the required attributes in the **Rule Configuration** dialog then click **Save Rule**. Note that this dialog can display different fields depending on the selected options. For example, if you select the **dual-nat** action type, you can view the **Dual NAT Pool** field to specify the action.

Table 21. Session Access Control List Rule Attributes

Parameter	Action/Description
Source Traffic Match:	Source of the traffic, which can be one of the following: • **alias**: Refers to using an alias for a ho st or network. You configure the alias by navigating to the **Configuration > Roles**& **Policies > Policies**tab. Select a policy created and click + to create a Rule. Select the Access Control optio n in the Rule Type. Select Alias from t he Destination drop - down list and th e alias name from the Destination alia

Parameter	Action/Description
	s drop - down list. Select a Source fro m the traffic Source drop - down list. • **any**: Acts as a wildcard and applies to any source address. • **host**: Refers to traffic from a specific host. When this option is chosen, you must configure the IP address of the h ost. • **network**: Refers to a traffic that has a source IP from a subnet of IP addres ses. When this option is chosen, you must configure the IP address and net work mask of the subnet. • **user**: Refers to traffic from the wireles s client.
Destination Traffic Match:	Destination of the traffic, which can be c onfigured in the same manner as **Source Traffic Match**.
Service Type:	Type of traffic, which can be one of the fo llowing: • **any**: This option specifies that this rul e applies to any type of traffic. • **protocol**: Enter a protocol number (0 - 255) to apply this rule to a specific protocol. • **service**: Select a Netservice configur ation. • **tcp**: Using this option, you configure a range of TCP port(s) to match for th e rule to be applied. • **udp**: Using this option, you configure a range of UDP port(s) to match for th e rule to be applied.

Parameter	Action/Description
Action:	The action that you want to perform on a packet that matches the specified criteria. This can be one of the following: • **deny**: Drops packets matching this rule. • **dst - nat**: This option redirects traffic to the IP address and destination port defined in the **Destination NAT IP Address** and **Destination NAT Port** fields. • **dual - nat**: This option performs both source and destination NAT on packets matching the rule. If you select this option, you are prompted to select a NAT pool in the **Dual NAT Pool** field. For more information, see NAT Pool Configuration • **permit**: Permits traffic matching this rule. • **redirect**: This option redirects traffic into a GRE tunnel specified in the **Tunnel ID** field. • **src - nat**: Performs NAT on packets matching the rule. When this option is selected, you need to select a NAT pool in the **Source NAT Pool** field. For more information, see NAT Pool Configuration • **permit**: Permits traffic matching this rule.
Denylist user if ACL gets applied	Automatically adds to a blocked list any client that is the source or destination of traffic matching this rule. This option is recommended for rules that indicate a security breach where the option to add to a blocked list can be used to prevent access

Parameter	Action/Description
	to clients that are attempting to breach t he security.
802.1p Priority (0 - 7)	When this parameter is enabled, the valu e of 802.1p priority bits are marked in the frame of a packet matching this rule whe n it leaves the managed device. 0 is the l owest priority (background traffic) and 7 i s the highest (network control).
Log if ACL is applied	Logs a match to this rule. This is recomm ended when a rule indicates a security b reach, such as a data packet on a policy t hat is meant only to be used for voice cal ls.
Mirror	Mirrors session packets to datapath or re mote destination.
Position (1 - 2000):	Use this field to select a position for this r ule in the Session Access Control List. Th e rule in position 1 is executed first.
Queue Priority	Select the queue in which a packet matc hing this rule should be placed. Select **High** for higher priority data, such as voice, and **Low** for lower priority traffic .
Time Range	Click the dropdown menu and select a ti me range. You can create an absolute tim e range with a single fixed start and end date and time, or create a periodic (recur ring) time range that starts and ends at a specified time on a weekday, weekend, or selected day. For more information, see Ti me Range Configuration
TOS (0 - 63)	Select the value of the TOS bits in the IP header of a packet,

Ethertype/MAC Access Control Lists

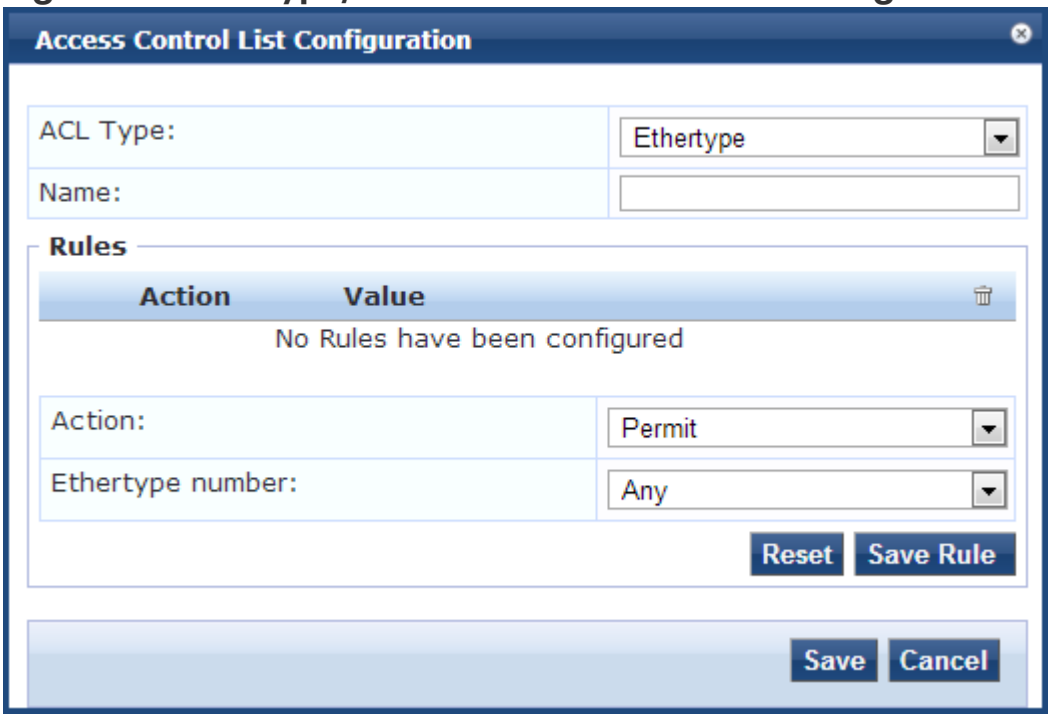
About this task

Ethertype ACLs are used to filter based on the **Ethertype** field in the frame header. These ACLs can be used to permit IP while blocking other non-IP protocols, such as IPX or AppleTalk. To add an Ethernet/MAC Access Control List to a downloadable role enforcement profile for a mobility controller or mobility access switch:

Procedure

1. From the **Role Configuration** tab, click the **Add Ethertype/MAC Access Control List** link. The **Access Control List Configuration** dialog opens. The ACL Type is set to **Ethertype**.

Figure 14. Ethertype/MAC Access Control List Configuration Profile



2. Enter a name for the Ethertype/MAC Access Control List.
3. Enter the required attributes in the **Rules** section of the page and click **Reset**, then click **Save Rule**.
4. When finished, click **Save**.

Table 22. Ethertype/MAC Access Control List Attributes

Parameter	Action/Description
Action	The action that you want to perform on a packet that matches the specified criteri a. This can be one of the following: • **deny**: Drops packets matching this rul e. • **permit**: Permits traffic matching this r ule.
Ethertype number	Select **Any** to apply the access control li st to any Ethertype value, or select **Enter Value** to specify a ethernet value (0 - 655 35) and subnet bits (0 - 65535)

Role Configuration: Advanced Mode

When you set **Role Configuration Mode** to **Advanced**, the Enforcement Profile page displays the **Attributes** tab (see Figure 18 below).

> **NOTE**
> In Advanced mode, a validation check is not available for downloadable role names that are greater than 64 characters. This is due to a limitation on the switch. Thus, if a downloadable role name configured on the Policy Manager server exceeds 64 characters, the enforcement profile may fail on the switch.

In **Advanced** mode, the **Aruba Downloadable Role Enforcement** profile provides two dictionaries and two attributes.

- Mobility Controllers, and Mobility Access Switches and AOS-CX devices use the **Aruba** dictionary and the **Aruba-CPPM-Role** and Aruba-**UBT-Gateway-CPPM-Role** attributes.
- ArubaOS-Switch devices use the **Hewlett-Packard-Enterprise** dictionary and the **HPE-CPPM-Role** and **HPE-CPPM-Secondary-Role** attributes. The **HPE-CPPM-Secondary-Role** attribute adds support for a downloadable secondary role that can be used with Per User Tunneled Node (PUTN). When the attribute is added to the

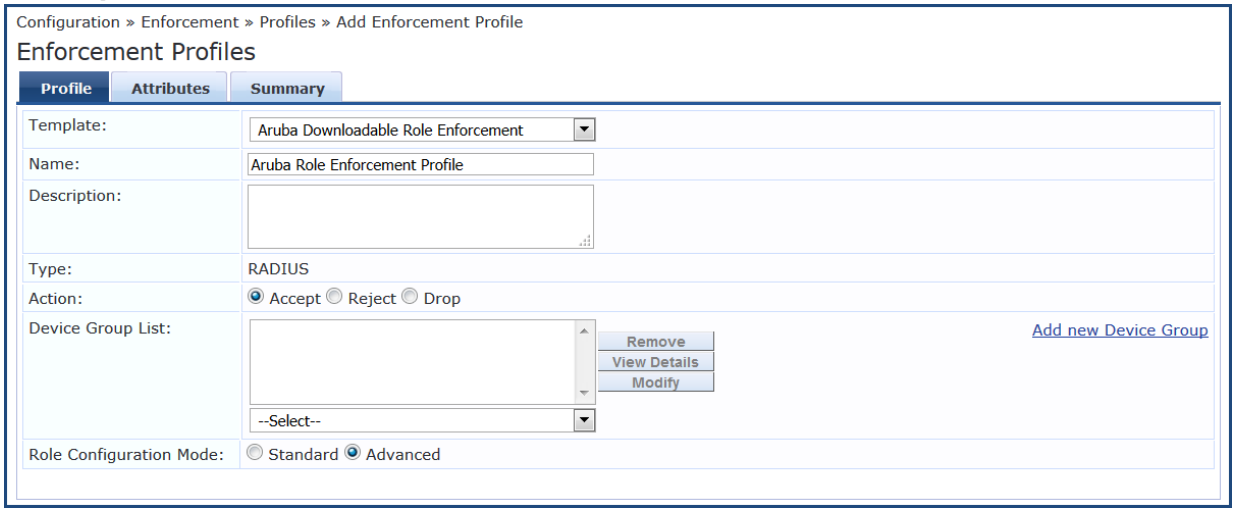
enforcement profile, Policy Manager can send the controller role for the ArubaOS switch.

> **NOTE**
>
> You can use only one of the Advanced mode dictionaries at any given time; these dictionaries can't both be used at the same time.

To configure the **Aruba Downloadable Role Enforcement > Advanced** attributes:

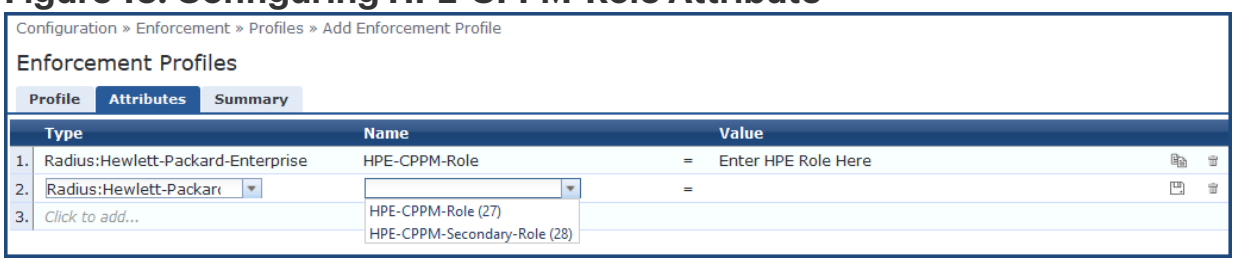
1. Navigate to **Configuration > Enforcement > Profiles > Add.**
2. Select **Aruba Downloadable Role Enforcement** from the **Template** drop-down list.
3. Configure the settings on the **Profile** tab as described in Configuring Roles: Standard Mode, but select the **Advanced** option for the **Role Configuration Mode** setting.

Figure 15. Downloadable Role Enforcement > Profile Tab (Advanced Mode)



4. Next, select the **Attributes** tab. The appropriate RADIUS dictionary for the selected device type is enabled by default.

Figure 16. Configuring HPE-CPPM-Role Attribute



5. In the **Value** field, enter the appropriate commands.

   a. For RADIUS dictionary type **Radius: Hewlett-Packard Enterprise**, select **HPE-CPPM-Role (27)** or **HPE-CPPM-Secondary-Role (28)**, then enter a value in the **Value** field.

   b. For RADIUS dictionary type **Radius: Aruba** select **Aruba-CPPM-Role (23)** or **Aruba-UBT-Gateway-CPPM-Role (59)**, then enter a value in the **Value** field.

6. Click **Save**.

Summary Information

For a profile in **Standard** Role configuration mode, the **Summary** tab summarizes the parameters configured in the **Profile** and **Role Configuration** tabs. For a profile in **Advanced** role configuration mode, the **Summary** tab summarizes the parameters configured in the **Profile** and **Attribute** tabs.

Aruba RADIUS Enforcement Profile

This section describes the following Aruba RADIUS Enforcement profile features:

Profile Configuration

Use the **Profile** tab to configure the template, type of the profile, and device group list. The following figure displays the **Aruba RADIUS Enforcement > Profile** tab:

Figure 1. ArubaRADIUS Enforcement > Profile Tab

The following table describes the **Aruba RADIUS Enforcement > Profile** tab parameters:

Table 1. Aruba RADIUS Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select Aruba RADIUS Enforcement.
Name	Enter the name of the profile.

Parameter	Action/Description
	The name is displayed on the **Configuration > Enforcement > Profiles** page.
Description	Enter a description that provides additional information about the profile. This description is displayed in on the **Configuration > Enforcement > Profiles** page.
Type	This field is populated automatically.
Action	Click **Accept**, **Reject**, or **Drop** to define the action taken on the request.
Device Group List	Select a device group from the drop - down list. The list displays all configured device groups. All configured device groups are listed in the **Configuration > Network > Device Groups** page. After adding one or more device group(s), you can select a group and take one of the following actions: • To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**. • To change the parameters of the selected device group, click **Modify**.
Add New Device Group	Click this link to add a new device group, For more information, see Adding and Modifying Device Groups .

Attributes Configuration

Use the **Attribute** tab to configure the attribute type, name, and value for the enforcement profile.

The following table describes the **Aruba RADIUS Enforcement > Attributes** parameters:

Table 2. Aruba RADIUS Enforcement > Attributes Parameters

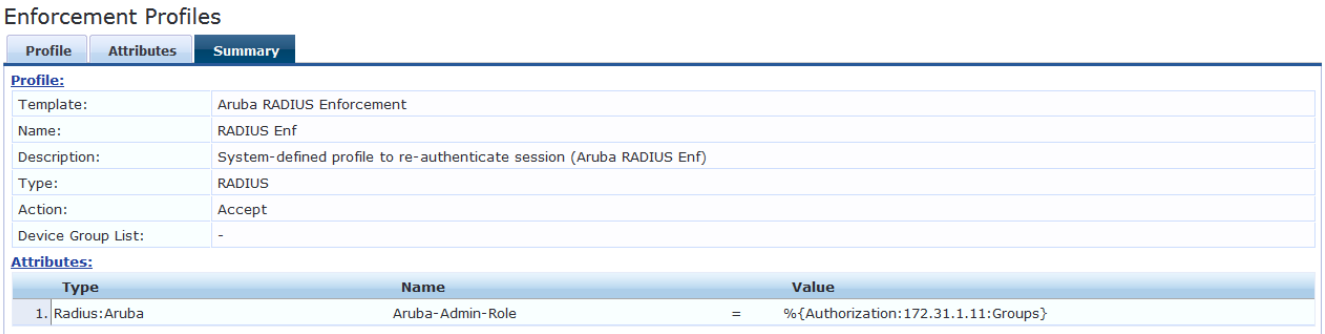
Attribute	Action/Description
Type	Select one of the following a

Attribute	Action/Description
	• Radius:Aruba • Radius:IETF • Radius:Cisco • Radius: Hewlett - Packare • Radius: Lucent - Alcatel - • Radius:Microsoft • Radius:Avenda For more information, see RA
Name	Select the appropriate **Name** The options provided for the on the **Type** attribute selecte
Value	Specify the appropriate **Valu** The options provided for the on the **Type** and **Name** attrib

Summary Information

The **Summary** tab summarizes the parameters configured in the **Profile** and **Attributes** tab.

Figure 2. Aruba RADIUS Enforcement > Summary Tab



Cisco Downloadable ACL Enforcement Profile

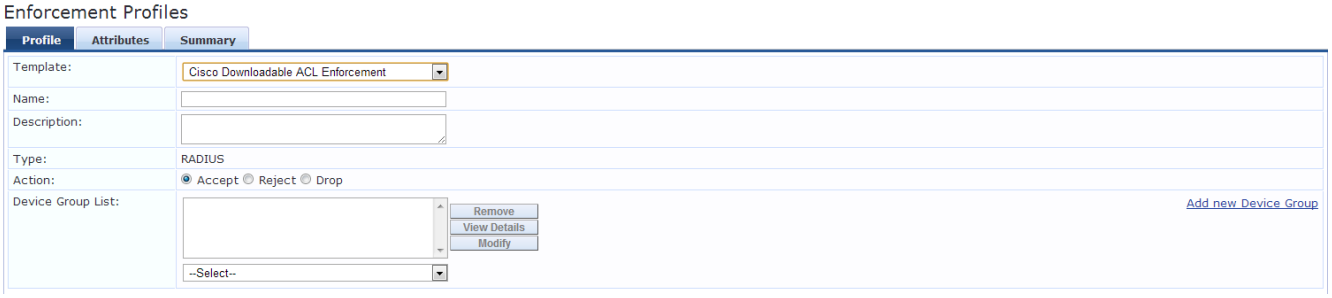
Use this page to configure the Cisco Downloadable ACL Enforcement profile.

Profile Configuration

Use the **Profile** tab to configure the Cisco Downloadable ACL Enforcement profile.

The following figure displays the **Cisco Downloadable ACL Enforcement > Profile** dialog:

Figure 1. Cisco Downloadable ACL Enforcement > Profile Dialog



Specify the **Cisco Downloadable ACL Enforcement > Profile** parameters as described in the following table:

Table 1. Cisco Downloadable ACL Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select the **Cisco Downloadable ACL Enforcement** template.
Name	Enter the name of the profile. The name is displayed in the **Name** column on the **Configuration > Enforcement > Profiles** page.
Description	Enter a description of the profile. The description is displayed in the **Description** column on the **Configuration > Enforcement > Profiles** page.
Type	The field is populated automatically with Type: **RADIUS**.
Action	To define the action to take on the request, click **Accept**, **Reject**, or **Drop**.
Device Group List	Select a Device Group from the drop - down list. The list displays all configured device groups. All configured device groups are listed in the **Configuration > Network > Device Groups** page. After adding one or more device

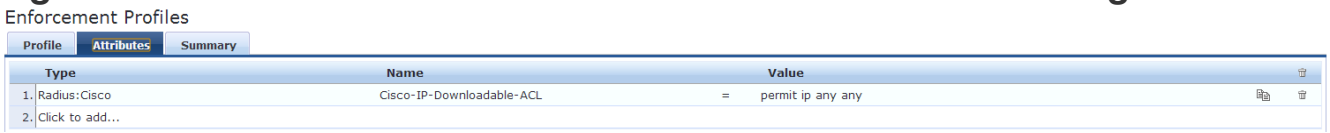
Parameter	Action/Description
	group(s), you can select a group and take one of the following actions: • To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**. • To change the parameters of the selected device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Group** link and see Adding and Modifying Device Groups .

Attributes Configuration

About this task

Use the **Attribute** tab to configure the attribute type, name, and value for the enforcement profile. The following figure displays the **Cisco Downloadable ACL Enforcement > Attributes** dialog:

Figure 2. Cisco Downloadable ACL Enforcement > Attributes Dialog



Procedure

1. Specify the **Cisco Downloadable ACL Enforcement > Attributes** parameters as described in the following table:

Table 2. Cisco Downloadable ACL Enforcement > Attributes Parameters

Parameter	Action/Description
Type	The attribute type **Radius: Cisco** is selected by default, though you can click this option and select any of the available attribute types:

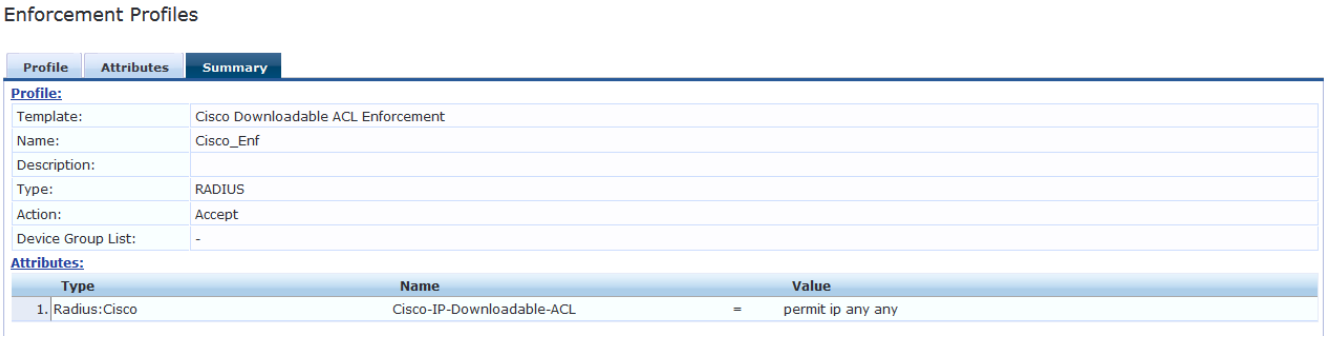
Parameter	Action/Description
	• Radius:Aruba • Radius:IETF • Radius:Cisco • Radius: Hewlett - Packard - Enterprise • Radius: Lucent - Alcatel - Enterprise • Radius:Microsoft • Radius:Avenda For more information, see RADIUS Names paces .
Name	If you retained the default **Radius: Cisc o** type, the default value for this setting i s **Cisco - IP - Downloadable - ACL**, altho ugh the options displayed for the **Name** a ttribute depend on the **Type** attribute tha t was selected.
Value	By default, the ACL for the Cisco - IP - D ownloadable - ACL is **permit IP any any**.

2. (Optional) To create another attribute, select the **Click to Add** field to define another new attribute type for that profile.
3. Click **Save**.

Summary Information

The **Summary** tab summarizes the parameters configured in the **Cisco Downloadable ACL Enforcement profile**.

Figure 3. Cisco Downloadable ACL Enforcement > Summary Tab



Cisco Web Authentication Enforcement Profile

Use this page to configure profile and attribute parameters for the **Cisco Web Authentication Enforcement** profile.

Profile Configuration

Use the **Profile** tab to configure the template, type of the profile, and device group list.

Figure 1. Cisco Web Authentication Enforcement > Profile Tab

Specify the **Cisco Web Authentication Enforcement > Profile** tab parameters as described in the following table:

Table 1. Cisco Web Authentication Enforcement > Profile Tab Parameters

Parameter	Action/Description
Template	Select the **Cisco Web Authentication Enforcement** template.
Name	Enter the name of the profile.
Description	Enter a description that provides additional information about the profile (recommended).
Type	This field is populated automatically.

Parameter	Action/Description
Action	Click **Accept**, **Reject**, or **Drop** to define the action taken on the request.
Device Group List	Select a device group from the drop - dow n list. The list displays all configured devic e groups. All configured device groups are listed in the **Configuration > Network > D evice Groups** page. After adding one or m ore device group(s), you can select a group and take one of the following actions: • To delete the selected Device Group Lis t entry, click **Remove**. • To see the device group parameters, cli ck **View Details**. • To change the parameters of the select ed device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Gro up** link and see Adding and M odifying Device Groups .

Attributes Configuration

About this task

Use the **Attributes** tab to configure the attribute name and attribute value. The following figure displays the **Cisco Web Authentication Enforcement > Profile** tab:

Figure 2. Cisco Web Authentication Enforcement > Attributes Tab

Procedure

1. Specify the **Cisco Web Authentication Enforcement > Attributes** parameters as described in the following table:

Table 2. Cisco Web Authentication Enforcement > Attributes Parameters

Parameter	Description
Type	The attribute type **Radius: Cisco** is selected by default, though you can click this option and select any of the available attribute types: • Radius:Aruba • Radius:IETF • Radius:Cisco • Radius: Hewlett - Packard - Enterprise • Radius: Lucent - Alcatel - Enterprise • Radius:Microsoft • Radius:Avenda For more information, see RADIUS Namespaces .
Name	If you retained the default **Radius: Cisco** type, select any of the Cisco vendor - specific RADIUS attributes. The options displayed for the **Name** attribute depend on the **Type** attribute that was selected.
Value	The options displayed for the **Value** attribute depend on the **Type** and **Name** attributes that were selected.

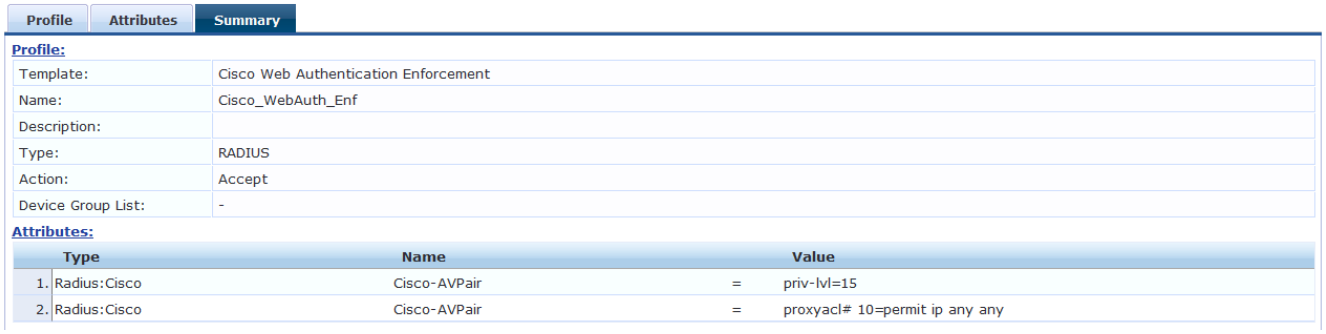
2. (Optional) To create another attribute, select the **Click to Add** field to define another new attribute type for that profile.
3. Click **Save**.

Summary Information

The **Summary** tab summarizes the parameters configured in the **Profile** and **Attribute** tabs.

Figure 3. Cisco Web Authentication Enforcement > Summary Tab

Enforcement Profiles



Policy Manager Entity Update Enforcement Profile

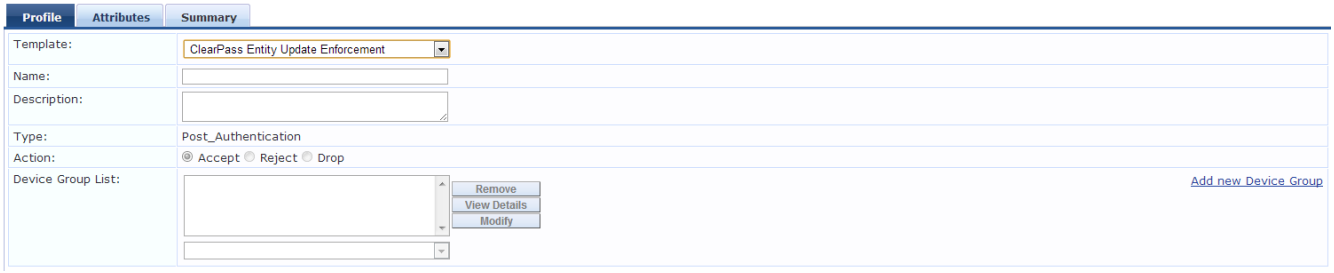
Use this page to configure profile and attribute parameters for the **Policy Manager Entity Update Enforcement** profile.

Profile Configuration

Use the **Profile** tab to configure the template, type of the profile, and device group list.

Figure 1. Policy Manager Entity Update Enforcement > Profile Tab

Enforcement Profiles



Specify the **Policy Manager Entity Update Enforcement > Profile** parameters as described in the following table:

Table 1. Policy Manager Entity Update Enforcement > Profile Parameters

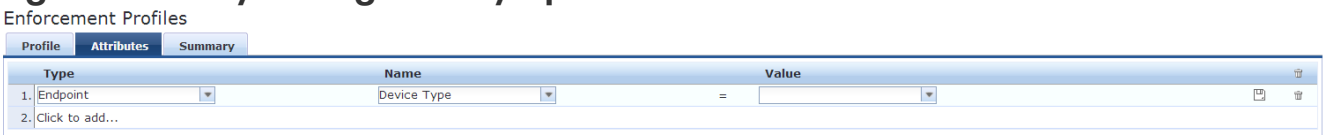
Parameter	Action/Description
Template	Select the template from the drop - down l ist. In this context, select Policy Manager E ntity Update Enforcement.
Name	Enter the name of the profile. The name is displayed in the **Name** column on the **Conf iguration > Enforcement > Profiles** page.

Parameter	Action/Description
Description	Enter a description that provides additiona l information about the profile. This descrip tion is displayed in the **Description** column on the **Configuration > Enforcement > Prof iles** page.
Type	This field is populated automatically with t he Post_Authentication type.
Action / Device Group List	This parameters are disabled because they are not applicable to the Agent Script Enfo rcement Profile.

Attributes Configuration

Use the **Attribute** tab to configure the attribute type, name, and value for the enforcement profile. The following figure displays the **Policy Manager Entity Update Enforcement > Attributes** tab:

Figure 2. Policy ManagerEntity Update Enforcement Attributes tab



Specify the **ClearPass Entity Update Enforcement > Attributes** parameters as described in the following table:

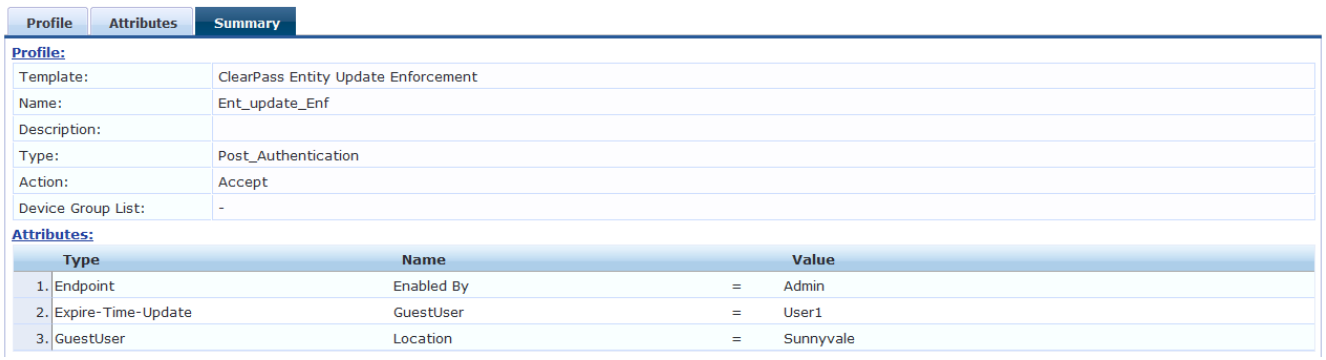
Table 2. Policy Manager Entity Update Enforcement > Attributes Parameters

Attribute	Description
Type	Select one of the followin • Endpoint • Expire - Time - Update • GuestUser • Status - Update
Name	The options displayed for on the **Type** attribute that
Value	The options displayed for on the **Type** and **Name** at

Summary Information

The **Summary** tab summarizes the parameters configured in the **Profile** and **Attributes** tab.

Figure 3. Policy ManagerEntity Update Enforcement > Summary Tab



CLI-Based Enforcement Profile

CLI-based Enforcement profiles allow Policy Manager to execute CLI commands on endpoint devices as part of agent enforcement. Use the **Command** attribute on the **Attributes** tab to enter the command you want to run. The **CLI-Based Enforcement** profile contains the following tabs:

Profile Configuration

About this task

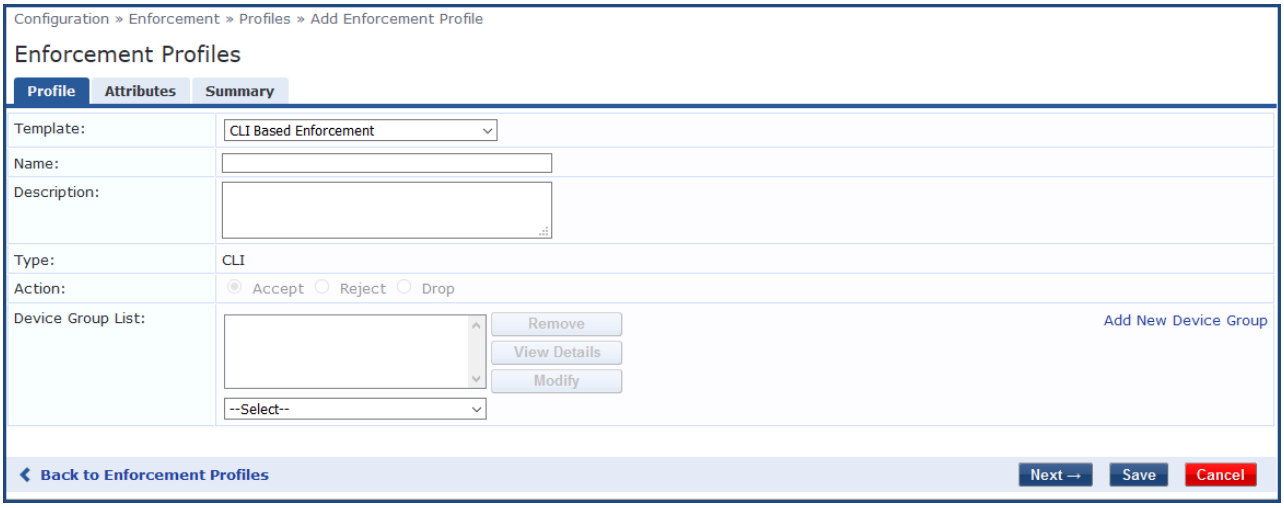
To configure a CLI-based enforcement profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**.
2. From the Enforcement Profiles page, click the **Add** link.
3. From the Add Enforcement Profile page **Template** drop-down, select **CLI Based Enforcement**.

   The following screen is displayed:

   Figure 1. CLI-Based Enforcement > Profile Tab



Use the **Profile** tab to configure the template, type of the profile, and device group list.

4. Specify the **CLI-Based Enforcement > Profile** tab parameters as described in the following table:

Table 1. CLI Based Enforcement > Profile Parameters

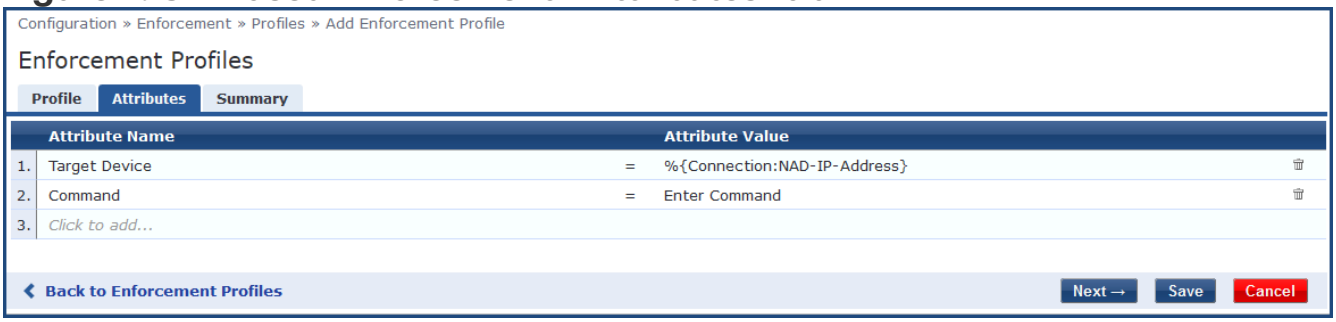
Parameter	Action/Description
Template	Select the **CLI Based Enforcement** template.
Name	Enter the name of the profile.
Description	Enter a description that provides additional information about the enforcement profile.
Type	This field is populated automatically with **CLI.**
Action	The **Action** option for this enforcement profile is preset to **Accept**.
Device Group List	Select a device group from the drop - down list. The list displays all configured device groups. All configured device groups are listed in the **Device Groups** (**Configuration > Network > Device Groups**) page.

Parameter	Action/Description
	After adding one or more device group(s) , you can select a group and take one of t he following actions: • To delete the selected **Device Group List** entry, click **Remove**. • To see the device group parameters, c lick **View Details**. • To change the parameters of the sele cted device group, click **Modify**. **NOTE** To add a new a device grou p, click the **Add New Devic e Group** link and see Adding and Modifying Device Group s .

Attributes Configuration

Use the **Attribute** tab to configure the attribute type, name, and value for the enforcement profile.

Figure 2. CLI Based Enforcement > Attributes Tab



Specify the **CLI Based Enforcement > Attributes** parameters as described in the following table:

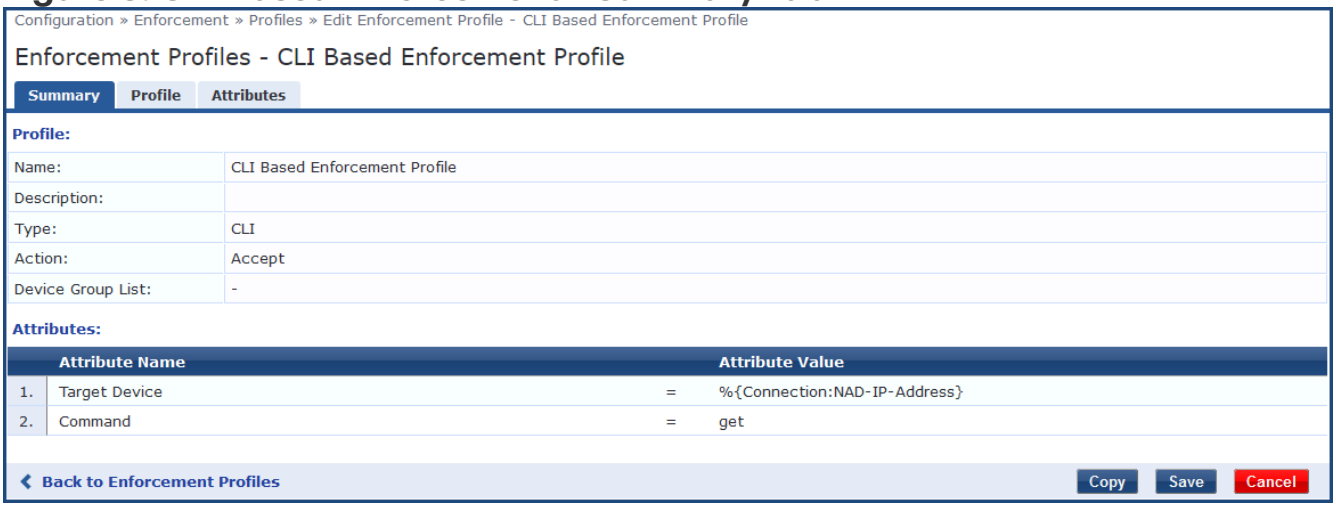
Table 2. CLI Based Enforcement > Attributes Parameters

Attribute	Action/Parameter
Attribute Name	Select **Command** or **Target Device**. A CLI enforcement profile must contain at least one **command** attribute and one **target device** attribute.
Attribute Value	The default value for the **target device** attribute is **%{Connection:NAD - IP - Address}**. Click in the **Enter Command** field to enter a command to be executed for the **command** attribute

Summary Information

The **Summary** tab summarizes the parameters configured in the **Profile** and **Attributes** tab.

Figure 3. CLI-Based Enforcement > Summary Tab



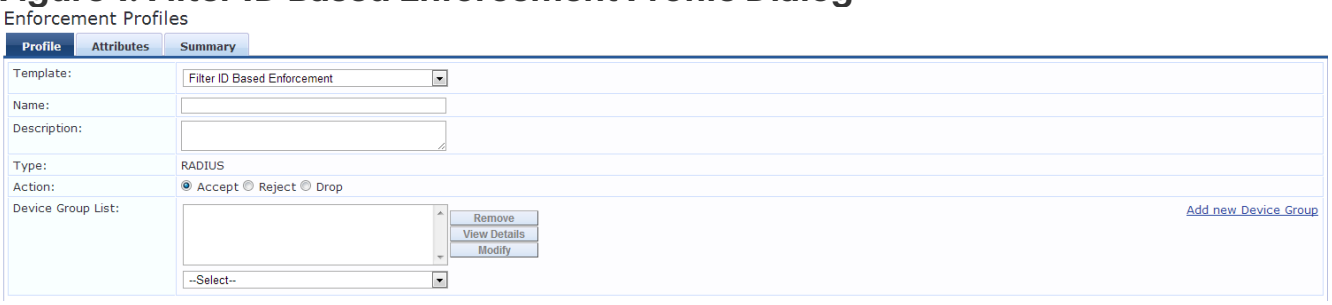
Filter ID Based Enforcement Profile

Use this page to configure profile and attribute parameters for the Filter ID based enforcement profile. The **Filter ID Based Enforcement** profile contains the following tabs:

Profile Configuration

The following figure displays the **Filter ID Based Enforcement > Profile** dialog:

Figure 1. Filter ID Based Enforcement Profile Dialog



Specify the **Filter ID Based Enforcement Profile** parameters as described in the following table:

Table 1. Filter ID Based Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select the **Filter ID Based Enforcement** te mplate.
Name	Enter the name of the profile. The name is displayed in the Name column on the **Conf iguration > Enforcement > Profiles** page.
Description	Enter a description of the profile. The Desc ription is displayed in the Description colu mn on the **Configuration > Enforcement > Profiles** page.
Type	**RADIUS**. The field is populated automatica lly.
Action	Enabled. Click **Accept**, **Reject**, or **Drop** to d efine the action taken on the request.
Device Group List	Select a Device Group from the drop - dow n list. The list displays all configured Devic e Groups. All configured device groups are listed in the **Device Groups** page: **Configuration > Network > Device Groups**. After you add o ne or more device group(s), you can select a group and take one of the following actio ns: • To delete the selected Device Group Lis t entry, click **Remove**.

Parameter	Action/Description
	• To see the device group parameters, cli ck **View Details**. • To change the parameters of the select ed device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Gro up** link and see Adding and M odifying Device Groups .

Attributes Configuration

The following figure displays the **Filter ID Based Enforcement Profile > Attributes** dialog:

Specify the **Filter ID Based Enforcement > Attributes** parameters as described in the following table:

Table 2. Filter ID Based Enforcement Profile > Attributes Tab Parameters

Parameter	Description
Type	By default, the Filter ID Based Enforcement Profile is prepopulated with the **RADIUS: IT EF** attribute type. To select another attribut e type, click the Type drop - down menu a nd select another of the following RADIUS options: • Radius:Aruba • Radius:Cisco • Radius: Hewlett - Packared - Enterprise • Radius: Lucent - Alcatel - Enterprise • Radius:Microsoft • Radius:Avenda For more information, see RADIUS Namesp aces .
Name	By default, the Filter ID Based Enforcemen t Profile is prepopulated with the **Filter - ID** RADIUS attribute type. To seelct another R

Parameter	Description
	ADIUS attribute, click the name dropdown menu and select from the list of available a ttributes. The options displayed for the **Na me** attribute depend on the attribute that was selected.
Value	Specify the appropriate value for the RADI US attrbute in the **Value**. Depending upon the RADIUS attribute sele cted in the Name field, you must either ma nually enter a value in the **Value** field, or cl ick the **Value** dropdown menu and one of t he available options.

Generic Application Enforcement Profile

Use this page to configure profile and attribute parameters for the **Generic Application Enforcement** profile. The **Generic Application Enforcement** profile contains the following tabs:

Profile Configuration

Use the **Profile** tab to configure the template, type of the profile, and device group list. The following figure displays the **Generic Application Enforcement > Profile** tab:

Figure 1. Generic Application Enforcement > Profile Tab

Specify the **Generic Application Enforcement > Profile** parameters as described in the following table:

Table 1. Generic Application Enforcement > Profile Tab Parameters

Parameter	Action/Description
Template	Select the template from the drop - down l ist. In this context, select Generic Applicati on Enforcement.
Name	Enter the name of the profile. The name is displayed in the **Name** column on the **Conf iguration > Enforcement > Profiles** page.
Description	Enter a description that provides additiona l information about the profile. This descrip tion is displayed in the **Description** column on the **Configuration > Enforcement > Prof iles** page.
Type	This field is populated automatically with t he **Application** type.
Action	Click **Accept**, **Reject**, or **Drop** to define the action taken on the request.
Device Group List	This setting is not applicable to the Gener ic Application Enforcement profile templat e, and is disabled.

Attributes Configuration

Use the **Attribute** tab to configure the attribute type, name, and value for the enforcement profile. The following figure displays the **Generic Application Enforcement > Attributes** tab:

Figure 2. Generic Application Enforcement > Attributes Tab

Specify the **Generic Application Enforcement > Attributes** parameters as described in the following table:

Table 2. Generic Application Enforcement > Attributes Parameters

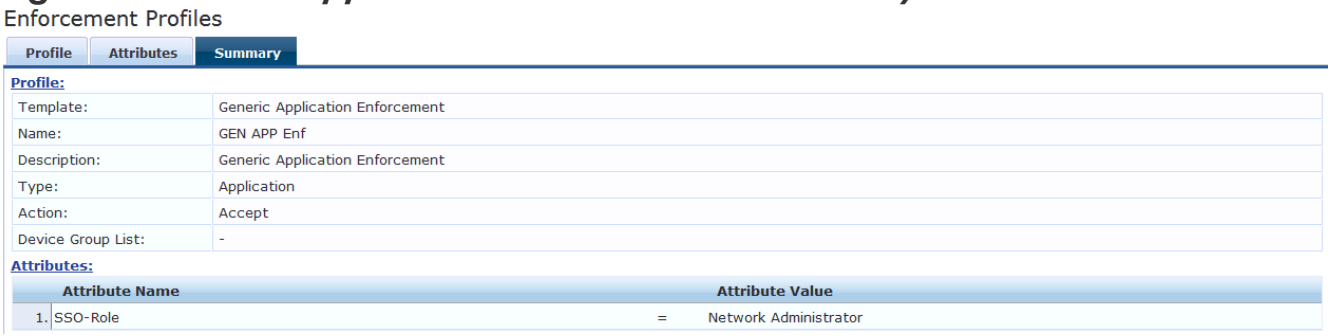
Parameter	Action/Description
Attribute Name	Select an attribute name from the d has multiple names.

Parameter	Action/Description
Attribute Value	Enter the appropriate value for the **e**.

Summary Information

The **Summary** tab summarizes the parameters configured in the **Profile** and **Attributes** tab.

Figure 3. *Generic Application Enforcement > Summary Tab*



HTTP Based Enforcement Profile

Use this page to configure the HTTP based Enforcement Profile.

Subtopics

Profile Configuration
Attributes Configuration

Profile Configuration

The following figure displays the **HTTP Based Enforcement > Profile** dialog:

Figure 1. HTTP Based Enforcement Profile Dialog

Specify the **HTTP Based Enforcement > Profile** parameters as described in the following table:

Table 1. HTTP Based Enforcement Profile Parameters

Parameter	Action/Description
Template	Select the **HTTP Based Enforcement** template.
Name	Enter the name of the profile. The name is displayed in the **Name** column on the **Configuration > Enforcement > Profiles** page.
Description	Enter a description of the profile. The description is displayed in the **Description** column on the **Configuration > Enforcement > Profiles** page.
Type	This field is populated automatically with **HTTP**.
Action	This setting is not configurable.
Device Group List	Select a Device Group from the drop - down list. The list displays all configured Device Groups. All configured device groups are listed in the **Configuration > Network > Device Groups**.page. After you add one or more device group(s), you can select a group and take one of the following actions: • To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**.

Parameter	Action/Description
	• To change the parameters of the select ed device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Gro up** link and see Adding and M odifying Device Groups .

Attributes Configuration

Figure 1. HTTP Based Enforcement Attributes Dialog

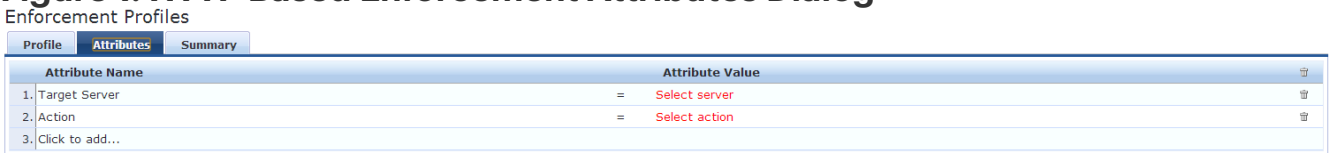


Table 1. HTTP Based Enforcement Attributes Parameters

Parameter	Action/Description
Attribute Name	Select the attribute name: **Target Server** or **Action**. An HTTP - based enforcement profi le must contain at least one target server a nd one action attribute.
Attribute Value	For the **Target Server** attribute, click the d ropdown list and choose a server. For the **Action** attribute, specify one of the followi ng values. • Handle AirGroup Time Sharing • Nmap Scan • Nmap Scan Device Profiler • Check Point Login - AD User • Check Point Logout • Fortinet Login

Parameter	Action/Description
	• Fortinet Logout • SNMP Scan • Infoblox Login • Check Point Login - Guest User • OnDemand Endpoint Scan • Store Attributes in zone cache

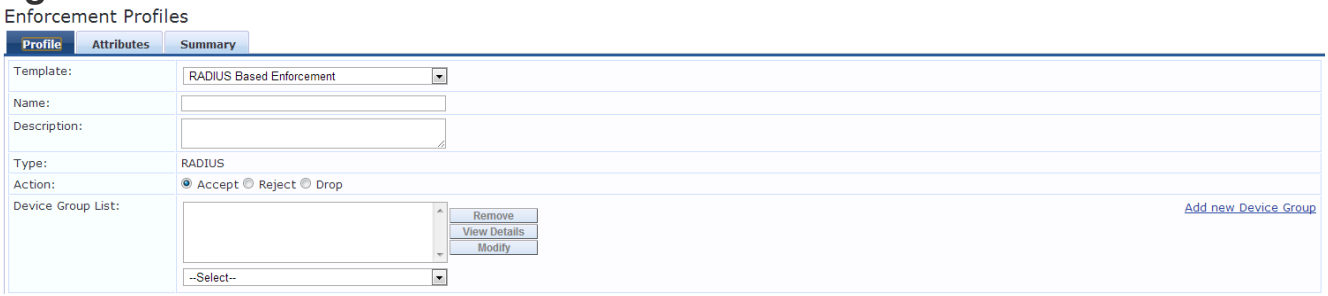
RADIUS Based Enforcement Profile

Use this page to configure profile and attribute parameters for the RADIUS-based enforcement profiles.

Profile Configuration

The following figure displays the **RADIUS Based Enforcement Profile** tab:

Figure 1. RADIUS Based Enforcement > Profile Tab



Specify the **RADIUS Based Enforcement Profile** parameters as described in the following table:

Table 1. RADIUS Based Enforcement Profile Parameters

Parameter	Action/Description
Template	Select the **RADIUS Based Enforcement** te mplate.
Name	Enter the name of the profile.
Description	Enter a description of the profile.

Parameter	Action/Description
Type	**RADIUS**. This field is populated automatica lly.
Action	Enabled. Click **Accept**, **Reject**, or **Drop**.
Device Group List	Select a Device Group from the drop - dow n list. All configured device groups are listed in t he **Configuration > Network > Device Grou ps** page. After you add one or more device group(s), you can select a group and take o ne of the following actions: • To delete the selected Device Group Lis t entry, click **Remove**. • To see the device group parameters, cli ck **View Details**. • To change the parameters of the select ed device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Gro up** link and see Adding and M odifying Device Groups .

Attributes Tab

Specify the **RADIUS Based Enforcement > Attributes** parameters as described in the following table:

Table 2. RADIUS Based Enforcement > Attributes Parameters

Parameter	Description
Type	Select one of the following attribute types: • Radius:Aruba • Radius:IETF • Radius:Cisco • Radius: Hewlett - Packared - Enterprise

Parameter	Description
	• Radius: Lucent - Alcatel - Enterprise • Radius:Microsoft • Radius:Avenda The RADIUS - type attributes can be re - o rdered using the arrows in each attribute's row to move them up or down in the list as needed. For more information, see RADIUS Namesp aces .
Name	The options displayed for the **Name** attribu te depend on the **Type** attribute that was s elected.
Value	The options displayed for the **Value** attribut e depend on the **Type** and **Name** attributes that were selected.

NOTE

The **Aruba-PoE-Allocate-By-Method** attribute type is included with the RADIUS dictionary and available in enforcement profiles. Use this attribute with an AOS-CX switch to set the PoE power allocation method for device onboarding after authentication. Supported values include **Class** and **Usage**.

RADIUS Dynamic Authorization Enforcement Profile

About this task

To configure a RADIUS Dynamic Authorization enforcement profile:

Procedure

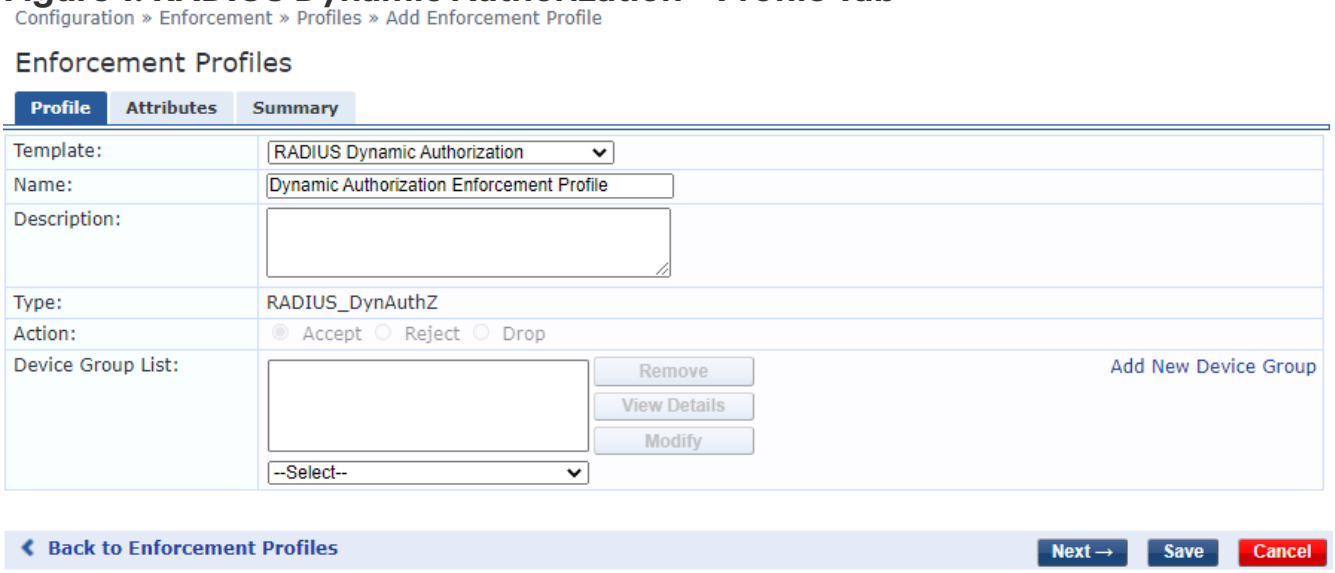
1. Navigate to **Configuration > Enforcement > Profiles**.
2. Click **Add**.
3. From the **Template** drop-down, select **RADIUS Dynamic Authorization**.

   The **RADIUS Dynamic Authorization** configuration dialog opens.

Profile Tab

About this task

Figure 1. RADIUS Dynamic Authorization > Profile Tab



Procedure

Specify the **RADIUS Dynamic Authorization > Profile** tab parameters as described in the following table:

Table 1. RADIUS Dynamic Authorization Profile Tab Parameters

Parameter	Action/Description
Template	Select the **RADIUS Dynamic Authorization** template.
Name	Enter the name of this enforcement profile.
Type	This is automatically populated with Type **RADIUS_DynAuthZ**
Action	These options are disabled.
Device Group List	Optionally, select a Device Group from the drop - down list. All configured device groups are listed on the **Device Groups** page: **Configuration > Network > Device Groups**. After you add one or more device group(s), you can select a group and take one of the following actions:

Parameter	Action/Description
	• To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**. • To change the parameters of the selected device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Group** link and see Adding and Modifying Device Groups .

Attributes Tab

About this task

The following table describes the **RADIUS Dynamic Authorization > Attributes** parameters:

Table 2. RADIUS Dynamic Authorization Attributes Parameters

Parameter	Action/Description
Select RADIUS Dynamic Authorization Template	From the drop - down, select one of the RADIUS Dynamic Authorization templates. When you select a template, the **Type**, **Name** and **Value** settings are pre - populated with RADIUS values appropriate for that template type.
Type	The default options displayed for the **Type** attribute depend on the **Template** and **Type** attributes that were selected.
Name	The default options displayed for the **Name** attribute depend on the **Template** and **Type** attributes that were selected.
Value	The default content for the **Value** attribute depends on the **Template**, **Type**, and **Name** attributes that were selected.

Procedure

1. To add an enforcement profile attribute, select **Click to add** on an open row and proceed to define the **Type**, **Name**, and **Value** for the new attribute.
2. Click **Next** to view the enforcement profile **Summary**.
3. Make any desired changes, then click **Save**.

Change of Authorization with TEAP

About this task

Change of authorization (CoA) can fail with tunnel-based EAP (TEAP) as the authentiation method. To ensure that CoA works properly with TEAP when using Username from Method-1 in the Access Tracker, Aruba recommends the following workaround:

Procedure

1. From the **Configuration > Enforcement > Profile** page, then click **Add** to create an enforcement profile called **Set-TEAP-Method1-Username** with the following attributes:
   - Type = Radius: IETF
   - Name = User-Name
   - Value = %{Authentication: TEAP-Method-1-Username}
2. Save the profile.
3. Go to Configuration > Services, then select the service.
4. In the **Enforcement** tab, add the profile to the enforcement policy that is associated with the service.
5. Click **Save**.

Session Notification Enforcement Profile

Use this page to configure the **Session Notification Enforcement** profile. You can send notification of a change in IP address to any external context server (such as a firewall) by configuring that server as a generic HTTP server and adding the appropriate generic HTTP context server actions.

The content of the payload to be posted by Policy Manager to the external server is based on the REST API defined by the external server.

Profile Tab

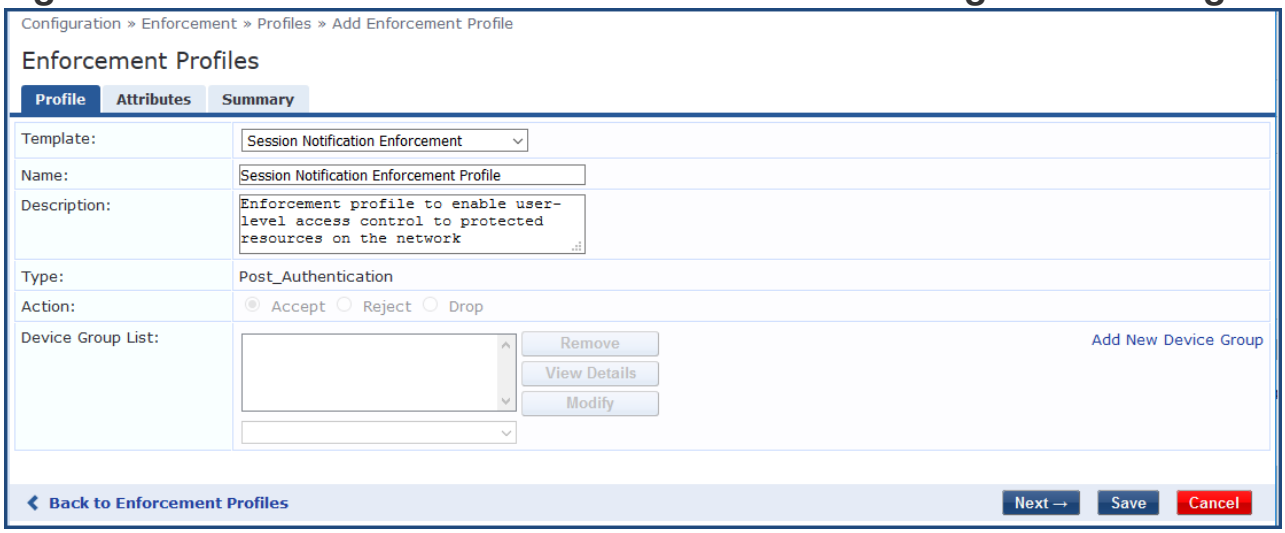
About this task

To configure a Session Notification Enforcement profile:

Procedure

1. On the publisher, navigate to **Configuration > Enforcement > Profiles**. The **Enforcement Profiles** page opens.
2. Click the **Add** link. The **Add Enforcement Profiles** dialog opens.
3. From the **Template** drop-down, select **Session Notification Enforcement**.

Figure 1. Session Notification Enforcement > Profile Configuration Dialog



4. Specify the **Session Notification Enforcement > Profile** parameters as described in the following table:

Table 1. Session Notification Enforcement Profile Parameters

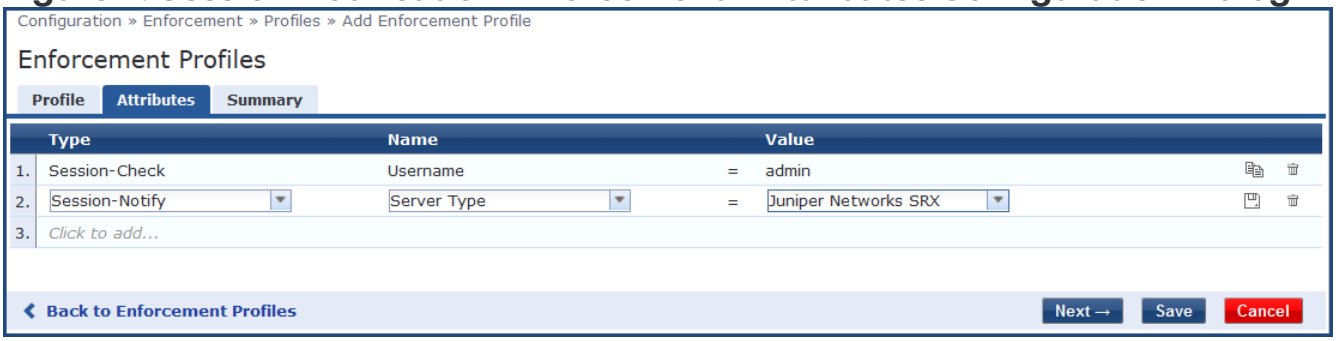
Parameter	Action/Description
Template	Select **Session Notification Enforcement** .
Name	Enter the name of the profile.
Description	Enter a description of the profile (recommended).
Type	The field is populated automatically with **Post_Authentication**.

Parameter	Action/Description
Action	This options is disabled.
Device Group List	Select a device group from the drop - down list. All configured device groups are listed in the **Device Groups Configuration > Network > Device Groups** page.
Add New Device Group	To add a new a device group, click the **Add New Device Group** link. See Adding and Modifying Device Groups for more information.

Attributes Tab

About this task

Figure 2. Session Notification Enforcement > Attributes Configuration Dialog



Procedure

Specify the **Session Notification Enforcement > Attributes** parameters as described in the following table:

Table 2. Session Notification Enforcement > Attributes Parameters

Parameter	Action/Description
Type	Select one of the following **Ty** • Session - Check • Session - Notify Palo Alto integration is extend n - Check attributes as follows

Parameter	Action/Description
	• **Session - Check::Usernan** **NOTE** Post authentication s n the user ID update • **Session - Notify**: The **Nam** ◦ Login Action ◦ Logout Action ◦ Server IP ◦ Server Type ◦ **Server Type** options: ◦ Generic HTTP ◦ Palo Alto Networks Pan ◦ Palo Alto Networks Fire ◦ **Server IP** options: a ch f server as **Value**. The **T** e the **Server IP** option. ◦ Once the server IP addi **ion**. The list of actions c hoices for **Value**. This enforcement type should servers.
Name	The options displayed for the cted.
Value	The options displayed for the hat were selected.

Summary Tab

About this task

This **Summary** tab summarizes the parameters configured for Session Notification Enforcement.

Figure 3. Session Notification Enforcement > Summary Tab

Procedure

Click **Save**.

Session Restrictions Enforcement Profile

If OnGuard Agent is disconnected, Policy Manager uses Keep-Alive messages to issue Disconnect Messages for a Session Restrictions Enforcement Profile for agent-based enforcement. (See below, Examples of Session-Check Enforcement Profile Configurations). For related information, see OnGuard Global Agent Settings .

> **NOTE**
> An active session count is tracked per username. This enforcement limits a user's simultaneous connections to a configurable number of devices. that OnGuard Agent is not required for enforcement based upon active session count.

Enforcement Profile Configuration

About this task

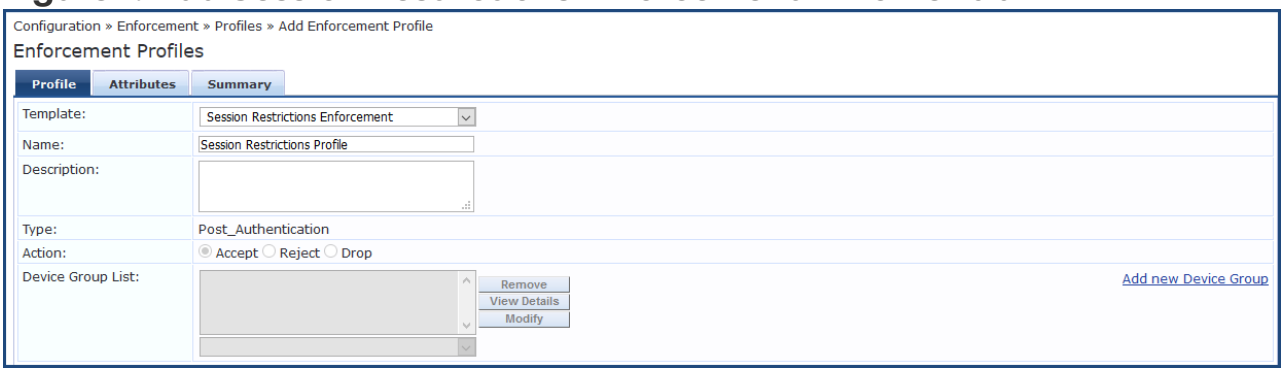
To configure Profile and Attributes parameters for a Session Restrictions Enforcement profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**. The **Enforcement Profiles** page opens.

2. Click **Add**. The **Add Enforcement Profiles > Profile** tab opens.
3. From the **Template** drop-down, select **Session Restrictions Enforcement**. The **Add Session Restrictions Enforcement > Profile** dialog opens:

Figure 1. Add Session Restrictions Enforcement > Profile Tab



4. Specify the **Session Restrictions Enforcement > Profile** parameters as described in the following table:

Table 1. Session Restrictions Enforcement Profile Parameters

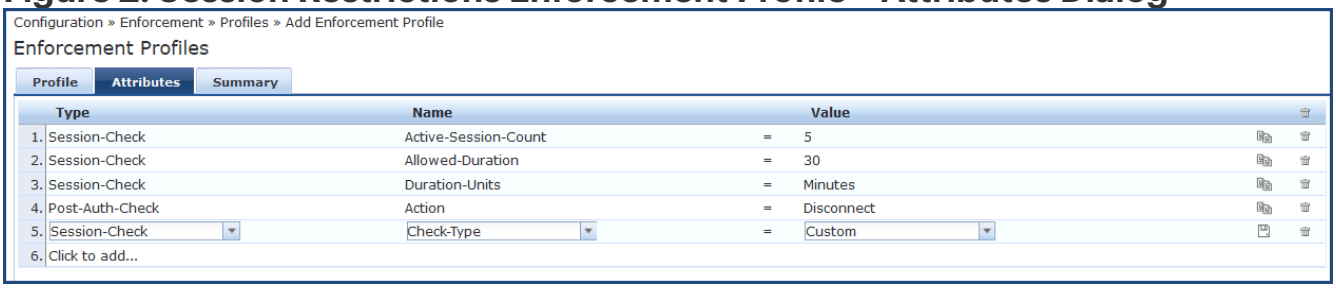
Parameter	Action/Description
Template	Select **Session Restrictions Enforcement** .
Name	Enter the name of the enforcement profil e.
Description	Optionally, enter a description of the en forcement profile (recommended).
Type	**Post_Authentication**. The **Type** field is populated automatically when you select the **Session Restrictions Enforcement** template.
Action / Device Group List	**These fields are not applicable to a sess ion restrictions enforcement profile and are disabled.**

Attributes Configuration

About this task

The following figure displays the **Session Restrictions Enforcement > Attributes** tab:

Figure 2. Session Restrictions Enforcement Profile > Attributes Dialog



Procedure

Specify the **Session Restrictions Enforcement > Attributes** parameters as described in Table 2:

Table 2. Session Restrictions Enforcement Attributes Parameters

Parameter	Description
Type	Select from the following attribute • Bandwidth - Check • Expiry - Check • Post - Auth - Check • Session - Check
Name	The options displayed for the **Nan** end on the **Type** attribute that wa **Type: Bandwidth - Check** • **Allowed - Limit**: Defines the tc mit to be allowed per user or e • **Check - Type**: Defines the per bandwidth - based checks. Ap th **Allowed - Limit**. • **Limit - Units**: Defines the metr h - based checks (KB, MB, GB, only with **Allowed - Limit**. • **Start - Date**: Defines the start idth - based checks. Applicabl **owed - Limit**.

Parameter	Description
	• **Start - Time**: Defines the start idth - based checks. Applicabl **owed - Limit**. • **Stop - Date**: Defines the stop idth - based checks. Applicabl **owed - Limit**. • **Stop - Time**: Defines the stop idth - based checks. Applicabl **owed - Limit**. ◦ For configuration examples ection, Examples of Bandw forcement Profile Configur **Type: Expiry - Check** • Expiry - Action **Type: Post - Auth - Check** • Action **Type: Session - Check** • **Active - Session - Count**: Defi of active sessions to be allowe dpoint. • **Agent - Connection**: Set the v o have Policy Manager check t Guard Agent is down. • **Allowed - Duration**: Defines th duration to be allowed per use • **Check - Type**: Defines the per for duration - based checks. A with **Allowed - Duration**. • **Duration - Units**: Defines the r tion - based checks. Applicabl **owed - Duration**.

Parameter	Description
	• **Start - Date**: Defines the start n - based checks. Applicable c **d - Duration**. • **Start - Time**: Defines the start n - based checks. Applicable c **d - Duration**. • **Stop - Date**: Defines the stop c n - based checks. Applicable c **d - Duration**. • **Stop - Time**: Defines the stop n - based checks. Applicable c **d Duration**. • **Username**: Defines the userna ession restrictions are enabled e client MAC address is to be c rname. ◦ For configuration examples n below, Examples of Sessi rcement Profile Configurati
Value	The options displayed for the **Valu** end on the **Type** and **Name** attribu elected.

Examples of Bandwidth-Check Enforcement Profile Configurations

About this task

The following are typical examples of **Session Restriction > Bandwidth-Check** enforcement profile configurations:

Procedure

1. **Allowed-Limit**: Users/Endpoints will be disconnected after exceeding the 50 MB daily limit:
   - Bandwidth-Check > Allowed-Limit = 50
   - Bandwidth-Check > Limit-Units = MB

- Bandwidth-Check > Check-Type = Daily
- Post-Auth-Check > Action = Disconnect

2. **Allowed-Limit**: Users/Endpoints will be disconnected after exceeding 1 GB total bandwidth consumption. Also, users are allowed access to the network only during the defined period (between 9:00 a.m. and 6:00 p.m.).
   - Bandwidth-Check > Allowed-Limit = 1
   - Bandwidth-Check > Limit-Units = GB
   - Bandwidth-Check > Check-Type = Total
   - Bandwidth-Check > Start-Time = 09:00:00
   - Bandwidth-Check > Stop-Time = 18:00:00
   - Post-Auth-Check > Action = Disconnect

Examples of Session-Check Enforcement Profile Configurations

About this task

The following are typical examples of **Session Restriction > Session-Check** enforcement profile configurations:

Procedure

1. **Active Session Count**: The Users/Endpoints active session count is set to **5**. Users/Endpoints connecting after the session count reaches 5 are disconnected:
   - Session-Check > Active-Session-Count = 5
   - Post-Auth-**Check > Action**= Disconnect
2. **Agent-Connection**: You can disconnect a session if OnGuard Agent is down:
   - Session-Check > Agent-Connection = Down
   - Post-Auth-Check > Action = Disconnect
3. **Session Duration**: The User/Endpoint is allowed access for 60 minutes daily. Users/Endpoints that exceed this session duration limit are disconnected:
   - Session-Check > Allowed-Duration = 60
   - Session-Check > Duration-Units = Minutes
   - Session-Check > Check-Type = Daily
   - Post-Auth-**Check > Action**= Disconnect

4. **Session Duration**: The User/Endpoint is allowed access to the network daily for three hours in a specified time period (between 9:00 a.m. and 5:00 p.m.)
    - Session-Check > Allowed-Duration = 3
    - Session-Check >Duration-Units = Hours
    - Session-Check > Check-Type = Daily
    - Session-Check > Start-Time = 09:00:00
    - Session-Check Stop-Time = 17:00:00
    - Post-Auth-**Check > Action**= Disconnect

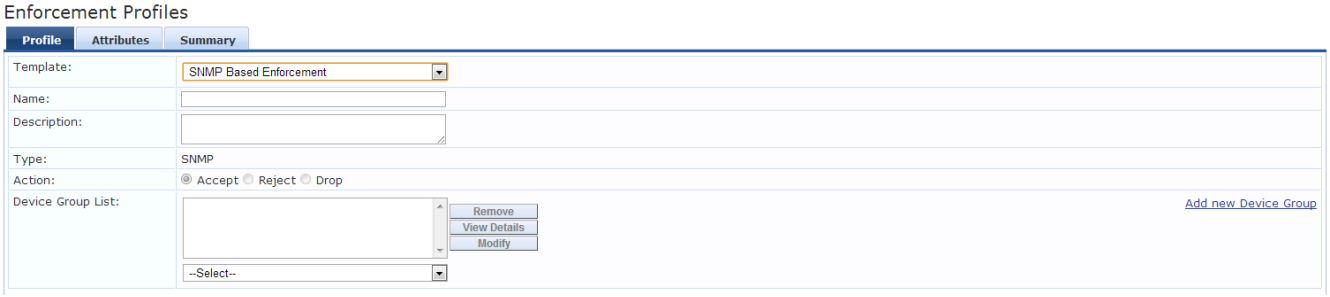
SNMP-Based Enforcement Profile

Use this page to configure the SNMP-Based Enforcement profile.

Profile Configuration

The following figure displays the **SNMP Based Enforcement > Profile** dialog:

Figure 1. SNMP Based Enforcement > Profile Dialog



Specify the **SNMP Based Enforcement > Profile** parameters as described in the following table:

Table 1. SNMP Based Enforcement > Profile Tab Parameters

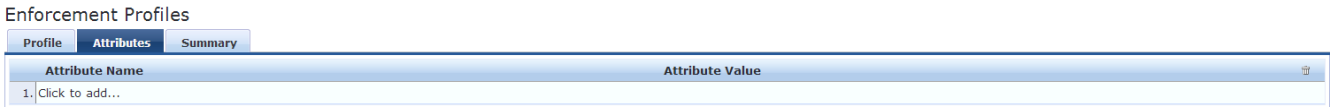
Parameter	Description
Template	Select the **SNMP Based Enforcement** tem plate.
Name	Enter the name of the profile.

Parameter	Description
	The name is displayed in the **Name** column on the **Configuration** > **Enforcement** > **Profiles** page.
Description	Enter a description of the profile (recommended). The description is displayed in the **Description** column on the **Configuration** > **Enforcement** > **Profiles** page.
Type	**SNMP**. This field is populated automatically when you select the **SNMP Based Enforcement** template.
Action	Disabled.
Device Group List	Select a Device Group from the drop - down list. All configured device groups are listed in the **Configuration > Network > Device Groups** page. After you add one or more device group(s), you can select a group and take one of the following actions: • To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**. • To change the parameters of the selected device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device Group** link and see Adding and Modifying Device Groups .

Attributes Configuration

The following figure displays the **SNMP Based Enforcement > Attributes** dialog:

Figure 2. SNMP Based Enforcement > Attributes Dialog



Specify the **SNMP Based Enforcement > Attributes** parameters as described in the following table:

Table 2. SNMP Based Enforcement Attributes Parameters

Parameter	Action/Description
Attribute Name	Select one or more of the following attributes: • VLAN ID • Session Timeout (in seconds) • Reset Connection (after the settings are applied)
Attribute Value	In the **Attribute Value** field, enter an integer between 0 and 4095 for the **VLAN ID** attribute value or the number of seconds for a session in the **Session Timeout** value field. To define the **Reset Connection** attribute value, click the dropdown list and select **Enabled** or **Disabled**.

TACACS+ Based Enforcement Profile

To review a sample TACACS profile configuration, refer to Sample TACACS Profile Configuration.

Enforcement Profile Configuration

About this task

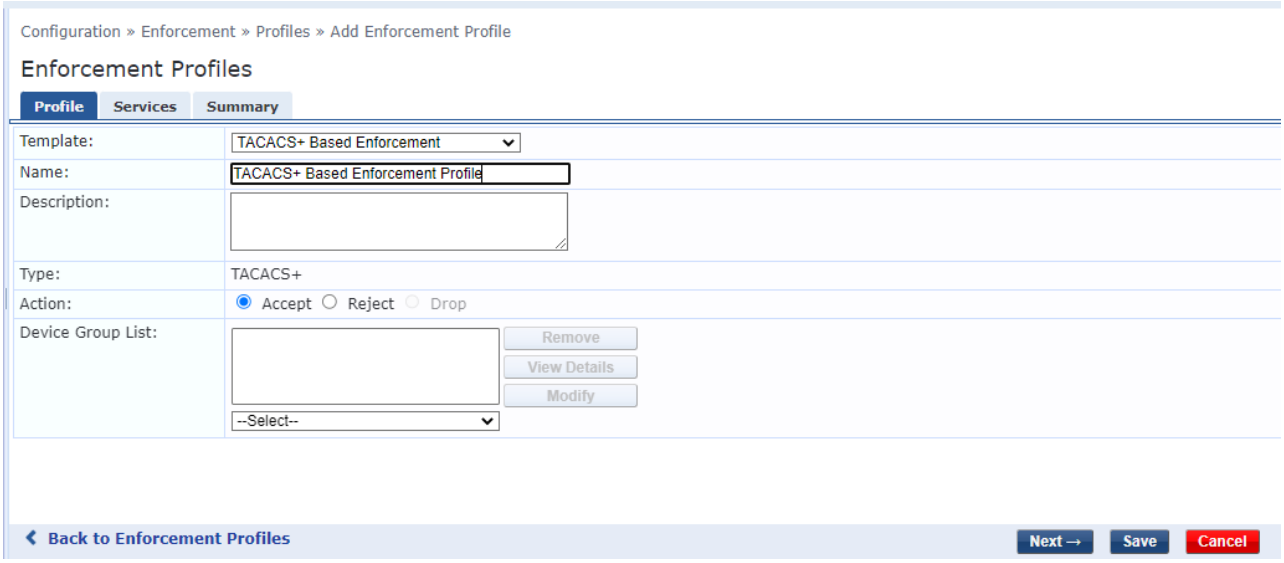
To create a TACACS+ Based Enforcement profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**.
2. From the **Enforcement Profiles** page, click **Add**.The **Add Enforcement Profiles** dialog opens.

3. From the **Template** drop-down, select **TACACS+ Based Enforcement.**

Figure 1. Add TACACS+ Based Enforcement Profile Dialog



4. Specify the **Add TACACS+ Based Enforcement Profile > Profile** tab parameters as described in the following table:

Table 1. TACACS+ Based Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select the **TACACS+ Based Enforcement** template.
Name	Enter the name of the profile.
Description	Enter a description of the profile (recommended).
Type	This field is populated automatically for TACACS+.
Action	A TACACS+ based Enforcement profile allows you to set the **Action** value to either **Accept** or **Reject**. The default value is **Accept**. If you select the **Reject** option for the **Action** setting, the TACACS+ authentication request is rejected when the enforcement profile is assigned. The **Drop** option is disabled and is not availab

Parameter	Action/Description
	le for a TACACS+ enforcement profile. ClearPass does not allow the modification of system defined Enforcement Profiles (“[TACACS+ Deny Enforcement Profile]” is set as Action = Accept, by default). HPE Aruba Networking recommends creating a new profile to modify the Action field. The TACACS+ protocol correctly returns an Accept event if the user authenticates correctly, but the defult Deny rule will then also return the privilege level 0 for the TACACS+ authorization. To change this behavior create a new profile to Reject in these circumstances.
Device Group List	Select a **Device Group** from the drop - down list. All configured device groups are listed in the **Configuration > Network > Device Groups** page. When one or more device group(s) have been added to Policy Manager, you can select a device group and take one of the following actions: • To delete the selected Device Group List entry, click **Remove**. • To see the device group parameters, click **View Details**. • To change the parameters of the selected device group, click **Modify**. **NOTE** To add a new a device group, click the **Add New Device**

Parameter	Action/Description
	Group link. See Adding and Modifying Device Groups .

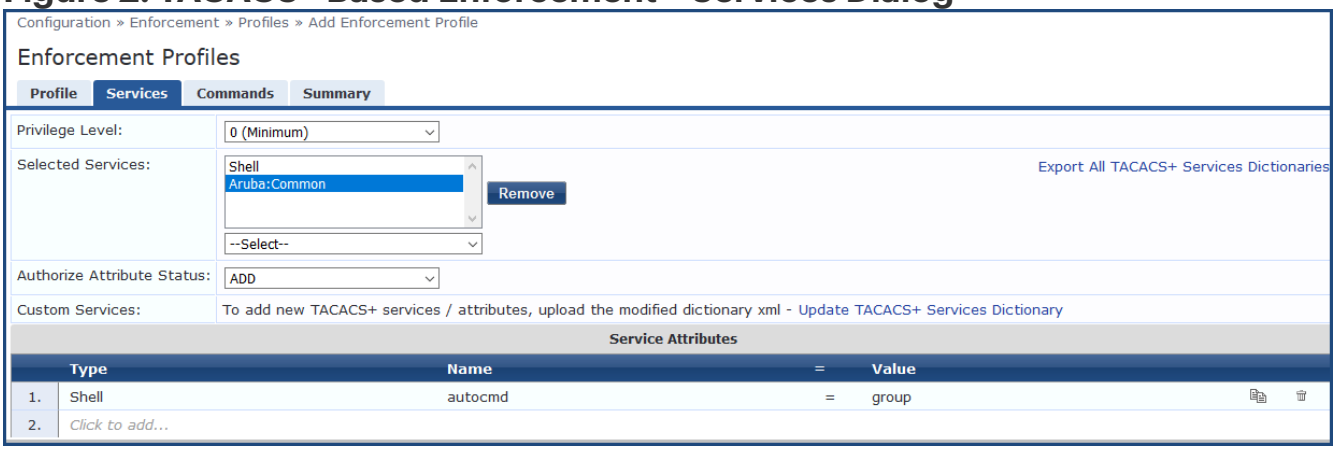
NOTE

In a TACACS+ workflow, if the tokenGroup attribute is used to filter a user's group membership list from Active Directory, the filter query fails and the error message "Failed to get value for attributes=Nested Groups" displays. Keep in mind, the tokenGroups attribute is not currently supported in TACACS+ workflows. Use the **memberOf** attribute instead to check membership.

Services Configuration

The following figure displays the **TACACS+ Based Enforcement > Services** dialog:

Figure 2. TACACS+ Based Enforcement > Services Dialog



Specify the **TACACS+ Based Enforcement Profile > Service** parameters as described in the following table:

Table 2. TACACS+ Based Enforcement > Services Parameters

Parameter	Action/Description
Privilege Level	Select a level between **0** and **15**, with **0** being the minimum privilege level and **15** being the highest. **NOTE** Use Privilege Level **0** for basic TACACS+ authentication.
Selected Services	Select one or more of the following services: • Shell • PIX Shell • PPP:IP • PPP:IPX • PPP:LCP • ARAP • cpass:HTTP • cpass:CLI • Wireless - WCS:HTTP • CiscoWLC:Common • Aruba:Common • junos - exec • AMP:https • NCS:HTTP • Infoblox **NOTE** When you select one or more services, the **Commands** tab is enable.

Parameter	Action/Description
Export All TACACS+ Services Dictionaries	Click this link to download the TACACS+ Services dictionary (*TacacsServiceDictionary.xml*) to the local system.
Authorize Attribute Status	Select one of the following options: • ADD • REPLACE • FAIL **NOTE** Specifying the **FAIL** option is an instruction to cause the authorization request to fail.
Custom Services	To add new TACACS+ services or attributes by uploading the modified dictionary, click the **Update TACACS+ Services Dictionary** link.
Service Attributes	
Type	Click the Type dropdown menu to select a service attribute type. For example, if one of the services you selected was **Shell**, **Shell** will be one of the available Service Attribute types. After the services have been selected, you can select the attributes to send from Policy Manager a drop - down list in the **Name** field). You can also add custom attributes in the *Name* field. Add service attributes corresponding to the services selected in **Selected Services.** Policy Manager by default provides attributes for some of the listed services.
Name	The options displayed for the **Name** attribute depend on the **Type** attribute that was specified.

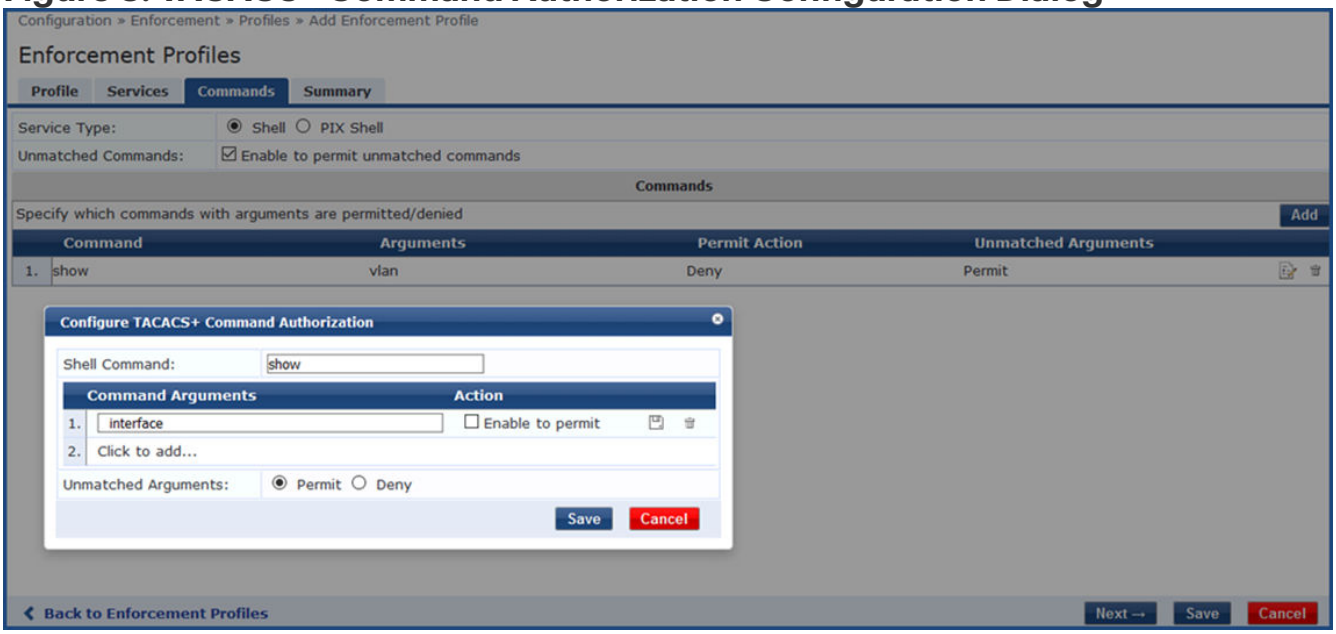
Parameter	Action/Description
Value	The options displayed for the **Value** attribute depend on the **Type** and **Name** attributes that were specified.

Command Authorization Configuration

About this task

From the **Commands** tab, you can configure commands and arguments allowed or disallowed for the selected **Service Type**.

Figure 3. TACACS+ Command Authorization Configuration Dialog



Procedure

1. In the **Service Type** field, select the **Shell** or **PIX shell** radio button. Subsequent selections in this tab configure commands and arguments allowed or disallowed for this selection.
2. (Optional) In the **Unmatched Commands** field, select the **Enable this option to permit unmatched commands** option to allow commands that are not explicitly entered in the **Commands** field.
3. Next, you must create a list of commands recognized for the specified service type. To add a command, click the **Add** button on the **Commands** tab. The **Configure TACACS+ Command Authorization dialog** opens.
4. In the **Configure TACACS+ Command Authorization** dialog, enter values for the following paramters:

- **Shell Command.** A string for the command (for example, **show**). This is followed by one or more command argument rows.
- **Command Arguments.** The arguments for the command.
- **Action.** To permit use of this command argument, click the **Enable to permit** check box. If this box is unchecked, the column shows **Deny** and the command argument is not allowed.

5. asIn the **Unmatched Arguments** field, select the **Permit** option to permit this command even if Policy Manager receives arguments for the command that it does not recognize.

   Select the **Deny** radio button to deny the command if Policy Manager receives unrecognized arguments.

6. To save and exit, click outside the row you are editing then click **Save**.

TACACS Profile Configuration Example

About this task

When setting up authentication for TACACS+, it can be a challenge to configure the correct attributes to send back to ExtraHop for authorization (or permission roles). In order to properly configure Policy Manager to know of these attributes, they must be added to the attribute dictionary.

To configure profile and attribute parameters for TACACS+ based enforcement profiles:

Procedure

1. Download the *ExtraHop_tacacsCPPMdictionary.txt* file to insert the correct attributes into the Policy Manager attribute.
2. Log in to Policy Manager.
3. In Policy Manager, navigate to **Administration > Dictionaries > TACACS+ Services**. The **TACACS+ Services Dictionaries** page opens.
4. To import the *ExtraHop_tacacsCPPMdictionary.txt* XML file, click the **Import** link.
5. Navigate to **Configuration > Network Devices**.

6. Add the ExtraHop device IP address as a Network Device (for details, see Adding a Network Device).

> **NOTE**
>
> Make sure to use the same **TACACS+ Shared Secret** here as is configured on the ExtraHop device.

7. Create a *TACACS+ Based Enforcement* profile as described above, using the ExtraHop TACACS+ attributes that you imported.

VLAN Enforcement Profile

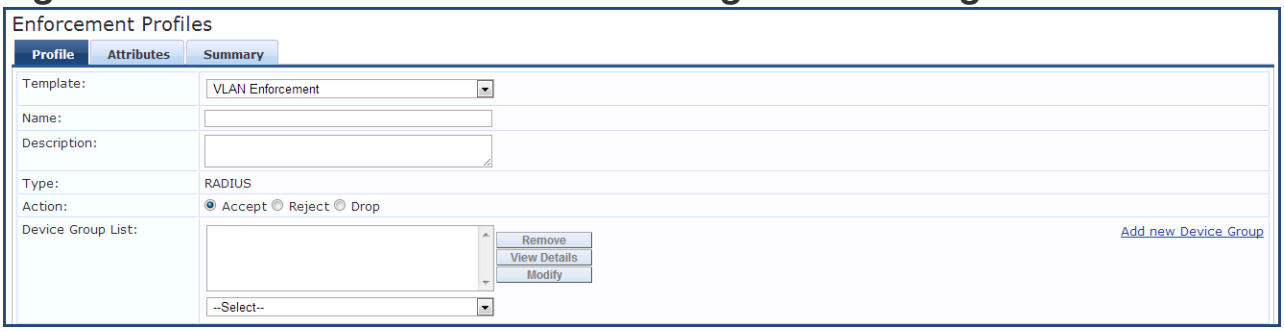
About this task

To configure profile and attribute parameters for a VLAN Enforcement profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**. The **Enforcement Profiles** page opens.
2. Click **Add**. The **Add Enforcement Profiles > Profile** tab opens.
3. From the **Template** drop-down, select **Agent Enforcement**. The **VLAN Enforcement > Profile** dialog opens.

Figure 1. VLAN Enforcement > Profile Configuration Dialog



4. Specify the **VLAN Enforcement > Profile** parameters as described in the following table:

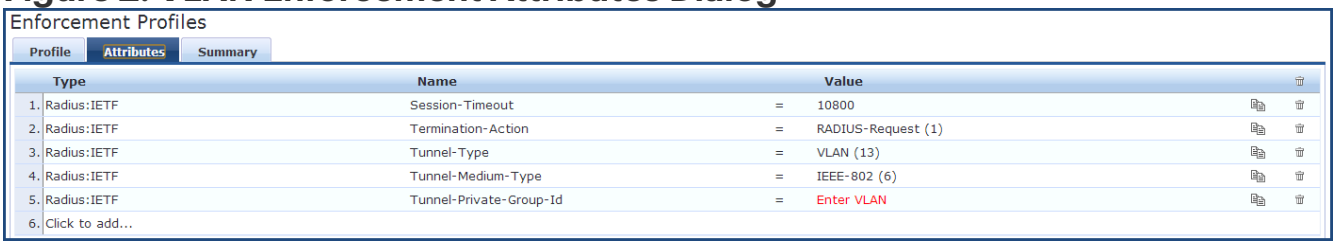
Table 1. VLAN Enforcement > Profile Parameters

Parameter	Action/Description
Template	Select **VLAN Enforcement** from the drop - down list.
Name	Enter the name of the enforcement profil e.
Description	Enter a description of theenforcement pr ofile.
Type	**RADIUS**. The field is populated automati cally.
Action	To define the action taken on the request, click **Accept**, **Reject**, or **Drop**.
Device Group List	Select a Device Group from the drop - d own list. All configured device groups are listed in the **Configuration > Network > Device G roups** page. After you add one or more d evice group(s), you can select a group an d take one of the following actions: • To delete the selected Device Group L ist entry, click **Remove**. • To see the device group parameters, c lick **View Details**. • To change the parameters of the sele cted device group, click **Modify**. **NOTE** To add a new a device grou p, click the **Add New Devic e Group** link and see Adding and Modifying Device Group s .

Attributes Configuration

The following figure displays the **VLAN Enforcement > Attributes** dialog:

Figure 2. VLAN Enforcement Attributes Dialog



Specify the **RADIUS Based Enforcement > Attributes** parameters as described in the following table:

Table 2. VLAN Enforcement Attributes Tab Parameters

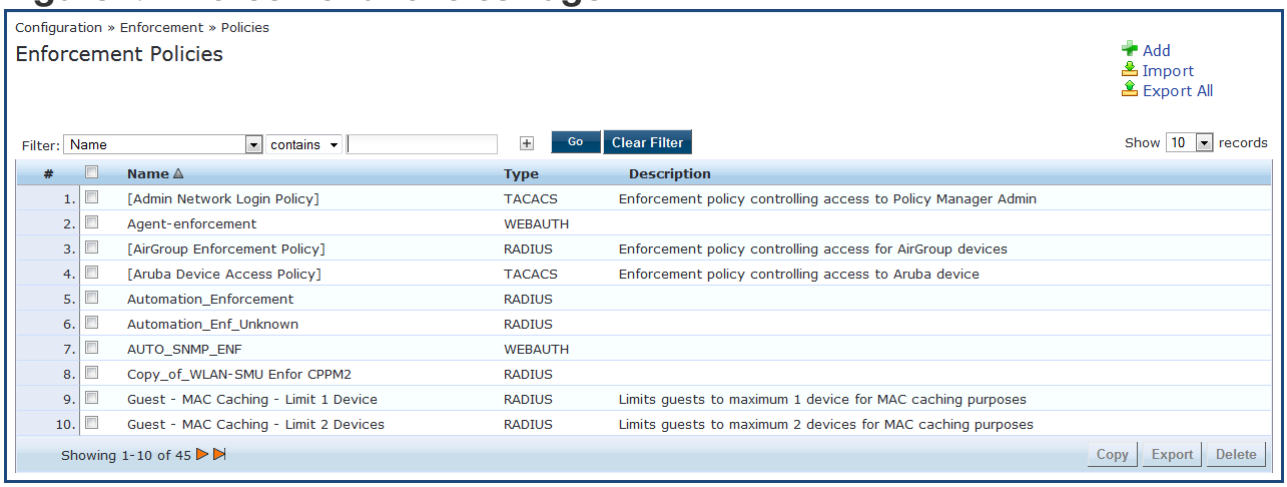
Parameter	Action/Description
Type	Select one of the following attribute types: • Radius:Aruba • Radius:IETF • Radius:Cisco • Radius: Hewlett - Packared - Enterprise • Radius: Alcatel - Lucent - Enterprise • Radius:Microsoft • Radius:Avenda For more information, see RADIUS Namespaces
Name	The options displayed for the **Name** attribute depend on the **Type** attribute that was selected.
Value	The options displayed for the **Value** attribute depend on the **Type** and **Name** attributes that were selected.

Configuring Enforcement Policies

Only one enforcement policy can be associated with each service. Enforcement policies can be added in one of two ways:

- From the **Configuration** > **Services** page as part of the flow of the **Add Service** wizard.
- From **Configuration** > **Enforcement** > **Enforcement Policies**.

Figure 1. Enforcement Policies Page



Subtopics

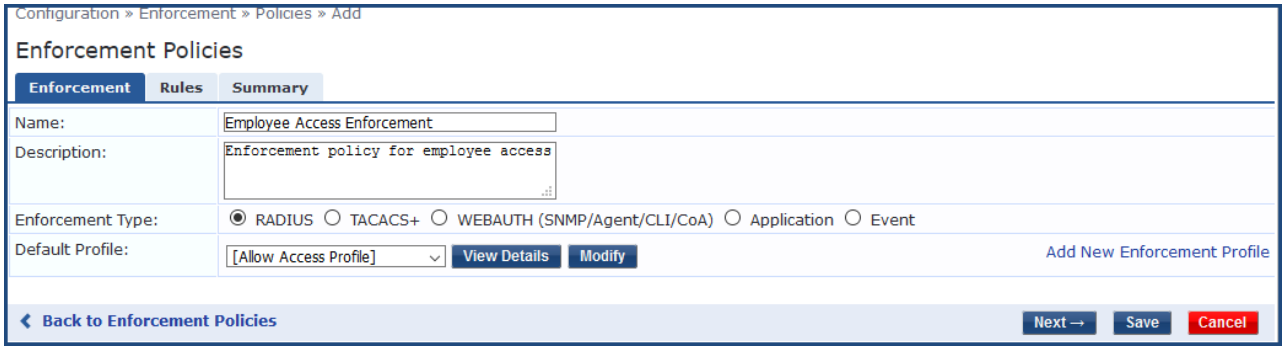
Adding an Enforcement Policy
Binding SNMP Enforcement for Ingress Events

Adding an Enforcement Policy

Procedure

1. To add a new enforcement policy from the **Enforcement Policies** page, navigate to **Configuration > Enforcement > Enforcement Policies**.
2. Click **Add**. The **Add Enforcement Policy** page opens to the **Enforcement** tab:

Figure 1. Add Enforcement Policy > Enforcement Tab



3. Specify the **Add Enforcement Policy > Enforcement** parameters as described in the following table:

Table 1. Add Enforcement Policy > Enforcement Tab Parameters

Parameter	Action/Description
Name	Enter the name of this enforcement poli cy.
Description	Enter a useful description of this enforce ment policy (recommended).
Enforcement Type	Select one of the following enforcement t ypes: • RADIUS • TACACS+ • WebAuth (SNMP/CLI)/CoA • Application • Event Based on this selection, the **Default Profi le** drop - down lists the associated enforc ement profiles.

NOTE

Policy Manager OnGuard. B oth SNMP - and CLI - (SSH /Telnet) based enforcement profiles can be sent to the n etwork device based on the

Parameter	Action/Description
	type of device and the use c ase.
Default Profile	Select the **Default Profile** from the drop - down menu. An enforcement policy ap plies conditions (roles, health, and time a ttributes) against specific values associat ed with those attributes to determine th e enforcement profile. If none of the rul es matches, Policy Manager applies the d efault profile. **NOTE** To add a new enforcement profile, click t he **Add New Enforcement Profile** link.

4. In the **Rules** tab, click **Add Rule** to display the **Rules Editor**:

Figure 2. Add Enforcement Policy > Rules Editor

Table 2. Add Enforcement Policy: Rules Editor Action Buttons

Button	Action/Description
Add Rule	Click the **Add Rule** action button to bring up the Rules Editor and add a new rule.
Copy Rule	Select the rule you want to copy, then cli ck the **Copy Rule** action button. The cop

Button	Action/Description
	ied rule is added to the existing list of rul es.
Move Up/ Move Down	To change the order that rules are execu ted in the enforcement policy, select an enforcement policy rule, then click **Move Up** or **Move Down** as desired.
Edit Rule	Select the rule you want to edit, then clic k the **Edit Rule** action button.
Remove Rule	Select the rule you want to delete, then c lick the **Remove Rule** action button.

5. Specify the **Add Enforcement Policy > Rules** tab parameters as described in the following table:

Table 3. Add Enforcement Policy: Rules Editor

Field	Description
Conditions/Enforcement Profiles	Select conditions for this rule. For each c ondition, select a matching action (enfo rcement profile).

NOTE

A condition in an enforceme nt policy rule can contain att ributes from the following na mespaces: *Tips:Role*, *Tips:Po sture*, and *Date*.

NOTE

The value field for the *Tips: Role* attribute can be a role defined in Policy Manager, o r a role fetched from the aut horization source. You can e nter role names fetched fro

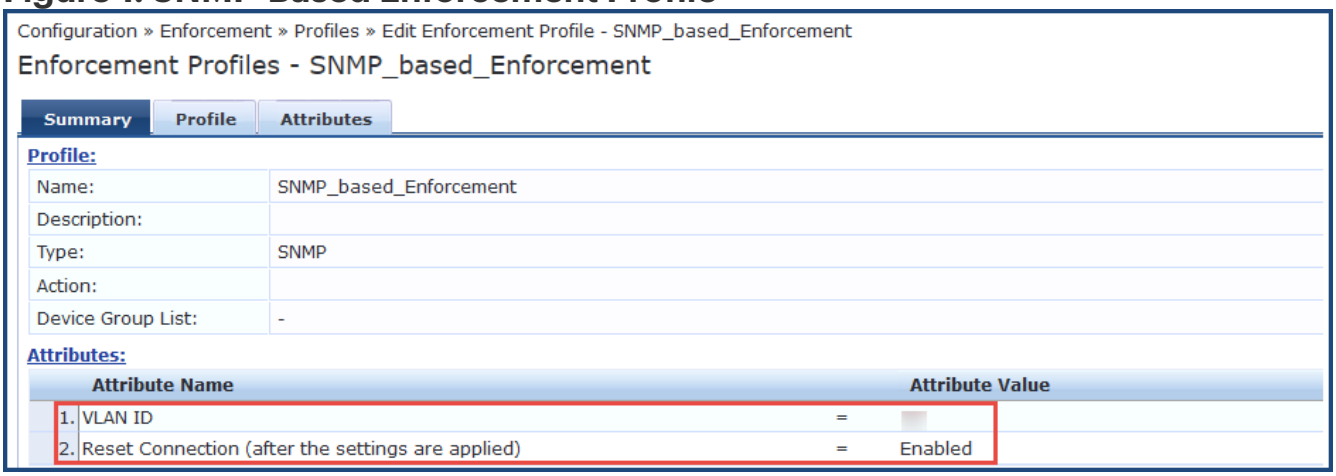
Field	Description
	m the authorization source f ree - form in the *Value* field . To commit the rule, click **S ave**.
Enforcement Profiles	• If the rule conditions match, attribut es from the selected enforcement pro files are sent to the Network Access D evice • If a rule matches and there are multi ple enforcement profiles, the enforce ment profile disambiguation rules app ly. • Refer to Configuring Enforcement Pro files for a list of the default profiles.

Binding SNMP Enforcement for Ingress Events

SNMP enforcement profiles can be bound to an event-based enforcement policy, which enables Policy Manager to trigger SNMP enforcement based on an Ingress event.

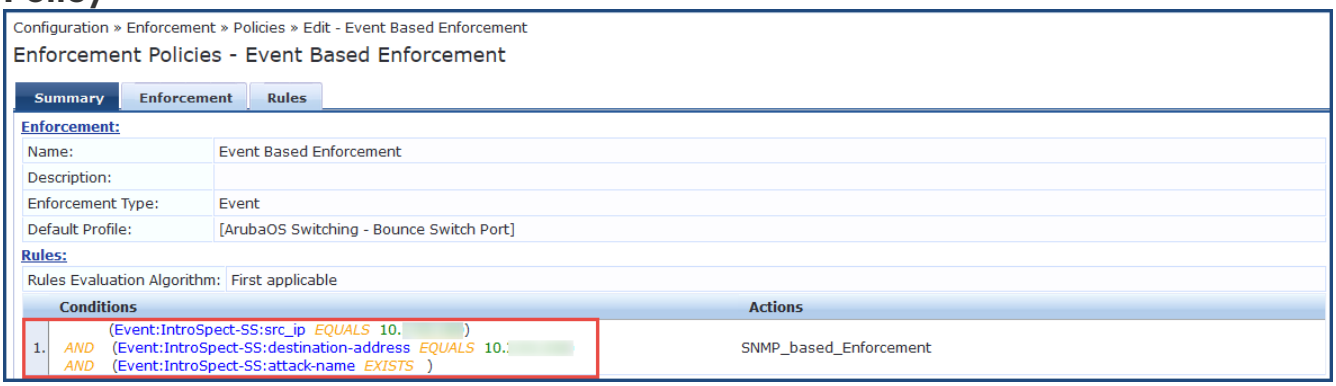
This section demonstrates how Policy Manager allows the binding of an SNMP-based enforcement profile to an event-based enforcement policy.

Figure 1. SNMP-Based Enforcement Profile



The configuration shown in Figure 5 demonstrates the binding of an SNMP-based enforcement to an event-based enforcement policy.

Figure 2. Binding an SNMP-Based Enforcement to an Event-Based Enforcement Policy



Configuring Deep-Nested Active Directory Queries

This section describes the method for configuring Policy Manager deep-nested Active Directory queries. This configuration prevents failed authentications due to lengthy look-up times.

- Configuring the Active Directory Authentication Source
- Using the LDAP Browser to Select the Group Information
- Configuring the Enforcement Policy for Deep-Nested AD Queries

Configuring the Active Directory Authentication Source

About this task

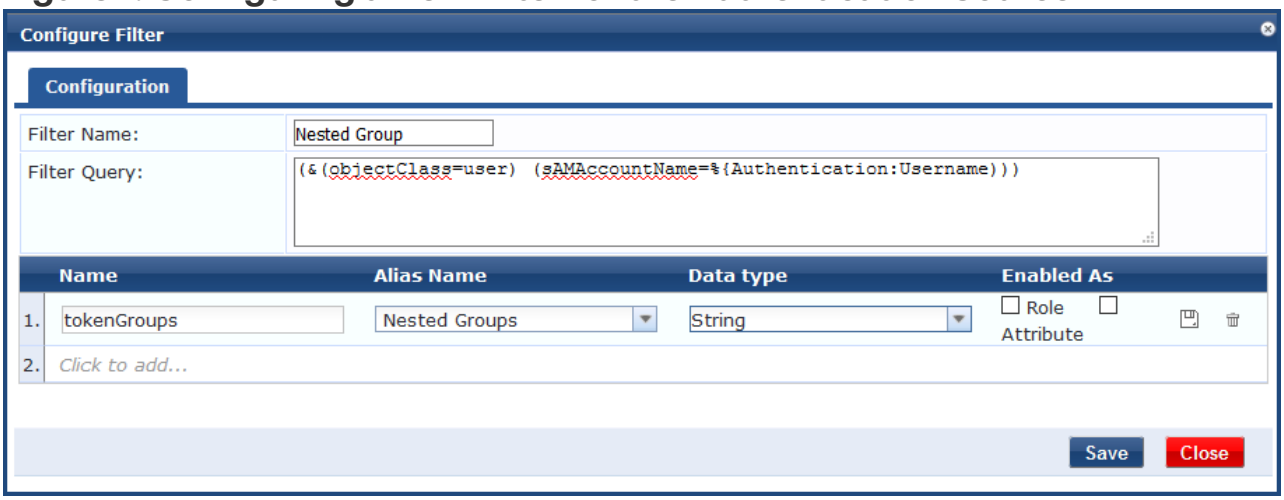
The first task in this procedure is to create an Active Directory authentication source and configure a new filter that will then be used in the enforcement policy.

To add a new Active Directory authentication source:

Procedure

1. Navigate to **Configuration > Authentication > Sources**. The **Authentication Sources** page opens.
2. Click **Add**. The **Add Authentication Sources** dialog opens.
3. For **Type**, select **Active Directory**. For details on configuration for an Active Directory authentication source, refer to Generic LDAP and Active Directory.
4. From the **Attributes** tab, click **Add More Filters**. The **Configure Filter** dialog opens.

Figure 1. Configuring a New Filter for the Authentication Source



5. Specify the **Authentication Source > Configure Filter** parameters as described in the following table:

Table 1. Authentication Source > Configure Filter Parameters

Parameter	Action/Description
Filter Name	Enter the name of this authentication source filter.
Filter Query	Enter the following filter query: *(&(objectClass=user) (sAMAccountName=%{Authentication:Username}))*
Name	Enter the following name: *tokenGroups*
Attributes	
Alias Name	Enter the following Alias Name: *Nested Groups* **NOTE** Take note of this Alias Name as you will use it when configuring the Rule for the enforcement policy later in this section (see Configuring the

Parameter	Action/Description
	Enforcement Policy for Deep - Nested AD Queries).
Data Type	From the drop - down, select *String*.

6. Click **Save**.

Using the LDAP Browser to Select the Group Information

About this task

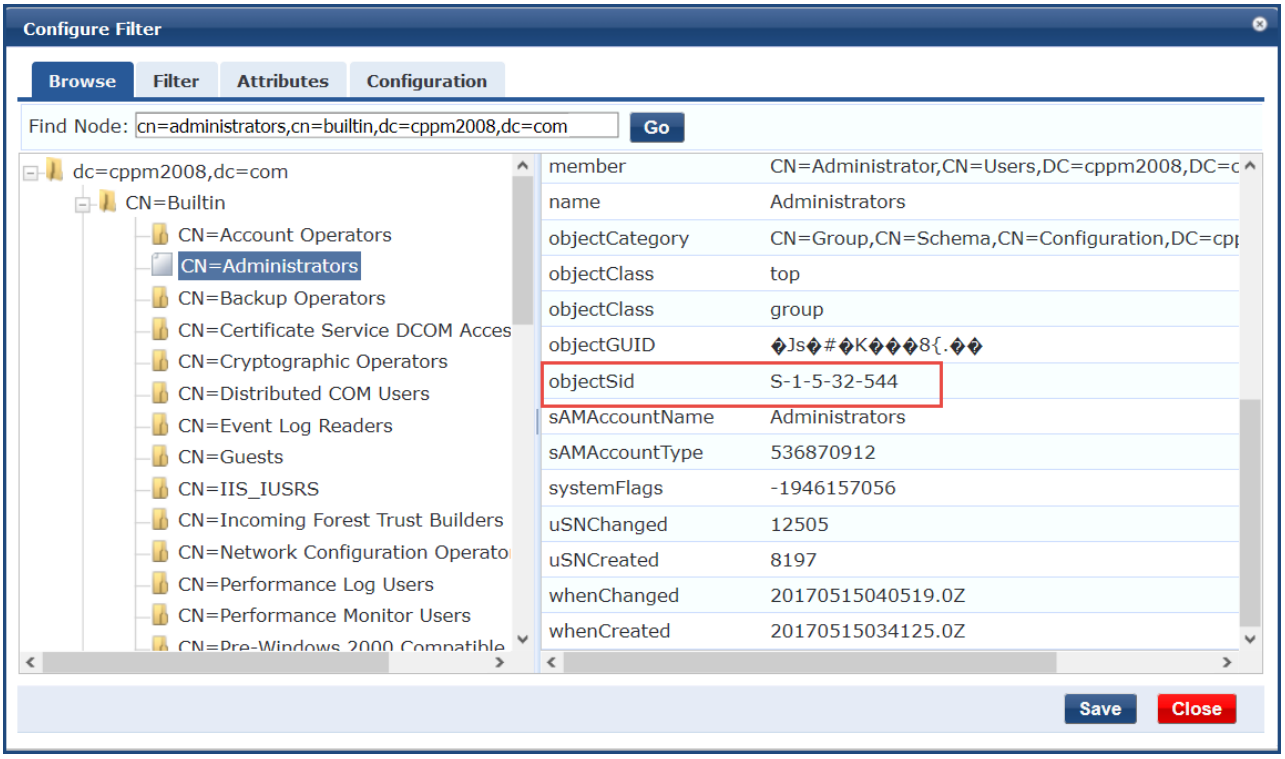
Use the LDAP **Browse** page to select the Group Name. Then copy the corresponding **objectSid** value. The **objectSid** value will be used for the Rule Configuration in the enforcement policy (as described in the next section).

To select the group information:

Procedure

1. From the **Configure Filter** configuration dialog, select the **Browse** tab. The **Browse** tab shows an LDAP browser from which you can browse the nodes in the Active Directory (or LDAP) directory, starting at the base DN. This is presented in read-only mode.

Figure 2. Selecting the Group Name and objectSid Value from the Browse Dialog



2. From the panel on the left select the Group Name (for example, *CN=Administration*).
3. Then, from the panel on the right, locate and copy the corresponding **objectSid** value.
4. Click **Close**.

Configuring the Enforcement Policy for Deep-Nested AD Queries

About this task

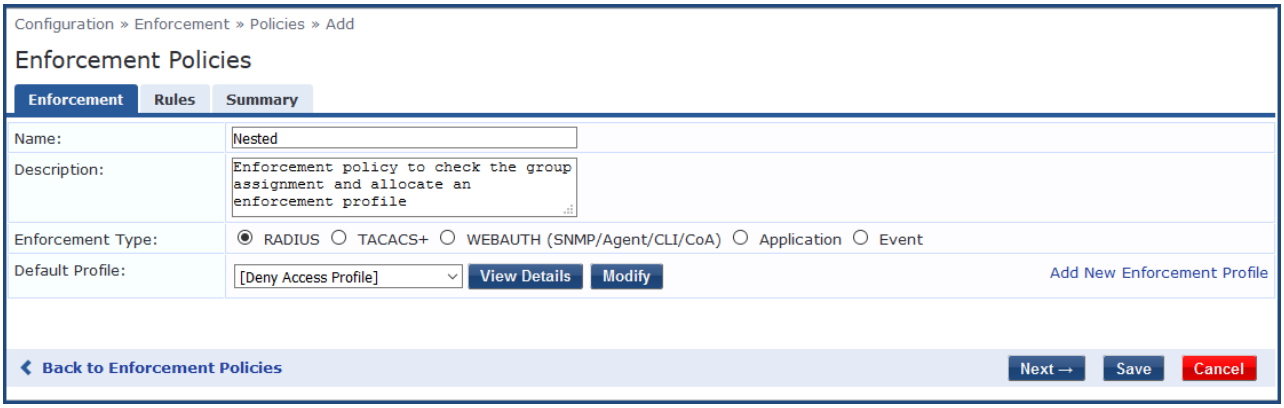
For details on configuring enforcement policies, see Configuring Enforcement Policies.

To configure the enforcement policy for deep-nested queries:

Procedure

1. Navigate to **Configuration > Enforcement > Policies**. The **Enforcement Policies** page opens.
2. Click the **Add** link. The **Add Enforcement Policies** page opens.

Figure 3. Enforcement Policy



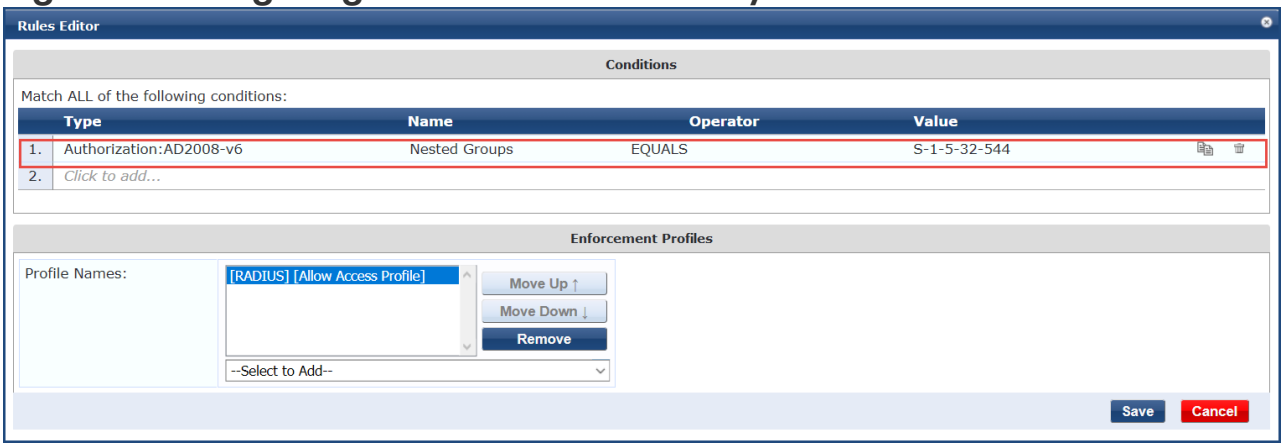
3. Specify the **Enforcement** tab parameters as described in the following table:

Table 2. Add Enforcement Policy > Enforcement Tab Parameters

Parameter	Action/Description
Name	Enter the name of this enforcement
Description	Enter a description of this enforcem
Enforcement Type	Select **RADIUS**.
Default Profile	From the drop - down, select any ap nt profile.

4. Click **Next** (or click the **Rules** tab). The **Rules Editor** opens.

Figure 4. Configuring the Enforcement Policy Rule



5. Select conditions for this rule. For each condition, select a matching action (enforcement profile).

6. Specify the **Rules Editor** conditions as described in the following table:

Table 3. Configuring the Nested Group Rule

Parameter	Action/Description
Type	Select the Active Directory u
Name	Enter the **Alias Name** specif urce (see Configuring the Ac n Source).
Operator	Select the appropriate oper
Value	Specify the **objectSid** value er (see Using the LDAP Brow mation).
Profile Name	From the drop - down, selec e.

7. Click **Save**. You return to the **Add Enforcement Policies > Rules** page.

Figure 5. Enforcement Policy Rule Configured

8. Click **Save**. If the rule conditions match, attributes from the selected enforcement profiles are sent to the Network Access Device. If a rule matches and there are multiple enforcement profiles, the enforcement profile disambiguation rules apply. Refer to Configuring Enforcement Profiles for a list of the default profiles.

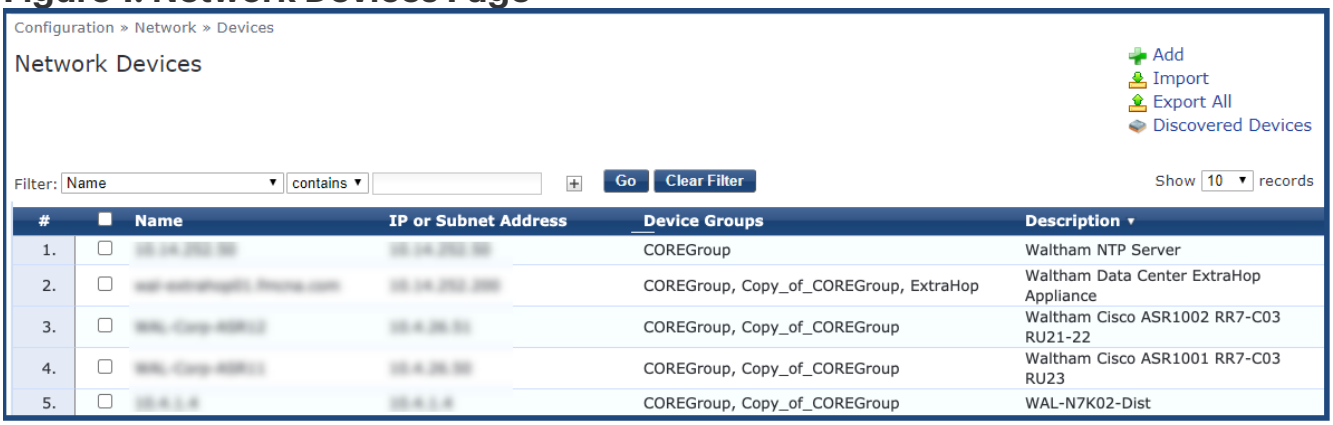
Adding and Modifying Network Devices

A Network Access Device (NAD) must belong to the global list of devices in the Policy Manager database in order to connect to Policy Manager using any of the supported

protocols. The Policy Manager **Network Devices** page displays the device name, IP address or subnet address, and a brief description of each configured device.

To view this page, navigate to **Configuration > Network > Devices**. The **Network Devices** page opens:

Figure 1. Network Devices Page



Subtopics

Integrate ClearPass with Internal Business Intelligence (BI) Tools
Adding a Network Device
Adding and Modifying Device Groups
Adding and Modifying Proxy Targets
Configuring the Ingress Event Source

Integrate ClearPass with Internal Business Intelligence (BI) Tools

If wanting to integrate ClearPass with internal business intelligence (BI) tools, the following **tips_nad_client** and **tips_nad_group** tables provide information about NAD clients and groups. Read - only access is allowed to some of the data in the following tables, enabling queries to ClearPass services for information often used in internal business intelligence reports or policies.

The fields available for **tips_nad_client** and **tips_nad_group** queries are as follows:

Table 1. Available tips_nad_client fields

Column	Details
name	Name of the NAD device
ipaddress	IP or subnet of the NAD device

Column	Details
description	User - defined description of the NAD
vendor_id	The vendor's unique identifier
coa_capable	Is RADIUS dynamic authentication ena
coa_port	Port used for RADIUS dynamic authen
attributes	Attributes configured under **Administ aries > dictionary** attributes
extras	Contains NAD groups the client is par
is_radsec_enabled	Is RadSec enabled

Table 2. Available tips_nad_group fields

Column	Details
name	Name of the NAD group
description	User defined description of the NAD grou
vendor_id	The vendor's unique identifier
members	NAD clients based on the member forma
members_fmt	Format of the NAD clients in the group

Adding a Network Device

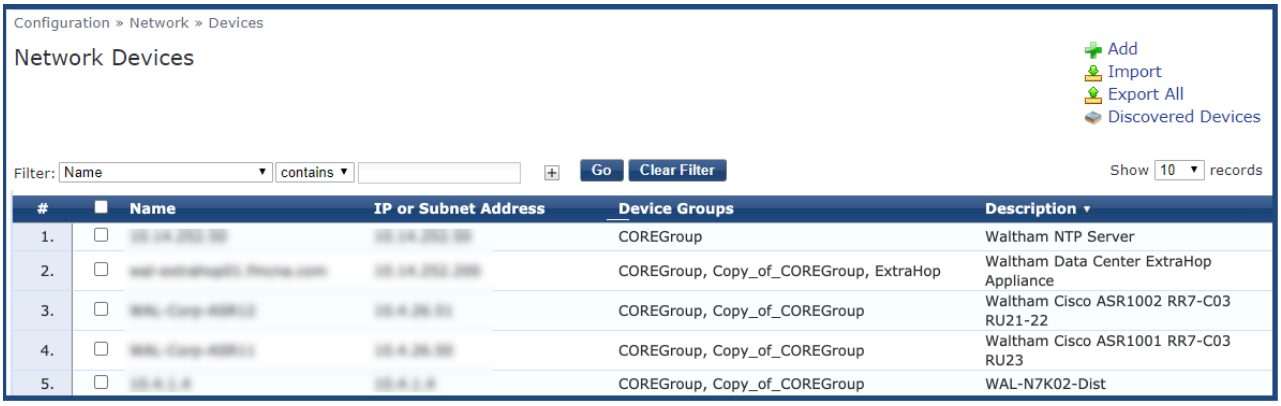
About this task

To add a network device:

Procedure

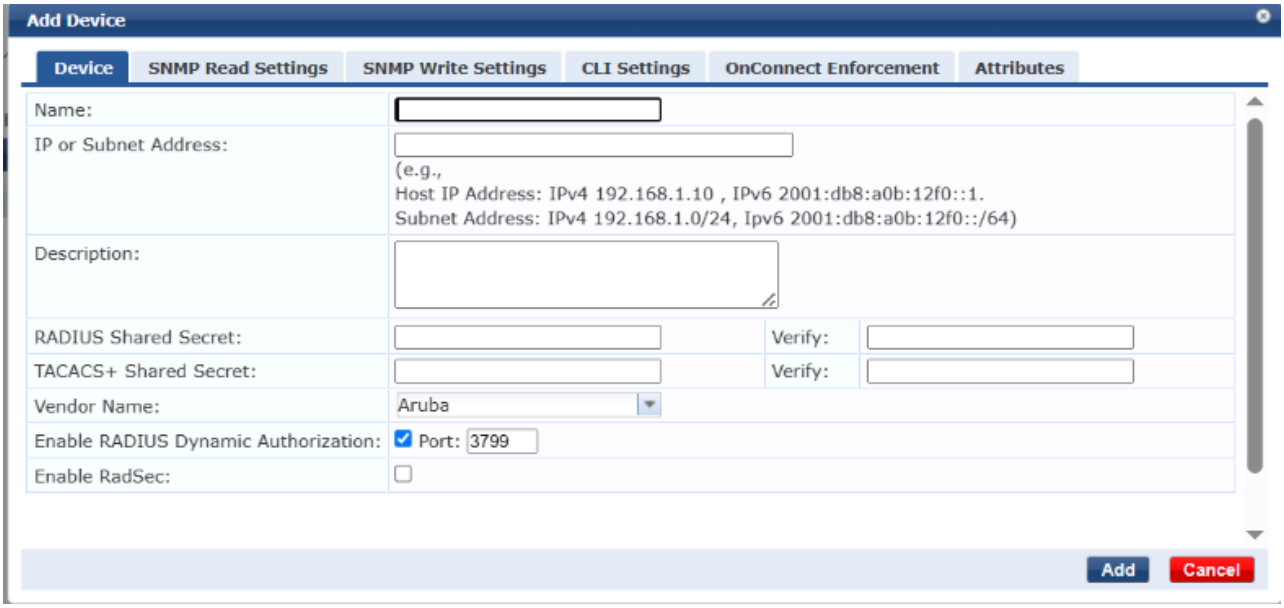
1. Navigate to the **Configuration > Network > Devices** page. The **Network Devices** page opens.

   Figure 1. Network Devices Page



2. Click the **Add** link. The **Add Device** page opens.

Figure 2. Add Device > Device Dialog



Subtopics

Configure Device Settings
Configure RadSec Settings
SNMP Read Settings Parameters
SNMP Write Settings Parameters
CLI Settings Parameters
Enabling OnConnect Enforcement on a Network Device
Attributes Parameters
Modifying a Network Device

Configure Device Settings

Click the **Device** tab and configure the parameters as described in Table 1:

Table 1. Add Device > Device Parameters

Parameter	Action/Description
Name	Enter the name of the device.
IP Address or Subnet	Specify the IPv4 or IPv6 address or the sub net of the device. You can use a hyphen to indicate the range of device IP addresses following the format **a.b.c.d - e**. For example, **192.168.1.1 - 20**. IP v6 addresses can be entered in the format s **2001:db8:a0b:12f0::1/64** or **2001:db8:a 0b:12f0::1fab - 20ff**. The prefix length can be 0 - 128 bits. Link l ocal, site local, loopback, and multicast ad dresses are not allowed. An IPv6 address o r subnet cannot be configured for network access devices (NADs) where RadSec is en abled. **NOTE** When a subnet is added to a d evice through the fields on thi s tab, the network devices bel onging to that subnet will only be read if traps are received f rom those devices
Description	Enter a description that provides additional information to identify the device.
RADIUS Shared Secret	Enter the RADIUS shared secret.
TACACS+ Shared Secret	Enter the TACACS+ shared secret.
Vendor Name	Specify the name of the vendor to load the dictionary associated with this vendor for t his device. **NOTE** **RADIUS:IETF**, the dictionary c ontaining the standard set of

Parameter	Action/Description
	RADIUS attributes, is always loaded. When you specify a vendor here, the RADIUS dictionary associated with this vendor is automatically enabled.
Enable RADIUS Dynamic Authorization	If RADIUS Dynamic Authorization has not been automatically enabled, click the check box to enable this option. RADIUS Dynamic Authorization allows dynamic changes to a user session, as implemented by network access server products. This includes support for disconnecting users and changing authorizations applicable to a user session.
Dynamic Authorization Port	The access point's UDP Port for Dynamic Authorization must be reachable from your RADIUS server. The Dynamic Authorization Port is set by default to **3799**.
Enable RadSec	To enable RadSec, click the **Enable RadSec** check box. When RadSec is enabled, the RADIUS shared secret is populated with a default shared secret with the string “radsec.” **NOTE** It is important that the controller is configured with the same shared secret. By default, RadSec communications use TCP port 2083. Therefore, when you enable RadSec, ClearPass automatically creates a policy rule to allow communication on port 2083.

Configure RadSec Settings

NOTE

Users should be aware that if a ClearPass 6.7.x system uses RadSec, during upgrade to 6.9.0, all RadSec functionality is lost. This issue occurs because the source override IP address feature that was introduced after 6.7.x must be configured, and because the Trust List usage for RadSec needs to be updated.

To configure the source override IP address:

1. Go to **Configuration > Network > Devices** and open the **Edit Device Details** form for the RadSec-enabled device.
2. On the **RadSec Settings** tab, in the **Source Override IP Address** field enter the same IP address as the network access device (NAD) IP for RadSec.
3. This will need to be done for all NAD definitions on upgraded appliances; however, it will not need to be done thereafter for new NAD definitions.

To update the certificate usage:

1. Go to **Administration > Certificates > Trust List**.
2. For each Certificate Authority (CA) root certificate that will be used for RadSec, open the **View Certificate Details** form and add RadSec in the **Usage** list.

If you selected the you select the **Enable RadSec** option on the **Device** tab, select the **RadSec Settings** tab and configure RadSec parameters as described in the following table. This tab does not appear unless you select the **Enable Radsec** option.

Figure 1. Add Device > RadSec Settings Tab

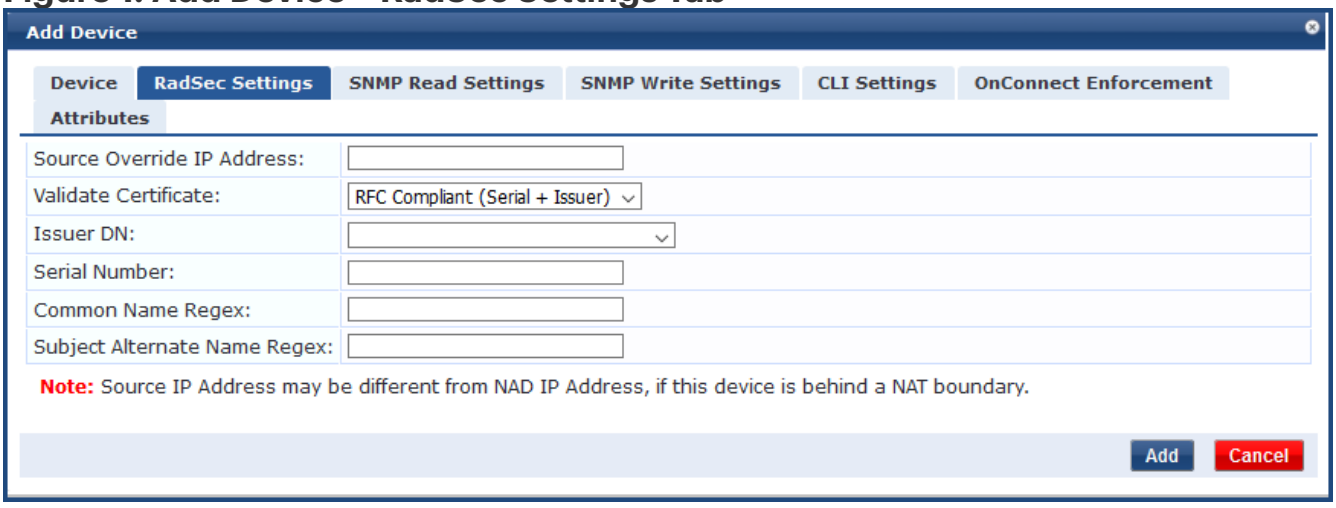


Table 1. Add Device > RadSec Settings Parameters

Parameter	Action/Description
Source Override IP address	The default value is the IP address or subn et of the device entered in the **IP Address or Subnet** field on the **Device** tab. If the NA D IP address is different from the source IP address, enter the source IP address or sub net in this field to override the NAD IP with the desired source IP address. **NOTE** Subnets must be defined by a slash, not a hyphen. • Supported format: *10.2.54/24* • Not supported format: *10.2.14.0 - 24* • IPv6 addresses are not allowed in NAD configurations when RadSec is enabled . **NOTE** When setting the Source Over ride IP address, keep in mind the following warning messag e displays if the address is se t to 0.0.0.0/0 or ::/0. "WARNING : Setting the Source Override I P address to 0.0.0.0/0 or ::/0 a llows radsec traffic from any d evice trying to establish conn ection on the configured rads ec port. It is recommended to enable certificate validation w hen setting the Source Overri de IP address to 0.0.0.0/0 or : :/0 to allow only authorized d evices."
Validate Certificate	If you do not want any validation or authori zation checks for this device, select the **No Authorization Checks** option. To validate the certificate with a common n ame (CN) or Subject Alternative Name (SA

Parameter	Action/Description
	N) select **Validate with CN or SAN** and ent er the following values • **Common Name Regex**: Enter the name associated with this entity. This can be a host name, IP address, or other name. • **Subject Alternative Name Regex**: Enter the Subject Alternative Name (SAN) for the specified Common Name in one of the following formats: ◦ email: *email_address* ◦ URI: *uri* ◦ IP: x.x.x.x ◦ dns: *dns_name* ◦ rid: *id*
	For RFC 6614 - compliant validation using t he issuer distinguished name (DN) and cer tificate serial number, select **RFC Complia nt (serial + Issuer)** and enter the following values: • Issuer DN • Serial Number • Common Name Regex • Subject Alternative Name Regex

RadSec Certification Validation

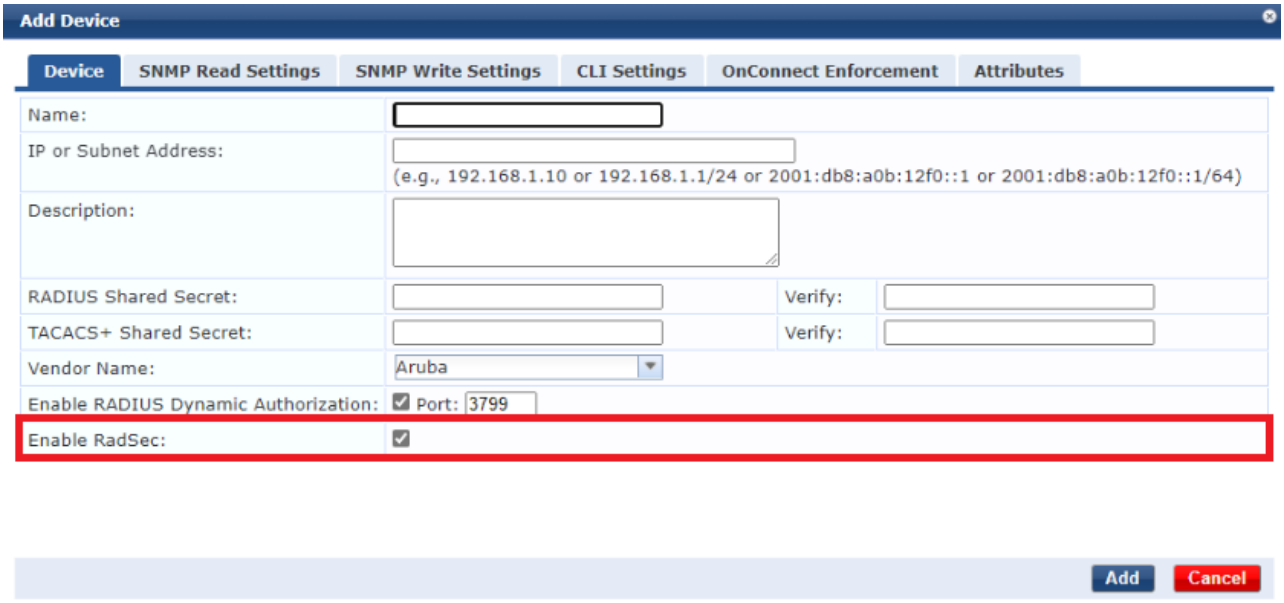
About this task

To configure RadSec settings for RFC compliance (Serial + Issuer):

Procedure

1. Navigate to **Configuration > Network > Devices** and select **Add**. Provide a default configuration for the device and be sure to select the **Enable Radsec** option to display the required **RadSec Settings** tab within the **Add Device** screen.

Figure 2. Add Device > Enable RadSec

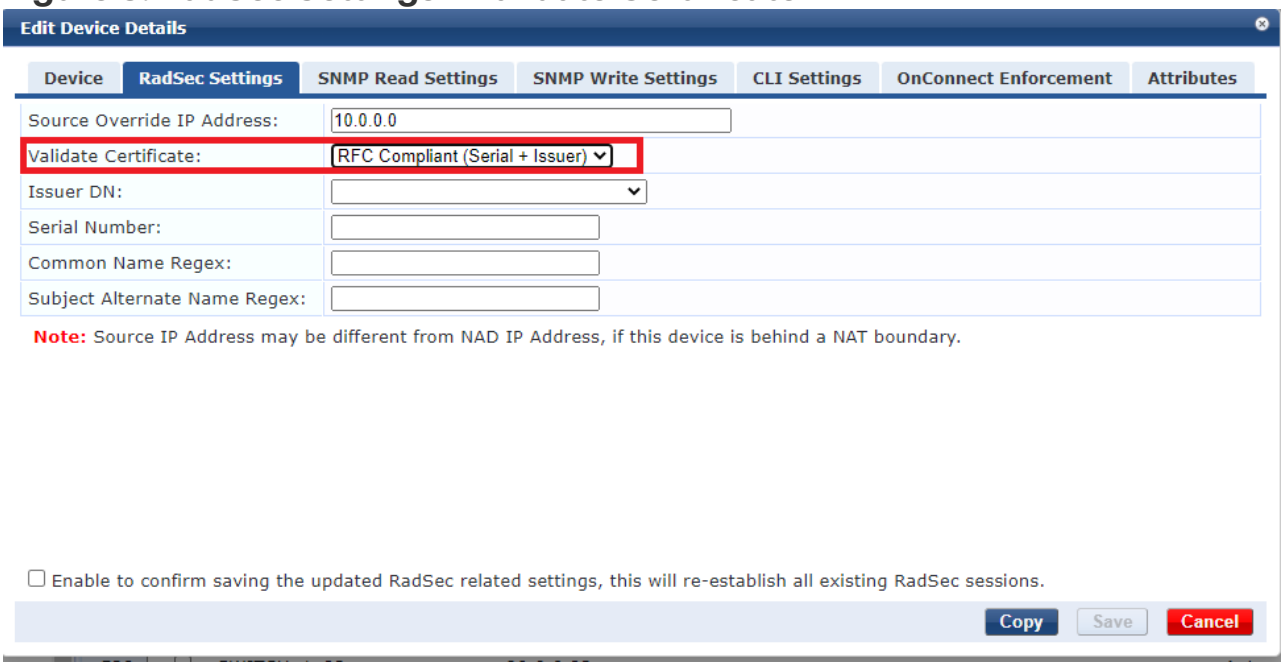


2. Select the newly added device from the **Network Devices** tab. From the **RadSec Settings** tab, ensure **No Authorization Checks** is selected from the **Validate Certificate** drop-down menu.

3. Navigate to **Administration > Certificates** and select **Trust List**. From the **Certificate Trust List**, locate the certificate made available in the output of the access point's **show cert all** command and enable its **Usage** as **RadSec**.

4. Modify the existing network access device (NAD) that was added for Instant AP (IAP) information. From the **Edit Device Details > RadSec Settings** screen, change the value of the **Validate Certificate** field to **RFC Compliant (Serial + Issuer)**.

Figure 3. RadSec Settings > Validate Certificate



5. Select the certificate returned by the **show cert all** command from within the **Device Certificate** section.
6. In the **Serial Number** field, enter the serial number displayed in the **Device Certificate** section, and returned by the **show cert all** command.
7. In the **Common Name Regex** field, enter the value displayed for the **Subject** in the **Device Certificate** section and returned by the **show cert all** command.
8. Select **Enable to confirm saving the updated RadSec related settings, this will re-establish all existing RadSec sessions**. Click **Save**.

SNMP Read Settings Parameters

Click the SNMP Read Settings tab to define values that allow ClearPass Policy Manager to read information from the device using SNMPv1, SNMPv2, or SNMPv3.

> **NOTE**
> Large or geographically-spread cluster deployments typically do not require each Policy Manager node to probe all SNMP configured devices.

Figure 1. Add Device > SNMP Read Settings Dialog

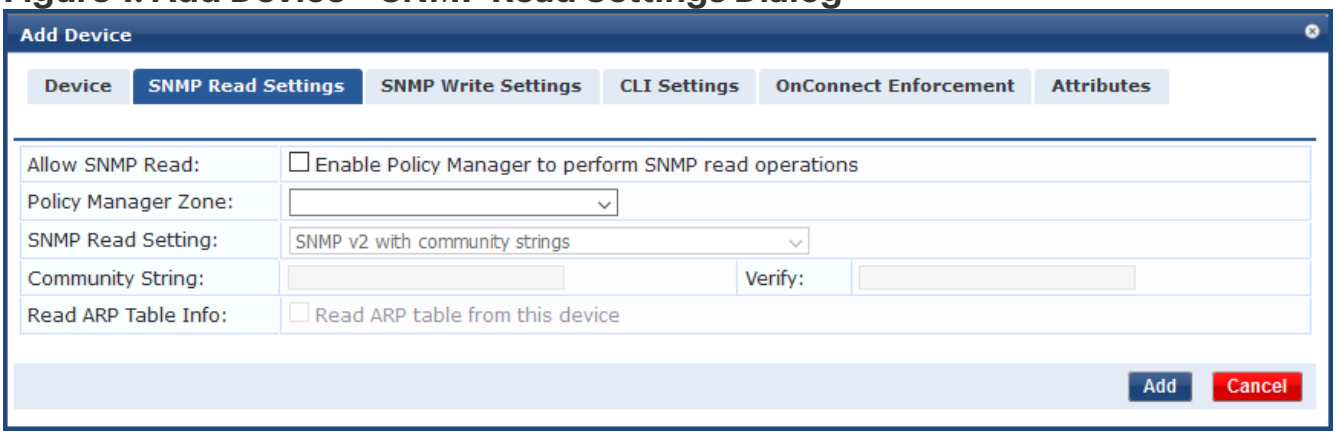


Table 1. Add Device > SNMP Read Settings Parameters

Parameter	Action/Description
Allow SNMP Read	Toggle to enable or disable SNMP Read operations. **NOTE**

Parameter	Action/Description
	Network device polling is not dependent on SNMP traps tha t are configured on NAD devi ces. In a cluster, Policy Manag er will automatically load - bal ance NAD SNMP Reads across all the nodes in a zone.
Policy Manager Zone	Use this field to can assign Network Access Devices to a zone, allowing the SNMP servi ce to poll or query only the NADs that are i n its zone. OnConnect Enforcement is triggered when a trap from a NAD is received by a Policy M anager node. If the zone assigned to a Poli cy Manager node is not the same as the zo ne configured here, OnConnect Enforceme nt is not triggered on that Policy Manager node. **NOTE** Assigning a Policy Manager Zo ne is mandatory for all devices if **SNMP Read** or **OnConnect enforcement** is enabled.
SNMP Read Setting	Specify one of the following SNMP Read S ettings: • SNMP v1 with community strings • SNMP v2 with community strings • SNMP v3 with no Authentication • SNMP v3 with Authentication using MD 5 and no Privacy • SNMP v3 with Authentication using MD 5 and with Privacy • SNMP v3 with Authentication using SH A and no Privacy

Parameter	Action/Description
	• SNMP v3 with Authentication using SH A and with Privacy **NOTE** The MD5 authentication type i s not supported when you use Policy Manager in **FIPS** mode.
Community String	Enter the community string for sending the traps. **NOTE** Available in SNMP v2 only.
Verify	Reenter the community string for sending t he traps.
Read ARP Table Info	Enable the **Read ARP table on this device** check box on a Layer - 3 device if you inte nd to use the ARP table on this device to di scover endpoints in the network. **NOTE** When this option is selected, all ARP entries read during pe riodic Network Access Device reads are added to Policy Man ager endpoints. SNMP, WMI, N Map, and SSH scans are not u sed in this process. When **Dev ice Insight Integration** is enab led, this field is hidden (for mo re information, see Device Insi ght Integration Page).
Username	Specify the Admin user name to use for SN MP read operations. **NOTE** Available in SNMP v3 only.

Parameter	Action/Description
Authentication Key	Specify the SNMP v3 with authentication option (SHA or MD5). **NOTE** The **EAP - MD5** authentication type is not supported if you run Policy Manager in FIPS mode. **NOTE** Authentication Key is available in SNMP v3 only.
Privacy Key	Specify the SNMP v3 with privacy option. **NOTE** Available in SNMP v3 only.
Privacy Protocol	Choose one of the available privacy protocols: • DES - CBC • AES - 128 **NOTE** This option is available in *SNMP v3 with Privacy* only. Privacy allows for encryption of SNMP v3 messages to ensure confidentiality of data.

SNMP Write Settings Parameters

Click the SNMP Write Settings tab to define values that allow Policy Manager to write to (manage) the device using SNMPv1, SNMPv2, or SNMPv3. Available parameters are described in the following table.

Figure 1. Add Device > SNMP Write Settings Dialog

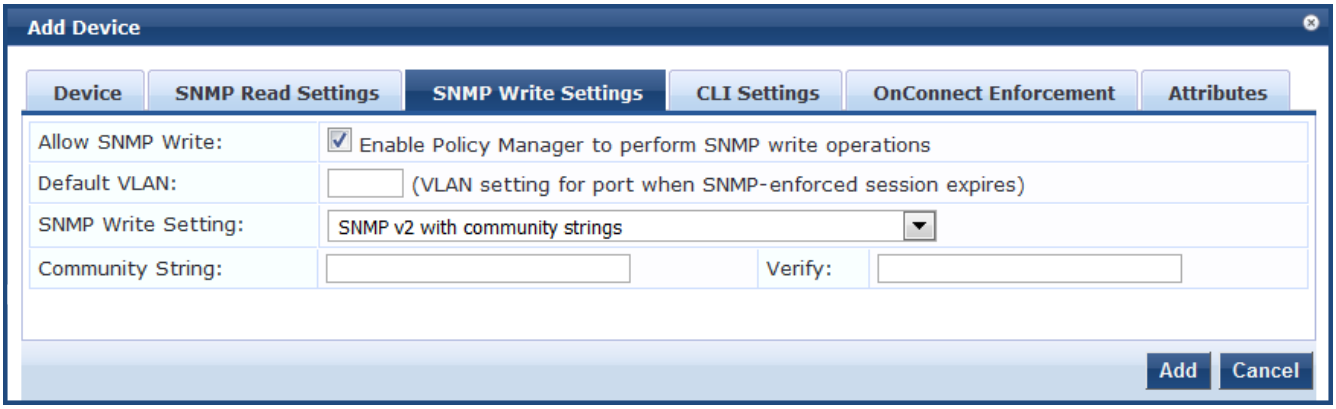


Table 1. Add Device > SNMP Write Settings Parameters

Parameter	Action/Description
Allow SNMP Write	Toggle to enable or disable SNMP write.
Default VLAN	Specify the VLAN port setting after the SN MP - enforced session expires.
SNMP Write Setting	Specify the SNMP Write setting for the devi ce. You can set any of the following options : • SNMP v1 with community strings • SNMP v2 with community strings • SNMP v3 with no Authentication • SNMP v3 with Authentication using MD 5 and no Privacy • SNMP v3 with Authentication using MD 5 and with Privacy • SNMP v3 with Authentication using SH A and no Privacy • SNMP v3 with Authentication using SH A and with Privacy **NOTE** The MD5 authentication type i s not supported if you use C learPass Policy Manager in **FI PS** mode.

Parameter	Action/Description
Community String	Enter the community string for sending the traps.
Verify	Re - enter the community string for sendin g the traps.

CLI Settings Parameters

From the **Configuration > Network > Devices** page, use the **CLI Settings** tab to enable or disable the CLI, and define user names, passwords, and port settings for accessing the CLI. Available parameters are described in the following table.

Figure 1. Add Device > CLI Settings Dialog

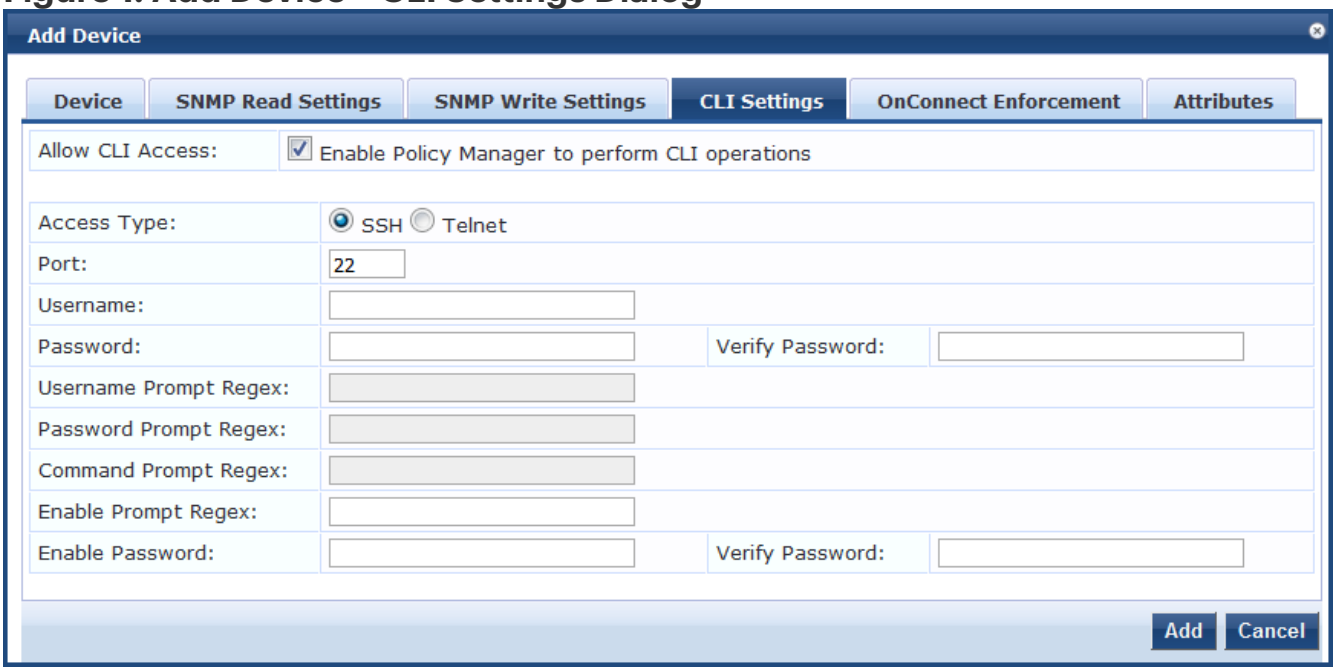


Table 1. Add Device > CLI Parameters

Parameter	Action/Description
Allow CLI Access	Toggle to enable or disable CLI access.
Access Type	Select **SSH** or **Telnet**. Policy Manager uses the selected access method to log into the device CLI.

Parameter	Action/Description
Port	Specify the **SSH** or Telnet **TCP** port numbe r.
Username	Enter the username to log into the CLI.
Password	Enter the password to log into the CLI.
Username Prompt Regex	Specify the regular expression for the user name prompt. Policy Manager looks for this pattern to rec ognize the Telnet username prompt.
Password Prompt Regex	Specify the regular expression for the pas sword prompt. Policy Manager looks for this pattern to rec ognize the Telnet password prompt.
Command Prompt Regex	Specify the regular expression for the com mand line prompt. Policy Manager looks for this pattern to rec ognize the Telnet command - line prompt.
Enable Prompt Regex	Specify the regular expression for the com mand line in the **enable** prompt. Policy Manager looks for this pattern to rec ognize the Telnet command - line prompt.
Enable Password	Enter then reenter the credentials for the **E nable** password in the CLI.

Enabling OnConnect Enforcement on a Network Device

OnConnect Enforcement is an enforcement model that allows you to use non-802.1X methods for device scans, VLAN placement, and so on. **OnConnect Enforcement** allows enforcement in non-802.1X environments without the need for an agent (such as OnGuard) on the endpoint.

NOTE

Assigning a Policy Manager Zone is mandatory for all devices if **SNMP Read** or **OnConnect enforcement** is enabled.

When this feature is enabled, Policy Manager performs the following actions:

- Detects when a new endpoint connects to the network.
- Scans the endpoint to identify the logged-in user and other device-specific information.
- Triggers a Web-based authentication (WebAuth) for the device.
- Performs SNMP-based enforcement to change the network access profile for the device.

  Use the **OnGuard Enforcement** tab on the **Configuration > Network > Devices** page to configure the settings described in Table 6.

Figure 1. Add Device > OnConnect Enforcement Dialog

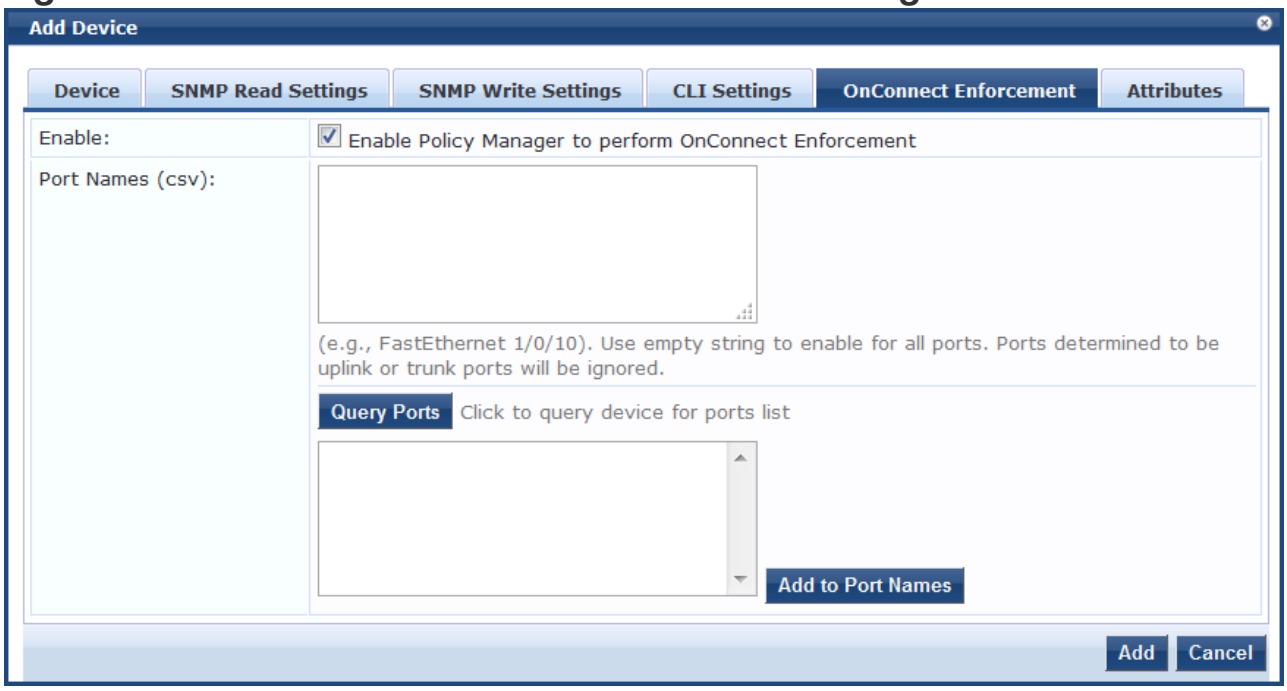


Table 1. Add Device > OnConnect Enforcement Parameters

Parameter	Action/Description
Enable	Select this check box to enable Policy Manager OnConnect on the network access device being added.
Port Names	Specify the names and descriptions of the ports to be enabled for OnConnect Enforcement (see the next section for details). You can do so in two ways:

Parameter	Action/Description
	◦ Click **Query Ports**. ◦ Manually enter port names the the **Port Names** field as a comma - separated list. Only the ports added in the **Port Names** field will have OnConnect Enforcement enabled. For example, if you add the port names **Fa1/0/3,Fa1/0/5**, when clients connect to any of these ports on the specified network device, OnConnect Enforcement is triggered on that network device. **NOTE** An empty string will enable OnConnect on all ports. Policy Manager will attempt to determine the uplink or upstream trunk ports; however, it is recommended to explicitly remove those ports.
Query Ports	Click **Query Ports t**o display the list of ports on the current server. Select the ports to use, then click **Add to Port Names**.The selected port names are added to the **Port Names** list. Only the ports added in the **Port Names** field will have OnConnect Enforcement enabled. **NOTE** **Allow SNMP Read: Enable Policy Manager to perform SNMP read operations** setting on the **SNMP Read Settings** tab.

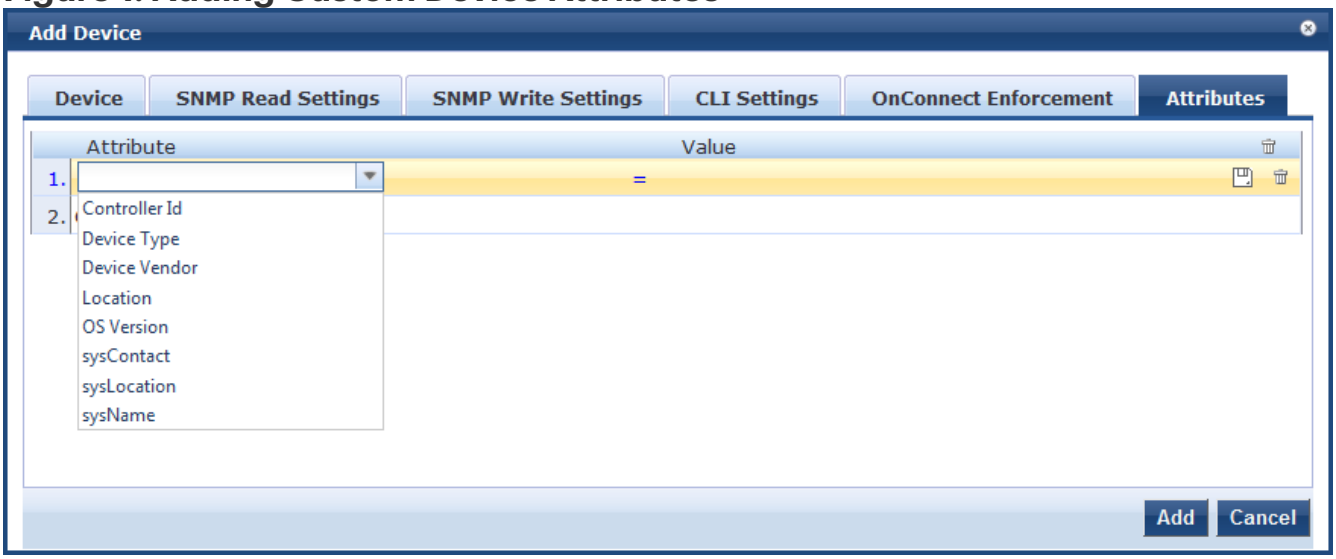
Parameter	Action/Description
Add to Port Names	Once a query displays the list of ports, select the desired ports from the list, then click **Add to Port Names**. The selected ports are added to the **Port Names** field.

Attributes Parameters

About this task

Use the **OnGuard Enforcement** tab on the **Configuration > Network > Devices** page to add custom attributes for this device.

Figure 1. Adding Custom Device Attributes



Procedure

1. From the *Attribute* field, click **Click to add...**.

   By default, the following custom attributes appear in the **Attribute** drop down:

   - Controller ID
   - Device Type
   - Device Vendor
   - Location

- OS Version
- sysContact
- sysLocation
- sysName

2. Select one of the default attributes or enter a new attribute. You can enter any name in the *Attribute* field. All attributes are of *string* datatype.
3. Specify the attribute's value. You can populate the *Value* field with any string.
4. Repeat this procedure as necessary.
5. When finished adding custom attributes, click **Add**. All attributes entered for a device are available in the role-mapping Rules Editor under the Device namespace.

Modifying a Network Device

About this task

To modify a Policy Manager network device:

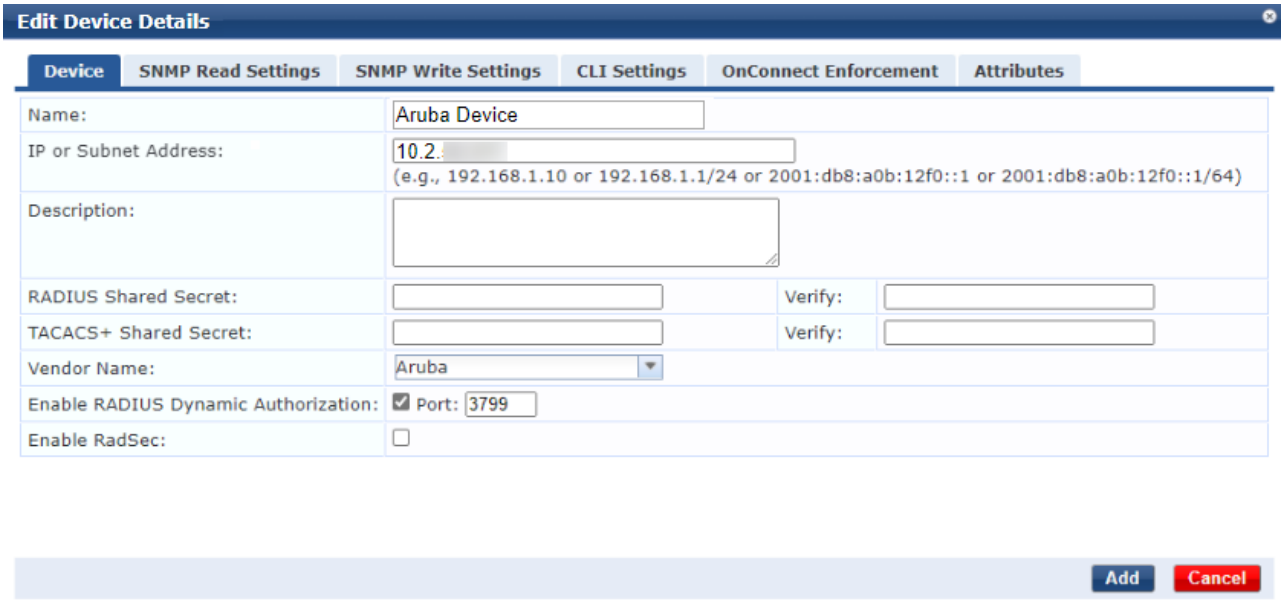
Procedure

1. Navigate to the **Configuration > Network > Devices** page. The **Network Devices** page opens.

Figure 1. Network Devices Page

2. In the **Network Devices** table, click the name of the network device you want to modify. The **Edit Device Details** dialog opens.

Figure 2. Modifying a Network Device



3. Modify any device settings as necessary. For details about all of the Network Device tabs and parameters, refer to the previous section, Adding a Network Device.

> **NOTE**
>
> If you disable **RadSec**, the shared secret is removed, and you must re-enter the original shared secret.

4. Click **Save**.

Adding and Modifying Device Groups

Policy Manager groups devices into **Device Groups**, which function as a component in service and role-mapping rules. Device groups can also be associated with enforcement profiles; Policy Manager sends the attributes associated with these profiles only if the request originated from a device that belongs to a device group. Administrators configure device groups at the global level. Device groups can contain the members of the IP address of a specified subnet, regular expression-based variation, or devices that are previously configured in the Policy Manager database.

Policy Manager lists all configured device groups in the **Device Groups** page .

> **NOTE**
> The Device Group list displays device names together with their corresponding IP addresses, making it easier to locate and select devices. When adding or modifying a Device Group list, you can also select Network Access Server (NAS) devices based on both the device name and IP address.

When adding a device group, you must perform all tasks from a single browser tab. The Policy Manager WebUI uses server-side session caching during add or edit workflows, so performing add or edit actions on the same device group from different tabs of the same browser can lead to data loss and impact network access.

Subtopics

Adding a Device Group
Modifying a Device Group

Adding a Device Group

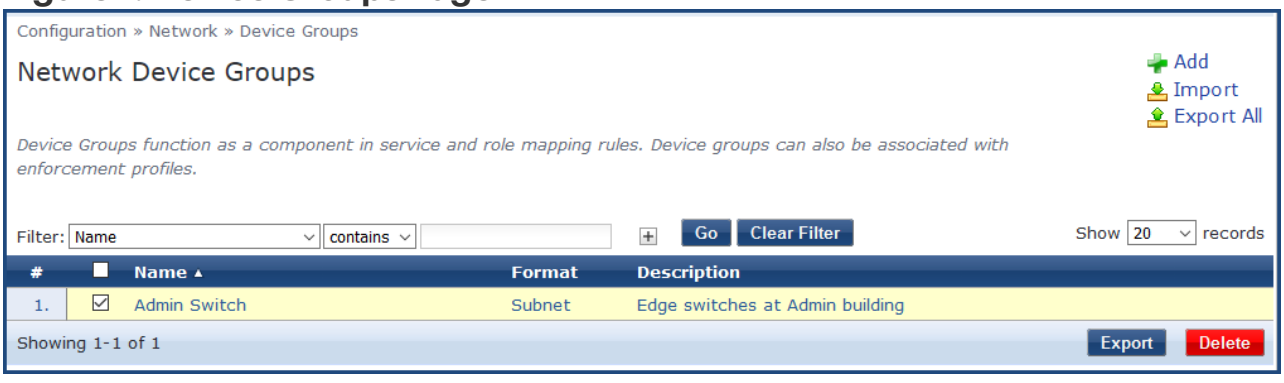
About this task

To add a device group:

Procedure

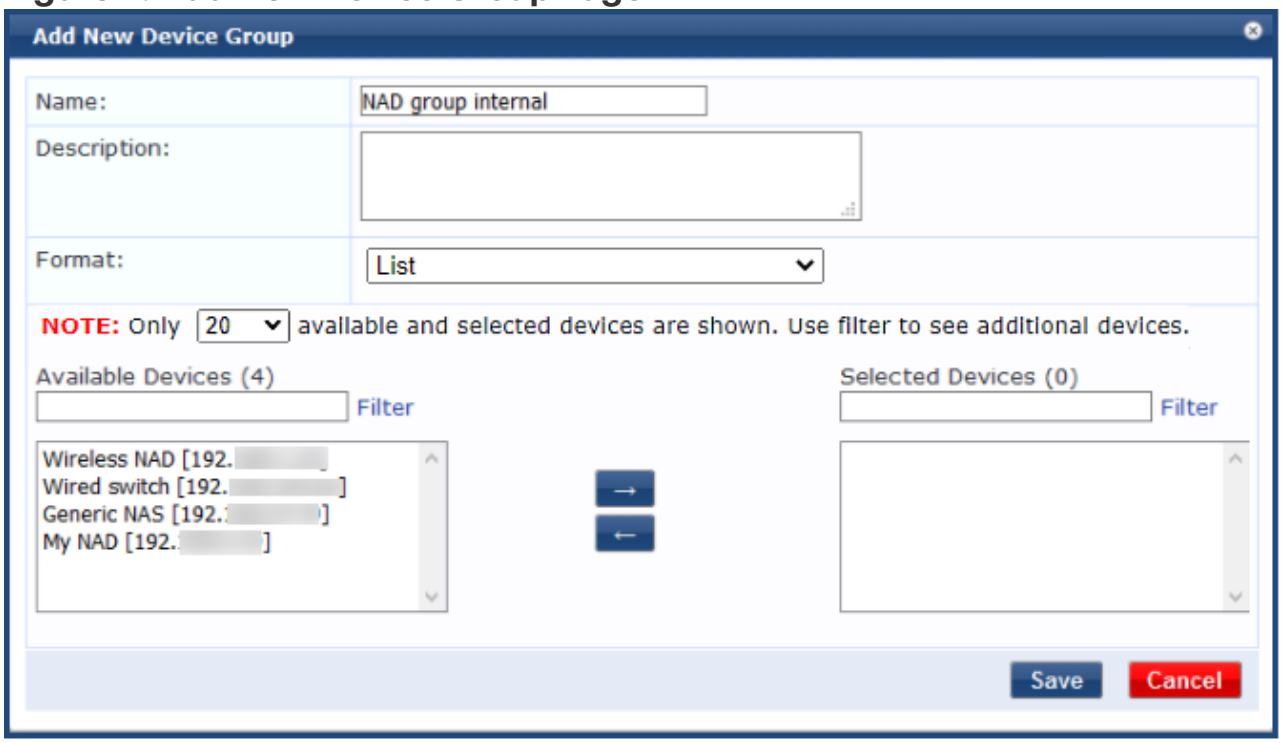
1. Navigate to **Configuration > Network > Device Groups** . The **Network Device Groups** page opens:

Figure 1. Device Groups Page



2. Click **Add**.
3. Complete the fields in the **Add New Device Group** page as described in the following figure:

Figure 2. Add New Device Group Page



4. Specify the **Add New Device Group** page parameters as described in the following table:

Table 1. Add New Device Group Page

Parameter	Action/Description
Name	Enter the name of the device group.
Description	Optionally, enter the description that p rovides additional information about the device group.
Format	Click the **Format** menu and select a form at: *Subnet*, *Regular Expression*, or *List*. • **Subnet**: Enter a subnet consisting of I Pv4 network address and the network suffix (CIDR notation). For example, *19 2.168.5.0/24*. This parameter does not support IPv6 subnets. • **Regular Expression**: Specify a regular expression that represents all IPv4 or I

Parameter	Action/Description
	Pv6 addresses matching that expressi on. For example, *^192(.[0 - 9]*){3}$*. • **List**: You can use the **List** format to c reate a group of IPv4 or IPv6 devices. If you select this option, the **Add New Device Group** dialog displays the foll owing additional fields: ◦ **Available and selected devices sh own**: this drop - down list to spec ify the maximum number of devic es to display in the **Available Devi ces** and **Selected Devices** lists. O ptions include 20, 50, 100, 500, an d 1000. ◦ **Filter**: To filter the list based on the text in the associated text box, clic k **Filter**. The filter fields are case - i nsensitive. ◦ **Available Devices** and S**elected D evices**: Use the widgets to move d evice identifiers between **Availabl e** and **Selected**.

5. Click **Save**.

Modifying a Device Group

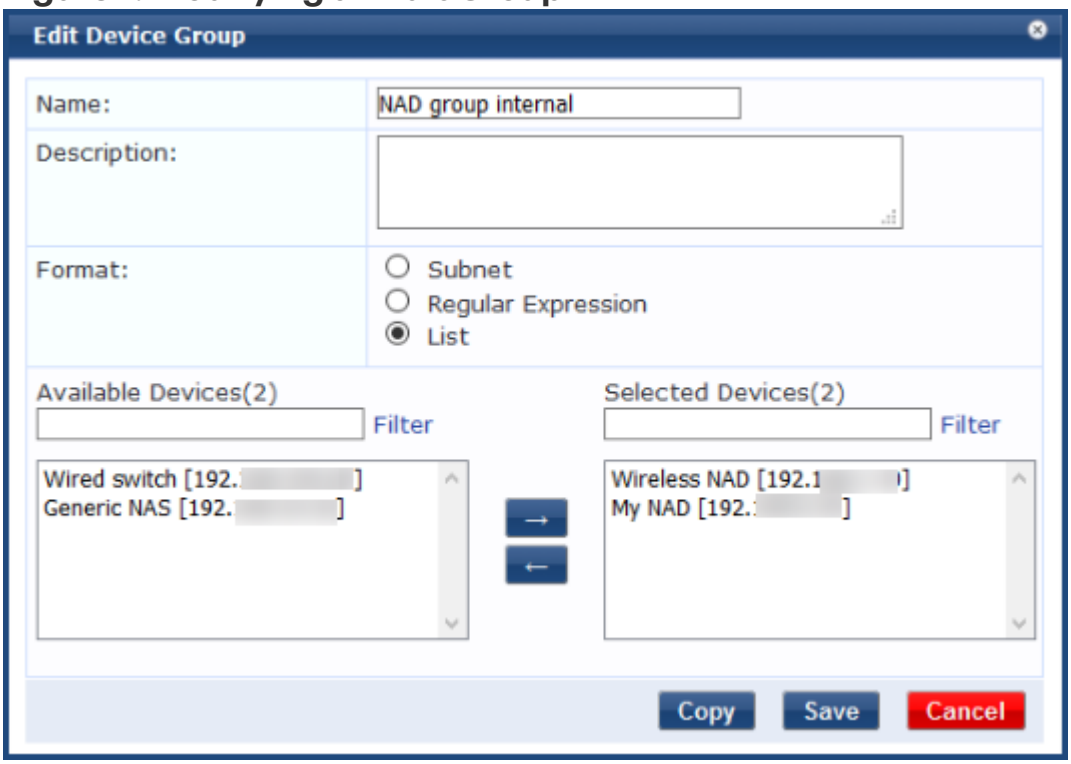
About this task

To modify a device group:

Procedure

1. From the **Network Device Groups** page, select the device group you wish to modify. The **Edit Device Group** dialog opens.

Figure 1. Modifying an Edit Group



2. Modify the Device Group as necessary, then click **Save**.

Adding and Modifying Proxy Targets

A *proxy server* is a dedicated computer or a software system running on a computer that acts as an intermediary between an endpoint device, such as a computer, and another server from which a user or client is requesting a service. The proxy server can exist in the same machine as a firewall server or it can be on a separate server, which forwards requests through the firewall. In Policy Manager, a *proxy target* represents a RadSec-enabled server or RADIUS server (Policy Manager or a third party) that is the target of a proxied RADIUS request. For example, when a branch office employee visits a main office and logs into the network, Policy Manager assigns the request to the first service in priority order that contains a service rule for RADIUS proxy services and appends the domain to the username. Proxy targets are configured at a global level. They can be used in configuring RADIUS or RadSec proxy services. For more information, refer to Configuring Policy Manager Services .

Adding a Proxy Target

About this task

To add a proxy target:

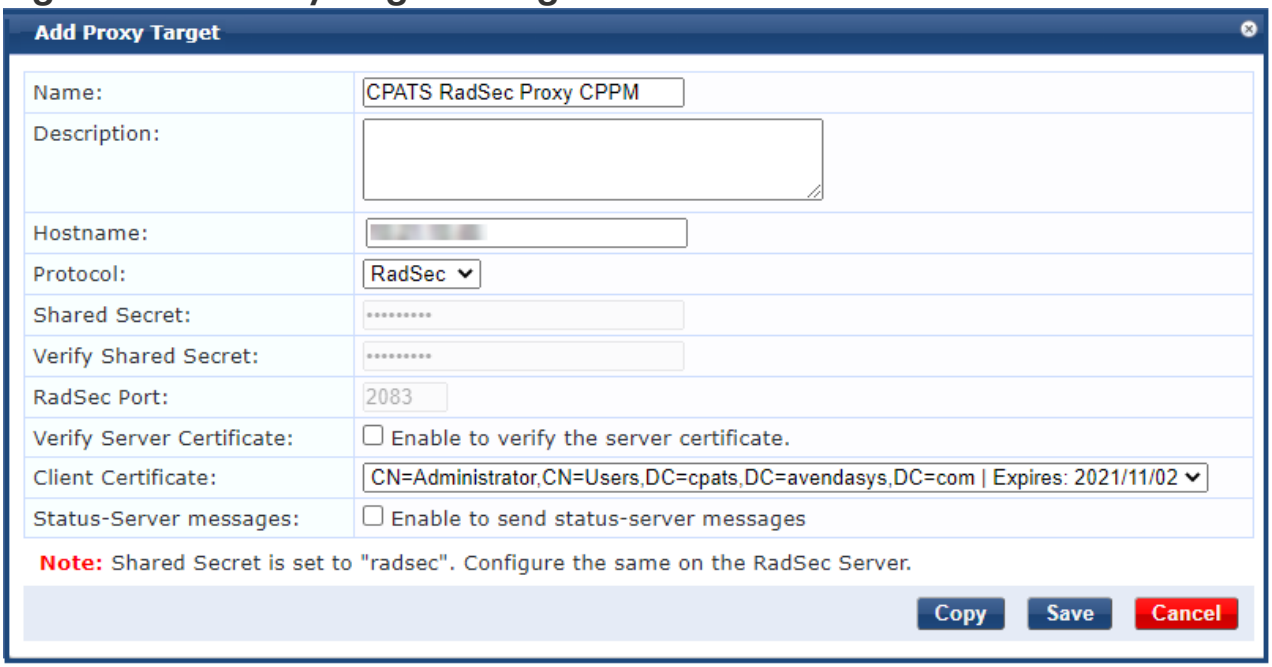
Procedure

1. Navigate to **Configuration > Network > Proxy Targets**.

   The **Proxy Targets** page opens. Policy Manager lists all configured proxy servers in the **Proxy Targets** page.

2. Click **Add**. The **Add Proxy Target** dialog opens.

Figure 1. Add Proxy Target Dialog



3. Specify the **Add Proxy Target** parameters as described in the following table, then click **Save**. Note that the available parameters in this dialog depend upon whether you are creating a **RADIUS** or **RadSec** proxy target.

Table 1. Add Proxy Target Parameters

Parameter	Action/Description
Name	Enter the name of the proxy target.
Description	Enter the description that provides additi onal information about the proxy target.
Hostname/Shared Secret	Specify the proxy hostname.
Protocol	Select either **RADIUS** or **RadSec**

Parameter	Action/Description
Shared Secret Verify Shared Secret	Enter the shared secret, then verify it. **NOTE** Use the same shared secre t that you entered on the p roxy target (refer to your RA DIUS server configuration).
RADIUS Authentication Port	If you selected the RADIUS protocol, ent er the UDP port to send the RADIUS requ est. The default value for this port is **1812**.
RADIUS Accounting Port	If you selected the RADIUS protocol, ente r the UDP port to send the RADIUS accou nting request. The default value for this p ort is **1813**.
RadSec Port	If you selected the RadSec protocol, ente r the TCP port to send the request. The d efault value for this port is **2083**.
Verify Server Certificate	When enabled, the RadSec proxy perfor ms checks to validate the AAA server cer tificate, including checks for valid start a nd end dates, and checks to verify the ce rtificate chain of validity back to the issui ng CA.
Client Certificate	If you selected the RadSec protocol, click this drop - down list and select a client c ertificate for RADIUS authentication from the list of client certificates uploaded via the **Administration > Certificates > Certi ficate** Store page, then defined as a Rad Sec certificate via the **Administration > C ertificates > Trust** List page**.** This drop - d own list displays both the subject name a nd certificate expiry date, allowing you t o distingush between certificates with th

Parameter	Action/Description
	e same name but with different expiratio n dates.
Status - Server Messages	This setting is available when you select the RadSec protocol option. When enabl ed, the proxy will send regular status - se rver messages to the server during idle p eriods to verify that the proxy is still alive and active. This option is enabled by defa ult.

Configuring the Ingress Event Source

About this task

An event source is a device that sends Syslog events to Policy Manager.

> **NOTE**
> Any events sent that are not from configured event sources are ignored.

To configure an Event Source:

Procedure

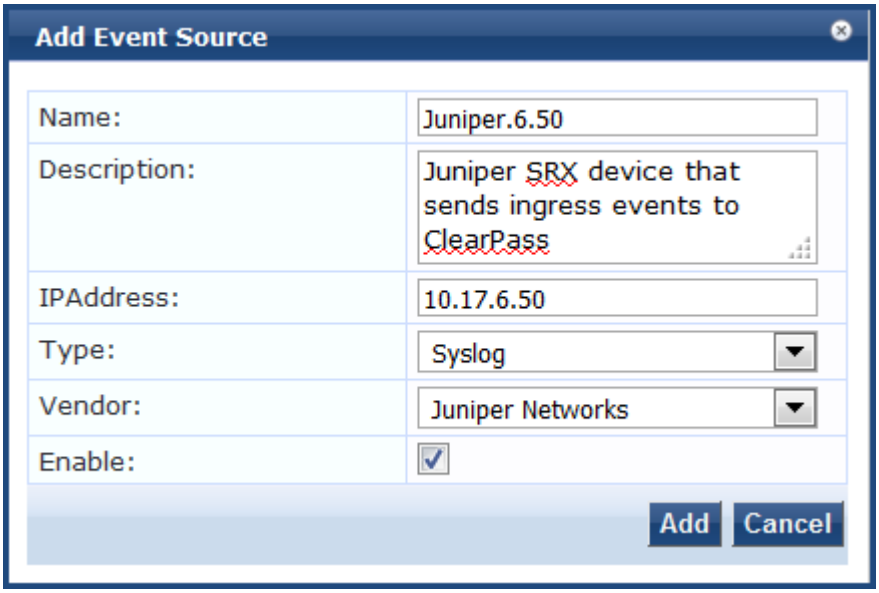
1. Navigate to **Configuration > Network > Event Sources**.

   The **Event Sources** page opens.

2. To add the Event Source for the desired vendor, click **Add**.

   The **Add Event Source** dialog opens.

   Figure 1. Adding an Event Source



3. Populate the **Add Event Source** parameters as described in Table 1.

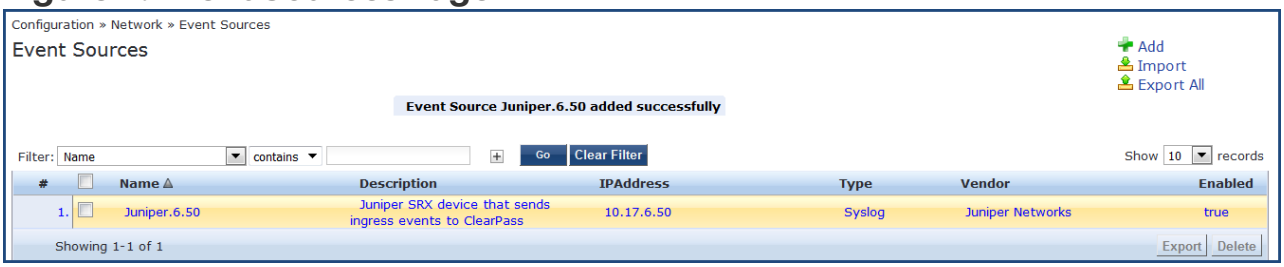
Table 1. Configuring the Add Event Source Parameters

Parameter	Action/Description
Name	Enter the name or the IP address of the device t d Syslog events to Policy Manager.
Description	Optionally, enter a description of this Event Sou
IP Address	Enter the IP address of the device that will send nts to Policy Manager.
Type	From the drop - down, select **Syslog**.
Vendor	From the drop - down, select the event source **V** • Aruba Introspect • Check Point • Infoblox • Juniper Networks • Palo Alto Networks
Enable	Select this check box to enable the device as an rce.

4. When finished, click **Add**.

   The **Event Sources** page now displays the new Event Sources (see Figure 2).

Figure 2. Event Sources Page



Note that the IP address displayed in Figure 2 is the IP address and host name of the Juniper SRX gateway that sends Syslog events to Policy Manager.

Configuring Network and Subnet Scans

This chapter provides the following information:

- Configuring SNMP, SSH, and WMI Credentials
- Configuring Network Scans and Subnet Scans

Subtopics

Configuring SNMP, SSH, and WMI Credentials
Configuring Network Scans and Subnet Scans

Configuring SNMP, SSH, and WMI Credentials

Subtopics

About Network Scan
WMI Credentials Configuration
SNMP Credentials Configuration
SSH Credentials Configuration

About Network Scan

Network Scan uses a configured *seed network device* (typically a switch, router, or controller) to discover endpoints and network devices. You can schedule network scans and subnet scans (see).

The following information is read from the seed device:

- SNMP information

  An SNMP description is necessary for discovering and profiling the network devices. For more information, see SNMP Credentials Configuration.

- SSH credentials

  For Linux server or network device discovery, specify SSH configuration credentials. For more information, see SSH Credentials Configuration.

- WMI credentials

  For Windows device discovery, specify WMI (Windows Management Instrumentation) credentials. For more information, see WMI Credentials Configuration.

- Connected endpoints

  Information about endpoints connected to the network device (typically MAC addresses of endpoints connected to switch ports). These are added as discovered endpoints. For more information, see Configuring SNMP, SSH, and WMI Credentials.

- ARP table

  Policy Manager supports Address Resolution Protocol (ARP) probes for network discovery scans. When the **Read ARP Table Info** option is enabled, the scan also probes all available ARP entries. The ARP table provides information about MAC **address > IP**associations for endpoints that were recently seen by this device. These endpoints are probed further in an attempt to profile those devices. For more information, see Adding a Network Device.

- Neighbor network devices

  Other network devices connected to the seed device as determined by neighbor discovery protocols such as Cisco Discovery Protocol (CDP) and Link Layer Discovery Protocol (LLDP) (if enabled in your network).

  Each of the discovered neighbor network devices are further queried as seed devices; this is repeated for multiple levels in your network up to a specified scan depth. For more information, see Monitoring Discovered Devices).

- Services and processes running on an Endpoint

During the subnet scan, Network Discovery and the OnConnect domain-joined Windows client will be queried to retrieve all the services and processes running on the endpoint. This information will be displayed in the Policy Manager Insight Endpoint reports.

Subtopics

Network Scan High-Level Tasks

Network Scan High-Level Tasks

About this task

Configuring Network Scan consists of these major tasks:

Procedure

1. Add the Domain/WMI, SNMP, or SSH configurations needed to query all the devices in the target network.

   You must configure SNMP, SSH, and WMI credentials for the devices that you want to discover as part of the network scan. These credentials are used during a network scan or a subnet scan to profile Linux servers and machines (SSH credentials), Windows servers and machines (WMI credentials), and network devices (SNMP).

2. Configure a network scan or subnet scan (see).
3. Import the discovered network devices into Policy Manager (see Importing Network Devices).
4. Review the set of discovered devices and view the connected endpoints and neighbors (see Viewing Details on a Discovered Device).

WMI Credentials Configuration

For Windows device discovery, specify **WMI** (Windows Management Instrumentation) configuration credentials. WMI configuration is necessary to discover Windows systems and device fingerprint details.

WMI a key part of the Windows operating system. It's used to gather system statistics, monitor system health, and manage system components. To work properly, WMI relies on the WMI service. This service must be running and properly configured for your environment.

For WMI, the login format for **username** is usually **domain\username**. Whatever domain you provide, it will be prepended to the username before logging into that machine.

Suppose you have provided an IP subnet address:

- Policy Manager first checks to see if WMI is configured for that subnet/IP address.
- If WMI is configured, Policy Manager checks to see if **port 135** is open.
- If **port 135** is open, Policy Manager attempts the WMI login with those credentials.

  If you provide just one IP address, the WMI login is performed for that particular IP address only.

  To configure WMI credentials for a network scan or subnet scan:

  1. If you have not already done so, you must create a **Domain/WMI External Account**. For details on that procedure, refer to Adding a Domain/WMI External Account.
  2. From the **Administration > External Accounts** page, click the **Profiler Subnet Mappings** link.

     The **Profiler Subnet Mappings** page is displayed.
  3. Click the **Add** link.

     The **Profiler Subnet Mappings** configuration dialog opens:

Figure 1. Configuring WMI Subnet Mapping

SNMP Credentials Configuration

About this task

For querying hosts discovered by a subnet scan, specify **SNMP** configuration. An SNMP-based scan sends an SNMP request to retrieve network device information.

To add the SNMP configuration for a network scan or subnet scan:

Procedure

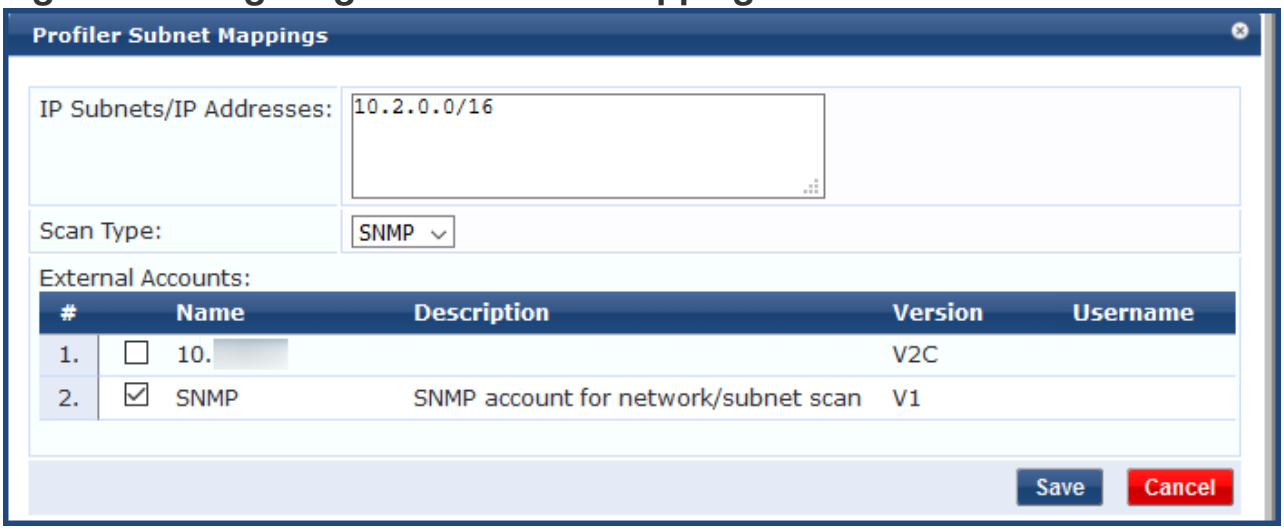
1. If you have not already done so, you must create an **SNMP External Account**. For details on that procedure, refer to Adding an SNMP External Account.
2. From the **Administration > External Accounts** page, click the **Profiler Subnet Mappings** link.

   The **Profiler Subnet Mappings** page opens.
3. Click the **Add** link.

   The **Profiler Subnet Mappings** page is displayed.
4. Click the **Add** link.

   The **Profiler Subnet Mappings** configuration dialog opens.

Figure 1. Configuring SNMP Subnet Mapping



5. Specify the **Subnet Mappings** parameters as described in the following table:

Table 1. SNMP Subnet Mapping Parameters

Parameter	Action/Description
IP Subnets/ IP Addresses	Enter either one or more IP subnets or on e or more IP addresses.

Parameter	Action/Description
	For multiple entries, separate multiple IP addresses or subnets with commas. When you configure the network scan, Policy Manager will use the SNMP config uration to fetch the network device infor mation for discovered devices.
Scan Type	From the **Scan Type** drop - down, select **SNMP**:
External Accounts	Select the check boxes for the correspon ding SNMP accounts.

6. Click **Save**.

   You return to the **Profiler Subnet Mappings** dialog, where the new SNMP configuration has been added and the following message is displayed:

   SNMP configuration added successfully

7. Click **Add** to add another subnet mapping; or click **Close** to exit.

SSH Credentials Configuration

About this task

For Linux servers or network device discovery, specify **SSH** (Secure Shell) configuration credentials. When SSH is found for an IP address or subnet, Network Scan looks for any Linux server or machine associated with that IP address or subnet.

You can configure multiple user names and passwords. These credentials are organized in the order in which they were created. You can configure multiple user names and passwords. These credentials are organized in the order in which they were created.

To add the SSH configuration for a network scan or subnet scan:

> **NOTE**
> The SSH configuration can be for a single IP address or a subnet. These credentials are used when an SSH scan is initiated.

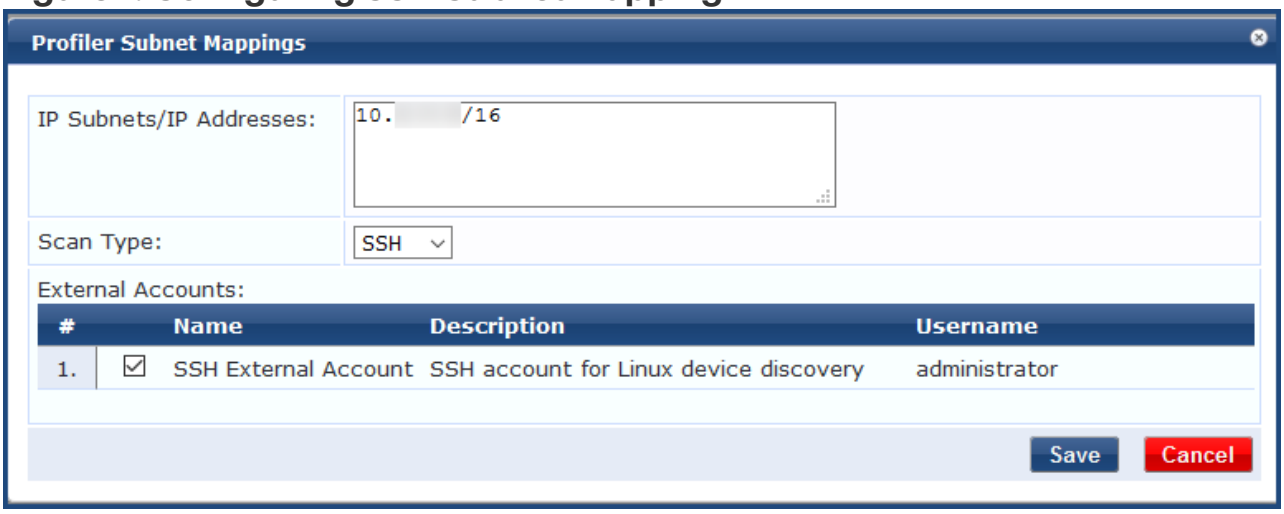
Procedure

1. If you have not already done so, you must create an **SSH External Account**. For details on that procedure, refer to Adding an SSH External Account.
2. From the **Administration > External Accounts** page, click the **Profiler Subnet Mappings** link.

   The **Profiler Subnet Mappings** page is displayed.
3. Click the **Add** link.

   The **Profiler Subnet Mappings** configuration dialog opens:

Figure 1. Configuring SSH Subnet Mapping



4. Specify the SSH Subnet Mapping parameters as described in the following table:

Table 1. SSH Subnet Mapping Parameters

Parameter	Action/Description
IP Subnets/ IP Addresses	Enter either one or more subnets or one or more IP addresses. For multiple entries, separate multiple IP addresses or subnets with commas. When you configure the network scan, Policy Manager will use the SSH configur ation to fetch the network device informa tion for discovered devices.

Parameter	Action/Description
Scan Type	From the **Scan Type** drop - down, select **SSH**:
External Accounts	Select the check boxes for the correspon ding SSH accounts.

5. Click **Save**.

   You return to the **Profiler Subnet Mappings** dialog, where the new SSH configuration has been added.

6. Click **Add** to add another subnet mapping; or click **Close** to exit.

Configuring Network Scans and Subnet Scans

About this task

Configuring Network Scan operations consist of these major tasks:

Procedure

1. Create the external accounts of Domain/WMI, SNMP, or SSH type for network hosts and devices. For details, see Adding External Accounts.
2. Add the scan configurations (Domain/WMI, SNMP, or SSH) needed to query all the devices in the target network. For details, see:
   - Configuring the Networks/Subnets for WMI Scan Type
   - Configuring the Networks/Subnets for SNMP Scan Type
   - Configuring the Networks/Subnets for SSH Scan Type

   The Domain/WMI, SNMP, and SSH credentials are used during a network scan or a subnet scan to profile Windows servers and machines (WMI credentials), Linux servers and machines (SSH credentials), and network devices (SNMP).
3. After running a network scan, import the discovered network devices into Policy Manager (see Monitoring Discovered Devices).
4. Review the set of discovered devices and view the connected endpoints and neighbors (see Monitoring Discovered Devices).

Subtopics

Configuring a Network Scan
Configuring a Subnet Scan
Viewing Details About a Subnet Scan
Configuring Nmap-Based Endpoint Port Scans

Configuring a Network Scan

About this task

Seed devices are the initial IP addresses provided by the network administrator to start the network scan. When you initiate a network scan and specify the seed devices, network discovery uses SNMP to:

- Find any other devices connected to the seed devices.
- Profile the connected devices.
- Policy Manager uses that information to detect more devices in the network. The network discovery scan will proceed to the network depth specified by the **Scan Depth** parameter (described in Table 1 below).
- You can go to those devices and see their neighbor devices.

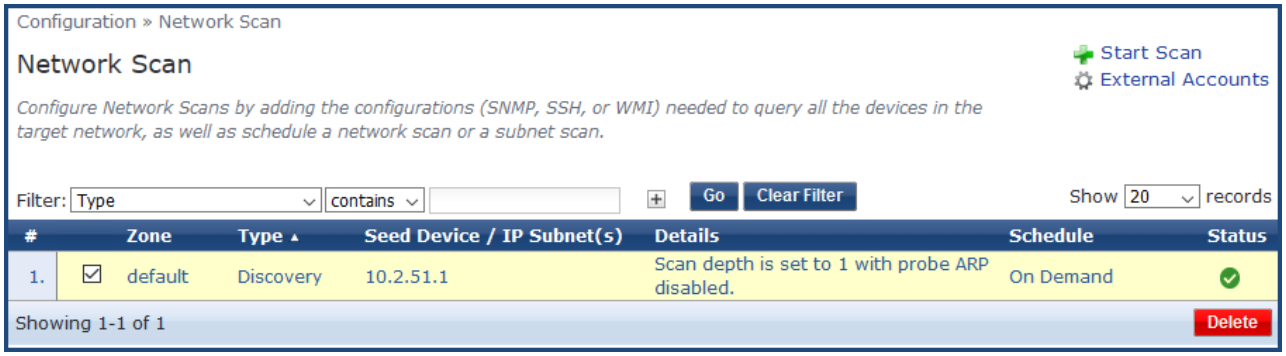
> **CAUTION**
>
> Running a network scan on seed devices is a time- and resource-consuming operation. Depending on the number of devices associated with the seed device, a complete scan can take more than an hour.
>
> It is recommended that the network scan should be done outside of normal business hours or performed on a Policy Manager node that is not servicing core authentications.

To configure a network scan:

Procedure

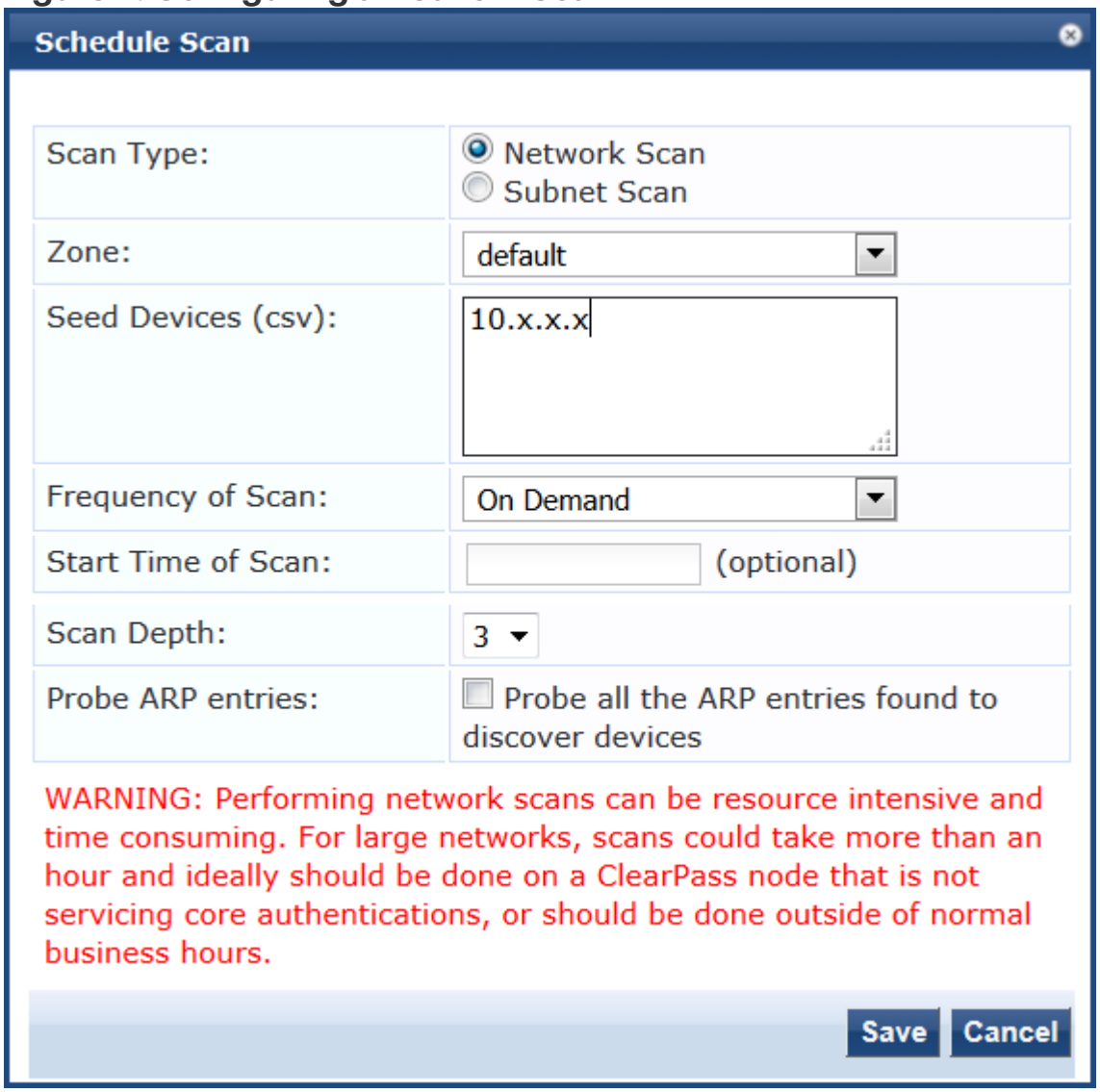
1. Navigate to **Configuration > Network Scan**. The **Network Scan** page opens.

Figure 1. Network Scan Page



2. Click the **Start Scan** link. The **Schedule Scan** dialog opens.

Figure 2. Configuring a Network Scan



3. Specify the **Schedule Scan** parameters as described in the following table:

Table 1. Configuring Network Scan Parameters

Parameter	Action/Description
Scan Type	Select **Network Scan**. **NOTE** For scan operations to be op erable, the **Primary Server i n Zone** parameter must be c onfigured. The primary Clea rPass server in a zone distri butes the load among other Policy Manager nodes in the zone. For details, see Master Server in Zone.
Zone	Specify the ClearPass Zone (for more in formation, see Managing Policy Manager Zones). If Policy Manager Zones have not yet been set up, accept the **default** zone. **NOTE** The primary server in the zo ne forwards the scan reques t to the other Policy Manage r nodes in the zone, dependi ng on the seed device. Each scan configuration added is distributed by the **primary** s erver to a different node in t he zone. If one scan configu ration has multiple seed devi ces, all the seed devices are forwarded to one node.
Seed Devices	Enter the IP addresses of one or more se ed devices from which the network scan should proceed. Separate multiple devic e IP addresses with commas.

Parameter	Action/Description
Frequency of Scan	Specify the frequency of the network sca n: • On Demand ◦ This option runs the network scan once, either immediately if no sta rt time is specified, or at the time s pecified by the **Start Time of Scan** parameter. • Hourly • Daily
Scan Depth	Specify the **Scan Depth** by selecting the desired number from **1** to **5**. The Scan De pth numbers indicate the levels of the ne twork you want to scan. The default is **Sc an Depth 3**. The seed devices are, by default, at **Scan Depth 1**. Starting from the seed device, t he next device level is **Scan Depth 2**, and so on, until the scan reaches the scan de pth specified here.
Probe ARP entries	The ARP (Address Resolution Protocol) ta ble provides information about MAC add ress and IP address associations for endp oints that were discovered by this Policy Manager server. The ARP entries for the s pecified network(s) are read irrespective of whether this check box is enabled. When this option is enabled, Policy Mana ger uses the ARP entries to discover net work access devices (NADs), then procee ds to perform a network scan to the confi gured scan depth.

4. Click **Save**. You return to the *Network Scan* page. The status of the network scan operation shows initially as *Running*, and finally, *Completed*. The green status indicator indicates that the scan operation was successful.
5. To restart a completed scan, click the green **Status** button.

Configuring a Subnet Scan

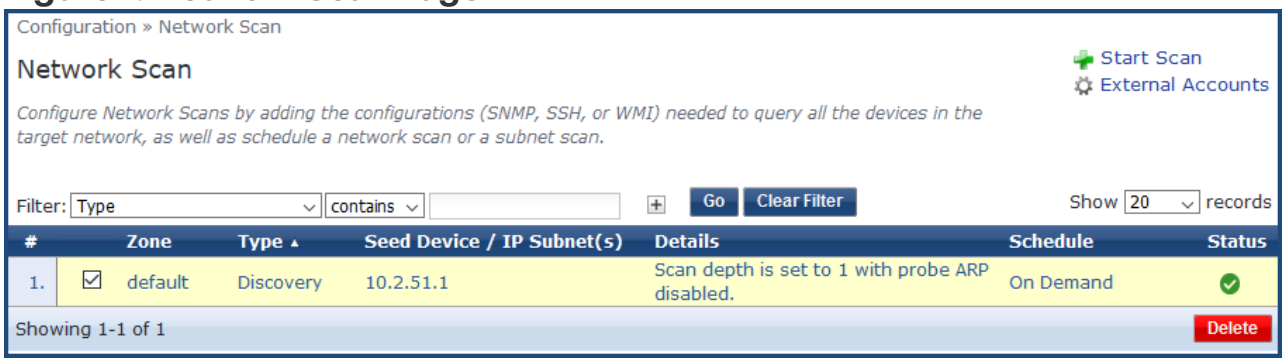
About this task

To configure a subnet scan:

Procedure

1. Navigate to **Configuration > Network Scan**. The **Network Scan** page opens.

Figure 1. Network Scan Page

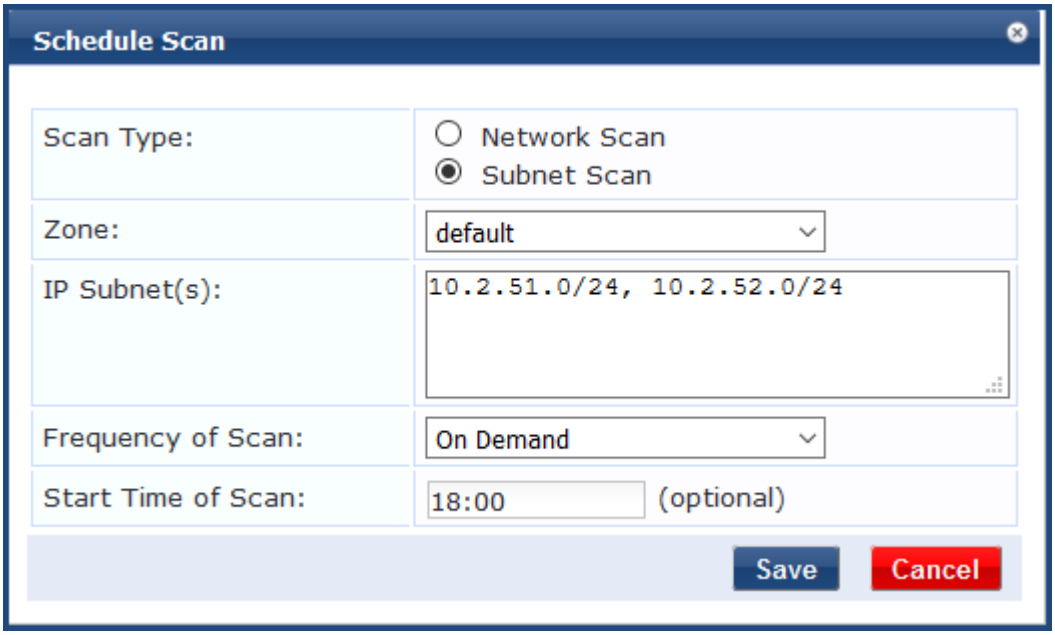


2. Click the **Start Scan** link.

   The **Schedule Scan** dialog opens.

> **NOTE**
>
> When running an IPv6 subnet scan, an IPv6 subnet might have a very large IP space. If a scan is triggered for a large IPv6 subnet (for example, /64 or more), it will take a very long time to complete. Aruba recommends configuring smaller subnets so that the number of the host space is small and the scan will finish within a reasonable amount of time.

Figure 2. Configuring a Subnet Scan



3. Specify the **Schedule Scan** parameters as described in the following table:

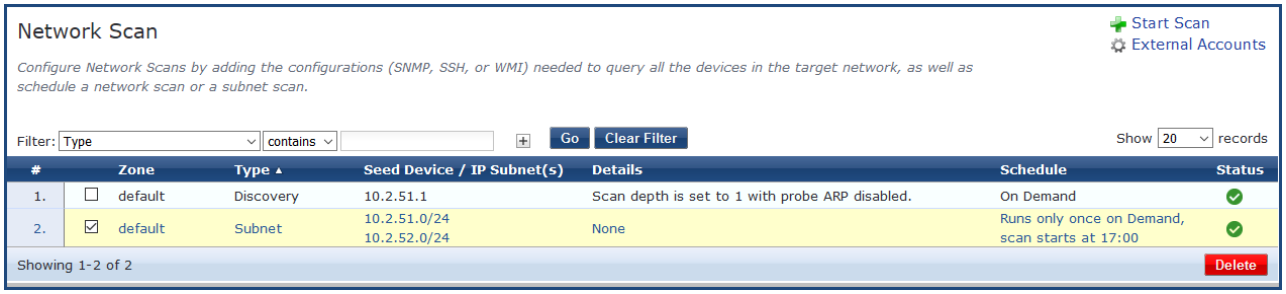
Table 1. Configuring Subnet Scan Parameters

Parameter	Action/Description
Scan Type	Select **Subnet Scan**. • You can schedule multiple subnet scans per zone. • To monitor subnet scan progress, navigate to **Monitoring > Profiler and Network Scan > Network Scan Results**. **NOTE** For scan operations to be operable, the **Primary Server in Zone** parameter must be configured. The primary ClearPass server in a zone distributes the load among other Policy Manager nodes in the zone. For details, see Master Server in Zone.

Parameter	Action/Description
Zone	Specify the Policy Manager Zone. If a Zon e is not configured, accept the default.
IP Subnet(s)	If you selected the **Subnet Scan** option for the **Scan Type**, enter the IP addresses of one or more IP subnets from which the subnet scan should proceed. Separate m ultiple IP subnets with commas. **NOTE** An IPv6 subnet scan only su pport Nmap (Network Mappi ng), and the device classific ation for an IPv4 subnet sca n and IPv6 subnet can vary based on what Nmap can fet ch from the device.
Frequency of Scan	Specify the frequency of the scan: • On Demand ◦ This option runs the subnet scan o nce, either immediately if no start time is specified, or at the time sp ecified by the **Start Time of Scan** parameter. • Hourly • Daily
Start Time of Scan	Select the time you want the subnet scan to start.

4. Click **Save**. You return to the *Network Scan* page. The status of the subnet scan operation shows initially as *Running*, and finally, *Completed*. The green status indicator indicates that the scan operation was successful.

Figure 3. Seed Devices Successfully Scanned



Viewing Details About a Subnet Scan

Additional information is available for the configured subnet scans. To view details about a subnet scan, navigate to **Monitoring > Profiler and Network Scan > Network Scan Results**.

Configuring Nmap-Based Endpoint Port Scans

About this task

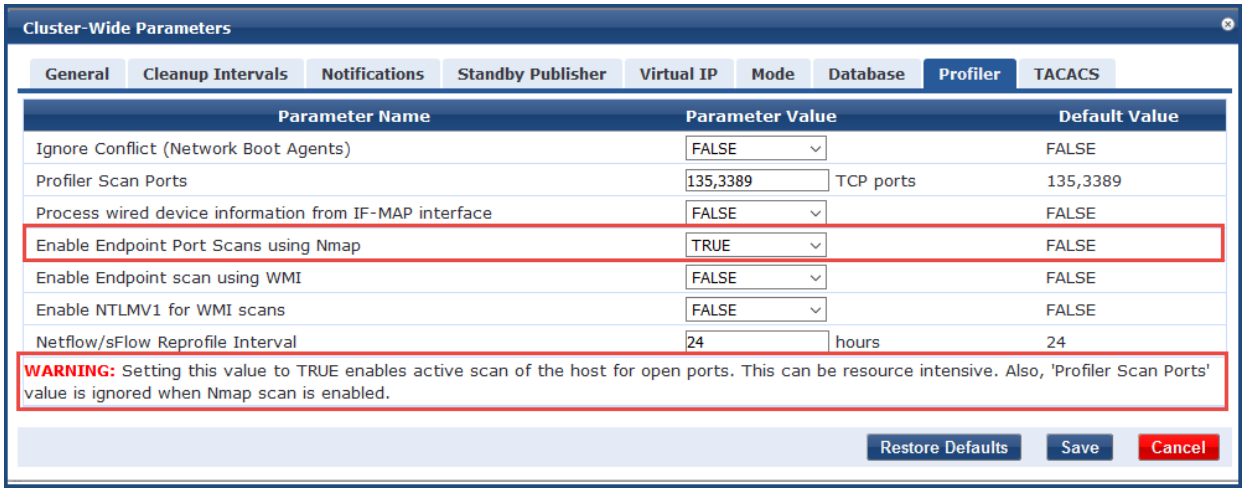
The network scan feature supports running an Nmap (Network Mapper)-based scan on a host to detect open ports and also to fingerprint the services running behind those ports. This information is used in the device profile.

To configure endpoint port scans using Nmap:

Procedure

1. Enable Nmap-based endpoint port scans.
   a. Navigate to **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters**. The **Cluster-Wide Parameters** page opens.
   b. Select the **Profiler** tab.

Figure 1. Cluster-Wide Parameters > Profiler Dialog



c. Set the **Enable Endpoint Port Scans using Nmap** parameter to **TRUE**, then click **Save**.

   For more information, see Profiler Parameters.

> **CAUTION**
>
> Setting this value to TRUE enables active scan of the host for open ports. This can be resource intensive. Also, the **Profiler Scan Ports** value is ignored when Nmap scan is enabled.

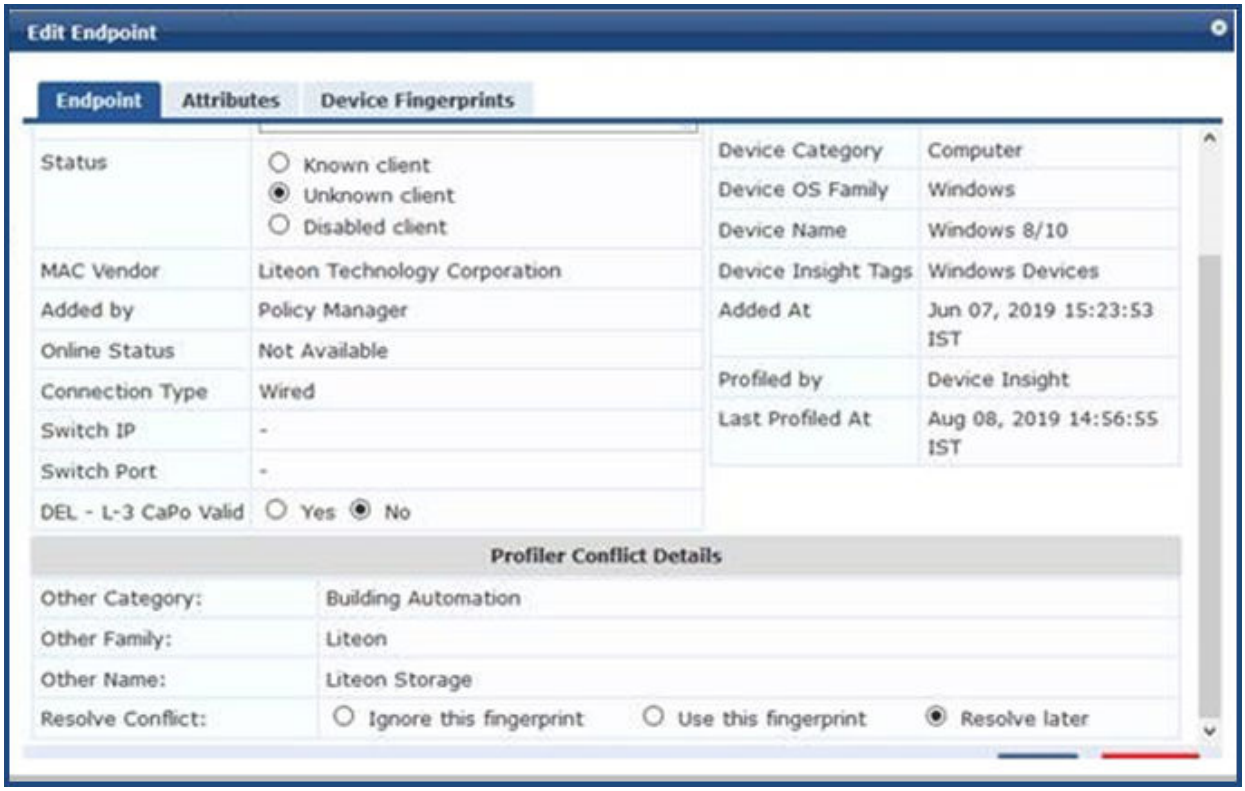
2. Schedule a network scan configuring a seed device with **Probe ARP** entries enabled (see).

3. When the network scan is completed, select an endpoint (see Adding and Modifying Endpoints).

   a. Navigate to **Configuration > Identity > Endpoints**.

   b. From the **Endpoints** page, click the endpoint of interest. The **Edit Endpoint** page opens.

Figure 2. Edit Endpoint Page



4. To view the list of host services and the list of open ports returned by the network scan for the selected host or endpoint, select the **Device Fingerprints** tab.

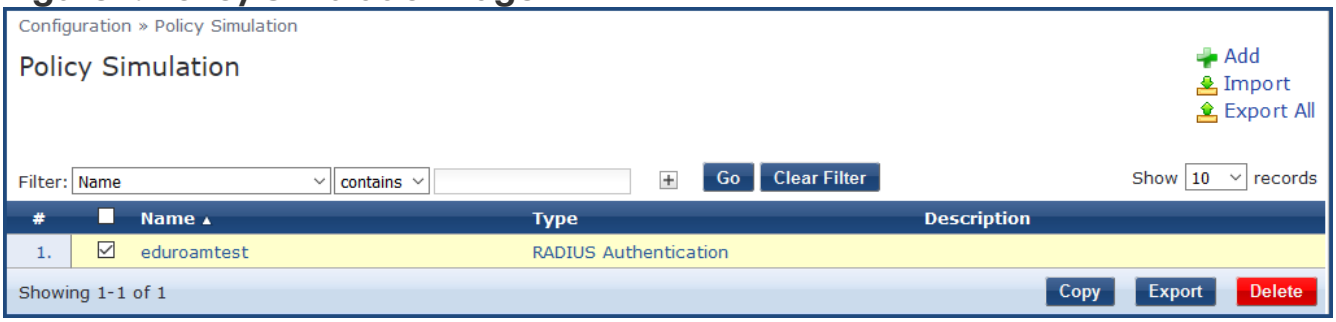
   The **Edit Endpoint > Fingerprints** page shows the fingerprint details that include Nmap scan data.

Configuring Policy Simulation

After creating policies, use the **Policy Simulation** feature to evaluate those policies before deployment. The **Policy Simulation** feature applies a set of request parameters as input against a given policy component and displays the outcome.

Policy simiations are created on the **Configuration > Policy Simulation** page.

Figure 1. Policy Simulation Page



- Active Directory Authentication Simulation
- Application Authentication Simulation
- Audit Simulation
- Chained Simulation
- Enforcement Policy Simulation
- RADIUS Authentication Simulation
- Role Mapping Simulation
- Service Categorization Simulation

Subtopics

Active Directory Authentication Simulation
Application Authentication Simulation
Audit Simulation
Chained Simulation
Enforcement Policy Simulation
RADIUS Authentication Simulation
Role Mapping Simulation
Service Categorization Simulation
Import and Export Simulations

Active Directory Authentication Simulation

- Adding an Active Directory Simulation

  verify that the ClearPass Policy Manager domain membership is valid.

> **NOTE**
>
> The **Attributes** tab is not available for this simulation type.

Subtopics

Adding an Active Directory Simulation
Viewing the Simulation Results

Adding an Active Directory Simulation

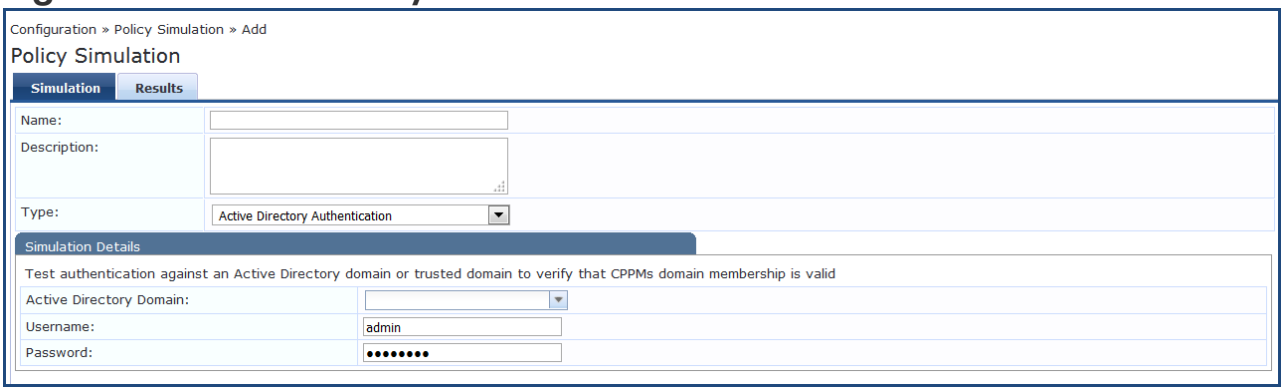
About this task

To add an Active Directory simulation for the authentication test:

Procedure

1. Navigate to the **Configuration > Policy Simulation > Add** page. The **Add Policy Simulation** dialog opens.

Figure 1. Active Directory Authentication > Simulation Tab



2. Specify the **Active Directory Authentication > Simulation** tab parameters as described in the following table:

Table 1. Active Directory Authentication Simulation Tab Parameters

Parameter	Action/Description
Name	Enter the **name** of the simulation
Description	Provide a description of this simulation.
Type	**Active Directory Authentication** is specified by default.
Simulation Details	
Active Directory Domain	Select the Active Directory domain to which the node is joined.
Username	Enter the username to log in to the domain.
Password	Enter the password to log in to the domain.

Viewing the Simulation Results

The **Results** tab for the **Active Directory Authentication** simulation displays a summary of the Authentication test and provides a status message. The following figure displays the **Active Directory Authentication > Results** tab:

Figure 1. Active Directory Authentication Results Tab

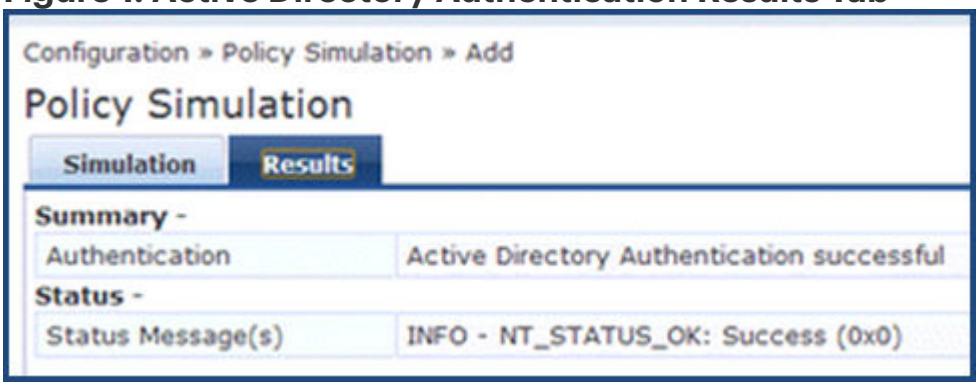


Table 1. Active Directory Authentication Results Tab Parameters

Parameter	Description
Summary	Displays the results of the Active Directory A on simulation.
Status	Displays the status message.

Application Authentication Simulation

This simulation tests authentication requests generated from Policy Manager Guest. The following figure displays the **Application Authentication** policy simulation settings available on the **Configuration > Policy Simulation > Add** page:

Subtopics

Simulation Tab
Attributes Tab
Results tab

Simulation Tab

Figure 1. Application Authentication - Simulation Tab

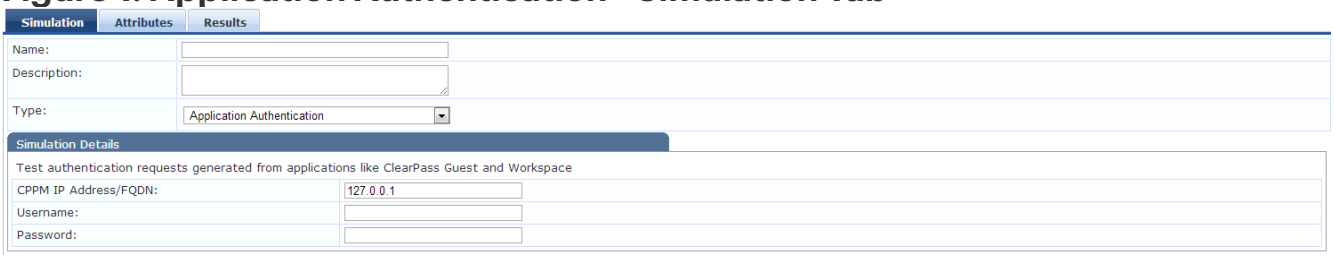


Table 1. Application Authentication Simulation Tab Parameters

Parameter	Description
CPPM IP Address/FQDN	Enter the IP Address or FQDN of the domai n(s) to which the node is joined.
Username	Enter the username.
Password	Enter the password.

Attributes Tab

Enter the attributes of the policy component to be tested. The following figure displays the **Application Authentication - Attributes** tab:

Figure 1. Application Authentication - Attributes Tab

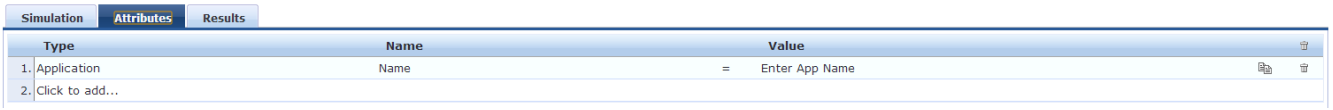


Table 1. Application Authentication - Attributes Tab Parameters

Attribute	Parameter
Type	Select Application or select Appli Application Namespace
Name	The options displayed for the Nar the Type Attribute that was select
Value	The options displayed for the Valu the Type Attribute and Name Attr d.

Results tab

The Results tab of the Application Authentication simulation displays the outcome of the **Authentication Result** and the **Application Authentication Output Attributes**. The following figure displays the **Application Authentication Results** tab:

Figure 1. Application Authentication Results Tab

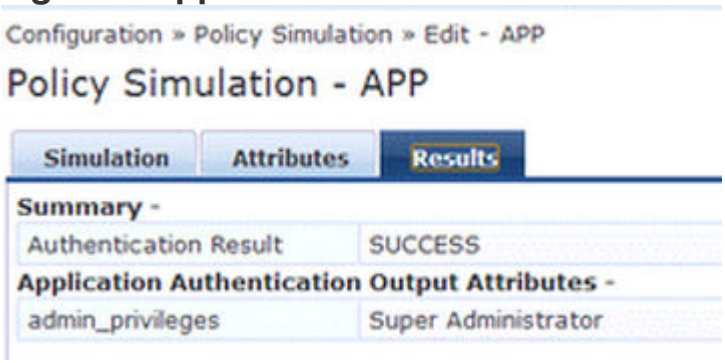


Table 1. Application Authentication Results Tab Parameters

Parameter	Description
Summary	Displays the results of the Active ion simulation.
Application Authentication Output Attributes	Displays the output attributes, su rator.

Audit Simulation

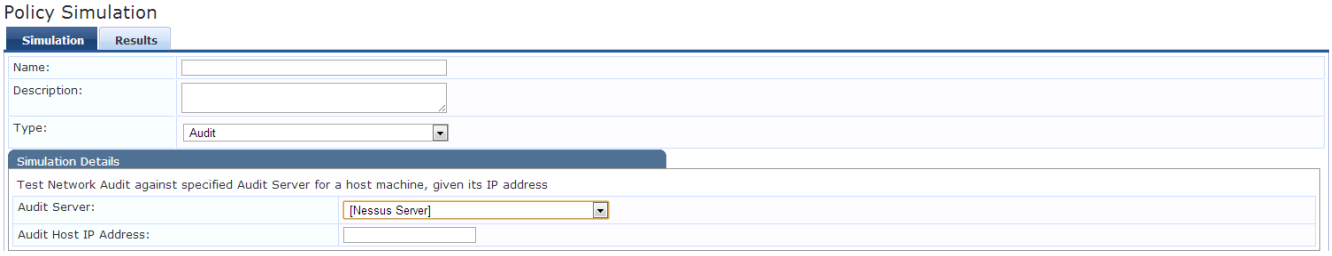
This simulation allows you to specify an audit against a Nessus Server or Nmap Server with its IP address.

NOTE
The **Attributes** tab is not available for this simulation type.

NOTE
Audit simulations can take more than 30 minutes. An *AuditinProgress* status message is displayed until the audit is completed.

The following figure displays the **Audit Simulation** tab:

Figure 1. Audit Simulation - Simulation Tab



The following table describes the **Audit Simulation - Simulation** tab parameters:

Table 1. Audit Simulation Tab Parameters

Parameter	Description
Audit Server	Select [Nessus Server] or [Nmap Audit].
Audit Host IP Address	Enter the host IP address of the audit host.

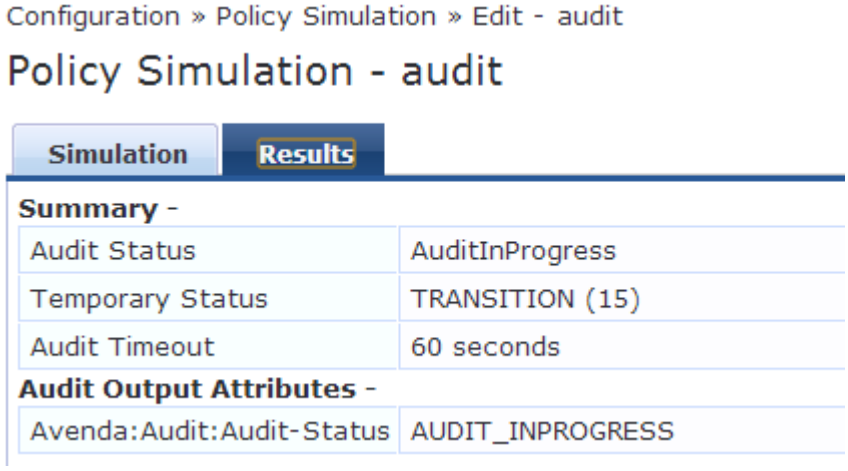
Subtopics

Results Tab

Results Tab

The following figure displays the **Audit Simulation - Results** tab:

Figure 1. Audit Simulation Results Tab



The following table describes the **Audit Simulation - Results** tab parameters:

Table 1. Audit Results Tab Parameters

Parameter	Description
Summary	Displays information about the Audit Status, Temporary Status, and Audit Timeout.
Audit Output Attributes	Displays the Audit - Status such as AUDIT_INPROGRESS.

Chained Simulation

Given the service name, authentication source, user name, and an optional date and time, the chained simulation combines the results of role mapping, posture validation and enforcement policy simulations and displays the corresponding results.

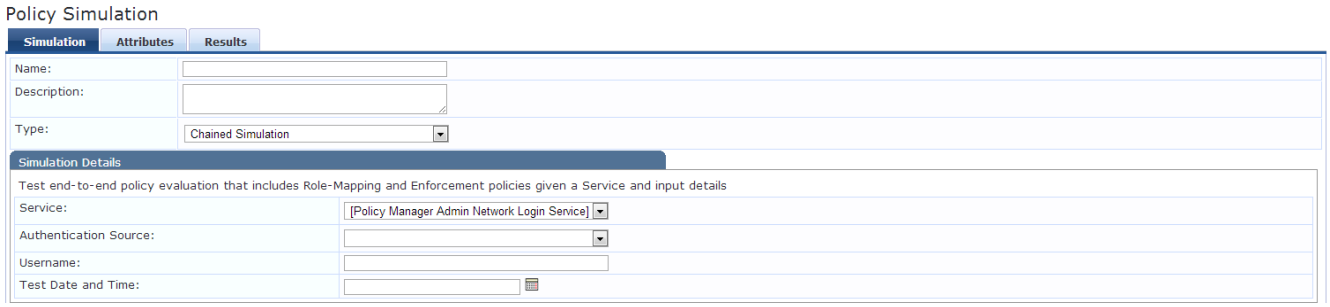
Subtopics

Simulation Tab
Attributes Tab
Results Tab

Simulation Tab

The following figure displays the **Chained Simulation Simulation** tab:

Figure 1. Chained Simulation Tab



The following table describes the **Chained Simulation - Results** tab parameters:

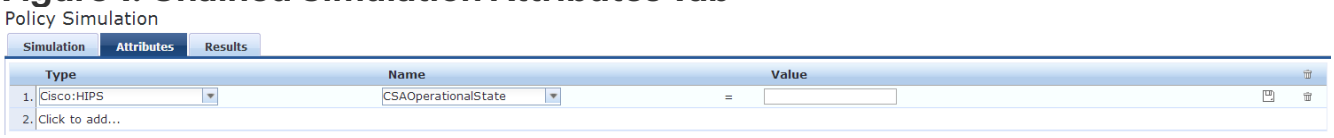
Table 1. Chained Simulation Tab Parameters

Parameters	Description
Service	Select from: • [Policy Manager Admin Network Login Service] • [AirGroup Authorization Service] • [Aruba Device Access Service] • [Guest Operator Logins] • Guest Access • Guest Access With MAC Caching
Authentication Source	Default Value = [Local User Repository] if you select: • [Policy Manager Admin Network Login Service] • [Aruba Device Access Service] Default Value = [Guest Device Repository] if you select: • [AirGroup Authorization Service] • Guest Access • Guest Access With MAC Caching Values = [Guest Device Repository] or [Local User Repository] if you select [Guest Operator Logins]
Username	Enter the username.
Test Date and Time	Click the calendar icon to select a start date and time for simulation test. For more information, see Date Namespaces

Attributes Tab

Enter the attributes of the policy component to be tested.

Figure 1. Chained Simulation Attributes Tab



The following table describes the **Chained Simulation Attributes - Results** tab parameters:

Table 1. Chained Simulation Attributes tab Parameters

Attribute	Parameter
Type	Select the type of attributes from the drop - down list.
Host	See Host Namespaces
Authentication	See Authentication Namespaces
Connection	See Connection Namespaces
Application	See Application Namespace
Certificate	See Certificate Namespaces
• Radius:IETF • Radius:Cisco • Radius:Microsoft • Radius:Avenda • Radius:Aruba • Trend:AV • Cisco: HIPS • Cisco:HOST • Cisco:PA • NAI:AV • Symantec:AV	See RADIUS Namespaces
Name	The options displayed for the **Name** attribute depend on the **Type** attribute that was selected.

Attribute	Parameter
Value	The options displayed for the **Value** attribute depend on the **Type** and **Name** attributes that were selected.

Results Tab

The following figure displays the **Chained Simulation - Results** tab:

Figure 1. Chained Simulation Results Tab

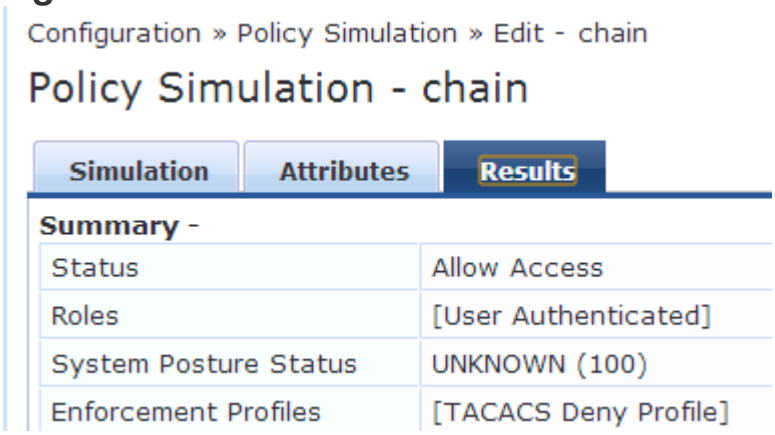


Table 1. Chained Simulation Results Tab Parameters

Parameter	Description
Summary	Provides the following information about the simulation: • Status • Roles • System Posture Status • Enforcement Profiles

Enforcement Policy Simulation

About this task

Given the service name (and the associated enforcement policy), a role or a set of roles, the system posture status, and an optional date and time, the enforcement policy simulation evaluates the rules in the enforcement policy and displays the resulting enforcement profiles and their contents. Authentication Source and User Name inputs are used to derive dynamic values in the enforcement profile that are retrieved from the authorization source. These inputs are optional. Dynamic roles are attributes that are enabled as a role retrieved from the authorization source. For an example of enabling attributes as a role, see Generic LDAP and Active Directory .

To configure an enforcement policy simulation:

Procedure

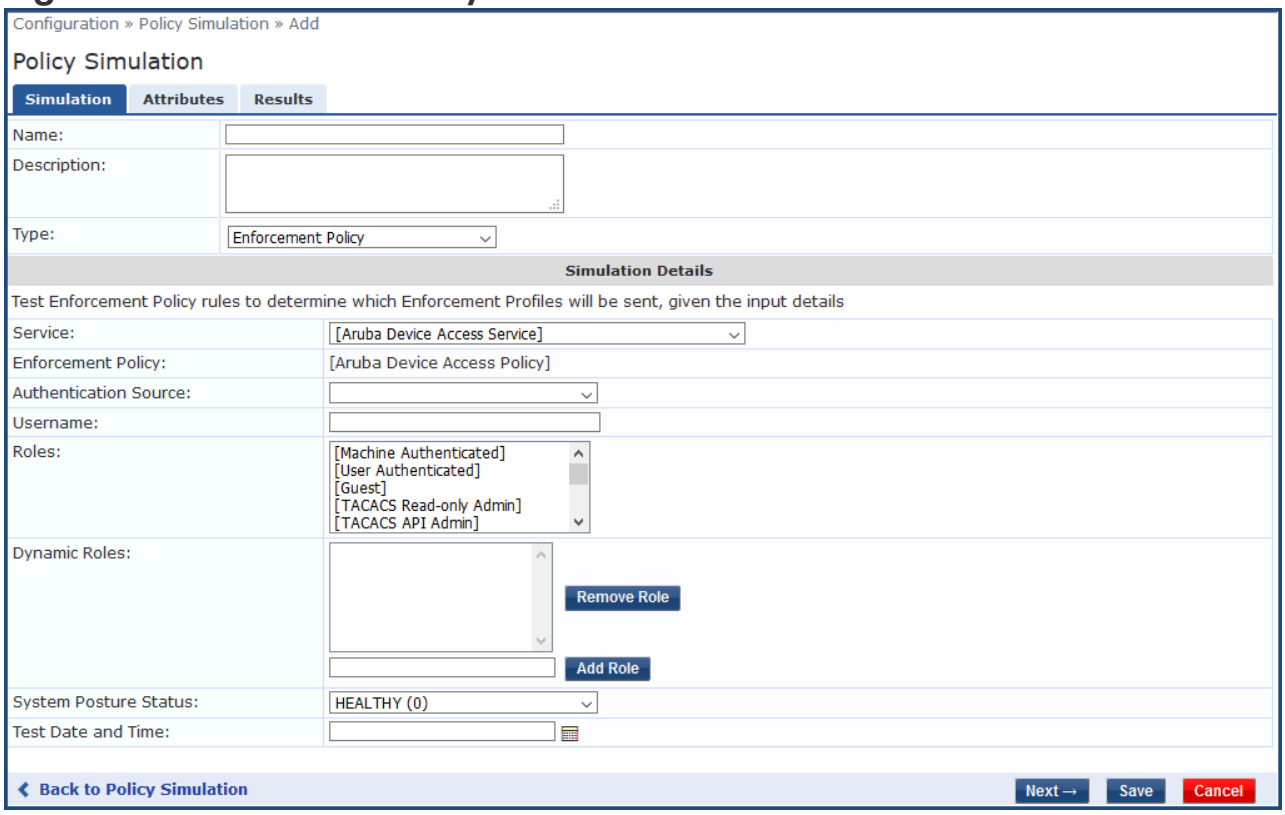
1. Navigate to **Configuration > Policy Simulation**.

   The **Policy Simulation** page opens.

2. Click the **Add** link.

   The **Add Policy Simulation** page opens to the **Simulation** tab.

Figure 1. Enforcement Policy Simulation Tab



3. Specify the **Policy Simulation** tab parameters as described in the following table:

Table 1. Enforcement Policy Simulation Tab Parameters

Parameter	Action/Description
Name	Enter the name of this policy simulation
Description	Enter a description of this policy simulati on
Type	From the **Type** drop - down, select **Enfo rcement Policy**.
Service	From the **Service** drop - down, select the appropriate service.
Enforcement Policy	Depending on which service you selecte d, the value for the **Enforcement Policy** will be filled automatically.
Authentication Source	Value = [Local User Repository] if you sele ct: • [Policy Manager Admin Network Login Service] • [Aruba Device Access Service] Value = [Guest Device Repository] if you s elect: • [AirGroup Authorization Service] • Guest Access • Guest Access With MAC Caching Values = [Local User Repository] or [Gues t Device Repository] if you select Guest Operator Logins
Username	Enter the user name.
Roles	From the **Roles** drop - down, select the a ppropriate role.
Dynamic Roles	To add a role:

Parameter	Action/Description
	a. Enter the name of a dynamic role in th e *Add Role* field. b. Click the **Add Role** button to populate the Dynamic Roles list. To remove a role, highlight the dynami c role of choice, then click **Remove Ro le**.
System Posture Status	Select from: • HEALTHY (0) • CHECKUP (10) • TRANSITION (15) • QUARANTINE (20) • INFECTED (30) • UNKNOWN (100)
Test Date and Time	Click the calendar icon to select the start date and time for the simulation test. For more information, see Date Namespaces

Subtopics

Attributes Tab
Results Tab

Attributes Tab

Enter the attributes of the policy component to be tested.

Table 1. Enforcement Policy Attributes Tab Parameters

Attribute	Description
Type	Select the type of attribute • Host (for more informati

Attribute	Description
	• Authentication (for more ation Namespaces). • Connection (for more in Namespaces). • Application (for more in amespace). • Radius:IETF • Radius:Cisco • Radius:Hewlett - Packar • Radius:Microsoft • Radius:Alcatel - Lucent • Radius:F5 • Radius:Avenda • Radius:Freeswitch • Radius:Aruba ◦ For more informatior es .
Name	The options displayed for th n the **Type** attribute that wa
Value	The options displayed for th n the **Type** and **Name** attrib

Results Tab

The following figure displays the **Policy Simulation > Results** tab:

Figure 1. Policy Simulation Results Tab

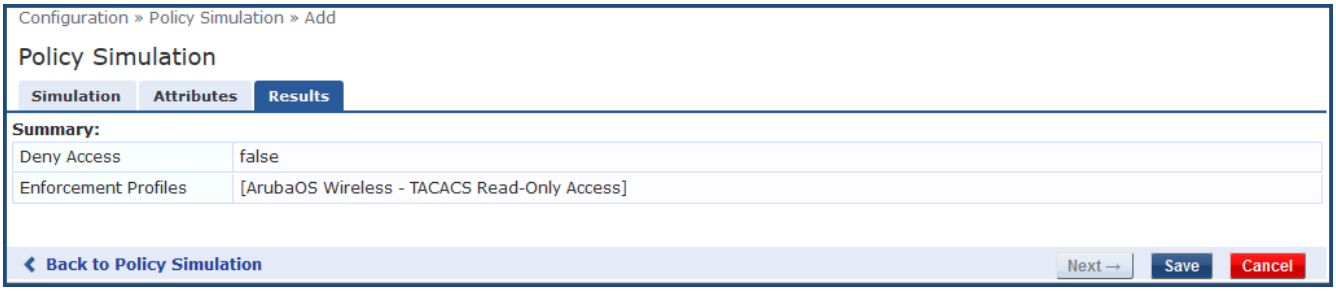


Table 1. Enforcement Policy Results Tab Parameters

Parameter	Description
Deny Access	Displays the output of the Deny Access test. • False: Device/network access is permitted. • True: Device/network access is denied.
Enforcement Profile	Displays the name of the Enforcement Profile that is applied to this policy.

RADIUS Authentication Simulation

- Adding a RADIUS Authentication Simulation
- Setting the Attributes to Be Tested
- Viewing the Simulation Results

Dictionaries in the RADIUS namespace come with the ClearPass Policy Manager. The administration interface does provide a way to add dictionaries into the system (see RADIUS Dictionary for more information). The RADIUS namespace uses the notation RADIUS:Vendor, where Vendor is the name of the company that has defined attributes in the dictionary. The same vendor can have multiple dictionaries, in which case the "Vendor" portion includes a suffix or some other unique string by the name of the device to differentiate the dictionaries.

Subtopics

Adding a RADIUS Authentication Simulation
Setting the Attributes to Be Tested
Viewing the Simulation Results

Adding a RADIUS Authentication Simulation

About this task

To add the RADIUS authentication server for the authentication test:

Procedure

1. Navigate to the **Configuration** > **Policy Simulation** > **Add** page.

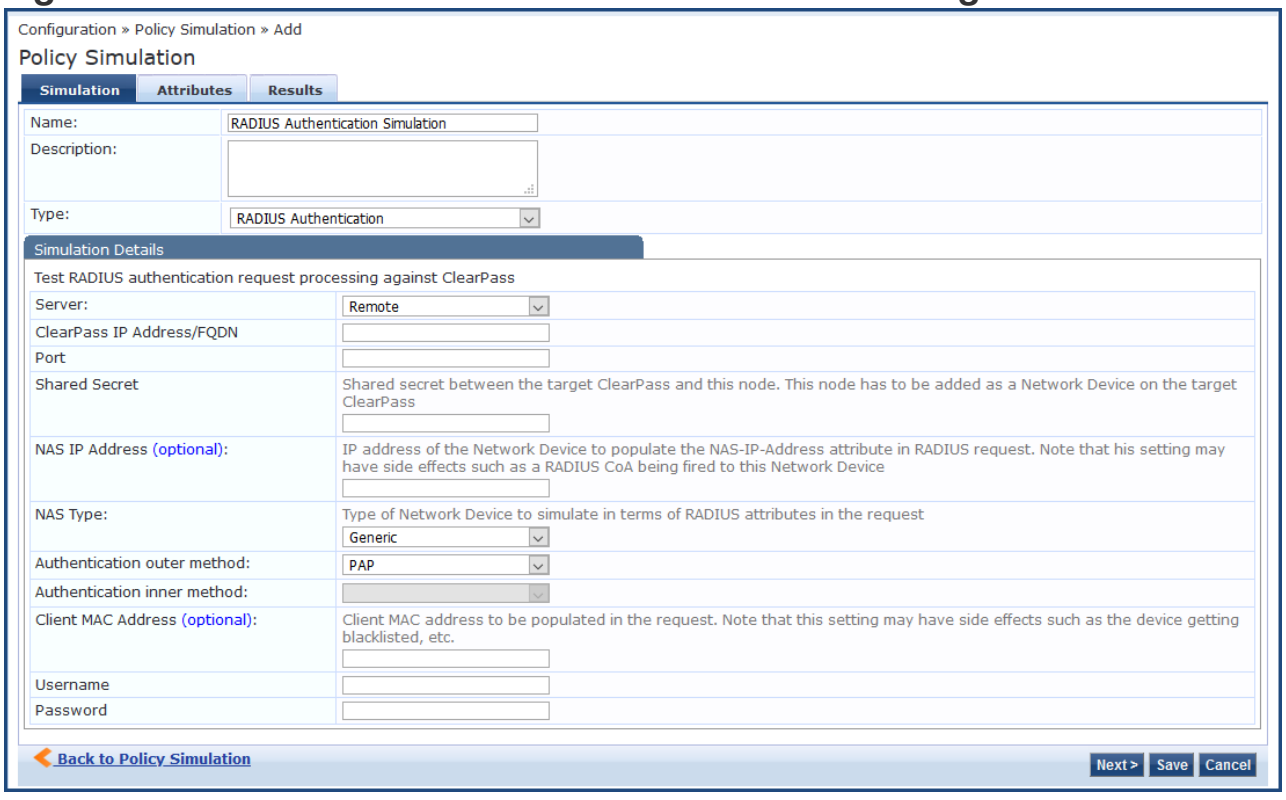
   The **Add Policy Simulation** dialog opens.

2. Enter the **Name** of the simulation.

3. From the **Type** drop-down list, select **RADIUS Authentication**.

   Figure 1 displays the **RADIUS Authentication Simulation Details** dialog, with the **Server** parameter set to **Remote**.

Figure 1. RADIUS Authentication Simulation Details Dialog



4. Enter the values for each of the **RADIUS Simulation** parameters as described in Table 1.

Table 1. RADIUS Simulation Tab Parameters

Parameter	Action/Description
Server	Specify **Local** or **Remote**.
CPPM IP Address or FQDN	This field is displayed only if **Remote Server** is selected. Enter the IP address or the fully qualified domain name (FQDN) of the remote ClearPass Policy Manager server.
Shared Secret	This field is displayed only if **Remote Server** is selected. Enter the shared secret between the target Policy Manager server and this node. You must add the node as a Network Device on the target Policy Manager server (for details, see Adding a Network Device) .
NAS IP Address (optional)	To populate the *NAS - IP - Address* attribute in a RADIUS request, enter the IP address of the network device.
NAS Type	Select the type of network device to simulate in terms of RADIUS attributes in the request. The network access server (NAS) types are: • Aruba Wireless Controller • Aruba Wired Switch • Cisco Wireless Controller • Generic
Authentication outer method	Specify one of the following authentication outer methods: • PAP • CHAP

Parameter	Action/Description
	• MSCHAPv2 • PEAP: Authentication inner method: *enabled*. ◦ Select one of the following PEAP Authentication inner methods: ◦ EAP - MSCHAPv2 ◦ EAP - GTC ◦ EAP - TLS • TTLS: Authentication inner method field: *enabled*. ◦ Select one of the following TTLS Authentication inner methods: ◦ PAP ◦ CHAP ◦ MSCHAPv2 ◦ EAP - MSCHAPv2 ◦ EAP - GTC ◦ EAP - TLS ◦ TLS
Client MAC Address (optional)	Enter the client MAC address of the network device to populate the NAS - IP address attribute in the RADIUS request.
Username	Enter the user name.
CA Certificate (optional)	This is the optional Root CA certificate needed to verify the RADIUS server's certificate. a. Click **Browse** and navigate to the optional Root CA certificate. b. Click **Open**, then click **Upload**.

Parameter	Action/Description
Client Certificate PKCS12 (PFX)*	This is the client certificate that is used fo r TLS in PKCS12 (Public - Key Cryptograp hy Standards). Formats: .pfx or .p12 a. Click **Browse** and navigate to the *Clien t Certificate PKCS12*. b. Click **Open**, then click **Upload**.
Passphrase for PFX file*	Enter the pass phrase for the selected PF X file.

* These fields are displayed only if you select **TTLS** or **PEAP** as the authentication ou ter method and you select **EAP - TLS** as the authentication inner method.

5. Click **Save**, or click **Next** to proceed to the **Attributes** tab.

Setting the Attributes to Be Tested

About this task

> **NOTE**
> The attributes that you can set depend on the **NAS Type** selected on the **Policy Simulation** page (see Figure 1).

To set the attributes to be tested:

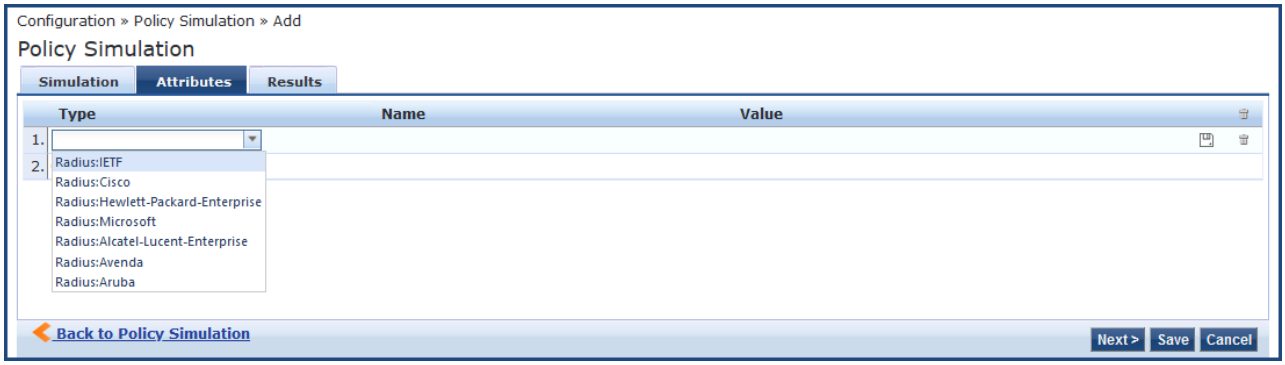
Procedure

1. From the **Attributes** tab, click **Click to add**.

   The **Add Policy Simulation Attributes** dialog opens.

2. From the **Type** drop-down, select the attribute **Type**.

Figure 1. Specifying Policy Simulation Attributes



3. Select the attribute **Name**.
4. Select the attribute **Value**.
5. Repeat these steps for each additional attribute you wish to add.
6. Click **Save**, or click **Next** to proceed to the **Results** tab.

Subtopics

NAS Type: Aruba Wireless Controller
NAS Type: Aruba Wired Switch
NAS Type: Cisco Wireless Switch

NAS Type: Aruba Wireless Controller

When you set the **NAS Type** to *Aruba Wireless Controller*, the available attributes are as follows:

Figure 1. NAS Type: Aruba Wireless Controller - Attributes

Table 1. NAS Type: Aruba Wireless Controller - Attribute Settings

Type	Name	Value
Radius:IETF	NAS - Port - Type	Wireless

Type	Name	Value
Radius:IETF	Service - Type	Login - L
Radius:Aruba	Aruba - Essid - Name	SSID

NAS Type: Aruba Wired Switch

When you set the **NAS Type** to *Aruba Wired Switch*, the available attributes are as follows:

Figure 1. NAS Type: Aruba Wired Switch Attributes Tab

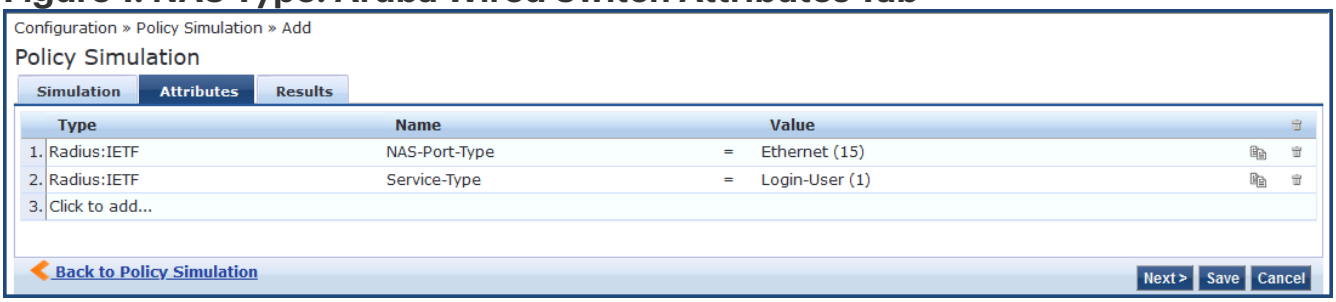


Table 1. NAS Type: Aruba Wired Switch Controller - Attribute Settings

Type	Name	Value
Radius:IETF	NAS - Port - Type	Ethernet
Radius:IETF	Service - Type	Login - L

NAS Type: Cisco Wireless Switch

Figure 1. NAS Type: Cisco Wireless Switch Attributes

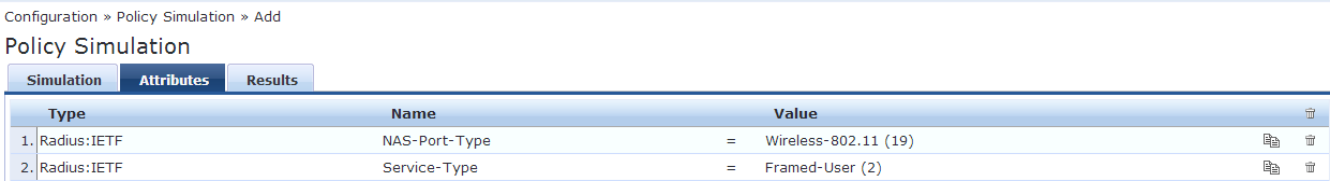


Table 1. NAS Type: Cisco Wireless Switch Required Attribute Settings

Attribute
Line 1: • Type = Radius:IETF • Name = NAS - Port - Type • Value = 802.11(19)
Line 2: Type = Radius:IETF Name = Service - Type Value = Framed - User(2)

Viewing the Simulation Results

The following figure displays the **Policy Simulation RADIUS - Results** dialog:

Figure 1. Results Tab

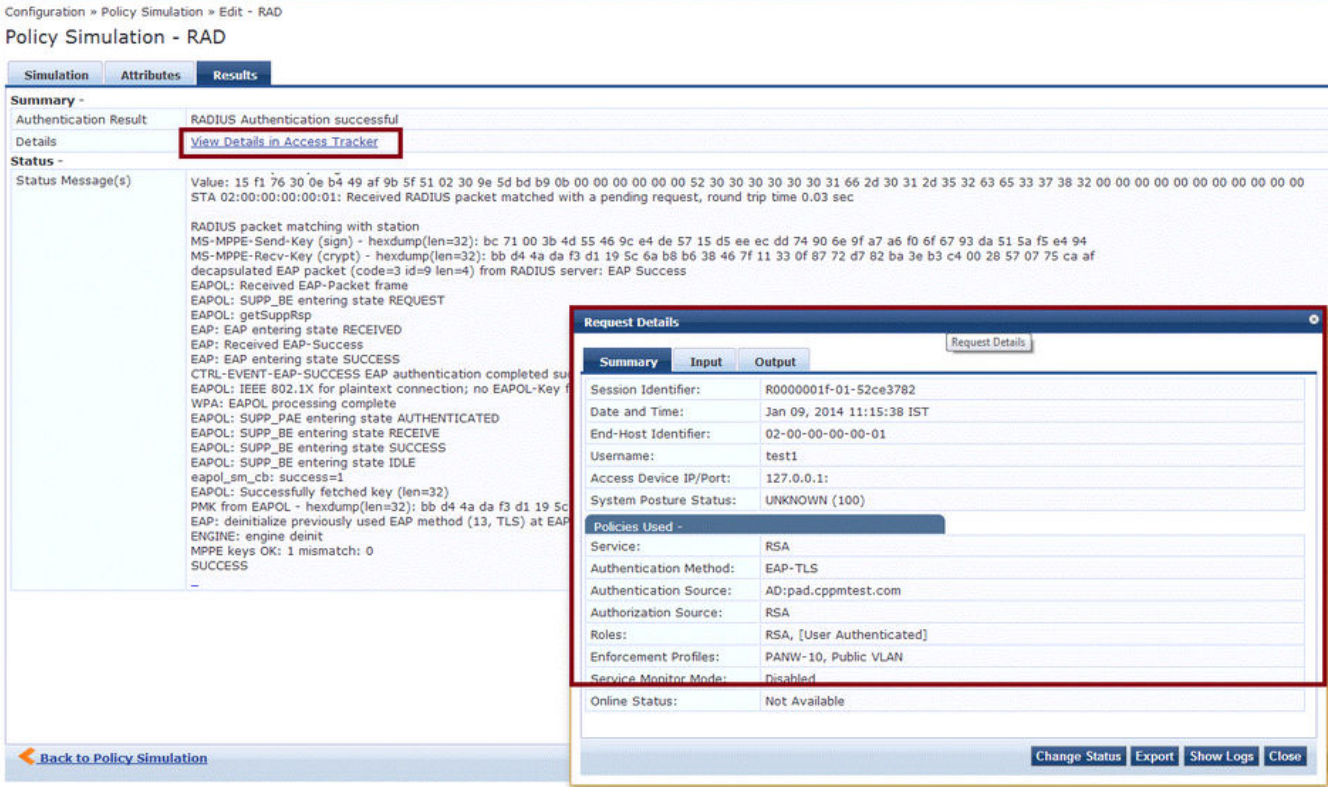


Table 1. RADIUS Authentication Results Tab Parameters

Parameter	Description
Summary	Displays a summary of the simulation.
Authentication Result	Displays the outcome of the Authenticatio n test.
Details	Click this link to open a dialog that provid es details about the Authentication test. Yo u can take the following actions: • Click the **Summary**, **Input**, or **Output** ta bs. • Click the **Change Status**, **Show Logs**, **Export**, or **Close** buttons.
Status Message(s)	Displays the status messages resulting fro m the test.

Role Mapping Simulation

The role mapping simulation tests Role-Mapping policy rules to determine which roles will be output, given the service name (and associated role mapping policy), the authentication source and the user name. You can also use role mapping simulation to test whether the specified authentication source is reachable.

Subtopics

Simulation Tab
Attributes Tab
Results Tab

Simulation Tab

The following figure displays the **Role Mapping Simulation** tab:

Figure 1. Role Mapping Simulation Tab

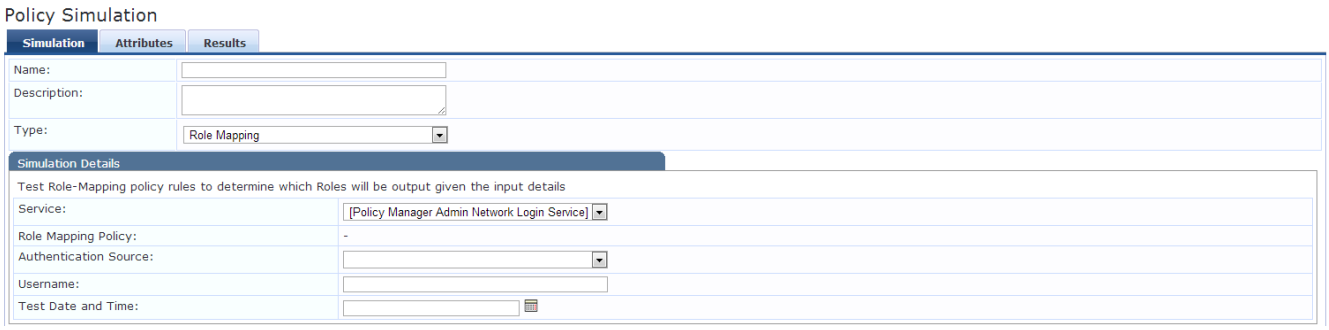


Table 1. Role Mapping Simulation Tab Parameters

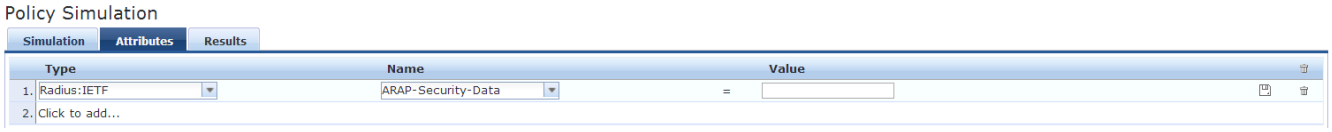
Parameter	Description
Service	Select from: • [Policy Manager Admin Network Login Service] • [AirGroup Authorization Service] • [Aruba Device Access Service] • [Guest Operator Logins] • Guest Access • Guest Access With MAC Caching
Role Mapping Policy	Field is disabled if you select: • [Policy Manager Admin Network Login Service] • [Aruba Device Access Service] • [Guest Operator Logins] • Field is auto - filled with **[AirGroup Version Match]** if you select [AirGroup Authorization Service] • Field is autofilled with **[Guest Roles]** if you select Guest Access • Field is autofilled with **Guest MAC Authentication Role Mapping** if you select Guest Access With MAC Caching
Authentication Source	Value = [Local User Repository] if you select:

Parameter	Description
	• [Policy Manager Admin Network Login Service] • [Aruba Device Access Service] Value = [Guest Device Repository] if you select: • [AirGroup Authorization Service] • Guest Access • Guest Access With MAC Caching Values = [Guest Device Repository] or [Local User Repository] if you select [Guest Operator Logins]
Username	Enter the user name.
Test Date and Time	Click calendar icon to select start date and time for simulation test. For more information, see Date Namespaces

Attributes Tab

Enter the attributes of the policy component to be tested. The following figure displays the **Role Mapping Simulation Attributes** tab:

Figure 1. Role Mapping Simulation Attributes Tab



The following table describes the **Role Mapping Simulation Attributes** tab parameters:

Table 1. Role Mapping Simulation Attributes Tab Parameters

Attribute	Parameter
Type	Select the type of attributes from the drop - down list.
Host	See Host Namespaces

Attribute	Parameter
Authentication	See Authentication Namespaces
Connection	See Connection Namespaces
Application	See Application Namespace
Certificate	See Certificate Namespaces
• Radius:IETF • Radius:Cisco • Radius:Microsoft • Radius:Avenda • Radius:Aruba	See RADIUS Namespaces
Name	The options displayed for the **Name** attribu te depend on the **Type** attribute that was s elected.
Value	The options displayed for the **Value** attribut e depend on the **Type** and **Name** attributes that were selected.

Results Tab

The following figure displays the **Role Mapping Simulation - Results** tab:

Figure 1. Results Tab

The following table describes the **Role Mapping Simulation - Results** tab parameters:

Table 1. Role Mapping Results Tab Parameters

Parameter	Description
Summary	Displays the results of the simulatio

Service Categorization Simulation

A service categorization simulation allows you to specify a set of attributes in the RADIUS or Connection namespace and test which configured service the request will be categorized into. The request attributes that you specify represent the attributes sent in the simulated request.

Subtopics

Simulation Tab
Attributes Tab
Results Tab

Simulation Tab

The following figure displays the **Service Categorization Simulation - Simulation** tab:

Figure 1. Service Categorization Simulation Tab

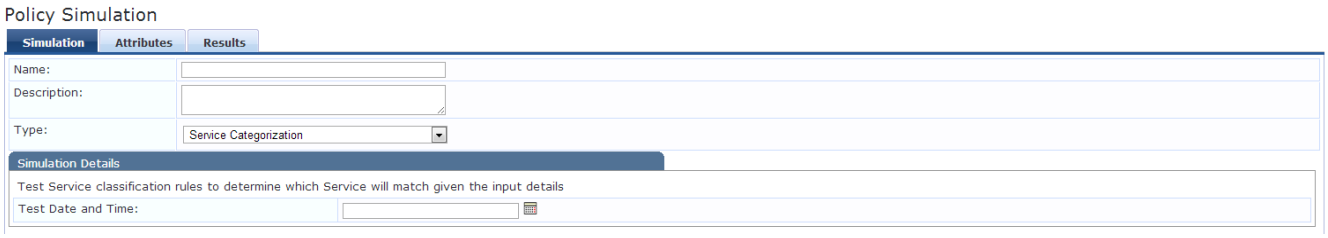


Table 1. Service Categorization Simulation Tab Parameter

Parameter Type	Namespace Details
Test Date and Time	Click calendar widget and select: • Test start date • Test start time

Attributes Tab

Enter the attributes of the policy component to be tested. The following figure displays the **Service Categorization Simulation - Attributes** tab:

Figure 1. Service Categorization Attributes Tab

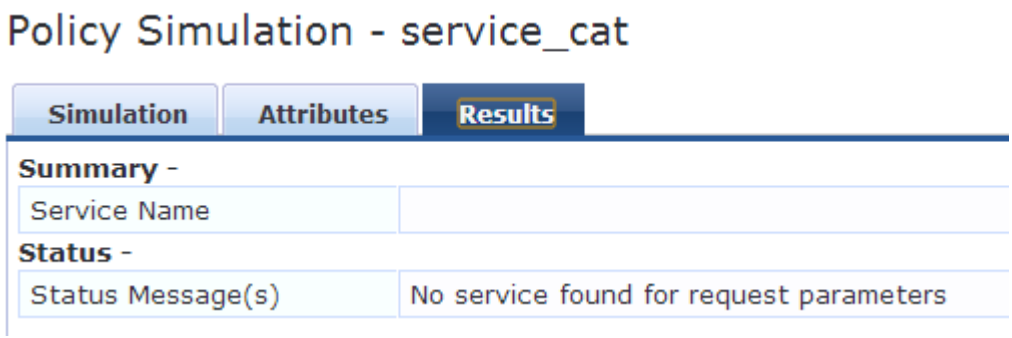
Table 1. Service Categorization Simulation Attributes Tab Parameters

Attribute	Parameter
Type	Select the type of attributes from the drop - down list.
Host	See Host Namespaces
Authentication	See Authentication Namespaces
Connection	See Connection Namespaces
Application	See Application Namespace
• Radius:IETF • Radius:Cisco • Radius:Microsoft • Radius:Aruba	See RADIUS Namespaces
Name	The options displayed for the **Name** attribute depend on the **Type** attribute that was selected.
Value	The options displayed for the **Value** attribute depend on the **Type** and **Name** attributes that were selected.

Results Tab

The following figure displays the **Service Categorization - Results** tab:

Figure 1. Results Tab



The following table describes the **Service Categorization Simulation Results** tab parameters:

Table 1. Service Configuration Results Tab Parameters

Parameter	Description
Summary	Gives the name of the service.

Import and Export Simulations

Navigate to **Configuration > Policy Simulation** and select the **Import** link. The following figure shows an example of the **Import from file** page.

Figure 1. Import Simulations

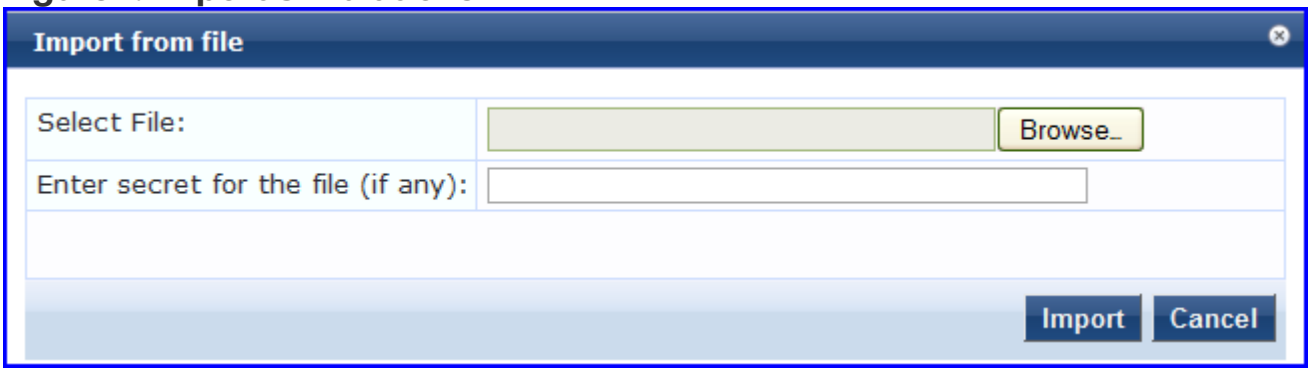


Table 1. Import from file page Parameters

Parameter	Description
Select file	Browse to select name of simulations to import.
Enter secret for the file (if any)	If the file was exported with a secret key for encryption, enter the same key here.

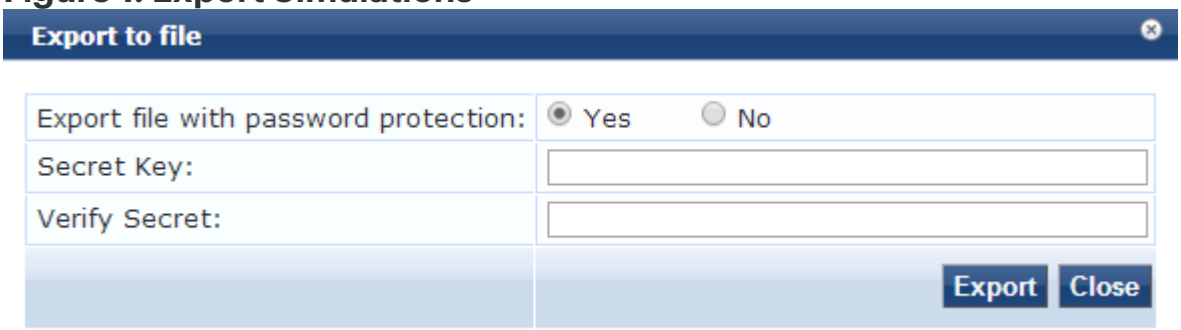
Subtopics

Export Simulations

Export Simulations

Click the **Export All** link to export all simulations. The browser displays the **Save As** dialog box in which you can enter the name of the XML file to export all simulations. The following image shows an example of the **Export** page to file page.

Figure 1. Export Simulations



To export a specific simulation, click **Export.** In the **Save As** dialog box, enter the name of the XML file to contain the export data.

Table 1. Export Simulations

Parameter	Description
Export file with password protection	Select **Yes** to export the file with password protection.
Secret Key	Enter the secret key in this field.
Verify Secret	Enter the same secret key to confirm and c omplete export.

Policy Manager Profiler

This section contains the following information:

- Policy Manager Device Profiler Overview

- About the Device Profile
- Endpoint Information Collectors

Subtopics

Policy Manager Device Profiler Overview
Device Profiler identifies and classifies network devices in Policy Manager to improve visibility, control, and policy enforcement.
About the Device Profile
Endpoint Information Collectors

Policy Manager Device Profiler Overview

Device Profiler identifies and classifies network devices in Policy Manager to improve visibility, control, and policy enforcement.

Subtopics

Enabling Endpoint Classification
Viewing Live Endpoint Information for a Specific Device
Configuring CoA for an Endpoint-Connected Device
Fingerprint Dictionaries
DHCPv6 Classification

Enabling Endpoint Classification

To enable endpoint classification, you must configure the **Primary Server in Zone** parameter by specifying the **main primary server** and **secondary primary server** (for more information, see Master Server in Zone). If you don't specify the main primary server in the zone, Policy Manager will automatically select the node in the zone with the lowest UUID (universally unique identifier). When you enable the Policy Manager server for endpoint classification, this associates each endpoint with a specific user or location and secures access for other devices.

Viewing Live Endpoint Information for a Specific Device

About this task

The Policy Manager Live Monitoring feature allows you to view endpoint information in graphic format for the device category, device family, and device name items you

selected. You can also examine the endpoint details and attributes about a specific device .

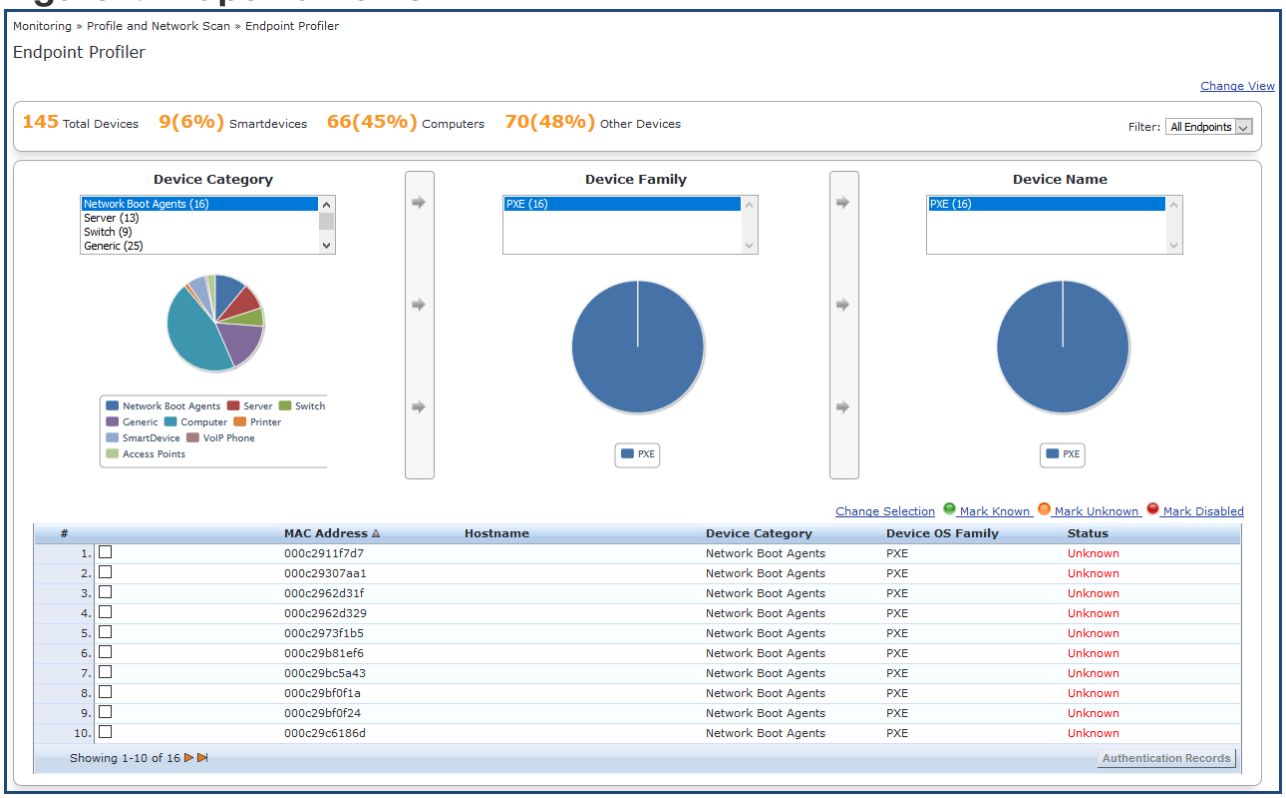
To access the Endpoint Profiler Live Monitoring information:

Procedure

1. Navigate to **Monitoring > Profile and Network Scan > Endpoint Profiler**.

   The **Endpoint Profiler** opens.

Figure 1. Endpoint Profiler



2. To view endpoint details about a specific device, click a device in the table below the graphs.
3. To return to the **Endpoint Profiler** page, click **Cancel**.

   For more information, see:

   - Monitoring Endpoint Profiler

Configuring CoA for an Endpoint-Connected Device

About this task

After profiling an endpoint, use the **Profiler** page to configure Change of Authorization (CoA) on the network device to which an endpoint is connected.

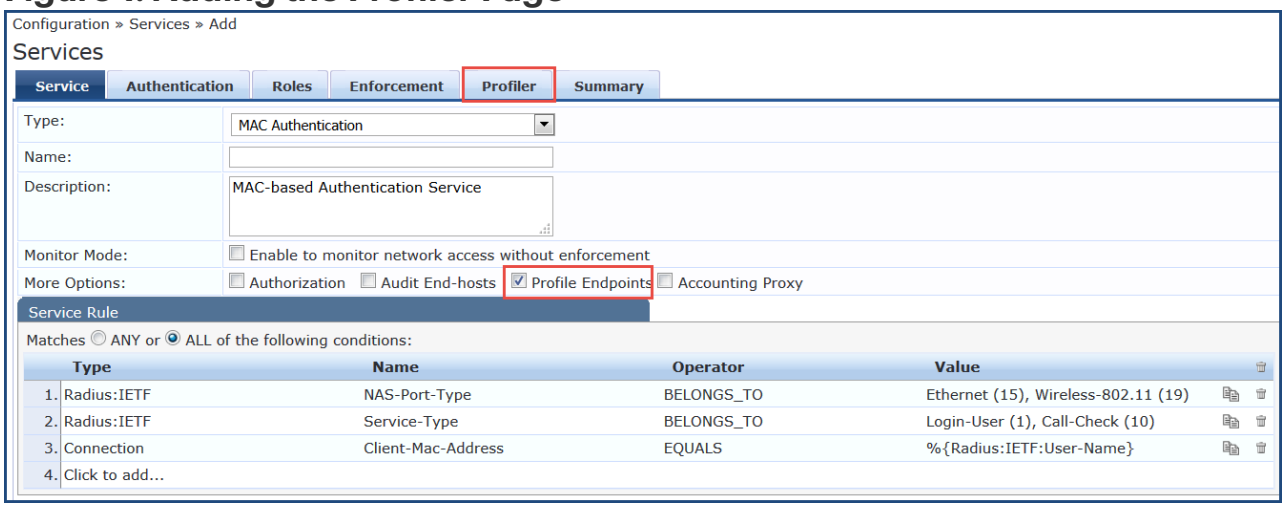
The **Profiler** tab is not displayed by default. To access the **Profiler** tab:

Procedure

1. Navigate to **Configuration > Services**, then click **Add**.
2. Enter the name of the service.
3. From the *More Options* field on the **Service** tab, enable the **Profile Endpoints** check box.

   The **Profiler** tab is added to the **Services** tabs:

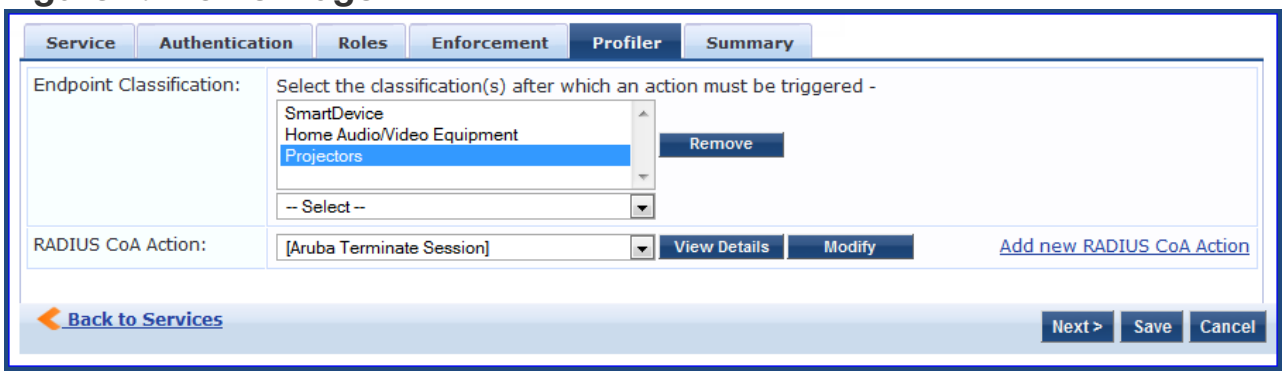
Figure 1. Adding the Profiler Page



4. Select the **Profiler** tab.

   The **Profiler** page opens.

Figure 2. Profiler Page



5. You can select a set of categories and a CoA profile to be applied when the profile matches one of the selected categories.

   CoA is triggered using the selected CoA profile. You can use any option from Endpoint Classification to invoke CoA on a change of any one of the fields (category, family, and name).

6. Specify the **Profiler** page parameters as described in Table 1:

Table 1. Profiler Page Parameters

Parameter	Action/Description
Endpoint Classification	Select one or more endpoint classificatio n items from the drop - down list. You can select a new action, or remove a current action.
RADIUS CoA Action	Select the RADIUS CoA action from the d rop - down list. • To view the **Policy Manager Entity De tails** page with the summary of enforc ement profile details, click **View Detai ls**. • To view the **Summary** tab with profile details, click **Modify**.
Add new RADIUS CoA Action	To create a new RADIUS CoA action, click the **Add New RADIUS CoA Action** link.

7. When finished, click **Save**.

Fingerprint Dictionaries

About this task

ClearPass Policy Manager uses a set of dictionaries and rules to perform device fingerprinting.

Because these dictionaries can change frequently, ClearPass Policy Manager provides a way to automatically update fingerprints from a hosted portal. The device fingerprints are updated from the Aruba Policy Manager Update Portal (for more information, see [%=aruba_style_guide.ShortPrdNm%] Guest Portal).

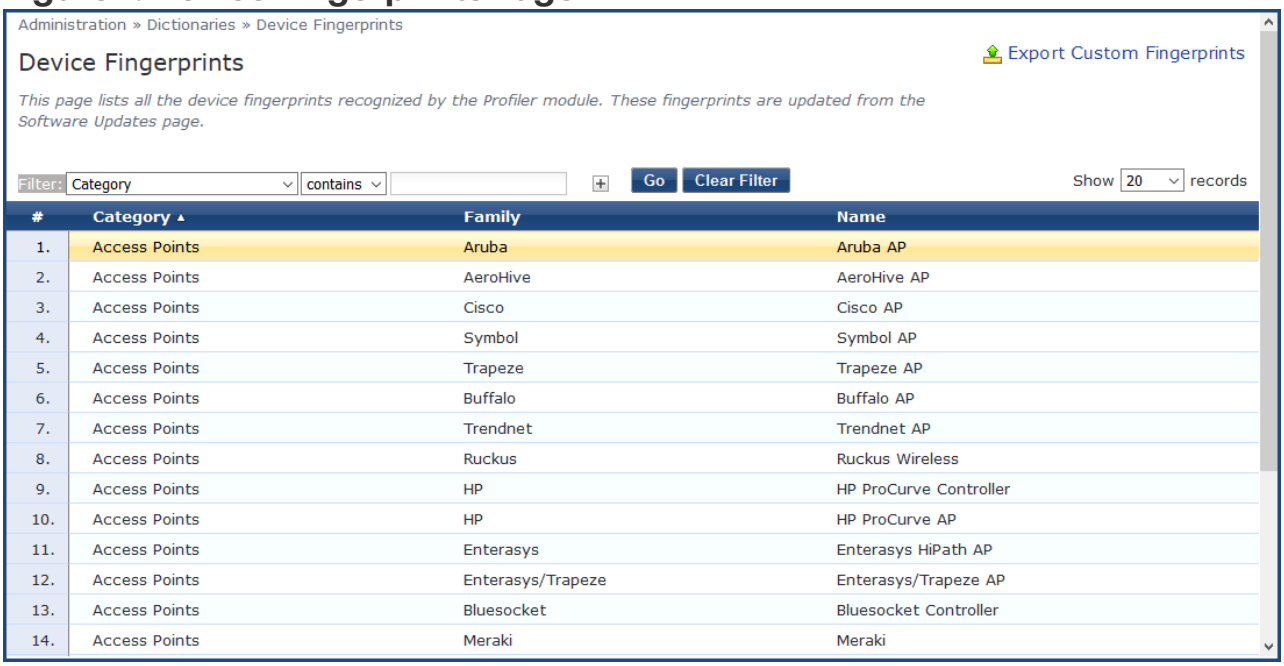
To view the contents of the fingerprints dictionary:

Procedure

1. Navigate to **Administration > Dictionaries > Fingerprints**.

   The **Device Fingerprints** page opens. This page lists all the device fingerprints recognized by the Device Profiler module.

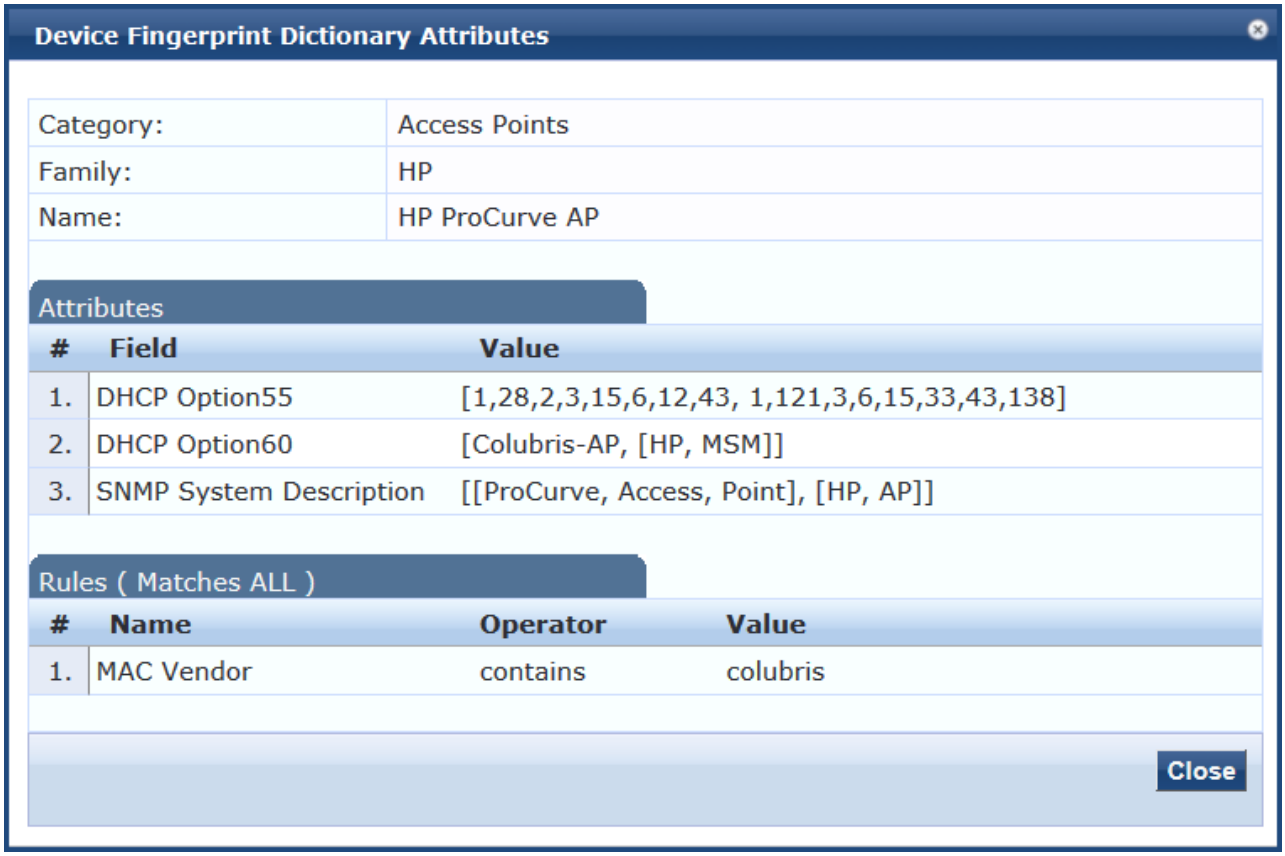
Figure 1. Device Fingerprints Page



2. To view the device fingerprint dictionary attributes, select the device fingerprint of interest.

   The attributes for the selected Device Fingerprint Dictionary are displayed:

Figure 2. Device Fingerprint Dictionary Attributes Page



3. To exit, click **Close**.

DHCPv6 Classification

About this task

ClearPass ClearPass now supports DHCPv6 profiling. When both DHCPv4 and DHCPv6 based fingerprints are received together in a profiling request, DHCPv4 has priority over DHCPv6. The classification of the endpoint is based on DHCPv4 in this instance. When a pure DHCPv6 profiling request is received, endpoint classification is based on DHCPv6 option 16. Currently, DHCPv6 based classification support is available only for the Windows OS.

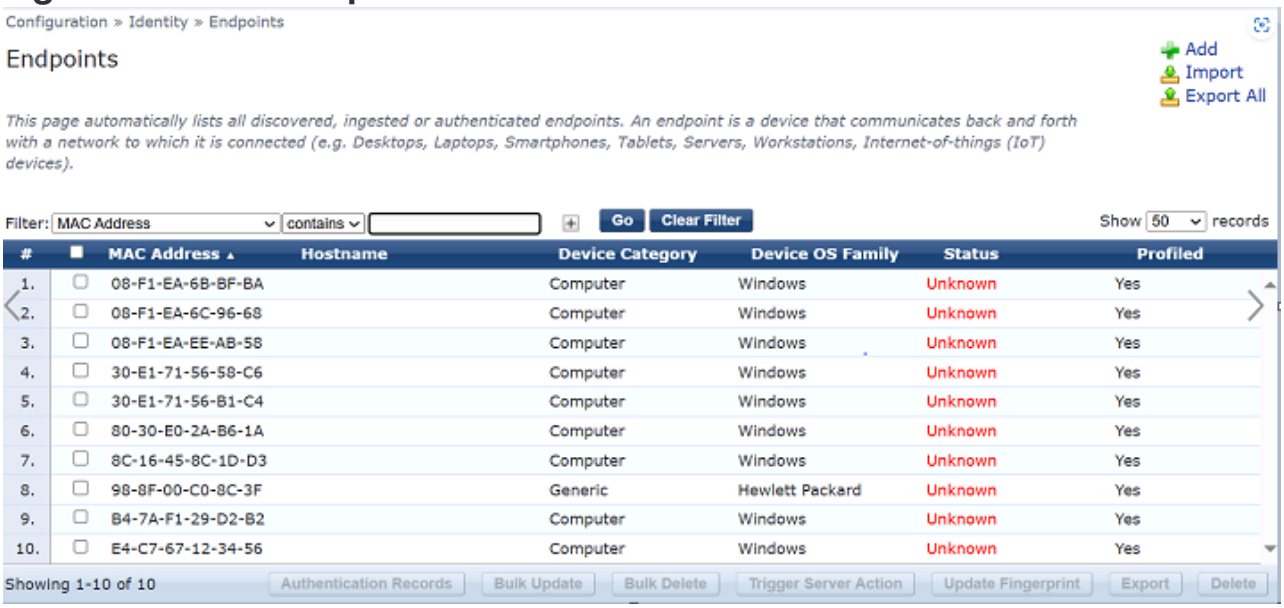
To review DHCPv6 profiling information:

Procedure

1. Navigate to **Configuration > Endpoints**.

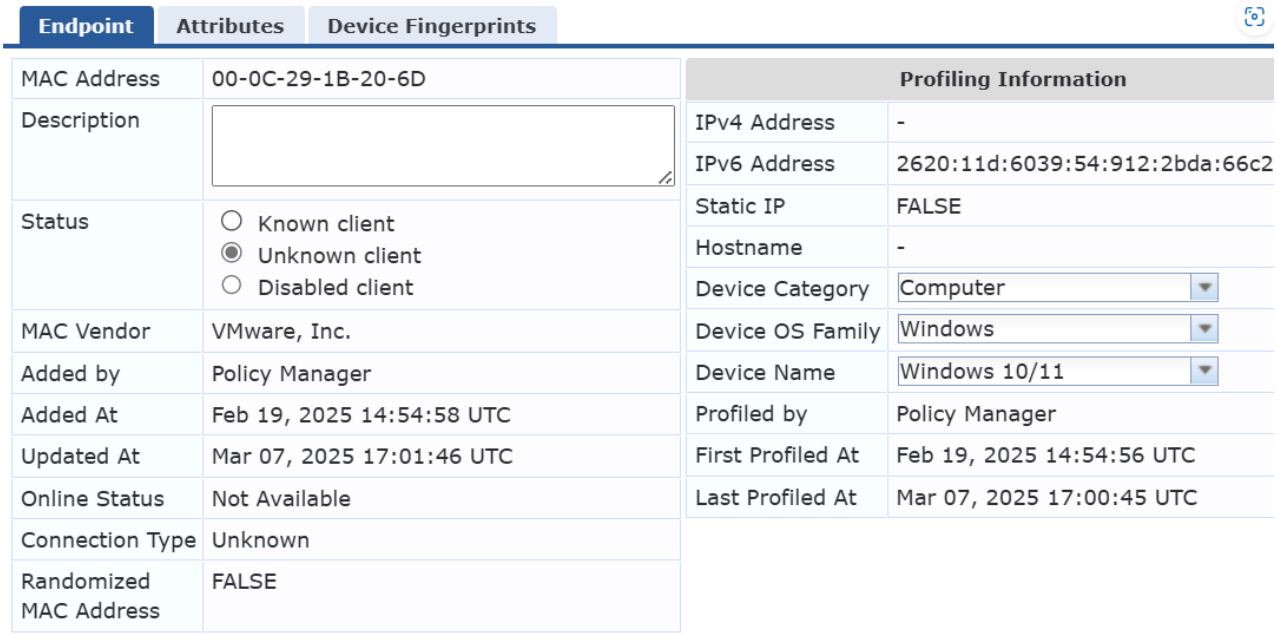
   The endpoints page displays. To review endpoint details about a specific device, click a device in the table.

Figure 1. Review Endpoint Details



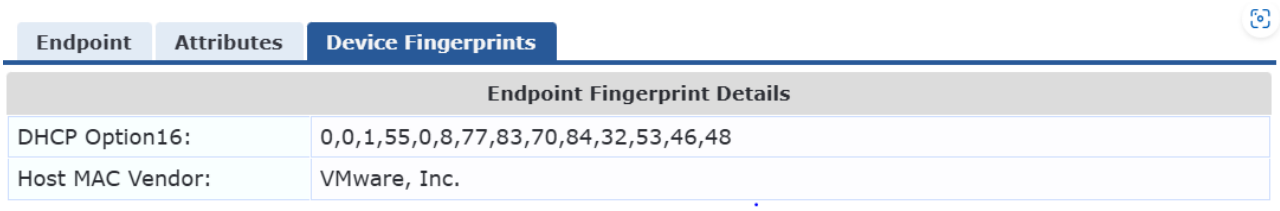
2. Open a specific device to display endpoint information for that device.

Figure 2. Review Device Endpoint Information



3. Select the **Device Fingerprints** tab to review fingerprint information for the selected device.

Figure 3. Review Device Fingerprint Information



About the Device Profile

The Device Profiler module produces a *Device Profile*, which is a hierarchical model consisting of three elements that are derived by the endpoint attributes—*DeviceCategory*, *DeviceFamily*, and *DeviceName*.

Table 1. Elements of a Device Profile

Endpoint Attributes	Description
DeviceCategory	Denotes the type of the device, for example, *Computer*, *Smart Device*, *Printer*, or *Access Point*.
DeviceFamily	Classifies devices based on the type of operating system or vendor. For example, when the category is *Computer*, ClearPass Policy Manager shows a device family of *Windows*, *Linux*, or *macOS*.
DeviceName	Denotes the name of the device. Devices in a family are organized based on characteristics such as their operating system version. For example, in a DeviceFamily of *Windows*, Policy Manager could show a DeviceName of *Windows 8.1* or *Windows Server 2019*.

This hierarchical model provides a structured view of all endpoints accessing the network. In addition to these, a Device Profile also collects and stores the following:

- IP address
- Host name
- Device vendor (via MAC OUI)
- Timestamp indicates when the device was first discovered and last seen

Endpoint Information Collectors

Collectors are the network elements that provide data in order to profile endpoints. This section provides the following information:

- DHCP Collector
- NetFlow Collector
- sFlow Collector
- Policy Manager Onboard Collector
- HTTP User-Agent Strings Collector
- MAC OUI Collector
- ActiveSync Plug-in Collector
- OnGuard Agent
- Subnet Scan Collector

Subtopics

DHCP Collector
NetFlow Collector
sFlow Collector
Policy Manager Onboard Collector
HTTP User-Agent Strings Collector
MAC OUI Collector
ActiveSync Plug-in Collector
OnGuard Agent
SNMP Collector
Subnet Scan Collector

DHCP Collector

Dynamic Host Configuration Protocol (DHCP) attributes such as *option 55* (parameter request list), *option 60* (vendor class), and the options list from the Discover and Request packets can uniquely fingerprint most devices that use the DHCP mechanism to acquire an IP address on the network. You can configure switches, controllers and gateways to forward DHCP Discover, Request, and Inform packets to Policy Manager. These DHCP packets are decoded by ClearPass Policy Manager to arrive at the

appropriate device category, OS family, and device name. In addition to fingerprints, DHCP also provides the host name and IP address.

Subtopics

Sending DHCP Traffic to the Policy Manager Server

Sending DHCP Traffic to the Policy Manager Server

To configure your Aruba controller and Cisco switch to send DHCP traffic to the Policy Manager server, enter the following CLI commands:

```
interface <VLAN_name>
ip address <IP_addr> <netmask>
ip helper-address <DHCP_server_IO>
ip helper-address <clearpass_IP> end
end
```

You can configure multiple *IP helper-address* statements to send DHCP packets to servers other than the DHCP server.

NetFlow Collector

NetFlow provides the ability to collect IP network traffic as it enters or exits an interface. By analyzing the data provided by NetFlow, a network administrator can determine things such as the source and destination of traffic, class of service, and the causes of congestion.

The ClearPass Policy Manager NetFlow collector provides the ability to identify the open ports of a device connected to a network by analyzing the received NetFlow packets.

NOTE

Policy Managercollects the source IP address and the source port address forall NetFlow versions—v1, v5, v6, v7, and v9—as well as IPFIX and sFlow (v5). Also, Policy Manager collects the source MAC address (if it exists) for NetFlow V9, IPFix, and sFlow (v5).

sFlow Collector

sFlow is a general purpose network-traffic measurement system technology. sFlow is embedded in network devices and provides continuous statistics on any protocol so

that all traffic throughout a network can be accurately characterized and monitored. The main difference between NetFlow and sFlow is that NetFlow is restricted to IP traffic only, whereas sFlow has the ability sample all traffic (that is, it is network-layer independent).

The sFlow collector has the following characteristics:

- In ClearPass Policy Manager, sFlow listens to UDP port **6343**.

  > **NOTE**
  >
  > UDP port **6343** must be opened to the firewall. The switch/ sFLow source should be configured to send sFlow packets to Policy Manager UDP port **6343**.

- sFlow identifies the source IP/port and destination IP/port (not the source MAC address) from the flow that reaches the Policy Manager server and updates the port information for endpoints existing in the Policy Manager server.
- When the source MAC address exists in the sFlow traffic, the behavior of the Policy Manager server is as follows:
  - If the source MAC address is present in the Policy Manager endpoints, Policy Manager updates that endpoint entry with the open ports and source IP address.
  - If the source MAC address is not present in the ClearPass endpoints, and if the source IP address is present in the endpoints, Policy Manager updates the MAC address and source port for that endpoint.
  - If both the MAC address and IP address are not present in the Policy Manager endpoints, a new entry is created.
  - The sFlow traffic received is sent to Profiler for further processing in batches of 10 in one-minute intervals.

Policy Manager Onboard Collector

Policy Manager Onboard collects authentic device information from all devices during the onboarding process. Onboard then posts this information to the Policy Manager Profile. Because the information collected is definitive, Policy Manager Profiler can directly classify these devices into their appropriate category, OS family, and name without having to rely on any other fingerprinting information.

HTTP User-Agent Strings Collector

In some cases, DHCP fingerprinting alone cannot fully classify a device. A common example is the Apple family of smart devices; for example, DHCP fingerprints cannot distinguish between an iPad and an iPhone. In these scenarios, user-agent strings sent by browsers in the HTTP protocol are useful to further refine classification results.

User-agent strings are collected from the following:

- Policy Manager Guest
- Policy Manager Onboard
- Aruba controller through an IF-MAP (Interface for Metadata Access Points) interface

MAC OUI Collector

The MAC OUI (Organization Unique Identifier) is expressed in the first 24 bits of a MAC address for a network-connected device. Thus, the MAC OUI indicates the specific vendor for that device. The MAC OUI is acquired through various authentication mechanisms, such as 802.1X and MAC address authentication. The MAC OUI can be useful to more accurately classify endpoints. An example is Android™ devices where DHCP fingerprints can only classify a device as generic Android, but it cannot provide more details regarding the vendor. Combining this information with MAC OUI, the Policy Manager Profiler can classify a device as HTC™ Android, Samsung™ Android, or Motorola® Droid, etc. The MAC OUI is also useful to profile devices such as printers that might be configured with static IP addresses.

ActiveSync Plug-in Collector

You can install the ActiveSync plug-in provided by Aruba on Microsoft Exchange servers. When a device communicates with an Exchange server using the Active Sync protocol, the device provides attributes such as device-type and user-agent. These attributes are collected by the ActiveSync plug-in and sent to the Policy Manager Profiler. Profiler uses dictionaries to derive profiles from these attributes.

OnGuard Agent

The Policy Manager OnGuard agent performs advanced endpoint posture assessment. This agent collects and sends operating system details from endpoints during

authentication. The Policy Manager Profiler uses the OnGuard **OSType** attribute to derive a profile (for more information, refer to Web-Based Authentication Service).

SNMP Collector

Endpoint information obtained by reading the Simple Network Management Protocol (SNMP) MIBs of network devices is used to discover and profile static IP devices in the network.

Table 1. SNMP MIBs Used by the SNMP Collector

MIB	Description
SysDescr	A textual description of the entity used both h for profiling switches, controllers, and routers configured in Policy Manager, and for profiling printers and other static IP devices discovered through SNMP or subnet scans (RFC1213).
cdpCacheTable	Provides the cached information obtained via receiving Cisco Discovery Protocol (CDP) messages from CDP - capable devices. This is used to discover neighbor devices connected to the switch or controller configured in Policy Manager.
lldpRemTable	This table contains one or more rows per physical network connection known to this agent read from LLDP (Link Layer Discovery Protocol) - capable devices. This is used to discover and profile neighbor devices connected to the switch or controller configured in Policy Manager.
ARPtable	Address Resolution Protocol (ARP) information is read from the network devices. This is used as a means to discover endpoints in the network.

Setting SNMP Community Attributes

About this task

The SNMP-based mechanism is capable of profiling devices only if they respond to SNMP, or if the device advertises its capability via LLDP (Link Layer Discovery Protocol). When performing SNMP reads for a device, Policy Manager uses SNMP Read credentials

configured in the network devices, or defaults to using SNMPv2 with public community strings specified.

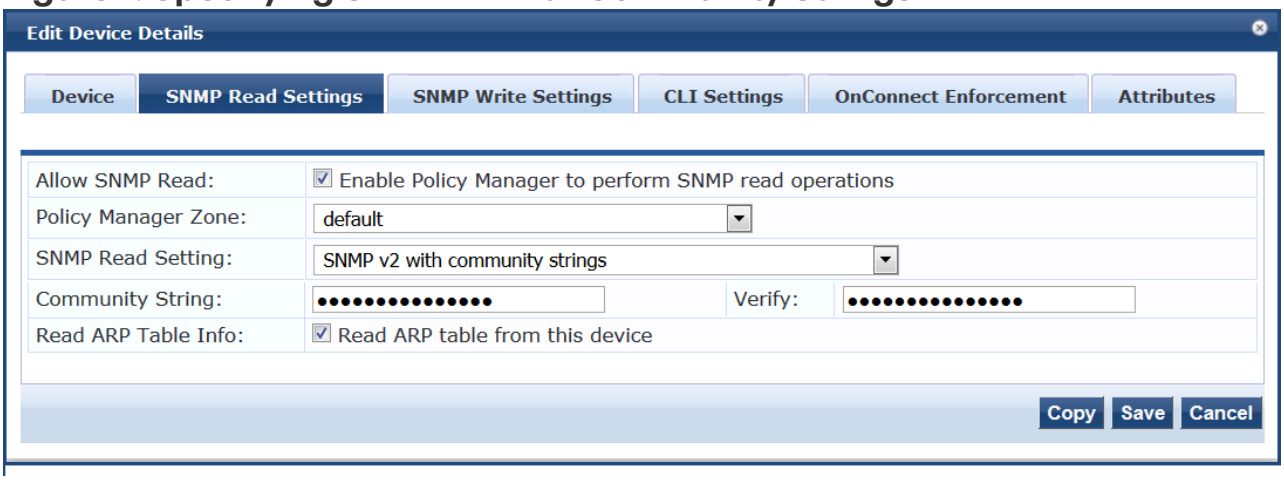
To specify SNMPv2 with community strings:

Procedure

1. Navigate to **Configuration > Network > Devices**.
2. From the **Network Devices** page, select the appropriate device.

   The **Edit Device Details** dialog opens.
3. Select the **SNMP Read Settings** tab.

Figure 1. Specifying SNMP v2 with Community Strings



4. Specify the SNMP Read Settings parameters as described in the following table, then click **Save**.

Table 2. SNMP Read Settings Parameters

Parameter	Action/Description
Allow SNMP Read	If not already enabled, enable the **Allow SNMP Read** check box. The SNMP Read Settings parameter field s are now enabled for configuration.
Policy Manager Zone	Select the Policy Manager Zone. If no Poli cy Manager Zone is configured, select **de fault**.
SNMP Read Setting	Select **SNMPv2 with community strings**.

Parameter	Action/Description
Community String	Enter the **Community String** value, then reenter the string to verify it.
Read ARP Table Info	If this is a Layer - 3 device, and you wan t to use the ARP table on this device as a way to discover endpoints in the network , enable the **Read ARP Table Info** check b ox. **NOTE** Static IP endpoints discove red in this way are further p robed via SNMP to profile th e device.

Configuring the Device Info Poll Interval

About this task

Network devices configured with **SNMP Read** enabled are polled periodically for updates based on the time interval configured for the **Device Info Poll Interval**.

To set the Device Info Poll Interval:

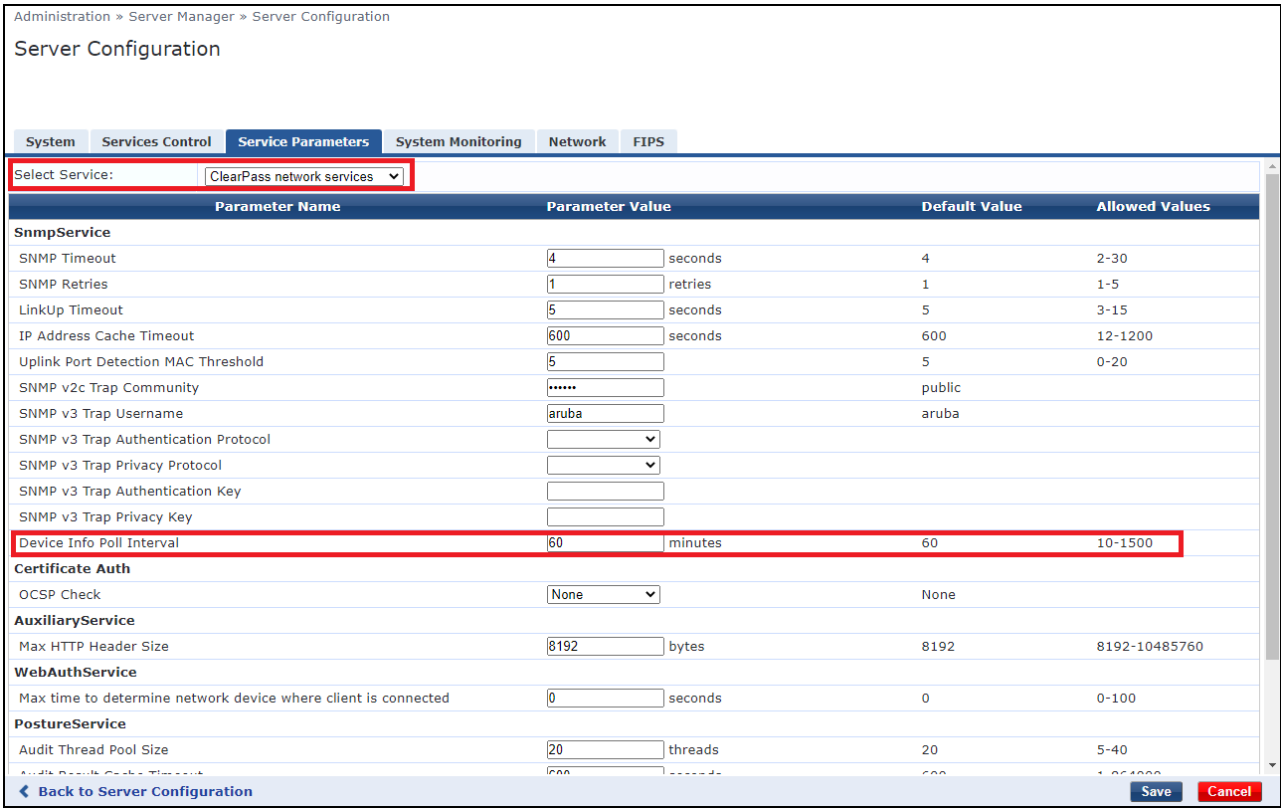
Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest.

   The **Server Configuration** page opens.
3. Select the **Service Parameters** tab.
4. From the **Select Service** drop-down, select **Policy Manager network services**.

   The **Policy Manager network services** page opens.

Figure 2. Specifying the Device Info Poll Interval



5. In the **minutes** field, enter the **Device Info Poll Interval**, then click **Save**.

Subnet Scan Collector

About this task

A network or subnet scan discovers the IP addresses of devices in the network. The devices discovered in this way are further probed using SNMP to fingerprint and assign a profile to the device. Network subnets to be scanned are configured per Policy Manager Zone.This is particularly useful in deployments that are geographically distributed. In such deployments, it is recommended that you complete the following tasks:

Procedure

1. Assign the ClearPass Policy Manager nodes in a cluster to multiple zones depending on the geographical area served by that node.

   To set up Policy Manager Zones, navigate to **Administration > Server Manager > Server Configuration > Manage Policy Manager Zones.**

2. Then enable the profile for a minimum of one node per zone.

   For more information, see Managing Policy Manager Zones .

3. Configure the SNMP, SSH, and WMI credentials required to query for network devices, Linux devices, and Windows devices respectively (for details, see Configuring SNMP, SSH, and WMI Credentials).
4. Configure a network scan or subnet scan.

Administration

You can access all Policy Manager administrative activities, including server configuration, license management, external accounts, software updates, support services, and administrator user account maintenance from the following Administration sections:

Subtopics

External Accounts
Policy Manager Portal
Server Manager
External Servers
Dictionaries
Agents and Software Updates
Cluster Update and Upgrade
Insight Report Tool
Support Services
ClearPass Guest Portal
Managing Admin Users
Managing Admin Privileges
Server Manager
External Servers
Adding External Accounts
Certificate Store
Using Policy Manager Dictionaries
Data and Management Port Interfaces
OnGuard Settings and Agent Library Updates
OnGuard Agent Installers
Agentless OnGuard Packages
OnGuard Global Agent Settings
Creating OnGuard Custom Web Pages
Software Updates
Support Services
SSH Timed Account Lockout

External Accounts

Adding External Accounts

Policy Manager Portal

- ClearPass Guest Portal

  Users and Privileges

  - Managing Admin Users
  - Managing Admin Privileges

Server Manager

- Server Configuration Operations
- Log Configuration
- Downloading Local Shared Folders
- Device Insight

External Servers

- SNMP Trap Receivers
- Syslog Targets
- Syslog Export Filters
- Messaging Setup
- Endpoint Context Servers
- Adding Vendor-Specific Endpoint Context Servers
- File Backup Servers

  Certificates

- Certificate Store
- Creating and Installing a Self-Signed Server Certificate
- Creating a Certificate Signing Request
- Importing a Server Certificate
- Exporting a Server Certificate
- Service and Client Certificates
- Associating a RADIUS Service Certificate with a Service
- Certificate Trust List
- Certificate Revocation Lists

Dictionaries

- RADIUS Dictionary
- RADIUS Dynamic Authorization Templates
- TACACS+ Services Dictionary
- Device Fingerprints Dictionary
- Dictionary Attributes
- Applications Dictionaries
- Configuring Endpoint Context Server Actions

Agents and Software Updates

- Software Updates
- OnGuard Settings and Agent Library Updates
- OnGuard Agent Installers
- Agentless OnGuard Packages
- Creating OnGuard Custom Web Pages

- OnGuard Global Agent Settings
- Upgrading From OnGuard Plugin Version 1.0 to 2.0

Cluster Update and Upgrade

- Updating Clusters
- Upgrading Clusters

  Ingress Events Processing

  - Configuring Processing for Ingress Events

Insight Report Tool

- Creating Reports

Support Services

- Contacting Support
- Remote Assistance
- Accessing the OnGuard Support Chart for Plugin Version 2.0

ClearPass Guest Portal

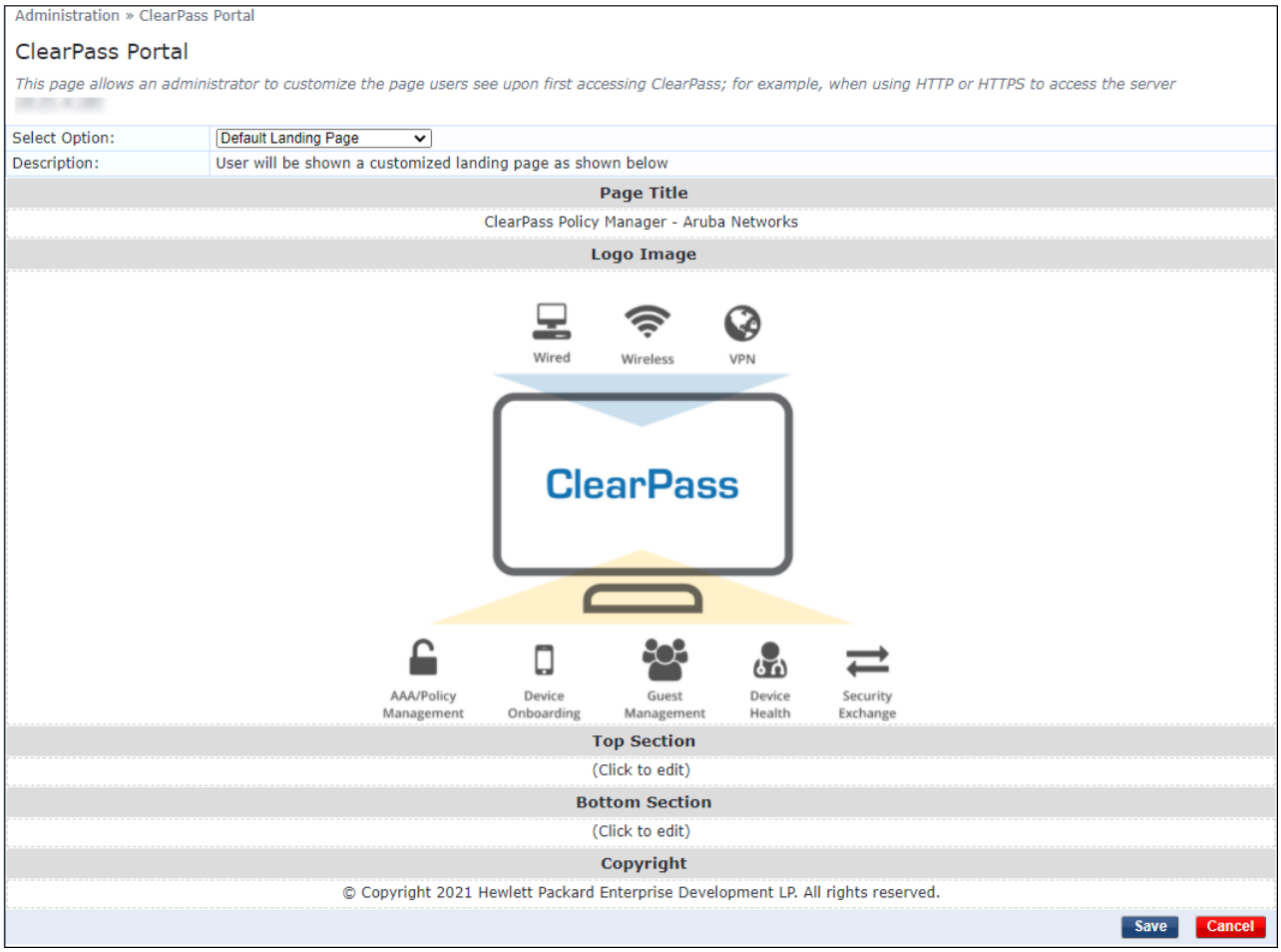
About this task

To customize the ClearPass Portal content for your enterprise:

Procedure

1. Navigate to the **Administration > ClearPass Portal** page. The ClearPass Portal page opens:

   Figure 1. Policy ManagerPortal



2. Specify the ClearPass Portal parameters as described in the following table, then click **Save**:

Table 1. ClearPass Portal Parameters

Parameter	Action/Description
Select Option	Select the page that the user first sees a fter logging in to ClearPass: • Default Landing Page • Application Login Page: • Guest Portal
Page Title	Click and enter the text to appear as the page title in the default landing page.
Logo Image	Click and browse to select an image for t he banner in the default landing page.

Parameter	Action/Description
Top section	Click and enter the text to appear as the header in the default landing page.
Bottom section	Click and enter the text to appear as the footer in the default landing page.
Copyright	Click and enter the copyright text to app ear in the default landing page.

NOTE

Both the HTTP and HTTPS protocols are supported for Portal redirection.

Managing Admin Users

Policy Manager provides three types of Administrator users by default:

- *Super Administrator*: Allowed read/write access to all configuration elements.
- *API Administrator*: Allowed API access to all configuration elements.
- *Read-only Administrator*: Allowed read-only access to all configuration elements.

In addition, Policy Manager allows you to add the following types of admin users:

- *Aruba User Role Download*: This admin role provides access to only download user roles.
- *Help Desk*: Allows the Help Desk to log in to troubleshoot problems reported by end users.
- *Network Administrator*: Allowed to configure all Policy Manager policies.
- *Receptionist*: Allowed to access the main monitoring screens.

Select any of the links below to view more information about these procedures.

Subtopics

Changing the Administration Password

Adding an Admin User
Importing and Exporting Admin Users
Setting Password Policy for Admin Users
Disabling Admin User Accounts

Changing the Administration Password

About this task

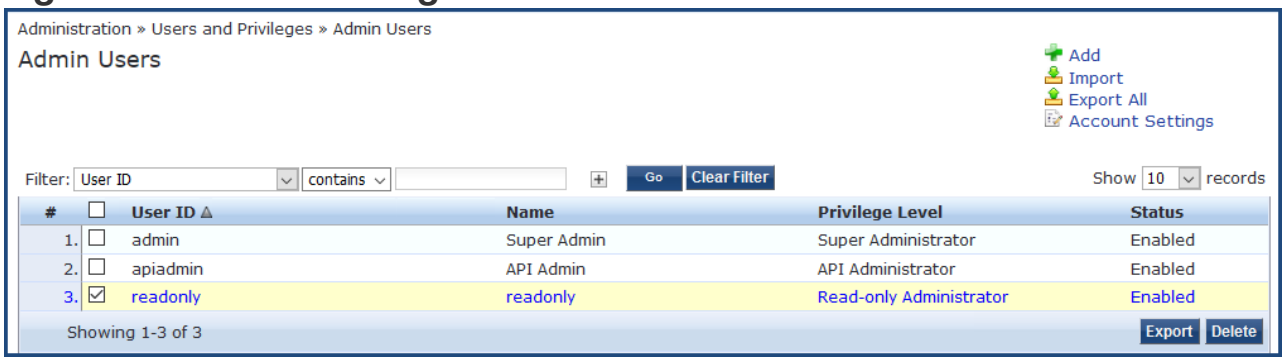
After activating the Policy Manager appliance, the recommended next task is to change the administration password for the newly-activated Policy Manager server.

To change the administration password:

Procedure

1. Navigate to **Administration > Users and Privileges > Admin Users**.The **Admin Users** page opens.

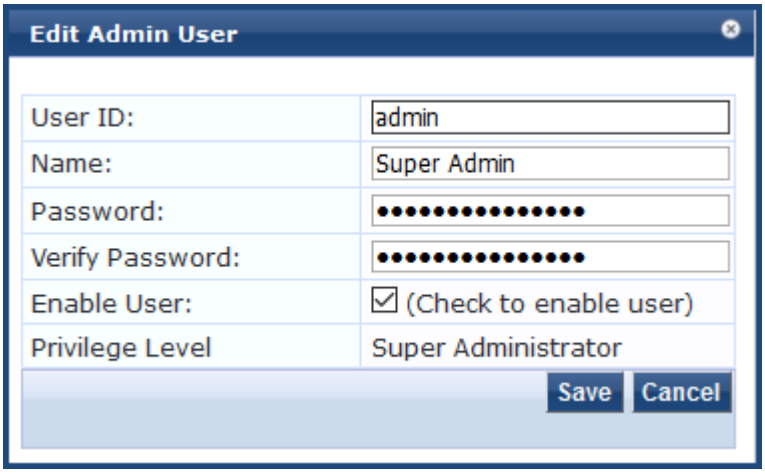
Figure 1. Admin Users Page



Review the displayed administrator details such as user ID, user name, and privilege level, and verify whether the admin user type is enabled or disabled. Optionally change the admin password, and add, import, export, set password policies for the admin users, and disable admin user accounts by using the links provided at the top-right corner of this page.

2. Select the Admin user you want to modify. The **Edit Admin User** dialog opens.

Figure 2. Changing the Administration User Password



3. Change the administration password, then click **Save**.

Adding an Admin User

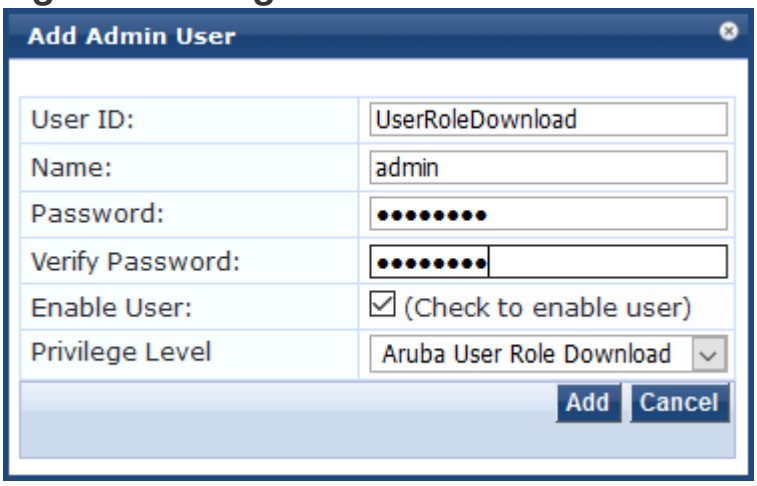
About this task

To add a new admin user:

Procedure

1. Navigate to **Administration > Users and Privileges > Admin Users**.
2. Click the **Add** link. The **Add Admin User** dialog opens.

Figure 1. Adding an Admin User



3. Specify the **Add Admin User** parameters as described in the following table, then click **Save**:

Table 1. Adding an Admin User Parameters

Parameter	Action/Description
User ID	Specify a user ID for this administrator. Valid characters allowed in the **User ID** fi eld are letters, numbers, and the special characters hyphen, underscore, period, a nd "at" character: - _ . @
Name	Specify the name for the admin user.
Password/ Verify Password	Specify a password for the local user, the n verify the password.
Enable User	You must enable this check box to enable the admin user account (is is enabled by default). Otherwise, the admin user acco unt is disabled.
Privilege Level	From the drop - down list, select one of t he following admin user privilege levels: • API Administrator • Aruba User Role Download • Help Desk • Network Administrator • Read - only Administrator • Receptionist • Super Administrator

Importing and Exporting Admin Users

You can import or export the admin user accounts by using the **Import** and **Export All** links at the top-right corner of the **Admin Users** page. You can also export specific admin user accounts by using the **Export** button that appears after selecting one or

more admin user accounts from the list. For more information, refer to Importing and Exporting Information.

> **NOTE**
> The admin user account passwords are not stored in clear text when exported to an XML file.

Setting Password Policy for Admin Users

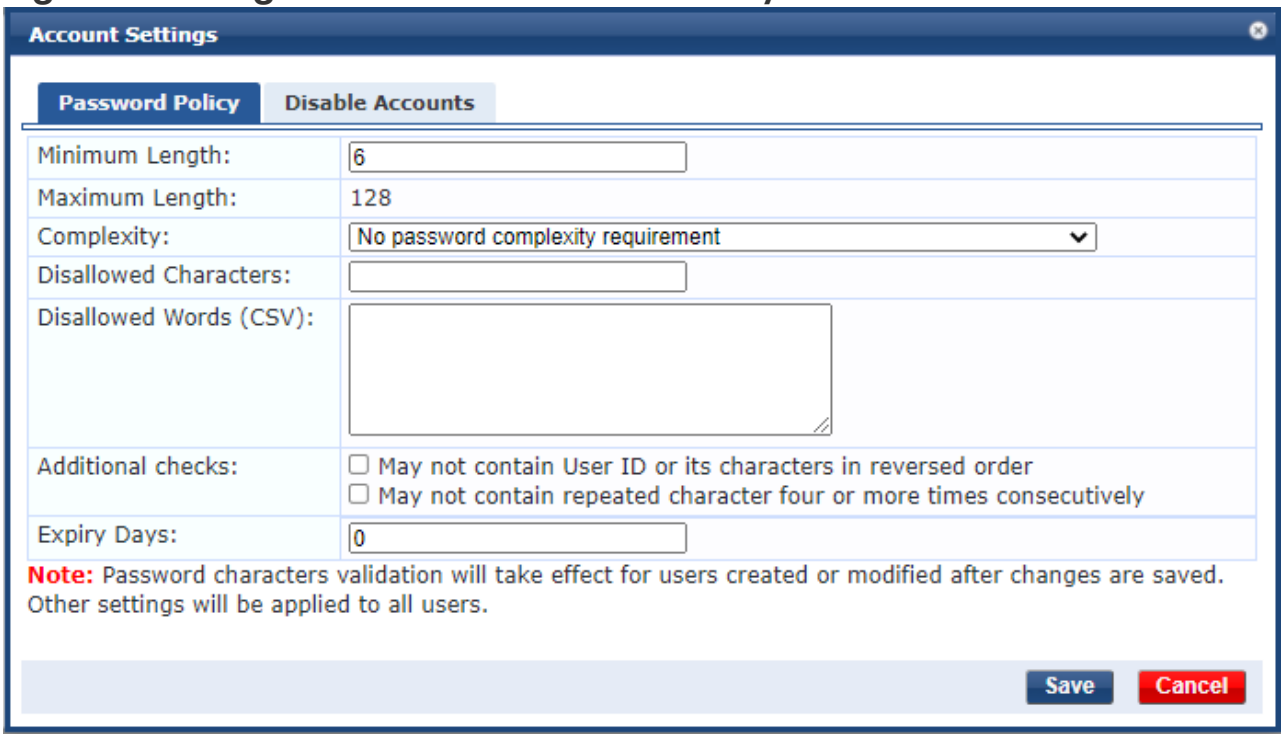
About this task

To set password policies for the administrators:

Procedure

1. Navigate to **Administration > Users and Privileges > Admin Users**.
2. Click the **Account Settings** link at the top-right corner of the **Admin Users** page. The **Password Policy Settings** dialog opens.

Figure 1. Setting Admin Users Password Policy



3. Specify the **Password Policy** parameters as described in Table 1, then click **Save**:

Table 1. Password Policy Parameters

Parameter	Action/Description
Minimum Length	Specify the minimum length required for the password.
Maximum Length	The maximum length of a password for a local or admin user is 128 characters.
Complexity	Select the complexity setting from the **C omplexity** drop - down list. The complexi ty settings can be one of the following: • No password complexity requirement • At least one uppercase and one lowe rcase letter • At least one digit • At lease one letter and one digit • At least one of each: uppercase letter, lowercase letter, digit • At least one symbol • At least one of each: uppercase letter, lowercase letter, digit, and symbol
Disallowed Characters	Specify the characters not to be allowed in the password.
Disallowed Words (CSV)	Specify the words not to be allowed in th e password
Additional Checks	Select any additional checks, if required. The options are: • May not contain User ID or its charact ers in reversed order. • May not contain repeated character four or more times consecutively.

Parameter	Action/Description
Expiry Days	Set the password expiry time for admin u sers. The allowed range is **0** to **500** days. The default value is **0**. **NOTE** If the value is set to **0**, the password never expires. For any other value, admin use rs must reset the expired pa ssword the next time they lo g in. Policy Manager alerts u sers five days before the pas sword expires.
History	Use the History field to restrict the repetit ive use of the same password. Enter a val ue from 1 - 99 to prohibit the reuse of the same password from previous passwords entered. A legacy password can only be reused once the number of new passwo rds exceeds the History value entered.

NOTE

Password Policy settings are effective only for the users created or modified after the changes are saved.

Disabling Admin User Accounts

The **Disable Account** check occurs every day at midnight. The Admin user account can be disabled in two ways:

- When the Admin user tries to log in with an invalid password for a configured number of times defined by the **Failed attempts count** parameter, the Admin user account is locked.

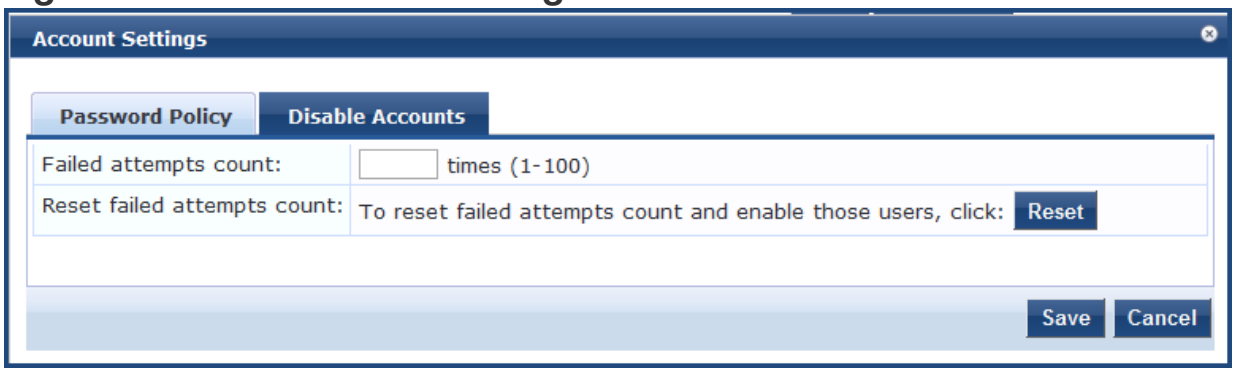
> **NOTE**
>
> If the mechanism for logging in to Policy Manager Policy Manager is Certificate + Password, the Admin user is allowed to enter the password even if the certificate is invalid.

- When the Admin user tries to log in with an invalid user certificate for a configured number of times defined by the **Failed attempts count** parameter, the Admin user account is disabled.
- To reset the **Failed attempts count** and enable a disabled Admin user account, click the **Reset** button (see Table 3).
- For Admin users whose accounts are locked due to account settings validations, and whose accounts are enabled again after being locked out, entries are logged in both the Audit Viewer (see Audit Viewer) and the Event Viewer (see Event Viewer).

To specify the conditions for disabling admin user accounts:

1. Navigate to **Administration > Users and Privileges > Admin Users**.
2. Click the **Account Settings** link at the top-right corner of the **Admin Users** page. The **Account Settings** page opens.
3. Select the **Disable Accounts** tab.The **Disable Accounts** dialog opens.

Figure 1. Disable Accounts Dialog



4. Specify the **Disable Accounts** parameters as described in Table 3, then click **Save**.

Table 1. Admin Users > Disable Accounts Parameters

Parameter	Action/Description
Failed attempts count	Specify the number of failed log - in att empts are allowed before the account i s disabled. The range is from **1** to **100** attempts.
Reset failed attempts count	To reset the failed attempts count to ze ro and reenable those admin users who were disabled after exceeding the faile d attempts count, click **Reset**.

Managing Admin Privileges

ClearPass Policy Manager provides the following default administrator privileges XML files:

> **NOTE**
> If a user privilege is changed, or a user account is disabled by an administrator, the changes do not take effect until the user's next login.

Table 1. Default Admin Privileges

Parameter	Description
API Administrator	An API administrator is allowed API access only with read/write privileges for all config uration elements.
Aruba User Role Download	This is the privilege level used for Aruba Us er Role Download API operations.
Help Desk	The Help Desk person logs in to troublesho ot problems reported by end users.
Help Desk - ASE	The Help Desk person logs in to troublesho ot problems reported by end users.

Parameter	Description
Network Administrator	Allowed to configure all the policies in Poli cy Manager.
Privileged user	Allowed to add Network Access Devices (N ADs), NAD groups, and edit endpoints, as well as monitoring operations.
Read - only Administrator	This administrator is allowed to read config uration elements but not modify them.
Receptionist	The receptionist is allowed access to the m ain monitoring screens.
STS - Operations	Allowed access to the STS - Operations 7 x 24 team.
Super Administrator	This administrator is allowed read/write ac cess to all configuration elements.

Each of these default admin privileges define the admin privileges for Policy Manager and Policy Manager Insight. The default set of admin privileges cannot be modified. You can export one or more default files and modify the file to create a customized administrator privileges file. Customized administrator privileges are defined in an XML file with a specific format and then imported into ClearPass Policy Manager on the **Admin Privileges** page.

Subtopics

Defining Custom Admin Privileges
Configuring Policy Manager Admin Privileges
Configuring Insight Admin Privileges
Creating Custom Administrator Privileges
Administrator Privilege XML File Structure
Administrator Privileges and Task IDs
Sample Administrator Privilege XML File

Defining Custom Admin Privileges

About this task

When a different set of admin privileges is needed (for example, if you require different admin privileges for the Report module than for the other Insight modules), you must create a new admin privileges administrator.

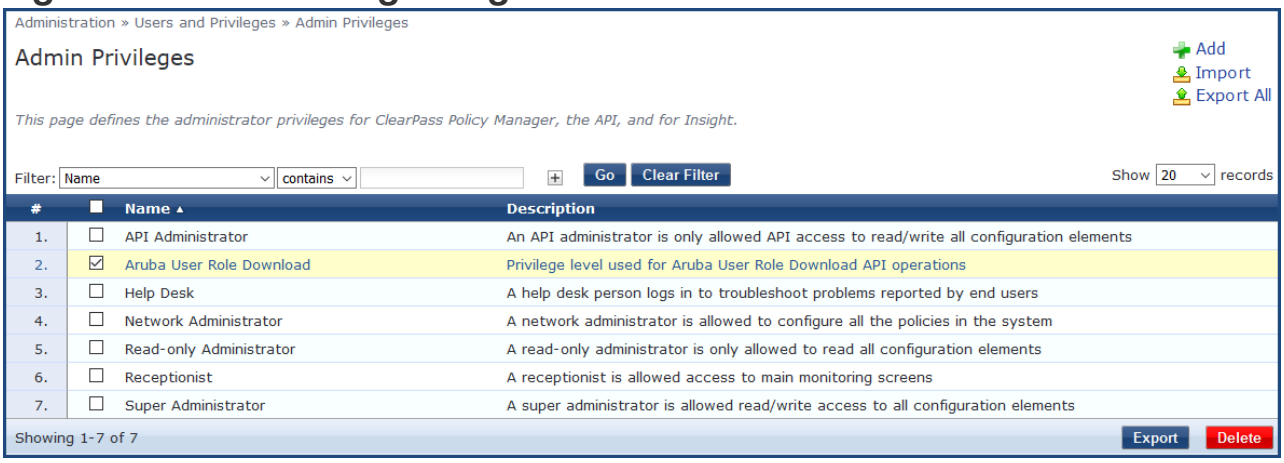
To define custom admin privileges for Policy Manager and Insight:

Procedure

1. Navigate to the **Administration > Users and Privileges > Admin Privileges** page.

   The **Admin Privileges** page opens:

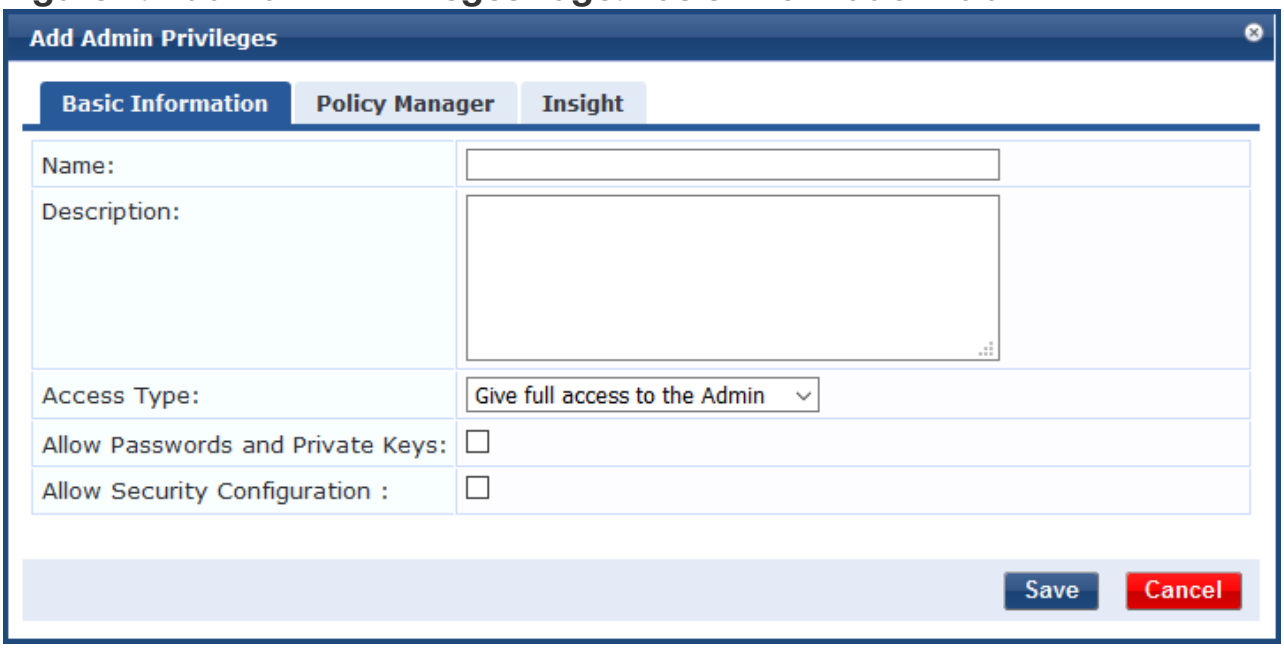
Figure 1. Admin Privileges Page



2. Click the **Add** link.

   The **Add Admin Privileges** dialog opens.

Figure 2. Add Admin Privileges Page: Basic Information Tab



3. Specify the parameters in the **Basic Information** tab.

Table 1. Add Admin Privileges Parameters: Basic Information Tab

Parameter	Action/Description
Name	Enter the name of the Admin Privileges a dministrator.
Description	Provide a description of this new admin p rivileges administrator.
Access Type	Select one of the following Access Types: • Give full access to the Admin • Give UI access to the Admin • Give API access to the Admin
Allow Passwords and Private Keys	Select this check box if you want to allow password access and control the ability t o export key pairs from Policy Manager.
Allow Security Configuration	Select this check box to allow Security c onfiguration. **NOTE** To be able to export the Se rver certificate and the Servi ce and Client certificates fro m the Certificate Store, you must enable both the **Allow Passwords and Private Keys** option and the **Allow Securi ty Configuration** option. **NOTE** The admin user must log out and log in for the changes to take effect.

Configuring Policy Manager Admin Privileges

About this task

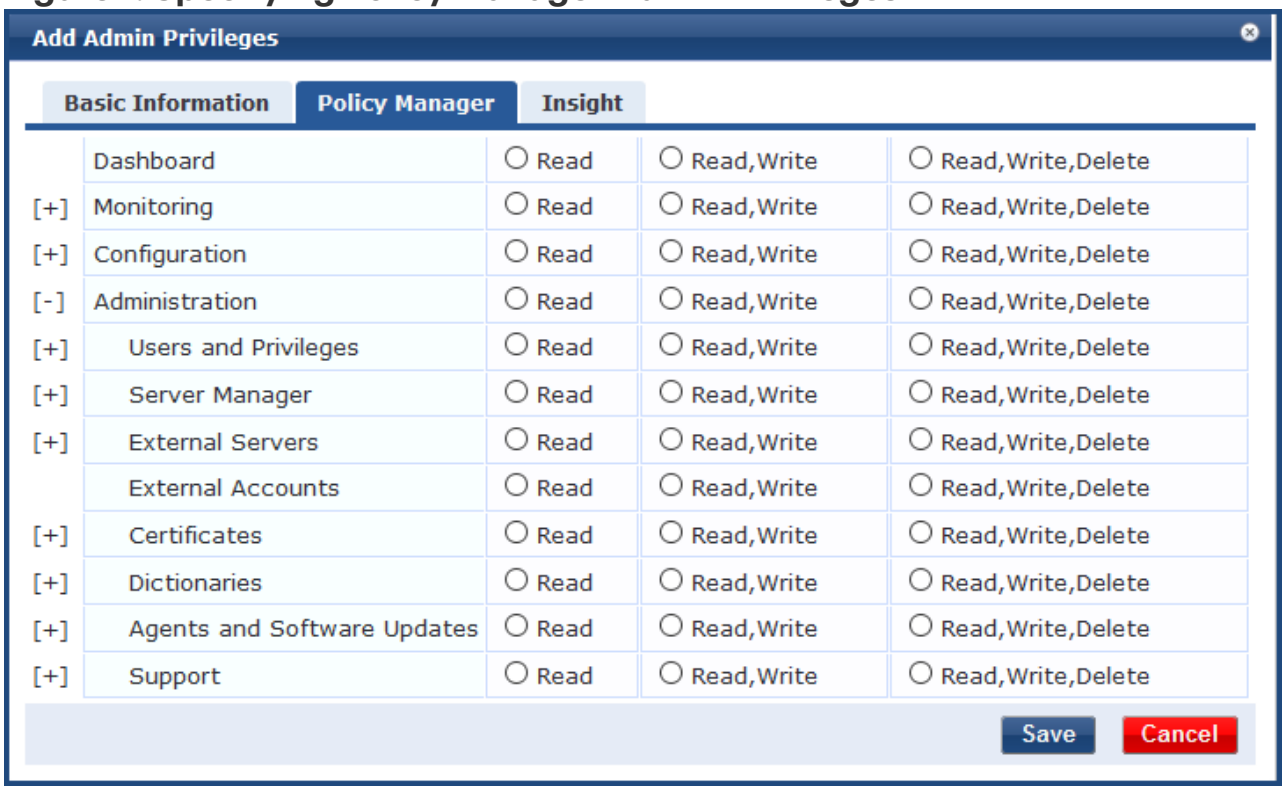
To configure the Policy Manager admin privileges:

Procedure

1. Select the **Policy Manager** tab.

   The following dialog opens:

Figure 1. Specifying Policy Manager Admin Privileges



2. Specify the admin privileges for each of the Policy Manager Policy Manager components, then click **Save**.

Configuring Insight Admin Privileges

About this task

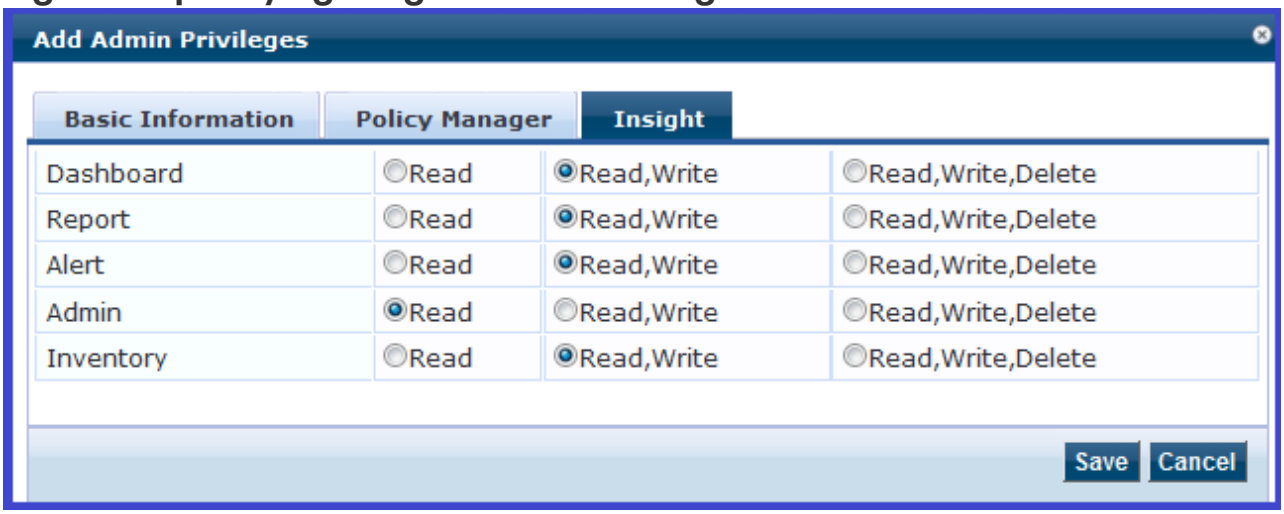
To configure the Insight admin privileges:

Procedure

1. Select the **Insight** tab.

   The following dialog opens:

Figure 1. Specifying Insight Admin Privileges



2. Specify the admin privileges for each of the Insight modules, then click **Save**.

Creating Custom Administrator Privileges

About this task

To create a custom admin privilege XML file, you must use a plain text or XML editor.

> **CAUTION**
> Do not use word-processing applications such as Microsoft Word. These applications can introduce tags and corrupt the XML file.

To create a custom administrator privilege:

Procedure

1. Create an XML file that defines a privilege.
2. Store the new file.
3. Navigate to **Administration > Users and Privileges > Admin Privileges**.
4. Click **Import Admin Privileges**.

5. Import the administrator privilege file you created in step 1.

   After you complete steps 1 through 5, the new administrator privileges document is displayed on the **Admin Privileges** page.

Administrator Privilege XML File Structure

Admin privilege files are XML files with a specific structure. It must have a header at the beginning of the file in the following format:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
```

The root tag is `TipsContents`. It is a container for the data in the XML file and must be in the following format:

```
<TipsContents xmlns="http://www.avendasys.com/tipsapiDefs/1.0">
⋮
</TipsContents>
```

An optional `TipsHeader` tag can follow the `TipsContents` tag. The actual admin privileges information is defined with the `AdminPrivilege` and `AdminTask` tags. You can use one `AdminPrivilege` tag for each admin privilege you want to define. The `AdminPrivilege` tag contains the following two attributes:

- `name`
- `description`

  You can have one or more `AdminTask` tags inside the `AdminPrivilege` tag. Each `AdminTask` tag defines a lace within the ClearPass Policy Manager application that a user with that privilege can view or change. The `AdminTask` tag contains one `taskid` attribute and a single `AdminTaskAction` tag. The `AdminTaskAction` tag contains an attribute, `type` which can take a value, `RO` (read only) or `RW` (read/write).

The following sample gives the basic structure of an admin privilege file:

```
<AdminPrivileges>
  <AdminPrivilege name="" description="">
    <AdminTask taskid="">
      <AdminTaskAction type=""/>
    </AdminTask>
    <AdminTask taskid="">
      <AdminTaskAction type=""/>
    </AdminTask>
```

```
  </AdminPrivilege>
</AdminPrivileges>
```

Administrator Privileges and Task IDs

Every element in the ClearPass Policy Manager user interface has a task ID associated with it. The users have access to the elements based on the permissions set for each task or element. By default, any permission provided for a task is applicable for all its sub-tasks.

For example, if you give RW (read-write) permissions for the task, *Enforcements* (con.en), it is automatically applied to its subtasks, *Policies* (con.en.epo) and *Profiles* (con.en.epr). You need not explicitly define the same permission for those subtasks.

The following table provides the tasks and subtasks of ClearPass Policy Manager and their associated task IDs:

Table 1. Administrator Privileges and Task IDs

Area (ClearPass Policy Manager Menu)	Task ID
Dashboard	did
Monitoring	mon
Live Monitoring	mon.li
• Access Tracker	mon.li.ad
• Accounting	mon.li.ac
• OnGuard Activity	mon.li.ag
• Analysis and Trending	mon.li.sp
• Endpoint Profiles	mon.li.ep
• System Monitor	mon.li.sy
Audit Viewer	mon.av
Denylist Users	mon.bl

Area (ClearPass Policy Manager Menu)	Task ID
Event Viewer	mon.ev
Data Filters	mon.df
Configuration	con
Start Here (Services Wizard)	con.sh
Services	con.se
Service Templates	con.st
Authentication	con.au
• Methods	con.au.am
• Sources	con.au.as
Identity	con.id
• Single Sign - On	con.id.sso
• Local Users	con.id.lu
• Endpoints	con.id.ep
• Static Host Lists	con.id.sh
• Roles	con.id.rs
• Role Mappings	con.id.rm
Posture	con.pv
• Posture Policies	con.pv.in
• Posture Servers	con.pv.ex

Area (ClearPass Policy Manager Menu)	Task ID
• Audit Servers	con.pv.au
Enforcements	con.en
• Policies	con.en.epo
• Profiles	con.en.epr
Network	con.nw
• Devices	con.nw.nd
• Device Groups	con.nw.ng
• Proxy Targets	con.nw.pr
Policy Simulation	con.ps
Profile Settings	con.prs
Administration	adm
User and Privileges	adm.us
• ClearPass Portal	adm.po.cp
• Admin Users	adm.us.au
• Admin Privileges	adm.us.ap
Server Manager	adm.mg
• Server Configuration	adm.mg.sc
• Log Configuration	adm.mg.ls
• Local Shared Folders	adm.mg.sf

Area (ClearPass Policy Manager Menu)	Task ID
• Licensing	adm.mg.li
External Servers	adm.xs
• SNMP Trap Receivers	adm.xs.st
• Syslog Targets	adm.xs.es
• Syslog Export Filters	adm.xs.sx
• Messaging Setup	adm.xs.me
• Endpoint Context Servers	adm.xs.cs
• Context Server Actions	adm.di.csa
Certificates	adm.cm
• Server Certificate	adm.cm.mc
• Trust List	adm.cm.ctl
• Revocation List	adm.cm.crl
Dictionaries	adm.di
• RADIUS	adm.di.rd
• Posture	adm.di.pd
• TACACS+ Services	adm.di.td
• Fingerprints	adm.di.df

Area (ClearPass Policy Manager Menu)	Task ID
• Attributes	adm.di.at
• Applications	adm.di.ad
Agents and Software Updates	adm.po
• OnGuard Settings	adm.po.aas
• Software Updates	adm.po.es
Support	adm.su
• Contact Support	adm.su.cs
• Remote Assistance	adm.su.ra
• Documentation	adm.su.doc

Sample Administrator Privilege XML File

This section provides sample XML files with different admin privileges for various user interface elements

Read Only (R) Privileges to All Sections

The following sample provides Read Only (R) privileges to all the sections (dnd, con, mon, adm):

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<TipsContents xmlns="http://www.avendasys.com/tipsapiDefs/1.0">
<TipsHeader exportTime="Thu Jul 26 17:57:50 IST 2016" version="6.6"/>
  <AdminPrivileges>
    <AdminPrivilege name="Read-only Administrator" description="A read-only
administrator is only allowed to read all configuration elements">
      <AdminTask taskid="con"> //Refers to Configuration
        <AdminTaskAction type="R"/>
```

```
      </AdminTask>
      <AdminTask taskid="dnd"> //Refers to DashBoard
        <AdminTaskAction type="R"/>
      </AdminTask>
      <AdminTask taskid="mon"> //Refers to Monitoring
        <AdminTaskAction type="R"/>
      </AdminTask>
      <AdminTask taskid="adm"> //Refers to Administration
        <AdminTaskAction type="R"/>
      </AdminTask>
    </AdminPrivilege>
  </AdminPrivileges>
</TipsContents>
```

Read/Write Access

The following sample provides Read/Write access only to Guest, Local and Endpoint Repository:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<TipsContents xmlns="http://www.avendasys.com/tipsapiDefs/1.0">
<TipsHeader exportTime="Thu Jul 26 17:57:50 IST 2016" version="6.6"/>
  <AdminPrivileges>
    <AdminPrivilege name="Read/Write Access to Guest, Local and Endpoint
Repository" description="A read-only administrator is only allowed to read
all configuration elements">
      <AdminTask taskid="con.id.lu"> //Refers to Local Users Section
        <AdminTaskAction type="RW"/>
      </AdminTask>
      <AdminTask taskid="con.id.gu"> //Refers to Guest Users Section
        <AdminTaskAction type="RW"/>
      </AdminTask>
      <AdminTask taskid="con.id.ep"> //Refers to Endpoints Section
        <AdminTaskAction type="RW"/>
      </AdminTask>
    </AdminPrivilege>
  </AdminPrivileges>
</TipsContents>
```

Read/Write Permissions

The following sample provides Read/Write permissions to DashBoard/ Monitoring and ReadOnly permissions to Server Configuration:

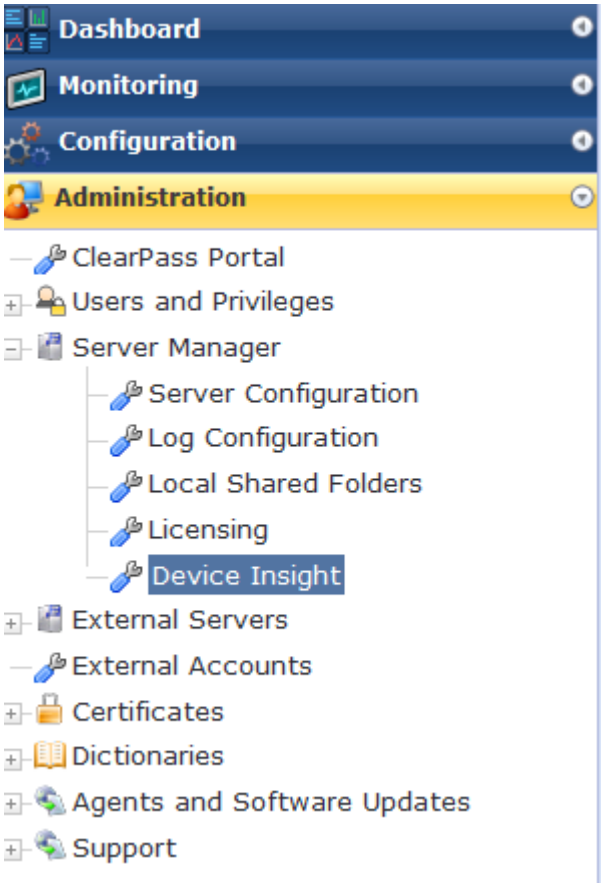
```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<TipsContents xmlns="http://www.avendasys.com/tipsapiDefs/1.0">
<TipsHeader exportTime="Thu Jul 26 17:57:50 IST 2016" version="6.6"/>
  <AdminPrivileges>
    <AdminPrivilege name="Limited access permission" description="A read-
only administrator is only allowed to read all configuration elements">
      <AdminTask taskid="dnd"> //Refers to DashBoard
        <AdminTaskAction type="RW"/>
      </AdminTask>
      <AdminTask taskid="mon"> //Refers to Monitoring
        <AdminTaskAction type="RW"/>
      </AdminTask>
      <AdminTask taskid="adm.mg.sc"> //Refers to Server Configuration
        <AdminTaskAction type="R"/>
      </AdminTask>
    </AdminPrivilege>
  </AdminPrivileges>
</TipsContents>
```

Server Manager

This section provides the following information:

- Server Configuration Operations
- Log Configuration
- Downloading Local Shared Folders
- Device Insight

Figure 1. Server Manager Menu



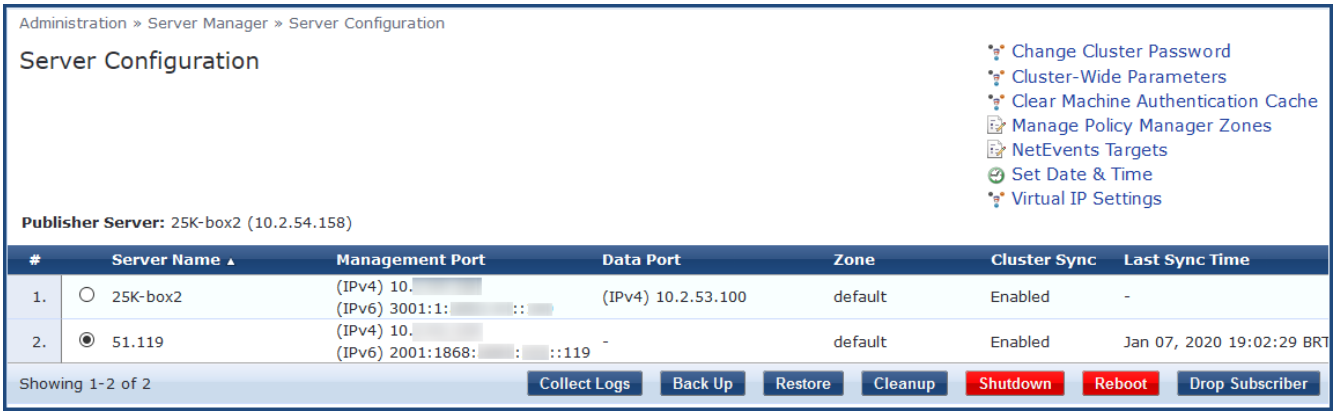
Subtopics

Server Configuration Operations
Log Configuration
Downloading Local Shared Folders
Device Insight

Server Configuration Operations

You can perform numerous server configuration tasks by navigating to **Administration > Server Manager > Server Configuration**.

Figure 1. Server Configuration Page



This section describes the following server configuration tasks:

- Configuring the Policy Manager Server
- Changing the Cluster-Wide Password
- Cluster-Wide Parameters
- Clearing Machine Authentication Cache
- Managing Policy Manager Zones
- Configuring NetEvents Targets
- Synchronizing the Cluster Date and Time with the NTP Server
- Configuring Virtual IP Settings
- Making a Subscriber Node
- Upload Nessus Plugins
- Collecting Logs
- Backing Up the Policy Manager Database
- Restoring Policy Manager Configuration Data
- Performing a System Cleanup
- Shutting Down or Rebooting the Server
- Configuration Tasks for Disabled Nodes in a Cluster

Subtopics

Configuring the Policy Manager Server
ClearPass Policy Manager Cluster Configuration Options
Collecting Logs
Backing Up the Policy Manager Database

Restoring Policy Manager Configuration Data
Performing a System Cleanup
Shutting Down or Rebooting the Server

Configuring the Policy Manager Server

About this task

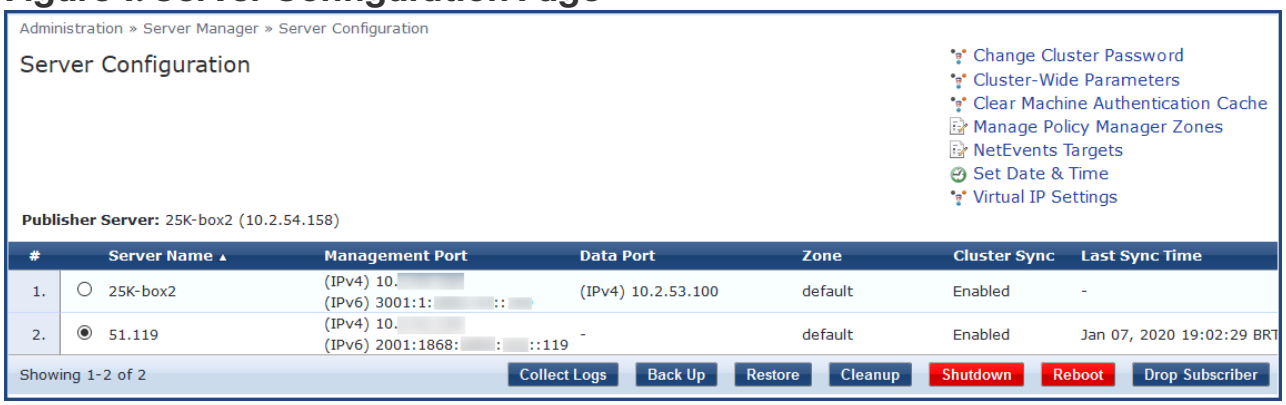
To modify the configuration settings of a Policy Manager server:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.

   The **Server Configuration** page opens:

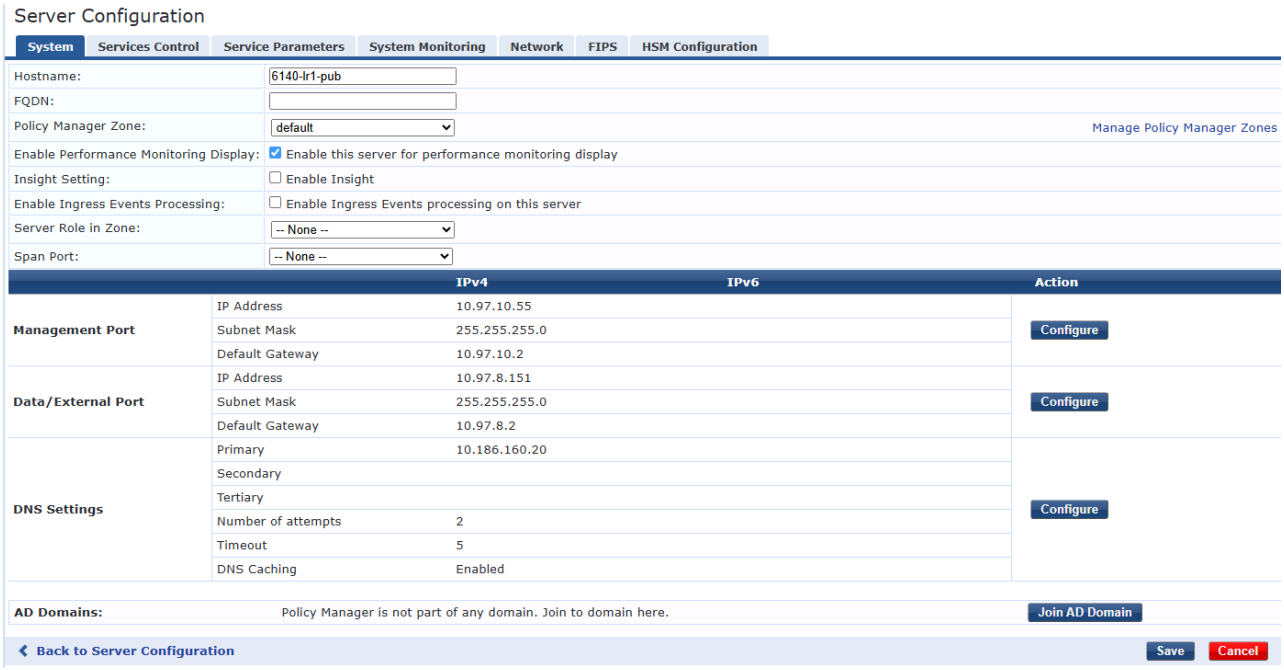
   Figure 1. Server Configuration Page



2. Select the Policy Manager server name of interest.

   The **Server Configuration** page for the selected server opens:

   Figure 2. Server Configuration Screen - System tab



Subtopics

Policy Manager Server Configuration Tabs
Cluster Configuration Options
System Page
Services Control Page
Service Parameters Page
System Monitoring Page
Network Page
FIPS Page

Policy Manager Server Configuration Tabs

For details on viewing or modifying Policy Manager server configuration settings, refer to the following sections:

- System Page
- Services Control Page
- Service Parameters Page
- System Monitoring Page
- Network Page

- FIPS Page

Cluster Configuration Options

For details on the cluster configuration options, see ClearPass Policy Manager Cluster Configuration Options.

System Page

About this task

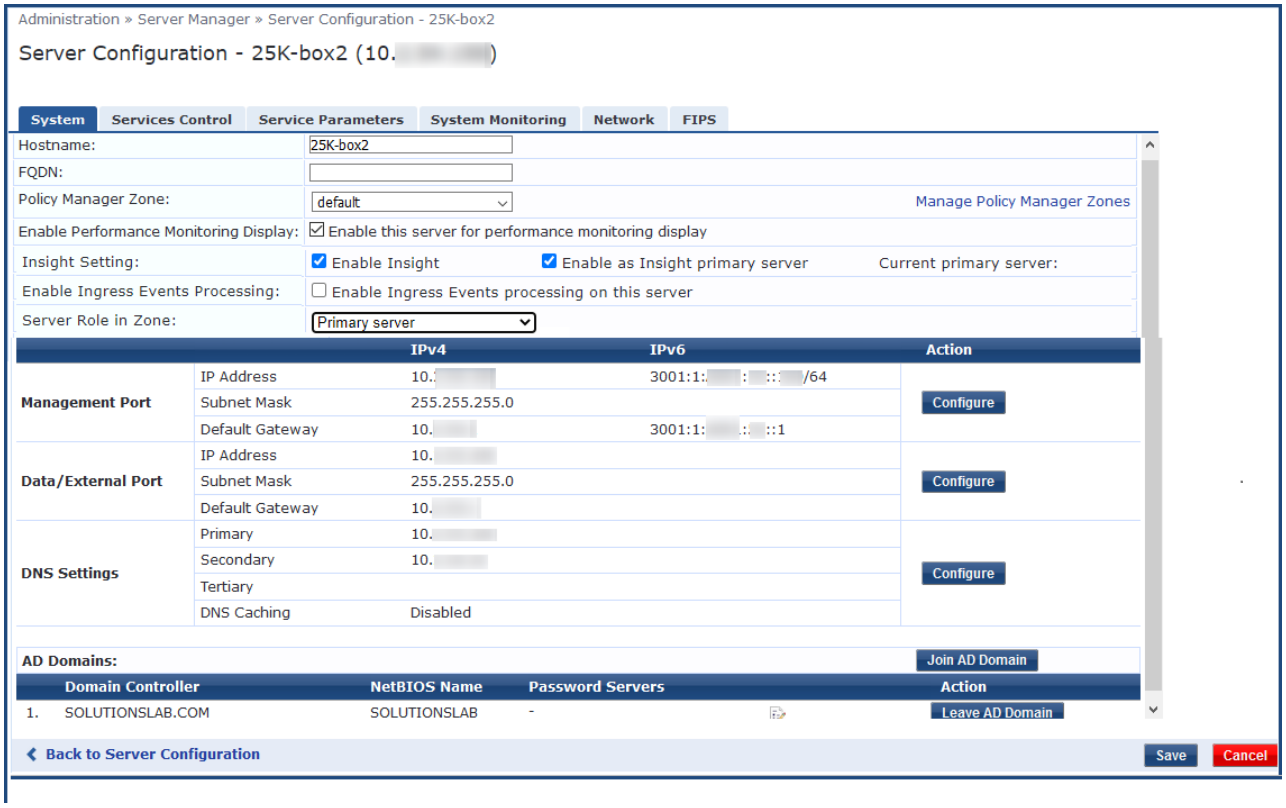
Use this page to manage port configurations, join an Active Directory domain, or set a server password. For more information on ClearPass port interfaces, see also Data and Management Port Interfaces

To configure the **Server Configuration > System** page parameters:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest. The **Server Configuration** page opens onto the **System** page (see Figure 1).

Figure 1. Server Configuration > System Page



3. Specify the **Server Configuration > System** page parameters as described in the following table, then click **Save**:

Table 1. Server Configuration > System Page Parameters

Parameter	Action/Description
Hostname	Specify the host name of the Policy Manager server. Although you do not need to enter the fully qualified domain name (FQDN) in this field, when you create a certificate signing request, the request uses the information in the hostname field as the common name (CN) by default. If the hostname field does not use a FQDN, the common name field in CSR requests must be manually updated to include a proper FQDN.

NOTE

Users should be aware that, when configuring a hostname that includes a period cha

Parameter	Action/Description
	racter (.), the substring be fore the first period charact er must be unique for each appliance. This is because a hostname field that includes a period character is interp reted to be a Fully Qualified Domain Name (FQDN), in w hich case the substring befo re the first period character i s the hostname. Examples of valid hostname configuratio ns: • cppm1.arubanetworks.com • cppm2.arubanetworks.com Examples of invalid hostname configurati ons: • cppm1.santaclara.arubanetworks.com • cppm1.bangalore.arubanetworks.com
FQDN	(Optional) Enter the Fully - Qualified Dom ain Name (FQDN) of the Policy Manager s erver.
Policy Manager Zone	To specify a Policy Manager zone, select a previously configured zone from the d rop - down list, then click the **Manage Po licy Manager Zones** link. If you need to add a Policy Manager zon e, click the **Manage Policy Manager Zon es** link. For more information on adding zones, see Adding Policy Manager Zones.
Enable Performance Monitoring Display	To enable the ClearPass Policy Manager s erver to perform performance monitorin

Parameter	Action/Description
	g, select the **Enable this server for performance monitoring display** check box.
Insight Setting	To enable the Insight reporting tool on this node, select the **Enable Insight** check box. **NOTE** • When you enable this check box for Insight on a node in a cluster, the [Insight Repository] configuration is updated automatically to point to the management IP address of that Policy Manager server. • When this check box is enabled for other servers in the cluster, they are added as backups for the same authentication source. • The order of the primary and backup servers in the [Insight Repository] is the same order in which the user enables Insight on the server.
Enable as Insight Primary Server	To specify the current server in the cluster as an Insight Primary server, select this check box. **NOTE**

Parameter	Action/Description
	This option is available only when **Insight Setting > Enable Insight** is enabled.
Enable Ingress Events Processing	Check this check box to enable ingress events processing on this server. For more information, see Enabling Ingress Events Processing.
Server Role in Zone	Use this option to specify the current Policy Manager server as the Primary or Secondary master Policy Manager server within a Policy Manager zone. **NOTE** If no Primary master server for a zone is configured, the Policy Manager server with the lowest UUID is designated as the Primary master server. To do so, select **Primary master** or **Secondary master** from the drop - down list. • The Primary Policy Manager server in the zone distributes the scan requests to all the nodes in the zone, depending on the number of seed devices (Network Discovery) or the number of networks/subnets (for subnet scans) configured. **NOTE** If the Primary server goes down, the secondary server assumes the role of the Primary server.

Parameter	Action/Description
	• Each scan configuration that is added is distributed by the master Policy Ma nager server to a different node in the zone. • If one scan configuration has multiple seed devices (Network Discovery), sc an requests are distributed to other n odes in the zone based on the numbe r of ARP entries received from the see d devices. • If one scan configuration has multiple subnets configured (Subnet Scan), sc an requests are distributed to other n odes in the zone.
Span Port	If necessary, select a port for DHCP span ning. On selecting a port, the **Enable TC P/ARP Fingerprinting** check box appears. This field is optional. **NOTE** Starting wiith Policy Manage r 6.10.2, the Policy Manager server no longer needs to h ave Device Insight integratio n disabled in order to define a span port on a Policy Mana ger server. In Policy Manage r 6.10.0 and 6.10.1, when **Dev ice Insight Integration** is en abled, the **Span Port** option is not available. (for more in formation, see Device Insigh t Integration Page).
Management Port	To configure the Management Port para meters, click **Configure.** The **Configure**

Parameter	Action/Description
	Management Port dialog opens. For deta ils, see Management Port Configuration.
Data/External Port	To configure the Data/External port, click **Configure.** For details, see Data/External Port Config uration.
DNS Settings	To configure the DNS settings, click **Conf igure**. For details, see DNS Settings Configurati on.
AD Domains	Displays a list of the joined Active Directo ry domains. To join an active directory domain, click **Join Domain**.

Management Port Configuration

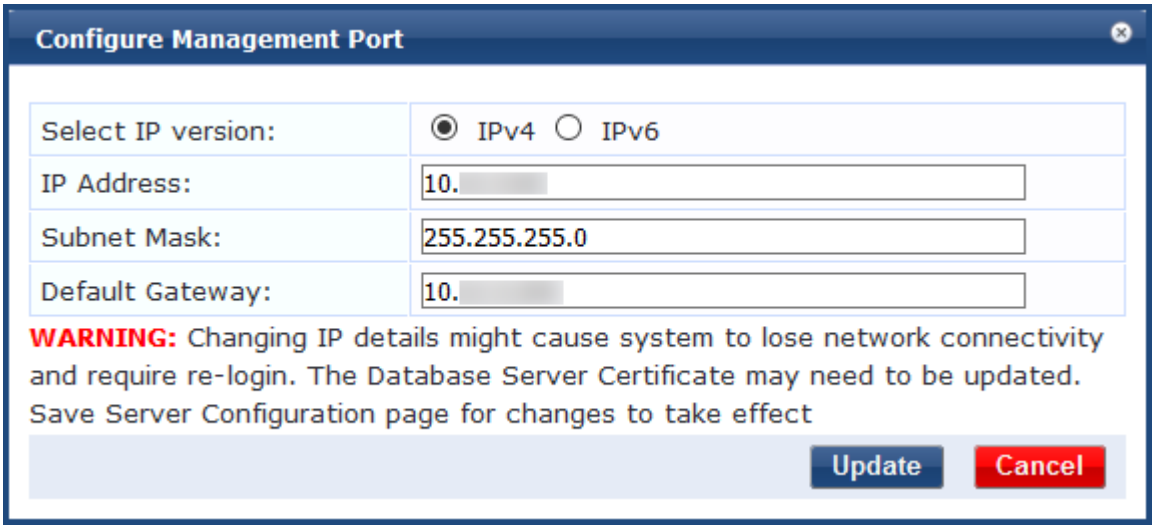
About this task

To configure the Policy Manager server's Management port:

Procedure

1. From the **Administration > Server Manager > Server Configuration > System > Management Port** section, click **Configure**. The **Configure Management Port** dialog opens.

Figure 2. Configure Management Port Dialog (IPv4)



2. Specify the **Configure Management Port** parameters for IPv4 as described in the following table:

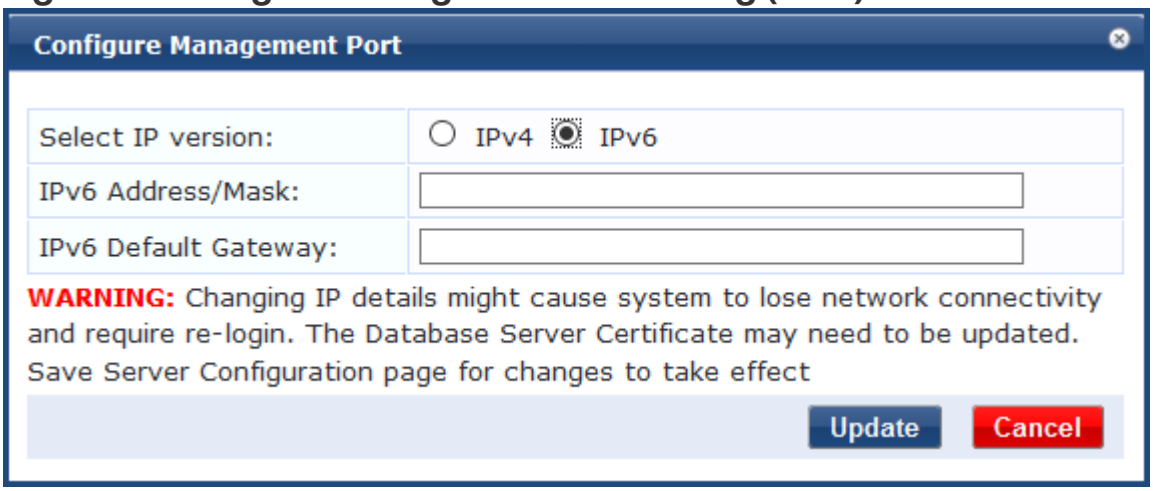
Table 2. Configure Management Port Parameters (IPv4)

Parameter	Action/Description
Select IP Version	Select **IPv4**.
IP Address	Specify the IPv4 address is used to acc ess the ClearPass Policy Manager server's management port. When a new hardware appliance is first powered on, it already s hows an existing management IP address . This IP address is set during the manufa cturing process, and is replaced when yo u enter your management IP address duri ng appliance configuration.
Subnet Mask	Specify the management interface subne t mask for an IPv4 address.
Default Gateway	Specify the default gateway for the mana gement interface. A default gateway serves as an access po int or IP router that a networked device u ses to send information to a device in an other network or the Internet.

Parameter	Action/Description
Warning	Changing IP details might cause the system to lose network connectivity and require re - login. The Database Server Certificate may need to be updated. Save the Server Configuration page for changes to take effect.

3. Click **Update**.

Figure 3. Configure Management Port Dialog (IPv6)



4. Specify the **Configure Management Port** parameters for IPv6 as described in the following table:

Table 3. Configure Management Port Parameters (IPv6)

Parameter	Action/Description
Select IP Version	Select **IPv6**.
IPv6 Address/Mask	Specify the IPv6 address that is used to access the ClearPass Policy Manager server's management port. To configure the IPv6 address, specify the IPv6address with prefix length: < *ipv6address\prefixlength* > **NOTE**

Parameter	Action/Description
	IPv6 addresses don't require a netmask as they use Class less Inter - Domain Routing (CIDR).
IPv6 Default Gateway	Specify the default gateway for the mana gement interface. A default gateway serves as an access po int or IP router that a networked device u ses to send information to a device in an other network or the Internet.
Warning	Changing IP details might cause the syst em to lose network connectivity and req uire re - login. The Database Server Certi ficate may need to be updated. Save the Server Configuration page for changes t o take effect.

Data/External Port Configuration

About this task

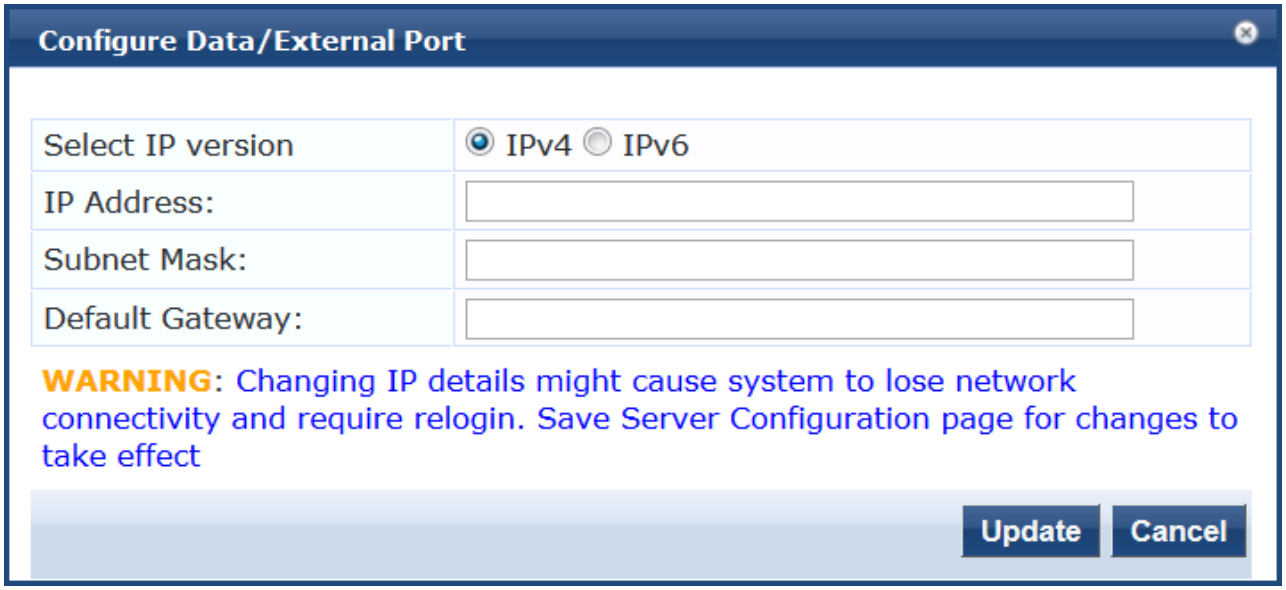
To configure the Policy Manager server's Data/External port:

Procedure

1. From the **Server Configuration > System > Data/External Port** section, click **Configure**.

   The **Configure Data/External Port** dialog opens.

   Figure 4. Configure Data/External Port Dialog (IPv4)



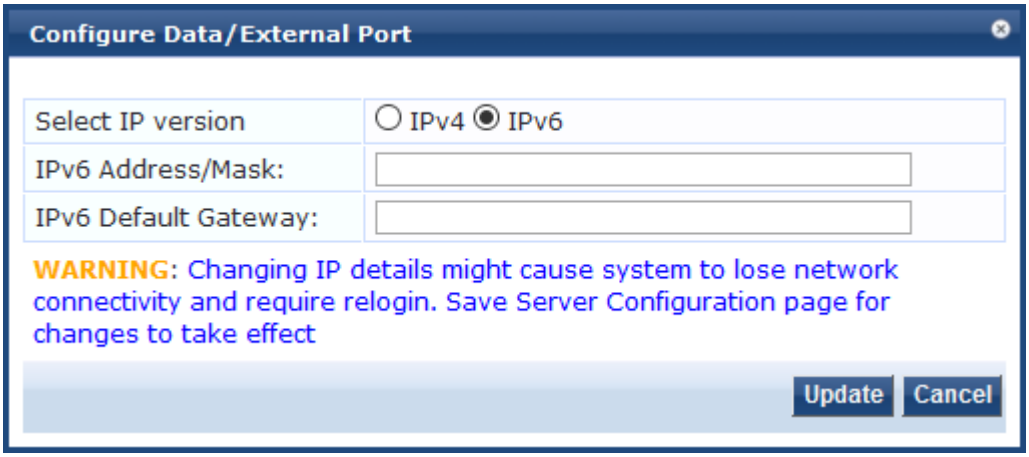
2. Specify the **Configure Data/External Port** parameters for IPv4 as described in the following table:

Table 4. Configure Data/External Port Parameters (IPv4)

Parameter	Action/Description
Select IP Version	Select **IPv4** .
IP Address	Specify the IPv4 address of the Policy Manager server's data interface.
Subnet Mask	Specify the management interface subnet mask for an IPv4 address.
Default Gateway	Specify the IPv4 address of the default gateway for the data interface.

3. Click **Update**.

Figure 5. Configure Data/External Port Dialog (IPv6)



4. Specify the **Configure Data/External Port** parameters for IPv6 as described in the following table:

Table 5. Configure Data/External Port Parameters (IPv6)

Parameter	Action/Description
Select IP Version	Select **IPv6**.
IPv6 Address/Mask	Specify the IPv6 address that is used to access the ClearPass Policy Manager server's data port. To configure the IPv6 address, specify the IPv6address with prefix length: < *ipv6address\prefixlength* > **NOTE** IPv6 addresses don't require a netmask as they use Classless Inter - Domain Routing (CIDR).
IPv6 Default Gateway	Specify the default gateway for the data /external interface. A default gateway serves as an access point or IP router that a networked device uses to send information to a device in another network or the Internet.

DNS Settings Configuration

About this task

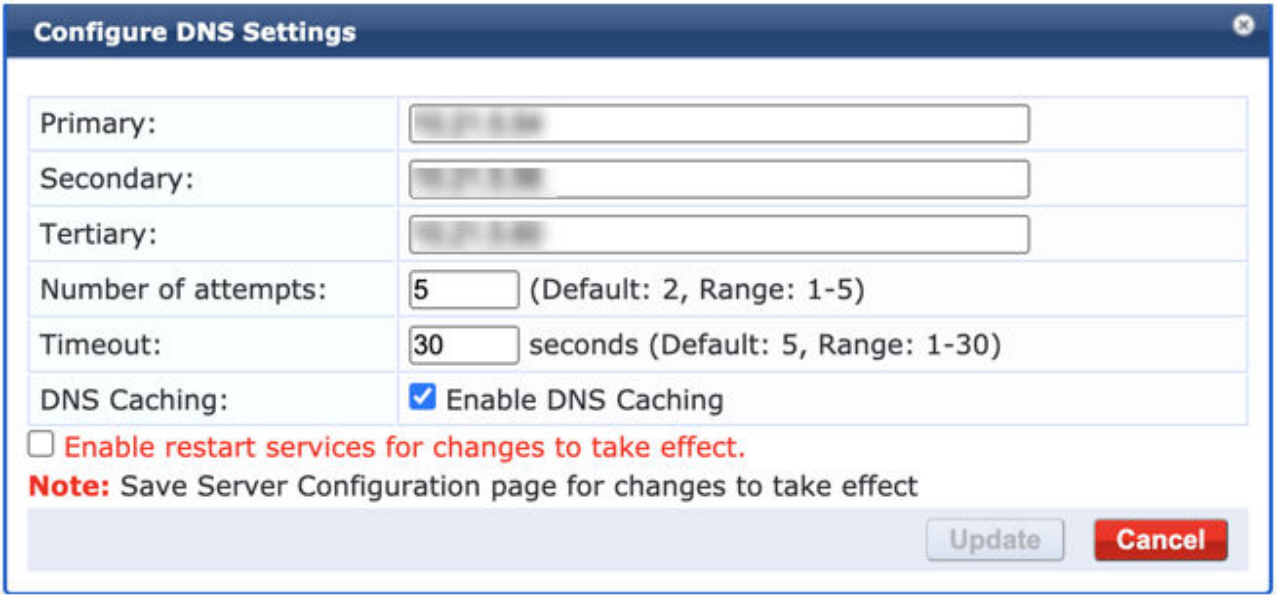
To configure the Policy Manager server's DNS (Domain Name System) settings:

Procedure

1. From the **Server Configuration** page > **System** tab > **DNS Settings**, click **Configure**.

   The **Configure DNS Setting** dialog opens.

Figure 6. Configure DNS Settings Dialog



2. Specify the **Configure DNS Settings** parameters as described in the following table:

Table 6. Configure DNS Settings Parameters

Parameter	Action/Description
Primary	Specify the primary DNS server for name look - up. You can specify an IPv4 or IPv6 address. **NOTE** A DNS server can be primary for one domain and seconda ry for another. Only one DNS server should be configured as primary for a domain, but

Parameter	Action/Description
	you can have any number of secondary DNS servers.
Secondary	Specify one or more secondary DNS se rvers for name look - up. You can specify an IPv4 or IPv6 address.
Tertiary	Optionally, in the rare event of both the p rimary and secondary DNS servers going down, you can configure a tertiary DNS s erver. You can specify an IPv4 or IPv6 add ress. **NOTE** The recommended practice is to configure the primary a nd secondary DNS servers o n separate machines, on se parate Internet connections, and in separate geographic l ocations.
Number of attempts	Maximum number of DNS retry attempts. The supported range is 1 - 15 attempts, a nd the default is 2.
Timeout:	DNS timeout duration, in seconds. The su pported range is 1 - 30 seconds, and the default is 5.
DNS Caching	If enabled, Policy Manager will cache DN S entries from the DNS server for the Tim e to live (TTL) value for each entry, reduc ing the number of queries to the DNS se rver. This feature is enabled by default.

3. To save your changes, select **Enable restart services for changes to take effect**, then click **Update**. Note that you also have to save changes to the server configuration page to update your DNS settings.

Join AD Domain Configuration

You can join Policy Manager to an Active Directory (AD) domain to authenticate users and computers that are members of an Active Directory domain. If you join Policy Manager to an Active Directory domain, it creates an account for the Policy Manager node in the Active Directory database. Users can then authenticate into the network using 802.1X and EAP methods, such as PEAP-MSCHAPv2, with their own their own Active Directory credentials.

If you need to authenticate users belonging to multiple Active Directory forests or domains in your network, and there is no trust relationship between these entities, then you must join Policy Manager to each of these untrusted forests or domains.

> **NOTE**
> Policy Manager is not required to join multiple domains belonging to the same Active Directory forest because a one-way trust relationship exists between those domains. In this case, Policy Manager can join the root domain.

Policy Manager can join or leave an Active Directory domain by using the following two buttons in the **Server Configuration** page > **System** tab:

- **Join Domain**: Click **Join Domain** to join this Policy Manager appliance to an Active Directory domain.
- **Leave Domain**: If the server is already part of multiple Active Directory domains, click **Leave Domain** to disassociate this Policy Manager appliance from an Active Directory domain.

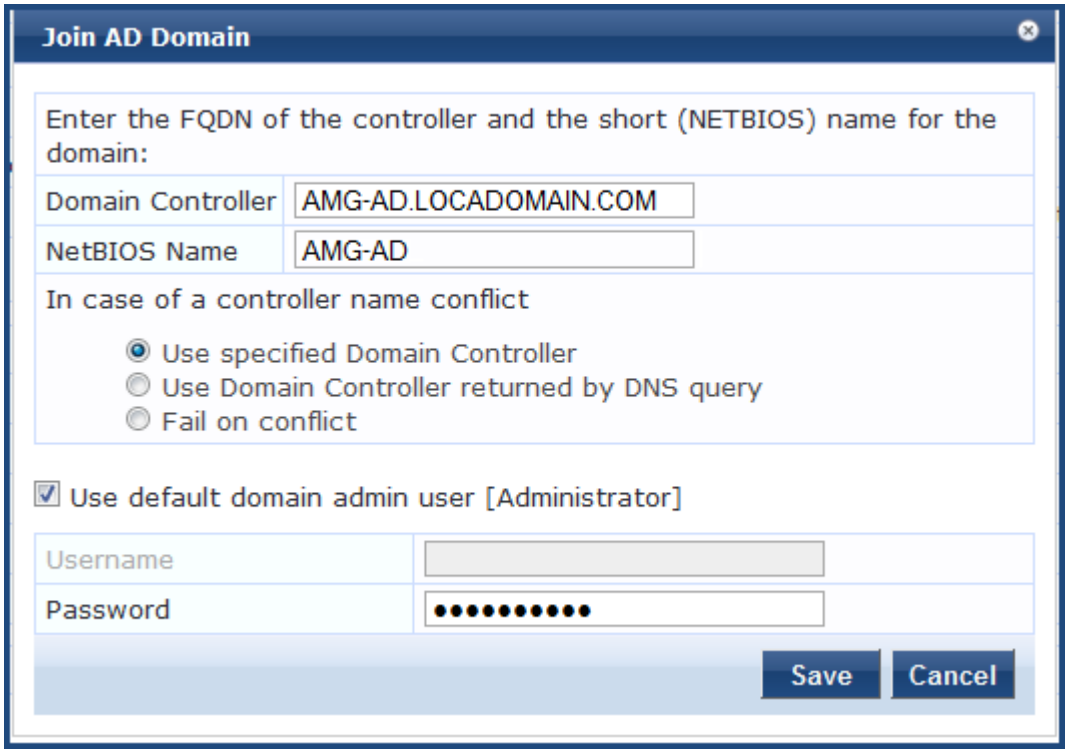
> **NOTE**
> For most use cases, if you have multiple nodes in the cluster, you must join each node to the same Active Directory domain. Always join to the domain controller that is physically closest to the Policy Manager server, otherwise authentication might fail or the performance might be impacted.

To join the selected Policy Manager server to an Active Directory domain:

1. From the **Server Configuration** page > **System** tab > **AD Domains**, click **Join AD Domain**.

   The **Join AD Domain** dialog opens:

   Figure 7. Join AD Domain Dialog



2. Specify the **Join AD Domain** parameters as described in the following table.

Table 7. Join AD Domain Parameters

Parameter	Action/Description
Domain Controller	Enter the Fully Qualified Domain Name (FQDN) of the domain controller, then p ress **Tab**. The following message is displayed: *Try ing to determine the NetBIOS name...* Policy Manager searches for the NetBI OS name for the domain. Policy Manager displays a warning mes sage if the domain controller is not the closest one. **NOTE** You should always join to the domain controller tha t is physically the closest t o the Policy Manager serve r, otherwise authentication

Parameter	Action/Description
	might fail or the controller 's performance might be i mpacted.
NetBIOS name	When you specify a valid domain cont roller, the NetBIOS name is automaticall y populated and it is not editable.
Domain Controller name conflict	Specify the action to take in the event o f a domain controller name conflict. In some deployments (especially if the re are multiple domain controllers, or if the domain name has been wrongly en tered in the last step), the domain cont roller FQDN returned by the DNS query can be different from what was entered . In this case, you can: ◦ **Use specified Domain Controller**: Continue to use the domain controll er name that you entered. ◦ **Use Domain Controller returned by DNS query**: Use the domain controll er name returned by the DNS query. ◦ **Fail on conflict**: Abort the Join Dom ain operation.
Use default domain admin user	Check this box to use the Administrator user name to join the domain. **NOTE** In a production environme nt, it is likely that an Admin istrative username that has permissions to join machin es to the domain would be used for the default domai n admin user. In that case, 1) disable (that is, uncheck

Parameter	Action/Description
	) the **Use default domain admin user [Administrator]** check box and 2) enter the Admin username and password in the fields provided.
Username	Enter the user ID of the domain administrator account. **NOTE** This field is disabled if the **Use default domain admin user** check box is selected.
Password	Enter the password for the user account that will join Policy Manager with the domain (for related information, see Table 9, which displays the characters that are allowed and not allowed for the Active Directory username and password).

Table 8. Active Directory Username and Password Characters Allowed and Not Allowed

Field	Characters Allowed	Not Allowed
Username	~ ! @ # $ % ^ * _ - + = { } , . ` & ( ) \ ' " ? /	
Password	! @ # $ % ^ & * ( ) _ - + = { } < , > . ? / ~ ` [ ] \ | ; : ' "	

3. Click **Save**.

   The **Join AD Domain** status screen opens. During the joining process, the screen displays the messages "Checking whether <name> domain controller is

co-located with this ClearPass Policy Manager server." and then "*Adding host to AD domain*," The screen displays status during the joining process.

When the joining process completes successfully, you see the message "*Added host to the domain*."

- If Policy Manager cannot identify the domain controller because Active Directory has not been configured with site information, this form displays the warning "Unable to identify which Active Directory site this Policy Manager server is located in. Authentications might fail or be impacted if the selected domain controller is not co-located."
- If Policy Manager determines that you are not joining the closest domain controller, the form displays both a warning message and a **Find Domain Controller** button that you can use to find the closest domain controllers.

4. To display a list of available controllers, click the **Find Domain Controller** button .

   You can then select a controller from the available list, or proceed with the domain controller you originally selected.

5. Click **Close**.

   You return to the **Server Configuration** page, and it now shows that the Policy Manager server is joined to the domain.

   Now that the Policy Manager server has joined the domain, the server can authenticate users with Active Directory.

   After an Active Directory Domain is added, the domain controller can be set up as a password server (as described in the next section).

Adding a Password Server

About this task

After Policy Manager successfully joins an Active Directory domain, you can configure a restricted list of domain controllers to be used for MSCHAP authentication. If this is not configured, then all available domain controllers obtained from DNS will be included.

> **NOTE**
> When an Active Directory password server is updated in the domain server configuration, the RADIUS service is automatically restarted.

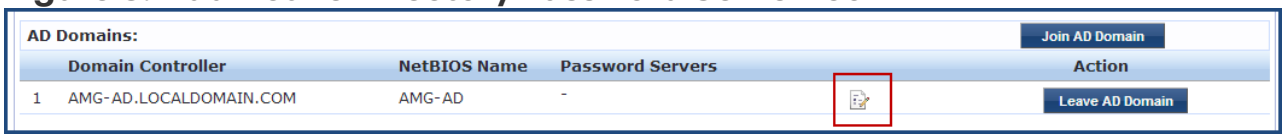
To add an Active Directory password server:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**, then select the Policy Manager server of interest.
2. In the **AD Domains** section of the **System** page, click the **Add Password Server** icon: .

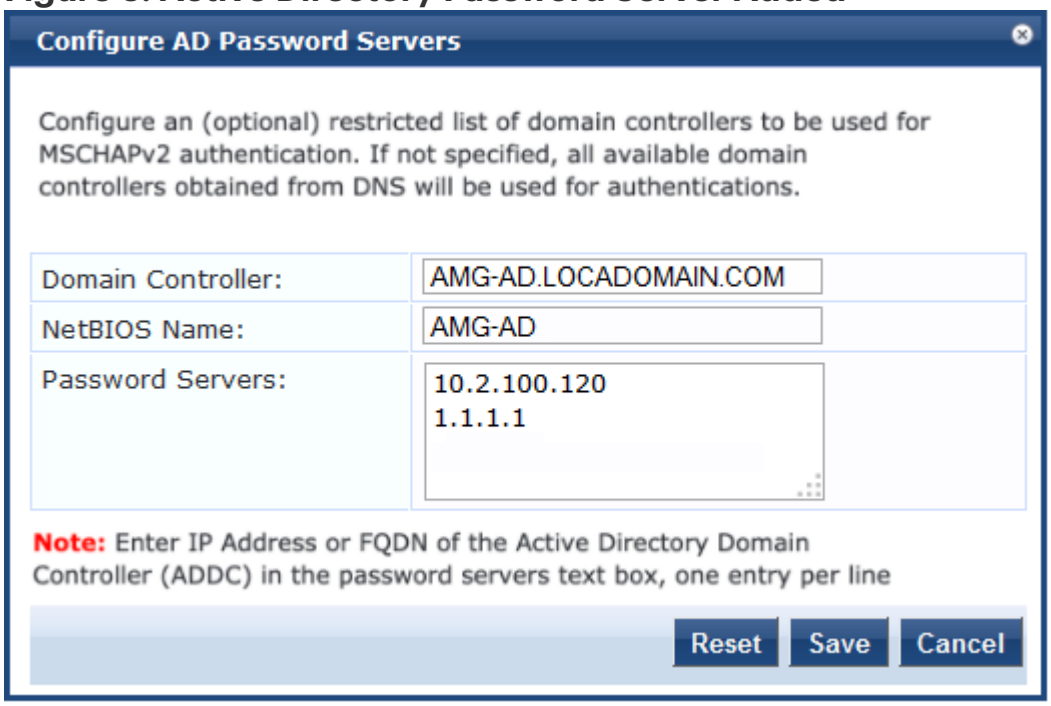
   This icon appears only after Policy Manager joins at least one Active Directory domain.

Figure 8. Add Active Directory Password Server Icon



The **Configure AD Password Servers** dialog opens.

Figure 9. Active Directory Password Server Added



3. Specify the following:
   - Domain controller
   - NetBIOS name
   - Password servers: Enter IP Address or FQDN of the Active Directory Domain Controller (ADDC) in the password servers text box, one entry per line.
4. To complete adding the Active Directory password servers, click **Save**.

Services Control Page

About this task

From the **Services Control** page, you can:

- View the status of all the services: Running or Stopped.
- Stop or start Policy Manager services, including any Active Directory domains that the server joins.

> **NOTE**
>
> When a service is stopped by an administrator from the **Services Control** screen, that particular service remains stopped until the administrator starts it again, or until ClearPass is rebooted. All other services are restarted after reboot as well. Date and time changes, as well as DNS setting updates also restart all the services.

To access the **Services Control** page:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. From the **Server Configuration** page, select the desired Policy Manager server.
3. Click the **Services Control** tab. The **Services Control** page opens.

> **NOTE**
>
> Grafana is now available as an additional Policy Manager service. Grafana is an open-source visualization resource that can be integrated with different data sources. Grafana can be used with Graphite to better visualize time series metrics. Grafana dashboards provide advanced and easy to use visualization and alerts. Grafana is not supported in CC mode. The Grafana service is stopped when CC mode is enabled in Policy Manager.

4. To start a stopped service, click the **Start** button; to stop a running service, click the **Stop** button.

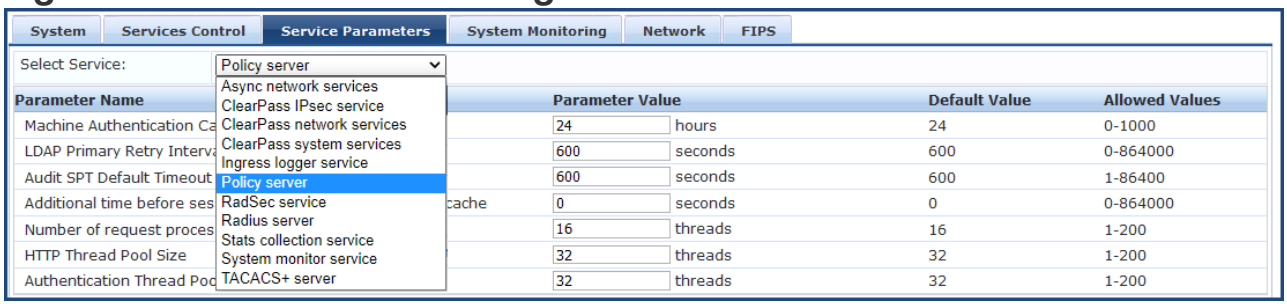
Service Parameters Page

To modify server service settings, navigate to the **Administration > Server Manager > Server Configuration > Service Parameters** page, and click the **Select Service** menu to select a service type.

The following topics describe system parameter settings for each service types.

- Async Network Services Options
- Policy Manager IPsec Service
- Policy Manager Network Services Options
- Policy Manager System Services Options
- Ingress Logger Service Ports
- Policy Server Options
- RadSec Service Options
- RADIUS Server Options
- Stats Collection Service Options
- System Monitor Service Options
- TACACS Server Options

Figure 1. Service Parameters Page



Subtopics

Async Network Services Options
Policy Manager IPsec Service
Policy Manager Network Services Options
Policy Manager System Services Options
Ingress Logger Service Ports
Policy Server Options

RadSec Service Options
RADIUS Server Options
Stats Collection Service Options
System Monitor Service Options
TACACS+ Server Options

Async Network Services Options

About this task

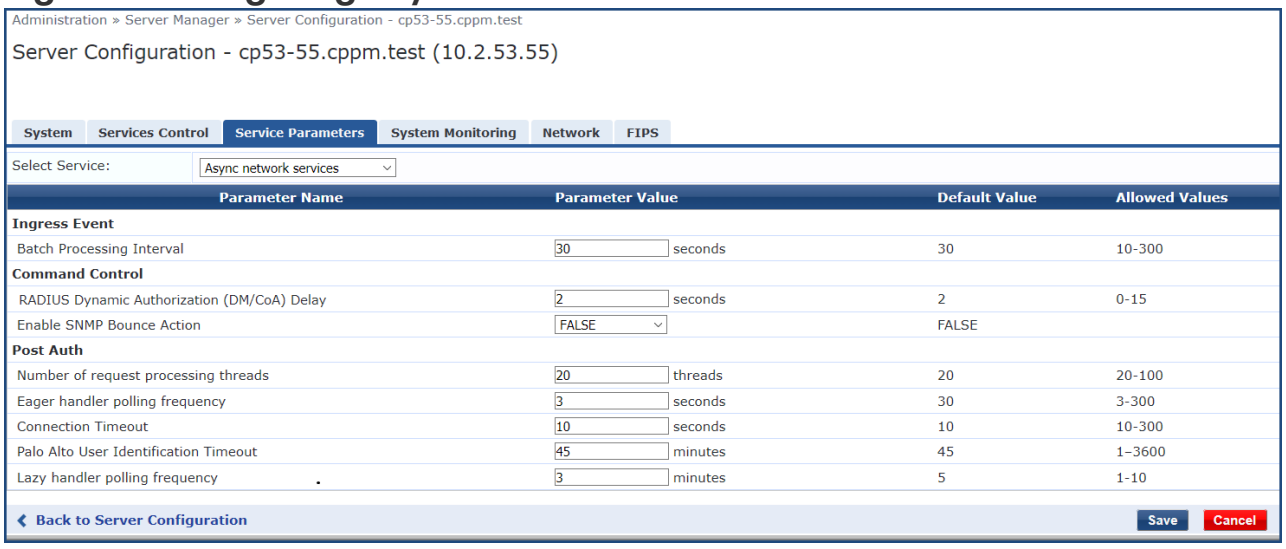
To configure the parameters for the Async network service:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server.
3. From the **Server Configuration** page, select the **Service Parameters** tab.

   The **Service Parameters > Async network services** dialog opens by default:

Figure 1. Configuring Async Network Services



Specify the **Service Parameters > Async Network Services** parameters as described in Table 1.

Table 1. Service Parameters > Async Network Services Parameters

Parameter	Action/Description
Ingress Event	
Batch Processing Interval	Specify the **Batch Processing Interval** for ingress event processing. The default interval is **30** seconds. The ra nge of values is **10** to **300** seconds. **NOTE** For changes to the Batch P rocessing Interval to take eff ect, you must restart the **A sync Network** service.
Command Control	
RADIUS Dynamic Authorization (DM/COA) Delay	Set the **RADIUS Dynamic Authorization delay** value (in seconds). The default value is **2**, and the allowed va lues are from **0** to **15** seconds.
Enable SNMP Bounce Action	Set the **Enable SNMP Bounce Action** val ue. The default value is **FALSE**.
Post Auth	
Number of request processing threads	Set the number of request processing th reads. The default value is **20** threads, and the range of values is between **20** and **100**.
Lazy handler polling frequency	Set the **Lazy handler polling frequency** (i n minutes).

Parameter	Action/Description
	The default value is **5** minutes, and the all owed values are from **1** to **10** minutes. **NOTE** If the value of the polling int erval is less than three minu tes, a warning message is di splayed that advises that set ting the Lazy - handler pollin g frequency to lower than **3** minutes can cause performa nce - related issues.
Eager handler polling frequency	Set the **Eager handler polling frequency** (in seconds). The default value is **30** seconds, and the allowed values are from **10** to **300** secon ds. **Eager handler polling** is employed when an attribute requires the freshest possibl e value.
Connection Timeout	Set the connection timeout in seconds. T he default value is **10**, and the range is f rom **1** to **3600** (one hour).
Palo Alto User Identification Timeout	Specify the Palo Alto endpoint context s erver user identification timeout in minut es. The default value is **45**; the range is f rom **1** to **3600**.

Policy Manager IPsec Service

About this task

When a network device requests an IPsec connection between the device and a Policy Manager server, Policy Manager uses the Online Certificate Status Protocol (OCSP) URI (uniform resource identifier) to contact a third-party server that checks to see if the certificate sent by the requesting device is valid. If the certificate is confirmed as valid, an IPsec connection between the Policy Manager server and the requesting network device is established.

> **NOTE**
> Common Criteria (CC) evaluation for ClearPass utilizes IPsec as the evaluated Syslog method for security.

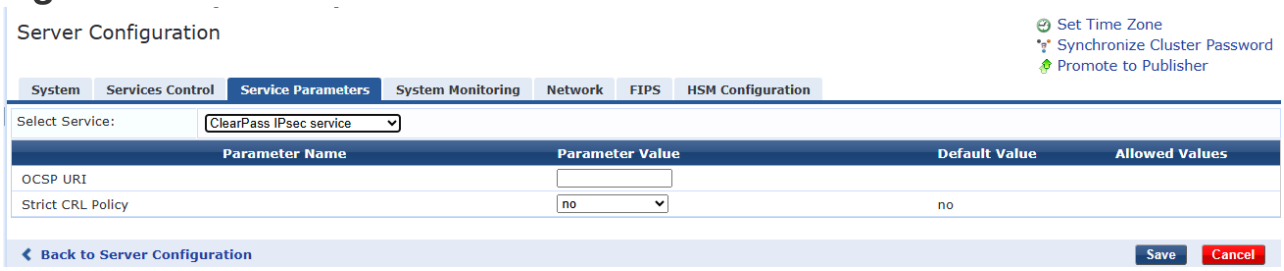
To configure the Policy Manager IPsec service:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. Select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **Policy Manager IPsec service**.

   The following dialog opens:

Figure 1. IPsec Service Parameters



Specify the **Service Parameters > Policy Manager IPsec Service** parameters as described in Table 1, then click **Save**.

Table 1. Service Parameters > Policy Manager IPsec Service Parameters

Parameter	Action/Description
OCSP URI	In the **Parameter Value** field, specify the **HTTP** or **HTTPS URI** (uniform resource id entifier) for the Online Certificate Status Protocol (OCSP).

Parameter	Action/Description
	OCSP enables the Policy Manager server to determine the revocation state of a ce rtificate presented by a peer—for exampl e a network device requesting an IPsec c onnection to the Policy Manager server. **NOTE** When you enter the OCSP U RI, Policy Manager checks th at 1) the URI is in the proper format (it must start with HT TP or HTTPS and be syntact ically correct), and 2) Policy Manager checks to see if th e specified OCSP server IP a ddress or host name is reach able from the Policy Manage r node. A descriptive error m essage will be displayed in t he event of an incorrect OC SP URI.
Strict CRL Policy	You can enable or disable a strict Certific ate Revocation List (CRL) policy. This para meter is disabled by default. • To enable **Strict CRL Policy**, select **Yes** from the **Parameter Value** drop - down. When this option is enabled, a fresh Cert ificate Revocation List must be available i n order for a peer connection to succeed . Whenever **Strict CRL Policy** is modified, existing IPsec tunnels that use Public Key Authentication are brought down and the n brought up again.

Policy Manager Network Services Options

The **Policy Manager Network Services** parameters aggregate service parameters from the following services:

- SNMP Service
- Certificate Authentication Service
- Web Authentication Service
- Posture Service
- DHCP Snooper Service

To configure the parameters for Policy Manager Network Services:

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **Policy Manager network services**.

   The **Service Parameters > Policy Manager network services** dialog opens:

   Figure 1. Service Parameters >Policy ManagerNetwork Services

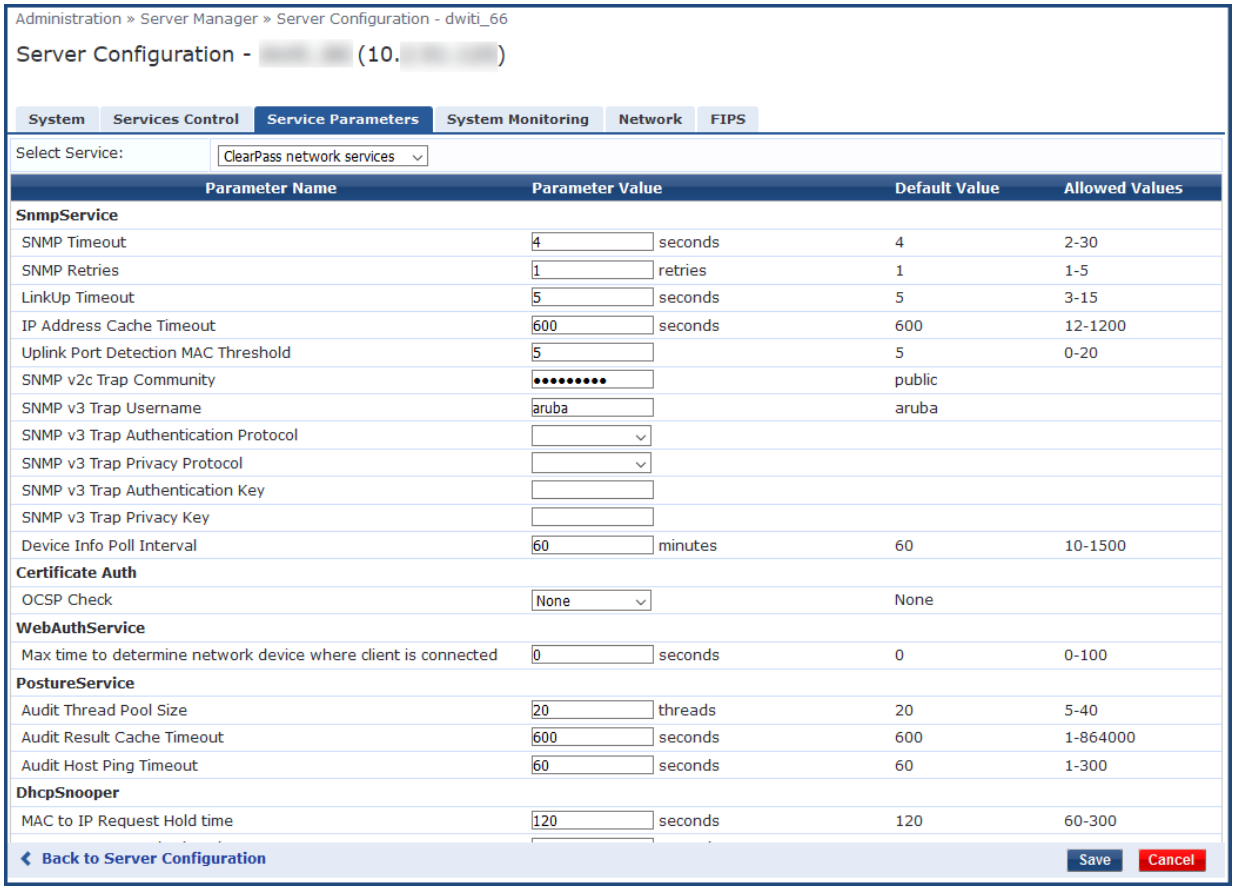


Figure 2. Service Parameters >Policy ManagerNetwork Services in FIPS Mode

4. Specify the **Policy Manager Network Services** parameters as described in the following table:

Table 1. Service Parameters > Policy Manager Network Services

Service Parameters	Action/Description
SnmpService	
SNMP Timeout	Specify the seconds to wait for an SNM P response from the network device.
SNMP Retries	Specify the number of retries for SNMP requests.
LinkUp Timeout	Specify the seconds to wait before proc essing link - up traps.

Service Parameters	Action/Description
	If a MAC notification trap arrives in this time, the SNMP service does not try to poll the switch for MAC addresses behi nd a port for link - up processing.
IP Address Cache Timeout	Specify the duration in seconds for whi ch MAC - to - IP lookup response is cac hed.
Uplink Port Detection Threshold	Specify the limit for the number of MAC addresses found behind a port after wh ich the port is considered an uplink po rt and not considered for SNMP lookup and enforcement. The default value is **5**, with a range fro m **0** to **20**.
SNMP v2c Trap Community	Specify the community string that must be checked in all incoming SNMP v2 tra ps.
SNMP v3 Trap Username	Specify the SNMP v3 Username to be u sed for all incoming traps.
SNMP v3 Trap Authentication Protocol	Specify the SNMP v3 Authentication p rotocol for traps. The options are: **MD5**, **SHA**, or **empty** (t o disable authentication). **NOTE** The **EAP - MD5** authentica tion type is not supported i n **FIPS** mode.
SNMP v3 Trap Privacy Protocol	Specify the SNMP v3 Privacy protocol for traps.

Service Parameters	Action/Description
	The options are: **DES_CBC**, **AES_128**, or **empty** (to disable privacy). **NOTE** The **DES_CBC** privacy protocol is not supported in **FIPS** mode. When FIPS or CC mode is enabled, the DES - CBC option is removed from the **Privacy Protocol** drop - down list. For configurations that used this option in FIPS mode, the DES - CBC setting is automatically updated to AES - 128. **NOTE** ClearPass 6.11 and 6.12 are the last Long Support Release (LSR) and Short Support Release (SSR) versions to support the use of the DES_CBC priviacy protocol in any operating mode, Customers who use these protocols chould plan to migrate to other avilable protocols before the next major release.
SNMP v3 Trap Authentication Key	Specify the SNMP v3 authentication key and privacy key for incoming traps.
SNMP v3 Trap Privacy Key	
Device Info Poll Interval	Specify the time (in minutes) between polling for device information.
Certificate Auth	

Service Parameters	Action/Description
OCSP Check	Specify one of the following options for initiating an Online Certificate Status P rotocol (OCSP) check: ◦ **None** (the default setting) ◦ **Optional** ◦ **Required**
AuxiliaryService	
Enable LDAP Connection Pool	Set this parameter to **FALSE** and restart **System Auxiliary Services** to use the a uthentication implementation available in earlier Policy Manager versions. Setti ng this parameter to false ignores the L DAP connection pool implementation, c reates new LDAP instances as needed, and closes them once completed. The default setting is **TRUE**, resulting in Policy Manager remaining in the 6.11.0 L DAP connection pool implementation, a llowing it to reuse the connection objec t for the next LDAP authentication durin g user lookups. This improves LDAP aut hentication flow performance and redu ces the time needed to search for a use r domain name.
WebAuthService	
Max time to determine network device where client is connected	Specifies the maximum time to wait for Policy Manager to determine the netwo rk device to which the client is connect ed. In some usage scenarios where the we b authentication request does not origi nate from the network device, Policy M anager has to determine the network d evice to which the client is connected

Service Parameters	Action/Description
	through an out - of - band SNMP mech anism. The network device deduction p rocess can take some time.
PostureService	
Audit Thread Pool Size	Specify the number of threads to use fo r connections to audit servers.
Audit Result Cache Timeout	Specify the time (in seconds) for which audit result entries are cached by Policy Manager.
Audit Host Ping Timeout	Specify the number of seconds for whi ch Policy Manager pings an end - host before giving up and deeming the host to be unreachable.
DhcpSnooper	
MAC to IP Request Hold time	Specify the number of seconds to wait before responding to a query to get an I P address corresponding to a MAC add ress. Any DHCP message received in this tim e period refreshes the MAC address - to - IP address binding. Typically, an audit service requests a M AC - to - IP mapping as soon the RADIU S request is received, but the client m ay take some more time to receive the I P address through DHCP. This wait perio d takes into account the latest DHCP IP address that the client received.
DHCP Request Probation Time	Specify the number of seconds to wai t before considering the MAC - to - IP b inding received in a DHCP REQUEST m essage as final.

Service Parameters	Action/Description
	This wait handles cases where a client receives a DHCPNAK for a DHCPREQUEST and receives a new IP address after going through the DHCPDISCOVER process again.

5. Click **Save**.

Policy Manager System Services Options

About this task

You can use the Policy Manager System Services parameters for PHP (Hypertext Preprocessor) configuration and for HTTP traffic flowing through a proxy server. ClearPass relies on an HTTP connection for the ClearPass Update Portal to download the latest information for system services.

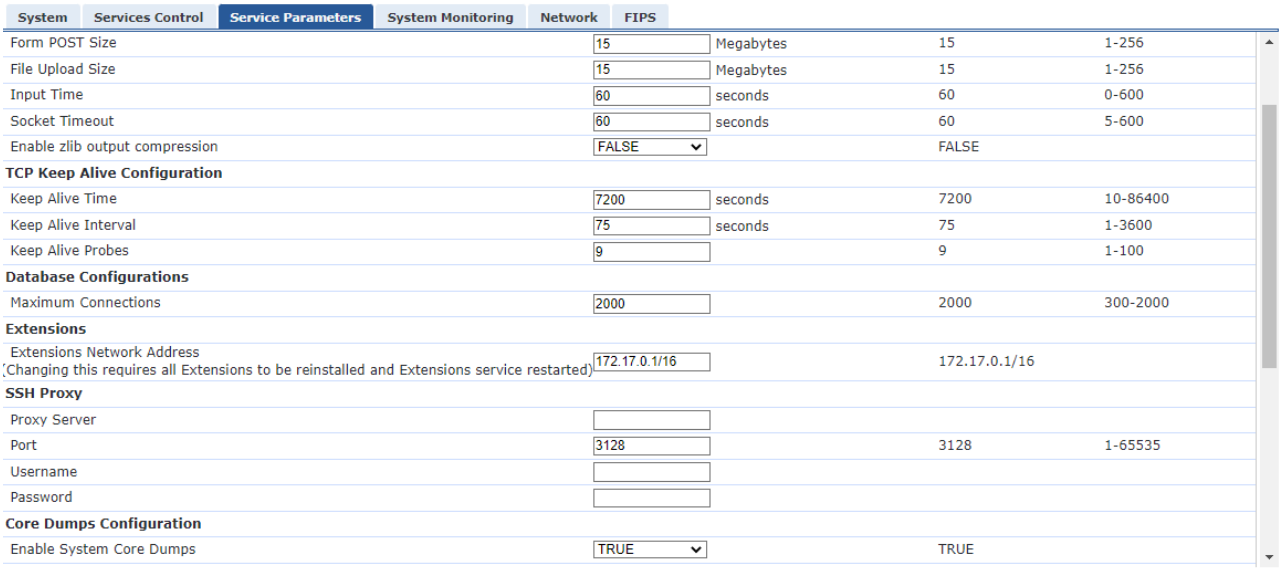
To configure ClearPass System Services:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **Policy Manager system services**.

   The following dialog displays:

Figure 1. Policy ManagerSystem Services Parameters



4. Specify the **Service Parameters** > **ClearPass system services** as described in the following table.

> **NOTE**
>
> The **HTTP Proxy** configuration service parameters described in the following table send both HTTP and HTTPS traffic to an external proxy server.

Table 1. Service Parameters > Policy Manager System Services

Service Parameter	Action/Description
PHP System Configuration	
Memory Limit	Specify the maximum memory that can be used by the PHP (Hypertext Preprocessor) applications.
Form POST Size	Specify the maximum HTTP POST content size that can be sent to the PHP application.
File Upload Size	Specify the maximum file size that can be uploaded into the PHP application.

Service Parameter	Action/Description
Input Time	Specify the time limit after which the Pol icy Manager server will detect no activity from the user and will take some action.
Socket Timeout	Specify the maximum time for any socket connections.
Enable zlib output compression	Specify the setting to compress the outp ut files.
TCP Keep Alive Configuration	
Keep Alive Time	Specify a value in seconds from 10 to 86400. TCP connections with an Active Directory (AD) domain controller are configurable u sing these system services **TCP Keep Ali ve Configuration** settings.
Keep Alive Interval	Specify a value in seconds from 1 to 360 0.
Keep Alive Probes	Specify a value from 1 to 100 for the num ber of probes.
Database Configurations	
Maximum connections	Specify a number between 300 and 200 0 for a maximum number of allowed con nections.
Extensions	
Extensions Network Address	You can use the **Extensions Network Add ress** parameter to configure the adapter i nterface and change the subnet used for the Extension. Specify the bridge IP address used for int ernal networking with local extensions.

Service Parameter	Action/Description
	• **Default** : 172.17.0.1/16 • **Format** : IPv4_ADDR/PREFIX_LEN • **Validation**: A valid, non - routable IPv 4 address must be supplied. The IPv 4 address must match 10.0.0.0/8, 192.1 68.0.0/16, or 172.16.0.0/12. • **Prefix Length**: A valid prefix length m ust be supplied: >= 16 and <= 29 **NOTE** If you change this field, yo u must uninstall and reinstal l all extensions and then res tart the extensions service. Extensions may be reinstalle d through ClearPass Guest, at **Guest > Administration > Extensions**.
SSH Proxy	
Proxy Server	Specify the hostname or IP address of th e proxy server. **NOTE** ClearPass Policy Manager su pports IPv6 SSH - proxy add resses for sending data to th ird - party context servers.
Port	Specify the port at which the proxy serve r listens for SSH traffic.
Username	Specify the user name to authenticate wi th the proxy server.

Service Parameter	Action/Description
Password	Specify the password to authenticate wit h the proxy server.
Core Dumps Configuration	
Enable System Core Dumps	System core dumps are enabled (set to **T RUE**) by default. A core dump includes a memory image taken at the time the ope rating system terminates an application. The memory image could contain sensiti ve data. Set this option to **FALSE** to avoid the risk of exposing sensitive data in a c ore dump, though setting this value to FA LSE does not debug a problems when a core dump occurs. Network administrato rs should consider local deployment requ irements and policies before temporarily enabling core dumps to support advance d debugging.
HTTP Proxy	
Proxy Server	Specify the hostname or IP address of th e proxy server. **NOTE** If utilizing IPv6 HTTP - proxy addresses for sending data t o third - party context serve rs, square brackets are provi ded to enclose the address i f not entered while configuri ng, as providing the address without the brackets results i n a polling failure.
Port	Specify the port at which the proxy serve r listens for HTTP traffic.

Service Parameter	Action/Description
Username	Specify the user name to authenticate wi th the proxy server.
Password	Specify the password to authenticate wit h the proxy server.
Web Server Configuration	
Timeout	Specify a server timeout value in second s from 1 to 60.
Keep Alive	To enable or disable keep - alive for the web server, select **TRUE** or **FALSE**.
Request Wait	Specify the request wait time in seconds from 1 to 60. The default value is **4** secon ds.
Enable Host Header check	Specify whether to enable the host head er check. The default value is **TRUE**. • When you set this value to **TRUE**, the **Host Header Restriction check** is ena bled and ClearPass can use only speci fically allowed host headers. • When you set this value to **FALSE**, ir respective of Host Headers in the http packet, ClearPass Policy Manager red irects to **https://<Policy Manager - se rver>/tips**.
Allowlist Host Names	When the **Enable Host Header check** va lue is set to **TRUE**, the web access is all owed for specified host names, IP addres ses, and VIP addresses in Policy Manager . Create a comma - separated of host na mes list to allow multiple hostnames. When the **Enable Host Header check** val ue is set to **TRUE** and the **Allowlist Host**

Service Parameter	Action/Description
	Names field is blank, the web access is a llowed only for host names, IP addresses, and VIP addresses in ClearPass Policy Ma nager.
Enable HTTP Strict Transport Security (H STS)	In the *Enable HTTP Strict Transport Secur ity (HSTS)* field, the value is set per node. A value of zero (0) indicates that HSTS is off, and any value greater than zero (0) e nables HSTS and indicates the timeout in seconds.
Set Default Request Read Timeout	Set this value to **YES** to enforce the client to have a maximum of 20 seconds to sta rt sending header data before it times ou t. The client must send header data at a transfer rate of 500 bytes per second, fo r a maximum of 40 seconds. Additionally, the client has a maximum of 20 seconds to start sending body data. The client m ust also send message body data at a tra nsfer rate of 500 bytes per second, and may do it for a maximum of 40 seconds. Leave this value set to **NO** (default settin g) to not enforce the Default Request Rea d Timeout.
Maximum Clients	The **Maximum Clients** value represents t he maximum number of concurrent http d processes that can be spawned to han dle requests managed by the Policy Man ager Web server. As an example, if this va lue is set to 500 and there are currently 1 0 active WebUI sessions and 490 API call s being served, then next request will be put into a queue, and will get served onc e an existing client (httpd process) is free d. If a client is not freed during the timeo ut period, the request times out and the request will report a Timeout error.

Service Parameter	Action/Description
	Specify a value from 10 to 20000 for the maximum number of clients allowed. The default value is **100**.
Maximum Requests	If the **Keep Alive** parameter is enabled, t he **Maximum Requests** value allows you to define the maximum number of back - to - back requests to the Policy Manag er Web server that can be made on a sing le connection within the Keep Alive timeo ut interval, without having to span a new httpd request. Specify a number betwee n 0 and 3000 for the maximum number of requests allowed. The default value is **500**.

5. Click **Save**.

Ingress Logger Service Ports

About this task

When Ingress Event Processing is enabled and configured on Policy Manager (see Enabling Ingress Events Processing), logging of ingress events occurs automatically. By default, the Policy Manager server listens for Ingress Events on *TCP port 514* and *UDP port 514*. If necessary, you can change these Syslog Ingress Logger ports.

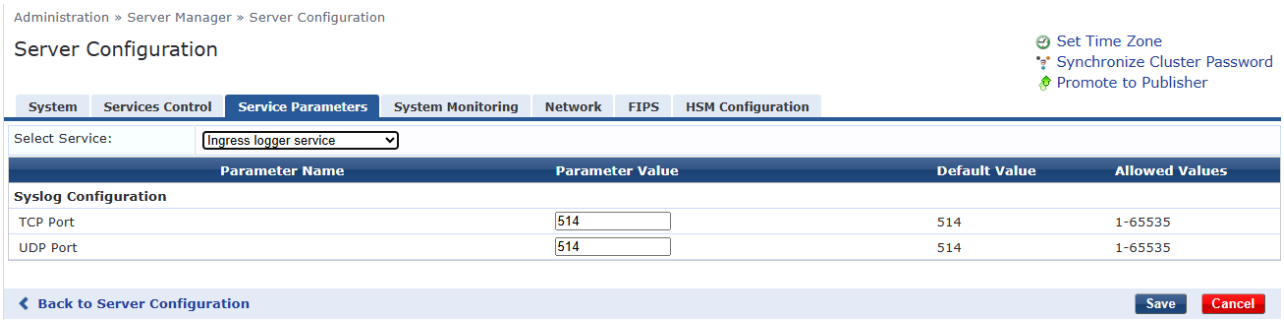
To change the Syslog Ingress Logger ports:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. Select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **Ingress Logger Service**.

   The following dialog opens:

Figure 1. Ingress Logger Service



4. To change the Ingress Logger **TCP Port**, enter the new port number in the *Parameter Value* field.
5. To change the Ingress Logger **UDP Port**, enter the new port number.
6. Click **Save**.

Policy Server Options

About this task

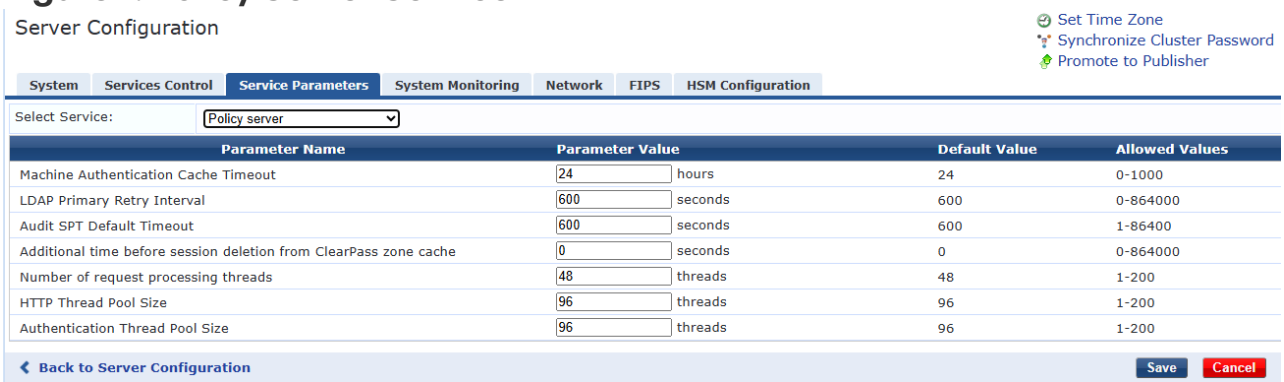
To configure the Policy Server service:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **Policy server**.

   The following dialog opens:

Figure 1. Policy Server Service



4. Specify the **Service Parameters > Policy Server** parameters as described in the following table.

Table 1. Service Parameters > Policy Server Service

Service Parameter	Action/Description
Machine Authentication Cache Timeout	Specify the time (in hours) for which mac hine authentication entries are cached by ClearPass Policy Manager. The default is **24 hours**.
LDAP Primary Retry Interval	After a primary LDAP server is down, the Policy Manager server connects to one of the backup servers. Specify how long the Policy Manager se rver waits (in seconds) before it tries to c onnect to the primary server again.
Audit SPT Default Timeout	Specify the time (in seconds) for which a n Audit success or error response is cach ed in the Policy server.
Additional time before session deletion f rom ClearPass zone cache	Specify the number of seconds the Poli cy server will wait before deleting the Cl earPass zone entry. The default value is **0**. This parameter handles roaming scenario s where an Accounting - Start occurs wit hout an authentication request. If the va lue for this parameter is **0**, the Policy se rver deletes the ClearPass zone entry wh en an Accounting - Stop is received. The RADIUS server updates the ClearPas s zone entry with attribute values from th e accounting request. These can be used in the Change of Authorization (CoA). In a roaming scenario, this NAS information update from the accounting request help

Service Parameter	Action/Description
	s Policy Manager send the CoA to the cor rect NAS.
Number of request processing threads	Specify the maximum number of threads used to process requests.
HTTP Thread Pool Size	Specify the number of threads allotted fo r the HTTP thread pool.
Authentication Thread Pool Size	Specify the number of threads to use for LDAP/Active Directory and SQL connectio ns.

RadSec Service Options

About this task

The parameters for the RadSec service pertain to Online Certificate Status Protocol (OCSP) support for RadSec and Elliptic Curve Cryptography (ECC) certificates.

> **NOTE**
> Policy Manager does not support RadSec services over IPv6 networks.

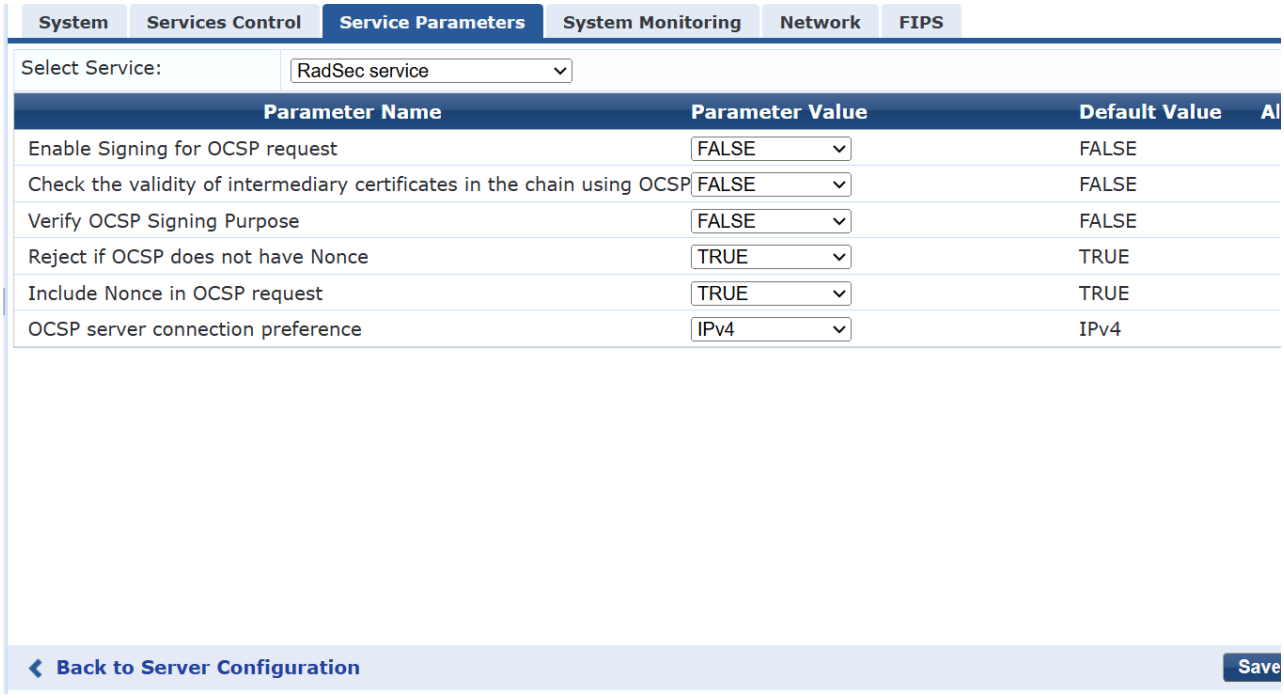
To configure the RadSec service:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **RadSec service**.

   The following dialog opens:

Figure 1. RadSec Service Configuration Dialog



4. Specify the **Service Parameters > Policy Server** parameters as described in the following table.

Table 1. Service Parameters > RadSec Service

Service Parameter	Action/Description
Enable Signing for OCSP request	This determines whether Policy Manager should sign an OCSP request with a RadSec server certificate. To enable signing for OCSP request, select **TRUE**. The default value is **FALSE**.
ECDH Curve	Elliptic - curve Diffie–Hellman (ECDH) is an anonymous key agreement protocol that allows two parties, each having an elliptic - curve public–private key pair to establish a shared secret over an insecure channel. Select one of the following options: • *X9.62/SECG curve over a 256 bit prime field*

Service Parameter	Action/Description
	• *NIST/SECG curve over a 384 bit prime field*.
Check the validity of intermediary certificates in the chain using OCSP	To check the validity of all certificates in the chain against Certificate Revocation Lists (CRLs), select **TRUE**. Otherwise, select **FALSE**.
Verify OCSP Signing Purpose	Specify one of the following: • When the value for this parameter is set to **TRUE**, Policy Manager verifies the purpose of the signed cert in the OCSP Response. If the purpose is not for OCSP sigining, then the verification fails. • If this parameter is left at the default setting of **FALSE**, Policy Manager does not verify the signing purpose.
Reject if OCSP does not have Nonce	To reject an OCSP response without a nonce, select **TRUE**. Otherwise, select **FALSE**. The Default value of the parameter is **True**. If OCSP check is enabled on Policy Manager: • Policy Manager always sends nonce in an OCSP request. • If nonce is present in the OCSP response, it should match the nonce in the OCSP request. If it does not, the certificate verification will fail and the request will be rejected. • If nonce is not present in the OCSP response, whether the certificate verification will fail or succeed will depend on the value of RADIUS service param

Service Parameter	Action/Description
	eter *Reject if OCSP response does no t have nonce*. ◦ If the value of the parameter is tru e, certificate verification will fail. ◦ If the value of the parameter is fals e, certificate verification will succe ed.
Include Nonce in OCSP request	Specify one of the following: • **TRUE**: Select if the OCSP (Online Cer tificate Status Protocol) request shoul d include the nonce. This is the defaul t value. • **FALSE**: To avoid the EAP - TLS authen tication failure, select if the OCSP se rver does not support the nonce.
OCSP server connection preference	Specify whether Policy Manager should u se an **IPv4** or **IPv6** network to connect to the OCSP server. The default value is IPv 4.

5. Click **Save**.

RadSec Proxy Failover Support

About this task

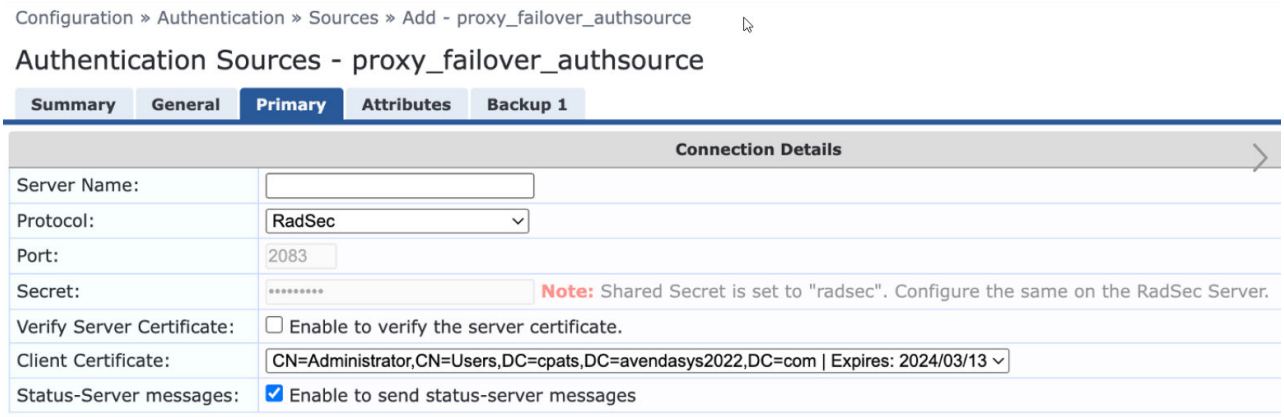
RadSec proxy failover is now supported when proxy targets are configured. Consider the following deployment use cases:

Procedure

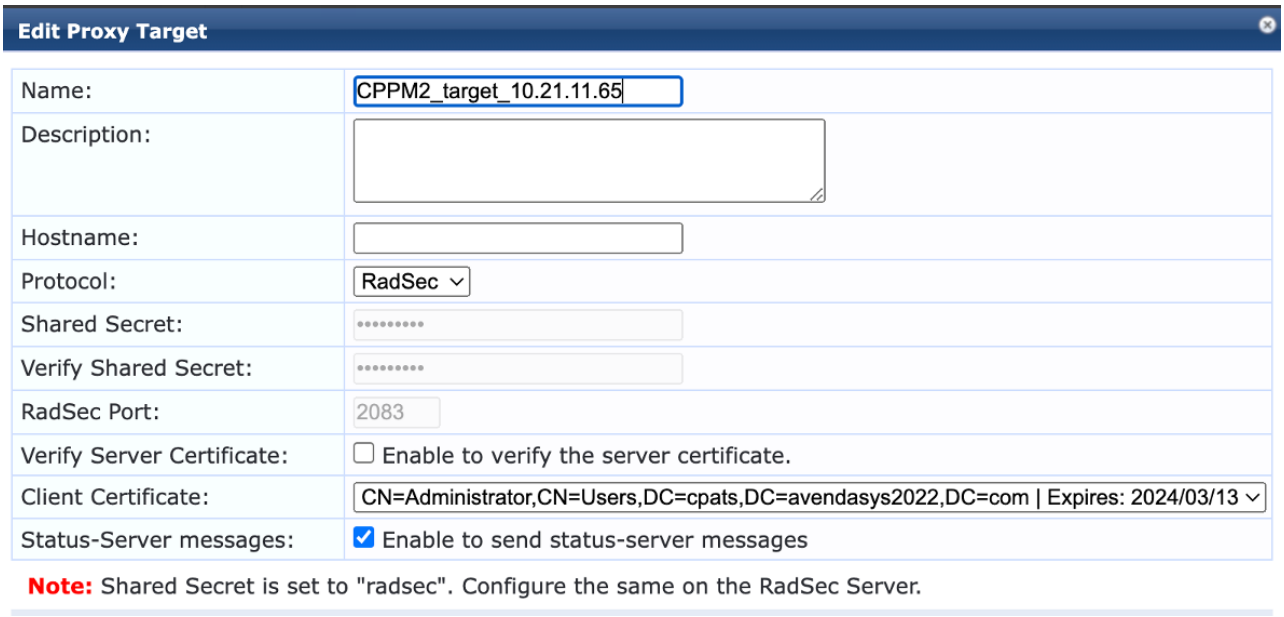
1. If the third-party RadSec server (proxy destination) does not support server status messages but the TLS connection is up, does this trigger the failover?

   Status server messages can be enabled or disabled in CPPM1 (proxy) for the Radius server. Refer to the authentication source and proxy target examples below:

   Authentication source:



Proxy target:



If the third-party RadSec server (proxy destination) does not support server status messages, the administrator is recommended to disable status-server for that server. In this case the status-server is not considered for triggering failover.

2. What if TLS is UP and the status-server messages are fine, but there is delay in response for actual authentication requests, does this trigger failover? How mush time does Policy Manager wait for a response before giving up on the request and triggering failover?

   Yes. It is based on number of messages without a response. But if there is only a single response message, this count is reset to 0.

   Currently, the RadSec default retry interval time is 255 seconds and the maximum retry count is 255. RadSec will wait 1083 minutes for response before giving up on the request.

3. Is a response to a server-status message mandatory for primary recovery, or will recovery work if the TLS connection is Up?

   Primary recovery will work if the TLS connection is UP and its lost request count is 0, or less than the backup's lost count. Lost request count will be incremented in the following cases:

   - If the status-server message is enabled, Proxy CPPM1 sent a status-server message and did not get reply from the primary for 1 status-server message.
   - The proxy CPPM1 sent a RadSec (RADIUS) message and did not receive a reply from the primary, even after the Max Retries retried in the Retry Interval.

   Currently, the RadSec default retry interval time is 255 seconds and maximum retry count is 255. RadSec will wait 1083 minutes for a response before giving up on the request.

RADIUS Server Options

About this task

To configure the RADIUS server service:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **RADIUS server**.
4. Specify the **Service Parameters > RADIUS server** parameters as described in the following table.

   Figure 1. RADIUS Server Parameters Dialog (Partial View)

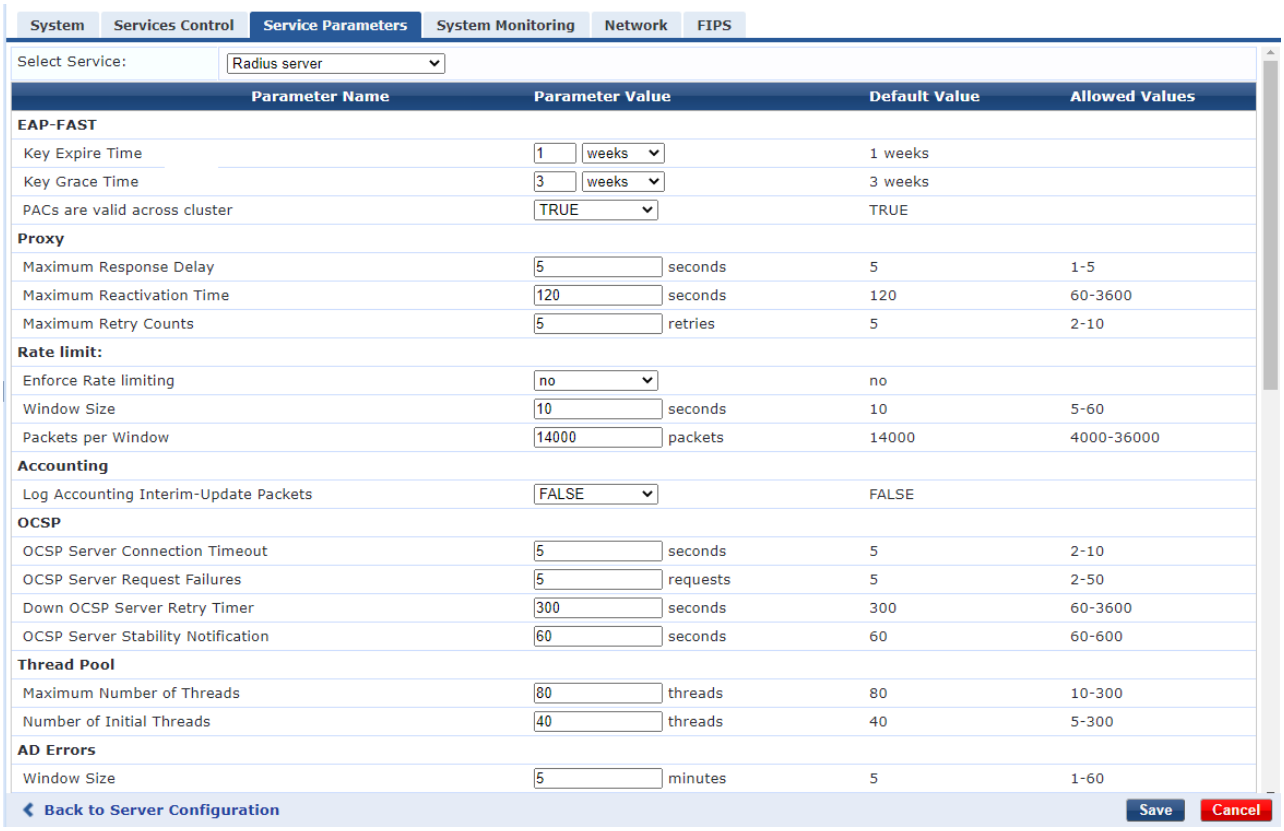


Table 1. Service Parameters > RADIUS Server Service

Service Parameter	Action/Description
EAP - FAST	
Key Expire Time	Specify the lifetime of a generated EAP - FAST key.
Key Grace Time	Specify the grace period for an EAP - FAST key after its lifetime expires. The default is **3 weeks**. If a client presents a PAC (Protected Access Credential) that is encrypted using the key in this period after its TTL (Time - to - Live), it is accepted and a new PAC encrypted with the latest key is provisioned on the client.
PACs are valid across cluster	If PACs (Protected Access Credentials) generated by this server are valid across the cluster, set to **TRUE** (the default setting).

Service Parameter	Action/Description
	If not, select **FALSE**.
Proxy	
Maximum Response Delay	If the target server has not responded, s pecify the delay time before retrying a p roxy request. The default is **5** seconds.
Maximum Reactivation Time	Specify the time to elapse before retryin g a dead proxy server.
Maximum Retry Counts	If the target server doesn't respond, spec ify the maximum number of times to ret ry a proxy request.
Rate Limit	
Enforce Rate limiting	Policy Manager can provide traffic throttl ing on the RADIUS server. In scenarios w here an occasional, sudden increase in a uthentications per second might put an excessive load on the policy server, Clear Pass Zone, or Post - Authentication modu le, this rate limiting can alleviate that loa d and prevent the RADIUS server from go ing into an unresponsive state. Set this o ption to **yes** to enable traffic rate limiting, or **no** to disable limiting. The default is **no** .
Window Size	Enter the number of seconds for the traff ic - limit window. The supported range is 5 - 60 seconds, and the default is ten sec onds.
Packets per Window	Enter the number of packets to allow wit hin the time window. The supported rang e is 4000 - 36000 packets, and the defa ult is 14,000 packets.
Accounting	

Service Parameter	Action/Description
Log Accounting Interim - Update Packets	To store the Interim - Update packets in s ession logs, select **TRUE**. **FALSE** is the de fault setting.
OCSP	
Use HTTP Proxy setting for OCSP Connec tion	Specifies the OCSP connection when **EA P - TLS with OCSP Enabled** is the authen tication method, and an HTTP proxy is co nfigured. Set this parameter to **TRUE** to a pply the HTTP proxy setting to all OCSP l ookups that are not localhost. This param eter is set to **FALSE** by default.
OCSP Server Connection Timeout	Specifies the time in seconds within whic h the TCP connection must complete (de fault 5, range 2 - 10).
OCSP Server Request Failures	Specifies the number of failed responses that must be timed out before the serve r is marked as down (default 5, range 2 - 50).
Down OCSP Server Retry Timer	Specifies the time in seconds that the se rver is "down" and ClearPass will not atte mpt to reach it again (default 300, range 60 - 3600).

NOTE

For a server marked with a "down" OCSP state after the **Down OCSP Server Retry Ti mer** elapses, Policy Manager will try only once to reach O CSP server. However, if durin g this time if a a configuratio n is reloaded on the server, t he OCSP connection state a nd cache is lost, resulting in Policy Manager trying to con

Service Parameter	Action/Description
	nect to OCSP for the config ured number of retries.
OCSP Server Stability Notification	Specifies the time window in seconds th at the OCSP Server Request Failures valu e is measured across (Default 60, range 60 - 600)
Thread Pool	
Maximum Number of Threads	Specify the maximum number of threads in the RADIUS server thread pool to proc ess requests.
Number of Initial Threads	Specify the initial number of threads in t he RADIUS server thread pool to process requests.
Active Directory Errors	
Window Size	Enter a duration during which Active Dire ctory errors are accumulated for possible action. The default is **5 minutes**.
Number of Errors	Enter a number to specify the number of Active Directory errors that can occur wit hin the defined Window Size and have th e self - healing Recovery Action taken. Th e default is **150**.
Recovery Action	Select one of the following recovery acti ons from the drop - down list: • **None**: To not initiate a self - recovery action. This is the default. • **Exit**: To restart the RADIUS server. (Th e monitoring daemon restarts it.) • **Restart Domain Service**: To restart th e Domain service.

Service Parameter	Action/Description
TLS	
Disable RSA - PSS Signature Suite in EAP - TLS	Set this value to **TRUE** to assist Windows 10 devices using a Trusted Platform Module (TPM) authenticate when certificates installed in the TPM fail with a **RSA_verify_PKCS1_PSS_mgf1:last octet invalid** error. This issue is reported in ClearPass 6.11.x and later deployments on some Windows 10 devices using TPM, especially TPM 1.16 and lower. This issue does not impact pre 6.11.x ClearPass versions. The default value is **FALSE**.
Security	
Enable Packet Rejection Delay	If set to **TRUE** (default setting), a rejection is not sent until a consecutive Radius client request is made. If set to **FALSE**, a failed authentication packet is rejected immediately. **NOTE** This field previously accepted an integer value. However, this parameter now accepts either True/False. If needed, manually update the value to reflect the required value change from a previous version of ClearPass.
Maximum Attributes	Specify the maximum number of RADIUS attributes allowed in a request. The default is **200**.
Process Server - Status Request	• **TRUE**: Send replies to Status - Server RADIUS packets.

Service Parameter	Action/Description
	• **FALSE**: Do not send replies to Status - Server RADIUS packets. This is the default setting.
Require Message - Authenticator from NAD	Set this parameter to **yes** for Policy Manager to check and validate the Message - Authenticator in each Access - Request. If the Message - Authenticator is not present, or the Message - Authenticator validation fails for the incoming packet, Policy Manager drops the packet and an Event Viewer alert is written. However, there is no Access Tracker entry. The default setting is **no**, with no operation performed.
Require Message - Authenticator from Proxy Server	Set this parameter to **yes** for Policy Manager to check and validate the Message - Authenticator in each Access - Challenge, Access - Accept, and Access - Reject when functioning as a Proxy Client. If the Message - Authenticator is not present or the Message - Authenticator validation fails for the incoming packet, Policy Manager drops the packet an Event Viewer alert is written. However, there is no Access Tracker entry. The default setting is **no**, with no operation performed.
Main	
Authentication Port	Specify the ports on which the RADIUS server listens for authentication requests. Default values are ports **1645** and **1812**. **NOTE** You can configure the Authentication Port to different values if desired.

Service Parameter	Action/Description
Accounting Port	Specify the ports on which the RADIUS s erver listens for accounting requests. The default values are **1646** and **1813**. **NOTE** You can configure the Accou nting Port to different values if desired.
Maximum Request Time	Specify the maximum time (in seconds) a llowed for processing a request after whi ch it is considered timed out. The default is **30 seconds**.
Cleanup Time	Specify the time to cache the response s ent to a RADIUS request after sending it. The range is from **2** to **10** seconds. The d efault is **5** seconds. If the RADIUS server gets a duplicate req uest for which the response is already se nt, and the duplicate request arrives with in this time period, the cached response i s resent.
Local DB Authentication Source Connecti on Count	Specify the maximum number of Local D B connections opened.
AD/LDAP Authentication Source Connecti on Count	Specify the maximum number of Active Directory and LDAP (Lightweight Directo ry Access Protocol) connections opened. The range is from **5** to **300**. The default is **64**.
SQL DB Authentication Source Connecti on Count	Specify the maximum number of SQL DB .
Kerberos Authentication Source Connecti on Count	Specify the maximum number of Kerbero s connections opened.

Service Parameter	Action/Description
Authentication Source Primary Fallback I nterval	Set a value from 60 - 86400 seconds to determine how long the Radius server wa its before it checks to see whether the pr imary server is back online. The default is 120 seconds.
EAP - TLS Fragment Size	Specify the maximum allowed size (in byt es) for the EAP - TLS fragment.
Use Inner Identity in Access - Accept Rep ly	To use the inner identity in the Access - Accept replies, select **TRUE**. The defaul t setting is **FALSE**.
Reject if OCSP response does not have N once	To reject an OCSP response without a no nce, select **TRUE**. Else, select **FALSE**.
Include Nonce in OCSP request	Specify one of the following: • **TRUE**: Select if the OCSP (Online Cer tificate Status Protocol) request shoul d include the nonce. This is the defaul t value. • **FALSE**: To avoid the EAP - TLS authen tication failure, select if the OCSP se rver does not support the nonce.
Enable signing for OCSP Request	To enable signing for OCSP request, sele ct **TRUE**. This determines whether Policy Manager should sign an OCSP request with a RADI US/EAP server certificate. The default val ue is **FALSE**.
Check the validity of all certificates in the chain against CRLs	To check the validity of all certificates in t he chain against Certificate Revocation Li sts (CRLs), select **TRUE**. Else, select **FALS E**.
Disable TLS 1.2	To disable Transport Layer Security 1.2 (T LS 1.2), select **TRUE**.

Service Parameter	Action/Description
	FALSE is the default setting—TLS 1.2 is en abled by default.
Check the validity of intermediary certific ates in the chain using OCSP	To check the validity of intermediary certi ficates in the chain using OCSP (Online C ertificate Status Protocol), select **TRUE**. The default setting is **FALSE**.
Maximum Number of AD Authentication Processes	To specify the maximum number of Acti ve Directory authentication processes, e nter a number between **1** and **5**. The default is **1**.
Verify OCSP Signing Purpose	Specify one of the following: • When the value for this parameter is s et to **TRUE**, EAP - TLS authentication will fail unless the OCSP signing certi ficate also has the OCSP signing purp ose set. • If this parameter is left at the default s etting of **FALSE**, the OCSP signing cer tificate does not need to have the OC SP signing purpose set.
OCSP server connection preference	Specify whether Policy Manager should u se an **IPv4** or **IPv6** network to connect to the OCSP server. The default value is **IPv 4**.
Disable Policy Manager Machine account password expiry in AD	Specify one of the following: • When the value for this parameter is s et to **TRUE**, it disables the Samba mac hine account password timeout settin g so that a change to the machine ac count password will not be made. • If this parameter is left at the default setting of **FALSE**, the Samba machine

Service Parameter	Action/Description
	account password timeout setting re mains in effect.
Skip Netevents for IPv6 Accounting Pac kets	This option is recommended in a dual - s tack (IPv4/IPv6) environment when Policy Manager is configured to send updates t o endpoint servers. Use this option to disable accounting net events if an IPv6 address is present in the accounting packet. When you enable this option (that is, set t o **TRUE**), Policy Manager does not send c lient IPv6 updates to endpoint servers. T he default setting is **FALSE**.
DH Param Size	Select one of the following sizes for the D iffie Hellman key exchange: • DH2048 : Use a 2046 - bit Diffie Hellm an key • DH4096: Use a 4096 - bit Diffie Hellm an key **NOTE** Support for Suite - B enable d clients using WPA3 requi res either the **NIST/SECG cu rve over a 384 bit prime fiel d** option, or the **4096 - bit** D H Param Size option.
Parse Cisco - AVPair to get device mac	Identify the device MAC address using Ci sco - supported RADIUS attribute - value (AV) pairs.
Kerberos DNS URI lookup	Starting with ClearPass 6.10.2, you can u se this setting to enable or disable Kerbe ros DNS URL lookup during the process t o join or authenticate users to your Acti

Service Parameter	Action/Description
	ve Directory domain controller. This setti ng is enabled (set to TRUE) by default. T he setting can be disabled (set to False) i f DNS servers don't response to Kerberos URI queries, causing ClearPass appliance to retry multiple times causing significant delay and fail to join or to authenticate us ers to Active Directory
Request Expire Time	Radius server requests are prone to time out when they take longer than 50 secon ds. Set the Request Expire Time paramet er from 30 - 300 seconds as needed to accommodate the anticipated time need ed to authenticate to the Radius server. C onfiguring a high value means more requ est state must be maintained, and will inc rease Radius process memory usage. The default is 45 seconds.
TLS Session Cache Limit	Specify the number of TLS sessions to c ache before purging the cache (used in T LS based 802.1X EAP Methods). The range is from **1,000** to **100,000**. The default is **10,000**.

Stats Collection Service Options

About this task

To configure the Statistics Collection service:

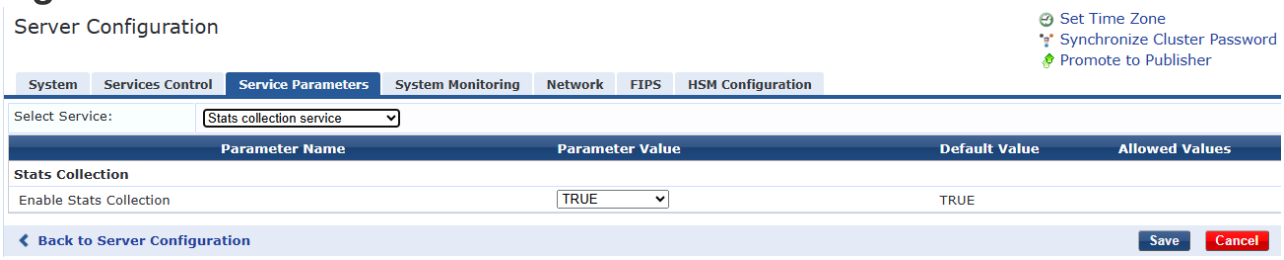
Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.

3. From the **Select Service** drop-down, select **Stats collection service**.

   The following dialog opens:

Figure 1. Statistics Collection Service



4. Specify the **Service Parameters** tab > **Stats Collection Service** parameter as described in the following table:

Table 1. Service Parameters > Statistics Collection Service

Service Parameter	Action/Description
Enable Stats Collection	Enable or disable statistics collection and aggregation. The **Statistics Collection Service** is enab led by default (**TRUE**). If this is not enabled, statistics collection and aggregation services will not run on t he node. In addition, if statistics collection and ag gregation is not enabled, the following er ror message is displayed if the administra tor attempts to start these services: **Failed to start Stats collection service - Ignoring service start request as Stats C ollection option is disabled on the node** **NOTE** Enabling or disabling this p arameter requires a restart of the *cpass - statsd - serve*

Service Parameter	Action/Description
	r and the *cpass - carbon - s erver*.

System Monitor Service Options

About this task

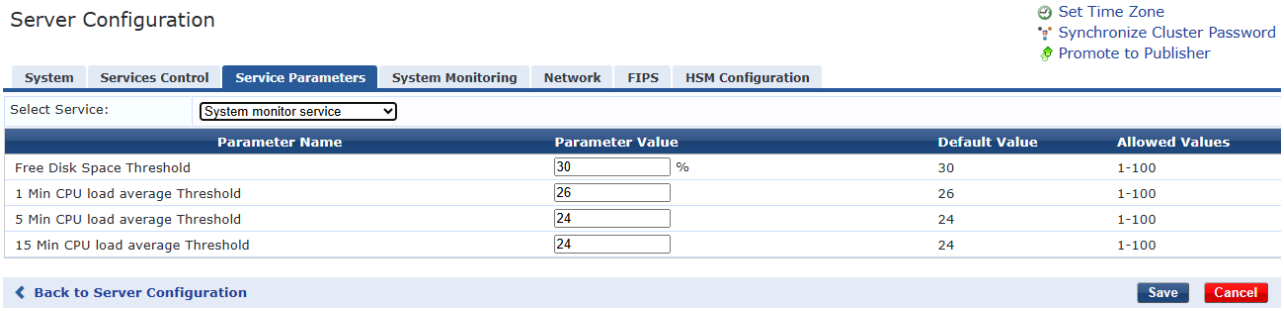
To configure the System Monitor service:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **System monitor service**.

   The following dialog opens:

Figure 1. System Monitor Service



4. Specify the **Service Parameters > System Monitor Service** parameters as described in the following table:

Table 1. Services Parameters > System Monitor Service Parameters

Service Parameter	Action/Description
Free Disk Space Threshold	This parameter monitors the available dis k space on the current Policy Manager se rver node. Specify the **Free Disk Space Threshold** (t he default is **30%**). If the available disk free space falls below the specified threshold, the Policy Manag er server sends SNMP traps to the config ured trap servers.
1 Min CPU load average Threshold 5 Min CPU load average Threshold 15 Min CPU load average Threshold	These parameters monitor the CPU load average of the system, specifying thresh olds for 1 - minute, 5 - minute, and 15 - m inute averages, respectively. If any of these loads exceed the associat ed maximum value, the Policy Manager s erver sends traps to the configured trap s ervers.

5. Click **Save**.

TACACS+ Server Options

About this task

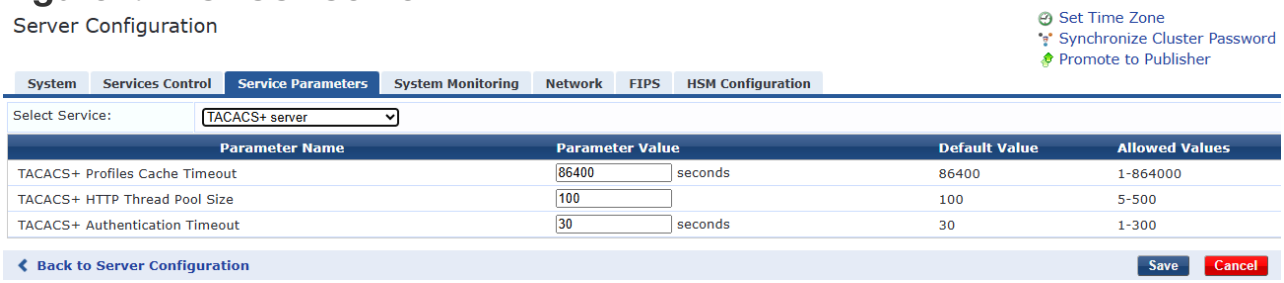
To configure the TACACS+ Server service:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **TACACS+ server**.

The following dialog opens:

Figure 1. TACACS+ Server



4. Specify the **Service Parameters > TACACS+ server** parameters as described in the following table:

Table 1. Service Parameters > TACACS+ Server

Service Parameter	Action/Description
TACACS+ Profiles Cache Timeout	Specify the time (in seconds) for which TACACS+ profile result entries are cached by ClearPass Policy Manager. The default is **86400** seconds (24 hours).
TACACS+ HTTP Thread Pool Size	Specify the maximum number of simulta neous requests the server can handle. The default value is **100**. The range is from **5** to **500**. When the server has reached the limit or request threads, it defers processing new requests until the number of active requ ests drops below the specified amount. I ncreasing this value reduces HTTP respo nse latency times.
TACACS+ Authentication Timeout	Specify the timeout value for the TACACS + server. The default value is **30** seconds. The range is from **1** to **300** seconds.

Service Parameter	Action/Description
	This parameter specifies the timeout bet ween the TACACS+ server (the Policy Man ager server) and the Authentication serve r (such as Active Directory, Token server, or any other authentication method). If the Authentication server does not repl y back to Policy Manager (TACACS+ serve r) within the configured TACACS+ Authen tication Timeout, then the TACACS+ serve r (Policy Manager) will timeout.

5. Click **Save**.

System Monitoring Page

About this task

By configuring the **System Monitoring** parameters, you can ensure that the external Management Information Base (MIB) browsers can browse the system-level MIB objects exposed by the ClearPass Policy Manager appliance. The options in this page vary based on the SNMP version that you select.

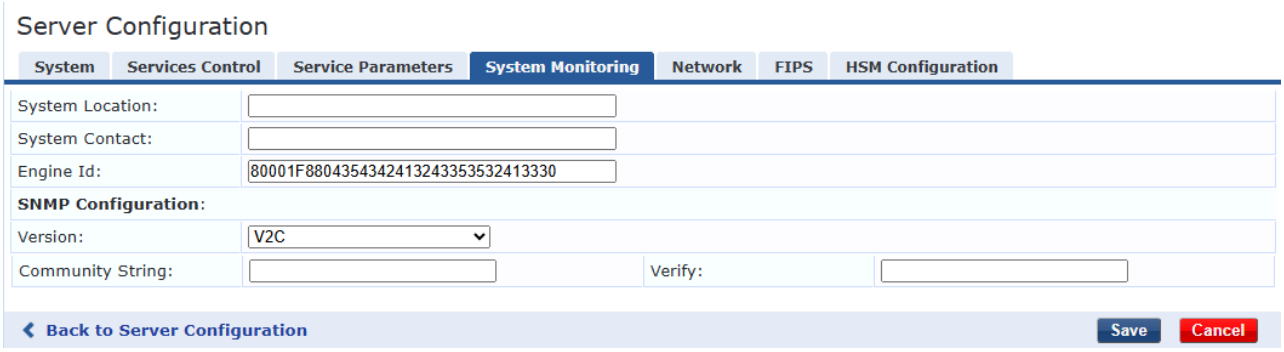
To configure the System Monitoring parameters:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Select the Policy Manager server of interest.
3. Select the **System Monitoring** tab.

   The **System Monitoring** configuration dialog opens:

Figure 1. System Monitoring



4. Specify the **System Monitoring** configuration parameters as described in the following table:

Table 1. System Monitoring Parameters

Parameter	Action/Description
System Location	Specify the location of the ClearPass Poli cy Manager appliance.
System Contact	Specify the contact information of the Cl earPass Policy Manager appliance.
Engine ID	A unique identifier for the SNMP v3 agent . The engine ID is used with a hashing fun ction to generate keys for authentication and encryption of SNMP v3 messages. The default value for the Engine ID is **66 20000004030662**. The Engine ID is automatically generated when you enable the stand - alone SNMP agent.
SNMP Configuration	
Version	Specify the SNMP version from the optio ns **V1**, **V2C**, or **V3**. The SNMP parameters on this page vary based on the SNMP version selected.
Community String	V1 and V2C: Enter and reenter the comm unity string for sending traps. This is appl

Parameter	Action/Description
	icable only for SNMP V1 and V2C versions .
Username	V3 only: Specify the user name to use for SNMP v3 communication.
Security Level	V3 only: Select any of the following optio ns: • **NOAUTH_NOPRIV** (No authentication or privacy): When you select this secu rity level, only the **SHA** authentication protocol is available. • **AUTH_NOPRIV** (Authentication but n o privacy): When you select this secur ity level, the **MD5** and **SHA** authentica tion protocols are available. • **AUTH _PRIV** (Authenticate and keep t he communication private): When you select this security level, the **MD5** and **SHA** authentication protocols are avai lable.
Authentication Protocol	V3 only: Select the authentication protoc ol from **MD5** or **SHA**. These protocols vary depending on the s ecurity level that you selected in the **Sec urity Level** field. **NOTE** The **MD5** authentication pro tocol is not supported in **FI PS** mode.
Authentication key	V3 only: Enter and reenter the authentic ation key. This field is available only if yo u selected **V3** as the SNMP version in the **Version** field.

Parameter	Action/Description
Privacy Protocol	V3 only: Select the privacy protocol from **DES** or **AES**.
Privacy Key	V3 only: Enter the privacy key.

Network Page

About this task

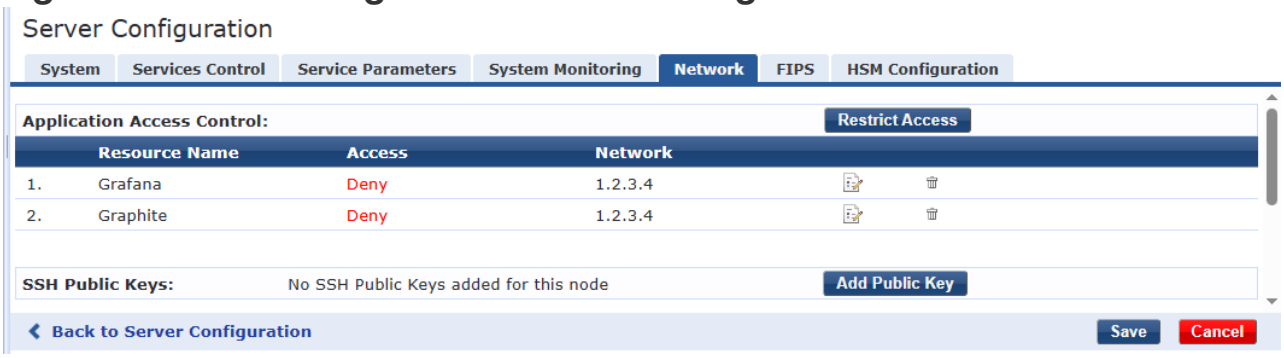
To configure the **Server Configuration > Network** parameters:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest.
3. Select the **Network** tab.

   The **Server Configuration > Network** page opens:

Figure 1. Server Configuration > Network Page



Defining Application Access Control Restrictions

About this task

Use this function to define specific network resources and allow or deny them access to specific applications. You can create multiple definitions.

To configure network application access control restrictions:

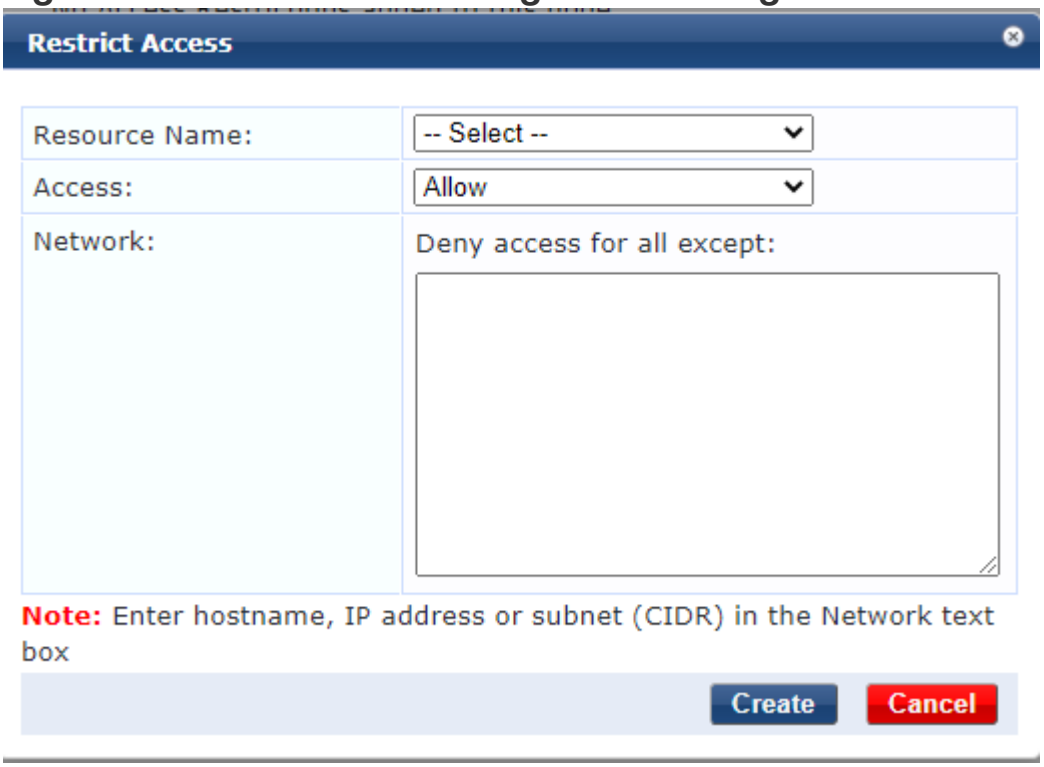
Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest.
3. From the **Server Configuration** page, select the **Network** tab.

   The **Server Configuration > Network** page opens.
4. From the **Application Access Control** option, click **Restrict Access**.

   The **Restrict Access** dialog opens.

Figure 2. Restrict Access Configuration Dialog



5. Specify the **Restrict Access** parameters as described in the following table, then click **Create**:

Table 1. Restrict Access Parameters

Parameter	Action/Description
Resource Name	Select the application for which you want to allow or deny access: • OnGuard • Policy Manager API

Parameter	Action/Description
	• CLI • Policy Manager • Grafana • Graphite • Guest Operator • Insight **NOTE** Restricting CLI access restri cts SSH access to the *appad min* user. The console acces s is still available through the *appadmin* user account.
Access	Select one of the access control options: • Allow: Allows access to the selected a pplication. ◦ When you configure the Access o ption as **Allow**, with a specific set of IP addresses or subnets, this se t becomes the allowed list. This m eans there is an implicit Deny Acc ess statement for all other IP add resses or subnets in the network. • Deny: Denies access to the selected a pplication. ◦ When you configure the Access o ption as **Deny**, with a specific set o f IP addresses or subnets, then eve ry other IP address and subnet in t he network has access except tho se specified here. This means the re is an implicit Allow Access state

Parameter	Action/Description
	ment for all other IP addresses or s ubnets in the network.
Network	Enter one or more host names, IP addres ses, or IP subnets (CIDR) per line. Startin g with ClearPass 6.10.2, this field accepts both IPv4 and IPv6 addresses. The devices defined by what you enter h ere will be either specifically allowed or s pecifically denied access to the applicati on you select.

Grafana Access Control

Refer to the following guidelines when configuring Grafana application access:

- Grafana is available on https://<CPPM_IP>/grafana. ClearPass API User credentials are used to login.
- To access Grafana, set the Application Access Control rule for the Grafana resource in **Administration > Server Manager > Server Configuration > Network** and ensure the Grafana server service is in a Running status in **Administration > Server Manager > Server Configuration > Services Control**.
- Grafana is initially configured with a Graphite datasource (Graphite-Default) pointing to Graphite running on a local node, and an example dashboard (Default Monitoring Dashboard). When configuring Graphite datasources pointing to other nodes, ensure port (:8081) is not used in the URL.
- Grafana is not supported in CC mode. The Grafana service is stopped when CC mode is enabled in Policy Manager.

Remove a Node from a Cluster Without Impacting OnGuard

This section describes how to remove a cluster node (SUB1 in this example) without impacting the cluster's exising OnGuard configuration. The tasks required include:

- Blocking OnGuard on node SUB1
- Moving OnGuard clients from SUB1 to a different node
- Dropping SUB1 from the cluster

To remove a node from a ClearPass cluster without impacting the OnGuard configuration:

> **NOTE**
>
> Complete the following steps before removing the intended node from the cluster

1. Navigate to **Administration > Server Manager > Server Configuration**. Select a listed server resource and then the **Network** tab. Select the **Restrict Access** button.
2. From the Restrict Access screen select **OnGuard** from the **Resource Name** drop-down menu.
3. Select **Deny** from the **Access** drop-down menu. With this restriction configured, OnGuard Agents will no longer send WebAuth requests to SUB1.
4. Select the **Create** button to save the deny access restriction for OnGuard.
5. Navigate back to the **Administration** > **Server Manager** > **Server Configuration** screen and select SUB1.
6. Select the **Services Control** tab.
7. Locate **Async network services** and select the **Stop** button to stop this service on SUB1

> **NOTE**
>
> When **Async network services** is stopped on SUB1, the control channel will close for all the OnGuard Agents connected to SUB1. Once the control channel is closed, and OnGuard Agents cannot establish a control channel with SUB1, the OnGuard Agents will send WebAuth requests to other nodes in the cluster. Wait approximately 30 minutes to allow OnGuard Agents from SUB1 to move to other cluster nodes before dropping SUB 1 from the cluster.

8. Drop SUB1 from the cluster.

Adding an SSH Public Key

About this task

Policy Manager supports public key-based SSH logins. This includes public key management and the ability to enable public key authentication in Policy Manager on a node-by-node basis.

When you add the SSH public key to the clients, Policy Manager allows passwordless SSH public key-based authentication to the appadmin Policy Manager console.

> **NOTE**
> SSH public key-based authentication will continue to work even when the cluster password or the appadmin password have been changed.

To add an SSH public key:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.

   The **Server Configuration** page opens.

2. Select the Policy Manager server for which passwordless SSH is needed.

   The **Server Configuration** dialog for the selected server opens.

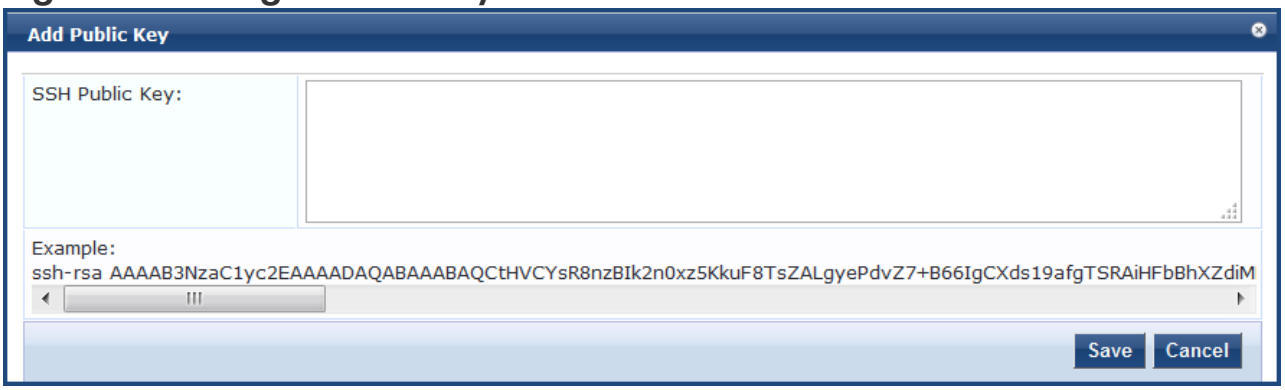
3. Select the **Network** tab.

   The **Server Configuration > Network** page opens.

4. From the **SSH Public Keys** option, click **Add Public Key**.

   The **Add Public Key** configuration page opens.

Figure 3. Adding a Public Key



5. In the **SSH Public Key** page, copy and paste the SSH public key of the client, then click **Save**.

> **NOTE**
>
> If the SSH public key is regenerated on the client, passwordless public key-based SSH authentication will cease to work. The existing entry for that client must be deleted. Then copy and paste the new SSH public key.

6. From the **Server Configuration** page, click **Save**.

   The SSH operation to the Policy Manager server using a public key is now active, and you can perform passwordless SSH to the Policy Manager server appadmin console.

Creating GRE Tunnels

About this task

You can use the Generic Routing Encapsulation (GRE) protocol to create a virtual point-to-point link over a standard IP network or the Internet.

To create a GRE tunnel:

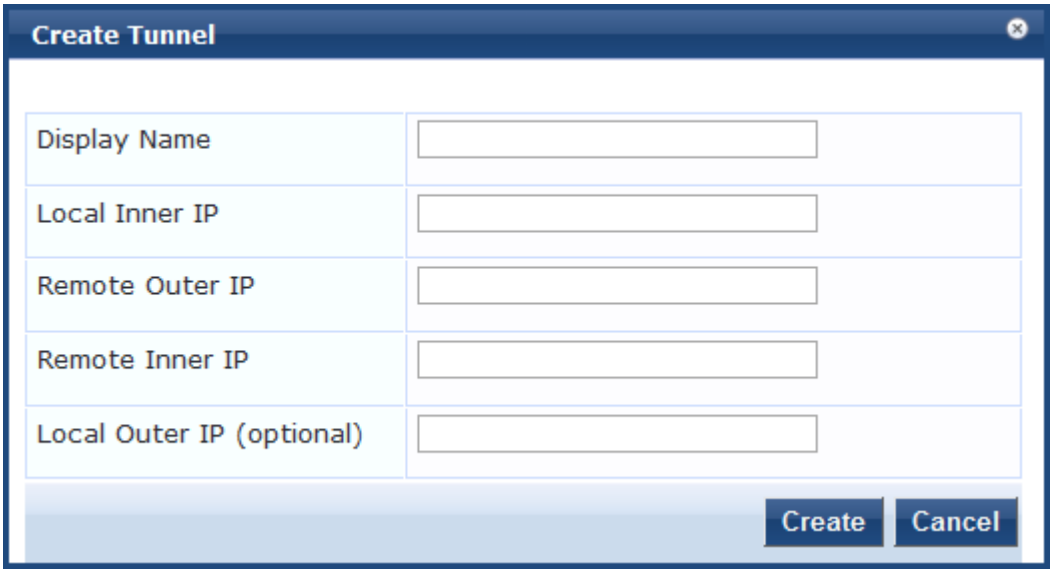
Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest.
3. From the **Server Configuration** page, select the **Network** tab.

   The **Server Configuration > Network** page opens.
4. From the **GRE Tunnels** option, click **Create Tunnel.**

   The **Create Tunnel** dialog opens:

Figure 4. Creating a GRE Tunnel



5. Specify the **Create Tunnel** parameters as described in the following table, then click **Create**:

Table 2. Create Tunnel Parameters

Parameter	Action/Description
Display Name	Specify the name for the tunnel interface. This name is used to identify the tunnel in the list of network interfaces.
Local Inner IP	Enter the local IP address of the tunnel network interface.
Remote Outer IP	Enter the IP address of the remote tunnel endpoint.
Remote Inner IP	Enter the remote IP address of the tunnel network interface. Enter a value to automatically create a route to this address through the tunnel.
Local Outer IP (Optional)	Optionally, enter the local IP address of the tunnel endpoint.

Creating IPsec Tunnels

About this task

Policy Manager provides the option to configure rules that can determine which IPsec traffic to tunnel, which traffic to drop, and which traffic to encrypt or bypass.

Thus, Policy Manager supports adding traffic selectors based on port number and protocol (TCP/UDP) with rule options *Bypass, Encrypt*, and *Drop*.

> **NOTE**
> To meet FIPS compliant requirements in RL9 FIPS mode, IPSec Tunnels configured with a Pre-Shared Key authentication type require the IKE Shared Secret (Pre-Shared Key) to be a length >=14 characters.

To create an IPsec tunnel:

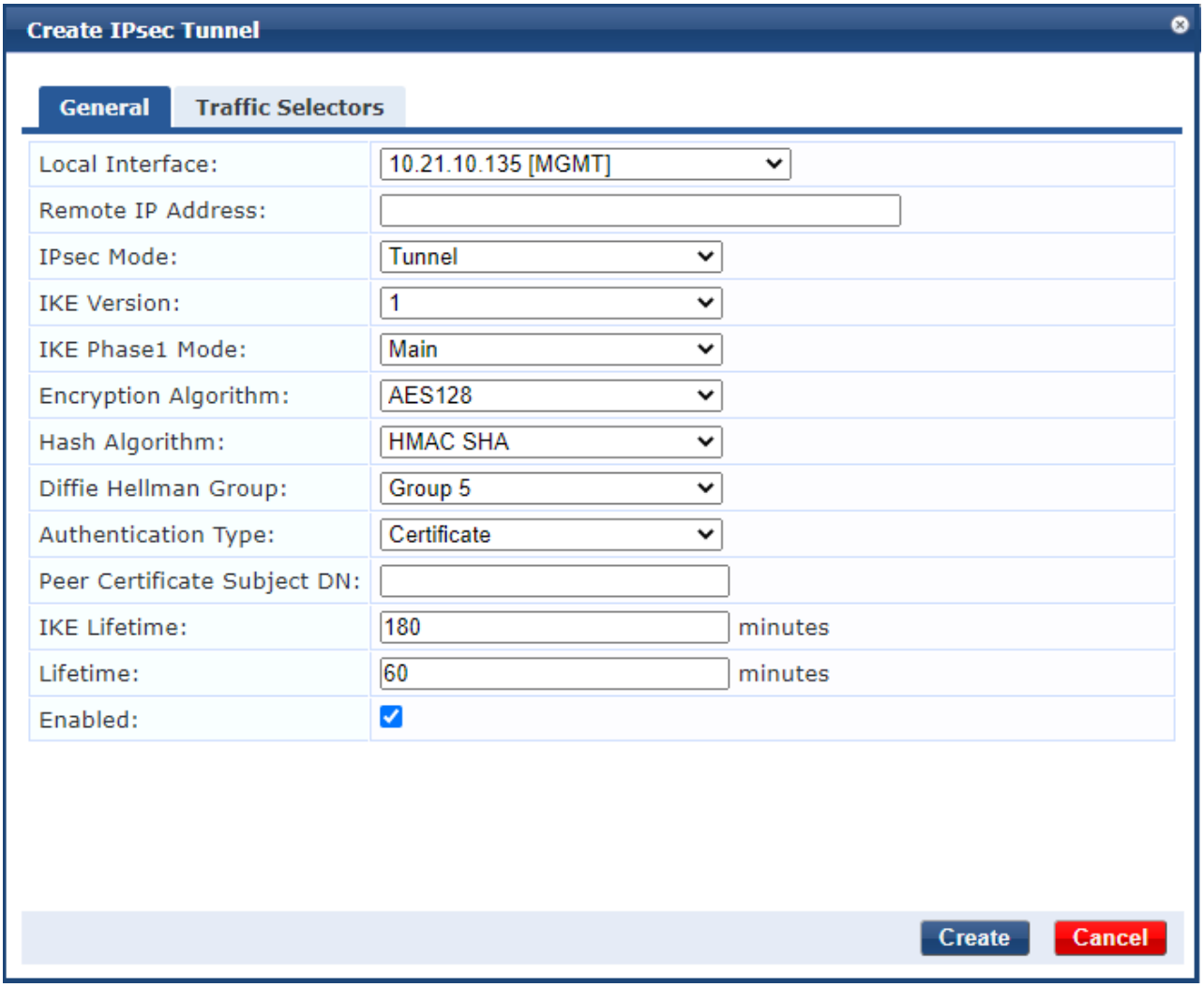
Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest.
3. From the **Server Configuration** page, select the **Network** tab.

   The **Server Manager > Configuration > Network** page opens.
4. Click **Create IPsec Tunnel**.

   The **Create IPsec Tunnel** dialog opens to the **General** tab.

Figure 5. Creating an IPsec Tunnel Dialog



5. Specify the **Create IPsec Tunnel** parameters as described in the following table, then click **Create**:

Table 3. Create IPSec Tunnel Parameters

Parameter	Action/Description
Local Interface	Specify the local Management interface. • IPv6 addresses are available in the dro p - down list and can be specified as t he remote IP address. • The local interface and remote IP add ress must either both be IPv4, or both be IPv6. **NOTE**

Parameter	Action/Description
	The IP addresses for the local and remote fields cannot be local link addresses starting with *fe80*.
Remote IP Address	Specify the IP address of the remote host. The remote IP address can be either IPv4 or IPv6.
IPsec Mode	Select one of the following IPsec modes: • **Tunnel** • **Transport**
IKE Version	Select the version of the Internet Key Exchange (IKE) protocol from the options: **1** or **2**.
IKE Phase 1 Mode	This parameter is enabled when you select **IKE Version 1**. **IKE Phase 1 Mode** is set by default to **Main**.
Encryption Algorithm	Select one of the following encryption algorithms: • AES128 • AES256
Hash Algorithm	Select one of the following hash algorithms: • HMAC SHA • HMAC SHA1 • HMAC SHA256 • HMAC SHA384 • HMAC SHA512

Parameter	Action/Description
	• HMAC MD5
PRF	The **PRF** (*pseudorandom function*) parameter is enabled when you select **IKE Version 2**. Select one of the following PRF options: • PRF - HMAC - SHA1 • PRF - HMAC - SHA256 • PRF - HMAC - SHA384 • PRF - HMAC - MD5
Diffie Hellman Group	Select one of the following Diffie Hellman groups: • Group 5 • Group 14 • Group 19 • Group 20
Authentication Type	Select one of the following authentication types: • **Pre - Shared Key** • **Certificate**
Peer Certificate Subject DN	Subject Distinguished Name (DN) of the peer certificate.
IKE Shared Secret Verify IKE Shared Secret	Enter the IKE Shared Secret, then verify the IKE secret key. The IKE Shared Secret can be up 128 characters in length. **NOTE**

Parameter	Action/Description
	This option appears only wh en the Authentication Type i s set to **Pre - Shared Key**.
IKE Lifetime	Specify the number of minutes for the li fetime of the IKE. The default is **180** minu tes. The valid range for **IKE Lifetime** is from 5 to 1440 minutes.
Lifetime	Specify the lifetime of the IPsec tunnel in minutes. The default is **60** minutes. The valid range for **Lifetime** is from 5 to 1440 minutes.
Enabled	To enable the IPsec tunnel, click the **Enab led** check box.

Traffic Selectors

A traffic selector (also known as a *proxy ID* in IKEv1) is an agreement between IKE peers to permit traffic through a tunnel if the traffic matches a specified pair of local and remote addresses. Only traffic that conforms to a traffic selector is permitted through the associated IPsec security association (SA).

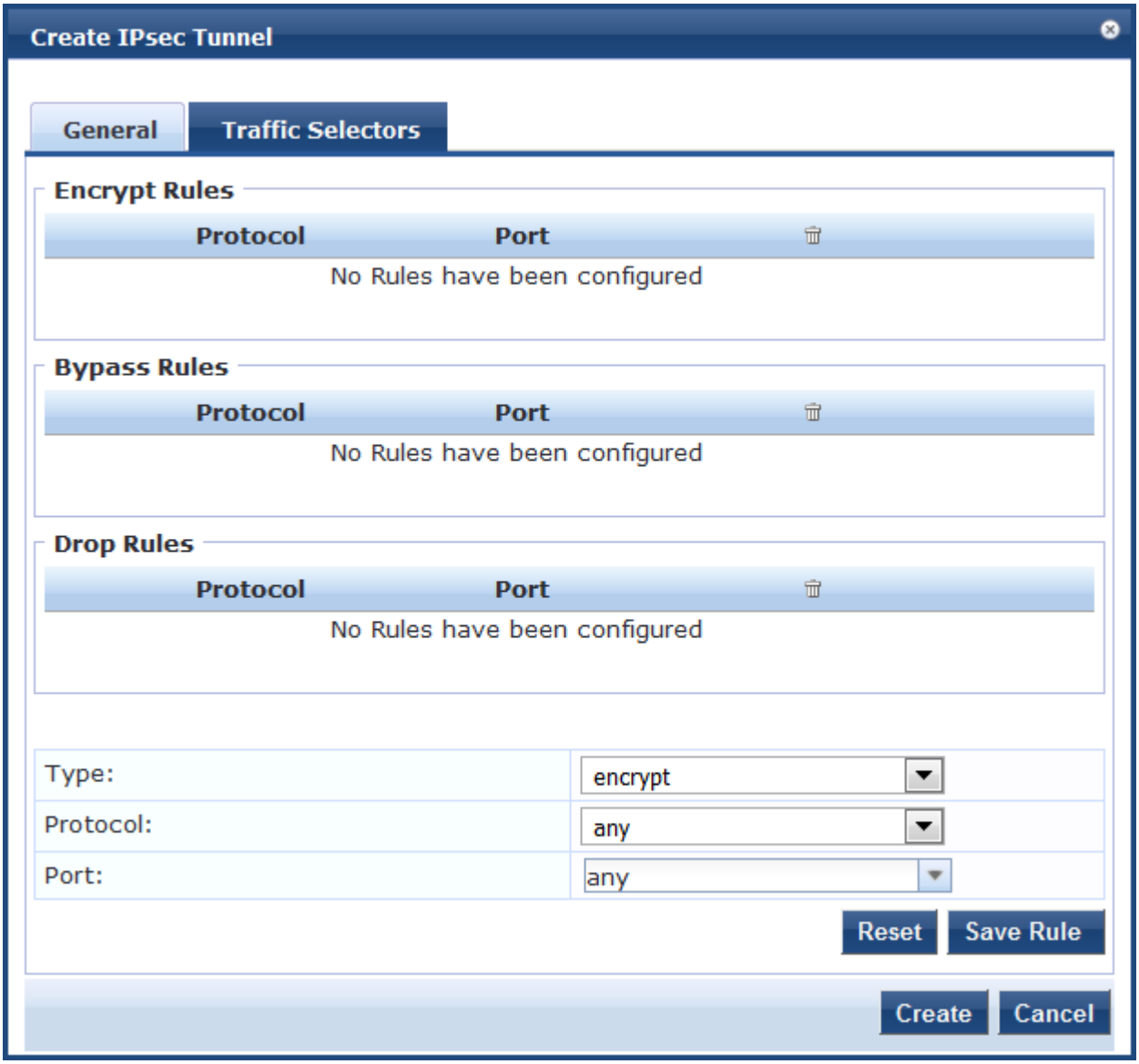
Traffic selectors are retained after a system restart, a service restart of network services, and a service restart of the IPsec service.

To configure the traffic selectors for this IPsec tunnel:

6. From the **Create IPsec Tunnel** dialog, select the **Traffic Selectors** tab.

   The **Traffic Selectors** dialog opens.

Figure 6. Create IPsec Tunnel > Traffic Selectors Dialog



7. Specify the **Traffic Selectors** parameters as described in the following table, then click **Create**.

Table 4. Create IPSec Tunnel > Traffic Selectors Parameters

Parameter	Action/Description
Encrypt Rules	Displays the IPsec tunnel encryption rul es configured for this IPsec tunnel.
Bypass Rules	Displays the IPsec tunnel bypass rules co nfigured for this IPsec tunnel.
Drop Rules	Displays the IPsec tunnel drop rules confi gured for this IPsec tunnel.

Parameter	Action/Description
Type	Select one of the following traffic selecto r types: • **Bypass** • **Encrypt** • **Drop**
Protocol	Select one of the following protocols: • **Any** • **TCP** • **UDP**
Port	From the **Port** drop - down list, select the port.
Reset	To reset the configuration settings to the defaults, click **Reset**.
Save Rule	To save the current Rule configuration, cli ck **Save Rule**.

Table 5. Create IPSec Tunnel > Traffic Selectors Parameters

Parameter	Action/Description
Encrypt Rules	Displays the IPsec tunnel encryption rul es configured for this IPsec tunnel.
Bypass Rules	Displays the IPsec tunnel bypass rules co nfigured for this IPsec tunnel.
Drop Rules	Displays the IPsec tunnel drop rules confi gured for this IPsec tunnel.
Type	Select one of the following traffic selecto r types: • **Bypass**

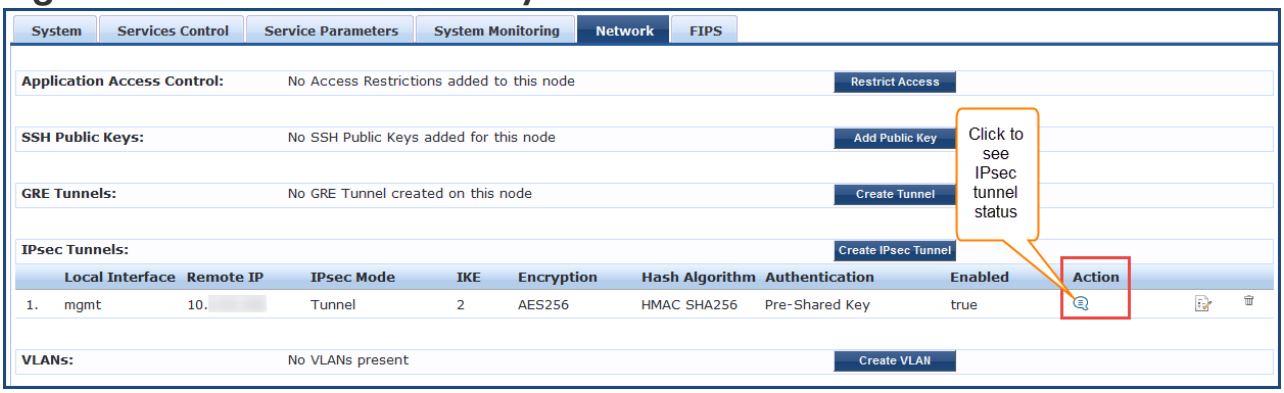
Parameter	Action/Description
	• **Encrypt** • **Drop**
Protocol	Select one of the following protocols: • **Any** • **TCP** • **UDP**
Port	From the **Port** drop - down list, select the port.
Reset	To reset the configuration settings to the defaults, click **Reset**.
Save Rule	To save the current Rule configuration, cli ck **Save Rule**.

Checking IPsec Tunnel Status

To check the status of an IPsec tunnel:

8. Navigate to the **Server Manager > Configuration > Network** page. The **IPsec Tunnels** section displays the configuration summary for each configured IPsec tunnel, along with an **Action** button to provide each IPsec tunnel's current status.

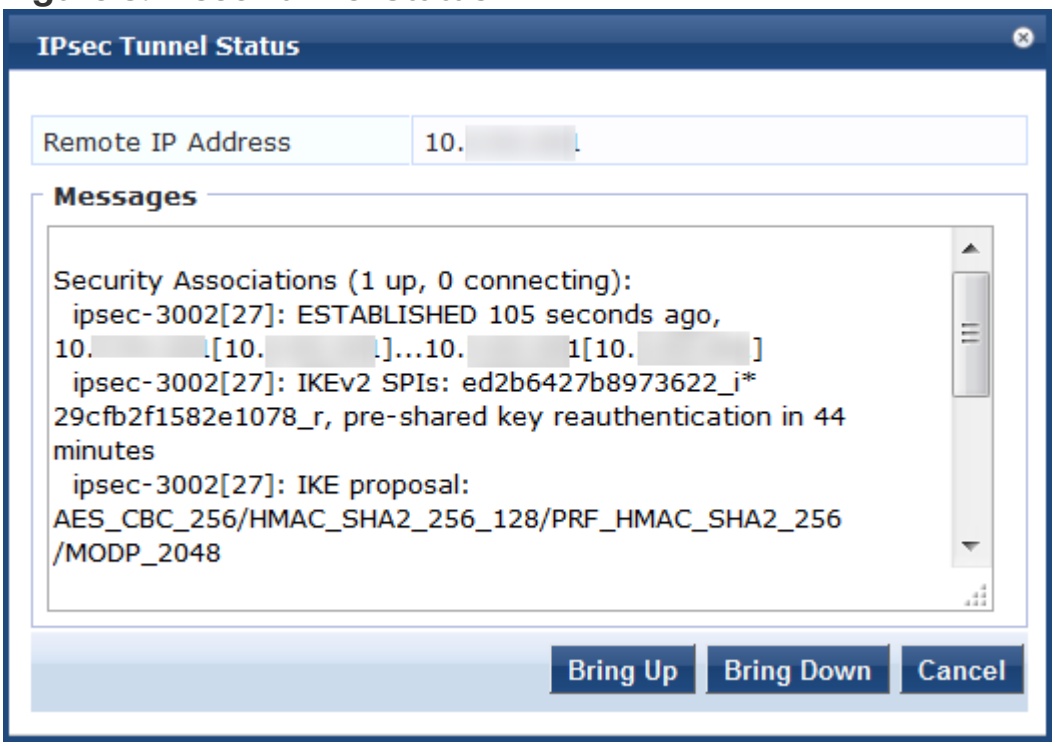
Figure 7. IPsec Tunnel Summary and Action Button to See Tunnel Status



9. To see the current status for an IPsec tunnel, click the **Action** button (see Figure 7).

   The **IPsec Tunnel Status** window for the selected tunnel opens:

Figure 8. IPsec Tunnel Status



- **Bring Up**

If the tunnel is down, **Bring Up** brings up the IPsec tunnel. If you select **Bring Up** when the tunnel is up, Policy Manager creates a new tunnel.

- **Bring Down**

If the tunnel is up, **Bring Down** tears down the IPsec tunnel.

If you select **Bring Down** when the tunnel is down (for example, when the tunnel is still negotiating), Policy Manager stops the tunnel from forming.

Understanding the IPsec Tunnel Status Information

A way to quickly decipher the IPsec tunnel status information is as follows:

- If the tunnel status shows **ESTABLISHED**, only IKE Phase 1 is complete.
- If the tunnel status shows **INSTALLED, Rekeying**, IKE Phase 2 is complete.

  Example 1

  If tunnel status displays as shown in Figure 9, Phase 1 is complete but Phase 2 is failing. Look at the Audit Viewer events (**Monitoring > Audit Viewer**) to find the root cause.

  Figure 9. IPsec Tunnel Status: Only IKE Phase 1 Complete

```
Security Associations (1 up, 0 connecting):
  ipsec-3001[21]: ESTABLISHED 2 minutes ago, 10.[redacted]1[10.[redacted].]...10.[redacted]L[10[redacted]]
  ipsec-3001[21]: IKEv2 SPIs: 601f3be10351483c_i* d033dd3590e120ff_r, pre-shared key reauthentication in 45
minutes
  ipsec-3001[21]: IKE proposal: AES_CBC_256/HMAC_SHA2_256_128/PRF_HMAC_SHA2_256/MODP_2048

Shunted Connections:
ipsec-bypass-3001-1:  10.[redacted]/32 === 10.[redacted]./32[udp/syslog] PASS

Shunted Connections:
ipsec-bypass-3001-2:  10.[redacted]/32[udp/syslog] === 10.[redacted]L/32 PASS
```

Example 2

When the tunnel status displays the information as shown in Figure 10, Phase 2 is also complete.

Figure 10. IPsec Tunnel Status: IKE Phase 1 and Phase 2 Complete

```
Security Associations (1 up, 0 connecting):
  ipsec-3001[24]: ESTABLISHED 10 seconds ago, 10.[redacted]1[10.[redacted]1]...10.[redacted]L[10.[redacted]]
  ipsec-3001[24]: IKEv2 SPIs: 032a75ba9e13b408_i 0bc85e45e5d7de2c_r*, pre-shared key reauthentication in 2 hours
  ipsec-3001[24]: IKE proposal: AES_CBC_256/HMAC_SHA2_256_128/PRF_HMAC_SHA2_256/MODP_2048
  ipsec-3001{20}:  INSTALLED, TUNNEL, ESP SPIs: c53510b1_i cb34ca38_o
  ipsec-3001{20}:  AES_CBC_256/HMAC_SHA2_256_128, 0 bytes_i, 0 bytes_o, rekeying in 44 minutes
  ipsec-3001{20}:   10.[redacted]/32 === 10.[redacted]/32
```

Creating VLANs

About this task

To create VLAN interfaces:

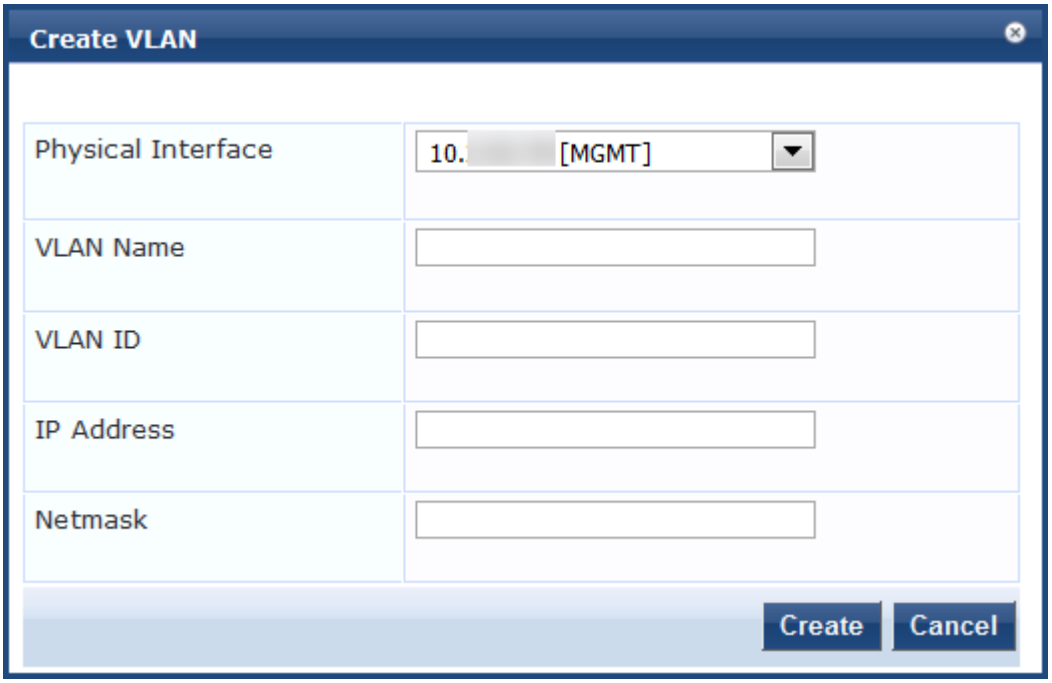
Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest.
3. From the **Server Configuration** page, select the **Network** tab.

   The **Server Configuration > Network** page opens.
4. From the **VLANS** option, click **Create VLAN.**

   The **Create VLAN** dialog opens:

   Figure 11. Creating a VLAN



5. Specify the **Create VLAN** parameters as described in the following table, then click **Create**:

Table 6. Server Configuration > Create VLAN Parameters

Parameter	Action/Description
Physical Interface	Enter the physical port on which to creat e the VLAN interface. This is the interface through which the V LAN traffic will be routed. **NOTE** Make sure your network sup ports tagged 802.1Q packets on the selected physical inte rface.
VLAN Name	Enter the name for the VLAN interface. This name is used to identify the VLAN in the list of network interfaces.
VLAN ID	Specify the 802.1Q VLAN identifier. Enter a value between 1 and 4094.

Parameter	Action/Description
	The VLAN ID cannot be changed after th e VLAN interface has been created. **NOTE** **VLAN ID 1** is often reserved for use by network manage ment components. Avoid usi ng this VLAN ID unless you k now it will not conflict with a VLAN already defined in you r network.
IP Address	Enter the IP address of the VLAN.
Netmask	Enter the netmask for the VLAN.

FIPS Page

This section provides information on using ClearPass Policy Manager in Federal Information Processing Standards (FIPS) 140-2 approved mode. The U. S. Government developed FIPS 140-2 to define procedures, architectures, cryptographic algorithms, and other security techniques for use in government applications and networks that use cryptography. When running in FIPS Approved mode, ClearPass Policy Manager utilizes a FIPS 140-2 validated cryptographic module. Support is not available for non-approved authentication methods such as EAP-MD5 and MD5 digest algorithms. Additionally, cipher suites that use RSA, DSA, or DH keys shorter than 2048 bits or ECC keys shorter than 224 bits are not supported in FIPS mode.

ClearPass 6.14 includes cryptography libraries that are FIPS 140-3 certified. The FIPS 140-3 standard mandates use the TLS Extended Master Secret (EMS) extension in order to be FIPS compliant. Starting with the 6.14 release, when ClearPass is in FIPS mode clients that connect by TLS must use cryptographic libraries that support the EMS extension or the connection will be rejected. This impacts RADIUS, Captive Portal authentications, RadSec connections, and HTTPS connections for UI logins from older clients. Affected clients must be upgraded to recent versions that support the EMS extension for TLS connections.

For details on the Validated FIPS 140-1 and FIPS 140-2 Cryptographic Modules, see:

http://csrc.nist.gov/groups/STM/cmvp/documents/140-1/140val-all.htm#2577

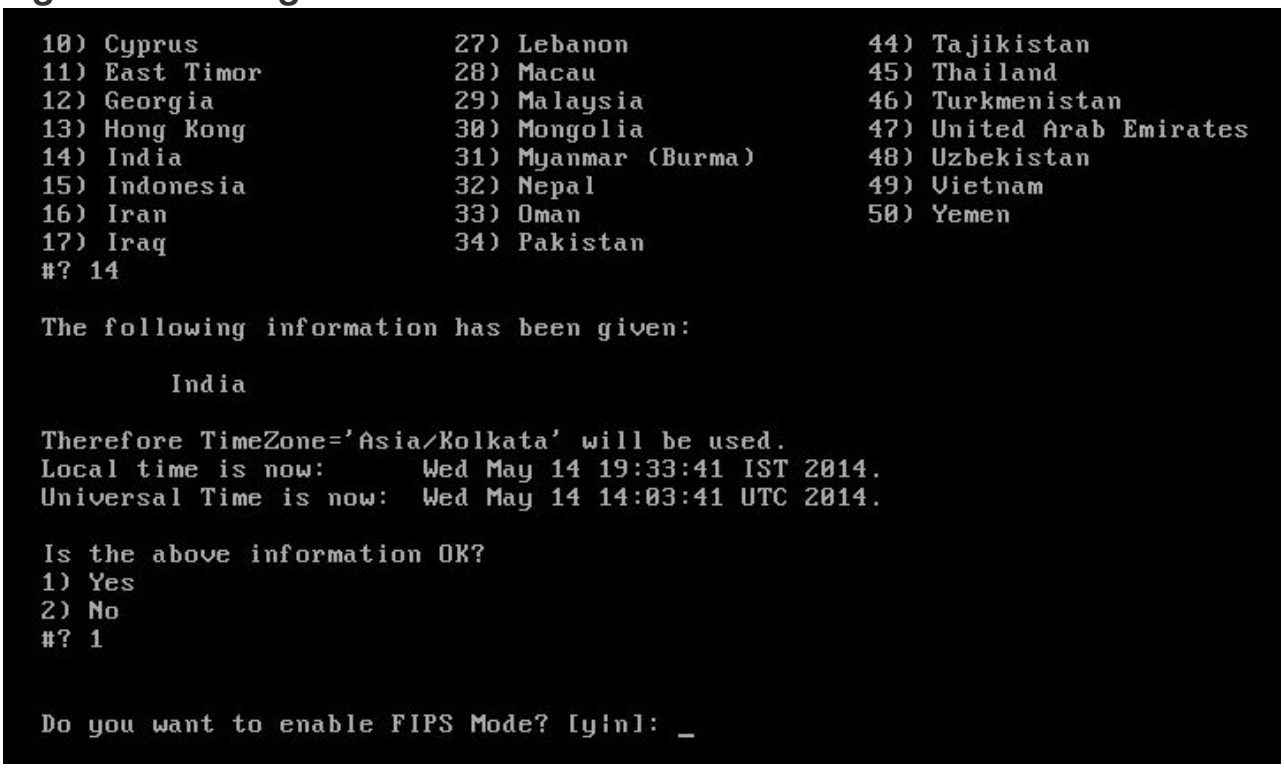
Enabling FIPS Mode Using CLI

Procedure

1. You can enable FIPS mode in Policy Manager during installation using the CLI or post-installation using the Web UI.

   The following figure displays the prompt to enable FIPS mode using the CLI:

   Figure 1. Enabling FIPS Mode



2. After enabling FIPS mode using the CLI commands, you can verify whether FIPS mode is enabled or not in the **Configuration Summary** page.

   Figure 2. FIPS Mode > Configuration Summary

```
=====================================================
              Configuration Summary
=====================================================
Hostname                        :  VM-582
Management Port IP Address      :  10.17.5.82
Management Port Subnet Mask     :  255.255.255.0
Management Port Gateway         :  10.17.5.254
Data Port IP Address            :  <not configured>
Data Port Subnet Mask           :  <not configured>
Data Port Gateway               :  <not configured>
Primary DNS                     :  10.17.4.10
Secondary DNS                   :  <not configured>
Primary NTP Server              :  pool.ntp.org
Secondary NTP Server            :  <not configured>
Timezone                        :  'Asia/Kolkata'
FIPS Mode                       :  True

=====================================================
```

Enabling FIPS Mode in the Policy Manager User Interface

About this task

Alternatively, you can enable or disable the FIPS mode in the Policy Manager user interface:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**, then select the server of interest. The **Server Configuration** dialog for the selected server opens.
2. Select the **FIPS** tab, then click **Enable**. On enabling FIPS mode, the WebUI displays the text [FIPS] in the version footer at the bottom of the page.

Figure 3. Server Configuration > FIPS Tab

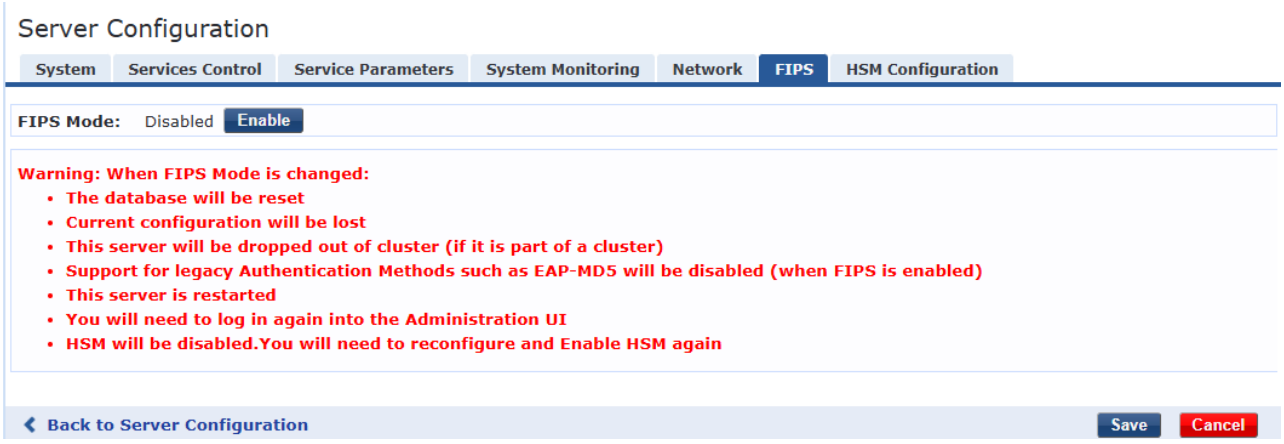


Figure 4. Server Configuration > FIPS Tab

Important Points to Remember

Note the following important points, when you enable FIPS mode in the ClearPass Policy Manager user interface:

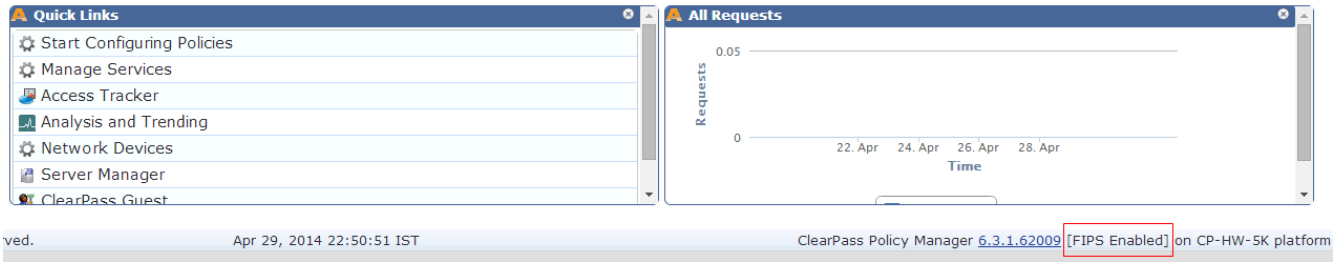
- The database is reset when you enable the FIPS mode in ClearPass Policy Manager.

> **CAUTION**
>
> Ensure that you backed up your database before enabling FIPS mode.

- Configuration backup file from the ClearPass Policy Manager in non-FIPS mode cannot be restored on ClearPass Policy Manager in FIPS mode. However, configuration backup file from the ClearPass Policy Manager in FIPS mode can be restored on the ClearPass Policy Manager in non-FIPS mode.
- The server will be removed from the cluster if FIPS mode is enabled.
- All nodes in a cluster must be either in FIPS or non-FIPS mode. The ClearPass Policy Manager nodes in FIPS mode cannot be connected to the cluster whose nodes are in the non-FIPS mode.
- The legacy authentication method such as EAP-MD5 and MD5 digest algorithm are not supported in FIPS mode. You cannot import the certificates that are created with the MD5 authentication type to the **Certificates Trust List** (**Administration > Certificates > Certificate Trust List**) page.
- The server reboots when you enable FIPS mode. You need to log in again to the Administration interface.
- Cipher suites that use RSA, DSA, or DH keys shorter than 2048 bits or ECC keys shorter than 224 bits are not supported in FIPS mode.

You can view the status of FIPS mode in the status bar. The following figure displays the **Status** bar with the status of FIPS mode:

Figure 5. FIPS Status



You can also view the status of the FIPS mode using the CLI commands. For more information, see Show Commands .

ClearPass Policy Manager Cluster Configuration Options

This section describes the cluster configuration options that are available from the **Administration > Server Manager > Server Configuration** page.

- Cluster-Wide Parameters
- Synchronizing the Cluster Date and Time with the NTP Server
- Changing the Cluster-Wide Password
- Managing Policy Manager Zones
- Configuring NetEvents Targets
- Clearing Machine Authentication Cache
- Configuring Virtual IP Settings
- Making a Subscriber Node

Subtopics

Cluster-Wide Parameters
Synchronizing the Cluster Date and Time with the NTP Server
Changing the Cluster-Wide Password
Managing Policy Manager Zones
Configuring NetEvents Targets
Clearing Machine Authentication Cache
Configuring Virtual IP Settings
Making a Subscriber

Cluster-Wide Parameters

This section describes the following **Cluster-Wide Parameters** features:

> **NOTE**
> In ClearPass 6.11.0 and later, the maximum single cluster size is 32 servers. This includes the publisher, standby publisher, subscribers, dedicated insight server, and standby insight servers.

Subtopics

General Parameters
Cleanup Intervals Parameters
Notifications Parameters
Standby Publisher Parameters
Common Criteria Mode Parameter
Database Parameters
Profiler Parameters
TACACS+ Parameters

General Parameters

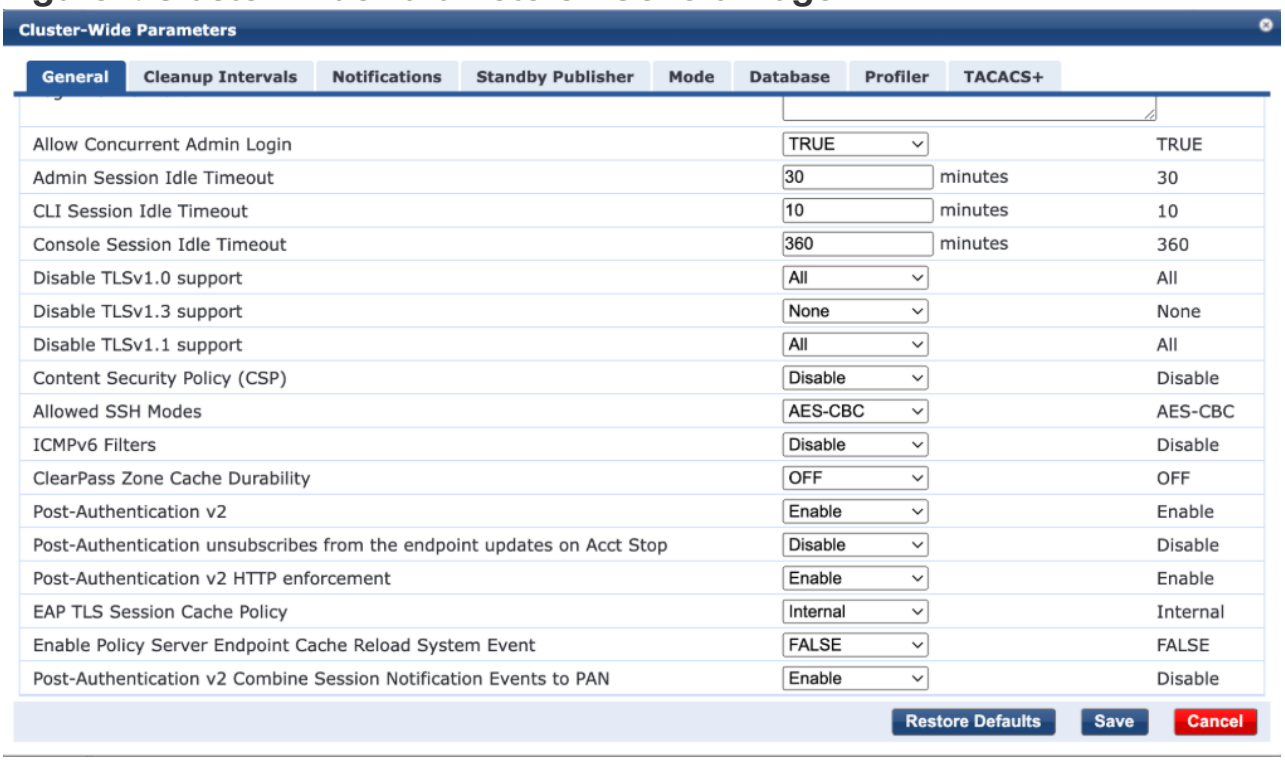
About this task

You can configure the parameters that apply to all the nodes in a Policy Manager cluster by configuring the **Cluster-Wide Parameters**. To configure the Cluster-Wide parameters:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Select the **Cluster-Wide Parameters** link. The **Cluster-Wide Parameters** page opens to the **General** page:

Figure 1. Cluster-Wide Parameters > General Page



3. Configure **Cluster-Wide Parameters > General** parameters as described in the following table, then click **Save**.

Table 1. Cluster - Wide Parameters > General Page Parameters

Parameter	Action/Description
Policy result cache timeout	Specify the duration allowed in minutes t o store the role mapping and posture res ults derived by the policy engine during a policy evaluation. A value of **0** disables caching. This result can then be used in subseque nt evaluation of policies associated with a service, if the **Use cached Roles and Po sture attributes from previous sessions** option is turned on for the service. **NOTE** The value of the **Policy resul t cache timeout** field must b e greater than the highest va lue set in the **Health Check Interval (in hours)** fields. Fo r example, if you have creat ed the profiles Student - En forcement - Profile and Staff - Enforcement - Profile with health check interval config ured, then the value of the **Policy result cache timeout** field must be greater than th e highest value of the **Healt h Check Quiet Period (in ho urs)** value configured amon g the following profiles: • **Global Agent Settings** • Student - Enforcement - Profile

Parameter	Action/Description
	• Staff - Enforcement - Profile
Free disk space threshold value (Aggressive cleanup starts at 10% below this value)	Specify the percentage below which disk usage warnings are issued in the **Monitoring > Event Viewer** page. For example, a default value of 30% indicates a warning is issued when the available disk space is 30% or lower. However, an aggressive cleanup is triggered when the disk space available drops below 20% (the 30% default value minus an additional 10%). In this instance, the additional 10% is the hard coded threshold delta value implemented as part of an aggressive cleanup operation.
Free memory threshold value	Specify the percentage below which RAM usage warnings are issued in the Policy Manager Event Viewer. For example, a value of **30** indicates that a warning is issued only when the available RAM is 30% or lower.
Endpoint Context Servers polling interval	Enter the interval in minutes between polling of endpoint context servers, between 1 and 10,500 minutes. (10,500 minutes is 175 hours, or a bit more than a week). The default interval is **60 minutes**.
Endpoint Context Servers polling start time	Specify the Endpoint Context Servers polling start time in *HH:mm:ss* format.
Syslog Export Interval	This parameter provides the ability to configure a Syslog messaging batch interval. The default value is 120 seconds. The supported interval value is between 30 seconds to 120 seconds. **NOTE**

Parameter	Action/Description
	The interval specified for thi s parameter is applied to al l appliances in a cluster and to all the Syslog Export Filte rs that are in the enabled sta te. For related information, s ee Syslog Export Filters.
Automatically check for available Softwa re Updates	Specify whether to enable automatic che cking for available software updates. The default it **TRUE**. For related informati on, see Software Updates.
Automatically download Posture Signatu re and Windows Hotfixes Updates	Specify whether to automatically downlo ad and install Posture Signature and Win dows Hotfixes Updates. The default is **FA LSE**. For related information, see Softwa re Updates.
Automatically download minor updates for Extensions	Set this parameter to **TRUE** to download micro - services that can run on top of th e ClearPass platform and provide access to some features not yet part of the curre nt ClearPass version. The default is **FALS E**.
Automatically download Endpoint Profile Fingerprints	Specify whether to automatically downlo ad Endpoint Profile Fingerprints. The defa ult is **FALSE**. For related information, see Software Updates.
Force Enable User Acknowledgment	Set this parameter to **TRUE** to require use r consent before login. Once set to True, a banner displays requiring the user to s elect **I agree to the terms and condition s** and then click **Proceed** before continui ng to the login.

Parameter	Action/Description
Login Banner Text	Customize the banner text that appears o n the Policy Manager login screen and CL I access window. **NOTE** By default, if the login bann er text and forced user ackn owledgment are configured for Policy Manager, they are also displayed in ClearPass Guest operator logins and g uest registrations. If you do not want these displayed in Guest logins and registratio ns, go to **Guest > Administra tion > Operator Logins > Log in Configuration**. Under **Ban ner Options**, deselect the **Lo gin Banner** check box.
Cluster Communication Mode	Policy Manager allows Cluster Communic ation Mode to be set to **IPv4** or **IPv6**. The default is **IPv4**. If the value of this parameter is set to **IPv 6**, all database and API calls will use IPv6 addresses for cluster communication. If t he value is set to ipv4, it will use IPv4 for database and API calls instead. The defau lt value of the cluster communication mo de will depend on the IP address configu red on the appliance during installation o r upgrade. If the appliance has only an IP v6 address, the default cluster communi cation mode will be IPv6. If the appliance has both IPv4 and IPv6 addresses config ured, or if only an IPv4 address is configu red, then the default cluster communicat ion mode will be IPv4.

Parameter	Action/Description
	Whenever the cluster communication mode is changed, it performs the following validations: • Configuration checks to verify an IP address in the correct format is configured for the interface. • Certificate checks to verify the database certificates have the correct IP address in the SAN field. • Certificate checks to verify the HTTPS certificates have the correct IP address in the SAN field. For more information on setting the Cluster Communication Mode and the conditions for using IPv4 and IPv6 formatted IP addresses, go to Setting the Cluster Communication Mode. **NOTE** You can only use the **network reset mgmt <v4|v6>** command to reset an IPv4 or IPv6 management port address if the cluster communication mode is set to a different IP address format. For example, you cannot reset a management port IPv4 address, unless the cluster communication mode is set to IPv6. **NOTE** In a cluster with both IPv4 and IPv6 addresses (dual stack) configured on the management interface of every appliance, the ClearPass Zone

Parameter	Action/Description
	Cache service always uses the IPv4 management address to keep the cache of each appliance synchronized across the cluster. The ClearPass Zone Cache service does not use the IPv6 management address for synchronization when the cluster communication mode parameter is set to IPv6 in a dual - stack scenario. **NOTE** In a cluster using self - signed certificates, if the management IP address is changed the database certificate does not need to be regenerated. The generation of the database certificate and the restart of backend services are automatic. There may be a delay of up to 10 minutes while backend services are re - started, and configuration updates and replication setup are re - established. This change only applies to clusters with self - signed certificates, not to clusters with CA - signed certificates.
Allow Concurrent Admin Login	• When this field is set to **TRUE** (which is the default setting), an Admin user can log in to the same Policy Manager server concurrently from either the same device or a different device.

Parameter	Action/Description
	• When this field is set to **FALSE**, and a n Admin user logs into a Policy Manag er server from one device, and then la ter logs into the same Policy Manager server from either the same device or a different device, the Admin user is l ogged out of all previous sessions on that Policy Manager server. When this occurs, the following message is displ ayed: ◦ *You have been logged out of previ ous active session(s)*
Admin Session Idle Timeout	Specify the maximum idle time permitte d for admin users, beyond which the sess ion times out. The default value is **30 minutes**. The all owed range is **5** to **1440** minutes (24 hou rs). HPE Aruba Networking recommends setti ng the **Admin Session Idle Timeout** as re quired to manage re-authentications.
CLI Session Idle Timeout	Specify the maximum idle time permitte d for CLI users, beyond which the session times out. The default value is **10 minutes**. The all owed range is **5** to **1440** minutes (24 hou rs). When this parameter is changed, the cha nges take effect when the client opens a new CLI session. Any active CLI sessions will continue to use the old timeout setti ng—the sessions have to be disconnecte d and reconnected for the updated timeo ut value to take effect. Before performing any operations like make subscriber, drop subscriber, backup, restore, update, or u

Parameter	Action/Description
	pgrade operations from the appadmin CL I that take more than 15 minutes, first ma ke sure to change the value of the CLI S ession Idle Timeout parameter to the am ount of time required for the operation. HPE Aruba Networking recommends setti ng the **CLI Session Idle Timeout** as requi red to manage re-authentications.
Console Session Idle Timeout	Specify the **Console Session Idle Timeou t** duration in minutes. The default value is **360 minutes**. The **Console Idle Session Timeout** does not begin counting down until the admin user exits the admin session, or when the admin session has timed out. If the user i s still logged in to the admin session, the **Console Session Idle Timeout** is not enfo rced. To impose an automatic log out fro m the Policy Manager server for an admi n user connected to the console, you mu st set the idle timeout for the **Admin Sess ion Idle Timeout** as well as the **Console S ession Idle Timeout**. HPE Aruba Networking recommends setti ng the **Console Session Idle Timeout** as required to manage re-authentications. The **Console Session Idle Timeout**: • Must have a valid integer value (for ex ample, a setting of *10.5* wouldn't be va lid). • The range of valid values is from 5 to 1,440 minutes (24 hours). • The Audit Viewer captures the details about changes in **Console Session Idl e Timeout**.

Parameter	Action/Description
Disable TLSv1.0 support	On new Policy Manager installations, TLS 1.0 is disabled by default, so the default s etting is **All**. To modify Transport Layer Se curity (TLS) v1.0 support, select one of th e following options: • None • Admin • Network • All **NOTE** Users cannot change into C ommon Criteria (CC) mode while the **Disable TLS v1.0** p arameter is set to **None**. This parameter must be set to **Al l** for CC mode. This paramet er is not configurable and d oes not appear in the Policy Manager WebUI when Policy Manager is in FIPs mode. **NOTE** Beginning with the ClearPas s 6.11 release, the ClearPass OpenSSL cryptography libra ry does not support TLS 1.0 when FIPS mode is enabled. As part of this change, the D isable TLSv1.0 support optio n is hidden within the **Cluste r - Wide Parameters** when F IPS mode is enabled. **NOTE**

Parameter	Action/Description
	ClearPass 6.11 and 6.12 are th e last Long Support Release (LSR) and Short Support Rel ease (SSR) versions to supp ort TLS v1.0 in any operating mode (they are already disa bled in FIPS and CC modes) . Customers can use Insight reports to identify clients th at might be using these TLS versions in RADIUS - related communications. After Clear Pass 6.12, the system - wide cryptographic policy will be set to DEFAULT in non - FIPS mode.
Disable TLSv1.3 support	To modify Transport Layer Security (TLS) v1.3 support, select one of the following options: **NOTE** TLS 1.3 removed renegotiati on, requiring the server to d ecide at the start of the ha ndshake whether client certi ficates are required. Now th e server cannot dynamically switch between certificate-r equired and certificate-optio nal modes within the same connection, so a single virtu al host can no longer suppo rt both certificate-required a nd certificate-optional conn ections. To support both sce narios (certificate-required v s. certificate-optional), a two virtual host configuration is required. As part of the fix to

Parameter	Action/Description
	support Mutual TLS authentication to request the client certificate, additional virtual host levels will get added automatically in the backend configuration once TLS1.3 is enabled from the Policy Manager UI. For information on configuring a SAML Service Provider (SP) Configuration, see SAML Service Provider (SP) Configuration. • **None**: Enables TLS v1.3 communication for all servers. On new ClearPass installations, this is the default setting. When TLS v1.3 is enabled it is used as the preferred connection for systems that support it, and can be used for post - handshake authentications. If a system does not support TLSv1.3, TLS v1.2 or v1.1 will automatically be used instead. • **Admin**: Disables TLSv1.3 for all servers. On upgraded ClearPass servers, this is the default setting. This setting must be used in cases where client certificate - based authentications with SSO, OnGuard, or downloadable user roles are configured, as those do not work with TLSv1.3. Having Admin (TLSv1.3 disabled) as the default setting for upgraded servers ensures that if any of those configurations are present, the TLS setting will not interfere with them.
Disable TLSv1.1 support	On new Policy Manager installations, TLS 1.1 is disabled by default, so the default setting is **All**. To modify Transport Layer Se

Parameter	Action/Description
	curity (TLS) v1.1 support, select one of the following options: • None • Admin • Network • All **NOTE** Clusters in Common Criteria (CC) mode can have a value of **None**, so TLS v1.1 is enabled on all devices. This is the default setting for clusters upgrading to Policy Manager 6.10. **NOTE** Beginning with the ClearPass 6.11 release, the ClearPass OpenSSL cryptography library does not support TLS 1.1 when FIPS mode is enabled. As part of this change, the Disable TLSv1.1 support option is hidden within the **Cluster - Wide Parameters** when FIPS mode is enabled. **NOTE** ClearPass 6.11 and 6.12 are the last Long Support Release (LSR) and Short Support Release (SSR) versions to support TLS v1.1 in any operating mode (they are already disabled in FIPS and CC modes)

Parameter	Action/Description
	. Customers can use Insight reports to identify clients th at might be using these TLS versions in RADIUS - related communications. After Clear Pass 6.12, the system - wide cryptographic policy will be set to DEFAULT in non - FIPS mode
Content Security Policy (CSP)	When enabled, Content Security Policy (CSP) has significant impact on the way b rowsers renders pages (e.g., inline JavaS cript is disabled by default and must be explicitly allowed in the Content Security Policy). CSP prevents a wide range of at tacks, including cross - site scripting and other cross - site injections.
Allowed SSH Modes	When you specify a particular cipher mo de, all Policy Manager servers in the clust er will accept only the SSH connections t hat use that cipher mode. Select one of the following to specify the SSH cipher mode: • AES - CBC • AES - CTR • AES - GCM • All **NOTE** Unlike legacy releases, Clear Pass now enables the selecti on of either **AES - CBC**, **AES - CTR**, **AES - GCM** or **All** to specify a specific SSH ciphe r mode in FIPS mode. With p

Parameter	Action/Description
	revious ClearPass releases, t he **Allowed SSH Modes** para meter was disabled when FI PS mode is enabled.
Performance Monitor Rendering Port	Specify the port for performance monitor rendering. The default value is **80**.
ICMPv6 Filters	When the **ICMPv6 Filters** parameter is se t to **Disable** (the default), Policy Manager responds to all ICMPv6 (Internet Control Message Protocol) traffic. When this para meter is set to **Enable**, the following restr ictions are applicable: • Policy Manager does not respond to I CMPv6 traffic sent to an anycast or m ulticast address. • Policy Manager does not transmit to I CMPv6 type - 3 (Destination Unreach able) messages.
ClearPass Zone Cache Durability	For the ClearPass Zone Cache to survive abrupt shutdowns, set this to **Normal** or **Full**. The default value is **OFF**. **NOTE** Enabling this feature might result in some performance degradation.
Post - Authentication v2	When **Post - Authentication v2** is enable d, it improves performance and scaling for post - authentication events when int eroperating with third - party systems. Th is setting is enabled by default in IPv4 - o nly, IPv6 - only, or dual - stack mode (IPv 4 and IPv6). Post - Authentication v2 can still be disabled when in IPv4 - only mod

Parameter	Action/Description
	e. However, in an IPv6 - only or dual stac k environment, disabling Post - Authentic ation v2 is not permitted. The following additional **Post - Authentic ation v2** restrictions apply: • When you either enable or disable **Po st - Authentication v2**, you must also manually restart the *Async Network* s ervice. For more information on starti ng or stopping services, refer to Servi ces Control Page • **Post - Authentication v2** does not su pport Policy Manager Proxy. • When defining a context server action , the following HTTP methods are sup ported: PATCH, DELETE, POST, GET, an d PUT.
Post - Authentication unsubscribes from the endpoint updates on Acct Stop	When using post authentication for NMA P trigger, enabling this option ensures en dpoints are unsubscribed on an accounti ng stop. This ensures a payload is not se nt to the NMAP service when the device reauthenticates and a different enforcem ent profile is applied. This option must be disabled for firewall i ntegration in general, and must be disabl ed for dual stack or pure IPv6 if multiple I P addresses are expected.
Post - Authentication v2 HTTP enforceme nt	Administrators have the option to enabl e or disable Post - Authentication v2 HTT P enforcement, when the **Post - Authenti cation v2** parameter is enabled. Selectin g **Post - Authentication v2 HTTP enforce ment** lets administrators disable the Post - Authentication HTTP enforcement profi le while the **Session Notification Enforce**

Parameter	Action/Description
	ment enforcement profile remains enabled. This can improve the scalability and performance of the Post - Authentication v2 HTTP enforcement in some deployments. If **Post - Authentication v2** is changed from Enable to Disable, **Post - Authentication v2 HTTP enforcement** automatically changes to Disable as well. Certificate Based Authentication is supported for VmWare AirWatch with Post - Authentication v2 HTTP Enforcement. To use this feature with AirWatch, both the **Post - Authentication v2** and **Post Authentication v2 HTTP enforcement** settings must be enabled from the Cluster - Wide **Parameters > General**tab. In addition, go to: **Administration > External Servers > Endpoint Context Servers > Server > Add**, and select **VMWare AirWatch** as the server type and **Certificate** as the authentication method. Lastly, go to: **Administration > Dictionaries > Context Server Actions > Add**, and select **VMWare AirWatch** as the server type and **Certificate** as the authentication method.
EAP TLS Session Cache Policy	ClearPass supports TLS session cache persistence for EAP - based authentications. Use the **EAP TLS Session Cache Policy** parameter to specify whether the session cache should persist in internal memory only, or in both memory and disk. The duration of the session persistence can also be configured. Select **Internal+Disk** for the user session to be read from disk, so re - authentication is not required before the configured session timeout. The default setting is **Internal**, which retains the p

Parameter	Action/Description
	revious behavior of periodically clearing t he session cache when roaming. **NOTE** If set to **Enable** and a subsc riber is OUT_OF_SYNC due t o a network outage and una ble to fetch the publisher's t ransactions after the outage is resolved, it may take some additional PITR Restore cycl es to recover and synchroni ze with publisher.
Enable Policy Server Endpoint Cache Rel oad System Event	Use this parameter to log system events when the policy server endpoint cache re loads. These event messages provide tro ubleshooting information if there are del ays in the reload process. This feature is disabled (set to FALSE) by default. When enabled (set to TRUE), a message is logg ed with the duration of the policy_server endpoint cache reload once it is complet ed.
Post - Authentication v2 Combine Sessio n Notification Events to PAN	Use this parameter to configure Post - Au thentication to combine API events when posting to the Palo Alto Networks Firewall (PAN). Enable this setting when multiple s ervers in a cluster make near - simultane ous API calls to PAN, to reduce the numb er of calls and help avoid exceeding the fi rewall's concurrency limit for an API sessi on. This parameter is disabled by default.
Override PITR Restore (When subscriber I n-sync with publisher)	When this parameter is set to Enable, the PITR restore job does not attempt to fetc h the most recent generated PITR backup on the publisher, if the subscriber is in-sy

Parameter	Action/Description
	nc with the publisher in the cluster. This parameter is set to Disable by default. Set this parameter as needed to save ba ndwidth on the publisher's network link resulting from the PITR Restore process o n subscriber fetching the PITR backup fro m the publisher. If set to Enable and a subscriber is OUT_ OF_SYNC due to a network outage and u nable to fetch the publisher's transaction s after the outage is resolved, it may take some additional PITR Restore cycles to re cover and synchronize with publisher.

Mutual TLS Authentication for SSO

Prerequisites

TLS 1.3 removed renegotiation, requiring the server to decide at the start of the handshake whether client certificates are required. Now the server cannot dynamically switch between certificate-required and certificate-optional modes within the same connection, so a single virtual host can no longer support both certificate-required and certificate-optional connections. To support both scenarios (certificate-required vs. certificate-optional), a two virtual host configuration is required.

As part of the fix to support Mutual TLS authentication to request the client certificate, additional virtual host levels will get added automatically in the backend configuration once TLS1.3 is enabled through the Policy Manager UI.

Pre-Condition: Enable a second FQDN/Hostname for the publisher node (DNS Mapping).

Example FQDNs

- FQDN1: cppm1.example.com
- FQDN2: cppm2.example.com

About this task

Switching from TLS1.2 to TLS1.3

This is a two-stage process from the Policy Manager UI. Complete the following steps in order:

Procedure

1. Navigate to the **Configuration > Identity > Single Sign-On (SSO) > SAML SP Configuration** tab.

   Enter the appropriate value in the **Identity Provider (IdP) URL** field. The DNS value should be the second (for example, FQDN2) fully-qualified domain name (FQDN) used for Mutual TLS (mTLS), to support TLS 1.3 certificate-based authentication.

2. Navigate to the **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters > General** tab.

   Change the value of the **Disable TLSv1.3 support** parameter to **None**. This confirms TLS1.3 is enabled.

   Switching back from TLS1.3 to TLS1.2 (process for backward compatibility)

   Complete the following steps in order:

   NOTE

   Switching back from TLS1.3 to TLS1.2 authentication makes the SAML handshake on the Primary Hostname/FQDN.

3. Navigate to the **Configuration > Identity > Single Sign-On (SSO) > SAML SP Configuration** tab

   Enter the appropriate value in the **Identity Provider (IdP) URL** field. The DNS value should be the primary (for example, FQDN1) fully qualified domain name (FQDN) for certificate-based authentication.

4. Navigate to the **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters > General** tab.

   Change the value of the **Disable TLSv1.3 support** parameter to **Admin**. This confirms that TLS1.2 is enabled.

Setting the Cluster Communication Mode

About this task

Cluster configuration operations (the CLI commands available under the cluster command section) support IPv4 or IPv6 communication depending on the cluster wide parameter called the **Cluster Communication Mode**.

```
[appadmin@pub_612]# cluster

 diagnostics                    Check network and database connectivity between cluster nodes
 drop-subscriber                Remove subscriber node from the cluster
 list                           List cluster nodes
 make-publisher                 Make this node publisher
 make-subscriber                Make this node subscriber
 reset-database                 Resets the local database and erases all the configuration
 set-cluster-passwd             Change the cluster password
 set-communication-mode         Set cluster communication mode to IPv4 or IPv6
 set-standby-publisher          Set standby publisher for the cluster
 sync-cluster-passwd            Synchronize the cluster password with Publisher

[appadmin@pub_612]# sh-4.4#
```

If set to IPv6, cluster replication, database connections and internal API calls specific to the cluster operations listed above will use an IPv6 address, or otherwise use IPv4.

The default Cluster Communication Mode depends on the IP address configured on the node at the time of an upgrade or install. If the node has only an IPv6 address, the default Cluster Communication Mode will be IPv6.

If both IPv4 and IPv6 addresses are configured, or if only an IPv4 address is configured, the default Cluster Communication Mode will be IPv4.

Keep in mind, the following modules are not supported for cluster communication using IPv6:

- Zone cache (Battery/Redis)
- Insight database communication
- Admin UI features (Access Tracker, API calls etc.)
- License usage computation on Publisher
- Async-netd Monitor (tracks the syncronization status of other modules)

Set the Cluster Communication Mode from either the ClearPass UI or command line interface (CLI). To set the Cluster Communication Mode from the UI, go to Setting Cluster Wide Parameters.

To review the cluster specific commands supported by ClearPass, go to . For information on using the ClearPass command line to set the Cluster Communication mode, go to Cluster Communication Mode.

The Cluster Communication Mode performs 3 pre-checks to ensure cluster operations do not fail if the mode is changed. These pre-checks include:

Procedure

1. Configuration checks - Ensure the IP address of a given format is configured on the interface.
2. Certificate checks for database - The database certs must have the correct IP address configured in the SAN field. Make sure the SAN field is updated to use the correct IP address format in the pattern DNS:<ip address>. Multiple IP addresses

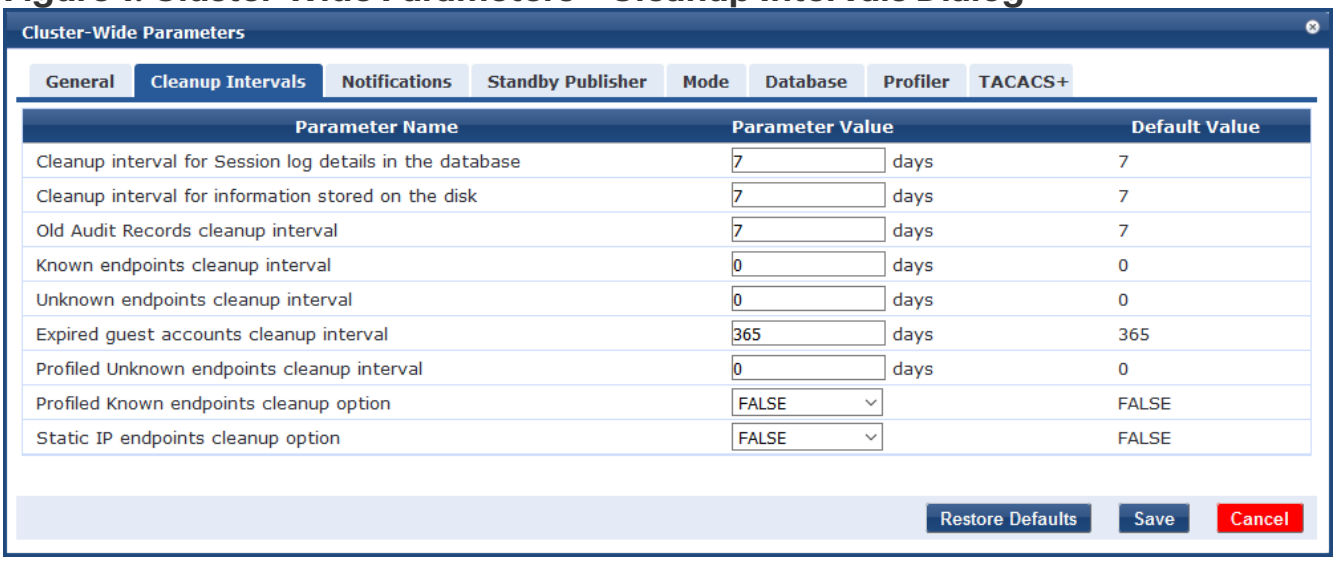
must be separated by a comma (,) delimiter. For example, DNS:<ipv6 address>, DNS:<ipv4 address>.

3. Certificate checks for HTTPS certs - The HTTPS certs must have the correct IP address configured in the SAN field. Make sure the SAN field is updated to use the correct IP address format in the pattern IP Address:<ip address>. Multiple IP addresses must be separated by a comma (,) delimiter. For example, DNS:<ipv6 address>, DNS:<ipv4 address>.

Cleanup Intervals Parameters

The following figure displays the **Cluster-Wide Parameters > Cleanup Intervals** dialog:

Figure 1. Cluster-Wide Parameters > Cleanup Intervals Dialog



Specify the **Cluster-Wide Parameters > Cleanup Intervals** parameters as described in the following table:

Table 1. Cluster - Wide Parameters > Cleanup Intervals Parameters

Parameter	Action/Description
Cleanup interval for Session log details in t he database	Specify the duration in number of days to keep the following data in the Policy Mana ger database: • Session logs (found on the **Monitoring > Live Monitoring > Access Tracker** pag e)

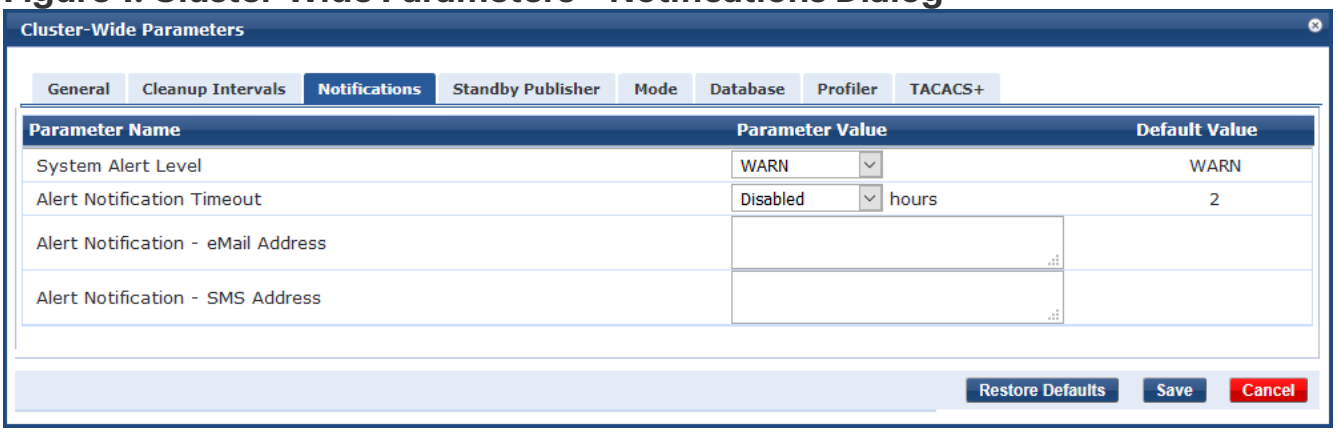
Parameter	Action/Description
	• Event logs (found on the **Monitoring > Event Viewer** page) • Machine authentication cache The default value is **7 days**.
Cleanup interval for information stored on t he disk	Specify the duration in number of days to keep log files that are written to the disk. The default value is **7 days**.
Cleanup interval for CSRs and private keys.	By default, ClearPass automaticaly perform s a cleanup for Certificate Signing Request s (CSRs) older than 15 days. use this param eter to extend the cleanup interval for old CSRs and associated private keys from 15 d ays to up to 90 days. Network administrato rs should extend the cleanup interval for C RS and private keys in cases where the CA takes more than 15 days to create a certifi cate, as the CSR generated on the system would become invalid with the default 15 d ay setting, and would be removed automati cally from the system.
Old Audit Records cleanup interval	Specify the cleanup interval in number of days that Policy Manager uses to determin e when to start deleting old audit records f rom the **Audit Viewer** page. The default val ue is **7 days**.
Known endpoints cleanup interval	Specify the duration in number of days tha t Policy Manager uses to determine when t o start deleting known or disabled entries f rom the Endpoint repository. Known entries are deleted based on the w hen the endpoint was last seen. For examp le, if this value is **7**, then known Endpoints that have not been seen within the last 7 d ays are deleted. The default value is **0 day**s. This indicates t hat no cleanup interval is specified.
Unknown endpoints cleanup interval	Specify the duration in number of days th at Policy Manager uses to determine when to start deleting unknown entries from the Endpoint repository. Unknown entries are d eleted based on the when the endpoint wa

Parameter	Action/Description
	s last seen. For example, if this value is **7**, t hen unknown Endpoints that have not bee n seen within the last 7 days are deleted. The default value is **0 days**. This indicates t hat no cleanup interval is specified.
Expired guest accounts cleanup interval	The cleanup interval for expired guest acc ounts indicates the number of days after expiry that the cleanup occurs. A value of **0** specifies no expired guest acc ounts cleanup interval. The default value is **365 days**.
Profiled Unknown endpoints cleanup inte rval	Specify the cleanup interval in number of days that Policy Manager uses to determin e when to start deleting profiled unknown entries from the Endpoint repository. Profil ed unknown entries are deleted based eith er the time the unknown Endpoint was last seen or the time it was last profiled, which ever is most recent. The default value is **0**,indicating that no cl eanup interval is specified.
Profiled Known endpoints cleanup option	Specify whether to enable the option to cl ean up profiled known endpoints. The default value is **FALSE**. Profiled known entries are deleted based either the time th e known Endpoint was last seen or the tim e it was last profiled, whichever is most rec ent.
Static IP endpoints cleanup option	Specify whether to enable the option to cl ean up static IP endpoints. The default option is **FALSE**.
Close Idle RadSec Tunnel from NAD after	Use this setting to configure a timeout valu e between 15 and 2880 minutes used to cl ose an idle RadSec tunnel once the set val ue is exceeded. The default setting is 15 mi nutes. If no server status or RADIUS messa ge is received from the RadSec client by th e time the configured interval is reached, t hen the RadSec tunnel is disconnected.

Notifications Parameters

The following figure displays the **Cluster-Wide Parameters > Notifications** dialog:

Figure 1. Cluster-Wide Parameters > Notifications Dialog



Specify the **Cluster-Wide Parameters > Notifications** parameters as described in the following table:

Table 1. Cluster - Wide Parameters > Notifications Parameters

Parameter	Action/Description
System Alert Level	Specify the alert notifications that are ge nerated for system events logged at this l evel or higher. • **INFO**: Alerts that provide Information, Warnings, and Error messages are gene rated. • **WARN**: Alerts that provide Warnings an d Error messages are generated. • **ERROR**: Alerts that provide Error messa ges only are generated. • The default value is **WARN**.
Alert Notification Timeout	Specify the timeout in hours that determin es how often alert messages are generated and distributed. If you select **Disabled**, alert generation is d isabled. The default value is **2 hours**.

Parameter	Action/Description
Alert Notification - eMail Address	Enter a comma - separated list of email ad dresses to which alert messages are sent.
Alert Notification - SMS Address	Enter a comma - separated list of phone n umbers to which alert messages are sent.

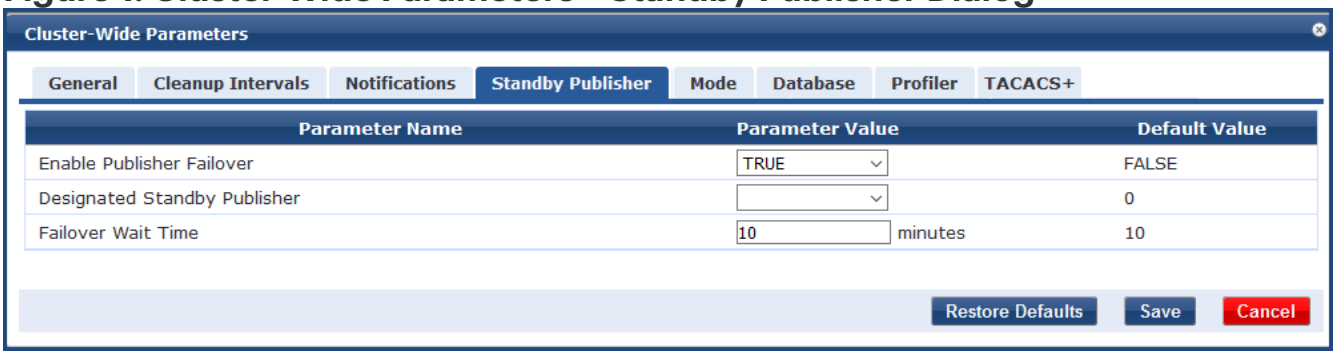
Standby Publisher Parameters

The standby publisher is the publisher in the cluster that is configured to come up in the event that the publisher is not reachable. The following figure displays the **Cluster-Wide Parameters > Standby Publisher** dialog:

NOTE

During a failover, the standby publisher does not have to wait for all the servers in the cluster to fully resynchronize, and instead is available with complete publisher functionality within moments of the triggering failover event.

Figure 1. Cluster-Wide Parameters > Standby Publisher Dialog



Specify the following **Cluster-Wide Parameters > Standby Publisher** parameters:

Table 1. Cluster - Wide Parameters > Standby Publisher Parameters

Parameter	Action/Description
Enable Publisher Failover	To authorize a node in a cluster on the syst em to act as a publisher if the primary publ isher fails, select **TRUE**. The default value is **FALSE**. **NOTE**

Parameter	Action/Description
	To avoid false failover triggers, the **Enable Publisher Failover** value should be set to **false** be fore starting a cluster update.
Designated Standby Publisher	Select the server in the cluster to act as th e standby publisher. The default value is **0**. **NOTE** If the standby publisher is on a different subnet than the pu blisher, then ensure that a relia ble connection between the t wo subnets is available to avo id unwanted network segmen tation and potential data loss f rom a false failover.
Failover Wait Time	Specify the time (in minutes) that the stan dby publisher must wait before it assumes the role of publisher after the primary publi sher becomes unreachable. The default fail over wait time is **10 minutes**. This parameter prevents the standby publis her from taking over when the publisher is temporarily unavailable during a restart. **NOTE** Failover wait times vary based on the size of the data needin g to synchronize, as well as th

Parameter	Action/Description
	e number of subscribers that need to be checked.

NOTE

If a cluster is configured with a standby publisher, add the HTTPS server certificate of the standby publisher to the Trust list and ensure sure all the servers in the cluster have this certificate in the Trust list. Similarly, before you promote a subscriber to publisher, add the HTTPS server certificate of the subscriber to the Trust list and ensure sure all the servers in the cluster have this certificate in the Trust list. This step is not required if the HTTPS server certificates for all the nodes in the cluster are signed by a certificate authority (CA).

NOTE

When a database backup is made with a subscriber configured as the standby publisher, and the backup is restored without using the **-s flag**, then after the backup is restored the cluster list command could fail with an error message displaying **Caught unexpected error while retrieving the Cluster Node List**. If this occurs, navigate to the **Standby Publisher** tab and remove the server from the **Designated Standby Publisher** field. You can designate it as the standby again once the restoration is complete. If the subscriber is configured as the standby in the backup file, pass the **-s flag** during the restoration to restore the cluster configuration.

Common Criteria Mode Parameter

The **Mode** tab in the **Cluster-Wide Parameters** page allows you to enable or disable **Common Criteria Mode** and choose the mode devices within the cluster will communicate. Common Criteria is an international standard for security certification. Use Common Criteria Mode for deployments that require strict compliance to Common Criteria requirements.

Common Criteria Mode has the following restrictions and requirements:

- When a user enables Common Criteria mode they must enter the existing cluster password, which is validated against existing Common Criteria password rules. If the existing cluster password meets the requirements, no action is needed for the password and Common Criteria mode can be enabled. If the existing password does

not meet the requirements, the user is prompted to change the password. If the new cluster password provided is not strong enough, the requirements display so it can be corrected. The **Change Cluster Password** screen also includes this validation step when the cluster is in Common Criteria mode. The cluster password strenghth minimum requirements for Common Criteria mode include:

> **NOTE**
>
> If a **Password did not meet the minimum requirements** error displays, change the password to include a stronger combination of letters, numbers and special characters.

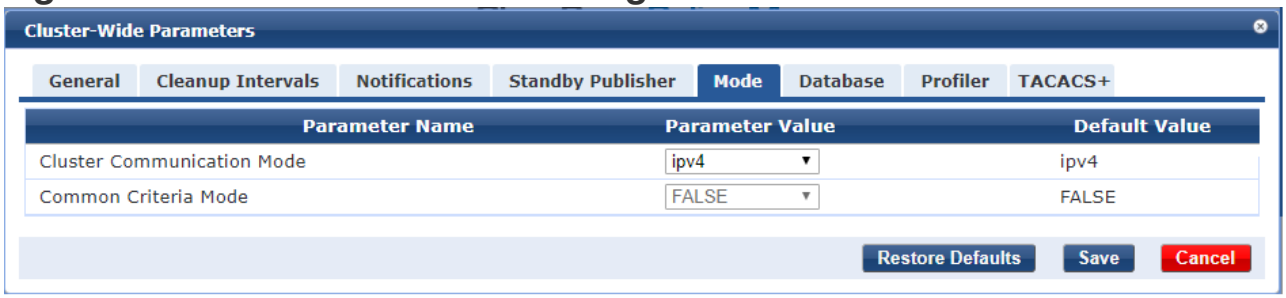
  - At least one uppercase letter
  - At least one lowercase letter
  - At least one number
  - At least one special character { } & ! @ # $ % ^ ? _ = . , + < > - *
  - A minimum length of 15 characters
  - The password cannot include a dictionary word
  - The password cannot contain more than four consecutive repeating characters
  - The password cannot contain more than four consecutive repeating characters from the same character class

- Common Criteria Mode cannot be enabled if there are less than three NTP servers configured (for information on configuring multiple NTP servers, see the **Add More NTP Servers** option in Table 1, *Changing Date and Time Parameters*).
- Common Criteria Mode requires that all the Policy Manager servers in the cluster must have FIPS mode enabled (see FIPS Page).
- Server certificates must be updated before you enable Common Criteria Mode (see Certificate Store).
- Only Certificate Authority (CA)-issued certificates can be used for Policy Manager server certificates.
- No self-signed certificates are allowed as trusted certificates.
- All X.509 v3 trusted Certificate Authority (CA) certificates must satisfy the basic constraints.

X.509 is an important standard for a public key infrastructure to manage digital certificates and public-key encryption. X.509 is a key part of the Transport Layer Security protocol used to secure web and email communication.

- All HTTPS communication to external services using X.509 v3 certificates must pass the basic constraint checks.
- Cipher suites that use RSA, DSA, or DH keys shorter than 2048 bits or ECC keys shorter than 224 bits are not supported in Common Criteria Mode.
- If a cluster is using CA-signed certificates, the replication service remains in a stopped state until after a cluster communication mode change is completed. This is expected behavior, introduced as a safety measure to keep the cluster secure while the operation completes. Reboot the individual cluster nodes after changing the cluster communication mode using ca-signed certificates.

Figure 1. Cluster-Wide Parameters Page > Common Criteria Mode Parameter



Specify the **Cluster-Wide Parameters > Mode** parameter as described in the following table:

Table 1. Cluster - Wide Parameters > Mode Parameter

Parameter	Action/Description
Cluster Communication Mode	Click the drop - down list and select either **IPv4** or **IPv6** as the mode of communication for all cluster operations. If the value of this parameter is set to IPv6, all database and API calls will use IPv6 addresses for cluster communication. If the value is set to ipv4, it will use IPv4 for database and API calls instead. The default value of the cluster communication mode will depend on the IP address configured on the appliance during installation or upgrade. If the appliance has only an I

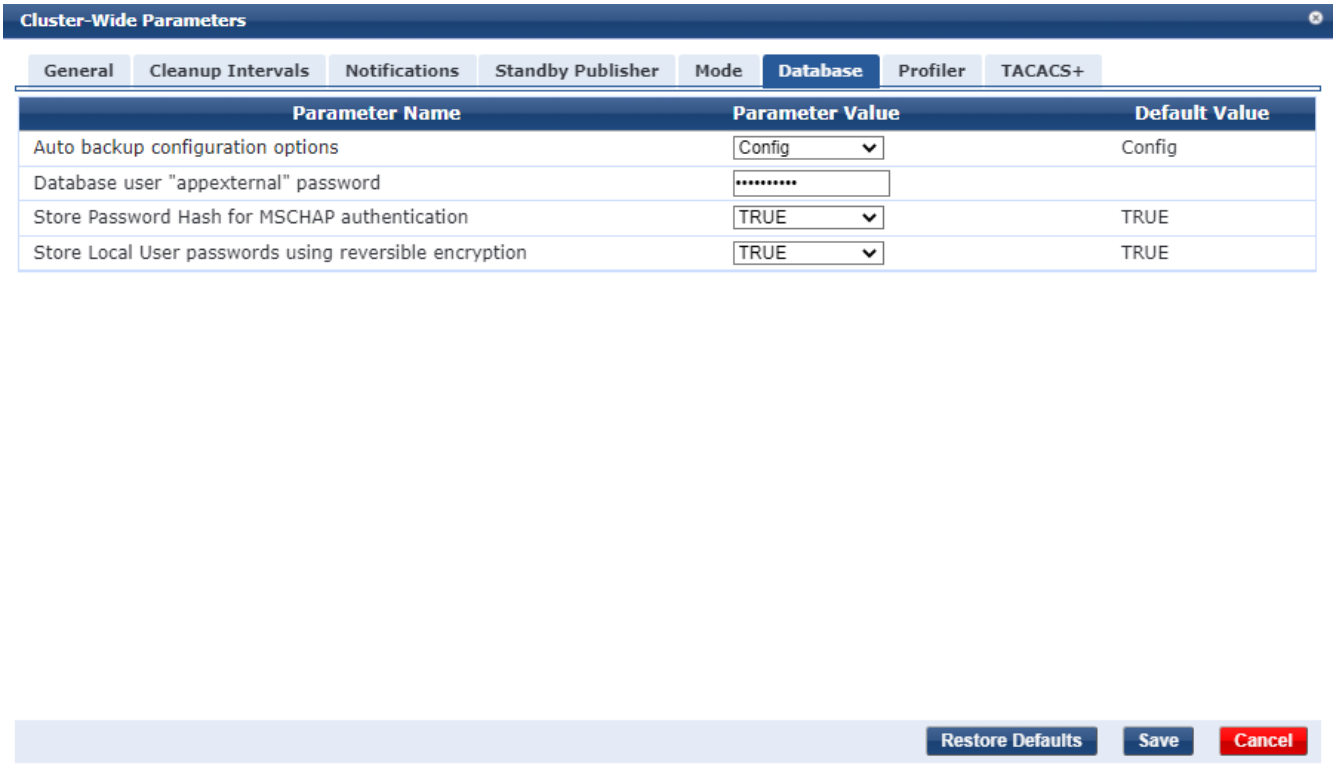
Parameter	Action/Description
	Pv6 address, the default cluster commun ication mode will be IPv6. If the applianc e has both IPv4 and IPv6 addresses confi gured, or if only an IPv4 address is config ured, then the default cluster communica tion mode will be IPv4. Whenever the clu ster communication mode is changed, it performs the following validations: ◦ Configuration checks to verify an IP a ddress in the correct format is config ured for the interface. ◦ Certificate checks to verify the databa se certificates have the correct IP add ress in the SAN field. ◦ Certificate checks to verify the HTTPS certificates have the correct IP addres s in the SAN field **NOTE** Before changing the cluster communication mode, caref ully review the displayed wa rning message stating that resetting the cluster commu nication mode can cause th e cluster to go out of sync by resetting the IP address format of the cluster commu nication interface. Resetting the certificates on each clus ter node along with rebootin g ensures the cluster is in sy nc.
Common Criteria Mode	**WARNING** The **Common Criteria Mode** is for specific deployments t

Parameter	Action/Description
	hat require strict compliance to Common Criteria requirements. To enable or disable **Common Criteria Mode**, select **TRUE** or **FALSE**. The default is **FALSE**. When you set **Common Criteria Mode to TRUE**, the following message appears: Setting this value to TRUE enables strict validation of Certificates and changes to modules in order to comply with Common Criteria requirements. When Policy Manager is in Common Criteria (CC) mode, TLS 1.2 is enforced with RadSec, and TLS 1.0 and TLS 1.1 are no longer supported. **NOTE** When Common Criteria Mode is enabled, the text **[CC]** is shown in the release version footer of the WebUI.

Database Parameters

The following figure displays the **Cluster-Wide Parameters > Database** dialog:

Figure 1. Cluster-Wide Parameters > Database Dialog



Configure the **Cluster-Wide Parameters > Database** parameters as described in the following table:

Table 1. Cluster - Wide Parameters > Database Parameters

Parameter	Action/Description
Auto backup configuration options	Select any of the following auto - backup c onfiguration options: • **Off**: Select this to *not* perform periodic backups. • Select **Off** before upgrading Policy Man ager to avoid the interference between Auto backup and migration process. • **Config**: Performs a periodic backup of t he configuration database that includes licensing information that would be req uired to restore a server to its original s tate. This is the default auto backup co nfiguration option. In a cluster deploym ent, the backup file on the publisher als o contains license information for all su bscribers.

Parameter	Action/Description
	• **Config|SessionInfo**: Performs a backup of the configuration database, certificates and licenses, and also the session log database. • **Config|Extensions**: Extensions are included in the backup, in addition to configuration data. • **Config|SessionInfo|Extensions**: Extensions are included in the backup, in addition to configuration data, session log, and Insight data. **NOTE** Keep in mind, if extensions are excluded from a backup file, and that backup file is restored on a system that already has extensions, that system's existing extensions remain and are not overwritten. **NOTE** It is recommended you set this option to **Off** or **Config** before starting an upgrade. This ensures the Auto Backup process does not interfere with migration post upgrade. If required, you can change this setting back to **Config|SessionInfo** 24 hours after upgrade completion. Evaluation and Subscription licenses which have expired will not be included in the backup.
Database user "appexternal" password	Enter the password for the **appexternal** username for this connection to the database.
Store Password Hash for MSCHAP authentication	To store passwords for admin and local users to Hash and NTLM hash formats (which enables RADIUS MSCHAP authentications a

Parameter	Action/Description
	gainst admin or local repositories), set this to **TRUE**. If you set this to **FALSE**, RADIUS MSCHAP a uthentications are not possible because th e NTLM hash passwords are removed for all the users. **NOTE** When you set this value to **T RUE**, you must reset all the p asswords to reenable RADIUS MSCHAP authentication again st the user repositories.
Store Local User Passwords using reversibl e encryption	To enable cleartext password comparison a gainst local users, set this to **TRUE**. If you set this to **FALSE**, cleartext password comparison against local users is not possi ble because the reversible passwords for lo cal users are removed. **NOTE** After setting this value to **TRU E**, you must reset all the local user passwords to reenable cl eartext password comparison against local users.

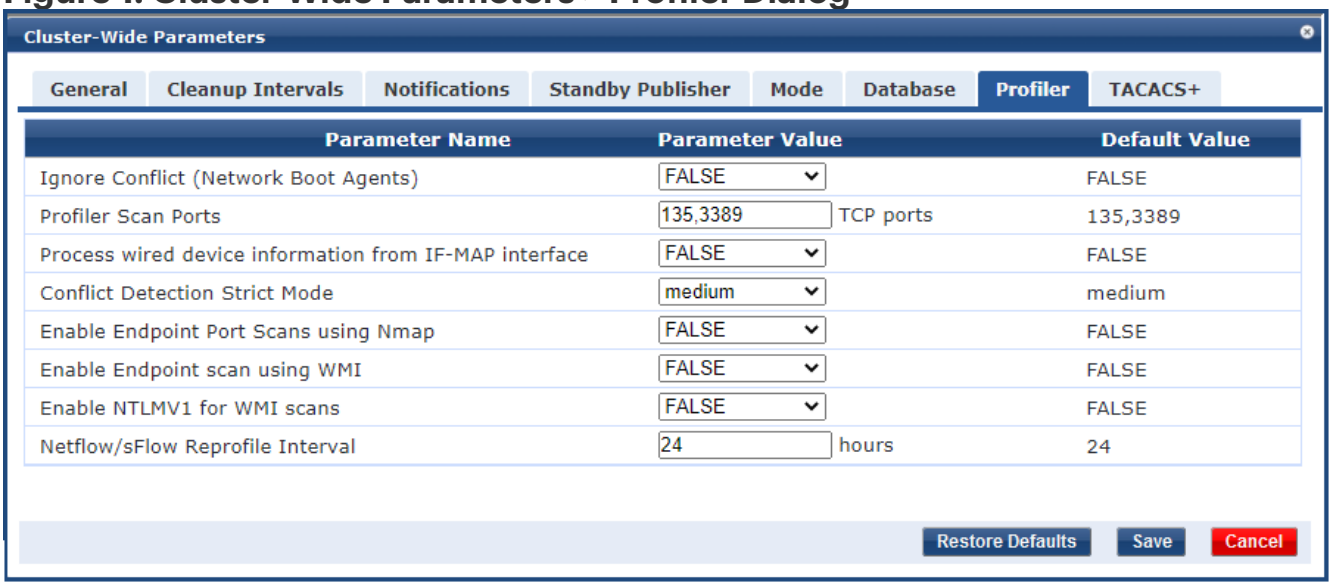
NOTE

The time it takes for subscribers to synchronize with the publisher in the event of downtime will depend on the duration of the subscriber's downtime and on when the replication recovery is next run on the subscriber after it is back online. It might take as much as 24 hours for the subscriber to synchronize with the publisher.

Profiler Parameters

The following figure displays the **Cluster-Wide Parameters > Profiler** dialog:

Figure 1. Cluster-Wide Parameters > Profiler Dialog



NOTE

Starting with Policy Manager 6.10.2, the Policy Manager server no longer needs to have Device Insight integration disabled in order to run other types of scans that use ClearPass Profilerk so users can seamlessly run any type of Profiler scan even while Device Insight's device discovery is active. In Policy Manager 6.10.0 and 6.10.1, when **Device Insight Integration** is enabled, the **Cluster-wide Parameters > Profiler** tab is hidden because Device Insight functionality replaces the Policy Manager functions available through the **Profiler** tab (for more information, see Device Insight Integration Page).

Configure the **Cluster-Wide Parameters > Profiler** parameters as described in the following table:

Table 1. Cluster - Wide Parameters > Profiler Tab Parameters

Parameter	Action/Description
Ignore Conflict (Network Boot Agents)	To enable the **Ignore Conflict (Network Bo ot Agents)** parameter, choose **TRUE**. The d efault is **FALSE**.
Profiler Scan Ports	*To change the list of ports to scan and add custom fingerprints to classify based on th em, enter the new TCP port numbers.* *The TCP ports scanner checks to see if the specified Profiler Scan Ports are open. The default TCP ports are 135 and 3389.*

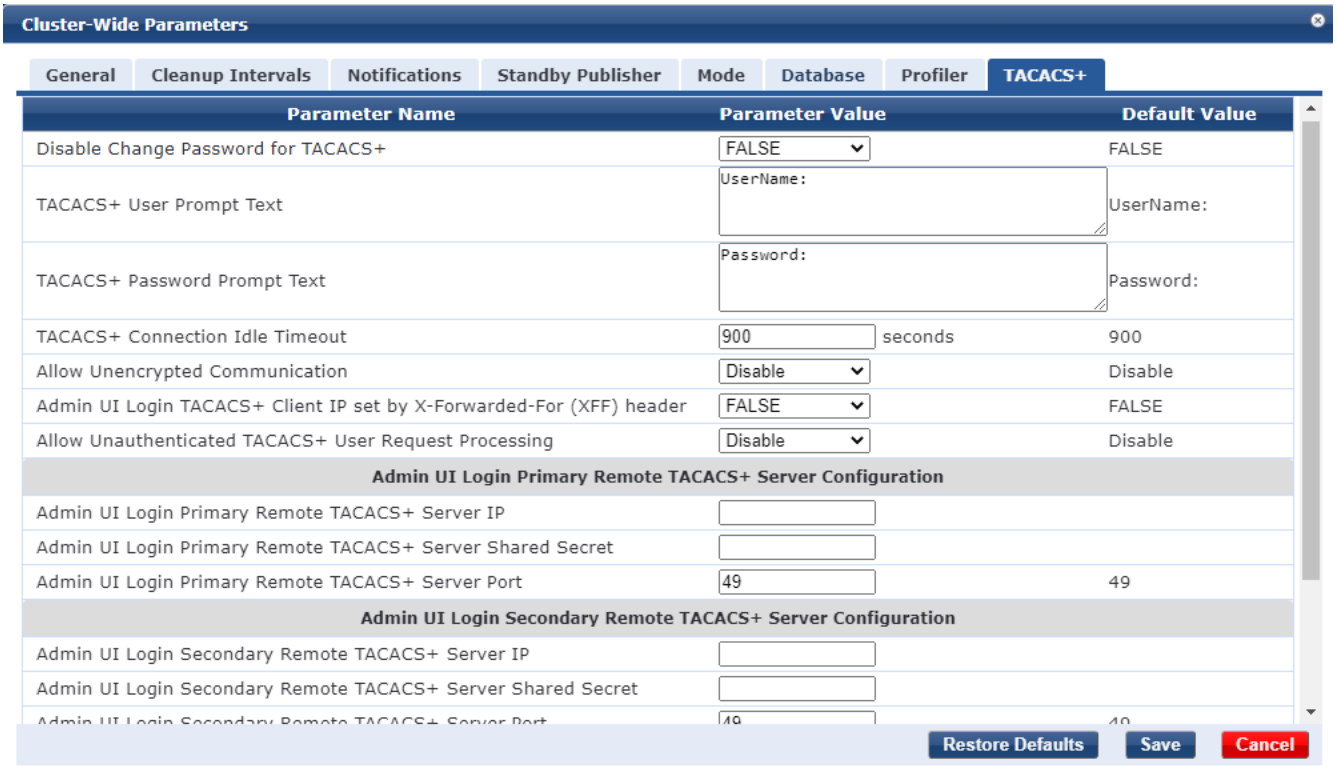
Parameter	Action/Description
Process wired device information from IF - MAP interface	Choose whether to process wired device information from the IF - MAP interface. The default is **FALSE**.
Conflict Detection Strict Mode	Select from the following settings. The default value is **medium**. **high** - Flag a Profiler conflict if the device category and the DHCP attributes or the hostname change. **medium** - Flag a Profiler conflict if the device category changes. **low** - Flag a Profiler conflict if the device category or the device family changes.
Enable Endpoint Port Scans using Nmap	Set this option to **TRUE** to enable Endpoint scans using Nmap (Network Mapper). **NOTE** The Open Ports scanner is disabled when Nmap - based port scanning is enabled. When Nmap scan is enabled, the following warning is displayed: **WARNING** Setting this value to **TRUE** enables active scan of the host for open ports. This can be resource intensive. Also, the **Profiler Scan Ports** value is ignored when Nmap scan is enabled.
Enable Endpoint Port Scans using WMI	Set this option to **TRUE** to enable Endpoint scans using WMI (Windows Management Instrumentation).
Enable NTLMv1 for WMI scans	Set this parameter to **TRUE** to enable NT LAN Manager v1 for WMI (Windows Management Instrumentation) scans. The default is **FALSE**. For more information, see WMI Credentials Configuration.

Parameter	Action/Description
Netflow/sFlow Reprofile Interval	Specify the interval after which endpoints will be reprofiled. The default value is 24 h ours. The minimum value is one hour. This setting requires the Access license. If you h ave an Entry license only, this field will not appear. For more information about the **Netflow** an d **sFlow** collectors, see Endpoint Informati on Collectors .
Netflow Decoder Inactive Timeout	Specify the interval after which the inacti ve NetFlow decoders can be released. Thi s can be used to prevent NetBridge proces s memory from steadily increasing proport ionally with incoming requests when NetF low is enabled on the firewall. The default value is 30 minutes. Allowed values are fro m 5 - 1440 minutes.

TACACS+ Parameters

The following figure displays the **Cluster-Wide Parameters > TACACS+** dialog:

Figure 1. Cluster-Wide Parameters > TACACS+ Dialog



Configure the **Cluster-Wide Parameters > TACACS+** parameters as described in the following table, then click **Save**.

> **NOTE**
>
> Clearpass will not save a Cluster-Wide TACACS+ configuration if the primary TACACS+ server is not configured, but secondary/ tertiary TACACS+ servers are configured. The primary TACACS+ server must be configured to configure the secondary TACACS+ server, and the secondary TACACS+ server must be configured to configure the tertiary TACACS+ server.

Table 1. Cluster - Wide Parameters > TACACS+ Parameters

Parameter	Action/Description
Disable Change Password for TACACS+	When logging in for TACACS+ user authenti cation: • If set to **FALSE** (the default setting), aft er entering a blank password, you are p resented with an option to change the TACACS+ user password. • If set to **TRUE**, the option to enter the TACACS+ user password is displayed. T

Parameter	Action/Description
	he option to change the TACACS+ pas sword is **not** displayed.
TACACS+ User Prompt Text	You can modify the text to be used for the TACACS+ username prompt as needed. The default TACACS+ username prompt is *User Name:*
TACACS+ Password Prompt Text	You can modify the text to be used for th e TACACS+ password prompt as needed. T he default TACACS+ password prompt is *Pa ssword:*
TACACS+ Connection Idle Timeout	An idle TACACS+ login session is one in wh ich the CLI operational mode prompt is dis played but there is no input from the keybo ard. To close idle sessions automatically, yo u must configure a time limit for each login class. Specify the TACACS+ Connection Idle Time out duration in seconds as needed. • The default value is **900 seconds** (15 mi nutes). • The minimum allowed value is **60 seco nds**. • The maximum allowed value is **172800 seconds** (two days).
Allow Unencrypted Communication	Enable this setting to allow unencrypted c ommunication in test environments. This p arameter is disabled by default. When it is enabled, Policy Manager displays the warn ing message **"Warning: Setting the "Allow Unencrypted Communication" value to " Enable" will allow unencrypted TACACS+ requests to be processed. This is not reco mmended for production environments."**
Admin UI Login TACACS+ Client IP set by X - Forwarded - For (XFF) header	The default setting for this parameter is **FA LSE** (disabled). If an HTTP proxy server will use the **X - Forwarded - For header** value as the remote client IP address for the adm in server or TACACS+ server, then the para meter should be set to **True** (enabled).

Parameter	Action/Description
Admin UI Login Remote TACACS+ Server IP	This option is used if you want the Policy Manager server to authenticate against a TACACS+ server. In this case, the Policy Ma nager server is acting as a TACACS+ client and talking to the TACACS+ server in order to authenticate users. Specify the IP address for the remote TACACS+ server.
Admin UI Remote TACACS+ Server Shared Secret	Use this option if you want the Policy Mana ger server to authenticate against a TACAC S+ server. Enter the Shared Secret for the remote TACACS+ server.
Admin UI Remote TACACS+ Server Port	Use this option to set a custom value for th e remote TACACS+ server port. The default server port value is **49**.
Allow Unauthenticated TACACS+ User Requ est Processing	If enabled, TACACS+ authorization and acc ounting requests will be processed using a n authenticated session based on usernam e. Users should be aware the authenticated session lifetime expires when the TACACS+ service restarts, or when the cache times o ut.

Synchronizing the Cluster Date and Time with the NTP Server

- Restarting Policy Manager Services
- Specifying the Time Zone on the Publisher Node

  Policy Manager supports both authenticated NTP (Network Time Protocol) and NTP without authentication.

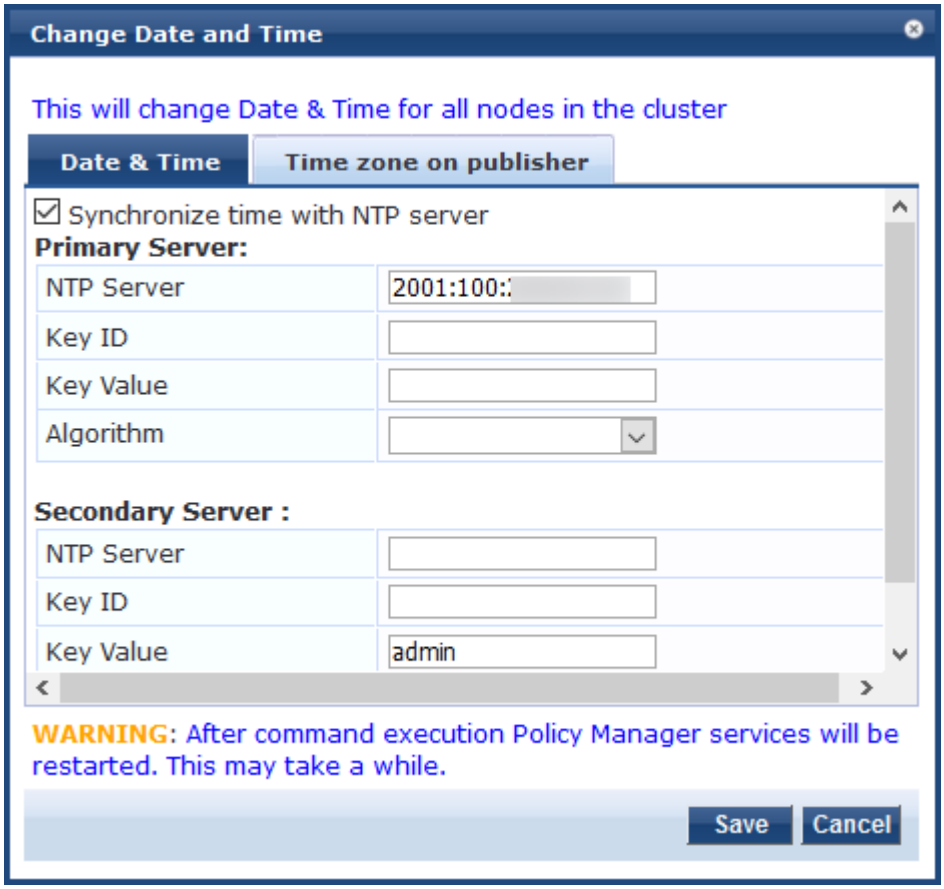
To synchronize the date and time on the nodes in a cluster with an NTP (Network Time Protocol) server.

> **NOTE**
>
> The option to change the date and time for the Policy Manager cluster is available only on the publisher. subscriber nodes in a cluster will synchronize the date and time from the publisher. Users should be aware that ClearPass can time sync with any of the configured NTP servers in any order. It does not need to prefer the primary NTP server first and then the secondary as per configuration.

1. Log in to the publisher.
2. Navigate to the **Administration > Server Manager > Server Configuration** page.
3. Select the **Set Date & Time** link.

   The **Change Date and Time** dialog opens.

   Figure 1. Change Date and Time > Date & Time Tab

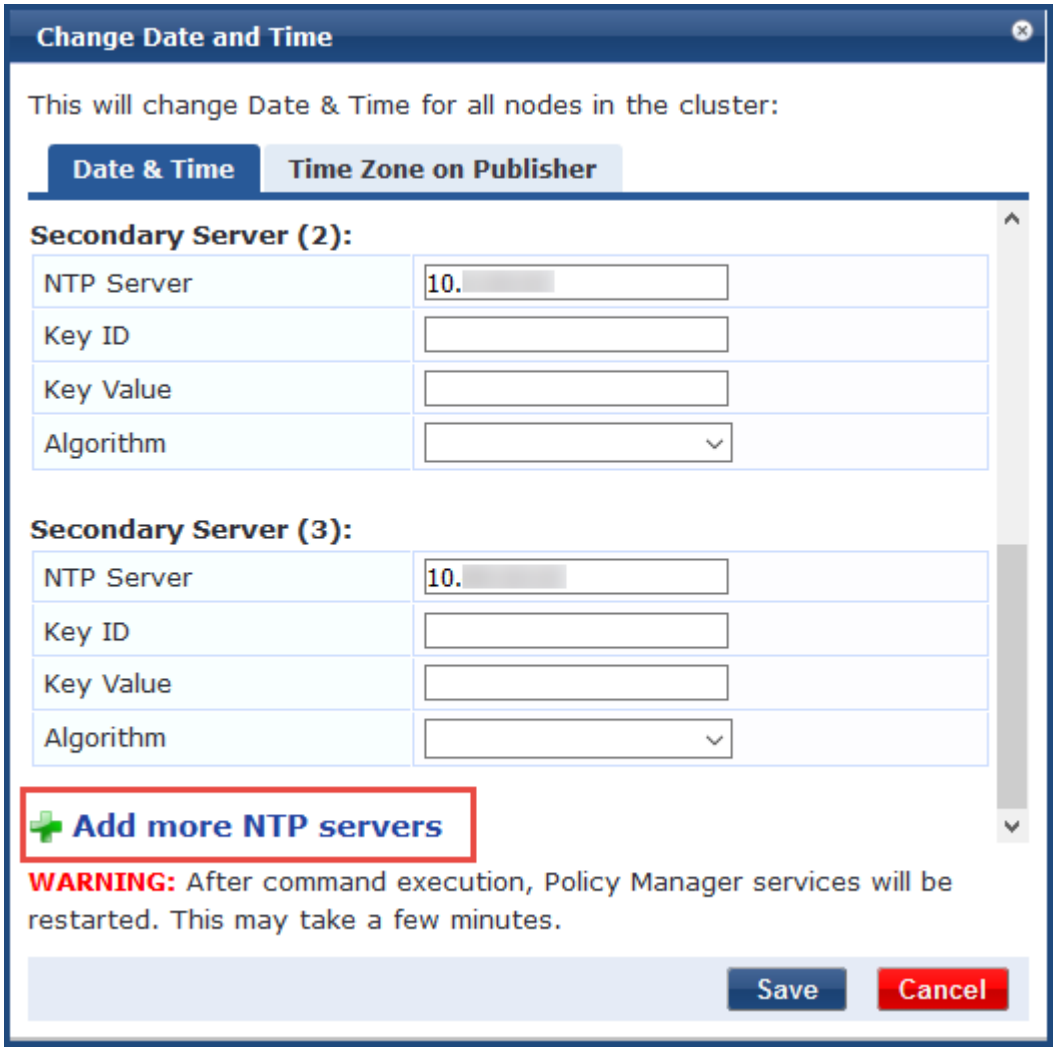


NOTE

The **Key ID**, **Key Value**, and **Algorithm** parameters apply only when using authenticated NTP. If you are employing unauthenticated NTP, you do not need to specify the NTP primary and secondary server parameters.

4. To add additional Secondary NTP servers, scroll down to see the **Add More NTP Servers** link:

Figure 2. Adding Additional NTP Servers

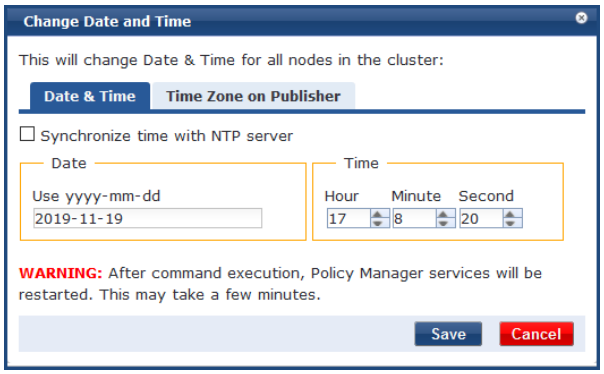


See Table 1 for configuration details on adding more NTP servers.

5. Specify the **Date & Time** parameters as described Table 1:

Table 1. Changing Date and Time Parameters

Parameter	Action/Description
Synchronize time with NTP server	To synchronize with a Network Time Pro tocol (NTP) server, enable this check b ox (enabled by default). *Manually Setting the Date and Time* **NOTE** **Synchronize time with NT P server** check box and en tering the current date and

Parameter	Action/Description
	time in the dialog provided : 
Primary Server and Secondary Server	
NTP Server	Specify the IP address or host name for the Primary NTP server and the Second ary NTP server. The IP address can be a n IPv4 or IPv6 address.
Key ID	The **Key ID** must be in the range from 1 to 65535. **NOTE** **Key ID** should match Key I D configured for the NTP s erver. **NOTE** **Key ID** applies only to auth enticated NTP.
Key Value	**Key Value** is a form of shared secret, wh ich both the client and the Policy Mana ger server use for authenticating NTP m essages. The **Key Value** ASCII string must start a nd end with one of the following charac ters:

Parameter	Action/Description
	◦ **-** (hyphen) ◦ **'** (apostrophe) ◦ **"** (quotation mark) The **Key Value** can be: ◦ Up to a 20 - character printable ASC II string ◦ Up to a 40 - character hex value When entering an ASCII string for the **Key Value**, note that it *cannot* contain t he following characters: ◦ **&** (ampersand) ◦ **;** (semicolon) ◦ **`** (grave accent) ◦ **|** (pipe) ◦ **<** (left angle bracket) ◦ **>** (right angle bracket) ◦ **(** (left parenthesis) ◦ **)** (right parenthesis) **NOTE** **Key Value** applies only to a uthenticated NTP.
Algorithm	ClearPass supports the **SHA** and **SHA1** encryption types. If upgrading to Clear Pass 6.11, keep in mind if there are any NTP servers configured with the SHA al gorithm, the algorithm corresponding t o those servers will be automatically ch anged to SHA1 as part of upgrade proc edure. Only SHA1 is supported as a clus

Parameter	Action/Description
	ter NTP authentication algorithm going forward from ClearPass 6.11. **NOTE** **Algorithm** applies only to authenticated NTP.
Add More NTP Servers	You can configure up to five NTP serve rs. To add additional NTP servers: a. Scroll to the bottom of the **Change D ate and Time** configuration dialog. b. Click the **Add More NTP Servers** link. c. Specify the newly added *Secondary NTP Server* fields as needed. **NOTE** Common Criteria Mode cannot be enabled if the re are less than three NT P servers configured.

6. Click **Save**.

Restarting Policy Manager Services

About this task

Once you have saved the **Date & Time** configuration, you must restart Policy Manager services.

NOTE
The Audit Viewer (**Monitoring > Audit Viewer**) tracks NTP configuration changes.

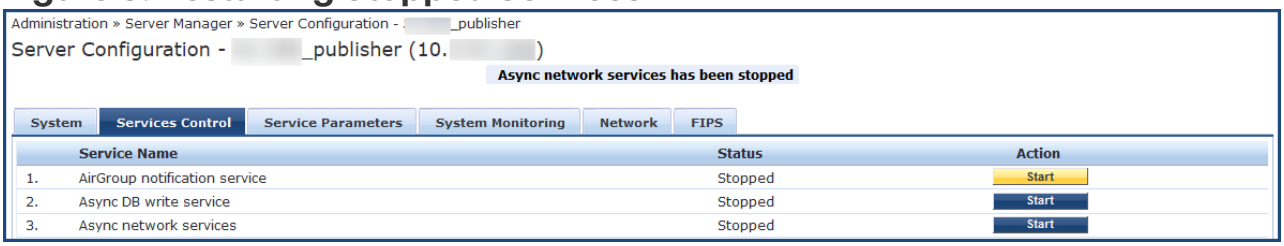
To restart Policy Manager services:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.

2. Select the Policy Manager publisher.
3. From the **Server Configuration** page, select the **Services Control** tab.

Figure 3. Restarting Stopped Services



4. From the **Action** column, click **Start** for each service that needs to be restarted.

   For each restarted service, the **Start** button is changed to **Stop**.

Specifying the Time Zone on the Publisher Node

About this task

To specify the time zone on the publisher:

Procedure

1. From the publisher, click the **Time Zone on Publisher** tab.

   The time zones are listed in alphabetical order.

Figure 4. Time Zone on Publisher Dialog

2. Select the time zone where the publisher resides, then click **Save**.

> **NOTE**
>
> This option is available only on the publisher. To set the time zone on a subscriber node, select the specific server and set the time zone from the server-specific page.

Changing the Cluster-Wide Password

About this task

> **NOTE**
>
> Cluster password changes require that all subscribers be reachable. If the cluster password change fails on any of the subscriber servers, the change will reverted back to the original password for all servers in the cluster.

To change the cluster-wide password:

Procedure

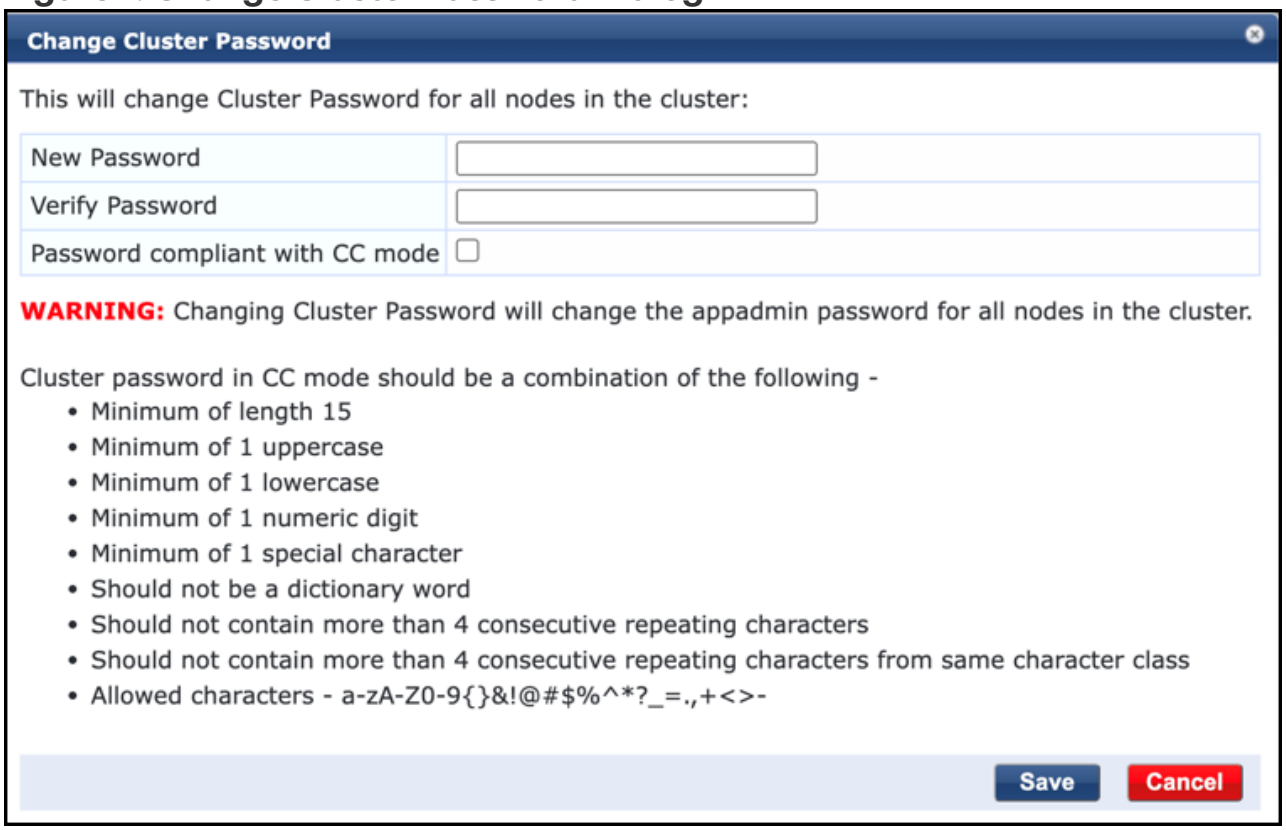
1. Navigate to **Administration > Server Manager > Server Configuration**.

   The **Server Configuration** page opens.

2. Click the **Change Cluster Password** link.

   The **Change Cluster Password** dialog opens.

Figure 1. Change Cluster Password Dialog



3. If the cluster password is intended for a CC mode deployment, select the **Password compliant with CC mode** checkbox. If selected, the provided password is validated against a set of CC mode password criteria before it can be saved and CC mode enabled.

4. Enter the new cluster password, then verify the password.

   Allowed characters for the cluster password should minimally include letters A through Z (both capitalized and lower case), numbers 0 through 9, and the following special characters:

   { } & ! @ # $ % ^ ? _ = . , + < > - *

The following two password examples meet the strength requirements for length and for combinations of uppercase and lowercase letters, numbers, and special characters:

- G!e#2-YR@w%Bu&o
- A6c@D&e7*g^H7Y3!6

The following two password examples do not meet the strength requirements, because the first one has no lowercase letters and the second one has no uppercase letters:

- ABCD1ABCD1ABCD@
- 1234a1234b1234c@

> **NOTE**
>
> If a **Password did not meet the minimum requirements** error displays, change the password to include a stronger combination of letters, numbers and special characters.

5. Click **Save**.

> **NOTE**
>
> Changing this password changes the password for the CLI user *appadmin* as well.

Managing Policy Manager Zones

Policy Manager shares a distributed cache of run-time states across all nodes in a cluster. These run-time states include:

- Roles and postures of connected entities
- Connection status of all endpoints running OnGuard
- Endpoint details gathered by OnGuard Agent

  Policy Manager uses this run-time state information to make policy decisions across multiple transactions. In a deployment where a cluster spans WAN boundaries and multiple geographic zones, it is not necessary to share all of this run-time

state across all nodes in the cluster. For example, when endpoints present in one geographical area are not likely to authenticate or be present in another area, it is more efficient from a network bandwidth usage and processing perspective to restrict the sharing of such run-time state to a given geographical area. You can configure zones in Policy Manager to match with the geographical areas in your deployment. There can be multiple zones per cluster, and each zone has a number of Policy Manager nodes that share their run-time state.

NOTE

OnGuard Agent does not work when Policy Manager is on an Azure instance that is configured with an Azure public IP address. As a workaround, configure the Policy Manager server's public IP address instead of its private IP address in the **Administration > Agents and Software Updates > OnGuard Settings > Policy Manager Zones > Override Server IPs** field.

Adding Policy Manager Zones

About this task

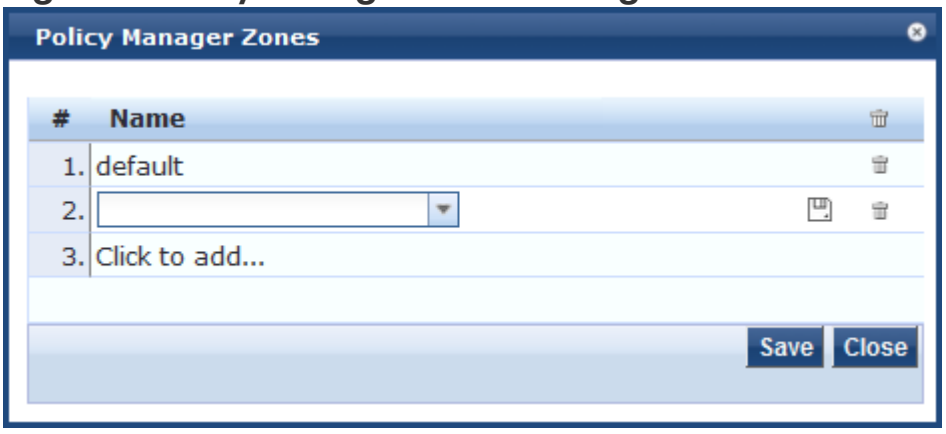
To add a Policy Manager Zone:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Click the **Manage Policy Manager Zones** link.

   The **Policy Manager Zones** dialog opens:

Figure 1. Policy Manager Zones Dialog



3. To add a new Policy Manager Zone, click **Click to add...** and enter the name of the Policy Manager Zone to be added.

4. Be sure to click the **Save** icon, then click **Save .**
5. To delete a zone, click the trash can icon (🗑).

Mapping Policy Manager Zones to OnGuard Clients

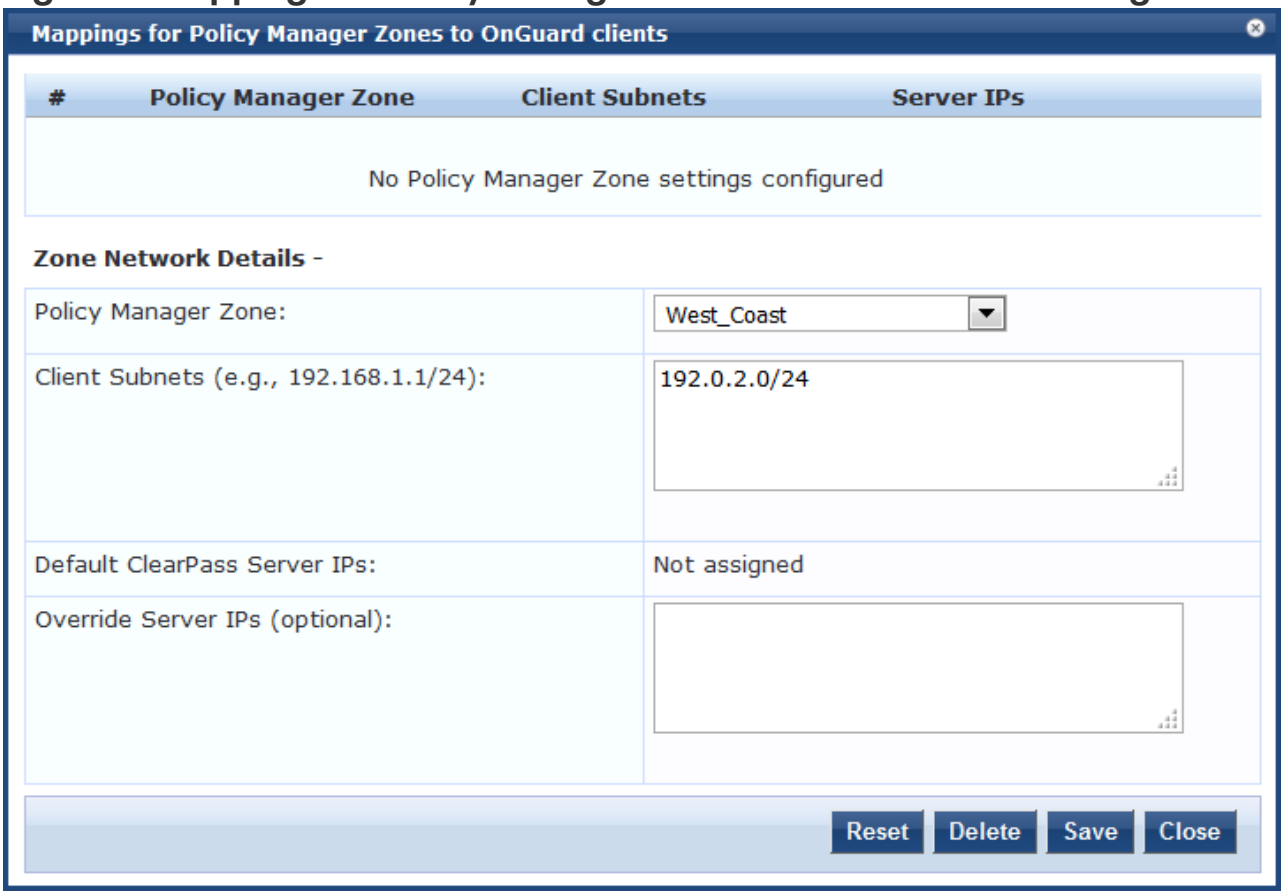
About this task

Use the following procedure to configure the Policy Manager Zone you created.

Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**. The **OnGuard Settings** page opens.
2. Click the **Policy Manager Zones** link. The **Mappings for Policy Manager Zones to OnGuard Clients** page opens.

Figure 2. Mappings forPolicy ManagerZones to OnGuard Clients Page



3. Specify the **Mappings for Policy Manager Zones to OnGuard Clients** parameters as described in the following table:

Table 1. OnGuard Settings > Policy Manager Zones Parameters

Parameter	Action/Description
Policy Manager Zone	Select the Policy Manager Zone from the drop - down list. If no Policy Manager zone is configured, the default Policy Manager zone is displ ayed in this field.
Client Subnets/Prefixes	Specify the client subnet addresses spec ified for the selected Policy Manager zon e.
Default ClearPass IPv4 Server IPs	This field displays the default ClearPass s erver IPv4 address specific to the selecte d Policy Manager zone.
Default ClearPass IPv6 server IPs	This field displays the default ClearPass s erver IPv6 address specific to the selecte d Policy Manager zone.
Override Server IPs (Optional)	Optionally, specify the IP addresses or th e Fully Qualified Domain Name (FQDN) to which you want the OnGuard agent to se nd the request in the sequence. You can specify the data port or load balancer IP address in this field. The IP addresses configured here will ove rride the IP address configured in the De fault ClearPass Server IPs field. For exam ple, if you have configured the IP address es 10.17.XXX.1, 10.17.XXX.2, and 10.17.XXX.3 OnGuard agent will send the request in t he same sequence.
Zone Network Details	
Policy Manager Zone	Select the Policy Manager zone from th e drop - down list that is created from th e **Administration > Server Manager > Se**

Parameter	Action/Description
	rver Configuration > Manage Policy Man ager Zones page.
Client Subnets	Specify the client subnets that are config ured for the selected zone. When adding a new subnet in the Policy Manager zone, the current domain ID is excluded from a database query. Previously, an error displ ayed stating "The configured client subn et is already present in the MDC - ongua rd zone. Please use a different client sub net." Now a new subnet can be added wi thin an existing Policy Manager zone with out an error message.
Default Policy Manager Server IPs	Specify the IP address of the default Poli cy Manager server.
Override Server IPs	Optionally, specify the IP addresses or th e Fully Qualified Domain Name (FQDN) to which you want the OnGuard agent to se nd the request in the sequence. You can specify the data port or load bal ancer IP address in this field. The IP addresses configured here will ove rride the IP address configured in the **De fault ClearPass Server IPs** field. For example, if you have configured the I P addresses 10.17.XXX.1, 10.17.XXX.2, and 1 0.17.XXX.3, OnGuard agent will send the request in the same sequence.

4. Click **Save**. The new Policy Manager Zone configuration settings are displayed.

Configuring NetEvents Targets

About this task

NetEvents are a collection of information regarding various ClearPass Policy Manager users, endpoints, guests, authentications, accounting details, and so on. This information is periodically posted to a server that is configured as the NetEvents target. If Policy Manager Insight is enabled on a ClearPass Policy Manager server (see Enabling Insight and Specifying a Master Insight Node), it will receive net events from all other server nodes within the same Policy Manager cluster.

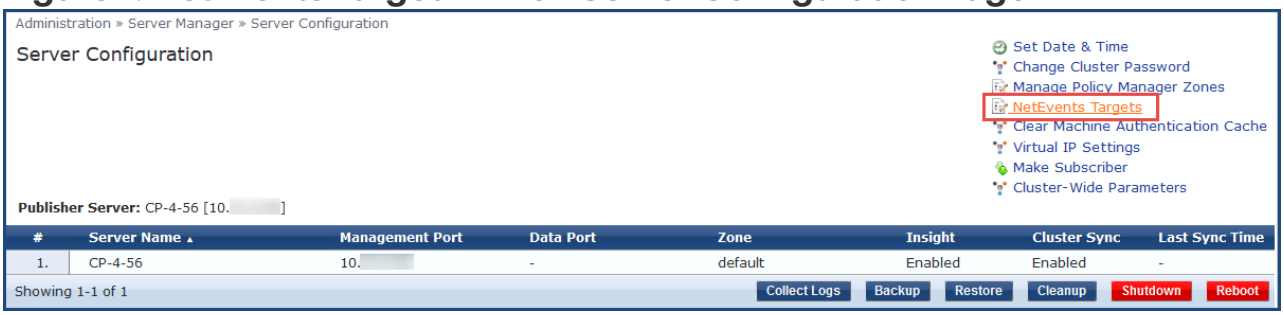
If you want to post these details to an external server that can aggregate these events or to an external dedicated Policy Manager Insight server for multiple Policy Manager clusters, you have to configure an external NetEvents Target.

To configure an external NetEvents Target:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.

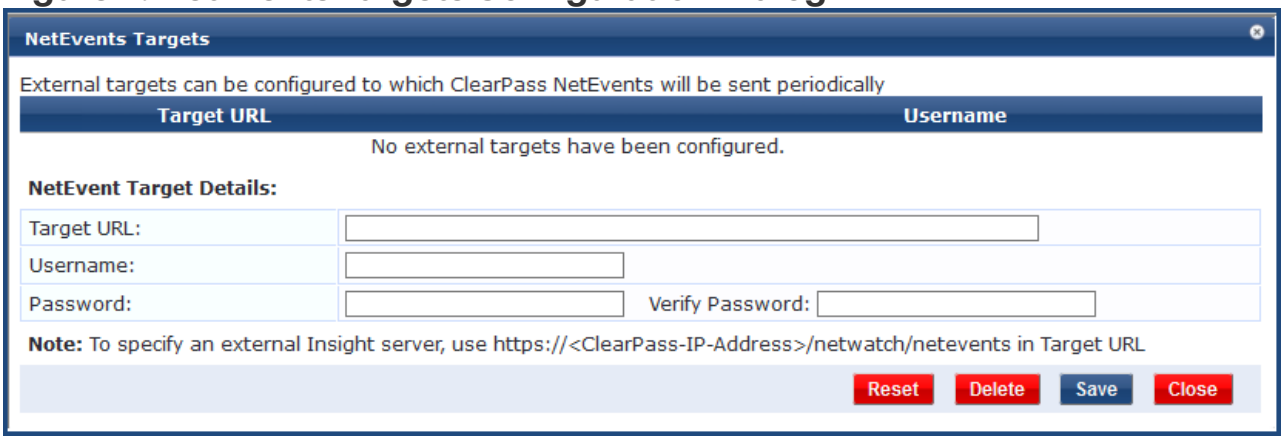
Figure 1. NetEvents Target Link on Server Configuration Page



2. Click the **NetEvents Targets** link.

   The **NetEvents Targets** configuration dialog opens.

Figure 2. NetEvents Targets Configuration Dialog



3. Specify the **NetEvents Targets** parameters as described in the following table, then click **Save**:

Table 1. NetEvents Targets Parameters

Parameter	Action/Description
Target URL	Enter the HTTP URL for the service that s upports posting to the NetEvents target and requires authentication using userna me and password. To specify an external Insight server, use t he following Target URL: `https://<Policy Manager-IP-addr ess>netwatch/netevents`
Username/Password	Enter the Policy Manager admin credenti als configured for authentication for the HTTP service that is provided in the Targe t URL.
Reset button	Resets the values entered in this configur ation dialog.
Delete button	Deletes the specified Target URL.

Clearing Machine Authentication Cache

About this task

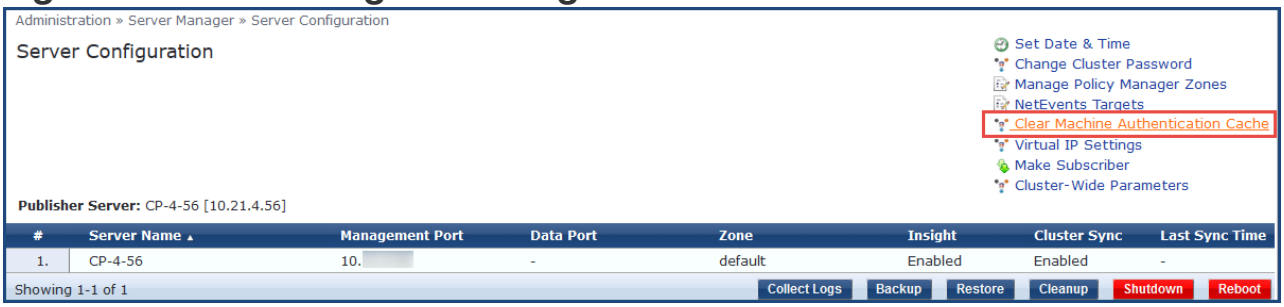
The **Clear Machine Authentication Cache** option clears the machine authentication cache from the local node; this operation is synced during zone cache replication. On confirmation, machine authentication cache is cleared from all nodes in the cluster. Once the machine authentication cache is cleared, it takes up to 5 seconds to resync the cache.

To clear machine authentication cache on all the nodes in a cluster:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page. The **Server Configuration** page opens:

Figure 1. Server Configuration Page > Clear Machine Authentication Cache



2. Click the **Clear Machine Authentication Cache** link. The following prompt is displayed:

   Are you sure you want to clear machine authentication cache?

3. To proceed with the operation, click **Yes**. The following message appears:

   Machine authentication cache cleared from all nodes

Configuring Virtual IP Settings

About this task

You can configure two nodes in a cluster to share a virtual IP address. Each virtual IP address is bound to the Primary node by default. The Secondary node takes over when the Primary node is unavailable.

> **NOTE**
>
> When configuring virtual IP on a hybrid cluster with both cloud hosted and on-premise nodes included, the details of the cloud hosted node will not display when configuring the virtual IP. In the previous releases, the **Virtual IP Settings** option was not displayed on the Server Configuration page for cloud hosted VMs. Going forward, this option is enabled for cloud hosted nodes.

> **NOTE**
>
> In a virtual machine deployment of ClearPass Policy Manager, you must enable forged transmits on the VMWare distributed virtual switch for the Virtual IP feature to be effective.

To configure a virtual IP address:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Click the **Virtual IP Settings** link.

   By default, the **Virtual IP Settings** for IPv4 dialog opens:

Figure 1. Virtual IP Settings for IPv4

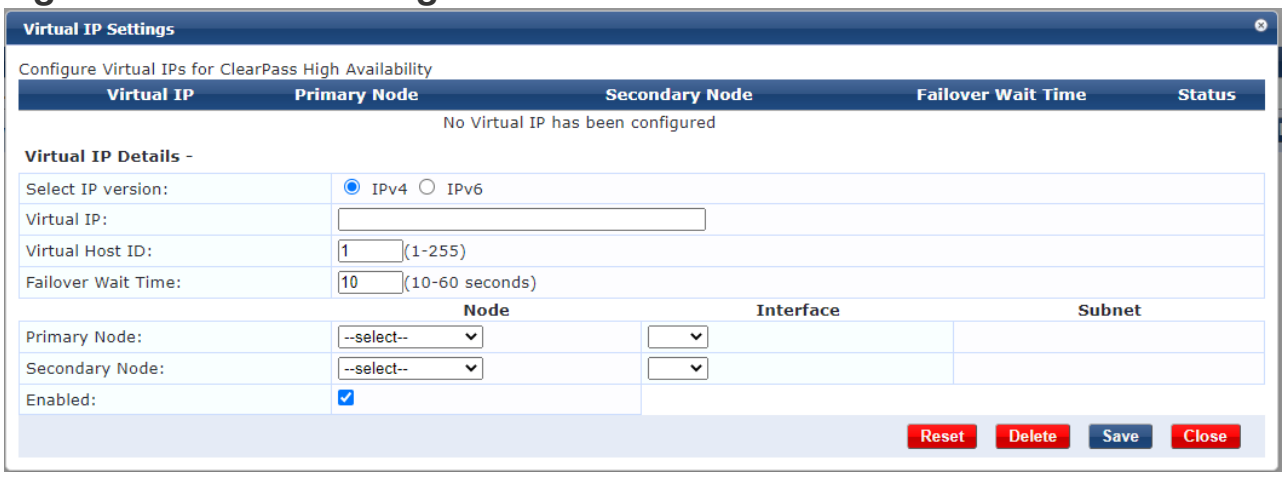
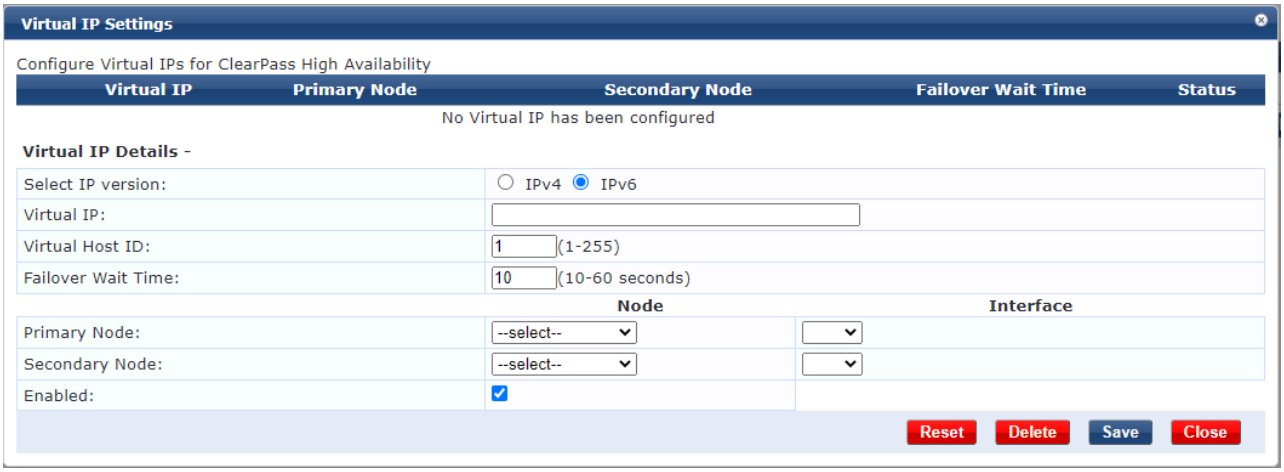


Figure 2. Virtual IP Settings for IPv6



3. Specify the **Virtual IP Settings** parameters as described in the following table, then click **Save**:

Table 1. Virtual IPv4 and IPv6 Settings Parameters

Parameter	Action/Description
Select IP Version	Select either the **IPv4** or **IPv6** addressing format.
Virtual IP	Enter the IPv4 or IPv6 address you want t o define as the virtual IP address.
Virtual Host ID	To specify a unique virtual Host for this Vi rtual IP, enter the virtual router ID (1 - 255) .
Failover Wait Time	In the **Failover Wait Time** field, enter the number of seconds for the secondary no de to wait after primary node failure befo re it acquires the virtual IP address. The default fail - over wait time is 10 seconds , in order for the secondary node to take over and respond quickly to authenticati on access requests.
Primary Node	Select the server to use as the primary n ode. **NOTE**

Parameter	Action/Description
	When IPv6 is selected, the **Primary Node** and **Secondary Node** drop - down lists are populated with the corresponding IPv6 addresses for selection. **NOTE** To add an IPv6 address as a Virtual IP, it must be in the same subnet as the Primary and Secondary nodes.
Secondary Node	Select the server to use as the secondary node.
Interface	When you select the primary node and the secondary node, the **Interface** field is populated with that node's management interface IP address. **NOTE** If the Data/External port is configured, the **Interface** field is populated with both the management interface IP address and the Data/External port IP address.
Subnet	The **Subnet** value for the management interface IP address is automatically populated when you select the primary node and secondary node.
Enabled	The virtual IP settings configured here are enabled by default.

Making a Subscriber

In the Policy Manager cluster environment, the publisher acts as the primary server. A Policy Manager cluster can contain only one publisher. Administration, configuration, and database write operations can occur only on the publisher.

The Policy Manager appliance defaults to a publisher unless it is made a subscriber. Cluster commands can be used to change the state of the server; for example, the publisher can be made a subscriber. When it is a subscriber, the **Make Subscriber** link is not displayed.

Caveats When Adding a Subscriber

About this task

Note the following caveats when adding a subscriber:

- As part of this operation, configuration changes are blocked on the publisher during the initial cluster sync process.
- If a ClearPass appliance is a subscriber in a cluster, all configuration options throughout the user interface specific to the publisher are grayed-out.
- All the application licenses on this server will be removed. To add and reactivate these application licenses, contact Aruba Support—navigate to **Administration > Support > Contact Support** for contact information.
- When an IPsec tunnel is already established between two ClearPass appliances, trying to add one of the appliances as a subscriber to the other fails. Users should be aware that ClearPass does not support adding an appliance to a cluster as a subscriber while an IPsec tunnel already exists between it and the publisher.
- Policy Manager indicates errors during the Make-Subscriber action if either of the following conditions are present:
    - The certificate chain used is not present on both systems for the HTTPS and database certificates.
    - An IP address is not included in the database certificate's subject or the *Subject Alternative Name* (SAN) field.

An error message is displayed on the **Administration > Server Manager > Server Configuration > Make Subscriber > Add Subscriber Node** configuration dialog and an alert is shown in the Dashboard if either of these requirements are missing.

> **NOTE**
>
> A subscriber's HTTPS trust list is required to include the certificate chain that signed the publisher's HTTPS server certificate. You do not have to reform the cluster when you upgrade. A database certificate is generated automatically for each node during the cluster upgrade process.

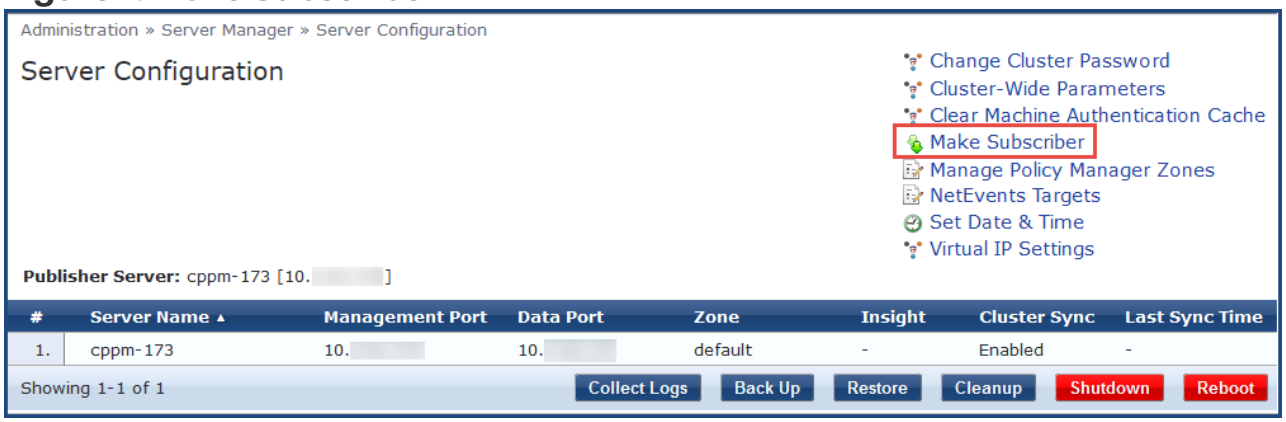
To convert a publisher to a subscriber:

Procedure

1. From the publisher, navigate to the **Administration > Server Manager > Server Configuration** page.

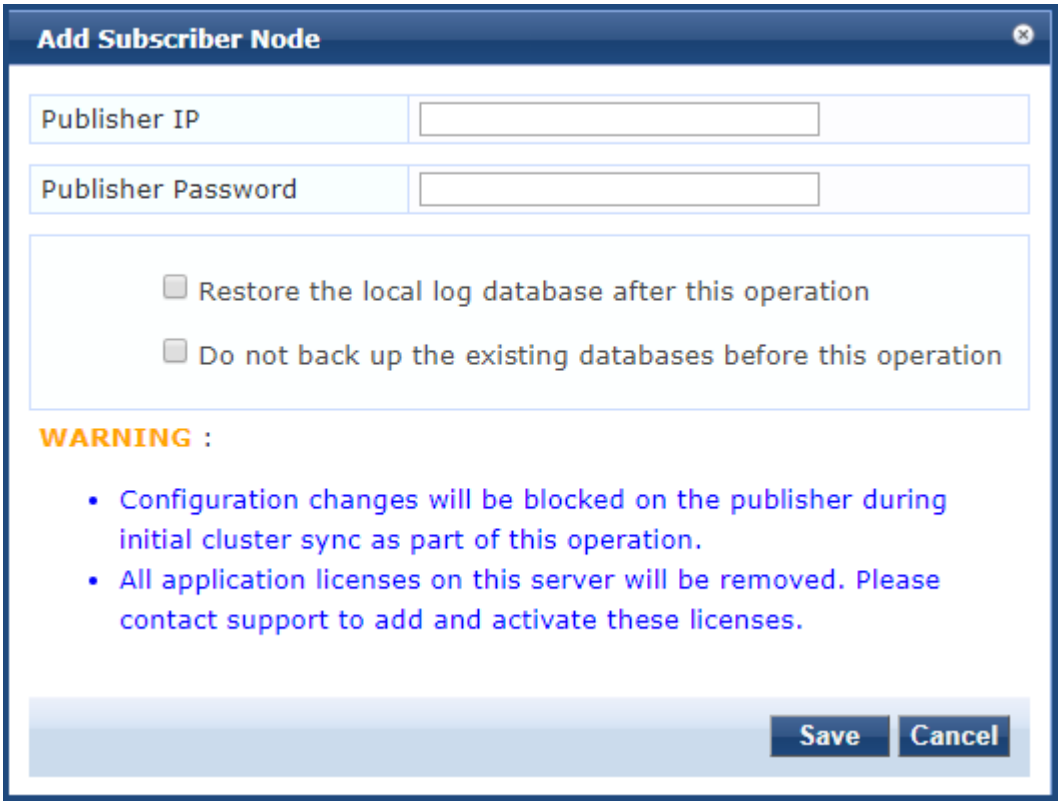
   The **Server Configuration** page opens.

   Figure 1. Make Subscriber Link



2. Click the **Make Subscriber** link. The **Add Subscriber Node** page opens:

   Figure 2. Adding a Subscriber



3. Specify the **Add Subscriber Node** parameters as described in the following table, then click **Proceed**.

Table 1. Add Subscriber Node Parameters

Parameter	Action/Description
Publisher IP	Enter the publisher's IP address.
Publisher Password	Specify the publisher's password. **NOTE** The password specified here is the password for the CLI u ser appadmin.
Restore the local log database after this o peration	To restore the log database after the subs criber node has been added, select the c heck box.

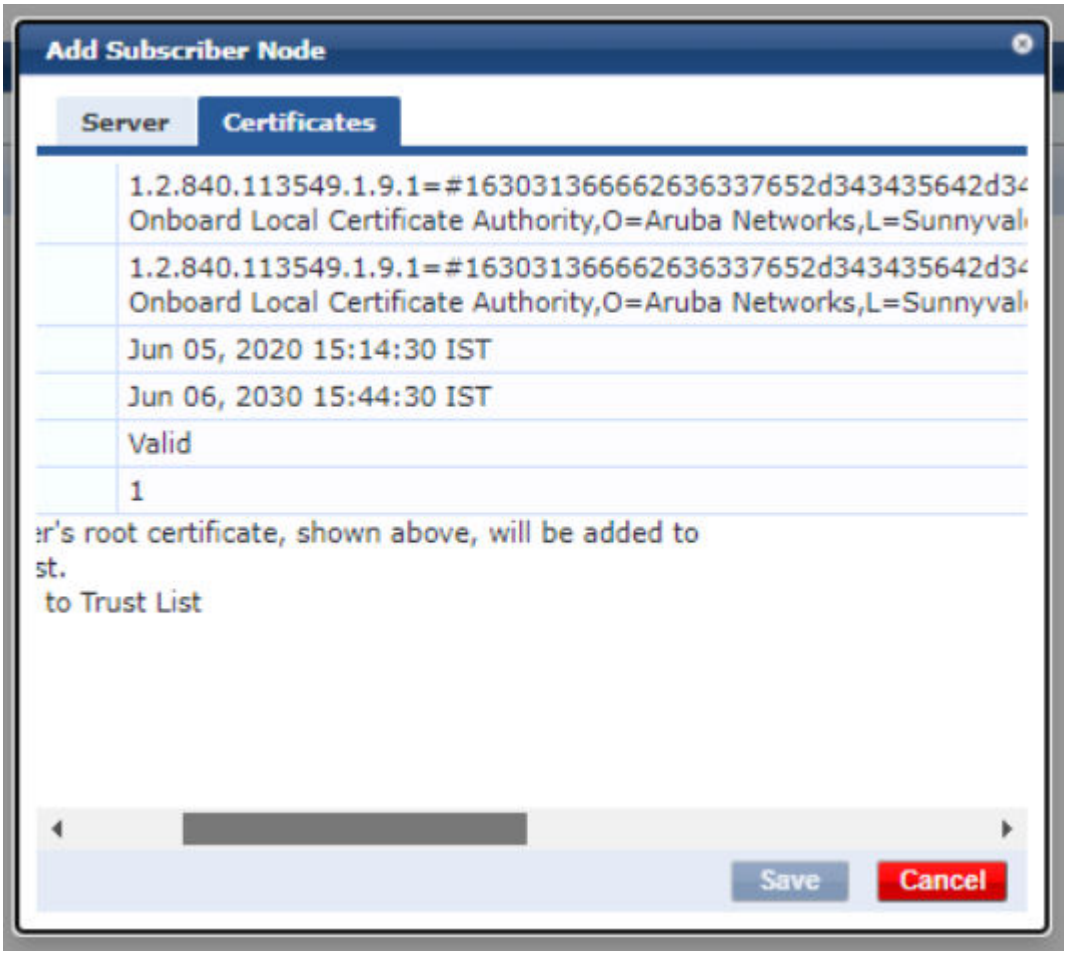
Parameter	Action/Description
Do not backup the existing databases before this operation	If you do not require a backup to the existing databases on this node, select the check box.

4. After you click **Proceed**, the Add Subscriber Node window displays the progress of the Add Subscriber process.

> **NOTE**
>
> Do not close the window until the process is complete. Closing this window could result in the failure of the cluster join operation.

Certificate details are displayed in a new tab in the **Add Subscriber Node** window. If you mark the check box to accept the certificate and click the **Save** button, the subscriber automatically adds the root server certificate of the publisher to the subscriber's certificate trust list and then proceeds with the make-subscriber process.

Figure 3. Automatically add a Certificate in the Add Subscriber Node Process



Dropping a Subscriber

About this task

If a cluster is broken apart to be all standalone appliances, then when those appliances contact the ClearPass server at night (phone home), each at a slightly different time, only the first one authenticates successfully with the HPE Passport software updates token. Authentication fails for the rest of the appliances.

Users should be aware that whenever appliances are taken out of a cluster, then on each resulting standalone appliance you will need to go to **Administration > Agents and Software Updates > Software Updates** and use the **Generate Token** button to generate a new software updates token specific to that appliance. For more information, refer to HPE Passport Credentials.

To drop a subscriber from the cluster:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Select the node you want to drop from the cluster.

3. Click the **Drop Subscriber** button.

> **NOTE**
> This option is not available in a single-node deployment.

Collecting Logs

About this task

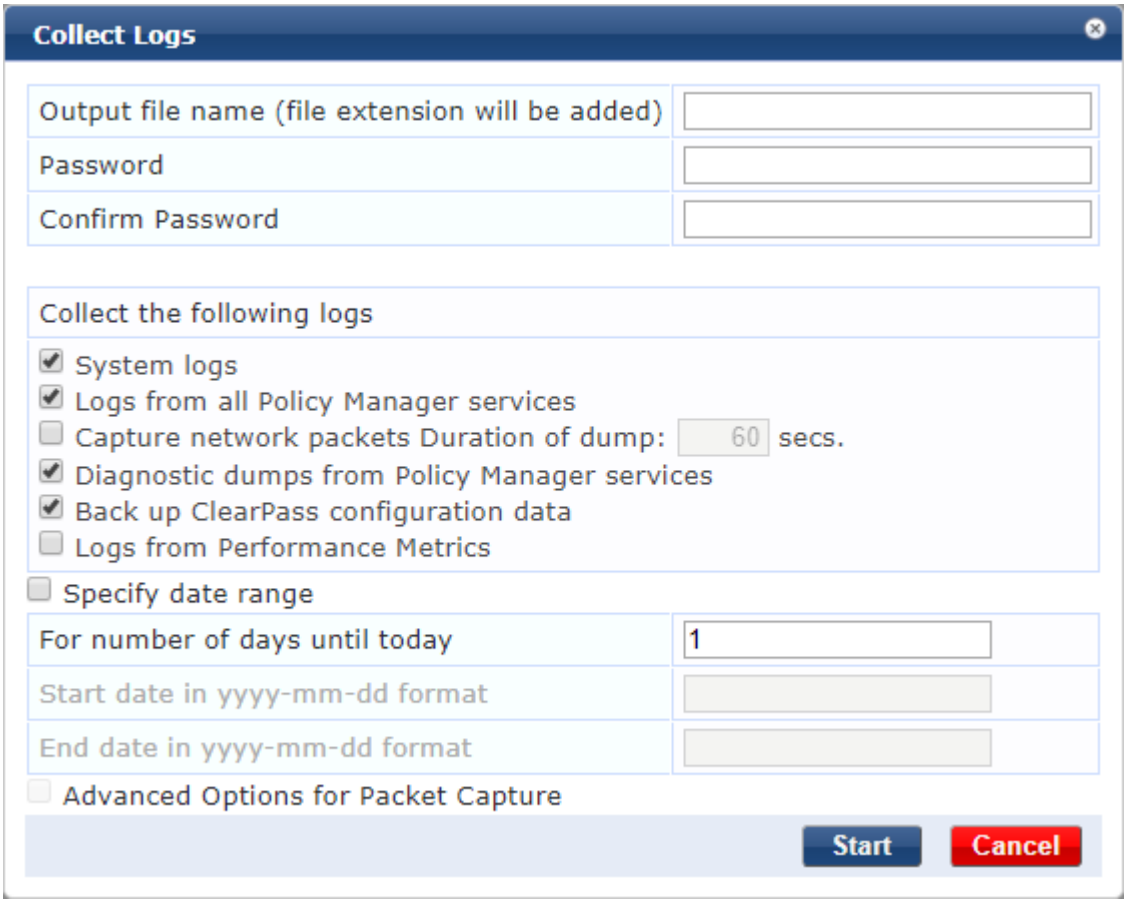
When you need to review performance or troubleshoot issues in detail, Policy Manager can compile and save transactional and diagnostic data into several log files secured with a password. These files are saved in Local Shared Folders and can be downloaded to your computer (see Downloading Local Shared Folders).

To collect logs:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**. The **Server Configuration** page opens.
2. Click **Collect Logs**. The **Collect Logs** dialog opens.

Figure 1. Collect Logs Dialog



3. In the **Password** and **Confirm Password** fields, enter and confirm the password used to collect logs and to zip the tar.gz log file. The password fields on this form must:
   - Contain only the following special characters: **& ; ` | < > ( )**
   - Start and end with one of the following special characters: ***- ' ***
4. Enter an output filename and add the .tar.gz extension to the filename.
5. Select the types of logging information you want to collect. The types of logging are:
   - System Logs
   - Logs from all Policy Manager services
   - Capture network packets Duration of dump in seconds.

     CAUTION

     Use this option only when you want to debug a problem. System performance can be severely impacted.
   - Diagnostic dumps from Policy Manager services

- Back up Policy Manager configuration data

6. Enter the time period for which you want to collect the information.
   - Specify a number to collect logs for the number of days until the current day.
   - To collect logs for the specified time period, select the **Specify date range** check box and enter a start date and end date in yyyy-mm-dd format in the respective fields.
7. Click **Start**.

   You'll see the progress of the information collection.
8. To finish, click **Close**
9. To save the log file to your computer, click **Download File**.

> **NOTE**
>
> If you are attempting to open a capture file (.cap or .pcap) using WireShark, untar or unzip the file (based on the file extension). When the entire file is extracted, navigate to the PacketCapture folder. In this folder, you will find a file with a .cap extension. WireShark can be used to open this file and study the network traffic.

Backing Up the Policy Manager Database

About this task

The backup file is automatically placed in the *Shared Local Folder* under folder type *Backup Files* (for details, see Downloading Local Shared Folders).

Backup files are in a zipped tar format (tar.gz extension).

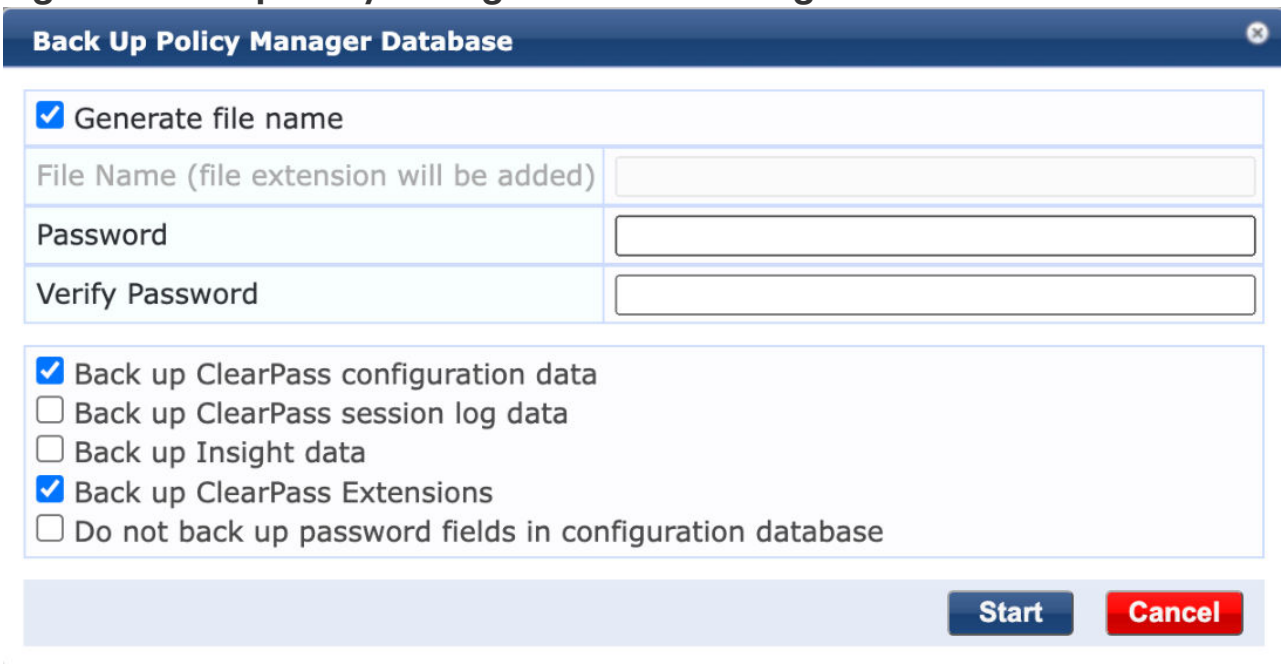
To back up the Policy Manager database:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Click the **Back Up** button.

   The **Back up Policy Manager Database** dialog opens:

Figure 1. BackupPolicy ManagerDatabase Dialog



3. Specify the **Back up Policy Manager Database** parameters as described in the following table, then click **Start**:

Table 1. Back up Policy Manager Database Parameters

Parameter	Action/Description
Generate file name	To enable Policy Manager to generate a filename for the database backup, select this check box. This option is enabled by default.
File Name	To manually specify the backup file name, clear the **Generate file name** check box,, then enter the desired file name.
Password	Optionally enter and verify a password to provide protection on backup files when backing up or restoring a ClearPass server. To restore the file, the same password must be provided in the **Restore Policy Manager Database** screen. Providing a password ensures no changes are made in the restored file. Password protection is not mandatory. If a password is not c

Parameter	Action/Description
	reated when backing up the file, then it is not required when restoring the file.
Back up ClearPass configuration data	The option to back up Policy Manager co nfiguration data is enabled by default.
Back up ClearPass session log data	To enable back up of Policy Manager sess ion log data, select this check box.
Back up Insight data	Select this option to enable the back up of Policy Manager Insight data.
Do not back up password fields in config uration database	If you don't want to backup the password fields in the configuration database, sele ct this check box.

Restoring Policy Manager Configuration Data

About this task

The automatic backup file (**auto-backup-tips*.tar.gz**) generated by Policy Manager contains plain text files with application and platform license information. Evaluation and Subscription licenses which have expired will not be included in the backup. In a cluster deployment, the backup file on the publisher also contains license information for all subscribers. For information about importing server certificates in PKCS12 format, refer to Importing a Server Certificate.

NOTE

Users should be aware that a backup made from an earlier version of ClearPass in Common Criteria (CC) mode (such as ClearPass 6.9.x or 6.10.x) should not be restored on a 6.11 system that is in non-FIPS mode.

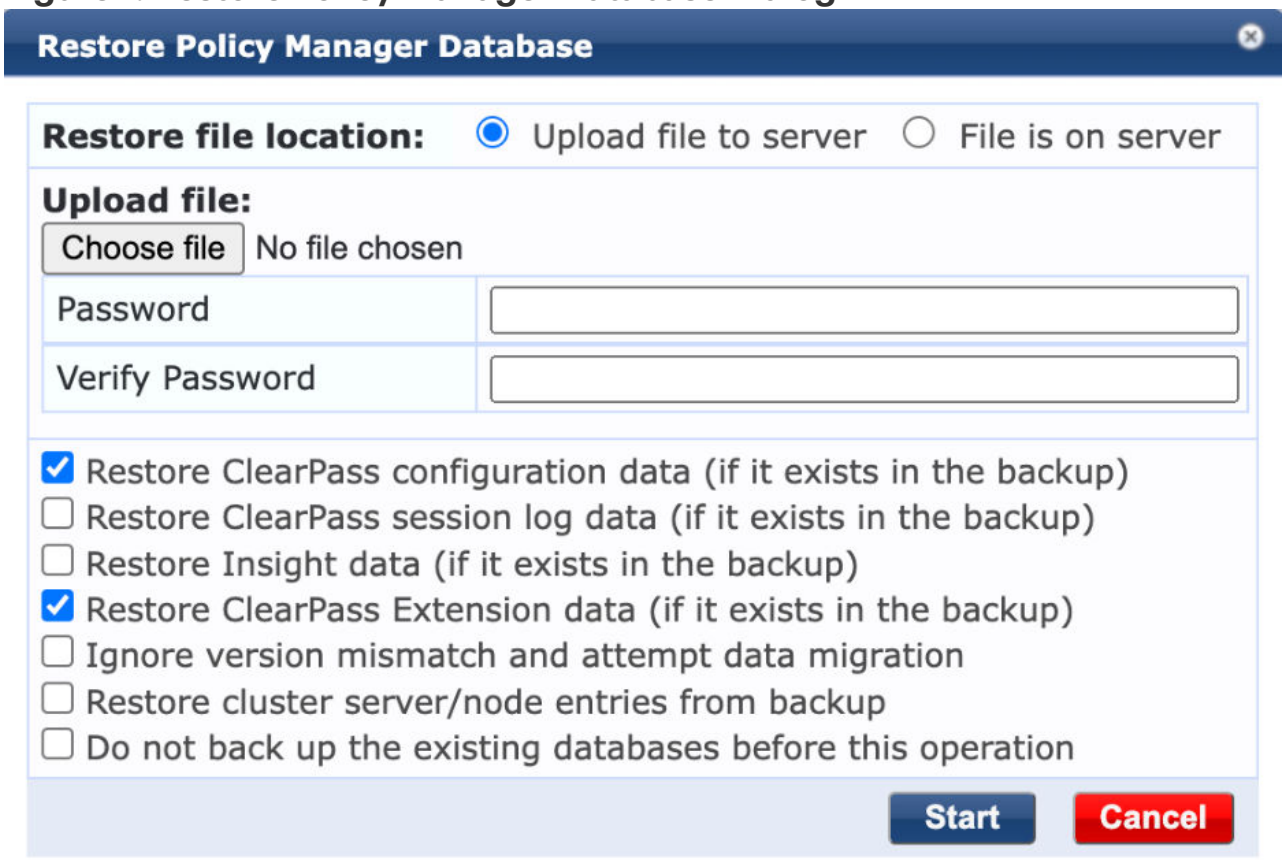
To restore the Policy Manager configuration data:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Click the **Restore** button.

The **Restore Policy Manager Database** dialog opens:

Figure 1. Restore Policy Manager Database Dialog



3. Specify the **Restore Policy Manager Database** parameters as described in the following table, then click **Start**:

Table 1. Restore Policy Manager Database Parameters

Parameter	Action/Description
Restore file location	Select either **Upload file to server** or **File is on server**.
Upload file	Browse to select name of backup file. **NOTE** This option is available only when the **Upload file to se rver** option is selected.

Parameter	Action/Description
Password	If a password was provided in the **Backup Policy Manager Database** screen it must also be provided and verified within the **Restore Policy Manager Database** screen. Providing a password ensures no changes are made in the restored file. Password protection is not mandatory. If a password is not created when backing up the file, then it is not required when restoring the file. **NOTE** Do not use this Password option unless your backup file is password - encrypted. This field should only be used for a password - encrypted password, if it exists. Otherwise, leave this Password field empty. If you enter an admin or cluster password, the restore operation will fail.
Shared backup files present on the server	If the files is on a server, select a file from the files in the local shared folders. (See Downloading Local Shared Folders.) **NOTE** This is displayed only when the **File on server** option is selected.
Restore ClearPass configuration data (if it exists in the backup)	Select the check box to include an existing configuration data in the restore.
Restore ClearPass session log data (if it exists in the backup)	Select the check box to include the log data in the restore.

Parameter	Action/Description
Restore Insight data (if it exists in the backup)	Select the check box to include Insight reporting data in the restore.
Restore ClearPass Extension data (if it exists in the backup)	Leave this option selected (default) to restore extensions during a backup. Unselect this option to exclude extensions from the restoration, as backups can be large and time consuming when multiple extensions are installed.
Ignore version mismatch and attempt data migration	Select the check box if you are migrating configuration and/or log data from a backup file that was created with a previous compatible version.
Restore cluster server/node entries from backup.	Select the check box to include the cluster server/node entries in the restore.
Do not backup the existing databases before this operation.	Select the check box if you do not want to backup the existing databases before performing a restore.

Performing a System Cleanup

You can perform a system cleanup operation to purge the following records:

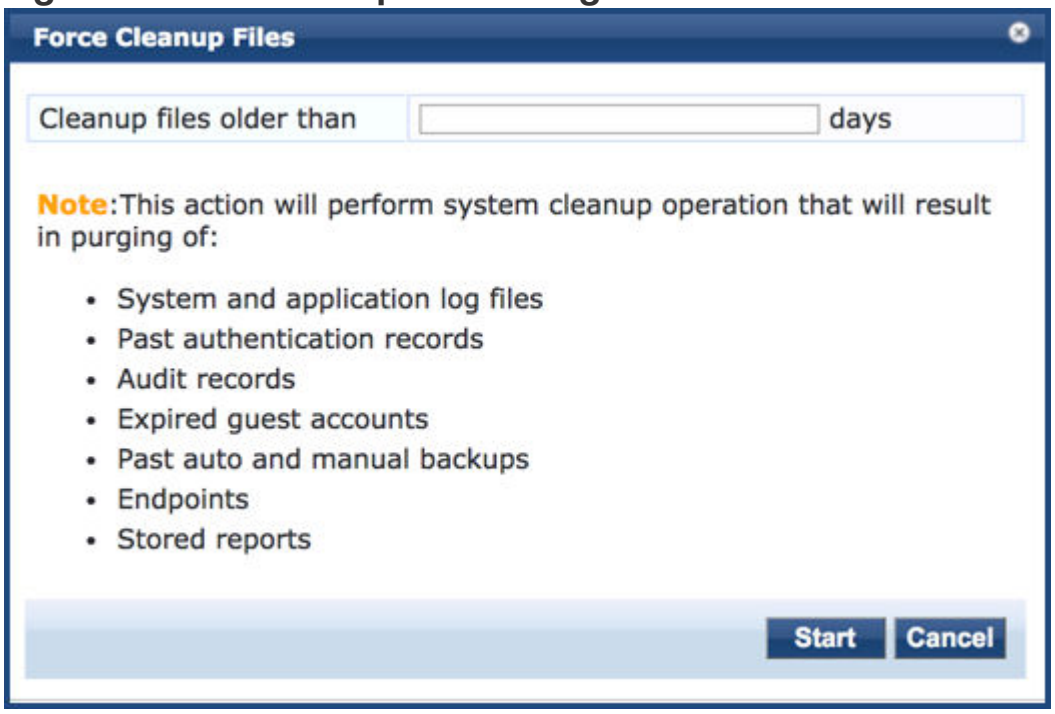
- System and application log files
- Past authentication records
- Audit records
- Expired guest accounts
- Past auto and manual backups
- Endpoints
- Stored reports

To perform a system cleanup:

1. Navigate to the **Administration > Server Manager > Server Configuration** page.
2. Select the Policy Manager server you wish to clean up.
3. Click the **Cleanup** button.

   The **Force Cleanup Files** dialog opens.

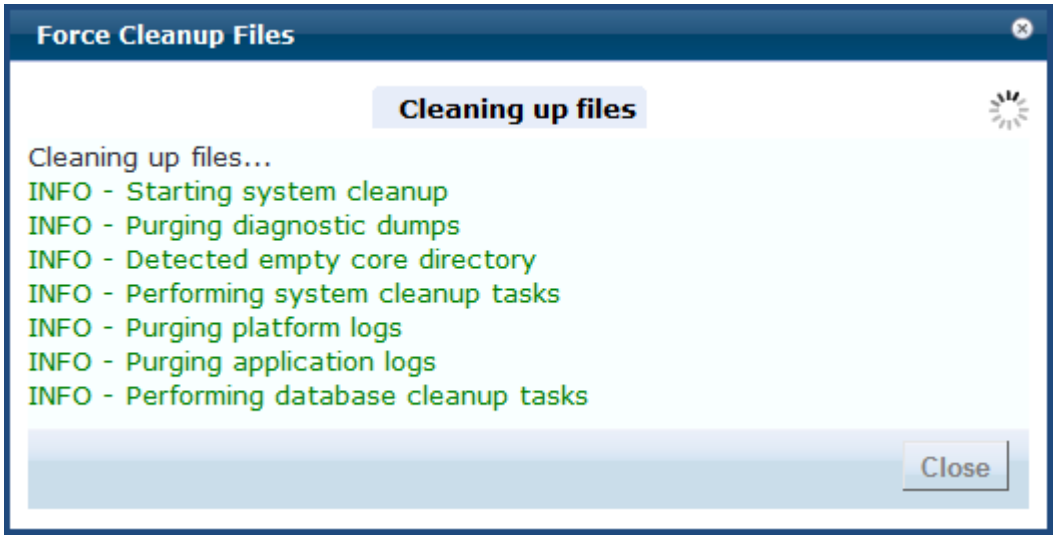
Figure 1. Force Cleanup Files Dialog



4. Enter the number of days system files can remain before they are removed. The allowed range is 0 to 15 days.
5. To initiate the cleanup process, click **Start**.

   The **Force Cleanup Files** status report opens:

Figure 2. Force Cleanup Files Status Report



Shutting Down or Rebooting the Server

Shutting Down the Server

About this task

To shut down the current Policy Manager server:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page .
2. Click the **Shutdown** button.

   You receive the message:

   Are you sure you want to shut down Policy Manager Server <IP_address>?
3. To proceed with the shut down operation, click **Yes**.

   You receive the message:

   Shut down of server initiated

   The Policy Manager server shuts down in an orderly manner.
4. To cancel the shut down operation, click **Cancel**.

Rebooting the Server

About this task

To reboot the current Policy Manager server:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page .
2. Click the **Reboot** button.

   You receive the message:

   Are you sure you want to reboot Policy Manager Server <IP_address>?
3. To proceed with the reboot, click **Yes**.
4. To cancel the reboot operation, click **Cancel**.

Log Configuration

- Service Log Configuration
- System Level Configuration
- Logging Best Practices

Service Log Configuration

About this task

Use the **Service Log Configuration** tab to specify the default log level for each service and associated module.

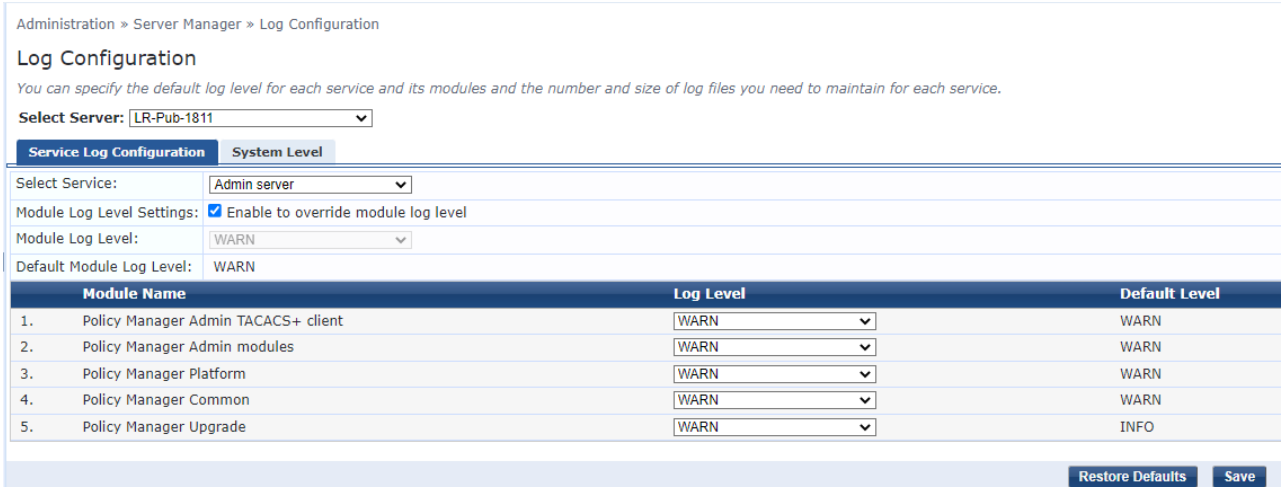
To configure logs for services and system level:

Procedure

1. Navigate to the **Administration > Server Manager > Log Configuration** page.

   The **Service Log Configuration** page opens:

   Figure 1. Log Configuration > Service Log Configuration Tab



2. Specify the following **Service Log Configuration** parameters:

Table 1. Log Configuration > Service Log Configuration Parameters

Parameter	Action/Description
Select Server	From the **Select Server** drop - down, specify the server for which you want to configure logs. All nodes in the cluster appear in the drop - down list.
Select Service	Specify the service for which you want to configure logs: • Admin server • AirGroup notification service • Async network service • Policy Manager network services • DB change notification server • DB replication service • Micros Fidelio FIAS • Zone cache • Policy server

Parameter	Action/Description
	• RadSec Service **NOTE** When you select the **RadSec service** and change the **Log Level**, it takes approximately 8 seconds for the Radsec service to restart. Therefore, Aruba recommends waiting 8 to 10 seconds after modifying the Log Level before collecting logs (**Administration > Server Manager > Server Configuration** > click the **Collect Logs** button). • Radius server • Syslog client service • TACACS+ server
Module Log Level Settings	Select the **Enable to override module log level** check box to set the log level for each module individually (listed in decreasing level of verbosity). For optimal performance you must run Policy Manager with the log level set to **ERROR** or **FATAL**. Available options include the following: • DEBUG • INFO • WARN • ERROR • FATAL

Parameter	Action/Description
	If this option is disabled, the log level for a selected service must be set using the **Module Log Level** drop - down menu.
Module Log Level	When the **Enable to override module log level** option is disabled, use the **Module Log Level** drop - down menu to define th e log level for the selected service. Availa ble log level options include: • DEBUG • INFO • WARN • ERROR • FATAL
Default Module Log Level	Lists the default module log level for the service. If the Log Levels are set to non - default value, an event viewer warning message displays once each day. Default log level options include: • DEBUG • INFO • WARN • ERROR • FATAL
Restore Defaults/Save	Click **Save** to save changes. To restore default log configuration settin gs, click **Restore Defaults**.

System Level Configuration

About this task

Use the **System Level** tab to specify the number and size of the log files to maintain for each supported service listed, and the server to which they can be sent.

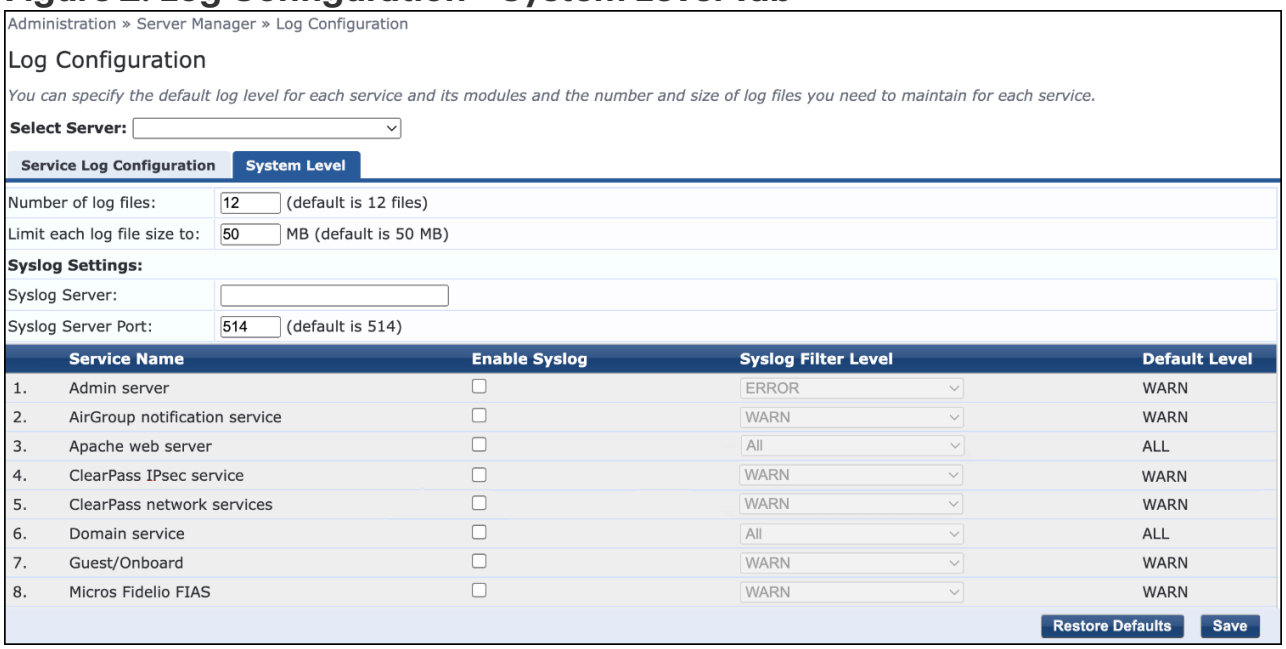
> **NOTE**
>
> If needed, enable a syslog export for a Async network service log to export Async network service system level (deviceprofiler.log, cmd-ctrl.log, mdm.log, agentcontroller.log, optikd.log, postauth.log, etc.) logs to a dedicated external syslog server for analysis and trending as required.

Procedure

1. From the **Log Configuration** page, select the **System Level** tab.

   The **System Level** page opens:

Figure 2. Log Configuration > System Level Tab



2. Specify the following **System Level** tab parameters:

> **NOTE**
>
> When enabling the **ClearPass IPSec service**, all the IPSec events from ClearPass Policy Manager and sent OUT to the Syslog target configured for the System Level.

Table 2. Log Configuration > System Level Parameters

Parameter	Action/Description
Select Server	Specify the server for which you want to configure logs.
Number of log files	Specify the number of log files of a specific module to keep at any given time. When a log file reaches the specified size (see **Limit each log file size to**), Policy Manager rolls the log over to another file until the specified number of log files is reached. Once the number of log files exceeds the specified value, Policy Manager overwrites the oldest file.
Limit each log file size to	Specify the size of each log file before the log rolls over to the next file. The default value is 50 MB.
Syslog Settings	
Syslog Server	Specify the name of the syslog server. ClearPass sends the configured module logs to this selected syslog server.
Syslog Server Port	Specify the syslog server port number. The default is **514**.
Enable Syslog	To override the **Syslog Filter Level** for a listed service, select the **Enable Syslog** option.
Syslog Filter Level	If desired, change the **Syslog Filter Level**. The current **Syslog Filter level** is based on the default log level specified on the **Service Log Configuration** tab.
Restore Defaults/Save	Click **Save** to save changes.

Parameter	Action/Description
	To restore default settings, click **Restore Defaults**.

Logging Best Practices

For optimal logging, HPE Aruba Networking recommends using the default log configuration for Policy Manager logging. Setting the Log levels to **Debug** is recommended only for debugging operations and must be changed back to default once completed.

Downloading Local Shared Folders

The supported Policy Manager folder types are:

- **Backup files**: Database backup files that are backed up manually.
- **Log files**: Log files backed up via the method described in Collecting Logs.
- **Automated Backup files**: Database backup files that are backed up automatically on a daily basis.

To download a local shared folder:

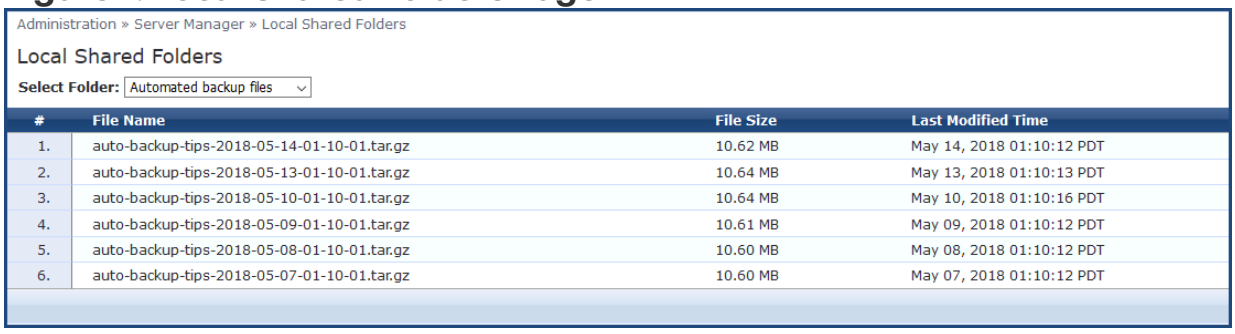
1. Navigate to **Administration > Server Manager > Local Shared Folders**.

   The **Local Shared Folders** page opens.

2. Choose a folder type from the **Select Folder** drop-down list.

   Optionally select **Service log** from the **Select Folder** drop-down menu to display an additional **Select Service** drop-down menu to view and download module specific logs.

Figure 1. Local Shared Folders Page



3. Select the local shared folder you want to download. The selected auto-backup file is downloaded to your system's **Download** folder.

Device Insight

This section describes the following topics:

- Device Insight Integration Page
- Launching Device Insight

Subtopics

About Device Insight
Device Insight Integration Page
Launching Device Insight

About Device Insight

Policy Manager supports the seamless integration between Policy Manager and Policy Manager Device Insight.

- Notable Processes and Concepts
- Prerequisites to Using Device Insight Integration
- Changes in ClearPass Behaviors When Device Insight Integration is Enabled
- Changes in the ClearPass User Interface When Device Insight Integration is Enabled

Subtopics

Introduction
Identifying Headless Devices
Device Insight and Aruba Central
About Device Insight Integration Mode in Policy Manager
Notable Processes and Concepts
Prerequisites to Using Device Insight Integration
Changes in ClearPass Behaviors When Device Insight Integration is Enabled
Changes in the ClearPass User Interface When Device Insight Integration is Enabled

Introduction

If you have Policy Manager Device Insight, you have the option to enable Device Insight Integration in Policy Manager. With integration enabled, bidirectional metadata is exchanged, providing enhanced classification of endpoints connected to the network. When Policy Manager Device Insight is combined with Policy Manager for real-time enforcement, visibility, security, and compliance are greatly increased for all devices connecting to the network.

Aruba's Policy Manager Device Insight solution is a cloud-hosted application that provides detailed device visibility, classification, and reporting. It uses active and passive discovery methods, deep-packet inspection, contextual information, behavioral analysis, and advanced machine learning to continuously discover, profile, and monitor devices. The Device Insight user interface provides granular visibility into the devices on your network.

Identifying Headless Devices

Identifying headless devices (commonly referred to as the *Internet of Things* or *IoT*) can be challenging for traditional discovery methods: New products rapidly released by emerging vendors may be difficult to identify with standard discovery and profiling techniques; or devices such as Raspberry Pi may be built with generic hardware and software, but the individual devices serve a variety of different uses. Using traditional discovery and profiling techniques in such cases could result in partial or inaccurate profiling, making it difficult to apply appropriate policies.

Device Insight and Aruba Central

Policy Manager Device Insight is part of the Aruba Central Platform. It uses data collectors deployed on your network to continuously gather metadata from devices and send them to Policy Manager Device Insight Analyzer, which analyzes the attributes and classifies the devices. Policy Manager Device Insight data collectors are available as hardware appliances or virtual appliances.

About Device Insight Integration Mode in Policy Manager

A simple setting in Policy Manager lets you integrate it with Device Insight (see Device Insight Integration Page). When Policy Manager Device Insight integration is enabled, a Policy Manager server also acts as a collector, and Policy Manager seamlessly switches

to using Device Insight instead of Profiler for all device discovery and classification functionality. Data is automatically transmitted in both directions between Policy Manager and Device Insight, providing enhanced profiling and reporting. Policy Manager Device Insight also updates Policy Manager in real time if it detects a change in device classification. Such a change could be an indication of device spoofing and should be viewed as a security threat.

You can view all the resulting device information and reports in the Policy Manager Device Insight application. Policy Manager provides direct links to Policy Manager Device Insight from the Policy Manager Menu, from the Dashboard device widgets, and from the Device Insight Integration page. In Policy Manager, integration with Policy Manager Device Insight is disabled by default. You can enable, disable, and re-enable integration as you wish. When you disable integration, Policy Manager returns to using Profiler for device profiling. While integration is disabled, the Policy Manager server is still listed as a collector in Policy Manager Device Insight, but its status is shown as unreachable.

Notable Processes and Concepts

Take note of the following processes and concepts:

Subtopics

Collector Service, Sync Service, and Device Insight Analyzer
Device Insight Integration in a ClearPass Policy Manager Cluster
Tags
OnGuard, Onboard, Endpoint Context Servers, and Extensions
When Policy Manager Is Running in FIPS or Common Criteria (CC) Mode

Collector Service, Sync Service, and Device Insight Analyzer

While integration is enabled, two subsystems in Policy Manager, the *collector service* and the *sync service*, handle data transfer between Policy Manager and Device Insight.

- The collector service receives profile requests from various services in Policy Manager and acquires the device data. It consolidates the events from the Policy Manager server or Policy Manager cluster and sends them to Policy Manager Device Insight Analyzer. The collector service listens on port 6180.
- The Policy Manager Device Insight Analyzer processes the device's characteristics and communication and behavior patterns. Using rules and machine learning, it uses

those attributes to match a device to a set of known fingerprints and classifies the device.

- The Sync service then fetches the endpoint's classification from Analyzer and updates Policy Manager's endpoint repository.
- Updates happen in real time, and are backed up by periodic polling. This polling interval is separate from the polling interval set in **Cluster-Wide Parameters**, and can be adjusted in the Policy Manager Device Insight Integration configuration (see Table 1, Device Insight Integration Page Parameters).

Device Insight Integration in a ClearPass Policy Manager Cluster

Device Insight Integration can be enabled on a single Aruba appliance, or on any appliance in a Policy Manager cluster:

- If only a single appliance is enabled, the collector service and sync service run on the same appliance, which handles all required data exchanges and updates.
- In a cluster, you designate one of the servers as the primary server. The collector service runs on all the appliances, and the subscribers forward their data to the primary server. The sync service runs only on the primary server in a cluster and is responsible for sending data to the Analyzer. All data exchanges with Policy Manager Device Insight go through the primary node.

Tags

In Policy Manager Device Insight, after you have filtered for devices that match your criteria, you can create a tag and use it to group similar devices together. Tags can only be applied on classified devices. The tag is added to each device in the group as another attribute, and is sent to Policy Manager along with the other endpoint attributes. These tags are then available for filtering in **Configuration > Identity > Endpoints** and in Policy Manager Guest API Explorer > **Identity > Endpoint**, and can be used in services for role-mapping or enforcement policies.

OnGuard, Onboard, Endpoint Context Servers, and Extensions

If the Policy Manager deployment uses OnGuard, OnGuard's posture and health-related data is also sent to Policy Manager Device Insight for reporting purposes. Similarly, if the deployment uses Onboard, the device information Onboard collects during the onboarding process is sent to Device Insight. Information collected through endpoint context servers or integrations with 3rd-party ClearPass Extensions is also sent to Policy Manager Device Insight.

When Policy Manager Is Running in FIPS or Common Criteria (CC) Mode

When Policy Manager is running in FIPS or Common Criteria (CC) mode, only FIPS or CC -approved ciphers are negotiated between Device Insight and Policy Manager. Device Insight is not included as part of Policy Manager's Common Criteria validations. When Policy Manager is in Common Criteria (CC) mode, TLS 1.2 is enforced with RadSec, and TLS 1.0 and TLS 1.1 are no longer supported.

Prerequisites to Using Device Insight Integration

About this task

> **NOTE**
> A Policy Manager Access License is required order to enable Device Insight Integration.

Before you enable Device Insight Integration in Policy Manager, complete the following prerequisites:

Procedure

1. Ensure that your Policy Manager server has an Access License. This license is required to enable Device Insight integration.
2. Have a subscription in the Aruba Cloud Platform for ClearPass Device Insight.
3. Configure Policy Manager Device Insight Integration (for details, see Device Insight Integration Page).
4. Generate your **Registration Token** and have it ready:

a. In Aruba Central, go to the **Account Home** page.

b. Click the **Data Collectors** link.

c. On the **Managed Collectors** page, click **Registration Token**.

5. Make the following changes to existing Policy Manager configurations:

> **NOTE**
>
> The Device Insight Integration page will provide a warning notification when you try to save it if any of these tasks have not been done.

a. Remove any custom fingerprints you have configured (for related information, see Updating Device Fingerprints From a Hosted Portal).

b. Remove any network scans you have configured (see).

c. Remove any enforcement profiles that are configured with Nmap scans, SNMP scans, or OnDemand scans (see Configuring Enforcement Profiles and).

d. Disable or remove any services that have audit servers enabled (see Configuring Services Using Service Templates & Wizards and Configuring Other [%=aruba_style_guide.ShortPrdNm2%] Services).

Changes in ClearPass Behaviors When Device Insight Integration is Enabled

About this task

When Device Insight Integration is enabled on a Policy Manager server, the functionality of a number of Policy Manager features are replaced by features in Device Insight. If you disable Device Insight Integration, these features are restored:

Procedure

1. Policy Manager stops using Policy Manager Profiler for fingerprinting and uses Device Insight Analyzer instead for in-depth data analysis. If you later disable integration, Policy Manager reverts to using Policy Manager Profiler for fingerprinting.

2. The following active scans are disabled:

- Network scans (physical): Policy Manager network scans where a seed switch is identified as the scan root are not supported.
- Subnet scans (logical): Policy Manager IP subnets' scans are not supported.
- On Demand scans (Context-Server-Action): OnDemand scans for hosts are not natively supported.
- SSH scans: Native SSH scans are not supported.
- WMI scans: Native WMI scans are not supported.
- SNMP scans:Native SNMP scans are not supported.
- Nmap scans: Native Nmap scans are not supported.
- Context Server Action (CSA) scans: On Demand context server actions are not allowed.

3. When Device Insight integration is enabled, the audit server is disabled. At **Configuration > Services**, any service that includes an audit server configuration must be disabled or have its audit server configuration removed before Device Insight Integration is enabled (see Configuring Services Using Service Templates & Wizards and Configuring Other [%=aruba_style_guide.ShortPrdNm2%] Services).

Changes in the ClearPass User Interface When Device Insight Integration is Enabled

About this task

When Device Insight Integration is enabled, several new options are added in the Policy Manager user interface and a number of standard features are removed:

Procedure

1. In the Policy Manager Dashboard, the lists of device categories in the *Device Category* and *Device Family* widgets are enhanced to include the variety of device types that Device Insight can identify (for example, medical devices, barcode scanners, network cameras, and more). Each widget also includes a link to the Device Insight application.
2. At **Administration > Server Manager > Server Configuration > NetEvents Targets**, a new NetEvents target for the Device Insight primary server is added on all appliances in the cluster.

3. When Device Insight Integration is enabled, links directly to the Device Insight application are available on the Device Insight Integration page; in the *Device Category*, *Device Family*, and *Device Insight Tags* widgets in the Dashboard; and in the Policy Manager Menu in the upper-right corner of every Policy Manager page.

4. Policy Manager Device Insight tags are exposed wherever endpoint attributes are exposed in ClearPass service rules. They are available in service, role-mapping, and enforcement policy Rules Editors.

5. On the **Configuration > Identity > Endpoints > Edit Endpoint > Endpoint** tab, if tags are available for an endpoint received from Device Insight, a *Device Insight Tags* field is added to the form to include them.

6. At **Configuration > Identity > Endpoints**, you can use the filter **Attribute: Device Insight Tags** to search for devices.

   The following changes are applicable to Policy Manager 6.10.0 and 6.10.1 only. Starting with Policy Manager 6.10.2, enabling Device Insight does not disable additional user interface elements.

7. The **Monitoring > Profiler and Network Scan** section is removed from Policy Manager, along with all three pages under it: **Endpoint Profiler**, **Network Scan Results**, and **Discovered Devices**. These device scans and the configuration options for them are available in the Device Insight application instead.

8. The **Configuration > Network Scan** page is removed.

9. At **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters**, the **Profiler** tab is removed, and is replaced by the **OnConnect** tab.

0. On the **Configuration > Network > Devices > Add > SNMP Read Settings** tab, the **Read ARP Table Info** and **Force Read** options are removed.

1. When Device Insight Integration is enabled, information collected by Policy Manager Insight is sent to Device Insight for analysis. Because some widgets and reports are then available in Device Insight instead, the corresponding dashboard widgets and report templates in Insight are removed.

2. On the **Configuration > Enforcement > Profiles > Attributes** tab for an HTTP action, values match the trigger server actions that are listed for endpoints, and disabled values are not listed.

3. At **Administration > Dictionaries > Context Server Actions**, the **On Demand Endpoint Scan Generic HTTP-type** action is removed.

Device Insight Integration Page

The **Device Insight Integration** page enables Policy Manager to integrate with Policy Manager Device Insight, a cloud-hosted application for comprehensive device visibility. When Device Insight is enabled, Policy Manager disables device profiling on the current Policy Manager server and passes the raw preprofiled information to the Device Insight-enabled node. This change occurs across all the Profiler-enabled nodes (see Changes in the ClearPass User Interface When Device Insight Integration is Enabled). A number of other Policy Manager features are removed from the user interface when Device Insight Integration is enabled. For details, see Changes in ClearPass Behaviors When Device Insight Integration is Enabled.

- Enabling Device Insight Integration
- Disabling Device Insight Integration

Enabling Device Insight Integration

About this task

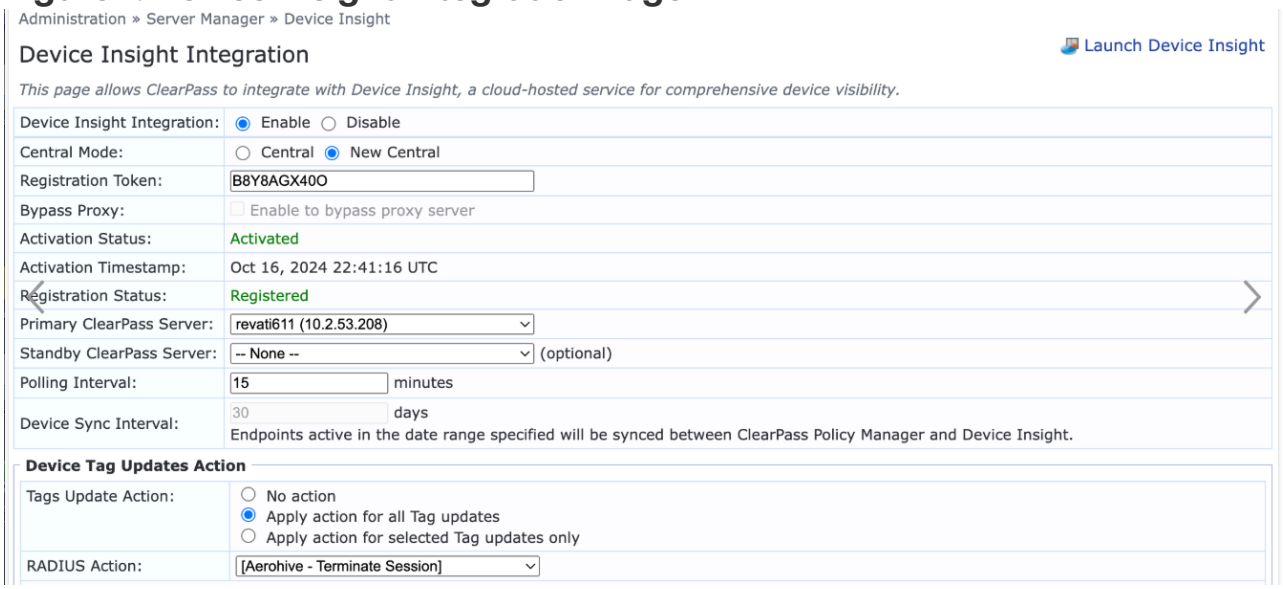
To enable Policy Manager Device Insight Integration:

Procedure

1. Complete the prerequisites described in Prerequisites to Using Device Insight Integration.
2. On the publisher, navigate to **Administration > Server Manager > Device Insight**.

   The **Device Insight Integration** page opens:

Figure 1. Device Insight Integration Page



3. Click **Enable**.

4. Specify the **Device Insight Integration** parameters.

Table 1. Device Insight Integration Page Parameters

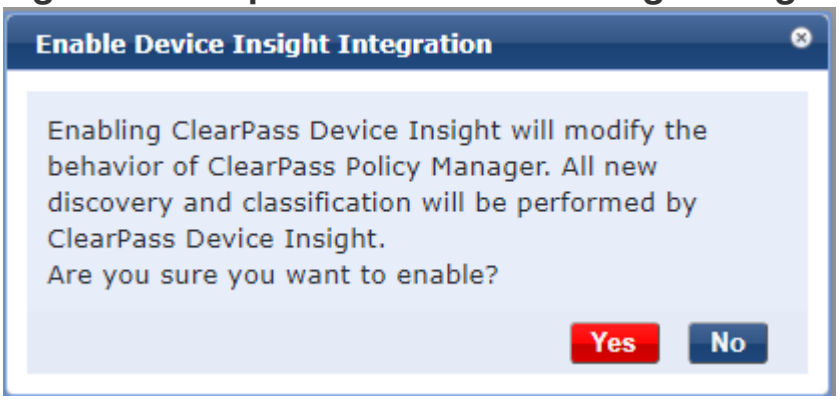
Parameter	Action/Description
Device Insight Integration	To enable Device Insight I **e** radio button. This feature is disabled by
Central Mode	If central - communicatio he Policy Manager CLI, se **Central** to define whether tworking Central interface ing New Central interface anager integration mode. e Policy Manager CLI to e ion, go to Configure Cent
Registration Token	To initiate the Activation p **on Token** that was genera Insight was deployed.
Activation Status	This is a read - only field. abled, this field displays t part of the Activation proc nabled node is provisione • If provisioning is succe status *SUCCESS.* • If an incorrect or an ex his field displays the st
Primary ClearPass Server	Select the ClearPass Polic nce designated as the pri If the Policy Manager insta e, this field is already pop er, you can select any app er publisher or subscriber anager server.

Parameter	Action/Description
Standby ClearPass Server	Use this option to designa t Integration - enabled se ng a standby is optional; h d. If the primary server is rver will take over functio unication failures and pro he standby server should nd certificate provisionin ry.
Polling Interval	Specify the **Polling Interv** utes to **2000** minutes (33 fault value is **15** minutes. The **Polling Interval** is a c serves as a backup in the becomes unavailable. If there is no communicat cy Manager and Device In nterval, Policy Manager p devices, classification det s.
Device Sync interval	Devices active within the ied in this date range will Pass Policy Manager and I ed range is 0 - 999 days, days.
Device Tag Action Updates	If there is no RADIUS or D es configured in a service **g Action Updates** options ge of authorization (CoA) following options: • **No action**—Actions are er's endpoint tables ar • **Apply action for all Ta** ion is selected, you ca drop - down menu an

Parameter	Action/Description
	will be applied to all ta ny endpoint, the selec action will be triggere • **Apply action for selec** en this option is select the RADIUS action usi , you can use the **Tags** cify a specific set of ta ction. The selected Co will only be triggered f the **Add** drop - down r ove a tag, select a tag field and click **Remove**

5. Click **Save**.The following prompt appears:

Figure 2. Prompt to Enable Device Insight Integration



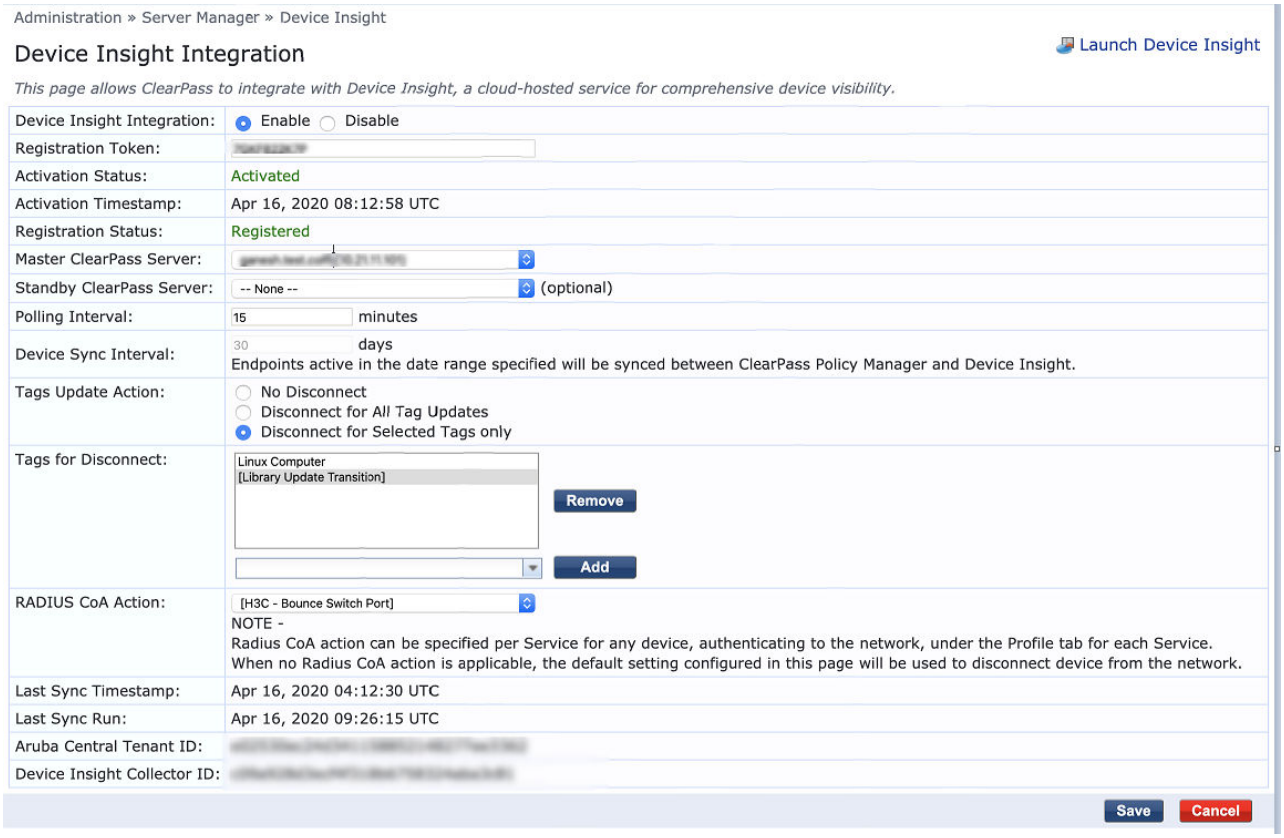
6. To proceed with enabling **Device Insight Integration**, click **Yes**; to cancel the operation, click **No**.
7. If activation is successful, the **Activation Status** field displays the status **Success**, and the form expands to include following read-only fields.

Table 2. Device Insight Integration Page Parameters

Parameter	Action/Description
Activation Timestamp	This is a read - only field that appears aft er successful activation. This field is auto

Parameter	Action/Description
	matically populated to indicate the time at which Device Insight was activated.
Registration Status	This is a read - only field that appears aft er successful activation. If a Websocket c onnection is successfully initiated betwe en the Device Insight - enabled node and Aruba Central, the registration appears a s *SUCCESS*.
Last Sync Timestamp	This is a read - only field that appears aft er successful activation. It indicates the t ime at which the last endpoint added to Policy Manager from Device Insight durin g synchronization. This field is automatic ally populated.
Last Sync Run	Time that the last sync operation started between Policy Manager and Device Insi ght.
Aruba Central Tenant ID	This is a read - only field that appears aft er successful activation. It indicates the *Aruba Central Tenant ID*.

Figure 3. Device Insight Activation Enabled



Disabling Device Insight Integration

About this task

Use the following procedure to disable **Device Insight Integration**,

> **NOTE**
> Be aware that if you disable Device Insight Integration at any point after it has been enabled, the Device Insight Registration Token and certificate are not retained. To enable integration again, you will need to enter the Registration Token again.

Procedure

1. Navigate to **Administration > Server Manager > Device Insight**. The **Device Insight Integration** page opens.
2. On the **Device Insight Integration** option, select **Disable**, then click **Save**. The following prompt is displayed.

Figure 4. Prompt to Disable Device Insight Integration

3. To proceed with disabling **Device Insight Integration**, click **Yes**; to cancel the operation, click **No**.

   Device Insight Integration is set to **Disabled**.

Launching Device Insight

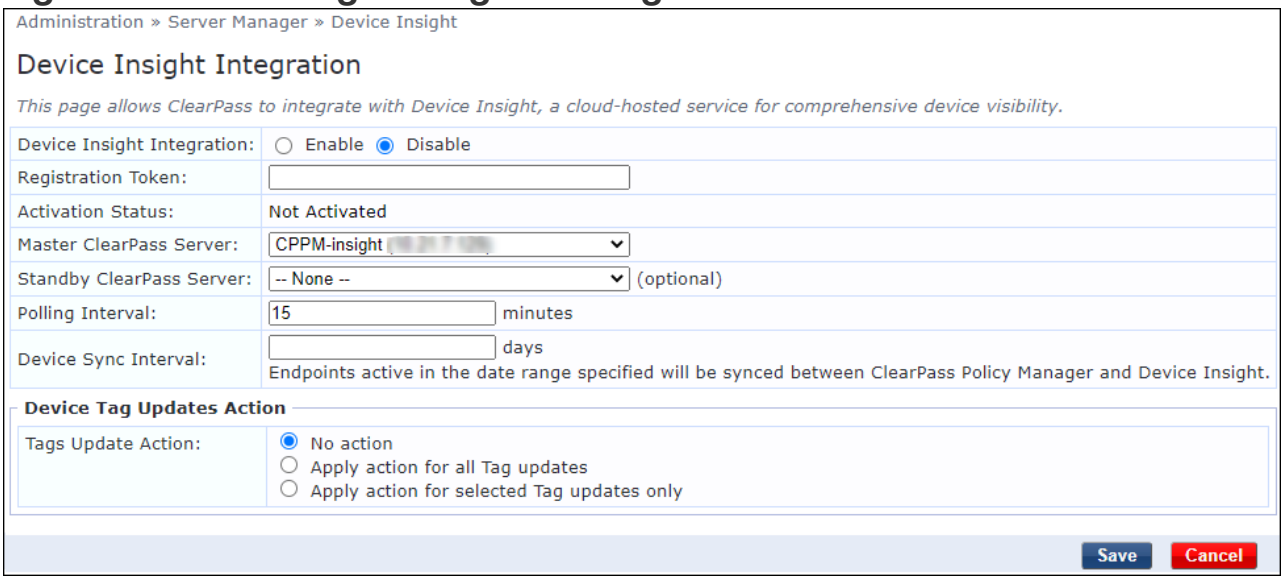
About this task

To launch Device Insight:

Procedure

1. Navigate to **Administration > Server Manager > Device Insight**. The **Device Insight Integration** page opens:

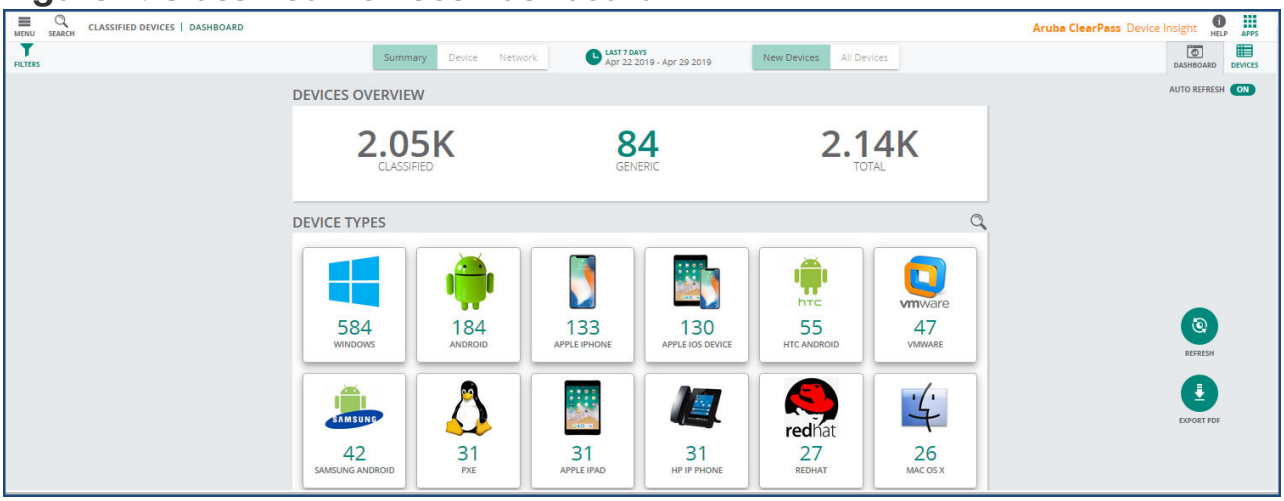
Figure 1. Device Insight Integration Page



2. If you have not already done so, configure the **Device Insight Integration** parameters as described in the previous section (see Device Insight Integration Page).

3. Click the **Launch Device Insight** link. The **Classified Devices Dashboard** opens:

Figure 2. Classified Devices Dashboard



External Servers

This section provides the following information:

- SNMP Trap Receivers
- Syslog Targets
- Syslog Export Filters
- Messaging Setup
- Endpoint Context Servers
- File Backup Servers

Subtopics

SNMP Trap Receivers
Syslog Targets
Syslog targets are servers that receive exported session, audit, and event data from ClearPass Policy Manager.
Syslog Export Filters
Messaging Setup
Endpoint Context Servers
File Backup Servers

SNMP Trap Receivers

- SNMP Trap Receivers Main Page
- Adding an SNMP Trap Server
- Importing an SNMP Trap Server
- Exporting All SNMP Trap Servers
- Exporting an SNMP Trap Server
- Deleting an SNMP Trap Server

Policy Manager sends SNMP traps that expose the following server information:

- **System up-time**: Provides information about how long the Policy Manager server has been running.
- **Network interface statistics [up/down]**: Provides information about whether the network interface is up or down.
- **Process monitoring information**: Checks for the processes that should be running, including maximum and minimum number of allowed instances. Sends traps if there is a change in value of the maximum and minimum numbers.
- **Disk usage**: Checks for disk space usage of a partition. The agent can check the amount of available disk space and make sure it's above the set limit. The value can be in percentage as well. Sends traps if there is a change in the value.
- **CPU load information**: Checks for unreasonable load average values. For example, if CPU load average for one minute exceeds the configured value (in percentage), the Policy Manager server sends a trap to the configured destination.
- **Memory usage**: Reports the Policy Manager server's memory usage.

SNMP Trap Receivers Main Page

To view a list of SNMP trap receivers configured on the ClearPass Policy Manager server, navigate to **Administration > External Servers > SNMP Trap Receivers**.

The following figure displays the **SNMP Trap Receivers** page:

Figure 1. SNMP Trap Receivers Page

For information about the Policy Manager SNMP Private MIB, see ClearPass SNMP Private MIB .

Adding an SNMP Trap Server

About this task

A trap is an SNMP message sent from one application to another (which is typically on a remote host). For SNMP trap server configuration, Policy Manager provides the **Type** parameter to specify whether the SNMP notification is a standard **Trap** notification or an **Inform** notification. An **Inform** notification is an acknowledged SNMP trap. When you send an **Inform** notification, Policy Manager uses an **SNMP Engine ID** when sending the message. The Engine ID is a unique identifier for the SNMP v3 agent. The engine ID is used with a hashing function to generate keys for authentication and encryption of SNMP v3 messages. The Engine ID is automatically generated when you enable the stand-alone SNMP agent. The default value for the SNMP Engine ID is *6620000004030662*. This value can be changed in the **Engine ID** field configured in the Policy Manager **Server Configuration > System Monitoring** page (for details, see System Monitoring Page).

NOTE

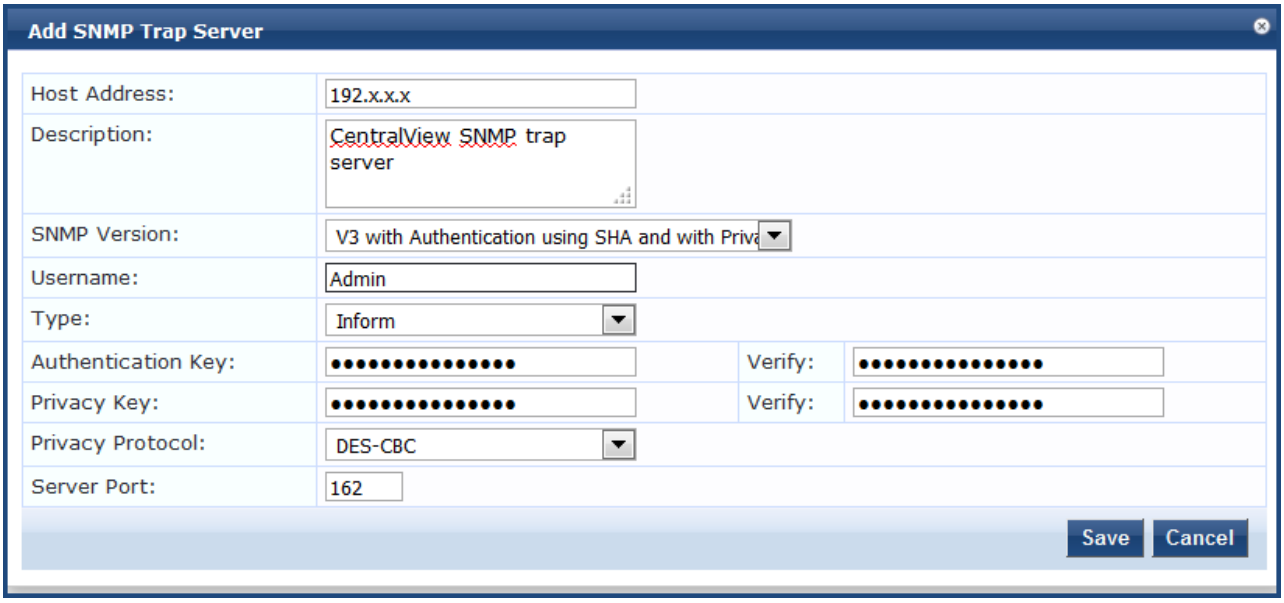
To receive traps, the same Engine ID value must be configured on the trap receiver side.

To add an SNMP trap server:

Procedure

1. Navigate to **Administration > External Servers > SNMP Trap Receivers**. The **SNMP Trap Receivers** page opens.
2. Click the **Add** link. The **Add SNMP Trap Server** dialog opens.

Figure 2. Add SNMP Trap Server Dialog



3. Specify the **Add SNMP Trap Server** parameters as described in the following table, then click **Save**:

Table 1. Add SNMP Trap Server Parameters

Parameter	Action/Description
Host Address	Enter the trap destination host address, formatted as a hostname, IP address or d omain name. **NOTE** This server must have an S NMP trap receiver or trap vi ewer installed.
Description	Enter a short description of the SNMP tra p server.
SNMP Version	Select one of the following SNMP version s: • SNMP v1 with community strings • SNMP v2 with community strings • SNMP v3 with no Authentication

Parameter	Action/Description
	• SNMP v3 with Authentication using MD5 and no Privacy • SNMP v3 with Authentication using MD5 and with Privacy • SNMP v3 with Authentication using SHA and no Privacy • SNMP v3 with Authentication using SHA and with Privacy **NOTE** The MD5 authentication type is not supported when you use Policy Manager in **FIPS** mode.
Username	Specify the Admin user name for SNMP operations. **NOTE** This parameter is available in SNMP v3 only.
Type	From the **Type** drop - down, select the type of SNMP notification: • Inform • Trap
Authentication Key	Specify the SNMP v3 with authentication option (SHA or MD5). **NOTE** The **EAP - MD5** authentication type is not supported if

Parameter	Action/Description
	you run Policy Manager in FIPS mode. **NOTE** Authentication Key is available in SNMP v3 only.
Privacy Key	Specify the SNMP v3 with privacy option. **NOTE** This parameter is available in SNMP v3 only.
Privacy Protocol	Choose one of the available privacy protocols: • DES - CBC • AES - 128 **NOTE** This parameter is available in SNMP v3 with Privacy only. Privacy allows for encryption of SNMP v3 messages to ensure confidentiality of data.
Server Port	Specify the port number for sending the traps. By default, the port number is **162**. **NOTE** Configure the trap server firewall for traffic on this port.

When SNMP Trap Server changes are saved, the system restarts the cpass-async-netd service to initialize the new trap receiver. As a result, all modules depending on the Asysnc Net demon service are momentarily severed.

For example, an OnGuard module experiences the following:

- The existing control channel connection from OnGuard Agent to ClearPass Server is momentarily severed.
- The OnGuard Agent detects this loss of connectivity, often resulting in a transient disconnect or a re-triggering of the health-check cycle for all active clients.

To avoid this situation, HPE Aruba Networking recommends performing SNMP trap add/updates during scheduled maintenance.

Importing an SNMP Trap Server

About this task

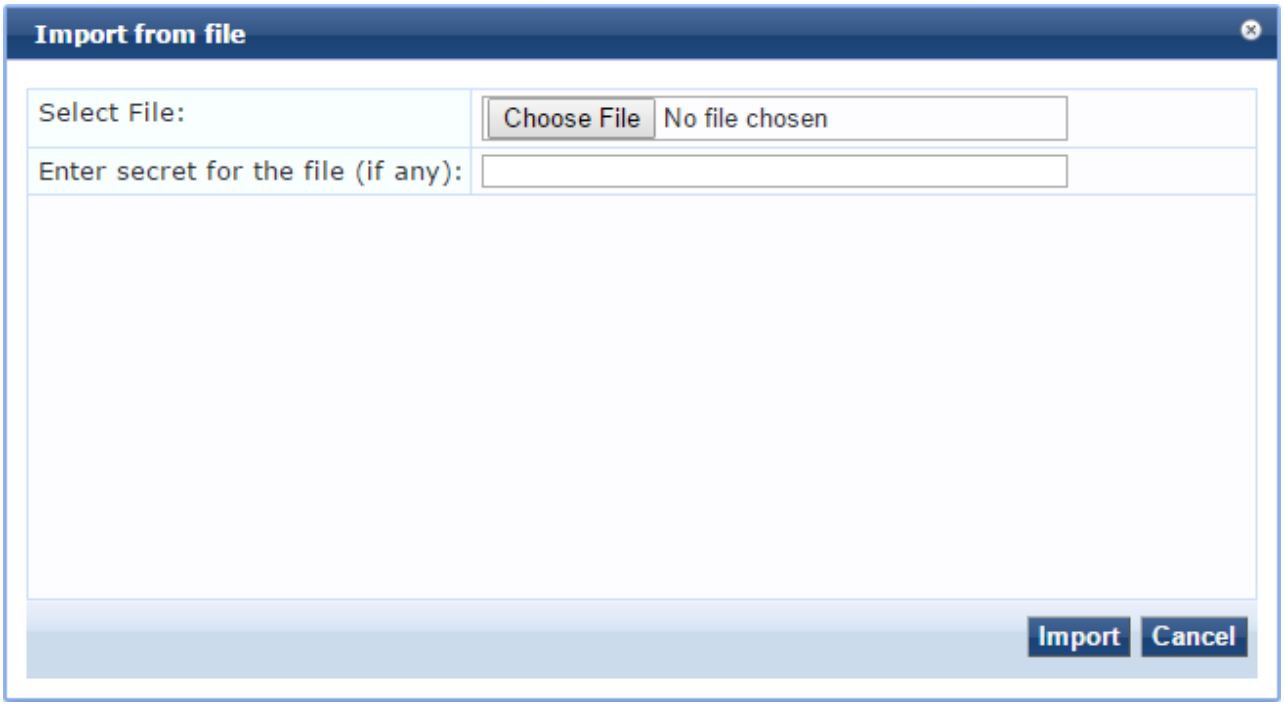
To import an SNMP trap server:

Procedure

1. Navigate to **Administration > External Servers > SNMP Trap Receivers**.
2. Click the **Import** link on the top right section of the page.
3. Click **Import**.

   The following figure displays the **Import from file** pop-up:

Figure 3. Import from file Pop-up



The following table describes the **Import from file** parameters:

Table 2. Import from file parameters

Parameter	Description
Select File	Browse to the SNMP Trap Server configuration file to be imported.
Enter secret for the file (if any)	If the file was exported with a secret key for encryption, enter the secret key here.

When SNMP Trap Server changes are saved, the system restarts the cpass-async-netd service to initialize the new trap receiver. As a result, all modules depending on the Asysnc Net demon service are momentarily severed.

For example, an OnGuard module experiences the following:

- The existing control channel connection from OnGuard Agent to ClearPass Server is momentarily severed.
- The OnGuard Agent detects this loss of connectivity, often resulting in a transient disconnect or a re-triggering of the health-check cycle for all active clients.

To avoid this situation, HPE Aruba Networking recommends performing SNMP trap add/updates during scheduled maintenance.

Exporting All SNMP Trap Servers

About this task

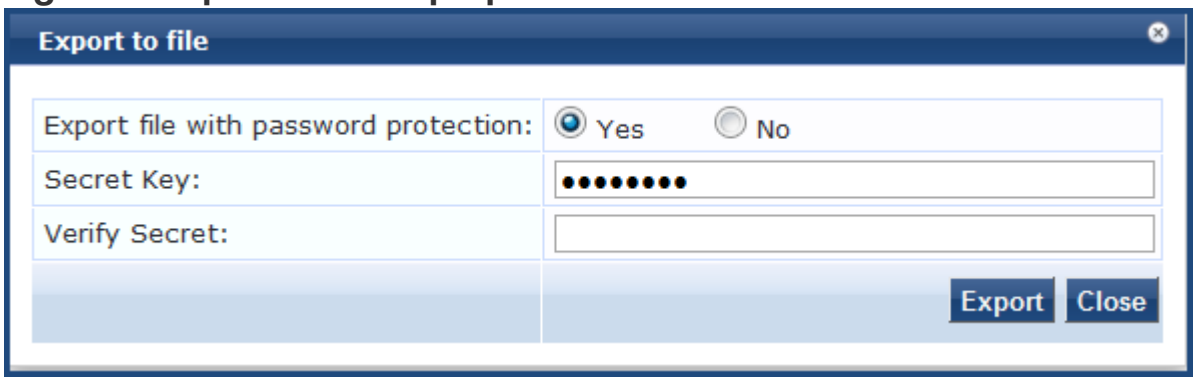
This link exports all configured SNMP Trap Receivers. To export all SNMP trap servers:

Procedure

1. Navigate to **Administration > External Servers > SNMP Trap Receivers**.
2. Click the **Export All** link on the top right section of the page.
3. Click **Export**.
4. Enter the XML file name in the **Save As** dialog box.
5. Click **Save**.

   The following figure displays the **Export to file** pop-up:

Figure 4. Export to file Pop-up



The following table describes the **Export to file** parameters:

Table 3. Export to file Parameters

Parameter	Description
Export file with password protection	Choose **Yes** to export the file with pas sword protection.
Secret Key	Enter the secret key.
Verify Secret	Re - enter the secret key.

Exporting an SNMP Trap Server

About this task

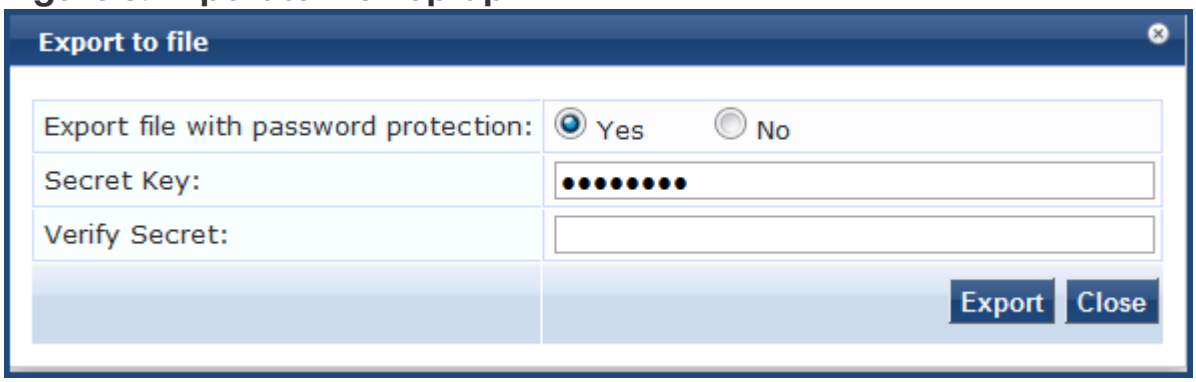
To export a single SNMP trap server:

Procedure

1. Navigate to **Administration > External Servers > SNMP Trap Receivers**.
2. Select the **Host Address** from the list of check boxes and click **Export**. Enter the details based on Table 1.
3. Enter the name of the XML file in the **Save As** dialog.
4. Click **Save**.

   The following figure displays the **Export to file** pop-up:

Figure 5. Export to file Pop-up



The following table describes the **Export to file** parameters:

Table 4. Export to file Parameters

Parameter	Description
Export file with password protection	Choose **Yes** to export the file with pas sword protection.
Secret Key	Enter the secret key.
Verify Secret	Re - enter the secret key.

Deleting an SNMP Trap Server

About this task

To delete a single SNMP trap server:

Procedure

1. Navigate to **Administration > External Servers > SNMP Trap Receivers**.
2. Click the check box next to the **Host Address** entry and click **Delete**.
3. Click **Yes**.

Syslog Targets

Syslog targets are servers that receive exported session, audit, and event data from ClearPass Policy Manager.

ClearPass Policy Manager can export session data (see Live Monitoring: Access Tracker), audit records (see Audit Viewer), and event records (see Event Viewer). You can send this information to one or more syslog targets (servers).

Adding a Syslog Target

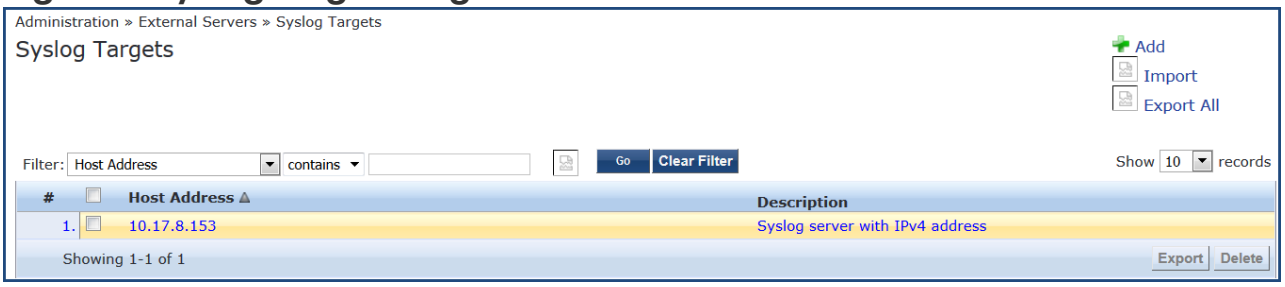
About this task

To add a syslog target:

Procedure

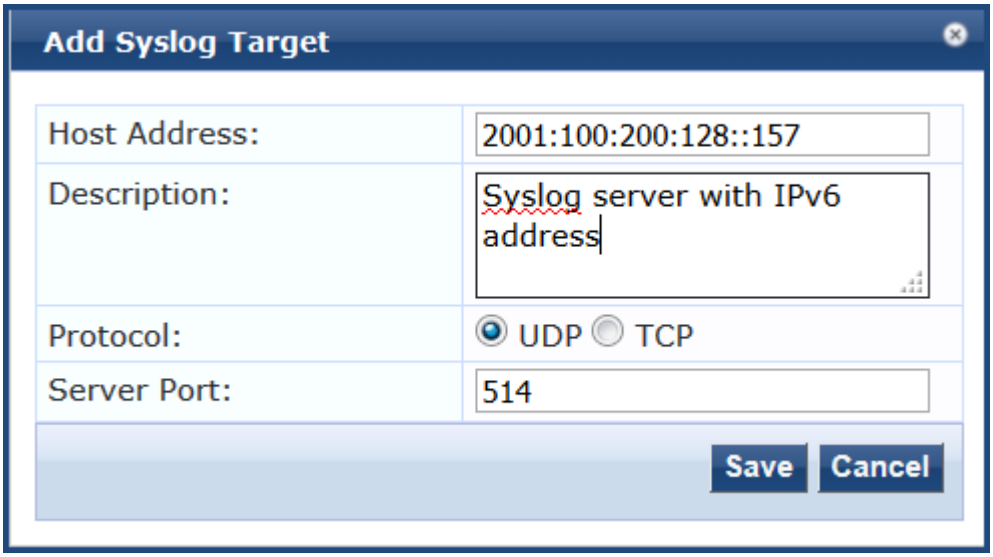
1. Navigate to **Administration > External Servers > Syslog Targets**. The **Syslog Targets** page opens.

Figure 1. Syslog Targets Page



2. Click the **Add** link. The **Add Syslog Target** dialog opens:

Figure 2. Add Syslog Target Dialog



3. Specify the **Add Syslog Target** parameters as described in the following table:

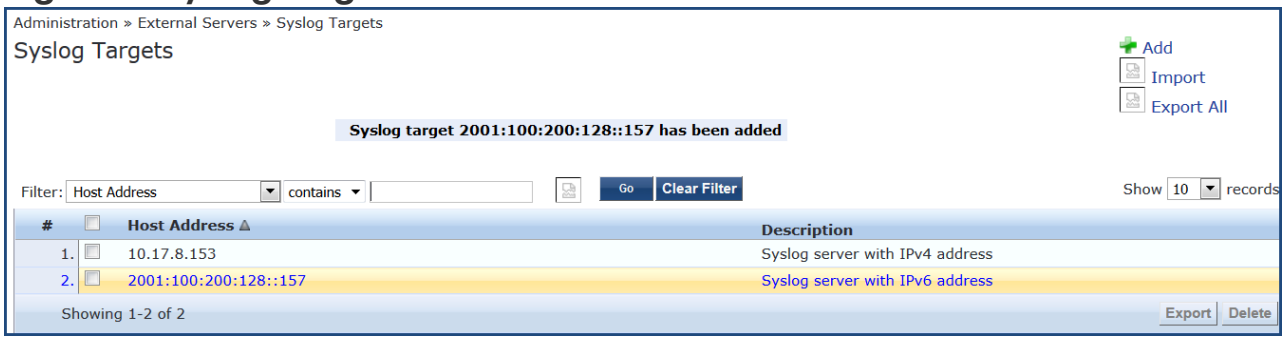
Table 1. Add Syslog Target Parameters

Parameter	Action/Description
Host Address	Enter the Syslog server hostname or IP a ddress. The IP address can be an IPv4 or IPv6 ad dress.
Description	Enter a short description of the Syslog se rver.
Protocol	Select one of the following options: • **UDP**: This option reduces overhead a nd latency. • **TCP**: This option provides error checki ng and packet delivery validation. • **TLS encrypted** This option encodes al l connections to the Syslog target ove r TLS using the RFC 5425 protocol/sta ndard. The TLS encrypted option is o nly applicable to TCP ports (not DTLS for UDP ports).
Server Port	The default port number is **514**.

Parameter	Action/Description
	If necessary, change the server port number.

4. Click **Save**. The new Syslog Target is now added to the list:

Figure 3. Syslog Target Added



Syslog Export Filters

- About Syslog Export Filters
- Syslog Export Filters Page
- Adding a Syslog Export Filter
- Importing a Syslog Filter
- Exporting All Syslog Filter
- Exporting a Syslog Filter
- Deleting a Syslog Filter

About Syslog Export Filters

Policy Manager can export session data (see Live Monitoring: Access Tracker), audit records (see Audit Viewer), and event records (see Event Viewer). You configure syslog export filters to instruct Policy Manager where to send this information and what kind of information should be sent through data filters.

Syslog messages are sent with the DEBUG log level from ClearPass. For example:

2015-01-20 16:50:05,210 [pool-6-thread-1] [R:] **DEBUG** com.avenda.tips.syslog.Syslogger - 2017-01-20 16:50:05,210 10.17.5.228 System Events 2 1 0 TimestampFormat=yyyy-MM-

dd HH:mm:ss,S,Description=Performed action stop on cpass-domain-server_CPATS,Category=stop,Action=Success,Level=INFO,src=10.17.5.228,Component=cpa domain-server_CPATS,Timestamp=Jan 20, 2017 16:48:57 IST

2015-01-20 16:50:05,211 [pool-6-thread-1] [R:] **DEBUG** com.avenda.tips.syslog.Syslogger - 2017-01-20 16:50:05,211 10.17.5.228 System Events 3 1 0 TimestampFormat=yyyy-MM-dd HH:mm:ss,S,Description=Performed action start on cpass-domain-server_CPATS,Category=start,Action=Success,Level=INFO,src=10.17.5.228,Component=cpa domain-server_CPATS,Timestamp=Jan 20, 2017 16:49:00 IST

Syslog Export Filters Page

The Syslog Export Filters page enables administrators to view, manage, and configure syslog export filters and related parameters.

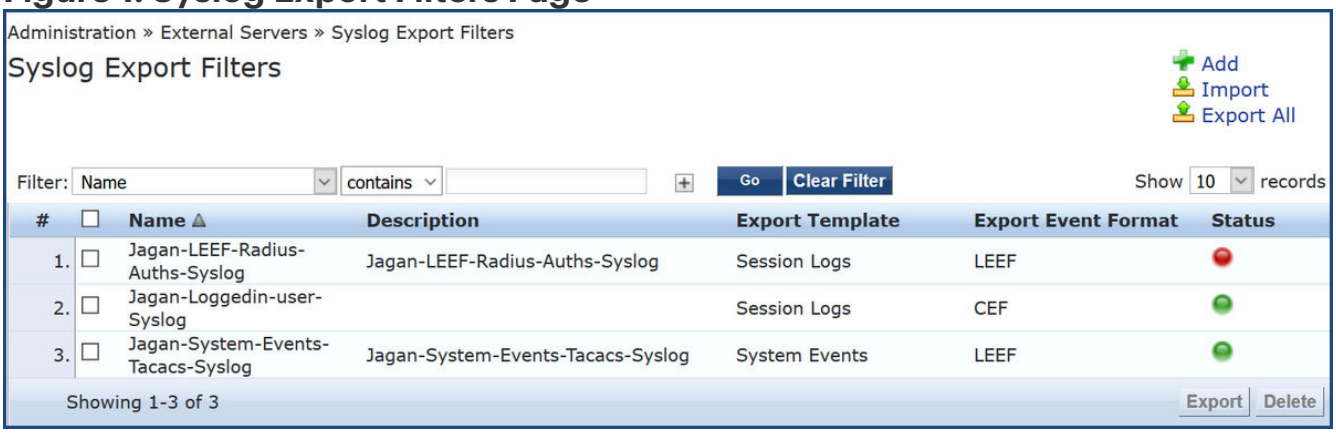
About this task

To configure syslog export filters:

Procedure

Navigate to **Administration > External Servers > Syslog Export Filters**. The **Syslog Export Filters** page opens.

Figure 1. Syslog Export Filters Page



The following table describes the **Syslog Export Filters** parameters:

Table 1. Syslog Export Filters Page Parameters

Parameter	Action/Description
Name	Displays the name of the syslog export filter.
Description	Displays the description of the syslog export filter.

Parameter	Action/Description
Export Template	Displays the name of the **Export Template** selected in the **Add Syslog Export Filter** di alog (see Adding a Syslog Export Filter).
Export Event Format	Displays the **Export Event Format Type** se lected in the **Add Syslog Export Filter** dial og; for example, LEEF (Log Event Extended Format) and CEF (Common Event Format).
Status	• A **red** icon indicates that the syslog exp ort filter is currently disabled. To enable a disabled filter, click the red icon. • A **green** icon indicates that the syslog export filter is currently enabled. To dis able an enabled filter, click the green ic on.
Export	To export a syslog export filter, select the c heck box for the appropriate filter, then clic k **Export**. The **Export to file** dialog opens. For more i nformation, see Importing and Exporting In formation.
Delete	To delete a syslog export filter, select the c heck box for the appropriate filter, then clic k **Delete**.

Importing a Syslog Filter

About this task

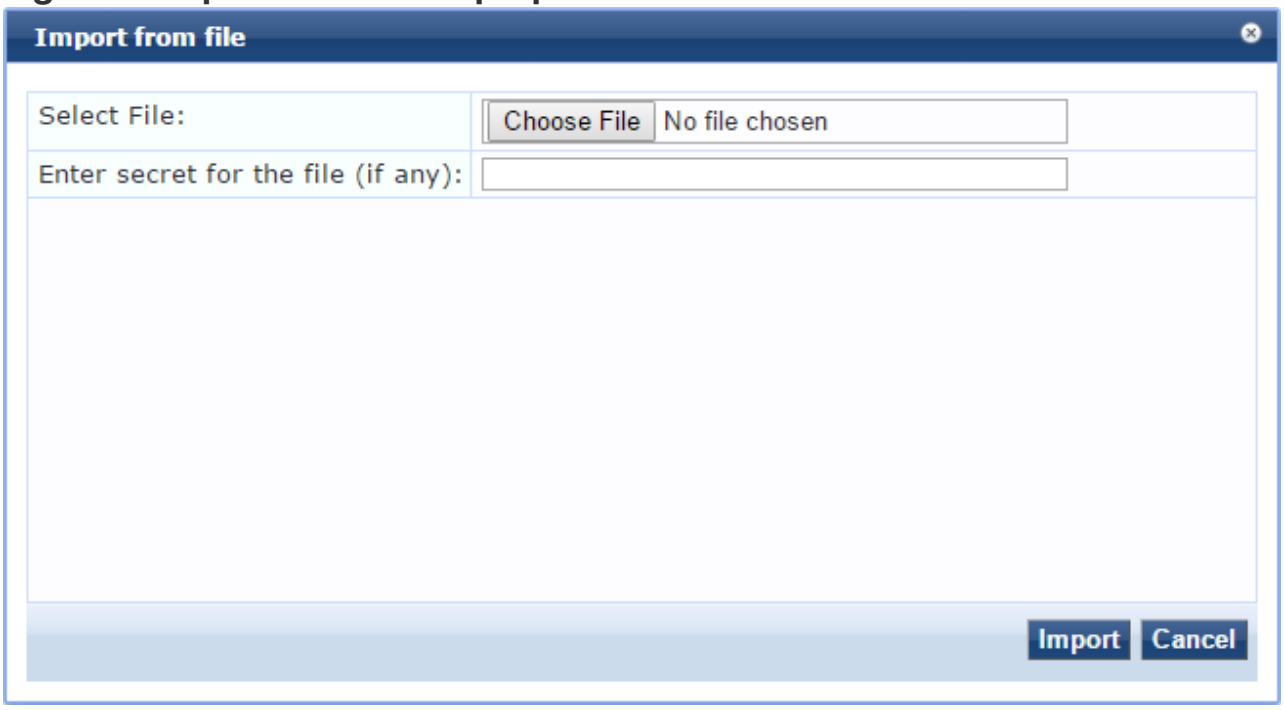
To import a syslog target:

Procedure

1. Navigate to **Administration > External Servers > Syslog Export Filters**.
2. Click the **Import** link on the top right section of the page. Enter the details based on Table 1.
3. Click **Import**.

   The following figure displays the **Import from file** pop-up:

Figure 2. Import from file Pop-up



The following table describes the **Import from file** parameters:

Table 2. Import from file Parameters

Parameter	Description
Select File	Browse to the Syslog Filter configuration file to be imported.
Enter secret for the file (if any)	If the file was exported with a secret key for encryption, enter the same key here.

Exporting All Syslog Filter

About this task

To export all syslog filters:

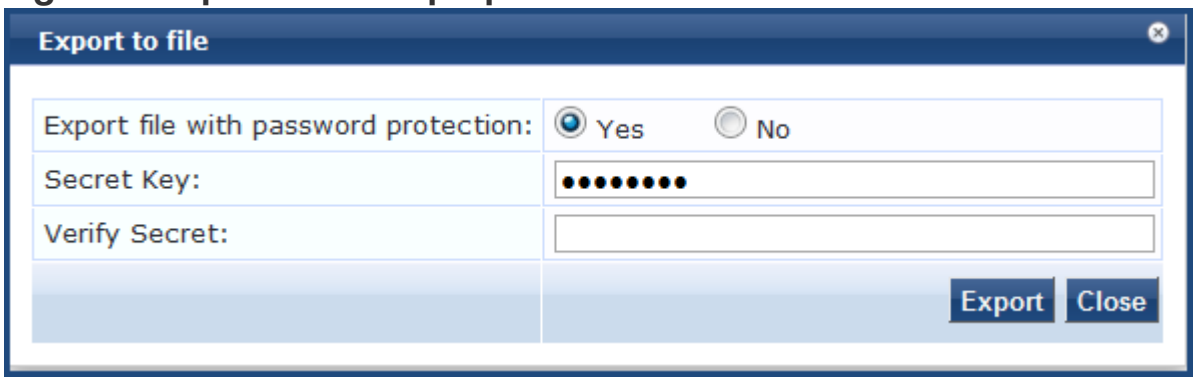
Procedure

1. Navigate to **Administration > External Servers > Syslog Export Filters**.
2. Click the **Export All** link on the top right section of the page. Enter the details based on Table 1.
3. Click **Export**.

4. Enter the XML file name in the **Save As** dialog box.
5. Click **Save**.

   The following figure displays the **Export to file** pop-up:

Figure 3. Export to file Pop-up



The following table describes the **Export to file** parameters:

Table 3. Export to file Parameters

Parameter	Description
Export file with password protection	Choose **Yes** to export the file with pas sword protection.
Secret Key	Enter the secret key.
Verify Secret	Re - enter the secret key.

Exporting a Syslog Filter

About this task

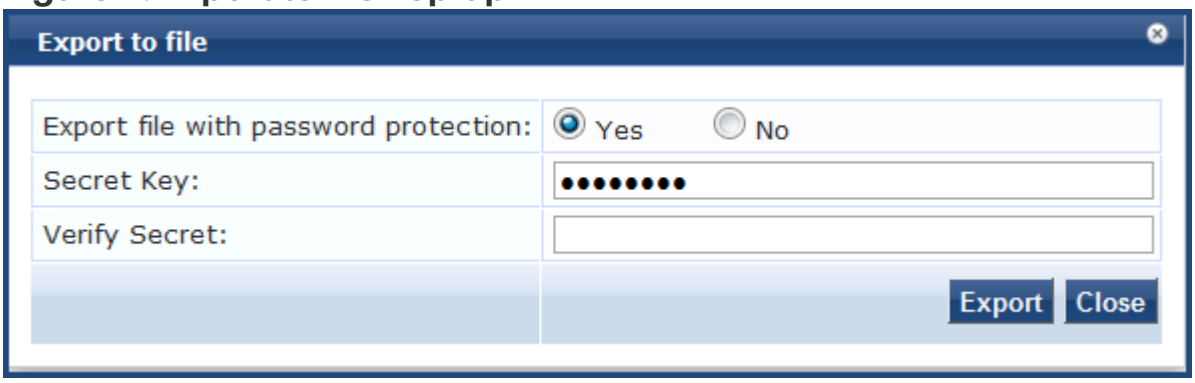
To export a syslog filter:

Procedure

1. Navigate to **Administration > External Servers > Syslog Export Filters**.
2. Select the **Host Address** from the list of check boxes and click **Export**. Enter the details based on Table 1.
3. Enter the name of the XML file in the **Save As** dialog.
4. Click **Save**.

The following figure displays the **Export to file** pop-up:

Figure 4. Export to file Pop-up



The following table describes the **Export to file** parameters:

Table 4. Export to file Parameters

Parameter	Description
Export file with password protection	Choose **Yes** to export the file with pas sword protection.
Secret Key	Enter the secret key.
Verify Secret	Re - enter the secret key.

Deleting a Syslog Filter

About this task

To delete a syslog filter:

Procedure

1. Navigate to **Administration > External Servers > Syslog Export Filters**.
2. Click the check box next to the syslog filter entry and click **Delete**.
3. Click **Yes**.

Messaging Setup

- Configuring Messaging

- Sending a Test Email Message
- Sending a Test SMS Message
- Resetting the SMTP Settings

Policy Manager messaging setup provides an interface to configure the Simple Mail Transfer Protocol (SMTP) server for email and SMS notifications.

Configuring Messaging

About this task

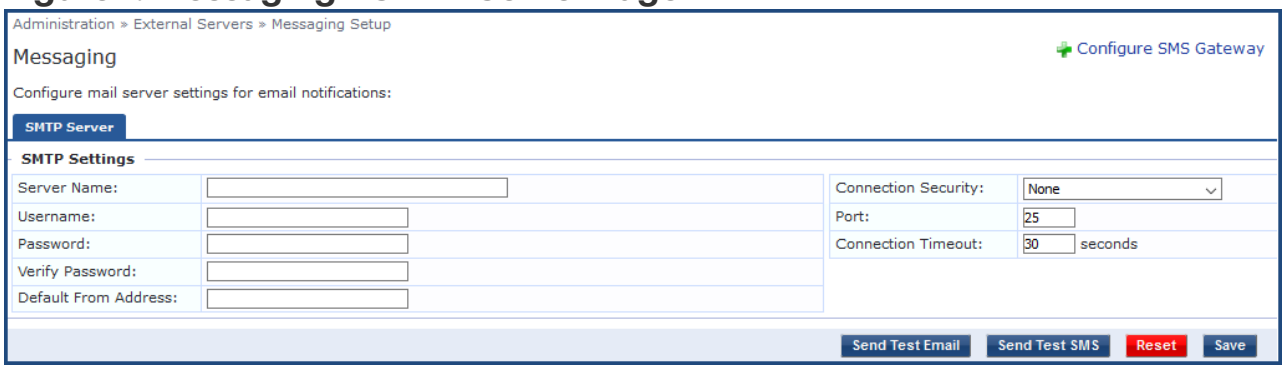
To configure messaging:

Procedure

1. Navigate to **Administration > External Servers > Messaging Setup**.

   The **Messaging > SMTP Server** page opens.

Figure 1. Messaging > SMTP Server Page



2. To configure a new SMS gateway using the Policy Manager Guest portal, click the **Configure SMS Gateway** link at the top right section of the page.

   The following table describes the **Messaging > SMTP Server** page parameters:

Table 1. Messaging > SMTP Server Page Parameters

Parameter	Action/Description
Server name	Enter the Fully Qualified Domain Name (FQDN) or the IP address of the SMTP se rver.
User Name	Enter the username if your email server requires authentication for sending email messages.

Parameter	Action/Description
Password	Enter the password for the specified user name, then verify the password.
Default From address	Enter the email address that must to be d isplayed as the sender's address in the m essage.
Connection Security	To establish the communication with the SMTP server, select from one of the foll owing options: • **None:** Select this option to disable se cure communication with the server. • **SSL**: Select this option to have a Secu red Socket Layer communication with the server. • **Start TLS**: Select this option to have a Transport Layer Security communicati on with the server.
Port	Enter the TCP port number that the SNM P server listens on. The default value of the port is **25**.
Connection timeout	Enter the timeout value for connection to the SMTP server (in seconds). The default value is **30** seconds.

Sending a Test Email Message

About this task

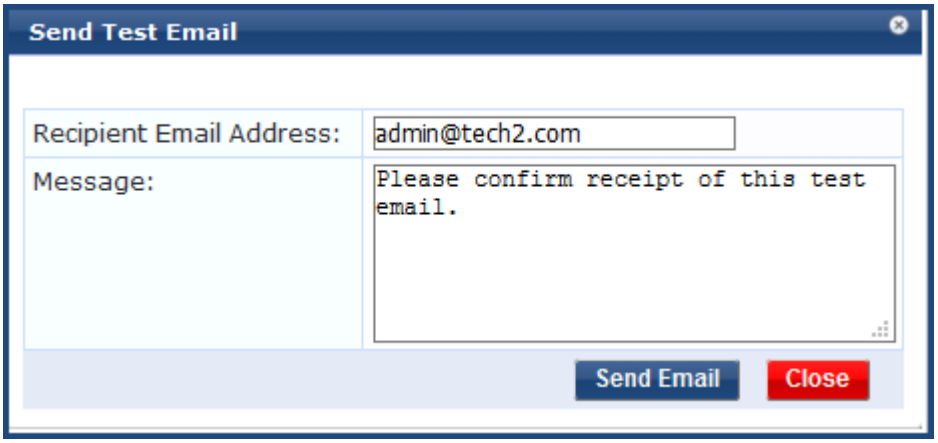
To send a test mail message to the preferred email address:

Procedure

1. Click **Send Test Email**.

   The **Send Test Email** dialog opens.

Figure 2. Send Test Email Dialog



2. **Recipient Email Address**: Enter the email address of the recipient.
3. **Message**: Enter the test message.
4. Click **Send Email**.

Sending a Test SMS Message

About this task

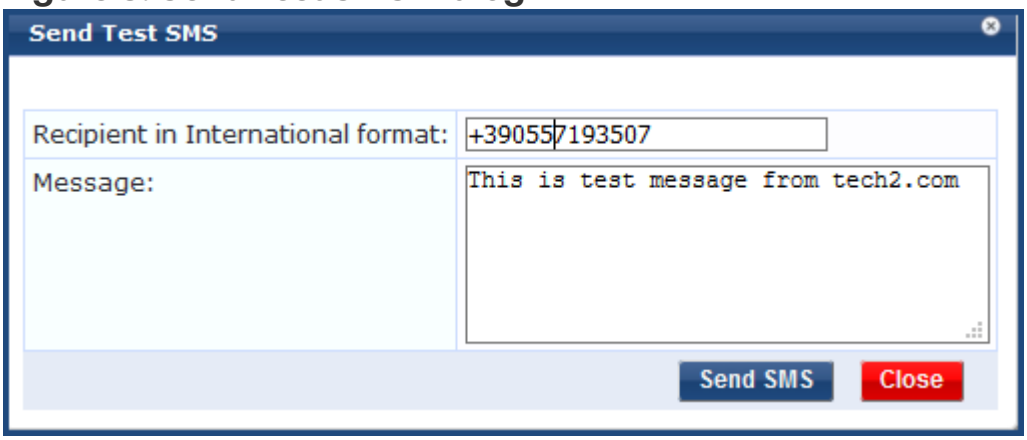
To send a test SMS message to the preferred email address:

Procedure

1. Click **Send Test SMS**.

   The **Send Test SMS** dialog opens.

Figure 3. Send Test SMS Dialog



2. **Recipient in International format**: Enter the mobile phone number of the recipient in international format.

The recipient's mobile number must be entered in the international format consisting of a **+** sign, followed by the country code and the mobile phone number (without the first '0' of the number).

3. **Message**: Enter the test message.
4. Click **Send SMS**.

Resetting the SMTP Settings

To remove the current SMTP settings specified in the **Messaging** dialog, click **Reset**.

Endpoint Context Servers

- Introduction
- Endpoint Context Servers Page
- Adding an Endpoint Context Server
- Importing an Endpoint Context Server
- Exporting All Endpoint Context Servers
- Modifying an Endpoint Context Server
- Polling an Endpoint Context Server
- Deleting an Endpoint Context Server
- Configuring Endpoint Context Server Actions
- Adding Vendor-Specific Endpoint Context Servers
- Endpoint Information Collectors

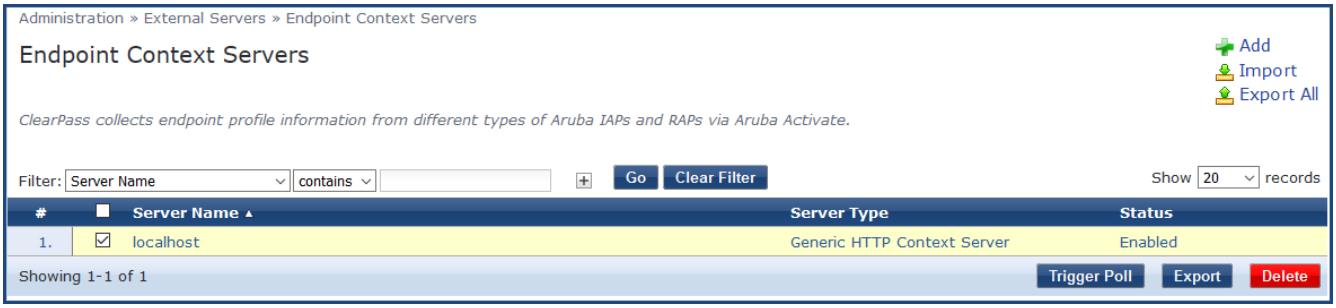
Endpoint Context Servers Page

Procedure

To access the Endpoint Context Servers page, navigate to **Administration > External Servers > Endpoint Context Servers**.

The **Endpoint Context Servers** page opens:

Figure 1. Endpoint Context Servers Page



The following table describes the **Endpoint Context Servers** categories:

Table 1. Endpoint Context Server Categories

Parameter	Description
Server Name	Displays the name of the endpoint context server.
Server Type	Displays the type of the endpoint context s erver.
Status	Displays the status of the endpoint context server: **Enabled** or **Disabled**. For non - MD M servers, the status is always displayed as **Disabled**.

Importing an Endpoint Context Server

About this task

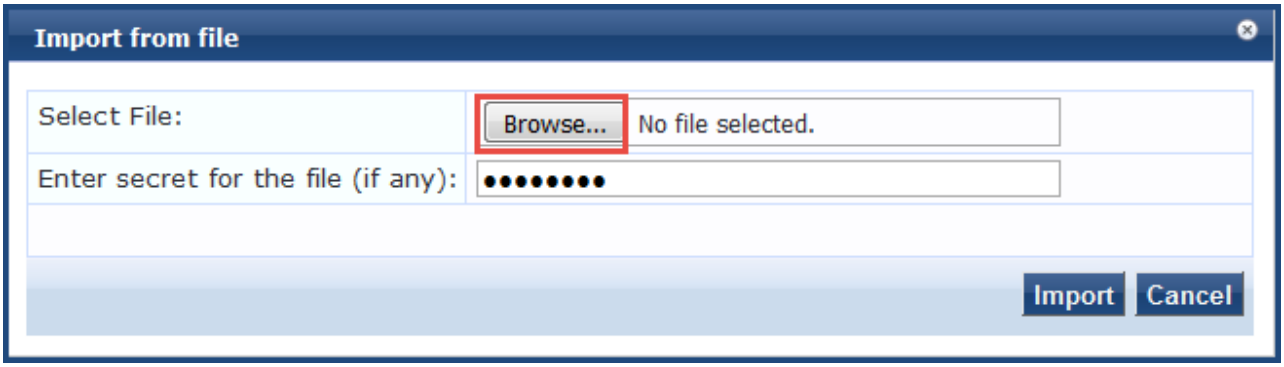
To import an endpoint context server:

Procedure

1. Navigate to **Administration > External Servers > Endpoint Context Servers**.
2. Click the **Import** link on the top right section of the page.
3. Enter the parameters based on the table below.
4. Click **Import**.

   The following figure displays the **Import from File** dialog:

   Figure 2. Import from File Dialog



The following table describes the **Import from file** parameters:

Table 2. Import from File Dialog Parameters

Parameter	Description
Select File	Browse to the Endpoint Context Server c onfiguration file to be imported.
Enter secret for the file (if any)	If the file was exported with a secret key for encryption, enter the same key here.

Exporting All Endpoint Context Servers

About this task

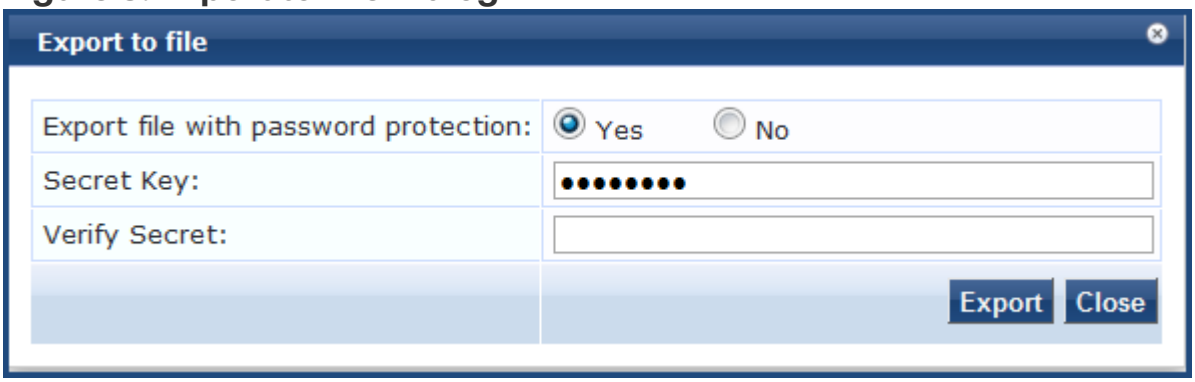
To export all endpoint context servers:

Procedure

1. Navigate to **Administration > External Servers > Endpoint Context Servers**.
2. Click the **Export All** link.

   The **Export to File** dialog opens.

Figure 3. Export to File Dialog



3. Enter the parameters as described in Table 4.
4. Click **Export**.
5. Enter the XML file name in the **Save As** dialog box.
6. Click **Save**.

   The following table describes the **Export to file** parameters:

Table 3. Export to File Dialog Parameters

Parameter	Action/Description
Export file with password protection	To export the file with password protection, choose **Yes**.
Secret Key	Enter the secret key, then verify it.

Polling an Endpoint Context Server

About this task

To poll an endpoint context server:

> **NOTE**
> You can poll only one server at a time. You cannot poll multiple server entries. Also, you can only poll MDM-type servers.

Procedure

1. Navigate to **Administration > External Servers > Endpoint Context Servers**.
2. In the **Endpoint Context Servers** page, click the check box next to the server name entry.

Figure 4. Selecting the Trigger Poll Option

3. Click **Trigger Poll**.

Deleting an Endpoint Context Server

About this task

Deleting an endpoint context server removes the configuration information from the Policy Manager server.

To save the endpoint context server configuration prior to deleting the server:

Procedure

1. Before you delete the endpoint context server, export the server.
2. Save the configuration so that you can import it in future if necessary.

   To delete an endpoint context server:

3. Navigate to **Administration > External Servers > Endpoint Context Servers**.
4. Select the check box next to the server name entry, then click **Delete**.
5. To confirm the delete operation, click **Yes**.

File Backup Servers

About this task

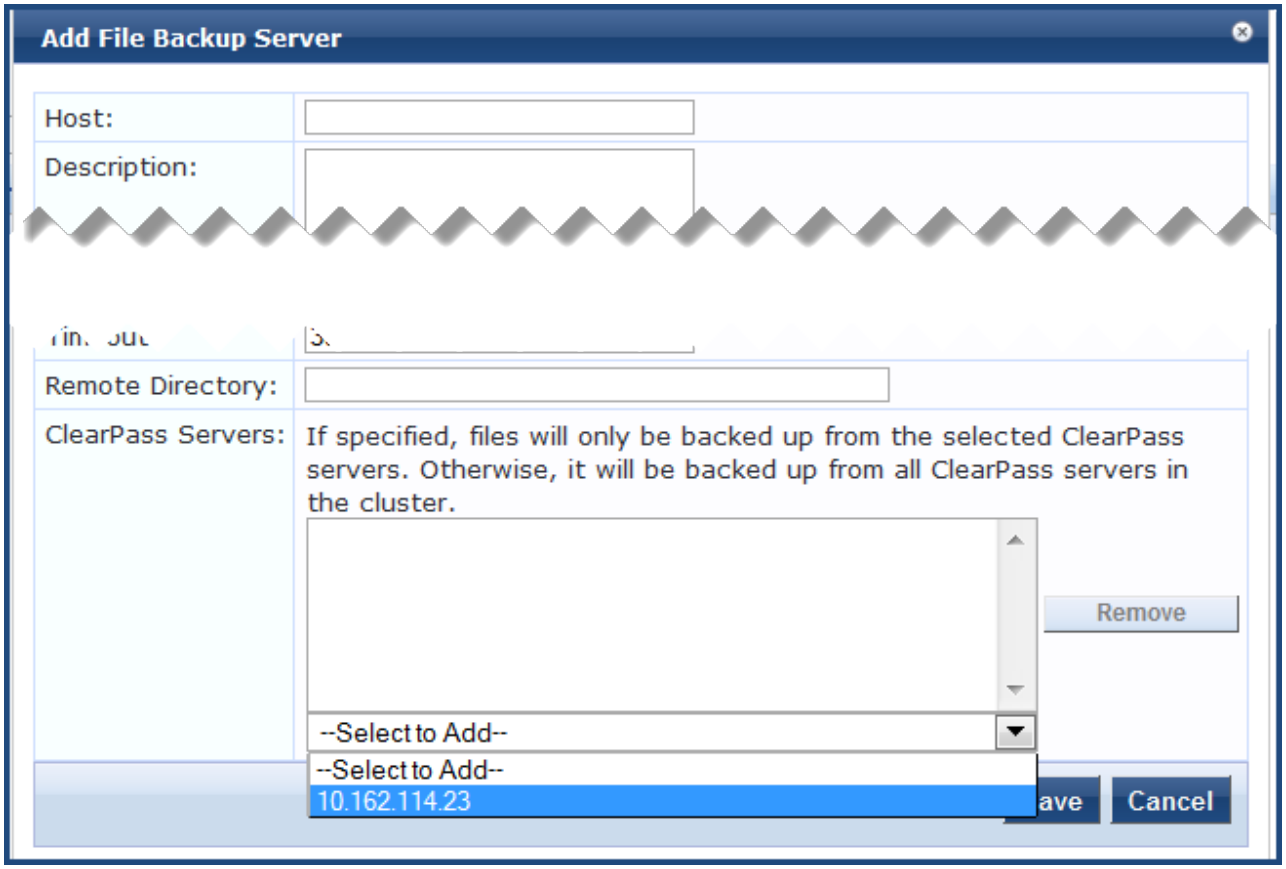
ClearPass Policy Manager provides the ability to push scheduled data securely to a File Backup server. You can push the data to the server using the SFTP (SSH File Transfer Protocol), SCP (Session Control Protocol), or NFS (Network File System) protocols. To add a File Backup Server:

> **NOTE**
> To automatically create a backup directory for a SCP configuration, a Windows path should use a backslash character (\) when the path is entered.

Procedure

1. Navigate to the **Administration > External Servers > File Backup Servers** page.
2. Click the **Add** link. The **Add File Backup Server** page opens.

Figure 1. Add File Backup Server Page



3. Specify the **Add File Backup Server** page parameters as described in the following table:

Table 1. Add File Backup Server Page Parameters

Parameter	Action/Description
Host	Enter the name or IP address of the host.
Description	Enter the description that provides additi onal information about the File Backup se rver.
Protocol	Specify the protocol to be used to uploa d the generated reports to a File Backup server. Select from the following protocol s: • SFTP (SSH File Transfer Protocol) • SCP (Session Control Protocol) • NFS 3 (Network File System version 3)

Parameter	Action/Description
	• NFS 4 (Network File System version 4) **NOTE**
Port	The default ports for each protocol are: • *SFTP*: port **22** • *SCP*: port **22** • *NFS*: port **2049**
Username Password	Enter the user name and password of the host server.
Timeout	Specify the timeout value in seconds. Th e default value is **30** seconds.
Remote Directory	Specify the location where the files are t o be copied. A folder will be automaticall y created in the file path that you specify based on the selected Policy Manager se rvers in the *ClearPass Servers* field.
ClearPass Servers	Specify the Policy Manager servers. If sp ecific servers are specified, files will only be backed up from the selected Policy M anager servers. Otherwise, it will be bac ked up from all ClearPass servers in the cl uster. You can select the servers from the **Select to Add** drop - down list.

4. Click **Save**.

Adding External Accounts

This section describes how to add multiple external accounts and use different external accounts for different subnets, zones, and so on. You can also configure the credentials to configure network scans and subnet scans for Windows devices (WMI), network devices (SNMP), and macOS or Linux devices (SSH). For related information, refer to Configuring Network and Subnet Scans.

Subtopics

Adding a Domain/WMI External Account
Configuring the Networks/Subnets for WMI Scan Type
Adding an SNMP External Account
Configuring the Networks/Subnets for SNMP Scan Type
Adding an SSH External Account
Configuring the Networks/Subnets for SSH Scan Type
Editing an External Account

Adding a Domain/WMI External Account

About this task

All the Domain/WMI accounts added here are available for Agentless OnGuard and subnet scans for Windows devices (WMI).

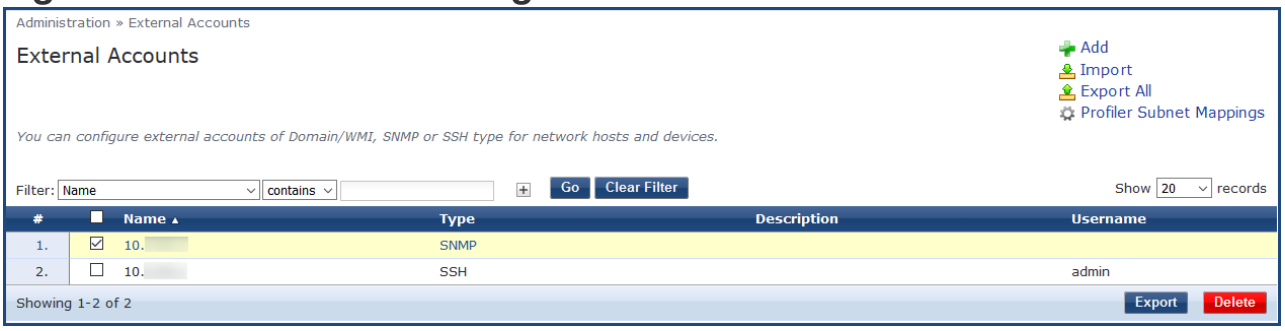
To add a Domain/WMI account:

Procedure

1. Navigate to **Administration > External Accounts**.

   The **External Accounts** page opens.

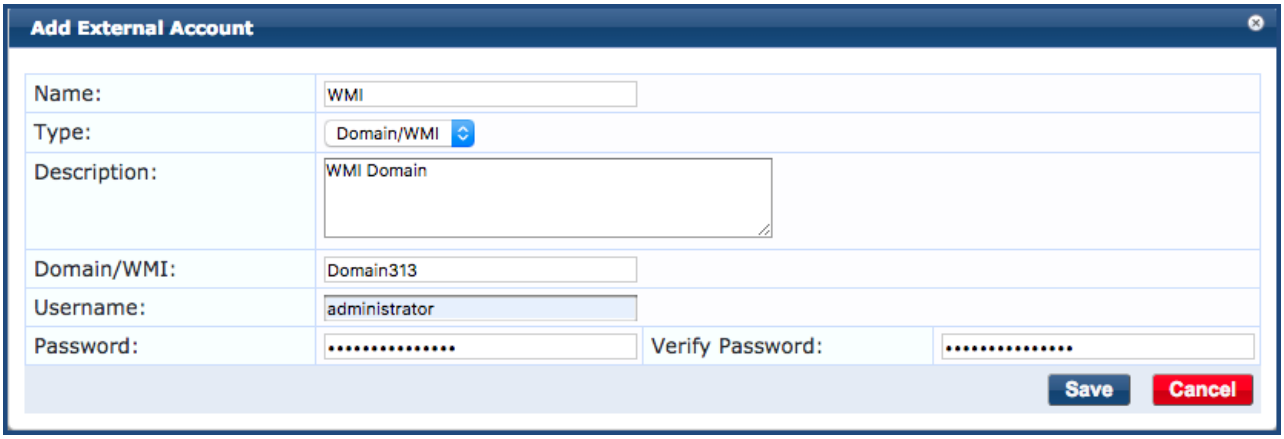
Figure 1. External Accounts Page



2. Click the **Add** link.

   The **Add External Account** dialog opens.

Figure 2. Adding a Domain/WMI Account



3. Specify the Domain/WMI external account parameters as described in the following table:

Table 1. Domain/WMI External Account Parameters

Parameter	Action/Description
Name	Enter a unique name to identify or select a domain account. This name will be disp layed in other Policy Manager configurati on pages to select the domain account.
Type	Select **Domain/WMI** (the default).
Description	Optionally, enter description and details of the domain account.
Domain/WMI	Enter the name of the Windows domain.
User Name	Enter the Windows domain user name for the domain - joined Windows client.
Password	Enter the Windows domain password; the n verify the password.

4. Click **Save**.

   You return to the **External Accounts** page, where the new Domain/WMI account is now listed.

Configuring the Networks/Subnets for WMI Scan Type

About this task

For Windows device discovery, specify WMI (Windows Management Instrumentation) credentials.

Procedure

1. From the **External Accounts** page, click the **Profiler Subnet Mappings** link, then click the **Add** link.

   The **Profiler Subnet Mappings** dialog opens.

Figure 1. Configuring the Subnet for the WMI Account/Scan Type

2. Specify the WMI Subnet Mapping parameters as described in the following table:

Table 1. Domain/WMI Subnet Mapping Parameters

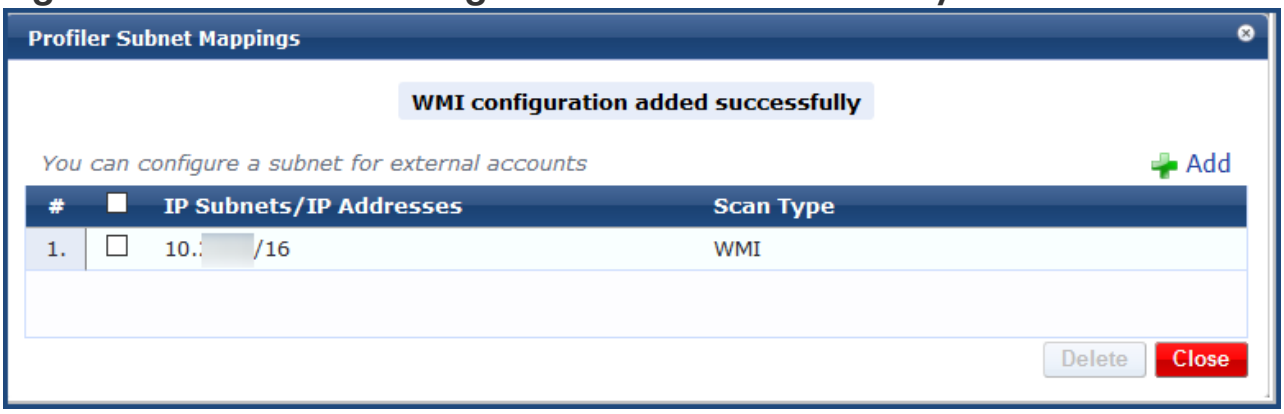
Parameter	Action/Description
IP Subnets/ IP Addresses	Enter either one or more IP subnets or on e or more IP addresses. For multiple entries, separate multiple IP addresses or subnets with commas. When you configure the network scan, Policy Manager will use the WMI configur ation to fetch the network device informa tion for discovered devices.

Parameter	Action/Description
Scan Type	From the **Scan Type** drop - down, select **WMI**.
External Accounts	Select the check boxes for the correspon ding Domain/WMI accounts.

3. Click **Save**.

   You return to the **Profiler Subnet Mappings** page, where the subnet for the WMI scan type has been added successfully.

Figure 2. WMI Subnet Configuration Added Successfully



4. Click **Add** to add another subnet mapping; or click **Close** to exit.
5. Click **Close**.

   You return to the External Accounts page, where the new WMI configuration has been added.

Adding an SNMP External Account

About this task

All the SNMP accounts added here are available for subnet scans for network devices (SNMP).

To add an SNMP external account:

Procedure

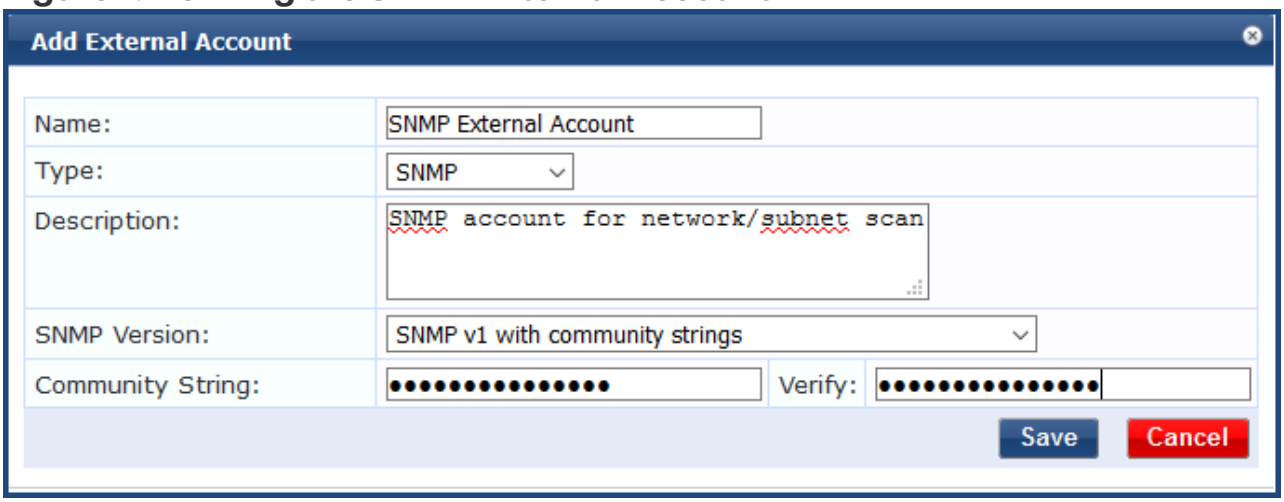
1. Navigate to **Administration > External Accounts**.

   The **External Accounts** page opens.

2. Click the **Add** link.

   The **Add External Account** dialog opens.

Figure 1. Defining the SNMP External Account



3. Specify the SNMP external account parameters as described in the following table:

Table 1. SNMP External Account Parameters

Parameter	Action/Description
Name	Enter a unique name to identify the SNMP account. This name will be displayed in other Policy Manager configuration pages to select the account.
Type	Select **SNMP**.
Description	Optionally, enter description and details of the account.
SNMP Version	From the drop - down, select the appropriate SNMP version: • SNMP v1 with community strings • SNMP v2 with community strings

Parameter	Action/Description
	• SNMP v3 with no Authentication • SNMP v3 with Authentication using M D5 and no privacy • SNMP v3 with Authentication using M D5 with privacy • SNMP v3 with Authentication using S HA and no privacy • SNMP v3 with Authentication using S HA with privacy
Community String	Enter the community string, then verify t he community string.

4. Click **Save**.

   You return to the **External Accounts** page, where the new SNMP account is now listed.

Configuring the Networks/Subnets for SNMP Scan Type

About this task

An SNMP description is necessary for discovering and profiling network devices.

Procedure

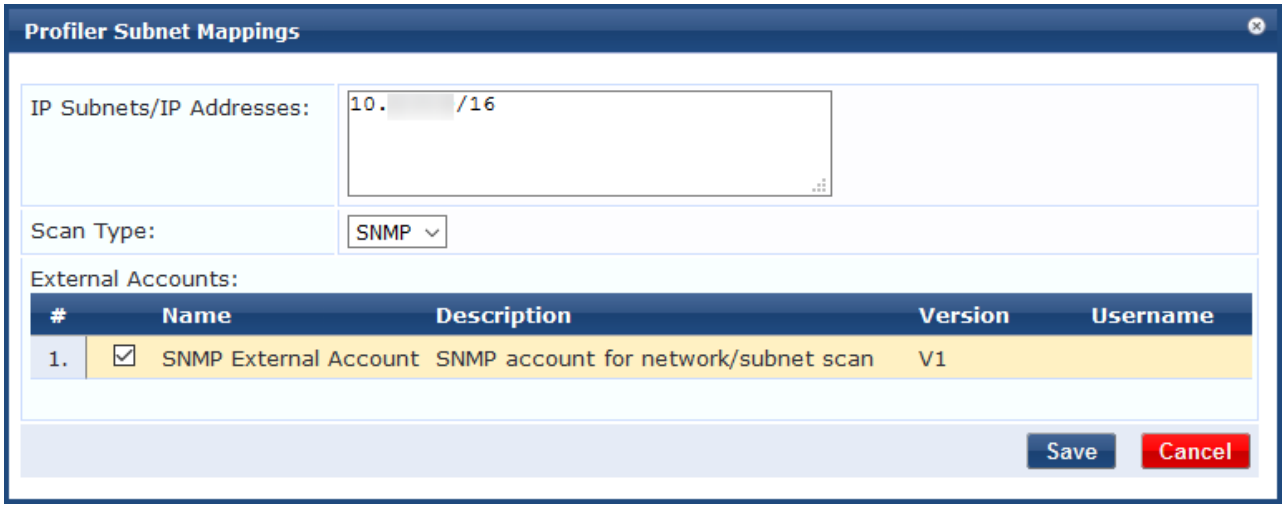
1. From the **External Accounts** page, click the **Profiler Subnet Mappings** link, then click **Add**.

   The **Profiler Subnet Mappings** page opens.

2. Click **Add**.

   The **Profiler Subnet Mappings** configuration dialog opens:

   Figure 1. Configuring SNMP Subnet Mapping



3. Specify the SNMP Subnet Mapping parameters as described in the following table:

Table 1. SNMP Subnet Mapping Parameters

Parameter	Action/Description
IP Subnets/ IP Addresses	Enter either one or more IP subnets or on e or more IP addresses. For multiple entries, separate multiple IP addresses or subnets with commas. When you configure the network scan, Policy Manager will use the SNMP config uration to fetch the network device infor mation for discovered devices.
Scan Type	From the **Scan Type** drop - down, select **SNMP**:
External Accounts	Select the check boxes for the correspon ding SNMP accounts.

4. Click **Save**.

   You return to the **Profiler Subnet Mappings** dialog, where the new SNMP configuration has been added.

5. Click **Add** to add another subnet mapping; or click **Close** to exit.

Adding an SSH External Account

An SSH account is necessary for discovering and profiling macOS and Linux devices. Note the following requirements for SSH accounts used to support Agentless OnGuard for posture analysis of MacOS and LInux clients.

- Linux and macOS endpoints must have a user account that Policy Manager will use to connect over SSH.
- SSH must be enabled on the Linux and macOS endpoints.
- Access to SSH port 22 should be allowed.

To add an SSH external account:

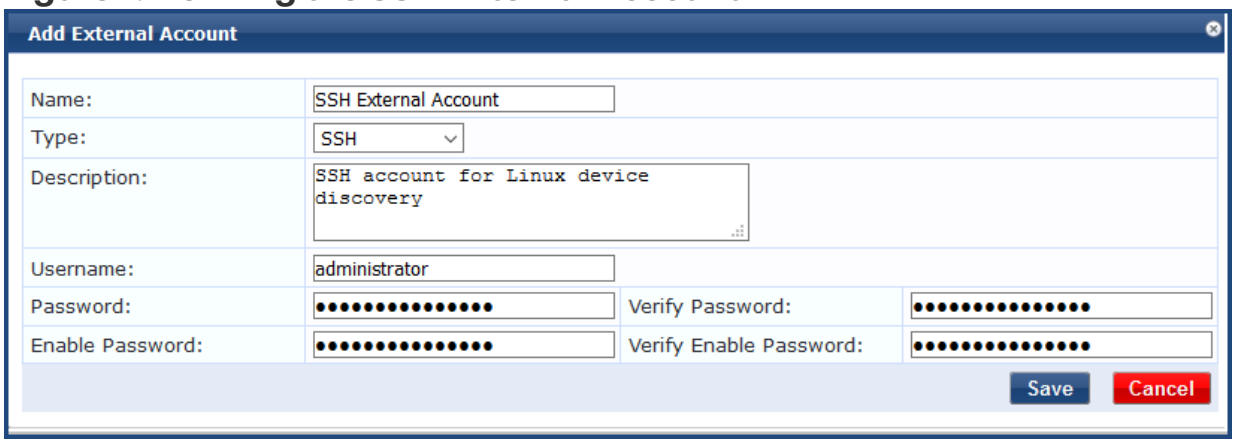
1. Navigate to **Administration > External Accounts**.

   The **External Accounts** page opens.

2. Click the **Add** link.

   The **Add External Account** dialog opens.

Figure 1. Defining the SSH External Account



3. Specify the SSH External Account parameters as described in the following table:

Table 1. SSH External Account Parameters

Parameter	Action/Description
Name	Enter a unique name to identify an SSH account. This name will be displayed in other Policy Manager configuration pag es to select the account.

Parameter	Action/Description
Type	Select **SSH**.
Description	Optionally, enter description and detail s of the SSH account.
Username	Enter the user name for the devices or s ervers on the network. Note that when creating an SSH extern al account for use with Agentless OnGu ard: ◦ The user account must be part of su doer group with NOPASSWD option set for sudo. ◦ The user account must have root pri vileges.
Password	Enter the password; then verify the pas sword.
Enable Password	Enter the Enable password; then verify t he Enable password.

4. Click **Save**.

   You return to the **External Accounts** page, where the new SSH account is now listed.

Configuring the Networks/Subnets for SSH Scan Type

About this task

For Linux servers or network device discovery, specify SSH configuration credentials.

Procedure

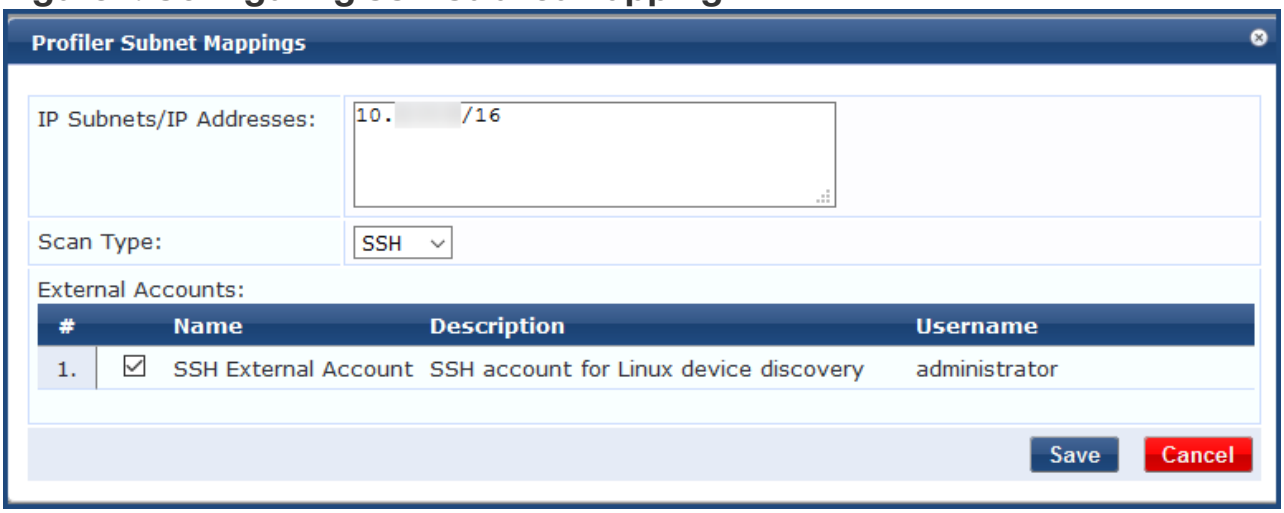
1. From the **External Accounts** page, click the **Profiler Subnet Mappings** link, then click **Add**.

   The **Profiler Subnet Mappings** page opens.

2. Click **Add**.

   The **Profiler Subnet Mappings** configuration dialog opens:

Figure 1. Configuring SSH Subnet Mapping



3. Specify the SSH Subnet Mapping parameters as described in the following table:

Table 1. SSH Subnet Mapping Parameters

Parameter	Action/Description
IP Subnets/ IP Addresses	Enter either one or more subnets or one or more IP addresses. For multiple entries, separate multiple IP addresses or subnets with commas. When you configure the network scan, Policy Manager will use the SSH configur ation to fetch the network device informa tion for discovered devices.
Scan Type	From the **Scan Type** drop - down, select **SSH**:
External Accounts	Select the check boxes for the correspon ding SSH accounts.

4. Click **Save**.

   You return to the **Profiler Subnet Mappings** dialog, where the new SSH configuration has been added.

5. Click **Add** to add another subnet mapping; or click **Close** to exit.

Editing an External Account

About this task

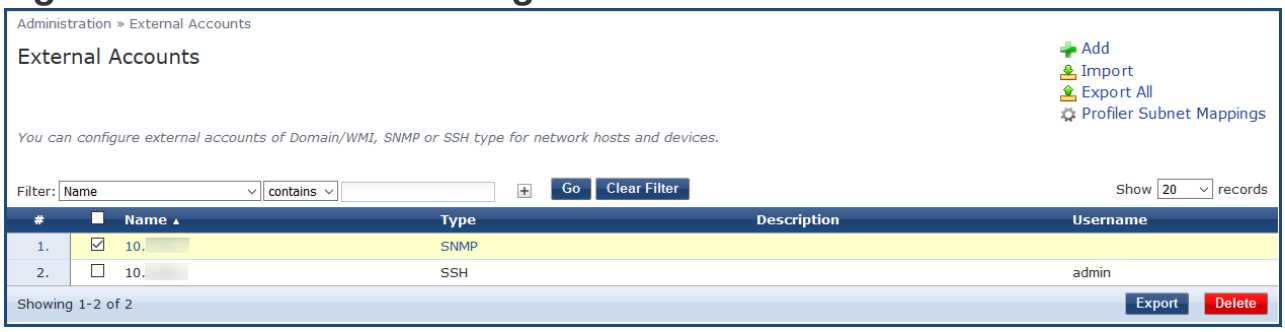
To modify an existing external account:

Procedure

1. Navigate to **Administration > External Accounts**.

   The **External Accounts** page opens.

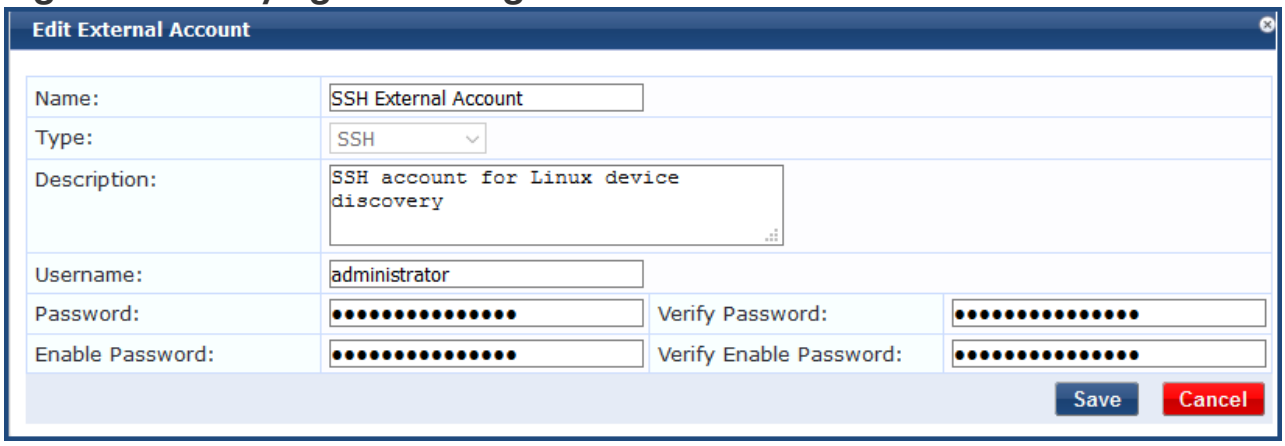
Figure 1. External Accounts Page



2. Click anywhere on the row for the external account you wish to modify.

   The **Edit External Account** configuration dialog opens.

Figure 2. Modifying an Existing External Account



3. Modify the parameters you wish to change, then click **Save**.

> **NOTE**
>
> The **Type** parameter is grayed out and is not editable.

Certificate Store

Server Certificates are small data files that digitally bind a cryptographic key to the details of an entity in order to ensure its authenticity, as well as the security and integrity of any connections with the entity's server. Policy Manager supports multiple EAP server certificates. You can tie different server certificates to different Policy Manager RADIUS-based services (for example, *Service A* can use *EAP server certificate A*, while *Service B* can use *EAP server certificate B*). The Certificate Store allows you to view the Server Certificates, create, modify, delete, and view Certificate Signing Requests (CSRs), as well as import and export CSRs. A *root certificate* is a public key certificate that identifies a root certificate authority (CA). A root certificate is the top-most certificate of the certificate tree structure.

The **Server Certificates** page displays the parameters configured when a self-signed certificate has been created and installed on a Policy Manager server. The **RADIUS/EAP Server Certificate** is selected by default.

The ClearPass Certificate Store provides five types of server certificates.

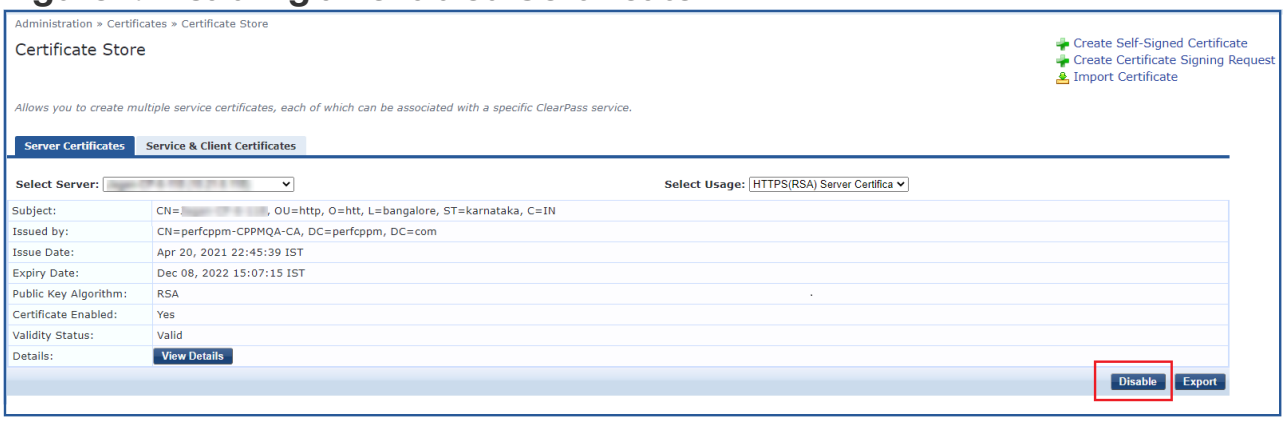
- RADIUS/EAP Server Certificates
- HTTPS (ECC) Server Certificates (HTTPS using Elliptic Curve Cryptography)
- HTTP(RSA) Server Certificates (HTTPS using RSA Cryptography)
- RadSec Server Certificates
- Database Server Certificates

  The availability of these certificate types (internally signed and publicly signed) provides deployment flexibility.

> **NOTE**
>
> Server Certificate expiration notices begin appearing in the Policy Manager user interface 30 days prior to the expiration date.

HTTPS(ECC) and HTTPS(RSA) certificates can be enabled or disabled by clicking the **Enable** or **Disable** buttons in the lower right corner of the **Certificates Store** window. A disabled certificate cannot be used, and you cannot disable both HTTPS(ECC) and HTTPS(RSA) simultaneously. If both HTTPS(ECC) and HTTPS(RSA) Certificates are enabled, any client that supports ECC ciphers will get HTTPS(ECC) certificates when contacting ClearPass. If you enable ECC certificates, the client trust lists should be updated accordingly.

Figure 1. Disabling an enabled Certificate



Subtopics

Viewing the Server Certificates
Creating and Installing a Self-Signed Server Certificate
Creating a Certificate Signing Request
Importing a Server Certificate
Exporting a Server Certificate
Server certificate exports let administrators save the current Policy Manager server certificate in PKCS#12 format for secure reuse.
Service and Client Certificates
Associating a RADIUS Service Certificate with a Service
Certificate Trust List
Certificate Revocation Lists

Viewing the Server Certificates

About this task

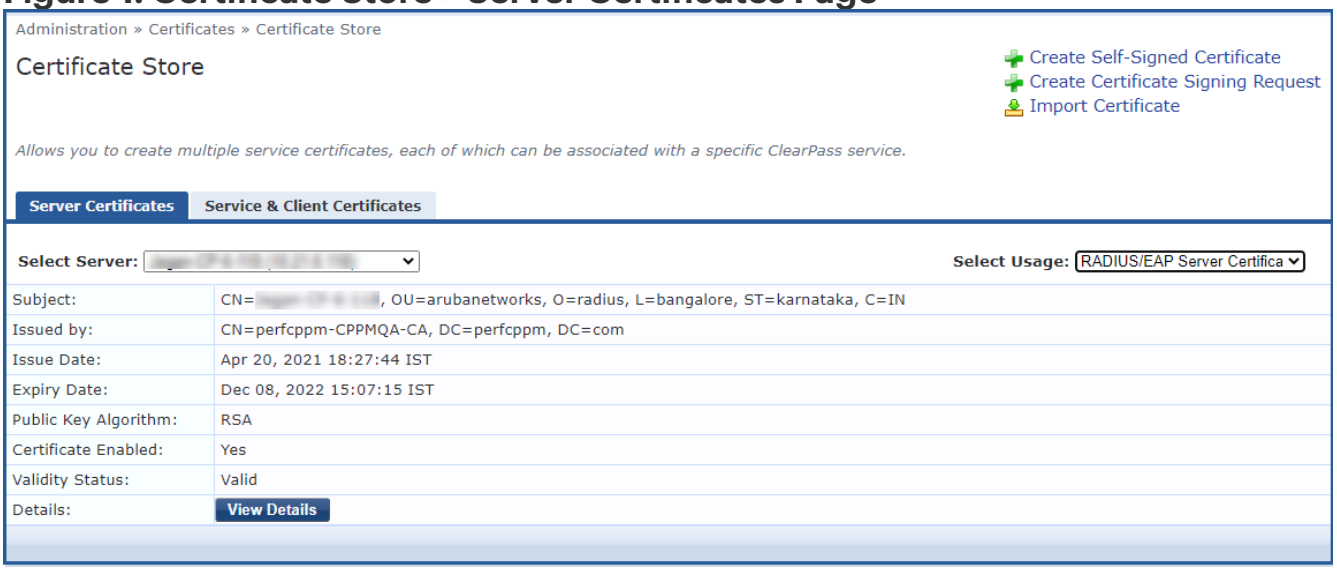
To view the Server Certificates available for the current Policy Manager server:

Procedure

Navigate to **Administration** > **Certificates** > **Certificate Store**.

The **Certificate Store** page opens to the **Server Certificate** tab:

Figure 1. Certificate Store > Server Certificates Page



Subtopics

RADIUS/EAP Server Certificate
HTTPS(ECC) Server Certificate
HTTPS(RSA) Server Certificate
RadSec Server Certificate
Database Server Certificate

RADIUS/EAP Server Certificate

The following table provides a summary of the **RADIUS/EAP Server Certificate** parameters:

Table 1. Summary of RADIUS/EAP Server Certificate Parameters

Parameter	Action/Description
Select Server	Select a Policy Manager server in the cluster for server certificate operations. **NOTE** From the publisher, you can select the publisher or any of the subscriber nodes.

Parameter	Action/Description
Select Usage	Select **RADIUS/EAP Server Certificate**.
Subject	Displays the Organization and Common Name.
Issued by	Displays the Organization and Common Name that issued this certificate.
Issue Date	Displays the date the self - signed certificate is installed.
Expiry Date	Displays the date (in days) when the self - signed certificate expires.
Public Key Algorithm	Public key algorithm type used by this certificate
Certificate enabled	Indicates if the certificate is or is not enabled.
Validity Status	Displays the validity status of the self - signed certificate: *Valid* or *Invalid*.

Subtopics

Viewing Server Certificate Details

Viewing Server Certificate Details

About this task

Click the **View Details** button to view details about the certificate, such as signature algorithm, subject public key Info, etc.

To view the Server Certificate details:

Procedure

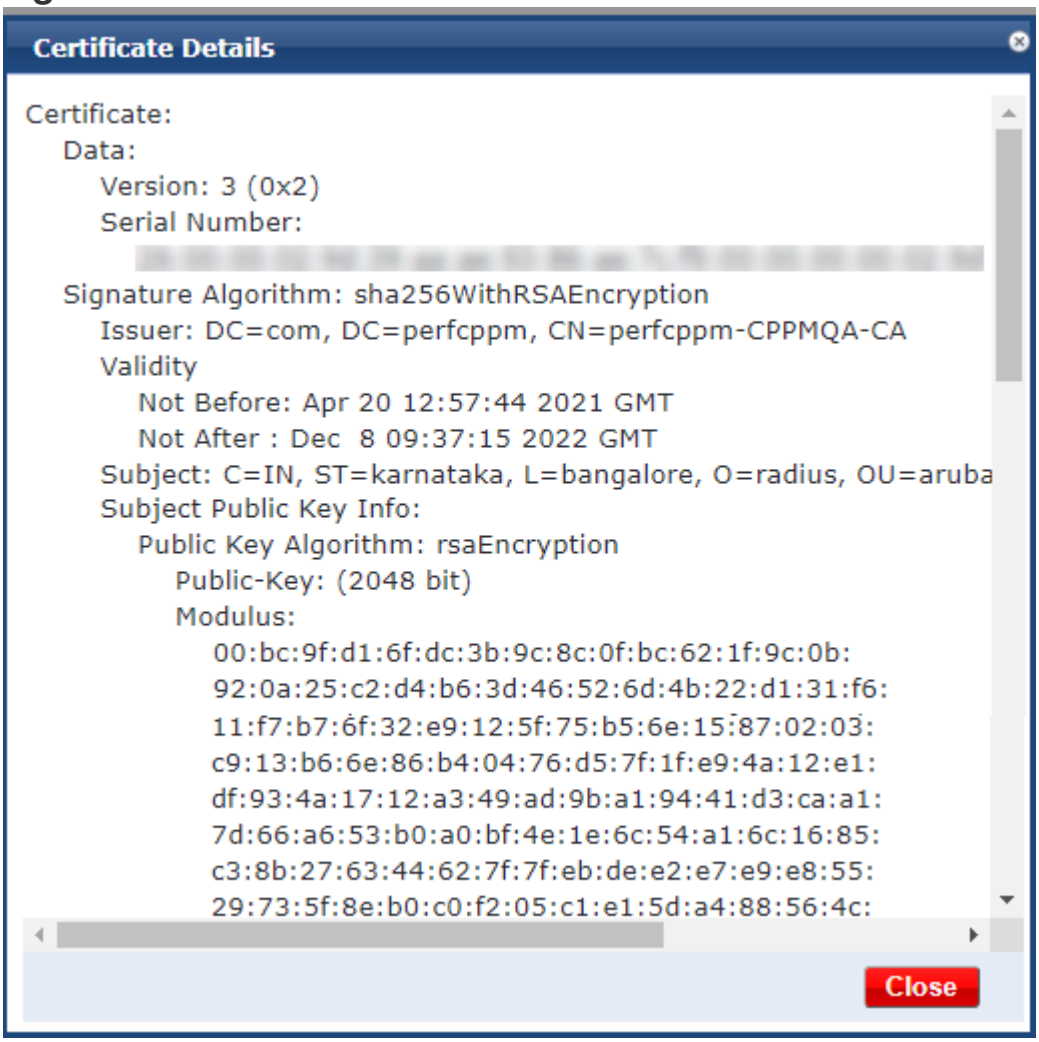
1. Navigate to **Administration Certificates > Certificates > Certificate Store**.

   The Server Certificate summary information is displayed.

2. Click **View Details**.

   The **Certificate Details** window opens.

Figure 1. Server Certificate Details



3. When finished viewing the information, click **Close**.

HTTPS(ECC) Server Certificate

About this task

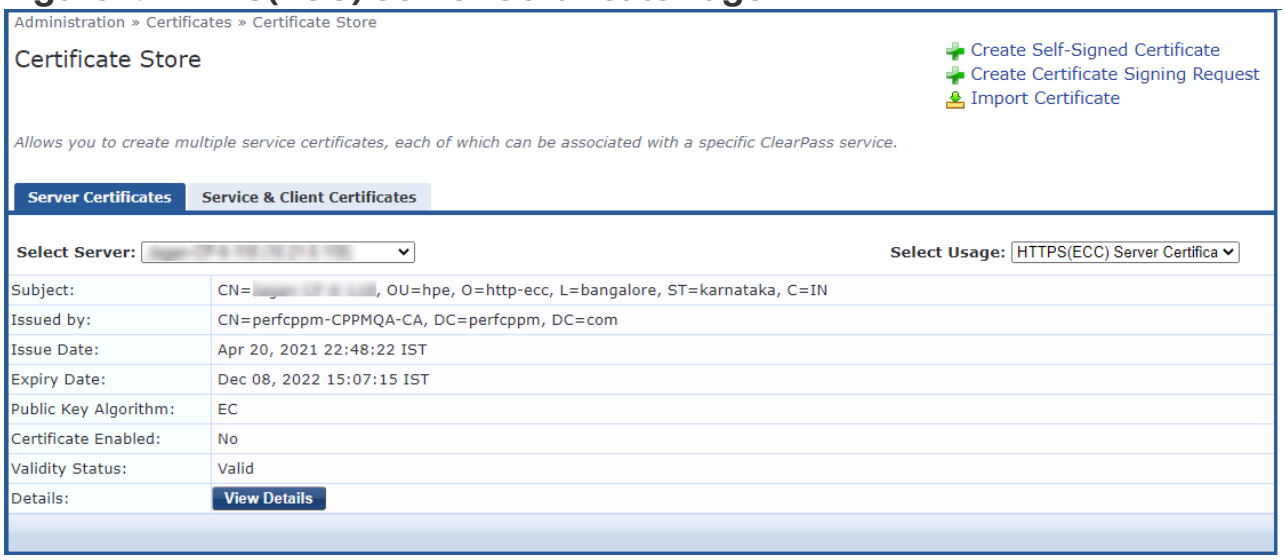
The **HTTPS(ECC) Server Certificate** page displays the parameters configured after a self-signed certificate with an HTTPS(ECC) Server Certificate using has been created and installed.

Procedure

1. Navigate to the **Administration > Certificates > Certificate Store > Server Certificates** tab.
2. From the **Select Usage** drop-down, choose **HTTPS(ECC) Server Certificate**.

The **HTTPS(ECC) Server Certificate** page opens.

Figure 1. HTTPS(ECC) Server Certificate Page



The following table describes the **HTTPS(ECC) Server Certificate** parameters:

Table 1. HTTPS(ECC) Server Certificate Parameters

Parameter	Action/Description
Subject	Displays the Organization and Common Name.
Issued by	Displays the Organization and Common Name that issued the server certificate.
Issue Date	Displays the date the self - signed certific ate is installed.
Expiry Date	Displays the date when the self - signed certificate expires.
Validity Status	Displays the validity status of the self - si gned certificate.
Public Key Algorithm	Algorithm type used by the certificate.
Certificate Enabled	Indicates whether or not the certificate is enabled in the certificate store.

Parameter	Action/Description
Details	To view details about the certificate, such as Signature Algorithm and Subject Publi c Key Info, click the **View Details** button.

HTTPS(RSA) Server Certificate

About this task

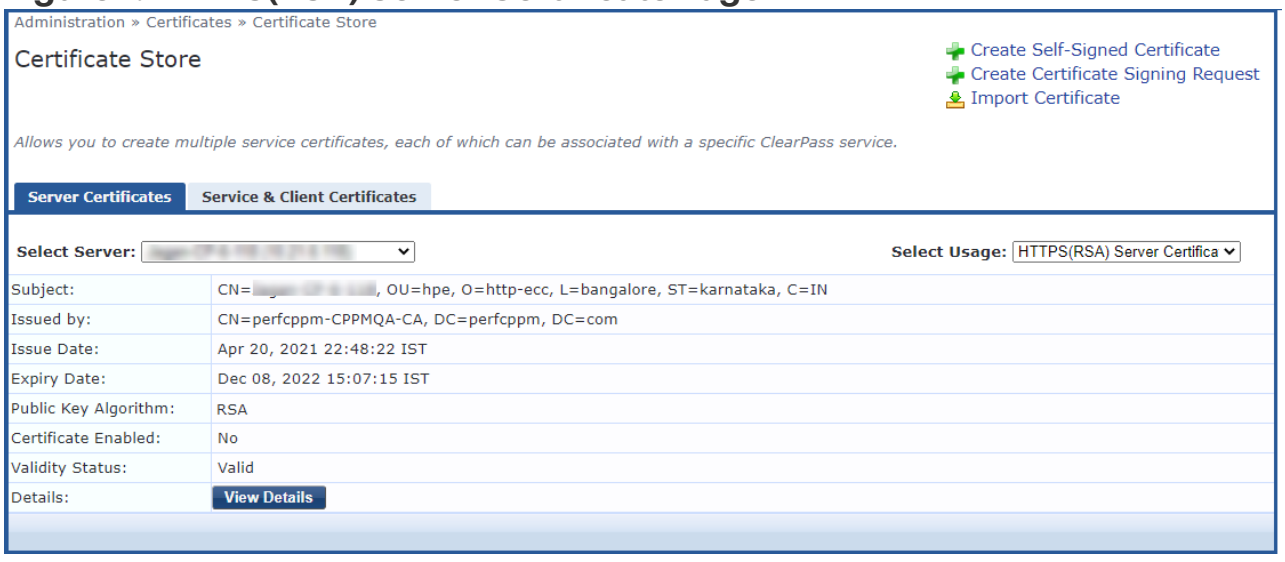
The **HTTPS(RSA) Server Certificate** page displays the parameters configured after a self-signed certificate with an HTTPS(RSA) Server Certificate using has been created and installed.

Procedure

1. Navigate to the **Administration > Certificates > Certificate Store > Server Certificates** tab.
2. From the **Select Usage** drop-down, choose **HTTPS(RSA) Server Certificate**.

   The **HTTPS(RSA) Server Certificate** page opens.

Figure 1. HTTPS(RSA) Server Certificate Page



The following table describes the **HTTPS(RSA) Server Certificate** parameters:

Table 1. HTTPS(RSA) Server Certificate Parameters

Parameter	Action/Description
Subject	Displays the Organization and Common Name.
Issued by	Displays the Organization and Common Name that issued the server certificate.
Issue Date	Displays the date the self - signed certific ate is installed.
Expiry Date	Displays the date when the self - signed certificate expires.
Validity Status	Displays the validity status of the self - si gned certificate.
Public Key Algorithm	Algorithm type used by the certificate.
Certificate Enabled	Indicates whether or not the HTTPS(RSA) cerificate is enabled in the certificate sto re.
Details	To view details about the certificate, such as Signature Algorithm and Subject Publi c Key Info, click the **View Details** button.

RadSec Server Certificate

About this task

RADIUS-over-TLS (Transport Layer Security), or RadSec, employs a TLS tunnel to enable secure communication between the controller and a Policy Manager server. Employing RADIUS communication over TLS increases the level of security for authentication. When configured, the RadSec protocol is used to safely transmit authentication and accounting data across the network.

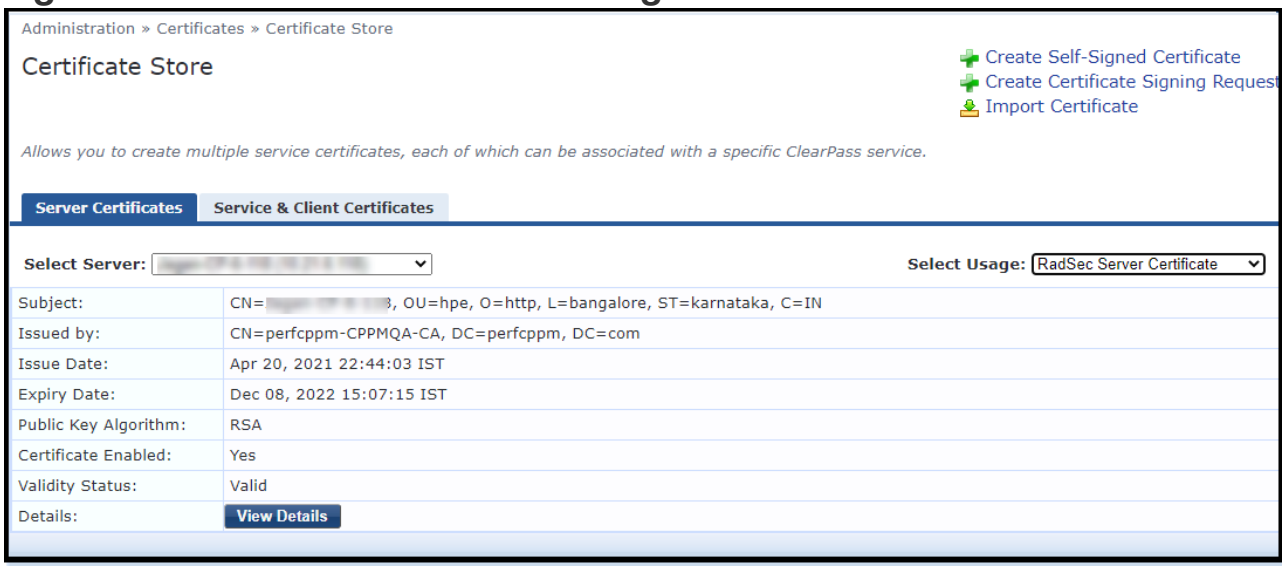
To access the RadSec Server Certificate:

Procedure

1. Navigate to the **Administration > Certificates > Certificate Store > Server Certificates** tab.
2. From the **Select Usage** drop-down, choose **RadSec Server Certificate**.

   The **RadSec Server Certificate** page opens.

Figure 1. RadSec Server Certificate Page



The following table describes the **RadSec Server Certificate** parameters:

Table 1. Summary of RadSec Server Certificate Parameters

Parameter	Action/Description
Select Server	Select a Policy Manager server in the clus ter for server certificate operations.
Select Usage	Select **RadSec Server Certificate**.
Subject	Displays the Organization and Common Name.
Issued by	Displays the Organization and Common Name that issued this certificate.
Issue Date	Displays the date the self - signed certific ate is installed.

Parameter	Action/Description
Expiry Date	Displays the date (in days) when the self - signed certificate expires.
Public Key Algorithm	Algorithm type used by the certificate.
Certificate Enabled	Indicates whether or not the RadSec server cerificate is enabled in the certificate store.
Validity Status	Displays the validity status of the self - signed certificate: *Valid* or *Invalid*.

Database Server Certificate

About this task

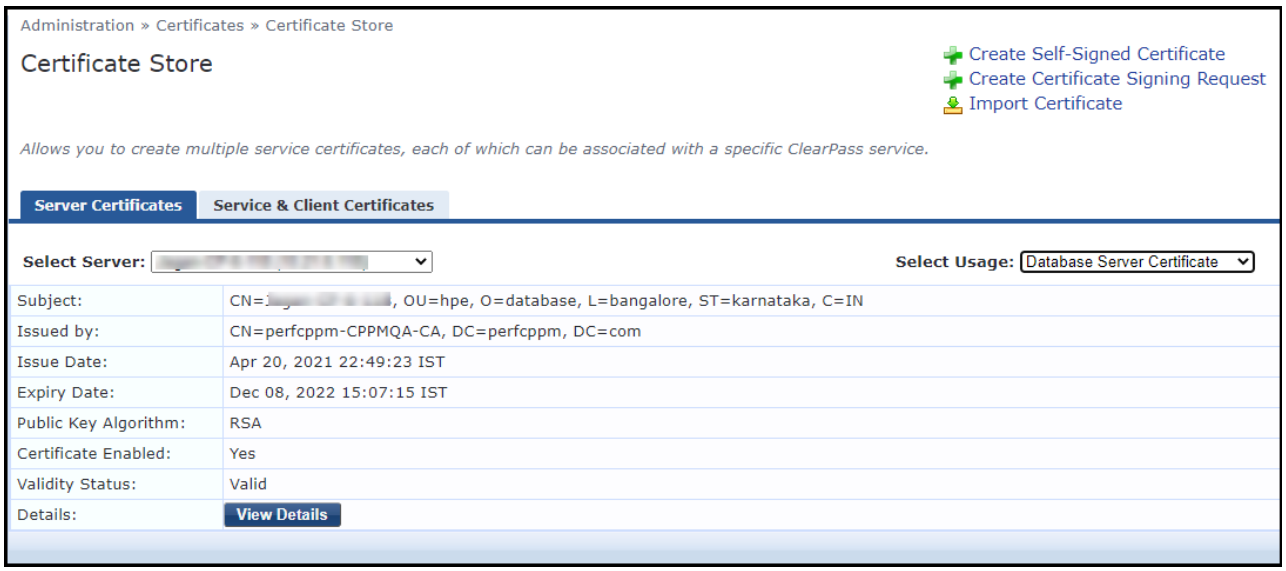
To access the Database Server Certificate:

> **NOTE**
> With self-signed certificates, database certificate regeneration and backend service restarts are automated and part of the update management IP script. However, after a re-login, the Database Server Certificate may need to be updated if CA-signed certificates are used.

Procedure

1. Navigate to the **Administration > Certificates > Certificate Store > Server Certificates** tab.
2. From the **Select Usage** drop-down, choose **Database Server Certificate**. The **Database Server Certificate** page opens.

Figure 1. Database Server Certificate Page



The following table describes the **Database Server Certificate** parameters:

Table 1. Summary of Database Server Certificate Parameters

Parameter	Action/Description
Select Server	Select a Policy Manager server in the clus ter for server certificate operations.
Select Usage	Select **Database Server Certificate**.
Subject	Displays the Organization and Common Name.
Issued by	Displays the Organization and Common Name that issued this certificate.
Issue Date	Displays the date the self - signed certific ate is installed.
Expiry Date	Displays the date (in days) when the self - signed certificate expires.
Public Key Algorithm	Algorithm type used by the certificate.
Certificate Enabled	Indicates whether or not the database se rver cerificate is enabled in the certificat e store.

Parameter	Action/Description
Validity Status	Displays the validity status of the self - si gned certificate: *Valid* or *Invalid*.

NOTE

Network administrators must restart the Policy Manager server after changing the Database certificate to insure that all client database connections are reestablished when the database comes back up.

Creating and Installing a Self-Signed Server Certificate

About this task

After you select a server and a certificate type, you can create and install a self-signed server certificate.

NOTE

When Common Criteria mode is enabled, the **Create-Self Signed Certificate** option for both HTTPS and RADIUS certificates is not available from the **Certificate Store** page (for more information, see Common Criteria Mode Parameter).

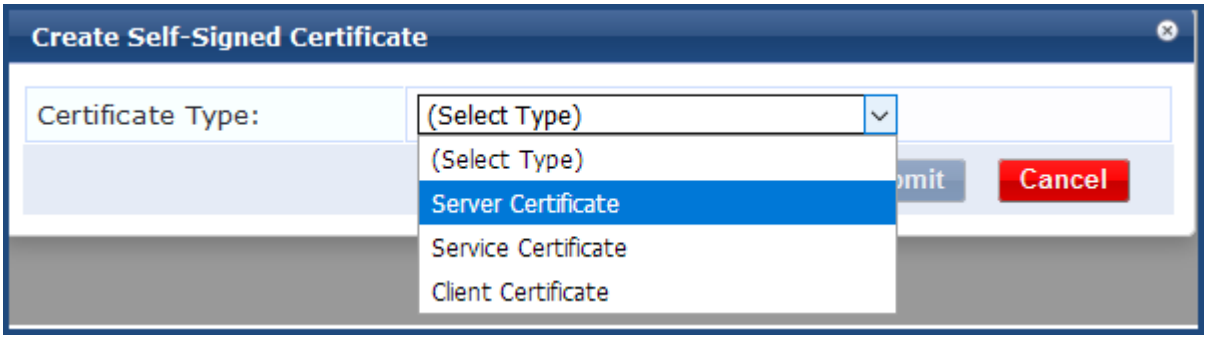
To create and install a self-signed server certificate:

Procedure

1. Navigate to **Administration > Certificates > Certificate Store**.
2. From the **Server Certificates** tab > **Select Server** drop-down, select a Policy Manager server.
3. Click the **Create Self-Signed Certificate** link.

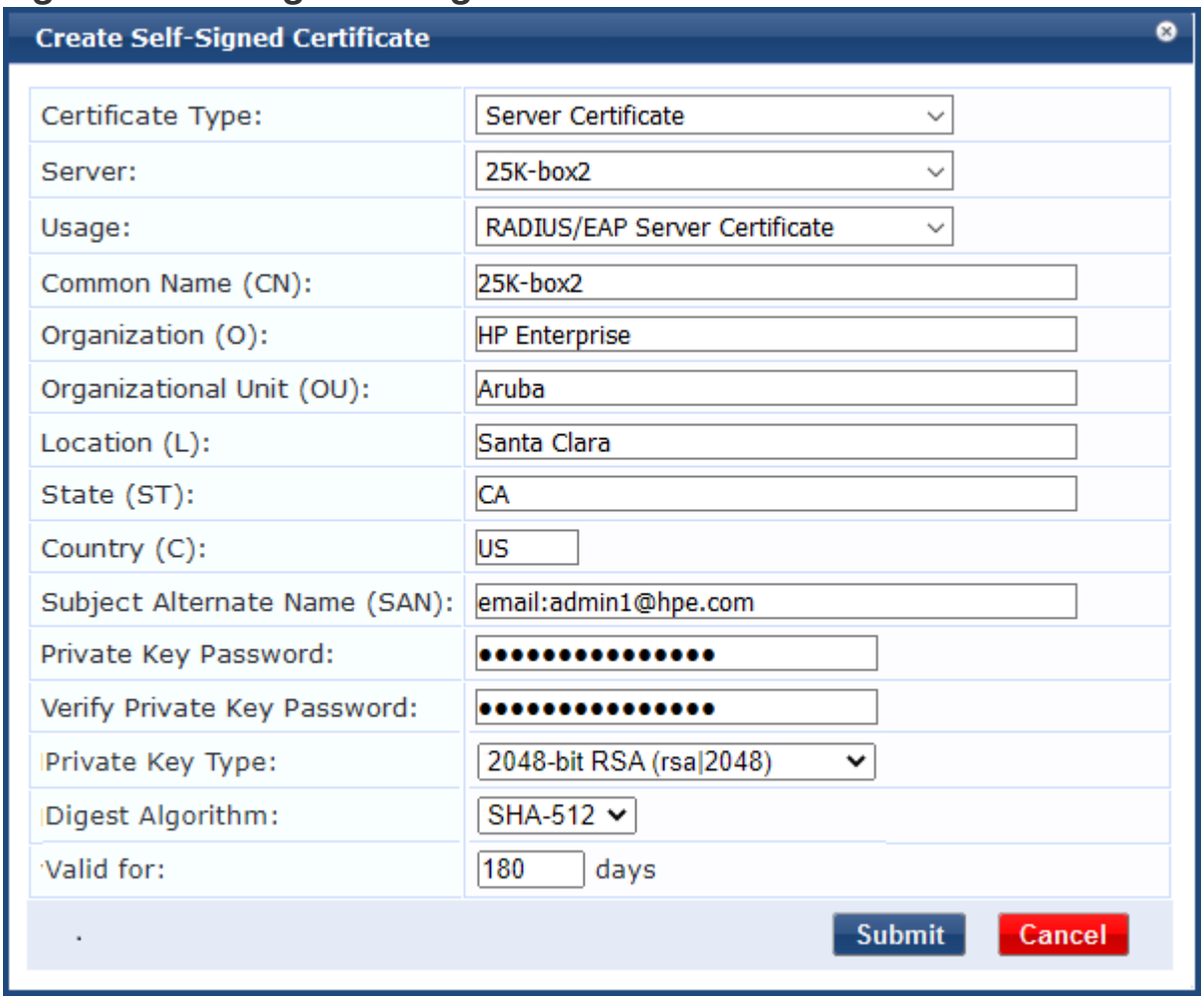
   The **Create Self-Signed Certificate** dialog opens.
4. From the **Certificate Type** drop-down menu, select **Server Certificate**.

Figure 1. Selecting the Server Certificate Type



The **Create Self-Signed Certificate** configuration dialog for a **Server Certificate** opens.

Figure 2. Creating a Self-Signed Server Certificate



5. Specify the **Create Self-Signed Certificate** parameters as described in Table 1.

Table 1. Specifying Self - Signed Server Certificate Parameters

Parameter	Action/Description
Certificate Type	Select **Server Certificate**.
Server	Displays the name of the selected Policy Manager server on the **Certificate Store** page.
Usage	Displays the selected server certificate us age for the server. The options are: • RADIUS/EAP Server Certificate • HTTPS Server Certificate • RadSec Server Certificate • Database Server Certificate
Common Name (CN)	Enter the name associated with this entit y. This can be a host name, IP address, or other meaningful name. This field is man datory. **NOTE** **Database Server Certificate** , either the *Common Name* or the *Subject Alternate Na me (SAN)* DNS name must b e set to the IP address (also, both fields can be set to th e IP address if desired). Whe n using IPv6 address, it shou ld be represented in the sho rtest possible format as defi ned in RFC 5952.
Organization (O)	Enter the name of the organization. This f ield is optional.

Parameter	Action/Description
Organizational Unit (OU)	Enter the name of the department, divisi on, section, or other meaningful organiza tional unit. This field is optional.
Location (L) State (ST) Country (C)	Enter the name of the location, state, co untry, and/or other meaningful location in formation. These fields are optional.
Subject Alternate Name (SAN)	Enter the alternative name for the specifi ed Common Name. This field is optional. Enter the Subject Alternate Name in one of the following formats: • email: *email_address* • URI: *URI* • IP: *IP_address* • dns: *DNS_name* or *IP_address* • rid: *ID* **NOTE** **Database Server Certificate** , either the *Common Name* or the *Subject Alternate Na me (SAN)* DNS name must b e set to the IP address (also, both fields can be set to th e IP address if desired). Whe n using IPv6 address, it shou ld be represented in the sho rtest possible format as defi ned in RFC 5952.
Private Key Password	Enter the Private Key password, then veri fy the password.
Private Key Type	Select the length for the generated priva te key types. The available options for pri

Parameter	Action/Description
	vate key type vary, depending upon the c ertificate type selected in the **Usage** field . Keys for **RADIUS/EAP Server Certificates** , **RadSec Server Certificates** and **Databa se Server Sertificates** • 2048 - bit RSA (rsa|2048) • 3072 - bit RSA (rsa|3072) • 4096 - bit RSA (rsa|4096) • NIST/SECG curve over a 256 bit prime field (ec|scep256r1) • NIST/SECG curve over a 384 bit prime field (ec|scep384r1) • NIST/SECG curve over a 521 bit prime field (ec|scep521r1) Keys for **HTTPS(ECC) Server Certificates** • NIST/SECG curve over a 256 bit prime field (ec|scep256r1) • NIST/SECG curve over a 384 bit prime field (ec|scep384r1) • NIST/SECG curve over a 521 bit prime field (ec|scep521r1) Keys for **HTTPS(RSA) Server Certificates** • 2048 - bit RSA (rsa|2048) • 3072 - bit RSA (rsa|3072) • 4096 - bit RSA (rsa|4096)
Digest Algorithm	Select the message digest algorithm. Th e available options for digest algorithm vary, depending upon the key type select ed in the **Private Key Type** field.

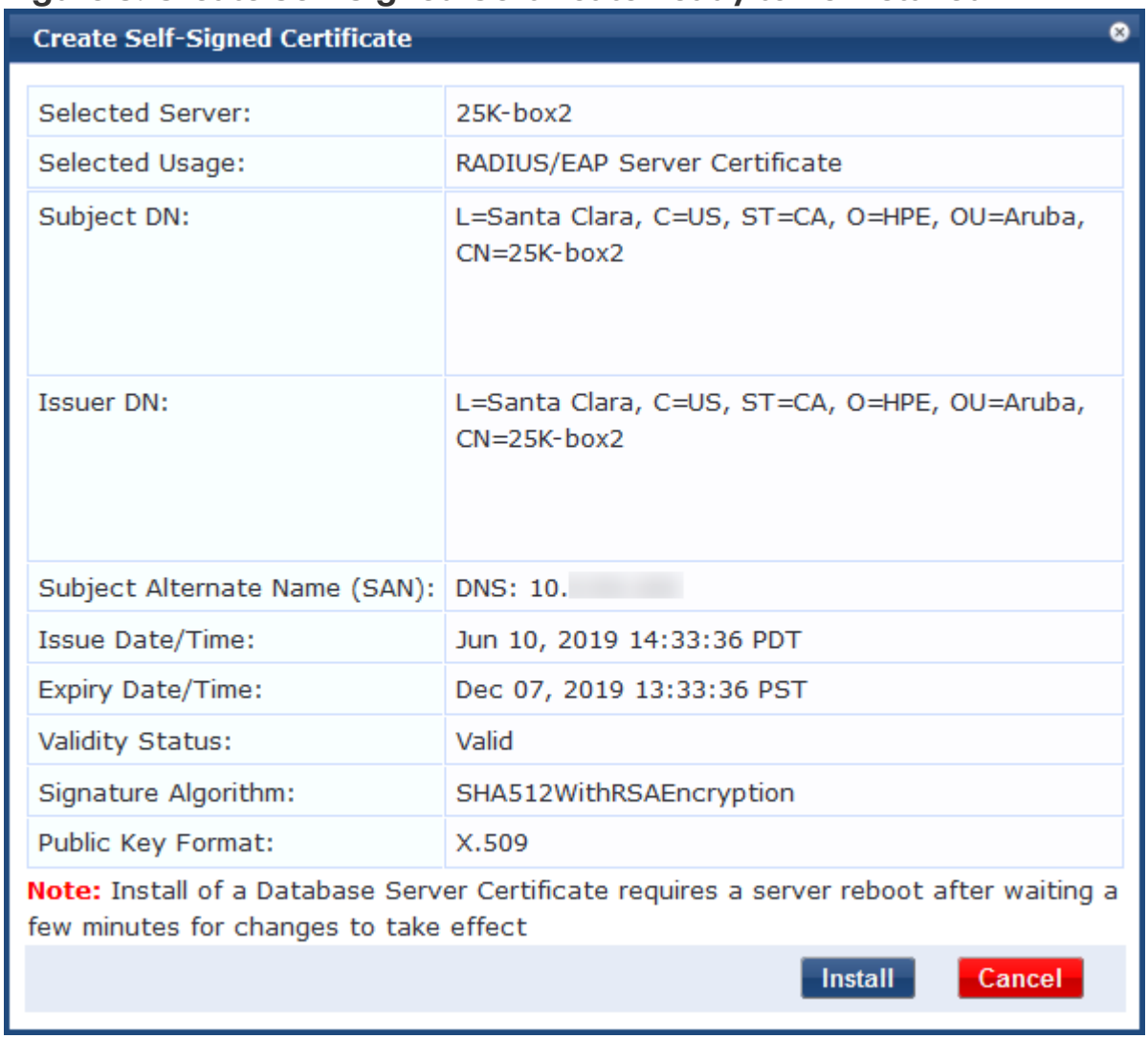
Parameter	Action/Description
Valid for	Enter the certificate duration in number o f days. The default is **180** days.

6. Click **Submit**.

   The completed **Create Self-Signed Certificate** page opens and is ready to be installed.

   This page displays a summary of the values specified in the **Create Self-Signed Certificate** page and provides the **Install** button to install the self-signed certificate.

Figure 3. Create Self-Signed Certificate Ready to Be Installed



7. Click **Install**.

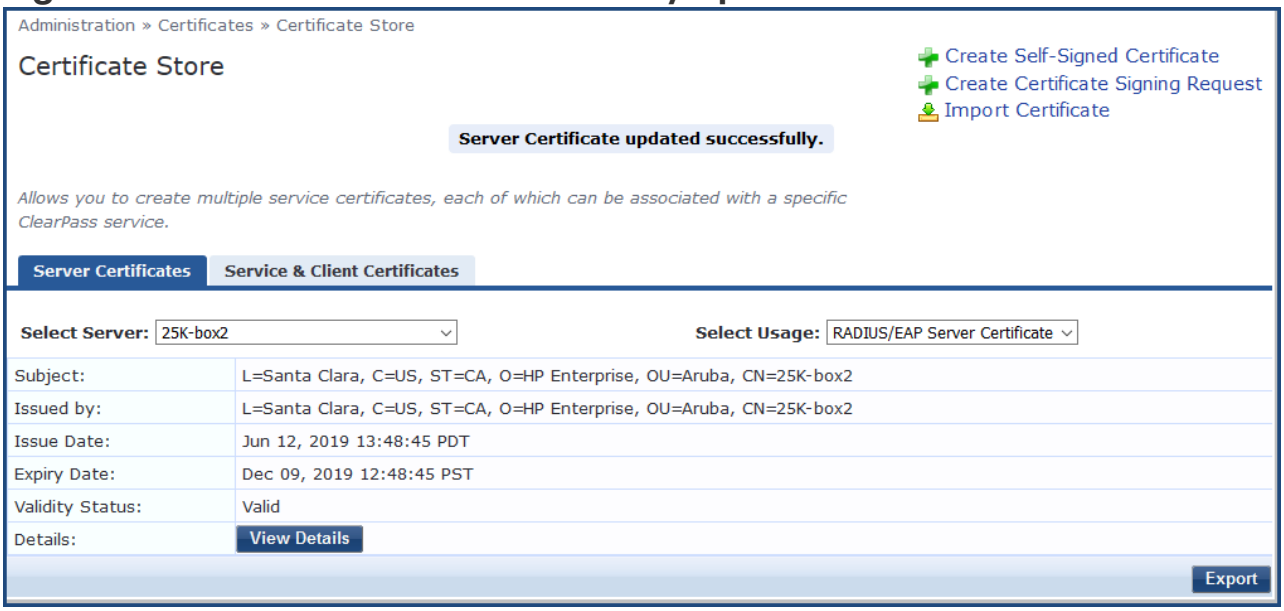
> **NOTE**
>
> Installing a **Database Server Certificate** requires a server reboot after waiting for a few minutes for the changes to take effect.

After you click **Install**, you return to the **Certificate Store** page and Policy Manager generates a message about the status of the certificate installation.

If the installation is successful the page displays the following message:

Server Certificate updated successfully.

Figure 4. Server Certificate Successfully Updated



8. Because all services are restarted after a successful certificate installation, you must log out (by clicking the **Menu > Logout** link), then log in to the Policy Manager client to continue.

Creating a Certificate Signing Request

About this task

After you select a certificate type (*Server Certificate* or *Service Certificate*), you can create a certificate signing request (CSR). When you generate certificate signing request, the private key is automatically stored on the current Policy Manager server.

Thus, you can then upload (import) the certificate alone, without having to include the Private Key in the import process.

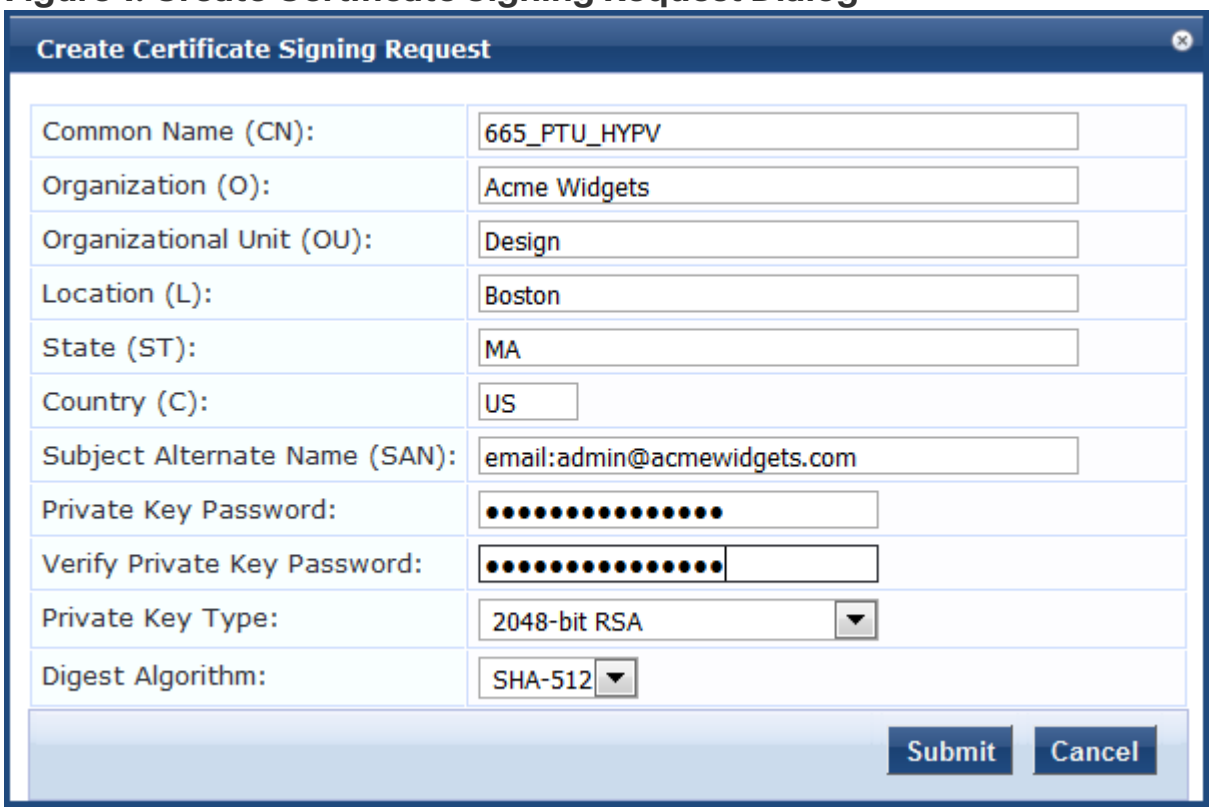
> **NOTE**
> A certificate signing request created in ClearPass is valid for only 15 days. After 15 days, the CSR expires and the certificates that were created by it cannot be installed. Also note that the stored private key is removed 15 days after the certificate signing request was created. If you don't import the certificate within 15 days, you must create the CSR and certificate again.

To create a certificate signing request:

Procedure

1. Navigate to **Administration > Certificates > Certificate Store**. The **Server Certificates** tab is displayed.
2. Click the **Create Certificate Signing Request** link. The **Create Certificate Signing Request** dialog opens:

Figure 1. Create Certificate Signing Request Dialog



3. Specify the **Create Certificate Signing Request** parameters, then click **Submit**.

Table 1. Create Certificate Signing Request Parameters

Parameter	Action/Description
Common Name (CN)	Enter the name associated with this entit y. This can be a hostname defined as a Fu lly Qualified Domain Name (FQDN), or an IP address. Best practices is to ensure th e CN is an FQDN when applicable. The default value for this field is the serve r hostname. If the hostname is not define d as a FQDN, then this field must be man ually updated to include an FQDN or IP a ddress. When using IPv6 address, it shoul d be represented in the shortest possible format as defined in RFC 5952.
Organization (O)	Optionally, enter the name of the organi zation.
Organizational Unit (OU)	Optionally, enter the name of the depart ment, division, section, or other meaning ful name.
Location (L) State (ST) Country (C)	Optionally, enter the name of the locatio n, state, country.
Subject Alternate Name (SAN)	Optionally, enter the alternative names fo r the specified Common Name in one of t he following formats: • email: *email_address* • URI: *uri* • IP: *ip_address* • dns: *dns_name* • rid: *id* **NOTE**

Parameter	Action/Description
	When configuring a Database Server Certificate, either the Common Name or the Subject Alternate Name (SAN) DNS name must be set to the IP address (also, both fields can be set to the IP address if desired). When using IPv6 address, it should be represented in the shortest possible format as defined in RFC 5952.
Private Key Password Verify Private Key Password	Enter the private key password, then verify it.
Private Key Type	Select the length for the generated private key types from the following options: • 2048 - bit RSA (rsa|2048) • 3072 - bit RSA (rsa|3072) • 4096 - bit RSA (rsa|4096) • NIST/SECG curve over a 256 bit prime field (ec|scep256r1) • NIST/SECG curve over a 384 bit prime field (ec|scep384r1) • NIST/SECG curve over a 521 bit prime field (ec|scep521r1)
Digest Algorithm	Select the message digest algorithm from the following options: • MD5 • SHA - 1 • SHA - 224 • SHA - 256

Parameter	Action/Description
	• SHA - 384 • SHA - 512

4. Click **Submit**.

   The generated certificate signing request is displayed.

Figure 2. Certificate Signing Request Generated

   Note that the Private Key is automatically stored on the current Policy Manager server. You can then upload (import) the certificate without having to include the Private Key as part of the import process (for related information, see Importing a Server Certificate).

5. Copy the certificate signing request content and paste it into the Web form as part of the enrollment process.

6. To save and download the *Certificate Signing Request* file, click **Download CSR**.

Importing a Server Certificate

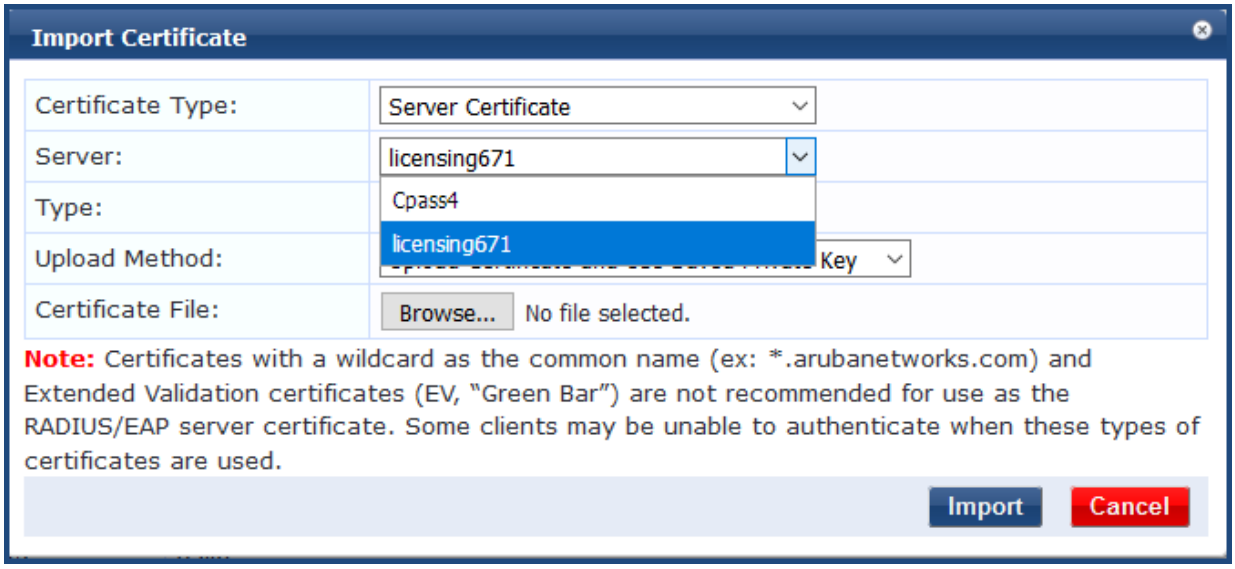
When you import the server certificate, you are provided with three upload options:

- **Upload Certificate and Use Saved Private Key**: This option allows the admin to upload only the certificate. The server certificate is then matched against the private key saved on the Policy Manager server.
- **Upload PKCS#12 Certificate (.pfx or .p12 only)**: With this option, the admin uploads the PKCS#12 file and provides a pass phrase.
- **Upload Certificate and Private Key Files**: The admin can choose to upload the private key file and password along with the server certificate file.

  To import a server certificate into the current Policy Manager server:

  1. Navigate to **Administration** > **Certificates** > **Certificate Store**.
  2. Click the **Import Certificate** link. The **Import Certificate** dialog opens.
  3. Click the **Certificate Type** menu and select **Server Certificate**.

     Figure 1. Import Server Certificate Dialog



NOTE

For security reasons, certificates signed using SHA1RSA are not recommended. Importing certificates signed with stronger keys, such as RSA with a length of more than 1024 bits, is recommended.

NOTE

Policy Manager does not support importing the HTTPS Server Certificate chain or RADIUS/EAP Server Certificate chain in P7b Base64 format. A P7B file contains only certificates and chain certificates (intermediate certificate authorities), not the private key.

4. Specify the **Import Certificate** parameters as described in the following table:

Table 1. Import Server Certificate Parameters

Parameter	Action/Description
Certificate Type	Select **Server Certificate** (selected by d efault).

Parameter	Action/Description
Server	Select the name of the Policy Manager server that the server certificate will be imported into. **NOTE** When importing a certificate to a subscriber node from the publisher, in the **Server** field, select the subscriber node.
Type	Select one of the following server certificate types: ◦ RADIUS/EAP Server Certificate ◦ HTTPS Server Certificate ◦ RadSec Server Certificate
Upload Method	Select one of the following methods to upload the certificate: ◦ **Upload Certificate and Use Saved Private Key** - This option allows the administrator to upload only the certificate. The certificate is then matched against the private key saved on the Policy Manager server. ◦ **Upload PKCS#12 Certificate (.pfx or .p12 only)** - With this option, the administrator uploads the PKCS#12 file and provides a passphrase. For cluster deployments, this is the cluster password. ◦ **Upload Certificate and Private Key Files**

Parameter	Action/Description
	- The administrator can choose to upload the private key file and password along with the certificate file.
Certificate File	Browse to the certificate file to be imported. **NOTE** Both certificates with a wild card as the common name and Extended Validation certificates are not recommended for use as the RADIUS/EAP server certificate. Some clients may be unable to authenticate when these types of certificates are used.

5. Click **Import**.

Exporting a Server Certificate

Server certificate exports let administrators save the current Policy Manager server certificate in PKCS#12 format for secure reuse.

About this task

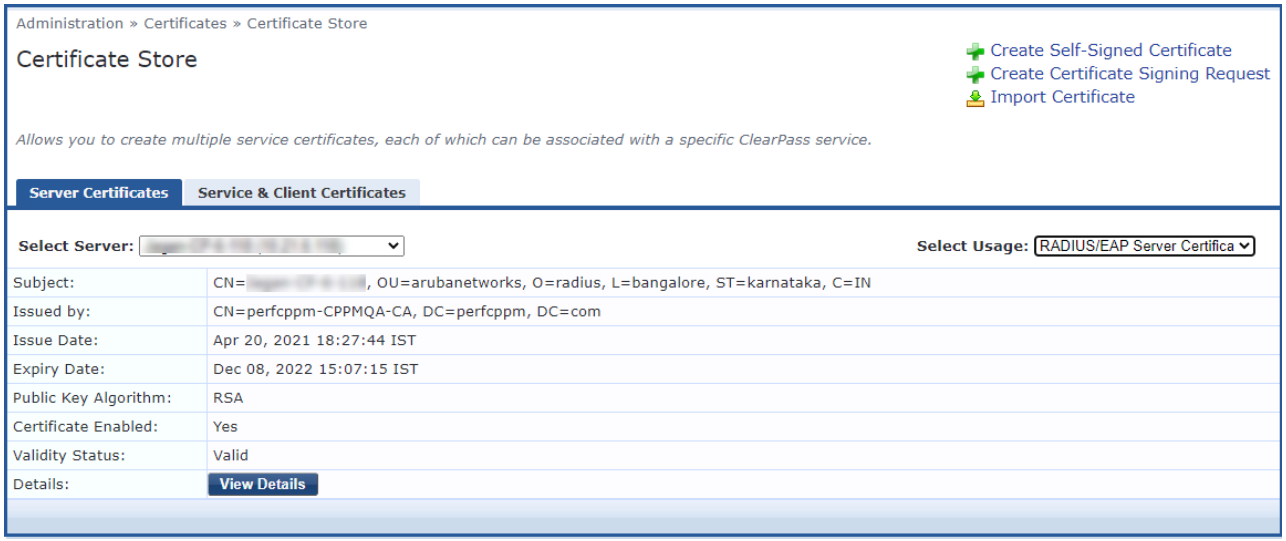
To export a server certificate from the current Policy Manager server:

Procedure

1. Navigate to **Administration > Certificates > Certificate Store**.

   The **Certificate Store** page opens.

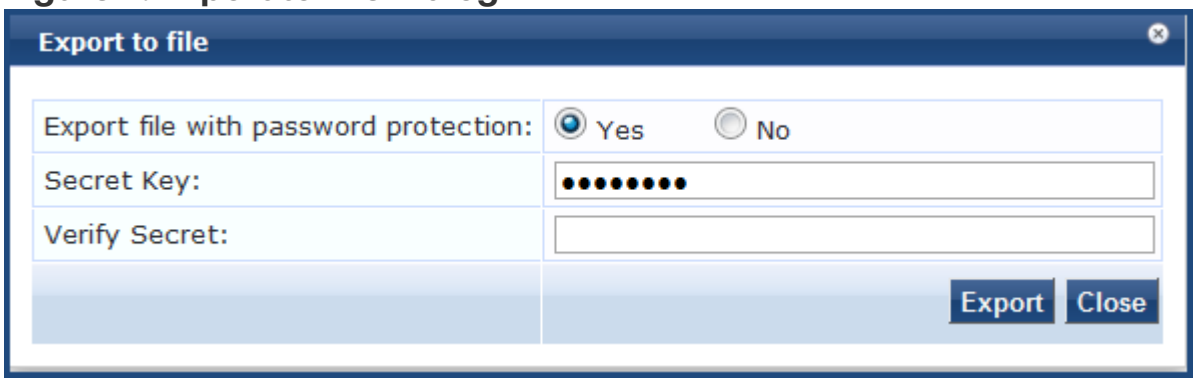
   Figure 1. Certificate Store Page



2. Click the **Export** link.

   The **Export to File** dialog opens.

Figure 2. Export to File Dialog



3. Specify the **Export to File** parameters as described in the following table.

Table 1. Export to File Parameters

Parameter	Action/Description
Export file with password protection	Select **Yes** (the default) or **No** to specify whether you want to require password p rotection.
Secret Key Verify Secret	Enter the secret key, then reenter the sec ret key to verify.

4. Click **Export**. Upon clicking **Export**, the certificate is exported in *PKCS#12* format.

The PKCS#12 (or PFX) format is a binary format for storing the server certificate, any intermediate certificates, and the private key in one encryptable file.

Service and Client Certificates

- Adding Service or Client Certificates
- Creating a Self-Signed Service or Client Certificate
- Creating a Certificate Signing Request
- Importing a Service or Client Certificate

Adding Service or Client Certificates

About this task

Service Certificates allow you to create multiple service certificates, each of which can be associated with a specific Policy Manager service. For details, see Associating a RADIUS Service Certificate with a Service.

> **NOTE**
> Be aware that if an expired service certificate or EAP server certificate is used with an enabled service, the RADIUS server will not start or reload and authentication requests will fail. To successfully authenticate, first renew or remove any expired service certificates.

> **NOTE**
> Service Certificates must always be RADIUS certificates.

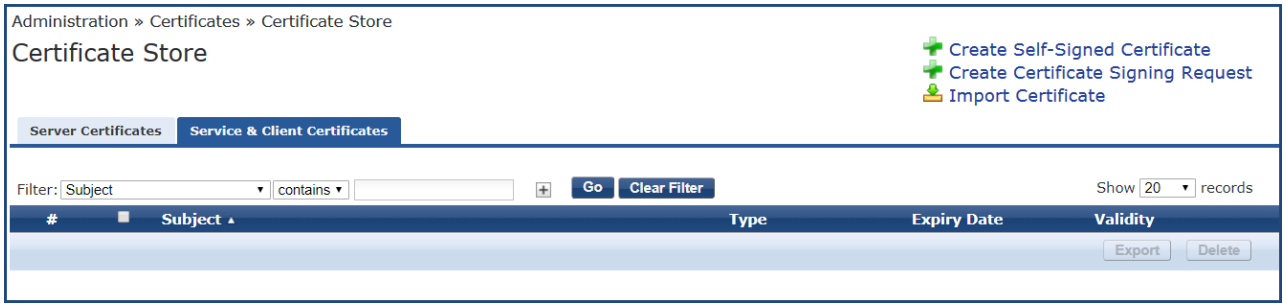
The **Client Certificates** feature allows you to import a client certificate and use that certificate to establish TLS connections with RadSec-enabled servers. Use client certificates to allow RadSec proxy to establish a connection with a remote server, such as an Eduroam (education roaming) server.

To access the **Service & Client Certificates** page:

Procedure

1. Navigate to **Administration > Certificates > Certificate Store**.
2. Select the **Service & Client Certificates** tab.

Figure 1. Service & Client Certificates Tab



Creating a Self-Signed Service or Client Certificate

About this task

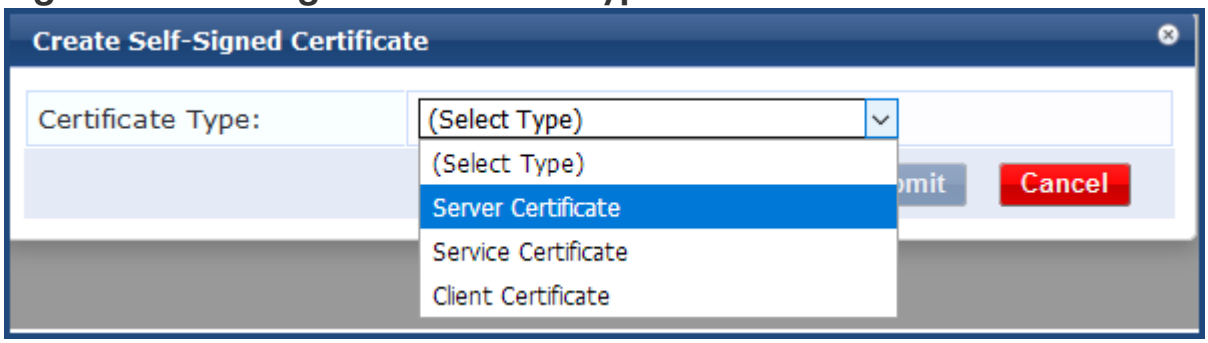
This task creates a self-signed certificate to be signed by a CA (Certificate Authority).

To create a Self-Signed Service or Client Certificate:

Procedure

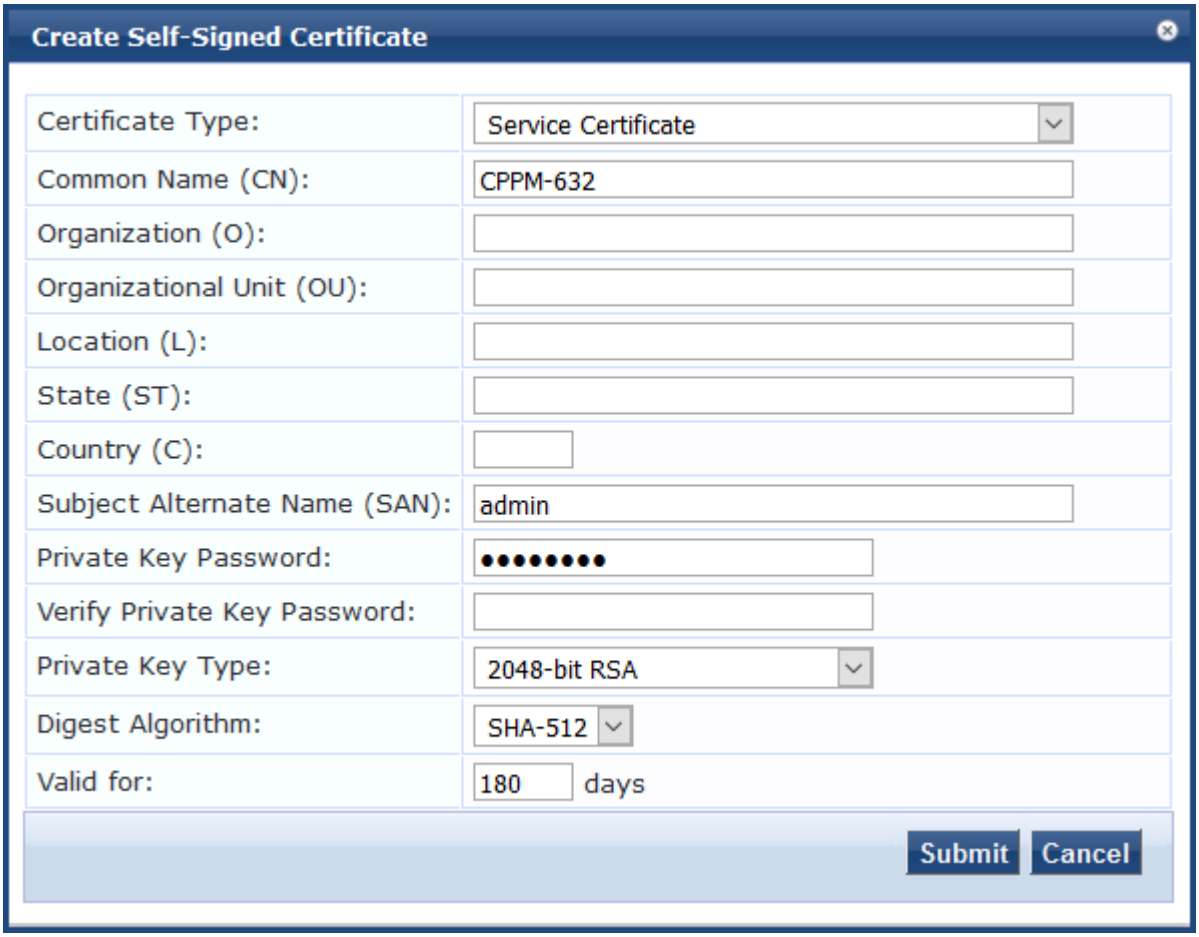
1. From the **Certificate Store > Service & Client Certificates** page, click the **Create Self-Signed Certificate** link. The **Create Self-Signed Certificate** dialog opens.

Figure 2. Selecting the Certificate Type



2. From the **Certificate Type** drop-down menu, select either **Service** or **Client** as the certificate type.

Figure 3. Creating a Self-Signed Service or Client Certificate



3. Specify the following parameters for the self-signed service or client certificate:

Table 1. Specifying Self - Signed Service Certificate Parameters

Parameter	Action/Description
Common Name (CN)	Enter the name associated with this entit y. Enter a host name, IP address, or othe r meaningful name. This field is mandato ry. When using IPv6 address, it should be represented in the shortest possible form at as defined in RFC 5952.
Organization (O)	Optionally, enter the name of the organi zation. This field is optional.
Organizational Unit (OU)	Enter the name of the department, divisi on, section, or other meaningful name. T his field is optional.

Parameter	Action/Description
Location (L) State (ST) Country (C)	Enter the name of the location, state, country, and/or other meaningful name. These fields are optional.
Subject Alternative Name (SAN)	Enter the alternative name for the specified Common Name. **NOTE** Enter the Subject Alternative Name in one of the following formats: • email: *email_address* • URI: *uri* • IP: x.x.x.x • dns: *dns_name* • rid: *id* This field is optional. When using IPv6 address, it should be represented in the shortest possible format as defined in RFC 5952.
Private Key Password Verify Private Key Password	Enter the Private Key password, then verify the password.
Private Key Type	Select the length for the generated private key types from the following options: • 1024 - bit RSA • 2048 - bit RSA. This is the default value. • 4096 - bit RSA • X9.62/SECG curve over a 256 bit prime field

Parameter	Action/Description
	• NIST/SECG curve over a 384 bit prime field
Digest Algorithm	Select the message digest algorithm fro m the following options: • MD5 • SHA - 1 • SHA - 224 • SHA - 256 • SHA - 384 • SHA - 512 **NOTE** The **MD5** algorithm is not available in **FIPS** mode.
Valid for	Enter the certificate duration in number o f days. The default is **180** days.

4. Click **Submit**. The new self-signed service certificate is added.

Figure 4. Self-Signed Service Certificate Added

5. Repeat these steps to add as many self-signed service or client certificates as required.

Creating a Certificate Signing Request

About this task

After you select a server and a certificate type, you can create a certificate signing request.

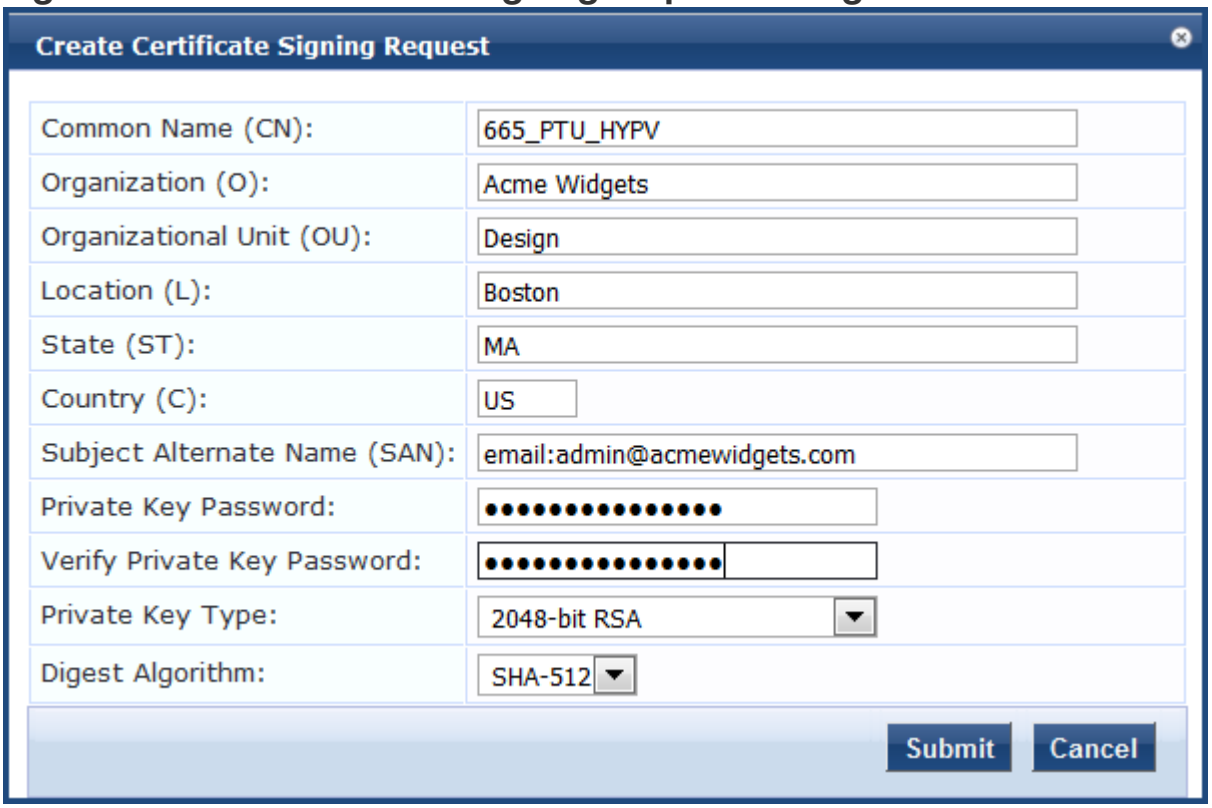
To create a certificate signing request:

Procedure

1. Navigate to **Administration** > **Certificates** > **Certificate Store**.
2. Click the **Create Certificate Signing Request** link.

   The **Create Certificate Signing Request** dialog opens:

Figure 5. Create Certificate Signing Request Dialog



3. Specify the **Create Certificate Signing Request** parameters as described in the following table:

Table 2. Create Certificate Signing Request Parameters

Parameter	Action/Description
Common Name (CN)	Enter the name associated with this entit y. This field is mandatory.

Parameter	Action/Description
	This can be a host name, IP address, or o ther name. The default is the fully - qualified domain name (FQDN).
Organization (O)	Optionally, enter the name of the organi zation.
Organizational Unit (OU)	Optionally, enter the name of the depart ment, division, section, or other meaning ful name.
Location (L) State (ST) Country (C)	Optionally, enter the name of the locatio n, state, country.
Subject Alternative Name (SAN)	Optionally, enter the Subject Alternative Name (SAN) for the specified Common N ame in one of the following formats: • email: *email_address* • URI: *uri* • IP: x.x.x.x • dns: *dns_name* • rid: *id*
Private Key Password Verify Private Key Password	Enter the private key password, then veri fy it.
Private Key Type	Select the length for the generated privat e key types from the following options: • 1024 - bit RSA • **2048 - bit RSA**. This is the default. • 4096 - bit RSA

Parameter	Action/Description
	• X9.62/SECG curve over a 256 bit prim e field • NIST/SECG curve over a 384 bit prime field
Digest Algorithm	Select the message digest algorithm fro m the following options: • MD5 • SHA - 1 • SHA - 224 • SHA - 256 • SHA - 384 • **SHA - 512**. This is the default.

4. Click **Submit**. The generated certificate signing request is displayed.

Figure 6. Certificate Signing Request Generated

5. Copy the certificate signing request content and paste it into the Web form as part of the enrollment process.
6. To save and download the *Certificate Signing Request* file, click **Download CSR**.

Importing a Service or Client Certificate

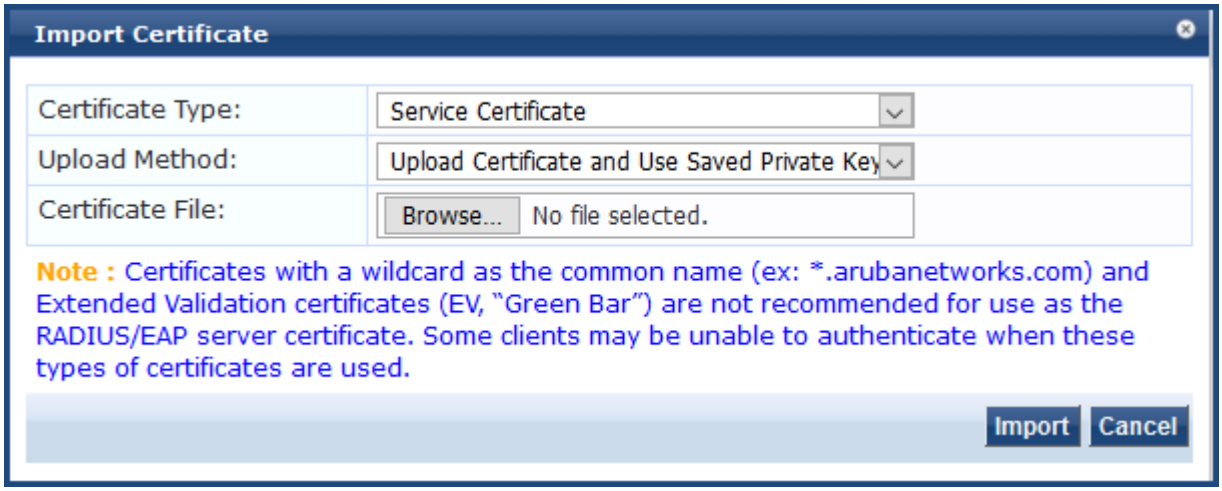
About this task

To import a service certificate or client certificate into the current Policy Manager server:

Procedure

1. Navigate to **Administration > Certificates > Certificate Store**.
2. Click the **Import Certificate** link.The **Import Certificate** dialog opens.
3. Click the **Certificate Type** menu and select a certificate type; **Service Certificate** or **Client Certificate**.
4. Click **Import**.

Figure 7. Import Service or Client Certificate Dialog



NOTE

For security reasons, certificates signed using SHA1RSA are not recommended. Importing certificates signed with stronger keys, such as RSA with a length of more than 1024 bits, is recommended.

NOTE

Policy Manager does not support importing the HTTPS Server Certificate chain or RADIUS Server Certificate chain in P7b Base64 format. A P7B file contains only certificates and chain certificates (intermediate certificate authorities), not the private key.

5. Specify the **Import Certificate** parameters as described in the following table:

Table 3. Import Service Certificate Parameters

Parameter	Action/Description
Certificate Type	The default value for this field is the certif icate type selected in Step 3. To import a nother certificate type, click the drop - d own menu and select any of the available certificate types: • Server Certificate • Service Certificate

Parameter	Action/Description
	• Client Certificate
Upload Method	Select one of the following upload methods: • Upload Certificate and Use Saved Private Key • Upload PKCS#12 Certificate (.pfx or .p12 only) • Upload Certificate and Private Key Files
Certificate File	Click **Browse** to specify the certificate file to be imported. **NOTE** Both certificates with a wildcard as the common name and Extended Validation certificates are not recommended for use as the RADIUS/EAP server certificate. Some clients may be unable to authenticate when these types of certificates are used.

6. Click **Import**.

Associating a RADIUS Service Certificate with a Service

About this task

The **Service Certificates** feature allows you to create multiple RADIUS service certificates (for details, see Service Certificates). Once you create the RADIUS service certificates you need, you can associate a service certificate with a specific Policy Manager service. This creates a virtual mapping between a Policy Manager service and a RADIUS service certificate.

To associate a service certificate with a specific Policy Manager service:

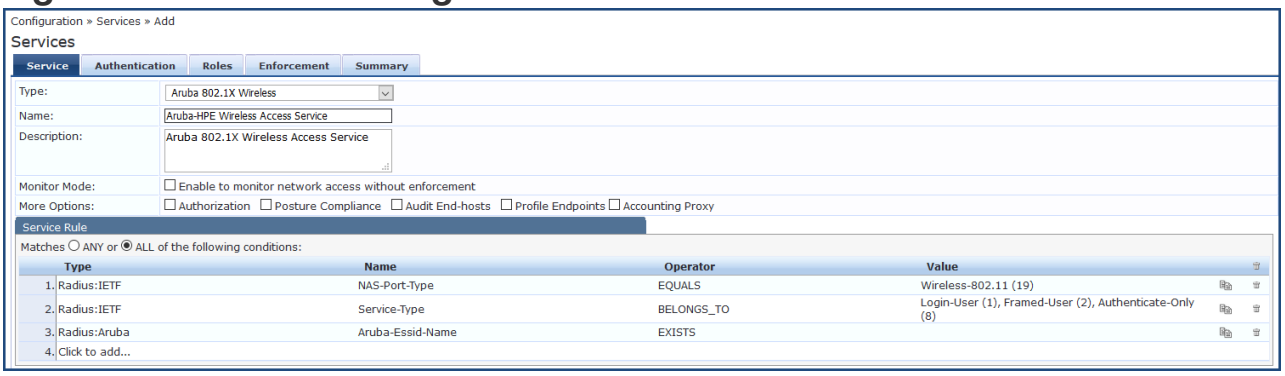
Procedure

1. Navigate to **Configuration > Services**. The **Services** page opens.

Figure 1. Services Page

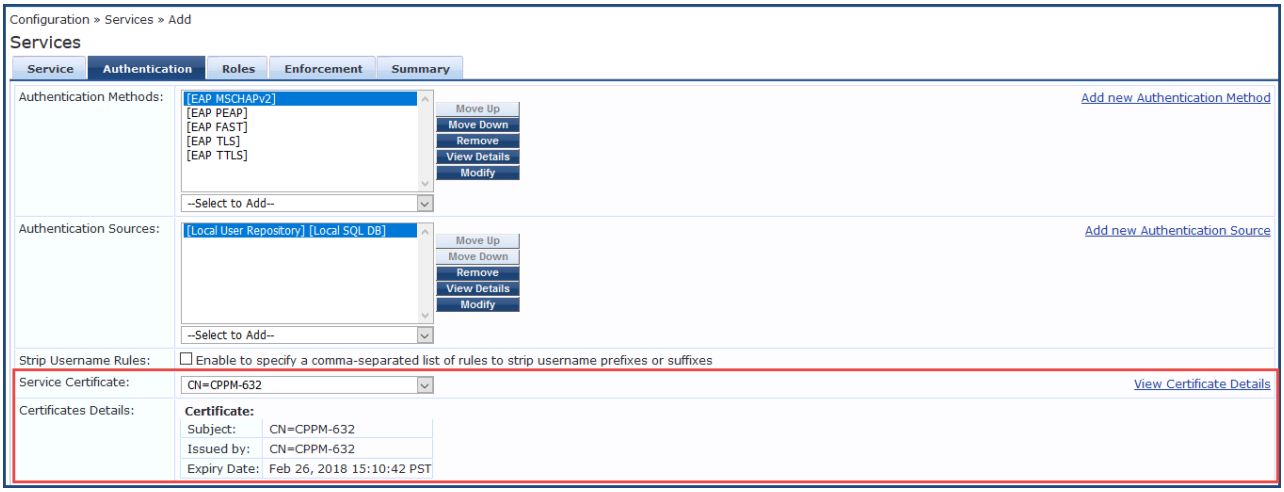
2. From the **Services** page, click **Add**. The **Add Services** page opens.

Figure 2. Add Services Page



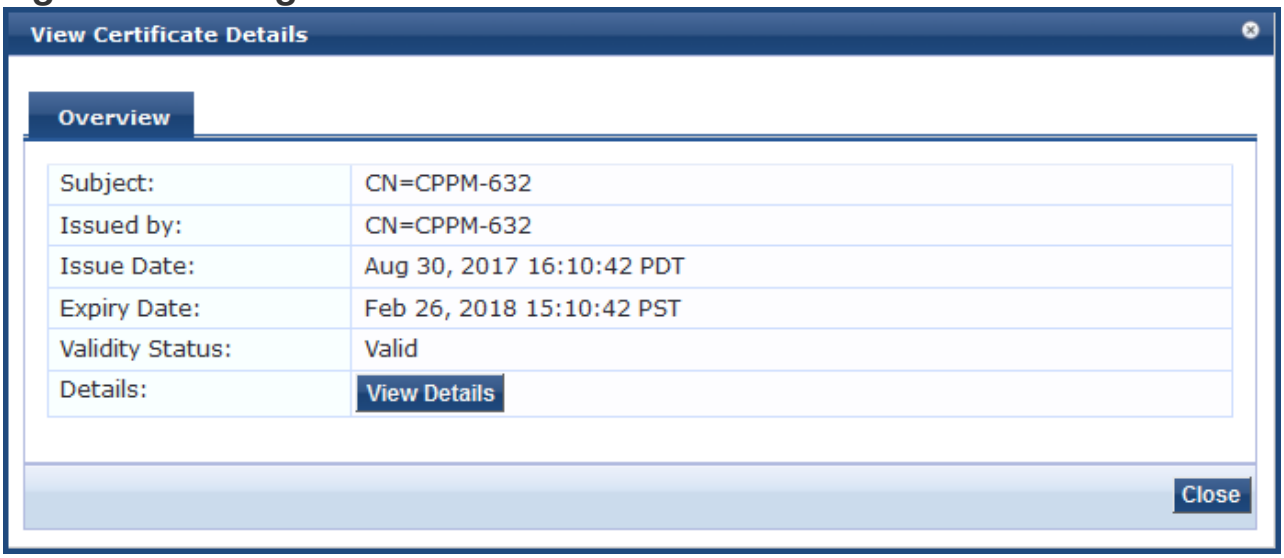
3. From the **Type** drop-down, select the type of service you want to add.
4. Enter the name of the new service.
5. Select the **Authentication** tab. The **Add Services > Authentication** tab opens.

Figure 3. Selecting a Service Certificate



6. Select the **Authentication Method**.
7. Specify the **Authentication Source**.
8. Select the **Service Certificate** you want to associate with this service.
9. To view the details, click **View Certificate Details**. The **View Certificate Details** window opens.

Figure 4. Viewing the Service Certificate Details



0. Click **Close**, then click **Save**. You return to the **Services** page. The following message is displayed:

   Service "<service_name>" has been added.

1. Repeat this procedure for each service with which you want associate a service certificate.

Certificate Trust List

The Certificate Trust List page displays a list of trusted Certificate Authorities (CA). On this page, you can add, view, or delete a certificate.

> **NOTE**
> When in FIPS mode, you cannot import the certificates that are created with the **MD5** digest algorithm to the **Certificate Trust List**. Users should be aware that, because importing a certificate to the Trust List with **Basic constraints CA = False** is not allowed in common criteria (CC) mode, SAML IdP encryption when ClearPass Policy Manager is acting as an IdP is not supported.

The following authority (CA) certificates are available in the Trust List. These certificates are disabled by default, and may be enabled and set as needed. These include:

- Aruba Networks Central Issuing CA 1.3
- Aruba Networks Central Issuing CA 1.4
- Aruba Networks Central Policy CA 1.1
- Aruba Networks Platform Root CA 1.0
- Aruba Networks Trusted Computing Issuing CA 5
- Aruba Networks Trusted Computing Issuing CA 4
- Aruba Networks Trusted Computing Issuing CA 6
- Dev2-Aruba Networks Trusted Computing CA 1

Subtopics

Certificate Trust List Page
Adding a Certificate
Enabling a Certificate
Deleting a Certificate

Certificate Trust List Page

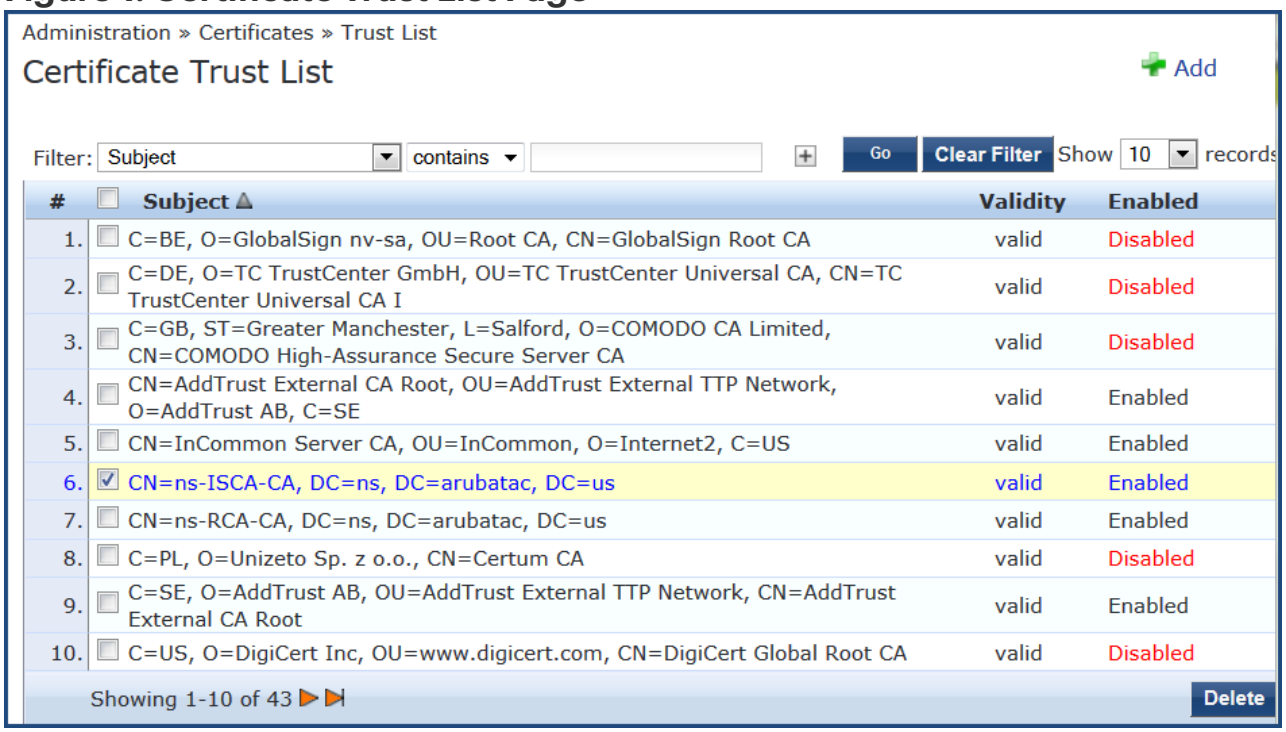
About this task

To display a list of trusted Certificate Authorities (CA):

Procedure

1. Navigate to **Administration** > **Certificates** > **Trust List**. The **Certificate Trust List** page displays.

Figure 1. Certificate Trust List Page



2. Use the **Filter** drop-down menu to filter the following **Certificate Trust List** display options:

   - Subject: Displays the Distinguished Name (DN) of the subject field in the CA certificate
   - Usage: Displays the Certificate Trust List usage type associated with the CA certificate
   - Validity: Indicates whether the CA certificate is valid or expired
   - Enabled: Indicates whether the CA certificate is enabled or disabled

> **NOTE**
>
> The maximum number of enabled CA trust lists that can be displayed is 200.

The USERTrust RSA Certificate Authority SHAS-2 root certificate authority (CA), Comodo RSA Certification Authority root certificate authority (CA), and Starfield Services Root Certificate Authority (CA) certificates are included by default in the

certificate trust list. The Comodo RSA Certification Authority certificate is enabled by default and is used by ClearPass Device Insight integrations to connect to Aruba Central. The Starfield Services root certificate is also enabled by default and is used for integration between ClearPass Policy manager and ClearPass Device Insight. The USERTrust RSA certificate is disabled by default. Policy Manager also includes the DigiCert root certificate authority (CA), which supports license activation.

The **Certificate Trust List** page can include the following certificates:

- *DoD (Department of Defense) certificates*: A DoD certificate enables a browser to trust web sites whose secure communications are authenticated by a DoD agency. These certificates are disabled by default.
- *Alcatel root certificate*: An Alcatel root certificate allows Alcatel-Lucent IP phones to authenticate using EAP-TLS. These certificates are disabled by default.
- ECC certificates - ECC certificates are SSL digitial certificates that use an elliptical curve cryptography (ECC) algorithm to encrypt and decrypt data between a client and web server.
- RSA certificates - RSA certificates are SSL digital certificates that use a RSA algorithm for data encryption to convert data into an undecipherable format so the original data cannot be compromised.

NOTE

The ECC and RSA certificates were added based on a partnership with Axis Communications.

Adding a Certificate

About this task

When you add a certificate to Policy Manager, the type of certificate you import must match the trust list usage type for the services specified by the *Usage* field in the **Add Certificate** dialog.

Table 1. Trust List Usage Types

Certificate Type	Trust List Usage Type
Secure Authentication and Authorization q ueries to AD/LDAP servers from ClearPass	**AD/LADP Servers** certificate trust list type

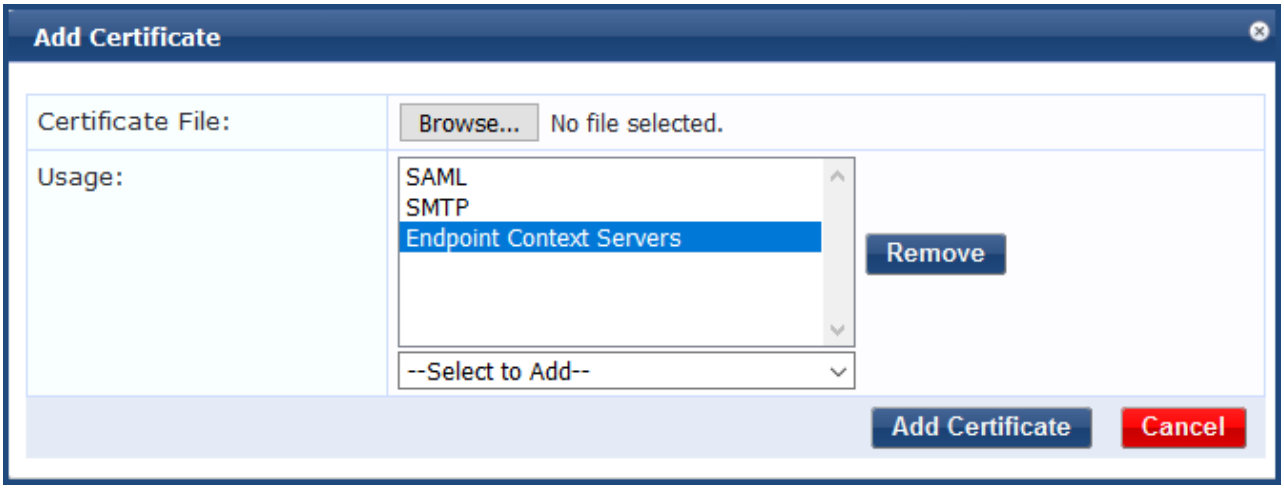
Certificate Type	Trust List Usage Type
CA certificate for the switches, gateways, and controllers from Aruba, used for certificate based authentication	**Aruba Infrastructure** trust list usage type
SSL certificate required to activate licenses from the Aruba license activation server **NOTE** The following certificate must be enabled in order to support license activation: CN=DigiCert Global Root G2,OU=www.digicert.com,O=DigiCert Inc,C=US	**Aruba Services** trust list usage type
Database Server Certificates	**Database** trust list usage type
RADIUS/EAP Server Certificates	**EAP** services trust list usage type
CA certificate for the server certificate used by MDM server or other Endpoint Context Server	**Endpoint Context Servers** trust list usage type
RadSec Server Certificates	**RadSec** services trust list usage type
Security Assertion Markup Language Certificates	**SAML** services trust list usage type
Simple Mail Transfer Protocol Certificates	**SMTP** services trust list usage type
CA certificates used to validate the client certificate for Onboard Enrollment Over Secure Transport (EST) client certificate authentication	**EST** services trust list usage type
HTTPS Server Certificates	**Others** services trust list usage type

To add a certificate to the Certificate Trust List:

Procedure

1. Navigate to **Administration > Certificates > Trust List**.
2. Click the **Add** link. The **Add Certificate** dialog opens.

Figure 1. Adding a Certificate to the Certificate Trust List



3. From the *Certificate File* field, click **Browse** to select the certificate file.
4. From the *Usage* field **Select to Add** drop-down menu, select one or more **Usage** types allowed with the certificate.

   The following **Usage** types are available:

   - AD/LDAP Servers
   - Aruba Infrastructure
   - Aruba Services
   - Database
   - EAP
   - Endpoint Context Servers
   - RadSec
   - SAML
   - SMTP
   - EST
   - Others
5. Click **Add Certificate**. The selected certificate is added to the Certificate Trust List.

Enabling a Certificate

About this task

When you enable a CA certificate, Policy Manager considers the entity whose certificate is signed by this CA to be trusted.

To enable a certificate that is currently disabled:

Procedure

1. Select the desired entry from the Certificate Trust List. The **View Certificate Details** page opens.

Figure 1. Viewing the Certificate Details

2. Click the **Enable** button. The following message is displayed:

   Certificate is enabled successfully

Deleting a Certificate

About this task

To delete a certificate:

Procedure

1. Navigate to **Administration > Certificates > Trust List**.
2. Select the check box to the left of the certificate.
3. Click **Delete**.

Certificate Revocation Lists

- About Certificate Revocation Lists
- Updating All Certificate Revocation Lists
- Adding a Certificate Revocation List
- Deleting a Certificate Revocation List

About Certificate Revocation Lists

A certificate revocation list (CRL) is a list of digital certificates that have been revoked by the issuing Certificate Authority (CA) before their scheduled expiration date and should no longer be trusted. Certificate revocation lists are a type of blocked list and they are used by various endpoints, including Web browsers, to verify whether a certificate is valid and trustworthy. Digital certificates are used in the encryption process to secure communications, often using the Transport Layer Security (TLS) or the Secure Sockets Layer (SSL) protocols. The certificate, which is signed by the issuing certificate authority, also provides proof of the identity of the certificate owner.

If both a base and delta CRL are available, both the base CRL URL and delta CRL URL need to be configured. The base CRL has the original set of revoked certificates, whereas the delta CRL contains only additional certificates that have been added over time.

Adding a Certificate Revocation List

About this task

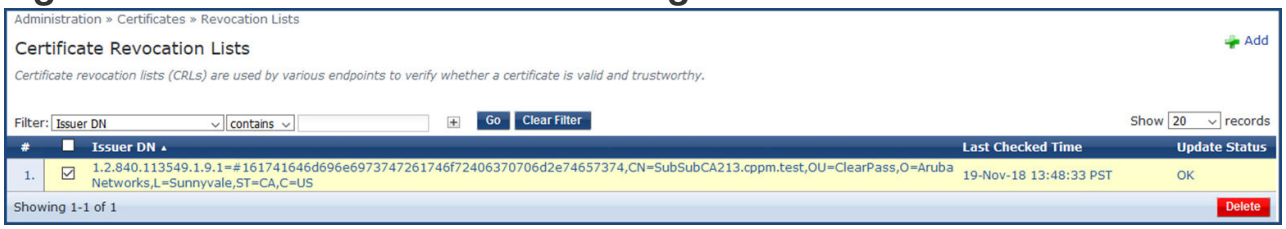
To add a certificate revocation list:

Procedure

1. Navigate to **Administration > Certificates > Revocation Lists**.

   The **Certificate Revocation Lists** page opens:

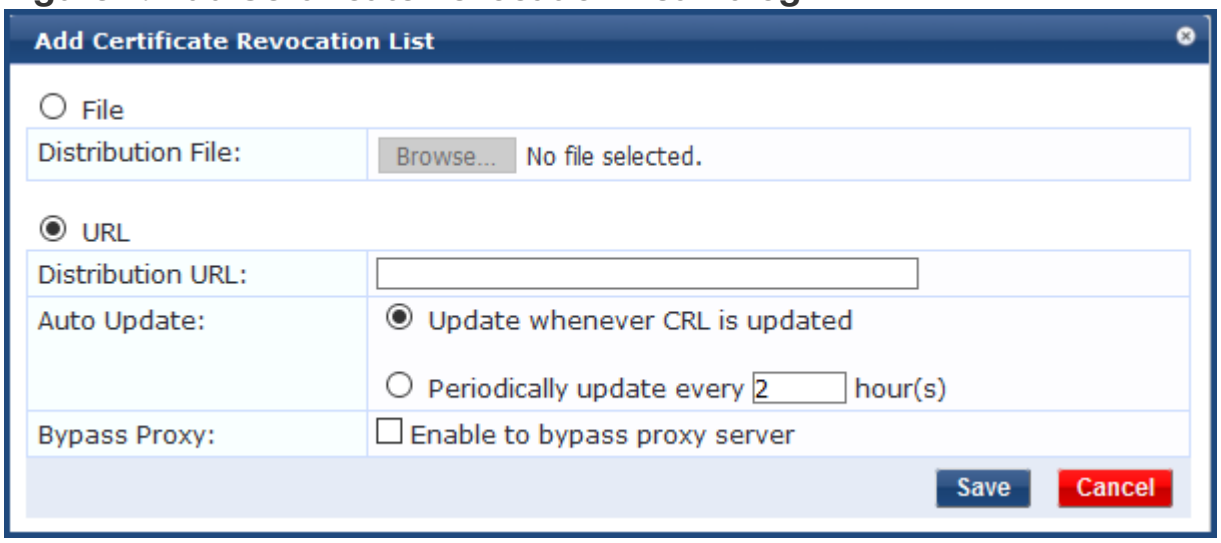
Figure 1. Certificate Revocation Lists Page



2. Click the **Add** link on the top-right section of the page.

   The **Add Certificate Revocation List** dialog opens:

Figure 2. Add Certificate Revocation List Dialog



3. Configure the **Add Certificate Revocation List** parameters as described in Table 2, then click **Save**.

Table 1. Add Certificate Revocation List Parameters

Parameter	Action/Description
File	Enable the **File** button to use a distributio n file as the Certificate Revocation List di stribution point.

Parameter	Action/Description
	File is enabled by default.
Distribution File	To select the distribution file to fetch the certificate revocation list, click **Browse** a nd select the CRL distribution file.
URL	Enable the **URL** button to use a URL as th e CRL distribution point. Selecting **URL** enables the **Distribution U RL** option.
Distribution URL	Specify the distribution URL to fetch the certificate revocation list.
Auto Update	Select **Update whenever CRL is updated** to update the CRL at intervals specified i n the list. To check periodically and at the specified frequency (in hours), select **Per iodically update every _____ hour(s)**. **NOTE** If the **Auto Update** value is s et to **Update whenever CRL is updated**, and the CRL file in CA server is updated befo re the next scheduled updat e, ClearPass will not update the CRL file.
Bypass Proxy	To bypass the proxy server, click the **Enab le to bypass proxy server** option.

Deleting a Certificate Revocation List

About this task

To delete a certificate revocation list:

Procedure

1. Navigate to **Administration > Certificates > Revocation Lists**.

2. Select the check box to the left of the certificate revocation list.
3. Click **Delete.**

Using Policy Manager Dictionaries

This section provides the following information:

- RADIUS Dictionary
- RADIUS Dynamic Authorization Templates
- TACACS+ Services Dictionaries
- Device Fingerprints Dictionary
- Dictionary Attributes
- Applications Dictionaries
- Configuring Endpoint Context Server Actions
- Enabling Ingress Event Dictionaries
- OnGuard Custom Scripts

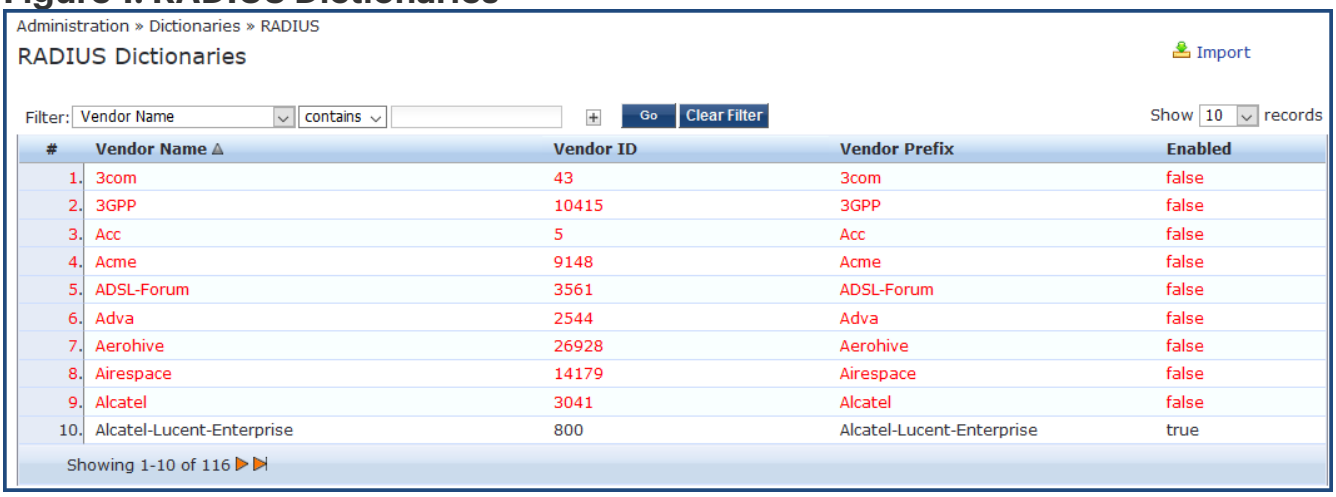
Subtopics

RADIUS Dictionary
RADIUS Dynamic Authorization Templates
TACACS+ Services Dictionaries
Device Fingerprints Dictionary
Dictionary Attributes
Applications Dictionaries
Configuring Endpoint Context Server Actions
Enabling Ingress Event Dictionaries
Managing Custom Windows Hotfixes
OnGuard Custom Scripts

RADIUS Dictionary

The **Administration** > **Dictionaries** > **RADIUS** page displays the list of available vendor dictionaries.

Figure 1. RADIUS Dictionaries

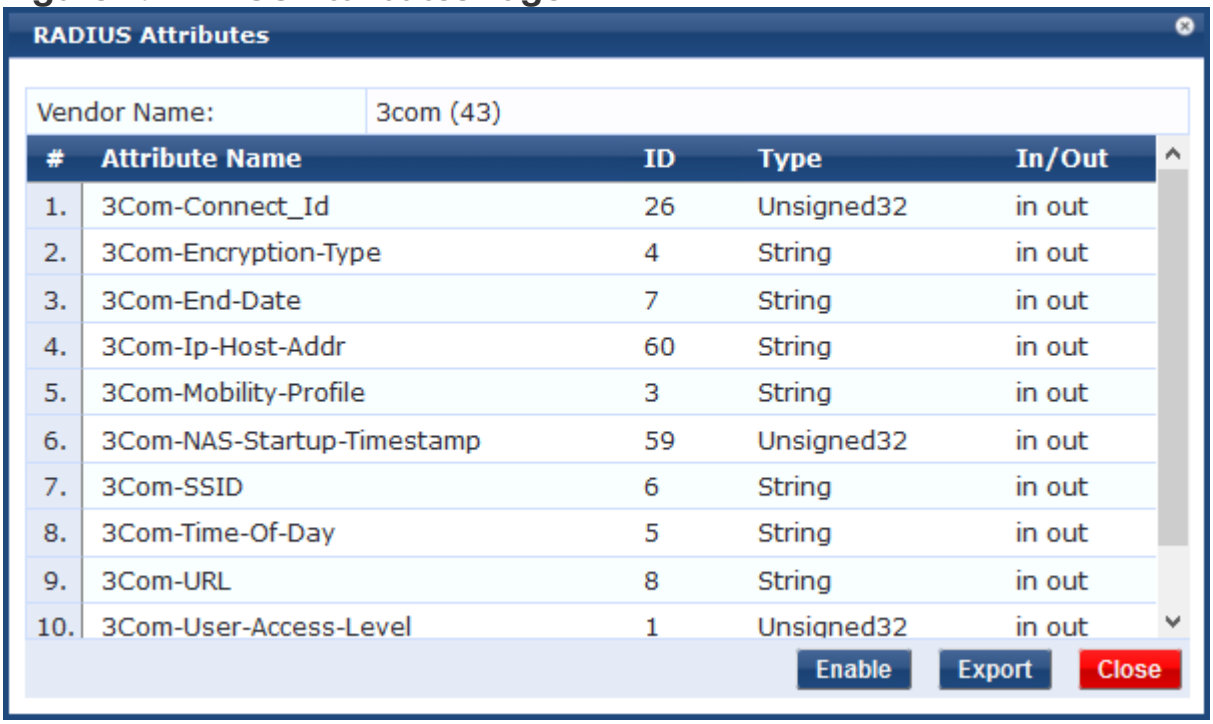


#	Vendor Name	Vendor ID	Vendor Prefix	Enabled
1.	3com	43	3com	false
2.	3GPP	10415	3GPP	false
3.	Acc	5	Acc	false
4.	Acme	9148	Acme	false
5.	ADSL-Forum	3561	ADSL-Forum	false
6.	Adva	2544	Adva	false
7.	Aerohive	26928	Aerohive	false
8.	Airespace	14179	Airespace	false
9.	Alcatel	3041	Alcatel	false
10.	Alcatel-Lucent-Enterprise	800	Alcatel-Lucent-Enterprise	true

Showing 1-10 of 116

To view the dictionary attributes, enable a dictionary, or export the dictionary, click on the row of interest. For example, click on vendor **3com** to see all its attributes and their data type. The following figure displays the RADIUS 3Com dictionary attributes page:

Figure 2. RADIUS Attributes Page



#	Attribute Name	ID	Type	In/Out
1.	3Com-Connect_Id	26	Unsigned32	in out
2.	3Com-Encryption-Type	4	String	in out
3.	3Com-End-Date	7	String	in out
4.	3Com-Ip-Host-Addr	60	String	in out
5.	3Com-Mobility-Profile	3	String	in out
6.	3Com-NAS-Startup-Timestamp	59	Unsigned32	in out
7.	3Com-SSID	6	String	in out
8.	3Com-Time-Of-Day	5	String	in out
9.	3Com-URL	8	String	in out
10.	3Com-User-Access-Level	1	Unsigned32	in out

Enabling a RADIUS Dictionary

About this task

To enable a RADIUS dictionary:

Procedure

1. From the RADIUS Dictionaries page, select the dictionary of interest.

2. Click the **Enable** button. You return to the RADIUS Dictionaries page. The following message is displayed:

   Dictionary has been successfully enabled.

Exporting a RADIUS Dictionary

About this task

To export a RADIUS dictionary:

Procedure

1. From the **RADIUS Dictionaries** page, select the dictionary of interest.
2. Click the **Export** button.
3. Open or save the XML file.
4. You can make modifications to the dictionary and import the file back into Policy Manager as needed.

Importing a RADIUS Dictionary

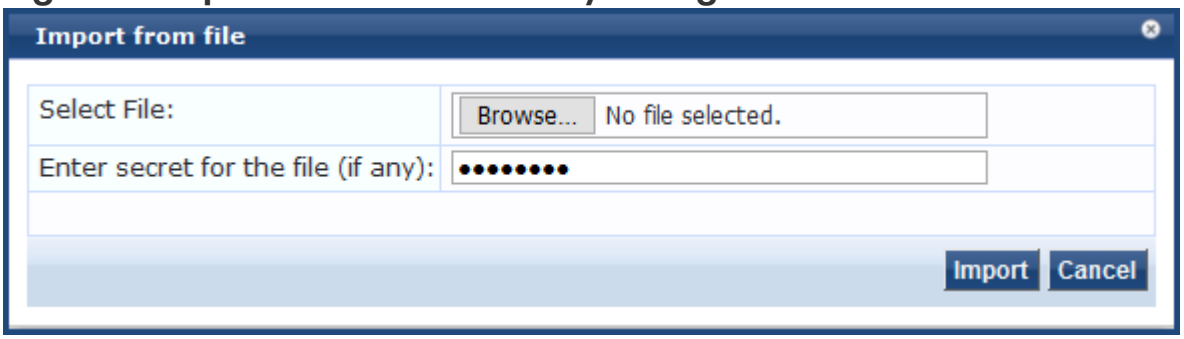
About this task

You can add additional dictionaries using the **Import** option. To add a new vendor dictionary:

Procedure

1. Navigate to **Administration > Dictionaries > RADIUS**.
2. Click the **Import** link. The **Import from file** dialog opens:

Figure 3. Import RADIUS Dictionary Dialog



3. Specify the **Import from file** parameters by browsing and selecting a file and providing any secret password parameters that may be required.

Editing an Existing RADIUS Dictionary

About this task

To edit an existing dictionary:

Procedure

1. Export an existing dictionary.
2. Edit the exported XML file.
3. Then import the dictionary.

RADIUS Dynamic Authorization Templates

About this task

The RADIUS Dynamic Authorization Templates allows users to add or update custom Change of Authorization (CoA) and Disconnect Message (DM) dictionaries for specific vendors. There is no user interface method for creating a RADIUS Dynamic Authorization Template—an existing template must be exported, modified as needed, and then reimported.

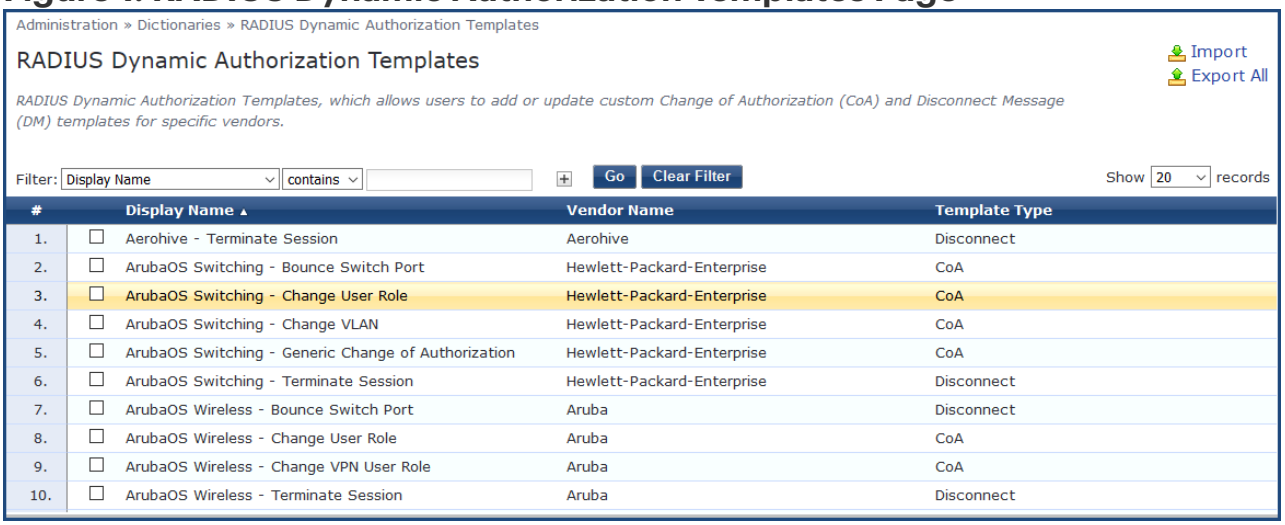
To access the RADIUS Dynamic Authorization Templates:

Procedure

1. Navigate to **Administration > Dictionaries > RADIUS Dynamic Authorization Templates**.

   The **RADIUS Dynamic Authorization Templates** page opens.

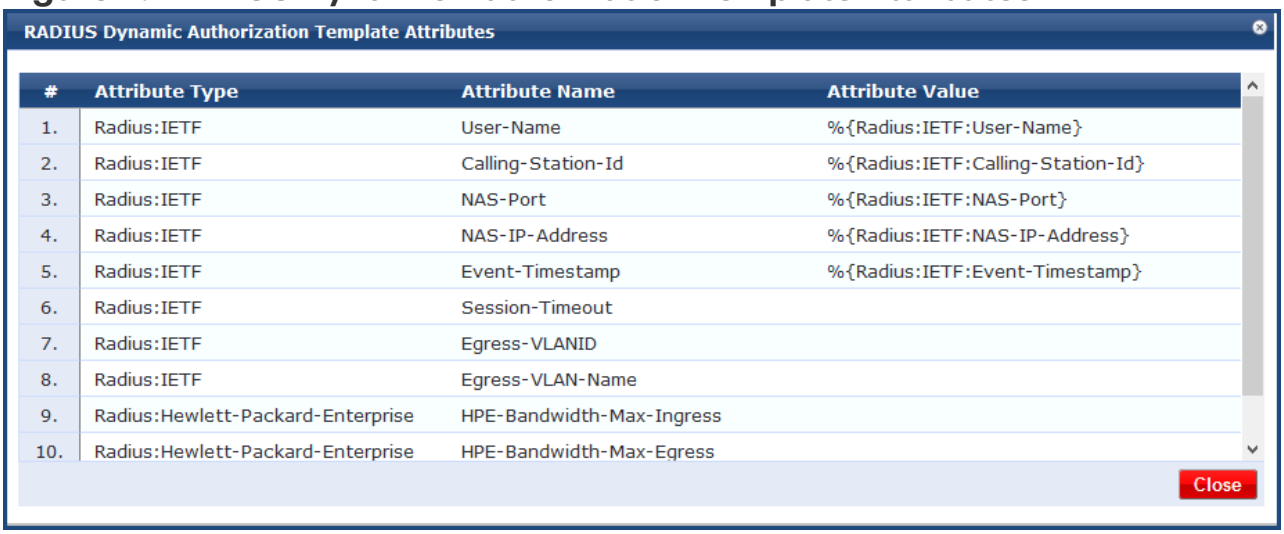
Figure 1. RADIUS Dynamic Authorization Templates Page



2. To enable or disable a template, and to export the template, click the desired row.

   The **RADIUS Dynamic Authorization Template Attributes** page opens.

Figure 2. RADIUS Dynamic Authorization Template Attributes



The following table describes the **RADIUS Dynamic Authorization Template Attributes** action buttons:

Table 1. RADIUS Dictionary Attributes Action Buttons

Parameter	Description
Export	Click the **Export** button to save the template file in XML format. You can make modifications to the template and import the file back into Policy Manager.
Enable/Disable	Enable or disable this CoA template. Enabling a template makes it appear in the Policy Manager Rules Editors (Service rules, Role mapping rules, etc.).

Editing a RADIUS Dynamic Authorization Template

About this task

To edit an existing template:

Procedure

1. From the **RADIUS Dynamic Authorization Templates** page, select the template you wish to modify, then click the **Export** button.

   The selected template is converted to XML format.

2. Edit the exported XML file as necessary.

3. Then import the RADIUS Dynamic Authorization template back into the Policy Manager server.

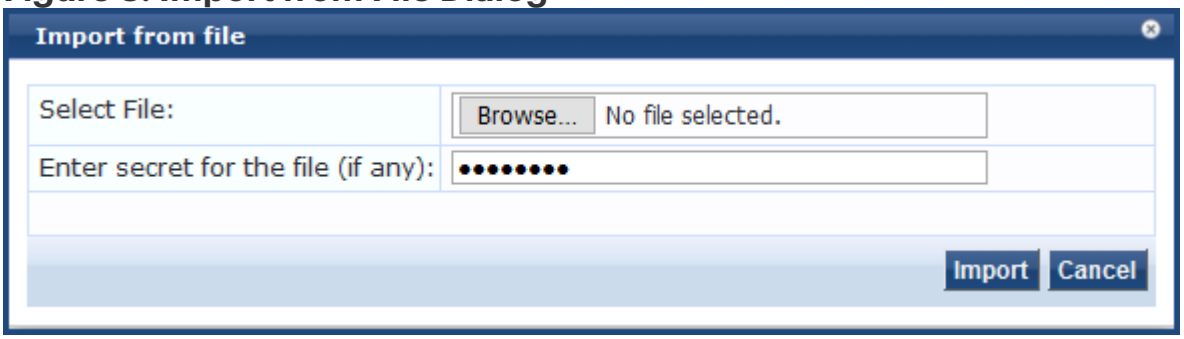
Importing a RADIUS Dynamic Authorization Template

To add additional templates:

From the **RADIUS Dynamic Authorization Templates** page, click the **Import** button.

The **Import from File** dialog opens.

Figure 3. Import from File Dialog



1. Browse to the template file you wish to import.

2. Enter the secret for the file if required

3. Click **Import**.

TACACS+ Services Dictionaries

About this task

This page allows admins to view the list of TACACS+ dictionaries, view their attributes, and enable or export dictionaries.

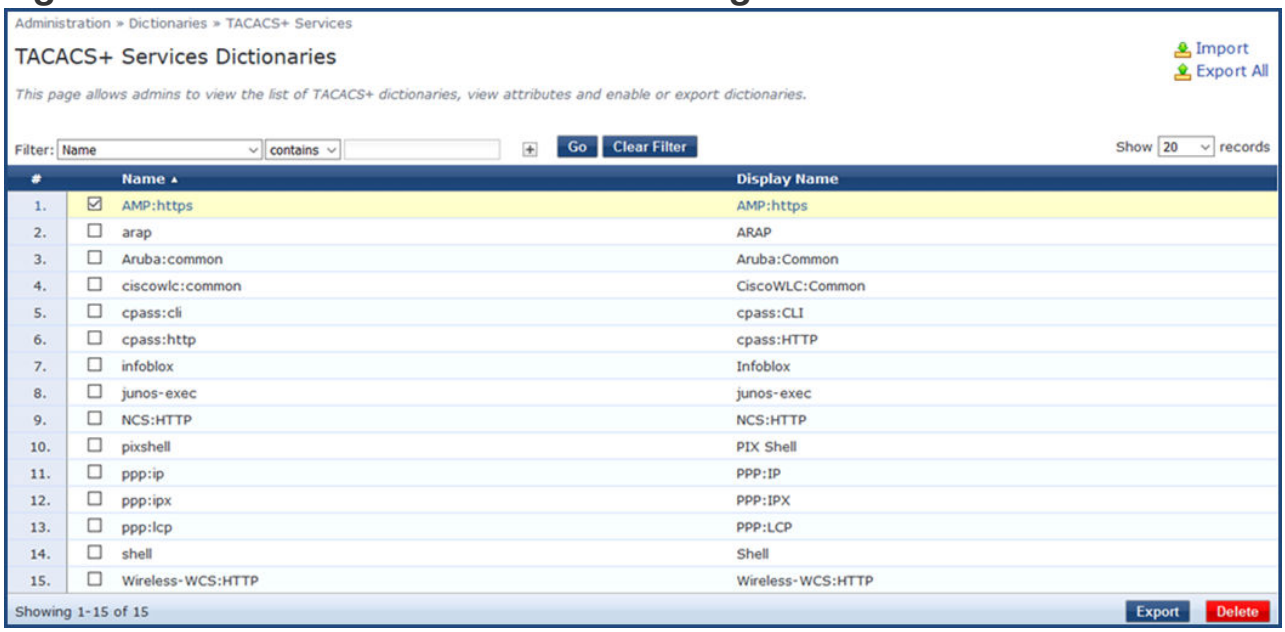
To view the TACACS+ service dictionaries:

Procedure

1. Navigate to **Administration > Dictionaries > TACACS+ Services**.

   The **TACACS+ Services Dictionaries** page opens.

Figure 1. TACACS+ Services Dictionaries Page



2. You can sort the dictionaries by using the Filter drop-down to sort by *Name* or *Display Name*.

Viewing Attributes and Data Types

About this task

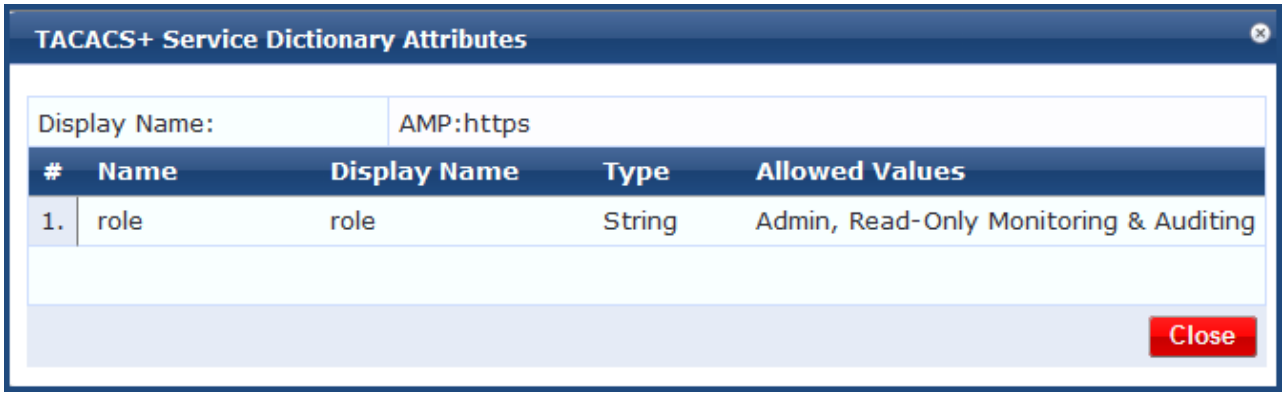
To see all the attributes and their data types:

Procedure

1. From the Services Dictionaries page, click the service row of interest.

   The **TACACS+ Service Dictionary Attributes** page opens.

Figure 2. TACACS+ Service Dictionary Attributes Page



2. Click **Close**.

Importing or Exporting TACACS+ Service Dictionaries

To import or export TACACS+ Service Dictionaries:

Table 1. TACACS+ Services Dictionaries Parameters

Parameter	Description
Import	Click to open the **Import Dictionary** dialog . Import the dictionary (XML file).
Export All	Export all TACACS+ services into one XML f ile containing multiple dictionaries.
Export a specific service dictionary	To export a specific service dictionary, sele ct a service and click **Export**.

Device Fingerprints Dictionary

Device fingerprints are composed of information collected about a network device in order to uniquely identify the device on subsequent sessions. A device fingerprint can fully or partially identify individual users or devices even when cookies and other tracking data are turned off. The **Device Fingerprints** page shows a listing of all the device fingerprints recognized by the Profile module. These fingerprints are updated from the Aruba Policy Manager Updates Portal (see Software Updates for more

information). Custom device fingerprints can be added and updated between Policy Manager servers.

> **NOTE**
> ClearPass now supports DHCPv6 profiling. When both DHCPv4 and DHCPv6 based fingerprints are received together in a profiling request, DHCPv4 has priority over DHCPv6. The Classification of the endpoint is based on DHCPv4 in this case. When pure DHCPv6 profiling request is received, endpoint classification is based on DHCPv6 option 16. Currently, DHCPv6 based classification support is added only for the Windows OS.

> **NOTE**
> Device fingerprinting is supported between ClearPass and Aruba AOS-CX switches. In a CX switch deployment, client protocol data (DHCP, HTTP, LLDP/CDP) is collected and processed by ClearPass from RADIUS accounting request packets. Keep in mind, the first time the device is profiled, the CoA is required to trigger re-authentication and associated accounting messages.

> **NOTE**
> If custom device fingerprints are configured, the custom rules are evaluated *before* the default rules are evaluated.

Subtopics

Viewing the Contents of a Device Fingerprints Dictionary
Exporting Custom Fingerprints

Viewing the Contents of a Device Fingerprints Dictionary

About this task

> **NOTE**
> When **Device Insight Integration** is enabled, the **Administration > Dictionaries > Device Fingerprints** option is hidden because Device Insight functionality replaces the Policy Manager functions available through the **Device Fingerprints** option (for more information, see Device Insight Integration Page).

The Device Fingerprints page lists all the device fingerprints recognized by the Profiler module. These fingerprints are updated from the Software Updates page (see Software Updates).

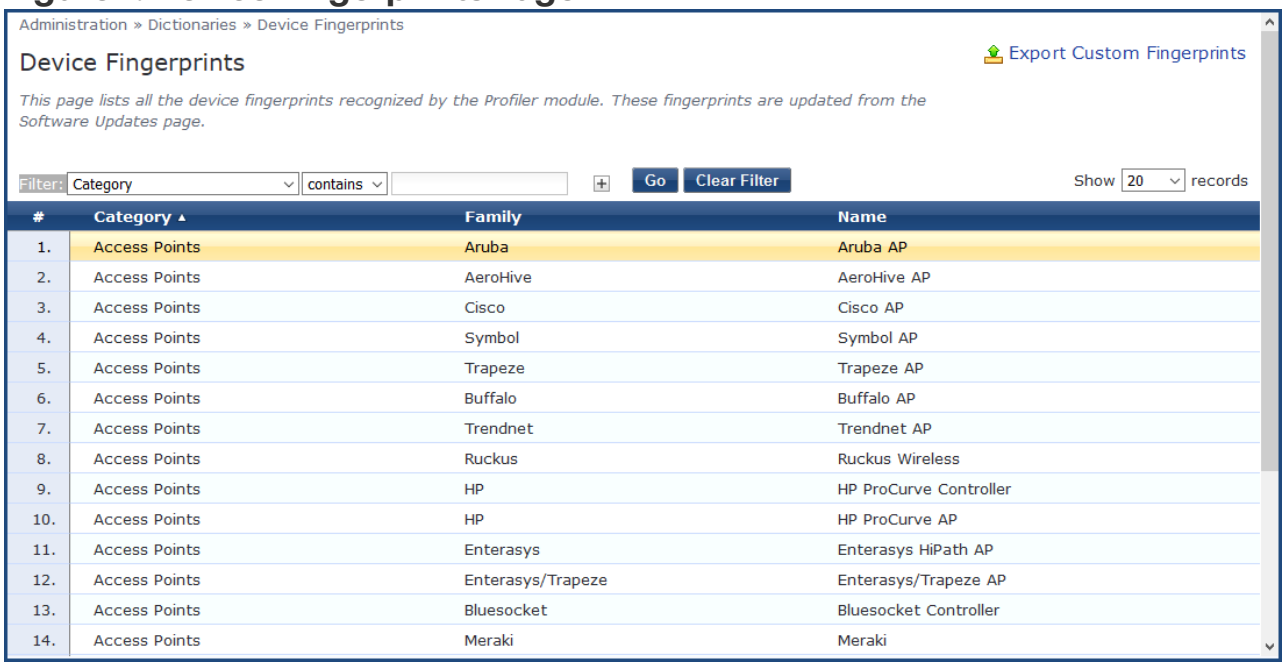
To view the contents of the Device Fingerprints Dictionary:

Procedure

1. Navigate to **Administration** > **Dictionaries** > **Device Fingerprints**.

   The **Device Fingerprints** page opens.

Figure 1. Device Fingerprints Page



2. To drill down and view additional details about the category, click on the line of interest in the Device Fingerprints list.

   The **Device Fingerprint Dictionary Attributes** page opens.

Figure 2. Device Fingerprint Dictionary Attributes Page

Exporting Custom Fingerprints

About this task

Policy Manager Policy Manager allows you to export configuration and administration-related information from the current Policy Manager server to an XML file. You can set this file to be password protected. You are not required to provide a new selection of rules when they are exported. All the associated rules for the Custom Created Fingerprints are exported.

To export custom fingerprints:

Procedure

1. From the **Device Fingerprints** page, click **Export Custom Fingerprints**.

   The **Export to file** dialog opens.

   Figure 1. Export to File Dialog

2. Specify the **Export to file** parameters as described in Table 1:

Table 1. Export to File Dialog Parameters

Parameter	Action/Description
Export file with password protection	To export the file with password protectio n, choose **Yes**.
Secret Key Verify Secret	You can choose to encrypt the exported data with a key. This protects data such a s shared secret from being visible in the exported file. To import it back, you speci fy the same key with which you exported the file. Enter the secret key. Then reenter the sec ret key.

3. Click **Export**.

The following figure shows Policy Manager an example of the exported custom fingerprints file:

Figure 2. Exported XML File

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<TipsContents xmlns="http://www.avendasys.com/tipsapiDefs/1.0">
  <TipsHeader exportTime="Mon Jan 29 11:07:52 IST 2018" version="6.7"/>
  <DeviceFingerprints>
    <DeviceFingerprint category="Spider" family="super" name="bat">
      <FingerprintRules>
        <FingerprintRule match-conditions="ALL">
          <RuleCondition name="mac_vendor" operator="contains" value="Super Micro Computer, Inc."/>
          <RuleCondition name="dhcp.option55" operator="contains">
            <valueList>1,3,6,12,15,28,42</valueList>
          </RuleCondition>
        </FingerprintRule>
      </FingerprintRules>
    </DeviceFingerprint>
    <DeviceFingerprint category="Earth" family="mars" name="venus">
      <FingerprintRules>
        <FingerprintRule match-conditions="ALL">
          <RuleCondition name="mac_vendor" operator="contains" value="Super Micro Computer, Inc."/>
          <RuleCondition name="dhcp.option60" operator="contains">
            <valueList>udhcp 1.12.0</valueList>
          </RuleCondition>
          <RuleCondition name="dhcp.options" operator="contains">
            <valueList>53,61,60,50,54,55</valueList>
          </RuleCondition>
        </FingerprintRule>
      </FingerprintRules>
    </DeviceFingerprint>
  </DeviceFingerprints>
</TipsContents>
```

Dictionary Attributes

Subtopics

About Dictionary Attributes
Adding a Dictionary Attribute
Modifying Dictionary Attributes
Importing Dictionary Attributes
Exporting All Dictionary Attributes
Exporting Selected Dictionary Attributes

About Dictionary Attributes

Use the **Attributes** dictionary page to specify unique sets of criteria for local users, guest users, endpoints, and devices. This information is used with role-based device policies for enabling appropriate network access. To view the contents of the attributes dictionary, navigate to **Administration > Dictionaries > Attributes**.

Figure 1. Dictionary Attributes Page

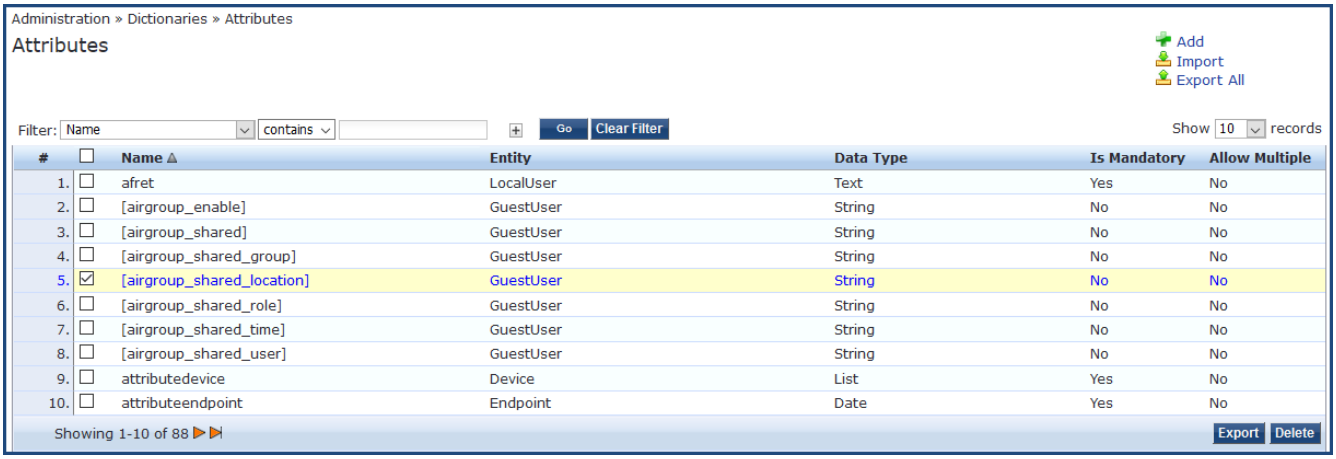


Table 1 describes the dictionary **Attributes**:

Table 1. Dictionary Attributes Parameters

Parameter	Action/Description
Filter	Use the Filter drop - down list to create a s earch based on the **Name**, **Entity**, **Data Typ e**, **Is Mandatory**, or **Allow Multiple** settings .
Name	The name of the attribute.
Entity	Indicates whether the attribute applies to a Local User, Guest User, Device, or Endpoint .
Data Type	Indicates whether the data type is String, Integer, Boolean, List, Text, Date, MAC add ress, or IPv4 address.
Is Mandatory	Indicates whether the attribute is required for a specific entity.
Allow Multiple	Indicates whether multiple attributes are al lowed for an entity.

Adding a Dictionary Attribute

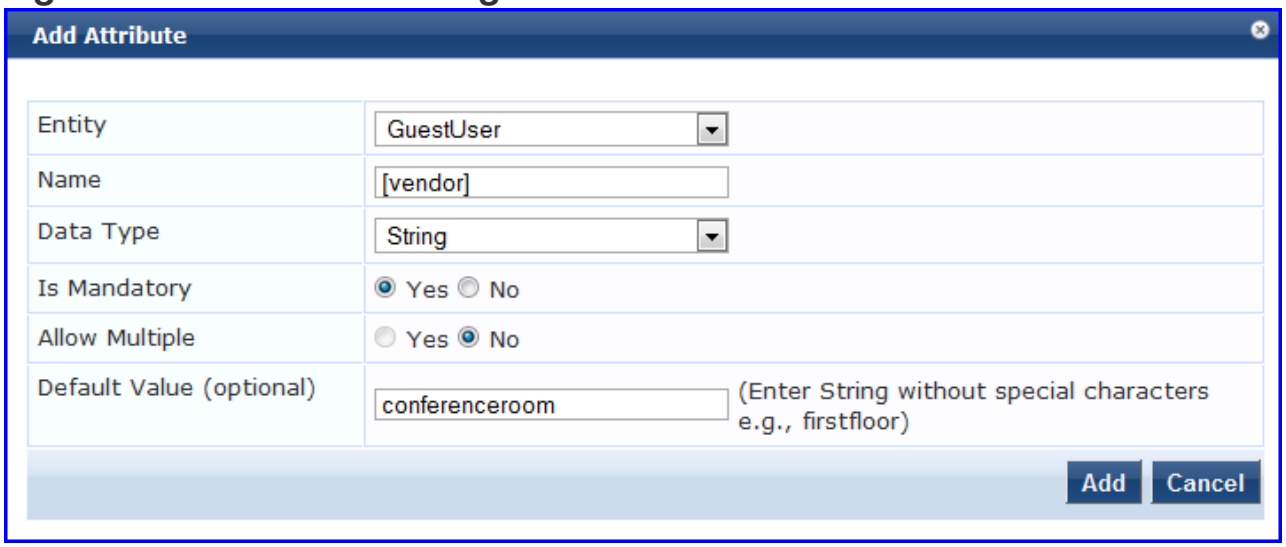
About this task

To add a dictionary attribute:

Procedure

1. From the **Attributes** page, click **Add**. The **Add Attribute** dialog appears.

Figure 1. Add Attribute Dialog



2. Specify the **Add Attribute** parameters as described in the following table, then click **Add**.

Table 1. Attribute Setting Parameters

Parameter	Action/Description
Entity	Specify whether the attribute applies to a Device, Endpoint, Guest User, Local User, or Onboard.
Name	Enter a unique ID for this dictionary attrib ute.
Data Type	From the drop - down, specify the data t ype.
Is Mandatory	Specify whether the attribute is required for a specific entity.
Allow Multiple	Specify whether multiple attributes are al lowed for an entity.

NOTE

Parameter	Action/Description
	Multiple attributes are not p ermitted if **Is Mandatory** is s pecified as **Yes**.
Default Value	Optionally, specify whether the default va lue is true or false.

Modifying Dictionary Attributes

About this task

To modify dictionary attributes in a service dictionary:

Procedure

1. From the **Attributes** page, select the dictionary attribute. The **Edit Attribute** page opens.
2. Make any necessary changes, then click **Save**.

Importing Dictionary Attributes

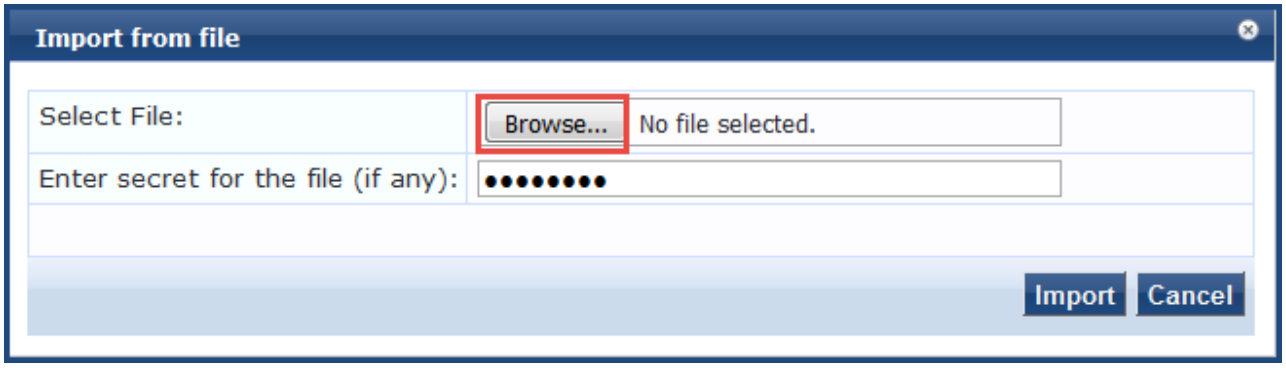
About this task

To import attributes:

Procedure

1. From the menu at the top-right section of the page, click **Import**. The **Import from File** dialog opens.

Figure 1. Importing Dictionary Attributes



2. Enter the **Import from File** parameters by browsing to the file to import and providing any required password requested.
3. When finished, click **Import**.

> **NOTE**
> The imported file is in XML format. To view a sample of this XML format, export a dictionary file and open it in an XML viewer.

Exporting All Dictionary Attributes

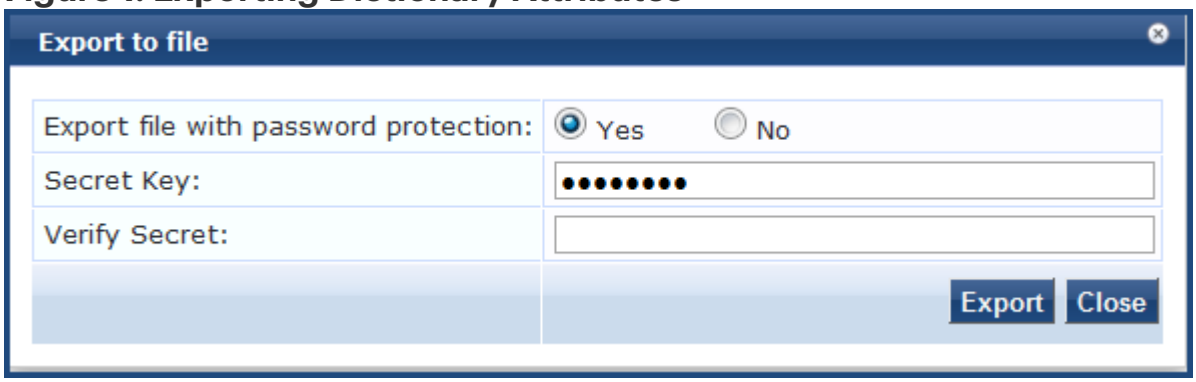
About this task

To export all the dictionary attributes at once:

Procedure

1. From the **Attributes** page, select **Export All**. The **Export to File** dialog opens.

Figure 1. Exporting Dictionary Attributes



2. Specify the **Export to File** parameters as described in Table 4.

Table 1. Export to File Parameters

Parameter	Action/Description
Export file with password protection	The **Yes** option is enabled by default. If you wish to disable password protectio n when exporting a file, select **No**.
Secret Key	If the file that you want to import is pas sword protected, enter the secret here. T hen verify the secret key.

3. When finished, click **Export**. The **TagDictionary.xml** file is created.
4. Download the file.

Exporting Selected Dictionary Attributes

About this task

To export selected dictionary attributes:

Procedure

1. On the **Attributes** dictionary page, select one or more attribute entries. The **Export** and **Delete** buttons on the lower right are now enabled.
2. Click **Export**. The **Export to File** dialog opens.
3. Specify the **Export to File** parameters as described in Table 4. When finished, click **Export**. The **TagDictionary.xml** file is created.
4. Download the file.

Applications Dictionaries

Application dictionaries define the attributes of the Onboard Policy Manager application and the type of each attribute. When Policy Manager is used as the Policy Definition

Point (PDP), it uses the information in these dictionaries to validate the attributes and data types sent in a WEB-AUTH request.

Viewing an Application Dictionary

About this task

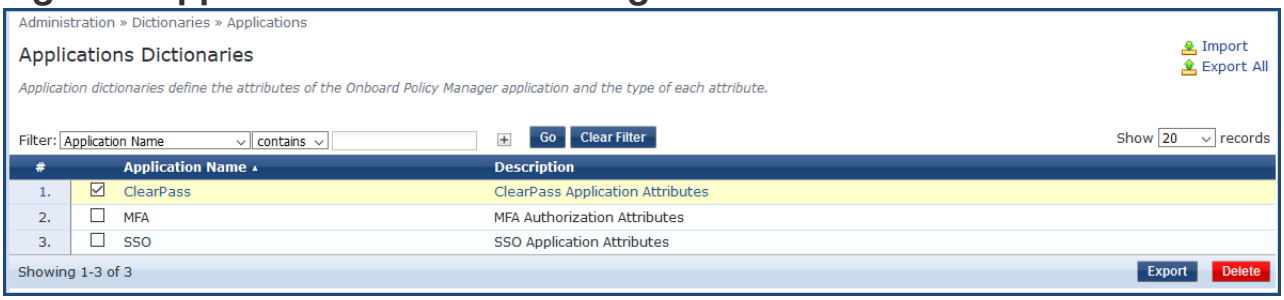
To view the contents of the application dictionary:

Procedure

1. Navigate to **Administration > Dictionaries > Applications**.

   The **Applications Dictionaries** page opens.

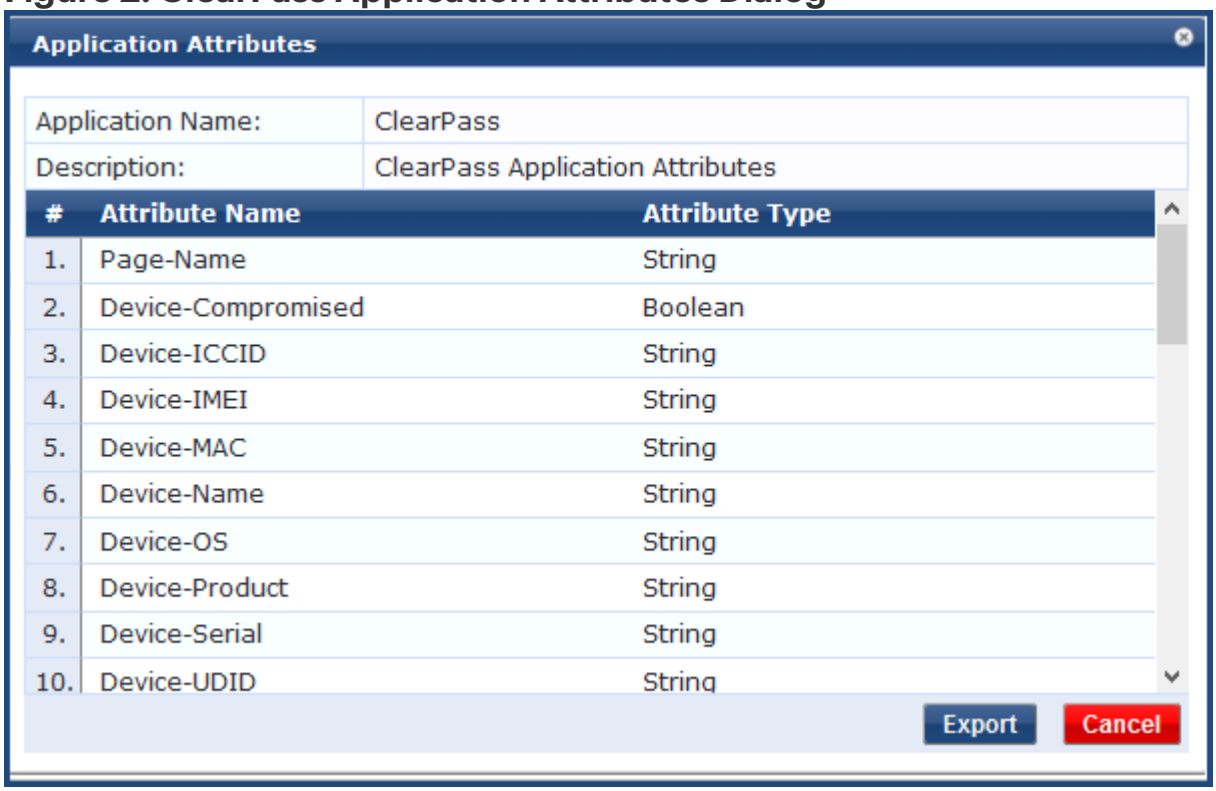
Figure 1. Applications Dictionaries Page



2. To see the application attributes, click the name of an application.

   The selected **Application Attributes** dialog box opens.

Figure 2. ClearPass Application Attributes Dialog



Deleting an Application Dictionary

About this task

In general, there is no need to delete an application dictionary. They have no effect on Policy Manager performance.

However, if you do need to delete an application dictionary:

Procedure

1. Navigate to **Administration > Dictionaries > Applications**.

   The **Applications Dictionaries** page opens.

2. Click the check box next to the application name you wish to delete.

3. Click **Delete**.

Configuring Endpoint Context Server Actions

About this task

Use the **Endpoint Context Server Actions** page to configure actions that are performed on endpoints, such as locking a device, triggering a remote, or enterprise wipe. The **Endpoint Context Server Actions** page displays a list of the context server actions that are available by default for each server type that Policy Manager supports.

> **NOTE**
> Only NMAP scans are supported for IPv6. All other scan types are ignored. For subnet scans, IPv6 is supported only for the NMAP scanning type. If an IPv6 subnet is configured, only the NMAP scan is run; no other scan types are run. For on-demand context server actions, even if other scan types are configured, only the NMAP scan is run if the IP address in the request is IPv6.

To configure endpoint context server actions:

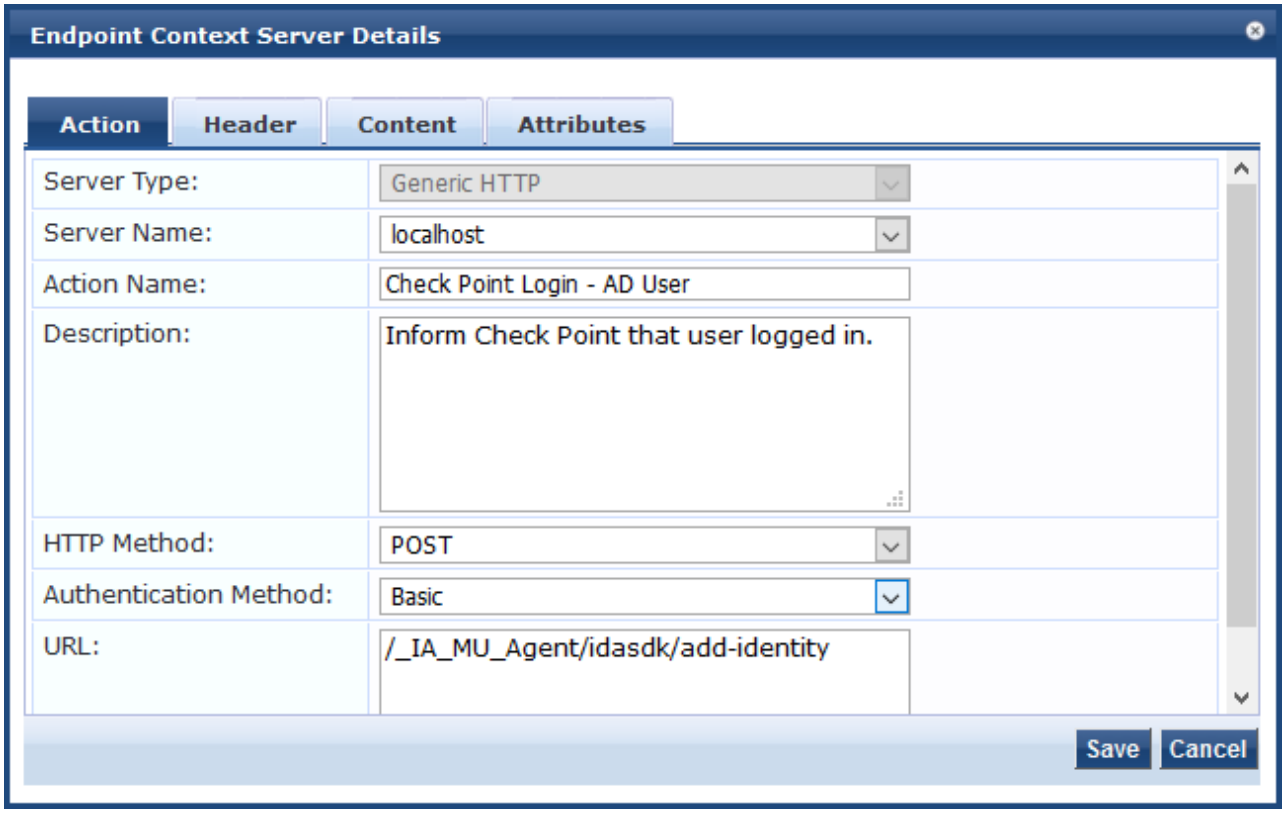
Procedure

1. Navigate to **Administration** > **Dictionaries** > **Context Server Actions**. The **Endpoint Context Server Actions** page opens.

   Figure 1. Endpoint Context Server Actions Page

2. From the **Endpoint Context Server Actions** page, click a row in the report. The **Endpoint Content Server Details** dialog opens.

Figure 2. Endpoint Context Server Details Dialog



NOTE

The **Is Sensitive** option is now available in the **Content** and **Header** tabs of **Endpoint Context Server Details**. When the **Is Sensitive** option is enabled, sensitive values such as tokens or passwords are masked in the UI. To allow values on the **Content** or **Header** tabs to be displayed, deselect the **Is Sensitive** option at **Administration > Dictionaries > Context Server Actions > Endpoint Context Server Actions > Endpoint Context Server Details**.

3. Click a tab to view details about the selected Endpoint Context Server action.
4. Make any changes required, then click **Save**.

Action Tab Parameters

Use the **Action** tab to specify the server type, action name, HTTP method, authentication method, and URL for the endpoint context server action. Specify the **Endpoint Context Server Details > Action** tab parameters as described in the following table, then click **Save**.

Table 1. Action Parameters—Endpoint Context Server Details

Parameter	Action/Description
Server Type	Specifies the server type configured when the server action was configured. You can select the server type from the drop - dow n list.
Server Name	Lists the context servers specific to the se rver type selected in the **Server Type** field. This field is visible only if you selected the service type **Generic HTTP**.
Action Name	Specifies the name of the action configure d.
Description	Provides additional information about the a ction specified.
HTTP Method	Select the HTTP method from the drop - d own: • DELETE • GET • POST (the default setting) • PUT • PATCH ◦ You can use the HTTP method **PATC H** to append content to existing end points/values.
Authentication Method	Select the authentication method: • **None**: Select **None** to disable the HTTP basic authentication for endpoint cont ext server actions. This exposes the con text server attributes to be used in cont ext server actions. • **Basic** • **OAuth2**: When you select **OAuth2** or **Ba sic & OAuth2**, the following fields are di splayed:

Parameter	Action/Description
	◦ *OAuth2 Client ID* ◦ *OAuth2 Client Secret* ◦ *OAuth2 Resource URL* ◦ *Validate OAuth2 Credentials*: **Validate** button: When you click **Validate**, Policy Manager retrieves a bearer token in order to validate whether the OAuth token server is reachable and the credentials are correct. If the credentials are not correct, an error message is displayed. • **Basic & OAuth2**: Selecting this option allows you to use both Basic and OAuth2 authentication. • **Certificate**: This option allows you to upload client certificates for certificate - based authentication of MDM context servers. When you select the **Certificate** authentication method, the following fields are displayed: **NOTE** When associating context server actions with an endpoint context server, be sure to specify the same authentication method that was configured for the endpoint context server.
URL	Indicates the URL for the selected HTTP method.

Header Tab Parameters

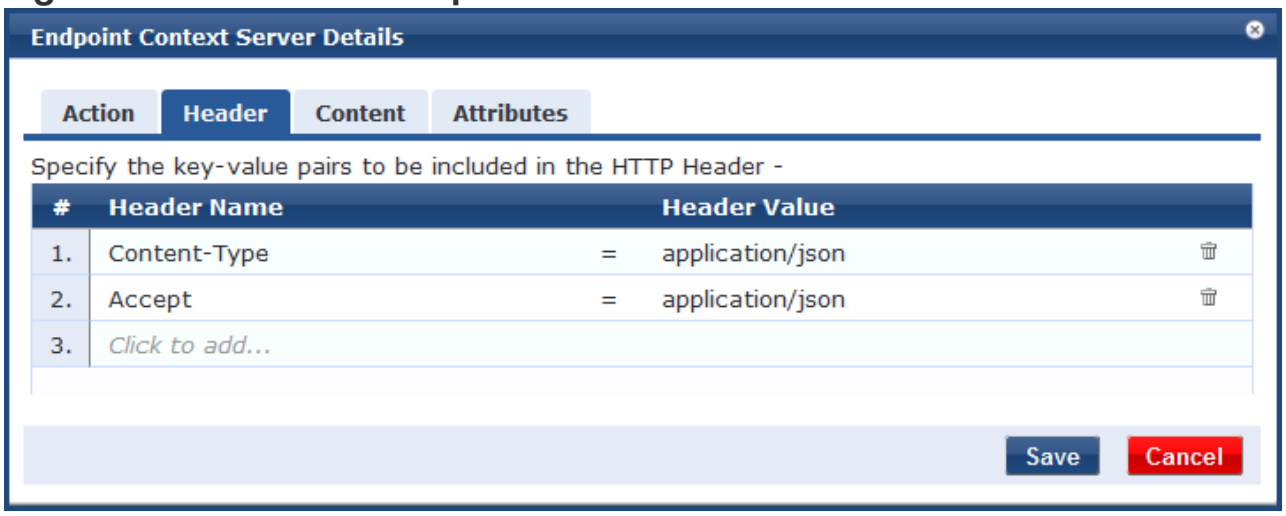
About this task

Use the **Header** tab to specify the key-value pairs to be included in the HTTP header.

Procedure

1. From the **Header** tab, select **Click to add**.

Figure 3. Header Tab—Endpoint Context Server Details



2. Specify the **Endpoint Context Server Details > Header** parameters as described in the following table:

Table 2. Header Parameters—Specifying Key Value Pairs

Parameter	Action/Description
Header Name	Enter the name of the heade e HTTP header.
Header Value	Specify the value of the hea me to be included in the HT

> **NOTE**
>
> If a Policy Manager server is used as an Endpoint context server to update the configuration using the REST API call, add the HTTP header, **Accept: application/json** in the context server action.

3. Repeat these steps for each key-value pairs you need to add.
4. Click **Save**.

Content Tab

Use the **Content** tab to specify a content type and add non-default context server attributes (see Figure 4).

The information in the Content window is the template of what will be posted to the server. The fields preceded by the % sign are replaced with their corresponding values.

Figure 4. Content Tab—Endpoint Context Server Details

Specify the **Endpoint Context Server Details—Content** parameters as described in the following table:

Table 3. Content Parameters—Endpoint Context Server Details

Parameter	Action/Description
Content - Type	Specify the type of the conte owing options: • CUSTOM • HTML • JSON • PLAIN • XML
Content	Specify the content. For example, { "mac": "%{Co - Address - NoDelim}","nma ECATEGORY}"}}.

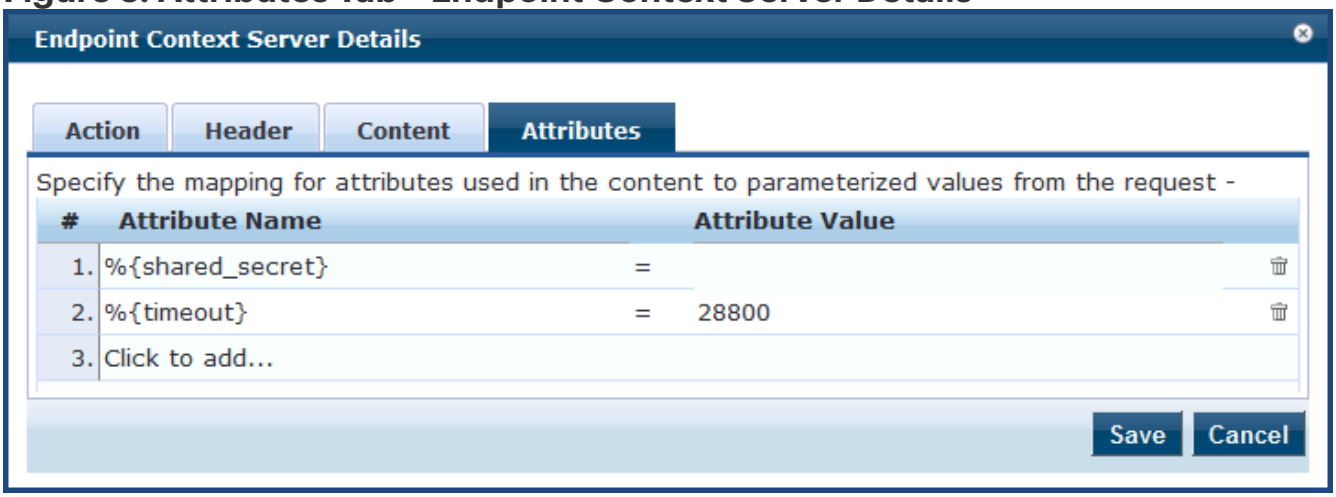
For related information, see Adding machine-os and host-type Endpoint Attributes).

Attributes Tab Parameters

About this task

Use the **Attributes** tab to specify the mapping for attributes used in the content to parameterized values from the request.

Figure 5. Attributes Tab—Endpoint Context Server Details



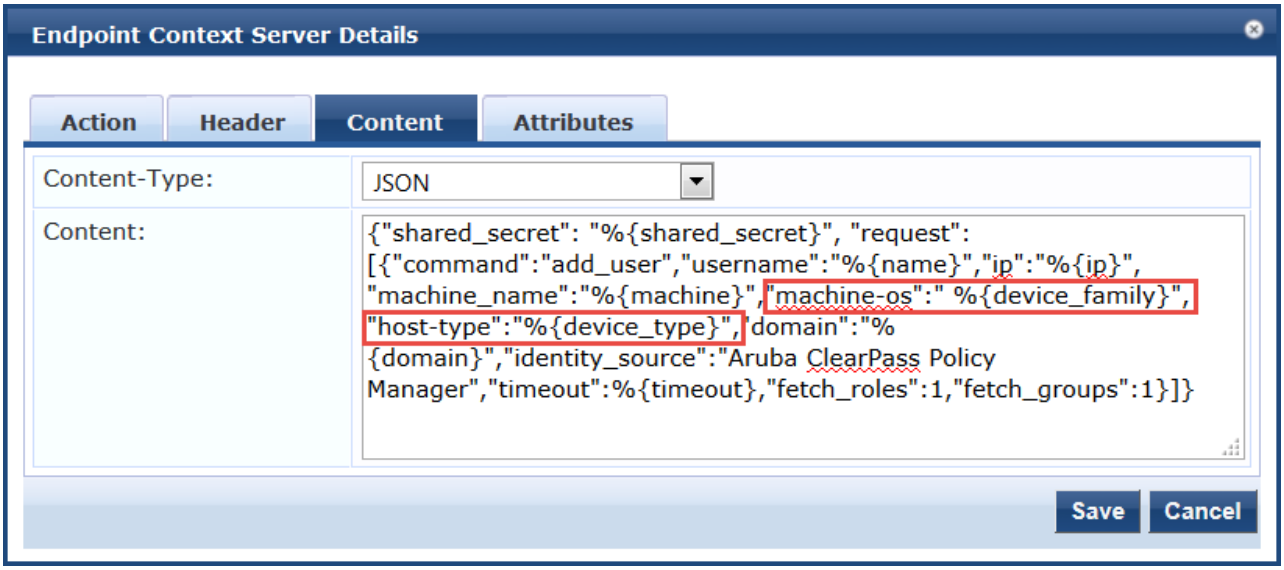
Procedure

1. Specify the **Endpoint Context Server Details > Attributes** parameters as described in the following table:

Table 4. Attributes Parameters—Endpoint Context Server Details

Parameter	Action/Description
Attribute Name	Enter attribute names and as names. These name/value pa ontext server actions.
Attribute Value	Enter the value for the select **bute Name** field.

Figure 6. Adding Endpoint Context Server Attributes



2. Click **Save**.

   You receive the following message:

   Context Server Action "Check Point Login (Generic HTTP)" updated successfully

Filtering an Endpoint Context Server Action Report

About this task

Use the **Filter** controls to configure a search for a subset of Endpoint Context Server Action items.

To filter an endpoint context server action report:

Procedure

1. Navigate to **Administration > Dictionaries > Context Server Actions.**

   The **Endpoint Context Server Actions** page opens (see Figure 1).

2. From the **Filter** drop-down, select a filter:

   - **ServerType**
   - **Action Name**
   - **HTTP method**

3. To add up to four new search fields, click the **Plus** icon.

4. Select a search argument.

   The search arguments are limited to **contains** or **equals**.

5. Click **Go**.

Storing Authentication Attributes in the Zone Cache

About this task

Policy Manager provides a context server action that serves as an example of how to store attributes from an authentication session into the Zone Cache for use in policy enforcement during an authentication session. The administrator can then clone it and customize the context server action to meet their requirements.

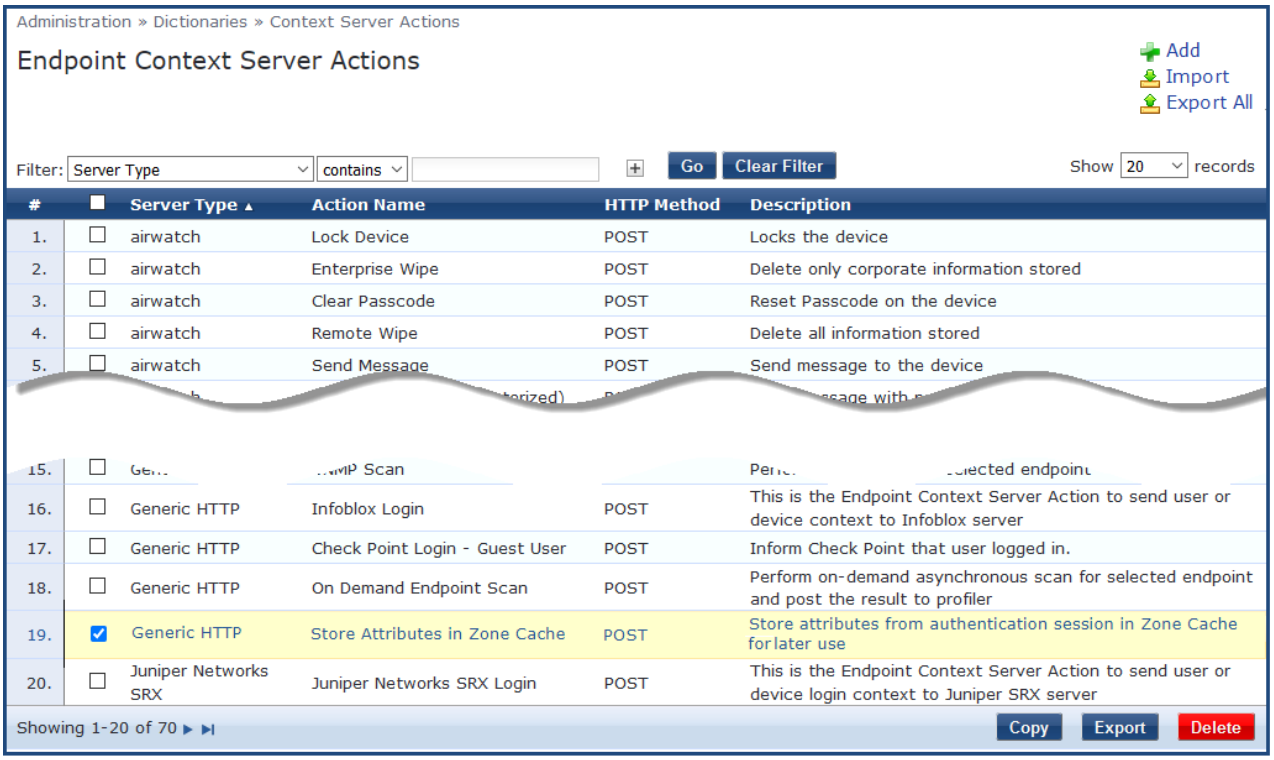
To access this context server action:

Procedure

1. Navigate to **Administration > Dictionaries > Context Server Actions**.

   The **Endpoint Context Server Actions** page opens.

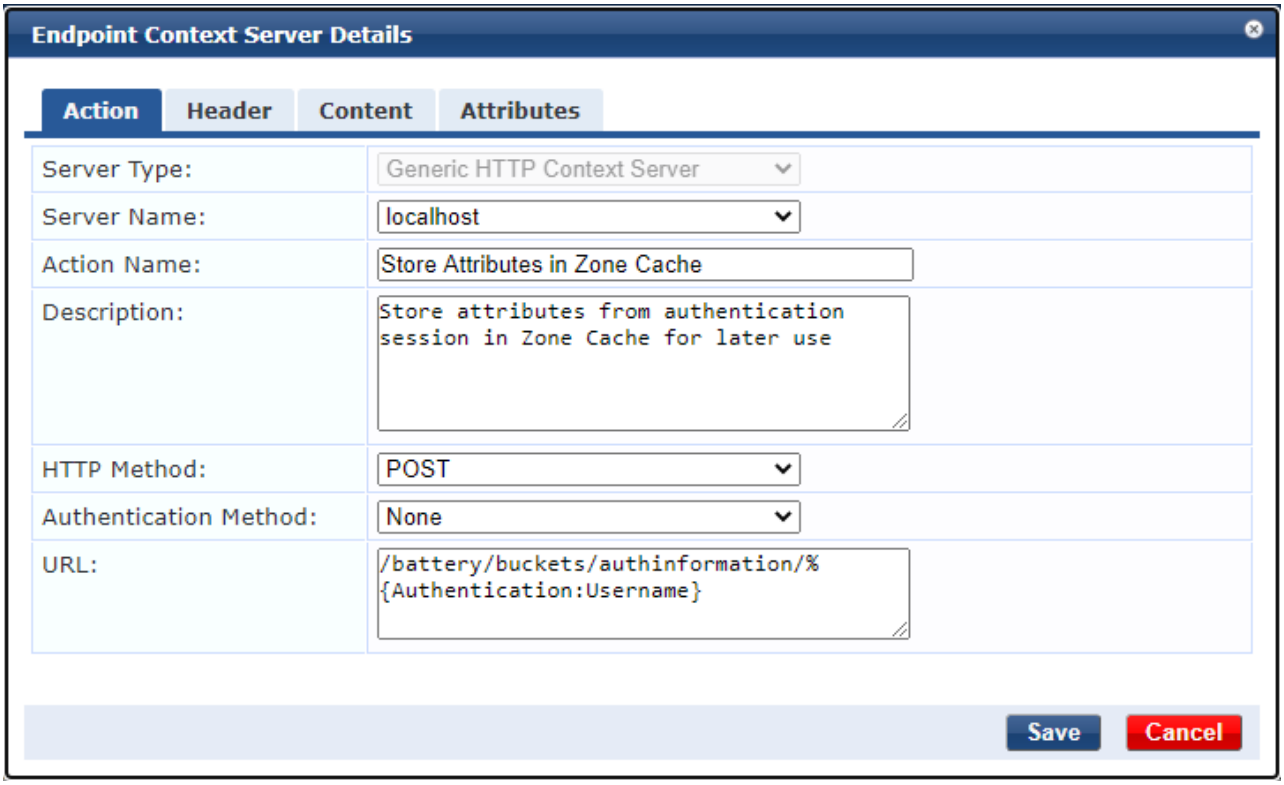
Figure 7. Endpoint Context Server Actions Page: Store Attributes inZone Cache



2. Scroll down and select the **Store Attributes in Zone Cache** endpoint context server action.

   The **Endpoint Context Server Details** page opens.

Figure 8. Endpoint Context Server Details: Store Attributes inZone Cache



3. Modify and customize the **Store Attributes in Zone Cache** endpoint context server action as needed.

   At **Configuration > Authentication > Sources**, the **Zone Cache Repository** authentication source will fetch the attributes stored in the Zone Cache and proceed with authentication.

Adding machine-os and host-type Endpoint Attributes

About this task

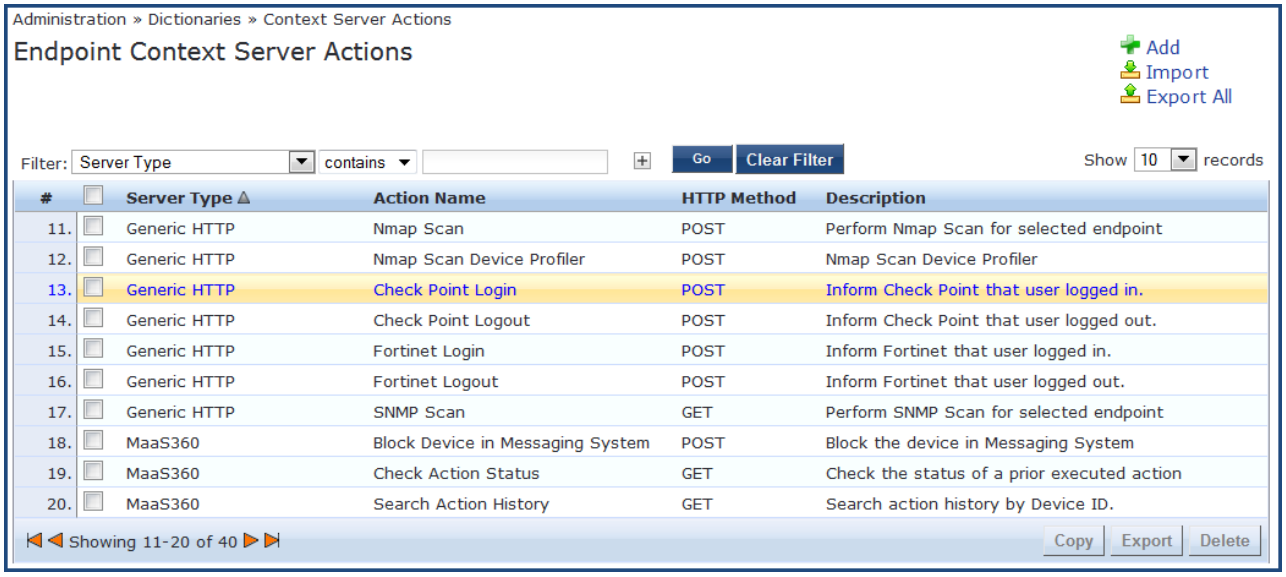
To be able to indicate the entire OS family (Android, Windows, Linux, etc.) and the type of device (iPad, iPhone, etc.), you can add the **machine-os** Device Family attribute and the **host-type** Device Type attribute to the default set of endpoint context attributes provided in the Content window:

To add the **machine-os** and **host-type** endpoint context attributes:

Procedure

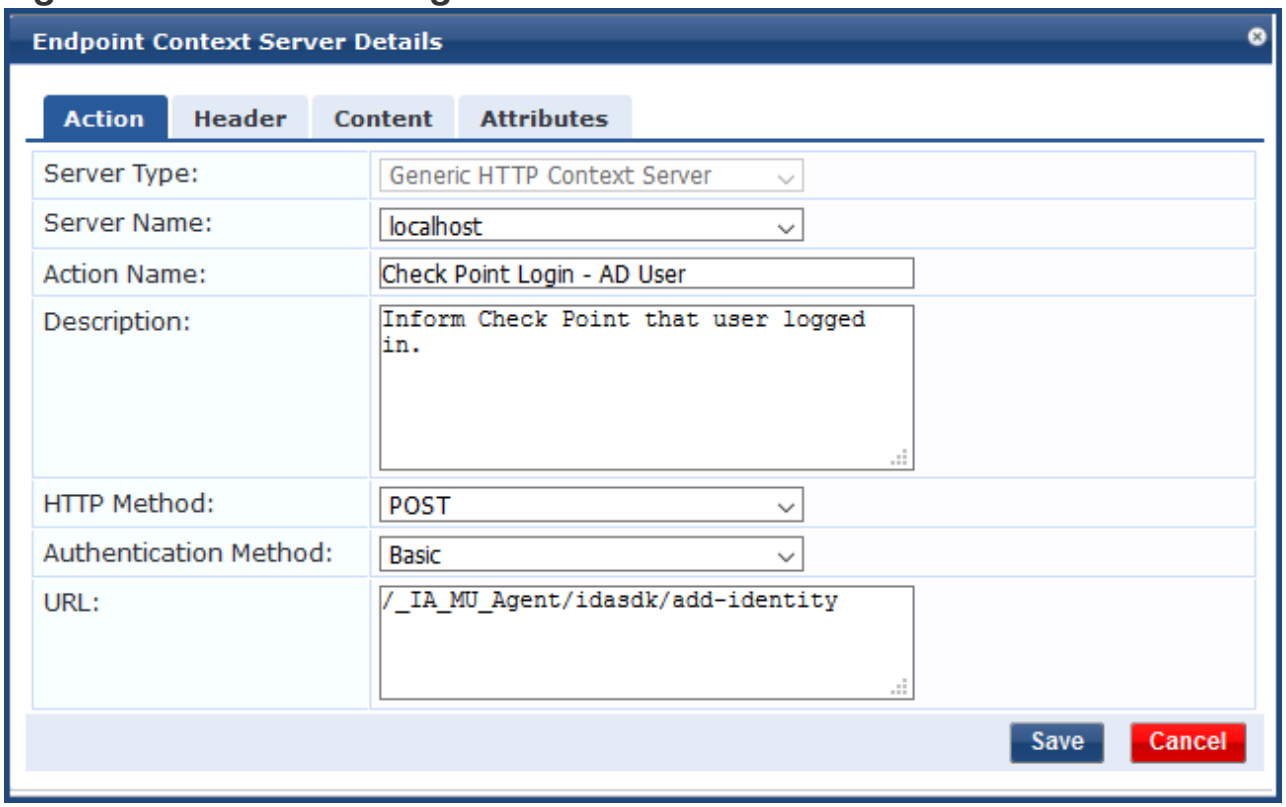
1. Navigate to **Administration > Dictionaries > Context Server Actions**. The **Endpoint Context Server Actions** page opens.
2. Scroll to and select the **Generic HTTP/Check Point Login - AD User** server action.

Figure 9. Selecting the Check Point Login Server Action



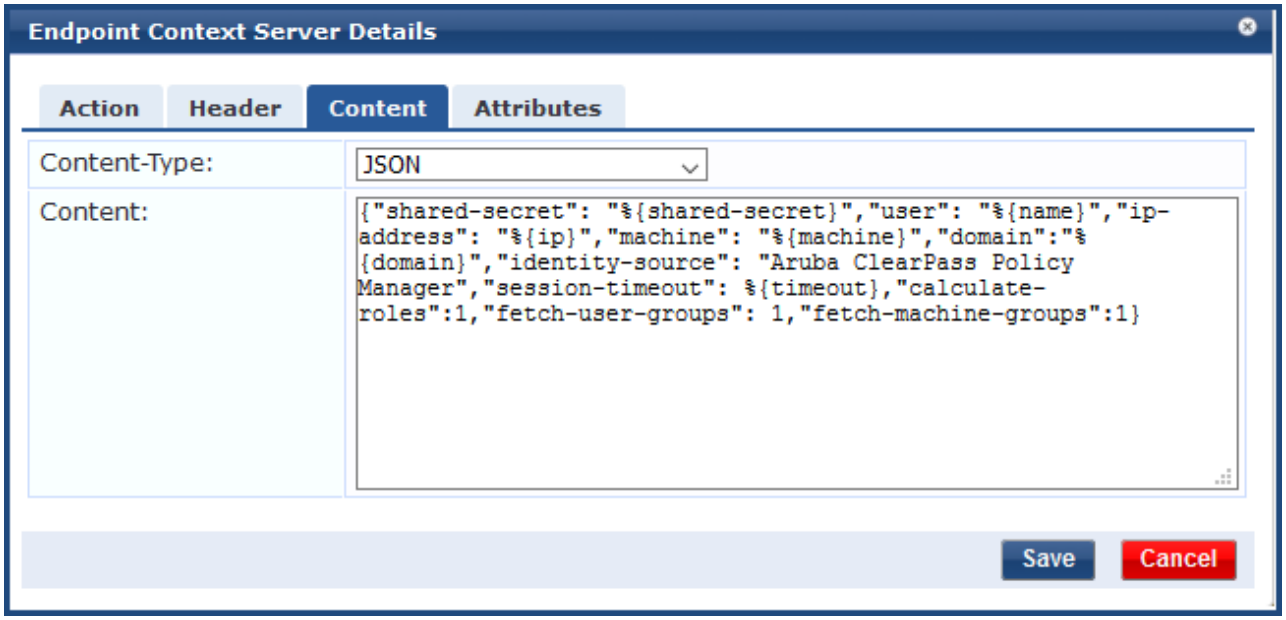
The **Endpoint Context Server Details** dialog opens.

Figure 10. Check Point Login - AD User Server Details



3. Select the **Content** tab.

Figure 11. Check Point Login - AD User Server Details > Content Tab



4. In the *Content* field, add the following attributes (see Figure 6):
   - `"machine-os":" %{device_family}"`
   - `"host-type":"%{device_type}"`
5. Click **Save**.

Enabling Ingress Event Dictionaries

About this task

By default, a set of ingress event dictionaries are available and initially set to disabled. You must enable the ingress event dictionaries before you proceed. IntroSpect sends Entity Syslog data and Alert Syslog data to the Policy Manager server using the Ingress Event engine to perform RADIUS COA (Change of Authorization) or Post Auth operations.

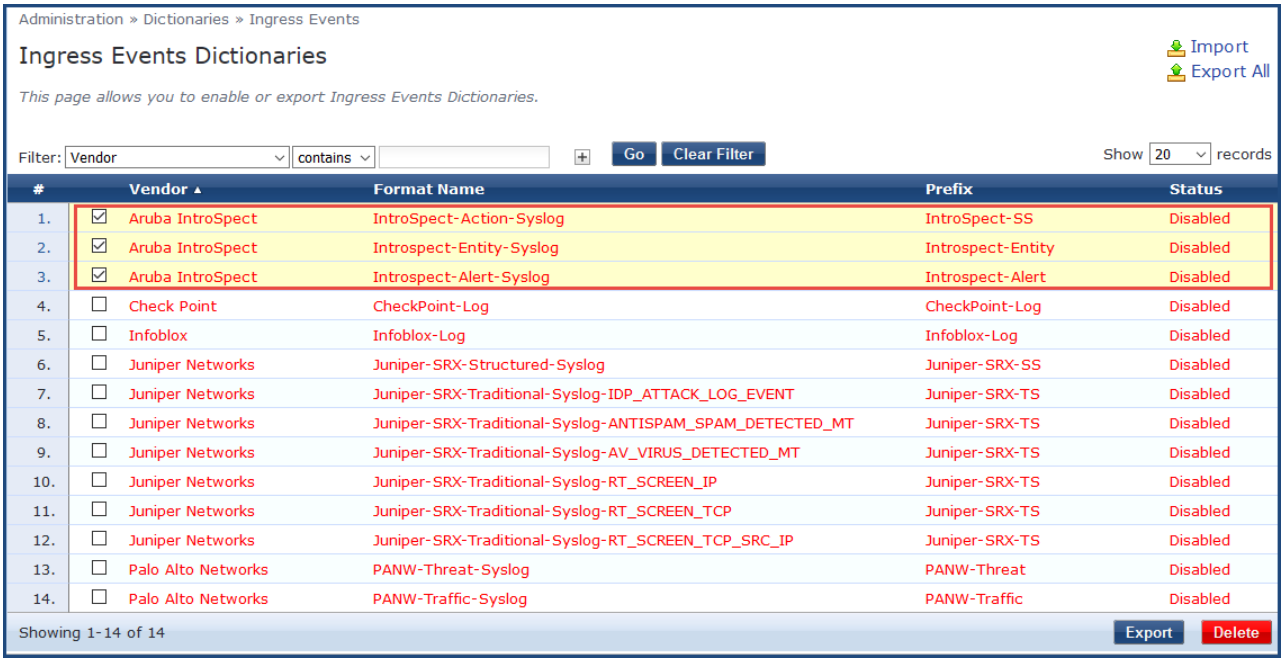
To enable an ingress event dictionary:

Procedure

1. Navigate to **Administration > Dictionaries > Ingress Events**.

   The **Ingress Events Dictionaries** page opens. The set of ingress event dictionaries are displayed. By default, they are disabled.

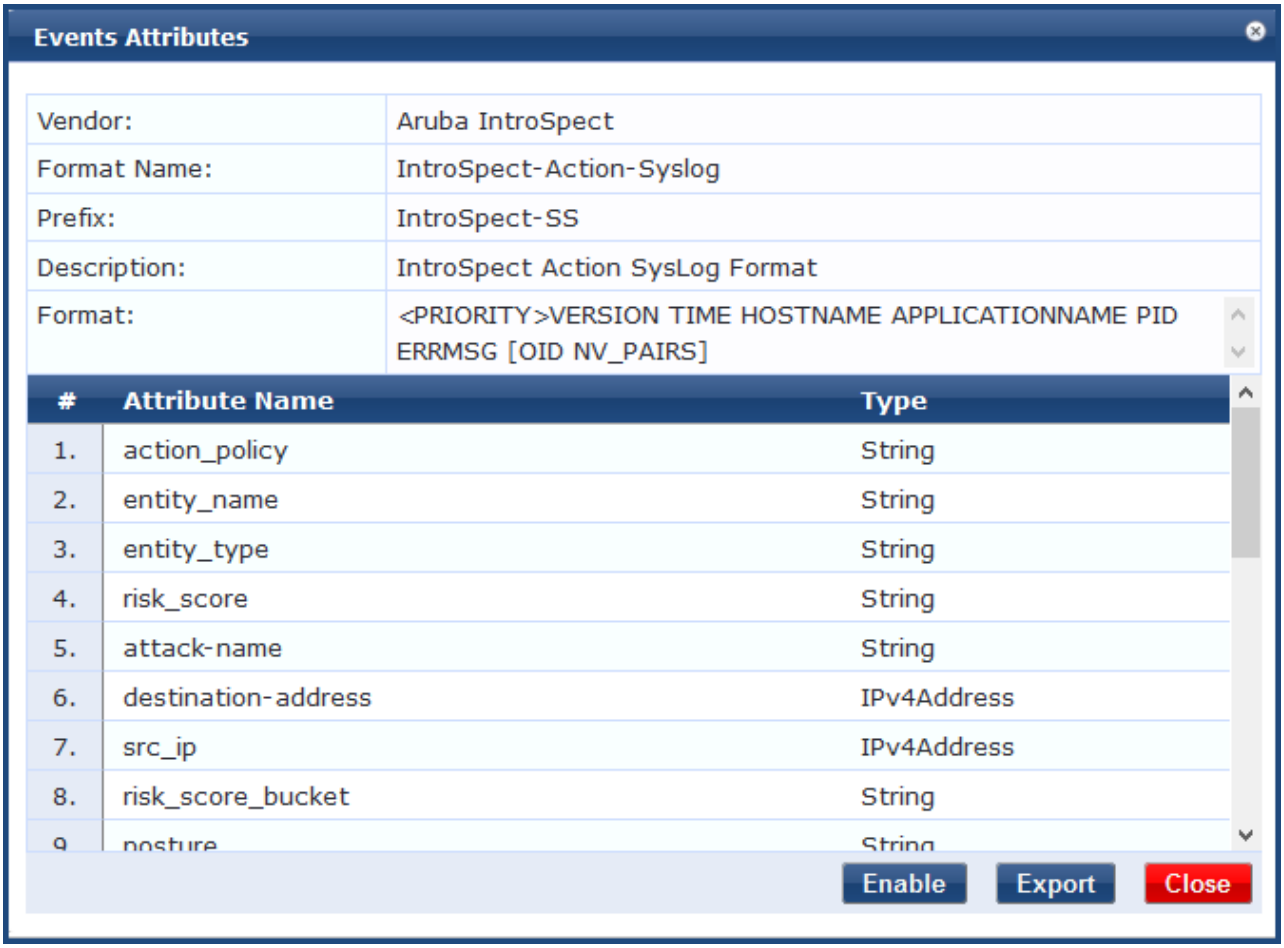
   Figure 1. Viewing Ingress Event Dictionaries



2. To enable a dictionary, select the disabled Ingress Events Dictionary for the appropriate vendor.

   The **Events Attributes** dialog opens.

Figure 2. Enabling an Ingress Events Dictionary



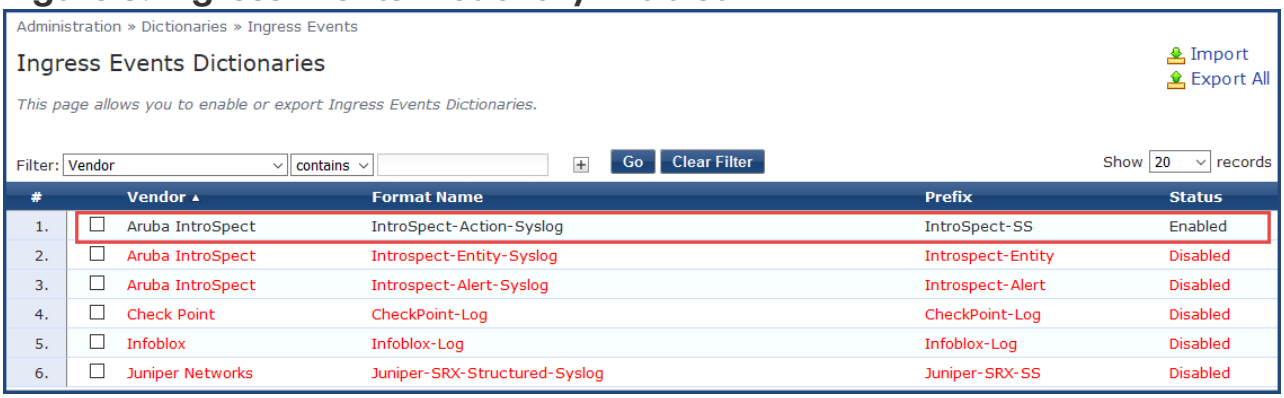
3. To enable the selected ingress events dictionary, click **Enable**. You receive the message:

   Dictionary has been successfully enabled

4. Click **Close**.

   You return to the **Ingress Events Dictionaries** page. The dictionary information is no longer displayed in red and the **Status** column is set to **Enabled** (see Figure 3).

Figure 3. Ingress Events Dictionary Enabled



Managing Custom Windows Hotfixes

Policy Manager provides an option to add hotfixes in the Windows Hotfixes health class that are not present in Windows Hotfixes Updates (**Administration > Agents and Software Updates > Software Updates**). This feature also allows you to edit an existing Windows Hotfix.

- Adding Custom Windows Hotfixes
- Updating an Existing Custom Windows Hotfix

Adding Custom Windows Hotfixes

About this task

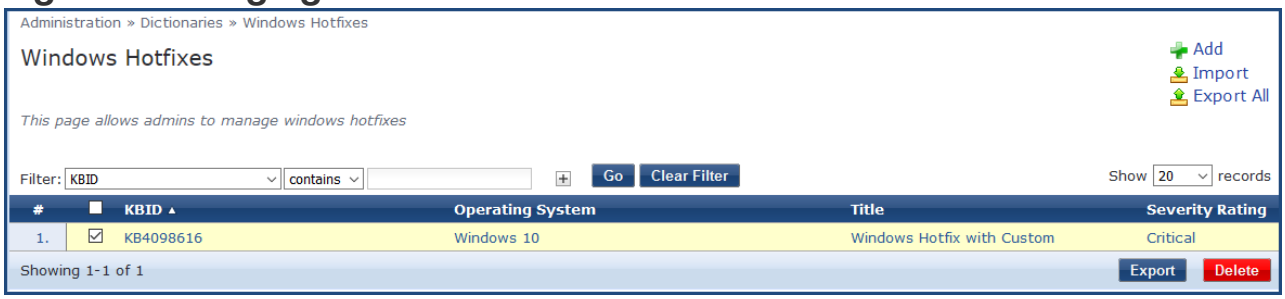
To add custom Windows Hotfixes:

Procedure

1. Navigate to **Administration > Dictionaries > Windows Hotfixes**.

   The **Windows Hotfixes** page opens.

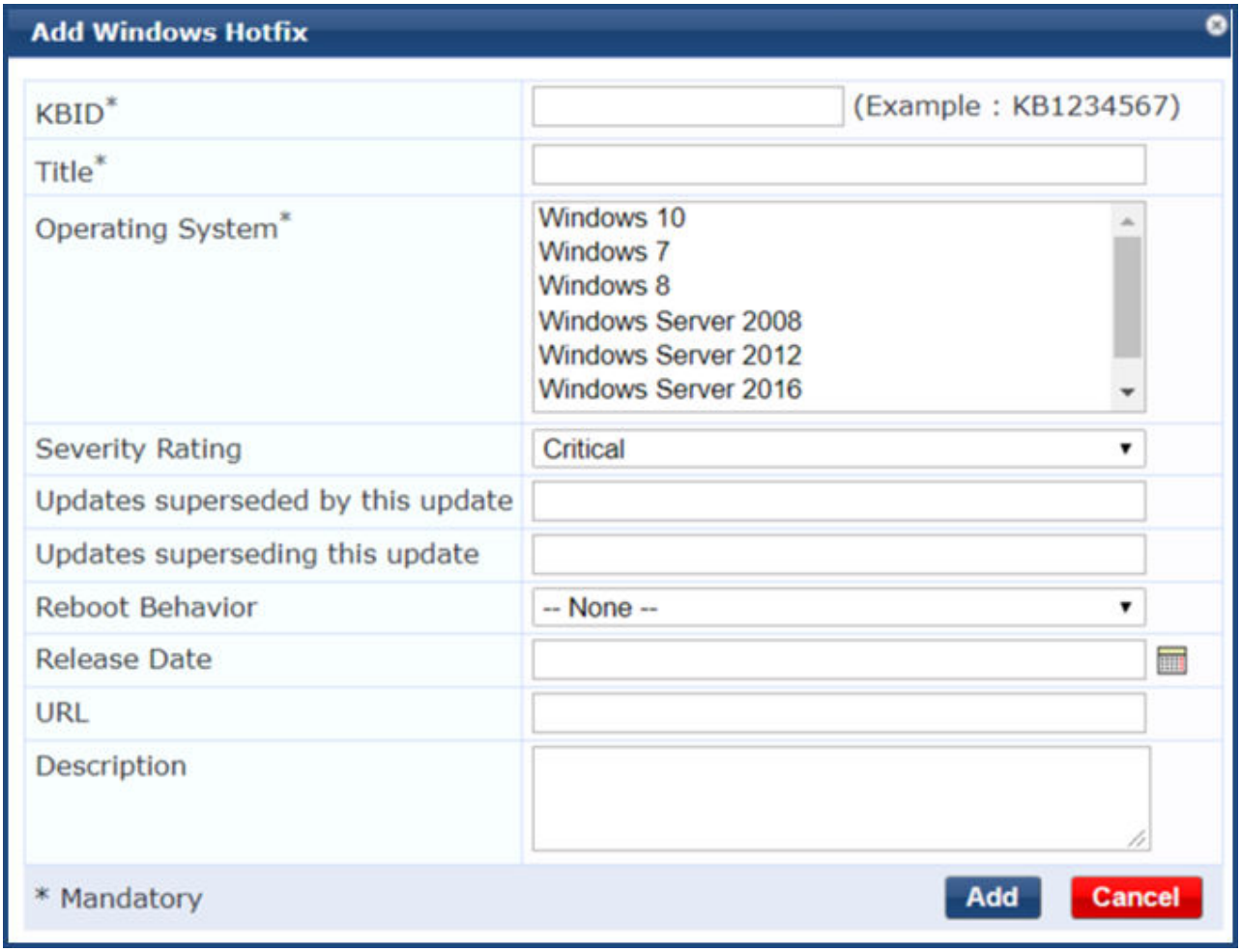
   Figure 1. Managing Windows Hotfixes



2. Click the **Add** link.

   The **Add Windows Hotfix** configuration page opens.

   Figure 2. Add Custom Windows Hotfix Configuration Page



3. Specify the **Add Custom Windows Hotfixes** parameters as described in the following table:

Table 1. Add Custom Windows Hotfixes Parameters

Parameter	Action/Description
KBID	Enter the KBID for this Windows Hotfix. **NOTE** This is a required field.
Title	Enter the title of this Windows Hotfix. **NOTE** This is a required field.

Parameter	Action/Description
Operating System	Select one or more Windows operating systems for this Windows Hotfix. This is a required field. **NOTE** As of January 14, 2020, Mic rosoft has reached End of Support for Windows 7, Win dows Server 2008, and Win dows Server 2008 R2. These operating systems are depre cated in with ClearPass 6.9.0 and later releases.
Severity Rating	Specify one of the following severity ratin gs: • Critical • Important • Low • Moderate • Unspecified
Updates superseded by this update	Optionally, enter the Windows Hotfixes s uperseded by this update.
Updates superseding this update	Optionally, enter the Windows Hotfixes th at supersede this Windows Hotfix.
Reboot Behavior	Specify one of the following reboot beh aviors: • None • Always Requires Reboot • Can Request Reboot • Never Reboots

Parameter	Action/Description
Release Date	Click the **Calendar** icon to bring up the c alendar, then select the release date for t his Windows Hotfix.
URL	Specify the URL for this Windows Hotfix.
Description	(Recommended) Enter a brief description of this Windows Hotfix.

4. Click **Add**.

Updating an Existing Custom Windows Hotfix

About this task

To update an existing custom Windows Hotfix:

Procedure

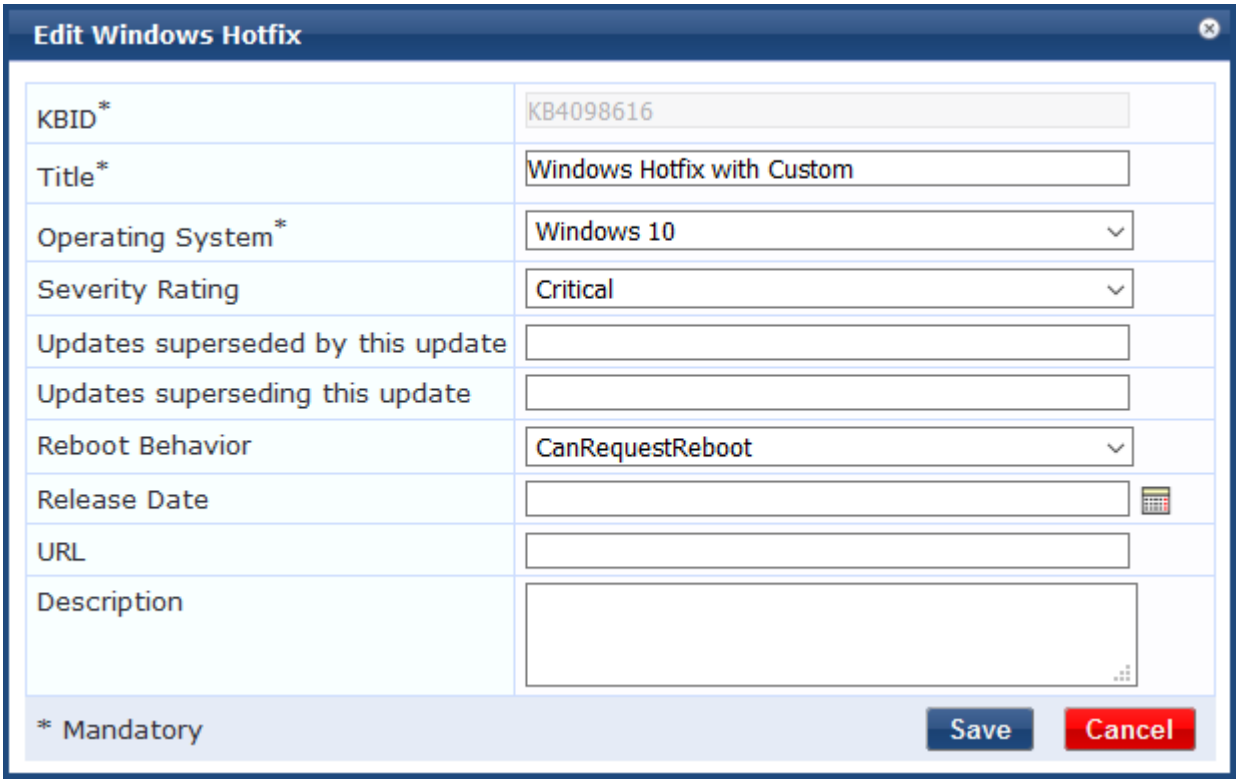
1. Navigate to **Administration > Dictionaries > Windows Hotfixes**.

   The **Windows Hotfixes** page opens.

2. Click the Windows Hotfix you wish to modify.

   The **Edit Windows Hotfix** configuration page opens.

Figure 3. Edit Custom Windows Hotfix Configuration Page



3. Modify the values for the parameters you wish to update, then click **Save**.

> **NOTE**
>
> The **KBID** value cannot be changed.

OnGuard Custom Scripts

Policy Manager allows you to configure three types of OnGuard custom scripts, **agent script enforcement profile scripts**, **health collection scripts,** and **auto remediation scripts**. Admin users can create and execute a script on endpoint devices as part of agent enforcement, or query for additional information that is not included in the existing health classes.

The responses to scripts can be used to determine the appropriate health of the endpoint. Admin users can also create remediation scripts to be executed if an endpoint does not pass health check. Custom scripts can be configured to appear to be executed by the current logged-on user, or as the system user with admin rights.

Scripts launched as a user do not support system operations that need root/admin privileges. Note that if a health collection script is launched as a user without privileges to complete the task, the script execution process can return an error code that can put the client in an unverified status for that check.

You can create custom scripts using the following approved language formats. Note that these formats may be platform-dependent based on available system capabilities.

- PowerShell (PS1)
- Executable Binary (exe for windows)
- Visual Basic Script (VBS)
- JScript (JS)
- Windows Script File (WSF)
- PerlScript (PLS)
- PScript (PS)
- PythonScript (PYS)
- Bash
- Python (2.7)
- Python (3.x)
- Perl
- PHP
- AppleScript
- sh
- csh
- SED/AWK
- Ruby

NOTE

The REST API for OnGuard Custom Scripts is located in theClearPass Guest API Explorer, at **API Explorer > Dictionaries > OnGuard Custom Script**.

Subtopics

Agent Script Enforcement Profile Scripts
Health Collection Scripts
Auto Remediation Scripts
Configuring OnGuard Custom Scripts

Agent Script Enforcement Profile Scripts

Policy Manager can use a custom script in agent script enforcement profiles to execute the script on endpoint devices as part of agent enforcement. When you associate an Agent Script Enforcement Profile script to the Enforcement profile, the details of the custom script, such as the path of the custom script, the command to be executed, and the execution level are configured in the OnGuard Custom Scripts section. You can then add one or more OnGuard custom scripts into an Agent Script Enforcement Profile that acts as a rule in an enforcement policy.

OnGuard Agent executes the OnGuard custom script in the Agent Script Enforcement profile after first applying Agent Enforcement profiles (that is, after **Agent Bounce** is executed, if configured). While applying an Agent Script Enforcement profile, OnGuard Agent does not check to see if a script is already running. It is possible for OnGuard Agent to launch the script multiple times if a previously launched script is still running.

This can occur if OnGuard Agent performs multiple health checks (either manually triggered or caused by a change in health status). The script exits after performing its task.

Health Collection Scripts

You can use the OnGuard Custom Scripts section to create health collection scripts for the Windows, Linux or macOS **Universal System Health Validator** posture plugins on the **Configuration > Posture > Posture Policies** page. Admin users can create a health collection script for Windows, Linux or macOS and associate it with the Universal System Health Validator posture plugin for the same operating system to query for additional information that is not included in the existing health classes. The responses of health collection scripts can be used to determine the appropriate health of the endpoint. Responses can be in JSON format or a key-value pair. The OnGuard Custom Script configuration allows you to create a set of rules that associate exit codes and key-values with a health status of **Healthy** or **Unhealthy**.

Auto Remediation Scripts

You can also use the OnGuard Custom Scripts section to create an auto remediation script for the Windows, Linux or macOS **Universal System Health Validator** posture plugins on the **Configuration > Posture > Posture Policies** page. If the posture plugin includes a health collection script and auto-remediation is enabled for that plugin, you can associate an additional auto remediation script to take an action by running a script on an endpoint that does not pass health checks.

Configuring OnGuard Custom Scripts

About this task

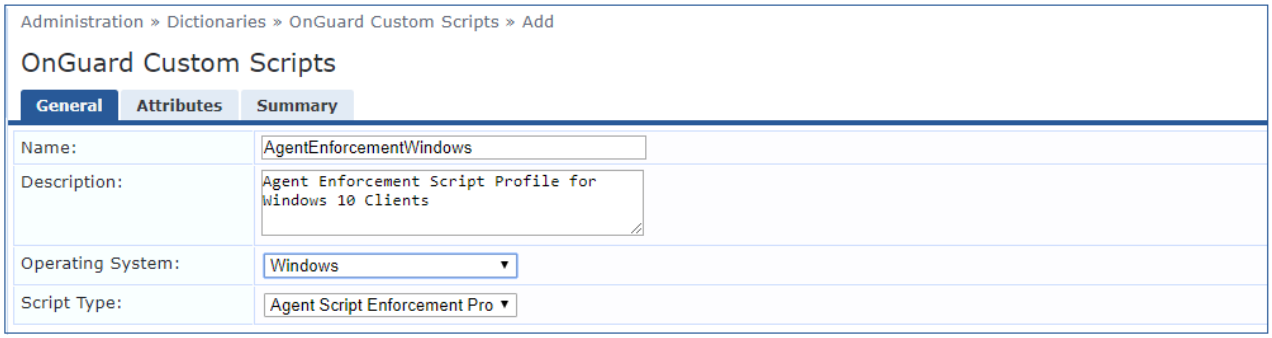
To configure a Custom Script:

Procedure

1. Navigate to **Administration > Dictionaries > OnGuard Custom Scripts**. The **OnGuard Custom Scripts** page opens.
2. Click the **Add** link. The **OnGuard Custom Scripts > Add** window opens.
3. In the **Name** and **Description** fields, enter a name and description of the script .

4. Click the **Operating System** drop-down list and select the operating system on which the script will run. OnGuard custom scripts are supported on Windows, macOS, and on all Linux operating systems that support OnGuard.
5. Click the **Script type** dropdown menu and identify how this custom script will be used:
   - **Agent Script Enforcement Profile**: Script will execute as part of a Agent Script Enforcement Profile.
   - **Health Collection**: Script will execute as part of a Universal Health Validator posture plugin for Windows, Linux or macOS devices to collect additional health data for devices of that type.
   - **Auto Remediation**: If you have associated a health collection script with the Universal Health Validator posture plugin for Windows, Linux or macOS devices, you can also associate a custom auto remediation script to run on non-compliant endpoints.

Figure 1. OnGuard Custom Scripts > Script Configuration Dialog



6. Click the **Attributes** tab. Mandatory attributes appear by default with information about the attribute. Select the **Click to Add** link to add other attributes.

Table 1. OnGuard Custom Scripts > Attributes

Attribute	Action/Description
Path of the Script	Required for all script types. Complete th e path of the script/program, including t he filename. This attribute checks for the existence of a file on an endpoint device and also verifies the SHA256 Checksum.
Command to Execute	Required for all script types. Specify the complete command that OnGuard Agent should execute. You can use the comman

Attribute	Action/Description
	d to launch scripts or pass command line arguments. For example, to launch VBScript (InstallHotfixes.vbs) and pass **All** as an argument, you would enter the following: `cscript /nologo C:\Test\InstallHotfixes.vbs All` If it is not required to pass arguments, set the value of this attribute to the same value specified for **Path of the Script**.
Execution Level	The attribute values are: **User** and **System**. • To launch the script/program as the current logged - on user, select **User**. • To launch the script/program as the system user with admin rights, select **System**.
SHA256 Checksum	Required for all script types. Specify the SHA256 checksum of the script or program. This attribute accepts comma - separated multiple SHA256 checksums to allow execution of different versions of same script or program.
Signer Name	Applicable only to Windows and macOS scripts that can be code signed. This attribute requires admins to enter a signer name to verify code signing of custom scripts. This field accepts multiple signer names separated by "|". **NOTE** The Signer Name should be the value of 'Issued to" in the Code Signing certificate that is used to sign the scrip

Attribute	Action/Description
	t/executable file. For exampl e, all OnGuard executable fi les on Windows and macOS are signed using Aruba's co de signing certificate. To ver ify code signing of any OnG uard executable using custo m scripts, use **Aruba Networ ks, Inc.** as the signer name.
Wait Time (Seconds) Before Executing Sc ript	Specify the time (in seconds) after which OnGuard Agent should launch the script /program. When **Wait Time Before Executing Script** is configured, the OnGuard Agent does n ot process events such as Interface Up/In terface Down and health changes during the wait time.
Pass Health Evaluation Results to Script	Check the check box (which sets the valu e to **true**) to enable OnGuard Agent to p ass health evaluation results to the scrip t/program as an argument. The default is **false**. • When the **Pass Health Evaluation Res ults to Script** attribute is set to **true**, OnGuard Agent passes health evaluat ion results to the script in a URL Enco ded JSson format. • URL Encode replaces double quotes, spaces, and Unicode characters with their ASCII value in %XX format. For example, spaces are replaced by %20 and double quotes are replaced by % 22.

Attribute	Action/Description
Success Message	Enter the message to be shown to the en d user when the script or program is laun ched successfully.
Failure Message	Enter the message to be shown to the en d user when execution of the script or p rogram fails.
Progress Message	This message will be shown on the OnG uard Progress Page (see Creating OnGu ard Custom Web Pages) while OnGuard Agent is executing the custom script and states what action is being performed.
Description	Provides a description of the custom use r interface window or script.
Download URL	If the script or program configured in the **Path of the Script** attribute is not presen t on the client machine, enter the URL o f the remote server from which OnGuard Agent can download the script or progra m. • OnGuard Agent supports downloadin g scripts only from HTTP and HTTPS U RLs. • For HTTPS URLs, OnGuard skips serve r certificate verification. **NOTE** OnGuard Agent does not s upport downloading files fro m URLs that require credenti als.
Script Execution Timeout (Seconds)	Specify the **Script Execution Timeout** att ribute value in the number of seconds. T

Attribute	Action/Description
	he value must be an integer between **60** and **3600**. When this attribute is configured, OnGua rd Agent waits for the specified time for t he custom script to finish execution. If th e custom script does not finish execution within the specified time, OnGuard Agen t returns failure for that script. When this attribute is not configured, On Guard Agent uses varying timeout values , depending upon whether or not custom remediation is enabled. • When Custom Remediation is enabled : the default value is 20 minutes. • When Custom Remediation is not ena bled: 1 minute.

Subtopics

Configuring Auto Remediation

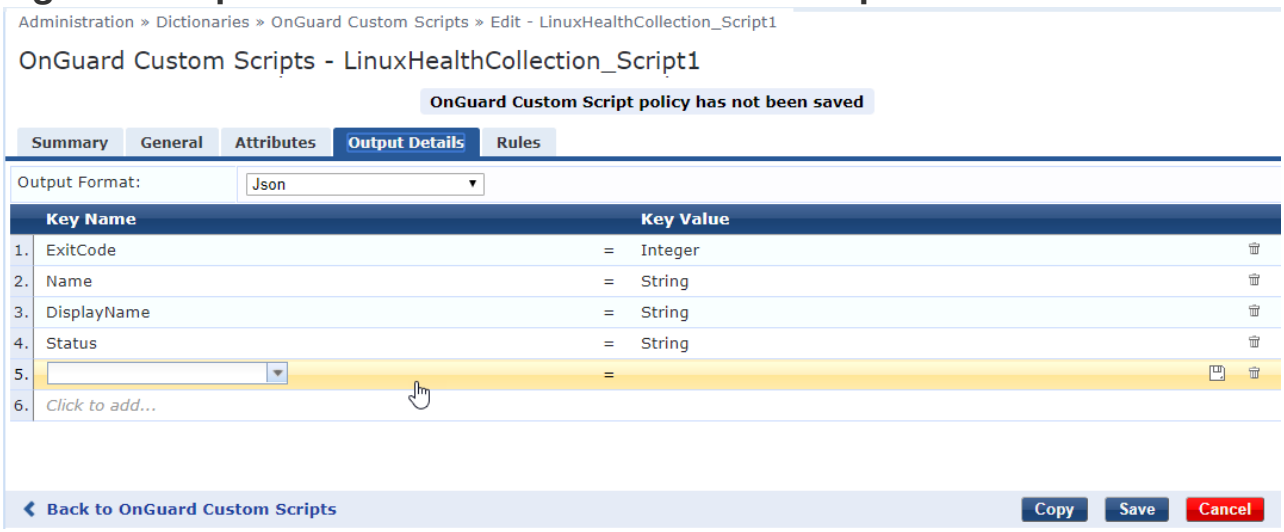
Configuring Auto Remediation

Procedure

If you are configuring auto remediation and Agent Scripts Enforcement Profile scripts, click **Save** to save your settings. For health collection scripts only, you must continue the configuration process to define output details and rules.

1. To configure output details for a health collection script, click the **Output Details** tab. (This tab appears only for health collection custom scripts.)
2. Click the **Output Format** drop-down menu and select either **Json** or **Key-Value Pair** to output the exit code integer in JSON format or as a key-value pair. Select the **Key-Value Pair** option for custom scripts do not have built-in support for JSON, such as batch files or VB scripts. For more information on exit codes, see OnGuard Custom Script Exit Codes.

Figure 1. Output Details for a Health Collection Script



3. The **Output Details** tab includes the ExitCode key by default. To create a new key-value, click the **Key Name** column and enter a custom key name, then click the **Key Value** column and select **String**, **Integer** or **Boolean** as the value for that key type. Click any blank row in the table to enter your key-value data into the table, as shown in Figure 2. If you do not click into another row, the data you just entered will not be saved.

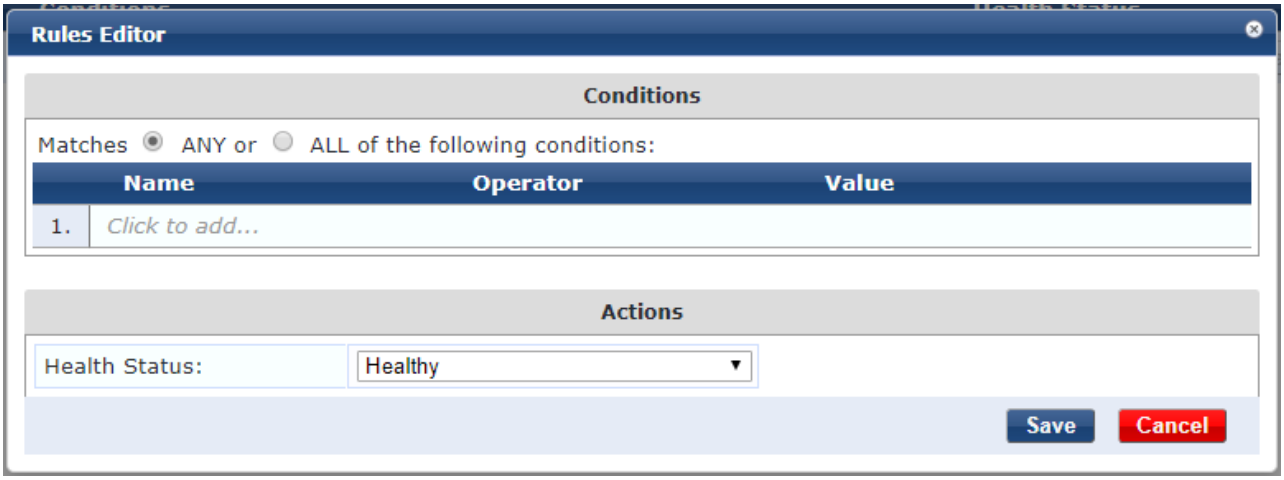
4. Next, click The **Rules** tab to define health evaluation rules based on the exit code and key-values returned by the health collection script.

Figure 2. Health Evaluation Rules

5. In the **Rules Evaluation Algorithm** field, click **Select first match** to return the health status for the first matched condition, or click **Select all matches** and return more than one health status value if more than one condition is matched.

6. Click **Add Rule** to open the **Rules Editor**.

Figure 3. Rules Editor



7. In the **Matches** field, select the match conditions for the health evaluation exit code. Select **ANY** to assign a health status to a rule that matches any of the conditions, or select **ALL** to assign the health status only if all the conditions are matched.
8. Click the **Name** field and select **ExitCode** or a key name, as defined in step c.
9. Click the **Operator** field and select any of the available operator types (for example, EQUALS, GREATER_THAN or LESS_THAN, or EQUALS, EXISTS or NOT_EXISTS).
0. Click the **Value** field and enter a value for the selected key.
1. In the **Actions** field, click the **Health Status** dropdown menu and assign a value or **Healthy** or **Unhealthy** to a rule that matches these conditions.
2. Click **Save** to exit the rule editor, and return to the OnGuard Custom Scripts dialog.
3. Click **Save** again to save your custom script.

Data and Management Port Interfaces

ClearPass supports multiple physical Ethernet Interfaces. including a data port interface and a management port interface.

You must configure the management port, but the data port is optional. Keep in mind, the management port cannot be changed if the data port is already configured.

NOTE

In a system with an existing data port configuration, changing the management port could fail, and an error message "Setting up routing table failed. Netmask and Gateway information are correct. Failed to set up routing" could display within the **Administration > Server Manager > Server Configuration > System** screen. This occurs when the data port is not connected to the network. Users should be aware that, before changing the management port IP address, the data port must be connected and able to ping the gateway.

Route selection

When ClearPass is configured with a management interface but not the optional data interface, then all traffic goes through management interface.

When ClearPass is configured with **both** interfaces:

- If the destination network/address is in the management subnet, then ClearPass uses the management interface.
- If the destination network/address is in the data subnet, then ClearPass uses the data interface.
- If the destination network is not in either management or data subnets, ClearPass uses the data interface by default.
- RADIUS Authentications can be sent to both interfaces. ClearPass will reply with the RADIUS response on the interface where the request was initially received.
- OnGuard communicates with ClearPass through the data interface if both the management and data interfaces are configured.

Cluster traffic interfaces

The management IP address of the publisher in a cluster needs to be accessible to all subscribers. However, the subscribers must reach the publisher's management IP only through the subscriber's management interface, either through a static route or other means. ClearPass cluster traffic uses the following TCP/UDP ports:

NOTE

ClearPass uses the Data or Management port for cluster traffic communication. However, ClearPass always reaches the cluster via the Data port if the Data port is configured.

- UDP Port 123 NTP (Subscriber to Publisher)
- TCP Port 443 HTTPS (Bi-directional)
- TCP Port 5432 PostgreSQL for DB replication (Subscriber to Publisher)
- TCP Port 4231 NetWatch (Post Authentication module and the node where Insight is enabled)
- 6658 TCP for OnGuard client to communicate with ClearPass Policy Manager. As mentioned above, if both interfaces are configured, OnGuard traffic goes to the data interface. Therefore if you have configured both interfaces, this port must be opened on the data interface.

ClearPass to Active Directory Ports

The following list of services and ports are used for Active Directory communication

- UDP Port 88 for Kerberos authentication
- UDP and TCP Port 135 for domain-controller - to - domain controller and client-to-domain-controller operations.
- UDP Port 389 for LDAP to handle normal queries from client computers to the domain controller.
- TCP and UDP Port 464 for Kerberos password change
- TCP Port 5433 must be open to access the Insight database. Port 5433 is also used for connecting to the tipsLogDb.

OnGuard Settings and Agent Library Updates

Use the **OnGuard Settings** page to configure the agent deployment packages. When you save the OnGuard configuration, Policy Manager creates agent deployment packages for the Windows, macOS, and Ubuntu operating systems and provides the packages at a fixed URL on the Policy Manager hardware or virtual appliance. You can then publish this URL to the user community or download the agent deployment packages to another location.

To view the OnGuard Support Charts, see Accessing the OnGuard Support Chart for Plugin Version 2.0.

Subtopics

Configuring OnGuard Settings
OnGuard Agent Library Updates

Configuring OnGuard Settings

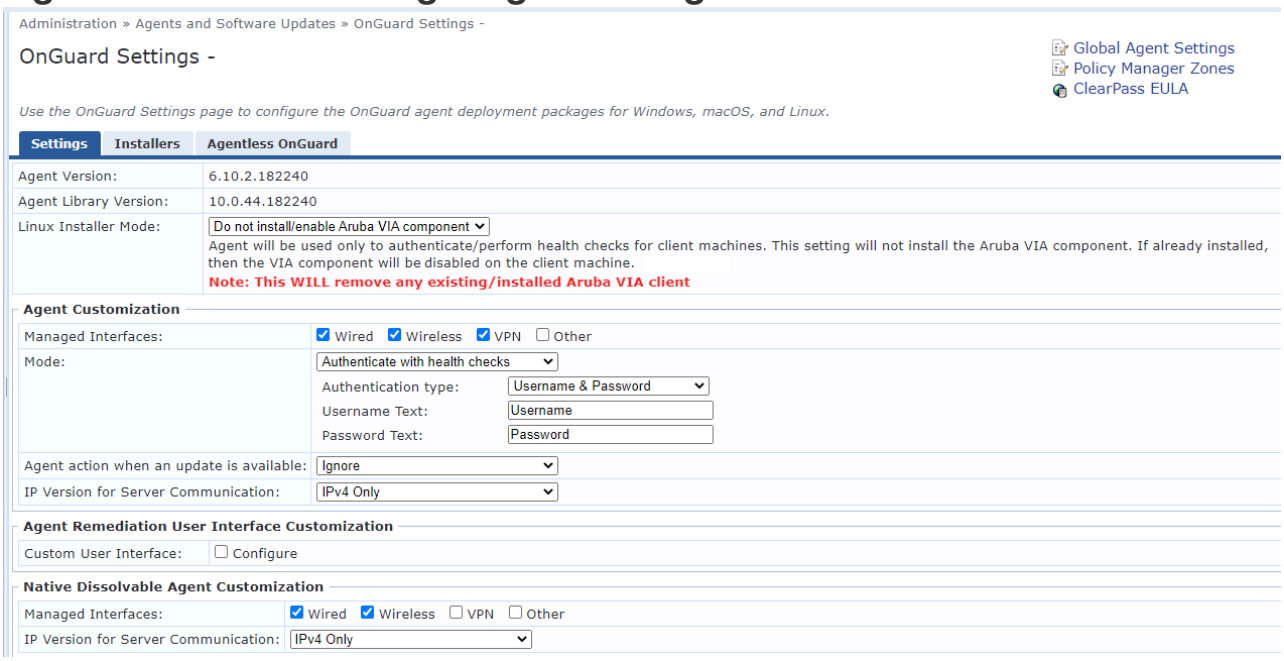
About this task

To configure the OnGuard settings:

Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**. The **OnGuard Settings** page opens to the **Settings** tab:

Figure 1. OnGuard Settings Page > Settings Tab



2. Configure the **OnGuard Settings** parameters as described in Table 1, then click **Save**.

Table 1. OnGuard Settings Parameters

Parameter	Action/Description
Global Agent Settings	Click the **Global Agent Settings** link at th e top of this page to configure the global agent settings for OnGuard agents.

Parameter	Action/Description
	For more information, see OnGuard Global Agent Settings.
Policy Manager Zones	Click the **Policy Manager Zones** link at the top of this page to configure the client subnets that comprise the Policy Manager Zone. For more information on configuring Policy Manager zones, see Managing Policy Manager Zones.
Agent Version	Indicates the current version of the OnGuard agent.
Agent Library Version	Indicates the version of the OnGuard Agent Library. OnGuard Agent uses this library for health collection and auto - remediation.
Linux Installer Mode	Specify the action to be taken from the following options when the Aruba VIA component is used to provide VPN - based access: • Selecting the **Do not install/enable Aruba VIA component** option will automatically disable any existing and installed Aruba VIA client software on the client machine. • Selecting the I**nstall and enable Aruba VIA component** option will automatically upgrade any existing and installed Aruba VIA client software. **NOTE** For related information, see Installing Standalone VIA with 6.8.x OnGuard Agent.

Parameter	Action/Description
Agent Customization	
Managed Interfaces	Select the type(s) of interfaces that OnGuard will manage on the endpoint. Select from the following options: • **Wired** • **Wireless** • **VPN** • **Other**
Mode	Select one of the following options: • **Authenticate with health checks**: OnGuard collects the username and password and also performs health checks on the endpoint. This is the default setting. • **Authenticate - no health checks**: OnGuard collects the username and password but does not perform health checks on the endpoint. • **Check health - no authentication**: OnGuard does not collect the username and password. When you select the **Authenticate with health checks** or **Authenticate - no health checks options,** OnGuard prompts you choose one of the following authentication types. • **Username & Password**. Select this option for standard username and password authentication. You are prompted to specify username and password text. • **Certificate**: When you select **Certificate**, the **Certificate Filter Criteria** opt

Parameter	Action/Description
	ion appears to allow you to show or s elect certificates based on specific cri teria. The **Certificate Filter Criteria** se tting is used by the OnGuard client. T he OnGuard client applies this filter to the list of installed user certificates o n the client machine prior to using th em for authentication. Policy Manage r OnGuard Agent supports certificate s that are stored under the personal c ertificate store of the user's system (fo r example, under *Certificates - Curre nt User* on Windows) and smart cards. For details, refer to Certificate - Based Authentication Using OnGuard.
Agent action when an update is available	Determines what the Policy Manager On Guard Agent does when an update of On Guard Agent is available. Select one of th e following options: • **Ignore**: Policy Manager OnGuard Age nt ignores the available OnGuard Age nt update. • **Notify User**: Policy Manager OnGuard Agent notifies the user that a new ve rsion of OnGuard Agent is available o n the Policy Manager server. • **Download and Install**: When this opti on is selected and a new version of O nGuard Agent is available on the Poli cy Manager server, Policy Manager On Guard Agent automatically download s and installs the new version of OnG uard Agent from the Policy Manager s erver.
IP version for Server Communication	Starting with ClearPass 6.10.2, use this op tion to select one of the following option s for the IP version used for communicati

Parameter	Action/Description
	on between clients using OnGuard agent s and the Policy Manager server. The defa ult setting is **IPv4**. • **IPv4 Only** — (Default) OnGuard will p erform health checks only for IPv4 ad dresses. If a client has an IPv6 addres s, OnGuard will not perform health ch ecks for the IPv6 address. • • **Dual (IPv4 Preferred)** — If an interfac e has both IPv4 and IPv6 addresses, O nGuard will perform health checks on ly for the IPv4 address and will not p erform health checks for the IPv6 add ress. OnGuard will perform health che cks for either an IPv4 or IPv6 address i f only one of them is present. • **Dual (IPv6 Preferred)** — If an interfac e has both IPv4 and IPv6 addresses, O nGuard will perform health checks on ly for the IPv6 address of that interfac e and will not perform health checks for the IPv4 address. OnGuard will pe rform health checks for either an IPv4 or IPv6 address if only one of them is present. • **Both (IPv4 and IPv6)** — OnGuard will perform health checks for both the IP v4 and IPv6 addresses if both are pres ent for an interface. If only one of the m is present, OnGuard will perform h ealth checks only for that address typ e. This option is not available for the **Native Dissolvable Agent**.
Agent Remediation User Interface Customization	

Parameter	Action/Description
Custom User Interface	When you select the **Configure** check box, the **Agent Remediation User Interface Customization** dialog opens. • **Web Pages**: To create the OnGuard custom web pages and define the properties for the web pages, click the **Create** link for the corresponding web page (for details, see Creating OnGuard Custom Web Pages.
Native Dissolvable Agent Customization	
Managed Interfaces	The Native Dissolvable Agent performs health checks for one of the selected interfaces. This feature ensures that, if both wired and wireless interfaces are connected, the OnGuard Agent will send health requests through the correct interface. Select the type(s) of managed interfaces that are supported for the Native Dissolvable Agent. Select from the following options: • **Wired** • **Wireless** • **VPN** • **Other**
IP version for Server Communication	Starting with ClearPass 6.10.2, use this option to select one of the following options for the IP version used for communication between clients using OnGuard dissolvable agents and the Policy Manager server. • **IPv4 Only** — (Default) OnGuard will perform health checks only for IPv4 ad

Parameter	Action/Description
	dresses. If a client has an IPv6 addres s, OnGuard will not perform health ch ecks for the IPv6 address. • **IPv6 Only** — OnGuard will perform h ealth checks only for IPv6 addresses. I f a client has an IPv4 address, OnGua rd will not perform health checks for t he IPv4 address. • **Dual (IPv4 Preferred)** — If an interfac e has both IPv4 and IPv6 addresses, O nGuard will perform health checks on ly for the IPv4 address and will not p erform health checks for the IPv6 add ress. OnGuard will perform health che cks for either an IPv4 or IPv6 address i f only one of them is present. • **Dual (IPv6 Preferred)** — If an interfac e has both IPv4 and IPv6 addresses, O nGuard will perform health checks on ly for the IPv6 address of that interfac e and will not perform health checks for the IPv4 address. OnGuard will pe rform health checks for either an IPv4 or IPv6 address if only one of them is present.

OnGuard Agent Library Updates

OnGuard modules, including detection libraries for client programs (V4 SDK), can be upgraded without having to upgrade your Policy Manager or OnGuard installations. To support this feature, *OnGuard Agent Library update* patches with new versions of the detection library/SDK will be released periodically. When they are released, the OnGuard Agent Library update patches will be available on the **Software Updates** page under **Firmware & Patch Updates** (see Software Updates).

When a new version of an OnGuard Agent Library patch is installed, it is immediately available in **OnGuard Settings**. OnGuard Agents can be configured to install the update

automatically, or administrators may install them manually instead. The OnGuard Agent Library update patches are also available on the Aruba Support Center for offline updates through the CLI. OnGuard Agent Library update patch versions are independent of Policy Manager and OnGuard versions, so newer versions of OnGuard Agent Library update patches can be installed on older versions of Policy Manager. This feature is available for both the persistent and dissolvable agents, and for Windows, macOS, and Ubuntu operating systems.

As OnGuard Agent Library update files become available, the files for the various operating systems are listed on the new **OnGuard Settings > Installers** tab. Administrators can download these files for manual installation or to push via Patch Management applications such as Microsoft System Center Configuration Manager (SCCM).

> **NOTE**
> While auto-remediation or ClearPassUSHARemediate.exe is running, the OnGuard Agent Library update will fail on Windows OS. Once auto-remediation has completed, you can install the OnGuard Agent Library update.

Subtopics

Important Points

Important Points

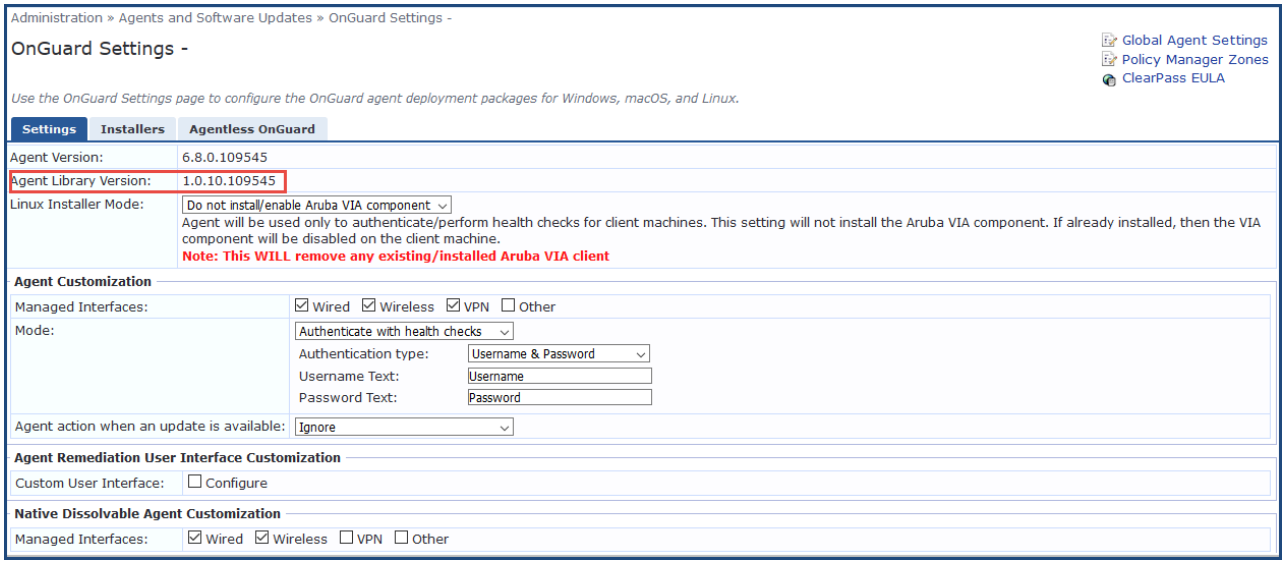
About this task

Please note the following important points:

Procedure

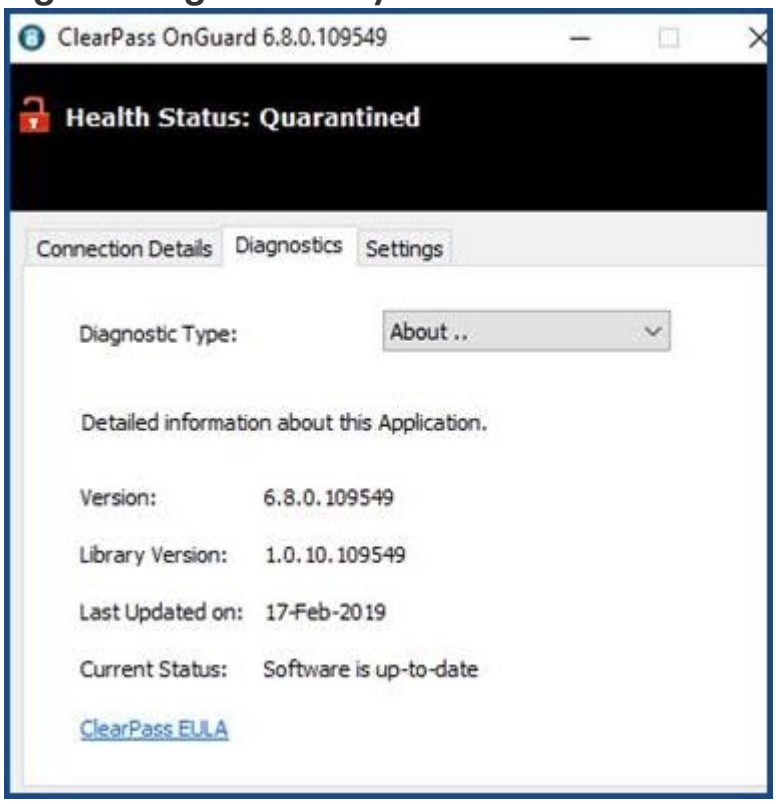
1. On a Policy Manager server, new versions of an OnGuard Agent Library Update patch must be installed manually from the **Software Updates** page.
2. OnGuard Agent Library Update patches are independent from Policy Manager server versions (for example, the same patch will be available for all 6.7.x versions).
3. The OnGuard Agent Library version is different than the Policy Manager server and OnGuard Agent versions. The currently installed version of OnGuard Agent Library is shown at **Administration > Agents and Software Updates > OnGuard Settings**:

Figure 1. OnGuard Settings > Agent Library Version



4. The current version of OnGuard Agent Library installed on the client is shown in the OnGuard Agent About box:

Figure 2. Agent Library Version and the OnGuard Agent About Box



5. On a Policy Manager server, the **Software Updates** page shows only those patches that have versions higher than the currently installed version. Installing a patch that is a lower version (that is, downgrading the version) on a Policy Manager server is not supported.

6. OnGuard Agent Library Updates are supported for both Persistent and Native Dissolvable Agents on the Windows, macOS, and Ubuntu operating systems.
7. Auto-updating OnGuard Agent Library is supported on Ubuntu clients having Policy Manager OnGuard Agent version 6.7.1 and higher.
8. Installing OnGuard Agent Library Update on clients having Policy Manager OnGuard version 6.7.1 or higher will restart OnGuard Agent and Services. Thirty to 60 seconds after installing OnGuard Agent Library Update, Policy Manager OnGuard Agent will be restarted.
9. OnGuard Agent Library Update patches will include only Library Update installers.
0. Installing a new version of the OnGuard Agent Library installs the new version of the V4 SDK library files.
1. While automatically updating the OnGuard Agent library, OnGuard Agent supports installing newer (higher) versions on clients only. If the Policy Manager server has a lower version than the version installed on the client, OnGuard Agent does not automatically downgrade the OnGuard Agent library.
2. Downgrading the OnGuard Agent Library version on a client is supported via manual installation; that is, an administrator can manually install a lower version of the OnGuard Agent Library on a client.
3. If auto-update of the OnGuard Agent Library is enabled and a new version of OnGuard Agent Library is available on the Policy Manager server, OnGuard Agent installs the new version after performing health checks as OnGuard Agent performs enforcement actions (such as applying the Agent Enforcement profile).
4. After successful installation of a new version of OnGuard Agent Library Update, OnGuard Agent performs health checks again, using the new version of the V4 SDK. This is applicable for both automatic and manual installations.
5. OnGuard Agent sends the installed version of OnGuard Agent Library update to the Policy Manager server in a WebAuth request. This information is available in **Access Tracker > Input > Computed Attributes > Host:AgentLibraryVersion**. This attribute is available in Service Rules and Role Mapping.
6. Third-party support charts under **Administration > Support > Documentation** will also get updated when a new version of the OnGuard Agent Library Update patch is installed on the Policy Manager server.
7. On Windows clients, installing the OnGuard Agent Library manually restarts OnGuard Agent and Services. Also, if multiple users are logged in at the time of installation,

OnGuard Agent is launched only for the current active user. For other, non-active users, OnGuard Agent must be restarted manually.

8. Uninstalling OnGuard Agent also uninstalls OnGuard Agent Library Update from the client. Reinstalling OnGuard Agent does not install the new version OnGuard Agent Library update automatically, as the new version of Library Update is not part of OnGuard Installer. If **Enable to auto update OnGuard Agent Library** is set to **True** in the Agent Enforcement profile (see Configuring Agent Enforcement Attributes), OnGuard Agent will install it again; otherwise, it must be installed manually.

Certificate-Based Authentication Using OnGuard

For certificate-based authentication on Windows, OnGuard agent uses the client certificate during the SSL handshake (the private key can be obtained from OS store/TPM/smart card). The Policy Manager server verifies the client certificate against the configured trusted CA list. On successful certificate verification, the client is authenticated. Additionally, Policy Manager can perform user-lookup for the identity fetched from the client certificate with the configured authentication source and display an alert in Access Tracker if lookup fails.

Subtopics

Prerequisites
Configuration
Certificate Filter Criteria Example
OnGuard Agent Behavior for Certificate-Based Authentication
Host:AuthType Attribute
Important Points

Prerequisites

About this task

Procedure

1. Open a Linux terminal on the device.
2. Navigate to /usr/share/clearpass-onguard/.
3. Run the command **./via-cli session start** to start the session

4. Run the command **./via-cli cert import --user --keypass <StoragePassword> --certpass <certPassword> <fileName>** to import the certificate

Configuration

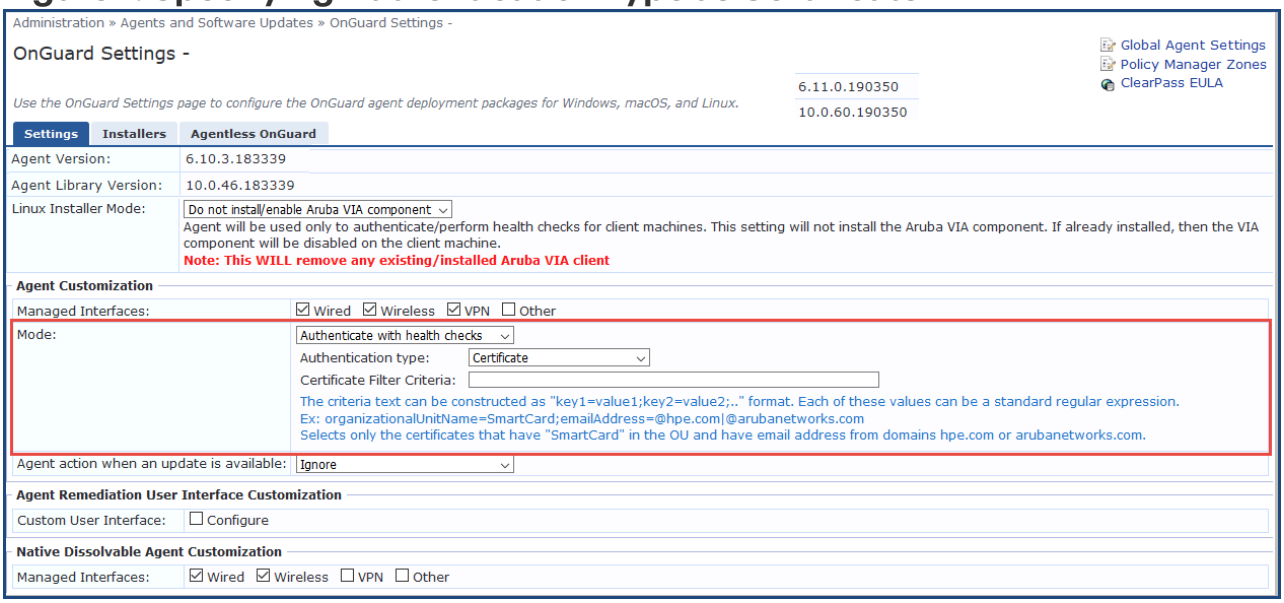
About this task

To configure certificate-based authentication:

Procedure

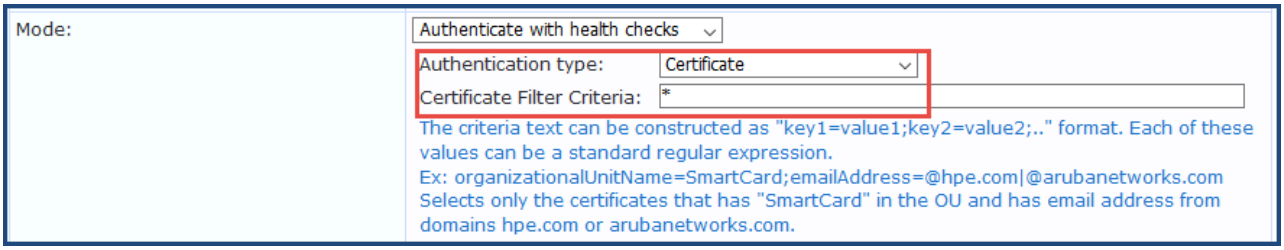
1. Navigate to **Administration > Agent and Software Updates > OnGuard Settings**. The **OnGuard Settings** page opens.
2. Under *Agent Customization* specify the **Authentication Type** as **Certificate**. The *Certificate Filter Criteria* field is displayed.

Figure 1. Specifying Authentication Type as Certificate



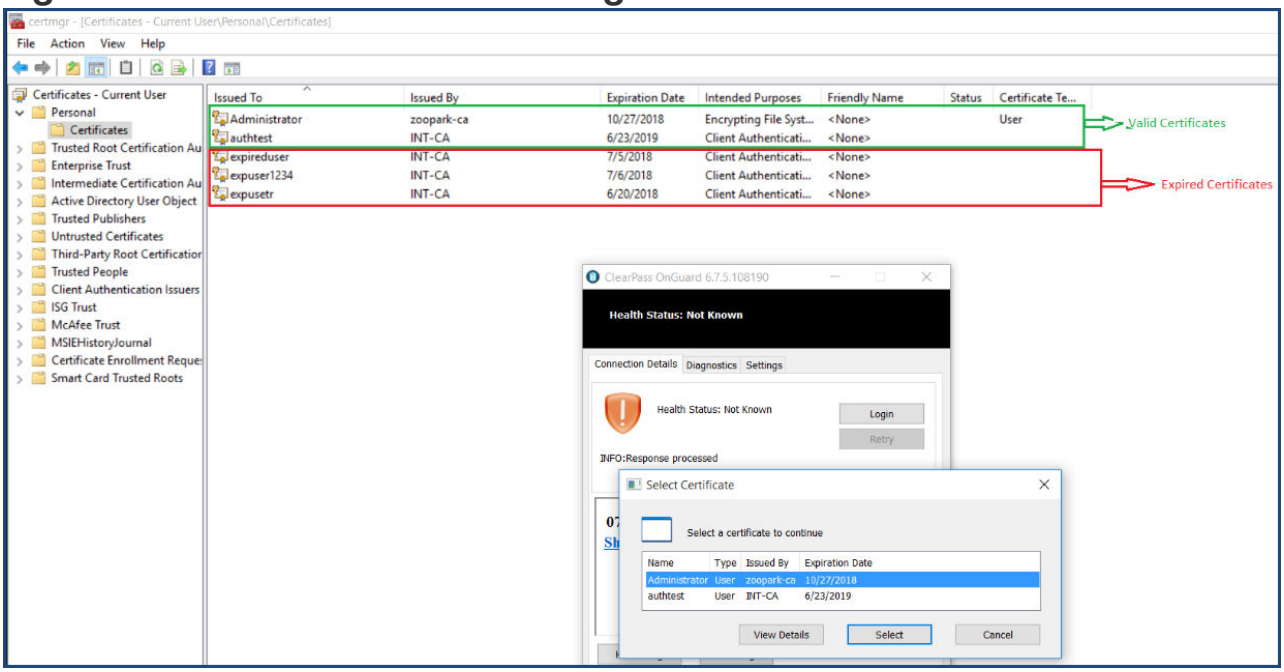
The **Configure Filter Criteria** parameter supports regular expressions with an exception of ***** (asterisk).

Figure 2. Asterisk (*) Specified as Certificate Filter Criteria



When * is configured for the certificate filter criteria, OnGuard Agent shows all the certificates that are stored under the personal certificate store of the user's system (for example, under *Certificates - Current User* on Windows) and smart cards in the Certificate Selection dialog:

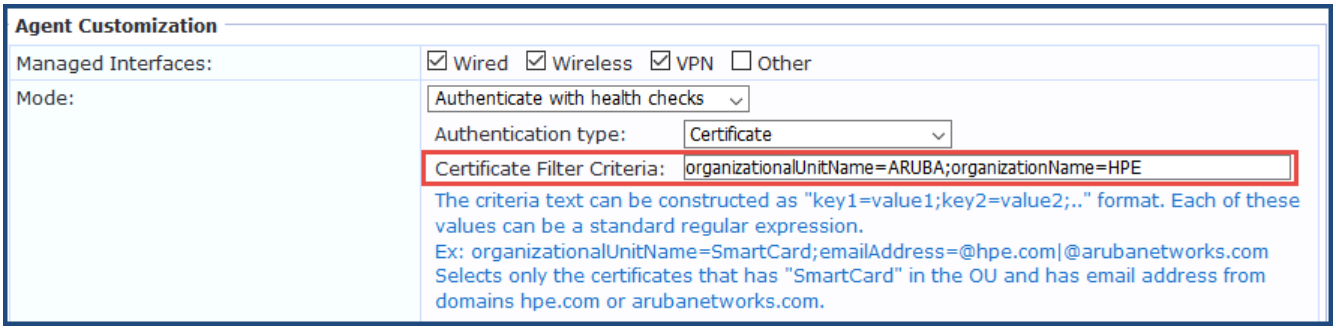
Figure 3. Certificate Selection Dialog



Certificate Filter Criteria Example

If **Certificate Filter Criteria** is configured as "*organizationalUnitName=ARUBA;organizationName=HPE*" OnGuard Agent selects the certificate that matches both the conditions:

Figure 1. Example of Certificate Filter Criteria with Two Conditions



NOTE

If only one certificate is present in the personal certificate store of the user's system that matches the filter criteria, OnGuard Agent uses that certificate without prompting the Certificate Selection dialog.

Table 1. Certificate Filter Criteria: Supported Attributes

Attribute	OID
commonName	2.5.4.3
organizationalUnitName	2.5.4.11
organizationName	2.5.4.10
certificateIssuer	2.5.29.29
userPrincipalName	1.3.6.1.4.1.311.20.2.3
emailAddress	1.2.840.113549.1.9.1
friendlyName	1.2.840.113549.1.9.2

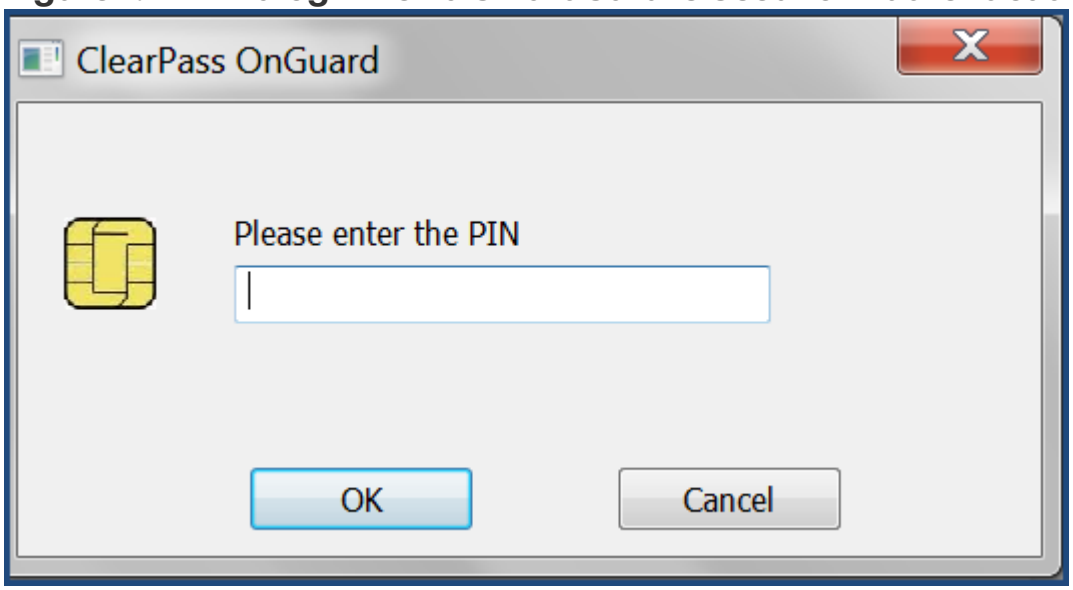
OnGuard Agent Behavior for Certificate-Based Authentication

When Certificate-based Authentication Mode is selected:

- OnGuard Agent prompts for the Certificate Selection dialog.
- OnGuard Agent shows the information of the selected certificate in Diagnostic logs.

- If a smart card is or PIV (Personal Identification Verification) system is used for authentication, OnGuard Agent prompts for the PIN dialog.
- OnGuard caches the certificate selection and PIN if the **Save Credential** setting is ON in the OnGuard Agent Preferences.

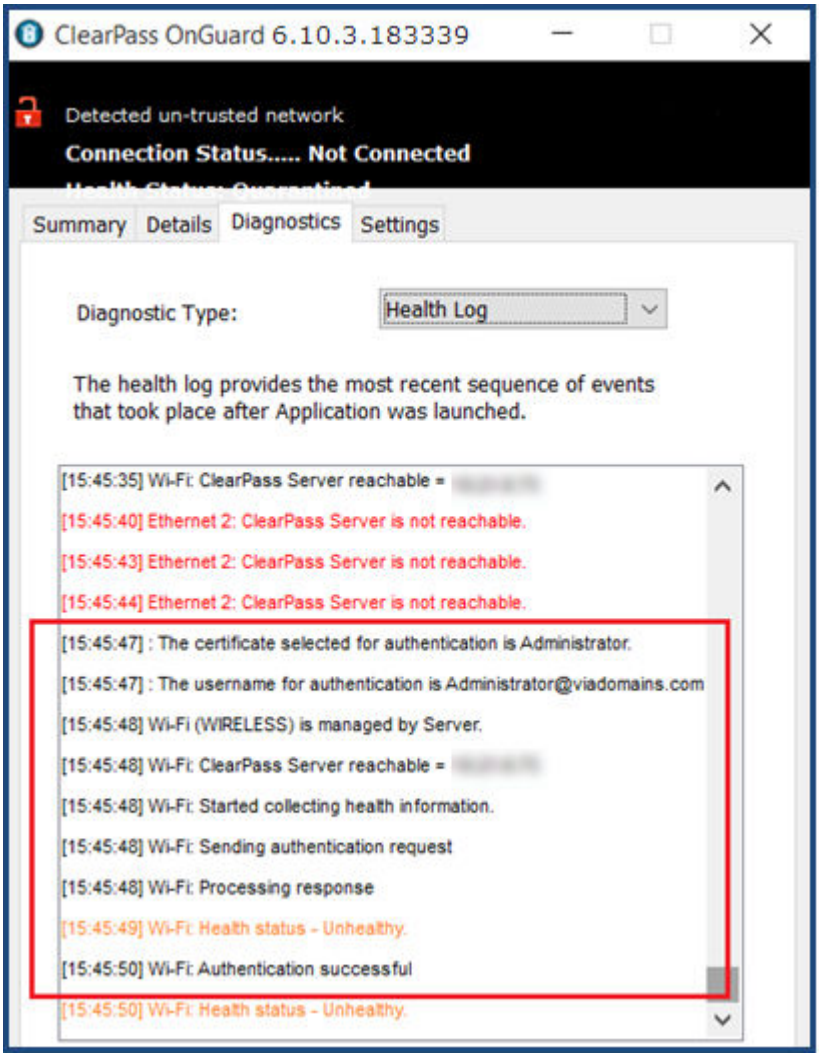
Figure 1. PIN Dialog When a Smart Card is Used for Authentication



NOTE

When reconnecting to the same running instance of the OnGuard Agent, the PIN may not be required again, due to internal caching supported by the operating system and the smart card provider.

Figure 2. Certificate Connection Status



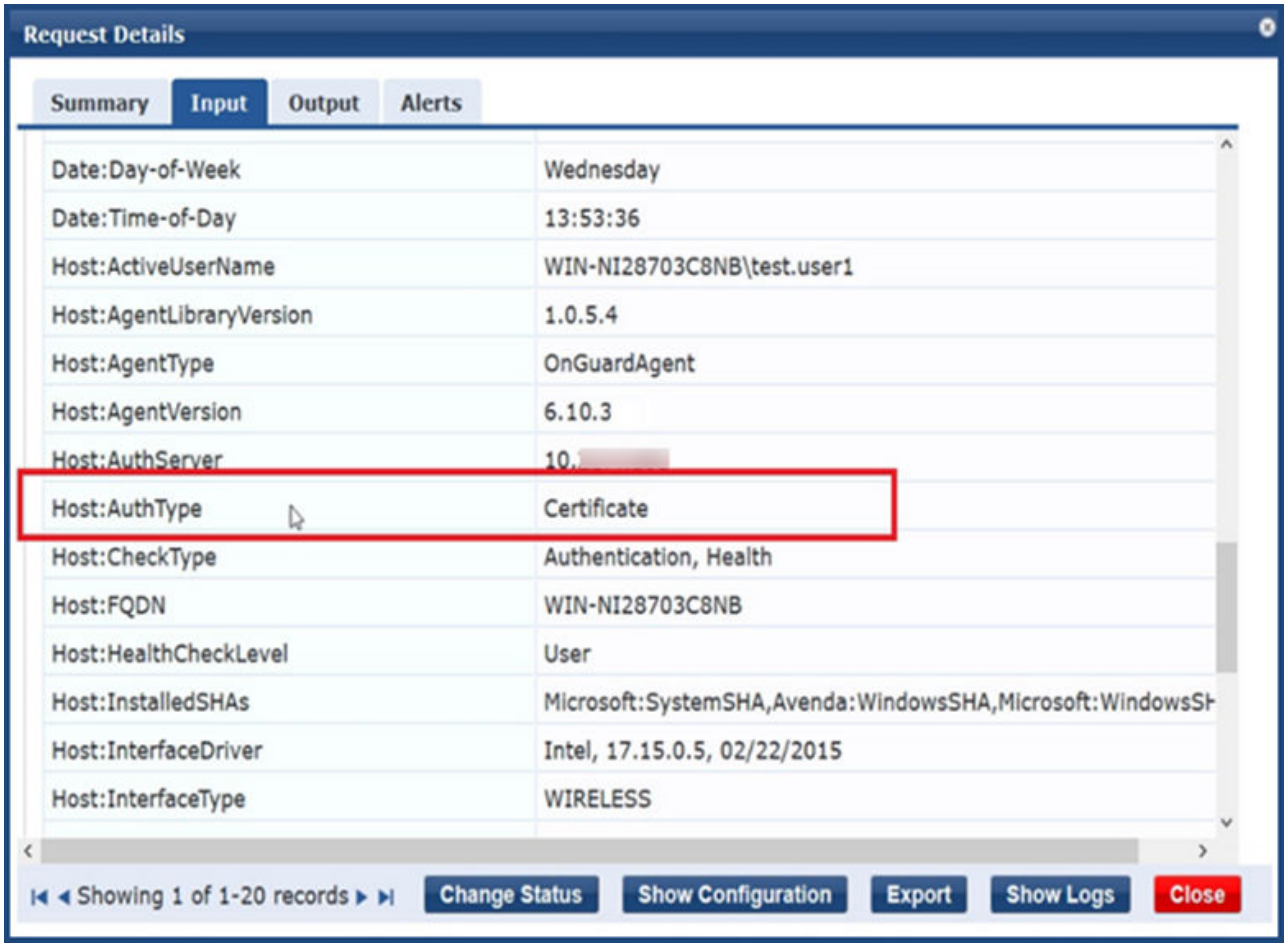
Host:AuthType Attribute

In the WebAuth request, OnGuard Agent sends a new attribute—**Host:AuthType**.

Host:AuthType has two values:

- **Certificate**: For Certificate-based authentication.
- **Password**: For username and password-based authentication.

Figure 1. Host:AuthType Attribute Displayed in Access Tracker



This attribute is available in Service Rules and Role Mapping.

Important Points

About this task

Please note the following important points:

Procedure

1. Certificate-based authentication is supported for Persistent Agent on the Windows operating system.
2. Certificates that are installed in the machine store are not displayed in the Certificate Selection dialog.
3. OnGuard Agent discards expired certificates.
4. The user's certificates that don't include the private key are displayed in the OnGuard Agent dialog, but these certificates don't work when used for authentication.

5. Policy Manager does not support *SubjectAltName* (SAN) in **Certificate Filter Criteria**. However, you can use values inside the SAN as certificate filter criteria (for example *email*, *IP*, *dns*).

Installing Standalone VIA with 6.11.x OnGuard Agent

Policy Manager supports standalone installation of the 6.11.x OnGuard Agent and VIA 3.2.x or later for Windows or later or VIA 3.1.3 or later for macOS. This allows administrators to use VIA functionality that is not yet available in the Unified Agent in conjunction with the OnGuard Agent.

To use this feature, no extra flag needs to be passed to the OnGuard and VIA installers in order for them to co-exist on the same system. Also, you can install VIA and OnGuard in any order; that is, you can choose to install VIA first or install OnGuard first.

- While using OnGuard with standalone VIA, the **Mode** parameter in **OnGuard Settings > Agent Customization** should be set to **Authenticate with health checks**.
- For authentication, end users should use the same **Username** in OnGuard and VIA. This is required because Policy Manager uses the **Username** value to correlate WebAuth and RADIUS requests for VPN interfaces.

OnGuard Agent Installers

About this task

To access the OnGuard Agent Installers:

Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**.
2. From the **OnGuard Settings** page, select the **Installers** tab.
3. In the **Show download URL with** field, select **FQDN**, **IPv4** or **IPV6** to display URLs with Fully Qualified Domain Names (FQDNs), or for IPv4 or IPv6 networks.

> **NOTE**
>
> The **FQDN** option is only available if FQDN is configured on Policy Manager.

The **Installers** tab provides all available installer files for OnGuard and for library updates for IPv4 or IPv6 networks. For each OS (Windows, macOS, and Ubuntu), you have the option of initiating a full installation or an update only, which installs only the OnGuard Agent for OPSWAT OESIS framework for the selected OS.

Figure 1. OnGuard Settings Page > Installers Tab

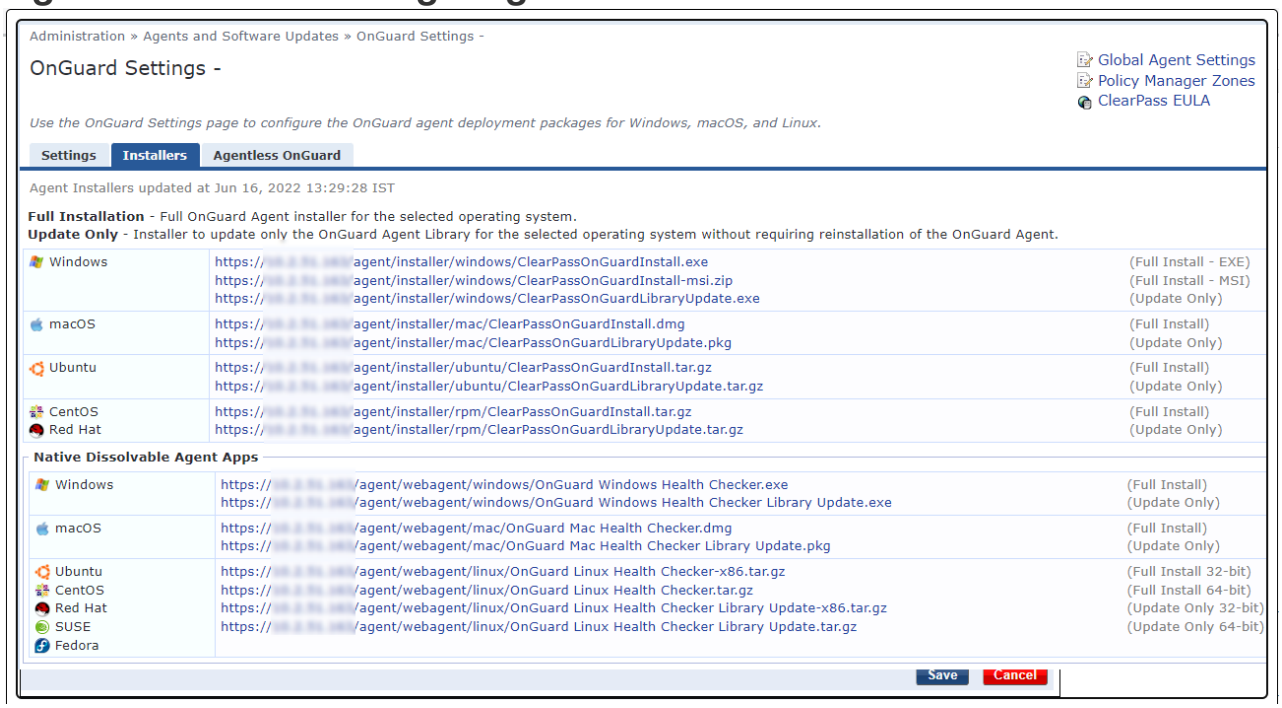


Table 1. OnGuard Settings > Installers Parameters

Parameter	Action/Description
Windows	Select the appropriate download link to d ownload OnGuard Agent for Windows. • Starting with Policy Manager 6.9.10, T he Policy ManagerOnGuardInstall bin ary file is provided in .exe format and also as a downloadable .zip file that c ontains both an MSI installer file and a ReadMe.txt file with instructions on us ing the MSI installer. • In Policy Manager 6.9.9 and earlier rel eases, the Policy ManagerOnGuardIns tall binary file is provided in .exe and . msi formats.

Parameter	Action/Description
	• The Policy ManagerOnGuardLibrary Update file updates only the OnGuard Agent Libary for Windows without req uiring reinstallation of OnGuard Agent .
macOS	Select the appropriate download link to d ownload OnGuard Agent for macOS. • The *ClearPassOnGuardInstall* binary fi le is provided in .DMG format. • The *ClearPassOnGuardLibraryUpdate* file updates only the OnGuard Agent Libary for macOS without requiring re installation of OnGuard Agent.
Ubuntu	Select the appropriate download link to d ownload OnGuard Ubuntu Agent for Linu x. • The Policy Manager OnGuardInstall bi nary file and the Policy Manager OnG uardUpdate file are provided in .tar.gz format. • The Policy ManagerOnGuardLibrary Update file updates only the OnGuard Ubuntu Agent for Linux without requir ing reinstallation of OnGuard Agent.
Native Dissolvable Agent Apps	
Windows	• Click the **OnGuard Windows Health C hecker** URL to install the Native Dissol vable Agent for Windows. • Click the **OnGuard Windows Health C hecker Library Update** URL to update only the Native Dissolvable Agent for Windows without requiring reinstallati on of OnGuard Agent.

Parameter	Action/Description
macOS	• Click the **OnGuard Mac Health Checker** URL to download Native Dissolvable Agent for macOS. • Click the **OnGuard Mac Checker Library Update** URL to update only the Native Dissolvable Agent for macOS without requiring reinstallation of OnGuard Agent.
Ubuntu	• Click the **OnGuard Ubuntu Health Checker** URL to download Native Dissolvable Agent for macOS. • Click the **OnGuard Ubuntu Checker Library Update** URL to update only the Native Dissolvable Agent for Ubuntu without requiring reinstallation of OnGuard Agent. **NOTE** **x86** in the filename) and 64 - bit systems.

Manually Installing a New Version of the OnGuard Agent Library

About this task

You can use the installers available on the **OnGuard Settings > Installers** page for offline or manual installation of OnGuard Agent Library updates on clients.

To manually install a new version of the OnGuard Agent Library:

Procedure

1. Download the applicable installer from the **OnGuard Settings > Installers** page.
2. Copy the installer file on the client and run it.
3. Wait for the installation to complete.
4. When the installation is complete:

a. On Windows clients, OnGuard Agent will be restarted.

b. On Ubuntu and macOS clients, OnGuard Agent will perform health checks.

Automatically Updating New Versions of OnGuard Agent Library from a ClearPass Server

About this task

> **NOTE**
> This feature is supported by Policy Manager OnGuard Agent for Windows and macOS.

ClearPass OnGuard Agents support automatically downloading and installing new versions of the OnGuard Agent Library from a ClearPass server when a new version is available.

> **NOTE**
> While auto-remediation or ClearPassUSHARemediate.exe is running, the OnGuard Agent Library update will fail on Windows OS. Once auto-remediation has completed, you can install the OnGuard Agent Library update.

To enable automatic updates of the OnGuard Agent Library:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**.
2. Configure a new Agent Enforcement profile or edit an existing Agent Enforcement profile.
3. Set the attribute **Enable to auto update OnGuard Agent Library** to **True** (for details, see Configuring Agent Enforcement Attributes).

Agentless OnGuard Packages

About this task

The **Agentless OnGuard** tab provides Agentless OnGuard wrapper files and and Agentless OnGuard Library Update files for Windows OS, macOS and Linux.

To access the **Agentless OnGuard** page:

Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**.
2. From the **OnGuard Settings** page, select the **Agentless OnGuard** tab.
3. In the **Show download URL with** field, select either **IPv4** or **IPV6** to display URLs for either IPv4 or IPv6 networks.

Figure 1. OnGuard Settings > Agentless OnGuard Page

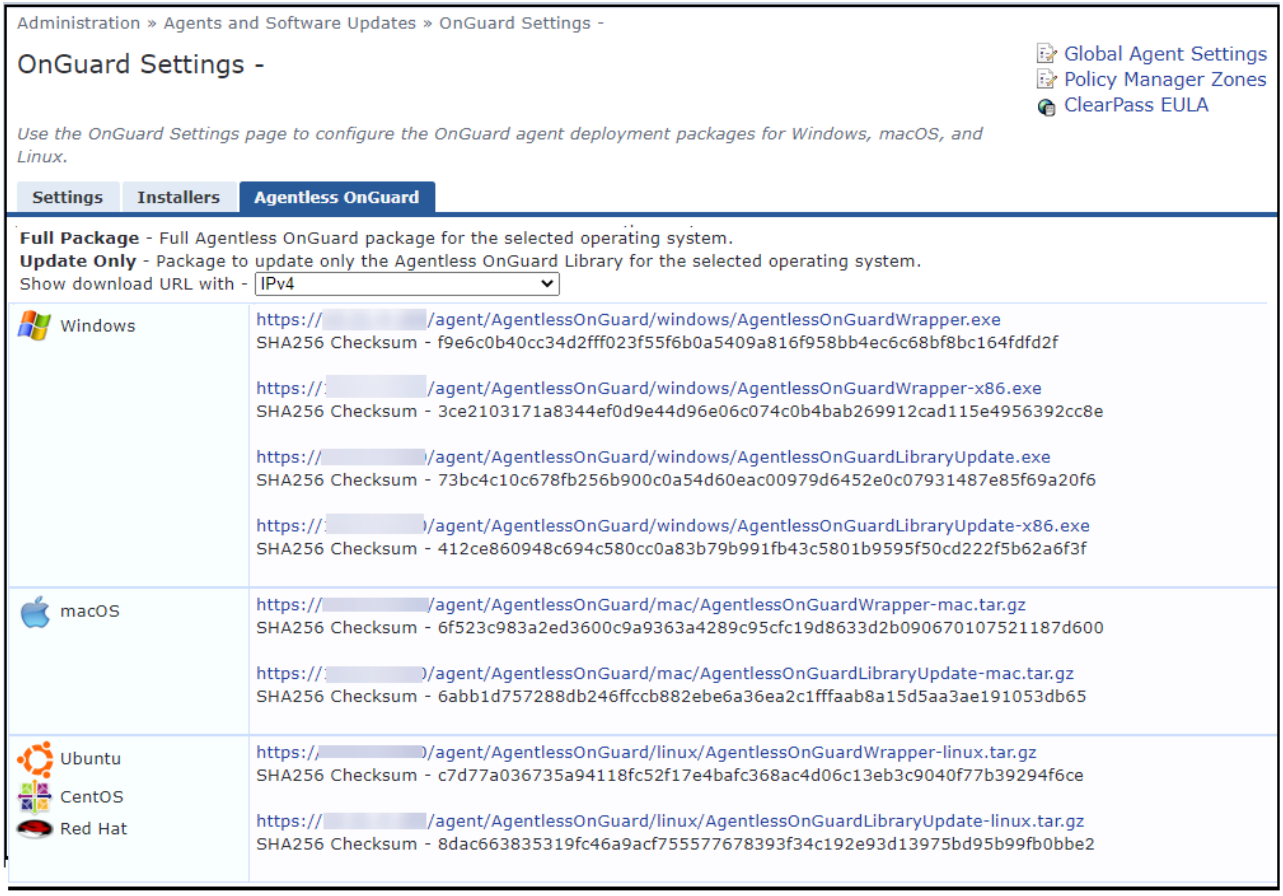


Table 1 describes the OnGuard packages available from the **OnGuard Settings > Agentless OnGuard** page:

Table 1. OnGuard Settings > Agentless OnGuard Packages

Operating System	OnGuard Package	Description
Windows	AgentlessOnGuardWrapper.exe	Agentless OnGuard Wrapper file for 64 - bit Windows OS SHA256 Checksum of Agentless OnGuard Wrapper fil

Operating System	OnGuard Package	Description
		e for 64 - bit Windows OS. This SHA256 Checksum ca n be used in **Agentless On Guard Settings** > *Agentles s OnGuard Checksum* field s.
	AgentlessOnGuardWrappe r - x86.exe	Agentless OnGuard Wrapp er file for 32 - bit Windows OS SHA256 Checksum of Age ntless OnGuard Wrapper fil e for 32 - bit Windows OS. This SHA256 Checksum ca n be used in **Agentless On Guard Settings** > *Agentles s OnGuard Checksum* field s.
	AgentlessOnGuardLibrary Update.exe	OnGuard Agent Library Up date file for Agentless OnG uard for 64 - bit Windows OS SHA256 Checksum of *OnG uard Agent Library Update* file for Agentless OnGuard for 64 - bit Windows OS.
	AgentlessOnGuardLibrary Update - x86.exe	OnGuard Agent Library Up date file for Agentless On Guard for 32 - bit Windows OS SHA256 Checksum of *OnG uard Agent Library Update* file for Agentless OnGuard for 32 - bit Windows OS.
macOS		Agentless OnGuard Wrapp er file for macOS

Operating System	OnGuard Package	Description
		This SHA256 Checksum can be used in **Agentless On Guard Settings** > *Agentless OnGuard Checksum* fields.
	AgentlessOnGuardLibraryUpdate - mac.tar.gz	OnGuard Agent Library Update file for Agentless OnGuard for macOS SHA256 Checksum of *OnGuard Agent Library Update* file for Agentless OnGuard for macOS
Ubuntu CentOS RedHat	AgentlessOnGuardWrapper - linux.tar.gz	Agentless OnGuard Wrapper file for supported Linux OS This SHA256 Checksum can be used in **Agentless On Guard Settings** > *Agentless OnGuard Checksum* fields.
	AgentlessOnGuardLibraryUpdate - linux.tar.gz	OnGuard Agent Library Update file for Agentless On Guard for supported Linux OS SHA256 Checksum of *OnGuard Agent Library Update* file for Agentless OnGuard for supported Linux OS

OnGuard Global Agent Settings

Global Agent Settings

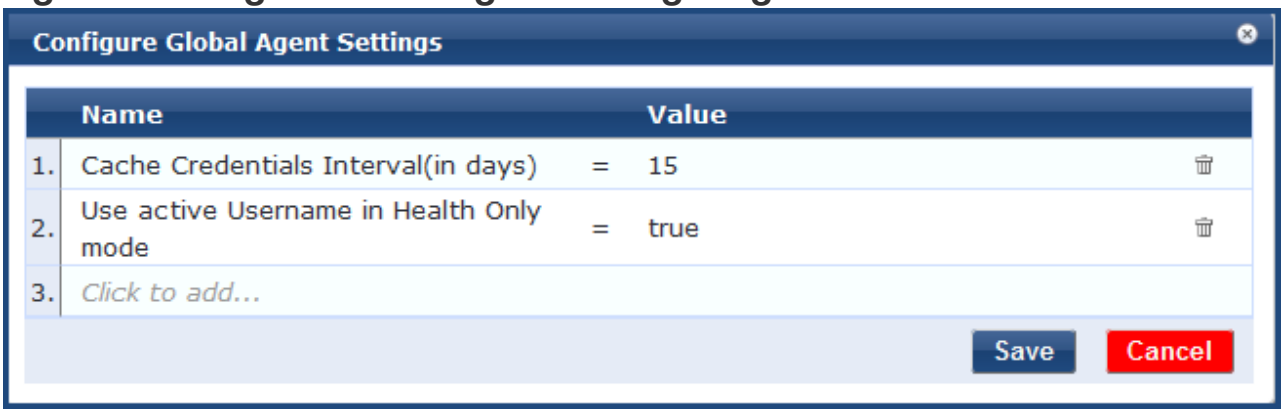
About this task

Use the **Global Agent Settings** page to configure the global parameters for OnGuard agents.

Procedure

1. Navigate to the **Administration > Agents and Software Updates > OnGuard Settings** page.
2. Click the **Global Agent Settings** link. The **Configure Global Agent Settings** page opens.

Figure 1. Configure Global Agent Settings Page



3. To add additional Global Agent Settings parameters, select **Click to add**.
   a. **Name**: Select the desired Global Agent Setting (see Table 1).
   b. **Value**: Specify the appropriate value.
4. Repeat these steps as necessary for each additional parameter, then click **Save**.

Global Agent Settings Parameters for OnGuard Agents

Table 1 describes the **Global Agent Settings** parameters for OnGuard agents:

Table 1. Configure Global Settings Parameters

Name Parameter	Action/Description
Cache Credentials Interval (in days)	Select the number of days the user credentials should be cached on OnGuard agents.
Delay to bounce after Logout (in minutes)	Specify the number of minutes that should elapse before OnGuard bounces the interface if OnGuard remains disconnected.

Name Parameter	Action/Description
Enable OnGuard requests load - balancing	Enable this option to balance the load of OnGuard authentication requests across ClearPass Policy Manager servers in a cluster.
Enable access over Remote Desktop Session	Enable this option to allow OnGuard access through a Remote Desktop session.
Enable to hide Logout button	Enable this option to hide the **Logout** button on OnGuard agent.
Enable to install VPN component	Enable this option to install the OnGuard VPN component.
Enable to use Windows Single - Sign On	Enable this option to allow use of a user's Windows credentials for authentication. **NOTE** For this feature to work on Windows 11 24H2 and higher versions, in the Windows device's system settings, the EnableMPRNotifications policy must be enabled in the group policy.
Keep - alive Interval (in seconds)	Specify a keep - alive interval for OnGuard agents. The connected OnGuard Agents periodically send heart - beat (Keep - Alive) messages to Policy Manager. This interval is defined by the **Keep - alive Interval (in seconds)** parameter. The default value is **60** seconds. Policy Manager uses Keep - Alive messages to update the status of OnGuard Agents regarding OnGuard Activity, and issue CoA (Change of Authorization) for a Session Restrictions Enforcement Profile if OnGuard Agent is disconnected: • Session - Check > Agent - Connection = Down • Post - Auth - Check > Action = Disconnect

Name Parameter	Action/Description
	◦ For related information, see Session Restrictions Enforcement Profile.
OnGuard Health Check Interval (in hours)	Specify the number of hours that OnGuard will skip health checks for healthy clients. **NOTE** the following information when you set the **OnGuard Health Check Interval** parameter: • Set this parameter if OnGuard mode is set to *health* only. • This parameter is valid only for wired and wireless interface types. • This parameter is not applicable for the *OnGuard Dissolvable Agent*, *VPN*, and *Other* interface types. You can also specify the health - check interval in the **Agent Enforcement** profile (**Configuration > Enforcement > Profiles > Add**) to create different Agent Enforcement Profiles for different users.
Run OnGuard As	For details, see the next section, Global Agent Settings: Run OnGuard As Parameter.
Server Certificate Validation	Enables the Policy Manager OnGuard Unified Agent to validate the Policy Manager Server Certificate when it sends a WebAuth health request to Policy Manager.
Server Communication Mode	Based on the value of this attribute, OnGuard Agent will use the *IP address*, *HostName*, or *FQDN* (Fully Qualified Domain Name) to communicate with the Policy Manager server. Select one of the following values: • IP • Hostname • FQDN

Name Parameter	Action/Description
Support Team Email Address	Enter an email address that automatically p opulates the **To** field in the user's email clie nt when they send logs.
Use active Username in Health Only mode	When this option is enabled (set to **True**) a nd the OnGuard Agent is running in *Health Only* mode, instead of using the MAC add ress as the username, OnGuard Agent sen ds the currently active user's username as t he username in the WebAuth request. • Username format on Windows: ◦ Local users: <*host_name*>\<*usernam e*> ◦ Domain users: <*domain_name*>\<*use rname*> • Username format on macOS and Ubunt u: ◦ Local users and Domain users: <*user name*> **NOTE** When this option is enabled, OnGuard Persistent Agent sen ds the attribute **Host:ActiveUs erName** in the WebAuth requ est. This attribute is available i n both Service Rules and Role Mapping. **NOTE** This option is not applicable for Native Dissolvable Agent a nd OnGuard running as *Servic e*.
Use Current OS Language (Windows Onl y)	Enables Policy Manager OnGuard Agent to use the current user's display language, if s upported, overriding the language selecte d at installation time. This parameter appli es only to Windows clients.

Name Parameter	Action/Description
Use OnGuard Credential in Service Mode	When this parameter is enabled, then after an admin user successfully authenticates t o OnGuard, those user credentials are save d to be used when OnGuard is running in **Service** mode. After the system reboots, t he OnGuard Service performs authenticati on using saved credentials, if the credentia ls are valid. If saved credentials are not va lid or missing then OnGuard Service does not perform authentication, and sends a H ealth - only WebAuth request with the MAC address as user name. **NOTE** This feature has the following prerequisites: • The Global Agent settings parameter Run OnGuard As must be set to **Both ServiceAndAgent**. • The OnGuard Mode setting must be ei ther **Authenticate - no health checks** or **Authenticate with health checks**.
VPN Device Names (Linux)	This parameter lets you specify a VPN ada pter (device) name and categorize the net work interface as a VPN on a Linux device. Enter a network interface name in the **Valu e** field. If the OnGuard Agent finds a netwo rk interface with a matching device name, i t classifies it as a VPN interface. Admins ca n specify multiple names by using a comm a separator and adding each name in doub le quotes. For example: **"VPN"**, **"F5"**, **"Chec kpoint"**. Admins can also use an asterisk (*****) as a wil d - card character in this field. For example : • ***VPN***: Specifies that any interface havi ng **VPN** in its device name will be set as a VPN interface.

Name Parameter	Action/Description
	• **F5***: Specifies that any interface that ha s a device name starting with **F5** will be set as a VPN interface. • ***Checkpoint**: Specifies that any interfa ce name that ends with **Checkpoint** will be set as a VPN interface. All the string comparisons are case - insen sitive.
VPN Device Names (macOS)	This parameter lets you specify a VPN ada pter (device) name and categorize the net work interface as a VPN on a macOS devic e. Enter a network interface name in the **Valu e** field. If the OnGuard Agent finds a netwo rk interface with a matching device name, i t classifies it as a VPN interface. Admins ca n specify multiple names by using a comm a separator and adding each name in doub le quotes. For example: **"VPN"**, **"F5"**, **"Chec kpoint"**. Admins can also use an asterisk (*) as a wil d - card character in this field. For example : • ***VPN***: Specifies that any interface havi ng **VPN** in its device name will be set as a VPN interface. • **F5***: Specifies that any interface that ha s a device name starting with **F5** will be set as a VPN interface. • ***Checkpoint**: Specifies that any interfa ce name that ends with **Checkpoint** will be set as a VPN interface. All the string comparisons are case - insen sitive.
VPN Device Names (Windows)	This parameter lets you specify a VPN ada pter (device) name and categorize the net work interface as a VPN on a Windows devi ce.

Name Parameter	Action/Description
	Enter a network interface name in the **Value** field. If the OnGuard Agent finds a network interface with a matching device name, it classifies it as a VPN interface. Admins can specify multiple names by using a comma separator and adding each name in double quotes. For example: **"VPN"**, **"F5"**, **"Checkpoint"**. Admins can also use an asterisk (*****) as a wild - card character in this field. For example: • ***VPN***: Specifies that any interface having **VPN** in its device name will be set as a VPN interface. • **F5***: Specifies that any interface that has a device name starting with **F5** will be set as a VPN interface. • ***Checkpoint**: Specifies that any interface name that ends with **Checkpoint** will be set as a VPN interface. All the string comparisons are case - insensitive.

Global Agent Settings: Run OnGuard As Parameter

About this task

You can configure OnGuard to run health checks even if a user is not logged in.

Procedure

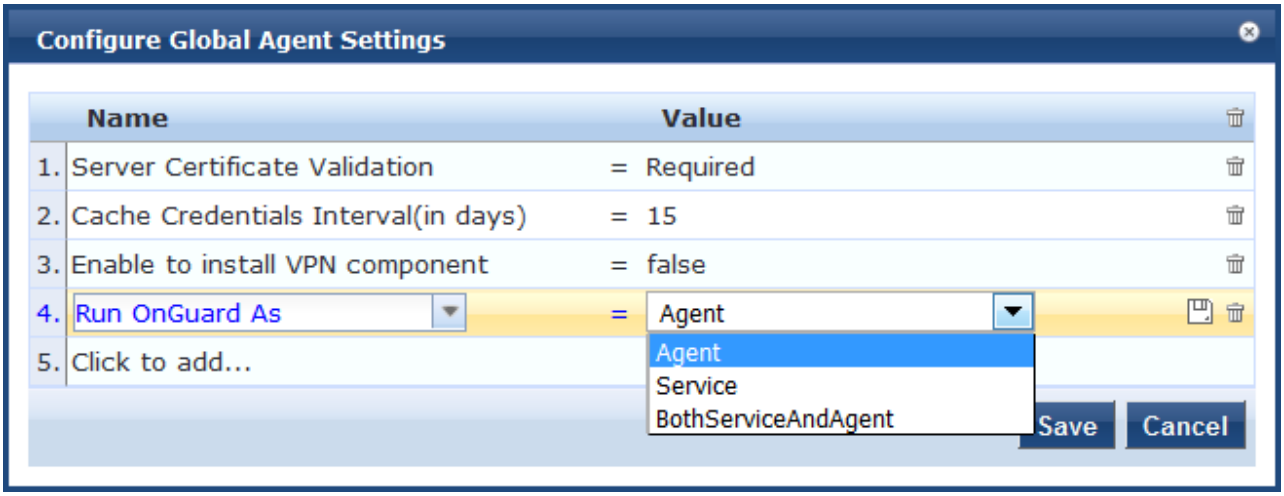
1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**.

   The OnGuard Settings page opens.

2. Click **Global Agent Settings**.

   The Global Agent Settings dialog opens.

Figure 2. Global Agent Settings Dialog



3. Click **Click to add...**.

   a. **Name**: Select **Run OnGuard As**.

   b. **Value**: Select the appropriate option as described in the following table.

      The following table describes available values for the **Run OnGuard As** parameter:

4. Click **Save**.

Table 2. Global Agent Settings: Run OnGuard As Parameters

Run OnGuard As Value	Description
Agent	Health checks are performed by the OnGuard Agent after the user logs in to the client.
Service	OnGuard Agent performs health checks as soon as the client boots up, that is, even before the user logs in to the client. When a user logs in to the client, the user can view the most recent health check results via the OnGuard Agent user interface. The user can perform health checks again by clicking the **Retry** button. For details, see the next section, Limitations for the Run OnGuard As Parameter.
Both Service And Agent	When the user is not logged in to the client, the Policy Manager OnGuard Agent service performs health checks. As soon as

Run OnGuard As Value	Description
	the user logs in to the client, the Policy M anager OnGuard Agent service stops hea lth checks and the OnGuard Agent user i nterface initiates health checks.

Limitations for the Run OnGuard As Parameter

About this task

When **the Run OnGuard As** parameter is set to **Service**, the following limitations pertain:

Procedure

1. In **Service** mode, OnGuard always runs in **Health Only** mode; that is, OnGuard always sends the client's MAC address as the User Name.
2. If a user is not logged in, some of the health checks and auto-remediation may fail in **Service** mode. These health checks are user-level checks, such as Registry Keys (HKCU), Processes, and Installed Applications (user applications).
3. When OnGuard Agent is running in **Service** mode, the OnGuard user interface is used only to display messages and provide the **Retry** button (to perform health checks).
4. The **Enable to Hide Quit Option** does not have any effect in **Service** mode as the **Quit** button is only for exiting the OnGuard user interface.

Creating OnGuard Custom Web Pages

OnGuard provides the ability to show end users a custom interface, or wizard, that guides them through the remediation process if their device is quarantined. When this feature is enabled and OnGuard needs to run a custom remediation script, the wizard tells the user why the device was denied network access, describes the tasks that are required to fix the problem, and lets the user choose whether to execute the remedial script or not. While the script is being executed and new health checks are run, progress messages are displayed.

The pages of the wizard are created using Policy Manager Guest's Web Pages configuration forms, and can be customized with logo, text, and images (for details, refer to the **Custom User Interface** parameter in Table 1, the *OnGuard Settings Parameters* table).

Subtopics

Enabling the Show Custom User Interface for Custom Scripts Attribute
Creating OnGuard Custom Web Pages
HTML Content for OnGuard Custom Web Pages
Important Points
OnGuard Custom Script Exit Codes

Enabling the Show Custom User Interface for Custom Scripts Attribute

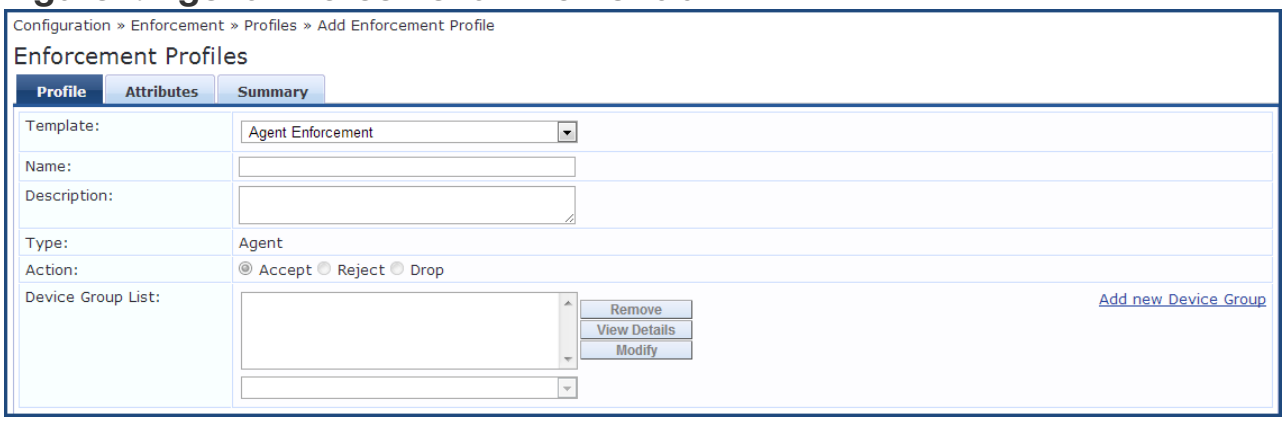
About this task

To configure the **Show Custom UI for Custom Scripts** attribute:

Procedure

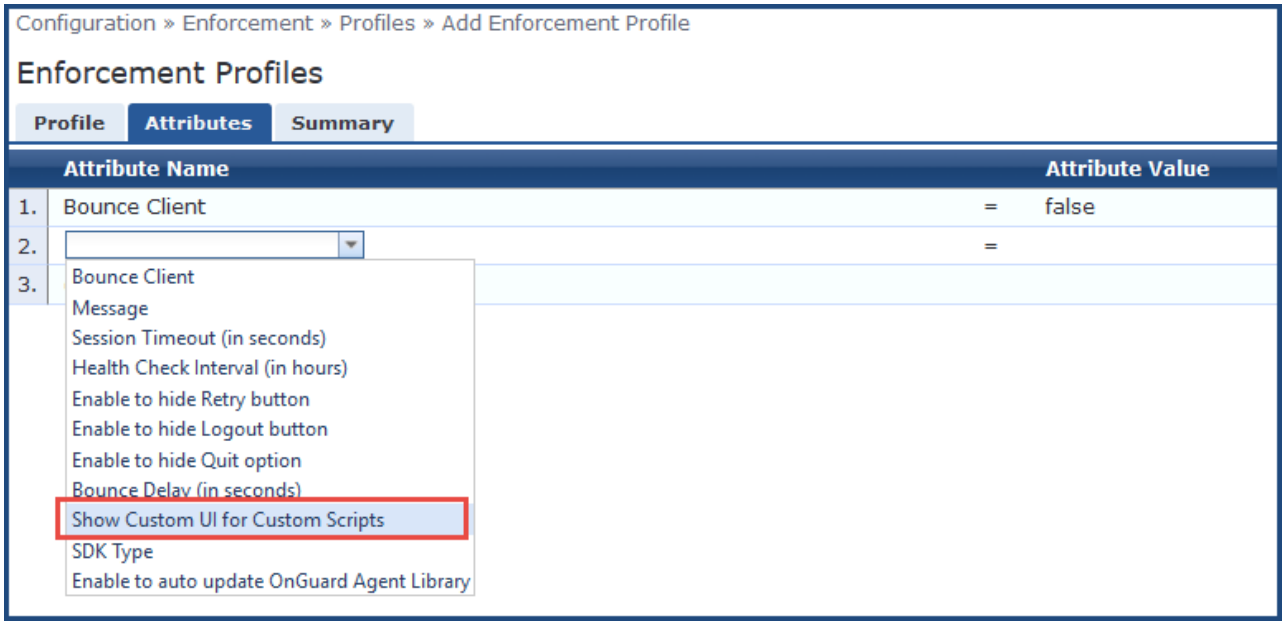
1. Navigate to **Configuration > Enforcement > Profiles**, then select or add the **Agent Enforcement** profile.

Figure 1. Agent Enforcement > Profile Tab



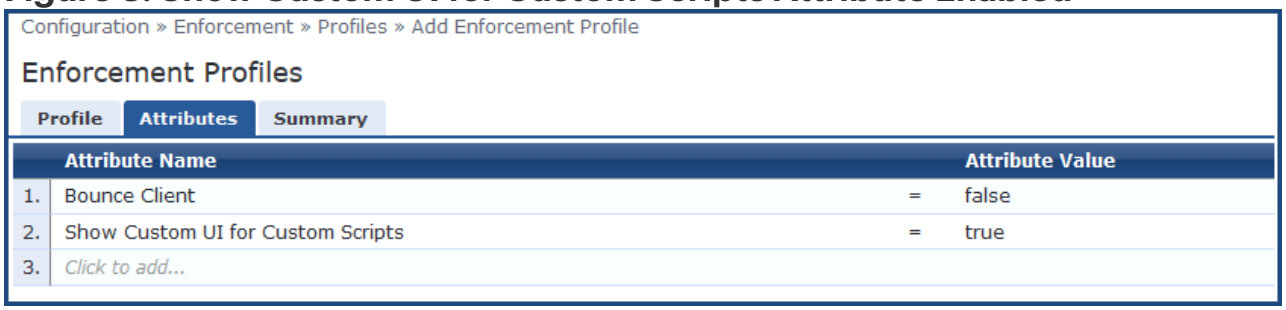
2. Configure the Agent Enforcement Profile as described in Agent Enforcement Profile.
3. Select the **Attributes** tab.
4. From the **Attribute Name** drop-down, select **Show Custom UI for Custom Scripts**.

Figure 2. Enabling the Show Custom UI for Custom Scripts Attribute



5. To set this attribute to **True**, click the **Attribute Value** check box.

Figure 3. Show Custom UI for Custom Scripts Attribute Enabled



6. If needed, complete the configuration for the Agent Enforcement Profile, then click **Save**.

7. Use the **Agent Script Enforcement Profile** to specify the *Success Message*, *Failure Message*, *Progress Message*, and *Description* parameters, as well as other script-related attributes (for configuration details, refer to Configuring Agent Script Enforcement Attributes).

   This step is required because the attributes in the Agent Script Enforcement Profile are used when the Custom UI for Custom Scripts is enabled.

Creating OnGuard Custom Web Pages

About this task

To create **the OnGuard custom web pages**:

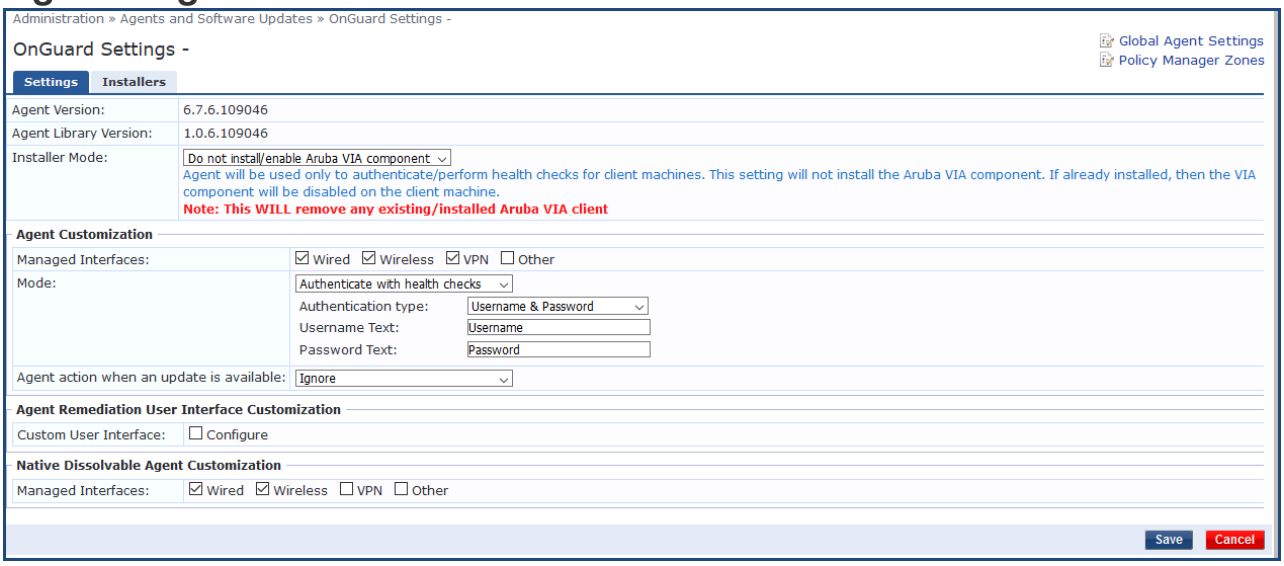
Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**.

   The **OnGuard Settings** page opens.

2. Scroll down to the **Agent Remediation User Interface Customization** section.

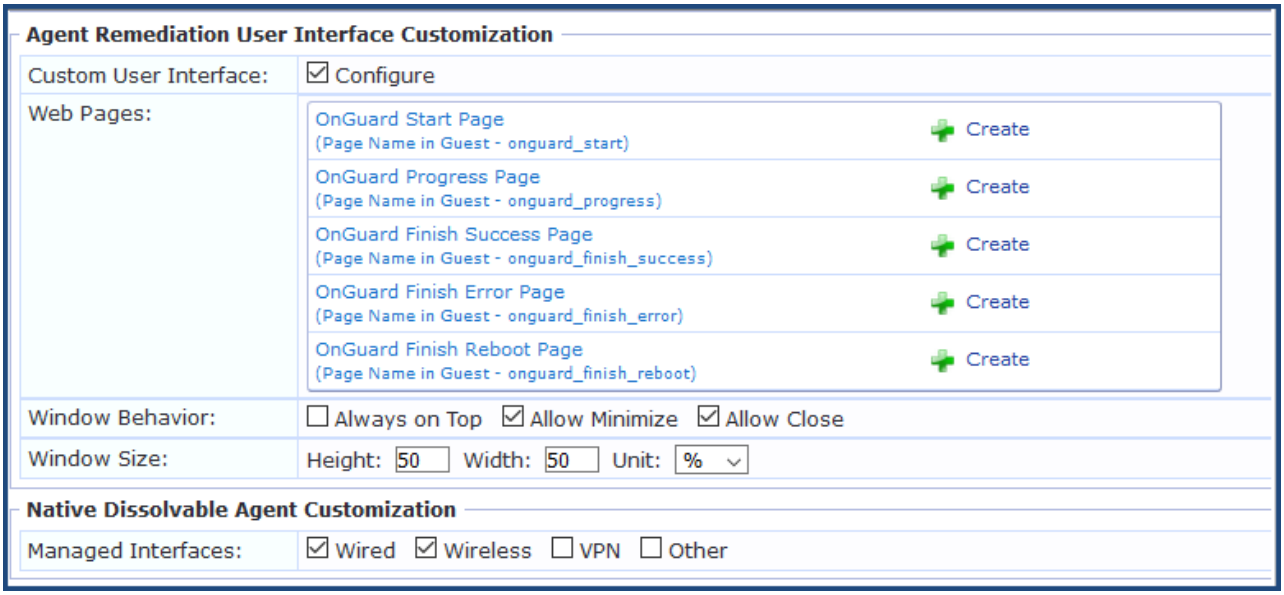
Figure 1. Agent Remediation User Interface Customization



3. To enable the **Custom User Interface** configuration, click (enable) the **Configure** check box.

   The dialog is expanded to show the list of custom web pages you can configure, as well as options to define window behavior and managed interfaces.

Figure 2. Agent Remediation User Interface Customization Dialog



4. Configure window behavior as described in the following table:

Table 1. Configuring the Custom Web Page Advanced Settings Parameters

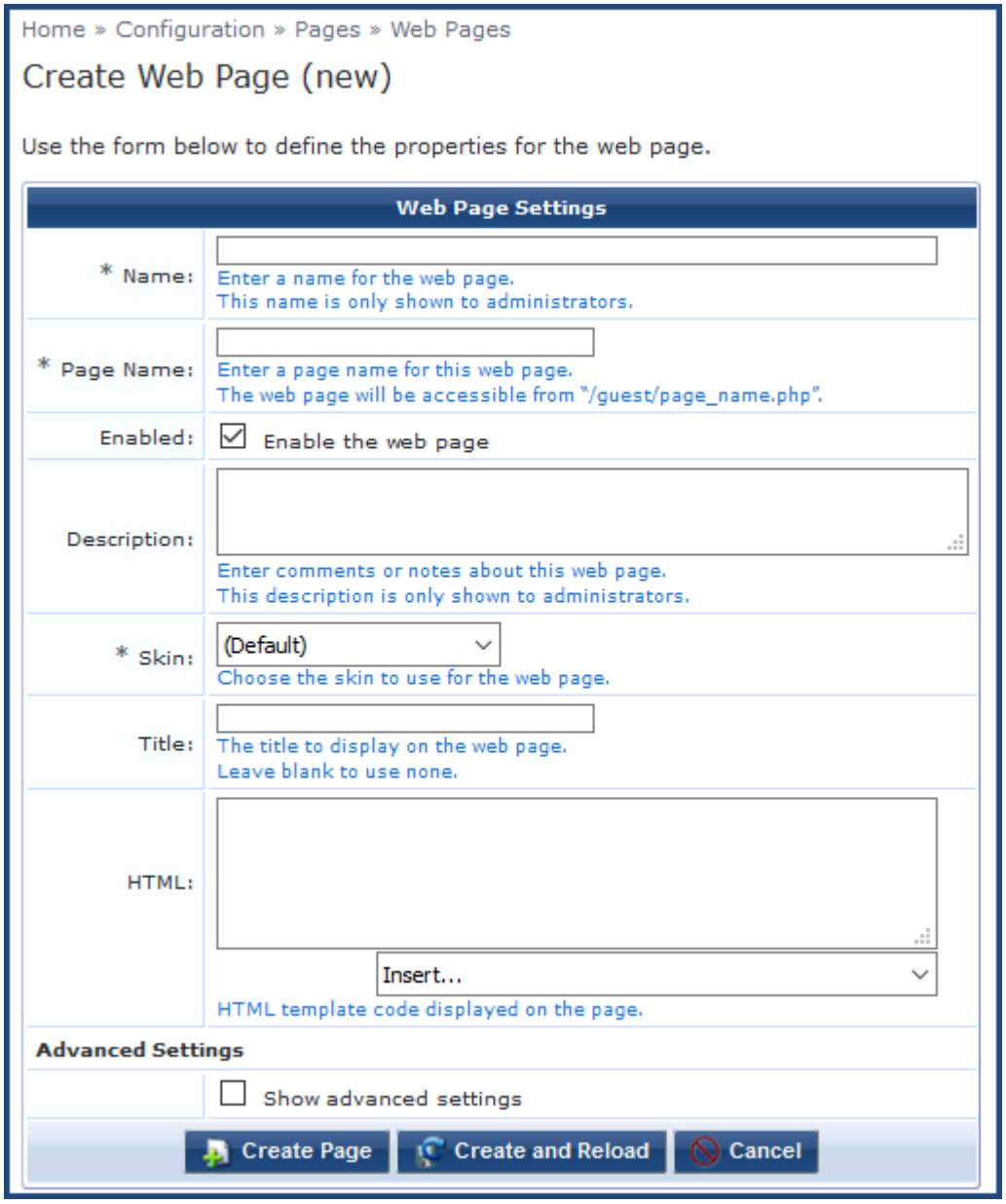
Parameter	Action/Description
Window Behavior	Enable one or more Window Behavior set tings: • **Always on Top**: The Custom User Inte rface window will always be on top of any other windows present. • **Allow Minimize**: When set to **True**, th e Custom User Interface window can be minimized. • **Allow Close**: Prevents users from clo sing the Custom User Interface wind ow. If set to **True**, users will be allowe d to close the Custom User Interface window; however, the execution of cu stom scripts will continue in the bac kground. **NOTE** Even if the **Allow Close** opti on is enabled, OnGuard Age nt disables the **Close** button

Parameter	Action/Description
	of the custom custom user i nterface while it is loading a page.
Window Size	Specify the window height and width, as well as whether the window size should b e a percentage of the the client's screen or defined by the size in pixels.
Native Dissolvable Agent Customization	
Managed Interfaces	The Native Dissolvable Agent performs h ealth checks for one of the selected inte rfaces. This feature ensures that, if both wired and wireless interfaces are connec ted, the OnGuard Agent will send health requests through the correct interface. Select the type(s) of managed interfaces that are supported for the Native Dissolva ble Agent. Select the type(s) of interfaces that are s upported for the Native Dissolvable Agen t: • Wired • Wireless • VPN • Other

5. Click the **Create** link for the OnGuard custom web page you want to create.

   The **Create Web Page (New) > Web Page Settings** configuration dialog opens.

Figure 3. Configuring a New OnGuard Custom Web Page



6. Specify the required Create Web Page parameters as described in the following table:

Table 2. Configuring OnGuard Create Web Page Parameters

Attribute Name	Action/Description
Name	Enter a name for the web page. This nam e is shown only to administrators.
Page Name	Enter a page name for this web page. The web page will be accessible from "*/guest /page_name.php*."

Attribute Name	Action/Description
Enabled	Click (enable) this check box to enable th is web page.
Description	Optionally, enter comments or notes ab out this web page. This description is sh own only to administrators.
Skin	From the **Skin** drop - down, select the ski n to apply to this web page.
Title	Enter the title to be displayed on the web page.
HTML	Enter the required names for each OnGu ard custom web page as well as the reco mmended HTML content. For details, see HTML Content for OnGuard Custom Web Pages.

7. Modify the **Agent Script Enforcement profile** to add the **Description** and **Progress Message** attributes and specify the appropriate text messages (for details, see Configuring Agent Script Enforcement Attributes).

Subtopics

Advanced Settings and Access Control

Advanced Settings and Access Control

About this task

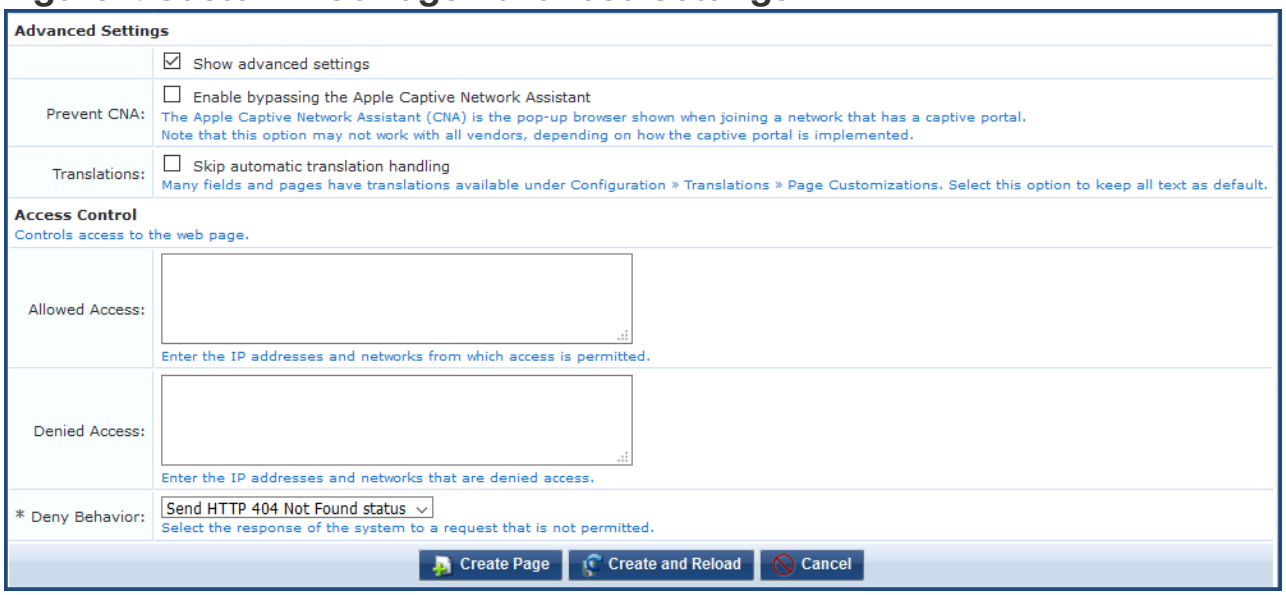
To configure the Custom Web Page Advanced Settings and Access Control settings:

Procedure

1. From the **Create Web Page (New)** dialog, check (enable) the **Show advanced settings** option.

   The Advanced Settings section expands to show the full set of advanced options.

Figure 1. Custom Web Page Advanced Settings



2. Specify the Advanced settings and Access Control settings as described in the following table:

Table 1. Configuring the Custom Web Page Advanced Settings and Access Control Parameters

Parameter	Action/Description
Advanced Settings	
Show advanced settings	To view the Advanced Settings, click the **Show advanced settings** check box.
Prevent CNA	The Apple Captive Network Assistant (CN A) is the pop - up browser shown when j oining a network that has a captive porta l. Note that this option may not work wit h all vendors, depending on how the cap tive portal is implemented.
Translations	Enable this check box to skip automatic t ranslation handling. Selecting this option keeps all the current text as the default. From Policy Manager Guest, many fields and pages have translations available un

Parameter	Action/Description
	der **Configuration > Translations > Page Customizations**.
Access Control	
Allowed Access	Enter the IP addresses and networks from which access is permitted.
Denied Access	Enter the IP addresses and networks that are denied access.
Deny Behavior	From the drop - down, select one of the following responses to a request that is n ot permitted: • Send HTTP 404 Not Found status • Show Access Denied page • Show a blank page

3. Click **Create Page**.

   The **OnGuard custom web page** is created.

HTML Content for OnGuard Custom Web Pages

This section provides the required names for each OnGuard custom web page as well as the recommended HTML content. Be sure to use the **Page Names** specified here as ClearPass Policy Manager and OnGuard Agent look for pages with these names. Text in *italics* should not be changed.

Subtopics

OnGuard Start Page
OnGuard Progress Page
OnGuard Finish Success Page
OnGuard Finish Error Page
OnGuard Finish Reboot Page

OnGuard Start Page

The OnGuard Start Page is the initial web page shown to the end user when script execution begins. This page might include a **Next** button.

Page Name: *onguard_start*

HTML:

Your device does not meet Minimum Specifications, which is required before you can connect to the Network.

The following is required:

* *

* *

Please click the **Next** button below to start the remediation needed.

You will be connected to the Network after verification that your device meets all Minimum Security Specifications.

*<button id="next_button" type="button" onclick=""/>*Next*</button>*

NOTE

If the **Next** button is missing on the OnGuard Start Page, OnGuard Agent will move to the OnGuard Progress Page after 30 seconds. This time duration is not configurable.

OnGuard Progress Page

The OnGuard Progress Page shows the progress and status of custom scripts that are being executed.

Page Name: *onguard_progress*

HTML:

Please do not disconnect your device.

* *

* *

OnGuard Finish Success Page

The OnGuard Finish Success Page is shown after all the scripts have executed successfully and a system reboot is not necessary. This page includes a **Close** button.

Page Name: *onguard_finish_success*

HTML:

We will now rescan your system to verify that it meets Minimum Security Specifications and then connect you to the Network.

If you are not connected in five minutes, please contact 12334 or click here .

<*button id="close_button" type="button" onclick=""/*>Close<*/button*>

OnGuard Finish Error Page

The OnGuard Finish Error Page is shown if at least one of the scripts returns *Failure* and a reboot is not required. This page includes a **Close** button.

Page Name: *onguard_finish_error*

HTML:

Remediating your device to meet Minimum Security Specifications was unsuccessful because:

* *

* *

Please visit this Support Page to get assistance.

You are not yet connected to the Network.

<*button id="close_button" type="button" onclick=""/*>Close<*/button*>

OnGuard Finish Reboot Page

Page Name: *onguard_finish_reboot*

The OnGuard Reboot Page is shown after all the scripts have executed successfully and a system reboot is necessary. This page includes a **Reboot** button.

HTML:

We will now rescan your system to verify that it meets Minimum Security Specifications and reboot your system, then connect you to the Intel Network.

If you are not connected in five minutes, please contact 12334 or visit this Support Page to get assistance.

<*button id="reboot_button" type="button" onclick=""/*>Reboot<*/button*>

Important Points

About this task

Note the following OnGuard Agent behaviors when using the Custom User Interface for Custom Scripts.

Procedure

1. OnGuard Agent checks the custom script's exit code to compute the custom script's status.
2. OnGuard Agent determines the final page based on the script's exit codes and the client's health status. For details, see the next section, OnGuard Custom Script Exit Codes.
3. This feature is not supported when OnGuard is running as a service.
4. The Custom User Interface loads a fresh web page from Policy Manager Guest every time. It does not cache the pages.
5. If the user closes the Custom User Interface while the script is executing, OnGuard Agent continues executing scripts without the Custom User Interface.
6. Administrators will have to refresh or open the **OnGuard Settings** page again after creating web pages in Policy Manager Guest (**Administration > Agents and Software Updates > OnGuard Settings**).
7. If the Policy Manager Server Certificate is not validated when Policy Manager loads the web page for the first time, the Custom User Interface displays the following security alert:

Figure 1. Server Certificate Not Validated Security Alert

8. A new option, **Server Communication Mode**, has been added in **Global Agent Settings** that you can configure to avoid receiving a Server Certificate security alert (see the **Server Communication Mode** parameter description in Global Agent Settings Parameters for OnGuard Agents).

OnGuard Custom Script Exit Codes

The OnGuard custom script exit codes are comprised of **Success Codes** and **Failure Codes** as described below:

Subtopics

Success Codes (0 to 63)
Failure Codes (65 to 255)
OnGuard Agent Codes (256 onwards):

Success Codes (0 to 63)

The range available to Administrators to define their own Success Codes = 3 (0x03) to 63 (0x3F).

- Script executed successfully = 0 (0x00)
- Reboot (Reboot is required) = 2 (0x02)

Failure Codes (65 to 255)

The range available to Administrators to define their own Failure Codes = 65 (0x41) to 255 (0xFF).

- Script executed successfully but its exit code indicates failure = 64 (0x40)

OnGuard Agent Codes (256 onwards):

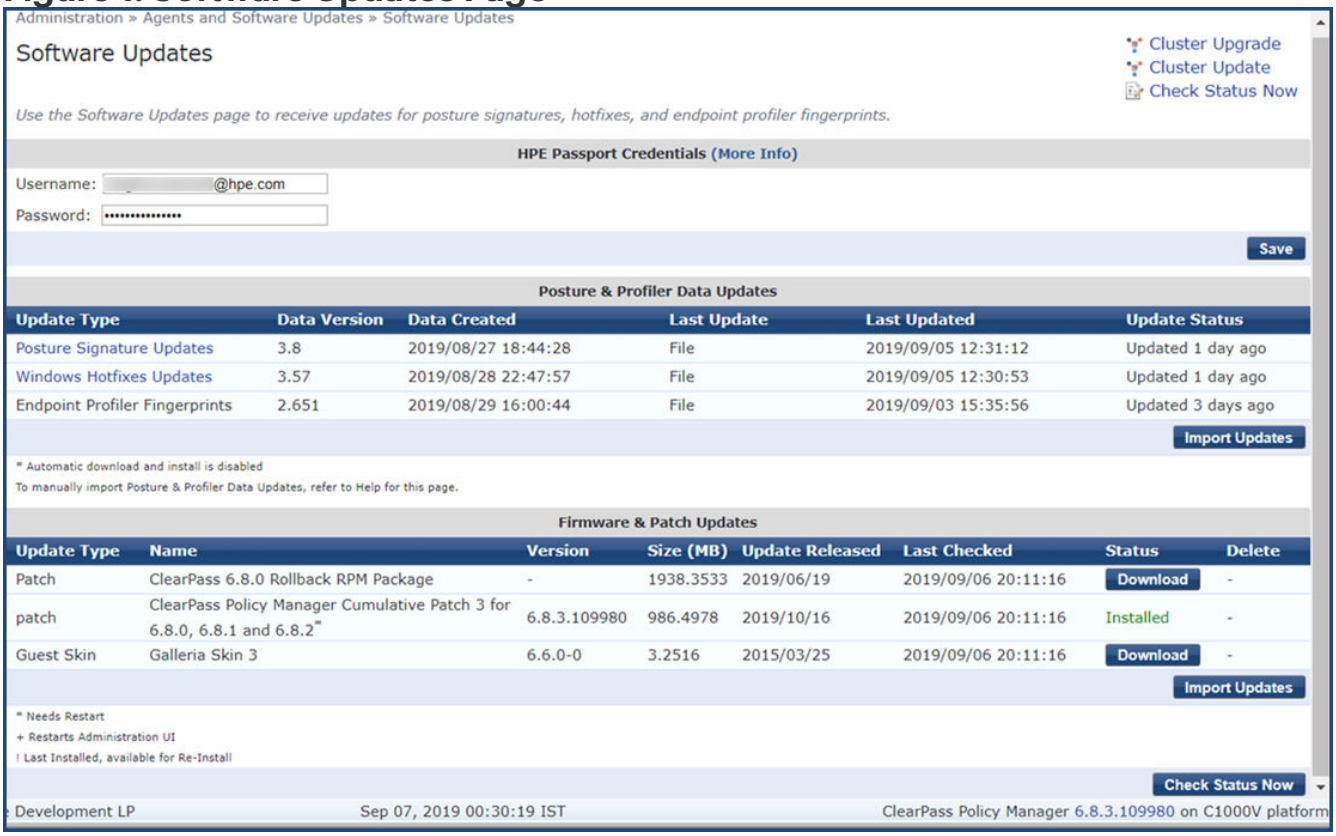
- Unknown error = 256 (0x100)
- Timeout: Script did not finish execution in expected time = 257 (0x101)
- Failed to read exit code of script = 258 (0x102)
- OnGuard failed to execute script = 259 (0x103
- Script file not found = 260 (0x104)
- Script file did not pass validation checks = 261 (0x105)
- Failed to download script file = 262 (0x106)
- Execution level is set to “User” but the user is not logged on, so OnGuard was not able to launch the script = 263 (0x107)

Software Updates

ClearPass Policy Manager regularly checks for available updates on the Policy Manager Webservice server. When new updates are available, a network administrator use the **Administration > Agents and Software Updates > Software Updates** page to download and install these updates. Firmware and patch updates and downloadable user role plugins must be manually downloaded and installed. Windows hotfixes, posture signature updates, and endoint profile fingerprint updates can be automatically downloaded and installed, but this behavior not enabled by default. To set these updates to be automatic, you must enable the **Automatically download Posture Signature and Windows Hotfixes Updates** and **Automatically download Endpoint Profile Fingerprints** settings on the Cluster-Wide Parameters options for the Policy Manager server. In a Policy Manager cluster, the **Import Updates** option is available on the publisher only.

Figure 1 displays the **Software Updates** page:

Figure 1. Software Updates Page



Policy Manager queries the Webservice server for the current list of firmware and patch updates every day at a random time every day. Policy Manager queries the Webservice server for the latest Posture Signature, Windows hotfixes and Fingerprints updates at random minute every hour. Fingerprint, firmware and patch data are refreshed as soon as new updates are available. A list of new firmware and update patches that are available for download and installation are noted by the Policy Manager server automatically and shown in the user interface. An event is generated and displayed in the Event Viewer with the list of new updates that are available. If the event affects an SMTP server, Alert Notification email addresses are configured, and an email from the publisher is sent with the list of downloaded images. System Events (as seen on the **Monitoring > Event Viewer** page) shows records for events, such as communication failures with Webservice, successful or failed update downloads, and successful or failed update installations.

Subtopics

HPE Passport Credentials
Posture, Profiler, and Windows Hotfix Updates
Downloadable User Role Plugin Updates
Firmware and Patch Updates

HPE Passport Credentials

This field is only available on a publisher mode. Software updates are authenticated using a token rather than username and password. Tokens are obtained by clicking the **Generate Token** button in the **HPE PassPort Credentials** section. A new browser page opens and is used to perform the Single Sign On action with the HPE Passport system using the current username/password. This prevents the problem of ClearPass Policy Manager updates attempting to authenticate with a password that has expired or been changed. If the token does not automatically renew, then approximately seven days prior to the expiration date the Ad**ministration > Agents and Software Updates > Software Updates** page will display a message indicating that the token will expire at a specific date and time. Administrators must then click the **Generate Token** button to obtain a new token again.You must authenticate to your HPE passport account to allow the Policy Manager server to automatically contact the Webservice server to check for any available updates. If you do not enter authenticate your credentials, you will have to manually upload and install any updates.

Users should be aware that whenever appliances are taken out of a cluster, then on each resulting standalone appliance you will need to go to **Administration > Agents and Software Updates > Software Updates** and use the **Generate Token** button to generate a new software updates token specific to that appliance.

Posture, Profiler, and Windows Hotfix Updates

Use this section of the **Software Updates** page to view or manually download and install the posture, endpoint profiler and Windows hotfix updates from the Webservice server to a Policy Manager publisher.

Figure 1. Posture & Profiler Data Updates

Posture & Profiler Data Updates

Update Type	Data Version	Data Created	Last Update	Last Updated	Update Status
Posture Signature Updates*	-	-	-	-	Needs Update
Windows Hotfixes Updates*	3.26	2026/01/14 22:07:23	File	2026/01/16 08:58:11	Needs Update
OS Version Updates*	2.23	2025/12/01 12:41:05	File	2026/01/16 08:58:15	Updated 58 days ago
Endpoint Profiler Fingerprints*	3.21	2025/12/06 03:12:46	File	2026/01/16 08:58:13	Updated 58 days ago
Posture Engine Updates*	2026.01.13.106	2026/01/13 14:30:52	File	2026/01/16 08:58:24	Updated 58 days ago

Import Updates

Subtopics

Viewing Available Posture Signature Updates
Viewing Available Windows Hotfixes
Viewing Available Endpoint Profiler Fingerprints
Downloading and Installing Updates

Viewing Available Posture Signature Updates

The Policy Manager server uses posture signature updates to check if the AntiVirus and the DAT files are the latest version. (for more details, refer to the Posture Signature Updates Page). Click the *Posture Hotfixes Updates* link on the **Administration > Agents and Software Updates > Software Updates** page to open an extensive list of all third-party antivirus products supported by Policy Manager. The top of this section displays a version number and a timestamp that identifies when it was last updated. For each product, information includes *Definition Version*, *Definition Date*, *Definition Signature*, and *Engine Version* for all supported versions.

Figure 1. Posture Signature Updates Data Displayed

Posture Signature Updates

Created On - 2019/10/09 16:00:32 UTC — Version - 2.5523

ESET

ESET NOD32 Antivirus

Definition Version	Definition Date	Definition Signature	Engine Version
19789 (20190802)	08/02/2019	-	-
19787 (20190802)	08/02/2019	-	-
19786 (20190801)	08/01/2019	-	-

Definition Version	Definition Date	Definition Signature	Engine Version
15734 (20170712)	-	-	-

ESET Endpoint Security

Definition Version	Definition Date	Definition Signature	Engine Version
19789 (20190802)	08/02/2019	-	-
19787 (20190802)	08/02/2019	-	-
19786 (20190801)	08/01/2019	-	-

ESET Smart Security

Definition Version	Definition Date	Definition Signature	Engine Version
19789 (20190802)	08/02/2019	-	-
19787 (20190802)	08/02/2019	-	-
19786 (20190801)	08/01/2019	-	-

Definition Version	Definition Date	Definition Signature	Engine Version
17207 (20180411)	04/11/2018	-	-
13452 (20160507)	05/06/2016	-	-

ESET Endpoint Antivirus

Viewing Available Windows Hotfixes

About this task

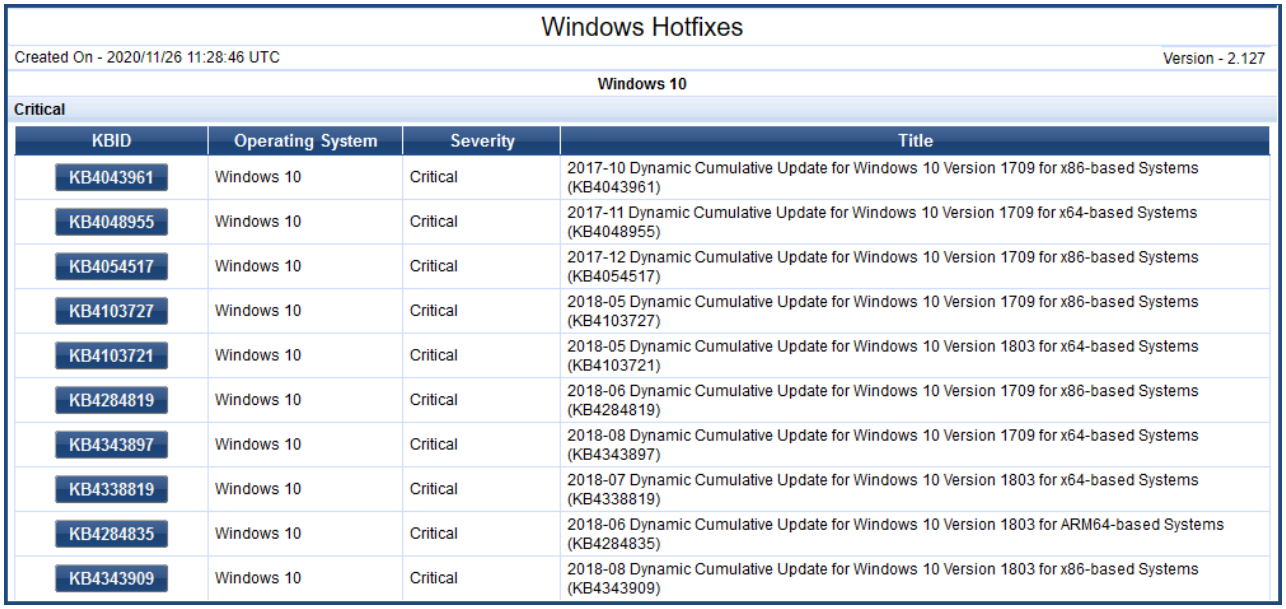
The **Administration > Agents and Software Updates > Software Updates** page includes a list of available Windows Hotfixes for supported Windows operating systems. To view a list of these updates:

Procedure

1. Click the *Windows Hotfixes Updates* link to open a list of all supported Windows OS versions. Then click any Windows version in the list to display the full list of all hotfixes issued for that version.

2. Click the link for any Windows version of interest to display detailed Windows Hotfixes information for the selected version of Windows. The information for each hotfix includes the *KBID* (Microsoft's ID number for the hotfix), *Operating System*, *Severity*, and *Title*.

Figure 1. Complete List of Windows Hotfixes for the Selected Version of Windows



Windows Hotfixes

Created On - 2020/11/26 11:28:46 UTC — Version - 2.127

Windows 10

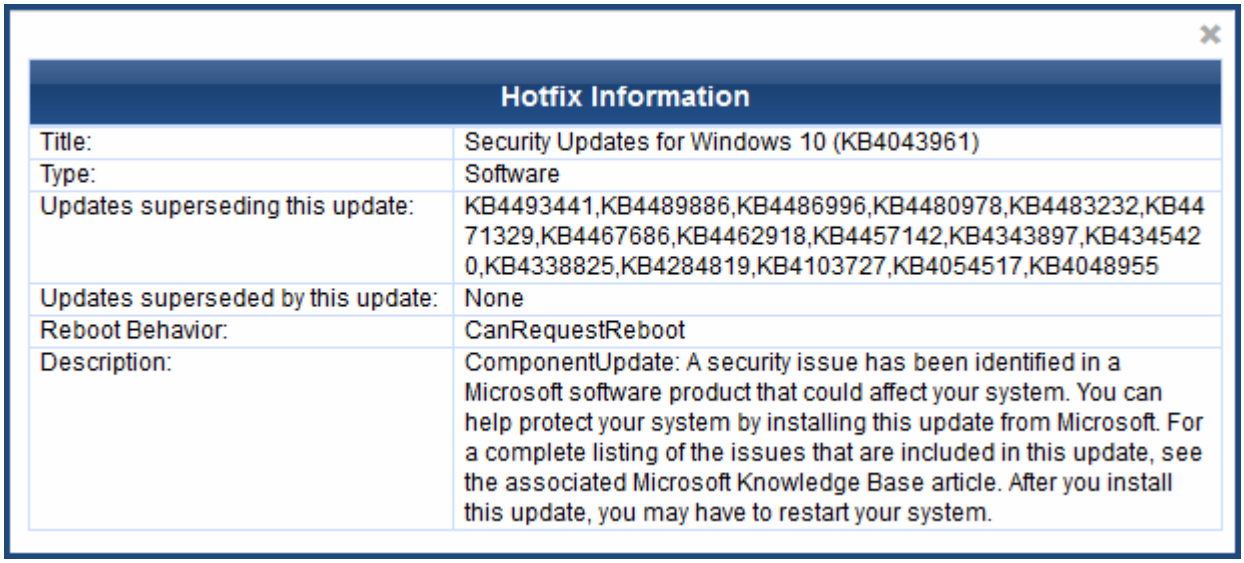
Critical

KBID	Operating System	Severity	Title
KB4043961	Windows 10	Critical	2017-10 Dynamic Cumulative Update for Windows 10 Version 1709 for x86-based Systems (KB4043961)
KB4048955	Windows 10	Critical	2017-11 Dynamic Cumulative Update for Windows 10 Version 1709 for x64-based Systems (KB4048955)
KB4054517	Windows 10	Critical	2017-12 Dynamic Cumulative Update for Windows 10 Version 1709 for x86-based Systems (KB4054517)
KB4103727	Windows 10	Critical	2018-05 Dynamic Cumulative Update for Windows 10 Version 1709 for x86-based Systems (KB4103727)
KB4103721	Windows 10	Critical	2018-05 Dynamic Cumulative Update for Windows 10 Version 1803 for x64-based Systems (KB4103721)
KB4284819	Windows 10	Critical	2018-06 Dynamic Cumulative Update for Windows 10 Version 1709 for x86-based Systems (KB4284819)
KB4343897	Windows 10	Critical	2018-08 Dynamic Cumulative Update for Windows 10 Version 1709 for x64-based Systems (KB4343897)
KB4338819	Windows 10	Critical	2018-07 Dynamic Cumulative Update for Windows 10 Version 1803 for x64-based Systems (KB4338819)
KB4284835	Windows 10	Critical	2018-06 Dynamic Cumulative Update for Windows 10 Version 1803 for ARM64-based Systems (KB4284835)
KB4343909	Windows 10	Critical	2018-08 Dynamic Cumulative Update for Windows 10 Version 1803 for x86-based Systems (KB4343909)

3. Click the **KBID** for a specific hotfix to open a Hotfix Information window that includes the following additional details:

   - Title
   - Type
   - Updates superseding this update
   - Updates superseded by this update
   - Reboot behavior
   - Description

Figure 2. Windows Hotfixes: KBID Information for Specific Hotfix



Hotfix Information	
Title:	Security Updates for Windows 10 (KB4043961)
Type:	Software
Updates superseding this update:	KB4493441,KB4489886,KB4486996,KB4480978,KB4483232,KB4471329,KB4467686,KB4462918,KB4457142,KB4343897,KB4345420,KB4338825,KB4284819,KB4103727,KB4054517,KB4048955
Updates superseded by this update:	None
Reboot Behavior:	CanRequestReboot
Description:	ComponentUpdate: A security issue has been identified in a Microsoft software product that could affect your system. You can help protect your system by installing this update from Microsoft. For a complete listing of the issues that are included in this update, see the associated Microsoft Knowledge Base article. After you install this update, you may have to restart your system.

Viewing Available Endpoint Profiler Fingerprints

The Endpoint Profiler Updates section of the **Posture & Profiler Data Updates** page displays details about the latest updates to fingerprints that are used by Policy Manager in profiling endpoints.

Downloading and Installing Updates

About this task

If you enabled automatic download and installation of updates in the Cluster-Wide Parameters configuration and entered your HPE passport credentials on this **Software Updates** page, the Policy Manager server automatically manages these tasks. Otherwise, to manually download and install a Windows hotfix, posture signature, or endpoint profiler update:

Procedure

1. Navigate to **https://clearpass.arubanetworks.com/cppm/appupdate/<apps_update_filename>**, where <apps_update_filename> is one of the following file names:
   - hpe_cppm_antivirus_updates.signed.tar (Posture Signature Updates)
   - hpe_cppm_fingerprints.signed.tar (Endpoint Profiler Fingerprints)
   - hpe_cppm_windowshotfixes_updates.signed.tar (Windows Hotfixes Updates)

- hpe_cppm_posture_engine_updates.signed.tar (Posture Engine Updates)
- hpe_cppm_apps_updates.signed.tar (File contains the four updates listed above. This file is updated once per day.)

2. When prompted for authentication credentials, enter your Subscription ID or or the Subscription ID 95jqdf-x6xvc4-gvvgy7-x288zb-vd7fjq for both the username and password. (The name of the downloaded file is cppm_apps_updates.signed.tar.)
3. Once you have downloaded the file, you can import it to the publisher by clicking the **Import Updates** button in the **Posture & Profiler Data Updates** section of the **Software Updates** page. The **Import from file** window opens.
4. Click **Choose File** to browse to and select the downloaded file.
5. Enter the shared secret for the file, (if any) then click **Import** to import the file.
6. Once the file is imported, click **Install** to install the update.

   The **Install Update** window can include buttons that perform the following actions:

   - **Reboot**: Initiate a reboot of the server. The **Reboot** button appears only for updates that require a reboot to complete the installation.
   - **Clear and Close**: Deletes the log messages and closes the dialog box.
   - **Close**: Closes the dialog box

     If the **Install Update** dialog is closed, you can bring it up again by any one of the following actions:

   - While the installation is in progress, click the **Install in progress...** link.
   - Click the **Installed > Install Error** link.
   - When the installation is completed, click the **Needs Restart** link.

Downloadable User Role Plugin Updates

About this task

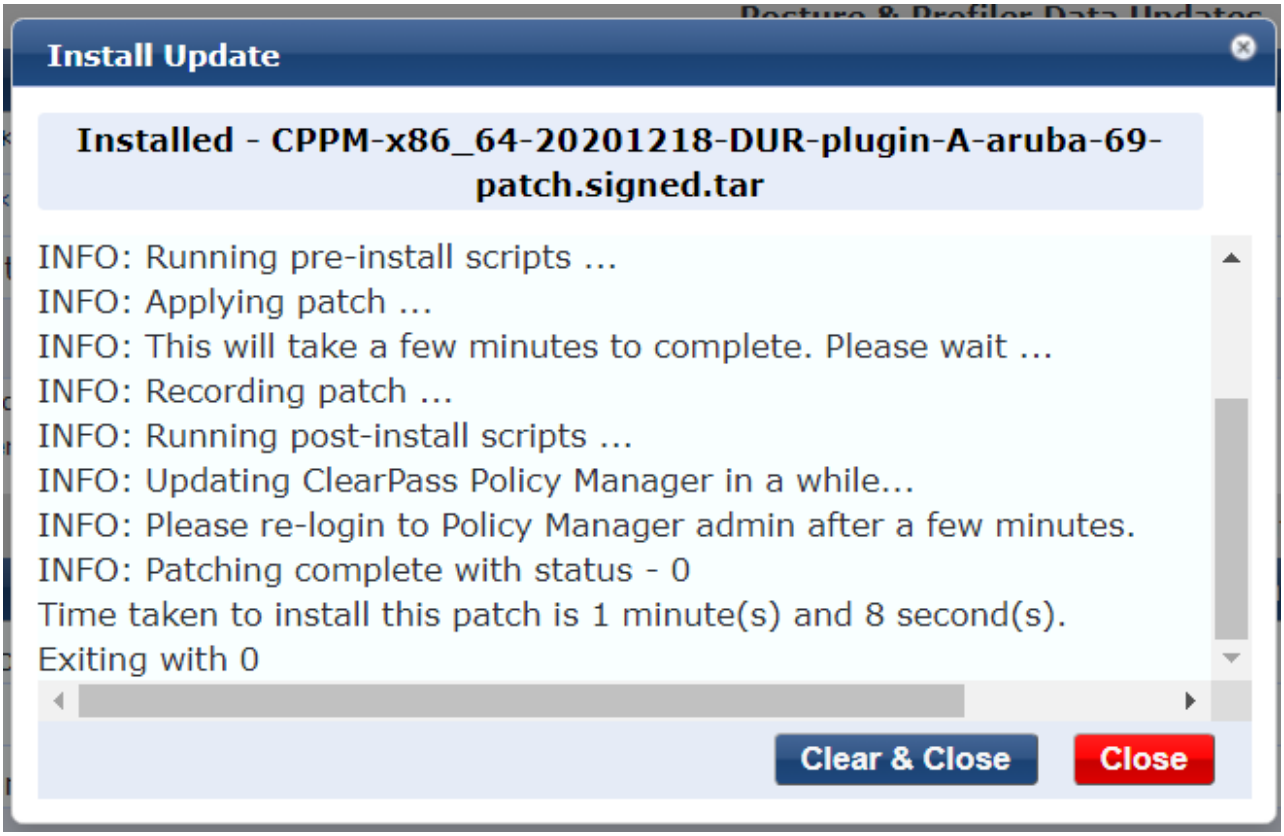
The procedure to install an downloadable user role plugin varies, depending upon whether your Policy Manager server can contact theClearPass Webservice server. If you have entered your HPE passport credentials and Policy Manager is able to reach the Webservice server,any applicable Downloadable User Role plugins are available for download using the following procedure:

Procedure

1. Click the **Download** button to download the Downloadable User Role (DUR) Plugin from the Webservice server.
2. Once the DUR Plugin Update is downloaded, click **Install** to install the update on your Policy Manager server.
3. Once the update is installed, you can click the **Installed** link by the update to display the **Install Update** dialog box shown in Figure 5 and view log messages generated during installation.

   If Policy Manager is not able to reach the Webservice server,
4. Click **Import Updates** to import a downloadable user role plugin obtained via support or other means.
5. You will be prompted to browse to and select the file, and to enter the shared secret for the file (if any).
6. Once the DUR Plugin Update is imported, click **Install** to install the update on your Policy Manager server.
7. Once the DUR Plugin Update is installed, you can click the **Installed** link by the update to display the **Install Update** dialog box and view log messages generated during installation.

Figure 1. Install Update Dialog Box



Firmware and Patch Updates

The **Firmware & Patch Updates** table shows only the data that is known to Webservice or imported using the **Import Updates** button. Patch residual files located in the */var/avenda/platform/backup, /var/avenda/platform/patches,* or */var/avenda/platform/store/updates* folders that are seven (or more) days old are automatically deleted daily.

Subtopics

Installing a Firmware or Patch Update Using the WebUI
Installing a Firmware or Patch Update Using the CLI
Reinstalling a Patch
Uninstalling a Skin

Installing a Firmware or Patch Update Using the WebUI

About this task

The procedure to install an firmware or patch update varies, depending upon whether Policy Manager can contact the Webservice server. If you have entered your HPE

passport credentials and Policy Manager is able to reach the Webservice server, the **Download** button appears by any new firmware or patch updates.

If Policy Manager is able to reach the Webservice server,

Procedure

1. Click the **Download** button to download the file.
2. Once the file is downloaded, click **Install** to install the file on your Policy Manager server.
3. Once the update is installed, you can click the **Installed** link by the update and open and the **Install Update** dialog box shown in Figure 5 and view the log messages generated during installation.

   If Policy Manager is not able to reach the Webservice server,
4. Click **Import Updates** to import a firmware or patch update file obtained via support or other means.
5. You will be prompted to browse to and select the file, and to enter the shared secret for the file (if any).
6. Once the file is imported, click **Install** to install the update.

   NOTE

   If a patch requires a prerequisite patch, that patch's **Install** button will not be enabled until the prerequisite patch is installed.
7. Once the update is installed, you can click the **Installed** link by the update and open and the **Install Update** dialog box shown in Figure 5 and view the log messages generated during installation.

   The **Firmware & Patch Updates** section also includes the following information:

   - **Needs Restart**: The **Needs Restart** link appears when an update needs a reboot of the server in order to complete the installation. Clicking this link displays the **Install Update** dialog box, which shows the log messages generated during the installation.
   - **Installed**: The **Installed** link appears when an update has been successfully installed. Clicking this link displays the Install Update dialog box, which shows the log messages generated during the installation.

- **Install Error**: This link appears when an update install encounters an error. Clicking this link displays the Install Update dialog box, which shows the log messages generated during the install.

Installing a Firmware or Patch Update Using the CLI

When logged in as *appadmin*, you can manually install the upgrade and patch binaries imported via the CLI using the following commands:

- **system update** (for patches)
- **system upgrade** (for upgrades)

Reinstalling a Patch

About this task

The **Reinstall Patch** feature allows the administrator to reinstall a patch in the event the previous attempt to install fails. You can only reinstall the last installed patch, which is indicated by a "!" symbol next to it in the Firmware & Patch Updates table on the **Administration > Agents and Software Updates > Software Updates** page.

To reinstall a patch or software update:

Procedure

1. From the **Firmware & Patch Updates** section of the **Administration > Agents and Software Updates > Software Updates** page, click the **Install Error** or **Needs Restart** link.
2. Click **Re-Install**. The **Install Update** screen closes and the reinstallation process begins. A window displays, showing the installation progress via log messages.

Uninstalling a Skin

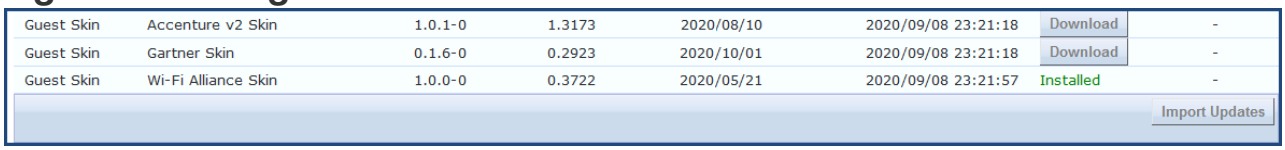
About this task

To uninstall a skin:

Procedure

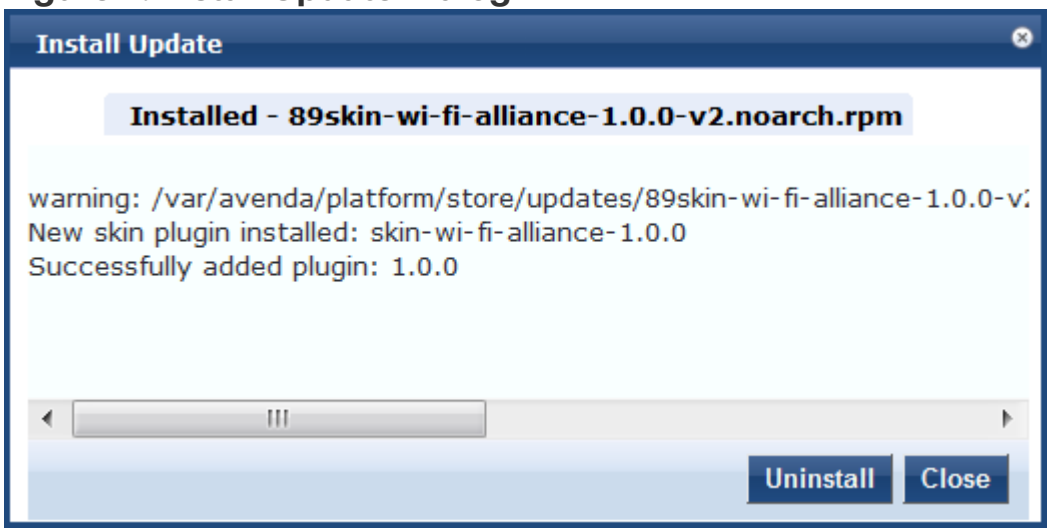
1. Navigate to **Administration > Agents and Software Updates > Software Updates**.
2. In the **Firmware & Patch Updates** section, select the installed skin that you want to uninstall.

Figure 1. Viewing the Installed Link for a Skin



3. Click the **Installed** link. The **Install Update** dialog opens.

Figure 2. Install Update Dialog



4. To uninstall the skin, click **Uninstall**. The **Install Update** screen closes and the software is uninstalled.

Support Services

The **Administration > Contact Support** page provide information for contacting Support, setting up a Remote Assistance session, and viewing Policy Manager documentation:

- Contacting Support
- Remote Assistance
- Accessing Policy Manager Documentation

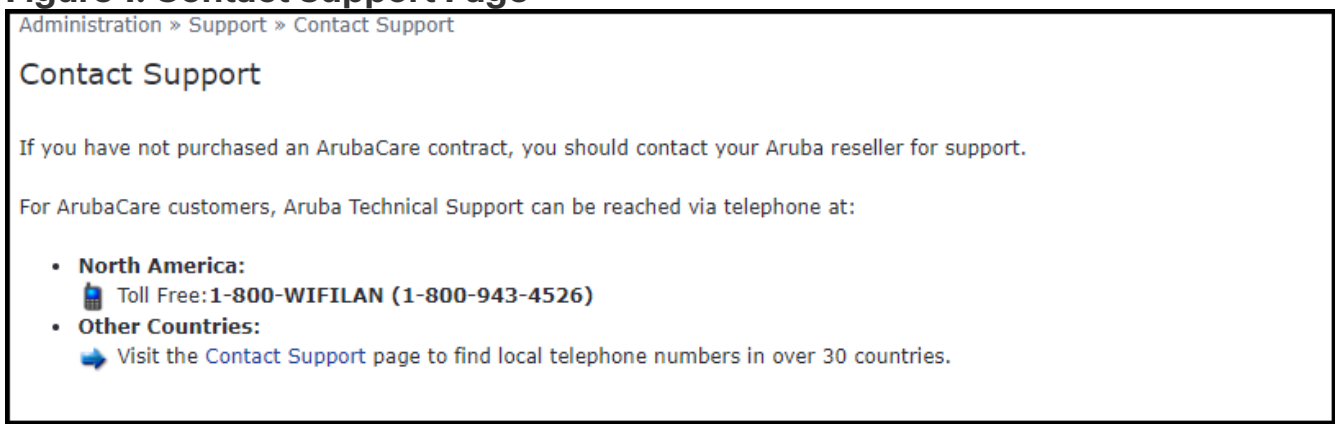
Subtopics

Contacting Support
Remote Assistance
Accessing Policy Manager Documentation

Contacting Support

The **Administration > Support > Contact Support** page provides y information on how to contact Aruba Technical Support.

Figure 1. Contact Support Page



Contacting North America Support

To contact HPE Networking Support Portal in North America (United States and Canada), call this number:

1-800-943-5426

Contacting Support in Other Countries:

About this task

You can call ArubaCare in the country of interest or create a support case by clicking the **HPE My Networking** website link.

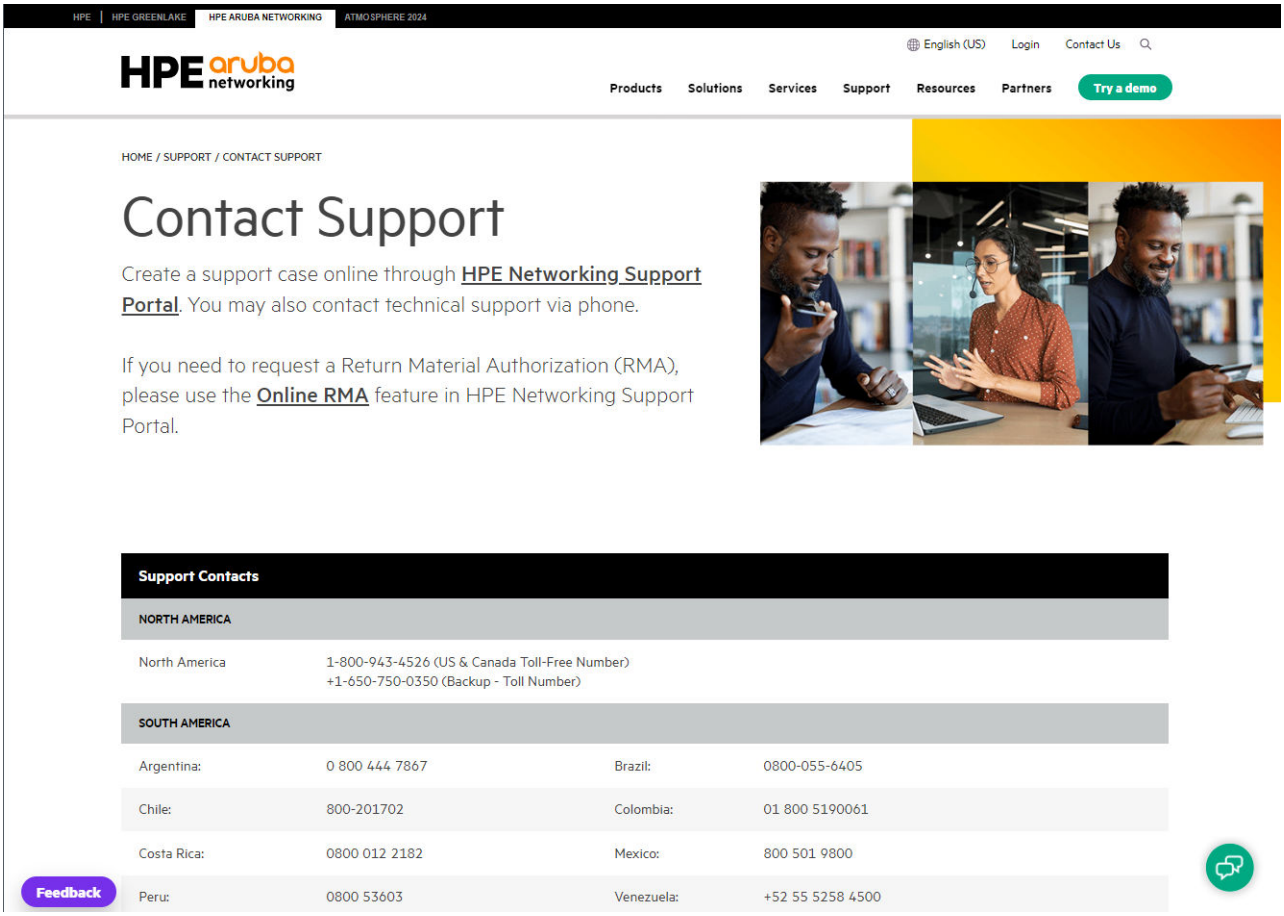
To contact ArubaCare Support in countries other than the United States and Canada:

Procedure

1. Click the **Contact Support** link.

   The **Contact Support** page for other countries opens.

   Figure 2. Contact Support Page for Foreign Countries



2. Scroll as needed to locate the country of interest.

Remote Assistance

- Remote Assistance Process
- Setting Up a Remote Assistance Session

The **Remote Assistance** feature allows a Support engineer to log in remotely using Secured Shell (SSH) to a customer's ClearPass Policy Manager server and perform the following tasks:

> **NOTE**
>
> Remote Assistance is deprecated in ClearPass 6.11, 6.12, and later versions. Although the Remote Assistance page still appears in the user interface in 6.11 and 6.12, the functionality is removed and cannot be used. The page will be removed in the 6.14 release.

- View and operate the ClearPass Policy Manager user interface to debug any issues the customer is experiencing.
- Perform proactive monitoring of the Policy Manager server.

Subtopics

Remote Assistance Process
Setting Up a Remote Assistance Session

Remote Assistance Process

About this task

The Remote Assistance process is as follows:

Procedure

1. The Administrator sets up a Remote Assistance session.
2. The Support contact receives an email with instructions and credentials to log in to the remote system.
3. The Remote Assistance session is terminated at the end of the specified duration.

> **NOTE**
>
> You can terminate a session before its stipulated duration.

4. If the ClearPass Policy Manager user interface at the customer site is inaccessible, configure a Remote Assistance session through the Policy Manager CLI. For more information, refer to the following System Commands.

- **system start-rasession**
- **system status-rasession**
- **system terminate-rasession**

Setting Up a Remote Assistance Session

About this task

You can set up a Remote Assistance session on a ClearPass Policy Manager server in either a single node or cluster setup. Sessions can be saved and deleted only from the publisher. Sessions can be terminated from publisher and subscriber nodes.

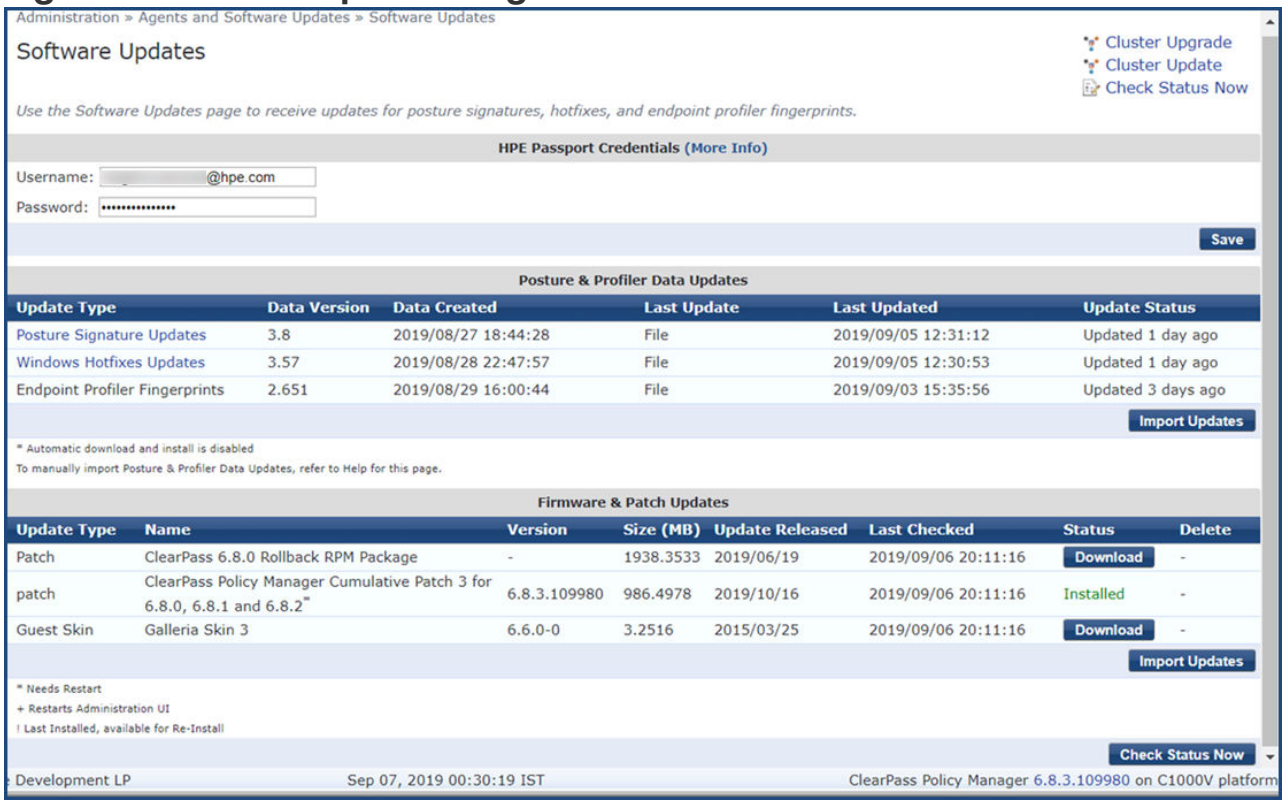
To set up a Remote Assistance session:

Procedure

1. Prior to setting up a Remote Assistance session, you must enter valid **HPE Passport Credentials**. To do so, navigate to **Administration > Agents and Software Updates > Software Updates**.

   The **Software Updates** page opens.

Figure 1. Software Updates Page

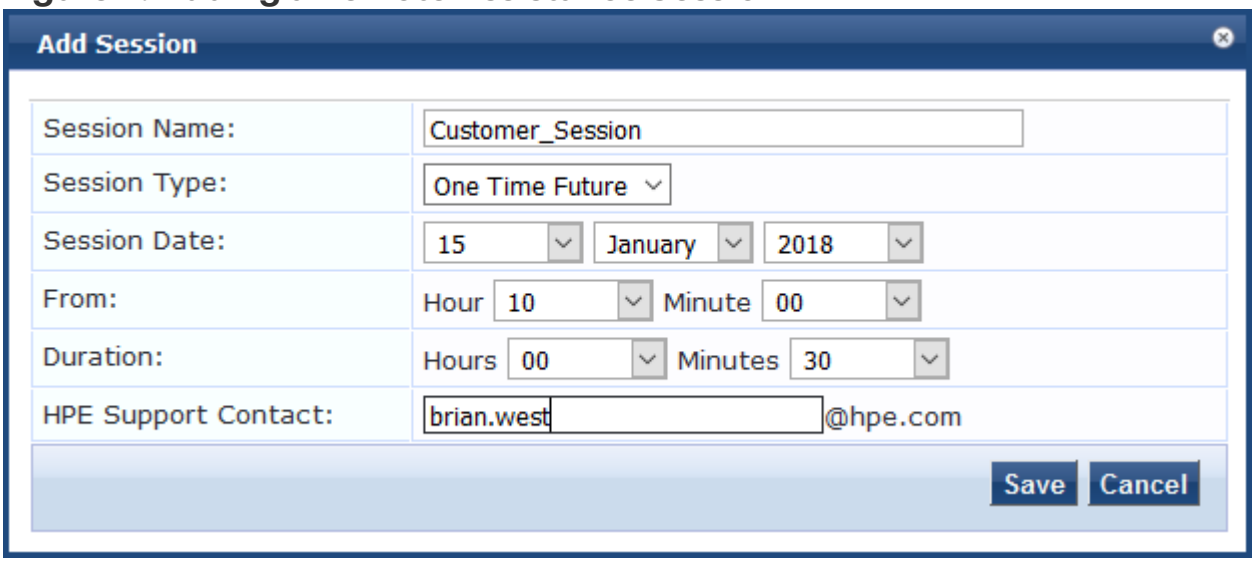


2. Enter the **HPE Passport Credentials**, then click **Save**.
3. Navigate to **Administration > Support > Remote Assistance**.

   The **Remote Assistance** page opens.
4. Click the **Add Session** link.

   The **Add Session** dialog opens:

Figure 2. Adding a Remote Assistance Session



5. Specify the **Add Session** parameters as described in Table 2.

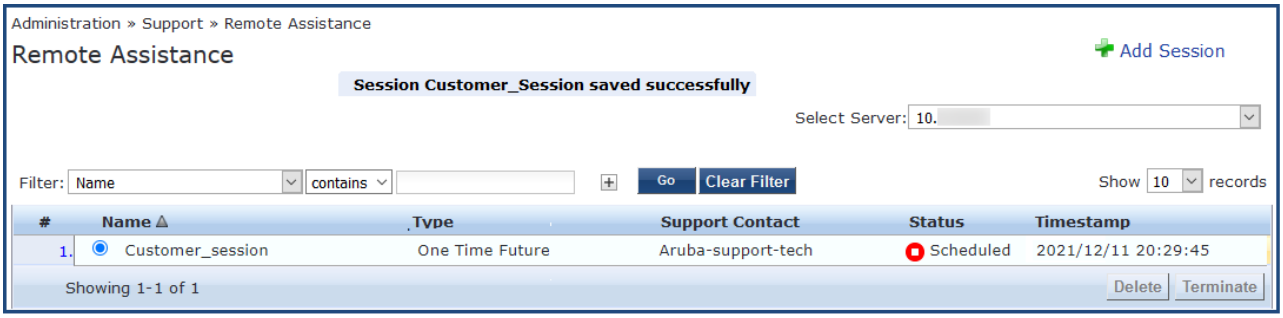
Table 1. . Add Session Parameters

Parameter	Action/Description
Session Name	Enter the name of the Remote Assistance session.
Session Type	Specify the **Session Type**: • **One Time Now** • **One Time Future**: Initiates a session a t a specified date and time. • **Weekly**: Initiates a session on a specif ied weekday at the selected time.

Parameter	Action/Description
	• **Monthly**: Initiates a session on a specified day of every month at the selected time.
Session Date	This fiield appears if you selected the **One Time Future** option in the **Session Type** menu. Select the date you want the session to begin.
Day of Week	This field appears if you selected the **Weekly** option in the **Session Type** menu. Select the day of the week you want to schedule a weekly remote session.
Day of Month	This field appears if you selected the **Monthly** option in the **Session Type** menu. Select the day of the month you want to schedule a momthly remote session.
From:	This fiield appears if you selected the **Weekly**, **Monthly**, **or One Time Future** option in the **Session Type** menu. Select the time you want the session to begin.
Duration	Specify the duration of the Remote Assistance session in hours and minutes. The "session begin" time is the time relative to the Policy Manager server's time, and is specified in a 24 - hour format.
Aruba Support Contact	Enter the email ID of the Aruba/HPE Support contact.

6. Click **Save**. You return to the **Remote Assistance** page where the new session has been added. The information on the **Remote Assistance** page automatically refreshes every few seconds to ensure that the session information remanis current.

Figure 3. Remote Assistance Session Added



The following table describes the information provided for configured Remote Assistance sessions:

Table 2. Remote Assistance Page Description

Parameter	Description
Name	Name of the session.
Type	Indicates if the session is a one - time s ession or a periodic session. To view the schedule of the session, move the cursor over the entry.
Support Contact	The email address of the support contact .
Status	Provides the session state. Available stat es are: • Saving • Scheduled • Initiated • Running • Terminated • Failed **NOTE** You can edit and save a sess ion in **Scheduled**, **Terminate d**, and **Failed** states.

Parameter	Description
	• To terminate a session in **Running** state, select that session and click **Terminate**. • To delete a session in **Scheduled**, **Terminated**, or **Failed** states, select that session and click **Delete**. • If a session fails, the Event Viewer indicates the cause of the failure.
Timestamp	Displays the server time when the Remote Assistance session status was last updated.

Accessing Policy Manager Documentation

About this task

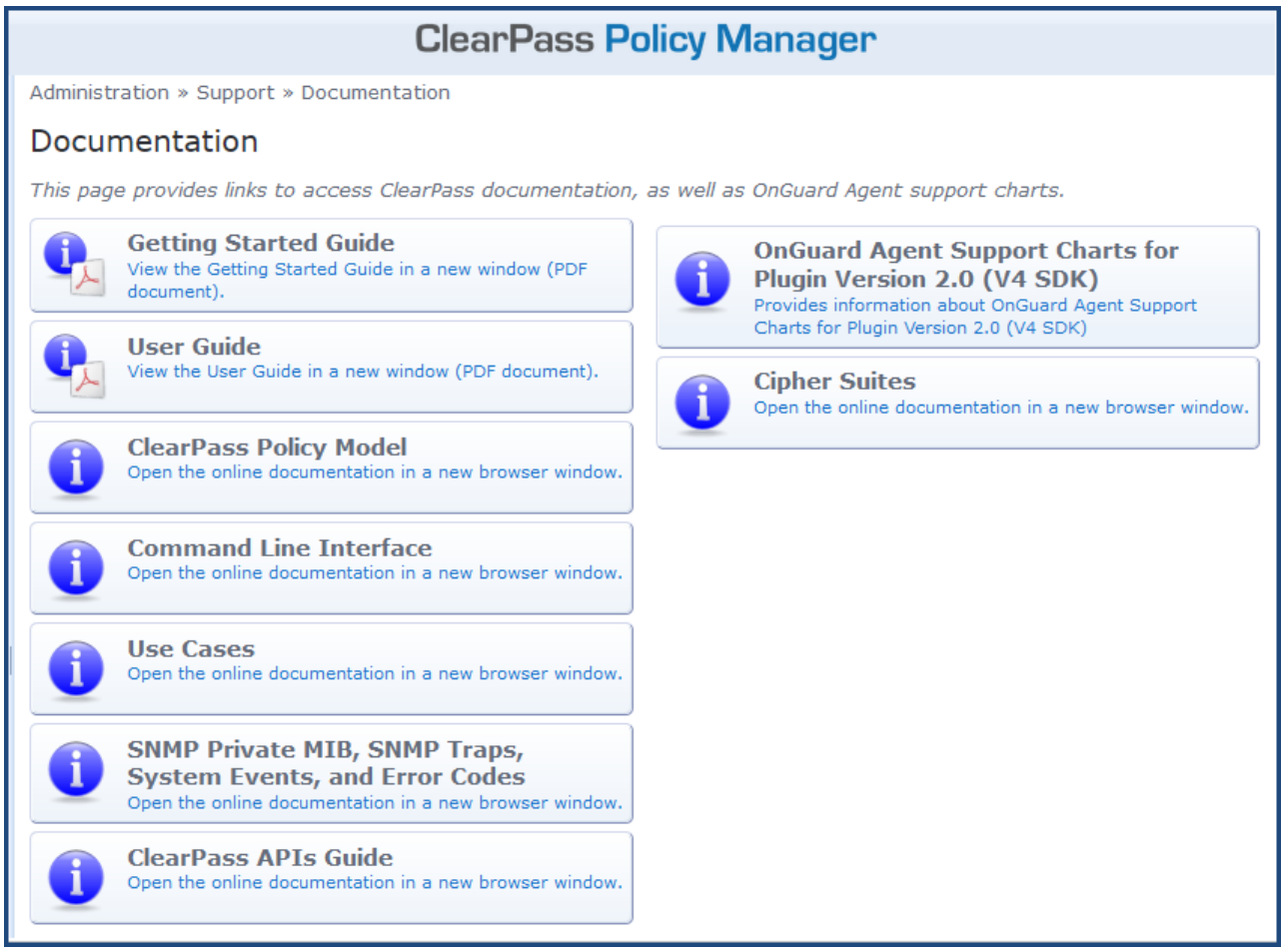
The ClearPass Policy Manager **Documentation** page includes links to useful sections of the ClearPass Policy Manager Online Help system. This page also provides links to the Policy Manager Deployment Guide, the OnGuard Agent Support Charts for Plugin Version 2.0 (V4 SDK), and supported cipher suites.

To access the **Documentation** page:

Procedure

1. Navigate to **Administration > Support > Documentation**. The **Documentation** page opens.
2. Click the Policy Manager documentation you wish to view.

Figure 1. Policy ManagerPolicy Manager Documentation Page



Note that on some browsers (for example, *Google Chrome*), the selected documentation is displayed on your screen. On other browsers (for example, *Mozilla Firefox*), the selected documentation is downloaded in PDF format to your *Downloads* folder.

SSH Timed Account Lockout

Subtopics

Introduction
SSH Account Lockout Configuration
SSH Account Lockout Alerts
SSH Account Lockout Behavior

Introduction

The **SSH Timed Account Lockout** feature provides an administrator with the ability to configure the number of successive unsuccessful authentication attempts for administrators attempting to authenticate remotely. The SSH timed account lockout feature configuration persists across reboots, updates and upgrades. The account lock status persists across reboots.

When the defined number of unsuccessful authentication attempts has been met, the CLI account is locked and administrators cannot log in to the system via the CLI until one of the following conditions are met:

- The offending remote administrator cannot successfully authenticate until an action is taken by a local administrator (the administrator issues the **ssh unlock** command).
- The offending remote administrator cannot successfully authenticate until a time period defined by the administrator has elapsed.

  This feature is node-specific. In a cluster with multiple nodes, SSH timed account lockout must be configured on each node in the cluster.

NOTE

The **cluster reset-database** command does not impact this feature.

NOTE

The SSH timed account lockout feature is disabled by default.

SSH Account Lockout Configuration

The **ssh lockout** commands provide ability to configure the SSH lockout configuration options.

- **SSH lockout count**: The **ssh lockout count** command sets the maximum number of failed login attempts before the account is locked out. The default is **5**.
- **SSH lockout duration**: The **ssh lockout duration** command sets the amount of time in minutes that the account will remain locked after the number of SSH password login attempts exceeds the SSH lockout count.
- **SSH lockout mode**: The **ssh lockout mode <basic|advanced>** command sets the mode for this feature. In basic mode, Login failures via SSH public key methods are

not counted towards the account lockout, while in advanced mode, login failures via SSH public key failures are cunted towards the account lockout.

- **ssh lockout reset**: The **ssh lockout reset** command resets the SSH lockout feture settings back to the default values.

SSH Account Lockout Alerts

Alerts for SSH lockout events are logged in to the Event Viewer when any of the following conditions are present:

- SSH lockout configurations are performed
- Account is locked
- Account is unlocked
- Failed SSH login attempts

SSH Account Lockout Behavior

The SSH account lockout feature is disabled by default. To enable SSH account lockout, issue the the **ssh lockout count** or **ssh lockout duration** commands to set the configuration options.

For example, if the SSH account lockout feature is configured with **failed attempts=3** and **unlock time = 5 minutes**:

- CLI access via SSH (password-based) authentication is locked on three consecutive failed login attempts.
- If a failed password attempt continues (even after the account is locked), it will not be tracked, and the unlock time will not be shifted. It will be the time of the last failed password attempt (third failure attempt), plus the lock out period.
- Successful password-based SSH logins are rejected during the lockout period.
- Console-based logins are allowed during the lockout period.
- SSH logins via public key methods are allowed during the lockout period.

Administrators can use any of the above options to reset the SSH account lockout by issuing the **ssh unlock** command.After the lockout period, successful SSH logins are accepted and the account is unlocked.

ClearPass Licensing Management

Subtopics

Licensing Overview
Activating a Platform License or Platform Activation Key
Updating an Evaluation Policy Manager Platform License
Adding an Application License
Activating an Application License
Managing and Monitoring Application Licenses
Backing Up Licenses
Monitoring License Usage
Offline License Activation
Licensing FAQs
Licensing Troubleshooting

Licensing Overview

The Policy Manager licensing structure is scalable for any size network, whether small or large. Almost all license management is available within the Policy Manager user interface, and up-to-the-minute usage statistics can be viewed at a granular level.

Subtopics

Permanent, Subscription and Evaluation License Types
Policy Manager Platform Licenses
Application License Types

Permanent, Subscription and Evaluation License Types

Policy Manager licenses are issued as *Permanent,Subscription,* or *Evaluation* types:

- **Permanent** licenses do not expire.
- **Subscription** licenses can be valid for one, three, or five years and expire after the specified subscription period.

- **Evaluation** licenses are valid for a shorter time period, typically between 90 and 180 days.

> **NOTE**
>
> When a Subscription or Evaluation license expires, Policy Manager continues to operate normally. However, administrators will not be able to make Policy Manager configuration or service changes and upgrades are not operable.
>
> A Policy Manager deployment cannot use both subscription and permanent licenses for a license type (for example, an Onboard Application License); they need to be of the same type.

Policy Manager Platform Licenses

The Policy Manager Platform License is the base-level license and enables Policy Manager on the appliance, including the Policy Manager and Guest user interface. You must have a Policy Manager Platform License for every hardware or virtual appliance. The Policy Manager Platform License is available as a permanent or evaluation license.

- If you are upgrading to Policy Manager 6.11 from Policy Manager 6.6.x or later, your existing Policy Manager License key will be automatically converted to a Policy Manager Platform Activation Key (PAK). You will not need to do anything to make the conversion happen, and the legacy Platform Activation Key is preactivated as a Policy Manager Platform License.
- If you are a new customer doing a fresh installation of Policy Manager 6.11.x, then you can use the *HPE Networking Support Portal* to receive a Platform Activation Key for each Policy Manager appliance and redeem your licenses. For details, refer to Activating a Platform License or Platform Activation Key .
- If you do not want to activate a platform license online through Policy Manager, you can activate the license offline by submitting a case through the My Networking portal. For details, refer to .
- If a Platform License is invalid, or does not have a valid Support Agreement Entitlement ID tied to it, an administrator cannot download the Update or Upgrade archive, or install Updates and Upgrades imported for offline systems.

Licensing requirements and procedures vary slightly between ClearPass virtual appliances and ClearPass hardware appliances. For details, see Platform License or Platform Activation Key Requirements.

Application License Types

ClearPass Policy Manager supports the following Application License types:

Subtopics

Entry Licenses
Access Licenses
Access Upgrade Licenses
OnGuard/Compliance Suite Licenses
Onboard Licenses
Application License Consumption
ClearPass 6.11 License Activation

Entry Licenses

An **Entry** license is a basic Application License that supports a limited number of core features, including:

- 802.1X Authentication
- MAC authentication
- Web-based user registration and authentication (such as self-registration, sponsor based, and social)
- Multi-Factor Authentication
- OnConnect
- Some 360 Security Exchange features, including local Endpoint Context Servers and Context Server Actions for the local host, and XML/REST APIs.

Entry licenses are available as permanent or evaluation licenses, and are supported on Policy Manager 6.8.0 or higher.

NOTE

The Entry license does not include support for the TACACS+ authentication and endpoint profiling features supported by the Access license. Entry licenses also do not support non-Local host endpoint context servers or Policy Manager extensions.

Access Licenses

About this task

The **Access** license enables the complete set of Policy Manager features and authentication types, including 802.1X, MAB, WebAuth, Oauth2, TACACS+, and Guest authentication. Guest functionality is also embedded in **Access** licenses. Access licenses are available as a Permanent, Subscription or Evaluation license. If your Policy Manager server does not have an Access license, the Network Scan feature is disabled, the Span port is set to the default setting. Note that the Network Scan feature is disabled when Device Insight Integration is enabled (see Device Insight Integration Page for more information).

When an Access license that was not activated expires, service creation on that Policy Manager server is blocked. For this reason, Aruba recommends that if multiple Access licenses are installed, activate them as soon as they are installed. This ensures that even if one activated Access license is expired, service configuration will be allowed if the other Access Licenses have not yet reached their expiration dates.

NOTE

If you have purchased Access licenses, the licensing system will not accept Entry licenses (or even Entry and Access Upgrade licenses together).

If you have purchased Access licenses, the ClearPass licensing system will not accept Entry licenses (or even Entry + Access Upgrade licenses). Once you have Access licenses, only Access will load for this function. If Entry and Access Upgrade are not applied in equal quantities, they are treated as Entry licenses only.

If Policy Manager service creation has been blocked due to an Access license that was allowed to expire while it was not activated:

Procedure

1. Activate the expired license.

2. Verify whether the other Access licenses are still within their valid date range.
3. Activate any other Access licenses that have expired.

Access Upgrade Licenses

An **Access Upgrade** license allows you to upgrade an Entry license to support the complete range of features supported by an Access license. This license is available as a permanent license only, and was introduced in Policy Manager 6.8.0. To upgrade Entry Licenses to Access Licenses, you must purchase an Access Upgrade license for every one of your Entry licenses. For example, if you have ten Entry licenses, you cannot purchase five Access Upgrade licenses and upgrade just five Entry licenses to Access Licenses. You must purchase ten Access Upgrade licenses for the ten Entry licenses for the upgrade to work.

NOTE

If Entry and Access Upgrade Licenses are not applied in equal quantities, they are treated as Entry licenses only.

OnGuard/Compliance Suite Licenses

The **OnGuard** and **Compliance Suite** licenses are responsible for all the activities related to Policy Manager OnGuard.

- An OnGuard license is a permanent license that does not expire.
- A Compliance Suite license is a subscription license available for one, three, or five years.
- A Compliance Suite license also allows Policy Manager to integrate with Device Insight.

  A Compliance Suite or OnGuard license is consumed for all OnGuard deployments (*Persistent* and *Dissolvable*) and any mode of operation (*Authentication with Health*

Checks, *Health Check only*, *Authentication only*). A Compliance Suite or OnGuard license is consumed on a device basis for a period of 24 hours.

> **NOTE**
>
> When a Compliance Suite license is installed without an Entry or Access license, the only services that can be configured are Device Insight Integration and OnGuard health checks.

Onboard Licenses

Onboard license usage is computed based on the number of users with Onboard-generated device certificates. It is available as either a Permanent, Subscription or Evaluation license. The minimum number of Onboard licenses is 100.

Application License Consumption

Policy Manager endpoints consume **Entry** and **Access** licenses based on active session counts. The active session is defined as a duration between:

- RADIUS accounting Start and Stop
- OnConnect SNMP MAC address learned and MAC address removed or aged

  If the session cannot be identified (due to a lack of accounting), the Access license is consumed based on the MAC address for 24 hours.

> **NOTE**
>
> On a Policy Manager system with a valid Access license, TACACS+ sessions are not counted towards Access license consumption.

> **NOTE**
>
> WebAuth (non-OnConnect), OAuth2, AppAuth, and API requests are not counted towards Access license consumption.

ClearPass 6.11 License Activation

About this task

If attempting to upgrade to 6.11.0, keep in mind license activation does not carry forward from 6.9.x and 6.10.x versions. Obtain a copy of all the license keys (Platform, Access, Onboard, OnGuard etc.) prior to moving to ClearPass 6.11. The keys can be viewed from the output of the command.

> **NOTE**
> The ClearPass 6.11.0 release has tight checks on the license format. Before a ClearPass system is installed with 6.11.0 software, Support should be contacted to replace each legacy license key with an updated license key in the correct format. Legacy license keys are those issued prior to ClearPass 6.8.0, and consist of a single line. The current license key format was introduced in ClearPass 6.8.0, and consists of several lines.

In case of a ClearPass cluster, run the command on publisher (to view all cluster-wide application licenses) and on each subscriber to view its Platform license.

ClearPass 6.11 license activation requires:

Procedure

1. Backing up existing licenses from ClearPass 6.9.X or 6.10.X systems.
2. Install ClearPass version 6.11.0 or deploy a new 6.11.0 VM. For more information, refer to the .
3. Add the existing license keys used in the 6.9.X or 6.10.X deployment.
4. Activate the license keys on the 6.11.0 system.

> **NOTE**
> Once licenses are re-applied in ClearPass 6.11.0, the system behaves as if the licenses are added for the first time, since the activation state is not ported from leagcy 6.9.X or 6.10.X versions to 6.11.0. In this case there is a 90 day period to activate licenses in ClearPass 6.11.0.

Activating a Platform License or Platform Activation Key

This topic includes the following information:

- Platform License or Platform Activation Key Requirements
- Activating the License Key for a Virtual Appliance
- Activating the Platform Activation Key for a Hardware Appliance

Subtopics

Platform License or Platform Activation Key Requirements
Activating the License Key for a Virtual Appliance
Activating the Platform Activation Key for a Hardware Appliance

Platform License or Platform Activation Key Requirements

Licensing requirements and procedures vary slightly between ClearPass virtual appliances and ClearPass hardware appliances.

- A ClearPass **Platform License** provides the Platform License Key that is installed on all the virtual appliances in a Policy Manager cluster.
- A ClearPass **Platform Activation Key** activates a Platform License Key for a ClearPass hardware appliance.

  You must have a Policy Manager Platform License or Product Activation Key for every instance of the product. Platform Licenses and Platform Activation Keys are available for purchase as a **Permanent** license (with no expiration date), or as an **Evaluation** license with a 90- or 180-day duration.

NOTE

You cannot have more than one Policy Manager Platform License or Platform Activation Key installed on a Policy Manager server.

Activating the License Key for a Virtual Appliance

To use a new license key for a virtual appliance, activate that license key on the Aruba support portal, then also activate it on the virtual appliance.

Subtopics

Activating the License in the HPE Networking Support Portal
Adding the License to the Virtual Appliance
Activating a License Key from the Virtual Appliance Login Page
Activating a License Key for the Virtual Appliance in the WebUI

Activating the License in the HPE Networking Support Portal

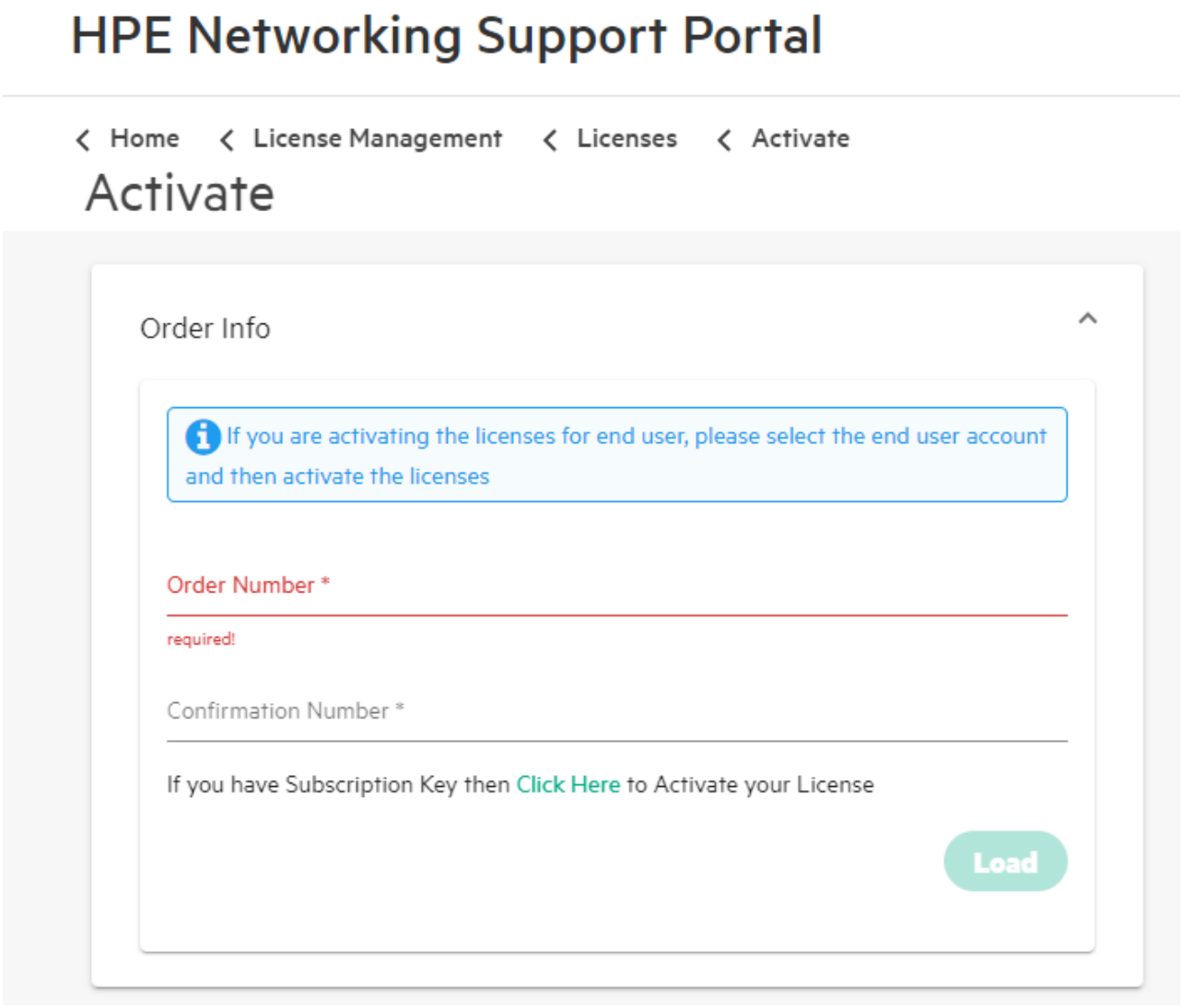
About this task

To activate a license key for a ClearPass virtual appliance on the HPE Networking Support Portal:

Procedure

1. Open the HPE Networking Support Portal at **https://networkingsupport.hpe.com/home**.
2. In the top navigation, go to **Service Management > License Management,** and then go to **Licenses > Activate.** The **License Activation** form opens (**https://lms.arubanetworks.com/licenses/activate**).
3. Enter the order number and confirmation number you received in an email from HPE.
4. Click **Load**. The HPE Networking Support Portal displays all licenses associated with that order.
5. Select a license from the table. In the **Redeem** column, enter the number of licenses you wish to activate.
6. In the **Friendly Name** field, enter a name for the license or VA you are activating.
7. In the **Software Version** field, select **6.8 & later** to indicate that this is a new license and not a license in the legacy license format used by ClearPass 6.7 and earlier releases.
8. Click **Activate** to activate that license.

Figure 1. Activating a License in the Support Portal



9. The HPE Networking Support Portal displays the activation key for the license. Copy this entire key. Be sure to include the "begin" and "end" lines, including all the hyphen characters (for example, "-----BEGIN ACCESS LICENSE KEY-----"); however, do not include any leading or trailing space characters.

Adding the License to the Virtual Appliance

Procedure

1. In a separate tab of your web browser, go to the ClearPass Policy Manager management interface:

   `https://x.x.x.x/tips/`, where **x.x.x.x** is the IP address of the management port defined for the ClearPass virtual appliance.

2. You will be prompted to enter the Platform License for that VM. Once entered, you have 90 days to activate the license.

   For ClearPass virtual appliances running version 6.10 or later, you can activate the license for that individual virtual appliance from the login screen of that VA or from the **Administration > Server Manager > Licensing** page of the WebUI. If your VA is running ClearPass 6.9.0, you must upgrade to a later version or download and install a licensing software patch before you can activate your license. For more information, refer to Upgrading the Appliance Prior to Activating the License.

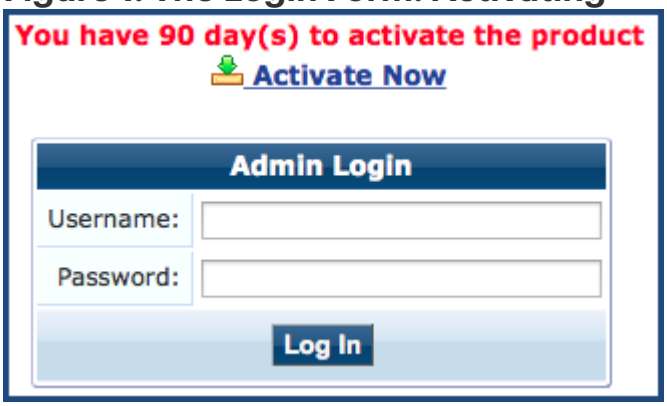
Activating a License Key from the Virtual Appliance Login Page

Procedure

1. In a separate tab of your web browser, go to the ClearPass Policy Manager management interface:

   `https://x.x.x.x/tips/`, where **x.x.x.x** is the IP address of the management port defined for the ClearPass virtual appliance. The ClearPass Policy Manager **Admin Login** page opens. It displays a message indicating that you have 90 days to activate the product, and includes an **Activate Now** link.

   Figure 1. The Login Form: Activating

2. Click the **Activate Now** link in the message on the login page. The **Activate License** form opens.

3. If you are connected to the internet, click the **Activate Now** button. ClearPass Policy Manager attempts to activate the product over the Internet with Aruba license activation servers. Otherwise, if the ClearPass Policy Manager virtual appliance does not have Internet access, follow the steps described in .

Activating a License Key for the Virtual Appliance in the WebUI

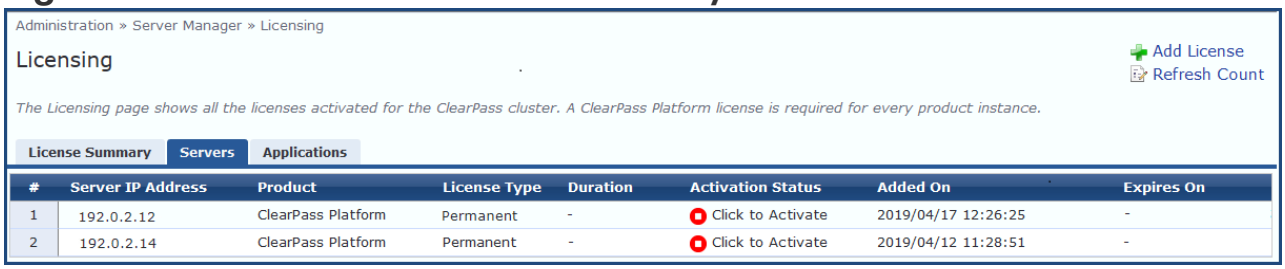
About this task

To activate the Platform License on a publisher or subscriber:

Procedure

1. Navigate to **Administration > Server Manager > Licensing**. The **Licensing** page opens to the **License Summary** tab.
2. Select the **Servers** tab.

Figure 1. ClearPass Platform License Ready to Be Activated

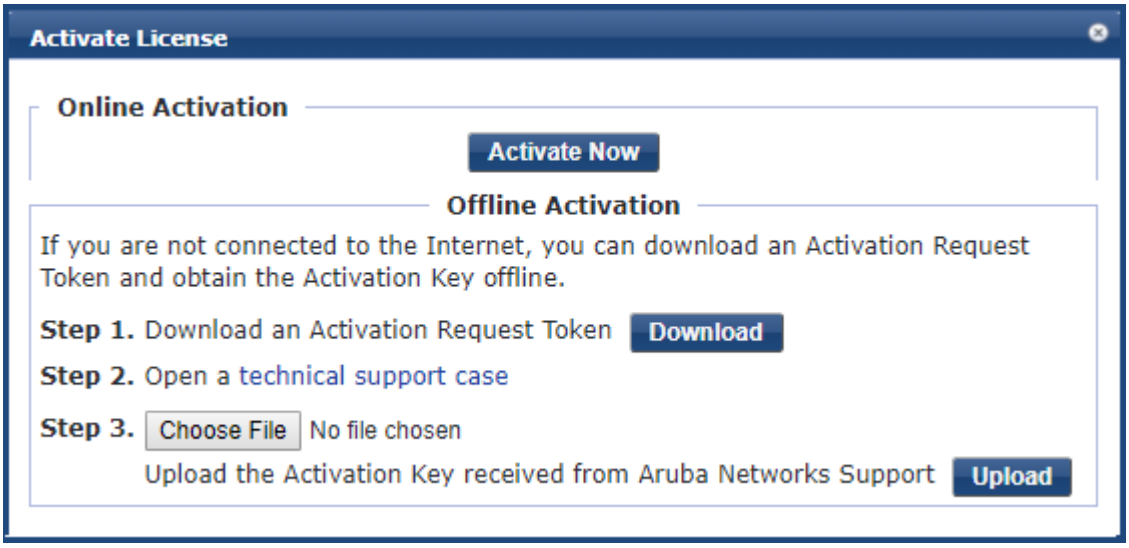


The Policy Manager Platform Licenses for the Policy Manager servers in the cluster are listed.

- If the Platform License is not activated, a red **Click to Activate** button appears on the **Administration > Licensing > Servers** tab. Continue to the next step.
- If the Policy Manager Platform License is already activated, it is shown with a green **Activated** button.

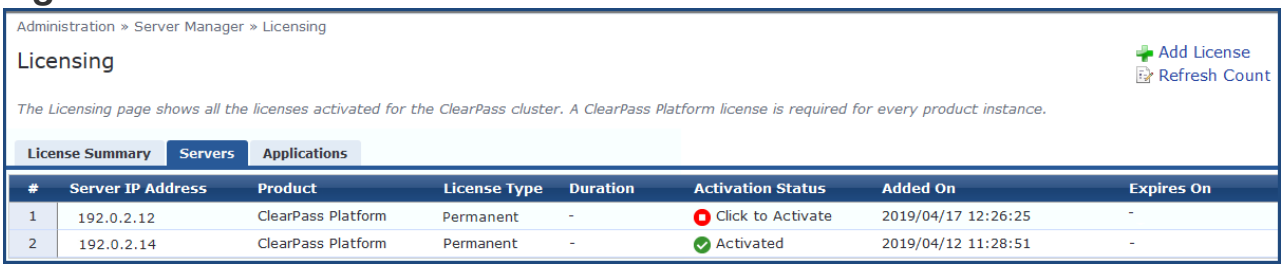
3. Select the **Click to Activate** icon. The **Activate License** form opens.

Figure 2. Activate License Form



4. In the **Online Activation** section, click **Activate Now**. The Policy Manager Platform License is now activated. On the **Servers** tab, the **Activation Status** column shows a green circle next to the keyword **Activated**.

Figure 3. ClearPass Platform License Activated



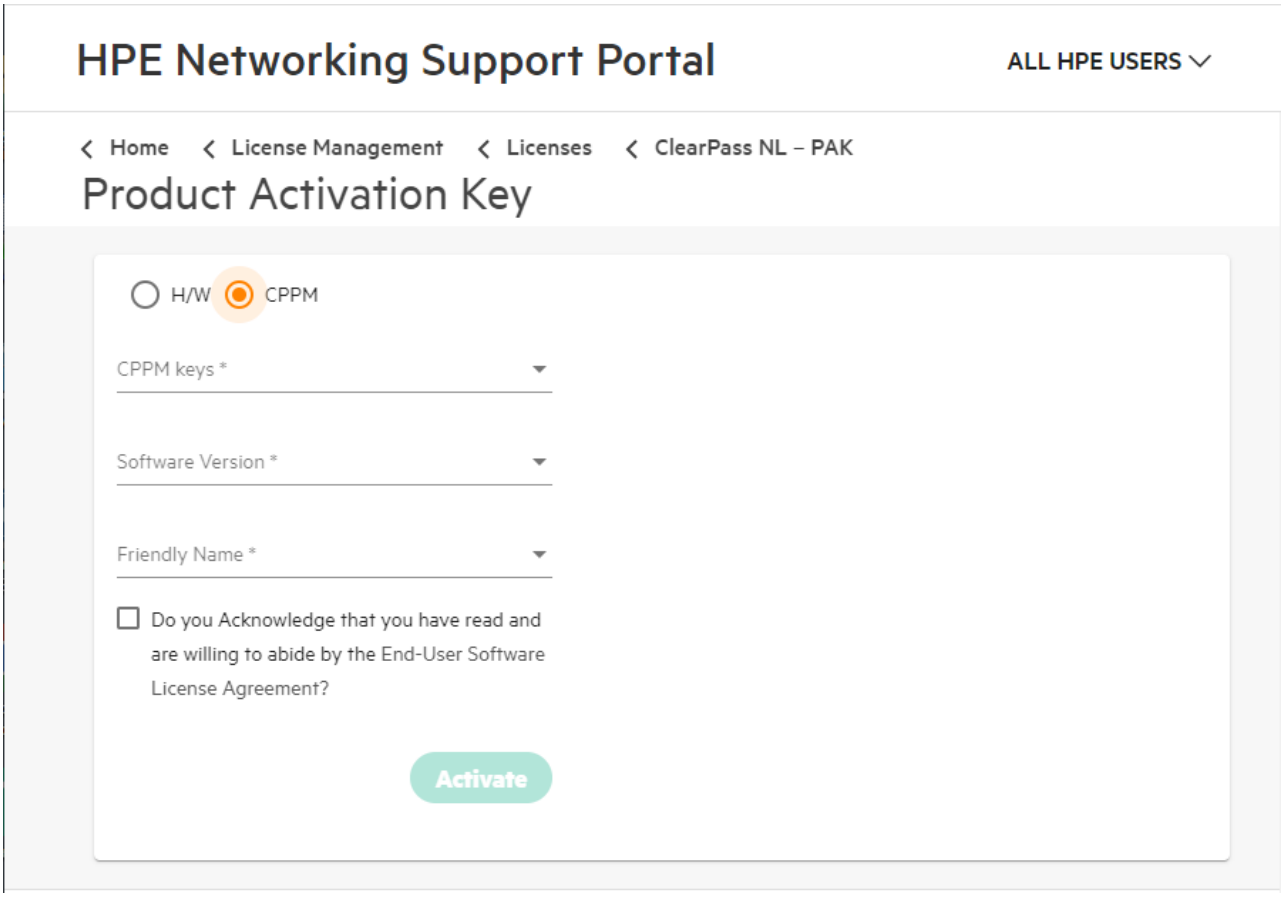
Activating the Platform Activation Key for a Hardware Appliance

Procedure

1. After the configuration for the hardware appliance is completed as described in the ClearPass Installation Guide, open the HPE Networking Support Portal at **https://networkingsupport.hpe.com/home**.

2. In the top navigation, go to **Service Management > License Management**, and then go to **Licenses > ClearPass NL – PAK**. The **Product Activation Key** form opens (**https://licensemanagement.hpe.com/**).

Figure 1. The Product Activation Key Form in License Management Services (LMS)



3. Enter the information for this hardware appliance in the **Serial Number** and **Software Version** fields.
4. In the **Friendly Name** field, enter a name that will clearly identify this Platform Activation Key as belonging to this harware appliance. This is included as a column in the list of licenses in the HPE Networking Support Portal.
5. Mark the check box to accept the End User Software License Agreement (EULA), and then click **Activate**. The Platform Activation Key (PAK) for the hardware appliance is displayed.
6. Copy and paste the PAK into a text file. Be sure to include the "begin" and "end" lines, including all the hyphen characters (for example, "-----BEGIN ACCESS LICENSE KEY-----"); however, do not include any leading or trailing space characters. Keep the key displayed in the HPE Networking Support Portal while you do the next steps, and save the text file for future reference.

Upgrading the Hardware Appliance Prior to Activating the License

A ClearPass hardware appliance is pre-provisioned with ClearPass versions 6.7.0 and 6.9.0. (The C1000 appliance includes version 6.7.0 only). Before you can activate a

license for a ClearPass hardware appliance running versions 6.7.0 or 6.9.0, you must install a software upgrade or update patch.

- To run ClearPass 6.9.0 on the hardware appliance, you first must download and install the ClearPass 6.9.x licensing patch **ClearPass Hotfix Patch for license activation failures after May 30, 2020** prior to activating the license. A known issue in the ClearPass 6.9.0 and 6.9.1 software prevents users from activating licenses, and this patch resolves this issue.
- To update the hardware appliance from ClearPass 6.9.0 to ClearPass 6.9.1, you first download and install both the ClearPass software update patch and the ClearPass 6.9.x licensing patch **ClearPass Hotfix Patch for license activation failures after May 30, 2020** prior to activating the license.
- To update the hardware appliance from ClearPass 6.9.0 to ClearPass 6.9.2 or later 6.9.x releases, or to upgrade to a later release (including 6.10.x), you first must download and install the ClearPass software update patch or upgrade image, and then activate the license on the appliance.
- To upgrade the hardware appliance from ClearPass 6.7.0 to 6.8.x, you first must download and install the ClearPass upgrade image, and then activate the license on the appliance.

  For information about downloading and installing software updates and patches, refer to Software Updates

Activating a License Key for the Hardware Appliance in the WebUI

About this task

To activate the Platform License on a publisher or subscriber:

Procedure

1. Navigate to **Administration > Server Manager > Licensing**. The **Licensing** page opens to the **License Summary** page.
2. Select the **Servers** tab.

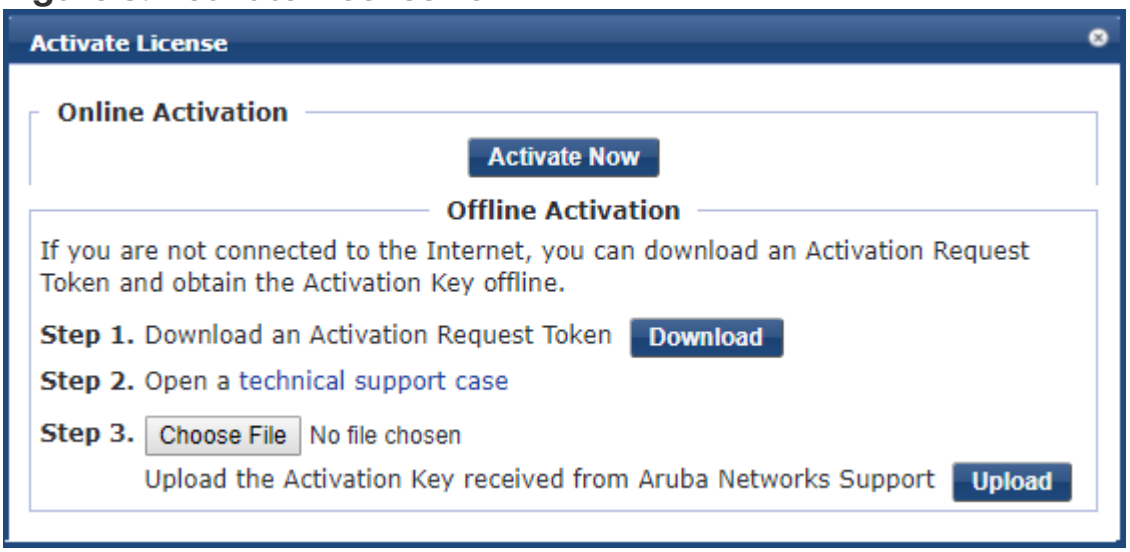
Figure 2. ClearPass Platform License Ready to Be Activated

The Policy Manager Platform Licenses for this Policy Manager servers in the cluster are listed.

- If the Platform License is not activated, a red **Click to Activate** button appears on the **Administration > Licensing > Servers** tab. Continue to the next step.
- If the Policy Manager Platform license is already activated, it is shown with a green **Activated** button.

3. Click the **Click to Activate** icon. The **Activate License** form opens.

Figure 3. Activate License Form



4. In the **Online Activation** section, click **Activate Now**. The Policy Manager Platform License is now activated. On the **Servers** tab, the **Activation Status** column shows a green circle next to the keyword **Activated**.

   If your ClearPass virtual appliance or hardware appliance is not connected to the Internet, you must submit a case through the HPE Networking Support Portal. For more information, refer to Offline License Activation .

Updating an Evaluation Policy Manager Platform License

About this task

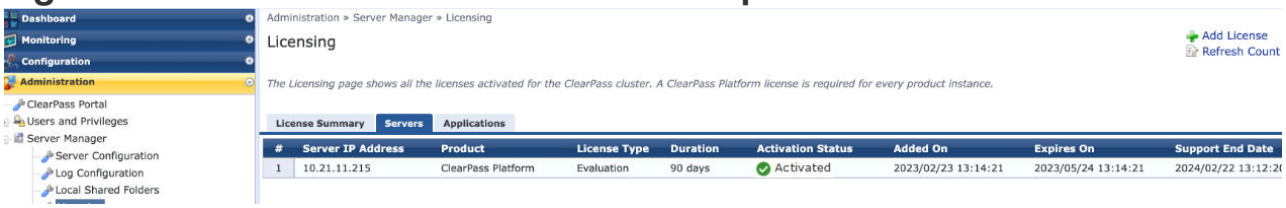
An evaluation ClearPass Platform License has an expiry date, after which it must be updated with a permanent Platform License. In the event that an incorrect Platform License Key is entered when activating the license, the Policy Manager Platform License would also have to be updated.

To update the Policy Manager Platform License:

Procedure

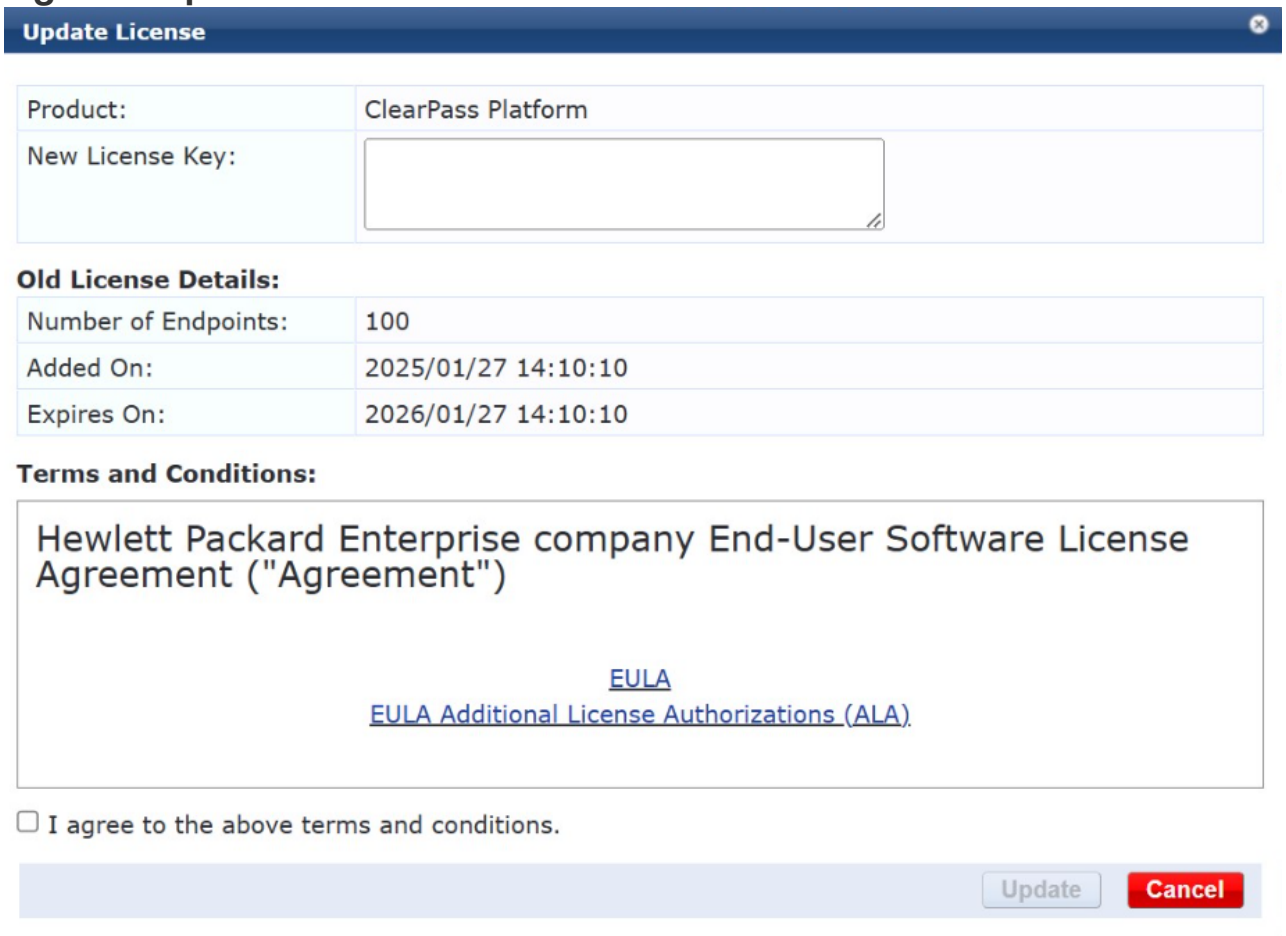
1. Navigate to **Administration > Server Manager > Licensing**. The **Licensing** page opens to the **License Summary** page.
2. Select the **Servers** tab. The Policy Manager Platform Licenses for the Policy Manager servers in the cluster are listed.

Figure 1. ClearPass Platform License to Be Updated



3. Click the **Activated** icon by the license you want to update. The **Update License** form opens.

Figure 2. Update License Form



4. Enter a new Platform License Key and click **Update**. The Policy Manager Platform License is now updated.

Adding an Application License

Application licenses must be installed on the publisher in a Policy Manager cluster. You can install multiple instances of each Application License. Guest functionality is included in Access licenses.

> **NOTE**
> License keys are encoded for a specific application type (for example, Platform, Onboard, Access). For this reason, you are not prompted to select the application type.

Activating a License Key in the HPE Networking Support Portal

Procedure

1. Open the HPE Networking Support Portal) at **https://networkingsupport.hpe.com/home.**
2. In the top navigation, go to Service **Management > License**Management, and then go to **Licenses > Activate**. The License Activation form opens (**https://lms.arubanetworks.com/licenses/activate**).
3. Enter the order number and confirmation number you received in an email from HPE.
4. Click **Load**. The HPE Networking Support Portal displays all licenses associated with that order.
5. Select a license from the table. In the **Redeem** column, enter the number of licenses you wish to activate.
6. In the **Friendly Name** field, enter a name for the license or VA you are activating.
7. In the **Software Version** field, select **6.8 & later** to indicate that this is a new license and not a license in the legacy license format used by ClearPass 6.7 and earlier releases.
8. Click **Activate** to activate that license.

   To add an Application License to your ClearPass server:

9. Navigate to **Administration > Server Manager > Licensing**. The **Licensing** page opens.
0. Click the **Add License** link. The **Add License** form opens.

Figure 1. Add License Form

1. Specify the **Add License** parameters as described in the following table:

Table 1. Add License Parameters

Parameter	Action/Description
License Key	Enter the license key for
Terms and Conditions	Click the **I agree to the a** heck box.

2. Click **Add License**. You return to the **License Summary** page, where the new Application License is now listed and the following message is displayed: *<Application name> license added successfully.*

   When you add an Application License, the **Administration > Server Manager > Licensing** page displays the **Applications** tab, allowing you to activate the application licenses.

Activating an Application License

About this task

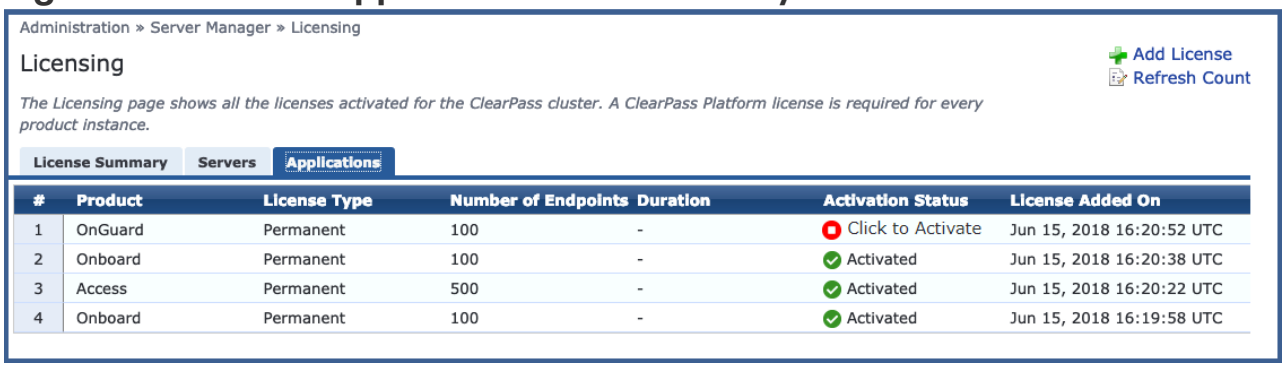
After you add or update an application license, it must be activated. The following procedure describes the steps to activate an Application License online.

To activate the application licenses on the current Policy Manager server:

Procedure

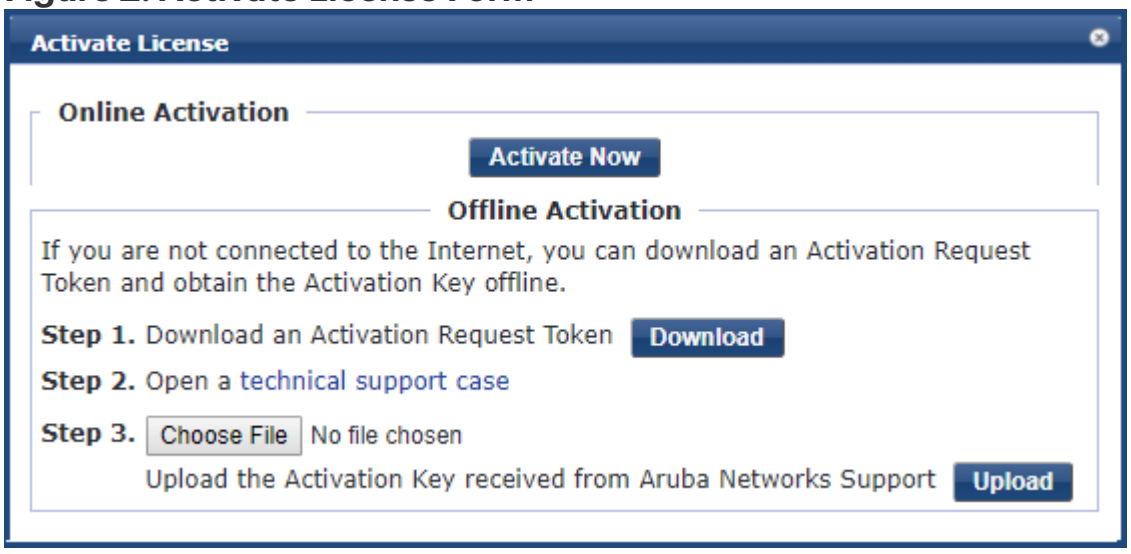
1. Navigate to **Administration > Server**Manager > Licensing. The **Licensing** page opens to the **License Summary** page.
2. Select the Applications tab. The new application licenses are listed. The **Activation Status** column shows a red circle next to the keyword **Activate**.

Figure 1. ClearPass Application Licenses Ready to Be Activated



3. Click Activate. The **Activate License** form opens.

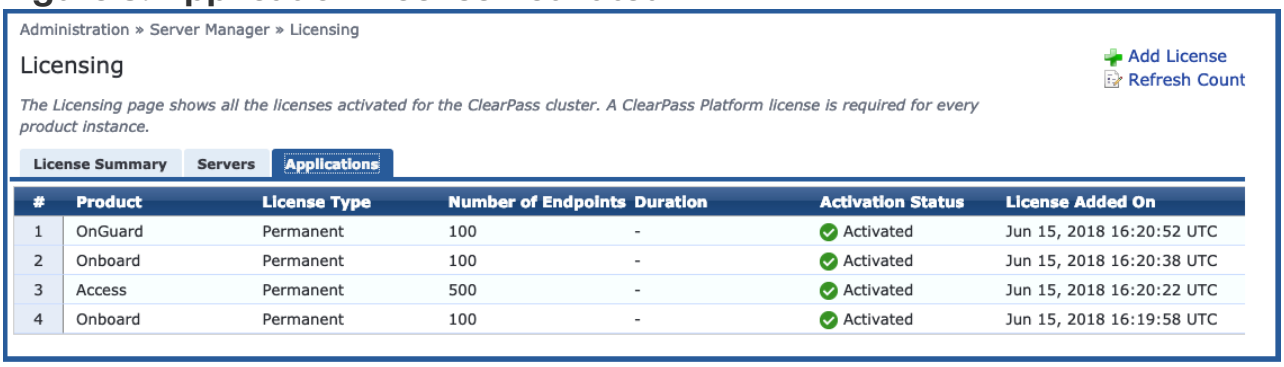
Figure 2. Activate License Form



4. In the **Online Activation** section, click Activate Now.

   The selected application license is now activated. The **Applications** tab > **Activation Status** column shows a green circle next to the keyword **Activated**.

Figure 3. Application License Activated



Managing and Monitoring Application Licenses

Policy Manager licensing computation is solely based on number of endpoints that successfully connect to the network over a period of time. The **Licensing** page shows all the licenses that are installed for the ClearPass Policy Manager cluster. There is one entry per server in the cluster. The **License Summary** page displays the number and type of purchased licenses, and the total license count and the number of licenses used.

In the Policy Manager dashboard, the pie chart in the **License Usage** widget shows the **Available Count** and **Used Count** for Entry, Access, Onboard, or OnGuard licenses. License usage is calculated every 15 minutes. The information banner on the Dashboard page is immediately updated with a message if the license limit for an Application License (Entry, Access, OnGuard, or Onboard) has been exceeded. The message is removed when the usage falls within the allowed number again. No banner messages are posted for CLABV systems or systems running in evaluation mode or with evaluation licenses.

NOTE

Entry licenses and Access Upgrade licenses combine to appear as a single Access license in the Policy Manager dashboard. For example, you have 100 Entry licenses and then add 50 Access Upgrade licenses, then license count in the License Usage dashboard widget will show 50 Entry licenses and 50 Access licenses.

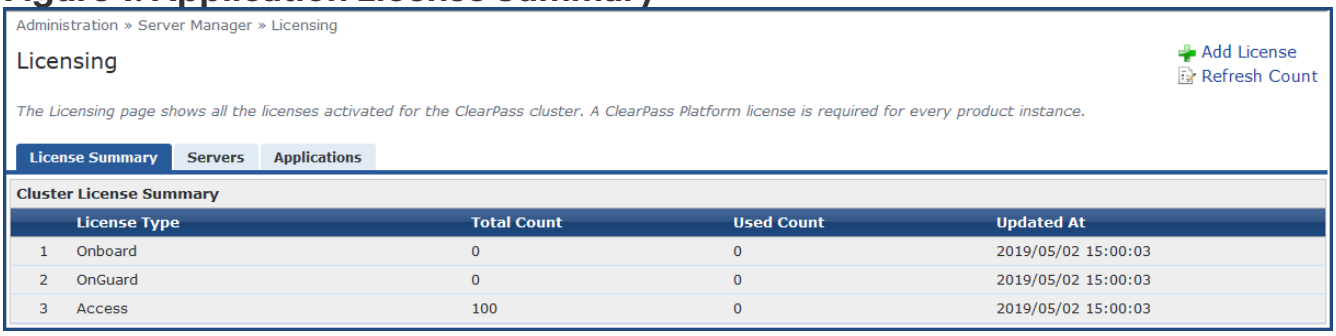
Subtopics

Viewing Current Licensing Usage
Viewing Application License Details
Viewing Notifications for Insufficient Licenses
Updating Evaluation Licenses

Viewing Current Licensing Usage

To view the Application Licenses that are currently installed for the cluster, navigate to **Administration > Server Manager > Licensing**. The **Licensing** page opens to the **License Summary** page.

Figure 1. Application License Summary



From the **Licensing** page, you can do the following:

- Add new Application Licenses (see Adding an Application License).
- Refresh the license usage count (see Refreshing the License Usage Count).

> **NOTE**
> The **Applications** tab on this page displays the number of licenses for each individual license type. However, the **License Summary** tab on this page combines pairs of Entry and Access Upgrade licenses and displays them as a single Access license.

In Insight, you can go to **Dashboard > Licensing** to open the **Licensing Dashboard** page. Use the three graphs on this page to view license information for the Entry, Access, Onboard, or OnGuard licenses.

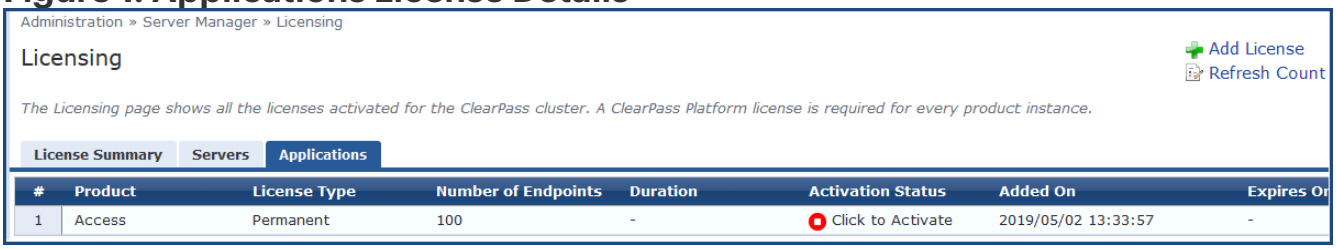
- **Current License Usage (15 minutes interval)** — For each type, this graph shows the **Total**, **Exceeds Limit**, and **Used** counts over the past 15 minutes.
- **License Usage In Last 24 Hours** — For each type, this graph shows the **Used Count** and **Available Count** for each hour over the past 24 hours.
- **Maximum License Usage** — For each type, this graph shows the maximum number of licenses used for a given time frame. This graph can be set for a look-back window of the last 24 hours, one week, or one month.

Viewing Application License Details

The **Licensing > Applications** page displays ClearPass Policy Manager Application License details. This information includes product type (*Access*, *OnGuard*, or *Onboard*), license type (*Perpetual*, *Subscription*, or *Evaluation*), number of endpoints, license duration, license activation status, and the date the Application License was added.

> **NOTE**
>
> The **Applications** tab is activated on adding an Application License (Access, Entry, OnGuard, Compliance Suite, or Onboard licenses).

Figure 1. Applications License Details



Viewing Notifications for Insufficient Licenses

License usage is calculated every 15 minutes. When Application license usage is greater than the number of available licenses, Policy Manager sends notifications that are displayed in the Policy Manager **Dashboard** and **Event Viewer**. When the license limit for an Application license (Access, Entry, OnGuard, Compliance Suite or Onboard) has been exceeded, the information banner on the **Dashboard** page is updated with message warning of excessive license use.

The message is removed as soon as license usage falls within the allowed threshold. Messages about the excessive license usage are also posted to the Event Viewer every 15 minutes for each Policy Manager application that exceeds its license limit.

The Event Viewer displays the number of licenses purchased and the current usage by application. When the appliance falls within the allowed usage, the Event Viewer displays a message to that effect. No banner messages are posted to the Dashboard for CLABV (evaluation) systems or for systems running in Evaluation mode or with Evaluation licenses. For CLABV systems or for applications running under an Evaluation license, messages are posted to the Event Viewer once each night at 11:55 P.M. local time.

Updating Evaluation Licenses

About this task

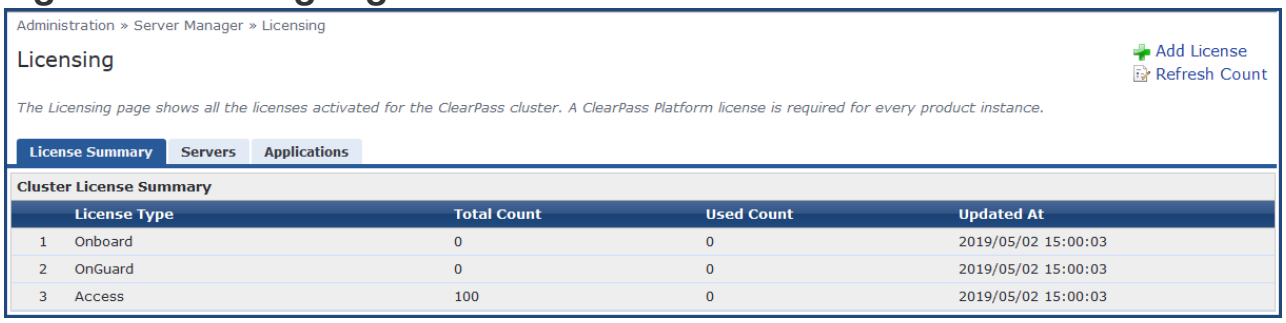
Application licenses typically require updating after they expire, for example, after the evaluation license expires, or when capacity exceeds its licensed number of unique endpoints (devices) that can be authenticated.

To update an Application License with a new License Key:

Procedure

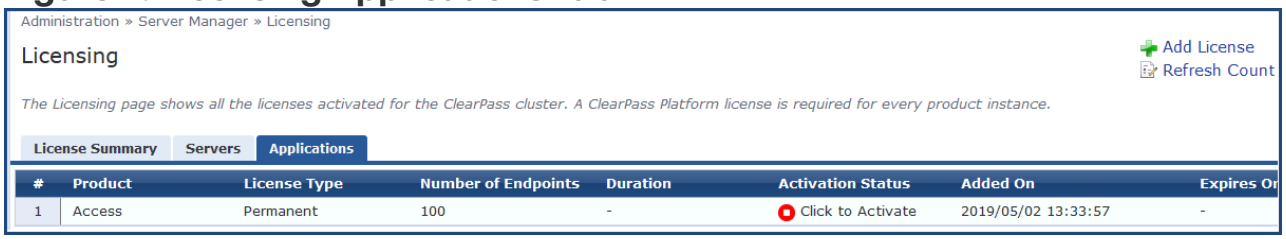
1. Navigate to **Administration > Server Manager > Licensing**. The **Licensing** page opens.

Figure 1. Licensing Page



2. Select the **Applications** tab.

Figure 2. Licensing Applications Tab



3. Click the table row containing the Application License that needs an update. The **Update License** form displays.

Figure 3. Update License Form

4. Enter the new License Key.
5. Click the **I agree to the above terms and conditions** check box. The **Update** button is now activated.
6. Click **Update**. The selected Application License is updated.

Backing Up Licenses

The automatic backup file (**auto-backup-tips*.tar.gz**) generated by Policy Manager contains plain text files with application and platform license information. This information can be used to restore a Policy Manager server in the event of a hardware failure or software upgrade issue. For more information on the auto-backup feature, refer to Database Parameters.

Monitoring License Usage

In the Policy Manager dashboard, the pie chart in the **License Usage** widget shows the **Available Count** and **Used Count** for Entry, Access, Onboard, or OnGuard licenses. License usage is calculated every 15 minutes, and the information banner on the Dashboard page is immediately updated with a message if the license limit for an Application License (Entry, Access, OnGuard or Onboard) has been exceeded. The message is removed when the usage falls within the allowed number again. No banner messages are posted for CLABV systems or for systems running in evaluation mode or with evaluation licenses.

NOTE

Entry licenses and Access Upgrade licenses combine to appear as a single Access license in the Policy Manager dashboard. If, for example, you have 100 Entry licenses and then add 50 Access Upgrade licenses, then license count in the License Usage dashboard widget will show 50 Entry licenses and 50 Access licenses.

In Policy Manager Insight, you can go to **Dashboard > Licensing** to open the **Licensing Dashboard** page. Three graphs on this page let you view license information for the Entry, Access, Onboard, or OnGuard license types.

- **Current License Usage (15 minutes interval)** — For each type, this graph shows the **Total**, **Exceeds Limit**, and **Used** counts over the past 15 minutes.
- **License Usage In Last 24 Hours** — For each type, this graph shows the **Used Count** and **Available Count** for each hour over the past 24 hours.
- **Maximum License Usage** — For each type, this graph shows the Max used count for a given time frame. This graph can be set for a look-back window of the last 24 hours, one week, or one month.

Monitoring Licensing Events

About this task

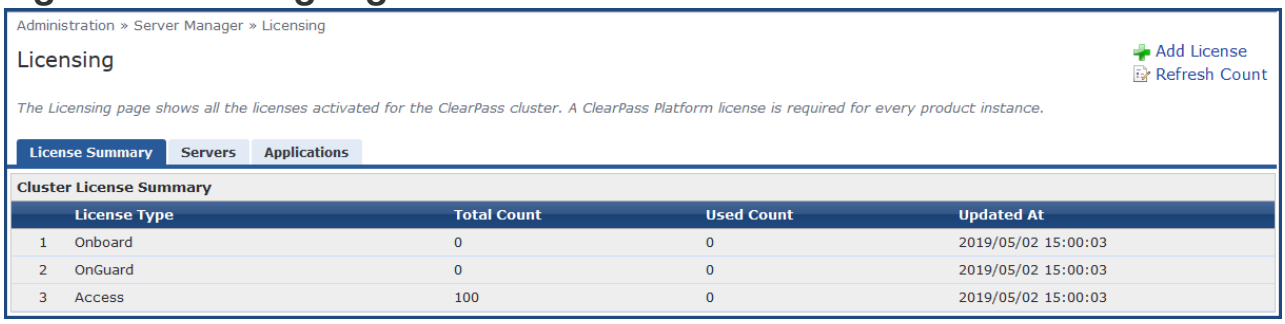
If a Policy Manager application exceeds its license limit, the **Monitoring > Event Viewer** page displays updated messages about that license every fifteen minutes (or whenever 'Refresh Count' is clicked) indicating the number of licenses purchased and the current usage by application. In addition, a UI banner displays messages about the licensing overrun and SNMP traps are sent to notify administrations about the issue. When the appliance falls within the allowed usage again, the Event Viewer displays an updated message indicating that Policy Manager again has adequate licenses. For CLABV systems or for applications running under an evaluation license, Event Viewer messages are not posted at 15-minute intervals but are instead posted once each night at 11:55 PM.

To update the latest usage count of all licenses in the Policy Manager cluster:

Procedure

1. Navigate to **Administration > Server Manager > Licensing**. The **Licensing** page opens.

Figure 1. Licensing Page



2. Click the **Refresh Count** link. When you click **Refresh Count**, the license usage count is refreshed immediately. You then receive the following confirmation: *License count refreshed successfully*

Offline License Activation

About this task

If your ClearPass virtual appliance or hardware appliance is not connected to the Internet, you must submit a case through the HPE Networking Support Portal to activate a Platform License or Platform Activation Key.

Procedure

1. Go to the ClearPass Policy Manager management interface:

   `https://x.x.x.x/tips/`, where **x.x.x.x** is the IP address of the management port defined for the ClearPass virtual appliance or hardware appliance. The ClearPass Policy Manager **Admin Login** page opens. It displays a message indicating that you have 90 days to activate the product, and includes an **Activate Now** link.

2. Click **Activate Now**.

3. In the **Offline Activation** section, click **Download** to download an activation request token from the ClearPass Policy Manager server.

4. Open the HPE Networking Support Portal at **https://networkingsupport.hpe.com/home** and log in to your account.

5. Navigate to **Service management > Case Management** and open a new service request.
6. The Support team will contact you and request the activation token.
7. The Support team will then activate the token and send you the activation key.

Licensing FAQs

Subtopics

How Are Licenses shared by servers within a Policy Manager cluster?
Does Policy Manager support SNMP traps for licensing status changes?
How does Policy Manager verify the SSL certificate of the license activation server?

How Are Licenses shared by servers within a Policy Manager cluster?

A cluster is a logical connection of any combination of Policy Manager hardware or virtual appliances. Policy Manager uses a publisher/subscriber model for multiple-box clustering, where the publisher functions as the primary controller in a cluster, and subscribers act as worker nodes. All license management operations for a cluster must be performed on the publisher. When you manage licenses through a cluster of Policy Manager devices, be aware of the following behaviors:

- Licenses are shared by a cluster. For example, if there are five Policy Manager appliances in the cluster and a 10K Access license is applied, that capacity is distributed across the cluster as needed.
- When you add an appliance to a cluster, it loses all of its licenses except for the Platform License. Any Application Licenses it had before it became a subscriber must then be added to the publisher.
- When you drop an appliance from a cluster, it loses all of its licenses except for the Platform License.
- When an appliance is manually promoted to publisher, all of its Application licenses must be reactivated.

- When an appliance is automatically promoted to publisher, there is no change in the status of any of its licenses.
- Systems without support can continue to operate without interruption, but are not able to update/upgrade.
- A warning displays if all cluster nodes are not compliant with PAK license checks, as each cluster node supporting a Policy Manager 6.11 cluster must be upgraded to the 6.11 baseline.

Does Policy Manager support SNMP traps for licensing status changes?

Policy Manager supports SNMP traps providing the status of **Entry**, **Access**, **OnGuard**, **Onboard** and **Compliance Suite** licenses. Traps provide information on activation expiry status, license expiration status, and license usage details.

How does Policy Manager verify the SSL certificate of the license activation server?

Starting with Policy Manager 6.10, the certificate trust list settings at **Administration > Certificates > Trust List**, includes an **Aruba Services** usage type is added to the **Usage** options at and is used to verify the SSL certificate of the License activation server. The following certificate must be enabled in order to support license activation. CN=DigiCert Global Root G2,OU=www.digicert.com,O=DigiCert Inc,C=US

In previous releases, the Policy Manager license activation workflow uses a hard-coded certificate in the path /usr/local/avenda/platform/etc/activation-server-cert.pem.

Licensing Troubleshooting

About this task

The following section describes common errors and solutions.

Platform License Key Cannot be Activated on the ClearPass Hardware Appliance

Activate Platform License Key on the Aruba Licensing Management website (**https://lms.arubanetworks.com/**) using the appliance serial number. After it is activated, the

Licensing Website will provide a License Key that you can enter in the ClearPass login screen to activate the license on that hardware appliance.

Platform Licenses Cannot be Activated on the ClearPass Virtual Appliance

Activate Platform licenses on the Aruba Licensing Management website (**https://lms.arubanetworks.com/**) using the order number and confirmation number you received in an email from HPE. After it is activated, the Licensing Website will provide a License Key that you can enter in the ClearPass login screen to activate the license on that virtual appliance. For details refer to Activating the License Key for a ClearPass Virtual Appliance

Cannot activate an Entry License

If you have purchased Access licenses, the ClearPass licensing system will not accept Entry licenses (or even Entry and Access Upgrade licenses together). Contact Aruba technical support if you require assistance to correct order issues.

Policy Manager Access Upgrade Licenses are note offering Access License Features

To upgrade Entry Licenses to Access Licenses, you must purchase an Access Upgrade license for every one of your Entry licenses. If Entry and Access Upgrade Licenses are not applied in equal quantities, they are treated as Entry licenses only. To resolve this issue, ensure that your deployment uses an equal number of Entry and Access licenses.

Compliance Suite License does not Support All Features

If you install a Compliance Suite license without an Entry or Access license, the only services that can be configured are Device Insight Integration and OnGuard health checks. Add these additional licenses to resolve this issue.

When adding a Platform Activation Key, the WebUI displays the error "Invalid license key detected. Please update the license key to continue."

When activating a Platform Activation Key (PAK) in the WebUI, be sure to include the begin and end lines, including all the hyphen characters. Do not include any leading or trailing space characters. Adding extra blank spaces at the beginning or end of a PAK can trigger this error.

NOTE

In some versions of Policy Manager, if there is an empty space before or after the "begin" or "end" lines, the UI might show an incorrect error message that says, "Services require an Entry or Access license. Please add a valid Entry or Access license and try again." If you receive this error, remove the empty characters. Starting in Policy Manager 6.10.4, this error is resolved, and an empty space in the key triggers the correct error message "Invalid license key detected. Please update the license key to continue."

Unable to download with valid PAK until AC license is added

After ClearPass 6.11.x is installed, and even though the system has a valid support contract for the Platform Activation Key (PAK), attempts to apply further updates or

upgrades fail with an error message System Update/Upgrade is not allowed as Platform License does not have valid Support Contract. Updates cannot be applied until a valid Access License is added.

Complete the following workaround instructions to address this issue:

Procedure

1. Install ClearPass 6.11.x.
2. Apply the PAK. The PAK does not need to be activated at this time.
3. Navigate to **Administration > Agents and Software Updates > Software Updates**. Generate an HPE passport token with the proper credentials tied to the Platform License.
4. Navigate to **Administration > Server Manager > Licensing page**. Add an Application License. The license does not need to be activated at this time.
5. Navigate back to the **Software Updates** screen. After 30 seconds the option to download patches should be available again, and the error message removed.
6. Download patches and apply them to the cluster servers as needed.
7. Once the system is updated to the intended version, restore the pre-6.11.1 configuration from the backup, as well as any additional files required.
8. Rejoin the cluster (optional).
9. Activate all licenses after the cluster is re-configured.

Cluster Update and Upgrade

This chapter covers the following topics:

- Updating Clusters
- Upgrading Clusters

Subtopics

Updating Clusters
Upgrading Clusters

Updating Clusters

This section includes the following information:

- Cluster Update Overview
- Before Updating the Cluster
- Updating the Cluster
- Viewing Subscriber Update Status

Subtopics

Cluster Update Overview
Before Updating the Cluster
Updating the Cluster
Viewing Subscriber Update Status

Cluster Update Overview

The **Cluster Update** page automates the process of updating the Policy Manager cluster with Patch and Skin releases. The cluster publisher is updated first. You can select one or more subscribers to be updated automatically after the publisher update is complete.

After you initiate the Cluster Update, no manual actions are required until the publisher and all the selected subscribers have been updated.

The Cluster Update performs the following actions:

- Copies the update image to the selected subscribers.

  subscribers copy the update image over an HTTPS connection from the publisher.

  If you want to avoid the copy operation on one or more subscribers, log in to the subscriber and trigger a download of the update image in the **Software Update** page. Alternatively, upload the update image through the **Software Update** page before initiating the cluster update.

- The publisher is updated and rebooted (reboot is initiated only if it is mandatory).

  After the publisher update completes, the **Cluster Update** page is accessible again to review progress and log messages.

- The Cluster Update is now initiated on the selected subscribers; after completion, the subscribers are rebooted (reboot is initiated only if it is mandatory).

- Where possible, multiple subscribers are updated in parallel.
- After all selected subscribers have been updated, you can select and trigger a Cluster Update for any additional subscribers.

Time Required for Subscriber Update

The time required for a subscriber update depends on the following factors:

- Hardware or virtual appliance model. In the case of virtual machine installations, update times vary significantly based on the I/O per second performance of your virtual machine infrastructure.
- For subscribers, bandwidth and latency of the network link between the subscriber and publisher.

Before Updating the Cluster

Before updating the Policy Manager cluster, review and complete the following tasks:

NOTE

Only patches listed under the **Software Updates** page are shown in the **Cluster Update** page.

- Plan for sufficient downtime and review the Release Notes for the current ClearPass Policy Manager release.
- Confirm that Relevant Patch updates are available under the **Firmware & Patch Updates** area of the **Software Updates** page before starting the cluster update. Please download the patches by clicking the **Download** button next to the patches on the **Software Updates** page or load them directly by clicking the **Import Updates** button on the **Software Updates** page.
- Confirm that your Cluster sync and replications are fine before starting the Cluster Update.
- When a particular subscriber's version information is set to "*UNKNOWN*," it means the publisher is not able to contact the remote subscriber. (If a subscriber has been disabled and gone out of sync, the Cluster Update page might not detect the disabled status until the patch failure has occurred, after which the failed/inaccessible subscriber is marked as *UNKNOWN*). Confirm the status of the cluster sync and service status of Async Network Services in such cases (for details, see Async Network Services Options).

- In virtual machine environments, ClearPass Policy Manager Virtual appliance host date and time settings should be in sync with that of the VMware vSphere, Microsoft Hyper-V, or KVM server that is hosting the instance. Otherwise, you might see inconsistent data in **Time Taken** columns of the **Cluster Update** page.
- To successfully update the cluster, you must have a current, valid Support Agreement Entitlement ID tied to the Platform license key on every ClearPass server in the cluster.

Updating the Cluster

About this task

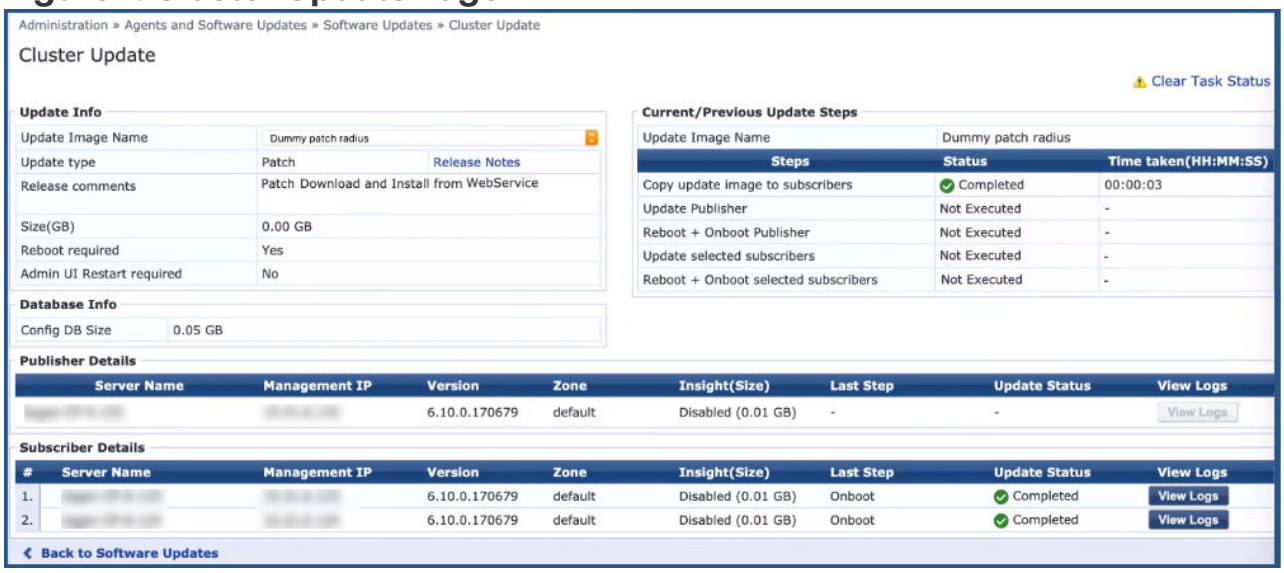
The Cluster Upgrade tool verifies if the upgrade image is installed before starting the upgrade process. If the upgrade Image already Installed, then the Cluster Upgrade tool will not install the upgrade Image, and it will directly reboot the Policy Manager server.

To update the cluster:

Procedure

1. From the publisher, navigate to **Administration** > **Agents and Software Updates** > **Software Updates**.
2. Click the **Cluster Update** link. The **Cluster Update** page opens.
3. Select the **Update Image Name** from the drop-down list.

Figure 1. Cluster Update Page



This page provides the information described below in Table 1.

Table 1. Information on the Cluster Update Page

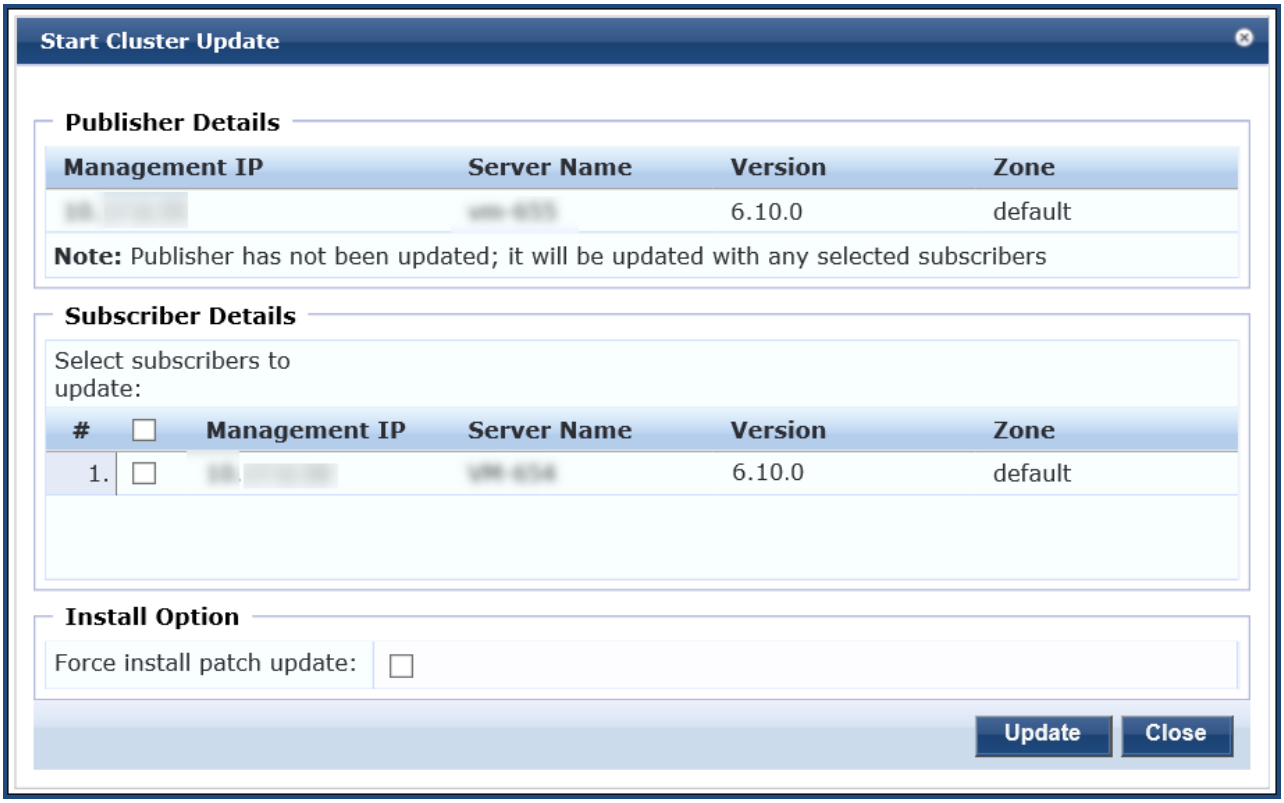
Field	Description
Update Info	Describes the patch update details, provides a link to the Release Notes, includes release - specific comments, and specifies if a reboot is required for the patch.
Database Info	Shows the size of the Configuration database.
Publisher Details	Information for the publisher and for all subscribers in the cluster, including the management IP address, version number, zone, Insight database size and whether Insight is enabled, last update step completed, and update status.
Subscriber Details	
Current/Previous Update Steps	During the cluster update, this area displays the status of each stage in the process. As each stage completes, it displays how long it took to complete.
View Logs	In each publisher and subscriber row, this button when clicked opens the Logs page, which provides detailed status and log messages for each cluster update stage.

When the update is available locally and all subscribers have been patched, the **Start Update** link is available in the upper-right corner of the page.

4. Click **Start Update**.

   The **Start Cluster Update** page opens.

Figure 2. The Start Cluster Update Page



NOTE

You can update the entire cluster or just a subset of subscribers.

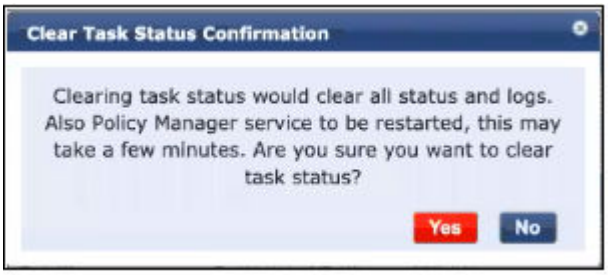
5. In the **Start Cluster Update** page, use the check boxes to select the subscriber nodes to update.
6. To force the update, select **Force install patch update** under the **Install Option** area.
7. Click **Update**. This initiates the automated update process. No further manual steps are required until all selected subscribers have been updated. The publisher is always updated and rebooted first.

   The **Cluster Update** page is not available while the publisher is rebooted. When the publisher update is complete, you can use the **Cluster Update** page to monitor update progress.

Clear Task Status

Cilck the **Clear Task Status** link at the top of the **Cluster Update** page to clear all logs related to the Cluster update process. It is not always necessary to clear update tasks, but can be useful in debugging update issues.

Figure 3. Clear Task Status Popup Window



Viewing Subscriber Update Status

About this task

After the publisher Update is complete, you can monitor the Update status of the subscriber nodes at **Administration > Agents and Software Updates > Software Updates > Cluster Update**.

The page provides two ways to monitor the progress of the update:

Procedure

1. On the **Cluster Update** page, progress indicators in the **Current/Previous Update Steps** area display the status of some of the main steps.

   Indicators in the **Publisher Details** and **Subscriber Details** areas also display when the publisher or each subscriber is in progress or completed.

   When the update completes successfully, these areas display a successful update status for the publisher and every subscriber.

2. For detailed progress information, click the **View Logs** button for a publisher or subscriber row.

   The **Logs** page opens. This page includes tabs for the **Download**, **Update**, **Reboot**, and **Onboot** logs.

   You can view detailed status in these logs during and after the update.

   NOTE

   This button is not available while the publisher is rebooted and data migration is in progress. It is available again when the publisher update is complete.

Upgrading Clusters

This section includes the following cluster upgrade information:

- Cluster Upgrade Overview
- Sample Times Required for Upgrade
- Before You Upgrade
- Upgrading the Cluster
- Viewing Upgrade Status
- Steps in the Cluster Upgrade Automated Workflow
- Moving to ClearPass 6.11
- Troubleshooting Tips

Subtopics

Cluster Upgrade Overview
Steps in the Cluster Upgrade Automated Workflow
Sample Times Required for Upgrade
Before You Upgrade
Upgrading the Cluster
Viewing Upgrade Status
Moving to ClearPass 6.11
Troubleshooting Tips

Cluster Upgrade Overview

The **Cluster Upgrade** page automates the upgrade procedure for a Policy Manager cluster. When the upgrade is initiated, no manual actions are required until the publisher and all selected subscriber nodes have been upgraded.

> **NOTE**
> In ClearPass 6.11.0 and later, the maximum single cluster size is 32 servers. This includes the publisher, standby publisher, subscribers, dedicated insight server, and standby insight servers.

> **NOTE**
> Policy Manager upgrades to 6.11 are not supported, but the Cluster Upgrade feature can be used for any future upgrades from 6.11.

If you have an earlier version of the Cluster Upgrade Tool already installed, you can install this version directly over the earlier version of the tool; no cleanup steps are needed.

Cluster Upgrade Process Overview

About this task

These tasks summarize the high-level cluster upgrade process:

Procedure

1. Download the upgrade image from the **Software Updates** page.
2. Initiate the **Upgrade** procedure (see Upgrading the Cluster).

   The Cluster Upgrade tool automatically performs the upgrade.
3. After the upgrade, verify that the publisher and all subscriber nodes in the cluster are back in sync and all services are accessible (see Viewing Upgrade Status).

> **NOTE**
> Cloning a virtual machine to facilitate a Policy Manager deployment is not recommended or supported.

Steps in the Cluster Upgrade Automated Workflow

About this task

This section describes the steps that are automatically completed by the Cluster Upgrade.

Procedure

1. To prepare the subscriber nodes for upgrade, a patch that provides required API support is automatically installed by the Cluster Upgrade on every Subscriber.

   The Cluster Upgrade uses remote API calls to control and monitor upgrade progress on the Subscribers.

   > **NOTE**
   >
   > To install the patch for API support on the subscribers, Subscribers must be able to access the publisher over HTTP, or they must be able to access the publisher over HTTPS using its hostname and validate the certificate that is presented (trust the issuer and match the hostname in the certificate CN).

2. After you select the subscriber nodes and click **Upgrade**, the upgrade image is copied to the Subscribers you selected.

   The Subscribers copy the upgrade image over an HTTPS connection from the publisher.

   If the upgrade image is already present on a subscriber node (assuming you have downloaded it from the **Software Updates** page, or uploaded it in the **Software Updates** page), the existing upgrade image on the Subscriber is used for the upgrade.

3. If the standby publisher settings were configured, they are temporarily disabled. This setting is restored after all Subscribers have been upgraded.

4. The publisher is the first to be upgraded and rebooted. Configuration database and Insight database migration is performed on reboot.

5. When the publisher upgrade is complete, you can use the **Cluster Upgrade** page to review log messages.

6. When the publisher upgrade is complete, upgrade is initiated on each selected Subscriber. When possible, multiple Subscribers are upgraded in parallel. When each Subscriber is complete, the subscriber node is rebooted.

7. During the parallel upgrade process, upgrade of the first Subscriber begins five minutes after the publisher upgrade is completed.

8. Upgrade of the second Subscriber begins five minutes after the upgrade of the first Subscriber begins. This pattern continues sequentially for all subscriber nodes in the cluster, with a five-minute delay between each start time.
9. When each Subscriber is rebooted, it is added back into the cluster. Insight data is migrated and restored.
0. When all selected Subscribers have been upgraded, you can select and trigger the upgrade operation for any additional subscriber nodes.
1. When all the Subscribers in the cluster have been upgraded, the standby publisher settings are restored.

   Detailed information for each of these steps is available in the **Logs** page during and after upgrade.

Sample Times Required for Upgrade

To help you estimate how much time the upgrade to Policy Manager 6.11 might take, Table 1 displays representative numbers for upgrade times under test conditions. Keep in mind that the figures here are only examples. The actual time required for your upgrade depends on several factors:

- Your hardware or virtual appliance model. In the case of virtual appliance installations, upgrade times vary significantly based on the I/O per second performance of your virtual appliance infrastructure.
- The size of the configuration database to be migrated.
- For Policy Manager Insight appliances, the size of the Insight database.
- For subscribers, the bandwidth and latency of the network link between the subscribers and the publisher.

NOTE

For the most current information about times required for upgrade, refer to the "Before Upgrade" section of the Policy Manager 6.11 Release Notes.

Table 1. Sample Times Required for Upgrade

Hardware Model	Configuration DB Size	Insight DB Size	Publisher Upgrade Time
C1000	100 MB	5 GB	50 minutes
	200 MB	5 GB	60 minutes
C2000	100 MB	5 GB	50 minutes
	200 MB	5 GB	60 minutes
C3000	200 MB	5 GB	30 minutes
	500 MB	10 GB	40 minutes

Before You Upgrade

About this task

Before you begin the cluster upgrade process, ensure that the following tasks have been completed:

Procedure

1. Review this section and the latest Release Notes for Policy Manager 6.11.
2. Plan for adequate downtime for the upgrade. Use the upgrade time estimates in Table 1 as a guide.
3. If the cluster password contains special characters, change it temporarily to only use alphanumeric characters (letters and numbers) before installing this patch. You can change the cluster password back to the old password after the cluster upgrade completes.
4. HTTP, HTTPS, and SSH port traffic must be allowed between the cluster appliances. This is required for the Cluster Upgrade to be able to communicate between appliances.

   Verify the following ports are in an open state between the cluster appliances:

   - Port 443 (HTTPS)

- Port 22 (SSH)

5. Confirm that the publisher and all subscriber nodes in the cluster are in sync before starting the upgrade.
6. On the **Software Updates** page, enter the **HPE Passport Credentials**.
7. On the publisher, download the Policy Manager 6.11 upgrade image from the **Software Updates** page (see Software Updates).

   The **Cluster Upgrade** page automates the process of copying over the upgrade image to the selected subscriber nodes in the cluster.
8. If you are upgrading on a reverted system (retrying an upgrade), you will need to replace the contents of certain directories first before triggering the new upgrade. Please contact HPE Networking Support (see Contacting Support), who will assist you with the following tasks:

   a. Copying the contents of the */var/avenda/platform/store/updates/backup/** directory to the */var/avenda/platform/store/updates/* directory.

   b. Clearing the contents of the */var/avenda/tips/upgrade/db/** directory.

   c. Restarting the **cpass-admin-server** on the publisher (for more information, see Service Commands).
9. When a particular subscriber's version information is set to "*UNKNOWN*," it indicates that the publisher is not able to contact the remote subscriber.

   If a subscriber has been disabled and gone out of sync, the Cluster Upgrade page might not detect the status until the patch failure has occurred, after which the failed or inaccessible subscriber is marked as *UNKNOWN*.

   In such cases, confirm the status of the cluster sync and the service status of **Async network services** (for details, see Async Network Services Options).

   a. In virtual machine environments, Policy Manager virtual machine host date and time settings should be in sync with that of the VMware vSphere Hypervisor server, Microsoft Hyper-V server, or KVM server that is hosting the instance. Otherwise, you might see inconsistent data in **Time Taken** columns of the **Cluster Upgrade** page.

Upgrading the Cluster

About this task

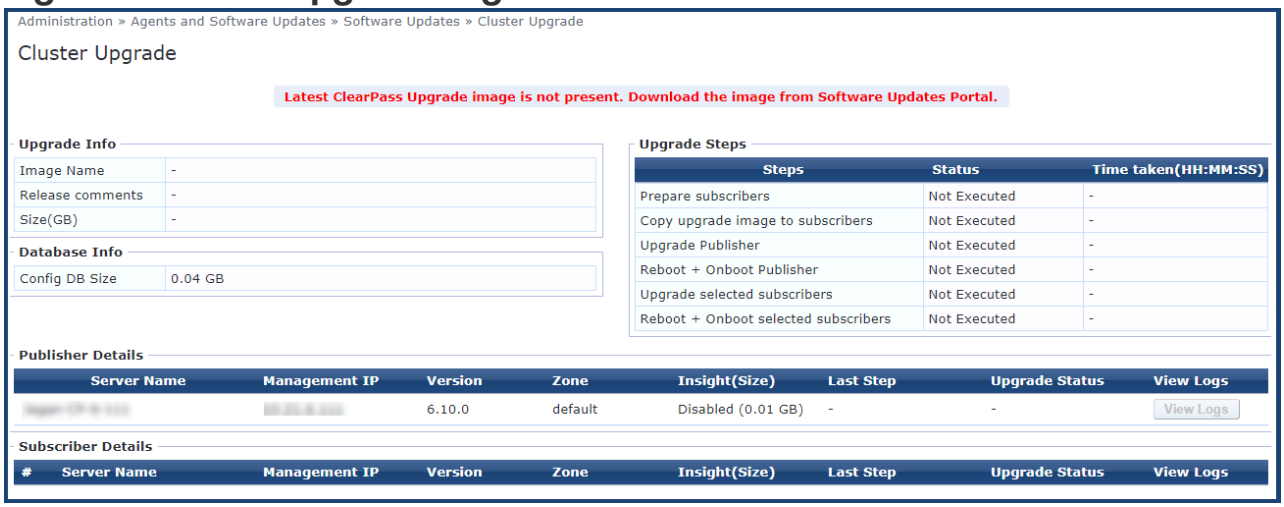
To upgrade the Policy Manager cluster:

Procedure

1. From the publisher, navigate to **Administration** > **Agents and Software Updates** > **Software Updates**.
2. Click the **Cluster Upgrade** link.

   The **Cluster Upgrade** page opens.

Figure 1. Cluster Upgrade Page



This page includes the information described below in Table 1.

Table 1. Information on the Cluster Upgrade Page

Field	Description
Upgrade Info	Describes the upgrade image's name and size, and includes release - specific com ments.
Database Info	Shows the size of the configuration data base.
Publisher Details	Provides information for the publisher an d for all Subscribers in the cluster. Infor mation includes the management IP add ress, version number, zone, Insight datab

Field	Description
	ase size, last upgrade step completed, an d upgrade status.
Upgrade Steps	During the cluster upgrade, this area disp lays the status of each stage in the proc ess. As each stage completes, it displays how long it took to complete. • Prepare subscribers • Copy upgrade image to subscribers • Upgrade Publisher • Reboot + Onboot Publisher • Upgrade selected subscribers • Reboot + Onboot selected subscribe rs
View Logs	In each publisher and Subscriber row, cli cking this button opens the **Logs** page w hich provides detailed status and log m essages for each upgrade stage.

When you open the **Cluster Upgrade** page, it immediately prepares the subscriber nodes for upgrade by automatically installing the required additional API support.

This is a background process and does not require any actions from the user. A progress indicator is displayed during this stage.

NOTE

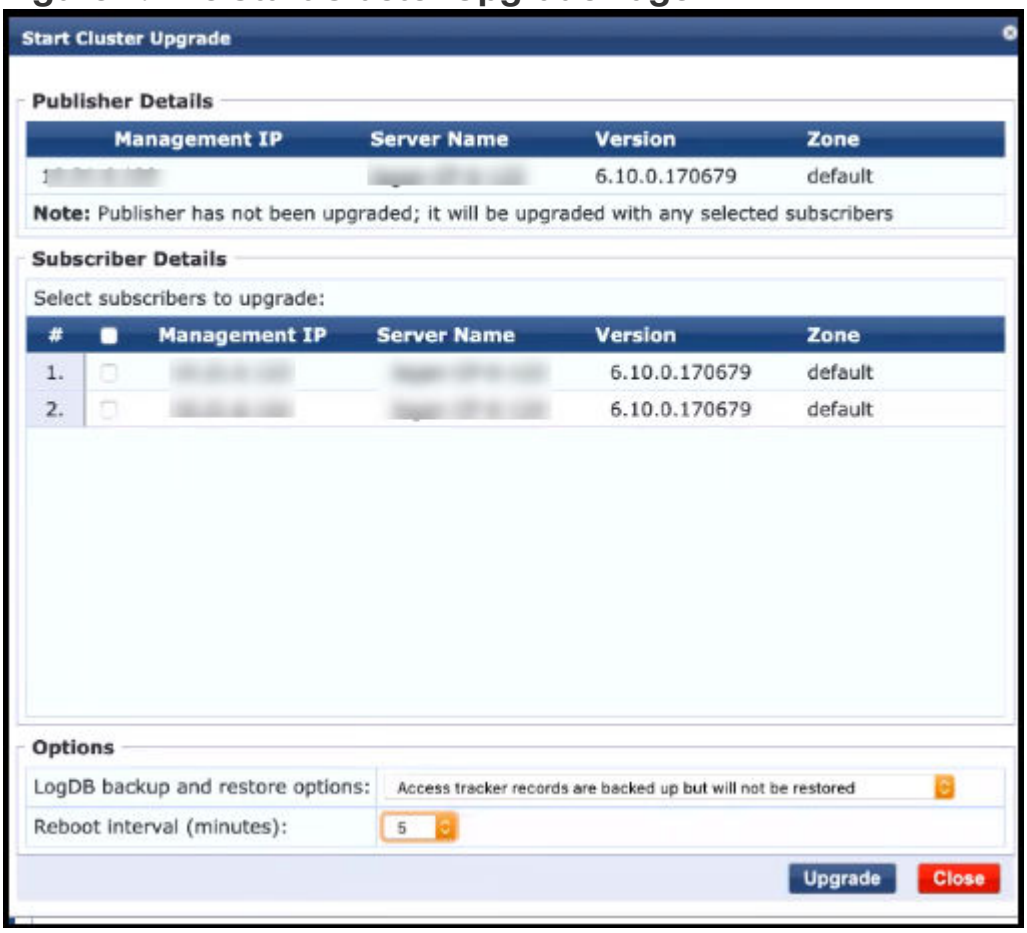
To install the patch for API support on subscribers, these subscribers must be able to access the publisher over HTTP, or they must be able to access the publisher over HTTPS using its host name and validate the certificate that is presented (that is, trust the issuer and match the host name in the certificate Common Name (CN)).

When the 6.11 upgrade image is available locally and all subscribers have been patched, the **Start Upgrade** link is available.

3. Click **Start Upgrade**.

   The **Start Cluster Upgrade** page opens.

Figure 2. The Start Cluster Upgrade Page



   You can upgrade the entire cluster or just a subset of Subscribers.

4. In the **Start Cluster Upgrade** window, use the check boxes to select the Subscribers to upgrade.

5. In the **LogDB backup and restore options** drop-down list, select one of these options:

   a. **Access tracker records are backed up but will not be restored:** Select if you need a backup of the access tracker records to potentially restore after upgrade.

      This option increases the overall upgrade time.

   b. **Do not backup access tracker records:** Select if you do not need a backup of the access tracker records.

6. The reboot interval determines the interval of time after the last subscriber reboots before the next subscriber can reboot. Available intervals are 5, 10, or 15 minutes. Five minutes is the default.
7. Click **Upgrade**.

   The Cluster Upgrade begins the automated upgrade process.

   No further manual steps are required until all selected Subscribers have been upgraded. For information on the automated process, see Steps in the Cluster Upgrade Automated Workflow .

   The publisher is always upgraded and rebooted first. The **Cluster Upgrade** page is not available while the publisher is rebooted and data migration is in progress.
8. When the publisher upgrade is complete, navigate to the **Cluster Upgrade** page to monitor upgrade progress, as described in Viewing Upgrade Status .
9. After a successful upgrade, confirm that all the subscriber nodes in the cluster are back in sync and all the services are accessible.
0. Verify that any preexisting standby publisher settings are restored:

   Navigate to **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters** link > **Standby Publisher** tab.

Viewing Upgrade Status

About this task

After the publisher upgrade is complete, you can monitor the upgrade status of the subscriber nodes at **Administration** > **Agents and Software Updates** > **Software Updates** > **Cluster Upgrade**.

This feature provides two ways to monitor the progress of the upgrade:

Procedure

1. On the **Cluster Upgrade** page, progress indicators in the **Upgrade Steps** area display the status of some of the main steps.

   Indicators in the **Publisher Details** and **Subscriber Details** areas also display when the publisher or each subscriber node is in progress or completed.

   When the upgrade is complete, these areas display a successful upgrade status for the publisher and every subscriber node.

2. For detailed progress information, click the **View Logs** button for the publisher or subscriber row.

   The **Logs** page opens. This page includes tabs for the **Patch**, **Download**, **Upgrade**, **Reboot**, and **Onboot** logs.

   You can view detailed status in these logs during and after the upgrade.

   > **NOTE**
   >
   > This option is not available while the publisher is rebooted and data migration is in progress. It is available again when the publisher upgrade is complete.

Moving to ClearPass 6.11

Subtopics

Version Considerations
Cluster Migration Considerations

Version Considerations

About this task

The migration to 6.11 is supported from either 6.9 or 6.10 ClearPass Policy Manager versions. With 6.9, migration is supported from 6.9.12 and higher. With 6.10, migration is supported from any 6.10 version.

> **NOTE**
>
> In ClearPass 6.11.0 and later, the maximum single cluster size is 32 servers. This includes the publisher, standby publisher, subscribers, dedicated insight server, and standby insight servers.

> **NOTE**
>
> After re-imaging the servers to 6.11.0, it is recommended to update to the latest 6.11 patch before restoring the configuration backup.

Refer to the following steps to restore your configuration and data, including licenses and certificates, on a new ClearPass 6.11.0 system:

> **NOTE**
> Default credentials applied in the clean install must be manually changed as passwords utilized from the previous version are not backed up and restored.

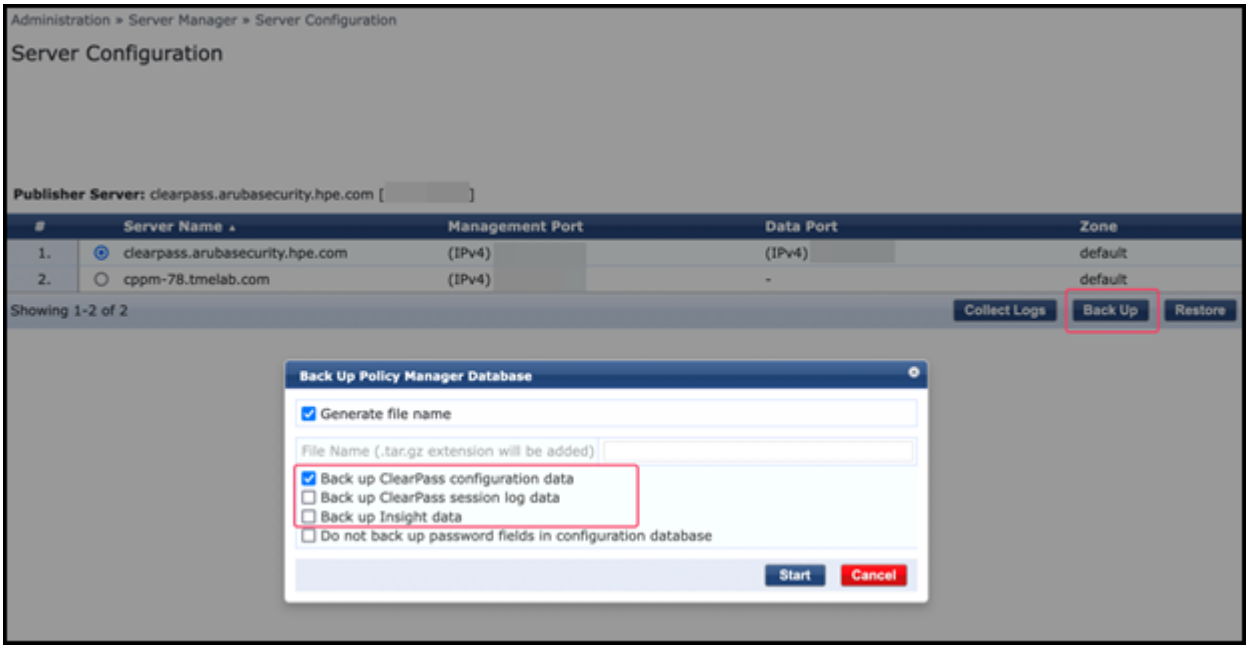
Procedure

1. Legacy license keys will not be accepted within the ClearPass 6.11.0 user interface. If using old (pre-6.8 version) license keys in a single line format, they must to be converted to the new multi-line format (starts with "-----BEGIN <license type> LICENSE KEY."). For more in formation, contact **Support**.
2. Save a backup of the configuration data (tipsdb and AppPlatform), Insight data (insightdb), and optionally the Session log data (tipsLogDb).

   Complete a configuration backup from the Publisher node by navigating to **Administration > Server Manager > Server Configuration**. Select the radio button fort the Publisher node, then select the **Back up** button.

   > **NOTE**
   > Extension and ClearPass Guest configurations are included within the Policy Manager backup and will also be restored. Custom skins are not included in the configuration backup and will need to be downloaded from the software updates portal and installed after the migration.

Figure 1. Backup the Policy Manager Database



If insight is enabled on any of the nodes, take a backup of insight module from the insight primary master. If Insight is enabled on the publisher, enable both the **Back up Insight data** and **Back up ClearPass configuration data** checkbox options. If Insight is enabled on any other cluster node, backup Insight data from the appropriate node.

If the session log database is used as part of any of the authorization workflows, back up the log database from each cluster node by selecting the **Back up ClearPass session log data** checkbox. This is not a common use case however, and the additional time required to backup and restore the log database will need to be accounted for.

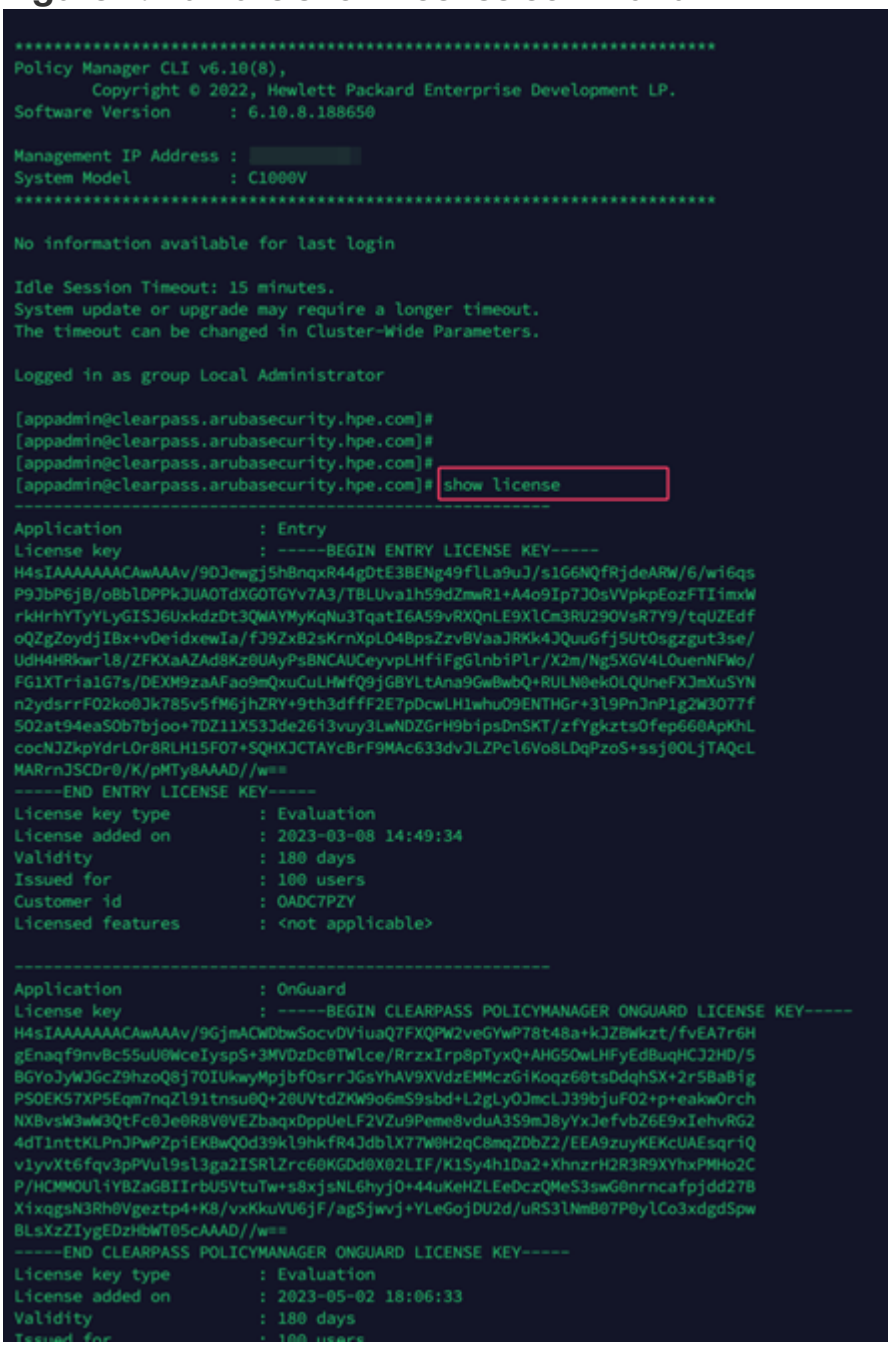
> **NOTE**
>
> Ensure port 5433 is open in order to access the Insight database. Port 5433 is also used for connecting to the tipsLogDb in ClearPass 6.11.0.

3. For migrations from ClearPass Policy Manager 6.9.12 and higher, backup the license keys manually as described below:

   Keep a copy of the Publisher license keys, as obtained using the show license command. Since the application licenses are cluster wide, the Publisher will have a list of all the application license keys used in the cluster. Additionally, backup the Platform license (PAK/PAC) from each Subscriber node. Platform licenses are specific to each node and need to be backed up separately from each node.

To obtain the licenses, log into the Publisher or Subscriber CLI as an appadmin and run the following show license command:

Figure 2. Run the show license command

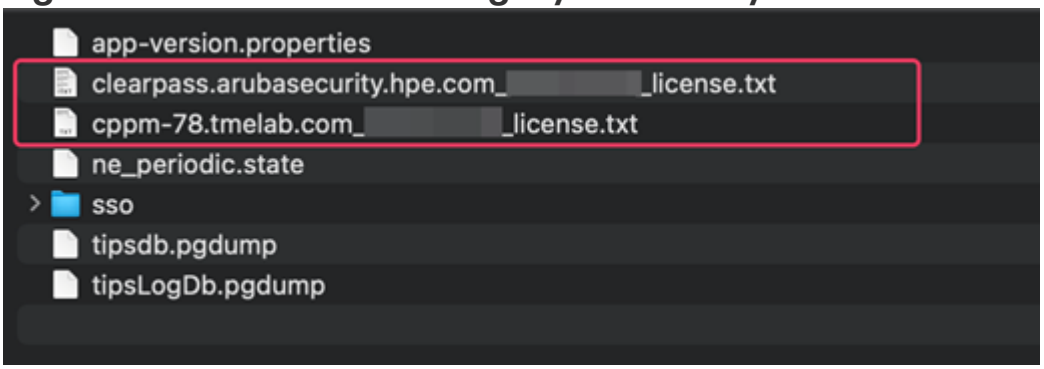


These steps are optional, since **Support** maintains license keys for each customer account. However, it is recommended to back up the license keys in case they are not registered correctly in the support portal.

> **NOTE**
>
> Backups taken in Publisher 6.10.0 and later have the Platform license (PAK/PAC) key of every reachable node in the cluster stored as plain text files. If a subscriber is not reachable, its Platform license will not be present in the publisher's backup. The activation status of the subscriber's PAK is not present in the backup.

For backups taken from ClearPass 6.10 nodes, the license keys from the Publisher, and all the reachable Subscriber nodes are available within the backup file under the PolicyManager folder as shown below:

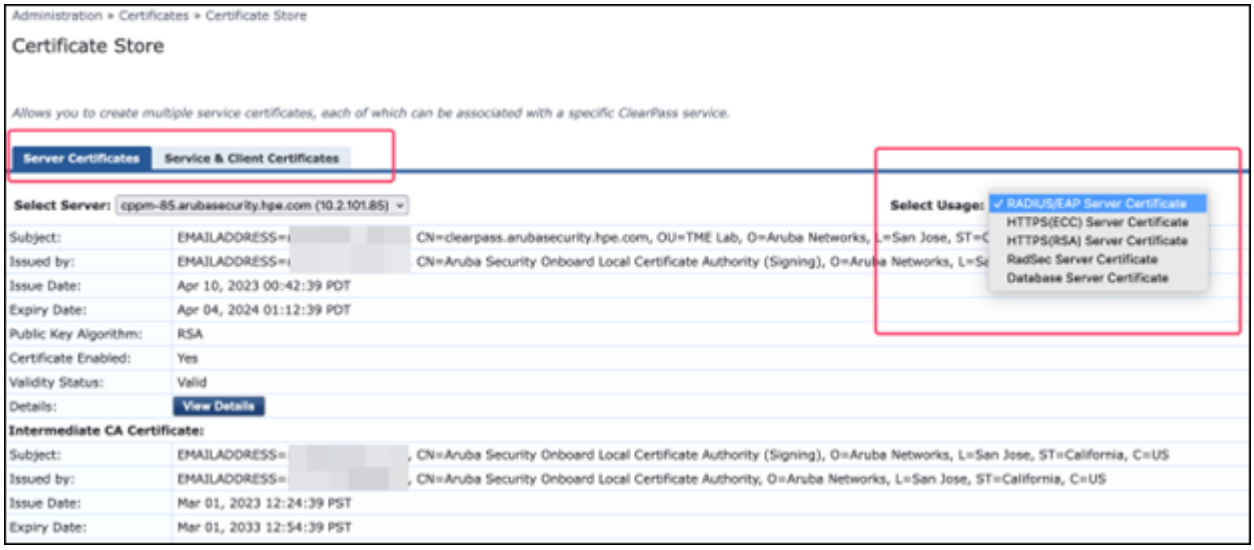
Figure 3. Locate ClearPass legacy license keys



4. Export and backup the (RADIUS / HTTPS / RadSec / Service / Client) Certificates as .p12 files with a password from **Administration > Certificates > Certificate Store**.

> **NOTE**
>
> Exporting certificates with a password will include the private key, along with the certificate file in .p12 format so it can be imported once the re-imaging process is complete.

Figure 4. Export and Backup Certificates



Export and backup the certificates for each node as a .p12 file if they are different from the certificates used on the Publisher.

5. Node specific parameters are not included in the backup. Document and archive the following settings for future reference:

   - AD Domain(s) that have been joined
   - The list of domains that ClearPass has joined can be found by running the show domain command from the CLI.

```
[appadmin@clearpass.arubasecurity.hpe.com]#
[appadmin@clearpass.arubasecurity.hpe.com]#
[appadmin@clearpass.arubasecurity.hpe.com]# show domain

------------------------------------------------------
                Domain Information
------------------------------------------------------
Domain Name               : TMELAB.COM
Domain NETBIOS Name       : TMELAB
Domain Server Ip Address  :
Domain Server Name        : dc-01.tmelab.com
Domain Status             : online
------------------------------------------------------
------------------------------------------------------
```

     - The list of password servers configured per domain can be found by running the ad passwd-server list -n <domain NETBIOS name> command.

```
[appadmin@clearpass.arubasecurity.hpe.com]#
[appadmin@clearpass.arubasecurity.hpe.com]# ad passwd-server list -n TMELAB
========================================
  Printing the list of password servers
----------------------------------------
 10.2.101.79
 10.2.101.80
 10.2.101.81
========================================
```

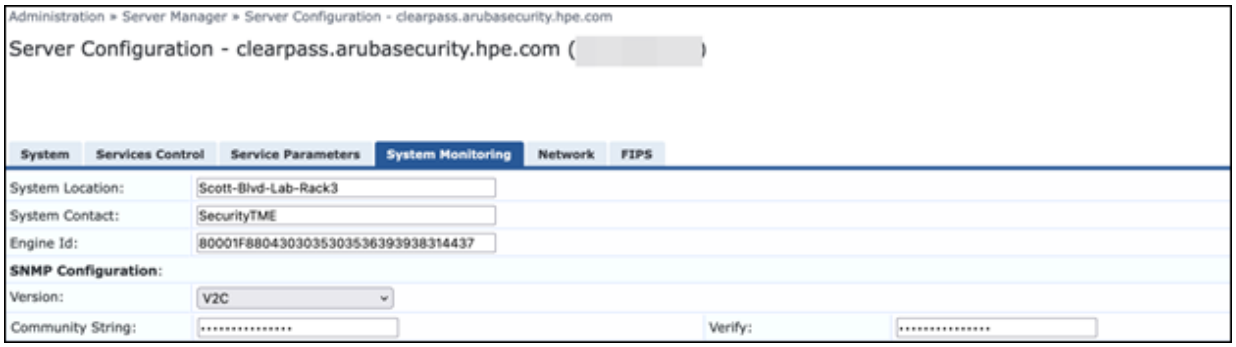
- Any static routes configured using ClearPass Policy Manager

- The List of static routes that have been configured can be found by running the network IP list command.

```
[appadmin@clearpass.arubasecurity.hpe.com]#
[appadmin@clearpass.arubasecurity.hpe.com]# network ip list
================================================================================
                        IP Rule Information
--------------------------------------------------------------------------------
0:      from all lookup local
220:    from all lookup 220
10020:  from all to 10.2.101.0/24 lookup mgmt
10040:  from 10.2.101.77 lookup mgmt
10060:  from 10.2.97.77 lookup data
12000:  from all to 10.162.78.0/24 lookup static
32766:  from all lookup main
32767:  from all lookup default

================================================================================
                  Route Information for Table main
--------------------------------------------------------------------------------
default via 10.2.97.1 dev eth1
10.2.97.0/24 dev eth1 proto kernel scope link src 10.2.97.77
10.2.101.0/24 dev eth0 proto kernel scope link src 10.2.101.77
127.17.0.0/16 dev docker0 proto kernel scope link src 127.17.0.1 linkdown
169.254.0.0/16 dev eth0 scope link metric 1002
169.254.0.0/16 dev eth1 scope link metric 1003
172.17.0.0/16 dev br-076ffb2df367 proto kernel scope link src 172.17.0.1

================================================================================
                 Route Information for Table static
--------------------------------------------------------------------------------
10.162.78.0/24 via 10.2.101.1 dev eth0

================================================================================
                  Route Information for Table mgmt
--------------------------------------------------------------------------------
default via 10.2.101.1 dev eth0
10.2.101.0/24 dev eth0 scope link src 10.2.101.77

================================================================================
                  Route Information for Table data
--------------------------------------------------------------------------------
default via 10.2.97.1 dev eth1
10.2.97.0/24 dev eth1 scope link src 10.2.97.77

================================================================================
               Route Information for Table IPSec(220)
--------------------------------------------------------------------------------

================================================================================
```

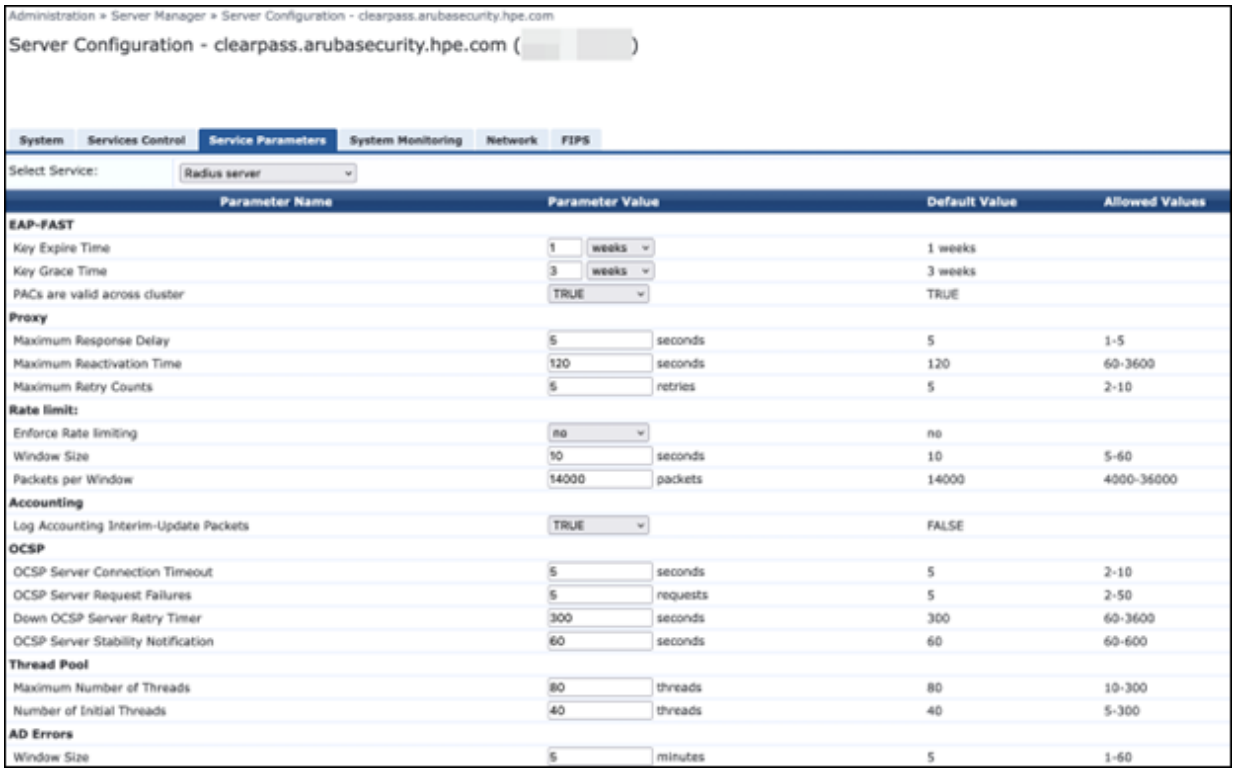
- Application ACLs that have been configured

- The application ACLs that have been applied per server can be found by navigating to **Administration > Server Manager > Server Configuration**. Click on a server in order to view its settings, then select the **Network** tab.



  - Custom service parameters

- Any changes made to the service parameters have to be documented before the upgrade and re-applied manually. Review the service parameters values by navigating to **Administration > Server Manager > Server Configuration**. Click on a server in order to view its settings, then select the **Service Parameters** tab.



6. Download the 6.11.0 ISO and/or VM image from the **HPE Networking Support Portal**. For more information, see .

7. Install the ClearPass 6.11.0 ISO or deploy the VM image.

> **NOTE**
>
> For information on deploying the VM image, go to: , or .

8. Complete the bootstrap configuration using the appadmin CLI command. This includes the hostname, IP address assignment for the management and data interfaces, NTP server, time zone and the FIPS mode configuration.

> **NOTE**
>
> The IP address should be the same to reuse DB server certificate exported in step 4. If the 6.9.X or 6.10.X backup is from a FIPS mode deployment, then FIPS mode should be enabled before restoring the backup.

9. Add the Platform License made available in backup described in step 2 to access the GUI. Add all other application licenses (Access, Onboard, OnGuard etc.) from the GUI as well. The following additional tasks also require completion:
    - Any static routes required need to be reconfigured
    - Join ClearPass Policy Manager to the AD domain
    - Configure application ACLs
    - Set the SNMP Monitoring configuration
    - Add Platform license to access the GUI (Subscriber only)
0. Restore the certificates exported in step 4. This includes restoring the Publisher's or Subscriber's server certificates.
1. Restore the backup taken in step 2 using the **-c** and **-r** command options, or by using the GUI.
    - -c restore the ClearPass Policy Manager configuration data
    - -r restore the Insight data

- -i ignore version mismatch and attempt data migration

> **NOTE**
>
> Do NOT restore the ClearPass Policy Manager session log data as part of the restore operation described in this step. Including the session log data in the restore adversely affects the time to complete the restore operation.

2. Join the subscriber nodes to the cluster as applicable. Repeat for each subscriber and configure Standby Publisher.
3. Download and install custom skins if needed from the software updates portal. Apply the skin to the appropriate guest pages.
4. Activate the licenses. There is a 90-day grace period to complete the license activation.
   - Activate the Publisher's PAK and each cluster-wide application license
   - Activate the Platform license on each Subscriber

Cluster Migration Considerations

For clustered ClearPass configurations, there are different ways to plan the migration. For ClearPass clusters running on virtual appliances, the migration process is easier compared to the hardware counterparts.

One option to consider for virtual appliances is to build a new 6.11cluster in parallel and restore the backup from the production cluster. If there is a need to retain existing IP address schema for the the VMs, power off the existing VMs and deploy new VMs running ClearPass 6.11 with the same IP addresses.

For hardware appliances, another option is to remove the standby publisher and a few subscriber nodes from the cluster, re-image them as ClearPass 6.11 versions and create a smaller parallel cluster running 6.11. This provides the opportunity to test different workflows against ClearPass 6.11. Once the validations are complete, move specific services from the old cluster to the new ClearPass 6.11 cluster. Additionally, instead of moving over all the services at once, move specific workflows from the old cluster to the new ClearPass 6.11 cluster.

Troubleshooting Tips

Here are troubleshooting tips when upgrading a cluster:

- If you encounter errors while upgrading a subscriber node, use a manual upgrade procedure to upgrade the Subscriber after the root cause for the upgrade failure has been fixed.
- If you need to revert to the previous version of Policy Manager, you can do so manually from the CLI for individual subscribers (see Cluster Commands).

  All status and progress information is reset when the publisher is reverted to a previous version. You can initiate the upgrade again from the **Cluster Upgrade** page.

Configuring Processing for Ingress Events

- Enabling Ingress Event Dictionaries
- Configuring the Event Source
- Configuring an Event-Based Enforcement Service
- Configuring the Ingress Receiving Ports
- Enabling Ingress Events Processing

  This section provides the procedures for configuring ClearPass Policy Manager to process ingress threat-related events.

  The Policy Manager Ingress Event Engine processes inbound threat-related events (which are Syslog events received from any third-party vendor device) and performs enforcements and actions based on the defined enforcement policies and services.

Subtopics

Enabling Ingress Event Dictionaries
Configuring the Ingress Event Source
Configuring the Ingress Receiving Ports
Configuring an Event-Based Enforcement Service
Enabling Ingress Events Processing

Enabling Ingress Event Dictionaries

About this task

By default, a set of ingress event dictionaries are available and initially set to disabled. You must enable the ingress event dictionaries before you proceed. IntroSpect sends Entity Syslog data and Alert Syslog data to the Policy Manager server using the Ingress Event engine to perform RADIUS COA (Change of Authorization) or Post Auth operations.

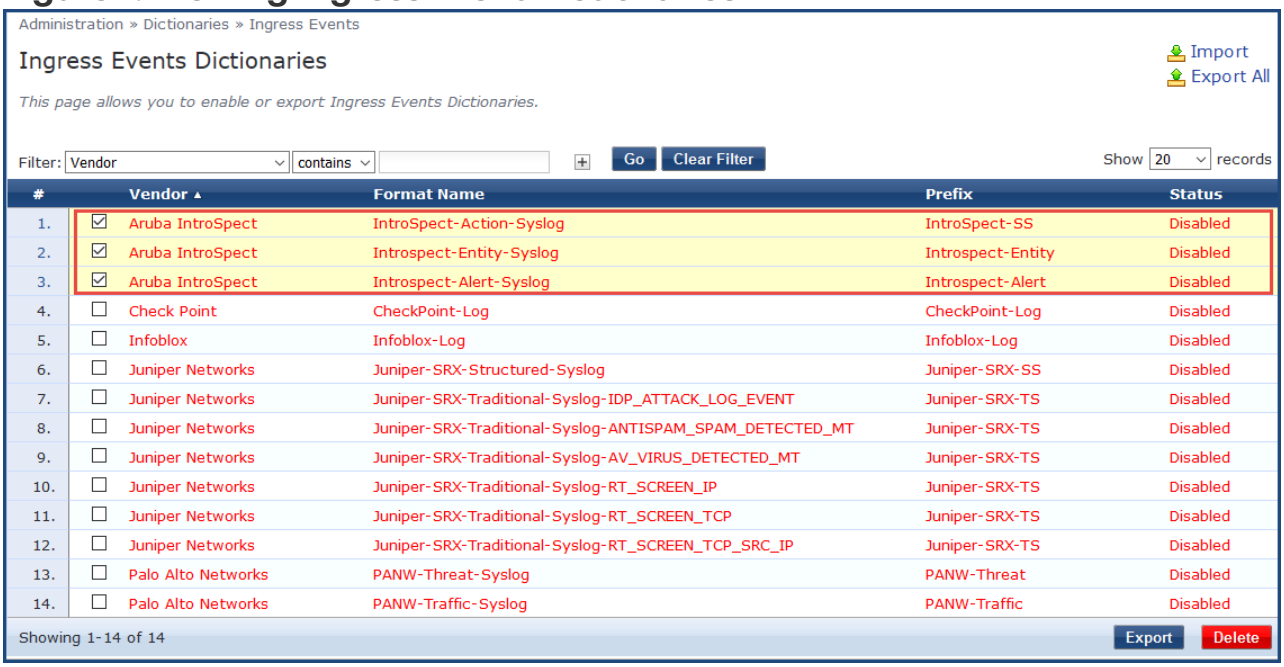
To enable an ingress event dictionary:

Procedure

1. Navigate to **Administration > Dictionaries > Ingress Events**.

   The **Ingress Events Dictionaries** page opens. The set of ingress event dictionaries are displayed. By default, they are disabled.

Figure 1. Viewing Ingress Event Dictionaries

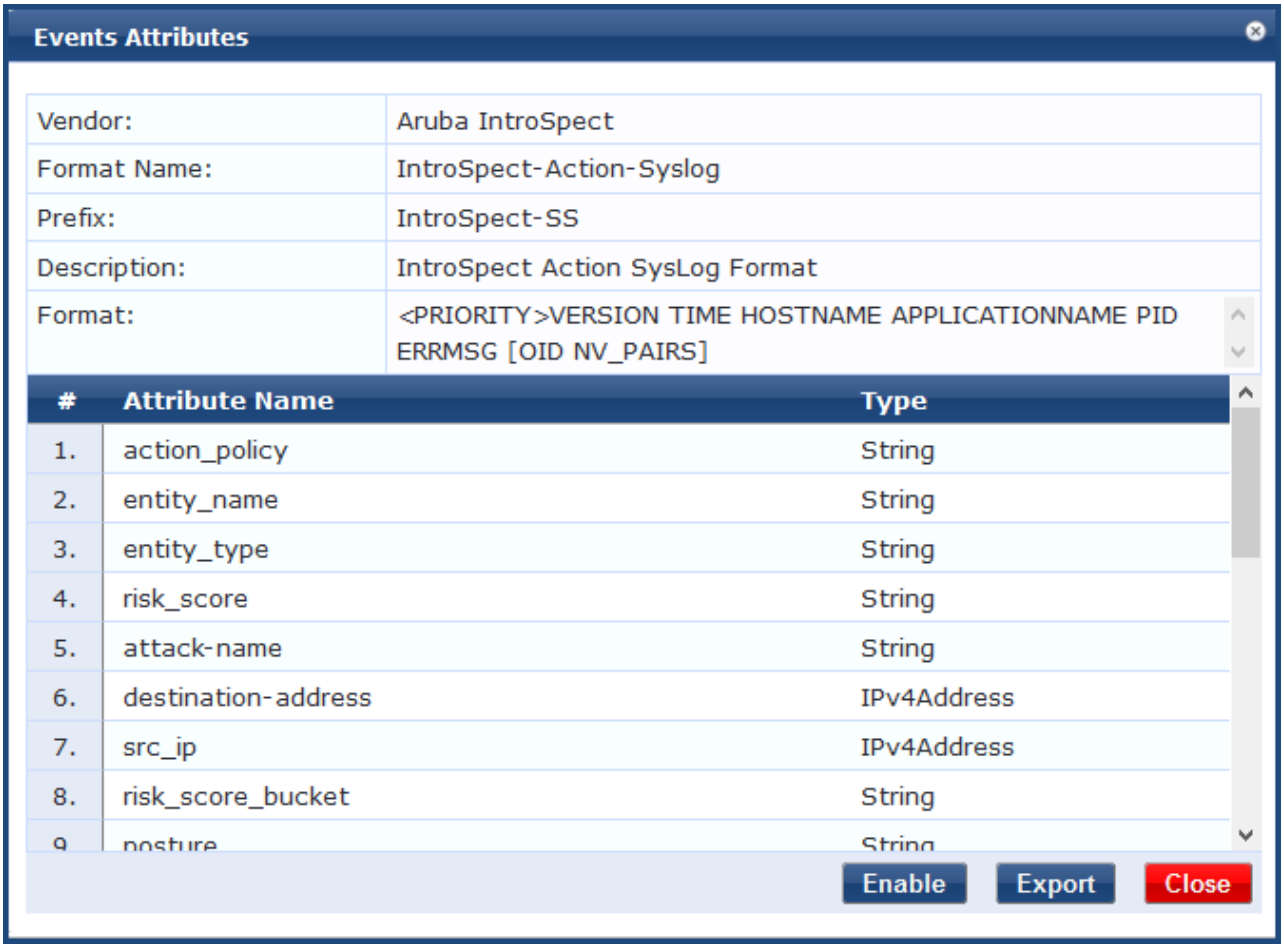


#	Vendor	Format Name	Prefix	Status
1.	Aruba IntroSpect	IntroSpect-Action-Syslog	IntroSpect-SS	Disabled
2.	Aruba IntroSpect	Introspect-Entity-Syslog	Introspect-Entity	Disabled
3.	Aruba IntroSpect	Introspect-Alert-Syslog	Introspect-Alert	Disabled
4.	Check Point	CheckPoint-Log	CheckPoint-Log	Disabled
5.	Infoblox	Infoblox-Log	Infoblox-Log	Disabled
6.	Juniper Networks	Juniper-SRX-Structured-Syslog	Juniper-SRX-SS	Disabled
7.	Juniper Networks	Juniper-SRX-Traditional-Syslog-IDP_ATTACK_LOG_EVENT	Juniper-SRX-TS	Disabled
8.	Juniper Networks	Juniper-SRX-Traditional-Syslog-ANTISPAM_SPAM_DETECTED_MT	Juniper-SRX-TS	Disabled
9.	Juniper Networks	Juniper-SRX-Traditional-Syslog-AV_VIRUS_DETECTED_MT	Juniper-SRX-TS	Disabled
10.	Juniper Networks	Juniper-SRX-Traditional-Syslog-RT_SCREEN_IP	Juniper-SRX-TS	Disabled
11.	Juniper Networks	Juniper-SRX-Traditional-Syslog-RT_SCREEN_TCP	Juniper-SRX-TS	Disabled
12.	Juniper Networks	Juniper-SRX-Traditional-Syslog-RT_SCREEN_TCP_SRC_IP	Juniper-SRX-TS	Disabled
13.	Palo Alto Networks	PANW-Threat-Syslog	PANW-Threat	Disabled
14.	Palo Alto Networks	PANW-Traffic-Syslog	PANW-Traffic	Disabled

2. To enable a dictionary, select the disabled Ingress Events Dictionary for the appropriate vendor.

   The **Events Attributes** dialog opens.

Figure 2. Enabling an Ingress Events Dictionary



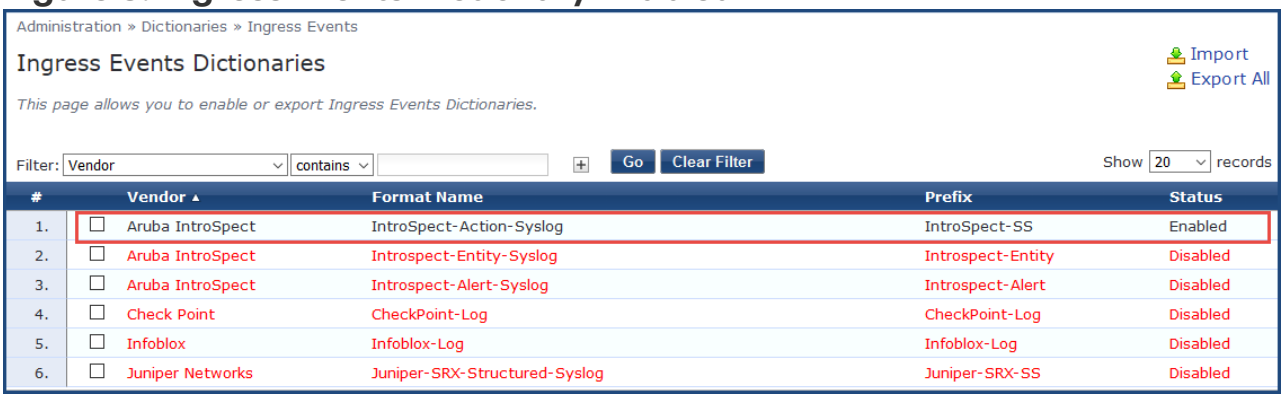
3. To enable the selected ingress events dictionary, click **Enable**. You receive the message:

   Dictionary has been successfully enabled

4. Click **Close**.

   You return to the **Ingress Events Dictionaries** page. The dictionary information is no longer displayed in red and the **Status** column is set to **Enabled** (see Figure 3).

Figure 3. Ingress Events Dictionary Enabled



Configuring the Ingress Event Source

About this task

An event source is a device that sends Syslog events to Policy Manager.

> **NOTE**
> Any events sent that are not from configured event sources are ignored.

To configure an Event Source:

Procedure

1. Navigate to **Configuration > Network > Event Sources**.

   The **Event Sources** page opens.

2. To add the Event Source for the desired vendor, click **Add**.

   The **Add Event Source** dialog opens.

Figure 1. Adding an Event Source

3. Populate the **Add Event Source** parameters as described in Table 1.

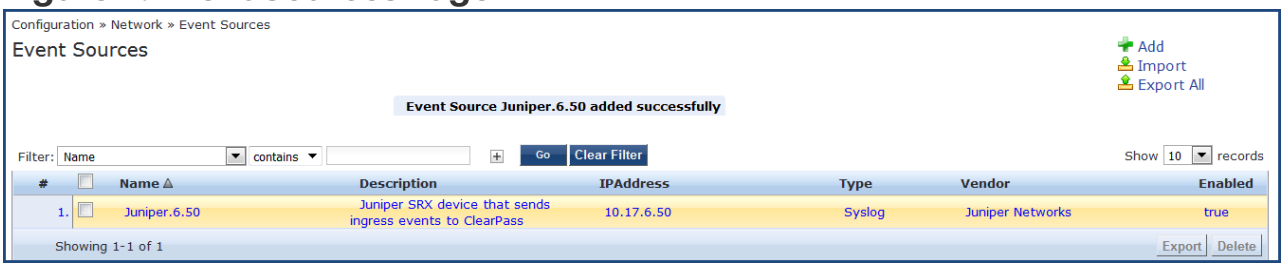
Table 1. Configuring the Add Event Source Parameters

Parameter	Action/Description
Name	Enter the name or the IP address of the device t d Syslog events to Policy Manager.
Description	Optionally, enter a description of this Event Sou
IP Address	Enter the IP address of the device that will send nts to Policy Manager.
Type	From the drop - down, select **Syslog**.
Vendor	From the drop - down, select the event source **** • Aruba Introspect • Check Point • Infoblox • Juniper Networks • Palo Alto Networks
Enable	Select this check box to enable the device as an rce.

4. When finished, click **Add**.

The **Event Sources** page now displays the new Event Sources (see Figure 2).

Figure 2. Event Sources Page



Note that the IP address displayed in Figure 2 is the IP address and host name of the Juniper SRX gateway that sends Syslog events to Policy Manager.

Configuring the Ingress Receiving Ports

About this task

The ingress receiving ports are the TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) ports on the Policy Manager server where the events source sends ingress-related events. By default, the ingress receiving port is **514** for both TCP and UDP. You can modify the ingress receiving ports with a custom value as necessary.

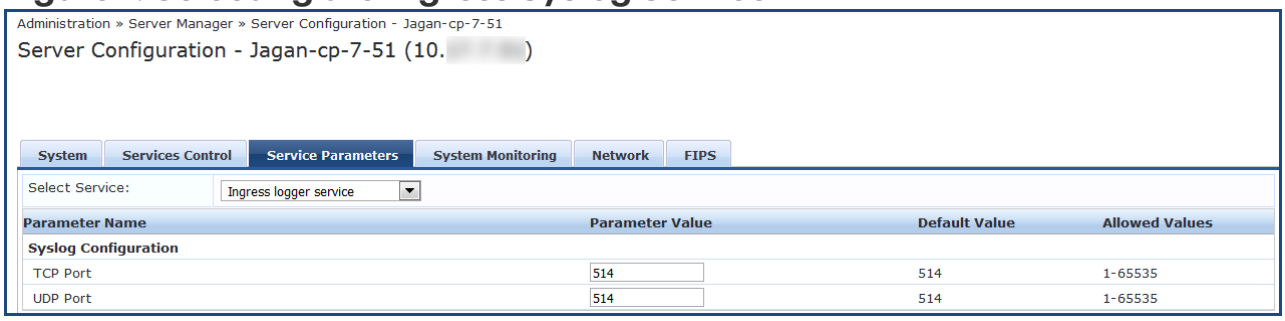
To confirm or change the ingress receiving ports on the Policy Manager server:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. From the list of Policy Manager servers, select the appropriate server.

   The **Server Configuration** page opens.
3. Select the **Service Parameters** tab.
4. From the **Select Service** drop-down, choose **Ingress syslog service** (see Figure 1).

Figure 1. Selecting the Ingress Syslog Service



As you can see in Figure 1, the parameter value for both the TCP and UDP receiving ports is set to the default value of **514**.

5. If you wish to modify the port numbers for one or both of the receiving ports, enter the new value(s).
6. When satisfied with the settings, click **Save**.

Configuring an Event-Based Enforcement Service

Introduction

This section describes how to add the **Event-Based Enforcement** service that manages enforcement actions in response to threat-event processing. When there is a suspicious user, this user could represent a common DOS attack or some other threat. When a threat is detected, Policy Manager performs enforcement operations as configured; for example, executing a change of authorization (COA) to disconnect a suspicious user from the network.

Adding an Event-Based Enforcement Service

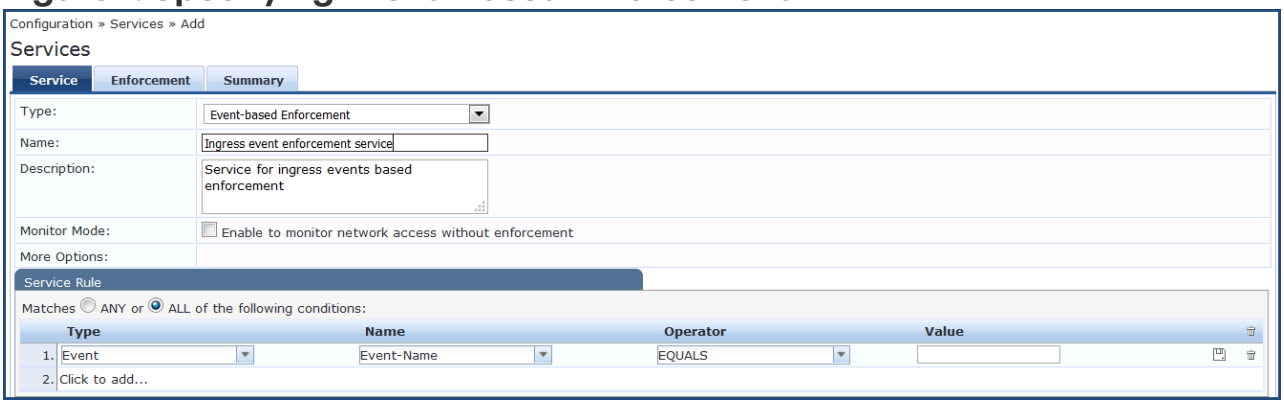
About this task

To add an event-based enforcement service:

Procedure

1. Navigate to **Configuration > Services**. The **Services** page opens.
2. To add the service, click **Add**. The **Add Services** dialog opens.
3. From the **Type** drop-down list, select **Event-based Enforcement** (see Figure 1).

Figure 1. Specifying Event -Based Enforcement



4. Specify the **Service** tab parameters for this service as described in the following table:

Table 1. Event - Based Enforcement > Service Parameters

Parameter	Action/Description
Type	Select **Event - based Enforcement**.
Name	Enter the name or label for the service you want to create.
Description	Enter the description that provides additional infor mation to identify the service. This field is optional but recommended.

Parameter	Action/Description
Monitor Mode	Optionally, check **Enable to monitor network acc ess without enforcement** to allow authentication a nd health validation exchanges to take place betwe en the endpoint and Policy Manager, but without e nforcement. In **Monitor Mode**, no enforcement profiles (and ass ociated attributes) are sent to the network device.
Service Rule	
Matches	Select one of the following: • **ANY**: Matches one of the configured conditions . • **ALL**: Specifies to match all of the configured co nditions.
Type	Select the type of data filter: • **Date** • **Event** Then configure the **Name**, **Operator**, and **Value** for the selected type accordingly.
Name	Select the name of the attribute. The **Name** list varies according to which **Type** you s elected.
Operator	Select any subset of string data type operators fro m the **Operator** drop - down list.
Value	Enter or select the value of the attribute.

Associating the Service with an Enforcement Policy

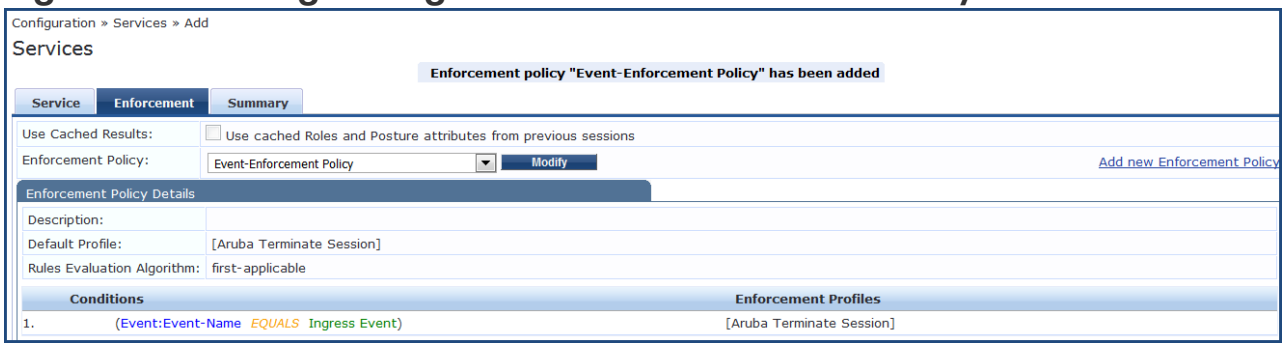
About this task

After you create the event-based enforcement service, you must associate the service with an enforcement policy. To associate the service with an enforcement policy:

Procedure

1. From the **Add Services > Service** page, click **Next**. The **Services > Enforcement** page opens.

Figure 2. Selecting the Ingress Events Enforcement Policy



From the **Add Services > Enforcement** page, you can either select an existing enforcement policy or create a new one.

2. Specify the **Add Services > Enforcement** parameters as described in the following table, then click **Save**.

Table 2. Add Service > Enforcement Page Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles and posture attributes from previous sess ions.
Enforcement Policy	From the drop - down list, select the prec onfigured enforcement policy. This is ma ndatory. If you do not have any preconfigured en forcement policies, click the **Add New En forcement Policy** link to create a new en forcement policy.
Enforcement Policy Details	
Description	Displays additional information about the selected enforcement policy.

Parameter	Action/Description
Default Profile	Displays the name of the profile applied by the selection of the enforcement poli cy.
Rules Evaluation Algorithm	Shows the first matched rule and returns the role; or selects all matched rules and return a set of roles.
Conditions	Displays the conditions configured for th e selected enforcement policy rules confi guration.
Enforcement Profiles	Displays the enforcement profile(s) specif ied in the selected enforcement policy.

Enabling Ingress Events Processing

About this task

The final task is to enable ingress events processing.

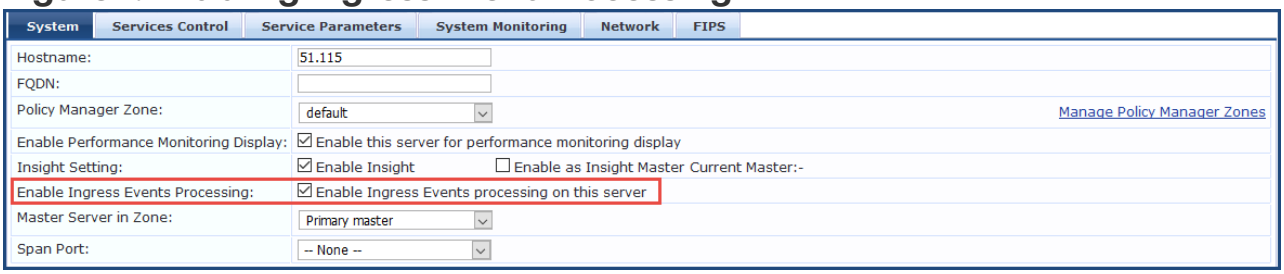
To enable ingress events processing on the Policy Manager server:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. From the list of Policy Manager servers, select the appropriate server.

   The **Server Configuration** page opens to the **System** tab.

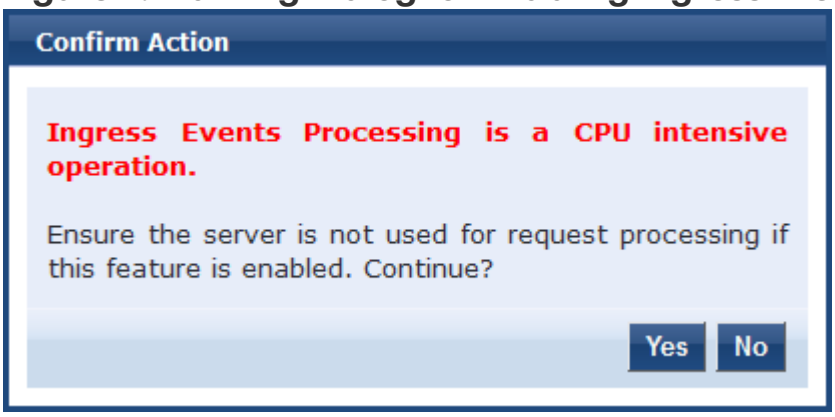
Figure 1. Enabling Ingress Event Processing



3. Click the **Enable Ingress Events Processing** check box.

The following warning dialog is displayed, alerting you to the impact on system performance that may occur when you enable ingress events processing.

Figure 2. Warning Dialog for Enabling Ingress Events Processing



4. To proceed with ingress events processing on this server, click **Yes**.

   For details on the **Server Configuration > System Tab** parameters, see System Tab.

Agentless OnGuard

Subtopics

Prerequisites
Configuring Agentless OnGuard
Monitoring Agentless OnGuard
Agentless OnGuard Tips and Troubleshooting
Using OnGuard with Aruba Virtual Intranet Access (VIA)
Agentless OnGuard Important Points and Limitations
Feature Comparison: Persistent OnGuard, Agentless OnGuard, and Native Dissolvable Agent for Windows, macOS and Supported Linux OS

Prerequisites

Note the following prerequisites for the Policy Manager server, as well as for Windows, macOS, and LInux clients.

Policy Manager Server Prerequisites

About this task

ClearPass server prerequisites include:

Procedure

1. ClearPass Insight should be enabled. For information on enabling Insight, see Table 1.
2. Device Profiling should be enabled (see Policy Manager Device Profiler Overview).
3. The ClearPass server should be joined to a Domain (for more information, see Join AD Domain Configuration).
4. Agentless OnGuard requires either an *OnGuard License* or a *Compliance Suite License* (for more information, see OnGuard / Compliance Suite Licenses).

> **NOTE**
>
> ClearPass does not support running both Agentless OnGuard and the Persistent Agent on the same client.

Windows Endpoint Prerequisites

About this task

The prerequisites for endpoints that run managed Windows OS are as follows:

Procedure

1. Windows endpoints should be joined to a domain.
2. *Administrative Shares* must be enabled on Windows endpoints.
3. *Server Message Block (SMB) v2* should be enabled on Windows endpoints.
4. Agentless OnGuard files should be specifically allowed in AntiVirus and Firewall applications.
5. In ClearPass 6.9.3 and earlier releases, PowerShell scripts must be allowed on Windows endpoints. Agentless OnGuard for Windows uses an .exe file instead of a PowerShell script to launch Agentless OnGuard on Windows endpoints.

   To configure network environment prerequisites settings:

6. Allow access to TCP Ports 135, 139, and 445.
7. For DHCP profiling, the Policy Manager server IP address should be configured as *IPHelper* in **Configuration > Network > Devices**:

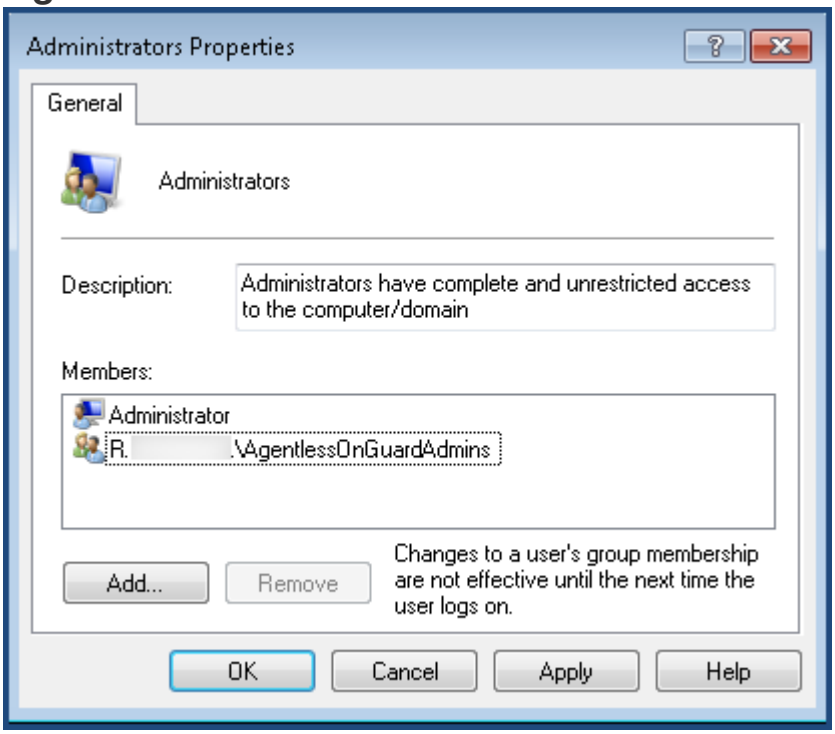
   Issue the following CLI command on the Network Access Device.

```
[appadmin]#ip helper-address <ClearPass IP> end
```

The Domain Account that is used for Agentless OnGuard must be part of the local Administrator's group on the Windows client.

Admins can create a new *Domain Group* (for example, *AgentlessOnGuardAdmins*) and add the newly created Domain Group to the local Administrators group on a domain-joined Windows client. Admins can then create a new service account for Agentless OnGuard and add it to the newly created Domain Group by configuring it in **Agentless OnGuard Settings**.

Figure 1. Domain Account in a Custom Domain Group



Linux and macOS Endpoints Prerequisites

The prerequisites for endpoints that run Linux or macOS include:

- Enable SSHon the Linux and macOS endpoints.
- Linux and macOS endpoints should have a user account ClearPass uses to connect over SSH.
- Allow access to SSH port 22.
- The user account must be part of sudoer group with the NOPASSWD option set for sudo.
- User account must have root privileges.

Configuring Agentless OnGuard

To use this feature, navigate to **Configuration > Posture > Agentless OnGuard**. See Configuring Agentless OnGuard for more information. To view a list of available Agentless OnGuard Packages, navigate to **Administration > Agents and Software Updates > OnGuard Settings** page and select the **Agentless OnGuard** tab .

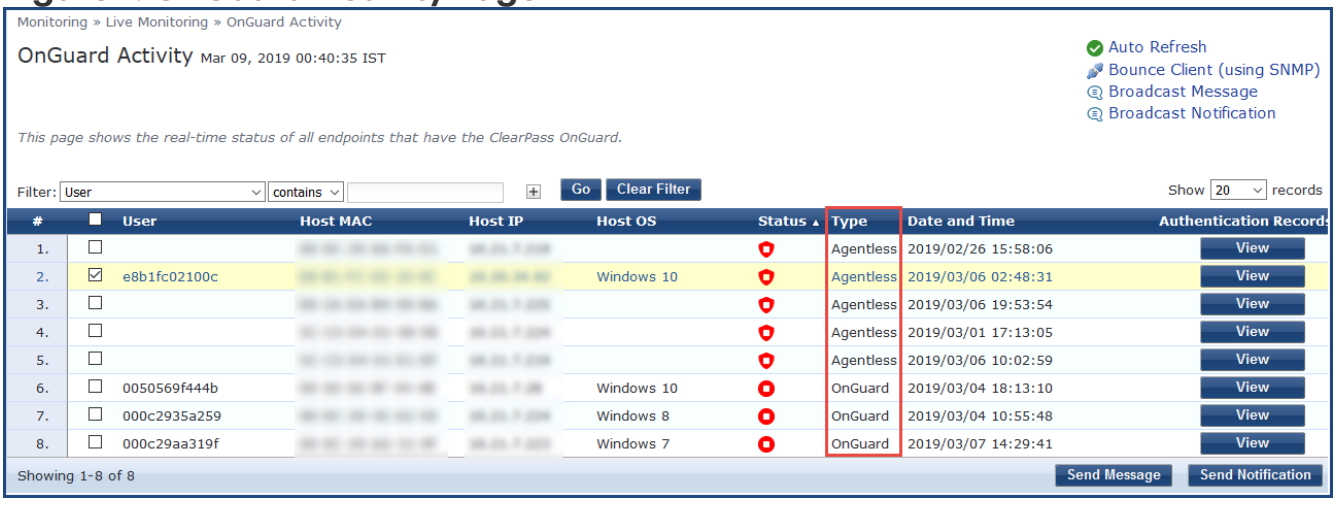
> **NOTE**
>
> ClearPass 6.9.3 and earlier releases support Agentless OnGuard only on managed Windows OS Devices. Policy Manager 6.9.4 introduced support for Agentless OnGuard for macOS and Linux devices. Unmanaged devices (such as printers) are not supported.

Monitoring Agentless OnGuard

The **OnGuard Activity** page shows the real-time status of all endpoints that have Policy Manager OnGuard. This page also presents configuration tools to bounce an endpoint, restart a session, and send unicast or broadcast messages to all endpoints running the OnGuard agent.

To access the **OnGuard Activity** page, navigate to **Monitoring > Live Monitoring > OnGuard Activity**, as shown in the figure below. For complete details on monitoring OnGuard activity, see Live Monitoring: OnGuard Activity.

Figure 1. OnGuard Activity Page



Agentless OnGuard Tips and Troubleshooting

This section provides useful tips for using Agentless OnGuard.

Subtopics

Add a Service of type Web-based Health Check Only for Agentless OnGuard
Create Separate Agentless OnGuard Sservices
Verify that both Agentless OnGuard and Persistent Agent are not Running on the Same Client
Verify Agentless OnGuard files are Copied to the Correct Folders
Verify AntiVirus and Firewall products do not Impact Agentless OnGuard
Use Agent Script enforcement profile to open a browser or any other application to show messages
Removing Agentless OnGuard from an endpoint
Troubleshooting Agentless OnGuard

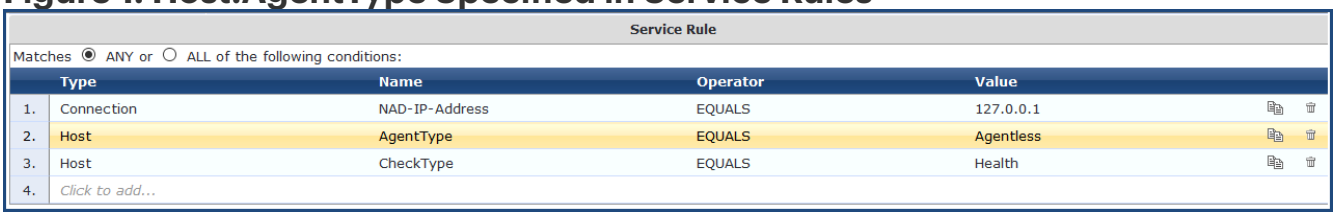
Add a Service of type Web-based Health Check Only for Agentless OnGuard

Agentless OnGuard does not support authentication, and runs exclusively in *Health Only* mode. Aruba recommends Administrators add the **Web-based Health Check Only** service for Agentless OnGuard.

Create Separate Agentless OnGuard Sservices

Administrators can create a separate Agentless OnGuard service by using **Host:AgentType** in Service Rules. For Agentless OnGuard, the value of **Host:AgentType** is **Agentless**. This is helpful in cases where OnGuard Persistent Agent is installed on some clients and on other clients Agentless OnGuard is required to run.

Figure 1. Host:AgentType Specified in Service Rules



Verify that both Agentless OnGuard and Persistent Agent are not Running on the Same Client

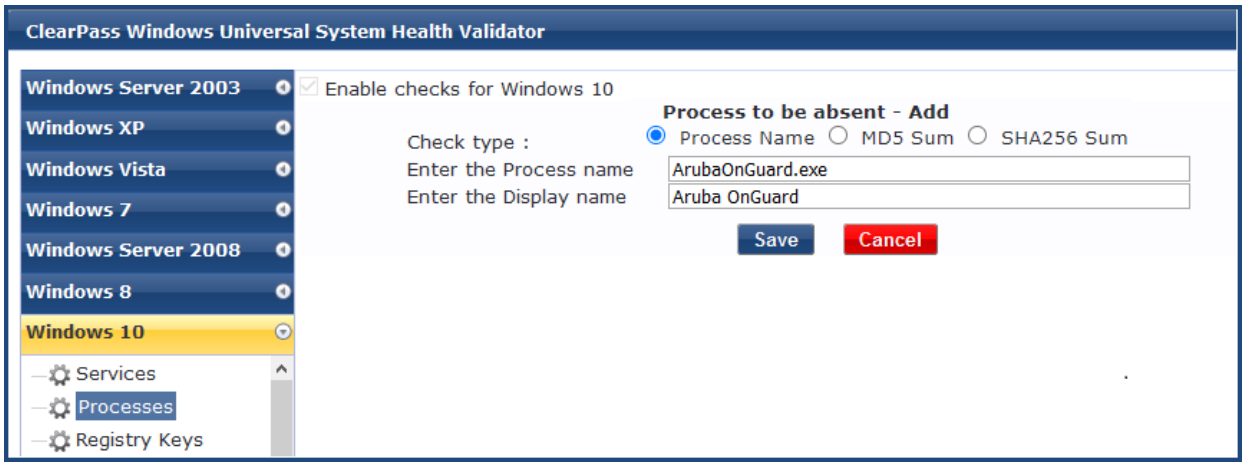
About this task

Policy Manager does not support running OnGuard and Persistent Agent on the same client. To make sure that only one Agent is running on a client, create separate services and posture policies for Persistent Agent and Agentless OnGuard (see Creating a New Posture Policy).

Procedure

1. *Posture Policy for Persistent Agent*:
   a. Posture Plugin: **ClearPass Windows Universal System Health Validator**.
   b. Select the desired version of Windows.
   c. From the list of health classes, select **Processes**.

Figure 1. Posture Policy for Persistent Agent: Processes to Be Absent



   d. Under *Processes to be absent*, enter the process name: *ArubaOnGuard.exe*. Click **Save**, then enable **Auto Remediation**.

      With this posture policy, Persistent Agent will close Agentless OnGuard if it is running.

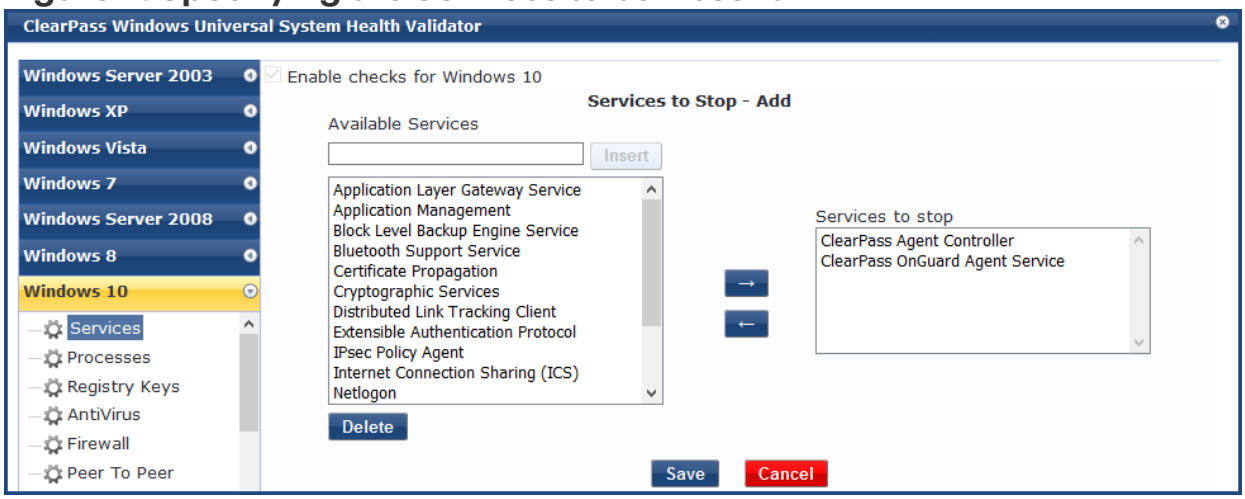
2. *Posture Policy for Agentless OnGuard to close Persistent Agent*:
   a. Posture Plugin: **ClearPass Windows Universal System Health Validator**.
   b. Select the desired version of Windows.
   c. From the list of health classes, select **Processes**.

d. Under *Processes to be absent*, enter the process name: *ClearPassOnGuard.exe*. Click **Save**, then enable **Auto Remediation**.

e. From the list of health classes, select **Services**.

f. Under *Service Groups for Services to Stop*, click **Add**, select the service names to be absent: *ClearPass Agent Controller* and *ClearPass OnGuard Agent Service*. **Note**: You may need to insert these services into the **Available Services** list.

Figure 2. Specifying the Services to be Absent



g. Click **Save**, enter the **Service Group Name**, then click **Save** again and enable **Auto Remediation**.

h. Click **Save** to save the configuration.

Verify Agentless OnGuard files are Copied to the Correct Folders

Agentless OnGuard files are copied to the following folders.

- On Windows endpoints: **C:\ProgramData\Aruba Networks\AgentlessOnGuard**
- On Linux endpoints: **/usr/share/clearpass-agentless-onguard/** and **/usr/bin/**
- On macOS endpoints: **/Library/Application Support/AgentlessOnGuard/** and **/Library/Logs/AgentlessOnGuard**

Verify AntiVirus and Firewall products do not Impact Agentless OnGuard

Some AntiVirus and Firewall products could block Agentless OnGuard files, resulting in failure to launch. To avoid such issues, the following Agentless OnGuard files should be specifically allowed in AntiVirus and Firewall products:

On Windows Endpoints:

- C:\ProgramData\Aruba Networks\AgentlessOnGuard*ArubaOnGuard.exe*
- C:\ProgramData\Aruba Networks\AgentlessOnGuard*ArubaOnGuardHelper.exe*
- C:\ProgramData\Aruba Networks\AgentlessOnGuard*wa_3rd_party_host_32.exe*
- C:\ProgramData\Aruba Networks\AgentlessOnGuard*wa_3rd_party_host_64.exe*
- <Working Directory>\ AgentlessLauncher.exe, where <*Working Directory*> refers to the working directory path configured in **Agentless OnGuard Settings**.

On Linux Endpoints:

- /usr/bin/clearpass-agentless-onguard-service
- <Working Directory>/clearpass-agentless-onguard-launcher-linux-x86_64, where <*Working Directory*> refers to the working directory path configured in **Agentless OnGuard Settings**.

On macOS Endpoints:

- /Library/Application Support/AgentlessOnGuard/AgentlessOnGuard
- /Library/Application Support/AgentlessOnGuard/ClearPassAgentHelper
- <Working Directory>/AgentlessLauncher, where <*Working Directory*> refers to the working directory path configured in **Agentless OnGuard Settings**.

Use Agent Script enforcement profile to open a browser or any other application to show messages

Since Agentless OnGuard does not have a user interface to show messages, admins can use the Agent Script enforcement profile to open a browser or any other application to show messages as part of enforcement (for details, see Agent Script Enforcement Profile).

Removing Agentless OnGuard from an endpoint

To remove Agentless OnGuard from an endpoint:

- On Windows, run **C:\ProgramData\Aruba Networks\AgentlessOnGuard\AgentlessOnGuardRemover.exe**.
- On Linux, Run **/usr/share/clearpass-agentless-onguard/clearpass-agentless-onguard-remover**.
- On macOS, Run **/Library/Application Support/AgentlessOnGuard/AgentlessOnGuardRemover**.

  Next, manually delete the <Working Directory> folder.

Troubleshooting Agentless OnGuard

This section provides Agentless OnGuard troubleshooting recommendations.

Subtopics

Policy Manager Server
Debugging Endpoint/Agentless OnGuard

Policy Manager Server

Procedure

1. The **Agentless OnGuard Log** tab shows status and details of the last five attempts to launch Agentless OnGuard on a selected endpoint (for details, see Agentless OnGuard Log).

2. If a Policy Manager server fails to launch Agentless OnGuard on a Windows endpoint with an *NT_STATUS_BAD_NETWORK_NAME* error, check and enable **Administrative shares on Windows endpoint**.

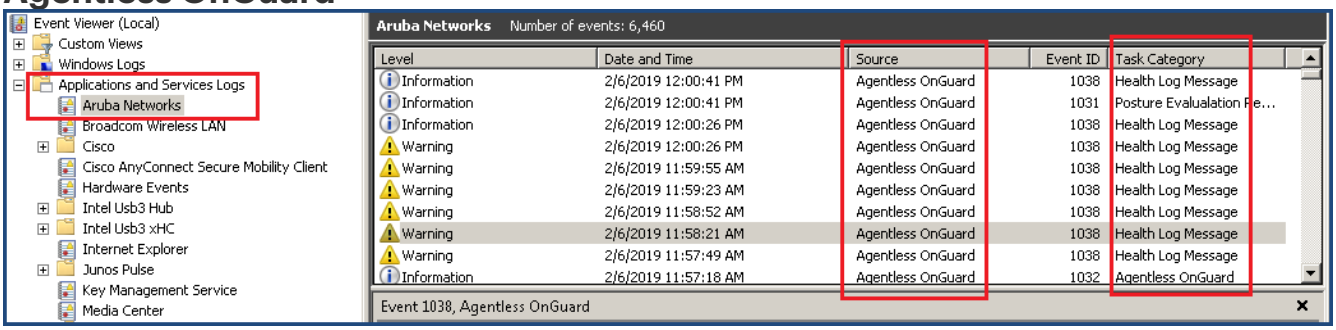
> **NOTE**
>
> To enable Administrative shares, set the value of the following two registry keys under *HKEY_LOCAL_MACHINE|SYSTEM|CurrentControlSet|services|LanmanServer* to **1**:

- autoshareserver
- AutoShareWks

Debugging Endpoint/Agentless OnGuard

To collect logs for Agentless OnGuard on macOS, run **/Library/Application Support/AgentlessOnGuard/ClearPassAgentHelper**. For Windows devices, Agentless OnGuard writes Windows Event viewer logs under **Applications and Services Logs > Aruba Networks**:

Figure 1. Aruba Networks Applications and Services Logs for Debugging Agentless OnGuard



Agentless OnGuard makes log entries in the following folders:

- On Windows Endpoints: *%ProgramData%|Aruba Networks|AgentlessOnGuard|logs*. All the processes that run under the SYSTEM account write logs to this folder.
- On Linux Endpoints: /usr/share/clearpass-agentless-onguard/logs/.
- On macOS Endpoints: /Library/Logs/AgentlessOnGuard/.
- <*Working Directory*>, where <*Working Directory*> refers to the working directory path configured in **Agentless OnGuard Settings**. See Table 1 (Agentless OnGuard Agent

Settings Parameters) in Configuring Agentless OnGuard. OnGuard Agent runs some helper processes in the current active user's context. Logs location for such helper processes can be found in this folder.

> **NOTE**
>
> Since Agentless OnGuard for Windows and Linux does not have a tool or option to collect logs, logs from the above folders must be collected manually.

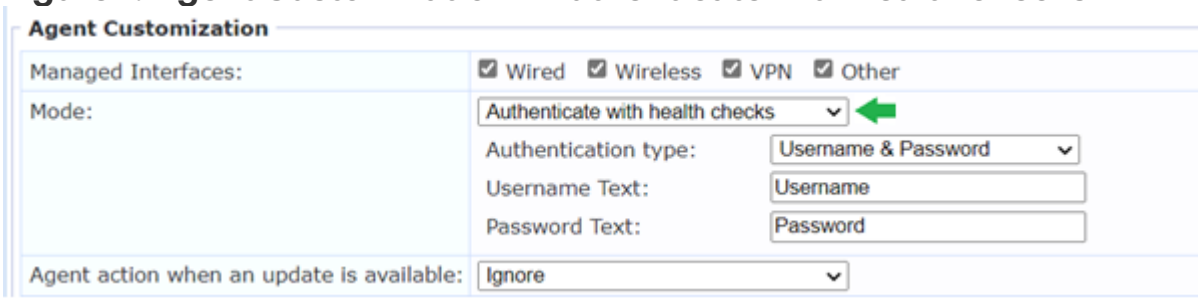
Using OnGuard with Aruba Virtual Intranet Access (VIA)

This section describes important configurations in ClearPass and ArubaOS when using OnGuard with Aruba VIA.

ClearPass/OnGuard Configuration

Ensure OnGuard is running in Auth + health mode by verifying **Authenticate with health checks** is selected for the OnGuard mode within the OnGuard Agent settings page.

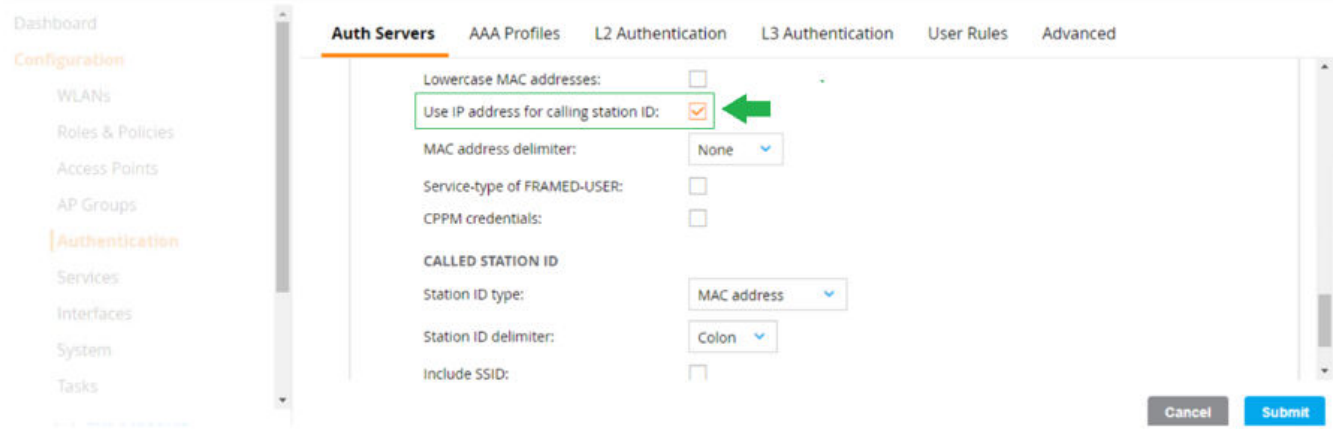
Figure 1. Agent Customization > Authenticate with health checks



ArubaOS Configuration

Select the **Use IP address for Calling Station ID:** option from within the **Auth Servers** tab to ensure ArubaOS sends the endpoint's IP address in the Radius:IETF:Calling-Station-Id field.

Figure 2. Authentication > Use IP address for Calling Station ID



Agentless OnGuard Important Points and Limitations

Deployment Considerations

About this task

Important points related to employing Agentless OnGuard:

Procedure

1. Agentless OnGuard is not supported in FIPS or CC mode.
2. Agentless OnGuard is supported only on managed (domain-joined) Windows clients.
3. For Agentless OnGuard to work, the Policy Manager server should also be joined to a domain. In case of multiple domains, Policy Manager should be joined to any one domain.
4. Agentless OnGuard always runs in Health Only mode.
5. Agentless OnGuard sends the current active user name in the **Host:ActiveUserName** attribute.
    a. Agentless OnGuard keeps track of the current active user and performs health checks whenever it detects a change in the current active user. This can be used to perform health checks after manual remediation (similar to the **Retry** option in Persistent Agent) by keeping the client locked for ~20 seconds and then logging in.
    b. When there is no active user (that is, all users are logged off or the client is locked), Agentless OnGuard does not send the **Host:ActiveUserName** attribute. Instead, it sends **Host:HealthCheckLevel** with the value **System** to indicate that Agentless OnGuard will perform System-level health checks.

6. As in Persistent Agent, Agentless OnGuard also establishes a Control Channel with the Policy Manager server and periodically sends *Keep-alive messages* (for details, refer to the **Keep-alive Interval (in seconds)** parameter in Global Agent Settings Parameters for OnGuard Agents).

   The status of Agentless OnGuard is shown in OnGuard Activity, based on launch results and keep-alive messages (refer to Live Monitoring: OnGuard Activity for details).

7. Only one instance of Agentless OnGuard is allowed to run on an endpoint. The behavior when Policy Manager tries to launch Agentless OnGuard on an endpoint is as follows:

   a. If Agentless OnGuard is already running and the Policy Manager server address is not changed, then launch status is returned as *Success* , with a message indicating Agentless OnGuard is already running.

   b. If Agentless OnGuard is already running and the Policy Manager server address has changed (as in roaming cases), the instance that is already running is closed and a new instance of Agentless OnGuard is launched with a new Policy Manager server address.

8. Once started, Agentless OnGuard runs until it is shut down or the endpoint is restarted.

9. Agentless OnGuard automatically updates without any user intervention whenever it detects a newer version on the Policy Manager server.

0. The Policy Manager server checks for new endpoints in the Policy Manager Insight database every two minutes. Since it takes about five minutes to add or update an endpoint in the Insight database, Policy Manager may take around seven minutes to launch Agentless OnGuard.

   a. To avoid a high CPU load, the Policy Manager server launches Agentless OnGuard on endpoints in batches. The size of the endpoint batches depends on the Policy Manager hardware appliance model:

      - Policy Manager C1000/ C1000V: 200 endpoints per batch
      - Policy Manager C2000/C2000V: 400 endpoints per batch
      - Policy Manager C3000/3010/C3000V: 1000 endpoints per batch

   b. The Policy Manager server ignores profiled endpoints that are non-Windows.

c. For unprofiled endpoints, the Policy Manager server tries to launch Agentless OnGuard. To do so, it must be able to connect to the endpoint on the required TCP ports: *135*, *139*, and *445*.

1. In a cluster, Policy Manager servers in a Zone distribute the load among themselves. For example, here is a case of a cluster with four nodes:

   - Number of Zones : 2
   - Zone1: 2 nodes (Policy Manager1 and Policy Manager2)
   - Zone2: 2 nodes (Policy Manager3 and Policy Manager4)
   - Zone1 has 500 endpoints, so Policy Manager1 and Policy Manager2 will process approximately 250 endpoints each.
   - Zone2 has 2,000 endpoints, so Policy Manager3 and Policy Manager4 will process approximately 1,000 endpoints each.

   NOTE

   This load distribution method is for launching Agentless OnGuard on endpoints only—not for WebAuth requests.

Agentless OnGuard Limitations

About this task

Compared to Persistent Agent, Agentless OnGuard has the following limitations:

Procedure

1. Agentless OnGuard does not have a user interface and does not show any pop-ups or messages to end users.
2. By default, Agentless OnGuard sends the client's MAC address as the username and also supports the **Use active Username in Health Only mode** Global Agent Setting option. For details, refer to the description of the **Use active Username in Health Only mode** parameter in Global Agent Settings Parameters for OnGuard Agents.
3. The following parameters on the **OnGuard Settings** page are not applicable to Agentless OnGuard (for more information, see OnGuard Settings and Agent Library Updates):

   - **Linux Installer Mode**: This option is only for Linux Persistent Agent.

- **Mode**: Agentless OnGuard does not support Authentication and always runs in Health Only mode irrespective of the value of **Mode**. Admins should add a service of type *Web-based Health Check Only* for Agentless OnGuard.
- **Agent action when an update is available**: Agentless OnGuard updates automatically to a newer version whenever a newer version is available on a Policy Manager server.
- **Agent Remediation User Interface Customization**: Agentless OnGuard does not support this OnGuard setting.

4. Agentless OnGuard does not support following **Global Agent Settings** parameters (see OnGuard Global Agent Settings):
   - **Cache Credentials Interval (in days)** and **Enable to use Windows Single-Sign On**: Agentless OnGuard does not support these two parameters as both of them are applicable for Authentication with OnGuard Agent.
   - **Enable to hide Logout button** and **Use Current OS Language (Windows only)**: Agentless OnGuard does not support these two settings as Agentless OnGuard does not have a user Interface.
   - **Enable to install VPN component**: This option is not applicable to Agentless OnGuard.
   - **Run OnGuard As**: Agentless OnGuard always runs under the SYSTEM account (similar to **Run OnGuard As Service**).
5. Agentless OnGuard does not support the following Agent Enforcement profile attributes:
   - **Messages**: Agentless OnGuard does not show messages configured in the Agent Enforcement profile to the end user. For debugging purposes, these messages are added to logs and the Windows Event Viewer or Windows devices. To collect logs for Agentless OnGuard on macOS devices, run /Library/ApplicationSupport/AgentlessOnGuard/ClearPassAgentHelper.
   - **Enable to hide Retry/Logout/Quit button**: Agentless OnGuard does not support hiding the **Retry**, **Logout**, and **Quit** buttons as Agentless OnGuard does not have user interface.
   - **Agent Enforcement > Attributes > Show Custom UI for Custom Scripts**: Agentless OnGuard does not support the Custom Remediation user interface.
6. Agentless OnGuard does not support the following **Agent Script Enforcement** profile attributes:

- Success Message
- Progress Message
- Failure Message
- Description

Agentless OnGuard ignores these attributes as it does not support the Customer Remediation user interface.

7. If a user is not logged in, some of the health checks and auto-remediation may fail for Agentless OnGuard. These health checks are user-level checks, such as Registry Keys (HKCU), Processes, and Installed Applications (user applications). Admins can use the following attributes in Service Rules and Role Mapping to check if a user is logged in:

   - **Host:HealthCheckLevel**: If a user is logged in, the value of this attribute is **User**; a value of **System** for this attribute indicates a system-level check.
   - **Host:ActiveUserName**: If a user is not logged in, WebAuth requests from Agentless OnGuard will not contain this attribute.

   > **NOTE**
   >
   > Agentless OnGuard ignores all the attributes that it does not support by allowing admins to configure these attributes for Persistent Agent.

Feature Comparison: Persistent OnGuard, Agentless OnGuard, and Native Dissolvable Agent for Windows, macOS and Supported Linux OS

This section provides comparisons between Persistent OnGuard, Agentless OnGuard, and Native Dissolvable Agent in categories such as health classes, and OnGuard Activity.

Subtopics

Health Classes
OnGuard Activity
OnGuard Settings and Global Agent Settings
Agent Enforcement
Agent Script Enforcement
Access Tracker Computed Attributes

Health Classes

Table 1. Health Classes Comparison

Agent Type	Persistent Agent	Agentless OnGuard
AntiVirus	Yes*	Yes*
Custom Scripts	Yes*	Yes*
Disk Encryption	Yes	Yes
File Check	Yes	Yes
Firewall	Yes	Yes
Installed Applications	Yes	Yes
Network Connections	Yes	Yes
Patch Management	Yes	Yes
Peer To Peer	Yes	Yes
Processes	Yes	Yes
Registry Keys	Yes**	Yes**
Services	Yes*	Yes*
USB Devices	Yes	Yes
Virtual Machine	Yes	Yes
Windows Hotfixes	Yes**	Yes**

* Health classes supported on the Linux OS

** Health classes supported only on Windows OS

OnGuard Activity

Table 1. OnGuard Activity Comparison

Agent Type	Persistent OnGuard	Agentless OnGuard
Status (Online / Offline)	Yes	Yes
Status (Success / Failure)	No	Yes
Agent Type	Yes	Yes
Broadcast Message	Yes	No
Broadcast Notification	Yes	Yes*
Send Message	Yes	No
Send Notification	Yes	Yes*
Posture Info	Yes	Yes
Agentless OnGuard Log	No	Yes

* Notification is set to Yes for a Bounce, Restart Session, or Upload Logs operation only. A Display Message or URL link is not sent to the end user.

OnGuard Settings and Global Agent Settings

Table 1. OnGuard Settings and Global Agent Settings Comparison

Agent Type	Persistent OnGuard	Agentless OnGuard	Native Dissolvable Agent
Managed Interfaces	Yes	Yes	Yes*
Mode (Auth, Health Only, Auth + Health)	Yes	No	No
Agent action when a n update is available	Yes	No**	No**
Custom Remediation User Interface	Yes	No	No
Cache Credentials In terval(in days)	Yes	No	No

Agent Type	Persistent OnGuard	Agentless OnGuard	Native Dissolvable Agent
Delay to bounce afte r Logout (in minutes)	Yes	Yes	No
Enable OnGuard req uests load - balancin g	Yes	Yes	No
Enable to hide Logo ut button	Yes	No	No
Enable to use Wind ows Single - Sign On	Yes	No	No
Keep - alive Interval (in seconds)	Yes	Yes	No
OnGuard Health Che ck Interval (in hours)	Yes	Yes	No
Run OnGuard As	Yes	No	No
Server Certificate Va lidation	Yes	Yes	No
Server Communicati on Mode	Yes	Yes	No
Support Team Email Address	Yes	No	No
Use Current OS Lang uage (Windows only)	Yes	No	No
Use active Username in Health Only mode	Yes	Yes	No
VPN Device Names (Windows)	Yes, for Windows OS only	Yes, for Windows OS only	Yes, for Windows OS only
VPN Device Names (macOS)	Yes, for macOS only	Yes, for macOS only	Yes, for macOS only

* **Managed Interface** under the **Native Dissolvable Agent Customization** section is applicable for Native Dissolvable Agent.

** Agentless OnGuard and Native Dissolvable Agent support auto-update by default.

Agent Enforcement

Table 1. Agent Enforcement Comparison

Agent Type	Persistent OnGuard	Agentless OnGuard
Bounce Client	Yes	Yes
Message	Yes	No
Session Timeout (in seconds)	Yes	Yes
Health Check interval (in hours)	Yes	Yes
Enable to hide Retry/ Logout/Q uit button	Yes	No
Bounce Delay (in seconds)	Yes	Yes
Show Custom UI for Custom S cripts	Yes	No
Enable to auto update OnGua rd Agent Library	Yes	Yes

Agent Script Enforcement

Table 1. Agent Script Enforcement Comparison

Agent Type	Persistent OnGuard	Agentless OnGuard
Path of the Script	Yes	Yes
Command To Execute	Yes	Yes
SHA256 Checksum	Yes	Yes
Execution Level	Yes	Yes

Agent Type	Persistent OnGuard	Agentless OnGuard
Wait Time (Seconds) Before Executing Script	Yes	Yes
Pass Health Evaluation Results to Script	Yes	Yes
Success Message	Yes	No
Failure Message	Yes	No
Progress Message	Yes	No
Description	Yes	No
Download URL	Yes	Yes
Script Execution Timeout (Seconds)	Yes	Yes

Access Tracker Computed Attributes

Table 1. Access Tracker Computed Attributes Comparison

Agent Type	Persistent OnGuard	Agentless OnGuard
Host:ActiveUserName	Yes	Yes
Host:AgentLibraryVersion	Yes	Yes
Host:AgentType	Yes	Yes
Host:AgentVersion	Yes	Yes
Host:AuthServer	Yes	Yes
Host:AuthType	Yes	No
Host:FQDN	Yes	Yes
Host:HealthCheckLevel	Yes	Yes
Host:InterfaceDriver	Yes	Yes

Agent Type	Persistent OnGuard	Agentless OnGuard
Host:InterfaceType	Yes	Yes
Host:MachineType	Yes	Yes
Host:OSArch	Yes	Yes
Host:OSName	Yes	Yes
Host:OSNameVersion	Yes	Yes
Host:OSType	Yes	Yes
Host:SDKType	Yes	Yes
Host:SDKVersion	Yes	Yes
Host:ServerCertificateCheck	Yes	Yes
Host:SSID	Yes	Yes
Host:UniqueID	Yes	Yes

Insight Reports

This chapter describes how to use the Policy Manager Insight Reporting tool.

This chapter includes the following information:

- About Insight
- About the Insight Dashboard
- Searching the Insight Database
- Creating Alerts
- Creating Reports
- Insight Report Categories Reference
- Insight Inventory
- Administration Operations
- Managing Insight Admin Privileges

Subtopics

About Policy Manager Insight
About the Insight Dashboard
Searching the Insight Database
Creating Alerts
Creating Reports
Insight Report Categories
Insight Inventory
Administration Operations
Managing Insight Admin Privileges

About Policy Manager Insight

This section presents an overview of Policy Manager Insight.

Subtopics

Introduction
Enabling Insight and Specifying a Primary Insight Node
Launching Insight
Special Characters Supported in Passwords for All ClearPass Modules

Introduction

ClearPass Insight is an application for use with ClearPass Policy Manager that is capable of aggregating data from multiple Policy Manager appliances that contain archived network access logs.

You can access each application within the Policy Manager suite with a single login. You need only sign in once for access to Policy Manager, Insight, Onboard, and Guest. For more information, see Launching Insight below. Insight supports the following features:

- Customizable dashboard "widgets" allow you view the specific kinds of information you need to monitor and understand what is occurring on the network. You can create customized dashboards to track detailed authentication records, audit trails, and details on network-access trends (see About the Insight Dashboard).
- The Insight Search feature allows you to search for clients, users, Policy Manager servers, and network access devices (see Searching the Insight Database).

- Customized reports analyze authentication information, device profiling, client health, licensing and posture data, as well as guest and BYOD use cases (for details, see Creating Reports and Insight Report Categories Reference).
- Alerts allow you to receive near-real-time messages regarding anomalous network activity. Alerts can be delivered via SMS or email notification to multiple recipients. Monitor a user watchlist (a list of VIPs, executives or devices that warrant special tracking) for authentication failures or other key events (see Creating Alerts).

  This chapter also provides information on how to configure operational elements about file transfers, as well as database and report data retention (see Administration Operations).

Enabling Insight and Specifying a Primary Insight Node

About this task

Before you can use Insight, you must enable it on the current Policy Manager server. If multiple nodes in a cluster have Insight enabled, one node should be configured as an Insight MasterPrimary server.

> **NOTE**
> Insight Reports, Alerts, and Administration settings can be configured on a primary Insight node only. To be able to generate a report, enabling the node as an Insight primary server(even in a single-node cluster) is mandatory.

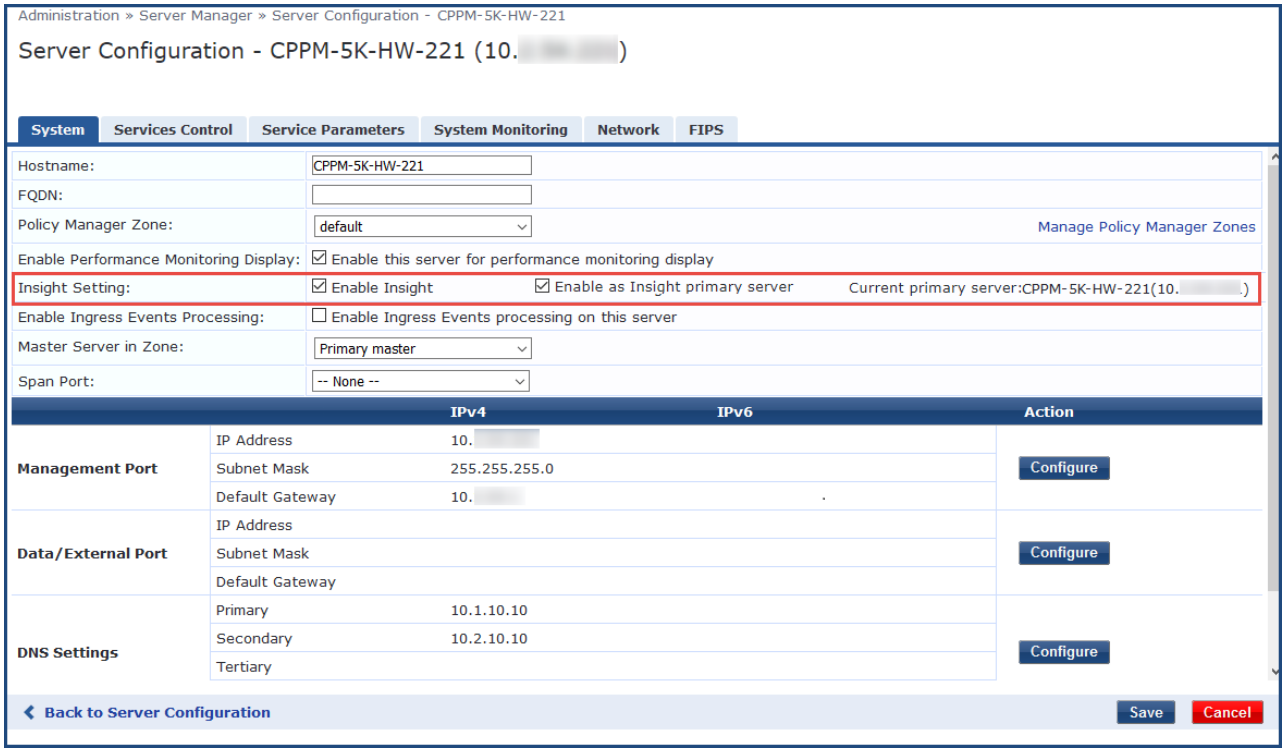
To enable Insight :

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. From the list of Policy Manager servers, click the server on which you want to enable Insight.

   The **Server Configuration > System** page opens.

   Figure 1. Server Configuration > System Page



a. **Enable Insight**: Select this check box to enable Policy Manager Insight on the current server.

b. **Enable as Insight Primary Server**: Select this check box to specify this server as an Insight primary server.

> **NOTE**
>
> To enable replication of Insight configurations across a cluster, you must configure one Policy Manager server in the cluster as an Insight primarynode.

3. Click **Save**.

Launching Insight

About this task

To launch Policy Manager Insight:

Procedure

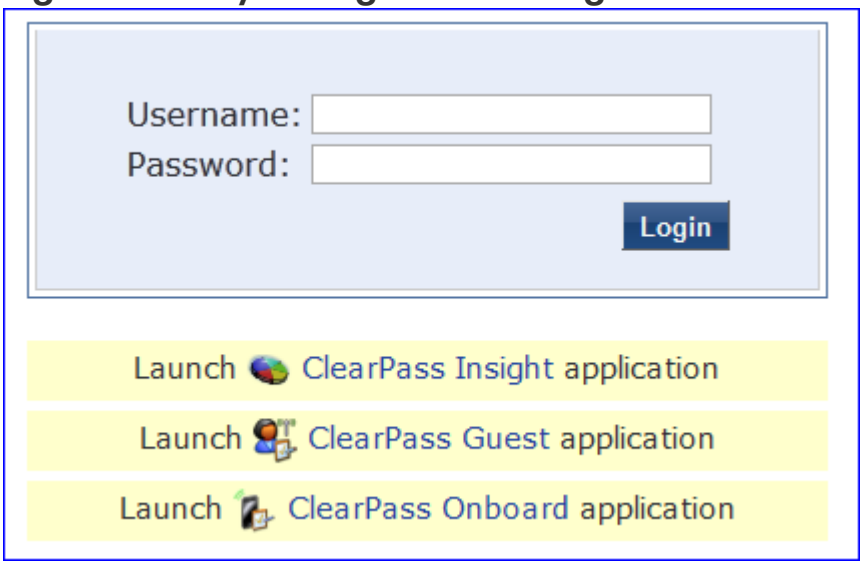
1. Use one of the following methods to launch Policy Manager Insight.

- Log in to Policy Manager, and then select **Insight** in the **Dashboard > Applications** widget. This opens Insight in a new tab.
- Access Policy Manager by pointing the browser to **https://<Policy Manager-host-name>/tips**, then select the **ClearPass Insight** link (see Figure 2).
- Point the browser to **https://<Policy Manager-host-name>/insight**.

2. Enter the default username and password, then click **Login** to launch Insight.

   Figure 2 displays the Policy Manager Access page:

Figure 1. Policy ManagerAccess Page



Special Characters Supported in Passwords for All ClearPass Modules

The special characters that are supported in passwords for all ClearPass modules (Policy Manager, ClearPass Guest, ClearPass OnGuard, ClearPass Onboard, and ClearPass Insight) are described in the following table:

Table 1. Special Characters for Module Passwords

Special Character	Description
+	Plus sign
,	Comma

Special Character	Description
-	Hyphen
.	Period
;	Semicolon
=	Equal sign
?	Question mark
_	Underscore

About the Insight Dashboard

- Dashboard Overview
- Adding a Report Widget to the Dashboard Landing Page
- Removing a Report Widget from the Dashboard Landing Page
- Creating a Report or Alert From the Dashboard
- Specifying the Date Range for Data Collection
- Authentication Dashboard
- Endpoints Dashboard
- Guest Dashboard
- Licensing Dashboard
- Network Dashboard
- Posture Dashboard
- System Monitor Dashboard

Subtopics

Dashboard Overview
Adding a Report Widget to the Dashboard Landing Page
Removing a Report Widget from the Dashboard Landing Page
Creating a Report or Alert From the Dashboard
Specifying the Date Range for Data Collection

Authentication Dashboard
Endpoints Dashboard
Guest Dashboard
Licensing Dashboard
Network Dashboard
Posture Dashboard
System Monitor Dashboard

Dashboard Overview

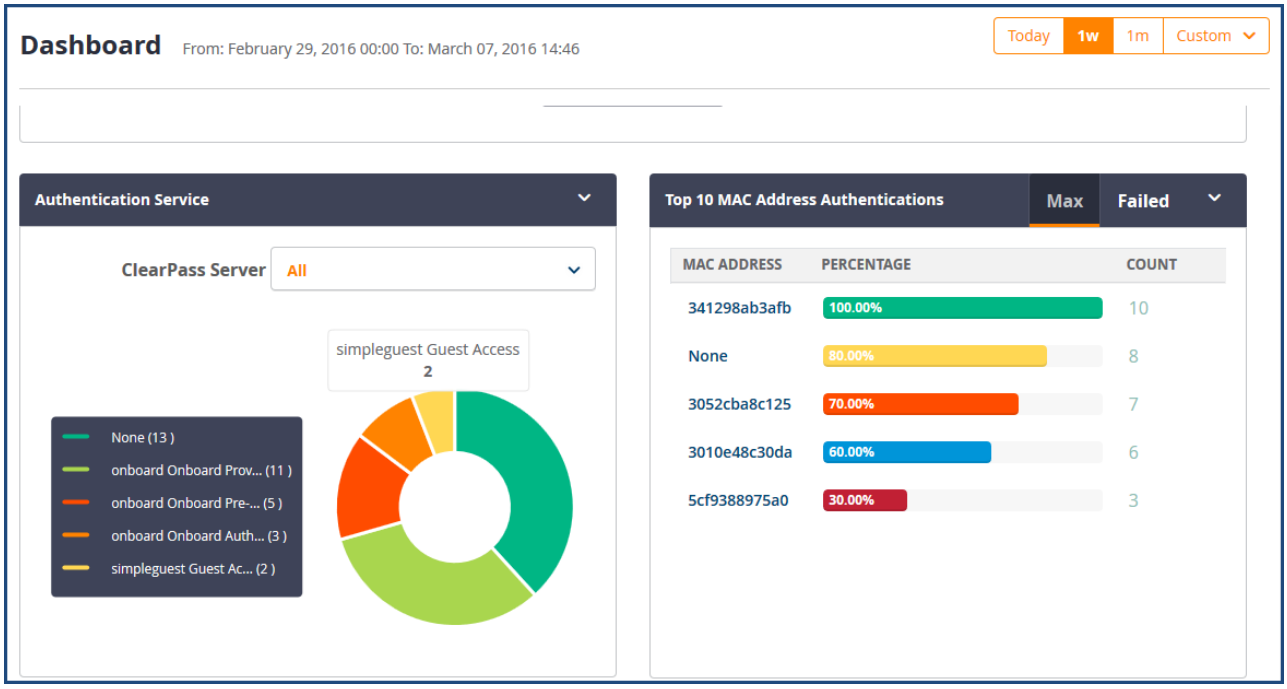
The **Dashboard Landing Page** opens immediately when you successfully log in to Policy Manager Insight. The Dashboard includes report widgets that provide a summarized and graphical view of your network analytics. The Insight header statistics at the top of the dashboard display authentication, endpoints and user statistics collected over the previous 24 hours. You can click any of these values to display a dashboard page with additional information. You can customize the Dashboard to display the report widgets that you use most often by adding widgets to the Dashboard Landing Page; you can also remove any report widget from the Dashboard Landing Page as needed.

Figure 1. Insight Header

The following report widgets are included by default on the **Dashboard Landing** page:

- Authentication Trend
- Authentication Distribution
- Authentication Status
- Authentication Service
- Top 10 MAC Address Authentications
- Top 20 NAD Authentications
- Top 10 Authentication Errors
- Latest 10 Authentication Alerts

Figure 2. Insight Dashboard Landing Page



Adding a Report Widget to the Dashboard Landing Page

About this task

When you add a report widget to the Dashboard Landing page, that widget will appear in the Landing page, and the widget will also continue to be available on its Dashboard category page; for example, if you added the **Top 10 Restarted Services** widget from the System Monitor Dashboard, the **Top 10 Restarted Services** widget would be present in both the Dashboard Landing page and the System Monitor Dashboard.

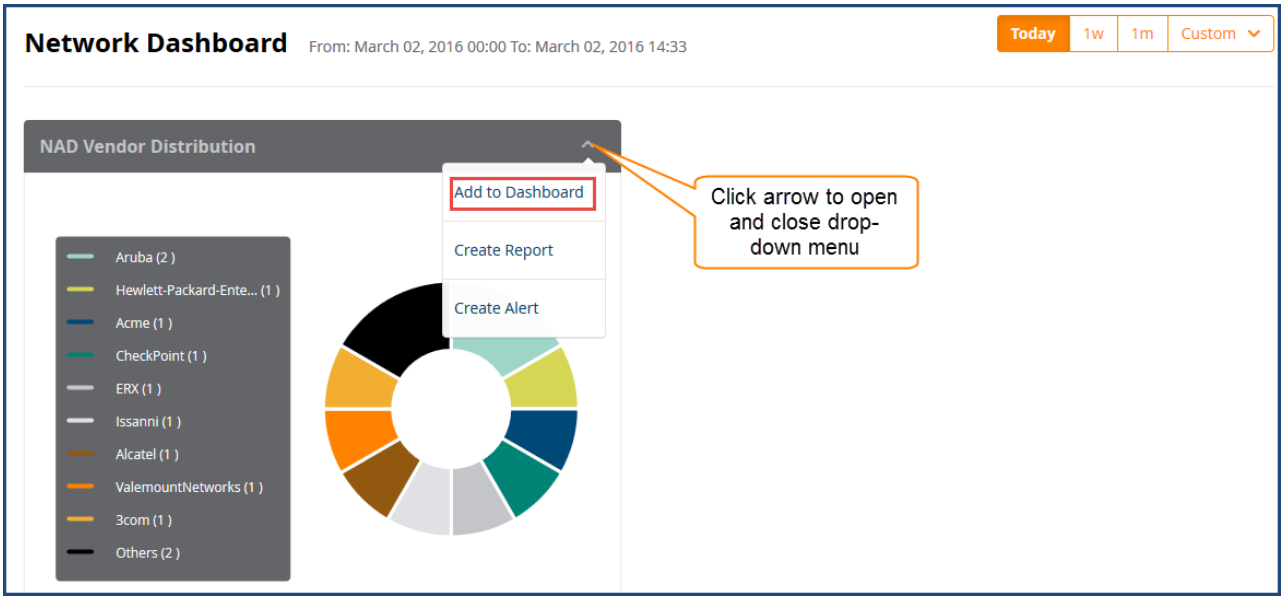
To add a report widget to the Dashboard Landing page:

Procedure

1. From any of the Dashboard category pages, click the arrow icon in the right corner of the widget title bar.
2. Select **Add to Dashboard** (see Figure 3).

   That report widget will appear when you return to the Dashboard Landing page.

Figure 1. Adding a Widget to the Dashboard Landing Page



3. To view the newly-added widget, return to the Dashboard Landing page.

Removing a Report Widget from the Dashboard Landing Page

About this task

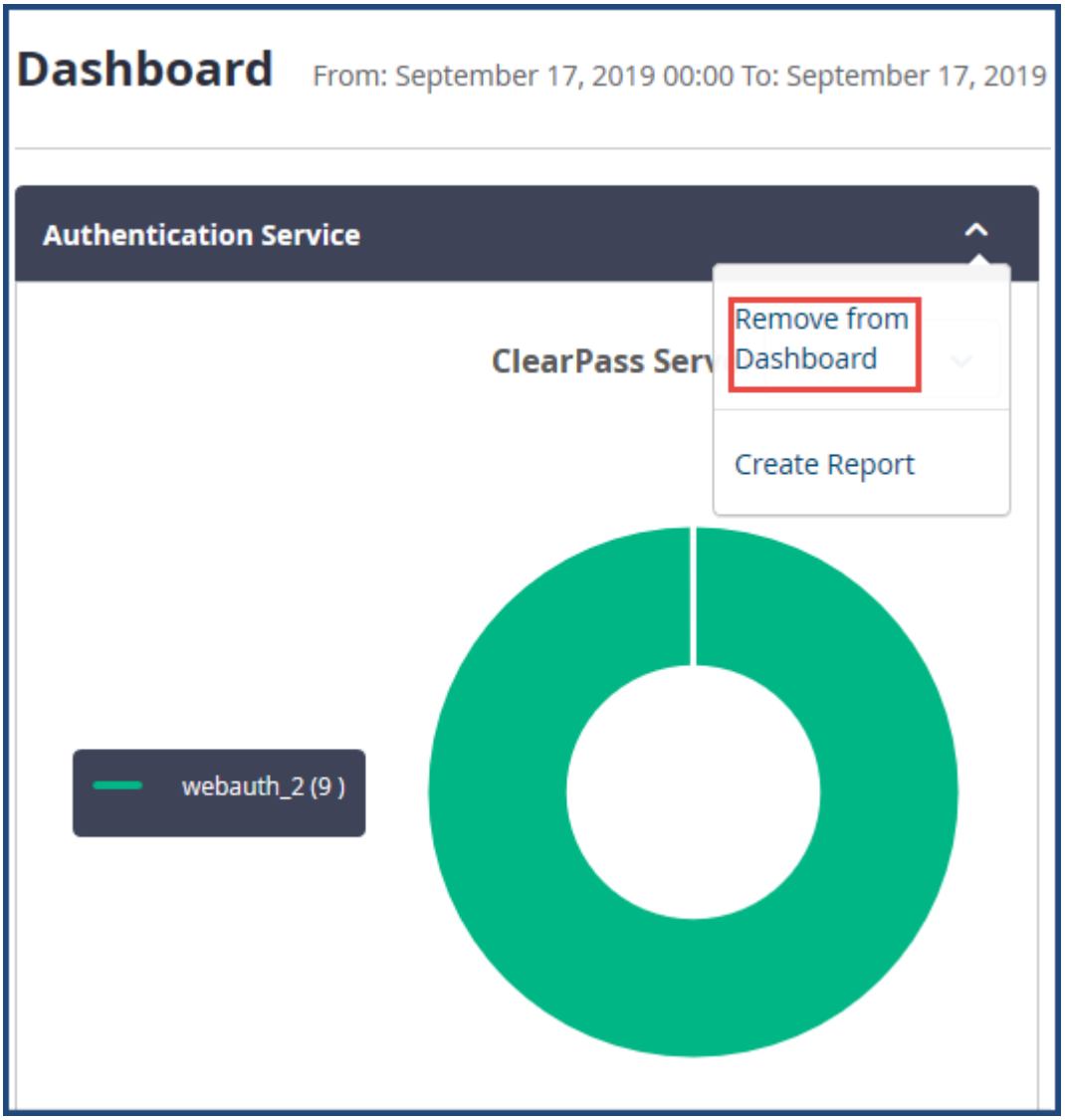
You can only remove a report widget from the **Dashboard Landing Page**. Report widgets cannot be deleted from Dashboard category pages (for example, if you choose to remove the Top 10 MAC Address Authentications widget from the Landing page, it will remain in the set of report widgets provided in the Authentication Dashboard).

To remove a report widget from the Dashboard Landing page:

Procedure

1. From the **Dashboard Landing Page**, locate the widget you want to remove.
2. Click the arrow icon in the right corner of the widget title bar.
3. From the menu, select **Remove from Dashboard** (see Figure 4).

Figure 1. Removing a Widget From the Dashboard



When you refresh the page, that widget will disappear from the Dashboard.

Creating a Report or Alert From the Dashboard

The widgets on the **Dashboard** include links to the **Create Reports** and **Create Alerts** pages.

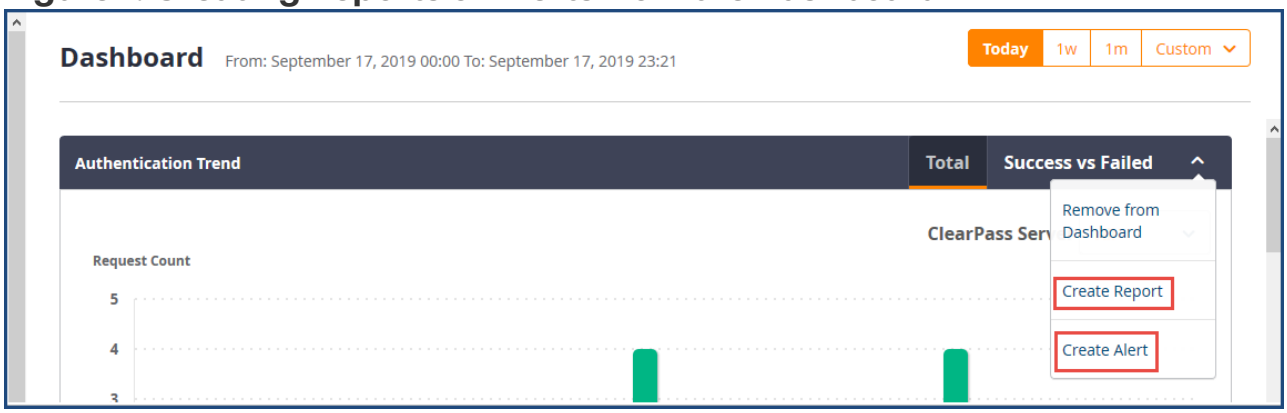
To define and to receive a regular report of data for that Dashboard:

- To open the **Create Reports** wizard from the Dashboard, click the down-arrow icon in the widget title bar and select **Create Report**.

  To define and to receive alerts when customized thresholds are reached:

- To open the **Create Alerts** wizard from the Dashboard, click the down-arrow icon in the widget title bar and select **Create Alert**.

Figure 1. Creating Reports or Alerts from the Dashboard



For detailed procedures to create reports and alerts, see Creating Reports and Creating Alerts.

Specifying the Date Range for Data Collection

About this task

By default, the Insight widgets, including those on the **Dashboard** page as well as all the other Insight widgets, such as **Endpoints**, **Guest**, **Posture**, and so on, display information collected over the previous seven days. The **System Monitor** widget is an exception as it displays data for the previous two hours. You can modify the Authentication, Endpoints, Guest, Posture, and System widgets to display widget data for today, one week, one month, or a custom date and time range.

To specify the date range to have data collected for a Dashboard widget:

Procedure

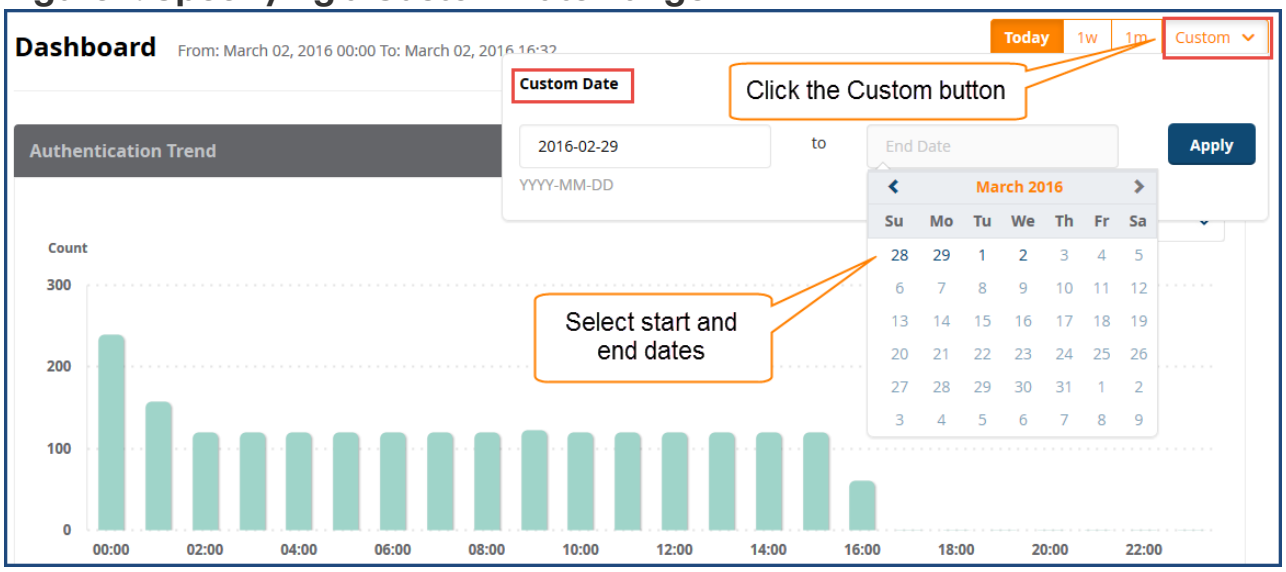
1. To specify data collection for today, one week, or one month, from the upper right corner of the Dashboard, select **Today**, **1w** (for one week), or **1m** (for one month) as desired.

   The Dashboard widgets then display the information for the specified number of days.

2. To specify a customized period for Insight data collection, click the **Custom** button.

   You are prompted to specify the start and end dates for the date range, as shown in the following figure:

Figure 1. Specifying a Custom Date Range



3. Select the **Start Date** and **End Date** from the calendar, then click **Apply**.

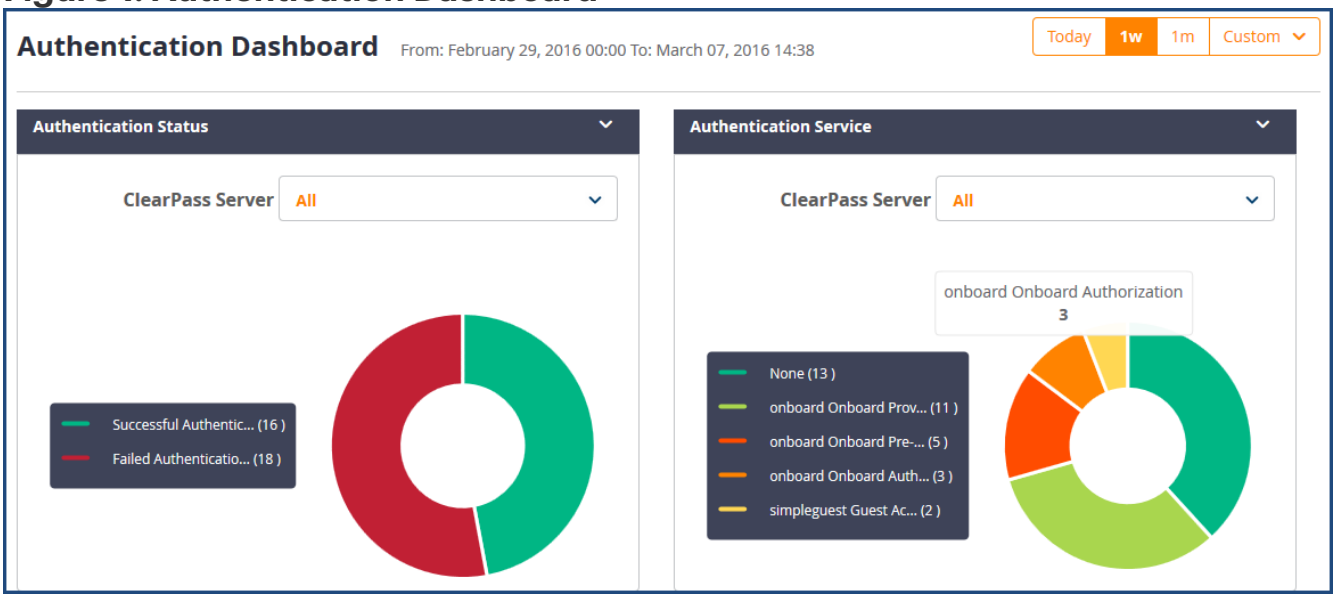
   The Dashboard widgets then display the information for the specified range of dates.

Authentication Dashboard

Authentication Dashboard widgets focus on authentication analytics and include widgets on trends, distribution, status, service, alerts, and statistics.

To access the Authentication Dashboard, navigate to **Dashboard > Authentication**.

Figure 1. Authentication Dashboard



The following widgets are included by default on the **Authentication Dashboard:**

- Authentication Trend
- Authentication Distribution
- TACACS Statistics for Authentication and Authorization
- Authentication Service
- Authentication Status
- Top 10 MAC Address Authentications
- Top 20 NAD Authentications
- Top 10 Authentication Errors
- Latest 10 Authentication Alerts

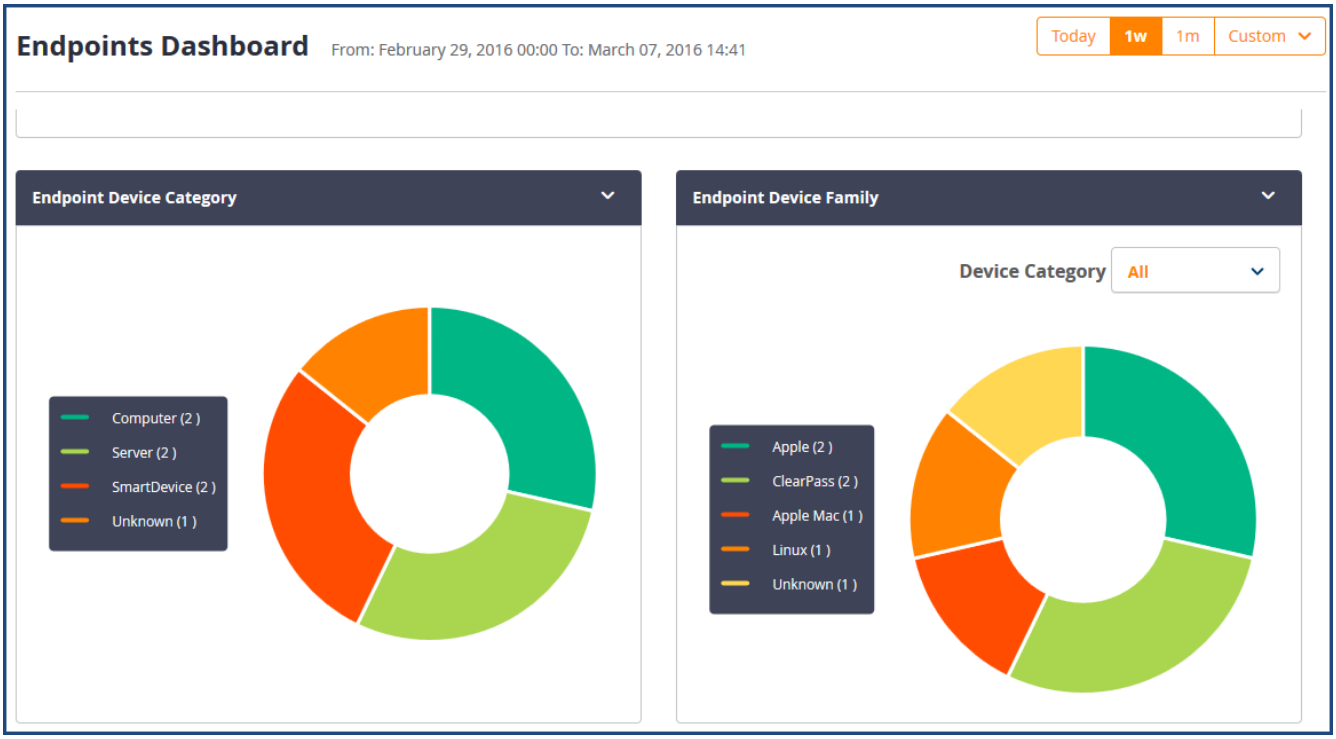
  For more information about the Authentication reports and the widgets provided for each report, see Authentication Category Reports.

Endpoints Dashboard

The Endpoints Dashboard widgets provide analytics that focus on Endpoint trends, distribution, device profile, and bandwidth usage.

To access the Authentication Dashboard, navigate to **Dashboard > Endpoints**.

Figure 1. Endpoints Dashboard



The following widgets are included by default on the **Endpoints Dashboard:**

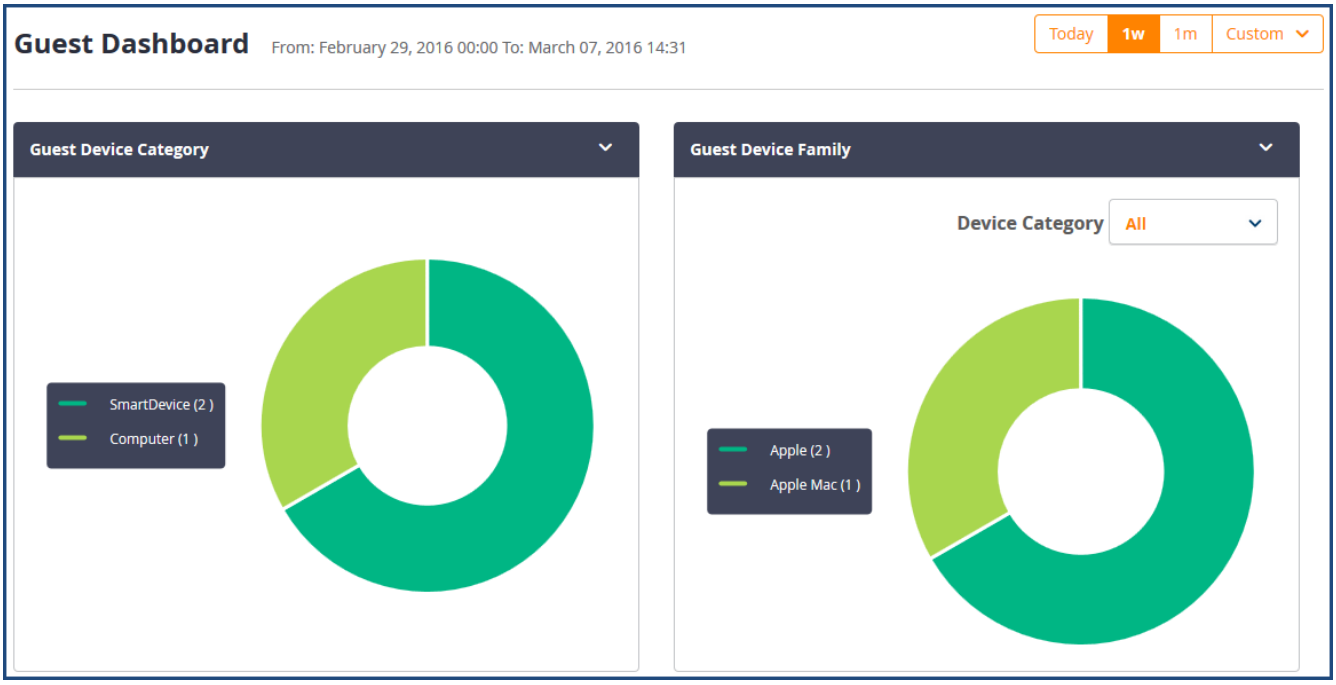
- Endpoint Unique Trend
- Endpoint Device Profiling
- Endpoint Device Categories
- Endpoint Device Families
- Endpoint Device Names
- Top 20 Bandwidth Users
- Endpoint Randomized MACs

  For more information about the Endpoints reports and the widgets provided for each report, see Endpoint Category Reports.

Guest Dashboard

To access the Guest Dashboard, navigate to **Dashboard > Guest**.

Figure 1. Guest Dashboard



The following widgets are included by default on the **Guest Dashboard**:

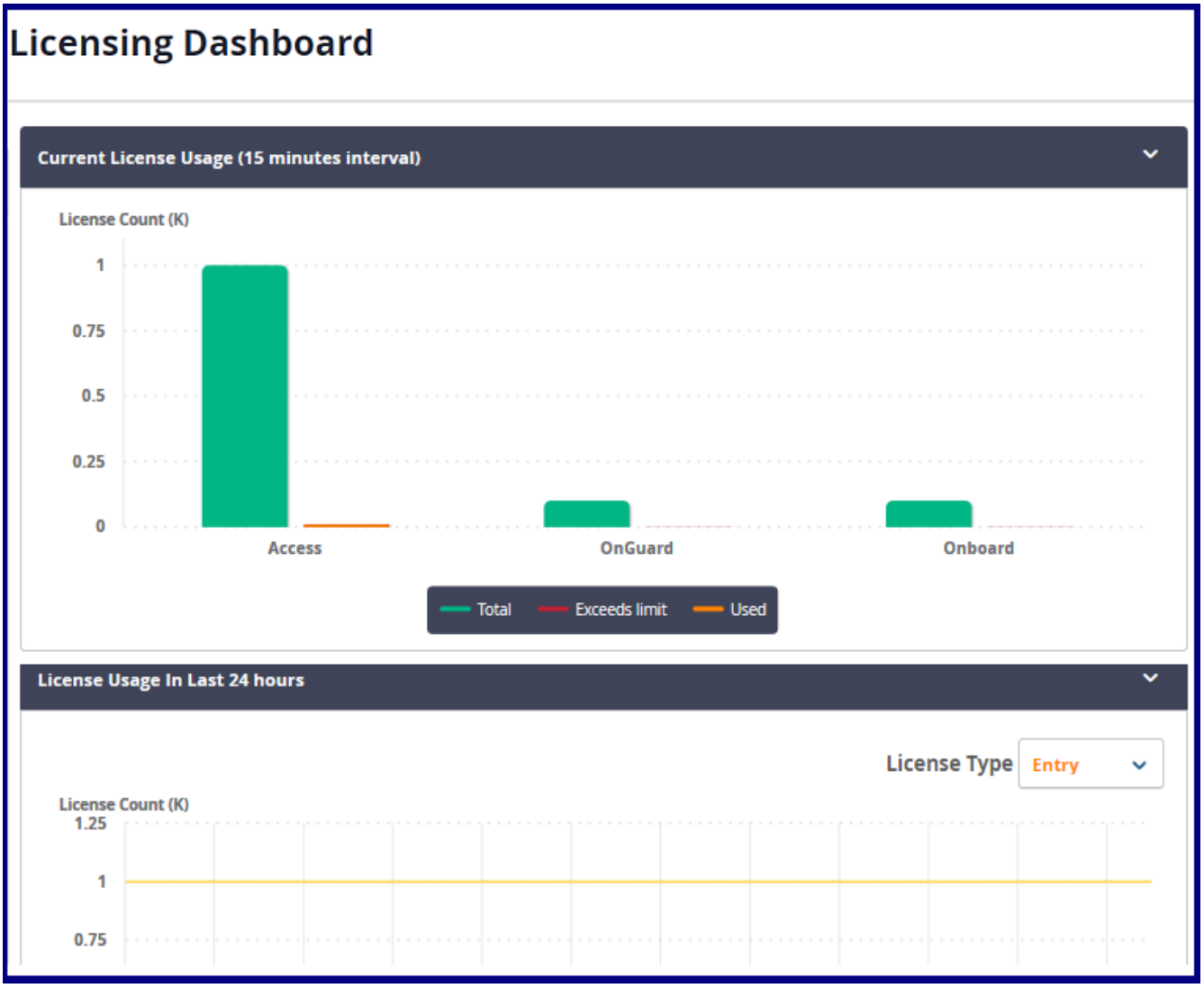
- Guests Authentication Trend
- Unique Guest Authentication
- Guests Provisioned
- Guest Device Category
- Guest Device Families
- Guest Device Names
- Top 20 Bandwidth Guest Users

  For more information about the Guest reports and the widgets provided for each report, see Guest Authentication Category Reports.

Licensing Dashboard

To access the **Licensing** dashboard, navigate to **Dashboard > Licensing**.

Figure 1. Licensing Dashboard



The following widgets are included by default on the **Guest Licensing** form:

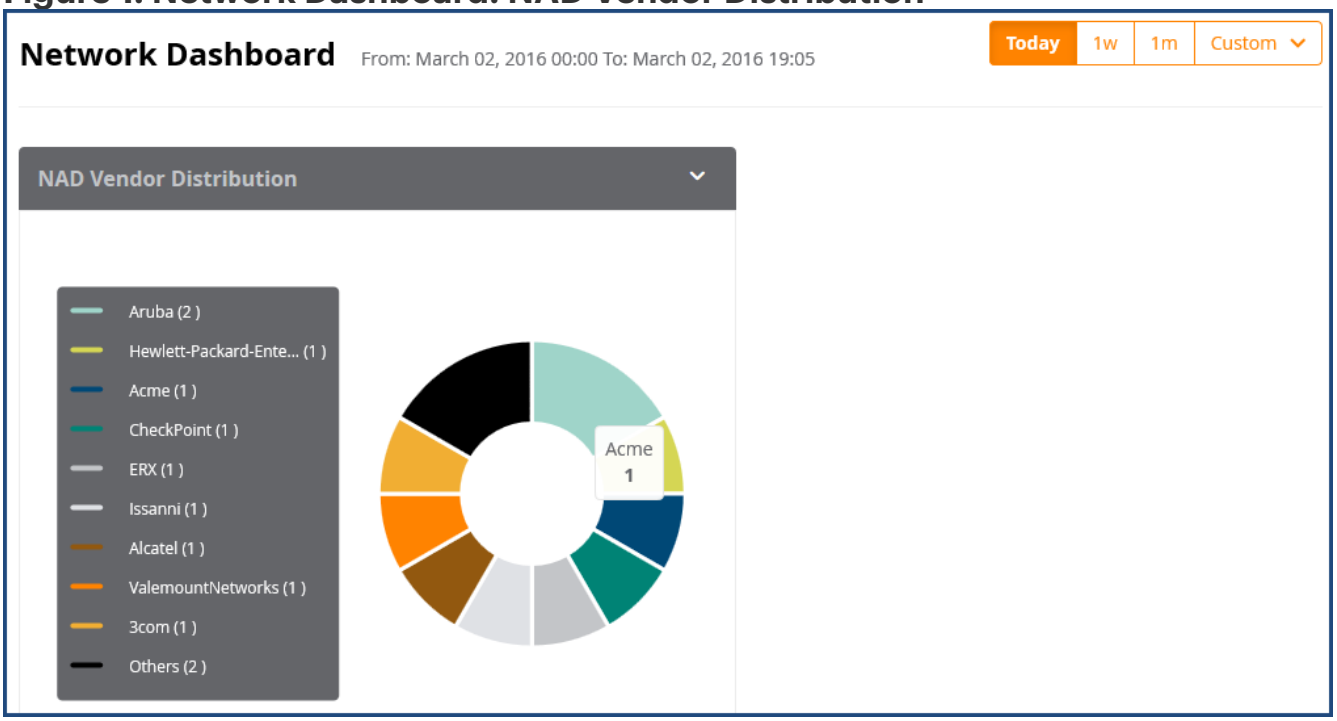
- Current License Usage
- License Usage in the Last 24 Hours
- Maximum License Usage

  This page displays the current license usage for OnGuard, Onboard, and Entry or Access licenses over the previous 15 minutes and previous 24 hour interval, and the maximum usage for each of these license types in the previous day, week or month. For more information, see Licensing Reports.

Network Dashboard

To access the Network Dashboard, navigate to **Dashboard > Network**.

Figure 1. Network Dashboard: NAD Vendor Distribution



The following widget is included on the **Network Dashboard**:

- NAD Vendor Distribution

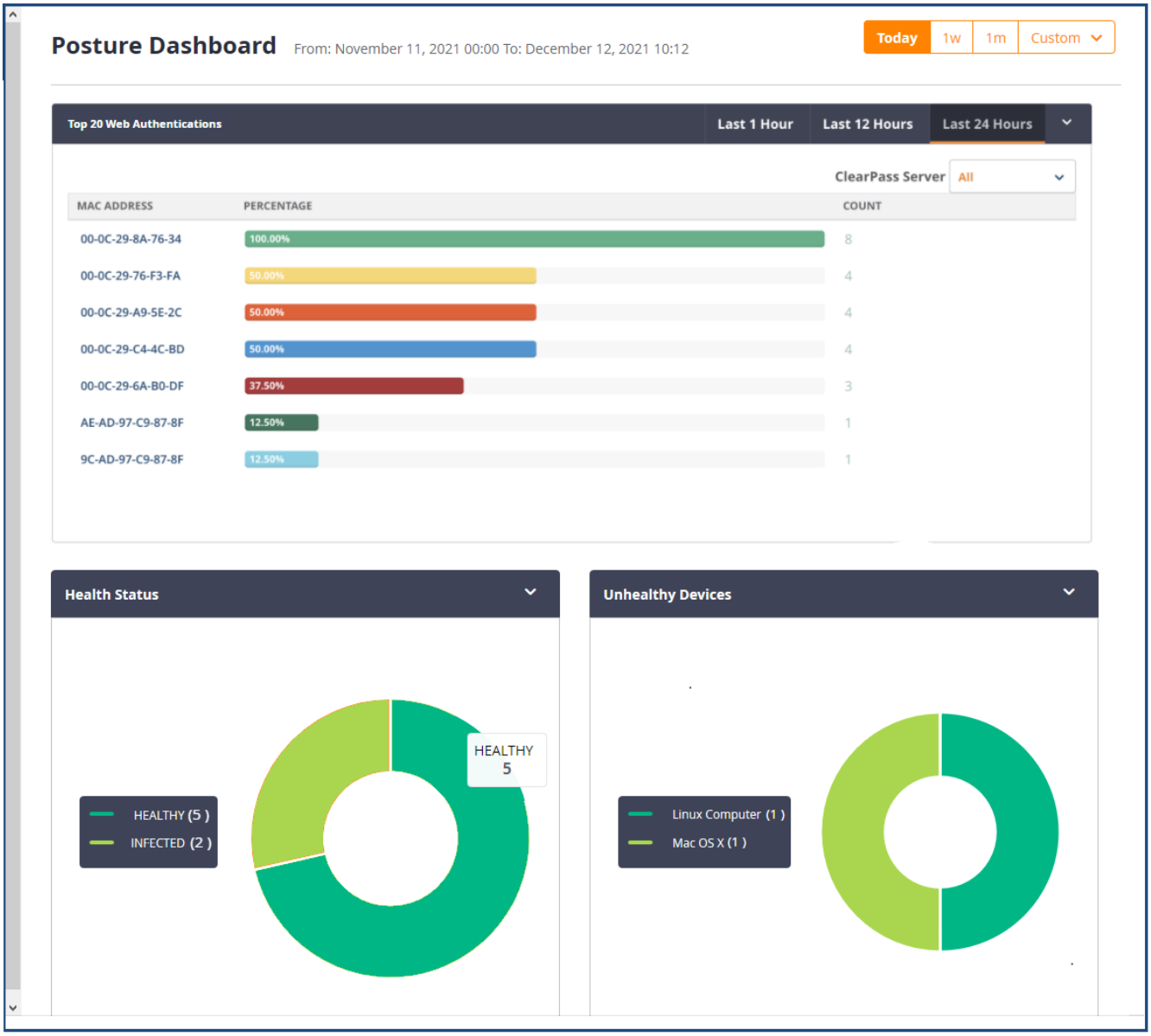
  This widget displays the list of all the NAD (Network Access Device) vendors, including the number of NADs by each vendor. Each vendor is associated with a unique color, and those colors are reflected in the circle graph that displays the distribution percentage each NAD vendor represents.

  For more information about the Network reports, see Network Category Reports.

Posture Dashboard

The Posture Dashboard widgets focus on device health status and device profiles. To access the Posture Dashboard, navigate to **Dashboard > Posture**.

Figure 1. Posture Dashboard



The following widgets are included by default on the **Posture Dashboard**:

- Top 20 Web Authentications

  The number of WEBAUTH health requests from each of the twenty MAC addresses with the most posture requests over the selected time period, and the percentage of requests for each MAC address that succeeded. This widget identifies devices sending frequent health requests as compared with the Onguard Polling time interval, and helps isolate issues where a control channel on the network is blocked.

- Health Status

  The numbers of devices identified as infected or healthy over the selected time period.

- Unhealthy Devices

The operating systems of devices identified as unhealthy during the selected time period.

For more information about the Posture-related reports, see OnGuard Category Reports.

System Monitor Dashboard

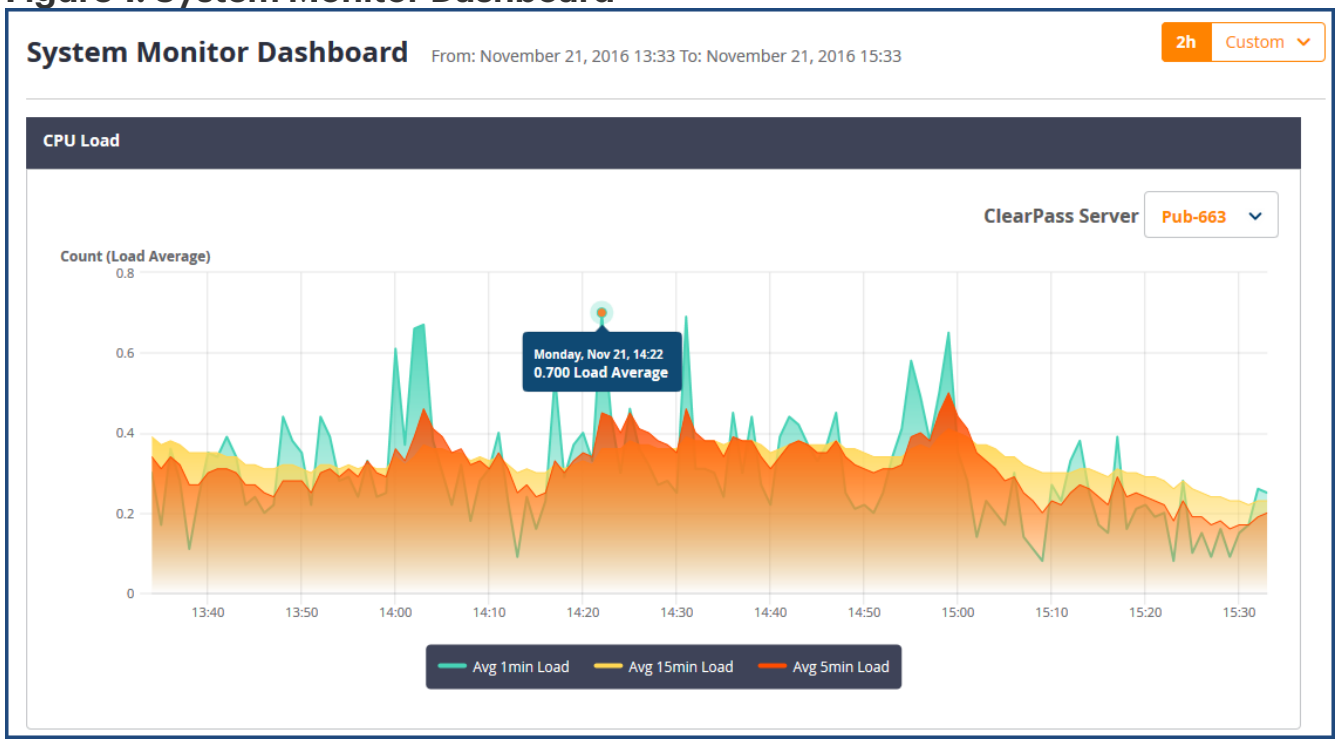
About this task

The System Monitor Dashboard widgets focus on health, including Authentication health, processing time, and CPU, memory, and disk usage.

> **NOTE**
> You cannot pin System Monitor widgets to the Dashboard.

To access the System Monitor Dashboard, navigate to **Dashboard > System Monitor**.

Figure 1. System Monitor Dashboard



The following widgets are included by default on the **System Monitor Dashboard**:

- Authentication Health

  The Authentication Health data graph can show data from any 24-hour period in the past 90 days.

- End-to-End Request Processing Time
- Memory Usage
- Swap Memory Usage
- Disk Usage
- CPU Usage
- CPU Load
- Top 10 Restarted Services
- Insight Disk usage

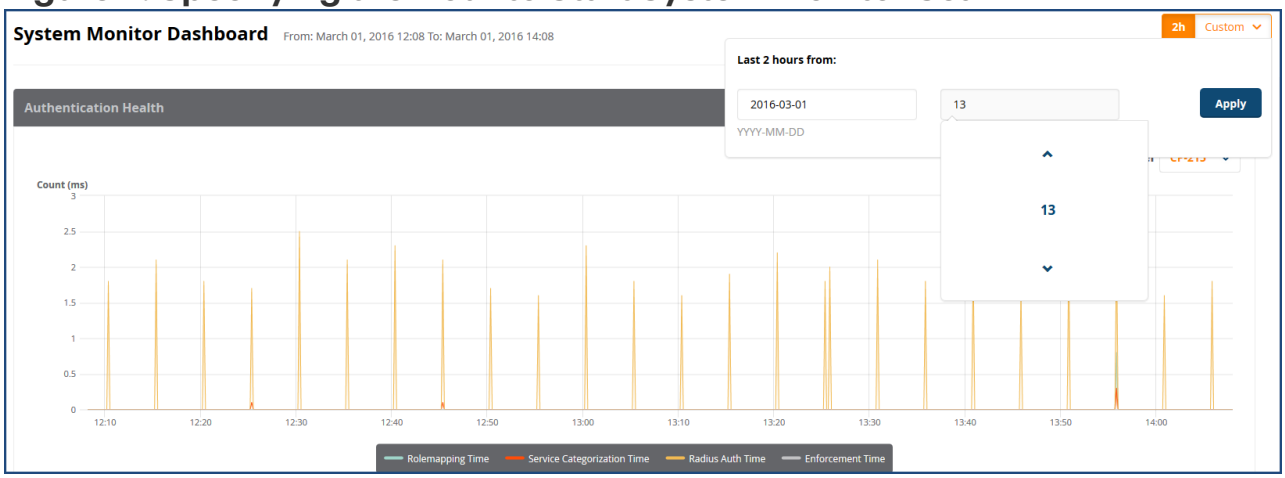
  The **System Monitor Dashboard** differs from the other Dashboard pages in that it can show data for two hours only (**2h**).

To define a custom two-hour time slot:

Procedure

1. Click the **Custom** drop-down list.

Figure 2. Specifying the Hour to Start System Monitor Scan



2. Select the starting date.
3. Click the **HH** field, then use the up- and down-arrows to specify the hour to start the system monitor scan.

   For example, specifying **13** in the **HH** field indicates that the start time for the two-hour period is **1:00 PM**.

4. Click **Apply**.

Searching the Insight Database

Subtopics

About Insight Search
Search Example

About Insight Search

Use the **Insight Search** feature to query the Insight database.

You can search for the following entities:

- Endpoint IP address (Framed-IP-Address)
- Clients by MAC address, hostname, or IP address
- User name
- Policy Manager servers by name or IP address
- Network access devices by name or IP address

  You can add clients and users to the Watchlist from Search results. For details, see Adding or Removing Users from the Watchlist . The Insight Search window is always available at the top of every page.

> **NOTE**
>
> Search works on all pages except the Report Configuration and Alert Configuration pages.

Figure 1. Search Window

Search Example

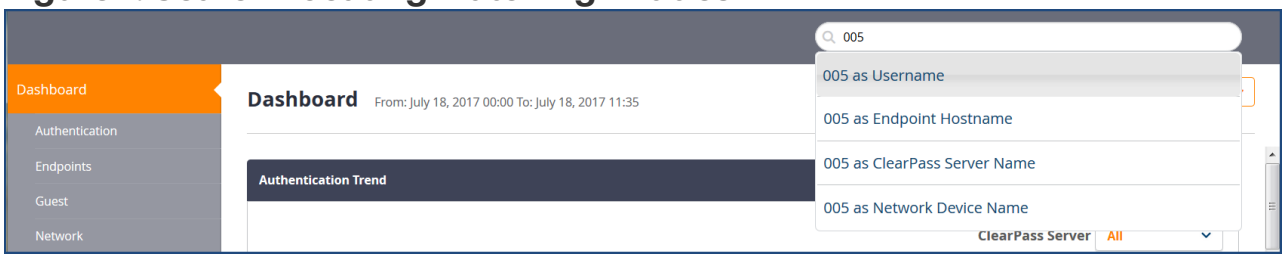
About this task

Let's take the example of searching for an endpoint hostname:

Procedure

1. Start entering the endpoint hostname into the **Search** window. As you type in the endpoint hostname, Insight Search may discover that there are multiple entities with the same initial characters (see Figure 2).

Figure 1. Search Locating Matching Entities



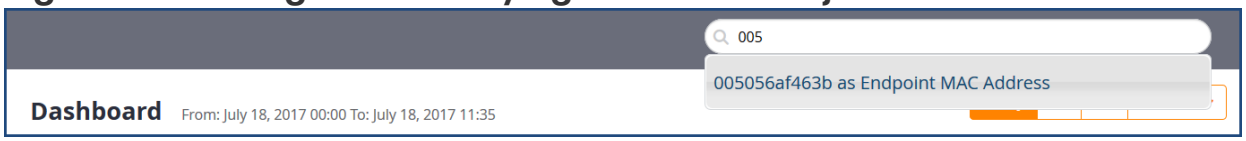
2. To locate the desired entity:

   a. Click on the suggestion and see which entity you are looking for from the list displayed.

   b. Or refine your search by typing more characters to further specify the search entity.

      In this example, the MAC address is identified as an Endpoint.

Figure 2. Locating and Identifying the Search Object



3. Select the search object. The endpoint hostname report is automatically displayed in an endpoint details page(see Figure 4). For more information about the data displayed in an endpoint details page, see Viewing Endpoints Details .

Figure 3. Report of Search Result

Creating Alerts

- Introduction
- Creating New Alerts
- Modifying the User Watchlist
- Adding or Removing Users from the Watchlist

Subtopics

Introduction
Creating New Alerts
Modifying the User Watchlist
Adding or Removing Users from the Watchlist

Introduction

Alerts provide network managers with near-real-time messages about anomalous network activity. Such activity could consist of:

- Irregular authentication activity
- Irregular network device access activity
- Users attempting privileged commands on network devices
- Irregular activity on the Policy Manager servers

Reports and alerts include templates for easy configuration. These templates allow you to quickly configure and monitor network activity. In addition to email notifications, you can also send alerts to mobile devices via SMS, providing the capability to receive mission-critical information on the go.

> **NOTE**
>
> Any Error-level System Event/Event Viewer entries in Policy Manager servers are notified with a System Alert Notification.

Creating New Alerts

About this task

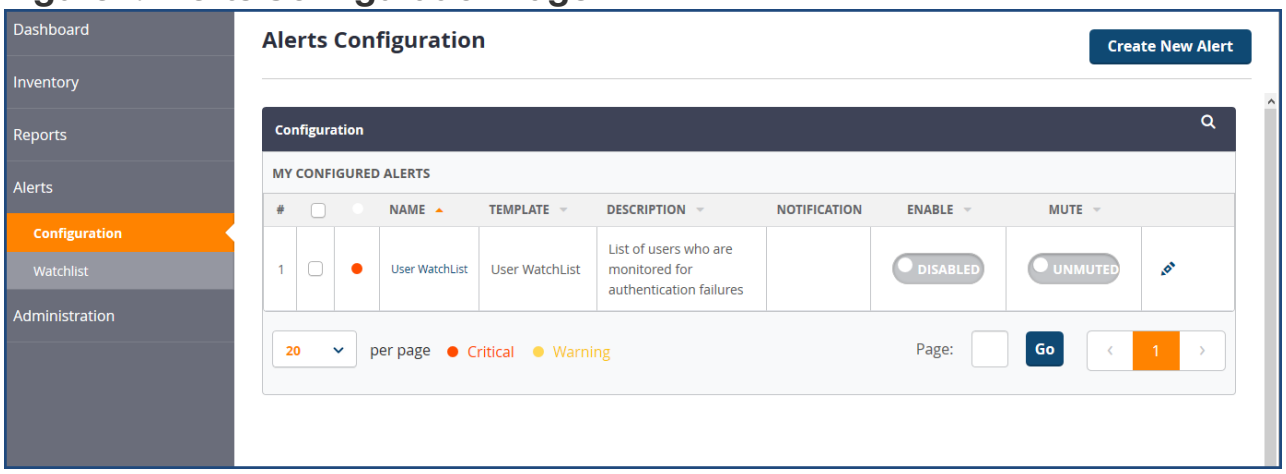
To create a new alert:

Procedure

1. Navigate to the **Alerts** page.
2. Click **Configuration**.

   The **Alerts Configuration** page opens.

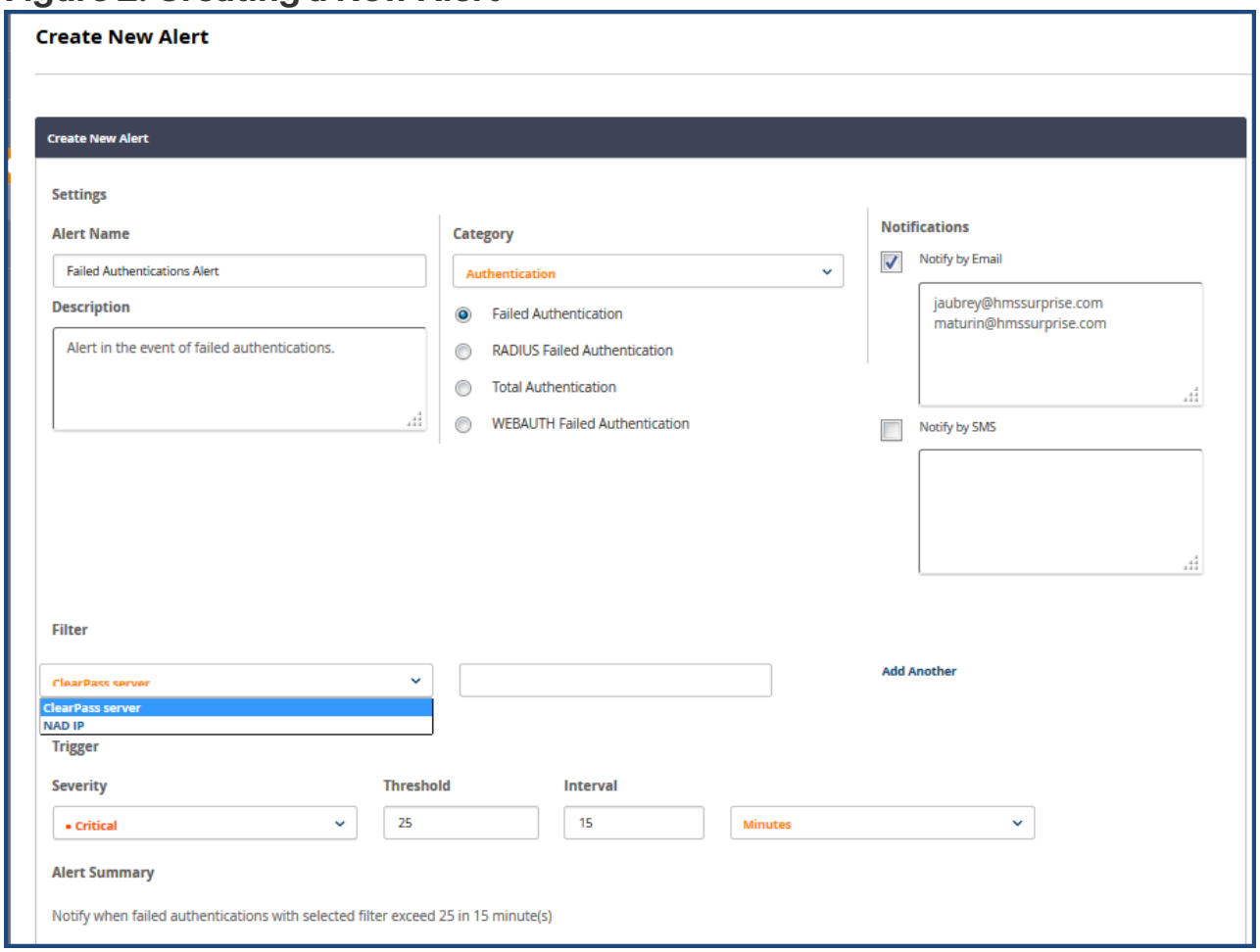
Figure 1. Alerts Configuration Page



- **Enable** button: From the switch, you can enable or disable the selected alert.
- **Mute** button: Allows you to mute alert output while you work to address the alert.

3. Click **Create New Alert**.

Figure 2. Creating a New Alert



4. Enter the information for each **Create New Alert** parameter as described in Table 1, then click **Save**.

Table 1. Create New Alert Parameters

Alert Field	Action/Description
Alert Name	Enter the name of the alert.
Description	Optionally, enter a summary description of the alert.

Alert Field	Action/Description
Category	Select the alert **Category**, then specify t he desired alert type in the selected cate gory: • Authentication ◦ Failed Authentication ◦ RADIUS Failed Authentication ◦ Total Authentication ◦ WEBAUTH Authentication • System • TACACS+ ◦ TACACS+ Commands ◦ TACACS+ Failures
Notifications	Specify report notifications: • **Notify by Email**. When you select this option, enter the list of email address es to be notified. The alert notification is sent whenever the trigger threshold is met. **NOTE** Enabling **Notify by Email** is mandatory. • **Notify by SMS**. When you select this option, enter the phone numbers of each recipient. The alert notification i s sent whenever the trigger threshold is met. **NOTE** If you have not configured t he SMTP mail server for ema il notifications, you will be u

Alert Field	Action/Description
	nable to configure Alert Noti fications. For the procedure, see Messaging Setup .
Filter	To configure an Alert filter: a. From the **Filter** drop - down, select the type of filter; for example, *Access Poin t* or *ClearPass Server*. b. Specify the operator: • Equals • Not_Equals • Starts_With • Contains • Not_Contains c. Enter the value. To create another filter, click the **Add Another** link. To remove a filter, click the trashcan lo go.
Trigger Severity	From the **Trigger Severity** drop - down, s elect one of the following: • *Critical* • *Warning*
Trigger Threshold	Specify **Threshold** and **Interval** values as criteria for determining whether an alert i s necessary. For example, if you specify the **threshold** as **25** and the **interval** as **15 minutes**, onc e the threshold of 25 is met within 15 min utes, an alert is triggered.

Alert Field	Action/Description
Trigger Interval	Specify the **Interval**, then select **Minutes** or **Hours**.
Alert Summary	When you have configured the alert setti ngs, the **Alert Summary** displays the setti ngs for your review.

Modifying the User Watchlist

A Watchlist is a list of VIPs, executives, and devices known to be problematic that are monitored for authentication failures. Policy Manager collects all user authentication status. When Policy Manager finds a user defined in the Watchlist that both fails to authenticate and also matches the Watchlist triggers (severity, threshold, and interval), an alert notification is sent to the notification list via email or to mobile devices via SMS. This allows the authentication failure to be resolved proactively before the problem is reported by the user. The Watchlist generates an alert only when an unsuccessful authentication for a specific device occurs.

Subtopics

Default Watchlist Trigger Settings

Default Watchlist Trigger Settings

About this task

The default Watchlist trigger settings are as follows:

- **Severity** = Critical
- **Threshold** = 1
- **Interval** = 30 seconds

> **NOTE**
> You cannot edit the The Watchlist trigger settings.

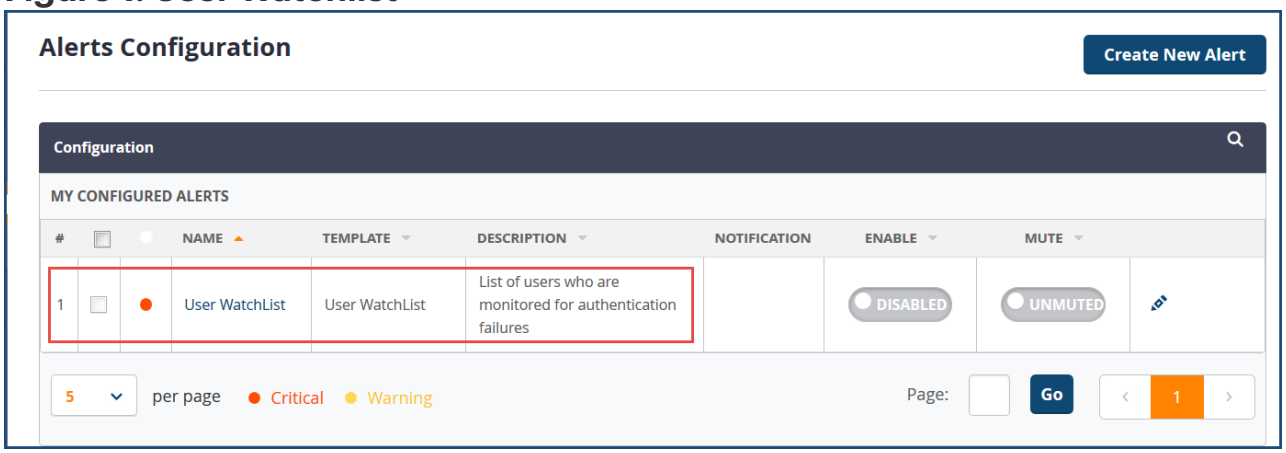
To modify the User Watchlist:

Procedure

1. From the Insight navigation panel, choose **Alerts**, then select **Configuration**.

   The Alerts Configuration is displayed, which shows the default **User Watchlist** (see Figure 3).

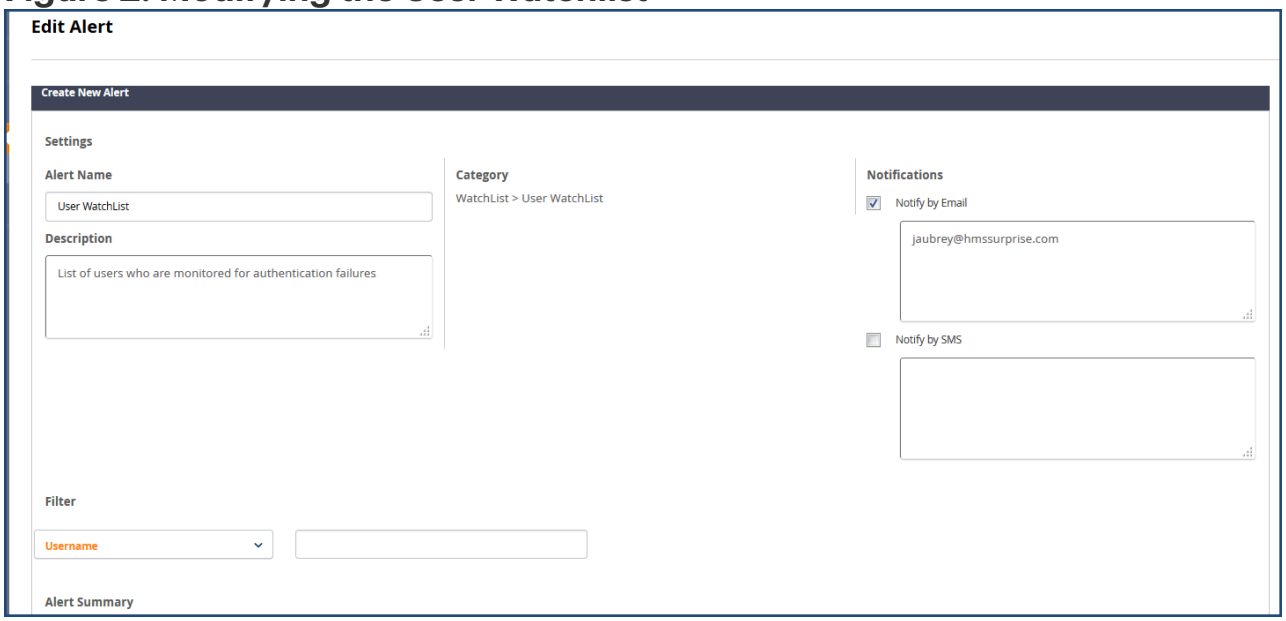
Figure 1. User Watchlist



2. Click **Watchlist**, then click **Modify Watchlist**.

   The **Edit Alert** page for the **User Watchlist** opens.

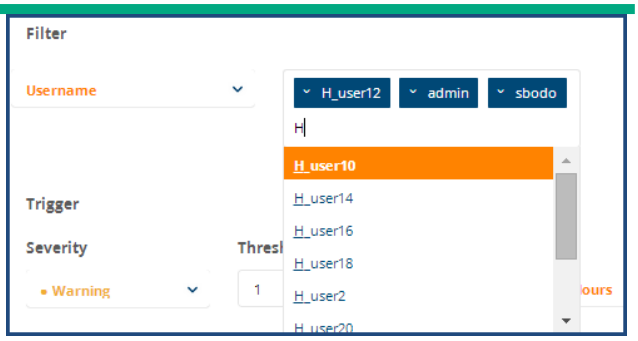
Figure 2. Modifying the User Watchlist



3. Enter the desired settings for each **User Watchlist** parameter as described in Table 2, then click **Save**.

Table 1. Modify User Watchlist Parameters

Alert Field	Action/Description
Alert Name	Optionally, you can modify the name of t he User Watchlist.
Description	Optionally (and recommended), enter a s ummary description of the User Watchlist .
Category	The **Category** is set to **Alert > User Watc hlist**. This is not an editable field.
Notifications	Specify Watchlist notifications. • **Notify by Email**. When you select this option, enter the list of email address es to be notified. The alert notification is sent whenever the threshold is met. • **Notify by SMS**. When you select this option, enter the phone numbers of each recipient. An SMS message is se nt with an alert notification whenever threshold is met. **NOTE** A warning message appears if you have not configured th e SMTP mail server for email notifications (for details, see Messaging Setup).
Filter: Username	The User Watchlist has only one filter: **Us ername**. From the Username drop - down, select one or more users to add to the Watchlist .

Alert Field	Action/Description
	
Alert Summary	When you have configured the Watchlist settings, the **Alert Summary** displays the settings for your review.

Adding or Removing Users from the Watchlist

You can use the Insight Search function to add users to or remove users from the Watchlist.

Subtopics

Adding a User to the Watchlist
Removing a User from the Watchlist

Adding a User to the Watchlist

About this task

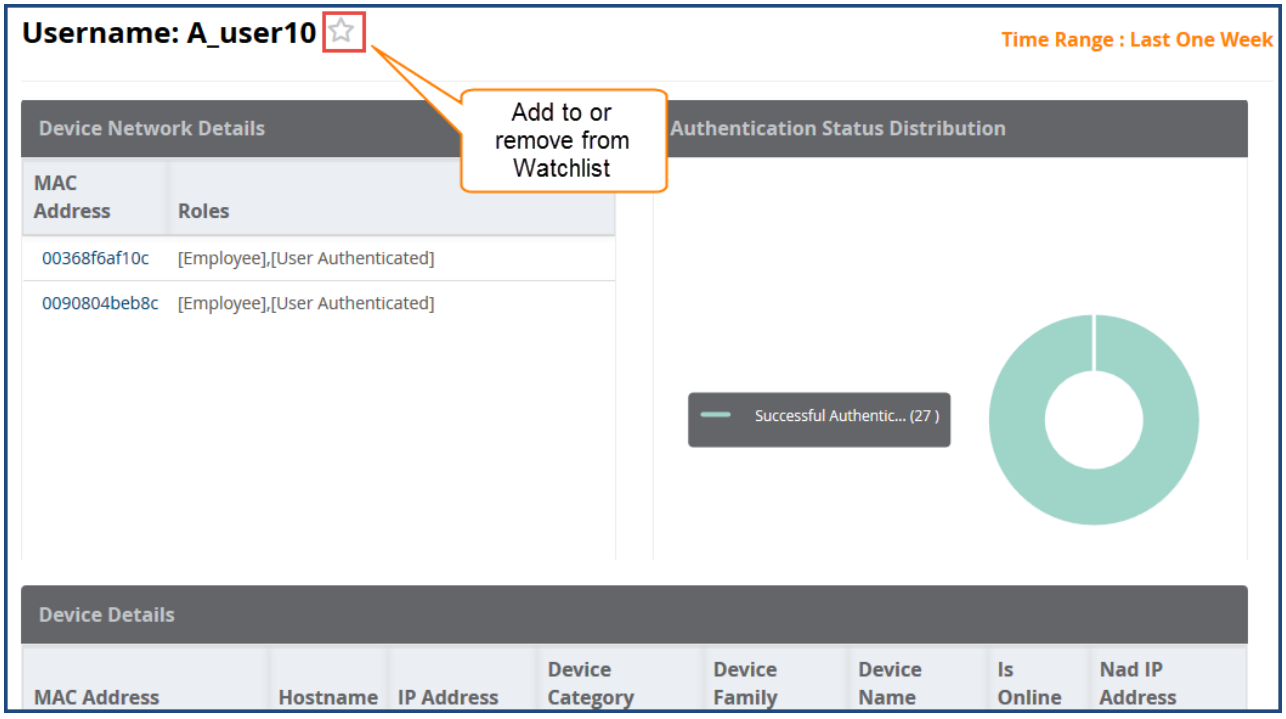
To add a user to the Watchlist:

Procedure

1. In the Insight Search window, enter the name of the user.

   The Insight **User Information** page for the selected user is displayed.

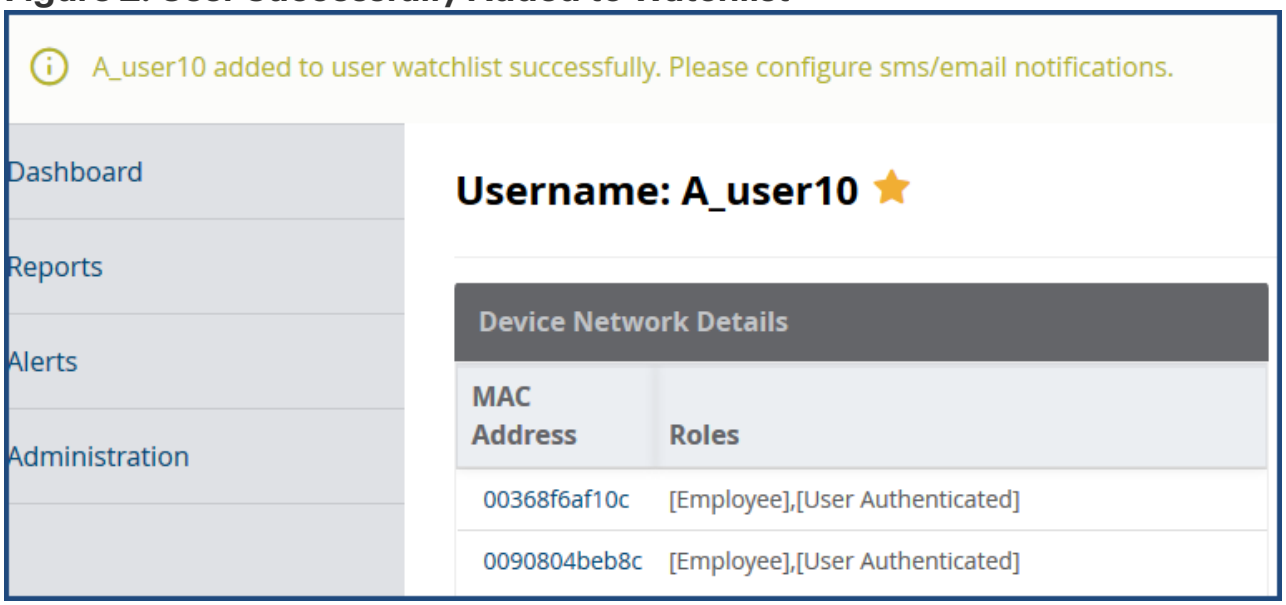
   Figure 1. Insight User Information Page



2. To add a user to the Watchlist, click the star icon next to the username as shown in Figure 5.

   The **User Information** page now displays the following information:

Figure 2. User Successfully Added to Watchlist



The star icon color is now set to orange, indicating the user has been added to the Watchlist.

The following message is displayed:

<User> added to User Watchlist successfully. Please configure SMS and email notifications.

Removing a User from the Watchlist

About this task

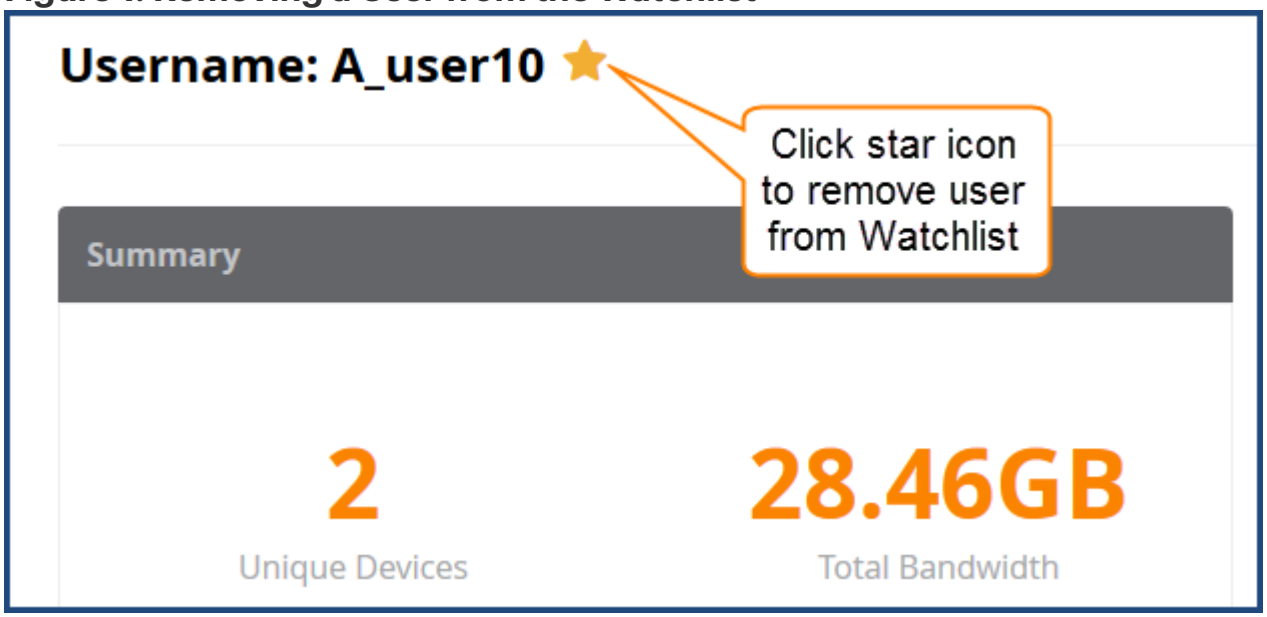
To remove a user from the Watchlist:

Procedure

1. In the Insight Search window, enter the name of the user.

   The Insight **User Information** page for the selected user opens.

Figure 1. Removing a User from the Watchlist



2. Click the orange star icon next to the username.

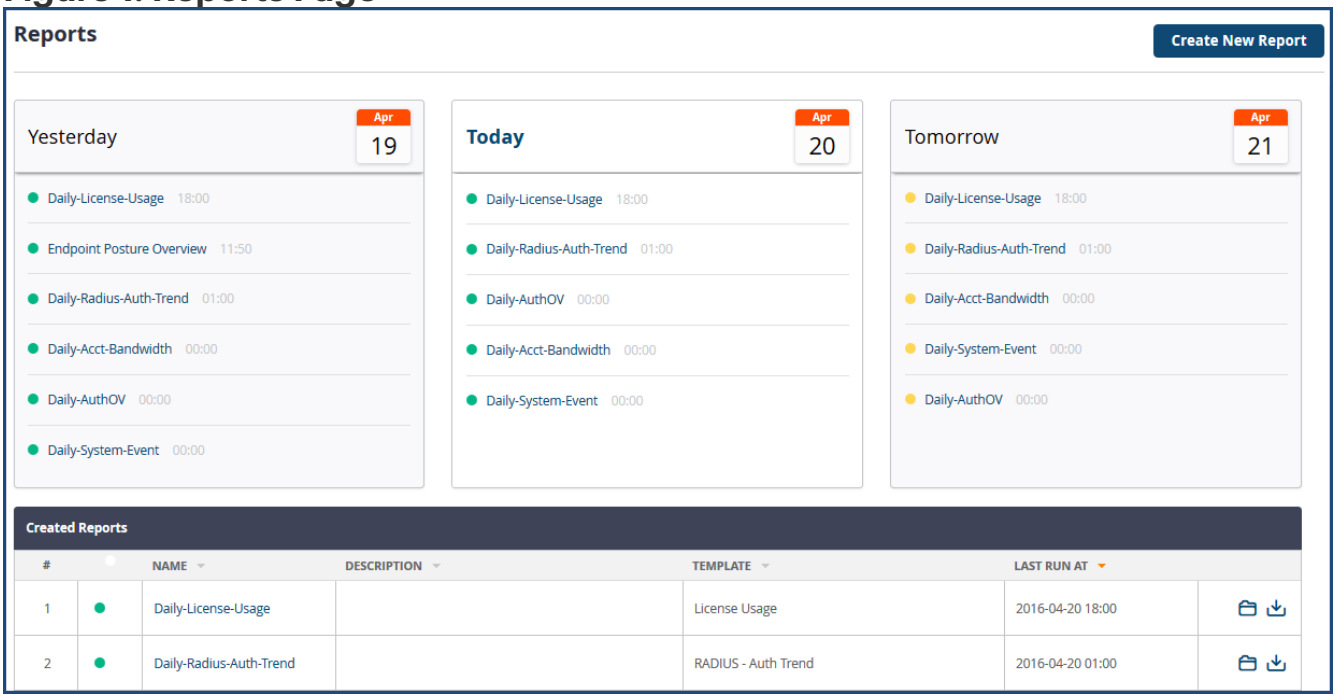
   The user is removed from the Watchlist. The star icon is now white. You receive the following message:

   <User> removed from User Watchlist successfully.

Creating Reports

The **Reports** page provides a method for creating reports with data filters and customized time ranges up to the previous two months.

Figure 1. Reports Page



Subtopics

Run Reports Now or on a Specified Schedule
Select Report Filters
PDF, CSV, and HTML Report Formats Are Available
Creating a New Report: Settings Configuration
Uploading a Custom Report Template
Configured Reports Page
Importing and Exporting Reports
Viewing Generated Reports
Regenerating Existing Reports

Run Reports Now or on a Specified Schedule

You can set up reports to run immediately or you can schedule a report to run on a daily, weekly, or monthly basis. Although Insight reports show data over the previous two-month period, Insight can retain data for up to two years.

Select Report Filters

Many reports allow you to select filters that include a simple AND condition. For example, you can use filters to create a report that displays data for RADIUS Authentications from the Active Directory AND the Guest User Repository source.

> **NOTE**
> For Compliance Reports in Insight, the Filters for Audit records are not allowed in the UI.

PDF, CSV, and HTML Report Formats Are Available

After a report is configured and run, the report is available for download in PDF and CSV formats. You can also open a report and view it in HTML format.

> **NOTE**
> Deleting a configured report deletes both the report configuration and all related report output.

Creating a New Report: Settings Configuration

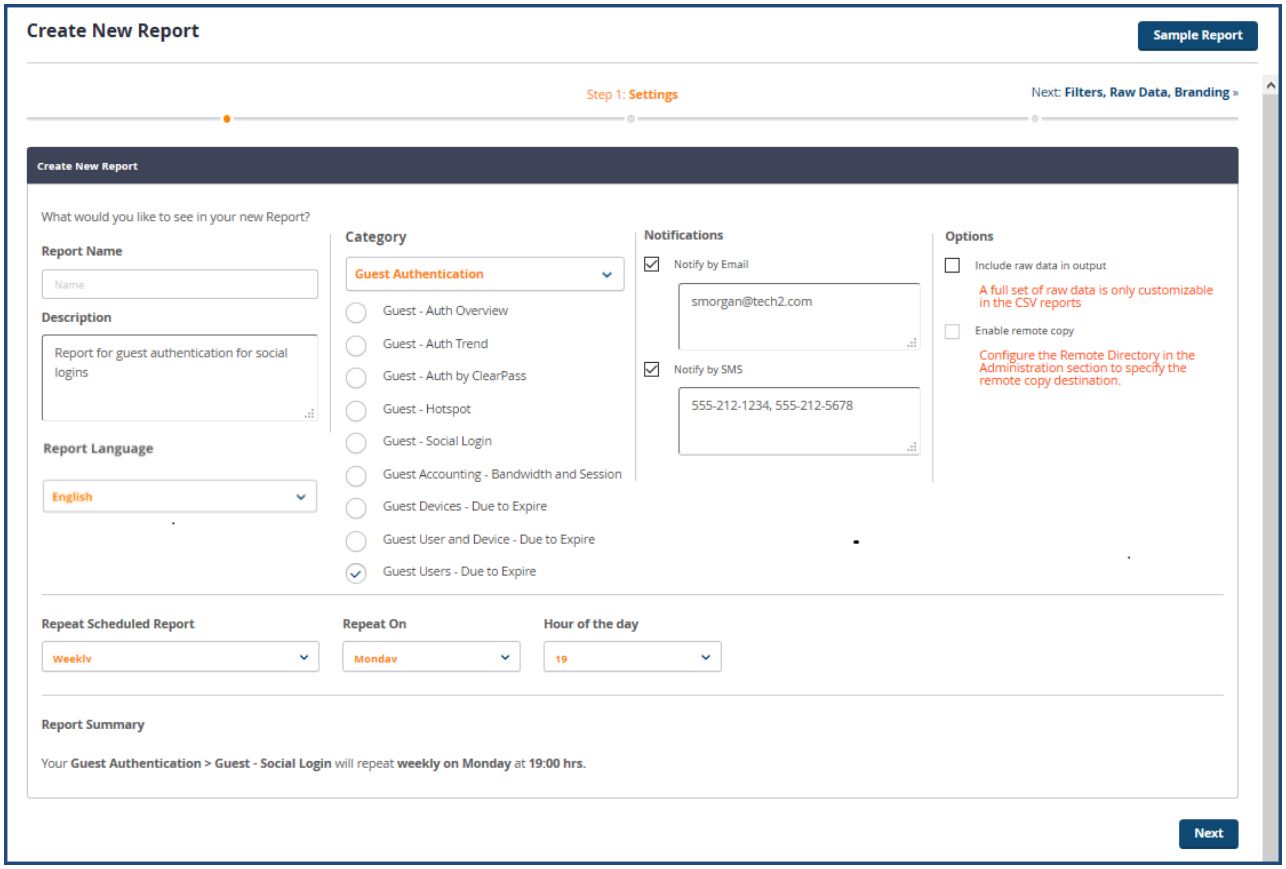
About this task

To create a new report:

Procedure

1. From the Insight navigation panel, click **Reports**.
2. Select **Create New Report**. The **Settings** page of the **Create New Report Wizard** opens.

Figure 1. Create New Report Wizard: Settings



3. Enter the appropriate information as described in Table 1, then click **Next**.

> **NOTE**
>
> For detailed information about what report types are provided for each report category, refer to Insight Report Categories .

Table 1. Specifying the Report Settings Parameters

Report Parameter	Action/Description
Report Name	Enter the name of the report.
Description	Optionally, enter a summary description of the report.
Report Language	Select the language in which the report i s displayed. Options are English (the defa ult), Korean, Chinese, and Japanese.

Report Parameter	Action/Description
Category	Select the report **Category**, then specify the desired report type in the selected ca tegory. • Authentication • Endpoint • Guest Authentication • Licensing • Network • OnGuard • Onboard • RADIUS Authentication • System • TACACS+
Notifications	Optionally, specify report notifications. T hese report notification options require t hat you first configure the SMTP mail se rver for email notifications. For details, se e Messaging Setup . • **Notify by Email**. When you select this option, enter the list of email address es (separated by commas) to be notifi ed. When Insight Reports are sent via email to notification recipients, the re cipients will receive an email with an HTML version of the report and a zip f ile containing the report in CSV and P DF formats. If the zip file is larger than 2 MB, the CSV file will not be included in the zip file, but the email will includ e a link to download that file. • **Notify by SMS**. When you select this option, enter the phone numbers of

Report Parameter	Action/Description
	each recipient (separated by commas).
Options	**NOTE** Before you can enable one o r both of these two options, you must configure the **File Transfer Settings** (including the Remote Directory) in th e **Administration** section. Fo r more information, see File Transfer Settings Configurati on . • **Include raw data in output** ◦ A full set of raw data is customizab le in the CSV reports only. • **Enable remote copy** ◦ This option lets you copy reports t o the location specified in the **Ad ministration > Remote Directory** s etting.
Repeat Scheduled Report	Specify whether you want to generate thi s report **Daily**, **Weekly**, or **Monthly**. The d efault is **No Repeat**. • To rerun a **No Repeat** report or a stati c report, edit and save the report. Insi ght will then automatically run the rep ort. • When you create a report with the **No Repeat** option selected, the report ru ns when you click **Save**. • When you create a periodic report (**D aily**, **Weekly**, or **Monthly**), the report i

Report Parameter	Action/Description
	s run according to the specified sche dule.
Preset Date Range	If you select the **No Repeat** option in the **Repeat Scheduled Report** field, you can choose one of the following **Preset Date Range** settings this report: • Custom Date: When you select **Custo m Date**, specify the **Start Date** and **Ti me** and the **End Date** and **Time**. • Today • Since Yesterday • This Week • Within Last Week • Within Last 2 Weeks • This Month • Within Last Month When you select one of these date range options (with the exception of **Custom D ate**), Insight automatically populates the **Start Date/Time** and **End Date/Time** setti ngs
Report Summary	When you have configured the report se ttings, the **Report Summary** displays the m for your review.

Subtopics

Report Filters Configuration
Specifying the Logo and Branding
Report Summary Page

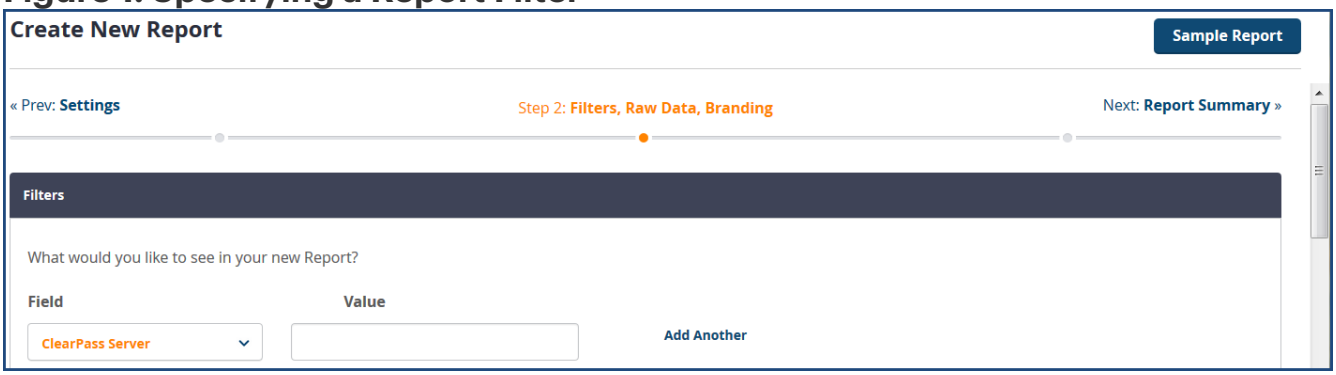
Report Filters Configuration

About this task

When you complete the **Settings** page in the **Create New Report** wizard and click **Next**, the page that opens allows you to configure the filters for your report. For maximum flexibility in defining your reports, the complete set of filters are available for each report.

Report filters apply the data fetched from the database, then Insight displays the result in the report. The filters that are available depend on the report category you specify. If you don't apply a filter, Insight includes all the data in the generated report that matches the report category.

Figure 1. Specifying a Report Filter



Procedure

Specify the report filter parameters as described in the following table:

Table 1. Specifying the Filters Parameters

Filter Parameter	Action/Description
Field	Click the **Field** drop - down list and select the p ish to use as a filter. The available parameters v o the report type.
Operator	Select the operator to determine how Insight fi in the **Value** field. Supported options are: • *Equals*: The field is an exact match with the • *Not_Equals*: The field does not exactly matc g • *Starts_with*: The field starts with the specifie • *Contains*: The field parameter contains the nywhere within the field parameter.

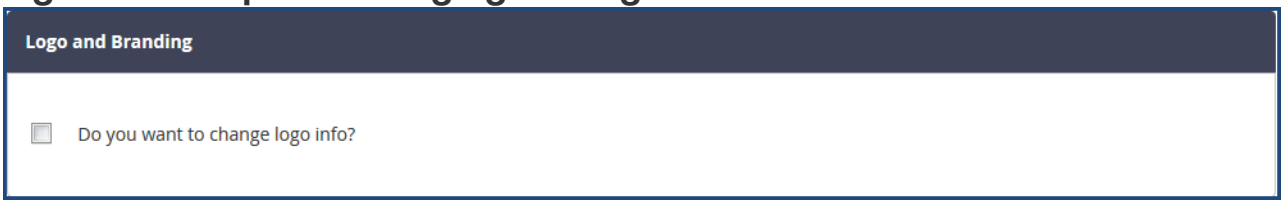
Filter Parameter	Action/Description
	• *Not_Contains*: The field parameter does not **ue** parameter
Value	Select the appropriate value.
Add Another	Click this link to add another filter.

Specifying the Logo and Branding

Procedure

1. When you complete the report filters configuration, scroll to the **Logo and Branding** section on the same page. The initial Logo and Branding screen presents a prompt, asking if you want to change the logo:

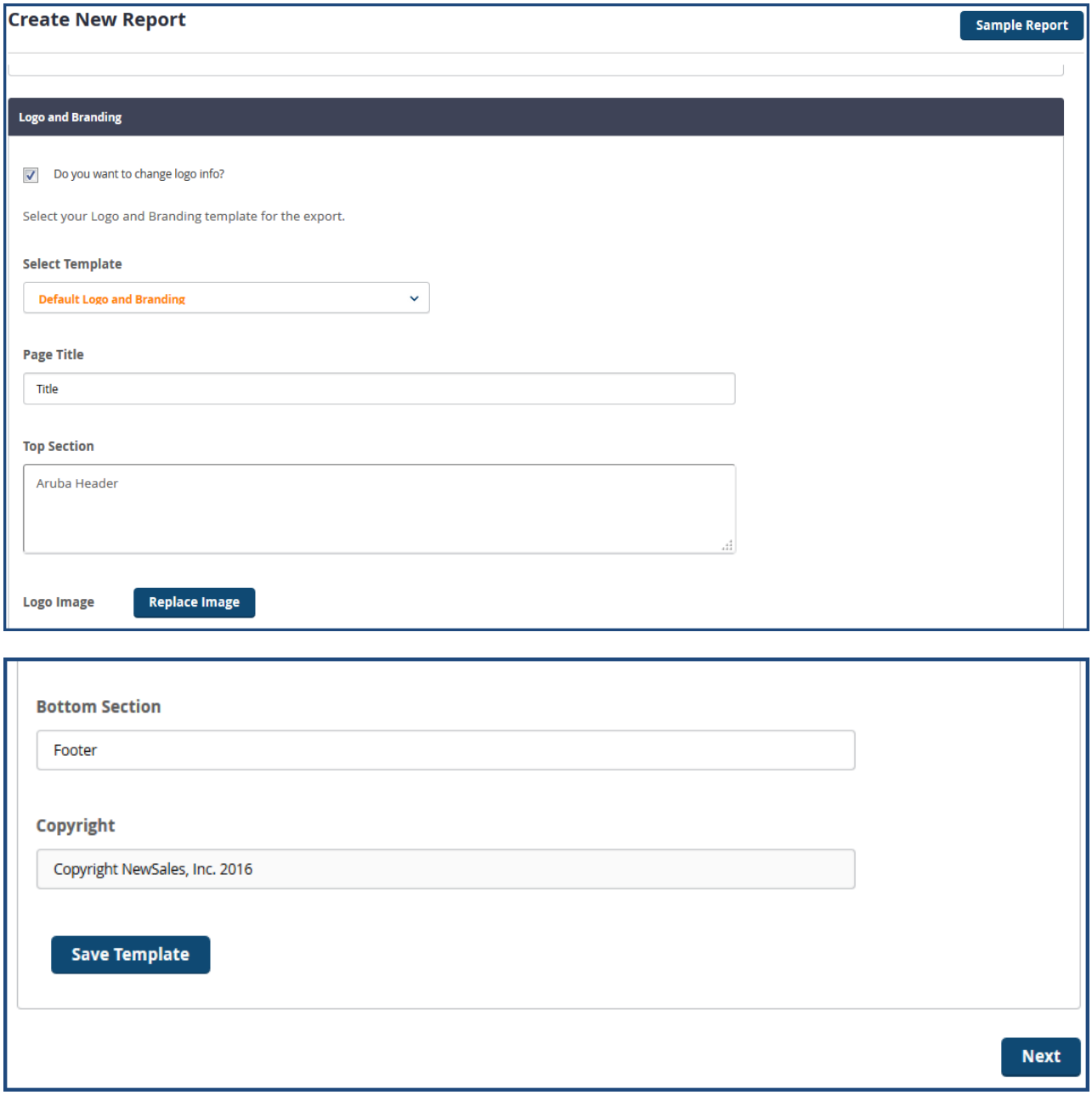
   Figure 1. Prompt for Changing the Logo

2. If you don't wish to change the logo, simply click **Next** to proceed.
3. If you do want to change the logo, click the check box.

   The **Logo and Branding** configuration section opens:

   Figure 2. Logo and Branding Section



4. To specify the logo and branding information, enter the information as described in Table 3, then click **Next**.

Table 1. Specifying Logo and Branding Parameters

Report Parameter	Action/Description
Select Template	From the drop - down, select the logo and
Page Title	Enter the page title.
Top Section	Enter the header for the top of the page.

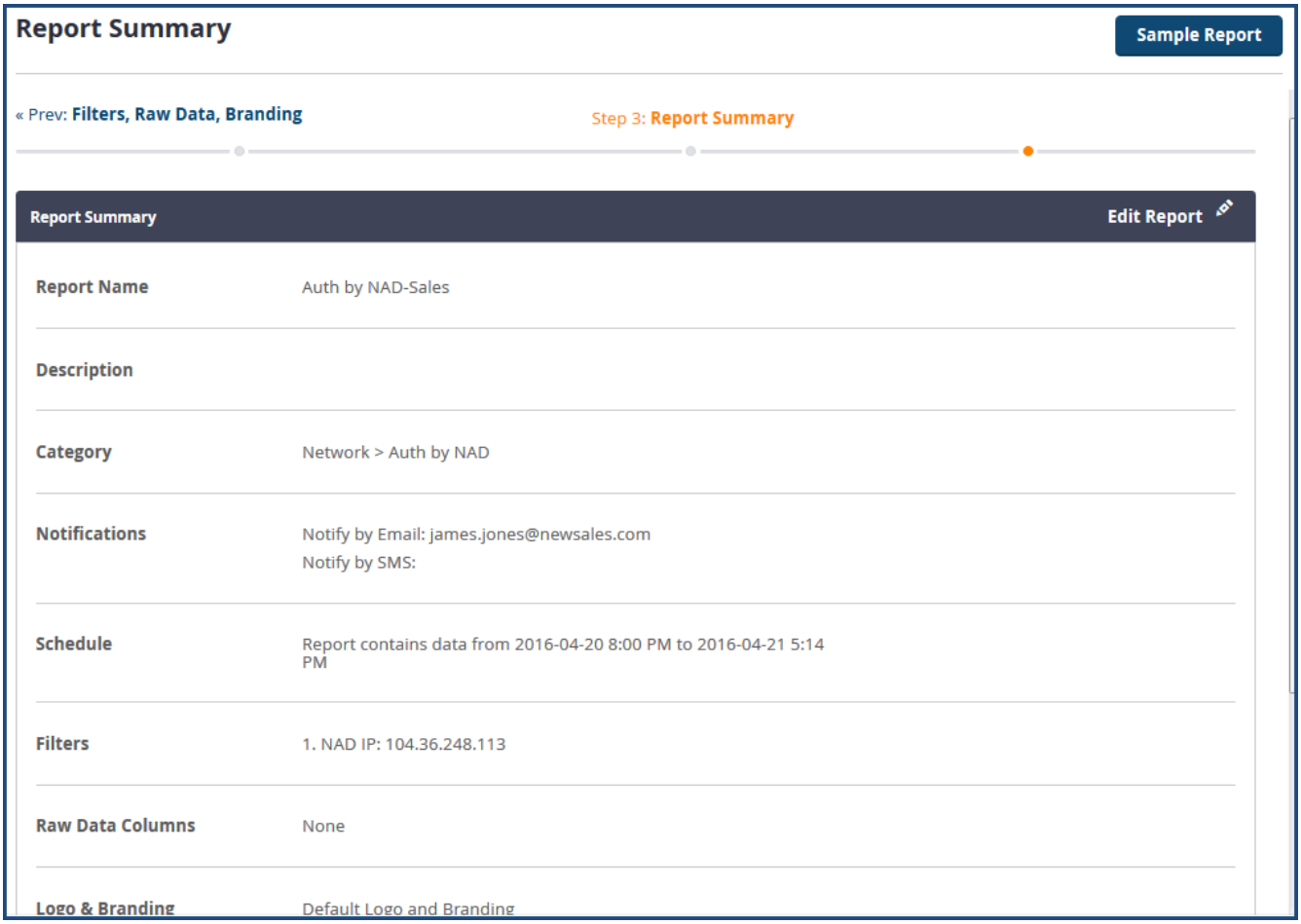
Report Parameter	Action/Description
Logo Image	To browse to the appropriate logo image,
Bottom Section	Enter the footer text.
Copyright	Enter the copyright information. For exam ewSales, Inc."
Save Template	To save the new branding and logo setting **e**.

Report Summary Page

About this task

When you complete the **Logo and Branding** section, the **Report Summary** is displayed.

Figure 1. Report Summary



Procedure

1. Review the report summary.
   a. If you wish to change any aspect of the report, click **Edit Report**. The **Report Summary** dialog opens. You can edit the current report settings as needed.
   b. Make any necessary changes, then click **Save**.
2. When the report settings are satisfactory, click **Save**. Insight generates the report. You return to the **Configured Reports** page.

Uploading a Custom Report Template

About this task

ou can upload a custom report template using the **Import** option on the **Reports > Custom Reports** page. To create a new report based on a custom report template:

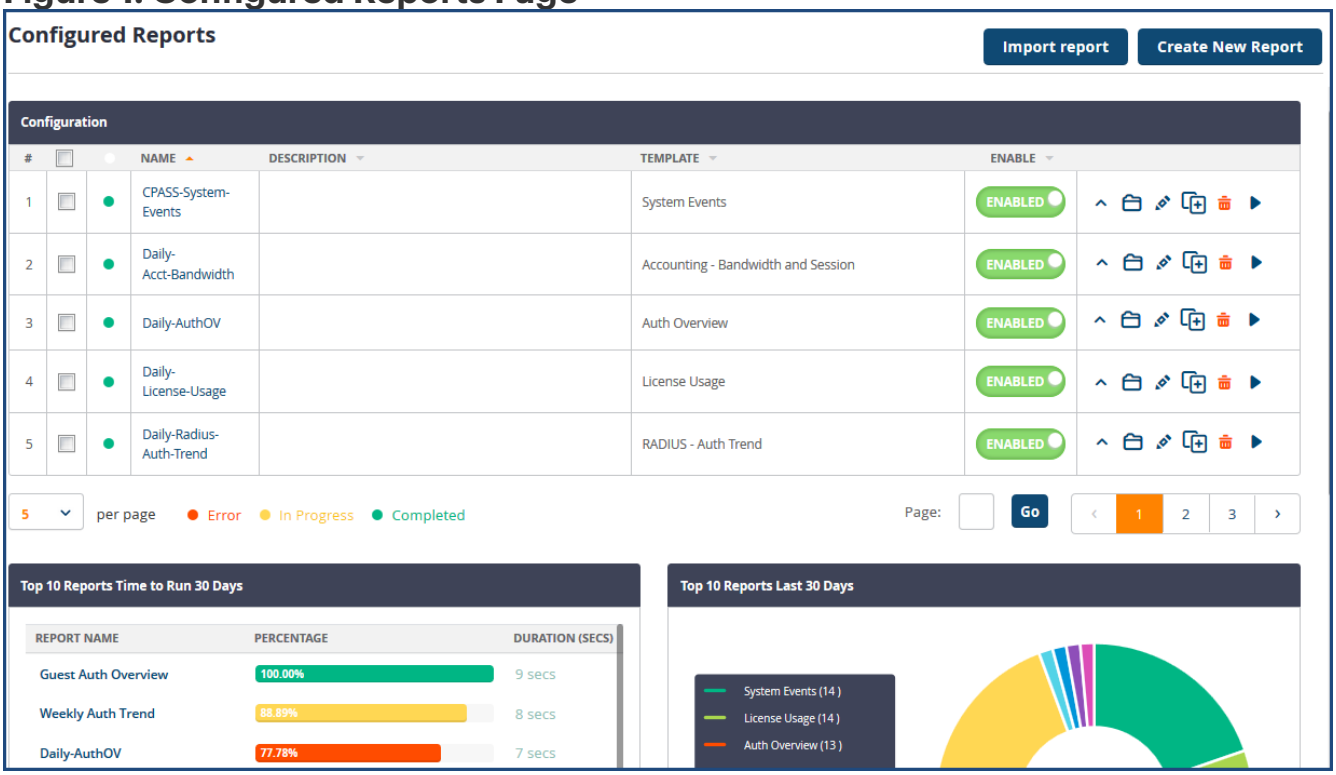
Procedure

1. Navigate to **Reports > Configuration.**
2. Select the **Custom** option in the **Category** drop-down list.

Configured Reports Page

To see the set of configured reports, select **Reports > Configuration**. From this view, you can edit or copy report settings, run or delete a configured report, and import or export reports between Policy Manager servers.

Figure 1. Configured Reports Page



A green dot next to a report name indicates that the report generation is complete, while a yellow dot indicates that Insight is still creating the report. A red dot by the report name indicates that the report failed to generate. To display a tooltip with additional information about why a report fail to generate, hover your mouse over red dot.

This page also provides two report widgets:

- **Top 10 Reports Time to Run 30 Days**

  This widget lists the ten reports that took the longest (in seconds) to run over the last 30 days.

- **Top 10 Reports Last 30 Days**

  This widget lists the ten most frequently run reports over the last 30 days.

Importing and Exporting Reports

About this task

Insight allows you to import or export individual reports, or import and export multiple report definitions at once.

To export a single report definition:

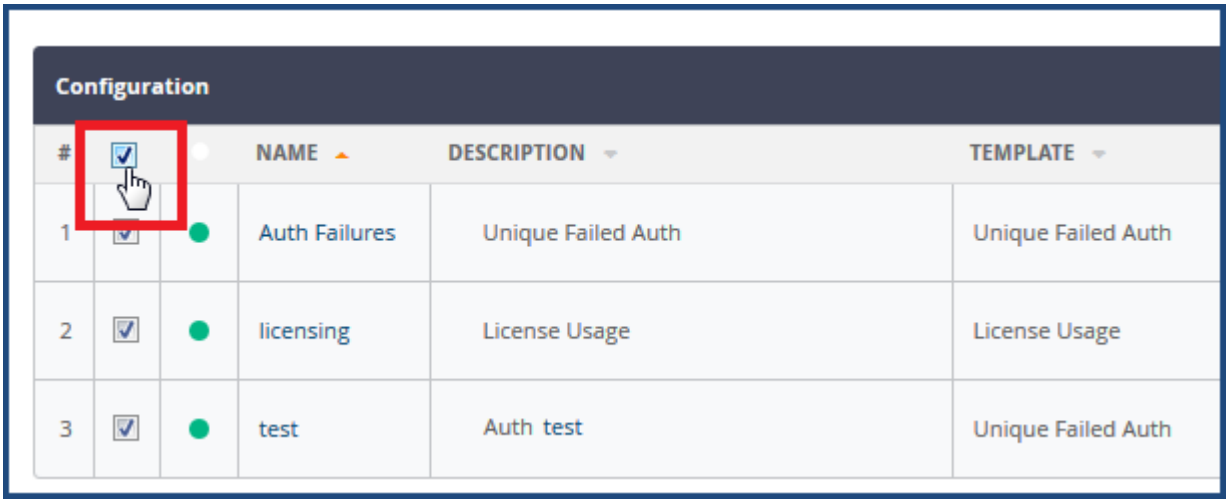
Procedure

1. Navigate to the **Reports > Configuration** page.
2. In the **Configuration** table, click the checkbox to the left of a report name to select that report.
3. to export the selected report definition as a JavaScript Object Notation (JSON) file, click the Export (^) icon to the right of the report name in the **Configuration** table, or click the **Export** button above the **Configuration** table.

   To export multiple report definitions at once (bulk export):
4. Navigate to the **Reports > Configuration** page.
5. In the **Configuration** table, click the check box to the left of each report name you want to export, or click the check box in the header of the **Configuration** table to select all reports.
6. Click the **Export** button above the **Configuration** table.

   All selected reports are exported in a single JSON file.

Figure 1. Selecting Multiple Report Definitions for Bulk Export



To import one or more report definitions:

7. Click the **Import Report** button.
8. Browse to and select a JSON report definition file.
9. If you select a file with multiple report definitions, the import process will add all reports in that file, and Insight will display a status message listing all the report definitions that have been added or updated.

> **NOTE**
>
> If you upload a report definition that uses an existing custom report template name, Insight displays a warning stating that the template name already exists, and that uploading a new report will overwrite the previous uploaded custom report configuration.

Viewing Generated Reports

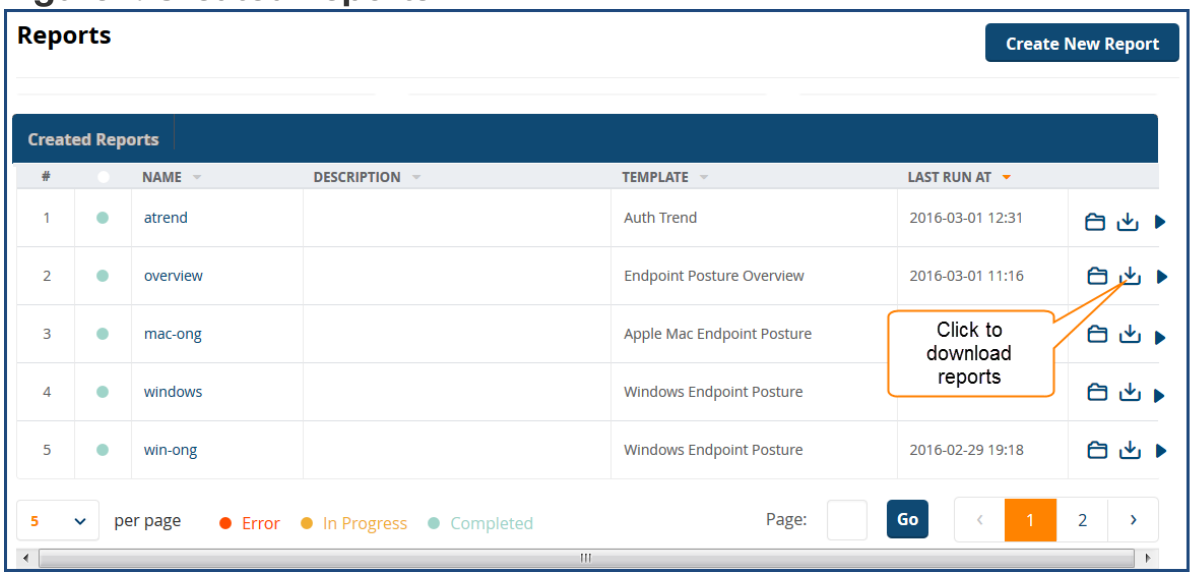
About this task

You can view or download an existing report on the Reports page of the Insight user interface. To view a generated report:

Procedure

1. From the navigation panel, click **Reports**.
2. Scroll to the **Created Reports** section.

Figure 1. Created Reports



3. To download the zip file that contains the reports in PDF and CSV formats, click the **Download** icon (as shown in Figure 9). When you download a file, this page displays banner messages that provide information about the file's download progress and when it is ready for download—for example:

   - Report <*filename*> will be generated now. Retry download after a few minutes.
   - Report <*filename*> file generation in progress.
   - Report <*filename*> file is ready for download.

4. To view the desired report in HTML format (which opens in new tab), click the name of the report.

   The generated report is displayed (see Figure 10).

Figure 2. Report Displayed in HTML Format

Regenerating Existing Reports

To regenerate an updated version of an existing report definition, click the run (▶) icon beside any report definition on the **Reports** or **Reports > Configured Reports** pages.

Insight Report Categories

This section provides detailed information about each of the report types and their associated widgets available for each Insight Report category. The Insight report templates are organized into categories, where each category has multiple report types that each contain a unique set of report data.

The following sections describe each report category, including the available reports within each category, and the contents of each report.

Subtopics

Custom Reports
Authentication Category Reports
Endpoint Category Reports
Guest Authentication Category Reports

Licensing Reports
Network Category Reports
OnGuard Category Reports
Onboard Category Report
RADIUS Authentication Category Reports
System Monitor Category Reports
TACACS+ Category Reports

Custom Reports

If you imported a custom report template via the **Reports > Custom Reports** page, you can select the **Custom Reports** option on the **Reports > Configuration** page to create a new custom report based upon that template. You can edit a custom report definition to add filters supported by that custom template type, and select raw data columns to appear in the CSV format output for that report. Note that the **Custom** option does not appear in the **Reports > Configuration** page unless you have previously uploaded a custom report template.

NOTE
Log files for custom report events are separate from other Insight event log files, and are located at /var/avenda/tips/insight/admin/customreports/upload_logs.txt.

Authentication Category Reports

The reports available in the Authentication category described in Table 1 provide the list of authentications that occurred during the report duration.

Additional authentication statistics are displayed on the **Authentication Dashboard**. For more information, see Authentication Dashboard.

NOTE
Insight can store RadSec accounting information, and produce a report of the commands executed on a AOS-CX switch. The commands executed by the AOS-CX switch are sent via RADIUS accounting packets in Insight.

Table 1. Authentication Category Reports

Report Type	Report Widgets
Accounting—Bandwidth and Session Provides the statistics using the accounting data generated during report duration. This report allows you to filter the report data by: • Policy Manager server • Network access device IP address • Device category • Device family • Device name • SSID • Endpoint IP address • User name	This report type includes the following bar • Bandwidth Statistics: Total Bandwidth, ximum Upstream Bandwidth, Maximun Duration, Users, Endpoints • Upstream Bandwidth and Downstream • Total Bandwidth and Average Bandwid • Average Session Time Trend • Unique Session Trend • Top 10 Device Categories with Most Ba • Top 10 Device Categories with Most Se • Top 10 Device Categories with Most Du • Top 10 Device Families with Most Band • Top 10 Device Families with Most Sessi • Top 10 Device Families with Most Durat • Top 10 Endpoints with Most Bandwidth • Top 10 Endpoints with Most Sessions • Top 10 Endpoints with Most Duration • Top 20 Users with Most Bandwidth Cor • Top 10 Users with Most Sessions • Top 10 Users with Most Duration • Domain Summary: Provides an overvie
Authentication by Authentication Source Provides the statistics for successful and fai led authentications per authentication sourc e.	This report type includes the following info • Authentication Statistics • Total Authentication Trend • Failed Authentication Trend • Authentication Distribution Across Autl • Authentication Distribution Across Autl

Report Type	Report Widgets
	• Authentication Distribution Across Autl **NOTE** This report allows you to filter the
Authentication by Policy Manager Provides the statistics for successful and failed authentications per Policy Manager servers in a cluster.	This report type includes the following info • Authentication Statistics • Total Authentication Trend • Failed Authentication Trend • Authentication Distribution—Error Type • Authentication Distribution Across Serv • Top 10 Policy Manager with Most Authe • Top 10 Policy Manager with Most Failed • Top 10 Policy Manager with Most MAC • Top 10 Policy Manager with Most Users **NOTE** This report allows you to filter the server.
Authentication Overview Provides statistics in general for the report duration, such as total authentications per day, unique devices authentications trend by day, unique users authentication trend by day, authentication distribution based on authentication status, service, Policy Manager server, SSID, VLAN, enforcement profile, authentication source, and top 10 users with most authentications, and so on.	This report type includes the following info • Authentication Statistics • Total Authentication Trend • Authentication Status Trend • Unique Devices Authentication Trend • Unique Users Authentication Trend • Authentication Distribution Across Autl • Authentication Distribution Across Clus • Authentication Distribution Across Serv • Authentication Distribution Across VLA • Authentication Distribution Across SSI

Report Type	Report Widgets
	• Authentication Distribution Across Enfc • Authentication Distribution Across Role • Authentication Distribution Across Autl • Top 10 Users with Most Authentications • Top 10 MAC Addresses with Most Auth • Top 10 Services with Most Authenticati • Top 10 Authentication Sources with Mc • Top 10 Policy Manager Roles Assigned • Top 10 Authorization Sources • Top 20 NADs with Most Authentication • Top 10 Enforcement Profiles Users **NOTE** This report allows you to filter the host name, Network Attached De e.
Authentication Trend Provides authentication trend statistics for to day and yesterday, today and the same day a week ago, and so on.	This report type includes the following infc • Authentication Statistics • Total Authentication Trend • Authentication Trend for Today and Yes • Authentication Trend for Today and Sai • Total Authentication for 1 Month (per m **NOTE** This report allows you to filter the host name, Network Access Devi
Failed Authentication Provides statistics based on failed authentica tions.	This report type includes the following infc • Error Statistics • Failed Authentication Trend

Report Type	Report Widgets
	• Authentication Distribution—Error Type • Failed Authentication Distribution acro • Failed Authentication Distribution acro • Top 10 Errors with Most Failed Authenti • Top 20 NADs with Most Failed Authenti • Top 10 Policy Manager Servers with Mc • Top 10 Users with Most Failed Authenti • Top 10 Endpoints with Most Failed Auth • Top 10 Services with Most Failed Authe **NOTE** ClearPass Policy Manager host na ress, SSID, and Error Code.
Unique Failed Authentication Provides statistics based on failed authentica tions.	This report type includes the following info • Error Statistics • Failed Authentication Trend • Authentication Distribution—Error Type • Failed Authentication Distribution acro • Failed Authentication Distribution acro • Top 10 Errors with Most Failed Authenti • Top 20 NADs with Most Failed Authenti • Top 10 Policy Manager Servers with Mc • Top 10 Users with Most Failed Authenti • Top 10 Endpoints with Most Failed Auth • Top 10 Services with Most Failed Authe **NOTE** ClearPass Policy Manager host na ress, SSID, and Error Code.

Endpoint Category Reports

The Endpoint category provides information on endpoints discovered during the report duration. The reports available in the Endpoint category contain data that can also be found in the endpoints widgets on the **Endpoints Dashboard** page. For additional information about the Endpoints Dashboard, see Endpoints Dashboard.

Table 1. Endpoint Category Reports

Report Type	Report Widgets
Active Endpoint Security	This report contains the following informati on: • Endpoint Status • Endpoint Spoofing count • IP Statistics • Device Category • Device category breakdown for devices that are active within a given timestamp
Endpoint Authentication Overview	This report type includes the following info rmation for all endpoint types: • Endpoint Statistics • Endpoints Distribution Across Device Category • Endpoints Distribution Across Device Fa mily • Endpoints Distribution Across Device N ame • Top 10 Users with Most Endpoints • Top 10 Device Categories with Most En dpoints • Top 10 Device Names with Most Endpoi nts

Report Type	Report Widgets
	• Top 10 Device Families with Most Endpoints **NOTE** This report also allows you to filter the report data by Network Access Device (NAD) IP address, Device Category, Device Family, Device name, and SSID.
Endpoint Overview	This report type includes the following information for all endpoint types: • Top 10 Reports Time to Run 30 Days • Top 10 Reports Last 30 Days
Guest—Endpoint Overview	This report type includes the following information for endpoints using Guest Authentication: • Endpoint Statistics • Endpoints Distribution Across Device Category • Endpoints Distribution Across Device Family • Endpoints Distribution Across Device Name • Top 10 Users with Most Endpoints • Top 10 Device Categories with Most Endpoints • Top 10 Device Names with Most Endpoints • Top 10 Device Families with Most Endpoints **NOTE** This report also allows you to filter the report data by Netwo

Report Type	Report Widgets
	rk Access Device (NAD) IP ad dress, Device Category, Devic e Family, Device name, and SS ID.
Idle Endpoint Security	This report includes the known endpoints t hat are not updated within a given timesta mp.
RADIUS—Endpoint Overview	This report type includes the following info rmation for endpoints using RADIUS authe ntication: • Endpoint Statistics • Endpoints Distribution Across Device Category • Endpoints Distribution Across Device Fa mily • Endpoints Distribution Across Device N ame • Top 10 Users with Most Endpoints • Top 10 Device Categories with Most En dpoints • Top 10 Device Names with Most Endpoi nts • Top 10 Device Families with Most Endpo ints **NOTE** This report also allows you to f ilter the report data by Netwo rk Access Device (NAD) IP ad dress, Device Category, Devic e Family, Device name, and SS ID.

Guest Authentication Category Reports

The reports available in the Guest Authentication category provide statistics based on Guest authentications from the Guest database. The statistics for authentication trend and usage for guest users are drawn from the accounting data.

Additional authentication statistics are displayed on the **Guest Dashboard**. For additional information about the Guest Dashboard, see Guest Dashboard.

Table 1. Guest Authentication Category Reports

Report Type	Report Widgets
Guest—Auth Overview	This report includes the following report widgets: • Authentication Statistics • Total Authentication Trend • Authentication Status Trend • Unique Devices Authentication Trend • Unique Guests Authentication Trend • Authentication Distribution Across Authentication Status • Authentication Distribution Across Cluster • Authentication Distribution Across Service • Authentication Distribution Across VLAN • Authentication Distribution Across SSID • Authentication Distribution Across Enforcement Profiles • Authentication Distribution Across Roles • Authentication Distribution Across Authentication Source • Top 10 Guests with Most Authentications

Report Type	Report Widgets
	• Top 10 MAC Addresses with Most Authentications • Top 10 IPs with Most Authentications • Top 10 Services with Most Authentications • Top 10 Authentication Sources with Most Authentications • Top 10 ClearPass Roles Assigned • Top 10 Authorization Sources with Most Authentications • Top 20 NADs with Most Authentications • Top 10 Enforcement Profiles Used **NOTE** This report also allows you to filter the report data by Policy Manager host name and Network Access Device (NAD) IP address.
Guest—Auth Trend	This report includes the following report widgets: • Authentication Statistics • Total Authentication Trend • Authentication Trend for Today and Yesterday • Authentication Trend for Today and Same Day Week Ago • Total Authentication for One Month • Sponsor List **NOTE** This report also allows you to filter the report data by Policy Manager host name and Net

Report Type	Report Widgets
	work Access Device (NAD) IP address.
Guest—Auth by Policy Manager	This report type includes the following info rmation guest authentications by Policy Ma nager: • Authentication Statistics • Total Authentication Trend • Failed Authentication Trend • Authentication Distribution—Error Typ es • Authentication Distribution Across Servi ce • Top 10 Policy Manager with Most Authe ntications • Top 10 Policy Manager with Most Failed Authentications • Top 10 Policy Manager with Most MAC Authentications • Top 10 Policy Manager with Most Guest s **NOTE** This report also allows you to filter the report data by Policy Manager host name.
Guest - Hotspot	This report includes the following informati on about hotspot user logins recorded ove r the selected time frame: • Hotspot plan distribution (for example, free vs hourly paid access) • Purchase amount distribution: purchas e totals per plan type, in either dollars o

Report Type	Report Widgets
	r the currency used to purchase the pla n.
Guest—Social Login	This report type includes the following info rmation for guest authentication for Social Logins: • Social Authentication Trend • Endpoint Distribution Across Social P roviders • Authentication Distribution Across Auth entication Source **NOTE** This report also allows you to filter the report data by Policy Manager host name and Net work Access Device (NAD) IP address.
Guest Accounting—Bandwidth and Sessio n This report allows you to filter the report da ta by: • Policy Manager server • Network access device IP address • Device category • Device family • Device name • SSID • Endpoint IP address • User name	This report type includes the following ban dwidth and session information: • Bandwidth Statistics: Total Bandwidth, Average Bandwidth, Maximum Bandwid th, Maximum Upstream Bandwidth, Max imum Downstream Bandwidth, Session s, Maximum Duration, Guests, Endpoint s • Upstream Bandwidth and Downstream Bandwidth Trend • Total Bandwidth and Average Bandwidt h Trend • Average Session Time Trend • Unique Session Trend • Top 10 Device Categories with Most Ba ndwidth Consumed

Report Type	Report Widgets
	• Top 10 Device Categories with Most Sessions • Top 10 Device Categories with Most Duration • Top 10 Device Families With Most Bandwidth Consumed • Top 10 Device Families With Most Sessions • Top 10 Device Families With Most Duration • Top 10 Endpoints with Most Bandwidth Consumed • Top 10 Endpoints with Most Sessions • Top 10 Endpoints with Most Duration • Top 20 Guests with Most Bandwidth Consumed • Top 10 Guests with Most Sessions • Top 10 Guests with Most Duration
Guest Devices—Due to Expire	The *Guest Devices—Expired* report lets you view information about devices that are due to expire within the selected report period. This report type includes the following report widgets: • Guest Expiry Statistics • Guest Expiry List (MAC Address and Device Name)
Guest Users and Devices—Due to Expire	The *Guest Users and Devices—Expired* report lets you view information about guest accounts and devices that are due to expire within the selected report period. This report type includes the following report widgets:

Report Type	Report Widgets
	• Guest Expiry Statistics • Guest Expiry List (MAC Address and Visitor Name)
Guest Users—Due to Expire	The *Guest Users—Expired* report lets you view information about guest accounts due to expire within the selected report period. This report type includes the following report widgets: • Guest Expiry Statistics • Guest Expiry List

Licensing Reports

The licensing report category allows you to generate a report about license usage over the selected time period. Similar information can also be found in the Licensing widgets on the **Licensing Dashboard**. For additional information, see Licensing Dashboard.

Table 1. Licensing Report Content

Report Type	Report Widgets
License Usage	This report type includes the following licensing information: • License statistics, including the total number of licenses, and the number of licenses used for each license type • Number of unique endpoints on the network over the selected time period • Number of Policy Manager Access, Entry, OnGuard or Onboard licenses used over the selected time period. • Policy Manager licenses distribution (available vs used) • Policy Manager license usage per host

Network Category Reports

The reports available in the Network category contain data about network access devices and gives details on authentication trends such as successful and failed authentications on a per-day basis.

Similar information can also be found in the Network widgets on the **Network Dashboard**. For additional information, see Licensing Dashboard.

Table 1. Network Category Reports

Report Type	Report Widgets
Authentication by NAD	This report type includes the following information for Network Access Devices (NADs) using guest authentication. • Authentication Statistics • Total Authentication Trend • Failed Authentication Trend • Authentication Distribution Across NAD Ports • Top 20 NADs with Most Authentication • Top 10 Services with Most Authentications • Top 20 NADs with Most Failed Authentications • Top 20 NADs with Most MAC Addresses • Top 20 NADs with Most Users **NOTE** This report also allows you to filter the report data by NAD IP address.
Guest—Authentication by NAD	This report type includes the following information for Network Access Devices (NADs) using guest authentication

Report Type	Report Widgets
	• Authentication Statistics • Total Authentication Trend • Failed Authentication Trend • Authentication Distribution Across NAD Ports • Top 20 NADs with Most Authentication • Top 10 Services with Most Authentications • Top 20 NADs with Most Failed Authentications • Top 20 NADs with Most MAC Addresses • Top 20 NADs with Most Guests **NOTE** This report also allows you to filter the report data by NAD IP address.
RADIUS—Auth by NAD	This report type includes the following information for Network Access Devices (NADs) using guest authentication: • Authentication Statistics • Authentication Distribution Across NAD Ports • Top 20 NADs with Most Authentication • Top 10 Services with Most Authentications • Top 20 NADs with Most Failed Authentications • Top 20 NADs with Most MACs • Top 20 NADs with Most Users **NOTE**

Report Type	Report Widgets
	This report also allows you to f ilter the report data by NAD IP address.

OnGuard Category Reports

The reports available in the OnGuard category provide analysis on the devices' posture and health status. These widgets contain data that can also be found in the Posture widgets on the **Posture Dashboard**. For additional information, see Posture Dashboard.

Table 1. OnGuard Category Reports

Report Type	Report Widgets
Apple Mac Endpoint Posture	This report type includes the following pos ture information for Apple/Macintosh endp oints: • OnGuard Statistics • OnGuard Device Authentication Trend • OnGuard Device Distribution Across H ealth Status • Antispyware Product Name • Antispyware Dat File Version • Antispyware Engine Version • OnGuard Device Distribution Across An tispyware Real - Time Protection Status • Antispyware Version • Antivirus Product Name • Antivirus Dat File Version • Antivirus Engine Version • OnGuard Device Distribution Across An tivirus RealTimeProtection Status • Antivirus Version

Report Type	Report Widgets
	• Disk Encryption Product Name • Disk Encryption Version • Firewall Product Name • OnGuard Device Distribution Across Firewall Status • Firewall Version • OnGuard Device Distribution Across Network Connection Type • OnGuard Device Distribution Across P2P Application Name • OnGuard Device Distribution Across P2P Status • OnGuard Device Distribution Across Patch Agent Name • Missing Patches Count • OnGuard Device Distribution Across Patch Agent Status • OnGuard Device Distribution Across Client Operating System • OnGuard Device Distribution Across Client Running as VM **NOTE** This report also allows you to filter the report data by System Posture Token (SPT).
Endpoint Posture Overview	This report type includes the following endpoint posture information: • OnGuard Statistics • OnGuard Device Distribution Across Health Status

Report Type	Report Widgets
	• Unhealthy OnGuard Device Distribution Across Device Family • OnGuard Device Distribution Across Ag ent Type • OnGuard Device Distribution Across Ag ent Version • Health Class • Missing Hotfixes **NOTE** This report also allows you to f ilter the report data by System Posture Token (SPT).
Linux Endpoint Posture	This report type includes the following pos ture information for endpoints using a Linu x operating system: • OnGuard Statistics • OnGuard Device Authentication Trend • OnGuard Device Distribution Across H ealth Status • Antivirus Product Name • Antivirus Dat File Version • Antivirus Engine Version • OnGuard Device Distribution Across An tivirus RealTimeProtection Status • Antivirus Version **NOTE** This report also allows you to f ilter the report data by System Posture Token (SPT).

Report Type	Report Widgets
Windows Endpoint Posture	This report type includes the following pos ture information for endpoints using a Win dows operating system: • OnGuard Statistics • OnGuard Device Authentication Trend • OnGuard Device Distribution Across H ealth Status • Antispyware Product Name • Antispyware Dat File Version • Antispyware Engine Version • OnGuard Device Distribution Across An tispyware Real - Time Protection Status • Antispyware Version • Antivirus Product Name • Antivirus Dat File Version • Antivirus Engine Version • OnGuard Device Distribution Across An tivirus RealTimeProtection Status • Antivirus Version • Disk Encryption Product Name • Disk Encryption Version • Firewall Product Name • OnGuard Device Distribution Across Fi rewall Status • Firewall Version • OnGuard Device Distribution Across Ne twork Connection Type • OnGuard Device Distribution Across P2 P Application Name • OnGuard Device Distribution Across P2 P Status

Report Type	Report Widgets
	• OnGuard Device Distribution Across Patch Agent Name • Missing Patches Count • OnGuard Device Distribution Across Patch Agent Status • OnGuard Device Distribution Across Client Operating System • OnGuard Device Distribution Across Client Running as VM **NOTE** This report also allows you to filter the report data by System Posture Token (SPT).

Onboard Category Report

The reports available in the Onboard category provides analysis on onboarded devices during the report period, such as the active users and devices count, revoked devices count, onboarded devices distribution based on device type, and Onboard enrollment details.

Table 1. Onboard Report Content

Report Type	Report Widgets
Onboard Certificate	This report type includes the following certificate information: • Onboard statistics for numbers of revoked devices, active devices, and users • Latest Onboard Device Distribution • Active Onboard Device Distribution • Top 10 Users with Most Active Devices
Onboard Certificate - Due to Expire	This report type includes the following information for manually created certificates d

Report Type	Report Widgets
	ue to expire within the selected time frame : • Total number of devices of each device type with certificates due to expire. • Top 10 Users with Most Active Devices
Onboard Enrollment	This report type provides the following info rmation: • Total Devices Onboarded • Onboarded Devices Enrollment Trend • Onboarded Devices • Unique Users and Their Associated Tota l Number of Devices • Unique Onboarded Devices

RADIUS Authentication Category Reports

The reports available in the RADIUS Authentication provide detailed analysis on authentication trends on successful and failed RADIUS authentication.

Additional authentication statistics are displayed on the **Authentication Dashboard**. For additional information, see Authentication Dashboard.

Table 1. RADIUS Authentication Category Reports

Report Type	Report Widgets
RADIUS—Authentication by Authentication Source	This report type includes the following info rmation for RADIUS authentication: • Authentication statistics for numbers a nd percentages of authentications succ esses and failures • Total Authentication Trend • Failed Authentication Trend

Report Type	Report Widgets
	• Authentication Distribution Across Authentication Source • Authentication Distribution Across Authorization Source • Failed Authentication Distribution Across Authentication Source **NOTE** Policy Manager host name.
RADIUS—Authentication by Policy Manager	This report type includes the following information for RADIUS authentication: • Authentication Statistics, including numbers and percentages of authentications successes and failures • Total Authentication Trend • Failed Authentication Trend • Authentication Distribution Error Types • Authentication Distribution Across Service • Top 10 Policy Manager with Most Authentications • Top 10 Policy Manager with Most Failed Authentications • Top 10 Policy Manager with Most MAC Addresses • Top 10 Policy Manager with Most Users **NOTE** This report also allows you to filter the report data by authentication source.
RADIUS—Authentication Overview	This report type includes the following information for RADIUS authentication:

Report Type	Report Widgets
	• Authentication statistics, including numbers and percentages of authentications successes and failures, and numbers of users, endpoints, network devices, roles, Policy Manager servers and enforcement profiles • Total Authentication Trend • Authentication Status Trend • Unique Devices Authentication Trend • Unique Users Authentication Trend • Authentication Distribution Across Auth Status • Authentication Distribution Across Cluster • Authentication Distribution Across Service • Authentication Distribution Across VLAN • Authentication Distribution Across SSID • Authentication Distribution Across Enforcement Profiles • Authentication Distribution Across Role • Authentication Distribution Across Auth Source • Top 10 Users with Most Authentications • Top 10 MACs with Most Authentications • Top 10 Services with Most Authentications • Top 10 Policy Manager Roles Assigned • Top 10 Authorization Sources • Top 20 NADs with Most Authentications

Report Type	Report Widgets
	• Top 10 Enforcement Profiles Used **NOTE** This report also allows you to filter the report data by Clear Pass Policy Manager host nam e, Network Access Device (NA D) IP address, SSID and authe ntication service name.
RADIUS—Authentication Trend	This report type includes the following info rmation: • Authentication Statistics, including aut hentication data for the previous day an d week • Total Authentication Trend • Authentication Trend for Today and Yest erday • Authentication Trend for Today and Sam e Day Week Ago • Total Authentication for 1 Month (per m onth) **NOTE** This report also allows you to filter the report data by Clear Pass Policy Manager name, Ne twork Access Device (NAD) IP address, and SSID.
RADIUS—Failed Authentication	This report type includes the following info rmation: • Error Statistics • Failed Authentication Trend • Authentication Distribution—Error Typ es

Report Type	Report Widgets
	• Failed Authentication Distribution Acros s Service • Failed Authentication Distribution Acros s Authentication Sources • Top 10 Errors with Most Failed Authentic ations • Top 10 Policy Manager Servers with Mos t Failed Authentications • Top 20 NADs with Most Failed Authentic ations • Top 10 Users with Most Failed Authentic ations • Top 10 Endpoints with Most Failed Auth entications • Top 10 Services with Most Failed Authe ntications **NOTE** This report also allows you to filter the report data by Clear Pass Policy Manager host nam e, Network Access Device (NA D) IP, SSID, and Error Code.

System Monitor Category Reports

The reports available in the **System Monitor** category provide information about system-level events, such as configuration changes performed on the Policy Manager server (configuration audit), license usage, and system events.

Additional system statistics are displayed on the **System Monitor Dashboard**. For additional information about the System Dashboard, see System Monitor Dashboard.

Table 1. System Monitor Category Reports

Report Type	Report Widgets
Configuration Audit	This report type includes the following info rmation for each configuration audit reco rd: • Name of change • Action (for example, modify, add, or del ete) • Category • Updated by • Update timestamp
System Events	This report type includes the following info rmation for each system event : • Policy Manager host name • Source of Event • Event Category • Event Level • Timestamp • Description **NOTE** This report also allows you to filter the report data by Policy Manager host name.

TACACS+ Category Reports

The reports available in the TACACS+ category provide TACACS+ authentication trends such as successful and failed TACACS+ authentication and command authorizations.

Table 1. TACACS+ Reports Content

Report Type	Report Widgets
TACACS+—Authentication	This report type includes the following TACACS+ information: • TACACS+ statistics, including the numb ers and percentages of successful and failed authentications, and the numbers of users, Policy Manager servers, and n etwork devices. • Total Authentication Trend • Authentication Status Trend • Authentication Trend For Today and Yest erday • Command List showing Authentication and Authorization commands • Authentication Distribution Across Auth entication Status • Authentication Distribution Across Clus ter • Top 10 Errors with Most Failed Authentic ations • Top 20 NADs with Most Authentication • Top 10 Users with Most Authentications **NOTE** This report also allows you to filter the report data by Policy Manager server and NAD IP ad dress.

Insight Inventory

- About the Insight Inventory
- Sorting and Filtering Inventory Table Data
- Viewing Endpoints Details

Subtopics

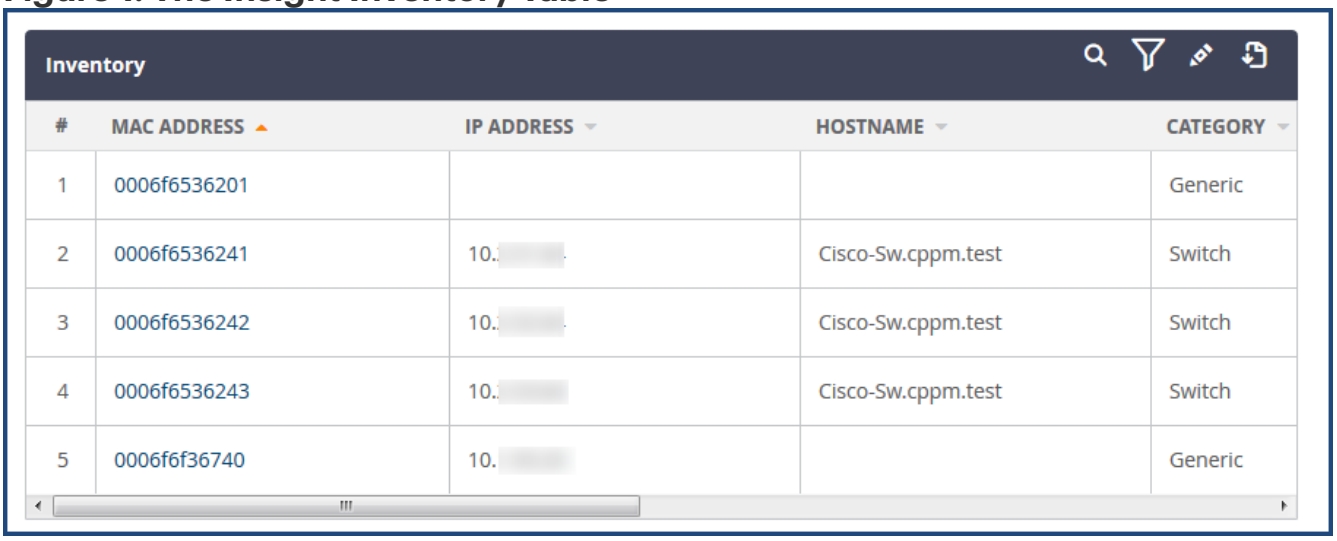
About the Insight Inventory
Sorting and Filtering Inventory Table Data
Viewing Endpoints Details

About the Insight Inventory

Insight lists all authenticated endpoints on the network on the **Inventory** page.

View inventory information for all endpoints by navigating to **Insight > Inventory**, or click a graph widget on the **Endpoints**, **Posture** or **Authentication** Insight dashboards and drill down to display inventory data related to that graph.

Figure 1. The Insight Inventory Table



The endpoints information in the **Inventory** table is described in Table 1.

Table 1. Endpoints Table Parameters

Column	Description
MAC Address	MAC address of the endpoint.
IP address	IP address of the endpoint.

Column	Description
Hostname	Host name of the endpoint.
Category	Category assigned to the profi example, *Access Points*, *Comp* *c*e, or *VoIP phone*.
Family	The operating system (OS) fan t device. For example, when th , Insight can show a device OS *x*, or Apple Mac.
Device Name	The **Device Name** column des evice type within a device fam uter in the **Windows** device fa ype of **Windows 8/10**.

Sorting and Filtering Inventory Table Data

By default, each page of the **Inventory** table displays endpoints sorted by MAC address.

Click any column heading to resort the table by that column data type.

Click the filter ( )icon on the **Inventory** table header to select one or more filter options and modify the table display only selected endpoint types.

The available filter types are described in Table 2 below.

Table 1. Inventory Table Filter Parameters

Column	Action/Description
Device Category	The **Device Category** drop - down list disp lays a list of the categories of devices foun d on your network, such as a server or prin ter. By default, the **All** option is selected, a nd the inventory displays devices in all cat egories. Click this drop - down list and select any o ther device category to filter the table and display only devices of the selected type. The contents of this drop - down list may vary, depending upon the types of devices in your network.

Column	Action/Description
Device Family	The **Device Family** drop - down list displays a list of the device families found on your network, such as a Hewlett Packard, Aruba, or Intel. By default, the **All** option is selected, and the inventory displays devices in all categories. Click this drop - down list and select any other device category to filter the table and display only devices of the selected type. The contents of this drop - down list may vary, depending upon the types of devices in your network, and the selected device category.
Device Name	The **Device Name** drop - down list displays a list of the device names found on your network, such as an Apple iPhone. By default, the **All** option is selected, and the inventory displays devices in all categories. Click this drop - down list and select any other device category to filter the table and display only devices of the selected type. The contents of this drop - down list may vary, depending upon the types of devices in your network, and the selected device category.
Static IP	Click the **Static IP** drop - down list and select one of the following options: • **All**: Display endpoints with static and non - static IP addresses. • **True**: Display endpoints with a static IP address. • **False**: Display endpoints with a non - static IP address.
Conflict Status	Click the **Conflict Status** drop - down list and select one of the following options: • **All**: Display all endpoints.

Column	Action/Description
	• **True**: Display endpoints with a conflict (device is profiled into two or more diffe rent categories.) • **False**: Display endpoints with no conflic t.
Fingerprint Type	Insight can obtain fingerprint information for network devices through a variety of so urces (for example, obtained directly from the host, or via DHCP or SNMP) The **finger print** drop - down list displays a list of fing erprint information sources for the devices found on your network. By default, the **All** option is selected, and the inventory displ ays devices identified by all fingerprinting t ypes. Click this drop - down list and select any other fingerprinting type to filter the ta ble and display only devices identified by t he selected fingerprinting method. **NOTE** The contents of this drop - d own list may vary, depending upon the fingerprinting metho ds used to identify devices on your network.
Posture	Click the **Posture** option select one of the following options to filter endpoints that h ave a posture status as a result of having u ndergone a health check. • **All**: Display all endpoints. • **True**: Display endpoints that have postu re information. • **False**: Display endpoints that do not h ave posture information
System Posture Status	Click this option to filter the **Inventory** tabl e by one or more of the posture status typ es for the endpoints on your network that h ave undergone a health check and have po sture status information. This field will only return information for endpoints that have

Column	Action/Description
	a posture status of **True**. The available para meters in this filter may vary, depending up on the posture status of the devices on you r network.
Added at	Enter a start date and an end date to displ ay devices added within a specified time ra nge. You can also click the calendar (🗓) ico n in the **Start Date** and **End Date** fields to s elect a start and end date from a calendar.
Last Updated	Enter a start date and an end date to displ ay devices updated within a specified time range. You can also click the calendar (🗓) i con in the **Start Date** and **End Date** fields t o select a start and end date from a calend ar.
Authentication At	Enter a start date and an end date to displ ay devices authenticated within a specified time range. You can also click the calendar (🗓) icon in the **Start Date** and **End Date** fiel ds to select a start and end date from a cal endar.

Viewing Endpoints Details

The **Endpoint Details** page appears when you select any MAC address in the **Inventory** table, or when you enter and search for a search string in the Insight search bar. This data can include information in the following categories:

- **Summary**: Total number of successful authentications for that device over the selected time period.
- **Overview**: This section displays endpoint information such as the device IP address, MAC address and host name, and network details such as the AP , controller or switch port to which the endpoint is connected.
- **Device Profile**: Conflict information, and device profile data, such as the device category, family and name, and the time the device was first added or last updated.
- **OnGuard**: This section displays OnGuard health check data, if available.

- **Endpoint Details**: This section displays authentication details, endpoint role and policy information, the IP address of the NAD (Network Access Device), and the current authentication status.
- **Fingerprint Details**: This section displays fingerprint details and information about fingerprint information sources for that endpoint.
- **Authentication Trend**: A line chart showing the number of successful authentications during each day of the selected time period
- **Bandwidth Usage**: A bar chart showing bandwidth used by the endpoint during each day of the selected time period.

> **NOTE**
>
> If the endpoint is seen on the trunk port instead of an access port, the information shown in the *Switch Port* field of the *Network Details* section displays the word **(Trunk)** after the network switch port number.

Administration Operations

- Overview
- File Transfer Settings Configuration
- Database Settings Configuration

Subtopics

Overview
File Transfer Settings Configuration
Testing File Transfer Configuration
Database Settings Configuration

Overview

You can use the **Administration** page to do the following tasks:

- Specify the number of days to retain information in the database.
- Test the new notification settings to review Insight log files.

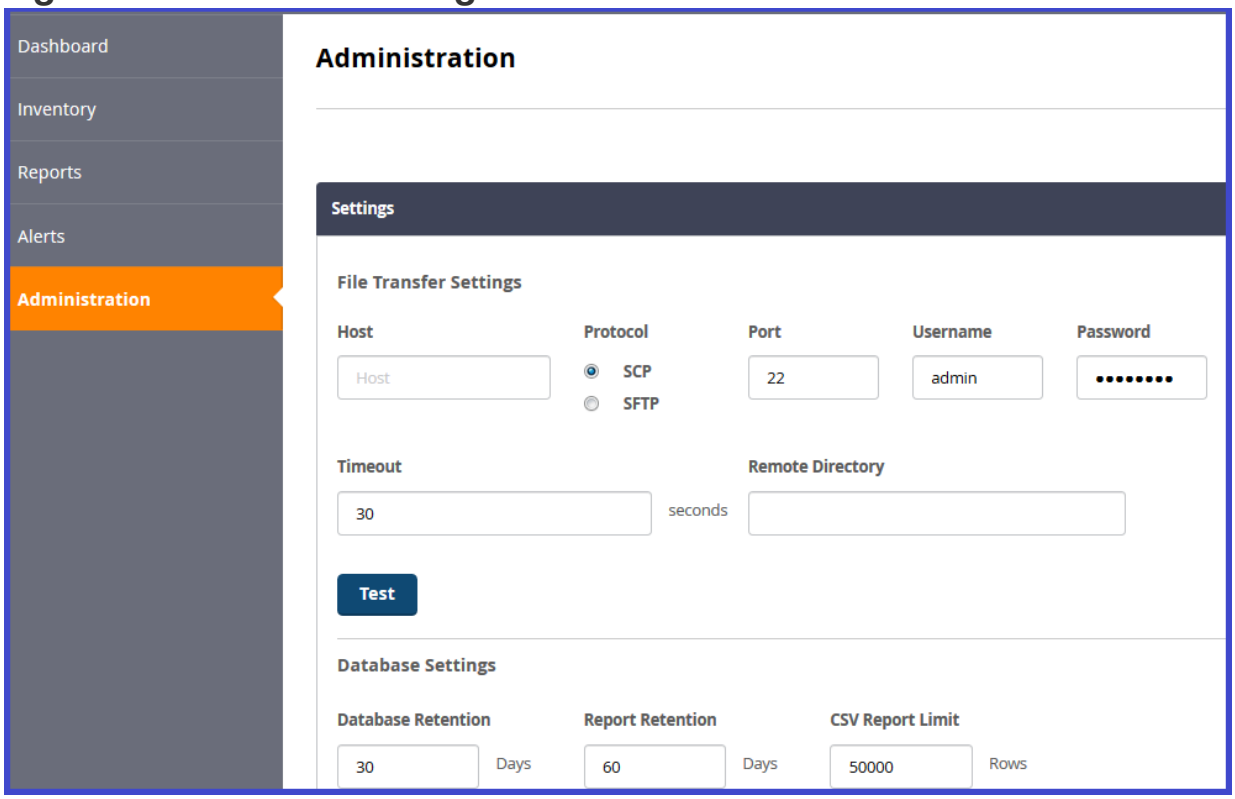
- Store reports offline using the SCP (Secure Copy Protocol) or SFTP (Secure File Transfer Protocol) file transfer protocols.

  To access the **Administration** page:

  1. From the Insight navigation pane, click **Administration**.

     The **Administration** page appears.

Figure 1. Administration Page



Subtopics

Support Information

Support Information

- Insight database migration is supported.
- Configuration migration is not supported.
- Database retention default: 30 days
- Report retention default: 60 days
- CSV report limit: 5,000,000 rows

File Transfer Settings Configuration

About this task

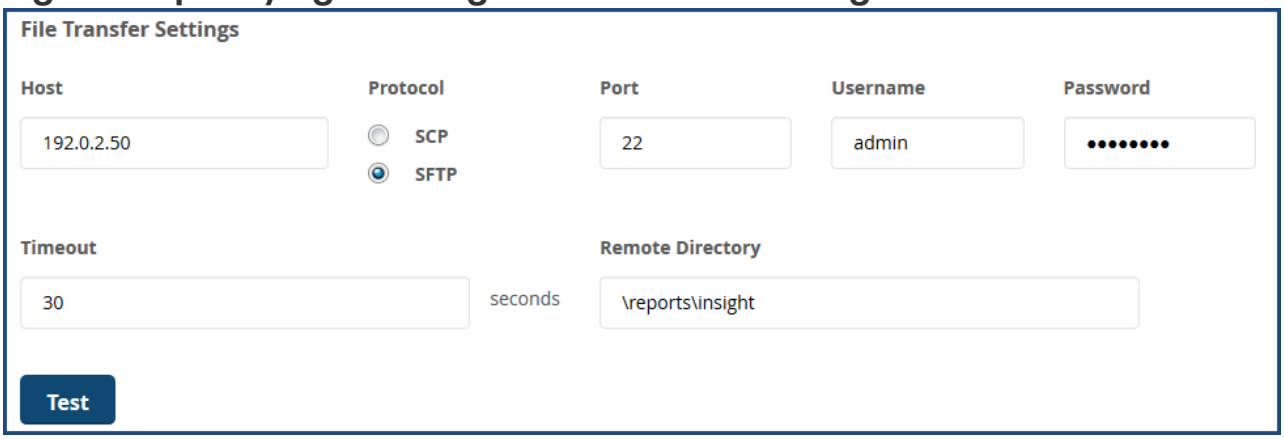
You can specify the file transfer settings for uploading generated Insight reports to a FileStore.

To configure the file transfer settings:

Procedure

1. Navigate to the **Administration** page.

Figure 1. Specifying the Insight File Transfer Settings



2. In the **File Transfer Settings** section, enter the appropriate values as described in Table 1.
3. When finished, click **Save**.

Table 1. Insight File Transfer Parameters

Parameter	Action/Description
Host	Specify the IP address of the destination host FTP server.
Protocol	Specify the protocol to be used to upload the generated reports to a FileStore. You can select from the following protocols: • **SCP** (Session Control Protocol)

Parameter	Action/Description
	• **SFTP** (SSH File Transfer Protocol)
Port	Specify the destination port number. The default destination port is **22**.
Username/Password	Enter the username and password of the host FTP server.
Timeout	Specify the timeout value in seconds. The default value is **30 seconds**.
Remote Directory	Specify the location where the generated reports are to be copied. If the remote directory location is same a s default root of FTP, you can leave this fi eld blank. **NOTE** To copy reports to a remote directory, you must enable t he **Reports > Create New Re port > Enable remote copy** o ption.

Testing File Transfer Configuration

About this task

When you have configured the Insight file transfer settings, you can then test to see if file transfer is operational.

To test the Insight file transfer configuration:

Procedure

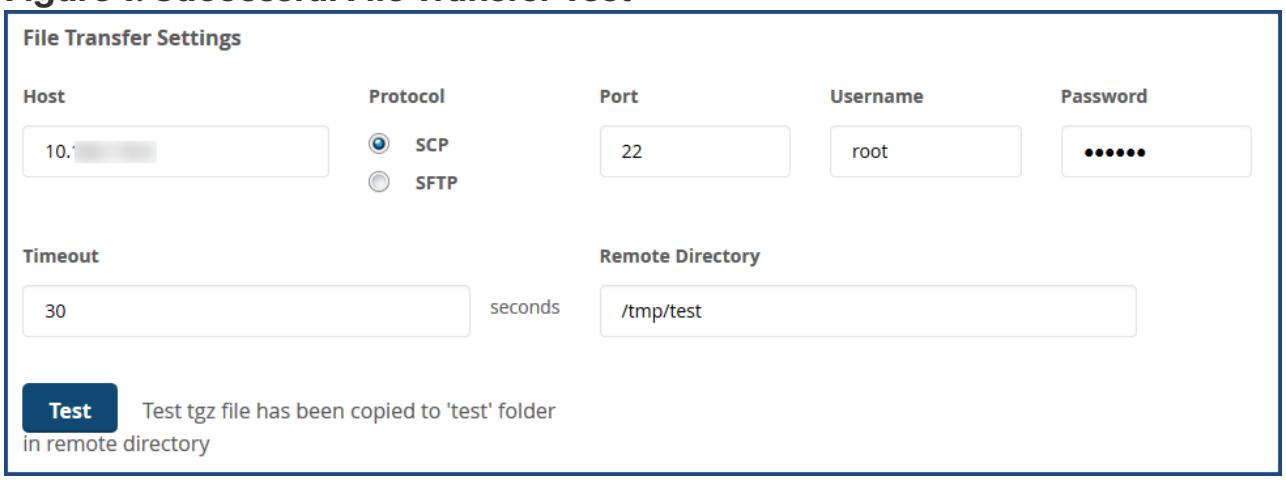
1. Review the **File Transfer Settings** to ensure they are correct.

2. Click the **Test** button.

   You see the message: *File Transfer Settings testing in progress...*

   Then the following screen opens:

Figure 1. Successful File Transfer Test



You are now ready to commence transferring Insight files to the FTP server as needed.

Database Settings Configuration

About this task

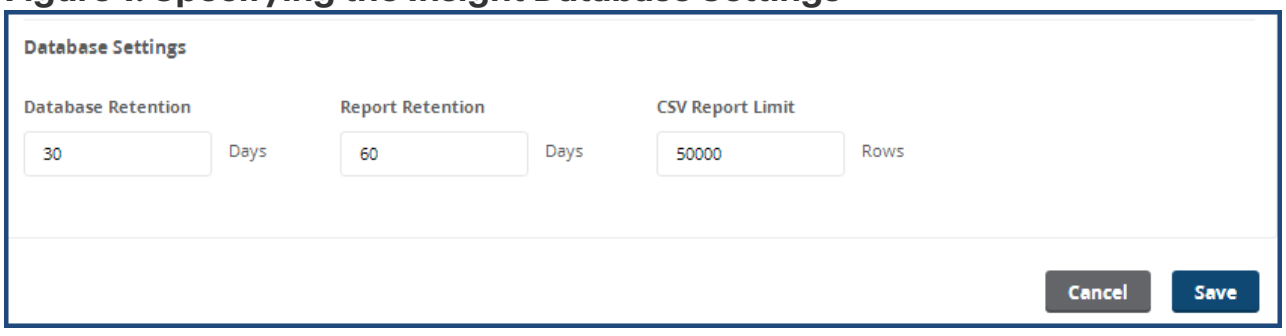
To configure the Insight database parameters:

Procedure

1. Navigate to the **Administration** page.

   The **Database Settings** section is at the bottom of the **Administration** page.

Figure 1. Specifying the Insight Database Settings



2. Enter appropriate values in the **Database Settings** section.
3. When finished, click **Save**.

Table 1. Insight Database Parameters

Parameter	Action/Description
Database Retention	Specify the number of days to re e. The supported range is from **1** to efault value is **30 days**.
Report Retention	Specify the number of days to re ed reports. The supported range is from **1** to efault value is **60 days**.
CSV Report Limit	Specify the number of rows for a The supported range is from **1** to . The default value is **50000** row

Managing Insight Admin Privileges

- Overview
- Viewing the Default Insight Admin Privileges
- Defining Custom Insight Admin Privileges
- Insight User Interface Differences for Read-Only Users

Subtopics

Overview
Viewing the Default Insight Admin Privileges
Defining Custom Insight Admin Privileges
Insight User Interface Differences for Read-Only Users

Overview

Policy Manager supports multilevel Insight administrators, each with a different level of administrative access to Insight. Policy Manager provides a default Admin Privileges Read-only Administrator. The default sets of admin privileges cannot be modified.

Each of the Insight modules (Dashboard, Reports, Alerts, Administration, and Inventory) can have three privilege levels or no privileges:

- Read-only
- Read and Write
- Read, Write, and Delete

  In the case of a user with no Insight admin privileges, the navigation panel on the left side of the Insight user interface is not displayed.

Viewing the Default Insight Admin Privileges

About this task

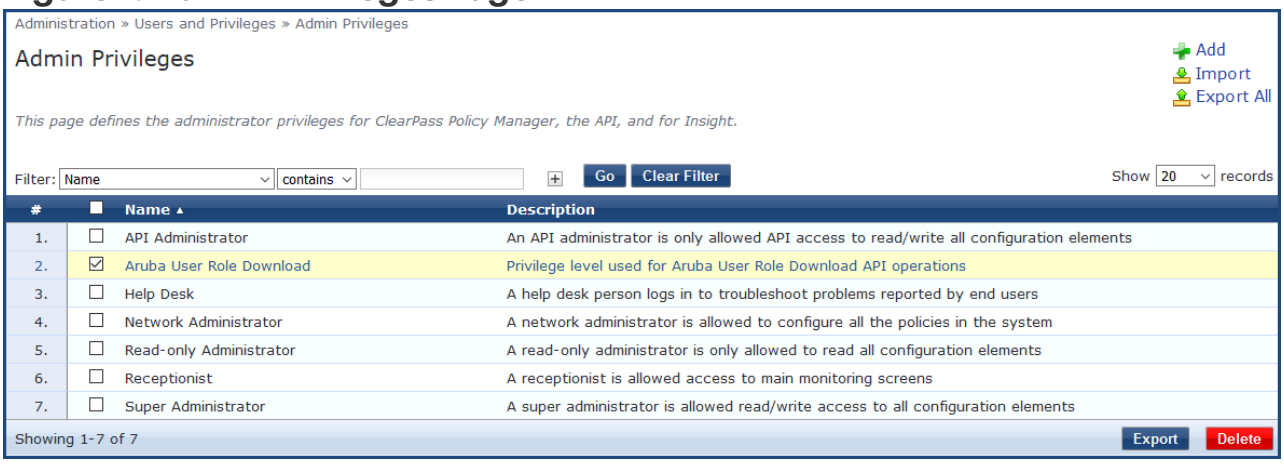
The settings for the default admin privileges cannot be modified. To view the default Insight admin privileges defined in Policy Manager:

Procedure

1. From the Policy Manager user interface, navigate to **Administration > Users and Privileges > Admin Privileges**.

   The **Admin Privileges** page opens.

   Figure 1. Admin Privileges Page



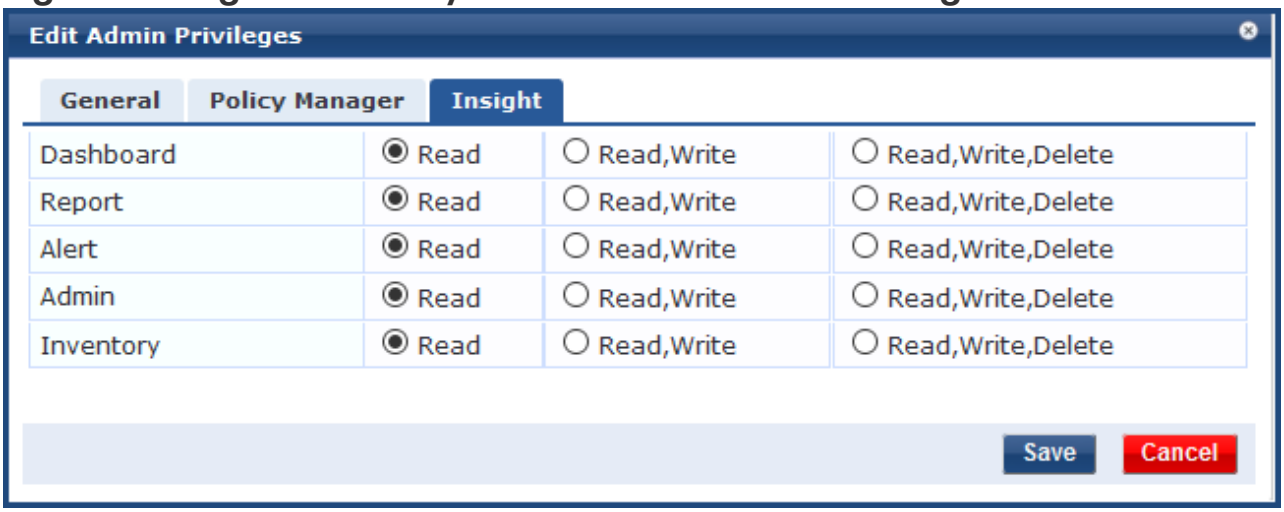
2. To view the Read-only admin privileges for Insight, select **Read-only Administrator**.

The **Edit Admin Privileges** dialog opens.

3. Select the **Insight** tab.

   The default Insight admin privileges for the Read-only Administrator are displayed.

Figure 2. Insight Read-Only Administrator Admin Privileges



As shown in Figure 2, the default admin privileges for the Insight Read-only Administrator specifies Read-only access to all of the Insight modules:

- Dashboard
- Reports
- Alerts
- Administration
- Inventory

Defining Custom Insight Admin Privileges

As described above, Policy Manager provides a default Read-only Administrator. The default sets of admin privileges cannot be modified. When a different set of admin privileges is needed (for example, if you require different admin privileges for the Report module than the admin privileges defined for the other Insight modules), you must create a new admin privileges administrator.

Insight privileges can be defined from two locations:

- Operator Profiles in Policy Manager Guest

- Admin Privileges in Policy Manager

  To define custom admin privileges for Insight:

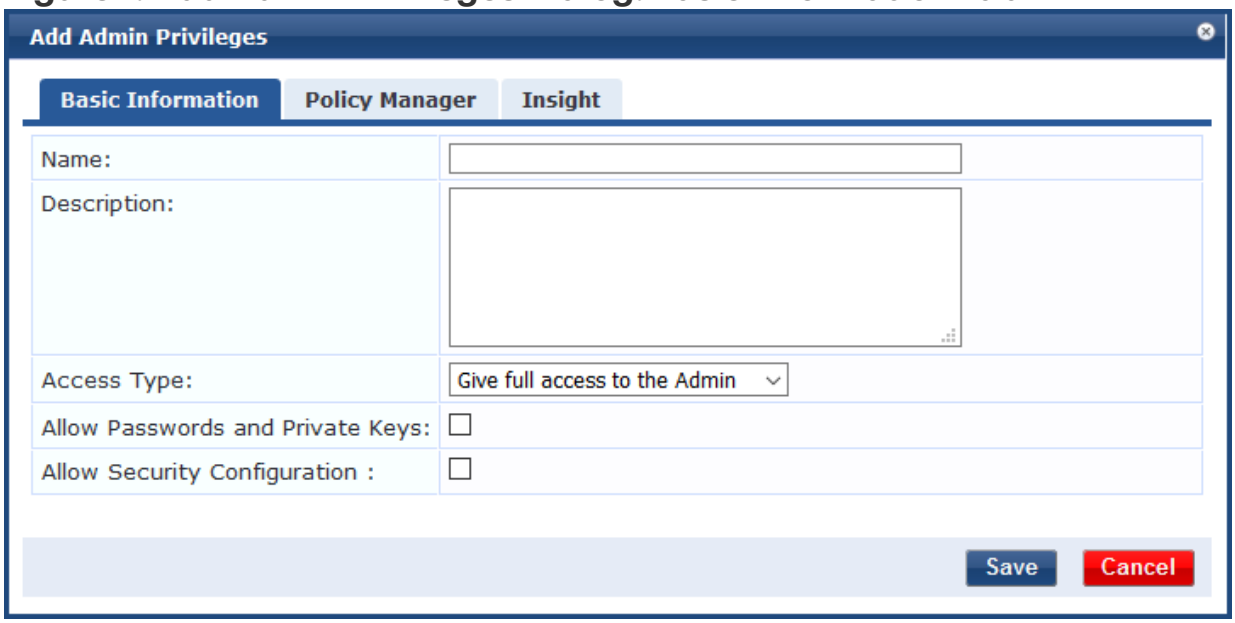
  1. Navigate to **Administration > Users and Privileges > Admin Privileges**.

     The **Admin Privileges** page opens.

  2. Click the **Add** link.

     The **Add Admin Privileges** dialog opens.

Figure 1. Add Admin Privileges Dialog: Basic Information Tab



  3. Specify the parameters in the **Basic Information** tab as described in Table 1.

Table 1. Add Admin Privileges Parameters: Basic Information Tab

Parameter	Action/Description
Name	Enter the name of the Admin Privileges administrator.
Description	Provide a description of this new admin privileges administrator.
Access Type	Select one of the following Access Typ es: ◦ Give full access to the Admin ◦ Give UI access to the Admin

Parameter	Action/Description
	◦ Give API access to the Admin
Allow Passwords	Select this check box if you want to all ow password access.

Subtopics

Specifying Insight Admin Privileges

Specifying Insight Admin Privileges

About this task

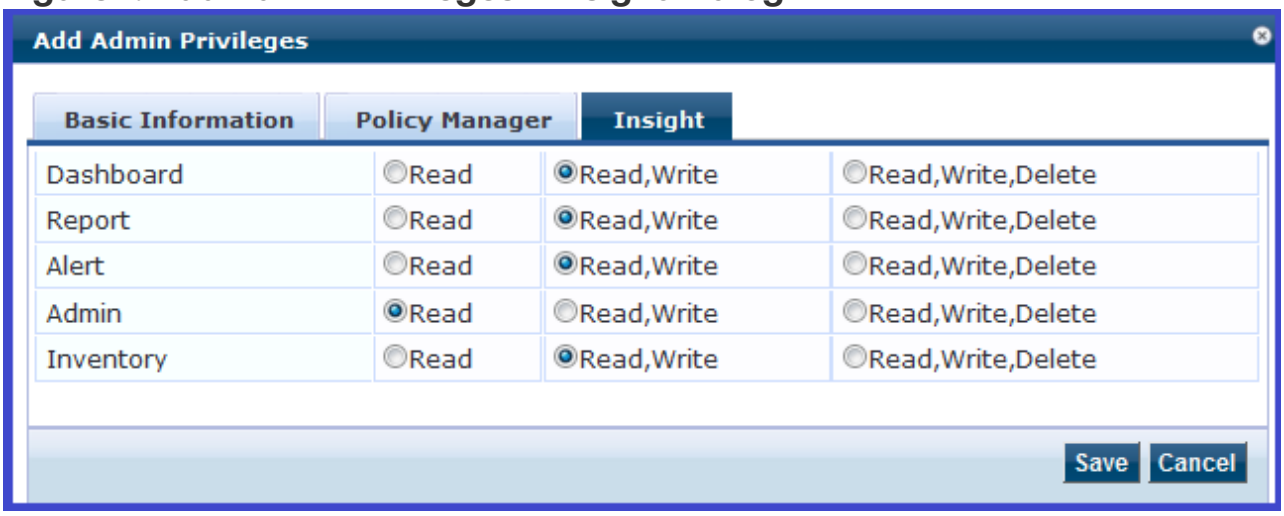
To specify the Insight admin privileges for the new administrator:

Procedure

1. When you complete the **Basic Information** parameters, select the **Insight** tab.

   The **Add Admin Privileges > Insight** dialog opens.

Figure 1. Add Admin Privileges > Insight Dialog



NOTE

You must configure the admin privileges for Policy Manager also, otherwise the changes to the Insight admin privileges cannot be saved.

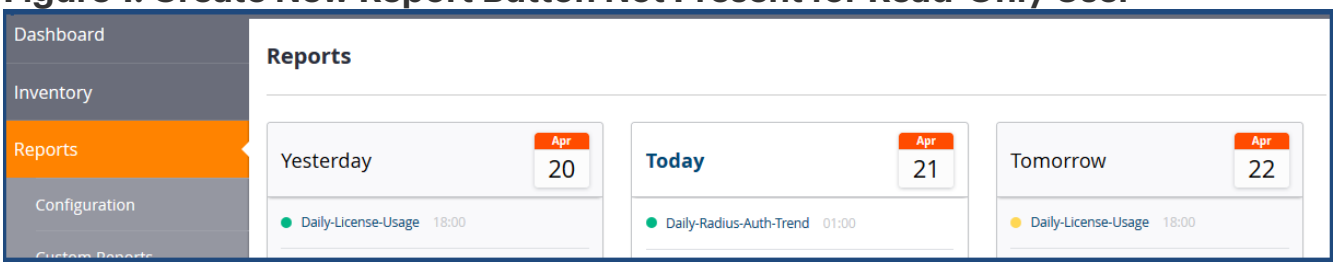
2. Specify the desired admin privileges for each of the Insight modules, then click **Save**.

Insight User Interface Differences for Read-Only Users

When Insight is accessed by a user who has Read-only privileges for all five Insight modules (Dashboard, Reports, Alerts, Inventory and Administration), that user is not allowed to create or delete reports. As shown in Figure 5, when a Read-only administrator logs in to Insight, the **Create New Report** button is not visible.

Likewise, the **Delete** icon on the **Configured Reports** table is not visible for a Read-only administrator.

Figure 1. Create New Report Button Not Present for Read-Only User



Various action buttons, icons, and so on throughout the Insight user interface are shown only to users who are allowed to execute the actions provided by their admin privilege level.

Command Line Interface

Refer to the following sections to perform configuration tasks using the Policy Manager Command Line Interface (CLI):

- Cluster Commands
- Configure Commands
- Misscellaneous Commands
- Network Commands
- Service Commands
- Show Commands
- SSH Timed Account Lockout Commands

- System Commands

Subtopics

Cluster Commands
Configure Commands
Miscellaneous Commands
Network Commands
Service Commands
Show Commands
System Commands

Cluster Commands

Subtopics

cluster diagnostics
cluster drop-subscriber
cluster list
cluster make-publisher
cluster make-subscriber
cluster set-communication-mode
cluster reset-database
cluster set-cluster-passwd
cluster set-standby-publisher
cluster sync-cluster-passwd

`cluster diagnostics`

Syntax

```
cluster diagnostics
-s [-6]
-c [-6]
-p
-d
-r
-q
-i
-t
```

Description

Use the cluster diagnostics command to run diagnostics on a Policy Manager cluster. This command supports both IPv4 and IPv6 management port addresses and uses port number 7432 to collect diagnostics data. Cluster diagnostics for both IPv4 and Ipv6 environments also include metrics for how much time was taken to establish a database connection to the publisher, and how much time was taken for HTTPS API calls to the publisher. These Publisher database connection check and HTTPS connection check to host duration metrics are in hour:min:sec.microsec format.

The command syntax for an IPv4 management address is:

- On subscriber: **appadmin# cluster diagnostics -c**
- On publisher: **appadmin# cluster diagnostics -s**

Parameter	Description
`-s`	Starts server using management IPv4 address for cl uster diagnostics.
`-c`	Starts client for cluster diagnostics with an IPv4 add ress..
`-6`	When used with the **- s** parameter, starts cluster dia gnostics on a server using a management IPv6 add ress. When used with the **- c** parameter, starts clust er diagnostics on a client using a IPv6 address.
`-p`	Runs pgmetrics.
`-d`	Specifies the database on which to run pgmetrics o r reset statistics.
`-r`	Resets database statistics.
`-q`	Find and terminate queries.
`-i`	Use with the - q flag to view or terminate a query by its process ID (PID).
`-t`	Use with the - q flag and a time interval (for examp le, **10 minutes**) to display queries running longer th an the specified time frame. The user then can term inate these queries by specifying their PIDs, or can

Parameter	Description
	enter the **all** option at the CLI prompt to terminate a ll of the queries.

NOTE

Database diagnostics initiated on the server using pgmetrics may show idle queries that hold locks on public tables in the server's tipsdb database. These may delay or block a make-subscriber operation. Use the cluster diagnostics -p -d tips db command to verify the queries, then use the DbRecoverFromHungQueries script to resolve them.

Example

```
[appadmin]# cluster diagnostics -c 192.0.2.21
Enter Cluster Password for 192.0.2.21
Throughput 130914.19 kbps/sec
Configured MTU:     1500
Ping Lagency : 0.323333333333 ms
Publisher database connection check:  [OK]
Publisher database connection time (HH:MM:SS.µs): 0:00:00.018555
Verify HTTPS connection to host - 192.0.2.21
HTTPS connection check to host :   [OK]
Publisher API connection time (HH:MM:SS.µs): 0:00:00.274626
MTU check for 1400 payload size :  [OK]
[appadmin]# cluster diagnostics -q -t 1 minutes
The following 1 queries were found within the time interval provided
PID         Query
1950565     SELECT * from ( SELECT timestamp,ct,(SELECT text '192.0.2.1')
as server ip FROM dblink ('host=192.0.2.1 port=5432 dbname=tip
sLogDb user=appuser password=123456789', 'SELECT date
trunc(''DAY''Tl.timestamp AT TIME ZONE INTERVAL''+05:30'')as date,
count(T1.id) as count
FROM tips dashboard summary Tl WHERE ((((T1.id != ''0'''))) AND
(Tl.timestamp >=''2021-12-22 00:00:00.000+530'') AND (Tl.timestamp <=
'2021-12-29 23:09
:24:944+530''))GROUP BY date ORDER BY date DESC' ) as tmpt ( timestamp
timestamp with time zone, ct int) as t ORDER BY timestamp DESC
Type "all" to terminate all found queries, or the query
Type "exit" to end query termination.
all
Terminated PG backend 1950565 based on user input
```

Command History

Version	Modification
Policy Manager 6.10.4	The - q, - i, and - t flags are introduced.
Policy Manager 6.10.0	Command Introduced

cluster drop-subscriber

Syntax

```
cluster drop-subscriber [-f] [-i <IP address>] -s
```

Description

Use the **drop-subscriber** command to remove a specific subscriber node from a cluster.

NOTE

The time taken by this operation can be longer than the default CLI Session Idle Timeout of 10 min, and depends on network latency, the size of the configuration database, and the load on the Publisher node. Consider increasing this timeout to an adequately high value to ensure the cluster operation does not terminate due to the CLI Session Idle Timeout.

NOTE

This command does not support Stateless Address Auto-configuration (SLAAC) IPv6 addresses.

Parameter	Description
`-f`	Enter the **- f** parameter to force Policy Manager to drop even the nodes that are down.
`-i <IP Address>`	Specify the Management IP address of the node. If t his IP address is not specified and the current node is a subscriber, Policy Manager drops the current no de.

NOTE

Parameter	Description
	The IP address of the subscriber to be dropped must be passed in the correct format. The IP address format depends on the cluster communication mode: If the mode is ipv4, use the subscriber's IPv4 address, otherwise if the mode is ipv6, use the subscriber's IPv6 address. To verify which cluster communication mode is configured, use the cluster list command.
`-s`	Restricts resetting the database on the dropped node. By default, Policy Manager drops the current node—if it's a subscriber—from the cluster.

Example

The following example removes the IP address 192.0.2.1 from the cluster:

```
[appadmin]#   cluster drop-subscriber -f -i 192.0.2.1 -s
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

cluster list

Syntax

```
cluster list
```

Description

Use the **cluster list** command to list all the cluster nodes, and show if cluster high availability is enabled or disabled. The **cluster list** also displays the cluster communication mode (IPv4 or IPv6), and lists whether a standby publisher and failover wait time has been defined. The output also includes the replication status, and replication last update timestamps.

Example

The following example lists the nodes in a sample cluster:

```
[appadmin@pd-190674]# cluster list

Cluster Communication Mode : ipv4

Cluster high-availability : DISABLED

+-------------+-------------+-----------+-------------+---------------------------+--------------------------+----------+
| MGMT IP     |   MGMT IPV6 |   DATA IP |   DATA IPV6 | TYPE                      | REP_LAST_UPDATE_TS       | STATUS   |
+=============+=============+===========+=============+===========================+==========================+==========+
| 10.0.00.000 |             |           |             | Publisher   [local machine] |                          | ENABLED  |
+-------------+-------------+-----------+-------------+---------------------------+--------------------------+----------+
| 10.0.00.000 |             |           |             | Subscriber                | 2022-06-03 15:26:59-07:00 | ENABLED  |
+-------------+-------------+-----------+-------------+---------------------------+--------------------------+----------+
REP_LAST_UPDATE_TS - Last timestamp when Replication delay was updated
```

Command History

Version	Modification
Policy Manager 6.11.0	New table format for command output examples
Policy Manager 6.10.0	Command description updated

`cluster make-publisher`

Syntax

```
cluster make-publisher
```

Description

Use the **cluster make-publisher** command to promote a specific subscriber to be the publisher in the same cluster.

> **NOTE**
> Before you promote a subscriber to publisher, add the HTTPS server certificate of the subscriber to the Trust list and ensure sure all the servers in the cluster have this certificate in the Trust list. This step is not required if the HTTPS server certificates for all the servers in the cluster are signed by a certificate authority (CA)

> **NOTE**
> When running this command, do not close the shell or interrupt the command execution.

Example

The following example promotes a subscriber to publisher status:

```
[appadmin]#   cluster make-publisher
******************************************************
* WARNING: Executing this command will promote the     *
* current machine (which must be a subscriber in the   *
* cluster) to the cluster publisher. Do not close the  *
* shell or interrupt this command execution.           *
******************************************************
Continue? [y|n]: y
To continue the make-publisher operation, enter y.
```

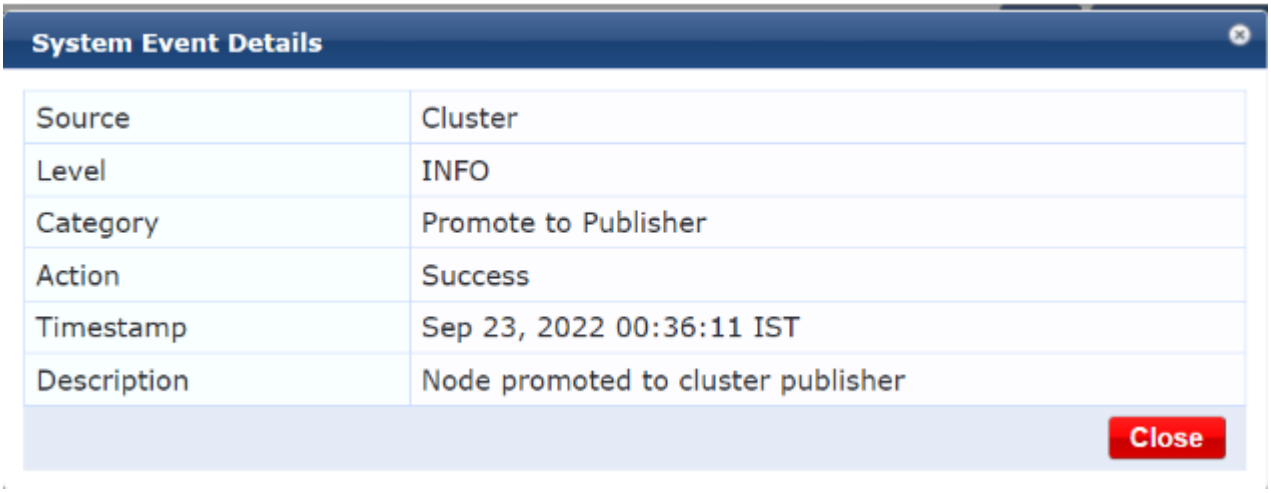
Example

The following is a ClearPass 6.11 example of the make-publisher operation:

```
[appadmin]# cluster make-publisher
******************************************************
* WARNING: Executing this command will promote the     *
* current machine (which must be a subscriber in the   *
* cluster) to the cluster publisher.                   *
*                                                      *
* If the original publisher is not available,          *
* all applicable licenses will be deactivated.         *
* Please contact support to reactivate the licenses.   *
*                                                      *
* Do not close the shell or interrupt this command     *
```

```
* execution.                                                  *
*************************************************************
Continue? [y|n]:
INFO - Check if current node is Publisher
INFO - Starting failover checks
INFO - Check connectivity of current node with management IP 10.1.1.1
INFO - Check connectivity of current node with management IP 10.1.1.2
INFO - Check connectivity of current node with management IP 10.1.1.3
INFO - Validate membership of promotion candidate(currect node) in cluster
INFO - Failover checks passed. Continue with failover
INFO - Sync current node with publisher
INFO - Check promotion of current node to Publisher
INFO - Promote checks passed. Promote current node to cluster
INFO - Transferring ownership from node with management IP=10.1.1.1
INFO - Switch replication on on 10.1.1.3 to follow 10.1.1.2
INFO - Old Publisher node to follow new Publisher as subscriber
INFO - Rejoin 10.1.1.1 to follow 10.1.1.2
INFO - Database service restarted on old publisher with node ID 1
INFO - Promote publisher operation completed
[appadmin]#
```

Once the make-publisher operation completes successfully, system events are logged in the event viewer.



Similarly, system events and UI messages are logged for failures in the make-publisher operation.

Command History

Version	Modification
Policy Manager 6.11.0	New table format for command output

Version	Modification
Policy Manager 6.10.0	Command description updated

`cluster make-subscriber`

Syntax

```
cluster make-subscriber -i <ip-address> [-l] [-b] [-V]
```

Description

Run the **cluster make-subscriber** command on a standalone publisher to make the standalone node a subscriber node and add it to the cluster.

> **NOTE**
> This command does not support Stateless Address Auto-configuration (SLAAC) IPv6 addresses.

Parameter	Description
`-b`	Instructs Policy Manager to skip making a backup of the publisher before it is made a subscriber.
`-i <ip-address>`	Specify the publisher's IP address. **NOTE** This parameter allows both IPv4 and IP v6 addresses.
`-l`	Restores the local log database after this operation. This parameter is optional.
`-V`	Instructs Policy Manager to *not* verify the publisher certificate.

Keep the following caveats in mind when adding a subscriber:

- All the application licenses on this server will be removed. To add and reactivate these application licenses, contact Aruba Support—navigate to **Administration > Support > Contact Support** for contact information.

- When an IPsec tunnel is already established between two ClearPass appliances, trying to add one of the appliances as a subscriber to the other fails. Users should be aware that ClearPass does not support adding an appliance to a cluster as a subscriber while an IPsec tunnel already exists between it and the publisher.
- Policy Manager indicates errors during the Make-Subscriber action if either of the following conditions are present:
- The certificate chain used is not present on both systems for the HTTPS and database certificates.
- An IP address is not included in the database certificate's subject or the *Subject Alternative Name* (SAN) field.

Example

The following example converts the node with IP address 192.0.2.1 to a subscriber node, and restores the local log database:

```
[appadmin]#  cluster make-subscriber -i 192.0.2.1 -l
```

Example

The following example shows the ClearPass 6.11 make subscriber command:

```
Enter Publisher Password
Setting up local machine as a subscriber to 10.1.1.1
INFO - Check publisher connection passed
INFO - Local checks before adding subscriber passed
INFO - 10.1.1.1 - Subscriber node added successfully for host=2010-844
INFO - Subscriber node entry added in publisher
INFO - Backup databases for AppPlatform
INFO - Backup databases for PolicyManager
INFO - Backup Server config
INFO - Stopping services
INFO - Subscriber password changed
INFO - Setup subscriber replication
INFO - Check if publisher database can be cloned on current node
INFO - Subscriber clone checks passed
INFO - Prepare subscriber for publisher database clone
INFO - Prior database configuration cleared
INFO - Clone publisher database
INFO - Publisher database successfully cloned
INFO - Register subscriber node for replication
INFO - Retaining local node certificate
INFO - Subscriber replication and node setup complete
INFO - Notify publisher that adding subscriber is complete
```

```
INFO - Subscriber added successfully
INFO - Restarting Policy Manager admin server
[appadmin@c2010-844]#
```

Some of the messages for the make subscriber operation have changed. The new meaning of these messages is as follows:

- INFO - Check if the publisher database can be cloned on the current node - database connection and permissions check made from subscriber to publisher node to ensure publisher's database can be cloned and restored on subscriber.
- INFO - Prior database configuration cleared - clear subscriber's old config data. This is similar to the message - Dropped existing databases for Policy Manager stage in older versions.
- INFO - Clone publisher database - clone publisher's config db and download to subscriber node.
- INFO - Register subscriber node for replication - add subscriber to cluster management system.

Command History

Version	Modification
Policy Manager 6.11.0	New table format for command output
Policy Manager 6.10.0	Command description updated

cluster set-communication-mode

Syntax

```
cluster set-communication-mode <ipv4/ipv6>
```

Description

This command sets the cluster communication mode to either IPv4 or IPv6.

Cluster configuration operations support IPv4 or IPv6 communication depending on the defined Cluster Communication Mode.

```
[appadmin@pub_612]# cluster

 diagnostics                    Check network and database connectivity between cluster nodes
 drop-subscriber                Remove subscriber node from the cluster
 list                           List cluster nodes
 make-publisher                 Make this node publisher
 make-subscriber                Make this node subscriber
 reset-database                 Resets the local database and erases all the configuration
 set-cluster-passwd             Change the cluster password
 set-communication-mode         Set cluster communication mode to IPv4 or IPv6
 set-standby-publisher          Set standby publisher for the cluster
 sync-cluster-passwd            Synchronize the cluster password with Publisher

[appadmin@pub_612]# sh-4.4#
```

If set to IPv6, cluster replication, database connections and internal API calls specific to the cluster operations listed above will use an IPv6 address, or otherwise use IPv4.

The default Cluster Communication Mode depends on the IP address configured on the node at the time of an upgrade or install. If the node has only an IPv6 address, the default Cluster Communication Mode will be IPv6.

In case both IPv4 and IPv6 addresses are configured, or if only an IPv4 address is configured, the default Cluster Communication Mode is IPv4.

Parameter	Description
Cluster Communication Mode	IPv4 or IPv6

Example

The following example sets the cluster communication mode to IPv6:

```
[appadmin]# cluster set-communication-mode ipv6
```

Command History

Version	Modification
ClearPass Policy Manager 6.9.0	Command Introduced

cluster reset-database

Syntax

```
cluster reset-database
```

Description

Use the **reset-database** command to reset the local database and erase its configuration.

> **NOTE**
> When running the reset-database command, do not close the shell or interrupt the command execution.

> **NOTE**
> Once the cluster reset-database command is complete, the login banner message within the CLI is updated.

> **NOTE**
> Some hardware appliances may take more than 20 minutes to complete a make-subscriber operation. If your particular appliance takes more than 20 minutes to complete a make subscriber operation, increase the timeout to a higher value such as 200 or 300 minutes.

Example

The following example resets the database:

```
[appadmin]#  cluster reset-database
**********************************************************
* WARNING: Running this command will erase the Policy Manager    *
* configuration and leave the database with default     *
* configuration. You will lose all the configured data. *
* Do not close the shell or interrupt this command       *
* execution.                                            *
**********************************************************
Continue? [y|n]: y
To continue the reset-database operation, enter y.
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

cluster set-cluster-passwd

Syntax

```
cluster set-cluster-passwd <password>
```

Description

Use the **cluster set-cluster-passwd** command to change the cluster password on each node in the cluster. You can only issue this command from the publisher.

Allowed characters for the cluster password include letters A through Z (both capitalized and lower case), numbers 0 through 9, and the following special characters:

{ } & ! @ # $ % ^ ? _ = . , + < > - *

The following two password examples meet the strength requirements for length and for combinations of uppercase and lowercase letters, numbers, and special characters:

- G!e#2-YR@w%Bu&o
- A6c@D&e7*g^H7Y3!6

The following two password examples do not meet the strength requirements, because the first one has no lowercase letters and the second one has no uppercase letters:

- ABCD1ABCD1ABCD@
- 1234a1234b1234c@

NOTE
Setting the cluster password changes the **appadmin** password for each node in the cluster.

Parameter	Description
`<password>`	New password for the cluster.

Example

The following example changes the cluster password on the publisher:

```
[appadmin]#  cluster set-cluster-passwd
cluster set-cluster-passwdContinue? [y|n]:   y
Enter Cluster Passwd: newpassword.123
Re-enter Cluster Passwd: newpassword.123
INFO - Password changed on local (publisher) node
Cluster password changed
```

Command History

Version	Modification
Policy Manager 6.11.3	Cluster password requirements updated
Policy Manager 6.10.0	Command Introduced

cluster set-standby-publisher

Syntax

```
cluster set-standby-publisher [-i <IP Address>]|[-t <Failover wait time>]
```

Description

Use the **cluster set-standby-publisher** command to set a standby publisher for cluster high availability. You may only issue this command from the publisher.

NOTE

If a cluster is configured with a standby publisher, add the HTTPS server certificate of the standby publisher to the Trust list and ensure sure all the servers in the cluster have this certificate in the Trust list. This step is not required if the HTTPS server certificates for all the servers in the cluster are signed by a certificate authority (CA)

Parameter	Description
- i <*IP Address*>	Management IP Address of the server to be configured as s tandby publisher. **NOTE** This parameter accepts both IPv4 and IPv6 add resses.
- t <Failover wait time>	Specify the time (in minutes) that the standby publisher mu st wait before it assumes the role of publisher after the prim ary publisher becomes unreachable. This parameter preven ts the standby publisher from taking over when the publish

Parameter	Description
	er is temporarily unavailable during a restart. The default fail over wait time is **10 minutes**.

Example

The following example defines a standby publisher and sets the failover wait time to five minutes:

```
[appadmin]# cluster set-standby-publisher -i 10.21.4.33 -t 5
failover time = 5
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

cluster sync-cluster-passwd

Syntax

```
cluster sync-cluster-passwd
```

Description

Use the **cluster sync-cluster-passwd** command to synchronize the cluster (**appadmin**) password currently set on the publisher with all the subscriber nodes in the cluster.

NOTE

Synchronizing the cluster password changes the **appadmin** password for all the nodes in the cluster.

Example

The following example synchronizes the cluster password:

```
[appadmin]#  cluster sync-cluster-passwd
Continue? [y|n]:   y
Enter Password: password.123
Re-enter Password: password.123
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

Configure Commands

Subtopics

configure date
configure dns
configure fips-mode
configure hostname
configure ip
configure ip6
configure mtu
configure port
configure timezone
configure central-communication

configure date

Syntax

```
configure date -p <ntp_server1> [-a <key-index> -v <key-value> -t
<encryption-type>] [-s <ntp_server2> [-a <key-index> -v <key-value> -t
<encryption-type>]] [-z <timezone>]
```

Description

Use the **configure date** command to specify the cluster's primary and secondary NTP (Network Time Protocol) servers, the key index, key value, encryption type, and optionally, the time zone for the publisher.

NOTE

The Audit Viewer (**Monitoring > Audit Viewer**) tracks NTP configuration changes.

The following table describes the parameters for the **configure date** command:

Parameter	Description
`-p <ntp_server1>`	Specify the primary NTP server name or IP address . Policy Manager can support up to five NTP serve rs; one primary server, and up to four secondary se rvers. Users should be aware that ClearPass can tim e sync with any of the configured NTP servers in any order. It does not need to prefer the primary NTP se rver first and then the secondary as per configuratio n. **NOTE** You can specify a destination node wit h an IPv6 address enabled.
`-s <ntp_server2>`	Use this parameter create a list of up to four second ary NTP servers. For example, **configure date - p us .pool.ntp.org - s ntp - b.nist.gov - s ntp - c.colora do.edu** When Policy Manager is in CC mode, you must confi gure a primary server and at least two secondary se rvers. Failure to define at least two secondary serve rs will trigger a warning message. **NOTE** Do not use the **- A** and **- s** parameters together in a single command, as this c an trigger an **Invalid Syntax** error mess age.
`-A <ntp_server3>`	Add an additional NTP server name or IP address to an existing list of secondary NTP servers. **NOTE** Do not use the **- A** and **- s** parameters together in a single command, as this c an trigger an **Invalid Syntax** error mess age.
`-a <key-index>`	The **Key Index** (also referred to as the **Key ID**) is a nu mber that specifies the index for key values.

Parameter	Description
	The **key - index** value can be from 1 to 65534 inclusive. Typically an NTP client and server have to trust the same key index and key value pair for authentication to succeed.
- v <*key - value*>	*The Key Value* is a form of shared secret, which both the client and server use for authenticating NTP messages. The **Key Value** can be: • Up to 20 - character printable ASCII string • Up to 40 - character hex value When entering an ASCII string for the **Key Value**, note that it *cannot* contain the following characters: • **&** (ampersand) • **;** (semicolon) • ` (grave accent) • **|** (pipe) • **<** (left angle bracket) • **>** (right angle bracket) • **(** (left parenthesis) • **)** (right parenthesis) Finally, the **Key Value** ASCII string must start and end with one of the following characters: • **-** (hyphen) • **'** (apostrophe) • **"** (quote)
- t <*encryption - type*>	Policy Manager supports SHA1as the **Encryption Type.**
- z <*timezone*>	Specify the time zone on the publisher. To view the list of supported time zones, enter **show all - timezones**. This field is optional.

Examples

The following example configures the *key-index*, *key-value*, and *encryption type* for the primary and secondary NTP servers:

```
[appadmin]#  configure date -p ntp1.cppm.main -a 24 -v cp1234567890 -t SHA
-s ntp2.cppm.main -a 16 -v cp53.56 -t SHA1
```

This example synchronizes with the primary NTP server. Note that in this example, the *key-value* is a hex code. Using a hex code for the *key-value* is supported only in the CLI, not in the user interface.

```
[appadmin]# configure date -p ntp1.cppm.main -a 96 -v
6bf60ca1876b57248311aa07c7783d391be95d6c -t SHA1
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure dns

Syntax

```
configure dns -p <dnsserver> [-s <dnsserver>] [-t <dnsserver>] [-n
<attempts>] [-o <timeout>]
```

Description

Use the **configure dns** command to configure DNS servers. You must specify a minimum of one DNS server; you can specify a maximum of three DNS servers.

Parameter	Description
`-p <dnsserver>`	Define the primary DNS server**,** where **<dnsserver>** i s the IPv4 or IPv6 address of a DNS server.
`-s <dnsserver>`	(Optional) Define the second DNS server, where <**dn sserver >** is the IPv4 or IPv6 address of a DNS serve r.
`-t <dnsserver>`	(Optional) Define the third DNS server, where <**dnss erver >** is the IPv4 or IPv6 address of a DNS server.

Parameter	Description
`-n <attempts>`	(Optional) Specify the maximum number of attempt s. The value can be from 1 to 5, and the default is 2.
`-o <timeout>`	(Optional Specify the timeout duration in seconds. Value can be from 1 to 30. Default is 5 seconds.

Examples

The following example configures a single primary DNS server.

```
[appadmin]# configure dns -p 192.168.14.1
```

The following example configures the primary and secondary DNS servers with the maximum number of attempts set at 5, and with a 30 second timeout. You can configure an IPv6 address as described in this example.

```
[appadmin]# configure dns -p 192.168.14.1 -s 2001:4860:4860::8888 -n 5 -o 30
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure fips-mode

Syntax

```
configure fips-mode [0|1]
```

> **CAUTION**
> Running this command erases the ClearPass Policy Manager configuration settings and returns the database to the default configuration. All configured data will be lost. This command also shuts down all running applications and reboots the system.

Description

Use the **configure fips-mode** command to enable or disable **FIPS** (Federal Information Processing Standard) mode.

Parameter	Description
`0`	To disable **FIPS** mode, enter **0**. Read the caution message carefully before enabling or disabling **FIPS** mode.
`1`	To enable **FIPS** mode, enter **1.**

Example

The following example disables **FIPS** mode:

```
[appadmin]# configure fips-mode 0
*********************************************************************
*                                                                   *
* WARNING: Running this command will erase the Policy Manager      *
* configuration and leave the database with default                 *
* configuration. You will lose all the configured data.             *
*                                                                   *
* This command will also shutdown all applications and reboot       *
* the system.                                                       *
*                                                                   *
* Do not close the shell or interrupt this command execution.      *
*                                                                   *
*********************************************************************
Continue? [y|n]: y
Clicking y in this example disables FIPS mode.
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure hostname

Syntax

```
configure hostname <hostname>
```

Description

Use the **configure hostname** command to configure the host name of the Policy Manager server. When configuring a host name that includes a period character (.), the substring before the first period character must be unique for each device. This is because a hostname field that includes a period character is interpreted to be a Fully Qualified Domain Name (FQDN), in which case the substring before the first period character is the hostname.

NOTE

In Policy Manager

version 6.14.0 and above, hostname validation follows strict RFC standards (such as RFC 952 and RFC 1123), which do not permit an underscore (_) character in hostnames. Although Fully Qualified Domain Names (FQDNs) are sometimes informally considered “free-form,” this flexibility applies only in certain DNS record contexts (e.g., SRV or TXT records) and not when configuring system hostnames within ClearPass. As a result, starting from version 6.14.0, any attempt to configure a hostname or FQDN containing an underscore is rejected as invalid. All hostnames and FQDNs used at the platform level must adhere to RFC-compliant naming conventions, allowing only letters, digits, hyphens (-), and dots (.).

Valid hostname configurations:

- cppm1.arubanetworks.com
- cppm2.arubanetworks.com

Invalid hostname configurations:

- cppm1.santaclara.arubanetworks.com
- cppm1.bangalore.arubanetworks.com

Parameter	Description
`<hostname>`	Host name of the Policy Manager server

Example

The following example configures a hostname:

```
[appadmin]#  configure hostname sun..com
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure ip

Syntax

```
configure ip <mgmt|data> <ip_address> netmask <netmask address> gateway
<gateway address>
```

Description

Use the **configure ip** command to configure the IPv4 address of the management interface or the data interface, netmask, and gateway address.

Parameter	Description
`ip <mgmt|data> <IP address>`	Specify the network interface type: *management port interface* or *data point interface*. **<ip address>** specifies the IPv4 address of the host.
`netmask <netmask>`	Specify the netmask for the IPv4 address.
`gateway <gateway address>`	Specify the IP address for the network gateway.

Example

The following example configures the IP address for the data interface, the netmask for that address, and the gateway address:

```
[appadmin]#  configure ip data 192.168.xx.12 netmask 255.255.255.0 gateway
192.168.xx.1
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure ip6

Syntax

```
configure ip6 <mgmt|data> <IPv6 address>/<PrefixLen> gateway <gateway
address>
```

Description

Use the **configure ip6** command to configure the IPv6 management or data interface, IPv6 address/PrefixLength, and the gateway address.

Policy Manager uses Stateless Address Auto-Configuration (SLAAC) to obtain IPv6 addresses when IPv6 router-advertisements are enabled. If a Policy Manager instance is configured with an IPv4 and an IPv6 address (dual-stack), then after upgrade the static IPv6 address is retained and, if IPv6 router-advertisements are enabled, a SLAAC IPv6 address will also be obtained. If no static IPv6 is configured, addresses obtained by SLAAC are shown in the server details on the **Administration > Server Manager > Server Configuration** page.

If a Policy Manager sever's network settings are configured to have only an IPv6 address for the management port, the **Admin Server** service and **Async network service** service connect to an IPv6 address for an external server's hostname, even if that hostname resolves to both an IPv4 and IPv6 address (dual-stack). If the Policy Manager management port has a IPv4 address, then the IPv4 address of the external server will be preferred for connectivity by the **Admin server** and **Async network service** services.

NOTE

During bootstrap configuration, anycast, multicast, and site-local addresses (Unique Local Addresses) are not allowed as management or data port interface address.

Parameter	Description
`<mgmt|data>`	Specify the network interface type: • **mgmt**: management interface • **data**: data interface
`<IP address>`	Enter the IPv6 IP address for the interface.
`<prefixLen>`	Enter the prefix length. The prefix length in IPv6 is t he equivalent of the subnet mask in IPv4.

Parameter	Description
`gateway <gateway address>`	Specify the gateway (IPv4 or IPv6) address .

Example

The following example configures the IPv6 management or data interface, IPv6 address/PrefixLength, and gateway address:

```
[appadmin]# configure ip6 mgmt 2001:100:200:128::157/64 gateway
2001:100:200:128::129
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure mtu

Syntax

```
configure mtu <mgmt|data> <mtu-value>
```

Description

Use the **configure mtu** command to set the MTU (Maximum Transmission Unit) for the management and data port interfaces.

CAUTION

Running this command can cause the Policy Manager server to lose network connectivity.

Parameter	Description
`<mgmt|data>`	Specify the network interface type: • **mgmt**: management interface • **data**: data interface

Parameter	Description
`<MTU value>`	Specify the MTU value in bytes. The default value is **1500** bytes.

Examples

The following example configures the MTU management interface:

```
[appadmin] # configure mtu mgmt 1498
*********************************************************
*                                                       *
* WARNING: Running this command might cause system      *
* to lose network connectivity and may require relogin.*
*                                                       *
*********************************************************
Continue? [y|n]: y
INFO: Restarting network services
INFO: Successfully applied MTU settings
```

The following example configures the MTU data port value:

```
[appadmin]# configure mtu data 1498
*********************************************************
*                                                       *
* WARNING: Running this command might cause system      *
* to lose network connectivity and may require relogin.*
*                                                       *
*********************************************************
Continue? [y|n]: y
INFO: Restarting network services
INFO: Successfully applied MTU settings
```

Use the **show ip** command to display the settings of the MTU management and data port interfaces:

```
[appadmin]# show ip
=========================================
Device Type      :   Management Port
-----------------------------------------
IPv4 Address     :   10.2.xx.86
Subnet Mask      :   255.255.255.0
Gateway          :   10.2.xx.1
IPv6 Address     :   2607:f0d0:1002:0011:0000:0000:0000:0002
Subnet Mask      :   ffff:ffff:ffff:ffff:0000:0000:0000:0000
Gateway          :   2607:f0d0:1002:0011:0000:0000:0000:0001
Hardware Address :   00:0C:29:70:27:40
MTU              :   1499
```

```
==========================================
Device Type      :   Data Port
------------------------------------------
IPv4 Address     :   <not configured>
Subnet Mask      :   <not configured>
Gateway          :   <not configured>
IPv6 Address     :   fe80:0000:0000:0000:020c:29ff:fe70:274a
Subnet Mask      :   ffff:ffff:ffff:ffff:0000:0000:0000:0000
Gateway          :   fe80:0000:0000:0000:020c:29ff:fe70:2741
Hardware Address :   00:0C:29:70:27:4A
MTU              :   1498
==========================================
DNS Information
------------------------------------------
Primary   DNS  :   10.2.xx.3
Secondary DNS  :   10.1.xx.50
Tertiary  DNS  :   10.1.xx.200
==========================================
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure port

Syntax

configure port <direction> <protocol> <port> <action>

Description

This command is used to filter inbound or outbound traffic on a selected port.

Parameter	Description
<direction>	Specify the access control rule direction. Allowed values are: • **input**

Parameter	Description
	• **output**
`<protocol>`	Specify the access protocol. Allowed values are: • **tcp** • **udp**
`<port>`	Specify the port number (1 - 65535)
`<action>`	Select filtering action Allowed values are: • **accept** • **reject**

Example

The following example configures the Policy Manager server to block SSH conection requests to the Policy Manager server by rejecting SSH connections to port 22.

```
[appadmin]#  configure port input tcp 22 reject
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

`configure timezone`

Syntax

```
configure timezone
```

Description

Use the **configure timezone** command to interactively configure the time zone. The Policy Manager command-line interface prompts you to enter a continent or ocean, a country, and for countries with more than one time zone, also the region where your Policy Manager server is located.

Example

The following example interactively configures the time zone for a Policy Manager server in the United States/Pacific time zone:

```
[appadmin]#  configure timezone
**********************************************************************
*                                                                    *
* WARNING: When the command is completed Policy Manager services *
* are restarted to reflect the changes.                              *
*                                                                    *
**********************************************************************
Continue? [y|n]: y
Please identify a location so that time zone rules can be set correctly.
Please select a continent or ocean.
1) Africa
2) Americas
3) Antarctica
4) Arctic Ocean
5) Asia
6) Atlantic Ocean
7) Australia
8) Europe
9) Indian Ocean
10) Pacific Ocean
11) quit
#? 2
Please select a country.
1) Anguilla              19) Dominican Republic    37) Peru
2) Antigua & Barbuda     20) Ecuador               38) Puerto Rico
3) Argentina             21) El Salvador           39) St Barthelemy
4) Aruba                 22) French Guiana         40) St Kitts & Nevis
5) Bahamas               23) Greenland             41) St Lucia
6) Barbados              24) Grenada               42) St Maarten (Dutch)
7) Belize                25) Guadeloupe            43) St Martin (French)
8) Bolivia               26) Guatemala             44) St Pierre & Miquelon
9) Brazil                27) Guyana                45) St Vincent
10) Canada                28) Haiti                 46) Suriname
11) Caribbean NL          29) Honduras              47) Trinidad & Tobago
12) Cayman Islands        30) Jamaica               48) Turks & Caicos Is
13) Chile                 31) Martinique            49) United States
14) Colombia              32) Mexico                50) Uruguay
15) Costa Rica            33) Montserrat            51) Venezuela
16) Cuba                  34) Nicaragua             52) Virgin Islands (UK)
17) Curaçao               35) Panama                53) Virgin Islands (US)
```

```
18) Dominica                36) Paraguay
#? 49
Please select one of the following time zone regions.
1) Eastern (most areas)                16) Central - ND (Morton rural)
2) Eastern - MI (most areas)           17) Central - ND (Mercer)
3) Eastern - KY (Louisville area)      18) Mountain (most areas)
4) Eastern - KY (Wayne)                19) Mountain - ID (south); OR (east)
5) Eastern - IN (most areas)           20) MST - Arizona (except Navajo)
6) Eastern - IN (Da, Du, K, Mn)        21) Pacific
7) Eastern - IN (Pulaski)              22) Alaska (most areas)
8) Eastern - IN (Crawford)             23) Alaska - Juneau area
9) Eastern - IN (Pike)                 24) Alaska - Sitka area
10) Eastern - IN (Switzerland)          25) Alaska - Annette Island
11) Central (most areas)                26) Alaska - Yakutat
12) Central - IN (Perry)                27) Alaska (west)
13) Central - IN (Starke)               28) Aleutian Islands
14) Central - MI (Wisconsin border)     29) Hawaii
15) Central - ND (Oliver)
#? 21
The following information has been given:
United States
Pacific
Therefore TimeZone='America/Los_Angeles' will be used.
Local time is now:      Tue Jun  8 15:19:12 PDT 2021.
Universal Time is now:  Tue Jun  8 22:19:12 UTC 2021.
Is the above information OK?
1) Yes
2) No
#?1
```

Related Commands

Command	Description
show all - timezones	View all available time zones.
show timezone	Interactively configure the time zone.

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

configure central-communication

Syntax

```
configure central-communication <enable - 1> <disable - 0>
```

Description

Use the **configure central-communication** command to enable the option to integrate with the New Central interface of HPE's Aruba Networking Central. When this command is enabled, the ClearPass Policy Manager UI displays an option to allow Policy Manager to select either the existing Central interface or the New Central interface. Only one interface can be selected for integration with Policy Manager at any one time.

Parameter	Description
`1 <enable>`	When enabled, the Policy Manager UI displays a option to select either the existing HPE Aruba Networking Central interface or the HPE Aruba Networking New Central interface as the Policy Manager integration option.
`0 <disable>`	When disabled, the Policy Manager UI does not display the Policy Manager integration options.

Examples

The following example enables central communication:

```
[appadmin]# configure central-communication 1
Central Communication Mode is Enabled.
```

The following example disables central communication:

```
[appadmin]# configure central-communication 0
Central Communication Mode is Disabled.
```

Command History

Version	Modification
Policy Manager 6.11.10	Command Introduced

Miscellaneous Commands

Subtopics

ad auth
ad netjoin
ad netleave
ad passwd-server
ad testjoin
alias
backup
dump certchain
dump logs
dump servercert
exit
help
krb auth
krb list
ldapsearch
quit
restore
ssh lockout

`ad auth`

Syntax

```
ad auth -u <username> -n <NetBIOS domain name>
```

Description

Use the **ad auth** command to authenticate the user against Active Directory. This command manually checks against Active Directory to indicate whether or not a username and password are valid.

Parameter	Description
`-u <username>`	Specifies the username of the authenticating user. T his is a mandatory parameter.
`-n <domain NetBIOS name>`	Specifies the domain name. This field is optional.

Description

```
[appadmin]#  ad auth -u jbrown -n sanfranedu
```

You are prompted to enter the password. If the username and password you provide in this command are correct, the following message is displayed:

```
INFO - NT_STATUS_OK: Success (0x0)
```

This message indicates that NT LAN Manager (NTLM) authentication (NTLM being the mechanism that Policy Manager uses to authenticate users) has succeeded.

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

ad netjoin

Syntax

```
ad netjoin <domain-controller.domain-name> [domain NetBIOS name] [domain
REALM name] [ou=<object container>]
```

Description

Use the **ad netjoin** command to join the host to the domain.

Parameter	Description
`<domain-controller. domain-name>`	Specify the complete Fully Qualified Domain Name (FQDN) of the domain controller, including its hostn ame.

Parameter	Description
	For example, if **atlas.org** is the Domain FQDN and **DC01.atlas.org** is one of its domain controllers, then this argument would be correctly expressed as **DC01.atlas.org** This field is mandatory.
`[domain NetBIOS name]`	Specify the NetBIOS name of the domain (optional argument). You can specify this argument if the derived NetBIOS name is different from the actual name. This is an optional argument.
`[domain REALM name]`	You can specify this argument if the derived REALM is different from the actual. This is an optional argument.
`[ou=<object container>]`	If the computer account must be created in a different OU, this argument specifies the Object Container . For example '`ou=Domain Computer`' OR '`ou=Domain Computer+Linux Hosts`'. Note the usage of the separator '**+**' to specify the OU hierarchy.

Example

The following example joins the host to the domain:

```
[appadmin]#  ad netjoin DC01.atlas.org.arubanetworks.com
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

`ad netleave`

Syntax

```
ad netleave <domain NetBIOS name> [-f]
```

Description

Use the **ad netleave** command to remove the host from the domain.

Parameter	Description
`<domain NetBIOS name>`	Specifies the host to be joined to the domain. This field is mandatory.
`-f`	Forces the removal of Active Directory domain membership even if the operation fails.

Example

The following example removes the host from the domain **examplecollege.edu**.

```
[appadmin]#  ad netleave examplecollege.edu -f
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

ad passwd-server

Syntax

```
ad passwd-server
   set -n <domain NETBIOS name> -s <Server1> [Server2 Server3 Server4 ...]
   list -n <domain NETBIOS name>
   reset -n <domain NETBIOS name>
```

Description

Use the **ad passwd-server** command to do the following tasks:

- Set the Active Directory password servers.
- List the configured Active Directory password servers.

- Reset the Active Directory password servers.

> **NOTE**
> When an Active Directory password server is updated in the domain server configuration, the RADIUS service is restarted.

Parameter	Description
`set -n <domain NetBIOS name> -s <Server1> [Server2 Server3 Server4 ...]`	Set the password server(s). The **-n** parameter specifies the domain name. The **-s** parameter specifies one or more password server names.
`list -n <domain NetBIOS name>`	List the configured password servers for the specified domain name.
`reset -n <domain NetBIOS name>`	Reset the password servers for the specified domain name.

Example

The following example sets a password server for the domain **examplecollege.edu**.

```
[appadmin]#  ad passwd-server set -n examplecollege.edu -s cppm.campus1
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

ad testjoin

Syntax

```
ad testjoin <domain NetBIOS name>
```

Description

Use the **ad testjoin** command to test if the **ad netjoin** command succeeded. This command also tests whether Policy Manager is a member of the Active Directory domain.

Parameter	Description
`<domain NetBIOS name>`	Specifies the host to be joined to the domain. This field is mandatory.

Example

The following example tests if the **ad testjoin** command succeeded:

```
[appadmin]#  ad testjoin balsamcollege.edu
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

`alias`

Syntax

```
alias <name>=<command>
```

Description

Use the **alias** command to create or remove aliases.

Parameter	Description
`<name>=<command>`	Sets **<name>** as the alias for **<command>**.
`<name>=`	Removes the association.

Examples

This example set the alias **sh** for the **show** command:

```
[appadmin]# alias sh=show
```

This example removes the alias "**sh**":

```
[appadmin]# alias sh=show
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

backup

Syntax

```
backup [-f <filename>] [-c] [-l] [-r] [-e] [-w] [-P]
            [-k]
```

Description

Use the **backup** command to create a backup of ClearPass configuration data. If no arguments are provided, the system automatically generates a filename and backs up the configuration to this file.

NOTE

The time taken by this operation can be longer than the default CLI Session Idle Timeout of 10 min, and depends on network latency, the size of the configuration database, and the load on the system. Consider increasing the timeout to an adequately high value to ensure this operation does not terminate due to the CLI Session Idle Timeout.

Parameter	Description
`[-f <filename>]`	Specify the backup target by defining a file name. If not specified, Policy Manager automatically generates a file name.
`-c`	Back up Policy Manager configuration data.
`-l`	Back up Policy Manager session log data.

Parameter	Description
`-r`	Back up Insight data.
`-e`	Back up Policy Manager extensions.
`-P`	Do not include password fields from the configurati on database in the backup.
`-w`	Back up only the most recent records from the log d atabase (the last one week).

Example

```
[appadmin]#  backup -f PolicyManager-data.tar.gz
Continue? [y|Y]:  y
```

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced
Policy Manager 6.11.0	Introduced - k flag for backup file password protection

dump certchain

Syntax

```
dump certchain <hostname:port-number>
```

Description

Use the **dump certchain** command to view the certificate chain of a SSL-secured server.

Parameter	Description
`<hostname:port-number>`	Specifies the hostname and SSL port number.

Example

The following example dumps the certificate chain of an SSL-secured server:

```
[appadmin]# dump certchain ldap.acme.com:636
```

Example

The following example shows how an input value should be enclosed within double quotes, and the IP address should be enclosed in square brackets when an **IPv6** address and **port number** are provided as input.

```
[appadmin]# dump certchain "[0:0:0:0:0:0:0:1]:443"
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

dump logs

Syntax

```
dump logs
   -f <output-file-name> [-s yyyy-mm-dd] [-e yyyy-mm-dd] [-n <days>] [-t
<log-type> [<filter>]]
   -o [-r] [-v] [-q "yyyy-mm-dd HH:MM"] [-u HH:MM] [-i <tac-case-id>] [-m
"<policy-manager-service>"] [-n <days>] [-h]
   -f <output-file-name> -w <scp|sftp> --user=<username> --
password=<password> --host=<hostname> --remote_dir=<directory>
   -h
```

Description

Use the **dump logs** command to remove Policy Manager application log files. If you are working with HPE Networking Support, you can use the **dump logs** command to tag a .tar file of log messages with a date, and server and company information, then upload that file to a log server.

If you use the **dump logs** command to schedule the collection of log files, be aware of the following caveats:

- When you schedule log collection, Policy Manager services collects logs in DEBUG mode. After the log files are collected, the collected services reset from DEBUG mode to their default logging level.

- You can schedule log collection without specifying the debug duration and list of services names. This result in directly collecting the logs at specified date and time.
- If log levels are not yet changed, you can delete the log collection schedule without using the **-z** force delete option.
- You can delete the schedule using the **-z force delete** option only when log levels are changed to DEBUG mode but the log collection has not yet started. Once log collection started, you can't delete the schedule to delete it using the **-z** force delete option, and must wait until log collection completes.
- Optionally enter **-c** followed by the password string to password protect the ZIP file generated for log collection. Packet Capture is only allowed with password protection.

Parameter	Description
`-f <output-file-name>`	Output file to generate with the collected logs
`-s`	Start date for the date range (default is today)
`-e`	End date for the date range (default is today)
`-n`	Define the date range as number of days from today
`-o`	Schedule log collection with other options
`-t`	Type of logs to collect (can be specified multiple ti mes)
`-w <scp|sftp>`	Send collected logs to backup server using the spe cified protocol: • scp: send logs using Secure Copy Protocol (SCP) • sftp: Send logs using Secure File Transfer Protoc ol (SFTP)
`-r`	Delete the schedule for log collection, use - z optio n with - r to try force delete
`-v`	View the schedule for log collection and status of la st log collection
`-q`	Start date and time to set Policy Manager services l og level to DEBUG mode

Parameter	Description
`-u`	Duration to set Policy Manager services log level in DEBUG mode
`-i`	Aruba TAC Case ID. The case ID is typically 10 digits long.
`-m ["<policy-manager-service>"]`	Policy Manager service name to set log level to DEB UG mode Use the following Policy Manager service names wi th the**<policy - manager - service>** parameter. Note that the service names need to be entered inside q uotation marks, for example, "ClearPass Network Se rvices" • ClearPass network services • Syslog client service • TACACS+ server • Admin serve r • Micros Fidelio FIAS • AirGroup notification service • DB change notification server • Async network services • Radius server • RadSec service • Zone cache • Policy server • DB replication service
`--user=<username>`	Specify a user name to authenticate to a remote di rectory and upload log files.
`--password=<password>`	Specify a password to authenticate to a remote dire ctory and upload log files.
`--host=<hostname>`	Specify the host name or IP address of a the server where the log files will be uploaded

Parameter	Description
`--remote_dir=<directory>`	Specify the name of the remote directory where the log files will be uploaded.
`<log-type> [<filter>]`	Specify one of the following log types: • **SystemLogs**: Collects system logs • **PerformanceMetricsLogs**: Collects performanc e metrics logs • **AirGroupLogs**: Collects logs from AirGroup noti fication service • **ClearPassGuestLogs**: Collects logs from Clear Pass Guest application • **ConfigBackup:** Collects configuration backup (without passwords) • **DiagnosticDumps**: Collects diagnostic dumps f rom ClearPass services • **PacketCapture** : Capture packets for a fixed du ration. Default is 60 seconds (set using - d 60). Filter Options • - **a**: Sets Source Port • - **A**: Sets Destination Port • - **b**: Sets Source IP • - **B**: Sets Destination IP • - **d**: Sets number of seconds for packet capture. Default is 60 seconds • - **p**: Sets Protocol for packet capture • - **c**: Sets number of packets to be captured • - **C**:Sets size limit of log file
`-h =`	Print help with the types of logs available

Subtopics

To collect logs for a give date range (in days)

To schedule log collection
To upload log files to a log server to assist with a technical support issue.

To collect logs for a give date range (in days)

```
dump logs -f <output-file-name> [-s yyyy-mm-dd] [-e yyyy-mm-dd] [-n <days>]
[-t <log-type>] [-o <schedule-log-collection>] [-b <file-copy-settings>] [-
h]
where,
-f = the output file to generate with the logs collected
-s = the start date for the date range (default is today)
-e = the end date for the date range (default is today)
-n = use to define the date range as number of days from today
-o = use to schedule log collection with other options
-t = the type of logs to collect (can be specified multiple times)
-b = collect logs to backup server
-h = print help with the types of logs available
```

To schedule log collection

```
dump logs -o [-r] [-v] [-q "yyyy-mm-dd HH:MM"] [-u HH:MM] [-i <tac-case-
id>] [-m "<policy-manager-service>"] [-n <days>] [-h]
where:
-r = delete the schedule for log collection, use -z option with -r to try
force delete
-v = view the schedule for log collection and status of last log collection
-q = start date and time to set Policy Manager services log level to DEBUG
mode
-u = duration to set Policy Manager services log level in DEBUG mode
-i = Aruba TAC Case Id. Should be 10 digits long
-m = Policy Manager service name to set log level to DEBUG mode
-h = Display help information
```

Use the following Policy Manager service names with the [**-m "<policy-manager-service>"]** parameter described in the syntax above.

- ClearPass network services
- Syslog client service
- TACACS+ server

- Admin serve r
- Micros Fidelio FIAS
- AirGroup notification service
- DB change notification server
- Async network services
- Radius server
- RadSec service
- Zone cache
- Policy server
- DB replication service

To upload log files to a log server to assist with a technical support issue.

Syntax

```
[appadmin]#dump logs -f <output-file-name> -w <scp|sftp> --user=username --
password=password --host=hostname --remote_dir=directory
where:
-w <scp|sftp> : Specify protocol to copy logs archive to remote server
--host = Hostname or IP address of the remote server
--user = Username allowed to copy the file to remote server
--password = Password configured for the Username, allowed on the remote
server
--remote_dir = Remote directory to which logs archive will be copied into
tmp folder
```

- This command sends the following log types:
- System logs
- Performance metrics logs
- AirGroup notification service
- Logs from ClearPass Guest application
- Configuration backup (without passwords)

- Diagnostic dumps from ClearPass services
- Logs from all PolicyManager services
- PacketCapture

Examples

The following example uploads log files to a log server to assist with a technical support issue.

```
[appadmin]#dump logs -f tips-policy-manager-logs.tgz -t PolicyManagerLogs
-w SCP --user=arubasupport --password=mypassword --host=192.0.2.112 --
remote_dir=testCase_tgz
```

The following example dumps Policy Manager application log files:

```
[appadmin]#  dump logs –f tips-system-logs.tgz -s 2007-10-06 –e 2007-10-17
–t SystemLogs
```

The following example prints help for the available log types:

```
[appadmin]#  dump logs -h
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

dump servercert

Syntax

```
dump servercert <hostname:port-number>
```

Description

```
[appadmin]#  dump servercert ldap.acme.com:636
```

Use the **dump servercert** command to view the server certificate of an SSL-secured server.

Parameter	Description
`<hostname:port-number>`	Specify the hostname and an SSL port number, separated by a colon. For example, **example.com:636**.

Example

The following example retrieves the server certificate of the specified SSL-secured server:

```
[appadmin]# dump servercert ldap.acme.com:636
```

Example

The following example shows how an input value should be enclosed within double quotes and an IP address should be enclosed in square brackets when an **IPv6** address and **port number** are provided as input.

```
[appadmin]# dump servercert "[0:0:0:0:0:0:0:1]:443"
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

exit

Syntax

```
exit
```

Description

Use the **exit** command to exit the Policy Manager shell.

Example

The following example exits the shell:

```
[appadmin]#  exit
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

help

Syntax

```
help
```

Description

Use the **help** command to display the list of supported Policy Manager commands:

Example

The following example displays the list of supported commands:

```
[appadmin]#  help
ad                      Domain Controller set of commands
alias                   Create aliases
backup                  Backup Policy Manager data
cluster                 Policy Manager cluster related commands
configure               Configure the system parameters
dump                    Dump Policy Manager information
exit                    Exit the shell
help                    Display the list of supported commands
krb                     Kerberos authentication commands
ldapsearch              Search entries in the LDAP repository
network                 Network troubleshooting commands
quit                    Exit the shell
restore                 Restore Policy Manager database
service                 Control Policy Manager services
show                    Show configuration details
ssh                     SSH Lockout related commands
system                  System commands
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

krb auth

Syntax

```
krb auth <user@domain>
```

Description

Use the **krb auth** command to perform a Kerberos authentication against a Kerberos server (such as Microsoft Active Directory).

Parameter	Description
`<user@domain>`	Specify the user name and domain in the format **username@domain.**

Example

The following example performs a kerberos authentication against a kerberos server:

```
[appadmin]#  krb auth mike@example.com
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

krb list

Syntax

```
krb list
```

Description

Use the **krb list** command to list the cached Kerberos tickets.

Example

The following example lists the cached Kerberos tickets:

```
[appadmin]#  krb list
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: laura@EXAMPLE.EDU
Valid starting Expires Service principal
11/06/20 09:07:51 11/06/20 19:07:59 krbtgt/EXAMPLE.EDU@EXAMPLE.EDU
renew until 11/07/20 09:07:51
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

ldapsearch

Syntax

```
ldapsearch -B <user@hostname>
```

Description

Use the Linux **ldapsearch** command to find objects in an LDAP directory. Note that only the Policy Manager-specific command line arguments are listed. For other command line arguments, refer to **ldapsearch** man pages on the Internet.

Parameter	Description
`-B`	Find the bind DN (Distinguished Name) of the LDAP directory.
`<user@hostname>`	Specify the user name and fully qualified domain na me of the host in the format **username@domain**.

Description

The following example finds objects in an LDAP directory:

```
[appadmin]#  ldapsearch -B admin@example.com
**********************************************************
This command provides limited functionality. For *
unsupported query options or complex query executions *
use administration UI. *
**********************************************************
ldap_initialize( ldap://example.com )
Enter LDAP Password: ******
filter: (&(cn=Administrator)(objectclass=*))
requesting: dn
extended LDIF
#
LDAPv3
base <dc=example,dc=com> with scope subtree
filter: (&(cn=Administrator)(objectclass=*))
requesting: dn
#
Administrator, Users, soltest.com
dn: CN=Administrator,CN=Users,DC=example,DC=com
ufn: Administrator, Users, example.com
search reference
ref: ldap://ForestDnsZones.example.com/DC=ForestDnsZones,DC=example,DC=com
search reference
ref: ldap://DomainDnsZones.example.com/DC=DomainDnsZones,DC=example,DC=com
search reference
ref: ldap://example.com/CN=Configuration,DC=example,DC=com
search result
search: 2
result: 0 Success
numResponses: 5
numEntries: 1
numReferences: 3
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

quit

Syntax

```
quit
```

Description

Use the **quit** command to exit the Policy Manager command shell.

Example

The following command quits the shell:

```
[appadmin]# quit
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

restore

Syntax

```
restore [{<user@hostname>:/}|{http://<hostname>/}]<backup-filename> [-l] [-i] [-b] [-c] [-r] [-n|-N] [-s] [-k]
```

Description

Use the **restore** command to restore Policy Manager configuration data from the backup file.

NOTE

The time taken by this operation can be longer than the default CLI Session Idle Timeout of 10 min, and depends on network latency, the size of the configuration database, and the load on the system. Consider increasing the timeout to an adequately high value to ensure this operation does not terminate due to the CLI Session Idle Timeout.

Parameter	Description
<user@hostname>:/	Specify a user and a hostname used with t he <backup - filename> parameter to speci fy the filepath of the restore source.
<hostname>	Specify hostname to be used with the <ba ckup - filename> parameter to specify the f ilepath of the restore source.
<backup - filename>	Name of the backup file to be restored.
- b	Do not backup the current configuration d ata before the restore operation starts.
- c	Restore Policy Manager configuration data.
- l	If it exists in the backup file, restore the Pol icy Manager log database. This field is opti onal.
- i	Ignores version mismatch errors and attem pts data migration. This field is optional.
- k	Enter the command with a - k flag to enter the password required to restore the backu p file.
- n	Retain local node configuration data, such as certificates, after the restore operation (default).
- N	Do not retain local node configuration data after the restore operation.
- r	Restore Insight data if it exists in the backu p.
- s	Restore cluster server/node entries from th e backup file. Node entries are in a disable d state upon restore. This field is optional.

Example

The following example restores Policy Manager configuration data from the backup file:

```
[appadmin]#  restore user@hostname:/tmp/cppm1-backup.tgz -l -i -c -s
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced
Policy Manager 6.11	Introduced k flag to enter the password required to restore the b ackup file

ssh lockout

Syntax

```
ssh lockout
   count <n>
   duration <N mins>
   mode <basic|advanced>
   reset
```

Description

Configure the SSH timed lockout feature. This feature provides an administrator with the ability to configure the number of successive unsuccessful authentication attempts for administrators attempting to authenticate remotely. The SSH timed account lockout feature configuration persists across reboots, updates and upgrades. The account lock status persists across reboots.

When the defined number of unsuccessful authentication attempts has been met, the CLI account is locked and administrators cannot log in to the system via the CLI until one of the following conditions are met:

- The offending remote administrator cannot successfully authenticate until an action is taken by a local administrator (the administrator issues the **ssh unlock** command).
- The offending remote administrator cannot successfully authenticate until a time period defined by the administrator has elapsed.

This feature is node-specific. In a cluster with multiple nodes, SSH timed account lockout must be configured on each node in the cluster.

NOTE

The **cluster reset-database** command does not impact this feature.

NOTE

The SSH timed account lockout feature is disabled by default.

Subtopics

SSH Account Lockout Behavior
SSH Account Lockout Alerts

SSH Account Lockout Behavior

The SSH account lockout feature is disabled by default. To enable SSH account lockout, issue the the **ssh lockout count** or **ssh lockout duration** commands to set the configuration options.

- CLI access via SSH (password-based) authentication is locked on three consecutive failed login attempts.
- If a failed password attempt continues (even after the account is locked), it will not be tracked, and the unlock time will not be shifted. It will be the time of the last failed password attempt (third failure attempt), plus the lock out period.
- Successful password-based SSH logins are rejected during the lockout period.
- Console-based logins are allowed during the lockout period.
- SSH logins via public key methods are allowed during the lockout period.

`SSH Account Lockout Alerts`

Syntax

Alerts for SSH lockout events are logged in to the Event Viewer when any of the following conditions are present:

- SSH lockout configurations are performed
- Account is locked
- Account is unlocked
- Failed SSH login attempts

Parameter	Description
`count`	The maximum failed SSH password login attempts before the account attempting the logins is locked out. Supported values are **1 - 1000**. The default val ue is **5** attempts.
`duration`	The number of minutes account will remain locke d after the account exceeds the maximum number of SSH login attempts. Supported values are **1 - 10 080**. The default value is **15** minutes.
`mode <basic|advanced>`	**mode <basic|advanced>** command sets the mode for this feature. In basic mode, login failures via SS H public key methods are not counted towards the account lockout, while in advanced mode, login fail ures via SSH public key failures are counted towards the account lockout.
`reset`	Resets SSH lockout settings back to the default valu es.

Example

The following displays an example of the **show date** command output:

```
[appadmin]#  show date
Wed Feb 27 13:30039 UTC 2018
```

Related Commands

Command	Description
show ssh	Show ssh timed lockout settings.

Command History

Version	Modification
Policy Manager 6.10.0	Command Introduced

Network Commands

Subtopics

network ip
network ip6
network ping
network ping6
network reset
network traceroute
network traceroute6
network nslookup

network ip

Syntax

```
network ip
   add <mgmt|data|greN|vlanN> [-i <id>] <[-s <SrcAddr>] [-d <DestAddr>]>
[-g <ViaAddr>]
   [ext <mgmt|data> -o <add|del>] del [-i <id>] list   reset
```

Description

Use the **network ip** command to add, delete, or list custom routes to the data or management interface routing table for IPv4 networks. Note that network IP routing commands are disabled for ClearPass cloud deployments (such as deployments hosted in Azure or AWS) because:

- The network IP assignments are managed by the cloud networking vendor.
- The process through which the cloud deployment DHCP and IP assignments are not managed by ClearPass, so there is no way to define static IP addresses

- The routing configuration defined in the cloud deployment will no longer be valid if the system renews its IP address in the next DHCP refresh cycle.

Parameter	Description
`add <mgmt | data| greN | vlanN>`	Add a custom route by specifying the management interface, data interface, the name of the GRE tunn el, or the VLAN number. • <greN>: N identifies the GRE tunnel number rang ing from 1,2,3...N. • <vlanN>: N identifies the VLAN number.
`-i <id>`	Specify the ID of the network IP rule. If this ID is not specified, the system generates an ID automatically. **NOTE** This ID determines the priority in the o rdered list of rules in the routing table.
`-s <SrcAddr>`	Specify the IPv4 address or network. For example, 192.168.xx.0/24 or 0/0 (for all traffic) of traffic origina tor. You must specify only one source IP address. Th is parameter is optional.
`-d <DestAddr>`	Specify the destination IPv4 address or network. Fo r example, 192.168.xx.0/24 or 0/0 (for all traffic). You must specify only one destination IP address. This p arameter is optional.
`-g <ViaAddr>`	Specify the via or gateway IPv4 address through wh ich the network traffic should flow. A valid IP addres s is allowed. This parameter is optional.
`ext <mgmt|data> -o <add| del>`	Allows the addition of custom routes for extension traffic. Specify either the data or management <mg mt|data> interface to define which interface the rou te should be added or deleted using the -o flag.
`del -i <id>`	Delete the network IPv4 route with the specified rou te ID.
`list`	Display a list of all routing rules.
`reset`	Reset the routing table to the factory default setting s. All custom routes are removed.

Subtopics

Example:

Example:

Syntax

The following example adds a custom route:

```
[appadmin]# network ip add data -s 192.0.20/24
INFO - New ip rule created with the id = 20066
```

The following example lists all custom routes:

```
[appadmin]# network ip list
================================================
            IP Rule Information
------------------------------------------------
0:      from all lookup local
10020:  from all to 10.xx.4.0/24 lookup mgmt
10040:  from 10.xx.4.200 lookup mgmt
10060:  from 10.xx.5.200 lookup data
32766:  from all lookup main
32767:  from all lookup default
================================================
```

Related Commands

Command	Description
network ip6	Add, delete, or list custom routes to the data or management int erface routing table for IPv6 networks.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

network ip6

Syntax

```
network ip6
   add <mgmt|data|greN|vlanN> [-i <id>] <[-s <SrcAddr>] [-d <DestAddr>]>
[-g <ViaAddr>]
   del [-i <id>]
   list
   reset
```

Description

Add, delete, or list custom routes to the data or management interface routing table for IPv6 networks. Note that network IP routing commands are disabled for ClearPass cloud deployments (such as deployments hosted in Azure or AWS) because:

- The network IP assignments are managed by the cloud networking vendor.
- The process through which the cloud deployment DHCP and IP assignments are not managed by ClearPass, so there is no way to define static IP addresses
- The routing configuration defined in the cloud deployment will no longer be valid if the system renews its IP address in the next DHCP refresh cycle.

Parameter	Description
`add <mgmt | data| greN | vlanN>`	Add a custom route by specifying the management interface, data interface, the name of the GRE tunn el, or the VLAN number. • <greN>: N identifies the GRE tunnel number rang ing from 1,2,3...N. • <vlanN>: N identifies the VLAN number.
`-i <id>`	Specify the ID of the network IP rule. If this ID is not specified, the system generates an ID automatically. **NOTE** This ID determines the priority in the o rdered list of rules in the routing table.
`-s <SrcAddr>`	Specify the IPv6 address or network. For example, 2001:DB8:441:1020::0/32 or 0/0 (for all traffic) of traf

Parameter	Description
	fic originator. You must specify only one source IP a ddress. This parameter is optional.
`-d <DestAddr>`	Specify the destination IPv6 address or network. Fo r example, 2001:DB8:441:1020::0/32 or 0/0 (for all tr affic). You must specify only one destination IP add ress. This parameter is optional.
`-g <ViaAddr>`	Specify the via or gateway IPv6 address through wh ich the network traffic should flow. A valid IP addres s is allowed. This parameter is optional.
`del -i <id>`	Delete the network IPv6 route with the specified rou te ID.
`list`	Display a list of all routing rules.
`reset`	Reset the routing table to the factory default setting s. All custom routes are removed.

Subtopics

Example:

Example:

Syntax

The following example adds a custom route:

```
[appadmin]# network ip6 add mgmt -i 20067 -s 2001:DB8:441:1020::0/32
INFO - New ip rule created with the id = 20067
```

The following example lists all custom routes:

```
[appadmin]# network ip6 list
================================================
IP Rule Information
------------------------------------------------
0: from all lookup local
220: from all lookup 220
20020: from all to 2001:DB8:441:13a0::5:8/64 lookup mgmt
20040: from 2001:DB8:441:13a0::5:8 lookup mgmt
20060: from 2001:DB8:441:13e1::1:a1 lookup data
```

```
20067: from 2001:DB8:441:1020::/32 lookup mgmt
20099: from all lookup data
32766: from all lookup main
================================================
```

Related Commands

Command	Description
network ip	Add, delete, or list custom routes to the data or management int erface routing table for IPv4 networks.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

network ping

Syntax

```
network ping [-i <SrcIPv6Addr>] [-t] <host>
```

Description

Test the reachability of the host on an IPv4 network

Parameter	Description
`-i <SrcIPv6Addr>`	The originating IPv4 address for the ping. This para meter is optional.
`-t`	Use this parameter to ping indefinitely. This parame ter is optional.
`<host>`	The host to be pinged.

Example

The following example pings an IPv4 network host to test its reachability:

```
[appadmin]#  network ping6 -i 192.0.2.11 -t or.us.example.com
```

Related Commands

Command	Description
network ping6	Test the reachability of the host on an IPv6 network

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

network ping6

Syntax

```
network ping6 [-i <SrcIPv6Addr>] [-t] <host>
```

Description

Test the reachability of the host on an IPv6 network

Parameter	Description
`-i <SrcIPv6Addr>`	The originating IPv6 address for the ping. This parameter is optional.
`-t`	Use this parameter to ping indefinitely. This parameter is optional.
`<host>`	The host to be pinged.

Example

The following example pings an IPv6 network host to test its reachability:

```
[appadmin]#  network ping6 -i f2001:db8:: -t ca.us.example.com
```

Related Commands

Command	Description
network ping	Test the reachability of the host on an IPv4 network

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

network reset

Syntax

```
network reset data|mgmt [v4|v6]
```

Description

Use the **network reset** command to reset the network data and management ports. You can use this command to reset both IPv4 and IPv6 addresses. Before resetting an IPv4 address for the port, ensure than an IPv6 address is set for the port and that the cluster communication mode is set to IPv6. Conversely, before resetting an IPv6 address for the port, ensure that an IPv4 address is set for it and that the cluster communication mode is set to IPv4. Before Policy Manager resets an IPv4 or IPv6 address, it displays the warning message "This command erases network management port configuration and reconfigures the network. This might cause the system to lose network connectivity and require you to log in again."

Parameter	Description
`data`	Reset the network data port.
`mgmt`	Reset the network management port.
`[v4|v6]`	Indicate whether the port has an IPv4 or IPv6 add ress.

Examples

The following example resets the IPv4 network data port.

```
[appadmin]# network reset data v4
```

The following example resets the IPv6 network management port.

```
[appadmin]# network reset mgmt v6
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

network traceroute

Syntax

```
network traceroute <host>
```

Description

Print the route taken to reach the IPv6 network host.

Table 1. Network Traceroute Command Parameters

Parameter	Description
`<host>`	The IPv4 address or FQDN of the network host.

Example

The following example prints the route taken to reach the network host:

```
[appadmin]#  network traceroute6 ca.us.example.com
```

Related Commands

Command	Description
network traceroute6	Print the route taken to reach the IPv6 network host.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

network traceroute6

Syntax

```
network traceroute6 <host>
```

Description

Print the route taken to reach the IPv6 network host.

Table 1. Network Traceroute Command Parameters

Parameter	Description
`<host>`	The IPv6 address of the network host.

Example

The following example prints the route taken to reach the network host:

```
[appadmin]#  network traceroute6 2001:DB8:441:1020::3
```

Related Commands

Command	Description
network traceroute	Print the route taken to reach the IPv4 network host.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

network nslookup

Syntax

```
network nslookup [-q <query-option>] <host>
```

Description

Use the **network nslookup** command to get the IP address of the host using DNS.

Parameter	Description
`-q <query-option>`	The <query - option> parameter specifies a type of DNS record. The supported record types are: • **A**: Address record • **AAAA**: IPv6 address record • **CNAME**: Canonical name record • **PTR**: Pointer resource record • **SRV**: Service Locator
`<host>`	The host or domain name to be queried.

Examples

The following example obtains the IPv4 address of the host or domain using DNS:

```
[appadmin]#

network nslookup ca.us.









.com
```

The following examples perform network nslookups for a destination with an IPv6 address:

```
[appadmin]# network nslookup 2001:db8::
Server:   2001:db8::
Address:  2001:db8::#53
3.9.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.1.0.0.2.ip6.arpa
name = ipv6test-n1.cppmipv6.com
[appadmin]# network nslookup -q AAAA ipv6test-n1.cppmipv6.com
Server:   2001:db8::
Address:  22001:db8::#53
ipv6test-n1.cppmipv6.com has AAAA address 2001:db8::
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

Service Commands

Subtopics

service

service

Syntax

```
service <action> <service-name>
```

Description

Use the **service** <*action*> <*service-name*> command to control the specified Policy Manager service.

Table 1. Service Action Command Parameters

Service Parameter	Description
<action>	Specify an action: • list

Service Parameter	Description
	• restart • start • status • stop
<service - name>	Specify a service: • *airgroup - notify* • cpass - admin - server • *cpass - async - netd* • *cpass - carbon - server* • *cpass - dbcn - server* • *cpass - dbwrite - server* • *cpass - domain - server_<NetBIOS_name>* • *cpass - igssyslog - server* • *cpass - igslogger - server* • *cpass - igslogrepo - server* • *cpass - ipsec - service* • *cpass - zone - cache - server* • cpass - policy - server • cpass - radius - server • *cpass - radsec* • cpass - repl - server • *cpass - statsd - server* • cpass - sysmon - server • cpass - system - auxiliary - server • *cpass - tacacs - server* • *cpass - vip* • *fias_server*

Example

The following example lists all Policy Manager services:

```
[appadmin#] service list
Policy server [ cpass-policy-server ]
TACACS server [ cpass-tacacs-server ]
Radius server [ cpass-radius-server ]
Async DB write service [ cpass-dbwrite-server ]
DB replication service [ cpass-replication ]
DB change notification server [ cpass-dbcn-server ]
System monitor service [ cpass-sysmon-server ]
System auxiliary service [ cpass-system-auxiliary-server ]
Admin server [ cpass-admin-server ]
Async netd service [ cpass-async-netd ]
Zone cache [ cpass-zone-cache-server ]
Domain Server [  cpass-domain-server@CPATSAD2012 ]
Domain Server [  cpass-domain-server@CPATSAD2016 ]
Stats collection service [ cpass-statsd-server ]
Stats aggregation service [ cpass-carbon-server ]
Ingress logger service [ cpass-igslogger-server ]
Ingress logrepo service [ cpass-igslogrepo-server ]
RadSec Service [ cpass-radsec ]
AirGroup notification service [ airgroup-workqueue ]
ClearPass Guest background service [ cpg-background ]
ClearPass Guest cache [ cpg-redis-cache ]
Extensions service [ cpass-extensions ]
Micros Fidelio FIAS [ fias-server ]
ClearPass Virtual IP service [ cpass-vip ]
ClearPass IPsec service [ cpass-ipsec ]
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

Show Commands

Subtopics

show all-timezones

show date
show domain
show dns
show fipsmode
show hostname
show ip
show license
show ntp
show sysinfo
show ssh
show timezone
show version

show all-timezones

Syntax

```
show all-timezones
```

Description

View all available time zones.

Example

The following displays an example of the **show all-timezones** command output.

```
[appadmin]# show all-timezones
Africa/Abidjan
Africa/Accra
Africa/Addis_Ababa
Africa/Algiers
Africa/Asmara
Africa/Asmera
Africa/Bamako
Africa/Bangui
[More]
```

Related Commands

Command	Description
configure timezone	Interactively configure the time zone.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show date

Syntax

```
show date
```

Description

View the system date, time, and time zone information.

Example

The following displays an example of the **show date** command output:

```
[appadmin]#  show date
Wed Feb 27 13:30039 UTC 2018
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show domain

Syntax

Use the **show domain** command to view the Active Directory Domain controller information.

> **NOTE**
> The **show domain** command is operational only when the current Policy Manager server is joined to an Active Directory domain.

Description

```
show domain
```

Example

The following displays an example of the **show domain** command output:

```
[appadmin]#  show domain
=========================================================
Domain Information
---------------------------------------------------------
Domain Name               : COLLEGE152.COM
Domain NETBIOS Name       : COLLEGE152
Domain Server IP Address  : 10.xx.110
Domain Server Name        : balsam.college152.com
Domain Status             : online
---------------------------------------------------------
=========================================================
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show dns

Syntax

Use the **show dns** command to view DNS (Domain Name System) servers.

Description

```
show dns
```

Example

The following example of **show dns** command output displays the DNS servers configured for the current Policy Manager server:

```
[appadmin]#  show dns
==========================================
DNS Information
------------------------------------------
Primary   DNS  :   192.0.2.16
Secondary DNS  :   <not configured>
Tertiary  DNS  :   <not configured>
==========================================
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show fipsmode

Syntax

```
show fipsmode
```

Description

Determine whether **FIPS** (Federal Information Processing Standard) mode is enabled or disabled.

Example

The following example shows that **FIPS** mode is enabled:

```
[appadmin]# show fipsmode
FIPS Mode: Enabled
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show hostname

Syntax

```
show hostname
```

Description

View the hostname of the current Policy Manager server.

Example

The following displays an example of the **show hostname** command:

```
[appadmin]#  show hostname
cppm_135
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show ip

Syntax

```
show ip
```

Description

View the IPv4, IPv6, and DNS information of the host.

Example

The following example of the **show ip** command displays the IPv4, IPv6, and DNS information of the host:

```
[appadmin]#  show ip
==========================================
Device Type      :   Management Port
------------------------------------------
IPv4 Address     :   192.0.2.15
Subnet Mask      :   255.255.255.0
Gateway          :   192.0.2.1
IPv6 Address     :   2001:db8:1:2:020c:29ff:fe0c:4703
Subnet Mask      :   ffff:ffff:ffff:ffff:0000:0000:0000:0000
Gateway          :   2001:db8:1:2:020c:29ff:fe0c:0001
Hardware Address :   00:0C:39:10:27:20
MTU              :   1499
==========================================
Device Type      :   Data Port
------------------------------------------
IPv4 Address     :   <not configured>
Subnet Mask      :   <not configured>
Gateway          :   <not configured>
IPv6 Address     :   2001:db8:1:2:020c:29ff:fe0c:4704
Subnet Mask      :   ffff:ffff:ffff:ffff:0000:0000:0000:0000
Gateway          :   2001:db8:1:2:020c:29ff:fe0c:0001
Hardware Address :   01:0C:29:70:22:4A
MTU              :   1498
==========================================
DNS Information
------------------------------------------
Primary   DNS  :   192.0.2.30
Secondary DNS  :   192.0.2.50
Tertiary  DNS  :   192.0.2.200
==========================================
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show license

Syntax

```
show license
```

Description

View the Policy Manager license information.

Example

The following displays an example of the **show license** command output.

```
[appadmin]#  show license
show license
-----------------------------------------------------
Application              : Entry
License key              : -----BEGIN ENTRY LICENSE KEY-----
XXXXfKksJGYkjpMvgjNLmrcdb/
dZHa+yfW7b2pxSh9MUKPZyMy6VxFhH7ARboTm4vsx226DqfdEAmuh
PMoCX1OTi4dTskbN2v0AAAD//w==
-----END ENTRY LICENSE KEY-----
License key type         : Permanent
License added on         : 2021-04-06 10:45:06
Validity                 : <not applicable>
Issued for               : 25 users
Customer id              : 6FTK5UTJ
Licensed features        : <not applicable>
-----------------------------------------------------
Application              : AccessUpgrade
License key              : -----BEGIN ACCESSUPGRADE LICENSE KEY-----
XXXXH4sIAAAAAAACAwAAAv/9cZ6NB2Mk1kIonwbXukdAcC/khM3mKovp2lgjOYyLdq/
M4I5VGtvp9tRj
8QoEI6/6PODrSElBwvAAAAD//w==
-----END ACCESSUPGRADE LICENSE KEY-----
License key type         : Permanent
License added on         : 2021-04-26 13:37:25
Validity                 : <not applicable>
Issued for               : 25 users
Customer id              : D7GLYVBD
Licensed features        : <not applicable>
-----------------------------------------------------
Application              : Access
License key              : -----BEGIN ACCESS LICENSE KEY-----
XXXXH4sIAAAAAAACAwAAAv/9GqXsQreo8/
```

```
O4+Iex+Vb+xRBXqnhD0LxeMH2JfGJ7/2BDREvw6G0du//d
-----END ACCESS LICENSE KEY-----
License key type          : Permanent
License added on          : 2021-04-06 10:45:11
Validity                  : <not applicable>
Issued for                : 25 users
Customer id               : 6FTK5UTJ
Licensed features         : <not applicable>
------------------------------------------------------
Application               : ClearPassPlatform
License key               : -----BEGIN CLEARPASS PLATFORM LICENSE KEY-----
XXXXH4sIAAAAAAACAwAAAv/
9JFRruUEgSUeqO8GWmFwIvxNMYSLo0oSdfnkNKVQpvAg93QJCxBV64GR6
rtbIACCaLpSb+T6X+hwAAAD//w==
-----END CLEARPASS PLATFORM LICENSE KEY-----
License key type          : Permanent
License added on          : 2021-04-06 10:45:02
Validity                  : <not applicable>
Customer id               : 6FTK5UTJ
Licensed features         : <not applicable>
======================================================
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show ntp

Syntax

```
show ntp
```

Description

View the IP addresses of the primary and secondary Network Time Protocol (NTP) servers configured for the current Policy Manager server. The **show ntp** command also displays information such as the NTP authentication key details corresponding to the NTP server configured (for example, Key ID and hash algorithm).

Example

The following displays an example of the **show ntp** command output:

```
[appadmin]#  show ntp
==========================================
NTP Server Information
------------------------------------------
Primary   NTP  :    192.0.2.1
Key ID: 24
Algorithm: SHA1
Secondary NTP  :    192.0.2.2
Key ID: 48
Algorithm: SHA1
==========================================
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show sysinfo

Syntax

```
show sysinfo
```

Description

Displays the node uptime, disk utilization, and memory utilization information.

Example

The following is an example of the show sysinfo command output:

```
[appadmin]# show sysinfo
System Uptime :  1 day, 23:29:15.510000
==========================================
Disk Utilization
------------------------------------------
Total         :    344.00 GB
Free          :    334.84 GB (97%)
```

```
==========================================
Memory Utilization
------------------------------------------
Total          :    16.00 GB
Free           :    10.53 GB (66%)
==========================================
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show ssh

Syntax

```
show ssh
```

Description

View SSH timed lockout settings.

Example

The following displays an example of the **show ssh** command output for non-default SSH lockout settings.

```
[appadmin]# show show ssh
=======================================================
SSH lockout details
-------------------------------------------------------
SSH lockout count     : 6 attempts
SSH lockout duration  : 900 secs
SSH lockout mode      : basic
-------------------------------------------------------
SSH sessions
-------------------------------------------------------
Client IP Address =  192.0.2.15 : Session Count =  1
=======================================================
```

If the SSH lockout feature is not configured and is currently using all default values, the **ssh lockout details** section of the output will show only **<not-configured>.**

```
[appadmin]# show show ssh
=========================================================
SSH lockout details
---------------------------------------------------------
SSH lockout options    :    <not-configured>
---------------------------------------------------------
SSH sessions
---------------------------------------------------------
Client IP Address =  192.0.2.15 : Session Count =  1
=========================================================
```

Related Commands

Command	Description
ssh lockout	Configure the number of successive unsuccessful authentication attempts for administrators attempting to authenticate remotely

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show timezone

Syntax

```
show timezone
```

Description

Use the **show timezone** command to view the current system time zone.

Example

The following displays an example of the **show timezone** command output:

```
[appadmin]#  show timezone
Timezone is set to 'Asia/Kolkata'
```

Related Commands

Command	Description
configure timezone	Interactively configure the time zone.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

show version

Syntax

```
show version
```

Description

Use the **show version** command to view the Policy Manager software version and the hardware model.

Example

The following displays an example of the **show version** command output:

```
[appadmin]#  show version
=====================================
Policy Manager software version : 6.10.0.180009
Policy Manager model number     : C1000V
=====================================
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

System Commands

Select any command from the left navigation menu.

> **NOTE**
> The system `install-image` command is not supported in ClearPass 6.14.

Subtopics

system admin-passwd-reset
system apps-access-reset
system boot-image
system cleanup
system create-api-client
system export-endpoints-csv
system factory-reset
system gen-recovery-key
system gen-support-key
system install-image
system morph-vm
system patch-rollback
system refresh-license
system refresh-network
system restart
system reset-server-certificate
system shutdown
system start-rasession
system sso-reset
system status-rasession
system terminate-rasession
system update
system update-luks-key
system upgrade

`system admin-passwd-reset`

Syntax

```
system admin-passwd-reset
```

Description

Resets the admin password for the Policy Manager WebUI back to the default setting of the eth0 MAC address.

The eth0 MAC address is only applied to AWS deployments as the default password. The standard default account credential is applied to all other deployments.

Command History

Version	Modification
ClearPass 6.11	Default admin password change
ClearPass 6.10	Command Introduced

system apps-access-reset

Syntax

```
system apps-access-reset
```

Description

Reset the access control restrictions for Policy Manager.

Example

The following example resets the access control restrictions for Policy Manager:

```
[appadmin]# system apps-access-reset
Policy Manager application access is restored
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system boot-image

Syntax

```
system boot-image {-l}|{-a <version>}
```

Description

Use the **system boot-image** command to set system boot image control options.

Parameter	Description
`-l`	List the boot images installed on the system.
`-a <version>`	Set the active boot image version.

Example

The following example lists the system boot images. The index number for each entry is displayed in front of the image version.

```
[appadmin]#  system boot-image -l
0) Aruba ClearPass Platform 6.10.0.181036
1) Aruba ClearPass Platform 6.10.1.181039 [Active]
```

The following example sets the active boot image to the image in index 0

```
[appadmin]# system boot-image -a 0
Boot-image with index 0 is set to active
Restart the system to boot from the updated image
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system cleanup

Syntax

```
system cleanup <num_days>
```

Description

Use the **system cleanup** command to perform a system cleanup operation that purges the following records:

- System and application log files
- Past authentication records
- Audit records
- Expired guest accounts
- Past auto and manual backups
- Stored reports

Parameter	Description
<num_days>	This is the cleanup interval that specifies the numbe r of days to retain the data. This field is mandatory.

Example

The following example performs a system cleanup operation that retains records for four days:

```
[appadmin]#  system cleanup 4
******************************************************
*                                                    *
* WARNING: This command will perform system cleanup  *
* operation that will result in purging of:          *
*    [*] system and application log files            *
*    [*] past authentication records                 *
*    [*] audit records                               *
*    [*] expired guest accounts                      *
*    [*] past auto and manual backups                *
*    [*] endpoints                                   *
*    [*] stored reports etc...                       *
*                                                    *
******************************************************
Are you sure you want to continue? [y|n]: y
INFO - Starting system cleanup
INFO - Purging diagnostic dumps
INFO - Detected empty core directory
INFO - Performing system cleanup tasks
INFO - Purging platform logs
```

```
INFO - Purging application logs
INFO - Performing database cleanup tasks
INFO - Completed system cleanup
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system create-api-client

Syntax

```
system create-api-client <Client_ID> <Client_Secret>
```

Description

Create a new API client.

Parameter	Description
<Client_ID>	A unique string used to identify the client.
<Client_Secret>	A string used as the client password.

Example

The following example creates an API client by specifying the client ID and client secret.

```
[appadmin]#system create-api-client Win.139 college52
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system export-endpoints-csv

Syntax

```
system export-endpoints-csv
```

Description

use this command to export endpoints and endpoint profile details to a zip file that can be downloaded from **Admin UI - Backup files** under **Administration > Server Manager > Local Shared Folders.** When using an XML file to export or import a very large number of endpoints, performance is sometimes degraded.

When using an XML file to export a very large number of endpoints (> 250 K), performance is sometimes degraded or the user interface hangs and out-of-memory error messages are logged. Although exporting CSV files through the CLI is still supported, users should be aware that importing ZIP files that contain CSV files of endpoints and endpoint profiles is not currently allowed through either the CLI or the user interface (UI).

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system factory-reset

Syntax

```
system factory-reset
```

Description

The **system factory-reset** command restores a Policy Manager hardware appliance to factory defaults. This command is available only to the *appadmin* user on a physical appliance. It is not available on a virtual machine.

> **CAUTION**
>
> The **system factory-reset** command is inherently a destructive one as it wipes out data, including any licenses on the current partition and any backups currently stored on the server. Hence, the user should create data backups outside of the target Policy Manager server before running this command. This command is not available on Policy Manager installations hosted on a cloud services platform such as Amazon Web Services (AWS) or Azure.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system gen-recovery-key

Syntax

```
system gen-recovery-key
```

Description

Support engineers use this password to generate the recovery key for the Policy Manager server.

Example

The following example generates the recovery key for the system:

```
[appadmin]# system gen-recovery-key
******************************************************
This command is intended for support engineers only *
******************************************************
Username: support@hpe.com
Token is c2ab-97a8-bf3f-49f6
Use the above token and Generate password from support server
```

```
Password: ***********
Password is valid.
Use the Password to login as user 'arubasupport'
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system gen-support-key

Syntax

```
system gen-support-key
```

Description

The **system gen-support-key** command uses the Support Engineer's email ID to generate a token that the Support Engineer can use to generate a password that allows privileged access to the Policy Manager server.

Example

The following example generates the support key for the system:

```
[appadmin]#  system gen-support-key
*******************************************************
This command is intended for support engineers only *
*******************************************************
Username: support@hpe.com
Token is c2ab-97a8-bf3f-49f6
Use the above token and Generate password from support server
Password: ***********
Password is valid.
Use the Password to login as user 'arubasupport'
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system install-image

Syntax

```
system install-image
   {http|https}://<hostname>/<filename>
   <user@hostname>:/<filename>
   <filename>
```

Description

The **system install-image** command installs a fresh image of the major product version specified in the second partition of a Policy Manager hardware appliance.

This command is available only for the *appadmin* user on a physical appliance. It is not available on a virtual appliance.

NOTE

This command is not available on Policy Manager installations hosted on a cloud services platform such as Amazon Web Services (AWS) or Azure.

After successful execution of the **system install-image** command, the system will reboot and you will return to the installed image. After successful configuration and reboot, you will be presented with the bootstrap configuration screen, where you will have to reset all the ClearPass parameters.

CAUTION

Any data present in the second partition prior to the execution of the **system install-image** command will be wiped out. Also, no licensing information from where the command is executed is carried forward.

You can apply the **system install-image** command in the following ways:

Parameter	Description
{http|https}://<hostname>/<filename>	Install the specified Policy Manager image through or **HTTPS.**
< user@hostname>:/<filename>	Specify the username and hostname to install the selected file through SCP (Secure Copy Protocol).
<filename>	Install the specified image imported to the Policy Manager server and available locally (offline install - image).

Example

```
[appadmin]#  system install-image http://cppm.example.com/downloads/6.8.0/
CPPM-x86_64-6.8.0.109568-upgrade.signed.tar
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system morph-vm

Syntax

```
system morph-vm C1000V|C2000V|C3000V
```

Description

Use the **system morph-vm** command to convert an evaluation virtual appliance (also called a virtual machine, or VM) to a production virtual appliance. With this command, licenses must still be installed after the morph operation is completed.

When you use the **system morph-vm** command to morph a virtual appliance (VA) to a larger size, all the licenses are deleted. This issue does not affect configuration data.

After the upgrade, contact the Technical Assistance Center (TAC) to have the licenses activated again.

NOTE

Keep in mind, users cannot remove existing hard disks once the **system morph-vm** command is run in deployments supported by ClearPass Policy Manager version 6.11 and above. Existing disk resources are used to extend server disk space and should be taken into account when computing the required capacity of an additional hard disk to be attached to the target virtual appliance. For example, an existing CLABV instance is deployed with 400GB disk space. If the deployment intends to morph to C2000v, it needs to attach a 600GB hard disk to meet the 1000GB disk space recommended for C2000v, instead of attaching a new disk with 1000GB of disk space.

NOTE

ClearPass uses Linux Unified Key Setup (LUKS) for 256-bit AES data at-rest encryption on all physical or upgraded virtual appliances. New virtual appliance installations can otionally use LUKS. When you run the **system morph-vm** command during virtual appliance installation, excluding AMI images, administrators are prompted to enable or disable LUKS. If you know the hypervisor or hardware you are deploying already has disk encryption, you do not need to add additional disk encryption. If you are not sure whether the hypervisor has underlying disk encryption, then always select this option for encryption. This operation cannot be changed after installation.

Parameter	Description
<vm - version>	Select a virtual appliances type. The following optio ns are available: • C1000V • C2000V • C3000V

To convert an evaluation virtual appliance to a production virtual appliance:

1. Determine the type of the appliance to which you want to morph your evaluation virtual appliance .
2. Procure the license for the target virtual appliance.

3. Shut down the virtual appliance.
4. Determine the required capacity of an additional hard disk and attach it to the target virtual appliance.
5. Adjust the CPU and Memory settings for the evaluation virtual appliance to match the target virtual appliance.
6. Boot the virtual appliance.
7. Execute the **system morph-vm** command. The configuration data from the evaluation virtual appliance will migrate to the newly-attached disk. The node will reboot as a virtual appliance of the selected appliance model.
8. Log in to the user interface and enter the permanent license. The evaluation virtual appliance is now a production virtual appliance.

   Example

   The following example converts an evaluation virtual appliance to a production C3000V virtual appliance:

```
[appadmin]# system morph-vm C3000V
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system patch-rollback

Syntax

```
system patch-rollback
```

Description

The **system patch-rollback** command allows a user with *appadmin* credentials to revert to the most recent installed version of Policy Manager. For example, if a Policy Manager system is at 6.9.6 and cumulative update 6.9.x is applied, Policy Manager can be reverted to 6.9.6 through the **system patch-rollback** command.

This command can also be used if there is a problem that occurs after the patch update process—for example, if an issue is identified in production that was not identified during testing, resulting in a degradation of capabilities.

> **NOTE**
> Before using this command to revert command to revert from 6.10.x to 6.10.0, you must first download the **6.10.0_source-rollback-package** from the **Software Updates** page and install it .

When issuing the **system patch-rollback** command, keep in mind the following points:

- Patch-rollback is supported only for Policy Manager versions 6.7 and above.
- The **system patch-rollback** command reverts only the most recently installed cumulative patch update within the major version. After the cumulative patch is reverted, the user will be in the patch version that was installed prior to the patch update.

> **NOTE**
> The **system patch-rollback** command cannot be used after an upgrade to revert to an earlier major version.

- Although you can only roll back to the last version that was installed, if multiple hotfix patches are included within the cumulative patch version you are rolling back from, then you can roll back multiple hotfix patches, one at a time, to a specific hotfix within the current version. To roll back to the previously installed version, you must first roll back each intervening hotfix patch.
- As best practice, users should always back up all data before proceeding with an update.
- This command can also be used at the cluster level. In this case, **system patch-rollback** must be run individually on each appliance in the cluster within 24 hours after the rollback in order to maintain the cluster status. For patch rollback across a cluster, the appadmin user must go to each Policy Manager server in the cluster to rollback the last applied patch.
- Any custom skins that are installed in the current version are retained after the rollback to the earlier version.
- System rollback events are logged in the Event Viewer.

Example

```
[appadmin]# system patch-rollback
**********************************************************************
```

```
************
* WARNING: This command is recommended to be executed from local console
unless otherwise instructed by TAC * Execution through SSH console may
result in system instability.*
* WARNING: This command will undo software changes done by the currently
installed patch. Configuration
*changes should not  be affected by this action.
* As a best practice, please be sure to back-up this system before starting
the operation. *
* Are you sure you want to continue?  y
********************************************************************************
**************
INFO: Preparing for rollback
INFO: 2018022-clearpass-6.8-updates-2 will be rolled back
INFO: This will take a few minutes to complete. Please wait.
INFO: Running pre-rollback scripts
INFO: Executing rollback
INFO: Running post-rollback scripts
INFO: Please reboot now for the changes to take effect.
********************************************************************************
**************
```

For example, if Policy Manager has been installed in the order 6.11.0 > 6.11.1 > 6.11.2, when the appadmin user executes the **system patch-rollback** command, the system would revert to a time just before Policy Manager 6.11.2 was installed.

If, in this example, the installed 6.11.2 patch added an rpm-X, **system patch-rollback** deletes rpm-Y, and updates rpm-Z to rpm-Z+1 version. Then **system patch-rollback** deletes rpm-X, adds rpm-Y, and restores rpm-Z.

Also note that if, for example, a system was at 6.11.0 and cumulative update 6.11.3 is applied, the system can only be reverted to 6.11.0 because that was the last installed version. It cannot be reverted to 6.11.2.

NOTE

For more information, refer to the "After You Update: Performing a Patch Rollback" section in the most recent version of the *ClearPass Release Notes*.

NOTE

The **system patch-rollback** command also removes any configuration and database changes that were done as part of post-installation during the patch update.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system refresh-license

Syntax

```
system refresh-license
```

Description

Refresh the license count information.

Example

The following example refreshes the license count information.

```
[appadmin]# system refresh-license
INFO: Refreshing license count information
INFO: Successfully refreshed license count information
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system refresh-network

Syntax

```
system refresh-network
```

Description

Refresh the newly added or removed network adapters in Policy Manager so that they are reflected in the system. This command also enforces network adapter ordering and

associates the lower-order MAC address to *eth0* and the next higher-order MAC address to *eth1*, and so on. Ensure that you have the console session available.

The **system refresh-network** command is useful when you bring up a virtual appliance without one or more of the network interface cards (NICs) and you then add them at a later stage. This command is required when you delete NICs and add them back into the system (VMware ESXi may generate new MAC addresses as a result).

For the network refresh to take effect, you must reboot the Policy Manager server.

> **CAUTION**
> Using this command may result in loss of network connectivity.

Example

```
[appadmin]# system refresh-network
*********************************************************
*
WARNING: The command will refresh the network *
adapters which may result in loss of network *
connectivity. *
*
*********************************************************
Are you sure you want to continue? [y|Y]: y
INFO: Associating "2001:DB8:441:1020::3" with eth0
INFO: Associating "2001:DB8:441:1022::4" with eth1
INFO: Command execution completed successfully.
INFO: Reboot the system to reflect the changes.
WARNING: System may lose network connectivity after the reboot.
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system restart

Syntax

```
system restart
```

Description

Restart the Policy Manager system.

> **CAUTION**
> Executing this command shuts down all running applications and reboots the system.

Example

The following example restarts the system with a confirmation before proceeding:

```
[appadmin]#  system restart
system restart
******************************************************

* WARNING: This command will shut down all applications *
* and reboot the system                                  *
******************************************************
Are you sure you want to continue? [y|Y]: y
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system reset-server-certificate

Syntax

```
system reset-server-certificate
```

Description

Reset the HTTPS(RSA), HTTPS(ECC), RADIUS/EAP or Database Server Certificates, or all of them.

After executing the command, the Policy Manager services are restarted to reflect the changes.

Example

The following example resets resets the HTTPS, RADIUS/EAP and Database Server Certificates.:

```
[appadmin]# system reset-server-certificate
**********************************************************************
*                                                                    *
* WARNING: When the command is completed Policy Manager services *
* are restarted to reflect the changes.                              *
*                                                                    *
**********************************************************************
Continue? [y|n]: y
0: Reset HTTPS, RADIUS and Database Server Certificates
1: Reset RADIUS Server Certificate
2: Reset HTTPS (ECC) Server Certificate
3: Reset Database Server Certificate
4: Reset HTTPS (RSA) Server Certificate
5: Quit
0
Updating server certificate(s)...
Update of server certificate(s) complete
INFO - Stopping Policy Manager services...
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system shutdown

Syntax

```
system shutdown
```

Description

Shut down the current Policy Manager server.

> **CAUTION**
> Executing this command shuts down all running applications and powers off the system.

Example

The following example shuts down the system with a confirmation before proceeding:

```
[appadmin]#  system shutdown
******************************************************

* WARNING: This command will shut down all applications *
* and power off the system                               *
******************************************************
Are you sure you want to continue? [y|N]: y
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system start-rasession

Syntax

```
system start-rasession <duration_hours 0-12>  <duration_mins 0-59>
<contact_id>  <cppm_server_ip>
```

Description

Start a Remote Assistance (RA) session.

Parameter	Description
`<duration_hours 0-12>`	Specify the session duration in hours. You can specify values from 0 to 12.

Parameter	Description
`<duration_mins 0-59>`	Specify the session duration in minutes. You can specify values from 0 to 59.
`<contact_id>`	Enter the username ID of the Aruba TAC or Engineering contact.
`<cppm_server_ip>`	Specify the Policy Manager server IP address.

Example

The following example starts a 30-minute remote session on a Policy Manager server with the IP address **198.0.2.14**.

```
[appadmin]# system start-rasession 0 30 tacperson 198.0.2.14
Successfully scheduled a RemoteAssist session with id[3001], Please use
this id to check status and terminate the session.
```

Related Commands

Command	Description
system status - rasession	Check the status of a Remote Assistance (RA) session.
system terminate - rasession	Terminate a a Remote Assistance (RA) session.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system sso-reset

Syntax

```
system sso-reset
```

Description

Reset the Policy Manager Single Sign-On (SSO) configuration.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system status-rasession

Syntax

```
system status-rasession <session_id>
```

Description

Check the status of a Remote Assistance (RA) session.

Parameter	Description
`<session_id>`	The Remote Session ID. This session ID is generated in the output of the command system start - rasession.

Example

The following example displays the status of an active remote assistance session with the session ID **3001**.

```
[appadmin]# system status-rasession 3001
Session is running.
```

Related Commands

Command	Description
system start - rasession	Start a Remote Assistance (RA) session.

Command	Description
system terminate - rasession	Terminate a a Remote Assistance (RA) session.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system terminate-rasession

Syntax

```
system terminate-rasession <session_id>
```

Description

Terminate a running Remote Assistance session.

Parameter	Description
<session_id>	The Remote Session ID. This session ID is generated in the output of the command system start - rasession.

Example

The following example terminates a running Remote Assist session with session ID 3001.

```
[appadmin]# system terminate-rasession 3001
```

Related Commands

Command	Description
system start - rasession	Start a Remote Assistance (RA) session.

Command	Description
system status - rasession	Check the status of a Remote Assistance (RA) session.

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system update

Syntax

```
system update
   -i [-f] <user@hostname:/<filename> | http://hostname/<filename>>
   -f
   -l
```

Description

The **system update** command provides options to manage system patch updates.

> **CAUTION**
> The time taken by this operation can be longer than the default CLI Session Idle Timeout of 10 min, and depends on network latency, size of the software update, and the resources on the system (CPU, Memory and disk IOPS). Consider increasing the timeout to an adequately high value to ensure this operation does not terminate due to the CLI Session Idle Timeout.

Parameter	Description
`-i user@hostname:/<filename>| http://hostname/<filename>`	Installs the specified patch on the system. • **user@hostname:/<filename>**: Install the update from a Linux server

Parameter	Description
	• `http://hostname/<filename>`: Install the up date from a Web server.
`-f`	Reinstalls the patch in the event of a problem with t he initial installation attempt.
`-l`	Lists the patches installed on the system.

NOTE

This command supports Secure Copy (SCP), HTTPS, HTTP, and local uploads.

Example

The following system update command example installs the selected Policy Manager 6.10 patch.

```
[appadmin# system update -i http://198.0.2.12/downloads/6.10.0/Upgrade/
latest/CPPM-x86_64-6.10.0.180055-upgrade.signed.tar
```

The following system update command example lists the patches currently installed on the Policy Manager server:

```
[appadmin]#  system update -l
Update         : 20201119-clearpass-6.9-updates-5
Installed Date : Thu Feb 11 18:33:28 2021
Description    : ClearPass Policy Manager Cumulative Patch 5 for 6.9.0,
6.9.1, 6.9.2, 6.9.3 and 6.9.4
Packages       : tips-,avenda-quick1x,aruba-pgtools,clearpass-
guest,cppm-in,platform-,radsecproxy-,mod_php,php-cli,php-common,php-
gd,php-ldap,php-mysql,php-opcache,php-pdo,php-pgsql,php-process,php-
soap,php-xml,php-xmlrpc,updateverify_v4,avenda-tomcat,mod_jk,oracle-
instantclient19.6-basic,oracle-instantclient19.6-odbc,libsmbclient,samba-
client-libs,samba-libs,libwbclient,samba-common,samba-winbind,samba,samba-
common-libs,samba-winbind-clients,samba-client,samba-common-tools,samba-
winbind-modules,dnsmasq,PhantomJS,openssl-perl-aruba-w,openssl-aruba-
w,openssl-libs-aruba-w,pycparser,zulu8.48.0.51-ca-jdk8.0.262
Affects        : tips-ntp,platform-system-setup,tips-async-netd,tips-system-
monitor
Update         : 6.9.0_source-rollback-package
Installed Date : Thu Feb 11 21:50:58 2021
Description    : Optional ClearPass 6.9.x package
required to rollback to 6.9.0 version
```

```
Packages       : clearpass
Affects        :
Update         : 20210204-snmp-cdp-fix
Installed Date : Sat Feb 13 08:37:43 2021
Description    : This hotfix provides the enhancements
to network discovery profiling to fetch MAC address from CDP.
Packages       : tips-network-services
Affects        : backend-tomcat
Update         : 20210204-clearpass-6.9-updates-6
Installed Date : Sat Feb 13 08:49:33 2021
Description    : ClearPass Policy Manager Cumulative
Patch 6 for 6.9.0, 6.9.1, 6.9.2, 6.9.3, 6.9.4 and 6.9.5
Packages       : tips-,avenda-quick1x,aruba-pgtools,clearpass-guest,cppm-
in,platform-,radsecproxy-,mod_php,php-cli,php-common,php-gd,php-ldap,php-
mysql,php-opcache,php-pdo,php-pgsql,php-process,php-soap,php-xml,php-
xmlrpc,updateverify_v4,avenda-tomcat,mod_jk,oracle-instantclient19.6-
basic,oracle-instantclient19.6-odbc,libsmbclient,samba-client-libs,samba-
libs,libwbclient,samba-common,samba-winbind,samba,samba-common-
libs,samba-winbind-clients,samba-client,samba-common-tools,samba-winbind-
modules,dnsmasq,PhantomJS,openssl-perl-aruba-x,openssl-aruba-x,openssl-
libs-aruba-x,pycparser,redis,zulu8.48.0.51-ca-jdk8.0.262,httpd,httpd-
tools,mod_ssl
Affects        : tips-ntp,platform-system-setup,tips-async-netd,tips-system-
monitor
4 installed updates.
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system update-luks-key

Syntax

```
system update-luks-key
```

Description

This command allows an administrator to specify the passphrase (key) for LUKS encryption. The LUKS passphrase has a minimum length of eight characters. When this command is used, the new passphrase replaces the default LUKS key in ClearPass. Each time Policy Manager is rebooted or a new session is initiated, the administrator will be prompted to enter the specified LUKS passphrase before access is granted.

If the LUKS passphrase is long, and depending on the size of the console window, the administrator might be prompted to enter the passphrase on the next line. In this case, the administrator only needs to enter the remainder of the characters in the next line and does not need to enter the entire passphrase again.

If the system update-luks-key command has been used to specify a LUKS key, then when the Cluster Update page is used to update to a 6.10.x version, the key must be manually entered in the publisher and in each subscriber after the reboot stage before the cluster update can proceed.

CAUTION

After a LUKS key has been specified, if the administrator forgets the LUKS key and a reboot of ClearPass Policy Manager is initiated, the system cannot be recovered. System administrators must be extremely careful to not forget this key after it is created.

Example

The following example updates the existing LUKS passphrase.

```
[appadmin@test]# system update-luks-key
WARN: VM will be encrypted with new passphrase.
Once initiated you cannot revert.Upon reboot, enter new LUKS passphrase on
the prompt.
Press 'y' or 'Y' to proceed: y
Enter LUKS passphrase to be changed (Press Enter if None): *********
Enter new LUKS passphrase: ********
Re-enter new LUKS passphrase: ********
Updating LUKS passphrase...
Upon reboot, enter new LUKS passphrase on the prompt.
```

Command History

Version	Modification
Policy Manager 6.10	Command Introduced

system upgrade

Syntax

```
system upgrade
   user@hostname:/<filepath>
   http://hostname/<filepath>
   <filepath>
   -W
   -l
   -L
```

Description

The **system upgrade** command upgrades the Policy Manager system. This command provides the following system upgrade options:

> **CAUTION**
> The time taken by this operation can be longer than the default CLI Session Idle Timeout of 10 min, and depends on network latency (when downloading from a remote server), the size of the upgrade image, and the resources on the system (CPU, Memory and disk IOPS). Consider increasing the timeout to an adequately high value to ensure this operation does not terminate due to the CLI Session Idle Timeout.

Parameter	Description
<user@hostname>:/<filepath>	Specify a user name, host name and file path to up grade to the specified file on a Linux server
http://<hostname>/<filepath>	Specify a host name and file path to upgrade to the specified file on a Web server
<filepath>	Specify the path of a local file to perform an offline upgrade.
- w	Restores last (one) week of access tracker records a fter the upgrade.
- l	Restores all access tracker records from this version .

Parameter	Description
- L	Does not backup or restore access tracker records f rom this version.

NOTE

This command supports Secure Copy (SCP), HTTPS, HTTP, and local uploads.

NOTE

If none of these **system upgrade** command options are specified, Access Tracker records are backed up, but they are not restored by default.

Examples

Subtopics

Upgrading from a Web Server
Performing an Offline Upgrade

Upgrading from a Web Server

About this task

To upgrade the Policy Manager image from a Web server:

Procedure

1. Upload the upgrade image to a Web server.
2. Use the following syntax to upload the upgrade image:

```
system upgrade http://hostname/<filepath> [-w] [-l] [-L]
```

For example:

```
[appadmin]#  system upgrade http://192.0.2.15/downloads/master/Upgrade/
CPPM-x86_64-6.9.0.170156-upgrade.signed.tar
```

Performing an Offline Upgrade

About this task

To perform an offline upgrade:

Procedure

1. Navigate to the **ClearPass > Policy Manager** > *Current Release folder* **Upgrade** folder.

   The **Upgrade** page opens.

2. In the **Description/Remarks** section, click the link for the appropriate upgrade.

   The upgrade file is uploaded to the local system.

3. Navigate to the Policy Manager **Software Updates** page at **Administration > Agents and Software Updates > Software Updates**.

4. In the **Firmware & Patch Updates** section of the **Software Updates** page, click the **Import Updates** button.

   The **Import from File** dialog opens.

5. Browse to the location of the upgrade file on your system, then click **Import**.

   The selected upgrade file is uploaded to the Policy Manager.

6. Log in to the Policy Manager command line interface (CLI) with the following user name: *appadmin*.

7. Initiate the upgrade process by entering the following command:

   ```
   system upgrade <filepath> [-w] [-l] [-L]
   ```

   For example:

   ```
   [appadmin]# system upgrade CPPM-upgradeimage.bin
   ```

8. After the upgrade process is complete, restart the appliance by issuing the system restart command.The Policy Manager restarts and boots up to the most recent version of Policy Manager.

   Command History

Version	Modification
Policy Manager 6.10	Command Introduced

SNMP, System Events & Error Codes

This appendix contains the following information:

- SNMP Private MIB
- SNMP Trap Details
- Important System Events
- Error Codes

> **NOTE**
>
> To view ClearPass API information, access the ClearPass Guest WebUI and navigate to **Administration > API Services > API Explorer**.

Subtopics

Policy Manager SNMP Private MIB
Policy Manager SNMP Traps and OIDs
SNMP Trap Details
Important System Events
Error Codes

Policy Manager SNMP Private MIB

- System MIB Entries
- RADIUS Server MIB Entries
- Policy Server MIB Entries
- Web Authentication Server MIB Entries
- TACACS+ Server MIB Entries
- Network Traffic MIB Entries

Subtopics

Introduction

System MIB Entries
RADIUS Server MIB Entries
Policy Server MIB Entries
Web Authentication Server MIB Entries
TACACS+ Server MIB Entries
Network Traffic MIB Entries

Introduction

A MIB (Management Information Base) is a collection of definitions that define the properties of the managed object within the device to be managed. The various pieces of information are accessed by a protocol such as SNMP. To download the MIB, go to **Support Portal.**

This section describes the MIB objects exposed and traps sent through the Policy Manager Private SNMP MIB.

System MIB Entries

Table 1 describes the *CPPMSystemTableEntry* MIB objects.

Table 1. CPPMSystemTableEntry System MIB Objects

MIB Object	Description
cppmClusterNodeType	Policy Manager cluster node type indic e node is a publisher or Subscriber
cppmNwDataPortIPAddress	Policy Manager server data port IP adc
cppmNwDataPortMACAddress	Policy Manager server data port MAC a
cppmNwMgmtPortIPAddress	Policy Manager server management p
cppmNwMgmtPortMACAddress	Policy Manager server management p
cppmSystemDiskSpaceFree	Amount of disk space free (in bytes) in ger server
cppmSystemDiskSpaceTotal	Total amount of disk space available (i licy Manager server
cppmSystemHostname	Policy Manager server host name

MIB Object	Description
cppmSystemMemoryFree	Amount of memory free (in bytes) in th r server
cppmSystemMemoryTotal	Total amount of memory available (in b cy Manager server
cppmSystemModel	Model of the Policy Manager server
cppmSystemNumCPUs	Total number of CPUs in the Policy Ma
cppmSystemSerialNumber	Serial number of the Policy Manager s
cppmSystemUptime	Amount of time the Policy Manager se
cppmSystemVersion	Product version of the Policy Manager

RADIUS Server MIB Entries

RadiusServerTableEntry

The following table describes the *RadiusServerTableEntry* objects:

Table 1. RadiusServerTableEntry Objects

MIB Object	Description
radAuthRequestTime	Total time taken for an end - to - end RADI US request
radPolicyEvalTime	Time taken for policy evaluation from the R ADIUS server perspective
radServerCounterCounts	Total number of successful RADIUS authen tications
radServerCounterFailure	Total number of failed RADIUS authenticati ons
radServerCounterSuccess	Total number of successful RADIUS authen tications

RadiusServerAuthTableEntry

RadiusServerAuthTableEntry exposes the following counters that refer to *authSourceName* wherever applicable. Counters and delays reflect details that are logged into Graphite.

Table 2. RadiusServerAuthEntry MIB MIB Objects

MIB Object	Description
radAuthCounterCount	Total number of RADIUS authentications
radAuthCounterFailure	Total number of failed RADIUS authentications
radAuthCounterSuccess	Total number of successful RADIUS authentications
radAuthCounterTime	Time taken to perform RADIUS authentications
radAuthSourceName	Name of the RADIUS server authentication source

Policy Server MIB Entries

PolicyServerTableEntry

PolicyServerTableEntry exposes the following MIB objects. Counters and delays reflect details logged into Graphite.

Table 1. PolicyServerTableEntry Objects

MIB Object	Description
psAuditPolicyEvalCount	Audit policy evaluation count
psAuditPolicyEvalTime	Audit policy evaluation time
psAuthCounterFailure	Number of failed Policy Server authentications
psAuthCounterSuccess	Number of successful Policy Server authentications
psAuthCounterTotal	Total number of Policy Server authentications

MIB Object	Description
psEnforcementPolicyEvalCount	Enforcement policy evaluation count
psEnforcementPolicyEvalTime	Enforcement policy evaluation time
psPosturePolicyEvalCount	Posture policy evaluation count
psRestrictionPolicyEvalCount	Authorization restriction policy evaluation count
psRolemappingPolicyEvalCount	Role mapping policy evaluation count
psRolemappingPolicyEvalTime	Role mapping policy evaluation time
psPosturePolicyEvalTime	Posture policy evaluation time
psRestrictionPolicyEvalTime	Restriction policy evaluation time
psServicePolicyEvalCount	Service policy evaluation count
psServicePolicyEvalTime	Service policy evaluation time
psSessionlogTime	Policy Server session logging time
dailySuccessAuthCount	Total daily number of successful authentic ations
dailyFailedAuthCount	Total daily number of failed authentications
dailyTotalAuthCount	Total daily number of authentications

PolicyServerProtoTableEntry

PolicyServerProtoTableEntry exposes MIB objects for the counter values for the RADIUS, TACACS+, WEBAUTH, and APPLICATION protocols.

Table 2. PolicyServerProtoTableEntry MIB Objects

MIB Object	Description
psPolicyEvalTime	Policy evaluation time for the protocol
psProtocolName	Name of the protocol

PolicyServerAutzTableEntry

PolicyServerAutzTableEntry exposes MIB objects for authorization counters.

Table 3. PolicyServerAutzTableEntry MIB Objects

MIB Object	Description
psAutzCounterCount	Total number of Policy Server authorizations
psAutzCounterFailure	Number of failed Policy Server authorizations
psAutzCounterSuccess	Number of successful Policy Server authorizations
psAutzCounterTime	Time taken to perform Policy Server authorizations
psAutzAuthSourceName	Name of the Policy Server authorization source

Web Authentication Server MIB Entries

WebAuthProtoTableEntry exposes MIB objects for the WebLogin, AppLogin, SamlIdp, and SamlSp web authentication protocols.

Table 1. WebAuthProtoTableEntry MIB Objects

MIB Object	Description
waAuthCounterAuthTime	Time taken for web authentication
waAuthCounterCount	Total number of web authentications
pwaAuthCounterFailure	Number of failed web authentications
waAuthCounterSuccess	Number of successful web authentications
waAuthCounterTime	Total time taken for web login
waPolicyEvalTime	Time taken to perform policy evaluation
waProtocolName	Name of the protocol
pwaServicePolicyEvalTime	Time taken to perform service policy evaluation

TACACS+ Server MIB Entries

TacacsAuthTableEntry

TacacsAuthTableEntry exposes MIB objects for TACACS+ authentication counters.

Table 1. TacacsAuthTableEntry Objects

MIB Object	Description
tacAuthCounterAuthTime	Time taken for TACACS+ authentications
tacAuthCounterCount	Total number of TACACS+ server authentications
tacAuthCounterFailure	Number of failed TACACS+ server authentications
tacAuthCounterSuccess	Number of successful TACACS+ server authentications
tacAuthCounterTime	Total time taken for TACACS+ login
tacPolicyEvalTime	Time taken to perform policy evaluation
tacServicePolicyEvalTime	Time taken to perform service policy evaluation

TacacsAutzTableEntry

TacacsAutzTableEntry exposes MIB objects for TACACS+ authorization counters.

Table 2. TacacsAuthTableEntry Objects

MIB Object	Description
tacAutzCounterCount	Total number of TACACS+ server authorizations
tacAutzCounterFailure	Number of failed TACACS+ server authorizations
tacAutzCounterSuccess	Number of successful TACACS+ server authorizations

MIB Object	Description
tacAutzCounterTime	Total time taken for TACACS+ authorization s

Network Traffic MIB Entries

NetworkTrafficTableEntry exposes MIB objects for network protocol and applications. These MIB objects cover the following:

- agent_controller (6658)
- db (5432)
- http (80)
- https (443)
- ntp (123)
- radius (1645, 1646, 1812, 1813)
- ssh (22)
- tacacs (49)

Table 1. TacacsAuthTableEntry Objects

MIB Object	Description
nwAppPort	Application port
nwAppName	Application name
nwTrafficTotal	Total network traffic in bytes

Policy Manager SNMP Traps and OIDs

Subtopics

Hardware Monitoring OIDs

Interface Statistics Monitoring OIDs
Daily Authentication Statistics OIDs
Authentication Count OIDs
Policy Manager SNMP Traps
SNMP Poll Disk Space Monitoring

Hardware Monitoring OIDs

Clearpass 6.10, the Policy Manager MIB **aruba-cppm.my** does not include hardware monitoring OIDs. The MIBs for Policy Manager hardware monitoring are included in the HPE System Insight manager MIB kit, which can be downloaded from the HPE website at **https://support.hpe.com/hpesc/public/docDisplay?docId=emr_na-c04272529**. To query HPE hardware-specific metrics. Include the following MIB files in the NMS agent:

- cpqhlth.mib
- cpqida.mib
- cpqsinfo.mib
- cpqhost.mib

  The hardware metrics for HPE appliances can be polled using the OID -

- Fan - 1.3.6.1.4.1.232.6.2.9
- Power - 1.3.6.1.4.1.232.6.2.9.3.1.4
- Disk - 1.3.6.1.4.1.232.3.2.

  The hardware monitoring OIDs allow admins to monitor the fan, power and disk status of the Policy Manager C2000, C2010, C3000, and C3010 hardware appliances using **snmpget** and **snmpwalk** commands. The hardware monitoring OIDs enable sending hardware traps in the event of a fan, power, or disk failure in a hardware appliance. SNMP v1, v2, and v3 queries are supported. Traps for hardware failure are not Policy Manager enterprise-specific traps, but native traps sent by the HP agent running on the hardware appliances. Hardware monitoring IDs not available for the Policy Manager C1000 hardware appliance, nor are they available for virtual appliances.

Policy Manager obtains the underlying hardware configuration from the **dmidecode** (Linux) system manufacturer property. A limited number of early generations of Policy Manager hardware appliances did not identify themselves properly and these traps may not work in this situation. If the vendor details are not correctly set under the system manufacturer, Policy Manager assumes that the underlying configuration does not

correspond to the current HP/Aruba hardware, which would lead to incorrect installation and set up.

Interface Statistics Monitoring OIDs

These OIDs monitor interface statistics through SNMP polling.

Table 1. Interface Statistics Monitoring OIDs

Trap Name	OID	Desc
ifInOctets	.1.3.6.1.2.1.2.2.1.10	Tota the i arac
ifOutOctets	.1.3.6.1.2.1.2.2.1.16	Tota out ing
ifInDiscards	.1.3.6.1.2.1.2.2.1.13	Num were thou d, tc e to poss h a p er sp
ifOutDiscards	.1.3.6.1.2.1.2.2.1.19	Num were thou d, tc d. O ing s up b
ifInErrors	.1.3.6.1.2.1.2.2.1.14	Num onta rom - lay
ifOutErrors	.1.3.6.1.2.1.2.2.1.20	Num coul of er

Daily Authentication Statistics OIDs

These OIDs monitor daily authentication statistics on Policy Manager. These OIDs show the total number of authenticationss on the Policy Manager server for the given day. There are no traps associated with these OIDs.

Table 1. Daily Authentication Statistics OIDs

OID	Description
.1.3.6.1.4.1.14823.1.6.1.1.2.3.1.17	Daily successful authenticatic
.1.3.6.1.4.1.14823.1.6.1.1.2.3.1.18	Daily failed authentications
.1.3.6.1.4.1.14823.1.6.1.1.2.3.1.19	Total number of authenticatic

Authentication Count OIDs

This section describes the SNMP trap OIDs in the *policyServerTable* of the Policy Manager MIB (CPPM-MIB) that show the total number of authentications and authorizations on the Policy Manager server over the past 24 hours.

Authentication counts are calculated from 00:00:00 through 23:59:59 each day. If polls are done several times a day, by subtracting a previous poll's count from the latest poll's count, the number of authentications within a polling interval can be calculated. The authentication count traps and their OIDs are described in the following table:

Table 1. Authentication Count OIDs

Trap Name	OID	Descr
dailyFailedAuthCount	.1.3.6.1.4.1.14823.1.6.1.1.2.3.1.18	Total (ticatic
dailySuccessAuthCount	1.3.6.1.4.1.14823.1.6.1.1.2.3.1.17	Total (uthen
dailyTotalAuthCount	.1.3.6.1.4.1.14823.1.6.1.1.2.3.1.19	Total (ns

Policy Manager SNMP Traps

The following table provides the description and OID for each Policy Manager SNMP trap.

Table 1. SNMP Traps Supported by the SNMP Private MIB

Trap Name	OID	Desc
cppmActivationExpiry	.1.3.6.1.4.1.14823.1.6.1.1.200.1002	Indic tivati *deAp* Serve *aysRe*
cppmClusterLicense Usage	.1.3.6.1.4.1.14823.1.6.1.1.200.1016	Indic ense • < in n. • < *t*> us • < *ni* at
cppmClusterNode AddNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1006	Indic ger n <*cpp* addre er.
cppmClusterNode DbldNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1009	Indic the c <*cpp* addre
cppmClusterNode DelNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1007	Indic as be <*cpp* addre cluste

Trap Name	OID	Desc
cppmClusterNode NSyncNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1010	Indic he cl tate. • <c e oc • <c nc at c.
cppmClusterNode PromNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1008	Indic nage <*cpp* addre sher.
cppmClusterLicenseOverrun	.1.3.6.1.4.1.14823.1.6.1.1.200.1023	Indic r has
cppmClusterPwd ChangeNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1011	Indic been
cppmConfigReset	.1.3.6.1.4.1.14823.1.6.1.1.200.1012	Indic e's cc
cppmConfigRestore	.1.3.6.1.4.1.14823.1.6.1.1.200.1013	Indic e's cc
cppmHighCPULoadNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1021	Indic h am CPU
cppmLicenseExpiry	.1.3.6.1.4.1.14823.1.6.1.1.200.1001	Indic ociat ion <i he Pc <*cpp*
cppmLowDiskSpace	.1.3.6.1.4.1.14823.1.6.1.1.200.1004	Indic ow o *mDis* speci

Trap Name	OID	Descr
cppmLowMemory	.1.3.6.1.4.1.14823.1.6.1.1.200.1005	Indic ow o *Mem* ified i
cppmNodeCertExpiry	.1.3.6.1.4.1.14823.1.6.1.1.200.1003	Indic iated *ionN* expir ays.
cppmServiceRestartNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1019	Indic rted.
cppmServiceStartNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1018	Indic d
cppmServiceStopNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1017	Indic manu is, an fer <F tails. s refe not a smor round he, c -igslc
cppmUpdateNotification	.1.3.6.1.4.1.14823.1.6.1.1.200.1014	Indic ion h
cppmUpgrade Notification	.1.3.6.1.4.1.14823.1.6.1.1.200.1015	Indic ion h

SNMP Poll Disk Space Monitoring

These OIDs monitor total and available disk space across system.

Table 1. SNMP Poll Disk Space Monitoring

OID	Description
1.3.6.1.4.1.14823.1.6.1.1.1.1.1.14.0	Total Disk Space

OID	Description
1.3.6.1.4.1.14823.1.6.1.1.1.1.15.0	Available Disk Space

SNMP Trap Details

ClearPass Policy Manager leverages native SNMP support from the UC Davis 'net-SNMP' MIB package to send trap notifications for the following events. In these trap OIDs, the value of X varies from 1 through N, depending on the number of process states that are being checked. Details about specific OIDs associated with the processes are listed in this section.

For more information, see:

- Policy Manager Processes Stop and Start Events
- CPU Load Average Exceed Events
- CPU Load Average Traps
- Disk Space Threshold Traps
- Disk Utilization Threshold Exceed Events
- Network Interface Link Up and Link Down Events
- Network Interface Status Traps
- Process Status Traps
- SNMP Daemon Trap Events

Subtopics

Policy Manager Processes Stop and Start Events
CPU Load Average Exceed Events
CPU Load Average Traps
Disk Utilization Threshold Exceed Events
Network Interface Status Traps
Network Interface Link Up and Link Down Events
Process Status Traps
SNMP Daemon Trap Events

Policy Manager Processes Stop and Start Events

The OIDs for the start and stop events for Policy Manager processes are described in the following table:

Table 1. Processes Stop and Start Events

OID	Description
.1.3.6.1.4.1.2021.8.1.2.X	Process Name
.1.3.6.1.4.1.2021.8.1.101.X	Process Status Message.

CPU Load Average Exceed Events

The OIDs for the CPU load average exceed events for 1, 5, and 15 minute threshold are described in the following table:

Table 1. CPU Load Average Exceed Events or 1, 5, and 15 Minute Threshold

OID	Description
.1.3.6.1.4.1.2021.10.1.100.1	Error flag on the CPU load - 1
.1.3.6.1.4.1.2021.10.1.2.1	Name of CPU load - 1 averag
.1.3.6.1.4.1.2021.10.1.100.2	Error flag on the CPU load - 5
.1.3.6.1.4.1.2021.10.1.2.2	Name of CPU load - 5 averag
.1.3.6.1.4.1.2021.10.1.100.3	Error flag on the CPU load - 1
.1.3.6.1.4.1.2021.10.1.2.3	Name of CPU load - 15 avera

CPU Load Average Traps

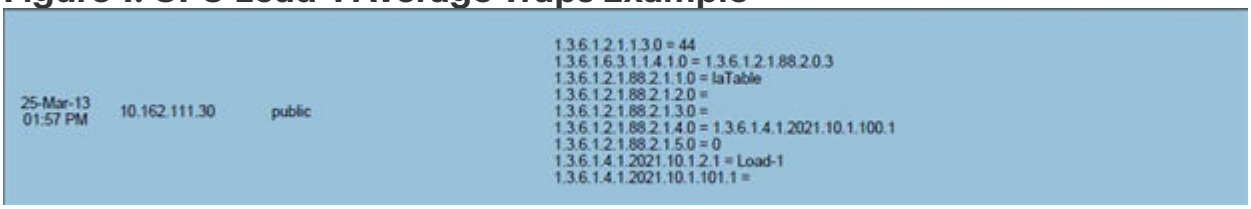
The OIDs for the CPU load average traps are described in the following tables:

CPU Load-1 Average Traps

Table 1. CPU Load - 1 Average Traps

OID	Description
.1.3.6.1.4.1.2021.10.1.100.1	Error flag on the CPU load - 1 average. Value of **1** indicates the load - 1 has crossed its threshold; a value of **0** indicates otherwise.
.1.3.6.1.4.1.2021.10.1.2.1	Name of CPU load - 1 average.

Figure 1. CPU Load-1 Average Traps Example

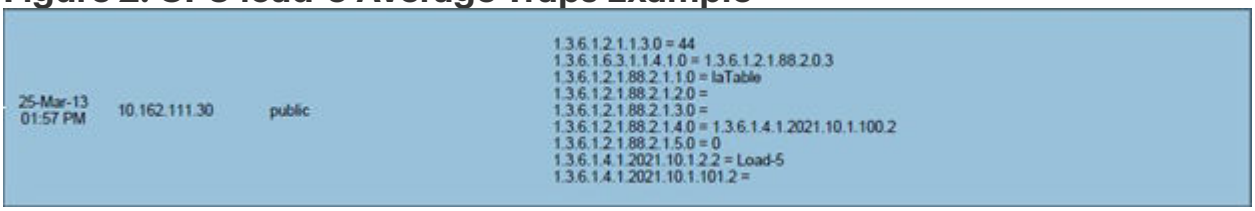


CPU Load-5 Average Traps

Table 2. CPU Load - 5 Average Traps

OID	Description
.1.3.6.1.4.1.2021.10.1.100.2	Error flag on the CPU load - 5 average. Value of **1** indicates the load - 5 has crossed its threshold; a value of **0** indicates otherwise.
.1.3.6.1.4.1.2021.10.1.2.2	Name of CPU load - 5 average.

Figure 2. CPU load-5 Average Traps Example



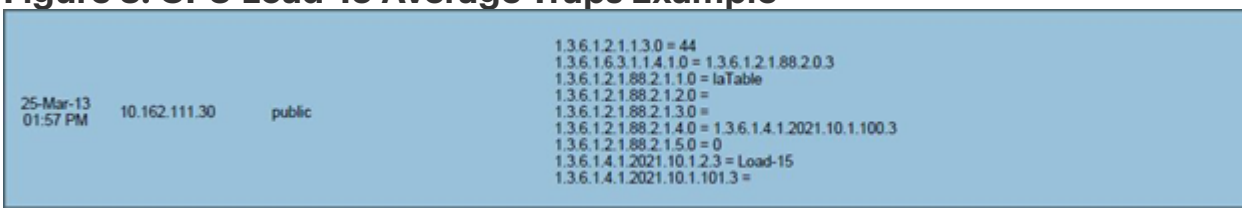
CPU Load-15 Average Traps

Table 3. CPU Load - 5 Average Traps

OID	Description
.1.3.6.1.4.1.2021.10.1.100.3	Error flag on the CPU load - 15 average. Value of **1** indicates the load - 15 has crossed i

OID	Description
	ts threshold; a value of **0** indicates otherwi se.
.1.3.6.1.4.1.2021.10.1.2.3	Name of CPU load - 15 average.

Figure 3. CPU Load-15 Average Traps Example



Disk Utilization Threshold Exceed Events

The OIDs for the disk utilization threshold exceed events are described in the following table:

Table 1. Disk Utilization Threshold Exceed Events

OID	Description
1.3.6.1.4.1.14823.1.6.1.1.200.1004	Indicates the appliance is ru k partitions , as determined l d under Cluster-Wide Param

Network Interface Status Traps

The OIDs for the network interface status traps are described in the following table:

Table 1. Network Interface Status Traps

OID	Description
.1.3.6.1.6.3.1.1.5.3	Indicates the link down trap w and **ifOperStatus** values set t
.1.3.6.1.6.3.1.1.5.4	Indicates the link up trap with **ifOperStatus** values set to **1**.

In each case, the **ifIndex** value is set to **2** for the management interface and **3** for the data port interface.

Network Interface Link Up and Link Down Events

The network interface up and down events are described in the following table:

Table 1. Network Interface Link Up and Link Down OIDs

OID	Description
.1.3.6.1.6.3.1.1.5.3	Link Down.
.1.3.6.1.6.3.1.1.5.4	Link Up

Process Status Traps

All process status traps are sent by snmpd, the Policy Managernet-snmp daemon

Subtopics

RADIUS server stop SNMP trap
RADIUS server start SNMP trap
Admin Server stop SNMP trap
Admin Server start SNMP trap
System Auxiliary server stop SNMP trap
System Auxiliary server start SNMP trap
Policy server stop SNMP trap
Policy server start SNMP trap
Async DB write service stop SNMP trap
Async DB write service start SNMP trap
DB replication service stop SNMP trap
DB replication service start SNMP trap
DB Change Notification server stop SNMP trap
DB Change Notification server start SNMP trap
Async netd service stop SNMP trap
Async netd service start SNMP trap
Cache service stop SNMP trap
Cache service start SNMP trap
AirGroup Notification service stop SNMP trap
AirGroup Notification service start SNMP trap

Micros Fidelio FIAS service stop SNMP trap
Micros Fidelio FIAS service start SNMP trap
TACACS server stop SNMP trap
TACACS server start SNMP trap
Virtual IP service stop SNMP trap
Virtual IP service start SNMP trap
Statistics Collection service stop SNMP trap
Statistics Collection service start SNMP trap
Statistics Aggregation service stop SNMP trap
Statistics Aggregation service start SNMP trap

RADIUS server stop SNMP trap

NOTE
The below trap is sent only on the abnormal stop of the Radius server.

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.5
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.5: cpass-radius-server
- .1.3.6.1.4.1.2021.8.1.101.5: Radius server [cpass-radius-server] is stopped

RADIUS server start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:

- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.5
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.5: cpass-radius-server
- .1.3.6.1.4.1.2021.8.1.101.5: Radius server [cpass-radius-server] is running

Admin Server stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.1
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.1: cpass-admin-server
- .1.3.6.1.4.1.2021.8.1.101.1: Admin server [cpass-admin-server] is stopped

Admin Server start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.1
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.1: cpass-admin-server
- .1.3.6.1.4.1.2021.8.1.101.1: Admin server [cpass-admin-server] is running

System Auxiliary server stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.2
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.2: cpass-system-auxiliary-server
- .1.3.6.1.4.1.2021.8.1.101.2: System auxiliary service [cpass-system-auxiliary-server] is stopped

System Auxiliary server start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.2
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.2: cpass-system-auxiliary-server
- .1.3.6.1.4.1.2021.8.1.101.2: System auxiliary service [cpass-system-auxiliary-server] is running

Policy server stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable

- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.3
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.3: cpass-policy-server
- .1.3.6.1.4.1.2021.8.1.101.3: Policy server [cpass-policy-server] is stopped

Policy server start SNMP trap

snmpTrapOID: .1.3.6.1.2.1.88.2.0.3

.1.3.6.1.2.1.88.2.1.1.0: extTable

.1.3.6.1.2.1.88.2.1.2.0:

.1.3.6.1.2.1.88.2.1.3.0:

.1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.3

.1.3.6.1.2.1.88.2.1.5.0: 0

.1.3.6.1.4.1.2021.8.1.2.3: cpass-policy-server

.1.3.6.1.4.1.2021.8.1.101.3: Policy server [cpass-policy-server] is running

Async DB write service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.6
- .1.3.6.1.2.1.88.2.1.5.0: 1
- .1.3.6.1.4.1.2021.8.1.2.6: cpass-dbwrite-server
- .1.3.6.1.4.1.2021.8.1.101.6: Async DB write service [cpass-dbwrite-server] is stopped

Async DB write service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.6
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.6: cpass-dbwrite-server
- .1.3.6.1.4.1.2021.8.1.101.6: Async DB write service [cpass-dbwrite-server] is running

DB replication service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.7
- .1.3.6.1.2.1.88.2.1.5.0: 1
- .1.3.6.1.4.1.2021.8.1.2.7: cpass-repl-server
- .1.3.6.1.4.1.2021.8.1.101.7: DB replication service [cpass-repl-server] is stopped

DB replication service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable

- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.7
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.7: cpass-repl-server
- .1.3.6.1.4.1.2021.8.1.101.7: DB replication service [cpass-repl-server] is running

DB Change Notification server stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.8
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.8: cpass-dbcn-server
- .1.3.6.1.4.1.2021.8.1.101.8: DB change notification server [cpass-dbcn-server] is stopped

DB Change Notification server start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.8
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.8: cpass-dbcn-server

- .1.3.6.1.4.1.2021.8.1.101.8: DB change notification server [cpass-dbcn-server] is running

Async netd service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.9
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.9: cpass-async-netd
- .1.3.6.1.4.1.2021.8.1.101.9: Async netd service [cpass-async-netd] is stopped

Async netd service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.9
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.9: cpass-async-netd
- .1.3.6.1.4.1.2021.8.1.101.9: Async netd service [cpass-async-netd] is running

Cache service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2

- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.10
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.10: cpass--cache-server
- .1.3.6.1.4.1.2021.8.1.101.10: cache [cpass--cache-server] is stopped

Cache service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.10
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.10: cpass--cache-server
- .1.3.6.1.4.1.2021.8.1.101.10: cache [cpass--cache-server] is running

AirGroup Notification service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.11
- .1.3.6.1.2.1.88.2.1.5.0: 3

- .1.3.6.1.4.1.2021.8.1.2.11: airgroup-notify
- .1.3.6.1.4.1.2021.8.1.101.11: AirGroup notification service [airgroup-notify] is stopped

AirGroup Notification service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.11
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.11: airgroup-notify
- .1.3.6.1.4.1.2021.8.1.101.11: AirGroup notification service [airgroup-notify] is running

Micros Fidelio FIAS service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.12
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.12: fias_server
- .1.3.6.1.4.1.2021.8.1.101.12: Micros Fidelio FIAS [fias_server] is stopped

Micros Fidelio FIAS service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.12
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.12: fias_server
- .1.3.6.1.4.1.2021.8.1.101.12: Micros Fidelio FIAS [fias_server] is running

TACACS server stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.4
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.4: cpass-tacacs-server
- .1.3.6.1.4.1.2021.8.1.101.4: TACACS server [cpass-tacacs-server] is stopped

TACACS server start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.4

- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.4: cpass-tacacs-server
- .1.3.6.1.4.1.2021.8.1.101.4: TACACS server [cpass-tacacs-server] is running

Virtual IP service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.13
- .1.3.6.1.2.1.88.2.1.5.0: 1
- .1.3.6.1.4.1.2021.8.1.2.13: cpass-vip
- .1.3.6.1.4.1.2021.8.1.101.13: ClearPass Virtual IP service [cpass-vip] is stopped

Virtual IP service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0:
- .1.3.6.1.2.1.88.2.1.3.0:
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.13
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.13: cpass-vip
- .1.3.6.1.4.1.2021.8.1.101.13: ClearPass Virtual IP service [cpass-vip] is running

Statistics Collection service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0
- .1.3.6.1.2.1.88.2.1.3.0
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.15
- .1.3.6.1.2.1.88.2.1.5.0: 3
- .1.3.6.1.4.1.2021.8.1.2.15: cpass-statsd-server
- .1.3.6.1.4.1.2021.8.1.101.15: Stats collection service [cpass-statsd-server] is stopped

Statistics Collection service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0
- .1.3.6.1.2.1.88.2.1.3.0
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.15
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.15: cpass-statsd-server
- .1.3.6.1.4.1.2021.8.1.101.15: Stats collection service [cpass-statsd-server] is running

Statistics Aggregation service stop SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.2
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0
- .1.3.6.1.2.1.88.2.1.3.0

- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.14
- .1.3.6.1.2.1.88.2.1.5.0: 1
- .1.3.6.1.4.1.2021.8.1.2.14: cpass-carbon-server
- .1.3.6.1.4.1.2021.8.1.101.14: Stats aggregation service [cpass-carbon-server] is stopped

Statistics Aggregation service start SNMP trap

- snmpTrapOID: .1.3.6.1.2.1.88.2.0.3
- .1.3.6.1.2.1.88.2.1.1.0: extTable
- .1.3.6.1.2.1.88.2.1.2.0
- .1.3.6.1.2.1.88.2.1.3.0
- .1.3.6.1.2.1.88.2.1.4.0: .1.3.6.1.4.1.2021.8.1.100.14
- .1.3.6.1.2.1.88.2.1.5.0: 0
- .1.3.6.1.4.1.2021.8.1.2.14: cpass-carbon-server
- .1.3.6.1.4.1.2021.8.1.101.14: Stats aggregation service [cpass-carbon-server] is running.

SNMP Daemon Trap Events

This section contains OIDs for various trap events that are sent from ClearPass Policy Manager.

Table 1. SNMP Daemon Trap Events

OID	Description
.1.3.6.1.6.3.1.1.5.1	Cold Start trap indicates th **tsnmp** daemon and that its ave been altered.
.1.3.6.1.6.3.1.1.5.2	Warm Start trap indicates t **etsnmp** daemon and that i t been altered.

Important System Events

- Admin User Interface Events
- Admin Server Events
- Async Service Events
- Domain Controller Events
- System Configuration Events
- Update Events
- Cluster Events
- Command Line Events
- Database Replication Services Events
- Licensing Events
- Policy Server Events
- RADIUS/TACACS+ Server Events
- Service Names
- SNMP Events
- Support Shell Events
- System Auxiliary Service Events
- System Monitor Events

This topic describes the important System Events logged by Policy Manager. These messages are available for consumption on the administrative interface, and in the form of a syslog stream. The events below are in the following format

<*Source*>, <*Level*>, <*Category*>, <*Message*>

Elements listed below within angle brackets (for example, *<content>*) are variable, and are substituted by Policy Manager as applicable (such as an IP address). For the list of available service names, refer to Service Names .

Admin User Interface Events

Critical Events

"Admin UI", "ERROR" "Email Failed", "Sending email failed"

“Admin UI”, “ERROR” “SMS Failed”, “Sending SMS failed”

“Admin UI”, “WARN”, “Login Failed”, “User:<X>”

"Admin UI", "WARN", "Login Failed", description

Info Events

"Admin UI", "INFO", "Logged out"

"Admin UI", "INFO", "Session destroyed"

"Admin UI", "INFO", "Logged in", description

"Admin UI", "INFO", "Clear Authentication Cache", “Cache is cleared for authentication source <X>"

"Admin UI", "INFO", "Clear Denylist User Cache", “Denylist Users cache is cleared for authentication source <X>"

"Admin UI", "INFO", "Server Certificate", "Subject:<X>“, "Updated"

"Install Update", "INFO", "Installing Update", "File: <X>", "Success"

"Admin UI", “INFO” “Email Successful”, “Sending email succeeded”

"Admin UI", “INFO” “SMS Successful”, “Sending SMS succeeded”

Admin Server Events

Info Events

“Admin server”, “INFO”, “Performed action start on Admin server”

Async Service Events

Info Events

“Async DB write service”, “INFO”, “Performed action start on Async DB write service”

“Zone cache”, “INFO”, “Performed action start on Zone cache”

“Async netd service”, “INFO”, “Performed action start on Async netd service”

Domain Controller Events

Critical Events

“netleave”, “ERROR”, “Failed to remove <HOSTNAME> from the domain <DOMAIN_NAME>”

“netjoin”, “WARN”, “configuration”, “<HOSTNAME> failed to join the domain <DOMAIN NAME> with domain controller as <DOMAIN CONTROLLER>”

Info Events

“Netjoin”, “INFO”, "<HOSTNAME> joined the domain <REALM>"

“Netjoin”, “INFO”, “<HOSTNAME> removed from the domain <DOMAIN_NAME>“

Policy Manager System Configuration Events

Critical Events

“DNS”, “ERROR”, “Failed configure DNS servers = <X>”

“datetime”, “ERROR”, “Failed to change system datetime.”

“hostname”, “ERROR”, “Setting hostname to <X> failed”

“ipaddress”, “ERROR”, “Testing cluster node connectivity failed”

“System TimeCheck “, “ WARN ,” , “Restarting CPPM services as the system detected time drift , Current system time= 2016-07-13 17:00:01, System time 5 mins back = 2016-06-20 16:55:01”

Info Events

“Cluster”, “INFO”, “Setup”, “Database initialized”

“hostname”, “INFO”, “configuration”, “Hostname set to <X>”

“ipaddress”, “INFO”, “configuration”, Management port information updated to - IpAddress = <X>, Netmask = <X>, Gateway = <X>”

“IpAddress”, “INFO”, "Data port information updated to - IpAddress = <X>, Netmask = <Y>, Gateway = <Z>"

“DNS”, “INFO”, “configuration”, “Successfully configured DNS servers - <X>”

“Time Config”, “INFO”, “Remote Time Server”, “Old List: <X>\nNew List: <Y>”

“timezone”, “INFO”, “configuration”, “”

“datetime”, “INFO”, “configuration”, “Successfully changed system datetime.\nOld time was <X>”

Policy Manager Update Events

Critical Events

“Install Update”, “ERROR”, “Installing Update”, “File: <X>”, “Failed with exit status - <Y>”

“ClearPass Firmware Update Checker”, “ERROR”, “Firmware Update Checker”, “No HPE Passport Credentials were supplied. To find new plugins, you must provide your HPE Passport Credentials in the application configuration”

Info Events

“ClearPass Updater”, “INFO”, “Hotfixes Updates”, “Updated Hotfixes from File”

“ClearPass Updater”, “INFO”, “Fingerprints Updates”, “Updated fingerprints from File”

“ClearPass Updater”, “INFO”, “Updated AV/AS from ClearPass Portal (Online)”

“ClearPass Updater”, “INFO”,” Updated Hotfixes from ClearPass Portal (Online)”

Cluster Events

Critical Events

“Cluster”, “ERROR”, “SetupSubscriber”, “Failed to add subscriber node with management IP=<IP>“

Info Events

"AddNode", “INFO”, "Added subscriber node with management IP=<IP>"

"DropNode", “INFO”, "Dropping node with management IP=<IP>, hostname=<Hostname>"

Command Line Events

Info Events

"Command Line”, “INFO”, “User:appadmin"

Database Replication Services Events

Info Events

"DB replication service”, “INFO”, “Performed action start on DB replication service”

"DB replication service”, “INFO”, “Performed action stop on DB replication service”

“DB change notification server”, “INFO”, “Performed action start on DB change notification server”

“DB replication service”, “INFO”, “Performed action start on DB replication service”

Licensing Events

Critical Events

“Admin UI”, “WARN”, “Activation Failed”, “Action Status: This Activation Request Token is already in use by another instance\nProduct Name: Policy Manager\nLicense Type: <X>\nUser Count: <Y>”

Info Events

“Admin UI”, “INFO”, “Add License”, “Product Name: Policy Manager\nLicense Type: <X>\nUser Count: <Y>”

Policy Server Events

Info Events

“Policy Server”, “INFO”, “Performed action start on Policy server”

“Policy Server”, “INFO”, “Performed action stop on Policy server”

RADIUS/TACACS+ Server Events

Critical Events

“TACACSServer”, “ERROR”, “Request”, “Nad Ip=<X> not configured”

“RADIUS”, “WARN”, “Authentication”, “Ignoring request from unknown client <IP>:<PORT>”

“RADIUS”, “ERROR”, “Authentication”, “Received packet from <IP> with invalid Message-Authenticator! (Shared secret is incorrect.)”

“RADIUS”, “ERROR”, “Received Accounting-Response packet from client <IP Address> port 1813 with invalid signature (err=2)! (Shared secret is incorrect.)”

“RADIUS”, “ERROR”, “Received Access-Accept packet from client <IP Address> port 1812 with invalid signature (err=2)! (Shared secret is incorrect.)”

Info Events

“RADIUS”, “INFO”, “Performed action start on Radius server”

“RADIUS”, “INFO”, “Performed action restart on Radius server

“TACACS server”, “INFO”, “Performed action start on TACACS server”

“TACACS server”, “INFO”, “Performed action stop on TACACS server”

Service Names

- AirGroup notification service
- Async DB write service
- Async network services
- DB change notification server
- DB replication service
- Micros Fidelio FIAS
- Zone cache
- Policy server
- RADIUS server
- System auxiliary services
- System monitor service

- TACACS server
- Virtual IP service
- [YourServerName] Domain service

SNMP Events

Critical Events

“SNMPService”, “ERROR”, “ReadDeviceInfo”, “SNMP GET failed for device <X> with error=No response received\nReading sysObjectId failed for device=<X>\nReading switch initialization info failed for <X>”

"SNMPService","ERROR", "Error fetching table snmpTargetAddr. Request timed out. Error reading SNMP target table for NAD=10.1.1.1 Maybe SNMP target address table is not supported by device? Allow NAD update. SNMP GET failed for device 10.1.1.1 with error=No response received Reading sysObjectId failed for device=10.1.1.1 Reading switch initialization info failed for 10.1.1.1”

Info Events

“SNMPService”, “INFO”, “Device information not read for <Ip Address> since no traps are configured to this node”

Support Shell Events

Info Events

“Support Shell” , “INFO”, “User:customersupport”

System Auxiliary Service Events

Info Events

“System auxiliary service”, “INFO”, “Performed action start on System auxiliary service”

System Monitor Events

Critical Events

“Sysmon”, “ERROR”, “System”, “System is running with low memory. Available memory = <X>%”

“Sysmon”, “ERROR”, “System”, “System is running with low disk space. Available disk space = <X>%”

“System TimeCheck”, “WARN”, “Restart Services”, “Restarting CPPM services as the system detected time drift. Current system time= <X>, System time 5 mins back = <Y>”

Info Events

“<Service Name>”, “INFO”, “restart”, “Performed action restart on <Service Name>”

“SYSTEM”, “INFO”, “<X> restarted”, “System monitor restarted <X>, as it seemed to have stopped abruptly”

"SYSTEM", "ERROR", "Updating CRLs failed", "Could not retrieve CRL from <URL>."

“System monitor service”, “INFO”, “Performed action start on System monitor service”

"Shutdown” “INFO” system "System is shutting down" Success

Error Codes

Table 1 describes the ClearPass Policy Manager error codes:

Table 1. Policy Manager Policy Manager Error Codes

Code	Description	Type
0	Success	Success
101	Failed to perform service classification	Internal Error
102	Failed to perform policy evaluation	Internal Error
103	Failed to perform posture notification	Internal Error
104	Failed to query authstatus	Internal Error
105	Internal error in performing authentication	Internal Error
106	Internal error in RADIUS server	Internal Error
201	User not found	Authentication failure
202	Password mismatch	Authentication failure
203	Failed to contact Authentication Source	Authentication failure
204	Failed to classify request to service	Authentication failure

Code	Description	Type
205	Authentication Source not c onfigured for service	Authentication failure
206	Access denied by policy	Authentication failure
207	Failed to get client MAC Add ress in order to perform We b authentication	Authentication failure
208	No response from home se rver	Authentication failure
209	No password in request	Authentication failure
210	Unknown CA in client certifi cate	Authentication failure
211	Client certificate not valid	Authentication failure
212	Client certificate has expire d	Authentication failure
213	Certificate comparison faile d	Authentication failure
214	No certificate in authenticati on source	Authentication failure
215	TLS session error	Authentication failure
216	User authentication failed	Authentication failure
217	Search failed due to insuffici ent permissions	Authentication failure
218	Authentication source timed out	Authentication failure
219	Bad search filter	Authentication failure
220	Search failed	Authentication failure
221	Authentication source error	Authentication failure
222	Password change error	Authentication failure

Code	Description	Type
223	Username not available in request	Authentication failure
224	CallingStationID not available in request	Authentication failure
225	User account disabled	Authentication failure
226	User account expired or not active yet	Authentication failure
227	User account needs approval	Authentication failure
228	User account has exceeded bandwidth limit	Authentication failure
229	User account has exceeded session duration limit	Authentication failure
230	User account has exceeded session count limit	Authentication failure
5001	Internal Error	Command and Control
5002	Invalid MAC Address	Command and Control
5003	Invalid request received	Command and Control
5004	Insufficient parameters received	Command and Control
5005	Query - No MAC address record found	Command and Control
5006	Query - No supported actions	Command and Control
5007	Query - Cannot fetch MAC address details	Command and Control
5008	Request: MAC address not online	Command and Control
5009	Request: No MAC address record found	Command and Control

Code	Description	Type
6001	Unsupported TACACS parameter in request	TACACS Protocol
6002	Invalid sequence number	TACACS Protocol
6003	Sequence number overflow	TACACS Protocol
6101	Not enough inputs to perform authentication	TACACS Authentication
6102	Authentication privilege level mismatch	TACACS Authentication
6103	No enforcement profiles matched to perform authentication	TACACS Authentication
6201	Authorization failed as session is not authenticated	TACACS Authorization
6202	Authorization privilege level mismatch	TACACS Authorization
6203	Command not allowed	TACACS Authorization
6204	No enforcement profiles matched to perform command authorization	TACACS Authorization
6301	New password entered does not match	TACACS Change Password
6302	Empty password	TACACS Change Password
6303	Change password allowed only for local users	TACACS Change Password
6304	Internal error in performing change password	TACACS Change Password
9001	Wrong shared secret	RADIUS Protocol
9002	Request timed out	RADIUS Protocol
9003	Phase 2 PAC failure	RADIUS Protocol

Code	Description	Type
9004	Client rejected after PAC p rovisioning	RADIUS Protocol
9005	Client does not support pos ture request	RADIUS Protocol
9006	Received error TLV from clie nt	RADIUS Protocol
9007	Received failure TLV from cli ent	RADIUS Protocol
9008	Phase 2 PAC not found	RADIUS Protocol
9009	Unknown Phase 2 PAC	RADIUS Protocol
9010	Invalid Phase 2 PAC	RADIUS Protocol
9011	PAC verification failed	RADIUS Protocol
9012	PAC binding failed	RADIUS Protocol
9013	Session resumption failed	RADIUS Protocol
9014	Cached session data error	RADIUS Protocol
9015	Client does not support con figured EAP methods	RADIUS Protocol
9016	Client did not send Cryptobi nding TLV	RADIUS Protocol
9017	Failed to contact OCSP Se rver	RADIUS Protocol
9018	RADIUS protocol error	RADIUS Protocol
9019	Client sent conflicting identi ties	RADIUS Protocol

Cipher Suites

This section describes the cipher suites supported by Policy Manager.

Cipher suites are sets of instructions on how to secure a network through SSL (Secure Sockets Layer) or TLS (Transport Layer Security). As such, cipher suites provide essential information on how to communicate secure data when using HTTPS, FTPS, SMTP and other network protocols. This information takes the form of algorithms and protocols that help determine how a web server secures a client's web traffic.

- EAP (RADIUS) Cipher Suites
- OnGuard Agent Cipher Suites
- RadSec Cipher Suites
- SSH Cipher Suites
- SSL Cipher Suites

Subtopics

EAP (RADIUS) Cipher Suites
OnGuard Agent Cipher Suites
RadSec Cipher Suites
SSH Cipher Suites
SSL Cipher Suites

EAP (RADIUS) Cipher Suites

The following tables provides the list of available EAP (Extensible Authentication Protocol) RADIUS cipher suites that are supported in non-FIPS mode, FIPS mode, and Common Criteria (CC) mode :

Table 1. TLS 1.3 EAP/RADIUS Cipher Suites Available in Non - FIPS Mode

OpenSSL Cipher Name	IANA Cipher Name
TLS_AES_256_GCM_SHA384	TLS_AES_256_GCM_SHA384
TLS_CHACHA20_POLY1305_SHA256	TLS_CHACHA20_POLY1305_SHA2
TLS_AES_128_GCM_SHA256	TLS_AES_128_GCM_SHA256
TLS_AES_128_CCM_SHA256	TLS_AES_128_CCM_SHA256

Table 2. TLS 1.2 EAP/RADIUS Cipher Suites Available in Non - FIPS Mode

OpenSSL Cipher Name	IANA Cipher Name
ECDHE-ECDSA-AES256-GCM-SHA384	TLS_ECDHE_ECDSA_WITH_AES_25
ECDHE-RSA-AES256-GCM-SHA384	TLS_ECDHE_RSA_WITH_AES_256_C
ECDHE-ECDSA-CHACHA20-POLY1305	TLS_ECDHE_ECDSA_WITH_CHACH
ECDHE-RSA-CHACHA20-POLY1305	TLS_ECDHE_RSA_WITH_CHACHA2
ECDHE-ECDSA-AES256-CCM	TLS_ECDHE_ECDSA_WITH_AES_25
ECDHE-ECDSA-AES128-GCM-SHA256	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-RSA-AES128-GCM-SHA256	TLS_ECDHE_RSA_WITH_AES_128_G
ECDHE-ECDSA-AES128-CCM	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-ECDSA-AES256-SHA384	TLS_ECDHE_ECDSA_WITH_AES_25
ECDHE-RSA-AES256-SHA384	TLS_ECDHE_RSA_WITH_AES_256_C
ECDHE-ECDSA-AES128-SHA256	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-RSA-AES128-SHA256	TLS_ECDHE_RSA_WITH_AES_128_C
ECDHE-ECDSA-AES256-SHA	TLS_ECDHE_ECDSA_WITH_AES_25
ECDHE-RSA-AES256-SHA	TLS_ECDHE_RSA_WITH_AES_256_C
ECDHE-ECDSA-AES128-SHA	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-RSA-AES128-SHA	TLS_ECDHE_RSA_WITH_AES_128_C
AES256-GCM-SHA384	TLS_RSA_WITH_AES_256_GCM_SH
AES256-CCM	TLS_RSA_WITH_AES_256_CCM
AES128-GCM-SHA256	TLS_RSA_WITH_AES_128_GCM_SH
AES128-CCM	TLS_RSA_WITH_AES_128_CCM
AES256-SHA256	TLS_RSA_WITH_AES_256_CBC_SH
AES128-SHA256	TLS_RSA_WITH_AES_128_CBC_SHA
AES256-SHA	TLS_RSA_WITH_AES_256_CBC_SH

OpenSSL Cipher Name	IANA Cipher Name
AES128-SHA	TLS_RSA_WITH_AES_128_CBC_SHA
DHE-RSA-AES128-GCM-SHA256	TLS_DHE_RSA_WITH_AES_128_GCM
DHE-RSA-AES256-GCM-SHA384	TLS_DHE_RSA_WITH_AES_256_GCI
DHE-RSA-CHACHA20-POLY1305	TLS_DHE_RSA_WITH_CHACHA20_I
DHE-RSA-AES256-CCM	TLS_DHE_RSA_WITH_AES_256_CCI
DHE-RSA-AES128-CCM	TLS_DHE_RSA_WITH_AES_128_CCM
DHE-RSA-AES256-SHA256	TLS_DHE_RSA_WITH_AES_256_CBC
DHE-RSA-AES128-SHA256	TLS_DHE_RSA_WITH_AES_128_CBC
DHE-RSA-AES256-SHA	TLS_DHE_RSA_WITH_AES_256_CBC
DHE-RSA-AES128-SHA	TLS_DHE_RSA_WITH_AES_128_CBC

Table 3. TLS 1.3 EAP/RADIUS Cipher Suites Available in FIPS Mode

OpenSSL Cipher Name	IANA Cipher Name
TLS_AES_256_GCM_SHA384	TLS_AES_256_GCM_SHA384
TLS_AES_128_GCM_SHA256	TLS_AES_128_GCM_SHA256
TLS_AES_128_CCM_SHA256	TLS_AES_128_CCM_SHA256

Table 4. TLS 1.2 EAP/RADIUS Cipher Suites Available in FIPS Mode

OpenSSL Cipher Name	IANA Cipher Name
ECDHE-ECDSA-AES256-GCM-SHA384	TLS_ECDHE_ECDSA_WITH_AES_25
ECDHE-RSA-AES256-GCM-SHA384	TLS_ECDHE_RSA_WITH_AES_256_C
ECDHE-ECDSA-AES256-CCM	TLS_ECDHE_ECDSA_WITH_AES_25
ECDHE-ECDSA-AES128-GCM-SHA256	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-RSA-AES128-GCM-SHA256	TLS_ECDHE_RSA_WITH_AES_128_G
ECDHE-ECDSA-AES128-CCM	TLS_ECDHE_ECDSA_WITH_AES_128

OpenSSL Cipher Name	IANA Cipher Name
ECDHE-ECDSA-AES128-SHA256	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-ECDSA-AES256-SHA384	TLS_ECDHE_ECDSA_WITH_AES_25
DHE-RSA-AES256-GCM-SHA384	TLS_DHE_RSA_WITH_AES_256_GCI
DHE-RSA-AES256-CCM	TLS_DHE_RSA_WITH_AES_256_CCI
DHE-RSA-AES128-GCM-SHA256	TLS_DHE_RSA_WITH_AES_128_GCM
DHE-RSA-AES128-CCM	TLS_DHE_RSA_WITH_AES_128_CCM
DHE-RSA-AES128-SHA	TLS_DHE_RSA_WITH_AES_128_CBC
DHE-RSA-AES256-SHA	TLS_DHE_RSA_WITH_AES_256_CBC
DHE-RSA-AES128-SHA256	TLS_DHE_RSA_WITH_AES_128_CBC
DHE-RSA-AES256-SHA256	TLS_DHE_RSA_WITH_AES_256_CBC

Table 5. TLS 1.3 EAP/RADIUS Cipher Suites Available in CC Mode

OpenSSL Cipher Name	IANA Cipher Name
TLS_AES_256_GCM_SHA384	TLS_AES_256_GCM_SHA384
TLS_AES_128_GCM_SHA256	TLS_AES_128_GCM_SHA256
TLS_AES_128_CCM_SHA256	TLS_AES_128_CCM_SHA256

Table 6. TLS 1.2 EAP/RADIUS Cipher Suites Available in CC Mode

OpenSSL Cipher Name	IANA Cipher Name
ECDHE-ECDSA-AES256-GCM-SHA384	TLS_ECDHE_ECDSA_WITH_AES_25
ECDHE-ECDSA-AES128-GCM-SHA256	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-ECDSA-AES128-SHA256	TLS_ECDHE_ECDSA_WITH_AES_128
ECDHE-ECDSA-AES256-SHA384	TLS_ECDHE_ECDSA_WITH_AES_25
DHE-RSA-AES128-SHA	TLS_DHE_RSA_WITH_AES_128_CBC
DHE-RSA-AES256-SHA	TLS_DHE_RSA_WITH_AES_256_CBC

OpenSSL Cipher Name	IANA Cipher Name
DHE-RSA-AES128-SHA256	TLS_DHE_RSA_WITH_AES_128_CBC
DHE-RSA-AES256-SHA256	TLS_DHE_RSA_WITH_AES_256_CBC

OnGuard Agent Cipher Suites

The following table provides the list of cipher suites available for the OnGuard agent:

Cipher Suites Available for OnGuard Agent	Hex Code
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256	0x00,0x9E
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384	0x00,0x9F
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256	0xC0,0x2B
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	0xC0,0x2C
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	0xC0,0x2F
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384	0xC0,0x30

RadSec Cipher Suites

The following tables provides the list of available RadSec cipher suites that are supported in non-FIPS mode, FIPS mode and CC mode:

Table 1. RadSec Cipher Suites Available in Non - FIPS Mode

Cipher Suites Supported in Non - FIPS Mode	Hex Code
TLS_AES_256_GCM_SHA384	0x1302
TLS_CHACHA20_POLY1305_SHA256	0x1303
TLS_AES_128_GCM_SHA256	0x1301
TLS_AES_128_CCM_SHA256	0x1304
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	0xc02c
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384	0xc030
TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256	0xcca9
TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256	0xcca8
TLS_ECDHE_ECDSA_WITH_AES_256_CCM	0xc0ad
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256	0xc02b
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	0xc02f
TLS_ECDHE_ECDSA_WITH_AES_128_CCM	0xc0ac
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256	0xc023
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256	0xc027
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA	0xc00a
TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA	0xc014
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA	0xc009
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA	0xc013
TLS_RSA_WITH_AES_256_GCM_SHA384	0x009d
TLS_RSA_WITH_AES_256_CCM	0xc09d
TLS_RSA_WITH_AES_128_GCM_SHA256	0x009c
TLS_RSA_WITH_AES_128_CCM	0xc09c
TLS_RSA_WITH_AES_256_CBC_SHA256	0x003d

Cipher Suites Supported in Non - FIPS Mode	Hex Code
TLS_RSA_WITH_AES_128_CBC_SHA256	0x003c
TLS_RSA_WITH_AES_256_CBC_SHA	0x0035
TLS_RSA_WITH_AES_128_CBC_SHA	0x002f
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384	0x009f
TLS_DHE_RSA_WITH_CHACHA20_POLY1305_SHA256	0xccaa
TLS_DHE_RSA_WITH_AES_256_CCM	0xc09f
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256	0x009e
TLS_DHE_RSA_WITH_AES_128_CCM	0xc09e
TLS_DHE_RSA_WITH_AES_256_CBC_SHA256	0x006b
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256	0x0067
TLS_DHE_RSA_WITH_AES_256_CBC_SHA	0x0039
TLS_DHE_RSA_WITH_AES_128_CBC_SHA	0x0033

Table 2. RadSec Cipher Suites Available in FIPS Mode

Cipher Suites Supported in FIPS Mode	Hex Code
TLS_AES_256_GCM_SHA384	0x1302
TLS_AES_128_GCM_SHA256	0x1301
TLS_AES_128_CCM_SHA256	0x1304
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	0xc02c
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384	0xc030
TLS_ECDHE_ECDSA_WITH_AES_256_CCM	0xc0ad
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256	0xc02b
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	0xc02f
TLS_ECDHE_ECDSA_WITH_AES_128_CCM	0xc0ac
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384	0x009f

Cipher Suites Supported in FIPS Mode	Hex Code
TLS_DHE_RSA_WITH_AES_256_CCM	0xc09f
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256	0x009e
TLS_DHE_RSA_WITH_AES_128_CCM	0xc09e

Table 3. RadSec Cipher Suites Available in CC Mode

Cipher Suites Supported in CC Mode	Hex Code
TLS_AES_256_GCM_SHA384	0x1302
TLS_AES_128_GCM_SHA256	0x1301
TLS_AES_128_CCM_SHA256	0x1304
TLS_DHE_RSA_WITH_AES_128_CBC_SHA	0x0033
TLS_DHE_RSA_WITH_AES_256_CBC_SHA	0x0039
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256	0x0067
TLS_DHE_RSA_WITH_AES_256_CBC_SHA256	0x006b
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256	0xc023
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384	0xc024
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256	0xc02b
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	0xc02c

SSH Cipher Suites

The following tables provide the lists of available cipher suites that Policy Manager operating as an SSH server or as an SSH client can use in Non-FIPS mode or FIPS mode.

- **Cipher Suites for ClearPass as SSH Client** lists the cipher suites that are available when Policy Manager Policy Manager acts as an SSH client.

- **Cipher Suites for ClearPass as SSH Server** lists the cipher suites supported when Policy Manager acts as an SSH server. Any SSH client connecting to the Policy Manager server can negotiate these ciphers with Policy Manager.

ClearPass as SSH Client in Non - FIPS Mode	
3des - cbc	aes256 - cbc
blowfish - cbc	aes128 - ctr
cast128 - cbc	aes192 - ctr
arcfour	aes256 - ctr
arcfour128	aes128 - gcm@openssh.com
arcfour256	aes256 - gcm@openssh.com
aes128 - cbc	chacha20 - poly1305@openssh.com
aes192 - cbc	—

ClearPass as SSH Server in Non - FIPS Mode	
aes128 - cbc	aes256 - ctr
aes256 - cbc	aes128 - gcm@openssh.com
es128 - ctr	aes256 - gcm@openssh.com

ClearPass as SSH Client in FIPS Mode and CC Mode	
3des - cbc	aes128 - ctr
aes128 - cbc	aes192 - ctr
aes192 - cbc	aes256 - ctr
aes256 - cbc	rijndael - cbc@lysator.liu.se

ClearPass as SSH Server in FIPS Mode and CC Mode	
aes128 - cbc	aes256 - ctr
aes256 - cbc	aes128 - gcm@openssh.com
aes128 - ctr	aes256 - gcm@openssh.com

SSL Cipher Suites

Refer to the following SSL Cipher Suite availability information:

- SSL Cipher Suites Available for PostgresSQL
- SSL Cipher Suites Available for Webserver
- SSL Cipher Suites Available in Non-FIPS Mode
- Cipher Suites for ClearPass as SSH Client in Non-FIPS Mode
- Cipher Suites for ClearPass as SSH Server in Non-FIPS Mode
- Cipher Suites for ClearPass as SSH Client in FIPS Mode and CC Mode
- Cipher Suites for ClearPass as SSH Server in FIPS Mode and CC Mode

SSL Cipher Suites Available for PostgresSQL

SSL Cipher Suites Available for PostgreSQL	
ECDHE - ECDSA - AES256 - GCM - SHA384:	DHE - RSA - AES128 - SHA:
ECDHE - ECDSA - AES128 - GCM - SHA256:	AES256 - SHA256:
ECDHE - ECDSA - AES256 - SHA384:	AES128 - SHA256:
ECDHE - ECDSA - AES128 - SHA256:	AES256 - SHA:
ECDHE - ECDSA - AES256 - SHA:	AES128 - SHA:
ECDHE - ECDSA - AES128 - SHA:	AES128 - GCM - SHA256:

SSL Cipher Suites Available for PostgreSQL	
ECDHE - RSA - AES256 - GCM - SHA384:	AES256 - GCM - SHA384:
ECDHE - RSA - AES128 - GCM - SHA256:	DHE - RSA - AES128 - GCM
ECDHE - RSA - AES256 - SHA:	DHE - RSA - AES256 - GCM
ECDHE - RSA - AES128 - SHA:	ECDHE - RSA - AES128 - SH
DHE - RSA - AES256 - SHA256:	ECDHE - RSA - AES256 - SH
DHE - RSA - AES128 - SHA256:	!CAMELLIA:+DSA:!DES:!3DE
DHE - RSA - AES256 - SHA:	-

SSL Cipher Suites Available for Webserver

SSL Cipher Suites Available for Webserver	
ECDHE - ECDSA - AES256 - GCM - SHA384:	DHE - RSA - AES128 - SHA:
ECDHE - ECDSA - AES128 - GCM - SHA256:	AES256 - SHA256:
ECDHE - ECDSA - AES256 - SHA384:	AES128 - SHA256:
ECDHE - ECDSA - AES128 - SHA256:	AES256 - SHA:
ECDHE - ECDSA - AES256 - SHA:	AES128 - SHA:
ECDHE - ECDSA - AES128 - SHA:	AES128 - GCM - SHA256:
ECDHE - RSA - AES256 - GCM - SHA384:	AES256 - GCM - SHA384:
ECDHE - RSA - AES128 - GCM - SHA256:	DHE - RSA - AES128 - GCM
ECDHE - RSA - AES256 - SHA:	DHE - RSA - AES256 - GCM
ECDHE - RSA - AES128 - SHA:	ECDHE - RSA - AES128 - SH
DHE - RSA - AES256 - SHA256:	ECDHE - RSA - AES256 - SH
DHE - RSA - AES128 - SHA256:	!CAMELLIA:+DSA:!DES:!3DE
DHE - RSA - AES256 - SHA:	-

SSL Cipher Suites Available in Non-FIPS Mode

SSL Cipher Suites Available in Non - FIPS Mode
ECDHE - ECDSA - AES256 - GCM - SHA384
ECDHE - ECDSA - AES128 - GCM - SHA256
ECDHE - ECDSA - AES256 - SHA384
ECDHE - ECDSA - AES128 - SHA256
ECDHE - ECDSA - AES256 - SHA
ECDHE - ECDSA - AES128 - SHA
ECDHE - RSA - AES256 - GCM - SHA384
ECDHE - RSA - AES128 - GCM - SHA256
ECDHE - RSA - AES256 - SHA
ECDHE - RSA - AES128 - SHA
DHE - RSA - AES256 - SHA256
DHE - RSA - AES128 - SHA256
DHE - RSA - AES256 - SHA
DHE - RSA - AES128 - SHA
AES256 - SHA256
AES128 - SHA256
AES256 - SHA
AES128 - SHA
AES128 - GCM - SHA256
AES256 - GCM - SHA384
DHE - RSA - AES128 - GCM - SHA256
DHE - RSA - AES256 - GCM - SHA384

SSL Cipher Suites Available in Non - FIPS Mode
ECDHE - RSA - AES128 - SHA256
ECDHE - RSA - AES256 - SHA384
!CAMELLIA:+DSA:!DES:!3DES:+AES:!aNULL:!ADH:!eNULL:!EXP:!RC4:!RC2:!kDHE

Cipher Suites for ClearPass as SSH Client in Non-FIPS Mode

Cipher Suites for ClearPass as SSH Client in Non - FIPS Mode
3des - cbc
aes128 - cbc
aes192 - cbc
aes256 - cbc
rijndael - cbc@lysator.liu.se
aes128 - ctr
aes192 - ctr
aes256 - ctr
aes128 - gcm@openssh.com
aes256 - gcm@openssh.com
chacha20 - poly1305@openssh.com

Cipher Suites for ClearPass as SSH Server in Non-FIPS Mode

Cipher Suites for ClearPass as SSH Server in Non - FIPS Mode	
Ciphers	aes128 - cbc,aes256 - cbc,aes128 - ctr,aes256 - ctr,aes128 - gcm@openssh.com,aes256 - gcm@openssh.com
MACs	hmac - sha1,hmac - sha2 - 256,hmac - sha2 - 512

Cipher Suites for ClearPass as SSH Server in Non - FIPS Mode	
KexAlgorithm	ecdh - sha2 - nistp256,ecdh - sha2 - nistp 384,ecdh - sha2 - nistp521

Cipher Suites for ClearPass as SSH Client in FIPS Mode and CC Mode

Cipher Suites for ClearPass as SSH Client in FIPS Mode and CC Mode
3des - cbc
aes128 - cbc
aes192 - cbc
aes256 - cbc
rijndael - cbc@lysator.liu.se
aes128 - ctr
aes192 - ctr
aes256 - ctr
aes128 - gcm@openssh.com
aes256 - gcm@openssh.com
chacha20 - poly1305@openssh.com

Cipher Suites for ClearPass as SSH Server in FIPS Mode and CC Mode

Cipher Suites for ClearPass as SSH Server in Non - FIPS Mode	
Ciphers	aes128 - cbc,aes256 - cbc,aes128 - ctr,aes 256 - ctr,aes128 - gcm@openssh.com,aes2 56 - gcm@openssh.com
MACs	hmac - sha1,hmac - sha2 - 256,hmac - sha 2 - 512

Cipher Suites for ClearPass as SSH Server in Non - FIPS Mode	
KexAlgorithm	ecdh - sha2 - nistp256,ecdh - sha2 - nistp 384,ecdh - sha2 - nistp521

Common Criteria (CC)

This document contains configuration examples from ClearPass Policy Manager. When possible, all examples will be shown using the graphical user interface (Web UI) rather than command line interface (CLI) commands. Instances where no Web UI can be used to configure a setting will use CLI commands.

When applicable, the document will direct back to the official User Guide. Common Criteria evaluation was performed against the 6.11 version of the document. Once submitted, this document will be available at: **HPE Networking Support Portal - Software and Documents**.

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Subtopics

Copyright/Trademark Information
Configuration
AGD_OPE.1
Baseline Setup Requirements
Enable Common Criteria Mode
Configure Notifications
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Appendix A - FAU_GEN.1 AUDITABLE EVENTS
Appendix B - Rules

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Configuration

Configuration of ClearPass Policy Manager (herein referred to as ClearPass) to conform to Common Criteria evaluated configuration is broken into two primary sections. The first section is for initial configuration and entering into the high-level Common Criteria Mode. This will establish the primary configuration requirements of NDcPP version 3.0e. The second section will outline any remaining configurations or individual notes from Common Criteria configuration for individual settings, to perform optional configurations.

AGD_OPE.1

ClearPass has been evaluated for compliance with Common Criteria Collaborative Protection Profile for Network Devices (CPP_ND_V3.0e). The limits of this evaluation are documented in the Security Target (ST) as submitted during certification.

The cryptographic limits outlined in this document ensure the ClearPass appliance is configured to use only approved ciphers and algorithms. Without these configurations, there are additional capabilities that are capable of being used that were not evaluated as part of the Common Criteria process. To ensure that only approved cryptographic functionality is enabled, ClearPass must be configured to use both FIPS140-2 and Common Criteria Mode when operating to limit functionality to evaluated capabilities.

The Aruba ClearPass Access Management System™ includes several components. The Policy Manager component has been evaluated by Common Criteria for all the security functions indicated by the protection profiles. Many of the other components were outside of scope, including features that require additional licenses.

ClearPass includes a reporting system known as Insight. Insight does not perform any security functions that were within scope for Common Criteria evaluation. The interface has been evaluated as part of Common Criteria only due to the same functionality being shared between Insight and Policy Manager.

The Guest functionality provides workflows for allowing guest users to access networks. Guest functionality was not within Common Criteria evaluation scope. The interface has been evaluated as part of Common Criteria only due to the same functionality being shared between it and Policy Manager.

The add-on Onboard functionality provides a certificate authority (CA) for use with device authentication. Onboard functionality was not within Common Criteria evaluation scope. The interface has been evaluated as part of Common Criteria only due to

the same functionality being shared between it and Policy Manager. No Onboard CA functionality should be considered evaluated by Common Criteria.

The add-on OnGuard functionality provides endpoint posture checking capabilities for use with Policy Manager. OnGuard policy is configured within Policy Manager but has not been evaluated by Common Criteria in any capacity.

ClearPass includes the ability to actively or passively profile endpoints and network devices. This functionality is configured within Policy Manager but has not been evaluated by Common Criteria in any capacity.

ClearPass makes use of a digital signature whenever updates/upgrades are applied to the system, regardless of the package size or intent. All ClearPass systems store a copy of the package-signing public key. When a new package is installed, the server will load the package on the serve, then validate the signing key against the stored copy of the public key. If the cryptographic signatures are identical, then the update process is allowed to proceed. If the signatures do not match, then the package update will fail with an error message indicating that the package has failed to validate.

To reduce errors in systems downloading packages manually from **https://networkingsupport.hpe.com**/, it is also recommended to validate the package hash and compare against the published values from the download site prior to loading onto ClearPass. While this process was not evaluated as part of Common Criteria evaluation, it is helpful in updates to systems without direct internet connections.

Applying patches to ClearPass can be performed by direct connection or manual upload of the patch for non-Internet connected systems. Navigate to **Administration > Agents and Software Updates > Software Update**s to install patches. To manually install patches, first the patch must be loaded to the ClearPass server by clicking the **Import Updates** button under **Firmware & Patch Updates**. The interface box will upload the patch to the appropriate directory for installation. Installation will then proceed as the Internet connected systems once the patch has been downloaded to the system.

Internet connected systems may download the patches through the **Firmware & Patch Updates** section by clicking the **Download** button to download the patch, then **Install** to install the patch. Most patches will require a reboot once installed. ClearPass has been evaluated for Common Criteria using a single node. The patching of clusters is outside the scope of the evaluation and should follow regular documentation processes for applying patches to clusters.

Baseline Setup Requirements

Subtopics

Passwords and Accounts
FCS_CKM.1 - Enable FIPS 140-2 Mode
FCS_CKM.4 – Cryptographic Key Destruction
Configure System Time
Configure Audit Export

Establish Password Policy Enforcement
FIA_X509_EXT.1/Rev (Install Certificates)
Enable Ingress Events Processing
Verify Local User Repository is Available

Passwords and Accounts

During initial setup, administrators are allowed to specify the initial password for use with the CLI and Web UI accounts. While minimum complexity and length requirements exist, they are not considered strong or secure passwords for ongoing use. It is recommended that the following guidelines be followed for establishing a more secure password to be used:

- Require a minimum password length of at least 15 characters
- Make use of upper case, lower case, numerical values, and allowed special characters in all passwords
- Passwords are not based on dictionary words (unless passphrases longer than 22 characters are used)
- Secure common passwords (such as CLI users) in a secure location with restricted access.

  Examples of special characters include: ! @ # $ % ^ & * ( )

  Initial setup will create two accounts: appadmin for CLI/SSH access and admin for Web UI access. Both use the same password initially. Aruba recommends securing the appadmin account for emergency access if the core authentication services become unavailable.

  After initial setup, the administrator should create individual accounts for all the administrators and no longer use the default Web UI account or password. Directions to perform this can be found in the Managing Admin Users section of the ClearPass Policy Manager User Guide. Navigate to **Administration > Users and Privileges > Admin Users** to create and modify administrator accounts. Administrator accounts should always have strong passwords set.

  Administrator permissions are limited to users with appropriate roles. In compliance with FMT_MTD.1, only administrators should have access to the security management functionality on the system. General users are not required to have local accounts defined.

FCS_CKM.1 - Enable FIPS 140-2 Mode

Placing the TOE into FIPS 140-2 mode is all that is necessary to enable a complaint DRBG. No further configuration is needed.

As noted in AGD_OPE.1, the evaluated configuration requires FIPS 140-2 mode to be enabled. Configurations that do not apply this requirement may use cryptographic capabilities that were not evaluated or tested during the Common Criteria evaluation process.

Enabling the FIPS 140-2 mode may be accomplished during installation or after installation. Performing the transition after installation will reset the system configuration, and is not recommended.

During initial setup through the command line interface (CLI), the administrator is prompted with the following option:

Do you want to enable FIPS mode? [y|n]: _

Answering yes to this question enables the system to operate using FIPS 140-2 algorithms from initial configuration.

After installation, to enable FIPS mode, open ClearPass Policy Manager. Navigate to **Administration > Server Manager > Server Configuration** and select the server in the list. Select the **FIPS** tab, and then click the **Enable** button in the **FIPS Mode** field, as shown below.

Post-installation conversions require a reboot when enabling FIPS mode before continuing with the configuration process.

FCS_CKM.4 – Cryptographic Key Destruction

Cryptographic key destruction is performed automatically. There are no administrator prerequisites to meet this requirement. There are no circumstances that do not strictly conform to the key destruction requirement and there are no situations where key destruction may be delayed at the physical layer.

Configure System Time

It is important to set the system date and time prior to continuing. Certificates are based off validity durations that can be affected by changes to the date/time. To manually configure time with ClearPass, navigate to **Administration > Server Manager > Server Configuration**, and select the option **Set Date & Time** in the upper right corner.

Time and date settings can be entered using the **Date & Time** tab. The **Time zone on publisher** tab can be used to set the time zone of the server. Time can be set manually, or using the **Synchronize time with NTP server** option.

NTP was configured during Common Criteria evaluation in compliance with FCS_NTP_EXT.1 requirements. NTP servers must support NTPv4 to work. The Web UI allows a minimum of one (1) NTP server to be used, but it is recommended to specify at least three (3) NTP servers. The Web UI allows the specification of 1-5 NTP servers. When configuration is performed with NTP the communication between appliance and NTP server should be configured to a secure key and hash algorithm to ensure the communication is not modified. Only SHA-1 is allowed for a CC evaluated configuration.

The NTP service does not accept multicast or broadcast NTP information, there are no configuration options to change this behavior.

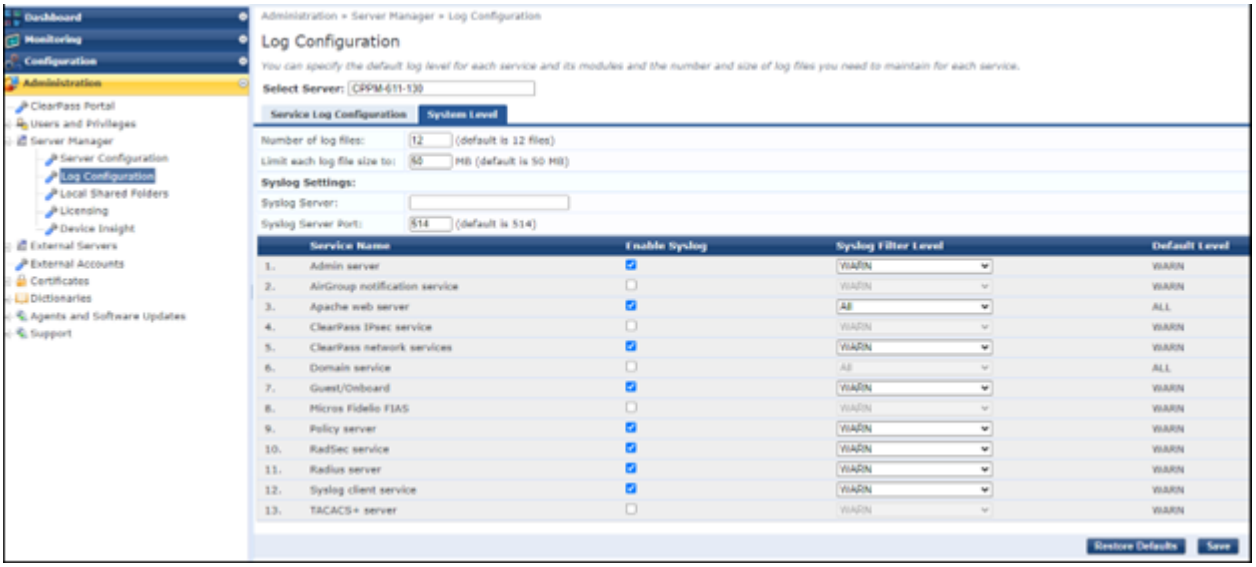
In cases where NTP servers cannot support secure hash algorithms IPsec encapsulation is recommended.

When the date and/or time is modified, the system will restart services and require a re-login.

Configure Audit Export

ClearPass has limited storage available to retain logs. Aruba recommends exporting all the audit logs to an external source. The recommended process to accomplish this is via syslog export. Because log information may be sent to multiple syslog receivers, there are two places that syslog export must be configured on ClearPass.

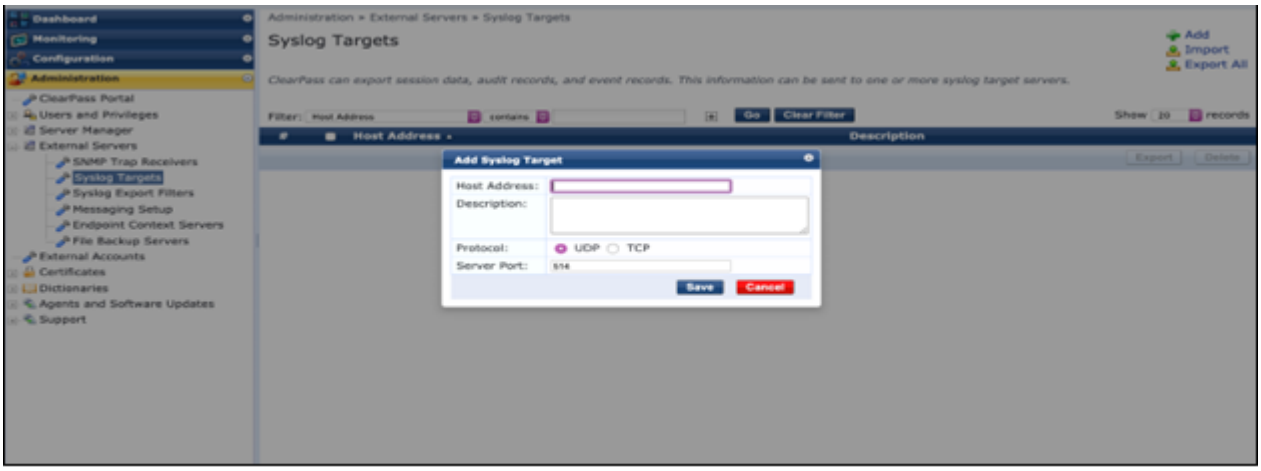
Exporting all ClearPass audit information begins with specifying the configuration at the system level. Navigate to **Administration > Server Manager > Log**Configuration, and select the System Level tab. Specify the IP address of the syslog server in the appropriate space.



Select the components desired to export by selecting the **Enable Syslog** option for the appropriate services. To ensure maximum audit compliance, it is recommended to enable syslog for all services. To capture all Common Criteria related audit messages, the RADIUS server should be configured to display audits to the DEBUG level.

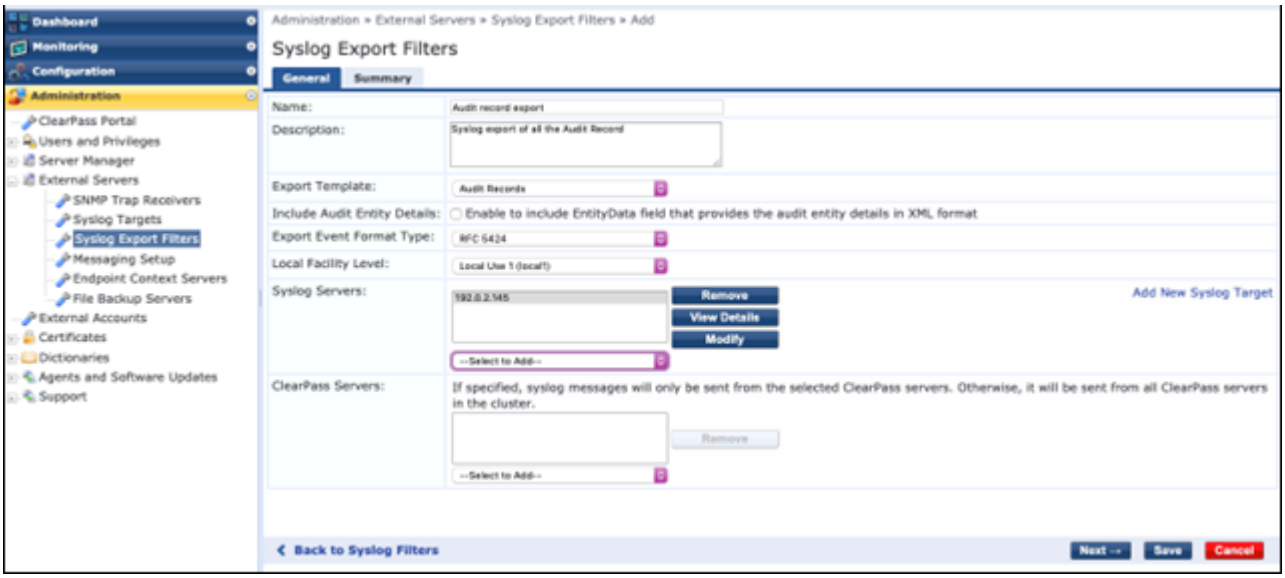
> **NOTE**
>
> The ClearPass IPsec service is now included in the list of services whose service logs can be sent to a syslog server, and is available in the Service Name list on the **Administration > Server Manager > Log Configuration > System Level** tab.

At least one syslog receiver must be defined for general use. Navigate to **Administration > External Servers > Syslog Targets** and click **Add** in the upper right corner.



The syslog target IP address should be specified, along with the protocol and port to send to. The default value for syslog is to use UDP port 514. Further information on Common Criteria recommended deployments of syslog is available in section FTP_ITC.1.1(1).

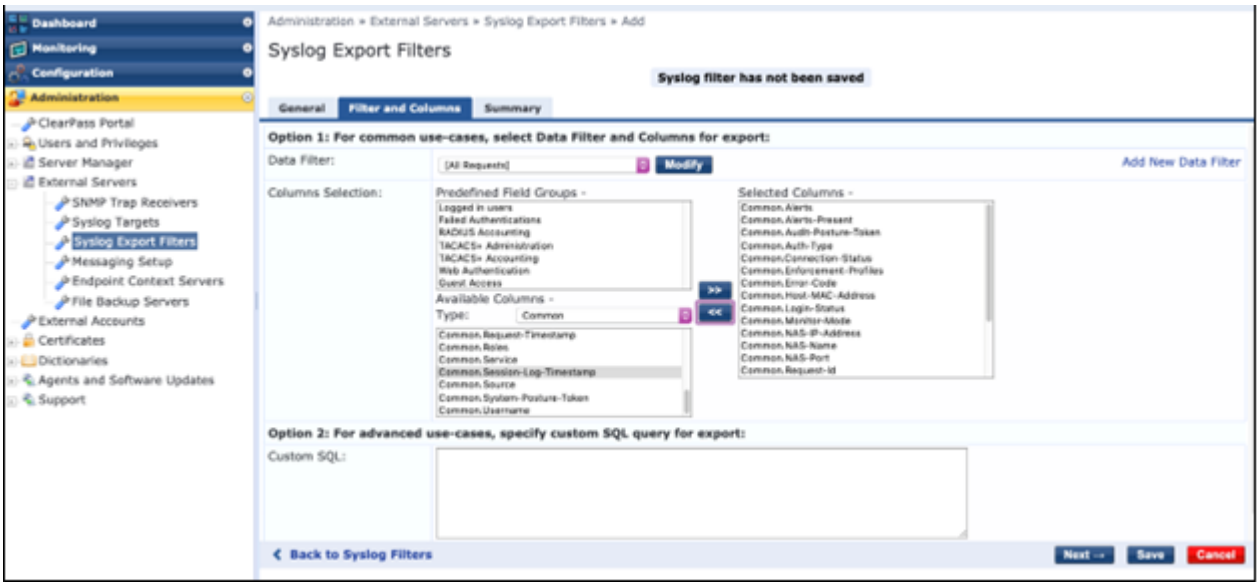
Once the target is defined, the data to be transmitted needs to be specified. Navigate to **Administration > External Servers > Syslog Export Filters**, and click **Add** in the upper right corner. A total of three (3) filters are required to send the data to the syslog server(s).



The first filter will need to use the Audit Records **Export Template**. Specify the syslog target from the available list to note the receiver. The **Export Event Format Type** offers the choice between Standard, LEEF, CEF, and RFC 5424. It is not required to specify the ClearPass Servers that this filter will be applied to unless using a cluster. Clusters were not evaluated by Common Criteria.

The second filter must use the System Events **Export Template**. Specify the syslog target from the available list to note the receiver.

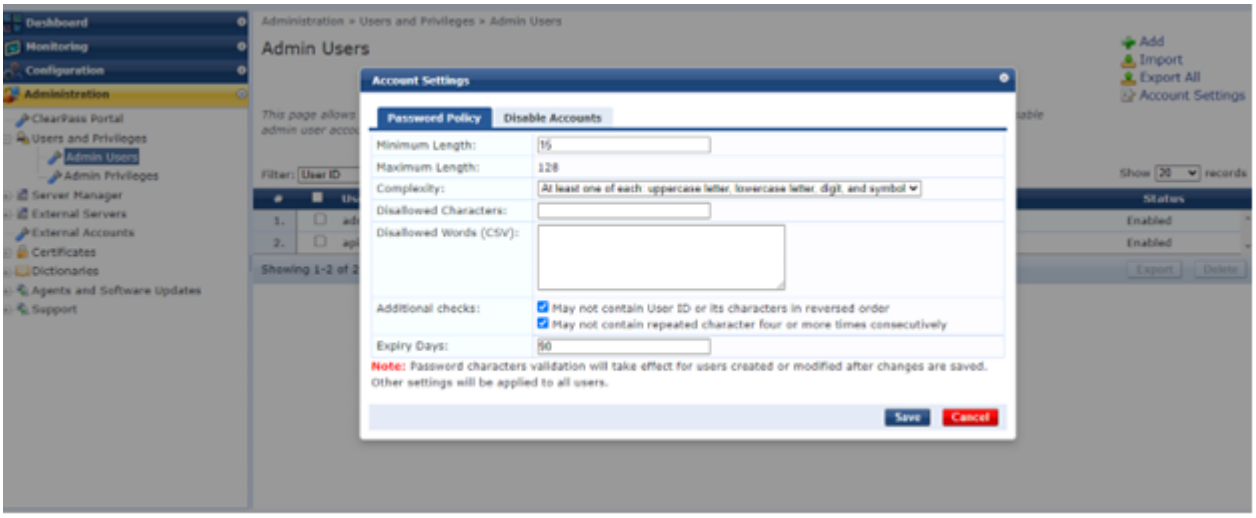
The final filter will need to use the Session Logs **Export Template**. Specify the syslog target from the available list to note the receiver. Unlike the first two filters, session logs require a second set of information to be included.



The **Filters and Columns** tab allows the two options to be specified when selecting the information to export. It is recommended to use the first option. Specifying the **Data Filter** of [All Requests] will capture all session related information. The recommended deployment is to select all available columns from the Common type selection.

Establish Password Policy Enforcement

ClearPass uses a default password policy that requires only a six (6) character password length with no password complexity requirements. The password policy allows passwords with six (6) to one hundred (100) characters for Web UI accounts and six (6) to one hundred twenty-eight (128) characters for SSH/CLI access. This may create confusion to administrators that may attempt to use different password length maximums, it is recommended that a maximum of 100-character password length be used. A future release will align this maximum to two hundred fifty-six (256) characters as are enforced maximums for all passwords. To ensure compliance with Common Criteria evaluated configuration, the defaults should be changed to have a higher security setting. Navigate to **Administration > Users and Privileges > Admin Users**, and then select the option **Account Settings** in the upper right corner.

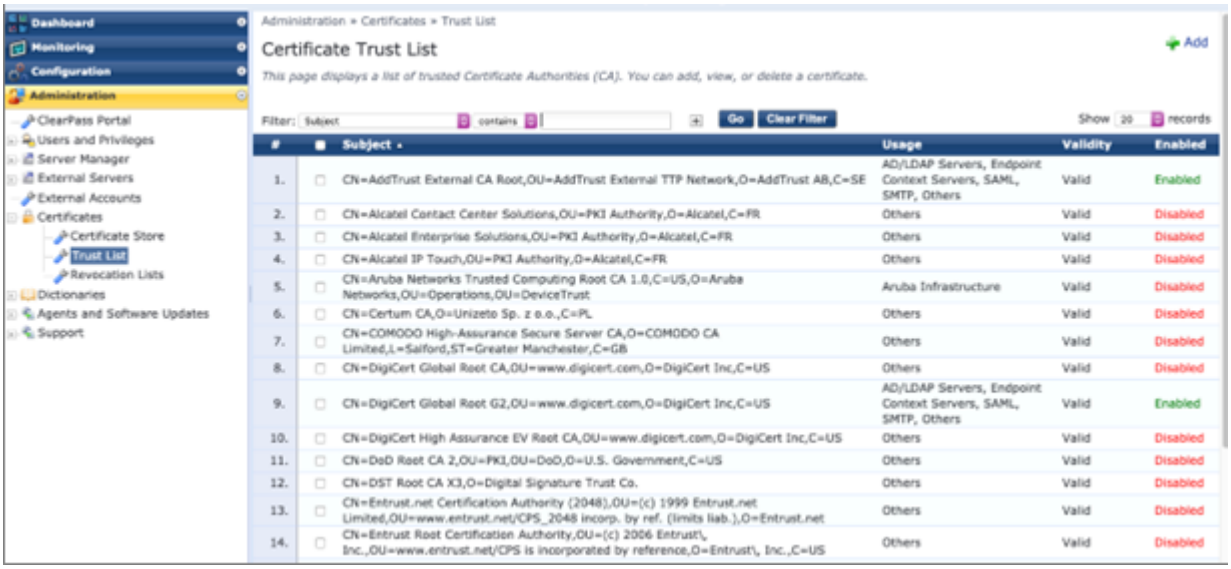


The **Minimum Length** value has been modified to fifteen (**15**) characters. **Complexity** is set to require **At least one of each: uppercase letter, lowercase letter, digit, and symbol**. The **Additional Checks** have both been selected to prevent user ID or reversed user ID, or repeating characters four (4) or more times in the password. The **Expiry Days** have been set to ninety (90) days to force administrative users to change their passwords regularly.

FIA_X509_EXT.1/Rev (Install Certificates)

The use of self-signed certificates is not allowed in Common Criteria configurations. It is recommended to use certificates from trusted issuers in all cases, but rigidly enforced when enabling Common Criteria mode. ClearPass will not allow administrators to enable Common Criteria Mode without externally, certificate authority (CA) signed HTTPS and RADIUS certificates installed.

By default, ClearPass generates self-signed certificates for the RADIUS, HTTPS, and Database servers. All certificates will need to be replaced with certificates that are signed by a trusted certificate authority (CA). Begin the process by navigating to **Administration > Certificates > Trust List**. Ensure the CA root is listed and enabled in the available list.



To enable a CA, click its row in the list to open the **View Certificate Details** window. Select the **Enable** button to enable a trusted CA. If the required CA certificate is not loaded in ClearPass, it can be manually imported by selecting the Add button in the top right of the **Certificate Trust List** screen.

Certificate usage must be enabled for the CA certificate to be used. If the certificate is enabled, but not allowed for use with the specific system it will not be considered valid for those services. Common Criteria evaluated services were limited to functions required for validation: HTTPS (for Web UI administration), Database (required to enable CC mode), , SAML (for testing with FCS_TLSS_EXT.2 only), and other (for IPsec).

When using a CA that is not listed in the available trust list, the CA's public certificate must be imported. Imported CAs will automatically be enabled during the import process. Imported CA certificates cannot be self-signed when using Common Criteria mode.

Then, to update a ClearPass certificate, navigate to **Administration > Certificates > Server Certificate** and select the desired server certificate from the **Select Type** drop down list. The new certificate can then be imported by using the **Import Server Certificate** link, or a new Certificate Signing Request (CSR) can be made by using the **Create Certificate Signing Request** link.

When this process is completed for one certificate, the other can be completed. After the **RADIUS/EAP Server Certificate, HTTPS Server Certificate**, and **Database Server Certificat**e are not self-signed, the process can continue.

NOTE

The type of certificate used will influence which ciphers are available later. For example, RSA certificates will not be able to perform ECDSA based ciphers, so those encryption options will automatically be disabled.

The following list is all the allowed hash and encryption types that may be used for either HTTPS server certificates when operating in CC Mode:

```
Encryption: RSA
Size: 2048-bit, 3072-bit, or 4096-bit
Hash: SHA256, SHA384, or SHA512
Encryption: ECDSA
Size: NIST/SECG curve over 256-bit prime field
Size: NIST/SECG curve over 384-bit prime field
Size: NIST/SECG curve over 521-bit prime field
Hash: SHA256, SHA384, or SHA512
```

NOTE

While listed as possible, the RSA 4096-bit value was not evaluated.

The type of key will be used to automatically determine the available cipher suites. Cipher suites cannot be manually modified for use from those listed in FCS_TLSS_EXT.2.1 later in this document.

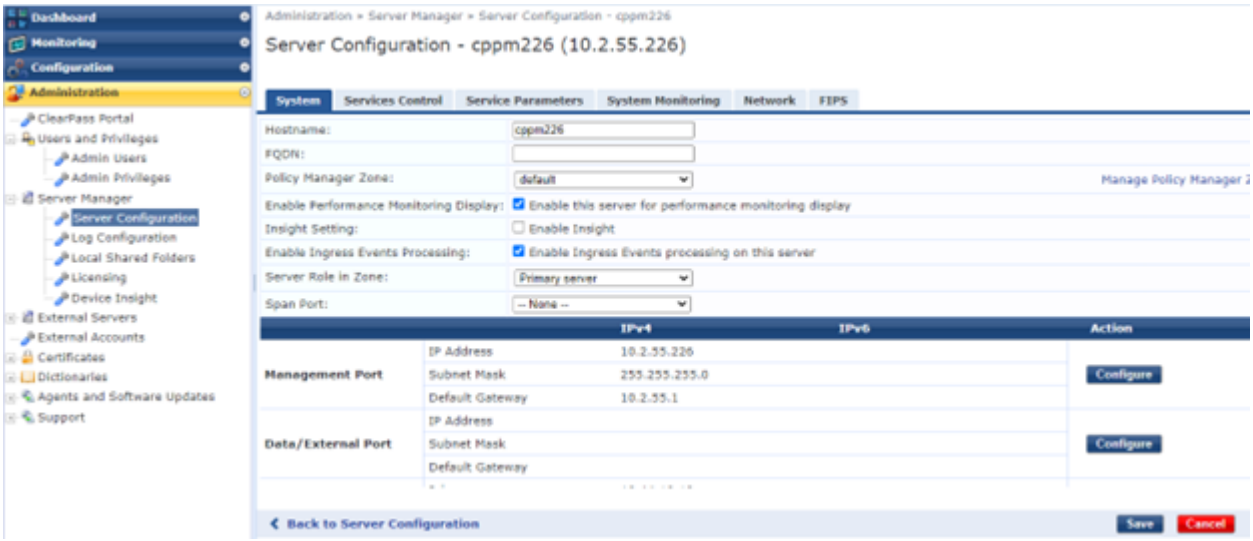
Attempts to generate a CSR or load a certificate with sizes below the specified thresholds will fail. The UI will fail to complete the CSR generation, it will continue to spin in the waiting state.

The ClearPass system must be restart after configuring the Database Server Certificate. Navigate to **Administration > Server Manager > Server Configuration** and select the option **Reboot** in the lower right area.

Enable Ingress Events Processing

To properly track events related to IPsec processing or HTTP daemon logging, ClearPass must be configured to process these events. Each node within a cluster (if applicable) must repeat the following process.

Navigate to **Administration > Server Manager > Server Configuration** and select the server/node from the list. From the System tab, enable the **Enable Ingress Events Processing** on this server option.

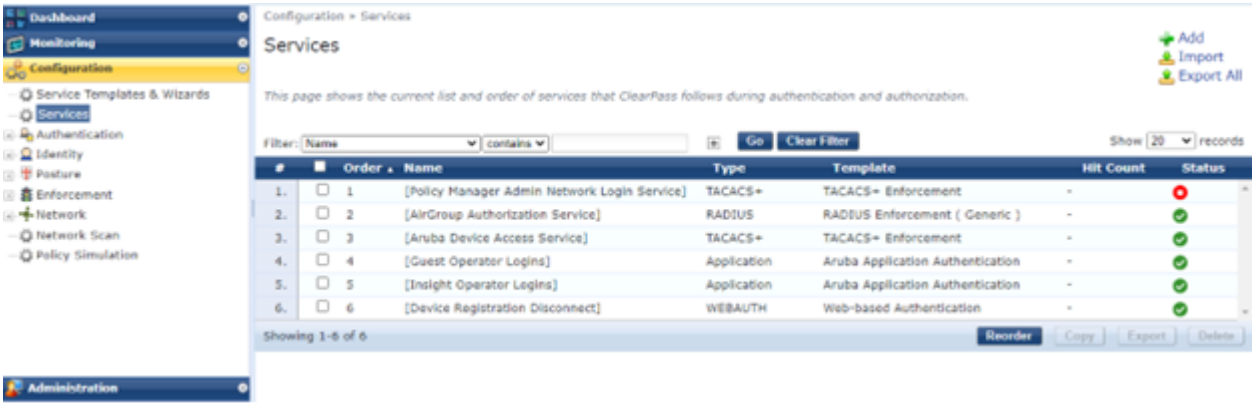


A warning message appears when enabling this option indicating the process is a CPU-intensive operation. The impact of this engine for these events is within acceptable limits; click **Yes** to continue. Without this, several later components will be impacted. This includes FCS_IPSEC_EXT.1, FCS_SSH_EXT.1, FCS_SSHS_EXT.1.

After enabling ingress events processing on the server/node, open the **Services Control** tab and validate the **Ingress logger service and Ingress logrepo service** (position 11) services are both running. If they have not automatically started, click the **Start** button to complete the process.

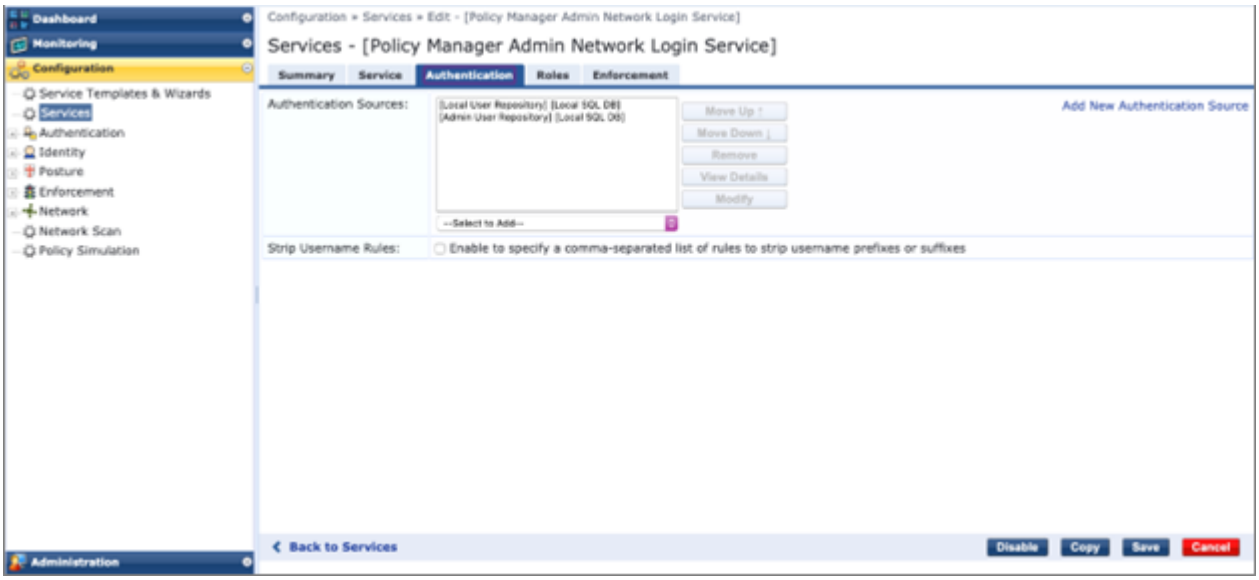
Verify Local User Repository is Available

From **Configuration > Services**, the **Policy Manager Admin Network Login Service** is enabled by default in position one (1). Ensure the Local User Repository is available when performing initial deployment until all remote authentication sources are validated.



Ensure the rule **[Policy Manager Admin Network Login Service]** is listed at the top. Select the rule's row in the list to view its details.

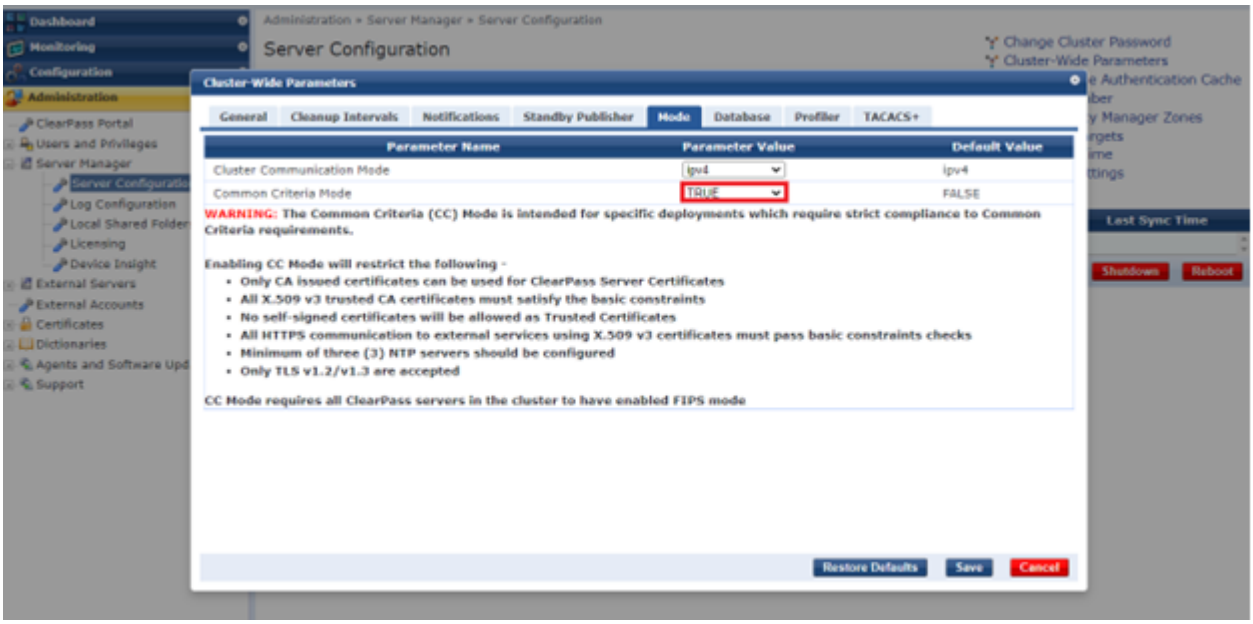
Within the **Summary** tab, the **Authentication Sources** field must include **[Local User Repository]** prior to enabling Common Criteria mode or an administrator may be locked out.



From the **Authentication** tab, if the default service is not used, ensure the **[Local User Repository]** value has been added to the service. In the **Authentication Sources** field, use the drop-down menu and buttons to add to or reorder the list of available authentication sources.

Enable Common Criteria Mode

To enable Common Criteria mode through the Web UI, navigate to **Administration > Server Manager > Server Configuration**, select the **Cluster-Wide Parameters** link, and then select the **Mode** tab, as shown below.



While Common Criteria mode is supported by ClearPass clusters, it has been evaluated as a single, non-clustered server for certification.

Subtopics

FMT_SMR.2.3
FIA_AFL.1.2
Disable Admin User and Local User Account

FMT_SMR.2.3

Once Common Criteria mode is enabled, the list of ciphers available for use is limited to those specified within the Security Target (ST). ClearPass console access does not require further changes to access it in this mode. Most modern Web browsers support the available ciphers without further configuration. SSH clients that are not configured to support only FIPS 140-2 approved cryptographic ciphers will need to have ciphers re-prioritized to use the ones allowed by ClearPass or connections will not establish.

FIA_AFL.1.2

SSH access can be locked after a specified number of failed attempts for a configurable length of time. By default, SSH lockout is not enabled. To enable SSH lockout, one of the following commands should be executed:

```
ssh lockout count <N>
ssh lockout duration <N minutes>
```

Where the value of <N> is the number of failed login attempts, or the value of <N minutes> is the length of time the lockout will be enabled for. Example: To trigger a lockout after 3 failed attempts for a 30-minute window, the following commands would be executed:

```
ssh lockout count 3
ssh lockout duration 30
```

nlocking the SSH account can be accomplished only from the console, or from another SSH session that is authorized using public key authentication. To reset the SSH lockout, the following command must be executed:

ssh unlock

By default, when the account is locked, you can perform this operation by logging in to the system via the console or from a host that is enabled for SSH public key authentication with ClearPass. The lockout capability can be extended to include SSH public key authentication by executing the following command:

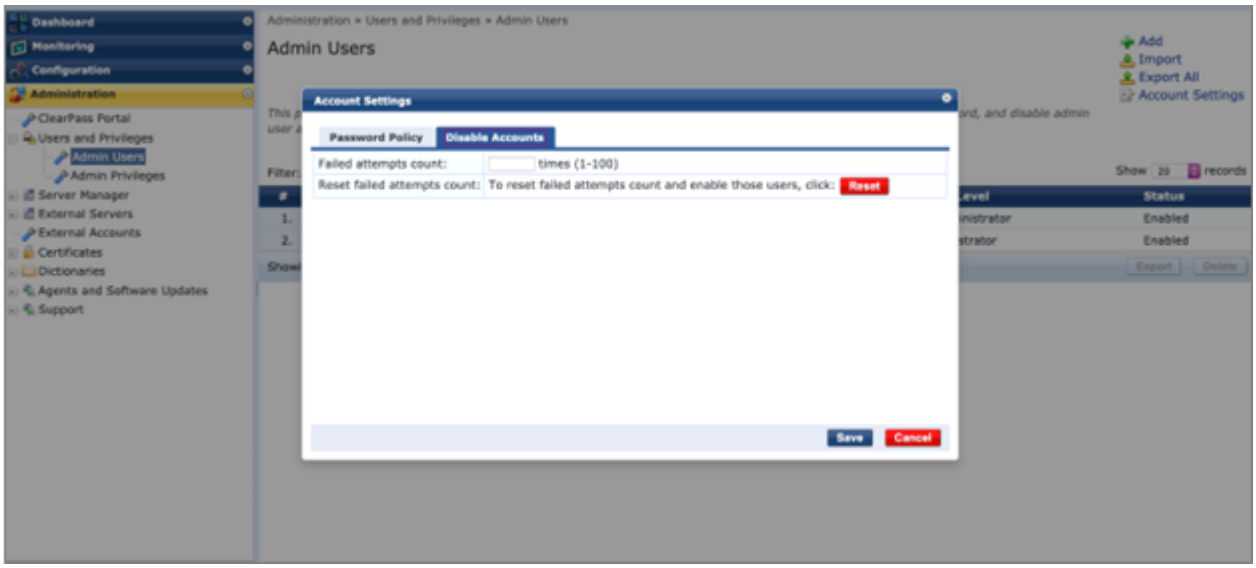
ssh lockout mode advanced

Advanced mode will apply the same conditions to both username/password authentication and SSH public key authentication. When Advanced mode is enabled, the only way to unlock the account is by waiting for the duration to expire or to execute the unlock command from the console or previously established SSH session.

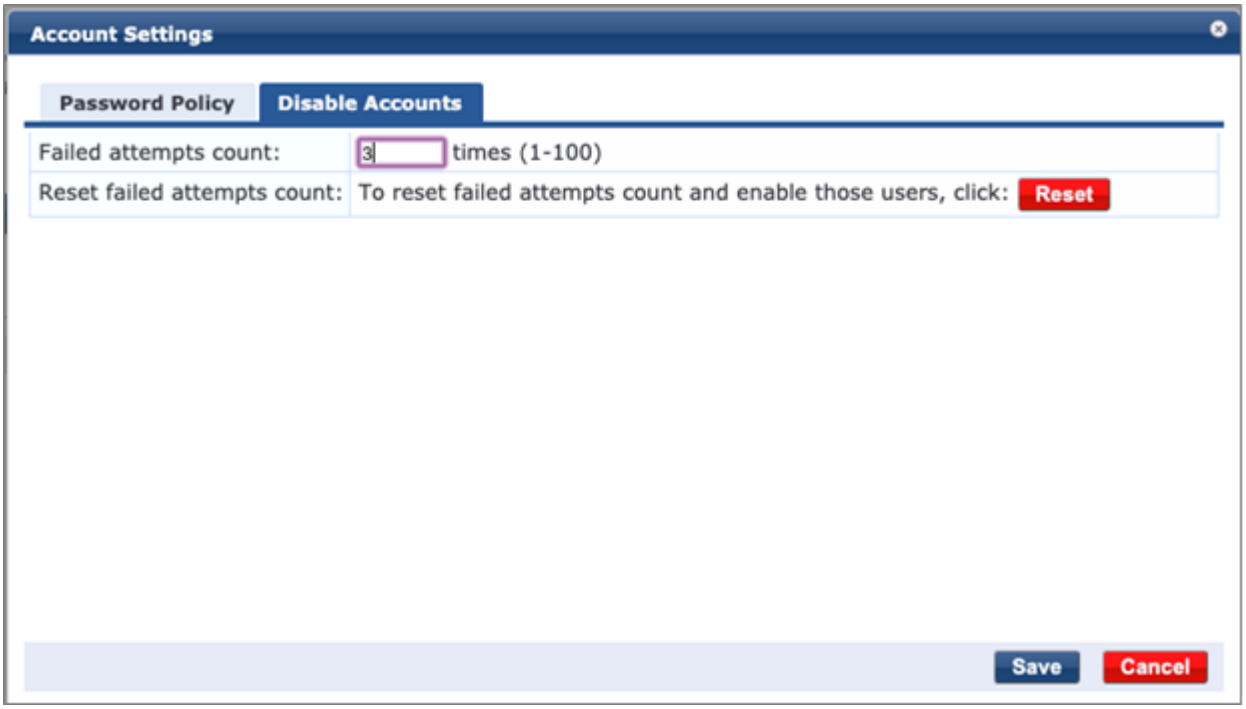
To avoid an Administrator being completely locked out of the TOE, the defined Administrator accounts are always available for use, with the correct credentials, over the Local Console. Despite their lockout status for SSH or the Web UI, even if remote administration via SSH and the Web UI are blocked as a result of the accounts being locked, an Administrator will always be able to access the TOE over the local console.

Disable Admin User and Local User Account

Web UI access can be locked after a specified number of failed attempts. The time duration for these events is permanent until unlocked by another administrator. The number of failed attempts can be configured through the Web UI. Navigate to **Administration > Users and Privileges > Admin Users**, select the **Account Settings** link, and then select the **Disable Accounts** tab. The **Failed attempts count** field can be populated with the desired number of failed login attempts.



Re-enable the account by clicking the **Reset** button.



If the **Reset** button is selected, a message displays notifying you of the number of accounts being unlocked. Accounts may also be individually unlocked directly from the **Admin Users** screen by selecting individual administrators and re-enabling their account.

If the Web UI access is lost, the following steps can help resolve the issue:

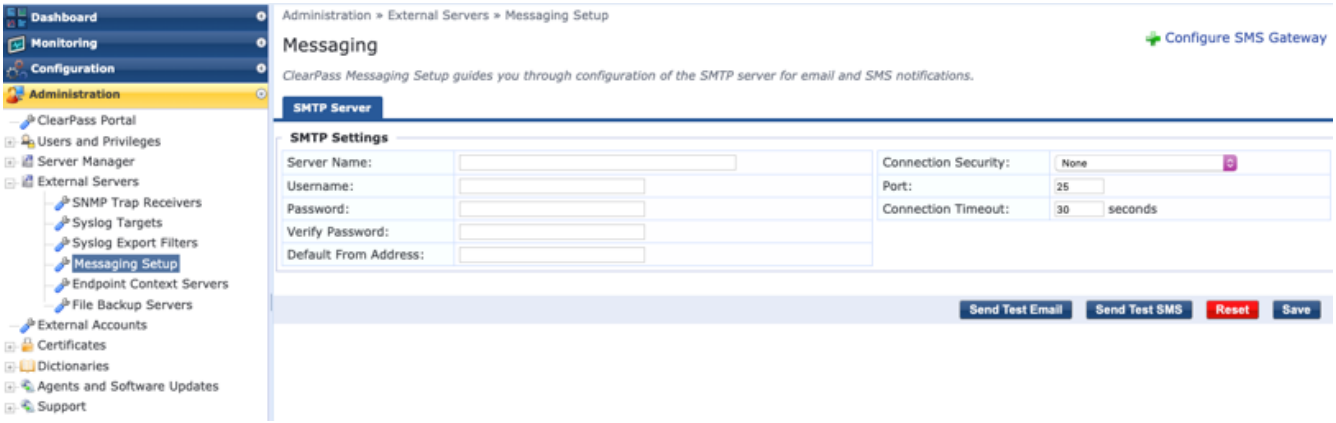
Issue	
Login fails	Incorrect username and credentials Attempt with another user
Web UI blocked by b rowser due to HTTPS certificate expired	The ‘cpass - admin - server’ service has stopped Execute the CLI command: service start cpass - admin - server Verify the server is restarted with the CLI command: service status cpass - admin - server Admin server [cpass - admin - server] is running
Web UI blocked by b rowser due to HTTPS certificate expired	View audit on syslog server, look for “SSL_ERROR_EXPIRED CERT_ ALERT” with “error:140800FF:SSL routines:ssl3_accept:unknown s tate Client IP Address” (including client IP address) Temporarily regenerate a self - signed certificate to return to acc ess on the system with the following CLI commands: Cluster reset - database

Issue	
	system reset - server - certificate. Select option 2 (Reset HTTP Server Certificate) This will reset the system to initial configuration. Log in through t he UI, restore the last known configuration backup, import valid c ertificate(s) and re - enable CC Mode.

Configure Notifications

ClearPass notifies administrators when specific alerts and alarms occur. These alerts are triggered by email, SNMP, or SMS notifications, depending on configuration. SNMP and SMS notifications were not validated during Common Criteria validation.

To configure email notification events, navigate to **Administration > External Servers > Messaging Setup**.



Specify the appropriate information to transmit SMTP messages to your server. When completed, it is recommended to click the button **Send Test Email** to validate that the configuration works. ClearPass does support TLS encoded SMTP delivery or message delivery may be secured over IPsec to ensure security. Common Criteria evaluation was performed using the IPsec security.

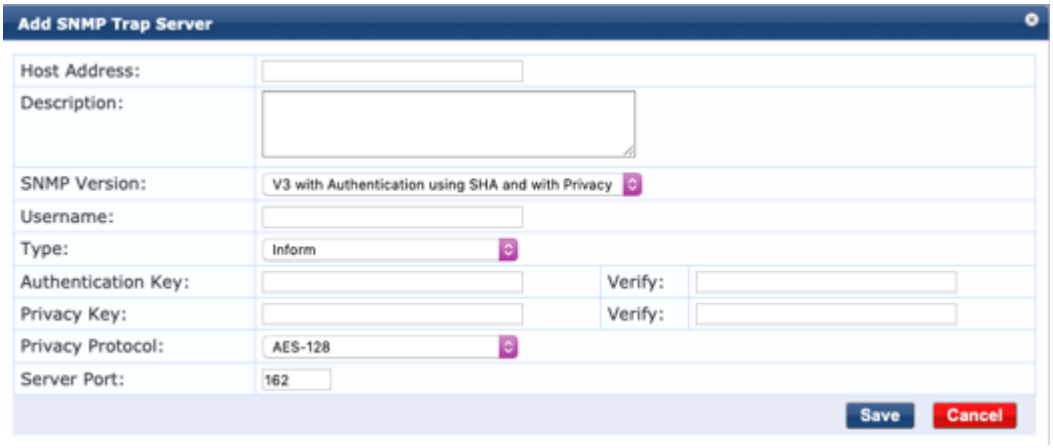
To configure SMS notification events, navigate to **Administration > External Servers > Messaging Setup**. Select the **Configure SMS Gateway** option. This will open a new browser tab in the Guest Web UI, as if you navigated to **Guest > Configuration > SMS Services > Gateways.**

Specify the appropriate information to transmit SMTP messages to your server. When completed, it is recommended to click the **Send Test Email** button to validate the configuration works. ClearPass does support TLS encoded SMTP delivery or message delivery may be secured over IPsec to ensure security. Common Criteria evaluation was performed using the IPsec security.

To configure SMS notification events, navigate to **Administration > External Servers > Messaging Setup**. Select **Configure SMS Gateway**. This will open a new browser tab in the Guest Web UI, as if you navigated to **Guest > Configuration > SMS Services > Gateways**.

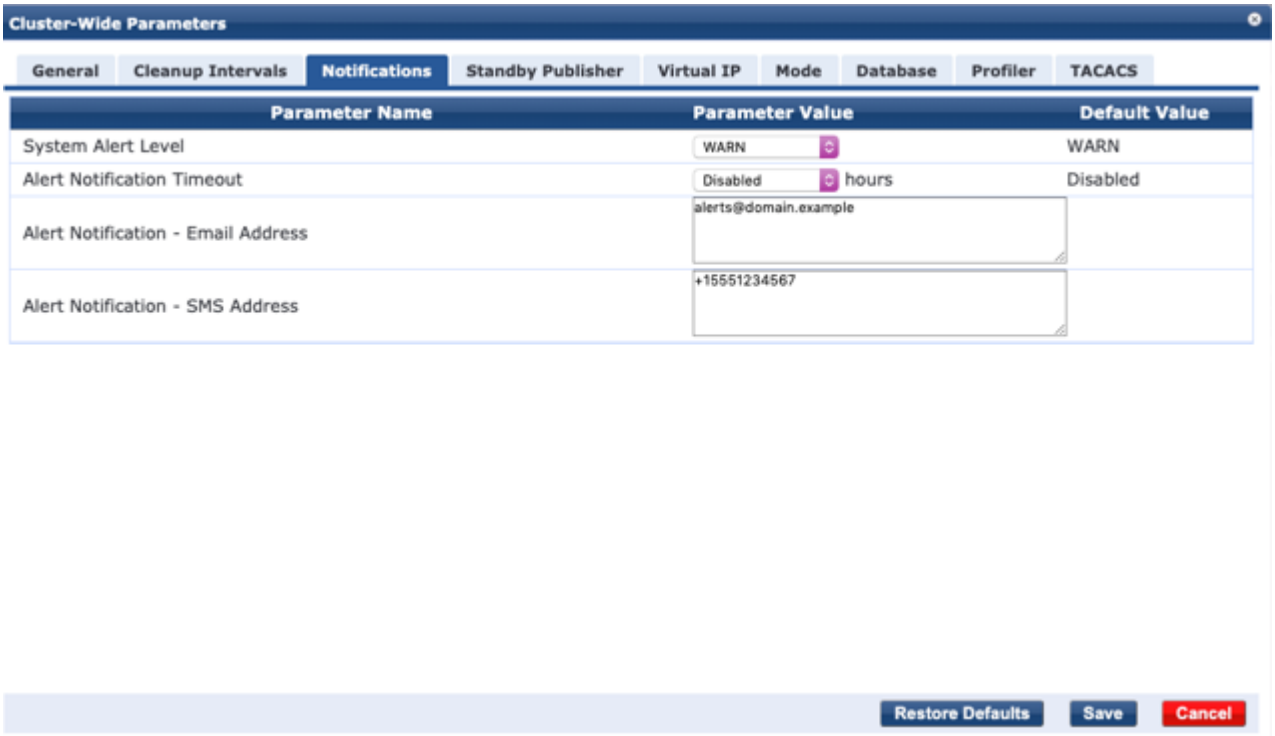
If using a new SMS provider, click the option **Create new SMS gateway** and specify the appropriate information. Once the SMS gateways are specified correctly, return to the Policy Manager **Messaging Setup** screen and test the configuration using the **Send Test SMS** button.

To configure SNMP notification events, navigate to **Administration > External Servers > SNMP Trap Receivers**. Select **Add** to input a new SNMP destination.



Enter the appropriate information for the required SNMP version. Monitoring the SNMP receiver will indicate that info is being received after a ten (10) minute window. Aruba recommends importing the ClearPass SNMP MIBs to the SNMP receiver to ensure accurate data is displayed.

When notifying via email and/or SMS alerts, the recipients must be specified. This can be accomplished by navigating to **Administration > Server Manager > Server Configuration** and selecting the **Cluster-Wide Parameters** link.



Email and/or SMS recipients may be specified in the provided fields of the **Notifications** tab.

Continued Guidance Configuration

This section describes guidance criteria for ClearPass Policy Manager Common Criteria acceptance.

Subtopics

FIA_UIA_EXT.1
FIA_X509_EXT.1/Rev
FIA_X509_EXT.2.2
FIA_X509_EXT.3.1
FPT_TUD_EXT.1.3
FMT_SMF.1.1
FTA_SSL.3 / FTA_SSL.4 / FTA_SSL_EXT.1.1
FTA_TAB.1
FTP_ITC.1
FCS_SSH_EXT.1.2
FCS_SSH_EXT.1.4
FCS_SSH_EXT.1.5
FCS_SSH_EXT.1.6
FCS_SSH_EXT.1.8
FCS_SSHS_EXT.1.1

FCS_TLSS_EXT.2.1
FCS_TLSS_EXT.2.2
FCS_TLSS_EXT.1.2
FCS_IPSEC_EXT.1
FCS_IPSEC_EXT.1.3
FCS_IPSEC_EXT.1.4
FCS_IPSEC_EXT.1.5
FCS_IPSEC_EXT.1.6
FCS_IPSEC_EXT.1.7
FCS_IPSEC_EXT.1.8
FCS_IPSEC_EXT.1.11
FCS_IPSEC_EXT.1.14
FAU_STG_EXT.1
FPT_TST_EXT.1 (self-tests)

FIA_UIA_EXT.1

ClearPass includes support for clustering multiple systems together. If ClearPass is being deployed in a stand-alone environment, one (1) additional port must be blocked to prevent inbound connections. This is accomplished by administrators logging in to the console directly and entering the following command:

configure port input tcp 5432 reject configure port input tcp 5433 reject

FIA_X509_EXT.1/Rev

Valid certificates (including intermediate Certificate Authorities) must be installed prior to enabling Common Criteria mode, as previously noted.

The TOE performs a check for validity of certificates, which takes place at the time of connection for each respective protocol.

The TOE supports the following extendedKeyUsage fields:

- Certificates used for trusted updates and executable code integrity verification shall have the Code Signing purpose (id-kp 3 with OID 1.3.6.1.5.5.7.3.3) in the extendedKeyUsage field.
- Server certificates presented for DTLS/TLS shall have the Server Authentication purpose (id-kp 1 with OID 1.3.6.1.5.5.7.3.1) in the extendedKeyUsage field.
- Client certificates presented for DTLS/TLS shall have the Client Authentication purpose (id-kp 2 with OID 1.3.6.1.5.5.7.3.2) in the extendedKeyUsage field.

- OCSP certificates presented for OCSP responses shall have the OCSP Signing purpose (id-kp 9 with OID 1.3.6.1.5.5.7.3.9) in the extendedKeyUsage field.

  Certificates are checked to ensure the appropriate extendedKeyUsage field value for the purpose of the certificate. Certificates that do not have the appropriate purpose for their use case are rejected during the certificate validation portion of the handshake.

  Certificate revocation checking takes place when a connection is being initiated using a User or Device certificate. This is the case for both IPsec and Mutually-Authenticated TLS. In the evaluated configuration, OCSP must be used. If the revocation status cannot be determined due to the inaccessibility of the appropriate OCSP responder for the certificate, the connection will be rejected by the TOE. Every certificate in the chain, except the Root CA, is checked for revocation.

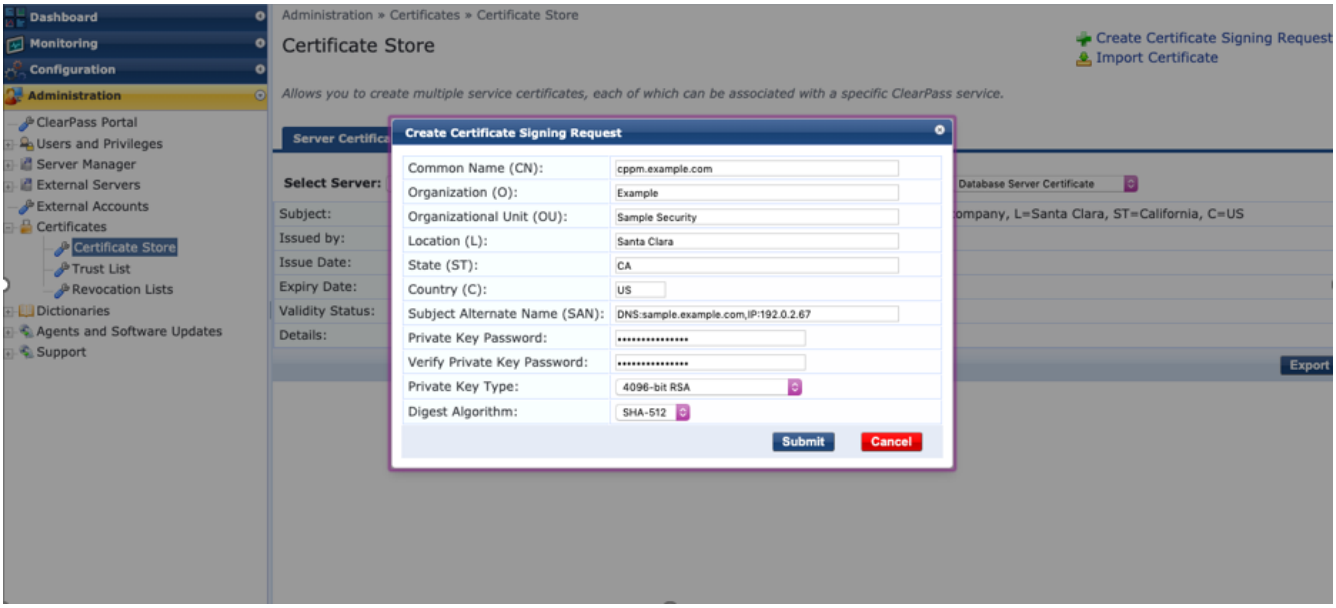
FIA_X509_EXT.2.2

If the validity of the certificate cannot be established, the default configuration is to not accept the certificate.

FIA_X509_EXT.3.1

The minimum required selection of a Certificate Request Message is the Common Name. It is recommended to include all relevant information (Common Name, Organization, Organizational Unit, and Country) when generating certificates or certificate signing requests (CSR) for ClearPass.

Generating a CSR on ClearPass can be found by navigating to **Administration > Certificates > Certificate Store**. Select the desired certificate type from the drop-down list **Select Usage** and selecting the **Create Certificate Signing Request** link. This will generate a new CSR of the selected type. The default will be RADIUS/EAP Server Certificate. Other valid selections include HTTPS Server Certificate, RadSec Server Certificate, and Database Server Certificate use. Individual **Service and Client Certificates** may also be generated from the specified tab but were not part of the evaluated configuration.



Specify the Common Name that the certificate will use, the Organization name, Organizational Unit, and two (2) letter Country code for all certificates to be used in Common Criteria evaluated configurations. The use of the Locality and Subject Alternative Name (SAN) are optional and were not evaluated as part of Common Criteria evaluation. Though not required, when specifying SAN, the values must be indicated with the appropriate type (DNS or IP) and a colon (:) to indicate the desired values.

Specify the Private Key Type as an approved CC evaluated type (2048-bit RSA, 3072-bit RSA, 4096-bit RSA, NIST/SECG curve over 384-bit prime field, or NIST/SECG curve over 521-bit prime field, X9.62/SECG curve over a 256-bit field). While listed as possible, the X9.62/SECG curve over 256-bit prime field is not a CC approved encryption type and should not be used. The NIST/SECG curve over 521-bit prime field was not evaluated. Specify the Digest Algorithm as an approved CC evaluated type (SHA-1, SHA-256, SHA-384, or SHA-512). The use of SHA-224 is not approved for use in CC evaluations. Specify the Private Key Password and verify.

FPT_TUD_EXT.1.3

The currently active version of the TOE can be queried by issuing the command “show version” in an SSH session. This will report both the currently running software version and the TOE model number. Additionally, the Web UI displays the currently running software version in right side of the footer. This footer visible on all pages in a Web UI session.

When an update is loaded onto the TOE, in **Administration > Agents and Software Updates > Software Updates**, the page displays a column for “Version” which shows the software version of the update file, as well as a “Status” column which will display an **Installed** label for an active update and an **Install** button for an inactive update. The software version of loaded but inactive update can be queried on this page.

ClearPass makes use of a digital signature whenever updates/upgrades are applied to the system, regardless of the package size or intent. When a new package is to be installed on ClearPass, it will initially be loaded to the server. Package signatures are verified after the package is loaded, but prior to the installation process. The signature is verified using a locally stored copy of the public key. If the cryptographic signatures are identical, then the update process is allowed to proceed. If the signatures do not match, the package update will fail with an error message indicating that the package has failed validation prior to installation.

To reduce error potential when manually downloading packages, such as for a non-internet connected system, it is also recommended to validate the package hash and compare it against the published values from the ClearPass download site prior to loading onto ClearPass. While this process was not evaluated as part of Common Criteria evaluation, it is helpful in pre-validating that downloads have not been tampered with when updating systems without direct internet connections.

The evaluator shall verify that the guidance documentation describes how to query the currently active version. If a trusted update can be installed on the TOE with a delayed activation, the guidance documentation needs to describe how to query the loaded but inactive version.

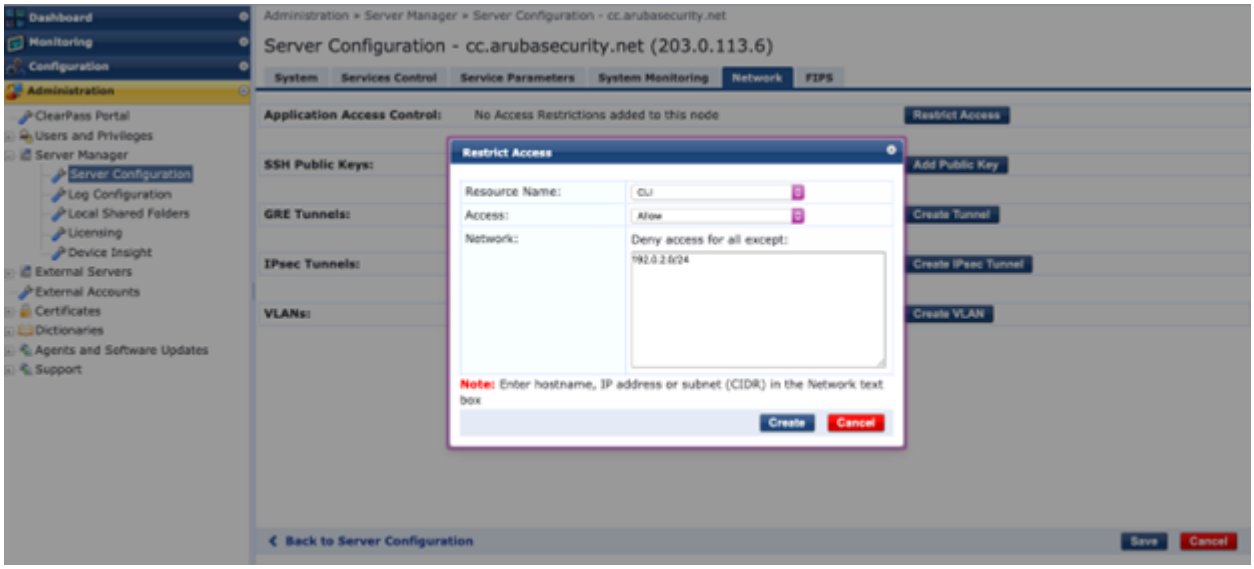
FMT_SMF.1.1

Network Time Protocol (NTP) is recommended to maintain reliable time stamps. The Common Criteria evaluation was performed with NTP enabled.

Local administration is available on appliances using the following interfaces:

- Peripherals (monitor and keyboard directly attached
- RS-232 terminal (serial console)
- Management Ethernet port

  Because the Ethernet port may also be used for other communications, it is recommended to restrict the access for both CLI (secure shell) and Administrative Web UI. This is accomplished by navigating to **Administration > Server Manager > Server Configuration**. Select the server and then select **the Network** tab. Click the **Restrict Access** button near **Application Access Control** to create the desired controls.

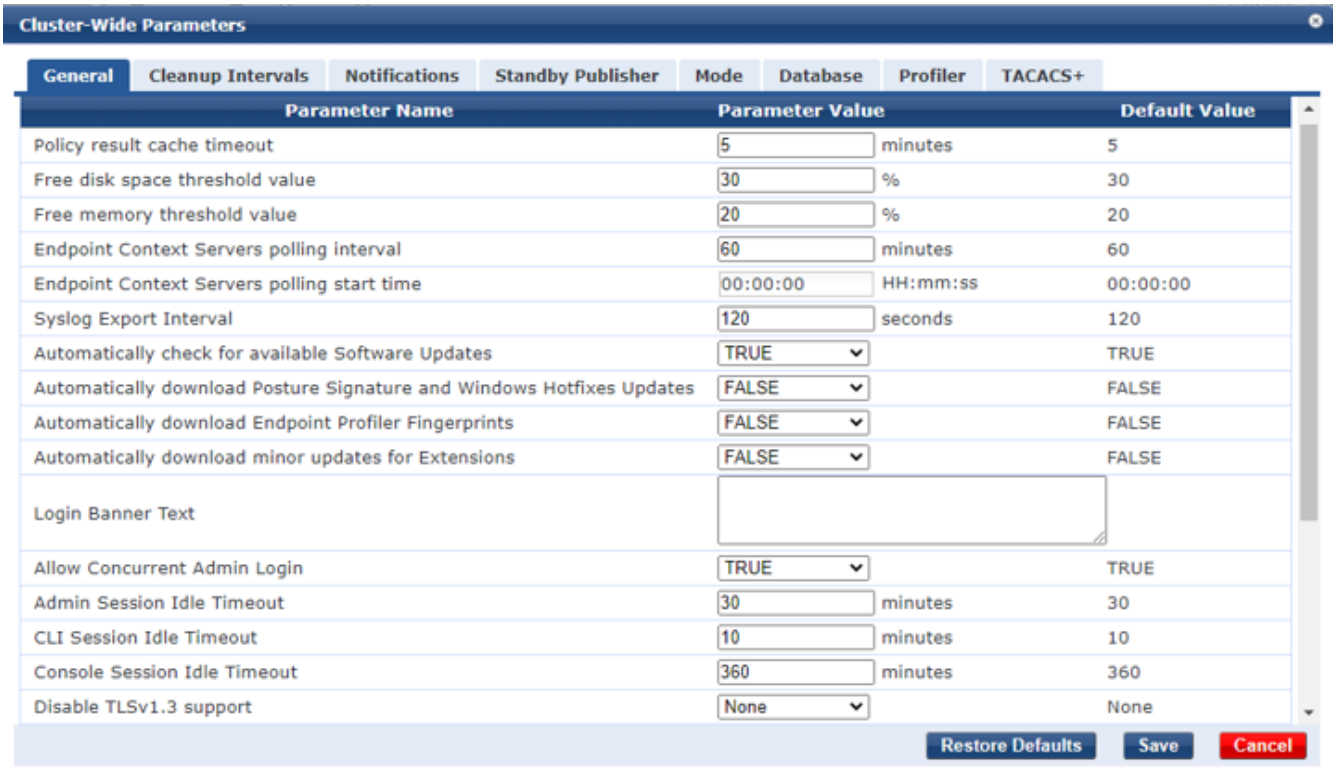


Definitions of the Resource Names may be found in the . Note that restricting the CLI will only apply to SSH connections. Console connections (including serial connections) are not impacted by these network restrictions.

Multiple application access controls may be specified to restrict service availability. When selecting Policy Manager as the Resource Name, similar restrictions should be applied to the Insight and Guest Operator nodes to ensure all interfaces are restricted equally.

FTA_SSL.3 / FTA_SSL.4 / FTA_SSL_EXT.1.1

Both CLI (console and SSH) and Web UI sessions can be configured to timeout sessions after inactivity. This setting is available through the Web UI by navigating to **Administration > Server Manager > Server Configuration**. Select **Cluster-Wide Parameters**.



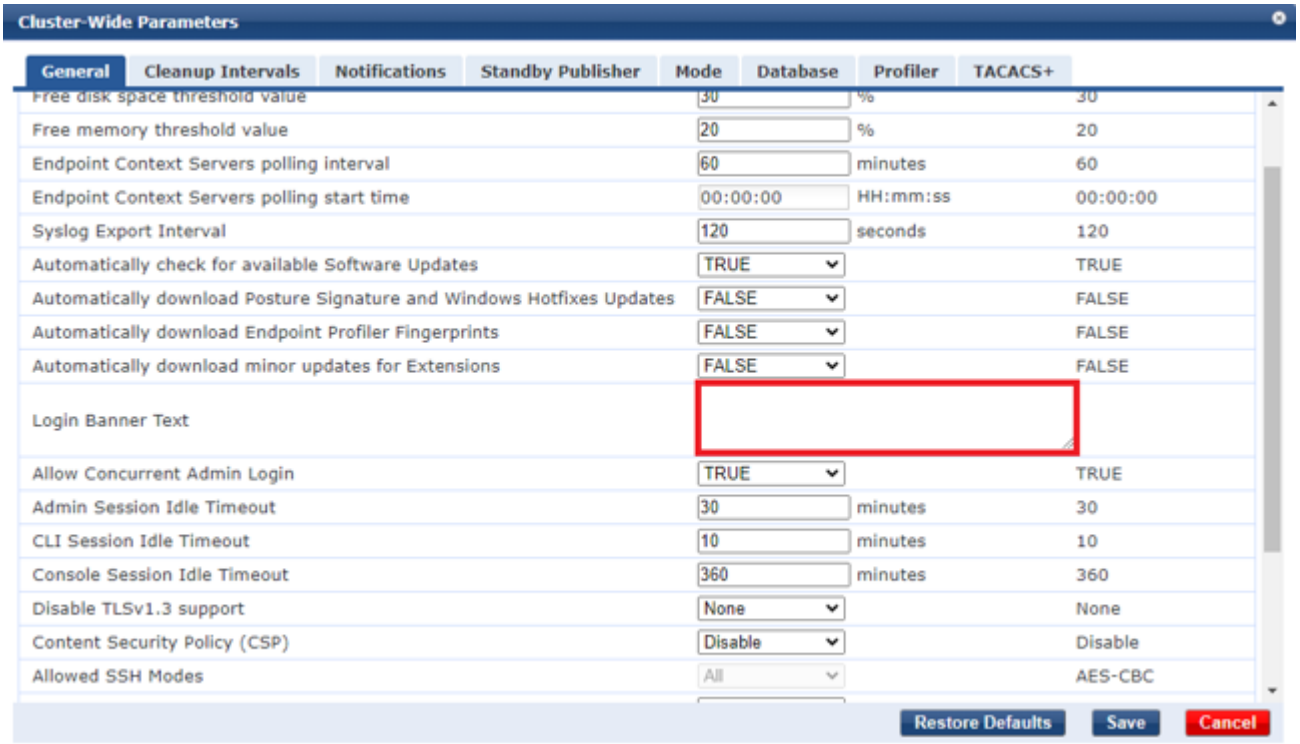
Modify the **Admin Session Idle Timeout** (default value **30**) to the desired time in minutes to change the Web UI settings. SSH sessions will timeout based on the **CLI Session Idle Timeout** (default value **15**) time in minutes. Console sessions will timeout based on the **Console Session Idle** Timeout (default value **360**) time in minutes.

The Web UI screens available under **Monitoring > Live Monitoring** will automatically refresh by default.

Termination of local console or CLI (SSH) sessions by the administrator is accomplished by entering the “exit” command to log out before idle session timeout. Web UI screens can be triggered from the menu in the upper-right corner by selecting **Logout**.

FTA_TAB.1

Configure an access banner with appropriate text by navigating t**o Administration > Server Manager > Server Configuration**. Select **Cluster-Wide Parameters.**



Modify the **Login Banner Text** field to include the information desired. This text will be applied to both the local console, Web UI, and SSH login events prior to the user logging in.

FTP_ITC.1

Most communication is already performed over encrypted channels, but some protocols do not support TLS encryption to ensure confidentiality and integrity. An example of this could be Syslog. In use cases where trusted communications are required to interact with these external devices, the use of IPsec is recommended.

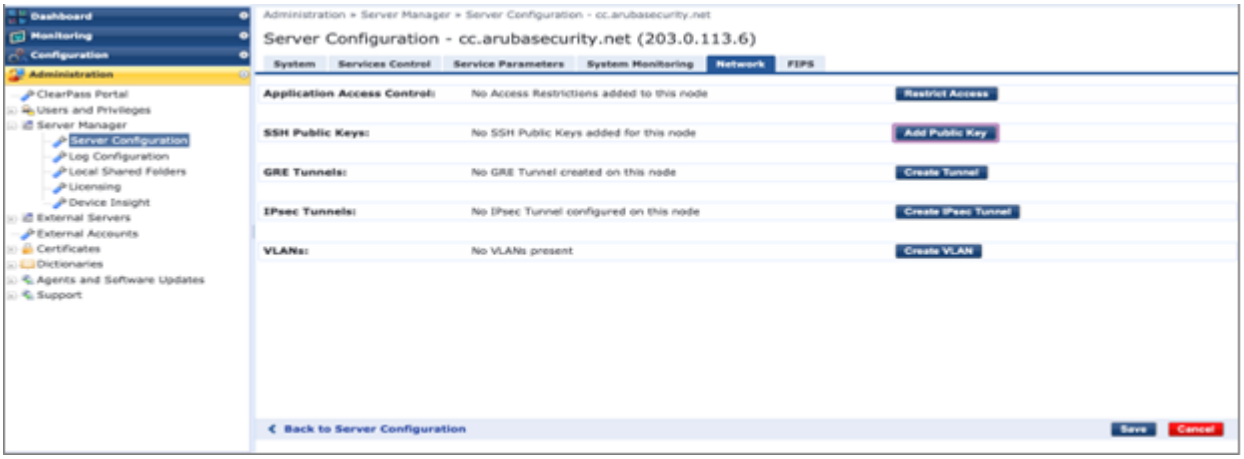
To comply with Common Criteria evaluated status, all syslog communications should be encrypted between ClearPass and the remote syslog system(s). Section FCS_IPSEC_EXT.1 details the basic information to establish IPsec tunnels. It is recommended to restrict the traffic to only the syslog traffic (default UDP port 514) unless additional services are required on the same remote server.

FCS_SSH_EXT.1.2

Configure SSH public key authentication by navigating to **Administration > Server Manager > Server Configuration**. Each node within a cluster (if applicable) must repeat the following process. Select the server/node to enable SSH public keys. Navigate to the

Network tab. Select the **Add Public Key** button and paste the desired key information in the **SSH Public Key** text field.

ClearPass supports SSH Public Key Authentication when using ssh-rsa, rsa-sha2-256, rsa-sha2-512, and ecdsa-sha2-nistp256 key types only, regardless of operating modes. Attempting to import an unsupported SSH key type will result in the UI error indicating the 'SSH Public key is invalid'.



Additional keys for different users can be added as needed.

FCS_SSH_EXT.1.4

The modification of the Allowed SSH Modes option is not permitted when FIPS/CC mode is enabled. When FIPS/CC mode is enabled, aes256-gcm@openssh.com, aes256-ctr, aes256-cbc, aes128-gcm@openssh.com, aes128-ctr and aes128-cbc ciphers are supported.

FCS_SSH_EXT.1.5

The SSH transport uses hmac-sha1, hmac-sha2-256, or hmac-sha2-512 MAC algorithms. No administrator settings are configurable.

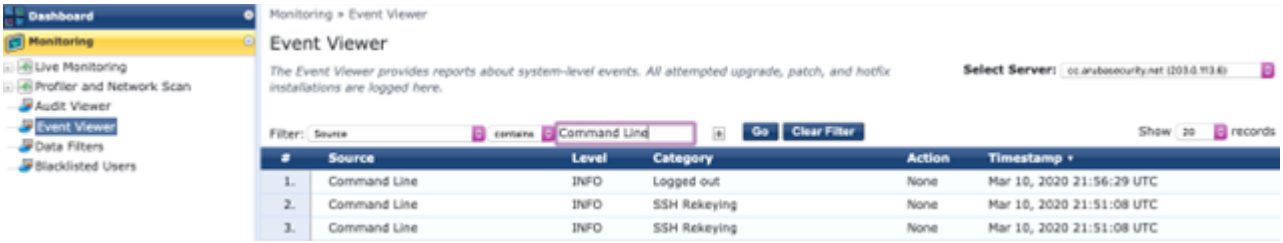
FCS_SSH_EXT.1.6

The SSH key exchange methods available are ecdh-sha2-nistp256, ecdh-sha2-nistp384 and ecdh-sha2-nistp521.

No administrator settings are configurable.

FCS_SSH_EXT.1.8

SSH rekey events are initiated for every 128 MB of data sent over the connection, or every sixty (60) minutes (1 hour). These events can be monitored in the Web UI by navigating to **Monitoring > Event Viewer**. Applying the filter **Category contains SSH Rekeying** will show all rekey events. Below is an example event.



Two (2) events will occur for rekey events. The first is ClearPass sending clients updated keys. The second is ClearPass receiving updated client keys. SSH rekey events will occur for either one (1) hour or 128 megabyte (MB) of data transferred, whichever event occurs first.

FCS_SSHS_EXT.1.1

The SSH Host Key algorithms available include ecdsa-sha2-nistp256, rsa-sha2-256, and rsa-sha2-512.

No administrator settings are configurable.

FCS_TLSS_EXT.2.1

The following is the complete list of evaluated cipher suites available on ClearPass in configured Common Criteria mode (includes functional limits of FIPS mode when enabled). When Common Criteria mode is enabled, these suites are automatically enabled without further administrator action.

```
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
```

```
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
TLS_RSA_WITH_AES_256_CBC_SHA256
TLS_RSA_WITH_AES_128_CBC_SHA256
TLS_RSA_WITH_AES_256_CBC_SHA
TLS_RSA_WITH_AES_128_CBC_SHA
TLS_RSA_WITH_AES_256_GCM_SHA384
TLS_RSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
```

The following cipher suites are available only when an ECDSA certificate is installed on ClearPass:

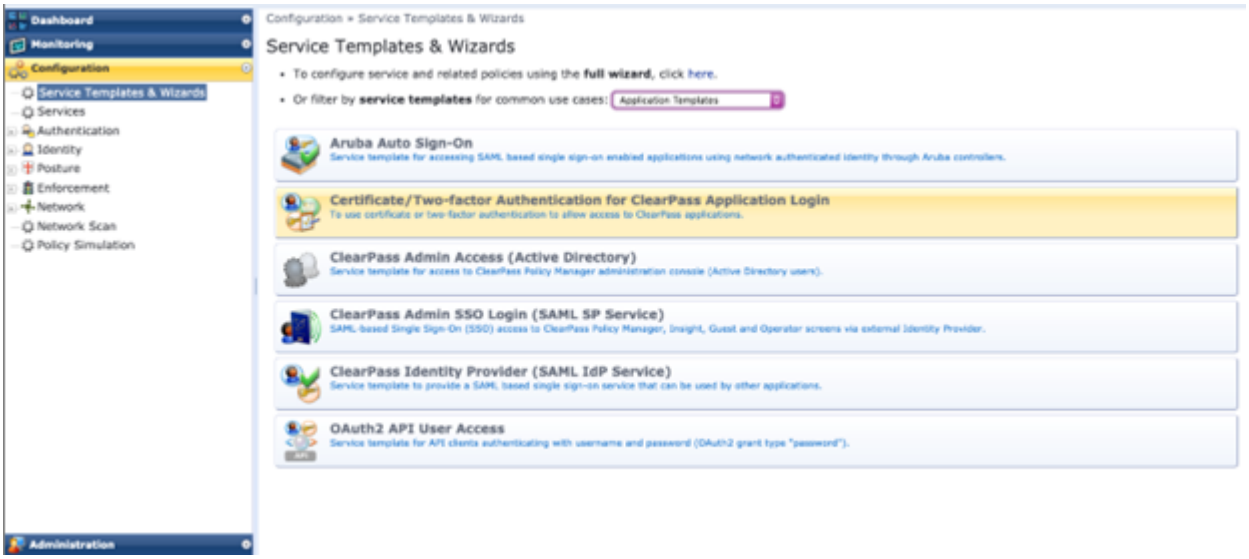
```
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
```

To configure this OCSP revocation checking, navigate to **Administration > Server Manager > Server Configuration**, select the server in the list, and then select the **Service Parameters** tab. From the **Select Service** drop-down menu, select **ClearPass network services**. In the **Certificate Auth** section, set the **OCSP Check** value to **Required**. When this configuration is enabled, OCSP checks will be performed on certificates received for mutual TLS authentication.

FCS_TLSS_EXT.2.2

Web UI sessions may use certificate identification through mutual TLS authentication. This process requires that all DNS entries be configured correctly prior to establishment. It is critical to ensure that fully qualified domain names (FQDN) are resolvable from the client. Additionally, client systems will need to have the ClearPass Web UI public certificate available locally, along with any required CA intermediate certificates. That process is outside the scope of this document.

To aid in this process, a setup wizard is available to administrators in the Web UI. Begin by navigating to **Configuration > Service Templates & Wizards**. Select the **Certificate/ Two-factor Authentication for ClearPass Application Login** service template.



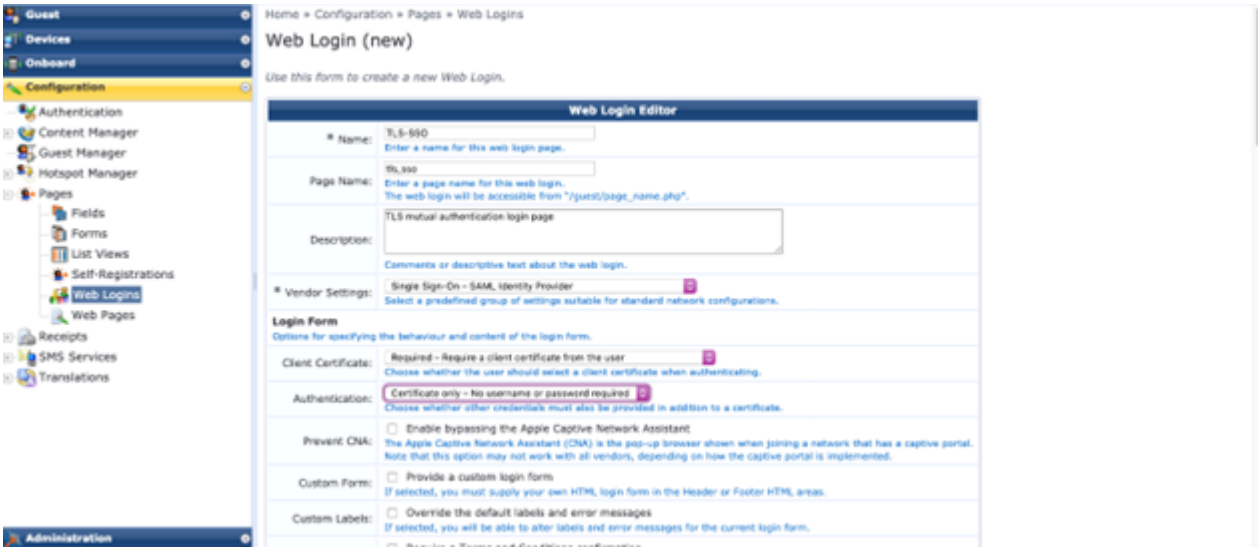
Within the **General** tab, the **Name Prefix** specifies is used to identify all components the wizard generates when used. For reference, the name “TLS-SSO” is used in later screen examples. Select the **Next** button to advance through the wizard.

The Service Role tab allows the selection of Web UI components that will be configured to use TLS mutual authentication. The default includes the **PolicyManager** component, which controls the policy pieces of the system. Additional components may be selected as desired.

By default, the **Authentication** tab includes all the authentication sources that are already defined. If one has not been created, a new one may be created. This will default to an Active Directory (AD). Select or create the appropriate source and click **Next**.

The **IdP Details** tab allows selection of the appropriate Web login page. By default, only the device provisioning page is available. A new page must be created. Select the **Add New Guest Web Login page** link. A new browser tab displays.

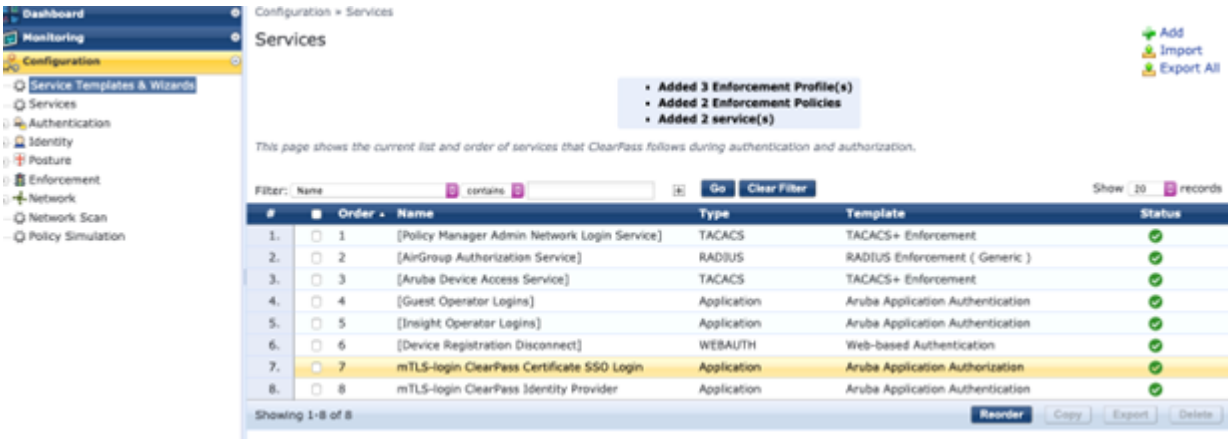
From the **Guest > Configuration > Pages > Web Logins** page, select the **Create a new web login page** link. The Web Login (new) page opens. In the Vendor Settings drop-down list, select **Single Sign-On – SAML Identify Provider**. From the **Client Certificate** drop-down list, select **Required – Require a client certificate from the user**. To allow certificate-only authentication, the default value may be used in the Authentication field (Certificate only – No username or password required).



Additional edits can be made as needed. When completed, select **Save Changes** at the bottom of the Web Login (new) page. Return to the other browser tab where the **Policy Manager > Configuration > Service Template** wizard is displayed. In the **IdP Details** tab, click the blue arrow. This refreshes the **Page Name** drop-down list to include the newly generated page name. Select the new page name in the list and then click **Next**.

The **Enforcement Details** tab lets you select attributes from the certificate to match against enforcements. A wide variety of components may be selected based on the certificate attribute or attributes.

When the **Add Service** button is selected, the appropriate services are created within the system. By default, two services will be created that have the prefix provided in the **Name Prefix** step.



After completion of the Service Template & Wizard, select **Configuration > Identity > Single Sign-On (SSO)** and select the **SAML IdP Configuration** tab. The Web Login Configuration & Metadata section must be configured to indicate the previously created page and specify the Identity Provider (IdP) Signing Certificate. The Identity Provider (IdP) Encryption Certificate is not required.

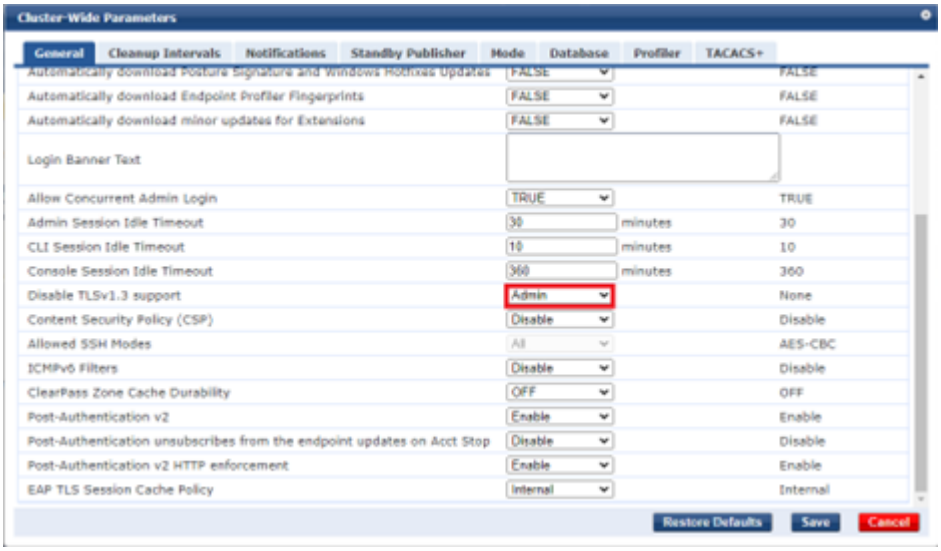
Certificates are not available for the IdP Signing Certificate if the certificate under Service & Client Certificates is not created. Root and Intermediate CA certificates used

to sign the user certificate must be configured with **Others** in the certificate trust list for the certificate authentication to work.

Additional information on modifications and on troubleshooting this process can be obtained by contacting technical support.

FCS_TLSS_EXT.1.2

ClearPass supports a flexible TLS model for backwards compatibility with older devices. Support for older SSL-based protocols (SSL 1.0, SSL 2.0, or SSL 3.0) is no longer available in any ClearPass configuration. In Common Criteria deployments TLS v1.0 and TLS v1.1 are disabled and hidden in the UI by default. Navigate to **Administration > Server Manager > Server Configuration** and select the **Cluster-Wide Parameters** link.



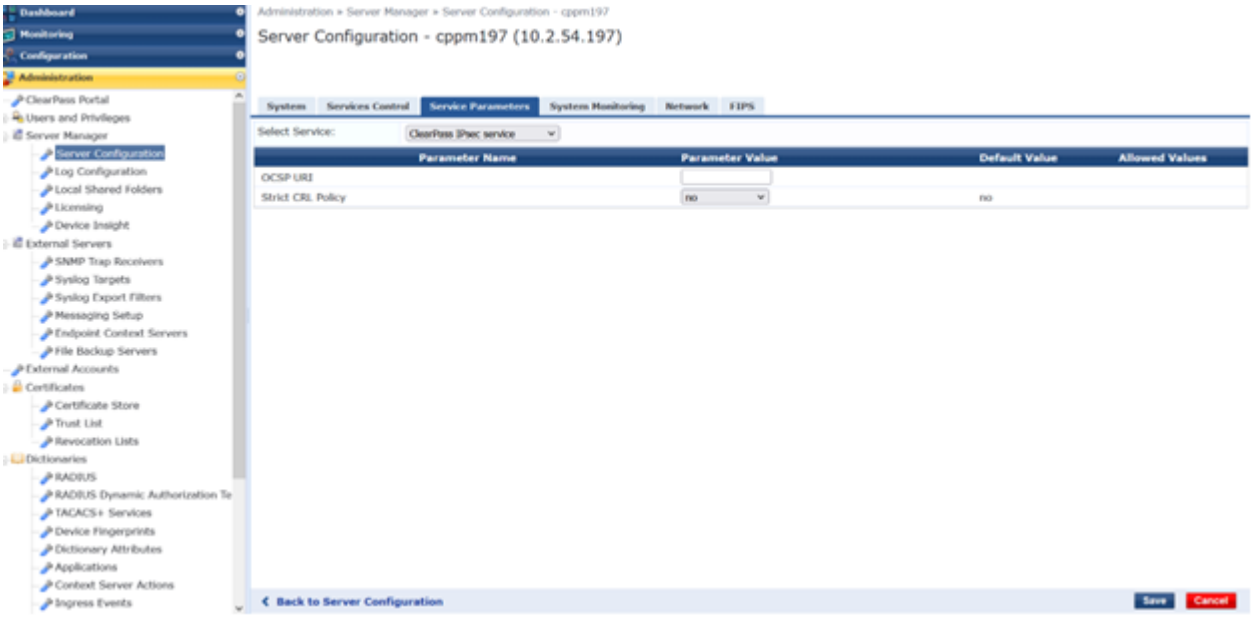
Within the **General** tab, **Disable TLSv1.3 support** is set to **None** by default, as TLSv1.3 is disabled in the evaluated configuration. This prevents legacy TLS versions prior to v1.2 from using components such as RADIUS, RadSec, or Web UI. TLSv1.3 is disabled in the evaluated configuration. This setting cannot be modified in CC operating mode.

FCS_IPSEC_EXT.1

When situations require additional encryption and integrity, an IPsec VPN tunnel may be established between ClearPass and a remote device. The IPsec tunnel cannot be used as a gateway to or from ClearPass. Remote endpoints should be configured to accept the ClearPass appliance's address exclusively.

While using Online Certificate Status Protocol (OCSP), the Uniform Resource Indicator (URI) should be specified in the **OCSP URI** field, beginning with HTTPS or HTTP. This is only required if connection to a remote VPN device does not transfer a certificate with the OCSP URI encoded. To configure this OCSP URI value, navigate to **Administration >**

Server Manager > Server Configuration, select the server in the list, and then select the **Service Parameters** tab. In the **Select Service** drop-down list, select **ClearPass IPsec service.**



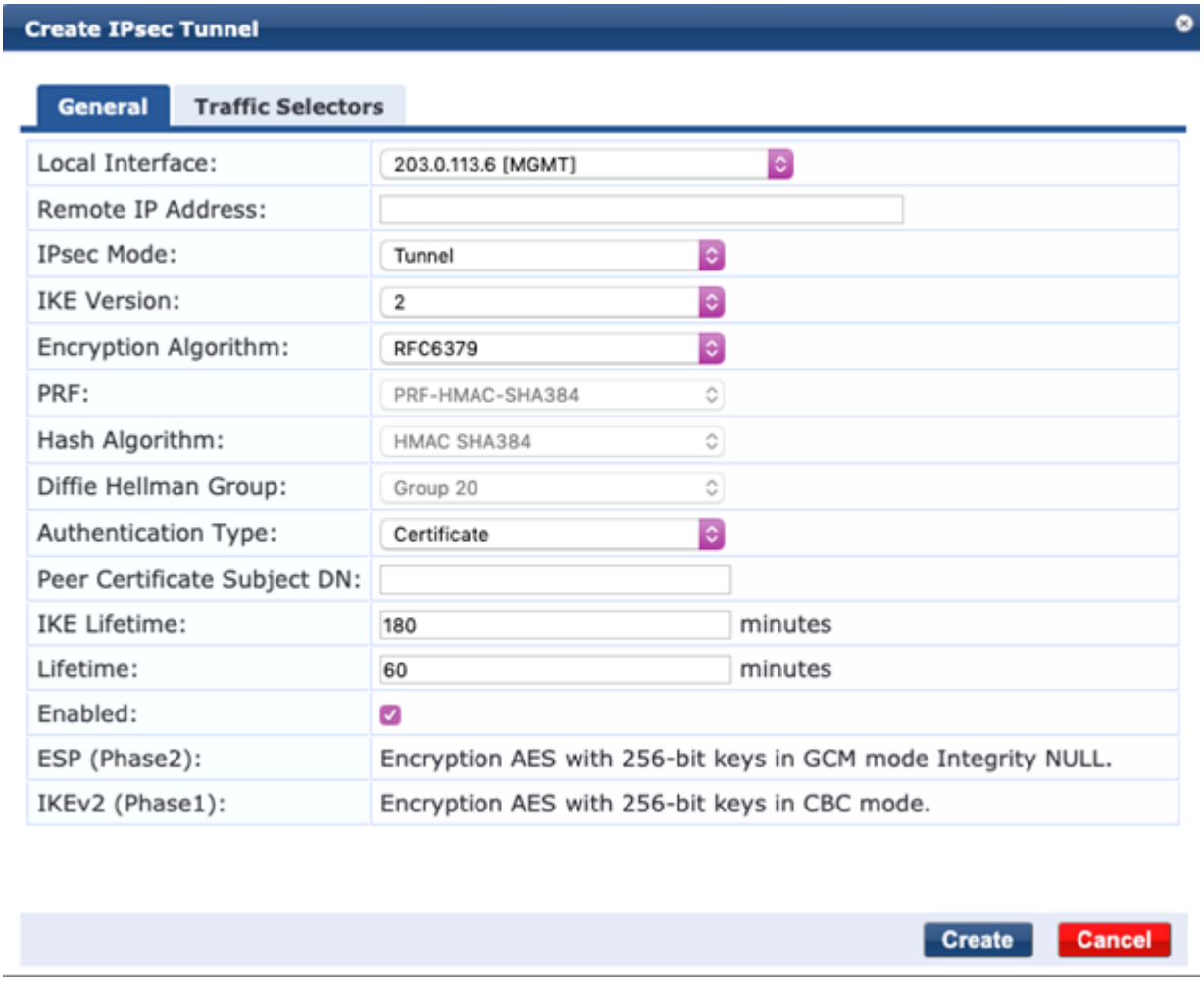
After clicking **Save**, select the **Network** tab. The **Create IPsec Tunnel** button may be used to generate a new IPsec tunnel. Existing entries may also be directly deleted or modified from this location.

Adding a new IPsec tunnel allows the specification of either Pre-Shared Key (PSK) or certificate- based systems. Select the values required for connection with the remote IPsec device.

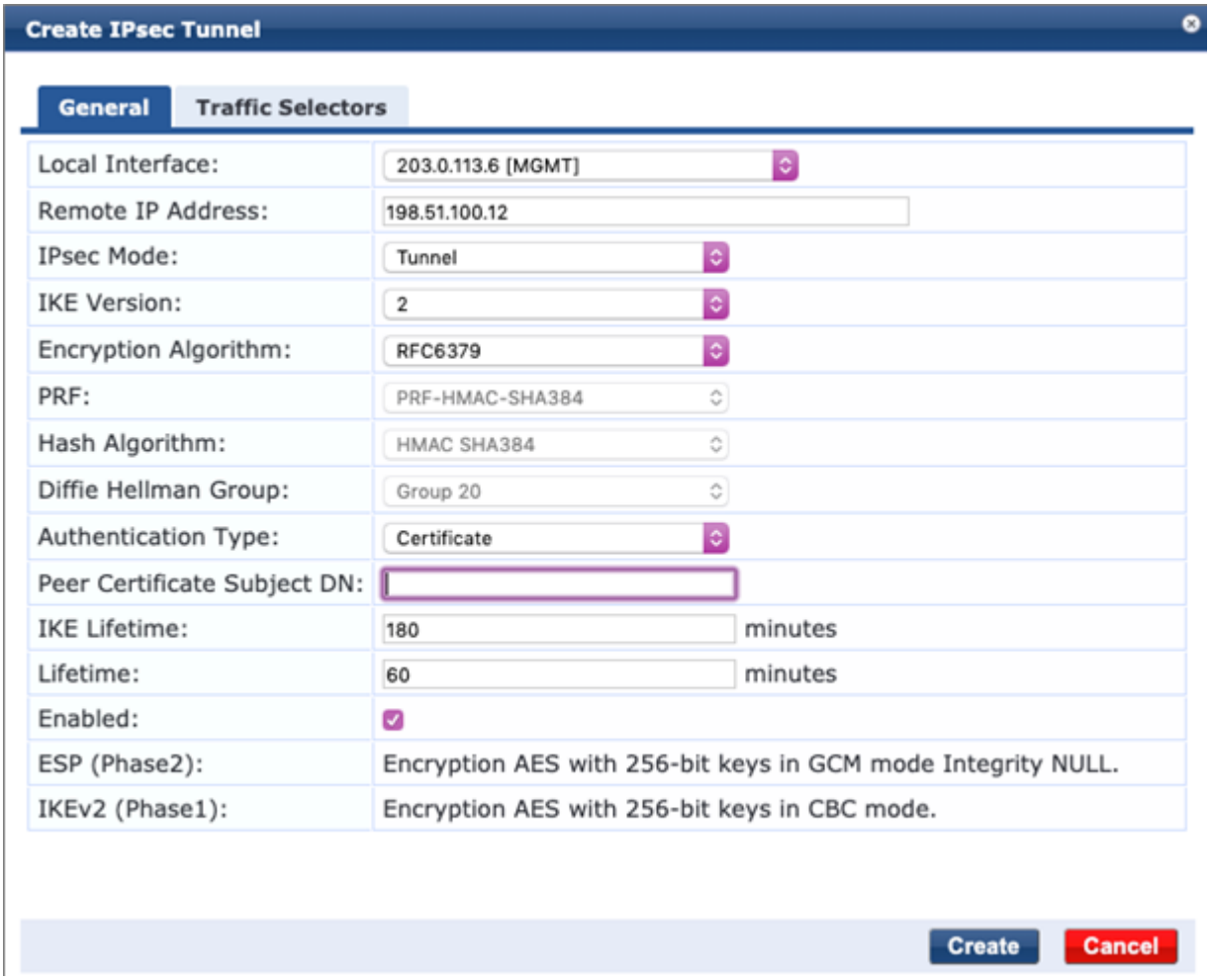
NOTE

The TOE does not generate Pre-Shared Keys

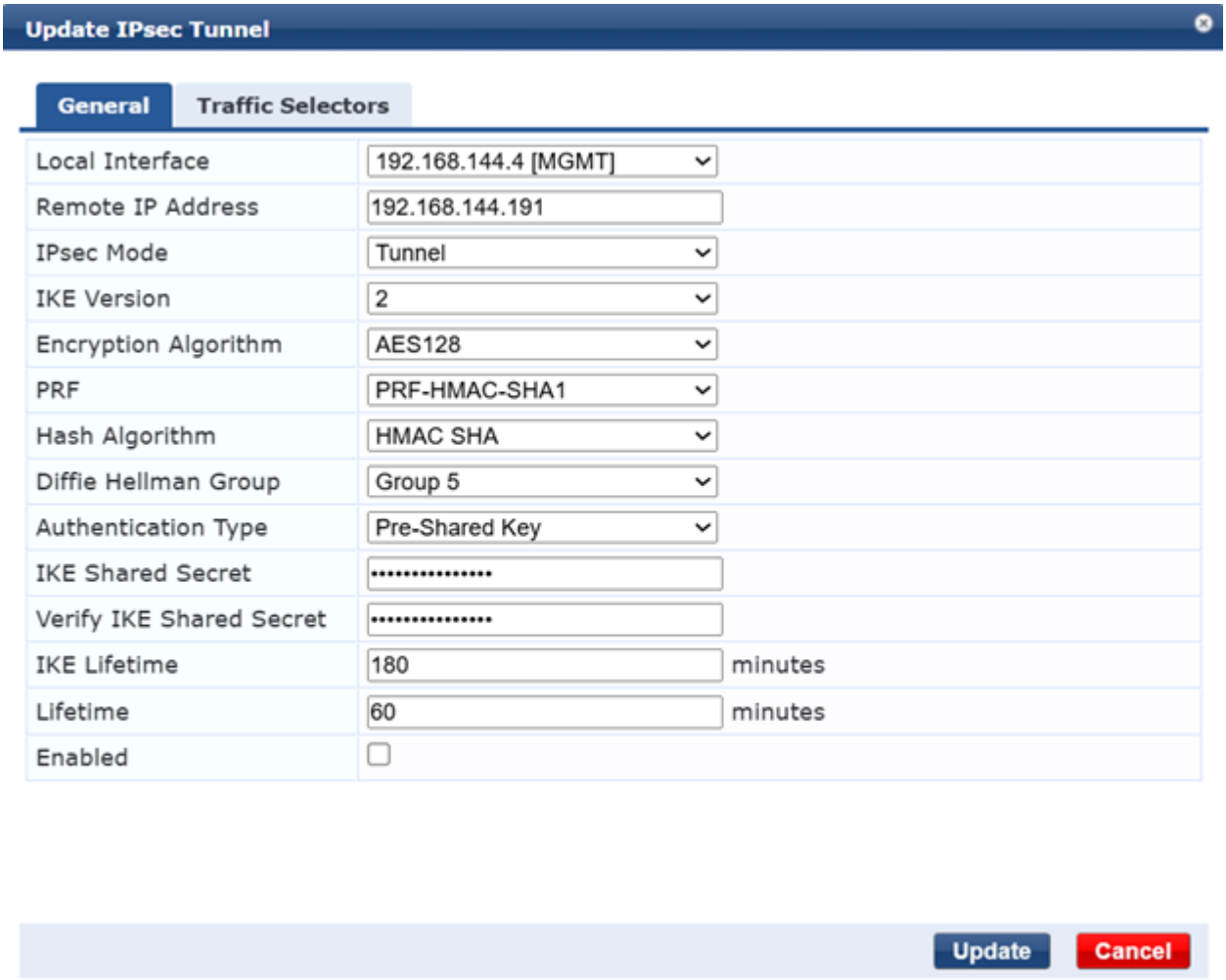
To reduce the likelihood of configuration errors where weaker algorithms are used in Phase 2 rather than in Phase 1 negotiations, the encryption algorithm and hash algorithms are selected only once and applied across the Security Association (SA). These values will also apply to child SAs. Remote peers should be configured to accept the same settings.



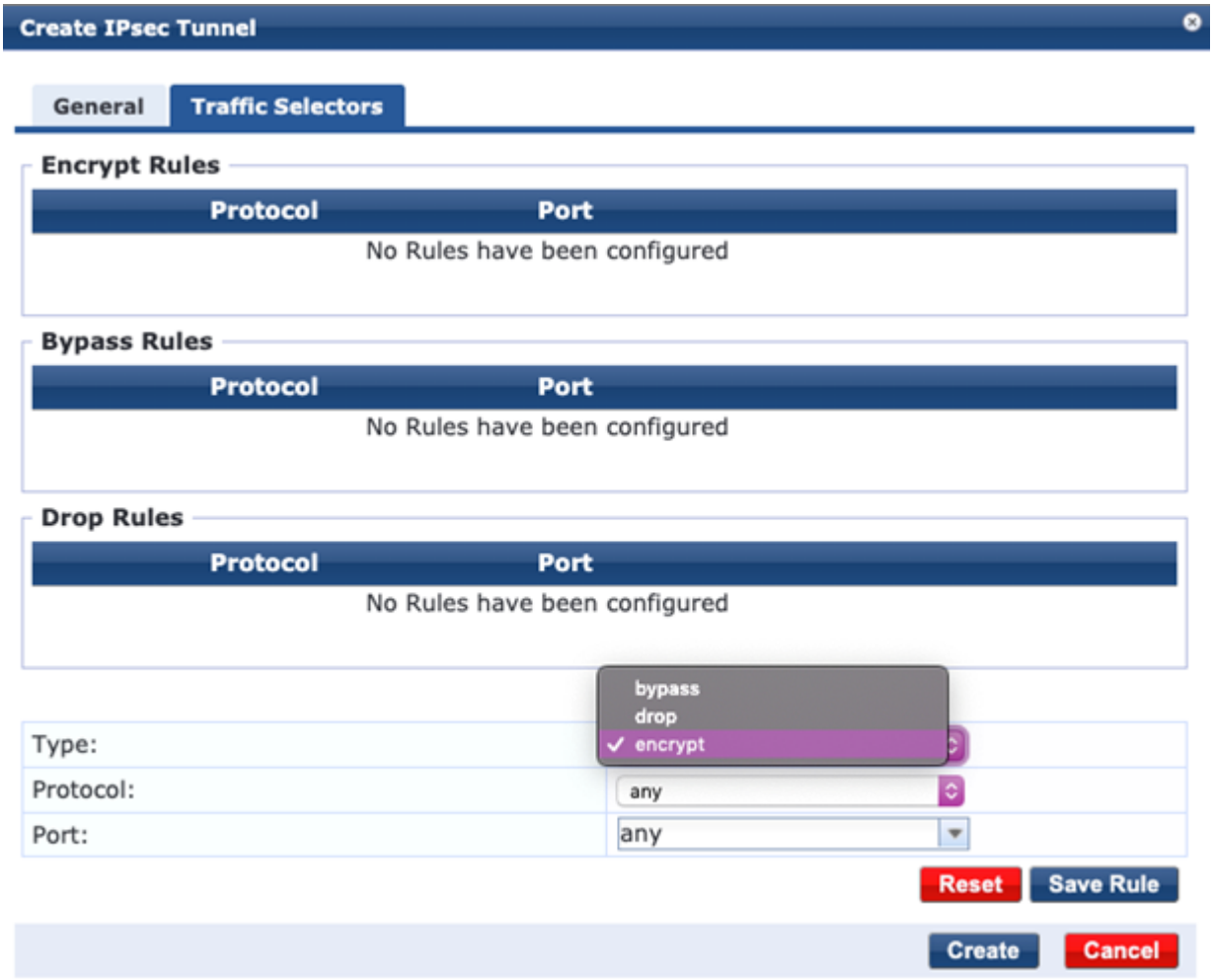
If **Certificate** is selected as the **Authentication Typ**e, then when specifying the value of the **Peer Certificate Subject DN**, the specified distinguished name must be an exact match to the certificate that the remote device is using. If this is not exactly matched, the tunnel will fail to negotiate. ClearPass will use its HTTPS certificate for IPsec identity, but the CA from the remote peer must also be included in the ClearPass trust list or validation will not occur.



If **Pre-Shared Key** is selected as the **Authentication Type**, then the pre-shared key should be entered into the "IKE Shared Secret" field and verified by the administrator by being entered again into the "Verify IKE Shared Secret" field.



If only specific traffic is sent to the remote host over a VPN, the **Traffic Selectors** tab can also be configured. This will default to encryption of all traffic (protocol and port) between the two hosts. Additional traffic rules can be applied to bypass the traffic, as noted in Appendix B.



In the event IPsec VPNs unexpectedly drop the following steps may be taken to resolve the issue. IPsec tunnels may be alerted on when they change status to down. This will aid in identifying that the IPsec session has failed.

Issue	Likely Problem and Resolution
Certificate failure (expired)	Replace the HTTPS Server certificate on ClearPass or the remote peer certificate device.
Tunnel will not establish	Ensure parameters have not been changed by the remote peer
IPsec indicates it is active (up), but traffic is not passing	Ensure the tunnel status is up Validate the traffic selectors are not restricting access as expected Validate intermediate devices such as firewalls are not preventing traffic from passing

When configuring IPsec tunnels with remote peers that change the peer certificate, the IPsec service on ClearPass should be restarted to clear the previous certificate from the cache. This is accomplished when administrators log into the console directly and enter the following command:

service restart cpass-ipsec

IPsec VPNs may be configured to use various settings. The selected settings determine the options available. When using IKEv1 in either Tunnel or Transport mode, the following settings can be selected:

- IKE Phase 1 Mode: Main
- Encryption Algorithm: AES128, AES256
- Hash Algorithm: HMAC SHA, HMAC SHA256, HMAC SHA384
- Diffie Hellman Group: Group 14, Group 19, Group 20

  When using IKEv2 in either Tunnel or Transport mode, the following settings can be selected:

- Encryption Algorithm: AES128, AES256, AES128GCM16, AES256GCM16, RFC6379
- PRF: PRF-HMAC-SHA1, PRF-HMAC-SHA256, PRF-HMAC-SHA384
- Hash Algorithm: HMAC SHA, HMAC SHA256, HMAC SHA384
- Diffie Hellman Group: Group 14, Group 19, Group 20

  The RFC6379 Encryption Algorithm is available for use exclusively under IKEv2. This will utilize AES256 in CBC mode for Phase 1 and AES256 in GCM with Integrity NULL for Phase 2. This is the only condition where Encryption Algorithms are not the same for both phases. Selection will also set PRF to PRF-HMAC-SHA384, the Hash Algorithm to HMAC SHA384, and Diffie Hellman Group to utilize Group 20.

  As noted in FCS_IPSEC_EXT.1.4 and FCS_IPSEC_EXT.1.11, the UI will offer options not allowed under CC evaluated criteria.

FCS_IPSEC_EXT.1.3

IPsec VPNs can be configured to use either Transport or Tunnel by selecting the IPsec Mode. Tunnel mode is the default IPsec Mode.

FCS_IPSEC_EXT.1.4

Hash Algorithms are limited to HMAC SHA1, HMAC SHA256 and HMAC SHA384. HMAC SHA should not be selected. The selected hash algorithms are applied to both Phase 1 and Phase 2 for all configurations.

FCS_IPSEC_EXT.1.5

Support for NAT traversal is included in IPsec.

FCS_IPSEC_EXT.1.6

Encrypted payloads will be encrypted using the selected IKE version and cryptographic algorithms selected. The selected cryptographic algorithms are applied to both Phase 1 and Phase 2 for all configurations except RFC6379.

FCS_IPSEC_EXT.1.7

SA lifetimes are specified in minutes for both IKEv1 and IKEv2. To specify the Phase 1 lifetime, the value **IKE Lifetime** parameter should be set, the default value is 180 minutes. Valid times are 5-1440 minutes for Phase 1 lifetimes.

FCS_IPSEC_EXT.1.8

SA lifetimes are specified in minutes for both IKEv1 and IKEv2. To specify the Phase 2 lifetime, the **Lifetime** value should be set, the default value is 60 minutes. Valid times are 5-1440 minutes for Phase 2 lifetimes.

FCS_IPSEC_EXT.1.11

Diffie Hellman (DH) Groups are limited to group 14, group 19, and group 20. Group 24 is not available. Group 5 should not be selected.

FCS_IPSEC_EXT.1.14

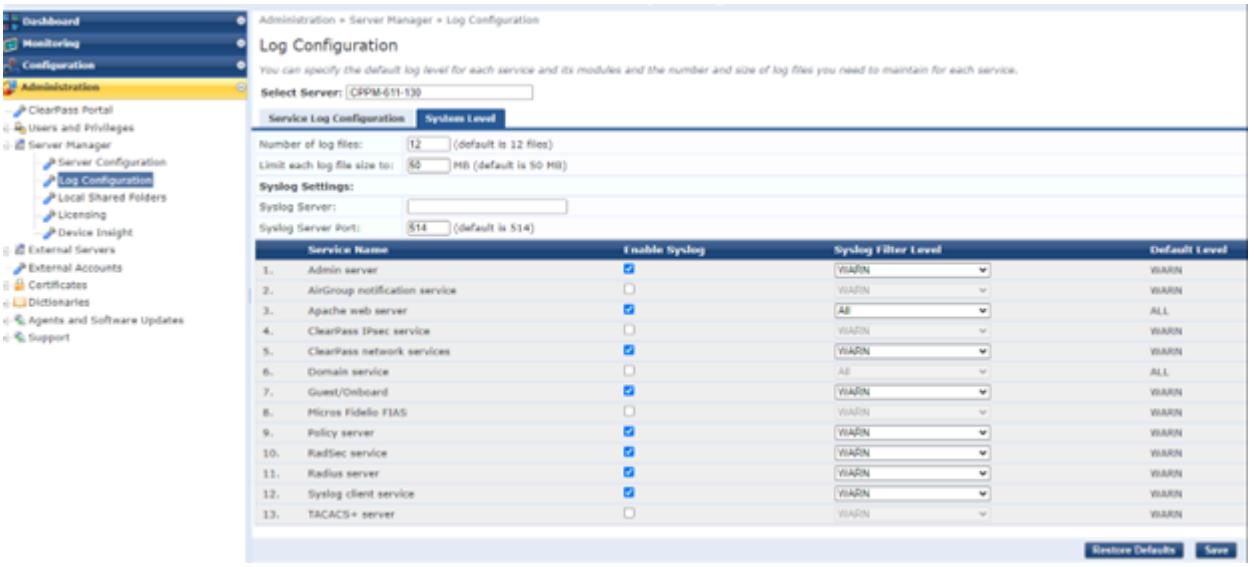
If **Certificate** is selected as the **Authentication Type**, then when specifying the value of the **Peer Certificate Subject DN**, the specified distinguished name must be an exact match to the certificate the remote device is using. If not exactly matched, the tunnel will fail to negotiate. The peer certificate should be specified as stated in the client certificate beginning with the CN= field until the end of the DN is met. When applied to IPsec VPN configurations, the SAN extension in the certificate is not used to match against.

FAU_STG_EXT.1

Audit integrity is crucial to ClearPass. As such, any modifications to the audit records themselves by anyone is not possible. The only action that an administrator may take involving modification of the logs is to configure the log file size limit and retention numbers in the FAU_STG_EXT.1 section. These setting will affect the on-box log retention settings. The ability to modify or delete records is not a function supported by ClearPass.

ClearPass is not intended to be a long-term audit storage system. The use of syslog to export data is recommended to transfer data to another system that has been built for the purpose of long-term audit record storage. Local audit records are stored for seven (7) days prior to automatic cleanup (deletion). To extend the local audit record storage, navigate to **Administration > Server Manager > Server Configuration** and select the **Cluster-Wide Parameters**. The settings can be adjusted by modifying the value on the **Cleanup Intervals** tab. The Access Tracker events can be modified by adjusting the **Cleanup interval for Session log details in the database** parameter (default value is 7 days). The general audit (such as the Accounting) events can be modified by adjusting the **Old Audit Records cleanup interval**(default value is 7 days). Event Viewer records are stored for seven (7) days prior to automatic cleanup. There is no user configurable setting to modify the Event Viewer log storage. Audit records that exceed cleanup intervals will be deleted from the file system and the space reclaimed to write new audit events.

ClearPass log file storage is limited by drive space. The typical storage duration for on-system log storage is seven (7) days. Navigate to **Administration > Server Manager > Log Configuration** and select the **System Level** tab.



The number and size of log files may be specified based on observed logging levels. The number and size limits apply to all log file settings. Modifying these values will affect the log files that contain information created by services. Reducing the capacity may decrease the information available to less than seven (7) days. Increasing may cause issues with system free disk space thresholds.

The IP address of the external syslog server that will receive audit messages from ClearPass should be specified, along with all the appropriate audit events to be sent. The default setting does not select any services to enable syslog. It is recommended at least one service be selected. All audit messages equal or higher in priority to the **Syslog Filter Level** setting will be sent to the specified syslog server.

ClearPass does not transfer syslog messages in real time. Messages are queued to a syslog buffer that then transfers all messages to the syslog server every 120 seconds. This value may be reduced to a minimum of every 30 seconds, but will default to every 120 seconds. The potential delay in message queue and receipt by the remote server should be noted to comply with Common Criteria evaluated settings.

NOTE

The ClearPass IPsec service is now included in the list of services whose service logs can be sent to a syslog server, and is available in the Service Name list on the **Administration > Server Manager > Log Configuration > System Level** tab.

FPT_TST_EXT.1 (self-tests)

ClearPass will execute self-tests on the cryptographic core when operating in Common Criteria mode. These tests are also executed as part of the FIPS operating mode. To ensure the integrity of the module and the correctness of the cryptographic functionality at start up, self-tests are run. In the event of a self-test error, the module will

log the error and will halt, resulting in a failure to boot ClearPass. The module must be initialized into memory to resume function.

Power - on self - tests are executed automatically when the module is loaded into memory. The module verifies the integrity of the runtime executable using a HMAC-SHA-256 digest computed at build time. If the fingerprints match, the power-up self-tests are then performed. If the power-up self-test is successful, a flag is set to place the module in FIPS mode.

TYPE	DETAIL
Software Integrity C heck	• HMAC - SHA1 on all module components
Known Answer Tests	• AES ECB • AES GCM • AES CCM • AES XTS • AES CMAC • Triple - DES ECB • Triple - DES CMAC • Diffie - Hellman • EC Diffie - Hellman • HMAC - SHA - 1, HMAC - SHA - 224, HMAC - SHA - 256, HM AC - SHA - 384, • HMAC - SHA - 512 • RSA • SHA - 1, SHA - 256 and SHA - 512 • SHA3 - 256, SHA3 - 512, SHAKE - 128 and SHAKE - 256 • SP 800 - 90 DRBG (CTR_DRBG) • TLS KDF • SSH KDF • PBKDF KDF • HKDF KDF

TYPE	DETAIL
	• KDKDF KDF
Pair - wise Consistency Tests	• RSA • ECDSA • DSA

Power-on self-tests include capabilities not available in Common Criteria mode.

Input, output, and cryptographic functions cannot be performed while the module is in a self-test or error state because the module is single-threaded and will not return to the calling application until the power-up self-tests are complete. If the power-up self-tests fail, subsequent calls to the module will also fail - thus no further cryptographic operations are possible.

The module implements the following conditional self-tests upon key generation or upon random number generation, respectively:

The module verifies the integrity of the runtime executable using a HMAC-SHA1 digest which is computed at build time. If this computed HMAC-SHA1 digest matches the stored, known digest, then the power-up self-test (consisting of the algorithm-specific Pairwise Consistency and Known Answer tests) is performed. If any component of the power-up self-test fails, an internal global error flag is set to prevent subsequent invocation of any cryptographic function calls. Any such power-up self-test failure is a hard error that can only be recovered by reinstalling the module. The power-up self-tests may be performed at any time by reloading the module. Additionally, the pair-wise consistency tests are run as a conditional test each time a key pair is generated.

No operator intervention is required during the running of the self-tests.

Appendix A - FAU_GEN.1 AUDITABLE EVENTS

Many implementations use external syslog servers instead of locally hosted audit messages. ClearPass supports four (4) export event formats: Standard, Log Enhanced Event Format (LEEF), Common Event Format (CEF), and the RFC 5424 compliant format (RFC 5424). The default export syslog format is standard, sometimes referred to as raw.

List of auditable events by Common Criteria requirement. Events that include audit by ClearPass will specify the location to observe the audit message. These will be specified as “Audit Observed in” and specify the Web UI location messages of this type are located. Some events are logged in more than one observable location and will have examples specified for each event.

Audit events located in the **Monitoring > Audit Viewer** location will be noted based upon the tab the event is notified in. Most events in the Audit Viewer will have the ability to

note three (3) tabs: Old Data, New Data, Inline Difference. This allows the administrator to see the original value (Old Data), the vale that was set (New Data), and the single view to note both old and new together (Inline Difference).

The ClearPass version information displayed in syslog entries will update according to the operating ClearPass release. The message content will not change between versions.

Format entry examples are as follows:

NDcPP30e: FAU_GEN.1	
Auditable Events	Administrative login and logout (name of Administrator account s hall be logged if individual user accounts are required for admini strators) Changes to TSF data related to configuration changes (in additio n to the information that a change occurred it shall be logged wh at has been changed) Generating/import of, changing, or deleting of cryptographic key s (in addition to the action itself a unique key name or key referen ce shall be logged) Start - up and shut - down of the audit functions
Additional Content	None
syslog examples(s)	**Shutdown of the Audit Function (All TOE services stopped):** **The TOE logs shutdown of all of its services one by one. Each service is responsible for its own logging. This is two captured examples**: 2024 - 09 - 03 14: 37:22,383 192.168.144.3 Audit Records 703 1 0 Timestamp=Sep 03 2024 14:37:02.243 PDT,Component=Radius se rver,Level=INFO,Category=stop,Action=Success,Description=Perfo rmed action stop on Radius server 2024 - 09 - 03 14: 37:22,383 192.168.144.3 Audit Records 705 1 0 Timestamp=Sep 03 2024 14:37:12.160 PDT,Component=ClearPass I Psec service,Level=INFO,Category=stop,Action=Success,Descripti on=Performed action stop on ClearPass IPsec service] **Startup of the Audit Function (all TOE services startup)** **The TOE logs startup of all of its services one by one. Each serv ice is responsible for its own logging. This is two captured exa mples:** 2024 - 09 - 03 14: 37:22,383 192.168.144.3 Audit Records 704 1 0 Timestamp=Sep 03 2024 14:37:09.058 PDT,Component=Radius se rver,Level=INFO,Category=start,Action=Success,Description=Perfo rmed action start on Radius server

NDcPP30e: FAU_GEN.1

	2024 - 09 - 03 14: 37:22,383 192.168.144.3 Audit Records 706 1 0 Timestamp=Sep 03 2024 14:37:14.297 PDT,Component=ClearPass IPsec service,Level=INFO,Category=start,Action=Success,Description=Performed action start on ClearPass IPsec service * All TOE services generate an auditable record upon starting and stopping. The above are provided as examples.

NDcPP30e: FAU_GEN.2	Auditable Events	None	Additional Content	None

NDcPP30e:FAU_STG_EXT.1

Auditable Events	Configuration of local audit settings.
Additional Content	Identity of account making changes to the audit configuration.
Audit Observed In	**Monitoring > Audit**Viewer
syslog examples(s)	2025 - 03 - 18T02:52:59.690Z 192.168.144.3 ClearPass 18202 34 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.11.261850" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="MODIFY" Category="Log Service Configuration" User="admin" EntityName="Admin server" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2025 - 03 - 18T02:52:42.941Z"]

NDcPP30e: FCS_CKM.1	Auditable Events	None	Additional Content	None

NDcPP30e: FCS_CKM.2	Auditable Events	None	Additional Content	None

NDcPP30e: FCS_CKM.4	Auditable Events	None	Additional Content	None

NDcPP30e: FCS_COP.1/DATAENCRYPTION	Auditable Events	None	Additional Content	None

NDcPP30e: FCS_COP.1/SIGGEN	Auditable Events	None	Additional Content	None

NDcPP30e: FCS_COP.1/HASH	Auditable Events	None	Additional Content	None

NDcPP30e: FCS_COP.1/KEYEDHASH	Auditable Events	None	Additional Content	None

NDcPP30e: FCS_HTTPS_EXT.1

Auditable Events	Failure to stablish a HTTPS session
Additional Content	Reason for failure
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	Refer to FCS_TLSS_EXT.1 for TLS protocol failures, FCS_TLSS_EXT.2 for cert auth related failures, and NDcPP30e:FIA_UIA_EXT.1 for username/password failure.

NDcPP30e: FCS_IPSEC_EXT.1

Auditable Events	Failure to establish an IPsec SA.

NDcPP30e: FCS_IPSEC_EXT.1

Additional Content	Reason for failure.
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**Authentication Failure:** 2024 - 09 - 03T15:22:18 - 07:00 192.168.144.3 cppm.ipsec[11192]: Sep 3 15:22:18 05[ENC] <22> generating IKE_AUTH response 1 [N(AUTH_FAILED)] **No matching IKE proposal:** 2024 - 08 - 27T11:19:55 - 07:00 192.168.144.3 cppm.ipsec[11192]: Aug 27 11:19:55 14[IKE] <10> received proposals unacceptable **No matching ESP proposal:** 2024 - 08 - 26T10:06:10 - 07:00 192.168.144.3 cppm.ipsec[13638] : Aug 26 10:06:10 11[IKE] <ipsec - 3031|189> no acceptable propo sal found **Peer using aggressive mode:** 2024 - 08 - 26T11:56:38 - 07:00 192.168.144.3 cppm.ipsec[13638]: Aug 26 11:56:38 13[ENC] <35> parsed AGGRESSIVE request 0 [SA KE No ID V V V V] **No matching peer config:** 2024 - 08 - 26T12:16:34 - 07:00 192.168.144.3 cppm.ipsec[13638]: Aug 26 12:16:34 01[IKE] <49> no IKE config found for 192.168.144.3. ..192.168.144.91, sending NO_PROPOSAL_CHOSEN

NDcP302e: FCS_NTP_EXT.1

Auditable Events	Configuration of a new time server Removal of configured time s erver
Additional Content	Identity if new/removed time server
syslog examples(s)	**Add:** 2024 - 10 - 16T21:11:48.498Z 192.168.144.3 ClearPass 561654 0 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="ADD" Category="Remote Time Server" User="admin" EntityName="192.1 68.144.91" CppmNode.CPPM - Node="192.168.144.3" Timestamp=" 2024 - 10 - 16T21:08:08.936Z"] **Remove:**

NDcP302e: FCS_NTP_EXT.1	
	2024 - 10 - 17T14:40:53.661Z 192.168.144.3 ClearPass 4141570 0 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="REM OVE" Category="Remote Time Server" User="admin" EntityName ="192.168.144.91" CppmNode.CPPM - Node="192.168.144.3" Timesta mp="2024 - 10 - 17T14:37:21.414Z"]

NDcPP30e: FCS_RBG _EXT.1	Auditable Events	None	Additional Conte nt	None

SSH10	
Auditable Events	Failure to establish an SSH session. Successful SSH rekey. **NOTE**
Additional Content	Reason for failure. Non - TOE endpoint of connection (IP Address) .
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**No Matching Encryption Algorithm:** 2024 - 09 - 25T21:48:12.074Z 192.168.144.3 ClearPass 86813 47 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="Failu re" Category="Cipher Mismatch" Description="No matching ciphe r found. Client IP Address : 192.168.144.92:no matching cipher fou nd. Their offer: 3des - cbc [preauth\]" Level="ERROR" Component ="Command Line" CppmNode.CPPM - Node="192.168.144.3" Times tamp="2024 - 09 - 25T21:47:40.780Z"] **No matching MAC Algorithm:** 2024 - 09 - 25T21:48:42.077Z 192.168.144.3 ClearPass 86813 4 8 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="Failu re" Category="MAC Mismatch" Description="No matching MAC fo und. Client IP Address : 192.168.144.92:no matching MAC found . Their offer: hmac - md5 [preauth\]" Level="ERROR" Component

SSH10

	="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 09 - 25T21:48:32.797Z"] **No matching Kex Algorithm:** 2024 - 09 - 25T21:49:42.083Z 192.168.144.3 ClearPass 86813 4 9 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="Failure" Category="Kex Mismatch" Description="No matching Key exchange algorithm found. Unable to negotiate a key exchange method. Client IP Address : 192.168.144.92" Level="ERROR" Component ="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 09 - 25T21:49:24.811Z"] **Failed Authentication (Password):** 2024 - 09 - 29T18:57:59.487Z 192.168.144.3 ClearPass 3041719 54 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="Failure" Category="Login Failed" Description="Failed SSH password login attempt using appadmin account. Last login attempt from the remote host 192.168.144.92" Level="WARN" Component="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp= "2024 - 09 - 29T18:57:49.500Z"] **Failed Authentication (Pubkey):** 2024 - 09 - 03 14: 10:22,302 192.168.144.3 Audit Records 696 1 0 Timestamp=Sep 03 2024 14:09:57.214 PDT,Component=Command Line,Level=WARN,Category=Login Failed,Action=Failure,Description=Failed SSH public key login attempt using appadmin account. Last login attempt from the remote host 192.168.144.91

NDcPP30e: FCS_TLSS_EXT.1

Auditable Events	Failure to establish a TLS Session.
Additional Content	Reason for failure.
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**No Shared Cipher:** Sep 29 19:45:37 192.168.144.3 cppm.apache[3042157]: [Sun Sep 29 19:45:37.709959 2024] [ssl:warn] [pid 3056115] [client 192.168.144.92:59460] SSL Library Error: error:1417A0C1:SSL routines:tls_po

NDcPP30e: FCS_TLSS_EXT.1	
	st_process_client_hello:no shared cipher - - Too restrictive SSLCipherSuite or using DSA server certificate? **Modified Finish Message:** Sep 29 19:23:12 192.168.144.3 cppm.apache[3042157]: [Sun Sep 29 19:23:12.576141 2024] [ssl:warn] [pid 3056116] [client 192.168.144.92:60288] SSL Library Error: error:1416C095:SSL routines:tls_process_finished:digest check failed **Mismatched version:** Sep 29 19:51:08 192.168.144.3 cppm.apache[3042157]: [Sun Sep 29 19:51:08.003759 2024] [ssl:warn] [pid 3057672] [client 192.168.144.92:43922] SSL Library Error: error:14209102:SSL routines:tls_early_post_process_client_hello:unsupported protocol **Wrong Curve:** Sep 29 19:55:19 192.168.144.3 cppm.apache[3042157]: [Sun Sep 29 19:55:19.787308 2024] [ssl:warn] [pid 3056114] [client 192.168.144.92:47412] SSL Library Error: error:140940F4:SSL routines:ssl3_read_bytes:unexpected message

NDcPP30e: FCS_TLSS_EXT.2	
Auditable Events	Failure to authenticate the client
Additional Content	Reason for failure.
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**Invalid Chain / No Client Certificate / Imposter CA:** 2024 - 10 - 29T20:13:04Z 192.168.144.3 cppm.apache[9015]: [Tue Oct 29 20:13:04.017227 2024] [ssl:warn] [pid 231516] [client 192.168.144.241:49645] SSL Library Error: error:14094416:SSL routines:ssl3_read_bytes:sslv3 alert certificate unknown (SSL alert number 46) **Invalid EKU:** 2024 - 09 - 13T19:22:43Z 192.168.144.3 cppm.apache[408487]: [Fri Sep 13 19:22:43.695892 2024] [ssl:error] [pid 809893] [client 192.168.144.92:60622] AH02039: Certificate Verification: Error (26): unsupported certificate purpose **Modified Certificate Verify signature block:** Sep 30 07:36:05 192.168.144.5 cppm.apache[1972815]: [Mon Sep 30 07:36:05.324652 2024] [ssl:error] [pid 197940] SSL Library Er

NDcPP30e: FCS_TLSS_EXT.2

	ror: error:1417B07B:SSL routines:tls_process_cert_verify:bad signature **Invalid reference identifier:** 2024 - 10 - 23T21:42:24.247Z 192.168.144.3 ClearPass 10541 0 - 2 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3002" Common.Auth - Type="" Common.Host - MAC - Address="" Common.Service="TLS - SSO ClearPass Identity Provider" Common.Source="Application" Common.Enforcement - Profiles="TLS - SSO ClearPass Identity Provider Enforcement Profile" Common.Monitor - Mode="Disabled" Common.Login - Status="ACCEPT" Common.Roles="[User Authenticated\]" CppmNode.CPPM - Node="192.168.144.3" Common.System - Posture - Token="UNKNOWN" Common.Request - Id="W00000002 - 05 - 67196da8" Common.Request - Timestamp="2024 - 10 - 23 21:42:01.309+00" Common.Session - Log - Timestamp="2024 - 10 - 23 21:42:01.297+00" Common.Username="client - TOE - 00 - ecdsa" Common.Alerts="WebAuthService: User 'client - TOE - 00 - ecdsa' not present in [Local User Repository\](localhost)\\nUser 'client - TOE - 00 - ecdsa' not present in [Admin User Repository\](localhost)\\nFailed to

NDcPP30e: FCS_AFL.1

Auditable Events	Unsuccessful login attempt limit is met or exceeded
Additional Content	Reason for failure.
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**SSH:** Oct 23 21:49:23 n3000 [] [R:] DEBUG com.avenda.tips.syslog.Syslogger - 1 2024 - 10 - 23T21:49:23.498Z 192.168.144.3 ClearPass 10541 25 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="Failure" Category="Account Locked" Description="Failed SSH login attempts 3 exceeded the configured threshold of 2. SSH access via appadmin account locked for 60 secs.\\nUser: appadmin" Level="ERROR" Component="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T21:48:59.981Z"] **Web UI:** 2024 - 10 - 23T22:01:23.545Z 192.168.144.3 ClearPass 10541 36 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259

NDcPP30e: FCS_AFL.1	
	680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="Non e" Category="Admin User Disable" Description="User IDs disabled by Account - Settings:Attempts - Exceeded for configured thresh old of 1 - testadmin" Level="INFO" Component="User Account S ettings" CppmNode.CPPM - Node="192.168.144.3" Timestamp="20 24 - 10 - 23T22:00:56.036Z"] *The origin of the attempt (IP address) is presented along with au dits shown in FIA_UIA_EXT.1 (for WebUI) and SSH10 (for SSH) for th e failed password attempt.

NDcPP30e: FIA_PMG_ EXT.1	Auditable Events	None	Additional Conte nt	None

NDcPP30e: FIA_UAU.7	Auditable Events	None	Additional Conte nt	None

NDcPP30e: FIA_UIA_EXT.1 & NDcPP30e:FIA_UAU_EXT.2	
Auditable Events	All use of identification and authentication mechanism.
Additional Content	Origin of the attempt (e.g., IP address).
Audit Observed In	**Monitoring > Event**Viewer
Audit Observed In	**Monitoring > Live Monitoring > Access**Tracker
syslog examples(s)	**CLI Login Succes and Failure:** Oct 23 18:11:51 n3000 [] [R:] DEBUG com.avenda.tips.sy slog.Syslogger - 1 2024 - 10 - 23T18:11:51.542Z 192.168.14 4.3 ClearPass 10580 58 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyMana ger" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clea rPass@14823 eventId="3003" Action="None" Category=" Logged in" Description="User: appadmin\\nGroup: Loca l Administrator\\nClient IP Address:" Level="INFO" Com ponent="Command Line" CppmNode.CPPM - Node="192.1 68.144.3" Timestamp="2024 - 10 - 23T18:11:33.004Z"]

Oct 23 18:11:51 n3000 [] [R:] DEBUG com.avenda.tips.syslog.Syslogger - 1 2024 - 10 - 23T18:11:51.541Z 192.168.144.3 ClearPass 10580 57 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="Failure" Category="Console Login Failed" Description="Failed console login using account appadmin" Level="WARN" Component="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T18:11:30.152Z"]

SSH Login Success and Failure:

Oct 23 18:25:51 n3000 [] [R:] DEBUG com.avenda.tips.syslog.Syslogger - 1 2024 - 10 - 23T18:25:51.594Z 192.168.144.3 ClearPass 10580 74 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="None" Category="Logged in" Description="User: testadmin\\nGroup: Local Administrator\\nClient IP Address: 192.168.144.91" Level="INFO" Component="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T18:25:29.015Z"]

For failures, see the audits for SSH10.

Web UI Password Auth Success and Failure:

Oct 23 18:05:51 n3000 [] [R:] DEBUG com.avenda.tips.syslog.Syslogger - 1 2024 - 10 - 23T18:05:51.518Z 192.168.144.3 ClearPass 10580 43 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="None" Category="Logged in" Description="User: admin\\nRole: Super Administrator\\nAuthentication Source: Policy Manager Local Admin Users\\nSession ID: ce5e5cfd8a3e48e18b6af52f14901e0\\nClient IP Address: 192.168.144.241\\nSession Inactive Expiry Time: 359 mins" Level="INFO" Component="Policy Manager UI" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T18:05:39.271Z"]

Oct 23 18:06:21 n3000 [] [R:] DEBUG com.avenda.tips.syslog.Syslogger - 1 2024 - 10 - 23T18:06:21.520Z 192.168.144.3 ClearPass 10580 45 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="None" Category="Login Failed" Description="User: admin\\nClient IP Address: 192.168.144.241" Level="WARN" Component="Policy

NDcPP30e: FIA_UIA_EXT.1 & NDcPP30e:FIA_UAU_EXT.2	
	Manager UI" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T18:06:16.777Z"] **Web UI Cert Auth Success:** Oct 23 14:13:23 2024 - 10 - 23 18: 13:31,314 [R:W0000 0008 - 05 - 67193cc9] DEBUG com.avenda.tips.webauth service.policy.ChainedPolicyClient - ChainedResponse= {statusCode=StatusSuccess, statusMsgList=null, denyAccess=false, roles=[Employee],[User Authenticated], spt=UNKNOWN(100), enfProfiles=TLS - SSO ClearPass Certificate SSO Login Profile1, opaqueAuthState=OpaqueAuthState [hostMac=, service=TLS - SSO ClearPass Certificate SSO Login, username=superadmin, roles=[Employee],[User Authenticated]], enfParams={Application:SSORole=Super Administrator, }} **Web UI Cert Auth Failure:** See NDcPP30e:FCS_TLSS_EXT.2.

NDcPP30e: FIA_X509_EXT.1/Rev (IPsec and TLS)	
Auditable Events	Unsuccessful attempt to validate a certificate.
Additional Content	Reason for Failure
Audit Observed In	**Monitoring > Live Monitoring > Access**Tracker or **Monitoring > Event**Viewer
syslog examples(s)	**IPsec** **'Trust issue' with the certificate, e.g. due to failed path validation:** 2024 - 10 - 02T22:12:12Z 192.168.144.3 cppm.ipsec[14808]: Oct 2 22:12:12 09[CFG] <ipsec - 3016|59> no issuer certificate found for "C=US, ST=MD, L=Catonsville, O=GSS, CN=tl4116x.example.com" **Absence of basic Constraints extension or CA flag not set for a certificate presented as a CA:** TG:2024 - 10 - 02T19:13:21.211918 - 04:00 HN:192.168.144.3 TR:2024 - 10 - 02T23:13:15Z MSG:cppm.ipsec[145176]:<142>2024 - 10 - 02T23:13:15Z 192.168.144.3 cppm.ipsec[145176]: Oct 2 23:13:15 15[IKE] <ipsec - 3016|23> no trusted ECDSA public key found for 'C=US, ST=MD, L=Catonsville, O=GSS, CN=tl4116x.example.com' **Use of an 'expired certificate':**

2024 - 09 - 03 14: 59:22,448 192.168.144.3 Audit Records 731 1 0 Timestamp=Sep 03 2024 14:59:06.567 PDT,Component=ClearPass IPsec Tunnel,Level=ERROR,Category=Tunnel Action,Action=,Description=Tunnel (Remote IP : 192.168.144.91):\nsubject certificate invalid (valid from Aug 22 10:47:18 2024 to Aug 22 10:52:00 2024)

Revoked Certificate:

2024 - 09 - 03 15: 01:22,454 192.168.144.3 Audit Records 740 1 0 Timestamp=Sep 03 2024 15:01:12.609 PDT,Component=ClearPass IPsec Tunnel,Level=ERROR,Category=Tunnel Action,Action=,Description=Tunnel (Remote IP : 192.168.144.91):\ncertificate was revoked on Aug 22 17:48:11 UTC 2024, reason: unspecified

Failure to establish revocation status:

2024 - 09 - 03T15:02:14 - 07:00 192.168.144.3 cppm.ipsec[11192]: Sep 3 15:02:14 14[CFG] <ipsec - 3032|9> ocsp request to http://192.168.144.191 failed

Signature validation failure for any certificate in the certificate path:

2024 - 09 - 03T15:05:27 - 07:00 192.168.144.3 cppm.ipsec[11192]: Sep 3 15:05:27 10[LIB] <15> OpenSSL X.509 parsing failed

TLS/HTTPS

'Trust issue' with the certificate, e.g. due to failed path validation:

2024 - 09 - 27T19:43:56Z 192.168.144.3 cppm.apache[85489]: [Fri Sep 27 19:43:56.291590 2024] [ssl:error] [pid 96153] [client 192.168.144.92:37936] AH02039: Certificate Verification: Error (24): invalid CA certificate

Absence of basicConstraints extension or CA flag not set for a certificate presented as a CA:

2024 - 09 - 27T19:43:56Z 192.168.144.3 cppm.apache[85489]: [Fri Sep 27 19:43:56.291590 2024] [ssl:error] [pid 96153] [client 192.168.144.92:37936] AH02039: Certificate Verification: Error (24): invalid CA certificate

Use of an 'expired certificate':

Sep 27 19:35:15 192.168.144.3 cppm.apache[85489]: [Fri Sep 27 19:35:15.162551 2024] [ssl:error] [pid 86665] [client 192.168.144.92:45664] AH02039: Certificate Verification: Error (10): certificate has expired

Revoked Certificate or Failure to establish revocation status:

2024 - 10 - 23T18:35:48.831Z 192.168.144.3 ClearPass 10580 5 - 2 - 1 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.

	14823"]certificate OCSP verification failed" Common.Error - Code="211" Common.Audit - Posture - Token="UNKNOWN"] **Signature validation failure for any certificate in the certificate path:** 2024 - 09 - 13T19:23:41Z 192.168.144.3 cppm.apache[408487]: [Fri Sep 13 19:23:41.323447 2024] [ssl:error] [pid 809893] SSL Library Error: error:1417B07B:SSL routines:tls_process_cert_verify:bad signature **Addition and Removal of trust anchors:** 2024 - 10 - 23T18:44:51.454Z 192.168.144.3 ClearPass 10580 9 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="ADD" Category="Certificate Trust List" User="testadmin" EntityName="CN=subsubca - rsa,O=GSS,L=Catonsville,ST=MD,C=US" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T18:44:37.567Z"] 2024 - 10 - 23T18:42:51.447Z 192.168.144.3 ClearPass 10580 8 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="REMOVE" Category="Certificate Trust List" User="testadmin" EntityName="CN=subsubca - rsa,O=GSS,L=Catonsville,ST=MD,C=US" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T18:42:26.950Z"] The TOE does not offer the capability for "replacement" of a trust anchor. The only way to do this is to go through the removal and additional processes in sequence. Those audits are presented above. Furthermore, the TOE does not use separate trust stores for IPsec and TLS, so the audits above cover both protocols.

NDcPP30e: FIA_X509_EXT.2	Auditable Events	None	Additional Content	None

NDcPP30e: FIA_X509_EXT.3	Auditable Events	None	Additional Content	None

NDcPP30e: FMT_MOF.1/AutoUpdate	Auditable Events	None	Additional Content	None

NDcPP30e: FMT_MOF.1/Functions	Auditable Events	None	Additional Content	None

NDcPP30e: FMT_MOF.1/ManualUpdate

Auditable Events	Any attempt to initiate a manual update.
Additional Content	None
Audit Observed In	**Monitoring > Event**Viewer
syslog example(s)	See NDcPP30e:FPT_TUD_EXT.1 below

NDcPP30e: FMT_MOF.1/Services	Auditable Events	None	Additional Content	None

NDcPP30e: FMT_MTD.1/CoreData	Auditable Events	None	Additional Content	None

NDcPP30e: FMT_MTD.1/CryptoKeys	Auditable Events	None	Additional Content	None

NDcPP30e:FMT_SMF.1

Auditable Events	All management activities of TSF data.
Additional Content	None

NDcPP30e:FMT_SMF.1

Audit Observed In	**Configuration > Audit**Viewer
syslog example(s)	**Ability to administer the TOE remotely:** See audits for SSH10 and the SSH/Web UI audits for FIA_UIA_EXT.1. **Ability to configure the access banner**: 2024 - 10 - 07T20:23:00.491Z 192.168.144.3 ClearPass 9299 48 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="13001" Action="MODIFY" Category="Cluster - wide Parameter" User="admin" EntityName="Login Banner Text" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 07T20:22:51.291Z"] **Ability to configure the remote session inactivity time before session termination:** 2024 - 10 - 07T20:24:30.497Z 192.168.144.3 ClearPass 9299 50 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="13001" Action="MODIFY" Category="Cluster - wide Parameter" User="admin" EntityName="Admin Session Idle Timeout" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 07T20:24:04.342Z"] **Ability to update the TOE, and to verify the updates using digital signature capability prior to installing those updates:** See audits for FPT_TUD_EXT.1. **Ability to configure audit behavior (e.g. changes to storage locations for audit; changes to behavior when local audit storage space is full):** 2024 - 10 - 23T19:31:12.212Z 192.168.144.3 ClearPass 10580 2 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="MODIFY" Category="Cluster - wide Parameter" User="testadmin" EntityName="Old Audit Records cleanup interval" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T19:30:47.624Z"] **Ability to modify the behavior of the transmission of audit data to an external IT entity,** 2024 - 10 - 23T19:28:05Z 192.168.144.3 cppm.apache[9054]: 192.168.144.241 - - [23/Oct/2024:19:28:05 +0000] "GET /tips/editExtSyslog.action HTTP/1.1" 200 741 "https://n3000/tips/tipsContent.action" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/5

37.36 (KHTML, like Gecko) Chrome/130.0.0.0 Safari/537.36" TLSv1.2 ECDHE - ECDSA - AES256 - GCM - SHA384 3194µs

Ability to manage the cryptographic keys,

See audits for FIA_X509_EXT.3, and also the audits in this same cell for "Ability to manage the trusted public keys database".

WebUI: Cryptographic Functionality is configured entirely by entering CC mode. No other administrator configuration is necessary once in CC mode. For the purposes of the auditable event, the evaluator exited CC mode and changed the TLS version setting manually to generate the required event:

2024 - 10 - 23T19:47:42.268Z 192.168.144.3 ClearPass 10580 5 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="MODIFY" Category="Cluster - wide Parameter" User="testadmin" EntityName="Disable TLSv1.3 support" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T19:47:18.678Z"]

Ability to configure the lifetime for IPsec SAs

2024 - 10 - 23T19:40:24Z 192.168.144.3 cppm.ipsec[9054]: 2024 - 10 - 23 19:40:24,405 DEBUG Platform.IPSec UpdateIPSecConfig Updated IPSec configuration: status 0, output Updating strong Swan IPsec configuration...

Ability to enable or disable automatic checking for updates or automatic updates,

2024 - 10 - 23T19:57:12.304Z 192.168.144.3 ClearPass 10580 9 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="MODIFY" Category="Cluster - wide Parameter" User="testadmin" EntityName="Automatically check for available Software Updates" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 23T19:56:46.754Z"]

Ability to set the time which is used for time - stamps

See FPT_STM_EXT.1.

Ability to configure NTP, Ability to configure the reference identifier for the peer,

See audits for FCS_NTP_EXT.1.

Ability to manage the TOE's trust store and designate X509.v3 certificates as trust anchors,

See audits for FIA_X509_EXT.3.

Ability to generate Certificate Signing Request (CSR) and proc ess CA certificate response

CSR Generated: 2024 - 10 - 23T19:29:42.208Z 192.168.144.3 Clear Pass 10580 1 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion= "6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enter priseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action ="ADD" Category="Certificate Signing Request" User="testadmin" EntityName="Certificate Signing Request" CppmNode.CPPM - No de="192.168.144.3" Timestamp="2024 - 10 - 23T19:29:33.238Z"]

CA Response Processesed, certificate Imported: 2024 - 09 - 09 T13:40:33.701Z 192.168.144.3 ClearPass 47754 475 - 1 - 0 [time Quality tzKnown="1"][origin swVersion="6.11.9.259680" software= "PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"] [clearPass@14823 eventId="3003" Action="Added" Category="S ervice Certificate" Description="Subject: EMAILADDRESS=server - N3000 - 85174 - ecdsa@gossamersec.com,CN=server - N3000 - 85174 - ecdsa.example.com,O=GSS,L=Catonsville,ST=MD,C=US" Level="INFO" Component="Admin UI" CppmNode.CPPM - Node= "192.168.144.3" Timestamp="2024 - 09 - 09T13:40:26.021Z"]

Ability to administer the TOE locally

See the local console audits for FIA_UIA_EXT.1.

Ability to configure the local session inactivity time before session termination or locking,

2024 - 10 - 23T19:54:42.295Z 192.168.144.3 ClearPass 10580 8 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="MODI FY" Category="Cluster - wide Parameter" User="testadmin" Entity Name="CLI Session Idle Timeout" CppmNode.CPPM - Node="192.1 68.144.3" Timestamp="2024 - 10 - 23T19:54:19.388Z"]

Ability to configure the authentication failure parameters for F IA_AFL.1,

WebUI: 2024 - 10 - 23T19:53:12.289Z 192.168.144.3 ClearPass 10 580 6 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.2 59680" software="PolicyManager" ip="192.168.144.3" enterpriseId ="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="MOD IFY" Category="Account Settings" User="testadmin" EntityName= "Admin Users" CppmNode.CPPM - Node="192.168.144.3" Timestam p="2024 - 10 - 23T19:52:45.883Z"]

SSH: 2024 - 10 - 16T18:50:02.409Z 192.168.144.3 ClearPass 2819 6 45 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.2 59680" software="PolicyManager" ip="192.168.144.3" enterpriseId ="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="Suc cess" Category="Account Lockout Configuration" Description="SS

NDcPP30e:FMT_SMF.1

	H lockout details updated to - Lockout count = 2, Unlock time = 60 secs\\nUser: appadmin" Level="INFO" Component="Comma nd Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="20 24 - 10 - 16T18:49:32.794Z"] **Ability to manage the trusted public keys database** 2024 - 10 - 11T01:47:29.297 - 04:00 192.168.144.3 ClearPass 1235 2 15 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.2 59680" software="PolicyManager" ip="192.168.144.3" enterpriseId ="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="REM OVE" Category="SSH Public Key" User="admin" EntityName="roo t@tl41 - 22x" CppmNode.CPPM - Node="192.168.144.3" Times tamp="2024 - 10 - 11T01:47:11.596 - 04:00"] 2024 - 10 - 11T01:47:29.297 - 04:00 192.168.144.3 ClearPass 1235 2 16 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.2 59680" software="PolicyManager" ip="192.168.144.3" enterpriseId ="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3001" Action="ADD" Category="SSH Public Key" User="admin" EntityName="root&#x40 ;tl41 - 22x" CppmNode.CPPM - Node="192.168.144.3" Timestamp=" 2024 - 10 - 11T01:47:13.979 - 04:00"]

NDcPP30e: FMT_SMR.2	Auditable Events	None	Additional Content	None

NDcPP30e: FPT_APW_EXT.1	Auditable Events	None	Additional Content	None

NDcPP30e: FPT_SKP_EXT.1	Auditable Events	None	Additional Content	None

NDcPP30e: FPT_TUD_EXT.1

Auditable Events	Initiation of update; result of the update attempt (success or failure).
Additional Content	None

NDcPP30e: FPT_TUD_EXT.1

Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**Success (Test 1):** 2024 - 11 - 27T18:45:13.686Z [] [R:] DEBUG com.avenda.tips.syslog .Syslogger - 1 2024 - 11 - 27T18:45:13.686Z 192.168.144.3 ClearPa ss 28096 3595 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion ="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" ente rpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Actio n="None" Category="Update Status" Description="User:admin Cli ent IP Address:192.168.144.241 System update using image file C PPM - x86_64 - 20240801 - clearpass - 6.11 - updates - 10 - aruba - 611 - patch.signed.tar." Level="INFO" Component="Update" Cpp mNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 11 - 27T1 8:44:44.475Z"] **Failure (Test 2):** 2024 - 11 - 27T20:43:42.836Z 192.168.144.3 ClearPass 70291 16 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.10.26078 5" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="Non e" Category="Image File Status" Description="User:admin Clien t IP Address:192.168.144.241 Invalid signed file: bad - iamge.tar." Level="ERROR" Component="Image File" CppmNode.CPPM - Nod e="192.168.144.3" Timestamp="2024 - 11 - 27T20:43:33.430Z"]

NDcPP30e: FTA_SSL.3

Auditable Events	The termination of a remote session by the session locking mech anism.
Additional Content	None
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**SSH:** 2024 - 10 - 14T01:55:16.201 - 04:00 192.168.144.3 ClearPass 16620 109472 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9. 259680" software="PolicyManager" ip="192.168.144.3" enterprisel d="1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="No ne" Category="Session Inactivity" Description="Disconnecting SS H session due to session inactivity. Client IP Address : 192.168.14 4.91" Level="WARN" Component="Command Line" CppmNode.CP PM - Node="192.168.144.3" Timestamp="2024 - 10 - 14T01:55:01.96 3 - 04:00"]

NDcPP30e: FTA_SSL.3

	Web UI: 2024 - 10 - 23T21:26:23.399Z 192.168.144.3 ClearPass 10541 14 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="3003" Action="Non e" Category="Session destroyed" Description="Session ID: 2f7b 670f4ac495ab4f0cc341391fea6a\\nClient IP Address: 192.168.144. 241\\nSession Inactive Expiry Time: 5 minutes" Level="INFO" Com ponent="Policy Manager UI" CppmNode.CPPM - Node="192.168.14 4.3" Timestamp="2024 - 10 - 23T21:25:52.937Z"]

NDcPP30e: FTA_SSL.4

Auditable Events	The termination of an interactive session.
Additional Content	None
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**Console Logout:** 2024 - 09 - 30T13:45:05.049Z 192.168.144.3 ClearPass 3041719 1 81 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="Non e" Category="Logged out" Description="User: appadmin\\nClient IP Address:" Level="INFO" Component="Command Line" CppmNo de.CPPM - Node="192.168.144.3" Timestamp="2024 - 09 - 30T13:4 4:44.419Z"] **SSH Logout:** 2024 - 09 - 30T13:07:04.886Z 192.168.144.3 ClearPass 3041719 1 79 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="Non e" Category="Logged out" Description="User: appadmin\\nClient IP Address: 192.168.144.92" Level="INFO" Component="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 09 - 30T13:06:51.854Z"] **WebUI Logout:** 2024 - 09 - 18T16:21:05.577Z 192.168.144.3 ClearPass 1999323 59 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="Non e" Category="Logged out" Description="User: admin\\nRole: Supe

NDcPP30e: FTA_SSL.4	
	r Administrator\\nSession ID: d6f07a46ba03a9afbca637d85d53faef\\nClient IP Address: 192.168.144.241" Level="INFO" Component="Policy Manager UI" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 09 - 18T16:20:46.517Z"]

NDcPP30e: FTA_SSL_EXT.1	
Auditable Events	If lock the session is selected - Any attempts to unlock an interactive session. If terminate the session is selected - The termination of a local session by the session locking mechanism.
Additional Content	None
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	2024 - 09 - 18T15:49:10.315Z 192.168.144.3 ClearPass 9268 86 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259680" software="PolicyManager" ip="192.168.144.3" enterpriseId="1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="None" Category="Session Inactivity" Description="Disconnecting console session due to session inactivity.\\nUser: appadmin" Level="WARN" Component="Command Line" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 09 - 18T15:49:01.616Z"]

NDcPP30e: FTA_TAB.1	Auditable Events	None	Additional Content	None

NDcPP30e: FTP_ITC.1	
Auditable Events	Initiation of the trusted channel. Termination of the trusted channel. Failure of the trusted channel functions.
Additional Content	Identification of the initiator and target of failed trusted channels establishment attempt.
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**Initiation:**

NDcPP30e: FTP_ITC.1	
	2024 - 10 - 02T21:38:45Z 192.168.144.3 cppm.ipsec[14808]: Oct 2 21:38:45 07[IKE] <ipsec - 3016|37> IKE_SA ipsec - 3016[37] established between 192.168.144.3[CN=n3000]...192.168.144.91[C=US, ST=MD, L=Catonsville, O=GSS, CN=tl4116x.example.com] 2024 - 10 - 02T21:38:45Z 192.168.144.3 cppm.ipsec[14808]: Oct 2 21:38:45 07[IKE] <ipsec - 3016|37> CHILD_SA ipsec - 3016{19} established with SPIs c31b52c8_i cf2717ae_o and TS 192.168.144.3/32 === 192.168.144.91/32 **Termination:** 2024 - 10 - 02T21:38:12Z 192.168.144.3 cppm.ipsec[14808]: Oct 2 21:38:12 12[IKE] <ipsec - 3016|35> deleting IKE_SA ipsec - 3016[35] between 192.168.144.3[CN=n3000]...192.168.144.91[C=US, ST=MD, L=Catonsville, O=GSS, CN=tl4116x.example.com] 2024 - 10 - 04T12:34:05.106Z 192.168.144.3 ClearPass 14951 2 93 - 1 - 0 [timeQuality tzKnown="1"][origin swVersion="6.11.9.259 680" software="PolicyManager" ip="192.168.144.3" enterpriseId= "1.3.6.1.4.1.14823"][clearPass@14823 eventId="13003" Action="" Category="Down" Description="Tunnel (Remote IP : 192.168.144.91):\\ nclosing CHILD_SA ipsec - 3016{1} with SPIs c8bab01c_i (572 bytes) c014208a_o (420 bytes) and TS 192.168.144.3/32 === 192.168.144.91/32" Level="WARN" Component="ClearPass IPsec Tunnel" CppmNode.CPPM - Node="192.168.144.3" Timestamp="2024 - 10 - 04T 12:33:37.783Z"] **Failure:** See NDcPP30E:FCS_IPSEC_EXT.1

NDcPP30e: FTP_TRP.1/Admin	
Auditable Events	Initiation of the trusted path. Termination of the trusted path. Failure of the trusted path functions.
Additional Content	Identification of the claimed user identity.
Audit Observed In	**Monitoring > Event**Viewer
syslog examples(s)	**Initiation:** See NDcPP30e:FIA_UIA_EXT.1 for successful WebUI and SSH login **Termination:** See NDcPP30e:FTA_SSL.4 for logout of WebUI and SSH **Failure:**

	Refer to FCS_TLSS_EXT.2 for WebUI cert auth related failures. See the SSH10 row for protocol related failures and NDcPP30e:FIA_UIA_EXT.1 for username/password failures.

Appendix B - Rules

Subtopics

IPsec Traffic Selector Rules
Encryption Rules
Bypass Rules
Drop Rules
Final Rule
Processing Order

IPsec Traffic Selector Rules

The default behavior for IPsec rules is to encrypt all traffic between ClearPass and the VPN peer. Traffic can be separated on a per-port and/or per-protocol level for encrypt, bypass, or drop actions. When implementing IKEv1, only one (1) rule of each type may be created. When implementing IKEv2, a maximum of ten (10) rules can be created for each IPsec tunnel.

Encryption Rules

All outbound packets matching these rules will be encrypted through the IPsec tunnel. When no subordinate actions are specified, this is the default for all traffic between hosts.

Bypass Rules

All outbound packets matching these rules bypass the IPsec tunnel and flow to the remote peer outside of the VPN. This is commonly known as traffic “in the clear”, even though it may already be encrypted.

When using bypass rules, both peers must be configured to bypass selected traffic, or the remote peer does not correctly process the packets.

Drop Rules

All outbound packets matching these rules are dropped.

Final Rule

An implicit rule is created with all IPsec traffic selection that will drop any outbound traffic not processed. This rule creates a behavior where all traffic that should be encrypted or dropped between peers is always blocked when the VPN is inactive. Bypass traffic is unaffected by tunnel status.

Processing Order

IPsec rules are processed using both order and specificity. Order is established beginning by rule position #1 and descending within a rule group.

Specificity is established based on the exactness of a rule to match against. Rules with specific ports and protocols will be evaluated prior to more general rules that apply to all ports or protocols prior to rules that catch “any” traffic.

A series of rules defined in the following scenarios will have the appropriate results:

Encrypt	Bypass	Deny	Result
123	443	22	Encrypt TCP/UDP 123, Bypass TCP/UDP 443, Deny all other traffic
Any	123	22	Bypass TCP/UDP 123, Deny TCP/UDP 2 2, Encrypt all other traffic
22	Any	123	Deny TCP/UDP 123, Encrypt TCP/UDP 2 2, Bypass all other traffic
123	443	Any	Encrypt TCP/UDP 123, Bypass TCP/UDP 443, Deny all other traffic
22	-	-	Encrypt TCP/UDP 22, Deny all other traf fic (Bypass none)

Encrypt	Bypass	Deny	Result
-	22	-	Bypass TCP/UDP 22, Encrypt all other traffic (Deny none)
-	-	22	Deny TCP/UDP 22, Encrypt all other traffic (Bypass none)

Use Cases

This section contains several specific ClearPass Policy Manager use cases. Each use case explains the common usage scenarios, and then describes how to configure Policy Manager for that use case.

- 802.1X Wireless Use Case
- Web Based Authentication for Guests Use Case
- MAC Authentication Use Case
- TACACS+ Use Case
- Single Port Use Case

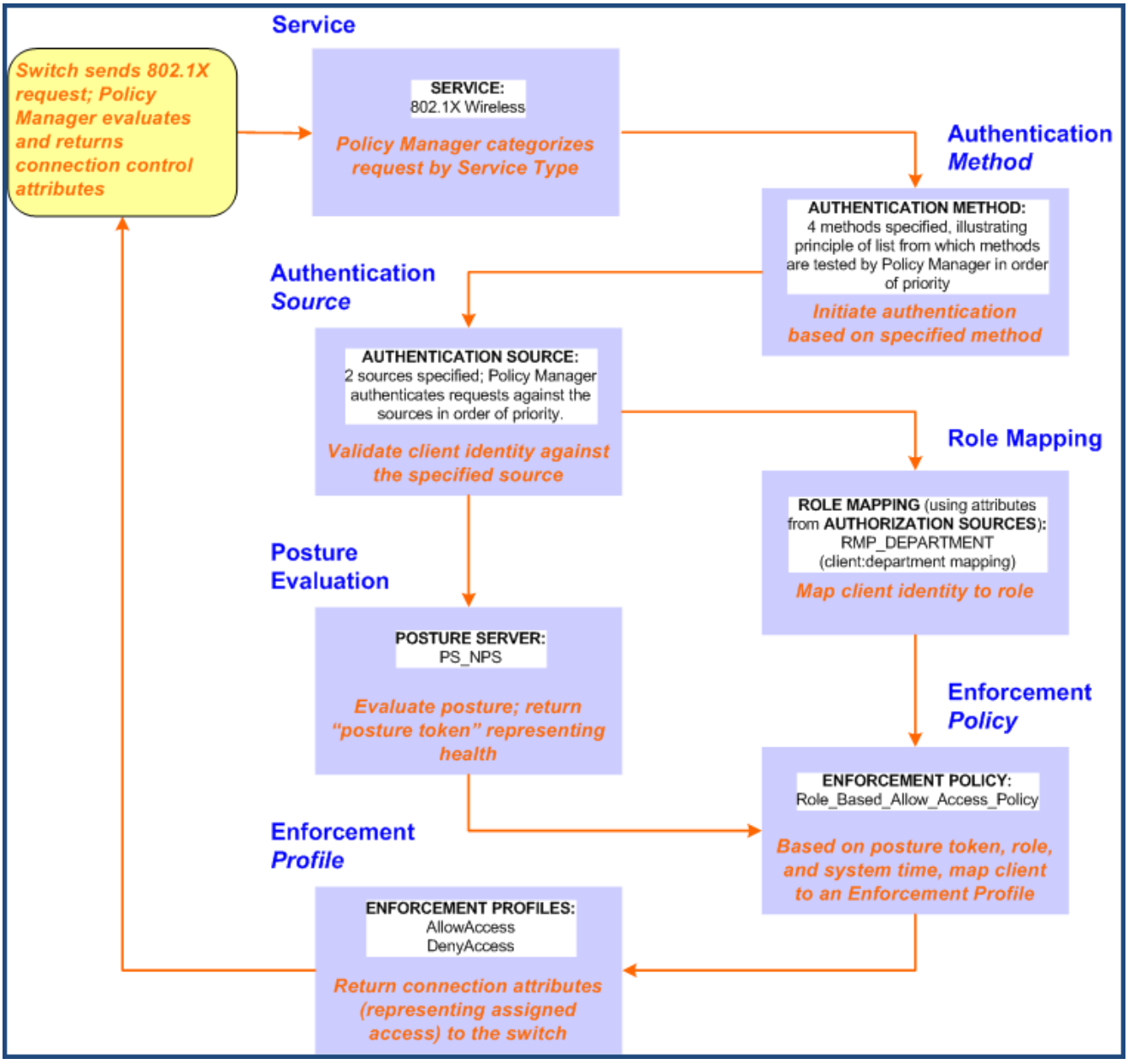
Subtopics

802.1X Wireless Use Case
Web Based Authentication for Guests Use Case
MAC Authentication Use Case
TACACS+ Use Case
Single Port Use Case

802.1X Wireless Use Case

The basic Policy Manager Use Case configures a Policy Manager Service to identify and evaluate an 802.1X request from a user logging into a wireless access device. The following image illustrates the flow of control for this service:

Figure 1. Flow of Control: Basic 802.1X Configuration Use Case



Policy Manager ships with more than twenty preconfigured services. In this use case, you select a service that supports 802.1X wireless requests. Follow the steps below to configure this basic 802.1X service that uses **[EAP FAST]**, one of the preconfigured Policy Manager authentication methods, and **Active Directory Authentication Source (AD)**, an external authentication source within your existing enterprise.

NOTE

Policy Manager fetches attributes used for role mapping from the authorization sources that are associated with the authentication source. In this example, the authentication and authorization source are one and the same.

Policy Manager tests client identity against role-mapping rules, appending any match (multiple roles are acceptable) to the request for use by the enforcement policy. In

the event of a role-mapping failure, Policy Manager assigns a default role. This use case creates the role-mapping policy **RMP_DEPARTMENT** that distinguishes clients by department and the corresponding roles **ROLE_ENGINEERING** and **ROLE_FINANCE**, to which the role-mapping policy maps.

Policy Manager can be configured for a third-party posture server to evaluate client health based on vendor-specific credentials, typically credentials that cannot be evaluated internally by Policy Manager (that is, not in the form of internal posture policies). Currently, Policy Manager supports the following posture server interface: **Microsoft NPS (RADIUS)**.

> **NOTE**
> For purposes of posture evaluation, you can configure a posture policy (internal to Policy Manager), a posture server (external), or an audit server (internal or external). Each of the first three use cases demonstrates one of these options; here, we use the posture server.

Subtopics

Configuring an 802.1X Wireless Service
Creating a New Role-Mapping Policy
Enabling Posture Checking for the 802.1X Service
Assigning an Enforcement Policy

Configuring an 802.1X Wireless Service

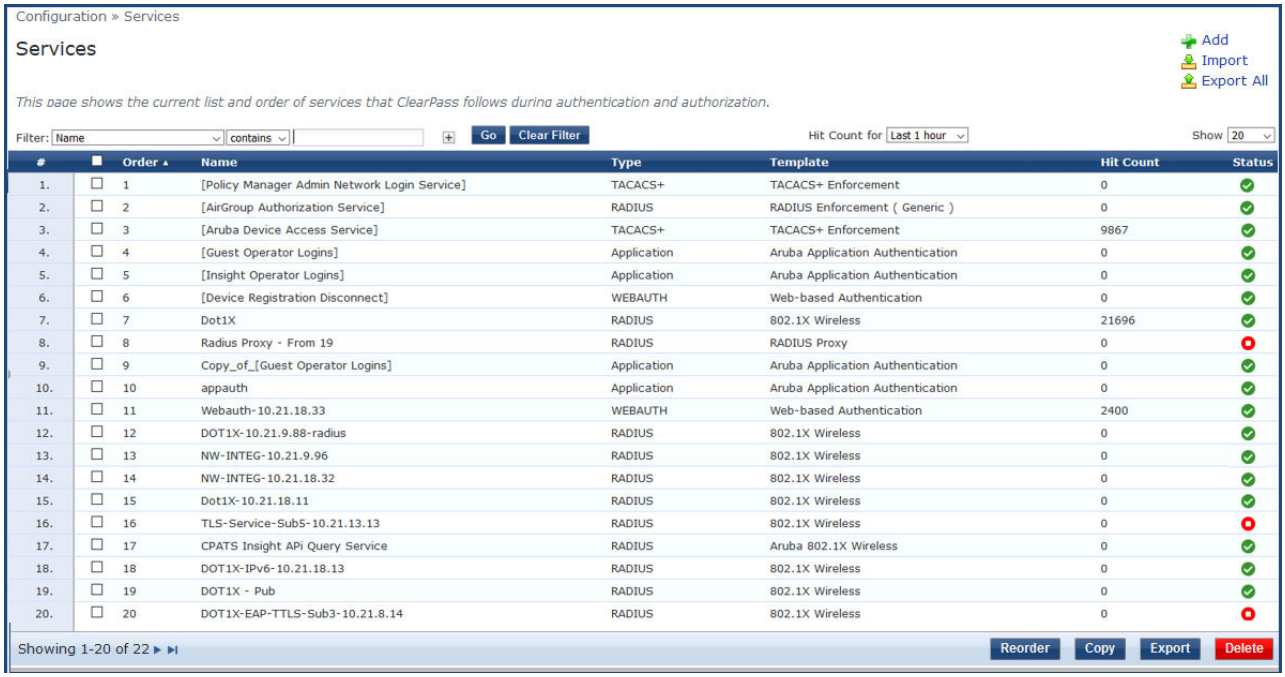
About this task

To create an 802.1X wireless service:

Procedure

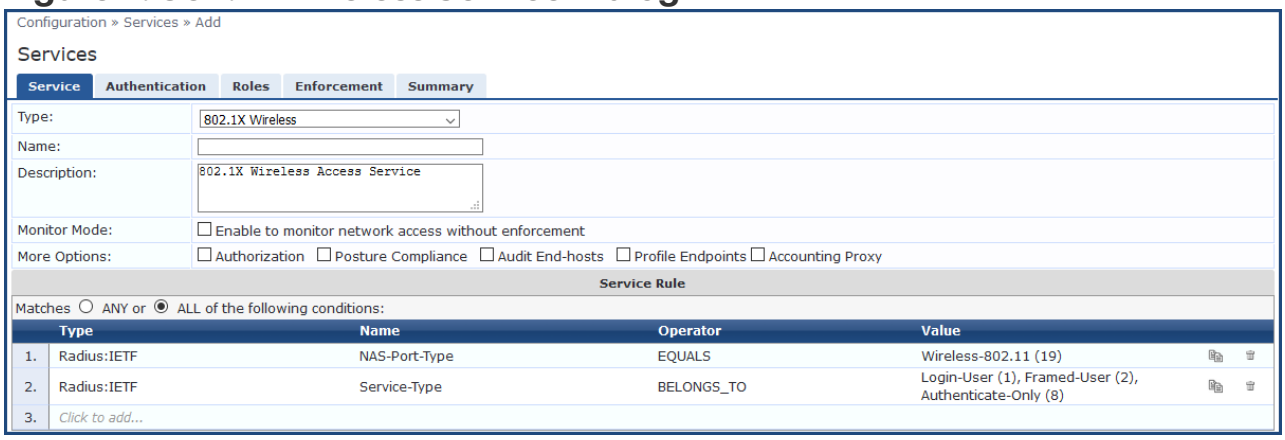
1. Navigate to **Configuration > Services.** The **Services** page opens, which lists the current list and order of services that Policy Manager follows during authentication and authorization:

Figure 1. Services Page



2. Click the Add link to add a service. The **Add Services** page opens. The **Service** tab is displayed; this tab defines basic service information.

Figure 2. 802.1X Wireless Service Dialog



3. Specify the **Service** tab parameters as described in the following table:

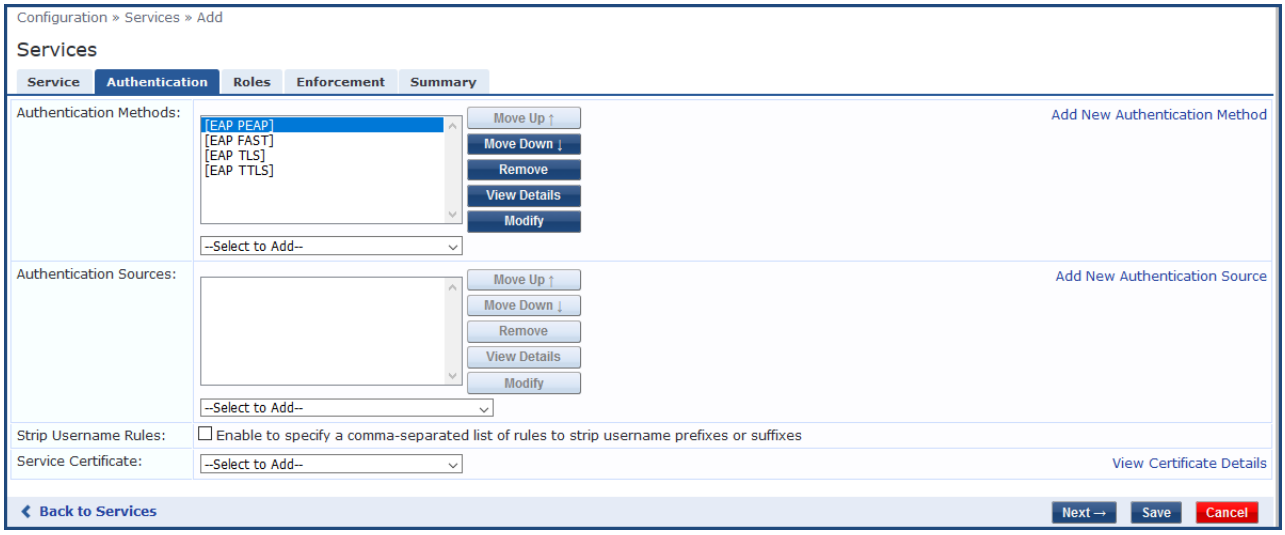
Table 1. Add 802.1X Wireless Service Tab Parameters

Parameter	Action/Description
Type	Select **802.1X Wireless**.
Name	Enter the name of this service.
Description	Policy Manager autofills the *Description* field with "802.1X Wireless Access Service.

Parameter	Action/Description
	" You can change the description if you w ish.
Monitor Mode	Select this check box to monitor network access activity without enforcement.
More Options	Click the **Posture Compliance** check box . The **Posture** tab is added.
Service Rule	
Matches	Select the match condition for this servic e: • Matches ANY • Matches ALL of the following conditio ns
Type	Select **Click to add** to select the service r ule type.
Name	Select the name of the service rule from t he drop - down list.
Operator	Select an appropriate operator from the l ist of operators for the data type of the at tribute.
Value	Enter the value or select the value from t he drop - down list. The value list depend s on the **Type** and **Operator** selected.

4. Click **Next** to view the **Authentication** tab.

Figure 3. Add 802.1X Wireless Service > Authentication Dialog



5. Specify the **Authentication** tab parameters as described in the following table:

Table 2. Add Aruba 802.1X Wireless Service > Authentication Parameters

Parameter	Action/Description
Authentication Methods	Select authentication methods using the **Select to Add** drop - down. For this use case, select **[EAP FAST]**. For more information, see Adding and Modifying Authentication Methods .
Authentication Sources	Specify one or more instances of the authentication sources using the **Select to Add** drop - down. • Admin User Repository • Denylist Repository • Endpoints Repository • Guest Device Repository • Guest User Repository • Insight Repository • Local User Repository • Zone Cache Repository • Onboard Devices Repository

Parameter	Action/Description
	• Social Login Repository • Time Source For more information, see Adding and Configuring Authentication Sources .
Strip Username Rules	Select the check box to preprocess the username (that is, remove prefixes and suffixes) before authenticating and authorizing against the authentication source.
Service Certificate	Optionally, select the Service Certificate. For more information, refer to Service Certificates.

Creating a New Role-Mapping Policy

About this task

To create a new Role-Mapping policy:

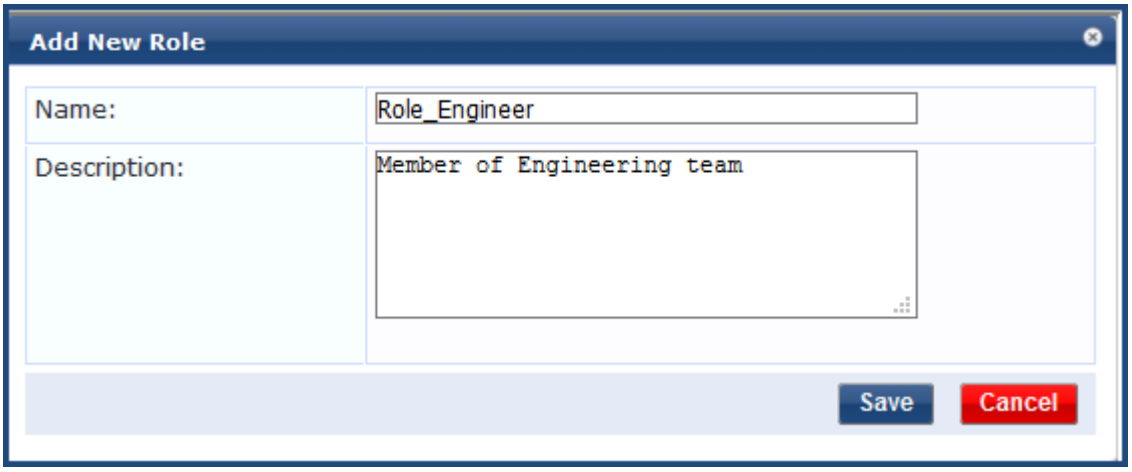
Procedure

1. Select the **Roles** tab, then click the **Add New Role Mapping Policy** link.

Figure 1. Role-Mapping Policy for 802.1X Wireless Use Case

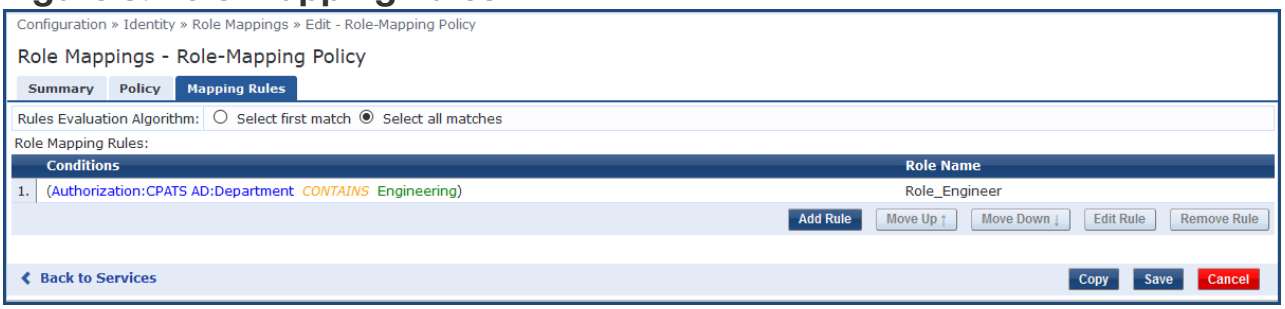
2. From the **Policy** tab, enter the **Policy Name**, for example, *Role-Mapping Policy*.
3. Click the **Add New Role** link.

Figure 2. Adding New Role to Role-Mapping Policy



4. Click **Save**. You return to the **Role Mappings** page. The message "*Role Role_Engineer added successfully*" is displayed.
5. To create rules to map client identity to a role, select the **Mapping Rules** tab.

Figure 3. Role-Mapping Rules



6. To add additional rules, click the **Add Rule** button. The **Rules Editor** dialog opens.
7. Select the **Select all matches** radio button.
8. Match the conditions with the role name.
9. Upon completion of each rule, click the **Save** button in the **Rules Editor**.
0. Click the **Save** button.

Enabling Posture Checking for the 802.1X Service

About this task

To enable posture checking for this service:

The **Posture** tab is not enabled by default.

Procedure

1. From the **Service** tab, select the **More Options** > **Posture Compliance** check box. The **Posture** tab is displayed.

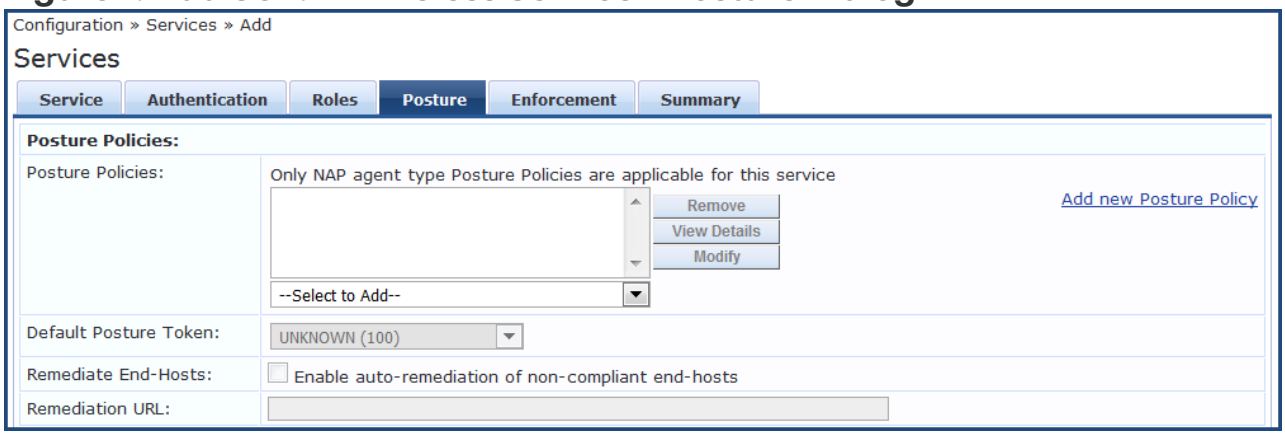
   You can enable the posture checking for this kind of service if you deploy any of the following:

   - ClearPass Policy Manager in a Microsoft Network Access Protection (NAP) environment
   - Cisco Network Admission Control (NAC) Framework environment
   - Aruba hosted captive portal that performs posture checks through a dissolvable agent

2. Select the **Posture** tab.

   The **Posture** dialog opens:

Figure 1. Add 802.1X Wireless Service > Posture Dialog



3. Specify the 802.1X Wireless Service **Posture** parameters as described in Table 3:

Table 1. Add 802.1X Wireless Service > Posture Parameters

Parameter	Action/Description
Posture Policies	
Posture Policies	Select the posture policy from the **Select to Add** drop - down list. If you do not have any preconfigured pos ture policies, click **Add New Posture Poli cy** to create a new posture policy. For mo re information on configuring posture pol

Parameter	Action/Description
	ices, see Configuring Posture Policy Agents and Hosts . **NOTE** Only NAP agent type posture policies are applicable for this service.
Default Posture Token	Select the default posture token from the drop - down list.
Remediate End - Hosts	To perform remediation action, when a client is quarantined, select the **Enable auto - remediation of non - compliant end - hosts** check box.
Remediation URL	To perform the remediation, enter the web link of a server resource.

For more information on configuring posture polices, see Configuring Posture Policy Agents and Hosts .

Assigning an Enforcement Policy

About this task

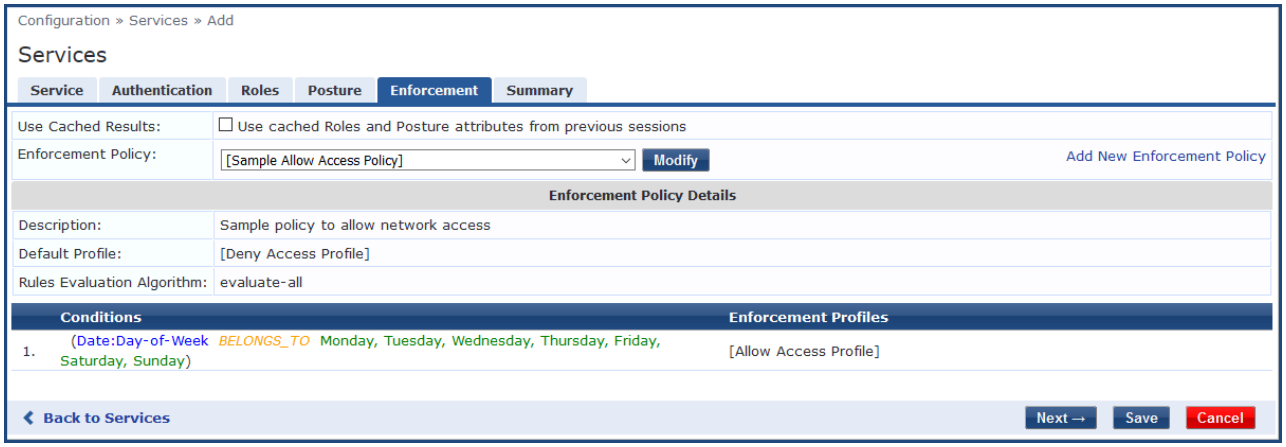
Enforcement policies contain dictionary-based rules for evaluation of Role, Posture Tokens, and System Time to evaluation profiles. Policy Manager applies all matching enforcement profiles to the request. In the case of no match, Policy Manager assigns a default enforcement profile.

To assign an enforcement policy to the 802.1X wireless service:

Procedure

1. Select the **Enforcement** tab. The **Enforcement** dialog opens:

   Enforcement Policy Navigation and Settings



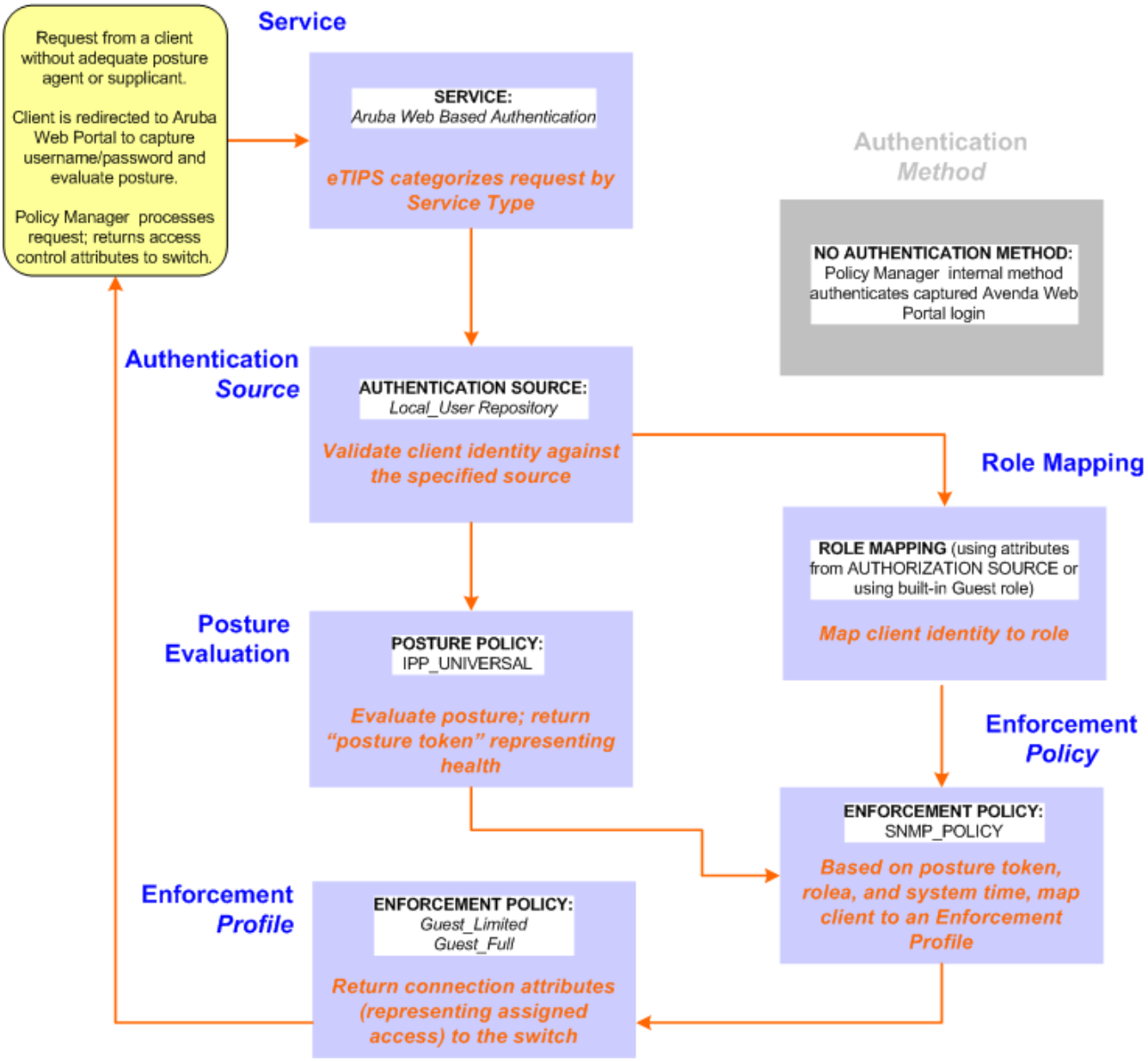
2. From the **Enforcement** tab, select the **Enforcement Policy** or click the **Add New Enforcement Policy** to create a new enforcement policy. For instructions about how to build an enforcement policy, refer to Configuring Enforcement Policies.

3. Save the service.

Web Based Authentication for Guests Use Case

This Service supports known Guests with inadequate 802.1X supplicants or posture agents. The following figure illustrates the overall flow of control for this Policy Manager service.

Figure 1. Flow-of-Control of Web-Based Authentication for Guests



Configuring a Service

About this task

Perform the following steps to configure Policy Manager for WebAuth-based Guest access.

Procedure

1. Prepare the switch to pre-process WebAuth requests for the Policy Manager Aruba WebAuth service.

   Refer to your Network Access Device documentation to configure the switch such that it redirects HTTP requests to the Aruba Guest Portal, which captures username and password and optionally launches an agent that returns posture data.

2. Create a WebAuth-based Service.

   a. Navigate to **Configuration > Services**and select Add Service

   b. Name the Service and select a pre-configured Service Type:

      - **Service** (tab) >
      - **Type** (selector): Web-Based Authentication >
      - **Name/Description** (freeform) >

   c. Upon completion, click **Next**.

3. Configure a Posture Policy. Use this procedure to create and add the posture policy IPP_Universal , which (as you will configure it in this Use Case, checks any Windows XP clients to verify the most current Service Pack). Select the local Policy Manager database:

   a. **Authentication** (tab) >

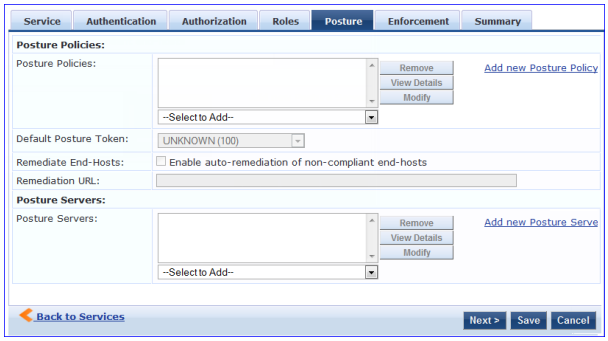
   b. **Sources**(**Select** from the list): **[Local User Repository]** >

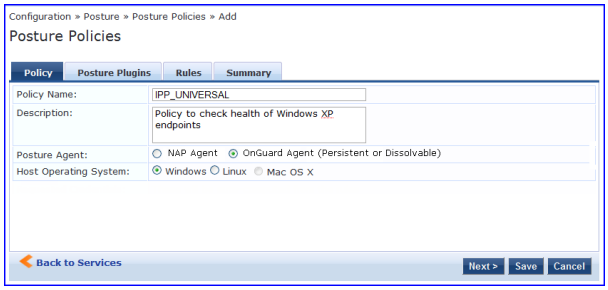
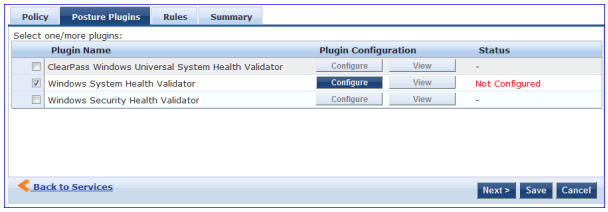
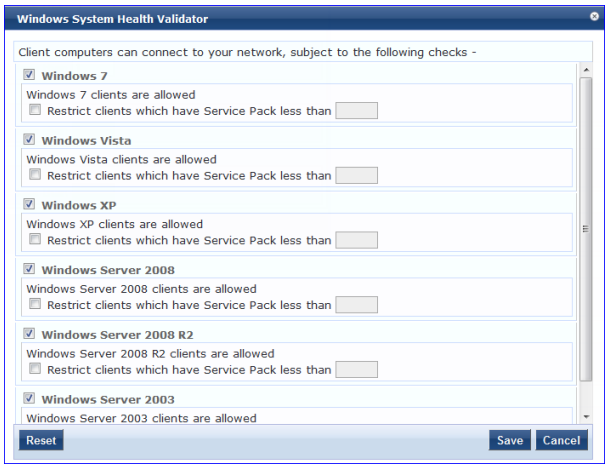
   c. Select **Enable to specify a comma-separated list of rules to strip username prefixes or suffixes** (check box) >

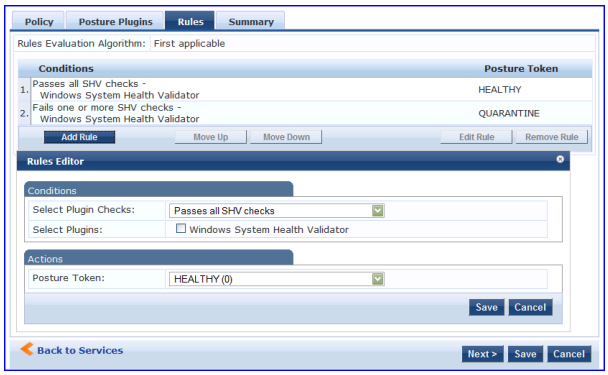
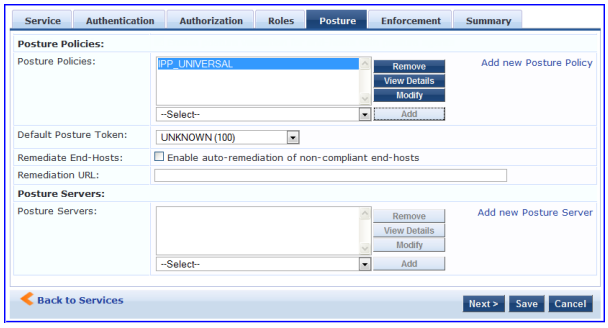
   d. Enter an example of preceding or following separators (if any), with the phrase "user" representing the username to be returned. For authentication, Policy Manager strips the specified separators and any paths or domains beyond them.

   e. Upon completion, click **Next** (until you reach Enforcement Policy).

Table 1. Posture Policy Navigation and Settings

Navigation	Setting
Create a Posture Policy: • **Posture** (tab) > • Enable **Validation Check** (check box) > • **Add new Internal Policy** (link) >	

Navigation	Setting
Name the Posture Policy and specify a general class of operating system: • **Policy** (tab) > • **Policy Name** (freeform): IPP_Universal • **Host Operating System** (radio buttons): **Windows** > • When finished working in the **Policy** tab, click **Next** to open the Posture Plugins tab	
Select a Validator: • **Posture Plugins** (tab) > • Enable **Windows Health System Validator** > • **Configure** (button) >	
Configure the Validator: • **Windows System Health Validator** (popup) > • Select the desired operating system. • Enable Service Pack for levels the desired Windows version. • **Save** (button) > • When finished working in the **Posture Plugin** tab click **Next** to move to the Rules tab)	

Navigation	Setting
Set rules to correlate validation results with posture tokens: • **Rules** (tab) > • **Add Rule** (button opens popup) > • **Rules Editor** (popup) > • **Conditions/ Actions:** match Conditi ons (Select Plugin/ Select Plugin che cks) to Actions (Posture Token)> • In the **Rules Editor,** upon completion of each rule, click the **Save** button > • When finished working in the **Rules** tab, click the **Next** button.	
Add the new Posture Policy to the Servi ce: Back in **Posture** (tab) > **Internal Policies** (selector): **IPP_Univers al**, then click the **Add** button	

The following fields deserve special mention:

- **Default Posture Token.** Value of the posture token to use if health status is not available.
- **Remediate End-Hosts.** When a client does not pass posture evaluation, redirect to the indicated server for remediation.
- **Remediation URL.** URL of remediation server.

4. Create an Enforcement Policy. Because this Use Case assumes the Guest role, and the Aruba Web Portal agent has returned a posture token, it does not require configuration of Role Mapping or Posture Evaluation.
   - **Enforcement** (tab) >
   - Enforcement Policy (selector): **SNMP_POLICY**

- Upon completion, click **Save**.

> **NOTE**
>
> The SNMP_POLICY selected in this step provides full guest access to a Role of [Guest] with a Posture of Healthy, and limited guest access.

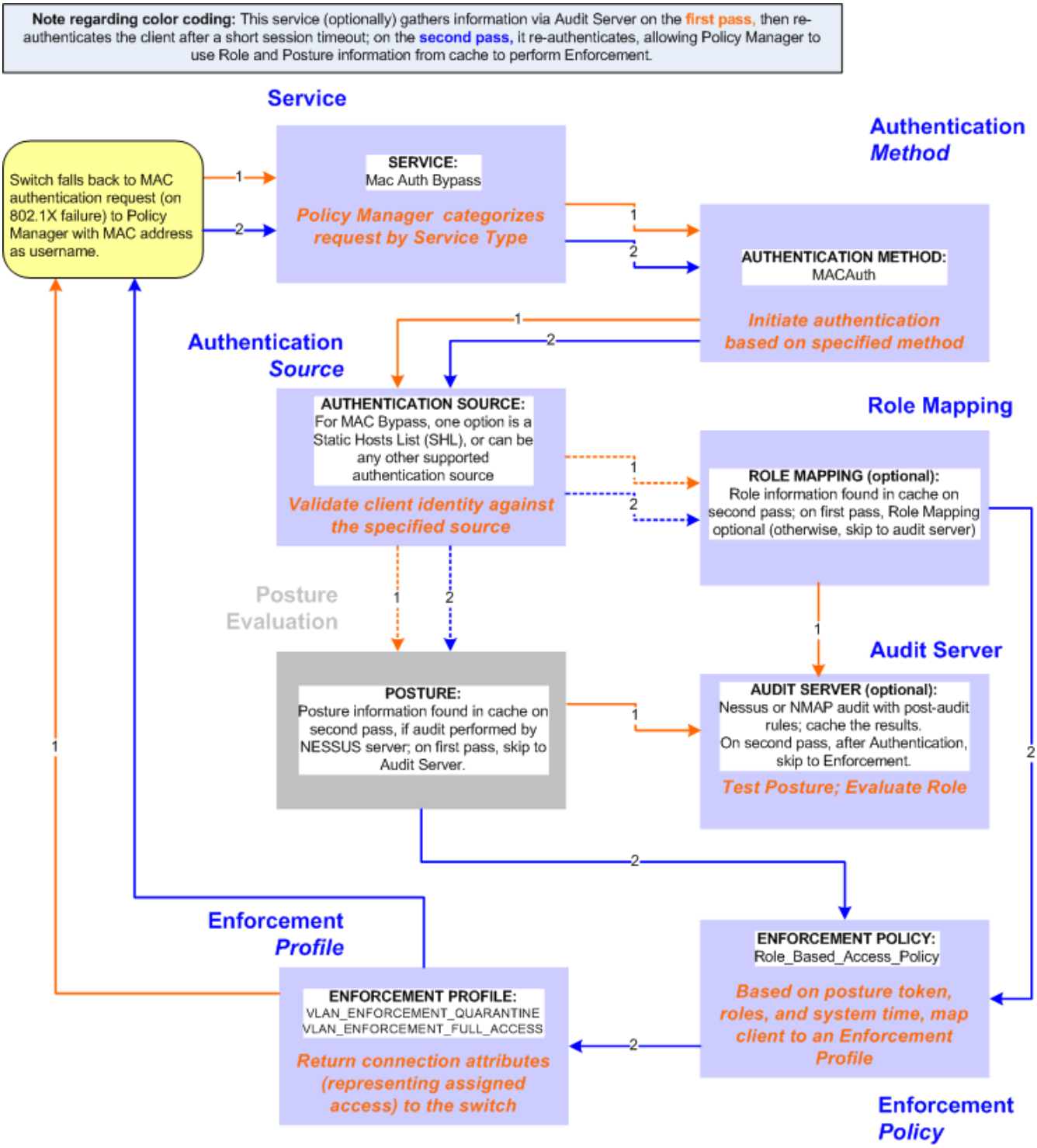
5. Save the Service. Click **Save**. The Service now appears at the bottom of the **Services** list.

MAC Authentication Use Case

This service supports Network Devices, such as printers or hand-helds. In this service, an audit is initiated on receiving the first MAC Authentication request. A subsequent MAC Authentication request (triggered after the audit, or triggered after a short session timeout) uses the cached results from the audit to determine the posture and role(s) for the device.

The following diagram illustrates the overall flow of control for this Policy Manager service.

Figure 1. Flow-of-Control of MAC Authentication for Network Devices



Configuring the Service

About this task

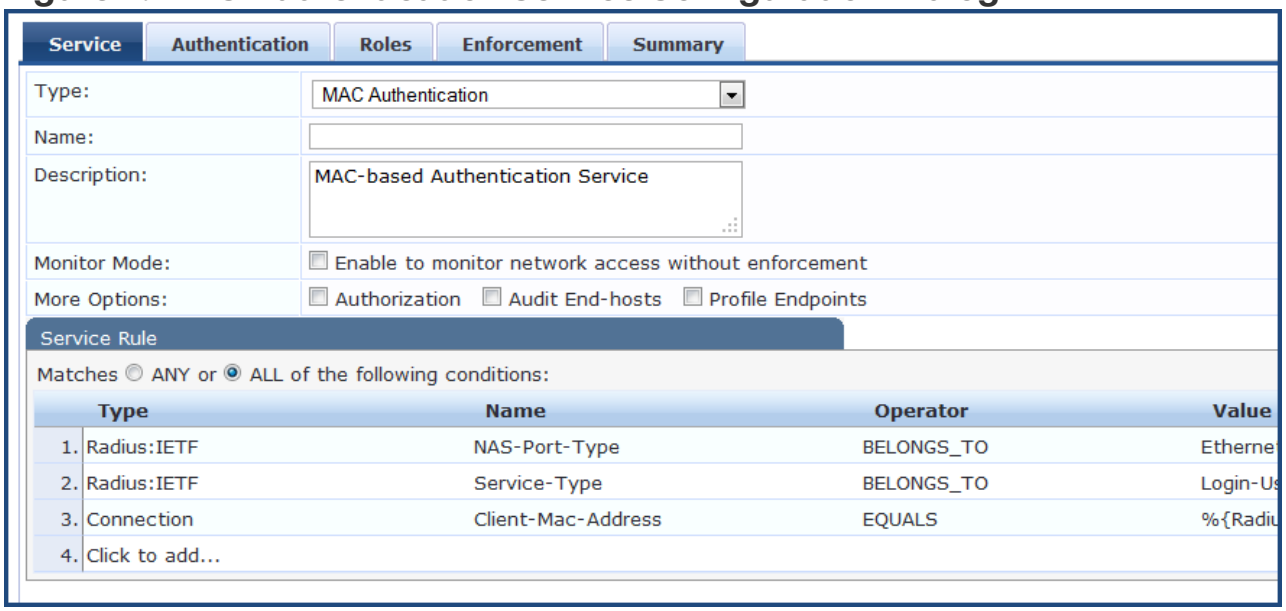
To configure Policy Manager for MAC-based network device access:

Procedure

1. First create a MAC Authentication Service by navigating to **Configuration > Services**. The **Services** page opens.

2. Click the **Add** link. The **Add Services** dialog opens.

Figure 2. MAC Authentication Service Configuration Dialog



a. Create a new Service:

- **Services** >
- **Add Service** (link) >

b. Name the Service and select a pre-configured Service Type:

- **Service** (tab) >
- **Type** (selector): **MAC Authentication** >
- **Name/Description** (freeform) >
- Upon completion, click **Next** to configure Authentication

3. Set up Authentication. You can select any type of authentication/authorization source for a MAC Authentication service. Only a static host list of type **MAC Address List** or **MAC Address Regular Expression** shows up in the list of authentication sources (of type **Static Host List**). For more information on static host list, see Managing Static Host Lists . You can also select any other supported type of authentication source.

   Select an Authentication Method and two authentication sources—one of type **Static Host List** and the other of type **Generic LDAP** server (that you have already configured in Policy Manager):

   - **Authentication** (tab) >

- **Methods** (This method is automatically selected for this type of service): **[MAC AUTH]** >
- **Add** >
- **Sources** (**Select** drop-down list): **Handhelds [Static Host List]** and Policy Manager Clients White List [Generic LDAP] >
- **Add** >
- Upon completion, **Next** (to Audit)

4. Configure an Audit Server.

   This step is optional if no Role Mapping Policy is provided, or if you want to establish health or roles using an audit. For more information, see Configuring Audit Servers . An audit server determines health by performing a detailed system and health vulnerability analysis (Nessus). You can also configure the audit server (Nmap or Nessus) with post-audit rules that enable Policy Manager to determine client identity. Configure the Audit Server:

   - **Audit** (tab) >
   - **Audit End Hosts** (enable) >
   - **Audit Server** (selector): **NMAP**
   - **Trigger Conditions** (radio button): **For MAC authentication requests**
   - **Reauthenticate client** (check box): **Enable**

   Upon completion of the audit, Policy Manager caches Role (Nmap and Nessus) and Posture (Nessus), then resets the connection (or the switch reauthenticates after a short session timeout), triggering a new request, which follows the same path until it reaches Role Mapping/Posture/Audit; this appends cached information for this client to the request for passing to Enforcement.

5. Select the Enforcement Policy Sample_Allow_Access_Policy:

   - **Enforcement** (tab) >
   - **Enforcement** (tab) >
   - **Use Cached Results** (check box): Select **Use cached Roles and Posture attributes from previous sessions** >
   - **Enforcement Policy** (selector): UnmanagedClientPolicy
   - When you are finished with your work in this tab, click **Save**.

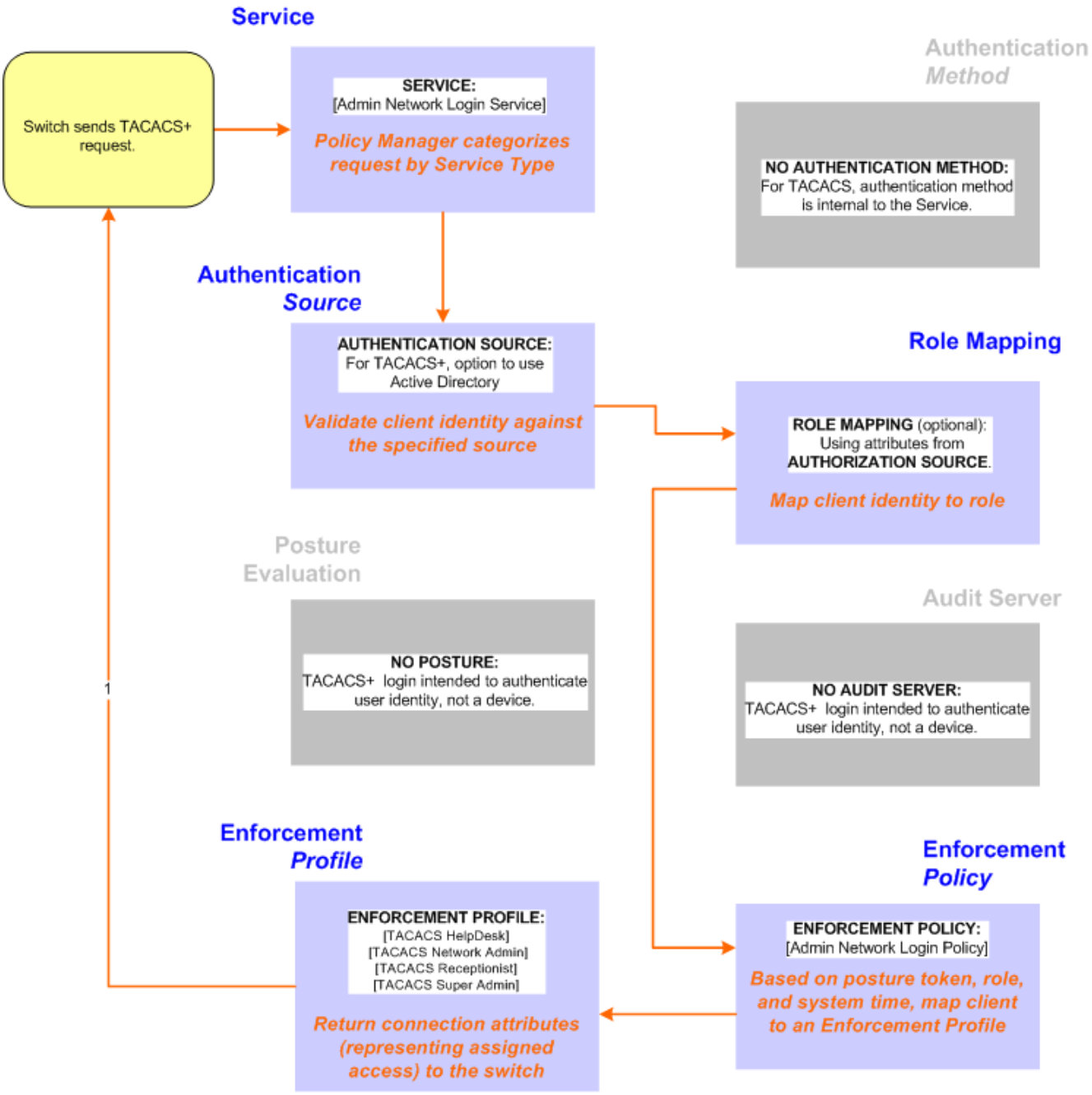
Unlike the 802.1X service, which uses the same Enforcement Policy (but uses an explicit Role Mapping Policy to assess Role), in this use case, Policy Manager applies post-audit rules against attributes captured by the Audit server to infer Role(s).

6. Click **Save**. The service now appears at the bottom of the **Services** list.

TACACS+ Use Case

This Service supports Administrator connections to Network Access Devices via TACACS+. The following image illustrates the overall flow of control for this Policy Manager Service.

Figure 1. Administrator connections to Network Access Devices via TACACS+



Configuring the Service

About this task

Perform the following steps to configure Policy Manager for TACACS+-based access:

Procedure

1. Navigate to **Configuration > Services.**
2. Click the Add icon to add a service. The **Configuration > Services > Add** window opens.

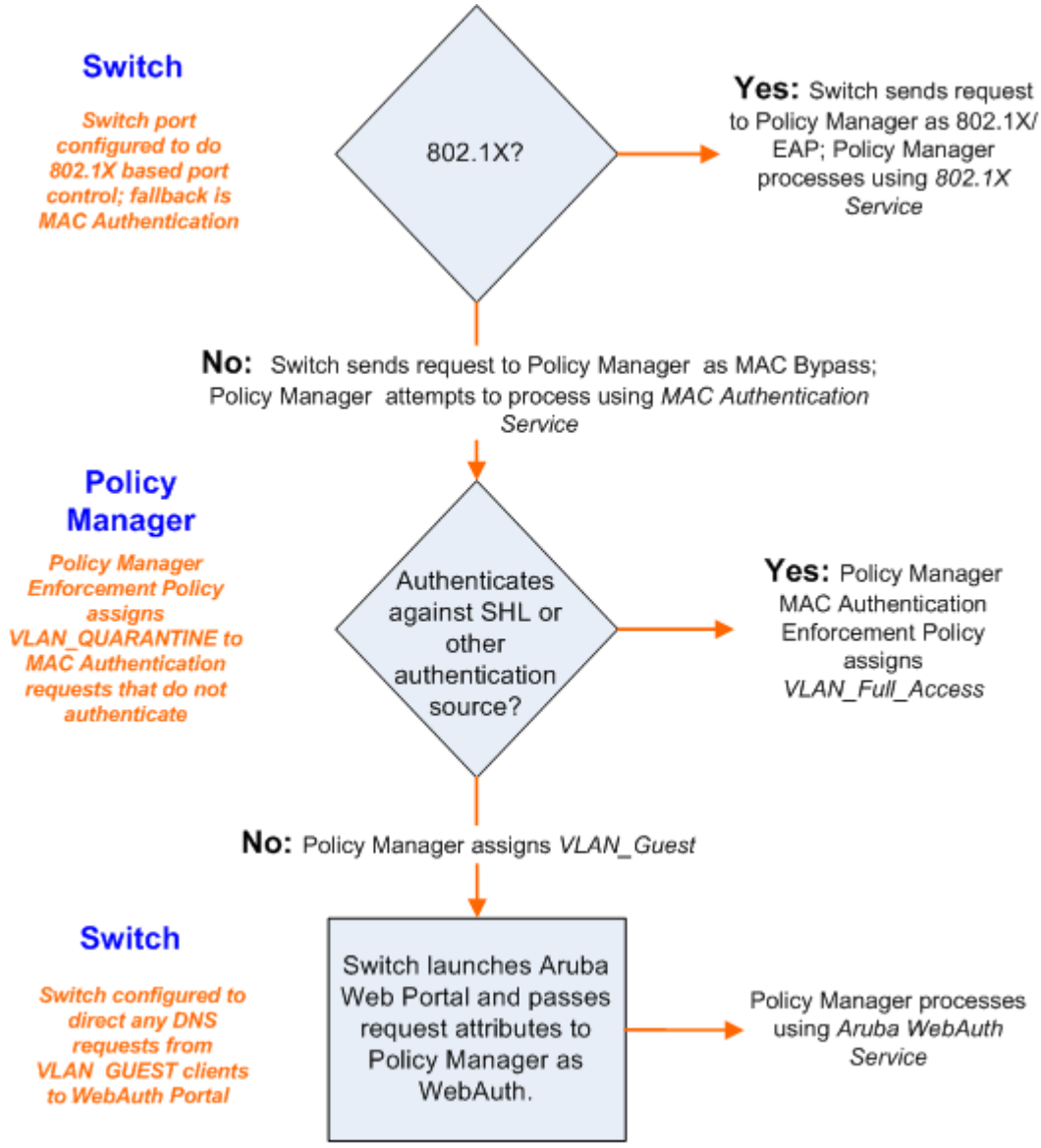
3. If it is not already selected, click the **Service** tab and define basic service information.
   a. Enter a name for the service in the **Name** field.
   b. Click the **Type** drop-down list and select the preconfigured service type that matches your Policy Manager Admin Network Login Service.
   c. Click **Next** to display the **Authentication** tab.
4. Define the Authentication settings for the service. Authentication methods can be left to their default values, as the Policy Manager TACACS+ service authenticates TACACS+ requests internally.
   a. In the **Authentication Sources** section, click the **Select to Add** drop-down list.
   b. Select **AD (Active Directory)**. For this use case example, Network Access Device authentication data will be stored in the Active Directory.
5. Click the **Enforcement** tab and select an Enforcement Policy.
   a. Click the Enforcement Policy drop-down list and select the Enforcement Policy **[Admin Network Login Policy]** that distinguishes the two allowed roles (**Net Admin Limited** and **Device SuperAdmin)**.
6. Click **Save.** The Service now appears at the bottom of the **Services** list.

Single Port Use Case

This Service supports all three types of connections on a single port.

The following figure illustrates both the overall flow of control for this hybrid service, in which complementary switch and Policy Manager configurations allow all three types of connections on a single port:

Figure 1. Flow of the Multiple Protocol Per Port Case



Rules Editing and Namespaces

The Policy Manager administration User Interface allows you to create different types of objects:

- Service rules
- Role mapping policies
- Internal user policies
- Enforcement policies
- Enforcement profiles

- Post-audit rules
- Proxy attribute pruning rules
- Filters for Access Tracker and activity reports
- Attributes editing for policy simulation

When editing all these elements, you are presented with a tabular interface with the same column headers:

- **Type**: Type is the namespace from which these attributes are defined. This is a drop-down list that contains namespaces defined in the system for the current editing context.
- **Name**: Name is the name of the attribute. This is a drop-down list with the names of the attributes present in the namespace.
- **Operator**: Operator is a list of operators appropriate for the data type of the attribute. The drop-down list shows the operators appropriate for data type on the left (that is, the attribute).
- **Value**: The value is the value of the attribute. Again, depending on the data type of the attribute, the value field can be a free-form one-line edit box, a free-form multi-line edit box, a drop-down list containing pre-defined values (enumerated types), or a time or date widget.

  In some editing interfaces (for example, enforcement profile and policy simulation attribute editing interfaces) the operator does not change; it is always the EQUALS operator. Providing a uniform tabular interface to edit all these elements enables you to use the same steps while configuring these elements. Also, providing a context-sensitive editing experience (for names, operators and values) takes the guess-work out of configuring these elements.

The following sections describe namespaces, variables, and operators:

- Namespaces
- Variables
- Operators

Subtopics

Namespaces
Variables
Operators
OnGuard Agents

Namespaces

Multiple namespaces are displayed in the rules-editing interfaces, depending upon what you are editing.

For detailed information about the available namespaces, see the following topics:

- Application Namespace
- Audit Namespaces
- Authentication Namespaces
- Authorization Namespaces
- Certificate Namespaces
- Connection Namespaces
- Date Namespaces
- Device Namespaces
- Endpoint Namespaces
- Guest User Namespaces
- Host Namespace
- Local User Namespaces
- RADIUS Namespaces
- Tacacs Namespaces
- Tips Namespaces

Subtopics

Application Namespace
Audit Namespaces
Authentication Namespaces
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Date Namespaces
Device Namespaces
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Local User Namespaces

RADIUS Namespaces
TACACS Namespaces
Tips Namespaces

Application Namespace

The Application namespace has one name attribute. This attribute is an enumerated type currently containing the following string values:

- Guest
- Insight
- PolicyManager
- Onboard
- Policy Manager

The **Application:ClearPass** namespace has the following string values available for the *Name* field:

- Onboard-Request-type
  - Value is **New** when a user onboards the device for the first time.
  - Value is **Renewal** a user reprovisions the device
- AssertionConsumerUrl
- Configuration-Profile-ID
- Device-Compromised
- Device-ICCID
- Device-IMEI
- Device-MAC
- Device-MDM-Managed
- Device-Name
- Device-OS
- Device-PRODUCT
- Device-SERIAL

- Device-UDID
- Device-Version
- IDDP-COOKIE-TIMEOUT-MINS
- IDPURL
- MDM-Data-Roaming
- MDM-Voice-Roaming
- Onboard-Max-Devices
- Page-Name
- Provisioning-Settings-ID
- SAMLRequest
- SAMLResponse
- Session-Timeout
- User-Email-Address

Audit Namespaces

The dictionaries in the audit namespace are provided with Policy Manager. The Audit namespace has the notation Vendor:*Audit,* where Vendor is the name of the company that has defined attributes in the dictionary. Examples of dictionaries in the audit namespace are *AvendaSystems:Audit* or *Qualys:Audit*. The Audit namespace appears when editing post-audit rules (see Audit Service Flow Control for more information). The *Avenda Systems:Audit* namespace appears when editing post-audit rules for NESSUS and NMap audit servers.

The following table displays the Audit Namespace attributes:

Table 1. Audit Namespace Attributes

Attribute Name	Values
Audit - Status	• AUDIT_ERROR • AUDIT_INPROGRESS • AUDIT_SUCCESS

Attribute Name	Values
Device - Type	Type of device returned by an NMap port scan.
Output - Msgs	The output message returned by Nessus plugin after a vulnerability scan.
Network - Apps	String representation of the open network ports (HTTP, Telnet, etc.).
Mac - Vendor	Vendor associated with the MAC address of the host.
OS - Info	OS information string returned by NMap.
Open - Ports	The port numbers of open applications on the h ost.

Authentication Namespaces

The authentication namespace can be used in role mapping policies to define roles based on the type of authentication method used or the status of the authentication.

Authentication Namespace Editing Context

The following table describes the **Authentication Namespace Attributes** parameters:

Table 1. Authentication Namespace Attributes

Attribute Name	Values
Error Code	Enter error code value.
Full - Username	Enter full username.
Full - Username - Normalized	Enter full username.
Hotspot2 - Remediation - URL	Enter the remediation URL.
InnerMethod	• Authorize • CHAP • EAP - AKA

Attribute Name	Values
	• EAP - AKA - PRIME • EAP - GTC • EAP - MD5 • EAP - MSCHAPv2 • EAP - SIM • EAP - TLS • Anonymous - EAP - TLS • EAP - PWD • EAP - LEAP • MSCHAP • PAP • MAC - AUTH **NOTE** The **EAP - MD5** authentication type is not supported if you use the ClearPass Policy Manager in FIPS mode.
MFA - Status	• SUCCESS • FAIL • IN_PROGRESS • NOT_REQUIRED
MacAuth	• NotApplicable • KnownClient • UnknownClient
NetBIOS - Name	Enter NetBIOS name.
OuterMethod	• CHAP

Attribute Name	Values
	• EAP - FAST • EAP - MD5 • EAP - PEAP • EAP - TLS • EAP - TTLS • MSCHAP • PAP • TEAP **NOTE** The **EAP - MD5** authentication type is not supported if you use the ClearPass Policy Manager in FIPS mode.
Phase1PAC	• **None**: No Protected Access Credential (PAC) was used to establish the outer tunnel in the EAP - FAST authentication method • **Tunnel**: A tunnel PAC was used to establish the outer tunnel in the EAP - FAST authentication method • **Machine**: A machine PAC was used to establish the outer tunnel in the EAP - FAST authentication method; machine PAC is used for machine authentication (see EAP - FAST in Adding and Modifying Authentication Methods).
Phase2PAC	• **None**: No Protected Access Credential (PAC) was used instead of an inner method handshake in the EAP - FAST authentication method • **UserAuthPAC** : A user authentication PAC was used instead of the user authe

Attribute Name	Values
	ntication inner method handshake in th e EAP - FAST authentication method • **PosturePAC**: A posture PAC was used in stead of the posture credential handsha ke in the EAP - FAST authentication met hod
Posture	• **Capable** : The client is capable of provi ding posture credentials • **Collected**: Posture credentials were col lected from the client • **Not - Capable**: The client is not capabl e of providing posture credentials • **Unknown**: It is not known whether the client is capable of providing credential s
SSO - Role	Enter value.
Status	• **None**: No authentication took place • **User**: The user was authenticated • **Machine**: The machine was authenticat ed • **Failed**: Authentication failed • **AuthSource - Unreachable**: The authe ntication source was unreachable
MacAuth	• **NotApplicable**: Not a MAC Auth reques t • **Known Client**: Client MAC address was found in an authentication source • **Unknown Client**: Client MAC address was not found in an authentication sou rce

Attribute Name	Values
Username	The username as received from the client (after the strip user name rules are applied).
Full - Username	The username as received from the client (before the strip user name rules are applie d).
Source	The name of the authentication source use d to authenticate the user.
Status	• **None**: No authentication took place • **User**: The user was authenticated • **Machine**: The machine was authenticat ed • **Failed**: Authentication failed • **AuthSource - Unreachable**: The authe ntication source was unreachable
TEAP - Method - 1	Enter value.
TEAP - Method - 1 - Id - Type	• **User** • **Machine**
TEAP - Method - 1 - Status	• **Success** • **Failure**
TEAP - Method - 1 - User - DN	Enter User - DN (Distinguished Name).
TEAP - Method - 1 - Username	Enter username.
TEAP - Method - 2	Enter value.
TEAP - Method - 2 - Id - Type	• **User** • **Machine**
TEAP - Method - 2 - Status	• **Success** • **Failure**

Attribute Name	Values
TEAP - Method - 2 - User - DN	Enter User - DN (Distinguished Name).
TEAP - Method - 1 - Username	Enter username.
Type	**SSO**
Username	Enter username.

Authorization Namespaces

Policy Manager supports multiple types of authorization sources. Authorization sources from which values of attributes can be retrieved to create role mapping rules have their own separate namespaces (prefixed with Authorization).

Authorization editing context

Role mapping policies

AD Instance Namespace

For each instance of an Active Directory authentication source, there is an AD instance namespace that appears in the rules editing interface. The AD instance namespace consists of all the attributes that were defined when the authentication source was created. These attribute names are pre-populated. For Policy Manager to fetch the values of attributes from Active Directory, you need to define filters for that authentication source (see Adding and Modifying Authentication Sources for more information).

Authorization

The authorization namespace has one attribute: sources. The values are pre-populated with the authorization sources defined in Policy Manager. Use this to check for the authorization source(s) from which attributes were extracted for the authenticating entity.

LDAP Instance Namespace

For each instance of an LDAP authentication source, there is an LDAP instance namespace that appears in the rules editing interface. The LDAP instance namespace consists of all the attributes that were defined when the authentication source was created. These attribute names are pre-populated. For Policy Manager to fetch the values of attributes from an LDAP-compliant directory, you need to define filters for that authentication source (see Adding and Modifying Authentication Sources).

RSAToken Instance Namespace

For each instance of an RSA Token Server authentication source, there is an RSA Token Server instance namespace that appears in the rules editing interface. The RSA Token Server instance namespace consists of attributes names defined when you created an instance of this authentication source. The attribute names are pre-populated for administrative convenience.

Sources

This is the list of the authorization sources from which attributes were fetched for role mapping. Authorization namespaces appear in Role mapping policies.

SQL Instance Namespace

For each instance of an SQL authentication source, there is an SQL instance namespace that appears in the rules editing interface. The SQL instance namespace consists of attributes names defined when you created an instance of this authentication source. The attribute names are pre-populated for administrative convenience. For Policy Manager to fetch the values of attributes from a SQL-compliant database, you need to define filters for that authentication source.

Certificate Namespaces

The certificate namespace can be used in role mapping policies to define roles based on attributes in the client certificate presented by the end host. Client certificates are presented in mutually authenticated 802.1X EAP methods (EAP-TLS, PEAP/TLS, EAP-FAST/TLS).

Certificate Namespace Editing Context

Role mapping policies

Table 1. Certificate Namespace Attributes

Attribute Name	Values
Version	Certificate version
Serial - Number	Certificate serial number
• Subject - C • Subject - CN • Subject - DC	Attributes associated with the subject (use r or machine, in this case). Not all of these f ields are populated in a certificate.

Attribute Name	Values
• Subject - DN • Subject - emailAddress • Subject - GN • Subject - L • Subject - O • Subject - OU • Subject - SN • Subject - ST • Subject - UID	
• Issuer - C • Issuer - CN • Issuer - DC • Issuer - DN • Issuer - emailAddress • Issuer - GN • Issuer - L • Issuer - O • Issuer - OU • Issuer - SN • Issuer - ST • Issuer - UID	Attributes associated with the issuer (Certif icate Authorities or the enterprise CA). Not all of these fields are populated in a certific ate.
• Subject - AltName - DirName • Subject - AltName - DNS • Subject - AltName - EmailAddress • Subject - AltName - IPAddress • Subject - AltName - msUPN	Attributes associated with the subject (use r or machine, in this case) alternate name. Not all of these fields are populated in a ce rtificate.

Attribute Name	Values
• Subject - AltName - RegisterdID • Subject - AltName - URI	

Connection Namespaces

The connection namespace can be used in role mapping policies to define roles based on where the protocol request originated from and where it terminated.

Connection Namespace Editing Contexts

- Role mapping policies
- Service rules

The following table describes the **Connection Namespace Pre-defined Attributes** parameters:

Table 1. Connection Namespace Pre - defined Attributes

Attribute	Description
Src - IP - Address	Src - IP - Address and Src - Port are the IP address and port from which the request (RADIUS) originated.
Src - Port	
Dest - IP - Address	Dst - IP - Address and Dst - Port are the IP address and port at which Policy Manager received the request (RADIUS).
Dest - Port	
Protocol	Request protocol: RADIUS, TACACS+, Web Auth.
NAD - IP - Address	IP address of the network device from whic h the request originated.
Client - Mac - Address	MAC address of the client.
• Client - Mac - Address - Colon • Client - Mac - Address - Dot • Client - Mac - Address - Hyphen	Client MAC address in different formats.

Attribute	Description
• Client - Mac - Address - Nodelim	
Client - IP - Address	IP address of the client (if known).

Date Namespaces

The date namespace has three pre-defined attributes:

- Day-of-Week
- Date-of-Year
- Time-of-Day

  For Day-of-Week, the supported operators are BELONG_TO and NOT_BELONGS_TO, and the value field shows a multi-select list box with days from Monday through Sunday.

  The Time-of-Day attribute shows a time icon in the value field.

  The Date-of-Year attribute shows a date, month and year icon in the value field.

  The operators supported for Date-of-Year and Time-of-Day attributes are the similar to the ones supported for the integer data type.

Date Namespace Editing Contexts

- Enforcement policies
- Filter rules for Access Tracker and Activity Reports
- Role mapping policies
- Service rules

Device Namespaces

The Device namespace has four pre-defined attributes:

- Location

- OS-Version
- Device-Type
- Device-Vendor

  Custom attributes also appear in the attribute list if they are defined as custom tags for the device.

> **NOTE**
>
> These attributes can be used only if you have pre-populated the values for these attributes when a network device is configured.

Endpoint Namespaces

Use these attributes to look for attributes of authenticating endpoints, which are present in the Policy Manager endpoints list. The Endpoint namespace has the following attributes:

- Disabled By
- Disabled Reason
- Enabled By
- Enabled Reason
- Info URL

Guest User Namespaces

The Guest User namespace has the attributes associated with the guest user (resident in the Policy Manager guest user database) who authenticated in this session. This namespace is only applicable if a guest user is authenticated. The Guest User namespace has six pre-defined attributes:

- Company-Name
- Designation
- Email
- Location

- Phone
- Sponsor

  Custom attributes also appear in the attribute list if they are defined as custom tags for the guest user.

> **NOTE**
>
> These attributes can be used only if you have pre-populated the values for these attributes when a guest user is configured in Policy Manager.

Host Namespaces

The Host namespace has the following predefined attributes:

- Name*
- OSType*
- FQDN*
- UserAgent**
- CheckType**
- UniqueID
- AgentType*
- InstalledSHAs*

  * Only populated when request is originated by a Microsoft NAP-compatible agent.

  ** Only present if Policy Manager acts as a Web authentication portal.

Local User Namespaces

The LocalUser namespace has the attributes associated with the local user (resident in the Policy Manager local user database) who authenticated in this session. This namespace is only applicable if a local user is authenticated.

The LocalUser namespace has four pre-defined attributes:

- Designation
- Email
- Phone
- Sponsor

  Custom attributes also appear in the attribute list if they are defined as custom tags for the local user.

> **NOTE**
>
> These attributes can be used only if you have pre-populated the values for these attributes when a local user is configured in Policy Manager.

RADIUS Namespaces

Dictionaries in the RADIUS namespace come pre-packaged with the product. The administration interface does provide a way to add dictionaries into the system (See RADIUS Dictionary for more information). RADIUS namespace has the notation RADIUS:Vendor, where Vendor is the name of the Company that has defined attributes in the dictionary. Sometimes, the same vendor has multiple dictionaries, in which case the "Vendor" portion has the name suffixed by the name of device or some other unique string.

IETF is a special vendor for the dictionary that holds the attributes defined in the RFC 2865 and other associated RFCs. Policy Manager comes pre-packaged with a number of vendor dictionaries.

Some examples of dictionaries in the RADIUS namespace are:

- RADIUS:Aruba
- RADIUS:IETF
- RADIUS:Juniper
- RADIUS:Microsoft

RADIUS Namespace Editing Contexts

- Filter rules for Access Tracker and Activity Reports
- Policy simulation attributes

- Post-proxy attribute pruning rules
- RADIUS Enforcement profiles: All RADIUS namespace attributes that can be sent back to a RADIUS client (the ones marked with the OUT or INOUT qualifier)
- Role mapping policies
- Service rules: All RADIUS namespace attributes that can appear in a request (the ones marked with the IN or INOUT qualifier)

TACACS Namespaces

The TACACS (Terminal Access Controller Access-Control System) namespace has the attributes associated with attributes available in a TACACS+ request. Available attributes are:

- AuthSource
- AvendaAVPair
- UserName

Tips Namespaces

The pre-defined attributes for the Tips namespace are Role and Posture. Values are assigned to these attributes at run-time after Policy Manager evaluates role mapping and posture related policies.

Role

The value for the Role attribute is a set of roles assigned by either the role mapping policy or the post-audit policy. The value of the Role attribute can also be a dynamically fetched “Enable as role” attribute from the authorization source. The posture value is computed after Policy Manager evaluates internal posture policies, and gets posture status from posture servers or audit servers.

Posture

The value for the Posture attribute is one of the following:

- CHECKUP
- HEALTHY

- INFECTED
- QUARANTINE
- TRANSITION
- UNKNOWN

Tips Namespace Editing Context

Enforcement policies

Variables

Variables are populated with the connection-specific values. Variable names (prefixed with % and enclosed in curly braces; for example, %{Username}") can be used in filters, role mapping, enforcement rules, and enforcement profiles. Policy Manager does in-place substitution of the value of the variable during run-time rule evaluation.

The following built-in variables are supported in Policy Manager:

Table 1. Policy Manager Variables Supported by Policy Manager

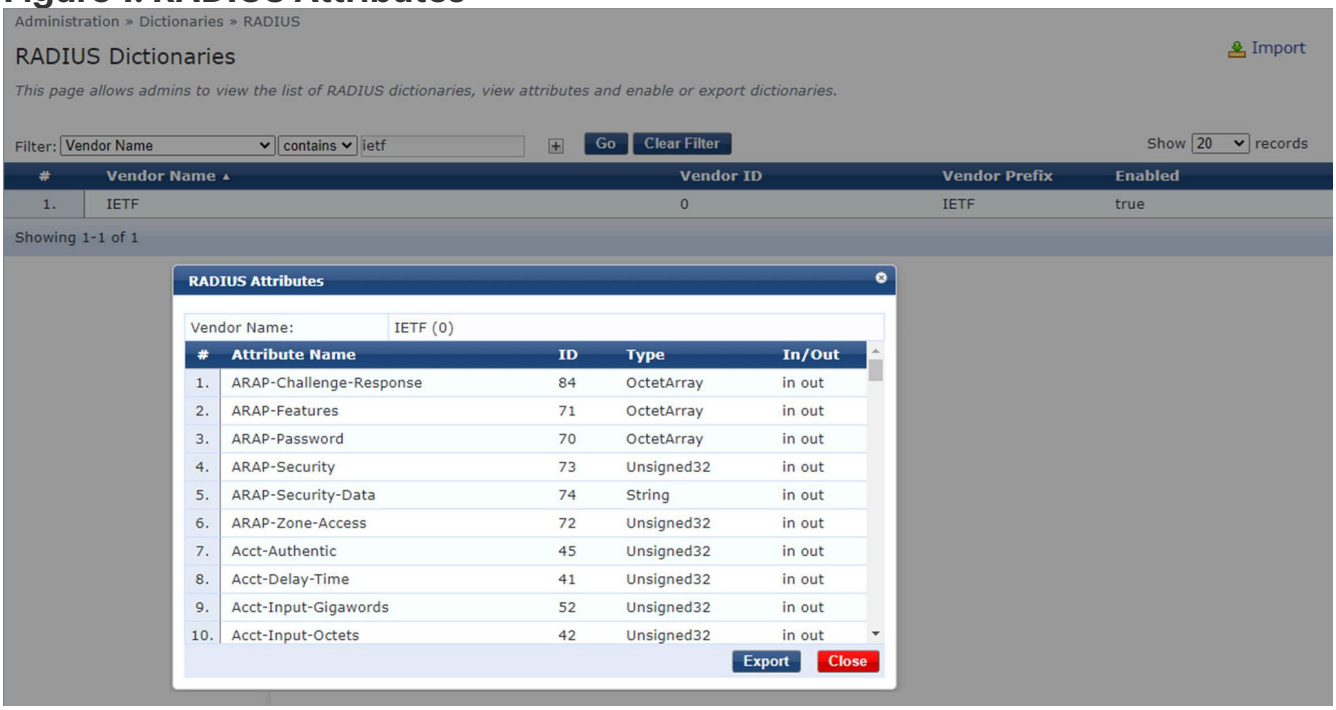
Variable	Description
%{`attribute-name`}	`attribute-name` is the alias name for an attribute that you have configured to be retrieved from an authentication source. See Adding and Modifying Authentication Sources.
%{Radius:IETF:MAC - Address - Colon}	MAC address of client in aa:bb:cc:dd:ee:ff format
%{Radius:IETF:MAC - Address - Hyphen}	MAC address of client in aa - bb - cc - dd - ee - ff format
%{RadiusS:IETF:MAC - Address - Dot}	MAC address of client in aabb.ccdd.eeff format
%{Radius:IETF:MAC - Address - NoDelim}	MAC address of client in aabbccddeeff format

You can also use any other dictionary-based attributes (or namespace attributes) as variables in role-mapping rules, enforcement rules, enforcement profiles, and

LDAP or SQL filters. For example, you can use %{Radius:IETF:Calling-Station-ID}or %{Radius:Airespace:Airespace-Wlan-Id} in rules or filters.

To view available dictionary attributes, refer to the **Administration > Dictionaries > RADIUS** and **Administration > Dictionaries > Dictionary Attributes** pages of the Policy Manager WebUI. The following image shows some of the RADIUS ITEF attribute names that can be used as variables.

Figure 1. RADIUS Attributes



Operators

The Rules editing interface in Policy Manager supports a rich set of operators. The type of operators presented are based on the data type of the attribute for which the operator is being used. Where the data type of the attribute is not known, the attribute is treated as a string type.

The following table lists the operators presented for common attribute data types:

Attribute Type	Operators
String	• BELONGS_TO • NOT_BELONGS_TO • BEGINS_WITH

Attribute Type	Operators
	• NOT_BEGINS_WITH • CONTAINS • NOT_CONTAINS • ENDS_WITH • NOT_ENDS_WITH • EQUALS • NOT_EQUALS • EQUALS_IGNORE_CASE • NOT_EQUALS_IGNORE_CASE • EXISTS • NOT_EXISTS • MATCHES_REGEX • NOT_MATCHES_REGEX
Integer	• BELONGS_TO • NOT_BELONGS_TO • EQUALS • NOT_EQUALS • EXISTS • NOT_EXISTS • GREATER_THAN • GREATER_THAN_OR_EQUALS • LESS_THAN • LESS_THAN_OR_EQUALS
Time or Date	• EQUALS • NOT_EQUALS • GREATER_THAN

Attribute Type	Operators
	• GREATER_THAN_OR_EQUALS • LESS_THAN • LESS_THAN_OR_EQUALS • IN_RANGE
Day	• BELONGS_TO • NOT_BELONGS_TO
List (Example: Role)	• EQUALS • NOT_EQUALS • MATCHES_ALL • NOT_MATCHES_ALL • MATCHES_ANY • NOT_MATCHES_ANY • MATCHES_EXACT • NOT_MATCHES_EXACT
Group (Example: Calling - Station - Id, NAS - IP - Address)	• BELONGS_TO_GROUP • NOT_BELONGS_TO_GROUP

The following table describes all operator types:

Table 1. Operator Descriptions

Operator	Description
BEGINS_WITH	For string data type, true if the run - time value of the attribute begins with the confi gured value. Example: `Radius:IETF:NAS-Identifie r BEGINS_WITH "SJ-"`
BELONGS_TO	For string data type, true if the run - time value of the attribute matches a set of conf igured string values.

Operator	Description
	Example: `Radius:IETF:Service-Type BELONGS_TO Login-User,Framed-User,Authenticate-Only` For integer data type, true if the run - time value of the attribute matches a set of conf igured integer values. Example: `Radius:IETF:NAS-Port BELONGS_TO 1,2,3` For day data type, true if run - time value o f the attribute matches a set of configured days of the week. Example: `Date:Day-of-Week BELONGS_TO MONDAY,TUESDAY,WEDNESDAY` When Policy Manager is aware of the valu es that can be assigned to BELONGS_TO o perator, it populates the value field with th ose values in a multi - select list box; you c an select the appropriate values from the p resented list. Otherwise, you must enter a comma separated list of values.
BELONGS_TO_GROUP	For group data types, true if the run - time value of the attribute belongs to the config ured group (either a static host list or a ne twork device group, depending on the attri bute). Example: `Radius:IETF:Calling-Station-Id BELONGS_TO_GROUP Printers`.
CONTAINS	For string data type, true if the run - time value of the attribute is a substring of the c onfigured value. Example: `Radius:IETF:NAS-Identifier CONTAINS "VPN"`
ENDS_WITH	For string data type, true if the run - time value of the attribute ends with the configu red value. Example: `Radius:IETF:NAS-Identifier ENDS_WITH "DEVICE"`
EQUALS	True if the run - time value of the attribute matches the configured value. For string d

Operator	Description
	ata type, this is a case - sensitive comparis on. Example: `Radius:IETF:NAS-Identifie r EQUALS "SJ-VPN-DEVICE"`
EQUALS_IGNORE_CASE	For string data type, true if the run - time value of the attribute matches the configur ed value, regardless of whether the string i s upper case or lower case. Example: `Radius:IETF:NAS-Identifie r EQUALS_IGNORE_CASE "sj-vpn-devi ce"`
EXISTS	For string data type, true if the attribute exi sts. Checks only if the attribute exits or not , not the run - time value. This is a unary op erator. Example: `Radius:IETF:NAS-Identifie r EXISTS`
GREATER_THAN	For integer, time and date data types, true if the run - time value of the attribute is g reater than the configured value. Example: `Radius:IETF:NAS-Port GREA TER_THAN 10`
GREATER_THAN_OR_EQUALS	For integer, time and date data types, true if the run - time value of the attribute is g reater than or equal to the configured valu e. Example: `Radius:IETF:NAS-Port GREA TER_THAN_OR_EQUALS 10`
IN_RANGE	For time and date data types, true if the ru n - time value of the attribute is less than o r equal to the first configured value and les s than equal to the second configured valu e. Example: `Date:Date-of-Year IN_RANG E 2007-06-06,2007-06-12`
LESS_THAN	For integer, time and date data types, true i f the run - time value of the attribute is less than the configured value.

Operator	Description
	Example: `Radius:IETF:NAS-Port LESS_THAN 10`
LESS_THAN_OR_EQUALS	For integer, time and date data types, true i f the run - time value of the attribute is less than or equal to the configured value. Example: `Radius:IETF:NAS-Port LESS_THAN_OR_EQUALS 10`
MATCHES_ALL	For list data types, true if all of the run - ti me values in the list are found in the config ured values. Example: `Tips:Role MATCHES_ALL HR, ENG,FINANCE`. In this example, if the run - time values of Tips:Role are HR,ENG,FINA NCE,MGR,ACCT the condition evaluates to true.
MATCHES_ANY	For list data types, true if any of the run - ti me values in the list match one of the confi gured values. Example: `Tips:Role MATCHES_ANY HR, ENG,FINANCE`
MATCHES_EXACT	For list data types, true if all of the run - ti me values of the attribute match all of the configured values. Example: `Tips:Role MATCHES_ALL HR, ENG,FINANCE`. In this example, if the run - time values of Tips:Role are HR,ENG,FINA NCE,MGR,ACCT the condition evaluates to false, because there are some values in the configured values that are not present in th e run - time values.
MATCHES_REGEX	For string data type, true if the run - time value of the attribute matches the regular expression in the configured value. Example: `Radius:IETF:NAS-Identifie r MATCHES_REGEX sj-device[1-9]-de v*`

OnGuard Agents

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- Evil Twin Detection with OnGuard
- Authentication in Guest Web Login Using OnGuard Native Dissolvable Agent
- Configuring Workflow in Native Agents Only Mode
- Accessing the OnGuard Support Chart for Plugin Version 2.0
- Upgrading From OnGuard Plugin Version 1.0 to 2.0
- Configuring Web Agent Flow—Native Agent
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- Using OnGuard with Aruba Virtual Intranet Access (VIA)

Subtopics

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Introduction

The HPE Aruba Networking ClearPass OnGuard agent controls compromised devices by detecting and blocking access to unsecure or unhealthy devices. The client is denied access to network resources across wired, wireless, and remote networks when it is determined to be unsecure, which is accomplished by running an extensive posture assessment. The OnGuard Agent is supported by Windows, Linux, and macOS devices. For information on the supported browsers for OnGuard Native agent, refer to the Policy Manager Release Notes for the current release. You can configure the OnGuard Dissolvable Agent flow in different modes to perform health scans on endpoints. This section provides information on the end-to-end flow and how to configure

OnGuard Dissolvable Agent in Native Agents Only mode. The Native Dissolvable Agent communicates with Policy Manager Guest to send information about endpoints such as status, health status, remediation messages and so on. This communication is independent of the operating systems and browsers (for details, see Configuring Workflow in Native Agents Only Mode).

> **NOTE**
> For information on Agentless OnGuard, see About Agentless OnGuard.

Updating the OnGuard Agent

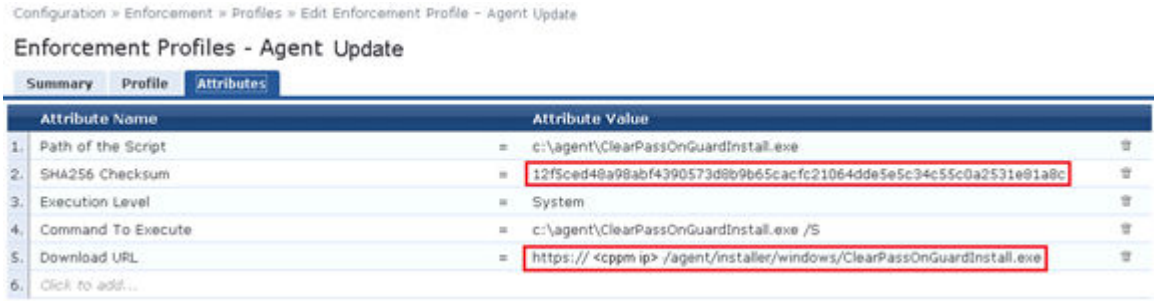
About this task

The Agent Script Enforcement Profile can be used to update the OnGuard Agent.

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**, then click **Add.**
2. In the **Template** field, select **Agent Script Enforcement** to create a new Agent Script Enforcement profile.
3. Click the **Attributes** tab.
4. Select **Click to add...**, add the **Download URL** field, and enter the IP address of the ClearPass Server from which to download OnGuard Agent installer.
5. Select **Click to add...**, add the **SHA256 Checksum** field, and update it with the checksum of the installer from ClearPass Server.

Figure 1. Agent Script Enforcement Profile can be used to update the OnGuard Agent



6. Next, navigate to **Configuration > Enforcement > Policies** and create a policy that uses **Host: AgentVersion** rules to check the OnGuard Version and apply the Agent

Update enforcement profile. The rule should be edited periodically to update from earlier versions on OnGuard.

Figure 2. Enforcement Rule to check the OnGuard Version

OnGuard Deployment Considerations

Keep the following considerations in mind prior to configuring an OnGuard deployment:

- Keep your OnGuard Agent versions up-to-date and comparable to deployed ClearPass version(s).
- The OnGuard Agents now support TLS 1.3 for communication on Linux, macOS and Windows. This applies to all 3 OnGuard Agent types, Persistent, Agentless and Native DA.
- HPE Aruba Networking recommends utilizing the Guest Device Repository as opposed to Static Host Lists, since using too many Static Host Lists is known to cause performance issues by adding load on the policy server that may result in higher request processing times and the policy server running out of threads.
- Add enough ClearPass server capacity to handle peak OnGuard traffic projections. For example: 9-10 am when users come to office and begin to consume resources.
- Use a load balancer to distribute the OnGuard load evenly among the nodes in cluster.
- Use Override IP to configure the Load balancer IP.
- Allow-list ClearPass OnGuard processes in the deployment's Antivirus/Firewall configuration.
- If the **Policy Cache** is configured, then configure the **Session Timeout** attribute in the Agent enforcement profile with a value that is slightly more than the **Policy**

Cache to keep the cache up-to date. For example: If the Policy Cache is set to 8-hours, then configure the **Session Timeout** to 7 hours.

- Before upgrading or moving to new 3rd Party Apps like Antivirus, Firewall etc., test them in test/lab environment with OnGuard Agent to confirm OnGuard Agent works as expected with new version/product.
- Use an OnGuard custom script for updating the OnGuard Agent or Agent Library to perform updates in batches or during projected non-peak times.

Evil Twin Detection with OnGuard

Policy Manager OnGuard Agent supports Evil Twin Detection for Wired Interfaces.

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- Configuring Evil Twin Detection for Enforcement Policy
- Evil Twin OnGuard WebAuth Request Attribute in Access Tracker

Subtopics

Introduction
Configuring Evil Twin Detection for Agent Enforcement Profile
Configuring Evil Twin Detection for Enforcement Policy
Evil Twin OnGuard WebAuth Request Attribute in Access Tracker

Introduction

To enable OnGuard Agent to check for the presence of an "evil twin" with Wired interface, a new option, **Evil Twin Detection**, has been added to the Agent Enforcement Profile attributes (for details, see Agent Enforcement Profile). When OnGuard Agent detects an Evil Twin for Wired interface, it sends a WebAuth request with the *Host:EvilTwin* attribute having the value "*MayExist*," indicating that an Evil Twin may exist. The *Host:EvilTwin* attribute is available in Service rules, Role Mapping, and

Enforcement profiles. Admins can use this attribute to take action by applying Policy Manager enforcement profiles.

> **NOTE**
> This feature is supported only with wired interfaces; it is not available for wireless interfaces. Additionally, this feature is only supported with a network hub, not an unmanaged switch.

Configuring Evil Twin Detection for Agent Enforcement Profile

About this task

Policy Manager administrators can use the *Host:EvilTwin* attribute to apply other enforcement profiles to send email, issue a disconnect request, update endpoint attributes, and/or send updates to third-party firewalls.

To enable the *Evil Twin Detection* attribute for an Agent Enforcement profile:

Procedure

1. Navigate to **Configuration > Enforcement > Profiles**.

   The **Enforcement Profiles** page opens.

2. Click **Add**.

   The **Add Enforcement Profiles > Profile** tab opens.

3. From the **Template** drop-down, select **Agent Enforcement**.

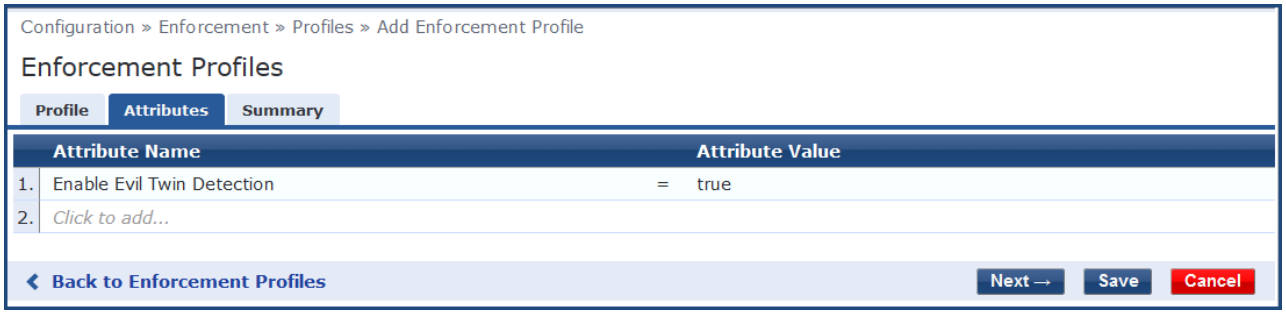
   The **Agent Enforcement > Profile** dialog opens.

4. Specify the **Profile** tab parameters as described in Agent Enforcement Profile.

5. Select the **Attributes** tab.

6. Add the *Enable Evil Twin Detection* attribute as shown in Figure 1:

   Figure 1. Agent Enforcement Profile > Attributes Tab



7. Click **Save**.

Configuring Evil Twin Detection for Enforcement Policy

About this task

To configure an enforcement policy with the *Host:EvilTwin* condition:

Procedure

1. Navigate to **Configuration** > **Enforcement** > **Enforcement Policies**.

   The **Enforcement Policies** page opens:

2. Click **Add**.

   The **Add Enforcement Policy** page opens to the **Enforcement** tab.

3. Specify the **Add Enforcement Policy** > **Enforcement** tab parameters as described in Configuring Enforcement Policies.

4. In the **Rules** tab, click **Add Rule** to display the **Rules Editor**.

5. Configure the *Host:EvilTwin EQUALS MayExist* condition as shown in Figure 2:

Figure 1. Configuring Enforcement Policy with Evil Twin May Exist Condition

6. Click **Save**.

Evil Twin OnGuard WebAuth Request Attribute in Access Tracker

About this task

The **Input** tab shows protocol-specific attributes that Policy Manager received in a transaction request, including authentication and posture details (if available). The **Input** tab also shows computed attributes that Policy Manager derived from the request attributes. All of these attributes can be used in role-mapping rules.

Access Tracker records the presence of the *Host:EvilTwin* attribute on the **WebAuth Request Details > Input** page.

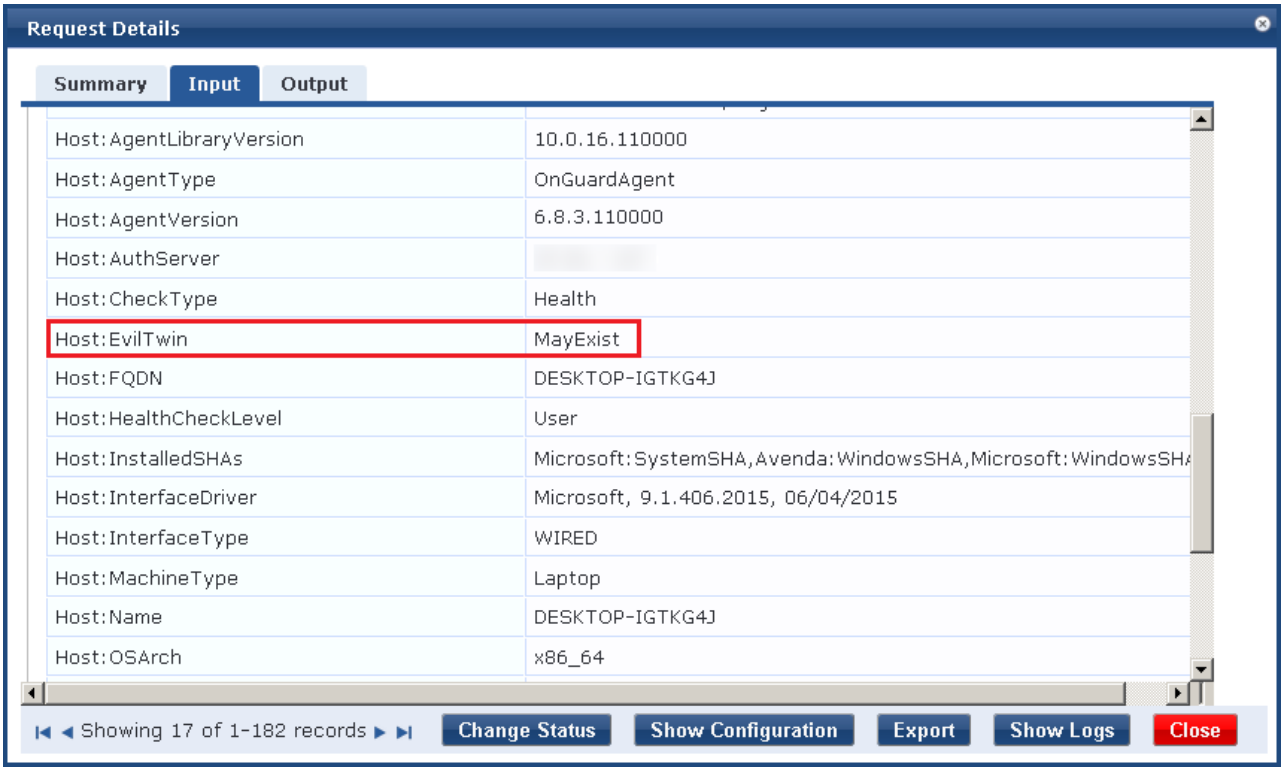
Procedure

1. Navigate to **Live Monitoring > Access Tracker**.
2. From the **Access Tracker** page, select the pertinent WebAuth session.

   The **Request Details** page opens.
3. Click the **Input** tab and select the **Computed Attributes** section.

   The *Host:EvilTwin* attribute is displayed as shown in Figure 3.

   Figure 1. Host:EvilTwin Attribute Displayed in Access Tracker > Request Details



Authentication in Guest Web Login Using OnGuard Native Dissolvable Agent

This section describes how to enable application authentication in Guest Web Login pages while using the OnGuard Native Dissolvable Agent to also identify the username.

> **NOTE**
> The Dissolvable Agent, by default, does not perform end-user authentication.

The OnGuard Dissolvable Agent can be supported with VPN interfaces when:

- The same username is used to identify the client in both RADIUS VPN authentication and application authentication.
- The **Pre-Auth Check** is configured using one of the following values (refer to Table 1 for details):
  - App Authentication
  - Local
  - RADIUS

- Single Sign On

Subtopics

Creating a Guest Web Login Page for Native Dissolvable Agent
Creating a RADIUS Service in Policy Manager for VPN Authentication
Configuring the Guest Access - Web Login Service Template
Configuring a Web-Based Health Check Only Service
Enabling the VPN Interface
Access Tracker Requests Reflect Credentials

Creating a Guest Web Login Page for Native Dissolvable Agent

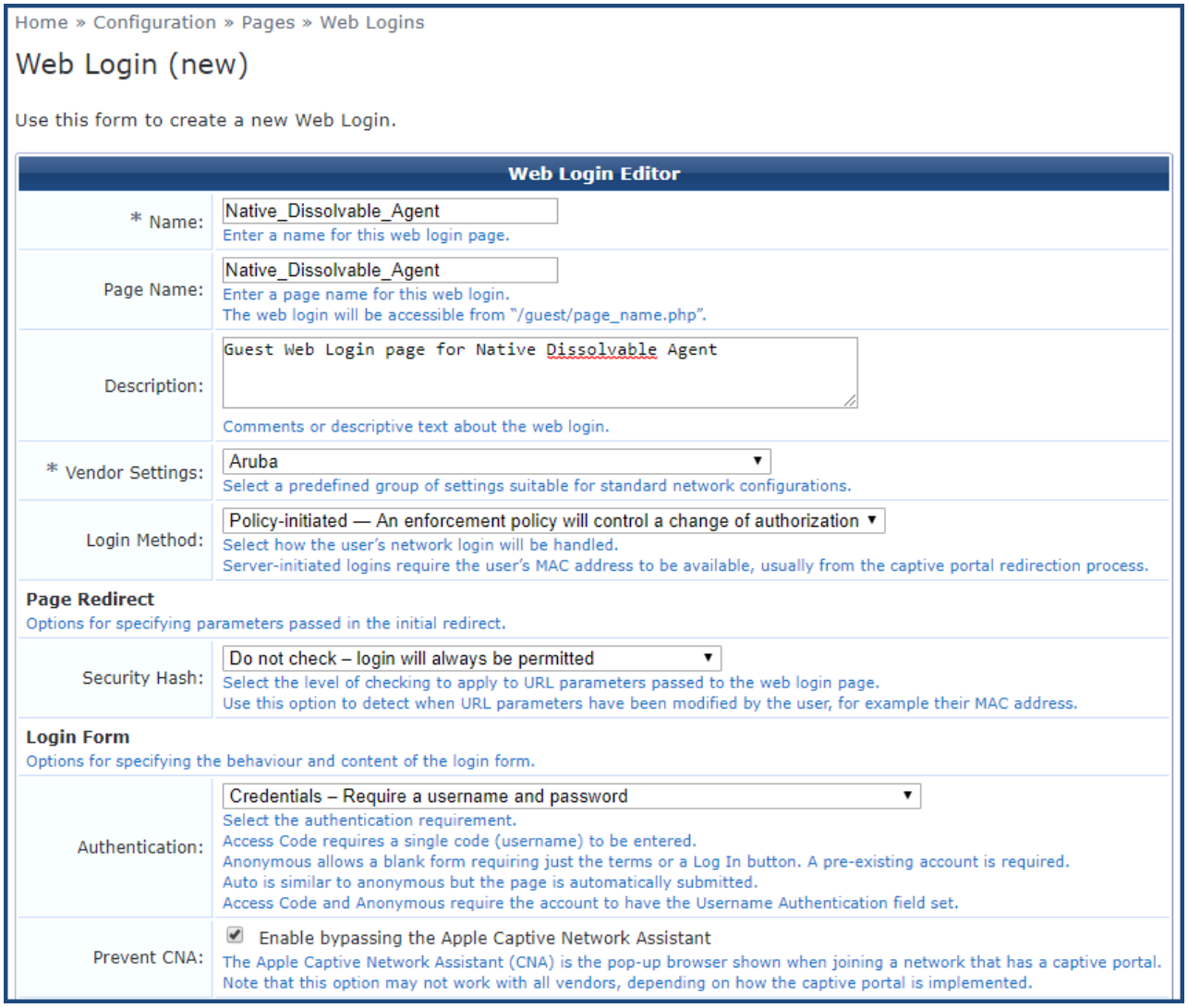
About this task

To create a Guest Web Login page for Native Dissolvable Agent:

Procedure

1. Log in to Policy Manager Guest, then navigate to **Configuration > Pages > Web Logins**. The **Web Logins** page opens.
2. Click the **Create a new web login page** link. The **Guest Web Login Editor** opens. For details about the fields and settings available through the Web Login Editor, refer to the "Creating and Editing Web Login Pages" section in the Policy Manager Guest Online Help.

Figure 1. Guest Web Login Editor



3. Specify the **Web Login Editor** parameters as described in the following table:

> **NOTE**
>
> For details about all the settings available in the Web Login Editor, see the "Creating and Editing Web Login Page" section in the Policy Manager 6.11 Guest User Guide.

Table 1. Web Login Editor Properties for Native Dissolvable Agent Page

Field	Action/Description
General Properties	
Name	(Required) Enter the name for the page.

Field	Action/Description
Page Name	Enter the Identifier page name that will a ppear in the URL—for example, "/guest/pa ge_name.php". The Page Name must consist of letters, n umbers, underscore (_), or dash (-).
Description	Enter additional information or comment s about the page.
Vendor Settings	(Required) Specifies vendor - specific set tings for network configuration. This drop - down list includes a list of **vendors** you can select from and a list of **Other** settin gs. Select **Aruba**.
Login Method	Specifies how the user's network login s hould be handled. Select **Policy Initiated —An enforcement policy will control a c hange of authorization**. This option should be selected if you are using OnGuard health checks.
Address	(Required) Enter the IP address or hostna me of the vendor's product.
Secure Login	Specifies the security option to use for th e Web login process. Options include: • *Use vendor default* • *Secure login using HTTPS* • *Send cleartext passwords over HTTP*
Allowed Dynamic	Enter the IP addresses and networks that will be allowed.
Denied Dynamic	Enter the IP addresses and networks that will be denied.
Dynamic Address	For multi - controller deployments, if sele cted, this option enables sending the IP a

Field	Action/Description
	ddress to submit credentials. When enabl ed, the *Allowed Dynamic* and *Denied Dyn amic* fields are added to the form.
Page Redirect	
Security Hash	Specifies the level of checking to apply t o URL parameters passed to the Web Log in page. • Select **Do not check —login will al ways be permitted**.
Login Form	
Authentication	Specifies the authentication requirement method to use. • Select **Credentials — Require a usern ame and password**.
Prevent CNA	If selected, this option enables bypassin g the Apple Captive Network Assistant (C NA). The CNA is the pop - up browser sh own when joining a network that has a ca ptive portal. • Click this check box to enable the **P revent CNA** option.
Custom Form	If selected, indicates you will provide a c ustom login form. You must supply your own HTML login form for the header and footer HTML areas.
Custom Labels	If selected, this option enables altering th e default labels and error messages. Whe n enabled, this option adds the **Pre - Aut h Error** field to the form.
Pre - Auth Error	Customized label text to display if userna me and password lookup fails. Leave blan

Field	Action/Description
	k to use the default (Invalid username or password).
Pre - Auth Check	Specifies how the username and passwo rd should be checked before authenticati on. • Select **App Authentication – check u sing Aruba Application Authenticatio n**. When **App Authentication** is not enabled , Native Dissolvable Agent uses the MAC address as the user name. When **App Authentication** is enabled (wit h **Health Check** also enabled), instead of using the MAC address as the username i n the WebAuth Request, Native Dissolvab le Agent sends the username specified in the **App Authentication** option. If **Pre - Auth Check** has one of the follow ing values, Native Dissolvable Agent send s the username specified in **App Authent ication**: • *App Authentication*: Check using Aru ba Application Authentication. • *Local*: Match a local account. • *RADIUS*: Check using a RADIUS requ est. • *Single Sign On*: Enable Single Sign - On for this web login. When this option is selected, guests a re redirected to the Identity Provider (IdP) configured in Policy Manager, w here they authenticate themselves. Th ey are redirected back to Policy Mana ger, which verifies the login was succ essful and uses the same credentials t

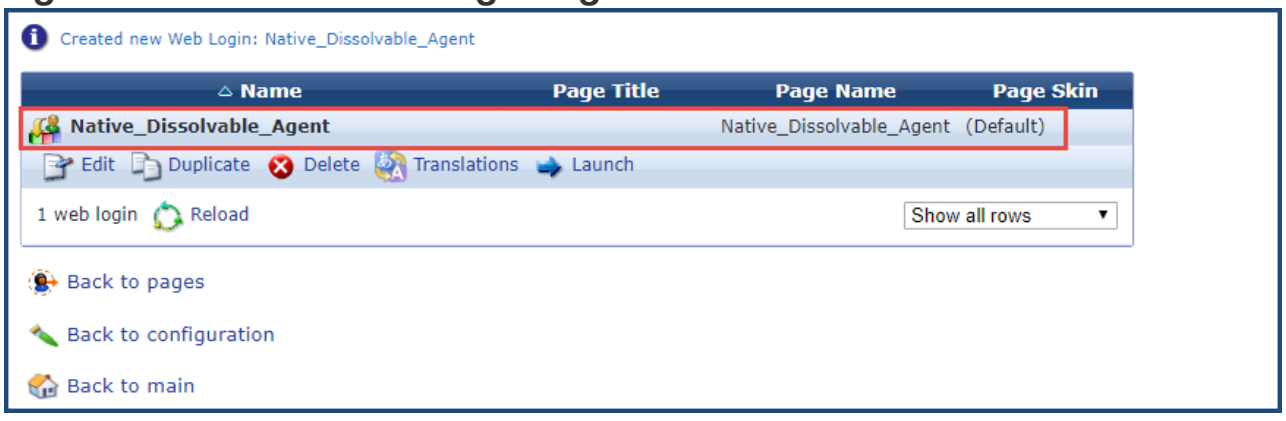
Field	Action/Description
	o redirect to the actual Web login flow . (To enable SSO support, go to **Policy Manager > Configuration > Identity > Single Sign - On**).
Terms	Select this check box to require the user to mark a check box to accept a Terms an d Conditions agreement.
Default Destination	**Default Destination**
Default URL	(Required) Specify the default URL for the redirect page. For external domains, this must include t he `http://` prefix.
Override Destination	Select this check box to force the defaul t destination for all clients, overriding any default value already set on the client.
Login Page	
For descriptions and details about the **Login Page** options, refer to the "Creating an d Editing Web Login Pages" section in the Policy Manager Guest Online **Help > Table** 4: Web Login Editor, Login Page Parameters.	
Advertising Services	
Advertising	The **Enable Advertising Services content** check box lets you selectively enable Ad vertising Services for each Web Login pa ge on which you want to display advertisi ng.
Cloud Identity	
Options in the **Cloud Identity** area let you present guests with various cloud identity or social network login options. For descriptions and details about the **Cloud Identity** options, refer to the "Creating and Editing Web Login Pages" section in the Policy Manager Guest Online Help> Tab	

Field	Action/Description
le 5: Web Login Editor Form, Cloud Identity Area: Enabling Cloud Identity or Social N etwork Logins and *Table 6: Web Login Editor Form, Cloud Identity Area, Authenticati on Providers*.	
Multi - Factor Authentication	
Options in the **Multi - Factor Authentication** area let you require a secondary factor when authenticating. When the guest enters their username to log in, they are sent a code which they must successfully retrieve and enter to complete the login. For descriptions and details about the **Multi - Factor Authentication** options, refer t o the "Creating and Editing Web Login Pages" section in the Policy Manager Guest O nline **Help > Table**7: Web Logins Editor, Multi - Factor Authentication.	
Network Login Access	
Allowed Access	Specify the IP addresses and networks f rom which logins will be allowed. (IPv4 a nd IPv6 are both supported.)
Denied Access	Specify the IP addresses and networks f rom which logins will be denied. (IPv4 an d IPv6 are both supported.)
Deny Behavior	(Required) Specify the response shown t o the user if their login request is denied. Options in this drop - down list include: • *Send HTTP 404 Not Found status* • *Show Access Denied page* • *Show a blank page*
Post - Authentication	
Health Check	Select this check box to require the gues t to pass a health check before they can access the network. The health check is done automatically through the OnGuard Dissolvable Agent. • *Header HTML*: Enter the HTML conten t to display above the health check t

Field	Action/Description
	ext. The default content is shown, and can be modified. You can also use the drop - down list to add images or oth er content items. • Footer HTML: Enter the HTML content to display below the health check text . The default content is shown, and ca n be modified. You can also use the d rop - down list to add images or other content items.
Update Endpoint	To have the endpoint's attributes updated with other details from the user account, enable (select) the **Mark the user's MAC address as a known endpoint** check box. Specify the list of name - value pairs to p ass as custom attributes. Follow the form at **user_field | Endpoint Attribute**. Examp les are shown in this text box.

4. Click **Save Changes**. You return to the **Web Logins** page, where the name of the new page is now displayed.

Figure 2. New Guest Web Login Page Added



Creating a RADIUS Service in Policy Manager for VPN Authentication

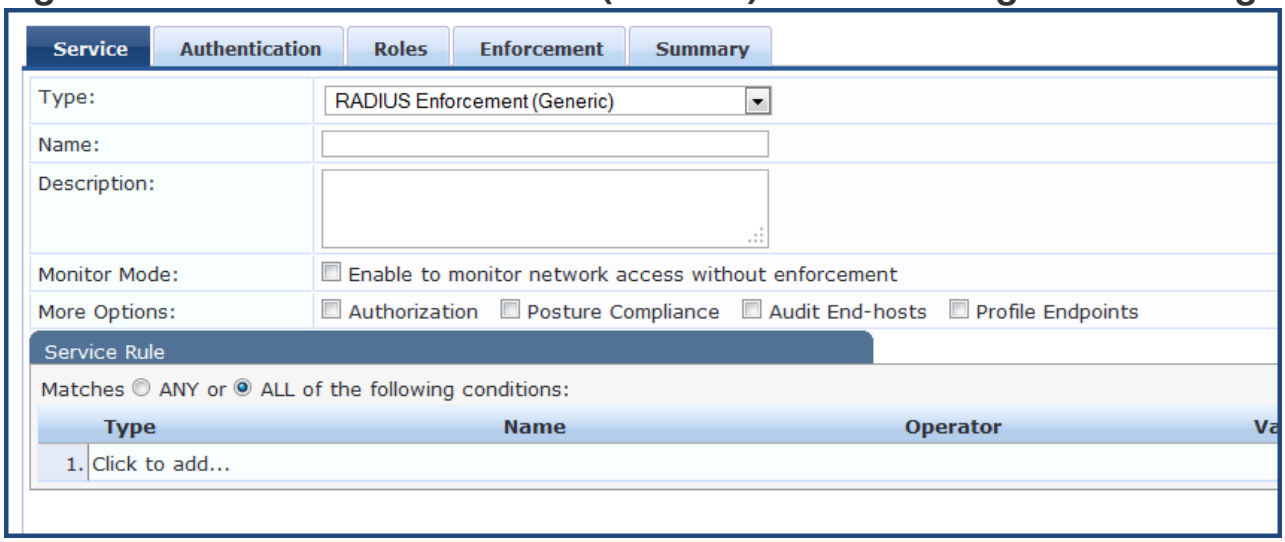
About this task

To create a RADIUS service in Policy Manager for VPN Authentication:

Procedure

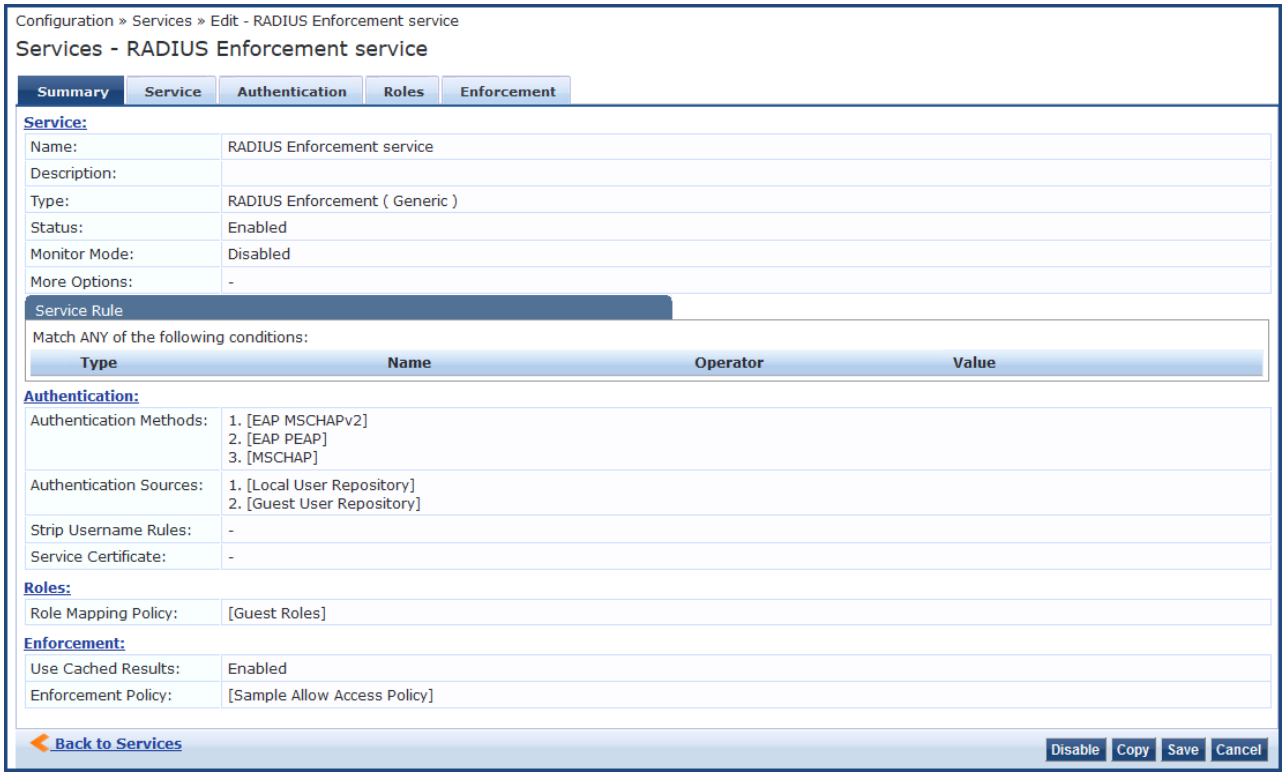
1. In Policy Manager, navigate to **Configuration > Services**.The **Services** page opens.
2. Click **Add**. The **Add Services** dialog opens.
3. From the **Type** drop-down, select **RADIUS Enforcement (Generic)**. The **RADIUS Enforcement (Generic)** service configuration dialog opens:

Figure 1. Add RADIUS Enforcement (Generic) Service Configuration Dialog



4. Proceed to configure the RADIUS Enforcement service to match the service settings displayed in Figure 4. For details on configuring RADIUS Enforcement Services (Generic), refer to RADIUS Enforcement (Generic) Service .

Figure 2. Summary of the RADIUS Enforcement Service



5. Specify the RADIUS Enforcement Service (Generic) service parameters as described in the table below.

Table 1. Aruba RADIUS Enforcement (Generic) Service Parameters

Parameter	Action/Description
Service	
Name	Enter the name of the service.
Description	Provide additional information that helps to identify the service.
Type	Select **RADIUS Enforcement (Generic)** .
Status	Enabled
Monitor Mode	Disabled
More Options	Not required.
Authentications	

Parameter	Action/Description
Authentication Methods	Specify the following authentication methods: • EAP - MSCHAPV2 • MSCHAPV2 • EAP - PEAP Add additional authentication methods as needed for this service.
Authentication Sources	Select the following authentication sources: • **[Local User Repository] [Local SQL DB]** • **[Guest User Repository] [Local SQL DB]** **NOTE** You can also add Active Directory (AD) as an authentication source.
Strip Username Rules	Not required.
Service Certificate	Not required.
Roles	
Role Mapping Policy	Not required.
Enforcement	
User Cached Results	Select this check box to use cached roles and posture attributes from previous sessions.
Enforcement Policy	Select **[Sample Allow Access Policy]**.

6. Click **Save**.

Configuring the Guest Access - Web Login Service Template

About this task

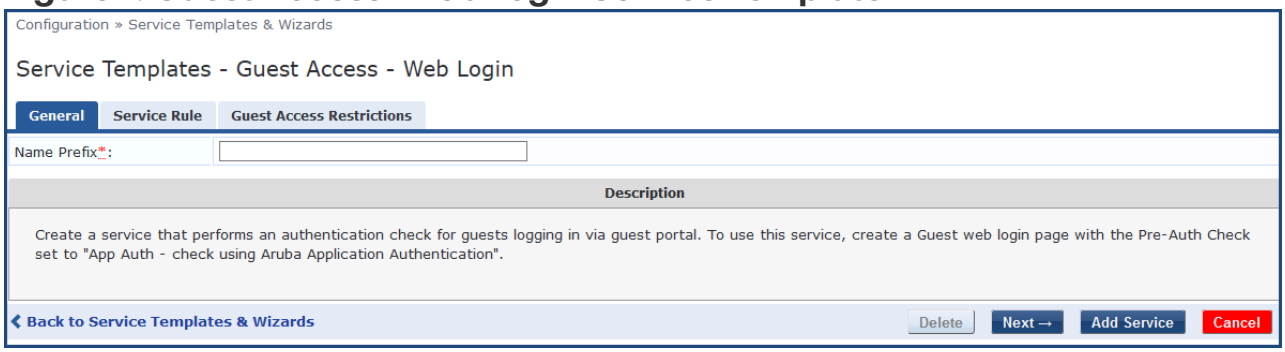
In this web login service, you associate the *Guest Web Login page for Native Dissolvable Agent* created earlier (see Creating a Guest Web Login Page for Native Dissolvable Agent above) with the **Guest Access - Web Login** service.

To configure the **Guest Access - Web Login** service:

Procedure

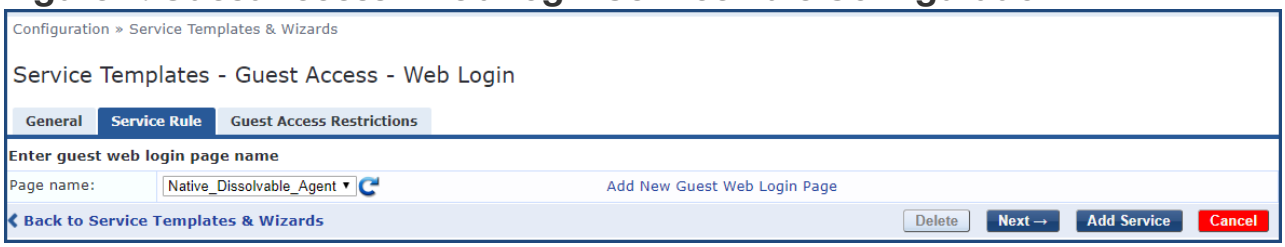
1. Navigate to **Configuration > Service Templates & Wizards**. The list of Policy Manager Service Templates and Wizards opens.
2. Select **Guest Access - Web Login**. The **Guest Access Web Login** service template opens:

Figure 1. Guest Access- Web Login Service Template



3. In the **Name Prefix** field, enter the prefix. This prefix is appended to services that use the Guest Access - Web Login template.
4. Click Next. The **Service Rule** dialog opens.

Figure 2. Guest Access - Web Login Service Rule Configuration



5. Select the **Guest Access Restrictions** tab to specify the days of the week that guest users are allowed network access.
6. Click **Add Service**. You return to the **Services** page. The following message is displayed:

   Added 1 Enforcement Policies

   Added 1 Service(s)

Configuring a Web-Based Health Check Only Service

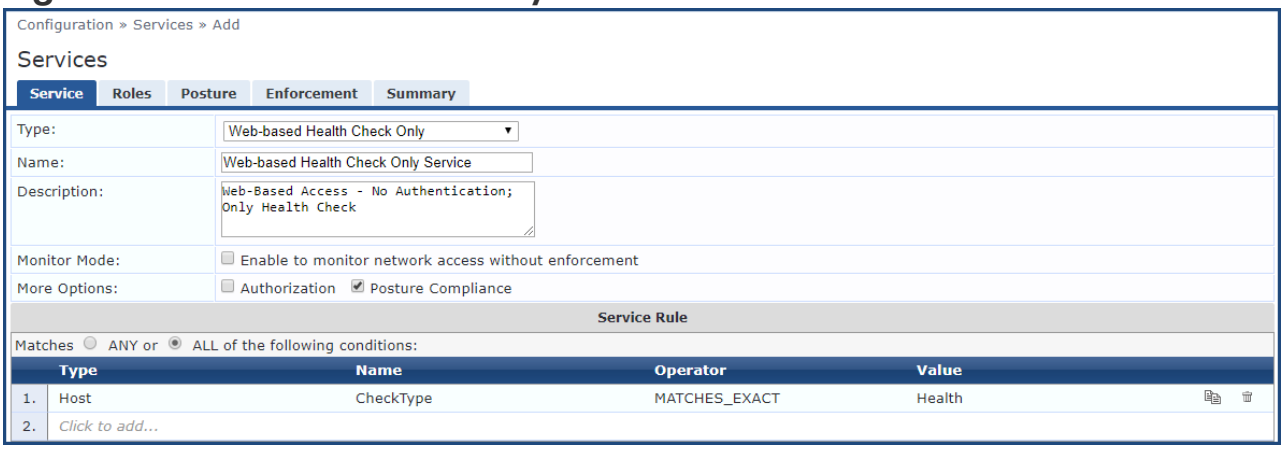
About this task

To create a Web-Based Health Check Only service:

Procedure

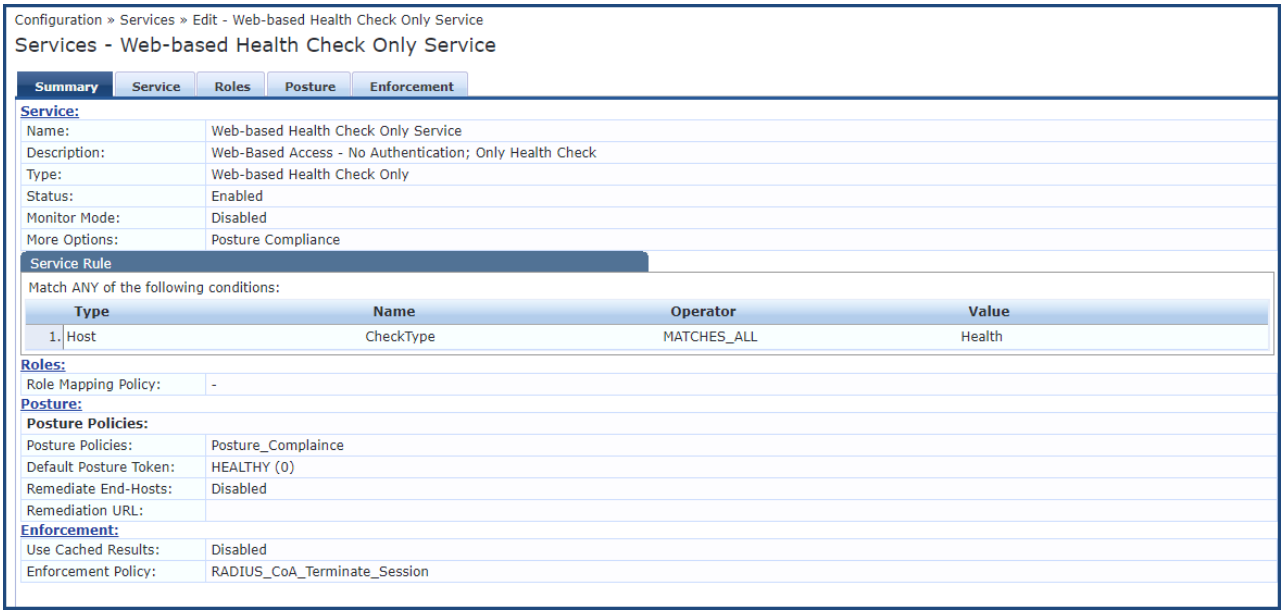
1. Navigate to Configuration > Services, then click the **Add** link. The **Add Services** dialog opens.
2. From the **Type** drop-down, select **Web-based Health Check Only**. The **Web-based Health Check Only** service configuration dialog opens:

Figure 1. Web-based Check Only Service > Service Tab



3. Proceed to configure the Web-based Health Check Only service to match the service settings displayed in Figure 8.

Figure 2. Summary of Web-based Health Check Only Service



4. Specify the Web-based Health Check Only service parameters as described in the following table.

Table 1. Web - based Check Only Service Parameters

Parameter	Action/Description
Service	
Name	Enter the name of the service.
Description	Provide additional information that helps to identify the service.
Type	Select **Web - based Health Check Only** .
Status	Enabled
Monitor Mode	Disabled
More Options	Click the check box for **Posture Compliance**. The **Posture** tab is added.
Service Rule	
Type	Select **Host**.

Parameter	Action/Description
Name	Select **CheckType**.
Operator	Select **MATCHES_ALL**.
Value	Select Health.
Roles	
Role Mapping Policy	Not required.
Posture	
Posture Policies	Select **Posture_Compliance**.
Default Posture Token	**HEALTHY**
Remediate End - Hosts	Disabled
Remediation URL	N/A
Enforcement	
User Cached Results	Disabled
Enforcement Policy	Select **[RADIUS_CoA_Terminate Session]**.

5. Click Save.

   For details on configuring a Web-based Health Only service, refer to Web-based Health Check Only Service .

Enabling the VPN Interface

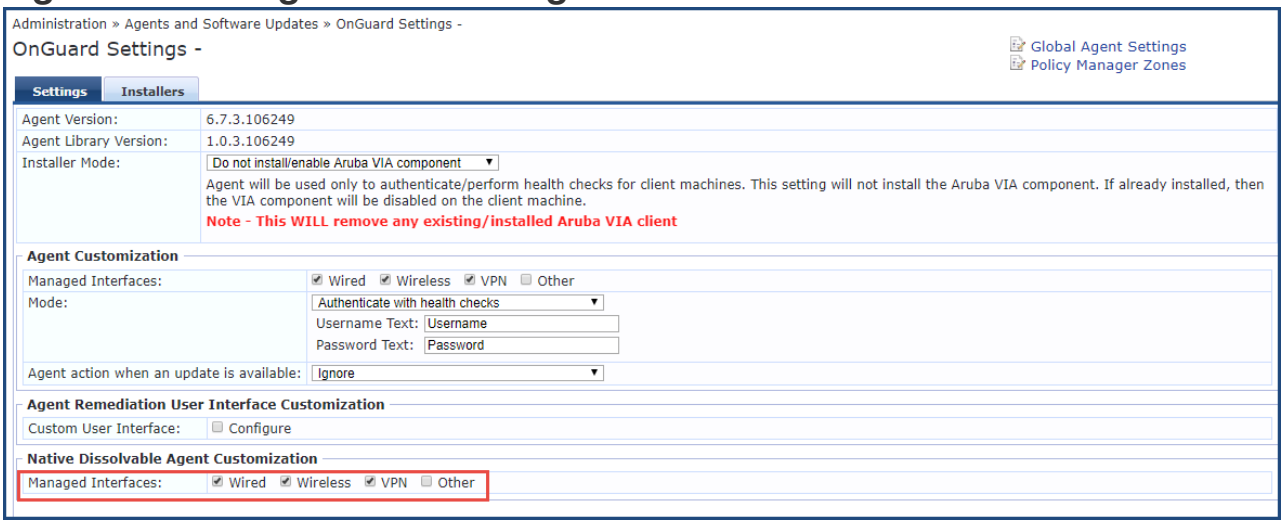
About this task

To enable the VPN interface in the set of managed interfaces:

Procedure

1. Navigate to **Administration > Agents and Software Updates > OnGuard Settings**. The **OnGuard Settings** page opens.

Figure 1. Enabling the VPN Managed Interface



2. In the **Native Dissolvable Agent Customization > Managed Interfaces** section, check (enable) the **VPN** check box. (The **Wired** and **Wireless** interfaces are enabled by default.)

3. Click **Save**.

Access Tracker Requests Reflect Credentials

As shown in Figure 10, the Policy Manager Access Tracker shows that the credentials configured in the RADIUS Enforcement service are the same credentials used in the Guest Web Login and the Webauth service.

Figure 1. Access Tracker Information

10.	RADIUS	user1	RADIUS Enforcement Service	ACCEPT	2018/05/10 14:27:51
10.	Application	user1	Guest Web Login	ACCEPT	2018/05/10 14:30:40
10.	WEBAUTH	user1	Web-based Health check Only Service	ACCEPT	2018/05/10 14:32:37

- The **RADIUS** request is obtained when connected to a VPN.
- The **Application** authentication is obtained when logged into Guest Web Login.
- The **Webauth** request is obtained after health checks have been performed.

Configuring Workflow in Native Agents Only Mode

The Native Dissolvable Agent communicates with Policy Manager Guest portal to send information about endpoints, such as status, health status, remediation messages, and so on. This communication is independent of the operating systems and browsers.

Native Dissolvable Agent supports the following browsers and operating systems:

Table 1. Native Dissolvable Agent: Supported Operating Systems and Browsers

OS	Browsers
Windows	• Internet Explorer • Google Chrome • FireFox (versions less than
macOS	• Safari • Google Chrome • FireFox (versions less than • macOS 14 (Sonoma)
Linux	• FireFox (versions less than

- ClearPass Policy Manager hosts the Native Dissolvable Agent binary files with OnGuard Persistent Agent installers.
- You can use the links to download the binaries in the **OnGuard Settings** page for Windows (.exe) and macOS (.DMG).

Subtopics

Configuring Workflow in Native Agents Only Mode

Configuring Workflow in Native Agents Only Mode

About this task

In Policy Manager Guest, the web login page is enhanced to avoid an additional web authentication service and simplifies the configuration of dissolvable agent flow with the *policy-initiated guest web login method*.

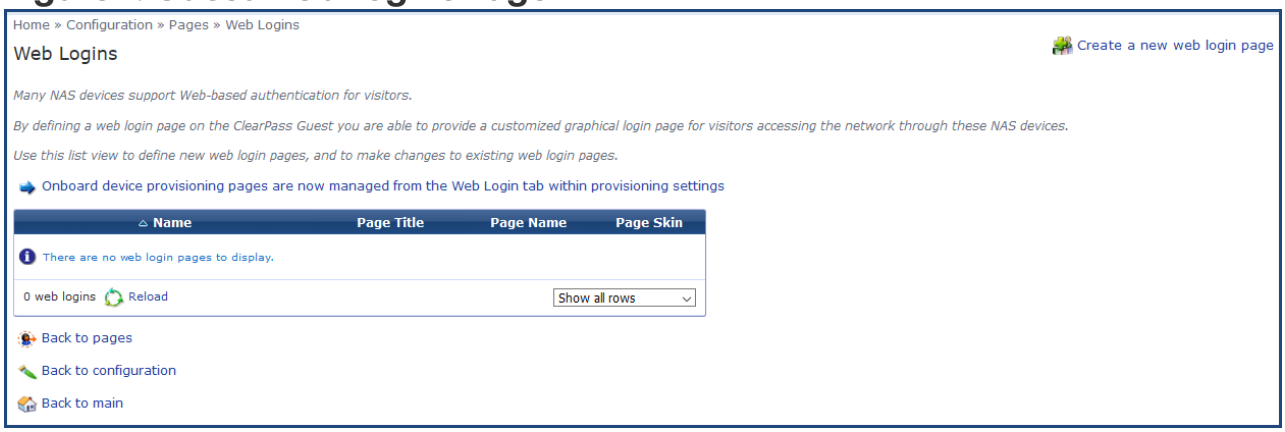
To configure the OnGuard Dissolvable Agent in **Native agents only** mode:

Procedure

1. In Policy Manager Guest, navigate to **Configuration > Pages > Web Logins**.

   The **Web Logins** page opens.

Figure 1. Guest Web Logins Page



2. To edit an existing Web Login page, select the page from the list and click **Edit**.
3. To create new Web Login page, click the **Create a new web login page** link.

   The **Web Login (new)** page opens.

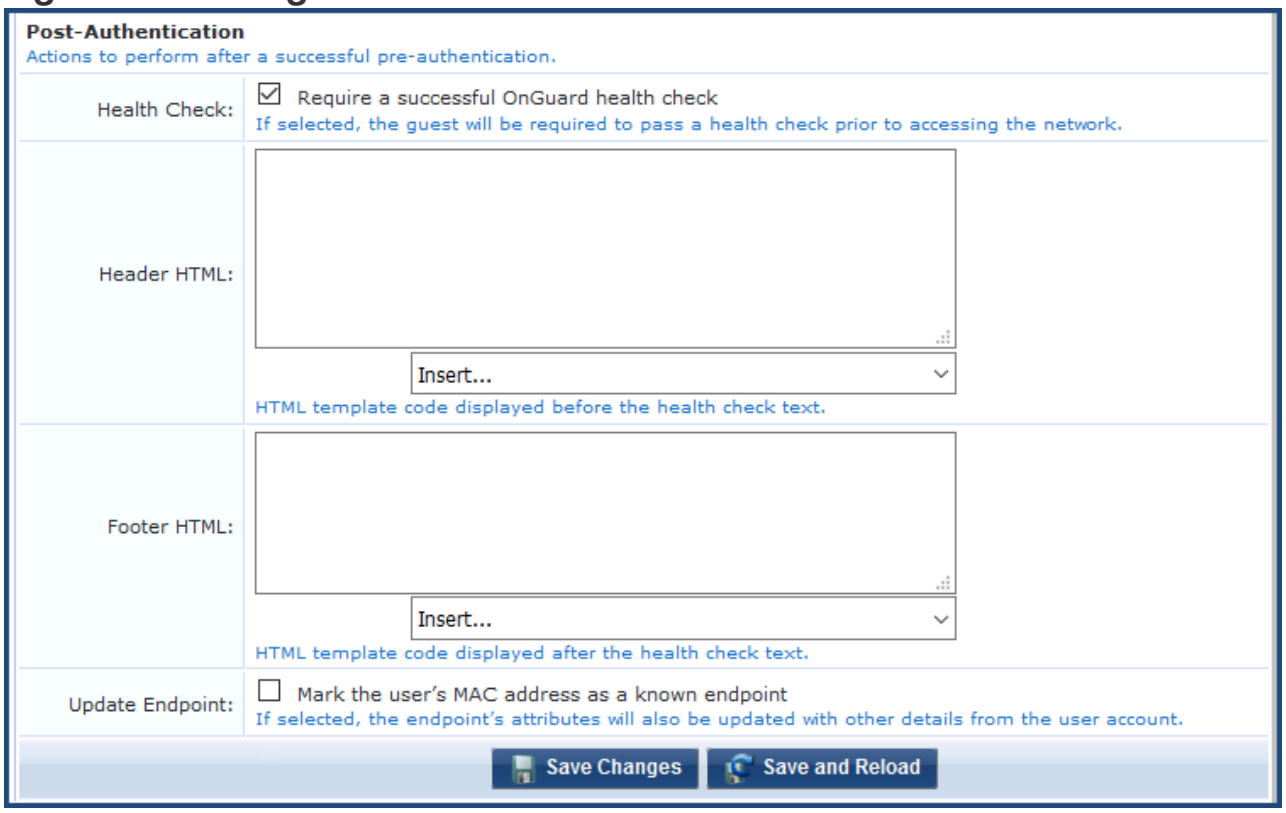
4. As shown in Figure 2, in the *Login Method* field, select the **Policy-initiated - An enforcement policy will control a change of authorization** option from the drop-down.

Figure 2. Specifying the Policy-Initiated Guest Web Login Method

5. Scroll down to the **Post-Authentication** section of the Web Login page and, in the *Health Check* field, select the **Require a successful OnGuard health check** check box.

   Select this check box to require the guest to pass a health check before they can access the network. The health check is done automatically through the OnGuard Dissolvable Agent.

Figure 3. Enabling OnGuard Health Checks



6. Click **Save Changes**.

Configuring Web Agent Flow—Native Agent

To perform health scans on endpoints, you can configure web agent flow in Policy Manager.

Subtopics

Configuring Web Agent Flow in ClearPass Policy Manager
Creating the 802.1X Wireless Service
Creating the Web-based Health Check Only Service
Configuring Web Agent Flow in ClearPass Guest
End-to-End Flow in Native Agents Only Mode

Configuring Web Agent Flow in ClearPass Policy Manager

About this task

To configure a new web agent flow in Policy Manager you will:

Procedure

1. Create an 802.1X wireless service to perform RADIUS authentication and enforce restricted or full access based on end-point posture assessments on the Policy Manager server.
2. Create a **Web-based Health Check Only** service on the Policy Manager server.

Creating the 802.1X Wireless Service

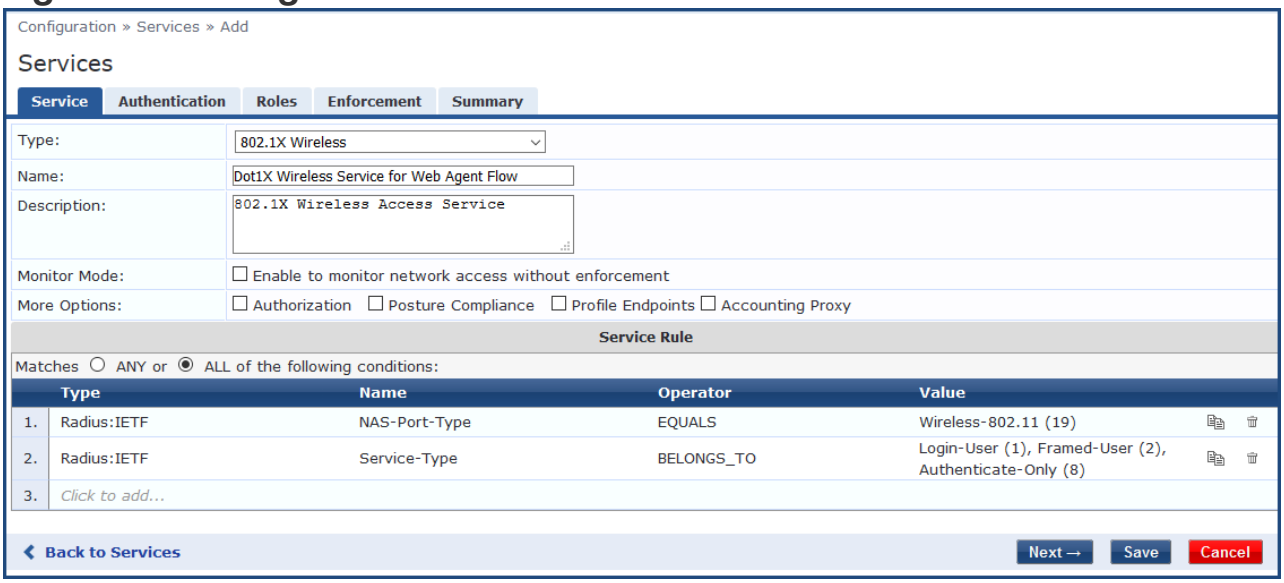
About this task

To create an 802.1X wireless service to perform RADIUS authentication and enforce restricted or full access based on end-point posture assessments (for details on configuring an 802.1X service, see 802.1X Wired, 802.1X Wireless, and 802.1X Wireless Service Template):

Procedure

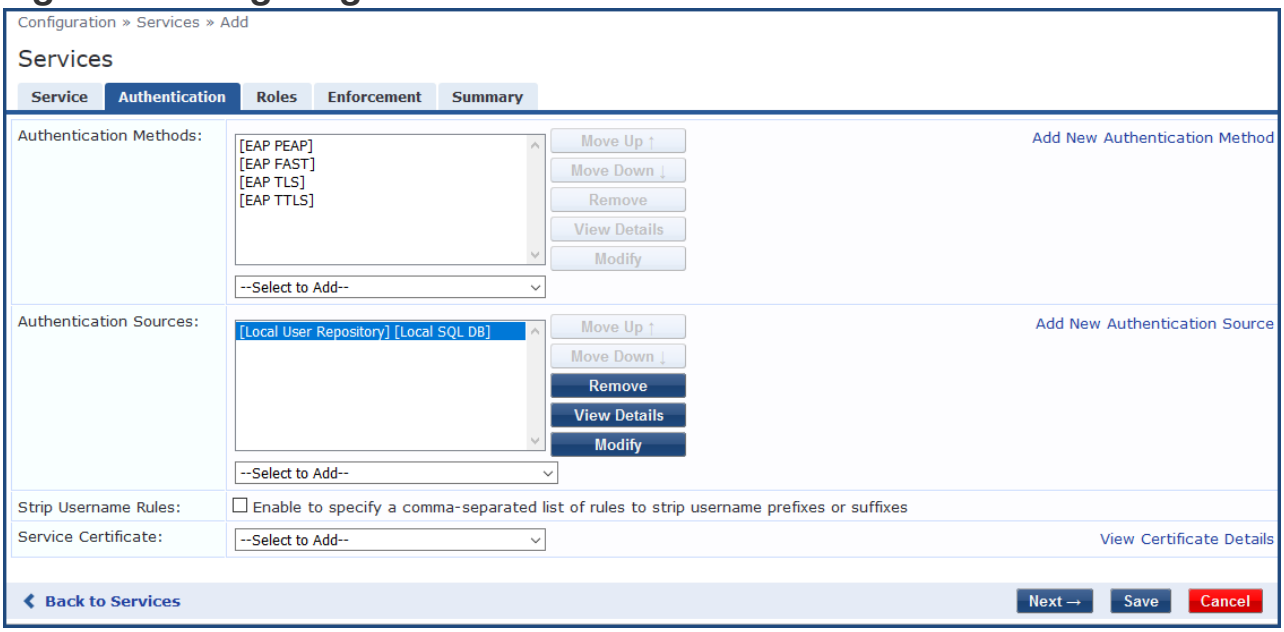
1. In Policy Manager, navigate to **Configuration > Services**. The **Services** page opens.
2. Click **Add**. The **Add Services** page opens.
3. From the **Services** tab **Type** drop-down, select **802.1X Wireless**. The **Service** tab configuration dialog opens.

Figure 1. Creating the 802.1X Wireless Service



4. Click **Next**. The **Authentication** tab configuration dialog opens.

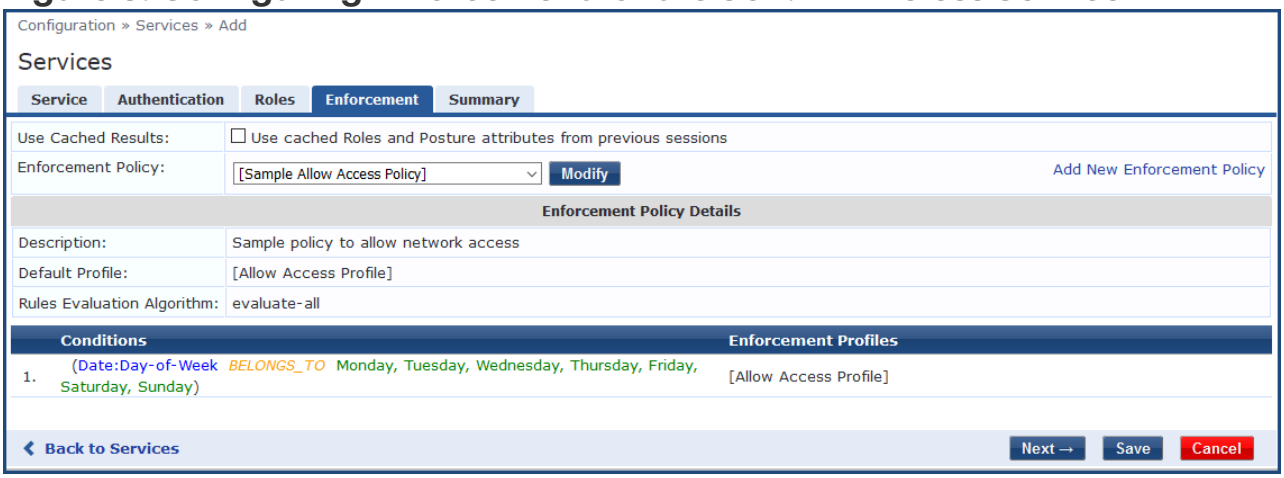
Figure 2. Configuring Authentication for the 802.1X Wireless Service



5. From the **Authentication Sources** drop-down, select **[Local User Repository] [Local SQL DB]**.

6. Click the **Enforcement** tab. The **Enforcement** tab configuration dialog opens.

Figure 3. Configuring Enforcement for the 802.1X Wireless Service



7. Specify the **Enforcement** parameters as described in the following table:

Table 1. 802.1X Wireless Service Enforcement Parameters

Parameter	Action/Description
Use Cached Results	Select this check box to use cached roles and posture attributes from previous sess ions.
Enforcement Policy	Select the preconfigured enforcement p olicy from the drop - down list. This is ma ndatory. If you do not have any preconfigured en forcement policies, click **Add New Enforc ement Policy** to create a new enforceme nt policy.
Enforcement Policy Details	
Description	Displays additional information about the selected enforcement policy.
Default Profile	Displays a default profile applied by Poli cy Manager.
Rules Evaluation Algorithm	Shows the first matched rule.

8. Click **Save**. You return to the **Services** page. This service is added to the set of Policy Manager services.

Creating the Web-based Health Check Only Service

About this task

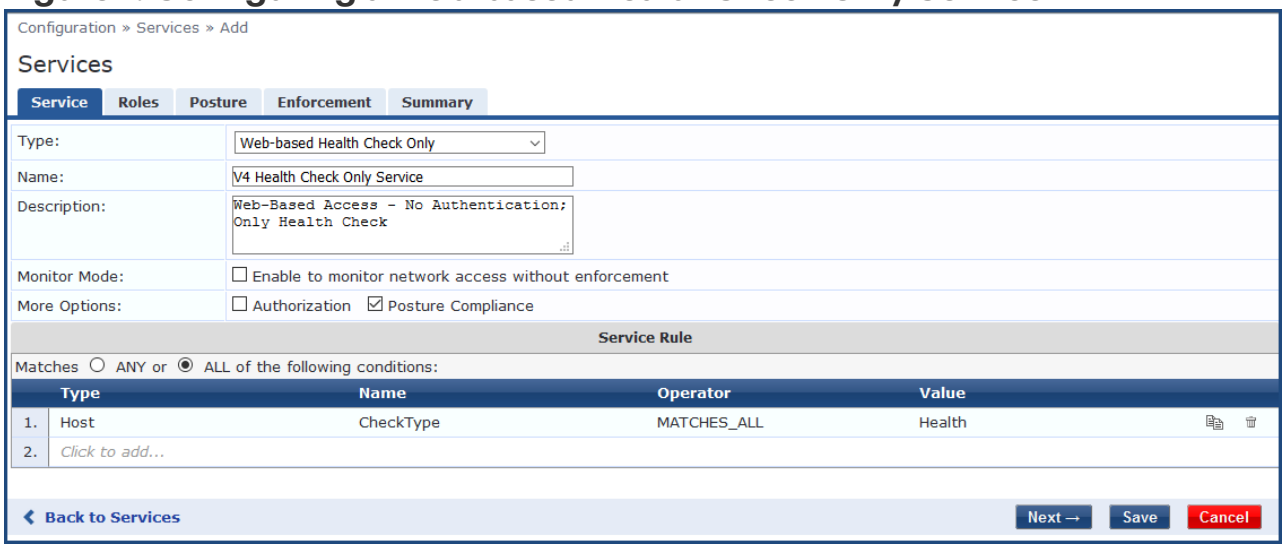
The next task in configuring web agent flow in Policy Manager is to create a Web-based Health Check Only service.

To create a Web-based Health Check Only service:

Procedure

1. Navigate to **Configuration > Services**. The **Services** page opens.
2. Click **Add**. The **Add Services** page opens.
3. From the **Services** tab **Type** drop-down, select **Web-based Health Check Only**. The **Service** tab configuration dialog opens.

Figure 1. Configuring a Web-based Health Check Only Service



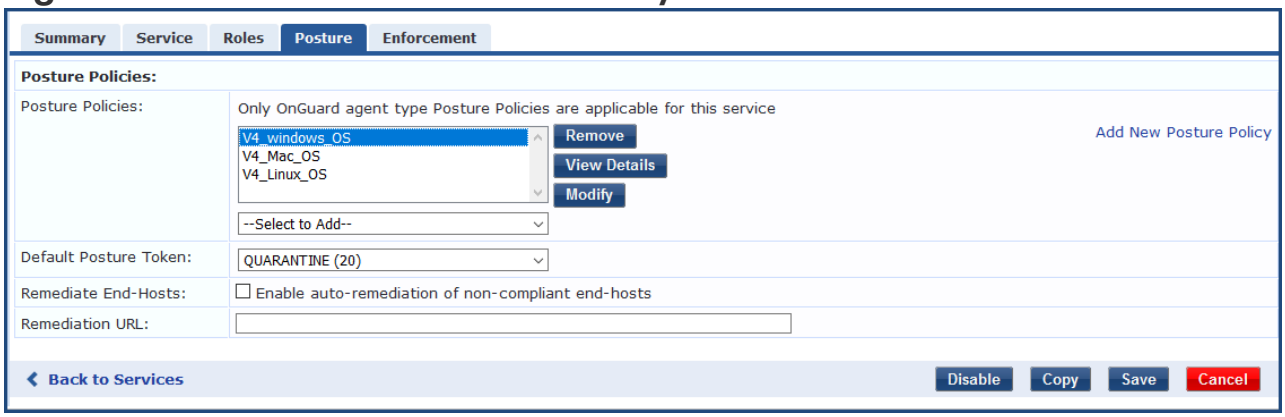
4. Specify the **Service** tab parameters as described in the following table:

Table 1. Web - based Health Check Only > Service Tab Parameters

Parameter	Action/Description
Type	From the **Type** drop - down, select **Web - based Health Check Only**.
Name	Enter the name for this service.
Description	Enter a brief description of this service.
More Options	Select the **Posture Compliance** check box.

5. Select the **Posture** tab. The **Web-based Health Check Only > Posture** configuration tab opens. This service requires that you have one or more OnGuard agent type posture policies configured on the Policy Manager server.

Figure 2. Web-based Health Check Only > Posture Tab



6. To add a new posture policy, click the **Add New Posture Policy** link (for details, see Creating a New Posture Policy).

7. Specify the **Web-based Health Check Only > Posture** tab parameters as described in the following table:

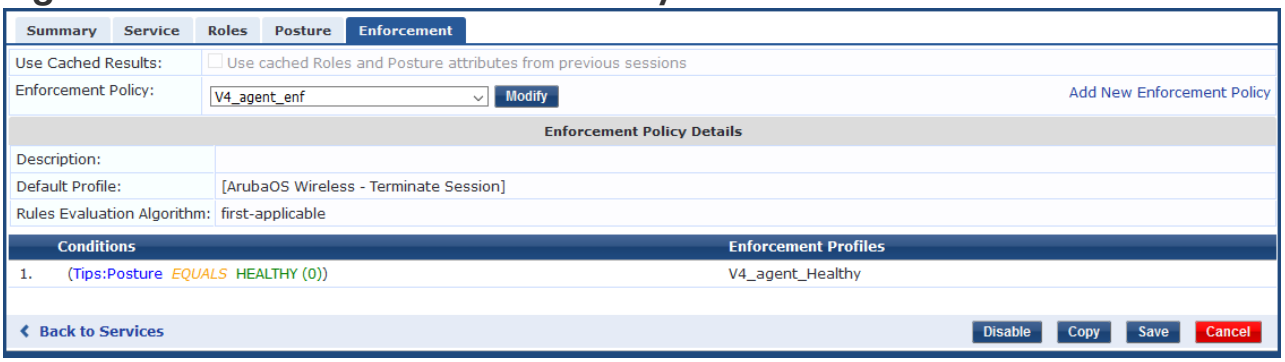
Table 2. Web - based Health Check Only > Posture Tab Parameters

Parameter	Action/Description
Posture Policies	From the **Select to Add**drop - down, select one or more posture policies.

Parameter	Action/Description
Default Posture Token	Select **QUARANTINE (20)**
Remediate End - Hosts	Do not enable this option: Native Dissol vable Agent does not support auto - rem ediation.

8. Click **Next**. The **Web-based Health Check Only > Enforcement** configuration tab opens.

Figure 3. Web-based Health Check Only > Enforcement Tab



9. Click **Save**.

Configuring Web Agent Flow in ClearPass Guest

About this task

In this section, you create a simple Web Authentication service to authenticate users against the ClearPass Guest user database to accept or perform an App authentication request after completing a sandwich flow.

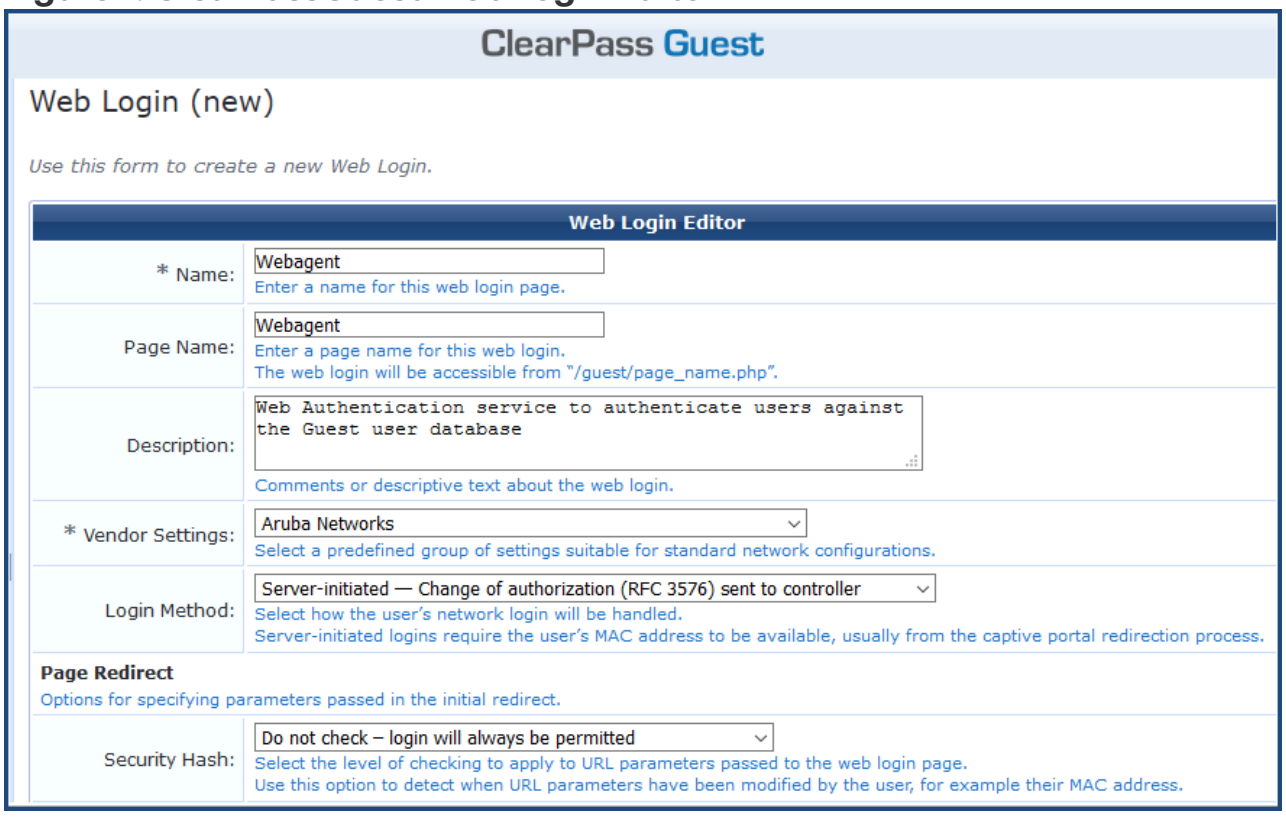
For more information, refer to ClearPass Guest Online Help.

To create a web agent flow in ClearPass Guest:

Procedure

1. In ClearPass Guest, navigate to **Configuration > Pages > Web Logins**. The **Web Logins** page opens.
2. Click the **Create a new web login page** link (in the upper right-hand corner). The **Web Login Editor** page opens:

Figure 1. ClearPassGuest Web Login Editor



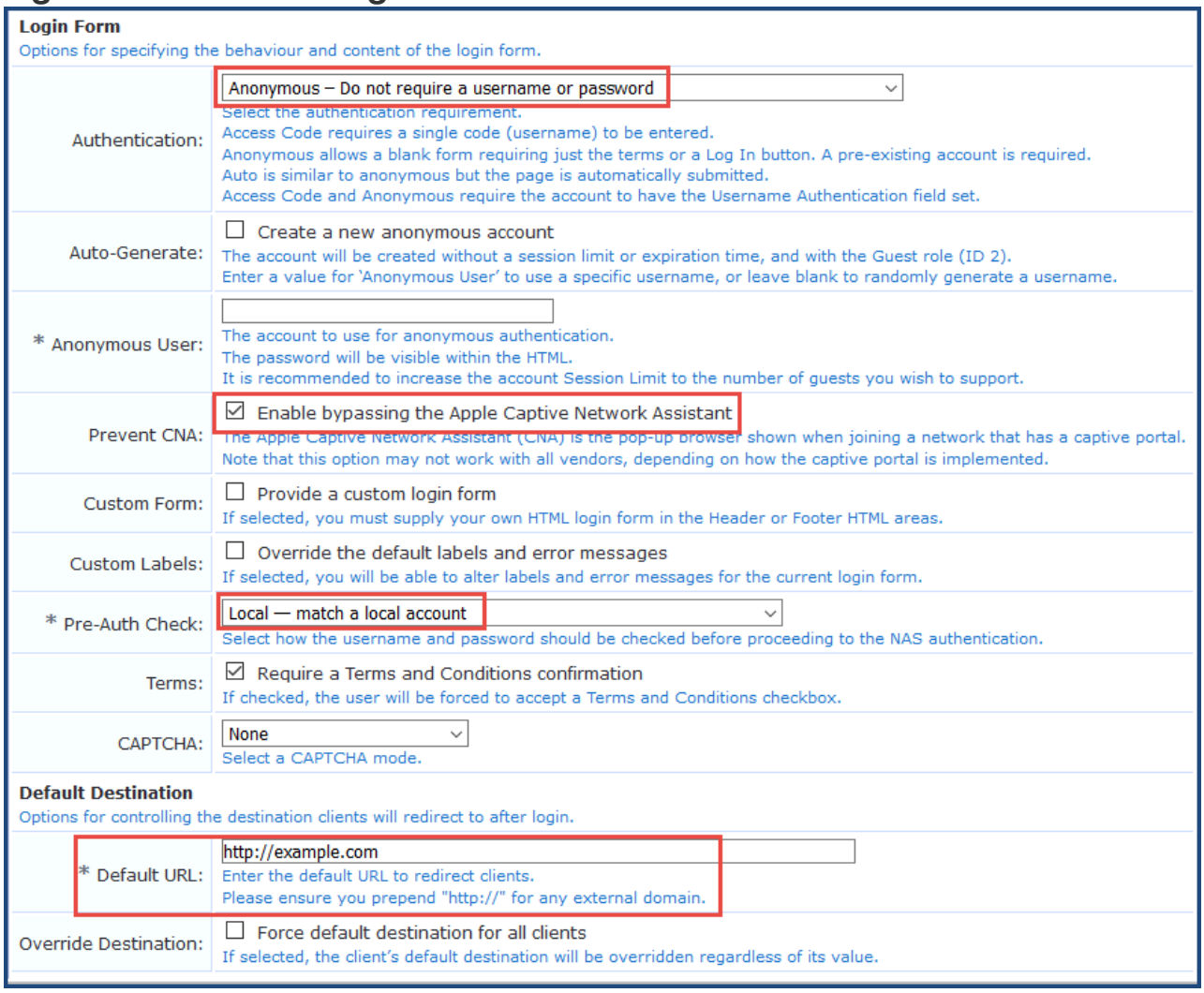
3. Specify the **Web Login Editor** settings as described in the following table:

Table 1. Guest Web Login Editor

Login Field	Action/Description
Name	Enter the name of this login page. This is a required field.
Page Name	Enter the page name for this web login. This page will be accessible from */guest/page_name.php*.
Description	Provide a brief description of this login p age.
Vendor Settings	From the drop - down, select the vendor. This is a required field.
Login Method	Select **Server initiated—Change of auth orization (RFC 3576) sent to controller**.

Login Field	Action/Description
Security Hash	Select **Do not check—login will always b e permitted**.

Figure 2. Guest Web Login Form



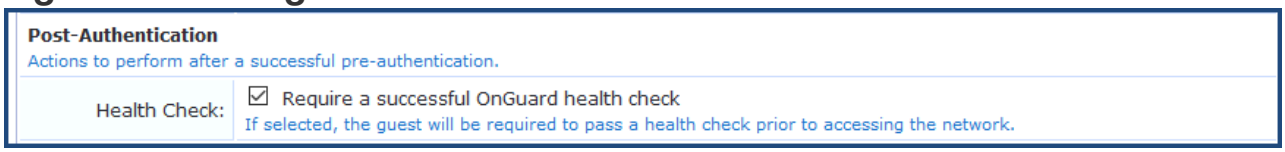
4. Specify the **Login Form**settings as described in the following table:

Table 2. Guest Web Login Form

Login Field	Action/Description
Authentication	From the drop - down, select **Anonymou s — Do not require a username or pas sword**.

Login Field	Action/Description
Prevent CNA	Check the **Enable bypassing the Apple Captive Network Assistant** check box.
Pre - Auth Check	Select **Local - match a local account**.
Default Destination	Specify the **Default URL** to which the clie nt will be redirected after health checks a re completed.

Figure 3. Web Login Post-Authentication



5. Select the **Require a successful OnGuard health check** check box.
6. Click **Save Changes**.

End-to-End Flow in Native Agents Only Mode

About this task

This section describes the end-to-end flow of the OnGuard Dissolvable Agent running in **Native agents only** mode.

Procedure

1. After configuring workflow in Native Agents Only Mode (see Configuring Workflow in Native Agents Only Mode), you are redirected to the Policy Manager Guest Portal where you can download the Native Agent installer.
2. After accepting the terms and conditions for collecting end point posture assessment scan checks and performing remediation actions, run the Native Agent Installer.

   The following figure shows an example of the **Native Dissolvable Agent Login** page:

Figure 1. Native Dissolvable Agent Login Page

> **NOTE**
>
> The Terms specified in the **Login** page are optional. You can configure this optionally by selecting the **Require a Terms and Conditions confirmation** check box in the *Terms* field in the Policy Manager Guest Login Form.

3. Click the **Terms** check box, then click **Log In**. The following OnGuard Agent download prompt appears when you log in to the Native Dissolvable Agent for the first time:

Figure 2. Native Dissolvable Agent Installer Prompt

> **NOTE**
>
> The download options are available only when you log in for the first time. Alternatively, you can download the OnGuard agent by clicking the **Download Policy Manager OnGuard Agent** link.

4. To proceed to download the OnGuard Agent, click **OK**. The **OnGuard Windows Health Checker** binary download dialog opens:

> **NOTE**
>
> If you are running Windows OS, Internet Explorer provides options to **Run** or **Save**. Firefox and Chrome browsers provide the option to only save the .exe files.

Figure 3. Native Dissolvable Agent Binary Downloader

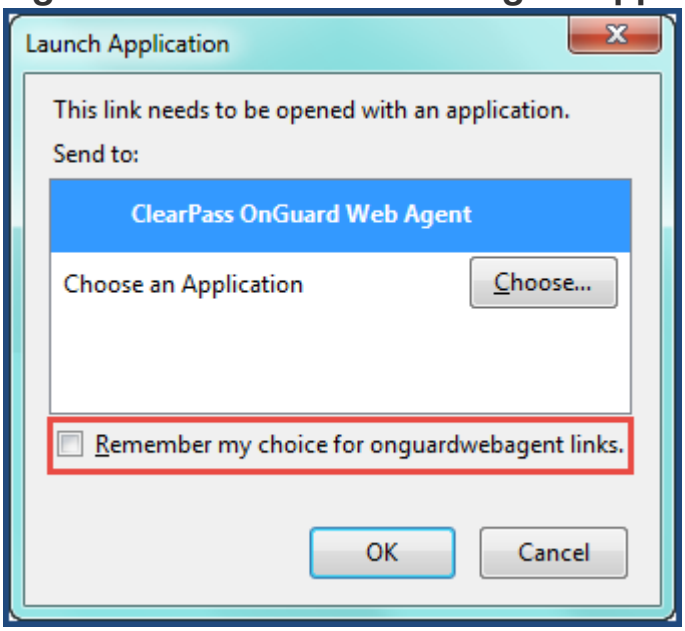
5. To download the OnGuard agent, click **Save File**. The *OnGuard Windows Health Checker.exe* file is downloaded to your system.
6. To install the OnGuard agent, click **Run**.

Figure 4. Native Dissolvable Agent Installation

> **NOTE**
>
> If you are running macOS, Firefox provides options to open the binary file with **DiskImageMounter** or save the .DMG files. Safari and Google Chrome browsers provide the option to **Save** only.

7. From the **Launch Application** page, select the **Policy Manager OnGuard Web Agent** application.

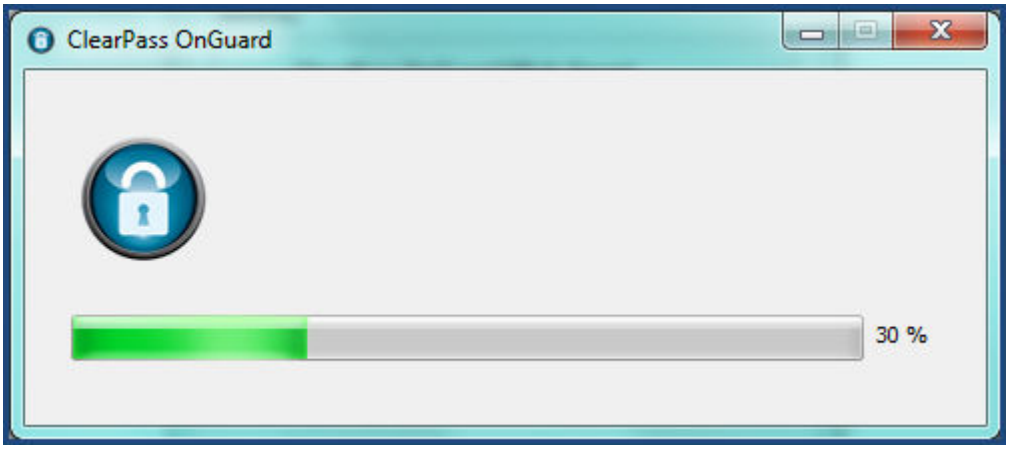
Figure 5. Native Dissolvable Agent Application Launcher



8. To register and perform auto-launch of native OnGuard agent on successive logins, select **Remember my choice for onguardwebagent links**, then click **OK**.

   The following screen shows the installation progress:

Figure 6. Native Dissolvable Agent Installation Progress

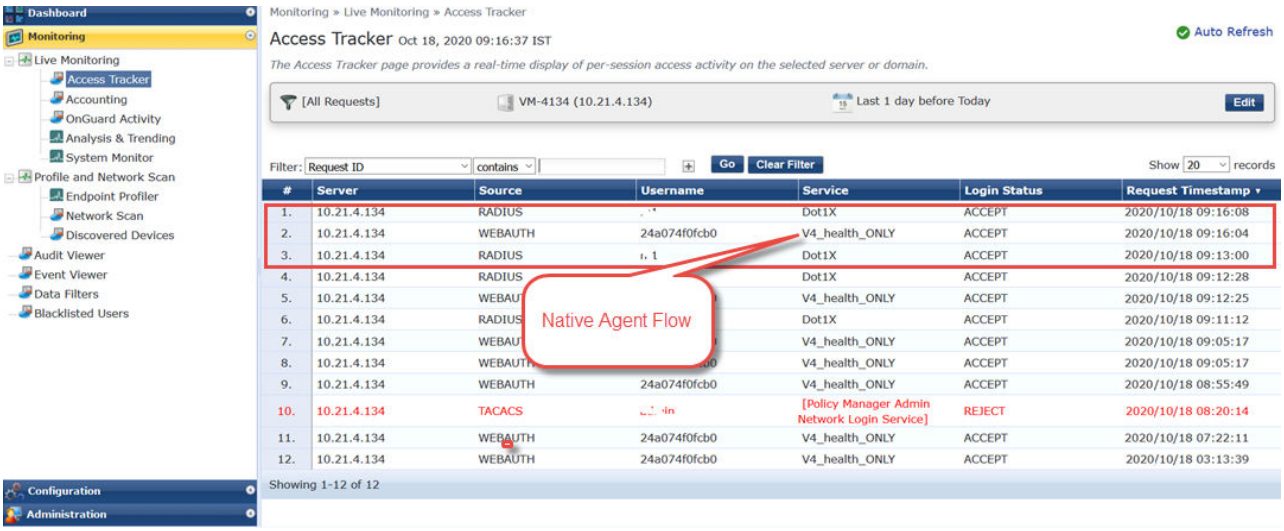


After the successful installation, the health check scanning is initiated. After the health-check scanning is completed, a message screen similar to the following example appears if the client is unhealthy:

Figure 7. Health Check Results

9. Take the appropriate actions to fix the issues listed in remediation and agent enforcement messages, then click **Scan Again**.
0. Repeat this step until the client becomes healthy. Once the client is healthy, you can access the destination URL.
1. You can track the events with the end-to-end flow in the **Access Tracker**. The following figure is an example of the **Access Tracker** page showing the Native Dissolvable Agent flow:

Figure 8. Access Tracker Page



NOTE

The Auto-launch feature works in the *Native agents only* and *Java Only* modes without user intervention, except for needing to configure Terms on the Guest Login page.

Subtopics

About Auto-Login
Accessing Native Dissolvable Agent Flow Logs

About Auto-Login

The Native Dissolvable Agent supports the **Auto-Login** method, which eliminates the **Require a Terms and Conditions confirmation** check box in the **Guest Web Login** page by avoiding the web page and submitting automatically.

Accessing Native Dissolvable Agent Flow Logs

Accessing the Native Dissolvable Agent flow logs can help with troubleshooting issues.

- In *Windows*, Native Dissolvable Agent flow logs are available at:

  %appdata%Aruba Networks/ ClearPassOnguard Temp/Logs

- In *macOS*, the Native dissolvable agent flow logs are available at:

~/Library/Logs/ClearPassOnGuardTemp/logs

Client-Side Settings for OnGuard Agents

- Setting to Prevent OnGuard Agent From Accessing Internet URLs for CRL Checks
- Setting to Disable Network Interface Bounce After Wake-up From Sleep
- Setting to Disable Usage of PowerShell by OnGuard Agents During Health Collection
- Setting to relaunch the OnGuard Agent when the Policy Manager Server is not reachable

Subtopics

Setting to Prevent OnGuard Agent From Accessing Internet URLs for CRL Checks
Setting to Disable Network Interface Bounce
Setting to Disable Usage of PowerShell by OnGuard Agents During Health Collection
Setting to relaunch the OnGuard Agent when the Policy Manager Server is not reachable

Setting to Prevent OnGuard Agent From Accessing Internet URLs for CRL Checks

As part of integrity checks, Policy Manager OnGuard Agents for Windows performs sign verification of OnGuard Agent DLLs before loading them. During sign verification, Windows OS can access CRL (Certificate Revocation List) servers/Internet to update the local CRL Cache. Policy Manager OnGuard Agents for Windows can instruct Windows OS to perform CRL checks without accessing CRL servers/Internet (that is, using the existing local CRL cache) using the value of the **CacheOnlyCRLCheck** registry key in the following locations:

- OnGuard Persistent Agent: *HKLM|Software|Aruba Networks|ClearPassOnGuard*
- Agentless OnGuard: *HKLM|Software|Aruba Networks|AgentlessOnGuard*

Subtopics

Registry Key Details

Registry Key Details

- If the value of *CacheOnlyCRLCheck = 0*, or if the key is not present, during sign verification, OnGuard Agents will instruct Windows OS to update the CRL Cache by accessing CRL servers/Internet, if required (default behavior).
- If the value of *CacheOnlyCRLCheck = 1*, during sign verification, OnGuard Agents will skip going online to update CRL Cache and instead it will do CRL checks using local cache.

  The type for **CacheOnlyCRLCheck** should be **REG_DWORD**.

Setting to Disable Network Interface Bounce

By default, Policy Manager OnGuard Agents bounce network interfaces when a client comes out of sleep mode, but admins can also bounce network interfaces in other cases, such as when the OnGuard UI is closed on a Windows client.

Subtopics

Windows OS
macOS

Windows OS

Policy Manager OnGuard for Windows can bounce the network interface after waking up from sleep mode, or completely disable network interface bounce.

Subtopics

To bounce the network interface after waking up from sleep mode:
To completely disable network interface bounce

To bounce the network interface after waking up from sleep mode:

Use the value of the **DisableBounceAfterWakeup** registry key in following locations:

- OnGuard Persistent Agent: *HKLM|Software|Aruba Networks|ClearPassOnGuard*

- Agentless OnGuard: *HKLM|Software|Aruba Networks|AgentlessOnGuard*

  Registry Key Details:

- If the value of *DisableBounceAfterWakeup = 0*, or if the key is not present, OnGuard Agent will bounce the network interface after waking up from sleep mode (default behavior).
- If the value of *DisableBounceAfterWakeup = 1*, OnGuard Agent will not bounce the network interface after waking up from sleep mode.
- The type for **DisableBounceAfterWakeup** should be **REG_DWORD**.

To completely disable network interface bounce

Admins can completely disable the bouncing of network interfaces on clients like Windows AD\DNS servers using the value of the DisableInterfaceBounce registry key in the following locations:

- OnGuard Persistent Agent: *HKLM|Software|Aruba Networks|ClearPassOnGuard*
- Agentless OnGuard: *HKLM|Software|Aruba Networks|AgentlessOnGuard*

  Registry Key Details:

- If the value of *DisableInterfaceBounce = 0*, or if the key is not present, OnGuard Agent will bounce the network interface (default behavior).
- If the value of *= 1*, OnGuard Agent will not bounce the network interface in any case, even when Bounce is enabled in the Agent Enforcement Profile.
- The type for **DisableInterfaceBounce** should be **REG_DWORD**.

macOS

The Policy Manager OnGuard Agent for macOS can bounce the network interface after waking up from sleep mode by checking the presence of the */Library/Application Support/ClearPassOnGuard/.disableBounceAfterWakeup* file.

> **NOTE**
> A period (.) is included at the beginning of this file name.

Subtopics

File Details

File Details

- If the *.disableBounceAfterWakeup* file is not present, OnGuard Agent will bounce the network interface after waking up from sleep mode (default behavior).
- If the *.disableBounceAfterWakeup* file is present, OnGuard Agent will not bounce the network interface after waking up from sleep mode.

Setting to Disable Usage of PowerShell by OnGuard Agents During Health Collection

Policy Manager OnGuard on Windows uses Microsoft PowerShell™ to get the list of virtual machines that were created using Hyper-V, and to get the Real-Time Protection (RTP) status of Windows Defender. Usage of PowerShell for these functions can be disabled using the value of the **DisablePowerShell** registry key in following locations:

- OnGuard Persistent Agent: *HKLM|Software|Aruba Networks|ClearPassOnGuard*
- Agentless OnGuard: *HKLM|Software|Aruba Networks|AgentlessOnGuard*

Subtopics

Registry Key Details

Registry Key Details

- If the value of *DisablePowerShell = 0*, or if the key is not present, OnGuard Agents will use PowerShell to get the list of virtual machines created using Hyper-V, and to get the RTP status of Windows Defender (default behavior).
- If the value of *DisablePowerShell = 1*, OnGuard Agents will not use PowerShell to get the list of virtual machines created using Hyper-V, nor to get the RTP status of Windows Defender.

  The type for **DisablePowerShell** should be **REG_DWORD.**

Setting to relaunch the OnGuard Agent when the Policy Manager Server is not reachable

About this task

The OnGuard Agent on MacOS will restart/relaunch when following conditions are met:

Procedure

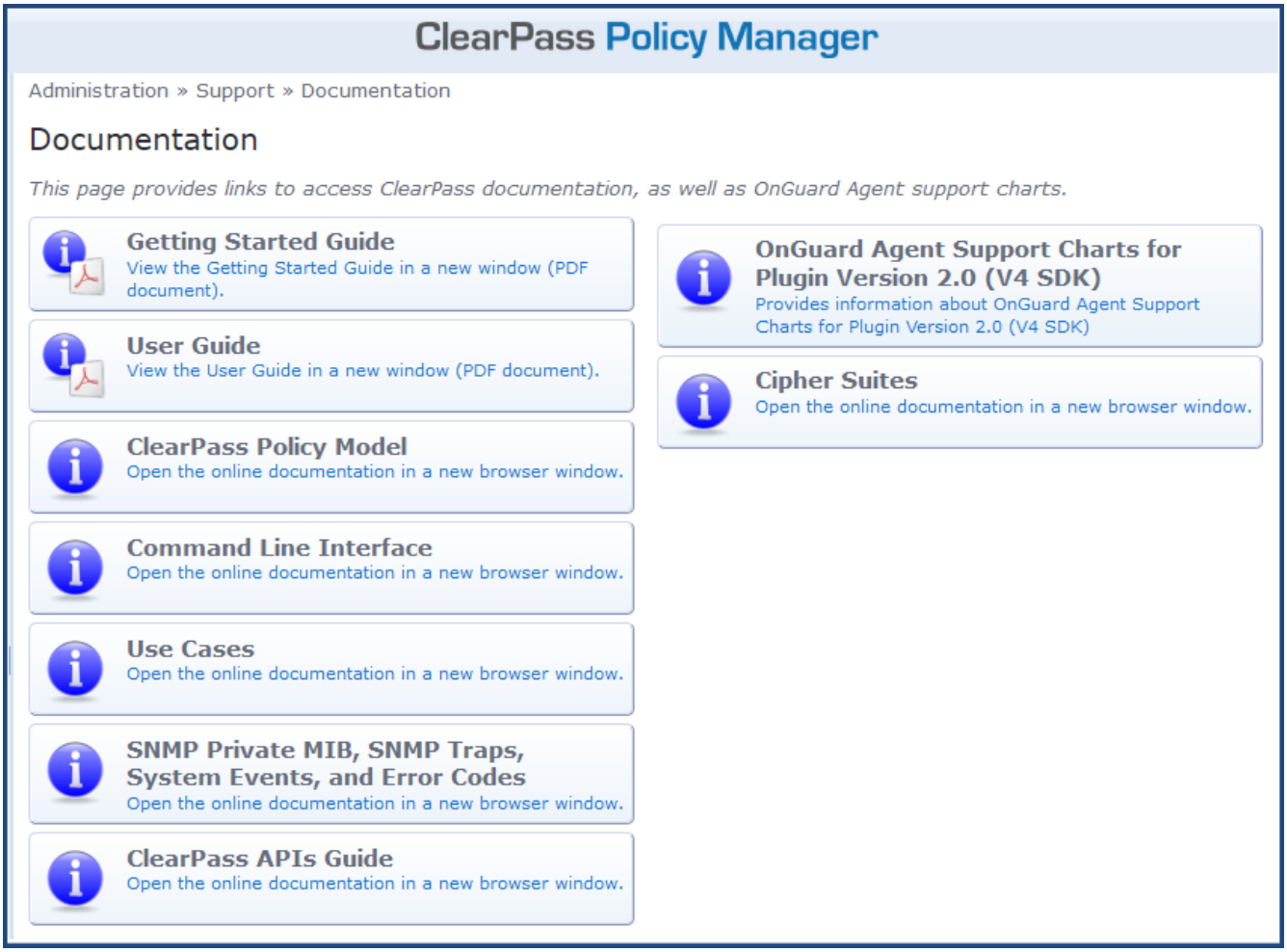
1. The "~/Library/Application\ Support/ClearPassOnGuard/.relaunchOnGuardOnError" file is present on the client. By default, the "~/Library/Application\ Support/ClearPassOnGuard/.relaunchOnGuardOnError" file is not present on a client. The file can be created using an Agent Script Enforcement profile.
2. The Policy Manager Server is not reachable from the OnGuard Agent, but it is reachable from the OnGuard Agent Helper.
3. The client is not in sleep mode.

> **NOTE**
>
> This is applicable only for the Persistent Agent on MacOS.

Accessing the OnGuard Support Chart for Plugin Version 2.0

For information about the OnGuard Agent Support Chart for Plugin Version 2.0 (OnGuard Detection V4 SDK) that is included with ClearPass Policy Manager, navigate to **Administration > Support > Documentation > OnGuard Agent Support Charts**

Figure 1. OnGuard Agent Support Chart for Plugin Version 2.0 (V4 SDK)



Upgrading From OnGuard Plugin Version 1.0 to 2.0

This section contains the following information:

- Overview
- Creating a New Enforcement Profile to Set the SDK Type
- Modifying an Existing Enforcement Policy for OnGuard Plugin v2.0
- Creating a New Posture Policy for OnGuard Plugin v2.0 Agents
- Creating a WebAuth Service for OnGuard Plugin v2.0 Agents
- Important Points

Subtopics

Overview
Creating a New Enforcement Profile to Set the SDK Type

Modifying an Existing Enforcement Policy for OnGuard Plugin v2.0
Creating a New Posture Policy for OnGuard Plugin v2.0 Agents
Creating a WebAuth Service for OnGuard Plugin v2.0 Agents
Important Points

Overview

The Policy Manager OnGuard Agents for Windows and macOS support OnGuard plugin version 2.0, which provides enhanced product detection. The new OnGuard plugin version 2.0 is based on the **OESIS V4 SDK**, while the earlier plugin version 1.0 is based on the **OESIS V3 SDK**. OnGuard continues to use plugin version 1.0 and your existing V3 SDK policies until you explicitly upgrade to plugin version 2.0, as described in this section.

> **CAUTION**
> Customers who use Policy Manager OnGuard must upgrade to the OnGuard Plugin version 2.0 (V4 SDK) in order to maintain application signature and virus definition updates. As of May 1, 2018, the V3 SDK and AV Updates for V3 SDK are no longer supported by OPSWAT. Since virus definitions are updated at least once a day, and sometimes several times a day, it is important to maintain regular automatic updates.

To upgrade to plugin version 2.0, you will first upgrade the OnGuard agents, after which you create a new enforcement profile, enforcement policy, and a Web Auth service. Finally, you will need to modify any existing V3 SDK enforcement policies to use the V4 SDK.

Creating a New Enforcement Profile to Set the SDK Type

About this task

The first task in upgrading to the OnGuard plugin version 2.0 is to create a new enforcement profile where you set the attribute **SDK Type** to **V4**.

To create the OnGuard plugin version 2.0 enforcement profile:

Procedure

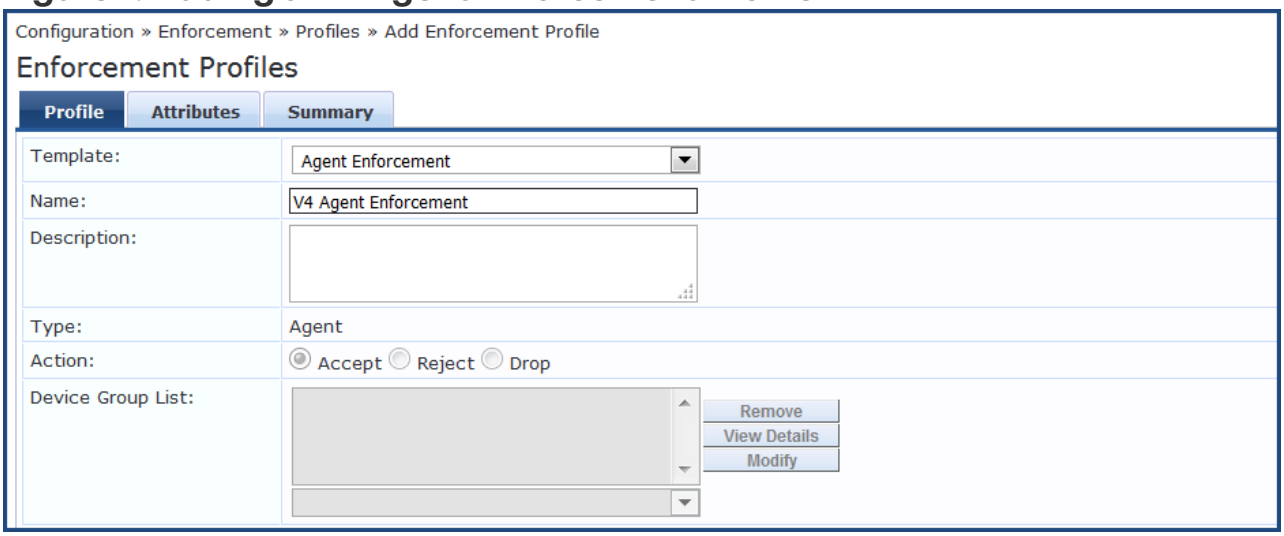
1. Navigate to **Configuration > Enforcement > Profiles**.

   The **Enforcement Profiles** page opens.

2. Click the **Add** link.

   The **Add Enforcement Profile** dialog opens.

Figure 1. Adding a V4 Agent Enforcement Profile



3. Specify the **Add V4 Agent Enforcement Profile** parameters as described in the following table:

Table 1. Add V4 Agent Enforcement Profile Parameters

Parameter	Action/Description
Template	Select **Agent Enforcement**.
Name	Enter a name for this enforcement profile .
Description	Optionally (but recommended), add a d escription of this enforcement profile.
Type	When you select the *Agent Enforcement* template, the enforcement profile **Type** is set automatically to **Agent**.
Action	Keep the default action: **Accept**.
Device Group List	The **Device Group List** is no longer pertin ent and this option is grayed out.

4. Click **Next**.

The **Attributes** dialog opens.

Figure 2. Specifying the SDK Type Attribute to V4

5. Optionally (but recommended), specify a message in the **Message** attribute.
6. Select **Click to add**, then make the following selections:
   - Attribute Name: **SDK Type**
   - Attribute Value: **V4**
7. Click **Save**.

   The new enforcement profile is added.

Modifying an Existing Enforcement Policy for OnGuard Plugin v2.0

About this task

If you have an existing enforcement policy of the WebAuth service that is being used for OnGuard plugin version 1.0: V3 SDK, you must modify the enforcement policy to support OnGuard plugin version 2.0.

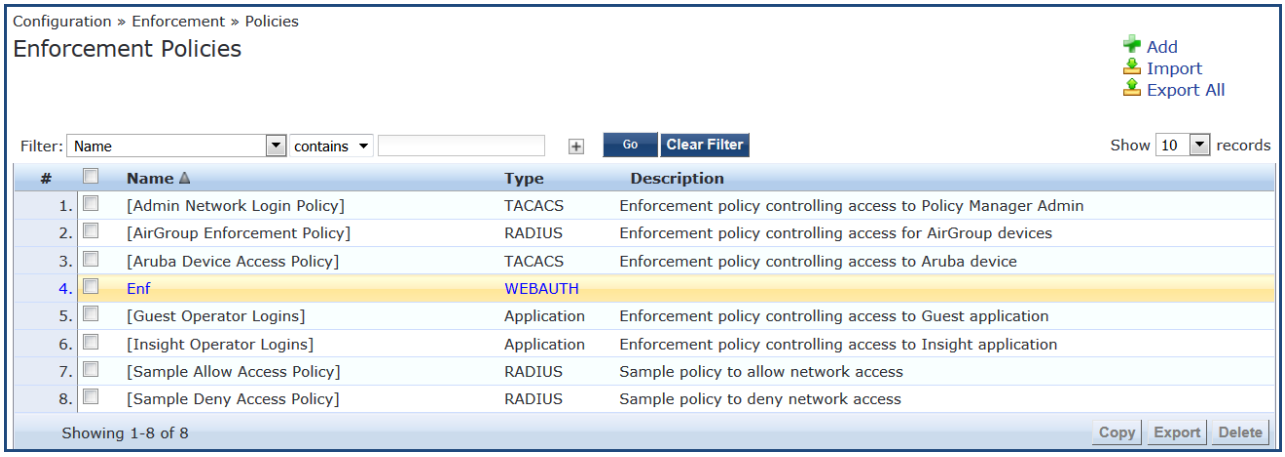
To modify an existing enforcement policy to support OnGuard plugin version 2.0:

Procedure

1. Navigate to **Configuration > Enforcement > Policies**.

   The **Enforcement Policies** page opens.

   Figure 1. Enforcement Policies Page

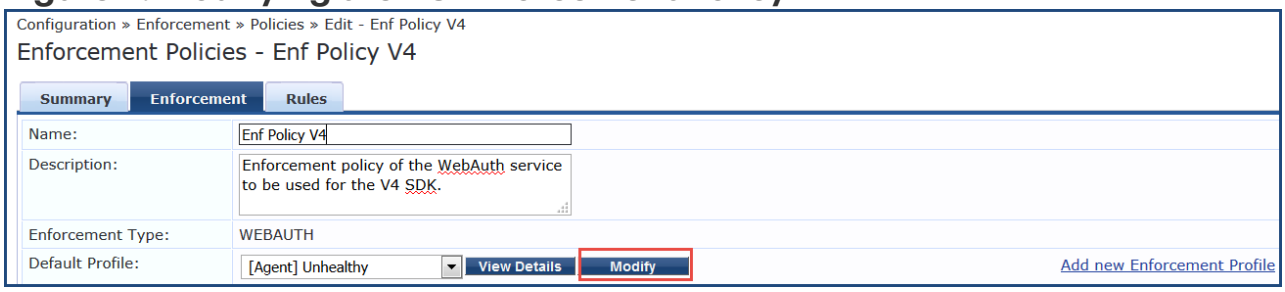


2. Select the enforcement policy of the WebAuth service that is being used for the OnGuard plugin version 1.0.

   The **Edit Enforcement Policies** page opens.

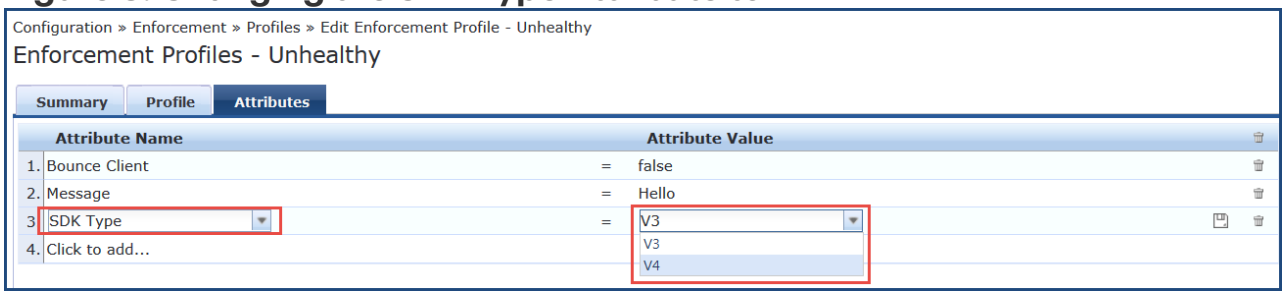
3. Select the **Enforcement** tab.

Figure 2. Modifying the V3 Enforcement Policy



4. Update the **Name** and **Description** if necessary.
5. Click **Modify**.
6. Select the **Attributes** tab.

Figure 3. Changing the SDK Type Attribute to V4



7. Change the **SDK Type > Attribute Value** to **V4**, then click **Save**.

The Enforcement Policy has been updated to support the OnGuard plugin version 2.0: V4 SDK. When the agent next performs a health check, it picks OnGuard plugin version 2.0.

Creating a New Posture Policy for OnGuard Plugin v2.0 Agents

About this task

The supported posture policy for the OnGuard plugin version 2.0 is required because many third-party products that were not supported by OnGuard plugin version 1.0 are supported by OnGuard plugin version 2.0. Also, the names of some of the antivirus products that are recognized by the OnGuard plugin version 1.0 are changed in OnGuard plugin version 2.0. When you create a new posture policy, by default the new posture policy uses V4 support charts (see Accessing the OnGuard Support Charts).

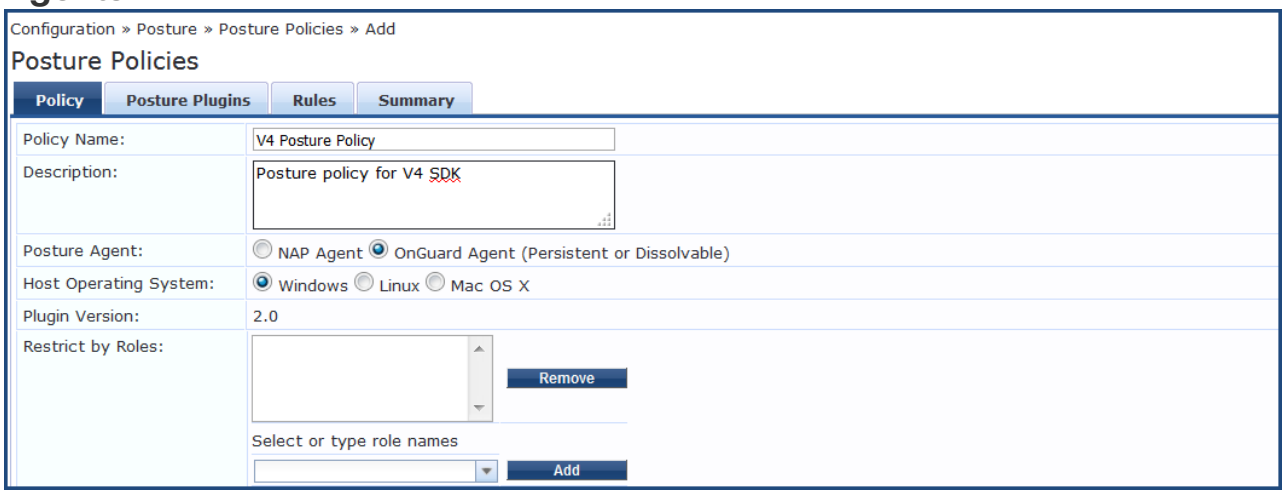
To create a posture policy for OnGuard plugin version 2.0 V4 SDK agents:

Procedure

1. Navigate to **Configuration > Posture > Posture Policies**.

   The **Add Posture Policies** page opens.

   Figure 1. Creating a Posture Policy for OnGuard Plugin Version 2.0 V4 SDK Agents



2. Specify the **Add V4 Posture Policy** parameters as described in the following table:

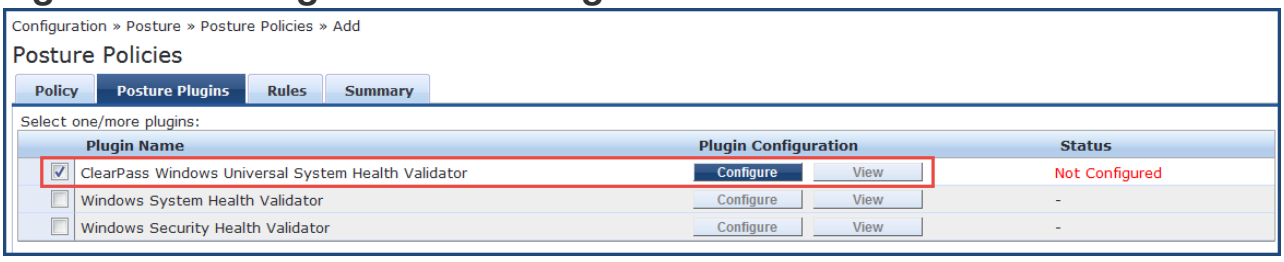
Table 1. Adding V4 Posture Policy Parameters

Parameter	Action/Description
Policy Name	Enter the name of this posture policy.
Description	Optionally (but recommended), add a d escription of this posture policy.
Posture Agent	Specify **OnGuard Agent** (the default).
Host Operating System	Specify **Windows** (the default).
Plugin Version	Plugin version **2.0** is specified by default. This is the plugin version required by the V4 SDK.
Restrict by Roles	Configure the roles as required by your in stallation. For more information on role configuratio n, see Adding and Modifying Roles.

3. Click **Next**.

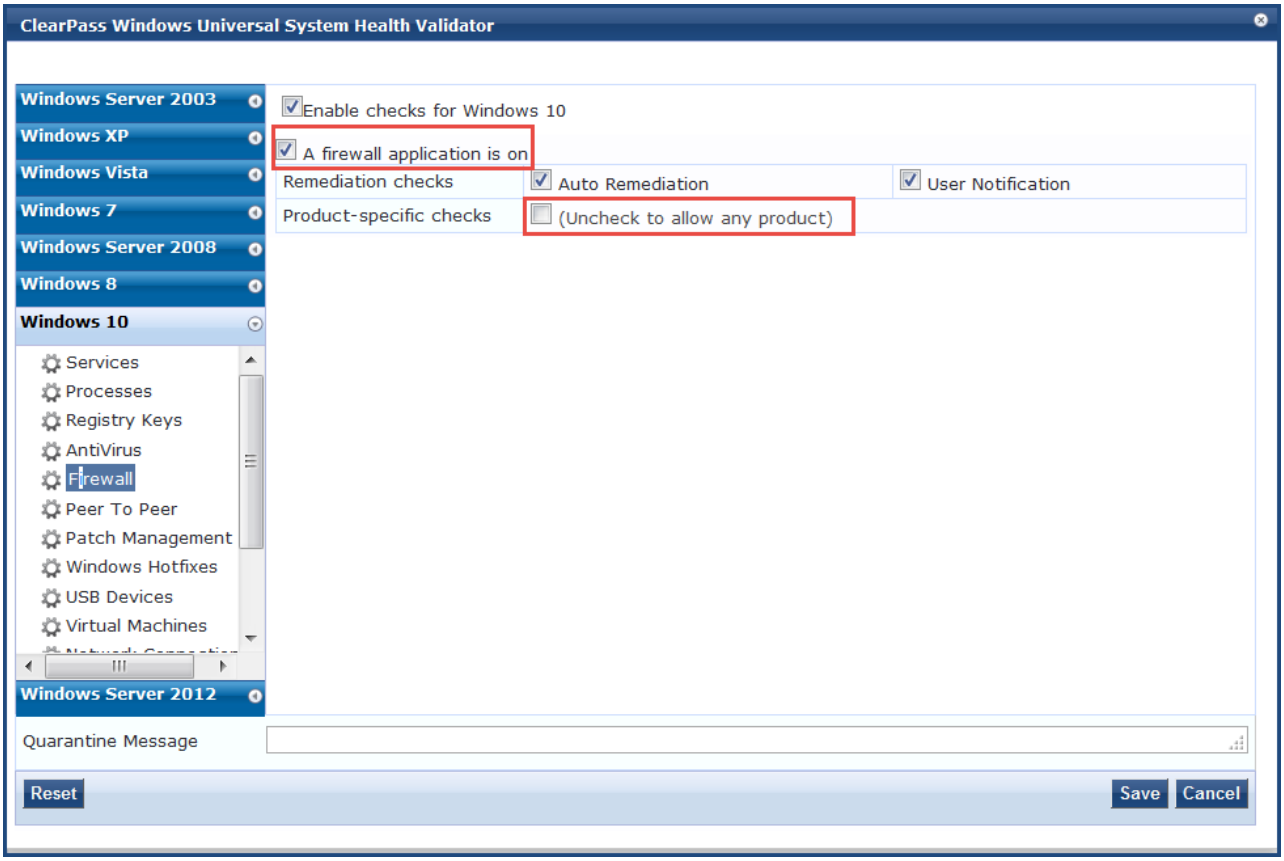
   The **Posture Plugins** page opens.

Figure 2. Selecting the Posture Plugin



4. Click the check box for the **Policy Manager Windows Universal System Health Validator**, then click **Configure**:

Figure 3. Configuring the V4 Posture Plugin



5. Specify the **Policy Manager Windows Universal System Health Validator** parameters as described in the following table:

Table 2. Add V4 Posture Plugin Parameters

Parameter	Action/Description
Windows OS list	Select the Windows version of choice.
Enable checks for Windows <*version*>	Select the check box for **Enable checks** for the selected version of Windows.
Firewall	From the list of Windows checks, select **Firewall**.
Firewall application	Select the check box for **A firewall application is on**.
Product - specific checks	To allow any firewall product, uncheck **Product - specific checks**.

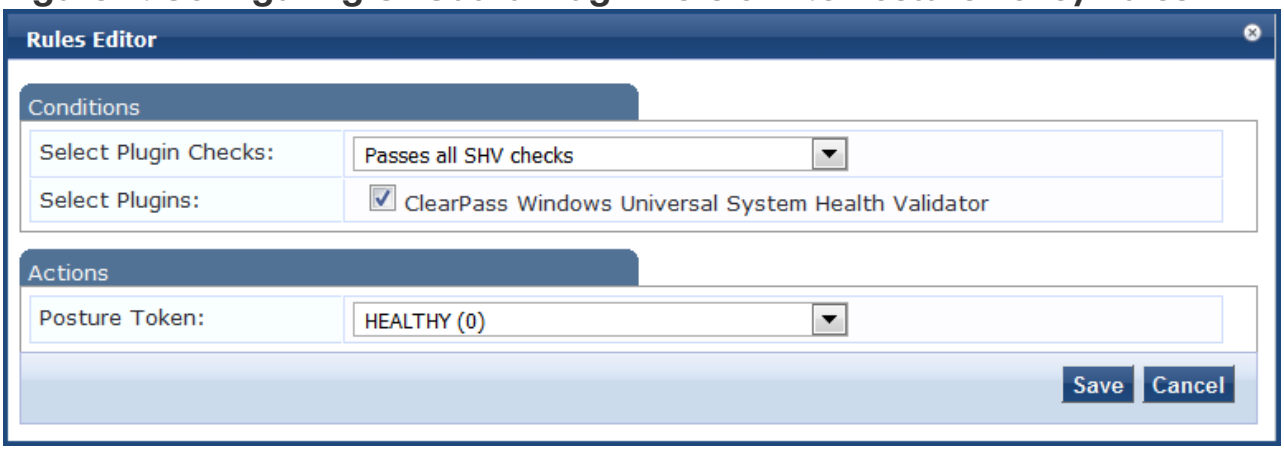
6. Click **Save**.

Once you have defined the posture hosts, agents, and plugins, you must configure the rules for the posture policy.

7. Select the **Rules** tab, then click **Add Rule**.

   The **Rules Editor** opens.

Figure 4. Configuring OnGuard Plugin Version 2.0 Posture Policy Rules



8. Specify the **Rules Editor** parameters as described in the following table, then click **Save**:

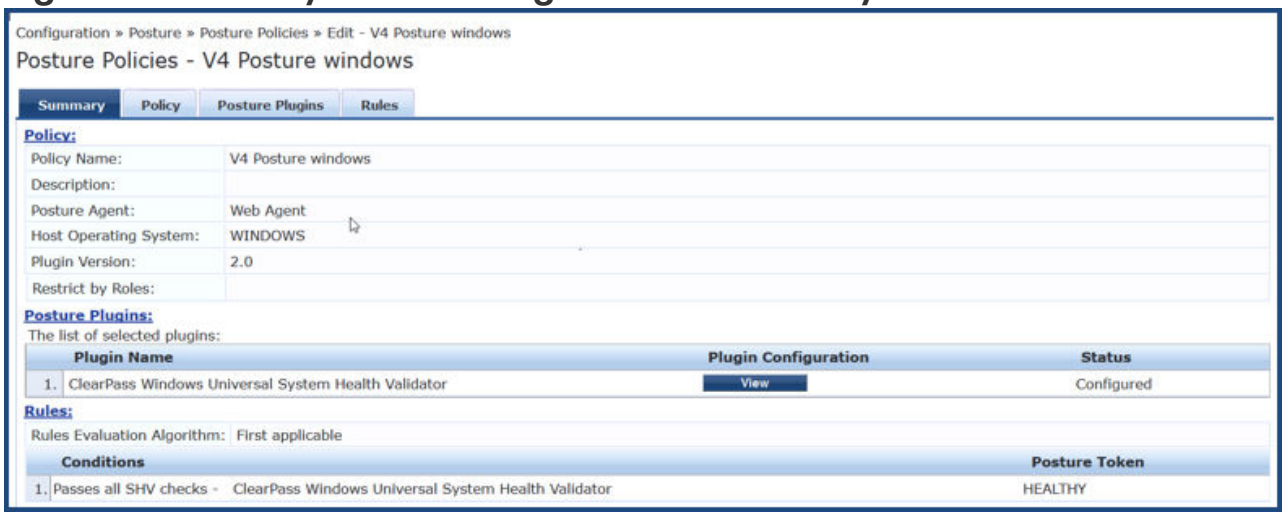
Table 3. Rules Editor Parameters

Parameter	Action/Description
Conditions	
Select Plugin Checks	Select **Passes all SHV** checks (the default setting). The following plugin check types are avai lable for System Health Validators (SHVs): • Passes all SHV checks • Passes one or more SHV checks • Fails all SHV checks • Fails one or more SHV checks
Select Plugins	Select the plugin to which the plugin che cks should apply.

Parameter	Action/Description
	In this case, the **Policy Manager Windows Universal System Health Validator** plug in has been automatically selected.
Actions	
Posture Token	Select **HEALTHY (0)** (the default setting). The following Posture Token settings are available: • HEALTHY (0) • CHECKUP (10) • TRANSITION (15) • QUARANTINE (20) • INFECTED (30) • UNKNOWN (100)

The following figure displays a summary of all the settings for this posture policy:

Figure 5. Summary of V4 SDK Agents Posture Policy



Creating a WebAuth Service for OnGuard Plugin v2.0 Agents

About this task

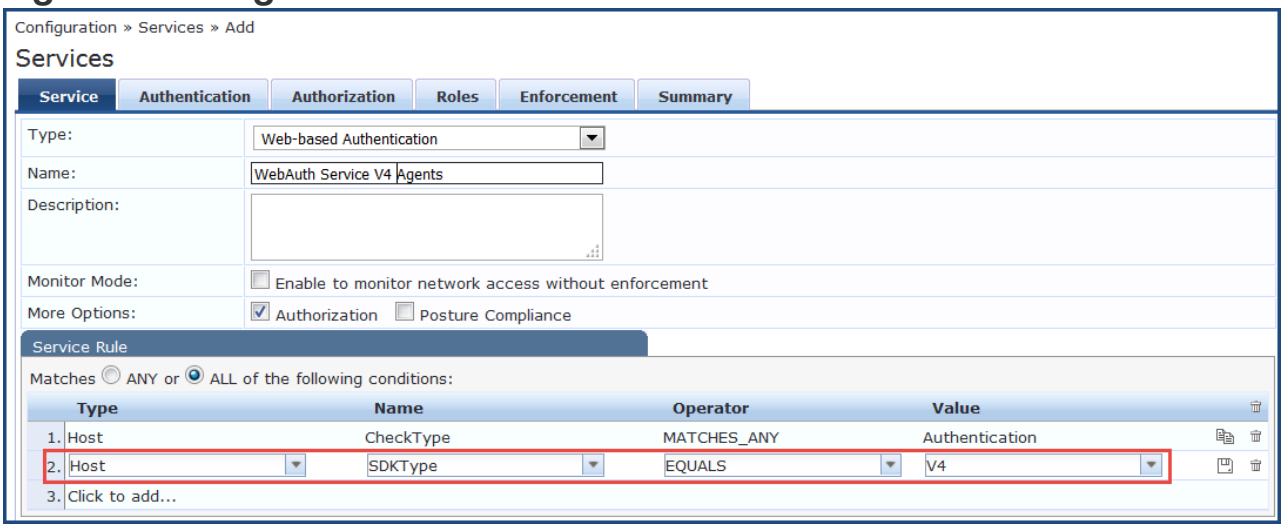
The final task is to create a WebAuth service for OnGuard plugin version 2.0 V4 SDK Agents. To do so:

Procedure

1. Navigate to **Configuration > Services**.
2. Click **Add**.

   The **Add Services** page opens.

Figure 1. Adding a Web-Based Authentication Service



3. **Type**: Select **Web-based Authentication**.
4. Name: Enter the name for this service.
5. Service Rule:
   a. Matches: Leave the default setting, **ALL of the following conditions**.
   b. Select **Click to add...** and specify the following attributes:
   c. Type: Select **Host**.
   d. Name: Select **SDKType**.
   e. Operator: Select **EQUALS**.
   f. Value: Select **V4**.

6. Select the Authentication tab and specify the authentication source(s).
7. Select the Enforcement tab and select the enforcement policy created in the previous section.
8. Click Save.
9. From the **Services** page, click **Reorder**, then place the service for the V4 SDK *before* the service for the V3 SDK.

   This ensures that WebAuth requests with the V4 SDK are evaluated by the service configured for the V4 SDK.

Important Points

About this task

This section provides the important points to keep in mind and, in some cases, follow up on, when upgrading to OnGuard plugin version 2.0.

Procedure

1. After installing Policy Manager7.0, OnGuard Agent is configured to use the OnGuard plugin version 2.0: OESIS V4 SDK by default. Thus, to fully configure the OnGuard plugin version 2.0, you must follow the procedure described above in Creating a WebAuth Service for OnGuard Plugin v2.0 Agents.
2. To locate the support charts for OnGuard plugin version 2.0, navigate to **Administration > Support > Documentation > OnGuard Agent Support Charts for Plugin Version 2.0**.
3. The *Plugin Version* field in the **Posture Policy > Summary** tab indicates the version and related SDK as follows (see Figure 10):
   - Plugin Version 1.0: OESIS V3 SDK
   - Plugin Version 2.0: OESIS V4 SDK
4. The names of some of the third-party products (for example, *AntiVirus*, *Firewall*, and *Patch Management*) have changed in the OnGuard plugin version 2.0, so be sure to test the OnGuard plugin version 2.0 Service and Posture policies in your lab before applying them in a production environment.
5. In OnGuard plugin version 2.0 Posture Policy for Windows and macOS, **AntiVirus** and **AntiSpyware** health classes are merged into the **AntiVirus** health class.

6. The following features/checks are *not* supported with the OnGuard plugin version 2.0:
    - **AntiVirus health class**: *Engine Version Check*, *Display Update URL*, *Disable RTP Check* (see Antivirus Health Check).
    - **Patch Management health class**: *Selected On Server* and *Security* options in the **Install Level Check Type** (see Patch Management).
7. Note that new posture policies created on ClearPass Policy Manager7.0 and later will be for OnGuard plugin version 2.0: OESIS V4 SDK.
8. ClearPass Policy Manager7.0 and later does not allow creating a new Posture Policy for OnGuard plugin version 1.0: OESIS V4 SDK. However, you can import a Posture Policy for OnGuard plugin version 1.0 from the previously released versions of Policy Manager.
9. Make sure that the Agent Enforcement profile has the required SDK Type configured —**V3** or **V4** (see Modifying an Existing Enforcement Policy for OnGuard Plugin v2.0).
0. Make sure to use the posture policy having Plugin Version 2.0, if V4 SDK is enabled. Similarly for V3 SDK, use the posture policy with Plugin Version 1.0 (see Creating a New Posture Policy for OnGuard Plugin v2.0 Agents).
1. You can change the **SDK Typ**e from **V4** to **V3** by modifying the Agent Enforcement profile to have **V3** as the **SDK Type**. In this case, be sure to configure the service posture policy is set to **Plugin Version 1.0** (see Creating a New Enforcement Profile to Set the SDK Type).
2. OnGuard Agent sends two WebAuth requests when the **SDK Type** is changed on a client. The first request uses the previously configured **SDK Type**, and the second request is for the new **SDK Type**.
3. If an Agent Enforcement Profile without the **SDK Type** attribute is applied, it will not reset the **SDK Type** on the client; that is, once the **SDK Type** is changed on the client by the Agent Enforcement profile, it will not change until a new Agent Enforcement profile having a different **SDK Type** is applied.
4. If the **Host:SDK Type** attribute is missing in WebAuth Requests, it indicates that OnGuard Agent is using the **V3** SDK as versions prior to OnGuard Agents version 6.7.0 do not send the **Host:SDK Type** attribute.
5. You can check the value of the **Host:SDKType** attribute in **Monitoring > Access Tracker > Input > Computed Attributes**.

Unified Agent System Tray Status Icons

Policy Manager OnGuard agent is integrated with Aruba VIA (Virtual Internet Access) to support both VIA functionality and OnGuard agent system health status collection. Thus, the integrated product is the called the *Unified Agent*. NAP (Network Access Protection) is a framework to collect system health status on Microsoft clients.Policy Manager supports health status collection for both NAP and OnGuard.

The Unified Agent System Tray icons display the following states of the Unified Agent status:

- OnGuard health status
- Trusted or untrusted network status
- VPN connectivity status
- Error conditions

Subtopics

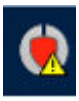
Unified Agent System Tray Icons
OnGuard-Only System Tray Icons

Unified Agent System Tray Icons

Table 1 describes that icons that indicate the possible states for the Unified Agent.

Table 1. Unified Agent System Tray Icons

OnGuard Status	Network Type	VPN Status	Icon
Healthy	Trusted	Connected	
Healthy	Trusted	Disconnected	
Unhealthy	Trusted	Connected	
Unhealthy	Trusted	Disconnected	

OnGuard Status	Network Type	VPN Status	Icon
Healthy	Untrusted	Connected	
Healthy	Untrusted	Disconnected	
Unhealthy	Untrusted	Connected	
Unhealthy	Untrusted	Disconnected	
Healthy	N/A	Error	
Unhealthy	N/A	Error	
Logged Out: No Health Status	N/A	Error	
Error	Trusted	Connected	
Error	Trusted	Disconnected	
Error	Untrusted	Connected	
Error	Untrusted	Disconnected	
Error	No Profile	N/A	
Error	N/A	Error	
Healthy	No Profile	N/A	

OnGuard Status	Network Type	VPN Status	Icon
Unhealthy	No Profile	N/A	
Logged Out: No Health Status	No Profile	N/A	
Logged Out: No Health Status	Trusted	Connected	
Logged Out: No Health Status	Untrusted	Disconnected	

OnGuard-Only System Tray Icons

Table 2 describes that icons that indicate the possible states for OnGuard-only.

Table 1. OnGuard - Only System Tray Icons

OnGuard Status	Network Type	VPN Status	Icon
Healthy	N/A	N/A	
Unhealthy	N/A	N/A	
Logged Out: No Healt h Status	N/A	N/A	
Error	N/A	N/A	

ClearPass Policy Manager Hardening Guide

Subtopics

Introduction

ClearPass Security Overview
External Security Testing and Accreditation
Internal Security Testing
Vulnerability Management Process
Ports
Locking Down Services
RADIUS Protocol
Locking Down Administrative Access
Enhanced Security for Admin and Local User Accounts
Rate Limiting
Monitoring
Locking Down User Access
User access lockdown restricts permissions to protect systems and data from unauthorized use.
Remote Support
WebUI and CLI Accounts
For More Information

Introduction

This document is intended to assist Aruba customers and partners in securely configuring Aruba ClearPass within their unique deployment environment. It should be noted that security recommendations often involve tradeoffs; not every recommendation in this document will be appropriate for every situation. Recommendations in this document represent security best practices and should be followed wherever network security is a priority.

ClearPass Security Overview

ClearPass provides numerous security checks and balances.

- All management and configuration actions (including create, delete and modify operations) are recorded in an audit log. The Audit Viewer provides real-time searchable access that allows an administrator to review all policy-level actions.
- Policy simulations allow all policies to be verified before they are deployed.
- A Monitor Only mode allows administrators to deploy policies without actually enforcing any access control. This enables administrators to fine-tune their policies and resolve policy exceptions before enabling enforcement.

- Once the policies are deployed, ClearPass provides administrators with multiple ways to track authentications and authorizations:
  - Access Tracker is a real-time searchable log that shows which policies are being applied and what actions are being taken. Access Tracker also shows all exceptions and failures.
  - Insight is a reporting, analytics, and alerting tool that can be configured to generate historical reports on user and device authentication, applied policies, and enforcement actions. Insight also allows administrators to specify alert thresholds and conditions for policy exceptions or other system-level failures. When the conditions and thresholds are satisfied, the system sends out email and SMS text alerts.
- At a system level, ClearPass has been hardened in numerous ways.
  - ClearPass runs on a hardened Linux- based operating system.
  - All sensitive data directories are protected using AES-256 encryption.
  - CLI access into the operating system is granted through a restricted shell.
  - Only required services are allowed to run.
  - The appliance is firewalled internally to allow only limited traffic.
  - ClearPass utilizes separate management and data interfaces, and provides the ability to restrict access to the management interface to just authorized end stations.
  - ClearPass provides timely security patches for critical and high-level operating system and application-level security advisories.
- Before every ClearPass release, system level vulnerability scans are performed using tools such as QualysGuard, and IBM AppScan.

External Security Testing and Accreditation

Aruba invests heavily in independent third-party security testing of its products. While the majority of this testing is relevant to (and required by) government agencies, it has value to all types of users. In some cases, organizations may choose to rely on recognized security testing authorities rather than conducting their own product testing.

Subtopics

Common Criteria
FIPS 140-2
Suite B cryptographic support

Common Criteria

ClearPass was awarded Common Criteria certification under both the Network Device collaborative Protection Profile (NDcPP) and the Authentication Server Extended Package.

FIPS 140-2

The Federal Information Processing Standard (FIPS) 140-2 is a system for testing and certifying cryptographic modules. As part of this testing, a laboratory accredited by the US and Canadian governments examines design documentation, source code, and development practices, in addition to conducting extensive testing of cryptographic functions.

Products that implement FIPS 140-2 validated cryptography are assured to be using cryptography correctly. **http://csrc.nist.gov/groups/STM/cmvp/standards.html**

When operating in FIPS mode, ClearPass Policy Manager, Guest and Onboard are FIPS 140-2 compliant because they incorporate a FIPS-validated module, which provides all cryptography functions for the application. ClearPass incorporates the Aruba Linux Cryptographic Module which implements full and approved cryptographic algorithm support, including Suite B algorithm compliance, for Aruba products. It provides secure key management, data integrity, data at rest encryption, and secure communications.

http://csrc.nist.gov/groups/STM/cmvp/documents/140-1/140val-all.htm#2577Suite B cryptograph

Suite B cryptographic support

ClearPass Policy Manager and RADIUS server include Suite B cryptographic support.

Suite B cryptographic algorithms are specified by the National Institute of Standards and Technology (NIST) and are used by NSA's Information Assurance Directorate in solutions approved for protecting National Security Systems (NSS). Suite B includes cryptographic algorithms for encryption, key exchange, digital signature, and hashing.

Algorithm	Function	Specification
Advanced Encryption Standard (AES)	Encryption	FIPS Pub 197
Elliptic Curve Diffie - Hellman (ECDH)	Key Exchange	NIST SP 800 - 56A
Elliptic Curve Digital Signature Algorithm (ECDSA)	Digital Signature	FIPS Pub 186 - 4
Secure Hash Algorithm (SHA)	Hashing	FIPS Pub 180 - 4

Internal Security Testing

Each ClearPass release goes through extensive quality assurance testing. As part of the testing process, several commercial vulnerability scanners are used, including QualysGuard, Nessus, Nexpose, ZAP and BurpSuite. Any findings returned by these scanners are examined to determine if they are genuine vulnerabilities or false positives. Actual vulnerabilities will result in a defect being opened for prompt resolution. In addition to quality assurance testing, an internal group known as HPE Aruba Networking Threat Labs provides advanced vulnerability research against HPE Aruba Networking products.

HPE Aruba Networking Threat Labs runs a public bug bounty program which includes ClearPass. More information can be found at **https://bugcrowd.com/engagements/aruba-product-public**. From time to time, HPE Aruba Networking also contracts with external third-party penetration testing firms to conduct targeted testing. HPE Aruba Networking maintains a database of all the security bulletins published for ClearPass. This database is available at **https://support.hpe.com/connect/s/securitybulletinlibrary**.

Vulnerability Management Process

Aruba publishes a vulnerability response policy at **http://www.arubanetworks.com/support-services/security-bulletins/**. This location also hosts security advisories published by Aruba.

Ports

Most ClearPass Policy Manager communication for updates is through HTTPS (TCP 443) to **clearpass.arubanetworks.com**. To allow devices to communicate over a network firewall, ensure communication between the (publisher/servers) is allowed to that URL over port 443.

	Between		Protocol
ClearPass UI	Management Station	ClearPass	TCP
Secure Shell	Client	ClearPass	TCP
RadSec	Radius Server	ClearPass	TCP
Session Logs	Event Server	ClearPass	TCP
Guest Portal	Controller	ClearPass	TCP
Update service	ClearPass	Update Server	TCP
OnGuard Agent	Endpoints	ClearPass	TCP
	Endpoints	ClearPass	TCP
NAS Devices - AAA Services	NAS Devices	ClearPass	TCP/UDP
	NAS Devices	ClearPass	TCP/UDP
	NAS Devices	ClearPass	TCP/UDP
	NAS Devices	ClearPass	TCP/UDP
	ClearPass	NAS Devices	TCP/UDP
	NAS Devices	ClearPass	TCP/UDP
ClearPass to Active Directory	ClearPass	AD Servers	ICMP echo (8) and echo
	ClearPass	AD Servers	TCP/UDP
	ClearPass	AD Servers	TCP/UDP
	ClearPass	AD Servers	TCP/UDP
	ClearPass	AD Severs	TCP

	Between		Protocol
	ClearPass	AD Severs	TCP
	ClearPass	AD Servers	TCP/UDP
	ClearPass	AD Servers	TCP
	ClearPsss	AD Servers	TCP
	ClearPsss	AD Servers	TCP
SNMP	ClearPass	Endpoint	UDP
SNMP	Endpoint	ClearPass	UDP
WMI	ClearPass	Endpoint	TCP
Cluster	Publisher	Subscriber	TCP
	Subscriber	Publisher	TCP
	Subscriber	Subscriber	TCP
	Publisher	Subscriber	TCP
	Subscriber	Publisher	TCP
	Subscriber	Publisher	UDP
ClearPass Misc Services	ClearPass	NTP Servers	UDP
	ClearPass	SMTP Servers	TCP
	ClearPass	SMTP Servers	TCP
	ClearPass	DNS Servers	TCP
	Network	ClearPass	UDP
	Network	ClearPass	UDP
	Network	ClearPass	UDP

	Between		Protocol
Ingress Event Engine	Network	ClearPass	UDP

NOTE
Ports required will vary based on features configured.

SMB Port Range Note

SMBv2 and SMBv3 protocols use an increased remote procedure call (RPC) port range for Windows Server 2008 and later. The following AD deployments determine the ports:

- Active Directory deployments that use only Windows Server 2008 or later use the high port range of 49152 through 65535.
- Active Directory deployment that use Windows Server 2008 or later and earlier versions of Windows Server use both the low port range of 1025 through 5000 and the high port range of 49152 through 65535.
- Active Directory deployment that use only versions earlier than Windows Server 2008 use the low port range of 1025 through 5000.

Locking Down Services

Subtopics

Cryptography
AD over SSL
Disk Encryption for Virtual Appliances
FIPS Mode
Enabling FIPS
HTTP Strict Transport Security
OCSP
NTP Authentication
SMBv2/v3 support
SNMP
SNMP Traps
System Cleanup Options
SNMP Private Enterprise MIB
SNMP Trap Receivers
External syslog

LEEF and CEF Format Syslog

Cryptography

Aruba ClearPass uses cryptography as just one of several services, including HTTPS, SSH, and IPsec. ClearPass utilizes cryptography to keep network managed information secret and safe by transforming it in a manner that unintended recipients cannot understand.

AD over SSL

When using Active Directory as an authentication source with connection security "AD over SSL," the following cipher suites are supported;

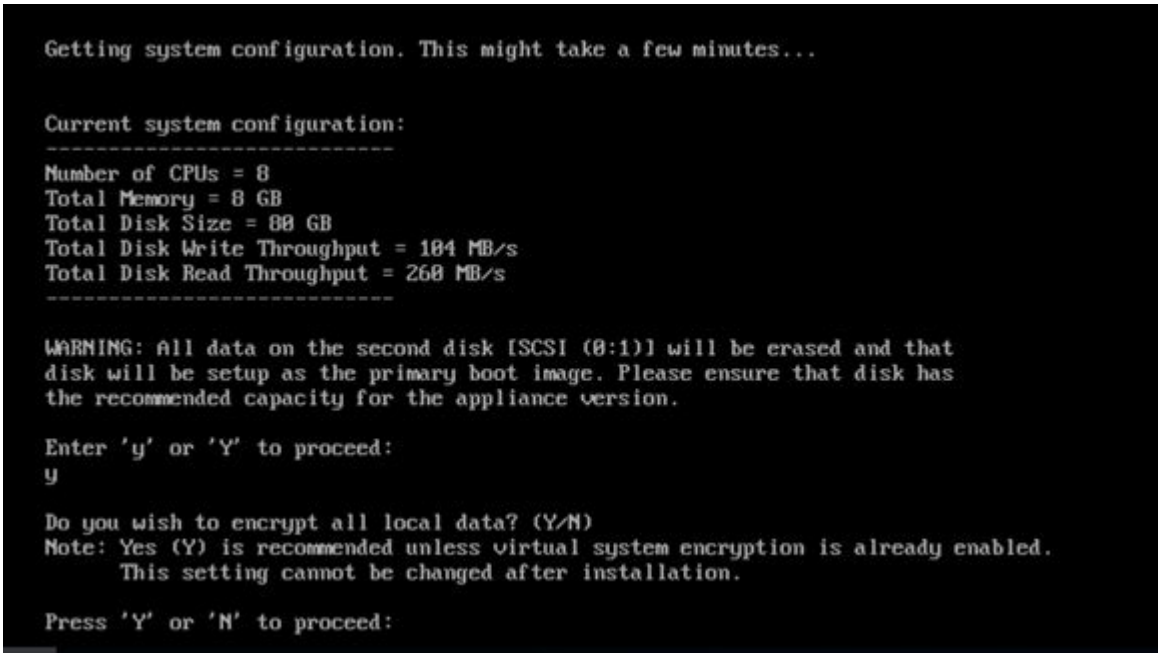
- TLS_AES_128_GCM_SHA256 (0x1301)
- TLS_AES_256_GCM_SHA384 (0x1302)
- TLS_CHACHA20_POLY1305_SHA256 (0x1303)
- TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (0xc02b)
- TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)
- TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 (0xc02c)
- TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (0xc030)
- TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca9)
- TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca8)
- TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (0xc013)
- TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (0xc014)
- TLS_RSA_WITH_AES_128_GCM_SHA256 (0x009c)
- TLS_RSA_WITH_AES_256_GCM_SHA384 (0x009d)
- TLS_RSA_WITH_AES_128_CBC_SHA (0x002f)
- TLS_RSA_WITH_AES_256_CBC_SHA (0x0035)
- TLS_RSA_WITH_AES_128_CBC_SHA (0x002f)

- TLS_RSA_WITH_AES_256_CBC_SHA (0x0035)

Disk Encryption for Virtual Appliances

ClearPass supports Linux Unified Key Setup-on-disk-format (LUKS) disk encryption for both hardware and virtual appliances. LUKS is enabled by default on all ClearPass hardware appliances for disk encryption. For virtual appliances, ClearPass provides the administrator the option to choose to encrypt the disk partition while setting up a VM appliance.

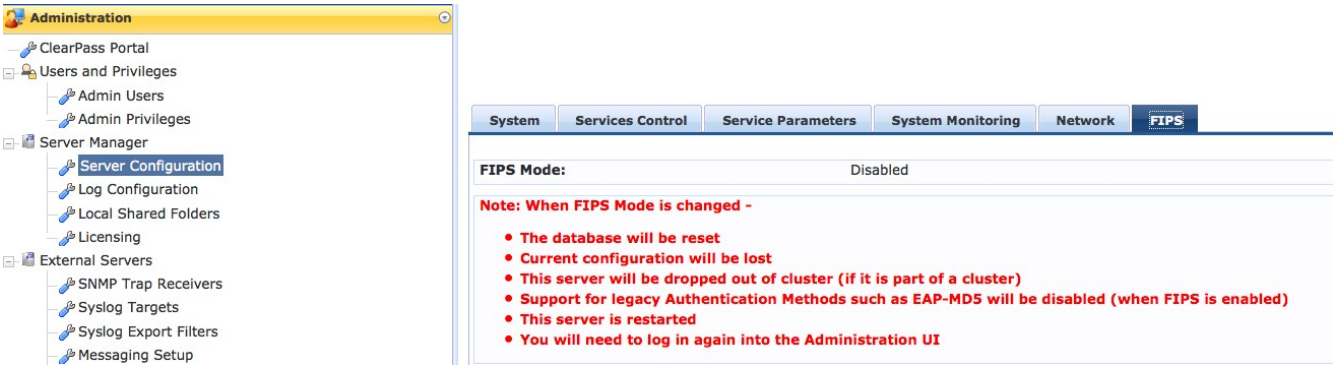
If the administrator knows that the hypervisor or the hardware already has an underlying disk encryption, then no additional disk encryption for ClearPass is required. If you are not sure whether the hypervisor has underlying disk encryption or not, then it is recommended to choose the **Do you wish to encrypt all local data?** option for encryption, as shown in the following image of the VM setup process.



FIPS Mode

In FIPS mode, all cryptographic services provide a minimum strength of 112 bits as mandated by FIPS 140-2. Services which provide less than 112 bits of security (such as RSA-1024, SHA1 for digital signatures, MD5, DES) may not be configured. In non-FIPS mode, there are no restrictions on minimum security strength. Algorithms such as DES (56 bits of strength) and MD5 (<64 bits of strength) are permitted, although this is not the default configuration.

Enabling FIPS



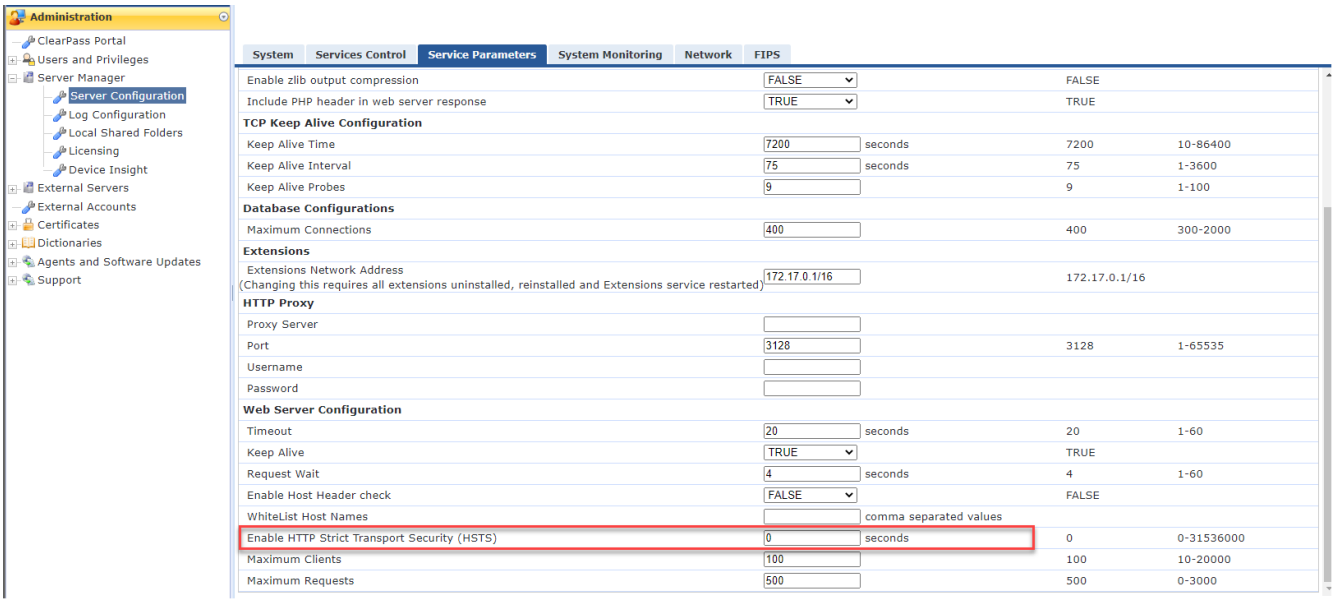
- Review the following important points, before enabling FIPS mode in ClearPass Policy Manager:
- The database is reset when FIPS mode is enabled. Ensure that a secure current back-up of the ClearPass database exists before enabling FIPS mode.
- Configuration backup files from Policy Manager in non-FIPS mode cannot be restored to Policy Manager in FIPS mode.
- Configuration backup files from Policy Manager in FIPS mode can be restored to Policy Manager in the non- FIPS mode.
- The server will be removed from the cluster when FIPS mode is enabled.
- All nodes in a cluster must be either in FIPS or non-FIPS mode.
- Legacy authentication methods such as EAP-MD5 and MD5 message digest algorithm are not supported in FIPS mode.
- Certificates that are created with MD5 authentication cannot be imported to the Certificates Trust List (**Administration > Certificates > Certificate** Trust List).
- The server reboots when FIPS mode is enabled.

HTTP Strict Transport Security

HSTS (HTTP Strict Transport Security) lets Policy Manager inform browsers that it should only be accessed using HTTPS, instead of using the less secure HTTP.

This feature can be enabled as a server-wide setting on a ClearPass Server. To configure this feature, navigate to **Administration > Server Manager > Server Configuration** and select a server, then select the **Service Parameters** tab. In the **Select Service** drop-down menu, select ClearPass **system services**, then scroll down to the **Web Server Configuration** section. By default, the **Enable HTTP Strict Transport Security (HSTS)**

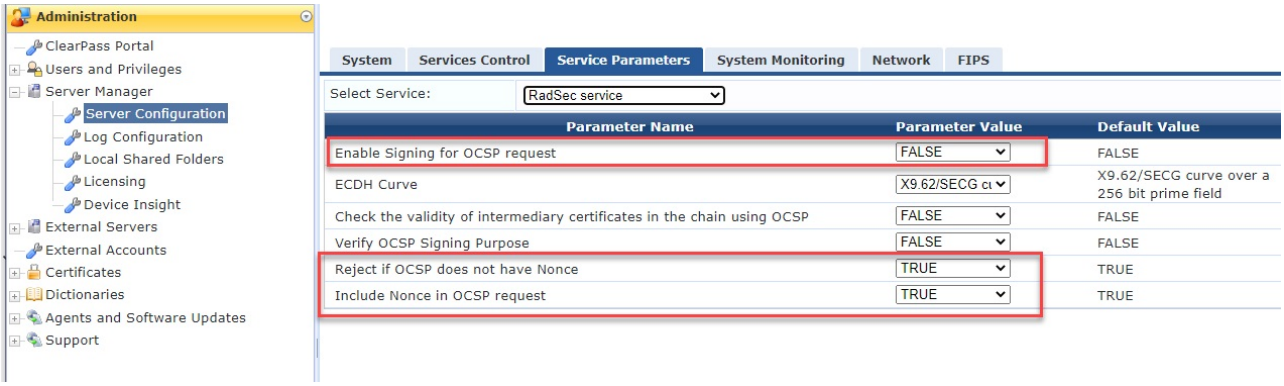
field is set to zero (0), which disables this feature. Set the timeout value to any value from 1-31536000 seconds (one year) to enable this feature and define the timeout value. Once the timeout value has elapsed, a browser is allowed to make another HTTP request, then gets immediately re-directed to an HTTPS connection.



OCSP

Online Certificate Status Protocol (OCSP) can be used to obtain the revocation status of an X.509 digital certificate. To configure OCSP settings, navigate to **Administration > Server Manager > Server Configuration** and select a server, then select the **Service Parameters** tab. In the **Select Service** drop-down menu, select **RadSec service**.

- **Include Nonce in OCSP request**: A nonce is a cryptographic value used to protect against record and replay attacks. If the OCSP server does not support the nonce, set this value to FALSE to avoid an EAP-TLS authentication failure.
- **Enable Signing for OSCP Requests.** Enables ClearPass to sign the OCSP request with the RADIUS server certificate. The default value for this parameter is set to FALSE to disable the signing process. Signing verifies the integrity of the data and the identity of the sender.

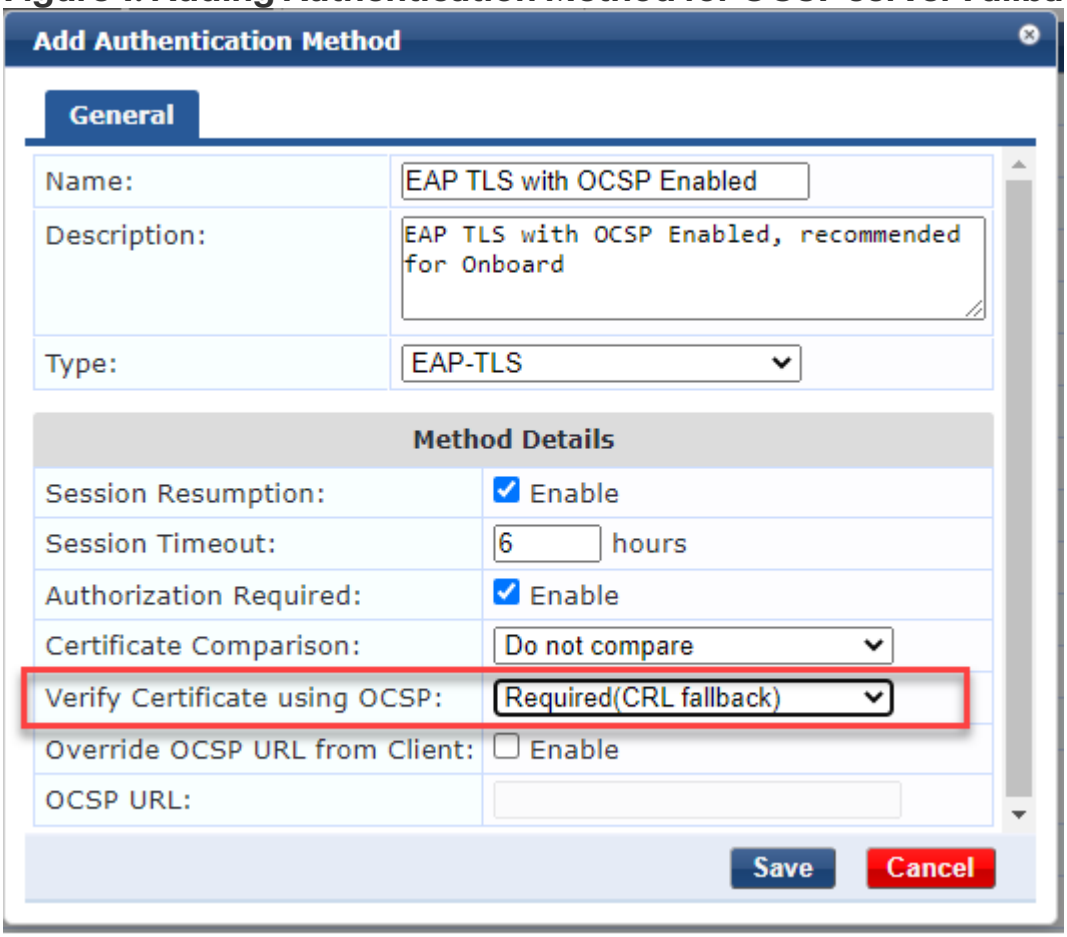


OCSP Server Fail-Open to CRL

For EAP-TLS authentication, If the OCSP server is not accessible to perform certificate validation, ClearPass provides a means to validate against a CRL instead (fallback).

To add or modify the EAP-TLS authentication method, navigate to **Configuration > Authentication > Methods**. Select an existing authentication method to modify it, or click the **Add** link to add a new authentication method.

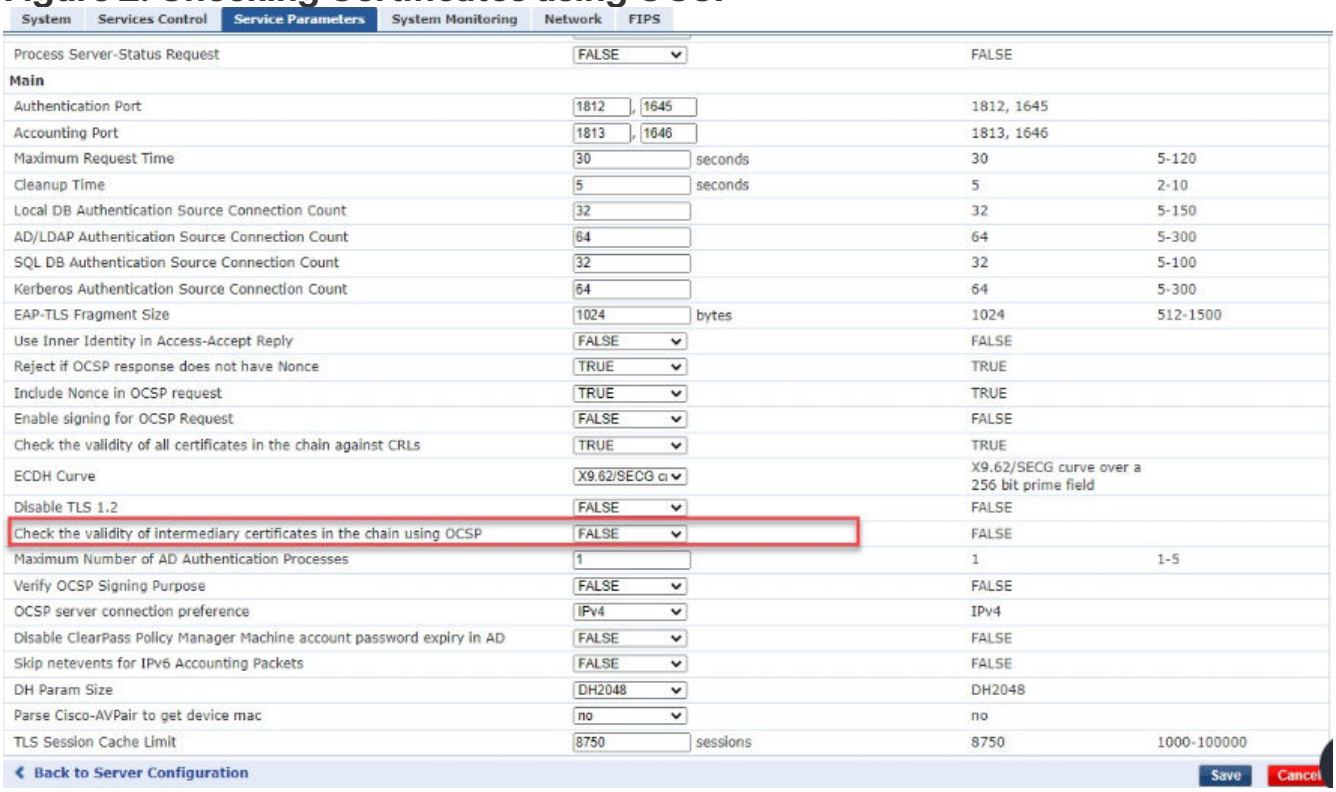
Figure 1. Adding Authentication Method for OCSP server Fallback



OCSP validation check Intermediate Certificates

To enhance certificate security the RADIUS service parameter, enable the **Check the validity of intermediary certificates in the chain using OCSP** server configuration setting. Enabling this option validates the entire certificate chain, but will put greater load on the system and is not intended for all customer use cases. To modify this setting, navigate to **Administration > Server Manager > Server Configuration**, select a server from the **Servers** list, then click the **Service Parameters** tab.

Figure 2. Checking Certificates using OCSP



OCSP/CRL Status Messages

The Event Viewer at **Monitoring > Event Viewer** provides a notification when:

- The connection to an OCSP server times out
- No response is received from an OCSP server
- CRL has expired
- CRL download fails

NTP Authentication

About this task

If the Network Time Protocol is not authenticated, an attacker can introduce a rogue NTP server. This rogue server can then be used to send incorrect time information to otherwise secure network devices, which will make log timestamps inaccurate and affect scheduled actions. NTP authentication is used to prevent such tampering by authenticating the dedicated time resource.

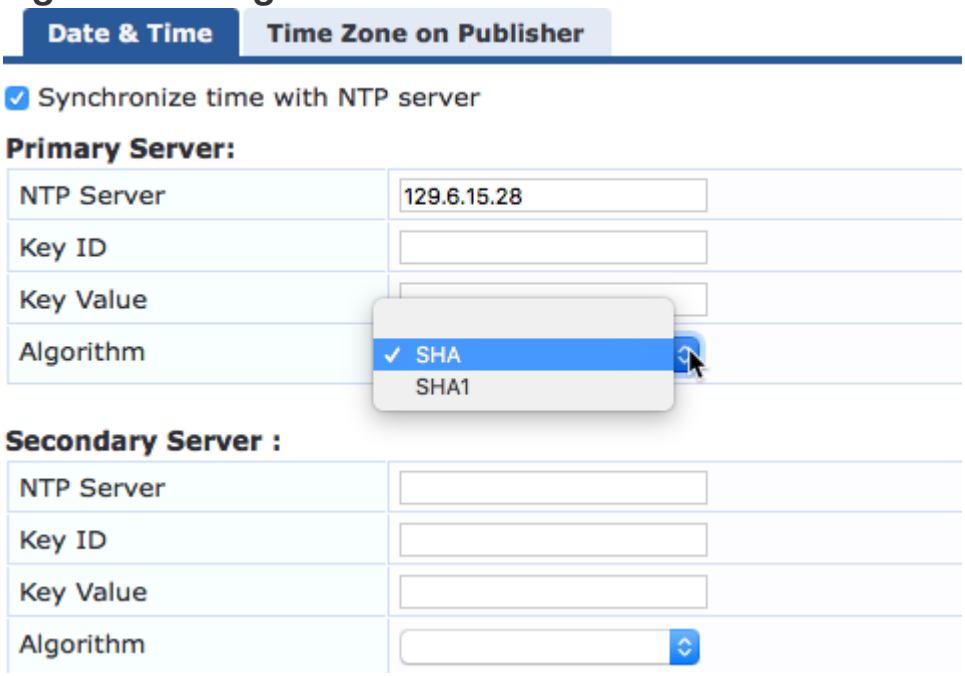
ClearPass supports authenticating its network time sources using SHA or SHA1. In FIPS mode, only SHA1 is supported. The authentication users the Linux NTP service. In a cluster, the publisher is the time source for all subscribers and subscribers do not authenticate the publisher.

To set NTP Authentication settings:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration** page,
2. Select the **Set Date & Time** link.
3. Select the **Synchronize time with NTP server** option, then enter the values your server settings.

Figure 1. Setting Authentication for NTP Servers



SMBv2/v3 support

ClearPass supports SMBv2/v3 for PEAPv0/EAP-MSCHAPv2 and Microsoft Active Directory Domain Services. ClearPass will use the highest version available on the controller;

- SMBv3 will be automatically used by default for AD joins and any requests that use PEAPv0/EAP-MSCHAPv2.
- If SMBv3 is not enabled, ClearPass will then automatically failover to SMBv2.
- If SMBv2 is also not enabled, ClearPass will then fail over to use SMBv1.

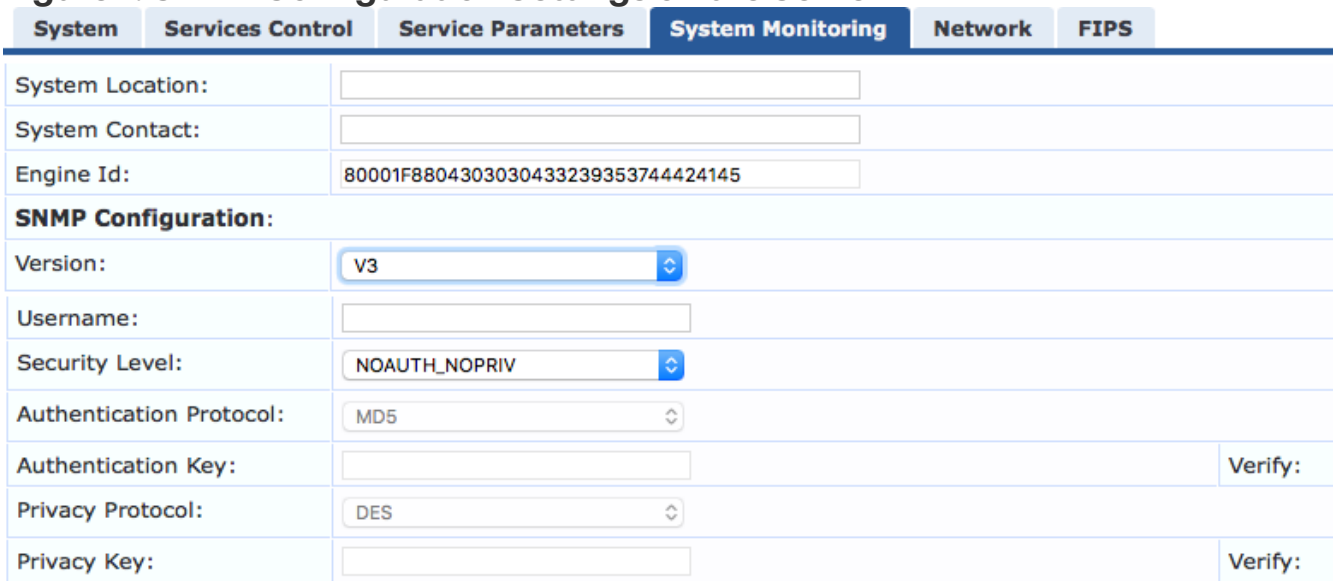
  If higher SMB versions are later enabled on the client, ClearPass will then detect the changes and attempt to use the highest available SMB version automatically.

SNMP

The Simple Network Management Protocol is commonly used by network management systems to poll devices for information such as port configuration, status, and interface counters. SNMP versions 1 and 2 provide very little security beyond the community string. If an attacker has network access to a device and can guess the community string, it may lead to disclosure of sensitive information. Aruba strongly recommends the use of SNMPv3, which includes much stronger security through authentication and encryption.

Navigate to **Administration > Server Manager > Server Configuration**, select a server then click the **System Monitor** tab to configure the SNMP configuration parameters. This ensures that external Management Information Base (MIB) browsers can browse the system level MIB objects exposed by the Policy Manager appliance. The options in this page vary based on the SNMP version selected.

Figure 1. SNMP Configuration Settings on the Server



SNMP Traps

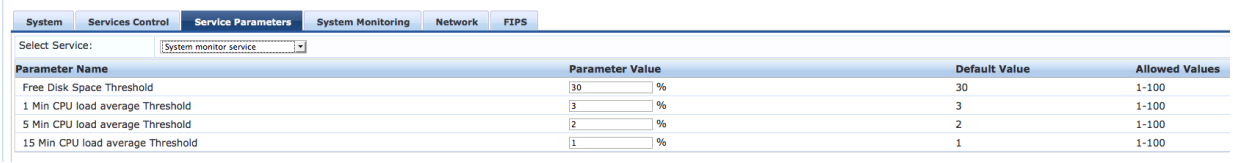
SNMP Trap Receivers can be configured to receive traps for system events. Policy Manager sends SNMP traps that expose the following server information:

- System uptime. Conveys information about how long the system is running.
- Network interface statistics [up/down]. Provides information if the network interface is up or down.
- Process monitoring information. Check for the processes that should be running. Maximum and minimum number of allowed instances. Sends traps if there is a change in value of maximum and minimum numbers.
- Disk usage. Check for disk space usage of a partition. The agent can check the amount of available disk space, and make sure it is above a set limit. The value can be in % as well. Sends traps if there is a change in the value.
- CPU load information. Check for unreasonable load average values. For example, if a 1 minute CPU load average exceeds the configured value [in percentage], then the system would send the trap to the configured destination.
- Memory usage. Report the memory usage of the system.

  To configure free disk and CPU load thresholds:

  1. Navigate to A**dministration > Server Manager > Server Configuration**,
  2. Select a server, then click the **Service Parameters** tab.
  3. In the **Select Service** menu, select s**ystem monitor service**.

Figure 1. Free Disk and CPU Load Threshold Configuration

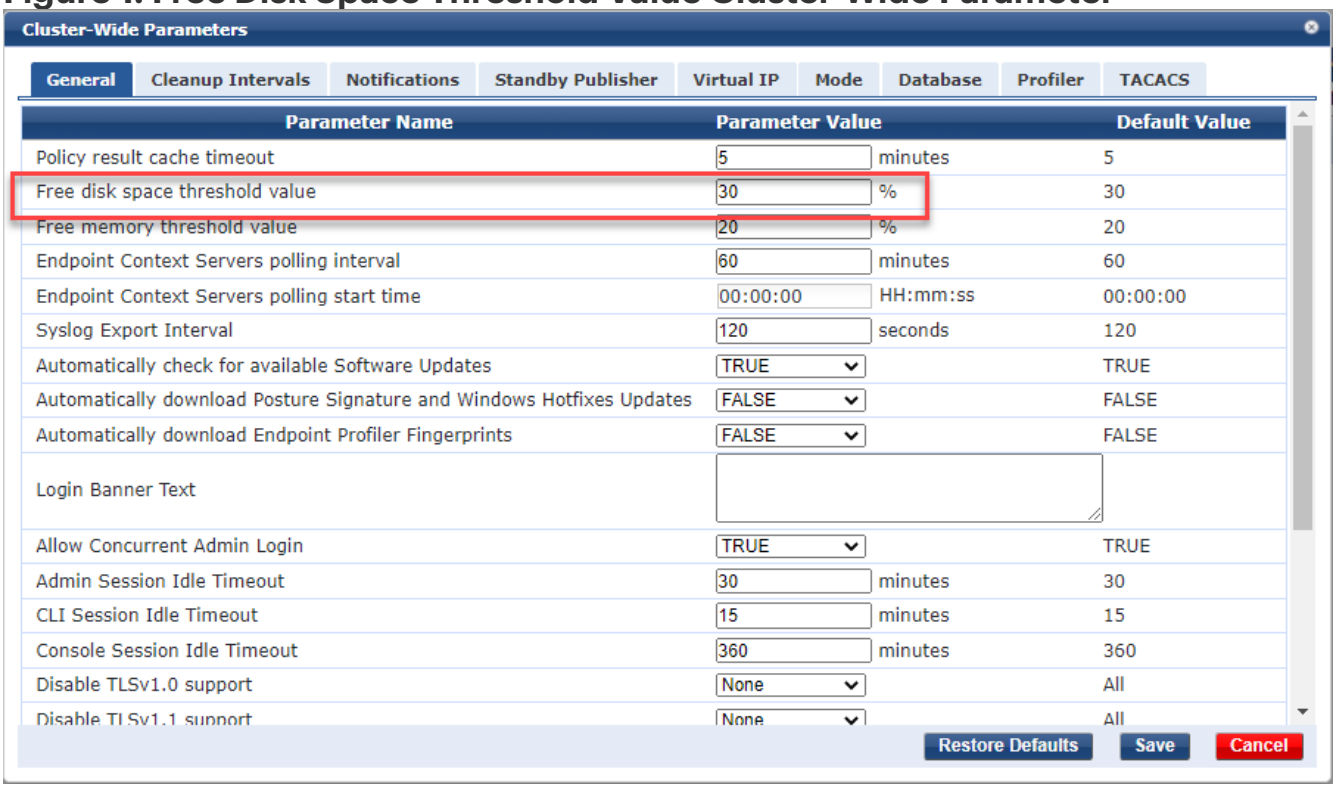


System Cleanup Options

To prevent disk space exhaustion, ClearPass includes the **Free disk space threshold value** cluster-wide parameter setting used to trigger a disk cleanup when the threshold is exceeded. The default value is 30% free disk space, anything below which triggers

a disk cleanup. To change this setting, navigate to **Administration > Server Manager > Server Configuration**, select a server then click the **System Monitor** tab.

Figure 1. Free Disk Space Threshold Value Cluster-Wide Parameter



Once an hour ClearPass checks the free disk space, if it's below the defined threshold an alert is logged and an aggressive cleanup operation is run. The job cleans up any records older then one day from the following;

- Log database records
- Core files
- System load monitor files
- Application and system log files
- Auto and manual backup files
- Stored reports
- Expired guest accounts
- Audit records

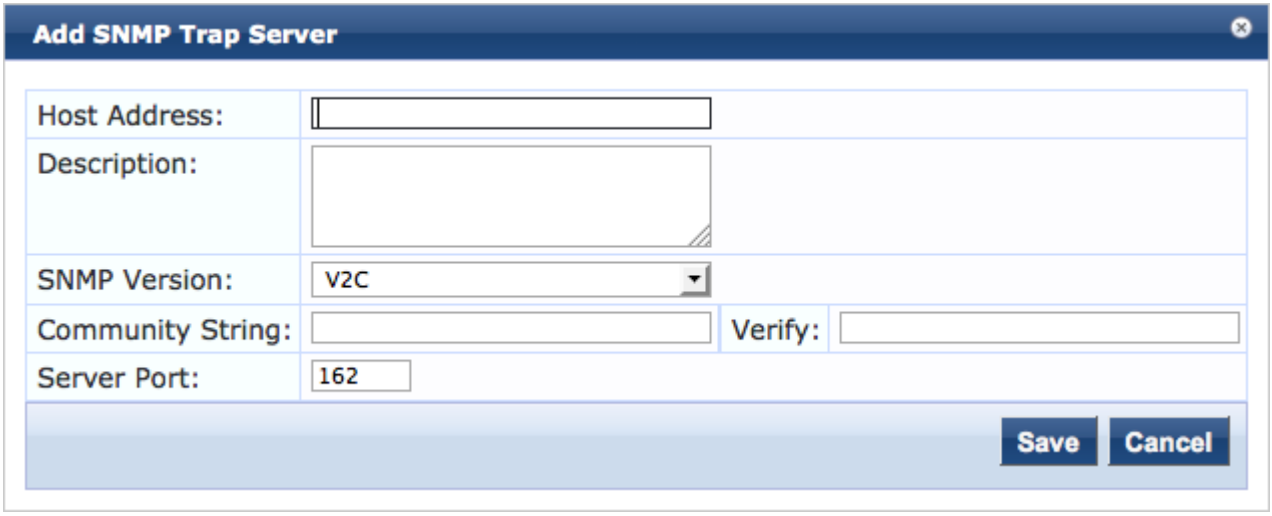
SNMP Private Enterprise MIB

ClearPass includes a Private Enterprise MIB that exposes over 70 OID's. Information and traps include

- Performance counters
  - Authentication counters
  - Authorization counters
  - Request processing time/delays
  - Authorization time/delays
- System statistics
  - Disk statistics (available, total, used)
  - Memory statistics (available, total, used)
  - CPU load averages
- Network traffic counters
  - Application name
  - Application port
  - Total network traffic in bytes
- SNMP Traps
  - Free disk space is lower than the configured threshold
  - Low system memory
  - High CPU utilization
  - License expiration
  - Certificate expiration
  - Cluster node add
  - Cluster node promote
  - Cluster node delete
  - Cluster password change
  - Cl\uster license utilization

SNMP Trap Receivers

External trap receivers are added at: **Administration > External Servers > SNMP Trap Receivers**.

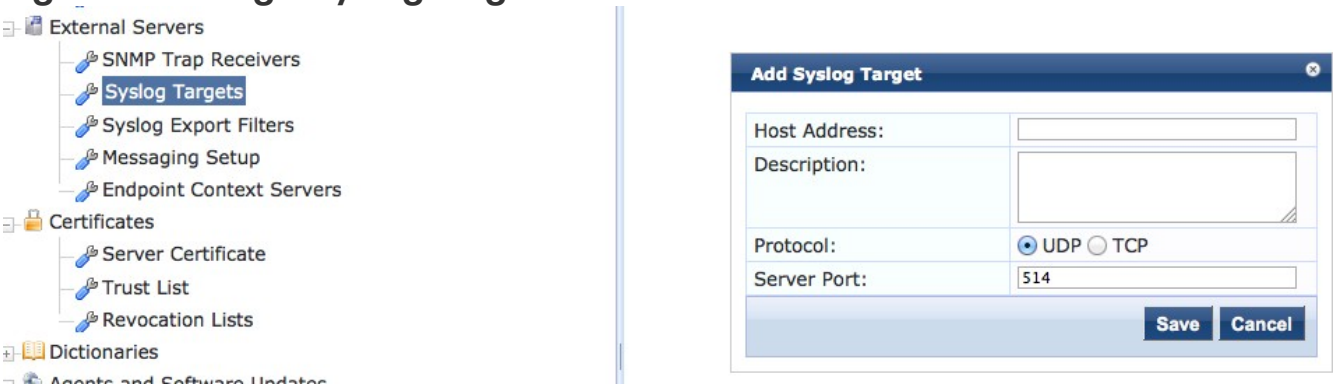


External syslog

When a system is compromised, one of the first things an attacker will do is to remove evidence of the intrusion from the system logs. For this reason, it is important to send logs to a secure external system – preferably one with automated log analysis tools that can identify and flag unusual activity. ClearPass supports the syslog standards for log distribution. Log information can be sent to one or more syslog targets (servers).

To configure Syslog targets, navigate to **Administration > External servers > Syslog Targets**, then click **Add** to create a new syslog target. Export filters are configured at **Administration > External servers > Syslog Export Filters.**

Figure 1. Adding a Syslog Target



Policy Manager uses Syslog to export session data from the access tracker, audit records from the audit viewer, event records from the event viewer and Insight logs. Syslog Export Filters are configured to tell Policy Manager where to send log information, and

what information should be included in the logs sent to each Syslog target. If desired, different information can be uniquely sent to each Syslog target.

You can define Syslog export filters using the **Audit Records**, **Insight Logs**, **Session Logs**, or **System Events** export templates. If you use the **Insight Logs** export template, you have the option to select predefined groups of fields or to select individual fields. If you use the **Session Logs** export template, the **Active Sessions** filter adds the option for customized SQL Queries.

Figure 2. Syslog Export Filter Settings with theAudit RecordsExport Template

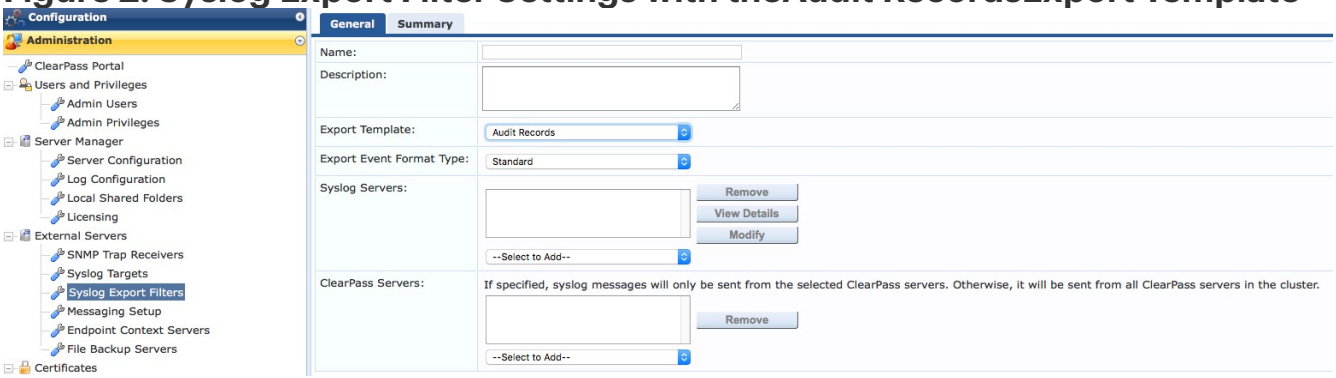
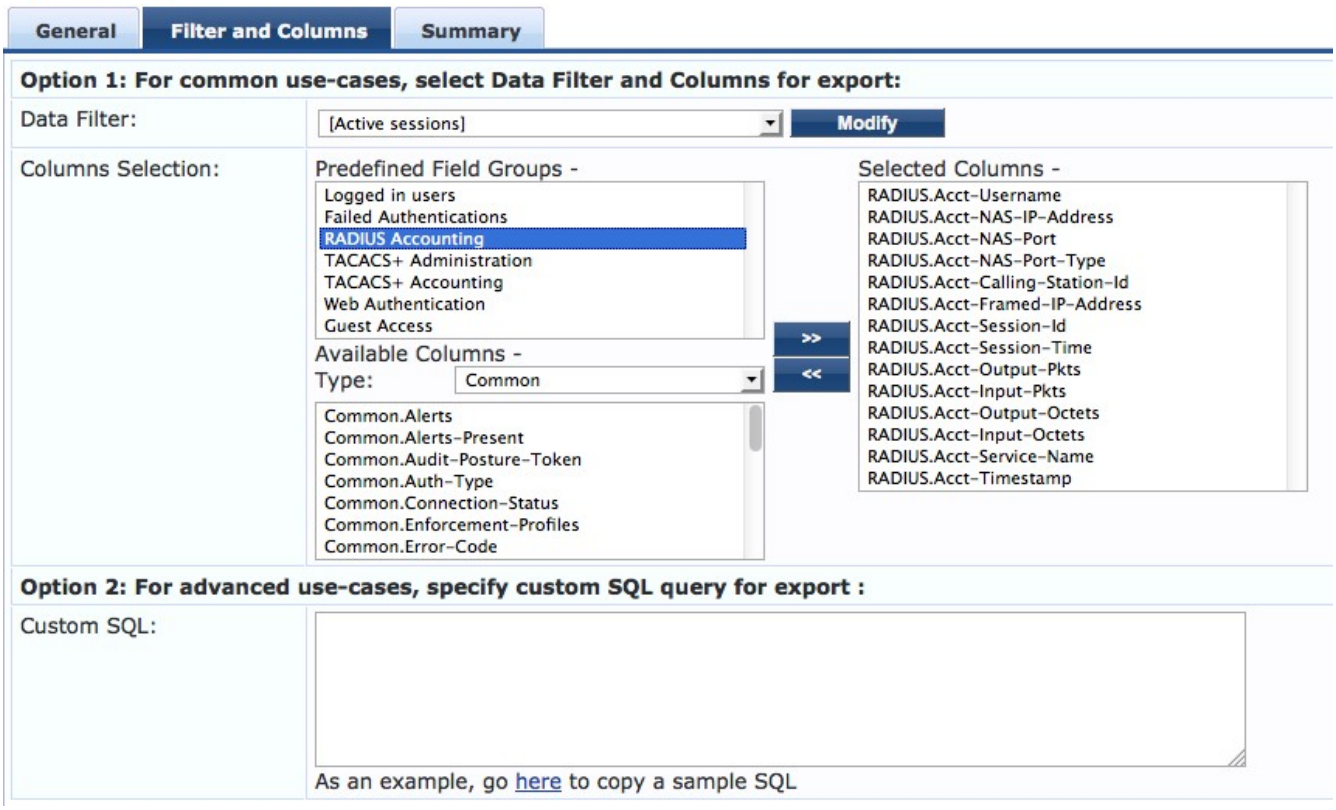


Figure 3. Syslog Export Filter Settings with the Session Logs Export Template



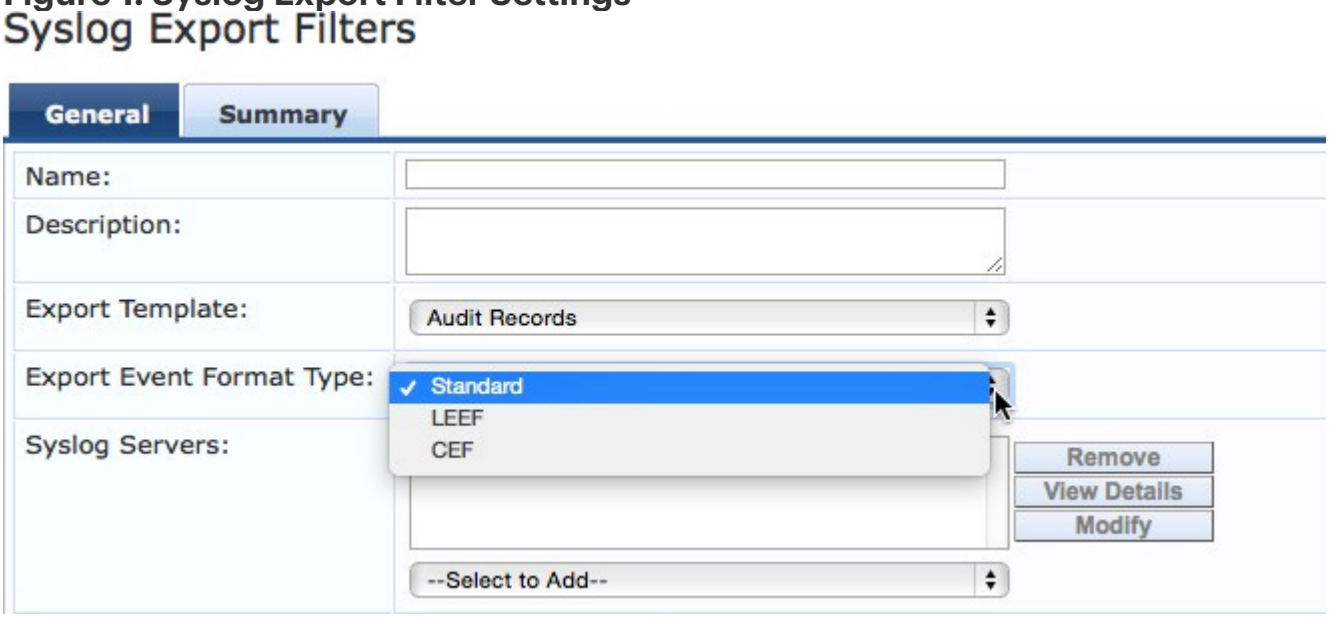
LEEF and CEF Format Syslog

ClearPass supports both the CEF (Common Event Format) and LEEF (Log Event Extended Format) Syslog formats. To choose a syslog format for an export filter, navigate to **Administration > External Servers > Syslog Export Filters** and click **Add**, or select a existing filter from the Syslog Export Filters table.

NOTE

The maximum CEF header length has increased from 4 KB to 64 KB to ensure the complete CEF header is captured even when fragmentation occurs.

Figure 1. Syslog Export Filter Settings



RADIUS Protocol

ClearPass uses RADIUS to exchange authentication information with Network Access Devices. The RADIUS protocol provides a weak form of encryption, which uses a static RADIUS shared secret as the basis for the encryption key.

To maximize security, the RADIUS shared secret should be both long and complex. Since there is no need for this secret to be memorable by a human, the use of a service such as **http://www.random.org/** to generate a truly random string is recommended. To minimize the damage from a compromised shared secret, each Network Access Device that communicates with ClearPass should be configured to use a different RADIUS shared secret.

Subtopics

Disabling TLSv1.0 and TLSv1.1 in the WebUI and the RADIUS Server
Enabling RadSec

Disabling TLSv1.0 and TLSv1.1 in the WebUI and the RADIUS Server

For enhanced security, two cluster-wide settings allow network administrators to enable or disable TLSv1.0 and TLSv1. On new Policy Manager installations, TLSv1.0 is disabled by default, so the default setting is **All**.

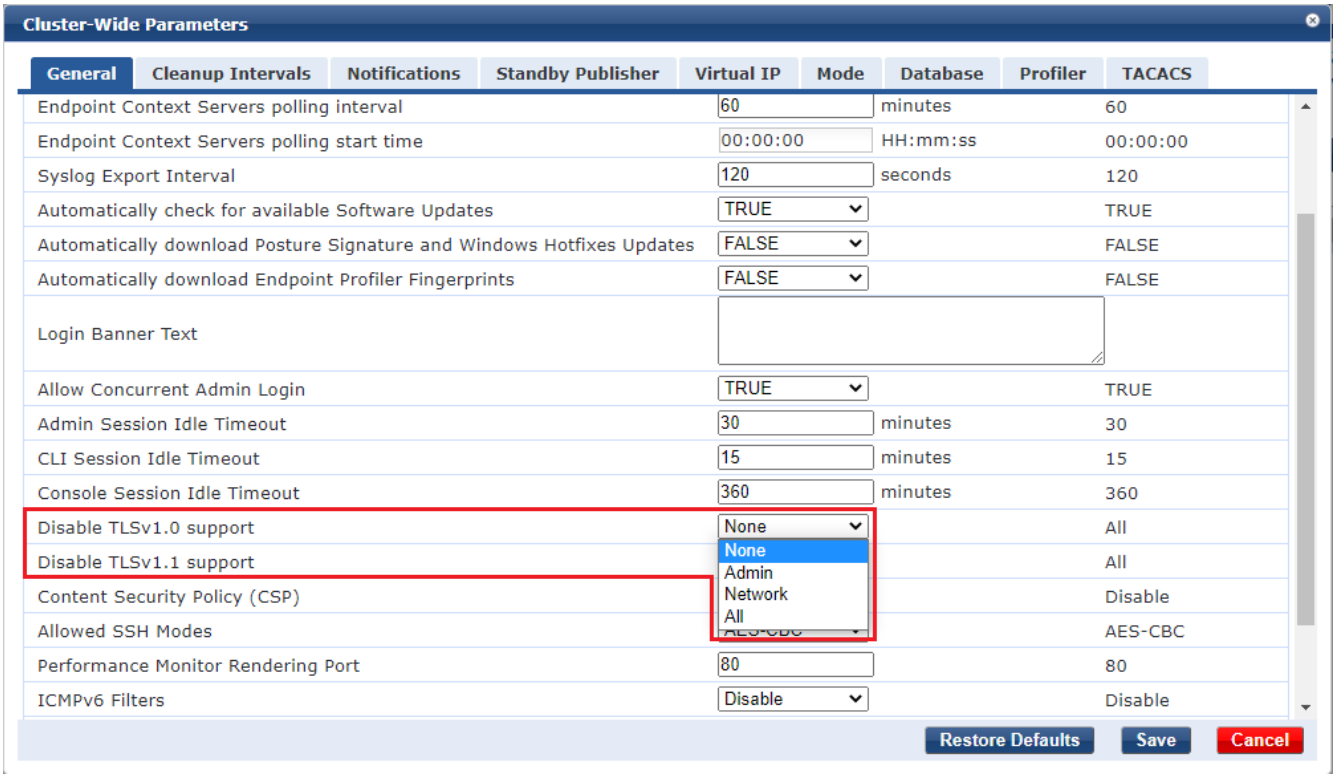
Users cannot put clusters into Common Criteria (CC) mode while the Disable TLS v1.0 parameter is set to None. This parameter must be set to **All** for clusters in CC mode. The **Disable TLS v1.1** setting can have a value of **None** for clusters in Common Criteria (CC) mode, so TLS v1.1 is enabled on all devices. This is the default setting for clusters upgrading to Policy Manager 6.10 or later releases.

ClearPass 6.11 and 6.12 are the last Long Support Release (LSR) and Short Support Release (SSR) versions to support TLS v1.0 or TLS v1.1 in any operating mode (they are already disabled in FIPS and CC modes). Customers can use Insight reports to identify clients that might be using these TLS versions in RADIUS-related communications. After ClearPass 6.12, the system-wide cryptographic policy will be set to DEFAULT in non-FIPS mode. Additionally, the following cryptographic algorithms and ciphers will be deprecated after ClearPass 6.12:

- SHA-1 signature algorithm in digital signatures and certificates
- 3DES, RC4 encryption algorithms
- RSA and Diffie-Hellman params less than 2048 bits
- DSA signature algorithm
- CBC mode ciphers in SSH
- IKEv1 IPsec protocol

To modify these settings navigate to **Administration > Server Manager > Server Configuration** and select **Cluster-Wide Parameters**.

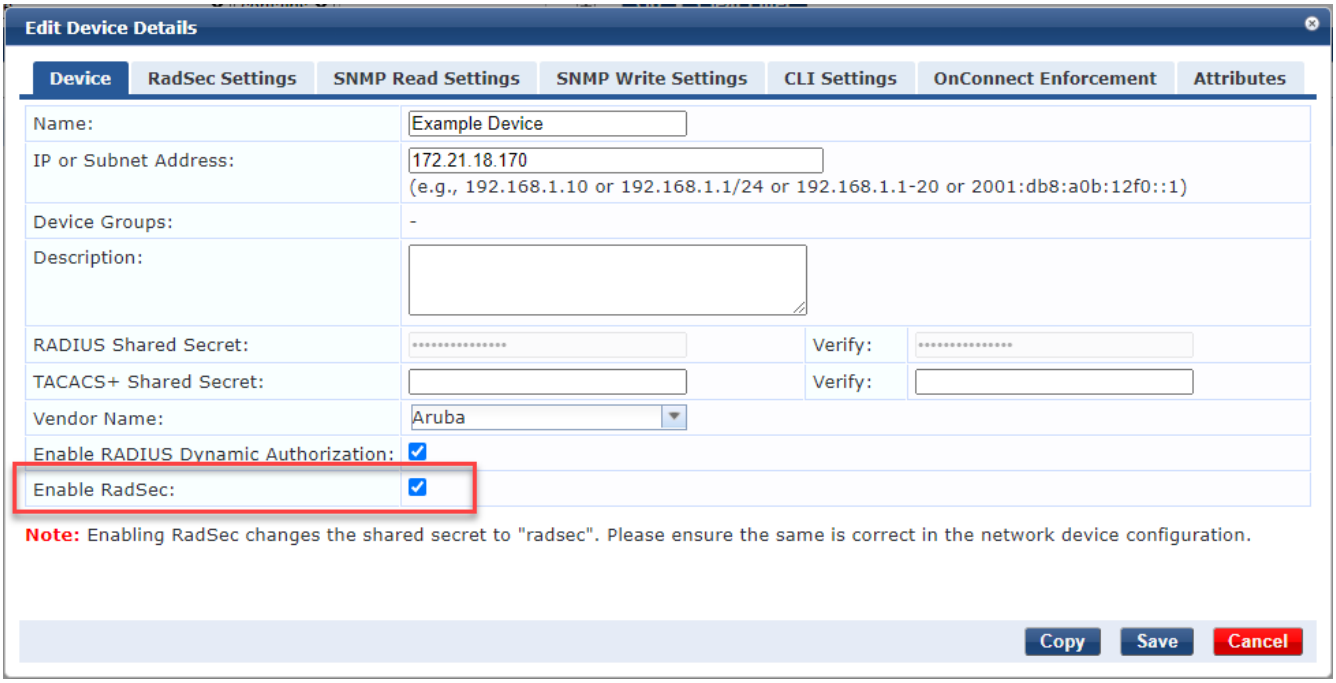
Figure 1. Disabling TLSv1.0 and TLSv1.1



Enabling RadSec

Existing network devices that appear in the Devices list at **Configuration > Network > Devices** can be enabled with the RadSec option to securely transport RADIUS data over TLS between the NAD and the ClearPass appliance. When you enable this option, ClearPass will stop accepting RADIUS traffic from that device. Be aware, you must enable RadSec on the device as well, or ClearPass will not receive traffic from that host.

Figure 1. Enabling RadSec



Locking Down Administrative Access

A primary intrusion attack vector used against network devices is the device's administrative console. The ClearPass Admin Web UI and command line interface (CLI) should be made as secure as possible to minimize the chances of a successful compromise.

Subtopics

Management Access Control
Allowed SSH Modes
Restrict Concurrent Admin Logins
Content Security Policy (CSP) Option
IPsec Tunnel Support
Application Access Control
Smart card and Certificate BasedLogin
Restricting Administrator Privileges
Password Policy
Password Policy Enforcement
Centralized Authentication and Authorization

Management Access Control

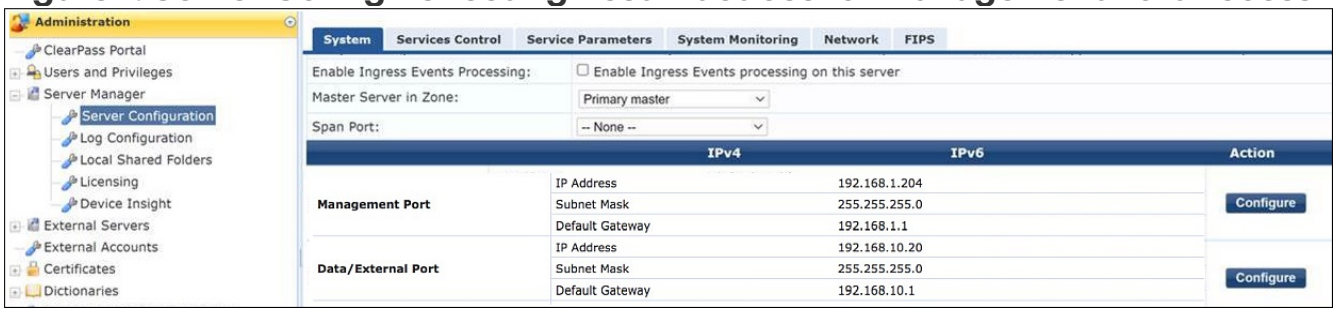
Aruba recommends permitting administrative access only from authorized end systems. If the network design permits, separate management and user-facing services (data traffic) by creating a dedicated management network and attaching the ClearPass management interface to that network.

In this type of deployment, the ClearPass management interface provides WebUI and CLI access for server and cluster administration and configuration. The management interface also handles internal cluster (Publisher/Subscriber) communication.

The data interface provides point of contact for all user- facing services including authentication and authorization requests using RADIUS, TACACS+ and Web authentication. Data port security is enhanced by restricting the SSH protocol. SSH is not permitted to the data port and is denied by default internal firewall rules.

To view management port settings for your Policy Manager server, navigate to **Administration > Server manager > Server Configuration**. Select a server from the **Servers** list, then click the **System** tab. A portion of the tab is shown in the figure below.

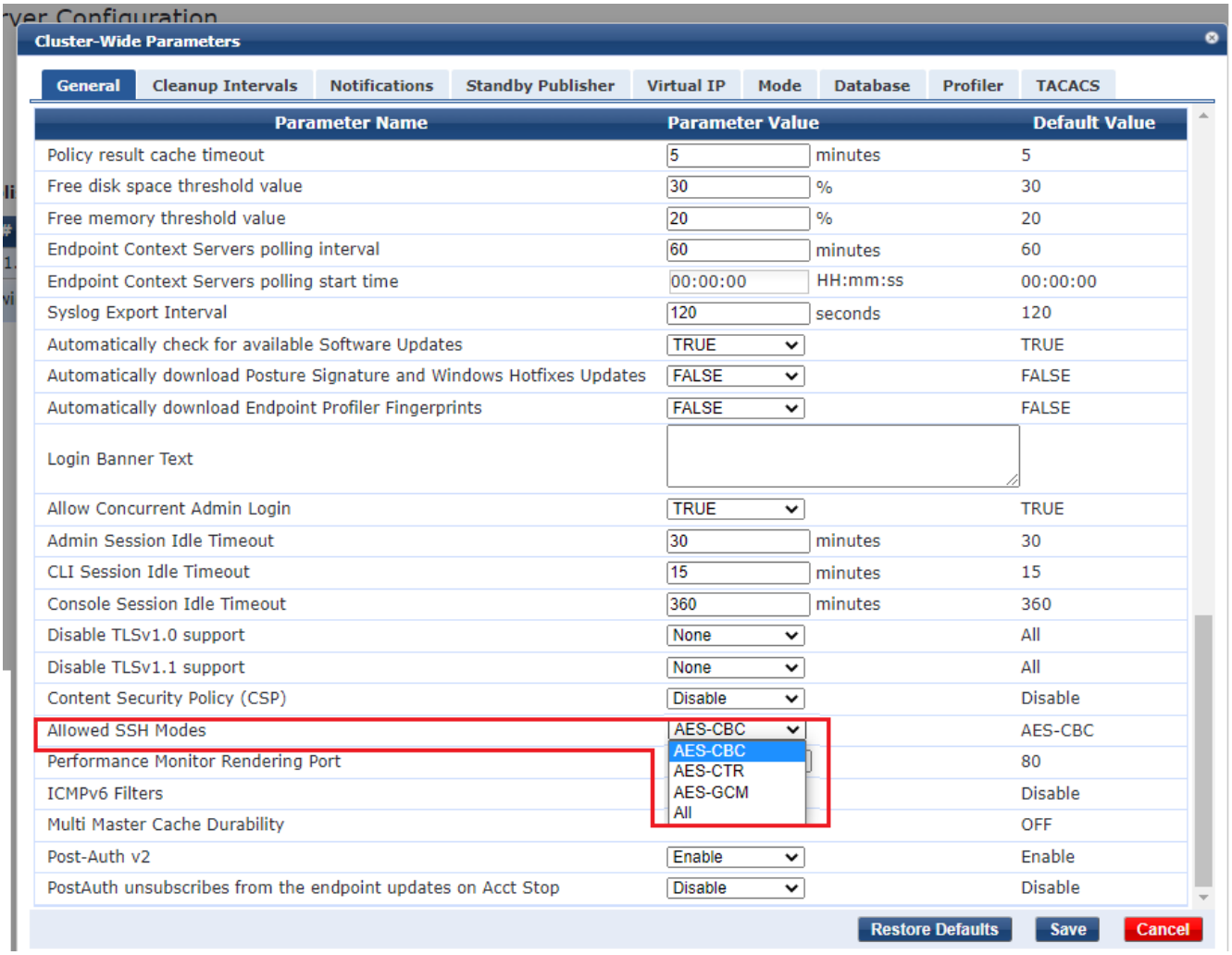
Figure 1. Server Config Reflecting Best Practices for Management Port Access



Allowed SSH Modes

A cluster-wide option to accept SSH connections only on the selected cipher mode is available on the **General** tab at **Administration > Server Manager > Server Configuration > Cluster-Wide Parameter**s. The **Allowed SSH Modes** drop-down list includes options for **AES-CBC** (the default value), **AES-CTR, AES-GCM,** or **All**.

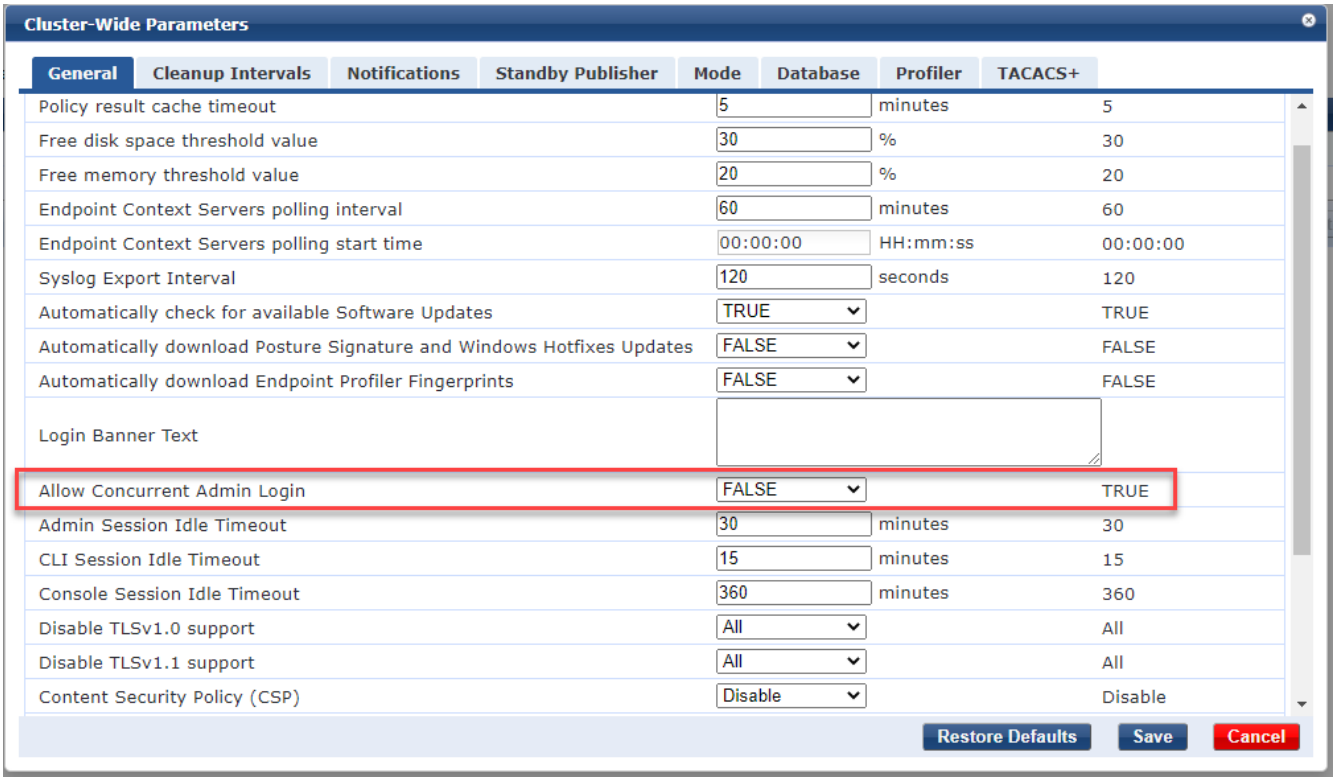
Figure 1. The Allowed SSH Modes Cluster-Wide Parameters



Restrict Concurrent Admin Logins

When the **Allow Concurrent Admin Logins** cluster-wide parameter is set to FALSE, and a new user logs in as an admin user, earlier sessions using the same credentials that are still active on other cluster appliances will automatically be logged out.

Figure 1. The Allow Concurrent Admin Logins Cluster-Wide Parameter



Content Security Policy (CSP) Option

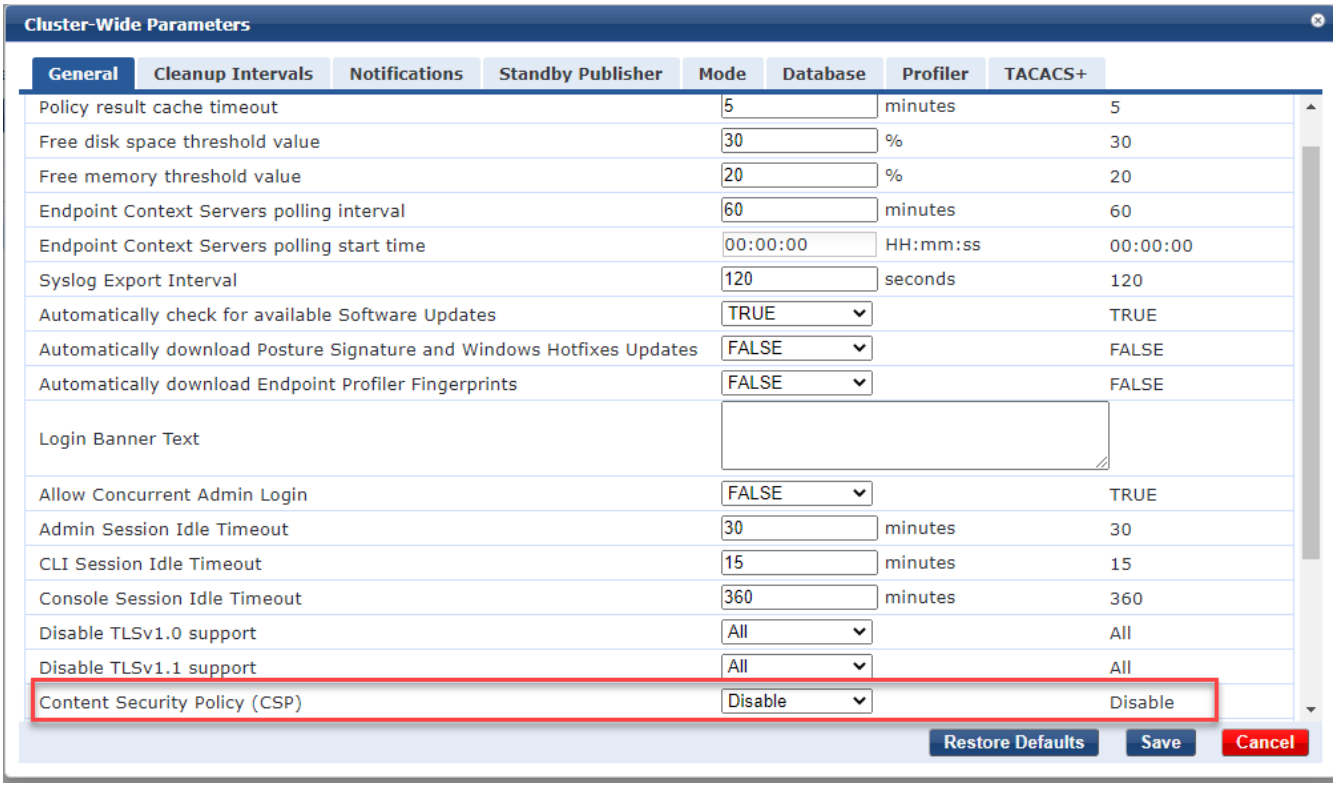
When enabled, the Content Security Policy (CSP) option helps reduce the cross-site scripting (XSS) risks in browsers by declaring which dynamic resources can be loaded via an HTTP Header. This setting is disabled by default.

Administrators should be aware when this parameter is enabled, it can negatively affect any customized HTML code customers might have for skins, captive portals, and self-registration workflows in ClearPass Guest. If the configuration includes customized HTML code that references images, media, scripts, or other resources on servers outside ClearPass, the CSP parameter should not be enabled or a different approach for accessing these resources should be used.

NOTE

Referrer Policy and Permission Policy headers are optional, and are not added to ClearPass. For the Referrer Policy, ClearPass relies on default Browser settings, which is "strict-origin-when-cross-origin." The Permission Policy is not relevant to ClearPass.

Figure 1. The Content Security Policy (CSP) Cluster-Wide Parameter



IPsec Tunnel Support

About this task

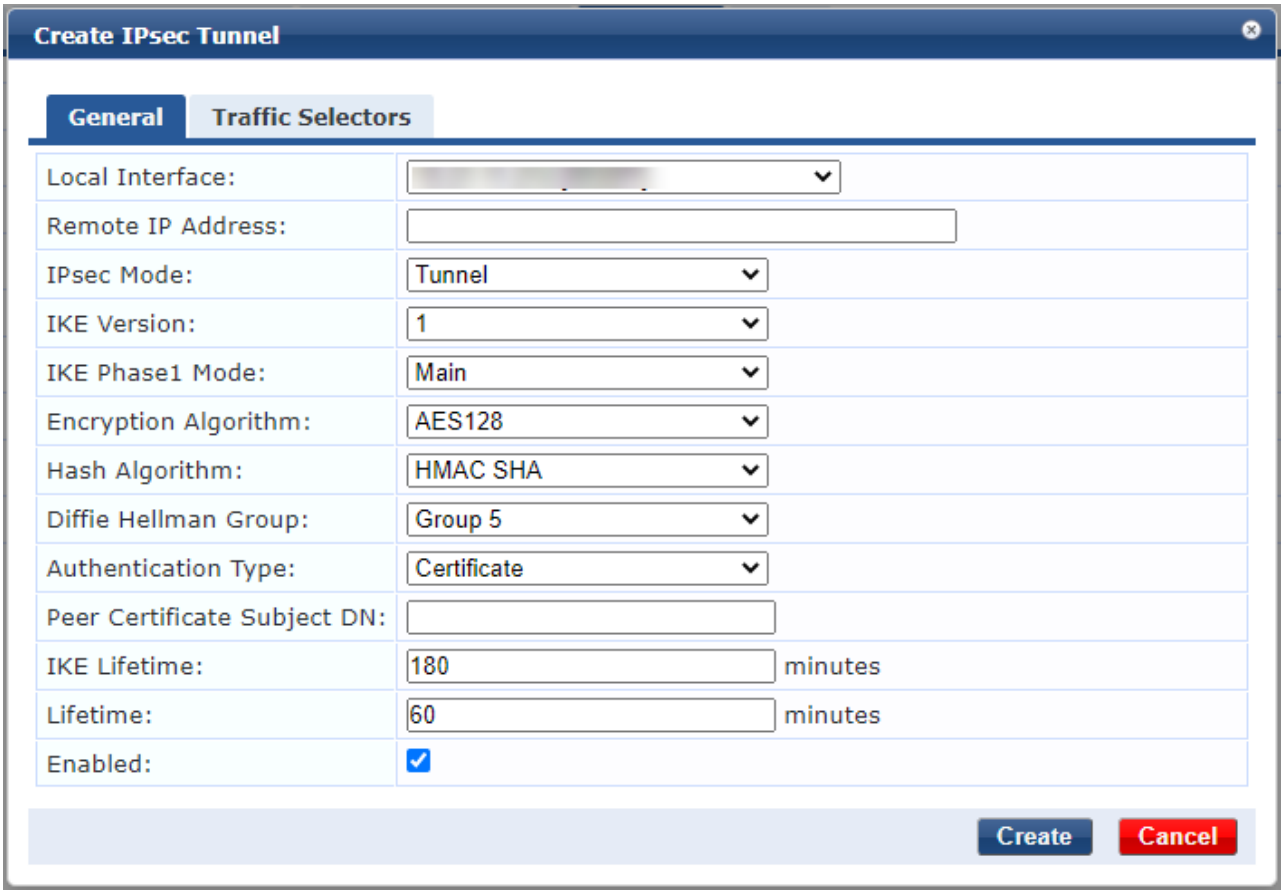
ClearPass supports IPsec tunnels for the management and data interfaces. IPsec provides encrypted tunnels that guarantee the confidentiality of the communications and the identity of the endpoints. This is critical in high security environments or when the communications path crosses a public network.

To create an IPsec tunnel:

Procedure

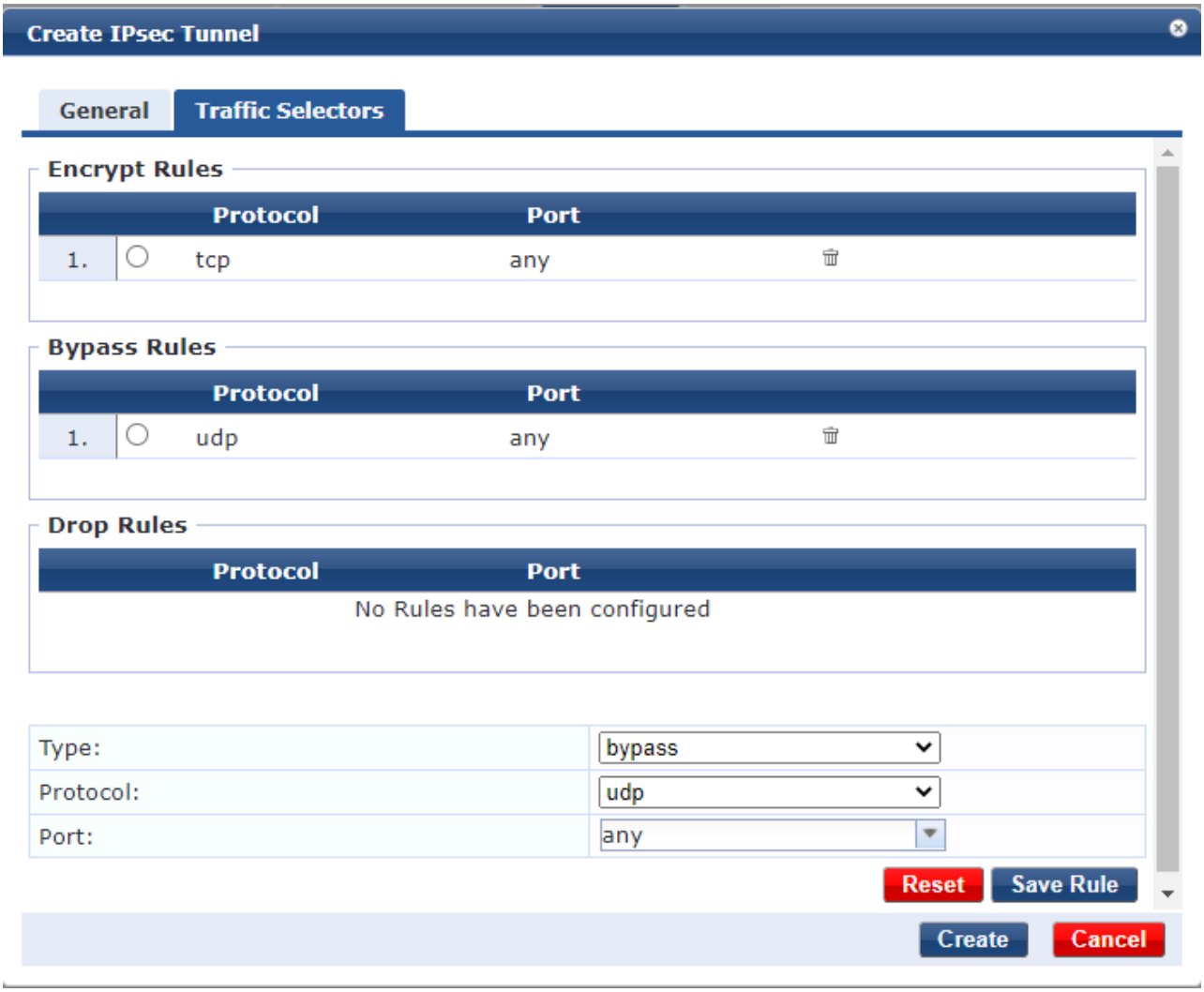
1. Navigate to the **Administration > Server Manager > Server Configuration**.
2. Select the Policy Manager server of interest.
3. From the **Server Configuration** page, select the **Network** tab.
4. Click **Create IPsec Tunnel**.

Figure 1. Creating an IPsec tunnel



Traffic selector rules can be used to control the IPSec tunnel traffic. Selector type options include **Encrypt**, **Drop** and **Bypass**.

Figure 2. Traffic Rule Options for IPsec Tunnels

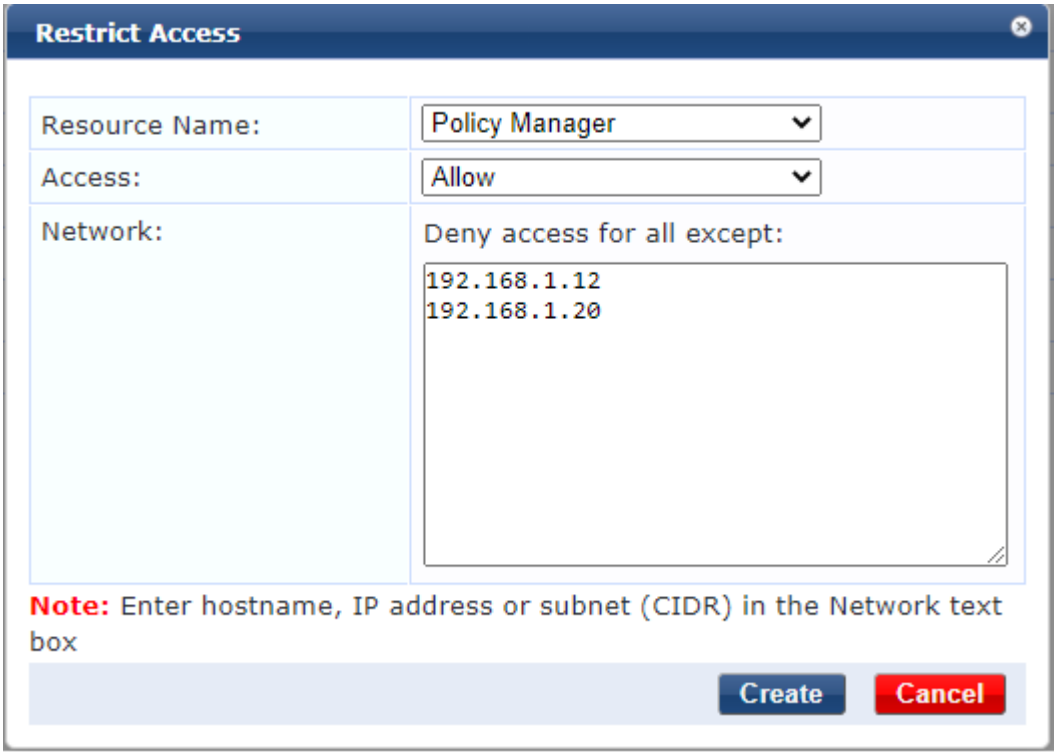


Application Access Control

ClearPass provides application-level restrictions that can define networks / end systems and allow or deny access to specific applications. Applications include; Policy Manager, OnGuard, Graphite, Guest Operator and Insight. ClearPass application restrictions do not extend to captive portal configurations.

To configure these restrictions, navigate to **Administration > Server Manager > Server Configuration**, click on the server, go to the **Network** tab and select the option **Application access control**. In a cluster, restrictions need to be configured on each server.

Figure 1. Restricting Application Access



In this example only the defined IP Addresses, 192.168.1.12 and 192.168.1.20, will be able to access ClearPass Policy Manager.

Smart card and Certificate BasedLogin

ClearPass supports smart card and TLS certificate-based login for all ClearPass applications; ClearPass Policy Manager, ClearPass Guest, ClearPass Onboard and ClearPass Insight.

- The certificate can come from a smart card or certificate store
- The certificate can be mandatory or optional
- The certificate can be in addition to username/password or standalone

  When the user attempts to log in to the ClearPass application they are prompted to select a certificate before moving to the login screen.

  The login process may require both a valid certificate and a valid password.

Restricting Administrator Privileges

Admin users should be assigned privileges appropriate to their job responsibilities. By default, there are seven levels of Administrative privilege. The **Administration > Users and Privileges > Admin Privileges** page describes these administrator levels.

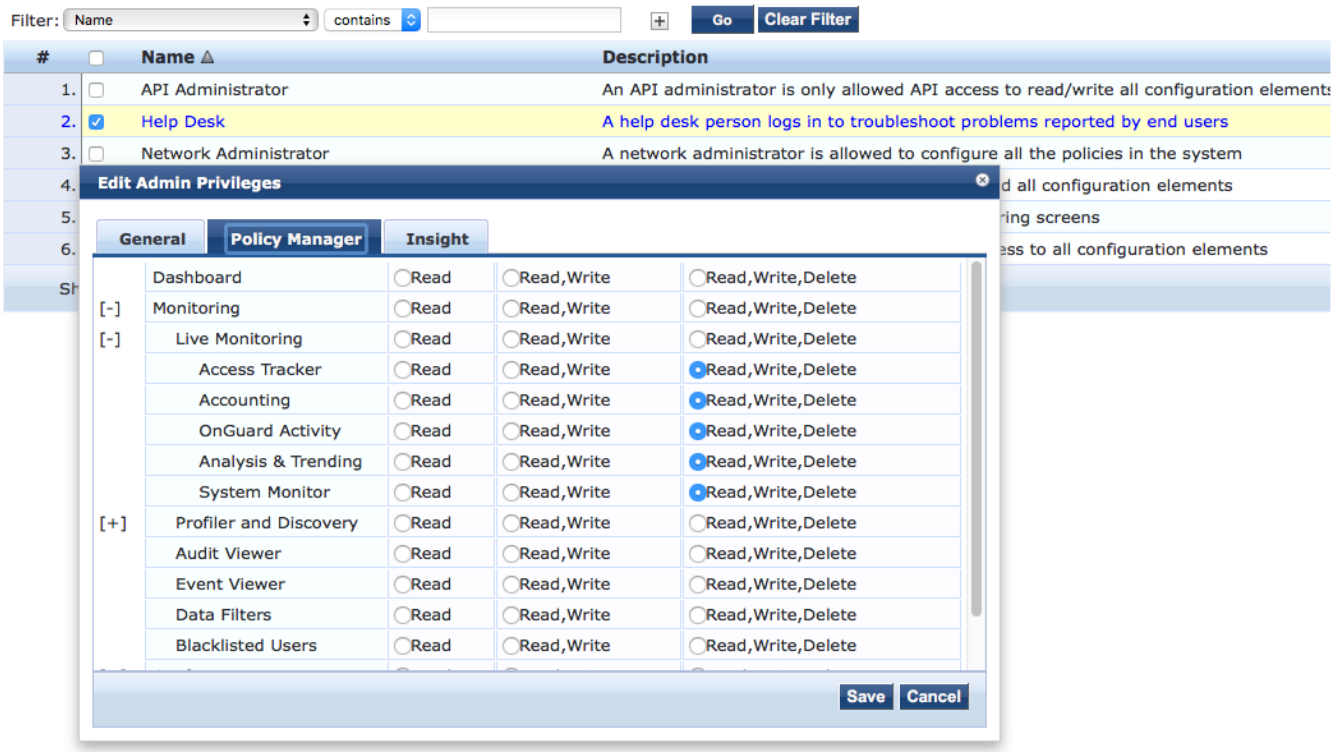
Figure 1. Administrator Privilege Levels

#	Name ▲	Description
1.	API Administrator	An API administrator is only allowed API access to read/write all configuration elements
2.	Aruba User Role Download	Privilege level used for Aruba User Role Download API operations
3.	Help Desk	A help desk person logs in to troubleshoot problems reported by end users
4.	Network Administrator	A network administrator is allowed to configure all the policies in the system
5.	Read-only Administrator	A read-only administrator is only allowed to read all configuration elements
6.	Receptionist	A receptionist is allowed access to main monitoring screens
7.	Super Administrator	A super administrator is allowed read/write access to all configuration elements

For API access, the **API Administrator** privilege level should be used. This privilege level allows programmatic access but prevents the user from logging in through the WebUI.

If necessary, admin access can be further restricted by creating custom privileges. These policies can be tailored to provide fine-grained control of access to ClearPass components and services. Admin privileges can be customized for both Policy Manager and Insight.

Figure 2. Editing Admin Privileges



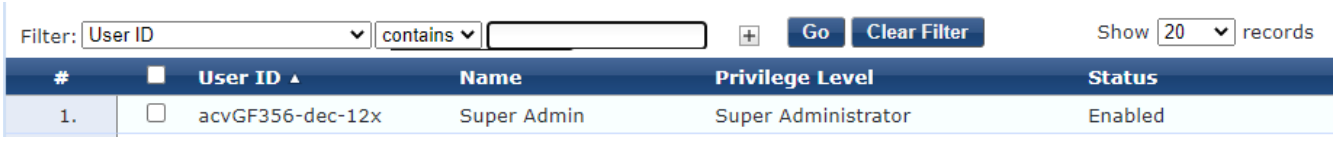
Password Policy

Authentication with a username/password does not provide the strongest security, yet it is extremely common. To strengthen administrator access, both the Admin account

username and password should be changed. Changing the Admin account username means an attacker would have to hack not only the password but also the username, increasing the complexity of the attack. The account name should not be too descriptive or easily guessed.

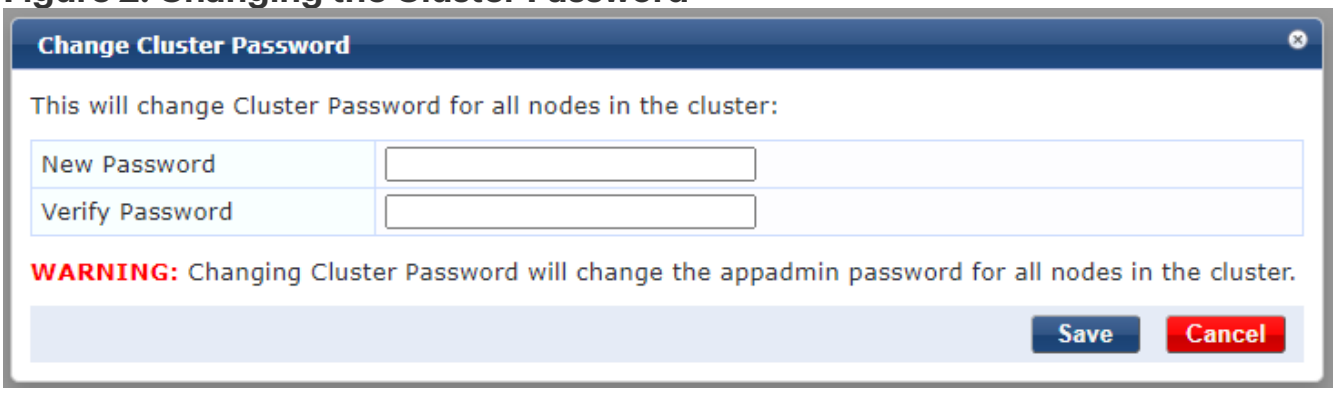
Apply the same strong password policy, mixed case, mixed alphanumeric characters and special characters for both the username and the password. Only the dash (–) and underscore (_) special characters are permitted in the username. To provide audit control, every administrator should have their own account and accounts should never be shared between users.

Figure 1. Admin Users list at Administration > Users and Privileges > Admin Users



The cluster (appadmin) password should also be changed to a strong value. To modify this password, navigate to **Administration > Server Manager > Server Configuration** and click **Change Cluster Password**.

Figure 2. Changing the Cluster Password



Password Policy Enforcement

The password policy enforcement feature llows administrators to set enforcement rules for Admin and Local user account passwords. Separate policies can be set for Admin and Local users, and can include:

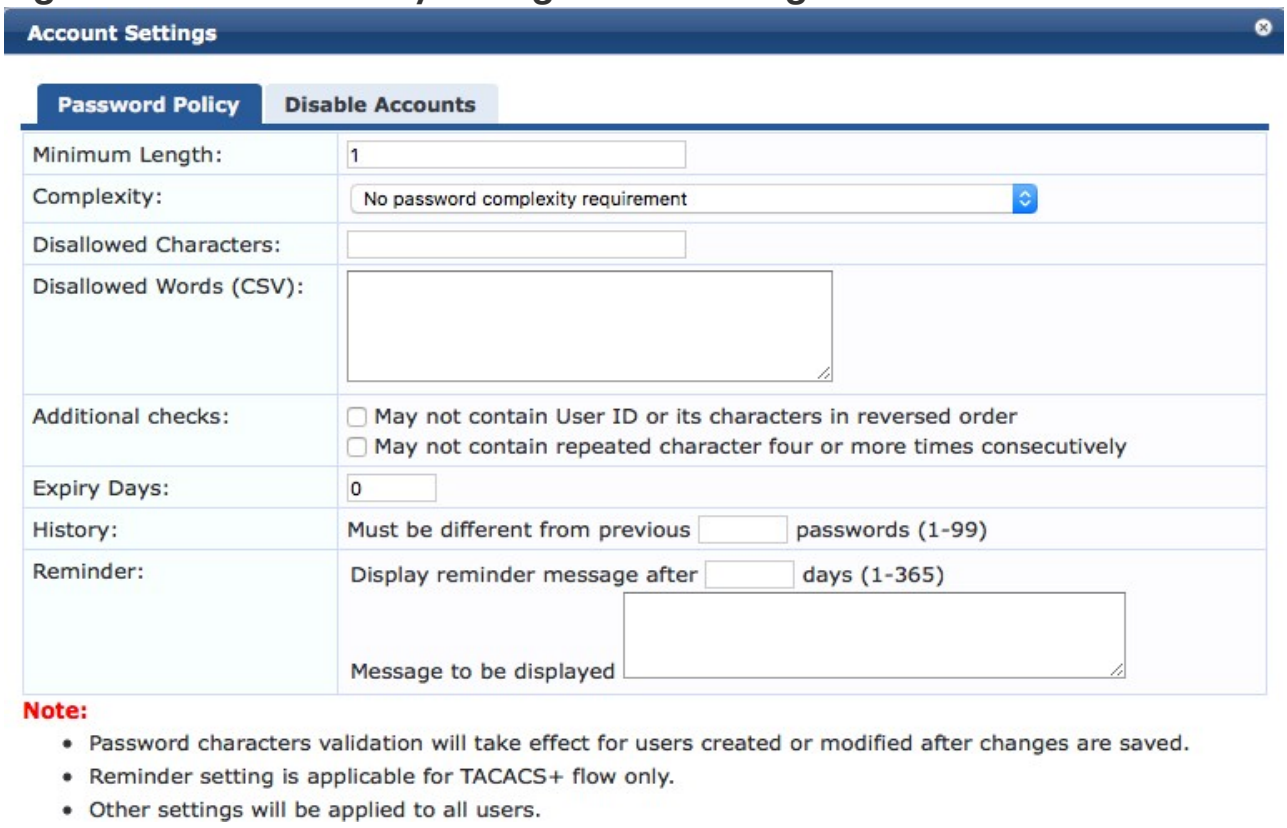
- Minimum password length
- Password complexity
- Additionalhecks
- Passworde xpiration

- History
- Reminder (TACACS+ only)
- Disable settings

To configure this feature, navigate to **Administration > Users and Privileges > Admin Users**, and click **Password Policy**.

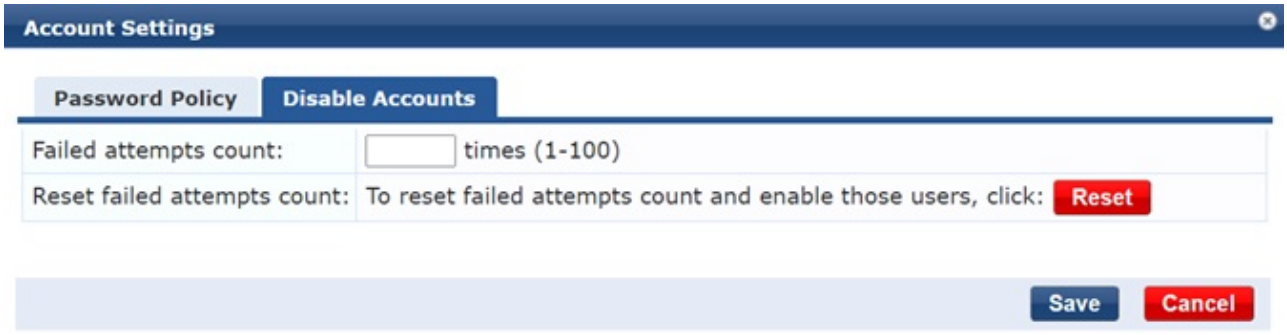
Figure 1. Link to configure Password Policy Enforcement

Figure 2. Password Policy Configuration Settings



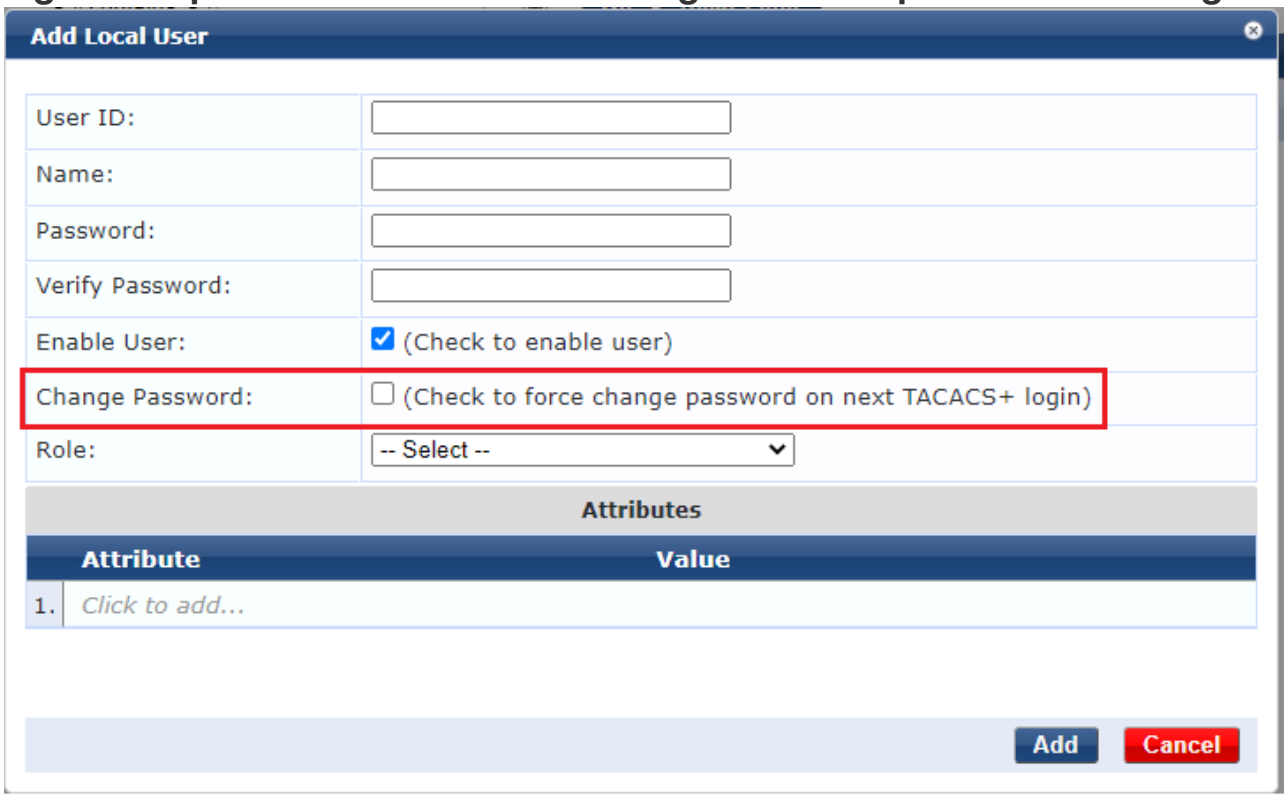
Accounts can be automatically disabled based on failed authentication attempts.

Figure 3. Configure Policy to Disable User Acounts



Additionally, when you access the **Configuration > Identity > Local Users** page to create a new local user, you can enable the **Change Password** option to require a password change upon that user's first login using TACACS+ authentication.

Figure 4. Option Force a Password Change on Subsequent TACACS+ Logins



Centralized Authentication and Authorization

In an organization with multiple administrators, the use of centralized authentication helps to prevent insider attacks. With centralized authentication, ClearPass does not need multiple local administrative accounts. Instead, administrative users log in with credentials that are authenticated remotely by an Active Directory or LDAP server. The remote server should return both authentication and authorization information. After

authenticating the user, attribute information such as group membership or primary security affiliation should be used to assign the correct administrative privilege level.

The following example enforcement policy assigns Super Admin access if the user authenticates successfully and is a member of the Active Directory group CP Admin. Enforcement Policies for services (such a TACACS+ Enforcement Policy Manager admin network login service) are configured at **Configuration > Services** when you click the **Add** link or select a service in the **Services** table, then click the **Enforcement** tab on the **Add Services** or **Edit Services** page.

Figure 1. Policy Assigning Super Admin Access

Enhanced Security for Admin and Local User Accounts

Subtopics

Store hash for Admin and Local User Passwords
Session Timeouts
Enable Public Key Authentication
DoS Attack Prevention

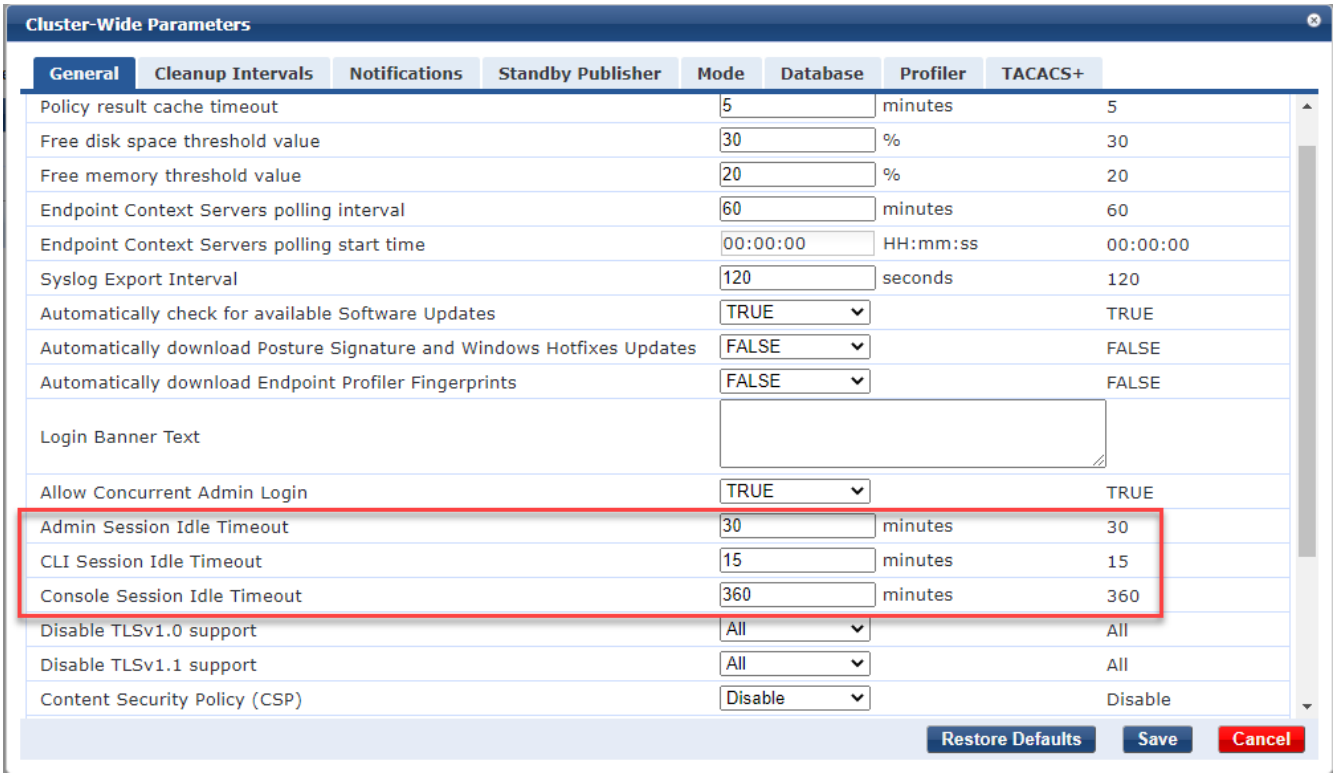
Store hash for Admin and Local User Passwords

Admin and local user passwords are stored in PBKDF2_SHA1-based password hashes. ClearPass provides a global setting to optionally also store the NTLM hash of the password. This is required for MSCHAPv2-based authentications against the local database. The Guest user password fields in the database are individually encrypted and the database itself is stored encrypted (where the data is at reset encryption).

Session Timeouts

Optionally eliminate session timeouts to terminate stale sessions. To modify these settings, navigate to **Administration > Server Manager > Server Configuration**, and select **Cluster-Wide Parameters.**

Figure 1. Configuring Session Timeout settings on the Cluster-Wide Parameters Page.



Subtopics

Admin Session Timeout
CLI Session Timeout (SSH)
Console Session Timeout

Admin Session Timeout

The cluster-wide **Admin Session Idle Timeout** setting allows administrators to configure the maximum idle time permitted for admin access, This admin timeout limit applies to Policy Manger, Guest, and Insight. The default is 30 minutes.

WARNING

There are several monitoring screens in the WebUI (including the Dashboard, Access Tracker, and OnGuard Activity pages) that automatically auto refresh every two minutes. If there is constant activity on these pages, the WebUI session will never time out. Administrators should not to leave the Web browser open on these pages for an extended period. Some portions of the WebUI have an **Auto Refresh** option that can be enabled or disabled, (such as the **Monitoring > Profiler and Network Scan > Network Scan** page). Consider disabling the **Auto Refresh** option on these pages to allow the session timeout period to take effect.

CLI Session Timeout (SSH)

The cluster-wide **CLI Session Idle Timeout** setting allows administrators to control how long a CLI session remains idle before it automatically terminates. If this parameter is changed, the changes take effect when the client opens a new CLI session. Any active CLI sessions will continue to use the old timeout value, as CLI sessions have to be disconnected and reconnected for the new changes to take effect.

Console Session Timeout

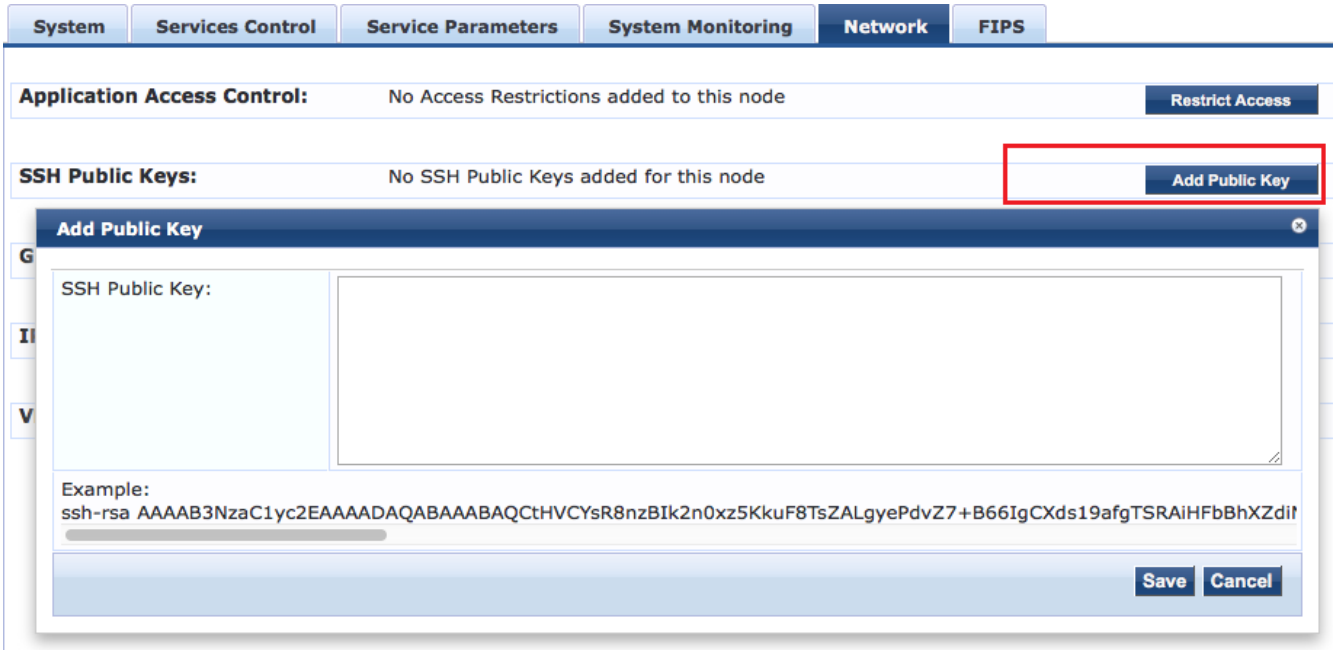
The cluster-wide parameter **Console Session Idle Timeout** allows administrators to control how long a console session may be idle before it is automatically terminated.

> **NOTE**
> Since background processes are not counted as a part of active session, setting a short console timeout value may lead to a console session automatically logging out during a system upgrade.

Enable Public Key Authentication

ClearPass supports public key-based SSH logins on a per-appliance basis. The **SSH Public Keys** option is available within the **Network** tab at **Administration > Server Manager > Server Configuration**.

Figure 1. Adding an SSH Public Key



DoS Attack Prevention

About this task

A Denial-of-Service (DoS) attack is a malicious attempt to shut down a machine or network, making it inaccessible to intended users. DoS attacks flood the target device with traffic, or send the target specific information to trigger a crash.

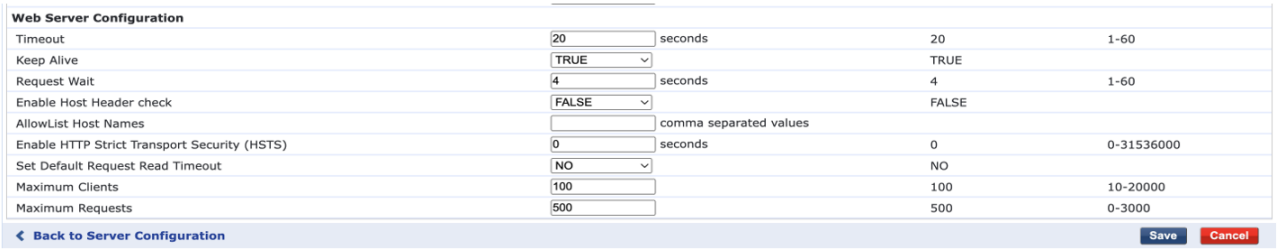
ClearPass includes several mechanisms to mitigate Denial-of-Service (DoS) attacks and ensure service disruptions are promptly remedied before network traffic is compromised.

In particular, setting the **Maximum Clients**, **Maximum Requests** and **Set Default Request Read Timeout** parameters correctly help deter to DoS attacks.

The configure these ClearPass service parameters:

Procedure

1. Navigate to the **Administration > Server Manager > Server Configuration**, then select the Policy Manager server.
2. From the **Server Configuration** page, select the **Service Parameters** tab.
3. From the **Select Service** drop-down, select **Policy Manager system services**.
4. Refer to the **Web Server Configuration** field and configure the following DoS attack prevention specific settings:



Web Server Configuration				
Timeout	20	seconds	20	1-60
Keep Alive	TRUE		TRUE	
Request Wait	4	seconds	4	1-60
Enable Host Header check	FALSE		FALSE	
AllowList Host Names		comma separated values		
Enable HTTP Strict Transport Security (HSTS)	0	seconds	0	0-31536000
Set Default Request Read Timeout	NO		NO	
Maximum Clients	100		100	10-20000
Maximum Requests	500		500	0-3000

Back to Server Configuration | Save | Cancel

Table 1. DoS Attack Prevention Parameters

Service Parameter	Action/Description
Web Server Configuration	
Set Default Request Read Timeout	Set this value to **YES** to enforce the client to have a maximum of 20 seconds to sta rt sending header data before it times ou t. The client must send header data at a transfer rate of 500 bytes per second, fo r a maximum of 40 seconds. Additionally, the client has a maximum of 20 seconds to start sending body data. The client m ust also send message body data at a tra nsfer rate of 500 bytes per second, and may do it for a maximum of 40 seconds. Leave this value set to **NO** (default settin g) to not enforce the Default Request Rea d Timeout.
Maximum Clients	The **Maximum Clients** value represents t he maximum number of concurrent http d processes that can be spawned to han dle requests managed by the Policy Man ager Web server. As an example, if this va lue is set to 500 and there are currently 1 0 active WebUI sessions and 490 API call s being served, then the next request wil l be put into a queue, and will get served once an existing client (httpd process) is freed. If a client is not freed during the ti meout period, the request times out and the request will report a Timeout error.

Service Parameter	Action/Description
	Specify a value from 10 to 20000 for the maximum number of clients allowed. The default value is **100**.
Maximum Requests	If the parameter is enabled, refer to the **Maximum Requests** value to define the maximum number of back - to - back req uests to the Policy Manager Web server t hat can be made on a single connection within the Keep Alive timeout interval, wit hout having to span a new request. Speci fy a number between 0 and 3000 for the maximum number of requests allowed. T he default value is **500**.

NOTE

The Apache web server has some additional DoS-protection requirements (connection limits, etc.).

For information on setting the other parameters that can be configured from the **Administration > Server Manager > Server Configuration** menu, go to:

> **NOTE**
>
> For information on limiting network access to reduce the potential device attack surface, go to: Restricting Administrative Access.

> **NOTE**
>
> For information on restricting ClearPass access to a known subnet/client, go to: Locking Down Administrative Access.

> **NOTE**
>
> For information on disabling an admin user account after a number of failed login attempts, go to: Disabling Admin User Accounts.

Rate Limiting

A Radius server can become unresponsive when impacted by a sudden burst of Radius requests. A sudden burst of traffic can have an adverse impact, since the policy server's processing and authorization times increase when confronted by a burst of requests.

Additionally, with a sudden burst of Radius requests, there could be delays in authorization and policy evaluation, which in-turn could manipulate the availability of Radius server threads.

Enable Rate Limiting

About this task

Beginning with ClearPass 6.9.4, you can enable a Radius server traffic limiting option to help avoid the Radius server being stuck in an unresponsive state. Enabling rate limiting helps ensure the Radius server only processes an administrator defined number of packets within a defined period.

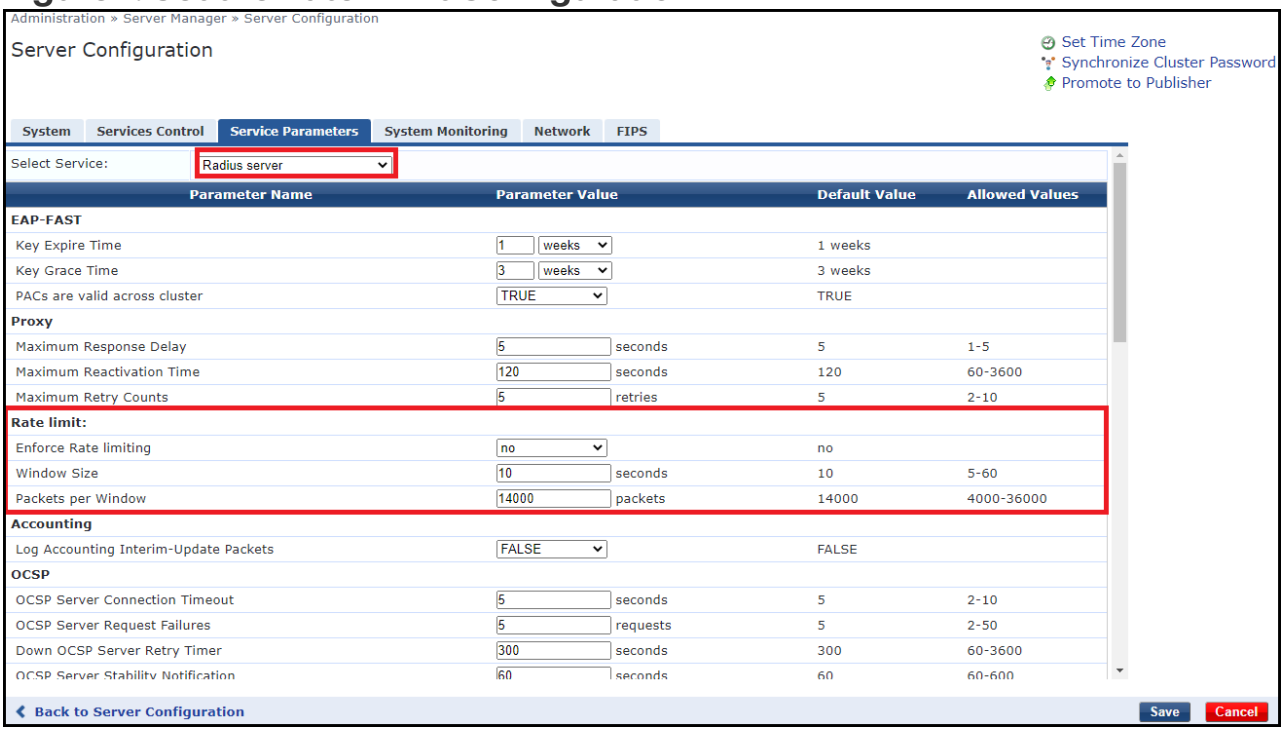
Rate limiting can help ensure the Radius server does not run out of threads, even when confronted with a sudden burst of authentication flow request packets. Rate limiting also helps avoid assigning more load on the policy server, since only a limited number of Radius requests are processed/forwarded to a policy server for additional processing.

To enable and define a rate limiting configuration:

Procedure

1. Navigate to **Administration > Server Manager > Server Configuration**.
2. Select **Radius server** from the **Select Service** drop-down menu.

Figure 1. Set the Rate Limit Configuration



3. Refer to the **Rate Limit** portion of the screen and select **yes** to enable rate limiting on Radius packets.
4. Set the **Window Size** from 5 - 60 seconds. This configures the Radius server to only process Radius packets during the specified time window. The default setting is 10 seconds.
5. Use the **Packets per Window** setting to define the number of Radius packets that can be processed within the defined processing window. The default **Packets per Window** is 14000, and is derived by Authentications per second (200) x Transactions per Authentication (7) x default Window size (10) = 14000.

   Keep in mind:

   - Authentication per second = Number of unique authentication requests generated by NAS/NAD.
   - Transaction per Authentication = Number of incoming Radius packets per authentication. For EAP-PEAP ~ 7 packets (It may differ based on different authentication methods)

- Window size = Number of seconds mentioned in Window size.

So, if the **Packets per Window** are set to 500, and the **Window Size** is the default 10 seconds, then ClearPass only accepts the first 500 Radius packets in the defined 10 second window.

This should help the Radius server's response time should it become negatively impacted by NAS reload/Authentication re-tries due to a ClearPass failure or delay.

NOTE

This is applicable only on incoming Radius packets that are Access-Requests from NAS/NAD/Proxy. The rate limit is not applied on Access-Challenge, Access-Reject, and Access-Accept packets, or CoA-Requests sent from ClearPass.

Monitoring

Subtopics

Excessive Failed Admin Login Attempts
Insight Alert
Last Admin Login Notification – WebUI
Last admin login notification – CLI
Access tracker
Audit trail
Exporting Audit Records
Automated backup recovery
Storing Backups Externally

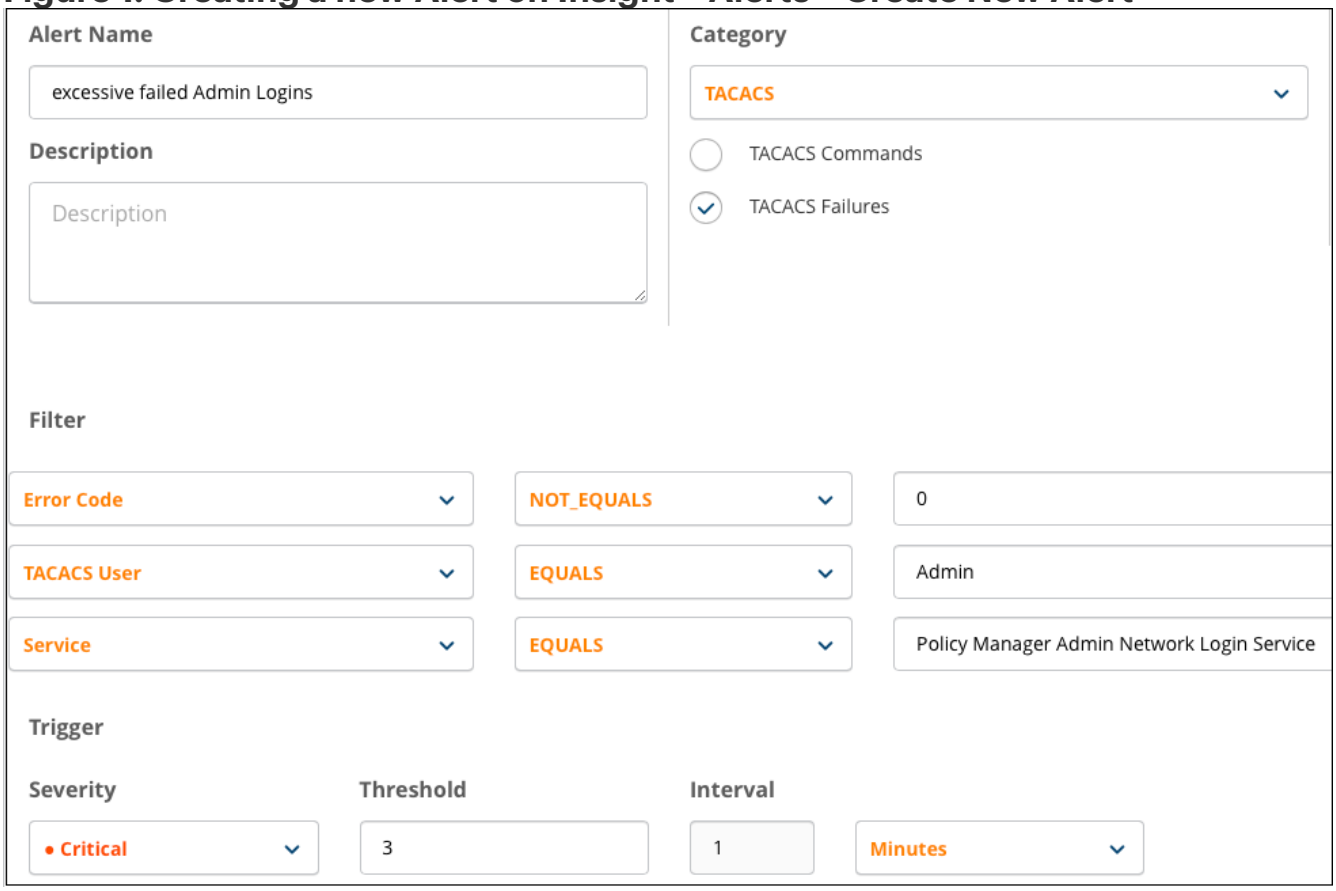
Excessive Failed Admin Login Attempts

One of the most common attack vectors is password guessing. The attacker attempts to gain access to Administrator and privileged accounts by first trying the default Admin password, then trying the most commonly used passwords. Finally the attacker uses a brute force tool that utilizes a large number of potential passwords from an attack dictionary. The indicator of this type of attack is usually a large number of failed authentications in a short period of time to the same account.

Insight Alert

Compliance and local security policies often require logging and alerting for potential brute force attacks on administrator and privileged accounts. The example Insight alert shown below sends an Email and SMS text message when three login failures for the Admin username occur in a one-minute interval. These values should be adjusted to meet local deployment security policy requirements.

Figure 1. Creating a new Alert on Insight > Alerts > Create New Alert



Last Admin Login Notification – WebUI

The admin WebUI shows when the most recent successful login occurred and the number of failed attempts that were made since the most recent successful login. The messages are displayed on the **Dashboard** or **Home** page of:

- Policy Manager
- Guest
- Onboard

- Insight

Figure 1. ClearPass Login Notifications

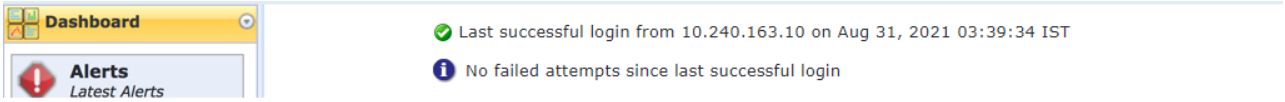
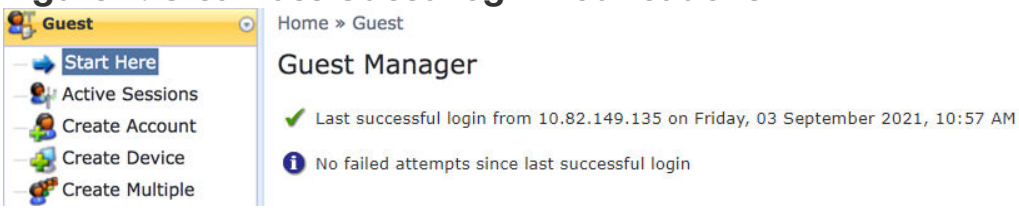


Figure 2. ClearPass Guest Login Notifications



NOTE

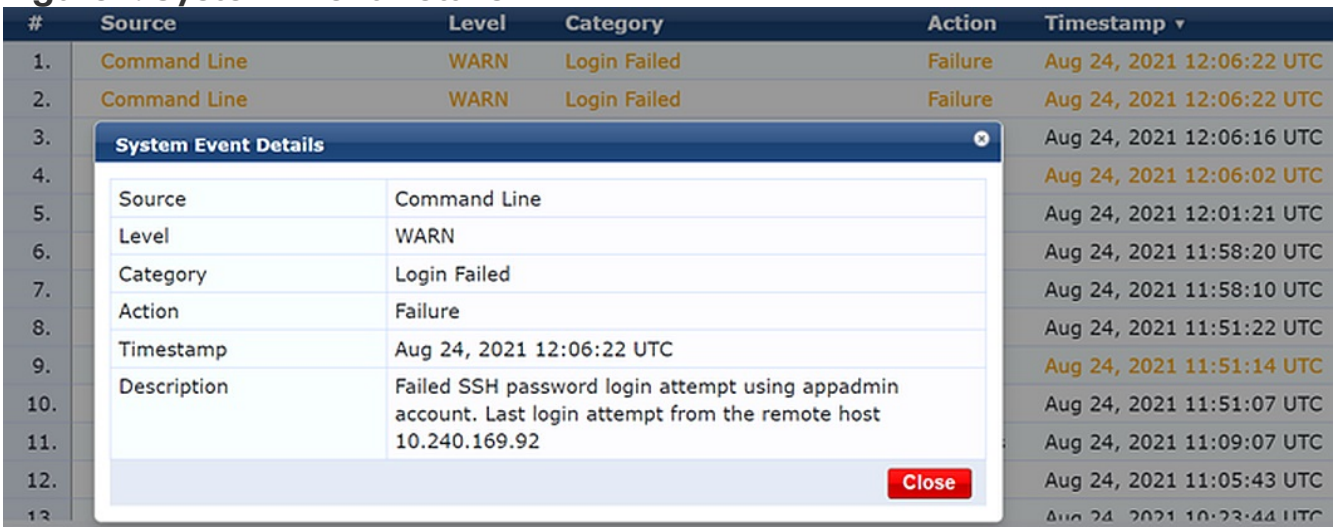
These login notification messages appear when you log into any hese modules directly from the ClearPass Welcome page. If you log into one module then use the WebUI to switch to a different module, that second module will not display login information.

Last admin login notification – CLI

The Console and SSH now show a message with the source and timestamp of the most recent successful login and the number of failed attempts that were made through both SSH and the console since the most recent successful login.

Event details are also displayed in the Event Viewer at **Monitoring > Event Viewer**.

Figure 1. System Event Details

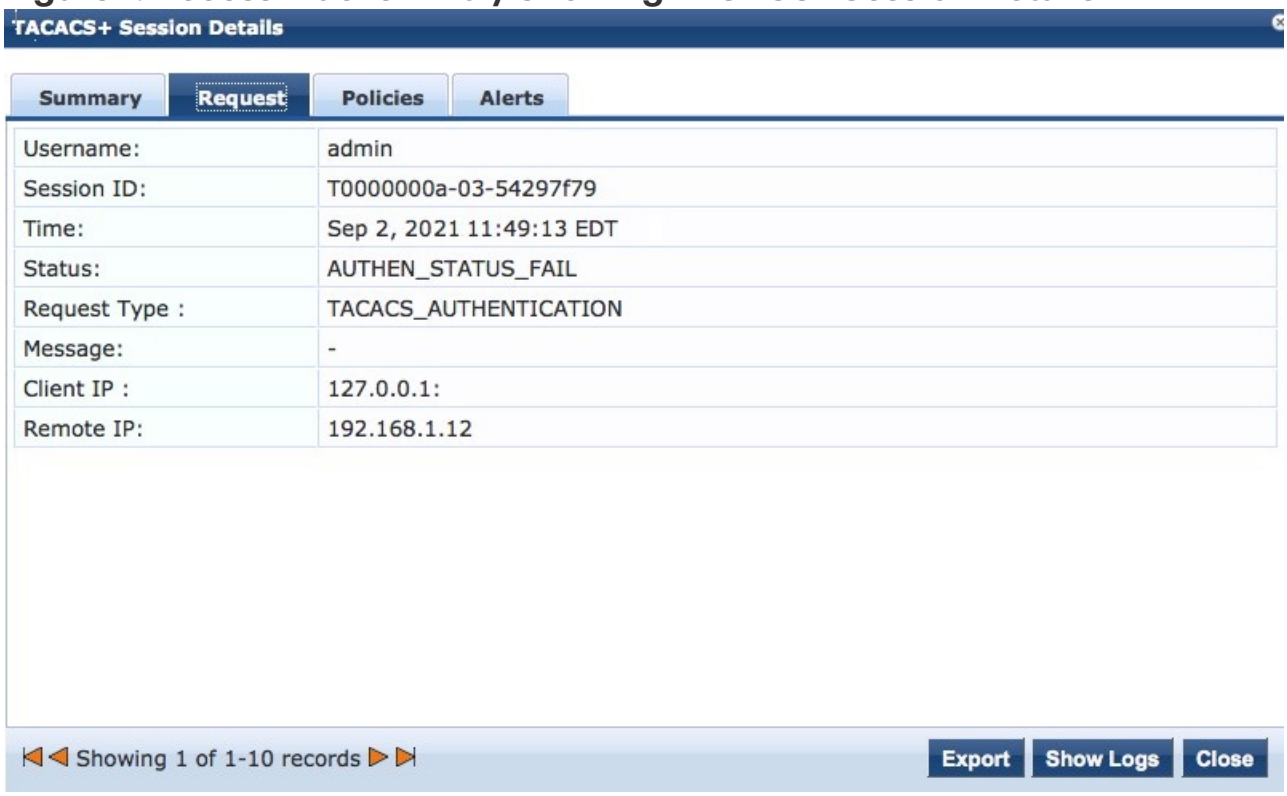


Access tracker

ClearPass will log authentication failures to the Access Tracker. View logged Access Tracker information by navigating to **Live Monitoring > Access Tracker** and then clicking on any entry in the table to view details for that session access activity. An Access Tracker entry includes valuable information that can be used for analysis;

- Username
- Date and Time
- Status
- Client IP (typically 127.0.0.1)
- Remote IP

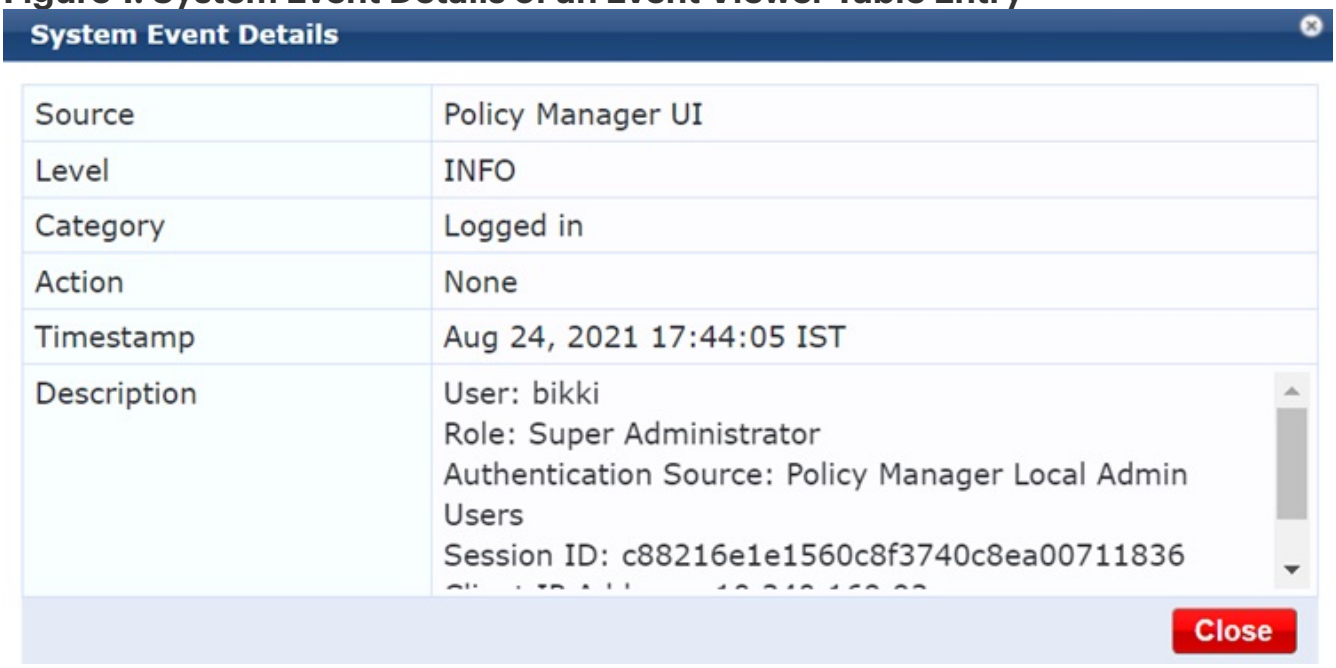
Figure 1. Access Tracker Entry Showing TACACS+ Session Details



Audit trail

An Audit Trail is a sequential record of which users have accessed the Admin UI and what changes they made. The Access Tracker table at **Monitoring > Live Monitoring > Access Tracker**, and the Event Viewer table at **Monitoring > Event Viewer** display each successful and unsuccessful login attempt. The event detail below shows user **bikki** successfully logging into the Admin UI in the role of Super Administrator on Aug 24 at 17:44:05 IST.

Figure 1. System Event Details of an Event Viewer Table Entry

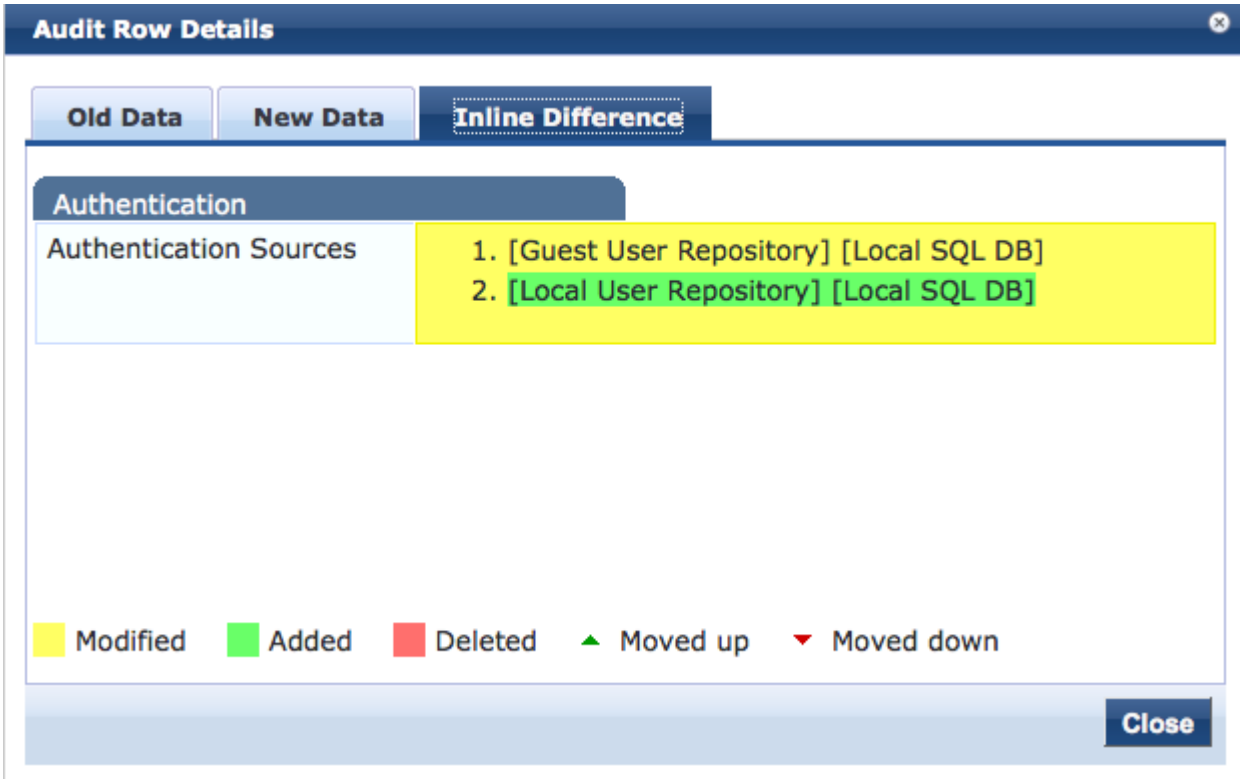


To view the Audit Viewer table, navigate to the **Monitoring > Audit Viewer** page. The entry in the figure below shows user **bikki** modified the lab guest access service at 18:07:42.

Figure 2. Audit Viewer Table Entry

Click on any entry in this table to display a detailed Audit record, including a complete record of the old data, the new data, and as shown below, the inline differences. In this case, user bikki added the local user repository as an authentication source for the lab guest access service.

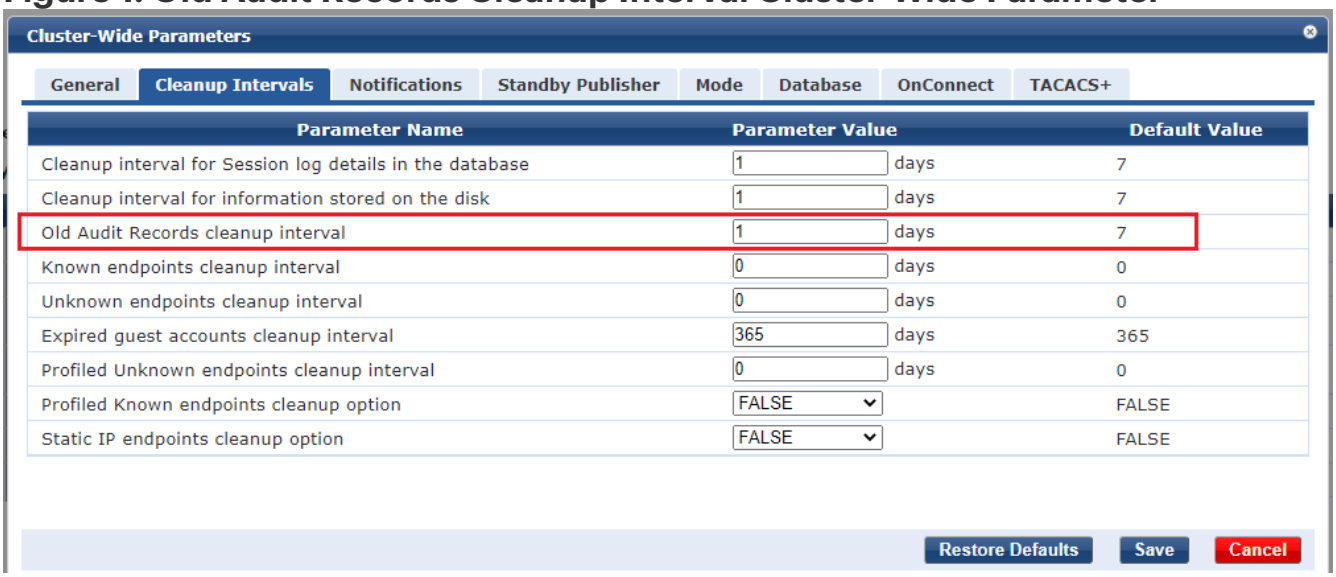
Figure 3. Audit Row Details Showing Authentication Source Changes



Exporting Audit Records

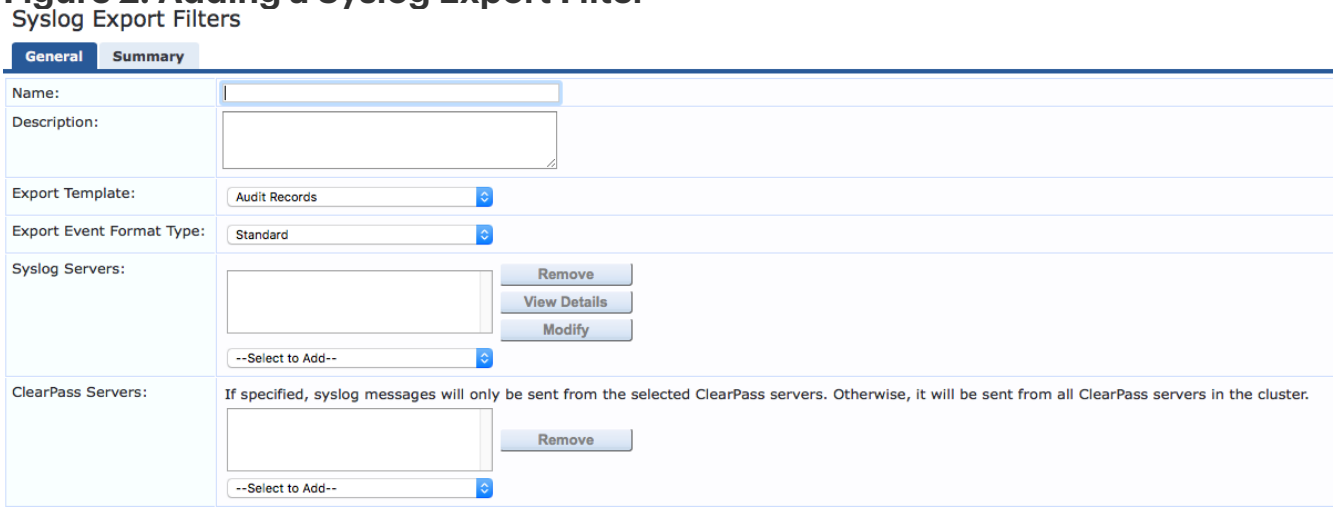
By default, the ClearPass server only retains Audit records for seven days. This value is configurable at **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters**.

Figure 1. Old Audit Records Cleanup Interval Cluster-Wide Parameter



If a longer audit trail is required, audit records can be sent to an external syslog target for security and long-term storage. Syslog export filters are configurable at **Administration > External Servers > Syslog Export Filters > Add**.

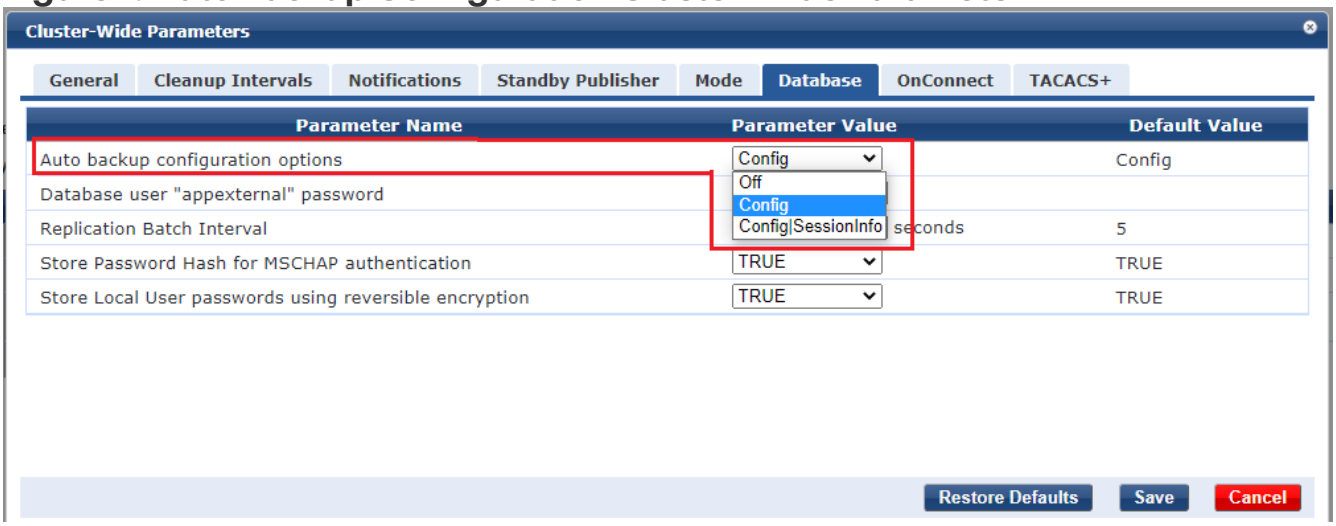
Figure 2. Adding a Syslog Export Filter



Automated backup recovery

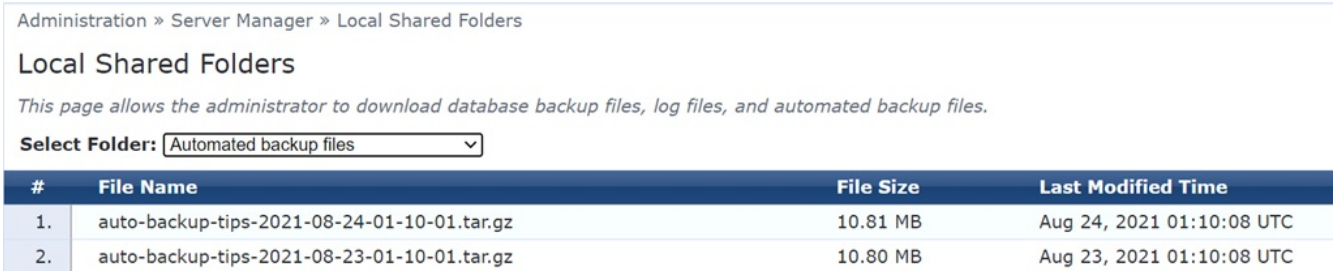
By default, ClearPass automatically creates configuration backups at 1:00 am every night. The **Auto backup configuration options** cluster-wide parameter allows you to backup configuration seettings or configuration and session information.

Figure 1. Auto Backup Configuration Cluster-Wide Parameter



The backup files can be downloaded from **Administration > Server manager > Local Shared Folders.**

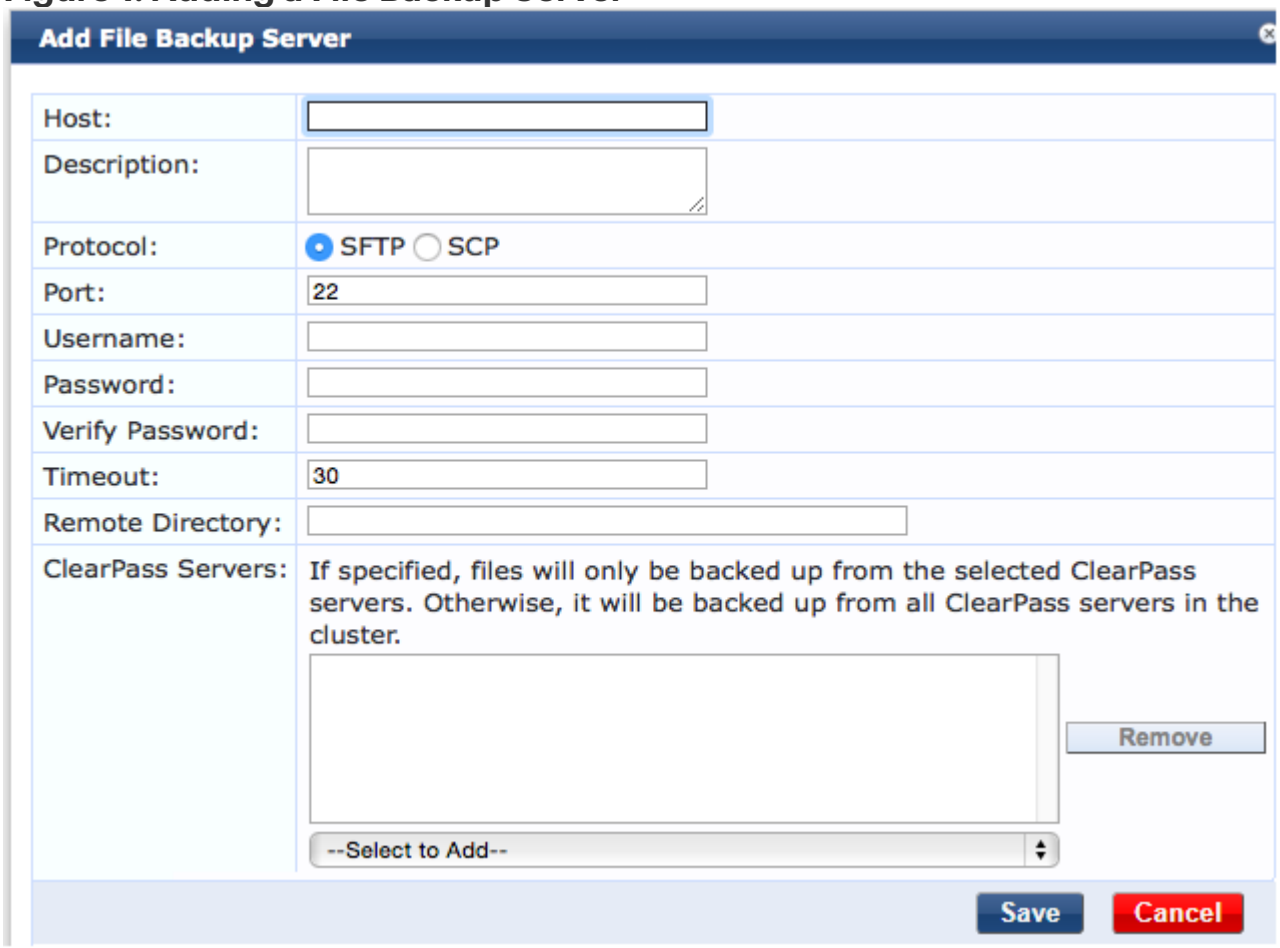
Figure 2. Backup Files in the Automated Backup Files folder



Storing Backups Externally

To guarantee business continuity and disaster recovery, Aruba recommends backup files be stored on an external server. ClearPass can be configured to automatically push backup files to a remote server defined at **Administration > External Servers > File Backup servers**.

Figure 1. Adding a File Backup Server



Locking Down User Access

User access lockdown restricts permissions to protect systems and data from unauthorized use.

User Roles and Firewall Policies

Aruba recommends the deployment of role-based access controls for all wired and wireless users. Rather than granting one-size-fits-all access to the network once they have authenticated, users are only granted access appropriate for that user's role in the organization. For example, only ClearPass administrators should be assigned roles that permit access to the ClearPass management Interface. Roles are enforced on the Network Access Device so the wireless controller or switch would apply a role to ClearPass administrators that permits access to the ClearPass management interface while all other users would be assigned roles that deny access to the management interface.

To enable devices to communicate over a network firewall, ensure communication between the (publisher/servers) is allowed to that URL over port 443. For more information on the ports utilized by ClearPass, go to: Ports.

Remote Support

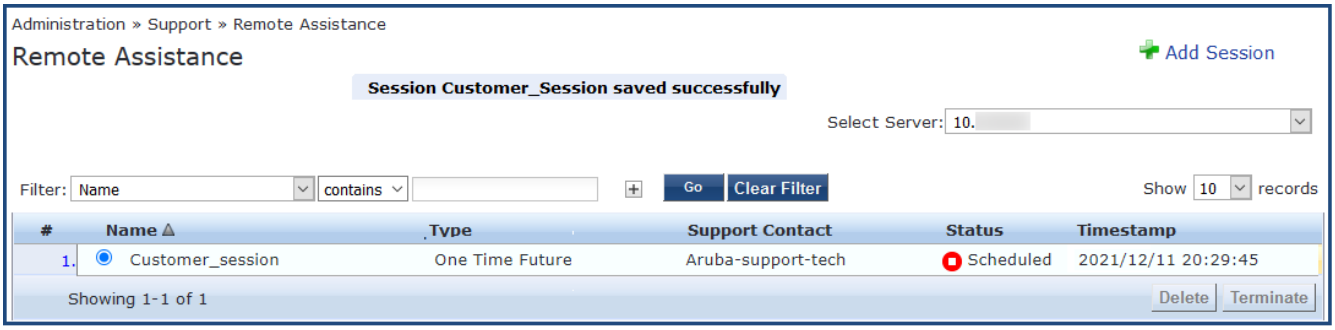
ClearPass includes two remote support interfaces that allow Aruba customer support engineers to access the ClearPass server to help resolve customer issues.

Remote Assistance

The Remote Assistance feature allows an Aruba Networks support engineer to log in remotely using SSH to the ClearPass Policy Manager server, view the Administration WebUI to debug any issues the customer is facing, or perform proactive monitoring of the server. This feature can be enabled and configured at **Administration > Support > Remote Assistance**.

The Remote Assistance account can specify time as a one time session lasting anywhere from for 1 to 24 hours, or as weekly or monthly occurrences. The Administrator should make sure that remote sessions are terminated once the debugging session is done.

Figure 1. Defining a Remote Session



Support Shell

The ClearPass support shell provides technical support engineers CLI root access to the ClearPass server. The customer uses the CLI to generate a one-time key for the support engineer. This key provides the engineer root-level access.

Use the **gen-support-key** command to generate the support key for the system.

`Syntax`

Syntax

```
system gen-support-key
```

Example

The following example generates the support key for the system:

```
[appadmin]# system gen-support-key system gen-support-key
Support key='01U2FsdGVkX1+/WS9jZKQajRyzXhM8mF6zAKrzxrHvaM='
```

The Aruba support account is valid for 24 hours by default. Aruba recommends administrators deactivate this account as soon as the debugging session is over.

WebUI and CLI Accounts

ClearPass uses several specialized accounts. Aruba recommends changing passwords from their default settings as soon as possible.

Subtopics

WebUI Admin User
CLI Appadmin User
Database Appexternal User
API Apiadmin User
Internal User Accounts

WebUI Admin User

The **admin** user has full access to the CPPM WebUI. The WebUI admin user also has full access to the ClearPass API. Navigate to **Administrator > Users and Privileges > Admin Users** to modify the WebUI admin user password.

CLI Appadmin User

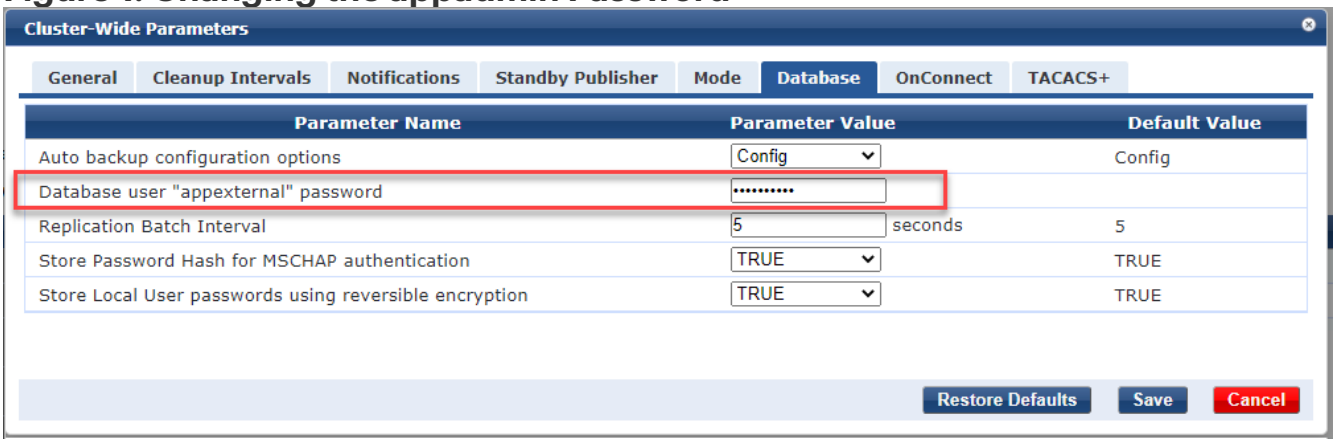
The **appadmin** user can access the CPPM command-line interface to run CLI commands and update their values. The appadmin password can be changed from the **Change Cluster Password** link on the **Administration > Server manager > Server Configuration** page.

Database Appexternal User

This user has read-only access to the **tipsLogDb** database, including system events, session logs, RADIUS accounting, and alerts, the **insightdb** database, which contains information Insight uses to generate reports, and some tables in the **tipsdb** database. The password can be changed using the **Database user "appexternal" password** cluster-wide parameter at **Administration > Server Manager > Server Configuration > Cluster-Wide Parameters**.

Database connections are made over TCP port 5432 to the Policy Manager management port. Remote database connections are allowed on the Policy Manager data port. Ensure port 5432 is open when attempting to remotely connect to the Policy Manager database. For more information about ClearPass port utilization, go to: Ports.

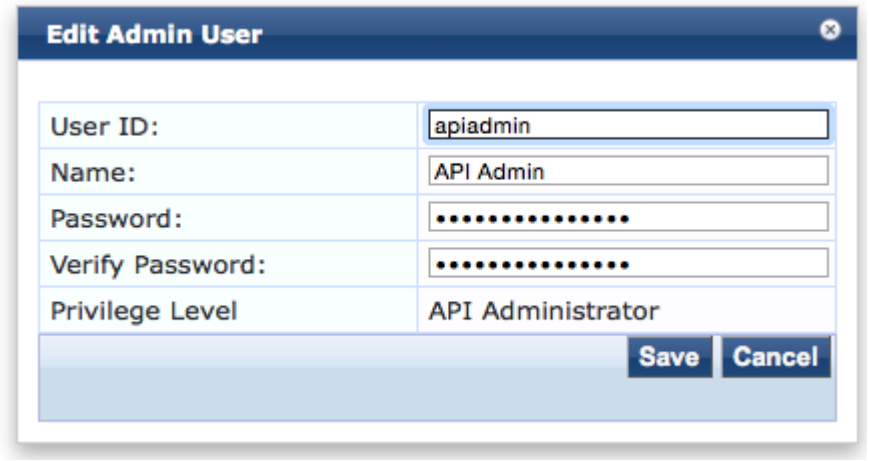
Figure 1. Changing the appadmin Password



API Apiadmin User

This user has read and write access to APIs only. Navigate to **Administrator > Users and Privileges > Admin Users** to modify the apiadmin password.

Figure 1. Changing the apiadmin User Password



Internal User Accounts

The following accounts are for ARUBA INTERNAL USE ONLY and are used for ClearPass troubleshooting.

Subtopics

ArubaSupport
AppSuperUser
AppUser

ArubaSupport

This user account provides full shell access to the ClearPass CLI. This is strictly for ARUBA INTERNAL purposes. The **system gen-support-key** command resets the support password and provides a token from which Aruba customer support can recover the password and login. Running the command again will reset the support password. There is also a periodic scheduled job that resets this password everyday. The Remote Assistance feature, which is customer triggered, builds on this to provide remote access capability to customer support.

AppSuperUser

Full access to the postgres db on ClearPass after logging in as ArubaSupport. This is strictly for ARUBA INTERNAL purposes.

AppUser

This user has read-only access to postgresql after logging in as ArubaSupport. This is strictly for ARUBA INTERNAL purposes.

NOTE

The appuser and appsuperuser accounts are defined as critical internal accounts that do not permit remote logins (ssh or console).

For More Information

The best source of information on Aruba products, outside of official documentation, is the Airheads Social community. For security-related discussions, please visit the "Security" forum at **http://community.arubanetworks.com/.**